

PERRY'S CHEMICAL ENGINEERS' HANDBOOK

8TH EDITION



DON W. GREEN
ROBERT H. PERRY

**Perry's
Chemical
Engineers'
Handbook**

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PERRY'S CHEMICAL ENGINEERS' HANDBOOK

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under the editorial direction of**

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Preface to the Eighth Edition

Perry's has been an important source of information related to the fundamentals and practice of chemical engineering since it was first published in 1934, with John H. Perry both the initiator and editor. Several chemical engineers, serving as editor- or coeditor-in-chief, have guided the preparation of the different editions over the years. These include John H. Perry (first to third editions), Robert H. Perry (fourth to sixth editions), Cecil H. Chilton (fourth and fifth editions), Sidney D. Kirkpatrick (fourth edition), Don W. Green (sixth to eighth editions) and James O. Maloney (sixth and seventh editions). Robert H. Perry was also listed as an editor for the seventh edition, and is listed again as an editor for the current edition, although his tragic death occurred during the preparation of the sixth edition. Many of the ideas developed through his leadership during preparation of earlier editions carried over to the seventh and eighth editions. I owe much to the friendship and mentoring of Bob Perry.

The organization of this eighth edition is much the same as for the seventh edition, although content changes are extensive. The first group of sections includes comprehensive tables with units conversions and fundamental constants, physical and chemical data, methods to predict properties, and fundamentals of mathematics most useful to engineers. The second group, comprising the fourth through the ninth sections, covers fundamentals of chemical engineering. The third and largest group of sections deals with processes, such as heat-transfer operations, distillation, gas-liquid processes, chemical reactors, and liquid-liquid processes. The last group covers auxiliary information including waste management, safety and the handling of hazardous materials, energy sources, and materials of construction. All sections have been updated to cover the latest advances in technology related to chemical engineering. As there are a significant number of new section editors, the material in the *Handbook* has been extensively revised. Section 2, which covers physical and chemical data, has been expanded by well over 100 pages to include, among other new information, data from the AIChE Design Institute for Physical Properties.

A large number of section editors and contributors worked on this eighth edition, and these persons and their affiliations are listed as a part of the front material. Many of these authors are Fellows of the AIChE. I would like to recognize two of these colleagues, Dr. Michael M. Abbott and Dr. James Y. Oldshue, who passed away while this work was being prepared. They will be missed. A number of chemical engineering students at the University of Kansas assisted in the preparation of the index. They are Jonathan Ashley, Andrew Becker, Jonathan Bunn, Don Claus, Andrew Duncan, Meghan Easter, Bill Eckman, Justin Ellrich, Mehrdad Hosni, Kaitlyn Kelly, Jennifer Lawrence, Casey Morris, Chris Roatch, Chris Sharpe, Jeremy Steeley, Daniel Theimer, and Nick Willis. In addition, Maxine Younes, Susan Bolton, and my wife Patricia Green provided extensive secretarial assistance.

DON W. GREEN
Editor-in-Chief
University of Kansas

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SECTION 1

Conversion
Factors and
Mathematical
Symbols

PERRY'S CHEMICAL ENGINEERS' HANDBOOK

8TH EDITION



JAMES O. MALONEY

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Conversion Factors and Mathematical Symbols*

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CONVERSION FACTORS

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CONVERSION OF VALUES FROM U.S. CUSTOMARY UNITS TO SI UNITS

* Much of the material was taken from Sec. 1. of the fifth edition. The contribution of Cecil H. Chilton in developing that material is acknowledged.

TABLE 1-1 SI Base and Supplementary Quantities and Units

Quantity or “dimension”	SI unit	SI unit symbol (“abbreviation”); Use roman (upright) type
Base quantity or “dimension”		
length	meter	m
mass	kilogram	kg
time	second	s
electric current	ampere	A
thermodynamic temperature	kelvin	K
amount of substance	mole ^o	mol
luminous intensity	candela	cd
Supplementary quantity or “dimension”		
plane angle	radian	rad
solid angle	steradian	sr

^o When the mole is used, the elementary entities must be specified; they may be atoms, molecules, ions, electrons, other particles, or specified groups of such particles.

TABLE 1-2a Derived Units of SI that Have Special Names

Quantity	Unit	Symbol	Formula
frequency (of a periodic phenomenon)	hertz	Hz	1/s
force	newton	N	(kg·m)/s ²
pressure, stress	pascal	Pa	N/m ²
energy, work, quantity of heat	joule	J	N·m
power, radiant flux	watt	W	J/s
quantity of electricity, electric charge	coulomb	C	A·s
electric potential, potential difference, electromotive force	volt	V	W/A
capacitance	farad	F	C/V
electric resistance	ohm	Ω	V/A
conductance	siemens	S	A/V
magnetic flux	weber	Wb	V·s
magnetic-flux density	tesla	T	Wb/m ²
inductance	henry	H	Wb/A
luminous flux	lumen	lm	cd·sr
illuminance	lux	lx	lm/m ²
activity (of radionuclides)	becquerel	Bq	1/s
absorbed dose	gray	Gy	J/kg

TABLE 1-2b Additional Common Derived Units of SI

Quantity	Unit	Symbol
acceleration	meter per second squared	m/s ²
angular acceleration	radian per second squared	rad/s ²
angular velocity	radian per second	rad/s
area	square meter	m ²
concentration (of amount of substance)	mole per cubic meter	mol/m ³
current density	ampere per square meter	A/m ²
density, mass	kilogram per cubic meter	kg/m ³
electric-charge density	coulomb per cubic meter	C/m ³
electric-field strength	volt per meter	V/m
electric-flux density	coulomb per square meter	C/m ²
energy density	joule per cubic meter	J/m ³
entropy	joule per kelvin	J/K
heat capacity	joule per kelvin	J/K
heat-flux density, irradiance	watt per square meter	W/m ²
luminance	candela per square meter	cd/m ²
magnetic-field strength	ampere per meter	A/m
molar energy	joule per mole	J/mol
molar entropy	joule per mole-kelvin	J/(mol·K)
molar-heat capacity	joule per mole-kelvin	J/(mol·K)
moment of force	newton-meter	N·m
permeability	henry per meter	H/m
permittivity	farad per meter	F/m
radiance	watt per square-meter-steradian	W/(m ² ·sr)
radiant intensity	watt per steradian	W/sr
specific-heat capacity	joule per kilogram-kelvin	J/(kg·K)
specific energy	joule per kilogram	J/kg
specific entropy	joule per kilogram-kelvin	J/(kg·K)
specific volume	cubic meter per kilogram	m ³ /kg
surface tension	newton per meter	N/m
thermal conductivity	watt per meter-kelvin	W/(m·K)
velocity	meter per second	m/s
viscosity, dynamic	pascal-second	Pa·s
viscosity, kinematic	square meter per second	m ² /s
volume	cubic meter	m ³
wave number	1 per meter	1/m

TABLE 1-3 SI Prefixes

Multiplication factor	Prefix	Symbol
1 000 000 000 000 000 000 = 10 ¹⁸	exa	E
1 000 000 000 000 000 = 10 ¹⁵	peta	P
1 000 000 000 000 000 = 10 ¹²	tera	T
1 000 000 000 = 10 ⁹	giga	G
1 000 000 = 10 ⁶	mega	M
1 000 = 10 ³	kilo	k
100 = 10 ²	hecto ^a	h
10 = 10 ¹	deka ^a	da
0.1 = 10 ⁻¹	deci ^a	d
0.01 = 10 ⁻²	centi	c
0.001 = 10 ⁻³	milli	m
0.000 001 = 10 ⁻⁶	micro	μ
0.000 000 001 = 10 ⁻⁹	nano	n
0.000 000 000 001 = 10 ⁻¹²	pico	p
0.000 000 000 000 001 = 10 ⁻¹⁵	femto	f
0.000 000 000 000 000 001 = 10 ⁻¹⁸	atto	a

^a Generally to be avoided.

TABLE 1-4 Conversion Factors: U.S. Customary and Commonly Used Units to SI Units

Quantity	Customary or commonly used unit	SI unit	Alternate SI unit	Conversion factor; multiply customary unit by factor to obtain SI unit	
Space,† time					
Length	naut mi	km		1.852°	E + 00
	mi	km		1.609 344°	E + 00
	chain	m		2.011 68°	E + 01
	link	m		2.011 68°	E − 01
	fathom	m		1.828 8°	E + 00
	yd	m		9.144°	E − 01
	ft	m		3.048°	E − 01
		cm		3.048°	E + 01
	in	mm		2.54°	E + 01
	in	cm		2.54	E + 00
	mil	μm		2.54°	E + 01
Length/length	ft/mi	m/km		1.893 939	E − 01
Length/volume	ft/U.S. gal	m/m ³		8.051 964	E + 01
	ft/ft ³	m/m ³		1.076 391	E + 01
	ft/bbl	m/m ³		1.917 134	E + 00
Area	mi ²	km ²		2.589 988	E + 00
	section	ha		2.589 988	E + 02
	acre	ha		4.046 856	E − 01
	ha	m ²		1.000 000°	E + 04
	yd ²	m ²		8.361 274	E − 01
	ft ²	m ²		9.290 304°	E − 02
	in ²	mm ²		6.451 6°	E + 02
		cm ²		6.451 6°	E + 00
Area/volume	ft ² /in ³	m ² /cm ³		5.699 291	E − 03
	ft ² /ft ³	m ² /m ³		3.280 840	E + 00
Volume	cubem	km ³		4.168 182	E + 00
	acre-ft	m ³		1.233 482	E + 03
		ha-m		1.233 482	E − 01
	yd ³	m ³		7.645 549	E − 01
	bbl (42 U.S. gal)	m ³		1.589 873	E − 01
	ft ³	m ³		2.831 685	E − 02
		dm ³	L	2.831 685	E + 01
	U.K. gal	m ³		4.546 092	E − 03
		dm ³	L	4.546 092	E + 00
	U.S. gal	m ³		3.785 412	E − 03
		dm ³	L	3.785 412	E + 00
	U.K. qt	dm ³	L	1.136 523	E + 00
	U.S. qt	dm ³	L	9.463 529	E − 01
	U.S. pt	dm ³	L	4.731 765	E − 01
	U.K. fl oz	cm ³		2.841 307	E + 01
	U.S. fl oz	cm ³		2.957 353	E + 01
	in ³	cm ³		1.638 706	E + 01
Volume/length (linear displacement)	bbl/in	m ³ /m		6.259 342	E + 00
	bbl/ft	m ³ /m		5.216 119	E − 01
	ft ³ /ft	m ³ /m		9.290 304°	E − 02
	U.S. gal/ft	m ³ /m		1.241 933	E − 02
		L/m		1.241 933	E + 01
Plane angle	rad	rad		1	
	deg (°)	rad		1.745 329	E − 02
	min (')	rad		2.908 882	E − 04
	sec (")	rad		4.848 137	E − 06
Solid angle	sr	sr		1	
Time	year	a		1	
	week	d		7.0°	E + 00
	h	s		3.6°	E + 03
		min		6.0°	E + 01
	min	s		6.0°	E + 01
		h		1.666 667	E − 02
	mμs	ns		1	
Mass, amount of substance					
Mass	U.K. ton	Mg	t	1.016 047	E + 00
	U.S. ton	Mg	t	9.071 847	E − 01
	U.K. cwt	kg		5.080 234	E + 01
	U.S. cwt	kg		4.535 924	E + 01
	lbm	kg		4.535 924	E − 01
	oz (troy)	g		3.110 348	E + 01
	oz (av)	g		2.834 952	E + 01
	gr	mg		6.479 891	E + 01

TABLE 1-4 Conversion Factors: U.S. Customary and Commonly Used Units to SI Units (Continued)

Quantity	Customary or commonly used unit	SI unit	Alternate SI unit	Conversion factor; multiply customary unit by factor to obtain SI unit	
Amount of substance	lbm-mol	kmol		4.535 924	E - 01
	std m ³ (0°C, 1 atm)	kmol		4.461 58	E - 02
	std ft ³ (60°F, 1 atm)	kmol		1.195 30	E - 03
Enthalpy, calorific value, heat, entropy, heat capacity					
Calorific value, enthalpy (mass basis)	Btu/lbm	MJ/kg		2.326 000	E - 03
		kJ/kg	J/g	2.326 000	E + 00
		kWh/kg		6.461 112	E - 04
	cal/g cal/lbm	kJ/kg	J/g	4.184°	E + 00
		J/kg		9.224 141	E + 00
Caloric value, enthalpy (mole basis)	kcal/(g-mol)	kJ/kmol		4.184°	E + 03
	Btu/(lb-mol)	kJ/kmol		2.326 000	E + 00
Calorific value (volume basis—solids and liquids)	Btu/U.S. gal	MJ/m ³	kJ/dm ³	2.787 163	E - 01
		kJ/m ³		2.787 163	E + 02
		kWh/m ³		7.742 119	E - 02
	Btu/U.K. gal	MJ/m ³	kJ/dm ³	2.320 800	E - 01
		kJ/m ³		2.320 800	E + 02
	Btu/ft ³	kWh/m ³		6.446 667	E - 02
		MJ/m ³	kJ/dm ³	3.725 895	E - 02
		kJ/m ³		3.725 895	E + 01
	cal/mL (ft-lbf)/U.S. gal	kWh/m ³		1.034 971	E - 02
		MJ/m ³		4.184°	E + 00
		kJ/m ³		3.581 692	E - 01
Calorific value (volume basis—gases)	cal/mL	kJ/m ³	J/dm ³	4.184°	E + 03
	kcal/m ³	kJ/m ³	J/dm ³	4.184°	E + 00
	Btu/ft ³	kJ/m ³	J/dm ³	3.725 895	E + 01
		kWh/m ³		1.034 971	E - 02
Specific entropy	Btu/(lbm·°R)	kJ/(kg·K)	J/(g·K)	4.186 8°	E + 00
	cal/(g·K)	kJ/(kg·K)	J/(g·K)	4.184°	E + 00
	kcal/(kg·°C)	kJ/(kg·K)	J/(g·K)	4.184°	E + 00
Specific-heat capacity (mass basis)	kWh/(kg·°C)	kJ/(kg·K)	J/(g·K)	3.6°	E + 03
	Btu/(lbm·°F)	kJ/(kg·K)	J/(g·K)	4.186 8°	E + 00
	kcal/(kg·°C)	kJ/(kg·K)	J/(g·K)	4.184°	E + 00
Specific-heat capacity (mole basis)	Btu/(lb-mol·°F)	kJ/(kmol·K)		4.186 8°	E + 00
	cal/(g-mol·°C)	kJ/(kmol·K)		4.184°	E + 00
Temperature, pressure, vacuum					
Temperature (absolute)	°R	K		5/9	
	K	K		1	
Temperature (traditional)	°F	°C		5/9(°F - 32)	
Temperature (difference)	°F	K, °C		5/9	
Pressure	atm (760 mmHg at 0°C or 14,696 psi)	MPa		1.013 250°	E - 01
		kPa		1.013 250°	E + 02
		bar		1.013 250°	E + 00
	bar	MPa		1.0°	E - 01
		kPa		1.0°	E + 02
		MPa		6.894 757	E - 03
	mmHg (0°C) = torr	kPa		6.894 757	E + 00
		bar		6.894 757	E - 02
		kPa		3.376 85	E + 00
	μmHg (0°C)	kPa		2.488 4	E - 01
	μ bar	kPa		1.333 224	E - 01
	mmHg = torr (0°C)	kPa		9.806 38	E - 02
	cmH ₂ O (4°C)	kPa		4.788 026	E - 02
	lb/ft ² (psf)	Pa		1.333 224	E - 01
	mHg (0°C)	Pa		1.0°	E + 05
	bar	Pa		1.0°	E - 01
	dyn/cm ²	Pa			
Vacuum, draft	inHg (60°F)	kPa		3.376 85	E + 00
	inH ₂ O (39.2°F)	kPa		2.490 82	E - 01
	inH ₂ O (60°F)	kPa		2.488 4	E - 01
	mmHg (0°C) = torr	kPa		1.333 224	E - 01
	cmH ₂ O (4°C)	kPa		9.806 38	E - 02
Liquid head	ft	m		3.048°	E - 01
	in	mm		2.54°	E + 01
		cm		2.54°	E + 00
Pressure drop/length	psi/ft	kPa/m		2.262 059	E + 01

TABLE 1-4 Conversion Factors: U.S. Customary and Commonly Used Units to SI Units (Continued)

Quantity	Customary or commonly used unit	SI unit	Alternate SI unit	Conversion factor; multiply customary unit by factor to obtain SI unit
Density, specific volume, concentration, dosage				
Density	lbm/ft ³	kg/m ³		1.601 846 E + 01
		g/m ³		1.601 846 E + 04
	lbm/U.S. gal	kg/m ³		1.198 264 E + 02
		g/cm ³		1.198 264 E - 01
	lbm/U.K. gal	kg/m ³		9.977 633 E + 01
	lbm/ft ³	kg/m ³		1.601 846 E + 01
		g/cm ³		1.601 846 E - 02
Specific volume	g/cm ³	kg/m ³		1.0° E + 03
	lbm/ft ³	kg/m ³		1.601 846 E + 01
	ft ³ /lbm	m ³ /kg		6.242 796 E - 02
		m ³ /g		6.242 796 E - 05
	ft ³ /lbm	dm ³ /kg		6.242 796 E + 01
Specific volume (mole basis)	U.K. gal/lbm	dm ³ /kg	cm ³ /g	1.002 242 E + 01
	U.S. gal/lbm	dm ³ /kg	cm ³ /g	8.345 404 E + 00
Specific volume	L/(g·mol)	m ³ /kmol		1
	ft ³ /(lb·mol)	m ³ /kmol		6.242 796 E - 02
Specific volume	bbl/U.S. ton	m ³ /t		1.752 535 E - 01
	bbl/U.K. ton	m ³ /t		1.564 763 E - 01
Yield	bbl/U.S. ton	dm ³ /t	L/t	1.752 535 E + 02
	bbl/U.K. ton	dm ³ /t	L/t	1.564 763 E + 02
	U.S. gal/U.S. ton	dm ³ /t	L/t	4.172 702 E + 00
	U.S. gal/U.K. ton	dm ³ /t	L/t	3.725 627 E + 00
Concentration (mass/mass)	wt %	kg/kg		1.0° E - 02
		g/kg		1.0° E + 01
	wt ppm	mg/kg		1
Concentration (mass/volume)	lbm/bbl	kg/m ³	g/dm ³	2.853 010 E + 00
	g/U.S. gal	kg/m ³		2.641 720 E - 01
	g/U.K. gal	kg/m ³	g/L	2.199 692 E - 01
	lbm/1000 U.S. gal	g/m ³	mg/dm ³	1.198 264 E + 02
	lbm/1000 U.K. gal	g/m ³	mg/dm ³	9.977 633 E + 01
	gr/U.S. gal	g/m ³	mg/dm ³	1.711 806 E + 01
	gr/ft ³	mg/m ³		2.288 351 E + 03
	lbm/1000 bbl	g/m ³	mg/dm ³	2.853 010 E + 00
	mg/U.S. gal	g/m ³	mg/dm ³	2.641 720 E - 01
	gr/100 ft ³	mg/m ³		2.288 351 E + 01
Concentration (volume/volume)	ft ³ /ft ³	m ³ /m ³		1
	bbl/(acre-ft)	m ³ /m ³		1.288 931 E - 04
	vol%	m ³ /m ³		1.0° E - 02
	U.K. gal/ft ³	dm ³ /m ³	L/m ³	1.605 437 E + 02
	U.S. gal/ft ³	dm ³ /m ³	L/m ³	1.336 806 E + 02
	mL/U.S. gal	dm ³ /m ³	L/m ³	2.641 720 E - 01
	mL/U.K. gal	dm ³ /m ³	L/m ³	2.199 692 E - 01
	vol ppm	cm ³ /m ³		1
		dm ³ /m ³	L/m ³	1.0° E - 03
	U.K. gal/1000 bbl	cm ³ /m ³		2.859 403 E + 01
	U.S. gal/1000 bbl	cm ³ /m ³		2.380 952 E + 01
	U.K. pt/1000 bbl	cm ³ /m ³		3.574 253 E + 00
Concentration (mole/volume)	(lb·mol)/U.S. gal	kmol/m ³		1.198 264 E + 02
	(lb·mol)/U.K. gal	kmol/m ³		9.977 644 E + 01
	(lb·mol)/ft ³	kmol/m ³		1.601 846 E + 01
	std ft ³ (60°F, 1 atm)/bbl	kmol/m ³		7.518 21 E - 03
Concentration (volume/mole)	U.S. gal/1000 std ft ³ (60°F/60°F)	dm ³ /kmol	L/kmol	3.166 91 E + 00
	bbl/million std ft ³ (60°F/60°F)	dm ³ /kmol	L/kmol	1.330 10 E - 01
Facility throughput, capacity				
Throughput (mass basis)	U.K. ton/year	t/a		1.016 047 E + 00
	U.S. ton/year	t/a		9.071 847 E - 01
	U.K. ton/day	t/d		1.016 047 E + 00
		t/h		4.233 529 E - 02
	U.S. ton/day	t/d		9.071 847 E - 01
		t/h		3.779 936 E - 02
	U.K. ton/h	t/h		1.016 047 E + 00
	U.S. ton/h	t/h		9.071 847 E - 01
	lbm/h	kg/h		4.535 924 E - 01

TABLE 1-4 Conversion Factors: U.S. Customary and Commonly Used Units to SI Units (Continued)

Quantity	Customary or commonly used unit	SI unit	Alternate SI unit	Conversion factor; multiply customary unit by factor to obtain SI unit
Throughput (volume basis)	bbl/day	t/a		5.803 036 E + 01
		m ³ /d		1.589 873 E - 01
	ft ³ /day	m ³ /h		1.179 869 E - 03
	bbl/h	m ³ /h		1.589 873 E - 01
	ft ³ /h	m ³ /h		2.831 685 E - 02
	U.K. gal/h	m ³ /h		4.546 092 E - 03
		L/s		1.262 803 E - 03
	U.S. gal/h	m ³ /h		3.785 412 E - 03
		L/s		1.051 503 E - 03
	U.K. gal/min	m ³ /h		2.727 655 E - 01
Throughput (mole basis)		L/s		7.576 819 E - 02
		m ³ /h		2.271 247 E - 01
		L/s		6.309 020 E - 02
	(lbm-mol)/h	kmol/h		4.535 924 E - 01
		kmol/s		1.259 979 E - 04
Flow rate				
Flow rate (mass basis)	U.K. ton/min	kg/s		1.693 412 E + 01
	U.S. ton/min	kg/s		1.511 974 E + 01
	U.K. ton/h	kg/s		2.822 353 E - 01
	U.S. ton/h	kg/s		2.519 958 E - 01
	U.K. ton/day	kg/s		1.175 980 E - 02
	U.S. ton/day	kg/s		1.049 982 E - 02
	million lbm/year	kg/s		5.249 912 E + 00
	U.K. ton/year	kg/s		3.221 864 E - 05
	U.S. ton/year	kg/s		2.876 664 E - 05
	lbm/s	kg/s		4.535 924 E - 01
	lbm/min	kg/s		7.559 873 E - 03
	lbm/h	kg/s		1.259 979 E - 04
Flow rate (volume basis)	bbl/day	m ³ /d		1.589 873 E - 01
		L/s		1.840 131 E - 03
	ft ³ /day	m ³ /d		2.831 685 E - 02
		L/s		3.277 413 E - 04
	bbl/h	m ³ /s		4.416 314 E - 05
		L/s		4.416 314 E - 02
	ft ³ /h	m ³ /s		7.865 791 E - 06
		L/s		7.865 791 E - 03
	U.K. gal/h	dm ³ /s	L/s	1.262 803 E - 03
	U.S. gal/h	dm ³ /s	L/s	1.051 503 E - 03
	U.K. gal/min	dm ³ /s	L/s	7.576 820 E - 02
	U.S. gal/min	dm ³ /s	L/s	6.309 020 E - 02
	ft ³ /min	dm ³ /s	L/s	4.719 474 E - 01
	ft ³ /s	dm ³ /s	L/s	2.831 685 E + 01
Flow rate (mole basis)	(lb-mol)/s	kmol/s		4.535 924 E - 01
	(lb-mol)/h	kmol/s		1.259 979 E - 04
	million scf/D	kmol/s		1.383 45 E - 02
Flow rate/length (mass basis)	lbm/(s-ft)	kg/(s-m)		1.488 164 E + 00
	lbm/(h-ft)	kg/(s-m)		4.133 789 E - 04
Flow rate/length (volume basis)	U.K. gal/(min-ft)	m ² /s	m ³ /(s-m)	2.485 833 E - 04
	U.S. gal/(min-ft)	m ² /s	m ³ /(s-m)	2.069 888 E - 04
	U.K. gal/(h-in)	m ² /s	m ³ /(s-m)	4.971 667 E - 05
	U.S. gal/(h-in)	m ² /s	m ³ /(s-m)	4.139 776 E - 05
	U.K. gal/(h-ft)	m ² /s	m ³ /(s-m)	4.143 055 E - 06
	U.S. gal/(h-ft)	m ² /s	m ³ /(s-m)	3.449 814 E - 06
Flow rate/area (mass basis)	lbm/(s-ft ²)	kg/(s-m ²)		4.882 428 E + 00
	lbm/(h-ft ²)	kg/(s-m ²)		1.356 230 E - 03
Flow rate/area (volume basis)	ft ³ /(s-ft ²)	m/s	m ³ /(s-m ²)	3.048° E - 01
	ft ³ /(min-ft ²)	m/s	m ³ /(s-m ²)	5.08° E - 03
	U.K. gal/(h-in ²)	m/s	m ³ /(s-m ²)	1.957 349 E - 03
	U.S. gal/(h-in ²)	m/s	m ³ /(s-m ²)	1.629 833 E - 03
	U.K. gal/(min-ft ²)	m/s	m ³ /(s-m ²)	8.155 621 E - 04
	U.S. gal/(min-ft ²)	m/s	m ³ /(s-m ²)	6.790 972 E - 04
	U.K. gal/(h-ft ²)	m/s	m ³ /(s-m ²)	1.359 270 E - 05
	U.S. gal/(h-ft ²)	m/s	m ³ /(s-m ²)	1.131 829 E - 05

TABLE 1-4 Conversion Factors: U.S. Customary and Commonly Used Units to SI Units (Continued)

Quantity	Customary or commonly used unit	SI unit	Alternate SI unit	Conversion factor; multiply customary unit by factor to obtain SI unit
Energy, work, power				
Energy, work	therm	MJ		1.055 056 E + 02
		kJ		1.055 056 E + 05
		kWh		2.930 711 E + 01
	U.S. tonf·mi hp·h	MJ		1.431 744 E + 01
		MJ		2.684 520 E + 00
		kJ		2.684 520 E + 03
	ch·h or CV·h	kWh		7.456 999 E − 01
		MJ		2.647 780 E + 00
		kJ		2.647 780 E + 03
	kWh	kWh		7.354 999 E − 01
		MJ		3.6° E + 00
		kJ		3.6° E + 03
	Chu	kJ		1.899 101 E + 00
		kWh		5.275 280 E − 04
		kJ		1.055 056 E + 00
	Btu	kWh		2.930 711 E − 04
		kJ		4.184° E + 00
		kJ		4.184° E − 03
	cal	kJ		1.355 818 E − 03
		kJ		1.355 818 E − 03
		kJ		1.0° E − 03
Impact energy	J (lbf·ft ³)/s ² erg	kJ		4.214 011 E − 05
		kJ		1.0° E − 07
		J		9.806 650° E + 00
	kgf·m lbf·ft	J		1.355 818 E + 00
		J		1.0° E + 00
		J		1.0° E + 00
	erg/cm ²	mJ/m ²		1.0° E + 00
		mJ/m ²		1.0° E + 00
		mJ/m ²		1.0° E + 00
	(kgf·m)/cm ² (lbf·ft)/in ²	J/cm ²		9.806 650° E − 02
		J/cm ²		2.101 522 E − 03
		J/cm ²		2.101 522 E − 03
Power	million Btu/h ton of refrigeration Btu/s kW	MW		2.930 711 E − 01
		kW		3.516 853 E + 00
		kW		1.055 056 E + 00
	hydraulic horsepower—hhp hp (electric) hp [(550 ft·lbf)/s] ch or CV	kW		1
		kW		7.460 43 E − 01
		kW		7.46° E − 01
	Btu/min (ft·lbf)/s kcal/h Btu/h (ft·lbf)/min	kW		7.456 999 E − 01
		kW		7.354 999 E − 01
		kW		1.758 427 E − 02
	(ft·lbf)/s kcal/h Btu/h (ft·lbf)/min	kW		1.355 818 E − 03
		W		1.162 222 E + 00
		W		2.930 711 E − 01
	(ft·lbf)/min	W		2.259 697 E − 02
		W		2.259 697 E − 02
		W		2.259 697 E − 02
Power/area	Btu/(s·ft ²)	kW/m ²		1.135 653 E + 01
	cal/(h·cm ²)	kW/m ²		1.162 222 E − 02
	Btu/(h·ft ²)	kW/m ²		3.154 591 E − 03
Heat-release rate, mixing power	hp/ft ³	kW/m ³		2.633 414 E + 01
	cal/(h·cm ³)	kW/m ³		1.162 222 E + 00
	Btu/(s·ft ³)	kW/m ³		3.725 895 E + 01
	Btu/(h·ft ³)	kW/m ³		1.034 971 E − 02
Cooling duty (machinery)	Btu/(bhp·h)	W/kW		3.930 148 E − 01
Specific fuel consumption (mass basis)	lbm/(hp·h)	mg/J		1.689 659 E − 01
		kg/kWh		6.082 774 E − 01
Specific fuel consumption (volume basis)	m ³ /kWh	dm ³ /MJ	mm ³ /J	2.777 778 E + 02
	U.S. gal/(hp·h)	dm ³ /MJ	mm ³ /J	1.410 089 E + 00
	U.K. pt/(hp·h)	dm ³ /MJ	mm ³ /J	2.116 806 E − 01
Fuel consumption	U.K. gal/mi	dm ³ /100 km	L/100 km	2.824 807 E + 02
	U.S. gal/mi	dm ³ /100 km	L/100 km	2.352 146 E + 02
	mi/U.S. gal	km/dm ³	km/L	4.251 437 E − 01
	mi/U.K. gal	km/dm ³	km/L	3.540 064 E − 01

TABLE 1-4 Conversion Factors: U.S. Customary and Commonly Used Units to SI Units (Continued)

Quantity	Customary or commonly used unit	SI unit	Alternate SI unit	Conversion factor; multiply customary unit by factor to obtain SI unit	
Velocity (linear), speed	knot	km/h		1.852°	E + 00
	mi/h	km/h		1.609 344°	E + 00
	ft/s	m/s		3.048°	E - 01
		cm/s		3.048°	E + 01
	ft/min	m/s		5.08°	E - 03
	ft/h	mm/s		8.466 667	E - 02
	ft/day	mm/s		3.527 778	E - 03
		m/d		3.048°	E - 01
	in/s	mm/s		2.54°	E + 01
	in/min	mm/s		4.233 333	E - 01
Corrosion rate	in/year (ipy)	mm/a		2.54°	E + 01
	mil/year	mm/a		2.54°	E - 02
Rotational frequency	r/min	r/s		1.666 667	E - 02
		rad/s		1.047 198	E - 01
Acceleration (linear)	ft/s ²	m/s ²		3.048°	E - 01
		cm/s ²		3.048°	E + 01
Acceleration (rotational)	rpm/s	rad/s ²		1.047 198	E - 01
Momentum	(lbm·ft)/s	(kg·m)/s		1.382 550	E - 01
Force	U.K. tonf	kN		9.964 016	E + 00
	U.S. tonf	kN		8.896 443	E + 00
	kgf (kp)	N		9.806 650°	E + 00
	lbf	N		4.448 222	E + 00
	dyn	mN		1.0	E - 02
Bending moment, torque	U.S. tonf·ft	kN·m		2.711 636	E + 00
	kgf·m	N·m		9.806 650°	E + 00
	lbf·ft	N·m		1.355 818	E + 00
	lbf·in	N·m		1.129 848	E - 01
Bending moment/length	(lbf·ft)/in	(N·m)/m		5.337 866	E + 01
	(lbf·in)/in	(N·m)/m		4.448 222	E + 00
Moment of inertia	lbm·ft ²	kg·m ²		4.214 011	E - 02
Stress	U.S. tonf/in ²	MPa	N/mm ²	1.378 951	E + 01
	kgf/mm ²	MPa	N/mm ²	9.806 650°	E + 00
	U.S. tonf/ft ²	MPa	N/mm ²	9.576 052	E - 02
	lbf/in ² (psi)	MPa	N/mm ²	6.894 757	E - 03
	lbf/ft ² (psf)	kPa		4.788 026	E - 02
	dyn/cm ²	Pa		1.0°	E - 01
Mass/length	lbm/ft	kg/m		1.488 164	E + 00
Mass/area structural loading, bearing capacity (mass basis)	U.S. ton/ft ²	Mg/m ²		9.764 855	E + 00
	lbm/ft ²	kg/m ²		4.882 428	E + 00
Miscellaneous transport properties					
Diffusivity	ft ² /s	m ² /s		9.290 304°	E - 02
	m ² /s	mm ² /s		1.0°	E + 06
	ft ² /h	m ² /s		2.580 64°	E - 05
Thermal resistance	(°C·m ² ·h)/kcal	(K·m ²)/kW		8.604 208	E + 02
	(°F·ft ² ·h)/Btu	(K·m ²)/kW		1.761 102	E + 02
Heat flux	Btu/(h·ft ²)	kW/m ²		3.154 591	E - 03
Thermal conductivity	(cal·cm)/(s·cm ² ·°C)	W/(m·K)		4.184°	E + 02
	(Btu·ft)/(h·ft ² ·°F)	W/(m·K)		1.730 735	E + 00
		(kJ·m)/(h·m ² ·K)		6.230 646	E + 00
	(kcal·m)/(h·m ² ·°C)	W/(m·K)		1.162 222	E + 00
	(Btu·in)/(h·ft ² ·°F)	W/(m·K)		1.442 279	E - 01
	(cal·cm)/(h·cm ² ·°C)	W/(m·K)		1.162 222	E - 01
Heat-transfer coefficient	cal/(s·cm ² ·°C)	kW/(m ² ·K)		4.184°	E + 01
	Btu/(s·ft ² ·°F)	kW/(m ² ·K)		2.044 175	E + 01
	cal/(h·cm ² ·°C)	kW/(m ² ·K)		1.162 222	E - 02
	Btu/(h·ft ² ·°F)	kW/(m ² ·K)		5.678 263	E - 03
		kJ/(h·m ² ·K)		2.044 175	E + 01
	Btu/(h·ft ² ·°R)	kW/(m ² ·K)		5.678 263	E - 03
	kcal/(h·m ² ·°C)	kW/(m ² ·K)		1.162 222	E - 03

TABLE 1-4 Conversion Factors: U.S. Customary and Commonly Used Units to SI Units (Continued)

Quantity	Customary or commonly used unit	SI unit	Alternate SI unit	Conversion factor; multiply customary unit by factor to obtain SI unit	
Volumetric heat-transfer coefficient	Btu/(s·ft ³ ·°F)	kW/(m ³ ·K)		6.706 611	E + 01
	Btu/(h·ft ³ ·°F)	kW/(m ³ ·K)		1.862 947	E – 02
Surface tension	dyn/cm	mN/m		1	
Viscosity (dynamic)	(lbf·s)/in ²	Pa·s	(N·s)/m ²	6.894 757	E + 03
	(lbf·s)/ft ²	Pa·s	(N·s)/m ²	4.788 026	E + 01
	(kgf·s)/m ²	Pa·s	(N·s)/m ²	9.806 650*	E + 00
	lbm/(ft·s)	Pa·s	(N·s)/m ²	1.488 164	E + 00
	(dyn·s)/cm ²	Pa·s	(N·s)/m ²	1.0*	E – 01
	cP	Pa·s	(N·s)/m ²	1.0*	E – 03
	lbm/(ft·h)	Pa·s	(N·s)/m ²	4.133 789	E – 04
Viscosity (kinematic)	ft ² /s	m ² /s		9.290 304*	E – 02
	in ² /s	mm ² /s		6.451 6*	E + 02
	m ² /h	mm ² /s		2.777 778	E + 02
	ft ² /h	m ² /s		2.580 64*	E – 05
	cSt	mm ² /s		1	
Permeability	darcy	μm ²		9.869 233	E – 01
	millidarcy	μm ²		9.869 233	E – 04
Thermal flux	Btu/(h·ft ²)	W/m ²		3.152	E + 00
	Btu/(s·ft ²)	W/m ²		1.135	E + 04
	cal/(s·cm ²)	W/m ²		4.184	E + 04
Mass-transfer coefficient	(lb·mol)/[h·ft ² (lb·mol/ft ³)]	m/s		8.467	E – 05
	(g·mol)/[s·m ² (g·mol/L)]	m/s		1.0	E + 01
Electricity, magnetism					
Admittance	S	S		1	
Capacitance	μF	μF		1	
Charge density	C/mm ³	C/mm ³		1	
Conductance	S	S		1	
	Ω (mho)	S		1	
Conductivity	S/m	S/m		1	
	Ω/m	S/m		1	
	m Ω/m	mS/m		1	
Current density	A/mm ²	A/mm ²		1	
Displacement	C/cm ²	C/cm ²		1	
Electric charge	C	C		1	
Electric current	A	A		1	
Electric-dipole moment	C·m	C·m		1	
Electric-field strength	V/m	V/m		1	
Electric flux	C	C		1	
Electric polarization	C/cm ²	C/cm ²		1	
Electric potential	V	V		1	
	mV	mV		1	
Electromagnetic moment	A·m ²	A·m ²		1	
Electromotive force	V	V		1	
Flux of displacement	C	C		1	
Frequency	cycles/s	Hz		1	
Impedance	Ω	Ω		1	
Linear-current density	A/mm	A/mm		1	
Magnetic-dipole moment	Wb·m	Wb·m		1	
Magnetic-field strength	A/mm	A/mm		1	
	Oe	A/m		7.957 747	E + 01
	gamma	A/m		7.957 747	E – 04
Magnetic flux	mWb	mWb		1	

TABLE 1-4 Conversion Factors: U.S. Customary and Commonly Used Units to SI Units (Continued)

Quantity	Customary or commonly used unit	SI unit	Alternate SI unit	Conversion factor; multiply customary unit by factor to obtain SI unit	
Magnetic-flux density	mT G gamma	mT T nT		1 1.0° 1	E - 04
Magnetic induction	mT	mT		1	
Magnetic moment	A·m ²	A·m ²		1	
Magnetic polarization	mT	mT		1	
Magnetic potential difference	A	A		1	
Magnetic-vector potential	Wb/mm	Wb/mm		1	
Magnetization	A/mm	A/mm		1	
Modulus of admittance	S	S		1	
Modulus of impedance	Ω	Ω		1	
Mutual inductance	H	H		1	
Permeability	μH/m	μH/m		1	
Permeance	H	H		1	
Permittivity	μF/m	μF/m		1	
Potential difference	V	V		1	
Quantity of electricity	C	C		1	
Reactance	Ω	Ω		1	
Reluctance	H ⁻¹	H ⁻¹		1	
Resistance	Ω	Ω		1	
Resistivity	Ω·cm Ω·m	Ω·cm Ω·m		1 1	
Self-inductance	mH	mH		1	
Surface density of charge	mC/m ²	mC/m ²		1	
Susceptance	S	S		1	
Volume density of charge	C/mm ³	C/mm ³		1	
Acoustics, light, radiation					
Absorbed dose	rad	Gy		1.0°	E - 02
Acoustical energy	J	J		1	
Acoustical intensity	W/cm ²	W/m ²		1.0°	E + 04
Acoustical power	W	W		1	
Sound pressure	N/m ²	N/m ²		1.0°	
Illuminance	fc	lx		1.076 391	E + 01
Illumination	fc	lx		1.076 391	E + 01
Irradiance	W/m ²	W/m ²		1	
Light exposure	fc·s	lx·s		1.076 391	E + 01
Luminance	cd/m ²	cd/m ²		1	
Luminous efficacy	lm/W	lm/W		1	
Luminous exitance	lm/m ²	lm/m ²		1	
Luminous flux	lm	lm		1	
Luminous intensity	cd	cd		1	
Radiance	W/m ² ·sr	W/m ² ·sr		1	
Radiant energy	J	J		1	
Radiant flux	W	W		1	
Radiant intensity	W/sr	W/sr		1	
Radiant power	W	W		1	

TABLE 1-4 Conversion Factors: U.S. Customary and Commonly Used Units to SI Units (Concluded)

Quantity	Customary or commonly used unit	SI unit	Alternate SI unit	Conversion factor; multiply customary unit by factor to obtain SI unit	
Wavelength	Å	nm		1.0*	E - 01
Capture unit	10 ⁻³ cm ⁻¹	m ⁻¹		1.0*	E + 01
	m ⁻¹	m ⁻¹	10 ⁻³ cm ⁻¹	1 1	
Radioactivity	Ci	Bq		3.7*	E + 10

* An asterisk indicates that the conversion factor is exact.

† Conversion factors for length, area, and volume are based on the international foot. The international foot is longer by 2 parts in 1 million than the U.S. Survey foot (land-measurement use).

NOTE: The following unit symbols are used in the table:

Unit symbol	Name	Unit symbol	Name
A	ampere	lm	lumen
a	annum (year)	lx	lux
Bq	becquerel	m	meter
C	coulomb	min	minute
cd	candela	'	minute
Ci	curie	N	newton
d	day	naut mi	U.S. nautical mile
°C	degree Celsius	Oe	oersted
°	degree	Ω	ohm
dyn	dyne	Pa	pascal
F	farad	rad	radian
fc	footcandle	r	revolution
G	gauss	S	siemens
g	gram	s	second
gr	grain	"	second
Gy	gray	sr	steradian
H	henry	St	stokes
h	hour	T	tesla
ha	hectare	t	tonne
Hz	hertz	V	volt
J	joule	W	watt
K	kelvin	Wb	weber
L, ℓ, l	liter		

NOTE: Copyright SPE-AIME, *The SI Metric System of Units and SPE's Tentative Metric Standard*, Society of Petroleum Engineers, Dallas, 1977.

TABLE 1-5 Metric Conversion Factors as Exact Numerical Multiples of SI Units

The first two digits of each numerical entry represent a power of 10. For example, the entry “-02 2.54” expresses the fact that 1 in = 2.54×10^{-2} m.

To convert from	To	Multiply by	To convert from	To	Multiply by
abampere	ampere	+01 1.00	fluid ounce (U.S.)	meter ³	-05 2.957 352
abcoulomb	coulomb	+01 1.00	foot	meter	-01 3.048
abfarad	farad	+09 1.00	foot (U.S. survey)	meter	-01 3.048 006
abhenry	henry	-09 1.00	foot of water (39.2°F)	newton/meter ²	+03 2.988 98
abmho	mho	+09 1.00	footcandle	lumen/meter ²	+01 1.076 391
abohm	ohm	-09 1.00	footlambert	candela/meter ²	+00 3.426 259
abvolt	volt	-08 1.00	furlong	meter	+02 2.011 68
acre	meter ²	+03 4.046 856	gal (galileo)	meter/second ²	-02 1.00
ampere (international of 1948)	ampere	-01 9.998 35	gallon (U.K. liquid)	meter ³	-03 4.546 087
angstrom	meter	-10 1.00	gallon (U.S. dry)	meter ³	-03 4.404 883
are	meter ²	+02 1.00	gallon (U.S. liquid)	meter ³	-03 3.785 411
astronomical unit	meter	+11 1.495 978	gamma	tesla	-09 1.00
atmosphere	newton/meter ²	+05 1.013 25	gauss	tesla	-04 1.00
bar	newton/meter ²	+05 1.00	gilbert	ampere turn	-01 7.957 747
barn	meter ²	-28 1.00	gill (U.K.)	meter ³	-04 1.420 652
barrel (petroleum 42 gal)	meter ³	-01 1.589 873	gill (U.S.)	meter ³	-04 1.182 941
barye	newton/meter ²	-01 1.00	grad	degree (angular)	-01 9.00
British thermal unit (ISO/TC 12)	joule	+03 1.055 06	grad	radian	-02 1.570 796
British thermal unit (International Steam Table)	joule	+03 1.055 04	grain	kilogram	-05 6.479 891
British thermal unit (mean)	joule	+03 1.055 87	gram	kilogram	-03 1.00
British thermal unit (thermochemical)	joule	+03 1.054 350	hand	meter	-01 1.016
British thermal unit (39°F)	joule	+03 1.059 67	hectare	meter ²	+04 1.00
British thermal unit (60°F)	joule	+03 1.054 68	henry (international of 1948)	henry	+00 1.000 495
bushel (U.S.)	meter ³	-02 3.523 907	hogshead (U.S.)	meter ³	-01 2.384 809
cable	meter	+02 2.194 56	horsepower (550 ft lbf/s)	watt	+02 7.456 998
caliber	meter	-04 2.54	horsepower (boiler)	watt	+03 9.809 50
calorie (International Steam Table)	joule	+00 4.1868	horsepower (electric)	watt	+02 7.46
calorie (mean)	joule	+00 4.190 02	horsepower (metric)	watt	+02 7.354 99
calorie (thermochemical)	joule	+00 4.184	horsepower (U.K.)	watt	+02 7.457
calorie (15°C)	joule	+00 4.185 80	horsepower (water)	watt	+02 7.460 43
calorie (20°C)	joule	+00 4.181 90	hour (mean solar)	second (mean solar)	+03 3.60
calorie (kilogram, International Steam Table)	joule	+03 4.186 8	hour (sidereal)	second (mean solar)	+03 3.590 170
calorie (kilogram, mean)	joule	+03 4.190 02	hundredweight (long)	kilogram	+01 5.080 234
calorie (kilogram, thermochemical)	joule	+03 4.184	hundredweight (short)	kilogram	+01 4.535 923
carat (metric)	kilogram	-04 2.00	inch	meter	-02 2.54
Celsius (temperature)	kelvin	$t_K = t_c + 273.15$	inch of mercury (32°F)	newton/meter ²	+03 3.386 389
centimeter of mercury (0°C)	newton/meter ²	+03 1.333 22	inch of mercury (60°F)	newton/meter ²	+03 3.376 85
centimeter of water (4°C)	newton/meter ²	+01 9.806 38	inch of water (39.2°F)	newton/meter ²	+02 2.490 82
chain (engineer's)	meter	+01 3.048	inch of water (60°F)	newton/meter ²	+02 2.4884
chain (surveyor's or Gunter's)	meter	+01 2.011 68	joule (international of 1948)	joule	+00 1.000 165
circular mil	meter ²	-10 5.067 074	kayser	1/meter	+02 1.00
cord	meter ³	+00 3.624 556	kilocalorie (International Steam Table)	joule	+03 4.186 74
coulomb (international of 1948)	coulomb	-01 9.998 35	kilocalorie (mean)	joule	+03 4.190 02
cubit	meter	-01 4.572	kilocalorie (thermochemical)	joule	+03 4.184
cup	meter ³	-04 2.365 882	kilogram mass	kilogram	+00 1.00
curie	disintegration/second	+10 3.70	kilogram-force (kgf)	newton	+00 9.806 65
day (mean solar)	second (mean solar)	+04 8.64	kilopond-force	newton	+00 9.806 65
day (sidereal)	second (mean solar)	+04 8.616 409	kip	newton	+03 4.448 221
degree (angle)	radian	-02 1.745 329	knot (international)	meter/second	-01 5.144 444
denier (international)	kilogram/meter	-07 1.111 111	lambert	candela/meter ²	+04 1/π
dram (avoirdupois)	kilogram	-03 1.771 845	lambert	candela/meter ²	+03 3.183 098
dram (troy or apothecary)	kilogram	-03 3.887 934	langley	joule/meter ²	+04 4.184
dram (U.S. fluid)	meter ³	-06 3.696 691	lbf (pound-force, avoirdupois)	newton	+00 4.448 221
dyne	newton	-05 1.00	lbm (pound-mass, avoirdupois)	kilogram	-01 4.535 923
electron volt	joule	-19 1.602 10	league (British nautical)	meter	+03 5.559 552
erg	joule	-07 1.00	league (international nautical)	meter	+03 5.556
Fahrenheit (temperature)	kelvin	$t_K = (5/9)(t_F + 459.67)$	league (statute)	meter	+03 4.828 032
Fahrenheit (temperature)	Celsius	$t_c = (5/9)(t_F - 32)$	light-year	meter	+15 9.460 55
farad (international of 1948)	farad	-01 9.995 05	link (engineer's)	meter	-01 3.048
faraday (based on carbon 12)	coulomb	+04 9.648 70	link (surveyor's or Gunter's)	meter	-01 2.011 68
faraday (chemical)	coulomb	+04 9.649 57	liter	meter ³	-03 1.00
faraday (physical)	coulomb	+04 9.652 19	lux	lumen/meter ²	+00 1.00
fathom	meter	+00 1.828 8	maxwell	weber	-08 1.00
fermi (femtometer)	meter	-15 1.00	meter	wavelengths Kr 86	+06 1.650 763
			micrometer	meter	-06 1.00
			mil	meter	-05 2.54
			mile (U.S. statute)	meter	+03 1.609 344
			mile (U.K. nautical)	meter	+03 1.853 184
			mile (international nautical)	meter	+03 1.852
			mile (U.S. nautical)	meter	+03 1.852
			millibar	newton/meter ²	+02 1.00
			millimeter of mercury (0°C)	newton/meter ²	+02 1.333 224

TABLE 1-5 Metric Conversion Factors as Exact Numerical Multiples of SI Units (Concluded)

The first two digits of each numerical entry represent a power of 10. For example, the entry “–02 2.54” expresses the fact that 1 in = 2.54×10^{-2} .

To convert from	To	Multiply by	To convert from	To	Multiply by
minute (angle)	radian	–04 2.908 882	second (ephemeris)	second	+00 1.000 000
minute (mean solar)	second (mean solar)	+01 6.00	second (mean solar)	second (ephemeris)	Consult
minute (sidereal)	second (mean solar)	+01 5.983 617		American	
month (mean calendar)	second (mean solar)	+06 2.628		Ephemeris	
nautical mile (international)	meter	+03 1.852		and Nautical	
nautical mile (U.S.)	meter	+03 1.852		Almanac	
nautical mile (U.K.)	meter	+03 1.853 184	second (sidereal)	second (mean solar)	–01 9.972 695
oersted	ampere/meter	+01 7.957 747	section	meter ²	+06 2.589 988
ohm (international of 1948)	ohm	+00 1.000 495	scruple (apothecary)	kilogram	–03 1.295 978
ounce-force (avoirdupois)	newton	–01 2.780 138	shake	second	–08 1.00
ounce-mass (avoirdupois)	kilogram	–02 2.834 952	skein	meter	+02 1.097 28
ounce-mass (troy or apothecary)	kilogram	–02 3.110 347	slug	kilogram	+01 1.459 390
ounce (U.S. fluid)	meter ³	–05 2.957 352	span	meter	–01 2.286
pace	meter	–01 7.62	statampere	ampere	–10 3.335 640
parsec	meter	+16 3.083 74	statcoulomb	coulomb	–10 3.335 640
pascal	newton/meter ²	+00 1.00	statfarad	farad	–12 1.112 650
peck (U.S.)	meter ³	–03 8.809 767	stathenry	henry	+11 8.987 554
pennyweight	kilogram	–03 1.555 173	statmho	mho	–12 1.112 650
perch	meter	+00 5.0292	statohm	ohm	+11 8.987 554
phot	lumen/meter ²	+04 1.00	statute mile (U.S.)	meter	+03 1.609 344
pica (printer's)	meter	+03 4.217 517	statvolt	volt	+02 2.997 925
pint (U.S. dry)	meter ³	–04 5.506 104	stere	meter ³	+00 1.00
pint (U.S. liquid)	meter ³	–04 4.731 764	stillb	candela/meter ²	+04 1.00
point (printer's)	meter	–04 3.514 598	stoke	meter ² /second	–04 1.00
poise	(newton-second)/meter ²	–01 1.00	tablespoon	meter ³	–05 1.478 676
pole	meter	+00 5.0292	teaspoon	meter ³	–06 4.928 921
pound-force (lbf)	newton	+00 4.448 221	ton (assay)	kilogram	–02 2.916 666
avoirdupois)			ton (long)	kilogram	+03 1.016 046
pound-mass (lbm)	kilogram	–01 4.535 923	ton (metric)	kilogram	+03 1.00
avoirdupois)			ton (nuclear equivalent of TNT)	joule	+09 4.20
pound-mass (troy or	kilogram	–01 3.732 417	ton (register)	meter ³	+00 2.831 684
apothecary)			ton (short, 2000 lb)	kilogram	+02 9.071 847
poundal	newton	–01 1.382 549	tonne	kilogram	+03 1.00
quart (U.S. dry)	meter ³	–03 1.101 220	torr (0°C)	newton/meter ²	+02 1.333 22
quart (U.S. liquid)	meter ³	–04 9.463 529	township	meter ²	+07 9.323 957
rad (radiation dose	joule/kilogram	–02 1.00	unit pole	weber	–07 1.256 637
absorbed)			volt (international of 1948)	volt	+00 1.000 330
Rankine (temperature)	kelvin	$t_K = (5/9)t_R$	watt (international of 1948)	watt	+00 1.000 165
rayleigh (rate of photon	1/second-meter ²	+10 1.00	yard	meter	–01 9.144
emission)			year (calendar)	second (mean solar)	+07 3.1536
rhe	meter ² /(newton-second)	+01 1.00	year (sidereal)	second (mean solar)	+07 3.155 815
			year (tropical)	second (mean solar)	+07 3.155 692
rod	meter	+00 5.0292	year 1900, tropical, Jan., day	second (ephemeris)	+07 3.155 692
roentgen	coulomb/kilogram	–04 2.579 76	0, hour 12		
rutherford	disintegration/second	+06 1.00	year 1900, tropical, Jan., day	second	+07 3.155 692
second (angle)	radian	–06 4.848 136	0, hour 12		

TABLE 1-6 Alphabetical Listing of Common Conversions

To convert from	To	Multiply by	To convert from	To	Multiply by
Acres	Square feet	43,560	Drams (avoirdupois)	Grams	1.7719
Acres	Square meters	4074	Dynes	Newtons	1×10^{-5}
Acres	Square miles	0.001563	Ergs	Joules	1×10^{-7}
Acre-feet	Cubic meters	1233	Faradays	Coulombs (abs.)	96,500
Ampere-hours (absolute)	Coulombs (absolute)	3600	Fathoms	Feet	6
Angstrom units	Inches	3.937×10^{-9}	Feet	Meters	0.3048
Angstrom units	Meters	1×10^{-10}	Feet per minute	Centimeters per second	0.5080
Angstrom units	Microns	1×10^{-4}	Feet per minute	Miles per hour	0.011364
Atmospheres	Millimeters of mercury at 32°F	760	Feet per (second) ²	Meters per (second) ²	0.3048
Atmospheres	Dynes per square centimeter	1.0133×10^6	Feet of water at 39.2°F.	Newtons per square meter	2989
Atmospheres	Newtons per square meter	101,325	Foot-pounds	B.t.u.	3.995×10^{-5}
Atmospheres	Feet of water at 39.1°F	33.90	Foot-pounds	Joules	0.04214
Atmospheres	Grams per square centimeter	1033.3	Foot-pounds	Liter-atmospheres	4.159×10^{-4}
Atmospheres	Inches of mercury at 32°F	29.921	Foot-pounds	B.t.u.	0.0012856
Atmospheres	Pounds per square foot	2116.3	Foot-pounds	Calories, gram	0.3239
Atmospheres	Pounds per square inch	14.696	Foot-pounds	Foot-pounds	32.174
Bags (cement)	Pounds (cement)	94	Foot-pounds	Horsepower-hours	5.051×10^{-7}
Barrels (cement)	Pounds (cement)	376	Foot-pounds	Kilowatt-hours	3.766×10^{-7}
Barrels (oil)	Cubic meters	0.15899	Foot-pounds	Liter-atmospheres	0.013381
Barrels (oil)	Gallons	42	Foot-pounds force	Joules	1.3558
Barrels (U.S. liquid)	Cubic meters	0.11924	Foot-pounds per second	Horsepower	0.0018182
Barrels (U.S. liquid)	Gallons	31.5	Foot-pounds per second	Kilowatts	0.0013558
Bars	Gallons per minute	0.02917	Furlongs	Miles	0.125
Bars	Atmospheres	0.9869	Gallons (U.S. liquid)	Barrels (U.S. liquid)	0.03175
Bars	Newtons per square meter	1×10^5	Gallons	Cubic meters	0.003785
Board feet	Pounds per square inch	14.504	Gallons	Cubic feet	0.13368
Boiler horsepower	Cubic feet	$\frac{1}{12}$	Gallons	Gallons (Imperial)	0.8327
Boiler horsepower	B.t.u. per hour	33,480	Gallons	Liters	3.785
B.t.u.	Kilowatts	9.803	Gallons	Ounces (U.S. fluid)	128
B.t.u.	Calories (gram)	252	Gallons per minute	Cubic feet per hour	8.021
B.t.u.	Centigrade heat units (c.h.u. or p.c.u.)	0.55556	Gallons per minute	Cubic feet per second	0.002228
B.t.u.	Foot-pounds	777.9	Grains	Grams	0.06480
B.t.u.	Horsepower-hours	3.929×10^{-4}	Grains	Pounds	$\frac{1}{7000}$
B.t.u.	Joules	1055.1	Grains per cubic foot	Grams per cubic meter	2.2884
B.t.u.	Liter-atmospheres	10.41	Grains per gallon	Parts per million	17.118
B.t.u.	Pounds carbon to CO ₂	6.88×10^{-5}	Grams	Drams (avoirdupois)	0.5644
B.t.u.	Pounds water evaporated from and at 212°F	0.001036	Grams	Drams (troy)	0.2572
B.t.u.	Cubic foot-atmospheres	0.3676	Grams	Grains	15.432
B.t.u.	Kilowatt-hours	2.930×10^{-4}	Grams	Kilograms	0.001
B.t.u. per cubic foot	Joules per cubic meter	37,260	Grams	Pounds (avoirdupois)	0.0022046
B.t.u. per hour	Watts	0.29307	Grams	Pounds (troy)	0.002679
B.t.u. per minute	Horsepower	0.02357	Grams per cubic centimeter	Pounds per cubic foot	62.43
B.t.u. per pound	Joules per kilogram	2326	Grams per cubic centimeter	Pounds per gallon	8.345
B.t.u. per pound per degree Fahrenheit	Calories per gram per degree centigrade	1	Grams per liter	Grains per gallon	58.42
B.t.u. per pound per degree Fahrenheit	Joules per kilogram per degree Kelvin	4186.8	Grams per liter	Pounds per cubic foot	0.0624
B.t.u. per second	Watts	1054.4	Grams per square centimeter	Pounds per square foot	2.0482
B.t.u. per square foot per hour	Joules per square meter per second	3.1546	Grams per square centimeter	Pounds per square inch	0.014223
B.t.u. per square foot per minute	Kilowatts per square foot	0.1758	Hectares	Acres	2.471
B.t.u. per square foot per second	Calories, gram (15°C.), per square centimeter per second for a temperature gradient of 1°C. per centimeter	1.2405	Hectares	Square meters	10,000
for a temperature gradient of 1°F. per inch			Horsepower (British)	B.t.u. per minute	42.42
			Horsepower (British)	B.t.u. per hour	2545
			Horsepower (British)	Foot-pounds per minute	33,000
			Horsepower (British)	Foot-pounds per second	550
			Horsepower (British)	Watts	745.7

B.t.u. (60°F) per degree Fahrenheit	Calories per degree centigrade	453.6	Horsepower (British)	Horsepower (metric)	1.0139
Bushels (U.S. dry)	Cubic feet	1.2444	Horsepower (British)	Pounds carbon to CO ₂ per hour	0.175
Bushels (U.S. dry)	Cubic meters	0.03524	Horsepower (British)	Pounds water evaporated per hour at 212°F	2.64
Calories, gram	B.t.u.	3.968×10^{-3}	Horsepower (metric)	Foot-pounds per second	542.47
Calories, gram	Foot-pounds	3.087	Horsepower (metric)	Kilogram-meters per second	75.0
Calories, gram	Joules	4.1868	Hours (mean solar)	Seconds	3600
Calories, gram	Liter-atmospheres	4.130×10^{-2}	Inches	Meters	0.0254
Calories, gram	Horsepower-hours	1.5591×10^{-6}	Inches of mercury at 60°F	Newtons per square meter	3376.9
Calories, gram, per gram per degree C.	Joules per kilogram per degree Kelvin	4186.8	Inches of water at 60°F	Newtons per square meter	248.84
Calories, kilogram	Kilowatt-hours	0.0011626	Joules (absolute)	B.t.u. (mean)	9.480×10^{-4}
Calories, kilogram per second	Kilowatts	4.185	Joules (absolute)	Calories, gram (mean)	0.2389
Candle power (spherical)	Lumens	12.556	Joules (absolute)	Cubic foot-atmospheres	0.3485
Carats (metric)	Grams	0.2	Joules (absolute)	Foot-pounds	0.7376
Centigrade heat units	B.t.u.	1.8	Joules (absolute)	Kilowatt-hours	2.7778×10^{-7}
Centimeters	Angstrom units	1×10^8	Joules (absolute)	Liter-atmospheres	0.009869
Centimeters	Feet	0.03281	Kilocalories	Joules	4186.8
Centimeters	Inches	0.3937	Kilograms	Pounds (avoirdupois)	2.2046
Centimeters	Meters	0.01	Kilograms force	Newtons	9.807
Centimeters	Microns	10,000	Kilograms per square centimeter	Pounds per square inch	14.223
Centimeters of mercury at 0°C.	Atmospheres	0.013158	Kilometers	Miles	0.6214
Centimeters of mercury at 0°C.	Feet of water at 39.1°F.	0.4460	Kilowatt-hours	B.t.u.	3414
Centimeters of mercury at 0°C.	Newtons per square meter	1333.2	Kilowatts	Foot-pounds	2.6552×10^6
Centimeters of mercury at 0°C.	Pounds per square foot	27.845	Knots (international)	Horsepower	1.3410
Centimeters of mercury at 0°C.	Pounds per square inch	0.19337	Knots (nautical miles per hour)	Meters per second	0.5144
Centimeters per second	Feet per minute	1.9685	Lamberts	Miles per hour	1.1516
Centimeters of water at 4°C.	Newtons per square meter	98.064	Liter-atmospheres	Candles per square inch	2.054
Centistokes	Square meters per second	1×10^{-6}	Liter-atmospheres	Cubic foot-atmospheres	0.03532
Circular mils	Square centimeters	5.067×10^{-6}	Liters	Foot-pounds	74.74
Circular mils	Square inches	7.854×10^{-7}	Liters	Cubic feet	0.03532
Circular mils	Square mils	0.7854	Liters	Cubic meters	0.001
Cords	Cubic feet	128	Lumens	Gallons	0.26418
Cubic centimeters	Cubic feet	3.532×10^{-5}	Micromicrons	Watts	0.001496
Cubic centimeters	Gallons	2.6417×10^{-4}	Microns	Microns	1×10^{-6}
Cubic centimeters	Ounces (U.S. fluid)	0.03381	Microns	Angstrom units	1×10^4
Cubic centimeters	Quarts (U.S. fluid)	0.0010567	Miles (nautical)	Meters	1×10^{-6}
Cubic feet	Bushels (U.S.)	0.8036	Miles (nautical)	Feet	6080
Cubic feet	Cubic centimeters	28,317	Miles	Miles (U.S. statute)	1.1516
Cubic feet	Cubic meters	0.028317	Miles	Feet	5280
Cubic feet	Cubic yards	0.03704	Miles	Meters	1609.3
Cubic feet	Gallons	7.481	Miles per hour	Feet per second	1.4667
Cubic feet	Liters	28.316	Miles per hour	Meters per second	0.4470
Cubic foot-atmospheres	Foot-pounds	2116.3	Milliliters	Cubic centimeters	1
Cubic foot-atmospheres	Liter-atmospheres	28.316	Millimeters	Meters	0.001
Cubic feet of water (60°F.)	Pounds	62.37	Millimeters of mercury at 0°C.	Newtons per square meter	133.32
Cubic feet per minute	Cubic centimeters per second	472.0	Millimicrons	Microns	0.001
Cubic feet per minute	Gallons per second	0.1247	Mils	Inches	0.001
Cubic feet per second	Gallons per minute	448.8	Mils	Meters	2.54×10^{-5}
Cubic feet per second	Million gallons per day	0.64632	Minims (U.S.)	Cubic centimeters	0.06161
Cubic inches	Cubic meters	1.6387×10^{-5}	Minutes (angle)	Radians	2.909×10^{-4}
Cubic yards	Cubic meters	0.76456	Minutes (mean solar)	Seconds	60
Curies	Disintegrations per minute	2.2×10^{12}	Newtons	Kilograms	0.10197
Curies	Coulombs per minute	1.1×10^{12}	Ounces (avoirdupois)	Kilograms	0.02835
Degrees	Radians	0.017453	Ounces (avoirdupois)	Ounces (troy)	0.9115
Drams (apothecaries' or troy)	Grams	3.888	Ounces (U.S. fluid)	Cubic meters	2.957×10^{-5}
			Ounces (troy)	Ounces (apothecaries')	1.000

TABLE 1-6 Alphabetical Listing of Common Conversions (Concluded)

To convert from	To	Multiply by	To convert from	To	Multiply by
Pints (U.S. liquid)	Cubic meters	4.732×10^{-4}	Square centimeters	Square feet	0.0010764
Poundals	Newtons	0.13826	Square feet	Square meters	0.0929
Pounds (avoirdupois)	Grains	7000	Square feet per hour	Square meters per second	2.581×10^{-5}
Pounds (avoirdupois)	Kilograms	0.45359	Square inches	Square centimeters	6.452
Pounds (avoirdupois)	Pounds (troy)	1.2153	Square inches	Square meters	6.452×10^{-4}
Pounds per cubic foot	Grams per cubic centimeter	0.016018	Square yards	Square meters	0.8361
Pounds per cubic foot	Kilograms per cubic meter	16.018	Stokes	Square meters per second	1×10^{-4}
Pounds per square foot	Atmospheres	4.725×10^{-4}	Tons (long)	Kilograms	1016
Pounds per square foot	Kilograms per square meter	4.882	Tons (long)	Pounds	2240
Pounds per square inch	Atmospheres	0.06805	Tons (metric)	Kilograms	1000
Pounds per square inch	Kilograms per square centimeter	0.07031	Tons (metric)	Pounds	2204.6
Pounds per square inch	Newtons per square meter	6894.8	Tons (metric)	Tons (short)	1.1023
Pounds force	Newtons	4.4482	Tons (short)	Kilograms	907.18
Pounds force per square foot	Newtons per square meter	47.88	Tons (short)	Pounds	2000
Pounds water evaporated from and at 212°F.	Horsepower-hours	0.379	Tons (refrigeration)	B.t.u. per hour	12,000
Pound-centigrade units (p.c.u.)	B.t.u.	1.8	Tons (British shipping)	Cubic feet	42.00
Quarts (U.S. liquid)	Cubic meters	9.464×10^{-4}	Tons (U.S. shipping)	Cubic feet	40.00
Radians	Degrees	57.30	Torr (mm. mercury, 0°C.)	Newtons per square meter	133.32
Revolutions per minute	Radians per second	0.10472	Watts	B.t.u. per hour	3.413
Seconds (angle)	Radians	4.848×10^{-6}	Watts	Joules per second	1
Slugs	Gee pounds	1	Watts	Kilogram-meters per second	0.10197
Slugs	Kilograms	14.594	Watt-hours	Joules	3600
Slugs	Pounds	32.17	Yards	Meters	0.9144

TABLE 1-7 Common Units and Conversion Factors*

Mass (M)	1 pound mass = 453.5924 grams = 0.45359 kilograms = 7000 grains 1 slug = 32.174 pounds mass 1 ton (short) = 2000 pounds mass 1 ton (long) = 2240 pounds mass 1 ton (metric) = 1000 kilograms = 2204.62 pounds mass 1 pound mole = 453.59 gram moles	1 atm = 760 millimeters of mercury at 0°C (density 13.5951 g/cm ³) = 29.921 inches of mercury at 32°F = 14.696 pounds force/square inch = 33.899 feet of water at 39.1°F = 1.01325 × 10 ⁶ dynes/square centimeter = 1.01325 × 10 ⁵ Newtons/square meter
Length (L)	1 foot = 30.480 centimeters = 0.3048 meters 1 inch = 2.54 centimeters = 0.0254 meters 1 mile (U.S.) = 1.60935 kilometers 1 yard = 0.9144 meters	Density (M/L ³) 1 pound mass/cubic foot = 0.01601846 grams/cubic centimeter = 16.01846 kilogram/cubic meter
Area (L ²)	1 square foot = 929.0304 square centimeters = 0.09290304 square meters 1 square inch = 6.4516 square centimeters 1 square yard = 0.836127 square meters	Energy (H or FL) 1 British thermal unit = 251.98 calories = 1054.4 joules = 777.97 foot-pounds force = 10.409 liter-atmospheres = 0.2930 watt-hour
Volume (L ³)	1 cubic foot = 28,316.85 cubic centimeters = 0.02831685 cubic meters = 28.31685 liters = 7.481 gallons (U.S.) 1 gallon = 3.7853 liters = 231 cubic inches	Diffusivity (L ² /θ) 1 square foot/hour = 0.258 cm ² /s = 2.58 × 10 ⁻⁵ m ² /s
Time (θ)	1 hour = 60 minutes = 3600 seconds	Viscosity (M/Lθ) 1 pound mass/foot hour = 0.00413 g/cm s = 0.000413 kg/m s 1 centipoise = 0.01 poise = 0.01 g/cm s = 0.001 kg/m s = 0.000672 lbm/ft s = 0.0000209 lbf _s /ft ²
Temperature (T)	1 centigrade or Celsius degree = 1.8 Fahrenheit degree Temperature, Kelvin = T°C + 273.15 Temperature, Rankine = T°F + 459.7 Temperature, Fahrenheit = 9/5 T°C + 32 Temperature, centigrade or Celsius = 5/9 (T°F - 32) Temperature, Rankine = 1.8 T K	Thermal conductivity [H/θL ² (T/L)] 1 Btu/hr ft ² (°F/ft) = 0.00413 cal/s cm ² (°C/cm) = 1.728 J/s m ² (°C/m)
Force (F)	1 pound force = 444,822.2 dynes = 4.448222 Newtons = 32.174 poundals	Heat transfer coefficient 1 Btu/hr ft ² °F = 5.678 J/s m ² °C Heat capacity (H/MT) 1 Btu/lbm °F = 1 cal/g °C = 4184 J/kg °C
Pressure (F/L ²)	Normal atmospheric pressure	Gas constant 1.987 Btu/lbm mole °R = 1.987 cal/mol K = 82.057 atm cm ³ /mol K = 0.7302 atm ft ³ /lb mole °F = 10.73 (lb _f /in. ²) (ft ³)/lb mole °R = 1545 (lb _f /ft ²) (ft ³)/lb mole °R = 8.314 (N/m ²) (m ³)/mol K
		Gravitational acceleration g = 9.8066 m/s ² = 32.174 ft/s ²

NOTE: U.S. customary units, or British units, on left and SI units on right.

*Adapted from Faust et al., *Principles of Unit Operations*, John Wiley and Sons, 1980.**TABLE 1-8 Kinematic-Viscosity Conversion Formulas**

Viscosity scale	Range of <i>t</i> , sec	Kinematic viscosity, stokes
Saybolt Universal	32 < <i>t</i> < 100 <i>t</i> > 100	0.00226 <i>t</i> - 1.95/ <i>t</i> 0.00220 <i>t</i> - 1.35/ <i>t</i>
Saybolt Furol	25 < <i>t</i> < 40 <i>t</i> > 40	0.0224 <i>t</i> - 1.84/ <i>t</i> 0.0216 <i>t</i> - 0.60/ <i>t</i>
Redwood No. 1	34 < <i>t</i> < 100 <i>t</i> > 100	0.00260 <i>t</i> - 1.79/ <i>t</i> 0.00247 <i>t</i> - 0.50/ <i>t</i>
Redwood Admiralty Engler		0.027 <i>t</i> - 20/ <i>t</i> 0.00147 <i>t</i> - 3.74/ <i>t</i>

TABLE 1-9 Values of the Gas-Law Constant

Temp. scale	Press. units	Vol. units	Wt. units	Energy units*	<i>R</i>
Kelvin	atm.	cm ³	g-moles	calories	1.9872
			g-moles	joules (abs)	8.3144
			g-moles	joules (int)	8.3130
			g-moles	atm cm ³	82.057
			g-moles	atm liters	0.08205
			mm Hg-liters	mm Hg-liters	62.361
			g-moles	bar-liters	0.08314
			g-moles	kg/(cm ³)(liters)	0.08478
			lb-moles	atm-ft ³	1.314
			lb-moles	mm Hg-ft ³	998.9
Rankine	atm	ft ³	lb-moles	chu or pcu	1.9872
			lb-moles	Btu	1.9872
			lb-moles	hp-hr	0.0007805
			lb-moles	kw-hr	0.0005819
			lb-moles	atm-ft ³	0.7302
			lb-moles	in Hg-ft ³	21.85
			lb-moles	mm Hg-ft ³	555.0
			lb-moles	(lb)/(ft ²)/in ²	10.73
			lb-moles	ft-lb	1545.0
			lb-moles		

*Energy units are the product of pressure units and volume units.

TABLE 1-10 United States Customary System of Weights and Measures

Linear Measure	
12 inches (in) or (") = 1 foot (ft) or (')	
3 feet = 1 yard (yd)	
16.5 feet 5.5 yards	= 1 rod (rd)
5280 feet 320 rods	= 1 mile (mi)
1 mil = 0.001 inch	
<i>Nautical:</i>	
6080.2 feet = 1 nautical mile	
6 feet = 1 fathom	
120 fathoms = 1 cable length	
1 knot = 1 nautical mile per hour	
60 nautical miles = 1° of latitude	
Square Measure	
144 sq. inches (sq. in) or (in ²) or (□") = 1 sq. foot (ft ²) or (□')	
9 sq. feet (ft ²) (□') = 1 sq. yard (yd ²)	
30.25 sq. yards = 1 sq. rod, pole, or perch	
160 sq. rods = $\left\{ \begin{array}{l} 10 \text{ sq. chains} \\ 43,560 \text{ sq. ft} \end{array} \right\}$	= 1 acre
640 acres = 1 sq. mile = 1 section	
1 circular inch (area of circle of 1 inch diameter) = 0.7854 sq. inch	
1 sq. inch = 1.2732 circular inch	
1 circular mil = area of circle of 0.001 inch diameter	
1,000,000 circular mils = 1 circular inch	
Circular Measure	
60 seconds (") (sec) = 1 minute (min) or (')	
60 minutes (') = 1 degree (°)	
90 degrees (°) = 1 quadrant	
360 degrees (°) = 1 circumference	
57.29578 degrees $\left\{ \begin{array}{l} = 1 \text{ radian (rad.)} \\ = 57^{\circ} 17' 44.81'' \end{array} \right.$	
Volume Measure	
<i>Solid:</i>	
1728 cubic in (cu. in) (in ³) = 1 cubic foot (cu. ft)(ft ³)	
27 cu. ft = 1 cubic yard (cu. yd)	
<i>Dry Measure:</i>	
2 pints = 1 quart	
8 quarts = 1 peck	
4 pecks = 1 bushel	
1 United States Winchester bushel = 2150.42 cubic inches	
<i>Liquid:</i>	
4 gills = 1 pint (pt)	
2 pints = 1 quart (qt)	
4 quarts = 1 gallon (gal)	
7.4805 gallons = 1 cubic foot	
<i>Apothecaries' Liquid:</i>	
60 minims (min. or ℥) = 1 fluid dram or drachm	
8 drams (℥) = 1 fluid ounce	
16 ounces (oz. ℥) = 1 pint	
Avoirdupois Weight	
16 drams = 437.5 grains = 1 ounce (oz)	
16 ounces = 7000 grains = 1 pound (lb)	
100 pounds = 1 hundredweight (cwt)	
2000 pounds = 1 short ton: 2240 pounds = 1 long ton	
Troy Weight	
24 grains = 1 pennyweight (dwt)	
20 pennyweights = 1 ounce (oz)	
12 ounces = 1 pound (lb)	
Apothecaries' Weight	
20 grains (gr) = 1 scruple (℥)	
3 scruples = 1 dram (℥)	
8 drams = 1 ounce (℥)	
12 ounces = 1 pound (lb)	

TABLE 1-11 Temperature Conversion Formulas

$^{\circ}\text{F} = (^{\circ}\text{C} \times 5/9) + 32$
$^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times 5/9$
$^{\circ}\text{R} = ^{\circ}\text{F} + 459.69$
$^{\circ}\text{K} = ^{\circ}\text{C} + 273.15$
$^{\circ}\text{K} = ^{\circ}\text{R} \times 5/9$
Temperature difference, ΔT
$^{\circ}\text{F} = ^{\circ}\text{C} \times 9/5$

NOTE: An extensive table of temperature conversions may be found in the sixth edition of the *Handbook* (Table 1-12).

TABLE 1-12 Greek Alphabet

Alpha = A, α = A, a	Nu = N, ν = N, n
Beta = B, β = B, b	Xi = Ξ, ξ = X, x
Gamma = Γ, γ = G, g	Omicron = O, ο = O, o
Delta = Δ, δ = D, d	Pi = Π, π = P, p
Epsilon = Ε, ε = E, e	Rho = Ρ, ρ = R, r
Zeta = Ζ, ζ = Z, z	Sigma = Σ, σ = S, s
Eta = Η, η = E, e	Tau = Τ, τ = T, t
Theta = Θ, θ = Th, th	Upsilon = Υ, υ = U, u
Iota = Ι, ι = I, i	Phi = Φ, φ = Ph, ph
Kappa = Κ, κ = K, k	Chi = Χ, χ = Ch, ch
Lambda = Λ, λ = L, l	Psi = Ψ, ψ = Ps, ps
Mu = Μ, μ = M, m	Omega = Ω, ω = O, o

TABLE 1-13 Specific Gravity, Degrees Baumé, Degrees API, Degrees Twaddell, Pounds per Gallon, Pounds per Cubic Foot*

$$^{\circ}\text{Bé} = 145 - \frac{145}{\text{sp gr}} \quad (\text{heavier than H}_2\text{O}); \quad ^{\circ}\text{Bé} = \frac{140}{\text{sp gr}} - 130 \quad (\text{lighter than H}_2\text{O}); \quad ^{\circ}\text{Tw} = \frac{\text{sp gr } 60^{\circ}/60^{\circ}\text{F} - 1}{0.005} \quad ^{\circ}\text{API} = \frac{141.5}{\text{sp gr}} - 131.5$$

Sp gr 60°/ 60°	°Bé	°API	Lb per gal at 60°F wt in air	Lb per ft³ at 60°F wt in air	Sp gr 60°/ 60°	°Bé	°API	Lb per gal at 60°F wt in air	Lb per ft³ at 60°F wt in air	Sp gr 60°/ 60°	°Bé	°API	Lb per gal at 60°F wt in air	Lb per ft³ at 60°F wt in air	Sp gr 60°/ 60°	°Bé	°API	Lb per gal at 60°F wt in air	Lb per ft³ at 60°F wt in air
0.600	103.33	104.33	4.9929	37.350	0.700	70.00	70.64	5.8268	43.587	0.800	45.00	45.38	6.6606	49.825	0.900	25.56	25.72	7.4944	56.062
.605	101.40	102.38	5.0346	37.662	.705	68.58	69.21	5.8685	43.899	.805	43.91	44.28	6.7023	50.137	.905	24.70	24.85	7.5361	56.374
.610	99.51	100.47	5.0763	37.973	.710	67.18	67.80	5.9101	44.211	.810	42.84	43.19	6.7440	50.448	.910	23.85	23.99	7.5777	56.685
.615	97.64	98.58	5.1180	38.285	.715	65.80	66.40	5.9518	44.523	.815	41.78	42.12	6.7857	50.760	.915	23.01	23.14	7.6194	56.997
.620	95.81	96.73	5.1597	38.597	.720	64.44	65.03	5.9935	44.834	.820	40.73	41.06	6.8274	51.072	.920	22.17	22.30	7.6612	57.310
.625	94.00	94.90	5.2014	39.910	.725	63.10	63.67	6.0352	45.146	.825	39.70	40.02	6.8691	51.384	.925	21.35	21.47	7.7029	57.622
.630	92.22	93.10	5.2431	39.222	.730	61.78	62.34	6.0769	45.458	.830	38.67	38.98	6.9108	51.696	.930	20.54	20.65	7.7446	57.934
.635	90.47	91.33	5.2848	39.534	.735	60.48	61.02	6.1186	45.770	.835	37.66	37.96	6.9525	52.008	.935	19.73	19.84	7.7863	58.246
.640	88.75	89.59	5.3265	39.845	.740	59.19	59.72	6.1603	46.082	.840	36.67	36.95	6.9941	52.320	.940	18.94	19.03	7.8280	58.557
.645	87.05	87.88	5.3682	40.157	.745	57.92	58.43	6.2020	46.394	.845	35.68	35.96	7.0358	52.632	.945	18.15	18.24	7.8697	58.869
.650	85.38	86.19	5.4098	40.468	.750	56.67	57.17	6.2437	46.706	.850	34.71	34.97	7.0775	52.943	.950	17.37	17.45	7.9114	59.181
.655	83.74	84.53	5.4515	40.780	.755	55.43	55.92	6.2854	47.018	.855	33.74	34.00	7.1192	53.255	.955	16.60	16.67	7.9531	59.493
.660	82.12	82.89	5.4932	41.092	.760	54.21	54.68	6.3271	47.330	.860	32.79	33.03	7.1609	53.567	.960	15.83	15.90	7.9947	59.805
.665	80.53	81.28	5.5349	41.404	.765	53.01	53.47	6.3688	47.642	.865	31.85	32.08	7.2026	53.879	.965	15.08	15.13	8.0364	60.117
.670	78.96	79.69	5.5766	41.716	.770	51.82	52.27	6.4104	47.953	.870	30.92	31.14	7.2443	54.191	.970	14.33	14.38	8.0780	60.428
.675	77.41	78.13	5.6183	42.028	.775	50.65	51.08	6.4521	47.265	.875	30.00	30.21	7.2860	54.503	.975	13.59	13.63	8.1197	60.740
.680	75.88	76.59	5.6600	42.340	.780	49.49	49.91	6.4938	48.577	.880	29.09	29.30	7.3277	54.815	.980	12.86	12.89	8.1615	61.052
.685	74.38	75.07	5.7017	42.652	.785	48.34	48.75	6.5355	48.889	.885	28.19	28.39	7.3694	55.127	.985	12.13	12.15	8.2032	61.364
.690	72.90	73.57	5.7434	42.963	.790	47.22	47.61	6.5772	49.201	.890	27.30	27.49	7.4111	55.438	.990	11.41	11.43	8.2449	61.676
.695	71.44	72.10	5.7851	43.275	.795	46.10	46.49	6.6189	49.513	.895	26.42	26.60	7.4528	55.750	.995	10.70	10.71	8.2866	61.988
															1.000	10.00	10.00	8.3283	62.300
Sp gr 60°/ 60°	°Bé	°Tw	Lb per gal at 60°F wt in air	Lb per ft³ at 60°F wt in air	Sp gr 60°/ 60°	°Bé	°Tw	Lb per gal at 60°F wt in air	Lb per ft³ at 60°F wt in air	Sp gr 60°/ 60°	°Bé	°Tw	Lb per gal at 60°F wt in air	Lb per ft³ at 60°F wt in air	Sp gr 60°/ 60°	°Bé	°Tw	Lb per gal at 60°F wt in air	Lb per ft³ at 60°F wt in air
1.005	0.72	1	8.3700	62.612	1.255	29.46	51	10.4546	78.206	1.505	48.65	101	12.5392	93.800	1.755	62.38	151	14.6238	109.394
1.010	1.44	2	8.4117	62.924	1.260	29.92	52	10.4963	78.518	1.510	48.97	102	12.5809	94.112	1.760	62.61	152	14.6655	109.705
1.015	2.14	3	8.4534	63.236	1.265	30.38	53	10.5380	78.830	1.515	49.29	103	12.6226	94.424	1.765	62.85	153	14.7072	110.017
1.020	2.84	4	8.4950	63.547	1.270	30.83	54	10.5797	79.141	1.520	49.61	104	12.6643	94.735	1.770	63.08	154	14.7489	110.329
1.025	3.54	5	8.5367	63.859	1.275	31.27	55	10.6214	79.453	1.525	49.92	105	12.7060	95.047	1.775	63.31	155	14.7906	110.641
1.030	4.22	6	8.5784	64.171	1.280	31.72	56	10.6630	79.765	1.530	50.23	106	12.7477	95.359	1.780	63.54	156	14.8323	110.953
1.035	4.90	7	8.6201	64.483	1.285	32.16	57	10.7047	80.077	1.535	50.54	107	12.7894	95.671	1.785	63.77	157	14.8740	111.265
1.040	5.58	8	8.6618	64.795	1.290	32.60	58	10.7464	80.389	1.540	50.84	108	12.8310	95.983	1.790	63.99	158	14.9157	111.577
1.045	6.24	9	8.7035	65.107	1.295	33.03	59	10.7881	80.701	1.545	51.15	109	12.8727	96.295	1.795	64.22	159	14.9574	111.889
1.050	6.91	10	8.7452	65.419	1.300	33.46	60	10.8298	81.013	1.550	51.45	110	12.9144	96.606	1.800	64.44	160	14.9990	112.200
1.055	7.56	11	8.7869	65.731	1.305	33.89	61	10.8715	81.325	1.555	51.75	111	12.9561	96.918	1.805	64.67	161	15.0407	112.512
1.060	8.21	12	8.8286	66.042	1.310	34.31	62	10.9132	81.636	1.560	52.05	112	12.9978	97.230	1.810	64.89	162	15.0824	112.824
1.065	8.85	13	8.8703	66.354	1.315	34.73	63	10.9549	81.948	1.565	52.35	113	13.0395	97.542	1.815	65.11	163	15.1241	113.136
1.070	9.49	14	8.9120	66.666	1.320	35.15	64	10.9966	82.260	1.570	52.64	114	13.0812	97.854	1.820	65.33	164	15.1658	113.448
1.075	10.12	15	8.9537	66.978	1.325	35.57	65	11.0383	82.572	1.575	52.94	115	13.1229	98.166	1.825	65.55	165	15.2075	113.760
1.080	10.74	16	8.9954	67.290	1.330	35.98	66	11.0800	82.884	1.580	53.23	116	13.1646	98.478	1.830	65.77	166	15.2492	114.072
1.085	11.36	17	9.0371	67.602	1.335	36.39	67	11.1217	83.196	1.585	53.52	117	13.2063	98.790	1.835	65.98	167	15.2909	114.384
1.090	11.97	18	9.0787	67.914	1.340	36.79	68	11.1634	83.508	1.590	53.81	118	13.2480	99.102	1.840	66.20	168	15.3326	114.696
1.095	12.59	19	9.1204	68.226	1.345	37.19	69	11.2051	83.820	1.595	54.09	119	13.2897	99.414	1.845	66.41	169	15.3743	115.007
1.100	13.18	20	9.1621	68.537	1.350	37.59	70	11.2467	84.131	1.600	54.38	120	13.3313	99.725	1.850	66.62	170	15.4160	115.318
1.105	13.78	21	9.2038	68.849	1.355	37.99	71	11.2884	84.443	1.605	54.66	121	13.3730	100.037	1.855	66.83	171	15.4577	115.630
1.110	14.37	22	9.2455	69.161	1.360	38.38	72	11.3301	84.755	1.610	54.94	122	13.4147	100.349	1.860	67.04	172	15.4993	115.943
1.115	14.96	23	9.2872	69.473	1.365	38.77	73	11.3718	85.067	1.615	55.22	123	13.4564	100.661	1.865	67.25	173	15.5410	116.255
1.120	15.54	24	9.3289	69.785	1.370	39.16	74	11.4135	85.379	1.620	55.49	124	13.4981	100.973	1.870	67.46	174	15.5827	116.567
1.125	16.11	25	9.3706	70.097	1.375	39.55	75	11.4552	85.691	1.625	55.77	125	13.5398	101.285	1.875	67.67	175	15.6244	116.879
1.130	16.68	26	9.4123	70.409	1.380	39.93	76	11.4969	86.003	1.630	56.04	126	13.5815	101.597	1.880	67.87	176	15.6661	117.191
1.135	17.25	27	9.4540	70.721	1.385	40.31	77	11.5386	86.315	1.635	56.32	127	13.6232	101.909	1.885	68.08	177	15.7078	117.503
1.140	17.81	28	9.4957	71.032	1.390	40.68	78	11.5803	86.626	1.640	56.59	128							

TABLE 1-14 Fundamental Physical Constants

1 sec = 1.00273791 sidereal seconds	sec = mean solar second
$g_0 = 9.80665 \text{ m/sec}^2$	Definition: g_0 = standard gravity
1 liter = 0.001 cu. m	
1 atm = 101,325 newtons/sq m	Definition: atm = standard atmosphere
1 mm Hg (pressure) = $(1/760) \text{ atm}$ = 133.3224 newtons/sq m	mm Hg (pressure) = standard millimeter mercury
1 int ohm = 1.000495 ± 0.000015 abs ohm	int = international; abs = absolute
1 int amp = 0.999835 ± 0.000025 abs amp	amp = ampere
1 int coul = 0.999835 ± 0.000025 abs coul	coul = coulomb
1 int volt = 1.000330 ± 0.000029 abs volt	
1 int watt = 1.000165 ± 0.000052 abs watt	
1 int joule = 1.000165 ± 0.000052 abs joule	
$T_{0^\circ\text{C}} = 273.150 \pm 0.010^\circ\text{K}$	Absolute temperature of the ice point, 0°C
$(PV)_{0^\circ\text{C}}^{P=0} = (RT)_{0^\circ\text{C}} = 2271.16 \pm 0.04$ abs joule/mole = 22,414.6 ± 0.4 cu. cm atm/mole = 22.4146 ± 0.0004 liter atm/mole	PV product for ideal gas at 0°C
$R = 8.31439 \pm 0.00034$ abs joule/deg mole = 1.98719 ± 0.00013 cal/deg mole = 82.0567 ± 0.0034 cu. cm atm/deg mole = 0.0820567 ± 0.0000034 liter atm/deg mole	R = gas constant per mole
$\ln 10 = 2.302585$	$\ln$ = natural logarithm (base e)
$R \ln 10 = 19.14460 \pm 0.00078$ abs joule/deg mole = 4.57567 ± 0.00030 cal/deg mole	
$N = (6.02283 \pm 0.0022) \times 10^{23}$ /mole	N = Avogadro number
$h = (6.6242 \pm 0.0044) \times 10^{-34}$ joule sec	h = Planck constant
$c = (2.99776 \pm 0.00008) \times 10^8$ m/sec	c = velocity of light
$(h^2/8\pi^2k) = (4.0258 \pm 0.0037) \times 10^{-39}$ g sq cm deg	Constant in rotational partition function of gases
$(h/8\pi^2c) = (2.7986 \pm 0.0018) \times 10^{-39}$ g cm	Constant relating wave number and moment of inertia
$Z = Nhc = 11.9600 \pm 0.0036$ abs joule cm/mole = 2.85851 ± 0.0009 cal cm/mole	Z = constant relating wave number and energy per mole
$(Z/R) = (hc/k) = c_2 = 1.43847 \pm 0.00045$ cm deg	c_2 = second radiation constant
$\mathcal{F} = 96,501.2 \pm 10.0$ int coul/g-equiv or int joule/int volt g-equiv = 96,485.3 ± 10.0 abs coul/g-equiv or abs joule/abs volt g-equiv = 23,068.1 ± 2.4 cal/int volt g-equiv = 23,060.5 ± 2.4 cal/abs volt g-equiv	$\mathcal{F}$ = Faraday constant
$e = (1.60199 \pm 0.00060) \times 10^{-19}$ abs coul = $(1.60199 \pm 0.00060) \times 10^{-20}$ abs emu = $(4.80239 \pm 0.00180) \times 10^{-10}$ abs esu	e = electronic charge
1 int electron-volt/molecule = 96,501.2 ± 10 int joule/mole = 23,068.1 ± 2.4 cal/mole	
1 abs electron-volt/molecule = 96,485.3 $\pm 10.$ abs joule/mole = 23,060.5 ± 2.4 cal/mole	
1 int electron-volt = $(1.60252 \pm 0.00060) \times 10^{-12}$ erg	
1 abs electron-volt = $(1.60199 \pm 0.00060) \times 10^{-12}$ erg	
$hc = (1.23916 \pm 0.00032) \times 10^{-4}$ int electron-volt cm = $(1.23957 \pm 0.00032) \times 10^{-4}$ abs electron-volt cm	Constant relating wave number and energy per molecule
$k = (8.61442 \pm 0.00100) \times 10^{-5}$ int electron-volt/deg = $(8.61727 \pm 0.00100) \times 10^{-5}$ abs electron-volt/deg = $(R/N) = (1.38048 \pm 0.00050) \times 10^{-23}$ joule/deg	k = Boltzmann constant
1 IT cal = $(1/860) = 0.00116279$ int watt-hr = 4.18605 int joule = 4.18674 abs joule = 1.000654 cal	Definition of IT cal: IT = International steam tables
1 cal = 4.1840 abs joule = 4.1833 int joule = 41.2929 ± 0.0020 cu. cm atm = 0.0412929 ± 0.0000020 liter atm	cal = thermochemical calorie Definition: cal = thermochemical calorie
1 IT cal/g = 1.8 Btu/lb	
1 Btu = 251.996 IT cal = 0.293018 int watt-hr = 1054.866 int joule = 1055.040 abs joule = 252.161 cal	Definition of Btu: Btu = IT British Thermal Unit
1 horsepower = 550 ft-lb (wt)/sec = 745.578 int watt = 745.70 abs watt	
1 in = $(1/0.3937) = 2.54$ cm	
1 ft = 0.304800610 m	
1 lb = 453.5924277 g	
1 gal = 231 cu. in = 0.133680555 cu. ft = 3.785412×10^{-3} cu. m = 3.785412 liter	cal = thermochemical calorie Definition of horsepower (mechanical): lb (wt) = weight of 1 lb at standard gravity Definition of in: in = U.S. inch ft = U.S. foot (1 ft = 12 in) Definition: lb = avoirdupois pound Definition: gal = U.S. gallon

CONVERSION OF VALUES FROM U.S. CUSTOMARY UNITS TO SI UNITS

American engineers are probably more familiar with the magnitude of physical entities in U.S. customary units than in SI units. Consequently, errors made in the conversion from one set of units to the other may go undetected. The following six examples will show how to convert the elements in six dimensionless groups. Proper conversions will result in the same numerical value for the dimensionless number. The dimensionless numbers used as examples are the Reynolds, Prandtl, Nusselt, Grashof, Schmidt, and Archimedes numbers.

Table 1-7 provides a number of useful conversion factors. To make a conversion of an element in U.S. customary units to SI units, one multiplies the value of the U.S. customary unit, found on the left side in the table, by the equivalent value on the right side. For example, to convert 10 British thermal units to joules, one multiplies 10 by 1054.4 to obtain 10544 joules.

In each example, the initial values of the factors are expressed in U.S. customary units, and the dimensionless value is calculated. Then the factors are converted to SI units, and the dimensionless value is recalculated. The two dimensionless values will be approximately the same. (Small variations occur due to the number of significant figures carried in the solution.)

Example 1. Calculation of a Reynolds Number

$$N_{Re} = \frac{DV\rho}{\mu}$$

U.S. customary units

$$D = 3 \text{ in.} = \frac{3}{12} \text{ ft}$$

$$V = 6 \text{ ft/s}$$

$$\rho = 0.08 \text{ lbm/ft}^3$$

$$\mu = 0.015 \text{ cp} = (0.015)(0.000672) \text{ lbm/ft-s}$$

$$N_{Re} = \frac{(3/12)(6)(0.08)}{(0.015)(0.000672)} = 11,904$$

SI units

$$D = (3)(0.0254) \text{ m}$$

$$V = (6)(0.3048) \text{ m/s}$$

$$\rho = (0.08)(16.018) \text{ kg/m}^3$$

$$\mu = (0.015)(0.001) \text{ kg/m-s}$$

$$N_{Re} = \frac{(3 \times 0.0254)(6 \times 0.3048)(0.08 \times 16.018)}{(0.015)(0.001)} = 11,904$$

Example 2. Calculation of a Prandtl Number

$$N_{Pr} = \frac{C_p \mu}{k}$$

U.S. customary units

$$\gamma_p = 0.47 \text{ Btu/lbm}^\circ\text{F}$$

$$\mu = 15 \text{ centipoise} = (15)(0.000672)(3600) \text{ lbm/ft-hr}$$

$$k = 0.065 \text{ Btu/hr-ft}^2 (^\circ\text{F/ft})$$

$$N_{Pr} = \frac{(0.47)(15 \times 0.000672 \times 3600)}{0.065} = 262.4$$

SI units

$$\gamma = (0.47)(4184) \text{ J/kg}^\circ\text{C}$$

$$\mu = (15)(0.001) \text{ kg/m-s}$$

$$k = (0.065)(1.728) \text{ J/s-m}^2 (^\circ\text{C/m})$$

$$N_{Pr} = \frac{(0.47)(4184)(15)(0.001)}{(0.065)(1.728)} = 262.6$$

(Difference due to rounding)

Example 3. Calculation of a Nusselt Number

$$N_{Nu} = \frac{hD}{k}$$

U.S. customary units

$$h = 200 \text{ Btu/hr-ft}^2\text{-}^\circ\text{F}$$

$$D = 1.5 \text{ in.} = 1.5/12 \text{ ft}$$

$$k = 0.07 \text{ Btu/hr-ft}^2 (^\circ\text{F/ft})$$

$$N_{Nu} = \frac{(200)(1.5/12)}{0.07} = 357.1$$

SI units

$$h = (200)(5.678) \text{ J/(s-m}^2\text{-}^\circ\text{C)}$$

$$D = (1.5)(0.0254) \text{ m}$$

$$k = (0.07)(1.728) \text{ J/s-m}^2 (^\circ\text{C/m})$$

$$N_{Nu} = \frac{(200)(5.678)(1.5)(0.0254)}{(0.07)(1.728)} = 357.7$$

(Difference due to rounding)

Example 4. Calculation of a Grashof Number

$$N_{Gr} = L^3 \rho^2 g \beta (\Delta T) / \mu^2$$

U.S. Customary units

$$L = 3 \text{ ft}$$

$$\rho = 0.0725 \text{ lbm/ft}^3$$

$$g = 32.174 \text{ ft/s}^2$$

$$\beta = 0.00168/^\circ\text{R}$$

$$\Delta T = 99^\circ\text{R}$$

$$\mu = 0.019 \text{ centipoise} = 0.019 \times 0.000672 \text{ lbm/ft-s}$$

$$= 1.277 \times 10^{-5} \text{ lbm/ft-s}$$

$$N_{Gr} = \frac{(3)^3 (0.0725)^2 (32.174) (0.00168) (99)}{(1.277 \times 10^{-5})^2} = 4.66 \times 10^9$$

SI units

$$L = (3)(0.3048) = 0.9144 \text{ m}$$

$$\rho = (0.0725)(16.018) = 1.1613 \text{ kg/m}^3$$

$$g = 9.807 \text{ m/s}^2$$

$$\beta = (0.00168)/(1.8) = 0.000933/^\circ\text{K}$$

$$\Delta T = (99)(1.8) = 178.2^\circ\text{K}$$

$$\mu = (0.019)(0.001) = 1.9 \times 10^{-5} \text{ kg/m-s}$$

$$N_{Gr} = \frac{(0.9144)^3 (1.1613)^2 (9.807) (0.000933) (178.2)}{(1.9 \times 10^{-5})^2} = 4.66 \times 10^9$$

Example 5. Calculation of a Schmidt Number

$$N_{Sc} = \frac{\mu}{\rho D}$$

U.S. customary units

$$\mu = 0.02 \text{ centipoise} = (0.02)(2.42) \text{ lbm/ft-hr}$$

$$\rho = 0.08 \text{ lbm/ft}^3$$

$$D = 1.0 \text{ ft}^2/\text{hr (diffusivity)}$$

$$N_{Sc} = \frac{(0.02)(2.42)}{(0.08)(1.0)} = 0.605$$

SI units

$$\mu = (0.02)(0.001) \text{ kg/m-s}$$

$$\rho = (0.08)(16.02) \text{ kg/m}^3$$

$$D = (1.0)(2.58 \times 10^{-5}) \text{ m}^2/\text{s}$$

$$N_{Sc} = \frac{(0.02)(0.001)}{(0.08)(16.02)(1.0)(2.58 \times 10^{-5})} = 0.605$$

Example 6. Calculation of an Archimedes Number

$$N_{Ar} = \frac{d^3 \rho (\rho_p - \rho_f) g}{\mu^2}$$

U.S. customary units

$$d = 2 \text{ mm} = 2/[(1000)(0.3048)] = 0.00656 \text{ ft}$$

$$\rho_f = 0.0175 \text{ lbm/ft}^3$$

$$\rho_p = 168.5 \text{ lbm/ft}^3$$

$$g = 32.174 \text{ ft/s}^2$$

$$\mu = 0.04 \text{ centipoise} = 0.04 \times 0.000672 = 2.688 \times 10^{-5} \text{ lbm/ft-s}$$

$$N_{Ar} = \frac{(0.00656)^3 (0.0175) (168.5 - 0.017) (32.174)}{(2.688 \times 10^{-5})^2} = 37,064$$

SI units

$$d = 2/1000 \text{ m}$$

$$\rho_p = 168.5 \times 16.02 = 2699.37 \text{ kg/m}^3$$

$$\rho_f = 0.0175 \times 16.02 = 0.2804 \text{ g/m}^3$$

$$g = 9.807 \text{ m/s}^2$$

$$\mu = 0.04 \times 0.001 = 4 \times 10^{-5} \text{ kg/m-s}$$

$$N_{Ar} = \frac{(2/1000)^3 (0.2804) (2699.37 - 0.28) (9.807)}{(4 \times 10^{-5})^2} = 37,118$$

(Difference due to rounding)

SECTION 2

Physical and
Chemical Data

PERRY'S CHEMICAL ENGINEERS' HANDBOOK

8TH EDITION



BRUCE E. POLING, GEORGE H. THOMSON
DANIEL G. FRIEND, RICHARD L. ROWLEY
W. VINCENT WILDING

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GENERAL REFERENCES

Considerations of reader interest, space availability, the system or systems of units employed, copyright considerations, etc., have all influenced the revision of material in previous editions for the present edition. Reference is made at numerous places to various specialized works and, when appropriate, to more general works. A listing of general works may be useful to readers in need of further information.

ASHRAE Handbook—Fundamentals, SI edition, ASHRAE, Atlanta, 2005; Benedek, P., and F. Olti, *Computer-Aided Chemical Thermodynamics of Gases and Liquids*, Wiley, New York, 1985; Brule, M. R., L. L. Lee, and K. E. Starling, *Chem. Eng.*, **86**, 25, Nov. 19, 1979, pp. 155–164; Cox, J. D., and G. Pilcher, *Thermochemistry of Organic and Organometallic Compounds*, Academic Press, New York, 1970; Cox, J. D., D. D. Wagman, and V. A. Medvedev, *CODATA Key Values for Thermodynamics*, Hemisphere Publishing Corp., New York, 1989; Daubert, T. E., R. P. Danner, H. M. Sibel, and C. C. Stebbins, *Physical and Thermodynamic Properties of Pure Chemicals: Data Compilation*, Taylor & Francis, Washington, 1997; Domalski, E. S., and E. D. Hearing, Heat capacities and entropies of organic compounds in the condensed phase, vol. 3, *J. Phys. Chem. Ref. Data* 25(1):1–525, Jan-Feb 1996; Dykij, J., and M. Repas, Saturated vapor pressures of organic compounds, *Veda*, Bratislava, 1979 (Slovak); Dykij, J., M. Repas, and J. Svoboda, Saturated vapor pressures of organic compounds, *Veda*, Bratislava, 1984 (Slovak); Glushko, V. P., Ed., *Thermal Constants of Compounds*, Issues 1–X, Moscow, 1965–1982 (Russian only); Gmehling, J., *Azeotropic Data*, 2 vols., VCH Weinheim, Germany, 1994; Gmehling, J., and U. Onken, *Vapor-Liquid Equilibrium Data Collection*, Dechema Chemistry Data Series, Frankfurt, 1977–1978; International Data Series, *Selected Data on Mixtures*, Series A: Thermodynamics Research Center, National Institute of Standards and Technology, Boulder, Colo.; Kaye, S. M., *Encyclopedia of Explosives and Related Items*, U.S. Army R&D command, Dover, N.J., 1980; King, M. B., *Phase Equilibrium in Mixtures*, Pergamon, Oxford, 1969; Landolt-Boernstein, Numerical Data and Functional Relationships in Science and Technology (New Series), <http://www.springeronline.com/sgw/cda/frontpage/0,11855,4-10113-2-95859-0,00.html>; Lide, D. R., *CRC Handbook of Chemistry and Physics*, 86th ed., CRC Press, Boca Raton, Fla., 2005; Lyman, W. J., W. F. Reehl, and D. H. Rosenblatt, *Handbook of Chemical Property Estimation Methods*, McGraw-Hill, New York, 1990; Majer, V., and V. Svoboda, *Enthalpies of Vaporization of Organic Compounds: A Critical Review and Data Compilation*, Blackwell Science, 1985; Majer V., V. Svoboda, and J. Pick, *Heats of Vaporization of Fluids*, Elsevier, Amsterdam, 1989 (general discussion); Marsh, K. N., *Recommended Reference Materials for the Realization of Physicochemical Properties*, Blackwell Science, 1987; NIST-IUPAC Solubility Data Series, Pergamon Press, <http://www.iupac.org/publications/ci/1999/march/solubility.html>; Ohse, R. W., and H. von Tippelskirch, *High Temp.—High Press.*, **9**:367–385, 1977; Ohse, R. W., *Handbook of Thermodynamic and Transport Properties of Alkali Metals*, Blackwell Science Pubs., Oxford, England, 1985; Pedley, J. B., R. D. Naylor, and S. P. Kirby, *Thermochemical Data of Organic Compounds*, Chapman and Hall, New York, 1986; *Physical Property Data for the Design Engineer*, Hemisphere, New York, 1989; Poling, B. E., J. M. Prausnitz, and J. P. O'Connell, *The Properties of*

Gases and Liquids, 5th ed., McGraw-Hill, New York, 2001; Rothman, D., et al., Max Planck Inst. f. Stromungsforschung, Ber 6, 1978; Smith, B. D., and R. Srivastava, *Thermodynamic Data for Pure Compounds*, Part A: *Hydrocarbons and Ketones*, Elsevier, Amsterdam, 1986, Physical sciences data 25, http://www.elsevier.com/wps/find/bookseriesdescription.librarians/BS_PSD/description; Sterbacek, Z., B. Biskup, and P. Tausk, *Calculation of Properties Using Corresponding States Methods*, Elsevier, Amsterdam, 1979; Stull, D. R., E. F. Westrum, and G. C. Sink, *The Chemical Thermodynamics of Organic Compounds*, Wiley, New York, 1969; *TRC Thermodynamic Tables—Hydrocarbons*, Thermodynamics Research Center, National Institute of Standards and Technology, Boulder, Colo.; *TRC Thermodynamic Tables—Non-Hydrocarbons*, Thermodynamics Research Center, National Institute of Standards and Technology, Boulder, Colo.; Young, D. A., “Phase Diagrams of the Elements,” UCRL Rep. 51902, 1975 republished in expanded form by the University of California Press, 1991; Zabransky, M., V. Ruzicka, Jr., V. Majer, and E. S. Domalski, Heat Capacity of Liquids: Critical Review and Recommended Values, *J. Phys. Chem. Ref. Data*, Monograph No. 6, 1996.

CRITICAL DATA ARE COMPILED IN:

Ambrose, D., “Vapor-Liquid Critical Properties,” N. P. L. Teddington, Middlesex, Rep. 107, 1980; Kudchaker, A. P., G. H. Alani, and B. J. Zwolinski, *Chem. Revs.* **68**:659–735, 1968; Matthews, J. F., *Chem. Revs.* **72**:71–100, 1972; Simmrock, K., R. Janowsky, and A. Ohnsorge, *Critical Data of Pure Substances*, Parts 1 and 2, Dechema Chemistry Data Series, 1986; Other recent references for critical data can be found in Lide, D. R., *CRC Handbook of Chemistry and Physics*, 86th ed., CRC Press, Boca Raton, Fla., 2005.

PUBLICATIONS ON THERMOCHEMISTRY

Pedley, J. B., *Thermochemical Data and Structures of Organic Compounds*, 1, Thermodynamic Research Center, Texas A&M Univ., 1994 (976 pp., 3000 cpds.); Frenkel, M., et al., *Thermodynamics of Organic Compounds in the Gas State*, 2 vols., Thermodynamic Research Center, Texas A&M Univ., 1994 (1825 pp., 2000 cpds.); Barin, I., *Thermochemical Data of Pure Substances*, 2 vols., 2d ed., VCH Weinheim, Germany, 1993 (1834 pp., 2400 substances); Gurvich, L.V., et al., *Thermodynamic Properties of Individual Substances*, 3 vols., 4th ed., Hemisphere, New York, 1989, 1990, and 1993 (2520 pp.); Lide, D. R., and G. W. A. Milne, *Handbook of Data on Organic Compounds*, 7 vols., 3d ed., Chemical Rubber, Miami, 1993 (7000 pp.); Daubert, T. E., et al., *Physical and Thermodynamic Properties of Pure Chemicals: Data Compilation*, extant 1995, Taylor & Francis, Bristol, Pa., 1995; Database 11, NIST, Gaithersburg, Md. U.S. Bureau of Mines publications include Bulletins 584, 1960 (232 pp.); 592, 1961 (149 pp.); 595, 1961 (68 pp.); 654, 1970 (26 pp.); Chase, M. W., et al., *JANAF Thermochemical Tables*, 3d ed., *J. Phys. Chem. Ref. Data* 14 suppl 1, 1986 (1896 pp.); *Journal of Physical and Chemical Reference Data* is available online at <http://listserv.nd.edu/cgi-bin/wa?A2=ind0501&L=pamnet&F=&S=&P=8490> and at <http://www.nist.gov/srd/reprints.htm>

PHYSICAL PROPERTIES OF PURE SUBSTANCES

TABLE 2-1 Physical Properties of the Elements and Inorganic Compounds*

Abbreviations Used in the Table

a., acid	atm., atmosphere or 760 mm. of mercury pressure	d. 50, decomposes at 50°C; 50 d., melts at 50°C with decomposition	hyg., hygroscopic	pl., plates	trig., trigonal
A., specific gravity with reference to air = 1	bk., black	delq., deliquescent	i., insoluble	pr., prisms or prismatic	v., very
abs., absolute	brn., brown	dil., dilute	ign., ignites	pyr., pyridine	vac., in vacuo
ac., acetic acid	bz., benzene	dk., dark	lq., liquid	rhb., rhombic (orthorhombic)	vl., violet
act., acetone	c., cold	eff., effloresces or efflorescent	lt., light	s., soluble	volt., volatile or volatilizes
al., 95 percent ethyl alcohol	cb., cubic	et., ethyl ether	m. al., methyl alcohol	satd., saturated	wh., white
alk, alkali (i.e., aq. NaOH or KOH)	cc, cubic centimeter	expl., explodes	mn., monoclinic	sl., slightly	yel., yellow
am., amyl (C ₅ H ₁₁)	chl., chloroform	gel., gelatinous	nd., needles	soln., solution	∞, soluble in all proportions
amor., amorphous	col., colorless or white	gly., glycerol (glycerin)	NH ₃ , liquid ammonia	subl., sublimates	<, less than
anh., anhydrous	conc., concentrated	gn., green	NH ₄ OH, ammonium hydroxide	sulf., sulfides	>, greater than
aq., aqueous or water	cr., crystals or crystalline	h., hot	oct., octahedral	tart. a., tartaric acid	42±, about or near 42
aq. reg., aqua regia	d., decomposes	hex., hexagonal	or., orange	tet., tetragonal	–3H ₂ O, 100, loses 3 moles of water per formula weight at 100°C
	D., specific gravity with reference to hydrogen = 1		pd., powder	tri., triclinic	

Formula weights are based upon the International Atomic Weights in "Atomic Weights of the Elements 2001," *Pure Appl. Chem.*, **75**, 1107, 2003, and are computed to the nearest hundredth.

Refractive index, where given for a uniaxial crystal, is for the ordinary (ω) ray; where given for a biaxial crystal, the index given is for the median (β) value. Unless otherwise specified, the index is given for the sodium D-line (λ = 589.3 mμ).

Specific gravity values are given at room temperatures (15 to 20 °C) unless otherwise indicated by the small figures which follow the value: thus, "5.6^{18°}" indicates a specific gravity of 5.6 for the substance at 18 °C referred to water at 4°C. In this table the values for the specific gravity of gases are given with reference to air (A) = 1, or hydrogen (D) = 1.

Melting point is recorded in a certain case as "82 d." and in some other case as "d. 82," the distinction being made in this manner to indicate that the former is a melting point with decomposition at 82°C, while in the latter decomposition only occurs at 82 °C. Where a value such as "–2H₂O, 82" is given it indicates loss of 2 moles of water per formula weight of the compound at a temperature of 82 °C.

Boiling point is given at atmospheric pressure (760 mm. of mercury) unless otherwise indicated; thus, "82^{15mm}" indicates the boiling point is 82°C when the pressure is 15 mm.

Solubility is given in parts by weight (of the formula shown at the extreme left) per 100 parts by weight of the solvent; the small superscript indicates the temperature. In the case of gases the solubility is often expressed in some manner as "5^{10°} cc" which indicates that at 10 °C, 5 cc. of the gas are soluble in 100 g of the solvent. The symbols of the common mineral acids: H₂SO₄, HNO₃, HCl, etc., represent dilute aqueous solutions of these acids. See also special tables on Solubility.

REFERENCES: The information given in this table has been collected mainly from the following sources: Mellor, *A Comprehensive Treatise on Inorganic and Theoretical Chemistry*, Longmans, New York, 1922. Abegg, *Handbuch der anorganischen Chemie*, S. Hirzel, Leipzig, 1905. Gmelin-Kraut, *Handbuch der anorganischen Chemie*, 7th ed., Carl Winter, Heidelberg; 8th ed., Verlag Chemie, Berlin, 1924. Friend, *Textbook of Inorganic Chemistry*, Griffin, London, 1914. Winchell, *Microscopic Character of Artificial Inorganic Solid Substances or Artificial Minerals*, Wiley, New York, 1931. *International Critical Tables*, McGraw-Hill, New York, 1926. *Tables annuelles internationales de constantes et donnees numeriques*, McGraw-Hill, New York. *Annual Tables of Physical Constants and Numerical Data*, National Research Council, Princeton, N.J., 1943. Comey and Hahn, *A Dictionary of Chemical Solubilities*, Macmillan, New York, 1921. Seidell, *Solubilities of Inorganic and Metal Organic Compounds*, Van Nostrand, New York, 1940.

Name	Formula	Formula weight	Color, crystalline form and refractive index	Specific gravity	Melting point, °C	Boiling point, °C	Solubility in 100 parts		
							Cold water	Hot water	Other reagents
Aluminum	Al	26.98	silv., cb.	2.70 ^{20°}	660	2056	i.	i.	s. HCl, H ₂ SO ₄ , alk.
acetate, normal	Al(C ₂ H ₃ O ₂) ₃	204.11	wh. pd.		d. 200		s.	d.	
acetate, basic	Al(OH)(C ₂ H ₃ O ₂) ₂	162.08	wh., amor.		d.		i.		s.a.; i. NH ₄ salts
bromide	AlBr ₃	266.69	trig.	3.01 ^{25°} ₄	97.5	268	s.		s.al., act., CS ₂
bromide	AlBr ₃ ·6H ₂ O	374.78	col., delq. cr.		d. 100		s.	s.	s. al., CS ₂
carbide	Al ₄ C ₃	143.96	yel., hex., 2.70	2.95	d. >2200		d. to CH ₄		s. a.; i. act.
chloride	AlCl ₃	133.34	wh., delq., hex.	2.44 ^{25°} ₄	194 ^{5.2atm.}	182.7 ^{752mm} ; subl. 178	69.87 ^{13°}	s. d.	s. et., chl., CCl ₄ ; i. bz.
chloride	AlCl ₃ ·6H ₂ O	241.43	col., delq., trig., 1.560				400	v. s.	50 al.; s. et.
fluoride (fluellite)	AlF ₃ ·H ₂ O	101.99	col., rhb., 1.490	2.17	d.		sl. s.		
fluoride	Al ₂ F ₆ ·7H ₂ O	294.06	wh., cr. pd.		–4H ₂ O, 120	–6H ₂ O, 250	i.	sl. s.	
hydroxide	Al(OH) ₃	78.00	wh., mn.	2.42	–2H ₂ O, 300		0.000104 ^{18°}	i.	s. a., alk.; i. a.
nitrate	Al(NO ₃) ₃ ·9H ₂ O	375.13	rhb., delq.		73	d. 134	v. s.	v. s. d.	s. al., CS ₂
nitride	Al ₂ N ₂	81.98	yel., hex.	3.05 ^{25°} ₄	2150 ^{4atm.}	d. >1400	d. slowly		s. alk. d.
oxide	Al ₂ O ₃	101.96	col., hex., 1.67–8	3.99	1999 to 2032		i.	i.	v. sl. s. a., alk.
oxide (corundum)	Al ₂ O ₃	101.96	wh., trig., 1.768	4.00	1999 to 2032	2210	i.	i.	v. sl. s. a., alk.
phosphate	AlPO ₄	121.95	col., hex.	2.59			i.	i.	s. a., alk.; i. ac.

*By N. A. Lange, Ph.D., Handbook Publishers, Inc., Sandusky, Ohio. Abridged from table of Physical Constants of Inorganic Compounds in *Lange's Handbook of Chemistry*.

TABLE 2-1 Physical Properties of the Elements and Inorganic Compounds (Continued)

Name	Formula	Formula weight	Color, crystalline form and refractive index	Specific gravity	Melting point, °C	Boiling point, °C	Solubility in 100 parts		
							Cold water	Hot water	Other reagents
Aluminum (<i>Cont.</i>)									
potassium silicate (muscovite)	$3\text{Al}_2\text{O}_3 \cdot \text{K}_2\text{O} \cdot 6\text{SiO}_2 \cdot 2\text{H}_2\text{O}$	796.61	mn., 1.590	2.9	d.		i.		
potassium silicate (orthoclase)	$\text{Al}_2\text{O}_3 \cdot \text{K}_2\text{O} \cdot 6\text{SiO}_2$	556.66	col., mn., 1.524	2.56	1450 (1150)		i.		
Aluminum potassium tartrate	$\text{AlK}(\text{C}_4\text{H}_4\text{O}_6)_2$	362.22	col.				s.	s.	
sodium fluoride (cryolite)	$\text{AlF}_3 \cdot 3\text{NaF}$	209.94	wh., mn., 1.3389	2.90	1000		sl. s.		i. HCl
sodium silicate	$\text{Al}_2\text{O}_3 \cdot \text{Na}_2\text{O} \cdot 6\text{SiO}_2$	524.44	col., tri., 1.529	2.61	1100		i.	i.	d. a.
sulfate	$\text{Al}_2(\text{SO}_4)_3$	342.15	wh. cr.	2.71	d. 770		31.3 ^{0°}	89 ^{100°}	
Alum, ammonium (tschermigite)	$\text{Al}_2(\text{SO}_4)_3 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 24\text{H}_2\text{O}$	906.66	col., oct., 1.4594	1.64 $\frac{20^\circ}{4}$	93.5	-20H ₂ O, 120; -24H ₂ O, 200	3.9 ^{0°}	∞ 100°	i. al.
ammonium chrome	$\text{Cr}_2(\text{SO}_4)_3 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 24\text{H}_2\text{O}$	956.69	gn. or vl., oct., 1.4842	1.72	100 d.		21.2 ^{25°}		s. al.
ammonium iron	$\text{Fe}_2(\text{SO}_4)_3 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 24\text{H}_2\text{O}$	964.38	vl., oct., 1.485	1.71	40		124 ^{25°}		i. al.
potassium (kalinite)	$\text{Al}_2(\text{SO}_4)_3 \cdot \text{K}_2\text{SO}_4 \cdot 24\text{H}_2\text{O}$	948.78	col., mn., 1.4564	1.76 $\frac{26^\circ}{4}$	92	-18H ₂ O, 64.5	5.7 ^{0°}	∞ ^{93°}	
potassium chrome	$\text{Cr}_2(\text{SO}_4)_3 \cdot \text{K}_2\text{SO}_4 \cdot 24\text{H}_2\text{O}$	998.81	red or gn., cb., 1.4814	1.83	89		20	50	i. al.
sodium	$\text{Al}_2(\text{SO}_4)_3 \cdot \text{Na}_2\text{SO}_4 \cdot 24\text{H}_2\text{O}$	916.56	col., oct., 1.4388	1.675 $\frac{20^\circ}{4}$	61		106.4 ^{0°}	121.7 ^{45°}	i. al.
Ammonia†	NH_3	17.03	col. gas, 1.325 (lq.)	0.817 ^{-79°} 0.5971 (A)	-77.7	-33.4	89.9 ^{0°}	7.4 ^{96°}	14.8 ^{20°} al.; s. et.
Ammonium acetate	$\text{NH}_4\text{C}_2\text{H}_3\text{O}_2$	77.08	wh., hyg. cr.	1.073	114	d.	148 ^{4°}		s. al.; sl. s. act.
auricyanide	$\text{NH}_4\text{CN} \cdot \text{Au}(\text{CN})_3 \cdot \text{H}_2\text{O}$	337.09	pl.		d. 200		s.	v. s.	i. al.
bicarbonate	NH_4HCO_3	79.06	mn. or rhb., 1.5358	1.573	d. 35–60		11.9 ^{0°}	27 ^{30°}	i. al.
bromide	NH_4Br	97.94	col., cb., 1.7108	2.327 $\frac{15^\circ}{4}$	subl. 542		68 ^{10°}	145.6 ^{100°}	s. al., et., act.
carbonate	$(\text{NH}_4)_2\text{CO}_3 \cdot \text{H}_2\text{O}$	114.10	col. pl.		d. 58		100 ^{15°}		i. al., CS ₂ , NH ₃
carbonate, carbamate	NH_4HCO_3	157.13	wh. cr.		subl.		25 ^{15°}	67 ^{65°}	
carbonate, sesqui-	$\text{NH}_4\text{CO}_2\text{NH}_4\ddagger$ $(\text{NH}_4)_2\text{CO}_3$	272.21	wh.		d.		20 ^{15°}	50 ^{49°}	
chloride (salammoniac)	NH_4Cl	53.49	wh., cb., 1.639, 1.6426	1.53 ^{17°}	d. 350	subl. 520	29.4 ^{0°}	77.3 ^{100°}	s. NH ₃ ; sl. s. al., m. al.
chloroplatinate	$(\text{NH}_4)_2\text{PtCl}_6$	443.87	yel., cb.	3.065	d.		0.7 ^{15°}	1.25 ^{100°}	0.005 al.
chloroplatinite	$(\text{NH}_4)_2\text{PtCl}_4$	372.97	tet.		d.		s.		
chlorostannate	$(\text{NH}_4)_2\text{SnCl}_6$	367.50	pink., cb.	2.4			33.3 ^{15°}		
chromate	$(\text{NH}_4)_2\text{CrO}_4$	152.07	yel., mn.	1.917 ^{12°}	d. 180		40.5 ^{30°}	d.	sl. s. act., NH ₃ ; i. al.
cyanide	NH_4CN	44.06	col., cb.	0.79 ^{100°} (A)	36		s.	v. s.	s. al.
dichromate	$(\text{NH}_4)_2\text{Cr}_2\text{O}_7$	252.06	or., mn.	2.15	d. 185		47.2 ^{30°}	v. s.	s. al.; i. act.
ferrocyanide	$(\text{NH}_4)_4\text{Fe}(\text{CN})_6 \cdot 6\text{H}_2\text{O}$	392.19	mn.		d.		s.		i. al.
fluoride	NH_4F	37.04	wh., hex.				v. s.	d.	s. al.; i. NH ₃
fluoride, acid	$\text{NH}_4\text{F} \cdot \text{HF}$	57.04	wh., rhb., 1.390	2.21 $\frac{12^\circ}{12}$			v. s.		
formate	HCO_2NH_4	63.06	col., mn., delq.	1.266	114–116	d. 180; subl. in vac. subl. 120	102 ^{0°}	531 ^{80°}	s. al.
hydrosulfide	NH_4HS	51.11	col., rhb.		d.		v. s.		s. al.
hydroxide	NH_4OH	35.05	in soln. only				s.		
molybdate	$(\text{NH}_4)_2\text{MoO}_4$	196.01	mn.	2.27	d.		d.	d.	i. al., NH ₃
molybdate, hepta-	$(\text{NH}_4)_6\text{Mo}_7\text{O}_{24} \cdot 4\text{H}_2\text{O}\ddagger$	1235.86	col., mn.				44 ^{25°}		i. al.
nitrate (α), stable -16° to 32°	NH_4NO_3	80.04	col., tet., 1.611	1.66 $\frac{25^\circ}{4}$	169.6	d. 210	118.3 ^{0°}	241.8 ^{30°}	3.8 ^{20°} al., 17.1 ^{20°} m. al.;
nitrate (β), stable 32° to 84°	NH_4NO_3	80.04	col., rhb. or mn.	1.725 $\frac{25^\circ}{4}$		d. 210	365.8 ^{35°}	580 ^{80°}	v. s. NH ₃
nitrite	NH_4NO_2	64.04	wh. nd.	1.69	expl.		s.	d.	s. al.
osmochloride	$(\text{NH}_4)_2\text{OsCl}_6$	439.02	cb.	2.93 $\frac{20^\circ}{4}$					
oxalate	$(\text{NH}_4)_2\text{C}_2\text{O}_4 \cdot \text{H}_2\text{O}$	142.11	col., rhb.	1.501			2.5 ^{0°}	11.8 ^{50°}	sl. s. al.; i. NH ₃
oxalate, acid	$\text{NH}_4\text{HC}_2\text{O}_4 \cdot \text{H}_2\text{O}$	125.08	col., trimetric	1.556	d.		s.		
perchlorate	NH_4ClO_4	117.49	col., rhb., 1.4833	1.95	d.		10.9 ^{0°}	46.9 ^{100°}	2 ^{20°} al.; s. act.; i. et.
persulfate	$(\text{NH}_4)_2\text{S}_2\text{O}_8$	228.20	wh., mn., 1.5016	1.98	d. 120		58.2 ^{0°}	d.	
phosphate, monobasic	$\text{NH}_4\text{H}_2\text{PO}_4$	115.03	col., tet., 1.5246	1.803 $\frac{19^\circ}{4}$			22.7 ^{0°}	173.2 ^{100°}	i. ac.

phosphate, dibasic	(NH ₄) ₂ HPO ₄	132.06	col., mn., 1.53	1.619			131 ^{15°}		i. act.
phosphate, meta-	(NH ₄) ₄ P ₂ O ₁₂	388.04	col., mn.	2.21			s.		
Ammonium phosphomolybdate	(NH ₄) ₃ PO ₄ ·12MoO ₃ ·3H ₂ O (P)	1930.39	yel.		d.		0.03 ^{15°}	i.	s. alk.; i. al., HNO ₃
silicofluoride	(NH ₄) ₂ SiF ₆	178.15	cb., 1.3696	2.01		subl.	18.5 ^{17.5°}	55.5	s. al.; i. act.
sulfamate	NH ₄ ·SO ₃ NH ₂	114.12	col. pl.		132	d. 160	134 ^{0°}	357 ^{50°}	
sulfate (mascagnite)	(NH ₄) ₂ SO ₄	132.14	col., rhb., 1.5230	1.769 ^{20°} ₄	235 d.		70.6 ^{0°}	103.3 ^{100°}	i. al., act., CS ₂
sulfate, acid	NH ₄ HSO ₄	115.11	col., rhb., 1.480	1.78	146.9	490	100		v. sl. s. al.; i. act.
sulfide	(NH ₄) ₂ S	68.14	yel.-wh.		d.		v. s.		120 ^{25°} NH ₃
sulfide, penta-	(NH ₄) ₂ S ₅	196.40	or.-red pr.				s.		
sulfite	(NH ₄) ₂ SO ₃ ·H ₂ O	134.16	col., mn.	1.41	d.		100 ^{12°}		i. al., act.
sulfite, acid	NH ₄ HSO ₃	99.11	rhb.	2.03 ^{12°} ₄	d.		s.		
tartrate	(NH ₄) ₂ C ₄ H ₄ O ₆	184.15	col., mn.	1.60	d.		45 ^{0°}	87 ^{60°}	sl. s. al.
thiocyanate	NH ₄ CNS	76.12	col., mn., 1.685±	1.305	149.6	d. 170	120 ^{0°}	170 ^{20°}	s. al., act., NH ₃ , SO ₂
vanadate, meta-	NH ₄ VO ₃	116.98	col. cr.	2.326	d.		0.44 ^{16°}	3.05 ^{70°}	i. al., NH ₄ Cl
Antimony	Sb	121.76	tin wh., trig.	6.684 ^{25°}	630.5	1380	i.	i.	s. aq. reg., h. conc.
chloride, tri- (butter of antimony)*	SbCl ₃	228.12	col., rhb., delq.	3.14 ^{20°} ₄	73.4	220.2	601.6 ^{0°}	∞ ^{72°}	H ₂ SO ₄ s. al., HCl, HBr
oxide, tri- (valentinite)	Sb ₂ O ₃	291.52	rhb., 2.35	5.67	656	1570	v. sl. s.	sl. s.	H ₂ C ₄ H ₄ O ₆ s. HCl, KOH, H ₂ C ₄ H ₄ O ₆
oxide, tri- (senarmonite)	Sb ₂ O ₃	291.52	cb., 2.087	5.2	652				
sulfide, tri- (stibnite)	Sb ₂ S ₃	339.72	bk., rhb., 4.046	4.64	550		0.00017 ^{18°}	d.	s. HCl; alk., NH ₄ HS, K ₂ S; i. ac.
sulfide, penta-	Sb ₂ S ₅	403.85	golden	4.120 ^{0°}	-2S, 135		i.	i.	s. HCl, alk., NH ₄ HS
telluride, tri-	Sb ₂ Te ₃	626.32	gray		629				
Antimonyl potassium tartrate (tartar emetic)	(SbO)KC ₄ H ₄ O ₆ ·½H ₂ O	333.94	wh., rhb.	2.60	-½H ₂ O, 100		5.26 ^{8.7°}	35.7 ^{100°}	s. gly.; i. al.
sulfate, normal	(SbO) ₂ SO ₄	371.58	wh. pd.	4.89			d.	d.	
sulfate, basic	(SbO) ₂ SO ₄ ·Sb ₂ (OH) ₄	683.20	wh. pd.				i.	d.	5.15 ^{15°} gly.
Argon	Ar	39.95	col. gas	1.65 ^{-288°} 1.402 ^{-185.7°} 1.38 (A)	-189.2	-185.7	5.6 ^{0°} cc	2.23 ^{50°} cc	24 ^{25°} cc al.
Arsenic (crystalline) (α)	As ₄	299.69	met., hex.	5.727 ^{14°}	814 ^{36atm.}	subl. 615	i.	i.	s. HNO ₃
Arsenic (black) (β)	As ₄	299.69	bk., amor.	4.7 ^{20°}			i.	i.	s. HNO ₃ , aq. reg., aq. Cl ₂ , h. alk.
Arsenic (yellow)(γ)	As ₄	299.69	yel., cb.	2.0 ^{20°}	d. 358				
acid, ortho-	H ₃ AsO ₄ ·½H ₂ O	150.95	col., hyg.	2.0-2.5	35.5	-H ₂ O, 160	16.7	50	s. alk.
acid, meta-	HAsO ₃	123.93	wh., hyg.		d.		d. to form	H ₃ AsO ₄	
acid, pyro-	H ₄ As ₂ O ₇	265.87	col.		d. 206		d. to form	H ₃ AsO ₄	
pentoxide	As ₂ O ₅	229.84	wh., amor.	4.086		d.	59.5 ^{0°}	76.7 ^{100°}	s. alk., al.
sulfide, di- (realgar)	As ₂ S ₂	213.97	red, mn., 2.68	(α)3.506 ^{19°} ; (β)3.254 ^{19°}	(α)tr. 267; (β)307	565	i.	d.	s. K ₂ S, NaHCO ₃
sulfide, penta-	As ₂ S ₅	310.17	yel.			d. 500	0.000136 ^{0°}	i.	s. HNO ₃ , alk.
Arsenious chloride (butter of arsenic)	AsCl ₃	181.28	oily lq.	lq. 2.163	-18	130	d.	d.	s. HCl, HBr, PCl ₃
hydride (arsine)	AsH ₃	77.95	col. gas	2.695 (A)	-113.5	-55; d. 230	20 cc	sl. s.	sl. s. alk.
oxide (arsenolite)	As ₂ O ₃	197.84	col., cb., fibrous, 1.755	3.865 ^{25°} ₄	subl.		sl. s.	sl. s.	i. al., et.
oxide (claudeite)	As ₂ O ₃	197.84	col., mn., 1.92	3.85	subl.		sl. s.	sl. s.	i. al., et.
oxide	As ₂ O ₃	197.84	amor. or vitreous	3.738	315		1.21 ^{0°}	2.93 ^{40°}	s. HCl, alk., Na ₂ CO ₃ ; i. al., et.
Auric chloride	AuCl ₃ ·2H ₂ O	339.36	or. cr.		d.		v. s.	v. s.	s. HCl, al., et.; sl. s. NH ₃
cyanide	Au(CN) ₃ ·6H ₂ O	383.11			d. 50		v. s.	v. s.	s. al.
Aurous chloride	AuCl	232.42	yel. cr.	7.4	AuCl ₃ , 170	d. 290	d.	d.	s. HCl, HBr; d. al.
cyanide	AuCN	222.98	yel. cr.		d.		i.	i.	s. KCN; i. al., et.
<i>Cf. also under Gold</i>									
Barium	Ba	137.33	silv. met.	3.5	850	1140	d.	d.	s. a.; d. al.
acetate	Ba(C ₂ H ₃ O ₂) ₂	255.42	col.	2.468			58.8 ^{0°}	75.0 ^{100°}	
acetate	Ba(C ₂ H ₃ O ₂) ₂ ·H ₂ O	273.43	wh., tri. pr., 1.517	2.19	-H ₂ O, 41		75 ^{30°} (anh.)	79 ^{40°} (anh.)	i. al.
bromide	BaBr ₂	297.14	col.	4.781 ^{24°} ₄	847	d.	98 ^{0°}	149 ^{100°}	v. s. m. al.; v. sl. s. act.

*Usually the solution.

†See special tables.

‡Usual commercial form.

TABLE 2-1 Physical Properties of the Elements and Inorganic Compounds (Continued)

Name	Formula	Formula weight	Color, crystalline form and refractive index	Specific gravity	Melting point, °C	Boiling point, °C	Solubility in 100 parts		
							Cold water	Hot water	Other reagents
Barium (<i>Cont.</i>)									
bromide	BaBr ₂ ·2H ₂ O	333.17	col., mn., 1.7266	3.69	−2H ₂ O, 100	d.	v. s.	v. s.	s. al.
carbonate (witherite)	BaCO ₃	197.34	wh., rhb., 1.676	4.29	tr. 811 to α	d. 1450	0.0022 ^{18°}	0.0065 ^{100°}	s. a.; i. al.
carbonate (α)	BaCO ₃	197.34	wh., hex.		tr. 982 to β				
carbonate (β)	BaCO ₃	197.34	wh.		1740 ^{90atm}		0.0022 ^{18°}	0.0065 ^{100°}	s. a.; i. al.
Barium chlorate	Ba(ClO ₃) ₂	304.23	col.		414		20.35 ^{0°}	84.8 ^{80°}	
chlorate	Ba(ClO ₃) ₂ ·H ₂ O ^o	322.24	col., mn., 1.577	3.179	d. 120		s.		sl. s. al., act.
chloride	BaCl ₂	208.23	col., mn., 1.7361	3.856 ^{34°}	tr. 925	1560	31 ^{0°}	59 ^{100°}	sl. s. HCl, HNO ₃ ; i. al.
chloride	BaCl ₂	208.23	col., cb.		962	1560			
chloride	BaCl ₂ ·2H ₂ O†	244.26	col., mn., 1.646	3.097 ^{34°}	−2H ₂ O, 100		39.3 ^{0°}	76.8 ^{100°}	sl. s. HCl, HNO ₃ ; i. al.
hydroxide	Ba(OH) ₂	171.34	col., mn.	4.495			1.67 ^{0°}	101.4 ^{80°}	
hydroxide	Ba(OH) ₂ ·8H ₂ O	315.46	col., mn., 1.5017	2.188 ^{16°}	77.9	−8H ₂ O, 550	5.6 ^{15°}		v. sl. s. al.; i. et.
nitrate (nitrobarite)	Ba(NO ₃) ₂	261.34	col., cb., 1.572	3.244 ^{25°}	592	d.	5.0 ^{0°}	34.2 ^{100°}	sl. s. a.; i. al.
oxalate	BaC ₂ O ₄	225.35	wh. cr.	2.658			0.0016 ^{8°}	0.0024 ^{24°}	s. a., NH ₄ Cl; i. al.
oxide	BaO	153.33	col., cb., 1.98	5.72	1923	2000 ±	1.5 ^{0°}	90.8 ^{80°}	s. HCl, HNO ₃ , abs. al.; i. NH ₃ , act.
peroxide	BaO ₂ ^o	169.33	gray or wh. pd.	4.958	−O, 800		v. sl. s.	d.	s. dil. a.; i. act.
peroxide	BaO ₂ ·8H ₂ O	313.45	pearly sc.		−8H ₂ O, 100		0.168	d.	s. dil. a.; i. al., et., act.
phosphate, monobasic	BaH ₄ (PO ₄) ₂	331.30	tri.	2.9 ^{4°}			d.	d.	s. a.
phosphate, dibasic	BaHPO ₄	233.31	wh., rhb. nd., 1.635	4.165 ^{15°}			0.015		s. a., NH ₄ salts
phosphate, tribasic	Ba ₃ (PO ₄) ₂	601.92	wh., cb.	4.1 ^{16°}			i.		s. a.
phosphate, pyro-	Ba ₂ P ₂ O ₇	448.60	wh., rhb.	3.9 ^{30°}			0.01		s. a., NH ₄ salts
silicofluoride	BaSiF ₆	279.40	pr.	4.279 ^{15°}			0.026 ^{17°}	0.09 ^{100°}	sl. s. HCl, NH ₄ Cl; i. al.
sulfate (barite, barytes)	BaSO ₄	233.39	col., rhb., 1.636	4.499 ^{15°}	1580 d.	tr. to mn. 1149	0.000115 ^{0°}	0.000285 ^{30°}	s. conc. H ₂ SO ₄ ; 0.006, 3% HCl
sulfide, mono-	BaS	169.39	col., cb., 2.155	4.25 ^{15°}			d.	d.	d. HCl; i. al.
sulfide, tri-	BaS ₃	233.52	yel.-gn.		d. 400		s.	s.	
sulfide, tetra-	BaS ₃ ·2H ₂ O	301.62	red, rhb.	2.988 ^{20°}	d. 200		41 ^{15°}	v. s.	i. al., CS ₂
Beryllium (glucinum)	Be(Cl)	9.01	gray, met., hex.	1.816	1284	2767	i.	sl. s. d.	s. dil. a., alk.
Bismuth	Bi	208.98	silv. wh. or reddish, hex.	9.80 ^{20°}	271	1450	i.	i.	s. aq. reg., conc. H ₂ SO ₄ , HNO ₃
carbonate, sub-	Bi ₂ O ₃ ·CO ₂ ·H ₂ O	527.98	wh. pd.	6.86	d.		i.	i.	s. a.
chloride, di-	BiCl ₃ (?)	279.89	bk. nd.	4.86	163	d. 300	d.		
chloride, tri-	BiCl ₃ ^o	315.34	wh. cr.	4.75	230	447	d.		s. al.
nitrate	Bi(NO ₃) ₃ ·5H ₂ O	485.07	col., tri.	2.82	d. 30	−5H ₂ O, 80	d.		42 ^{19°} act.; s. a.; i. al.
nitrate, sub-	BiONO ₃ ·H ₂ O	305.00	hex. pl.	4.928 ^{15°}	d. 260		i.	i.	s. a.
oxide, tri-	Bi ₂ O ₃	465.96	yel., rhb.	8.9	820	1900 ±	i.	i.	s. a.
oxide, tri-	Bi ₂ O ₃	465.96	yel., tet.	8.55	860		i.	i.	s. a.
oxide, tri-	Bi ₂ O ₃	465.96	yel., cb.	8.20	tr. 704		i.	i.	s. a.
oxychloride	BiOCl	260.43	wh., amor.	7.72 ^{15°}			sl. s.	sl. s.	s. a.; i. act., NH ₃ , H ₂ C ₂ H ₄ O ₆
Boric acid	H ₃ BO ₃	61.83	wh., tri.	1.435 ^{15°}	185 d.		2.66 ^{0°}	40.2 ^{100°}	22.2 ^{20°} gly., 0.24 ^{25°} et.; s. al.
Boron	B	10.81	gray or bk., amor. or mn.	2.32	2300	2550	i.	i.	s. HNO ₃ ; i. al.
carbide	B ₄ C	55.25	bk. cr.	2.54	2450	>3500	i.	i.	i. a.
oxide	B ₂ O ₃	69.62	col. glass, 1.459	1.85	577	>1500	1.1 ^{0°}	15.7 ^{100°}	s. a., al., gly.
oxide (sassolite)	B ₂ O ₃ ·3H ₂ O	123.67	tri., 1.456	1.49	d.		sl. s.	s.	
Bromic acid	HBrO ₃	128.91	col.; in soln. only		d. 100		v. s.	d.	
Bromine	Br ₂	159.81	rhb., or red liq.	3.119 ^{30°} ; 5.87 (A)	−7.2	58.78	4.22 ^{0°}	3.13 ^{30°}	s. al., et., alk., CS ₂
hydrate	Br ₂ ·10H ₂ O	339.96	red, oct.		d. 6.8		s.		
Cadmium	Cd	112.41	silv. met., hex.	8.65 ^{20°}	320.9	767	i.	i.	s. a., NH ₄ NO ₃
acetate	Cd(C ₂ H ₃ O ₂) ₂	230.50	col.	2.341	256	d.	v. s.		s. m. al.
acetate	Cd(C ₂ H ₃ O ₂) ₂ ·2H ₂ O ^o	266.53	col., mn.	2.41	−H ₂ O, 130		v. s.		s. al.
carbonate	CdCO ₃	172.42	wh., trig.	4.258 ^{4°}	d. <500		i.	i.	s. a., KCN, NH ₄ salts; i. NH ₃
chloride	CdCl ₂	183.32	wh., cb.	4.047 ^{25°}	568	960	90 ^{0°}	147 ^{100°}	1.52 ^{15°} al.; i. et., act.

chloride	$\text{CdCl}_2 \cdot 2\frac{1}{2}\text{H}_2\text{O}$	228.36	col., mn., 1.6513	3.327	tr. 34		168^{20°	180^{100°	2.05^{15° m. al.
cyanide	$\text{Cd}(\text{CN})_2$	164.45			d. >200		0.0247^{18°		s. a.; NH_4OH , KCN
hydroxide	$\text{Cd}(\text{OH})_2$	146.43	wh., trig.	4.79^{15°	d. 300		0.00026^{25°		s. a., NH_4 salts; i. alk.
nitrate	$\text{Cd}(\text{NO}_3)_2$	236.42	col.		350		109.7^{9°	$326^{59.5^\circ}$	v. s. a.
nitrate	$\text{Cd}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}^*$	308.48	col. nd.	2.455^{17°	59.4	132	215^{0°		s. al., NH_3 ; i. HNO_3
oxide	CdO	128.41	brn., cb.	8.15			i.	i.	s. a., NH_4 salts; i. alk.
oxide	CdO	128.41	brn., amor. 2.49	6.95	d. 900–1000		i.	i.	s. a., NH_4 salts; i. alk.
oxide, sub-	Cd_2O	240.82	gn., amor.	8.192^{18°	d.				d. a., alk.
Cadmium sulfate	CdSO_4	208.47	rhb.	4.691^{24°	1000		76.5^{0°	60.8^{100°	i.act., NH_3
sulfate	$\text{CdSO}_4 \cdot \text{H}_2\text{O}$	226.49	mn.	3.786^{20°	tr. 108		s.	s.	
sulfate	$3\text{CdSO}_4 \cdot 8\text{H}_2\text{O}^*$	769.54	col., mn., 1.565	3.09	tr. 41.5		114.2^{0°	127.6^{60°	i. al.
sulfate	$\text{CdSO}_4 \cdot 4\text{H}_2\text{O}$	280.53	col.	3.05			s.	s.	i. al.
sulfate	$\text{CdSO}_4 \cdot 7\text{H}_2\text{O}$	334.58	mn.	2.48^{20°	tr. 4		350^{-5°		i. al.
sulfide (greenockite)	CdS	144.48	yel.-or., hex., 2.506	4.58	$1750^{100\text{atm}}$	subl. in N_2 , 980	0.000001	Colloidal	s. a.; v. s. NH_4OH
Calcium	Ca	40.08	silv. met., cb.	1.55^{20°	810	1200 ± 30	d.	d.	s. a.; sl. s. al.
acetate	$\text{Ca}(\text{C}_2\text{H}_3\text{O}_2)_2 \cdot \text{H}_2\text{O}$	176.18	wh. nd.		d.		52^{0°	45.5^{90°	sl. s. al.
aluminate	$\text{Ca}(\text{AlO}_2)_2$	158.04	col., rhb. or mn.	3.67^{20°	1600		d.		s. HCl
aluminum silicate (anorthite)	$\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2$	278.21	tri., 1.5832	2.765	1551				
arsenate	$\text{Ca}_3(\text{AsO}_4)_2$	398.07	wh. pd.				0.013^{25°	i.	s. dil. a.
bromide	CaBr_2	199.89	delq. nd.	3.353^{25°	760	1810	125^{0°	312^{105°	s. al., act.; sl. s. NH_3
carbonate (aragonite)	CaCO_3	100.09	col., rhb., 1.6809	2.93	d. 825		$0.0012^{20^\circ} \dagger$	0.002^{100°	s. a., NH_4Cl
carbonate (calcite)	CaCO_3	100.09	col., hex., 1.550	2.711^{25°	$1339^{103\text{atm}}$		0.0014^{25°	0.002^{100°	s. a., NH_4Cl
chloride (hydrophilite)	CaCl_2^*	110.98	wh., delq., cb. 1.52	2.152^{15°	772	>1600	59.5^{0°	347^{260°	s. al.
chloride	$\text{CaCl}_2 \cdot \text{H}_2\text{O}$	129.00	col., delq.				s.	s.	s. al.
chloride	$\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$	219.08	col., trig., 1.417	1.68^{17°	29.92	$-6\text{H}_2\text{O}$, 200	v. s.	v. s.	s. al.
citrate	$\text{Ca}_3(\text{C}_6\text{H}_5\text{O}_7)_2 \cdot 4\text{H}_2\text{O}$	570.49	col. nd.		$-2\text{H}_2\text{O}$, 130	$-4\text{H}_2\text{O}$, 185	0.085^{18°	0.096^{26°	0.0065^{18° al.
cyanamide	CaCN_2	80.10	col., rhombohedral				s. d.	d.	
ferrocyanide	$\text{Ca}_2\text{Fe}(\text{CN})_6 \cdot 12\text{H}_2\text{O}$	508.29	yel., tri., 1.5818	1.7			s.	150^{90°	i. al.
fluoride (fluorite)	CaF_2	78.07	wh., cb., 1.4339	3.180^{30°	1330		0.0016^{18°	0.0017^{26°	sl. s. a.
formate	$\text{Ca}(\text{HCO}_2)_2$	130.11	col., rhb.	2.015	d.		16.1^{0°	18.4^{100}	i. al., et.
hydride	CaH_2	42.09	wh. cr. or pd.	1.7	d. 675		d.		d. a.; i. bz.
hydroxide	$\text{Ca}(\text{OH})_2$	74.09	col., hex., 1.574	2.2	$-\text{H}_2\text{O}$, 580		0.185^{0°	0.077^{100°	s. NH_4Cl
hypochlorite	$\text{Ca}(\text{ClO})_2 \cdot 4\text{H}_2\text{O}$	215.04	wh., feathery cr.		d.		delq.; d.	d.	d. a.
hypophosphate	$\text{Ca}_3\text{P}_2\text{O}_6 \cdot 2\text{H}_2\text{O}$	274.13	granular		$-2\text{H}_2\text{O}$, 200		i.		s. HCl, $\text{H}_4\text{P}_2\text{O}_6$
lactate	$\text{Ca}(\text{C}_3\text{H}_5\text{O}_3)_2 \cdot 5\text{H}_2\text{O}$	308.29	col., eff.		$-3\text{H}_2\text{O}$, 100		10.5	∞	∞ h. al.; i. et.
magnesium carbonate (dolomite)	$\text{CaO} \cdot \text{MgO} \cdot 2\text{CO}_2$	184.40	trig., 1.68174	2.872	d. 730–760		0.032^{18°		
magnesium silicate (diopside)	$\text{CaO} \cdot \text{MgO} \cdot 2\text{SiO}_2$	216.55	wh., mn.	3.3	1391		i.	i.	
nitrate (nitrocalcite)	$\text{Ca}(\text{NO}_3)_2$	164.09	col., cb.	2.36	561		102^{0°	376^{151°	14^{15° al.; s. amyl al., NH_3
nitrate	$\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}^*$	236.15	col., mn., 1.498	1.82	42.7		266^{0°	v. s.	
nitride	Ca_3N_2	148.25	brn. cr.	2.63^{17°	900		d.	d.	s. dil. a.; i. abs. al.
nitrite	$\text{Ca}(\text{NO}_2)_2 \cdot \text{H}_2\text{O}$	150.10	delq., hex.	2.23^{34°			77^{0°	417^{90°	s. 90% al.
oxalate	CaC_2O_4	128.10	col., cb.	2.2^{4°	d.		0.00067^{13°	0.0014^{95°	s. a.; i. ac.
oxalate	$\text{CaC}_2\text{O}_4 \cdot \text{H}_2\text{O}$	146.11	col.	2.2	$-\text{H}_2\text{O}$, 200		i.	i.	s. a.; i. ac
oxide	CaO	56.08	col., cb., 1.837	3.32	2570	2850	Forms $\text{Ca}(\text{OH})_2$		s. a.; i. al.
peroxide	$\text{CaO}_2 \cdot 8\text{H}_2\text{O}$	216.20	pearly, tet.		$-8\text{H}_2\text{O}$, 100	expl. 275	sl. s.	d.	s. a. d.; i. al., et.
phosphate, monobasic	$\text{CaH}_4(\text{PO}_4)_2 \cdot \text{H}_2\text{O}$	252.07	wh., tri.	2.220^{16°	$-\text{H}_2\text{O}$, 100	d. 200		d.	
phosphate, dibasic	$\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$	172.09	wh., mn. pl.	2.306^{16°	d.		$0.02^{24.5^\circ}$	0.075^{100°	
phosphate, tribasic	$\text{Ca}_3(\text{PO}_4)_2$	310.18	wh., amor.	3.14	1670		0.0025	d.	s. a.; i. al., ac.
phosphate, meta-	$\text{Ca}(\text{PO}_3)_2$	198.02	wh., tet., 1.588	2.82	975		i.	i.	i. a.
phosphate, pyro-	$\text{Ca}_2\text{P}_2\text{O}_7$	254.10	col., biaxial, 1.60	3.09	1230		i.		s. a.
phosphate, pyro- (brushite)	$\text{Ca}_2\text{P}_2\text{O}_7 \cdot 5\text{H}_2\text{O}$	344.18	wh., mn.	2.25			sl. s.		s. a.; i. NH_4Cl
phosphide	Ca_3P_2	182.18	red cr.	2.51^{15°	>1600		d.		s. dil. a.; i. al., et.
silicate (α) (pseudowollastonite)	CaSiO_3	116.16	col., pseudo hex., 1.6150 or mn.(?)	2.905	1540		0.0095^{17°		s. HCl
silicate (β) (wollastonite)	CaSiO_3	116.16	col., mn., 1.610	2.915	tr. 1190 to α				
sulfate (anhydrite)	CaSO_4	136.14	col., rhb., 1.576, or mn. 1.50	2.96	1450(mn.)	tr. 1193 to rhb.	0.298^{20°	0.1619^{100°	s. a., $\text{Na}_2\text{S}_2\text{O}_3$, NH_4 salts

*Usual commercial form.

†The solubility of CaCO_3 in H_2O is greatly increased by increasing the amount of CO_2 in the H_2O .

TABLE 2-1 Physical Properties of the Elements and Inorganic Compounds (Continued)

Name	Formula	Formula weight	Color, crystalline form and refractive index	Specific gravity	Melting point, °C	Boiling point, °C	Solubility in 100 parts		
							Cold water	Hot water	Other reagents
Calcium (<i>Cont.</i>) sulfate (gypsum)	CaSO ₄ ·2H ₂ O	172.17	col., mn., 1.5226	2.32	−1½H ₂ O, 128	−2H ₂ O, 163	0.223 ^{0°}	0.257 ^{50°}	s. a., gly., Na ₂ S ₂ O ₃ , NH ₄ salts
sulphhydrate	Ca(SH) ₂ ·6H ₂ O	214.32	col. pr.				v. s.	v. s.	s. al.
sulfide (oldhamite)	CaS	72.14	col., cb.	2.8 ^{15°}	d. 15		d.	d.	s. a.
sulfite	CaSO ₃ ·2H ₂ O	156.17	wh., cr., 1.595		−2H ₂ O, 100	d. 650	0.0043 ^{18°}	0.0027 ^{90°}	s. H ₂ SO ₃
tartrate	CaC ₄ H ₄ O ₆ ·4H ₂ O	260.21	col., rhb.		d.		0.037 ^{0°}	0.22 ^{35°}	sl. s. al.
thiocyanate	Ca(CNS) ₂ ·3H ₂ O	210.29	wh., delq. cr.				s.	v. s.	v. s. al.
thiosulfate	CaS ₂ O ₃ ·6H ₂ O	260.30	col., tri., 1.56	1.873 ^{16°}	d.		71.2 ^{9°}	d.	i. al.
tungstate (scheelite)	CaWO ₄	287.92	wh., tet., 1.9200	6.06			0.2		s. NH ₄ Cl; i. a.
Carbon, <i>cf.</i> table of organic compounds									
Carbon, amorphous	C	12.01	bk., amor.	1.8–2.1	>3500	4200	i.	i.	i. a., alk.
Carbon, diamond	C	12.01	col., cb., 2.4195	3.51 ^{20°}	>3500	4200	i.	i.	i. a., alk.
Carbon, graphite	C	12.01	bk., hex.	2.26 ^{20°}	>3500	4200	i.	i.	i. a., alk.
dioxide	CO ₂	44.01	col. gas	lq. 1.101 ^{−87°} ; 1.53 (A); solid 1.56 ^{−79°}	−56.6 ^{5.2atm.}	subl. −78.5	179.7 ^{0°} cc	90.1 ^{20°} cc	s. a., alk.
disulfide	CS ₂	76.14	col. lq.	lq. 1.261 ^{22°} ; 2.63 (A)	−108.6	46.3	0.2 ^{0°}	0.014 ^{50°}	s. al.; et.
monoxide	CO	28.01	col., poisonous, odorless gas	lq. 0.814 ^{−195°} ; 0.968 (A)	−207	−192	0.0044 ^{0°} ; 3.5 ^{0°} cc	0.0018 ^{50°} ; 2.32 ^{20°} cc	s. al., Cu ₂ Cl ₂
oxychloride (phosgene)	COCl ₂	98.92	poisonous gas	1.392 ^{19°}	−104	8.2 ^{756mm}	v. s. sl. d.	d.	s. ac., CCl ₄ , bs.; d. a.
oxysulfide	COS	60.08	gas	lq. 1.24 ^{−87°} ; 2.10 (A)	−138.2	−50.2 ^{760mm}	133 ^{0°} cc	40.3 ^{30°} cc	v. s. alk., al.
suboxide	C ₃ O ₂	68.03	gas	lq. 1.114 ^{0°}	−107	7 ^{761mm}	d.		s. et.
thionyl chloride	CSCl ₂	114.98	yel.-red lq.	1.509 ^{15°}		73.5			
Ceric hydroxide	2CeO ₂ ·3H ₂ O	398.28	yel., gelatinous						s. a.; sl. s. alk. carb.; i. alk
hydroxynitrate	Ce(OH)(NO ₃) ₃ ·3H ₂ O	397.18	red, mn.				d.		
oxide	CeO ₂	172.11	wh. or pa. yel., cb.	7.3	1950		i.	i.	s. H ₂ SO ₄ , HCl
sulfate	Ce(SO ₄) ₂ ·4H ₂ O	404.30	yel., rhb.	3.91			s. d.		s. dil. H ₂ SO ₄
Cerium	Ce	140.12	steel gray, cb. or hex.	6.9 ^{30°} cb.; 6.7 hex.	645	1400	i.	Slowly oxidized	s. dil. a.; i. al.
Cerous sulfate	Ce ₂ (SO ₄) ₃	568.42	wh., mn. or rhb.	3.91			18.98 ^{0°}	0.4 ^{100°}	
sulfate	Ce ₂ (SO ₄) ₃ ·8H ₂ O	712.54	tri.	2.886 ^{17°}	−8H ₂ O, 630		25 ^{0°}	7.6 ^{40°}	
Cesium	Cs	132.91	silv. met., hex.	1.90 ^{20°}	28.5	670	d.		s. a., al., NH ₃
Chloric acid	HClO ₃ ·7H ₂ O	210.57	lq.	1.282 ^{14.2°}	<−20	d. 40	v. s.		
Chlorine	Cl ₂	70.91	rhb., or gn.-yel. gas	lq. 1.56 ^{−33.6°} ; 2.49 ^{0°} (A)	−101.6	−34.6	1.46 ^{0°} ; 310 ^{10°} cc	0.57 ^{30°} ; 177 ^{30°} cc	s. alk.
hydrate	Cl ₂ ·8H ₂ O	215.03	rhb.	1.23	d. 9.6		s.		s. alk.
Chloroplatinic acid	H ₂ PtCl ₆ ·6H ₂ O	517.90	red-brn., delq.	2.431	60		v. s.	v. s.	s. al., et.
Chlorostannic acid	H ₂ SnCl ₆ ·6H ₂ O	441.54	delq.	1.971 ^{38°}	19.2		s.		
Chlorosulfonic acid	HO·SO ₂ ·Cl	116.52	col. lq.	1.787 ^{25°}	−80	151.5 ^{765mm}	d.		d. al.; i. CS ₂
Chromic acetate	Cr ₂ (C ₂ H ₃ O ₂) ₆ ·2H ₂ O	494.29	gn.				s.		4.76 ^{15°} m. al.
chloride	CrCl ₃	158.36	pink, trig.	2.757 ^{15°}		1200–1500 d.	i. §	sl. s.	i. a., act., CS ₂
chloride	CrCl ₃ ·6H ₂ O*	266.45	vl. or gn., hex. pl.	1.835 ^{25°}	subl. 83		v. s. d.		s. al.; i. et.
fluoride	CrF ₃	108.99	gn., rhb.	3.8	>1000	d.	i.		sl. s. a.; i. al., NH ₃
hydroxide	Cr(OH) ₃	103.02	gn. or blue, gelatinous				i.		s. a., alk.; sl. s. NH ₃
hydroxide	Cr(OH) ₃ ·2H ₂ O	139.05	gn.		−2H ₂ O, 100		i.	i.	s. a., alk.
nitrate	Cr(NO ₃) ₃ ·9H ₂ O*	400.15	purple pr.		36.5	d. 100	s.	s.	s. a., alk., al., act.
nitrate	Cr(NO ₃) ₃ ·7½H ₂ O	373.13	purple, mn.		100	d.	s.	s.	
oxide	Cr ₂ O ₃	151.99	dark gn., hex.	5.21	1900		i.	i.	sl. s. a.
sulfate	Cr ₂ (SO ₄) ₃	392.18	rose pd.	3.012			i. †		i. a.
sulfate	Cr ₂ (SO ₄) ₃ ·5H ₂ O	482.26	gn.				s.		s. al., H ₂ SO ₄
sulfate	Cr ₂ (SO ₄) ₃ ·15H ₂ O	662.41	vl.	1.867 ^{17°}	100	−10H ₂ O, 100	s.	d. 67°	sl. s. al.
sulfate	Cr ₂ (SO ₄) ₃ ·18H ₂ O	716.46	vl., cb., 1.564	1.7 ^{22°}		−12H ₂ O, 100	120 ^{20°}	d.	s. al.
sulfide	Cr ₂ S ₃	200.19	brn.-bk. pd.	3.77 ^{19°}	−S, 1350		i.	d.	s. h. HNO ₃

Chromium	Cr	52.00	gray, met., cb.	7.1	1615	2200	i.	i.	s. HCl, dil. H ₂ SO ₄ ; i. HNO ₃
trioxide (chromic acid)	CrO ₃	99.99	red, rhb.	2.70	197 d.		164.9 ^{0°}	206.7 ^{100°}	s. H ₂ SO ₄ , al., et.
Chromous chloride	CrCl ₂	122.90	wh., delq.	2.75			v. s.	v. s.	sl. s. al.; i. et.
hydroxide	Cr(OH) ₂	86.01	yel.-brn.		d.		d.		s. conc. a.
oxide	CrO	68.00	bk. pd.				i.	i.	i. dil. HNO ₃
sulfate	CrSO ₄ ·7H ₂ O	274.17	blue				12.35 ⁰		sl. s. al.
sulfide (daubrelite)	CrS	84.06	bk. pd.	3.97	1550		i.		v. s. a.
Chromyl chloride	CrO ₂ Cl ₂	154.90	dark red lq.	1.92	-96.5	117.6	d.		s. et.
Cobalt	Co	58.93	silv. met., cb.	8.9 ^{20°}	1480	2900	i.	i.	s. a.
carbonyl	Co(CO) ₄	170.97	or. cr.	1.73 ^{18°}	51	d. 52	i.	d.	s. al., et., CS ₂
sulfide, di-	CoS ₂	123.06	bk., cb.	4.269			i.		s. HNO ₃ , aq. reg.
Cobaltic chloride	CoCl ₃	165.29	red cr.	2.94	subl.		s.	s.	
chloride, dichro	Co(NH ₃) ₃ Cl ₃ ·H ₂ O	234.40					s.		s. a.; al.
chloride, luteo	Co(NH ₃) ₆ Cl ₃	267.48	or., mn.	1.7016 ^{20°}			4.26 ^{0°}	12.74 ^{46.5°}	i. al., NH ₄ OH
chloride, praseo	Co(NH ₃) ₄ Cl ₃ ·H ₂ O	251.43	gn., rhb.	1.847			v. s.		s. a.; i. al.
Cobaltic chloride, purpureo	Co(NH ₃) ₅ Cl ₃	250.44	rhb.	1.819 ^{25°} ₂₅			0.232 ^{0°}	1.031 ^{46.5°}	i. al.
chloride, roseo	Co(NH ₃) ₅ Cl ₃ ·H ₂ O	268.46	brick red		d. 100		16.12 ^{0°}	24.87 ^{16°}	sl. s. HCl
hydroxide	Co(OH) ₃	109.96	bk.		-1½H ₂ O, 100		i.	i.	s. a.; i. al.
oxide	Co ₂ O ₃	165.86	bk.	5.18	d. 900		i.	i.	s. a.
sulfate	Co ₂ (SO ₄) ₃	406.05	blue cr.				d.		s. H ₂ SO ₄
sulfide	Co ₂ S ₃	214.06	bk. cr.	4.8			i.		d. a.
Cobalto-cobaltic oxide	Co ₃ O ₄	240.80	bk., cb.	6.07			i.	i.	s. H ₂ SO ₄ ; i. HCl, HNO ₃
Cobaltous acetate	Co(C ₂ H ₃ O ₂) ₂ ·4H ₂ O	249.08	red-vl., mn., 1.542	1.7053 ^{18.7°}	-4H ₂ O, 140		s.	s.	s. a., al.
chloride	CoCl ₂	129.84	blue cr.	3.356	subl.	1049	45 ^{7°}	105 ^{96°}	31 al.; 8.6 act.
chloride	CoCl ₂ ·6H ₂ O*	237.93	red, mn.	1.924 ^{25°} ₂₅	86	-6H ₂ O, 110	116.5 ^{0°}	177 ^{80°}	v. s. et., act.
nitrate	Co(NO ₃) ₂ ·6H ₂ O	291.03	red, mn., 1.4	1.883 ^{25°} ₂₅	<100	d.	84.03 ^{0°} (anh.)	334.9 ^{90°} (anh.)	100 ^{12.5°} al.; s. act.; sl. s. NH ₃
oxide	CoO	74.93	brn., cb.	5.68	d. 1800		i.	i.	s. a., NH ₄ OH; i. al.
sulfate	CoSO ₄	155.00	red pd.	3.710 ^{25°}	d. 880		25.6 ^{0°}	83 ^{100°}	1.04 ¹⁸ m. al.; i. NH ₃
sulfate	CoSO ₄ ·H ₂ O	173.01	red pd., mn.(?), 1.639	3.13	d.		s.	s.	
sulfate (biebeorite)	CoSO ₄ ·7H ₂ O*	281.10	red, mn., 1.483	1.948 ^{25°} ₂₅	96.8	-7H ₂ O, 420	33 ^{80°}	s.	2.5 ^{8°} al.
sulfide (syeporite)	CoS	91.00	brn. nd.	5.45 ^{18°}	>1100		0.00038 ^{18°}		s. a., aq. reg.
Copper	Cu	63.55	yel.-red met., cb.	8.92 ^{20°}	1083	2300	i.	i.	s. HNO ₃ , h. H ₂ SO ₄
Cupric acetate	Cu(C ₂ H ₃ O ₂) ₂	181.63		1.930 ^{20°} ₄			s.		
acetate	Cu(C ₂ H ₃ O ₂) ₂ ·H ₂ O	199.65	dark gn., mn.	1.882	115	240 d.	7.2	20	7 al.; s. et.; gly.
aceto-arsenite (Paris green)	(CuOAs ₂ O ₃) ₃ · Cu(C ₂ H ₃ O ₂) ₂ *	1013.79	gn.				i.		s. a., NH ₄ OH
ammonium chloride	CuCl ₂ ·2NH ₄ Cl·2H ₂ O	277.47	blue, tet., 1.670, 1.744	1.98	d. 110		33.8 ^{0°}	99.3 ^{80°}	s. a.
ammonium sulfate	CuSO ₄ ·4NH ₃ ·H ₂ O	245.75	blue, rhb.	1.81	d. 150		18.05 ^{21.5°}	d.	i. al.
carbonate, basic (azurite)	2CuCO ₃ ·Cu(OH) ₂	344.67	blue, mn., 1.758	3.88	d. 220		i.	d.	s. NH ₄ OH, h. aq. NaHCO ₃
carbonate, basic (malachite)	CuCO ₃ ·Cu(OH) ₂	221.12	dark gn., mn., 1.875	3.9	d.		i.	d.	s. KCN; 0.03 aq. CO
chloride (eriochalcite)	CuCl ₂	134.45	brn.-yel. pd.	3.054	498	Forms Cu ₂ Cl ₂ 993	70.7 ^{0°}	107.9 ^{100°}	53 ^{15°} al.; 68 ^{15°} m. al.
chloride	CuCl ₂ ·2H ₂ O	170.48	gn., rhb., 1.684	2.39 ^{22.4°}	-2H ₂ O, 110	d.	110.4 ^{0°}	192.4 ^{100°}	s. al.; et., NH ₄ Cl
chromate, basic	CuCrO ₄ ·2CuO·2H ₂ O	374.66	yel.-brn.		-2H ₂ O, 260		i.		s. HNO ₃ , NH ₄ OH
cyanide	Cu(CN) ₂	115.58	yel.-gn.		d.		i.		s. KCN, C ₅ H ₅ N
dichromate	CuCr ₂ O ₇ ·2H ₂ O	315.56	bk., tri.	2.286 ^{18°}	-2H ₂ O, 100		sl. s.	d.	s. a.; NH ₄ OH
ferricyanide	Cu ₃ [Fe(CN) ₆] ₂	614.54	yel.-gn.				i.		s. NH ₄ OH; i. HCl
ferrocyanide	Cu ₂ Fe(CN) ₆ ·7H ₂ O	465.15	red-brn.				i.	i.	s. NH ₄ OH; i. a., NH ₃
formate	Cu(HCO ₂) ₂	153.58	blue, mn.	1.831			12.5	d.	0.25 al.
hydroxide	Cu(OH) ₂	97.56	blue, gelatinous	3.368	-H ₂ O		i.		s. a., NH ₄ OH, KCN, al.
lactate	Cu(C ₃ H ₃ O ₂) ₂ ·2H ₂ O	277.72	dark blue, mn.				16.7	45 ^{100°}	sl. s. al.
nitrate	Cu(NO ₃) ₂ ·3H ₂ O*	241.60	blue, delq.	2.047 ^{3.9°}	114.5	-HNO ₃ , 170	381 ^{40°}	666 ^{80°}	100 ^{12.5°} al.
nitrate	Cu(NO ₃) ₂ ·6H ₂ O	295.65	blue, rhb.	2.074	-3H ₂ O, 26.4		243.7 ^{0°}	∞	s. al.

*Usual commercial form.

†Also a soluble modification.

TABLE 2-1 Physical Properties of the Elements and Inorganic Compounds (Continued)

Name	Formula	Formula weight	Color, crystalline form and refractive index	Specific gravity	Melting point, °C	Boiling point, °C	Solubility in 100 parts		
							Cold water	Hot water	Other reagents
Cupric acetate (<i>Cont.</i>)									
oxide (paramelaconite)	CuO	79.55	bk., cb.	6.40	d. 1026		i.	i.	s. a.; KCN, NH ₄ Cl
oxide (tenorite)	CuO	79.55	bk., tri., 2.63	6.45	d. 1026		i.	i.	s. a.; KCN, NH ₄ Cl
oxychloride	CuCl ₂ ·2CuO·4H ₂ O	365.60	blue-gn.		-3H ₂ O, 140		i.		s. a.
phosphide	Cu ₃ P ₂	252.59	bk.	6.35	d.		i.		s. HNO ₃ ; i. HCl
sulfate (hydrocyanite)	CuSO ₄	159.61	gn.-wh., rhb., 1.733	3.606 ^{15°}	d. >600	Forms CuO, 650	14.3 ^{0°}	75.4 ^{100°}	i. al.
sulfate (blue vitriol or chalcantithite)	CuSO ₄ ·5H ₂ O*	249.69	blue, tri., 1.5368	2.286 ^{15.6°} / ₄	-4H ₂ O, 110	-5H ₂ O, 250	24.3 ^{0°}	205 ^{100°}	1.1 ^{8°} al.
sulfide (covellite)	CuS	95.61	blue, hex. or mn., 1.45	4.6	tr. 103	d. 220	0.000033 ^{18°}		s. HNO ₃ , KCN
tartate	Cu ₂ C ₄ H ₄ O ₆ ·3H ₂ O	265.66	1 gn. pd.		d.		0.02 ^{15°}	0.14 ^{85°}	s. a., KOH
Cuprous ammonium iodide	CuI·NH ₄ I·H ₂ O	353.41	rhb. pl.				d.		s. NH ₄ I
carbonate	Cu ₂ CO ₃	187.10	yel.	4.4	d.		i.	i.	s. a., NH ₄ OH
chloride (nantokite)	Cu ₂ Cl ₂	198.00	wh., cb., 1.973	3.53	422	1366	1.52 ^{25°}		s. HCl, NH ₄ OH, al.
cyanide	Cu ₂ (CN) ₂	179.13	wh., mn.	2.9	474.5	d.	i.	i.	s. KCN, HCl, NH ₄ OH; sl. s. NH ₃
ferricyanide	Cu ₃ Fe(CN) ₆	402.59	brn.-red				i.		s. NH ₄ OH; i. HCl
ferrocyanide	Cu ₄ Fe(CN) ₆	466.13	brn.-red				i.		s. NH ₄ OH; i. NH ₄ Cl
fluoride	Cu ₂ F ₂	165.09	red cr.		908	subl. 1100	i.		s. HF, HCl, HNO ₃ ; i. al.
hydroxide	CuOH	80.55	yel.	3.4	-½H ₂ O, 360		i.	i.	s. a., NH ₄ OH
oxide (cuprite)	Cu ₂ O	143.09	red, cb., 2.705	6.0	1235	-O, 1800	i.	i.	s. HCl, NH ₄ Cl, NH ₄ OH
Cuprous phosphide	Cu ₆ P ₂	443.22	gray-bk.	6.4 to 6.8			i.		s. HNO ₃ ; i. HCl
sulfide (chalcocite)	Cu ₂ S	159.16	bk., rhb.	5.6	1100		0.0005 ^{18°}		s. HNO ₃ , NH ₄ OH; i. act.
sulfide	Cu ₂ S	159.16	bk., cb.	5.80	1130		0.0005 ^{18°}		s. HNO ₃ , NH ₄ OH; i. act.
Cyanogen	C ₂ N ₂	52.03	poisonous gas	lq. 0.866 ^{-17.2°} ; 1.806 (A)	-34.4	-20.5	450 ^{30°} cc		2300 ^{20°} cc al.; 500 ^{18°} cc et.
Cyanogen compounds, <i>cf.</i> table of organic compounds									
Ferric acetate, basic ammonium sulfate, <i>cf.</i> Alum chloride (molysite)	Fe(OH)(C ₂ H ₃ O ₂) ₂	190.94	brn., amor.				i.		s. a.; al.
chloride	FeCl ₃	162.20	bk.-brn., hex. delq.	2.804 ^{11°}	282	315	74.4 ^{0°}	535.8 ^{100°}	v. s. al.; et. +HCl
chloride	FeCl ₃ ·6H ₂ O*	270.30	red-yel., delq.		37	280	246 ^{0°}	∞	s. al., act., gly.
ferrocyanide (Prussian blue)	Fe ₄ [Fe(CN) ₆] ₃	859.23	dark blue		d.		i.	d.	s. HCl, conc. H ₂ SO ₄ ; i. al., et.
hydroxide	Fe(OH) ₃	106.87	red-brn.	3.4 to 3.9	-½H ₂ O, 500		i.	i.	s. a.; i. al., et.
lactate	Fe(C ₃ H ₅ O ₃) ₃	323.06	brn., amor., delq.				v. s.	v. s.	i. et.
nitrate	Fe(NO ₃) ₃ ·6H ₂ O	349.95	rhb., delq.	1.684 ^{20°}	35	d.	150 ^{0°}	∞	s. al., act.
oxide (hematite)	Fe ₂ O ₃	159.69	red or bk., trig., 3.042	5.12	1560 d.		i.		s. HCl
sulfate	Fe ₂ (SO ₄) ₃	399.88	rhb., 1.814	3.097 ^{18°}	d. 480		sl. s.	d.	i. H ₂ SO ₄ , NH ₃
sulfate (coquimbite)	Fe ₂ (SO ₄) ₃ ·9H ₂ O	562.02	yel., trig.	2.1			440	d.	s. abs. al.
Ferroso-ferric chloride	FeCl ₂ ·2FeCl ₃ ·18H ₂ O	775.43	yel., delq.		d. 50		s.	s.	
ferricyanide (Prussian green)	Fe ^{III} Fe ^{II} [Fe(CN) ₆] ₆	1662.61	gn.		d. 180		i.		s. d. h. HCl
oxide (magnetite; magnetic iron oxide)	Fe ₃ O ₄	231.53	bk., cb., 2.42	5.2	1538 d.		i.	i.	i. al.
oxide, hydrated	Fe ₃ O ₄ ·4H ₂ O	303.59	bk.		d.		i.	i.	s. a.
Ferrous ammonium sulfate	FeSO ₄ ·(NH ₄) ₂ SO ₄ ·6H ₂ O	392.14	blue-gn., mn., 1.4915	1.864	d.		18 ^{0°}	100 ^{75°}	i. al.
chloride (lawrencite)	FeCl ₂	126.75	gn.-yel., hex., 1.567	2.7		delq.	64.4 ^{10°}	105.7 ^{100°}	100 al.; s. act.; i. et.
chloroplatinate	FePtCl ₆ ·6H ₂ O	571.73	yel., hex.	2.714			v. s.	v. s.	
ferricyanide (Turnbull's blue)	Fe ₃ [Fe(CN) ₆] ₂	591.43	dark blue		d.		i.		i. dil. a., al.
ferrocyanide	Fe ₂ Fe(CN) ₆	323.64	blue-wh., amor.				i.		
formate	Fe(HCO ₂) ₂ ·2H ₂ O	181.91			d.		sl. s.		
hydroxide	Fe(OH) ₂	89.86	lt. gn.	3.4			0.00067		s. a., NH ₄ Cl
nitrate	Fe(NO ₃) ₂ ·6H ₂ O	287.95	cr.		60.5		200 ^{0°}	300 ^{25°}	
oxide	FeO	71.84	bk.	5.7	1420		i.	i.	s. a.; i. alk.

phosphate (vivianite)	$\text{Fe}_3(\text{PO}_4)_2 \cdot 8\text{H}_2\text{O}$	501.60	blue, mn., 1.592, 1.603	2.58			i.	i.	s. a.; i. ac.
silicate	FeSiO_3	131.93	mn.	3.5	1550				
sulfate (siderotilite)	$\text{FeSO}_4 \cdot 5\text{H}_2\text{O}$	241.98	gn., tri., 1.536	2.2		$-5\text{H}_2\text{O}, 300$	s.	s.	i. al.
sulfate (copperas)	$\text{FeSO}_4 \cdot 7\text{H}_2\text{O}^*$	278.01	blue-gn., mn.	1.899 ^{14.8°}	64	$-7\text{H}_2\text{O}, 300$	32.8 ^{0°}	149 ^{50°}	i. al.
sulfide	FeS	87.91	bk., hex.	4.84	1193	d.	0.000616 ^{18°}		s. a.; i. NH_3
<i>cf.</i> also under iron									
Fluoboric acid	HBF_4	87.81	col. lq.			130 d.	∞	∞	s. al.
Fluorine	F_2	38.00	gn.-yel. gas	lq. 1.51 ^{-187°} ; 1.31 ^{15°} (A)	-223	-187	d.		
Fluosilicic acid	H_2SiF_6	144.09					s.	s.	
Gadolinium	Gd	157.25							
Gallium bromide	GaBr_3	309.44	delq. cr.				s.	s.	
Glucinum <i>cf.</i> Beryllium									
Gold	Au	196.97	yel. met., cb.	19.3 ^{20°}	1063	2600	i.	i.	s. aq. reg., KCN; i. a.
Gold, colloidal	Au	196.97	blue to vl.				s.		s. aq. reg., KCN; i. a.
Gold salts <i>cf.</i> under Auric and Aurous									
Hafnium	Hf	178.49	hex.	12.1	>1700	>3200(?)			
Helium	He	4.00	col. gas	0.1368 (A)	<-272.2	-268.9	0.97 ^{0°} cc	1.08 ^{50°} cc	Absorbed by Pt
Hydrazine	N_2H_4	32.05	col. lq.	1.011 ^{12°}	1.4	113.5	∞	∞	s. al.
formate	$\text{N}_2\text{H}_4 \cdot 2\text{HCO}_2\text{H}$	124.10	cb.		128		s.		
hydrate	$\text{N}_2\text{H}_4 \cdot \text{H}_2\text{O}$	50.06	col.	1.03 ^{21°}	-40	118.5 ^{739.5mm}	∞	∞	∞ al.; i. et.
hydrochloride	$\text{N}_2\text{H}_4 \cdot \text{HCl}$	68.51	yel. lq.				v. s.	v. s.	sl. s. al.
hydrochloride, di-	$\text{N}_2\text{H}_4 \cdot 2\text{HCl}$	104.97	wh., cb.	1.42	198		s.	v. s.	s. al.
nitrate	$\text{N}_2\text{H}_4 \cdot \text{HNO}_3$	95.06	cr.		70.7	subl. 140	174.9 ^{10°}	v. s.	
nitrate, di-	$\text{N}_2\text{H}_4 \cdot 2\text{HNO}_3$	158.07	nd.		104	d.	v. s.		
sulfate	$\text{N}_2\text{H}_4 \cdot \frac{1}{2}\text{H}_2\text{SO}_4$	81.08	delq. pl.		85		v. s.		i. al.
sulfate	$\text{N}_2\text{H}_4 \cdot \text{H}_2\text{SO}_4$	130.12	rhb.	1.378	254		3.055 ^{22°}	27.65 ^{60°}	v. sl. s. abs. al.
Hydrazoic acid (azoimide)	HN_3	43.03	col. lq.		-80	37	∞	∞	∞ al.
Hydriodic acid	HI	127.91	col. gas	4.4 ^{0°} (A)	-50.8	-35.5	42500 ^{10°} cc	v. s.	s. al.
Hydriodic acid	$\text{HI} \cdot \text{H}_2\text{O}$	145.93	col. lq.	1.7 ^{15°}		127 ^{774mm}	∞	∞	∞ al.
Hydriodic acid	$\text{HI} \cdot 2\text{H}_2\text{O}$	163.94	col. lq.		-43		∞	∞	s. al.
Hydriodic acid	$\text{HI} \cdot 3\text{H}_2\text{O}$	181.96	col. lq.		-48		∞	∞	s. al.
Hydriodic acid	$\text{HI} \cdot 4\text{H}_2\text{O}$	199.97	col. lq.		-36.5		∞	∞	s. al.
Hydrobromic acid	HBr	80.91	col. gas; 1.325 (lq.)	2.71 ^{0°} (A)	-86	-67	221 ^{0°}	130 ^{100°}	s. al.
Hydrobromic acid	$\text{HBr} \cdot \text{H}_2\text{O}$	98.93	col. lq.	1.78					Stable at -15.5° and 1 atm., and at -11.3° and 2.5 atm.
Hydrobromic acid	HBr (47.8% in H_2O)	80.91	col. lq.	1.486		126	∞		s. al.
Hydrobromic acid	$\text{HBr} \cdot 2\text{H}_2\text{O}$	118.96	wh. cr.	2.11 ^{-15°}	-11		s.	s.	
Hydrochloric acid	$\text{HCl}^\dagger$	36.46	col. gas; 1.256 (lq.)	1.268 ^{0°} (A)	-111	-85	82.3 ^{0°}	56.1 ^{60°}	s. al., et.
Hydrochloric acid	HCl (45.2% in H_2O)	36.46	col. lq.	1.48	-15.35		∞		s. al.
Hydrochloric acid	$\text{HCl} \cdot 2\text{H}_2\text{O}$	72.49	col. lq.	1.46 ^{-18.3°}	0	d.	∞	∞	s. al.
Hydrochloric acid	$\text{HCl} \cdot 3\text{H}_2\text{O}$	90.51	col. lq.		-24.4	d.	∞	∞	s. al.
Hydrocyanic acid (prussic acid)	HCN	27.03	poisonous gas or col. lq., 1.254	0.697 ^{18°}	-14	26	∞	∞	∞ al., et.
Hydrofluoric acid	HF	20.01	gas or col. lq.	0.988 ^{13.6°}	-83	19.4	∞ 0° to 19.4°	v. s.	
Hydrofluoric acid	HF (35.35% in H_2O)	20.01	col. lq.	1.15	-35	120	v. s.		
Hydrogen	H_2	2.02	col. gas or cb.	lq. 0.0709 ^{-252.7°} 0.06948 (A)	-259.1	-252.7	2.1 ^{0°} cc	0.85 ^{90°} cc	sl. s. Fe, Pd, Pt
peroxide	$\text{H}_2\text{O}_2^\ddagger$	34.01	col. lq., 1.333	1.438 ^{30°}	-0.89	151.4 ^{760mm}	∞		s. a., et.; i. petr. et
selenide	H_2Se	80.98	col. gas	2.12 ^{-42°}	-64	-42	377 ^{4°} cc	270 ^{22.5°} cc	s. CS_2 , COCl_2
sulfide	H_2S	34.08	col. gas	1.1895 (A)	-82.9	-59.6	437 ^{0°} cc	186 ^{40°} cc	9.54 ^{15°} cc al.; s. CS_2
Hydroxylamine	NH_2OH	33.03	rhb., delq.	1.35 ^{18°}	34	56.5 ^{22mm}	s.	d.	s. a., al.
hydrochloride	$\text{NH}_2\text{OH} \cdot \text{HCl}$	69.49	col., mn.	1.67 ^{17°}	151	d.	83.3 ^{17°}	v. s.	s. al.; i. et.
nitrate	$\text{NH}_2\text{OH} \cdot \text{HNO}_3$	96.04	col. cr.		48	d. <100	v. s.	d.	v. s. abs. al.
sulfate	$\text{NH}_2\text{OH} \cdot \frac{1}{2}\text{H}_2\text{SO}_4$	82.07	col., mn.		170 d.		32.9 ^{0°}	68.5 ^{90°}	v. sl. s. al.; i. et., abs. al.

*Usual commercial form.

†Usual commercial form about 31 percent.

‡Usual commercial forms 3 or 30 percent.

TABLE 2-1 Physical Properties of the Elements and Inorganic Compounds (Continued)

Name	Formula	Formula weight	Color, crystalline form and refractive index	Specific gravity	Melting point, °C	Boiling point, °C	Solubility in 100 parts		
							Cold water	Hot water	Other reagents
Hypobromous acid	HBrO	96.91	yel.			40 ^{50mm}	s.	d.	
Indium	In	114.82	soft, tet. met.	7.3 ^{20°}	155	1450	i.	i.	s. a.
Iodic acid	HIO ₃	175.91	col., rhb.	4.629 ^{0°}	110 d.		286 ^{0°}	576 ^{101°}	v. s. 87% al.; i. abs. al. et., chl.
Iodine	I ₂	253.81	blue-bk., rhb.	4.93 ^{20°}	113.5	184.35	0.0162 ^{0°}	0.09566 ^{60°}	s. al., KI, et.
oxide, penta-	I ₂ O ₅	333.81	wh., trimetric	4.799 ^{25°} ₄	d. 300		187.4 ^{12°}		i. abs. al., et., chl.
Iodoplatinic acid	H ₂ PtI ₆ ·9H ₂ O	1120.66	brn., delq. mn.				s. d.		
Iridium	Ir	192.22	wh. met., cb.	22.4 ^{20°}	2350	>4800	i.	i.	sl. s. aq. reg., aq. Cl ₂
Iron, cast†	Fe	55.85	gray	7.03	1275		i.	i.	s. a.; i. alk.
pure	Fe	55.85	silv. met., cb.	7.86 ^{20°}	1535	3000	i.	i.	s. a.; i. alk.
steel	Fe	55.85	silv. gray	7.6 to 7.8	1375		i.	i.	s. a.; i. alk.
white pig	Fe	55.85	gray	7.6 to 7.8	1075		i.	i.	s. a.; i. alk.
wrought	Fe	55.85	gray	7.86	1505		i.	i.	s. a.; i. alk.
carbide (cementite)	Fe ₃ C	179.55	pseudo hex.	7.4	1837		i.	i.	s. a.
carbonyl	Fe(CO) ₅	195.90	pa. yel. lq.	1.457 ^{21°}	-21	102.5 ^{760mm}	i.		s. al., H ₂ SO ₄ , alk.
nitride	Fe ₂ N	125.70	gray	6.35	d. >560		d.		s. HCl, H ₂ SO ₄
silicide	FeSi	83.93	yel.-gray, oct.	6.1 ^{20°} ₄			i.	i.	i. aq. reg.
sulfide, di- (marcasite)	FeS ₂	119.98	yel., rhb.	4.87	tr. 450	d.	0.00049		i. dil. a.
sulfide, di- (pyrite)	FeS ₂	119.98	yel., cb.	5.0	1171	d.	0.0005		i. dil. a.
sulfide (pyrrhotite)	Fe ₇ S ₈	647.44	hex.	4.6 ^{20°} ₄	d. >700		i.		
Cf. also under ferric and ferrous									
Krypton	Kr	83.80	col. gas	2.818 (A)	-169	-151.8	11.05 ^{0°} cc	3.57 ^{60°} cc	sl. s. al., bz.
Lanthanum	La	138.91	lead gray	6.15 ^{20°}	826	1800	d.		s. a.
Lead	Pb	207.20	silv. met., cb.	11.337 ^{20°} ₂₀	327.5	1620	i.	i.	s. HNO ₃ ; i. c. HCl, H ₂ SO ₄
acetate	Pb(C ₂ H ₃ O ₂) ₂	325.29	wh. cr.	3.251 ^{20°} ₄	280		19.7 ^{0°}	221 ^{50°}	s. gly.; v. sl. s. al.
acetate (sugar of lead)	Pb(C ₂ H ₃ O ₂) ₂ ·3H ₂ O†	379.33	wh., mn.	2.55	-3H ₂ O, 75		45.64 ^{15°}	200 ^{100°}	s. gly.; sl. s. al.
acetate	Pb(C ₂ H ₃ O ₂) ₂ ·10H ₂ O	505.44	wh., rhb.	1.689	22		s.	s.	
acetate, basic	Pb ₂ (C ₂ H ₃ O ₂) ₃ OH	608.54	wh.				v. s.		sl. s. al.
acetate, basic	Pb(C ₂ H ₃ O ₂) ₂ ·Pb(OH) ₂ ·H ₂ O	584.52	wh. nd.				v. s.		s. al.
acetate, basic	Pb(C ₂ H ₃ O ₂) ₂ ·2Pb(OH) ₂	807.72	wh. nd.				5.55	18.2	s. al.
arsenate, monobasic	PbH ₂ (AsO ₄) ₂	489.07	tri., 1.82	4.46 ^{15°}	d. 140		d.		s. HNO ₃
arsenate, dibasic (schultenite)	PbHAsO ₄	347.13	wh., mn., 1.9097	5.94	d. >200	-H ₂ O, 280	i.	sl. s.	s. HNO ₃ , NaOH
arsenate, meta-	Pb(AsO ₃) ₂	453.04	hex.	6.42 ^{15°}			d.		s. HNO ₃
arsenate, pyro-	Pb ₃ As ₂ O ₇	676.24	rhb., 2.03	6.85 ^{15°} ₁₅	802		i.	d.	s. HCl, HNO ₃ ; i. sc.
Lead azide	PbN ₆	291.24	col. nd.		expl. 350		i.	0.05 ^{100°}	v. s. ac.; i. NH ₄ OH
bromide	PbBr ₂	367.01	col., rhb.	6.66	373	918	0.4554 ^{0°}	4.75 ^{100°}	s. a., KBr; sl. s. NH ₃ ; i. al.
carbonate (cerussite)	PbCO ₃	267.21	wh., rhb., 2.0763	6.6	d. 315		0.00011 ^{20°}	d.	s. a., alk.; i. NH ₃ , al.
carbonate, basic (hydrocerussite; white lead)	2PbCO ₃ ·Pb(OH) ₂ †	775.63	wh., hex.	6.14	d. 400		i.	i.	s. ac.; sl. s. aq. CO ₂
chloride (cotunnite)	PbCl ₂	278.11	wh., rhb., 2.2172	5.80	501	954 ^{760mm}	0.673 ^{0°}	3.34 ^{100°}	sl. s. dil. HCl, NH ₃ , i. al.
chromate (crocoite)	PbCrO ₄	323.19	yel., mn., 2.42	6.12	844	d.	0.000007 ^{20°}	i.	s. a., alk.; i. NH ₃ , ac.
chromate, basic	PbCrO ₄ ·PbO	546.39	or.-yel. nd.				i.	i.	s. a., alk.
formate	Pb(HCO ₂) ₂	297.23	wh., rhb.	4.56	d. 190		1.6 ^{16°}	18 ^{100°} d.	i. al.
hydroxide	3PbO·H ₂ O	687.61	cb.	7.592	-H ₂ O, 130		0.014		s. a., alk.
nitrate	Pb(NO ₃) ₂	331.21	col., cb. or mn., 1.7815	4.53	d. 470		38.8 ^{0°}	138.8 ^{100°}	8.8 ^{23°} al.
oxide, sub-	Pb ₂ O	430.40	bk., amor.	8.34	d. red heat		i.	i.	s. a., alk.
oxide, mono- (litharge)	PbO	223.20	yel., tet.	9.53	888		0.0068 ^{18°}		s. alk., PbAc, NH ₄ Cl, CaCl ₂
oxide, mono (massicotite)	PbO	223.20	yel., rhb., 2.61	8.0					

oxide, mono-	PbO	223.20	amor.	9.2 to 9.5			i.	i.	s. alk., PbAc, NH ₄ Cl, CaCl ₂
oxide, red (minium)	Pb ₃ O ₄	685.60	red, amor.	9.1	d. 500		i.	i.	s. ac., h. HCl
oxide, sesqui-	Pb ₂ O ₃	462.40	red-yel., amor.		d. 360		i.	i.	s. a., alk.
oxide, di- (plattnerite)	PbO ₂	239.20	brn., tet., 2.229	9.375	d. 290		i.	i.	s. ac., h. alk.; i. al.
silicate	PbSiO ₃	283.28	col., mn., 1.961	6.49	766		i.		s. a.
sulfate (anglesite)	PbSO ₄	303.26	wh., mn. or rhb., 1.8823	6.2	1170		0.0028 ⁹⁰	0.0056 ⁴⁰	s. conc. a., NH ₄ salts; i. al.
sulfate, acid	Pb(HSO ₄) ₂ ·H ₂ O	419.36	cr.		d.		0.0001 ¹⁸⁰		sl. s. H ₂ SO ₄
sulfate, basic (lanarkite)	PbSO ₄ ·PbO	526.46	col., mn.	6.92	977		0.0044 ¹⁸⁰		sl. s. H ₂ SO ₄
sulfide (galena)	PbS	239.27	lead gray, cb., 3.912	7.5	1120		0.00009 ¹⁸⁰	i.	s. a.; i. alk.
thiocyanate	Pb(CNS) ₂	323.36	col., mn.	3.82	d. 190		0.05 ²⁰	s.	s. KCNS, HNO ₃
Lithium	Li	6.94	silv. met. cb.	0.53 ²⁰	186	1336 ± 5	d.	d.	s. a., NH ₃
benzoate	LiC ₇ H ₅ O ₂	128.05	wh. leaflets	3.464 ²⁵ ₄	547	1265	33 ²⁵	40 ¹⁰⁰	7.7 ²⁵ , 10 ⁷⁸ al.
bromide	LiBr	86.85	wh., delq., cb., 1.784				143 ⁰ (2H ₂ O)	266 ¹⁰⁰	s. al., act.
bromide	LiBr·2H ₂ O	122.88	wh. pr.		44		246 ²⁰	(1H ₂ O)	s. al.
carbonate	Li ₂ CO ₃	73.89	col., mn., 1.567	2.11 ⁰	618	d.	1.54 ⁰	0.72 ¹⁰⁰	s. dil. a.; i. al., act., NH ₃
chloride	LiCl	42.39	wh., delq., cb., 1.662	2.068 ²⁵ ₄	614	1360	67 ⁰	127.5 ¹⁰⁰	2.48 ¹⁵ al.; s. et.
citrate	Li ₃ C ₆ H ₅ O ₇ ·4H ₂ O	281.98	wh. cr.		d.		61.2 ¹⁵	66.7 ¹⁰⁰	sl. s. al., et.
fluoride	LiF	25.94	wh., cb., 1.3915	2.295 ^{21.5}	870	1670	0.27 ¹⁵	0.135 ³⁵	s. HF; i. act.
formate	LiHCO ₂ ·H ₂ O	69.97	col., rhb.	1.46	-H ₂ O, 94		49.2 ⁰	346.6 ¹⁰⁴	sl. s. al., et.
hydride	LiH	7.95	wh., cb.	0.820	680		d.		i. et.
hydroxide	LiOH	23.95	wh. cr.	2.54	445	925 ±	12.7 ⁰	17.5 ¹⁰⁰	sl. s. al.
hydroxide	LiOH·H ₂ O	41.96	col., mn.	1.83		d.	22.3 ¹⁰	26.8 ⁹⁰	sl. s. al.
nitrate	LiNO ₃	68.95	col., trig., 1.735	2.38	261		53.4 ⁰	194 ⁷⁰	s. al., NH ₃
nitrate	LiNO ₃ ·3H ₂ O	122.99	col.		29.88		v. s.	∞	
oxide	Li ₂ O	29.88	col., 1.644	2.013 ²⁵ ₄		subl. <1000	forms LiOH		
phosphate, monobasic	LiH ₂ PO ₄	103.93	col.	2.461	>100				
phosphate, tribasic	Li ₃ PO ₄	115.79	wh., rhb.	2.537 ^{17.5}	837		0.034 ¹⁸⁰	v. sl. s.	s. a., NH ₄ Cl; i. act.
phosphate, tribasic	Li ₃ PO ₄ ·12H ₂ O	331.98	wh., trig.	1.645	100		v. sl. s.	v. sl. s.	
salicylate	LiC ₇ H ₅ O ₃	144.05	col.		d.		128 ²⁶		v. s. al.
sulfate	Li ₂ SO ₄	109.94	col., mn., 1.465	2.22	860		35.34 ⁰	29.9 ¹⁰⁰	i. act., 80% al.
sulfate	Li ₂ SO ₄ ·H ₂ O†	127.96	col., mn., 1.477	2.06	-H ₂ O, 130		43.6 ⁰	35 ¹⁰⁰	i. 80% al.
sulfate, acid	LiHSO ₄	104.01	pr.	2.123 ¹³	170.5		d.		
Lutecium	Lu	174.97							
Magnesium	Mg	24.31	silv. met., hex.	1.74 ²⁰	651	1110	i.	sl. s. d.	s. a., NH ₄ salts
acetate	Mg(C ₂ H ₃ O ₂) ₂	142.39	wh.	1.42	323		v. s.	v. s.	5.25 ¹⁵ m. al.
acetate	Mg(C ₂ H ₃ O ₂) ₂ ·4H ₂ O†	214.45	wh., mn. pr., 1.491	1.454	80		v. s.	v. s.	v. s. al.
aluminate (spinel)	MgO·Al ₂ O ₃	142.26	col. cb., 1.718–23	3.6	2135		i.		v. sl. s. dil. HCl; i. dil. HNO ₃
ammonium chloride	MgCl ₂ ·NH ₄ Cl·6H ₂ O	256.79	wh., rhb., delq.	1.456	-4H ₂ O, 195		16.7	s.	s. a.; i. al.
ammonium phosphate (struvite)	MgNH ₄ PO ₄ ·6H ₂ O	245.41	col., rhb., 1.496	1.715	d. 100		0.0231 ⁰	0.0195 ⁹⁰	
ammonium sulfate (boussingaultite)	MgSO ₄ ·(NH ₄) ₂ SO ₄ ·6H ₂ O	360.60	col., mn.	1.72	>120		16.86 ⁰	130 ¹⁰⁰	
benzoate	Mg(C ₇ H ₅ O ₂) ₂ ·3H ₂ O	320.58	wh. pd.		-3H ₂ O, 110		4.5 ²⁵ (anh.)	s.	s. act.
carbonate (magnesite)	MgCO ₃	84.31	wh., trig. 1.700	3.037	d. 350		0.0106		s. a., aq. CO ₂ ; i. act., NH ₃
carbonate (nesquehonite)	MgCO ₃ ·3H ₂ O	138.36	col., rhb., 1.501	1.852	-H ₂ O, 100		0.1518 ¹⁹	d.	s. a., aq. CO ₂
carbonate, basic (hydromagnesite)	3MgCO ₃ ·Mg(OH) ₂ ·3H ₂ O	365.31	wh., rhb., 1.530	2.16	d.		0.04	0.011	s. a., NH ₄ salts; i. al.
Magnesium chloride (chloromagnesite)	MgCl ₂	95.21	col., hex., 1.675	2.325 ²⁵	712	1412	52.8 ⁰	73 ¹⁰⁰	50 al.
chloride (bischofite)	MgCl ₂ ·6H ₂ O†	203.30	wh., delq., mn., 1.507	1.56	118 d.	d.	281 ⁰	918 ¹⁰⁰	50 al.
hydroxide (brucite)	Mg(OH) ₂	58.32	wh., trig., 1.5617	2.4	d.		0.0009 ¹⁸⁰		s. NH ₄ salts, dil. a.
nitride	Mg ₃ N ₂	100.93	gn.-yel., amor.		d.		i.	d.	s. a.; i. al.
oxide (magnesia; periclase)	MgO	40.30	col., cb., 1.7364	3.65	2800	3600	0.00062		s. a., NH ₄ salts; i. al.
perchlorate	Mg(ClO ₄) ₂ †	223.21	wh., delq.	2.60 ²⁵	d.		99.6 ²⁵	v. s.	24 ²⁵ al., 51.8 ³⁵ m. al.; 0.29 et.

*See also a table of alloys.

†Usual commercial form.

TABLE 2-1 Physical Properties of the Elements and Inorganic Compounds (Continued)

Name	Formula	Formula weight	Color, crystalline form and refractive index	Specific gravity	Melting point, °C	Boiling point, °C	Solubility in 100 parts		
							Cold water	Hot water	Other reagents
Magnesium chloride (<i>Cont.</i>)									
peroxide	MgO ₂	56.30	wh. pd.		expl. 275		i.	i.	s. a.
phosphate, pyro-	Mg ₂ P ₂ O ₇	222.55	col., mn., 1.604	2.598 ^{22°}	1383		i.	i.	s. a.; i. alk.
phosphate, pyro-	Mg ₃ P ₃ O ₇ ·3H ₂ O	276.60	wh., amor.	2.56	−3H ₂ O, 100		i.	sl. s.	s. a.; i. al.
potassium chloride (carnallite)	MgCl ₂ ·KCl·6H ₂ O	277.85	delq., rhb., 1.475	1.60 ^{19.4°}	265		64.5 ^{19°} d.	d.	d. al.
potassium sulfate (picromerite)	MgSO ₄ ·K ₂ SO ₄ ·6H ₂ O	402.72	mn., 1.4629	2.15	d. 72		19.26 ^{0°}	81.7 ^{75°}	
silicofluoride	MgSiF ₆ ·6H ₂ O	274.47	col., trig., 1.3439	1.788 ^{17.5°}	d.		64.8 ^{17.5°}	s.	d. HF
sodium chloride	MgCl ₂ ·NaCl·H ₂ O	171.67	col.				s.	s.	
sulfate	MgSO ₄	120.37	col.	2.66	1185		26.9 ^{0°}	68.3 ^{100°}	s. al.
sulfate (epsom salt; epsomite)	MgSO ₄ ·7H ₂ O*	246.47	col., rhb., 1.4554	1.68	70 d.		72.4 ^{0°}	178 ^{40°}	s. al.
Manganese	Mn	54.94	gray-pink met.	7.2 ^{20°}	1260	1900	d.	s.	s. dil. a.
acetate	Mn(C ₂ H ₃ O ₂) ₂	173.03		1.74 ^{20°}			s.	s.	
acetate	Mn(C ₂ H ₃ O ₂) ₂ ·4H ₂ O*	245.09	pa. pink, mn.	1.589			s.	64.5 ^{50°}	s. al., m. al.
carbonate (rhodocrosite)	MnCO ₃	114.95	rose, trig., 1.817	3.125	d.		0.0065 ^{25°}		s. aq. CO ₂ , dil. a.; l. NH ₃ , al.
chloride (scacchite)	MnCl ₂	125.84	rose, delq., cb.	2.977 ^{25°}	650	1190	63.4 ^{0°}	123.8 ^{100°}	s. al.; i. et., NH ₃
chloride	MnCl ₂ ·4H ₂ O*	197.91	rose red, delq., mn. 1.575	2.01	58.0	−H ₂ O, 106; −4H ₂ O, 200	151 ^{8°}	∞	s. al.; i. et.
chloride, per-	MnCl ₄	196.75	gn.				s.	s.	s. al., et.
hydroxide (ous) (pyrochroite)	Mn(OH) ₂	88.95	wh., trig.	3.258 ^{18°}	d.		0.002 ^{20°}	i.	s. a., NH ₄ salts; i. alk.
hydroxide (ic) (manganite)	Mn ₂ O ₃ ·H ₂ O	175.89	brn., rhb., 2.24	3.258	d.		i.	i.	s. h. H ₂ SO ₄
nitrate	Mn(NO ₃) ₂ ·6H ₂ O	287.04	rose red, mn.	1.82 ^{21°}	25.8	129.5	426 ^{0°}	∞	v. s. al.
oxide (ous) (manganosite)	MnO	70.94	gray-gn., cb., 2.16	5.18	1650		i.	i.	s. a., NH ₄ Cl
oxide (ic)	Mn ₂ O ₃	157.87	brn.-bk., cb.	4.81	−0, 1080		i.	i.	s. a.; i. act.
oxide, di- (pyrolusite; polianite)	MnO ₂ *	86.94	bk., rhb.	5.026	−0, >230		i.	i.	s. HCl; i. HNO ₃ , act.
sulfate (ous)	MnSO ₄	151.00	red-wh.	3.235	700	d. 850	53 ^{0°}	73 ^{50°}	s. al.; i. et.
sulfate (ous) (szmikite)	MnSO ₄ ·H ₂ O	169.02	pa. pink, mn., 1.595	2.87	Stable 57 to 117		98.47 ^{48°}	79.77 ^{100°}	
sulfate (ous)	MnSO ₄ ·2H ₂ O	187.03		2.526 ^{15°}	Stable 40 to 57		85.27 ^{35°}	106.8 ^{55°}	
sulfate (ous)	MnSO ₄ ·3H ₂ O	205.05		2.356 ^{15°}	Stable 30 to 40		74.22 ^{5°}	99.31 ^{57°}	
sulfate (ous)	MnSO ₄ ·4H ₂ O*	223.06	pink, rhb. or mn., 1.518	2.107	Stable 18 to 30	−4H ₂ O, 450	136 ^{16°}	169 ^{50°}	i. al.
sulfate (ous)	MnSO ₄ ·5H ₂ O	241.08	pink, tri., 1.508	2.103 ^{15°}	Stable 8 to 18		142 ^{5°}	200 ^{35°}	
sulfate (ous)	MnSO ₄ ·6H ₂ O	259.09			Stable −5 to +8		204 ^{0°}	247 ^{9°}	
sulfate (ous)	MnSO ₄ ·7H ₂ O	277.11	pink, mn. or rhb.	2.092	Stable −10 to −5; 19 d.	−7H ₂ O, 280	176 ^{0°}	251 ^{14°}	
sulfate (ic)	Mn ₂ (SO ₄) ₃	398.06	gn., delq. cr.	3.24	d. 160		v. s.	d.	s. HCl, dil. H ₂ SO ₄ ; l. conc. H ₂ SO ₄ , HNO ₃
Mercuric acetate	Hg(C ₂ H ₃ O ₂) ₂	318.68	wh. pl.	3.270	d.		25 ^{10°}	100 ^{100°}	s. al. sl. d.
bromide	HgBr ₂	360.40	wh., rhb.	6.053	237	322	0.5 ^{20°}	25 ^{100°}	25.2 ^{0°} al.; v. sl. s. et.
carbonate, basic	HgCO ₃ ·2HgO	693.78	brn.-red				i.		s. aq. CO ₂ , NH ₄ Cl
chloride (corrosive sublimate)	HgCl ₂	271.50	wh., rhb., 1.859	5.44	277	304	3.6 ^{0°}	61.3 ^{100°}	33 ^{25°} 99% al.; 33 et.
fulminate	Hg(CNO) ₂	284.62	cb.	4.42	expl.		sl. s.		s. NH ₄ OH, al.
hydroxide	Hg(OH) ₂	234.60			−H ₂ O, 175		i.	i.	s. a.
oxide (montroydite)	HgO	216.59	yel. or red, rhb., 2.5	11.14	d. 100		0.0052 ^{25°}	0.041 ^{100°}	s. a.; i. al.
oxychloride (kleinite)	HgCl ₂ ·3HgO	921.26	yel., hex.	7.93	d. 260		i.	d.	s. HCl
silicofluoride, basic	HgSiF ₆ ·HgO·3H ₂ O	613.30	yel. nd.				d.		s. a.
sulfate	HgSO ₄	296.65	wh., rhb.	6.47	d.		d.		s. a.; i. al., act., NH ₃
sulfate, basic (turpeth)	HgSO ₄ ·2HgO	729.83	yel., tet.	6.44			0.005	0.167 ^{100°}	s. a.; i. al.
Mercurous acetate	Hg ₂ (C ₂ H ₃ O ₂) ₂	259.63	wh. sc.		d.		0.75 ^{13°}	d.	s. H ₂ SO ₄ , HNO ₃ ; i. al.
bromide	HgBr	280.49	wh., tet.	7.307	subl. 345		7 × 10 ^{−9}	i.	s. a.; i. al., act.
carbonate	Hg ₂ CO ₃	461.19	yel. pd.		d. 130		i.	d.	s. NH ₄ Cl

chloride (calomel)	HgCl	236.04	wh., tet., 1.9733	7.150	302	383.7	0.0014 ⁹⁰	0.0007 ⁴³⁰	s. aq. reg., Hg(NO ₃) ₂ ; sl. s. HNO ₃ , HCl; i. al., etc.
iodide	HgI	327.49	yel., tet.	7.70	290 d.	subl. 140; 310d.	2 × 10 ⁻⁸	v. sl. s.	s. KI; i. al.
nitrate	HgNO ₃ ·H ₂ O	280.61	wh. mn.	4.785 ^{3,90}	70	expl.	v. s.	d.	s. HNO ₃ ; i. al., et.
Mercurous oxide	Hg ₂ O	417.18	bk.	9.8	d. 100		i.	0.0007	s. h. ac.; i. alk., dil. HCl, NH ₃
sulfate	Hg ₂ SO ₄	497.24	wh., mn.	7.56	d.		0.055 ^{16,50}	0.092 ¹⁰⁰⁰	s. H ₂ SO ₄ , HNO ₃
Mercury†	Hg	200.59	silv. lq. or hex.(?)	13.546 ²⁰⁰	-38.87	356.9	i.	i.	s. HNO ₃ ; i. HCl
Molybdenum	Mo	95.94	gray, cb.	10.2	2620 ± 10	3700	i.	i.	s. h. conc. H ₂ SO ₄ ; i. HCl, HF, NH ₃ , dil. H ₂ SO ₄ , Hg
chloride, di-	MoCl ₂	166.85	yel., amor.	3.714 ²⁵⁰ ₄	d.		i.	i.	s. HCl, H ₂ SO ₄ , NH ₄ OH, al., et.
chloride, tri-	MoCl ₃	202.30	dark red pd.	3.578 ²⁵⁰ ₄	d.		i.	d.	s. HNO ₃ , H ₂ SO ₄ ; v. sl. s. al., et.
chloride, tetra-	MoCl ₄	237.75	brn., delq.		volt.	d.	s.	d.	s. HNO ₃ , H ₂ SO ₄ ; sl. s. al., et.
chloride, penta-	MoCl ₅	273.21	bk. cr.	2.928 ²⁵⁰ ₄	194	268	s.	d.	s. HNO ₃ , H ₂ SO ₄ ; i. abs. al., et.
oxide, tri- (molybdite)	MoO ₃	143.94	col., rhb.	4.50 ^{19,50}	795	subl.	0.107 ¹⁸⁰	2.106 ⁷⁹⁰	s. a., NH ₄ OH
sulfide, di- (molybdenite)	MoS ₂	160.07	bk., hex., 4.7	4.801 ¹⁴⁰	1185		i.	i.	s. H ₂ SO ₄ , aq. reg.
sulfide, tri-	MoS ₃	192.14	red-brn.	d.	d.		sl. s.	s.	s. alk. sulfides
sulfide, tetra-	MoS ₄	224.20	brn. pd.	d.	d.		i.	i.	s. alk. sulfides; i. NH ₃
Molybdic acid	H ₂ MoO ₄	161.95	yel-wh., hex.	d. 115	d. 115		v. sl. s.	sl. s.	s. NH ₄ OH, H ₂ SO ₄ ; i. NH
Molybdic acid	H ₂ MoO ₄ ·H ₂ O	161.95	yel., mn.	3.124 ¹⁵⁰	-H ₂ O, 70	-2H ₂ O, 200	0.133 ¹⁸⁰	2.13 ⁷⁰⁰	s. a., NH ₄ OH, NH ₄ salts
Neodymium	Nd	144.24	yellowish	6.9 ²⁰⁰	840		d.		
Neon	Ne	20.18	col. gas	lq. 1.204 ^{-245,90} 0.674 (A)	-248.67	-245.9	2.6 ⁰⁰ cc	1.1 ⁴⁵⁰ cc	s. lq. O ₂ , al., act., bz.
Neptunium	Np ²³⁹	239.05			Produced by Neutron bombardment of U ²³⁸				
Nickel	Ni	58.69	silv. met., cb.	8.90 ²⁰	1452	2900	i.	i.	s. dil. HNO ₃ ; sl. s. H ₂ SO ₄ , HCl; i. NH ₃
acetate	Ni(C ₂ H ₃ O ₂) ₂	176.78	gn. pr.	1.798	d.		16.6		i. al.
ammonium chloride	NiCl ₂ ·NH ₄ Cl·6H ₂ O	291.18	gn., delq., mn.	1.645			150 ²⁵⁰	v. s.	
ammonium sulfate	NiSO ₄ ·(NH ₄) ₂ SO ₄ · 6H ₂ O	394.99	blue-gn., mn., 1.5007	1.923			2.5 ^{3,50}	39.2 ⁸⁸⁰	v. sl. s. (NH ₄) ₂ SO ₄
bromate	Ni(BrO ₃) ₂ ·6H ₂ O	422.59	gn., cb.	2.575	d.		28		s. NH ₄ OH
bromide	NiBr ₂	218.50	yel., delq.	4.64 ²⁸⁰ ₄	d.		112.8 ⁰⁰	156 ¹⁰⁰⁰	s. al., et., NH ₄ OH
bromide	NiBr ₂ ·3H ₂ O	272.55	gn., delq.		-3H ₂ O, 200		199 ⁰⁰	316 ¹⁰⁰⁰	s. al., et., NH ₄ OH
bromide, ammonia	NiBr ₂ ·6NH ₃	320.68	vl. pd.	1.837			v. s.	d.	i. c. NH ₄ OH
bromoplatinate	NiPtBr ₆ ·6H ₂ O	841.29	trig.	3.715					
carbonate	NiCO ₃	118.70	lt. gn., rhb.		d.		0.0093 ²⁵⁰	i.	s. a.
carbonate, basic	2NiCO ₃ ·3Ni(OH) ₂ · 4H ₂ O	587.59	lt. gn.		d.		i.	d.	s. a., NH ₄ salts
carbonyl	Ni(CO) ₄	170.73	lq.	1.31 ¹⁷⁰	-25	43 ^{751mm}	0.018 ^{9,80}	i.	s. aq. reg., HNO ₃ , al., et.
chloride	NiCl ₂	129.60	yel., delq.	3.544	subl.	973	53.8 ⁰⁰	87.6 ¹⁰⁰⁰	s. NH ₄ OH, al.; i. NH ₃
chloride	NiCl ₂ ·6H ₂ O*	237.69	gn., delq., mn., 1.57 ±				180	v. s.	v. s. al.
chloride, ammonia	NiCl ₂ ·6NH ₃	231.78					s.	d.	s. NH ₄ OH; i. al.
cyanide	Ni(CN) ₂ ·4H ₂ O	182.79	gn. pl.		-4H ₂ O, 200	d.	i.	i.	s. KCN; i. dil. KCl
dimethylglyoxime	NiC ₈ H ₁₄ O ₄ N ₄	288.91	scarlet red cr.		subl. 250		i.	i.	s. abs. al., a.; i. ac., NH ₄ OH
formate	Ni(HCO ₂) ₂ ·2H ₂ O	184.76	gn. cr.	2.154	d.		s.		
hydroxide (ic)	Ni(OH) ₃	109.72	bk.		d.		i.	i.	s. a., NH ₄ OH, NH ₄ Cl
hydroxide (ous)	Ni(OH) ₂ ·¼H ₂ O	97.21	lt. gn.	4.36	d.		v. sl. s.	v. sl. s.	s. a., NH ₄ OH; i. alk.
nitrate	Ni(NO ₃) ₂ ·6H ₂ O	290.79	gn., mn.	2.05	56.7	136.7	243.0 ⁰⁰	∞ ^{56,70}	s. NH ₄ OH; i. abs. al.
nitrate, ammonia	Ni(NO ₃) ₂ ·4NH ₃ ·2H ₂ O	286.86					v. s.	i. al.	i. al.
oxide, mono- (bunsenite)	NiO	74.69	gn.-bk., cb., 2.37	7.45	Forms Ni ₂ O ₃ at 400		i.	i.	s. a., NH ₄ OH
potassium cyanide	Ni(CN) ₂ ·2KCN·H ₂ O	258.97	red yel., mn.	1.875 ¹¹⁰	-H ₂ O, 100		s.		d. a.
sulfate	NiSO ₄	154.76	yel., cb.	3.68	-SO ₃ , 840		27.2 ⁰⁰	76.7 ¹⁰⁰⁰	i. al., et., act.

*Usual commercial form.

†See also Tables 2-28 and 2-280.

TABLE 2-1 Physical Properties of the Elements and Inorganic Compounds (Continued)

Name	Formula	Formula weight	Color, crystalline form and refractive index	Specific gravity	Melting point, °C	Boiling point, °C	Solubility in 100 parts		
							Cold water	Hot water	Other reagents
Nickel (<i>Cont.</i>) sulfate	NiSO ₄ ·6H ₂ O ^o	262.85	gn. mn. or blue, tet., 1.5109	2.07	tr. 53.3	-6H ₂ O, 280	131 ^{50°}	280 ^{100°}	v. s. NH ₄ OH, al.
sulfate (morenosite)	NiSO ₄ ·7H ₂ O	280.86	gn., rhb., 1.4893	1.948	98-100	-6H ₂ O, 103	63.5 ^{0°}	117.8 ^{30°}	s. al.
Nitric acid	HNO ₃	63.01	col. lq.	1.502	-42	86	∞	∞	expl. with al.
Nitric acid	HNO ₃ ·H ₂ O	81.03	col. lq.		-38		∞	∞	d. al.
Nitric acid	HNO ₃ ·3H ₂ O	117.06	col. lq.		-18.5		263 ^{-20°}	∞	d. al.
Nitro acid sulfite	NO ₂ HSO ₃	127.08	col., rhb.		73 d.		d.		s. H ₂ SO ₄
Nitrogen	N ₂	28.01	col. gas or cb. cr.	1.026 ^{-252.5°} 0.808 ^{-195.8°} 12.5 ^{0°} (D)	-209.86	-195.8	2.35 ^{0°} cc	1.55 ^{20°} cc	sl. s. al.
Nitrogen oxide, mono- (ous)	N ₂ O	44.01	col. gas	lq. 1.226 ^{-89°} 1.530 (A)	-102.3	-90.7	130.52 ^{0°} cc	60.82 ^{24°} cc	s. H ₂ SO ₄ , al.
oxide, di- (ic)	NO or (NO) ₂	30.01 60.01	col. gas	lq. 1.269 ^{-150.2°} 1.0367 (A)	-161	-151	7.34 ^{0°} cc	0.0 ^{100°}	26.6 cc al.; 3.5 cc H ₂ SO ₄ ; s. aq. FeSO ₄
oxide, tri-	N ₂ O ₃	76.01	red-brn. gas or blue lq. or solid	1.447 ^{2°}	-102	3.5	s.		s. a., et.
oxide, tetra- (per- or di-)	NO ₂ or (NO ₂) ₂	46.01 92.01	yel. lq., col. solid, red-brn. gas	1.448 ^{20°}	-9.3	21.3	d.		s. HNO ₃ , H ₂ SO ₄ , chl., CS ₂
oxide, penta-	N ₂ O ₅	108.01	wh., rhb.	1.63 ^{18°}	30	47	s.	Forms HNO ₃	
oxybromide	NOBr	109.91	brn. lq.	>1.0	-55.5	-2	d.		
oxychloride	NOCl	65.46	red-yel. lq. or gas	1.417 ^{-12°} 2.31 (A)	-64.5	-5.5	d.		s. fuming H ₂ SO ₄
Nitroxyl chloride	NO ₂ Cl	81.46	yel.-brn. gas	lq. 1.32 ^{14°}	<-30	5	d		
Osmium	Os	190.23	blue, hex.	22.48 ^{20°}	2700	>5300	i.	i.	sl. s. aq. reg., HNO ₃ ; i. NH ₃
chloride, di-	OsCl ₂	261.14	gn., delq.				s. d.		s. NaCl, al., et.
chloride, tri-	OsCl ₃	296.59	brn., cb.		d. 560-600		sl. s.		s. a., alk., al.; sl. s. et.
chloride, tetra-	OsCl ₄	332.04	red-yel. nd.				s. d.		s. HCl, al.
Oxygen	O ₂	32.00	col. gas or hex. solid	1.14 ^{-185°} 1.426 ^{-252.5°} 1.1053 (A)	-218.4	-183	4.89 ^{0°} cc	2.6 ^{30°} cc 1.7 ^{100°} cc	sl. s. al., s. fused Ag
Ozone	O ₃	48.00	col. gas	1.71 ^{-183°} 3.03 ^{-80°} 1.658 (A)	-251	-112	0.494 ^{0°} cc	0 ^{60°} cc	s. oil turp., oil cinn.
Palladium	Pd	106.42	silv. met., cb.	12.0 ^{20°} 11 ^{1550°}	1555	2200	i.	i.	s. aq. reg., h. H ₂ SO ₄ ; i. NH ₃
bromide (ous)	PdBr ₂	266.23	brn.				i.	i.	s. HBr
chloride	PdCl ₂	177.33	brn., cb.		500 d.		s.	s.	s. HCl, act., al.
chloride	PdCl ₂ ·2H ₂ O	213.36	brn. pr.				s.	s.	s. HCl, act., al.
cyanide	Pd(CN) ₂	158.45	yel.		d.		i.	i.	s. HCN, KCN, NH ₄ OH; i. dil. a.
hydride	Pd ₃ H	213.85	met.	11.06	d.				
Palladous dichlorodiammine	Pd(NH ₃) ₂ Cl ₂	211.39	red or yel., tet.	2.5			s.		s. a., NH ₄ OH
Perchloric acid	HClO ₄	100.46	unstable, col. lq	1.768 ^{22°} 4	-112	16 ^{18mm}	s.		
Perchloric acid	HClO ₄ ·H ₂ O	118.47	fairly stable nd.	1.88	50	d.	s.		
Perchloric acid	HClO ₄ ·2H ₂ O ^o 73.6% anh.	136.49	stable lq., col.	1.71 ^{25°} 4	-17.8	200	v. s.		s. al.
Periodic acid	HIO ₄	191.91	wh. cr.		d. 138	subl. 110	s.		
Periodic acid	HIO ₄ ·2H ₂ O	227.94	delq., mn.		d. 110		v. s.	v. s.	sl. s. al., et.
Permanganic acid	HMnO ₄	119.94	exists only in solution				v. s.	d.	d. al.
Permolybdic acid	HMoO ₄ ·2H ₂ O	196.98	wh. cr.				v. s.	v. s.	
Persulfuric acid	H ₂ S ₂ O ₈	194.14	hyg. cr.		<60		v. s.	d.	
Phosphamic acid	PONH ₂ ·(OH) ₂	97.01	cb.		d.		v. s.	v. s. d.	i. al.
Phosphatomolybdic acid	H ₂ P(Mo ₂ O ₇) ₆ ·28H ₂ O	2365.71	yel. cb.		78	-25H ₂ O, 140	s.		s. HNO ₃
Phosphine	PH ₃	34.00	col. gas	lq. 0.746 ^{-90°} 1.146 (A)	-132.5	-85	26 ^{17°} cc	i. ^{100°}	s. Cu ₂ Cl ₂ , al., et.
Phosphonium chloride	PH ₄ Cl	70.46	wh., cb.		28 ^{46atm}	subl.	d.		

Phosphoric acid, hypo-	H ₃ P ₂ O ₆	161.98	cr.		55	d. 70	s.	450 ^{62°}	
Phosphoric acid, meta-	HPO ₃	79.98	vitreous, delq.	2.2–2.5	subl.		s.	Forms H ₃ PO ₄	i. lq. CO ₂
Phosphoric acid, ortho-	H ₃ PO ₄ †	98.00	col., rhb.	1.834 ^{18.2°}	42.35	–½H ₂ O, 213	2340 ^{26°}	v. s.	s. al.
Phosphoric acid, pyro-	H ₄ P ₂ O ₇	177.98	wh. nd.		61		800 ^{28°}	Forms H ₃ PO ₄	v. s. al., et.
Phosphorous acid, hypo-	H ₃ PO ₃	66.00	syrupey	1.493 ^{18.8°}	26.5	d.	∞	∞	
Phosphorous acid, ortho-	H ₃ PO ₃	82.00	col.	1.651 ^{21.2°}	74	d. 200	307.3 ^{9°}	730 ^{40°}	
Phosphorous acid, pyro-	H ₄ P ₂ O ₅	145.98	nd.		38	d. 130	d.		
Phosphorus, black	P ₄	123.90	rhombohedral	2.69		ign. in air, 400	i.	i.	i. CS ₂
Phosphorus, red	P ₄	123.90	red, cb.	2.20 ^{20°}	590 ^{43atm}	ign. in air, 725	i.	i.	s. alk.; i. CS ₂ , NH ₃ , et.
Phosphorus, yellow	P ₄	123.90	yel., hex., 2.1168	1.82 ^{20°} ; lq. 1.745 ^{44.5°}	44.1; ign. 34		280	0.0003	0.4 al.; 1000 ^{10°} CS ₂ ; 1.5 ^{0°} , 10 ^{81°} bs.; s. NH ₃
chloride, tri-	PCl ₃	137.33	col., fuming lq.	1.574 ^{20.8°}	–111.8	75.95 ^{760mm}	d.		s. et., chl., CS ₂
chloride, penta-	PCl ₅	208.24	delq., tet.	solid 1.6; 3.60 ^{295°} (A)	148 under pressure	subl. 160	d.		s. CS ₂ , C ₆ H ₅ COCl
oxide, penta-	P ₂ O ₅	141.94	wh., delq., amor.	2.387	subl. 250			Forms H ₃ PO ₄	v. s.
oxychloride	POCl ₃	153.33	col., fuming lq.	1.675	2	107.2 ^{760mm}	d.		s. H ₂ SO ₄ ; i. NH ₃ , act.
Phosphotungstic acid	H ₃ PO ₄ ·12WO ₃ ·xH ₂ O	2880.05	yel.-gn. cr.				s.		d. al.
Platinum	Pt	195.08	silv. met., cb.	21.45 ^{20°} lq. 19 ^{1755°}	1755	4300	i.	i.	s. al., et.
chloride (ic)	PtCl ₄	336.89	brn.		d. 370		140 ^{25°}	v. s.	s. al., act.; sl. s. NH ₂ ;
chloride (ous)	PtCl ₂	265.98	brn.	5.87 ^{11°}	d. 581		i.	i.	i. et.
chloride (ic)	PtCl ₄ ·8H ₂ O	481.01	red, mn.	2.43	–4H ₂ O, 100		v. s.	v. s.	s. HCl, NH ₄ OH; sl. s. NH ₃ ; i. al., et.
cyanide (ous)	Pt(CN) ₂	247.11	yel.-brn.				i.	i.	s. al., et.
Plutonium	Pu	238.05		Produced by deuteron bombardment on U ²³⁸					i. alk.
Plutonium	Pu	239.05		Produced by neutron bombardment on U ²³⁸					
Potassium	K	39.10	silv. met., cb.	0.86 ^{20°} lq. 0.83 ^{42°}	62.3	760	d.	Forms KOH	s. a., al., Hg
acetate	KC ₂ H ₃ O ₂	98.14	wh. pd.	1.8	292		217 ^{0°}	396 ^{90°}	33 al.; i. et.
acetate, acid	KH(C ₂ H ₃ O ₂) ₂	158.19	delq. nd. or pl.		148	d. 200	d.		s. ac.
aluminate	K ₂ (AlO ₂) ₂ ·3H ₂ O	250.20	cr.				s.	d.	s. alk.; i. al.
amide	KNH ₂	55.12	yel.-grm.		338	subl. 400	d.		d. al.; 3.6 ^{25°} NH ₃
arsenate (monobasic)	KH ₂ AsO ₄	180.03	col., tet., 1.5674	2.867	288		18.87 ^{6°}	v. s.	i. al.
auricyanide	KAu(CN) ₄ ·1.5H ₂ O	367.16	pl.		d. 200		s.	v. s.	s. al.
aurocyanide	KAu(CN) ₂	288.10	rhb.				14.3	200 ^{100°}	sl. s. al.; i. et.
bicarbonate	KHCO ₃	100.12	mn., 1.482	2.17	d. 100–200		22.4 ^{0°}	60 ^{90°}	i. satd. K ₂ CO ₃ , al.
bisulfate	KHSO ₄	136.17	rhb., or mn., 1.480	2.35	210	d.	36.3 ^{0°}	121.6 ^{100°}	d. al.
bromate	KBrO ₃	167.00	trig.	3.27 ^{17.5°}	370 d.		3.11 ^{0°}	49.75 ^{100°}	sl. s. al.; i. act.
bromide	KBr	119.00	col., cb., 1.5594	2.75 ^{25°}	730	1380	53.5 ^{0°}	104 ^{100°}	sl. s. al., et.
carbonate	K ₂ CO ₃	138.21	wh., delq. pd., 1.531	2.29	891	d.	105.5 ^{0°}	156 ^{100°}	i. al.
carbonate	K ₂ CO ₃ ·2H ₂ O	174.24	rhb.	2.043			183 ^{0°}	331 ^{100°}	
carbonate	2K ₂ CO ₃ ·3H ₂ O	330.46	mn.	2.13			129.4 ^{0°}	268 ^{100°}	
chlorate	KClO ₃	122.55	col., mn., 1.5167	2.32	368	d. 400	3.3 ^{0°}	57 ^{100°}	0.83 al.; s. alk.
chloride (sylvite)	KCl	74.55	col., cb., 1.4904	1.988	790	1500	27.6 ^{0°}	56.7 ^{100°}	s. al., alk.
chloroplatinate	K ₂ PtCl ₆	485.99	yel., cb., 1.825±	3.499	d. 250		0.74 ^{0°}	5.2 ^{100°}	i. al., et.
chromate (tarapacait)	K ₂ CrO ₄	194.19	yel., rhb., 1.7261	2.732 ^{18°}	975		58.0 ^{0°}	75.6 ^{100°}	i. al.
cyanate	KCN	81.12	wh., tet.	2.048			s.	d.	v. sl. s. al.
cyanide	KCN	65.12	wh., cb., delq., 1.410	1.52 ^{16°}	634.5		s.	122.2 ^{108.8°}	s. gly.; 0.9 ^{19.5°} al.; 1.3 h. al.
dichromate	K ₂ Cr ₂ O ₇	294.18	red, tri.	2.69	398	d.	4.9 ^{0°}	80 ^{100°}	i. al.
ferricyanide	K ₃ Fe(CN) ₆	329.24	red, mn. pr., 1.5689	1.84	d.		334 ^{4°}	77.5 ^{100°}	s. act.; sl. s. al.; i. NH ₃
ferrocyanide	K ₄ Fe(CN) ₆ ·3H ₂ O	422.39	yel., mn., 1.5772	1.853 ^{17°}	–3HO ₂ , 70		27.8 ^{12.2°}	90.6 ^{96.8°}	s. act.; i. NH ₃ , al., et.
formate	KHCO ₂	84.12	col., rhb.	1.91	167.5	d.	331 ^{18°}	657 ^{90°}	sl. s. al.; i. et.
hydride	KH	40.11	cb., 1.453	0.80	d.		d.		i. et., bz., CS ₂
hydrosulfide	KHS	72.17	wh., delq., rhb.	2.0	455		s.	s. d.	s. al.
hydroxide	KOH	56.11	wh., delq., rhb.	2.044	380	1320	97 ^{0°}	178 ^{100°}	v. s. al., et.; i. NH ₃
iodate	KIO ₃	214.00	col., mn.	3.89	560		4.73 ^{0°}	32.2 ^{100°}	s. KI; i. al., NH ₃
iodide	KI	166.00	wh., cb., 1.6670	3.13	723	1330	127.5 ^{0°}	208 ^{100°}	4 ^{20°} al.; s. NH ₃ ; sl. s. et.

*One commercial form 70 to 72 per cent.

†Common commercial form 85 per cent H₃PO₄ in aqueous solution.

TABLE 2-1 Physical Properties of the Elements and Inorganic Compounds (Continued)

Name	Formula	Formula weight	Color, crystalline form and refractive index	Specific gravity	Melting point, °C	Boiling point, °C	Solubility in 100 parts		
							Cold water	Hot water	Other reagents
Potassium (<i>Cont.</i>)									
iodide, tri-	KI ₃	419.81	dark blue, delq., mn.	3.498	45	d. 225	v. s.		s. KI, al.
iodoplatinate	K ₂ PtI ₆	1034.70	cb.	5.18			s.		
manganate	K ₂ MnO ₄	197.13	gn., rhb.		d. 190		d.		s. KOH
metabisulfite	K ₂ S ₂ O ₅	222.32	mn., pl.		d. 150		25 ^{0°}	120 ^{0°}	sl. s. al.; i. et.
nitrate (saltpeter)	KNO ₃	101.10	col., rhb., 1.5038	2.11 ^{10.6°}	tr. 129; 333	d. 400	13.3 ^{0°}	246 ^{100°}	0.1 ^{30°} al.; i. et.
nitrite	KNO ₂	85.10	pr.	1.915	297	d. 350	281 ^{0°}	413 ^{100°}	v. s. NH ₃ ; sl. s. al.
oxalate	K ₂ C ₂ O ₄ ·H ₂ O	184.23	wh., mn.	2.13	d.		28.7 ^{0°}	83.2 ^{100°}	
oxalate, acid	KHC ₂ O ₄ °	128.13	mn., 1.545	2.0	d.		14.3 ^{30°}	48.1 ^{100°}	
oxalate, acid	KHC ₂ O ₄ ·½H ₂ O	137.13	trimetric		d.		2.2 ^{0°}	51.5 ^{100°}	
oxide	K ₂ O	94.20	wh., cb.	2.32 ^{30°}			Forms KOH	v. s.	s. al., et.
perchlorate	KClO ₄	138.55	col., rhb., 1.4737	2.524 ^{11°}	d. 400		0.75 ^{0°}	21.8 ^{100°}	0.105 ^{20°} m. al.; i. et.
permanganate	KMnO ₄	158.03	purple, rhb.	2.703	d. <240		2.83 ^{0°}	32.35 ^{75°}	s. H ₂ SO ₄ ; d. al.
persulfate	K ₂ S ₂ O ₈	190.32	wh., tri., 1.4669		d. <100		1.77 ^{0°}	10 ^{40°}	i. al.
phosphate, monobasic	KH ₂ PO ₄	136.09	col., delq., tet., 1.5095	2.338	256		14.8 ^{0°}	83.5 ^{90°}	i. al.
phosphate, dibasic	K ₂ HPO ₄	174.18	wh., delq.		d.		33 ^{25°}	v. s.	sl. s. al.
phosphate, tribasic	K ₃ PO ₄	212.27	wh., rhb.	2.564 ^{17°}	1340		193.1 ^{25°}	v. s.	i. al.
phosphate, meta-	KPO ₃	118.07	wh. pd.	2.258 ^{14.5°}	tr. 450; 798	1320	s.	s.	
phosphate, meta-	K ₄ P ₂ O ₇ ·2H ₂ O	508.31	amor.	2.264 ^{14.5°}	-2H ₂ O, 100	d.	s.	83	s. a.
phosphate, pyro-	K ₄ P ₂ O ₇ ·3H ₂ O	384.38	delq.	2.33	-2H ₂ O, 180	-3H ₂ O, 300	s.	v. s.	i. al.
phthalate, acid	KHC ₈ H ₄ O ₄	204.22	wh. cr.	1.63	d.		10.2 ^{25°}	36	
platinocyanide	K ₂ Pt(CN) ₄ ·3H ₂ O	431.39	yel., rhb., 1.62±	2.45 ^{16°}			sl. s.	v. s.	s. al., et.
silicate	K ₂ SiO ₃	154.28	hyg. 1.521±		976		s.	s.	i. al.
silicate, tetra-	K ₄ Si ₃ O ₉ ·H ₂ O	352.55	rhb., 1.530	2.417	d. 400		s.	s.	i. al.
sulfate (arcenite)	K ₂ SO ₄	174.26	col., rhb., 1.4947	2.662	tr. 588		7.35 ^{0°}	24.1 ^{100°}	i. al., act., CS ₂
Potassium sulfate, pyro-	K ₂ S ₂ O ₇	254.32	col.	2.277	300		s.	d.	
sulfide, mono-	K ₂ S·5H ₂ O	200.34	rhb., delq.		60	-3H ₂ O, 150	s.		s. al., gly.; i. et.
sulfite	K ₂ SO ₃ ·2H ₂ O	194.29	wh., rhb.		d.		100	>100	sl. s. al.; i. NH ₃
sulfite, acid	KHSO ₃	120.17	wh., mn.		d. 190		45.5 ^{15°}	91.5 ^{75°}	i. abs. al.
tartrate	K ₂ C ₄ H ₄ O ₆ ·½H ₂ O	235.28	col., mn., 1.526	1.98		d.	12.5 ^{17.5°}	278 ^{100°}	sl. s. al.
tartrate, acid	KHC ₄ H ₄ O ₆ °	188.18	col., rhb.	1.956			0.37 ^{0°}	6.1 ^{100°}	s. a., alk.; i. al., ac.
thiocyanate	KCNS	97.18	col., delq., mn., 1.660±	1.886	172.3	d. 500	177 ^{0°}	217 ^{20°}	20.8 ^{22°} act.; s. al.
thiosulfate	K ₂ S ₂ O ₃	190.32	col., cb.		d. 400		96.1 ^{0°}	311.2 ^{90°}	
thiosulfate	3K ₂ S ₂ O ₃ ·H ₂ O	588.99	delq., mn.	2.23	-H ₂ O, 180	d.			i. al.
Praseodymium	Pr	140.91	yel.	6.5 ^{20°}	940		d.		
Radium	Ra	226.03	wh., met.	5?	960	1140	d. +H ₂		d. a.
bromide	RaBr ₂	385.83	wh., mn.	5.79	728	subl. 900	70 ^{20°}	s.	s. al.
Radon (Niton)	Rn	222.02	gas	lq. 5.5; 111 (D)	-71	-62	51 ^{0°} cc	8.5 ^{60°} cc	
Rhenium	Re	186.21	hex.		3440				i. HF, HCl; s. H ₂ SO ₄ ; HNO ₃
Rhodium	Rh	102.91	gray-wh., cb.	12.5	1955	>2500	i.	i.	sl. s. aq. reg., a.
chloride	RhCl ₃	209.26	red		d. 450	subl. 800±	i.	i.	v. sl. s. alk.; i. aq. reg., a.
chloride	RhCl ₃ ·4H ₂ O	281.33	dark red				v. s.		s. HCl, al.; i. et.
Rubidium	Rb	85.47	silv. wh.	lq. 1.475 ^{88.5°} ; 1.53 ^{20°}	38.5	700	d.		s. a., al.
Ruthenium	Ru	101.07	bk., porous	8.6	>1950		i.	i.	sl. s. aq. reg., a.
Ruthenium	Ru	101.07	gray, hex.	12.2 ^{20°}	2450	>2700	i.	i.	
Samarium	Sm (also Sa)	150.36		7.7	>1300				
Scandium	Sc	44.96		2.5?	1200	2400			
Selenic acid	H ₂ SeO ₄	144.97	hex. pr.	2.950 ^{15°}	58	260	1300 ^{30°}	∞ ^{60°}	s. H ₂ SO ₄ ; d. al.; i. NH ₃
Selenic acid	H ₂ SeO ₄ ·H ₂ O	162.99	nd.	2.627 ^{15°}	26	205	v. s.		
Selenium	Se ₈	631.68	red pd., amor., 2.92	4.26 ^{25°}	50	688	i.	i.	s. CS ₂ , H ₂ SO ₄ , CH ₂ I ₂
Selenium	Se ₈	631.68	gray, trig., 3.00; red, hex.	4.80; 4.50	220	688	i.	i.	s. CS ₂ , H ₂ SO ₄

Selenium	Se ₈	631.68	steel gray	4.8 ^{25°}	217	688	i.	i.	i. CS ₂ ; s. H ₂ SO ₄
Selenous acid	H ₂ SeO ₃	128.97	hex.	3.004 ^{15°} ₄	d.		90 ^{0°}	400 ^{90°}	v. s. al.; i. NH ₃
Silicic acid, meta-	H ₂ SiO ₃	78.10	amor., 1.41	2.1–2.3			i.	i.	s. alk.; i. NH ₄ Cl
Silicic acid, ortho-	H ₄ SiO ₄	96.11	amor.	1.576 ^{17°}			sl. s.	sl. s.	s. alk.; i. NH ₄ Cl
Silicon, crystalline	Si	28.09	gray, cb., 3.736	2.4 ^{20°}	1420	2600	i.	i.	s. HNO ₃ + HF, Ag; sl. s. Pb, Zn; i. HF
Silicon, graphitic	Si	28.09	cr.	2.0–2.5		2600	i.	i.	s. HNO ₃ + HF, fused alk.; i. HF
Silicon, amorphous	Si	28.09	brn., amor.	2		2600	i.	i.	s. HF, KOH
carbide	SiC	40.10	blue-bk., trig., 2.654	3.17	>2700	subl. 2200	i.	i.	s. fused alk.; i. a.
chloride, tri-	Si ₂ Cl ₆	268.89	lf. or lq.	1.58 ^{0°}	–1	144 ^{760mm}	d.	d.	d. alk.
chloride, tetra-	SiCl ₄	169.90	col., fuming lq., 1.412	1.50	–70	57.6	d.	d.	d. conc. H ₂ SO ₄ , al.
fluoride	SiF ₄	104.08	gas	3.57 (A)	–95.7	–65 ^{1810mm}	v. s. d.		s. HNO ₃ , al., et.
hydride (silane)	SiH ₄	32.12	col. gas	lq. 0.68 ^{–185°}	–185	–112 ^{760mm}	i.		i. al., et.; d. KOH
oxide, di- (opal)	SiO ₂ ·xH ₂ O		iridescent, amor.	2.2	1600–1750	subl. 1750	i.	i.	s. HF, h. alk., fused CaCl ₂
oxide, di- (cristobalite)	SiO ₂	60.08	col., cb. or tet., 1.487	2.32	1710	2230	i.	i.	s. HF; i. alk.
oxide, di- (lechatelierite)	SiO ₂	60.08	hex., 1.5442	2.20		2230	i.	i.	s. HF; i. alk.
oxide, di- (quartz)	SiO ₂	60.08	trig., rhb., 1.469	2.26	tr. <1425	2230	i.	i.	s. HF; i. alk.
oxide, di- (tridymite)	SiO ₂	60.08	trig., rhb., 1.469	2.26	tr. 1670	2230	i.	i.	s. HNO ₃ , h. H ₂ SO ₄ ; i. alk.
Silver	Ag	107.87	silv. met., cb.	10.5 ^{20°}	960.5	1950	i.	i.	0.51 ^{18°} NH ₄ OH; s. KCN, Na ₂ S ₂ O ₃
bromide (bromyrite)	AgBr	187.77	pa. yel., cb., 2.252	6.473 ^{25°} ₄	434	d. 700	0.00002 ^{20°}	0.00037 ^{100°}	s. NH ₄ OH, Na ₂ S ₂ O ₃ ; i. al.
carbonate	Ag ₂ CO ₃	275.75	yel. pd.	6.077	218 d.		0.003 ^{20°}	0.05 ^{100°}	s. NH ₄ OH, KCN; sl. s.
chloride (cerargyrite)	AgCl	143.32	wh., cb., 2.071	5.56	455	1550	0.000089 ^{10°}	0.00217 ^{100°}	HCl
cyanide	AgCN	133.89	wh., 1.685±	3.95	–(CN) ₂ , 320		0.000022 ^{20°}		s. NH ₄ OH, KCN, HNO ₃
nitrate (lunar caustic)	AgNO ₃	169.87	col., rhb., 1.744	4.352 ^{19°} ₄	212	444 d.	122 ^{0°}	952 ^{100°}	s. gly.; v. sl. s. al.
Sodium	Na	22.99	silv. met. cb.	0.97 ^{20°}	97.5	880	d., forms NaOH		i. bz.; d. al.
acetate	NaC ₂ H ₃ O ₂	82.03	wh., mn., 1.464	1.528	324		46.5 ^{20°}	170 ^{100°}	2.1 ^{18°} al.
acetate	NaC ₂ H ₃ O ₂ ·3H ₂ O	136.08	wh., mn.	1.45	58	–3H ₂ O, 120	v. s.	v. s.	7.8 ^{25°} abs. al.
aluminate	NaAlO ₂	81.97	amor.		1650		s.	v. s.	i. al.
amide	NaNH ₂	39.01	olive gn.		210	400	d.	d.	d. al.
Sodium ammonium phosphate	NaNH ₄ HPO ₄ ·4H ₂ O	209.07	col., mn.	1.574	79 d.		16.7	100	i. al.
antimonate, meta-	2NaSbO ₃ ·7H ₂ O	511.60	cb.				0.031 ^{12.8°}		sl. s. al., NH ₄ salts; i. ac.
arsenate	Na ₃ AsO ₄ ·12H ₂ O	424.07	hex., 1.4589	1.759	86.3		26.7 ^{17°}		1.67 al., 50 ^{15°} gly.
arsenate, acid (monobasic)	NaH ₂ AsO ₄ ·H ₂ O	181.94	rhb., 1.5535	2.535	d. 100		s.		
arsenate, acid (dibasic)	Na ₂ HAsO ₄ ·7H ₂ O°	312.01	col., mn., 1.4658	1.871	125	–7H ₂ O, 100	61 ^{15°}	v. s.	sl. s. al.
arsenate, acid (dibasic)	Na ₂ HAsO ₄ ·12H ₂ O	402.09	mn., 1.4496	1.72	28	–12H ₂ O, 100	5.59 ^{0.1°}	140.7 ^{30°}	sl. s. al.
arsenite, acid	Na ₂ HAsO ₃	169.91	col.	1.87			v. s.		
benzoate	NaC ₇ H ₅ O ₂	144.10	col. cr.				62.5 ^{25°}	76.9 ^{100°}	2.3 ^{25°} , 8.3 ^{78°} al.
bicarbonate	NaHCO ₃	84.01	wh., mn., 1.500	2.20	–CO ₂ , 270		6.9 ^{0°}	16.4 ^{60°}	i. al.
bifluoride	NaHF ₂	61.99	col. cr.		d.		3.7 ^{20°}	s.	
bisulfate	NaHSO ₄	120.06	col., tri.	2.742	>315	d., –H ₂ O	50 ^{0°}	100 ^{100°}	d. al.; i. NH ₃
bisulfite	NaHSO ₃	104.06	col., mn., 1.526	1.48	d.		sl. s.	s.	i. al., act.
borate, tetra-	Na ₂ B ₄ O ₇	201.22		2.367	741		1.3 ^{0°}	8.79 ^{40°}	i. al.
borate, tetra	Na ₂ B ₄ O ₇ ·5H ₂ O	291.30	col., rhb., 1.461	1.815			22 ^{92°} (anh.)	52.3 ^{100°} (anh.)	
borate, tetra- (borax)	Na ₂ B ₄ O ₇ ·10H ₂ O°	381.37	wh., mn., 1.4694	1.73	75	–10H ₂ O, 200	1.3 ^{0.5} (anh.)	20.3 ^{90°} (anh.)	s. gly.; i. abs. al.
bromate	NaBrO ₃	150.89	col., cb.	3.339 ^{17.5°}	381		27.5 ^{0°}	90.9 ^{100°}	i. al.
bromide	NaBr	102.89	col., cb., 1.6412	3.205 ^{17.5°}	755	1390	90 ^{20°}	121 ^{100°}	sl. s. al.
bromide	NaBr·2H ₂ O	138.92	col., mn.	2.176	50.7		79.5 ^{0°} (anh.)	118.3 ^{90°} (anh.)	sl. s. al.
carbonate (soda ash)	Na ₂ CO ₃	105.99	wh. pd., 1.535	2.533	851	d.	7.1 ^{0°}	48.5 ^{104°}	i. al., et.
carbonate	Na ₂ CO ₃ ·H ₂ O	124.00	wh., rhb., 1.506–1.509	1.55	–H ₂ O, 100		s.	s.	s. gly.; i. al., et.
carbonate	Na ₂ CO ₃ ·7H ₂ O	232.10	rhb. or trig.	1.51	d. 35.1		s.	s.	
carbonate (sal soda)	Na ₂ CO ₃ ·10H ₂ O	286.14	wh., mn., 1.425	1.46			21.5 ^{0°}	238 ^{30°}	i. al.

°Usual commercial form.

TABLE 2-1 Physical Properties of the Elements and Inorganic Compounds (Continued)

Name	Formula	Formula weight	Color, crystalline form and refractive index	Specific gravity	Melting point, °C	Boiling point, °C	Solubility in 100 parts		
							Cold water	Hot water	Other reagents
Sodium ammonium phosphate (<i>Cont.</i>)									
carbonate, sesqui- (trona)	$\text{Na}_3\text{H}(\text{CO}_3)_2 \cdot 2\text{H}_2\text{O}$	226.03	wh., mn., 1.5073	2.112	d.		13°	42 ^{100°}	
chlorate	NaClO_3	106.44	wh., cb., or trig., 1.5151	2.490 ^{15°}	248	d.	79°	230 ^{100°}	s. al.
chloride	NaCl	58.44	col., cb., 1.5443	2.163	800.4	1413	35.7°	39.8 ^{100°}	sl. s. al.; i. conc. HCl
chromate	Na_2CrO_4	161.97	yel., rhb.	2.723	392		32°	126 ^{100°}	
chromate	$\text{Na}_2\text{CrO}_4 \cdot 10\text{H}_2\text{O}$	342.13	yel., delq., mn.	1.483	19.9		v. s.	∞	sl. s. al.
citrate	$2\text{Na}_3\text{C}_6\text{H}_5\text{O}_7 \cdot 11\text{H}_2\text{O}$	714.31	wh., rhb.	1.857 ^{23.5°} ₄	-11H ₂ O, 150	d.	91 ^{25°}	250 ^{100°}	i. al.
cyanide	NaCN	49.01	wh., cb., 1.452		563.7	1496	48 ^{10°}	82 ^{35°}	s. NH ₃ ; sl. s. al.
dichromate	$\text{Na}_2\text{Cr}_2\text{O}_7 \cdot 2\text{H}_2\text{O}$	298.00	red, mn., 1.6994	2.52 ^{18°}	-2H ₂ O, 84.6; 356 (anh.)	d. 400	238°	508 ^{80°}	
ferricyanide	$\text{Na}_3\text{Fe}(\text{CN})_6 \cdot \text{H}_2\text{O}$	298.93	red, delq.				18.9°	67 ^{100°}	i. al.
ferrocyanide	$\text{Na}_4\text{Fe}(\text{CN})_6 \cdot 10\text{H}_2\text{O}$	484.06	yel., mn.	1.458			17.9 ^{20°} (anh.)	63 ^{28.5°} (anh.)	
fluoride (villiumite)	NaF	41.99	tet., 1.3258	2.79	992		4°	5 ^{100°}	i. al.
formate	NaHCO_2	68.01	wh., mn.	1.919	253		44°	160 ^{100°}	v. sl. s. al.
hydride	NaH	24.00	silv. nd., 1.470	0.92	d. 800		d.		sl. s. al.; i. et.
hydrosulfide	$\text{NaSH} \cdot 2\text{H}_2\text{O}$	92.09	col., delq., nd.		d.		s.	s.	i. bz., CS ₂ , CCl ₄ , NH ₃ ;
hydrosulfide	$\text{NaSH} \cdot 3\text{H}_2\text{O}$	110.11	rhb.		22	d.	s.	s.	s. molten metal
hydrosulfite	$\text{Na}_2\text{S}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$	210.14	col. cr.		d.		22 ^{20°}	d.	s. al.; d. a.
hydroxide	NaOH	40.00	wh., delq.	2.130	318.4	1390	42°	347 ^{100°}	i. al.
hydroxide	$\text{NaOH} \cdot 3\frac{1}{2}\text{H}_2\text{O}$	103.05	mn.		15.5		s.	v. s.	v. s. al., et., gly.; i. act.
hypochlorite	NaOCl	74.44	pa. yel., in soln. only		d.		26°	158 ^{56°}	
iodide	NaI^*	149.89	col., cb., 1.7745	3.667°	651	1300	158.7°	302 ^{100°}	v. s. al., act.
iodide	$\text{NaI} \cdot 2\text{H}_2\text{O}$	185.92	col., mn.	2.448			v. s.	v. s.	v. s. NH ₃
lactate	$\text{NaC}_3\text{H}_5\text{O}_3$	112.06	col., amor.		d.		v. s.	v. s.	s. al.; i. et.
nitrate (soda niter)	NaNO_3	84.99	col., trig., 1.5874	2.257	308	d. 380	73°	180 ^{100°}	s. NH ₃ ; sl. s. gly., al.
nitrite	NaNO_2	69.00	pa. yel., rhb.	2.168°	271	d. 320	72.1°	163.2 ^{100°}	0.3 ^{20°} et.; 0.3 abs. al.;
oxide	Na_2O	61.98	wh., delq.	2.27	subl.			4.4 ^{20°} m. al.; v. s. NH ₃	d. al.
perborate	$\text{NaBO}_3 \cdot \text{H}_2\text{O}$	99.81	wh. pd.		d. 40		Forms NaOH		
perchlorate	NaClO_4	122.44	rhb., 1.4617		482 d.		sl. s.	d.	s. gly., alk.
perchlorate	$\text{NaClO}_4 \cdot \text{H}_2\text{O}$	140.46	hex.	2.02	d. 130		170°	320 ^{100°}	s. al.; 51 m. al.; 52 act.; i. et.
peroxide	Na_2O_2^*	77.98	yel.-wh. pd.	2.805	d.		209 ^{15°}	284 ^{50°}	s. al.
peroxide	$\text{Na}_2\text{O}_2 \cdot 8\text{H}_2\text{O}$	222.10	wh., hex.		d. 30		s. d.	d.	s. dil. a.
phosphate, monobasic	$\text{NaH}_2\text{PO}_4 \cdot \text{H}_2\text{O}^*$	137.99	col., rhb., 1.4852	2.040	-H ₂ O, 100	d. 200	71°	390 ^{83°}	i. al.
phosphate, monobasic	$\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$	156.01	col., rhb., 1.4629	1.91	60		91.1°	308 ^{40°}	
phosphate, dibasic	$\text{Na}_2\text{HPO}_4 \cdot 7\text{H}_2\text{O}$	268.07	col., mn., 1.4424	1.679	d.		185 ^{40°}	2000 ^{100°}	
phosphate, dibasic	$\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O}$	358.14	col., mn., 1.4361	1.52	34.6	-12H ₂ O, 180	4.3°	76.7 ^{30°}	i. al.
phosphate, tribasic	Na_3PO_4	163.94	wh.	2.537 ^{17.5°}	1340		4.5°	77 ^{100°}	
phosphate, tribasic	$\text{Na}_3\text{PO}_4 \cdot 12\text{H}_2\text{O}^*$	380.12	wh., trig., 1.4458	1.62	73.4	-11H ₂ O, 100	28.3 ^{15°}	∞	i. CS ₂
phosphate, meta-	$\text{Na}_4\text{P}_2\text{O}_{12}$	407.85	col.	2.476	616 d.		s.	s. a., alk.	
phosphate, pyro-	$\text{Na}_4\text{P}_2\text{O}_7^*$	265.90	wh.	2.45	988		2.26 ^{60°}	45 ^{96°}	d. a.
phosphate, pyro-	$\text{Na}_4\text{P}_2\text{O}_7 \cdot 10\text{H}_2\text{O}$	446.06	mn., 1.4525	1.82	d.		5.4°	93 ^{100°}	i. al., NH ₃
phosphate (pyrodisodium)	$\text{Na}_2\text{H}_2\text{P}_2\text{O}_7$	221.94	col., mn., 1.510	1.862	d. 220		4.5°	21 ^{40°}	
phosphate (pyrodisodium)	$\text{Na}_2\text{H}_2\text{P}_2\text{O}_7 \cdot 6\text{H}_2\text{O}$	330.03	col., mn., 1.4645	1.848			6.9°	36 ^{40°}	
potassium tartrate	$\text{NaKC}_4\text{H}_4\text{O}_6 \cdot 4\text{H}_2\text{O}$	282.22	rhb., 1.493	1.790	70 to 80	-4H ₂ O, 215	26°	66 ^{36°}	sl. s. al.
silicate, meta-	Na_2SiO_3	122.06	col., rhb., 1.520		1088		s.	s. d.	i. Na or K salts, al.
Sodium silicate, meta-	$\text{Na}_2\text{SiO}_3 \cdot 9\text{H}_2\text{O}$	284.20	rhb.		47	-6H ₂ O, 100	v. s.	v. s.	29 ^{18°} , $\frac{1}{2}\text{N}$ NaOH
silicate, ortho-	Na_4SiO_4	184.04	col., hex., 1.530		1018		s.	s.	
silicofluoride	Na_2SiF_6	188.06	wh., hex., 1.312	2.679	d.		0.44°	2.45 ^{100°}	i. al.
stannate	$\text{Na}_2\text{SnO}_3 \cdot 3\text{H}_2\text{O}$	266.73	hex. tablets		d. 140		50°	67 ^{50°}	i. al., act.
sulfate (thenardite)	Na_2SO_4	142.04	col., rhb., 1.477	2.698	tr. 100 to mn.		5°	42 ^{100°}	i. al.
sulfate	Na_2SO_4	142.04	col., mn.		tr. 500 to hex.		48.8 ^{40°}	42.5 ^{100°}	d. HI; s. H ₂ SO ₄

sulfate	Na ₂ SO ₄	142.04	col., hex.		884		19.4 ^{20°}	45.3 ^{60°}	
sulfate	Na ₂ SO ₄ ·7H ₂ O	268.15	tet.				44.9 ^{9°}	202.6 ^{26°}	
sulfate (Glauber's salt)	Na ₂ SO ₄ ·10H ₂ O	322.19	col., mn., 1.396	1.464	32.4	−10H ₂ O, 100	36 ^{15°}	412 ^{34°}	i. al.
sulfide, mono-	Na ₂ S	78.04	pink or wh., amor.	1.856			15.4 ^{10°}	57.3 ^{90°}	sl. s. al.; i. et.
sulfide, tetra-	Na ₂ S ₄	174.24	yel., cb.		275		s.	s.	s. al.
sulfide, penta-	Na ₂ S ₅	206.30	yel.		251.8		s.	s.	s. al.
sulfite	Na ₂ SO ₃	126.04	hex. pr., 1.565	2.633 ^{15°} ₄	d.		13.9 ^{0°}	28.3 ^{84°}	i. al., NH
sulfite	Na ₂ SO ₃ ·7H ₂ O	252.15	mn.	1.561	−7H ₂ O, 150	d.	34.7 ^{2°}	67.8 ^{15°}	i. al.
tartrate	Na ₂ C ₄ H ₄ O ₆ ·2H ₂ O	230.08	rhb.	1.818			29 ^{9°}	66 ^{33°}	i. al.
thiocyanate	NaCNS	81.07	delq., rhb., 1.625±		287		110 ^{10°}	225 ^{100°}	v. s. al.
thiosulfate	Na ₂ S ₂ O ₃	158.11	mn.	1.667			50 ^{9°}	231 ^{80°}	
thiosulfate (hypo)	Na ₂ S ₂ O ₃ ·5H ₂ O*	248.18	mn. pr., 1.5079	1.685	d. 48.0		74.7 ^{0°}	301.8 ^{60°}	s. NH ₃ ; v. sl. s. al.
tungstate	Na ₂ WO ₄	293.82	wh., rhb.	4.179	692		57.58 ^{0°}	97 ^{100°}	
tungstate	Na ₂ WO ₄ ·2H ₂ O*	329.85	wh., rhb.	3.245	−2H ₂ O, 100		88 ^{0°}	123.5 ^{100°}	sl. s. NH ₃ ; i. a., al.
tungstate, para-	Na ₆ W ₇ O ₂₄ ·16H ₂ O	2097.05	wh., tri.	3.987 ^{14°}	−16H ₂ O, 300		8	d.	
uranate	Na ₂ UO ₄	348.01	yel.				i.	i.	s. alk. carb., dil. a.
vanadate	Na ₃ VO ₄ ·16H ₂ O	472.15	col. nd.		866 (anh.)		v. s.	d.	i. al.
vanadate, pyro-	Na ₄ V ₂ O ₇	305.84	hex.		654		s.	i. al.	
Stannic chloride	SnCl ₄	260.52	col., fuming lq.	2.226	−30.2	114.1	s.	d.	s. abs. al., act., NH ₃ ; s. ∞ CS ₂
oxide (cassiterite)	SnO ₂	150.71	wh., tet., 1.9968	7.0	1127		i.	i.	s. conc. H ₂ SO ₄ ; i. alk.; NH ₄ OH, NH ₃
sulfate	Sn(SO ₄) ₂ ·2H ₂ O	346.87	col., delq., hex.				v. s.	d.	s. dil. H ₂ SO ₄ , HCl; d. abs. al.
Stannous bromide	SnBr ₂	278.52	yel., rhb.	5.12 ^{17°}	215.5	620	s.	d.	s. C ₆ H ₅ N
chloride	SnCl ₂	189.62	wh., rhb.		246.8	623	83.9 ^{0°}	269.8 ^{15°}	s. alk., abs. al., et.
chloride (tin salt)	SnCl ₂ ·2H ₂ O*	225.65	wh., tri.	2.71 ^{15.5°}	37.7	d.	118.7 ^{0°}	∞	s. tart. a., alk., al.
sulfate	SnSO ₄	214.77	wh. cr.		−SO ₂ , 360		19 ^{19°}	18 ^{100°}	s. H ₂ SO ₄
Strontium	Sr	87.62	silv. met.	2.6	800	1150	d.	Forms Sr(OH) ₂	s. al., a.
acetate	Sr(C ₂ H ₃ O ₂) ₂	205.71	wh. cr.	2.099		d.	36.9 ^{0°}	36.4 ^{97°}	0.26 ^{15°} m. al.
carbonate (strontianite)	SrCO ₃	147.63	wh., rhb., 1.664	3.70	149 ^{760atm}	−CO ₂ , 1350	0.0011 ^{18°}	0.065 ^{100°}	s. a., NH ₄ salts, aq. CO ₂
chloride	SrCl ₂	158.53	wh., cb., 1.6499	3.052	873		43.5 ^{0°}	100.8 ^{100°}	v. sl. s. act., abs. al.; i. NH ₃
chloride	SrCl ₂ ·6H ₂ O*	266.62	wh., rhb., 1.5364	1.933 ^{17°}	−4H ₂ O, 61	−6H ₂ O, 100	104 ^{0°}	198 ^{40°}	
hydroxide	Sr(OH) ₂	121.63	wh., delq.	3.625	375		0.41 ^{0°}	21.83 ^{100°}	s. NH ₄ Cl
hydroxide	Sr(OH) ₂ ·8H ₂ O*	265.76	col., tet., 1.499	1.90	−7H ₂ O in dry air		0.90 ^{0°}	47.7 ^{100°}	s. NH ₄ Cl; i. act.
nitrate	Sr(NO ₃) ₂ *	211.63	col., cb., 1.5878	2.986	570		40 ^{0°}	100 ^{80°}	s. NH ₃ ; 0.012 abs. al.
nitrate	Sr(NO ₃) ₂ ·4H ₂ O	283.69	wh., mn.	2.2			62.2 ^{0°}	124 ^{20°}	i. HNO ₃
oxide (strontia)	SrO	103.62	col., cb., 1.870	4.7	2430		Forms Sr(OH) ₂		sl. s. al.; i. et.
peroxide	SrO ₂	119.62	wh. pd.		d.		0.008 ^{20°}	d.	s. al., NH ₄ Cl; i. act.
peroxide	SrO ₂ ·8H ₂ O	263.74	wh. cr.		−8H ₂ O, 100	d.	0.018 ^{20°}	d.	s. al.; i. NH ₄ OH
sulfate (celestite)	SrSO ₄	183.68	col., rhb., 1.6237	3.96	1580 d.		0.0113 ^{0°}	0.0114 ^{32°}	sl. s. a.; i. dil. H ₂ SO ₄ , al.
sulfate, acid	Sr(HSO ₄) ₂	281.76	col., granular		d.		d.		14 ^{70°} H ₂ SO ₄
Sulfamic acid	NH ₂ SO ₃ H	97.09	wh., rhb.	2.03 ^{13°} ₄	205 d.		20 ^{0°}	40 ^{70°}	sl. s. al., act.; i. et.
Sulfur, amorphous	S	32.07	pa. yel. pd., 2.0–2.9	2.046	120	444.6	i.	i.	sl. s. CS ₂
Sulfur, monoclinic	S ₈	256.52	pa. yel., mn.	1.96	119.0	444.6	i.	i.	s. CS ₂ , al.
Sulfur, rhombic	S ₈	256.52	pa. yel., rhb.	2.07	112.8	444.6	i.	i.	24 ^{0°} , 181 ^{55°} CS ₂
Sulfur bromide, mono-	S ₂ Br ₂	223.94	red, fuming lq.	2.635	−46	54 ^{0.18mm}	d.		
chloride, mono-	S ₂ Cl ₂	135.04	red-yel. lq.	1.687	−80	138	d.		s. CS ₂ , et., bz.
chloride, di-	SCl ₂	102.97	dark red fuming lq.	1.621 ^{15°} ₁₅	−78	59	d.		d. al.
chloride, tetra-	SCl ₄	173.88	yel.-bm. lq.		−30	d. > −20	d.		
oxide, di-	SO ₂	64.06	col. gas	lq., 1.434 ^{0°} ; 2.264 (A)	−75.5	−10.0	22.8 ^{0°}	4.5 ^{50°}	s. H ₂ SO ₄ ; al., ac.
oxide, tri-(α)	SO ₃	80.06	col. pr.	lq., 1.923; 2.75 (A)	16.83	44.6	d.		s. H ₂ SO ₄
oxide, tri-(β)	(SO ₃) ₂	160.13	col., silky, nd.	1.97 ^{20°}	50		Forms H ₂ SO ₄		s. H ₂ SO ₄
Sulfuric acid	H ₂ SO ₄ *	98.08	col., viscous lq.	1.834 ^{18°} ₄	10.49	d. 340	∞	∞	d. al.
Sulfuric acid	H ₂ SO ₄ ·H ₂ O	116.09	pr. or lq.	1.842 ^{15°} ₄	8.62	290	∞	∞	d. al.

*Usual commercial form.

TABLE 2-1 Physical Properties of the Elements and Inorganic Compounds (Concluded)

Name	Formula	Formula weight	Color, crystalline form and refractive index	Specific gravity	Melting point, °C	Boiling point, °C	Solubility in 100 parts		
							Cold water	Hot water	Other reagents
Sulfuric acid	H ₂ SO ₄ ·2H ₂ O	134.11	col. lq.	1.650 ^{40°}	−38.9	167	∞	∞	d. al.
Sulfuric acid, pyro-	H ₂ S ₂ O ₇	178.14	cr.	1.9 ^{20°}	35	d.	d.		d. al.
Sulfuric oxychloride	SO ₂ Cl ₂	134.97	col. lq.	1.667 ^{20°} ₄	−54.1	69.1 ^{760mm}	d.		s. ac.; d. al.
Sulfurous oxybromide	SOBr ₂	207.87	or.-yel. lq.	2.68 ^{18°}	−50	68 ^{40mm}	d.		s. bz., CS ₂ , CCl ₄ ; d. act.
Sulfurous oxychloride	SOCl ₂	118.97	col. lq.	1.638	−104.5	78.8	d.		s. bz., chl.
Tantalum	Ta	180.95	bk.-gray, cb.	16.6	2850	>4100	i.	i.	s. fused alk., HF; i. HCl, HNO ₃ , H ₂ SO ₄
Tellurium	Te	127.60	met., hex.	(α) 6.24; (β) 6.00	452	1390	i.	i.	s. H ₂ SO ₄ , HNO ₃ , KCN, KOH, aq. reg.; i. CS ₂
Terbium	Tb	158.93							
Thallium	Tl	204.38	blue-wh., tet.	11.85	303.5	1650	i.	i.	s. HNO ₃ , H ₂ SO ₄ ; i. NH ₃
acetate	TlC ₂ H ₃ O ₂	263.43	silky nd.	3.68	110		v. s.		v. s. al.
chloride, mono-	TlCl	239.84	wh., cb.	7.00	430	806	0.21 ^{0°}	1.8 ^{100°}	sl. s. HCl; i. al., NH ₄ OH
chloride, sesqui-	Tl ₂ Cl ₃	515.13	yel., hex.	5.9	400–500	d.	0.26 ^{15°}	1.9 ^{100°}	
chloride, tri-	TlCl ₃	310.74	hex. pl.		25	d.	v. s.		s. al., et.
chloride, tri-	TlCl ₃ ·4H ₂ O	382.80	nd.		37	−4H ₂ O, 100	86.2 ^{17°}	d.	s. al., et.
sulfate (ic)	Tl ₂ (SO ₄) ₃ ·7H ₂ O	823.06	lf.		−6H ₂ O, 200	d.	d.	d.	s. dil. H ₂ SO ₄
sulfate (ous)	Tl ₂ SO ₄	504.83	col., rhb., 1.8671	6.77	632	d.	2.70 ^{0°}	18.45 ^{100°}	
sulfate, acid	TlHSO ₄	301.45	trimorphous		115 d.	d.			v. sl. s. dil. H ₂ SO ₄
Thio, <i>cf.</i> sulfo or sulfur									
Thorium	Th	232.04	cb.	11.2	1845	>3000	i.	i.	s. HCl, H ₂ SO ₄ ; sl. s. HNO ₃ ; i. HF, alk.
oxide, di- (thorianite)	ThO ₂	264.04	wh., cb.	9.69	>2800	4400	i.		s. h. H ₂ SO ₄ ; i. alk.
sulfate	Th(SO ₄) ₂	424.16		4.225 ^{17°}			0.74 ^{0°}	5.22 ^{50°}	
sulfate	Th(SO ₄) ₂ ·9H ₂ O	586.30	mn. pr.	2.77	−9H ₂ O, 400		sl. s.	sl. s.	
Thulium	Tm	168.93					i.	i.	
Tin	Sn	118.71	silv. met., tet.	7.31	231.85	2260	i.	i.	s. HCl, H ₂ SO ₄ , dil. HNO ₃
Tin	Sn	118.71	gray, cb.	5.750	Stable −163 to +18	2260	i.	i.	h. aq. KOH
Tin salts, <i>cf.</i> stannic and stannous									s. a., h. alk. solns.
Titanic acid	H ₂ TiO ₃	97.88	wh. pd.				i.	i.	s. alk.; v. sl. s. dil. a.; i. al.
Titanium	Ti	47.87	dark gray, cb.	4.50 ^{17.5°}	1800	>3000	i.	d.	s. a.
chloride, di-	TiCl ₂	118.77	bk., delq.		Unstable in air		d.		i. CS ₂ , et., chl.
chloride, tri-	TiCl ₃	154.23	vl., delq.		d. 440		s.	s.	
chloride, tetra-	TiCl ₄ ^o	189.68	col. lq.	lq., 1.726	−30	136.4	s.	d.	s. dil. HCl
oxide, di- (anatase)	TiO ₂	79.87	brn. or bk., tet., 2.534–2.564	3.84			i.	i.	sl. s. alk.
oxide, di- (brookite)	TiO ₂	79.87	brn. or bk., rhb., 2.586	4.17			i.	i.	
oxide, di- (rutile)	TiO ₂	79.87	col. if pure, tet., 2.615	4.26	1640 d.	<3000	i.	i.	s. H ₂ SO ₄ , alk.
Tungsten	W	183.84	gray-bk., cb.	19.3	3370	5900	i.	i.	s. h. conc. KOH; sl. s. NH ₃ , HNO ₃ , aq. reg.
carbide	WC	195.85	gray pd., cb.	15.7 ^{18°}	2777	6000	i.	i.	s. F ₂ ; i. a.
carbide	W ₂ C	379.69	iron gray	16.06 ^{18°}	2877	6000	i.	i.	s. h. HNO ₃ ; sl. s. HCl, H ₂ SO ₄
oxide, tri-	WO ₃	231.84	yel., rhb.	7.16	>2130		i.	i.	s. alk.; i. a.
Tungstic acid (tungstite)	H ₂ WO ₄	249.85	yel., rhb. 2.24	5.5	−½H ₂ O, 100; 1473		i.	sl. s.	s. HF, alk., NH ₃
Uranic acid	H ₂ UO ₄	304.04	yel. pd.	5.926 ^{15°}	−H ₂ O, 250 to 300		i.	i.	s. a., alk. carb.; i. alk.
Uranium	U	238.03	wh. cr.	18.485 ^{13°} ₄	1133	3500	i.	i.	s. a.; i. alk.
carbide	U ₂ C ₃	512.09	cr.	11.28	2400		d.		d. a.
oxide, di- (uraninite)	UO ₂	270.03	bk., rhb.	10.9	2176		i.	i.	s. HNO ₃ , conc. H ₂ SO ₄

oxide (pitchblende)	U ₃ O ₈	842.08	olive gn.	7.31	d.		i.	i.	s. HNO ₃ , H ₂ SO ₄
sulfate (ous)	U(SO ₄) ₂ ·4H ₂ O	502.22	gn., rhb.	424.15	–4H ₂ O, 300		23 ^{11°}	9 ^{63°}	s. dil. a.
Uranyl acetate	UO ₂ (C ₂ H ₃ O ₂) ₂ ·2H ₂ O	424.15	yel., rhb.	2.89 ^{15°}	–2H ₂ O, 110		9.2 ^{17°}	d.	s. al., act.
carbonate (rutherfordine)	UO ₂ CO ₃	330.04	tet.	5.6					
nitrate	UO ₂ (NO ₃) ₂ ·6H ₂ O	502.13	yel., rhb., 1.4967	2.807	60.2	118	170.3 ^{0°}	∞ ^{60°}	v. s. ac., al., et.; i. dil., alk.
sulfate	UO ₂ SO ₄ ·3H ₂ O	420.14	yel. cr.	3.28 ^{16.5°}	d. 100		18.9 ^{13.2°}	230 ^{25°}	4 al.; s. a.
Vanadic acid, meta-	HVO ₃	99.95	yel. scales				i.		s. a., alk.; i. NH ₃
Vanadic acid, pyro-	H ₄ V ₂ O ₇	217.91	pa. yel., amor.				i.		s. a., alk., NH ₄ OH
Vanadium	V	50.94	lt. gray, cb.	5.96	1710	3000	i.	i.	s. HNO ₃ , H ₂ SO ₄ ; i. aq., alk.
chloride, di-	VCl ₂	121.85	gn., hex., delq.	3.23 ^{18°}			s.	d.	s. al., et.
chloride, tri-	VCl ₃	157.30	pink, tabular, delq.	3.00 ^{18°}	d.		s.	d.	s. abs. al., et.
chloride, tetra-	VCl ₄	192.75	red lq.	1.816 ^{30°}	–109	148.5 ^{75mm}	s. d.		s. abs. al., et., chl., ac.
oxide, di-	V ₂ O ₃	133.88	lt. gray cr.	3.64	ign.		i.	i.	s. a.
oxide, tri-	V ₂ O ₃	149.88	bk. cr.	4.87 ^{18°}	1970		sl. s.	s.	s. HNO ₃ , HF, alk.
oxide, tetra-	V ₂ O ₄	165.88	blue cr.	4.399	1967		i.	i.	s. a., alk.
oxide, penta-	V ₂ O ₅	181.88	red-yel., rhb.	3.357 ^{18°}	800	d. 1750	0.8 ^{20°}		s. a., alk.; i. abs. al.
oxychloride, mono-	VOCl	86.39	brn. pd.	2.824			i.		v. s. HNO ₃
Vanadyl chloride	(VO) ₂ Cl	169.33	yel. cr.	3.64	d. in air		i.		s. HNO ₃
chloride, di-	VOCl ₂	137.85	gn., delq.	2.88 ^{13°}			d.		s. abs. al., dil. HNO ₃
chloride, tri-	VOCl ₃	173.30	yel. lq.	1.829	<–15	127.19	s. d.		s. al., et., ∞Br ₂
Water†	H ₂ O	18.02	col. lq., 1.33300 ^{20°} ; hex. solid, 1.309 col. lq., 1.32844 ^{20°}	1.00 ^{4°} (lq.); 0.915 ^{0°} (ice)	0	100			∞ al.; sl. s. et.
Water, heavy	D ₂ O	20.029		1.107 ^{20°}	3.82	101.42	∞	∞	∞ al.; sl. s. et.
Xenon	Xe	131.29	col. gas	lq., 3.06 ^{–109.1} 2.7 ^{–140°} 4.53 (A)	–140	–109.1	24.2 ^{0°} cc	7.3 ^{50°} cc	
Ytterbium	Yb	173.04		5.51	1490	2500	sl. d.	d.	v. s. dil. a., h. KOH
Yttrium	Y	88.91	dark gray, hex.	7.140	419.4	907	i.	i.	s. a., ac., alk.
Zinc	Zn	65.41	silv. met., hex.	1.840	242	subl. in vac.	30 ^{25°}	44.6 ^{100°}	2.8 ^{25°} , 166 ^{70°} al.
acetate	Zn(C ₂ H ₃ O ₂) ₂	183.50	mn.	1.735	237	–2H ₂ O, 100	40 ^{25°}	66.6 ^{100°}	v. s. al.
acetate	Zn(C ₂ H ₃ O ₂) ₂ ·2H ₂ O*	219.53	wh., mn., 1.494	4.219 ^{4°}	394	650	390 ^{0°}	670 ^{100°}	v. s. NH ₄ OH, al., et.
bromide	ZnBr ₂	225.22	rhb.	4.42	–CO ₂ , 300		0.001 ^{15°}		s. a., alk., NH ₄ salts;
carbonate	ZnCO ₃	125.42	wh., trig., 1.818						i. act., NH ₃
chloride	ZnCl ₂	136.32	wh., delq., 1.687, uniaxial	2.91 ^{35°}	283	732	432 ^{25°}	615 ^{100°}	100 ^{12.5°} al.; v. s. et.; i.
cyanide	Zn(CN) ₂	117.44	col., rhb.		d. 80		0.0005 ^{18°}	sl. s.	s. KCN, NH ₃ , alk.; i. al.
hydroxide	Zn(OH) ₂	99.42	col., rhb.	3.053	d. 125		0.00052 ^{18°}		s. a., alk., NH ₄ OH
iodide	ZnI ₂	319.22	cb.	4.666 ^{14.2°}	446	624	430 ^{0°}	510 ^{100°}	s. a., al., NH ₃ , aq.
nitrate	Zn(NO ₃) ₂ ·6H ₂ O	297.51	col., tet.	2.065 ^{14°}	36.4	–6H ₂ O, 105	324.5	∞ ^{36.4°}	(NH ₄) ₂ CO ₃
oxide (zincite)	ZnO	81.41	wh., hex., 2.004	5.606	>1800		0.00042 ^{18°}		v. s. al.
oxide	ZnO	81.41	wh., amor.	5.47	>1800		0.00042 ^{18°}		s. a., alk., NH ₄ Cl; i. NH ₃
peroxide	ZnO ₂	97.41	yel.	1.571	expl. 212		0.0022		i. NH ₄ OH; d. a.
phosphide	Zn ₃ P ₂	258.17	steel gray, cb.	4.55 ^{13°}	>420	1100	i.		s. dil. a.
silicate	ZnSiO ₃	141.49	hex. or rhb.; glass, 1.650	3.52	1437		i.		
sulfate (zincosite)	ZnSO ₄	161.47	wh., rhb., 1.669	3.74 ^{15°}	d. 740		42 ^{0°}	61 ^{100°}	sl. s. al.; s. gly.
sulfate	ZnSO ₄ ·H ₂ O	179.49	col.	3.28 ^{15°}	d. 238		s.	89.5 ^{100°}	
sulfate	ZnSO ₄ ·6H ₂ O	269.56	mn.	2.072 ^{15°}	–5H ₂ O, 70		s.	s.	sl. s. al.; i. act.; NH ₃
sulfate (goslarite)	ZnSO ₄ ·7H ₂ O*	287.58	rhb., 1.4801	1.966 ^{16.5°}	tr. 39	–7H ₂ O, 280	115.2 ^{0°}	653.6 ^{100°}	sl. s. al.; i. act.; NH ₃
sulfide (α) (wurzite)	ZnS	97.47	wh., hex., 2.356	4.087	1850 ^{150atm}	subl. 1185	0.00069 ^{18°}	i.	v. s. a.; i. ac.
sulfide (β) (sphalerite)	ZnS	97.47	wh., cb.; glass (?) 2.18–2.25	4.102 ^{25°}	tr. 1020		i.	i.	s. a.
sulfide (blende)	ZnS	97.47	wh., granular	4.04			i.	i.	v. s. a.; i. ac.
sulfite	ZnSO ₃ ·2½H ₂ O	190.51	mn.		–2½H ₂ O, 100	d. 200	0.16	d.	s. H ₂ SO ₃ , NH ₄ OH; i. al.
Zirconium	Zr	91.22	cb., pd. ign. easily	6.4	1700	>2900	i.	i.	s. HF, aq. reg.; sl. s. a.
oxide, di- (baddeleyite)	ZrO ₂	123.22	yel. or brn., mn., 2.19	5.49	2700		i.	i.	s. H ₂ SO ₄ , HF
oxide, di- (free from Hf)	ZrO ₂	123.22	wh., mn.	5.73		4300	i.	i.	s. H ₂ SO ₄ , HF

*Usual commercial form.

†Cf. special tables on water and steam, Tables 2-3, 2-4, 2-5, 2-185, 2-186 and 2-351 through 2-357.

NOTE: °F = ½ °C + 32.

TABLE 2-2 Physical Properties of Organic Compounds*

Abbreviations Used in the Table										
(A), density referred to air	cr., crystalline	i-, iso-, containing the group	nd., needles	s-, sec-, secondary	v. s., very soluble					
al., ethyl alcohol	d., decomposes	(CH ₃) ₂ CH-	o-, ortho	silv., silvery	v. sl. s., very slightly soluble					
amor., amorphous	d-, dextrorotatory	i., insoluble	or., orange	sl., slightly	wh., white					
aq., aqua, water	dl-, dextro-laevorotatory	ign., ignites	p-, para	subl., sublimates	yel., yellow					
brn., brown	et., ethyl ether	l-, laevorotatory	pd., powder	sym., symmetrical	(+), right rotation					
bz., benzene	expl., explodes	lf., leaflets	pet., petroleum ether	t-, tertiary	>, greater than					
c., cubic	gn., green	lq., liquid	pl., plates	tet., tetragonal	<, less than					
cc., cubic centimeter	h., hot	m-, meta	pr., prisms	tri., triclinic	∞, infinitely					
chl., chloroform	hex., hexagonal	mn., monoclinic	rhb., rhombic	uns., unsymmetrical						
col., colorless		n-, normal	s-, soluble	v., very						

This table of the physical properties includes the organic compounds of most general interest. For the properties of other organic compounds, reference must be made to larger tables in Lange's *Handbook of Chemistry* (Handbook Publishers), *Handbook of Chemistry and Physics* (Chemical Rubber Publishing Co.), Van Nostrand's *Chemical Annual*, *International Critical Tables* (McGraw-Hill), and similar works.

The **molecular weights** are based on the atomic weight values in "Atomic weights of the Elements 2001," *PURE Appl. Chem.*, **75**, 1107, 2003. The **densities** are given for the temperature indicated and are usually referred to water at 4°C, *e.g.*, 1.028^{95/4} a density of 1.028 at 95° C referred to water at 4° C, the 4 being omitted when it is not clear whether the reference is to water at 4° C or at the temperature indicated by the upper figure. The melting and boiling points given have been selected from available data as probably the most accurate. The **solubility** is given in grams of the substance in 100 of the solvent. In the case of gases, the solubility is often expressed in some manner as "5¹⁰ cc." which indicates that, at 10° C, 5 cc. of the gas are soluble in 100 of the solvent.

Name	Synonym	Formula	Formula weight	Form and color	Specific gravity	Melting point, °C	Boiling point, °C	Solubility in 100 parts		
								Water	Alcohol	Ether
Abietic acid	sylvic acid, abietinic acid	C ₂₀ H ₃₀ O ₂	302.45	lf.		182		i.	v. s.	v. s.
Acenaphthene	naphthylene ethylene	C ₁₀ H ₈ (CH ₂) ₂	154.21	rhb./al.	1.069 ^{95/95}	95	278-9	i.	s. h.	s. chl.
Acetal	acetaldehyde diethylacetal	CH ₃ CH(OC ₂ H ₅) ₂	118.17	lq.	0.821 ^{22/4}		102.2	6 ²⁵	∞	∞
Acet-aldehyde	ethanal	CH ₃ CHO	44.05	col. lq.	0.783 ^{18/4}	-123.5	20.2	∞	∞	∞
-aldehyde, par-	paraldehyde	(C ₂ H ₄ O) ₃	132.16	col. cr.	0.994 ^{20/4}	10.5-12	124.4 ⁷³²	12 ¹³	∞	∞
-aldehyde ammonia		CH ₃ CHOHNH ₂	61.08	col. cr.		97	100-10 d.	v. s.	v. s.	sl. s.
-amide	ethanamide	CH ₃ CONH ₂	59.07	col. cr.	1.159	81(69.4)	222	s.	v. sl. s.	
-anilide	antifebrin	C ₆ H ₅ NHCOCH ₃	135.16	rhb./al.	1.21 ⁴	113-4	305	0.5 ⁶	21 ²⁰	7 ²⁵
-phenetidine (o-)	o-ethoxycetanilide	CH ₃ CONHC ₆ H ₄ OC ₂ H ₅	179.22	lf./al.		79	>250	i.	s.	
(m-)	acetyl-m-phenetidine	CH ₃ CONHC ₆ H ₄ OC ₂ H ₅	179.22	lf./al.		96-7		sl. s.	s.	
-toluidide (o-)	N-tolylacetamide	CH ₃ C ₆ H ₄ NHCOCH ₃	149.19	rhb.	1.168 ¹⁵	110	296	0.86 ¹⁹	s.	s.
(p-)	N-tolylacetamide	CH ₃ C ₆ H ₄ NHCOCH ₃	149.19	rhb. or mn.	1.212 ¹⁵	153	306-7	0.09 ²²	10 ²⁵	s.
Acetic acid	ethanoic acid, vinegar acid	CH ₃ CO ₂ H	60.05	col. lq.	1.049 ^{20/4}	16.7	118.1	∞	∞	∞
anhydride	acetyl oxide, acetic oxide	(CH ₃ CO) ₂ O	102.09	col. lq.	1.082 ^{20/4}	-73	139.6	12 c.	∞	∞
nitrile	methyl cyanide	CH ₃ CN	41.05	col. lq.	0.783 ^{20/4}	-41	81.6-2.0	∞	∞	∞
Acetone	propanone, dimethyl ketone	CH ₃ COCH ₃	58.08	col. lq.	0.792 ^{20/4}	-94.6	56.5	∞	∞	∞
Acetonyl urea	dimethyl hydantoin	<NHCONHCOC>(CH ₃) ₂	128.13	tri./al.		175	subl.	s.	s.	s.
Acetophenone benzoyl hydride	methyl-phenyl ketone	CH ₃ COC ₆ H ₅	120.15	lf.	1.033 ^{15/15}	20.5	202.3 ⁷⁴⁹	i.	s.	s.
Acetyl-chloride	ethanoyl chloride	CH ₃ COCl	78.50	col. lq.	1.105 ^{20/4}	-112.0	51-2	d.	d.	∞
-phenylenediamine (p)	amino-acetanilide (p)	C ₆ H ₅ ONHC ₆ H ₄ NH ₂	150.18	nd./aq.		162		s. h.	v. s.	v. s.
Acetylene	ethyne; ethine	HC:CH	26.04	col. gas	(A) 0.906	-81.5 ⁹⁰¹	-84 ⁷⁶⁰	100 cc. ¹⁸	600 cc. ¹⁸	
dichloride (cis)	1,2-dichloroethene	CHCl:CHCl	96.94	col. lq.	1.291 ^{15/4}	-80.5	60.3	0.35 ²⁰	∞	∞
(trans)	dioform	CHCl:CHCl	96.94	col. lq.	1.265 ^{15/4}	-50	48.4	0.63 ²⁰	∞	∞
Aconitic acid	equisetic acid; citridic acid	C ₆ H ₄ (CO ₂ H) ₃	174.11	cr./aq.		192 d.		33 ¹⁵	sl. s.	v. sl. s.
Acridine		C ₆ H ₄ < (CH)(N) > C ₆ H ₄	179.22	rhb./aq. al.		110-1	346	sl. s. h.	s.	s.
Acrolein ethylene aldehyde	acrylic aldehyde, propenal	CH ₂ :CH-CHO	56.06	col. lq.	0.841 ^{20/4}	-87.7	52.5	40	s.	s.
Acrylic acid	propenoic acid	CH ₂ :CH-CO ₂ H	72.06	col. lq.	1.062 ^{16/4}	12-13	141-2	∞	∞	∞
nitrile	vinyl cyanide	CH ₂ :CH-CN	53.06	col. lq.	0.811 ²⁰	-82	78-9	s.		
Adipic acid	hexandioic acid, adipinic acid	(CH ₂ CH ₂ CO ₂ H) ₂	146.14	mn. pr.	1.360 ^{25/4}	151-3	265 ¹⁰	1.4 ¹⁵	v. s.	0.6 ¹⁵
amide		(CH ₂ CH ₂ CONH ₂) ₂	144.17	cr. pd.		226-7		0.4 ¹²		
nitrile	tetramethylene	(CH ₂ CH ₂) ₄	108.14	col. oil	0.951 ^{19/19}	1	295	v. sl. s.	s.	v. sl. s.
Adrenaline (1-) (3,4,1)	1-suprarenine	C ₉ H ₇ (OH) ₂ (CHOHCH ₂ NHCH ₃)	183.20	col. pd.		d. 207-11		0.03 ²⁰	v. sl. s.	i.
Alanine (α) (dl-)		CH ₃ CH(NH ₂)CO ₂ H	89.09	nd./aq.		295 d.	subl. >200	22 ¹⁷	v. sl. s.	i.
Aldol acetaldol	2-hydroxybutyraldehyde	CH ₃ CH(OH)CH ₂ COH	88.11	col. lq.	1.103 ^{20/4}		83 ²⁰	∞	∞	s.
Alizarin	Anthraquinone acid	C ₆ H ₄ (CO) ₂ C ₆ H ₃ (OH) ₂	240.21	red rhb.		289-90	430	0.03 ¹⁰⁰	v. s.	v. s.
Allyl alcohol	propen-1-ol-3,propenyl alcohol	CH ₂ :CH-CH ₂ OH	58.08	col. lq.	0.854 ^{20/4}	-129	96.6	∞	∞	∞
bromide	3-bromo-propene-1	CH ₂ :CH-CH ₂ Br	120.98	lq.	1.398 ^{20/4}	-119.4	70-1 ⁷³³	i.	∞	∞
chloride	3-chloro-propene-1	CH ₂ :CH-CH ₂ Cl	76.52	col. lq.	0.938 ^{20/4}	-136.4	44.6	<0.1	∞	∞
thiocyanate (i)	mustard oil	CH ₂ :CH-CH ₂ NCS	99.15	col. oil	1.013 ^{20/4}	-80	152	0.2	∞	∞
thiourea	thiosinamide	CH ₂ :CH-CH ₂ NHCSNH ₂	116.18	col. pr.	1.219 ^{20/20}	77-8		3 ⁹	s.	v. sl. s.
Aluminum ethoxide		Al(OCH ₂ CH ₃) ₃	162.16	pd.	1.142 ^{20/20}	150-60	200-5 ¹⁰	d.	i.	v. sl. s.
Amino-anthraquinone (α)		C ₆ H ₄ (CO) ₂ C ₆ H ₃ NH ₂	223.23	red nd.		256	subl.	i.	s.	s.
(β)		C ₆ H ₄ (CO) ₂ C ₆ H ₃ NH ₂	223.23	red nd.		302	subl.	i.	s.	i.
-azobenzene		C ₆ H ₅ :N=N-C ₆ H ₄ NH ₂	197.24	yel. mn.		126-7	225 ¹²⁰	sl. s. h.	s. h.	s.
-benzoic acid (m-)		H ₂ N-C ₆ H ₄ CO ₂ H	137.14	nd./aq.	1.511 ^{4*}	173-4		v. sl. s.	2 ¹⁰	1.8 ⁶
(p-)	aminodracylic acid	H ₂ N-C ₆ H ₄ CO ₂ H	137.14	mn. pr.		187-8		0.3 ¹³	11 ¹⁰	8.2 ⁶

Amino-diphenylamine (<i>p</i> -)		$\text{H}_2\text{N}\cdot\text{C}_6\text{H}_4\text{NH}\cdot\text{C}_6\text{H}_5$	184.24	nd./aq. al.		67	354	sl. s.		s.	s.
-G-acid (2-)(6-,8-), Na_2 salt		$\text{C}_{10}\text{H}_5(\text{NH}_2)(\text{SO}_3\text{Na})_2$	347.28					v. sl. s.			
-mono-potassium salt		$\text{C}_{10}\text{H}_5(\text{NH}_2)\text{S}_2\text{O}_6\text{HK}$	341.40					12.8 ²⁰			
-sodium salt		$\text{C}_{10}\text{H}_5(\text{NH}_2)\text{S}_2\text{O}_6\text{HNa}$	325.29					2.7 ¹⁸			
-J-acid (2-)(5-,7-)		$\text{C}_{10}\text{H}_5(\text{NH}_2)(\text{SO}_3\text{H})_2$	303.31					10.0 ²⁰			
-mono-potassium salt		$\text{C}_{10}\text{H}_5(\text{NH}_2)\text{S}_2\text{O}_6\text{HK}$	341.40					3.4 ¹⁸			
-naphthol sulfonic (1-,2-,4-)(α -)		$\text{C}_{10}\text{H}_5\text{OHNNH}_2\text{SO}_3\text{H}\cdot\frac{1}{2}\text{H}_2\text{O}$	248.26					v. s.			
(1-,8-,4-)		$\text{NH}_2(\text{OH})\text{C}_{10}\text{H}_5\text{SO}_3\text{H}$	239.25					v. sl. s.			
-phenol (<i>o</i> -)	2-aminophenol	$\text{H}_2\text{N}\cdot\text{C}_6\text{H}_4\cdot\text{OH}$	109.13	col. nd.		173	subl.	1.7 ⁹		4.3 ⁹	v. s.
(<i>m</i> -)	3-aminophenol	$\text{H}_2\text{N}\cdot\text{C}_6\text{H}_4\cdot\text{OH}$	109.13	pr.		122-3		2.6 ⁹		s.	sl. s.
(<i>p</i> -)	<i>p</i> -hydroxyaniline	$\text{H}_2\text{N}\cdot\text{C}_6\text{H}_4\cdot\text{OH}$	109.13	lf.		184-6 d.	subl.	1.1 ⁹		4 ⁹	sl. s.
-toluene sulfonic acid (1-,2-,3-)		$\text{C}_6\text{H}_4(\text{CH}_3)(\text{NH}_2)\text{SO}_3\text{H}$	187.22	nd.				0.97 ¹¹			i. bz.
(1-,4-,2-)		$\text{C}_6\text{H}_4(\text{CH}_3)(\text{NH}_2)\text{SO}_3\text{H}\cdot\text{H}_2\text{O}$	205.23	mn.		d.		0.5 ²⁰		i.	
(1-,4-,3-)		$\text{C}_6\text{H}_3(\text{CH}_3)(\text{NH}_2)\text{SO}_3\text{H}\cdot\frac{1}{2}\text{H}_2\text{O}$	196.22	nd.				0.47			
(1-,2-,5-)		$\text{C}_6\text{H}_3(\text{CH}_3)(\text{NH}_2)\text{SO}_3\text{H}\cdot\text{H}_2\text{O}$	205.23	tri./aq.		$-\text{H}_2\text{O}$, 120		3 ¹¹		i.	
Amyl acetate (<i>n</i> -)		$\text{CH}_3\text{CO}_2\text{CH}_2(\text{CH}_2)_4\text{CH}_3$	130.18	col. lq.	0.879 ^{20/20}	-70.8	148.4 ⁷³⁷	v. sl. s.		∞	∞
(<i>i</i> -)	common amyl acetate	$\text{CH}_3\text{CO}_2\text{CH}_2\text{CH}_2\text{CH}(\text{CH}_3)_2$	130.18	col. lq.	0.876 ^{15/4}		142 ⁷³⁷	0.3 ¹⁵		∞	∞
		$\text{CH}_3\text{CO}_2\text{CH}_2\text{CH}(\text{CH}_3)\text{C}_2\text{H}_5$	130.18	col. lq.	0.880 ¹³		141-2	v. sl. s.		∞	∞
(<i>s</i> -)	α -Me-Bu-acetate	$\text{CH}_3\text{CO}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{C}_2\text{H}_5$	130.18	col. lq.	0.922 ⁹		133.5	sl. s.		∞	∞
(<i>s</i> -)	di Et-carbinol acetate	$\text{CH}_3\text{CO}_2\text{CH}(\text{C}_2\text{H}_5)_2$	130.18	col. lq.	0.871 ^{20/4}		133	sl. s.		∞	∞
(<i>t</i> -)		$\text{CH}_3\text{CO}_2\text{C}(\text{CH}_3)_2\text{C}_2\text{H}_5$	130.18	col. lq.	0.874 ¹⁹		124.5 ⁷⁴⁰	v. sl. s.		∞	∞
alcohol (<i>n</i> -) fusel oil,	pentanol-1	$\text{CH}_3(\text{CH}_2)_4\text{CH}_2\text{OH}$	88.15	col. lq.	0.817 ^{20/20}	-78.5	137.9	2.7 ²²		∞	∞
(<i>s</i> - <i>n</i> -) methyl-propyl carbinol,	pentanol-2	$\text{C}_2\text{H}_5\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$	88.15	col. lq.	0.810 ^{20/20}		119.5	4 ²⁰		∞	∞
(prim.-, <i>i</i> -) isobutyl carbinol,	2-methyl-butanol-4	$(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{OH}$	88.15	col. lq.	0.813 ^{15/4}	-117.2	132.0	2 ¹⁴		∞	∞
(<i>s</i> - <i>i</i> -)	2-methyl-butanol-3	$(\text{C}_2\text{H}_5)_2\text{CHOH}$	88.15	col. lq.	0.815 ^{25/4}		115.6	5.5 ³⁰		∞	∞
(<i>t</i> -)	2-methyl-butanol-2	$(\text{CH}_3)_2\text{CCH}(\text{OH})\text{CH}_3$	88.15	col. lq.	0.819 ¹⁹	-11.9	113-4	2.8 ³⁰		∞	∞
		$(\text{CH}_3)_2\text{C}(\text{OH})\text{C}_2\text{H}_5$	88.15	col. lq.	0.809 ^{20/4}	52-3	102	sl. s.		s.	s.
(<i>d</i> -)	active amyl alcohol	$(\text{CH}_3)_3\text{CCH}_2\text{OH}$	88.15	cr.			113-4	sl. s.		∞	∞
-amine (<i>n</i> -)		$\text{C}_5\text{H}_9\text{CH}(\text{CH}_3)\text{CH}_2\text{OH}$	88.15	col. lq.	0.816 ^{20/4}		128	3.6 ³⁰		∞	∞
(<i>s</i> - <i>n</i> -)		$\text{CH}_3(\text{CH}_2)_4\text{NH}_2$	87.16	col. lq.	0.766 ¹⁹	-55	103-4	s.		s.	s.
(<i>i</i> -)		$(\text{C}_2\text{H}_5)_2\text{CH}(\text{CH}_3)\text{CHNH}_2$	87.16	col. lq.	0.749 ^{20/4}		91-2	∞		∞	∞
(<i>t</i> -)		$(\text{CH}_3)_2\text{CH}(\text{CH}_2)_2\text{NH}_2$	87.16	col. lq.	0.751 ^{15/4}		95	∞		∞	∞
		$(\text{C}_2\text{H}_5)_2\text{CH}(\text{CH}_2)_2\text{CNH}_2$	87.16	col. lq.	0.731 ^{25/4}	-105	77-8	∞		∞	∞
	1-NH ₂ -2-Me-butane	$\text{C}_2\text{H}_5\text{CH}(\text{CH}_3)\text{CH}_2\text{NH}_2$	87.16	col. lq.	0.755 ¹⁸		95-6	∞		∞	∞
	3-amino pentane	$(\text{C}_2\text{H}_5)_2\text{CHNH}_2$	87.16	col. lq.	0.749 ^{20/4}		90-1	∞		∞	∞
	3-NH ₂ -2-Me-butane	$(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)\text{NH}_2$	87.16	col. lq.	0.757 ¹⁸		83-4	∞		∞	∞
		$\text{C}_6\text{H}_5\text{NHC}_5\text{H}_{11}$	163.26	lq.	0.928 ^{15/4}		254.5	i.		∞	∞
aniline (<i>i</i> -)		$\text{C}_6\text{H}_5\text{CO}_2\text{C}_5\text{H}_{11}$	192.25	col. lq.	0.992 ^{4/14}		261 ⁷⁴⁶	i.		∞	∞
benzoate (<i>i</i> -)	1-bromopentane	$\text{CH}_3(\text{CH}_2)_4\text{Br}$	151.04	col. lq.	1.218 ^{8/4}	-95	129.7	i.		s.	s.
bromide (<i>n</i> -)	4-Br-2-Me-butane	$(\text{CH}_3)_2\text{CH}(\text{CH}_2)_2\text{Br}$	151.04	col. lq.	1.220 ^{77/15}		120 ⁷⁴⁵	0.02 ¹⁶		s.	s.
(<i>i</i> -)	2-Br-2-Me-butane	$(\text{CH}_3)_2\text{C}(\text{Br})\text{C}_2\text{H}_5$	151.04	lq.	1.216 ^{10/10}		108 ⁷⁶⁵	i.		s.	s.
(<i>t</i> -)		$\text{C}_2\text{H}_5\text{CH}_2\text{CO}_2(\text{CH}_2)_4\text{CH}_3$	158.24	col. lq.	0.871 ^{15/4}	-73.2	186.4	0.05 ³⁰		∞	∞
<i>n</i> -butyrate (<i>n</i> -)		$\text{C}_2\text{H}_5\text{CH}_2\text{CO}_2\cdot\text{C}_2\text{H}_{11}$	158.24	col. lq.	0.866 ^{19/15}		178.6	i.		∞	∞
(<i>i</i> -)		$\text{C}_2\text{H}_5\text{CO}_2\text{C}(\text{CH}_3)_2\text{C}_2\text{H}_5$	158.24	col. lq.	0.865 ^{15/10}		164	sl. s.		∞	∞
(<i>t</i> -)		$(\text{CH}_3)_2\text{CHCO}_2\text{C}_5\text{H}_{11}$	158.24	lq.	0.876 ^{10/4}		168.8	i.		s.	s.
<i>i</i> -butyrate (<i>i</i> -)	1-chloropentane	$\text{CH}_3(\text{CH}_2)_4\text{CH}_2\text{Cl}$	106.59	col. lq.	0.878 ^{20/4}	-99	108.4	i.		s.	s.
chloride (<i>n</i> -)	2-chloropentane	$\text{C}_2\text{H}_5\text{CH}_2\text{CH}_2\text{CHClCH}_3$	106.59	lq.	0.870 ^{20/4}		96.7	i.		s.	s.
(<i>s</i> -)	3-chloropentane	$(\text{C}_2\text{H}_5)_2\text{CHCl}$	106.59	col. lq.	0.895 ²¹		97.3	i.		∞	∞
(<i>i</i> -)	4-Cl-2-Me-butane	$(\text{CH}_3)_2\text{CH}(\text{CH}_2)_2\text{Cl}$	106.59	col. lq.	0.893 ^{20/4}		99.7 ⁷⁵⁸	i.		s.	∞
(<i>s</i> - <i>i</i> -)	3-Cl-2-Me-butane	$(\text{CH}_3)_2\text{CHCHClCH}_3$	106.59	lq.	0.883 ⁹		91 ⁷⁵³	i.		s.	s.
(<i>t</i> -)	2-Cl-2-Me-butane	$(\text{CH}_3)_2\text{CClC}_2\text{H}_5$	106.59	lq.	0.871 ^{20/4}	-72.9	85.7	i.		s.	s.
	1-Cl-2-Me-butane	$(\text{CH}_3)_2\text{C}(\text{H}_2)\text{CHCH}_2\text{Cl}$	106.59	lq.	0.881 ^{17/5}		98-9	i.		s.	s.
<i>i</i> -cyanide (<i>i</i> -)	iso-caproic iso-nitrile	$(\text{CH}_3)_2\text{CH}(\text{CH}_2)_5\text{NC}$	97.16	lq.			137-9	i.		s.	s.
formate (<i>n</i> -)		$\text{HCO}_2\text{CH}_2(\text{CH}_2)_4\text{CH}_3$	116.16	lq.	0.902 ⁹	-73.5	132	v. sl. s.		∞	∞
(<i>i</i> -)		$\text{HCO}_2\text{CH}_2\text{CH}_2\text{CH}(\text{CH}_3)_2$	116.16	lq.	0.882 ^{20/4}		123.5	0.3 ²²		∞	∞
iodide (<i>n</i> -)	1-iodopentane	$\text{CH}_3(\text{CH}_2)_4\text{CH}_2\text{I}$	198.05	lq.	1.510 ^{20/4}	-86	157.0	i.		s.	∞
(<i>i</i> -)	4-I-2-Me-butane	$(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{I}$	198.05	lq.	1.515 ^{15/4}		147 ⁷⁶⁵	i.		∞	∞
(<i>s</i> - <i>n</i> -)	2-iodopentane	$\text{C}_2\text{H}_5\text{CH}_2\text{CHICH}_3$	198.05	lq.	1.507 ^{17/4}		144.5	i.		∞	∞
(<i>t</i> -)	2-I-2-Me-butane	$(\text{CH}_3)_2\text{CIC}_2\text{H}_5$	198.05	lq.	1.471 ^{19/15}		127 ⁷⁶⁵	i.		∞	∞
		$\text{C}_2\text{H}_5\text{CH}(\text{CH}_3)\text{CH}_2\text{I}$	198.05	lq.	1.524 ^{20/4}		148	i.		∞	∞
mercaptan (<i>n</i> -)	pentanthiol-1	$\text{CH}_3(\text{CH}_2)_4\text{CH}_2\text{SH}$	104.21	lq.	0.857 ²⁰		126 ⁷⁶⁷	i.		∞	∞
(<i>n</i> -)	pentanthiol-3	$(\text{C}_2\text{H}_5)_2\text{CHSH}$	104.21	col. lq.			105	i.		∞	∞
(<i>i</i> -)	2-Me-butanethiol-4	$(\text{CH}_3)_2\text{CH}(\text{CH}_2)_2\text{SH}$	104.21	lq.	0.835 ^{20/4}		120	i.		∞	∞
phenol (<i>t</i> -)(<i>p</i> -)	pentaphen	$\text{C}_5\text{H}_{11}\cdot\text{C}_6\text{H}_4\cdot\text{OH}$	164.24	cr.		93	265-7	sl. s.		s.	s.
propionate (<i>n</i> -)		$\text{C}_2\text{H}_5\text{CO}_2(\text{CH}_2)_4\text{CH}_3$	144.21	lq.	0.876 ^{15/4}	-73.1	168.7	i.		∞	∞
(<i>i</i> -)		$\text{C}_2\text{H}_5\text{CO}_2(\text{CH}_2)_2\text{CH}(\text{CH}_3)_2$	144.21	col. lq.	0.870 ^{20/4}		160.2	0.1 ²⁵		∞	∞
(act.)		$\text{C}_2\text{H}_5\text{CO}_2\text{C}_5\text{H}_{11}$	144.21	col. lq.	0.866 ^{20/4}		158 ¹⁶	v. sl. s.		∞	∞
salicylate (<i>n</i> -)		$\text{HOC}_6\text{H}_4\text{CO}_2\text{C}_5\text{H}_{11}$	208.25	lq.	1.065 ¹⁵		265	v. s.		∞	∞
Amyl <i>i</i> -valerate (<i>i</i> -)		$\text{C}_4\text{H}_9\text{CO}_2\text{C}_5\text{H}_{11}$	172.26	col. lq.	0.858 ^{20/15}		194	v. sl. s.		∞	∞
		$\text{C}_4\text{H}_9\text{CO}_2\text{C}_5\text{H}_{11}$	172.26	col. lq.	0.861 ^{14/10}		173-4	sl. s.		s.	s.

*By N. A. Lange, Ph.D., Handbook Publishers, Inc., Sandusky, Ohio. Abridged from table of Physical Constants of Organic Compounds in *Lange's Handbook of Chemistry*.

TABLE 2-2 Physical Properties of Organic Compounds (Continued)

Name	Synonym	Formula	Formula weight	Form and color	Specific gravity	Melting point, °C	Boiling point, °C	Solubility in 100 parts		
								Water	Alcohol	Ether
Amylene (<i>n</i> -)(α -)	pentene-1	C ₅ H ₈ CH ₂ CH:CH ₂	70.13	lq.	0.644 ²⁰		30–1	i.	∞	∞
(<i>i</i> -)	2-methyl-butene-3	(CH ₃) ₂ CHCH:CH ₂	70.13	col. lq.	0.632 ¹⁵	–135	20.5 ⁷⁷¹	i.	∞	∞
(α -)	2-methyl-butene-1	(C ₆ H ₅)(CH ₃)C:CH ₂	70.13	col. lq.	0.667 ⁴⁰		31–2 ⁷⁸⁵	i.	∞	∞
(<i>n</i> -)(β -)	pentene-2	C ₅ H ₈ CH:CHCH ₃	70.13	col. lq.	0.650 ^{20/4}	–139	36.4	v. sl. s.	∞	∞
(<i>i</i> -)(β -)	2-methyl-butene-2	(CH ₃) ₂ C:CHCH ₃	70.13	col. lq.	0.663 ^{19/4}	–124	37–8	i.	s.	∞
Anethole (<i>p</i> -)	<i>p</i> -propenyl anisole	CH ₃ CH ₂ CH:C ₆ H ₄ OCH ₃	148.20	lf./al.	0.991 ^{20/20}	22.5	235.3	v. sl. s.	s.	∞
Anhydroformal-aniline	methylene aniline	(CH ₂ NC ₆ H ₅) ₂	315.41	pr./al.		143	185	i.	sl. s.	s.
Aniline	amino benzene, phenyl amine, cyanol	C ₆ H ₅ NH ₂	93.13	col. oil	1.022 ^{20/4}	–6.2	184.4	3.6 ¹⁸	∞	∞
hydrochloride	aniline salt, aniline chloride	C ₆ H ₅ NH ₂ ·HCl	129.59	cr.	1.222 ⁴	198	245	18 ¹⁵	s.	i.
nitrate		C ₆ H ₅ NH ₂ ·HNO ₃	156.14	rhb.	1.356 ⁴	d. 190		s.	s.	sl. s.
sulfate		(C ₆ H ₅ NH ₂) ₂ ·H ₂ SO ₄	284.33	lf./al.	1.377 ⁴	d.		5 ¹⁴	sl. s.	i.
Anisal-acetone (<i>p</i> -)	MeO-benzalacetone	CH ₃ OC ₆ H ₄ CH:CHCOCH ₃	176.21	lf./et.		73–4		i.	v. s.	v. s.
Anisic acid (<i>p</i> -)		CH ₃ OC ₆ H ₄ CO ₂ H	152.15	mm./aq.	1.385 ⁴	184.2	275–80	0.03 ¹⁹	v. s.	v. s.
aldehyde (<i>p</i> -)		CH ₃ OC ₆ H ₄ CHO	136.15	col. oil	1.123 ^{20/4}	2.5	247–8	v. sl. s.	∞	∞
Anisidine (<i>o</i> -)	2-amino-anisole	CH ₃ OC ₆ H ₄ NH ₂	123.15	col. lq.	1.098 ^{15/15}	5.2	225	v. sl. s.	∞	∞
(<i>m</i> -)	MeO-aniline(<i>m</i>)	CH ₃ OC ₆ H ₄ NH ₂	123.15	oil	1.096 ^{20/4}	<–12	251	v. sl. s.	s.	s.
(<i>p</i> -)	4-amino anisole	CH ₃ OC ₆ H ₄ NH ₂	123.15	pl./aq.	1.089 ^{35/55}	57.2	243	s. h.	s.	s.
Anisole	methyl phenyl ether	CH ₃ OC ₆ H ₅	108.14	col. lq.	0.990 ^{22/4}	–37.3	154–5	i.	s.	s.
Anthracene	paranaphthalene, anthracin green oil	C ₆ H ₄ :(CH ₂) ₂ :C ₆ H ₄	178.23	col. mn.	1.25 ^{27/4}	217–8	340–2	i.	1.5 ²⁰	
Anthramine (α)	α -amino-anthracene	C ₆ H ₄ :(CH ₂) ₂ :C ₆ H ₃ NH ₂	193.24	yel./al.		130±		i.	s.	
(β)	β -amino-anthracene	C ₆ H ₄ :(CH ₂) ₂ :C ₆ H ₃ NH ₂	193.24	yel./al.		238	subl.	i.	sl. s.	sl. s.
Anthranil		C ₆ H ₄ :(NH)CO	119.12	col. oil	1.187 ^{15/4}	<–18	d. >215	sl. s. h.	s.	
Anthranilic acid (<i>o</i> -)		H ₂ NC ₆ H ₄ CO ₂ H	137.14	col. rhb.		144–5	subl.	0.35 ¹⁴	11 ¹⁰	16 ⁷
Anthrapurpurin (1-,2-,7-)		C ₁₄ H ₅ O ₃ (OH) ₃	256.21	or. nd./al.		369	462	v. s. h.	v. s. h.	sl. s.
Anthraquinone	diphenyleneketone, dihydrodiketoanthracene	C ₆ H ₄ :(CO) ₂ :C ₆ H ₄	208.21	yel. rhb.	1.438 ^{20/4}	286	379–81	i.	0.05 ¹⁸	v. sl. s.
disulfonate Na ₂ (1-,5-)	<i>p</i> -anthraquinone disulfonate	C ₁₄ H ₆ O ₃ (SO ₃ Na) ₂ ·5H ₂ O	502.38	yel. lf.				v. s.	i.	i.
(1-,8-)	<i>x</i> -anthraquinone disulfonate	C ₁₄ H ₆ O ₃ (SO ₃ Na) ₂ ·4H ₂ O	484.36	yel. pr.				sl. s.		
(2-,6-)		C ₁₄ H ₆ O ₃ (SO ₃ Na) ₂ ·7H ₂ O	538.41	col. cr.				3.9 ²⁰		
(2-,7-)		C ₁₄ H ₆ O ₃ (SO ₃ Na) ₂ ·4H ₂ O	484.36	cr.				30.5 ²⁰	v. sl. s.	i.
sulfonate Na (1-)		C ₁₄ H ₇ O ₃ SO ₃ Na	310.26	yel. lf.				0.53 ²⁰		i.
(2-)		C ₁₄ H ₇ O ₃ SO ₃ Na	310.26	slv. lf.				0.84 ²⁵	i.	i.
Anthrarufin (1-,5-)		C ₁₄ H ₆ O ₃ (OH) ₂	240.21	yel. lf.		280	subl.	i.	sl. s.	s.
Antipyrène	1-ph-2,3-diMepyrzalone-5	C ₁₁ H ₁₂ ON ₂	188.23	mm./aq.	1.088 ^{113/4}	113(109)	319 ¹⁷⁴	100 ²⁵	100	sl. s.
Apiole	1-allyl-2,5-diMeO-3,4-methylenedioxybenzene	C ₁₂ H ₁₄ O ₄	222.24	col. nd.	1.02 ^{20/4}	30	294	i	s.	s.
Arabinose (α)(<i>d</i> - or <i>l</i> -)		CH ₂ OH(CHOH) ₃ CHO	150.13	rhb. pr.	1.585 ^{20/4}	159.5		46 ⁰	0.5 ⁹⁷	i.
(<i>dl</i> -)		CH ₂ OH(CHOH) ₃ CHO	150.13			164.5		16.9 ¹⁰		
Arachidic acid	eicosanoic acid	CH ₃ (CH ₂) ₁₈ CO ₂ H	312.53	col. lf.		77	328	i.	s. h.	v. s.
Arsanilic acid (<i>p</i> -)		H ₂ N·C ₆ H ₄ ·AsO ₃ H ₂	217.05	nd./aq.		232		v. s. h.	v. s. h.	
Asparagine (<i>l</i> -)		HO ₂ C·C ₂ H ₄ (NH ₂)·CONH ₂	132.12	rhb.	1.543 ^{15/4}	227–35	d. 235	3.1 ²⁸	i. c.	
Aspirin (<i>o</i> -)		CH ₃ CO ₂ ·C ₆ H ₄ ·CO ₂ H	180.16	nd./aq.		135–6		1 ³⁷	s.	5 ²⁰
Atropic acid	α -phenyl acrylic acid	C ₆ H ₅ C(CH ₃)·CO ₂ H	148.16	nd./aq.		106–7	267 d.	0.1 c.	s.	s.
Auramine	4,4'-dimethylaminobenzo-phenomide	[(CH ₃) ₂ NC ₆ H ₄] ₂ C:NH	267.37	col./al.		136		i.	7 ²⁰	2.3 ²⁰
Aurine, coralline (4-,4'-)		(HOC ₆ H ₄) ₂ C:C ₆ H ₄ O	290.31	red		310 d.		i.	s.	s.
Azo-anisole (2-,2'-)	diMeO-azobenzene	(CH ₃ O·C ₆ H ₄ N·) ₂	242.27	or. pr.		153		i.	s.	s.
benzene	diphenyldiimide	C ₆ H ₅ N·N·C ₆ H ₅	182.22	or. mn.	1.203 ^{20/4}	68	297	i.	4.2 ²⁰	
Azoxybenzene		(C ₆ H ₅) ₂ N ₂ O	198.22	yel. rhb.	1.248 ^{20/20}	36	d.	i.	11.4 ¹⁵	
Barbituric acid	malonyl urea	CO ₂ (NHCO) ₂ ·CH ₂ ·2H ₂ O	164.12	col./aq.		d. 245		s. h.	sl. s.	s.
Benzal acetone	Me-cinnamyl ketone	C ₆ H ₅ CH:CHCOCH ₃	146.19	pl.	1.035 ^{20/20}	41–2	260–2	i.	s.	s.
Benzaldehyde	artificial almond oil	C ₆ H ₅ CHO	106.12	col. lq.	1.046 ^{20/4}	–26	179	0.3	∞	∞
Benzamide		C ₆ H ₅ CONH ₂	121.14	col. pr.	1.341	130	290	1.35 ²⁵	17 ²⁵	sl. s.
Benzanilide		C ₆ H ₅ CONHC ₆ H ₅	197.23	lf./al.	1.31 ⁴	163	117–9 ¹⁰	i.	4 ³⁰	sl. s.
Benzene	benzol, phenyl hydride, cyclohexatriene	C ₆ H ₆	78.11	col. lq.	0.879 ^{20/4}	5.5	80.1	0.07 ²²	s.	∞
sulfonic acid		C ₆ H ₅ SO ₃ H	142.18	pr./aq.		83–4	d. > 100	v. s. h.	v. s.	v. s.
sulfonic acid		C ₆ H ₅ SO ₃ H	158.18	col. nd.		65–6	d.	v. s.	v. s.	i.
sulfonic amide	benzene sulfonamide	C ₆ H ₅ SO ₂ NH ₂	157.19	mm./aq.		156		0.43 ¹⁶	v. s.	v. s.
sulfonic chloride	benzene sulfonyl chloride	C ₆ H ₅ SO ₂ Cl	176.62	cr.	1.384 ^{15/15}	14.5	251.5	i.	v. s.	
Benzidine (4-,4'-)		NH ₂ ·C ₆ H ₄ ·C ₆ H ₄ ·NH ₂	184.24	cr./aq.		128–9	400 ⁷⁴⁰	1 h.	1 h.	2
disulfonic acid (2-,2'-)		(·C ₆ H ₃ (NH ₂)SO ₃ H) ₂ ·3H ₂ O	398.41	pr./aq.		d. >175		0.09 ²⁵	i.	i.
(3-,3'-)		(·C ₆ H ₃ (NH ₂)SO ₃ H) ₂	344.36					v. sl. s.		
Benzil	dibenzoyl	C ₆ H ₅ CO·COC ₆ H ₅	210.23	pr.	1.23 ¹⁵	95	348 d.	i.	v. s.	v. s.
Benzoic acid		C ₆ H ₅ CO ₂ H	122.12	mm. pr.	1.266 ^{15/4}	121.7	249.2	0.2 ¹⁷	46 ¹⁵	66 ¹⁵
anhydride		(C ₆ H ₅ CO) ₂ O	226.23	rhb./et.	1.199 ^{15/4}	42	360	i.	s.	
nitrile	phenyl cyanide	C ₆ H ₅ CN	103.12	col. lq.	1.001 ^{25/6}	–12.9	190.7	1 ¹⁰⁰	∞	∞

Benzoin (<i>dl</i> -)		$C_6H_5CO \cdot CHOHC_6H_5$	212.24	mn.		133-7	344 ⁷⁶⁸	v. sl. s.	s. h.	sl. s.
Benzophenone	diphenyl ketone	$C_6H_5COC_6H_5$	182.22	col. rhb.	1.083 ⁵⁴	48.5	305.4	i.	6.5 ¹⁵	15 ¹³
Benzotrichloride	phenyl chloroform	$C_6H_5CCl_3$	195.47	col. lq.	1.380 ¹⁴	-4.75	220.7	i.	s.	s.
Benzoyl-benzoic acid (<i>o</i> -)		$C_6H_5COC_6H_4CO_2H \cdot H_2O$	244.24	tri./aq.		93(128)		sl. s.		
-chloride		C_6H_5COCl	140.57	col. lq.	1.212 ^{20/4}	-0.5	197.2	d.	d. h.	∞
-peroxide		$(C_6H_5CO)_2O_2$	242.23	rhb./et.		108 d.	expl.	i.	s. h.	s.
Benzyl acetate	phenyl carbinol	$CH_3CO_2CH_2C_6H_5$	150.17	col. lq.	1.057 ¹⁷	-51.5	213.5	i.	∞	∞
alcohol	ω-amino toluene	$C_6H_5CH_2OH$	108.14	col. lq.	1.043 ^{20/4}	-15.3	204.7	i.	∞	∞
amine	phenyl-benzylamine	$C_6H_5CH_2NH_2$	107.15	lq.	0.982 ^{20/4}		184.5	∞	∞	∞
aniline		$C_6H_5CH_2NHC_6H_5$	183.25	mn. pr.	1.065 ^{25/25}	37-8	306 ⁷⁵⁰	i.		s.
benzoate		$C_6H_5CO_2CH_2C_6H_5$	212.24	nd.	1.12 ^{20/4}	21	323-4	i.	∞	∞
butyrate		$C_2H_5CH_2CO_2CH_2C_6H_5$	178.23	col. lq.	1.016 ^{16/18}	238-40	i.	v. s.	v. s.	
chloride	ω-chlorotoluene	$C_6H_5CH_2Cl$	126.58	col. lq.	1.100 ^{20/20}	-39	179.4	i.	∞	∞
ether	dibenzyl ether	$(C_6H_5CH_2)_2O$	198.26	lq.	1.036 ¹⁶		295-8	i.	s. h.	s.
formate		$HCO_2CH_2C_6H_5$	136.15	col. lq.	1.081 ²³	3.6	202-3 ⁷⁴⁷	i.	s.	∞
propionate		$C_2H_5CO_2CH_2C_6H_5$	164.20	lq.	1.036 ^{16/17}		220-2	i.		
Berberonic acid (2-,4-,5-)		$C_5H_5N(CO_2H)_3 \cdot 2H_2O$	247.16	tri.		243		v. sl. s.	sl. s. h.	i.
Biuret	allophanamide	$NH(CONH_2)_2$	103.08	nd./al.		192-3 d.	1.3 ⁹	v. sl. s.		
Borneol (<i>dl</i> -)		$C_{10}H_{17}OH$	154.25	col. cr.	1.011 ^{20/4}	210.5	subl.	v. sl. s.		
(<i>d</i> - or <i>l</i> -)		$C_{10}H_{17}OH$	154.25	col. cr.	1.011 ^{20/4}	208-9	212-3	v. sl. s.	v. s.	v. s.
(iso-)		$C_{10}H_{17}OH$	154.25	col. cr.		212		i.		
Bornyl acetate (<i>d</i> -)		$CH_3CO_2C_{10}H_{17}$	196.29	rhb./pet.	0.991 ¹⁵	29	226-7	i.	s.	s.
Bromo-aniline (<i>p</i> -)		$BrC_6H_4NH_2$	172.02	rhb.	1.8 ²⁰	63-4		i. c.	v. s.	v. s.
-benzene	phenyl bromide	C_6H_5Br	157.01	col. lq.	1.495 ^{20/4}	-30.6	156.2	i.	s.	∞
-camphor (3-)(<i>d</i> -)	α-bromocamphor	$BrC_{10}H_{15}O$	231.13	cr.	1.449 ^{20/4}	77-8	274	i.	20 ²⁶	v. s.
-diphenyl (<i>p</i> -)		$BrC_6H_4 \cdot C_6H_5$	233.10	cr./al.		90-1	310	i.	s.	34 ²⁵
-naphthalene (α-)	α-naphthyl bromide	$C_{10}H_7Br$	207.07	col. oil	1.482 ^{20/4}	5-6	281.1	i.	s.	∞
(β-)	β-naphthyl bromide	$C_{10}H_7Br$	207.07	lf./al.	1.605 ⁹	59	281-2	i.	6 ²⁰	v. s.
-phenol (<i>o</i> -)		BrC_6H_4OH	173.01	col. lq.	1.553 ³⁰	5.6	194-5	s.		
(<i>m</i> -)		BrC_6H_4OH	173.01	cr.		32-3	236-7		s.	s.
(<i>p</i> -)		BrC_6H_4OH	173.01	tet. cr.	1.588 ³⁰	63.5	238	v. s.	v. s.	v. s.
-styrene (ω)(1)		$C_6H_5CH=CHBr$	183.05	lq.	1.422 ^{20/4}	7	221	i.	∞	∞
(2)		$C_6H_5CH=CHBr$	183.05	lq.	1.427 ^{20/4}	-7.5	108 ²⁶	i.	∞	∞
-toluene (<i>o</i> -)	<i>o</i> -tolyl bromide	$CH_3 \cdot C_6H_4Br$	171.03	col. lq.	1.422 ^{20/4}	-28	181.8	i.	s.	∞ ²⁵
(<i>m</i> -)		$CH_3 \cdot C_6H_4Br$	171.03	col. lq.	1.410 ^{20/4}	-39.8	183.7	i.	s.	s.
(<i>p</i> -)		$CH_3 \cdot C_6H_4Br$	171.03	cr./al.	1.390 ^{20/4}	28.5	184-5	i.	s.	∞ ²⁵
Bromoform	tribromo-methane	$CHBr_3$	252.73	col. lq.	2.890 ^{20/4}	8-9	150.5	0.1 c.	∞	∞
Butadiene (1-,2-)	methyl-allene	$CH_3CH:C:CH_2$	54.09	lq.			18-9	i.	∞	∞
(1-,3-)	erythrene	$CH_2:CHCH:CH_2$	54.09	col. gas	0.621 ^{20/4}	-108.9	-4.41	i.	∞	∞
Butadienyl acetylene		$CH_2:(CH)_2CH:C:CH$	78.11	col. lq.	0.773 ^{20/4}		83-6	i.		
Butane	diethyl	$CH_3CH_2CH_2CH_3$	58.12	col. gas	0.60 ⁰	-135	-0.6	i.	s.	s.
(<i>i</i> -)	trimethyl-methane	$(CH_3)_3CHCH_3$	58.12	col. gas	0.60 ⁰	-145	-10	i.	s.	s.
Butyl acetate (<i>n</i> -)		$CH_3CO_2(CH_2)_3C_2H_5$	116.16	col. lq.	0.882 ²⁰	-76.3	125 ⁷⁴⁰	0.7	∞	∞
(<i>s</i> -)		$CH_3CO_2CH(CH_3)C_2H_5$	116.16	col. lq.	0.865 ^{25/4}		112 ⁷⁴⁴	i.	∞	∞
(<i>i</i> -)		$CH_3CO_2CH_2CH(CH_3)_2$	116.16	col. lq.	0.871 ^{20/4}	-98.9	118	0.6 ²⁵	∞	∞
(<i>tert</i> -)		$CH_3CO_2C(CH_3)_3$	116.16	col. lq.	0.866 ^{20/4}		95-6 ⁷⁶⁰	i.	∞	∞
alcohol (<i>n</i> -)	butanol-1	$C_2H_5CH_2CH_2OH$	74.12	col. lq.	0.810 ^{20/4}	-79.9	117	9 ¹⁵	∞	∞
(<i>s</i> -)	butanol-2	$C_2H_5CH(OH)CH_3$	74.12	col. lq.	0.808 ^{20/4}	-114.7	99.5	12.5 ²⁰	∞	∞
(<i>i</i> -)	2-methyl-propanol-1	$(CH_3)_2CHCH_2OH$	74.12	col. lq.	0.805 ^{17.5}	-108	107-8	10 ¹⁵	∞	∞
(<i>tert</i> -)	2-methyl-propanol-2	$(CH_3)_3COH$	74.12	lq.	0.779 ²⁶	25.5	82.9	∞	∞	∞
amine (<i>n</i> -)		$C_2H_5CH_2CH_2NH_2$	73.14	col. lq.	0.739 ^{25/4}	-50	77.8	∞	∞	∞
(<i>s</i> -)		$C_2H_5CH(NH_2)CH_3$	73.14	col. lq.	0.724 ^{20/4}	-104	66 ⁷⁷²	∞	∞	∞
(<i>i</i> -)		$(CH_3)_2CHCH_2NH_2$	73.14	col. lq.	0.732 ^{20/20}	-85	68-9	∞	∞	∞
(<i>t</i> -)		$(CH_3)_3CNH_2$	73.14	col. lq.	0.698 ^{18/4}	-67.5	45.2		∞	
<i>p</i> -aminophenol (<i>N</i>)(<i>n</i>)		$C_4H_9NH \cdot C_6H_4 \cdot OH$	165.23			71		i.		
(<i>N</i>)(<i>i</i> -)		$C_4H_9NH \cdot C_6H_4 \cdot OH$	165.23			79		i.		
aniline (<i>n</i> -)		$C_6H_5NHC_6H_5$	149.23	lq.			235 ⁷²⁰	i.		
(<i>i</i> -)		$C_6H_5NHC_6H_5$	149.23	oil	0.940 ^{20/4}		231-2	0.01 ¹⁵	v. s.	v. s.
arsonic acid (<i>n</i> -)		$C_6H_5AsO(OH)_2$	182.05	col. lf.		158-9		s.	s.	i.
benzoate (<i>n</i> -)		$C_6H_5CO_2C_6H_5$	178.23	col. oil	1.005 ^{25/25}	-22	249-50	i.	s.	s.
(<i>i</i> -)		$C_6H_5CO_2C_6H_5$	178.23	col. oil	0.997 ^{25/25}		241.5	i.	∞	∞
bromide (<i>n</i> -)	1-bromo-butane	$C_2H_5CH_2CH_2Br$	137.02	lq.	1.277 ^{20/4}	-112.4	101.6	0.06 ¹⁶	∞	∞
(<i>s</i> -)	2-bromo-butane	$C_2H_5CH(Br)CH_3$	137.02	lq.	1.251 ^{25/4}	-112	91.3	i.		
(<i>i</i> -)	1-Br-2-Me-propane	$(CH_3)_2CHCH_2Br$	137.02	lq.	1.258 ^{25/4}	-118.5	91.5	0.06 ¹⁸	∞	∞
(<i>t</i> -)	2-Br-2-Me-propane	$(CH_3)_3CBr$	137.02	lq.	1.211 ^{20/4}	-16.2	73.3	i.	∞	∞
butyrate (<i>n</i>)(<i>n</i> -)		$C_2H_5CH_2CO_2CH_2C_2H_5$	144.21	col. lq.	0.872 ^{20/20}		165.7 ⁷³⁶	i.	∞	∞
(<i>n</i>)(<i>i</i> -)		$C_2H_5CH_2CO_2CH_2CH(CH_3)_2$	144.21	col. lq.	0.863 ^{18/4}		156.9	i.	∞	∞
(<i>i</i>)(<i>i</i> -)		$(CH_3)_2CHCO_2CH_2CH(CH_3)_2$	144.21	col. lq.	0.875 ^{18/4}	-80.7	148-9	i.	∞	∞
caproate		$CH_3(CH_2)_4CO_2C_6H_5$	172.26	col. lq.	0.882 ²⁰		204.3	i.		
carbamate (<i>i</i> -)		$NH_2CO_2CH_2CH(CH_3)_2$	117.15	col. lf.	0.956 ^{26/4}	65	206-7	i.	s.	s.
cellosolve (<i>n</i> -)	2-BuO-ethanol-1	$C_4H_9OCH_2CH_2OH$	118.17	col. lq.	0.903 ^{20/4}		171.2	∞	∞	∞

TABLE 2-2 Physical Properties of Organic Compounds (Continued)

Name	Synonym	Formula	Formula weight	Form and color	Specific gravity	Melting point, °C	Boiling point, °C	Solubility in 100 parts		
								Water	Alcohol	Ether
chloride (<i>n</i> -)	1-chloro-butane	C ₄ H ₉ CH ₂ CH ₂ Cl	92.57	col. lq.	0.887 ²⁰	-123.1	77.9 ⁷⁶³	0.07 ¹⁸	∞	∞
(<i>s</i> -)	2-chloro-butane	C ₄ H ₉ :CHCl:CH ₃	92.57	col. lq.	0.871 ^{20/4}	-131	67.8 ⁷⁶⁷	i.	∞	∞
(<i>i</i> -)	1-Cl ₂ -2-Me-propane	(CH ₃) ₂ CHCH ₂ Cl	92.57	col. lq.	0.884 ¹⁵	-131.2	68.9	i.	∞	∞
(<i>t</i> -)	2-Cl ₂ -2-Me-propane	(CH ₃) ₃ CCl	92.57	col. lq.	0.847 ¹⁵	-26.5	51-2	i.	∞	∞
dimethylbenzene (<i>t</i> -)(1-,3-,5-)		(CH ₃) ₂ C:C ₆ H ₄ :(CH ₃) ₂	162.27	col. lq.			200-2 ¹⁴⁷	i.		
formate (<i>n</i> -)		HCO ₂ CH ₂ CH ₂ C ₂ H ₅	102.13	lq.	0.911 ⁹		106.9	v. sl. s.	∞	∞
(<i>s</i> -)		HCO ₂ CH(CH ₃)C ₂ H ₅	102.13	lq.	0.882 ^{20/4}		97	sl. s.	∞	∞
(<i>i</i> -)		HCO ₂ CH ₂ CH(CH ₃) ₂	102.13	lq.	0.885 ^{20/4}	-95.3	98.2	1.1 ²²	∞	∞
furoate (<i>n</i> -)		OC ₄ H ₇ CO ₂ C ₄ H ₉	168.19	col. lq.	1.056 ^{20/4}		118-20 ²⁵	i.	∞	∞
iodide (<i>n</i> -)	1-iodo-butane	C ₄ H ₉ CH ₂ CH ₂ I	184.02	lq.	1.617 ^{20/4}	-103.5	129.9	i.	∞	∞
(<i>s</i> -)	2-iodo-butane	C ₄ H ₉ CHICH ₃	184.02	lq.	1.595 ²⁰	-104	118-9	i.	∞	∞
(<i>i</i> -)	1-iodo-2-Me-propane	(CH ₃) ₂ CHCH ₂ I	184.02	lq.	1.606 ^{20/4}	-90.7	120	i.	∞	∞
(<i>t</i> -)	2-iodo-2-Me-propane	(CH ₃) ₃ CI	184.02	lq.	1.370 ^{19/15}	-34	99	i.	∞	∞
lactate (<i>n</i> -)		CH ₃ CH(OH)CO ₂ C ₄ H ₉	146.18	col. lq.	0.968		75-6 ⁵	sl. s.	∞	∞
mercaptan (<i>n</i> -)	butanthiol-1	C ₄ H ₉ CH ₂ CH ₂ SH	90.19	col. lq.	0.837 ^{25/4}	-116	97-8	sl. s.	v. s.	v. s.
(<i>i</i> -)	2-Me-propanthiol-1	(CH ₃) ₂ CHCH ₂ SH	90.19	lq.	0.836 ^{20/4}	<-79	88	v. sl. s.	s.	s.
(<i>t</i> -)		(CH ₃) ₃ CSH	90.19	lq.			65-7			
methacrylate (<i>n</i> -)		CH ₂ :C(CH ₃)CO ₂ C ₄ H ₉	142.20	lq.	0.889 ^{15,6}		155	i.		
(<i>i</i> -)		CH ₂ :C(CH ₃)CO ₂ C ₄ H ₉	142.20	lq.	0.889 ^{15,6}		155	i.		
phenol (<i>p</i> -)(<i>t</i> -)		(CH ₃) ₂ C:C ₆ H ₄ :OH	150.22	nd./aq.	0.908 ^{112/4}	99	236-8	sl. s.	s.	s.
propionate (<i>n</i> -)		C ₂ H ₅ CO ₂ C ₄ H ₉	130.18	col. lq.	0.883 ¹⁵	-89.55	146	i.	∞	∞
(<i>s</i> -)		C ₂ H ₅ CO ₂ C ₄ H ₉	130.18	col. lq.	0.866 ^{20/4}		132.5	i.	∞	∞
(<i>i</i> -)		C ₂ H ₅ CO ₂ C ₄ H ₉	130.18	col. lq.	0.888 ^{20/4}	-71.4	136.8	i.	∞	∞
stearate (<i>n</i> -)		CH ₃ (CH ₂) ₁₆ CO ₂ C ₄ H ₉	340.58	col. lq.	0.855 ^{25/25}	27.5	220-5 ²⁵	0.3 ²⁵	s.	s.
(<i>i</i> -)		CH ₃ (CH ₂) ₁₆ CO ₂ C ₄ H ₉	340.58	wax		25		i.		
iso-thiocyanate (<i>n</i> -)	butyl mustard oil	C ₄ H ₉ CH ₂ CH ₂ :N:CS	115.20	lq.	0.956 ¹¹		165 ⁷²⁴	i.	s.	s.
(<i>i</i> -)	iso-Bu mustard oil	(CH ₃) ₂ CHCH ₂ :N:CS	115.20	lq.	0.964 ^{14/4}		162	i.	s.	s.
(<i>s</i> -)(<i>d</i> -)		C ₄ H ₉ :N:CS	115.20	lq.	0.943 ^{20/4}		159-63	i.	s.	s.
(<i>t</i> -)		(CH ₃) ₂ C:N:CS	115.20	lq.	0.919 ¹⁰	10.5	140 ⁷⁷⁰	i.	s.	s.
valerate (<i>n</i> -)(<i>n</i> -)		CH ₃ (CH ₂) ₃ CO ₂ (CH ₂) ₃ CH ₃	158.24	lq.	0.870 ^{15/4}	-93	186	v. sl. s.	∞	∞
(<i>i</i> -)(<i>n</i> -)		(CH ₃) ₂ CHCH ₂ CO ₂ (CH ₂) ₃ CH ₃	158.24	lq.	0.862 ^{25/4}		168.8	i.	∞	∞
(<i>i</i> -)(<i>s</i> -)		(CH ₃) ₂ CHCH ₂ CO ₂ C ₄ H ₉	158.24	col. lq.	0.848 ^{20/4}		163-4 ⁷³²	i.	∞	∞
(<i>i</i> -)(<i>i</i> -)		C ₄ H ₉ CO ₂ C ₄ H ₉	158.24	col. lq.	0.874 ^{20/4}		168.7	i.	∞	∞
Butylene (α-)	butene-1	C ₄ H ₈ CH:CH ₂	56.11	col. gas	0.6 ⁹	-130	-5 ⁷³⁵	i.	v. s.	v. s.
(β-)	butene-2	CH ₃ CH:CHCH ₃	56.11	col. gas		-127	3 ⁷⁴⁶			
Butyraldehyde (<i>n</i> -)		CH ₃ CH ₂ CH ₂ CHO	72.11	col. lq.	0.817 ^{20/4}	-99	75.7	4	∞	∞
(<i>i</i> -)	2-Me-propanol	(CH ₃) ₂ CHCHO	72.11	col. lq.	0.794 ^{20/4}	-65.9	64 ⁷³⁷	11 ²⁰	∞	∞
Butyric acid (<i>n</i> -)	butanoic acid	C ₄ H ₈ CH ₂ CO ₂ H	88.11	col. lq.	0.964 ^{20/4}	-4.7	163.5 ⁷³⁷	∞	∞	∞
(<i>i</i> -)	2-Me-propanoic acid	(CH ₃) ₂ CHCO ₂ H	88.11	col. lq.	0.949 ^{20/4}	-47	154.5	20 ³⁰	∞	∞
amide (<i>n</i> -)	<i>n</i> -butyramide	C ₄ H ₉ CH ₂ CONH ₂	87.12	rhb.	1.032	115-6	216	16.3 ¹⁵	s.	sl. s.
(<i>i</i> -)	iso-butyramide	(CH ₃) ₂ CHCONH ₂	87.12	mn. pl.	1.013	129-30	216-20	v. s.	s.	sl. s.
anhydride (<i>n</i> -)		(C ₄ H ₇ CH ₂ CO) ₂ O	158.19	col. lq.	0.968 ^{20/20}	-75	199.5	d.	d	∞
(<i>i</i> -)		[(CH ₃) ₂ CHCO] ₂ O	158.19	col. lq.	0.950 ^{25/4}	-53.5	181.5 ⁷³⁴	d.	d	∞
anilide (<i>n</i> -)	<i>n</i> -butyranilide	C ₈ H ₇ CONHC ₄ H ₉	163.22	mn. pr.	1.134	92	189 ¹⁵	i.	s.	s.
Caffeic acid (3-,4-)		(HO) ₂ C ₆ H ₃ C ₂ H ₄ CO ₂ H	180.16	yel./aq.		195-213	d.	s. h.	s.	sl. s.
Caffeine		C ₈ H ₁₀ O ₂ N ₄ ·H ₂ O	212.21	nd./al.	1.23 ¹⁹	237	subl.	2	2	0.3
Camphene (<i>dl</i> -)		C ₁₀ H ₁₆	136.23	cr.	0.822 ⁷⁸	50	160	i.	s.	s.
(<i>d</i> - or <i>l</i> -)		C ₁₀ H ₁₆	136.23	cr.	0.845 ^{20/4}	42.7	159.6	i.	s.	s.
Camphor (<i>d</i> -)		C ₁₅ H ₁₆ O	152.23	trig.	0.999 ^{9/9}	178-9	209.1 ⁷⁵⁹	0.1	120 ¹²	v. s.
Camphoric acid (<i>d</i> -)		C ₈ H ₁₄ (CO ₂ H) ₂	200.23	mn.	1.186	187		0.6 ¹²	s.	
Cantharidine		C ₁₀ H ₁₂ O ₄	196.20	cr.		212		0.003		
Capric acid	decanoic acid	CH ₃ (CH ₂) ₈ CO ₂ H	172.26	col. nd.	0.889 ⁹⁷	31.5	268-70	0.003	s.	s.
Caproic acid (<i>n</i> -)	hexanoic acid	CH ₃ (CH ₂) ₄ CO ₂ H	116.16	oil. lq.	0.922 ^{20/4}	-1.5	202 ⁷⁶¹	1.1 ²⁰	s.	s.
(<i>i</i> -)	2-Me-pentanoic-5 acid	(CH ₃) ₂ CH(CH ₂) ₂ CO ₂ H	116.16	col. oil	0.925 ^{20/4}	-35	207.7	v. sl. s.	s.	s.
Caprylic acid (<i>n</i> -)	octanoic acid	CH ₃ (CH ₂) ₆ CO ₂ H	144.21	col. lf.	0.910 ^{20/4}	16	237.5	0.07 ¹⁵	s.	s.
Carbazole	diphenylenelimine, dibenzopyrrole	(C ₆ H ₄) ₂ NH	167.21	lf.		244.8	354.8	i.	0.92 ¹⁴	sl. s.
Carbitol	diethylene glycol mono-Et ether	C ₂ H ₅ O(CH ₂) ₂ O(CH ₂) ₂ OH	134.17	col. lq.	0.990 ^{20/20}		201.9	∞	v. s.	s.
Carbon disulfide		CS ₂	76.14	col. lq.	1.263 ^{20/4}	-108.6	46.3	0.2 ⁹	∞	∞
monoxide		CO	28.01	col. gas	0.81-1 ^{195/4}	-207	-192	3.5 ⁹ cc.	s.	
suboxide		OC:C:CO	68.03	gas	1.114 ⁹	-107	7 ⁷⁶¹	d.		s.
tetrabromide	tetrabromomethane	CBBr ₄	331.63	col. mn.	3.42	90.1(48)	189.5	0.02 ³⁰	s.	s.
tetrachloride	tetrachloromethane	CCl ₄	153.82	col. lq.	1.595 ^{20/4}	-22.6	76.8	0.08 ³⁰	∞	∞
tetrafluoride	tetrafluoromethane	CF ₄	88.00	gas			-128	sl. s.		
Carbonyl sulfide		COS	60.08	col. gas	1.24-5 ⁷	-138.2	-50.2 ⁷⁶⁰	80 ¹⁴ cc.	s.	s.
Carminic acid		C ₂₂ H ₂₀ O ₁₃	492.39	red pd.		d. 136		s.	s.	v. sl. s.
Carvacrol (1-,2-,4-)		CH ₃ C ₆ H ₃ (OH)CH(CH ₃) ₂	150.22	col. lq.	0.977 ^{20/4}	0.5	238	v. sl. s.	∞	∞

Carvacrylamine (2-,1-,4-)	$\text{H}_2\text{NC}_6\text{H}_7(\text{CH}_3)\text{C}_3\text{H}_7$	149.23	oil	0.994 ²⁰	−16	241	v. sl. s.	s.	s.
Carvone (<i>d</i> -)	$\text{C}_{10}\text{H}_{14}\text{O}$	150.22	col. lq.	0.961 ^{20/4}		230 ⁷⁶⁶	i.	∞	∞
Cellosolve	$\text{C}_2\text{H}_5\text{O}(\text{CH}_2)_2\text{OH}$	90.12	col. lq.	0.931 ^{20/4}	−70	135.1	∞	∞	∞
acetate	$\text{CH}_3\text{CO}_2\text{CH}_2\text{CH}_2\text{OC}_2\text{H}_5$	132.16	col. lq.	0.975 ^{20/4}		156.3	22	∞	∞
Cellulose	$(\text{C}_6\text{H}_{10}\text{O}_5)_x$	162.14	amor.	1.3−1.4			i.	i.	
Cetyl acetate	$\text{CH}_3\text{CO}_2(\text{CH}_2)_{15}\text{CH}_3$	284.48	nd.	0.858 ²⁰	22−3	200 ¹⁵	i.	v. sl. s. c.	
alcohol	$\text{CH}_3(\text{CH}_2)_{14}\text{CH}_2\text{OH}$	242.44	lf.	0.818 ^{20/4}	49−50	189.5 ¹⁵	i.	s.	s.
Chloral	$\text{CCl}_3\cdot\text{CHO}$	147.39	col. lq.	1.505 ^{25/4}	−57	97.6 ⁷⁶⁸	v. s.	∞	∞
hydrate	$\text{CCl}_3\cdot\text{CH}(\text{OH})_2$	165.40	mn. pr.	1.619 ^{20/4}	51.7	474 ¹⁷	d. 98	v. s.	s.
Chloranil	$\text{OC}_2(\text{CCl}_2\text{CCl})_2\text{CO}$	245.88	yel./bz.		290		subl.	i. c.	i. c.
Chloretonc	$\text{Cl}_3\text{C}\cdot\text{C}(\text{OH})(\text{CH}_3)_2$	177.46	col. cr.		97	167	0.8 c.	111	
Chloro-acetanilide (<i>p</i> -)	$\text{CH}_3\text{CONHC}_6\text{H}_4\text{Cl}$	169.61	rhb.	1.385 ²²	175−6		sl. s.	s.	v. s.
-acetic acid	$\text{ClCH}_2\text{CO}_2\text{H}$	94.50	col. cr.	1.58 ^{20/20}	61.2	189.5	v. s.	s.	s.
-acetone	$\text{CH}_3\text{COCH}_2\text{Cl}$	92.52	col. lq.	1.162 ¹⁶	−44.5	121	∞	∞	∞
-acetophenone (<i>o</i> -)	$\text{C}_6\text{H}_5\text{COCH}_2\text{Cl}$	154.59	rhb.	1.324 ¹⁵	58−9	245−7	0.11	v. s.	v. s.
-acetyl chloride	ClCH_2COCl	112.94	col. lq.	1.498 ^{20/20}		105	d.	d.	
-aniline (<i>o</i> -)	$\text{ClC}_6\text{H}_4\text{NH}_2$	127.57	lq.	1.213 ^{20/4}	0	210.5	i.		s.
(<i>m</i> -)	$\text{ClC}_6\text{H}_4\text{NH}_2$	127.57	lq.	1.216 ^{20/4}	−10.4	230 ⁷⁶⁷	i.		s.
(<i>p</i> -)	$\text{ClC}_6\text{H}_4\text{NH}_2$	127.57	rhb.	1.427 ¹⁹	70−1	230−1	s. h.	s.	s.
-anthraquinone (1-)	$\text{C}_{10}\text{H}_6(\text{CO})_2\text{C}_6\text{H}_4\text{Cl}$	242.66	yel. nd.		162	subl.	i.	sl. s. h.	
(2-)	$\text{C}_6\text{H}_4(\text{CO})_2\text{C}_6\text{H}_5\text{Cl}$	242.66	nd./al.		208−9		i.		
-benzaldehyde (<i>o</i> -)	$\text{ClC}_6\text{H}_4\text{CHO}$	140.57	nd.	1.29 ⁸	11	208 ⁷⁴⁸	v. sl. s.	v. s.	v. s.
(<i>m</i> -)	$\text{ClC}_6\text{H}_4\text{CHO}$	140.57	pr.	1.250 ¹⁵	17−8	213−4	v. sl. s.	v. s.	v. s.
(<i>p</i> -)	$\text{ClC}_6\text{H}_4\text{CHO}$	140.57	pr.	1.196 ⁶¹	47.8	213 ⁷⁴⁸	s. h.	v. s.	v. s.
-benzene	$\text{C}_6\text{H}_6\text{Cl}$	112.56	col. lq.	1.107 ^{20/4}	−45.2	132.1	0.049 ²⁰	∞	∞
-benzoic acid (<i>o</i> -)	$\text{ClC}_6\text{H}_4\text{CO}_2\text{H}$	156.57	mn./aq.	1.544 ^{25/4}	141−2		0.208 ²⁵	s.	s.
(<i>m</i> -)	$\text{ClC}_6\text{H}_4\text{CO}_2\text{H}$	156.57	pr.	1.496 ^{25/4}	158		0.041 ²⁵	s.	s.
(<i>p</i> -)	$\text{ClC}_6\text{H}_4\text{CO}_2\text{H}$	156.57	tri.	1.541 ²⁴	242−3	subl.	0.008 ²⁵	s.	s.
-buta-1,3-diene (2-)	$\text{CH}_2\cdot\text{CCl}\cdot\text{CH}\cdot\text{CH}_2$	88.54	col. lq.	0.958 ^{20/20}		59.4	v. sl. s.	∞	∞
(1-)	$\text{CH}_2\cdot\text{CH}\cdot\text{CH}\cdot\text{CHCl}$	88.54	col. lq.	0.965 ^{20/20}		69	v. sl. s.	∞	∞
-buta-1,2-diene (4-)	$\text{CH}_3\cdot\text{C}\cdot\text{CH}\cdot\text{CH}_2\text{Cl}$	88.54	col. lq.	0.991 ^{20/20}		88	d.		
-dimethylhydantoin	$\text{—C}(\text{CH}_3)_2\text{N}(\text{Cl})\text{CON}(\text{Cl})\text{CO—}$	197.02		1.5 ^{20/20}	130		0.21 ²⁵		
-dinitrobenzene (<i>o</i>)(1,2-)(4-)	$\text{ClC}_6\text{H}_3(\text{NO}_2)_2$	202.55	cr./et.		39(36)	315 d.	i.	v. s. h.	v. s.
(<i>o</i>)(1,3-)(4-)	$\text{ClC}_6\text{H}_3(\text{NO}_2)_2$	202.55	rhb./et.	1.697 ²²	53(43)	315 d.	i.	s. h.	s.
-diphenyl (<i>o</i> -)	$\text{C}_6\text{H}_5\cdot\text{C}_6\text{H}_4\text{Cl}$	188.65	cr.		34	267−8	i.		
(<i>m</i> -)	$\text{C}_6\text{H}_5\cdot\text{C}_6\text{H}_4\text{Cl}$	188.65	cr.		89	284−5	i.		
(<i>p</i> -)	$\text{C}_6\text{H}_5\cdot\text{C}_6\text{H}_4\text{Cl}$	188.65	lf.		77.5	282	i.		
-hydroquinone	$\text{ClC}_6\text{H}_4(\text{OH})_2$	144.56	mn.		106	263 sl. d.	v. s.	v. s.	v. s.
-naphthalene (<i>α</i> -)	$\text{C}_{10}\text{H}_7\text{Cl}$	162.62	col. lq.	1.194 ^{20/4}	−20	259.3	i.	s.	∞
(<i>β</i> -)	$\text{C}_{10}\text{H}_7\text{Cl}$	162.62	lf./al.	1.266 ¹⁶	56−7	264 ⁷⁵¹	i.	v. s.	v. s.
-nitrobenzene (<i>o</i> -)	$\text{ClC}_6\text{H}_4\text{NO}_2$	157.55	mn. nd.	1.305 ^{20/4}	32.5	245.5 ⁷⁵³	i.	s. h.	s.
(<i>m</i> -)	$\text{ClC}_6\text{H}_4\text{NO}_2$	157.55	yel./al.	1.343 ^{20/4}	44.4(24)	235.6	i.	v. s. h.	v. s.
(<i>p</i> -)	$\text{ClC}_6\text{H}_4\text{NO}_2$	157.55	mn. pr.	1.298 ⁹¹	83−4	242 ⁷⁶¹	i.	v. s. h.	v. s.
-nitrotoluene (2-,4-)	$\text{CH}_3\text{C}_6\text{H}_4(\text{NO}_2)(\text{Cl})$	171.58	cr.	1.256 ⁹⁰	38.2	240 ⁷¹⁸	i.		
(2-,6-)	$\text{CH}_3\text{C}_6\text{H}_3(\text{NO}_2)(\text{Cl})$	171.58	cr.		37.5	238	i.		
-phenol (<i>o</i> -)	$\text{ClC}_6\text{H}_4\text{OH}$	128.56	col. lq.	1.241 ^{18/15}	7(0)	175−6	2.85 ²⁰	s.	s.
(<i>m</i> -)	$\text{ClC}_6\text{H}_4\text{OH}$	128.56	nd.	1.268 ²⁵	32−3	214	2.60 ²⁰	s.	s.
(<i>p</i> -)	$\text{ClC}_6\text{H}_4\text{OH}$	128.56	nd.	1.306 ^{20/4}	41−3	217	2.71 ²⁰	v. s.	v. s.
-propionic acid (<i>α</i>)(dl-)	$\text{CH}_3\cdot\text{CHCl}\cdot\text{CO}_2\text{H}$	108.52	col. lq.	1.306 ⁹	<−20	186	∞	∞	∞
-toluene (<i>o</i> -)	$\text{CH}_3\cdot\text{C}_6\text{H}_4\text{Cl}$	126.58	col. lq.	1.082 ^{20/4}	−34	159.5	i.	s.	∞
(<i>m</i> -)	$\text{CH}_3\cdot\text{C}_6\text{H}_4\text{Cl}$	126.58	col. lq.	1.072 ^{20/4}	−47.8	161.6	i.	s.	∞
(<i>p</i> -)	$\text{CH}_3\cdot\text{C}_6\text{H}_4\text{Cl}$	126.58	col. lq.	1.070 ^{20/4}	7.5	162.2	i.	s.	∞
Chloroform	CHCl_3	119.38	col. lq.	1.489 ²⁰	−63.5	61.2	0.82 ²⁰	∞	∞
Chlorophyll (<i>α</i> -)	$\text{C}_{55}\text{H}_{72}\text{O}_5\text{N}_4\text{Mg}$	893.49			d.		i.	s.	
Chloropicrin	Cl_3CNO_2	164.38	lq.	1.651 ^{23/4}	−64	112.3 ⁷⁶⁶	0.17 ¹⁸	s.	s.
Cholesterol	$\text{C}_{27}\text{H}_{45}\text{OH}\cdot\text{H}_2\text{O}$	404.67	rhb./al.	1.067	149−51	subl.	0.26 ²⁰	1.1 ¹⁷	18
Chrysene	$\text{C}_{18}\text{H}_{12}$	228.29	col. rhb.		253−4	448	i.	0.1 ¹⁶	v. sl. s.
Chrysoidine (2-,4-)	$\text{C}_6\text{H}_5\cdot\text{N}\cdot\text{N}\cdot\text{C}_6\text{H}_3(\text{NH}_2)_2$	212.25	yel. cr.		117.5		sl. s. h.	s.	s.
Chrysophanic acid	$\text{C}_{14}\text{H}_5(\text{OH})_2(\text{CH}_3)_2\text{O}_2$	254.24	yel./al.		195	subl.	i. c.	s. h.	sl. s.
Cinchomeric acid (3-,4-)	$\text{C}_5\text{H}_5\text{N}(\text{CO}_2\text{H})_2$	167.12	cr./HCl		258−9 d.	subl. d.	v. sl. s.	sl. s.	i.
Cineole, eucalyptole	$\text{C}_{10}\text{H}_{18}\text{O}$	154.25	col. oil	0.927 ²⁰	1.5	176−7	1.9 ¹⁵	∞	∞
Cinnamic acid (<i>cis</i> -)	$\text{C}_6\text{H}_5\text{CH}\cdot\text{CHCO}_2\text{H}$	148.16	mn. pr.	1.284 ⁴	68	125 ¹⁹			
(<i>trans</i> -)	$\text{C}_6\text{H}_5\text{CH}\cdot\text{CHCO}_2\text{H}$	148.16	mn. pr.	1.245	133	300	0.04 ¹⁸	24 ²⁰	v. s.
aldehyde	$\text{C}_6\text{H}_5\text{CH}\cdot\text{CHCHO}$	132.16	lq.	1.110 ^{20/20}	−7.5	252 sl. d.	v. sl. s.	s.	∞
Cinnamyl alcohol	$\text{C}_6\text{H}_5\text{CH}\cdot\text{CHCH}_2\text{OH}$	134.18	nd.	1.040 ^{25/35}	33	257.5	sl. s.	v. s.	v. s.
cinnamate	$\text{C}_6\text{H}_5\cdot\text{CO}_2\text{C}_6\text{H}_5$	264.32	nd. or pr.	1.085 ^{16.5}	44		i.	4 c.	33
Citraconic acid (<i>cis</i> -)	$\text{CH}_3\text{C}(\text{CO}_2\text{H})\cdot\text{CHCO}_2\text{H}$	130.10	nd.	1.617	92−3		360 ²⁵	∞	s.
Citral (<i>α</i>)	$\text{C}_9\text{H}_{15}\text{CHO}$	152.23	col. oil	0.890 ^{17/4}		229	i.	∞	∞
Citric acid	$\text{C}_3\text{H}_4(\text{OH})(\text{CO}_2\text{H})_3$	192.12	cr.	1.542 ^{17.5}	153	d.	207.7 ²⁵	76 ¹⁵	2 ¹⁵
Citronellal (<i>d</i> -)	$\text{C}_9\text{H}_{17}\cdot\text{CHO}$	154.25	col. oil	0.855 ^{17.5}		204−8	v. sl. s.	∞	∞
Citronellol (<i>d</i> -)	$\text{C}_{10}\text{H}_{20}\text{O}$	156.27	col. oil	0.848 ^{20/4}		224−5	v. sl. s.	∞	∞
Coniine (<i>d</i> -)(2-)	$\text{C}_3\text{H}_7\cdot\text{C}_7\text{H}_{10}\text{N}$	127.23	col. lq.	0.847 ¹⁷	−2	166−7	1.1	v. s.	v. s.

TABLE 2-2 Physical Properties of Organic Compounds (Continued)

Name	Formula	Formula weight	Form and color	Specific gravity	Melting point, °C	Boiling point, °C	Solubility in 100 parts		
							Water	Alcohol	Ether
Coumaric acid (<i>o</i> -)	$\text{HOCH}_2\text{CH}=\text{CHCO}_2\text{H}$	164.16	nd./aq.		207–8	subl.	sl. s. c.	s.	v. sl. s.
(<i>p</i> -)	$\text{HOC}_6\text{H}_4\text{CH}=\text{CHCO}_2\text{H}$	164.16	cr./aq.		206–7 d.		s. h.	v. s. h.	v. s.
Coumarin	$\text{C}_9\text{H}_6\text{O}_2$	146.14	rhb./et.	0.935 ^{20/4}	70	290–1	0.3 c.	v. s.	s.
Coumarone	$\text{C}_9\text{H}_6\text{O}$	118.13	oil	1.078 ^{15/15}	<–18	173–4	i.		s.
Creatine	$\text{C}_4\text{H}_7\text{N}_3\text{O}_2 \cdot \text{H}_2\text{O}$	149.15	mn./aq.		295		1.4 ¹⁸	0.01 ¹⁷	i.
Creatinine	$\text{C}_4\text{H}_7\text{N}_3\text{O}$	113.12	mn.		260 d.		8.7 ¹⁶	1 ¹⁶	
Creosol (3-,1-,4-)	$\text{CH}_3\text{O}-\text{C}_6\text{H}_3(\text{CH}_3)\text{OH}$	138.16	pr.	1.092 ^{20/20}	5.5	221–2 ⁷⁶⁵	v. sl. s.	∞	∞
Cresidine (1-,2-,4-)	$\text{CH}_3(\text{NH}_2)\text{C}_6\text{H}_3 \cdot \text{OCH}_3$	137.18	nd./pet.		93–4	235	v. sl. s.	s.	s.
Cresol (<i>o</i> -)	$\text{CH}_3\text{C}_6\text{H}_4\text{OH}$	108.14	cr.	1.048 ^{20/4}	30.8	190.8	2.5	∞ ³⁰	∞ ³⁰
(<i>m</i> -)	$\text{CH}_3\text{C}_6\text{H}_4\text{OH}$	108.14	lq.	1.034 ^{20/4}	10.9	202.8	0.5	∞	∞
(<i>p</i> -)	$\text{CH}_3\text{C}_6\text{H}_4\text{OH}$	108.14	pr.	1.035 ^{20/4}	35–6	202	1.8	∞ ³⁶	∞ ³⁶
Cresyl benzoate (<i>o</i> -)	$\text{C}_6\text{H}_5\text{CO}_2\text{C}_6\text{H}_4\text{CH}_3$	212.24	lq.			308	i.		
(<i>m</i> -)	$\text{C}_6\text{H}_5\text{CO}_2\text{C}_6\text{H}_4\text{CH}_3$	212.24	cr.		55	314	i.		
(<i>p</i> -)	$\text{C}_6\text{H}_5\text{CO}_2\text{C}_6\text{H}_4\text{CH}_3$	212.24	cr.		71.5	316	i.		
Crotonic acid (α -)	$\text{CH}_3\text{CH}=\text{CHCO}_2\text{H}$	86.09	col. mn.	0.964 ^{79,7}	72	189	8.3 ¹⁵		
acid (β -)(<i>cis</i> -)	$\text{CH}_3\text{CH}=\text{CHCO}_2\text{H}$	86.09	nd.	1.031 ^{15/4}	15.5	170–1 d.	∞ ²⁵	s.	∞
aldehyde (α)	$\text{CH}_3\text{CH}=\text{CHCHO}$	70.09	col. lq.	0.853 ^{20/20}	–69	102.2	18	∞	∞
Cumene	$\text{C}_6\text{H}_5\text{CH}(\text{CH}_3)_2$	120.19	col. lq.	0.862 ^{20/4}	–96.9	152.5	i.	∞	∞
Cumic acid (<i>p</i> -)	$(\text{CH}_3)_2\text{CH}-\text{C}_6\text{H}_4\text{CO}_2\text{H}$	164.20	tri.	1.162 ⁴	116–7	subl.	0.02 ²⁵	s.	s.
Cumidine (<i>p</i> -)	$(\text{CH}_3)_2\text{CH}-\text{C}_6\text{H}_4\text{NH}_2$	135.21	lq.	0.953	<–20	225 ⁷⁰¹	i.		
Cyanamide	$\text{H}_2\text{N}-\text{CN}$	42.04	col. nd.	1.073 ^{8/4}	44–5	140 ¹⁹	v. s.	v. s.	v. s.
Cyanoic acid	HOCN or HNCO	43.02	gas	1.140 ⁹	–80	–64 ⁰	sl. s.		
Cyanoacetic acid	$\text{CH}_2(\text{CN})\text{CO}_2\text{H}$	85.06	col. lq.		65–6	108 ^{0,2}	s.	s.	
Cyanogen	$(\text{CN})_2$	52.03	col. gas	0.866 ¹⁷	–34.4	–21	450 ²⁰ cc.	2300 ²⁰ cc.	500 ²⁰ cc.
bromide	BrCN	105.92	nd.	2.015 ^{20/4}	52	61.3 ⁷⁵⁰	s.	s.	s.
chloride	ClCN	61.47	gas	1.222 ⁹	–6.5	12.5–13	2500 ²⁰ cc.	v. s.	5000 ²⁰ cc.
Cyanuric acid	$\text{C}_3\text{H}_3\text{O}_3\text{N}_3 \cdot 2\text{H}_2\text{O}$	165.10	mn./aq.	1.768 ^{9/4}	>360	d.	0.27 ¹⁷	0.1 ²²	
Cyclo-butane	$\text{CH}_2 < (\text{CH}_2)_2 > \text{CH}_2$	56.11	col. gas	0.703 ^{9/4}	–50	11–12 ⁷³⁶	i.	v. s.	
-heptane	$\text{CH}_2 < (\text{CH}_2\text{CH}_2\text{CH}_2)_2 >$	98.19	oil	0.819 ^{20/4}	–12	118–20	i.		
-hexane	$\text{CH}_2 < (\text{CH}_2\text{CH}_2)_2 > \text{CH}_2$	84.16	col. lq.	0.779 ^{20/4}	6.5	80–1	i.	∞	∞
-hexanol	$\text{CH}_2 < (\text{CH}_2\text{CH}_2)_2 > \text{CHOH}$	100.16	col. nd.	0.962 ^{20/4}	23.9	160–1	3.6 ²⁰	s.	s.
-hexanone	$\text{CH}_2 < (\text{CH}_2\text{CH}_2)_2 > \text{CO}$	98.14	col. oil	0.947 ^{19/4}	–45	155–6	s.	s.	s.
-hexene	$(-\text{CH}_2-\text{CH}_2\text{CH}_2-)_2$	82.14	lq.	0.810 ^{20/4}	–103.7	83.3	v. sl. s.	v. s.	v. s.
-hexyl acetate	$\text{CH}_3\text{CO}_2\text{C}_6\text{H}_{11}$	142.20	oil	0.985 ^{9/4}		174 ⁷⁵⁰	i.	∞	∞
amine	$\text{CH}_2 < (\text{CH}_2\text{CH}_2)_2 > \text{CHNH}_2$	99.17	col. lq.	0.865 ^{20/20}		134	i.	∞	
bromide	$\text{CH}_2 < (\text{CH}_2\text{CH}_2)_2 > \text{CHBr}$	163.06	col. lq.	1.324 ^{20/20}		165 ⁷¹⁴	i.	s.	s.
chloride	$\text{CH}_2 < (\text{CH}_2\text{CH}_2)_2 > \text{CHCl}$	118.60	col. lq.	0.977 ^{15/4}	–43.9	142	i.		
-pentadiene (1-,3-)	$\text{CH}_2 < (\text{CH}=\text{CH})_2 >$	66.10	col. lq.	0.805 ^{19/4}	–85	41–2	i.	∞	∞
-pentane	$\text{CH}_2 < (\text{CH}_2\text{CH}_2)_2 >$	70.13	col. oil	0.745 ^{20/4}	–93.3	49–50	i.		
-pentanone	$< (\text{CH}_2\text{CH}_2)_2 > \text{CO}$	84.12	col. oil	0.948 ²⁰	–58.2	129–30	v. sl. s.		
-propane	$< \text{CH}_2\text{CH}_2\text{CH}_2 >$	42.08	col. gas	0.720 ^{–79}	–126.6	–34 ⁷³⁹	i.	s.	s.
Cymene (<i>o</i> -)	$\text{CH}_3-\text{C}_6\text{H}_4\text{CH}(\text{CH}_3)_2$	134.22	col. lq.	0.875 ^{20/4}		177	i.	s.	s.
(<i>m</i> -)	$\text{CH}_3-\text{C}_6\text{H}_4\text{CH}(\text{CH}_3)_2$	134.22	col. lq.	0.862 ^{20/4}	<–25	175–6	i.	s.	s.
(<i>p</i> -)	$\text{CH}_3-\text{C}_6\text{H}_4\text{CH}(\text{CH}_3)_2$	134.22	col. lq.	0.857 ^{20/4}	–73.5	176–7	i.	s.	s.
Cystine (<i>L</i> -)	$[\text{SCH}_2\text{CH}(\text{NH}_2)\text{CO}_2\text{H}]_2$	240.30	pl.		d. 258–61		0.01 ¹⁹	i.	
Dambose	$\text{C}_6\text{H}_6(\text{OH})_6$	180.16	mn./aq.	1.752	253	319 ¹⁵	2 ¹²	i.	i.
Decahydronaphthalene (<i>cis</i> -)	$\text{C}_{10}\text{H}_{18}$	138.25	lq.	0.895 ^{15/4}	–51	193.3	i.	s.	s.
(<i>trans</i> -)	$\text{C}_{10}\text{H}_{18}$	138.25	lq.	0.872 ^{20/4}	–32	185.3	i.	s.	s.
Decane (<i>n</i> -)	$\text{CH}_3(\text{CH}_2)_8\text{CH}_3$	142.28	col. lq.	0.730 ²	–29.7	174.0	i.	∞	∞
Decyl alcohol	$\text{CH}_3(\text{CH}_2)_9\text{CH}_2\text{OH}$	158.28	col. oil	0.830 ^{20/4}	7	232.9	i.	s.	
Dextrin	$(\text{C}_6\text{H}_{10}\text{O}_5)_x$	162.14	amor.	1.038			s.	i.	i.
Diacetone alcohol	$(\text{CH}_3)_2\text{C}(\text{OH})-\text{CH}_2\text{COCH}_3$	116.16	lq.	0.931 ²⁵	–47	167.9	∞	∞	∞
Diamino-benzophenone (4-,4'-)	$\text{H}_2\text{NC}_6\text{H}_4\text{COC}_6\text{H}_4\text{NH}_2$	212.25	vel. nd.		237–9		sl. s. h.	∞	s.
-diphenylamine (4-,4'-)	$\text{H}_2\text{NC}_6\text{H}_4\text{NHC}_6\text{H}_4\text{NH}_2$	199.25	lf./aq.		158	d.	sl. s.	s.	s.
-diphenylmethane (4-,4'-)	$\text{H}_2\text{NC}_6\text{H}_4\text{CH}_2\text{C}_6\text{H}_4\text{NH}_2$	198.26	nd./aq.		93–4	249–53 ¹⁵	sl. s. c.	s.	s.
-diphenylurea (4-,4'-)	$(\text{H}_2\text{NC}_6\text{H}_4\text{NH})_2\text{CO}$	242.28	cr.		subl. 310		v. sl. s.		
Diamyl-amine (<i>i</i> -)	$[(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2]_2\text{NH}$	157.30	col. lq.	0.767 ^{21/4}	–44	188–90	sl. s.	s.	∞
ether (<i>n</i> -)	$(\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{CH}_2)_2\text{O}$	158.28	col. lq.	0.774 ^{20/4}	–69	190	i.	∞	∞
(<i>i</i> -)	$[(\text{CH}_3)_2\text{CH}(\text{CH}_2)_2]_2\text{O}$	158.28	col. lq.	0.777 ^{20/4}		173.4	i.	∞	∞
Diamyl ketone (<i>i</i> -)	$[(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2]_2\text{CO}$	170.29	yel. oil	0.821 ^{25/4}	14.6	228	i.	s.	s.
phthalate (<i>n</i> -)	$\text{C}_6\text{H}_4(\text{CO}_2\text{C}_5\text{H}_{11})_2$	306.40	col. lq.			204–6 ¹¹	i.	s.	s.
(<i>i</i> -)	$\text{C}_6\text{H}_4(\text{CO}_2\text{C}_5\text{H}_{11})_2$	306.40	col. lq.	1.03		225 ⁴⁰	i.		
tartrate (<i>i</i> -)	$(\text{HOCH}-\text{CO}_2\text{C}_5\text{H}_{11})_2$	290.35	lq.	1.063 ^{15/4}		195 ¹⁶	i.		
Dianisidine (<i>o</i> -)(4-,3-) ₂	$[\text{NH}_2(\text{OCH}_3)\text{C}_6\text{H}_3]_2$	244.29	col. lf.		131.5		i.	s.	s.
Diazo-aminobenzene	$\text{C}_6\text{H}_5\text{N}=\text{N}-\text{NHC}_6\text{H}_5$	197.24	yel. lf.		96–8	expl.	i.	s. h.	v. s.
-aminotoluene (2-,2'-)	$\text{C}_7\text{H}_7\text{N}=\text{N}-\text{NHC}_7\text{H}_7$	225.29	or. cr.		51		0.05		
-methane	$\text{CH}_2=\text{N}_2$	42.04	gas		–145		d.		s.

Dibenzothiazyl-disulfide (2-,2'-)	(C ₆ H ₄ NSC) ₂ S ₂	332.49	cr.	1.50	180	d.	i.			
Dibenzoyl methane	(C ₆ H ₅ CO) ₂ CH ₂	224.25	rhb./al.		78	219–21 ¹⁸	i.	4.4 ²⁰	s.	
Dibenzyl-amine	(C ₆ H ₅ CH ₂) ₂ NH	197.28	col. oil	1.028 ^{25/25}	–26	265–71 ²⁵⁰	i.	s.	s.	
-aniline	C ₆ H ₅ N(CH ₂ C ₆ H ₅) ₂	273.37	pr./al.		70–1	>300	i.	v. s. h.	s.	
ketone	(C ₆ H ₅ CH ₂) ₂ CO	210.27	cr.		34–5	330.6	i.			
phthalate (o-)	C ₆ H ₄ (CO ₂ CH ₂ C ₆ H ₅) ₂	346.38	pr./al.		42–3	274 ¹²	v. sl. s.	s.	s.	
succinate	(-CH ₂ CO ₂ CH ₂ C ₆ H ₅) ₂	298.33	lf./al.		45–6	238 ¹⁴	i.	s.	s.	
Dibromo-benzene (o-)	C ₆ H ₃ Br ₂	235.90	col. lq.	1.956 ^{20/4}	1.8	221–2	i.	s.	s.	
(m-)	C ₆ H ₃ Br ₂	235.90	col. lq.	1.952 ^{20/4}	–6.9	219 ⁷⁵⁵	i.	s.	s.	
(p-)	C ₆ H ₃ Br ₂	235.90	pl./al.	2.261 ¹⁸	87–8	218.6 ⁷⁵⁵	i.	1.6	71 ²⁵	
-diphenyl (4-,4'-)	BrC ₆ H ₄ -C ₆ H ₄ Br	312.00	mn. pr.	1.897	164–5	355–60	i.	v. sl. s. h.	∞	
Dibutyl-adipate (n-)	(-CH ₂ CH ₂ CO ₂ C ₄ H ₉) ₂	258.35	col. lq.	0.965 ^{20/4}	–38	183 ¹⁴	i.	∞	∞	
(i-)	(-CH ₂ CH ₂ CO ₂ C ₄ H ₉) ₂	258.35	col. lq.	0.950 ²⁵	–20	278–80	i.	∞	∞	
-amine (n-)	(C ₂ H ₅ CH ₂ CH ₂) ₂ NH	129.24	col. lq.	0.768 ^{20/20}		159 ⁷⁰¹	∞	∞	∞	
(i-)	[(CH ₃) ₂ CHCH ₂] ₂ NH	129.24	col. lq.	0.741 ^{25/4}	–70	139–40	v. sl. s.	s.	s.	
-p-aminophenol (s-)	(C ₂ H ₅) ₂ N-C ₆ H ₄ OH	221.34	lq.			170 ¹⁰	i.			
-aniline (n-)	C ₆ H ₅ N(C ₄ H ₉) ₂	205.34	lq.			262.8	i.	∞	∞	
carbonate (n-)	CO(OC ₄ H ₉) ₂	174.24	col. lq.	0.924 ^{20/4}		207 ⁷⁴⁰	i.	s.		
(i-)	CO(OC ₄ H ₉) ₂	174.24	col. lq.	0.919 ¹⁵		190	i.			
(s-)	CO(OC ₄ H ₉) ₂	174.24	col. lq.			178–80				
ether (n-)	(C ₂ H ₅ CH ₂ CH ₂) ₂ O	130.23	lq.	0.769 ^{20/20}	–98	142.4	<0.05	∞	∞	
(i-)	[(CH ₃) ₂ CHCH ₂] ₂ O	130.23	lq.	0.762 ¹⁵		122.5	i.	∞	∞	
(s-)	[C ₂ H ₅ (CH ₂) ₂ CH] ₂ O	130.23	lq.	0.756 ²¹		121	i.	∞	∞	
ketone (n-)	(C ₂ H ₅ CH ₂ CH ₂) ₂ CO	142.24	lq.	0.827 ^{15/4}	–5.9	187.7	i.	s.	v. s.	
(i-)	[(CH ₃) ₂ CHCH ₂] ₂ CO	142.24	oil	0.805 ^{21/4}		168.1	<0.06	∞	∞	
malate (l-)(n-)	C ₂ H ₄ O(CO ₂ C ₄ H ₉) ₂	246.30	lq.	1.038 ^{20/4}		170–1 ¹⁸	v. sl. s.			
oxalate (n-)	(-CO ₂ C ₄ H ₉) ₂	202.25	col. lq.	0.986 ^{20/4}	–29.6	245.5	i.	s.	s.	
phthalate (n-)	C ₆ H ₄ (CO ₂ C ₄ H ₉) ₂	278.34	col. lq.	1.045 ²¹		340	0.04 ²⁵	∞	∞	
tartrate (d-)(n-)	(CHOHCO ₂ C ₄ H ₉) ₂	262.30	pr.	1.098 ¹⁵	22–2.5	200–3 ¹⁸	i.			
(d-)(i-)	(CHOHCO ₂ C ₄ H ₉) ₂	262.30	cr.	1.031 ^{15/25}	73–4	323–5	v. sl. s.			
Dichloro-acetic acid	Cl ₂ CH-CO ₂ H	128.94	lq.	1.560 ^{25/25}	9.7(–4)	194.4	∞	∞	∞	
-acetone (αα-)	Cl ₂ CHCOCH ₃	126.97	lq.	1.234 ¹⁵		120	v. sl. s.	s.	s.	
-aniline (2-,5-)	Cl ₂ C ₆ H ₃ NH ₂	162.02	nd.		50	251	v. sl. s.	s.	s.	
-anthraquinone (1-,3-)	C ₆ H ₄ (CO) ₂ C ₆ H ₂ Cl ₂	277.10	yel. nd.		208–9		i.			
(1-,4-)	C ₆ H ₂ (CO) ₂ C ₆ H ₂ Cl ₂	277.10	yel. nd.		187.5		i.	v. sl. s.	v. sl. s.	
(1-,5-)	C ₆ H ₂ Cl ₂ (CO) ₂ C ₆ H ₃ Cl	277.10	yel. nd.		251		i.	sl. s.		
(1-,6-)	C ₆ H ₂ Cl ₂ (CO) ₂ C ₆ H ₃ Cl	277.10	yel. nd.		203–4		i.			
(1-,8-)	C ₆ H ₂ Cl ₂ (CO) ₂ C ₆ H ₃ Cl	277.10	yel. nd.		202–3		i.	sl. s.		
(2-,3-)	C ₆ H ₄ (CO) ₂ C ₆ H ₂ Cl ₂	277.10	yel. nd.		268–70		i.	sl. s.		
(2-,6-)	C ₆ H ₂ Cl ₂ (CO) ₂ C ₆ H ₃ Cl	277.10	yel. nd.		282		i.			
(2-,7-)	C ₆ H ₂ Cl ₂ (CO) ₂ C ₆ H ₃ Cl	277.10	yel. nd.		210–11		i.			
-benzene (o-)	C ₆ H ₄ Cl ₂	147.00	col. lq.	1.305 ^{20/4}	–17.6	179	i.	∞	∞	
(m-)	C ₆ H ₄ Cl ₂	147.00	col. lq.	1.288 ^{20/4}	–24.8	172 ⁷⁶⁶	i.	s.	s.	
(p-)	C ₆ H ₄ Cl ₂	147.00	col. mn.	1.458 ²¹	53	174 ⁷⁶⁴	i.	v. s.	v. s.	
-butane (n-)(1-,4-)	ClCH ₂ (CH ₂) ₂ CH ₂ Cl	127.01	lq.		–38.7	161–3				
-diphenyl (4-,4'-)	ClC ₆ H ₄ -C ₆ H ₄ Cl	223.10	pr.	1.442 ^{0/4}	148	315–9	i.	v. sl. s.	4 ²⁵	
-ethane (1-,2-)	ClCH ₂ -CH ₂ Cl	98.96	col. lq.	1.256 ^{20/20}	–35.3	83.7	0.9 ⁰	∞	∞	
-naphthalene (β-)(1-,4-)	C ₁₀ H ₆ Cl ₂	197.06	nd./al.	1.300 ^{76/4}	67–8	286–7 ⁷⁴⁰	i.	v. sl. s.		
(γ-)(1-,5-)	C ₁₀ H ₆ Cl ₂	197.06	lf./al.		107	subl.	i.	s.	s.	
-nitrobenzene (2-,5-)	Cl ₂ C ₆ H ₃ NO ₂	192.00	tri./al.	1.669 ²²	54.6	266	i.	v. s. h.		
-pentane (1-,5-)	ClCH ₂ (CH ₂) ₃ CH ₂ Cl	141.04	col. lq.	1.094 ^{25/4}		180–1	i.	s.	s.	
-phenol (2-,4-)	Cl ₂ C ₆ H ₃ OH	163.00	nd.	1.383 ^{90/25}	45	209–10	0.45 ²⁰	v. s.	v. s.	
Dichloramine T (p-)	CH ₂ C ₆ H ₄ SO ₂ NCN ₂	240.11	cr.		83		sl. s.			
Dicyandiamide	H ₂ N-C(=NH)-NH-CN	84.08	mn. pl.	1.40 ¹⁴	207–8	d.	2.3 ¹⁸	1.3 ¹⁸	0.01 ¹⁸	
Diethanolamine	HN(CH ₂ CH ₂ OH) ₂	105.14	pr.	1.097 ^{20/4}	28	270 ⁷⁴⁸	∞	∞	v. sl. s.	
Diethyl adipate	(-CH ₂ CH ₂ CO ₂ C ₂ H ₅) ₂	202.25	col. lq.	1.009 ^{20/4}	–21	239–41 ⁷⁶¹	0.43 ⁸⁰	s.	s.	
-amine	(C ₂ H ₅) ₂ NH	73.14	col. lq.	0.712 ^{15/15}	–38.9	55.5 ⁷⁵⁹	v. s.	∞	∞	
-aminophenol (m-)	(C ₂ H ₅) ₂ N-C ₆ H ₄ -OH	165.23	rhb.		78	276–80	s.			
-aniline	(C ₂ H ₅) ₂ NC ₆ H ₅	149.23	oil	0.934 ^{20/4}	–34.4	216	1.4 ¹²	s.	s.	
sulfonic acid (m-)	(C ₂ H ₅) ₂ NC ₆ H ₄ SO ₃ H	229.30	cr.		270 d.		s.			
carbonate	OC(OC ₂ H ₅) ₂	118.13	col. lq.	0.975 ^{20/4}	–43	126 ⁷⁵⁹	i.	∞	∞	
diethyl malonate	(C ₂ H ₅) ₂ C(CO ₂ C ₂ H ₅) ₂	216.27	col. lq.	0.985 ^{20/4}		230	i.	∞	∞	
Diethyl dimethyl malonate	(CH ₃) ₂ C(CO ₂ C ₂ H ₅) ₂	188.22	col. lq.	0.994 ^{25/25}		196.7	i.	∞	∞	
glutarate	CH ₂ (CH ₂ CO ₂ C ₂ H ₅) ₂	188.22	syrup	1.025 ²¹	–24	237	0.88 ²⁰	v. s.	s.	
ketone	(C ₂ H ₅) ₂ CO	86.13	col. lq.	0.816 ^{19/4}	–42	101.7	4.7 ²⁰	∞	∞	
malonate	CH ₂ (CO ₂ C ₂ H ₅) ₂	160.17	col. lq.	1.055 ^{20/4}	–49.8	198.9	2.08 ²⁰	∞	∞	
-malonic acid	(C ₂ H ₅) ₂ C(CO ₂ H) ₂	160.17	pr./aq.		125	d. 170–80	65 ¹⁸	v. s.	v. s.	
-naphthylamine (α-)	C ₁₀ H ₇ N(C ₂ H ₅) ₂	199.29	col. oil	1.005		285–90	i.	∞	∞	
(β-)	C ₁₀ H ₇ N(C ₂ H ₅) ₂	199.29	col. oil	1.026		318	i.	∞	∞	
oxalate	(-CO ₂ C ₂ H ₅) ₂	146.14	col. lq.	1.079 ^{20/4}	–40.6	186	v. sl. s.	∞	∞	
phthalate (o-)	C ₆ H ₄ (CO ₂ C ₂ H ₅) ₂	222.24	col. lq.	1.121 ^{25/25}		298–9	i.	∞	∞	
sulfate	O ₂ S(OC ₂ H ₅) ₂	154.18	col. lq.	1.172 ^{25/4}	–25	210	i.	s.	∞	
sulfide	(C ₂ H ₅) ₂ S	90.19	col. lq.	0.837 ^{20/4}	–99.5	92–3 ⁷⁵⁴	0.31 ²⁰	∞	∞	

TABLE 2-2 Physical Properties of Organic Compounds (Continued)

Name	Formula	Formula weight	Form and color	Specific gravity	Melting point, °C	Boiling point, °C	Solubility in 100 parts		
							Water	Alcohol	Ether
tartrate (<i>d</i> -)	(CHOH·CO ₂ C ₂ H ₅) ₂	206.19	lq.	1.204 ^{20/4}	17	280	sl. s.	∞	∞
-toluidine (<i>o</i> -)	CH ₃ ·C ₆ H ₄ ·N(C ₂ H ₅) ₂	163.26	lq.			208–9 ⁷⁵⁵	i.	s.	s.
(<i>m</i> -)	CH ₃ ·C ₆ H ₄ ·N(C ₂ H ₅) ₂	163.26	lq.			231–2	i.	s.	s.
(<i>p</i> -)	CH ₃ ·C ₆ H ₄ ·N(C ₂ H ₅) ₂	163.26	lq.	0.924 ^{15.5}		228–9	i.		
Diethyleneglycol dinitrate	O(CH ₂ CH ₂ ONO ₂) ₂	196.12	lq.	1.377 ^{25/4}	–11.3		i.		
Diffuorodichloromethane	F ₂ CCl ₂	120.91	gas	1.486 ^{–30}	–155	–29.2	5.7 cc. ²⁶	s.	s.
Diglycerol	[(HO) ₂ C ₃ H ₅] ₂ O	166.17	lq.			220–30 ¹⁰	s. h.		i.
Dihydroxy-dinaphthyl (<i>α</i> -)	(HO·C ₁₀ H ₆) ₂	286.32	pl./al.		300		i.	s.	v. s.
(–2,–2',–1,–1')	(HO·C ₁₀ H ₆) ₂	286.32	nd./al.		218	subl.	i.	s.	v. s.
-diphenyl (4,–4')	(HO·C ₆ H ₄) ₂	186.21	rhb./al.	1.25	270–2	subl.	sl. s.	v. s.	v. s.
-ethyl formal (β-)	CH ₃ (OCH ₂ CH ₂ OH) ₂	136.15	lq.	1.154 ²⁵	–5.3	264	∞		
-naphthalene (1,–5)	C ₁₀ H ₆ (OH) ₂	160.17	pr./aq.		258–60	d.	sl. s.	s.	v. s.
(1,–8)	C ₁₀ H ₆ (OH) ₂	160.17	nd.		140		sl. s. h.		v. s.
Dimethoxy-benzene (<i>p</i> -)	(CH ₃ O) ₂ C ₆ H ₄	138.16	lf.	1.053 ^{25/55}	56	212.6	v. sl. s.	v. s.	v. s.
-diphenylamine (4,–4')	HN(C ₆ H ₅ OCH ₃) ₂	229.27	cr.		103		i.		
-ethyl adipate	(CH ₂) ₄ (CO ₂ C ₂ H ₅ OCH ₃) ₂	262.30	lq.	1.075 ^{15.6}		145–50 ²	5		
Dimethyl adipate	[(CH ₃) ₂ CO ₂ CH ₃] ₂	174.19	col. lq.	1.063 ^{20/4}	10–1	115 ¹⁸	i.		
-amine	(CH ₃) ₂ NH	45.08	col. lq.	0.680 ^{2/4}	–96	7.4	v. s.	s.	s.
-aminoasobenzene (<i>p</i> -)	C ₆ H ₅ N=N·C ₆ H ₄ N(CH ₃) ₂	225.29	yel./al.		116–7	d.	i.	s.	s.
-aminoethanol	(CH ₃) ₂ NCH ₂ CH ₂ OH	89.14	col. lq.	0.887 ^{20/4}		135 ⁷⁹⁶	∞		
-aminophenol (<i>m</i> -)	(CH ₃) ₂ NCH ₂ CH ₂ OH	137.18	nd.		85	265–8	sl. s. h.	s.	s.
-aniline	(CH ₃) ₂ NCH ₂ CH ₂ OH	121.18	yel. lq.	0.956 ^{20/4}	2.5	193	i.	s.	s.
sulfonic acid (<i>m</i> -)	(CH ₃) ₂ NCH ₂ CH ₂ SO ₃ H	201.24	cr.		d. 266		s.		
(<i>p</i> -)	(CH ₃) ₂ NCH ₂ CH ₂ SO ₃ H·H ₂ O	219.26	pr.		257		s. h.	v. sl. s.	v. sl. s.
carbonate	OC(OCH ₃) ₂	90.08	col. lq.	1.070 ^{20/4}	0.5	89–90	i.	∞	∞
ether	CH ₃ OCH ₃	46.07	gas		–138.5	–23.7	3700 cc. ¹⁸	s.	s.
-formamide	HCON(CH ₃) ₂	73.09	lq.	0.945 ²⁵	–58.3	152.8	∞		
fumarate	(·CHCO ₂ CH ₃) ₂	144.13	col. tri.		102	192	i.	sl. s.	sl. s.
glutarate	(CH ₂) ₂ (CO ₂ CH ₃) ₂	160.17	lq.	1.089 ^{15.6}	–37	130 ⁵⁰			
glyoxime	(CH ₃ ·C:NOH) ₂	116.12	col. cr.		240–6		0.06 ²⁰	v. s.	v. s.
-naphthalene (1,–4)	C ₁₀ H ₆ (CH ₃) ₂	156.22	lq.	1.016 ^{20/4}	<–18	264–6	i.		
(2,–3)	C ₁₀ H ₆ (CH ₃) ₂	156.22	lf./al.		104	265 ⁷⁶⁷	i.	sl. s.	
-naphthylamine (<i>α</i> -)	C ₁₀ H ₇ N(CH ₃) ₂	171.24	col. oil	1.042 ²⁰		274.5 ⁷¹¹	i.	s.	s.
(β-)	C ₁₀ H ₇ N(CH ₃) ₂	171.24	col. cr.	1.039 ^{70/70}	46	304–5	i.	s.	s.
oxalate	(·CO ₂ CH ₃) ₂	118.09	col. mn.	1.148 ^{5/4}	54	163.3	6	s.	s.
phthalate (<i>o</i> -)	C ₆ H ₄ (CO ₂ CH ₃) ₂	194.18	col. lq.	1.189 ^{25/25}		280 ⁷³⁴	0.43		
sulfate	(CH ₃ O) ₂ SO ₂	126.13	col. oil	1.352 ^{20/4}	–26.8	188.3	v. sl. s.	∞	∞
sulfide	(CH ₃) ₂ S	62.13	oil	0.846 ^{21/4}	–83.2	37.3	i.	s.	s.
tartrate (<i>d</i> -)	(CHOH·CO ₂ CH ₃) ₂	178.14	cr.	1.328 ^{20/4}	61.5	280	s.	200 ¹⁵	
-vinyl-ethenyl carbinol	(CH ₃) ₂ COH·C≡C·CH:CH ₂	110.15	lq.	0.887 ^{20/4}		150	6 ²⁰		
Dinaphthyl (αα')	C ₁₀ H ₇ ·C ₁₀ H ₇	254.33	lf./al.		160	240–4 ¹²	i.	s. h.	s.
-methane (αα'-)	(C ₁₀ H ₇) ₂ CH ₂	268.35	pr./al.		109	>360	i.	0.8 c.	v. s.
(β,β')	(C ₁₀ H ₇) ₂ CH ₂	268.35	nd./al.		92		i.	s.	
Dinitro-anisole (1-)(2,–4-)	CH ₃ OC ₆ H ₃ (NO ₂) ₂	198.13	col. mn.	1.341 ²⁰	94–5		sl. s. h.	1.5 ²⁰	
-benzene (<i>o</i> -)	C ₆ H ₄ (NO ₂) ₂	168.11	col. mn.	1.59 ¹⁸	117–8	319 ⁷⁷⁴	0.01 c.	1.9 ²¹	
(<i>m</i> -)	C ₆ H ₄ (NO ₂) ₂	168.11	col. rhb.	1.575 ^{20/4}	89.8	300–2	0.3 ²⁰		
(<i>p</i> -)	C ₆ H ₄ (NO ₂) ₂	168.11	col. mn.	1.625 ¹⁸	173–4	299 ⁷⁷⁷	0.18 ¹⁰⁰	0.18 ²¹	
sulfonic acid (2,–4)(1-)	(NO ₂) ₂ C ₆ H ₃ SO ₃ H·3H ₂ O	302.22	pr.		106–8		s.		v. sl. s.
-benzoic acid (2,–4)	(NO ₂) ₂ C ₆ H ₃ CO ₂ H	212.12	cr./aq.		179–80		1.85 ²⁵	s.	
(3,–5)	(NO ₂) ₂ C ₆ H ₃ CO ₂ H	212.12	mn. pr.		204–5	subl.	s. h.	v. s.	sl. s.
-benzophenone (4,–4')	(NO ₂ C ₆ H ₄) ₂ CO	272.21	col. nd.		189		i.		
-diphenyl (4,–4')	(NO ₂ C ₆ H ₄) ₂	244.20	nd./al.	1.445	233		i.	1.5 ²⁰	
(2,–4')	(NO ₂ C ₆ H ₄) ₂	244.20	mn.	1.474	93.5		i.	v. s. h.	
-naphthalene (1,–5)	C ₁₀ H ₆ (NO ₂) ₂	218.17	nd.		216	subl.	i.		
(1,–8)	C ₁₀ H ₆ (NO ₂) ₂	218.17	rhb.		170–2	d.	i.	0.2 c.	
Dinitro-phenol (2,–3)	(NO ₂) ₂ C ₆ H ₄ OH	184.11	yel. mn.	1.681 ²⁰	144–5		sl. s.	v. s. h.	v. s.
(2,–4)	(NO ₂) ₂ C ₆ H ₄ OH	184.11	yel. rhb.	1.683 ²⁴	114–5	subl.	0.5 c.	4 ²⁰	v. s. h.
(2,–6)	(NO ₂) ₂ C ₆ H ₄ OH	184.11	yel. rhb.		63–4		s. h.	s. h.	s.
-salicylic acid (3,–5)	(NO ₂) ₂ C ₆ H ₃ (OH)CO ₂ H·H ₂ O	246.13	pl./aq.		173 d.		s. c.	v. s.	v. s.
-stilbene (4,–4')	(NO ₂ C ₆ H ₄ CH=)	270.24	yel. lf.		210–6		i.	v. sl. s.	v. sl. s.
-toluene (2,–4)	(NO ₂) ₂ C ₆ H ₃ CH ₃	182.13	nd.	1.321 ⁷¹	70	300	0.03 ²²	1.2 ¹⁵	9 ¹⁶
(3,–4)	(NO ₂) ₂ C ₆ H ₃ CH ₃	182.13	nd.	1.259 ¹¹¹	60–1		i.		
(3,–5)	(NO ₂) ₂ C ₆ H ₃ CH ₃	182.13	mn. pr.	1.277 ¹¹¹	92–3	subl.	sl. s.	s. h.	s.
Dioxane	O < (CH ₂) ₂ CH ₂ > O	88.11	col. lq.	1.033 ^{20/4}	9.5–10.5	101.1	∞	s.	s.
Dipentene	C ₁₀ H ₁₆	136.23	col. lq.	0.865 ¹⁸		178	i.		

Diphenyl	$C_6H_5 \cdot C_6H_5$	154.21	col. mn.	0.992 ^{73/4}	69–70	254.9	i.	10 ²⁰	6.6 ²⁰
-amine	$C_6H_5NHC_6H_5$	169.22	col. mn.	1.160 ^{20/20}	52.9	302	0.03 ²⁵	56 ^{10.5}	s.
carbonate	$CO(OC_6H_5)_2$	214.22	nd./al.	1.272 ¹⁴	80	302–6	i.	v. s.	s.
-chloroarsine	$(C_6H_5)_2AsCl$	264.58	rhb.	1.583 ⁴⁰	43–4	d. 327	0.2 d.	20	s.
-ethane	$(C_6H_5CH_2)_2$	182.26	col. pr.	0.978 ^{20/50}	52–3	284	i.	s.	v. s.
ether	$C_6H_5OC_6H_5$	170.21	col. rhb.	1.073 ²⁰	27	259	v. sl. s.	s.	∞
guanidine	$(C_6H_5NH)_2C:NH$	211.26	mn./al.		147–8	d. > 170	v. sl. s.	9 ²⁰	sl. s.
-methane	$(C_6H_5)_2CH_2$	168.23	col. pr.	1.001 ^{26/4}	26–7	265	i.	v. s.	v. s.
phenylenediamine (<i>p</i> -)	$(C_6H_5NH)_2C_6H_4$	260.33	cr.		152		i.		
succinate	$(-CH_2CO_2C_6H_5)_2$	270.28	lf./al.		122–3	330	i.	s.	s.
sulfide	$(C_6H_5)_2S$	186.27	col. lq.	1.119 ^{15/15}	<–40	296–7	sl. s. h.	s. h.	∞
sulfone	$(C_6H_5)_2SO_2$	218.27	nd./aq.	1.248 ^{25/4}	128–9	379	v. sl. s.	s.	s.
urea (uns.)	$(C_6H_5)_2NCONH_2$	212.25	rhb.	1.276	189		i.	s. h.	v. s.
Diphenylene oxide	$< (C_6H_4)_2O$	168.19	lf./al.		86–7	287–8	i.	s. h.	v. s.
Dipropyl adipate (<i>n</i> -)	$(-CH_2CH_2CO_2C_3H_7)_2$	230.30	col. lq.	0.979 ^{20/4}	–20.3	143–5 ¹⁰	i.	s.	s.
-amine (<i>n</i> -)	$(C_3H_7CH_2)_2NH$	101.19	col. lq.	0.739 ^{20/4}	–39.6	110–1	s.	∞	∞
(<i>i</i> -)	$[(CH_3)_2CH]_2NH$	101.19	col. lq.	0.722 ²²	–61	83.5 ⁷⁴³	s.	s.	
aniline (<i>n</i> -)	$C_6H_5N(C_3H_7)_2$	177.29	yel. oil	0.910 ²⁰		245.4	i.	s.	s.
carbonate (<i>n</i> -)	$CO(OCH_2C_6H_5)_2$	146.18	col. lq.	0.968 ²²		168.2	v. sl. s.		
ether (<i>n</i> -)	$(C_3H_7CH_2)_2O$	102.17	col. lq.	0.744 ^{21/0}	–122	91	sl. s.	∞	∞
(<i>i</i> -)	$[(CH_3)_2CH]_2O$	102.17	col. lq.	0.725 ^{21/0}	–60	69	0.2	∞	∞
ketone (<i>n</i> -)	$(C_3H_7CH_2)_2CO$	114.19	col. lq.	0.822 ^{20/4}	–32.6	144.2	0.43	∞	∞
(<i>i</i> -)	$[(CH_3)_2CH]_2CO$	114.19	col. lq.	0.806 ^{20/4}		123.7	v. sl. s.	∞	∞
oxalate (<i>n</i> -)	$CO_2CH_2C_3H_5_2$	174.19	col. lq.	1.038 ^{9/0}	–51.7	213.5	d. h.		
(<i>i</i> -)	$[CO_2CH(CH_3)_2]_2$	174.19	col. lq.			190			
Disalicylal ethylenediamine	$[HOC_6H_4CH:CHNCH_2]_2$	268.31	cr.	1.34	125–6		0.03 ²⁸		
Ditolyl guanidine (<i>o</i> -)	$(C_7H_7NH)_2C:NH$	239.32	cr.	1.10 ^{20/4}	178–9		v. sl. s.	s. h.	s.
Divinyl acetylene	$(H_2C:C:CH \cdot C:)_2$	78.11	lq.	0.776 ^{20/4}		85	i.		
Docosane (<i>n</i> -)	$CH_3(CH_2)_{20}CH_3$	310.60	cr.	0.778 ^{44/4}	44.5	224.5 ¹⁵	i.	4 h.	v. s.
Dodecane (<i>n</i> -)	$CH_3(CH_2)_{10}CH_3$	170.33	lq.	0.751 ^{20/4}	–9.6	214.5	i.	v. s.	v. s.
Dulcitol	$CH_3OH(CHOH)_4CH_2OH$	182.17	mn.	1.466 ¹⁵	189	290–5 ³	3.2 ¹⁵	v. sl. s.	i.
Durene (1-,2-,4-,5-)	$(CH_3)_4C_6H_2$	134.22	mn.	0.838 ^{21/4}	79–80	193–5	i.	s.	s.
Elaidic acid	$C_8H_{17}CH:CH(CH_2)_7CO_2H$	282.46	lf./al.	0.851 ^{79/4}	51–2	288 ¹⁰⁰	i.	v. s.	v. s.
Eosine	$C_{20}H_{12}O_2Br_4$	647.89	col. cr.				i.	s.	
Ephedrine (<i>L</i> -)	$C_6H_5CHOHCH(CH_3)NHCH_3$	165.23	cr./et.		40	255	5	500	s.
Epichlorhydrin (α -)	$C_2H_5O \cdot CH_2Cl$	92.52	lq.	1.183 ^{25/25}	–25.6	117 ⁷³⁶	<5	∞	∞
Epidichlorohydrin (α -)	$CH_2Cl \cdot CCl \cdot CH_2Cl$	110.97	col. lq.	1.204 ²⁵		94	i.	∞	∞
Erythritol (<i>dl</i> -)	$CH_2OH(CHOH)_2CH_2OH$	122.12	tet. pr.	1.451 ^{20/4}	126	329–31	60	sl. s. c.	i.
tetranitrate	$C_4H_6(ONO_2)_4$	302.11	lf./al.		61	expl.	i. c.	s.	s.
Ethane	CH_3CH_3	30.07	col. gas	0.546 ^{–88}	–172	–88.6	4.7 cc. ²⁰	150 cc.	
Ethanol-amine	$HOCH_2CH_2NH_2$	61.08	col. oil	1.022 ²⁰	10.5	171 ⁷⁵⁷	∞	∞	1
formamide	$HCONHCH_2CH_2OH$	89.09	lq.	1.169 ²⁵	<–40	d.	∞		
Ether	$(CH_3CH_2)_2O$	74.12	col. lq.	0.708 ^{25/4}	–116.3	34.6	7.5 ²⁰	∞	
Ethyl abietate	$C_{19}H_{30}CO_2C_2H_5$	330.50	lq.	1.020 ^{20/20}		200 ⁴	i.		
acetate	$CH_3CO_2C_2H_5$	88.11	col. lq.	0.901 ^{20/4}	–82.4	77.1	8.5 ¹⁵	∞	∞
acetacetate	$CH_3COCH_2CO_2C_2H_5$	130.14	col. lq.	1.025 ^{20/4}	–45	180 ⁷⁵⁵	13 ¹⁷	∞	∞
alcohol	CH_3CH_2OH	46.07	col. lq.	0.789 ^{20/4}	–112	78.4	∞	∞	∞
-amine	$C_2H_5NH_2$	45.08	col. lq.	0.689 ^{15/15}	–80.6	16.6	∞	∞	∞
hydrochloride	$C_2H_5NH_2 \cdot HCl$	81.54	mn.	1.216	108–9		240 ¹⁷	v. s.	i.
aniline	$C_6H_5NHC_2H_5$	121.18	lq.	0.963 ^{20/4}	–63.5	204	i.	∞	∞
sulfonic acid (<i>m</i> -)	$C_2H_5NHC_6H_4SO_3H$	201.24	nd./aq.		d. 294		2.15 ¹⁵		
anisate (<i>p</i> -)	$CH_3OC_6H_4CO_2C_2H_5$	180.20	lq.	1.103 ^{25/25}	7–8	269–70	i.	s.	s.
anthranilate (<i>o</i> -)	$NH_2C_6H_4CO_2C_2H_5$	165.19	cr.	1.117 ^{20/4}	13	266–8	v. sl. s.	s.	s.
benzene	C_6H_6	106.17	col. lq.	0.867 ^{20/4}	–94.4	136.2	0.01 ¹⁵	∞	∞
benzoate	$C_6H_5CO_2C_2H_5$	150.17	col. lq.	1.052 ^{15/15}	–34.6	211–2	0.08 ²⁰	∞	∞
-benzyl-aniline	$C_6H_5N(C_2H_5)CH_2C_6H_5$	211.30	yel. oil	1.034 ^{18.5}		285 ¹⁰	i.	18	∞
bromide	C_2H_5Br	108.97	col. lq.	1.431 ^{20/4}	–117.8	38.4	1.06 ⁰	∞	∞
butyrate (<i>n</i> -)	$C_2H_5CH_2CO_2C_2H_5$	116.16	col. lq.	0.879 ^{20/4}	–93.3	120–1	0.68 ²⁵	∞	∞
(<i>i</i> -)	$(CH_3)_2CHCO_2C_2H_5$	116.16	col. lq.	0.871 ^{20/4}	–88.2	110–1	sl. s.	∞	∞
caprate (<i>n</i> -)	$CH_3(CH_2)_8CO_2C_2H_5$	200.32	lq.	0.859 ²⁵	–20	244.6 ⁷⁵⁸	0.002 ²⁰	∞	∞
Ethyl caproate (<i>n</i> -)	$CH_3(CH_2)_6CO_2C_2H_5$	144.21	col. lq.	0.873 ^{20/20}	–67.5	165–6 ⁷³⁶	i.	∞	∞
caprylate (<i>n</i> -)	$CH_3(CH_2)_8CO_2C_2H_5$	172.26	col. lq.	0.878 ¹⁷	–45	207–8 ⁷⁵³	i.	∞	∞
chloride	CH_3CH_2Cl	64.51	col. lq.	0.917 ^{6/6}	–139	13	0.45 ⁰	∞	∞
chloroacetate	$ClCH_2CO_2C_2H_5$	122.55	col. lq.	1.159 ^{20/4}	–26	144	i.	∞	∞
chlorocarbonate	$ClCO_2CH_2CH_3$	108.52	col. lq.	1.138 ^{20/4}	–80.6	94–5	d.	∞	∞
cinnamate (<i>trans</i> -)	$C_6H_5CH:CHCO_2C_2H_5$	176.21	col. lq.	1.049 ^{20/4}	12	271	i.	∞	∞
cyanoacetate	$CH_2(CN)CO_2C_2H_5$	113.11	col. lq.	1.062 ^{20/4}	–22.5	208 ⁷⁵³	2 ²⁵	∞	∞
formate	$HCO_2CH_2CH_3$	74.08	col. lq.	0.923 ^{20/4}	–79	54 ⁷⁸⁰	11 ¹⁸	∞	∞
furoate (α)	$OC_4H_3CO_2C_2H_5$	140.14	lf.	1.117 ^{20/4}	34	195 ⁷⁶⁶	i.	∞	∞
heptoate	$CH_3(CH_2)_5CO_2C_2H_5$	158.24	col. lq.	0.872 ^{20/20}	–66.1	187–8	0.029 ²⁰	∞	∞
hypochlorite	$ClOCH_2CH_3$	80.51	yel. lq.	1.013 ^{6/4}	expl.			∞	∞
iodide	CH_3CH_2I	155.97	col. lq.	1.933 ^{20/4}	–105	72.4	0.4 ²⁰	∞	∞
lactate	$CH_3CH(OH)CO_2C_2H_5$	118.13	oil	1.030 ^{25/4}		155	∞	∞	∞

TABLE 2-2 Physical Properties of Organic Compounds (Continued)

Name	Formula	Formula weight	Form and color	Specific gravity	Melting point, °C	Boiling point, °C	Solubility in 100 parts		
							Water	Alcohol	Ether
laurate	$\text{CH}_3(\text{CH}_2)_{10}\text{CO}_2\text{C}_2\text{H}_5$	228.37	oil	0.868 ^{13/4}	-10.7	269	i.	s.	∞
mercaptan	$\text{CH}_3\text{CH}_2\text{SH}$	62.13	lq.	0.839 ^{20/4}	-121	36-7	1.5	s.	s.
methacrylate	$\text{CH}_2=\text{C}(\text{CH}_3)\text{CO}_2\text{C}_2\text{H}_5$	114.14	col. lq.	0.913 ^{15.6}		118	i.	s.	s.
naphthylamine (α -)	$\text{C}_{10}\text{H}_7\text{NH}_2\text{C}_6\text{H}_5$	171.24	oil	1.060 ^{20/4}		303 ⁷²³	i.	s.	s.
naphthyl ether (α -)	$\text{C}_{10}\text{H}_7\text{OC}_6\text{H}_5$	172.22	cr.	1.061 ^{20/20}	5.5	276.4	i.	s.	s.
nitrate	$\text{C}_2\text{H}_5\text{ONO}_2$	91.07	col. lq.	1.100 ^{25/4}	-102	87-8	1.3 ⁵⁵	∞	∞
nitrite	$\text{C}_2\text{H}_5\text{ONO}$	75.07	lq.	0.900 ^{15.5}		17	v. sl. s.	∞	∞
oleate	$\text{C}_{17}\text{H}_{33}\text{CO}_2\text{C}_2\text{H}_5$	310.51	oil	0.867 ²⁵	<-15	216-8 ¹⁵	i.	∞	∞
palmitate	$\text{CH}_3(\text{CH}_2)_{14}\text{CO}_2\text{C}_2\text{H}_5$	284.48	col. nd.	0.858 ^{25/4}	24-5	191 ¹⁰	i.	s.	s.
pelargonate	$\text{CH}_3(\text{CH}_2)_7\text{CO}_2\text{C}_2\text{H}_5$	186.29	col. lq.	0.866 ^{17.5}	-44.5	227-8 ⁷³⁷	i.	∞	∞
propionate	$\text{CH}_3\text{CH}_2\text{CO}_2\text{C}_2\text{H}_5$	102.13	col. lq.	0.891 ^{20/4}	-72.6	99.1	2.4 ²⁰	∞	∞
salicylate (o -)	$\text{HOOC}_6\text{H}_4\text{CO}_2\text{C}_2\text{H}_5$	166.17	col. lq.	1.136 ^{15/4}	1.3	233-4	i.	∞	∞
stearate	$\text{CH}_3(\text{CH}_2)_{16}\text{CO}_2\text{C}_2\text{H}_5$	312.53	col. cr.	0.848 ^{36.3}	33.4(31)	201 ¹⁰	i.	s.	s.
toluate (o -)	$\text{CH}_3-\text{C}_6\text{H}_4\text{CO}_2\text{C}_2\text{H}_5$	164.20	lq.	1.032 ^{25/25}	<-10	227	i.	∞	∞
toluate (m -)	$\text{CH}_3-\text{C}_6\text{H}_4\text{CO}_2\text{C}_2\text{H}_5$	164.20	lq.	1.030 ^{20/20}		231 ⁷⁵⁰	i.	∞	∞
toluene sulfonate (p -)	$\text{CH}_3-\text{C}_6\text{H}_4\text{SO}_3\text{C}_2\text{H}_5$	200.25	pr./al.	1.166 ^{48/4}	33-4	221.3	i.	s.	s.
toluidine (o -)	$\text{CH}_3-\text{C}_6\text{H}_4\text{NHC}_2\text{H}_5$	135.21	lq.	0.948 ^{25/4}	<-15	215-6	i.		
toluidine (p -)	$\text{CH}_3-\text{C}_6\text{H}_4\text{NHC}_2\text{H}_5$	135.21	lq.	0.942 ^{25/4}		217	i.		
urea	$\text{C}_2\text{H}_5\text{NH}-\text{CO}-\text{NH}_2$	88.11	nd.	1.213 ¹⁸	92		v. s.	80	i.
valerate (n -)	$\text{CH}_3(\text{CH}_2)_3\text{CO}_2\text{C}_2\text{H}_5$	130.18	col. lq.	0.877 ²⁰	-91.2	145.5	0.24 ²⁵	∞	∞
valerate (i -)	$(\text{CH}_3)_2\text{CH}(\text{CH}_2)\text{CO}_2\text{C}_2\text{H}_5$	130.18	col. lq.	0.867 ^{20/4}	-99.3	135	0.17 ²⁰	∞	∞
Ethylal	$\text{CH}_3(\text{OC}_2\text{H}_5)_2$	104.15	lq.	0.824 ^{25/4}	-66.5	89	9 ¹⁸	∞	∞
Ethylene	$\text{H}_2\text{C}=\text{CH}_2$	28.05	col. gas	0.57 ^{-102/4}	-169	-103.9	26 cc. ⁰	360 cc.	s.
bromide	$\text{BrCH}_2-\text{CH}_2\text{Br}$	187.86	col. lq.	2.180 ^{20/4}	10	131.5	0.43 ⁸⁰	∞	∞
bromohydrin	$\text{BrCH}_2-\text{CH}_2\text{OH}$	124.96	col. lq.	1.772 ^{20/4}		150.3	sl. s.	s.	
chlorobromide	$\text{ClCH}_2-\text{CH}_2\text{Br}$	143.41	lq.	1.689 ¹⁹	-16.6	106.7	0.69 ⁸⁰		
chlorohydrin	$\text{ClCH}_2-\text{CH}_2\text{OH}$	80.51	col. lq.	1.213 ^{20/4}	-69	128.8	∞	∞	∞
diamine	$\text{H}_2\text{NCH}_2-\text{CH}_2\text{NH}_2$	60.10	col. lq.	0.900 ^{20/20}	8.5	117.2	∞	∞	0.3
oxide	$<(\text{CH}_3)_2>\text{O}$	44.05	lq.	0.887 ^{7/4}	-111.3	13.5 ⁴⁷	∞	∞	v. s.
Ethylidene diacetate	$\text{CH}_3\text{CH}(\text{O}_2\text{CCH}_3)_2$	146.14	col. lq.	1.061 ¹²	18.85	168 ⁷⁴⁰	sl. s.	∞	
Eugenol (1,4,3-)	$\text{C}_9\text{H}_7-\text{C}_6\text{H}_3(\text{OH})\text{OCH}_3$	164.20	oil	1.070 ^{15/15}	10.3	253.5	v. sl. s.	∞	∞
(i -)(1,3,4-)	$\text{C}_9\text{H}_7-\text{C}_6\text{H}_3(\text{OCH}_3)\text{OH}$	164.20	oil	1.091 ^{15/15}	-10	267.5	v. sl. s.	∞	∞
Fenchyl alcohol (dl -)	$\text{C}_{10}\text{H}_{17}\text{OH}$	154.25	col. cr.	0.935 ⁴⁰	35	201	sl. s.		
(d -)(α -)	$\text{C}_{10}\text{H}_{17}\text{OH}$	154.25	col. pr.	0.964 ^{20/4}	45-7	201-2	sl. s.	s.	s.
(i -)(l -)	$\text{C}_{10}\text{H}_{17}\text{OH}$	154.25	col. cr.	0.961	61-2	201-2	i.		
Ferric dimethyl-dithiocarbamate	$\text{Fe}[\text{SSCN}(\text{CH}_3)_2]_3$	416.49	cr.		d. 100-30	ign. >150	v. sl. s.		
Fluorene	$(\text{C}_6\text{H}_4)_2>\text{CH}_2$	166.22	cr./al.	1.203 ^{9/4}	115-6	293-5	i.	s. h.	s.
Fluorescein	$\text{C}_{20}\text{H}_{12}\text{O}_5$	332.31	yel. red		d. > 290		v. sl. s. h.	s. h.	
Fluoro-dichloromethane	FCHCl_2	102.92	gas	1.426 ⁶	-127	14.5	i.	s.	s.
-trichloromethane	Cl_3CF	137.37	col. lq.	1.494 ^{17.2}		24.9	i.	∞	∞
Formaldehyde	HCHO	30.03	gas	0.815 ⁻²⁰	-92	-21	v. s.	v. s.	v. s.
(m -)	$(\text{CH}_2\text{O})_3$	90.08	wh.	1.17 ⁶⁵	64	114.5 ⁷⁵⁹	21 ²⁵	s.	s.
(p -)	$(\text{CH}_2\text{O})_x\cdot x\text{H}_2\text{O}$	(30.03)	amor.		150-60	subl.	20-30 ¹⁸	i.	i.
Formamide	HCONH_2	45.04	lq.	1.139 ^{20/4}	2	193	∞	∞	v. sl. s.
Formanilide	$\text{HCONHC}_6\text{H}_5$	121.14	mn.	1.147 ^{15/15}	47	216 ¹²⁰	sl. s.	v. s.	s.
Formic acid	HCO_2H	46.03	col. lq.	1.220 ^{20/4}	8.6	100.8	∞	∞	
Fructose	$\text{CH}_2\text{OH}(\text{CHOH})_3\text{COCH}_2\text{OH}$	180.16	nd./aq.	1.669 ^{17.5}	95-105		v. s.	8 ¹⁸	
Fuchsin	$\text{C}_{20}\text{H}_{19}\text{N}_3\text{HCl}$	337.85	red	1.22	d. >200		0.3	s.	i.
Fulminic acid	C_2NOH	43.02							
Fumaric acid ($trans$ -)	$\text{HO}_2\text{CCH}=\text{CHCO}_2\text{H}$	116.07	col. pr.	1.635 ^{20/4}	286-7	290	0.7 ¹⁷	5.8 ³⁰	0.7 ²⁵
Furfural	$\text{C}_4\text{H}_3\text{O}=\text{CHO}$	96.08	lq.	1.159 ^{20/4}	-38.7	161.7 ⁷⁶⁰	9.1 ¹³	∞	
Furfuran	$\text{C}_4\text{H}_4\text{O}$	68.07	col. lq.	0.937 ^{20/4}		31-2 ⁷⁵⁶	i.	s.	s.
Furfuryl acetate	$\text{CH}_3\text{CO}_2\text{CH}_2\text{C}_4\text{H}_3\text{O}$	140.14	col. oil	1.118 ^{20/4}		175-7	i.	s.	s.
alcohol	$\text{C}_4\text{H}_3\text{O}=\text{CH}_2\text{OH}$	98.10	oil	1.129 ^{25/4}		169.5 ⁷⁵²	∞	s.	∞
butyrate	$\text{C}_3\text{H}_7\text{CO}_2\text{CH}_2-\text{C}_4\text{H}_3\text{O}$	168.19	col. lq.	1.053 ^{20/4}		212-3	v. sl. s.	s.	s.
propionate	$\text{C}_2\text{H}_5\text{CO}_2\text{CH}_2-\text{C}_4\text{H}_3\text{O}$	154.16	col. lq.	1.109 ^{20/4}		195-6	v. sl. s.	s.	∞
Furoic acid	$\text{C}_4\text{H}_3\text{O}=\text{CO}_2\text{H}$	112.08	mn. pr.		133-4	230-2	3.6 ¹⁵	s.	s.
G-acid, K salt (2-)(6-8-)	$\text{HOC}_{10}\text{H}_7(\text{SO}_3\text{K})_2$	380.48	cr.				8 ²⁵		
Na salt (2-)(6-8-)	$\text{HOC}_{10}\text{H}_7(\text{SO}_3\text{Na})_2$	348.26	cr.				34 ²⁰		
Galactose (d -)(α -)	$\text{C}_6\text{H}_{11}\text{O}_5\cdot\text{CHO}$	180.16	pr.		165.5		10.3 ³⁰	0.6 ⁴⁰	
Gallic acid (3,4,5-)	$(\text{HO})_3\text{C}_6\text{H}_2\text{CO}_2\text{H}\cdot\text{H}_2\text{O}$	188.13	mn./aq.	1.694 ^{4/4}	d. 220		1 ¹³	28 ¹⁵	2.5 ¹⁵
Gamma acid (2-,8-,6-)	$\text{C}_{10}\text{H}_5(\text{NH}_2)(\text{OH})\text{SO}_3\text{H}$	239.25	cr.						
Geraniol	$\text{C}_{10}\text{H}_{15}\text{CH}_2\text{OH}$	154.25	col. lq.	0.883 ¹⁵	<-15	230	i.	∞	∞
Glucose (d -)(α -)	$\text{C}_6\text{H}_{11}\text{O}_5\cdot\text{CHO}$	180.16	rhb.	1.544 ²⁵	146		82 ^{17.5}	sl. s.	i.
(d -)(β -)	$\text{C}_6\text{H}_{12}\text{O}_6\cdot\text{H}_2\text{O}$	198.17	cr.	1.562 ^{16/4}	150		154 ¹⁵		
Glucuronic acid	$\text{CHO}(\text{CHOH})_4\text{CO}_2\text{H}$	194.14	cr.		154		v. s.		
Glutaminic acid (dl -)	$[\cdot\text{CHNH}_2(\text{CH}_2)_2\cdot](\text{CO}_2\text{H})_2$	147.13	cr./aq.	1.460	199 d.		1.5 ²⁰	v. sl. s.	v. sl. s.

Glutaric acid	$\text{CH}_2(\text{CH}_2\text{CO}_2\text{H})_2$	132.11	col. cr.	1.429 ¹⁵	97.5	200 ³⁰	63.9 ³⁰	v. s.	v. s.
Glycerol	$\text{CH}_2\text{OH}\cdot\text{CHOH}\cdot\text{CH}_2\text{OH}$	92.09	col. lq.	1.260 ^{90/4}	17.9	290	∞	∞	i.
acetate (mono-)	$\text{C}_5\text{H}_{10}\text{O}_4$	134.13	col. oil	1.20 ^{20/4}		158 ¹⁶⁵	v. s.	v. s.	sl. s.
(di-)	$(\text{CH}_3\text{CO}_2)_2\text{C}_3\text{H}_5\text{OH}$	176.17	col. lq.	1.178 ^{15/15}	40	175-6 ⁴⁰	s.	s.	sl. s.
nitrate (mono-) (α -)	$\text{CH}_2\text{OH}\cdot\text{CHOH}\cdot\text{CH}_2\text{NO}_3$	137.09	col. pr.	1.40 ¹⁵	58-9	155-60	70 ¹⁵	v. s.	v. sl. s.
(β -)	$\text{CH}_2\text{OH}\cdot\text{CHNO}_2\cdot\text{CH}_2\text{OH}$	137.09	lf.	1.40 ¹⁵	54	155-60		v. s.	sl. s.
dinitrate (1-,3-)	$\text{CHOH}(\text{CH}_2\text{ONO}_2)_2$	182.09	oil	1.47 ¹⁵	<-30	146-8 ¹⁵		v. s.	v. s.
Glyceryl triacetate	$(\text{CH}_3\text{CO}_2)_3\text{C}_3\text{H}_5$	218.20	col. lq.	1.161 ^{17/4}	-78	258-9	7.17 ¹⁵	∞	∞
tribenzoate	$(\text{C}_6\text{H}_5\text{CO}_2)_3\text{C}_3\text{H}_5$	404.41	nd.	1.228 ¹²	75-6	d.	i.	s. h.	s.
tributyrate	$(\text{C}_4\text{H}_9\text{CH}_2\text{CO}_2)_3\text{C}_3\text{H}_5$	302.36	col. lq.	1.032 ^{20/4}	<-75	305-9	i.	s.	s.
tricaprate	$[\text{CH}_3(\text{CH}_2)_8\text{CO}_2]_3\text{C}_3\text{H}_5$	554.84	col. cr.	0.921 ^{40/4}	31(25)		i.	s. h.	v. s.
tricaproate	$[\text{CH}_3(\text{CH}_2)_4\text{CO}_2]_3\text{C}_3\text{H}_5$	386.52	col. lq.	0.987 ^{20/4}	-25		i.	s.	v. s.
tricaprylate	$[\text{CH}_3(\text{CH}_2)_6\text{CO}_2]_3\text{C}_3\text{H}_5$	470.68	col. lq.	0.954 ^{20/4}	8.3(-21)		i.	s.	s.
trilaurate	$[\text{CH}_3(\text{CH}_2)_{10}\text{CO}_2]_3\text{C}_3\text{H}_5$	639.00	col. nd.	0.894 ^{90/4}	45-6		i.	sl. s. c.	v. s.
trimyristate	$[\text{CH}_3(\text{CH}_2)_{12}\text{CO}_2]_3\text{C}_3\text{H}_5$	723.16	lf.	0.885 ^{90/6}	56.5		i.	s.	
trinitrate	$\text{CH}_2\text{NO}_2\cdot\text{CHNO}_2\cdot\text{CH}_2\text{NO}_2$	227.09	yel. oil	1.601 ¹⁵	13.3(2)	160 ¹⁵	0.18 ²⁰	50 ²⁰	∞
trinitrite	$\text{CH}_2\text{NO}_2\cdot\text{CHNO}_2\cdot\text{CH}_2\text{NO}_2$	179.09	yel. lq.	1.291 ^{10/16}		150 sl. d.	d.	d.	s.
trioleate	$(\text{C}_{17}\text{H}_{33}\text{CO}_2)_3\text{C}_3\text{H}_5$	885.43	col. oil	0.915 ¹⁵	-4	240 ¹⁸	i.	sl. s.	v. s.
tripalmitate	$[\text{CH}_3(\text{CH}_2)_{14}\text{CO}_2]_3\text{C}_3\text{H}_5$	807.32	col. nd.	0.866 ^{90/4}	65.1	310-20 ^{0.1}	i.	0.004 ²¹	v. s.
tristearate	$[\text{CH}_3(\text{CH}_2)_{16}\text{CO}_2]_3\text{C}_3\text{H}_5$	891.48	col. pr.	0.862 ^{90/4}	70.8(55)		i.	s. h.	s. h.
Glycide	$\text{C}_3\text{H}_4\text{O}\cdot\text{CH}_2\text{OH}$	74.08	col. lq.	1.114 ^{16/16}		166 sl. d.	∞	∞	∞
Glycine, Glycocol	$\text{NH}_2\text{CH}_2\cdot\text{CO}_2\text{H}$	75.07	mn.	1.161	232-6 d.		23 c.	0.1 c.	i.
Glycol	$\text{CH}_2\text{OH}\cdot\text{CH}_2\text{OH}$	62.07	col. lq.	1.113 ^{19/4}	-15.6	197.4	∞	∞	1.0
diacetate	$(\text{CH}_3\text{CO}_2\text{CH}_2)_2$	146.14	col. lq.	1.109 ^{14/4}	-31	190.5	14.3 ²²	∞	∞
dibenzoate	$(\text{C}_6\text{H}_5\text{CO}_2\text{CH}_2)_2$	270.28	rhb./et.			>360	i.	s.	s.
dibutyrate	$(\text{C}_4\text{H}_9\text{CO}_2\text{CH}_2)_2$	202.25	col. lq.	1.024 ⁹		240	i.	v. s.	v. s.
dicaprylate	$(\text{C}_{15}\text{H}_{31}\text{CO}_2\text{CH}_2)_2$	314.46	lq.		22		i.		
diformate	$(\text{HCO}_2\text{CH}_2)_2$	118.09	lq.			174	v. sl. s.		
dilaurate	$(\text{C}_{11}\text{H}_{23}\text{CO}_2\text{CH}_2)_2$	426.67	amor.		52-4	188 ³⁰	i.	v. s.	v. s.
dinitrate	$(\text{O}_2\text{NO}\cdot\text{CH}_2)_2$	152.06	yel. lq.	1.482 ^{21/2}	-20	expl. 114	0.92 ²⁵	s.	∞
dinitrite	$(\text{ONO}\cdot\text{CH}_2)_2$	120.06	lq.	1.216 ⁹	<-15	96-8	i.	s. d.	s.
dipalmitate	$(\text{C}_{16}\text{H}_{33}\text{CO}_2\text{CH}_2)_2$	538.89	nd.		71-2	260 ^{0.1}	i.	s.	s.
dipropionate	$(\text{C}_3\text{H}_7\text{CO}_2\text{CH}_2)_2$	174.19	lq.	1.045 ²⁵		211-2	sl. s.	∞	∞
ether	$(\text{HO}\cdot\text{CH}_2\text{CH}_2)_2\text{O}$	106.12	lq.	1.118 ^{20/20}	-10.5	244.8	∞	∞	i.
formal	$<\text{O}\cdot\text{CH}_2\text{CH}_2\text{OCH}_2>$	74.08	lq.	1.060 ^{30/4}		75-6	∞		
formate (mono-)	$\text{HCO}_2\text{CH}_2\text{CH}_2\text{OH}$	90.08	lq.	1.199 ^{15/4}		180	∞		
Glycolic acid	$\text{HOCH}_2\text{CO}_2\text{H}$	76.05	nd./aq.		79(63)	d.	∞	90 ²⁵	v. s.
Guaiacol (o-)	$\text{CH}_3\text{O}\cdot\text{C}_6\text{H}_4\text{OH}$	124.14	pr.	1.140 ^{15/15}	28.3	205	1.7 ¹⁵	v. s.	v. s.
Guanidine	$\text{NH}_2\text{C}(\text{NH}_2)_2$	59.07	col. cr.		50		v. s.	s.	
H-acid, Na salt (1-,8-,3-,6-)	$\text{C}_{10}\text{H}_{18}\text{O}_2\text{NS}_2\text{Na}\cdot\frac{1}{2}\text{H}_2\text{O}$	368.32	cr.				0.17 ²⁰		
Heptacosane (n-)	$\text{CH}_3(\text{CH}_2)_{25}\text{CH}_3$	380.73	col. cr.	0.780 ^{90/4}	59.5	270 ¹⁵	i.		
Heptane (n-)	$(\text{CH}_3)_2\text{CH}(\text{CH}_2)_5\text{CH}_3$	100.20	col. lq.	0.684 ^{20/4}	-90.6	98.4 ⁶⁰	0.005 ¹⁵	sl. s.	∞
(i-)	$(\text{CH}_3)_2\text{CH}(\text{CH}_2)_3\text{CH}_3$	100.20	col. lq.	0.679 ^{20/4}	-118.2	90.0	i.	s.	∞
	$\text{C}_3\text{H}_7\cdot\text{CH}(\text{CH}_3)\cdot\text{C}_3\text{H}_5$	100.20	col. lq.	0.687 ^{20/4}	-119.4	91.8	i.	s.	∞
	$(\text{CH}_3)_3\text{C}\cdot\text{CH}_2\cdot\text{C}_2\text{H}_5$	100.20	col. lq.	0.674 ^{20/4}	-125	79.1	i.	s.	∞
	$[(\text{CH}_3)_2\text{CH}]_2\text{CH}_2$	100.20	col. lq.	0.675 ^{20/4}	-119.4	80.8	i.	s.	∞
	$(\text{CH}_3)_2\text{C}(\text{C}_2\text{H}_5)_2$	100.20	col. lq.	0.693 ^{20/4}	-135.0	86.0	i.	s.	∞
	$(\text{C}_2\text{H}_5)_3\text{CH}$	100.20	col. lq.	0.698 ^{20/4}	-118.7	93.5	i.	s.	∞
	$(\text{CH}_3)_3\text{C}\cdot\text{CH}(\text{CH}_3)_2$	100.20	col. lq.	0.690 ^{20/4}	-25	80.8	i.	s.	∞
Heptioic acid	$\text{CH}_3(\text{CH}_2)_5\text{CO}_2\text{H}$	130.18	col. lq.	0.918 ²⁰	-10	221-2	0.25 ¹⁵	s.	s.
aldehyde	$\text{CH}_3(\text{CH}_2)_5\text{CHO}$	114.19	col. lq.	0.850 ^{20/4}	-42	155	0.02 ²⁰	∞	∞
Heptyl acetate (n-)	$\text{CH}_3\text{CO}_2\text{CH}_2(\text{CH}_2)_5\text{CH}_3$	158.24	col. lq.	0.874 ^{16/16}		191.5 ⁷⁵⁰	i.	s.	s.
alcohol (n-)	$\text{CH}_3(\text{CH}_2)_6\text{CH}_2\text{OH}$	116.20	col. lq.	0.824 ^{20/4}	-34	175 ⁷⁵⁶	0.18 ²⁵	∞	∞
	$[(\text{CH}_3)_2\text{CH}]_2\text{CHOH}$	116.20	col. lq.	0.829 ^{20/4}		140	v. sl. s.	∞	∞
	$(\text{C}_2\text{H}_5)_3\text{CHOH}$	116.20	lq.	0.820 ^{20/4}	-37	156	i.	s.	s.
mercaptan	$\text{CH}_3\text{CH}(\text{SH})\cdot\text{C}_5\text{H}_{11}$	132.27	lq.	0.835 ²⁰		174-5 ⁷⁶⁵	i.		
Hexachloro-benzene	C_6Cl_6	284.78	mn.	2.044 ²⁴	228-31	309 ⁷⁴²	i.	v. sl. s. h.	s. h.
-ethane	$\text{CCl}_3\cdot\text{CCl}_3$	236.74	rhb.	2.091 ^{20/4}	186-7	186 ⁷⁷⁷	0.005 ²²	v. s.	v. s.
Hexacosane (n-)	$\text{CH}_3(\text{CH}_2)_{24}\text{CH}_3$	366.71	cr.	0.779 ^{27/4}	56.6	262 ¹⁵	i.	v. sl. s.	∞
Hexadecane (n-)	$\text{CH}_3(\text{CH}_2)_{14}\text{CH}_3$	226.44	lf.	0.774 ^{20/4}	18.5	287.5	i.	∞	∞
Hexaethylbenzene	$\text{C}_6(\text{C}_2\text{H}_5)_6$	246.43	pr./al.	0.831 ^{130/4}	130	298.3	i.	0.75 ²⁵	8 ²⁵
Hexamethylbenzene	$\text{C}_6(\text{CH}_3)_6$	162.27	pl./al.		166	265	i.	0.2 ⁹	v. s.
Hexamethylene-diamine	$\text{NH}_2(\text{CH}_2)_6\text{NH}_2$	116.20	lf.		42	204-5	v. s.	s.	
-diisocyanate	$\text{OCN}(\text{CH}_2)_6\text{NCO}$	168.19	lq.	1.04 ²⁵		143-4 ²⁰	d.	d.	
-glycol	$\text{HO}(\text{CH}_2)_6\text{OH}$	118.17	nd./aq.		42	250	s.	s.	sl. s. h.
tetramine	$(\text{CH}_2)_6\text{N}_4$	140.19	col. rhb.		subl.		81 ¹²	3	v. sl. s.
Hexane (n-)	$\text{CH}_3(\text{CH}_2)_4\text{CH}_3$	86.18	col. lq.	0.659 ^{20/4}	-94	69	0.014 ¹⁵	50 ³³	∞
(i-)	$(\text{CH}_3)_2\text{CH}(\text{CH}_2)_2\text{CH}_3$	86.18	lq.	0.654 ^{20/4}	-153.7	60.2	i.	s.	s.
(neo-)	$(\text{CH}_3)_3\text{C}\cdot\text{C}_2\text{H}_5$	86.18	lq.	0.649 ^{20/20}	-98.2	49.7	i.	s.	s.
	$(\text{CH}_3)_2\text{CH}\cdot\text{CH}(\text{CH}_3)_2$	86.18	lq.	0.662 ^{20/4}	-129.8	58.0 ⁷⁶⁰	i.	s.	s.
	$(\text{C}_2\text{H}_5)_2\text{CHCH}_3$	86.18	lq.	0.664 ^{20/4}	-118	63.2	i.	s.	s.

TABLE 2-2 Physical Properties of Organic Compounds (Continued)

Name	Formula	Formula weight	Form and color	Specific gravity	Melting point, °C	Boiling point, °C	Solubility in 100 parts		
							Water	Alcohol	Ether
Hexyl acetate (<i>n</i> -)	$\text{CH}_3\text{CO}_2(\text{CH}_2)_5\text{CH}_3$	144.21	col. lq.	0.890 ⁹⁰		169.2	i.	v. s.	v. s.
alcohol (<i>n</i> -)	$\text{CH}_3(\text{CH}_2)_4\text{CH}_2\text{OH}$	102.17	col. lq.	0.820 ^{20/20}	-51.6	157.2	0.6 ²⁰	∞	∞
	$(\text{CH}_3)_2\text{CH}-\text{C}(\text{CH}_3)_2\text{OH}$	102.17	lq.	0.821 ²⁰⁰	-14	120-1	v. sl. s.	∞	∞
	$(\text{CH}_3)_2\text{COH}-\text{CH}_2\text{C}_6\text{H}_5$	102.17	lq.	0.809 ²⁰⁴	-107	123 ⁷⁶²	v. sl. s.	∞	∞
formate (<i>n</i> -)	$\text{HCO}_2\text{CH}_2(\text{CH}_2)_4\text{CH}_3$	130.18	lq.	0.898 ⁹		153.6		∞	∞
resorcinol (2-,4-)	$\text{C}_6\text{H}_3(\text{CH}_2)_2\text{C}_6\text{H}_3(\text{OH})_2$	194.27	col. nd.		68-70	179 ⁷	0.05	v. s.	s.
Hippuric acid	$\text{C}_6\text{H}_5\text{CONHCH}_2\text{CO}_2\text{H}$	179.17	rhb.	1.371 ²⁰⁴	187-8	d.	0.4 ²⁰	s. h.	0.25 ¹⁸
Histidine (<i>L</i> -)	$\text{C}_6\text{H}_6\text{O}_3\text{N}_3$	155.15	lf./aq.		d. 287		s.	v. sl. s.	i.
Homophthalic acid (<i>o</i> -)	$\text{HO}_2\text{C}\cdot\text{C}_6\text{H}_4\cdot\text{CH}_2\text{CO}_2\text{H}$	180.16	cr./aq.		175-80		s. h.	v. s.	sl. s.
Hydracrylic acid	$\text{HOCH}_2\text{CH}_2\text{CO}_2\text{H}$	90.08	syrup			d.			
Hydro-cyanic acid	HCN	27.03	lq.	0.697 ¹⁸	-12	25-6	∞	∞	∞
-quinone (<i>p</i> -)	$\text{C}_6\text{H}_4(\text{OH})_2$	110.11	cr.	1.332 ¹⁵	170.3	285 ⁷³⁰	6 ¹⁵	v. s.	v. s.
Hydroxy-benzaldehyde (<i>p</i> -)	$\text{HO}\cdot\text{C}_6\text{H}_4\cdot\text{CHO}$	122.12	nd./aq.	1.129 ³⁰	116-7	subl.	1.36 ³¹		
-benzanilide (<i>o</i> -)	$\text{HO}\cdot\text{C}_6\text{H}_4\cdot\text{CONHC}_6\text{H}_5$	213.23	pr./al.		135	d.	v. sl. s. h.	s.	s.
-quinoline (2-)(α -)	$\text{C}_9\text{H}_6\text{N}\cdot\text{OH}$	145.16	pr./al.		199-200	subl.	s. h.	v. s.	v. s.
(8-)(<i>o</i> -)	$\text{C}_9\text{H}_6\text{N}\cdot\text{OH}$	145.16	pr.		75-6	266.6 ⁷³²	v. sl. s. c.	s.	sl. s.
Indigo	$[\text{C}_8\text{H}_4(\text{CO})(\text{NH})\text{C}]_2$	262.26	cr.	1.35	390-2	subl.	i.	i.	i.
White	$\text{C}_{16}\text{H}_{12}\text{O}_2\text{N}_2$	264.28	gray				i.	s.	s.
Indole	$\text{C}_8\text{H}_7\text{N}$	117.15	lf./aq.		52	253-4	s. h.	s. h.	s.
Indoxyl	$\text{C}_8\text{H}_6\text{NOH}$	133.15	yel. pr.		85	110	s.	s.	s.
Iodo-benzene	$\text{C}_6\text{H}_5\text{I}$	204.01	col. lq.	1.824 ²⁵⁴	-28.5	188.6	0.034 ²⁰	s.	∞
-phenol (<i>p</i> -)	$\text{IC}_6\text{H}_4\text{OH}$	220.01	nd./aq.	1.857 ¹¹²	93-4	d.	sl. s.	v. s.	v. s.
Iodoform	CHI_3	393.73	yel. hex.	4.008 ¹⁷	119	subl.	0.01 ²⁵	1.5 ¹⁷	13.6 ²⁵
Ionone (α -)	$\text{C}_{10}\text{H}_{16}\cdot\text{CHCOCH}_3$	192.30	col. oil	0.930 ²⁰		136.1 ¹⁷	sl. s.	∞	∞
(β -)	$\text{C}_{10}\text{H}_{16}\cdot\text{CHCOCH}_3$	192.30	col. oil	0.944 ²⁰		140 ¹⁸	sl. s.	∞	∞
Irone (β -)	$\text{C}_{14}\text{H}_{22}\text{O}$	206.32	col. oil	0.939 ²⁰		144 ¹⁶	v. sl. s.	v. s.	v. s.
Isatin	$\text{C}_6\text{H}_4 < (\text{CO})(\text{N}) > \text{COH}$	147.13	yel. red		200-1	subl.	s. h.	v. s. h.	sl. s.
Isoprene	$\text{CH}_2=\text{CH}\cdot\text{C}(\text{CH}_3)=\text{CH}_2$	68.12	col. lq.	0.681 ²⁰⁴	-120	34	i.	∞	∞
Ketene	$\text{H}_2\text{C}=\text{CO}$	42.04	col. gas		-151	-56	d.	d.	s.
Koch acid (1-)(3,6-,8-)	$\text{C}_{10}\text{H}_4(\text{NH}_2)_3\text{S}_3\text{OHNH}_2$	427.34	cr.				7.2 ²⁰		
Lactic acid (<i>dl</i> -)	$\text{CH}_3\text{CH}(\text{OH})\text{CO}_2\text{H}$	90.08	hyg.	1.249 ¹⁵⁴	16.8	122 ¹⁴	∞	∞	∞
anhydride	$\text{C}_6\text{H}_{10}\text{O}_5$	162.14	yel. oil			d. 250	v. sl. s.	s.	s.
Lactide (<i>dl</i> -)	$\text{C}_6\text{H}_8\text{O}_4$	144.13	tri./al.	0.862 ¹⁰⁴	124.5	255 ⁷⁵⁷	v. sl. s.	v. sl. s. c.	
Lactose	$\text{C}_{12}\text{H}_{22}\text{O}_{11}\cdot\text{H}_2\text{O}$	360.31	col. rhb.	1.525 ²⁰	202	d.	17 ¹⁰	i.	i.
Lauric acid	$\text{CH}_3(\text{CH}_2)_{10}\text{CO}_2\text{H}$	200.32	col. nd.	0.869 ²⁰⁴	48(44)	225 ¹⁰⁰	i.	s.	s.
Laurone	$[\text{CH}_3(\text{CH}_2)_{10}]_2\text{CO}$	338.61	pl.	0.809 ²⁰⁴	69-70		i.	i. c.	
Lauryl alcohol	$\text{CH}_3(\text{CH}_2)_{10}\text{CH}_2\text{OH}$	186.33	lf.	0.831 ²⁴⁴	24	255-9	i.	s.	s.
Lead tetraethyl	$\text{Pb}(\text{CH}_2\text{CH}_3)_4$	323.44	col. lq.	1.659 ¹⁸⁴	-136	152 ²⁰¹	i.	sl. s.	
tetramethyl	$\text{Pb}(\text{CH}_3)_4$	267.34	col. lq.	1.995 ²⁰⁴	-27.5	110 ⁷⁰⁰	i.	∞	∞
Lepidine (<i>py</i> -4)	$\text{C}_6\text{H}_6\text{N}\cdot\text{CH}_3$	143.19	lq.	1.086 ²⁰	9-10	261-3	sl. s.	∞	∞
Leucine (<i>L</i> -)	$(\text{CH}_3)_2\text{CHCH}_2\text{CH}(\text{NH}_2)\text{CO}_2\text{H}$	131.17	cr.	1.293 ¹⁸	295	subl.	2.2 ¹⁸		
Levulinic acid	$\text{CH}_3\text{CO}(\text{CH}_2)_2\text{CO}_2\text{H}$	116.12	lf.	1.140 ^{20/20}	33.5	245-6	v. s.	v. s.	v. s.
Limonene (<i>d</i> - or <i>L</i> -)	$\text{C}_{10}\text{H}_{16}$	136.23	lq.	0.842 ²⁰⁴	-96.9	177	i.	∞	∞
Linalool (<i>d</i> - or <i>L</i> -)	$\text{C}_{10}\text{H}_{17}\text{OH}$	154.25	col. oil	0.868 ²⁰		198-200	v. sl. s.	s.	∞
Linalyl acetate	$\text{CH}_3\text{CO}_2\text{C}_{10}\text{H}_{17}$	196.29	col. lq.	0.895 ²⁰		220 ⁷⁶² d.	v. sl. s.	∞	∞
Linoleic acid	$\text{C}_{17}\text{H}_{31}\text{CO}_2\text{H}$	280.45	yel. oil	0.903 ¹⁵⁴	-9.5	229-30 ¹⁶	i.	∞	∞
Maleic acid	$\text{HO}_2\text{C}\cdot\text{CH}=\text{CH}\cdot\text{CO}_2\text{H}$	116.07	mm.	1.609	130.5	135 d.	79 ²⁵	70 ³⁰	8 ²⁵
anhydride	$< (\cdot\text{CHCO})_2 > \text{O}$	98.06	cr.	1.5	57-60	202	16.3 ³⁰		
Malic acid (<i>dl</i> -)	$\text{HO}_2\text{CCH}_2\text{CH}(\text{OH})\text{CO}_2\text{H}$	134.09	col. cr.	1.601 ²⁰⁴	128-9	150 d.	144 ²⁶	v. s.	v. s.
(<i>d</i> - or <i>L</i> -)	$\text{HO}_2\text{CCH}_2\text{CH}(\text{OH})\text{CO}_2\text{H}$	134.09	col. cr.	1.595 ²⁰⁴	99-100	140 d.	v. s.	v. s.	8.4 ¹⁵
Malonic acid	$\text{H}_2\text{C}(\text{CO}_2\text{H})_2$	104.06	col. tri.	1.631 ¹⁵	130-5 d.		138 ¹⁶	42 ²⁵	8 ¹⁵
Maltose	$\text{C}_{12}\text{H}_{22}\text{O}_{11}\cdot\text{H}_2\text{O}$	360.31	col. nd.	1.540 ¹⁷	d.		108 ²⁵	v. sl. s. c.	i.
Mandelic acid (<i>dl</i> -)	$\text{C}_6\text{H}_5\text{CH}(\text{OH})\text{CO}_2\text{H}$	152.15	rhb./aq.	1.300 ²⁰⁴	118.1	d.	16 ²⁰	s.	s.
Mannitol (<i>d</i> -)	$\text{CH}_2\text{OH}(\text{CHOH})_4\text{CH}_2\text{OH}$	182.17	col. rhb.	1.489 ²⁰⁴	166	290-5 ³	13 ¹⁴	0.01 ¹⁴	i.
Mannose (<i>d</i> -)	$\text{CH}_2\text{OH}(\text{CHOH})_4\text{CHO}$	180.16	rhb.	1.539 ²⁰⁴	132		248 ¹⁷	v. sl. s.	i.
Margaric acid	$\text{CH}_3(\text{CH}_2)_{15}\text{CO}_2\text{H}$	270.45	col. pl.	0.853 ⁹⁰	60-1	227 ¹⁰⁰	i.	32 ²⁸	v. s.
Mellitic acid	$\text{C}_6(\text{CO}_2\text{H})_6$	342.17	nd./al.		286-8	d.	v. s.	v. s.	
Menthol (1-)(α -)	$\text{C}_{10}\text{H}_{18}\text{OH}$	156.27	col. cr.	0.890 ^{15/15}	42-3	212	0.04 c.	v. s.	v. s.
Mercapto-benzothiazole (2-)	$< \text{C}_6\text{H}_4\text{N}:\text{C}(\text{SH})\text{S} >$	167.25	nd.	1.420 ²⁴⁴	179	d.	i.	v. s.	sl. s.
-thiazoline (2-)	$< \text{CH}_2\text{N}:\text{C}(\text{SH})\text{SCH}_2 >$	119.21	cr.	1.50	106		1.6 ⁶⁰		
Mercuric cyanide	$\text{Hg}(\text{CN})_2$	252.62	cr.	4.003 ²²	d. 320		12.5 ¹⁵		
fulminate	$\text{Hg}(\text{ONC})_2\cdot\frac{1}{2}\text{H}_2\text{O}$	293.63	cr./aq.	4.4	expl.		0.07 ¹²	s.	
Mesityl oxide	$(\text{CH}_3)_2\text{C}=\text{CHCOCH}_3$	98.14	lq.	0.858 ²⁰⁴	-59	130 ⁷⁵⁰	3 ²⁰	∞	∞
Mesitylene (1-,3-,5-)	$\text{C}_6\text{H}_3(\text{CH}_3)_3$	120.19	col. lq.	0.865 ²⁰⁴	-45(-52)	164.8	i.	s.	∞
Metamlic acid (<i>m</i> -)	$\text{H}_2\text{NC}_6\text{H}_4\text{SO}_3\text{H}$	173.19	col. nd.		d.		2 ¹⁵	v. sl. s.	v. sl. s.
Methane	CH_4	16.04	gas	0.415 ⁻¹⁶⁴	-182.6	-161.4	0.4 ²⁰ cc.	47 ²⁰ cc.	104 ¹⁰ cc.

Methoxy-methoxyethanol	$\text{CH}_3(\text{OCH}_2)_2\text{CH}_2\text{OH}$	106.12	lq.	1.038 ²⁵	<-70	167.5	∞		
Methyl acetate	$\text{CH}_3\text{CO}_2\text{CH}_3$	74.08	col. lq.	0.924 ^{20/4}	-98.7	57.1	33 ²²	∞	∞
acrylic acid (α -)	$\text{CH}_2=\text{C}(\text{CH}_3)\text{CO}_2\text{H}$	86.09	pr.	1.015 ^{20/4}	15-16	161-3	s. h.	∞	∞
alcohol	CH_3OH	32.04	col. lq.	0.792 ^{20/4}	-97-8	64.7	∞	∞	∞
-amine	CH_3NH_2	31.06	col. gas	0.699 ⁻¹¹	-92.5	-6.7 ⁷⁵⁸	v. s.	v. s.	
-amine hydrochloride	$\text{CH}_3\text{NH}_2\cdot\text{HCl}$	67.52	pl./al.	1.23	226-8	230 ¹⁵	v. s.	23 h.	i.
aniline	$\text{C}_6\text{H}_5\text{NHCH}_3$	107.15	lq.	0.989 ^{20/4}	-57	195.5	0.01 ²⁵	∞	∞
anthracene (α -)	$\text{C}_6\text{H}_4:(\text{CH})_2\text{C}_6\text{H}_5\text{CH}_3$	192.26	lf./al.	1.047 ^{90/4}	86		i.		
(β -)	$\text{C}_6\text{H}_4:(\text{CH})_2\text{C}_6\text{H}_5\text{CH}_3$	192.26	col. lf.	1.181 ^{9/4}	207		i.	v. sl. s.	v. sl. s.
anthranilate (o -)	$\text{NH}_2\text{C}_6\text{H}_4\text{CO}_2\text{CH}_3$	151.16	col. lq.	1.168 ^{19/4}	24	135.5 ¹⁵	sl. s.	s.	s.
anthraquinone (2-)	$\text{C}_6\text{H}_7:(\text{CO})_2\text{C}_6\text{H}_5\text{CH}_3$	222.24	col. nd.	subl.	176-7		s.	s.	s.
benzoate	$\text{C}_6\text{H}_5\text{CO}_2\text{CH}_3$	136.15	col. lq.	1.087 ^{25/25}	-12.5	198-9	0.02 ³⁰	∞	∞
benzylaniline	$\text{C}_6\text{H}_5\text{N}(\text{CH}_3)\text{CH}_2\text{C}_6\text{H}_5$	197.28	lq.		9.2	305-6	i.	s.	s.
bromide	CH_3Br	94.94	gas	1.732 ^{9/0}	-93	4.5 ⁷⁵⁸	v. sl. s.	s.	s.
butyrate (n -)	$\text{CH}_3(\text{CH}_2)_2\text{CO}_2\text{CH}_3$	102.13	col. lq.	0.898 ^{20/4}	<-95	102.3	1.7	∞	∞
(i -)	$(\text{CH}_3)_2\text{CHCO}_2\text{CH}_3$	102.13	col. lq.	0.891 ^{20/4}	-84.7	92.6	v. sl. s.	∞	∞
caprate	$\text{CH}_3(\text{CH}_2)_8\text{CO}_2\text{CH}_3$	186.29	lq.		-18	223-4	i.	∞	∞
caproate (n -)	$\text{CH}_3(\text{CH}_2)_4\text{CO}_2\text{CH}_3$	130.18	col. lq.	0.904 ^{9/0}		149.5	i.	∞	∞
caprylate	$\text{CH}_3(\text{CH}_2)_6\text{CO}_2\text{CH}_3$	158.24	col. lq.	0.887 ¹⁸	-40	192-4	i.	∞	∞
cellosolve	$\text{CH}_3\text{OCH}_2\text{CH}_2\text{OH}$	76.09	col. lq.	0.965 ^{20/4}		124-5	∞	∞	∞
chloride	CH_2Cl	50.49	gas	0.952 ⁹	-97.7	-24	280 ¹⁶ cc.	v. s.	v. s.
chloroacetate	$\text{ClCH}_2\text{CO}_2\text{CH}_3$	108.52	col. lq.	1.236 ^{20/4}	-32.7	130 ^{7/40}	v. sl. s.	∞	∞
chloroformate	ClCO_2CH_3	94.50	col. lq.	1.236 ¹⁵		71-2	d.	∞	∞
cinnamate	$\text{C}_6\text{H}_5\text{CH}=\text{CHCO}_2\text{CH}_3$	162.19	cr.	1.042 ^{36/0}	33.4	263	i.	v. s.	v. s.
cyclohexane	$\text{CH}_2 < (\text{CH}_2\text{CH}_2)_2 > \text{CHCH}_3$	98.19	col. lq.	0.769 ^{20/4}	-126.3	101	i.	s.	s.
ethyl carbonate	$\text{CH}_3\text{O}-\text{CO}-\text{OC}_2\text{H}_5$	104.10	lq.	1.002 ²⁷	-14.5	109.2	i.	∞	∞
ethyl ketone	$\text{CH}_3\text{CO}-\text{C}_2\text{H}_5$	72.11	col. lq.	0.805 ^{20/4}	-85.9	79.6	35 ¹⁰	∞	∞
ethyl oxalate	$\text{CH}_3\text{OCO}-\text{CO}_2\text{C}_2\text{H}_5$	132.11	lq.	1.156 ^{9/0}		173.7	i.	v. s.	v. s.
formate	HCO_2CH_3	60.05	lq.	0.974 ^{20/4}	-99.8	32	30 ³⁰	∞	∞
furoate	$\text{C}_4\text{H}_3\text{O}-\text{CO}_2\text{CH}_3$	126.11	col. lq.	1.179 ^{21/4}		181.3	i.	∞	∞
glucamine	$\text{CH}_2\text{OH}(\text{CHOH})_4\text{CH}_2\text{NHCH}_3$	195.21							
glycolate	$\text{HOCH}_2\text{CO}_2\text{CH}_3$	90.08	lq.	1.168 ¹⁸		151.2			
heptate	$\text{CH}_3(\text{CH}_2)_5\text{CO}_2\text{CH}_3$	144.21	lq.	0.881 ^{15/4}		172-3	i.	∞	∞
hypochlorite	ClOCH_3	66.49	gas			12 ⁷²⁶			
iodide	CH_3I	141.94	col. lq.	2.279 ^{20/4}	-64.4	42.4	1.8 ¹⁵	∞	∞
lactate	$\text{CH}_3\text{CH}(\text{OH})\text{CO}_2\text{CH}_3$	104.10	lq.	1.090 ¹⁹		144.8	∞	s.	s.
laurate	$\text{CH}_3(\text{CH}_2)_{10}\text{CO}_2\text{CH}_3$	214.34	lq.			148 ¹⁵	i.	∞	∞
mercaptan	CH_3SH	48.11	gas	0.896 ⁹	-121	5.8 ⁷⁵²	s.	v. s.	v. s.
methacrylate	$\text{CH}_2=\text{C}(\text{CH}_3)\text{CO}_2\text{CH}_3$	100.12	lq.	0.950 ^{15/6}	-48	100.3	i.	∞	∞
myristate	$\text{CH}_3(\text{CH}_2)_{12}\text{CO}_2\text{CH}_3$	242.40	cr./al.		18-9	295 ⁷¹⁵	i.	∞	∞
naphthalene (α -)	$\text{C}_{10}\text{H}_7\text{CH}_3$	142.20	oil	1.025 ^{14/4}	-19	244.6	i.	v. s.	v. s.
(β -)	$\text{C}_{10}\text{H}_7\text{CH}_3$	142.20	mn.	0.994 ^{40/4}	35-6	241-2	i.	v. s.	v. s.
nitrate	CH_3ONO_2	77.04	lq.	1.203 ²⁵	expl.	65	sl. s.	s.	s.
nitrite	CH_3ONO	61.04	gas	0.991 ¹⁵		-12	s.	s.	s.
nonyl ketone (n -)	$\text{CH}_3(\text{CH}_2)_8\text{COCH}_3$	170.29	col. oil	0.828 ^{20/20}	13.5	228	i.	s.	s.
oleate	$\text{C}_{17}\text{H}_{33}\text{CO}_2\text{CH}_3$	296.49	oil	0.879 ¹⁸		190-1 ¹⁰	i.	∞	∞
orange	$(\text{CH}_3)_2\text{NC}_6\text{H}_4\text{N}_2\text{C}_6\text{H}_4\text{SO}_3\text{Na}$	327.33	red pd.				0.2 c.		
palmitate	$\text{CH}_3(\text{CH}_2)_{14}\text{CO}_2\text{CH}_3$	270.45	col. cr.		30-1	196 ¹⁸	i.	s.	s.
phosphine	CH_3PH_2	48.02	gas			-14 ⁷⁸⁹	i.	sl. s.	
propionate	$\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_3$	88.11	col. lq.	0.915 ^{20/4}	-87.5	79.7	0.5 ²⁰	∞	∞
propyl ketone (n -)	$\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3$	86.13	col. lq.	0.812 ^{15/15}	-77.8	102	v. sl. s.	∞	∞
salicylate (o -)	$\text{HO}-\text{C}_6\text{H}_4\text{CO}_2\text{CH}_3$	152.15	col. lq.	1.182 ^{25/25}	-8.3	222.2	0.07 ³⁰	∞	∞
stearate	$\text{CH}_3(\text{CH}_2)_{16}\text{CO}_2\text{CH}_3$	298.50	col. cr.		38-9	215 ¹⁵	i.	s.	s.
toluate (o -)	$\text{CH}_3-\text{C}_6\text{H}_4\text{CO}_2\text{CH}_3$	150.17	col. lq.	1.073 ¹⁵	<-50	213	i.	∞	∞
(m -)	$\text{CH}_3-\text{C}_6\text{H}_4\text{CO}_2\text{CH}_3$	150.17	col. lq.	1.066 ¹⁵		215	i.	∞	∞
(p -)	$\text{CH}_3-\text{C}_6\text{H}_4\text{CO}_2\text{CH}_3$	150.17	cr.		33-4	217	i.	v. s.	v. s.
Methyl toluidine (o -)	$\text{CH}_3-\text{C}_6\text{H}_4\text{NHCH}_3$	121.18	lq.	0.973 ¹⁵		206-7	i.	∞	∞
(m -)	$\text{CH}_3-\text{C}_6\text{H}_4\text{NHCH}_3$	121.18	lq.			206-7	i.	∞	∞
(p -)	$\text{CH}_3-\text{C}_6\text{H}_4\text{NHCH}_3$	121.18	lq.	0.935 ^{25/4}		211 ⁷⁶¹	i.	∞	∞
valerate (n -)	$\text{CH}_3(\text{CH}_2)_3\text{CO}_2\text{CH}_3$	116.16	lq.	0.895 ^{15/4}	-91	127.3	v. sl. s.	∞	∞
(i -)	$(\text{CH}_3)_2\text{CHCH}_2\text{CO}_2\text{CH}_3$	116.16	col. lq.	0.881 ^{20/4}		116.7 ⁷⁶⁴	v. sl. s.	∞	∞
vinyl ketone	$\text{CH}_3\text{COCH}=\text{CH}_2$	70.09	lq.	0.836 ^{20/4}		81	>85	∞	∞
Methylal	$\text{HCH}(\text{OCH}_3)_2$	76.09	col. lq.	0.866 ^{15/4}	-104.8	42-3	33	∞	∞
Methylene-bis-(phenyl-4-isocyanate)	$(\text{OCN}-\text{C}_6\text{H}_4)_2\text{CH}_2$	250.25	lq.	1.222 ³⁰		210-2 ¹³	d.	∞	∞
bromide	CH_2Br_2	173.83	col. lq.	2.495 ^{20/4}	-52.8	98.5 ⁷⁵⁶	1.17 ⁰	∞	∞
chloride	CH_2Cl_2	84.93	col. lq.	1.336 ^{20/4}	-96.7	40-1	2 ³⁰	∞	∞
dianiline	$(\text{C}_6\text{H}_5\text{NH})_2\text{CH}_2$	198.26	cr.		65	208-9 d.	i.	s.	s.
iodide	CH_2I_2	267.84	col. lq.	3.325 ^{20/4}	5.7	180 d.	1.4 ²⁰	∞	∞
Michler's hydrol (p -, p' -)	$[(\text{CH}_3)_2\text{NC}_6\text{H}_4]_2\text{CHOH}$	270.37	gn.		96-7		i.	s. h.	s.
ketone	$[(\text{CH}_3)_2\text{NC}_6\text{H}_4]_2\text{CO}$	268.35	lf./al.		174	>360 d.	i.	sl. s.	v. sl. s.
Morphine	$\text{C}_{17}\text{H}_{19}\text{O}_3\text{N}\cdot\text{H}_2\text{O}$	303.35	pr./al.	1.317	254 d.		0.02 ²⁰	sl. s.	s.
Mucic acid	$(-\text{CHOHCH}(\text{OHCO}_2\text{H})_2)$	210.14	pd.		206-14		0.33 ¹⁴	i.	i.

TABLE 2-2 Physical Properties of Organic Compounds (Continued)

Name	Formula	Formula weight	Form and color	Specific gravity	Melting point, °C	Boiling point, °C	Solubility in 100 parts		
							Water	Alcohol	Ether
Mustard gas	(ClCH ₂ ·CH ₂) ₂ S	159.08	oil	1.275 ^{20/4}	13–4	217	0.07 ²⁵	s.	s.
Myrcyl alcohol	C ₁₀ H ₁₆ OH(2)	452.84	cr.	0.777 ⁹⁵	88		i.	v. sl. s.	v. s.
Myristic acid	CH ₃ (CH ₂) ₁₂ CO ₂ H	228.37	col. lf.	0.853 ^{20/4}	57–8	250.5 ¹⁰⁰	i.	v. s.	v. s.
Myristyl alcohol	CH ₃ (CH ₂) ₁₂ CH ₂ OH	214.39	cr.	0.824 ^{25/4}	38	167 ¹⁵	<0.02	sl. s.	s.
Naphthalene	C ₁₀ H ₈	128.17	pl./al.	1.145 ^{20/4}	80.2	217.9	0.003 ²⁵	9.5 ²⁰	v. s.
disulfonic acid (1-,5-)	C ₁₀ H ₆ (SO ₃ H) ₂	288.30	lf.		d.		102 ³⁰	s.	i.
(1-,6-)	C ₁₀ H ₆ (SO ₃ H) ₂	288.30	cr.		d. 125		164 ²⁰	s.	i.
sulfonic acid (α-)	C ₁₀ H ₇ SO ₃ H·2H ₂ O	244.26	cr.		90		v. s.	v. s.	sl. s.
(β-)	C ₁₀ H ₇ SO ₃ H·H ₂ O	226.25	cr.		125		77 ³⁰		
Naphthasultam (1-,8-)	C ₁₀ H ₇ O ₂ NS	205.23	nd.		177–8		s. h.	sl. s.	s.
disulfonate Na (1-,8-)	C ₁₀ H ₅ O ₆ NS ₂ Na ₂ ·2H ₂ O	445.35	cr.				v. s.		
(2-,4-)	C ₁₀ H ₄ O ₆ NS ₂ Na ₃ ·8½H ₂ O	584.43	lf.				v. s.	sl. s.	
Naphthoic acid (α-)	C ₁₀ H ₇ CO ₂ H	172.18	nd.		160–1	300	v. sl. s. h.	sl. s.	s.
(β-)	C ₁₀ H ₇ CO ₂ H	172.18	mm.	1.077 ^{100/4}	184	>300	0.007 ²⁵	s.	v. s.
Naphthol (α-)	C ₁₀ H ₇ OH	144.17	mm.	1.224 ⁴	96	278–80	sl. s. h.	v. s.	v. s.
(β-)	C ₁₀ H ₇ OH	144.17	mm.	1.217 ⁴	122–3	285–6	0.074 ²⁵	v. s.	v. s.
sulfonic acid (α-)(1-,2-)	HO·C ₁₀ H ₆ SO ₃ H	224.23	pl./aq.		>250		v. s. h.		i.
(β-)(2-,6-)	HO·C ₁₀ H ₆ SO ₃ H	224.23	lf.		125		v. s.	v. s.	
Naphthyl acetate (α-)	CH ₃ CO ₂ C ₁₀ H ₇	186.21	nd./al.		46–9		sl. s. h.	s.	s.
(β-)	CH ₃ CO ₂ C ₁₀ H ₇	186.21	nd./al.		69–70		i.	s.	s.
amine (α-)	C ₁₀ H ₇ NH ₂	143.19	rhb.	1.123 ^{25/25}	50	300.8	0.17 c.	v. s.	v. s.
(β-)	C ₁₀ H ₇ NH ₂	143.19	lf./aq.	1.061 ^{98/4}	111–2	306.1	v. s. h.	s.	s.
amine hydrochloride (α-)	C ₁₀ H ₇ NH ₂ ·HCl	179.65	nd.			subl.	3.8 ²⁰	s.	s.
(β-)	C ₁₀ H ₇ NH ₂ ·HCl	179.65	lf.				v. s.	v. s.	
amine sulfonic acid (1-,4-)	NH ₂ ·C ₁₀ H ₆ ·SO ₃ H	223.25	nd.		d.		0.2 ¹⁰⁰	i.	i.
(1-,5-)	NH ₂ ·C ₁₀ H ₆ ·SO ₃ H·H ₂ O	241.26	cr.				sl. s.		
(1-,7-)	NH ₂ ·C ₁₀ H ₆ ·SO ₃ H·H ₂ O	241.26	cr.				0.46 ²⁵		
(1-,8-)	NH ₂ ·C ₁₀ H ₆ ·SO ₃ H·H ₂ O	241.26	cr.				0.42 ¹⁰⁰		
(2-,5-)	NH ₂ ·C ₁₀ H ₆ ·SO ₃ H	223.25	cr.				0.08		
(2-,6-)	NH ₂ ·C ₁₀ H ₆ ·SO ₃ H·H ₂ O	241.26	cr.				0.38 ¹⁰⁰		
(2-,7-)	NH ₂ ·C ₁₀ H ₆ ·SO ₃ H·H ₂ O	241.26	cr.				0.28 ¹⁰⁰		
isocyanate (α-)	C ₁₀ H ₇ N:CO	169.18	col. lq.	1.18		269–70	d.	s.	s.
Nicotine	C ₁₀ H ₁₄ N ₂	162.23	oil	1.009 ^{20/4}	<–80	246 ⁷³⁰	s.	∞	∞
Nicotinic acid (3-)	C ₆ H ₅ NCO ₂ H	123.11	nd./al.		235.2	subl.	s. h.	s. h.	v. sl. s.
(i-)(4-)	C ₆ H ₅ NCO ₂ H	123.11	nd./aq.		317	d.	sl. s. h.	v. sl. s.	
Nitro-acetanilide (p-)	CH ₃ CONHC ₆ H ₄ NO ₂	180.16	rhb.		215–6		s. h.	s.	s.
-acetophenone (m-)	CH ₃ COC ₆ H ₄ NO ₂	165.15	nd.		80–1	202	i.	s.	
-aminoanisole (4-,1-,2-)	NO ₂ ·C ₆ H ₃ (OCH ₃)NH ₂	168.15	red nd.	1.207 ¹⁵⁶	118		i.	s.	
(5-,1-,2-)	NO ₂ ·C ₆ H ₃ (OCH ₃)NH ₂	168.15	yel. nd.	1.211 ¹⁵⁶	139–40		sl. s.	s.	s.
(3-,1-,4-)	NO ₂ ·C ₆ H ₃ (OCH ₃)NH ₂	168.15	red		123		sl. s. c.	v. s.	v. s.
-aminophenol (4-,2-,1-)	NO ₂ ·C ₆ H ₃ (NH ₂)OH	154.12	or. pr.		142–3		s. h.	v. s.	v. s.
-aniline (o-)	NO ₂ ·C ₆ H ₄ NH ₂	138.12	yel. rhb.	1.442 ¹⁵	71.5	284.1	v. s.	v. s.	v. s.
(m-)	NO ₂ ·C ₆ H ₄ NH ₂	138.12	yel. rhb.	1.43	114	306.4	0.11 ²⁰	7.1 ²⁰	6.1 ²⁰
(p-)	NO ₂ ·C ₆ H ₄ NH ₂	138.12	yel. mn.	1.437 ¹⁴	146–7	331.7	0.08 ¹⁹	5.8 ²⁰	
-anisole (o-)	CH ₃ OC ₆ H ₄ NO ₂	153.14	col. cr.	1.254 ^{20/4}	9.4	272–3	0.17 ³⁰	∞	∞
(p-)	CH ₃ OC ₆ H ₄ NO ₂	153.14	pr./al.	1.233 ³⁰	54	274	0.06 ³⁰	v. s.	v. s.
-anthraquinone (α-)	C ₁₄ H ₈ (CO) ₂ ·C ₆ H ₃ NO ₂	253.21	nd.		230	270 ⁷	i.	sl. s.	v. sl. s.
-anthraquinone sulfonic acid (1-,5-)	NO ₂ ·C ₁₄ H ₆ O ₂ ·SO ₃ H	333.27	yel. cr.				s.	i.	i.
-benzal chloride (m-)	NO ₂ ·C ₆ H ₄ ·CHCl ₂	206.03	mn.		65		i.	v. s. h.	v. s.
-benzaldehyde (m-)	NO ₂ ·C ₆ H ₄ CHO	151.12	nd./aq.		58	164 ²³	1.95 ¹¹²	v. s. h.	v. s.
Nitro-benzene	C ₆ H ₅ NO ₂	123.11	yel. lq.	1.205 ^{18/4}	5.7	210.9	0.19 ²⁰	v. s.	∞
-benzidine (2-)	NH ₂ ·C ₆ H ₄ ·C ₆ H ₃ (NH ₂)NO ₂	229.23	red nd.		143		sl. s. h.		
-benzoic acid (o-)	NO ₂ ·C ₆ H ₄ ·CO ₂ H	167.12	tri./aq.	1.575 ^{4/4}	147.5		0.65 ²⁰	28 ¹¹	22 ¹¹
(m-)	NO ₂ ·C ₆ H ₄ ·CO ₂ H	167.12	mn.	1.494 ^{4/4}	140–1		0.24 ¹⁶⁵	31 ¹²	25 ¹⁰
(p-)	NO ₂ ·C ₆ H ₄ ·CO ₂ H	167.12	yel. mn.	1.550 ^{22/4}	240–2	subl.	0.02 ¹⁵	0.9 ¹⁰	2.2 ¹⁸
-benzyl alcohol (m-)	NO ₂ ·C ₆ H ₄ ·CH ₂ OH	153.14	cr.		27	175–80 ³	i.	2 ¹⁹	v. s.
-benzyl bromide (p-)	NO ₂ ·C ₆ H ₄ CH ₂ Br	216.03	nd./al.		99–100		i.		
-chlorotoluene (1-,2-,6-)	CH ₃ ·C ₆ H ₃ (NO ₂)Cl	171.58	cr.		37.5	238	i.		
-cresol (1-,3-,4-)	CH ₃ ·C ₆ H ₃ (NO ₂)OH	153.14	yel.	1.240 ^{39/4}	32	125 ²²	v. sl. s.	v. s.	v. s.
-cymene (1-,2-,4-)	CH ₃ ·C ₆ H ₃ (NO ₂)CH(CH ₃) ₂	179.22	oil	1.067 ^{20/4}		152 ¹⁵	i.		
-dimethylaniline (o-)	NO ₂ ·C ₆ N ₂ N(CH ₃) ₂	166.18	yel. oil	1.179 ^{20/4}		151–3 ⁸⁰	v. sl. s.	v. s.	v. s.
(m-)	NO ₂ ·C ₆ H ₄ N(CH ₃) ₂	166.18	red mn.	1.313 ¹⁷	60–1	280–5	i.	s.	s.
(p-)	NO ₂ ·C ₆ H ₄ N(CH ₃) ₂	166.18	yel. nd.		163–4		i.	s. h.	
-diphenyl (o-)	C ₆ H ₅ ·C ₆ H ₄ NO ₂	199.21	rhb.	1.44	37	320	i.	s.	v. s.
(p-)	C ₆ H ₅ ·C ₆ H ₄ NO ₂	199.21	nd./al.		113–4	340	i.	sl. s. c.	v. s.
-diphenylamine (o-)	C ₆ H ₅ ·NH·C ₆ H ₄ NO ₂	214.22	or. cr.		75–6				
-guanidine	H ₂ NC(NH)NHNO ₂	104.07	nd./aq.		246–7		9 ¹⁰⁰	sl. s.	v. sl. s.

-naphthalene (α -)	$C_{10}H_7NO_2$	173.17	yel/al.	1.223 ⁶²	59-60	304	i.	s.	s.
(β -)	$C_{10}H_7NO_2$	173.17	col/al.		79	165 ¹⁵	i.	v. s.	v. s.
-phenol (o -)	$NO_2 \cdot C_6H_4 \cdot OH$	139.11	yel. mn.	1.295 ⁴⁵	44-5	214.5	1.08 ¹⁰⁰	v. s.	v. s.
(m -)	$NO_2 \cdot C_6H_4 \cdot OH$	139.11	col. mn.	1.485 ³⁰	96-7	194 ⁴⁰	1.35 ²⁰	v. s.	s.
(p -)	$NO_2 \cdot C_6H_4 \cdot OH$	139.11	yel. pr.	1.479 ³⁰	113-4	subl.	1.6 ²⁵	v. s.	v. s.
-phenol sulfonic acid (1-,4-,2-)	$HO \cdot C_6H_3(NO_2)SO_3H \cdot 3H_2O$	273.22	nd.		d. 110		v. s.	v. s.	sl. s.
(1-,2-,4-)	$HO \cdot C_6H_3(NO_2)SO_3H \cdot 3H_2O$	273.22	nd./aq.		51.5		v. s.	v. s.	
-phthalic acid (3-)	$NO_2 \cdot C_6H_3(CO_2H)_2$	211.13	yel/aq.		222		2.05 ²⁵	v. s. h.	sl. s.
(4-)	$NO_2 \cdot C_6H_3(CO_2H)_2$	211.13	yel. cr.		164-5		v. s.	v. s.	s.
-toluene (o -)	$CH_3 \cdot C_6H_4NO_2$	137.14	yel. lq.	1.163 ^{20/4}	-4.1	222.3	0.07 ⁸⁰	∞	∞
(m -)	$CH_3 \cdot C_6H_4NO_2$	137.14	lq.	1.160 ^{18/4}	15-16	230-1	0.05 ⁸⁰	∞	∞
(p -)	$CH_3 \cdot C_6H_4NO_2$	137.14	rhb.	1.139 ^{95/55}	51.9	237.7	0.04 ⁸⁰	8.6 ¹⁵	80.8 ¹⁵
-toluene sulfonic acid (1-,4-,2-)	$CH_3 \cdot C_6H_3(NO_2)SO_3H \cdot 2H_2O$	253.23	pl/aq.		130		47.7 ²⁸	v. s.	v. s.
-toluidine (4-,1-,2-)	$NO_2 \cdot C_6H_3(CH_3)NH_2$	152.15	yel. mn.	1.365 ¹⁵	105-7		v. sl. s.	s.	s.
(3-,1-,4-)	$NO_2 \cdot C_6H_3(CH_3)NH_2$	152.15	red mn.	1.312 ¹⁷	116-7		sl. s. h.	s.	
Nitron	$C_{20}H_{16}N_4$	312.37	yel. lf.		189-90 d.		i.	s. h.	v. sl. s.
Nitroso-dimethylaniline (p -)	$ON \cdot C_6H_4N(CH_3)_2$	150.18	gn. tri.		86-7		i.	s.	s.
-naphthol (β -)(1-)	$ON \cdot C_{10}H_6OH$	173.17	brn. pr.		109.5		0.1 ²⁰	s.	
Nonadecane (n -)	$CH_3(CH_2)_{17}CH_3$	268.52	cr.	0.777 ^{32/4}	32	330	i.	sl. s.	s.
Nonane (n -)	$CH_3(CH_2)_7CH_3$	128.26	col. lq.	0.718 ^{20/4}	-53.7	150.5 ⁷⁵⁰	i.	sl. s.	v. s.
Octadecane (n -)	$CH_3(CH_2)_{16}CH_3$	254.49	cr.	0.775 ^{25/4}	28	317	i.	sl. s.	s.
Octane (n -)	$CH_3(CH_2)_6CH_3$	114.23	col. lq.	0.703 ^{20/4}	-56.5	125.7	0.002 ¹⁶	sl. s.	s.
(iso-)	$(CH_3)_2CCH_2CH(CH_3)_2$	114.23	col. lq.	0.692 ^{20/4}	-107.4	99.3 ⁷⁶⁰	i.	sl. s.	s.
Octyl acetate (n -)	$CH_3CO_2CH_2(CH_2)_6CH_3$	172.26	col. lq.	0.885 ^{9/4}	-35.5	210	i.	s.	s.
(sec-)	$CH_3CO_2CH(CH_3)(CH_2)_6CH_3$	172.26	col. lq.	0.863 ^{14/4}		195	i.	s.	s.
alcohol (n -)	$CH_3(CH_2)_6CH_2OH$	130.23	col. lq.	0.827 ^{27/20/4}	-16	194-5	0.054 ²⁵	∞	∞
(sec-)	$CH_3(CH_2)_5CH(OH)CH_3$	130.23	col. lq.	0.822 ^{20/4}	-35.6	179-80	0.096 ²⁵	∞	∞
Octylene (n -)	$CH_3(CH_2)_6CH=CH_2$	112.21	lq.	0.721 ^{18/4}		126	i.	∞	∞
Oleic acid	$C_{18}H_{33}CH=CH(CH_2)_7CO_2H$	282.46	col. nd.	0.854 ^{78/4}	14	285-6 ¹⁰⁰	i.	∞	∞
Orcinol (1-,3-,5-)	$(HO)_3C_6H_3 \cdot CH_3$	124.14	pr./bz.	1.290 ⁴	107-8	287-90	v. s.	v. s.	v. s.
Oxalic acid	$HO_2C \cdot CO_2H \cdot 2H_2O$	126.07	col. mn.	1.653 ^{19/4}	101.5	subl.	s.	s.	1.3
Palmitic acid	$CH_3(CH_2)_{14}CO_2H$	256.42	col. pl.	0.849 ^{70/4}	63-4	271.5 ¹⁰⁰	i.	9 ³⁰	s.
Pelargonic acid	$CH_3(CH_2)_7CO_2H$	158.24	col. oil	0.906 ^{20/4}	12.5	253-4	v. sl. s.	s.	s.
Penta-chloroethane	$CHCl_2 \cdot CCl_3$	202.29	col. lq.	1.671 ^{25/4}	-22	162	0.05 ²⁰	∞	∞
-decane (n -)	$CH_3(CH_2)_{13}CH_3$	212.41	col. lq.	0.770 ^{20/4}	10	270.5	i.	v. s.	v. s.
-erythritol	$C(CH_2OH)_4$	136.15	cr.		262	276 ³⁰	5.6 ¹⁵	v. sl. s.	i.
Pentandiol	$HOCH_2(CH_2)_3CH_2OH$	104.15	lq.	0.994 ^{20/4}		239.4	∞	∞	∞
Pentane (n -)	$CH_3(CH_2)_3CH_3$	72.15	col. lq.	0.630 ^{18/4}	-129.7	36.3	0.036 ¹⁶	∞	∞
(i -)	$(CH_3)_2CHCH_2CH_3$	72.15	col. lq.	0.621 ¹⁹	-160.0	27.95	i.	∞	∞
(neo-)	$(CH_3)_3CC(CH_3)_2$	72.15	col. lq.	0.613 ^{20/4}	-20	9.5	i.	s.	s.
Phenacetin	$C_9H_9OC_2H_5NHCOCCH_3$	179.22	col. mn.		134-5	d.	0.7 ²⁰	40 h.	1.6 ²⁵
Phenanthrene	$< (C_6H_4)CH_2 >$	178.23	pl/al.	1.179 ²⁵	99-100	340	i.	10 h.	v. s.
Phenetidine (o -)	$C_8H_9O \cdot C_6H_4 \cdot NH_2$	137.18	oil		<-21	228-9	i.	s.	s.
(p -)	$C_8H_9O \cdot C_6H_4 \cdot NH_2$	137.18	lq.	1.061 ¹⁵	3-4	254-5	i.	s.	s.
Phenetole	$C_8H_9O \cdot C_6H_5$	122.16	col. lq.	0.967 ^{20/4}	-30.2	172	i.	∞	∞
Phenol	C_6H_5OH	94.11	col. nd.	1.071 ^{25/4}	42-3	181.4	8.2 ¹⁵	∞	∞
-phthalein	$C_{20}H_{14}O_4$	318.32	col. rhb.	1.299 ^{25/4}	261-2		0.2 ²⁰	10 ²⁵	5.9 c.
-sulfonic acid (o -)	$HO \cdot C_6H_4SO_3H \cdot 3/4H_2O$	187.69	cr.		50 d.		v. s.	v. s.	
Phenyl acetaldehyde	$C_6H_5CH_2CHO$	120.15	lq.	1.025 ³⁰		193-4	v. sl. s.	∞	∞
acetic acid	$C_6H_5CH_2CO_2H$	136.15	lf.	1.081 ^{30/4}	76-7	265.5	1.66 ²⁰	v. s.	v. s.
-acetylene	$C_6H_5C \cdot CH$	102.13	col. lq.	0.930 ^{20/4}	-43	142-3	i.	∞	∞
aniline (o -)	$C_6H_5 \cdot C_6H_4 \cdot NH_2$	169.22	cr.		45-6	299 ⁷⁶⁰	v. sl. s.	s.	s.
(p -)	$C_6H_5 \cdot C_6H_4 \cdot NH_2$	169.22	lf.		50-2	302	s. h.	s.	s.
Phenyl-ethyl alcohol	$C_6H_5CH_2CH_2OH$	122.16	col. oil	1.023 ^{18/4}		219-21 ⁷⁵⁰	1.6 ²⁰	s.	sl. s.
-glycine	$C_6H_5NHCH_2CO_2H$	151.16	cr.		127		s.	s.	∞
-hydrazine	$C_6H_5NH \cdot NH_2$	108.14	yel. oil	1.097 ^{23/4}	19.6	243.5	sl. s. h.	∞	∞
-hydrazine sulfonic acid (p -)	$H_2NNHC_6H_4SO_3H$	188.20	cr/al.		286		0.6 ¹²	sl. s.	
isocyanate	$C_6H_5N \cdot CO$	119.12	lq.	1.096 ^{20/4}		166 ⁷⁶⁰	d.	d.	v. s.
-methylpyrazolone (3-)(N -)	$C_8H_5ON_2 \cdot C_6H_5$	174.20	pr/aq.		128	191 ¹⁷	1 ²⁰	v. s. h.	v. sl. s.
-mustard oil	$C_6H_5N \cdot CS$	135.19	col. lq.	1.138 ^{15/15}	-21	219-20	i.	s.	s.
naphthalene (α -)	$C_{10}H_7 \cdot C_6H_5$	204.27	waxy		45	336-7	i.	v. s.	v. s.
(β -)	$C_{10}H_7 \cdot C_6H_5$	204.27	lf/al.		102.5	345-6	i.	sl. s.	sl. s.
naphthylamine (α -)	$C_{10}H_7NHC_6H_5$	219.28	pr/al.	1.17	62	335 ²⁵⁸	0.08 ⁶⁰	s.	s.
(β -)	$C_{10}H_7NHC_6H_5$	219.28	rhb.	1.18	107-8	399.5	0.4 ⁶⁰	v. s. h.	v. s. h.
phenol (o -)	$C_6H_5 \cdot C_6H_4OH$	170.21	nd.		56-7	275	i.	s.	s.
(p -)	$C_6H_5 \cdot C_6H_4OH$	170.21	nd.		164-5	305-8	i.	s.	s.
propyl alcohol (γ -)	$C_6H_5(CH_2)_3OH$	136.19	oil	1.008 ^{20/4}	<-18	235-7	sl. s.	∞	∞
quinoline (2-)(α -)	$C_6H_5 \cdot C_6H_6N$	205.25	nd.		86	363	sl. s.	s. h.	s.
(8-)(0-)	$C_6H_5 \cdot C_6H_6N$	205.25	lq.			283 ¹⁸⁷	sl. s.	s.	s.
salicylate, salol	$HO \cdot C_6H_4CO_2C_6H_5$	214.22	rhb/al.	1.250 ^{20/4}	42-3	172-3 ¹²	0.015 ²⁵	v. s.	s.
stearate	$CH_3(CH_2)_{16}CO_2C_6H_5$	360.57	cr.		52	267 ¹⁵	i.		
urethane	$C_6H_5NHCOC_2H_5$	165.19	pl/al.	1.106 ^{20/4}	52-3	237-8	i. c.	s.	s.

TABLE 2-2 Physical Properties of Organic Compounds (Continued)

Name	Formula	Formula weight	Form and color	Specific gravity	Melting point, °C	Boiling point, °C	Solubility in 100 parts		
							Water	Alcohol	Ether
Phenylene-diamine (<i>o</i> -)	$C_6H_4(NH_2)_2$	108.14	lf./aq.		103–4	256–8	733 ⁸¹	v. s.	v. s.
(<i>m</i> -)	$C_6H_4(NH_2)_2$	108.14	rhb.	1.139 ^{15/15}	62.8	284–7	35.1 ²⁵	v. s.	s.
(<i>p</i> -)	$C_6H_4(NH_2)_2$	108.14	mn.		140	267	669 ¹⁰⁷	s.	s.
Phloroglucinol (1-,3-,5-)	$C_6H_3(OH)_3 \cdot 2H_2O$	162.14	rhb.		117	subl.	1.13 ²⁵	v. s.	v. s.
Phorone	$[(CH_3)_2C \cdot CH]_2CO$	138.21	yel. pr.	0.885 ^{20/4}	28	197.2 ⁷⁴³	0.1 ³⁰	s.	s.
Phosgene	$OCCl_2$	98.92	gas	1.392 ^{19/4}	–104	8.2 ⁷⁵⁶	v. sl. s.		
Phthalic acid (<i>o</i> -)	$C_6H_4(CO_2H)_2$	166.13	mn./aq.	1.593 ^{20/4}	208	d.	0.70 ²⁵	12 ¹⁸	0.68 ¹⁵
(<i>m</i> -)(iso-)	$C_6H_4(CO_2H)_2$	166.13	nd./aq.		330	subl.	0.2 ¹⁰⁰	s.	
anhydride (<i>o</i> -)	$C_6H_4 < (CO)_2 > O$	148.12	rhb.	1.527 ⁴	130.8	284.5	v. sl. s.	s.	sl. s.
nitride (<i>o</i> -)	$C_6H_4(CN)_2$	128.13	cr.		141		sl. s. c.		
Phthalide	$C_6H_4(CH_2)(CO) > O$	134.13	nd./aq.	1.164 ^{99/4}	73(65)	290	v. sl. s.	s.	
Phthalimide (<i>o</i> -)	$C_6H_4 < (CO)_2 > NH$	147.13	cr./et.		238	subl.	0.04 ²⁵	5	s. h.
Picoline (α -)	$C_5H_4N \cdot CH_3$	93.13	col. lq.	0.950 ^{15/4}	–70	128.8	v. s.	∞	∞
(β -)	$C_5H_4N \cdot CH_3$	93.13	col. lq.	0.961 ^{15/4}		143.5	∞	∞	∞
(γ -)	$C_5H_4N \cdot CH_3$	93.13	lq.	0.957 ^{15/4}		143.1	∞	∞	∞
Picramic acid (1-,2-,4-,6-)	$HO \cdot C_6H_3(NH_2)(NO_2)_2$	199.12	red nd.		169		0.14 ²²	s.	sl. s.
Picric acid (2-,4-,6-)	$HO \cdot C_6H_3(NO_2)_3$	229.10	yel. rhb.	1.763 ^{20/4}	121.8	expl.	1.23 ²⁰	6 ²⁰	1 ¹³
Picryl chloride (2-,4-,6-)	$ClC_6H_2(NO_2)_3$	247.55	yel. mn.	1.797 ²⁰	83	d.	0.018 ¹⁵	4.8 ¹⁷	7 ¹⁷
Pinacol	$[(CH_3)_2C \cdot OH]_2$	118.17	col. nd.	0.967 ¹⁵	43(38)	171–2 ⁷⁶⁹	sl. s. c.	v. s.	v. s.
Pinacolone	$CH_3COC(CH_3)_3$	100.16	col. lq.	0.800 ¹⁶	–52.5	106.2	2.5 ¹⁵	s.	s.
Pinene (α -)(<i>dl</i> -)	$C_{10}H_{16}$	136.23	col. lq.	0.878 ^{20/4}	–55	154–6	v. sl. s.	s.	∞
hydrochloride	$C_{10}H_{17}Cl$	172.69	lf.		131–2	207–8	i.	33	s.
Pinol (<i>dl</i> -)	$C_{10}H_{16}O$	152.23	lq.	0.953 ^{20/20}		183–4	s.	s.	s.
Piperidine	$CH_2 < (CH_2CH_2)_2 > NH$	85.15	lq.	0.860 ^{20/4}	–9	106	∞	∞	
carboxylic acid (α -)(<i>dl</i> -)	$HO_2C \cdot CH < (CH_2CH_2)_2 > NH$	129.16	cr.		264		s.		
Piperidinium pentamethylene dithiocarbamate	$(CH_2)_5CS_2H \cdot HN(CH_2)_5$	232.43	cr.	1.13	175		6 ²⁸		
Propane	$CH_3CH_2CH_3$	44.10	gas	0.585 ^{–45/4}	–187.1	–42.2	6.5 ¹⁸ cc.	s.	v. s.
Propionic acid	$CH_3CH_2CO_2H$	74.08	col. lq.	0.992 ^{20/4}	–22	141.1	∞	∞	∞
aldehyde	CH_3CH_2CHO	58.08	col. lq.	0.807 ^{20/4}	–81	49.5 ⁷⁴⁰	20 ³⁰	∞	∞
anhydride	$(CH_3CH_2CO)_2O$	130.14	col. lq.	1.012 ^{20/4}	–45	168.8 ⁷⁸⁰	d.	d.	
Propyl acetate (<i>n</i> -)	$CH_3CO_2CH_2CH_2CH_3$	102.13	col. lq.	0.886 ^{20/4}	–92.5	101.6	1.6 ¹⁶	∞	∞
(<i>i</i> -)	$CH_3CO_2CH(CH_3)_2$	102.13	col. lq.	0.874 ^{20/20}	–73.4	88.4	3 ²⁰	∞	∞
alcohol (<i>n</i> -)	$CH_3CH_2CH_2OH$	60.10	col. lq.	0.804 ^{20/4}	–127	97.8	∞	∞	∞
(<i>i</i> -)	$(CH_3)_2CHOH$	60.10	col. lq.	0.789 ^{20/4}	–85.8	82.5	∞	∞	∞
amine (<i>n</i> -)	$CH_3CH_2CH_2NH_2$	59.11	col. lq.	0.718 ^{20/20}	–83	49–50 ⁷⁶¹	∞	∞	∞
(<i>i</i> -)	$(CH_3)_2CHNH_2$	59.11	col. lq.	0.694 ^{15/4}	–101	33–4	∞	∞	∞
aniline (<i>n</i> -)	$C_6H_5NHCH_2CH_2CH_3$	135.21	lq.	0.949 ¹⁸	222	i.	v. s.	v. s.	v. s.
benzoate (<i>n</i> -)	$C_6H_5CO_2CH_2CH_2CH_3$	164.20	col. lq.	1.021 ^{25/25}	–51.6	231	i.	s.	s.
(<i>i</i> -)	$C_6H_5CO_2CH(CH_3)_2$	164.20	col. lq.	1.010 ^{25/25}		218.5	i.	s.	s.
bromide (<i>n</i> -)	$CH_3CH_2CH_2Br$	122.99	col. lq.	1.353 ^{20/4}	–109.9	70.8	0.25 ²⁰	∞	∞
(<i>i</i> -)	$(CH_3)_2CHBr$	122.99	col. lq.	1.310 ^{20/4}	–89	60	0.32 ²⁰	∞	∞
<i>n</i> -butyrate (<i>n</i> -)	$C_3H_7CH_2CO_2CH_2C_2H_5$	130.18	col. lq.	0.879 ¹⁵	–95.2	142.7	0.17 ¹⁷	∞	∞
<i>i</i> -butyrate (<i>n</i> -)	$(CH_3)_2CHCO_2CH_2C_2H_5$	130.18	col. lq.	0.884 ^{9/4}		134–5	v. sl. s.		
<i>n</i> -butyrate (<i>i</i> -)	$C_3H_7CH_2CO_2CH(CH_3)_2$	130.18	col. lq.	0.865 ¹⁵		128	v. sl. s.		
<i>i</i> -butyrate (<i>i</i> -)	$(CH_3)_2CHCO_2CH(CH_3)_2$	130.18	col. lq.	0.869 ^{9/4}		120.8	v. sl. s.		
chloride (<i>n</i> -)	$CH_3CH_2CH_2Cl$	78.54	col. lq.	0.890 ^{20/4}	–122.8	46.4	0.27 ²⁰	∞	∞
(<i>i</i> -)	$(CH_3)_2CHCl$	78.54	col. lq.	0.859 ²⁰	–117	36.5	0.31 ²⁰	∞	∞
Propyl formate (<i>n</i> -)	$HCO_2CH_2CH_2CH_3$	88.11	col. lq.	0.901 ^{20/4}	–92.9	81.3	12.2 ²²	∞	∞
(<i>i</i> -)	$HCO_2CH(CH_3)_2$	88.11	col. lq.	0.873 ^{20/4}		68–71 ⁷⁵¹	2.1 ²²	∞	∞
furoate (<i>n</i> -)	$C_4H_5O \cdot CO_2C_3H_7$	154.16	col. lq.	1.075 ^{26/4}		211	v. sl. s.	s.	∞
lactate (<i>n</i> -)	$CH_3CH(OH)CO_2CH_2C_2H_5$	132.16	col. lq.			122–3 ¹⁵⁰	s.	s.	s.
(<i>i</i> -)	$CH_3CH(OH)CO_2CH(CH_3)_2$	132.16	col. lq.			167.5	s.	s.	s.
mercaptopan (<i>n</i> -)	$CH_3CH_2CH_2SH$	76.16	lq.	0.836 ^{25/4}	–112	67–8	v. sl. s.	s.	s.
(<i>i</i> -)	$(CH_3)_2CHSH$	76.16	lq.	0.809 ^{25/4}	–130.7	58–60	v. sl. s.	∞	∞
propionate (<i>n</i> -)	$C_3H_7CO_2CH_2C_2H_5$	116.16	col. lq.	0.883 ^{20/4}	–76	122–3	0.56 ²⁵	∞	∞
(<i>i</i> -)	$C_3H_7CO_2CH(CH_3)_2$	116.16	col. lq.	0.893 ⁹		109–11 ⁷⁵⁰	0.6 ²⁵	∞	∞
thiocyanate (<i>i</i> -)	$(CH_3)_2CH \cdot CNS$	101.17	lq.	0.963 ²⁰		152–3 ⁷⁵⁴	i.	∞	∞
<i>n</i> -valerate (<i>n</i> -)	$CH_3(CH_2)_3CO_2CH_2C_2H_5$	144.21	lq.	0.874 ¹⁵	–70.7	67.5	i.	∞	∞
<i>i</i> -valerate (<i>n</i> -)	$(CH_3)_2CHCH_2CO_2C_2H_5$	144.21	col. lq.	0.863 ^{20/4}		155.9	i.	∞	∞
<i>i</i> -valerate (<i>i</i> -)	$(CH_3)_2CHCH_2CO_2CH_2C_2H_5$	144.21	col. lq.	0.854 ¹⁷		142 ⁷⁵⁶			
Propylene	$CH_3CH=CH_2$	42.08	gas	0.609 ^{–47/4}	–185	–48 ⁷⁴⁹	44.6 cc.	1200 cc.	
bromide	$CH_3CHBrCH_2Br$	201.89	col. lq.	1.933 ^{20/4}	–55.5	141.6	0.25 ²⁰	s.	v. s.
chlorohydrin	$CH_3CHClCH_2OH$	94.54	col. lq.	1.103 ²⁰		133–4	s.	s.	s.
chloride	$CH_3CHClCH_2Cl$	112.99	col. lq.	1.159 ^{20/20}	<–70	96.8	0.27 ²⁰	v. s.	v. s.
glycol	$CH_3CH(OH)CH_2OH$	76.09	col. oil	1.040 ^{19/4}		188–9	∞	∞	8
oxide	$CH_3(CHCH_2)_2O$	58.08	col. lq.	0.831 ^{20/20}		35	33 ²⁰	∞	∞
Protocatechuic acid (3-,4-)	$(HO)_2C_6H_3CO_2H \cdot H_2O$	172.14	nd./aq.	1.542 ^{1/4}	199 d.		1.82 ¹⁴	v. s.	s.

Pulegol (iso-)(<i>d</i> -)	C ₁₀ H ₁₇ OH	154.25	col. lq.	0.911 ^{20/4}		86-9 ¹⁰	v. sl. s.		
Pulegone	C ₁₀ H ₁₆ O	152.23	col. lq.	0.932 ^{20/20}		224 ⁷⁵⁴	i.	∞	∞
Pyrazole	—NH·N:CH·CH·CH—	68.08	nd./et.		70	186-8	s.	s.	s.
Pyrazoline	—NH·N:CH·CH ₂ CH ₂ —	70.09	lq.			144	∞	∞	sl. s.
Pyrazolone	—NH·CO·CH ₂ CH ₂ N—	84.08	nd.			165	subl. d.	v. s.	v. sl. s.
Pyrene	C ₁₆ H ₁₀	202.25	vel. pr.	1.277 ^{70/4}	149-50	>360	i.	3 h.	v. s.
Pyridazine	N ₂ < (CHCH) ₂ >	80.09	lq.	1.107 ^{20/4}	-8	208	∞		s.
Pyridine	CH < (CHCH) ₂ > N	79.10	col. lq.	0.982 ^{20/4}	-42	115-6	∞	∞	s.
Pyrocatechol (<i>o</i> -)	C ₆ H ₄ (OH) ₂	110.11	nd./aq.	1.344 ⁴	104-5	240-5	45.1 ²⁰	v. s.	v. s.
Pyrogallol (1-,2-,3-)	C ₆ H ₃ (OH) ₃	126.11	nd.	1.453 ⁴	133-4	309	40 ¹³	s.	s.
Pyrone	CO < (CHCH) ₂ > O	96.08	cr.	1.190 ^{40,3}	32.5	215-7	v. sl. s.	s.	v. s.
Pyrrrole	< (CH·CH) ₂ > NH	67.09	lq.	0.948 ^{20/4}		131	i.	s.	s.
Pyrrolidine	< (CH ₂ ·CH ₂) ₂ > NH	71.12	lq.	0.852 ^{22,5}		87-8	∞	∞	∞
Pyroline	< (CH·CH ₂) ₂ > NH	69.11	lq.	0.910 ^{20/4}		90-1	v. s.	∞	∞
Pyruvic acid	CH ₃ COCO ₂ H	88.06	col. lq.	1.267 ^{20/4}	13.6	165	∞	∞	∞
Quercitrin	C ₂₁ H ₃₀ O ₁₁ ·5H ₂ O	484.41	vel. nd.		182-5		0.04 ²⁰	s.	sl. s.
Quinaldine (<i>py</i> -2)	CH ₃ ·C ₉ H ₆ N	143.19	lq.	1.059 ^{20/4}	-1	244-5 ⁷⁵⁰	v. sl. s.		s.
Quinoline	C ₉ H ₇ N	129.16	lq.	1.095 ²⁰	-15	237.1 ⁷⁴⁷	6	∞	∞
(iso-)	C ₉ H ₇ N	129.16	pl.	1.099 ^{21/4}	24.6	240.5 ⁷⁶³	sl. s.		s.
-diol (1-,3-)	—C ₆ H ₄ CH·C(OH)(N):C(OH)—	161.16	cr.		237		v. sl. s.		
Quinone (<i>p</i> -)	CO < (CHCH) ₂ > CO	108.09	yel. mn.	1.318 ^{20/4}	115.7	subl.	sl. s. h.	s.	s.
R-acid Ca salt (2-)(3-,6-)	HOC ₁₀ H ₄ (SO ₃) ₂ Ca	342.36	cr.				30.6 ²⁵		
K salt	HOC ₁₀ H ₄ (SO ₃ K) ₂	380.48	cr.				29.5 ²⁵		
Na salt	HOC ₁₀ H ₄ (SO ₃ Na) ₂	348.26	cr.				25.2 ²⁵		
Raffinose	C ₁₈ H ₃₂ O ₁₆ ·5H ₂ O	594.51	cr./aq.	1.465 ⁹	119	d. 130	14.3 ³⁰	0.1 ³⁰	
Resorcinol (<i>m</i> -)	C ₆ H ₄ (OH) ₂	110.11	col. rhb.	1.272 ¹⁵	110.7	276.5	147 ¹²		v. s.
Retene	C ₁₈ H ₁₈	234.34	lf./al.	1.13 ¹⁶	98-9	390-4	i.	69 h.	v. s. h.
Rhamnose (β-)	CH ₃ (CHOH) ₄ CHO·H ₂ O	182.17	col. mn.	1.471 ^{20/4}	126		60.8 ²¹		i.
Ricinoleic acid	C ₁₇ H ₃₂ (OH)CO ₂ H	298.46	lq.	0.954 ¹⁶	4-5	226-8 ¹⁰	i.	∞	∞
Rosaniline	C ₂₀ H ₂₁ ON ₃	319.40	col. nd.			186 d.	v. sl. s.	sl. s.	i.
Rosolic acid	C ₂₀ H ₁₆ O ₃	304.34	red lf.			308-10 d.	0.12 ²⁵	v. s. h.	sl. s.
Saccharin	C ₆ H ₄ (CO)(SO ₂) > NH	183.18	mn.			225-8	0.4 ²⁵	3.1 c.	1.05 c.
Safrole (1-,3-,4-)	CH ₃ ·CHCH ₃ ·C ₆ H ₃ O ₂ CH ₂	162.19	col. mn.	1.100 ^{20/4}	11.2	subl.	233-4		
(iso-)(1-,3-,4-)	CH ₃ ·CH·CH·C ₆ H ₃ O ₂ CH ₂	162.19	col. lq.	1.122 ^{20/4}	6-7		252-3		
Salicylic acid (<i>o</i> -)	HO·C ₆ H ₄ ·CO ₂ H	138.12	mn.	1.443 ^{20/4}	159	211 ³⁰	0.2 ²³	49 ¹⁵	51 ¹⁵
aldehyde (<i>o</i> -)	HO·C ₆ H ₄ ·CHO	122.12	col. oil	1.153 ^{25/4}	7	196.5	1.7 ⁹⁶	∞	∞
Saligenin	HO·C ₆ H ₄ ·CH ₂ OH	124.14	rhb./aq.	1.161 ²⁵	86-7	subl.	6.6 ¹⁵	v. s.	v. s.
Schaeffer's salt, Ca	(HOC ₁₀ H ₆ SO ₃) ₂ Ca·5H ₂ O	576.60	cr.				4.76 ²⁰		
K	HOC ₁₀ H ₆ SO ₃ K	262.32	cr.				3.46 ²⁵		
Na	HOC ₁₀ H ₆ SO ₃ Na	246.21	cr.				6.29 ²⁵		
Semicarbazide	NH ₂ ·CO·NH·NH ₂	75.07	pr./al.		96		v. s.	v. s.	i.
hydrochloride	NH ₂ ·CO·NH·NH ₂ Cl	111.53	pr.		173 d.		v. s.	sl. s.	i.
Skatole (3-)	CH ₃ ·C ₆ H ₆ N	131.17	lf.		95	265-6 ⁷⁵⁵	0.05 c.	s.	s.
Sodium methylate	CH ₃ ONa	54.02	pd.		d. 300		d.		
Sorbitol	[CH ₂ OH(CHOH) ₂] ₂	182.17	cr.		110-2		v. s.	v. s. h.	
Sorbose (<i>d</i> - or <i>l</i> -)	C ₆ H ₁₂ O ₆	180.16	rhb.	1.654 ¹⁵	165		55 ¹⁷	sl. s.	
Starch	(C ₆ H ₁₀ O ₅) _x	162.14	amor.	1.50 ²¹	d.		i.	i.	i.
Stearic acid	CH ₃ (CH ₂) ₁₆ CO ₂ H	284.48	mn.	0.847 ^{69,3}	70-1	291 ¹¹⁰	0.03 ²⁵	2 ²⁰	6 ⁸
amide	CH ₃ (CH ₂) ₁₆ CONH ₂	283.49	col. cr.		108-9		i.	s. h.	s. h.
Styrene	C ₆ H ₅ CH=CH ₂	104.15	col. lq.	0.903 ^{20/4}	-31	145-6	v. sl. s.	∞	∞
Suberic acid	HO ₂ C(CH ₂) ₆ CO ₂ H	174.19	nd./aq.	1.266 ^{25/4}	140-4	279 ¹⁰⁰	0.14 ¹⁶	s.	0.8 ¹⁵
Succinic acid	HO ₂ C(CH ₂) ₂ CO ₂ H	118.09	col. mn.	1.572 ^{25/4}	189-90	235 d.	6.8 ²⁰	9.9 ¹⁵	1.2 ¹⁵
Sucrose	C ₁₂ H ₂₂ O ₁₁	342.30	col. mn.	1.588 ¹⁵	170-86 d.		179 ⁹	0.9	i.
Sulfanilic acid (<i>p</i> -)	H ₂ N·C ₆ H ₄ ·SO ₃ H	173.19	col. cr.		d. > 280		0.8 ¹⁰	v. sl. s.	v. sl. s.
Sylvestrene (<i>d</i> -)	C ₁₀ H ₁₆	136.23	lq.	0.863 ^{20/4}		176-7			
Tartaric acid (meso-)	(CHOHCO ₂ H) ₂	150.09	cr.	1.737	159-60		120 ¹⁵		
(racemic)	(CHOHCO ₂ H) ₂ ·H ₂ O	168.10	tri.	1.697 ^{20/4}	205-6		20.6 ²⁰	2 ⁹	0.09
(<i>d</i> - or <i>l</i> -)	(CHOHCO ₂ H) ₂	150.09	mn.	1.760 ^{20/4}	168-70	d.	139 ²⁰	25 ¹⁵	0.4 ¹⁵
Tartronic acid	CH(OH)(CO ₂ H) ₂ ·½H ₂ O	129.07	pr./aq.		d. 155-8	subl.	v. s.	v. s.	i.
Terephthalic acid (<i>p</i> -)	C ₆ H ₄ (CO ₂ H) ₂	166.13	cr.	1.510			0.001 c.	sl. s. h.	i.
Terpin hydrate (<i>cis</i> -)	C ₁₀ H ₂₀ O ₂ ·H ₂ O	190.28	rhb.		117	d.	0.4 ¹⁵	10 ¹⁵	1 ¹⁵
Terpineol (α-)(<i>d</i> - or <i>l</i> -)	C ₁₀ H ₁₈ O	154.25	col. cr.	0.935 ¹⁵	38-40	219-21	i.	v. s.	v. s.
(<i>dl</i> -)	C ₁₀ H ₁₈ O	154.25	col. cr.	0.935 ^{20/20}	35	218-9 ⁷⁵²	i.	v. s.	v. s.
Terpinyl acetate (α-)(<i>dl</i> -)	CH ₃ CO ₂ ·C ₁₀ H ₁₇	196.29	lq.	0.966 ^{20/4}	< -50	220 d.	i.	20	
Tetrabromo-ethane (sym)	Br ₂ CH·CHBr ₂	345.65	col. lq.	2.964 ^{20/4}	-1.0	151 ⁵⁴	i.	∞	∞
(uns)	Br ₂ C·CH ₂ Br	345.65	col. lq.	2.875 ^{20/4}	0	104 ¹³	i.	s.	
Tetrachloro-ethane (sym)	Cl ₂ CH·CHCl ₂	167.85	col. lq.	1.600 ^{20/4}	-36	146.3	0.29 ²⁰	∞	∞
(uns)	Cl ₂ C·CH ₂ Cl	167.85	lq.	1.588 ^{20/4}		129-30	i.	∞	∞
-ethylene	Cl ₂ C·CCl ₂	165.83	col. lq.	1.624 ^{15/4}	-19	120.8	0.02 ²⁰	∞	∞
Tetracosane (<i>n</i> -)	CH ₃ (CH ₂) ₂₂ CH ₃	338.65	cr.	0.779 ^{11/4}	51.1	324			s.
Tetradecane (<i>n</i> -)	CH ₃ (CH ₂) ₁₂ CH ₃	198.39	col. lq.	0.765 ^{20/4}	5.5	252.5	i.	v. s.	v. s.
Tetraethyl-thiuram disulfide	[(C ₂ H ₅) ₂ NCS] ₂ S ₂	296.54	cr.	1.17	70		i.		

TABLE 2-2 Physical Properties of Organic Compounds (Concluded)

Name	Formula	Formula weight	Form and color	Specific gravity	Melting point, °C	Boiling point, °C	Solubility in 100 parts		
							Water	Alcohol	Ether
Tetrafluoro-ethylene	$F_2C:CF_2$	100.02	gas	1.58 ⁻⁷⁸	-142.5	-76.3	0.01 ³⁰		
Tetrahydro-furan	$—CH_2(CH_2)_3CH_2O—$	72.11	col. lq.	0.888 ^{21/4}	-65	65-6	s.	s.	s.
-furfuryl alcohol	$C_4H_3O:CH_2OH$	102.13	col. lq.	1.050 ^{20/4}		177-8 ⁷⁴³	∞	∞	∞
-pyran	$—CH_2(CH_2)_3CH_2O—$	86.13	lq.	0.881 ^{20/4}		88	s.		
Tetralin	$—C_{10}H_7CH_2(CH_2)_2CH_2—$	132.20	col. lq.	0.973 ^{15/4}	-31	206 ⁷⁶⁴	i.	s.	s.
Tetramethyl-thiuram disulfide	$[(CH_3)_2NCS]_2S_2$	240.43	cr.	1.29	155-6		i.		
Tetryl (2-,4-,6-)	$(NO_2)_3C_6H_2:N(CH_3)NO_2$	287.14	yel. mn.	1.57 ¹⁹	130.5	expl.	i.	s. h.	s.
Theobromine	$C_7H_5O_3N_4$	180.16	rhb.		330		0.06 ¹⁵	0.06 c.	0.03 h.
Thio-acetic acid	$CH_3\cdot CO\cdot SH$	76.12	yel. lq.	1.074 ¹⁰	< -17	93	s.	∞	∞
-aniline (4-, 4'-)	$(NH_2\cdot C_6H_4)_2S$	216.30	nd./aq.		108		sl. s. h.	s.	s.
-carbanilide	$(C_6H_5\cdot NH)_2CS$	228.31	rhb./al.	1.3 ²⁴	154	d.	i.	v. s.	v. s.
-naphthol (β-)	$C_{10}H_7\cdot SH$	160.24	cr./al.		81	286-8	v. sl. s.	v. s.	v. s.
-phenol	$C_6H_5\cdot SH$	110.18	col. lq.	1.074 ^{23/4}		168-9	v. sl. s.	v. s.	∞
-salicylic acid (o-)	$HS\cdot C_6H_4\cdot CO_2H$	154.19	yel. nd.		164	subl.	sl. s. h.	s.	
-urea	$NH_2\cdot CS\cdot NH_2$	76.12	rhb./al.	1.405 ^{20/4}	180-2	d.	9.2 ¹³	s.	sl. s.
Thiophene	$<(CH\cdot CH)_2>S$	84.14	col. lq.	1.070 ^{15/4}	-30	84	i.	s.	
Thymol (5-,2-,1-)	$(CH_3)(C_3H_7)C_6H_3OH$	150.22	cr.	0.972 ^{25/5}	51.5	232 ⁷⁵²	0.09 ¹⁹	v. s.	v. s.
Tolidine (0-)/(3-,3',-4-,4'-)	$[CH_3(NH_2)C_6H_3]_2$	212.29	lf.		128-9		v. sl. s.	s.	v. s.
Toluene	$C_6H_5\cdot CH_3$	92.14	col. lq.	0.866 ^{20/4}	-95	110.8	0.05 ¹⁶	s.	∞
sulfonic acid (o-)	$CH_3\cdot C_6H_4SO_3H\cdot 2H_2O$	208.23	cr.		d.	128.8 ⁰	v. s.	s.	
(p-)	$CH_3\cdot C_6H_4SO_3H\cdot H_2O$	190.22	mn.		104-5		v. s.	s.	
sulfonic amide (p-)	$CH_3\cdot C_6H_4SO_2NH_2$	171.22	mn.		137	146-7 ⁰	0.2 ⁹	7.4 ⁵	
sulfonic chloride (p-)	$CH_3\cdot C_6H_4\cdot SO_2Cl$	190.65	tri.		69	134.5 ¹⁰	i.	s.	s.
Toluic acid (o-)	$CH_3\cdot C_6H_4\cdot CO_2H$	136.15	cr./aq.	1.062 ^{115/4}	104-5	259 ⁷⁵¹	v. s.	2.17 ¹⁰⁰	
(m-)	$CH_3\cdot C_6H_4\cdot CO_2H$	136.15	pr./aq.	1.054 ^{12/4}	110-1	263	1.6 ¹⁰⁰	v. s.	v. s.
(p-)	$CH_3\cdot C_6H_4\cdot CO_2H$	136.15	cr./aq.		179-80	274-5	1.3 ¹⁰⁰	v. s.	v. s.
Toluidine (o-)	$CH_3\cdot C_6H_4\cdot NH_2$	107.15	col. lq.	0.999 ^{20/4}	-16.3	199.7	1.5 ²⁵	∞	∞
(m-)	$CH_3\cdot C_6H_4\cdot NH_2$	107.15	col. lq.	0.989 ^{20/4}	-31.5	203.3	sl. s.	∞	∞
(p-)	$CH_3\cdot C_6H_4\cdot NH_2$	107.15	cr.	1.046 ^{20/4}	44-5	200.3	0.74 ²¹	v. s.	v. s.
hydrochloride (o-)	$CH_3\cdot C_6H_4\cdot NH_3Cl$	143.61	mn. pr.		218-20	242	s.	sl. s.	
sulfonic acid (1-,2-,3-)	$CH_3(NH_2)C_6H_3SO_3H$	187.22	cr.				0.97 ¹¹		
Tolylene diamine (1-,2-,4-)	$CH_3\cdot C_6H_3(NH_2)_2$	122.17	rhb.		99	283-5	s. h.	s.	s.
Tolylene diisocyanate (1-,2-,4-)	$CH_3\cdot C_6H_3(NCO)_2$	174.16	lq.	1.23 ²⁸		134.5 ²⁰	d.	d.	
Trehalose	$C_{12}H_{22}O_{11}\cdot 2H_2O$	378.33	rhb./al.		97		s. h.	sl. s. h.	i.
Triamylamine (n-)	$[CH_3(CH_2)_4CH_3]_3N$	227.43	lq.			240-5	i.		
(i-)	$[(CH_3)_2CH(CH_2)_2]_3N$	227.43	col. lq.	0.786 ^{20/4}		235	i.		
Tributyl-amine (n-)	$[CH_3(CH_2)_3CH_3]_3N$	185.35	col. lq.	0.778 ^{20/20}		216.5 ⁷⁶¹	i.	s.	∞
phosphite	$[CH_3(CH_2)_3O]_3P$	250.31	lq.	0.925 ^{20/4}		122-3 ¹²	i.		
Trichloro-acetic acid	$Cl_3C\cdot CO_2H$	163.39	cr.	1.617 ^{46/15}	58	195.5 ⁷⁵⁴	120 ²⁵	s.	s.
-benzene (s-)(1-,3-,5-)	$C_6H_2Cl_3$	181.45	nd.		63.5	208.5 ⁷⁶⁴	i.	sl. s.	
-ethane (1-,1-,1-)	$Cl_3C\cdot CH_3$	133.40	lq.	1.325 ^{26/4}		74.1	i.	∞	∞
-ethylene	$Cl_3C\cdot CHCl$	131.39	col. lq.	1.466 ^{29/20}	-73	87.2	0.1 ²⁵	∞	∞
-phenol	$Cl_3C\cdot H_2OH$	197.45	nd.	1.490 ^{75/4}	68-9	246	0.09 ²⁵	v. s.	v. s.
Tricosane (n-)	$CH_3(CH_2)_{21}CH_3$	324.63	lf.	0.779 ^{25/4}	47.7	234 ¹⁵	i.		
Tricresyl phosphate (o-)	$OP(OC_6H_4CH_3)_3$	368.36	lq.				i.		
Tridecane (n-)	$CH_3(CH_2)_{11}CH_3$	184.36	col. lq.	0.757 ^{20/4}	-6.2	234	i.	v. s.	v. s.
Triethanol amine	$(HOCH_2CH_2)_3N$	149.19	col. lq.	1.126 ^{20/20}	20-1	277-9 ¹⁵⁰	∞	∞	sl. s.
Triethyl-amine	$(CH_3CH_2)_3N$	101.19	col. oil	0.729 ^{20/20}	-114.8	89.4	∞ > 19 ⁰	∞	∞
-benzene (1-,3-,5-)	$(C_2H_5)_3C_6H_3$	162.27	lq.	0.861 ^{20/4}		215	i.	s.	s.
(1-,2-,4-)	$(C_2H_5)_3C_6H_3$	162.27	lq.	0.882 ^{17/4}		217-8 ⁷⁵⁵	i.	s.	s.
borate	$B(OCH_2CH_3)_3$	145.99	lq.	0.864 ^{20/20}		120	d.		
citrate	$HOC_2H_4(CO_2C_2H_5)_3$	276.28	oil	1.137 ^{20/4}		294	i.	∞	∞
Triethylene glycol	$(-CH_2OCH_2CH_2OH)_2$	150.17	col. lq.	1.125 ^{20/20}	-5	290	∞	∞	v. sl. s.
Trifluoro-chloromethane	CF_3Cl	104.46	gas	1.726 ⁻¹³⁰	-182	-80			
-chloroethylene	$F_3C\cdot CFCl$	116.47	gas		-157.5		d.		
-trichloroethane	$Cl_3CF\cdot CClF_2$	187.38	lq.	1.576 ^{20/4}	-35	47.6	i.	∞	∞
Trimethoxybutane (1-,3-,3-)	$CH_3(OCH_3)CH_2C(OCH_3)_2CH_3$	148.20	lq.	0.932		63-5 ²⁵	d.		
Trimethylamine	$(CH_3)_3N$	59.11	gas	0.662 ⁻⁵	-124	3.5	41 ¹⁹	s.	s.
Trimethylene bromide	$Br\cdot CH_2\cdot CH_2\cdot CH_2\cdot Br$	201.89	lq.	1.987 ^{15/4}	-34.4	167.5	0.17 ³⁰	s.	s.
chloride	$ClCH_2\cdot CH_2\cdot CH_2\cdot Cl$	112.99	lq.	1.201 ¹⁵		123-5	0.27 ²⁵	s.	s.
glycol	$HOCH_2CH_2CH_2OH$	76.09	oil	1.060 ^{20/4}		214	∞		
Trinitro-benzene (1-,3-,5-)	$C_6H_3(NO_2)_3$	213.10	col. rhb.	1.688 ^{20/4}	121	d.	0.03 ¹⁵	1.9 ¹⁸	1.5 ¹⁸
-benzoic acid (2-,4-,6-)	$(NO_2)_3C_6H_2CO_2H$	257.11	rhb./aq.		210-20 d.		2.05 ²⁴		
-tert-butylxylene	$(NO_2)_3C_6H_2(CH_3)_2C_4H_9$	297.26	nd./al.		110		i.	sl. s.	s.
-naphthalene (α-)(1-,3-,5-)	$C_{10}H_5(NO_2)_3$	263.16	rhb.		122-3		i.	s.	
(β-)(1-,3-,8-)	$C_{10}H_5(NO_2)_3$	263.16	cr./al.		218-9		0.02 ¹⁰⁰	0.05 ²³	0.13 ¹⁵
(γ-)(1-,4-,5-)	$C_{10}H_5(NO_2)_3$	263.16	yel. cr.		148-9		i.	0.11 ¹⁹	0.4 ¹⁹

-phenol (2-,3-,6-)	(NO ₂) ₂ C ₆ H ₃ OH	229.10	nd.		117-8		s. h.	v. s.	v. s.
-toluene (β-)(2-,3-,4-)	CH ₃ C ₆ H ₄ (NO ₂) ₃	227.13	cr.	1.620 ^{30/4}	112	expl.	i.	sl. s. c.	s.
(γ-)(2-,4-,5-)	CH ₃ C ₆ H ₂ (NO ₂) ₃	227.13	yel. pl.	1.620 ^{30/4}	104	expl.	i.	s. h.	v. s.
(α-)(2-,4-,6-)	CH ₃ C ₆ H ₃ (NO ₂) ₃	227.13	cr./al.	1.654	80.8	expl.	0.01 ³⁰	1.5 ³²	5 ³³
Trional	(C ₂ H ₅ SO ₂ C ₂ H ₄) ₂	242.36	pl./al.	1.199 ^{5/4}	76	d.	0.3 ¹⁵	5 ⁹	6.6 ¹⁵
Triphenyl-arsine	(C ₆ H ₅) ₃ As	306.23	pl.	1.306	59-60		i.	s.	v. s.
carbinol	(C ₆ H ₅) ₂ COH	260.33	cr.	1.188 ^{30/4}	162.5	>360	i.	v. s.	v. s.
guanidine (α-)	C ₆ H ₅ N:C(NHC ₆ H ₅) ₂	287.36	rhb./al.	1.13	144-5	d.	i.	4 ⁹	
methane	(C ₆ H ₅) ₃ CH	244.33	cr.	1.014 ^{90/4}	93.4	359 ^{75/4}	i.	v. s. h.	v. s.
methyl	(C ₆ H ₅) ₃ C . . .	243.32	col. cr.		145-7	d.	i.	sl. s. h.	
phosphate	OP(OC ₆ H ₅) ₃	326.28	pr./al.	1.206 ^{35/4}	49-50	245 ¹¹	i.	155 ²⁵	v. s.
Tripropylamine (n-)	(CH ₃ CH ₂ CH ₂) ₃ N	143.27	col. lq.	0.757 ^{20/4}	-93.5	156.5	v. sl. s.	∞	∞
Undecane (n-)	CH ₃ (CH ₂) ₉ CH ₃	156.31	col. lq.	0.741 ^{20/4}	-25.6	194.5	i.	∞	∞
Urea	H ₂ N·CO·NH ₂	60.06	col. pr.	1.335 ^{30/4}	132.7	d.	100 ¹⁷	20 ²⁰	sl. s.
nitrate	CO(NH ₂) ₂ ·HNO ₃	123.07	col. mn.		152 d.		v. s. h.	s.	
Uric acid	C ₅ H ₄ O ₃ N ₄	168.11	cr.	1.893 ³⁰	d.		0.06 h.	i.	i.
Valeric acid (n-)	C ₂ H ₅ CH ₂ CH ₂ CO ₂ H	102.13	col. lq.	0.939 ^{30/4}	-34.5	187	3.3 ¹⁶	∞	∞
(i-)	(CH ₃) ₂ CHCH ₂ CO ₂ H	102.13	col. lq.	0.931 ^{30/20}	-37.6	176	4.2 ²⁰	∞	∞
aldehyde (n-)	C ₂ H ₅ CH ₂ CH ₂ CHO	86.13	lq.	0.819 ¹¹	-92	103.4	v. sl. s.	s.	s.
(i-)	(CH ₃) ₂ CHCH ₂ CHO	86.13	col. lq.	0.803 ¹⁷	-51	92.5	sl. s.	s.	s.
amide (n-)	C ₃ H ₇ CH ₂ CH ₂ CONH ₂	101.15	mn. pl.	1.023	106		v. s.	v. s.	v. s.
(i-)	(CH ₃) ₂ CHCH ₂ CONH ₂	101.15	mn.	0.965 ^{20/4}	135-7	232	s.	s.	s.
Vanillic acid (3-,4-,1-)	CH ₃ O(OH)C ₆ H ₃ CO ₂ H	168.15	nd./aq.		207	subl.	0.12 ¹⁴	v. s.	v. s.
alcohol (3-,4-,1-)	CH ₃ O(OH)C ₆ H ₃ CH ₂ OH	154.16	mn./aq.		115	d.	v. s. h.	v. s.	v. s.
hyl-thiuram disulfide	[(C ₂ H ₅) ₂ NCS] ₂ S ₂	296.54	cr.	1.17	70		i.		
Vanillin (3-,4-,1-)	CH ₃ O(OH)C ₆ H ₃ CHO	152.15	mn.	1.056	81-2	285	1 ¹⁴	v. s.	v. s.
Veratrole (o-)	C ₆ H ₄ (OCH ₃) ₂	138.16	cr.	1.091 ^{15/15}	22.5	207.1	v. sl. s.	s.	s.
Vinyl acetate	CH ₃ CO ₂ CH:CH ₂	86.09	col. lq.	0.932 ^{20/4}	< -60	72-3	2 ²⁰	∞	∞
(poly-)	(CH ₃ CO ₂ CH:CH ₂) _x	(86.09)		1.19 ²⁰	100-25		i.		
acetic acid	CH ₃ :CH:CH ₂ CO ₂ H	86.09	col. lq.	1.013 ^{15/15}	-39	163	s.	∞	∞
acetylene	CH ₃ :CH:C:CH	52.07	gas	0.705 ^{1.5}		5.5	0.67 ^{0.6}		
alcohol	CH ₃ :CHOH	44.05							
(poly-)	(CH ₃ :CHOH) _x	(44.05)		1.3 ²⁰	d. >200		s.		
chloride	CH ₃ :CHCl	62.50	gas	0.908 ^{25/25}	-160	-12	sl. s.	s.	v. s.
propionate	C ₂ H ₅ CO ₂ CH:CH ₂	100.12	lq.			93-5	v. sl. s.		
Xylene (o-)	C ₆ H ₄ (CH ₃) ₂	106.17	col. lq.	0.881 ^{20/4}	-25	144	i.	s.	∞
(m-)	C ₆ H ₄ (CH ₃) ₂	106.17	col. lq.	0.867 ^{17/4}	-47.4	139.3	i.	s.	∞
(p-)	C ₆ H ₄ (CH ₃) ₂	106.17	col. lq.	0.861 ^{20/4}	13.2	138.5	i.	s.	v. s.
sulfonic acid (1-,4-,2-)	(CH ₃) ₂ C ₆ H ₃ SO ₃ H·2H ₂ O	222.26	col. lf.		86	149 ^{0.1}	s.		
Xylidine (1:2)(3-)	(CH ₃) ₂ C ₆ H ₃ NH ₂	121.18	lq.	0.991 ¹⁵	< -15	223	v. sl. s.	s.	s.
(1:2)(4-)	(CH ₃) ₂ C ₆ H ₃ NH ₂	121.18	pr.	1.076 ^{17.5}	49-50	224-6	v. sl. s.		
(1:3)(2-)	(CH ₃) ₂ C ₆ H ₃ NH ₂	121.18	lq.	0.980 ¹⁵		216-7	v. sl. s.		
(1:3)(4-)	(CH ₃) ₂ C ₆ H ₃ NH ₂	121.18	lq.	0.978 ^{20/4}		213-4	v. sl. s.		
(1:3)(5-)	(CH ₃) ₂ C ₆ H ₃ NH ₂	121.18	oil	0.972 ^{20/4}		221-2	v. sl. s.		
(1:4)(2-)	(CH ₃) ₂ C ₆ H ₃ NH ₂	121.18	oil	0.979 ^{21/4}	15.5	215 ⁷⁸⁰	v. sl. s.		
Xylose (l-)(+)	CH ₂ OH(CHOH) ₃ CHO	150.13	nd.	1.535 ⁹	153-4	117 ²⁰		v. sl. s.	i.
Xylylene dichloride (p-)	C ₆ H ₄ (CH ₂ Cl) ₂	175.06	mn.	1.417 ⁹	100.5	240-5 d.	i.	s.	v. sl. s.
Zinc diethyl	Zn(CH ₂ CH ₃) ₂	123.53	col. lq.	1.182 ¹⁸	-28	118	d.	d.	
dimethyl	Zn(CH ₃) ₂	95.48	col. lq.	1.386 ¹¹	-40	46	d.	d.	
dimethyl-dithiocarbamate	Zn[S ₂ CN(CH ₃) ₂] ₂	305.84		2.00 ^{40/4}	248-50		i.		

NOTE: °F = % °C + 32.

VAPOR PRESSURES OF PURE SUBSTANCES

TABLE 2-3 Vapor Pressure of Water Ice from 0 to -40 °C

t, °C	Vapor pressure		t, °C	Vapor pressure		t, °C	Vapor pressure	
	mmHg	kPa		mmHg	kPa		mmHg	kPa
0	4.584	0.6112	-13.5	1.423	0.1897	-27.0	0.3881	0.05174
-0.5	4.399	0.5865	-14.0	1.359	0.1812	-27.5	0.3688	0.04918
-1.0	4.220	0.5627	-14.5	1.298	0.1731	-28.0	0.3505	0.04673
-1.5	4.049	0.5398	-15.0	1.240	0.1653	-28.5	0.3330	0.04439
-2.0	3.883	0.5177	-15.5	1.184	0.1578	-29.0	0.3162	0.04216
-2.5	3.724	0.4965	-16.0	1.130	0.1507	-29.5	0.3003	0.04004
-3.0	3.571	0.4761	-16.5	1.079	0.1438	-30.0	0.2851	0.03801
-3.5	3.423	0.4564	-17.0	1.029	0.1372	-30.5	0.2706	0.03608
-4.0	3.281	0.4375	-17.5	0.9822	0.1310	-31.0	0.2568	0.03424
-4.5	3.145	0.4193	-18.0	0.9370	0.1249	-31.5	0.2437	0.03249
-5.0	3.013	0.4018	-18.5	0.8937	0.1191	-32.0	0.2311	0.03082
-5.5	2.887	0.3849	-19.0	0.8522	0.1136	-32.5	0.2192	0.02923
-6.0	2.766	0.3687	-19.5	0.8125	0.1083	-33.0	0.2078	0.02771
-6.5	2.649	0.3532	-20.0	0.7745	0.1033	-33.5	0.1970	0.02627
-7.0	2.537	0.3382	-20.5	0.7381	0.09841	-34.0	0.1867	0.02490
-7.5	2.429	0.3238	-21.0	0.7034	0.09377	-34.5	0.1769	0.02359
-8.0	2.325	0.3100	-21.5	0.6701	0.08934	-35.0	0.1676	0.02235
-8.5	2.225	0.2967	-22.0	0.6383	0.08510	-35.5	0.1587	0.02116
-9.0	2.130	0.2839	-22.5	0.6078	0.08104	-36.0	0.1503	0.02004
-9.5	2.038	0.2717	-23.0	0.5787	0.07716	-36.5	0.1423	0.01897
-10.0	1.949	0.2599	-23.5	0.5509	0.07345	-37.0	0.1347	0.01796
-10.5	1.865	0.2486	-24.0	0.5243	0.06991	-37.5	0.1274	0.01699
-11.0	1.783	0.2377	-24.5	0.4989	0.06652	-38.0	0.1206	0.01607
-11.5	1.705	0.2273	-25.0	0.4747	0.06329	-38.5	0.1140	0.01520
-12.0	1.630	0.2173	-25.5	0.4515	0.06020	-39.0	0.1078	0.01437
-12.5	1.558	0.2077	-26.0	0.4294	0.05725	-39.5	0.1019	0.01359
-13.0	1.489	0.1985	-26.5	0.4083	0.05443	-40.0	0.0963	0.01284

SOURCE: Formulation of Wagner, Saul, and Pruss, *J. Phys. Chem. Ref. Data*, **23**, 515 (1994), implemented in Harvey, Peskin, and Klein, *NIST/ASME Steam Properties*, NIST Standard Reference Database 10, Version 2.2, National Institute of Standards and Technology, Gaithersburg, Md., 2000. This source provides data down to 190 K (-83.15 °C). A formula extending to 110 K may be found in Murphy and Koop, *Q. J. R. Meteorol. Soc.*, **131**, 1539 (2005).

TABLE 2-4 Vapor Pressure of Supercooled Liquid Water from 0 to -40 °C*

t, °C	Vapor pressure		t, °C	Vapor pressure		t, °C	Vapor pressure	
	mmHg	kPa		mmHg	kPa		mmHg	kPa
0	4.584	0.6112	-13.5	1.623	0.2163	-27.0	0.5051	0.06734
-0.5	4.421	0.5894	-14.0	1.558	0.2077	-27.5	0.4824	0.06431
-1.0	4.262	0.5682	-14.5	1.495	0.1993	-28.0	0.4606	0.06141
-1.5	4.108	0.5477	-15.0	1.435	0.1913	-28.5	0.4397	0.05862
-2.0	3.959	0.5279	-15.5	1.377	0.1836	-29.0	0.4197	0.05595
-2.5	3.816	0.5087	-16.0	1.321	0.1761	-29.5	0.4005	0.05339
-3.0	3.676	0.4901	-16.5	1.267	0.1689	-30.0	0.3820	0.05094
-3.5	3.542	0.4722	-17.0	1.215	0.1620	-30.5	0.3644	0.04858
-4.0	3.411	0.4548	-17.5	1.165	0.1553	-31.0	0.3475	0.04633
-4.5	3.285	0.4380	-18.0	1.117	0.1489	-31.5	0.3313	0.04417
-5.0	3.163	0.4218	-18.5	1.070	0.1427	-32.0	0.3158	0.04210
-5.5	3.046	0.4061	-19.0	1.026	0.1367	-32.5	0.3009	0.04012
-6.0	2.932	0.3909	-19.5	0.9827	0.1310	-33.0	0.2867	0.03822
-6.5	2.822	0.3762	-20.0	0.9414	0.1255	-33.5	0.2731	0.03640
-7.0	2.715	0.3620	-20.5	0.9016	0.1202	-34.0	0.2600	0.03467
-7.5	2.612	0.3483	-21.0	0.8633	0.1151	-34.5	0.2476	0.03300
-8.0	2.513	0.3351	-21.5	0.8265	0.1102	-35.0	0.2356	0.03141
-8.5	2.417	0.3223	-22.0	0.7911	0.1055	-35.5	0.2242	0.02989
-9.0	2.324	0.3099	-22.5	0.7571	0.1009	-36.0	0.2133	0.02844
-9.5	2.235	0.2980	-23.0	0.7244	0.0965	-36.5	0.2029	0.02705
-10.0	2.149	0.2865	-23.5	0.6930	0.0923	-37.0	0.1929	0.02572
-10.5	2.065	0.2753	-24.0	0.6628	0.08836	-37.5	0.1834	0.02445
-11.0	1.985	0.2646	-24.5	0.6337	0.08449	-38.0	0.1743	0.02324
-11.5	1.907	0.2542	-25.0	0.6059	0.08078	-38.5	0.1656	0.02208
-12.0	1.832	0.2442	-25.5	0.5791	0.07721	-39.0	0.1573	0.02098
-12.5	1.760	0.2346	-26.0	0.5534	0.07379	-39.5	0.1494	0.01992
-13.0	1.690	0.2253	-26.5	0.5288	0.07050	-40.0	0.1419	0.01891

*SOURCE: Murphy and Koop, *Q. J. R. Meteorol. Soc.*, **131**, 1539 (2005). The formula in the reference extends down to 123 K (-150.15 °C), although in practice pure liquid water cannot be supercooled below about 235 K.

UNITS CONVERSIONS

For this subsection, the following units conversions are applicable:

$$^{\circ}\text{F} = \% ^{\circ}\text{C} + 32.$$

To convert millimeters of mercury to pounds-force per square inch, multiply by 0.01934.

ADDITIONAL REFERENCES

Additional vapor-pressure data may be found in major thermodynamic property databases, such as those produced by the AIChE's DIPPER program (www.aiche.org/dipper), NIST's Thermodynamics Research Center (trc.nist.gov), the Dortmund Databank (www.ddbst.de), and the Physical Property Data Service (www.ppps.co.uk). Additional sources include the *NIST Chemistry Webbook* (webbook.nist.gov/chemistry/); Boublik, Fried, and Hala, *The Vapor Pressures of Pure Substances*, 2d ed., Elsevier, Amsterdam, 1984; Poling, Prausnitz, and O'Connell, *The Properties of Gases and Liquids*, 5th ed., McGraw-Hill, New York, 2001; *Vapor Pressure of Chemicals* (Subvolumes A, B, and C), vol. IV/20 in Landolt-Bornstein: *Numerical Data and Functional Relationships in Science and Technology—New Series*, Springer-Verlag, Berlin, 1999–2001. The most recent work on water may be found at The International Association for the Properties of Water and Steam website <http://iapws.org>.

TABLE 2-5 Vapor Pressure (MPa) of Liquid Water from 0 to 100 °C

t, °C	P _{vp} , MPa	t, °C	P _{vp} , MPa	t, °C	P _{vp} , MPa
0.01	0.00061165	34	0.0053251	68	0.028599
1	0.00065709	35	0.0056290	69	0.029876
2	0.00070599	36	0.0059479	70	0.031201
3	0.00075808	37	0.0062823	71	0.032575
4	0.00081355	38	0.0066328	72	0.034000
5	0.00087258	39	0.0070002	73	0.035478
6	0.00093536	40	0.0073849	74	0.037009
7	0.0010021	41	0.0077878	75	0.038595
8	0.0010730	42	0.0082096	76	0.040239
9	0.0011483	43	0.0086508	77	0.041941
10	0.0012282	44	0.0091124	78	0.043703
11	0.0013130	45	0.0095950	79	0.045527
12	0.0014028	46	0.010099	80	0.047414
13	0.0014981	47	0.010627	81	0.049367
14	0.0015990	48	0.011177	82	0.051387
15	0.0017058	49	0.011752	83	0.053476
16	0.0018188	50	0.012352	84	0.055635
17	0.0019384	51	0.012978	85	0.057867
18	0.0020647	52	0.013631	86	0.060173
19	0.0021983	53	0.014312	87	0.062556
20	0.0023393	54	0.015022	88	0.065017
21	0.0024882	55	0.015762	89	0.067558
22	0.0026453	56	0.016533	90	0.070182
23	0.0028111	57	0.017336	91	0.072890
24	0.0029858	58	0.018171	92	0.075684
25	0.0031699	59	0.019041	93	0.078568
26	0.0033639	60	0.019946	94	0.081541
27	0.0035681	61	0.020888	95	0.084608
28	0.0037831	62	0.021867	96	0.087771
29	0.0040092	63	0.022885	97	0.091030
30	0.0042470	64	0.023943	98	0.094390
31	0.0044969	65	0.025042	99	0.097852
32	0.0047596	66	0.026183	100	0.10142
33	0.0050354	67	0.027368		

From E. W. Lemmon, M. O. McLinden, and D. G. Friend, "Thermophysical Properties of Fluid Systems" in *NIST Chemistry WebBook*, NIST Standard Reference Database Number 69, Eds. P. J. Linstrom and W. G. Mallard, June 2005, National Institute of Standards and Technology, Gaithersburg, Md. (<http://webbook.nist.gov>) and Wagner, W., and A., Pruss, "The IAPWS Formulation 1995 for the Thermodynamic Properties of Ordinary Water Substance for General and Scientific Use," *J. Phys. Chem. Ref. Data* **31**(2):387–535, 2002.

The website mentioned above allows users to generate their own tables of thermodynamic properties. The user can select the units as well as the temperatures and/or pressures for which properties are to be generated. The results can be then be copied into spreadsheets or other files.

TABLE 2-6 Substances in Tables 2-8, 2-32, 2-141, 2-150, 2-153, 2-155, 2-156, 2-179, 2-312, 2-313, 2-314, and 2-315 Sorted by Chemical Family

Name	Cmpd. No.	Formula	Name	Cmpd. No.	Formula
Paraffins			Acetylenes		
Methane	193	CH ₄	1-Octyne	273	C ₈ H ₁₄
Ethane	125	C ₂ H ₆	1-Nonyne	262	C ₉ H ₁₆
Propane	295	C ₃ H ₈	1-Decyne	79	C ₁₀ H ₁₈
Butane	31	C ₄ H ₁₀	Methyl acetylene	197	C ₃ H ₄
Pentane	279	C ₅ H ₁₂	Vinyl acetylene	339	C ₄ H ₄
Hexane	171	C ₆ H ₁₄	Dimethyl acetylene	105	C ₄ H ₆
Heptane	160	C ₇ H ₁₆	2-Methyl-1-butene-3-yne	207	C ₅ H ₆
Octane	265	C ₈ H ₁₈	3-Methyl-1-butyne	210	C ₅ H ₈
Nonane	256	C ₉ H ₂₀	Aromatics		
Decane	74	C ₁₀ H ₂₂	Benzene	16	C ₆ H ₆
Undecane	336	C ₁₁ H ₂₄	Toluene	325	C ₇ H ₈
Dodecane	123	C ₁₂ H ₂₆	Styrene	312	C ₈ H ₈
Tridecane	327	C ₁₃ H ₂₈	Ethylbenzene	129	C ₈ H ₁₀
Tetradecane	319	C ₁₄ H ₃₀	<i>m</i> -Xylene	343	C ₈ H ₁₀
Pentadecane	277	C ₁₅ H ₃₂	<i>o</i> -Xylene	344	C ₈ H ₁₀
Hexadecane	169	C ₁₆ H ₃₄	<i>p</i> -Xylene	345	C ₈ H ₁₀
Heptadecane	158	C ₁₇ H ₃₆	alpha-Methyl styrene	243	C ₉ H ₁₀
Octadecane	263	C ₁₈ H ₃₈	Cumene	62	C ₉ H ₁₂
Nonadecane	254	C ₁₉ H ₄₀	Propylbenzene	304	C ₉ H ₁₂
Eicosane	124	C ₂₀ H ₄₂	1,2,3-Trimethylbenzene	330	C ₉ H ₁₂
2-Methylpropane	236	C ₄ H ₁₀	1,2,4-Trimethylbenzene	331	C ₉ H ₁₂
2-Methylbutane	202	C ₅ H ₁₂	Naphthalene	246	C ₁₀ H ₈
2,3-Dimethylbutane	107	C ₆ H ₁₄	1,2,3,4-Tetrahydronaphthalene	321	C ₁₀ H ₁₂
2-Methylpentane	234	C ₆ H ₁₄	Butylbenzene	40	C ₁₀ H ₁₄
2,3-Dimethylpentane	114	C ₇ H ₁₆	Biphenyl	24	C ₁₂ H ₁₀
2,2,3,3-Tetramethylbutane	323	C ₈ H ₁₈	Phenanthrene	290	C ₁₄ H ₁₀
2,2,4-Trimethylpentane	332	C ₈ H ₁₈	<i>o</i> -Terphenyl	318	C ₁₈ H ₁₄
2,3,3-Trimethylpentane	333	C ₈ H ₁₈	Aldehydes		
Cyclopropane	71	C ₃ H ₆	Formaldehyde	153	CH ₂ O
Cyclobutane	64	C ₄ H ₈	Acetaldehyde	1	C ₂ H ₄ O
Cyclopentane	69	C ₅ H ₁₀	Propionaldehyde	299	C ₃ H ₆ O
Cyclohexane	65	C ₆ H ₁₂	Butyraldehyde	44	C ₄ H ₈ O
Methylcyclopentane	217	C ₆ H ₁₂	Pentanal	278	C ₅ H ₁₀ O
Ethylcyclopentane	134	C ₇ H ₁₄	Hexanal	170	C ₆ H ₁₂ O
Methylcyclohexane	213	C ₇ H ₁₄	Heptanal	159	C ₇ H ₁₄ O
1,1-Dimethylcyclohexane	108	C ₈ H ₁₆	Octanal	264	C ₈ H ₁₆ O
<i>cis</i> -1,2-Dimethylcyclohexane	109	C ₈ H ₁₆	Nonanal	255	C ₉ H ₁₈ O
<i>trans</i> -1,2-Dimethylcyclohexane	110	C ₈ H ₁₆	Decanal	73	C ₁₀ H ₂₀ O
Ethylcyclohexane	133	C ₈ H ₁₆	Ketones		
Olefins			Acrolein	8	C ₃ H ₄ O
Ethylene	135	C ₂ H ₄	Acetone	5	C ₃ H ₆ O
Propylene	305	C ₃ H ₆	Methylethyl ketone	222	C ₄ H ₈ O
1-Butene	36	C ₄ H ₈	Methylisopropyl ketone	229	C ₅ H ₁₀ O
<i>cis</i> -2-Butene	37	C ₄ H ₈	2-Pentanone	283	C ₅ H ₁₀ O
<i>trans</i> -2-Butene	38	C ₄ H ₈	3-Pentanone	284	C ₅ H ₁₀ O
Cyclopentene	70	C ₅ H ₈	Quinone	310	C ₆ H ₄ O ₂
1-Pentene	285	C ₅ H ₁₀	Cyclohexanone	67	C ₆ H ₁₀ O
Cyclohexene	68	C ₆ H ₁₀	Ethylisopropyl ketone	144	C ₆ H ₁₂ O
1-Hexene	177	C ₆ H ₁₂	2-Hexanone	175	C ₆ H ₁₂ O
1-Heptene	166	C ₇ H ₁₄	3-Hexanone	176	C ₆ H ₁₂ O
1-Octene	271	C ₈ H ₁₆	Methylisobutyl ketone	226	C ₆ H ₁₂ O
1-Nonene	260	C ₉ H ₁₈	Diisopropyl ketone	102	C ₇ H ₁₄ O
1-Decene	77	C ₁₀ H ₂₀	3-Heptanone	164	C ₇ H ₁₄ O
2-Methyl propene	238	C ₄ H ₈	2-Heptanone	165	C ₇ H ₁₄ O
2-Methyl-1-butene	205	C ₅ H ₁₀	2-Octanone	269	C ₈ H ₁₆ O
2-Methyl-2-butene	206	C ₅ H ₁₀	3-Octanone	270	C ₈ H ₁₆ O
1-Methylcyclopentene	218	C ₆ H ₁₀	Benzophenone	20	C ₁₃ H ₁₀ O
3-Methylcyclopentene	219	C ₆ H ₁₀	Heterocyclics		
Propenylcyclohexene	298	C ₉ H ₁₄	Furan	156	C ₄ H ₄ O
Propadiene	294	C ₃ H ₄	Thiophene	324	C ₄ H ₄ S
1,2-Butadiene	29	C ₄ H ₆	Tetrahydrofuran	320	C ₄ H ₈ O
1,3-Butadiene	30	C ₄ H ₆	Tetrahydrothiophene	322	C ₄ H ₈ S
3-Methyl-1,2-butadiene	201	C ₅ H ₈	Elements		
Acetylenes			Argon	14	Ar
Acetylene	7	C ₂ H ₂	Bromine	25	Br ₂
1-Butyne	43	C ₄ H ₆	Chlorine	52	Cl ₂
1-Pentyne	288	C ₅ H ₈	Deuterium	80	D ₂
2-Pentyne	289	C ₅ H ₈	Fluorine	149	F ₂
3-Hexyne	178	C ₆ H ₁₀			
1-Hexyne	180	C ₆ H ₁₀			
2-Hexyne	181	C ₆ H ₁₀			
1-Heptyne	168	C ₇ H ₁₂			

2-50 PHYSICAL AND CHEMICAL DATA

TABLE 2-6 Substances in Tables 2-8, 2-32, 2-141, 2-150, 2-153, 2-155, 2-156, 2-179, 2-312, 2-313, 2-314, and 2-315 Sorted by Chemical Family (Continued)

Name	Cmpd. No.	Formula	Name	Cmpd. No.	Formula
Elements			Acids		
Hydrogen	183	H ₂	Methacrylic acid	192	C ₄ H ₆ O ₂
Helium-4	157	He	Acetic anhydride	4	C ₄ H ₆ O ₃
Nitrogen	249	N ₂	Succinic acid	313	C ₄ H ₆ O ₄
Neon	247	Ne	Butyric acid	45	C ₄ H ₈ O ₂
Oxygen	275	O ₂	Isobutyric acid	189	C ₄ H ₈ O ₂
Alcohols			2-Methylbutanoic acid	203	C ₅ H ₁₀ O ₂
Methanol	194	CH ₃ O	Pentanoic acid	280	C ₅ H ₁₀ O ₂
Ethanol	126	C ₂ H ₆ O	2-Ethyl butanoic acid	131	C ₆ H ₁₂ O ₂
1-Propanol	296	C ₃ H ₈ O	Hexanoic acid	172	C ₆ H ₁₂ O ₂
2-Propanol	297	C ₃ H ₈ O	Benzoic acid	18	C ₇ H ₆ O ₂
1-Butanol	34	C ₄ H ₁₀ O	Heptanoic acid	161	C ₇ H ₁₄ O ₂
2-Butanol	35	C ₄ H ₁₀ O	Phthalic anhydride	293	C ₈ H ₆ O ₃
1-Pentanol	281	C ₅ H ₁₂ O	Terephthalic acid	317	C ₈ H ₆ O ₄
2-Pentanol	282	C ₅ H ₁₂ O	2-Ethyl hexanoic acid	141	C ₈ H ₁₆ O ₂
Cyclohexanol	66	C ₆ H ₁₂ O	Octanoic acid	266	C ₈ H ₁₆ O ₂
1-Hexanol	173	C ₆ H ₁₄ O	2-Methyloctanoic acid	233	C ₉ H ₁₈ O ₂
2-Hexanol	174	C ₆ H ₁₄ O	Nonanoic acid	257	C ₉ H ₁₈ O ₂
1-Heptanol	162	C ₇ H ₁₆ O	Decanoic acid	75	C ₁₀ H ₂₀ O ₂
2-Heptanol	163	C ₇ H ₁₆ O	Esters		
1-Octanol	267	C ₈ H ₁₈ O	Methyl formate	224	C ₂ H ₄ O ₂
2-Octanol	268	C ₈ H ₁₈ O	Ethyl formate	140	C ₃ H ₆ O ₂
1-Nonanol	258	C ₉ H ₂₀ O	Methyl acetate	196	C ₃ H ₆ O ₂
2-Nonanol	259	C ₉ H ₂₀ O	Methyl acrylate	198	C ₄ H ₆ O ₂
1-Decanol	76	C ₁₀ H ₂₂ O	Vinyl acetate	338	C ₄ H ₆ O ₂
1-Undecanol	337	C ₁₁ H ₂₄ O	Ethyl acetate	127	C ₄ H ₈ O ₂
2-Methyl-2-propanol	237	C ₄ H ₁₀ O	Methyl propionate	239	C ₄ H ₈ O ₂
3-Methyl-1-butanol	204	C ₅ H ₁₂ O	Propyl formate	306	C ₄ H ₈ O ₂
Benzyl alcohol	21	C ₇ H ₈ O	Methyl methacrylate	232	C ₅ H ₈ O ₂
1-Methylcyclohexanol	214	C ₇ H ₁₄ O	Ethyl propionate	146	C ₅ H ₁₀ O ₂
cis-2-Methylcyclohexanol	215	C ₇ H ₁₄ O	Methyl butyrate	211	C ₅ H ₁₀ O ₂
trans-2-Methylcyclohexanol	216	C ₇ H ₁₄ O	Propyl acetate	302	C ₅ H ₁₀ O ₂
Ethylene glycol	137	C ₂ H ₆ O ₂	Butyl acetate	39	C ₆ H ₁₂ O ₂
1,2-Propylene glycol	309	C ₃ H ₈ O ₂	Ethyl butyrate	132	C ₆ H ₁₂ O ₂
1,2-Butanediol	32	C ₄ H ₁₀ O ₂	Methyl benzoate	200	C ₈ H ₈ O ₂
1,3-Butanediol	33	C ₄ H ₁₀ O ₂	Ethyl benzoate	130	C ₉ H ₁₀ O ₂
Phenols			Dimethyl phthalate	115	C ₁₀ H ₁₀ O ₄
Phenol	291	C ₆ H ₆ O	Dimethyl terephthalate	119	C ₁₀ H ₁₀ O ₄
m-Cresol	59	C ₇ H ₈ O	Amines		
o-Cresol	60	C ₇ H ₈ O	Methyl amine	199	CH ₃ N
p-Cresol	61	C ₇ H ₈ O	Ethyleneimine	138	C ₂ H ₅ N
Ethers			Dimethyl amine	106	C ₂ H ₇ N
Dimethyl ether	112	C ₂ H ₆ O	Ethyl amine	128	C ₂ H ₇ N
Methyl vinyl ether	245	C ₃ H ₆ O	Ethylenediamine	136	C ₂ H ₆ N ₂
Methylethyl ether	221	C ₃ H ₈ O	Isopropyl amine	190	C ₃ H ₉ N
1,4-Dioxane	120	C ₄ H ₈ O ₂	Propyl amine	303	C ₃ H ₉ N
Diethyl ether	95	C ₄ H ₁₀ O	Trimethyl amine	329	C ₃ H ₉ N
Methylpropyl ether	240	C ₄ H ₁₀ O	Diethyl amine	94	C ₄ H ₁₁ N
Methylisopropyl ether	228	C ₄ H ₁₀ O	Diethanol amine	93	C ₄ H ₁₁ NO ₂
1,1-Dimethoxyethane	103	C ₄ H ₁₀ O ₂	Di-isopropyl amine	100	C ₆ H ₁₅ N
Methylbutyl ether	208	C ₅ H ₁₂ O	Dipropyl amine	122	C ₆ H ₁₅ N
Methylisobutyl ether	225	C ₅ H ₁₂ O	Triethyl amine	328	C ₆ H ₁₅ N
Methyl tert-butyl ether	244	C ₅ H ₁₂ O	Amides		
Ethylpropyl ether	147	C ₅ H ₁₂ O	Formamide	154	CH ₃ NO
Ethylisopropyl ether	143	C ₅ H ₁₂ O	Acetamide	2	C ₂ H ₅ NO
1,2-Dimethoxypropane	104	C ₅ H ₁₂ O ₂	N,N-Dimethyl formamide	113	C ₃ H ₇ NO
Di-isopropyl ether	101	C ₆ H ₁₄ O	N-Methyl acetamide	195	C ₃ H ₇ NO
Methyl pentyl ether	235	C ₆ H ₁₄ O	Benzamide	15	C ₇ H ₇ NO
Anisole	13	C ₇ H ₈ O	Nitriles		
Dibutyl ether	84	C ₈ H ₁₈ O	Acetonitrile	6	C ₂ H ₃ N
Ethylhexyl ether	142	C ₈ H ₁₈ O	Cyanogen	63	C ₂ N ₂
Benzyl ethyl ether	22	C ₉ H ₁₂ O	Acrylonitrile	10	C ₃ H ₃ N
Diphenyl ether	121	C ₁₂ H ₁₀ O	Propionitrile	301	C ₃ H ₅ N
Acids			Butyronitrile	46	C ₄ H ₇ N
Formic acid	155	CH ₂ O ₂	Benzonitrile	19	C ₇ H ₅ N
Oxalic acid	274	C ₂ H ₂ O ₄	Nitro Compounds		
Acetic acid	3	C ₂ H ₄ O ₂	Nitromethane	251	CH ₃ NO ₂
Acrylic acid	9	C ₃ H ₄ O ₂	Nitroethane	248	C ₂ H ₅ NO ₂
Malonic acid	191	C ₃ H ₄ O ₄			
Propionic acid	300	C ₃ H ₆ O ₂			

TABLE 2-6 Substances in Tables 2-8, 2-32, 2-141, 2-150, 2-153, 2-155, 2-156, 2-179, 2-312, 2-313, 2-314, and 2-315 Sorted by Chemical Family (Concluded)

Name	Cmpd. No.	Formula	Name	Cmpd. No.	Formula
Nitro Compounds			Halogenated Hydrocarbons		
1,3,5-Trinitrobenzene	334	C ₆ H ₃ N ₃ O ₆	1,2-Dichloroethane	89	C ₂ H ₄ Cl ₂
2,4,6-Trinitrotoluene	335	C ₇ H ₅ N ₃ O ₆	1,1-Difluoroethane	97	C ₂ H ₄ F ₂
Isocyanates			1,2-Difluoroethane	98	C ₂ H ₄ F ₂
Methyl isocyanate	227	C ₂ H ₃ NO	Bromoethane	27	C ₂ H ₅ Br
Phenyl isocyanate	292	C ₇ H ₅ NO	Chloroethane	54	C ₂ H ₅ Cl
Mercaptans			Fluoroethane	151	C ₂ H ₅ F
Methyl mercaptan	231	CH ₄ S	1,1-Dichloropropane	91	C ₃ H ₆ Cl ₂
Ethyl mercaptan	145	C ₂ H ₆ S	1,2-Dichloropropane	92	C ₃ H ₆ Cl ₂
Propyl mercaptan	308	C ₃ H ₈ S	1-Chloropropane	57	C ₃ H ₇ Cl
2-Propyl mercaptan	307	C ₃ H ₈ S	2-Chloropropane	58	C ₃ H ₇ Cl
Butyl mercaptan	41	C ₄ H ₁₀ S	<i>m</i> -Dichlorobenzene	85	C ₆ H ₄ Cl ₂
<i>sec</i> -Butyl mercaptan	42	C ₄ H ₁₀ S	<i>o</i> -Dichlorobenzene	86	C ₆ H ₄ Cl ₂
Pentyl mercaptan	287	C ₅ H ₁₂ S	<i>p</i> -Dichlorobenzene	87	C ₆ H ₄ Cl ₂
2-Pentyl mercaptan	286	C ₅ H ₁₂ S	Bromobenzene	26	C ₆ H ₅ Br
Benzenethiol	17	C ₆ H ₆ S	Chlorobenzene	53	C ₆ H ₅ Cl
Cyclohexyl mercaptan	72	C ₆ H ₁₂ S	Fluorobenzene	150	C ₆ H ₅ F
Hexyl mercaptan	179	C ₆ H ₁₄ S	Silanes		
Benzyl mercaptan	23	C ₇ H ₈ S	Methylsilane	242	CH ₄ Si
Heptyl mercaptan	167	C ₇ H ₁₆ S	Methylchlorosilane	212	CH ₃ ClSi
Octyl mercaptan	272	C ₈ H ₁₈ S	Methyldichlorosilane	220	CH ₃ Cl ₂ Si
Nonyl mercaptan	261	C ₉ H ₂₀ S	Vinyl trichlorosilane	341	C ₂ H ₃ Cl ₃ Si
Decyl mercaptan	78	C ₁₀ H ₂₂ S	Ethyltrichlorosilane	148	C ₂ H ₅ Cl ₃ Si
Sulfides			Dimethylsilane	116	C ₂ H ₆ Si
Dimethyl sulfide	117	C ₂ H ₆ S	Silicon tetrafluoride	311	F ₄ Si
Dimethyl disulfide	111	C ₂ H ₆ S ₂	Light Gases		
Methylethyl sulfide	223	C ₃ H ₈ S	Hydrogen cyanide	186	CHN
Diethyl sulfide	96	C ₄ H ₁₀ S	Carbon monoxide	49	CO
Methylisopropyl sulfide	230	C ₄ H ₁₀ S	Carbon dioxide	47	CO ₂
Methylpropyl sulfide	241	C ₄ H ₁₀ S	Carbon disulfide	48	CS ₂
Methylbutyl sulfide	209	C ₅ H ₁₂ S	Nitrogen trifluoride	250	F ₃ N
Halogenated Hydrocarbons			Sulfur hexafluoride	315	F ₆ S
Carbon tetrachloride	50	CCl ₄	Hydrogen bromide	184	HBr
Carbon tetrafluoride	51	CF ₄	Hydrogen chloride	185	HCl
Chloroform	55	CHCl ₃	Hydrogen fluoride	187	HF
Dibromomethane	83	CH ₂ Br ₂	Hydrogen sulfide	188	H ₂ S
Dichloromethane	90	CH ₂ Cl ₂	Ammonia	12	H ₃ N
Difluoromethane	99	CH ₂ F ₂	Hydrazine	182	H ₂ N ₂
Bromomethane	28	CH ₃ Br	Nitric oxide	253	NO
Chloromethane	56	CH ₃ Cl	Nitrous oxide	252	N ₂ O
Fluoromethane	152	CH ₃ F	Sulfur dioxide	314	O ₂ S
Vinyl chloride	340	C ₂ H ₃ Cl	Ozone	276	O ₃
1,1,2-Trichloroethane	326	C ₂ H ₃ Cl ₃	Sulfur trioxide	316	O ₃ S
1,1-Dibromoethane	81	C ₂ H ₄ Br ₂	Others		
1,2-Dibromoethane	82	C ₂ H ₄ Br ₂	Air	11	Mixture
1,1-Dichloroethane	88	C ₂ H ₄ Cl ₂	Ethylene oxide	139	C ₂ H ₄ O
			Dimethyl sulfoxide	118	C ₂ H ₆ OS
			Water	342	H ₂ O

2-52 PHYSICAL AND CHEMICAL DATA

TABLE 2-7 Formula Index of Substances in Tables 2-8, 2-32, 2-141, 2-150, 2-153, 2-155, 2-156, 2-179, 2-312, 2-313, 2-314, and 2-315

Formula	No.	Name	Formula	No.	Name
Ar	11	Air	C ₃ H ₆ Cl ₂	92	1,2-Dichloropropane
Br ₂	14	Argon	C ₃ H ₆ O	5	Acetone
CCl ₄	25	Bromine	C ₃ H ₆ O	245	Methyl vinyl ether
CF ₄	50	Carbon tetrachloride	C ₃ H ₆ O	299	Propionaldehyde
CHCl ₃	51	Carbon tetrafluoride	C ₃ H ₆ O ₂	140	Ethyl formate
CHN	55	Chloroform	C ₃ H ₆ O ₂	196	Methyl acetate
CH ₃ Br ₃	186	Hydrogen cyanide	C ₃ H ₆ O ₂	300	Propionic acid
CH ₃ Cl ₂	83	Dibromomethane	C ₃ H ₇ Cl	57	1-Chloropropane
CH ₃ F ₂	90	Dichloromethane	C ₃ H ₇ Cl	58	2-Chloropropane
CH ₃ O	99	Di fluoromethane	C ₃ H ₇ NO	113	N,N-Dimethyl formamide
CH ₃ O ₂	153	Formaldehyde	C ₃ H ₇ NO	195	N-Methyl acetamide
CH ₃ Br	155	Formic acid	C ₃ H ₈	295	Propane
CH ₃ Cl	28	Bromomethane	C ₃ H ₈ O	221	Methylethyl ether
CH ₃ F	56	Chloromethane	C ₃ H ₈ O	296	1-Propanol
CH ₃ NO	152	Fluoromethane	C ₃ H ₈ O	297	2-Propanol
CH ₃ NO ₂	154	Formamide	C ₃ H ₈ O ₂	309	1,2-Propylene glycol
CH ₄	251	Nitromethane	C ₃ H ₈ S	223	Methylethyl sulfide
CH ₄ Cl ₂ Si	193	Methane	C ₃ H ₈ S	308	Propyl mercaptan
CH ₄ O	220	Methyldichlorosilane	C ₃ H ₈ S	307	2-Propyl mercaptan
CH ₄ S	194	Methanol	C ₃ H ₉ N	190	Isopropyl amine
CH ₄ ClSi	231	Methyl mercaptan	C ₃ H ₉ N	303	Propyl amine
CH ₅ N	212	Methylchlorosilane	C ₃ H ₉ N	329	Trimethyl amine
CH ₆ Si	199	Methyl amine	C ₄ H ₄	339	Vinyl acetylene
CO	242	Methylsilane	C ₄ H ₄ O	156	Furan
CO ₂	49	Carbon monoxide	C ₄ H ₄ S	324	Thiophene
CS ₂	47	Carbon dioxide	C ₄ H ₆	29	1,2-Butadiene
C ₂ H ₂	48	Carbon disulfide	C ₄ H ₆	30	1,3-Butadiene
C ₂ H ₂ O ₄	7	Acetylene	C ₄ H ₆	43	1-Butyne
C ₂ H ₃ Cl	274	Oxalic acid	C ₄ H ₆	105	Dimethyl acetylene
C ₂ H ₃ Cl ₃	340	Vinyl chloride	C ₄ H ₆ O ₂	192	Methacrylic acid
C ₂ H ₃ Cl ₃ Si	326	1,1,2-Trichloroethane	C ₄ H ₆ O ₂	198	Methyl acrylate
C ₂ H ₃ Cl ₃ Si	341	Vinyl trichlorosilane	C ₄ H ₆ O ₂	338	Vinyl acetate
C ₂ H ₃ N	6	Acetonitrile	C ₄ H ₆ O ₃	4	Acetic anhydride
C ₂ H ₃ NO	227	Methyl Isocyanate	C ₄ H ₆ O ₄	313	Succinic acid
C ₂ H ₄	135	Ethylene	C ₄ H ₇ N	46	Butyronitrile
C ₂ H ₄ Br ₂	81	1,1-Dibromoethane	C ₄ H ₈	36	1-Butene
C ₂ H ₄ Br ₂	82	1,2-Dibromoethane	C ₄ H ₈	37	cis-2-Butene
C ₂ H ₄ Cl ₂	88	1,1-Dichloroethane	C ₄ H ₈	38	trans-2-Butene
C ₂ H ₄ Cl ₂	89	1,2-Dichloroethane	C ₄ H ₈	64	Cyclobutane
C ₂ H ₄ F ₂	97	1,1-Difluoroethane	C ₄ H ₈	238	2-Methyl propene
C ₂ H ₄ F ₂	98	1,2-Difluoroethane	C ₄ H ₈ O	44	Butyraldehyde
C ₂ H ₄ O	1	Acetaldehyde	C ₄ H ₈ O	222	Methylethyl ketone
C ₂ H ₄ O	139	Ethylene oxide	C ₄ H ₈ O	320	Tetrahydrofuran
C ₂ H ₄ O ₂	3	Acetic acid	C ₄ H ₈ O ₂	45	Butyric acid
C ₂ H ₄ O ₂	224	Methyl formate	C ₄ H ₈ O ₂	120	1,4-Dioxane
C ₂ H ₅ Br	27	Bromoethane	C ₄ H ₈ O ₂	127	Ethyl acetate
C ₂ H ₅ Cl	54	Chloroethane	C ₄ H ₈ O ₂	189	Isobutyric acid
C ₂ H ₅ Cl ₃ Si	148	Ethyltrichlorosilane	C ₄ H ₈ O ₂	239	Methyl propionate
C ₂ H ₅ F	151	Fluoroethane	C ₄ H ₈ O ₂	306	Propyl formate
C ₂ H ₅ N	138	Ethyleneimine	C ₄ H ₈ S	322	Tetrahydrothiophene
C ₂ H ₅ NO	2	Acetamide	C ₄ H ₁₀	31	Butane
C ₂ H ₅ NO ₂	248	Nitroethane	C ₄ H ₁₀	236	2-Methylpropane
C ₂ H ₆	125	Ethane	C ₄ H ₁₀ O	34	1-Butanol
C ₂ H ₆ O	112	Dimethyl ether	C ₄ H ₁₀ O	35	2-Butanol
C ₂ H ₆ O	126	Ethanol	C ₄ H ₁₀ O	95	Diethyl ether
C ₂ H ₆ O ₃	137	Ethylene glycol	C ₄ H ₁₀ O	237	2-Methyl-2-propanol
C ₂ H ₆ OS	118	Dimethyl sulfoxide	C ₄ H ₁₀ O	240	Methylpropyl ether
C ₂ H ₆ S	117	Dimethyl sulfide	C ₄ H ₁₀ O	228	Methylisopropyl ether
C ₂ H ₆ S ₂	145	Ethyl mercaptan	C ₄ H ₁₀ O ₂	32	1,2-Butanediol
C ₂ H ₇ N	111	Dimethyl disulfide	C ₄ H ₁₀ O ₂	33	1,3-Butanediol
C ₂ H ₇ N	106	Dimethyl amine	C ₄ H ₁₀ O ₂	103	1,1-Dimethoxyethane
C ₂ H ₇ N	128	Ethyl amine	C ₄ H ₁₀ S	41	Butyl mercaptan
C ₂ H ₈ N ₂	136	Ethylendiamine	C ₄ H ₁₀ S	42	sec-Butyl mercaptan
C ₂ H ₈ Si	116	Dimethylsilane	C ₄ H ₁₀ S	96	Diethyl sulfide
C ₂ N ₂	63	Cyanogen	C ₄ H ₁₀ S	230	Methylisopropyl sulfide
C ₃ H ₃ N	10	Acrylonitrile	C ₄ H ₁₀ S	241	Methylpropyl sulfide
C ₃ H ₄	197	Methyl acetylene	C ₄ H ₁₁ N	94	Diethyl amine
C ₃ H ₄	294	Propadiene	C ₄ H ₁₁ NO ₂	93	Diethanol amine
C ₃ H ₄ O	8	Acrolein	C ₅ H ₆	207	2-Methyl-1-butene-3-yne
C ₃ H ₄ O ₂	9	Acrylic acid	C ₅ H ₈	70	Cyclopentene
C ₃ H ₄ O ₄	191	Malonic acid	C ₅ H ₈	201	3-Methyl-1,2-butadiene
C ₃ H ₅ N	301	Propionitrile	C ₅ H ₈	210	3-Methyl-1-butyne
C ₃ H ₆	71	Cyclopropane	C ₅ H ₈	288	1-Pentyne
C ₃ H ₆	305	Propylene	C ₅ H ₈	289	2-Pentyne
C ₃ H ₆ Cl ₂	91	1,1-Dichloropropane	C ₅ H ₈ O ₂	232	Methyl methacrylate

TABLE 2-7 Formula Index of Substances in Tables 2-8, 2-32, 2-141, 2-150, 2-153, 2-155, 2-156, 2-179, 2-312, 2-313, 2-314, and 2-315 (Continued)

Formula	No.	Name	Formula	No.	Name
C ₅ H ₁₀	69	Cyclopentane	C ₇ H ₈	325	Toluene
C ₅ H ₁₀	205	2-Methyl-1-butene	C ₇ H ₈ O	13	Anisole
C ₅ H ₁₀	206	2-Methyl-2-butene	C ₇ H ₈ O	21	Benzyl alcohol
C ₅ H ₁₀	285	1-Pentene	C ₇ H ₈ O	59	<i>m</i> -Cresol
C ₅ H ₁₀ O	229	Methylisopropyl ketone	C ₇ H ₈ O	60	<i>o</i> -Cresol
C ₅ H ₁₀ O	278	Pentanal	C ₇ H ₈ O	61	<i>p</i> -Cresol
C ₅ H ₁₀ O	283	2-Pentanone	C ₇ H ₈ S	23	Benzyl mercaptan
C ₅ H ₁₀ O	284	3-Pentanone	C ₇ H ₁₂	168	1-Heptyne
C ₅ H ₁₀ O ₂	146	Ethyl propionate	C ₇ H ₁₄	134	Ethylcyclopentane
C ₅ H ₁₀ O ₂	203	2-Methylbutanoic acid	C ₇ H ₁₄	166	1-Heptene
C ₅ H ₁₀ O ₂	211	Methyl butyrate	C ₇ H ₁₄	213	Methylcyclohexane
C ₅ H ₁₀ O ₂	280	Pentanoic acid	C ₇ H ₁₄ O	102	Di-isopropyl ketone
C ₅ H ₁₀ O ₂	302	Propyl acetate	C ₇ H ₁₄ O	159	Heptanal
C ₅ H ₁₂	202	2-Methylbutane	C ₇ H ₁₄ O	164	3-Heptanone
C ₅ H ₁₂	279	Pentane	C ₇ H ₁₄ O	165	2-Heptanone
C ₅ H ₁₂ O	143	Ethylisopropyl ether	C ₇ H ₁₄ O	214	1-Methylcyclohexanol
C ₅ H ₁₂ O	147	Ethylpropyl ether	C ₇ H ₁₄ O	215	<i>cis</i> -2-Methylcyclohexanol
C ₅ H ₁₂ O	204	3-Methyl-1-butanol	C ₇ H ₁₄ O	216	<i>trans</i> -2-Methylcyclohexanol
C ₅ H ₁₂ O	208	Methylbutyl ether	C ₇ H ₁₄ O ₂	161	Heptanoic acid
C ₅ H ₁₂ O	225	Methylisobutyl ether	C ₇ H ₁₆	114	2,3-Dimethylpentane
C ₅ H ₁₂ O	244	Methyl <i>tert</i> -butyl ether	C ₇ H ₁₆	160	Heptane
C ₅ H ₁₂ O	281	1-Pentanol	C ₇ H ₁₆ O	162	1-Heptanol
C ₅ H ₁₂ O	282	2-Pentanol	C ₇ H ₁₆ O	163	2-Heptanol
C ₅ H ₁₂ O ₂	104	1,2-Dimethoxypropane	C ₇ H ₁₆ S	167	Heptyl mercaptan
C ₅ H ₁₂ S	209	Methylbutyl sulfide	C ₈ H ₄ O ₃	293	Therphthalic anhydride
C ₅ H ₁₂ S	286	2-Pentyl mercaptan	C ₈ H ₆ O ₄	317	Terephthalic acid
C ₅ H ₁₂ S	287	Pentyl mercaptan	C ₈ H ₈	312	Styrene
C ₆ H ₃ N ₃ O ₆	334	1,3,5-Trinitrobenzene	C ₈ H ₈ O ₂	200	Methyl benzoate
C ₆ H ₄ Cl ₂	85	<i>m</i> -Dichlorobenzene	C ₈ H ₁₀	129	Ethylbenzene
C ₆ H ₄ Cl ₂	86	<i>o</i> -Dichlorobenzene	C ₈ H ₁₀	343	<i>m</i> -Xylene
C ₆ H ₄ Cl ₂	87	<i>p</i> -Dichlorobenzene	C ₈ H ₁₀	344	<i>o</i> -Xylene
C ₆ H ₄ O ₂	310	Quinone	C ₈ H ₁₀	345	<i>p</i> -Xylene
C ₆ H ₅ Br	26	Bromobenzene	C ₈ H ₁₄	273	1-Octyne
C ₆ H ₅ Cl	53	Chlorobenzene	C ₈ H ₁₆	108	1,1-Dimethylcyclohexane
C ₆ H ₅ F	150	Fluorobenzene	C ₈ H ₁₆	109	<i>cis</i> -1,2-Dimethylcyclohexane
C ₆ H ₆	16	Benzene	C ₈ H ₁₆	110	<i>trans</i> -1,2-Dimethylcyclohexane
C ₆ H ₆ O	291	Phenol	C ₈ H ₁₆	133	Ethylcyclohexane
C ₆ H ₆ S	17	Benzenethiol	C ₈ H ₁₆	271	1-Octene
C ₆ H ₁₀	218	1-Methylcyclopentene	C ₈ H ₁₆ O	264	Octanal
C ₆ H ₁₀	68	Cyclohexene	C ₈ H ₁₆ O	269	2-Octanone
C ₆ H ₁₀	178	3-Hexyne	C ₈ H ₁₆ O	270	3-Octanone
C ₆ H ₁₀	180	1-Hexyne	C ₈ H ₁₆ O ₂	141	2-Ethyl hexanoic acid
C ₆ H ₁₀	181	2-Hexyne	C ₈ H ₁₆ O ₂	266	Octanoic acid
C ₆ H ₁₀	219	3-Methylcyclopentene	C ₈ H ₁₈	265	Octane
C ₆ H ₁₀ O	67	Cyclohexanone	C ₈ H ₁₈	323	2,2,3,3-Tetramethylbutane
C ₆ H ₁₂	65	Cyclohexane	C ₈ H ₁₈	332	2,2,4-Trimethylpentane
C ₆ H ₁₂	177	1-Hexene	C ₈ H ₁₈	333	2,3,3-Trimethylpentane
C ₆ H ₁₂	217	Methylcyclopentane	C ₈ H ₁₈ O	84	Dibutyl ether
C ₆ H ₁₂ O	66	Cyclohexanol	C ₈ H ₁₈ O	142	Ethylhexyl ether
C ₆ H ₁₂ O	144	Ethylisopropyl ketone	C ₈ H ₁₈ O	267	1-Octanol
C ₆ H ₁₂ O	170	Hexanal	C ₈ H ₁₈ O	268	2-Octanol
C ₆ H ₁₂ O	175	2-Hexanone	C ₈ H ₁₈ S	272	Octyl mercaptan
C ₆ H ₁₂ O	176	3-Hexanone	C ₉ H ₁₀	243	alpha-Methyl styrene
C ₆ H ₁₂ O	226	Methylisobutyl ketone	C ₉ H ₁₀ O ₂	130	Ethyl benzoate
C ₆ H ₁₂ O ₂	39	Butyl acetate	C ₉ H ₁₂	62	Cumene
C ₆ H ₁₂ O ₂	131	2-Ethyl butanoic acid	C ₉ H ₁₂	304	Propylbenzene
C ₆ H ₁₂ O ₂	132	Ethyl butyrate	C ₉ H ₁₂	330	1,2,3-Trimethylbenzene
C ₆ H ₁₂ O ₂	172	Hexanoic acid	C ₉ H ₁₂	331	1,2,4-Trimethylbenzene
C ₆ H ₁₂ S	72	Cyclohexyl mercaptan	C ₉ H ₁₂ O	22	Benzyl ethyl ether
C ₆ H ₁₄	107	2,3-Dimethylbutane	C ₉ H ₁₄	298	Propenylcyclohexene
C ₆ H ₁₄	171	Hexane	C ₉ H ₁₆	262	1-Nonyne
C ₆ H ₁₄	234	2-Methylpentane	C ₉ H ₁₈	260	1-Nonene
C ₆ H ₁₄ O	101	Di-isopropyl ether	C ₉ H ₁₈ O	255	Nonanal
C ₆ H ₁₄ O	173	1-Hexanol	C ₉ H ₁₈ O ₂	233	2-Methyloctanoic acid
C ₆ H ₁₄ O	174	2-Hexanol	C ₉ H ₁₈ O ₂	257	Nonanoic acid
C ₆ H ₁₄ O	235	Methyl pentyl ether	C ₉ H ₂₀	256	Nonane
C ₆ H ₁₄ S	179	Hexyl mercaptan	C ₉ H ₂₀ O	258	1-Nonanol
C ₆ H ₁₅ N	100	Di-isopropyl amine	C ₉ H ₂₀ O	259	2-Nonanol
C ₆ H ₁₅ N	122	Dipropyl amine	C ₉ H ₂₀ S	261	Nonyl mercaptan
C ₆ H ₁₅ N	328	Triethyl amine	C ₁₀ H ₈	246	Naphthalene
C ₇ H ₅ N	19	Benzonitrile	C ₁₀ H ₁₀ O ₄	115	Dimethyl phthalate
C ₇ H ₅ N ₃ O ₆	335	2,4,6-Trinitrotoluene	C ₁₀ H ₁₀ O ₄	119	Dimethyl terephthalate
C ₇ H ₅ NO	292	Phenyl isocyanate	C ₁₀ H ₁₂	321	1,2,3,4-Tetrahydronaphthalene
C ₇ H ₆ O ₂	18	Benzoic acid	C ₁₀ H ₁₄	40	Butylbenzene
C ₇ H ₇ NO	15	Benzamide	C ₁₀ H ₁₈	79	1-Decyne

2-54 PHYSICAL AND CHEMICAL DATA

TABLE 2-7 Formula Index of Substances in Tables 2-8, 2-32, 2-141, 2-150, 2-153, 2-155, 2-156, 2-179, 2-312, 2-313, 2-314, and 2-315 (Concluded)

Formula	No.	Name	Formula	No.	Name
C ₁₀ H ₂₀	77	1-Decene	D ₂	80	Deuterium
C ₁₀ H ₂₀ O	73	Decanal	F ₂	149	Fluorine
C ₁₀ H ₂₀ O ₂	75	Decanoic acid	F ₃ N	250	Nitrogen trifluoride
C ₁₀ H ₂₂	74	Decane	F ₄ Si	311	Silicon tetrafluoride
C ₁₀ H ₂₂ O	76	1-Decanol	F ₆ S	315	Sulfur hexafluoride
C ₁₀ H ₂₂ S	78	Decyl mercaptan	HBr	184	Hydrogen bromide
C ₁₁ H ₂₄	336	Undecane	HCl	185	Hydrogen chloride
C ₁₁ H ₂₄ O	337	1-Undecanol	HF	187	Hydrogen fluoride
C ₁₂ H ₁₀	24	Biphenyl	H ₂	183	Hydrogen
C ₁₂ H ₁₀ O	121	Diphenyl ether	H ₂ O	342	Water
C ₁₂ H ₂₆	123	Dodecane	H ₂ S	188	Hydrogen sulfide
C ₁₃ H ₁₀ O	20	Benzophenone	H ₃ N	12	Ammonia
C ₁₃ H ₂₈	327	Tridecane	H ₄ N ₂	182	Hydrazine
C ₁₄ H ₁₀	290	Phenanthrene	He	157	Helium-4
C ₁₄ H ₃₀	319	Tetradecane	NO	253	Nitric oxide
C ₁₅ H ₃₂	277	Pentadecane	N ₂	249	Nitrogen
C ₁₆ H ₃₄	169	Hexadecane	N ₂ O	252	Nitrous oxide
C ₁₇ H ₃₆	158	Heptadecane	Ne	247	Neon
C ₁₈ H ₁₄	318	<i>o</i> -Terphenyl	O ₂	275	Oxygen
C ₁₈ H ₃₈	263	Octadecane	O ₂ S	314	Sulfur dioxide
C ₁₉ H ₄₀	254	Nonadecane	O ₃	276	Ozone
C ₂₀ H ₄₂	124	Eicosane	O ₃ S	316	Sulfur trioxide
Cl ₂	52	Chlorine			

TABLE 2-8 Vapor Pressure of Inorganic and Organic Liquids, $\ln P = C1 + C2/T + C3 \ln T + C4 T^{C5}$, P in Pa

No.	Name	Formula	CAS no.	C1	C2	C3	C4	C5	$T_{\min}$, K	P at $T_{\min}$	$T_{\max}$, K	P at $T_{\max}$
1	Acetaldehyde	C ₂ H ₄ O	75-07-0	193.69	-8,036.7	-29.502	4.3678E-02	1	150.15	3.23E-01	466	5.565E+06
2	Acetamide	C ₂ H ₅ NO	60-35-5	125.81	-12,376	-14.589	5.0824E-06	2	353.33	3.36E+02	761	6.569E+06
3	Acetic acid	C ₂ H ₄ O ₂	64-19-7	53.27	-6,304.5	-4.2985	8.8865E-18	6	289.81	1.28E+03	591.95	5.739E+06
4	Acetic anhydride	C ₄ H ₆ O ₃	108-24-7	100.95	-8,873.2	-11.451	6.1316E-06	2	200.15	2.20E-02	606	3.970E+06
5	Acetone	C ₃ H ₆ O	67-64-1	69.006	-5,599.6	-7.0985	6.2237E-06	2	178.45	2.79E+00	508.2	4.709E+06
6	Acetonitrile	C ₂ H ₃ N	75-05-8	58.302	-5,385.6	-5.4954	5.3634E-06	2	229.32	1.87E+02	545.5	4.852E+06
7	Acetylene	C ₂ H ₂	74-86-2	39.63	-2,552.2	-2.78	2.3930E-16	6	192.4	1.27E+05	308.3	6.106E+06
8	Acrolein	C ₃ H ₄ O	107-02-8	138.4	-7,122.7	-19.638	2.6447E-02	1	185.45	1.03E+01	506	5.020E+06
9	Acrylic acid	C ₃ H ₄ O ₂	79-10-7	46.745	-6,587.1	-3.2208	5.2253E-07	2	286.15	2.57E+02	615	5.661E+06
10	Acrylonitrile	C ₃ H ₃ N	107-13-1	87.604	-6,392.7	-10.101	1.0891E-05	2	189.63	3.68E+00	535	4.480E+06
11	Air	Mixture	132259-10-0	21.662	-692.39	-0.392	4.7574E-03	1	59.15	5.64E+03	132.45	3.793E+06
12	Ammonia	H ₃ N	7664-41-7	90.483	-4,669.7	-11.607	1.7194E-02	1	195.41	6.11E+03	405.65	1.130E+07
13	Anisole	C ₇ H ₈ O	100-66-3	128.06	-9,307.7	-16.693	1.4919E-02	1	235.65	2.45E+00	645.6	4.273E+06
14	Argon	Ar	7440-37-1	42.127	-1,093.1	-4.1425	5.7254E-05	2	83.78	6.87E+04	150.86	4.896E+06
15	Benzamide	C ₇ H ₇ NO	55-21-0	85.474	-11,932	-8.3348	1.2850E-18	6	403	3.55E+02	824	5.047E+06
16	Benzene	C ₆ H ₆	71-43-2	83.107	-6,486.2	-9.2194	6.9844E-06	2	278.68	4.76E+03	562.05	4.875E+06
17	Benzenethiol	C ₆ H ₆ S	108-98-5	77.765	-8,455.1	-7.7404	4.3089E-18	6	258.27	7.68E+00	689	4.728E+06
18	Benzoic acid	C ₇ H ₆ O ₂	65-85-0	88.513	-11,829	-8.6826	2.3248E-19	6	395.45	7.96E+02	751	4.469E+06
19	Benzonitrile	C ₇ H ₅ N	100-47-0	138.5	-11,195	-17.085	9.5641E-06	2	260.4	3.08E+00	699.35	4.243E+06
20	Benzophenone	C ₁₃ H ₁₀ O	119-61-9	88.404	-11,769	-8.9014	1.9334E-18	6	321.35	1.49E+00	830	3.357E+06
21	Benzyl alcohol	C ₇ H ₈ O	100-51-6	100.68	-11,059	-10.709	3.0582E-18	6	257.85	1.88E-01	720.15	4.372E+06
22	Benzyl ethyl ether	C ₉ H ₁₂ O	539-30-0	68.541	-7,886.2	-6.5804	2.4285E-06	2	275.65	2.31E+01	662	3.113E+06
23	Benzyl mercaptan	C ₇ H ₈ S	100-53-8	118.02	-10,527	-13.91	6.4794E-06	2	243.95	2.98E-01	718	4.074E+06
24	Biphenyl	C ₁₂ H ₁₀	92-52-4	77.314	-9,910.4	-7.5079	2.2385E-18	6	342.2	9.42E+01	773	3.407E+06
25	Bromine	Br ₂	7726-95-6	108.26	-6,592	-14.16	1.6043E-02	1	265.85	5.85E+03	584.15	1.028E+07
26	Bromobenzene	C ₆ H ₅ Br	108-86-1	63.749	-7,130.2	-5.879	5.2136E-18	6	242.43	7.84E+00	670.15	4.520E+06
27	Bromoethane	C ₂ H ₅ Br	74-96-4	62.217	-5,113.3	-5.9761	4.7174E-17	6	154.55	3.72E-01	503.8	6.290E+06
28	Bromomethane	CH ₃ Br	74-83-9	72.586	-4,698.6	-7.9966	1.1553E-05	2	179.47	1.95E+02	467	7.997E+06
29	1,2-Butadiene	C ₄ H ₆	590-19-2	39.714	-3,769.9	-2.6407	6.9379E-18	6	136.95	4.47E-01	452	4.361E+06
30	1,3-Butadiene	C ₄ H ₆	106-99-0	75.572	-4,621.9	-8.5323	1.2269E-05	2	164.25	6.92E+01	425	4.303E+06
31	Butane	C ₄ H ₁₀	106-97-8	66.343	-4,363.2	-7.046	9.4509E-06	2	134.86	6.74E-01	425.12	3.770E+06
32	1,2-Butanediol	C ₄ H ₁₀ O ₂	584-03-2	103.28	-11,548	-10.925	4.2560E-18	6	220	2.93E-04	680	5.202E+06
33	1,3-Butanediol	C ₄ H ₁₀ O ₂	107-88-0	123.22	-12,620	-13.986	3.9260E-06	2	196.15	3.74E-07	676	4.033E+06
34	1-Butanol	C ₄ H ₁₀ O	71-36-3	106.295	-9,866.4	-11.655	1.0832E-17	6	183.85	2.90E-04	563.1	4.401E+06
35	2-Butanol	C ₄ H ₁₀ O	78-92-2	114.68	-9,850.2	-12.963	1.8738E-17	6	158.45	1.95E-06	535.9	4.182E+06
36	1-Butene	C ₄ H ₈	106-98-9	51.836	-4,019.2	-4.5229	4.8833E-17	6	87.8	6.94E-07	419.5	4.021E+06
37	cis-2-Butene	C ₄ H ₈	590-18-1	72.541	-4,691.2	-7.9776	1.0368E-05	2	134.26	2.72E-01	435.5	4.238E+06
38	trans-2-Butene	C ₄ H ₈	624-64-6	71.704	-4,563.1	-7.9053	1.1319E-05	2	167.62	7.45E+01	428.6	4.100E+06
39	Butyl acetate	C ₆ H ₁₂ O ₂	123-86-4	122.82	-9,253.2	-14.99	1.0470E-05	2	199.65	8.17E-02	575.4	3.087E+06
40	Butylbenzene	C ₁₀ H ₁₄	104-51-8	101.22	-9,255.4	-11.538	5.9208E-06	2	185.3	1.54E-04	660.5	2.882E+06
41	Butyl mercaptan	C ₄ H ₁₀ S	109-79-5	65.382	-6,262.4	-6.2585	1.4943E-17	6	157.46	2.35E-03	570.1	3.973E+06
42	sec-Butyl mercaptan	C ₄ H ₁₀ S	513-53-1	60.649	-5,785.9	-5.6113	1.5877E-17	6	133.02	3.40E-05	554	4.060E+06
43	1-Butyne	C ₄ H ₆	107-00-6	77.004	-5,054.5	-8.5665	1.0161E-05	2	147.43	1.18E+00	440	4.599E+06
44	Butyraldehyde	C ₄ H ₈ O	123-72-8	99.33	-7,083.6	-11.733	1.0027E-05	2	176.75	3.17E-01	537.2	4.323E+06
45	Butyric acid	C ₄ H ₈ O ₂	107-92-6	93.815	-9,942.2	-9.8019	9.3124E-18	6	267.95	6.78E+00	615.7	4.071E+06
46	Butyronitrile	C ₄ H ₇ N	109-74-0	66.32	-6,714.9	-6.3087	1.3516E-17	6	161.25	6.18E-04	582.25	3.787E+06
47	Carbon dioxide	CO ₂	124-38-9	140.54	-4,735	-21.268	4.0909E-02	1	216.58	5.19E+05	304.21	7.390E+06
48	Carbon disulfide	CS ₂	75-15-0	67.114	-4,820.4	-7.5303	9.1695E-03	1	161.11	1.49E+00	552	8.041E+06
49	Carbon monoxide	CO	630-08-0	45.698	-1,076.6	-4.8814	7.5673E-05	2	68.15	1.54E+04	132.92	3.494E+06
50	Carbon tetrachloride	CCl ₄	56-23-5	78.441	-6,128.1	-8.5766	6.8465E-06	2	250.33	1.12E+03	556.35	4.544E+06
51	Carbon tetrafluoride	CF ₄	75-73-0	61.89	-2,296.3	-7.086	3.4687E-05	2	89.56	1.08E+02	227.51	3.742E+06
52	Chlorine	Cl ₂	7782-50-5	71.334	-3,855	-8.5171	1.2378E-02	1	172.12	1.37E+03	417.15	7.793E+06
53	Chlorobenzene	C ₆ H ₅ Cl	108-90-7	54.144	-6,244.4	-4.5343	4.7030E-18	6	227.95	8.45E+00	632.35	4.529E+06
54	Chloroethane	C ₂ H ₅ Cl	75-00-3	65.988	-4,661.3	-6.8586	7.9404E-06	2	134.8	1.25E-01	460.35	5.327E+06
55	Chloroform	CHCl ₃	67-66-3	146.43	-7,792.3	-20.614	2.4578E-02	1	207.15	5.25E+01	536.4	5.554E+06
56	Chloromethane	CH ₃ Cl	74-87-3	64.697	-4,048.1	-6.8066	1.0371E-05	2	175.43	8.71E+02	416.25	6.691E+06
57	1-Chloropropane	C ₃ H ₇ Cl	540-54-5	79.24	-5,718.8	-8.789	8.4486E-06	2	150.35	6.96E-02	503.15	4.581E+06
58	2-Chloropropane	C ₃ H ₇ Cl	75-29-6	46.854	-4,445.5	-3.6533	1.3260E-17	6	155.97	9.08E-01	489	4.510E+06
59	m-Cresol	C ₇ H ₈ O	108-39-4	95.403	-10,581	-10.004	4.3032E-18	6	285.39	5.86E+00	705.85	4.522E+06
60	o-Cresol	C ₇ H ₈ O	95-48-7	210.88	-13,928	-29.483	2.5182E-02	1	304.19	6.53E+01	697.55	5.058E+06

TABLE 2-8 Vapor Pressure of Inorganic and Organic Liquids, $\ln P = C1 + C2/T + C3 \ln T + C4 T^{C5}$, P in Pa (Continued)

No.	Name	Formula	CAS no.	C1	C2	C3	C4	C5	$T_{\min}$, K	P at $T_{\min}$	$T_{\max}$, K	P at $T_{\max}$
61	<i>p</i> -Cresol	C ₇ H ₈ O	106-44-5	118.53	-11,957	-13.293	8.6988E-18	6	307.93	3.45E+01	704.65	5.151E+06
62	Cumene	C ₉ H ₁₂	98-82-8	102.81	-8,674.6	-11.922	7.0048E-06	2	177.14	4.71E-04	631	3.226E+06
63	Cyanogen	C ₂ N ₂	460-19-5	81.565	-4,808.9	-9.3748	1.3901E-05	2	245.25	7.39E+04	400.15	5.961E+06
64	Cyclobutane	C ₄ H ₈	287-23-0	85.899	-4,884.4	-10.883	1.4934E-02	1	182.48	1.80E+02	459.93	4.991E+06
65	Cyclohexane	C ₆ H ₁₂	110-82-7	51.087	-5,226.4	-4.2278	9.7554E-18	6	279.69	5.36E+03	553.8	4.094E+06
66	Cyclohexanol	C ₆ H ₁₂ O	108-93-0	189.19	-14,337	-24.148	1.0740E-05	2	296.6	7.65E+01	650.1	4.265E+06
67	Cyclohexanone	C ₆ H ₁₀ O	108-94-1	85.424	-7,944.4	-9.2862	4.9957E-06	2	242	6.80E+00	653	3.989E+06
68	Cyclohexene	C ₆ H ₁₀	110-83-8	88.184	-6,624.9	-10.059	8.2566E-06	2	169.67	1.04E-01	560.4	4.392E+06
69	Cyclopentane	C ₅ H ₁₀	287-92-3	66.341	-5,198.5	-6.8103	6.1930E-06	2	179.28	9.07E+00	511.7	4.513E+06
70	Cyclopentene	C ₅ H ₈	142-29-0	67.952	-5,187.5	-7.0785	6.8165E-06	2	138.13	1.28E-02	507	4.799E+06
71	Cyclopropane	C ₃ H ₆	75-19-4	40.608	-3,179.6	-2.8937	5.6131E-17	6	145.59	7.80E+01	398	5.494E+06
72	Cyclohexyl mercaptan	C ₆ H ₁₂ S	1569-69-3	85.146	-7,843.7	-9.2982	5.1788E-06	2	189.64	8.24E-03	664	3.970E+06
73	Decanal	C ₁₀ H ₂₀ O	112-31-2	201.64	-15,133	-26.264	1.4625E-05	2	267.15	4.86E-01	674.2	2.599E+06
74	Decane	C ₁₀ H ₂₂	124-18-5	112.73	-9,749.6	-13.245	7.1266E-06	2	243.51	1.39E+00	617.7	2.091E+06
75	Decanoic acid	C ₁₀ H ₂₀ O ₂	334-48-5	123.36	-14,680	-13.474	1.9491E-18	6	304.55	1.50E-01	722.1	2.233E+06
76	1-Decanol	C ₁₀ H ₂₂ O	112-30-1	156.239	-15,212	-18.424	8.5006E-18	6	280.05	1.51E-01	688	2.309E+06
77	1-Decene	C ₁₀ H ₂₀	872-05-9	68.401	-7,776.9	-6.4637	6.3750E-18	6	206.89	2.59E-02	616.6	2.223E+06
78	Decyl mercaptan	C ₁₀ H ₂₂ S	143-10-2	91.91	-10,565	-9.5957	5.7028E-18	6	247.56	2.59E-02	696	2.130E+06
79	1-Decyne	C ₁₀ H ₁₈	764-93-2	142.94	-11,119	-17.818	1.1020E-05	2	229.15	1.60E-01	619.85	2.363E+06
80	Deuterium	D ₂	7782-39-0	18.947	-154.47	-0.5723	3.8899E-02	1	18.73	1.72E+04	38.35	1.663E+06
81	1,1-Dibromoethane	C ₂ H ₄ Br ₂	557-91-5	62.711	-6,503.5	-5.7669	1.0427E-06	2	210.15	2.64E+00	628	6.034E+06
82	1,2-Dibromoethane	C ₂ H ₄ Br ₂	106-93-4	43.751	-5,587.7	-3.0891	8.2664E-07	2	282.85	7.53E+02	650.15	5.475E+06
83	Dibromomethane	CH ₂ Br ₂	74-95-3	86.295	-7,010.3	-9.5972	6.7794E-06	2	220.6	2.13E+01	611	7.170E+06
84	Dibutyl ether	C ₈ H ₁₈ O	142-96-1	72.227	-7,537.6	-7.0596	9.1442E-18	6	175.3	7.14E-04	584.1	2.459E+06
85	<i>m</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	541-73-1	53.187	-6,827.5	-4.3233	2.3112E-18	6	248.39	6.41E+00	683.95	4.070E+06
86	<i>o</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	95-50-1	77.105	-8,111.1	-7.8886	2.7267E-06	2	256.15	6.49E+00	705	4.074E+06
87	<i>p</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	106-46-7	88.31	-8,463.4	-9.6308	4.5833E-06	2	326.14	1.23E+03	684.75	4.070E+06
88	1,1-Dichloroethane	C ₂ H ₄ Cl ₂	75-34-3	66.611	-5,493.1	-6.7301	5.3579E-06	2	176.19	2.21E+00	523	5.106E+06
89	1,2-Dichloroethane	C ₂ H ₄ Cl ₂	107-06-2	92.355	-6,920.4	-10.651	9.1426E-06	2	237.49	2.37E+02	561.6	5.318E+06
90	Dichloromethane	CH ₂ Cl ₂	75-09-2	101.6	-6,541.6	-12.247	1.2311E-05	2	178.01	5.93E+00	510	6.093E+06
91	1,1-Dichloropropane	C ₃ H ₆ Cl ₂	78-99-9	83.495	-6,661.4	-9.2386	6.7652E-06	2	200	4.52E+00	560	4.239E+06
92	1,2-Dichloropropane	C ₃ H ₆ Cl ₂	78-87-5	65.955	-6,015.6	-6.5509	4.3172E-06	2	172.71	8.25E-02	572	4.232E+06
93	Diethanol amine	C ₄ H ₁₁ NO ₂	111-42-2	106.38	-13,714	-11.06	3.2645E-18	6	301.15	1.02E-01	736.6	4.260E+06
94	Diethyl amine	C ₄ H ₁₁ N	109-89-7	49.314	-4,949	-3.9256	9.1978E-18	6	223.35	3.74E+02	496.6	3.674E+06
95	Diethyl ether	C ₄ H ₁₀ O	60-29-7	136.9	-6,954.3	-19.254	2.4508E-02	1	156.85	3.95E-01	466.7	3.641E+06
96	Diethyl sulfide	C ₄ H ₁₀ S	352-93-2	46.705	-5,177.4	-3.5985	1.7147E-06	2	169.20	9.93E-02	557.15	3.961E+06
97	1,1-Difluoroethane	C ₂ H ₄ F ₂	75-37-6	73.491	-4,385.9	-8.1851	1.2978E-05	2	154.56	6.45E+01	386.44	4.507E+06
98	1,2-Difluoroethane	C ₂ H ₄ F ₂	624-72-6	84.625	-5,217.4	-9.871	1.3050E-05	2	215	2.83E+03	445	4.372E+06
99	Difluoromethane	CH ₂ F ₂	75-10-5	69.132	-3,847.7	-7.5868	1.5065E-05	2	136.95	5.43E+01	351.255	5.760E+06
100	Di-isopropyl amine	C ₆ H ₁₅ N	108-18-9	462.84	-18,227	-73.734	9.2794E-02	1	176.85	4.47E-03	523.1	3.199E+06
101	Di-isopropyl ether	C ₆ H ₁₄ O	108-20-3	41.631	-4,668.7	-2.8551	6.3693E-04	1	187.65	6.86E+00	500.05	2.869E+06
102	Di-isopropyl ketone	C ₇ H ₁₄ O	565-80-0	50.868	-6,036.5	-4.066	1.1326E-06	2	204.81	8.21E-01	576	3.017E+06
103	1,1-Dimethoxyethane	C ₃ H ₁₀ O ₂	534-15-6	53.637	-5,251.2	-4.5649	1.6754E-17	6	159.95	9.45E-02	507.8	3.773E+06
104	1,2-Dimethoxypropane	C ₃ H ₁₂ O ₂	7778-85-0	62.097	-6,174.9	-5.715	1.2323E-17	6	226.1	4.50E+01	543	3.447E+06
105	Dimethyl acetylene	C ₄ H ₆	503-17-3	66.592	-4,999.8	-6.8387	6.6793E-06	2	240.91	6.12E+03	473.2	4.870E+06
106	Dimethyl amine	C ₂ H ₇ N	124-40-3	71.738	-5,302	-7.3324	6.4200E-17	6	180.96	7.56E+01	437.2	5.258E+06
107	2,3-Dimethylbutane	C ₆ H ₁₄	79-29-8	77.161	-5,691.1	-8.501	8.0325E-06	2	145.19	1.52E-02	500	3.130E+06
108	1,1-Dimethylcyclohexane	C ₈ H ₁₆	590-66-9	81.184	-6,927	-8.8498	5.4580E-06	2	239.66	6.06E+01	591.15	2.939E+06
109	<i>cis</i> -1,2-Dimethylcyclohexane	C ₈ H ₁₆	2207-01-4	78.952	-7,075.4	-8.4344	4.5035E-06	2	223.16	6.41E+00	606.15	2.939E+06
110	<i>trans</i> -1,2-Dimethylcyclohexane	C ₈ H ₁₆	6876-23-9	78.429	-6,882.1	-8.4129	4.9831E-06	2	184.99	8.04E-02	596.15	2.938E+06
111	Dimethyl disulfide	C ₂ H ₆ S ₂	624-92-0	81.045	-6,941.3	-8.777	5.5501E-06	2	188.44	2.07E-01	615	5.363E+06
112	Dimethyl ether	C ₂ H ₆ O	115-10-6	44.704	-3,525.6	-3.4444	5.4574E-17	6	131.65	3.05E+00	400.1	5.274E+06
113	<i>N,N</i> -Dimethyl formamide	C ₃ H ₇ NO	68-12-2	82.762	-7,955.5	-8.8038	4.2431E-06	2	212.72	1.95E-01	649.6	4.365E+06
114	2,3-Dimethylpentane	C ₇ H ₁₆	565-59-3	78.335	-6,348.7	-8.5105	6.4311E-06	2	160	1.26E-02	537.3	2.882E+06
115	Dimethyl phthalate	C ₁₀ H ₁₀ O ₄	131-11-3	72.517	-10,415	-6.755	1.3269E-06	2	274.18	3.72E-02	766	2.780E+06
116	Dimethylsilane	C ₂ H ₆ Si	1111-74-6	63.08	-4,062.3	-6.425	1.5115E-16	6	122.93	4.15E-01	402	3.561E+06
117	Dimethyl sulfide	C ₂ H ₆ S	75-18-3	84.39	-5,740.6	-9.6454	1.0073E-05	2	174.88	7.86E+00	503.04	5.533E+06
118	Dimethyl sulfoxide	C ₂ H ₆ OS	67-68-5	56.273	-7,620.6	-4.6279	4.3819E-07	2	291.67	5.02E+01	729	5.648E+06
119	Dimethyl terephthalate	C ₁₀ H ₁₀ O ₄	120-61-6	43.541	-8,204.8	-2.7519	1.0466E-18	6	413.8	1.26E+03	772	2.778E+06
120	1,4-Dioxane	C ₄ H ₈ O ₂	123-91-1	44.494	-5,406.7	-3.1287	2.8913E-18	6	284.95	2.53E+03	587	5.158E+06

121	Diphenyl ether	C ₁₂ H ₁₀ O	101-84-8	59.969	-8,585.5	-5.1538	1.9983E-18	6	300.03	7.09E+00	766.8	3.097E+06
122	Dipropyl amine	C ₆ H ₁₅ N	142-84-7	54	-6,018.5	-4.4981	9.9684E-18	6	210.15	3.69E+00	550	3.111E+06
123	Dodecane	C ₁₂ H ₂₆	112-40-3	137.47	-11,976	-16.698	8.0906E-06	2	263.57	6.15E-01	658	1.822E+06
124	Eicosane	C ₂₀ H ₄₂	112-95-8	203.66	-19,441	-25.525	8.8382E-06	2	309.58	9.26E-03	768	1.175E+06
125	Ethane	C ₂ H ₆	74-84-0	51.857	-2,598.7	-5.1283	1.4913E-05	2	90.35	1.13E+00	305.32	4.852E+06
126	Ethanol	C ₂ H ₆ O	64-17-5	73.304	-7,122.3	-7.1424	2.8853E-06	2	159.05	4.96E-04	514	6.109E+06
127	Ethyl acetate	C ₄ H ₈ O ₂	141-78-6	66.824	-6,227.6	-6.41	1.7914E-17	6	189.6	1.43E+00	523.3	3.850E+06
128	Ethyl amine	C ₂ H ₇ N	75-04-7	81.56	-5,596.9	-9.0779	8.7920E-06	2	192.15	1.52E+02	456.15	5.594E+06
129	Ethylbenzene	C ₈ H ₁₀	100-41-4	89.063	-7,733.7	-9.917	5.9860E-06	2	178.2	3.91E-03	617.15	3.590E+06
130	Ethyl benzoate	C ₉ H ₁₀ O ₂	93-89-0	52.923	-7,531.7	-4.2347	1.1835E-06	2	238.45	1.69E-01	698	3.203E+06
131	2-Ethyl butanoic acid	C ₆ H ₁₂ O ₂	88-09-5	90.464	-10,243	-9.2836	5.2573E-18	6	258.15	4.63E-01	655	3.403E+06
132	Ethyl butyrate	C ₈ H ₁₂ O ₂	105-54-4	57.661	-6,346.5	-5.032	8.2534E-18	6	175.15	1.04E-02	571	2.935E+06
133	Ethylcyclohexane	C ₈ H ₁₆	1678-91-7	80.208	-7,203.2	-8.6023	4.5901E-06	2	161.84	3.57E-04	609.15	3.041E+06
134	Ethylcyclopentane	C ₇ H ₁₄	1640-89-7	88.671	-7,012.7	-10.045	7.4578E-06	2	134.71	3.71E-06	569.5	3.412E+06
135	Ethylene	C ₂ H ₄	74-85-1	53.963	-2,443	-5.5643	1.9079E-05	2	104	1.26E+02	282.34	5.032E+06
136	Ethylenediamine	C ₂ H ₆ N ₂	107-15-3	73.51	-7,572.7	-7.1435	1.2124E-17	6	284.29	6.78E+02	593	6.290E+06
137	Ethylene glycol	C ₂ H ₆ O ₂	107-21-1	84.09	-10,411	-8.1976	1.6536E-18	6	260.15	2.19E-01	720	8.257E+06
138	Ethyleneimine	C ₂ H ₅ N	151-56-4	66.51	-6,019.2	-6.3332	1.0394E-17	6	195.2	9.71E+00	537	6.850E+06
139	Ethylene oxide	C ₂ H ₄ O	75-21-8	91.944	-5,293.4	-11.682	1.4902E-02	1	160.65	7.79E+00	469.15	7.255E+06
140	Ethyl formate	C ₃ H ₆ O ₂	109-94-4	73.833	-5,817	-7.809	6.3200E-06	2	193.55	1.81E+01	508.4	4.708E+06
141	2-Ethyl hexanoic acid	C ₈ H ₁₆ O ₂	149-57-5	117.52	-12,991	-12.895	6.1306E-18	6	235	2.86E-04	674.6	2.788E+06
142	Ethylhexyl ether	C ₈ H ₁₈ O	5756-43-4	77.523	-7,978.8	-7.7757	1.0076E-17	6	180	7.60E-04	583	2.460E+06
143	Ethylisopropyl ether	C ₆ H ₁₂ O	625-54-7	57.723	-5,236.9	-5.2136	2.2998E-17	6	140	4.31E-03	489	3.415E+06
144	Ethylisopropyl ketone	C ₈ H ₁₂ O	565-69-5	57.459	-6,356.8	-4.9545	5.2015E-18	6	204.15	9.70E-01	567	3.293E+06
145	Ethyl mercaptan	C ₂ H ₆ S	75-08-1	65.551	-5,027.4	-6.6853	6.3208E-06	2	125.26	1.14E-03	499.15	5.492E+06
146	Ethyl propionate	C ₅ H ₁₀ O ₂	105-37-3	105.64	-8,007	-12.477	9.0000E-06	2	199.25	7.80E-01	546	3.337E+06
147	Ethylpropyl ether	C ₅ H ₁₂ O	628-32-0	86.898	-6,646.4	-9.5758	5.9615E-17	6	145.65	1.61E-03	500.23	3.372E+06
148	Ethyltrichlorosilane	C ₂ H ₅ Cl ₃ Si	115-21-9	62.614	-6,148.2	-5.84	1.0900E-17	6	167.55	1.85E-02	559.95	3.320E+06
149	Fluorine	F ₂	7782-41-4	42.393	-1,103.3	-4.1203	5.7815E-05	2	53.48	2.53E+02	144.12	5.167E+06
150	Fluorobenzene	C ₆ H ₅ F	462-06-6	51.915	-5,439	-4.2896	8.7527E-18	6	230.94	1.51E+02	560.09	4.544E+06
151	Fluoroethane	C ₂ H ₅ F	353-36-6	56.639	-3,576.5	-5.5801	9.8969E-06	2	129.95	8.37E+00	375.31	5.006E+06
152	Fluoromethane	CH ₃ F	593-53-3	59.123	-3,043.7	-6.1845	1.6637E-05	2	131.35	4.33E+02	317.42	5.875E+06
153	Formaldehyde	CH ₂ O	50-00-0	101.51	-4,917.2	-13.765	2.2031E-02	1	181.15	8.87E+02	408	6.594E+06
154	Formamide	CH ₃ NO	75-12-7	100.3	-10,763	-10.946	3.8503E-06	2	275.6	1.04E+00	771	7.751E+06
155	Formic acid	CH ₂ O ₂	64-18-6	50.323	-5,378.2	-4.203	3.4697E-06	2	281.45	2.40E+03	588	5.807E+06
156	Furan	C ₄ H ₄ O	110-00-9	74.738	-5,417	-8.0636	7.4700E-06	2	187.55	5.00E+01	490.15	5.550E+06
157	Helium-4	He	7440-59-7	11.533	-8.99	0.6724	2.7430E-01	1	1.76	1.46E+03	5.2	2.285E+05
158	Heptadecane	C ₁₇ H ₃₆	629-78-7	156.95	-15,557	-18.966	6.4559E-06	2	295.13	4.65E-02	736	1.344E+06
159	Heptanal	C ₇ H ₁₄ O	111-71-7	92.252	-8,349	-10.274	5.9252E-06	2	229.8	1.45E+00	616.8	3.155E+06
160	Heptane	C ₇ H ₁₆	142-82-5	87.829	-6,996.4	-9.8802	7.2099E-06	2	182.57	1.83E-01	540.2	2.719E+06
161	Heptanoic acid	C ₇ H ₁₄ O ₂	111-14-8	120.47	-13,106	-13.31	5.8384E-18	6	265.83	4.34E-02	677.3	3.039E+06
162	1-Heptanol	C ₇ H ₁₆ O	111-70-6	147.41	-13,466	-17.353	1.1284E-17	6	239.15	1.95E-02	632.3	3.013E+06
163	2-Heptanol	C ₇ H ₁₆ O	543-49-7	124.23	-11,637	-14.148	6.9486E-17	5.7	230	3.68E-02	608.3	2.995E+06
164	3-Heptanone	C ₇ H ₁₄ O	106-35-4	78.463	-8,077.2	-7.9062	8.0521E-18	6	234.15	2.30E+00	606.6	2.919E+06
165	2-Heptanone	C ₇ H ₁₄ O	110-43-0	75.494	-7,896.5	-7.5047	8.9130E-18	6	238.15	3.54E+00	611.4	2.946E+06
166	1-Heptene	C ₇ H ₁₄	592-76-7	65.922	-6,189	-6.3629	2.0091E-17	6	154.12	1.86E-03	537.4	2.921E+06
167	Heptyl mercaptan	C ₇ H ₁₆ S	1639-09-4	79.858	-8,501.8	-8.1043	8.1501E-18	6	229.92	3.05E-01	645	2.772E+06
168	1-Heptyne	C ₇ H ₁₂	628-71-7	59.083	-6,031.8	-5.3072	1.4357E-17	6	192.22	8.15E-01	547	3.209E+06
169	Hexadecane	C ₁₆ H ₃₄	544-76-3	156.06	-15,015	-18.941	6.8172E-06	2	291.31	9.23E-02	723	1.411E+06
170	Hexanal	C ₆ H ₁₂ O	66-25-1	81.507	-7,776.8	-8.4516	1.5143E-17	6	217.15	1.25E+00	591	3.461E+06
171	Hexane	C ₆ H ₁₄	110-54-3	104.65	-6,995.5	-12.702	1.2381E-05	2	177.83	9.02E-01	507.6	3.045E+06
172	Hexanoic acid	C ₆ H ₁₂ O ₂	142-62-1	114.05	-12,332	-12.45	5.6253E-18	6	269.25	2.43E-01	660.2	3.284E+06
173	1-Hexanol	C ₆ H ₁₄ O	111-27-3	135.421	-12,288	-15.732	1.2701E-17	6	228.55	2.25E-02	611.3	3.441E+06
174	2-Hexanol	C ₆ H ₁₄ O	626-93-7	109.42	-10,449	-12.051	2.6122E-46	16	223	7.44E-02	585.3	3.298E+06
175	2-Hexanone	C ₆ H ₁₂ O	591-78-6	107.44	-8,528.6	-12.679	8.4606E-06	2	217.35	1.45E+00	587.61	3.286E+06
176	3-Hexanone	C ₆ H ₁₂ O	589-38-8	73.155	-7,242.9	-7.2569	1.2741E-17	6	217.5	2.22E+00	582.82	3.322E+06
177	1-Hexene	C ₆ H ₁₂	592-41-6	51.024	-4,986.4	-4.2463	1.6768E-17	6	133.39	7.66E-04	504	3.212E+06
178	3-Hexyne	C ₆ H ₁₀	928-49-4	47.091	-5,104	-3.6371	5.1621E-04	1	170.05	2.20E-01	544	3.540E+06
179	Hexyl mercaptan	C ₆ H ₁₄ S	111-31-9	68.467	-7,390.5	-6.5456	7.7611E-18	6	192.62	1.31E-02	623	3.079E+06
180	1-Hexyne	C ₆ H ₁₀	693-02-7	133.2	-7,492.9	-18.405	2.2062E-02	1	141.25	3.92E-04	516.2	3.635E+06
181	2-Hexyne	C ₆ H ₁₀	764-35-2	123.71	-7,639	-16.451	1.6495E-02	1	183.65	5.40E-01	549	3.530E+06
182	Hydrazine	H ₂ N ₂	302-01-2	76.858	-7,245.2	-8.22	6.1557E-03	1	274.69	4.08E+02	653.15	1.473E+07
183	Hydrogen	H ₂	1333-74-0	12.69	-94.896	1.1125	3.2915E-04	2	13.95	7.21E+03	33.19	1.315E+06

TABLE 2-8 Vapor Pressure of Inorganic and Organic Liquids, $\ln P = C1 + C2/T + C3 \ln T + C4 T^{C5}$, P in Pa (Continued)

No.	Name	Formula	CAS no.	C1	C2	C3	C4	C5	$T_{\min}$, K	P at $T_{\min}$	$T_{\max}$, K	P at $T_{\max}$
184	Hydrogen bromide	HBr	10035-10-6	29.315	-2,424.5	-1.1354	2.3806E-18	6	185.15	2.95E+04	363.15	8.463E+06
185	Hydrogen chloride	HCl	7647-01-0	104.27	-3,731.2	-15.047	3.1340E-02	1	158.97	1.35E+04	324.65	8.356E+06
186	Hydrogen cyanide	HCN	74-90-8	36.75	-3,927.1	-2.1245	3.8948E-17	6	259.83	1.87E+04	456.65	5.353E+06
187	Hydrogen fluoride	HF	7664-39-3	59.544	-4,143.8	-6.1764	1.4161E-05	2	189.79	3.37E+02	461.15	6.487E+06
188	Hydrogen sulfide	H ₂ S	7783-06-4	85.584	-3,839.9	-11.199	1.8848E-02	1	187.68	2.29E+04	373.53	8.999E+06
189	Isobutyric acid	C ₄ H ₈ O ₂	79-31-2	110.38	-10,540	-12.262	1.4310E-17	6	227.15	7.82E-02	605	3.683E+06
190	Isopropyl amine	C ₃ H ₉ N	75-31-0	136.66	-7,201.5	-18.934	2.2255E-02	1	177.95	7.73E+00	471.85	4.540E+06
191	Malonic acid	C ₃ H ₄ O ₄	141-82-2	122.92	-16,258	-13.113	2.0609E-18	6	407.95	7.03E+01	805	5.652E+06
192	Methacrylic acid	C ₄ H ₆ O ₂	79-41-4	109.53	-10,410	-12.289	3.1990E-06	2	288.15	5.86E+01	662	4.812E+06
193	Methane	CH ₄	74-82-8	39.205	-1,324.4	-3.4366	3.1019E-05	2	90.69	1.17E+04	190.56	4.590E+06
194	Methanol	CH ₃ O	67-56-1	82.718	-6,904.5	-8.8622	7.4664E-06	2	175.47	1.11E-01	512.5	8.146E+06
195	N-Methyl acetamide	C ₃ H ₇ NO	79-16-3	79.128	-9,523.9	-7.7355	3.1616E-18	6	301.15	2.86E+01	718	4.997E+06
196	Methyl acetate	C ₃ H ₆ O ₂	79-20-9	61.267	-5,618.6	-5.6473	2.1080E-17	6	175.15	1.02E+00	506.55	4.695E+06
197	Methyl acetylene	C ₃ H ₄	74-99-7	50.242	-3,811.9	-4.2526	6.5326E-17	6	170.45	4.15E+02	402.4	5.619E+06
198	Methyl acrylate	C ₄ H ₆ O ₂	96-33-3	107.69	-7,027.2	-13.916	1.5185E-02	1	196.32	4.07E+00	536	4.277E+06
199	Methyl amine	CH ₃ N	74-89-5	75.206	-5,082.8	-8.0919	8.1130E-06	2	179.69	1.77E+02	430.05	7.414E+06
200	Methyl benzoate	C ₈ H ₈ O ₂	93-58-3	84.828	-9,334.7	-8.7063	6.1723E-18	6	260.75	1.81E+00	693	3.590E+06
201	3-Methyl-1,2-butadiene	C ₅ H ₈	598-25-4	66.575	-5,213.4	-6.7693	4.8106E-06	2	159.53	7.28E-01	490	3.831E+06
202	2-Methylbutane	C ₅ H ₁₂	78-78-4	71.308	-4,976	-7.7169	8.7271E-06	2	113.25	1.21E-04	460.4	3.366E+06
203	2-Methylbutanoic acid	C ₆ H ₁₀ O ₂	116-53-0	85.383	-9,575.4	-8.6164	5.6124E-18	6	193	6.94E-05	643	3.887E+06
204	3-Methyl-1-butanol	C ₅ H ₁₂ O	123-51-3	121.85	-10,976	-13.869	1.4283E-17	6	155.95	8.67E-09	577.2	3.916E+06
205	2-Methyl-1-butene	C ₅ H ₁₀	563-46-2	93.131	-5,525.4	-11.852	1.4205E-02	1	135.58	2.05E-02	465	3.465E+06
206	2-Methyl-2-butene	C ₅ H ₁₀	513-35-9	83.927	-5,640.5	-9.6453	1.1121E-05	2	139.39	1.94E-02	470	3.394E+06
207	2-Methyl-1-butene-3-yne	C ₅ H ₆	78-80-8	95.453	-5,448.8	-12.384	1.5643E-02	1	160.15	2.92E+00	492	4.469E+06
208	Methylbutyl ether	C ₅ H ₁₂ O	628-28-4	60.164	-5,621.7	-5.53	1.8629E-17	6	157.48	2.99E-02	512.74	3.377E+06
209	Methylbutyl sulfide	C ₅ H ₁₂ S	628-29-5	96.344	-7,856.3	-11.058	7.3080E-06	2	175.3	4.61E-03	593	3.464E+06
210	3-Methyl-1-butyne	C ₅ H ₈	598-23-2	69.459	-5,250	-7.1125	7.9289E-17	6	183.45	4.36E+01	462.3	4.199E+06
211	Methyl butyrate	C ₅ H ₁₀ O ₂	623-42-7	71.87	-6,885.7	-7.0944	1.4903E-17	6	187.35	1.34E-01	554.5	3.480E+06
212	Methylchlorosilane	CH ₃ ClSi	993-00-0	95.984	-5,401.7	-11.829	1.8092E-05	2	139.05	4.12E-01	442	4.170E+06
213	Methylcyclohexane	C ₇ H ₁₄	108-87-2	92.684	-7,080.8	-10.695	8.1366E-06	2	146.58	1.52E-04	572.1	3.486E+06
214	1-Methylcyclohexanol	C ₇ H ₁₄ O	590-67-0	134.63	-10,682	-16.511	8.4427E-06	2	299.15	2.57E+02	686	3.994E+06
215	cis-2-Methylcyclohexanol	C ₇ H ₁₄ O	7443-70-1	125.1	-10,288	-15.157	1.0918E-05	2	280.15	4.56E+01	614	3.808E+06
216	trans-2-Methylcyclohexanol	C ₇ H ₁₄ O	7443-52-9	54.179	-7,477.2	-4.22	3.5225E-18	6	269.15	1.62E+01	617	3.767E+06
217	Methylcyclopentane	C ₆ H ₁₂	96-37-7	55.368	-5,149.8	-5.0136	3.2220E-06	2	130.73	2.25E-04	532.7	3.759E+06
218	1-Methylcyclopentene	C ₆ H ₁₀	693-89-0	52.732	-5,286.9	-4.4509	1.0883E-17	6	146.62	3.98E-03	542	4.130E+06
219	3-Methylcyclopentene	C ₆ H ₁₀	1120-62-3	52.601	-5,120.3	-4.4554	1.3288E-17	6	115	2.12E-06	526	4.129E+06
220	Methyldichlorosilane	CH ₃ Cl ₂ Si	75-54-7	79.788	-5,420	-9.0702	1.1489E-05	2	182.55	2.58E+01	483	3.964E+06
221	Methylethyl ether	C ₃ H ₈ O	540-67-0	78.586	-5,176.3	-8.7501	9.1727E-06	2	160	7.85E+00	437.8	4.433E+06
222	Methylethyl ketone	C ₄ H ₈ O	78-93-3	72.698	-6,143.6	-7.5779	5.6476E-06	2	186.48	1.39E+00	535.5	4.120E+06
223	Methylethyl sulfide	C ₃ H ₈ S	624-89-5	79.07	-6,114.1	-8.631	6.5333E-06	2	167.23	2.25E-01	533	4.261E+06
224	Methyl formate	C ₂ H ₄ O ₂	107-31-3	77.184	-5,606.1	-8.392	7.8468E-06	2	174.15	6.88E+00	487.2	5.983E+06
225	Methylisobutyl ether	C ₆ H ₁₂ O	625-44-5	57.984	-5,339.6	-5.2362	2.0767E-17	6	150	2.13E-02	497	3.416E+06
226	Methylisobutyl ketone	C ₆ H ₁₂ O	108-10-1	80.503	-7,421.8	-8.379	1.8114E-17	6	189.15	6.99E-02	574.6	3.272E+06
227	Methyl Isocyanate	C ₂ H ₃ NO	624-83-9	57.612	-5,197.9	-5.1269	2.1702E-17	6	256.15	7.28E+03	488	5.480E+06
228	Methylisopropyl ether	C ₄ H ₁₀ O	598-53-8	53.867	-4,701	-4.7052	2.8791E-17	6	127.93	3.32E-03	464.48	3.764E+06
229	Methylisopropyl ketone	C ₅ H ₁₀ O	563-80-4	45.242	-5,324.4	-3.2551	3.0363E-18	6	180.15	2.95E-01	553.4	3.792E+06
230	Methylisopropyl sulfide	C ₄ H ₁₀ S	1551-21-9	52.82	-5,437.7	-4.442	9.5103E-18	6	171.64	1.80E-01	553.1	4.022E+06
231	Methyl mercaptan	CH ₃ S	74-93-1	54.15	-4,337.7	-4.8127	4.5000E-17	6	150.18	3.15E+00	469.95	7.231E+06
232	Methyl methacrylate	C ₅ H ₈ O ₂	80-62-6	107.36	-8,085.3	-12.72	8.3307E-06	2	224.95	1.91E+01	566	3.674E+06
233	2-Methyloctanoic acid	C ₉ H ₁₈ O ₂	3004-93-1	105.7	-12,458	-11.234	4.4629E-18	6	240	4.19E-04	694	2.545E+06
234	2-Methylpentane	C ₆ H ₁₄	107-83-5	53.579	-5,041.2	-4.6404	1.9443E-17	6	119.55	2.07E-05	497.7	3.044E+06
235	Methyl pentyl ether	C ₆ H ₁₄ O	628-80-8	61.907	-6,188.9	-5.706	1.1767E-17	6	176	6.33E-02	546.49	3.041E+06
236	2-Methylpropane	C ₄ H ₁₀	75-28-5	108.43	-5,039.9	-15.012	2.2725E-02	1	113.54	1.21E-02	407.8	3.630E+06
237	2-Methyl-2-propanol	C ₄ H ₁₀ O	75-65-0	172.27	-11,589	-22.113	1.3703E-05	2	298.97	5.88E+03	506.2	3.957E+06
238	2-Methyl propene	C ₄ H ₈	115-11-7	78.01	-4,634.1	-8.9575	1.3413E-05	2	132.81	6.45E-01	417.9	4.004E+06
239	Methyl propionate	C ₄ H ₈ O ₂	554-12-1	70.717	-6,439.7	-6.9845	2.0129E-17	6	185.65	6.34E-01	530.6	4.028E+06
240	Methylpropyl ether	C ₄ H ₁₀ O	557-17-5	67.942	-5,419.1	-6.8067	4.7778E-17	6	133.97	2.90E-03	476.25	3.802E+06
241	Methylpropyl sulfide	C ₄ H ₁₀ S	3877-15-4	83.711	-6,786.9	-9.2526	6.6666E-06	2	160.17	4.26E-03	565	3.972E+06
242	Methylsilane	CH ₃ Si	992-94-9	37.205	-2,590.3	-2.5993	6.0508E-06	2	116.34	1.43E+01	352.5	4.702E+06

243	α -Methyl styrene	C ₉ H ₁₀	98-83-9	56.485	-6.954.2	-4.7889	2.7753E-18	6	249.95	9.23E+00	654	3.341E+06
244	Methyl <i>tert</i> -butyl ether	C ₈ H ₁₈ O	1634-04-4	57.1511	-5.201.7	-5.1429	1.6529E-17	6	164.55	4.93E-01	497.1	3.285E+06
245	Methyl vinyl ether	C ₃ H ₆ O	107-25-5	51.085	-4.271	-4.307	3.0530E-17	6	151.15	3.37E+00	437	4.583E+06
246	Naphthalene	C ₁₀ H ₈	91-20-3	62.964	-8.137.5	-5.6317	2.2675E-18	6	353.43	9.91E+02	748.4	4.069E+06
247	Neon	Ne	7440-01-9	29.755	-271.06	-2.6081	5.2700E-04	2	24.56	4.38E+04	44.4	2.665E+06
248	Nitroethane	C ₂ H ₅ NO ₂	79-24-3	75.632	-7.202.3	-7.6464	1.8250E-17	6	183.63	3.18E-02	593	5.159E+06
249	Nitrogen	N ₂	7727-37-9	58.282	-1.084.1	-8.3144	4.4127E-02	1	63.15	1.25E+04	126.2	3.391E+06
250	Nitrogen trifluoride	F ₃ N	7783-54-2	68.149	-2.257.9	-8.9118	2.3233E-02	1	66.46	1.86E-01	234	4.500E+06
251	Nitromethane	CH ₃ NO ₂	75-52-5	57.278	-6.089	-4.9821	1.2154E-17	6	244.6	1.47E+02	588.15	6.309E+06
252	Nitrous oxide	N ₂ O	10024-97-2	96.512	-4.045	-12.277	2.8560E-05	2	182.3	8.69E+04	309.57	7.278E+06
253	Nitric oxide	NO	10102-43-9	72.974	-2.650	-8.261	9.7000E-15	6	109.5	2.20E+04	180.15	6.516E+06
254	Nonadecane	C ₁₉ H ₄₀	629-92-5	182.54	-17.897	-22.498	7.4008E-06	2	305.04	1.59E-02	758	1.208E+06
255	Nonanal	C ₉ H ₁₈ O	124-19-6	337.71	-18.506	-50.224	4.7345E-02	1	255.15	3.42E-01	658	2.743E+06
256	Nonane	C ₉ H ₂₀	111-84-2	109.35	-9.030.4	-12.882	7.8544E-06	2	219.66	4.31E-01	594.6	2.305E+06
257	Nonanoic acid	C ₉ H ₁₈ O ₂	112-05-0	137.6	-14.948	-15.618	5.5660E-18	6	285.55	4.71E-02	710.7	2.502E+06
258	1-Nonanol	C ₉ H ₂₀ O	143-08-8	162.854	-15.205	-19.424	1.0722E-17	6	268.15	8.55E-02	670.9	2.522E+06
259	2-Nonanol	C ₉ H ₂₀ O	628-99-9	146.46	-13.813	-17.158	8.6279E-40	14	238.15	4.32E-03	649.5	2.551E+06
260	1-Nonene	C ₉ H ₁₈	124-11-8	63.313	-7.040.4	-5.8055	7.5753E-18	6	191.91	2.04E-02	593.1	2.427E+06
261	Nonyl mercaptan	C ₉ H ₂₀ S	1455-21-6	106.2	-10.982	-11.696	8.8955E-18	6	253.05	1.47E-01	681	2.330E+06
262	1-Nonyne	C ₉ H ₁₆	3452-09-3	114.77	-9.430.8	-13.631	8.1918E-06	2	223.15	4.50E-01	598.05	2.620E+06
263	Octadecane	C ₁₈ H ₃₈	593-45-3	157.68	-16.093	-18.954	5.9272E-06	2	301.31	3.39E-02	747	1.256E+06
264	Octanal	C ₈ H ₁₆ O	124-13-0	83.601	-8.865.8	-8.5711	7.9446E-18	6	246	1.46E+00	638.9	2.951E+06
265	Octane	C ₈ H ₁₈	111-65-9	96.084	-7.900.2	-11.003	7.1802E-06	2	216.38	2.11E+00	568.7	2.467E+06
266	Octanoic acid	C ₈ H ₁₆ O ₂	124-07-2	140.16	-14.813	-16.004	6.4239E-18	6	289.65	1.83E-01	694.26	2.761E+06
267	1-Octanol	C ₈ H ₁₈ O	111-87-5	144.111	-13.667	-16.826	9.3666E-18	6	257.65	9.60E-02	652.3	2.782E+06
268	2-Octanol	C ₈ H ₁₈ O	123-96-6	133.41	-12.630	-15.369	2.9939E-41	14	241.55	4.04E-02	629.8	2.754E+06
269	2-Octanone	C ₈ H ₁₆ O	111-13-7	63.775	-7.711.3	-5.7359	3.0902E-18	6	252.85	4.68E+00	632.7	2.647E+06
270	3-Octanone	C ₈ H ₁₆ O	106-68-3	72.382	-8.054.8	-7.0002	5.8276E-18	6	255.55	7.84E+00	627.7	2.705E+06
271	1-Octene	C ₈ H ₁₆	111-66-0	74.936	-7.155.9	-7.5843	1.7106E-17	6	171.45	2.98E-03	566.9	2.663E+06
272	Octyl mercaptan	C ₈ H ₁₈ S	111-88-6	78.368	-8.855.4	-7.8202	5.6629E-18	6	223.95	3.05E-02	667.3	2.523E+06
273	1-Octyne	C ₈ H ₁₄	629-05-0	64.612	-6.802.5	-6.0261	1.1013E-17	6	193.55	1.04E-01	574	2.880E+06
274	Oxalic acid	C ₂ H ₂ O ₄	144-62-7	122.04	-16.050	-12.986	2.0871E-18	6	462.65	2.15E+03	804	7.060E+06
275	Oxygen	O ₂	7782-44-7	51.245	-1.200.2	-6.4361	2.8405E-02	1	54.36	1.48E+02	154.58	5.021E+06
276	Ozone	O ₃	10028-15-6	40.067	-2.204.8	-2.9351	7.7520E-16	6	80.15	7.35E-01	261	5.566E+06
277	Pentadecane	C ₁₅ H ₃₂	629-62-9	135.57	-13.478	-16.022	5.6136E-06	2	283.07	1.29E-01	708	1.474E+06
278	Pentanal	C ₅ H ₁₀ O	110-62-3	149.58	-8.890	-20.697	2.2101E-02	1	182	5.23E-02	566.1	3.969E+06
279	Pentane	C ₅ H ₁₂	109-66-0	78.741	-5.420.3	-8.8253	9.6171E-06	2	143.42	6.86E-02	469.7	3.364E+06
280	Pentanoic acid	C ₅ H ₁₀ O ₂	109-52-4	101.7	-10.955	-10.829	7.1880E-18	6	239.15	3.28E-02	639.16	3.589E+06
281	1-Pentanol	C ₅ H ₁₂ O	71-41-0	114.748	-10.643	-12.858	1.2491E-17	6	195.56	5.48E-04	588.1	3.896E+06
282	2-Pentanol	C ₅ H ₁₂ O	6032-29-7	122.26	-10.774	-13.943	1.0700E-42	15	200	4.15E-03	561	3.709E+06
283	2-Pentanone	C ₅ H ₁₀ O	107-87-9	84.635	-7.078.4	-9.3	6.2702E-06	2	196.29	7.52E-01	561.08	3.706E+06
284	3-Pentanone	C ₅ H ₁₀ O	96-22-0	44.286	-5.415.1	-3.0913	1.8580E-18	6	234.18	7.34E+01	560.95	3.699E+06
285	1-Pentene	C ₅ H ₁₀	109-67-1	46.994	-4.289.5	-3.7345	2.5424E-17	6	108.02	3.70E-05	464.8	3.562E+06
286	2-Pentyl mercaptan	C ₅ H ₁₂ S	2084-19-7	58.985	-6.193.1	-5.2746	7.3986E-18	6	160.75	1.77E-03	584.3	3.537E+06
287	Pentyl mercaptan	C ₅ H ₁₂ S	110-66-7	67.309	-6.880.8	-6.4449	1.0148E-17	6	197.45	2.01E-01	598	3.474E+06
288	1-Pentyne	C ₅ H ₈	627-19-0	82.805	-5.683.8	-9.4301	1.0767E-05	2	167.45	2.40E+00	481.2	4.170E+06
289	2-Pentyne	C ₅ H ₈	627-21-4	137.29	-7.447.1	-19.01	2.1415E-02	1	163.83	2.05E-01	519	4.020E+06
290	Phenanthrene	C ₁₄ H ₁₀	85-01-8	72.958	-10.943	-6.7902	1.0850E-18	6	372.38	2.93E+01	869	2.902E+06
291	Phenol	C ₆ H ₆ O	108-95-2	95.444	-10.113	-10.09	6.7603E-18	6	314.06	1.88E+02	694.25	6.059E+06
292	Phenyl isocyanate	C ₇ H ₅ NO	103-71-9	86.779	-8.101.8	-9.5303	6.1367E-06	2	243.15	4.33E+00	653	4.063E+06
293	Phthalic anhydride	C ₈ H ₄ O ₃	85-44-9	126.5	-12.551	-15.002	7.7521E-06	2	404.15	7.90E+02	791	4.734E+06
294	Propadiene	C ₃ H ₄	463-49-0	57.069	-3.682.7	-5.5662	6.5133E-06	2	136.87	1.83E+01	394	5.218E+06
295	Propane	C ₃ H ₈	74-98-6	59.078	-3.492.6	-6.0669	1.0919E-05	2	85.47	1.68E-04	369.83	4.214E+06
296	1-Propanol	C ₃ H ₈ O	71-23-8	84.6642	-8.307.2	-8.5767	7.5091E-18	6	146.95	4.28E-07	536.8	5.170E+06
297	2-Propanol	C ₃ H ₈ O	67-63-0	96.094	-8.575.4	-10.292	1.6665E-17	6	185.26	1.95E-02	508.3	4.783E+06
298	Propenylcyclohexene	C ₉ H ₁₄	13511-13-2	64.268	-7.298.9	-5.9109	4.8482E-18	6	199	2.48E-02	636	3.130E+06
299	Propionaldehyde	C ₃ H ₆ O	123-38-6	80.581	-5.896.1	-8.9301	8.2236E-06	2	170	1.31E+00	504.4	4.919E+06
300	Propionic acid	C ₃ H ₆ O ₂	79-09-4	54.552	-7.149.4	-4.2769	1.1843E-18	6	252.45	1.31E+01	600.81	4.608E+06
301	Propionitrile	C ₃ H ₅ N	107-12-0	82.699	-6.703.5	-9.1506	7.5424E-06	2	180.26	1.69E-01	564.4	4.191E+06
302	Propyl acetate	C ₅ H ₁₀ O ₂	109-60-4	115.16	-8.433.9	-13.934	1.0346E-05	2	178.15	1.71E-02	549.73	3.366E+06
303	Propyl amine	C ₃ H ₉ N	107-10-8	58.398	-5.312.7	-5.2876	1.9913E-06	2	188.36	1.30E+01	496.95	4.738E+06
304	Propylbenzene	C ₉ H ₁₂	103-65-1	91.379	-8.276.8	-10.176	5.6240E-06	2	173.55	1.81E-04	638.35	3.202E+06
305	Propylene	C ₃ H ₆	115-07-1	43.905	-3.097.8	-3.4425	9.9989E-17	6	87.89	1.17E-03	364.85	4.599E+06

TABLE 2-8 Vapor Pressure of Inorganic and Organic Liquids, $\ln P = C1 + C2/T + C3 \ln T + C4 T^{C5}$, P in Pa (Concluded)

No.	Name	Formula	CAS no.	C1	C2	C3	C4	C5	$T_{\min}$, K	P at $T_{\min}$	$T_{\max}$, K	P at $T_{\max}$
306	Propyl formate	$C_4H_8O_2$	110-74-7	104.08	-7,535.9	-12.348	9.6020E-06	2	180.25	2.11E-01	538	4.031E+06
307	2-Propyl mercaptan	C_3H_6S	75-33-2	60.43	-5,276.9	-5.6572	2.6039E-17	6	142.61	9.73E-03	517	4.752E+06
308	Propyl mercaptan	C_3H_6S	107-03-9	62.165	-5,624	-5.8595	2.0597E-17	6	159.95	6.51E-02	536.6	4.627E+06
309	1,2-Propylene glycol	$C_3H_8O_2$	57-55-6	212.8	-15,420	-28.109	2.1564E-05	2	213.15	9.29E-05	626	6.041E+06
310	Quinone	$C_6H_4O_2$	106-51-4	48.651	-7,289.5	-3.4453	1.0068E-18	6	388.85	1.17E+04	683	5.925E+06
311	Silicon tetrafluoride	F_4Si	7783-61-1	272.85	-9,548.9	-40.089	6.3699E-15	6	186.35	2.21E+05	259	3.748E+06
312	Styrene	C_8H_8	100-42-5	105.93	-8,685.9	-12.42	7.5583E-06	2	242.54	1.06E+01	636	3.823E+06
313	Succinic acid	$C_4H_6O_4$	110-15-6	128.65	-16,958	-13.872	2.1559E-18	6	460.65	8.85E+02	806	4.727E+06
314	Sulfur dioxide	O_2S	7446-09-5	47.365	-4,084.5	-3.6469	1.7990E-17	6	197.67	1.67E+03	430.75	7.860E+06
315	Sulfur hexafluoride	F_6S	2551-62-4	29.16	-2,383.6	-1.1342			223.15	2.30E+05	318.69	3.771E+06
316	Sulfur trioxide	O_3S	7446-11-9	180.99	-12,060	-22.839	7.2350E-17	6	289.95	2.09E+04	490.85	8.192E+06
317	Terephthalic acid	$C_8H_6O_4$	100-21-0	248.72	-32,238	-30.009	4.7950E-06	2	700.15	4.57E+03	1113	3.943E+06
318	<i>o</i> -Terphenyl	$C_{18}H_{14}$	84-15-1	110.52	-14,045	-11.861	2.2121E-18	6	329.35	4.14E-01	857	2.974E+06
319	Tetradecane	$C_{14}H_{30}$	629-59-4	140.47	-13,231	-16.859	6.5877E-06	2	279.01	2.53E-01	693	1.569E+06
320	Tetrahydrofuran	C_4H_8O	109-99-9	54.898	-5,305.4	-4.7627	1.4291E-17	6	164.65	1.96E-01	540.15	5.203E+06
321	1,2,3,4-Tetrahydronaphthalene	$C_{10}H_{12}$	119-64-2	137.23	-10,620	-17.908	1.4506E-02	1	237.38	1.33E-01	720	3.624E+06
322	Tetrahydrothiophene	C_4H_6S	110-01-0	75.881	-6,910.6	-7.9499	4.4315E-06	2	176.99	1.54E-02	631.95	5.117E+06
323	2,2,3,3-Tetramethylbutane	C_8H_{18}	594-82-1	57.963	-5,901.5	-5.2048	9.1301E-18	6	373.96	8.69E+04	568	2.871E+06
324	Thiophene	C_4H_4S	110-02-1	93.193	-7,001.5	-10.738	8.2308E-06	2	234.94	1.86E+02	579.35	5.702E+06
325	Toluene	C_7H_8	108-88-3	76.945	-6,729.8	-8.179	5.3017E-06	2	178.18	4.75E-02	591.75	4.080E+06
326	1,1,2-Trichloroethane	$C_2H_3Cl_3$	79-00-5	54.153	-6,041.8	-4.5383	4.9833E-18	6	236.5	4.47E+01	602	4.447E+06
327	Tridecane	$C_{13}H_{28}$	629-50-5	137.45	-12,549	-16.543	7.1275E-06	2	267.76	2.51E-01	675	1.679E+06
328	Triethyl amine	$C_6H_{15}N$	121-44-8	56.55	-5,681.9	-4.9815	1.2363E-17	6	158.45	1.06E-02	535.15	3.037E+06
329	Trimethyl amine	C_3H_9N	75-50-3	134.68	-6,055.8	-19.415	2.8619E-02	1	156.08	9.92E+00	433.25	4.102E+06
330	1,2,3-Trimethylbenzene	C_9H_{12}	526-73-8	78.341	-8,019.8	-8.1458	3.8971E-06	2	247.79	3.71E+00	664.5	3.447E+06
331	1,2,4-Trimethylbenzene	C_9H_{12}	95-63-6	85.301	-8,215.9	-9.2166	4.7979E-06	2	229.33	6.93E-01	649.1	3.212E+06
332	2,2,4-Trimethylpentane	C_8H_{18}	540-84-1	84.912	-6,722.2	-9.5157	7.2244E-06	2	165.78	1.71E-02	543.8	2.550E+06
333	2,3,3-Trimethylpentane	C_8H_{18}	560-21-4	83.105	-6,903.7	-9.1858	6.4703E-06	2	172.22	1.68E-02	573.5	2.812E+06
334	1,3,5-Trinitrobenzene	$C_6H_3N_3O_6$	99-35-4	506.33	-37,483	-69.22	2.7381E-05	2	398.4	8.50E+00	846	3.410E+06
335	2,4,6-Trinitrotoluene	$C_7H_5N_3O_6$	118-96-7	302	-24,324	-40.13	1.7403E-05	2	354	9.36E-01	828	3.019E+06
336	Undecane	$C_{11}H_{24}$	1120-21-4	131	-11,143	-15.855	8.1871E-06	2	247.57	4.08E-01	639	1.949E+06
337	1-Undecanol	$C_{11}H_{24}O$	112-42-5	182.571	-17,112	-22.125	1.12835E-17	6	288.45	1.26E-01	703.9	2.120E+06
338	Vinyl acetate	$C_4H_6O_2$	108-05-4	57.406	-5,702.8	-5.0307	1.1042E-17	6	180.35	7.06E-01	519.13	3.930E+06
339	Vinyl acetylene	C_4H_4	689-97-4	55.682	-4,439.3	-5.0136	1.9650E-17	6	173.15	6.69E+01	454	4.887E+06
340	Vinyl chloride	C_2H_3Cl	75-01-4	91.432	-5,141.7	-10.981	1.4318E-05	2	119.36	1.92E-02	432	5.750E+06
341	Vinyl trichlorosilane	$C_2H_3Cl_3Si$	75-94-5	54.571	-5,561.5	-4.712	1.0702E-17	6	178.35	3.54E-01	543.15	3.058E+06
342	Water	H_2O	7732-18-5	73.649	-7,258.2	-7.3037	4.1653E-06	2	273.16	6.11E+02	647.096	2.193E+07
343	<i>m</i> -Xylene	C_8H_{10}	108-38-3	85.099	-7,615.9	-9.3072	5.5643E-06	2	225.3	3.18E+00	617	3.528E+06
344	<i>o</i> -Xylene	C_8H_{10}	95-47-6	90.405	-7,955.2	-10.086	5.9594E-06	2	247.98	2.18E+01	630.3	3.741E+06
345	<i>p</i> -Xylene	C_8H_{10}	106-42-3	88.72	-7,741.2	-9.8693	6.0770E-06	2	286.41	5.76E+02	616.2	3.501E+06

Vapor pressure P_v is calculated by

$$P_v = \exp(C1 + C2/T + C3 \ln T + C4 T^{C5})$$

where P_v is in Pa and T is in K. All substances are listed by chemical family in Table 2-6 and by formula in Table 2-7.

Values in this table were taken from the Design Institute for Physical Properties (DIPPR) of the American Institute of Chemical Engineers (AIChE), copyright 2007 AIChE and reproduced with permission of AIChE and of the DIPPR Evaluated Process Design Data Project Steering Committee. Their source should be cited as R. L. Rowley, W. V. Wilding, J. L. Oscarson, Y. Yang, N. A. Zundel, T. E. Daubert, R. P. Danner, DIPPR® Data Compilation of Pure Chemical Properties, Design Institute for Physical Properties, AIChE, New York (2007).

The number of digits provided for values at $T_{\min}$ and $T_{\max}$ was chosen for uniformity of appearance and formatting; these do not represent the uncertainties of the physical quantities, but are the result of calculations from the standard thermophysical property formulations within a fixed format.

TABLE 2-9 Vapor Pressures of Inorganic Compounds, up to 1 atm*

Compound		Pressure, mmHg										Melting point, °C
Name	Formula	1	5	10	20	40	60	100	200	400	760	
		Temperature, °C										
Aluminum	Al	1284	1421	1487	1555	1635	1684	1749	1844	1947	2056	660
borohydride	Al(BH ₄) ₃		-52.2	-42.9	-32.5	-20.9	-13.4	-3.9	+11.2	28.1	45.9	-64.
bromide	AlBr ₃	81.3	103.8	118.0	134.0	150.6	161.7	176.1	199.8	227.0	256.3	97.
chloride	Al ₂ Cl ₆	100.0	116.4	123.8	131.8	139.9	145.4	152.0	161.8	171.6	180.2	192.4
fluoride	AlF ₃	1238	1298	1324	1350	1378	1398	1422	1457	1496	1537	1040
iodide	AlI ₃	178.0	207.7	225.8	244.2	265.0	277.8	294.5	322.0	354.0	385.5	
oxide	Al ₂ O ₃	2148	2306	2385	2465	2549	2599	2665	2766	2874	2977	2050
Ammonia	NH ₃	-109.1	-97.5	-91.9	-85.8	-79.2	-74.3	-68.4	-57.0	-45.4	-33.6	-77.7
heavy	ND ₃						-74.0	-67.4	-57.0	-45.4	-33.4	-74.0
Ammonium bromide	NH ₄ Br	198.3	234.5	252.0	270.6	290.0	303.8	320.0	345.3	370.8	396.0	
carbamate	N ₂ H ₆ CO ₂	-26.1	-10.4	-2.9	+5.3	14.0	19.6	26.7	37.2	48.0	58.3	
chloride	NH ₄ Cl	160.4	193.8	209.8	226.1	245.0	256.2	271.5	293.2	316.5	337.8	520
cyanide	NH ₄ CN	-50.6	-35.7	-28.6	-20.9	-12.6	-7.4	-0.5	+9.6	20.5	31.7	36
hydrogen sulfide	NH ₄ HS	-51.1	-36.0	-28.7	-20.8	-12.3	-7.0	0.0	+10.5	21.8	33.3	
iodide	NH ₄ I	210.9	247.0	263.5	282.8	302.8	316.0	331.8	355.8	381.0	404.9	
Antimony	Sb	886	984	1033	1084	1141	1176	1223	1288	1364	1440	630.5
tribromide	SbBr ₃	93.9	126.0	142.7	158.3	177.4	188.1	203.5	225.7	250.2	275.0	96.6
trichloride	SbCl ₃	49.2	71.4	85.2	100.6	117.8	128.3	143.3	165.9	192.2	219.0	73.4
pentachloride	SbCl ₅	22.7	48.6	61.8	75.8	91.0	101.0	114.1				2.8
triiodide	SbI ₃	163.6	203.8	223.5	244.8	267.8	282.5	303.5	333.8	368.5	401.0	167
trioxide	Sb ₄ O ₆	574	626	666	729	812	873	957	1085	1242	1425	656
Argon	A	-218.2	-213.9	-210.9	-207.9	-204.9	-202.9	-200.5	-195.6	-190.6	-185.6	-189.2
Arsenic	As	372	416	437	459	483	498	518	548	579	610	814
Arsenic tribromide	AsBr ₃	41.8	70.6	85.2	101.3	118.7	130.0	145.2	167.7	193.6	220.0	
trichloride	AsCl ₃	-11.4	+11.7	+23.5	36.0	50.0	58.7	70.9	89.2	109.7	130.4	-18
trifluoride	AsF ₃					-2.5	+4.2	13.2	26.7	41.4	56.3	-5.9
pentafluoride	AsF ₅	-117.9	-108.0	-103.1	-98.0	-92.4	-88.5	-84.3	-75.5	-64.0	-52.8	-79.8
trioxide	As ₂ O ₃	212.5	242.6	259.7	279.2	299.2	310.3	332.5	370.0	412.2	457.2	312.8
Arsine	AsH ₃	-142.6	-130.8	-124.7	-117.7	-110.2	-104.8	-98.0	-87.2	-75.2	-62.1	-116.3
Barium	Ba		984	1049	1120	1195	1240	1301	1403	1518	1638	850
Beryllium borohydride	Be(BH ₄) ₂	+1.0	19.8	28.1	36.8	46.2	51.7	58.6	69.0	79.7	90.0	123
bromide	BeBr ₂	289	325	342	361	379	390	405	427	451	474	490
chloride	BeCl ₂	291	328	346	365	384	395	411	435	461	487	405
iodide	BeI ₂	283	322	341	361	382	394	411	435	461	487	488
Bismuth	Bi	1021	1099	1136	1177	1217	1240	1271	1319	1370	1420	271
tribromide	BiBr ₃		261	282	305	327	340	360	392	425	461	218
trichloride	BiCl ₃		242	264	287	311	324	343	372	405	441	230
Diborane hydrobromide	B ₂ H ₅ Br	-93.3	-75.3	-66.3	-56.4	-45.4	-38.2	-29.0	-15.4	0.0	+16.3	-104.2
Borine carbonyl	BH ₃ CO	-139.2	-127.3	-121.1	-114.1	-106.6	-101.9	-95.3	-85.5	-74.8	-64.0	-137.0
triamine	B ₃ N ₃ H ₆	-63.0	-45.0	-35.3	-25.0	-13.2	-5.8	+4.0	18.5	34.3	50.6	-58.2
Boron hydrides												
dihydrodecaborane	B ₁₀ H ₁₄	60.0	80.8	90.2	100.0	117.4	127.8	142.3	163.8			99.6
dihydrodiborane	B ₂ H ₆	-159.7	-149.5	-144.3	-138.5	-131.6	-127.2	-120.9	-111.2	-99.6	-86.5	-169
dihydropentaborane	B ₅ H ₉		-40.4	-30.7	-20.0	-8.0	-0.4	+9.6	24.6	40.8	58.1	-47.0
tetrahydropentaborane	B ₅ H ₁₁	-50.2	-29.9	-19.9	-9.2	+2.7	10.2	20.1	34.8	51.2	67.0	
tetrahydrotriborane	B ₃ H ₁₀	-90.9	-73.1	-64.3	-54.8	-44.3	-37.4	-28.1	-14.0	+0.8	16.1	-119.9
Boron tribromide	BBr ₃	-41.4	-20.4	-10.1	+1.5	14.0	22.1	33.5	50.3	70.0	91.7	-45
trichloride	BCl ₃	-91.5	-75.2	-66.9	-57.9	-47.8	-41.2	-32.4	-18.9	-3.6	+12.7	-107
trifluoride	BF ₃	-154.6	-145.4	-141.3	-136.4	-131.0	-127.6	-123.0	-115.9	-108.3	-100.7	-126.8
Bromine	Br ₂	-48.7	-32.8	-25.0	-16.8	-8.0	-0.6	+9.3	24.3	41.0	58.2	-7.3
pentafluoride	BrF ₅	-69.3	-51.0	-41.9	-32.0	-21.0	-14.0	-4.5	+9.9	25.7	40.4	-61.4
Cadmium	Cd	394	455	484	516	553	578	611	658	711	765	320.9
chloride	CdCl ₂		618	656	695	736	762	797	847	908	967	568
fluoride	CdF ₂	1112	1231	1286	1344	1400	1436	1486	1561	1651	1751	520
iodide	CdI ₂	416	481	512	546	584	608	640	688	742	796	385
oxide	CdO	1000	1100	1149	1200	1257	1295	1341	1409	1484	1559	
Calcium	Ca		926	983	1046	1111	1152	1207	1288	1388	1487	851
Carbon (graphite)	C	3586	3828	3946	4069	4196	4273	4373	4516	4660	4827	
dioxide	CO ₂	-134.3	-124.4	-119.5	-114.4	-108.6	-104.8	-100.2	-93.0	-85.7	-78.2	-57.5
disulfide	CS ₂	-73.8	-54.3	-44.7	-34.3	-22.5	-15.3	-5.1	+10.4	28.0	46.5	-110.8
monoxide	CO	-222.0	-217.2	-215.0	-212.8	-210.0	-208.1	-205.7	-201.3	-196.3	-191.3	-205.0
oxyselenide	COSe	-117.1	-102.3	-95.0	-86.3	-76.4	-70.2	-61.7	-49.8	-35.6	-21.9	
oxysulfide	COS	-132.4	-119.8	-113.3	-106.0	-98.3	-93.0	-85.9	-75.0	-62.7	-49.9	-138.8
selenosulfide	CSeS	-47.3	-26.5	-16.0	-4.4	+8.6	17.0	28.3	45.7	65.2	85.6	-75.2
suboxide	C ₂ S ₂	14.0	41.2	54.9	69.3	85.6	96.0	109.9	130.8			+0.4
tetrabromide	CBr ₄					96.3	106.3	119.7	139.7	163.5	189.5	90.1
tetrachloride	CCl ₄	-50.0	-30.0	-19.6	-8.2	+4.3	12.3	23.0	38.3	57.8	76.7	-22.6
tetrafluoride	CF ₄	-184.6	-174.1	-169.3	-164.3	-158.8	-155.4	-150.7	-143.6	-135.5	-127.7	-183.7
Cesium	Cs	279	341	375	409	449	474	509	561	624	690	28.5
bromide	CsBr	748	838	887	938	993	1026	1072	1140	1221	1300	636
chloride	CsCl	744	837	884	934	989	1023	1069	1139	1217	1300	646
fluoride	CsF	712	798	844	893	947	980	1025	1092	1170	1251	683
iodide	CsI	738	828	873	923	976	1009	1055	1124	1200	1280	621

*Compiled from the extended tables published by D. R. Stull in *Ind. Eng. Chem.*, **39**, 517 (1947).

TABLE 2-9 Vapor Pressures of Inorganic Compounds, up to 1 atm (Continued)

Compound		Pressure, mmHg										Melting point, °C
Name	Formula	1	5	10	20	40	60	100	200	400	760	
		Temperature, °C										
Chlorine	Cl ₂	-118.0	-106.7	-101.6	-93.3	-84.5	-79.0	-71.7	-60.2	-47.3	-33.8	-100.7
fluoride	ClF		-143.4	-139.0	-134.3	-128.8	-125.3	-120.8	-114.4	-107.0	-100.5	-145
trifluoride	ClF ₃		-80.4	-71.8	-62.3	-51.3	-44.1	-34.7	-20.7	-4.9	+11.5	-83
monoxide	Cl ₂ O	-98.5	-81.6	-73.1	-64.3	-54.3	-48.0	-39.4	-26.5	-12.5	+2.2	-116
dioxide	ClO ₂		-59.0	-51.2	-42.8	-37.2	-29.4	-17.8	-4.0	+11.1	-59	-59
heptoxide	Cl ₂ O ₇	-45.3	-23.8	-13.2	-2.1	+10.3	+18.2	29.1	44.6	62.2	78.8	-91
Chlorosulfonic acid	HSO ₃ Cl	32.0	53.5	64.0	75.3	87.6	95.2	105.3	120.0	136.1	151.0	-80
Chromium	Cr	1616	1768	1845	1928	2013	2067	2139	2243	2361	2482	1615
carbonyl	Cr(CO) ₆	36.0	58.0	68.3	79.5	91.2	98.3	108.0	121.8	137.2	151.0	
oxychloride	CrO ₂ Cl ₂	-18.4	+3.2	13.8	25.7	38.5	46.7	58.0	75.2	95.2	117.1	
Cobalt chloride	CoCl ₂					770	801	843	904	974	1050	735
nitrosyl tricarboxyl	Co(CO) ₃ NO				-1.3	+11.0	18.5	29.0	44.4	62.0	80.0	-11
Columbium fluoride	CbF ₅			86.3	103.0	121.5	133.2	148.5	172.2	198.0	225.0	75.5
Copper	Cu	1628	1795	1879	1970	2067	2127	2207	2325	2465	2595	1083
Cuprous bromide	Cu ₂ Br ₂	572	666	718	777	844	887	951	1052	1189	1355	504
chloride	Cu ₂ Cl ₂	546	645	702	766	838	886	960	1077	1249	1490	422
iodide	Cu ₂ I ₂		610	656	716	786	836	907	1018	1158	1336	605
Cyanogen	C ₂ N ₂	-95.8	-83.2	-76.8	-70.1	-62.7	-57.9	-51.8	-42.6	-33.0	-21.0	-34.4
bromide	CNBr	-35.7	-18.3	-10.0	-1.0	+8.6	14.7	22.6	33.8	46.0	61.5	58
chloride	CNCl	-76.7	-61.4	-53.8	-46.1	-37.5	-32.1	-24.9	-14.1	-2.3	+13.1	-6.5
fluoride	CNF	-134.4	-123.8	-118.5	-112.8	-106.4	-102.3	-97.0	-89.2	-80.5	-72.6	
Deuterium cyanide	DCN	-68.9	-54.0	-46.7	-38.8	-30.1	-24.7	-17.5	-5.4	+10.0	26.2	-12
Fluorine	F ₂	-223.0	-216.9	-214.1	-211.0	-207.7	-205.6	-202.7	-198.3	-193.2	-187.9	-223
oxide	F ₂ O	-196.1	-186.6	-182.3	-177.8	-173.0	-170.0	-165.8	-159.0	-151.9	-144.6	-223.9
Germanium bromide	GeBr ₄		43.3	56.8	71.8	88.1	98.8	113.2	135.4	161.6	189.0	26.1
chloride	GeCl ₄	-45.0	-24.9	-15.0	-4.1	+8.0	16.2	27.5	44.4	63.8	84.0	-49.5
hydride	GeH ₄	-163.0	-151.0	-145.3	-139.2	-131.6	-126.7	-120.3	-111.2	-100.2	-88.9	-165
Trichlorogermane	GeHCl ₃	-41.3	-22.3	-13.0	-3.0	+8.8	16.2	26.5	41.6	58.3	75.0	-71.1
Tetramethylgermane	Ge(CH ₃) ₄	-73.2	-54.6	-45.2	-35.0	-23.4	-16.2	-6.3	+8.8	26.0	44.0	-88
Digermane	Ge ₂ H ₆	-88.7	-69.8	-60.1	-49.9	-38.2	-30.7	-20.3	-4.7	+13.3	31.5	-109
Trigermane	Ge ₃ H ₈	-36.9	-12.8	-0.9	+11.8	26.3	35.5	47.9	67.0	88.6	110.8	-105.6
Gold	Au	1869	2059	2154	2256	2363	2431	2521	2657	2807	2966	1063
Helium	He	-271.7	-271.5	-271.3	-271.1	-270.7	-270.6	-270.3	-269.8	-269.3	-268.6	
para-Hydrogen	H ₂	-263.3	-261.9	-261.3	-260.4	-259.6	-258.9	-257.9	-256.3	-254.5	-252.5	-259.1
Hydrogen bromide	HBr	-138.8	-127.4	-121.8	-115.4	-108.3	-103.8	-97.7	-88.1	-78.0	-66.5	-87.0
chloride	HCl	-150.8	-140.7	-135.6	-130.0	-123.8	-119.6	-114.0	-105.2	-95.3	-84.8	-114.3
cyanide	HCN	-71.0	-55.3	-47.7	-39.7	-30.9	-25.1	-17.8	-5.3	+10.2	25.9	-13.2
fluoride	H ₂ F ₂		-74.7	-65.8	-56.0	-45.0	-37.9	-28.2	-13.2	+2.5	19.7	-83.7
iodide	HI	-123.3	-109.6	-102.3	-94.5	-85.6	-79.8	-72.1	-60.3	-48.3	-35.1	-50.9
oxide (water)	H ₂ O	-17.3	+1.2	11.2	22.1	34.0	41.5	51.6	66.5	83.0	100.0	0.0
sulfide	H ₂ S	-134.3	-122.4	-116.3	-109.7	-102.3	-97.9	-91.6	-82.3	-71.8	-60.4	-85.5
disulfide	HSSH	-43.2	-24.4	-15.2	-5.1	+6.0	12.8	22.0	35.3	49.6	64.0	-89.7
selenide	H ₂ Se	-115.3	-103.4	-97.9	-91.8	-84.7	-80.2	-74.2	-65.2	-53.6	-41.1	-64
telluride	H ₂ Te	-96.4	-82.4	-75.4	-67.8	-59.1	-53.7	-45.7	-32.4	-17.2	-2.0	-49.0
Iodine	I ₂	38.7	62.2	73.2	84.7	97.5	105.4	116.5	137.3	159.8	183.0	112.9
heptafluoride	IF ₇	-87.0	-70.7	-63.0	-54.5	-45.3	-39.4	-31.9	-20.7	-8.3	+4.0	5.5
Iron	Fe	1787	1957	2039	2128	2224	2283	2360	2475	2605	2735	1535
pentacarbonyl	Fe(CO) ₅		-6.5	+4.6	16.7	30.3	39.1	50.3	68.0	86.1	105.0	-21
Ferric chloride	Fe ₂ Cl ₆	194.0	221.8	235.5	246.0	256.8	263.7	272.5	285.0	298.0	319.0	304
Ferrous chloride	FeCl ₂			700	737	779	805	842	897	961	1026	
Krypton	Kr	-199.3	-191.3	-187.2	-182.9	-178.4	-175.7	-171.8	-165.9	-159.0	-152.0	-156.7
Lead	Pb	973	1099	1162	1234	1309	1358	1421	1519	1630	1744	327.5
bromide	PbBr ₂	513	578	610	646	686	711	745	796	856	914	373
chloride	PbCl ₂	547	615	648	684	725	750	784	833	893	954	501
fluoride	PbF ₂		861	904	950	1003	1036	1080	1144	1219	1293	855
iodide	PbI ₂	479	540	571	605	644	668	701	750	807	872	402
oxide	PbO	943	1039	1085	1134	1189	1222	1265	1330	1402	1472	890
sulfide	PbS	852	928	975	1005	1048	1074	1108	1160	1221	1281	1114
Lithium	Li	723	838	881	940	1003	1042	1097	1178	1273	1372	186
bromide	LiBr	748	840	888	939	994	1028	1076	1147	1226	1310	547
chloride	LiCl	783	880	932	987	1045	1081	1129	1203	1290	1382	614
fluoride	LiF	1047	1156	1211	1270	1333	1372	1425	1503	1591	1681	870
iodide	LiI	723	802	841	883	927	955	993	1049	1110	1171	446
Magnesium	Mg	621	702	743	789	838	868	909	967	1034	1107	651
chloride	MgCl ₂	778	877	930	988	1050	1088	1142	1223	1316	1418	712
Manganese	Mn	1292	1434	1505	1583	1666	1720	1792	1900	2029	2151	1260
chloride	MnCl ₂		736	778	825	879	913	960	1028	1108	1190	650
Mercury	Hg	126.2	164.8	184.0	204.6	228.8	242.0	261.7	290.7	323.0	357.0	-38.9
Mercuric bromide	HgBr ₂	136.5	165.3	179.8	194.3	211.5	221.0	237.8	262.7	290.0	319.0	237
chloride	HgCl ₂	136.2	166.0	180.2	195.8	212.5	222.2	237.0	256.5	275.5	304.0	277
iodide	HgI ₂	157.5	189.2	204.5	220.0	238.2	249.0	261.8	291.0	324.2	354.0	259
Molybdenum	Mo	3102	3393	3535	3690	3859	3964	4109	4322	4553	4804	2622
hexafluoride	MoF ₆	-65.5	-49.0	-40.8	-32.0	-22.1	-16.2	-8.0	+4.1	17.2	36.0	17
oxide	MoO ₃	734	785	814	851	892	917	955	1014	1082	1151	795

TABLE 2-9 Vapor Pressures of Inorganic Compounds, up to 1 atm (Continued)

Compound		Pressure, mmHg										Melting point, °C
Name	Formula	1	5	10	20	40	60	100	200	400	760	
		Temperature, °C										
Neon	Ne	-257.3	-255.5	-254.6	-253.7	-252.6	-251.9	-251.0	-249.7	-248.1	-246.0	-248.7
Nickel	Ni	1810	1979	2057	2143	2234	2289	2364	2473	2603	2732	1452
carbonyl chloride	Ni(CO) ₄					-23.0	-15.9	-6.0	+8.8	25.8	42.5	-25
Nickel chloride	NiCl ₂	671	731	759	789	821	840	866	904	945	987	1001
Nitrogen	N ₂	-226.1	-221.3	-219.1	-216.8	-214.0	-212.3	-209.7	-205.6	-200.9	-195.8	-210.0
Nitric oxide	NO	-184.5	-180.6	-178.2	-175.3	-171.7	-168.9	-166.0	-162.3	-156.8	-151.7	-161
Nitrogen dioxide	NO ₂	-55.6	-42.7	-36.7	-30.4	-23.9	-19.9	-14.7	-5.0	+8.0	21.0	-9.3
Nitrogen pentoxide	N ₂ O ₅	-36.8	-23.0	-16.7	-10.0	-2.9	+1.8	7.4	15.6	24.4	32.4	30
Nitrous oxide	N ₂ O	-143.4	-133.4	-128.7	-124.0	-118.3	-114.9	-110.3	-103.6	-96.2	-85.5	-90.9
Nitrosyl chloride	NOCl					-60.2	-54.2	-46.3	-34.0	-20.3	-6.4	-64.5
fluoride	NOF	-132.0	-120.3	-114.3	-107.8	-100.3	-95.7	-88.8	-79.2	-68.2	-56.0	-134
Osmium tetroxide (yellow)	OsO ₄	3.2	22.0	31.3	41.0	51.7	59.4	71.5	89.5	109.3	130.0	56
(white)	OsO ₄	-5.6	+15.6	26.0	37.4	50.5	59.4	71.5	89.5	109.3	130.0	42
Oxygen	O ₂	-219.1	-213.4	-210.6	-207.5	-204.1	-201.9	-198.8	-194.0	-188.8	-183.1	-218.7
Ozone	O ₃	-180.4	-168.6	-163.2	-157.2	-150.7	-146.7	-141.0	-132.6	-122.5	-111.1	-251
Phosgene	COCl ₂	-92.9	-77.0	-69.3	-60.3	-50.3	-44.0	-35.6	-22.3	-7.6	+8.3	-104
Phosphorus (yellow)	P	76.6	111.2	128.0	146.2	166.7	179.8	197.3	222.7	251.0	280.0	44.1
(violet)	P	237	271	287	306	323	334	349	370	391	417	590
tribromide	PBr ₃	7.8	34.4	47.8	62.4	79.0	89.8	103.6	125.2	149.7	175.3	-40
trichloride	PCl ₃	-51.6	-31.5	-21.3	-10.2	+2.3	10.2	21.0	37.6	56.9	74.2	-111.8
pentachloride	PCl ₅	55.5	74.0	83.2	92.5	102.5	108.3	117.0	131.3	147.2	162.0	
Phosphine	PH ₃					-129.4	-125.0	-118.8	-109.4	-98.3	-87.5	-132.5
Phosphonium bromide	PH ₃ Br	-43.7	-28.5	-21.2	-13.3	-5.0	+0.3	7.4	17.6	28.0	38.3	
chloride	PH ₃ Cl	-91.0	-79.6	-74.0	-68.0	-61.5	-57.3	-52.0	-44.0	-35.4	-27.0	-28.5
iodide	PH ₃ I	-25.2	-9.0	-1.1	+7.3	16.1	21.9	29.3	39.9	51.6	62.3	
Phosphorus trioxide	P ₄ O ₆		39.7	53.0	67.8	84.0	94.2	108.3	129.0	150.3	173.1	22.5
pentoxide	P ₄ O ₁₀	384	424	442	462	481	493	510	532	556	591	569
oxychloride	POCl ₃			2.0	13.6	27.3	35.8	47.4	65.0	84.3	105.1	2
thiobromide	PSBr ₃	50.0	72.4	83.6	95.5	108.0	116.0	126.3	141.8	157.8	175.0	38
thiochloride	PSCl ₃	-18.3	+4.6	16.1	29.0	42.7	51.8	63.8	82.0	102.3	124.0	-36.2
Platinum	Pt	2730	3007	3146	3302	3469	3574	3714	3923	4169	4407	1755
Potassium	K	341	408	443	483	524	550	586	643	708	774	62.3
bromide	KBr	795	892	940	994	1050	1087	1137	1212	1297	1383	730
chloride	KCl	821	919	968	1020	1078	1115	1164	1239	1322	1407	790
fluoride	KF	885	988	1039	1096	1156	1193	1245	1323	1411	1502	880
hydroxide	KOH	719	814	863	918	976	1013	1064	1142	1233	1327	380
iodide	KI	745	840	887	938	995	1030	1080	1152	1238	1324	723
Radon	Rn	-144.2	-132.4	-126.3	-119.2	-111.3	-106.2	-99.0	-87.7	-75.0	-61.8	-71
Rhenium heptoxide	Re ₂ O ₇	212.5	237.5	248.0	261.0	272.0	280.0	289.0	307.0	336.0	362.4	296
Rubidium	Rb	297	358	389	422	459	482	514	563	620	679	38.5
bromide	RbBr	781	876	923	975	1031	1066	1114	1186	1267	1352	682
chloride	RbCl	792	887	937	990	1047	1084	1133	1207	1294	1381	715
fluoride	RbF	921	982	1016	1052	1096	1123	1168	1239	1322	1408	760
iodide	RbI	748	839	884	935	991	1026	1072	1141	1223	1304	642
Selenium	Se	356	413	442	473	506	527	554	594	637	680	217
dioxide	SeO ₂	157.0	187.7	202.5	217.5	234.1	244.6	258.0	277.0	297.7	317.0	340
hexafluoride	SeF ₆	-118.6	-105.2	-98.9	-92.3	-84.7	-80.0	-73.9	-64.8	-55.2	-45.8	-34.7
oxychloride	SeOCl ₂	34.8	59.8	71.9	84.2	98.0	106.5	118.0	134.6	151.7	168.0	8.5
tetrachloride	SeCl ₄	74.0	96.3	107.4	118.1	130.1	137.8	147.5	161.0	176.4	191.5	
Silicon	Si	1724	1835	1888	1942	2000	2036	2083	2151	2220	2287	1420
dioxide	SiO ₂			1732	1798	1867	1911	1969	2053	2141	2227	1710
tetrachloride	SiCl ₄	-63.4	-44.1	-34.4	-24.0	-12.1	-4.8	+5.4	21.0	38.4	56.8	-68.8
tetrafluoride	SiF ₄	-144.0	-134.8	-130.4	-125.9	-120.8	-117.5	-113.3	-107.2	-100.7	-94.8	-90
Trichlorofluorosilane	SiFCl ₃	-92.6	-76.4	-68.3	-59.0	-48.8	-42.2	-33.2	-19.3	-4.0	+12.2	-120.8
Iodosilane	SiH ₃ I		-53.0	-47.7	-33.4	-21.8	-14.3	-4.4	+10.7	27.9	45.4	-57.0
Diiodosilane	SiH ₂ I ₂		3.8	18.0	34.1	52.6	64.0	79.4	101.8	125.5	149.5	-1.0
Disiloxan	(SiH ₃) ₂ O	-112.5	-95.8	-88.2	-79.8	-70.4	-64.2	-55.9	-43.5	-29.3	-15.4	-144.2
Trisilane	Si ₃ H ₈	-68.9	-49.7	-40.0	-29.0	-16.9	-9.0	+1.6	17.8	35.5	53.1	-117.2
Trisilazane	(SiH ₃) ₃ N	-68.7	-49.9	-40.4	-30.0	-18.5	-11.0	-1.1	+14.0	31.0	48.7	-105.7
Tetrasilane	Si ₄ H ₁₀	-27.7	-6.2	+4.3	15.8	28.4	36.6	47.4	63.6	81.7	100.0	-93.6
Octachlorotrisilane	Si ₃ Cl ₈	46.3	74.7	89.3	104.2	121.5	132.0	146.0	166.2	189.5	211.4	
Hexachlorodisiloxane	(SiCl ₃) ₂ O	-5.0	17.8	29.4	41.5	55.2	63.8	75.4	92.5	113.6	135.6	-33.2
Hexachlorodisilane	Si ₂ Cl ₆	+4.0	27.4	38.8	51.5	65.3	73.9	85.4	102.2	120.6	139.0	-1.2
Tribromosilane	SiHBr ₃	-30.5	-8.0	+3.4	16.0	30.0	39.2	51.6	70.2	90.2	111.8	-73.5
Trichlorosilane	SiHCl ₃	-80.7	-62.6	-53.4	-43.8	-32.9	-25.8	-16.4	-1.8	+14.5	31.8	-126.6
Trifluorosilane	SiHF ₃	-152.0	-142.7	-138.2	-132.9	-127.3	-123.7	-118.7	-111.3	-102.8	-95.0	-131.4
Dibromosilane	SiH ₂ Br ₂	-60.9	-40.0	-29.4	-18.0	-5.2	+3.2	14.1	31.6	50.7	70.5	-70.2
Difluorosilane	SiH ₂ F ₂	-146.7	-136.0	-130.4	-124.3	-117.6	-113.3	-107.3	-98.3	-87.6	-77.8	
Monobromosilane	SiH ₃ Br		-85.7	-77.3	-68.3	-57.8	-51.1	-42.3	-28.6	-13.3	+2.4	-93.9
Monochlorosilane	SiH ₃ Cl	-117.8	-104.3	-97.7	-90.1	-81.8	-76.0	-68.5	-57.0	-44.5	-30.4	
Monofluorosilane	SiH ₃ F	-153.0	-145.5	-141.2	-136.3	-130.8	-127.2	-122.4	-115.2	-106.8	-98.0	
Tribromofluorosilane	SiFBr ₃	-46.1	-25.4	-15.1	-3.7	+9.2	17.4	28.6	45.7	64.6	83.8	-82.5
Dichlorodifluorosilane	SiF ₂ Cl ₂	-124.7	-110.5	-102.9	-94.5	-85.0	-78.6	-70.3	-58.0	-45.0	-31.8	-139.7
Trifluorobromosilane	SiF ₃ Br								-69.8	-55.9	-41.7	-70.5

TABLE 2-9 Vapor Pressures of Inorganic Compounds, up to 1 atm (Concluded)

Compound		Pressure, mmHg										Melting point, °C
Name	Formula	1	5	10	20	40	60	100	200	400	760	
		Temperature, °C										
Trifluorochlorosilane	SiF ₃ Cl	-144.0	-133.0	-127.0	-120.5	-112.8	-108.2	-101.7	-91.7	-81.0	-70.0	-142
Hexafluorodisilane	Si ₂ F ₆	-81.0	-68.8	-63.1	-57.0	-50.6	-46.7	-41.7	-34.2	-26.4	-18.9	-18.6
Dichlorofluorobromosilane	SiFCl ₂ Br	-86.5	-68.4	-59.0	-48.8	-37.0	-29.0	-19.5	-3.2	+15.4	35.4	-112.3
Dibromochlorofluorosilane	SiFClBr ₂	-65.2	-45.5	-35.6	-24.5	-12.0	-4.7	+6.3	23.0	43.0	59.5	-99.3
Silane	SiH ₄	-179.3	-168.6	-163.0	-156.9	-150.3	-146.3	-140.5	-131.6	-122.0	-111.5	-185
Disilane	Si ₂ H ₆	-114.8	-99.3	-91.4	-82.7	-72.8	-66.4	-57.5	-44.6	-29.0	-14.3	-132
Silver	Ag	1357	1500	1575	1658	1743	1795	1865	1971	2090	2212	960.5
chloride	AgCl	912	1019	1074	1134	1200	1242	1297	1379	1467	1564	455
iodide	AgI	820	927	983	1045	1111	1152	1210	1297	1400	1506	552
Sodium	Na	439	511	549	589	633	662	701	758	823	892	97.5
bromide	NaBr	806	903	952	1005	1063	1099	1148	1220	1304	1392	755
chloride	NaCl	865	967	1017	1072	1131	1169	1220	1296	1379	1465	800
cyanide	NaCN	817	928	983	1046	1115	1156	1214	1302	1401	1497	564
fluoride	NaF	1077	1186	1240	1300	1363	1403	1455	1531	1617	1704	992
hydroxide	NaOH	739	843	897	953	1017	1057	1111	1192	1286	1378	318
iodide	NaI	767	857	903	952	1005	1039	1083	1150	1225	1304	651
Strontium	Sr		847	898	953	1018	1057	1111	1192	1285	1384	800
Strontium oxide	SrO	2068	2198	2262	2333	2410						2430
Sulfur	S	183.8	223.0	243.8	264.7	288.3	305.5	327.2	359.7	399.6	444.6	112.8
monochloride	S ₂ Cl ₂	-7.4	+15.7	27.5	40.0	54.1	63.2	75.3	93.5	115.4	138.0	-80
hexafluoride	SF ₆	-132.7	-120.6	-114.7	-108.4	-101.5	-96.8	-90.9	-82.3	-72.6	-63.5	-50.2
Sulfuryl chloride	SO ₂ Cl ₂		-35.1	-24.8	-13.4	-1.0	+7.2	17.8	33.7	51.3	69.2	-54.1
Sulfur dioxide	SO ₂	-95.5	-83.0	-76.8	-69.7	-60.5	-54.6	-46.9	-35.4	-23.0	-10.0	-73.2
trioxide (α)	SO ₃	-39.0	-23.7	-16.5	-9.1	-1.0	+4.0	10.5	20.5	32.6	44.8	16.8
trioxide (β)	SO ₃	-34.0	-19.2	-12.3	-4.9	+3.2	8.0	14.3	23.7	32.6	44.8	32.3
trioxide (γ)	SO ₃	-15.3	-2.0	+4.3	11.1	17.9	21.4	28.0	35.8	44.0	51.6	62.1
Tellurium	Te	520	605	650	697	753	789	838	910	997	1087	452
chloride	TeCl ₄			233	253	273	287	304	330	360	392	224
fluoride	TeF ₆	-111.3	-98.8	-92.4	-86.0	-78.4	-73.8	-67.9	-57.3	-48.2	-38.6	-37.8
Thallium	Tl	825	931	983	1040	1103	1143	1196	1274	1364	1457	3035
Thallous bromide	TlBr		490	522	559	598	621	653	703	759	819	460
chloride	TlCl		487	517	550	589	612	645	694	748	807	430
iodide	TlI	440	502	531	567	607	631	663	712	763	823	440
Thionyl bromide	SOBr ₂	-6.7	+18.4	31.0	44.1	58.8	68.3	80.6	99.0	119.2	139.5	-52.2
Thionyl chloride	SOCl ₂	-52.9	-32.4	-21.9	-10.5	+2.2	10.4	21.4	37.9	56.5	75.4	-104.5
Tin	Sn	1492	1634	1703	1777	1855	1903	1968	2063	2169	2270	231.9
Stannic bromide	SnBr ₄		58.3	72.7	88.1	105.5	116.2	131.0	152.8	177.7	204.7	31.0
Stannous chloride	SnCl ₂	316	366	391	420	450	467	493	533	577	623	246.8
Stannic chloride	SnCl ₄	-22.7	-1.0	+10.0	22.0	35.2	43.5	54.7	72.0	92.1	113.0	-30.2
iodide	SnI ₄		156.0	175.8	196.2	218.8	234.2	254.2	283.5	315.5	348.0	144.5
hydride	SnH ₄	-140.0	-125.8	-118.5	-111.2	-102.3	-96.6	-89.2	-78.0	-65.2	-52.3	-149.9
Tin tetramethyl	Sn(CH ₃) ₄	-51.3	-31.0	-20.6	-9.3	+3.5	11.7	22.8	39.8	58.5	78.0	
trimethyl-ethyl	Sn(CH ₃) ₃ C ₂ H ₅	-30.0	-7.6	+3.8	16.1	30.0	38.4	50.0	67.3	87.6	108.8	
trimethyl-propyl	Sn(CH ₃) ₃ C ₃ H ₇	-12.0	+10.7	21.8	34.0	48.5	57.5	69.8	88.0	109.6	131.7	
Titanium chloride	TiCl ₄	-13.9	+9.4	21.3	34.2	48.4	58.0	71.0	90.5	112.7	136.0	-30
Tungsten	W	3990	4337	4507	4690	4886	5007	5168	5403	5666	5927	3370
Tungsten hexafluoride	WF ₆	-71.4	-56.5	-49.2	-41.5	-33.0	-27.5	-20.3	-10.0	+1.2	17.3	-0.5
Uranium hexafluoride	UF ₆	-38.8	-22.0	-13.8	-5.2	+4.4	10.4	18.2	30.0	42.7	55.7	69.2
Vanadyl trichloride	VOCl ₃	-23.2	+0.2	12.2	26.6	40.0	49.8	62.5	82.0	103.5	127.2	
Xenon	Xe	-168.5	-158.2	-152.8	-147.1	-141.2	-137.7	-132.8	-125.4	-117.1	-108.0	-111.6
Zinc	Zn	487	558	593	632	673	700	736	788	844	907	419.4
chloride	ZnCl ₂	428	481	508	536	566	584	610	648	689	732	365
fluoride	ZnF ₂	970	1055	1086	1129	1175	1207	1254	1329	1417	1497	872
diethyl	Zn(C ₂ H ₅) ₂	-22.4	0.0	+11.7	24.2	38.0	47.2	59.1	77.0	97.3	118.0	-28
Zirconium bromide	ZrBr ₄	207	237	250	266	281	289	301	318	337	357	450
chloride	ZrCl ₄	190	217	230	243	259	268	279	295	312	331	437
iodide	ZrI ₄	264	297	311	329	344	355	369	389	409	431	499

TABLE 2-10 Vapor Pressures of Organic Compounds, up to 1 atm*

Compound		Pressure, mmHg										Melting point, °C
		1	5	10	20	40	60	100	200	400	760	
Name	Formula	Temperature, °C										
Acenaphthalene	C ₁₂ H ₁₀		114.8	131.2	148.7	168.2	181.2	197.5	222.1	250.0	277.5	95
Acetal	C ₆ H ₁₄ O ₂	-23.0	-2.3	+8.0	19.6	31.9	39.8	50.1	66.3	84.0	102.2	
Acetaldehyde	C ₂ H ₄ O	-81.5	-65.1	-56.8	-47.8	-37.8	-31.4	-22.6	-10.0	+4.9	20.2	-123.5
Acetamide	C ₂ H ₅ NO	65.0	92.0	105.0	120.0	135.8	145.8	158.0	178.3	200.0	222.0	81
Acetanilide	C ₈ H ₉ NO	114.0	146.6	162.0	180.0	199.6	211.8	227.2	250.5	277.0	303.8	113.5
Acetic acid	C ₂ H ₄ O ₂	-17.2	+6.3	17.5	29.9	43.0	51.7	63.0	80.0	99.0	118.1	16.7
anhydride	C ₄ H ₆ O ₃	1.7	24.8	36.0	48.3	62.1	70.8	82.2	100.0	119.8	139.6	-73
Acetone	C ₃ H ₆ O	-59.4	-40.5	-31.1	-20.8	-9.4	-2.0	+7.7	22.7	39.5	56.5	-94.6
Acetonitrile	C ₂ H ₃ N	-47.0	-26.6	-16.3	-5.0	+7.7	15.9	27.0	43.7	62.5	81.8	-41
Acetophenone	C ₈ H ₈ O	37.1	64.0	78.0	92.4	109.4	119.8	133.6	154.2	178.0	202.4	20.5
Acetyl chloride	C ₂ H ₃ OCl	-50.0	-35.0	-27.6	-19.6	-10.4	-4.5	+3.2	16.1	32.0	50.8	-112.0
Acetylene	C ₂ H ₂	-142.9	-133.0	-128.2	-122.8	-116.7	-112.8	-107.9	-100.3	-92.0	-84.0	-81.5
Acridine	C ₁₃ H ₉ N	129.4	165.8	184.0	203.5	224.2	238.7	256.0	284.0	314.3	346.0	110.5
Acrolein (2-propenal)	C ₃ H ₄ O	-64.5	-46.0	-36.7	-26.3	-15.0	-7.5	+2.5	17.5	34.5	52.5	-87.7
Acrylic acid	C ₃ H ₄ O ₂	+3.5	27.3	39.0	52.0	66.2	75.0	86.1	103.3	122.0	141.0	14
Adipic acid	C ₆ H ₁₀ O ₄	159.5	191.0	205.5	222.0	240.5	251.0	265.0	287.8	312.5	337.5	152
Allene (propadiene)	C ₃ H ₄	-120.6	-108.0	-101.0	-93.4	-85.2	-78.8	-72.5	-61.3	-48.5	-35.0	-136
Allyl alcohol (propen-1-ol-3)	C ₃ H ₆ O	-20.0	+0.2	10.5	21.7	33.4	40.3	50.0	64.5	80.2	96.6	-129
chloride (3-chloropropene)	C ₃ H ₅ Cl	-70.0	-52.0	-42.9	-32.8	-21.2	-14.1	-4.5	10.4	27.5	44.6	-136.4
isopropyl ether	C ₆ H ₁₂ O	-43.7	-23.1	-12.9	-1.8	+10.9	18.7	29.0	44.3	61.7	79.5	
isothiocyanate	C ₃ H ₅ NS	-2.0	+25.3	38.3	52.1	67.4	76.2	89.5	108.0	129.8	150.7	-80
<i>n</i> -propyl ether	C ₆ H ₁₂ O	-39.0	-18.2	-7.9	+3.7	16.4	25.0	35.8	52.6	71.4	90.5	
4-Allylveratrole	C ₁₁ H ₁₄ O ₂	85.0	113.9	127.0	142.8	158.3	169.6	183.7	204.0	226.2	248.0	
iso-Amyl acetate	C ₇ H ₁₄ O ₂	0.0	+23.7	35.2	47.8	62.1	71.0	83.2	101.3	121.5	142.0	
<i>n</i> -Amyl alcohol	C ₆ H ₁₂ O	+13.6	34.7	44.9	55.8	68.0	75.5	85.8	102.0	119.8	137.8	
iso-Amyl alcohol	C ₆ H ₁₂ O	+10.0	30.9	40.8	51.7	63.4	71.0	80.7	95.8	113.7	130.6	-117.2
<i>sec</i> -Amyl alcohol (2-pentanol)	C ₆ H ₁₂ O	+1.5	22.1	32.2	42.6	54.1	61.5	70.7	85.7	102.3	119.7	
<i>tert</i> -Amyl alcohol	C ₆ H ₁₂ O	-12.9	+7.2	17.2	27.9	38.8	46.0	55.3	69.7	85.7	101.7	-11.9
<i>sec</i> -Amylbenzene	C ₁₁ H ₁₆	29.0	55.8	69.2	83.8	100.0	110.4	124.1	145.2	168.0	193.0	
iso-Amyl benzoate	C ₁₂ H ₁₆ O ₂	72.0	104.5	121.6	139.7	158.3	171.4	186.8	210.2	235.8	262.0	
bromide (1-bromo-3-methylbutane)	C ₆ H ₁₁ Br	-20.4	+2.1	13.6	26.1	39.8	48.7	60.4	78.7	99.4	120.4	
<i>n</i> -butyrate	C ₈ H ₁₆ O ₂	21.2	47.1	59.9	74.0	90.0	99.8	113.1	133.2	155.3	178.6	
formate	C ₆ H ₁₂ O ₂	-17.5	+5.4	17.1	30.0	44.0	53.3	65.4	83.2	102.7	123.3	
iodide (1-iodo-3-methylbutane)	C ₅ H ₁₁ I	-2.5	+21.9	34.1	47.6	62.3	71.9	84.4	103.8	125.8	148.2	
isobutyrate	C ₈ H ₁₆ O ₂	14.8	40.1	52.8	66.6	81.8	91.7	104.4	124.2	146.0	168.8	
Amyl isopropionate	C ₈ H ₁₆ O ₂	+8.5	33.7	46.3	60.0	75.5	85.2	97.6	117.3	138.4	160.2	
iso-Amyl isovalerate	C ₁₀ H ₂₀ O ₂	27.0	54.4	68.6	83.8	100.6	110.3	125.1	146.1	169.5	194.0	
<i>n</i> -Amyl levulinate	C ₁₀ H ₁₈ O ₃	81.3	110.0	124.0	139.7	155.8	165.2	180.5	203.1	227.4	253.2	
iso-Amyl levulinate	C ₁₀ H ₁₈ O ₃	75.6	104.0	118.8	134.4	151.7	162.6	177.0	198.1	222.7	247.9	
nitrate	C ₈ H ₁₁ NO ₃	+5.2	28.8	40.3	53.5	67.6	76.3	88.6	106.7	126.5	147.5	
4- <i>tert</i> -Amylphenol	C ₁₁ H ₁₆ O		109.8	125.5	142.3	160.3	172.6	189.0	213.0	239.5	266.0	93
Anethole	C ₁₀ H ₁₂ O	62.6	91.6	106.0	121.8	139.3	149.8	164.2	186.1	210.5	235.3	22.5
Angelonitrile	C ₈ H ₇ N	-8.0	+15.0	28.0	41.0	55.8	65.2	77.5	96.3	117.7	140.0	
Aniline	C ₆ H ₇ N	34.8	57.9	69.4	82.0	96.7	106.0	119.9	140.1	161.9	184.4	-6.2
2-Anilinoethanol	C ₈ H ₁₁ NO	104.0	134.3	149.6	165.7	183.7	194.0	209.5	230.6	254.5	279.6	
Anisaldehyde	C ₈ H ₈ O ₂	73.2	102.6	117.8	133.5	150.5	161.7	176.7	199.0	223.0	248.0	2.5
<i>o</i> -Anisidine (2-methoxyaniline)	C ₇ H ₉ NO	61.0	88.0	101.7	116.1	132.0	142.1	155.2	175.3	197.3	218.5	5.2
Anthracene	C ₁₄ H ₁₀	145.0	173.5	187.2	201.9	217.5	231.8	250.0	279.0	310.2	342.0	217.5
Anthraquinone	C ₁₄ H ₈ O ₂	190.0	219.4	234.2	248.3	264.3	273.3	285.0	314.6	346.2	379.9	286
Azelaic acid	C ₉ H ₁₆ O ₄	178.3	210.4	225.5	242.4	260.0	271.8	286.5	309.6	332.8	356.5	106.5
Azaldehyde	C ₉ H ₁₀ O	33.3	58.4	71.6	85.0	100.2	110.0	123.0	142.1	163.4	185.0	
Azobenzene	C ₁₂ H ₁₀ N ₂	103.5	135.7	151.5	168.3	187.9	199.8	216.0	240.0	266.1	293.0	68
Benzal chloride (α,α -Dichlorotoluene)	C ₇ H ₆ Cl ₂	35.4	64.0	78.7	94.3	112.1	123.4	138.3	160.7	187.0	214.0	-16.1
Benzaldehyde	C ₇ H ₆ O	26.2	50.1	62.0	75.0	90.1	99.6	112.5	131.7	154.1	179.0	-26
Benzanthrone	C ₁₇ H ₁₀ O	225.0	274.5	297.2	322.5	350.0	368.8	390.0	426.5			174
Benzene	C ₆ H ₆	-36.7	-19.6	-11.5	-2.6	+7.6	15.4	26.1	42.2	60.6	80.1	+5.5
Benzenesulfonylchloride	C ₆ H ₅ ClO ₂ S	65.9	96.5	112.0	129.0	147.7	158.2	174.5	198.0	224.0	251.5	14.5
Benzil	C ₁₄ H ₁₀ O ₂	128.4	165.2	183.0	202.8	224.5	238.2	255.8	283.5	314.3	347.0	95
Benzoic acid	C ₇ H ₆ O ₂	96.0	119.5	132.1	146.7	162.6	172.8	186.2	205.8	227.0	249.2	121.7
anhydride	C ₁₄ H ₁₀ O ₃	143.8	180.0	198.0	218.0	239.8	252.7	270.4	299.1	328.8	360.0	42
Benzoin	C ₁₄ H ₁₂ O ₂	135.6	170.2	188.1	207.0	227.6	241.7	258.0	284.4	313.5	343.0	132
Benzonitrile	C ₇ H ₅ N	28.2	55.3	69.2	83.4	99.6	109.8	123.5	144.1	166.7	190.6	-12.9
Benzophenone	C ₁₃ H ₁₀ O	108.2	141.7	157.6	175.8	195.7	208.2	224.4	249.8	276.8	305.4	48.5
Benzotrichloride (α,α,α -Trichlorotoluene)	C ₆ H ₂ Cl ₃	45.8	73.7	87.6	102.7	119.8	130.0	144.3	165.6	189.2	213.5	-21.2
Benzotrifluoride (α,α,α -Trifluorotoluene)	C ₆ H ₅ F ₃	-32.0	-10.3	-0.4	12.2	25.7	34.0	45.3	62.5	82.0	102.2	-29.3
Benzoyl bromide	C ₇ H ₅ BrO	47.0	75.4	89.8	105.4	122.6	133.4	147.7	169.2	193.7	218.5	0
chloride	C ₇ H ₅ ClO	32.1	59.1	73.0	87.6	103.8	114.7	128.0	149.5	172.8	197.2	-0.5
nitride	C ₈ H ₅ NO	44.5	71.7	85.5	100.2	116.6	127.0	141.0	161.3	185.0	208.0	33.5
Benzyl acetate	C ₉ H ₁₀ O ₂	45.0	73.4	87.6	102.3	119.6	129.8	144.0	165.5	189.0	213.5	-51.5
alcohol	C ₇ H ₈ O	58.0	80.8	92.6	105.8	119.8	129.3	141.7	160.0	183.0	204.7	-15.3

*Compiled from the extended tables published by D. R. Stull in *Ind. Eng. Chem.*, **39**, 517 (1947). For information on fuels see Hibbard, N.A.C.A. Research Mem. E56121, 1956. For methane see Johnson (ed.), WADD-TR-60-56, 1960.

TABLE 2-10 Vapor Pressures of Organic Compounds, up to 1 atm (Continued)

Compound		Pressure, mmHg										Melting point, °C
		1	5	10	20	40	60	100	200	400	760	
Name	Formula	Temperature, °C										
Benzylamine	C ₇ H ₉ N	29.0	54.8	67.7	81.8	97.3	107.3	120.0	140.0	161.3	184.5	
Benzyl bromide (α -bromotoluene)	C ₇ H ₇ Br	32.2	59.6	73.4	88.3	104.8	115.6	129.8	150.8	175.2	198.5	-4
chloride (α -chlorotoluene)	C ₇ H ₇ Cl	22.0	47.8	60.8	75.0	90.7	100.5	114.2	134.0	155.8	179.4	-39
cinnamate	C ₁₆ H ₁₄ O ₂	173.8	206.3	221.5	239.3	255.8	267.0	281.5	303.8	326.7	350.0	39
Benzylchlorosilane	C ₇ H ₈ Cl ₂ Si	45.3	70.2	83.2	96.7	111.8	121.3	133.5	152.0	173.0	194.3	
Benzyl ethyl ether	C ₉ H ₁₂ O	26.0	52.0	65.0	79.6	95.4	105.5	118.9	139.6	161.5	185.0	
phenyl ether	C ₁₃ H ₁₀ O	95.4	127.7	144.0	160.7	180.1	192.6	209.2	233.2	259.8	287.0	
isothiocyanate	C ₈ H ₇ NS	79.5	107.8	121.8	137.0	153.0	163.8	177.7	198.0	220.4	243.0	
Biphenyl	C ₁₂ H ₁₀	70.6	101.8	117.0	134.2	152.5	165.2	180.7	204.2	229.4	254.9	69.5
1-Biphenyloxy-2,3-epoxypropane	C ₁₅ H ₁₄ O ₂	135.3	169.9	187.2	205.8	226.3	239.7	255.0	280.4	309.8	340.0	
<i>d</i> -Bornyl acetate	C ₁₂ H ₂₀ O ₂	46.9	75.7	90.2	106.0	123.7	135.7	149.8	172.0	197.5	223.0	29
Bornyl <i>n</i> -butyrate	C ₁₄ H ₂₄ O ₂	74.0	103.4	118.0	133.8	150.7	161.8	176.4	198.0	222.2	247.0	
formate	C ₁₁ H ₁₈ O ₂	47.0	74.8	89.3	104.0	121.2	131.7	145.8	166.4	190.2	214.0	
isobutyrate	C ₁₀ H ₁₈ O ₂	70.0	99.8	114.0	130.0	147.2	157.6	172.2	194.2	218.2	243.0	
propionate	C ₁₃ H ₂₂ O ₂	64.6	93.7	108.0	123.7	140.4	151.2	165.7	187.5	211.2	235.0	
Brassicic acid	C ₂₂ H ₄₂ O ₂	209.6	241.7	256.0	272.9	290.0	301.5	316.2	336.8	359.6	382.5	61.5
Bromoacetic acid	C ₂ H ₃ BrO ₂	54.7	81.6	94.1	108.2	124.0	133.8	146.3	165.8	186.7	208.0	49.5
4-Bromoanisole	C ₇ H ₇ BrO	48.8	77.8	91.9	107.8	125.0	136.0	150.1	172.7	197.5	223.0	12.5
Bromobenzene	C ₆ H ₅ Br	+2.9	27.8	40.0	53.8	68.6	78.1	90.8	110.1	132.3	156.2	-30.7
4-Bromobiphenyl	C ₁₂ H ₉ Br	98.0	133.7	150.6	169.8	190.8	204.5	221.8	248.2	277.7	310.0	90.5
1-Bromo-2-butanol	C ₄ H ₉ BrO	23.7	45.4	55.8	67.2	79.5	87.0	97.6	112.1	128.3	145.0	
1-Bromo-2-butanone	C ₄ H ₇ BrO	+6.2	30.0	41.8	54.2	68.2	77.3	89.2	107.0	126.3	147.0	
<i>cis</i> -1-Bromo-1-butene	C ₄ H ₇ Br	-44.0	-23.2	-12.8	-1.4	+11.5	19.8	30.8	47.8	66.8	86.2	
<i>trans</i> -1-Bromo-1-butene	C ₄ H ₇ Br	-38.4	-17.0	-6.4	+5.4	18.4	27.2	38.1	55.7	75.0	94.7	-100.3
2-Bromo-1-butene	C ₄ H ₇ Br	-47.3	-27.0	-16.8	-5.3	+7.2	15.4	26.3	42.8	61.9	81.0	-133.4
<i>cis</i> -2-Bromo-2-butene	C ₄ H ₇ Br	-39.0	-17.9	-7.2	+4.6	17.7	26.2	37.5	54.5	74.0	93.9	-111.2
<i>trans</i> -2-Bromo-2-butene	C ₄ H ₇ Br	-45.0	-24.1	-13.8	-2.4	+10.5	18.7	29.9	46.5	66.0	85.5	-114.6
1,4-Bromochlorobenzene	C ₆ H ₄ BrCl	32.0	59.5	72.7	87.8	103.8	114.8	128.0	149.5	172.6	196.9	
1-Bromo-1-chloroethane	C ₂ H ₄ BrCl	-36.0	-18.0	-9.4	0.0	+10.4	17.0	28.0	44.7	63.4	82.7	16.6
1-Bromo-2-chloroethane	C ₂ H ₄ BrCl	-28.8	-7.0	+4.1	16.0	29.7	38.0	49.5	66.8	86.0	106.7	-16.6
2-Bromo-4,6-dichlorophenol	C ₆ H ₃ BrCl ₂ O	84.0	115.6	130.8	147.7	165.8	177.6	193.2	216.5	242.0	268.0	68
1-Bromo-4-ethyl benzene	C ₈ H ₉ Br	30.4	42.5	74.0	90.2	108.5	121.0	135.5	156.5	182.0	206.0	-45.0
(2-Bromoethyl)-benzene	C ₈ H ₉ Br	48.0	76.2	90.5	105.8	123.2	133.8	148.2	169.8	194.0	219.0	
2-Bromoethyl 2-chloroethyl ether	C ₆ H ₁₀ BrClO	36.5	63.2	76.3	90.8	106.6	116.4	129.8	150.0	172.3	195.8	
(2-Bromoethyl)-cyclohexane	C ₈ H ₁₅ Br	38.7	66.6	80.5	95.8	113.0	123.7	138.0	160.0	186.2	213.0	
1-Bromoethylene	C ₂ H ₃ Br	-95.4	-77.8	-68.8	-58.8	-48.1	-41.2	-31.9	-17.2	-1.1	+15.8	-138
Bromoform (tribromomethane)	CHBr ₃	22.0	34.0	48.0	63.6	73.4	85.9	106.1	127.9	150.5	175.0	8.5
1-Bromonaphthalene	C ₁₀ H ₇ Br	84.2	117.5	133.6	150.2	170.2	183.5	198.8	224.2	252.0	281.1	5.5
2-Bromo-4-phenylphenol	C ₁₂ H ₉ BrO	100.0	135.4	152.3	171.8	193.8	207.0	224.5	251.0	280.2	311.0	95
3-Bromopyridine	C ₅ H ₄ BrN	16.8	42.0	55.2	69.1	84.1	94.1	107.8	127.7	150.0	173.4	
2-Bromotoluene	C ₇ H ₇ Br	24.4	49.7	62.3	76.0	91.0	100.0	112.0	133.6	157.3	181.8	-28
3-Bromotoluene	C ₇ H ₇ Br	14.8	50.8	64.0	78.1	93.9	104.1	117.8	138.0	160.0	183.7	39.8
4-Bromotoluene	C ₇ H ₇ Br	10.3	47.5	61.1	75.2	91.8	102.3	116.4	137.4	160.2	184.5	28.5
3-Bromo-2,4,6-trichlorophenol	C ₆ H ₃ BrCl ₃ O	112.4	146.2	163.2	181.8	200.5	213.0	229.3	253.0	278.0	305.8	
2-Bromo-1,4-xylene	C ₈ H ₉ Br	37.5	65.0	78.8	94.0	110.6	121.6	135.7	156.4	181.0	206.7	+9.5
1,2-Butadiene (methyl allene)	C ₄ H ₆	-89.0	-72.7	-64.2	-54.9	-44.3	-37.5	-28.3	-14.2	+1.8	18.5	
1,3-Butadiene	C ₄ H ₆	-102.8	-87.6	-79.7	-71.0	-61.3	-55.1	-46.8	-33.9	-19.3	-4.5	-108.9
<i>n</i> -Butane	C ₄ H ₁₀	-101.5	-85.7	-77.8	-68.9	-59.1	-52.8	-44.2	-31.2	-16.3	-0.5	-135
iso-Butane (2-methylpropane)	C ₄ H ₁₀	-109.2	-94.1	-86.4	-77.9	-68.4	-62.4	-54.1	-41.5	-27.1	-11.7	-145
1,3-Butanediol	C ₄ H ₁₀ O ₂	22.2	67.5	85.3	100.0	117.4	127.5	141.2	161.0	183.8	206.5	77
1,2,3-Butanetriol	C ₄ H ₁₀ O ₃	102.0	132.0	146.0	161.0	178.0	188.0	202.5	222.0	243.5	264.0	
1-Butene	C ₄ H ₈	-104.8	-89.4	-81.6	-73.0	-63.4	-57.2	-48.9	-36.2	-21.7	-6.3	-130
<i>cis</i> -2-Butene	C ₄ H ₈	-96.4	-81.1	-73.4	-64.6	-54.7	-48.4	-39.8	-26.8	-12.0	+3.7	-138.9
<i>trans</i> -2-Butene	C ₄ H ₈	-99.4	-84.0	-76.3	-67.5	-57.6	-51.3	-42.7	-29.7	-14.8	+0.9	-105.4
3-Butenenitrile	C ₄ H ₅ N	-19.6	+2.9	14.1	26.6	40.0	48.8	60.2	78.0	98.0	119.0	
iso-Butyl acetate	C ₆ H ₁₂ O ₂	-21.2	+1.4	12.8	25.5	39.2	48.0	59.7	77.6	97.5	118.0	-98.9
<i>n</i> -Butyl acrylate	C ₈ H ₁₂ O ₂	-0.5	+23.5	35.5	48.6	63.4	72.6	85.1	104.0	125.2	147.4	-64.6
alcohol	C ₄ H ₁₀ O	-1.2	+20.0	30.2	41.5	53.4	60.3	70.1	84.3	100.8	117.5	-79.9
iso-Butyl alcohol	C ₄ H ₁₀ O	-9.0	+11.6	21.7	32.4	44.1	51.7	61.5	75.9	91.4	108.0	-108
<i>sec</i> -Butyl alcohol	C ₄ H ₁₀ O	-12.2	+7.2	16.9	27.3	38.1	45.2	54.1	67.9	83.9	99.5	-114.7
<i>tert</i> -Butyl alcohol	C ₄ H ₁₀ O	-20.4	-3.0	+5.5	14.3	24.5	31.0	39.8	52.7	68.0	82.9	25.3
iso-Butyl amine	C ₄ H ₁₁ N	-50.0	-31.0	-21.0	-10.3	+1.3	8.8	18.8	32.0	50.7	68.6	-85.0
<i>n</i> -Butylbenzene	C ₁₀ H ₁₄	22.7	48.8	62.0	76.3	92.4	102.6	116.2	136.9	159.2	183.1	-88.0
iso-Butylbenzene	C ₁₀ H ₁₄	14.1	40.5	53.7	67.8	83.3	93.3	107.0	127.2	149.6	172.8	-51.5
<i>sec</i> -Butylbenzene	C ₁₀ H ₁₄	18.6	44.2	57.0	70.6	86.2	96.0	109.5	128.8	150.3	173.5	-75.5
<i>tert</i> -Butylbenzene	C ₁₀ H ₁₄	13.0	39.0	51.7	65.6	80.8	90.6	103.8	123.7	145.8	168.5	-58
iso-Butyl benzoate	C ₁₁ H ₁₄ O ₂	64.0	93.6	108.6	124.2	141.8	152.0	166.4	188.2	212.8	237.0	
<i>n</i> -Butyl bromide (1-bromobutane)	C ₄ H ₉ Br	-33.0	-11.2	-0.3	+11.6	24.8	33.4	44.7	62.0	81.7	101.6	-112.4
iso-Butyl <i>n</i> -butyrate	C ₈ H ₁₆ O ₂	+4.6	30.0	42.2	56.1	71.7	81.3	94.0	113.9	135.7	156.9	
carbamate	C ₉ H ₁₁ NO ₂		83.7	96.4	110.1	125.3	134.6	147.2	165.7	186.0	206.5	65
Butyl carbitol (diethylene glycol butyl ether)	C ₈ H ₁₈ O ₃	70.0	95.7	107.8	120.5	135.5	146.0	159.8	181.2	205.0	231.2	
<i>n</i> -Butyl chloride (1-chlorobutane)	C ₄ H ₉ Cl	-49.0	-28.9	-18.6	-7.4	+5.0	13.0	24.0	40.0	58.8	77.8	-123.1
iso-Butyl chloride	C ₄ H ₉ Cl	-53.8	-34.3	-24.5	-13.8	-1.9	+5.9	16.0	32.0	50.0	68.9	-131.2

TABLE 2-10 Vapor Pressures of Organic Compounds, up to 1 atm (Continued)

Compound		Pressure, mmHg										Melting point, °C
		1	5	10	20	40	60	100	200	400	760	
Name	Formula	Temperature, °C										
sec-Butyl chloride (2-Chlorobutane)	C ₄ H ₉ Cl	-60.2	-39.8	-29.2	-17.7	-5.0	+3.4	14.2	31.5	50.0	68.0	-131.3
tert-Butyl chloride	C ₄ H ₉ Cl					-19.0	-11.4	-1.0	+14.6	32.6	51.0	-26.5
sec-Butyl chloroacetate	C ₆ H ₁₁ ClO ₂	17.0	41.8	54.6	68.2	83.6	93.0	105.5	124.1	146.0	167.8	
2-tert-Butyl-4-cresol	C ₁₁ H ₁₆ O	70.0	98.0	112.0	127.2	143.9	153.7	167.0	187.8	210.0	232.6	
4-tert-Butyl-2-cresol	C ₁₁ H ₁₆ O	74.3	103.7	118.0	134.0	150.8	161.7	176.2	197.8	221.8	247.0	
iso-Butyl dichloroacetate	C ₆ H ₁₀ Cl ₂ O ₂	28.6	54.3	67.5	81.4	96.7	106.6	119.8	139.2	160.0	183.0	
2,3-Butylene glycol (2,3-butanediol)	C ₄ H ₁₀ O ₂	44.0	68.4	80.3	93.4	107.8	116.3	127.8	145.6	164.0	182.0	22.5
2-Butyl-2-ethylbutane-1,3-diol	C ₁₀ H ₂₂ O ₂	94.1	122.6	136.8	151.2	167.8	178.0	191.9	212.0	233.5	255.0	
2-tert-Butyl-4-ethylphenol	C ₁₂ H ₁₈ O	76.3	106.2	121.0	137.0	154.0	165.4	179.0	200.3	223.8	247.8	
n-Butyl formate	C ₅ H ₁₀ O ₂	-26.4	-4.7	+6.1	18.0	31.6	39.8	51.0	67.9	86.2	106.0	
iso-Butyl formate	C ₅ H ₁₀ O ₂	-32.7	-11.4	-0.8	+11.0	24.1	32.4	43.4	60.0	79.0	98.2	-95.3
sec-Butyl formate	C ₅ H ₁₀ O ₂	-34.4	-13.3	-3.1	+8.4	21.3	29.6	40.2	56.8	75.2	93.6	
sec-Butyl glycolate	C ₆ H ₁₂ O ₃	28.3	53.6	66.0	79.8	94.2	104.0	116.4	135.5	155.6	177.5	
iso-Butyl iodide (1-iodo-2-methylpropane)	C ₄ H ₉ I	-17.0	+5.8	17.0	29.8	42.8	51.8	63.5	81.0	100.3	120.4	-90.7
isobutyrate	C ₈ H ₁₆ O ₂	+4.1	28.0	39.9	52.4	67.2	75.9	88.0	106.3	126.3	147.5	-80.7
isovalerate	C ₆ H ₁₂ O ₂	16.0	41.2	53.8	67.7	82.7	92.4	105.2	124.8	146.4	168.7	
levulinate	C ₉ H ₁₆ O ₃	65.0	92.1	105.9	120.2	136.2	147.0	160.2	181.8	205.5	229.9	
naphthylketone (1-isovaleronaphthone)	C ₁₅ H ₁₆ O	136.0	167.9	184.0	201.6	219.7	231.5	246.7	269.7	294.0	320.0	
2-sec-Butylphenol	C ₁₀ H ₁₄ O	57.4	86.0	100.8	116.1	133.4	143.9	157.3	179.7	203.8	228.0	
2-tert-Butylphenol	C ₁₀ H ₁₄ O	56.6	84.2	98.1	113.0	129.2	140.0	153.5	173.8	196.3	219.5	
4-iso-Butylphenol	C ₁₀ H ₁₄ O	72.1	100.9	115.5	130.3	147.2	157.0	171.2	192.1	214.7	237.0	
4-sec-Butylphenol	C ₁₀ H ₁₄ O	71.4	100.5	114.8	130.3	147.8	157.9	172.4	194.3	217.6	242.1	
4-tert-Butylphenol	C ₁₀ H ₁₄ O	70.0	99.2	114.0	129.5	146.0	156.0	170.2	191.5	214.0	238.0	99
2-(4-tert-Butylphenoxy)ethyl acetate	C ₁₄ H ₂₀ O ₃	118.0	150.0	165.8	183.3	201.5	212.8	228.0	250.3	277.6	304.4	
4-tert-Butylphenyl dichlorophosphate	C ₁₀ H ₁₃ Cl ₂ O ₃ P	96.0	129.6	146.0	164.0	184.3	197.2	214.3	240.0	268.2	299.0	
tert-Butyl phenyl ketone (pivalophenone)	C ₁₁ H ₁₄ O	57.8	85.7	99.0	114.3	130.4	140.8	154.0	175.0	197.7	220.0	
iso-Butyl propionate	C ₇ H ₁₄ O ₂	-2.3	+20.9	32.3	44.8	58.5	67.6	79.5	97.0	116.4	136.8	-71
4-tert-Butyl-2,5-xyleneol	C ₁₂ H ₁₈ O	88.2	119.8	135.0	151.0	169.8	180.3	195.0	217.5	241.3	265.3	
4-tert-Butyl-2,6-xyleneol	C ₁₂ H ₁₈ O	74.0	103.9	119.0	135.0	152.2	163.6	176.0	196.0	217.8	239.8	
6-tert-Butyl-2,4-xyleneol	C ₁₂ H ₁₈ O	70.3	100.2	115.0	131.0	148.5	158.2	172.0	192.3	214.2	236.5	
6-tert-Butyl-3,4-xyleneol	C ₁₂ H ₁₈ O	83.9	113.6	127.0	143.0	159.7	170.0	184.0	204.5	226.7	249.5	
Butyric acid	C ₄ H ₈ O ₂	25.5	49.8	61.5	74.0	88.0	96.5	108.0	125.5	144.5	163.5	-74
iso-Butyric acid	C ₄ H ₈ O ₂	14.7	39.3	51.2	64.0	77.8	86.3	98.0	115.8	134.5	154.5	-47
Butyronitrile	C ₄ H ₇ N	-20.0	+2.1	13.4	25.7	38.4	47.3	59.0	76.7	96.8	117.5	
iso-Valerophenone	C ₁₁ H ₁₄ O	58.3	87.0	101.4	116.8	133.8	144.6	158.0	180.1	204.2	228.0	
Camphene	C ₁₀ H ₁₆			47.2	60.4	75.7	85.0	97.9	117.5	138.7	160.5	50
Campholenic acid	C ₁₀ H ₁₆ O ₂	97.6	125.7	139.8	153.9	170.0	180.0	193.7	212.7	234.0	256.0	
d-Camphor	C ₁₀ H ₁₆ O	41.5	68.6	82.3	97.5	114.0	124.0	138.0	157.9	182.0	209.2	178.5
Camphylamine	C ₁₀ H ₁₉ N	45.3	74.0	83.7	97.6	112.5	122.0	134.6	153.0	173.8	195.0	
Capraldehyde	C ₁₀ H ₂₀ O	51.9	78.8	92.0	106.3	122.2	132.0	145.3	164.8	186.3	208.5	
Capric acid	C ₁₀ H ₂₀ O ₂	125.0	142.0	152.2	165.0	179.9	189.8	200.0	217.1	240.3	268.4	31.5
n-Caproic acid	C ₆ H ₁₂ O ₂	71.4	89.5	99.5	111.8	125.0	133.3	144.0	160.8	181.0	202.0	-1.5
iso-Caproic acid	C ₆ H ₁₂ O ₂	66.2	83.0	94.0	107.0	120.4	129.6	141.4	158.3	181.0	207.7	-35
iso-Caprolactone	C ₆ H ₁₀ O ₂	38.3	66.4	80.3	95.7	112.3	123.2	137.2	157.8	182.1	207.0	
Capronitrile	C ₆ H ₁₁ N	9.2	34.6	47.5	61.7	76.9	86.8	99.8	119.7	141.0	163.7	
Capryl alcohol (2-octanol)	C ₈ H ₁₈ O	32.8	57.6	70.0	83.3	98.0	107.4	119.8	138.0	157.5	178.5	-38.6
Caprylaldehyde	C ₈ H ₁₆ O	73.4	92.0	101.2	110.2	120.0	126.0	133.9	145.4	156.5	168.5	
Caprylic acid (octanoic acid)	C ₈ H ₁₆ O ₂	92.3	114.1	124.0	136.4	150.6	160.0	172.2	190.3	213.9	237.5	16
Caprylonitrile	C ₈ H ₁₅ N	43.0	67.6	80.4	94.6	110.6	121.2	134.8	155.2	179.5	204.5	
Carbazole	C ₁₂ H ₉ N						248.2	265.0	292.5	323.0	354.8	244.8
Carbon dioxide	CO ₂	-134.3	-124.4	-119.5	-114.4	-108.6	-104.8	-100.2	-93.0	-85.7	-78.2	-57.5
disulfide	CS ₂	-73.8	-54.3	-44.7	-34.3	-22.5	-15.3	-5.1	+10.4	28.0	46.5	-110.8
monoxide	CO	-222.0	-217.2	-215.0	-212.8	-210.0	-208.1	-205.7	-201.3	-196.3	-191.3	-205.0
oxyselenide (carbonyl selenide)	COSe	-117.1	-102.3	-95.0	-86.3	-76.4	-70.2	-61.7	-49.8	-35.6	-21.9	
oxysulfide (carbonyl sulfide)	COS	-132.4	-119.8	-113.3	-106.0	-98.3	-93.0	-85.9	-75.0	-62.7	-49.9	-138.8
tetrabromide	CBBr ₄					96.3	106.3	119.7	139.7	163.5	189.5	90.1
tetrachloride	CCl ₄	-50.0	-30.0	-19.6	-8.2	+4.3	12.3	23.0	38.3	57.8	76.7	-22.6
tetrafluoride	CF ₄	-184.6	-174.1	-169.3	-164.3	-158.8	-155.4	-150.7	-143.6	-135.5	-127.7	-183.7
Carvacrol	C ₁₀ H ₁₄ O	70.0	98.4	113.2	127.9	145.2	155.3	169.7	191.2	213.8	237.0	+0.5
Carvone	C ₁₀ H ₁₄ O	57.4	86.1	100.4	116.1	133.0	143.8	157.3	179.6	203.5	227.5	
Chavibetol	C ₁₀ H ₁₆ O ₂	83.6	113.3	127.0	143.2	159.8	170.7	185.5	206.8	229.8	254.0	
Chloral (trichloroacetaldehyde)	C ₂ HClCl ₃ O	-37.8	-16.0	-5.0	+7.2	20.2	29.1	40.2	57.8	77.5	97.7	-57
hydrate (trichloroacetaldehyde hydrate)	C ₂ H ₃ Cl ₃ O ₂	-9.8	+10.0	19.5	29.2	39.7	46.2	55.0	68.0	82.1	96.2	51.7
Chloranil	C ₆ Cl ₄ O ₂	70.7	89.3	97.8	106.4	116.1	122.0	129.5	140.3	151.3	162.6	290
Chloroacetic acid	C ₂ H ₃ ClO ₂	43.0	68.3	81.0	94.2	109.2	118.3	130.7	149.0	169.0	189.5	61.2
anhydride	C ₂ H ₂ Cl ₂ O ₃	67.2	94.1	108.0	122.4	138.2	148.0	159.8	177.8	197.0	217.0	46
2-Chloroaniline	C ₆ H ₆ ClN	46.3	72.3	84.8	99.2	115.6	125.7	139.5	160.0	183.7	208.8	0
3-Chloroaniline	C ₆ H ₆ ClN	63.5	89.8	102.0	116.7	133.6	144.1	158.0	179.5	203.5	228.5	-10.4
4-Chloroaniline	C ₆ H ₆ ClN	59.3	87.9	102.1	117.8	135.0	145.8	159.9	182.3	206.6	230.5	70.5
Chlorobenzene	C ₆ H ₅ Cl	-13.0	+10.6	22.2	35.3	49.7	58.3	70.7	89.4	110.0	132.2	-45.2
2-Chlorobenzotrithiolide (2-α,α,α-tetrachlorotoluene)	C ₇ H ₄ Cl ₄	69.0	101.8	117.9	135.8	155.0	167.8	185.0	208.0	233.0	262.1	28.7

TABLE 2-10 Vapor Pressures of Organic Compounds, up to 1 atm (Continued)

Compound		Pressure, mmHg										Melting point, °C
		1	5	10	20	40	60	100	200	400	760	
Name	Formula	Temperature, °C										
2-Chlorobenzotrifluoride (2-chloro- α,α,α -trifluorotoluene)	C ₇ H ₄ ClF ₃	0.0	24.7	37.1	50.6	65.9	75.4	88.3	108.3	130.0	152.2	-6.0
2-Chlorobiphenyl	C ₁₂ H ₉ Cl	89.3	109.8	134.7	151.2	169.9	182.1	197.0	219.6	243.8	267.5	34
4-Chlorobiphenyl	C ₁₂ H ₉ Cl	96.4	129.8	146.0	164.0	183.8	196.0	212.5	237.8	264.5	292.9	75.5
α -Chlorocrotonic acid	C ₅ H ₅ ClO ₂	70.0	95.6	108.0	121.2	135.6	144.4	155.9	173.8	193.2	212.0	
Chlorodifluoromethane	CHClF ₂	-122.8	-110.2	-103.7	-96.5	-88.6	-83.4	-76.4	-65.8	-53.6	-40.8	-160
Chlorodimethylphenylsilane	C ₈ H ₁₁ ClSi	29.8	56.7	70.0	84.7	101.2	111.5	124.7	145.5	168.6	193.5	
1-Chloro-2-ethoxybenzene	C ₈ H ₉ ClO	45.8	72.8	86.5	101.5	117.8	127.8	141.8	162.0	185.5	208.0	
2-(2-Chloroethoxy) ethanol	C ₄ H ₉ ClO ₂	53.0	78.3	90.7	104.1	118.4	127.5	139.5	157.2	176.5	196.0	
bis-2-Chloroethyl acetacetal	C ₈ H ₁₂ Cl ₂ O ₂	56.2	83.7	97.6	112.2	127.8	138.0	150.7	169.8	190.5	212.6	
1-Chloro-2-ethylbenzene	C ₈ H ₉ Cl	17.2	43.0	56.1	70.3	86.2	96.4	110.0	130.2	152.2	177.6	-80.2
1-Chloro-3-ethylbenzene	C ₈ H ₉ Cl	18.6	45.2	58.1	73.0	89.2	99.6	113.6	133.8	156.7	181.1	-53.3
1-Chloro-4-ethylbenzene	C ₈ H ₉ Cl	19.2	46.4	60.0	75.5	91.8	102.0	116.0	137.0	159.8	184.3	-62.6
2-Chloroethyl chloroacetate	C ₄ H ₆ Cl ₂ O ₂	46.0	72.1	86.0	100.0	116.0	126.2	140.0	159.8	182.2	205.0	
2-Chloroethyl 2-chloroisopropyl ether	C ₅ H ₁₀ Cl ₂ O	24.7	50.1	63.0	77.2	92.4	102.2	115.8	135.7	156.5	180.0	
2-Chloroethyl 2-chloropropyl ether	C ₅ H ₁₀ Cl ₂ O	29.8	56.5	70.0	84.8	101.5	111.8	125.6	146.3	169.8	194.1	
2-Chloroethyl α -methylbenzyl ether	C ₁₀ H ₁₃ ClO	62.3	91.4	106.0	121.8	139.6	150.0	164.8	186.3	210.8	235.0	
Chloroform (trichloromethane)	CHCl ₃	-58.0	-39.1	-29.7	-19.0	-7.1	+0.5	10.4	25.9	42.7	61.3	-63.5
1-Chloronaphthalene	C ₁₀ H ₇ Cl	80.6	104.8	118.6	134.4	153.2	165.6	180.4	204.2	230.8	259.3	-20
4-Chlorophenethyl alcohol	C ₈ H ₉ ClO	84.0	114.3	129.0	145.0	162.0	173.5	188.1	210.0	234.5	259.3	
2-Chlorophenol	C ₆ H ₅ ClO	12.1	38.2	51.2	65.9	82.0	92.0	106.0	126.4	149.8	174.5	7
3-Chlorophenol	C ₆ H ₅ ClO	44.2	72.0	86.1	101.7	118.0	129.4	143.0	164.8	188.7	214.0	32.5
4-Chlorophenol	C ₆ H ₅ ClO	49.8	78.2	92.2	108.1	125.0	136.1	150.0	172.0	196.0	220.0	42
2-Chloro-3-phenylphenol	C ₁₂ H ₉ ClO	118.0	152.2	169.7	186.7	207.4	219.6	237.0	261.3	289.4	317.5	+6
2-Chloro-6-phenylphenol	C ₁₂ H ₉ ClO	119.8	153.7	170.7	189.8	208.2	220.0	237.1	261.6	289.5	317.0	
Chloropicrin (trichloronitromethane)	CCl ₃ NO ₂	-25.5	-3.3	+7.8	20.0	33.8	42.3	53.8	71.8	91.8	111.9	-64
1-Chloropropene	C ₃ H ₅ Cl	-81.3	-63.4	-54.1	-44.0	-32.7	-25.1	-15.1	+1.3	18.0	37.0	-99.0
2-Chloropyridine	C ₅ H ₄ ClN	13.3	38.8	51.7	65.8	81.7	91.6	104.6	125.0	147.7	170.2	
3-Chlorostyrene	C ₈ H ₇ Cl	25.3	51.3	65.2	80.0	96.5	107.2	121.2	142.2	165.7	190.0	
4-Chlorostyrene	C ₈ H ₇ Cl	28.0	54.5	67.5	82.0	98.0	108.5	122.0	143.5	166.0	191.0	-15.0
1-Chlorotetradecane	C ₁₄ H ₂₉ Cl	98.5	131.8	148.2	166.2	187.0	199.8	215.5	240.3	267.5	296.0	+0.9
2-Chlorotoluene	C ₇ H ₇ Cl	+5.4	30.6	43.2	56.9	72.0	81.8	94.7	110.0	137.1	159.3	
3-Chlorotoluene	C ₇ H ₇ Cl	+4.8	30.3	43.2	57.4	73.0	83.2	96.3	116.6	139.7	162.3	
4-Chlorotoluene	C ₇ H ₇ Cl	+5.5	31.0	43.8	57.8	73.5	83.3	96.6	117.1	139.8	162.3	+7.3
Chlorotriethylsilane	C ₆ H ₁₅ ClSi	-4.9	+19.8	32.0	45.5	60.2	69.5	82.3	101.6	123.6	146.3	
1-Chloro-1,2,2-trifluoroethylene	C ₂ ClF ₃	-116.0	-102.5	-95.9	-88.2	-79.7	-74.1	-66.7	-55.0	-41.7	-27.9	-157.5
Chlorotrifluoromethane	CClF ₃	-149.5	-139.2	-134.1	-128.5	-121.9	-117.3	-111.7	-102.5	-92.7	-81.2	
Chlorotrimethylsilane	C ₃ H ₉ ClSi	-62.8	-43.6	-34.0	-23.2	-11.4	-4.0	+6.0	21.9	39.4	57.9	
<i>trans</i> -Cinnamic acid	C ₉ H ₈ O ₂	127.5	157.8	173.0	189.5	207.1	217.8	232.4	253.3	276.7	300.0	133
Cinnamyl alcohol	C ₉ H ₁₀ O	72.6	102.5	117.8	133.7	151.0	162.0	177.8	199.8	224.6	250.0	33
Cinnamylaldehyde	C ₉ H ₈ O	76.1	105.8	120.0	135.7	152.2	163.7	177.7	199.3	222.4	246.0	-7.5
Citraconic anhydride	C ₈ H ₆ O ₃	47.1	74.8	88.9	103.8	120.3	131.3	145.4	165.8	189.8	213.5	
<i>cis</i> - α -Citral	C ₁₀ H ₁₆ O	61.7	90.0	103.9	119.4	135.9	146.3	160.0	181.8	205.0	228.0	
<i>d</i> -Citronellal	C ₁₀ H ₁₈ O	44.0	71.4	84.8	99.8	116.1	126.2	140.1	160.0	183.8	206.5	
Citronellal acid	C ₁₀ H ₁₈ O ₂	99.5	127.3	141.4	155.6	171.9	182.1	195.4	214.5	236.6	257.0	
Citronellol	C ₁₀ H ₂₀ O	66.4	93.6	107.0	121.5	137.2	147.2	159.8	179.8	201.0	221.5	
Citronellyl acetate	C ₁₂ H ₂₂ O ₂	74.7	100.2	113.0	126.0	140.5	149.7	161.0	178.8	197.8	217.0	
Coumarin	C ₉ H ₆ O ₂	106.0	137.8	153.4	170.0	189.0	200.5	216.5	240.0	264.7	291.0	70
<i>o</i> -Cresol (2-cresol; 2-methylphenol)	C ₇ H ₈ O	38.2	64.0	76.7	90.5	105.8	115.5	127.4	146.7	168.4	190.8	30.8
<i>m</i> -Cresol (3-cresol; 3-methylphenol)	C ₇ H ₈ O	52.0	76.0	87.8	101.4	116.0	125.8	138.0	157.3	179.0	202.8	10.9
<i>p</i> -Cresol (4-cresol; 4-methylphenol)	C ₇ H ₈ O	53.0	76.5	88.6	102.3	117.7	127.0	140.0	157.3	179.4	201.8	35.5
<i>cis</i> -Crotonic acid	C ₄ H ₆ O ₂	33.5	57.4	69.0	82.0	96.0	104.5	116.3	133.9	152.2	171.9	15.5
<i>trans</i> -Crotonic acid	C ₄ H ₆ O ₂			80.0	93.0	107.8	116.7	128.0	146.0	165.5	185.0	72
<i>cis</i> -Crotononitrile	C ₄ H ₅ N	-29.0	-7.1	+4.0	16.4	30.0	38.5	50.1	68.0	88.0	108.0	
<i>trans</i> -Crotononitrile	C ₄ H ₅ N	-19.5	+3.5	15.0	27.8	41.8	50.9	62.8	81.1	101.5	122.8	
Cumene	C ₉ H ₁₂	+2.9	26.8	38.3	51.5	66.1	75.4	88.1	107.3	129.2	152.4	-96.0
4-Cumidene	C ₉ H ₁₃ N	60.0	88.2	102.2	117.8	134.2	145.0	158.0	180.0	203.2	227.0	
Cuminal	C ₁₀ H ₁₂ O	58.0	87.3	102.0	117.9	135.2	146.0	160.0	182.8	206.7	232.0	
Cuminy alcohol	C ₁₀ H ₁₄ O	74.2	103.7	118.0	133.8	150.3	161.7	176.2	197.9	221.7	246.6	
2-Cyano-2- <i>n</i> -butyl acetate	C ₇ H ₁₁ NO ₂	42.0	68.7	82.0	96.2	111.8	121.5	133.8	152.2	173.4	195.2	
Cyanogen	C ₂ N ₂	-95.8	-83.2	-76.8	-70.1	-62.7	-57.9	-51.8	-42.6	-33.0	-21.0	-34.4
bromide	CBr ₄ N	-35.7	-18.3	-10.0	-1.0	+8.6	14.7	22.6	33.8	46.0	61.5	58
chloride	CCl ₄ N	-76.7	-61.4	-53.8	-46.1	-37.5	-32.1	-24.9	-14.1	-2.3	+13.1	-6.5
iodide	CIN	25.2	47.2	57.7	68.6	80.3	88.0	97.6	111.5	126.1	141.1	
Cyclobutane	C ₄ H ₈	-92.0	-76.0	-67.9	-58.7	-48.4	-41.8	-32.8	-18.9	-3.4	+12.9	-50
Cyclobutene	C ₄ H ₆	-99.1	-83.4	-75.4	-66.6	-56.4	-50.0	-41.2	-27.8	-12.2	+2.4	
Cyclohexane	C ₆ H ₁₂	-45.3	-25.4	-15.9	-5.0	+6.7	14.7	25.5	42.0	60.8	80.7	+6.6
Cyclohexaneethanol	C ₈ H ₁₆ O	50.4	77.2	90.0	104.0	119.8	129.8	142.7	161.7	183.5	205.4	
Cyclohexanol	C ₆ H ₁₂ O	21.0	44.0	56.0	68.8	83.0	91.8	103.7	121.7	141.4	161.0	23.9
Cyclohexanone	C ₆ H ₁₀ O	+1.4	26.4	38.7	52.5	67.8	77.5	90.4	110.3	132.5	155.6	-45.0
2-Cyclohexyl-4,6-dinitrophenol	C ₁₂ H ₁₃ N ₂ O ₅	132.8	161.8	175.9	191.2	206.7	216.0	229.0	248.7	269.8	291.5	
Cyclopentane	C ₅ H ₁₀	-68.0	-49.6	-40.4	-30.1	-18.6	-11.3	-1.3	+13.8	31.0	49.3	-93.7
Cyclopropane	C ₃ H ₆	-116.8	-104.2	-97.5	-90.3	-82.3	-77.0	-70.0	-59.1	-46.9	-33.5	-126.6
Cymene	C ₁₀ H ₁₄	17.3	43.9	57.0	71.1	87.0	97.2	110.8	131.4	153.5	177.2	-68.2

TABLE 2-10 Vapor Pressures of Organic Compounds, up to 1 atm (Continued)

Compound		Pressure, mmHg										Melting point, °C
		1	5	10	20	40	60	100	200	400	760	
Name	Formula	Temperature, °C										
<i>cis</i> -Decalin	C ₁₀ H ₁₈	22.5	50.1	64.2	79.8	97.2	108.0	123.2	145.4	169.9	194.6	-43.3
<i>trans</i> -Decalin	C ₁₀ H ₁₈	-0.8	+30.6	47.2	65.3	85.7	98.4	114.6	136.2	160.1	186.7	-30.7
Decane	C ₁₀ H ₂₂	16.5	42.3	55.7	69.8	85.5	95.5	108.6	128.4	150.6	174.1	-29.7
Decan-2-one	C ₁₀ H ₂₀ O	44.2	71.9	85.8	100.7	117.1	127.8	142.0	163.2	186.7	211.0	+3.5
1-Decene	C ₁₀ H ₂₀	14.7	40.3	53.7	67.8	83.3	93.5	106.5	126.7	149.2	172.0	
Decyl alcohol	C ₁₀ H ₂₂ O	69.5	97.3	111.3	125.8	142.1	152.0	165.8	186.2	208.8	231.0	+7
Decyltrimethylsilane	C ₁₃ H ₃₀ Si	67.4	96.4	111.0	126.5	144.0	154.3	169.5	191.0	215.5	240.0	
Dehydroacetic acid	C ₈ H ₈ O ₄	91.7	122.0	137.3	153.0	171.0	181.5	197.5	219.5	244.5	269.0	
Desoxybenzoin	C ₁₄ H ₁₂ O	123.3	156.2	173.5	192.0	212.0	224.5	241.3	265.2	293.0	321.0	60
Diacetamide	C ₈ H ₇ NO ₂	70.0	95.0	108.0	122.6	138.2	148.0	160.6	180.8	202.0	223.0	78.5
Diacetylene (1,3-butadiyne)	C ₄ H ₂	-82.5	-68.0	-61.2	-53.8	-45.9	-41.0	-34.0	-20.9	-6.1	+9.7	-34.9
Diallyldichlorosilane	C ₆ H ₁₀ Cl ₂ Si	+9.5	34.8	47.4	61.3	76.4	86.3	99.7	119.4	142.0	165.3	
Diallyl sulfide	C ₆ H ₁₀ S	-9.5	+14.4	26.6	39.7	54.2	63.7	75.8	94.8	116.1	138.6	-83
Diisooamyl ether	C ₁₀ H ₂₂ O	18.6	44.3	57.0	70.7	86.3	96.0	109.6	129.0	150.3	173.4	
oxalate	C ₁₂ H ₂₂ O ₄	85.4	116.0	131.4	147.7	165.7	177.0	192.2	215.0	240.0	265.0	
sulfide	C ₁₀ H ₂₂ S	43.0	73.0	87.6	102.7	120.0	130.6	145.3	166.4	191.0	216.0	
Dibenzylamine	C ₁₄ H ₁₅ N	118.3	149.8	165.6	182.2	200.2	212.2	227.3	249.8	274.3	300.0	-26
Dibenzyl ketone (1,3-diphenyl-2-propanone)	C ₁₅ H ₁₄ O	125.5	159.8	177.6	195.7	216.6	229.4	246.6	272.3	301.7	330.5	34.5
1,4-Dibromobenzene	C ₆ H ₄ Br ₂	61.0	79.3	87.7	103.6	120.8	131.6	146.5	168.5	192.5	218.6	87.5
1,2-Dibromobutane	C ₄ H ₈ Br ₂	7.5	33.2	46.1	60.0	76.0	86.0	99.8	120.2	143.5	166.3	-64.5
<i>dl</i> -2,3-Dibromobutane	C ₄ H ₈ Br ₂	+5.0	30.0	41.6	56.4	72.0	82.0	95.3	115.7	138.0	160.5	
<i>meso</i> -2,3-Dibromobutane	C ₄ H ₈ Br ₂	+1.5	26.6	39.3	53.2	68.0	78.0	91.7	111.8	134.2	157.3	-34.5
1,2-Dibromodecane	C ₁₀ H ₂₀ Br ₂	95.7	123.6	137.3	151.0	167.4	177.5	190.2	209.6	229.8	250.4	
Di(2-bromoethyl) ether	C ₄ H ₈ Br ₂ O	47.7	75.3	88.5	103.6	119.8	130.0	144.0	165.0	188.0	212.5	
α,β -Dibromomaleic anhydride	C ₈ H ₆ Br ₂ O ₃	50.0	78.0	92.0	106.7	123.5	133.8	147.7	168.0	192.0	215.0	
1,2-Dibromo-2-methylpropane	C ₄ H ₈ Br ₂	-28.8	-3.0	+10.5	25.7	42.3	53.7	68.8	92.1	119.8	149.0	-70.3
1,3-Dibromo-2-methylpropane	C ₄ H ₈ Br ₂	14.0	40.0	53.0	67.5	83.5	93.7	107.4	117.8	150.6	174.6	
1,2-Dibromopentane	C ₅ H ₁₀ Br ₂	19.8	45.4	58.0	72.0	87.4	97.4	110.1	130.2	151.8	175.0	
1,2-Dibromopropane	C ₃ H ₆ Br ₂	-7.0	+17.3	29.4	42.3	57.2	66.4	78.7	97.8	118.5	141.6	-55.5
1,3-Dibromopropane	C ₃ H ₆ Br ₂	+9.7	35.4	48.0	62.1	77.8	87.8	101.3	121.7	144.1	167.5	-34.4
2,3-Dibromopropane	C ₃ H ₆ Br ₂	-6.0	+17.9	30.0	43.2	57.8	67.0	79.5	98.0	119.5	141.2	
2,3-Dibromo-1-propanol	C ₃ H ₆ Br ₂ O	57.0	84.5	98.2	113.5	129.8	140.0	153.0	173.8	196.0	219.0	
Diisobutylamine	C ₈ H ₁₉ N	-5.1	+18.4	30.6	43.7	57.8	67.0	79.2	97.6	118.0	139.5	-70
2,6-Ditert-butyl-4-cresol	C ₁₅ H ₂₄ O	85.8	116.2	131.0	147.0	164.1	175.2	190.0	212.8	237.6	262.5	
4,6-Ditert-butyl-2-cresol	C ₁₅ H ₂₄ O	86.2	117.3	132.4	149.0	167.4	179.0	194.0	217.5	243.4	269.3	
4,6-Ditert-butyl-3-cresol	C ₁₅ H ₂₄ O	103.7	135.2	150.0	167.0	185.3	196.1	211.0	233.0	257.1	282.0	
2,6-Ditert-butyl-4-ethylphenol	C ₁₈ H ₂₆ O	89.1	121.4	137.0	154.0	172.1	183.9	198.0	220.0	244.0	268.6	
4,6-Ditert-butyl-3-ethylphenol	C ₁₈ H ₂₆ O	111.5	142.6	157.4	174.0	192.3	204.4	218.0	241.7	264.6	290.0	
Diisobutyl oxalate	C ₁₀ H ₁₈ O ₄	63.2	91.2	105.3	120.3	137.5	147.8	161.8	183.5	205.8	229.5	
2,4-Ditert-butylphenol	C ₁₄ H ₂₂ O	84.5	115.4	130.0	146.0	164.3	175.8	190.0	212.5	237.0	260.8	
Dibutyl phthalate	C ₁₈ H ₂₂ O ₄	148.2	182.1	198.2	216.2	235.8	247.8	263.7	287.0	313.5	340.0	
sulfide	C ₈ H ₁₈ S	+21.7	51.8	66.4	80.5	96.0	105.8	118.6	138.0	159.0	182.0	-79.7
Diisobutyl <i>d</i> -tartrate	C ₁₂ H ₂₂ O ₆	117.8	151.8	169.0	188.0	208.5	221.6	239.5	264.7	294.0	324.0	73.5
Dicarvacryl-mono-(6-chloro-2-xenyl) phosphate	C ₃₂ H ₃₄ ClO ₄ P	204.2	234.5	249.3	264.5	280.5	290.7	304.9	323.8	342.0	361.0	
Dicarvacryl-2-tolyl phosphate	C ₂₇ H ₃₃ O ₄ P	180.2	209.3	221.8	237.0	251.5	260.3	272.5	290.0	309.8	330.0	
Dichloroacetic acid	C ₂ H ₂ Cl ₂ O ₂	44.0	69.8	82.6	96.3	111.8	121.5	134.0	152.3	173.7	194.4	9.7
1,2-Dichlorobenzene	C ₆ H ₄ Cl ₂	20.0	46.0	59.1	73.4	89.4	99.5	112.9	133.4	155.8	179.0	-17.6
1,3-Dichlorobenzene	C ₆ H ₄ Cl ₂	12.1	39.0	52.0	66.2	82.0	92.2	105.0	125.9	149.0	173.0	-24.2
1,4-Dichlorobenzene	C ₆ H ₄ Cl ₂			54.8	69.2	84.8	95.2	108.4	128.3	150.2	173.9	53.0
1,2-Dichlorobutane	C ₄ H ₈ Cl ₂	-23.6	-0.3	+11.5	24.5	37.7	47.8	60.2	79.7	100.8	123.5	
2,3-Dichlorobutane	C ₄ H ₈ Cl ₂	-25.2	-3.0	+8.5	21.2	35.0	43.9	56.0	74.0	94.2	116.0	-80.4
1,2-Dichloro-1,2-difluoroethylene	C ₂ Cl ₂ F ₂	-82.0	-65.6	-57.3	-48.3	-38.2	-31.8	-23.0	-10.0	+5.0	20.9	-112
Dichlorodifluoromethane	CCl ₂ F ₂	-118.5	-104.6	-97.8	-90.1	-81.6	-76.1	-68.6	-57.0	-43.9	-29.8	
Dichlorodiphenyl silane	C ₁₂ H ₁₀ Cl ₂ Si	109.6	142.4	158.0	176.0	195.5	207.5	223.8	248.0	275.5	304.0	
Dichlorodiisopropyl ether	C ₆ H ₁₂ Cl ₂ O	29.6	55.2	68.2	82.2	97.3	106.9	119.7	139.0	159.8	182.7	
Di(2-chloroethoxy) methane	C ₅ H ₁₀ Cl ₂ O ₂	53.0	80.4	94.0	109.5	125.5	135.8	149.6	170.0	192.0	215.0	
Dichloroethoxymethylsilane	C ₈ H ₈ Cl ₂ O ₂ Si	-33.8	-12.1	-1.3	+11.3	24.4	32.6	44.1	61.0	80.3	100.6	
1,2-Dichloro-3-ethylbenzene	C ₉ H ₈ Cl ₂	46.0	75.0	90.0	105.9	123.8	135.0	149.8	172.0	197.0	222.1	-40.8
1,2-Dichloro-4-ethylbenzene	C ₉ H ₈ Cl ₂	47.0	77.2	92.3	109.6	127.5	139.0	153.3	176.0	201.7	226.6	-76.4
1,4-Dichloro-2-ethylbenzene	C ₉ H ₈ Cl ₂	38.5	68.0	83.2	99.8	118.0	129.0	144.0	166.2	191.5	216.3	-61.2
<i>cis</i> -1,2-Dichloroethylene	C ₂ H ₂ Cl ₂	-58.4	-39.2	-29.9	-19.4	-7.9	-0.5	+9.5	24.6	41.0	59.0	-80.5
<i>trans</i> -1,2-Dichloroethylene	C ₂ H ₂ Cl ₂	-65.4	-47.2	-38.0	-28.0	-17.0	-10.0	-0.2	+14.3	30.8	47.8	-50.0
Di(2-chloroethyl) ether	C ₄ H ₈ Cl ₂ O	23.5	49.3	62.0	76.0	91.5	101.5	114.5	134.0	155.4	178.5	
Dichlorofluoromethane	CHCl ₂ F	-91.3	-75.5	-67.5	-58.6	-48.8	-42.6	-33.9	-20.9	-6.2	+8.9	-135
1,5-Dichlorohexamethyltrisiloxane	C ₆ H ₁₈ Cl ₂ O ₃ Si ₃	26.0	52.0	65.1	79.0	94.8	105.0	118.2	138.3	160.2	184.0	-53.0
Dichloromethylphenylsilane	C ₈ H ₈ Cl ₂ Si	35.7	63.5	77.4	92.4	109.5	120.0	134.2	155.5	180.2	205.5	
1,1-Dichloro-2-methylpropane	C ₄ H ₈ Cl ₂	-31.0	-8.4	+2.6	14.6	28.2	37.0	48.2	65.8	85.4	106.0	
1,2-Dichloro-2-methylpropane	C ₄ H ₈ Cl ₂	-25.8	-4.2	+6.7	18.7	32.0	40.2	51.7	68.9	87.8	108.0	
1,3-Dichloro-2-methylpropane	C ₄ H ₈ Cl ₂	-3.0	+20.6	32.0	44.8	58.6	67.5	78.8	96.1	115.4	135.0	
2,4-Dichlorophenol	C ₆ H ₄ Cl ₂ O	53.0	80.0	92.8	107.7	123.4	133.5	146.0	165.2	187.5	210.0	45.0
2,6-Dichlorophenol	C ₆ H ₄ Cl ₂ O	59.5	87.6	101.0	115.5	131.6	141.8	154.6	175.5	197.7	220.0	

2-70 PHYSICAL AND CHEMICAL DATA

TABLE 2-10 Vapor Pressures of Organic Compounds, up to 1 atm (Continued)

Compound		Pressure, mmHg										Melting point, °C
		1	5	10	20	40	60	100	200	400	760	
Name	Formula	Temperature, °C										
α,α -Dichlorophenylacetone	$C_8H_5Cl_2N$	56.0	84.0	98.1	113.8	130.0	141.0	154.5	176.2	199.5	223.5	
Dichlorophenylarsine	$C_6H_5AsCl_2$	61.8	100.0	116.0	133.1	151.0	163.2	178.9	202.8	228.8	256.5	
1,2-Dichloropropane	$C_3H_6Cl_2$	-38.5	-17.0	-6.1	+6.0	19.4	28.0	39.4	57.0	76.0	96.8	
2,3-Dichlorostyrene	$C_8H_6Cl_2$	61.0	90.1	104.6	120.5	137.8	149.0	163.5	185.7	210.0	235.0	
2,4-Dichlorostyrene	$C_8H_6Cl_2$	53.5	82.2	97.4	111.8	129.2	140.0	153.8	176.0	200.0	225.0	
2,5-Dichlorostyrene	$C_8H_6Cl_2$	55.5	83.9	98.2	114.0	131.0	142.0	155.8	178.0	202.5	227.0	
2,6-Dichlorostyrene	$C_8H_6Cl_2$	47.8	75.7	90.0	105.5	122.4	133.3	147.6	169.0	193.5	217.0	
3,4-Dichlorostyrene	$C_8H_6Cl_2$	57.2	86.0	100.4	116.2	133.7	144.6	158.2	181.5	205.7	230.0	
3,5-Dichlorostyrene	$C_8H_6Cl_2$	53.5	82.2	97.4	111.8	129.2	140.0	153.8	176.0	200.0	225.0	
1,2-Dichlorotetraethylbenzene	$C_{14}H_{20}Cl_2$	105.6	138.7	155.0	172.5	192.2	204.8	220.7	245.6	272.8	302.0	
1,4-Dichlorotetraethylbenzene	$C_{14}H_{20}Cl_2$	91.7	126.1	143.8	162.0	183.2	195.8	212.0	238.5	265.8	296.5	
1,2-Dichloro-1,1,2,2-tetrafluoroethane	$C_2Cl_2F_4$	-95.4	-80.0	-72.3	-63.5	-53.7	-47.5	-39.1	-26.3	-12.0	+3.5	-94
Dichloro-4-tolylsilane	$C_7H_5Cl_2Si$	46.2	71.7	84.2	97.8	113.2	122.6	135.5	153.5	175.2	196.3	
3,4-Dichloro- α,α,α -trifluorotoluene	$C_7H_3Cl_2F_3$	11.0	38.3	52.2	67.3	84.0	95.0	109.2	129.0	150.5	172.8	-12.1
Dicyclopentadiene	$C_{10}H_8$	34.1	47.6	62.0	77.9	88.0	101.7	121.8	144.2	166.6	192.9	32.9
Diethoxydimethylsilane	$C_6H_{16}O_2Si$	-19.1	+2.4	13.3	25.3	38.0	46.3	57.6	74.2	93.2	113.5	
Diethoxydiphenylsilane	$C_{16}H_{20}O_2Si$	111.5	142.8	157.6	174.3	193.2	205.0	220.0	243.8	259.7	296.0	
Diethyl adipate	$C_{10}H_{18}O_4$	74.0	106.6	123.0	138.3	154.6	165.8	179.0	198.2	219.1	240.0	-21
Diethylamine	$C_4H_{11}N$			-33.0	-22.6	-11.3	-4.0	+6.0	21.0	38.0	55.5	-38.9
N-Diethylaniline	$C_{10}H_{15}N$	49.7	78.0	91.9	107.2	123.6	133.8	147.3	168.2	192.4	215.5	-34.4
Diethyl arsinate	$C_{10}H_{16}As$											
	NO_3	38.0	62.6	74.8	88.0	102.6	111.8	123.8	141.9	161.0	181.0	
1,2-Diethylbenzene	$C_{10}H_{14}$	22.3	48.7	62.0	76.4	92.5	102.6	116.2	136.7	159.0	183.5	-31.4
1,3-Diethylbenzene	$C_{10}H_{14}$	20.7	46.8	59.9	74.5	90.4	100.7	114.4	134.8	156.9	181.1	-83.9
1,4-Diethylbenzene	$C_{10}H_{14}$	20.7	47.1	60.3	74.7	91.1	101.3	115.3	136.1	159.0	183.8	-43.2
Diethyl carbonate	$C_8H_{10}O_3$	-10.1	+12.3	23.8	36.0	49.5	57.9	69.7	86.5	105.8	125.8	-43
cis-Diethyl citraconate	$C_{10}H_{14}O_4$	59.8	88.3	103.0	118.2	135.7	146.2	160.0	182.3	206.5	230.3	
Diethyl dioxosuccinate	$C_8H_{10}O_6$	70.0	98.0	112.0	126.8	143.8	153.7	167.7	188.0	210.8	233.5	
Diethylene glycol	$C_4H_{10}O_3$	91.8	120.0	133.8	148.0	164.3	174.0	187.5	207.0	226.5	244.8	
Diethyleneglycol-bis-chloroacetate	$C_8H_{12}Cl_2O_5$	148.3	180.0	195.8	212.0	229.0	239.5	252.0	271.5	291.8	313.0	
Diethylene glycol dimethyl ether												
Di(2-methoxyethyl) ether	$C_6H_{14}O_3$	13.0	37.6	50.0	63.0	77.5	86.8	99.5	118.0	138.5	159.8	
glycol ethyl ether	$C_6H_{14}O_3$	45.3	72.0	85.8	100.3	116.7	126.8	140.3	159.0	180.3	201.9	
Diethyl ether	$C_4H_{10}O$	-74.3	-56.9	-48.1	-38.5	27.7	-21.8	-11.5	+2.2	17.9	34.6	-116.3
ethylmalonate	$C_9H_{16}O_4$	50.8	77.8	91.6	106.0	122.4	132.4	146.0	166.0	188.7	211.5	
fumarate	$C_8H_{12}O_4$	53.2	81.2	95.3	110.2	126.7	137.7	151.1	172.2	195.8	218.5	+0.6
glutarate	$C_8H_{16}O_4$	65.6	94.7	109.7	125.4	142.8	153.2	167.8	189.5	212.8	237.0	
Diethylhexadecylamine	$C_{20}H_{42}N$	139.8	175.8	194.0	213.5	235.0	248.5	265.5	292.8	324.6	355.0	
Diethyl itaconate	$C_9H_{14}O_4$	51.3	80.2	95.2	111.0	128.2	139.9	154.3	177.5	203.1	227.9	
ketone (3-pentanone)	$C_5H_{10}O$	-12.7	+7.5	17.2	27.9	39.4	46.7	56.2	70.6	86.3	102.7	-42
malate	$C_8H_{14}O_5$	80.7	110.4	125.3	141.2	157.8	169.0	183.9	205.3	229.5	253.4	
maleate	$C_8H_{12}O_4$	57.3	85.6	100.0	115.3	131.8	142.4	156.0	177.8	201.7	225.0	
malonate	$C_7H_{12}O_4$	40.0	67.5	81.3	95.9	113.3	123.0	136.2	155.5	176.8	198.9	-49.8
mesaconate	$C_8H_{14}O_4$	62.8	91.0	105.3	120.3	137.3	147.9	161.6	183.2	205.8	229.0	
oxalate	$C_8H_{10}O_4$	47.4	71.8	83.8	96.8	110.6	119.7	130.8	147.9	166.2	185.7	-40.6
phthalate	$C_{12}H_{14}O_4$	108.8	140.7	156.0	173.6	192.1	204.1	219.5	243.0	267.5	294.0	
sebacate	$C_{14}H_{26}O_4$	125.3	156.2	172.1	189.8	207.5	218.4	234.4	255.8	280.3	305.5	1.3
2,5-Diethylstyrene	$C_{12}H_{16}$	49.7	78.4	92.6	108.5	125.8	136.8	151.0	173.2	198.0	223.0	
Diethyl succinate	$C_8H_{14}O_4$	54.6	83.0	96.6	111.7	127.8	138.2	151.1	171.7	193.8	216.5	-20.8
isosuccinate	$C_8H_{14}O_4$	39.8	66.7	80.0	94.7	111.0	121.4	134.8	155.1	177.7	201.3	
sulfate	$C_8H_{10}O_4S$	47.0	74.0	87.7	102.1	118.0	128.6	142.5	162.5	185.5	209.5	-25.0
sulfide	$C_8H_{10}S$	-39.6	-18.6	-8.0	+3.5	16.1	24.2	35.0	51.3	69.7	88.0	-99.5
sulfite	$C_8H_{10}O_3S$	10.0	34.2	46.4	59.7	74.2	83.8	96.3	115.8	137.0	159.0	
d-Diethyl tartrate	$C_8H_{14}O_6$	102.0	133.0	148.0	164.2	182.3	194.0	208.5	230.4	254.8	280.0	17
dl-Diethyl tartrate	$C_8H_{14}O_6$	100.0	131.7	147.2	163.8	181.7	193.2	208.0	230.0	254.3	280.0	
3,5-Diethyltoluene	$C_{11}H_{16}$	34.0	61.5	75.3	90.2	107.0	117.7	131.7	152.4	176.5	200.7	
Diethylzinc	$C_4H_{10}Zn$	-22.4	0.0	+11.7	24.2	38.0	47.2	59.1	77.0	97.3	118.0	-28
1-Dihydrocarvone	$C_{10}H_{16}O$	46.6	75.5	90.0	106.0	123.7	134.7	149.7	171.8	197.0	223.0	
Dihydrocitronellol	$C_{10}H_{20}O$	68.0	91.7	103.0	115.0	127.6	136.7	145.9	160.2	176.8	193.5	
1,4-Dihydroxyanthraquinone	$C_{14}H_8O_2$	196.7	239.8	259.8	282.0	307.4	323.3	344.5	377.8	413.0	450.0	194
Dimethylacetylene (2-butyne)	C_4H_6	-73.0	-57.9	-50.5	-42.5	-33.9	-27.8	-18.8	-5.0	+10.6	27.2	-32.5
Dimethylamine	C_2H_7N	-87.7	-72.2	-64.6	-56.0	-46.7	-40.7	-32.6	-20.4	-7.1	+7.4	-96
N,N-Dimethylaniline	$C_8H_{11}N$	29.5	56.3	70.0	84.8	101.6	111.9	125.8	146.5	169.2	193.1	+2.5
Dimethyl arsinate	$C_8H_{12}AsNO_3$	15.0	39.6	51.8	65.0	79.7	88.6	101.0	119.8	140.3	160.5	
Di(α -methylbenzyl) ether	$C_{16}H_{18}O$	96.7	128.3	144.0	160.3	179.6	191.5	206.8	229.7	254.8	281.0	
2,2-Dimethylbutane	C_6H_{14}	-69.3	-50.7	-41.5	-31.1	-19.5	-12.1	-2.0	+13.4	31.0	49.7	-99.8
2,3-Dimethylbutane	C_6H_{14}	-63.6	-44.5	-34.9	-24.1	-12.4	-4.9	+3.4	21.1	39.0	58.0	-128.2
Dimethyl citraconate	$C_7H_{10}O_4$	50.8	78.2	91.8	106.5	122.6	132.7	145.8	165.8	188.0	210.5	
1,1-Dimethylcyclohexane	C_8H_{16}	-24.4	-1.4	+10.3	23.0	37.3	45.7	57.9	76.2	97.2	119.5	-34
cis-1,2-Dimethylcyclohexane	C_8H_{16}	-15.9	+7.3	18.4	31.1	45.3	54.4	66.8	85.6	107.0	129.7	-50.0
trans-1,2-Dimethylcyclohexane	C_8H_{16}	-21.1	+1.7	13.0	25.6	39.7	48.7	61.0	79.6	100.9	123.4	-88.0
trans-1,3-Dimethylcyclohexane	C_8H_{16}	-19.4	+3.4	14.9	27.4	41.4	50.4	62.5	81.0	102.1	124.4	-92.0
cis-1,3-Dimethylcyclohexane	C_8H_{16}	-22.7	0.0	+11.2	23.6	37.5	46.4	58.5	76.9	97.8	120.1	-76.2
cis-1,4-Dimethylcyclohexane	C_8H_{16}	-20.0	+3.2	14.5	27.1	41.1	50.1	62.3	80.8	101.9	124.3	-87.4
trans-1,4-Dimethylcyclohexane	C_8H_{16}	-24.3	-1.7	+10.1	22.6	36.5	45.4	57.6	76.0	97.0	119.3	-36.9

TABLE 2-10 Vapor Pressures of Organic Compounds, up to 1 atm (Continued)

Compound		Pressure, mmHg										Melting point, °C
		1	5	10	20	40	60	100	200	400	760	
Name	Formula	Temperature, °C										
Dimethyl ether	C ₂ H ₆ O	-115.7	-101.1	-93.3	-85.2	-76.2	-70.4	-62.7	-50.9	-37.8	-23.7	-138.5
2,2-Dimethylhexane	C ₈ H ₁₈	-29.7	-7.9	+3.1	15.0	28.2	36.7	48.2	65.7	85.6	106.8	
2,3-Dimethylhexane	C ₈ H ₁₈	-23.0	-1.1	+9.9	22.1	35.6	44.2	56.0	73.8	94.1	115.6	
2,4-Dimethylhexane	C ₈ H ₁₈	-26.9	-5.3	+5.2	17.2	30.5	39.0	50.6	68.1	88.2	109.4	
2,5-Dimethylhexane	C ₈ H ₁₈	-26.7	-5.5	+5.3	17.2	30.4	38.9	50.5	68.0	87.9	109.1	-90.7
3,3-Dimethylhexane	C ₈ H ₁₈	-25.8	-4.4	+6.1	18.2	31.7	40.4	52.5	70.0	90.4	112.0	
3,4-Dimethylhexane	C ₈ H ₁₈	-22.1	+0.2	11.3	23.5	37.1	45.8	57.7	75.6	96.0	117.7	
Dimethyl itaconate	C ₇ H ₁₀ O ₄	69.3	94.0	106.6	119.7	133.7	142.6	153.7	171.0	189.8	208.0	38
1-Dimethyl malate	C ₆ H ₁₀ O ₅	75.4	104.0	118.3	133.8	150.1	160.4	175.1	196.3	219.5	242.6	
Dimethyl maleate	C ₆ H ₈ O ₄	45.7	73.0	86.4	101.3	117.2	127.1	140.4	160.0	182.2	205.0	
malonate	C ₆ H ₈ O ₄	35.0	59.8	72.0	85.0	100.0	109.7	121.9	140.0	159.8	180.7	-62
<i>trans</i> -Dimethyl mesaconate	C ₇ H ₁₀ O ₄	46.8	74.0	87.8	102.1	118.0	127.8	141.5	161.0	183.5	206.0	
2,7-Dimethyloctane	C ₁₀ H ₂₂	+6.3	30.5	42.3	55.8	71.2	80.8	93.9	114.0	136.0	159.7	-52.8
Dimethyl oxalate	C ₆ H ₈ O ₄	20.0	44.0	56.0	69.4	83.6	92.8	104.8	123.3	143.3	163.3	
2,2-Dimethylpentane	C ₇ H ₁₆	-49.0	-28.7	-18.7	-7.5	+5.0	13.0	23.9	40.3	59.2	79.2	-123.7
2,3-Dimethylpentane	C ₇ H ₁₆	-42.0	-20.8	-10.3	+1.1	13.9	22.1	33.3	50.1	69.4	89.8	-135
2,4-Dimethylpentane	C ₇ H ₁₆	-48.0	-27.4	-17.1	-5.9	+6.5	14.5	25.4	41.8	60.6	80.5	-119.5
3,3-Dimethylpentane	C ₇ H ₁₆	-45.9	-25.0	-14.4	-2.9	+9.9	18.1	29.3	46.2	65.5	86.1	-135.0
2,3-Dimethylphenol (2,3-xyleneol)	C ₈ H ₁₀ O	56.0	83.8	97.6	112.0	129.2	139.5	152.2	173.0	196.0	218.0	75
2,4-Dimethylphenol (2,4-xyleneol)	C ₈ H ₁₀ O	51.8	78.0	91.3	105.0	121.5	131.0	143.0	161.5	184.2	211.5	25.5
2,5-Dimethylphenol (2,5-xyleneol)	C ₈ H ₁₀ O	51.8	78.0	91.3	105.0	121.5	131.0	143.0	161.5	184.2	211.5	74.5
3,4-Dimethylphenol (3,4-xyleneol)	C ₈ H ₁₀ O	66.2	93.8	107.7	122.0	138.0	148.0	161.0	181.5	203.6	225.2	62.5
3,5-Dimethylphenol (3,5-xyleneol)	C ₈ H ₁₀ O	62.0	89.2	102.4	117.0	133.3	143.5	156.0	176.2	197.8	219.5	68
Dimethylphenylsilane	C ₈ H ₁₂ Si	+5.3	30.3	42.6	56.2	71.4	81.3	94.2	114.2	136.4	159.3	
Dimethyl phthalate	C ₁₀ H ₁₀ O ₄	100.3	131.8	147.6	164.0	182.8	194.0	210.0	232.7	257.8	283.7	
3,5-Dimethyl-1,2-pyrone	C ₈ H ₆ O ₂	78.6	107.6	122.0	136.4	152.7	163.8	177.5	198.0	221.0	245.0	51.5
4,6-Dimethylresorcinol	C ₈ H ₁₀ O ₂	49.0	76.8	90.7	105.8	122.5	133.2	147.3	167.8	192.0	215.0	
Dimethyl sebacate	C ₁₂ H ₂₂ O ₄	104.0	139.8	156.2	175.8	196.0	208.0	222.6	245.0	269.6	293.5	38
2,4-Dimethylstyrene	C ₁₀ H ₁₂	34.2	61.9	75.8	90.8	107.7	118.0	132.3	153.2	177.5	202.0	
2,5-Dimethylstyrene	C ₁₀ H ₁₂	29.0	55.9	69.0	84.0	100.2	110.7	124.7	145.6	168.7	193.0	
α,α -Dimethylsuccinic anhydride	C ₆ H ₈ O ₃	61.4	88.1	102.0	116.3	132.3	142.4	155.3	175.8	197.5	219.5	
Dimethyl sulfide	C ₂ H ₆ S	-75.6	-58.0	-49.2	-39.4	-28.4	-21.4	-12.0	+2.6	18.7	36.0	-83.2
<i>d</i> -Dimethyl tartrate	C ₆ H ₁₀ O ₆	102.1	133.2	148.2	164.3	182.4	193.8	208.8	230.5	255.0	280.0	61.5
<i>dl</i> -Dimethyl tartrate	C ₆ H ₁₀ O ₆	100.4	131.8	147.5	164.0	182.4	193.8	209.5	232.3	257.4	282.0	89
<i>N,N</i> -Dimethyl-2-toluidine	C ₉ H ₁₃ N	28.8	54.1	66.2	80.2	95.0	105.2	118.1	138.3	161.5	184.8	-61
<i>N,N</i> -Dimethyl-4-toluidine	C ₉ H ₁₃ N	50.1	74.3	86.7	100.0	116.3	126.4	140.3	161.6	185.4	209.5	
Di(nitrosomethyl) amine	C ₂ H ₅ N ₂ O ₂	+3.2	27.8	40.0	53.7	68.2	77.7	90.3	110.0	131.3	153.0	
Diosphenol	C ₁₀ H ₁₆ O ₂	66.7	95.4	109.0	124.0	141.2	151.3	165.6	186.2	209.5	232.0	
1,4-Dioxane	C ₆ H ₈ O ₂	-35.8	-12.8	-1.2	+12.0	25.2	33.8	45.1	62.3	81.8	101.1	10
Dipentene	C ₁₀ H ₁₆	14.0	40.4	53.8	68.2	84.3	94.6	108.3	128.2	150.5	174.6	
Diphenylamine	C ₁₂ H ₁₁ N	108.3	141.7	157.0	175.2	194.3	206.9	222.8	247.5	274.1	302.0	52.9
Diphenyl carbinol (benzhydrol)	C ₁₃ H ₁₂ O	110.0	145.0	162.0	180.9	200.0	212.0	227.5	250.0	275.6	301.0	68.5
chlorophosphate	C ₁₂ H ₁₀ ClPO ₃	121.5	160.5	182.0	203.8	227.9	244.2	265.0	299.5	337.2	378.0	
disulfide	C ₁₂ H ₁₀ S ₂	131.6	164.0	180.0	197.0	214.8	226.2	241.3	262.6	285.8	310.0	61
1,2-Diphenylethane (dibenzyl)	C ₁₄ H ₁₄	86.8	119.8	136.0	153.7	173.7	186.0	202.8	227.8	255.0	284.0	51.5
Diphenyl ether	C ₁₂ H ₁₀ O	66.1	97.8	114.0	130.8	150.0	162.0	178.8	203.3	230.7	258.5	27
1,1-Diphenylethylene	C ₁₄ H ₁₂	87.4	119.6	135.0	151.8	170.8	183.4	198.6	222.8	249.8	277.0	
<i>trans</i> -Diphenylethylene	C ₁₄ H ₁₂	113.2	145.8	161.0	179.8	199.0	211.5	227.4	251.7	278.3	306.5	124
1,1-Diphenylhydrazine	C ₁₂ H ₁₂ N ₂	126.0	159.3	176.1	194.0	213.5	225.9	242.5	267.2	294.0	322.2	44
Diphenylmethane	C ₁₃ H ₁₂	76.0	107.4	122.8	139.8	157.8	170.2	186.3	210.7	237.5	264.5	26.5
Diphenyl sulfide	C ₁₂ H ₁₀ S	96.1	129.0	145.0	162.0	182.8	194.8	211.8	236.8	263.9	292.5	
Diphenyl-2-tolyl thiophosphate	C ₁₈ H ₁₇ O ₃ PS	159.7	179.8	201.6	215.5	230.6	240.4	252.5	270.3	290.0	310.0	
1,2-Dipropoxyethane	C ₈ H ₁₈ O ₂	-38.8	-10.3	+5.0	22.3	42.3	55.8	74.2	103.8	140.0	180.0	
1,2-Diisopropylbenzene	C ₁₂ H ₁₈	40.0	67.8	81.8	96.8	114.0	124.3	138.7	159.8	184.3	209.0	
1,3-Diisopropylbenzene	C ₁₂ H ₁₈	34.7	62.3	76.0	91.2	107.9	118.2	132.3	153.7	177.6	202.0	-105
Dipropylene glycol	C ₆ H ₁₄ O ₃	73.8	102.1	116.2	131.3	147.4	156.5	169.9	189.9	210.5	231.8	
Dipropylene glycol monobutyl ether	C ₁₀ H ₂₀ O ₃	64.7	92.0	106.0	120.4	136.3	146.3	159.8	180.0	203.8	227.0	
isopropyl ether	C ₉ H ₂₀ O ₃	46.0	72.8	86.2	100.8	117.0	126.8	140.3	160.0	183.1	205.6	
Di- <i>n</i> -propyl ether	C ₆ H ₁₄ O	-43.3	-22.3	-11.8	0.0	+13.2	21.6	33.0	50.3	69.5	89.5	-122
Diisopropyl ether	C ₆ H ₁₄ O	-57.0	-37.4	-27.4	-16.7	-4.5	+3.4	13.7	30.0	48.2	67.5	-60
Di- <i>n</i> -propyl ketone (4-heptanone)	C ₇ H ₁₄ O	23.0	44.4	55.0	66.2	78.1	85.8	96.0	111.2	127.3	143.7	-32.6
Di- <i>n</i> -propyl oxalate	C ₈ H ₁₄ O ₄	53.4	80.2	93.9	108.6	124.6	134.8	148.1	168.0	190.3	213.5	
Diisopropyl oxalate	C ₈ H ₁₄ O ₄	43.2	69.0	81.9	95.6	110.5	120.0	132.6	151.2	171.8	193.5	
Di- <i>n</i> -propyl succinate	C ₁₀ H ₁₈ O ₄	77.5	107.6	122.2	138.0	154.8	166.0	180.3	202.5	226.5	250.8	
Di- <i>n</i> -propyl <i>d</i> -tartrate	C ₁₀ H ₁₈ O ₆	115.6	147.7	163.5	180.4	199.7	211.7	227.0	250.1	275.6	303.0	
Diisopropyl <i>d</i> -tartrate	C ₁₀ H ₁₈ O ₆	103.7	133.7	148.2	164.0	181.8	192.6	207.3	228.2	251.8	275.0	
Divinyl acetylene (1,5-hexadiene-3-yne)	C ₆ H ₆	-45.1	-24.4	-14.0	-2.8	+10.0	18.1	29.5	46.0	64.4	84.0	
1,3-Divinylbenzene	C ₁₀ H ₁₀	32.7	60.0	73.8	88.7	105.5	116.0	130.0	151.4	175.2	199.5	-66.9
Docosane	C ₂₂ H ₄₆	157.8	195.4	213.0	233.5	254.5	268.3	286.0	314.2	343.5	376.0	44.5
<i>n</i> -Dodecane	C ₁₂ H ₂₆	47.8	75.8	90.0	104.6	121.7	132.1	146.2	167.2	191.0	216.2	-9.6
1-Dodecene	C ₁₂ H ₂₄	47.2	74.0	87.8	102.4	118.6	128.5	142.3	162.2	185.5	208.0	-31.5
<i>n</i> -Dodecyl alcohol	C ₁₂ H ₂₆ O	91.0	120.2	134.7	150.0	167.2	177.8	192.0	213.0	235.7	259.0	24
Dodecylamine	C ₁₂ H ₂₇ N	82.8	111.8	127.8	141.6	157.4	168.0	182.1	203.0	225.0	248.0	
Dodecyltrimethylsilane	C ₁₅ H ₃₄ Si	91.2	122.1	137.7	153.8	172.1	184.2	199.5	222.0	248.0	273.0	
Elaidic acid	C ₁₈ H ₃₄ O ₂	171.3	206.7	223.5	242.3	260.8	273.0	288.0	312.4	337.0	362.0	51.5

TABLE 2-10 Vapor Pressures of Organic Compounds, up to 1 atm (Continued)

Compound		Pressure, mmHg										Melting point, °C
		1	5	10	20	40	60	100	200	400	760	
Name	Formula	Temperature, °C										
Epichlorohydrin	C ₃ H ₅ ClO	-16.5	+5.6	16.6	29.0	42.0	50.6	62.0	79.3	98.0	117.9	-25.6
1,2-Epoxy-2-methylpropane	C ₄ H ₈ O	-69.0	-50.0	-40.3	-29.5	-17.3	-9.7	+1.2	17.5	36.0	55.5	
Erucic acid	C ₂₂ H ₄₂ O ₂	206.7	239.7	254.5	270.6	289.1	300.2	314.4	336.5	358.8	381.5	33.5
Estragole (<i>p</i> -methoxy allyl benzene)	C ₁₀ H ₁₂ O	52.6	80.0	93.7	108.4	124.6	135.2	148.5	168.7	192.0	215.0	
Ethane	C ₂ H ₆	-159.5	-148.5	-142.9	-136.7	-129.8	-125.4	-119.3	-110.2	-99.7	-88.6	-183.2
Ethoxydimethylphenylsilane	C ₁₀ H ₁₆ OSi	36.3	63.1	76.2	91.0	107.2	127.5	131.4	151.5	175.0	199.5	
Ethoxymethylphenylsilane	C ₉ H ₁₄ OSi	-50.9	-31.0	-20.7	-9.8	+3.7	11.5	22.1	38.1	56.3	75.7	
Ethoxytriphenylsilane	C ₂₀ H ₂₀ OSi	167.0	198.2	213.5	230.0	247.0	258.3	273.5	295.0	319.5	344.0	
Ethyl acetate	C ₄ H ₈ O ₂	-43.4	-23.5	-13.5	-3.0	+9.1	16.6	27.0	42.0	59.3	77.1	-82.4
acetoacetate	C ₆ H ₁₀ O ₃	28.5	54.0	67.3	81.1	96.2	106.0	118.5	138.0	158.2	180.8	-45
Ethylacetylene (1-butyne)	C ₄ H ₆	-92.5	-76.7	-68.7	-59.9	-50.0	-43.4	-34.9	-21.6	-6.9	+8.7	-130
Ethyl acrylate	C ₅ H ₈ O ₂	-29.5	-8.7	+2.0	13.0	26.0	33.5	44.5	61.5	80.0	99.5	-71.2
α-Ethylacrylic acid	C ₅ H ₈ O ₂	47.0	70.7	82.0	94.4	108.1	116.7	127.5	144.0	160.7	179.2	
α-Ethylacrylonitrile	C ₅ H ₇ N	-29.0	-6.4	+5.0	17.7	31.8	40.6	53.0	71.6	92.2	114.0	
Ethyl alcohol (ethanol)	C ₂ H ₆ O	-31.3	-12.0	-2.3	+8.0	19.0	26.0	34.9	48.4	63.5	78.4	-112
Ethylamine	C ₂ H ₇ N	-82.3	-66.4	-58.3	-48.6	-39.8	-33.4	-25.1	-12.3	+2.0	16.6	-80.6
4-Ethylaniline	C ₉ H ₁₁ N	52.0	80.0	93.8	109.0	125.7	136.0	149.8	170.6	194.2	217.4	-4
N-Ethylaniline	C ₈ H ₁₁ N	38.5	66.4	80.6	96.0	113.2	123.6	137.3	156.9	180.8	204.0	-63.5
2-Ethylanisole	C ₉ H ₁₂ O	29.7	55.9	69.0	83.1	98.8	109.0	122.3	142.1	164.2	187.1	
3-Ethylanisole	C ₉ H ₁₂ O	33.7	60.3	73.9	88.5	104.8	115.5	129.2	149.7	172.8	196.5	
4-Ethylanisole	C ₉ H ₁₂ O	33.5	60.2	73.9	88.5	104.7	115.4	128.4	149.2	172.3	196.5	
Ethylbenzene	C ₈ H ₁₀	-9.8	+13.9	25.9	38.6	52.8	61.8	74.1	92.7	113.8	136.2	-94.9
Ethyl benzoate	C ₉ H ₁₀ O ₂	44.0	72.0	86.0	101.4	118.2	129.0	143.2	164.8	188.4	213.4	-34.6
benzoylacetate	C ₁₁ H ₁₂ O ₃	107.6	136.4	150.3	166.8	181.8	191.9	205.0	223.8	244.7	265.0	
bromide	C ₂ H ₅ Br	-74.3	-56.4	-47.5	-37.8	-26.7	-19.5	-10.0	+4.5	21.0	38.4	-117.8
α-bromoisobutyrate	C ₆ H ₁₁ BrO ₂	10.6	35.8	48.0	61.8	77.0	86.7	99.8	119.7	141.2	163.6	
<i>n</i> -butyrate	C ₆ H ₁₂ O ₂	-18.4	+4.0	15.3	27.8	41.5	50.1	62.0	79.8	100.0	121.0	-93.3
isobutyrate	C ₆ H ₁₂ O ₂	-24.3	-2.4	+8.4	20.6	33.8	42.3	53.5	71.0	90.0	110.0	-88.2
Ethylcamphoronic anhydride	C ₁₁ H ₁₆ O ₅	118.2	149.8	165.0	181.8	199.8	211.5	226.6	248.5	272.8	298.0	
Ethyl isocaproate	C ₈ H ₁₆ O ₂	11.0	35.8	48.0	61.7	76.3	85.8	98.4	117.8	139.2	160.4	
carbamate	C ₉ H ₇ NO ₂		65.8	77.8	91.0	105.6	114.8	126.2	144.2	164.0	184.0	49
carbanilate	C ₉ H ₁₁ NO ₂	107.8	131.8	143.7	155.5	168.8	177.3	187.9	203.8	220.0	237.0	52.5
Ethylcetylamine	C ₁₈ H ₃₅ N	133.2	168.2	186.0	205.5	226.5	239.8	256.8	283.3	313.0	342.0	
Ethyl chloride	C ₂ H ₅ Cl	-89.8	-73.9	-65.8	-56.8	-47.0	-40.6	-32.0	-18.6	-3.9	+12.3	-139
chloroacetate	C ₂ H ₃ ClO ₂	+1.0	25.4	37.5	50.4	65.2	74.0	86.0	103.8	123.8	144.2	-26
chloroglyoxylate	C ₃ H ₃ ClO ₃	-5.1	+18.0	29.9	42.0	56.0	65.2	76.6	94.5	114.7	135.0	
α-chloropropionate	C ₄ H ₇ ClO ₂	+6.6	30.2	41.9	54.3	68.2	77.3	89.3	107.2	126.2	146.5	
<i>trans</i> -cinnamate	C ₁₁ H ₁₂ O ₂	87.6	108.5	134.0	150.3	169.2	181.2	196.0	219.3	245.0	271.0	12
3-Ethylcumene	C ₁₁ H ₁₆	28.3	55.5	68.8	83.6	99.9	110.2	124.3	145.4	168.2	193.0	
4-Ethylcumene	C ₁₁ H ₁₆	31.5	58.4	72.0	86.7	103.3	113.8	127.2	148.3	171.8	195.8	
Ethyl cyanoacetate	C ₅ H ₇ NO ₂	67.8	93.5	106.0	119.8	133.8	142.1	152.8	169.8	187.8	206.0	
Ethylcyclohexane	C ₈ H ₁₆	-14.5	+9.2	20.6	33.4	47.6	56.7	69.0	87.8	109.1	131.8	-111.3
Ethylcyclopentane	C ₇ H ₁₄	-32.2	-10.8	-0.1	+11.7	25.0	33.4	45.0	62.4	82.3	103.4	-138.6
Ethyl dichloroacetate	C ₄ H ₆ Cl ₂ O ₂	9.6	34.0	46.3	59.5	74.0	83.6	96.1	115.2	135.9	156.5	
<i>N,N</i> -diethyloxamate	C ₈ H ₁₅ NO ₃	76.0	106.3	121.7	137.7	154.4	166.0	180.3	202.8	226.5	252.0	
<i>N</i> -Ethyldiphenylamine	C ₁₆ H ₁₅ N	98.3	130.2	146.0	162.8	182.0	193.7	209.8	233.0	258.8	286.0	
Ethylene	C ₂ H ₄	-168.3	-158.3	-153.2	-147.6	-141.3	-137.3	-131.8	-123.4	-113.9	-103.7	-169
Ethylene-bis-(chloroacetate)	C ₆ H ₈ Cl ₂ O ₄	112.0	142.4	158.0	173.5	191.0	201.8	215.0	237.3	259.5	283.5	
Ethylene chlorohydrin (2-chloroethanol)	C ₂ H ₄ ClO	-4.0	+19.0	30.3	42.5	56.0	64.1	75.0	91.8	110.0	128.8	-69
diamine (1,2-ethanediamine)	C ₂ H ₈ N ₂	-11.0	+10.5	21.5	33.0	45.8	53.8	62.5	81.0	99.0	117.2	8.5
dibromide (1,2-dibromethane)	C ₂ H ₄ Br ₂	-27.0	+4.7	18.6	32.7	48.0	57.9	70.4	89.8	110.1	131.5	10
dichloride (1,2-dichloroethane)	C ₂ H ₄ Cl ₂	-44.5	-24.0	-13.6	-2.4	+10.0	18.1	29.4	45.7	64.0	82.4	-35.3
glycol (1,2-ethanediol)	C ₂ H ₆ O ₂	53.0	79.7	92.1	105.8	120.0	129.5	141.8	158.5	178.5	197.3	-15.6
glycol diethyl ether	C ₆ H ₁₄ O ₂	-33.5	-10.2	+1.6	14.7	29.7	39.0	51.8	71.8	94.1	119.5	
(1,2-diethoxyethane)												
glycol dimethyl ether	C ₄ H ₁₀ O ₂	-48.0	-26.2	-15.3	-3.0	+10.7	19.7	31.8	50.0	70.8	93.0	
(1,2-dimethoxyethane)												
glycol monomethyl ether	C ₃ H ₈ O ₂	-13.5	+10.2	22.0	34.3	47.8	56.4	68.0	85.3	104.3	124.4	
(2-methoxyethanol)												
oxide	C ₂ H ₄ O	-89.7	-73.8	-65.7	-56.6	-46.9	-40.7	-32.1	-19.5	-4.9	+10.7	-111.3
Ethyl α-ethylacetoacetate	C ₈ H ₁₄ O ₃	40.5	67.3	80.2	94.6	110.3	120.6	133.8	153.2	175.6	198.0	
fluoride	C ₂ H ₅ F	-117.0	-103.8	-97.7	-90.0	-81.8	-76.4	-69.3	-58.0	-45.5	-32.0	
formate	C ₂ H ₄ O ₂	-60.5	-42.2	-33.0	-22.7	-11.5	-4.3	-5.4	20.0	37.1	54.3	-79
2-furoate	C ₇ H ₆ O ₃	37.6	63.8	77.1	91.5	107.5	117.5	130.4	150.1	172.5	195.0	34
glycolate	C ₃ H ₄ O ₃	14.3	38.8	50.5	63.9	78.1	87.6	99.8	117.8	138.0	158.2	
3-Ethylhexane	C ₈ H ₁₈	-20.0	+2.1	12.8	25.0	38.5	47.1	58.9	76.7	97.0	118.5	
2-Ethylhexyl acrylate	C ₁₁ H ₂₀ O ₂	50.0	77.7	91.8	106.3	123.7	134.0	147.9	168.2	192.2	216.0	
Ethylidene chloride (1,1-dichloroethane)	C ₂ H ₂ Cl ₂	-60.7	-41.9	-32.3	-21.9	-10.2	-2.9	+7.2	22.4	39.8	57.4	-96.7
fluoride (1,1-difluoroethane)	C ₂ H ₄ F ₂	-112.5	-98.4	-91.7	-84.1	-75.8	-70.4	-63.2	-52.0	-39.5	-26.5	-117
Ethyl iodide	C ₂ H ₅ I	-54.4	-34.3	-24.3	-13.1	-0.9	+7.2	18.0	34.1	52.3	72.4	-105
Ethyl <i>l</i> -leucinate	C ₈ H ₁₇ NO ₂	27.8	57.3	72.1	88.0	106.0	117.8	131.8	149.8	167.3	184.0	
Ethyl levulinate	C ₇ H ₁₂ O ₃	47.3	74.0	87.3	101.8	117.7	127.6	141.3	160.2	183.0	206.2	
Ethyl mercaptan (ethanethiol)	C ₂ H ₆ S	-76.7	-59.1	-50.2	-40.7	-29.8	-22.4	-13.0	+1.5	17.7	35.0	-121
Ethyl methylcarbamate	C ₄ H ₉ NO ₂	26.5	51.0	63.2	76.1	91.0	100.0	112.0	130.0	149.8	170.0	
Ethyl methyl ether	C ₃ H ₈ O	-91.0	-75.6	-67.8	-59.1	-49.4	-43.3	-34.8	-22.0	-7.8	+7.5	

TABLE 2-10 Vapor Pressures of Organic Compounds, up to 1 atm (Continued)

Compound		Pressure, mmHg										Melting point, °C
		1	5	10	20	40	60	100	200	400	760	
Name	Formula	Temperature, °C										
1-Ethyl-naphthalene	C ₁₂ H ₁₂	70.0	101.4	116.8	133.8	152.0	164.1	180.0	204.6	230.8	258.1	-27
Ethyl α-naphthyl ketone (1-propionaphthone)	C ₁₃ H ₁₂ O	124.0	155.5	171.0	188.1	206.9	218.2	233.5	255.5	280.2	306.0	
Ethyl 3-nitrobenzoate	C ₉ H ₉ NO ₄	108.1	140.2	155.0	173.6	192.6	205.0	220.3	244.6	270.6	298.0	47
3-Ethylpentane	C ₇ H ₁₆	-37.8	-17.0	-6.8	+4.7	17.5	25.7	36.9	53.8	73.0	93.5	-118.6
4-Ethylphenetole	C ₁₀ H ₁₄ O	48.5	75.7	89.5	103.8	119.8	129.8	143.5	163.2	185.7	208.0	
2-Ethylphenol	C ₈ H ₁₀ O	46.2	73.4	87.0	101.5	117.9	127.9	141.8	161.6	184.5	207.5	-45
3-Ethylphenol	C ₈ H ₁₀ O	60.0	86.8	100.2	114.5	130.0	139.8	152.0	171.8	193.3	214.0	-4
4-Ethylphenol	C ₈ H ₁₀ O	59.3	86.5	100.2	115.0	131.3	141.7	154.2	175.0	197.4	219.0	46.5
Ethyl phenyl ether (phenetole)	C ₈ H ₁₀ O	18.1	43.7	56.4	70.3	86.6	95.4	108.4	127.9	149.8	172.0	-30.2
Ethyl propionate	C ₅ H ₁₀ O ₂	-28.0	-7.2	+3.4	14.3	27.2	35.1	45.2	61.7	79.8	99.1	-72.6
Ethyl propyl ether	C ₅ H ₁₂ O	-64.3	-45.0	-35.0	-24.0	-12.0	-4.0	+6.8	23.3	41.6	61.7	
Ethyl salicylate	C ₉ H ₁₀ O ₃	61.2	90.0	104.2	119.3	136.7	147.6	161.5	183.7	207.0	231.5	1.3
3-Ethylstyrene	C ₁₀ H ₁₂	28.3	55.0	68.3	82.8	99.2	109.6	123.2	144.0	167.2	191.5	
4-Ethylstyrene	C ₁₀ H ₁₂	26.0	52.7	66.3	80.8	97.3	107.6	121.5	142.0	165.0	189.0	
Ethylisothiocyanate	C ₅ H ₅ NS	-13.2	+10.6	22.8	36.1	50.8	59.8	71.9	90.0	110.1	131.0	-5.9
2-Ethyltoluene	C ₉ H ₁₂	9.4	34.8	47.6	61.2	76.4	86.0	99.0	119.0	141.4	165.1	
3-Ethyltoluene	C ₉ H ₁₂	7.2	32.3	44.7	58.2	73.3	82.9	95.9	115.5	137.8	161.3	-95.5
4-Ethyltoluene	C ₉ H ₁₂	7.6	32.7	44.9	58.5	73.6	83.2	96.3	116.1	136.4	162.0	
Ethyl trichloroacetate	C ₆ H ₅ Cl ₃ O ₂	20.7	45.5	57.7	70.6	85.5	94.4	107.4	125.8	146.0	167.0	
Ethyltrimethylsilane	C ₈ H ₁₄ Si	-60.6	-41.4	-31.8	-21.0	-9.0	-1.2	+9.2	25.0	42.8	62.0	
Ethyltrimethyltin	C ₈ H ₁₄ Sn	-30.0	-7.6	+3.8	16.1	30.0	38.4	50.0	67.3	87.6	108.8	
Ethyl isovalerate	C ₇ H ₁₄ O ₂	-6.1	+17.0	28.7	41.3	55.2	64.0	75.9	93.8	114.0	134.3	-99.3
2-Ethyl-1,4-xylene	C ₁₀ H ₁₄	25.7	52.0	65.6	79.8	96.0	106.2	120.0	140.2	163.1	186.9	
4-Ethyl-1,3-xylene	C ₁₀ H ₁₄	26.3	53.0	66.4	80.6	97.2	107.4	121.2	141.8	164.4	188.4	
5-Ethyl-1,3-xylene	C ₁₀ H ₁₄	22.1	48.8	62.1	76.5	92.6	103.0	116.5	137.4	159.6	183.7	
Eugenol	C ₁₀ H ₁₂ O ₂	78.4	108.1	123.0	138.7	155.8	167.3	182.2	204.7	228.3	253.5	
iso-Eugenol	C ₁₀ H ₁₂ O ₂	86.3	117.0	132.4	149.0	167.0	178.2	194.0	217.2	242.3	267.5	-10
Eugenyl acetate	C ₁₂ H ₁₄ O ₃	101.6	132.3	148.0	164.2	183.0	194.0	209.7	232.5	257.4	282.0	295
Fencholic acid	C ₁₀ H ₁₆ O ₂	101.7	128.7	142.3	155.8	171.8	181.5	194.0	215.0	237.8	264.1	19
d-Fenchone	C ₁₀ H ₁₆ O	28.0	54.7	68.3	83.0	99.5	109.8	123.6	144.0	166.8	191.0	5
dl-Fenchyl alcohol	C ₁₀ H ₁₈ O	45.8	70.3	82.1	95.6	110.8	120.2	132.3	150.0	173.2	201.0	35
Fluorene	C ₁₃ H ₁₀	129.3	146.0	164.2	185.2	197.8	214.7	240.3	268.6	295.0	313	
Fluorobenzene	C ₆ H ₅ F	-43.4	-22.8	-12.4	-1.2	+11.5	19.6	30.4	47.2	65.7	84.7	-42.1
2-Fluorotoluene	C ₇ H ₇ F	-24.2	-2.2	+8.9	21.4	34.7	43.7	55.3	73.0	92.8	114.0	-80
3-Fluorotoluene	C ₇ H ₇ F	-22.4	-0.3	+11.0	23.4	37.0	45.8	57.5	75.4	95.4	116.0	-110.8
4-Fluorotoluene	C ₇ H ₇ F	-21.8	+0.3	11.8	24.0	37.8	46.5	58.1	76.0	96.1	117.0	
Formaldehyde	CH ₂ O			-88.0	-79.6	-70.6	-65.0	-57.3	-46.0	-33.0	-19.5	-92
Formamide	CH ₃ NO	70.5	96.3	109.5	122.5	137.5	147.0	157.5	175.5	193.5	210.5	
Formic acid	CH ₂ O ₂	-20.0	-5.0	+2.1	10.3	24.0	32.4	43.8	61.4	80.3	100.6	8.2
trans-Fumaryl chloride	C ₄ H ₂ Cl ₂ O ₂	+15.0	38.5	51.8	65.0	79.5	89.0	101.0	120.0	140.0	160.0	
Furfural (2-furaldehyde)	C ₅ H ₄ O ₂	18.5	42.6	54.8	67.8	82.1	91.5	103.4	121.8	141.8	161.8	
Furfuryl alcohol	C ₅ H ₆ O ₂	31.8	56.0	68.0	81.0	95.7	104.0	115.9	133.1	151.8	170.0	
Geraniol	C ₁₀ H ₁₈ O	69.2	96.8	110.0	125.6	141.8	151.5	165.3	185.6	207.8	230.0	
Geranyl acetate	C ₁₂ H ₂₀ O ₂	73.5	102.7	117.9	133.0	150.0	160.3	175.2	196.3	219.8	243.3	
Geranyl n-butyrate	C ₁₄ H ₂₆ O ₂	96.8	125.2	139.0	153.8	170.1	180.2	193.8	214.0	235.0	257.4	
Geranyl isobutyrate	C ₁₄ H ₂₆ O ₂	90.9	119.6	133.0	147.9	164.0	174.0	187.7	207.6	228.5	251.0	
Geranyl formate	C ₁₁ H ₁₈ O ₂	61.8	90.3	104.3	119.8	136.2	147.2	160.7	182.6	205.8	230.0	
Glutaric acid	C ₅ H ₈ O ₄	155.5	183.8	196.0	210.5	226.3	235.5	247.0	265.0	283.5	303.0	97.5
Glutaric anhydride	C ₅ H ₈ O ₃	100.8	133.3	149.5	166.0	185.5	196.2	212.5	236.5	261.0	287.0	
Glutaronitrile	C ₅ H ₆ N ₂	91.3	123.7	140.0	156.5	176.4	189.5	205.5	230.0	257.3	286.2	
Glutaryl chloride	C ₅ H ₆ Cl ₂ O ₂	56.1	84.0	97.8	112.3	128.3	139.1	151.8	172.4	195.3	217.0	
Glycerol	C ₃ H ₈ O ₃	125.5	153.8	167.2	182.2	198.0	208.0	220.1	240.0	263.0	290.0	17.9
Glycerol dichlorohydrin (1,3-dichloro-2-propanol)	C ₃ H ₆ Cl ₂ O	28.0	52.2	64.7	78.0	93.0	102.0	114.8	133.3	153.5	174.3	
Glycol diacetate	C ₆ H ₁₀ O ₄	38.3	64.1	77.1	90.8	106.1	115.8	128.0	147.8	168.3	190.5	-31
Glycolide (1,4-dioxane-2,6-dione)	C ₄ H ₄ O ₄		103.0	116.6	132.0	148.6	158.2	173.2	194.0	217.0	240.0	97
Guaicol (2-methoxyphenol)	C ₇ H ₈ O ₂	52.4	79.1	92.0	106.0	121.6	131.0	144.0	162.7	184.1	205.0	28.3
Heneicosane	C ₂₁ H ₄₄	152.6	188.0	205.4	223.2	243.4	255.3	272.0	296.5	323.8	350.5	40.4
Heptacosane	C ₂₇ H ₅₆	211.7	248.6	266.8	284.6	305.7	318.3	333.5	359.4	385.0	410.6	59.5
Heptadecane	C ₁₇ H ₃₆	115.0	145.2	160.0	177.7	195.8	207.3	223.0	247.8	274.5	303.0	22.5
Heptaldehyde (enanthaldehyde)	C ₇ H ₁₄ O	12.0	32.7	43.0	54.0	66.3	74.0	84.0	102.0	125.5	155.0	-42
n-Heptane	C ₇ H ₁₆	-34.0	-12.7	-2.1	+9.5	22.3	30.6	41.8	57.7	78.0	98.4	-90.6
Heptanoic acid (enanthic acid)	C ₇ H ₁₄ O ₂	78.0	101.3	113.2	125.6	139.5	148.5	160.0	179.5	199.6	221.5	-10
1-Heptanol	C ₇ H ₁₆ O	42.4	64.3	74.7	85.8	99.8	108.0	119.5	136.6	155.6	175.8	34.6
Heptanoyl chloride (enanthyl chloride)	C ₇ H ₁₃ ClO	34.2	54.6	64.6	75.0	86.4	93.5	102.7	116.3	130.7	145.0	
2-Heptene	C ₇ H ₁₄	-35.8	-14.1	-3.5	+8.3	21.5	30.0	41.3	58.6	78.1	98.5	
Heptylbenzene	C ₁₃ H ₂₀	64.0	94.6	110.0	126.0	144.0	154.8	170.2	193.3	217.8	244.0	
Heptyl cyanide (enanthonitrile)	C ₇ H ₁₃ N	21.0	47.8	61.6	76.3	92.6	103.0	116.8	137.7	160.0	184.6	
Hexachlorobenzene	C ₆ Cl ₆	114.4	149.3	166.4	185.7	206.0	219.0	235.5	258.5	283.5	309.4	230
Hexachloroethane	C ₂ Cl ₆	32.7	49.8	73.5	87.6	102.3	112.0	124.2	143.1	163.8	185.6	186.6
Hexacosane	C ₂₆ H ₅₄	204.0	240.0	257.4	275.8	295.2	307.8	323.2	348.4	374.6	399.8	56.6
Hexadecane	C ₁₆ H ₃₄	105.3	135.2	149.8	164.7	181.3	193.2	208.5	231.7	258.3	287.5	18.5
1-Hexadecene	C ₁₆ H ₃₂	101.6	131.7	146.2	162.0	178.8	190.8	205.3	226.8	250.0	274.0	4
n-Hexadecyl alcohol (cetyl alcohol)	C ₁₆ H ₃₄ O	122.7	158.3	177.8	197.8	219.8	234.3	251.7	280.2	312.7	344.0	49.3

TABLE 2-10 Vapor Pressures of Organic Compounds, up to 1 atm (Continued)

Compound		Pressure, mmHg										Melting point, °C
		1	5	10	20	40	60	100	200	400	760	
Name	Formula	Temperature, °C										
<i>n</i> -Hexadecylamine (cetylamine)	C ₁₆ H ₃₅ N	123.6	157.8	176.0	195.7	215.7	228.8	245.8	272.2	300.4	330.0	
Hexaethylbenzene	C ₁₈ H ₃₀		134.3	150.3	168.0	187.7	199.7	216.0	241.7	268.5	298.3	130
<i>n</i> -Hexane	C ₆ H ₁₄	-53.9	-34.5	-25.0	-14.1	-2.3	+5.4	15.8	31.6	49.6	68.7	-95.3
1-Hexanol	C ₆ H ₁₄ O	24.4	47.2	58.2	70.3	83.7	92.0	102.8	119.6	138.0	157.0	-51.6
2-Hexanol	C ₆ H ₁₄ O	14.6	34.8	45.0	55.9	67.9	76.0	87.3	103.7	121.8	139.9	
3-Hexanol	C ₆ H ₁₄ O	+2.5	25.7	36.7	49.0	62.2	70.7	81.8	98.3	117.0	135.5	
1-Hexene	C ₆ H ₁₂	-57.5	-38.0	-28.1	-17.2	-5.0	+2.8	13.0	29.0	46.8	66.0	-98.5
<i>n</i> -Hexyl levulinate	C ₁₁ H ₂₀ O ₃	90.0	120.0	134.7	150.2	167.8	179.0	193.6	215.7	241.0	266.8	
<i>n</i> -Hexyl phenyl ketone (enantiofenone)	C ₁₃ H ₁₈ O	100.0	130.3	145.5	161.0	178.9	189.8	204.2	225.0	248.3	271.3	
Hydrocinnamic acid	C ₉ H ₁₀ O ₂	102.2	133.5	148.7	165.0	183.3	194.0	209.0	230.8	255.0	279.8	48.5
Hydrogen cyanide (hydrocyanic acid)	CHN	-71.0	-55.3	-47.7	-39.7	-30.9	-25.1	-17.8	-5.3	+10.2	25.9	-13.2
Hydroquinone	C ₆ H ₆ O ₂	132.4	153.3	163.5	174.6	192.0	203.0	216.5	238.0	262.5	286.2	170.3
4-Hydroxybenzaldehyde	C ₇ H ₆ O ₂	121.2	153.2	169.7	186.8	206.0	217.5	233.5	256.8	282.6	310.0	115.5
α-Hydroxyisobutyric acid	C ₄ H ₈ O ₃	73.5	98.5	110.5	123.8	138.0	146.4	157.7	175.2	193.8	212.0	79
α-Hydroxybutyronitrile	C ₅ H ₉ NO	41.0	65.8	77.8	90.7	104.8	113.9	125.0	142.0	159.8	178.8	
4-Hydroxy-3-methyl-2-butanone	C ₅ H ₁₀ O ₂	44.6	69.3	81.0	94.0	108.2	117.4	129.0	146.5	165.5	185.0	
4-Hydroxy-4-methyl-2-pentanone	C ₆ H ₁₂ O ₂	22.0	46.7	58.8	72.0	86.7	96.0	108.2	126.8	147.5	167.9	-47
3-Hydroxypropionitrile	C ₃ H ₅ NO	58.7	87.8	102.0	117.9	134.1	144.7	157.7	178.0	200.0	221.0	
Indene	C ₉ H ₈	16.4	44.3	58.5	73.9	90.7	100.8	114.7	135.6	157.8	181.6	-2
Iodobenzene	C ₆ H ₅ I	24.1	50.6	64.0	78.3	94.4	105.0	118.3	139.8	163.9	188.6	-28.5
Iodononane	C ₉ H ₁₉ I	70.0	96.2	109.0	123.0	138.1	147.7	159.8	179.0	199.3	219.5	
2-Iodotoluene	C ₇ H ₇ I	37.2	65.9	79.8	95.6	112.4	123.8	138.1	160.0	185.7	211.0	
α-Ionone	C ₁₃ H ₂₀ O	79.5	108.8	123.0	139.0	155.6	166.3	181.2	202.5	225.2	250.0	
Isoprene	C ₅ H ₈	-79.8	-62.3	-53.3	-43.5	-32.6	-25.4	-16.0	-1.2	+15.4	32.6	-146.7
Lauraldehyde	C ₁₂ H ₂₄ O	77.7	108.4	123.7	140.2	157.8	168.7	184.5	207.8	231.8	257.0	44.5
Lauric acid	C ₁₂ H ₂₄ O ₂	121.0	150.6	166.0	183.6	201.4	212.7	227.5	249.8	273.8	299.2	48
Levulinolaldehyde	C ₅ H ₈ O ₂	28.1	54.9	68.0	82.7	98.3	108.4	121.8	142.0	164.0	187.0	
Levulinic acid	C ₅ H ₈ O ₃	102.0	128.1	141.8	154.1	169.5	178.0	190.2	208.3	227.4	245.8	33.5
<i>d</i> -Limonene	C ₁₀ H ₁₆	14.0	40.4	53.8	68.2	84.3	94.6	108.3	128.5	151.4	175.0	-96.9
Linalyl acetate	C ₁₂ H ₂₀ O ₂	55.4	82.5	96.0	111.4	127.7	138.1	151.8	173.3	196.2	220.0	
Maleic anhydride	C ₄ H ₂ O ₃	44.0	63.4	78.7	95.0	111.8	122.0	135.8	155.9	179.5	202.0	58
Menthane	C ₁₀ H ₂₀	+9.7	35.7	48.3	62.7	78.3	88.6	102.1	122.7	146.0	169.5	
1-Menthol	C ₁₀ H ₂₀ O	56.0	83.2	96.0	110.3	126.1	136.1	149.4	168.3	190.2	212.0	42.5
Menthyl acetate	C ₁₂ H ₂₂ O ₂	57.4	85.8	100.0	115.4	132.1	143.2	156.7	178.8	202.8	227.0	
benzoate	C ₁₇ H ₂₄ O ₂	123.2	154.2	170.0	186.3	204.3	215.8	230.4	253.2	277.1	301.0	54.5
formate	C ₁₁ H ₂₀ O ₂	47.3	75.8	90.0	105.8	123.0	133.8	148.0	169.8	194.2	219.0	
Mesityl oxide	C ₈ H ₁₀ O	-8.7	+14.1	26.0	37.9	51.7	60.4	72.1	90.0	109.8	130.0	-59
Methacrylic acid	C ₄ H ₆ O ₂	25.5	48.5	60.0	72.7	86.4	95.3	106.6	123.9	142.5	161.0	15
Methacrylonitrile	C ₄ H ₅ N	-44.5	-23.3	-12.5	-0.6	+12.8	21.5	32.8	50.0	70.3	90.3	
Methane	CH ₄	-205.9	-199.0	-195.5	-191.8	-187.7	-185.1	-181.4	-175.5	-168.8	-161.5	-182.5
Methanethiol	CH ₃ S	-90.7	-75.3	-67.5	-58.8	-49.2	-43.1	-34.8	-22.1	-7.9	+6.8	-121
Methoxyacetic acid	C ₃ H ₆ O ₃	52.5	79.3	92.0	106.5	122.0	131.8	144.5	163.5	184.2	204.0	
<i>N</i> -Methylacetanilide	C ₉ H ₁₁ NO		103.8	118.6	135.1	152.2	164.2	179.8	202.3	227.4	253.0	102
Methyl acetate	C ₃ H ₆ O ₂	-57.2	-38.6	-29.3	-19.1	-7.9	-0.5	+9.4	24.0	40.0	57.8	-98.7
acetylene (propyne)	C ₂ H ₂	-111.0	-97.5	-90.5	-82.9	-74.3	-68.8	-61.3	-49.8	-37.2	-23.3	-102.7
acrylate	C ₃ H ₄ O ₂	-43.7	-23.6	-13.5	-2.7	+9.2	17.3	28.0	43.9	61.8	80.2	
alcohol (methanol)	CH ₃ O	-44.0	-25.3	-16.2	-6.0	+5.0	12.1	21.2	34.8	49.9	64.7	-97.8
Methylamine	CH ₃ N	-95.8	-81.3	-73.8	-65.9	-56.9	-51.3	-43.7	-32.4	-19.7	-6.3	-93.5
<i>N</i> -Methylaniline	C ₈ H ₉ N	36.0	62.8	76.2	90.5	106.0	115.8	129.8	149.3	172.0	195.5	-57
Methyl anthranilate	C ₈ H ₉ NO ₂	77.6	109.0	124.2	141.5	159.7	172.0	187.8	212.4	238.5	266.5	24
benzoate	C ₈ H ₈ O ₂	39.0	64.4	77.3	91.8	107.8	117.4	130.8	151.4	174.7	199.5	-12.5
2-Methylbenzothiazole	C ₈ H ₇ NS	70.0	97.5	111.2	125.5	141.2	150.4	163.9	183.2	204.5	225.5	15.4
α-Methylbenzyl alcohol	C ₈ H ₁₀ O	49.0	75.2	88.0	102.1	117.8	127.4	140.3	159.0	180.7	204.0	
Methyl bromide	CH ₃ Br	-96.3	-80.6	-72.8	-64.0	-54.2	-48.0	-39.4	-26.5	-11.9	+3.6	-93
2-Methyl-1-butene	C ₅ H ₁₀	-89.1	-72.8	-64.3	-54.8	-44.1	-37.3	-28.0	-13.8	+2.5	20.2	-135
2-Methyl-2-butene	C ₅ H ₁₀	-75.4	-57.0	-47.9	-37.9	-26.7	-19.4	-9.9	+4.9	21.6	38.5	-133
Methyl isobutyl carbinol (2-methyl-4-pentanol)	C ₆ H ₁₄ O	-0.3	+22.1	33.3	45.4	58.2	67.0	78.0	94.9	113.5	131.7	
<i>n</i> -butyl ketone (2-hexanone)	C ₆ H ₁₂ O	+7.7	28.8	38.8	50.0	62.0	69.8	79.8	94.3	111.0	127.5	-56.9
isobutyl ketone (4-methyl-2-pentanone)	C ₆ H ₁₂ O	-1.4	+19.7	30.0	40.8	52.8	60.4	70.4	85.6	102.0	119.0	-84.7
<i>n</i> -butyrate	C ₄ H ₈ O ₂	-26.8	-5.5	+5.0	16.7	29.6	37.4	48.0	64.3	83.1	102.3	
isobutyrate	C ₄ H ₈ O ₂	-34.1	-13.0	-2.9	+8.4	21.0	28.9	39.6	55.7	73.6	92.6	-84.7
caprate	C ₁₁ H ₂₂ O ₂	63.7	93.5	108.0	123.0	139.0	148.6	161.5	181.6	202.9	224.0	-18
caproate	C ₇ H ₁₄ O ₂	+5.0	30.0	42.0	55.4	70.0	79.7	91.4	109.8	129.8	150	
caprylate	C ₈ H ₁₆ O ₂	34.2	61.7	74.9	89.0	105.3	115.3	128.0	148.1	170.0	193.0	-40
chloride	CH ₃ Cl	-99.5	-92.4	-84.8	-76.0	-67.0	-63.0	-51.2	-38.0	-24.0	-9.7	-97.7
chloroacetate	C ₂ H ₃ ClO ₂	-2.9	19.0	30.0	41.5	54.5	63.0	73.5	90.5	109.5	130.3	-31.9
cinnamate	C ₁₀ H ₁₀ O ₂	77.4	108.1	123.0	140.0	157.9	170.0	185.8	209.6	235.0	263.0	33.4
α-Methylcinnamic acid	C ₁₀ H ₁₀ O ₂	125.7	155.0	169.8	185.2	201.8	212.0	224.8	245.0	266.8	288.0	
Methylcyclohexane	C ₇ H ₁₄	-35.9	-14.0	-3.2	+8.7	22.0	30.5	42.1	59.6	79.6	100.9	-126.4
Methylcyclopentane	C ₆ H ₁₂	-53.7	-33.8	-23.7	-12.8	-0.6	+7.2	17.9	34.0	52.3	71.8	-142.4
Methylcyclopropane	C ₄ H ₈	-96.0	-80.6	-72.8	-64.0	-54.2	-48.0	-39.3	-26.0	-11.3	+4.5	
Methyl <i>n</i> -decyl ketone (<i>n</i> -dodecan-2-one)	C ₁₂ H ₂₄ O	77.1	106.0	120.4	136.0	152.4	163.8	177.5	199.0	222.5	246.5	
dichloroacetate	C ₂ H ₃ Cl ₂ O ₂	3.2	26.7	38.1	50.7	64.7	73.6	85.4	103.2	122.6	143.0	
<i>N</i> -Methyldiphenylamine	C ₁₃ H ₁₃ N	103.5	134.0	149.7	165.8	184.0	195.4	210.1	232.8	257.0	282.0	-7.6

TABLE 2-10 Vapor Pressures of Organic Compounds, up to 1 atm* (Continued)

Compound		Pressure, mmHg										Melting point, °C
		1	5	10	20	40	60	100	200	400	760	
Name	Formula	Temperature, °C										
Methyl <i>n</i> -dodecyl ketone (2-tetradecanone)	C ₁₄ H ₂₈ O	99.3	130.0	145.5	161.3	179.8	191.4	206.0	228.2	253.3	278.0	
Methylene bromide (dibromomethane)	CH ₂ Br ₂	-35.1	-13.2	-2.4	+9.7	23.3	31.6	42.3	58.5	79.0	98.6	-52.8
chloride (dichloromethane)	CH ₂ Cl ₂	-70.0	-52.1	-43.3	-33.4	-22.3	-15.7	-6.3	+8.0	24.1	40.7	-96.7
Methyl ethyl ketone (2-butanone)	C ₄ H ₈ O	-48.3	-28.0	-17.7	-6.5	+6.0	14.0	25.0	41.6	60.0	79.6	-85.9
2-Methyl-3-ethylpentane	C ₈ H ₁₈	-24.0	-1.8	+9.5	21.7	35.2	43.9	55.7	73.6	94.0	115.6	-114.5
3-Methyl-3-ethylpentane	C ₈ H ₁₈	-23.9	-1.4	+9.9	22.3	36.2	45.0	57.1	75.3	96.2	118.3	-90
Methyl fluoride	CH ₃ F	-147.3	-137.0	-131.6	-125.9	-119.1	-115.0	-109.9	-99.9	-89.5	-78.2	
formate	C ₂ H ₄ O ₂	-74.2	-57.0	-48.6	-39.2	-28.7	-21.9	-12.9	+0.8	16.0	32.0	-99.8
α-Methylglutaric anhydride	C ₆ H ₈ O ₃	93.8	125.4	141.8	157.7	177.5	189.9	205.0	229.1	255.5	282.5	
Methyl glycolate	C ₃ H ₆ O ₃	+9.6	33.7	45.3	58.1	72.3	81.8	93.7	111.8	131.7	151.5	
2-Methylheptadecane	C ₁₈ H ₃₈	119.8	152.0	168.7	186.0	204.8	216.3	231.5	254.5	279.8	306.5	
2-Methylheptane	C ₈ H ₁₈	-21.0	+1.3	12.3	24.4	37.9	46.6	58.3	76.0	96.2	117.6	-109.5
3-Methylheptane	C ₈ H ₁₈	-19.8	+2.6	13.3	25.4	38.9	47.6	59.4	77.1	97.4	118.9	-120.8
4-Methylheptane	C ₈ H ₁₈	-20.4	+1.5	12.4	24.5	38.0	46.6	58.3	76.1	96.3	117.7	-121.1
2-Methyl-2-heptene	C ₈ H ₁₆	-16.1	+6.7	17.8	30.4	44.0	52.8	64.6	82.3	102.2	122.5	
6-Methyl-3-hepten-2-ol	C ₈ H ₁₆ O	41.6	65.0	76.7	89.3	102.7	111.5	122.6	139.5	156.6	175.5	
6-Methyl-5-hepten-2-ol	C ₈ H ₁₆ O	41.9	66.0	77.8	90.4	104.0	112.8	123.8	140.0	156.6	174.3	
2-Methylhexane	C ₇ H ₁₆	-40.4	-19.5	-9.1	+2.3	14.9	23.0	34.1	50.8	69.8	90.0	-118.2
3-Methylhexane	C ₇ H ₁₆	-39.0	-18.1	-7.8	+3.6	16.4	24.5	35.6	52.4	71.6	91.9	
Methyl iodide	CH ₃ I	-55.0	-45.8	-35.6	-24.2	-16.9	-7.0	+8.0	25.3	42.4	64.4	-64.4
laurate	C ₁₃ H ₂₆ O ₂	87.8	117.9	133.2	149.0	166.0	176.8	190.8				5
levulinate	C ₈ H ₁₀ O ₃	39.8	66.4	79.7	93.7	109.5	119.3	133.0	153.4	175.8	197.7	
methacrylate	C ₅ H ₈ O ₂	-30.5	-10.0	+1.0	11.0	25.5	34.5	47.0	63.0	82.0	101.0	
myristate	C ₁₅ H ₃₀ O ₂	115.0	145.7	160.8	177.8	195.8	207.5	222.6	245.3	269.8	295.8	18.5
α-naphthyl ketone (1-acetonaphthone)	C ₁₂ H ₁₀ O	115.6	146.3	161.5	178.4	196.8	208.6	223.8	246.7	270.5	295.5	
β-naphthyl ketone (2-acetonaphthone)	C ₁₂ H ₁₀ O	120.2	152.3	168.5	185.7	203.8	214.7	229.8	251.6	275.8	301.0	55.5
<i>n</i> -nonyl ketone (undecan-2-one)	C ₁₁ H ₂₂ O	68.2	95.5	108.9	123.1	139.0	148.6	161.0	181.2	202.3	224.0	15
palmitate	C ₁₇ H ₃₄ O ₂	134.3	166.8	184.3	202.0							30
<i>n</i> -pentadecyl ketone (2-heptadecanone)	C ₁₇ H ₃₄ O	129.6	161.6	178.0	196.4	214.3	226.7	242.0	265.8	291.7	319.5	
2-Methylpentane	C ₆ H ₁₄	-60.9	-41.7	-32.1	-21.4	-9.7	-1.9	+8.1	24.1	41.6	60.3	-154
3-Methylpentane	C ₆ H ₁₄	-59.0	-39.8	-30.1	-19.4	-7.3	+0.1	10.5	26.5	44.2	63.3	-118
2-Methyl-1-pentanol	C ₆ H ₁₄ O	15.4	38.0	49.6	61.6	74.7	83.4	94.2	111.3	129.8	147.9	
2-Methyl-2-pentanol	C ₆ H ₁₄ O	-4.5	+16.8	27.6	38.8	51.3	58.8	69.2	85.0	102.6	121.2	-103
Methyl <i>n</i> -pentyl ketone (2-heptanone)	C ₇ H ₁₄ O	19.3	43.6	55.5	67.7	81.2	89.8	100.0	116.1	133.2	150.2	
phenyl ether (anisole)	C ₇ H ₈ O	+5.4	30.0	42.2	55.8	70.7	80.1	93.0	112.3	133.8	155.5	-37.3
2-Methylpropene	C ₄ H ₈	-105.1	-96.5	-81.9	-73.4	-63.8	-57.7	-49.3	-36.7	-22.2	-6.9	-140.3
Methyl propionate	C ₄ H ₈ O ₂	-42.0	-21.5	-11.8	-1.0	+11.0	18.7	29.0	44.2	61.8	79.8	-87.5
4-Methylpropiophenone	C ₁₀ H ₁₀ O	59.6	89.3	103.8	120.2	138.0	149.3	164.2	187.4	212.7	238.5	
2-Methylpropionyl bromide	C ₄ H ₇ BrO	13.5	38.4	50.6	64.1	79.4	88.8	101.6	120.5	141.7	163.0	
Methyl propyl ether	C ₄ H ₁₀ O	-72.2	-54.3	-45.4	-35.4	-24.3	-17.4	-8.1	+6.0	22.5	39.1	
<i>n</i> -propyl ketone (2-pentanone)	C ₅ H ₁₀ O	-12.0	+8.0	17.9	28.5	39.8	47.3	56.8	71.0	86.8	103.3	-77.8
isopropyl ketone (3-Methyl-2-butanone)	C ₅ H ₁₀ O	-19.9	-1.0	+8.3	18.3	29.6	36.2	45.5	59.0	73.8	88.9	-92
2-Methylquinoline	C ₁₀ H ₉ N	75.3	104.0	119.0	134.0	150.8	161.7	176.2	197.8	211.7	246.5	-1
Methyl salicylate	C ₈ H ₈ O ₃	54.0	81.6	95.3	110.0	126.2	136.7	150.0	172.6	197.5	223.2	-8.3
α-Methyl styrene	C ₉ H ₁₀	7.4	34.0	47.1	61.8	77.8	88.3	102.2	121.8	143.0	165.4	-23.2
4-Methyl styrene	C ₉ H ₁₀	16.0	42.0	55.1	69.2	85.0	95.0	108.6	128.7	151.2	175.0	
Methyl <i>n</i> -tetradecyl ketone (2-hexadecanone)	C ₁₆ H ₃₂ O	109.8	151.5	167.3	184.6	203.7	215.0	230.5	254.4	279.8	307.0	
thiocyanate	C ₄ H ₉ NS	-14.0	+9.8	21.6	34.5	49.0	58.1	70.4	89.8	110.8	132.9	-51
isothiocyanate	C ₄ H ₉ NS	-34.7	-8.3	+5.4	20.4	38.2	47.5	59.3	77.5	97.8	119.0	35.5
undecyl ketone (2-tridecanone)	C ₁₃ H ₂₆ O	86.8	117.0	131.8	147.8	165.7	176.6	191.5	214.0	238.3	262.5	28.5
isovalerate	C ₆ H ₁₂ O ₂	-19.2	+2.9	14.0	26.4	39.8	48.2	59.8	77.3	96.7	116.7	
Monovinylacetylene (butenyne)	C ₄ H ₄	-93.2	-77.7	-70.0	-61.3	-51.7	-45.3	-37.1	-24.1	-10.1	+5.3	
Myrcene	C ₁₀ H ₁₆	14.5	40.0	53.2	67.0	82.6	92.6	106.0	126.0	148.3	171.5	
Myristaldehyde	C ₁₄ H ₂₈ O	99.0	132.0	148.3	166.2	186.0	198.3	214.5	240.4	267.9	297.8	23.5
Myristic acid (tetradecanoic acid)	C ₁₄ H ₂₈ O ₂	142.0	174.1	190.8	207.6	223.5	237.2	250.5	272.3	294.6	318.0	57.5
Naphthalene	C ₁₀ H ₈	52.6	74.2	85.8	101.7	119.3	130.2	145.5	167.7	193.2	217.9	80.2
1-Naphthoic acid	C ₁₁ H ₈ O ₂	156.0	184.0	196.8	211.2	225.0	234.5	245.8	263.5	281.4	300.0	160.5
2-Naphthoic acid	C ₁₁ H ₈ O ₂	160.8	189.7	202.8	216.9	231.5	241.3	252.7	270.3	289.5	308.5	184
1-Naphthol	C ₁₀ H ₈ O	94.0	125.5	142.0	158.0	177.8	190.0	206.0	229.6	255.8	282.5	96
2-Naphthol	C ₁₀ H ₈ O	128.6	145.5	161.8	177.1	193.7	209.8	234.0	260.6	288.0	312.5	122.5
1-Naphthylamine	C ₁₀ H ₉ N	104.3	137.7	153.8	171.6	191.5	203.8	220.0	244.9	272.2	300.8	50
2-Naphthylamine	C ₁₀ H ₉ N	108.0	141.6	157.6	175.8	195.7	208.1	224.3	249.7	277.4	306.1	111.5
Nicotine	C ₁₀ H ₁₁ N ₂	61.8	91.8	107.2	123.7	142.1	154.7	169.5	193.8	219.8	247.3	
2-Nitroaniline	C ₆ H ₆ N ₂ O ₂	104.0	135.7	150.4	167.7	186.0	197.8	213.0	236.3	260.0	284.5	71.5
3-Nitroaniline	C ₆ H ₆ N ₂ O ₂	119.3	151.5	167.8	185.5	204.2	216.5	232.1	255.3	280.2	305.7	114
4-Nitroaniline	C ₆ H ₆ N ₂ O ₂	142.4	177.6	194.4	213.2	234.2	245.9	261.8	284.5	310.2	336.0	146.5
2-Nitrobenzaldehyde	C ₇ H ₅ NO ₃	85.8	117.7	133.4	150.0	168.8	180.7	196.2	220.0	246.8	273.5	40.9
3-Nitrobenzaldehyde	C ₇ H ₅ NO ₃	96.2	127.4	142.8	159.0	177.7	189.5	204.3	227.4	252.1	278.3	58
Nitrobenzene	C ₆ H ₅ NO ₂	44.4	71.6	84.9	99.3	115.4	125.8	139.9	161.2	185.8	210.6	+5.7
Nitroethane	C ₂ H ₅ NO ₂	-21.0	+1.5	12.5	24.8	38.0	46.5	57.8	74.8	94.0	114.0	-90
Nitroglycerin	C ₃ H ₅ N ₃ O ₆	127	167	188	210	235	251					11
Nitromethane	CH ₃ NO ₂	-29.0	-7.9	+2.8	14.1	27.5	35.5	46.6	63.5	82.0	101.2	-29
2-Nitrophenol	C ₆ H ₅ NO ₃	49.3	76.8	90.4	105.8	122.1	132.6	146.4	167.6	191.0	214.5	45
2-Nitrophenyl acetate	C ₈ H ₇ NO ₄	100.0	128.0	142.0	155.8	172.8	181.7	194.1	213.0	233.5	253.0	

TABLE 2-10 Vapor Pressures of Organic Compounds, up to 1 atm (Continued)

Compound		Pressure, mmHg										Melting point, °C
		1	5	10	20	40	60	100	200	400	760	
Name	Formula	Temperature, °C										
1-Nitropropane	C ₃ H ₇ NO ₂	-9.6	+13.5	25.3	37.9	51.8	60.5	72.3	90.2	110.6	131.6	-108
2-Nitropropane	C ₃ H ₇ NO ₂	-18.8	+4.1	15.8	28.2	41.8	50.3	62.0	80.0	99.8	120.3	-93
2-Nitrotoluene	C ₇ H ₇ NO ₂	50.0	79.1	93.8	109.6	126.3	137.6	151.5	173.7	197.7	222.3	-4.1
3-Nitrotoluene	C ₇ H ₇ NO ₂	50.2	81.0	96.0	112.8	130.7	142.5	156.9	180.3	206.8	231.9	15.5
4-Nitrotoluene	C ₇ H ₇ NO ₂	53.7	85.0	100.5	117.7	136.0	147.9	163.0	186.7	212.5	238.3	51.9
4-Nitro-1,3-xylene (4-nitro- <i>m</i> -xylene)	C ₈ H ₉ NO ₂	65.6	95.0	109.8	125.8	143.3	153.8	168.5	191.7	217.5	244.0	+2
Nonacosane	C ₂₉ H ₆₀	234.2	269.8	286.4	303.6	323.2	334.8	350.0	373.2	397.2	421.8	63.8
Nonadecane	C ₁₉ H ₄₀	133.2	166.3	183.5	200.8	220.0	232.8	248.0	271.8	299.8	330.0	32
<i>n</i> -Nonane	C ₉ H ₂₀	+1.4	25.8	38.0	51.2	66.0	75.5	88.1	107.5	128.2	150.8	-53.7
1-Nonanol	C ₉ H ₂₀ O	59.5	86.1	99.7	113.8	129.0	139.0	151.3	170.5	192.1	213.5	-5
2-Nonanone	C ₉ H ₁₈ O	32.1	59.0	72.3	87.2	103.4	113.8	127.4	148.2	171.2	195.0	-19
Octacosane	C ₂₈ H ₅₈	226.5	260.3	277.4	295.4	314.2	326.8	341.8	364.8	388.9	412.5	61.6
Octadecane	C ₁₈ H ₃₈	119.6	152.1	169.6	187.5	207.4	219.7	236.0	260.6	288.0	317.0	28
<i>n</i> -Octane	C ₈ H ₁₈	-14.0	+8.3	19.2	31.5	45.1	53.8	65.7	83.6	104.0	125.6	-56.8
<i>n</i> -Octanol (1-octanol)	C ₈ H ₁₈ O	54.0	76.5	88.3	101.0	115.2	123.8	135.2	152.0	173.8	195.2	-15.4
2-Octanone	C ₈ H ₁₆ O	23.6	48.4	60.9	74.3	89.8	99.0	111.7	130.4	151.0	172.9	-16
<i>n</i> -Octyl acrylate	C ₁₁ H ₂₀ O ₂	58.5	87.7	102.0	117.8	135.6	145.6	159.1	180.2	204.0	227.0	
iodide (1-Iodoctane)	C ₈ H ₁₇ I	45.8	74.8	90.0	105.9	123.8	135.4	150.0	173.3	199.3	225.5	-45.9
Oleic acid	C ₁₈ H ₃₄ O ₂	176.5	208.5	223.0	240.0	257.2	269.8	286.0	309.8	334.7	360.0	14
Palmitaldehyde	C ₁₆ H ₃₂ O	121.6	154.6	171.8	190.0	210.0	222.6	239.5	264.1	292.3	321.0	34
Palmitic acid	C ₁₆ H ₃₂ O ₂	153.6	188.1	205.8	223.8	244.4	256.0	271.5	298.7	326.0	353.8	64.0
Palmitonitrile	C ₁₆ H ₃₁ N	134.3	168.3	185.8	204.2	223.8	236.6	251.5	277.1	304.5	332.0	31
Pelargonic acid	C ₉ H ₁₈ O ₂	108.2	126.0	137.4	149.8	163.7	172.3	184.4	203.1	227.5	253.5	12.5
Pentachlorobenzene	C ₆ HCl ₅	98.6	129.7	144.3	160.0	178.5	190.1	205.5	227.0	251.6	276.0	85.5
Pentachloroethane	C ₂ HCl ₅	+1.0	27.2	39.8	53.9	69.9	80.0	93.5	114.0	137.2	160.5	-22
Pentachloroethylbenzene	C ₈ H ₅ Cl ₅	96.2	130.0	148.0	166.0	186.2	199.0	216.0	241.8	269.3	299.0	
Pentachlorophenol	C ₆ HCl ₅ O				192.2	211.2	223.4	239.6	261.8	285.0	309.3	188.5
Pentacosane	C ₂₅ H ₅₂	194.2	230.0	248.2	266.1	285.6	298.4	314.0	339.0	365.4	390.3	53.3
Pentadecane	C ₁₅ H ₃₂	91.6	121.0	135.4	150.2	167.7	178.4	194.0	216.1	242.8	270.5	10
1,3-Pentadiene	C ₅ H ₈	-71.8	-53.8	-45.0	-34.8	-23.4	-16.5	-6.7	+8.0	24.7	42.1	
1,4-Pentadiene	C ₅ H ₈	-83.5	-66.2	-57.1	-47.7	-37.0	-30.0	-20.6	-6.7	+8.3	26.1	
Pentaethylbenzene	C ₁₆ H ₂₆	86.0	120.0	135.8	152.4	171.9	184.2	200.0	224.1	250.2	277.0	
Pentaethylchlorobenzene	C ₁₆ H ₂₅ Cl	90.0	123.8	140.7	158.1	178.2	191.0	208.0	230.3	257.2	285.0	
<i>n</i> -Pentane	C ₅ H ₁₂	-76.6	-62.5	-50.1	-40.2	-29.2	-22.2	-12.6	+1.9	18.5	36.1	-129.7
iso-Pentane (2-methylbutane)	C ₅ H ₁₂	-82.9	-65.8	-57.0	-47.3	-36.5	-29.6	-20.2	-5.9	+10.5	27.8	-159.7
neo-Pentane (2,2-dimethylpropane)	C ₅ H ₁₂	-102.0	-85.4	-76.7	-67.2	-56.1	-49.0	-39.1	-23.7	-7.1	+9.5	-16.6
2,3,4-Pentanetriol	C ₅ H ₁₂ O ₃	155.0	189.3	204.5	220.5	239.6	249.8	263.5	284.5	307.0	327.2	
1-Pentene	C ₅ H ₁₀	-80.4	-63.3	-54.5	-46.0	-34.1	-27.1	-17.7	-3.4	+12.8	30.1	
α-Phellandrene	C ₁₀ H ₁₆	20.0	45.7	58.0	72.1	87.8	97.6	110.6	130.6	152.0	175.0	
Phenanthrene	C ₁₄ H ₁₀	118.2	154.3	173.0	193.7	215.8	229.9	249.0	277.1	308.0	340.2	99.5
Phenethyl alcohol (phenyl cellosolve)	C ₈ H ₁₀ O ₂	58.2	85.9	100.0	114.8	130.5	141.2	154.0	175.0	197.5	219.5	
2-Phenetidine	C ₈ H ₁₁ NO	67.0	94.7	108.6	123.7	139.9	149.8	163.5	184.0	207.0	228.0	
Phenol	C ₆ H ₆ O	40.1	62.5	73.8	86.0	100.1	108.4	121.4	139.0	160.0	181.9	40.6
2-Phenoxyethanol	C ₈ H ₁₀ O ₂	78.0	106.6	121.2	136.0	152.2	163.2	176.5	197.6	221.0	245.3	11.6
2-Phenoxyethyl acetate	C ₁₀ H ₁₂ O ₃	82.6	113.5	128.0	144.5	162.3	174.0	189.2	211.3	235.0	259.7	-6.7
Phenyl acetate	C ₈ H ₈ O ₂	38.2	64.8	78.0	92.3	108.1	118.1	131.6	151.2	173.5	195.9	
Phenylacetic acid	C ₈ H ₈ O ₂	97.0	127.0	141.3	156.0	173.6	184.5	198.2	219.5	243.0	265.5	76.5
Phenylacetone	C ₈ H ₈ N	60.0	89.0	103.5	119.4	136.3	147.7	161.8	184.2	208.5	233.5	-23.8
Phenylacetyl chloride	C ₈ H ₇ ClO	48.0	75.3	89.0	103.6	119.8	129.8	143.5	163.8	186.0	210.0	
Phenyl benzoate	C ₁₀ H ₁₀ O ₂	106.8	141.5	157.8	177.0	197.6	210.8	227.8	254.0	283.5	314.0	70.5
4-Phenyl-3-buten-2-one	C ₁₀ H ₁₀ O	81.7	112.2	127.4	143.8	161.3	172.6	187.8	211.0	235.4	261.0	41.5
Phenyl isocyanate	C ₇ H ₅ NO	10.6	36.0	48.5	62.5	77.7	87.7	100.6	120.8	142.7	165.6	
isocyanide	C ₇ H ₅ N	12.0	37.0	49.7	63.4	78.3	88.0	101.0	120.8	142.3	165.0	
Phenylcyclohexane	C ₁₂ H ₁₆	67.5	96.5	111.3	126.4	144.0	154.2	169.3	191.3	214.6	240.0	+7.5
Phenyl dichlorophosphate	C ₆ H ₅ Cl ₂ O ₂ P	66.7	95.9	110.0	125.9	143.4	153.6	168.0	189.8	213.0	239.5	
<i>m</i> -Phenylenediamine	C ₆ H ₈ N ₂	99.8	131.2	147.0	163.8	182.5	194.0	209.9	233.0	259.0	285.5	62.8
Phenylglyoxal	C ₈ H ₆ O ₂		75.0	87.8	100.7	115.5	124.2	136.2	153.8	173.5	193.5	73
Phenylhydrazine	C ₆ H ₈ N ₂	71.8	101.6	115.8	131.5	148.2	158.7	173.5	195.4	218.2	243.5	19.5
<i>N</i> -Phenyliminodietanol	C ₁₀ H ₁₅ NO ₂	145.0	179.2	195.8	213.4	233.0	245.3	260.6	284.5	311.3	337.8	
1-Phenyl-1,3-pentanedione	C ₁₁ H ₁₆ O ₂	98.0	128.5	144.0	159.9	178.0	189.8	204.5	226.7	251.2	276.5	
2-Phenylphenol	C ₁₂ H ₁₀ O	100.0	131.6	146.2	163.3	180.3	192.2	205.9	227.9	251.8	275.0	56.5
4-Phenylphenol	C ₁₂ H ₁₀ O			176.2	193.8	213.0	225.3	240.9	263.2	285.5	308.0	164.5
3-Phenyl-1-propanol	C ₉ H ₁₂ O	74.7	102.4	116.0	131.2	147.4	156.8	170.3	191.2	212.8	235.0	
Phenyl isothiocyanate	C ₇ H ₅ NS	47.2	75.6	89.8	103.5	119.5	128.5	143.7	166.9	194.0	218.5	-21.0
Phorone	C ₉ H ₁₄ O	42.0	68.3	81.5	95.6	111.3	121.4	134.0	153.5	175.3	197.2	28
iso-Phorone	C ₉ H ₁₄ O	38.0	66.7	81.2	96.8	114.5	125.6	140.6	163.3	188.7	215.2	
Phosgene (carbonyl chloride)	CCl ₂ O	-92.9	-77.0	-69.3	-60.3	-50.3	-44.0	-35.6	-22.3	-7.6	+8.3	-104
Phthalic anhydride	C ₈ H ₄ O ₃	96.5	121.3	134.0	151.7	172.0	185.3	202.3	228.0	256.8	284.5	130.8
Phthalide	C ₈ H ₆ O ₂	95.5	127.7	144.0	161.3	181.0	193.5	210.0	234.5	261.8	290.0	73
Phthaloyl chloride	C ₈ H ₄ Cl ₂ O ₂	86.3	118.3	134.2	151.0	170.0	182.2	197.8	222.0	248.3	275.8	88.5
2-Picoline	C ₈ H ₇ N	-11.1	+12.6	24.4	37.4	51.2	59.9	71.4	89.0	108.4	128.8	-70
Pinelic acid	C ₇ H ₁₂ O ₄	163.4	196.2	212.0	229.3	247.0	258.2	272.0	294.5	318.5	342.1	103
α-Pinene	C ₁₀ H ₁₆	-1.0	+24.6	37.3	51.4	66.8	76.8	90.1	110.2	132.3	155.0	-55
β-Pinene	C ₁₀ H ₁₆	+4.2	30.0	42.3	58.1	71.5	81.2	94.0	114.1	136.1	158.3	

TABLE 2-10 Vapor Pressures of Organic Compounds, up to 1 atm (Continued)

Compound		Pressure, mmHg										Melting point, °C
		1	5	10	20	40	60	100	200	400	760	
Name	Formula	Temperature, °C										
Piperidine	C ₅ H ₁₁ N		-7.0	+3.9	15.8	29.2	37.7	49.0	66.2	85.7	106.0	-9
Piperonal	C ₈ H ₆ O ₃	87.0	117.4	132.0	148.0	165.7	177.0	191.7	214.3	238.5	263.0	37
Propane	C ₃ H ₈	-128.9	-115.4	-108.5	-100.9	-92.4	-87.0	-79.6	-68.4	-55.6	-42.1	-187.1
Propenylbenzene	C ₉ H ₁₀	17.5	43.8	57.0	71.5	87.7	97.8	111.7	132.0	154.7	179.0	-30.1
Propionamide	C ₃ H ₇ NO	65.0	91.0	105.0	119.0	134.8	144.3	156.0	174.2	194.0	213.0	79
Propionic acid	C ₃ H ₆ O ₂	4.6	28.0	39.7	52.0	65.8	74.1	85.8	102.5	122.0	141.1	-22
anhydride	C ₆ H ₁₀ O ₃	20.6	45.3	57.7	70.4	85.6	94.5	107.2	127.8	146.0	167.0	-45
Propionitrile	C ₃ H ₅ N	-35.0	-13.6	-3.0	+8.8	22.0	30.1	41.4	58.2	77.7	97.1	-91.9
Propiophenone	C ₉ H ₁₀ O	50.0	77.9	92.2	107.6	124.3	135.0	149.3	170.2	194.2	218.0	21
<i>n</i> -Propyl acetate	C ₅ H ₁₀ O ₂	-26.7	-5.4	+5.0	16.0	28.8	37.0	47.8	64.0	82.0	101.8	-92.5
iso-Propyl acetate	C ₅ H ₁₀ O ₂	-38.3	-17.4	-7.2	+4.2	17.0	25.1	35.7	51.7	69.8	89.0	
<i>n</i> -Propyl alcohol (1-propanol)	C ₃ H ₈ O	-15.0	+5.0	14.7	25.3	36.4	43.5	52.8	66.8	82.0	97.8	-127
iso-Propyl alcohol (2-propanol)	C ₃ H ₈ O	-26.1	-7.0	+2.4	12.7	23.8	30.5	39.5	53.0	67.8	82.5	-85.8
<i>n</i> -Propylamine	C ₃ H ₉ N	-64.4	-46.3	-37.2	-27.1	-16.0	-9.0	+0.5	15.0	31.5	48.5	-83
Propylbenzene	C ₉ H ₁₂	6.3	31.3	43.4	56.8	71.6	81.1	94.0	113.5	135.7	159.2	-99.5
Propyl benzoate	C ₁₀ H ₁₂ O ₂	54.6	83.8	98.0	114.3	131.8	143.3	157.4	180.1	205.2	231.0	-51.6
<i>n</i> -Propyl bromide (1-bromopropane)	C ₃ H ₇ Br	-53.0	-33.4	-23.3	-12.4	-0.3	+7.5	18.0	34.0	52.0	71.0	-109.9
iso-Propyl bromide (2-bromopropane)	C ₃ H ₇ Br	-61.8	-42.5	-32.8	-22.0	-10.1	-2.5	+8.0	23.8	41.5	60.0	-89.0
<i>n</i> -Propyl <i>n</i> -butyrate	C ₇ H ₁₄ O ₂	-1.6	+22.1	34.0	47.0	61.5	70.3	82.6	101.0	121.7	142.7	-95.2
isobutyrate	C ₇ H ₁₄ O ₂	-6.2	+16.8	28.3	40.6	54.3	63.0	73.9	91.8	112.0	133.9	
iso-Propyl isobutyrate	C ₇ H ₁₄ O ₂	-16.3	+5.8	17.0	29.0	42.4	51.4	62.3	80.2	100.0	120.5	
Propyl carbamate	C ₆ H ₉ NO ₂	52.4	77.6	90.0	103.2	117.7	126.5	138.3	155.8	175.8	195.0	
<i>n</i> -Propyl chloride (1-chloropropane)	C ₃ H ₇ Cl	-68.3	-50.0	-41.0	-31.0	-19.5	-12.1	-2.5	+12.2	29.4	46.4	-122.8
iso-Propyl chloride (2-chloropropane)	C ₃ H ₇ Cl	-78.8	-61.1	-52.0	-42.0	-31.0	-23.5	-13.7	+1.3	18.1	36.5	-117
iso-Propyl chloroacetate	C ₆ H ₉ ClO ₂	+3.8	28.1	40.2	53.9	68.7	78.0	90.3	108.8	128.0	148.6	
Propyl chloroglyoxylate	C ₆ H ₉ ClO ₃	9.7	32.3	43.5	55.6	68.8	77.2	88.0	104.7	123.0	150.0	
Propylene	C ₃ H ₆	-131.9	-120.7	-112.1	-104.7	-96.5	-91.3	-84.1	-73.3	-60.9	-47.7	-185
Propylene glycol (1,2-Propanediol)	C ₃ H ₈ O ₂	45.5	70.8	83.2	96.4	111.2	119.9	132.0	149.7	168.1	188.2	
Propylene oxide	C ₃ H ₆ O	-75.0	-57.8	-49.0	-39.3	-28.4	-21.3	-12.0	+2.1	17.8	34.5	-112.1
<i>n</i> -Propyl formate	C ₄ H ₈ O ₂	-43.0	-22.7	-12.6	-1.7	+10.8	18.8	29.5	45.3	62.6	81.3	-92.9
iso-Propyl formate	C ₄ H ₈ O ₂	-52.0	-32.7	-22.7	-12.1	-0.2	+7.5	17.8	33.6	50.5	68.3	
4,4'-iso-Propylidenebisphenol	C ₁₅ H ₁₆ O ₂	193.0	224.2	240.8	255.5	273.0	282.9	297.0	317.5	339.0	360.5	
<i>n</i> -Propyl iodide (1-iodopropane)	C ₃ H ₇ I	-36.0	-13.5	-2.4	+10.0	23.6	32.1	43.8	61.8	81.8	102.5	-98.8
iso-Propyl iodide (2-iodopropane)	C ₃ H ₇ I	-43.3	-22.1	-11.7	0.0	+13.2	21.6	32.8	50.0	69.5	89.5	-90
<i>n</i> -Propyl levulinate	C ₈ H ₁₄ O ₃	59.7	86.3	99.9	114.0	130.1	140.6	154.0	175.6	198.0	221.2	
iso-Propyl levulinate	C ₈ H ₁₄ O ₃	48.0	74.5	88.0	102.4	118.1	127.8	141.8	161.6	185.2	208.2	
Propyl mercaptan (1-propanethiol)	C ₃ H ₆ S	-56.0	-36.3	-26.3	-15.4	-3.2	+4.6	15.3	31.5	49.2	67.4	-112
2-iso-Propylnaphthalene	C ₁₃ H ₁₄	76.0	107.9	123.4	140.3	159.0	171.4	187.6	211.8	238.5	266.0	
iso-Propyl β-naphthyl ketone (2-isobutyronaphthone)	C ₁₄ H ₁₄ O	133.2	165.4	181.0	197.7	215.6	227.0	242.3	264.0	288.2	313.0	
2-iso-Propylphenol	C ₉ H ₁₂ O	56.6	83.8	97.0	111.7	127.5	137.7	150.3	170.1	192.6	214.5	15.5
3-iso-Propylphenol	C ₉ H ₁₂ O	62.0	90.3	104.1	119.8	136.2	146.6	160.2	182.0	205.0	228.0	26
4-iso-Propylphenol	C ₉ H ₁₂ O	67.0	94.7	108.0	123.4	139.8	149.7	163.3	184.0	206.1	228.2	61
Propyl propionate	C ₈ H ₁₆ O ₂	-14.2	+8.0	19.4	31.6	45.0	53.8	65.2	82.7	102.0	122.4	-76
4-iso-Propylstyrene	C ₁₁ H ₁₄	34.7	62.3	76.0	91.2	108.0	118.4	132.8	153.9	178.0	202.5	
Propyl isovalerate	C ₈ H ₁₆ O ₂	+8.0	32.8	45.1	58.0	72.8	82.3	95.0	113.9	135.0	155.9	
Pulegone	C ₁₀ H ₁₆ O	58.3	82.5	94.0	106.8	121.7	130.2	143.1	162.5	189.8	221.0	
Pyridine	C ₅ H ₅ N	-18.9	+2.5	13.2	24.8	38.0	46.8	57.8	75.0	95.6	115.4	-42
Pyrocatechol	C ₆ H ₆ O ₂		104.0	118.3	134.0	150.6	161.7	176.0	197.7	221.5	245.5	105
Pyrocaltechol diacetate (1,2-phenylene diacetate)	C ₁₀ H ₁₀ O ₄	98.0	129.8	145.7	161.8	179.8	191.6	206.5	228.7	253.3	278.0	
Pyrogallol	C ₆ H ₆ O ₃		151.7	167.7	185.3	204.2	216.3	232.0	255.3	281.5	309.0	133
Pyrotartaric anhydride	C ₈ H ₆ O ₃	69.7	99.7	114.2	130.0	147.8	158.6	173.8	196.1	221.0	247.4	
Pyruvic acid	C ₃ H ₄ O ₃	21.4	45.8	57.9	70.8	85.3	94.1	106.5	124.7	144.7	165.0	13.6
Quinoline	C ₈ H ₇ N	59.7	89.6	103.8	119.8	136.7	148.1	163.2	186.2	212.3	237.7	-15
iso-Quinoline	C ₈ H ₇ N	63.5	92.7	107.8	123.7	141.6	152.0	167.6	190.0	214.5	240.5	24.6
Resorcinol	C ₆ H ₆ O ₂	108.4	138.0	152.1	168.0	185.3	195.8	209.8	230.8	253.4	276.5	110.7
Safrole	C ₁₀ H ₁₀ O ₂	63.8	93.0	107.6	123.0	140.1	150.3	165.1	186.2	210.0	233.0	11.2
Salicylaldehyde	C ₇ H ₆ O ₂	33.0	60.1	73.8	88.7	105.2	115.7	129.4	150.0	173.7	196.5	-7
Salicylic acid	C ₇ H ₆ O ₃	113.7	136.0	146.2	156.8	172.2	182.0	193.4	210.0	230.5	256.0	159
Sebacic acid	C ₁₀ H ₁₈ O ₄	183.0	215.7	232.0	250.0	268.2	279.8	294.5	313.2	332.8	352.3	134.5
Selenophene	C ₄ H ₄ Se	-39.0	-16.0	-4.0	+9.1	24.1	33.8	47.0	66.7	89.8	114.3	
Skatole	C ₉ H ₉ N	95.0	124.2	139.6	154.3	171.9	183.6	197.4	218.8	242.5	266.2	95
Stearaldehyde	C ₁₈ H ₃₆ O	140.0	174.6	192.1	210.6	230.8	244.2	260.0	285.0	313.8	342.5	63.5
Stearic acid	C ₁₈ H ₃₆ O ₂	173.7	209.0	225.0	243.4	263.3	275.5	291.0	316.5	343.0	370.0	69.3
Stearyl alcohol (1-octadecanol)	C ₁₈ H ₃₆ O	150.3	185.6	202.0	220.0	240.4	252.7	269.4	293.5	320.3	349.5	58.5
Styrene	C ₈ H ₈	-7.0	+18.0	30.8	44.6	59.8	69.5	82.0	101.3	122.5	145.2	-30.6
Styrene dibromide [(1,2-dibromoethyl) benzene]	C ₈ H ₈ Br ₂	86.0	115.6	129.8	145.2	161.8	172.2	186.3	207.8	230.0	254.0	
Suberic acid	C ₈ H ₁₄ O ₄	172.8	205.5	219.5	238.2	254.6	265.4	279.8	300.5	322.8	345.5	142
Succinic anhydride	C ₄ H ₆ O ₃	92.0	115.0	128.2	145.3	163.0	174.0	189.0	212.0	237.0	261.0	119.6
Succinimide	C ₄ H ₅ NO ₂	115.0	143.2	157.0	174.0	192.0	203.0	217.4	240.0	263.5	287.5	125.5
Succinyl chloride	C ₄ H ₄ Cl ₂ O ₂	39.0	65.0	78.0	91.8	107.5	117.2	130.0	149.3	170.0	192.5	17
α-Terpineol	C ₁₀ H ₁₈ O	52.8	80.4	94.3	109.8	126.0	136.3	150.1	171.2	194.3	217.5	35
Terpenoline	C ₁₀ H ₁₆	32.3	58.0	70.6	84.8	100.0	109.8	122.7	142.0	163.5	185.0	

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TABLE 2-10 Vapor Pressures of Organic Compounds, up to 1 atm (Continued)

Compound		Pressure, mmHg										Melting point, °C
		1	5	10	20	40	60	100	200	400	760	
Name	Formula	Temperature, °C										
1,1,1,2-Tetrabromoethane	C ₂ H ₂ Br ₄	58.0	83.3	95.7	108.5	123.2	132.0	144.0	161.5	181.0	200.0	
1,1,2,2-Tetrabromoethane	C ₂ H ₂ Br ₄	65.0	95.5	110.0	126.0	144.0	155.1	170.0	192.5	217.5	243.5	
Tetraisobutylene	C ₁₀ H ₁₆	63.8	93.7	108.5	124.5	142.2	152.6	167.5	190.0	214.6	240.0	
Tetracosane	C ₂₄ H ₅₀	183.8	219.6	237.6	255.3	276.3	288.4	305.2	330.5	358.0	386.4	51.1
1,2,3,4-Tetrachlorobenzene	C ₆ H ₂ Cl ₄	68.5	99.6	114.7	131.2	149.2	160.0	175.7	198.0	225.5	254.0	46.5
1,2,3,5-Tetrachlorobenzene	C ₆ H ₂ Cl ₄	58.2	89.0	104.1	121.6	140.0	152.0	168.0	193.7	220.0	246.0	54.5
1,2,4,5-Tetrachlorobenzene	C ₆ H ₂ Cl ₄					146.0	157.7	173.5	196.0	220.5	245.0	139
1,1,2,2-Tetrachloro-1,2-difluoroethane	C ₂ H ₂ Cl ₂ F ₂	-37.5	-16.0	-5.0	+6.7	19.8	28.1	38.6	55.0	73.1	92.0	26.5
1,1,1,2-Tetrachloroethane	C ₂ H ₂ Cl ₄	-16.3	+7.4	19.3	32.1	46.7	56.0	68.0	87.2	108.2	130.5	-68.7
1,1,2,2-Tetrachloroethane	C ₂ H ₂ Cl ₄	-3.8	+20.7	33.0	46.2	60.8	70.0	83.2	102.2	124.0	145.9	-36
1,2,3,5-Tetrachloro-4-ethylbenzene	C ₈ H ₆ Cl ₄	77.0	110.0	126.0	143.7	162.1	175.0	191.6	215.3	243.0	270.0	
Tetrachloroethylene	C ₂ Cl ₄	-20.6	+2.4	13.8	26.3	40.1	49.2	61.3	79.8	100.0	120.8	-19.0
2,3,4,6-Tetrachlorophenol	C ₆ H ₂ Cl ₄ O	100.0	130.3	145.3	161.0	179.1	190.0	205.2	227.2	250.4	275.0	69.5
3,4,5,6-Tetrachloro-1,2-xylene	C ₈ H ₆ Cl ₄	94.4	125.0	140.3	156.0	174.2	185.8	200.5	223.0	248.3	273.5	
Tetradecane	C ₁₄ H ₃₀	76.4	106.0	120.7	135.6	152.7	164.0	178.5	201.8	226.8	252.5	5.5
Tetradecylamine	C ₁₄ H ₃₁ N	102.6	135.8	152.0	170.0	189.0	200.2	215.7	239.8	264.6	291.2	
Tetradecyltrimethylsilane	C ₁₇ H ₃₅ Si	120.0	150.7	166.2	183.5	201.5	213.3	227.8	250.0	275.0	300.0	
Tetraethoxysilane	C ₈ H ₂₀ O ₄ Si	16.0	40.3	52.6	65.8	81.1	90.7	103.6	123.5	146.2	168.5	
1,2,3,4-Tetraethylbenzene	C ₁₄ H ₂₂	65.7	96.2	111.6	127.7	145.8	156.7	172.4	196.0	221.4	248.0	11.6
Tetraethylene glycol	C ₈ H ₁₈ O ₅	153.9	183.7	197.1	212.3	228.0	237.8	250.0	268.4	288.0	307.8	
Tetraethylene glycol chlorohydrin	C ₈ H ₁₇ ClO ₄	110.1	141.8	156.1	172.6	190.0	200.5	214.7	236.5	258.2	281.5	
Tetraethyllead	C ₈ H ₂₀ Pb	38.4	63.6	74.8	88.0	102.4	111.7	123.8	142.0	161.8	183.0	-136
Tetraethylsilane	C ₈ H ₂₀ Si	-1.0	+23.9	36.3	50.0	65.3	74.8	88.0	108.0	130.2	153.0	
Tetralin	C ₁₀ H ₁₂	38.0	65.3	79.0	93.8	110.4	121.3	135.3	157.2	181.8	207.2	-31.0
1,2,3,4-Tetramethylbenzene	C ₁₀ H ₁₄	42.6	68.7	81.8	95.8	111.5	121.8	135.7	155.7	180.0	204.4	-6.2
1,2,3,5-Tetramethylbenzene	C ₁₀ H ₁₄	40.6	65.8	77.8	91.0	105.8	115.4	128.3	149.9	173.7	197.9	-24.0
1,2,4,5-Tetramethylbenzene	C ₁₀ H ₁₄	45.0	65.0	74.6	88.0	104.2	114.8	128.1	149.5	172.1	195.9	79.5
2,2,3,3-Tetramethylbutane	C ₈ H ₁₈	-17.4	+3.2	13.5	24.6	36.8	44.5	54.8	70.2	87.4	106.3	-102.2
Tetramethylene dibromide (1,4-dibromobutane)	C ₄ H ₈ Br ₂	32.0	58.8	72.4	87.6	104.0	115.1	128.7	149.8	173.8	197.5	-20
Tetramethyllead	C ₈ H ₁₂ Pb	-29.0	-6.8	+4.4	16.6	30.3	39.2	50.8	68.8	89.0	110.0	-27.5
Tetramethyltin	C ₈ H ₁₂ Sn	-51.3	-31.0	-20.6	-9.3	+3.5	11.7	22.8	39.8	58.5	78.0	
Tetrapropylene glycol monoisopropyl ether	C ₁₅ H ₃₂ O ₅	116.6	147.8	163.0	179.8	197.7	209.0	223.3	245.0	268.3	292.7	
Thioacetic acid (mercaptoacetic acid)	C ₂ H ₄ O ₂ S	60.0	87.7	101.5	115.8	131.8	142.0	154.0				-16.5
Thiodiglycol (2,2'-thiodiethanol)	C ₄ H ₁₀ O ₂ S	42.0	96.0	128.0	165.0	210.0	240.5	285				
Thiophene	C ₄ H ₄ S	-40.7	-20.8	-10.9	0.0	+12.5	20.1	30.5	46.5	64.7	84.4	-38.3
Thiophenol (benzenethiol)	C ₆ H ₆ S	18.6	43.7	56.0	69.7	84.2	93.9	106.6	125.8	146.7	168.0	
α-Thujone	C ₁₀ H ₁₆ O	38.3	65.7	79.3	93.7	110.0	120.2	134.0	154.2	177.8	201.0	
Thymol	C ₁₀ H ₁₄ O	64.3	92.8	107.4	122.6	139.8	149.8	164.1	185.5	209.2	231.8	51.5
Tiglaldehyde	C ₈ H ₈ O	-25.0	-1.6	+10.0	23.2	37.0	45.8	57.7	75.4	95.5	116.4	
Tiglic acid	C ₈ H ₁₄ O ₂	52.0	77.8	90.2	103.8	119.0	127.8	140.5	158.0	179.2	198.5	64.5
Tiglonitrile	C ₈ H ₇ N	-25.5	-2.4	+9.2	22.1	36.7	46.0	58.2	77.8	99.7	122.0	
Toluene	C ₇ H ₈	-26.7	-4.4	+6.4	18.4	31.8	40.3	51.9	69.5	89.5	110.6	-95.0
Toluene-2,4-diamine	C ₇ H ₁₀ N ₂	106.5	137.2	151.7	167.9	185.7	196.2	211.5	232.8	256.0	280.0	99
2-Toluic nitrile (2-tolunitrile)	C ₈ H ₇ N	36.7	64.0	77.9	93.0	110.0	120.8	135.0	156.0	180.0	205.2	-13
4-Toluic nitrile (4-tolunitrile)	C ₈ H ₇ N	42.5	71.3	85.8	101.7	109.5	130.0	145.2	167.3	193.0	217.6	29.5
2-Toluidine	C ₇ H ₉ N	44.0	69.3	81.4	95.1	110.0	119.8	133.0	153.0	176.2	199.7	-16.3
3-Toluidine	C ₇ H ₉ N	41.0	68.0	82.0	96.7	113.5	123.8	136.7	157.6	180.6	203.3	-31.5
4-Toluidine	C ₇ H ₉ N	42.0	68.2	81.8	95.8	111.5	121.5	133.7	154.0	176.9	200.4	44.5
2-Tolyl isocyanide	C ₈ H ₇ N	25.2	51.0	64.0	78.2	94.0	104.0	117.7	137.8	159.9	183.5	
4-Tolylhydrazine	C ₇ H ₁₀ N ₂	82.4	110.0	123.8	138.6	154.1	165.0	178.0	198.0	219.5	242.0	65.5
Tribromoacetaldehyde	C ₂ HBr ₃ O	18.5	45.0	58.0	72.1	87.8	97.5	110.2	130.0	151.6	174.0	
1,1,2-Tribromobutane	C ₄ H ₇ Br ₃	45.0	73.5	87.8	103.2	120.2	131.6	146.0	167.8	192.0	216.2	
1,2,2-Tribromobutane	C ₄ H ₇ Br ₃	41.0	69.0	83.2	98.6	116.0	127.0	141.8	163.5	188.0	213.8	
2,2,3-Tribromobutane	C ₄ H ₇ Br ₃	38.2	66.0	79.8	94.6	111.8	122.2	136.3	157.8	182.2	206.5	
1,1,2-Tribromoethane	C ₂ H ₃ Br ₃	32.6	58.0	70.6	84.2	100.0	110.0	123.5	143.5	165.4	188.4	-26
1,2,3-Tribromopropane	C ₃ H ₅ Br ₃	47.5	75.8	90.0	105.8	122.8	134.0	148.0	170.0	195.0	220.0	16.5
Triisobutylamine	C ₁₂ H ₂₇ N	32.3	57.4	69.8	83.0	97.8	107.3	119.7	138.0	157.8	179.0	-22
Triisobutylene	C ₁₂ H ₂₄	18.0	44.0	56.5	70.0	86.7	96.7	110.0	130.2	153.0	179.0	
2,4,6-Triterbutylphenol	C ₁₈ H ₃₀ O	95.2	126.1	142.0	158.0	177.4	188.0	203.0	226.2	250.6	276.3	
Trichloroacetic acid	C ₂ HCl ₃ O ₂	51.0	76.0	88.2	101.8	116.3	125.9	137.8	155.4	175.2	195.6	57
Trichloroacetic anhydride	C ₄ Cl ₆ O ₃	56.2	85.3	99.6	114.3	131.2	141.8	155.2	176.2	199.8	223.0	
Trichloroacetyl bromide	C ₂ BrCl ₃ O	-7.4	+16.7	29.3	42.1	57.2	66.7	79.5	98.4	120.2	143.0	
2,4,6-Trichloroaniline	C ₆ H ₃ Cl ₃ N	134.0	157.8	170.0	182.6	195.8	204.5	214.6	229.8	246.4	262.0	78
1,2,3-Trichlorobenzene	C ₆ H ₃ Cl ₃	40.0	70.0	85.6	101.8	119.8	131.5	146.0	168.2	193.5	218.5	52.5
1,2,4-Trichlorobenzene	C ₆ H ₃ Cl ₃	38.4	67.3	81.7	97.2	114.8	125.7	140.0	162.0	187.7	213.0	17
1,3,5-Trichlorobenzene	C ₆ H ₃ Cl ₃		63.8	78.0	93.7	110.8	121.8	136.0	157.7	183.0	208.4	63.5
1,2,3-Trichlorobutane	C ₄ H ₇ Cl ₃	+0.5	27.2	40.0	55.0	71.5	82.0	96.2	118.0	143.0	169.0	
1,1,1-Trichloroethane	C ₂ H ₃ Cl ₃	-52.0	-32.0	-21.9	-10.8	+1.6	9.5	20.0	36.2	54.6	74.1	-30.6
1,1,2-Trichloroethane	C ₂ H ₃ Cl ₃	-24.0	-2.0	+8.3	21.6	35.2	44.0	55.7	73.3	93.0	113.9	-36.7
Trichloroethylene	C ₂ HCl ₃	-43.8	-22.8	-12.4	-1.0	+11.9	20.0	31.4	48.0	67.0	86.7	-73
Trichlorofluoromethane	CCl ₃ F	-84.3	-67.6	-59.0	-49.7	-39.0	-32.3	-23.0	-9.1	+6.8	23.7	
2,4,5-Trichlorophenol	C ₆ H ₃ Cl ₃ O	72.0	102.1	117.3	134.0	151.5	162.5	178.0	201.5	226.5	251.8	62
2,4,6-Trichlorophenol	C ₆ H ₃ Cl ₃ O	76.5	105.9	120.2	135.8	152.2	163.5	177.8	199.0	222.5	246.0	68.5

TABLE 2-10 Vapor Pressures of Organic Compounds, up to 1 atm (Concluded)

Compound		Pressure, mmHg										Melting point, °C
		1	5	10	20	40	60	100	200	400	760	
Name	Formula	Temperature, °C										
Tri-2-chlorophenylthiophosphate	C ₁₈ H ₁₂ Cl ₃ O ₃ PS	188.2	217.2	231.2	246.7	261.7	271.5	283.8	302.8	322.0	341.3	
1,1,1-Trichloropropane	C ₃ H ₅ Cl ₃	-28.8	-7.0	+4.2	16.2	29.9	38.3	50.0	67.7	87.5	108.2	-77.7
1,2,3-Trichloropropane	C ₃ H ₅ Cl ₃	+9.0	33.7	46.0	59.3	74.0	83.6	96.1	115.6	137.0	158.0	-14.7
1,1,2-Trichloro-1,2,2-trifluoroethane	C ₂ Cl ₃ F ₃	-68.0	-49.4	-40.3	-30.0	-18.5	-11.2	-1.7	+13.5	30.2	47.6	-35
Tricosane	C ₂₃ H ₄₈	170.0	206.3	223.0	242.0	261.3	273.8	289.8	313.5	339.8	366.5	47.7
Tridecane	C ₁₃ H ₂₈	59.4	98.3	104.0	120.2	137.7	148.2	162.5	185.0	209.4	234.0	-6.2
Tridecanoic acid	C ₁₃ H ₂₆ O ₂	137.8	166.3	181.0	195.8	212.4	222.0	236.0	255.2	276.5	299.0	41
Triethoxymethylsilane	C ₆ H ₁₅ O ₃ Si	-1.5	+22.8	34.6	47.2	61.7	70.4	82.7	101.0	121.8	143.5	
Triethoxyphenylsilane	C ₁₂ H ₂₀ O ₃ Si	71.0	98.8	112.6	127.2	143.5	153.2	167.5	188.0	210.5	233.5	
1,2,4-Triethylbenzene	C ₁₂ H ₁₈	46.0	74.2	88.5	104.0	121.7	132.2	146.8	168.3	193.7	218.0	
1,3,4-Triethylbenzene	C ₁₂ H ₁₈	47.9	76.0	90.2	105.8	122.6	133.4	147.7	168.3	193.2	217.5	
Triethylborine	C ₆ H ₁₅ B			-148.0	-140.6	-131.4	-125.2	-116.0	-101.0	-81.0	-56.2	
Triethyl camphoronate citrate	C ₁₅ H ₂₆ O ₆		150.2	166.0	183.6	201.8	213.5	228.6	250.8	276.0	301.0	135
Triethyleneglycol	C ₆ H ₁₄ O ₄	107.0	138.7	144.0	171.1	190.4	202.5	217.8	242.2	267.5	294.0	
Triethylheptylsilane	C ₁₃ H ₃₀ Si	114.0	144.0	158.1	174.0	191.3	201.5	214.6	235.2	256.6	278.3	
Triethyloctylsilane	C ₁₄ H ₃₀ Si	70.0	99.8	114.6	130.3	148.0	158.2	174.0	196.0	221.0	247.0	
Triethyl orthoformate phosphate	C ₇ H ₁₆ O ₃	73.7	104.8	120.6	137.7	155.7	168.0	184.3	208.0	235.0	262.0	
Triethylthallium	C ₆ H ₁₅ OTl	+5.5	29.2	40.5	53.4	67.5	76.0	88.0	106.0	125.7	146.0	
Trifluorophenylsilane	C ₆ H ₅ TF ₃ Si	39.6	67.8	82.1	97.8	115.7	126.3	141.6	163.7	187.0	211.0	
Trimethyl phosphite	C ₃ H ₉ P	+9.3	37.6	51.7	67.7	85.4	95.7	112.1	136.0	163.5	192.1	-63.0
2,3,5-Trimethylacetophenone	C ₁₂ H ₂₀ O	-31.0	-9.7	+0.8	12.3	25.4	33.2	44.2	60.1	78.7	98.3	
Trimethylamine	C ₃ H ₉ N	93.7	131.0	149.8	169.8	192.0	207.0	225.7	255.0	288.5	324.0	
2,4,5-Trimethylaniline	C ₁₁ H ₁₄ N	79.0	108.0	122.3	137.5	154.2	165.7	179.7	201.3	224.3	247.5	
1,2,3-Trimethylbenzene	C ₉ H ₁₂	-97.1	-81.7	-73.8	-65.0	-55.2	-48.8	-40.3	-27.0	-12.5	+2.9	-117.1
1,2,4-Trimethylbenzene	C ₉ H ₁₂	68.4	95.9	109.0	123.7	139.8	149.5	162.0	182.3	203.7	234.5	67
1,3,5-Trimethylbenzene	C ₉ H ₁₂	16.8	42.9	55.9	69.9	85.4	95.3	108.8	129.0	152.0	176.1	-25.5
2,2,3-Trimethylbutane	C ₈ H ₁₈	13.6	38.3	50.7	64.5	79.8	89.5	102.8	122.7	145.4	169.2	-44.1
Trimethyl citrate	C ₉ H ₁₆ O ₇	9.6	34.7	47.4	61.0	76.1	85.8	98.9	118.6	141.0	164.7	-44.8
Trimethyleneglycol (1,3-propanediol)	C ₃ H ₈ O ₂		-18.8	-7.5	+5.2	13.3	24.4	41.2	60.4	80.9	102.0	-25.0
1,2,4-Trimethyl-5-ethylbenzene	C ₁₁ H ₁₆	106.2	146.2	160.4	177.2	194.2	205.5	219.6	241.3	264.2	287.0	78.5
1,3,5-Trimethyl-2-ethylbenzene	C ₁₁ H ₁₆	59.4	87.2	100.6	115.5	131.0	141.1	153.4	172.8	193.8	214.2	
2,2,3-Trimethylpentane	C ₈ H ₁₈	43.7	71.2	84.6	99.7	106.0	126.3	140.3	160.3	184.5	208.1	
2,2,4-Trimethylpentane	C ₈ H ₁₈	38.8	67.0	80.5	96.0	113.2	123.8	137.9	158.4	183.5	208.0	
2,3,3-Trimethylpentane	C ₈ H ₁₈	-29.0	-7.1	+3.9	16.0	29.5	38.1	49.9	67.8	88.2	109.8	-112.3
2,3,4-Trimethylpentane	C ₈ H ₁₈	-36.5	-15.0	-4.3	+7.5	20.7	29.1	40.7	58.1	78.0	99.2	-107.3
2,3,4-Trimethylpentane	C ₈ H ₁₈	-25.8	-3.9	+6.9	19.2	33.0	41.8	53.8	72.0	92.7	114.8	-101.5
2,2,4-Trimethyl-3-pentanone	C ₈ H ₁₆ O	-26.3	-4.1	+7.1	19.3	32.9	41.6	53.4	71.3	91.8	113.5	-109.2
Trimethyl phosphate	C ₃ H ₉ O ₄ P	14.7	36.0	46.4	57.6	69.8	77.3	87.6	102.2	118.4	135.0	
2,4,5-Trimethylstyrene	C ₁₁ H ₁₄	26.0	53.7	67.8	83.0	100.0	110.0	124.0	145.0	167.8	192.7	
2,4,6-Trimethylstyrene	C ₁₁ H ₁₄	48.1	77.0	91.6	107.1	124.2	135.5	149.8	171.8	196.1	221.2	
Trimethylsuccinic anhydride	C ₇ H ₁₀ O ₃	37.5	65.7	79.7	94.8	111.8	122.3	136.8	157.8	182.3	207.0	
Triphenylmethane	C ₁₉ H ₁₆	53.5	82.6	97.4	113.8	131.0	142.2	156.5	179.8	205.5	231.0	
Triphenylphosphate	C ₁₈ H ₁₅ O ₄ P	169.7	188.4	197.0	206.8	215.5	221.2	228.4	239.7	249.8	259.2	93.4
Tripropyleneglycol	C ₉ H ₂₀ O ₄	193.5	230.4	249.8	269.7	290.3	305.2	322.5	349.8	379.2	413.5	49.4
Tripropyleneglycol monobutyl ether	C ₁₃ H ₂₈ O ₄	96.0	125.7	140.5	155.8	173.7	184.6	199.0	220.2	244.3	267.2	
Tripropyleneglycol monoisopropyl ether	C ₁₃ H ₂₆ O ₄	101.5	131.6	147.0	161.8	179.8	190.2	204.4	224.4	247.0	269.5	
Tritolyl phosphate	C ₂₁ H ₂₄ O ₄ P	82.4	112.4	127.3	143.7	161.4	173.2	187.8	209.7	232.8	256.6	
Undecane	C ₁₁ H ₂₄	154.6	184.2	198.0	213.2	229.7	239.8	252.2	271.8	292.7	313.0	
Undecanoic acid	C ₁₁ H ₂₂ O ₂	32.7	59.7	73.9	85.6	104.4	115.2	128.1	149.3	171.9	195.8	-25.6
10-Undecenoic acid	C ₁₁ H ₂₀ O ₂	101.4	133.1	149.0	166.0	185.6	197.2	212.5	237.8	262.8	290.0	29.5
Undecan-2-ol	C ₁₁ H ₂₂ O	114.0	142.8	156.3	172.0	188.7	199.5	213.5	232.8	254.0	275.0	24.5
n-Valeric acid	C ₅ H ₁₀ O ₂	71.1	99.0	112.8	127.5	143.7	153.7	167.2	187.7	209.8	232.0	
iso-Valeric acid	C ₅ H ₁₀ O ₂	42.2	67.7	79.8	93.1	107.8	116.6	128.3	146.0	165.0	184.4	-34.5
γ-Valerolactone	C ₅ H ₈ O ₂	34.5	59.6	71.3	84.0	98.0	107.3	118.9	136.2	155.2	175.1	-37.6
Valeronitrile	C ₅ H ₉ N	37.5	65.8	79.8	95.2	101.9	122.4	136.5	157.7	182.3	207.5	
Vanillin	C ₈ H ₈ O ₃	-6.0	+18.1	30.0	43.3	57.8	66.9	78.6	97.7	118.7	140.8	
Vinyl acetate	C ₄ H ₆ O ₂	107.0	138.4	154.0	170.5	188.7	199.8	214.5	237.3	260.0	285.0	81.5
2-Vinylanisole	C ₉ H ₁₀ O	-48.0	-28.0	-18.0	-7.0	+5.3	13.0	23.3	38.4	55.5	72.5	
3-Vinylanisole	C ₉ H ₁₀ O	41.9	68.0	81.0	94.7	110.0	119.8	132.3	151.0	172.1	194.0	
4-Vinylanisole	C ₉ H ₁₀ O	43.4	69.9	83.0	97.2	112.5	122.3	135.3	154.0	175.8	197.5	
Vinyl chloride (1-chloroethylene)	C ₂ H ₃ Cl	45.2	72.0	85.7	100.0	116.0	126.1	139.7	159.0	182.0	204.5	
cyamide (acrylonitrile)	C ₃ H _{3.5} N	-105.6	-90.8	-83.7	-75.7	-66.8	-61.1	-53.2	-41.3	-28.0	-13.8	-153.7
fluoride (1-fluoroethylene)	C ₂ H ₃ F	-51.0	-30.7	-20.3	-9.0	+3.8	11.8	22.8	38.7	58.3	78.5	-82
Vinylidene chloride (1,1-dichloroethene)	C ₂ H ₂ Cl ₂	-149.3	-138.0	-132.2	-125.4	-118.0	-113.0	-106.2	-95.4	-84.0	-72.2	-160.5
4-Vinylphenetole	C ₁₀ H ₁₂ O	-77.2	-60.0	-51.2	-41.7	-31.1	-24.0	-15.0	-1.0	+14.8	31.7	-122.5
2-Xenyl dichlorophosphate	C ₁₂ H ₉ Cl ₂ PO	64.0	91.7	105.6	120.3	136.3	146.4	159.8	180.0	202.8	225.0	
2,4-Xyldaldehyde	C ₈ H ₁₀ O	138.2	171.1	187.0	205.0	223.8	236.0	251.5	275.3	301.5	328.5	
2-Xylene (2-xylene)	C ₈ H ₁₀	59.0	85.9	99.0	114.0	129.7	139.8	152.2	172.3	194.1	215.5	75
3-Xylene (3-xylene)	C ₈ H ₁₀	-3.8	+20.2	32.1	45.1	59.5	68.8	81.3	100.2	121.7	144.4	-25.2
4-Xylene (4-xylene)	C ₈ H ₁₀	-6.9	+16.8	28.3	41.1	55.3	64.4	76.8	95.5	116.7	139.1	-47.9
2,4-Xyldine	C ₈ H ₁₁ N	-8.1	+15.5	27.3	40.1	54.4	63.5	75.9	94.6	115.9	138.3	+13.3
2,6-Xyldine	C ₈ H ₁₁ N	52.6	79.8	93.0	107.6	123.8	133.7	146.8	166.4	188.3	211.5	
		44.0	72.6	87.0	102.7	120.2	131.5	146.0	168.0	193.7	217.9	

VAPOR PRESSURES OF SOLUTIONS

UNITS CONVERSIONS

For this subsection, the following units conversions are applicable:

$$^{\circ}\text{F} = \%^{\circ}\text{C} + 32$$

To convert millimeters of mercury to pounds-force per square inch, multiply by 0.01934.

To convert cubic feet to cubic meters, multiply by 0.02832.

To convert bars to pounds-force per square inch, multiply by 14.504.

To convert bars to kilopascals, multiply by 1×10^5 .

TABLE 2-11 Partial Pressures of Water over Aqueous Solutions of HCl*

$\log_{10} \text{ pmm} = A - B/T$, (T in K), which, however, agrees only approximately with the table. The table is more nearly correct.

Partial pressure of H_2O , mmHg, $^{\circ}\text{C}$

% HCl	A	B	0°	5°	10°	15°	20°	25°	30°	35°	40°	45°	50°	60°	70°	80°	90°	100°	110°
6	8.99156	2282	4.18	6.04	8.45	11.7	15.9	21.8	29.1	39.4	50.6	66.2	86.0	139	220	333	492	715	
10	8.99864	2295	3.84	5.52	7.70	10.7	14.6	20.0	26.8	35.5	47.0	61.5	80.0	130	204	310	463	677	960
14	8.97075	2300	3.39	4.91	6.95	9.65	13.1	18.0	24.1	31.9	42.1	55.3	72.0	116	185	273	425	625	892
18	8.98014	2323	2.87	4.21	5.92	8.26	11.3	15.4	20.6	27.5	36.4	47.9	62.5	102	162	248	374	550	783
20	8.97877	2334	2.62	3.83	5.40	7.50	10.3	14.1	19.0	25.1	33.3	43.6	57.0	93.5	150	230	345	510	729
22	9.02708	2363	2.33	3.40	4.82	6.75	9.30	12.6	17.1	22.8	30.2	39.8	52.0	85.6	138	211	317	467	670
24	8.96022	2356	2.05	3.04	4.31	6.03	8.30	11.4	15.4	20.4	27.1	35.7	46.7	77.0	124	194	290	426	611
26	9.01511	2390	1.76	2.60	3.71	5.21	7.21	9.95	13.5	18.0	24.0	31.7	41.5	69.0	112	173	261	387	555
28	8.97611	2395	1.50	2.24	3.21	4.54	6.32	8.75	11.8	15.8	21.1	27.9	36.5	60.7	99.0	154	234	349	499
30	9.00117	2422	1.26	1.90	2.73	3.88	5.41	7.52	10.2	13.7	18.4	24.3	32.0	53.5	87.5	136	207	310	444
32	9.03317	2453	1.04	1.57	2.27	3.25	4.55	6.37	8.70	11.7	15.7	21.0	27.7	46.5	76.5	120	184	275	396
34	9.07143	2487	0.85	1.29	1.87	2.70	3.81	5.35	7.32	9.95	13.5	18.1	24.0	40.5	66.5	104	161	243	355
36	9.11815	2526	0.68	1.03	1.50	2.19	3.10	4.41	6.08	8.33	11.4	15.4	20.4	34.8	57.0	90.0	140	212	311
38	9.20783	2579	0.53	0.81	1.20	1.75	2.51	3.60	5.03	6.92	9.52	13.0	17.4	29.6	49.1	77.5	120	182	266
40	9.33923	2647	0.41	0.63	0.94	1.37	2.00	2.88	4.09	5.68	7.85	10.7	14.5	25.0	42.1	67.3	105	158	230
42	9.44953	2709	0.31	0.48	0.72	1.06	1.56	2.30	3.28	4.60	6.45	8.90	12.1	21.2	35.8	57.2	89.2	135	195

*Uncertainty, ca. 2 percent for solutions of 15 to 30 percent HCl between 0 and 100°; for solutions of > 30 percent HCl the accuracy is ca. 5 percent at the lower temperatures and ca. 15 percent at the higher temperatures. Below 15 percent HCl, the uncertainty is ca. 5 percent at the lower temperatures and higher strengths to ca. 15 to 20 percent at the lower strengths and perhaps 15 to 20 percent at the higher temperatures and lower strengths.

International Critical Tables, vol. 3, p. 301.

TABLE 2-12 Partial Pressures of HCl over Aqueous Solutions of HCl*

$\log_{10} \text{ pmm} = A - B/T$, (T in K), which, however, agrees only approximately with the table. The table is more nearly correct. mmHg, $^{\circ}\text{C}$

% HCl	A	B	0°	5°	10°	15°	20°	25°	30°	35°	40°	45°	50°	60°	70°	80°	90°	100°	110°
2	11.8037	4736			0.0000117	0.000023	0.000044	0.000084	0.000151	0.000275	0.00047	0.00083	0.00140	0.00380	0.0100	0.0245	0.058	0.132	0.280
4	11.6400	4471	0.000018	0.000036	0.000069	0.000131	0.00024	0.00044	0.00077	0.00134	0.0023	0.00385	0.0064	0.0165	0.0405	0.095	.21	.46	.93
6	11.2144	4202	.000066	.000125	.000234	.000425	.00076	.00131	.00225	.0038	.0062	.0102	.0163	.040	.094	.206	.44	.92	1.78
8	11.0406	4042	.000118	.000323	.000583	.00104	.00178	.0031	.00515	.0085	.0136	.022	.0344	.081	.183	.39	.82	1.64	3.10
10	10.9311	3908	.00042	.00075	.00134	.00232	.00395	.0067	.0111	.0178	.0282	.045	.069	.157	.35	.73	1.48	2.9	5.4
12	10.7900	3765	.00099	.00175	.00305	.0052	.0088	.0145	.0234	.037	.058	.091	.136	.305	.66	1.34	2.65	5.1	9.3
14	10.6954	3636	.0024	.00415	.0071	.0118	.0196	.0316	.050	.078	.121	.185	.275	.60	1.25	2.50	4.8	9.0	16.0
16	10.6261	3516	.0056	.0095	.016	.0265	.0428	.0685	.106	.163	.247	.375	.55	1.17	2.40	4.66	8.8	16.1	28
18	10.4957	3376	.0135	.0225	.037	.060	.095	.148	.228	.345	.515	.77	1.11	2.3	4.55	8.6	15.7	28	48
20	10.3853	3245	.0316	.052	.084	.132	.205	.32	.48	.72	1.06	1.55	2.21	4.4	8.5	15.6	28.1	49	83
22	10.3172	3125	.0734	.119	.187	.294	.45	.68	1.02	1.50	2.18	3.14	4.42	8.6	16.3	29.3	52	90	146
24	10.2185	2995	.175	.277	.43	.66	1.00	1.49	2.17	3.14	4.5	6.4	8.9	16.9	31.0	54.5	94	157	253
26	10.1303	2870	.41	.64	.98	1.47	2.17	3.20	4.56	6.50	9.2	12.7	17.5	32.5	58.5	100	169	276	436
28	10.0115	2732	1.0	1.52	2.27	3.36	4.90	7.05	9.90	13.8	19.1	26.4	35.7	64	112	188	309	493	760
30	9.8763	2593	2.4	3.57	5.23	7.60	10.6	15.1	21.0	28.6	39.4	53	71	124	208	340	542	845	
32	9.7523	2457	5.7	8.3	11.8	16.8	23.5	32.5	44.5	60.0	81	107	141	238	390	623	970		
34	9.6061	2316	13.1	18.8	26.4	36.8	50.5	68.5	92	122	161	211	273	450	720				
36	9.5262	2229	29.0	41.0	56.4	78	105.5	142	188	246	322	416	535	860					
38	9.4670	2094	63.0	87.0	117	158	210	277	360	465	598	758	955						
40	9.2156	1939	130	176	233	307	399	515	627	830									
42	8.9925	1800	253	332	430	560	709	900											
44	8.8621	1681	510	655	840														
46			940																

*Uncertainty, ca. 2 percent for solutions of 15 to 30 percent HCl between 0 and 100°; for solutions of > 30 percent HCl the accuracy is ca. 5 percent at the lower temperatures and ca. 15 percent at the higher temperatures. Below 15 percent HCl, the uncertainty is ca. 5 percent at the lower temperatures and higher strengths to ca. 15 to 20 percent at the lower strengths and perhaps 15 to 20 percent at the higher temperatures and lower strengths.

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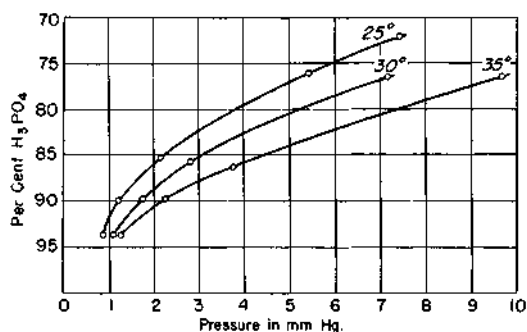


FIG. 2-1 Vapor pressures of H_3PO_4 aqueous: partial pressure of H_2O vapor. (Courtesy of Victor Chemical Works, Stauffer Chemical Company; measurements by W. H. Woodstock.)

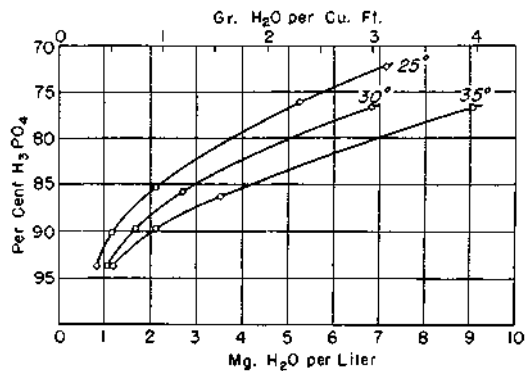


FIG. 2-2 Vapor pressures of H_3PO_4 aqueous: weight of H_2O in saturated air. (Courtesy of Victor Chemical Works, Stauffer Chemical Company; measurements by W. H. Woodstock.)

TABLE 2-13 Partial Pressures of H_2O and SO_2 over Aqueous Solutions of Sulfur Dioxide*

Partial pressures of H_2O and SO_2 , mmHg, °C

g SO_2 / 100 g H_2O	Temperature, °C								
	0	10	20	30	40	50	60	90	120
0.01	0.02	0.04	0.07	0.12	0.19	0.29	0.43	1.21	2.82
0.05	0.38	0.66	1.07	1.68	2.53	3.69	5.24	12.9	27.0
0.10	1.15	1.91	3.03	4.62	6.80	9.71	13.5	31.7	63.9
0.15	2.10	3.44	5.37	8.07	11.7	16.5	22.7	52.2	104
0.20	3.17	5.13	7.93	11.8	17.0	23.8	32.6	73.7	145
0.25	4.34	6.93	10.6	15.7	22.5	31.4	42.8	95.8	186
0.30	5.57	8.84	13.5	19.8	28.2	39.2	53.3	118	229
0.40	8.17	12.8	19.4	28.3	40.1	55.3	74.7	164	316
0.50	10.9	17.0	25.6	37.1	52.3	72.0	96.8	211	404
1.00	25.8	39.5	58.4	83.7	117	159	212	454	856
2.00	58.6	88.5	129	183	253	342	453	955	
3.00	93.2	139	202	285	393	530	700		
4.00	129	192	277	389	535	720			
5.00	165	245	353	496	679				
6.00	202	299	430	602	824				
8.00	275	407	585	818					
10.00	351	517	741						
15.00	542	796							
20.00	735								

*Extracted with permission from *J. Chem Eng. Data* 8, 1963: 333-336. Copyright 1963 American Chemical Society.

TABLE 2-14 Water Partial Pressure, bar, over Aqueous Sulfuric Acid Solutions*
Weight percent, H₂SO₄

°C	10.0	20.0	30.0	40.0	50.0	60.0	70.0	75.0	80.0	85.0
0	.582E-02	.534E-02	.448E-02	.326E-02	.193E-02	.836E-03	.207E-03	.747E-04	.197E-04	.343E-05
10	.117E-01	.107E-01	.909E-02	.670E-02	.405E-02	.180E-02	.467E-03	.175E-03	.490E-04	.952E-05
20	.223E-01	.205E-01	.174E-01	.130E-01	.802E-02	.367E-02	.995E-03	.388E-03	.115E-04	.245E-04
30	.404E-01	.373E-01	.319E-01	.241E-01	.151E-01	.710E-02	.201E-02	.811E-03	.253E-03	.589E-04
40	.703E-01	.649E-01	.558E-01	.427E-01	.272E-01	.131E-01	.387E-02	.162E-02	.531E-03	.133E-03
50	.117	.109	.939E-01	.725E-01	.470E-01	.232E-01	.715E-02	.309E-02	.106E-02	.286E-03
60	.189	.175	.152	.119	.782E-01	.395E-01	.127E-01	.565E-02	.204E-02	.584E-03
70	.296	.275	.239	.188	.126	.651E-01	.217E-01	.997E-02	.376E-02	.114E-02
80	.449	.417	.365	.290	.196	.104	.360E-01	.170E-01	.668E-02	.213E-02
90	.664	.617	.542	.434	.298	.161	.578E-01	.281E-01	.115E-01	.383E-02
100	.957	.891	.786	.634	.441	.244	.905E-01	.452E-01	.192E-01	.666E-02
110	1.349	1.258	1.113	.904	.638	.360	.138	.708E-01	.312E-01	.112E-01
120	1.863	1.740	1.544	1.264	.903	.519	.206	.108	.493E-01	.183E-01
130	2.524	2.361	2.101	1.732	1.253	.734	.301	.162	.760E-01	.291E-01
140	3.361	3.149	2.810	2.333	1.708	1.020	.481	.236	.115	.451E-01
150	4.404	4.132	3.697	3.090	2.289	1.392	.605	.339	.170	.682E-01
160	5.685	5.342	4.793	4.031	3.021	1.870	.837	.478	.246	.101
170	7.236	6.810	6.127	5.185	3.930	2.475	1.138	.662	.350	.147
180	9.093	8.571	7.731	6.584	5.045	3.233	1.525	.902	.489	.208
190	11.289	10.658	9.640	8.259	6.397	4.169	2.017	1.212	.673	.291
200	13.861	13.107	11.887	10.245	8.020	5.312	2.632	1.606	.913	.401
210	16.841	15.951	14.505	12.576	9.948	6.696	3.395	2.101	1.220	.542
220	20.264	19.225	17.529	15.287	12.217	8.354	4.331	2.714	1.609	.724
230	24.160	22.960	20.992	18.414	14.864	10.322	5.466	3.467	2.096	.952
240	28.561	27.188	24.927	21.992	17.929	12.641	6.831	4.381	2.699	1.237
250	33.494	31.939	29.364	26.056	21.452	15.351	8.458	5.480	3.435	1.587
260	38.984	37.240	34.334	30.642	25.472	18.496	10.382	6.788	4.326	2.012
270	45.055	43.116	39.865	35.784	30.030	22.121	12.640	8.333	5.395	2.525
280	51.726	49.590	45.984	41.514	35.168	26.274	15.269	10.142	6.663	3.136
290	59.015	56.681	52.715	47.865	40.926	31.003	18.311	12.242	8.155	3.857
300	66.934	64.407	60.081	54.868	47.346	36.360	21.808	14.665	9.897	4.701
310	75.495	72.781	68.100	62.553	54.470	42.395	25.804	17.438	11.912	5.680
320	84.705	81.816	76.792	70.947	62.337	49.164	30.343	20.591	14.227	6.806
330	94.567	91.518	86.172	80.077	70.988	56.721	35.473	24.153	16.867	8.093
340	105.083	101.894	96.252	89.969	80.463	65.123	41.240	28.154	19.855	9.551
350	116.251	112.946	107.043	100.646	90.802	74.426	47.692	32.622	23.217	11.193

*Vermeulen, Dong, Robinson, Nguyen, and Gmitro, AIChE meeting, Anaheim, Calif., 1982; and private communication from Prof. Theodore Vermeulen, Chemical Engineering Dept., University of California, Berkeley.

TABLE 2-14 Water Partial Pressure, bar, over Aqueous Sulfuric Acid Solutions (Concluded)
Weight percent, H₂SO₄

°C	90.0	92.0	94.0	96.0	97.0	98.0	98.5	99.0	99.5	100.0
0	.518E-06	.242E-06	.107E-06	.401E-07	.218E-07	.980E-08	.569E-08	.268E-08	.775E-09	.196E-09
10	.159E-05	.762E-06	.344E-06	.130E-06	.713E-07	.323E-07	.188E-07	.888E-08	.258E-08	.655E-09
20	.448E-05	.220E-05	.101E-05	.390E-06	.215E-06	.978E-07	.572E-07	.271E-07	.789E-08	.201E-08
30	.117E-04	.587E-05	.275E-05	.108E-05	.598E-06	.275E-06	.161E-06	.766E-07	.224E-07	.575E-08
40	.285E-04	.146E-04	.696E-05	.278E-05	.155E-05	.720E-06	.424E-06	.202E-06	.595E-07	.153E-07
50	.652E-04	.341E-04	.166E-04	.672E-05	.379E-05	.177E-05	.105E-05	.503E-06	.149E-06	.384E-07
60	.141E-03	.754E-04	.372E-04	.154E-04	.875E-05	.413E-05	.245E-05	.118E-05	.350E-06	.910E-07
70	.290E-03	.158E-03	.795E-04	.334E-04	.192E-04	.912E-05	.544E-05	.263E-05	.784E-06	.205E-06
80	.569E-03	.316E-03	.162E-03	.691E-04	.400E-04	.192E-04	.115E-04	.559E-05	.168E-05	.439E-06
90	.107E-02	.606E-03	.315E-03	.137E-03	.801E-04	.388E-04	.234E-04	.114E-04	.343E-05	.903E-06
100	.194E-02	.112E-02	.590E-03	.261E-03	.154E-03	.752E-04	.455E-04	.223E-04	.674E-05	.178E-05
110	.338E-02	.198E-02	.107E-02	.479E-03	.285E-03	.141E-03	.855E-04	.420E-04	.128E-04	.339E-05
120	.571E-02	.341E-02	.186E-02	.851E-03	.511E-03	.254E-03	.155E-03	.766E-04	.233E-04	.623E-05
130	.938E-02	.569E-02	.315E-02	.146E-02	.886E-03	.445E-03	.278E-03	.135E-03	.414E-04	.111E-04
140	.150E-01	.923E-02	.519E-02	.245E-02	.149E-02	.757E-03	.467E-03	.232E-03	.711E-04	.191E-04
150	.233E-01	.146E-01	.832E-02	.399E-02	.245E-02	.125E-02	.776E-03	.387E-03	.119E-03	.321E-04
160	.354E-01	.225E-01	.130E-01	.633E-02	.393E-02	.202E-02	.126E-02	.629E-03	.194E-03	.526E-04
170	.526E-01	.340E-01	.199E-01	.983E-02	.614E-02	.319E-02	.199E-02	.999E-03	.309E-03	.840E-04
180	.766E-01	.502E-01	.298E-01	.149E-01	.941E-02	.492E-02	.309E-02	.155E-02	.482E-03	.131E-03
190	.110	.729E-01	.438E-01	.222E-01	.141E-01	.744E-02	.469E-02	.236E-02	.735E-03	.201E-03
200	.154	.104	.631E-01	.325E-01	.208E-01	.110E-01	.698E-02	.352E-02	.110E-02	.300E-03
210	.213	.146	.894E-01	.467E-01	.300E-01	.161E-01	.102E-01	.516E-02	.161E-02	.442E-03
220	.290	.201	.125	.660E-01	.427E-01	.230E-01	.147E-01	.743E-02	.232E-02	.638E-03
230	.389	.273	.171	.918E-01	.598E-01	.325E-01	.208E-01	.105E-01	.329E-02	.906E-03
240	.514	.366	.232	.126	.825E-01	.451E-01	.290E-01	.147E-01	.460E-02	.127E-02
250	.673	.485	.310	.170	.112	.618E-01	.398E-01	.202E-01	.633E-02	.174E-02
260	.870	.635	.409	.227	.151	.835E-01	.540E-01	.274E-01	.858E-02	.237E-02
270	1.112	.822	.534	.300	.200	.111	.723E-01	.366E-01	.115E-01	.317E-02
280	1.407	1.052	.689	.391	.263	.147	.957E-01	.485E-01	.152E-01	.420E-02
290	1.763	1.335	.880	.505	.341	.192	.125	.634E-01	.199E-01	.548E-02
300	2.190	1.676	1.112	.646	.437	.248	.162	.820E-01	.257E-01	.708E-02
310	2.696	2.088	1.394	.817	.556	.316	.208	.105	.328E-01	.905E-02
320	3.292	2.578	1.732	1.025	.701	.400	.264	.133	.415E-01	.114E-01
330	3.990	3.159	2.133	1.274	.875	.502	.331	.167	.520E-01	.143E-01
340	4.801	3.843	2.608	1.571	1.083	.624	.413	.208	.646E-01	.178E-01
350	5.738	4.641	3.164	1.922	1.331	.770	.511	.256	.795E-01	.218E-01

TABLE 2-15 Sulfur Trioxide Partial Pressure, bar, over Aqueous Sulfuric Acid Solutions*
Weight percent, H₂SO₄

°C	10.0	20.0	30.0	40.0	50.0	60.0	70.0	75.0	80.0	85.0
0	.644E-29	.103E-27	.205E-26	.688E-25	.368E-23	.341E-21	.784E-19	.174E-17	.531E-16	.229E-14
10	.149E-27	.223E-26	.395E-25	.113E-23	.522E-22	.415E-20	.796E-18	.158E-16	.417E-15	.141E-13
20	.278E-26	.394E-25	.626E-24	.156E-22	.621E-21	.426E-19	.685E-17	.121E-15	.280E-14	.767E-13
30	.426E-25	.577E-24	.832E-23	.181E-21	.630E-20	.376E-18	.509E-16	.808E-15	.164E-13	.371E-12
40	.549E-24	.714E-23	.941E-22	.181E-20	.555E-19	.288E-17	.331E-15	.473E-14	.851E-13	.162E-11
50	.602E-23	.757E-22	.921E-21	.158E-19	.429E-18	.195E-16	.191E-14	.246E-13	.395E-12	.643E-11
60	.573E-22	.699E-21	.789E-20	.122E-18	.294E-17	.118E-15	.985E-14	.116E-12	.165E-11	.234E-10
70	.477E-21	.567E-20	.599E-19	.843E-18	.181E-16	.643E-15	.461E-13	.492E-12	.634E-11	.791E-10
80	.352E-20	.410E-19	.408E-18	.524E-17	.101E-15	.319E-14	.197E-12	.192E-11	.223E-10	.249E-09
90	.233E-19	.266E-18	.250E-17	.296E-16	.516E-15	.145E-13	.775E-12	.693E-11	.731E-10	.734E-09
100	.139E-18	.157E-17	.140E-16	.153E-15	.242E-14	.606E-13	.283E-11	.232E-10	.223E-09	.204E-08
110	.756E-18	.844E-17	.719E-16	.730E-15	.105E-13	.236E-12	.961E-11	.729E-10	.641E-09	.538E-08
120	.377E-17	.418E-16	.340E-15	.323E-14	.424E-13	.858E-12	.307E-10	.215E-09	.174E-08	.135E-07
130	.174E-16	.191E-15	.150E-14	.133E-13	.160E-12	.293E-11	.922E-10	.601E-09	.446E-08	.324E-07
140	.743E-16	.815E-15	.615E-14	.517E-13	.569E-12	.943E-11	.262E-09	.159E-08	.109E-07	.745E-07
150	.297E-15	.325E-14	.237E-13	.188E-12	.191E-11	.287E-10	.710E-09	.403E-08	.256E-07	.165E-06
160	.111E-14	.122E-13	.862E-13	.649E-12	.608E-11	.833E-10	.183E-08	.974E-08	.575E-07	.351E-06
170	.393E-14	.430E-13	.296E-12	.212E-11	.184E-10	.231E-09	.453E-08	.226E-07	.125E-06	.725E-06
180	.131E-13	.144E-12	.967E-12	.622E-11	.532E-10	.610E-09	.107E-07	.505E-07	.260E-06	.145E-05
190	.415E-13	.458E-12	.301E-11	.197E-10	.147E-09	.155E-08	.246E-07	.109E-06	.527E-06	.282E-05
200	.125E-12	.139E-11	.893E-11	.561E-10	.391E-09	.379E-08	.542E-07	.228E-06	.103E-05	.534E-05
210	.362E-12	.404E-11	.254E-10	.154E-09	.100E-08	.894E-08	.116E-06	.462E-06	.198E-05	.986E-05
220	.100E-11	.112E-10	.695E-10	.405E-09	.246E-08	.204E-07	.240E-06	.911E-06	.368E-05	.178E-04
230	.265E-11	.301E-10	.183E-09	.103E-08	.587E-08	.450E-07	.482E-06	.175E-05	.668E-05	.314E-04
240	.678E-11	.777E-10	.465E-09	.253E-08	.135E-07	.965E-07	.944E-06	.328E-05	.119E-04	.543E-04
250	.167E-10	.193E-09	.114E-08	.602E-08	.303E-07	.201E-06	.180E-05	.600E-05	.206E-04	.923E-04
260	.399E-10	.466E-09	.272E-08	.139E-07	.660E-07	.408E-06	.336E-05	.108E-04	.352E-04	.154E-03
270	.920E-10	.109E-08	.628E-08	.312E-07	.140E-06	.807E-06	.612E-05	.189E-04	.590E-04	.253E-03
280	.206E-09	.247E-08	.141E-07	.683E-07	.288E-06	.156E-05	.109E-04	.326E-04	.973E-04	.408E-03
290	.449E-09	.545E-08	.308E-07	.145E-06	.580E-06	.295E-05	.191E-04	.553E-04	.158E-03	.649E-03
300	.953E-09	.117E-07	.657E-07	.302E-06	.114E-05	.546E-05	.329E-04	.921E-04	.253E-03	.102E-02
310	.197E-08	.245E-07	.136E-06	.614E-06	.220E-05	.990E-05	.556E-04	.151E-03	.398E-03	.158E-02
320	.397E-08	.502E-07	.277E-06	.122E-05	.414E-05	.176E-04	.923E-04	.245E-03	.621E-03	.242E-02
330	.782E-08	.100E-06	.551E-06	.237E-05	.766E-05	.308E-04	.151E-03	.391E-03	.956E-03	.367E-02
340	.151E-07	.196E-06	.107E-05	.452E-05	.139E-04	.529E-04	.243E-03	.617E-03	.145E-02	.550E-02
350	.285E-07	.376E-06	.204E-05	.846E-05	.246E-04	.893E-04	.387E-03	.963E-03	.219E-02	.815E-02

*Vermeulen, Dong, Robinson, Nguyen, and Gmitro, AIChE meeting, Anaheim, Calif., 1982; and private communication from Prof. Theodore Vermeulen, Chemical Engineering Dept., University of California, Berkeley.

TABLE 2-15 Sulfur Trioxide Partial Pressure, bar, over Aqueous Sulfuric Acid Solutions (Concluded)
Weight percent, H₂SO₄

°C	90.0	92.0	94.0	96.0	97.0	98.0	98.5	99.0	99.5	100.0
0	.671E-13	.216E-12	.677E-12	.240E-11	.500E-11	.124E-10	.224E-10	.502E-10	.182E-09	.755E-09
10	.345E-12	.107E-11	.326E-11	.114E-10	.234E-10	.578E-10	.104E-09	.232E-09	.839E-09	.347E-08
20	.159E-11	.475E-11	.141E-10	.482E-10	.986E-10	.241E-09	.433E-09	.961E-09	.346E-08	.142E-07
30	.664E-11	.192E-10	.557E-10	.186E-09	.376E-09	.911E-09	.163E-08	.360E-08	.129E-07	.528E-07
40	.254E-10	.709E-10	.201E-09	.655E-09	.131E-08	.315E-08	.562E-08	.123E-07	.440E-07	.179E-06
50	.897E-10	.242E-09	.669E-09	.214E-08	.424E-08	.101E-07	.179E-07	.391E-07	.139E-06	.560E-06
60	.294E-09	.771E-09	.207E-08	.647E-08	.127E-07	.299E-07	.528E-07	.115E-06	.405E-06	.163E-05
70	.904E-09	.230E-08	.602E-08	.184E-07	.357E-07	.833E-07	.146E-06	.316E-06	.111E-05	.444E-05
80	.261E-08	.643E-08	.165E-07	.492E-07	.946E-07	.218E-06	.381E-06	.820E-06	.286E-05	.114E-04
90	.712E-08	.171E-07	.426E-07	.124E-06	.237E-06	.541E-06	.940E-06	.201E-05	.698E-05	.276E-04
100	.184E-07	.430E-07	.105E-06	.300E-06	.565E-06	.127E-05	.220E-05	.470E-05	.162E-04	.638E-04
110	.456E-07	.103E-06	.247E-06	.689E-06	.128E-05	.287E-05	.494E-05	.105E-04	.359E-04	.141E-03
120	.108E-06	.238E-06	.555E-06	.152E-05	.280E-05	.619E-05	.106E-04	.224E-04	.764E-04	.298E-03
130	.244E-06	.526E-06	.120E-05	.321E-05	.586E-05	.128E-04	.219E-04	.459E-04	.156E-03	.606E-03
140	.533E-06	.112E-05	.250E-05	.656E-05	.118E-04	.257E-04	.435E-04	.910E-04	.308E-03	.119E-02
150	.112E-05	.230E-05	.504E-05	.129E-04	.231E-04	.497E-04	.837E-04	.174E-03	.588E-03	.226E-02
160	.229E-05	.459E-05	.983E-05	.247E-04	.438E-04	.932E-04	.156E-03	.324E-03	.109E-02	.416E-02
170	.453E-05	.886E-05	.186E-04	.459E-04	.806E-04	.170E-03	.283E-03	.586E-03	.196E-02	.746E-02
180	.870E-05	.166E-04	.343E-04	.829E-04	.144E-03	.301E-03	.499E-03	.103E-02	.343E-02	.130E-01
190	.163E-04	.304E-04	.615E-04	.146E-03	.252E-03	.520E-03	.859E-03	.177E-02	.587E-02	.222E-01
200	.297E-04	.543E-04	.108E-03	.251E-03	.429E-03	.878E-03	.144E-02	.296E-02	.981E-02	.370E-01
210	.528E-04	.946E-04	.185E-03	.422E-03	.714E-03	.145E-02	.237E-02	.486E-02	.161E-01	.603E-01
220	.919E-04	.161E-03	.309E-03	.694E-03	.117E-02	.235E-02	.383E-02	.781E-02	.258E-01	.965E-01
230	.157E-03	.269E-03	.508E-03	.112E-02	.187E-02	.373E-02	.605E-02	.123E-01	.405E-01	.152
240	.261E-03	.441E-03	.819E-03	.178E-02	.293E-02	.582E-02	.939E-02	.191E-01	.627E-01	.234
250	.428E-03	.708E-03	.130E-02	.276E-02	.453E-02	.891E-02	.143E-01	.291E-01	.955E-01	.356
260	.690E-03	.112E-02	.202E-02	.423E-02	.688E-02	.134E-01	.215E-01	.437E-01	.143	.532
270	.109E-02	.174E-02	.309E-02	.638E-02	.103E-01	.200E-01	.319E-01	.646E-01	.212	.786
280	.170E-02	.266E-02	.466E-02	.948E-02	.152E-01	.293E-01	.465E-01	.943E-01	.309	1.144
290	.261E-02	.401E-02	.694E-02	.139E-01	.221E-01	.423E-01	.670E-01	.136	.444	1.646
300	.395E-02	.595E-02	.102E-01	.201E-01	.318E-01	.604E-01	.953E-01	.193	.632	2.339
310	.589E-02	.873E-02	.148E-01	.287E-01	.451E-01	.852E-01	.134	.272	.889	3.289
320	.868E-02	.126E-01	.211E-01	.405E-01	.632E-01	.119	.186	.378	1.236	4.575
330	.126E-01	.181E-01	.299E-01	.565E-01	.877E-01	.164	.256	.520	1.703	6.303
340	.181E-01	.255E-01	.418E-01	.780E-01	.120	.224	.348	.708	2.323	8.603
350	.258E-01	.357E-01	.578E-01	.107	.164	.303	.470	.956	3.142	11.640

TABLE 2-16 Sulfuric Acid Partial Pressure, bar, over Aqueous Sulfuric Acid*

°C	Weight percent, H ₂ SO ₄									
	10.0	20.0	30.0	40.0	50.0	60.0	70.0	75.0	80.0	85.0
0	.576E-21	.843E-20	.141E-18	.344E-17	.109E-15	.438E-14	.249E-12	.200E-11	.161E-10	.121E-09
10	.634E-20	.874E-19	.131E-17	.276E-16	.769E-15	.273E-13	.135E-11	.101E-10	.743E-10	.490E-09
20	.588E-19	.769E-18	.104E-16	.193E-15	.474E-14	.149E-12	.649E-11	.447E-10	.305E-09	.179E-08
30	.468E-18	.584E-17	.721E-16	.119E-14	.259E-13	.725E-12	.278E-10	.178E-09	.113E-08	.594E-08
40	.324E-17	.389E-16	.441E-15	.649E-14	.127E-12	.317E-11	.108E-09	.643E-09	.379E-08	.181E-07
50	.197E-16	.229E-15	.241E-14	.320E-13	.562E-12	.126E-10	.380E-09	.212E-08	.117E-07	.513E-07
60	.107E-15	.121E-14	.119E-13	.144E-12	.228E-11	.462E-10	.124E-08	.646E-08	.334E-07	.135E-06
70	.526E-15	.581E-14	.535E-13	.592E-12	.851E-11	.156E-09	.373E-08	.183E-07	.888E-07	.336E-06
80	.235E-14	.254E-13	.221E-12	.225E-11	.295E-10	.492E-09	.105E-07	.485E-07	.222E-06	.786E-06
90	.960E-14	.102E-12	.844E-12	.798E-11	.956E-10	.145E-08	.279E-07	.121E-06	.522E-06	.175E-05
100	.353E-13	.381E-12	.300E-11	.264E-10	.291E-09	.402E-08	.698E-07	.287E-06	.117E-05	.371E-05
110	.127E-12	.132E-11	.997E-11	.824E-10	.835E-09	.106E-07	.166E-06	.644E-06	.249E-05	.752E-05
120	.418E-12	.432E-11	.312E-10	.243E-09	.227E-08	.264E-07	.375E-06	.138E-05	.508E-05	.147E-04
130	.129E-11	.132E-10	.924E-10	.678E-09	.589E-08	.631E-07	.814E-06	.285E-05	.995E-05	.277E-04
140	.375E-11	.385E-10	.259E-09	.181E-08	.146E-07	.144E-06	.169E-05	.565E-05	.188E-04	.503E-04
150	.103E-10	.106E-09	.694E-09	.460E-08	.346E-07	.316E-06	.340E-05	.108E-04	.343E-04	.889E-04
160	.272E-10	.279E-09	.178E-08	.112E-07	.789E-07	.670E-06	.659E-05	.200E-04	.608E-04	.152E-03
170	.682E-10	.702E-09	.436E-08	.264E-07	.174E-06	.137E-05	.124E-04	.359E-04	.104E-03	.255E-03
180	.164E-09	.170E-08	.103E-07	.599E-07	.369E-06	.271E-05	.225E-04	.627E-04	.175E-03	.416E-03
190	.378E-09	.394E-08	.234E-07	.131E-06	.760E-06	.521E-05	.400E-04	.107E-03	.286E-03	.663E-03
200	.842E-09	.883E-08	.514E-07	.278E-06	.152E-05	.975E-05	.691E-04	.177E-03	.457E-03	.104E-02
210	.181E-08	.191E-07	.109E-06	.573E-06	.295E-05	.178E-04	.117E-03	.288E-03	.715E-03	.159E-02
220	.376E-08	.401E-07	.226E-06	.115E-05	.559E-05	.316E-04	.193E-03	.459E-03	.110E-02	.239E-02
230	.758E-08	.817E-07	.455E-06	.224E-05	.103E-04	.549E-04	.311E-03	.717E-03	.166E-02	.354E-02
240	.148E-07	.162E-06	.889E-06	.427E-05	.186E-04	.935E-04	.494E-03	.110E-02	.245E-02	.515E-02
250	.283E-07	.312E-06	.170E-05	.793E-05	.329E-04	.156E-03	.770E-03	.166E-02	.358E-02	.740E-02
260	.526E-07	.588E-06	.316E-05	.144E-04	.569E-04	.255E-03	.118E-02	.247E-02	.516E-02	.105E-01
270	.954E-07	.108E-05	.577E-05	.257E-04	.965E-04	.411E-03	.178E-02	.362E-02	.733E-02	.147E-01
280	.169E-06	.194E-05	.103E-04	.450E-04	.161E-03	.650E-03	.265E-02	.524E-02	.103E-01	.203E-01
290	.294E-06	.342E-05	.180E-04	.771E-04	.263E-03	.101E-02	.389E-02	.750E-02	.143E-01	.278E-01
300	.500E-06	.591E-05	.309E-04	.130E-03	.424E-03	.156E-02	.563E-02	.106E-01	.196E-01	.376E-01
310	.834E-06	.100E-04	.522E-04	.215E-03	.672E-03	.236E-02	.805E-02	.148E-01	.266E-01	.504E-01
320	.137E-05	.167E-04	.865E-04	.352E-03	.105E-02	.352E-02	.114E-01	.205E-01	.359E-01	.670E-01
330	.220E-05	.273E-04	.141E-03	.565E-03	.162E-02	.519E-02	.159E-01	.281E-01	.480E-01	.883E-01
340	.349E-05	.440E-04	.227E-03	.895E-03	.246E-02	.757E-02	.221E-01	.382E-01	.636E-01	.116
350	.544E-05	.698E-04	.360E-03	.140E-02	.369E-02	.109E-01	.303E-01	.516E-01	.836E-01	.150
°C	Weight percent, H ₂ SO ₄									
	90.0	92.0	94.0	96.0	97.0	98.0	98.5	99.0	99.5	100.0
0	.534E-09	.803E-09	.112E-08	.148E-08	.167E-08	.187E-08	.196E-08	.206E-08	.217E-08	.228E-08
10	.200E-08	.296E-08	.409E-08	.540E-08	.609E-08	.679E-08	.714E-08	.750E-08	.788E-08	.827E-08
20	.677E-08	.993E-08	.136E-07	.179E-07	.201E-07	.224E-07	.236E-07	.247E-07	.260E-07	.273E-07
30	.211E-07	.306E-07	.415E-07	.543E-07	.611E-07	.680E-07	.714E-07	.749E-07	.786E-07	.824E-07
40	.607E-07	.870E-07	.117E-06	.153E-06	.171E-06	.191E-06	.200E-06	.210E-06	.220E-06	.230E-06
50	.163E-06	.231E-06	.309E-06	.400E-06	.449E-06	.498E-06	.523E-06	.548E-06	.574E-06	.600E-06
60	.411E-06	.575E-06	.765E-06	.985E-06	.110E-05	.122E-05	.128E-05	.134E-05	.140E-05	.147E-05
70	.976E-06	.135E-05	.179E-05	.229E-05	.256E-05	.283E-05	.297E-05	.310E-05	.325E-05	.339E-05
80	.220E-05	.302E-05	.396E-05	.504E-05	.562E-05	.622E-05	.652E-05	.681E-05	.712E-05	.743E-05
90	.473E-05	.642E-05	.835E-05	.106E-04	.118E-04	.130E-04	.136E-04	.143E-04	.149E-04	.155E-04
100	.973E-05	.131E-04	.169E-04	.213E-04	.237E-04	.261E-04	.274E-04	.285E-04	.298E-04	.310E-04
110	.192E-04	.256E-04	.328E-04	.412E-04	.457E-04	.503E-04	.527E-04	.549E-04	.572E-04	.595E-04
120	.366E-04	.482E-04	.614E-04	.767E-04	.849E-04	.935E-04	.977E-04	.102E-03	.106E-03	.110E-03
130	.672E-04	.879E-04	.111E-03	.138E-03	.153E-03	.168E-03	.175E-03	.182E-03	.190E-03	.197E-03
140	.120E-03	.155E-03	.195E-03	.241E-03	.266E-03	.292E-03	.304E-03	.316E-03	.329E-03	.341E-03
150	.207E-03	.266E-03	.332E-03	.408E-03	.449E-03	.493E-03	.514E-03	.534E-03	.554E-03	.574E-03
160	.348E-03	.444E-03	.550E-03	.673E-03	.740E-03	.810E-03	.844E-03	.876E-03	.909E-03	.941E-03
170	.572E-03	.723E-03	.889E-03	.108E-02	.119E-02	.130E-02	.135E-02	.140E-02	.145E-02	.150E-02
180	.917E-03	.115E-02	.147E-02	.170E-02	.186E-02	.204E-02	.212E-02	.220E-02	.227E-02	.235E-02
190	.144E-02	.179E-02	.217E-02	.262E-02	.286E-02	.312E-02	.325E-02	.336E-02	.348E-02	.359E-02
200	.221E-02	.273E-02	.329E-02	.395E-02	.431E-02	.470E-02	.488E-02	.505E-02	.522E-02	.538E-02
210	.333E-02	.408E-02	.490E-02	.585E-02	.637E-02	.693E-02	.720E-02	.744E-02	.768E-02	.791E-02
220	.494E-02	.601E-02	.715E-02	.850E-02	.924E-02	.100E-01	.104E-01	.108E-01	.111E-01	.114E-01
230	.719E-02	.869E-02	.103E-01	.122E-01	.132E-01	.143E-01	.149E-01	.153E-01	.158E-01	.162E-01
240	.103E-01	.124E-01	.146E-01	.171E-01	.186E-01	.201E-01	.209E-01	.215E-01	.221E-01	.227E-01
250	.146E-01	.174E-01	.203E-01	.238E-01	.257E-01	.278E-01	.289E-01	.297E-01	.305E-01	.314E-01
260	.203E-01	.240E-01	.279E-01	.326E-01	.352E-01	.380E-01	.394E-01	.405E-01	.416E-01	.427E-01
270	.279E-01	.329E-01	.380E-01	.441E-01	.475E-01	.513E-01	.531E-01	.545E-01	.560E-01	.574E-01
280	.380E-01	.444E-01	.510E-01	.589E-01	.633E-01	.683E-01	.706E-01	.725E-01	.744E-01	.762E-01
290	.510E-01	.592E-01	.676E-01	.778E-01	.835E-01	.900E-01	.930E-01	.954E-01	.978E-01	.100
300	.678E-01	.782E-01	.888E-01	.102	.109	.117	.121	.124	.127	.130
310	.892E-01	.102	.115	.132	.141	.151	.156	.160	.164	.167
320	.116	.132	.149	.169	.180	.193	.199	.204	.209	.213
330	.150	.170	.190	.214	.228	.245	.252	.258	.263	.269
340	.192	.216	.240	.270	.287	.307	.317	.328	.330	.386
350	.243	.272	.301	.337	.358	.383	.394	.402	.410	.417

*Vermeulen, Dong, Robinson, Nguyen, and Gmitro, AIChE meeting, Anaheim, CA, 1982; and private communication from Prof. Theodore Vermeulen, Chemical Engineering Dept., University of California, Berkeley.

TABLE 2-17 Total Pressure, bar, of Aqueous Sulfuric Acid Solutions*

°C	Weight percent, H ₂ SO ₄									
	10.0	20.0	30.0	40.0	50.0	60.0	70.0	75.0	80.0	85.0
0	.582E-02	.534E-02	.448E-02	.326E-02	.193E-02	.836E-03	.207E-03	.747E-04	.197E-04	.343E-05
10	.117E-01	.107E-01	.909E-02	.670E-02	.405E-02	.180E-02	.467E-03	.175E-03	.490E-04	.952E-05
20	.223E-01	.205E-01	.174E-01	.130E-01	.802E-02	.367E-02	.995E-03	.388E-03	.115E-03	.245E-04
30	.404E-01	.373E-01	.319E-01	.241E-01	.151E-01	.710E-02	.201E-02	.811E-03	.253E-03	.589E-04
40	.703E-01	.649E-01	.558E-01	.427E-01	.272E-01	.131E-01	.387E-02	.162E-02	.531E-03	.134E-03
50	.117	.109	.939E-01	.725E-01	.470E-01	.232E-01	.715E-02	.309E-02	.106E-02	.286E-03
60	.189	.175	.152	.119	.782E-01	.395E-01	.127E-01	.565E-02	.204E-02	.584E-03
70	.296	.275	.239	.188	.126	.651E-01	.217E-01	.997E-01	.376E-02	.114E-02
80	.449	.417	.365	.290	.196	.104	.360E-01	.170E-01	.668E-02	.213E-02
90	.664	.617	.542	.434	.298	.161	.578E-01	.281E-01	.115E-01	.383E-02
100	.957	.891	.786	.634	.441	.244	.905E-01	.452E-01	.192E-01	.666E-02
110	1.349	1.258	1.113	.904	.638	.360	.138	.708E-01	.312E-01	.112E-01
120	1.863	1.740	1.544	1.264	.903	.519	.206	.108	.493E-01	.183E-01
130	2.524	2.361	2.101	1.732	1.253	.734	.301	.162	.760E-01	.291E-01
140	3.361	3.149	2.810	2.333	1.708	1.020	.431	.236	.115	.451E-01
150	4.404	4.132	3.697	3.090	2.289	1.392	.605	.339	.170	.683E-01
160	5.685	5.342	4.793	4.031	3.021	1.870	.837	.478	.246	.101
170	7.236	6.810	6.127	5.185	3.930	2.475	1.138	.662	.350	.147
180	9.093	8.571	7.731	6.584	5.045	3.233	1.525	.902	.489	.209
190	11.289	10.658	9.640	8.259	6.397	4.169	2.017	1.212	.673	.292
200	13.861	13.107	11.887	10.245	8.020	5.312	2.633	1.606	.913	.402
210	16.841	15.951	14.505	12.576	9.948	6.696	3.396	2.101	1.221	.544
220	20.264	19.225	17.529	15.287	12.217	8.354	4.331	2.715	1.610	.726
230	24.160	22.960	20.992	18.414	14.864	10.322	5.466	3.468	2.098	.956
240	28.561	27.188	24.927	21.992	17.929	12.641	6.832	4.382	2.701	1.242
250	33.494	31.939	29.364	26.056	21.452	15.351	8.459	5.481	3.439	1.594
260	38.984	37.240	34.334	30.642	25.472	18.496	10.384	6.791	4.332	2.023
270	45.055	43.116	39.865	35.784	30.030	22.122	12.642	8.337	5.402	2.540
280	51.726	49.590	45.984	41.514	35.168	26.275	15.272	10.147	6.673	3.157
290	59.015	56.681	52.715	47.866	40.926	31.004	18.315	12.250	8.170	3.886
300	66.934	64.407	60.081	54.869	47.347	36.361	21.814	14.675	9.916	4.740
310	75.495	72.781	68.101	62.553	54.470	42.398	25.812	17.453	11.939	5.732
320	84.705	81.816	76.792	70.947	62.338	49.168	30.355	20.611	14.264	6.876
330	94.567	91.518	86.172	80.078	70.990	56.727	35.489	24.182	16.916	8.185
340	105.083	101.894	96.252	89.970	80.466	65.130	41.262	28.193	19.920	9.672
350	116.251	112.947	107.043	100.647	90.806	74.437	47.723	32.674	23.303	11.351
°C	Weight percent, H ₂ SO ₄									
	90.0	92.0	94.0	96.0	97.0	98.0	98.5	99.0	99.5	100.0
0	.518E-06	.243E-06	.109E-06	.416E-07	.235E-07	.117E-07	.768E-08	.479E-08	.313E-08	.323E-08
10	.159E-05	.765E-06	.348E-06	.136E-06	.774E-07	.391E-07	.261E-07	.166E-07	.113E-07	.124E-07
20	.449E-05	.221E-05	.102E-05	.407E-06	.235E-06	.121E-06	.812E-07	.528E-07	.373E-07	.435E-07
30	.117E-04	.590E-05	.279E-05	.113E-05	.659E-06	.344E-06	.234E-06	.155E-06	.114E-06	.141E-06
40	.385E-04	.147E-04	.708E-05	.293E-05	.173E-05	.914E-06	.630E-06	.425E-06	.323E-06	.425E-06
50	.653E-04	.344E-04	.169E-04	.712E-05	.425E-05	.228E-05	.159E-05	.109E-05	.861E-06	.120E-05
60	.141E-03	.759E-04	.380E-04	.164E-04	.987E-05	.538E-05	.379E-05	.264E-05	.216E-05	.319E-05
70	.291E-03	.159E-03	.813E-04	.357E-04	.218E-04	.120E-04	.856E-05	.605E-05	.514E-05	.804E-05
80	.571E-03	.319E-03	.166E-03	.742E-04	.458E-04	.257E-04	.184E-04	.132E-04	.117E-04	.193E-04
90	.107E-02	.612E-03	.324E-03	.148E-03	.921E-04	.524E-04	.390E-04	.277E-04	.253E-04	.441E-04
100	.195E-02	.113E-02	.607E-03	.283E-03	.178E-03	.103E-03	.751E-04	.555E-04	.527E-04	.966E-04
110	.340E-02	.201E-02	.110E-02	.521E-03	.332E-03	.194E-03	.143E-03	.107E-03	.106E-03	.204E-03
120	.575E-02	.346E-02	.192E-02	.929E-03	.598E-03	.354E-03	.263E-03	.201E-03	.206E-03	.414E-03
130	.944E-02	.578E-02	.327E-02	.161E-02	.104E-02	.626E-03	.470E-03	.363E-03	.387E-03	.314E-03
140	.151E-01	.939E-02	.539E-02	.270E-02	.177E-02	.107E-02	.815E-03	.639E-03	.708E-03	.155E-02
150	.235E-01	.149E-01	.866E-02	.441E-02	.293E-02	.180E-02	.137E-02	.109E-02	.126E-02	.287E-02
160	.357E-01	.230E-01	.136E-01	.703E-02	.471E-02	.293E-02	.226E-02	.183E-02	.219E-02	.516E-02
170	.532E-01	.347E-01	.208E-01	.110E-01	.741E-02	.466E-02	.363E-02	.299E-02	.372E-02	.905E-02
180	.775E-01	.514E-01	.312E-01	.167E-01	.114E-01	.726E-02	.571E-02	.478E-02	.619E-02	.155E-01
190	.111	.747E-01	.460E-01	.250E-01	.172E-01	.111E-01	.880E-02	.749E-02	.101E-01	.260E-01
200	.156	.107	.665E-01	.367E-01	.255E-01	.166E-01	.133E-01	.115E-01	.161E-01	.427E-01
210	.216	.150	.944E-01	.530E-01	.371E-01	.245E-01	.198E-01	.175E-01	.253E-01	.687E-01
220	.295	.207	.132	.752E-01	.531E-01	.354E-01	.289E-01	.260E-01	.392E-01	.109
230	.396	.282	.182	.105	.749E-01	.505E-01	.417E-01	.382E-01	.596E-01	.169
240	.525	.379	.247	.145	.104	.710E-01	.592E-01	.553E-01	.895E-01	.258
250	.688	.503	.331	.197	.143	.985E-01	.830E-01	.790E-01	.132	.389
260	.881	.660	.439	.264	.193	.135	.115	.112	.193	.577
270	1.141	.856	.575	.351	.258	.153	.157	.156	.279	.846
280	1.447	1.099	.744	.460	.341	.245	.213	.215	.398	1.225
290	1.817	1.398	.954	.597	.446	.324	.285	.295	.562	1.751
300	2.261	1.761	1.211	.767	.578	.425	.379	.399	.785	2.476
310	2.791	2.199	1.524	.977	.742	.553	.498	.536	1.085	3.465
320	3.417	2.723	1.901	1.234	.944	.713	.649	.714	1.486	4.800
330	4.153	3.347	2.353	1.545	1.191	.911	.840	.944	2.018	6.586
340	5.011	4.084	2.889	1.919	1.491	1.156	1.078	1.239	2.718	8.957
350	6.006	4.949	3.523	2.366	1.852	1.456	1.374	1.614	3.631	12.079

*Vermeulen, Dong, Robinson, Nguyen, and Gmitro, AIChE meeting, Anaheim, Calif., 1982; and private communication from Prof. Theodore Vermeulen, Chemical Engineering Dept., University of California, Berkeley.

2-88 PHYSICAL AND CHEMICAL DATA

TABLE 2-18 Partial Pressures of HNO₃ and H₂O over Aqueous Solutions of HNO₃*

mmHg Percentages are weight % HNO ₃ in solution.														
°C	20%		25%		30%		35%		40%		45%		50%	
	HNO ₃	H ₂ O	HNO ₃	H ₂ O	HNO ₃	H ₂ O	HNO ₃	H ₂ O	HNO ₃	H ₂ O	HNO ₃	H ₂ O	HNO ₃	H ₂ O
0		4.1		3.8		3.6		3.3		3.0		2.6		2.1
5		5.7		5.4		5.0		4.6		4.2		3.6		3.0
10		8.0		7.6		7.1		6.5		5.8		5.0	0.12	4.2
15		10.9		10.3		9.7		8.9		8.0	0.10	6.9	.18	5.8
20		15.2		14.2		13.2		12.0		10.8	.15	9.4	.27	7.9
25		20.6		19.2		17.8		16.2	0.12	14.6	.23	12.7	.39	10.7
30		27.6		25.7		23.8	0.09	21.7	.17	19.5	.33	16.9	.56	14.4
35		36.5		33.8		31.1	.13	28.3	.25	25.5	.48	22.3	.80	19.0
40		47.5		44	0.11	41	.20	37.7	.36	33.5	.68	29.3	1.13	25.0
45		62	0.09	57.5	.17	53	.28	48	.52	43	.96	38.0	1.57	32.5
50		80	.13	75	.25	69	.42	63	.75	56	1.35	49.5	2.18	42.5
55	0.09	100	.18	94	.35	87	.59	79	1.04	71	1.83	62.5	2.95	54
60	.13	128	.28	121	.51	113	.85	102	1.48	90	2.54	80	4.05	70
65	.19	162	.40	151	.71	140	1.18	127	2.05	114	3.47	100	5.46	88
70	.27	200	.54	187	1.00	174	1.63	159	2.80	143	4.65	126	7.25	110
75	.38	250	.77	234	1.38	217	2.26	198	3.80	178	6.20	158	9.6	138
80	.53	307	1.05	287	1.87	267	3.07	243	5.10	218	8.15	195	12.5	170
85	.74	378	1.44	352	2.53	325	4.15	297	6.83	268	10.7	240	16.3	211
90	1.01	458	1.95	426	3.38	393	5.50	359	9.0	325	13.7	292	20.9	258
95	1.37	555	2.62	517	4.53	478	7.32	436	11.7	394	17.8	355	26.8	315
100	1.87	675	3.50	628	6.05	580	9.7	530	15.5	480	23.0	430	34.2	383
105	2.50	800	4.65	745	7.90	690	12.7	631	20.0	573	29.2	520	43.0	463
110							16.5	755	25.7	688	37.0	625	54.5	560
115									32.5	810	46	740	67	665
120													84	785

°C	55%		60%		65%		70%		80%		90%		100%
	HNO ₃	H ₂ O	HNO ₃	H ₂ O	HNO ₃	H ₂ O	HNO ₃	H ₂ O	HNO ₃	H ₂ O	HNO ₃	H ₂ O	HNO ₃
0		1.8	0.19	1.5	0.41	1.3	0.79	1.1	2		5.5		11
5	0.14	2.5	.28	2.1	.60	1.8	1.12	1.6	3		8		15
10	.21	3.5	.41	3.0	.86	2.6	1.58	2.2	4	1.2	11		22
15	.31	4.9	.59	4.1	1.21	3.5	2.18	3.0	6	1.7	15		30
20	.45	6.7	.84	5.6	1.68	4.9	3.00	4.1	8	2.4	20		42
25	.66	9.1	1.21	7.7	2.32	6.6	4.10	5.5	10.5	3.2	27	1	57
30	.93	12.2	1.66	10.3	3.17	8.8	5.50	7.4	14	4	36	1.3	77
35	1.30	16.1	2.28	13.6	4.26	11.6	7.30	9.8	18.5	5.5	47	1.8	102
40	1.82	21.3	3.10	18.1	5.70	15.5	9.65	12.8	24.5	7	62	2.4	133
45	2.50	28.0	4.20	23.7	7.55	20.0	12.6	16.7	32	9.5	80	3	170
50	3.41	36.3	5.68	31	10.0	26.0	16.5	21.8	41	12	103	4	215
55	4.54	46	7.45	39	12.8	33.0	21.0	27.3	52	15	127	5	262
60	6.15	60	9.9	51	16.8	43.0	27.1	35.3	67	20	157	6.5	320
65	8.18	76	13.0	64	21.7	54.5	34.5	44.5	85	25	192	8	385
70	10.7	95	16.8	81	27.5	68	43.3	56	106	31	232	10	460
75	13.9	120	21.8	102	35.0	86	54.5	70	130	38	282	13	540
80	18.0	148	27.5	126	43.5	106	67.5	86	158	48	338	16	625
85	23.0	182	34.8	156	54.5	131	83	107	192	60	405	20	720
90	29.4	223	43.7	192	67.5	160	103	130	230	73	480	24	820
95	37.3	272	55.0	233	83.5	195	125	158	278	89	570	29	
100	47	331	69.5	285	103	238	152	192	330	108	675	35	
105	58.5	400	84.5	345	124	288	183	231	392	129	790	42	
110	73	485	103	417	152	345	221	278	465	155			
115	90	575	126	495	181	410	262	330	545	185			
120	110	685	156	590	218	490	312	393	640	219			
125			187	700	260	580	372	469					

*International Critical Tables, vol. 3, pp. 304-305.

TABLE 2-19 Partial Pressures of H₂O and HBr over Aqueous Solutions of HBr at 20 to 55°C*

% HBr	mmHg							
	20 °C		25 °C		50 °C		55 °C	
	HBr	H ₂ O	HBr	H ₂ O	HBr	H ₂ O	HBr	H ₂ O
32			0.0016					
34			0.0022					
36			0.0033					
38			0.0061					
40			0.011					
42			0.023					
44			0.048					
46			0.10					
48	0.09	6.2	0.13	8.2	1.3	30.2	2.0	38
50	0.23	4.5	0.37	6.1	3.2	24.3	4.6	31
52	0.71	3.3	1.1	4.5	7.2	19.3	10.2	25
54	2.2	2.4	3.2	3.3	17	16.0	23.0	21
56	6.8	1.7	9.3	2.4	40	13.3	51	18
58	21	1.3	27	1.9	91	10.4	115	14
60							260	11.4

*International Critical Tables, vol. 3, p. 306.

TABLE 2-20 Partial Pressures of HI over Aqueous Solutions of HI at 25°C*

%HI	mmHg						
	44	46	48	50	52	54	56
p_{HI}	0.00064	0.0010	0.0022	0.0050	0.013	0.035	0.10

*International Critical Tables, vol. 3, p. 306.

TABLE 2-21 Vapor Pressures of the System: Water-Sulfuric Acid-Nitric AcidFor these data reference must be made to the graphs of *International Critical Tables*, vol. 3, pp. 306-308.**TABLE 2-22 Total Vapor Pressures of Aqueous Solutions of CH₃COOH***Percentages of weight % acetic acid in the solution
mmHg

°C	25%	50%	75%
20	16.3	15.7	15.3
25	22.1	21.4	20.8
30	29.6	28.8	27.8
35	39.4	38.3	36.6
40	51.7	50.2	48.1
45	67.0	65.0	62.0
50	87.2	85.0	80.1
55	110	107	102
60	141	138	130
65	178	172	162
70	223	216	203
75	277	269	251
80	342	331	310
85	419	407	376
90	510	497	458
95	618	602	550
100	743	725	666

*International Critical Tables, vol. 3, p. 306.

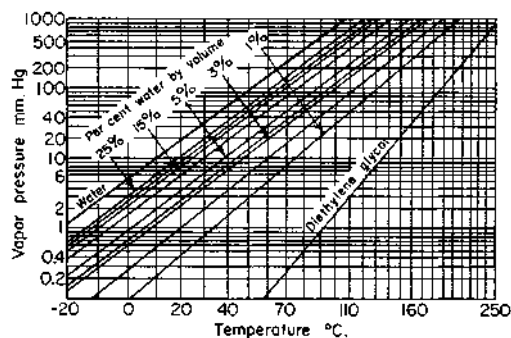
**FIG. 2-3** Vapor pressure of aqueous diethylene glycol solutions. (Courtesy of Carbide and Carbon Chemicals Corp.)

TABLE 2-23 Partial Pressure of H₂O over Aqueous Solutions of NH₃ (psia)

<i>t</i> , °F	Liquid mole percent NH ₃ (liquid weight percent NH ₃)																			
	0	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75	80	85	90	95
	(0)	(4.74)	(9.5)	(14.29)	(19.1)	(23.94)	(28.81)	(33.71)	(38.64)	(43.59)	(48.57)	(53.58)	(58.62)	(63.69)	(68.79)	(73.91)	(79.07)	(84.26)	(89.47)	(94.72)
32	0.089	0.083	0.077	0.071	0.063	0.055	0.047	0.039	0.031	0.025	0.019	0.014	0.011	0.008	0.006	0.004	0.003	0.002	0.002	0.001
40	0.122	0.115	0.106	0.097	0.087	0.077	0.065	0.054	0.044	0.035	0.027	0.021	0.016	0.012	0.009	0.007	0.005	0.004	0.002	0.001
50	0.178	0.168	0.156	0.143	0.129	0.113	0.097	0.081	0.066	0.053	0.041	0.032	0.025	0.019	0.014	0.011	0.008	0.006	0.004	0.002
60	0.256	0.242	0.225	0.207	0.186	0.164	0.142	0.119	0.098	0.079	0.062	0.049	0.038	0.030	0.023	0.018	0.014	0.010	0.007	0.004
70	0.363	0.343	0.320	0.294	0.266	0.235	0.204	0.172	0.143	0.116	0.093	0.073	0.058	0.045	0.036	0.028	0.022	0.016	0.011	0.006
80	0.507	0.479	0.448	0.413	0.374	0.332	0.289	0.245	0.205	0.168	0.136	0.109	0.087	0.069	0.055	0.043	0.034	0.025	0.018	0.010
90	0.699	0.661	0.618	0.571	0.518	0.462	0.403	0.345	0.290	0.240	0.196	0.159	0.128	0.103	0.083	0.066	0.052	0.040	0.028	0.015
100	0.951	0.899	0.843	0.780	0.710	0.634	0.556	0.479	0.405	0.338	0.279	0.228	0.186	0.152	0.123	0.100	0.079	0.061	0.043	0.024
110	1.277	1.209	1.135	1.052	0.960	0.861	0.758	0.656	0.559	0.470	0.392	0.324	0.268	0.220	0.181	0.148	0.119	0.092	0.065	0.036
120	1.695	1.607	1.510	1.402	1.283	1.154	1.021	0.889	0.763	0.647	0.544	0.455	0.380	0.316	0.263	0.217	0.176	0.137	0.099	0.056
130	2.226	2.112	1.988	1.850	1.696	1.532	1.361	1.192	1.030	0.881	0.747	0.632	0.532	0.448	0.376	0.313	0.257	0.202	0.147	0.083
140	2.893	2.748	2.591	2.415	2.221	2.012	1.796	1.582	1.376	1.186	1.016	0.867	0.738	0.628	0.532	0.448	0.371	0.295	0.216	0.124
150	3.723	3.540	3.343	3.122	2.879	2.618	2.347	2.078	1.821	1.582	1.367	1.177	1.013	0.870	0.746	0.634	0.529	0.425	0.314	0.183
160	4.747	4.519	4.273	4.000	3.698	3.374	3.039	2.706	2.387	2.090	1.821	1.584	1.376	1.194	1.033	0.887	0.748	0.607	0.453	0.267
170	6.000	5.717	5.416	5.079	4.709	4.312	3.902	3.493	3.101	2.736	2.405	2.110	1.851	1.622	1.418	1.229	1.047	0.858	0.647	0.386
180	7.520	7.174	6.807	6.397	5.947	5.465	4.968	4.472	3.995	3.551	3.148	2.787	2.468	2.184	1.928	1.688	1.451	1.201	0.917	0.555
190	9.350	8.931	8.488	7.994	7.452	6.873	6.275	5.680	5.107	4.573	4.086	3.650	3.262	2.914	2.598	2.297	1.994	1.669	1.290	0.793
200	11.538	11.035	10.504	9.916	9.270	8.580	7.869	7.160	6.479	5.842	5.262	4.740	4.275	3.856	3.470	3.098	2.718	2.300	1.802	1.129
210	14.136	13.538	12.910	12.213	11.449	10.635	9.796	8.962	8.160	7.410	6.725	6.110	5.559	5.061	4.598	4.146	3.675	3.147	2.502	1.600
220	17.201	16.496	15.758	14.941	14.047	13.095	12.115	11.141	10.205	9.331	8.534	7.817	7.175	6.592	6.045	5.504	4.932	4.277	3.455	2.262
230	20.796	19.971	19.111	18.162	17.124	16.020	14.886	13.760	12.679	11.672	10.754	9.930	9.192	8.522	7.889	7.255	6.573	5.777	4.751	3.196
240	24.986	24.029	23.037	21.943	20.748	19.479	18.179	16.889	15.654	14.506	13.463	12.530	11.696	10.938	10.221	9.496	8.703	7.759	6.508	4.520
250	29.844	28.744	27.607	26.358	24.996	23.549	22.070	20.608	19.212	17.917	16.748	15.708	14.783	13.946	13.153	12.346	11.452	10.369	8.891	6.413

The values in Tables 2-23 to 2-26 were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, Version 7.0, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002). The primary source for the properties of aqueous ammonia mixtures is R. Tillner-Roth and D. G. Friend, "A Helmholtz Free Energy Formulation of the Thermodynamic Properties of the Mixture [Water + Ammonia]," *J. Phys. Chem. Ref. Data* **27**:63-96 (1998).

TABLE 2-24 Mole Percentages of H₂O over Aqueous Solutions of NH₃

<i>t</i> , °F	Liquid mole percent NH ₃ (liquid weight percent NH ₃)																			
	0	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75	80	85	90	95
	(0)	(4.74)	(9.5)	(14.29)	(19.1)	(23.94)	(28.81)	(33.71)	(38.64)	(43.59)	(48.57)	(53.58)	(58.62)	(63.69)	(68.79)	(73.91)	(79.07)	(84.26)	(89.47)	(94.72)
32	100	32.046	14.173	7.263	3.959	2.202	1.226	0.679	0.375	0.207	0.116	0.067	0.039	0.024	0.015	0.010	0.007	0.005	0.003	0.001
40	100	33.233	15.064	7.842	4.321	2.427	1.365	0.765	0.428	0.240	0.137	0.080	0.048	0.030	0.019	0.013	0.009	0.006	0.004	0.002
50	100	34.709	16.192	8.588	4.797	2.727	1.554	0.883	0.502	0.287	0.167	0.100	0.061	0.039	0.026	0.017	0.012	0.008	0.005	0.003
60	100	36.172	17.334	9.359	5.299	3.050	1.760	1.015	0.587	0.342	0.203	0.123	0.077	0.050	0.034	0.023	0.016	0.011	0.007	0.004
70	100	37.619	18.489	10.154	5.828	3.396	1.985	1.162	0.683	0.405	0.245	0.152	0.097	0.064	0.044	0.030	0.021	0.015	0.010	0.005
80	100	39.047	19.653	10.974	6.382	3.765	2.231	1.325	0.791	0.478	0.294	0.186	0.121	0.081	0.056	0.040	0.028	0.020	0.013	0.007
90	100	40.455	20.827	11.816	6.963	4.160	2.497	1.505	0.913	0.561	0.351	0.226	0.149	0.102	0.072	0.051	0.037	0.026	0.017	0.009
100	100	41.840	22.007	12.681	7.571	4.580	2.786	1.703	1.049	0.655	0.418	0.273	0.184	0.127	0.091	0.066	0.048	0.034	0.023	0.012
110	100	43.201	23.194	13.567	8.205	5.027	3.098	1.920	1.201	0.763	0.494	0.328	0.224	0.158	0.114	0.084	0.062	0.044	0.030	0.016
120	100	44.537	24.385	14.475	8.867	5.501	3.434	2.158	1.370	0.884	0.582	0.393	0.273	0.195	0.142	0.106	0.079	0.057	0.039	0.021
130	100	45.849	25.580	15.404	9.557	6.002	3.796	2.419	1.558	1.020	0.682	0.467	0.329	0.238	0.177	0.133	0.100	0.073	0.050	0.027
140	100	47.136	26.779	16.353	10.273	6.532	4.184	2.702	1.766	1.173	0.796	0.554	0.396	0.290	0.218	0.166	0.126	0.094	0.064	0.035
150	100	48.398	27.979	17.322	11.017	7.091	4.600	3.010	1.994	1.344	0.925	0.653	0.473	0.352	0.268	0.206	0.158	0.118	0.082	0.045
160	100	49.636	29.182	18.310	11.789	7.680	5.044	3.344	2.246	1.534	1.071	0.767	0.563	0.425	0.327	0.254	0.197	0.149	0.104	0.058
170	100	50.849	30.385	19.317	12.587	8.298	5.517	3.705	2.521	1.746	1.236	0.897	0.668	0.509	0.396	0.312	0.245	0.187	0.132	0.074
180	100	52.038	31.588	20.341	13.413	8.948	6.021	4.094	2.822	1.981	1.421	1.045	0.788	0.609	0.479	0.381	0.302	0.233	0.166	0.094
190	100	53.204	32.792	21.383	14.266	9.628	6.557	4.513	3.151	2.240	1.628	1.212	0.926	0.724	0.576	0.463	0.371	0.288	0.209	0.120
200	100	54.347	33.994	22.442	15.146	10.340	7.124	4.963	3.508	2.526	1.859	1.402	1.083	0.857	0.690	0.561	0.453	0.357	0.261	0.153
210	100	55.468	35.196	23.517	16.052	11.083	7.725	5.445	3.896	2.840	2.116	1.615	1.263	1.011	0.823	0.676	0.552	0.439	0.326	0.195
220	100	56.567	36.395	24.607	16.985	11.858	8.359	5.961	4.316	3.184	2.401	1.854	1.468	1.187	0.977	0.811	0.669	0.539	0.406	0.248
230	100	57.645	37.592	25.711	17.943	12.665	9.028	6.512	4.769	3.560	2.716	2.122	1.699	1.390	1.156	0.969	0.809	0.660	0.506	0.317
240	100	58.702	38.787	26.830	18.927	13.503	9.732	7.098	5.258	3.970	3.064	2.421	1.960	1.621	1.363	1.155	0.975	0.806	0.629	0.406
250	100	59.740	39.978	27.962	19.936	14.374	10.471	7.722	5.784	4.416	3.447	2.754	2.254	1.884	1.601	1.371	1.171	0.982	0.783	0.524

The values in Tables 2-23 to 2-26 were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, Version 7.0, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002). The primary source for the properties of aqueous ammonia mixtures is R. Tillner-Roth and D. G. Friend, "A Helmholtz Free Energy Formulation of the Thermodynamic Properties of the Mixture [Water + Ammonia]," *J. Phys. Chem. Ref. Data* **27**:63–96 (1998).

TABLE 2-25 Partial Pressures of NH₃ over Aqueous Solutions of NH₃ (psia)

$t, ^\circ\text{F}$		Liquid mole percent NH_3 (liquid weight percent NH_3)																				
		0	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75	80	85	90	95	100
		(0)	(4.74)	(9.5)	(14.29)	(19.1)	(23.94)	(28.81)	(33.71)	(38.64)	(43.59)	(48.57)	(53.58)	(58.62)	(63.69)	(68.79)	(73.91)	(79.07)	(84.26)	(89.47)	(94.72)	(100)
32	0	0.177	0.468	0.901	1.533	2.456	3.797	5.710	8.358	11.868	16.279	21.480	27.206	33.087	38.745	43.900	48.431	52.368	55.846	59.055	62.277	
40	0	0.230	0.600	1.143	1.932	3.078	4.730	7.066	10.267	14.467	19.691	25.798	32.478	39.307	45.860	51.828	57.082	61.665	65.731	69.506	73.322	
50	0	0.316	0.808	1.522	2.552	4.036	6.154	9.117	13.129	18.328	24.721	32.120	40.150	48.315	56.128	63.243	69.520	75.020	79.931	84.523	89.205	
60	0	0.426	1.074	2.002	3.330	5.228	7.912	11.625	16.593	22.959	30.702	39.581	49.149	58.831	68.074	76.489	83.933	90.486	96.376	101.93	107.63	
70	0	0.568	1.410	2.603	4.296	6.696	10.056	14.656	20.742	28.456	37.744	48.307	59.615	71.008	81.861	91.745	100.51	108.26	115.29	121.94	128.85	
80	0	0.748	1.831	3.348	5.483	8.483	12.645	18.283	25.664	34.922	45.966	58.428	71.689	84.998	97.654	109.19	119.45	128.56	136.88	144.83	153.13	
90	0	0.973	2.351	4.261	6.926	10.639	15.741	22.582	31.447	42.460	55.485	70.073	85.514	100.96	115.62	128.99	140.93	151.60	161.39	170.82	180.76	
100	0	1.250	2.987	5.368	8.663	13.213	19.406	27.630	38.185	51.177	66.416	83.375	101.23	119.04	135.93	151.34	165.15	177.57	189.05	200.19	212.01	
110	0	1.590	3.758	6.700	10.735	16.258	23.709	33.509	45.971	61.177	78.881	98.461	118.98	139.39	158.73	176.40	192.30	206.69	220.07	233.16	247.19	
120	0	2.001	4.683	8.285	13.184	19.832	28.718	40.300	54.899	72.569	92.996	115.45	138.90	162.15	184.19	204.35	222.57	239.16	254.71	270.03	286.60	
130	0	2.494	5.784	10.158	16.055	23.989	34.503	48.086	65.064	85.455	108.87	134.49	161.13	187.48	212.44	235.36	256.14	275.20	293.18	311.05	330.54	
140	0	3.082	7.084	12.352	19.395	28.791	41.135	56.949	76.558	99.944	126.62	155.67	185.77	215.50	243.66	269.56	293.19	314.99	335.72	356.48	379.36	
150	0	3.774	8.604	14.902	23.250	34.295	48.685	66.972	89.472	116.12	146.35	179.11	212.96	246.33	277.95	307.13	333.87	358.75	382.56	406.60	433.38	
160	0	4.585	10.371	17.844	27.669	40.562	57.222	78.230	103.89	134.09	168.16	204.93	242.79	280.08	315.46	348.18	378.36	406.62	433.89	461.65	492.95	
170	0	5.527	12.408	21.216	32.700	47.649	66.816	90.803	119.90	153.93	192.14	233.19	275.37	316.87	356.26	392.85	426.79	458.82	489.95	521.92	558.45	
180	0	6.612	14.741	25.053	38.390	55.614	77.532	104.77	137.57	175.73	218.35	263.99	310.76	356.75	400.47	441.24	479.30	515.48	550.92	587.67	630.24	
190	0	7.856	17.395	29.393	44.786	64.514	89.432	120.18	156.99	199.56	246.89	297.40	349.06	399.82	448.16	493.43	535.98	576.75	616.99	659.12	708.74	
200	0	9.270	20.397	34.270	51.933	74.401	102.58	137.11	178.21	225.47	277.81	333.47	390.30	446.12	499.38	549.49	596.94	642.74	688.31	736.52	794.38	
210	0	10.869	23.769	39.721	59.876	85.326	117.02	155.62	201.28	253.53	311.15	372.25	434.50	495.68	554.15	609.47	662.22	713.55	765.01	820.08	887.64	
220	0	12.666	27.538	45.779	68.655	97.335	132.82	175.75	226.25	283.77	346.95	413.74	481.70	548.49	612.50	673.37	731.88	789.23	847.19	909.98	989.03	
230	0	14.673	31.727	52.477	78.310	110.47	150.00	197.54	253.17	316.23	385.23	457.98	531.89	604.55	674.39	741.21	805.90	869.79	934.86	1006.3	1099.1	
240	0	16.905	36.356	59.843	88.872	124.78	168.62	221.04	282.06	350.91	425.98	504.92	585.01	663.80	739.77	812.89	884.22	955.18	1028.0	1109.2	1218.7	
250	0	19.371	41.449	67.906	100.38	140.28	188.70	246.26	312.93	387.82	469.20	554.55	641.05	726.18	808.55	888.35	966.78	1045.3	1126.5	1218.3	1348.5	

The values in Tables 2-23 to 2-26 were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, Version 7.0, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002). The primary source for the properties of aqueous ammonia mixtures is R. Tillner-Roth and D. G. Friend, "A Helmholtz Free Energy Formulation of the Thermodynamic Properties of the Mixture {Water + Ammonia}," *J. Phys. Chem. Ref. Data* **27**:63–96 (1998).

TABLE 2-26 Total Vapor Pressures of Aqueous Solutions of NH₃ (psia)

	Liquid mole percent NH ₃ (liquid weight percent NH ₃)																				
	0	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75	80	85	90	95	100
<i>t</i> , °F	(0)	(4.74)	(9.5)	(14.29)	(19.1)	(23.94)	(28.81)	(33.71)	(38.64)	(43.59)	(48.57)	(53.58)	(58.62)	(63.69)	(68.79)	(73.91)	(79.07)	(84.26)	(89.47)	(94.72)	(100)
32	0.089	0.260	0.545	0.971	1.596	2.512	3.844	5.749	8.389	11.893	16.298	21.494	27.217	33.095	38.751	43.904	48.434	52.371	55.848	59.056	62.277
40	0.122	0.345	0.706	1.240	2.020	3.155	4.795	7.120	10.311	14.502	19.718	25.819	32.494	39.319	45.869	51.835	57.087	61.669	65.734	69.507	73.322
50	0.178	0.483	0.964	1.665	2.680	4.149	6.251	9.198	13.195	18.381	24.762	32.152	40.175	48.334	56.143	63.254	69.528	75.026	79.935	84.526	89.205
60	0.256	0.668	1.299	2.209	3.516	5.392	8.053	11.744	16.691	23.038	30.764	39.630	49.187	58.860	68.097	76.507	83.946	90.496	96.383	101.93	107.63
70	0.363	0.911	1.730	2.898	4.562	6.931	10.260	14.828	20.885	28.572	37.837	48.381	59.673	71.053	81.897	91.773	100.53	108.28	115.30	121.95	128.85
80	0.507	1.227	2.279	3.761	5.857	8.815	12.934	18.528	25.869	35.090	46.102	58.537	71.776	85.067	97.709	109.23	119.48	128.59	136.90	144.84	153.13
90	0.699	1.633	2.969	4.832	7.445	11.101	16.144	22.927	31.737	42.700	55.680	70.232	85.642	101.06	115.70	129.06	140.98	151.64	161.42	170.84	180.76
100	0.951	2.149	3.830	6.148	9.373	13.847	19.962	28.109	38.590	51.514	66.695	83.603	101.42	119.19	136.05	151.44	165.23	177.63	189.09	200.21	212.01
110	1.277	2.799	4.893	7.751	11.694	17.119	24.467	34.165	46.530	61.647	79.273	98.785	119.25	139.61	158.91	176.55	192.42	206.78	220.14	233.20	247.19
120	1.695	3.608	6.194	9.688	14.467	20.986	29.739	41.189	55.662	73.216	93.540	115.91	139.28	162.47	184.45	204.57	222.75	239.30	254.81	270.09	286.60
130	2.226	4.607	7.773	12.008	17.752	25.521	35.864	49.278	66.094	86.336	109.62	135.12	161.66	187.93	212.82	235.67	256.40	275.40	293.33	311.13	330.54
140	2.893	5.830	9.674	14.767	21.616	30.803	42.931	58.531	77.934	101.13	127.64	156.54	186.51	216.13	244.19	270.01	293.56	315.29	335.94	356.60	379.36
150	3.723	7.315	11.947	18.024	26.129	36.913	51.032	69.050	91.292	117.70	147.72	180.29	213.97	247.20	278.70	307.76	334.40	359.17	382.87	406.78	433.38
160	4.747	9.104	14.644	21.844	31.367	43.936	60.262	80.937	106.28	136.18	169.98	206.51	244.16	281.28	316.49	349.07	379.11	407.23	434.34	461.92	492.95
170	6.000	11.244	17.824	26.295	37.409	51.961	70.718	94.297	123.00	156.67	194.54	235.30	277.22	318.49	357.68	394.08	427.84	459.68	490.60	522.31	558.45
180	7.520	13.786	21.548	31.450	44.337	61.079	82.499	109.24	141.57	179.28	221.50	266.78	313.23	358.94	402.40	442.93	480.75	516.68	551.84	588.22	630.24
190	9.350	16.787	25.883	37.387	52.238	71.387	95.708	125.86	162.10	204.13	250.98	301.05	352.32	402.74	450.76	495.73	537.98	578.42	618.28	659.91	708.74
200	11.538	20.305	30.901	44.186	61.203	82.981	110.45	144.27	184.69	231.31	283.07	338.21	394.57	449.98	502.85	552.59	599.66	645.04	690.11	737.65	794.38
210	14.136	24.407	36.679	51.934	71.325	95.961	126.82	164.58	209.44	260.94	317.88	378.36	440.06	500.74	558.75	613.62	665.90	716.70	767.51	821.68	887.64
220	17.201	29.162	43.296	60.720	82.702	110.43	144.93	186.89	236.46	293.10	355.49	421.56	488.88	555.08	618.54	678.88	736.81	793.51	850.64	912.24	989.03
230	20.796	34.644	50.838	70.639	95.434	126.49	164.89	211.30	265.85	327.90	395.98	467.91	541.08	613.07	682.28	748.46	812.47	875.57	939.61	1009.5	1099.1
240	24.986	40.934	59.393	81.786	109.62	144.26	186.80	237.93	297.71	365.42	439.44	517.45	596.71	674.74	749.99	822.39	892.93	962.94	1034.5	1113.7	1218.7
250	29.844	48.115	69.056	94.264	125.38	163.83	210.77	266.87	332.14	405.74	485.95	570.25	655.83	740.12	821.71	900.70	978.24	1055.7	1135.4	1224.7	1348.5

The values in Tables 2-23 to 2-26 were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, Version 7.0, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002). The primary source for the properties of aqueous ammonia mixtures is R. Tillner-Roth and D. G. Friend, "A Helmholtz Free Energy Formulation of the Thermodynamic Properties of the Mixture (Water + Ammonia)," *J. Phys. Chem. Ref. Data* **27**:63–96 (1998).

2-94 PHYSICAL AND CHEMICAL DATA

TABLE 2-27 Partial Pressures of H₂O over Aqueous Solutions of Sodium Carbonate*

t, °C	mmHg						
	%Na ₂ CO ₃						
	0	5	10	15	20	25	30
0	4.5	4.5					
10	9.2	9.0	8.8				
20	17.5	17.2	16.8	16.3			
30	31.8	31.2	30.4	29.6	28.8	27.8	26.4
40	55.3	54.2	53.0	57.6	50.2	48.4	46.1
50	92.5	90.7	88.7	86.5	84.1	81.2	77.5
60	149.5	146.5	143.5	139.9	136.1	131.6	125.7
70	239.8	235	230.5	225	219	211.5	202.5
80	355.5	348	342	334	325	315	301
90	526.0	516	506	494	482	467	447
100	760.0	746	731	715	697	676	648

*International Critical Tables, vol. 3, p. 372.

TABLE 2-28 Partial Pressures of H₂O and CH₃OH over Aqueous Solutions of Methyl Alcohol*

Mole fraction CH ₃ OH	39.9 °C		Mole fraction CH ₃ OH	59.4 °C	
	P _{H₂O} , mmHg	P _{CH₃OH} , mmHg		P _{H₂O} , mmHg	P _{CH₃OH} , mmHg
0	54.7	0	0	145.4	0
14.99	39.2	66.1	22.17	106.9	210.1
17.85	38.5	75.5	27.40	102.2	240.2
21.07	37.2	85.2	33.24	96.6	272.1
27.31	35.8	100.6	39.80	91.7	301.9
31.06	34.9	108.8	47.08	84.8	335.6
40.1	32.8	127.7	55.5	76.9	373.7
47.0	31.5	141.6	69.2	57.8	439.4
55.8	27.3	158.4	78.5	43.8	486.6
68.9	20.7	186.6	85.9	30.1	526.9
86.0	10.1	225.2	100.0	0	609.3
100.0	0	260.7			

*International Critical Tables, vol. 3, p. 290.

TABLE 2-29 Partial Pressures of H₂O over Aqueous Solutions of Sodium Hydroxide*

Conc. g NaOH/100 g H ₂ O	mmHg											
	Temperature, °C											
	0	20	40	60	80	100	120	160	200	250	300	350
0	4.6	17.5	55.3	149.5	355.5	760.0	1,489	4,633	11,647	29,771	64,200	123,600
5	4.4	16.9	53.2	143.5	341.5	730.0	1,430	4,450	11,200	28,600	61,800	118,900
10	4.2	16.0	50.6	137.0	325.5	697.0	1,365	4,260	10,750	27,500	59,300	114,100
20	3.6	13.9	44.2	120.5	288.5	621.0	1,225	3,860	9,800	25,300	54,700	105,400
30	2.9	11.3	36.6	101.0	246.0	537.0	1,070	3,460	8,950	23,300	50,800	98,000
40	2.2	8.7	28.7	81.0	202.0	450.0	920	3,090	8,150	21,500	47,200	91,600
50		6.3	20.7	62.5	160.5	368.0	770	2,690	7,400	19,900	44,100	85,800
60		4.4	15.5	47.0	124.0	294.0	635	2,340	6,750	18,400	41,200	80,700
70		3.0	10.9	34.5	94.0	231.0	515	2,030	6,100	17,100	38,700	76,000
80		2.0	7.6	24.5	70.5	179.0	415	1,740	5,500	15,800	36,300	71,900
90		1.3	5.2	17.5	53.0	138.0	330	1,490	5,000	14,700	34,200	68,100
100		0.9	3.6	12.5	38.5	105.0	262	1,300	4,500	13,650	32,200	64,600
120			1.7	6.3	20.5	61.0	164	915	3,650	11,800	28,800	58,600
140				3.0	11.0	35.5	102	765	2,980	10,300	25,900	53,400
160				1.5	6.0	20.5	63	470	2,430	8,960	23,300	49,000
180					3.5	12.0	40	340	1,980	7,830	21,200	45,100
200					2.0	7.0	25	245	1,620	6,870	19,200	41,800
250					0.5	2.0	8	110	985	5,000	15,400	35,000
300					0.1	0.5	2.7	50	610	3,690	12,500	29,800
350							0.9	23	380	2,750	10,300	25,700
400								11	240	2,080	8,600	22,400
500									100	1,210	6,100	17,500
700										440	3,300	11,500
1000											1,470	6,800
2000											150	1,760
4000												120
8000												7

*International Critical Tables, vol. 3, p. 370.

WATER-VAPOR CONTENT OF GASES

CHART FOR GASES AT HIGH PRESSURES

The accompanying figure is useful in determining the water-vapor content of air at high pressure in contact with liquid water.

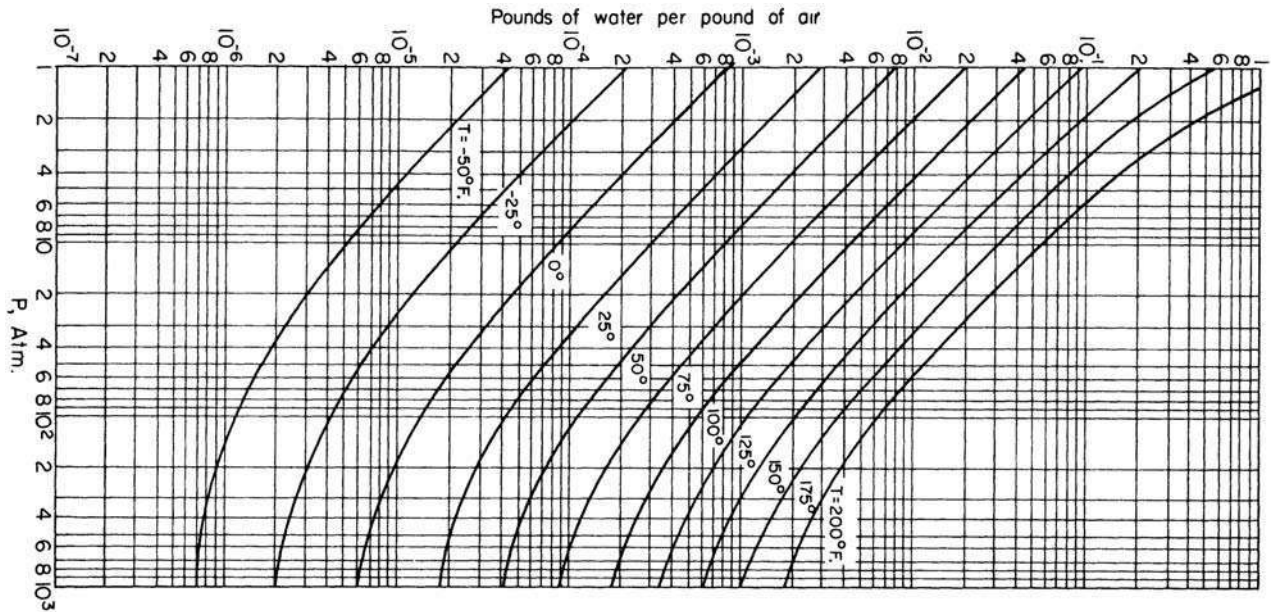


FIG. 2-4 Water content of air, $^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times \%$. (Landsbaum, Dadds, and Stutzman. Reprinted from vol. 47, January 1955 issue of Ind. Eng. Chem. [p. 192]. Copyright 1955 by the American Chemical Society and reproduced by permission of the copyright owner.)

DENSITIES OF PURE SUBSTANCES

TABLE 2-30 Density (kg/m³) of Saturated Liquid Water from the Triple Point to the Critical Point

T, K	ρ , kg/m ³	T, K	ρ , kg/m ³	T, K	ρ , kg/m ³	T, K	ρ , kg/m ³	T, K	ρ , kg/m ³
273.160*	999.793	352	972.479	432	908.571	512	814.982	592	669.930
274	999.843	354	971.235	434	906.617	514	812.164	594	664.974
276	999.914	356	969.972	436	904.645	516	809.318	596	659.907
278	999.919	358	968.689	438	902.656	518	806.441	598	654.722
280	999.862	360	967.386	440	900.649	520	803.535	600	649.411
282	999.746	362	966.064	442	898.624	522	800.597	602	643.97
284	999.575	364	964.723	444	896.580	524	797.629	604	638.38
286	999.352	366	963.363	446	894.519	526	794.628	606	632.64
288	999.079	368	961.984	448	892.439	528	791.594	608	626.74
290	998.758	370	960.587	450	890.341	530	788.527	610	620.65
292	998.392	372	959.171	452	888.225	532	785.425	612	614.37
294	997.983	374	957.737	454	886.089	534	782.288	614	607.88
296	997.532	376	956.285	456	883.935	536	779.115	616	601.15
298	997.042	378	954.815	458	881.761	538	775.905	618	594.16
300	996.513	380	953.327	460	879.569	540	772.657	620	586.88
302	995.948	382	951.822	462	877.357	542	769.369	622	579.26
304	995.346	384	950.298	464	875.125	544	766.042	624	571.25
306	994.711	386	948.758	466	872.873	546	762.674	626	562.81
308	994.042	388	947.199	468	870.601	548	759.263	628	553.84
310	993.342	390	945.624	470	868.310	550	755.808	630	544.25
312	992.610	392	944.030	472	865.997	552	752.308	632	533.92
314	991.848	394	942.420	474	863.664	554	748.762	634	522.71
316	991.056	396	940.793	476	861.310	556	745.169	636	510.42
318	990.235	398	939.148	478	858.934	558	741.525	638	496.82
320	989.387	400	937.486	480	856.537	560	737.831	640	481.53
322	988.512	402	935.807	482	854.118	562	734.084	641	473.01
324	987.610	404	934.111	484	851.678	564	730.283	642	463.67
326	986.682	406	932.398	486	849.214	566	726.425	643	453.14
328	985.728	408	930.668	488	846.728	568	722.508	644	440.73
330	984.750	410	928.921	490	844.219	570	718.530	645	425.05
332	983.747	412	927.157	492	841.686	572	714.489	646	402.96
334	982.721	414	925.375	494	839.130	574	710.382	647	357.34
336	981.671	416	923.577	496	836.549	576	706.206	647.096†	322
338	980.599	418	921.761	498	833.944	578	701.959		
340	979.503	420	919.929	500	831.313	580	697.638		
342	978.386	422	918.079	502	828.658	582	693.238		
344	977.247	424	916.212	504	825.976	584	688.757		
346	976.086	426	914.328	506	823.269	586	684.190		
348	974.904	428	912.426	508	820.534	588	679.533		
350	973.702	430	910.507	510	817.772	590	674.781		

*Triple point

†Critical point

From Wagner, W., and Pruss, A., "The IAPWS Formulation 1995 for the Thermodynamic Properties of Ordinary Water Substance for General and Scientific Use," *J. Phys. Chem. Ref. Data* **31**(2):387–535, 2002.

TABLE 2-31 Density (kg/m³) of Mercury from 0 to 350°C*

t, °C	Density, kg/m ³									
	0	1	2	3	4	5	6	7	8	9
0	13595.08	13592.61	13590.14	13587.68	13585.21	13582.75	13580.29	13577.82	13575.36	13572.90
10	13570.44	13567.98	13565.52	13563.06	13560.60	13558.14	13555.69	13553.23	13550.78	13548.32
20	13545.87	13543.41	13540.96	13538.51	13536.06	13533.61	13531.16	13528.71	13526.26	13523.81
30	13521.36	13518.91	13516.47	13514.02	13511.58	13509.13	13506.69	13504.25	13501.80	13499.36
40	13496.92	13494.48	13492.04	13489.60	13487.16	13484.72	13482.29	13479.85	13477.41	13474.98
50	13472.54	13470.11	13467.67	13465.24	13462.81	13460.38	13457.94	13455.51	13453.08	13450.65
60	13448.22	13445.80	13443.37	13440.94	13438.51	13436.09	13433.66	13431.23	13428.81	13426.39
70	13423.96	13421.54	13419.12	13416.69	13414.27	13411.85	13409.43	13407.01	13404.59	13402.17
80	13399.75	13397.34	13394.92	13392.50	13390.08	13387.67	13385.25	13382.84	13380.42	13378.01
90	13375.59	13373.18	13370.77	13368.36	13365.94	13363.53	13361.12	13358.71	13356.30	13353.89
100	13351.5	13349.1	13346.7	13344.3	13341.9	13339.4	13337.0	13334.6	13332.2	13329.8
110	13327.4	13325.0	13322.6	13320.2	13317.8	13315.4	13313.0	13310.6	13308.2	13305.8
120	13303.4	13301.0	13298.6	13296.2	13293.8	13291.4	13288.9	13286.6	13284.2	13281.8
130	13279.4	13277.0	13274.6	13272.2	13269.8	13267.4	13265.0	13262.6	13260.2	13257.8
140	13255.4	13253.0	13250.6	13248.2	13245.8	13243.4	13241.0	13238.7	13236.3	13233.9
150	13231.5	13229.1	13226.7	13224.3	13221.9	13219.5	13217.1	13214.7	13212.4	13210.0
160	13207.6	13205.2	13202.8	13200.4	13198.0	13195.6	13193.2	13190.8	13188.5	13186.1
170	13183.7	13181.3	13178.9	13176.5	13174.1	13171.7	13169.4	13167.0	13164.6	13162.2
180	13159.8	13157.4	13155.0	13152.6	13150.3	13147.9	13145.5	13143.1	13140.7	13138.3
190	13136.0	13133.6	13131.2	13128.3	13126.4	13124.0	13121.7	13119.3	13116.9	13114.5
200	13112.1	13109.7	13107.4	13105.0	13102.6	13100.2	13097.8	13095.4	13093.1	13090.7
210	13088.3	13085.9	13083.5	13081.1	13078.8	13076.4	13074.0	13071.6	13069.2	13066.8
220	13064.5	13062.1	13059.7	13057.3	13054.9	13052.6	13050.2	13047.8	13045.4	13043.0
230	13040.6	13038.3	13035.9	13033.5	13031.1	13028.7	13026.4	13024.0	13021.6	13019.2
240	13016.8	13014.5	13012.1	13009.7	13007.3	13004.9	13002.5	13000.2	12997.8	12995.4
250	12993.0	12990.6	12988.3	12985.9	12983.5	12981.1	12978.7	12976.3	12974.0	12971.6
260	12969.2	12966.8	12964.4	12962.0	12959.7	12957.3	12954.9	12952.5	12950.1	12947.7
270	12945.4	12943.0	12940.6	12938.2	12935.8	12933.4	12931.1	12928.7	12926.3	12923.9
280	12921.5	12919.1	12916.7	12914.4	12912.0	12909.6	12907.2	12904.8	12902.4	12900.0
290	12897.7	12895.3	12892.9	12890.5	12888.1	12885.7	12883.3	12880.9	12878.5	12876.2
300	12873.8	12871.4	12869.0	12866.6	12864.2	12861.8	12859.4	12857.0	12854.6	12852.2
310	12849.9	12847.5	12845.1	12842.7	12840.3	12837.9	12835.5	12833.1	12830.7	12828.3
320	12825.9	12823.5	12821.1	12818.7	12816.3	12813.9	12811.5	12809.1	12806.7	12804.3
330	12801.9	12799.5	12797.1	12794.7	12792.3	12789.9	12787.5	12785.1	12782.7	12780.2
340	12777.8	12775.4	12773.0	12770.6	12768.2	12765.8	12763.4	12761.0	12758.6	12756.1
350	12753.7									

*From "Mercury—Density and Thermal Expansion at Atmospheric Pressure and Temperatures from 0 to 350 °C," *Tables of Standard Handbook Data*, Standartov, Moscow, 1978. The density values obtainable from those cited for the specific volume of the saturated liquid in the "Thermodynamic Properties" subsection show minor differences. No attempt was made to adjust either set.

TABLE 2-32 Densities of Inorganic and Organic Liquids (mol/dm³)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1	C2	C3	C4	T _{min} , K	Density at T _{min}	T _{max} , K	Density at T _{max}
1	Acetaldehyde	C ₂ H ₄ O	75-07-0	44.053	1.6994	0.26167	466	0.2913	150.15	21.499	466.00	6.4944
2	Acetamide	C ₂ H ₅ NO	60-35-5	59.067	1.016	0.21845	761	0.26116	353.33	16.936	761.00	4.6509
3	Acetic acid	C ₂ H ₄ O ₂	64-19-7	60.052	1.4486	0.25892	591.95	0.2529	289.81	17.492	591.95	5.5948
4	Acetic anhydride	C ₄ H ₆ O ₃	108-24-7	102.089	0.86852	0.25187	606	0.31172	200.15	11.643	606.00	3.4483
5	Acetone	C ₃ H ₆ O	67-64-1	58.079	1.2332	0.25886	508.2	0.2913	178.45	15.683	508.20	4.7640
6	Acetonitrile	C ₂ H ₃ N	75-05-8	41.052	1.3064	0.22597	545.5	0.28678	229.32	20.628	545.50	5.7813
7	Acetylene	C ₂ H ₂	74-86-2	26.037	2.4507	0.27448	308.3	0.28752	192.40	23.692	308.30	8.9285
8	Acrolein	C ₃ H ₄ O	107-02-8	56.063	1.3261	0.26124	506	0.2489	185.45	16.822	506.00	5.0762
9	Acrylic acid	C ₃ H ₄ O ₂	79-10-7	72.063	1.2414	0.25822	615	0.30701	286.15	14.693	615.00	4.8075
10	Acrylonitrile	C ₃ H ₃ N	107-13-1	53.063	1.0816	0.2293	535	0.28939	189.63	17.265	535.00	4.7170
11	Air	Mixture	132259-10-0	28.960	2.8963	0.26733	132.45	0.27341	59.15	33.279	132.45	10.8340
12	Ammonia	H ₃ N	7664-41-7	17.031	3.5383	0.25443	405.65	0.2888	195.41	43.141	405.65	13.9070
13	Anisole	C ₇ H ₈ O	100-66-3	108.138	0.77488	0.26114	645.6	0.28234	235.65	9.668	645.60	2.9673
14	Argon	Ar	7440-37-1	39.948	3.8469	0.2881	150.86	0.29783	83.78	35.491	150.86	13.3530
15	Benzamide	C ₇ H ₇ NO	55-21-0	121.137	0.7371	0.25487	824	0.28571	403.00	8.938	824.00	2.8921
16	Benzene	C ₆ H ₆	71-43-2	78.112	1.0259	0.26666	562.05	0.28394	278.68	11.422	562.05	3.8472
17	Benzenethiol	C ₆ H ₆ S	108-98-5	110.177	0.83573	0.26326	689	0.30798	258.27	10.074	689.00	3.1745
18	Benzoic acid	C ₇ H ₆ O ₂	65-85-0	122.121	0.71587	0.24812	751	0.2857	395.45	8.894	751.00	2.8852
19	Benzonitrile	C ₇ H ₅ N	100-47-0	103.121	0.8552	0.26785	699.35	0.30523	260.40	10.011	699.35	3.1928
20	Benzophenone	C ₁₃ H ₁₀ O	119-61-9	182.218	0.43743	0.24833	830	0.27555	321.35	5.950	830.00	1.7615
21	Benzyl alcohol	C ₇ H ₈ O	100-51-6	108.138	0.59867	0.22849	720.15	0.23567	257.85	9.905	720.15	2.6201
22	Benzyl ethyl ether	C ₉ H ₁₂ O	539-30-0	136.191	0.60917	0.26925	662	0.2632	275.65	7.065	662.00	2.2625
23	Benzyl mercaptan	C ₇ H ₈ S	100-53-8	124.203	0.70797	0.25982	718	0.32144	243.95	8.862	718.00	2.7248
24	Biphenyl	C ₁₂ H ₁₀	92-52-4	154.208	0.52257	0.25833	773	0.27026	342.20	6.425	773.00	2.0229
25	Bromine	Br ₂	7726-95-6	159.808	2.1872	0.29527	584.15	0.3295	265.85	20.109	584.15	7.4075
26	Bromobenzene	C ₆ H ₅ Br	108-86-1	157.008	0.8226	0.26632	670.15	0.2821	242.43	9.909	670.15	3.0888
27	Bromoethane	C ₂ H ₅ Br	74-96-4	108.965	1.1908	0.25595	503.8	0.29152	154.55	15.833	503.80	4.6525
28	Bromomethane	CH ₃ Br	74-83-9	94.939	1.6762	0.26141	467	0.28402	179.47	20.640	467.00	6.4121
29	1,2-Butadiene	C ₄ H ₆	590-19-2	54.090	1.187	0.26114	452	0.3065	136.95	15.123	452.00	4.5455
30	1,3-Butadiene	C ₄ H ₆	106-99-0	54.090	1.2346	0.27216	425	0.28707	164.25	14.058	425.00	4.5363
31	Butane	C ₄ H ₁₀	106-97-8	58.122	1.0677	0.27188	425.12	0.28688	134.86	12.620	425.12	3.9271
32	1,2-Butanediol	C ₄ H ₁₀ O ₂	584-03-2	90.121	0.81696	0.24755	680	0.24535	220.00	11.734	680.00	3.3002
33	1,3-Butanediol	C ₄ H ₁₀ O ₂	107-88-0	90.121	0.81856	0.24967	676	0.22023	196.15	11.872	676.00	3.2786
34	1-Butanol	C ₄ H ₁₀ O	71-36-3	74.122	0.98279	0.26830	563.1	0.25488	183.85	12.035	563.10	3.6630
35	2-Butanol	C ₄ H ₁₀ O	78-92-2	74.122	0.9682	0.26244	535.9	0.26749	158.45	12.471	535.90	3.6892
36	1-Butene	C ₄ H ₈	106-98-9	56.106	1.0877	0.26454	419.5	0.2843	87.80	14.264	419.50	4.1117
37	cis-2-Butene	C ₄ H ₈	590-18-1	56.106	1.1591	0.27085	435.5	0.28116	134.26	13.894	435.50	4.2795
38	trans-2-Butene	C ₄ H ₈	624-64-6	56.106	1.1448	0.27154	428.6	0.28419	167.62	13.080	428.60	4.2160
39	Butyl acetate	C ₆ H ₁₂ O ₂	123-86-4	116.158	0.67794	0.2637	575.4	0.29318	199.65	8.337	575.40	2.5709
40	Butylbenzene	C ₁₀ H ₁₄	104-51-8	134.218	0.50812	0.25238	660.5	0.29373	185.30	7.026	660.50	2.0133
41	Butyl mercaptan	C ₄ H ₁₀ S	109-79-5	90.187	0.89458	0.27463	570.1	0.28512	157.46	10.585	570.10	3.2574
42	sec-Butyl mercaptan	C ₄ H ₁₀ S	513-53-1	90.187	0.89137	0.27365	554	0.2953	133.02	10.761	554.00	3.2573
43	1-Butyne	C ₄ H ₆	107-00-6	54.090	1.3409	0.27892	440	0.29661	147.43	14.901	440.00	4.8075
44	Butyraldehyde	C ₄ H ₈ O	123-72-8	72.106	1.0361	0.26731	537.2	0.28397	176.75	12.589	537.20	3.8760
45	Butyric acid	C ₄ H ₈ O ₂	107-92-6	88.105	0.88443	0.25828	615.7	0.248	267.95	11.087	615.70	3.4243
46	Butyronitrile	C ₄ H ₇ N	109-74-0	69.105	0.87533	0.24331	582.25	0.28586	161.25	13.047	582.25	3.5976
47	Carbon dioxide	CO ₂	124-38-9	44.010	2.768	0.26212	304.21	0.2908	216.58	26.828	304.21	10.5600
48	Carbon disulfide	CS ₂	75-15-0	76.141	1.7968	0.28749	552	0.3226	161.11	19.064	552.00	6.2500
49	Carbon monoxide	CO	630-08-0	28.010	2.897	0.27532	132.92	0.2813	68.15	30.180	132.92	10.5220
50	Carbon tetrachloride	CCl ₄	56-23-5	153.823	0.99835	0.274	556.35	0.287	250.33	10.843	556.35	3.6436
51	Carbon tetrafluoride	CF ₄	75-73-0	88.004	1.955	0.27884	227.51	0.28571	89.56	21.211	227.51	7.0112
52	Chlorine	Cl ₂	7782-50-5	70.906	2.23	0.27645	417.15	0.2926	172.12	24.242	417.15	8.0666
53	Chlorobenzene	C ₆ H ₅ Cl	108-90-7	112.557	0.8711	0.26805	632.35	0.2799	227.95	10.385	632.35	3.2498
54	Chloroethane	C ₂ H ₅ Cl	75-00-3	64.514	1.3	0.26019	460.35	0.27155	134.80	17.016	460.35	4.9963
55	Chloroform	CHCl ₃	67-66-3	119.378	1.0841	0.2581	536.4	0.2741	209.63	13.702	536.40	4.2003
56	Chloromethane	CH ₃ Cl	74-87-3	50.488	1.817	0.25877	416.25	0.2833	175.43	22.347	416.25	7.0217
57	1-Chloropropane	C ₃ H ₇ Cl	540-54-5	78.541	1.087	0.26832	503.15	0.28055	150.35	13.328	503.15	4.0511
58	2-Chloropropane	C ₃ H ₇ Cl	75-29-6	78.541	1.1202	0.27669	489	0.27646	155.97	12.855	489.00	4.0486
59	m-Cresol	C ₇ H ₈ O	108-39-4	108.138	0.9061	0.28268	705.85	0.2707	285.39	9.612	705.85	3.2054

60	<i>o</i> -Cresol	C ₇ H ₈ O	95-48-7	108.138	1.0861	0.30624	697.55	0.30587	304.19	9.575	697.55	3.5466
61	<i>p</i> -Cresol	C ₇ H ₈ O	106-44-5	108.138	1.1503	0.31861	704.65	0.30104	307.93	9.449	704.65	3.6104
62	Cumene	C ₉ H ₁₂	98-82-8	120.192	0.58711	0.25583	631	0.28498	177.14	7.939	631.00	2.2949
63	Cyanogen	C ₂ N ₂	460-19-5	52.035	1.0743	0.20948	400.15	0.20724	245.25	18.520	400.15	5.1284
64	Cyclobutane	C ₄ H ₆	287-23-0	56.106	1.3931	0.29255	459.93	0.24913	182.48	14.074	459.93	4.7619
65	Cyclohexane	C ₆ H ₁₂	110-82-7	84.159	0.88998	0.27376	553.8	0.28571	279.69	9.380	553.80	3.2509
66	Cyclohexanol	C ₆ H ₁₂ O	108-93-0	100.159	0.8243	0.26545	650.1	0.28495	296.60	9.469	650.10	3.1053
67	Cyclohexanone	C ₆ H ₁₀ O	108-94-1	98.143	0.86464	0.26888	653	0.29943	242.00	10.090	653.00	3.2157
68	Cyclohexene	C ₆ H ₁₀	110-83-8	82.144	0.92997	0.27056	560.4	0.28943	169.67	11.160	560.40	3.4372
69	Cyclopentane	C ₅ H ₁₀	287-92-3	70.133	1.0897	0.28356	511.7	0.25142	179.28	11.906	511.70	3.8429
70	Cyclopentene	C ₅ H ₈	142-29-0	68.117	1.1035	0.27035	507	0.28699	138.13	13.470	507.00	4.0817
71	Cyclopropane	C ₃ H ₆	75-19-4	42.080	1.7411	0.28205	398	0.29598	145.59	18.658	398.00	6.1730
72	Cyclohexyl mercaptan	C ₆ H ₁₂ S	1569-69-3	116.224	0.78578	0.27882	664	0.31067	189.64	8.905	664.00	2.8182
73	Decanal	C ₁₀ H ₂₀ O	112-31-2	156.265	0.46802	0.27146	674.2	0.26869	267.15	5.383	674.20	1.7241
74	Decane	C ₁₀ H ₂₂	124-18-5	142.282	0.41084	0.25175	617.7	0.28571	243.51	5.393	617.70	1.6320
75	Decanoic acid	C ₁₀ H ₂₀ O ₂	334-48-5	172.265	0.39348	0.2492	722.1	0.28571	304.55	5.181	722.10	1.5790
76	1-Decanol	C ₁₀ H ₂₂ O	112-30-1	158.281	0.38208	0.24645	688	0.26125	280.05	5.261	688.00	1.5503
77	1-Decene	C ₁₀ H ₂₀	872-05-9	140.266	0.43981	0.25661	616.6	0.29148	206.89	5.733	616.60	1.7139
78	Decyl mercaptan	C ₁₀ H ₂₂ S	143-10-2	174.347	0.44289	0.27636	696	0.27668	247.56	5.005	696.00	1.6026
79	1-Decyne	C ₁₀ H ₁₈	764-93-2	138.250	0.46877	0.25875	619.85	0.29479	229.15	5.895	619.85	1.8117
80	Deuterium	D ₂	7782-39-0	4.032	5.2115	0.315	38.35	0.28571	18.73	42.945	38.35	16.5440
81	1,1-Dibromoethane	C ₂ H ₄ Br ₂	557-91-5	187.861	0.95523	0.26364	628	0.29825	210.15	11.799	628.00	3.6232
82	1,2-Dibromoethane	C ₂ H ₄ Br ₂	106-93-4	187.861	1.0132	0.26634	650.15	0.28571	282.85	11.704	650.15	3.8042
83	Dibromomethane	CH ₂ Br ₂	74-95-3	173.835	1.1136	0.24834	611	0.27583	220.60	15.358	611.00	4.4842
84	Dibutyl ether	C ₈ H ₁₈ O	142-96-1	130.228	0.55941	0.27243	584.1	0.29932	175.30	6.607	584.10	2.0534
85	<i>m</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	541-73-1	147.002	0.74495	0.26147	683.95	0.31526	248.39	9.121	683.95	2.8491
86	<i>o</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	95-50-1	147.002	0.74404	0.26112	705	0.30815	256.15	9.166	705.00	2.8494
87	<i>p</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	106-46-7	147.002	0.74858	0.26276	684.75	0.30788	326.14	8.518	684.75	2.8489
88	1,1-Dichloroethane	C ₂ H ₄ Cl ₂	75-34-3	98.959	1.1055	0.26533	523	0.287	176.19	13.549	523.00	4.1665
89	1,2-Dichloroethane	C ₂ H ₄ Cl ₂	107-06-2	98.959	1.2591	0.27698	561.6	0.30492	237.49	13.462	561.60	4.5458
90	Dichloromethane	CH ₂ Cl ₂	75-09-2	84.933	1.3897	0.25678	510	0.2902	178.01	17.974	510.00	5.4120
91	1,1-Dichloropropane	C ₃ H ₆ Cl ₂	78-99-9	112.986	0.9551	0.27794	560	0.24132	200.00	10.862	560.00	3.4364
92	1,2-Dichloropropane	C ₃ H ₆ Cl ₂	78-87-5	112.986	0.89833	0.26142	572	0.2868	172.71	11.526	572.00	3.4363
93	Diethanol amine	C ₄ H ₁₁ NO ₂	111-42-2	105.136	0.68184	0.23796	736.6	0.2062	301.15	10.390	736.60	2.8654
94	Diethyl amine	C ₄ H ₁₁ N	109-89-7	73.137	0.85379	0.25675	496.6	0.27027	223.35	10.575	496.60	3.3254
95	Diethyl ether	C ₄ H ₁₀ O	60-29-7	74.122	0.9554	0.26847	466.7	0.2814	156.85	11.487	466.70	3.5587
96	Diethyl sulfide	C ₄ H ₁₀ S	352-93-2	90.187	0.82227	0.26314	557.15	0.27369	169.20	10.470	557.15	3.1248
97	1,1-Difluoroethane	C ₂ H ₄ F ₂	75-37-6	66.050	1.4345	0.25774	386.44	0.28178	154.56	18.006	386.44	5.5657
98	1,2-Difluoroethane	C ₂ H ₄ F ₂	624-72-6	66.050	1.173	0.22856	445	0.28571	215.00	17.424	445.00	5.1321
99	Diffuoromethane	CH ₂ F ₂	75-10-5	52.023	1.9973	0.24653	351.26	0.28153	136.95	27.399	351.26	8.6070
100	Di-isopropyl amine	C ₆ H ₁₅ N	108-18-9	101.190	0.6181	0.25786	523.1	0.271	176.85	8.054	523.10	2.3970
101	Di-isopropyl ether	C ₆ H ₁₄ O	108-20-3	102.175	0.69213	0.26974	500.05	0.28571	187.65	8.067	500.05	2.5659
102	Di-isopropyl ketone	C ₇ H ₁₄ O	565-80-0	114.185	0.64619	0.26881	576	0.28036	204.81	7.680	576.00	2.4039
103	1,1-Dimethoxyethane	C ₄ H ₁₀ O ₂	534-15-6	90.121	0.89368	0.26599	507.8	0.28571	159.95	11.029	507.80	3.3598
104	1,2-Dimethoxypropane	C ₅ H ₁₂ O ₂	7778-85-0	104.148	0.76327	0.26742	543	0.28571	226.10	8.843	543.00	2.8542
105	Dimethyl acetylene	C ₄ H ₆	503-17-3	54.090	1.1717	0.25895	473.2	0.27289	240.91	13.767	473.20	4.5248
106	Dimethyl amine	C ₂ H ₇ N	124-40-3	45.084	1.5436	0.27784	437.2	0.2572	180.96	16.964	437.20	5.5557
107	2,3-Dimethylbutane	C ₆ H ₁₄	79-29-8	86.175	0.7565	0.27305	500	0.27408	145.19	9.031	500.00	2.7706
108	1,1-Dimethylcyclohexane	C ₈ H ₁₆	590-66-9	112.213	0.55873	0.25143	591.15	0.27758	239.66	7.342	591.15	2.2222
109	<i>cis</i> -1,2-Dimethylcyclohexane	C ₈ H ₁₆	2207-01-4	112.213	0.52953	0.24358	606.15	0.26809	223.16	7.578	606.15	2.1739
110	<i>trans</i> -1,2-Dimethylcyclohexane	C ₈ H ₁₆	6876-23-9	112.213	0.54405	0.25026	596.15	0.2658	184.99	7.626	596.15	2.1739
111	Dimethyl disulfide	C ₂ H ₆ S ₂	624-92-0	94.199	1.1058	0.27866	615	0.31082	188.44	12.413	615.00	3.9683
112	Dimethyl ether	C ₂ H ₆ O	115-10-6	46.068	1.5693	0.2679	400.1	0.2882	131.65	18.950	400.10	5.8578
113	<i>N,N</i> -Dimethyl formamide	C ₃ H ₇ NO	68-12-2	73.094	0.89615	0.23478	649.6	0.28091	212.72	13.954	649.60	3.8170
114	2,3-Dimethylpentane	C ₇ H ₁₆	565-59-3	100.202	0.72352	0.28629	537.3	0.27121	160.00	7.874	537.30	2.5272
115	Dimethyl phthalate	C ₁₀ H ₁₀ O ₄	131-11-3	194.184	0.47977	0.25428	766	0.30722	274.18	6.233	766.00	1.8868
116	Dimethylsilane	C ₂ H ₆ Si	1111-74-6	60.170	1.0214	0.26351	402	0.28421	122.93	12.898	402.00	3.8761
117	Dimethyl sulfide	C ₂ H ₆ S	75-18-3	62.134	1.4029	0.27991	503.04	0.2741	174.88	15.556	503.04	5.0120
118	Dimethyl sulfoxide	C ₂ H ₆ OS	67-68-5	78.133	1.1096	0.25189	729	0.3311	291.67	14.111	729.00	4.4051
119	Dimethyl terephthalate	C ₁₀ H ₁₀ O ₄	120-61-6	194.184	0.50824	0.26885	772	0.2612	413.80	5.538	772.00	1.8904
120	1,4-Dioxane	C ₄ H ₈ O ₂	123-91-1	88.105	1.1819	0.2813	587	0.3047	284.95	11.838	587.00	4.2016
121	Diphenyl ether	C ₁₂ H ₁₀ O	101-84-8	170.207	0.52133	0.26218	766.8	0.31033	300.03	6.265	766.80	1.9884
122	Dipropyl amine	C ₆ H ₁₅ N	142-84-7	101.190	0.659	0.26428	550	0.2766	210.15	7.993	550.00	2.4936

TABLE 2-32 Densities of Inorganic and Organic Liquids (mol/dm³) (Continued)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1	C2	C3	C4	$T_{\min}$, K	Density at $T_{\min}$	$T_{\max}$, K	Density at $T_{\max}$
123	Dodecane	C ₁₂ H ₂₆	112-40-3	170.335	0.33267	0.24664	658	0.28571	263.57	4.521	658.00	1.3490
124	Eicosane	C ₂₀ H ₄₂	112-95-8	282.547	0.18166	0.23351	768	0.28571	309.58	2.729	768.00	0.7780
125	Ethane	C ₂ H ₆	74-84-0	30.069	1.9122	0.27937	305.32	0.29187	90.35	21.640	305.32	6.8447
126	Ethanol	C ₂ H ₆ O	64-17-5	46.068	1.6288	0.27469	514	0.23178	159.05	19.410	514.00	5.9296
127	Ethyl acetate	C ₄ H ₈ O ₂	141-78-6	88.105	0.8996	0.25856	523.3	0.278	189.60	11.478	523.30	3.4793
128	Ethyl amine	C ₂ H ₇ N	75-04-7	45.084	1.0936	0.22636	456.15	0.25522	192.15	17.588	456.15	4.8312
129	Ethylbenzene	C ₈ H ₁₀	100-41-4	106.165	0.70041	0.26162	617.15	0.28454	178.20	9.041	617.15	2.6772
130	Ethyl benzoate	C ₈ H ₁₀ O ₂	93-89-0	150.175	0.48864	0.23894	698	0.28421	238.45	7.291	698.00	2.0450
131	2-Ethyl butanoic acid	C ₈ H ₁₆ O ₂	88-09-5	116.158	0.66085	0.25707	655	0.31103	258.15	8.220	655.00	2.5707
132	Ethyl butyrate	C ₈ H ₁₆ O ₂	105-54-4	116.158	0.63566	0.25613	571	0.27829	175.15	8.491	571.00	2.4818
133	Ethylcyclohexane	C ₈ H ₁₆	1678-91-7	112.213	0.61587	0.26477	609.15	0.28054	161.84	7.868	609.15	2.3261
134	Ethylcyclopentane	C ₇ H ₁₄	1640-89-7	98.186	0.71751	0.26903	569.5	0.27733	134.71	9.018	569.50	2.6670
135	Ethylene	C ₂ H ₄	74-85-1	28.053	2.0961	0.27657	282.34	0.29147	104.00	23.326	282.34	7.5789
136	Ethylenediamine	C ₂ H ₈ N ₂	107-15-3	60.098	0.7842	0.20702	593	0.20254	284.29	15.055	593.00	3.7880
137	Ethylene glycol	C ₂ H ₆ O ₂	107-21-1	62.068	1.315	0.25125	720	0.21868	260.15	18.310	720.00	5.2338
138	Ethyleneimine	C ₂ H ₅ N	151-56-4	43.068	1.3462	0.23289	537	0.23357	195.20	21.450	537.00	5.7804
139	Ethylene oxide	C ₂ H ₄ O	75-21-8	44.053	1.836	0.26024	469.15	0.2696	160.65	23.477	469.15	7.0550
140	Ethyl formate	C ₃ H ₆ O ₂	109-94-4	74.079	1.1343	0.26168	508.4	0.2791	193.55	14.006	508.40	4.3347
141	2-Ethyl hexanoic acid	C ₈ H ₁₆ O ₂	149-57-5	144.211	0.47428	0.25028	674.6	0.25442	235.00	6.563	674.60	1.8950
142	Ethylhexyl ether	C ₈ H ₁₈ O	5756-43-4	130.228	0.55729	0.2714	583	0.29538	180.00	6.612	583.00	2.0534
143	Ethylisopropyl ether	C ₆ H ₁₄ O	625-54-7	88.148	0.8185	0.26929	489	0.30621	140.00	9.924	489.00	3.0395
144	Ethylisopropyl ketone	C ₈ H ₁₆ O	565-69-5	100.159	0.68162	0.25152	567	0.3182	204.15	8.975	567.00	2.7100
145	Ethyl mercaptan	C ₂ H ₆ S	75-08-1	62.134	1.3047	0.2694	499.15	0.27866	125.26	16.242	499.15	4.8430
146	Ethyl propionate	C ₅ H ₁₀ O ₂	105-37-3	102.132	0.7405	0.25563	546	0.2795	199.25	9.632	546.00	2.8968
147	Ethylpropyl ether	C ₆ H ₁₄ O	628-32-0	88.148	0.7908	0.266	500.23	0.292	145.65	9.847	500.23	2.9729
148	Ethyltrichlorosilane	C ₂ H ₅ Cl ₃ Si	115-21-9	163.506	0.58579	0.24246	559.95	0.29509	167.55	8.653	559.95	2.4160
149	Fluorine	F ₂	7782-41-4	37.997	4.2895	0.28587	144.12	0.28776	53.48	44.888	144.12	15.0050
150	Fluorobenzene	C ₆ H ₅ F	462-06-6	96.102	1.0146	0.27277	560.09	0.28291	230.94	11.374	560.09	3.7196
151	Fluoroethane	C ₂ H ₅ F	353-36-6	48.060	1.6525	0.27099	375.31	0.2442	129.95	19.785	375.31	6.0980
152	Fluoromethane	CH ₃ F	593-53-3	34.033	2.1854	0.24725	317.42	0.27558	131.35	29.526	317.42	8.8388
153	Formaldehyde	CH ₂ O	50-00-0	30.026	1.9415	0.22309	408	0.28571	181.15	30.945	408.00	8.7028
154	Formamide	CH ₃ NO	75-12-7	45.041	1.2486	0.20352	771	0.25178	275.60	25.488	771.00	6.1350
155	Formic acid	CH ₂ O ₂	64-18-6	46.026	1.938	0.24225	588	0.24435	281.45	26.806	588.00	8.0000
156	Furan	C ₄ H ₄ O	110-00-9	68.074	1.1339	0.24741	490.15	0.2612	187.55	15.702	490.15	4.5831
157	Helium-4	He	7440-59-7	4.003	7.2475	0.41865	5.2	0.24096	2.20	37.115	5.20	17.3120
158	Heptadecane	C ₁₇ H ₃₆	629-78-7	240.468	0.21897	0.23642	736	0.28571	295.13	3.219	736.00	0.9262
159	Heptanal	C ₇ H ₁₄ O	111-71-7	114.185	0.59006	0.25609	616.8	0.28384	229.80	7.600	616.80	2.3041
160	Heptane	C ₇ H ₁₆	142-82-5	100.202	0.61259	0.26211	540.2	0.28141	182.57	7.700	540.20	2.3371
161	Heptanoic acid	C ₇ H ₁₄ O ₂	111-14-8	130.185	0.53066	0.24729	677.3	0.28289	265.83	7.221	677.30	2.1459
162	1-Heptanol	C ₇ H ₁₆ O	111-70-6	116.201	0.55687	0.24725	632.3	0.31471	239.15	7.502	632.30	2.2523
163	2-Heptanol	C ₇ H ₁₆ O	543-49-7	116.201	0.57114	0.25534	608.3	0.26487	230.00	7.454	608.30	2.2368
164	3-Heptanone	C ₇ H ₁₄ O	106-35-4	114.185	0.59268	0.25663	606.6	0.27766	234.15	7.575	606.60	2.3095
165	2-Heptanone	C ₇ H ₁₄ O	110-43-0	114.185	0.58247	0.25279	611.4	0.29818	238.15	7.551	611.40	2.3042
166	1-Heptene	C ₇ H ₁₄	592-76-7	98.186	0.66016	0.26657	537.4	0.28571	154.12	8.226	537.40	2.4765
167	Heptyl mercaptan	C ₇ H ₁₆ S	1639-09-4	132.267	0.58622	0.2726	645	0.29644	229.92	6.728	645.00	2.1505
168	1-Heptyne	C ₇ H ₁₂	628-71-7	96.170	0.67304	0.26045	547	0.28388	192.22	8.492	547.00	2.5841
169	Hexadecane	C ₁₆ H ₃₄	544-76-3	226.441	0.23289	0.23659	723	0.28571	291.31	3.415	723.00	0.9844
170	Hexanal	C ₆ H ₁₂ O	66-25-1	100.159	0.71899	0.26531	591	0.27628	217.15	8.724	591.00	2.7100
171	Hexane	C ₆ H ₁₄	110-54-3	86.175	0.70824	0.26411	507.6	0.27537	177.83	8.747	507.60	2.6816
172	Hexanoic acid	C ₆ H ₁₂ O ₂	142-62-1	116.158	0.62833	0.25598	660.2	0.25304	269.25	8.096	660.20	2.4546
173	1-Hexanol	C ₆ H ₁₄ O	111-27-3	102.175	0.70093	0.26776	611.3	0.24919	228.55	8.456	611.30	2.6178
174	2-Hexanol	C ₆ H ₁₄ O	626-93-7	102.175	0.67393	0.25948	585.3	0.26552	223.00	8.518	585.30	2.5972
175	2-Hexanone	C ₆ H ₁₂ O	591-78-6	100.159	0.67816	0.25634	587.61	0.28365	217.35	8.732	587.61	2.6455
176	3-Hexanone	C ₆ H ₁₂ O	589-38-8	100.159	0.67666	0.25578	582.82	0.27746	217.50	8.763	582.82	2.6455
177	1-Hexene	C ₆ H ₁₂	592-41-6	84.159	0.76925	0.26809	504	0.28571	133.39	9.581	504.00	2.8694
178	3-Hexyne	C ₆ H ₁₀	928-49-4	82.144	0.78045	0.26065	544	0.28571	170.05	10.021	544.00	2.9942
179	Hexyl mercaptan	C ₆ H ₁₄ S	111-31-9	118.240	0.66372	0.27345	623	0.29185	192.62	7.773	623.00	2.4272
180	1-Hexyne	C ₆ H ₁₀	693-02-7	82.144	0.84427	0.27185	516.2	0.2771	141.25	10.230	516.20	3.1056
181	2-Hexyne	C ₆ H ₁₀	764-35-2	82.144	0.76277	0.25248	549	0.31611	183.65	10.133	549.00	3.0211

182	Hydrazine	H ₄ N ₂	302-01-2	32.045	1.0516	0.16613	653.15	0.1898	274.69	31.934	653.15	6.3300
183	Hydrogen	H ₂	1333-74-0	2.016	5.414	0.34893	33.19	0.2706	13.95	38.487	33.19	15.5160
184	Hydrogen bromide	HBr	10035-10-6	80.912	2.832	0.2832	363.15	0.28571	185.15	27.985	363.15	10.0000
185	Hydrogen chloride	HCl	7647-01-0	36.461	3.342	0.2729	324.65	0.3217	158.97	34.854	324.65	12.2460
186	Hydrogen cyanide	CHN	74-90-8	27.025	1.3413	0.18589	456.65	0.28206	259.83	27.202	456.65	7.2156
187	Hydrogen fluoride	HF	7664-39-3	20.006	2.5635	0.1766	461.15	0.3733	189.79	60.203	461.15	14.5160
188	Hydrogen sulfide	H ₂ S	7783-06-4	34.081	2.7672	0.27369	373.53	0.29015	187.68	29.130	373.53	10.1110
189	Isobutyric acid	C ₄ H ₈ O ₂	79-31-2	88.105	0.88575	0.25736	605	0.26265	227.15	11.420	605.00	3.4417
190	Isopropyl amine	C ₃ H ₉ N	75-31-0	59.110	1.2801	0.2828	471.85	0.2972	177.95	13.561	471.85	4.5265
191	Malonic acid	C ₃ H ₄ O ₄	141-82-2	104.061	0.84266	0.217	805	0.28571	407.95	13.533	805.00	3.8832
192	Methacrylic acid	C ₄ H ₆ O ₂	79-41-4	86.089	0.87025	0.24383	662	0.28571	288.15	11.834	662.00	3.5691
193	Methane	CH ₄	74-82-8	16.042	2.9214	0.28976	190.56	0.28881	90.69	28.180	190.56	10.0820
194	Methanol	CH ₃ O	67-56-1	32.042	2.3267	0.27073	512.5	0.24713	175.47	27.915	512.50	8.5942
195	N-Methyl acetamide	C ₃ H ₇ NO	79-16-3	73.094	0.88268	0.23568	718	0.27379	301.15	13.012	718.00	3.7452
196	Methyl acetate	C ₃ H ₆ O ₂	79-20-9	74.079	1.13	0.2593	506.55	0.2764	175.15	14.475	506.55	4.3579
197	Methyl acetylene	C ₃ H ₄	74-99-7	40.064	1.6085	0.26436	402.4	0.27987	170.45	19.031	402.40	6.0845
198	Methyl acrylate	C ₄ H ₆ O ₂	96-33-3	86.089	0.97286	0.26267	536	0.2508	196.32	12.203	536.00	3.7037
199	Methyl amine	CH ₃ N	74-89-5	31.057	1.39	0.21405	430.05	0.2275	179.69	25.378	430.05	6.4938
200	Methyl benzoate	C ₈ H ₈ O ₂	93-58-3	136.148	0.53382	0.23274	693	0.28147	260.75	8.220	693.00	2.2936
201	3-Methyl-1,2-butadiene	C ₅ H ₈	598-25-4	68.117	0.84623	0.24625	490	0.29041	159.53	11.994	490.00	3.4365
202	2-Methylbutane	C ₅ H ₁₂	78-78-4	72.149	0.91991	0.27815	460.4	0.28667	113.25	10.764	460.40	3.3072
203	2-Methylbutanoic acid	C ₅ H ₁₀ O ₂	116-53-0	102.132	0.72762	0.25244	643	0.28571	193.00	9.992	643.00	2.8823
204	3-Methyl-1-butanol	C ₅ H ₁₂ O	123-51-3	88.148	0.80828	0.26783	577.2	0.23588	155.95	10.254	577.20	3.0179
205	2-Methyl-1-butene	C ₅ H ₁₀	563-46-2	70.133	0.91619	0.26752	465	0.28164	135.58	11.332	465.00	3.4248
206	2-Methyl-2-butene	C ₅ H ₁₀	513-35-9	70.133	0.93391	0.27275	470	0.2578	139.39	11.216	470.00	3.4241
207	2-Methyl-1-butene-3-yne	C ₅ H ₆	78-80-8	66.101	1.1157	0.27671	492	0.30821	160.15	12.581	492.00	4.0320
208	Methylbutyl ether	C ₅ H ₁₂ O	628-28-4	88.148	0.8363	0.27514	512.74	0.27553	157.48	9.758	512.74	3.0395
209	Methylbutyl sulfide	C ₅ H ₁₂ S	628-29-5	104.214	0.75509	0.27183	593	0.29127	175.30	9.006	593.00	2.7778
210	3-Methyl-1-butyne	C ₅ H ₈	598-23-2	68.117	0.94575	0.26008	463.2	0.30807	183.45	11.519	463.20	3.6364
211	Methyl butyrate	C ₅ H ₁₀ O ₂	623-42-7	102.132	0.76983	0.26173	554.5	0.26879	187.35	9.764	554.50	2.9413
212	Methylchlorosilane	CH ₃ ClSi	993-00-0	80.589	1.0674	0.26257	442	0.26569	139.05	13.626	442.00	4.0652
213	Methylcyclohexane	C ₇ H ₁₄	108-87-2	98.186	0.73109	0.26971	572.1	0.29185	146.58	9.017	572.10	2.7107
214	1-Methylcyclohexanol	C ₇ H ₁₄ O	590-67-0	114.185	0.7013	0.266	686	0.28571	285.15	8.209	686.00	2.6365
215	cis-2-Methylcyclohexanol	C ₇ H ₁₄ O	7443-70-1	114.185	0.70973	0.26544	614	0.26016	280.15	8.293	614.00	2.6738
216	trans-2-Methylcyclohexanol	C ₇ H ₁₄ O	7443-52-9	114.185	0.72836	0.27241	617	0.2478	269.15	8.263	617.00	2.6738
217	Methylcyclopentane	C ₆ H ₁₂	96-37-7	84.159	0.84758	0.27037	532.7	0.28258	130.73	10.491	532.70	3.1349
218	1-Methylcyclopentene	C ₆ H ₁₀	693-89-0	82.144	0.88824	0.26914	542	0.27874	146.62	10.980	542.00	3.3003
219	3-Methylcyclopentene	C ₆ H ₁₀	1120-62-3	82.144	0.9109	0.276	526	0.26756	115.00	11.014	526.00	3.3004
220	Methyldichlorosilane	CH ₃ Cl ₂ Si	75-54-7	115.034	0.97608	0.28209	483	0.22529	182.55	10.789	483.00	3.4602
221	Methylethyl ether	C ₃ H ₈ O	540-67-0	60.095	1.2635	0.27878	437.8	0.2744	160.00	13.995	437.80	4.5322
222	Methylethyl ketone	C ₄ H ₈ O	78-93-3	72.106	0.93767	0.25035	535.5	0.29964	186.48	12.663	535.50	3.7454
223	Methylethyl sulfide	C ₃ H ₈ S	624-89-5	76.161	1.067	0.27102	533	0.29364	167.23	12.671	533.00	3.9370
224	Methyl formate	C ₂ H ₄ O ₂	107-31-3	60.052	1.525	0.2634	487.2	0.2806	174.15	18.811	487.20	5.7897
225	Methylisobutyl ether	C ₆ H ₁₂ O	625-44-5	88.148	0.84005	0.27638	497	0.27645	150.00	9.738	497.00	3.0395
226	Methylisobutyl ketone	C ₆ H ₁₂ O	108-10-1	100.159	0.71687	0.26453	574.6	0.28918	189.15	8.862	574.60	2.7100
227	Methyl isocyanate	C ₂ H ₃ NO	624-83-9	57.051	1.0228	0.20692	488	0.28571	256.15	17.666	488.00	4.9430
228	Methylisopropyl ether	C ₄ H ₁₀ O	598-53-8	74.122	0.97887	0.27017	464.48	0.28998	127.93	11.933	464.48	3.6232
229	Methylisopropyl ketone	C ₅ H ₁₀ O	563-80-4	86.132	0.86567	0.26836	553.4	0.28364	180.15	10.460	553.40	3.2258
230	Methylisopropyl sulfide	C ₄ H ₁₀ S	1551-21-9	90.187	0.78912	0.25915	553.1	0.26512	171.64	10.352	553.10	3.0450
231	Methyl mercaptan	CH ₃ S	74-93-1	48.107	1.9323	0.28018	469.95	0.28523	150.18	21.564	469.95	6.8966
232	Methyl methacrylate	C ₅ H ₈ O ₂	80-62-6	100.116	0.7761	0.25068	566	0.29773	224.95	10.176	566.00	3.0960
233	2-Methyloctanoic acid	C ₉ H ₁₈ O ₂	3004-93-1	158.238	0.4416	0.2521	694	0.28532	240.00	5.938	694.00	1.7517
234	2-Methylpentane	C ₆ H ₁₄	107-83-5	86.175	0.72701	0.26754	497.7	0.28268	119.55	9.204	497.70	2.7174
235	Methyl pentyl ether	C ₆ H ₁₄ O	628-80-8	102.175	0.71004	0.26981	546.49	0.29974	176.00	8.445	546.49	2.6316
236	2-Methylpropane	C ₄ H ₁₀	75-28-5	58.122	1.0631	0.27506	407.8	0.2758	113.54	12.574	407.80	3.8650
237	2-Methyl-2-propanol	C ₄ H ₁₀ O	75-65-0	74.122	0.92128	0.25442	506.2	0.27586	298.97	10.556	506.20	3.6211
238	2-Methyl propene	C ₄ H ₈	115-11-7	56.106	1.1446	0.2724	417.9	0.28172	132.81	13.507	417.90	4.2019
239	Methyl propionate	C ₄ H ₈ O ₂	554-12-1	88.105	0.9147	0.2594	530.6	0.2774	185.65	11.678	530.60	3.5262
240	Methylpropyl ether	C ₄ H ₁₀ O	557-17-5	74.122	0.96145	0.26536	476.25	0.30088	133.97	12.043	476.25	3.6232
241	Methylpropyl sulfide	C ₄ H ₁₀ S	3877-15-4	90.187	0.87496	0.26862	565	0.30259	160.17	10.689	565.00	3.2572
242	Methylsilane	CH ₃ Si	992-94-9	46.144	1.3052	0.26757	352.5	0.28799	116.34	15.791	352.50	4.8780
243	alpha-Methyl styrene	C ₉ H ₈	98-83-9	118.176	0.64856	0.25877	654	0.31444	249.95	8.010	654.00	2.5063
244	Methyl tert-butyl ether	C ₅ H ₁₂ O	1634-04-4	88.148	0.928	0.289	497.1	0.286	164.55	9.710	497.10	3.2111

TABLE 2-32 Densities of Inorganic and Organic Liquids (mol/dm³) (Concluded)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1	C2	C3	C4	$T_{\min}$, K	Density at $T_{\min}$	$T_{\max}$, K	Density at $T_{\max}$
245	Methyl vinyl ether	C ₃ H ₆ O	107-25-5	58.079	1.2587	0.26433	437	0.25819	151.15	15.691	437.00	4.7619
246	Naphthalene	C ₁₀ H ₈	91-20-3	128.171	0.6348	0.25838	748.4	0.27727	333.15	7.755	748.40	2.4568
247	Neon	Ne	7440-01-9	20.180	7.3718	0.3067	44.4	0.2786	24.56	61.796	44.40	24.0360
248	Nitroethane	C ₂ H ₅ NO ₂	79-24-3	75.067	1.0024	0.23655	593	0.278	183.63	15.556	593.00	4.2376
249	Nitrogen	N ₂	7727-37-9	28.013	3.2091	0.2861	126.2	0.2966	63.15	31.063	126.20	11.2170
250	Nitrogen trifluoride	F ₃ N	7783-54-2	71.002	2.3736	0.2817	234	0.29529	66.46	26.555	234.00	8.4260
251	Nitromethane	CH ₃ NO ₂	75-52-5	61.040	1.3728	0.23793	588.15	0.29601	244.60	19.632	588.15	5.7698
252	Nitrous oxide	N ₂ O	10024-97-2	44.013	2.781	0.27244	309.57	0.2882	182.30	27.928	309.57	10.2080
253	Nitric oxide	NO	10102-43-9	30.006	5.246	0.3044	180.15	0.242	109.50	44.487	180.15	17.2340
254	Nonadecane	C ₁₉ H ₄₀	629-92-5	268.521	0.19199	0.23337	758	0.28571	305.04	2.889	758.00	0.8227
255	Nonanal	C ₉ H ₁₈ O	124-19-6	142.239	0.49587	0.26135	658	0.30736	255.15	6.017	658.00	1.8973
256	Nonane	C ₉ H ₂₀	111-84-2	128.255	0.46321	0.25444	594.6	0.28571	219.66	6.043	594.60	1.8210
257	Nonanoic acid	C ₉ H ₁₈ O ₂	112-05-0	158.238	0.41582	0.24284	710.7	0.30036	285.55	5.759	710.70	1.7123
258	1-Nonanol	C ₉ H ₂₀ O	143-08-8	144.255	0.43682	0.25161	670.9	0.2498	268.15	5.850	670.90	1.7361
259	2-Nonanol	C ₉ H ₂₀ O	628-99-9	144.255	0.41687	0.24056	649.5	0.2916	238.15	6.031	649.50	1.7329
260	1-Nonene	C ₉ H ₁₈	124-11-8	126.239	0.48661	0.25722	593.1	0.28571	191.91	6.372	593.10	1.8918
261	Nonyl mercaptan	C ₉ H ₂₀ S	1455-21-6	160.320	0.47377	0.27052	681	0.30284	253.05	5.453	681.00	1.7513
262	1-Nonyne	C ₉ H ₁₆	3452-09-3	124.223	0.52152	0.25918	598.05	0.29177	223.15	6.537	598.05	2.0122
263	Octadecane	C ₁₈ H ₃₈	593-45-3	254.494	0.20448	0.23474	747	0.28571	301.31	3.042	747.00	0.8711
264	Octanal	C ₈ H ₁₆ O	124-13-0	128.212	0.53636	0.26174	638.9	0.26348	246.00	6.664	638.90	2.0492
265	Octane	C ₈ H ₁₈	111-65-9	114.229	0.5266	0.25693	568.7	0.28571	216.38	6.705	568.70	2.0500
266	Octanoic acid	C ₈ H ₁₆ O ₂	124-07-2	144.211	0.48251	0.25196	694.26	0.26842	289.65	6.311	694.26	1.9150
267	1-Octanol	C ₈ H ₁₈ O	111-87-5	130.228	0.48979	0.24931	652.3	0.27824	257.65	6.574	652.30	1.9646
268	2-Octanol	C ₈ H ₁₈ O	123-96-6	130.228	0.50726	0.25972	629.8	0.22	241.55	6.563	629.80	1.9531
269	2-Octanone	C ₈ H ₁₆ O	111-13-7	128.212	0.50006	0.24851	632.7	0.29942	252.85	6.648	632.70	2.0122
270	3-Octanone	C ₈ H ₁₆ O	106-68-3	128.212	0.5108	0.25386	627.7	0.26735	255.55	6.628	627.70	2.0121
271	1-Octene	C ₈ H ₁₆	111-66-0	112.213	0.55449	0.25952	566.9	0.28571	171.45	7.216	566.90	2.1366
272	Octyl mercaptan	C ₈ H ₁₈ S	111-88-6	146.294	0.52577	0.27234	667.3	0.30063	223.95	6.099	667.30	1.9306
273	1-Octyne	C ₈ H ₁₄	629-05-0	110.197	0.58945	0.26052	574	0.28532	193.55	7.483	574.00	2.2626
274	Oxalic acid	C ₂ H ₂ O ₄	144-62-7	90.035	1.0501	0.215	804	0.28571	462.65	16.271	804.00	4.8842
275	Oxygen	O ₂	7782-44-7	31.999	3.9143	0.28772	154.58	0.2924	54.35	40.770	154.58	13.6050
276	Ozone	O ₃	10028-15-6	47.998	3.3592	0.29884	261	0.28523	80.15	33.361	261.00	11.2410
277	Pentadecane	C ₁₅ H ₃₂	629-62-9	212.415	0.25142	0.23837	708	0.28571	283.07	3.642	708.00	1.0550
278	Pentanal	C ₅ H ₁₀ O	110-62-3	86.132	0.83871	0.26252	566.1	0.29444	182.00	10.534	566.10	3.1948
279	Pentane	C ₅ H ₁₂	109-66-0	72.149	0.84947	0.26726	469.7	0.27789	143.42	10.474	469.70	3.1784
280	Pentanoic acid	C ₅ H ₁₀ O ₂	109-52-4	102.132	0.73455	0.25636	639.16	0.25522	239.15	9.587	639.16	2.8653
281	1-Pentanol	C ₅ H ₁₂ O	71-41-0	88.148	0.81754	0.26732	588.1	0.25348	195.56	10.061	588.10	3.0583
282	2-Pentanol	C ₅ H ₁₂ O	6032-29-7	88.148	0.79324	0.25806	561	0.28571	200.00	10.147	561.00	3.0739
283	2-Pentanone	C ₅ H ₁₀ O	107-87-9	86.132	0.90411	0.27207	561.08	0.30669	196.29	10.398	561.08	3.3231
284	3-Pentanone	C ₅ H ₁₀ O	96-22-0	86.132	0.71811	0.24129	560.95	0.27996	234.18	10.102	560.95	2.9761
285	1-Pentene	C ₅ H ₁₀	109-67-1	70.133	0.89816	0.26608	464.8	0.28571	108.02	11.521	464.80	3.3755
286	2-Pentyl mercaptan	C ₅ H ₁₂ S	2084-19-7	104.214	0.65858	0.25367	584.3	0.28571	160.75	9.073	584.30	2.5962
287	Pentyl mercaptan	C ₅ H ₁₂ S	110-66-7	104.214	0.75345	0.27047	598	0.30583	197.45	8.858	598.00	2.7857
288	1-Pentyne	C ₅ H ₈	627-19-0	68.117	0.8491	0.2352	481.2	0.353	167.45	12.532	481.20	3.6101
289	2-Pentyne	C ₅ H ₈	627-21-4	68.117	0.92099	0.25419	519	0.31077	163.83	12.240	519.00	3.6232
290	Phenanthrene	C ₁₄ H ₁₀	85-01-8	178.229	0.45554	0.2523	869	0.24841	372.38	5.985	869.00	1.8055
291	Phenol	C ₆ H ₆ O	108-95-2	94.111	1.3798	0.31598	694.25	0.32768	314.06	11.244	694.25	4.3667
292	Phenyl isocyanate	C ₇ H ₅ NO	103-71-9	119.121	0.63163	0.23373	653	0.28571	243.15	9.647	653.00	2.7024
293	Phthalic anhydride	C ₈ H ₄ O ₃	85-44-9	148.116	0.5393	0.22704	791	0.248	404.15	8.222	791.00	2.3754
294	Propadiene	C ₃ H ₄	463-49-0	40.064	1.6087	0.26543	394	0.29895	136.87	19.479	394.00	6.0607
295	Propane	C ₃ H ₈	74-98-6	44.096	1.3757	0.27453	369.83	0.29359	85.47	16.583	369.83	5.0111
296	1-Propanol	C ₃ H ₈ O	71-23-8	60.095	1.2457	0.27281	536.8	0.23994	146.95	15.206	536.80	4.5662
297	2-Propanol	C ₃ H ₈ O	67-63-0	60.095	1.1799	0.2644	508.3	0.24653	185.26	14.663	508.30	4.4626
298	Propenylcyclohexene	C ₉ H ₁₄	13511-13-2	122.207	0.61255	0.26769	636	0.28571	199.00	7.476	636.00	2.2883
299	Propionaldehyde	C ₃ H ₆ O	123-38-6	58.079	1.296	0.26439	504.4	0.29471	170.00	15.929	504.40	4.9018
300	Propionic acid	C ₃ H ₆ O ₂	79-09-4	74.079	1.0969	0.25568	600.81	0.26857	252.45	13.935	600.81	4.2901
301	Propionitrile	C ₃ H ₅ N	107-12-0	55.079	1.0224	0.23452	564.4	0.2804	180.26	16.027	564.40	4.3595
302	Propyl acetate	C ₅ H ₁₀ O ₂	109-60-4	102.132	0.73041	0.25456	549.73	0.27666	178.15	9.794	549.73	2.8693
303	Propyl amine	C ₃ H ₉ N	107-10-8	59.110	0.9195	0.23878	496.95	0.2461	188.36	13.764	496.95	3.8508

304	Propylbenzene	C ₉ H ₁₂	103-65-1	120.192	0.57233	0.25171	638.35	0.29616	173.55	7.982	638.35	2.2738
305	Propylene	C ₃ H ₆	115-07-1	42.080	1.4403	0.26852	364.85	0.28775	87.89	18.070	364.85	5.3638
306	Propyl formate	C ₆ H ₈ O ₂	110-74-7	88.105	0.915	0.26134	538	0.28	180.25	11.590	538.00	3.5012
307	2-Propyl mercaptan	C ₆ H ₈ S	75-33-2	76.161	1.093	0.27762	517	0.29781	142.61	12.610	517.00	3.9370
308	Propyl mercaptan	C ₆ H ₈ S	107-03-9	76.161	1.0714	0.27214	536.6	0.29481	159.95	12.716	536.60	3.9369
309	1,2-Propylene glycol	C ₃ H ₈ O ₂	57-55-6	76.094	1.0923	0.26106	626	0.20459	213.15	14.363	626.00	4.1841
310	Quinone	C ₆ H ₄ O ₂	106-51-4	108.095	0.83228	0.25385	683	0.23658	388.85	10.082	683.00	3.2786
311	Silicon tetrafluoride	F ₄ Si	7783-61-1	104.079	1.1945	0.24128	259	0.16693	186.35	15.635	259.00	4.9507
312	Styrene	C ₈ H ₈	100-42-5	104.149	0.7397	0.2603	636	0.3009	242.54	9.109	636.00	2.8417
313	Succinic acid	C ₄ H ₆ O ₄	110-15-6	118.088	0.70284	0.22268	806	0.28571	460.65	10.261	806.00	3.1563
314	Sulfur dioxide	O ₂ S	7446-09-5	64.064	2.106	0.25842	430.75	0.2895	197.67	25.298	430.75	8.1495
315	Sulfur hexafluoride	F ₆ S	2551-62-4	146.055	1.3587	0.2701	318.69	0.2921	223.15	12.631	318.69	5.0304
316	Sulfur trioxide	O ₃ S	7446-11-9	80.063	1.4969	0.19013	490.85	0.4359	289.95	24.241	490.85	7.8730
317	Terephthalic acid	C ₈ H ₆ O ₄	100-21-0	166.131	0.42685	0.181	1113	0.28571	700.15	8.546	1113.00	2.3583
318	<i>o</i> -Terphenyl	C ₁₈ H ₁₄	84-15-1	230.304	0.3448	0.25116	857	0.29268	329.35	4.553	857.00	1.3728
319	<i>o</i> -Terphenyl [use Eq. (2)]	C ₁₈ H ₁₄	84-15-1	230.304	5.7136	-0.003474			288.15	4.713	313.19	4.6256
320	Tetradecane	C ₁₄ H ₃₀	629-59-4	198.388	0.27248	0.24007	693	0.28571	279.01	3.889	693.00	1.1350
321	Tetrahydrofuran	C ₄ H ₈ O	109-99-9	72.106	1.2543	0.28084	540.15	0.2912	164.65	13.998	540.15	4.4662
322	1,2,3,4-Tetrahydronaphthalene	C ₁₀ H ₁₂	119-64-2	132.202	0.67717	0.27772	720	0.2878	237.38	7.638	720.00	2.4383
323	Tetrahydrothiophene	C ₄ H ₆ S	110-01-0	88.171	1.1628	0.28954	631.95	0.28674	176.99	12.408	631.95	4.0160
324	2,2,3,3-Tetramethylbutane	C ₈ H ₁₈	594-82-1	114.229	0.58988	0.27201	568	0.27341	373.96	5.724	568.00	2.1686
325	Thiophene	C ₄ H ₄ S	110-02-1	84.140	1.2874	0.28194	579.35	0.30781	234.94	13.430	579.35	4.5662
326	Toluene	C ₇ H ₈	108-88-3	92.138	0.8792	0.27136	591.75	0.29241	178.18	10.487	591.75	3.2400
327	1,1,2-Trichloroethane	C ₂ H ₃ Cl ₃	79-00-5	133.404	0.9062	0.25475	602	0.31	236.50	11.478	602.00	3.5572
328	Tridecane	C ₁₃ H ₂₈	629-50-5	184.361	0.29934	0.2433	675	0.28571	267.76	4.182	675.00	1.2300
329	Triethyl amine	C ₆ H ₁₅ N	121-44-8	101.190	0.7035	0.27386	535.15	0.2872	158.45	8.284	535.15	2.5688
330	Trimethyl amine	C ₃ H ₉ N	75-50-3	59.110	1.0116	0.25683	433.25	0.2696	156.08	13.144	433.25	3.9388
331	1,2,3-Trimethylbenzene	C ₉ H ₁₂	526-73-8	120.192	0.6531	0.27002	664.5	0.26268	243.15	7.728	664.50	2.4187
332	1,2,4-Trimethylbenzene	C ₉ H ₁₂	95-63-6	120.192	0.60394	0.25956	649.1	0.27713	229.33	7.689	649.10	2.3268
333	2,2,4-Trimethylpentane	C ₈ H ₁₈	540-84-1	114.229	0.59059	0.27424	543.8	0.2847	165.78	6.915	543.80	2.1536
334	2,3,3-Trimethylpentane	C ₈ H ₁₈	560-21-4	114.229	0.6028	0.27446	573.5	0.2741	172.22	7.093	573.50	2.1963
335	1,3,5-Trinitrobenzene	C ₆ H ₃ N ₃ O ₆	99-35-4	213.105	0.48195	0.23093	846	0.28571	398.40	7.083	846.00	2.0870
336	2,4,6-Trinitrotoluene	C ₇ H ₅ N ₃ O ₆	118-96-7	227.131	0.37378	0.21379	828	0.29905	354.00	6.452	828.00	1.7484
337	Undecane	C ₁₁ H ₂₄	1120-21-4	156.308	0.36703	0.24876	639	0.28571	247.57	4.945	639.00	1.4750
338	1-Undecanol	C ₁₁ H ₂₄ O	112-42-5	172.308	0.33113	0.23676	703.9	0.2762	288.45	4.859	703.90	1.3986
339	Vinyl acetate	C ₄ H ₆ O ₂	108-05-4	86.089	0.9591	0.2593	519.13	0.27448	180.35	12.287	519.13	3.6988
340	Vinyl acetylene	C ₄ H ₄	689-97-4	52.075	1.2703	0.26041	454	0.297	173.15	15.664	454.00	4.8781
341	Vinyl chloride	C ₂ H ₃ Cl	75-01-4	62.498	1.5115	0.2707	432	0.2716	119.36	18.481	432.00	5.5837
342	Vinyl trichlorosilane	C ₂ H ₃ Cl ₃ Si	75-94-5	161.490	0.59595	0.24314	543.15	0.24856	178.35	8.824	543.15	2.4511
344	Water [use Eq. (2)]	H ₂ O	7732-18-5	18.015	-13.851	0.64038	-0.00191	1.8211E-06	273.16	55.497	353.15	54.0012
345	<i>m</i> -Xylene	C ₈ H ₁₀	108-38-3	106.165	0.68902	0.26086	617	0.27479	225.30	8.648	617.00	2.6413
346	<i>o</i> -Xylene	C ₈ H ₁₀	95-47-6	106.165	0.69962	0.26143	630.3	0.27365	247.98	8.623	630.30	2.6761
347	<i>p</i> -Xylene	C ₈ H ₁₀	106-42-3	106.165	0.67752	0.25887	616.2	0.27596	286.41	8.161	616.20	2.6172

Except for *o*-terphenyl and water, liquid density ρ is calculated by

$$\rho = C1/C2^{1 + (1-T/C3)^{C4}}$$

where ρ is in mol/dm³ and T is in K. The pressure is equal to the vapor pressure for pressures greater than 1 atm and equal to 1 atm when the vapor pressure is less than 1 atm.

Equation (2), used for the limited temperature ranges as noted for *o*-terphenyl and water, is

$$\rho = C1 + C2T + C3T^2 + C4T^3$$

For water over the entire temperature range of 273.16 to 647.096 K, use

$$\rho = 17.863 + 58.606\tau^{0.35} - 95.396\tau^{2/3} + 213.89\tau - 141.26\tau^{4/3}$$

where $\tau = 1 - T/647.096$.

All substances are listed by chemical family in Table 2-6 and by formula in Table 2-7.

Values in this table were taken from the Design Institute for Physical Properties (DIPPR) of the American Institute of Chemical Engineers (AIChE), copyright 2007 AIChE and reproduced with permission of AIChE and of the DIPPR Evaluated Process Design Data Project Steering Committee. Their source should be cited as R. L. Rowley, W. V. Wilding, J. L. Oscarson, Y. Yang, N. A. Zundel, T. E. Daubert, R. P. Danner, DIPPR® Data Compilation of Pure Chemical Properties, Design Institute for Physical Properties, AIChE, New York (2007).

The number of digits provided for values at $T_{\min}$ and $T_{\max}$ was chosen for uniformity of appearance and formatting; these do not represent the uncertainties of the physical quantities, but are the result of calculations from the standard thermophysical property formulations within a fixed format.

DENSITIES OF AQUEOUS INORGANIC SOLUTIONS AT 1 ATM

UNITS AND UNITS CONVERSIONS

Most densities are given in grams per cubic centimeter. To convert to pounds per cubic foot, multiply by 62.43. °F = % °C + 32. Compositions are weight percent unless otherwise stated.

ADDITIONAL REFERENCES

For more detailed data on densities see *International Critical Tables*: tabular index, vol. 3, p. 1; abrasives, vol. 2, p. 87; air, moist, vol. 1, p. 71; building stones, vol. 2, p. 52; clays, vol. 2, p. 56; coals, vol. 2, p. 135; compounds, vol. 1, pp. 106, 176, 313, 341; elements, vol. 1, pp. 102, 340; fibers, vol. 2, p. 237; gases and vapors, vol. 3, pp. 3, 345; glass, vol. 2, p. 93; liquids and vitreous solids, vol. 3, p. 22; vol. 1, pp. 102, 340; vol. 2,

pp. 456, 463; vol. 3, pp. 20, 35; liquid coolants and saturated vapors are available from WADC-TR-59-598, 1959; plastics are collected in the *Handbook of Chemistry and Physics*, Chemical Rubber Publishing Co.; solid helium, neon, argon, fluorine, and methane data are given by Johnson (ed.), WADD-TR-60-56, 1960; temperatures of maximum solubility, vol. 3, p. 107; metals, vol. 2, p. 463; oils, fats, and waxes, vol. 2, p. 201; orthobaric, vol. 3, pp. 202, 228, 237, 244; petroleum, vol. 2, pp. 137, 144; plastics, vol. 2, p. 296; porcelains, vol. 2, pp. 68, 75; refrigerating brines, vol. 2, p. 327; rubber, vol. 2, pp. 255, 259; soaps, vol. 5, p. 447; metallic solid solutions, vol. 2, p. 358; solids, vol. 3, pp. 43, 45; vol. 2, p. 456; vol. 3, p. 21; solutions and mixtures, vol. 3, pp. 17, 51, 95, 104, 107, 111, 125, 130; woods, vol. 2, p. 1. Also see the *Handbook of Chemistry and Physics*, Chemical Rubber Publishing Co., 86th ed., etc.

TABLE 2-33 Aluminum Sulfate [Al₂(SO₄)₃]*

%	<i>d</i> ₄ ¹⁵	%	<i>d</i> ₄ ¹⁵
1	1.0093	16	1.1770
2	1.0195	20	1.2272
4	1.0404	24	1.2803
8	1.0837	26	1.3079
12	1.1293		

*International Critical Tables, vol. 3, p. 70.

TABLE 2-34 Ammonia (NH₃)*

%	-15 °C	-10 °C	-5 °C	0 °C	5 °C	10 °C	20 °C	25 °C	%	<i>d</i> ₄ ¹⁵
1		0.9943	0.9954	0.9959	0.9958	0.9955	0.9939	0.993	32	0.889
2		.9906	.9915	.9919	.9917	.9913	.9895	.988	36	.877
4		.9834	.9840	.9842	.9837	.9832	.9811	.980	40	.865
8	0.970	.9701	.9701	.9695	.9686	.9677	.9651	.964	45	.849
12	.958	.9576	.9571	.9561	.9548	.9534	.9501	.948	50	.832
16	.947	.9461	.9450	.9435	.9420	.9402	.9362	.934	60	.796
20		.9353	.9335	.9316	.9296	.9275	.9229		70	.755
24		.9249	.9226	.9202	.9179	.9155	.9101		80	.711
28		.9150	.9122	.9094	.9067	.9040	.8980		90	.665
30		.9101	.9070	.9040	.9012	.8983	.8920		100	.618

*International Critical Tables, vol. 3, p. 59.

TABLE 2-35 Ammonium Acetate* (CH₃COONH₄)

%	<i>d</i> ₄ ²⁵
1	0.9992
2	1.0013
4	1.0055
8	1.0136
12	1.0216
16	1.0294
20	1.0368
24	1.0439
28	1.0507
30	1.0540
35	1.0618
40	1.0691
45	1.0760

*International Critical Tables, vol. 3, p. 62.

For data at 16 °C for 3(1/5) percent see Atack, *Handbook of Chemical Data*, p. 33, Reinhold, New York, 1957.

TABLE 2-36 Ammonium Bichromate [(NH₄)₂Cr₂O₇]*

%	<i>d</i> ₄ ¹²
1	1.0051
2	1.0108
4	1.0223
8	1.0463
12	1.0715
16	1.0981
20	1.1263

*International Critical Tables, vol. 3, p. 70.

TABLE 2-37 Ammonium Chloride (NH₄Cl)*

%	0 °C	10 °C	20 °C	30 °C	50 °C	80 °C	100 °C
1	1.0033	1.0029	1.0013	0.9987	0.9910	0.9749	0.9617
2	1.0067	1.0062	1.0045	1.0018	.9940	.9780	.9651
4	1.0135	1.0126	1.0107	1.0077	.9999	.9842	.9718
8	1.0266	1.0251	1.0227	1.0195	1.0116	.9963	.9849
12	1.0391	1.0370	1.0344	1.0310	1.0231	1.0081	.9975
16	1.0510	1.0485	1.0457	1.0422	1.0343	1.0198	1.0096
20	1.0625	1.0596	1.0567	1.0532	1.0454	1.0312	1.0213
24	1.0736	1.0705	1.0674	1.0641	1.0564	1.0426	1.0327

*International Critical Tables, vol. 3, p. 60.

TABLE 2-38 Ammonium Chromate [(NH₄)₂CrO₄]*

%	°C	<i>d</i> ₄
3.80	20	1.0219
10.52	13	1.0627
19.75	13.7	1.1189
28.04	19.6	1.1707

*International Critical Tables, vol. 3, p. 70.

TABLE 2-39 Ammonium Nitrate (NH₄NO₃)*

%	0 °C	10 °C	25 °C	40 °C	60 °C	80 °C
1.0	1.0043	1.0039	1.0011	0.9961	0.9870	0.9755
2.0	1.0088	1.0082	1.0051	1.0000	.9908	.9793
4.0	1.0178	1.0168	1.0132	1.0079	.9985	.9869
8.0	1.0358	1.0340	1.0297	1.0238	1.0142	1.0024
12.0	1.0539	1.0515	1.0464	1.0400	1.0301	1.0181
16.0	1.0721	1.0691	1.0633	1.0565	1.0462	1.0342
20.0	1.0905	1.0870	1.0806	1.0734	1.0627	1.0506
24.0	1.1090	1.1051	1.0982	1.0907	1.0796	1.0673
28.0	1.1277	1.1234	1.1161	1.1082	1.0968	1.0844
30.0	1.1371	1.1327	1.1252	1.1171	1.1055	1.0931
40.0	1.1862	1.1810	1.1727	1.1640	1.1515	1.1385
50.0	1.2380	1.2320	1.2229	1.2136	1.2006	1.1868

*International Critical Tables, vol. 3, p. 59.

TABLE 2-40 Ammonium Sulfate [(NH₄)₂SO₄]*

%	0 °C	20 °C	40 °C	80 °C	100 °C
1	1.0061	1.0041	0.9980	0.9777	0.9644
2	1.0124	1.0101	1.0039	.9836	.9705
4	1.0248	1.0220	1.0155	.9953	.9826
8	1.0495	1.0456	1.0387	1.0187	1.0066
12	1.0740	1.0691	1.0619	1.0421	1.0303
16	1.0980	1.0924	1.0849	1.0653	1.0539
20	1.1215	1.1154	1.1077	1.0883	1.0772
24	1.1448	1.1383	1.1304	1.1111	1.1003
28	1.1677	1.1609	1.1529	1.1338	1.1232
35	1.2072	1.2800	1.1919	1.1731	1.1629
40	1.2350	1.2277	1.2196	1.2011	1.1910
50	1.2899	1.2825	1.2745	1.2568	1.2466

*International Critical Tables, vol. 3, p. 60.

TABLE 2-41 Arsenic Acid (H₃AsO₄)*

%	<i>d</i> ₄ ¹⁵	%	<i>d</i> ₄ ¹⁵
1	1.0057	20	1.1447
2	1.0124	30	1.2331
6	1.0398	40	1.3370
10	1.0681	50	1.4602
16	1.1128	60	1.6070
		70	1.7811

*International Critical Tables, vol. 3, p. 61.

TABLE 2-42 Barium Chloride (BaCl₂)*

%	0 °C	20 °C	40 °C	60 °C	80 °C	100 °C
2	1.0181	1.0159	1.0096	1.0004	0.9890	0.9755
4	1.0368	1.0341	1.0275	1.0181	1.0066	.9931
8	1.0760	1.0721	1.0648	1.0551	1.0434	1.0299
12	1.1178	1.1128	1.1047	1.0948	1.0827	1.0692
16	1.1627	1.1564	1.1478	1.1373	1.1249	1.1113
20	1.2105	1.2031	1.1938	1.1828	1.1702	1.1563
24		1.2531	1.2430	1.2316	1.2186	1.2045
26		1.2793	1.2688	1.2571	1.2440	1.2298

*International Critical Tables, vol. 3, p. 75.

TABLE 2-43 Cadmium Nitrate [Cd(NO₃)₂]*

%	<i>d</i> ₄ ¹⁸	%	<i>d</i> ₄ ¹⁸
2	1.0154	20	1.1904
4	1.0326	25	1.2488
8	1.0683	30	1.3124
12	1.1061	40	1.4590
16	1.1468	50	1.6356

*International Critical Tables, vol. 3, p. 66.

TABLE 2-44 Calcium Chloride (CaCl₂)*

%	−5 °C	0 °C	20 °C	30 °C	40 °C	60 °C	80 °C	100 °C	120 °C†	140 °C
2		1.0171	1.0148	1.0120	1.0084	0.9994	0.9881	0.9748	0.9596	0.9428
4		1.0346	1.0316	1.0286	1.0249	1.0158	1.0046	0.9915	0.9765	0.9601
8	1.0708	1.0703	1.0659	1.0626	1.0586	1.0492	1.0382	1.0257	1.0111	0.9954
12	1.1083	1.1072	1.1015	1.0978	1.0937	1.0840	1.0730	1.0610	1.0466	1.0317
16	1.1471	1.1454	1.1386	1.1345	1.1301	1.1202	1.1092	1.0973	1.0835	1.0691
20	1.1874	1.1853	1.1775	1.1730	1.1684	1.1581	1.1471	1.1352	1.1219	1.1080
25		1.2376	1.2284	1.2236	1.2186	1.2079	1.1965	1.1846		
30		1.2922	1.2816	1.2764	1.2709	1.2597	1.2478	1.2359		
35			1.3373	1.3316	1.3255	1.3137	1.3013	1.2893		
40			1.3957	1.3895	1.3826	1.3700	1.3571	1.3450		

*International Critical Tables, vol. 3, pp. 72–73.

†Corrected to atmospheric pressure.

TABLE 2-45 Calcium Hydroxide [Ca(OH)₂]*

%	<i>d</i> ₄ ¹⁵	<i>d</i> ₄ ²⁵
0.05	0.99979	0.99773
.10	1.00044	.99838
.15	1.00110	.99904

*International Critical Tables, vol. 3, p. 72.

TABLE 2-46 Calcium Hypochlorite* (CaOCl₂)

% total salt	<i>d</i> ₄ ¹⁵
2	1.0169
4	1.0345
6	1.0520
8	1.0697
10	1.0876
12	1.1060

*International Critical Tables, vol. 3, p. 73.

CaOCl₂ = 89.15%

CaCl₂ = 7.31%

Ca(ClO₃)₂ = 0.26%

Ca(OH)₂ = 2.92%

TABLE 2-47 Calcium Nitrate [Ca(NO₃)₂]*

%	6 °C	18 °C	25 °C	30 °C
2*	1.0157	1.0137	1.0120	1.0105
4	1.0316	1.0291	1.0272	1.0256
8	1.0641	1.0608	1.0585	1.0565
12	1.0979	1.0937	1.0911	1.0887
16	1.1330	1.1279	1.1250	1.1224
20	1.1694	1.1636	1.1602	1.1575
25	1.2168	1.2106	1.2065	1.2032
30		1.260		
35		1.311		
40		1.365		
45		1.422		
68†		1.747	1.741	1.736

*International Critical Tables, vol. 3, pp. 73–74.

†Supercooled tetrahydrate (m.p. 41.4°C).

TABLE 2-48 Chromic Acid (CrO₃)*

%	<i>d</i> ₄ ¹⁵	%	<i>d</i> ₄ ¹⁵
1	1.006	20	1.163
2	1.014	26	1.220
6	1.045	30	1.260
10	1.076	40	1.371
16	1.127	50	1.505
		60	1.663

*International Critical Tables, vol. 3, p. 69.

TABLE 2-49 Chromium Chloride (CrCl₃)*

%	<i>d</i> ₄ ¹⁸		
	Violet	Green	Equilibrium mixture of violet and green
1	1.0076	1.0071	1.0075
2	1.0166	1.0157	1.0165
4	1.0349	1.0332	1.0347
8	1.0724	1.0691	1.0722
12	1.1114	1.1065	1.1111
14	1.1316		

*International Critical Tables, vol. 3, p. 69.

TABLE 2-50 Copper Nitrate [Cu(NO₃)₂]*

%	<i>d</i> ₄ ²⁰	%	<i>d</i> ₄ ²⁰
1	1.007	12	1.107
2	1.015	16	1.147
4	1.032	20	1.189
8	1.069	25	1.248

*International Critical Tables, vol. 3, p. 67.

TABLE 2-51 Copper Sulfate (CuSO₄)*

%	0 °C	20 °C	40 °C
1	1.0104	1.0086	1.0024
4	1.0429	1.0401	1.0332
8	1.0887	1.084	1.0764
12	1.1379	1.1308	1.1222
16		1.180	
18		1.206	

*International Critical Tables, vol. 3, p. 67.

TABLE 2-52 Cuprous Chloride (CuCl₂)*

%	0 °C	20 °C	40 °C
1	1.0095	1.0072	1.002
4	1.0387	1.036	1.0305
8	1.0788	1.0754	1.0682
12	1.1208	1.1165	1.1107
16	1.1653	1.1595	1.151
20	1.2121	1.2052	1.1953

*International Critical Tables, vol. 3, p. 66.

TABLE 2-53 Ferric Chloride (FeCl₃)*

%	0 °C	10 °C	20 °C	30 °C
1	1.0086	1.0084	1.0068	1.0040
2	1.0174	1.0168	1.0152	1.0122
4	1.0347	1.0341	1.0324	1.0292
8	1.0703	1.0692	1.0669	1.0636
12	1.1088	1.1071	1.1040	1.1006
16	1.1475	1.1449	1.1418	1.1386
20	1.1870	1.1847	1.1820	1.1786
25	1.2400	1.2380	1.2340	1.2290
30	1.2970	1.2950	1.2910	1.2850
35	1.3605	1.3580	1.3530	1.3475
40	1.4280	1.4235	1.4175	1.4115
45		1.4920	1.4850	
50		1.5610	1.5510	

*International Critical Tables, vol. 3, p. 68.

2-106 PHYSICAL AND CHEMICAL DATA

TABLE 2-54 Ferric Sulfate
[Fe₂(SO₄)₃]*

%	$d_4^{17.5}$
1	1.0072
2	1.0157
4	1.0327
8	1.0670
12	1.1028
16	1.1409
20	1.1811
30	1.3073
40	1.4487
50	1.6127
60	1.7983

*International Critical Tables,
vol. 3, p. 68.**TABLE 2-56 Ferrous Sulfate (FeSO₄)***

%	15 °C	18 °C	20 °C
0.2		1.00068	1.0002
0.4		1.00275	1.0022
0.8		1.00645	1.0062
1.0	1.0090	1.0085	1.0082
4.0	1.0380	1.0375	
8.0	1.0790	1.0785	
12.0	1.1235	1.1220	
16.0	1.1690	1.1675	
20.0	1.2150	1.2135	

*International Critical Tables,
vol. 3, p. 68.**TABLE 2-58 Hydrogen Cyanide (HCN)***

%	d_4^{15}
1	0.998
2	0.996
4	0.993
8	0.984
12	0.971
16	0.956
82	0.752
90	0.724
100	0.691

*International Critical Tables,
vol. 3, p. 61.**TABLE 2-59 Hydrogen Chloride (HCl)**

%	-5 °C	0 °C	10 °C	20 °C	40 °C	60 °C	80 °C	100 °C
1	1.0048	1.0052	1.0048	1.0032	0.9970	0.9881	0.9768	0.9636
2	1.0104	1.0106	1.0100	1.0082	1.0019	0.9930	0.9819	0.9688
4	1.0213	1.0213	1.0202	1.0181	1.0116	1.0026	0.9919	0.9791
6	1.0321	1.0319	1.0303	1.0279	1.0211	1.0121	1.0016	0.9892
8	1.0428	1.0423	1.0403	1.0376	1.0305	1.0215	1.0111	0.9992
10	1.0536	1.0528	1.0504	1.0474	1.0400	1.0310	1.0206	1.0090
12	1.0645	1.0634	1.0607	1.0574	1.0497	1.0406	1.0302	1.0188
14	1.0754	1.0741	1.0711	1.0675	1.0594	1.0502	1.0398	1.0286
16	1.0864	1.0849	1.0815	1.0776	1.0692	1.0598	1.0494	1.0383
18	1.0975	1.0958	1.0920	1.0878	1.0790	1.0694	1.0590	1.0479
20	1.1087	1.1067	1.1025	1.0980	1.0888	1.0790	1.0685	1.0574
22	1.1200	1.1177	1.1131	1.1083	1.0986	1.0886	1.0780	1.0668
24	1.1314	1.1287	1.1238	1.1187	1.1085	1.0982	1.0874	1.0761
26	1.1426	1.1396	1.1344	1.1290	1.1183	1.1076	1.0967	1.0853
28	1.1537	1.1505	1.1449	1.1392	1.1280	1.1169	1.1058	1.0942
30	1.1648	1.1613	1.1553	1.1493	1.1376	1.1260	1.1149	1.1030
32				1.1593				
34				1.1691				
36				1.1789				
38				1.1885				
40				1.1980				

*International Critical Tables, vol. 3, p. 54.

TABLE 2-55 Ferric Nitrate
[Fe(NO₃)₃]*

%	d_4^{18}
1	1.0065
2	1.0144
4	1.0304
8	1.0636
12	1.0989
16	1.1359
20	1.1748
25	1.2281

*International Critical Tables,
vol. 3, p. 68.**TABLE 2-57 Hydrogen Bromide (HBr)***

%	d_4^4	d_4^{10}	d_4^{25}
1.0	1.0073	1.0068	1.0041
2.0	1.0146	1.0139	1.0111
4.0	1.0295	1.0285	1.0255
6.0	1.0448	1.0435	1.0402
8.0	1.0604	1.0589	1.0552
10.0	1.0764	1.0747	1.0707
12.0	1.0928	1.0910	1.0867
14.0	1.1097	1.1078	1.1032
16.0	1.1272	1.1251	1.1202
18.0	1.1453	1.1430	1.1377
20.0	1.1640	1.1615	1.1557
22.0	1.1832	1.1806	1.1743
24.0	1.2030	1.2003	1.1935
26.0	1.2235	1.2206	1.2134
28.0	1.2446	1.2415	1.2340
30.0	1.2663	1.2630	1.2552
40.0	1.3877	1.3838	1.3736
50.0	1.5305	1.5257	1.5127
60.0	1.6950	1.6892	1.6731
65.0	1.7854	1.7792	1.7613

*International Critical Tables,
vol. 3, p. 55.**TABLE 2-60 Hydrogen Fluoride (HF)***

%	d_4^0	d_4^{20}
5	1.020	1.017
10	1.040	1.035
20	1.080	1.070
30	1.119	1.101
40	1.159	1.130
50	1.198	1.155
60	1.235	
70	1.258	
80	1.259	
90	1.178	
95	1.089	
100	1.0005	

*International Critical Tables,
vol. 3, p. 54.**TABLE 2-61 Hydrogen Peroxide (H₂O₂)***

%	d_4^{18}	%	d_4^{18}
1	1.0022	26	1.0959
2	1.0058	28	1.1040
4	1.0131	30	1.1122
6	1.0204	35	1.1327
8	1.0277	40	1.1536
10	1.0351	45	1.1749
12	1.0425	50	1.1966
14	1.0499	55	1.2188
16	1.0574	60	1.2416
18	1.0649	70	1.2897
20	1.0725	80	1.3406
22	1.0802	90	1.3931
24	1.0880	100	1.4465

*International Critical Tables,
vol. 3, p. 54.**TABLE 2-62 Hydrofluosilic Acid (H₂SiF₆)***

%	$d_4^{17.5}$	%	$d_4^{17.5}$
1	1.0080	16	1.1373
2	1.0161	20	1.1748
4	1.0324	25	1.2235
8	1.0661	30	1.2742
12	1.1011	34	1.3162

*O. Söhnel and P. Novotny, *Densities of Aqueous Solutions of Inorganic Substances*, Elsevier, 1985.**TABLE 2-63 Magnesium Chloride (MgCl₂)***

%	0 °C	20 °C	40 °C	60 °C	80 °C	100 °C
2	1.0168	1.0146	1.0084	0.9995	0.9883	0.9753
4	1.0338	1.0311	1.0248	1.0159	1.0050	0.9923
8	1.0683	1.0646	1.0580	1.0493	1.0388	1.0269
12	1.1035	1.0989	1.0921	1.0836	1.0735	1.0622
16	1.1395	1.1342	1.1272	1.1188	1.1092	1.0984
20	1.1764	1.1706	1.1635	1.1552	1.1460	1.1359
25	1.2246	1.2184	1.2111	1.2031	1.1942	1.1847
30	1.2754	1.2688	1.2614	1.2535	1.2451	1.2360

*International Critical Tables, vol. 3, p. 71.

TABLE 2-64 Magnesium Sulfate (MgSO₄)*

%	0 °C	20 °C	30 °C	40 °C	50 °C	60 °C	80 °C
2	1.0210	1.0186	1.0158	1.0123	1.0081	1.0032	0.9916
4	1.0423	1.0392	1.0362	1.0326	1.0283	1.0234	1.0118
8	1.0858	1.0816	1.0782	1.0743	1.0700	1.0650	1.0534
12	1.1309	1.1256	1.1220	1.1179	1.1135	1.1083	1.0968
16	1.1777	1.1717	1.1679	1.1637	1.1592		
20	1.2264	1.2198	1.2159	1.2117	1.2072		
26	1.3032	1.2961	1.2922	1.2879	1.2836		

*International Critical Tables, vol. 3, p. 72.

TABLE 2-65 Nickel Chloride (NiCl₂)*

%	d_4^{18}
1	1.0082
2	1.0179
4	1.0375
8	1.0785
12	1.1217
16	1.1674
20	1.2163
30	1.353

*International Critical Tables, vol. 3, p. 69.

TABLE 2-66 Nickel Nitrate [Ni(NO₃)₂]*

%	d_4^{30}
1	1.0065
2	1.0150
4	1.0325
8	1.0688
12	1.1070
16	1.1480
20	1.191
30	1.311
35	1.377

*International Critical Tables, vol. 3, p. 69.

TABLE 2-67 Nickel Sulfate (NiSO₄)*

%	d_4^{18}
1	1.0091
2	1.0198
4	1.0415
8	1.0852
12	1.1325
16	1.1825
18	1.2090

*International Critical Tables, vol. 3, p. 69.

TABLE 2-68 Nitric Acid (HNO₃)*

%	0 °C	5 °C	10 °C	15 °C	20 °C	25 °C	30 °C	40 °C	50 °C	60 °C	80 °C	100 °C
1	1.0058	1.00572	1.00534	1.00464	1.00364	1.00241	1.0009	0.9973	0.9931	0.9882	0.9767	0.9632
2	1.0117	1.01149	1.01099	1.01018	1.00909	1.00778	1.0061	1.0025	0.9982	0.9932	0.9816	0.9681
3	1.0176	1.01730	1.01668	1.01576	1.01457	1.01318	1.0114	1.0077	1.0033	0.9982	0.9865	0.9730
4	1.0236	1.02315	1.02240	1.02137	1.02008	1.01861	1.0168	1.0129	1.0084	1.0033	0.9915	0.9779
5	1.0296	1.02904	1.02816	1.02702	1.02563	1.02408	1.0222	1.0182	1.0136	1.0084	0.9965	0.9829
6	1.0357	1.03497	1.03397	1.03272	1.03122	1.02958	1.0277	1.0235	1.0188	1.0136	1.0015	0.9879
7	1.0418	1.0410	1.0399	1.0385	1.0369	1.0352	1.0333	1.0289	1.0241	1.0188	1.0066	0.9929
8	1.0480	1.0471	1.0458	1.0443	1.0427	1.0409	1.0389	1.0344	1.0295	1.0241	1.0117	0.9980
9	1.0543	1.0532	1.0518	1.0502	1.0485	1.0466	1.0446	1.0399	1.0349	1.0294	1.0169	1.0032
10	1.0606	1.0594	1.0578	1.0561	1.0543	1.0523	1.0503	1.0455	1.0403	1.0347	1.0221	1.0083
11	1.0669	1.0656	1.0639	1.0621	1.0602	1.0581	1.0560	1.0511	1.0458	1.0401	1.0273	1.0134
12	1.0733	1.0718	1.0700	1.0681	1.0661	1.0640	1.0618	1.0567	1.0513	1.0455	1.0326	1.0186
13	1.0797	1.0781	1.0762	1.0742	1.0721	1.0699	1.0676	1.0624	1.0568	1.0509	1.0379	1.0238
14	1.0862	1.0845	1.0824	1.0803	1.0781	1.0758	1.0735	1.0681	1.0624	1.0564	1.0432	1.0289
15	1.0927	1.0909	1.0887	1.0865	1.0842	1.0818	1.0794	1.0739	1.0680	1.0619	1.0485	1.0341
16	1.0992	1.0973	1.0950	1.0927	1.0903	1.0879	1.0854	1.0797	1.0737	1.0675	1.0538	1.0393
17	1.1057	1.1038	1.1014	1.0989	1.0964	1.0940	1.0914	1.0855	1.0794	1.0731	1.0592	1.0444
18	1.1123	1.1103	1.1078	1.1052	1.1026	1.1001	1.0974	1.0913	1.0851	1.0787	1.0646	1.0496
19	1.1189	1.1168	1.1142	1.1115	1.1088	1.1062	1.1034	1.0972	1.0908	1.0843	1.0700	1.0547
20	1.1255	1.1234	1.1206	1.1178	1.1150	1.1123	1.1094	1.1031	1.0966	1.0899	1.0754	1.0598
21	1.1322	1.1300	1.1271	1.1242	1.1213	1.1185	1.1155	1.1090	1.1024	1.0956	1.0808	1.0650
22	1.1389	1.1366	1.1336	1.1306	1.1276	1.1247	1.1217	1.1150	1.1083	1.1013	1.0862	1.0701
23	1.1457	1.1433	1.1402	1.1371	1.1340	1.1310	1.1280	1.1210	1.1142	1.1070	1.0917	1.0753
24	1.1525	1.1501	1.1469	1.1437	1.1404	1.1374	1.1343	1.1271	1.1201	1.1127	1.0972	1.0805
25	1.1594	1.1569	1.1536	1.1503	1.1469	1.1438	1.1406	1.1332	1.1260	1.1185	1.1027	1.0857
26	1.1663	1.1638	1.1603	1.1569	1.1534	1.1502	1.1469	1.1394	1.1320	1.1244	1.1083	1.0910
27	1.1733	1.1707	1.1670	1.1635	1.1600	1.1566	1.1533	1.1456	1.1381	1.1303	1.1139	1.0963
28	1.1803	1.1777	1.1738	1.1702	1.1666	1.1631	1.1597	1.1519	1.1442	1.1362	1.1195	1.1016
29	1.1874	1.1847	1.1807	1.1770	1.1733	1.1697	1.1662	1.1582	1.1503	1.1422	1.1251	1.1069
30	1.1945	1.1917	1.1876	1.1838	1.1800	1.1763	1.1727	1.1645	1.1564	1.1482	1.1307	1.1122
31	1.2016	1.1988	1.1945	1.1906	1.1867	1.1829	1.1792	1.1708	1.1625	1.1542	1.1363	1.1175
32	1.2088	1.2059	1.2014	1.1974	1.1934	1.1896	1.1857	1.1772	1.1687	1.1602	1.1419	1.1228
33	1.2160	1.2131	1.2084	1.2043	1.2002	1.1963	1.1922	1.1836	1.1749	1.1662	1.1476	1.1281
34	1.2233	1.2203	1.2155	1.2113	1.2071	1.2030	1.1988	1.1901	1.1812	1.1723	1.1533	1.1335
35	1.2306	1.2275	1.2227	1.2183	1.2140	1.2098	1.2055	1.1966	1.1876	1.1784	1.1591	1.1390
36	1.2375	1.2344	1.2294	1.2249	1.2205	1.2163	1.2119	1.2028	1.1936	1.1842	1.1645	1.1440
37	1.2444	1.2412	1.2361	1.2315	1.2270	1.2227	1.2182	1.2089	1.1995	1.1899	1.1699	1.1490
38	1.2513	1.2479	1.2428	1.2381	1.2335	1.2291	1.2245	1.2150	1.2054	1.1956	1.1752	1.1540
39	1.2581	1.2546	1.2494	1.2446	1.2399	1.2354	1.2308	1.2210	1.2112	1.2013	1.1805	1.1589
40	1.2649	1.2613	1.2560	1.2511	1.2463	1.2417	1.2370	1.2270	1.2170	1.2069	1.1858	1.1638
41	1.2717	1.2680	1.2626	1.2576	1.2527	1.2480	1.2432	1.2330	1.2229	1.2126	1.1911	1.1687
42	1.2786	1.2747	1.2692	1.2641	1.2591	1.2543	1.2494	1.2390	1.2287	1.2182	1.1963	1.1735
43	1.2854	1.2814	1.2758	1.2706	1.2655	1.2606	1.2556	1.2450	1.2345	1.2238	1.2015	1.1783
44	1.2922	1.2880	1.2824	1.2771	1.2719	1.2669	1.2618	1.2510	1.2403	1.2294	1.2067	1.1831
45	1.2990	1.2947	1.2890	1.2836	1.2783	1.2732	1.2680	1.2570	1.2461	1.2350	1.2119	1.1879
46	1.3058	1.3014	1.2955	1.2901	1.2847	1.2795	1.2742	1.2630	1.2519	1.2406	1.2171	1.1927
47	1.3126	1.3080	1.3021	1.2966	1.2911	1.2858	1.2804	1.2690	1.2577	1.2462	1.2223	1.1976
48	1.3194	1.3147	1.3087	1.3031	1.2975	1.2921	1.2867	1.2750	1.2635	1.2518	1.2275	1.2024
49	1.3263	1.3214	1.3153	1.3096	1.3040	1.2984	1.2929	1.2811	1.2693	1.2575	1.2328	1.2073
50	1.3327	1.3277	1.3215	1.3157	1.3100	1.3043	1.2987	1.2867	1.2748	1.2628	1.2377	1.2118
51	1.3391	1.3339	1.3277	1.3218	1.3160	1.3102	1.3045	1.2923	1.2802	1.2680	1.2425	1.2163
52	1.3454	1.3401	1.3338	1.3278	1.3219	1.3160	1.3102	1.2978	1.2856	1.2731	1.2473	1.2208
53	1.3517	1.3462	1.3399	1.3338	1.3278	1.3218	1.3159	1.3033	1.2909	1.2782	1.2521	1.2252
54	1.3579	1.3523	1.3459	1.3397	1.3336	1.3275	1.3215	1.3087	1.2961	1.2833	1.2568	1.2296
55	1.3640	1.3583	1.3518	1.3455	1.3393	1.3331	1.3270	1.3141	1.3013	1.2883	1.2615	1.2339
56	1.3700	1.3642	1.3576	1.3512	1.3449	1.3386	1.3324	1.3194	1.3064	1.2932	1.2661	1.2382
57	1.3759	1.3700	1.3634	1.3569	1.3505	1.3441	1.3377	1.3246	1.3114	1.2981	1.2706	1.2424
58	1.3818	1.3757	1.3691	1.3625	1.3560	1.3495	1.3430	1.3298	1.3164	1.3029	1.2751	1.2466
59	1.3875	1.3813	1.3747	1.3680	1.3614	1.3548	1.3482	1.3348	1.3213	1.3077	1.2795	1.2507
60	1.3931	1.3868	1.3801	1.3734	1.3667	1.3600	1.3533	1.3398	1.3261	1.3124	1.2839	1.2547
61	1.3986	1.3922	1.3855	1.3787	1.3719	1.3651	1.3583	1.3447	1.3308	1.3169	1.2881	1.2587
62	1.4039	1.3975	1.3907	1.3838	1.3769	1.3700	1.3632	1.3494	1.3354	1.3213	1.2922	1.2625
63	1.4091	1.4027	1.3958	1.3888	1.3818	1.3748	1.3679	1.3540	1.3398	1.3255	1.2962	1.2661
64		1.4078	1.4007	1.3936	1.3866	1.3795	1.3725					

2-108 PHYSICAL AND CHEMICAL DATA

TABLE 2-68 Nitric Acid (HNO₃) (Concluded)

%	0 °C	5 °C	10 °C	15 °C	20 °C	25 °C	30 °C	40 °C	50 °C	60 °C	80 °C	100 °C
65		1.4128	1.4055	1.3984	1.3913	1.3841	1.3770					
66		1.4177	1.4103	1.4031	1.3959	1.3887	1.3814					
67		1.4224	1.4150	1.4077	1.4004	1.3932	1.3857					
68		1.4271	1.4196	1.4122	1.4048	1.3976	1.3900					
69		1.4317	1.4241	1.4166	1.4091	1.4019	1.3942					
70		1.4362	1.4285	1.4210	1.4134	1.4061	1.3983					
71		1.4406	1.4328	1.4252	1.4176	1.4102	1.4023					
72		1.4449	1.4371	1.4294	1.4218	1.4142	1.4063					
73		1.4491	1.4413	1.4335	1.4258	1.4182	1.4103					
74		1.4532	1.4454	1.4376	1.4298	1.4221	1.4142					
75		1.4573	1.4494	1.4415	1.4337	1.4259	1.4180					
76		1.4613	1.4533	1.4454	1.4375	1.4296	1.4217					
77		1.4652	1.4572	1.4492	1.4413	1.4333	1.4253					
78		1.4690	1.4610	1.4529	1.4450	1.4369	1.4288					
79		1.4727	1.4647	1.4565	1.4486	1.4404	1.4323					
80		1.4764	1.4683	1.4601	1.4521	1.4439	1.4357					
81		1.4800	1.4718	1.4636	1.4555	1.4473	1.4391					
82		1.4835	1.4753	1.4670	1.4589	1.4507	1.4424					
83		1.4869	1.4787	1.4704	1.4622	1.4540	1.4456					
84		1.4903	1.4820	1.4737	1.4655	1.4572	1.4487					
85		1.4936	1.4852	1.4769	1.4686	1.4603	1.4518					
86		1.4968	1.4883	1.4799	1.4716	1.4633	1.4548					
87		1.4999	1.4913	1.4829	1.4745	1.4662	1.4577					
88		1.5029	1.4942	1.4858	1.4773	1.4690	1.4605					
89		1.5058	1.4970	1.4885	1.4800	1.4716	1.4631					
90		1.5085	1.4997	1.4911	1.4826	1.4741	1.4656					
91		1.5111	1.5023	1.4936	1.4850	1.4766	1.4681					
92		1.5136	1.5048	1.4960	1.4873	1.4789	1.4704					
93		1.5156	1.5068	1.4979	1.4892	1.4807	1.4722					
94		1.5177	1.5088	1.4999	1.4912	1.4826	1.4741					
95		1.5198	1.5109	1.5019	1.4932	1.4846	1.4761					
96		1.5220	1.5130	1.5040	1.4952	1.4867	1.4781					
97		1.5244	1.5152	1.5062	1.4974	1.4889	1.4802					
98		1.5278	1.5187	1.5096	1.5008	1.4922	1.4835					
99		1.5327	1.5235	1.5144	1.5056	1.4969	1.4881					
100		1.5402	1.5310	1.5217	1.5129	1.5040	1.4952					

*International Critical Tables, vol. 3, pp. 58–59.

TABLE 2-69 Perchloric Acid (HClO₄)*

%	d_4^{15}	d_4^{20}	d_4^{25}	d_4^{50}	%	d_4^{15}	d_4^{20}	d_4^{50}
1	1.0050		1.0020	0.9933	28	1.1900	1.1851	1.1645
2	1.0109		1.0070	0.9986	30	1.2067	1.2013	1.1800
4	1.0228		1.0169	0.9906	32	1.2239	1.2183	1.1960
6	1.0348		1.0270	1.0205	34	1.2418	1.2359	1.2130
8	1.0471		1.0372	1.0320	36	1.2603	1.2542	1.2310
10	1.0597		1.0475	1.0440	38	1.2794	1.2732	1.2490
12	1.0726			1.0560	40	1.2991	1.2927	1.2680
14	1.0589			1.0680	45	1.3521	1.3450	1.3180
16	1.0995			1.0810	50	1.4103	1.4018	1.3730
18	1.1135			1.0940	55	1.4733	1.4636	1.4320
20	1.1279			1.1070	60	1.5389	1.5298	1.4950
22	1.1428			1.1205	65	1.6059	1.5986	1.5620
24	1.1581			1.1345	70	1.6736	1.6680	1.6290
26	1.1738	1.1697		1.1490				

*International Critical Tables, vol. 3, p. 54.

TABLE 2-70 Phosphoric Acid (H₃PO₄)*

°C	2%	6%	14%	20%	26%	35%	50%	75%	100%
0	1.0113	1.0339	1.0811	1.1192					
10	1.0109	1.0330	1.0792	1.1167	1.1567	1.221	1.341		
20	1.0092	1.0309	1.0764	1.1134	1.1529	1.216	1.335	1.579	1.870
30	1.0065	1.0279	1.0728	1.1094	1.1484	1.211	1.329	1.572	1.862
40	1.0029	1.0241	1.0685	1.1048					

*International Critical Tables, vol. 3, p. 61.

TABLE 2-71 Potassium Bicarbonate (KHCO₃)*

°C	1%	2%	4%	6%	8%	10%
0	1.0066	1.0134	1.0270			
10	1.0064	1.0132	1.0268			
15	1.0058	1.0125	1.0260	1.0396	1.0534	1.0674
20	1.0049	1.0117	1.0252			
30	1.0024	1.0092	1.0228			
40	0.9990	1.0058	1.0195			
50	0.9949	1.0017	1.0154			
60	0.9901	0.9969	1.0106			
80	0.9786	0.9855	0.9993			
100	0.9653	0.9722	0.9860			

*International Critical Tables, vol. 3, p. 90.

TABLE 2-72 Potassium Bromide (KBr)*

%	d_4^{20}
1	1.0054
2	1.0127
6	1.0426
12	1.0903
20	1.1601
30	1.2593
40	1.3746

*International Critical Tables, vol. 3, p. 87.

TABLE 2-73 Potassium Carbonate (K_2CO_3)*

%	0 °C	10 °C	20 °C	40 °C	60 °C	80 °C	100 °C
1	1.0094	1.0089	1.0072	1.0010	0.9919	0.9803	0.9670
2	1.0189	1.0182	1.0163	1.0098	1.0005	0.9889	0.9756
4	1.0381	1.0369	1.0345	1.0276	1.0180	1.0063	0.9951
8	1.0768	1.0746	1.0715	1.0640	1.0538	1.0418	1.0291
12	1.1160	1.1131	1.1096	1.1013	1.0906	1.0786	1.0663
16	1.1562	1.1530	1.1490	1.1399	1.1290	1.1170	1.1049
20	1.1977	1.1941	1.1898	1.1801	1.1690	1.1570	1.1451
24	1.2405	1.2366	1.2320	1.2219	1.2106	1.1986	1.1869
28	1.2846	1.2804	1.2756	1.2652	1.2538	1.2418	1.2301
30	1.3071	1.3028	1.2979	1.2873	1.2759	1.2640	1.2522
35	1.3646	1.3600	1.3548	1.3440	1.3324	1.3206	1.3089
40	1.4244	1.4195	1.4141	1.4029	1.3913	1.3795	1.3678
45	1.4867	1.4815	1.4759	1.4644	1.4528	1.4408	1.4290
50	1.5517	1.5462	1.5404	1.5285	1.5169	1.5048	1.4928

*International Critical Tables, vol. 3, p. 90.

TABLE 2-74 Potassium Chromate (K_2CrO_4)*

%	d_4^{15}	d_4^{18}
1	1.0073	1.0066
2	1.0155	1.0147
4	1.0321	1.0311
8	1.0659	1.0647
12	1.1009	1.0999
16		1.1366
20		1.1748
24		1.2147
28		1.2566
30		1.2784

*International Critical Tables, vol. 3, p. 92.

TABLE 2-76 Potassium Chloride (KCl)*

%	0 °C	20 °C	25 °C	40 °C	60 °C	80 °C	100 °C
1.0	1.00661	1.00462	1.00342	0.99847	0.9894	0.9780	0.9646
2.0	1.01335	1.01103	1.00977	1.00471	0.9956	0.9842	0.9708
4.0	1.02690	1.02391	1.02255	1.01727	1.0080	0.9966	0.9634
8.0	1.05431	1.05003	1.04847	1.04278	1.0333	1.0219	1.0888
12.0	1.08222	1.07679	1.07506	1.06897	1.0592	1.0478	1.0350
16.0	1.11068	1.10434	1.10245	1.09600	1.0861	1.0746	1.0619
20.0	1.13973	1.13280	1.13072	1.12399	1.1138	1.1024	1.0897
24.0		1.16226	1.15995	1.15299	1.1425	1.1311	1.1185
28.0				1.18304	1.1723	1.1609	1.1483

%	110 °C	120 °C	130 °C	140 °C
3.79	0.9733	0.9663	0.9583	0.9502
7.45	0.9978	0.9899	0.9827	0.9745
13.62	1.0388	1.0313	1.0238	1.0159

*International Critical Tables, vol. 3, p. 87.

TABLE 2-77 Potassium Chrome Alum [$K_2Cr_2(SO_4)_4$]*

%	d_4^{15}
1	1.007
2	1.016
6	1.052
10	1.089
14	1.129
20	1.193
30	1.315
40	1.456
50	1.615

*International Critical Tables, vol. 3, p. 92.

TABLE 2-78 Potassium Hydroxide (KOH)*

%	d_4^{15}
1.0	1.0083
2.0	1.0175
4.0	1.0359
6.0	1.0544
8.0	1.0730
10.0	1.0918
15.0	1.1396
20.0	1.1884
25.0	1.2387
30.0	1.2905
35.0	1.3440
40.0	1.3991
45.0	1.4558
50.0	1.5143
51.7	1.5355 (sat'd. soln.)

*International Critical Tables, vol. 3, p. 86.

TABLE 2-79 Potassium Nitrate (KNO_3)*

%	0 °C	10 °C	20 °C	40 °C	60 °C	80 °C	100 °C
1	1.00654	1.00615	1.00447	0.99825	0.9890	0.9776	0.9641
2	1.01326	1.01262	1.01075	1.00430	0.9949	0.9834	0.9699
4	1.02677	1.02566	1.02344	1.01652	1.0068	0.9951	0.9816
8	1.05419	1.05226	1.04940	1.04152	1.0313	1.0192	1.0056
12	1.08221	1.07963	1.07620	1.06740	1.0567	1.0442	1.0304
16			1.10392	1.09432	1.0831	1.0703	1.0562
20			1.13261	1.12240	1.1106	1.0974	1.0831
24			1.16233	1.15175	1.1391	1.1256	1.1110

*International Critical Tables, vol. 3, p. 89.

TABLE 2-80 Potassium Dichromate ($K_2Cr_2O_7$)*

%	d_4^{20}
1	1.0052
2	1.0122
4	1.0264
6	1.0408
8	1.0554
10	1.0703

*International Critical Tables, vol. 3, p. 92.

TABLE 2-81 Potassium Sulfate (K_2SO_4)*

%	d_4^{20}
1	1.0063
2	1.0145
4	1.0310
6	1.0477
8	1.0646
10	1.0817

*International Critical Tables, vol. 3, p. 88.

TABLE 2-82 Potassium Sulfite (K_2SO_3)*

%	d_4^{15}
1	1.0073
2	1.0155
4	1.0322
8	1.0667
12	1.1026
16	1.1402
20	1.1793
24	1.2197
26	1.2404

*International Critical Tables, vol. 3, p. 87.

TABLE 2-83 Sodium Acetate ($NaC_2H_3O_2$)*

%	d_4^{20}
1	1.0033
2	1.0084
4	1.0186
8	1.0392
12	1.0598
18	1.0807
20	1.1021
26	1.1351
28	1.1462

*International Critical Tables, vol. 3, p. 83.

TABLE 2-84 Sodium Arsenate (Na_3AsO_4)*

%	d_4^{17}
1	1.0097
2	1.0207
4	1.0431
8	1.0892
10	1.1130
12	1.1373

*International Critical Tables, vol. 3, p. 82.

TABLE 2-85 Sodium Bichromate ($Na_2Cr_2O_7$)*

%	d_4^{15}
1	1.006
2	1.013
4	1.027
8	1.056
12	1.084
16	1.112
20	1.140
24	1.166
28	1.193
30	1.207
35	1.244
40	1.279
45	1.312
50	1.342

*International Critical Tables, vol. 3, p. 86.

TABLE 2-86 Sodium Bromide ($NaBr$)*

%	d_4^{20}
1	1.0060
2	1.0139
4	1.0298
8	1.0631
10	1.0803
12	1.0981
20	1.1745
30	1.2841
40	1.4138

*International Critical Tables, vol. 3, p. 80.

TABLE 2-87 Sodium Formate ($HCOONa$)*

%	d_4^{25}
1	1.003
2	1.009
4	1.022
8	1.048
12	1.074
16	1.100
20	1.127
24	1.155
28	1.184
30	1.199
35	1.236
40	1.274

*International Critical Tables, vol. 3, p. 83.

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TABLE 2-88 Sodium Carbonate (Na₂CO₃)*

%	0 °C	10 °C	20 °C	30 °C	40 °C	60 °C	80 °C	100 °C
1	1.0109	1.0103	1.0086	1.0058	1.0022	0.9929	0.9814	0.9683
2	1.0219	1.0210	1.0190	1.0159	1.0122	1.0027	0.9910	0.9782
4	1.0439	1.0423	1.0398	1.0363	1.0323	1.0223	1.0105	0.9980
8	1.0878	1.0850	1.0816	1.0775	1.0732	1.0625	1.0503	1.0380
12	1.1319	1.1284	1.1244	1.1200	1.1150	1.1039	1.0914	1.0787
14	1.1543	1.1506	1.1463	1.1417	1.1365	1.1251	1.1125	1.0996
16				1.1636				
18				1.1859				
20				1.2086				
24				1.2552				
28				1.3031				
30				1.3274				

*International Critical Tables, vol. 3, pp. 82–83.

TABLE 2-89 Sodium Chlorate (NaClO₃)*

%	d_4^{18}	%	d_4^{18}
1	1.0053	18	1.1288
2	1.0121	20	1.1449
4	1.0258	22	1.1614
6	1.0397	24	1.1782
8	1.0538	26	1.1953
10	1.0681	28	1.2128
12	1.0827	30	1.2307
14	1.0977	32	1.2491
16	1.1131	34	1.2680

*International Critical Tables, vol. 3, p. 80.

TABLE 2-90 Sodium Chloride (NaCl)*

%	0 °C	10 °C	25 °C	40 °C	60 °C	80 °C	100 °C
1	1.00747	1.00707	1.00409	0.99908	0.9900	0.9785	0.9651
2	1.01509	1.01442	1.01112	1.00593	0.9967	0.9852	0.9719
4	1.03038	1.02920	1.02530	1.01977	1.0103	0.9988	0.9855
8	1.06121	1.05907	1.05412	1.04798	1.0381	1.0264	1.0134
12	1.09244	1.08946	1.08365	1.07699	1.0667	1.0549	1.0420
16	1.12419	1.12056	1.11401	1.10688	1.0962	1.0842	1.0713
20	1.15663	1.15254	1.14533	1.13774	1.1268	1.1146	1.1017
24	1.18999	1.18557	1.17776	1.16971	1.1584	1.1463	1.1331
26	1.20709	1.20254	1.19443	1.18614	1.1747	1.1626	1.1492

*International Critical Tables, vol. 3, p. 79.

TABLE 2-91 Sodium Chromate (Na₂CrO₄)*

%	d_4^{18}
1	1.0074
2	1.0164
4	1.0344
8	1.0718
12	1.1110
16	1.1518
20	1.1942
24	1.2383
26	1.2611

*International Critical Tables, vol. 3, p. 86.

TABLE 2-95 Sodium Silicates*

Formula	Concentration, %												
	1	2	4	8	10	14	20	24	30	36	40	45	50
	d_4^{20}												
Na ₂ O/3.9SiO ₂	1.006	1.014	1.030	1.063	1.080	1.116	1.172	1.211	1.275				
Na ₂ O/3.36SiO ₂	1.006	1.014	1.030	1.065	1.083	1.120	1.179	1.222	1.290	1.365			
Na ₂ O/2.40SiO ₂	1.007	1.016	1.034	1.071	1.090	1.130							
Na ₂ O/2.44SiO ₂									1.309	1.387	1.445		
Na ₂ O/2.06SiO ₂	1.007	1.016	1.035	1.073	1.093	1.134	1.200	1.247	1.321	1.397	1.450	1.520	1.594
Na ₂ O/1.69SiO ₂	1.007	1.017	1.036	1.077	1.098	1.141	1.210	1.259	1.337	1.424			

*International Critical Tables, vol. 3, p. 85.

TABLE 2-92 Sodium Hydroxide (NaOH)*

%	0 °C	15 °C	20 °C	40 °C	60 °C	80 °C	100 °C
1	1.0124	1.01065	1.0095	1.0033	0.9941	0.9824	0.9693
2	1.0244	1.02198	1.0207	1.0139	1.0045	0.9929	0.9797
4	1.0482	1.04441	1.0428	1.0352	1.0254	1.0139	1.0009
8	1.0943	1.08887	1.0869	1.0780	1.0676	1.0560	1.0432
12	1.1399	1.13327	1.1309	1.1210	1.1101	1.0983	1.0855
16	1.1849	1.17761	1.1751	1.1645	1.1531	1.1408	1.1277
20	1.2296	1.22183	1.2191	1.2079	1.1960	1.1833	1.1700
24	1.2741	1.26582	1.2629	1.2512	1.2388	1.2259	1.2124
28	1.3182	1.3094	1.3064	1.2942	1.2814	1.2682	1.2546
32	1.3614	1.3520	1.3490	1.3362	1.3232	1.3097	1.2960
36	1.4030	1.3933	1.3900	1.3768	1.3634	1.3498	1.3360
40	1.4435	1.4334	1.4300	1.4164	1.4027	1.3889	1.3750
44	1.4825	1.4720	1.4685	1.4545	1.4405	1.4266	1.4127
48	1.5210	1.5102	1.5065	1.4922	1.4781	1.4641	1.4503
50	1.5400	1.5290	1.5253	1.5109	1.4967	1.4827	1.4690

*International Critical Tables, vol. 3, p. 79.

TABLE 2-93 Sodium Nitrate (NaNO₃)*

%	0 °C	20 °C	40 °C	60 °C	80 °C	100 °C
1	1.0071	1.0049	0.9986	0.9894	0.9779	0.9644
2	1.0144	1.0117	1.0050	0.9956	0.9840	0.9704
4	1.0290	1.0254	1.0180	1.0082	0.9964	0.9826
8	1.0587	1.0532	1.0447	1.0340	1.0218	1.0078
12	1.0891	1.0819	1.0724	1.0609	1.0481	1.0340
16	1.1203	1.1118	1.1013	1.0892	1.0757	1.0614
20	1.1526	1.1429	1.1314	1.1187	1.1048	1.0901
24	1.1860	1.1752	1.1629	1.1496	1.1351	1.1200
28	1.2204	1.2085	1.1955	1.1816	1.1667	1.1513
30	1.2380	1.2256	1.2122	1.1980	1.1830	1.1674
35	1.2834	1.2701	1.2560	1.2413	1.2258	1.2100
40	1.3316	1.3175	1.3027	1.2875	1.2715	1.2555
45		1.3683	1.3528	1.3371	1.3206	1.3044

*International Critical Tables, vol. 3, p. 82.

TABLE 2-94 Sodium Nitrite (NaNO₂)*

%	d_4^{15}
1	1.0058
2	1.0125
4	1.0260
8	1.0535
12	1.0816
16	1.1103
20	1.1394

*International Critical Tables, vol. 3, p. 82.

TABLE 2-96 Sodium Sulfate (Na₂SO₄)*

%	0 °C	20 °C	30 °C	40 °C	60 °C	80 °C	100 °C
1	1.0094	1.0073	1.0046	1.0010	0.9919	0.9805	0.9671
2	1.0189	1.0164	1.0135	1.0098	1.0007	0.9892	0.9758
4	1.0381	1.0348	1.0315	1.0276	1.0184	1.0068	0.9934
8	1.0773	1.0724	1.0682	1.0639	1.0544	1.0426	1.0292
12	1.1174	1.1109	1.1062	1.1015	1.0915	1.0795	1.0661
16	1.1585	1.1586	1.1456	1.1406	1.1299	1.1176	1.1042
20	1.2008	1.1915	1.1865	1.1813	1.1696	1.1569	
24	1.2443	1.2336	1.2292	1.2237			

*International Critical Tables, vol. 3, p. 81.

TABLE 2-97 Sodium Sulfide (Na₂S)*

%	d_4^{18}
1	1.0098
2	1.0211
4	1.0440
8	1.0907
12	1.1388
16	1.1885
18	1.2140

*International Critical Tables, vol. 3, p. 81.

TABLE 2-98 Sodium Sulfite (Na₂SO₃)*

%	d_4^{19}
1	1.0078
2	1.0172
4	1.0363
8	1.0751
12	1.1146
16	1.1549
18	1.1755

*International Critical Tables, vol. 3, p. 81.

TABLE 2-99 Sodium Thiosulfate (Na₂S₂O₃)*

%	d_4^{20}
1	1.0065
2	1.0148
4	1.0315
8	1.0654
12	1.1003
16	1.1365
20	1.1740
24	1.2128
28	1.2532
30	1.2739
35	1.3273
40	1.3827

*International Critical Tables, vol. 3, p. 81.

TABLE 2-100 Sodium Thiosulfate Pentahydrate (Na₂S₂O₃·5H₂O)

%	d_4^{19}
1	1.0052
2	1.0105
4	1.0211
8	1.0423
12	1.0639
16	1.0863
20	1.1087
24	1.1322
28	1.1558
30	1.1676
40	1.2297
50	1.2954

TABLE 2-101 Stannic Chloride (SnCl₄)*

%	d_4^{15}
1	1.007
2	1.015
4	1.031
8	1.064
12	1.099
16	1.135
20	1.173
24	1.212
28	1.255
30	1.278
35	1.337
40	1.403
45	1.475
50	1.555
55	1.644
60	1.742
65	1.851
70	1.971

*International Critical Tables, vol. 3, p. 63.

TABLE 2-102 Stannous Chloride (SnCl₂)*

%	d_4^{15}
1	1.0068
2	1.0146
4	1.0306
8	1.0638
12	1.0986
16	1.1353
20	1.1743
24	1.2159
28	1.2603
30	1.2837
35	1.3461
40	1.4145
45	1.4897
50	1.5729
55	1.6656
60	1.7695
65	1.8865

*International Critical Tables, vol. 3, p. 63.

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TABLE 2-103 Sulfuric Acid (H₂SO₄)*

%	0 °C	10 °C	15 °C	20 °C	25 °C	30 °C	40 °C	50 °C	60 °C	80 °C	100 °C
1	1.0074	1.0068	1.0060	1.0051	1.0038	1.0022	0.9986	0.9944	0.9895	0.9779	0.9645
2	1.0147	1.0138	1.0129	1.0118	1.0104	1.0087	1.0050	1.0006	0.9956	0.9839	0.9705
3	1.0219	1.0206	1.0197	1.0184	1.0169	1.0152	1.0113	1.0067	1.0017	0.9900	0.9766
4	1.0291	1.0275	1.0264	1.0250	1.0234	1.0216	1.0176	1.0129	1.0078	0.9961	0.9827
5	1.0364	1.0344	1.0332	1.0317	1.0300	1.0281	1.0240	1.0192	1.0140	1.0022	0.9888
6	1.0437	1.0414	1.0400	1.0385	1.0367	1.0347	1.0305	1.0256	1.0203	1.0084	0.9950
7	1.0511	1.0485	1.0469	1.0453	1.0434	1.0414	1.0371	1.0321	1.0266	1.0146	1.0013
8	1.0585	1.0556	1.0539	1.0522	1.0502	1.0481	1.0437	1.0386	1.0330	1.0209	1.0076
9	1.0660	1.0628	1.0610	1.0591	1.0571	1.0549	1.0503	1.0451	1.0395	1.0273	1.0140
10	1.0735	1.0700	1.0681	1.0661	1.0640	1.0617	1.0570	1.0517	1.0460	1.0338	1.0204
11	1.0810	1.0773	1.0753	1.0731	1.0710	1.0686	1.0637	1.0584	1.0526	1.0403	1.0269
12	1.0886	1.0846	1.0825	1.0802	1.0780	1.0756	1.0705	1.0651	1.0593	1.0469	1.0335
13	1.0962	1.0920	1.0898	1.0874	1.0851	1.0826	1.0774	1.0719	1.0661	1.0536	1.0402
14	1.1039	1.0994	1.0971	1.0947	1.0922	1.0897	1.0844	1.0788	1.0729	1.0603	1.0469
15	1.1116	1.1069	1.1045	1.1020	1.0994	1.0968	1.0914	1.0857	1.0798	1.0671	1.0537
16	1.1194	1.1145	1.1120	1.1094	1.1067	1.1040	1.0985	1.0927	1.0868	1.0740	1.0605
17	1.1272	1.1221	1.1195	1.1168	1.1141	1.1113	1.1057	1.0998	1.0938	1.0809	1.0674
18	1.1351	1.1298	1.1271	1.1243	1.1215	1.1187	1.1129	1.1070	1.1009	1.0879	1.0744
19	1.1430	1.1375	1.1347	1.1318	1.1290	1.1261	1.1202	1.1142	1.1081	1.0950	1.0814
20	1.1510	1.1453	1.1424	1.1394	1.1365	1.1335	1.1275	1.1215	1.1153	1.1021	1.0885
21	1.1590	1.1531	1.1501	1.1471	1.1441	1.1410	1.1349	1.1288	1.1226	1.1093	1.0957
22	1.1670	1.1609	1.1579	1.1548	1.1517	1.1486	1.1424	1.1362	1.1299	1.1166	1.1029
23	1.1751	1.1688	1.1657	1.1626	1.1594	1.1563	1.1500	1.1437	1.1373	1.1239	1.1102
24	1.1832	1.1768	1.1736	1.1704	1.1672	1.1640	1.1576	1.1512	1.1448	1.1313	1.1176
25	1.1914	1.1848	1.1816	1.1783	1.1750	1.1718	1.1653	1.1588	1.1523	1.1388	1.1250
26	1.1996	1.1929	1.1896	1.1862	1.1829	1.1796	1.1730	1.1665	1.1599	1.1463	1.1325
27	1.2078	1.2010	1.1976	1.1942	1.1909	1.1875	1.1808	1.1742	1.1676	1.1539	1.1400
28	1.2160	1.2091	1.2057	1.2023	1.1989	1.1955	1.1887	1.1820	1.1753	1.1616	1.1476
29	1.2243	1.2173	1.2138	1.2104	1.2069	1.2035	1.1966	1.1898	1.1831	1.1693	1.1553
30	1.2326	1.2255	1.2220	1.2185	1.2150	1.2115	1.2046	1.1977	1.1909	1.1771	1.1630
31	1.2409	1.2338	1.2302	1.2267	1.2232	1.2196	1.2126	1.2057	1.1988	1.1849	1.1708
32	1.2493	1.2421	1.2385	1.2349	1.2314	1.2278	1.2207	1.2137	1.2068	1.1928	1.1787
33	1.2577	1.2504	1.2468	1.2432	1.2396	1.2360	1.2289	1.2218	1.2148	1.2008	1.1866
34	1.2661	1.2588	1.2552	1.2515	1.2479	1.2443	1.2371	1.2300	1.2229	1.2088	1.1946
35	1.2746	1.2672	1.2636	1.2599	1.2563	1.2526	1.2454	1.2383	1.2311	1.2169	1.2027
36	1.2831	1.2757	1.2720	1.2684	1.2647	1.2610	1.2538	1.2466	1.2394	1.2251	1.2109
37	1.2917	1.2843	1.2805	1.2769	1.2732	1.2695	1.2622	1.2550	1.2477	1.2334	1.2192
38	1.3004	1.2929	1.2891	1.2855	1.2818	1.2780	1.2707	1.2635	1.2561	1.2418	1.2276
39	1.3091	1.3016	1.2978	1.2941	1.2904	1.2866	1.2793	1.2720	1.2646	1.2503	1.2361
40	1.3179	1.3103	1.3065	1.3028	1.2991	1.2953	1.2880	1.2806	1.2732	1.2589	1.2446
41	1.3268	1.3191	1.3153	1.3116	1.3079	1.3041	1.2967	1.2893	1.2819	1.2675	1.2532
42	1.3357	1.3280	1.3242	1.3205	1.3167	1.3129	1.3055	1.2981	1.2907	1.2762	1.2619
43	1.3447	1.3370	1.3332	1.3294	1.3256	1.3218	1.3144	1.3070	1.2996	1.2850	1.2707
44	1.3538	1.3461	1.3423	1.3384	1.3346	1.3308	1.3234	1.3160	1.3086	1.2939	1.2796
45	1.3630	1.3553	1.3515	1.3476	1.3437	1.3399	1.3325	1.3251	1.3177	1.3029	1.2886
46	1.3724	1.3646	1.3608	1.3569	1.3530	1.3492	1.3417	1.3343	1.3269	1.3120	1.2976
47	1.3819	1.3740	1.3702	1.3663	1.3624	1.3586	1.3510	1.3435	1.3362	1.3212	1.3067
48	1.3915	1.3835	1.3797	1.3758	1.3719	1.3680	1.3604	1.3528	1.3455	1.3305	1.3159
49	1.4012	1.3931	1.3893	1.3854	1.3814	1.3775	1.3699	1.3623	1.3549	1.3399	1.3253
50	1.4110	1.4029	1.3990	1.3951	1.3911	1.3872	1.3795	1.3719	1.3644	1.3494	1.3348
51	1.4209	1.4128	1.4088	1.4049	1.4009	1.3970	1.3893	1.3816	1.3740	1.3590	1.3444
52	1.4310	1.4228	1.4188	1.4148	1.4109	1.4069	1.3991	1.3914	1.3837	1.3687	1.3540
53	1.4412	1.4329	1.4289	1.4248	1.4209	1.4169	1.4091	1.4013	1.3936	1.3785	1.3637
54	1.4515	1.4431	1.4391	1.4350	1.4310	1.4270	1.4191	1.4113	1.4036	1.3884	1.3735
55	1.4619	1.4535	1.4494	1.4453	1.4412	1.4372	1.4293	1.4214	1.4137	1.3984	1.3834
56	1.4724	1.4640	1.4598	1.4557	1.4516	1.4475	1.4396	1.4317	1.4239	1.4085	1.3934
57	1.4830	1.4746	1.4703	1.4662	1.4621	1.4580	1.4500	1.4420	1.4342	1.4187	1.4035
58	1.4937	1.4852	1.4809	1.4768	1.4726	1.4685	1.4604	1.4524	1.4446	1.4290	1.4137
59	1.5045	1.4959	1.4916	1.4875	1.4832	1.4791	1.4709	1.4629	1.4551	1.4393	1.4240
60	1.5154	1.5067	1.5024	1.4983	1.4940	1.4898	1.4816	1.4735	1.4656	1.4497	1.4344
61	1.5264	1.5177	1.5133	1.5091	1.5048	1.5006	1.4923	1.4842	1.4762	1.4602	1.4449
62	1.5375	1.5287	1.5243	1.5200	1.5157	1.5115	1.5031	1.4950	1.4869	1.4708	1.4554
63	1.5487	1.5398	1.5354	1.5310	1.5267	1.5225	1.5140	1.5058	1.4977	1.4815	1.4660
64	1.5600	1.5510	1.5465	1.5421	1.5378	1.5335	1.5250	1.5167	1.5086	1.4923	1.4766

TABLE 2-103 Sulfuric Acid (H₂SO₄) (Concluded)

%	0 °C	10 °C	15 °C	20 °C	25 °C	30 °C	40 °C	50 °C	60 °C	80 °C	100 °C
65	1.5714	1.5623	1.5578	1.5533	1.5490	1.5446	1.5361	1.5277	1.5195	1.5031	1.4873
66	1.5828	1.5736	1.5691	1.5646	1.5602	1.5558	1.5472	1.5388	1.5305	1.5140	1.4981
67	1.5943	1.5850	1.5805	1.5760	1.5715	1.5671	1.5584	1.5499	1.5416	1.5249	1.5089
68	1.6059	1.5965	1.5920	1.5874	1.5829	1.5785	1.5697	1.5611	1.5528	1.5359	1.5198
69	1.6176	1.6081	1.6035	1.5989	1.5944	1.5899	1.5811	1.5724	1.5640	1.5470	1.5307
70	1.6293	1.6198	1.6151	1.6105	1.6059	1.6014	1.5925	1.5838	1.5753	1.5582	1.5417
71	1.6411	1.6315	1.6268	1.6221	1.6175	1.6130	1.6040	1.5952	1.5867	1.5694	1.5527
72	1.6529	1.6433	1.6385	1.6338	1.6292	1.6246	1.6155	1.6067	1.5981	1.5806	1.5637
73	1.6648	1.6551	1.6503	1.6456	1.6409	1.6363	1.6271	1.6182	1.6095	1.5919	1.5747
74	1.6768	1.6670	1.6622	1.6574	1.6526	1.6480	1.6387	1.6297	1.6209	1.6031	1.5857
75	1.6888	1.6789	1.6740	1.6692	1.6644	1.6597	1.6503	1.6412	1.6322	1.6142	1.5966
76	1.7008	1.6908	1.6858	1.6810	1.6761	1.6713	1.6619	1.6526	1.6435	1.6252	1.6074
77	1.7128	1.7026	1.6976	1.6927	1.6878	1.6829	1.6734	1.6640	1.6547	1.6361	1.6181
78	1.7247	1.7144	1.7093	1.7043	1.6994	1.6944	1.6847	1.6751	1.6657	1.6469	1.6286
79	1.7365	1.7261	1.7209	1.7158	1.7108	1.7058	1.6959	1.6862	1.6766	1.6575	1.6390
80	1.7482	1.7376	1.7323	1.7272	1.7221	1.7170	1.7069	1.6971	1.6873	1.6680	1.6493
81	1.7597	1.7489	1.7435	1.7383	1.7331	1.7279	1.7177	1.7077	1.6978	1.6782	1.6594
82	1.7709	1.7599	1.7544	1.7491	1.7437	1.7385	1.7281	1.7180	1.7080	1.6882	1.6692
83	1.7815	1.7704	1.7649	1.7594	1.7540	1.7487	1.7382	1.7279	1.7179	1.6979	1.6787
84	1.7916	1.7804	1.7748	1.7693	1.7639	1.7585	1.7479	1.7375	1.7274	1.7072	1.6878
85	1.8009	1.7897	1.7841	1.7786	1.7732	1.7678	1.7571	1.7466	1.7364	1.7161	1.6966
86	1.8095	1.7983	1.7927	1.7872	1.7818	1.7763	1.7657	1.7552	1.7449	1.7245	1.7050
87	1.8173	1.8061	1.8006	1.7951	1.7897	1.7842	1.7736	1.7632	1.7529	1.7324	1.7129
88	1.8243	1.8132	1.8077	1.8022	1.7968	1.7914	1.7809	1.7705	1.7602	1.7397	1.7202
89	1.8306	1.8195	1.8141	1.8087	1.8033	1.7979	1.7874	1.7770	1.7669	1.7464	1.7269
90	1.8361	1.8252	1.8198	1.8144	1.8091	1.8038	1.7933	1.7829	1.7729	1.7525	1.7331
91	1.8410	1.8302	1.8248	1.8195	1.8142	1.8090	1.7986	1.7883	1.7783	1.7581	1.7388
92	1.8453	1.8346	1.8293	1.8240	1.8188	1.8136	1.8033	1.7932	1.7832	1.7633	1.7439
93	1.8490	1.8384	1.8331	1.8279	1.8227	1.8176	1.8074	1.7974	1.7876	1.7681	1.7485
94	1.8520	1.8415	1.8363	1.8312	1.8260	1.8210	1.8109	1.8011	1.7914		
95	1.8544	1.8439	1.8388	1.8337	1.8286	1.8236	1.8137	1.8040	1.7944		
96	1.8560	1.8457	1.8406	1.8355	1.8305	1.8255	1.8157	1.8060	1.7965		
97	1.8569	1.8466	1.8414	1.8364	1.8314	1.8264	1.8166	1.8071	1.7977		
98	1.8567	1.8463	1.8411	1.8361	1.8310	1.8261	1.8163	1.8068	1.7976		
99	1.8551	1.8445	1.8393	1.8342	1.8292	1.8242	1.8145	1.8050	1.7958		
100	1.8517	1.8409	1.8357	1.8305	1.8255	1.8205	1.8107	1.8013	1.7922		

%	$d_4^{5.06}$	%	$d_4^{13.00}$	$d_4^{18.00}$
0.005	1.000 0140	0.05	0.999 810	0.999 028
.01	1.000 0576	0.1	1.000 185	0.999 400
.02	1.000 1434	0.2	1.000 912	1.000 119
.03	1.000 2276	0.3	1.001 623	1.000 820
.04	1.000 3104	0.4	1.002 326	1.001 512
.05	1.000 3920	0.5	1.003 023	1.002 197
.06	1.000 4726	0.6	1.003 716	1.002 877
.07	1.000 5523	0.8	1.005 090	1.004 227
.08	1.000 6313	1.0	1.006 452	1.005 570
.09	1.000 7098	1.2	1.007 807	1.006 909
.10	1.000 7880	1.4	1.009 159	1.008 247
.15	1.001 1732	1.6	1.010 510	1.009 583
.20	1.001 5514	1.8	1.011 860	1.010 918
.25	1.001 9254	2.0	1.013 209	1.012 252
.30	1.002 2961	2.2	1.014 557	1.013 586
.35	1.002 6639	2.4	1.015 904	1.014 919
.40	1.003 0292			
.45	1.003 3923			
.50	1.003 7534			

*International Critical Tables, vol. 3, pp. 56–57.

2-114 PHYSICAL AND CHEMICAL DATA

TABLE 2-104 Zinc Bromide (ZnBr₂)*

%	0 °C	20 °C	40 °C	60 °C	80 °C	100 °C
2	1.0188	1.0167	1.0102	1.0008	0.9890	0.9751
4	1.0381	1.0354	1.0285	1.0187	1.0065	0.9921
8	1.0777	1.0738	1.0660	1.0554	1.0422	1.0270
12	1.1186	1.1135	1.1046	1.0932	1.0789	1.0629
16	1.1609	1.1544	1.1445	1.1320	1.1169	1.1000
20	1.2043	1.1965	1.1855	1.1720	1.1560	1.1382
30	1.3288	1.3170	1.3030	1.2868	1.2688	1.2489
40	1.477	1.462	1.445	1.427	1.406	1.385
50	1.661	1.643	1.623	1.602	1.579	1.555
60	1.891	1.869	1.845	1.822	1.797	1.771
65	2.026	2.002	1.976	1.951	1.924	1.898

*International Critical Tables, vol. 3, p. 64.

TABLE 2-105 Zinc Chloride (ZnCl₂)*

%	0 °C	20 °C	40 °C	60 °C	80 °C	100 °C
2	1.0192	1.0167	1.0099	1.0003	0.9882	0.9739
4	1.0384	1.0350	1.0274	1.0172	1.0044	0.9894
8	1.0769	1.0715	1.0624	1.0508	1.0369	1.0211
12	1.1159	1.1085	1.0980	1.0853	1.0704	1.0541
16	1.1558	1.1468	1.1350	1.1212	1.1055	1.0888
20	1.1970	1.1866	1.1736	1.1590	1.1428	1.1255
30	1.3062	1.2928	1.2778	1.2614	1.2438	1.2252
40	1.4329	1.4173	1.4003	1.3824	1.3637	1.3441
50	1.5860	1.5681	1.5495	1.5300	1.5097	1.4892
60		1.749				
70		1.962				

*International Critical Tables, vol. 3, p. 64.

TABLE 2-106 Zinc Nitrate [Zn(NO₃)₂]*

%	18 °C	%	18 °C
2	1.0154	18	1.1652
4	1.0322	20	1.1865
6	1.0496	25	1.2427
8	1.0675	30	1.3029
10	1.0859	35	1.3678
12	1.1048	40	1.4378
14	1.1244	45	1.5134
16	1.1445	50	1.5944

*International Critical Tables, vol. 3, p. 65.

TABLE 2-107 Zinc Sulfate (ZnSO₄)*

%	20 °C
2	1.019
4	1.0403
6	1.0620
8	1.0842
10	1.1071
12	1.1308
14	1.1553
16	1.1806

*International Critical Tables, vol. 3, p. 65.

DENSITIES OF AQUEOUS ORGANIC SOLUTIONS*

UNITS AND UNITS CONVERSIONS

Unless otherwise noted, densities are given in grams per cubic centimeter. To convert to pounds per cubic foot, multiply by 62.43.

$$^{\circ}\text{F} = \% ^{\circ}\text{C} + 32$$

From *International Critical Tables*, vol. 3, pp. 115–129 unless otherwise stated. All compositions are in weight percent in vacuo. All density values are $d_4^t = \text{g/mL}$ in vacuo.

*For gasoline and aircraft fuels see Hibbard, NACA Res. Mem. E56121 (declassified 1958).

TABLE 2-108 Formic Acid (HCOOH)

%	0 °C	15 °C	20 °C	30 °C	%	0 °C	15 °C	20 °C	30 °C	%	0 °C	15 °C	20 °C	30 °C	%	0 °C	15 °C	20 °C	30 °C
0	0.9999	0.9991	0.9982	0.9957	25	1.0706	1.0627	1.0609	1.0540	50	1.1349	1.1225	1.1207	1.1098	75	1.1953	1.1794	1.1769	1.1636
1	1.0028	1.0019	1.0019	0.9980	26	1.0733	1.0652	1.0633	1.0564	51	1.1374	1.1248	1.1223	1.1120	76	1.1976	1.1816	1.1785	1.1656
2	1.0059	1.0045	1.0044	1.0004	27	1.0760	1.0678	1.0656	1.0587	52	1.1399	1.1271	1.1244	1.1142	77	1.1999	1.1837	1.1801	1.1676
3	1.0090	1.0072	1.0070	1.0028	28	1.0787	1.0702	1.0681	1.0609	53	1.1424	1.1294	1.1269	1.1164	78	1.2021	1.1859	1.1818	1.1697
4	1.0120	1.0100	1.0093	1.0053	29	1.0813	1.0726	1.0705	1.0632	54	1.1448	1.1318	1.1295	1.1186	79	1.2043	1.1881	1.1837	1.1717
5	1.0150	1.0124	1.0115	1.0075	30	1.0839	1.0750	1.0729	1.0654	55	1.1472	1.1341	1.1320	1.1208	80	1.2065	1.1902	1.1806	1.1737
6	1.0179	1.0151	1.0141	1.0101	31	1.0866	1.0774	1.0753	1.0676	56	1.1497	1.1365	1.1342	1.1230	81	1.2088	1.1924	1.1876	1.1758
7	1.0207	1.0177	1.0170	1.0125	32	1.0891	1.0798	1.0777	1.0699	57	1.1523	1.1388	1.1361	1.1253	82	1.2110	1.1944	1.1896	1.1778
8	1.0237	1.0204	1.0196	1.0149	33	1.0916	1.0821	1.0800	1.0721	58	1.1548	1.1411	1.1381	1.1274	83	1.2132	1.1965	1.1914	1.1798
9	1.0266	1.0230	1.0221	1.0173	34	1.0941	1.0844	1.0823	1.0743	59	1.1573	1.1434	1.1401	1.1295	84	1.2154	1.1985	1.1929	1.1817
10	1.0295	1.0256	1.0246	1.0197	35	1.0966	1.0867	1.0847	1.0766	60	1.1597	1.1458	1.1424	1.1317	85	1.2176	1.2005	1.1953	1.1837
11	1.0324	1.0281	1.0271	1.0221	36	1.0993	1.0892	1.0871	1.0788	61	1.1621	1.1481	1.1448	1.1338	86	1.2196	1.2025	1.1976	1.1856
12	1.0351	1.0306	1.0296	1.0244	37	1.1018	1.0916	1.0895	1.0810	62	1.1645	1.1504	1.1473	1.1360	87	1.2217	1.2045	1.1994	1.1875
13	1.0379	1.0330	1.0321	1.0267	38	1.1043	1.0940	1.0919	1.0832	63	1.1669	1.1526	1.1493	1.1382	88	1.2237	1.2064	1.2012	1.1893
14	1.0407	1.0355	1.0345	1.0290	39	1.1069	1.0964	1.0940	1.0854	64	1.1694	1.1549	1.1517	1.1403	89	1.2258	1.2084	1.2028	1.1910
15	1.0435	1.0380	1.0370	1.0313	40	1.1095	1.0988	1.0963	1.0876	65	1.1718	1.1572	1.1543	1.1425	90	1.2278	1.2102	1.2044	1.1927
16	1.0463	1.0405	1.0393	1.0336	41	1.1122	1.1012	1.0990	1.0898	66	1.1742	1.1595	1.1565	1.1446	91	1.2297	1.2121	1.2059	1.1945
17	1.0491	1.0430	1.0417	1.0358	42	1.1148	1.1036	1.1015	1.0920	67	1.1766	1.1618	1.1584	1.1467	92	1.2316	1.2139	1.2078	1.1961
18	1.0518	1.0455	1.0441	1.0381	43	1.1174	1.1060	1.1038	1.0943	68	1.1790	1.1640	1.1604	1.1489	93	1.2335	1.2157	1.2099	1.1978
19	1.0545	1.0480	1.0464	1.0404	44	1.1199	1.1084	1.1062	1.0965	69	1.1813	1.1663	1.1628	1.1510	94	1.2354	1.2174	1.2117	1.1994
20	1.0571	1.0505	1.0488	1.0427	45	1.1224	1.1109	1.1085	1.0987	70	1.1835	1.1685	1.1655	1.1531	95	1.2372	1.2191	1.2140	1.2008
21	1.0598	1.0532	1.0512	1.0451	46	1.1249	1.1133	1.1108	1.1009	71	1.1858	1.1707	1.1677	1.1552	96	1.2390	1.2208	1.2158	1.2022
22	1.0625	1.0556	1.0537	1.0473	47	1.1274	1.1156	1.1130	1.1031	72	1.1882	1.1729	1.1702	1.1573	97	1.2408	1.2224	1.2170	1.2036
23	1.0652	1.0580	1.0561	1.0496	48	1.1299	1.1179	1.1157	1.1053	73	1.1906	1.1751	1.1728	1.1595	98	1.2425	1.2240	1.2183	1.2048
24	1.0679	1.0604	1.0585	1.0518	49	1.1324	1.1202	1.1185	1.1076	74	1.1929	1.1773	1.1752	1.1615	99	1.2441	1.2257	1.2202	1.2061
															100	1.2456	1.2273	1.2212	1.2073

TABLE 2-109 Acetic Acid (CH₃COOH)

%	0 °C	10 °C	15 °C	20 °C	25 °C	30 °C	40 °C	%	0 °C	10 °C	15 °C	20 °C	25 °C	30 °C	40 °C
0	0.9999	0.9997	0.9991	0.9982	0.9971	0.9957	0.9922	50	1.0729	1.0654	1.0613	1.0575	1.0534	1.0492	1.0408
1	1.0016	1.0013	1.0006	0.9996	0.9987	0.9971	0.9934	51	1.0738	1.0663	1.0622	1.0582	1.0542	1.0499	1.0414
2	1.0033	1.0029	1.0021	1.0012	1.0000	0.9984	0.9946	52	1.0748	1.0671	1.0629	1.0590	1.0549	1.0506	1.0421
3	1.0051	1.0044	1.0036	1.0025	1.0013	0.9997	0.9958	53	1.0757	1.0679	1.0637	1.0597	1.0555	1.0512	1.0427
4	1.0070	1.0060	1.0051	1.0040	1.0027	1.0011	0.9970	54	1.0765	1.0687	1.0644	1.0604	1.0562	1.0518	1.0432
5	1.0088	1.0076	1.0066	1.0055	1.0041	1.0024	0.9982	55	1.0774	1.0694	1.0651	1.0611	1.0568	1.0525	1.0438
6	1.0106	1.0092	1.0081	1.0069	1.0055	1.0037	0.9994	56	1.0782	1.0701	1.0658	1.0618	1.0574	1.0531	1.0443
7	1.0124	1.0108	1.0096	1.0083	1.0068	1.0050	1.0006	57	1.0790	1.0708	1.0665	1.0624	1.0580	1.0536	1.0448
8	1.0142	1.0124	1.0111	1.0097	1.0081	1.0063	1.0018	58	1.0798	1.0715	1.0672	1.0631	1.0586	1.0542	1.0453
9	1.0159	1.0140	1.0126	1.0111	1.0094	1.0076	1.0030	59	1.0805	1.0722	1.0678	1.0637	1.0592	1.0547	1.0458
10	1.0177	1.0156	1.0141	1.0125	1.0107	1.0089	1.0042	60	1.0813	1.0728	1.0684	1.0642	1.0597	1.0552	1.0462
11	1.0194	1.0171	1.0155	1.0139	1.0120	1.0102	1.0054	61	1.0820	1.0734	1.0690	1.0648	1.0602	1.0557	1.0466
12	1.0211	1.0187	1.0170	1.0154	1.0133	1.0115	1.0065	62	1.0826	1.0740	1.0696	1.0653	1.0607	1.0562	1.0470
13	1.0228	1.0202	1.0184	1.0168	1.0146	1.0127	1.0077	63	1.0833	1.0746	1.0701	1.0658	1.0612	1.0566	1.0473
14	1.0245	1.0217	1.0199	1.0182	1.0159	1.0139	1.0088	64	1.0838	1.0752	1.0706	1.0662	1.0616	1.0571	1.0477
15	1.0262	1.0232	1.0213	1.0195	1.0172	1.0151	1.0099	65	1.0844	1.0757	1.0711	1.0666	1.0621	1.0575	1.0480
16	1.0278	1.0247	1.0227	1.0209	1.0185	1.0163	1.0110	66	1.0850	1.0762	1.0716	1.0671	1.0624	1.0578	1.0483
17	1.0295	1.0262	1.0241	1.0223	1.0198	1.0175	1.0121	67	1.0856	1.0767	1.0720	1.0675	1.0628	1.0582	1.0486
18	1.0311	1.0276	1.0255	1.0236	1.0210	1.0187	1.0132	68	1.0860	1.0771	1.0725	1.0678	1.0631	1.0585	1.0489
19	1.0327	1.0291	1.0269	1.0250	1.0223	1.0198	1.0142	69	1.0865	1.0775	1.0729	1.0682	1.0634	1.0588	1.0491
20	1.0343	1.0305	1.0283	1.0263	1.0235	1.0210	1.0153	70	1.0869	1.0779	1.0732	1.0685	1.0637	1.0590	1.0493
21	1.0358	1.0319	1.0297	1.0276	1.0248	1.0222	1.0164	71	1.0874	1.0783	1.0736	1.0687	1.0640	1.0592	1.0495
22	1.0374	1.0333	1.0310	1.0288	1.0260	1.0233	1.0174	72	1.0877	1.0786	1.0738	1.0690	1.0642	1.0594	1.0496
23	1.0389	1.0347	1.0323	1.0301	1.0272	1.0244	1.0185	73	1.0881	1.0789	1.0741	1.0693	1.0644	1.0595	1.0497
24	1.0404	1.0361	1.0336	1.0313	1.0283	1.0256	1.0195	74	1.0884	1.0792	1.0743	1.0694	1.0645	1.0596	1.0498
25	1.0419	1.0375	1.0349	1.0326	1.0295	1.0267	1.0205	75	1.0887	1.0794	1.0745	1.0696	1.0647	1.0597	1.0499
26	1.0434	1.0388	1.0362	1.0338	1.0307	1.0278	1.0215	76	1.0889	1.0796	1.0746	1.0698	1.0648	1.0598	1.0499
27	1.0449	1.0401	1.0374	1.0349	1.0318	1.0289	1.0225	77	1.0891	1.0797	1.0747	1.0699	1.0648	1.0598	1.0499
28	1.0463	1.0414	1.0386	1.0361	1.0329	1.0299	1.0234	78	1.0893	1.0798	1.0747	1.0700	1.0648	1.0598	1.0498
29	1.0477	1.0427	1.0399	1.0372	1.0340	1.0310	1.0244	79	1.0894	1.0798	1.0747	1.0700	1.0648	1.0597	1.0497
30	1.0491	1.0440	1.0411	1.0384	1.0350	1.0320	1.0253	80	1.0895	1.0798	1.0747	1.0700	1.0647	1.0596	1.0495
31	1.0505	1.0453	1.0423	1.0395	1.0361	1.0330	1.0262	81	1.0895	1.0797	1.0745	1.0699	1.0646	1.0594	1.0493
32	1.0519	1.0465	1.0435	1.0406	1.0372	1.0341	1.0272	82	1.0895	1.0796	1.0743	1.0698	1.0644	1.0592	1.0490
33	1.0532	1.0477	1.0446	1.0417	1.0382	1.0351	1.0281	83	1.0895	1.0795	1.0741	1.0696	1.0642	1.0589	1.0487
34	1.0545	1.0489	1.0458	1.0428	1.0392	1.0361	1.0289	84	1.0893	1.0793	1.0738	1.0693	1.0638	1.0585	1.0483
35	1.0558	1.0501	1.0469	1.0438	1.0402	1.0371	1.0298	85	1.0891	1.0790	1.0735	1.0689	1.0635	1.0582	1.0479
36	1.0571	1.0513	1.0480	1.0449	1.0412	1.0380	1.0306	86	1.0887	1.0787	1.0731	1.0685	1.0630	1.0576	1.0473
37	1.0584	1.0524	1.0491	1.0459	1.0422	1.0390	1.0314	87	1.0883	1.0783	1.0726	1.0680	1.0626	1.0571	1.0467
38	1.0596	1.0535	1.0501	1.0469	1.0432	1.0399	1.0322	88	1.0877	1.0778	1.0721	1.0675	1.0620	1.0564	1.0460
39	1.0608	1.0546	1.0512	1.0479	1.0441	1.0408	1.0330	89	1.0872	1.0773	1.0715	1.0668	1.0613	1.0557	1.0453
40	1.0621	1.0557	1.0522	1.0488	1.0450	1.0416	1.0338	90	1.0865	1.0766	1.0708	1.0661	1.0605	1.0549	1.0445
41	1.0633	1.0568	1.0532	1.0498	1.0460	1.0425	1.0346	91	1.0857	1.0758	1.0700	1.0652	1.0597	1.0541	1.0436
42	1.0644	1.0578	1.0542	1.0507	1.0469	1.0433	1.0353	92	1.0848	1.0749	1.0690	1.0643	1.0587	1.0530	1.0426
43	1.0656	1.0588	1.0551	1.0516	1.0477	1.0441	1.0361	93	1.0838	1.0739	1.0680	1.0632	1.0577	1.0518	1.0414
44	1.0667	1.0598	1.0561	1.0525	1.0486	1.0449	1.0368	94	1.0826	1.0727	1.0667	1.0619	1.0564	1.0506	1.0401
45	1.0679	1.0608	1.0570	1.0534	1.0495	1.0456	1.0375	95	1.0813	1.0714	1.0652	1.0605	1.0551	1.0491	1.0386
46	1.0689	1.0618	1.0579	1.0542	1.0503	1.0464	1.0382	96	1.0798		1.0632	1.0588	1.0535	1.0473	1.0368
47	1.0699	1.0627	1.0588	1.0551	1.0511	1.0471	1.0389	97	1.0780		1.0611	1.0570	1.0516	1.0454	1.0348
48	1.0709	1.0636	1.0597	1.0559	1.0518	1.0479	1.0395	98	1.0759		1.0590	1.0549	1.0495	1.0431	1.0325
49	1.0720	1.0645	1.0605	1.0567	1.0526	1.0486	1.0402	99	1.0730		1.0567	1.0524	1.0468	1.0407	1.0299
								100	1.0697		1.0545	1.0498	1.0440	1.0380	1.0271

TABLE 2-110 Oxalic Acid ($\text{H}_2\text{C}_2\text{O}_4$)

%	$d_4^{17.5}$	%	$d_4^{17.5}$
1	1.0035	8	1.0280
2	1.0070	10	1.0350
4	1.0140	12	1.0420

TABLE 2-111 Methyl Alcohol (CH_3OH)*

%	0°C	10°C	15.56°C	20°C	15°C	%	0°C	10°C	15.56°C	20°C	15°C	%	0°C	10°C	15.56°C	20°C	15°C
0	0.9999	0.9997	0.9990	0.9982	0.99913	35	0.9534	0.9484	0.9456	0.9433	0.94570	70	0.8869	0.8794	0.8748	0.8715	0.87507
1	0.9981	0.9980	0.9973	0.9965	0.99727	36	0.9520	0.9469	0.9440	0.9416	0.94404	71	0.8847	0.8770	0.8726	0.8690	0.87271
2	0.9963	0.9962	0.9955	0.9948	0.99543	37	0.9505	0.9453	0.9422	0.9398	0.94237	72	0.8824	0.8747	0.8702	0.8665	0.87033
3	0.9946	0.9945	0.9938	0.9931	0.99370	38	0.9490	0.9437	0.9405	0.9381	0.94067	73	0.8801	0.8724	0.8678	0.8641	0.86792
4	0.9930	0.9929	0.9921	0.9914	0.99198	39	0.9475	0.9420	0.9387	0.9363	0.93894	74	0.8778	0.8699	0.8653	0.8616	0.86546
5	0.9914	0.9912	0.9904	0.9896	0.99029	40	0.9459	0.9403	0.9369	0.9345	0.93720	75	0.8754	0.8676	0.8629	0.8592	0.86300
6	0.9899	0.9896	0.9889	0.9880	0.98864	41	0.9443	0.9387	0.9351	0.9327	0.93543	76	0.8729	0.8651	0.8604	0.8567	0.86051
7	0.9884	0.9881	0.9872	0.9863	0.98701	42	0.9427	0.9370	0.9333	0.9309	0.93365	77	0.8705	0.8626	0.8579	0.8542	0.85801
8	0.9870	0.9865	0.9857	0.9847	0.98547	43	0.9411	0.9352	0.9315	0.9290	0.93185	78	0.8680	0.8602	0.8554	0.8518	0.85551
9	0.9856	0.9849	0.9841	0.9831	0.98394	44	0.9395	0.9334	0.9297	0.9272	0.93001	79	0.8657	0.8577	0.8529	0.8494	0.85300
10	0.9842	0.9834	0.9826	0.9815	0.98241	45	0.9377	0.9316	0.9279	0.9252	0.92815	80	0.8634	0.8551	0.8503	0.8469	0.85048
11	0.9829	0.9820	0.9811	0.9799	0.98093	46	0.9360	0.9298	0.9261	0.9234	0.92627	81	0.8610	0.8527	0.8478	0.8446	0.84794
12	0.9816	0.9805	0.9796	0.9784	0.97945	47	0.9342	0.9279	0.9242	0.9214	0.92436	82	0.8585	0.8501	0.8452	0.8420	0.84536
13	0.9804	0.9791	0.9781	0.9768	0.97802	48	0.9324	0.9260	0.9223	0.9196	0.92242	83	0.8560	0.8475	0.8426	0.8394	0.84274
14	0.9792	0.9778	0.9766	0.9754	0.97660	49	0.9306	0.9240	0.9204	0.9176	0.92048	84	0.8535	0.8449	0.8400	0.8366	0.84009
15	0.9780	0.9764	0.9752	0.9740	0.97518	50	0.9287	0.9221	0.9185	0.9156	0.91852	85	0.8510	0.8422	0.8374	0.8340	0.83742
16	0.9769	0.9751	0.9738	0.9725	0.97377	51	0.9269	0.9202	0.9166	0.9135	0.91653	86	0.8483	0.8394	0.8347	0.8314	0.83475
17	0.9758	0.9739	0.9723	0.9710	0.97237	52	0.9250	0.9182	0.9146	0.9114	0.91451	87	0.8456	0.8367	0.8320	0.8286	0.83207
18	0.9747	0.9726	0.9709	0.9696	0.97096	53	0.9230	0.9162	0.9126	0.9094	0.91248	88	0.8428	0.8340	0.8294	0.8258	0.82937
19	0.9736	0.9713	0.9695	0.9681	0.96955	54	0.9211	0.9142	0.9106	0.9073	0.91044	89	0.8400	0.8314	0.8267	0.8230	0.82667
20	0.9725	0.9700	0.9680	0.9666	0.96814	55	0.9191	0.9122	0.9086	0.9052	0.90839	90	0.8374	0.8287	0.8239	0.8202	0.82396
21	0.9714	0.9687	0.9666	0.9651	0.96673	56	0.9172	0.9101	0.9065	0.9032	0.90631	91	0.8347	0.8261	0.8212	0.8174	0.82124
22	0.9702	0.9673	0.9652	0.9636	0.96533	57	0.9151	0.9080	0.9045	0.9010	0.90421	92	0.8320	0.8234	0.8185	0.8146	0.81849
23	0.9690	0.9660	0.9638	0.9622	0.96392	58	0.9131	0.9060	0.9024	0.8988	0.90210	93	0.8293	0.8208	0.8157	0.8118	0.81568
24	0.9678	0.9646	0.9624	0.9607	0.96251	59	0.9111	0.9039	0.9002	0.8968	0.89996	94	0.8266	0.8180	0.8129	0.8090	0.81285
25	0.9666	0.9632	0.9609	0.9592	0.96108	60	0.9090	0.9018	0.8980	0.8946	0.89781	95	0.8240	0.8152	0.8101	0.8062	0.80999
26	0.9654	0.9618	0.9595	0.9576	0.95963	61	0.9068	0.8998	0.8958	0.8924	0.89563	96	0.8212	0.8124	0.8073	0.8034	0.80713
27	0.9642	0.9604	0.9580	0.9562	0.95817	62	0.9046	0.8977	0.8936	0.8902	0.89341	97	0.8186	0.8096	0.8045	0.8005	0.80428
28	0.9629	0.9590	0.9565	0.9546	0.95668	63	0.9024	0.8955	0.8913	0.8879	0.89117	98	0.8158	0.8068	0.8016	0.7976	0.80143
29	0.9616	0.9575	0.9550	0.9531	0.95518	64	0.9002	0.8933	0.8890	0.8856	0.88890	99	0.8130	0.8040	0.7987	0.7948	0.79859
30	0.9604	0.9560	0.9535	0.9515	0.95366	65	0.8980	0.8911	0.8867	0.8834	0.88662	100	0.8102	0.8009	0.7959	0.7917	0.79577
31	0.9590	0.9546	0.9521	0.9499	0.95213	66	0.8958	0.8888	0.8844	0.8811	0.88433						
32	0.9576	0.9531	0.9505	0.9483	0.95056	67	0.8935	0.8865	0.8820	0.8787	0.88203						
33	0.9563	0.9516	0.9489	0.9466	0.94896	68	0.8913	0.8842	0.8797	0.8763	0.87971						
34	0.9549	0.9500	0.9473	0.9450	0.94734	69	0.8891	0.8818	0.8771	0.8738	0.87739						

*It should be noted that the values for 100 percent do not agree with some data available elsewhere, e.g., *American Institute of Physics Handbook*, McGraw-Hill, New York, 1957. Also, see Attack, *Handbook of Chemical Data*, Reinhold, New York, 1957. Also, see Tables 2-234 and 2-305 for pure component densities.

TABLE 2-112 Ethyl Alcohol (C₂H₅OH)*

%	10 °C	15 °C	20 °C	25 °C	30 °C	35 °C	40 °C	%	10 °C	15 °C	20 °C	25 °C	30 °C	35 °C	40 °C
0	0.99973	0.99913	0.99823	0.99708	0.99568	0.99406	0.99225	50	0.92126	0.91776	0.91384	0.90985	0.90580	0.90168	0.89750
1	785	725	636	520	379	217	034	51	0.91943	555	160	760	353	0.89940	519
2	602	542	453	336	194	031	0.98846	52	723	333	0.90936	534	125	710	258
3	426	365	275	157	014	0.98849	663	53	502	110	711	307	0.89896	479	056
4	258	195	103	0.98984	0.98839	672	485	54	279	0.90885	485	079	667	248	0.88823
5	098	032	0.98938	817	670	501	311	55	055	659	258	0.89850	437	016	589
6	0.98946	0.98877	780	656	507	335	142	56	0.90831	433	031	621	206	0.88784	356
7	801	729	627	500	347	172	0.97975	57	607	207	0.89803	392	0.88975	552	122
8	660	584	478	346	189	009	808	58	381	0.89980	574	162	744	319	0.87888
9	524	442	331	193	031	0.97846	641	59	154	752	344	0.88931	512	085	653
10	393	304	187	043	0.97875	685	475	60	0.89927	523	113	699	278	0.87851	417
11	267	171	047	0.97897	723	527	312	61	698	293	0.88882	446	044	615	180
12	145	041	0.97910	753	573	371	150	62	468	062	650	233	0.87809	379	0.86943
13	026	0.97914	775	611	424	216	0.96989	63	237	0.88830	417	0.87998	574	142	705
14	0.97911	790	643	472	278	063	829	64	006	597	183	763	337	0.86905	466
15	800	669	514	334	133	0.96911	670	65	0.88774	364	0.87948	527	100	667	227
16	692	552	387	199	0.96990	760	512	66	541	130	713	291	0.86863	429	0.85987
17	583	433	259	062	844	607	352	67	308	0.87895	477	054	625	190	747
18	473	313	129	0.96923	697	452	189	68	074	660	241	0.86817	387	0.85950	407
19	363	191	0.96997	782	547	294	023	69	0.87839	424	004	579	148	710	266
20	252	068	864	639	395	134	0.95856	70	602	187	0.86766	340	0.85908	470	025
21	139	0.96944	729	495	242	0.95973	687	71	365	0.86949	527	100	667	228	0.84783
22	024	818	592	348	087	809	516	72	127	710	287	0.85859	426	0.84986	540
23	0.96907	689	453	199	0.95929	643	343	73	0.86888	470	047	618	184	743	297
24	787	558	312	048	769	476	168	74	648	229	0.85806	376	0.84941	500	053
25	665	424	168	0.95895	607	306	0.94991	75	408	0.85988	564	134	698	257	0.83809
26	539	287	020	738	442	133	810	76	168	747	322	0.84891	455	013	564
27	406	144	0.95867	576	272	0.94955	625	77	0.85927	505	079	647	211	0.83768	319
28	268	0.95996	710	410	098	774	438	78	685	262	0.84835	403	0.83966	523	074
29	125	844	548	241	0.94922	590	248	79	442	018	590	158	720	277	0.82827
30	0.95977	686	382	067	741	403	055	80	197	0.84772	344	0.83911	473	029	578
31	823	524	212	0.94890	557	214	0.93860	81	0.84950	525	096	664	224	0.82780	329
32	665	357	038	709	370	021	662	82	702	277	0.83848	415	0.82974	530	079
33	502	186	0.94860	525	180	0.93825	461	83	453	028	599	164	724	279	0.81828
34	334	011	679	337	0.93986	626	257	84	203	0.83777	348	0.82913	473	027	576
35	162	0.94832	494	146	790	425	051	85	0.83951	525	095	660	220	0.81774	322
36	0.94986	650	306	0.93952	591	221	0.92843	86	697	271	0.82840	405	0.81965	519	067
37	805	464	114	756	390	016	634	87	441	014	583	148	708	262	0.80811
38	620	273	0.93919	556	186	0.92808	422	88	181	0.82754	323	0.81888	448	003	552
39	431	079	720	353	0.92979	597	208	89	0.82919	492	062	626	186	0.80742	291
40	238	0.93882	518	148	770	385	0.91992	90	654	227	0.81797	362	0.80922	478	028
41	042	682	314	0.92940	558	170	774	91	386	0.81959	529	094	655	211	0.79761
42	0.93842	478	107	729	344	0.91952	554	92	114	688	257	0.80823	384	0.79941	491
43	639	271	0.92897	516	128	733	332	93	0.81839	413	0.80983	549	111	669	220
44	433	062	685	301	0.91910	513	108	94	561	134	705	272	0.79835	393	0.78947
45	226	0.92852	472	085	692	291	0.90884	95	278	0.80852	424	0.79991	555	114	670
46	017	640	257	0.91868	472	069	660	96	0.80991	566	138	706	271	0.78831	388
47	0.92806	426	041	649	250	0.90845	434	97	698	274	0.79846	415	0.78981	542	100
48	593	211	0.91823	429	028	621	207	98	399	0.79975	547	117	684	247	0.77806
49	379	0.91995	604	208	0.90805	396	0.89979	99	094	670	243	0.78814	382	0.77946	507
								100	0.79784	360	0.78934	506	075	641	203

*For data from -78° to 78°C, see p. 2-142, Table 2N-5, *American Institute of Physics Handbook*, McGraw-Hill, New York, 1957. See Tables 2-214 and 2-305 for pure component densities.

2-118 PHYSICAL AND CHEMICAL DATA

TABLE 2-113 Densities of Mixtures of C₂H₅OH and H₂O at 20°C

g/mL																					
% alcohol by weight	Tenths of %										% alcohol by weight	Tenths of %									
	0	1	2	3	4	5	6	7	8	9		0	1	2	3	4	5	6	7	8	9
0	0.99823	804	785	766	748	729	710	692	673	655	50	0.91384	361	339	317	295	272	250	228	206	183
1	636	618	599	581	562	544	525	507	489	471	51	160	138	116	093	071	049	026	004	*981	*959
2	453	435	417	399	381	363	345	327	310	292	52	0.90936	914	891	869	846	824	801	779	756	734
3	275	257	240	222	205	188	171	154	137	120	53	711	689	666	644	621	598	576	553	531	508
4	103	087	070	053	037	020	003	*987	*971	*954	54	485	463	440	417	395	372	349	327	304	281
5	0.98938	922	906	890	874	859	843	827	811	796	55	258	236	213	190	167	145	122	099	076	054
6	780	765	749	734	718	703	688	673	658	642	56	031	008	*985	*962	*939	*917	*894	*871	*848	*825
7	627	612	597	582	567	553	538	523	508	493	57	0.89803	780	757	734	711	688	665	643	620	597
8	478	463	449	434	419	404	389	374	360	345	58	574	551	528	505	482	459	436	413	390	367
9	331	316	301	287	273	258	244	229	215	201	59	344	321	298	275	252	229	206	183	160	137
10	187	172	158	144	130	117	103	089	075	061	60	113	090	067	044	021	*998	*975	*951	*928	*905
11	047	033	019	006	*992	*978	*964	*951	*937	*923	61	0.88882	859	836	812	789	766	743	720	696	673
12	0.97910	896	883	869	855	842	828	815	801	788	62	650	626	603	580	557	533	510	487	463	440
13	775	761	748	735	722	709	696	683	670	657	63	417	393	370	347	323	300	277	253	230	206
14	643	630	617	604	591	578	565	552	539	526	64	183	160	136	113	089	066	042	019	*995	*972
15	514	501	488	475	462	450	438	425	412	400	65	0.87948	925	901	878	854	831	807	784	760	737
16	387	374	361	349	336	323	310	297	284	272	66	713	689	666	642	619	595	572	548	524	501
17	259	246	233	220	207	194	181	168	155	142	67	477	454	430	406	383	359	336	312	288	265
18	129	116	103	089	076	063	050	037	024	010	68	241	218	194	170	147	123	099	075	052	028
19	0.96997	984	971	957	944	931	917	904	891	877	69	004	*981	*957	*933	*909	*885	*862	*838	*814	*790
20	864	850	837	823	810	796	783	769	756	742	70	0.86766	742	718	694	671	647	623	599	575	551
21	729	716	702	688	675	661	647	634	620	606	71	527	503	479	455	431	407	383	359	335	311
22	592	578	564	551	537	523	509	495	481	467	72	287	263	239	215	191	167	143	119	095	071
23	453	439	425	411	396	382	368	354	340	326	73	047	022	*998	*974	*950	*926	*902	*878	*854	*830
24	312	297	283	269	254	240	225	211	196	182	74	0.85806	781	757	733	709	685	661	636	612	588
25	168	153	139	124	109	094	080	065	050	035	75	564	540	515	491	467	443	419	394	370	346
26	020	005	*990	*975	*959	*944	*929	*914	*898	*883	76	322	297	273	249	225	200	176	152	128	103
27	0.95867	851	836	820	805	789	773	757	742	726	77	079	055	031	006	*982	*958	*933	*909	*884	*860
28	710	694	678	662	646	630	613	597	581	565	78	0.84835	811	787	762	738	713	689	664	640	615
29	548	532	516	499	483	466	450	433	416	400	79	590	566	541	517	492	467	443	418	393	369
30	382	365	349	332	315	298	281	264	247	230	80	344	319	294	270	245	220	196	171	146	121
31	212	195	178	161	143	126	108	091	074	056	81	096	072	047	022	*997	*972	*947	*923	*898	*873
32	038	020	003	*985	*967	*950	*932	*914	*896	*878	82	0.83848	823	798	773	748	723	698	674	649	624
33	0.94860	842	824	806	788	770	752	734	715	697	83	599	574	549	523	498	473	448	423	398	373
34	679	660	642	624	605	587	568	550	531	512	84	348	323	297	272	247	222	196	171	146	120
35	494	475	456	438	419	400	382	363	344	325	85	095	070	044	019	*994	*968	*943	*917	*892	*866
36	306	287	268	249	230	211	192	172	153	134	86	0.82840	815	789	763	738	712	686	660	635	609
37	114	095	075	056	036	017	*997	*978	*958	*939	87	583	557	531	505	479	453	427	401	375	349
38	0.93919	899	879	859	840	820	800	780	760	740	88	323	297	271	245	219	193	167	140	114	088
39	720	700	680	660	640	620	599	579	559	539	89	062	035	009	*983	*956	*930	*903	*877	*850	*824
40	518	498	478	458	437	417	396	376	356	335	90	0.81797	770	744	717	690	664	637	610	583	556
41	314	294	273	253	232	212	191	170	149	129	91	529	502	475	448	421	394	366	339	312	285
42	107	086	065	044	023	002	*981	*960	*939	*918	92	257	230	203	175	148	120	093	066	038	010
43	0.92897	876	855	834	812	791	770	749	728	707	93	0.80983	955	928	900	872	844	817	789	761	733
44	685	664	642	621	600	579	557	536	515	493	94	705	677	649	621	593	565	537	509	480	452
45	472	450	429	408	386	365	343	322	300	279	95	424	395	367	338	310	281	253	224	195	166
46	257	236	214	193	171	150	128	106	085	063	96	138	109	080	051	022	*993	*963	*934	*905	*875
47	041	019	*997	*976	*954	*932	*910	*889	*867	*845	97	0.79846	816	787	757	727	698	668	638	608	578
48	0.91823	801	780	758	736	714	692	670	648	626	98	547	517	487	456	426	396	365	335	305	274
49	604	582	560	538	516	494	472	450	428	406	99	243	213	182	151	120	089	059	028	*997	*966
											100	0.78934									

*Indicates change in the first two decimal places.

TABLE 2-114 Specific Gravity {60°/60°F [(15.56°/15.56°C)]} of Mixtures by Volume of C₂H₅OH and H₂O

% alcohol by volume at 60°F	Tenths of %										% alcohol by volume at 60°F	Tenths of %									
	0	1	2	3	4	5	6	7	8	9		0	1	2	3	4	5	6	7	8	9
0	1.00000	*985	*970	*955	*940	*925	*910	*895	*880	865	50	0.93426	407	387	368	348	328	309	289	270	250
1	0.99850	835	820	806	791	776	761	747	732	717	51	230	210	190	171	151	131	111	091	071	051
2	703	688	674	659	645	630	616	602	587	573	52	031	011	*991	*971	*951	*931	*911	*890	*870	*850
3	559	545	531	516	502	488	474	460	446	432	53	0.92830	810	789	769	749	728	708	688	667	647
4	419	405	391	378	364	350	336	323	309	296	54	626	605	585	564	544	523	502	482	461	440
5	282	269	255	242	228	215	202	189	176	163	55	419	398	377	357	336	315	294	273	252	231
6	150	137	124	111	098	085	073	060	047	035	56	210	189	168	147	126	105	084	062	041	020
7	022	009	*997	*984	*972	*960	*947	*935	*923	*911	57	0.91999	978	956	935	914	892	871	849	827	806
8	0.98899	887	875	863	851	838	826	814	803	791	58	784	762	741	719	697	675	653	631	610	588
9	779	767	755	743	731	720	708	696	684	672	59	565	543	521	499	477	455	433	410	388	366
10	661	649	637	625	614	602	590	579	567	556	60	344	322	299	277	255	232	210	188	165	143
11	544	532	521	509	498	487	475	464	452	441	61	120	097	075	052	030	007	*984	*962	*939	*916
12	430	419	408	396	385	374	363	352	341	330	62	0.90893	870	847	825	802	779	756	733	710	687
13	319	308	297	286	275	264	254	243	232	221	63	664	641	618	595	572	549	526	503	480	457
14	210	200	190	179	168	157	147	136	125	115	64	434	411	388	365	341	318	295	272	249	225
15	104	093	083	072	062	051	040	030	019	009	65	202	179	155	132	108	085	061	038	014	*991
16	0.97998	988	977	967	956	946	936	925	915	905	66	0.89967	943	920	896	872	848	825	801	777	753
17	895	885	875	864	854	844	834	824	814	804	67	729	705	681	657	633	609	585	561	537	513
18	794	784	774	764	754	744	734	724	714	704	68	489	465	441	416	392	368	343	319	295	270
19	694	684	674	664	654	645	635	625	615	605	69	245	220	196	171	147	122	098	073	048	024
20	596	586	576	566	556	546	536	526	516	506	70	0.88999	974	950	925	900	875	850	825	801	776
21	496	486	476	466	456	446	436	425	415	405	71	751	725	700	675	650	625	600	574	549	524
22	395	385	375	365	354	344	334	324	313	303	72	499	474	448	423	397	372	346	321	296	270
23	293	283	272	262	252	241	231	221	210	200	73	244	218	193	167	141	116	090	064	039	013
24	189	179	168	158	147	137	126	116	105	095	74	0.87987	961	935	910	884	858	832	806	780	754
25	084	073	063	052	042	031	020	010	*999	*988	75	728	702	676	650	623	597	571	545	518	492
26	0.96978	967	957	946	935	924	914	903	892	881	76	465	439	412	386	359	332	306	279	252	226
27	870	859	848	837	826	815	804	793	782	771	77	199	172	145	118	092	065	038	011	*984	*957
28	760	749	738	727	715	704	693	682	671	659	78	0.86929	902	875	847	820	793	766	738	711	684
29	648	637	625	614	603	591	580	568	557	546	79	656	629	601	574	546	518	491	463	435	408
30	534	522	511	499	488	476	464	453	441	429	80	380	352	324	296	269	241	213	185	157	129
31	418	406	394	382	370	358	346	334	321	309	81	100	072	044	015	*987	*959	*931	*902	*874	*846
32	296	284	271	259	246	234	221	209	196	183	82	0.85817	789	760	732	703	674	646	617	588	560
33	170	157	144	132	119	106	093	080	067	054	83	531	502	473	444	415	386	357	328	299	270
34	041	028	015	002	*988	*975	*962	*948	*935	*921	84	240	211	181	152	122	093	063	033	004	*974
35	0.95908	894	881	867	854	840	826	812	798	784	85	0.84944	914	884	854	824	794	764	734	703	673
36	770	756	742	728	714	700	685	671	657	643	86	642	612	581	551	520	490	459	428	398	367
37	628	614	599	585	570	556	541	526	512	497	87	336	305	274	243	212	181	150	119	088	056
38	482	467	452	437	423	408	393	378	362	347	88	025	*994	*962	*930	*899	*867	*835	*803	*771	*739
39	332	317	302	286	271	256	240	225	209	194	89	0.83707	675	643	610	578	545	513	480	447	415
40	178	162	147	131	115	100	084	068	052	036	90	382	349	315	282	249	216	183	150	116	083
41	020	004	*988	*972	*956	*940	*923	*907	*891	*875	91	049	015	*981	*947	*913	*879	*845	*810	*776	*741
42	0.94858	842	825	809	792	776	759	743	726	710	92	0.82705	670	635	600	565	529	494	458	423	387
43	693	676	660	643	626	609	592	575	558	541	93	351	315	279	243	206	170	133	096	059	022
44	524	507	490	473	455	438	421	403	386	369	94	0.81984	947	909	871	834	796	757	719	681	642
45	351	334	316	298	281	263	245	228	210	192	95	603	564	525	486	446	407	367	327	287	247
46	174	156	138	120	102	084	066	048	030	011	96	206	165	125	084	042	001	*960	*918	*876	*834
47	0.93993	975	956	938	920	901	883	864	845	827	97	0.80792	750	707	664	620	577	533	489	445	401
48	808	789	771	752	733	714	695	676	657	638	98	356	311	265	219	173	127	080	033	*985	*937
49	619	600	581	562	543	523	504	485	465	446	99	0.79889	841	792	743	693	643	593	543	492	441
											100	389									

*Indicates change in first two decimal places.

2-120 PHYSICAL AND CHEMICAL DATA

TABLE 2-115 *n*-Propyl Alcohol (C₃H₇OH)

%	0°C	15°C	30°C	%	0°C	15°C	30°C	%	0°C	15°C	30°C	%	0°C	15°C	30°C	%	0°C	15°C	30°C
0	0.9999	0.9991	0.9957	20	0.9789	0.9723	0.9643	40	0.9430	0.9331	0.9226	60	0.9033	0.8922	0.8807	80	0.8634	0.8516	0.8394
1	0.9982	0.9974	0.9940	21	0.9776	0.9705	0.9622	41	0.9411	0.9310	0.9205	61	0.9013	0.8902	0.8786	81	0.8614	0.8496	0.8373
2	0.9967	0.9960	0.9924	22	0.9763	0.9688	0.9602	42	0.9391	0.9290	0.9184	62	0.8994	0.8882	0.8766	82	0.8594	0.8475	0.8352
3	0.9952	0.9944	0.9908	23	0.9748	0.9670	0.9583	43	0.9371	0.9269	0.9164	63	0.8974	0.8861	0.8745	83	0.8574	0.8454	0.8332
4	0.9939	0.9929	0.9893	24	0.9733	0.9651	0.9563	44	0.9352	0.9248	0.9143	64	0.8954	0.8841	0.8724	84	0.8554	0.8434	0.8311
5	0.9926	0.9915	0.9877	25	0.9717	0.9633	0.9543	45	0.9332	0.9228	0.9122	65	0.8934	0.8820	0.8703	85	0.8534	0.8413	0.8290
6	0.9914	0.9902	0.9862	26	0.9700	0.9614	0.9522	46	0.9311	0.9207	0.9100	66	0.8913	0.8800	0.8682	86	0.8513	0.8393	0.8269
7	0.9904	0.9890	0.9848	27	0.9682	0.9594	0.9501	47	0.9291	0.9186	0.9079	67	0.8894	0.8779	0.8662	87	0.8492	0.8372	0.8248
8	0.9894	0.9877	0.9834	28	0.9664	0.9576	0.9481	48	0.9272	0.9165	0.9057	68	0.8874	0.8759	0.8641	88	0.8471	0.8351	0.8227
9	0.9883	0.9864	0.9819	29	0.9646	0.9556	0.9460	49	0.9252	0.9145	0.9036	69	0.8854	0.8739	0.8620	89	0.8450	0.8330	0.8206
10	0.9874	0.9852	0.9804	30	0.9627	0.9535	0.9439	50	0.9232	0.9124	0.9015	70	0.8835	0.8719	0.8600	90	0.8429	0.8308	0.8185
11	0.9865	0.9840	0.9790	31	0.9608	0.9516	0.9418	51	0.9213	0.9104	0.8994	71	0.8815	0.8700	0.8580	91	0.8408	0.8287	0.8164
12	0.9857	0.9828	0.9775	32	0.9589	0.9495	0.9396	52	0.9192	0.9084	0.8973	72	0.8795	0.8680	0.8559	92	0.8387	0.8266	0.8142
13	0.9849	0.9817	0.9760	33	0.9570	0.9474	0.9375	53	0.9173	0.9064	0.8952	73	0.8776	0.8659	0.8539	93	0.8364	0.8244	0.8120
14	0.9841	0.9806	0.9746	34	0.9550	0.9454	0.9354	54	0.9153	0.9044	0.8931	74	0.8756	0.8639	0.8518	94	0.8342	0.8221	0.8098
15	0.9833	0.9793	0.9730	35	0.9530	0.9434	0.9333	55	0.9132	0.9023	0.8911	75	0.8736	0.8618	0.8497	95	0.8320	0.8199	0.8077
16	0.9825	0.9780	0.9714	36	0.9511	0.9413	0.9312	56	0.9112	0.9003	0.8890	76	0.8716	0.8598	0.8477	96	0.8296	0.8176	0.8054
17	0.9817	0.9768	0.9698	37	0.9491	0.9392	0.9289	57	0.9093	0.8983	0.8869	77	0.8695	0.8577	0.8456	97	0.8272	0.8153	0.8031
18	0.9808	0.9752	0.9680	38	0.9471	0.9372	0.9269	58	0.9073	0.8963	0.8849	78	0.8675	0.8556	0.8435	98	0.8248	0.8128	0.8008
19	0.9800	0.9739	0.9661	39	0.9450	0.9351	0.9247	59	0.9053	0.8942	0.8828	79	0.8655	0.8536	0.8414	99	0.8222	0.8104	0.7984
																100	0.8194	0.8077	0.7958

TABLE 2-116 Isopropyl Alcohol (C₃H₇OH)

	0°C	15°C°	15°C°	20°C	30°C	%	0°C	15°C°	15°C°	20°C	30°C	%	0°C	15°C°	15°C°	20°C	30°C	
0	0.9999	0.9991	0.99913	0.9982	0.9957	35	0.9557		0.9446	0.9419	0.9338	70	0.8761	0.8639	0.86346	0.8584	0.8511	
1	0.9980	0.9973	0.9972	0.9962	0.9939	36	0.9536		0.9424	0.9399	0.9315	71	0.8738	0.8615	0.8611	0.8560	0.8487	
2	0.9962	0.9956	0.9954	0.9944	0.9921	37	0.9514		0.9401	0.9377	0.9292	72	0.8714	0.8592	0.8588	0.8537	0.8464	
3	0.9946	0.9938	0.9936	0.9926	0.9904	38	0.9493		0.9379	0.9355	0.9269	73	0.8691	0.8568	0.8564	0.8513	0.8440	
4	0.9930	0.9922	0.9920	0.9909	0.9887	39	0.9472		0.9356	0.9333	0.9246	74	0.8668	0.8545	0.8541	0.8489	0.8416	
5	0.9916	0.9906	0.9904	0.9893	0.9871	40	0.9450			0.93333	0.9310	0.9224	75	0.8644	0.8521	0.8517	0.8464	0.8392
6	0.9902	0.9892	0.9890	0.9877	0.9855	41	0.9428			0.9311	0.9287	0.9201	76	0.8621	0.8497	0.8493	0.8439	0.8368
7	0.9890	0.9878	0.9875	0.9862	0.9839	42	0.9406			0.9288	0.9264	0.9177	77	0.8598	0.8474	0.8470	0.8415	0.8344
8	0.9878	0.9864	0.9862	0.9847	0.9824	43	0.9384			0.9266	0.9239	0.9154	78	0.8575	0.8450	0.8446	0.8391	0.8321
9	0.9866	0.9851	0.9849	0.9833	0.9809	44	0.9361			0.9243	0.9215	0.9130	79	0.8551	0.8426	0.8422	0.8366	0.8297
10	0.9856	0.9838	0.98362	0.9820	0.9794	45	0.9338			0.9220	0.9191	0.9106	80	0.8528	0.8403	0.83979	0.8342	0.8273
11	0.9846	0.9826	0.9824	0.9808	0.9778	46	0.9315			0.9197	0.9165	0.9082	81	0.8503	0.8379	0.8374	0.8317	0.8248
12	0.9838	0.9813	0.9812	0.9797	0.9764	47	0.9292			0.9174	0.9141	0.9059	82	0.8479	0.8355	0.8350	0.8292	0.8224
13	0.9829	0.9802	0.9800	0.9786	0.9750	48	0.9270			0.9150	0.9117	0.9036	83	0.8456	0.8331	0.8326	0.8268	0.8200
14	0.9821	0.9790	0.9788	0.9776	0.9735	49	0.9247			0.9127	0.9093	0.9013	84	0.8432	0.8307	0.8302	0.8243	0.8175
15	0.9814	0.9779	0.9777	0.9765	0.9720	50	0.9224			0.91043	0.9069	0.8990	85	0.8408	0.8282	0.8278	0.8219	0.8151
16	0.9806	0.9768	0.9765	0.9754	0.9705	51	0.9201			0.9081	0.9044	0.8966	86	0.8384	0.8259	0.8254	0.8194	0.8127
17	0.9799	0.9756	0.9753	0.9743	0.9690	52	0.9178			0.9058	0.9020	0.8943	87	0.8360	0.8234	0.8229	0.8169	0.8201
18	0.9792	0.9745	0.9741	0.9731	0.9675	53	0.9155			0.9035	0.8996	0.8919	88	0.8336	0.8209	0.8205	0.8145	0.8078
19	0.9784	0.9730	0.9728	0.9717	0.9658	54	0.9132			0.9011	0.8971	0.8895	89	0.8311	0.8184	0.8180	0.8120	0.8053
20	0.9777	0.9719	0.97158	0.9703	0.9642	55	0.9109			0.8988	0.8946	0.8871	90	0.8287	0.8161	0.81553	0.8096	0.8029
21	0.9768	0.9704	0.9703	0.9688	0.9624	56	0.9086			0.8964	0.8921	0.8847	91	0.8262	0.8136	0.8130	0.8072	0.8004
22	0.9759	0.9690	0.9689	0.9669	0.9606	57	0.9063			0.8940	0.8896	0.8823	92	0.8237	0.8110	0.8104	0.8047	0.7979
23	0.9749	0.9675	0.9674	0.9651	0.9587	58	0.9040			0.8917	0.8874	0.8800	93	0.8212	0.8085	0.8079	0.8023	0.7954
24	0.9739	0.9660	0.9659	0.9634	0.9569	59	0.9017			0.8893	0.8850	0.8777	94	0.8186	0.8060	0.8052	0.7998	0.7929
25	0.9727	0.9643	0.9642	0.9615	0.9549	60	0.8994			0.88690	0.8825	0.8752	95	0.8160	0.8034	0.8026	0.7973	0.7904
26	0.9714	0.9626	0.9624	0.9597	0.9529	61	0.8970			0.8845	0.8800	0.8728	96	0.8133	0.8008	0.7999	0.7949	0.7878
27	0.9699	0.9608	0.9605	0.9577	0.9509	62	0.8947	0.8829		0.8821	0.8776	0.8704	97	0.8106	0.7981	0.7972	0.7925	0.7852
28	0.9684	0.9590	0.9586	0.9558	0.9488	63	0.8924	0.8805		0.8798	0.8751	0.8680	98	0.8078	0.7954	0.7945	0.7901	0.7826
29	0.9669	0.9570	0.9568	0.9540	0.9467	64	0.8901	0.8781		0.8775	0.8727	0.8656	99	0.8048	0.7926	0.7918	0.7877	0.7799
30	0.9652	0.9551	0.95493	0.9520	0.9446	65	0.8878	0.8757		0.8752	0.8702	0.8631	100	0.8016	0.7896	0.78913	0.7854	0.7770
31	0.9634		0.9530	0.9500	0.9426	66	0.8854	0.8733		0.8728	0.8679	0.8607						
32	0.9615		0.9510	0.9481	0.9405	67	0.8831	0.8710		0.8705	0.8656	0.8583						
33	0.9596		0.9489	0.9460	0.9383	68	0.8807	0.8686		0.8682	0.8632	0.8559						
34	0.9577		0.9468	0.9440	0.9361	69	0.8784	0.8662		0.8658	0.8609	0.8535						

*Two different observers; see *International Critical Tables*, vol. 3, p. 120.

TABLE 2-117 Glycerol*

Glycerol, %	Density					Glycerol, %	Density					Glycerol, %	Density				
	15 °C	15.5 °C	20 °C	25 °C	30 °C		15 °C	15.5 °C	20 °C	25 °C	30 °C		15 °C	15.5 °C	20 °C	25 °C	30 °C
100	1.26415	1.26381	1.26108	1.15802	1.25495	65	1.17030	1.17000	1.16750	1.16475	1.16195	30	1.07455	1.07435	1.07270	1.07070	1.06855
99	1.26160	1.26125	1.25850	1.25545	1.25235	64	1.16755	1.16725	1.16475	1.16200	1.15925	29	1.07195	1.07175	1.07010	1.06815	1.06605
98	1.25900	1.25865	1.25590	1.25290	1.24975	63	1.16480	1.16445	1.16205	1.15925	1.15650	28	1.06935	1.06915	1.06755	1.06560	1.06355
97	1.25645	1.25610	1.25335	1.25030	1.24710	62	1.16200	1.16170	1.15930	1.15655	1.15375	27	1.06670	1.06655	1.06495	1.06305	1.06105
96	1.25385	1.25350	1.25080	1.24770	1.24450	61	1.15925	1.15895	1.15655	1.15380	1.15100	26	1.06410	1.06390	1.06240	1.06055	1.05855
95	1.25130	1.25095	1.24825	1.24515	1.24190	60	1.15650	1.15615	1.15380	1.15105	1.14830	25	1.06150	1.06130	1.05980	1.05800	1.05605
94	1.24865	1.24830	1.24560	1.24250	1.23930	59	1.15370	1.15340	1.15105	1.14835	1.14555	24	1.05885	1.05870	1.05720	1.05545	1.05350
93	1.24600	1.24565	1.24300	1.23985	1.23670	58	1.15095	1.15065	1.14830	1.14560	1.14285	23	1.05625	1.05610	1.05465	1.05290	1.05100
92	1.24340	1.24305	1.24035	1.23725	1.23410	57	1.14815	1.14785	1.14555	1.14285	1.14010	22	1.05365	1.05350	1.05205	1.05035	1.04850
91	1.24075	1.24040	1.23770	1.23460	1.23150	56	1.14535	1.14510	1.14280	1.14015	1.13740	21	1.05100	1.05090	1.04950	1.04780	1.04600
90	1.23810	1.23775	1.23510	1.23200	1.22890	55	1.14260	1.14230	1.14005	1.13740	1.13470	20	1.04840	1.04825	1.04690	1.04525	1.04350
89	1.23545	1.23510	1.23245	1.22935	1.22625	54	1.13980	1.13955	1.13730	1.13465	1.13195	19	1.04590	1.04575	1.04440	1.04280	1.04105
88	1.23280	1.23245	1.22975	1.22665	1.22360	53	1.13705	1.13680	1.13455	1.13195	1.12925	18	1.04335	1.04325	1.04195	1.04035	1.03860
87	1.23015	1.22980	1.22710	1.22400	1.22095	52	1.13425	1.13400	1.13180	1.12920	1.12650	17	1.04085	1.04075	1.03945	1.03790	1.03615
86	1.22750	1.22710	1.22445	1.22135	1.21830	51	1.13150	1.13125	1.12905	1.12650	1.12380	16	1.03835	1.03825	1.03695	1.03545	1.03370
85	1.22485	1.22445	1.22180	1.21870	1.21565	50	1.12870	1.12845	1.12630	1.12375	1.12110	15	1.03580	1.03570	1.03450	1.03300	1.03130
84	1.22220	1.22180	1.21915	1.21605	1.21300	49	1.12600	1.12575	1.12360	1.12110	1.11845	14	1.03330	1.03320	1.03200	1.03055	1.02885
83	1.21955	1.21915	1.21650	1.21340	1.21035	48	1.12325	1.12305	1.12090	1.11840	1.11580	13	1.03080	1.03070	1.02955	1.02805	1.02640
82	1.21690	1.21650	1.21380	1.21075	1.20770	47	1.12055	1.12030	1.11820	1.11575	1.11320	12	1.02830	1.02820	1.02705	1.02560	1.02395
81	1.21425	1.21385	1.21115	1.20810	1.20505	46	1.11780	1.11760	1.11550	1.11310	1.11055	11	1.02575	1.02565	1.02455	1.02315	1.02150
80	1.21160	1.21120	1.20850	1.20545	1.20240	45	1.11510	1.11490	1.11280	1.11040	1.10795	10	1.02325	1.02315	1.02210	1.02070	1.01905
79	1.20885	1.20845	1.20575	1.20275	1.19970	44	1.11235	1.11215	1.11010	1.10775	1.10530	9	1.02085	1.02075	1.01970	1.01835	1.01670
78	1.20610	1.20570	1.20305	1.20005	1.19705	43	1.10960	1.10945	1.10740	1.10510	1.10265	8	1.01840	1.01835	1.01730	1.01600	1.01440
77	1.20335	1.20300	1.20030	1.19735	1.19435	42	1.10690	1.10670	1.10470	1.10240	1.10005	7	1.01600	1.01590	1.01495	1.01360	1.01205
76	1.20060	1.20025	1.19760	1.19465	1.19170	41	1.10415	1.10400	1.10200	1.09975	1.09740	6	1.01360	1.01350	1.01255	1.01125	1.00970
75	1.19785	1.19750	1.19485	1.19195	1.18900	40	1.10145	1.10130	1.09930	1.09710	1.09475	5	1.01120	1.01110	1.01015	1.00890	1.00735
74	1.19510	1.19480	1.19215	1.18925	1.18635	39	1.09875	1.09860	1.09665	1.09445	1.09215	4	1.00875	1.00870	1.00780	1.00655	1.00505
73	1.19235	1.19205	1.18940	1.18650	1.18365	38	1.09605	1.09590	1.09400	1.09180	1.08955	3	1.00635	1.00630	1.00540	1.00415	1.00270
72	1.18965	1.18930	1.18670	1.18380	1.18100	37	1.09340	1.09320	1.09135	1.08915	1.08690	2	1.00395	1.00385	1.00300	1.00180	1.00035
71	1.18690	1.18655	1.18395	1.18110	1.17830	36	1.09070	1.09050	1.08865	1.08655	1.08430	1	1.00155	1.00145	1.00060	0.99945	0.99800
70	1.18415	1.18385	1.18125	1.17840	1.17565	35	1.08800	1.08780	1.08600	1.08390	1.08165	0	0.99913	0.99905	0.99823	0.99708	0.99568
69	1.18135	1.18105	1.17850	1.17565	1.17290	34	1.08530	1.08515	1.08335	1.08125	1.07905						
68	1.17860	1.17830	1.17575	1.17295	1.17020	33	1.08265	1.08245	1.08070	1.07860	1.07645						
67	1.17585	1.17555	1.17300	1.17020	1.16745	32	1.07995	1.07975	1.07800	1.07600	1.07380						
66	1.17305	1.17275	1.17025	1.16745	1.16470	31	1.07725	1.07705	1.07535	1.07335	1.07120						

*Bosart and Snoddy, *Ind. Eng. Chem.*, **20**, (1928): 1378.TABLE 2-118 Hydrazine (N₂H₄)*

%	d_4^{15}	%	d_4^{15}
1	1.0002	30	1.0305
2	1.0013	40	1.038
4	1.0034	50	1.044
8	1.0077	60	1.047
12	1.0121	70	1.046
16	1.0164	80	1.040
20	1.0207	90	1.030
24	1.0248	100	1.011
28	1.0286		

*International Critical Tables, vol. 3, p. 55.

2-122 PHYSICAL AND CHEMICAL DATA

TABLE 2-119 Densities of Aqueous Solutions of Miscellaneous Organic Compounds*

 d , d_w , and d_s are the density of the solution, pure water, and pure liquid solute, respectively, all in g/mL. p_s is the wt % solute. 0.0255 means 2.55×10^{-4} .

Section A $d = d_w + Ap_s + Bp_s^2 + Cp_s^3$						
Name	Formula	t , °C	Range, p_s	A	B	C
Acetaldehyde	C ₂ H ₄ O	18	0–30	+0.0255	–0.0516	
Acetamide	C ₂ H ₅ NO	15	0–6	+0.02639	+0.01171	
Acetone	C ₃ H ₆ O	0	0–100	–0.02856	–0.05449	–0.0588
		4	0–100	–0.027648	–0.041193	+0.0272
		15	0–100	–0.021009	–0.039682	–0.0624
		20	0–100	–0.021233	–0.033529	–0.05327
Acetonitrile	C ₂ H ₃ N	25	0–100	–0.021171	–0.05904	–0.056
		15	0–16	–0.021175	–0.02024	
		0	0–89	–0.03729	–0.01232	+0.02984
		25	0–0.6	+0.025615	–0.02117	
Allyl alcohol	C ₃ H ₆ O	0				
Benzenepentacarboxylic acid	C ₁₁ H ₆ O ₁₀	25	0–7.9	–0.021651	+0.02285	
Butyl alcohol (<i>n</i> -)	C ₄ H ₁₀ O	20				
Butyric acid (<i>n</i> -)	C ₄ H ₈ O ₂	18	0–10	+0.03414	+0.0131	
		25	0–62	+0.035135	–0.0166	+0.0611
		0	0–70	+0.024489	+0.02802	–0.01291
Chloral hydrate	C ₂ H ₃ Cl ₃ O ₂	15	0–78	+0.024455	+0.02198	+0.04366
		30	0–90	+0.024401	+0.01887	+0.06549
		20	0–32	+0.023648	+0.03302	
Chloroacetic acid	C ₂ H ₃ ClO ₂	25	0–86	+0.023602	+0.0552	+0.022
Citric acid (hydrate)	C ₆ H ₈ O ₇ + H ₂ O	18	0–50	+0.023824	+0.01141	+0.017
Dichloroacetic acid	C ₂ H ₂ Cl ₂ O ₂	20	0–30	+0.024427	+0.05537	+0.07534
		25	0–97	+0.024427	+0.05537	+0.07534
Diethylamine hydrochloride	C ₄ H ₁₂ ClN	21	0–36	+0.0334	+0.0676	
Ethylamine hydrochloride	C ₂ H ₅ ClN	21	0–65	+0.021193	–0.03307	–0.047
Ethylene glycol	C ₂ H ₆ O ₂	0	0–100	+0.021483	+0.02992	–0.05248
		15	0–6	+0.02133	–0.03108	
		20	0–5	–0.0221	+0.048	
Ethyl ether	C ₄ H ₁₀ O	25	0–4.5	–0.0221	+0.035	
		15	0–95	+0.02367	+0.0358	–0.06005
Formaldehyde	CH ₂ O	15	0–40	+0.02518	–0.0658	+0.0542
Formamide	CH ₃ NO	25	22–96	+0.021217	+0.03199	–0.02529
Furfural	C ₅ H ₄ O ₂	20	0–8	+0.021827	+0.0366	
		25	0–8	+0.021664	+0.021	
Isoamyl alcohol	C ₅ H ₁₂ O	20	0–2.5	+0.02155	+0.03	
Isobutyl alcohol	C ₄ H ₁₀ O	15	0–8	–0.02146	+0.06	
		20	0–8	–0.02169	+0.038	
		15	0–9	+0.0352		
Isobutyric acid	C ₄ H ₈ O ₂	18	0–9	+0.0345		
		25	0–12	+0.0337		
Isovaleric acid	C ₅ H ₁₀ O ₂	25	0–5	+0.0253	–0.0282	
Lactic acid	C ₃ H ₆ O	25	0–9	+0.0231	+0.03186	
Maleic acid	C ₄ H ₄ O ₄	25	0–40	+0.0234	+0.0575	
Malic acid	C ₄ H ₆ O ₅	20	0–40	+0.023933	+0.03957	
		25	0–40	+0.023736	+0.0175	
Malonic acid	C ₃ H ₄ O ₄	20	0–40	+0.02389	+0.01066	
Methyl acetate	C ₃ H ₆ O ₂	20	0–20	+0.0240	–0.0574	
		0	26–51	+0.023336	+0.03996	+0.01544
		30	26–51	+0.023151	+0.03975	+0.03978
Nicotine	C ₁₀ H ₁₄ N ₂	20	0–60	+0.03642	+0.03454	–0.0687
Nitrophenol (<i>p</i> -)	C ₆ H ₅ NO ₃	15	0–1.5	+0.023216	–0.055	
		0	0–4	+0.025898	–0.03185	+0.041
		15	0–4	+0.02494	–0.058	
Oxalic acid	C ₂ H ₂ O ₄	17.5	0–9	+0.02494	–0.058	
		20	0–4	+0.025264	–0.031996	+0.0254
		25	0–4	+0.025108	–0.031607	+0.0208
		15	0–5	+0.02111	–0.0283	
Phenol	C ₆ H ₆ O	80	0–65	+0.03462	–0.086	
Phenylglycolic acid	C ₈ H ₈ O ₃	25	0–11	+0.02207	+0.023	
Picoline (α -)	C ₆ H ₇ N	25	0–70	–0.0386	–0.031405	–0.04167
		25	0–60	–0.04683	–0.0313	
Propionic acid	C ₃ H ₆ O ₂	18	0–10	+0.0395	–0.0172	
		25	0–40	+0.029245	–0.0399	+0.0361
Pyridine	C ₅ H ₅ N	25	0–60	+0.0229	–0.0204	–0.028
Resorcinol	C ₆ H ₆ O ₂	18	0–52	+0.02201	+0.05519	–0.0819
Succinic acid	C ₄ H ₆ O ₄	25	0–5.5	+0.02304		
		15	0–15	+0.024482	+0.0185	
		17.5	0–50	+0.024455	+0.0185	
		20	0–50	+0.024432	+0.01837	
		30	0–50	+0.024335	+0.0185	
		40	0–50	+0.024265	+0.0185	
		50	0–50	+0.024205	+0.0185	
Tartaric acid (<i>d</i> , <i>l</i> , or <i>dl</i>)	C ₄ H ₆ O ₆	60	0–50	+0.024155	+0.0185	

*From *International Critical Tables*, vol. 3, pp. 111–114.

TABLE 2-119 Densities of Aqueous Solutions of Miscellaneous Organic Compounds (Concluded)

Section A $d = d_w + Ap_s + Bp_s^2 + Cp_s^3$ (Cont.)							
Name	Formula	$t, ^\circ\text{C}$	Range, p_s	A	B	C	
Tetraethyl ammonium chloride	$\text{C}_8\text{H}_{20}\text{ClN}$	21	0–63	+0.0 ₃ 1884	+0.0 ₅ 6	+0.0 ₇ 122	
	$\text{CH}_4\text{N}_2\text{S}$	15	0–7	+0.0 ₂ 2995	+0.0 ₃ 374		
Trichloroacetic acid	$\text{C}_2\text{HCl}_3\text{O}_2$	12.5	0–61	+0.0 ₂ 499	+0.0 ₁ 153		
		20	10–30	+0.0 ₂ 5053	+0.0 ₁ 1387		
		25	0–94	+0.0 ₂ 5051	+0.0 ₃ 6119	+0.0 ₆ 1038	
Triethylamine hydrochloride	$\text{C}_6\text{H}_{16}\text{ClN}$	21	0–54	+0.0 ₂ 6	+0.0 ₅ 558	–0.0 ₆ 69	
Trimethyl carbinol	$\text{C}_4\text{H}_{10}\text{O}$	20	0–100	–0.0 ₂ 117	–0.0 ₁ 1908	+0.0 ₂ 957	
		25	0–100	–0.0 ₂ 1286	–0.0 ₁ 176	+0.0 ₂ 887	
		14.8	0–12	+0.0 ₂ 3213	–0.0 ₄ 4802	+0.0 ₅ 1216	
Urea	$\text{CH}_4\text{N}_2\text{O}$	18	0–51	+0.0 ₂ 2718	+0.0 ₃ 1552	+0.0 ₂ 2573	
		20	0–35	+0.0 ₂ 2702	+0.0 ₃ 3712	–0.0 ₂ 2285	
		25	0–10	+0.0 ₂ 2728	–0.0 ₁ 1817	+0.0 ₅ 1379	
Urethane	$\text{C}_3\text{H}_7\text{NO}_2$	20	0–56	+0.0 ₂ 1278	–0.0 ₂ 245	–0.0 ₇ 3437	
Valeric acid (<i>n</i> -)	$\text{C}_5\text{H}_{10}\text{O}_2$	25	0–3	+0.0 ₃ 34	–0.0 ₂ 27		
Section B $d = d_s + Ap_w + Bp_w^2 + Cp_w^3$							
Name	Formula	d_s	$t, ^\circ\text{C}$	Range, p_w	A	B	C
Butyl alcohol (<i>n</i> -)	$\text{C}_4\text{H}_{10}\text{O}$	0.8097	20	0–20	+0.0 ₂ 2103	–0.0 ₁ 113	
Butyric acid (<i>n</i> -)	$\text{C}_4\text{H}_8\text{O}_2$	0.9534	25	0–38	+0.0 ₂ 1854	–0.0 ₂ 2314	
Ethyl ether	$\text{C}_4\text{H}_{10}\text{O}$	0.7077	25	0– 1.1	+0.0 ₂ 34	+0.0 ₃ 36	
Isobutyl alcohol	$\text{C}_4\text{H}_{10}\text{O}$	0.8170	0	0–14	+0.0 ₂ 2437	–0.0 ₂ 285	
		0.8055	15	0–16	+0.0 ₂ 224	–0.0 ₁ 129	
Isobutyric acid	$\text{C}_4\text{H}_8\text{O}_2$	0.9425	26	0–80	+0.0 ₂ 1808	–0.0 ₂ 2358	+0.0 ₆ 1253
Nicotine	$\text{C}_{10}\text{H}_{14}\text{N}_2$	1.0093	20	0–40	+0.0 ₂ 199	–0.0 ₁ 331	+0.0 ₃ 315
Picoline (α -)	$\text{C}_6\text{H}_7\text{N}$	0.9404	25	0–30	+0.0 ₂ 2715	–0.0 ₁ 393	
		0.9515	25	0–40	+0.0 ₂ 1925	–0.0 ₁ 352	+0.0 ₆ 25
Pyridine	$\text{C}_5\text{H}_5\text{N}$	0.9776	25	0–40	+0.0 ₂ 1157	–0.0 ₂ 536	–0.0 ₆ 2
Trimethyl carbinol	$\text{C}_4\text{H}_{10}\text{O}$	0.7856	20	0–20	+0.0 ₂ 2287	+0.0 ₂ 275	
Section C $d_t = d_o + At + Bt^2$							
Name	Formula	p_s	d_o	Range, $^\circ\text{C}$	A	B	
Allyl alcohol	$\text{C}_3\text{H}_6\text{O}$	76.60	0.9122	0–45	–0.0 ₃ 8	–0.0 ₂ 27	
Butyl alcohol (<i>n</i> -)	$\text{C}_4\text{H}_{10}\text{O}$	80.95	0.8614	0–43	–0.0 ₂ 7292	–0.0 ₆ 75	
Chloral hydrate	$\text{C}_2\text{H}_3\text{Cl}_3\text{O}_2$	2.00	1.0094	7–80	–0.0 ₂ 2597	–0.0 ₂ 4313	
		10.00	1.0476	7–80	–0.0 ₂ 7955	–0.0 ₅ 4253	
		5.00	1.0150	15–80	–0.0 ₂ 2103	–0.0 ₂ 2544	
Ethyl tartrate	$\text{C}_7\text{H}_{14}\text{O}_6$	10.00	1.0270	15–80	–0.0 ₂ 2116	–0.0 ₂ 2929	
		25.00	1.0665	15–80	–0.0 ₂ 401	–0.0 ₂ 23	
		4.62	1.0125	22–74	–0.0 ₂ 232	–0.0 ₂ 254	
Furfural	$\text{C}_5\text{H}_4\text{O}_2$	5.69	1.0140	22–74	–0.0 ₂ 221	–0.0 ₂ 268	
		6.56	1.0155	22–74	–0.0 ₂ 211	–0.0 ₂ 290	
		9.34	1.0055	11–73	–0.0 ₂ 171	–0.0 ₂ 3615	
Pyridine	$\text{C}_5\text{H}_5\text{N}$	21.20	1.0115	14–73	–0.0 ₂ 378	–0.0 ₂ 248	
		29.50	1.0145	12–72	–0.0 ₂ 463	–0.0 ₂ 235	
		40.40	1.0182	9–74	–0.0 ₂ 605	–0.0 ₂ 167	

DENSITIES OF MISCELLANEOUS MATERIALS

TABLE 2-120 Approximate Specific Gravities and Densities of Miscellaneous Solids and Liquids*

Water at 4°C and normal atmospheric pressure taken as unity. For more detailed data on any material, see the section dealing with the properties of that material.

Substance	Sp. gr.	Aver. density lb/ft ³	Substance	Sp. gr.	Aver. density lb/ft ³	Substance	Sp. gr.	Aver. density lb/ft ³
Metals, Alloys, Ores			Timber, Air-dry			Dry Rubble Masonry		
Aluminum, cast-hammered	2.55–2.80	165	Apple	0.66–0.74	44	Granite, syenite, gneiss	1.9–2.3	130
bronze	7.7	481	Ash, black	0.55	34	Limestone, marble	1.9–2.1	125
Brass, cast-rolled	8.4–8.7	534	white	0.64–0.71	42	Sandstone, bluestone	1.8–1.9	110
Bronze, 7.9 to 14% Sn	7.4–8.9	509	Birch, sweet, yellow	0.71–0.72	44			
phosphor	8.88	554	Cedar, white, red	0.35	22	Brick Masonry		
Copper, cast-rolled	8.8–8.95	556	Cherry, wild red	0.43	27	Hard brick	1.8–2.3	128
ore, pyrites	4.1–4.3	262	Chestnut	0.48	30	Medium brick	1.6–2.0	112
German silver	8.58	536	Cypress	0.45–0.48	29	Soft brick	1.4–1.9	103
Gold, cast-hammered	19.25–19.35	1205	Elm, white	0.56	35	Sand-lime brick	1.4–2.2	112
coin (U.S.)	17.18–17.2	1073	Fir, Douglas	0.48–0.55	32	Concrete Masonry		
Iridium	21.78–22.42	1383	balsam	0.40	25	Cement, stone, sand	2.2–2.4	144
Iron, gray cast	7.03–7.13	442	Hemlock	0.45–0.50	29	slag, etc.	1.9–2.3	130
cast, pig	7.2	450	Hickory	0.74–0.80	48	cinder, etc.	1.5–1.7	100
wrought	7.6–7.9	485	Locust	0.67–0.77	45	Various Building Materials		
spiegeleisen	7.5	468	Mahogany	0.56–0.85	44	Ashes, cinders	0.64–0.72	40–45
ferro-silicon	6.7–7.3	437	Maple, sugar	0.68	43	Cement, Portland, loose	1.5	94
ore, hematite	5.2	325	white	0.53	33	Lime, gypsum, loose	0.85–1.00	53–64
ore, limonite	3.6–4.0	237	Oak, chestnut	0.74	46	Mortar, lime, set	1.4–1.9	103
ore, magnetite	4.9–5.2	315	live	0.87	54	Portland cement	2.08–2.25	94–135
slag	2.5–3.0	172	red, black	0.64–0.71	42	Portland cement	3.1–3.2	196
Lead	11.34	710	white	0.77	48	Slags, bank slag	1.1–1.2	67–72
ore, galena	7.3–7.6	465	Pine, Norway	0.55	34	bank screenings	1.5–1.9	98–117
Manganese	7.42	475	white	0.51	32	machine slag	1.5	96
ore, pyrolusite	3.7–4.6	259	red	0.48	30	slag sand	0.8–0.9	49–55
Mercury	13.6	849	Southern	0.61–0.67	38–42	Earth, etc., Excavated		
Monel metal, rolled	8.97	555	white	0.43	27	Clay, dry	1.0	63
Nickel	8.9	537	Poplar	0.43	27	damp plastic	1.76	110
Platinum, cast-hammered	21.5	1330	Redwood, California	0.42	26	and gravel, dry	1.6	100
Silver, cast-hammered	10.4–10.6	656	Spruce, white, red	0.45	28	Earth, dry, loose	1.2	76
Steel, cold-drawn	7.83	489	Teak, African	0.99	62	dry, packed	1.5	95
machine	7.80	487	Indian	0.66–0.88	48	moist, loose	1.3	78
tool	7.70–7.73	481	Walnut, black	0.59	37	moist, packed	1.6	96
Tin, cast-hammered	7.2–7.5	459	Willow	0.42–0.50	28	mud, flowing	1.7	108
cassiterite	6.4–7.0	418	Various Liquids			mud, packed	1.8	115
Tungsten	19.22	1200	Alcohol, ethyl (100%)	0.789	49	Riprap, limestone	1.3–1.4	80–85
Zinc, cast-rolled	6.9–7.2	440	methyl (100%)	0.796	50	Riprap, sandstone	1.4	90
blende	3.9–4.2	253	Acid, muriatic, 40%	1.20	75	Riprap, shale	1.7	105
Various Solids			nitric, 91%	1.50	94	Sand, gravel, dry, loose	1.4–1.7	90–105
Cereals, oats, bulk	0.51	26	sulfuric, 87%	1.80	112	gravel, dry, packed	1.6–1.9	100–120
barley, bulk	0.62	39	Chloroform	1.500	95	gravel, wet	1.89–2.16	126
corn, rye, bulk	0.73	45	Ether	0.736	46	Excavations in Water		
wheat, bulk	0.77	48	Lye, soda, 66%	1.70	106	Clay	1.28	80
Cork	0.22–0.26	15	Oils, vegetable	0.91–0.94	58	River mud	1.44	90
Cotton, flax, hemp	1.47–1.50	93	mineral, lubricants	0.88–0.94	57	Sand or gravel	0.96	60
Fats	0.90–0.97	58	Turpentine	0.861–0.867	54	and clay	1.00	65
Flour, loose	0.40–0.50	28	Water, 4°C max. density	1.0	62.428	Soil	1.12	70
pressed	0.70–0.80	47	100°C	0.9584	59.830	Stone riprap	1.00	65
Glass, common	2.40–2.80	162	ice	0.88–0.92	56	Minerals		
plate or crown	2.45–2.72	161	snow, fresh fallen	0.125	8	Asbestos	2.1–2.8	153
crystal	2.90–3.00	184	sea water	1.02–1.03	64	Barytes	4.50	281
dint	3.2–4.7	247	Ashlar Masonry			Basalt	2.7–3.2	184
Hay and straw, bales	0.32	20	Bluestone	2.3–2.6	153	Bauxite	2.55	159
Leather	0.86–1.02	59	Granite, syenite, gneiss	2.4–2.7	159	Bluestone	2.5–2.6	159
Paper	0.70–1.15	58	Limestone	2.1–2.8	153	Borax	1.7–1.8	109
Potatoes, piled	0.67	44	Marble	2.4–2.8	162	Chalk	1.8–2.8	143
Rubber, caoutchouc	0.92–0.96	59	Sandstone	2.0–2.6	143	Clay, marl	1.8–2.6	137
goods	1.0–2.0	94	Rubble Masonry			Dolomite	2.9	181
Salt, granulated, piled	0.77	48	Bluestone	2.2–2.5	147	Feldspar, orthoclase	2.5–2.7	162
Saltpeter	1.07	67	Granite, syenite, gneiss	2.3–2.6	153	Gneiss	2.7–2.9	175
Starch	1.53	96	Limestone	2.0–2.7	147	Granite	2.6–2.7	165
Sulfur	1.93–2.07	125	Marble	2.3–2.7	156	Greenstone, trap	2.8–3.2	187
Wool	1.32	82	Sandstone	1.9–2.5	137	Gypsum, alabaster	2.3–2.8	159
						Hornblende	3.0	187
						Limestone	2.1–2.86	155
						Marble	2.6–2.86	170
						Magnesite	3.0	187
						Phosphate rock, apatite	3.2	200
						Porphyry	2.6–2.9	172

*From Marks' Standard Handbook for Mechanical Engineers, 10th ed., McGraw-Hill, 1996.

TABLE 2-120 Approximate Specific Gravities and Densities of Miscellaneous Solids and Liquids (Concluded)

Water at 4°C and normal atmospheric pressure taken as unity. For more detailed data on any material, see the section dealing with the properties of that material.

Substance	Sp. gr.	Aver. density lb/ft ³	Substance	Sp. gr.	Aver. density lb/ft ³	Substance	Sp. gr.	Aver. density lb/ft ³
Minerals (<i>Cont.</i>)			Bituminous Substances			Bituminous Substances (<i>Cont.</i>)		
Pumice, natural	0.37–0.90	40	Asphaltum	1.1–1.5	81	Petroleum	0.87	54
Quartz, flint	2.5–2.8	165	Coal, anthracite	1.4–1.8	97	refined (kerosene)	0.78–0.82	50
Sandstone	2.0–2.6	143	bituminous	1.2–1.5	84	benzine	0.73–0.75	46
Serpentine	2.7–2.8	171	lignite	1.1–1.4	78	gasoline	0.70–0.75	45
Shale, slate	2.6–2.9	172	peat, turf, dry	0.65–0.85	47	Pitch	1.07–1.15	69
						Tar, bituminous	1.20	75
Soapstone, talc	2.6–2.8	169	charcoal, pine	0.28–0.44	23			
Syenite	2.6–2.7	165	charcoal, oak	0.47–0.57	33	Coal and Coke, Piled		
			coke	1.0–1.4	75	Coal, anthracite	0.75–0.93	47–58
Stone, Quarried, Piled			Graphite	1.64–2.7	135	bituminous, lignite	0.64–0.87	40–54
Basalt, granite, gneiss	1.5	96	Paraffin	0.87–0.91	56	peat, turf	0.32–0.42	20–26
Greenstone, hornblende	1.7	107				charcoal	0.16–0.23	10–14
Limestone, marble, quartz	1.5	95				coke	0.37–0.51	23–32
Sandstone	1.3	82						
Shale	1.5	92						

NOTE: To convert pounds per cubic foot to kilograms per cubic meter, multiply by 16.02. °F = % °C + 32.

TABLE 2-121 Density (kg/m³) of Selected Elements as a Function of Temperature

Temperature, K°	Element symbol												
	Al	Be†	Cr	Cu	Au	Ir	Fe	Pb	Mo	Ni	Pt	Ag	Zn†
50	2736	3650	7160	9019	19,490	22,600	7910	11,570	10,260	8960	21,570	10,620	7280
100	2732	3640	7155	9009	19,460	22,580	7900	11,520	10,260	8950	21,550	10,600	7260
150	2726	3630	7150	8992	19,420	22,560	7890	11,470	10,250	8940	21,530	10,575	7230
200	2719	3620	7145	8973	19,380	22,540	7880	11,430	10,250	8930	21,500	10,550	7200
250	2710	3610	7140	8951	19,340	22,520	7870	11,380	10,250	8910	21,470	10,520	7170
300	2701	3600	7135	8930	19,300	22,500	7860	11,330	10,240	8900	21,450	10,490	7135
400	2681	3580	7120	8885	19,210	22,450	7830	11,230	10,220	8860	21,380	10,430	7070
500	2661	3555	7110	8837	19,130	22,410	7800	11,130	10,210	8820	21,330	10,360	7000
600	2639	3530	7080	8787	19,040	22,360	7760	11,010	10,190	8780	21,270	10,300	6935
800	2591	—	7040	8686	18,860	22,250	7690	10,430	10,160	8690	21,140	10,160	6430
1000	2365	—	7000	8568	18,660	22,140	7650	10,190	10,120	8610	21,010	10,010	6260
1200	2305	—	6945	8458	18,440	22,030	7620	9,940	10,080	8510	20,870	9,850	—
1400	2255	—	6890	7920	17,230	21,920	7520	—	10,040	8410	20,720	9,170	—
1600	—	—	6760	7750	16,950	21,790	7420	—	10,000	8320	20,570	8,980	—
1800	—	—	6700	7600	—	21,660	7320	—	9,950	7690	20,400	—	—
2000	—	—	7460	—	—	21,510	7030	—	9,900	7450	20,220	—	—

NOTE: Above the horizontal line the condensed phase is solid; below the line, it is liquid.

°R = % K.

† Polycrystalline form tabulated. Similar tables for an additional 45 elements appear in the *Handbook of Heat Transfer*, 2d ed., McGraw-Hill, New York, 1984.

SOLUBILITIES

UNITS CONVERSIONS

For this subsection, the following units conversions are applicable:

$$^{\circ}\text{F} = \% ^{\circ}\text{C} + 32.$$

To convert cubic centimeters to cubic feet, multiply by 3.532×10^{-5} .

To convert millimeters of mercury to pounds-force per square inch, multiply by 0.01934.

To convert grams per liter to pounds per cubic foot, multiply by 6.243×10^{-2} .A database containing solubilities originally published in the International Union for Pure and Applied Chemistry (IUPAC)-National Institute of Standards and Technology (NIST) Solubility Data Series is now available at no cost online at <http://srdata.nist.gov/solubility>.

TABLE 2-122 Solubilities of Inorganic Compounds in Water at Various Temperatures*

This table shows the grams of anhydrous substance that are soluble in 100 g of water at the temperature in degrees Celsius as indicated; when the name is followed by †, the value is expressed in grams of substance in 100 cm³ of saturated solution. Solid phase gives the hydrated form in equilibrium with the saturated solution.

	Substance	Formula	Solid phase	0 °C	10 °C	20 °C	30 °C	40 °C	50 °C	60 °C	70 °C	80 °C	90 °C	100 °C	
1	Aluminum chloride	AlCl ₃	6H ₂ O			69.86 ¹⁵⁰									1
2	sulfate	Al ₂ (SO ₄) ₃	18H ₂ O	31.2	33.5	36.4	40.4	46.1	52.2	59.2	66.1	73.0	80.8	89.0	2
3	Ammonium aluminum sulfate	(NH ₄) ₂ Al ₂ (SO ₄) ₄	24H ₂ O	2.1	4.99	7.74	10.94	14.88	20.10	26.70				109.7 ⁸⁰	3
4	bicarbonate	NH ₄ HCO ₃		11.9	15.8	21	27								4
5	bromide	NH ₄ Br		60.6	68	75.5	83.2	91.1	99.2	107.8	116.8	126	135.6	145.6	5
6	chloride	NH ₄ Cl		29.4	33.3	37.2	41.4	45.8	50.4	55.2	60.2	65.6	71.3	77.3	6
7	chloroplatinate	(NH ₄) ₂ PtCl ₆			0.7									1.25	7
8	chromate	(NH ₄) ₂ CrO ₄					40.4								8
9	chromium sulfate	(NH ₄) ₂ Cr ₂ (SO ₄) ₄	24H ₂ O			10.78 ²⁵⁰									9
10	dichromate	(NH ₄) ₂ Cr ₂ O ₇					47.17								10
11	dihydrogen phosphite	NH ₄ H ₂ PO ₃		171		190 ¹⁴⁵⁰	260 ³¹⁰								11
12	hydrogen phosphate	(NH ₄) ₂ HPO ₄				131 ¹⁵									12
13	iodide	NH ₄ I		154.2	163.2	172.3	181.4	190.5	199.6	208.9	218.7	228.8		250.3	13
14	magnesium phosphate	NH ₄ MgPO ₄	6H ₂ O	0.023		0.052		0.036	0.030	0.040	0.016	0.019			14
15	manganese phosphate	NH ₄ MnPO ₄	7H ₂ O			0		0		0	0.005	0.007			15
16	nitrate	NH ₄ NO ₃		118.3		192	241.8	297.0	344.0	421.0	499.0	580.0	740.0	871.0	16
17	oxalate	(NH ₄) ₂ C ₂ O ₄	1H ₂ O	2.2	3.1	4.4	5.9	8.0	10.3						17
18	perchlorate†	NH ₄ ClO ₄ †		11.56		20.85		30.58		39.05		48.19		57.01	18
19	persulfate	(NH ₄) ₂ S ₂ O ₈		58.2											19
20	sulfate	(NH ₄) ₂ SO ₄		70.6	73.0	75.4	78.0	81.0		88.0		95.3		103.3	20
21	thiocyanate	NH ₄ CNS		119.8	144	170	207.7								21
22	vanadate (meta)	NH ₄ VO ₃				0.48	0.84	1.32	1.78		3.05				22
23	Antimonious fluoride	SbF ₃		384.7		444.7	563.6								23
24	sulfide	Sb ₂ S ₃				0.000175 ¹⁸⁰									24
25	Arsenic oxide	As ₂ O ₅		59.5	62.1	65.8	69.5	71.2		73.0		75.1		76.7	25
26	Arsenious sulfide	As ₂ S ₃		5.17 × 10 ⁻⁵ at 18°											26
27	Barium acetate	Ba(C ₂ H ₃ O ₂) ₂	3H ₂ O	59	63	71									27
28	acetate	Ba(C ₂ H ₃ O ₂) ₂	1H ₂ O				75	79	77	74	74			75	28
29	carbonate	BaCO ₃			0.0016 ⁸⁰	0.0022 ¹⁸⁰	0.0024 at 24.2°								29
30	chlorate	Ba(ClO ₃) ₂	1H ₂ O	20.34	26.95	33.80	41.70	49.61		66.81		84.84		104.9	30
31	chloride	BaCl ₂	2H ₂ O	31.6	33.3	35.7	38.2	40.7	43.6	46.4	49.4	52.4		58.8	31
32	chromate	BaCrO ₄		0.0002	0.00028	0.00037	0.00046								32
33	hydroxide	Ba(OH) ₂	8H ₂ O	1.67	2.48	3.89	5.59	8.22	13.12	20.94		101.4			33
34	iodide	BaI ₂	6H ₂ O	170.2	185.7	203.1	219.6								34
35	iodide	BaI ₂	2H ₂ O					231.9		247.3		261.0		271.7	35
36	nitrate	Ba(NO ₃) ₂		5.0	7.0	9.2	11.6	14.2	17.1	20.3		27.0		34.2	36
37	nitrite	Ba(NO ₂) ₂	1H ₂ O			67.5						205.8		300	37
38	oxalate	BaC ₂ O ₄			0.0016 ⁸⁰	0.0022 ¹⁸⁰	0.0024 at 24.2°								38
39	perchlorate	Ba(ClO ₄) ₂	3H ₂ O	205.8		289.1		358.7	426.3		495.2		562.3		39
40	sulfate	BaSO ₄		1.15 × 10 ⁻⁴	2.0 × 10 ⁻⁴	2.4 × 10 ⁻⁴	2.85 × 10 ⁻⁴								40
41	Beryllium sulfate	BeSO ₄	6H ₂ O				52		60.67						41
42	sulfate	BeSO ₄	4H ₂ O				43.78	46.74			62		83	100	42
43	sulfate	BeSO ₄	2H ₂ O									84.76	98	110	43
44	Boric acid	H ₃ BO ₃		2.66	3.57	5.04	6.60	8.72	11.54	14.81	16.73	23.75	30.38	40.25	44
45	Boron oxide	B ₂ O ₃		1.1	1.5	2.2		4.0		6.2		9.5		15.7	45
46	Bromine	Br ₂		4.22	3.4	3.20	3.13								46
47	Cadmium chloride	CdCl ₂	4H ₂ O	97.59	125.1										47
48	chloride	CdCl ₂	2½H ₂ O	90.01			132.1								48
49	chloride	CdCl ₂	1H ₂ O		135.1	134.5		135.3		136.5		140.4		147.0	49
50	cyanide	Cd(CN) ₂				1.7 ¹⁵⁰									50
51	hydroxide	Cd(OH) ₂					2.6 × 10 ⁻⁴ at 25°								51
52	sulfate	CdSO ₄		76.48	76.00	76.60		78.54		83.68			63.13	60.77	52
53	Calcium acetate	Ca(C ₂ H ₃ O ₂) ₂	2H ₂ O	37.4	36.0	34.7	33.8	33.2		32.7		33.5			53
54	acetate	Ca(C ₂ H ₃ O ₂) ₂	1H ₂ O										31.1	29.7	54

1	Calcium bicarbonate	Ca(HCO ₃) ₂		16.15		16.60		17.05		17.50		17.95		18.40	1
2	chloride	CaCl ₂	6H ₂ O	59.5	65.0	74.5	102								2
3	chloride	CaCl ₂	2H ₂ O							136.8	141.7	147.0	152.7	159	3
4	fluoride	CaF ₂				0.0016 ¹⁸⁰	0.0017 ²⁶⁰								4
5	hydroxide	Ca(OH) ₂		0.185	0.176	0.165	0.153	0.141	0.128	0.116	0.106	0.094	0.085	0.077	5
6	nitrate	Ca(NO ₃) ₂	4H ₂ O	102.0	115.3	129.3	152.6	195.9							6
7	nitrate	Ca(NO ₃) ₂	3H ₂ O					237.5	281.5						7
8	nitrate	Ca(NO ₃) ₂										358.7		363.6	8
9	nitrite	Ca(NO ₂) ₂	4H ₂ O	62.07		76.68									9
10	nitrite	Ca(NO ₂) ₂	2H ₂ O							132.6	151.9		244.8		10
11	oxalate	CaC ₂ O ₄			6.7 × 10 ⁻⁴	6.8 × 10 ⁻⁴	9.5 × 10 ⁻⁴	14 × 10 ⁻⁴							11
					at 13°	at 25°	at 50°	at 95°							
12	sulfate	CaSO ₄	2H ₂ O	0.1759	0.1928		0.2090	0.2097		0.2047	0.1966			0.1619	12
13	Carbon dioxide, 760 mm †	CO ₂		0.3346	0.2318	0.1688	0.1257	0.0973	0.0761	0.0576				0	13
14	monoxide, 760 mm †	CO		0.0044	0.0035	0.0028	0.0024	0.0021	0.0018	0.0015	0.0013	0.0010	0.0006	0	14
15	Cesium chloride	CsCl		161.4	174.7	186.5	197.3	208.0	218.5	229.7	239.5	250.0	260.1	270.5	15
16	nitrate	CsNO ₃		9.33	14.9	23.0	33.9	47.2	64.4	83.8	107.0	134.0	163.0	197.0	16
17	sulfate	Cs ₂ SO ₄		167.1	173.1	178.7	184.1	189.9	194.9	199.9	205.0	210.3	214.9	220.3	17
18	Chlorine, 760 mm †	Cl ₂		1.46	0.980	0.716	0.562	0.451	0.386	0.324	0.274	0.219	0.125	0	18
19	Chromic anhydride	CrO ₃		164.9				174.0	182.1				217.5	206.8	19
20	Cuprio chloride	CuCl ₂	2H ₂ O	70.7	73.76	77.0	80.34	83.8	87.44	91.2		99.2		107.9	20
21	nitrate	Cu(NO ₃) ₂	6H ₂ O	81.8	95.28	125.1									21
22	nitrate	Cu(NO ₃) ₂	3H ₂ O					159.8		178.8		207.8			22
23	sulfate	CuSO ₄	5H ₂ O	14.3	17.4	20.7	25	28.5	33.3	40		55		75.4	23
24	sulfide	CuS				3.3 × 10 ⁻⁵									24
						at 18°									
						1.52 ²⁵⁰									
25	Cuprous chloride	CuCl													25
26	Ferric chloride	FeCl ₃		74.4	81.9	91.8			315.1			525.8		535.7	26
27	Ferrous chloride	FeCl ₂	4H ₂ O		64.5		73.0	77.3	82.5	88.7		100			27
28	chloride	FeCl ₂											105.3	105.8	28
29	nitrate	Fe(NO ₃) ₂	6H ₂ O	71.02		83.8				165.6					29
30	sulfate	FeSO ₄	7H ₂ O	15.65	20.51	26.5	32.9	40.2	48.6						30
31	sulfate	FeSO ₄	1H ₂ O								50.9	43.6	37.3		31
32	Hydrobromic acid, 760 mm	HBr		221.2	210.3	198			171.5					130	32
33	Hydrochloric acid, 760 mm	HCl		82.3			67.3	63.3	59.6	56.1					33
34	Iodine	I ₂				0.029	0.04	0.056	0.078						34
35	Lead acetate	Pb(C ₂ H ₃ O ₂) ₂	3H ₂ O				55.04 ²⁵⁰								35
36	bromide	PbBr ₂		0.4554		0.85	1.15	1.53	1.94	2.36		3.34		4.75	36
37	carbonate	PbCO ₃				0.00011									37
38	chloride	PbCl ₂		0.6728		0.99	1.20	1.45	1.70	1.98		2.62		3.34	38
39	chromate	PbCrO ₄				7 × 10 ⁻⁶									39
40	fluoride	PbF ₂			0.060	0.064	0.068								40
41	nitrate	Pb(NO ₃) ₂		38.8	48.3	56.5	66	75	85	95		115		38.8	41
42	sulfate	PbSO ₄		0.0028	0.0035	0.0041	0.0049	0.0056							42
43	Magnesium bromide	MgBr ₂	6H ₂ O	91.0	94.5	96.5	99.2	101.6	104.1	107.5		113.7		120.2	43
44	chloride	MgCl ₂	6H ₂ O	52.8	53.5	54.5		57.5		61.0		66.0		73.0	44
45	hydroxide	Mg(OH) ₂				0.0009 ¹⁸⁰									45
46	nitrate	Mg(NO ₃) ₂	6H ₂ O	66.55				84.74					137.0		46
47	sulfate	MgSO ₄	7H ₂ O		30.9	35.5	40.8	45.6							47
48	sulfate	MgSO ₄	6H ₂ O	40.8	42.2	44.5	45.3		50.4	53.5	59.5	64.2	69.0	74.0	48
49	sulfate	MgSO ₄	1H ₂ O									62.9		68.3	49
50	Manganous sulfate	MnSO ₄	7H ₂ O	53.23	60.01										50
51	sulfate	MnSO ₄	5H ₂ O		59.5	62.9	67.76								51
52	sulfate	MnSO ₄	4H ₂ O			64.5	66.44	68.8	72.6						52
53	sulfate	MnSO ₄	1H ₂ O						58.17	55.0	52.0	48.0	42.5	34.0	53
54	Mercurous chloride	HgCl		0.00014		0.0002		0.0007							54
55	Molybdc oxide	MoO ₃	2H ₂ O			0.138	0.264	0.476	0.687	1.206	2.055	2.106			55
56	Nickel chloride	NiCl ₂	6H ₂ O	53.9	59.5	64.2	68.9	73.3	78.3	82.2	85.2			87.6	56
57	nitrate	Ni(NO ₃) ₂	6H ₂ O	79.58		96.31		122.2							57
58	nitrate	Ni(NO ₃) ₂	3H ₂ O							163.1	169.1		235.1		58
59	sulfate	NiSO ₄	7H ₂ O	27.22	32		42.46								59
60	sulfate	NiSO ₄	6H ₂ O						50.15	54.80	59.44	63.17		76.7	60
61	Nitric oxide, 760 mm	NO		0.00984	0.00757	0.00618	0.00517	0.00440	0.00376	0.00324	0.00267	0.00199	0.00114	0	61
62	Nitrous oxide	N ₂ O			0.1705	0.1211									62

*By N. A. Lange; abridged from "Table of Solubilities of Inorganic Compounds in Water at Various Temperatures" in *Lange's Handbook of Chemistry*, 10th ed., McGraw-Hill, New York, 1961 (except for NaCl, which is from *CRC Handbook of Chemistry and Physics*, 86th ed., CRC Press, 2005). For tables of the solubility of gases in water at various temperatures, Atack (*Handbook of Chemical Data*, Reinhold, New York, 1957) gives values at closer temperature intervals, usually 1 or 5 °C, than are tabulated here. For materials marked by †, additional data are given in tables subsequent to this one. For the solubility of various hydrocarbons in water at high pressures see *J. Chem. Eng. Data*, **4**, 212 (1959).

TABLE 2-122 Solubilities of Inorganic Compounds in Water at Various Temperatures (Continued)

	Substance	Formula	Solid phase	0 °C	10 °C	20 °C	30 °C	40 °C	50 °C	60 °C	70 °C	80 °C	90 °C	100 °C	
1	Potassium acetate	KC ₂ H ₃ O ₂	1½H ₂ O	216.7	233.9	255.6	283.8	323.3							1
2	acetate	KC ₂ H ₃ O ₂	½H ₂ O						337.3	350	364.8	380.1	396.3		2
3	alum	K ₂ SO ₄ ·Al ₂ (SO ₄) ₃	24H ₂ O	3.0	4.0	5.9	8.39	11.70	17.00	24.75	40.0	71.0	109.0		3
4	bicarbonate	KHCO ₃		22.4	27.7	33.2	39.1	45.4		60.0					4
5	bisulfate	KHSO ₄		36.3		51.4		67.3						121.6	5
6	bitartrate	KHC ₄ H ₄ O ₆		0.32	0.40	0.53	0.90	1.32	1.83	2.46		4.6		6.95	6
7	carbonate	K ₂ CO ₃	2H ₂ O	105.5	108	110.5	113.7	116.9	121.2	126.8	133.1	139.8	147.5	155.7	7
8	chlorate	KClO ₃		3.3	5	7.4	10.5	14	19.3	24.5		38.5		57	8
9	chloride	KCl		27.6	31.0	34.0	37.0	40.0	42.6	45.5	48.3	51.1	54.0	56.7	9
10	chromate	K ₂ CrO ₄		58.2	60.0	61.7	63.4	65.2	66.8	68.6	70.4	72.1	73.9	75.6	10
11	dichromate	K ₂ Cr ₂ O ₇		5	7	12	20	26	34	43	52	61	70	80	11
12	ferricyanide	K ₃ Fe(CN) ₆		31	36	43	50	60		66				82.6 ¹⁰⁴	12
13	hydroxide	KOH	2H ₂ O	97	103	112	126								13
14	hydroxide	KOH	1H ₂ O						140					178	14
15	nitrate	KNO ₃		13.3	20.9	31.6	45.8	63.9	85.5	110.0	138	169	202	246	15
16	nitrite	KNO ₂		278.8		298.4		334.9						412.8	16
17	perchlorate	KClO ₄		0.75	1.05	1.80	2.6	4.4	6.5	9	11.8	14.8	18	21.8	17
18	permanganate	KMnO ₄		2.83	4.4	6.4	9.0	12.56	16.89	22.2					18
19	persulfate†	K ₂ S ₂ O ₈ †	†	1.62	2.60	4.49	7.19	9.89							19
20	sulfate	K ₂ SO ₄		7.35	9.22	11.11	12.97	14.76	16.50	18.17	19.75	21.4	22.8	24.1	20
21	thiocyanate	KCNS		177.0		217.5									21
22	Silver cyanide	AgCN				2.2 × 10 ⁻⁵									22
23	nitrate	AgNO ₃		122	170	222	300	376	455	525		669		952	23
24	sulfate	Ag ₂ SO ₄		0.573	0.695	0.796	0.888	0.979	1.08	1.15	1.22	1.30	1.36	1.41	24
25	Sodium acetate	NaC ₂ H ₃ O ₂	3H ₂ O	36.3	40.8	46.5	54.5	65.5	83	139					25
26	acetate	NaC ₂ H ₃ O ₂		119	121	123.5	126	129.5	134	139.5	146	153	161	170	26
27	bicarbonate	NaHCO ₃		6.9	8.15	9.6	11.1	12.7	14.45	16.4					27
28	carbonate	Na ₂ CO ₃	10H ₂ O	7	12.5	21.5									28
29	carbonate	Na ₂ CO ₃	1H ₂ O				50.5	48.5		46.4		45.8		45.5	29
30	chlorate	NaClO ₃		79	89	101	113	126	140	155	172	189		230	30
31	chloride	NaCl		35.65	35.72	35.89	36.09	36.37	36.69	37.04	37.46	37.93	38.47	38.99	31
32	chromate	Na ₂ CrO ₄	10H ₂ O	31.70	50.17	88.7									32
33	chromate	Na ₂ CrO ₄	4H ₂ O				88.7	95.96	104	114.6					33
34	chromate	Na ₂ CrO ₄									123.0	124.8		125.9	34
35	dichromate	Na ₂ Cr ₂ O ₇	2H ₂ O	163.0		177.8			244.8	316.7		376.2		426.3	35
36	dichromate	Na ₂ Cr ₂ O ₇													36
37	dihydrogen phosphate	NaH ₂ PO ₄	2H ₂ O	57.9	69.9	85.2	106.5	138.2							37
38	dihydrogen phosphate	NaH ₂ PO ₄	1H ₂ O						158.6						38
39	dihydrogen phosphate	NaH ₂ PO ₄								179.3	190.3	207.3	225.3	246.6	39
40	hydrogen arsenate	Na ₂ HAsO ₄	12H ₂ O	7.3	15.5	26.5	37	47		65		85			40
41	hydrogen phosphate	Na ₂ HPO ₄	12H ₂ O	1.67	3.6	7.7	20.8								41
42	hydrogen phosphate	Na ₂ HPO ₄	7H ₂ O					51.8							42
43	hydrogen phosphate	Na ₂ HPO ₄	2H ₂ O						80.2	82.9	88.1	92.4	102.9		43
44	hydrogen phosphate	Na ₂ HPO ₄												102.2	44
45	hydroxide	NaOH	4H ₂ O	42											45
46	hydroxide	NaOH	3½H ₂ O		51.5										46
47	hydroxide	NaOH	1H ₂ O			109	119	129	145	174					47
48	hydroxide	NaOH											313	347	48
49	nitrate	NaNO ₃		73	80	88	96	104	114	124		148		180	49
50	nitrite	NaNO ₂		72.1	78.0	84.5	91.6	98.4	104.1			132.6		163.2	50
51	oxalate	Na ₂ C ₂ O ₄				3.7								6.33	51
52	phosphate, tri-	Na ₃ PO ₄	12H ₂ O	1.5	4.1	11	20	31	43	55		81		108	52
53	pyrophosphate	Na ₄ P ₂ O ₇	10H ₂ O	3.16	3.95	6.23	9.95	13.50	17.45	21.83		30.04		40.26	53
54	sulfate	Na ₂ SO ₄	10H ₂ O	5.0	9.0	19.4	40.8								54
55	sulfate	Na ₂ SO ₄	7H ₂ O	19.5	30	44									55
56	sulfate	Na ₂ SO ₄						48.8	46.7	45.3		43.7		42.5	56
57	sulfide	Na ₂ S	9H ₂ O		15.42	18.8	22.5	28.5							57
58	sulfide	Na ₂ S	5½H ₂ O						39.82	42.69	45.73	51.40	59.23		58
59	sulfide	Na ₂ S	6H ₂ O						36.4	39.1	43.31	49.14	57.28		59
60	sulfite	Na ₂ SO ₃	7H ₂ O	13.9	20	26.9	36								60
61	sulfite	Na ₂ SO ₃						28	28.2	28.8		28.3			61
62	tetraborate	Na ₂ B ₄ O ₇	10H ₂ O	1.3	1.6	2.7	3.9		10.5	20.3					62
63	tetraborate	Na ₂ B ₄ O ₇	5H ₂ O								24.4	31.5	41	52.5	63
64	vanadate (meta)	NaVO ₃	2H ₂ O			15.3 ²⁵⁰		30.2		68.4					64

TABLE 2-122 Solubilities of Inorganic Compounds in Water at Various Temperatures (Concluded)

	Substance	Formula	Solid phase	0 °C	10 °C	20 °C	30 °C	40 °C	50 °C	60 °C	70 °C	80 °C	90 °C	100 °C	
1	Sodium vanadate (meta)	NaVO ₃				21.10 ^{25o}		26.23		32.97	36.9	38.8 ^{75o}			1
2	Stannous chloride	SnCl ₂		83.9		269.8 ^{15o}									2
3	sulfate	SnSO ₄				19								18	3
4	Strontium acetate	Sr(C ₂ H ₃ O ₂) ₂	4H ₂ O	36.9	43.61										4
5	acetate	Sr(C ₂ H ₃ O ₂) ₂	½H ₂ O		42.95	41.6	39.5		37.35		36.24	36.10		36.4	5
6	chloride	SrCl ₂	6H ₂ O	43.5	47.7	52.9	58.7	65.3	72.4	81.8					6
7	chloride	SrCl ₂	2H ₂ O								85.9	90.5		100.8	7
8	nitrate	Sr(NO ₃) ₂	1H ₂ O	52.7		64.0			83.8	97.2			130.4	139	8
9	nitrate	Sr(NO ₃) ₂	4H ₂ O	40.1		70.5									9
10	nitrate	Sr(NO ₃) ₂					88.6	90.1		93.8	96	98	100		10
11	sulfate	SrSO ₄		0.0113		0.0114	0.0114								11
12	Sulfur dioxide, 760 mm†	SO ₂		22.83	16.21	11.29	7.81	5.41	4.5						12
13	Thallium sulfate	Tl ₂ SO ₄		2.70	3.70	4.87	6.16		9.21	10.92	12.74	14.61	16.53	18.45	13
14	Thorium sulfate	Th(SO ₄) ₂	9H ₂ O	0.74	0.98	1.38	1.995	2.998	5.22						14
15	sulfate	Th(SO ₄) ₂	8H ₂ O	1.0	1.25	1.62									15
16	sulfate	Th(SO ₄) ₂	6H ₂ O	1.50		1.90	2.45			6.64					16
17	sulfate	Th(SO ₄) ₂	4H ₂ O					4.04	2.54	1.63	1.09				17
18	Zinc chlorate	ZnClO ₃	6H ₂ O	145.0	152.5										18
19	chlorate	ZnClO ₃	4H ₂ O			200.3	209.2	223.2	273.1						19
20	nitrate	Zn(NO ₃) ₂	6H ₂ O	94.78		118.3									20
21	nitrate	Zn(NO ₃) ₂	3H ₂ O					206.9							21
22	sulfate	ZnSO ₄	7H ₂ O	41.9	47	54.4									22
23	sulfate	ZnSO ₄	6H ₂ O					70.1	76.8						23
24	sulfate	ZnSO ₄	1H ₂ O									86.6	83.7	80.8	24

2-130 PHYSICAL AND CHEMICAL DATA

TABLE 2-123 Solubility as a Function of Temperature and Henry's Constant at 25°C for Gases in Water

Name	Formula	A	B	C	D	T range, K	H at 25°C, atm
Acetylene	C ₂ H ₂	-156.51	8,160.2	21.403	0	274-343	1,330
Carbon dioxide	CO ₂	-159.854	8,741.68	21.6694	-1.10261E-03	273-353	1,635
Carbon monoxide	CO	-171.764	8,296.9	23.3376	0	273-353	58,000
Ethane	C ₂ H ₆	-250.812	12,695.6	34.7413	0	275-323	29,400
Ethylene	C ₂ H ₄	-153.027	7,965.2	20.5248	0	287-346	11,726
Helium	He	-105.9768	4,259.62	14.0094	0	273-348	142,900
Hydrogen	H ₂	-125.939	5,528.45	16.8893	0	273-345	70,800
Methane	CH ₄	-338.217	13,282.1	51.9144	-0.0425831	273-523	39,200
Nitrogen	N ₂	-181.587	8,632.13	24.7981	0	273-350	84,600
Oxygen	O ₂	-171.2542	8,391.24	23.24323	0	273-333	43,400

The constants can be used to calculate solubility by the equation $\ln x = A + B/T + C \ln T + DT$, where T is in K and x is the mole fraction of the solute dissolved in water when the solute partial pressure is 1 atm. With the assumption that Henry's law is valid up to 1 atm, $H = 1/x$. Values of the constants are from P. G. T. Fogg and W. Gerrard, *Solubility of Gases in Liquids*, Wiley, 1991, New York, and *Solubility Data Series*, vol. 1, *Helium and Neon*, IUPAC, Pergamon Press, Oxford, 1979. For higher-temperature behavior and an up-to-date reference list, see R. Fernandez-Prini, J. L. Alvarez, and A. H. Harvey, *J. Phys. Chem. Ref. Data* **32**(2):903, 2003. To find H at temperatures other than 25°C, first find the solubility and then take the reciprocal.

TABLE 2-124 Henry's Constant H for Various Compounds in Water at 25°C

Compound	CAS no.	Formula	H , atm	Rating
<i>Paraffin Hydrocarbons</i>				
Methane	74828	CH ₄	36,600	4
Ethane	74840	C ₂ H ₆	26,700	3
Propane	74986	C ₃ H ₈	37,800	3
Butane	106978	C ₄ H ₁₀	51,100	3
Pentane	109660	C ₅ H ₁₂	70,000	3
Octane	111659	C ₈ H ₁₈	274,000	3
Nonane	111842	C ₉ H ₂₀	329,000	3
<i>Olefins</i>				
Ethylene	74851	C ₂ H ₄	11,700	3
Propylene	115071	C ₃ H ₆	11,700	4
<i>Aromatics</i>				
Benzene	71432	C ₆ H ₆	299	10
Toluene	108883	C ₇ H ₈	354	10
<i>o</i> -Xylene	95476	C ₈ H ₁₀	272	10
Cumene	98828	C ₉ H ₁₂	724	9
Phenol	108952	C ₆ H ₆ O	0.0394	7
<i>Aldehydes</i>				
Acetaldehyde	75,070	C ₂ H ₄ O	5.56	3
Propanal	123,386	C ₃ H ₆ O	4.36	4
<i>Ketones</i>				
Methyl ethyl ketone	78,933	C ₄ H ₈ O	2.59	5
<i>Esters</i>				
Methyl formate	107313	C ₂ H ₄ O ₂	13.6	3
Ethyl formate	109944	C ₃ H ₆ O ₂	13.6	3
Methyl acetate	79209	C ₃ H ₆ O ₂	5.04	3
Butyl acetate	123864	C ₆ H ₁₂ O ₂	13.6	3
<i>Chlorine Containing</i>				
Methyl chloride	74873	CH ₃ Cl	556	?
Chloroethane	75003	C ₂ H ₅ Cl	681	10
Chlorobenzene	108907	C ₆ H ₅ Cl	204	10
<i>Alcohols</i>				
Methanol	67561	CH ₃ O	0.272	4
Ethanol	64175	C ₂ H ₅ O	0.272	4
1-Propanol	71238	C ₃ H ₇ O	0.507	3
1-Butanol	71363	C ₄ H ₉ O	0.482	3
<i>Miscellaneous</i>				
Acrylonitrile	107131	C ₃ H _{3.5} N	5.54	3
Dimethyl sulfide	75183	C ₂ H ₆ S	121	3
Dimethyl disulfide	624920	C ₂ H ₆ S ₂	68.1	3
Methyl mercaptan	74931	CH ₃ S	177	3
Ethyl mercaptan	75081	C ₂ H ₅ S	161	3
Pyridine	110861	C ₅ H ₅ N	0.817	3

Values in this table were taken from the Design Institute for Physical Properties (DIPPR) of the American Institute of Chemical Engineers (AIChE), copyright 2007 AIChE and reproduced with permission of AIChE, the DIPPR Environmental Safety Property Data and Estimations Steering Committee and of the DIPPR Evaluated Process Design Data Project Steering Committee. Their source should be cited as T. N. Rogers, D. A. Zei, R. L. Rowley, W. V. Wilding, J. L. Oscarson, Y. Yang, N. A. Zundel, T. E. Daubert, R. P. Danner, DIPPR® Data Compilation of Pure Chemical Properties, Design Institute for Physical Properties, AIChE, New York (2007). Flammability limits are from B. Lewis and G. Von Elbe, *Combustion, Flames and Explosions of Gases*, New York: Harcourt Brace Jovanovich (1987). Flash point data are from N. I. Sax, *Dangerous Properties of Industrial Materials*, 6th ed, New York: Van Nostrand Reinhold (1984). Autoignition data are from I. Glassman, *Combustion*, 3d ed, New York: Academic Press (1996). The ratings reflect DIPPR® ESP's effort to provide a critical evaluation and quality assessment of each data point with 15

being the highest score possible. The rating is not directly correlated with the estimated experimental uncertainty. The ratings reflect DIPPR Project 911's effort to provide a critical evaluation and quality assessment of each data point, with 10 being the highest score possible. The rating is not directly correlated with the estimated experimental uncertainty. Henry's constant is a strong nonlinear function of temperature. A single value measured at one temperature, if used for calculation at a different temperature, can lead to serious errors. Procedures for extrapolation of single-point values over the ambient temperature range (4°C < T < 50°C) are presented in Sec. 22, p. 22-49, under "Estimating Henry's law constants." Estimation procedures for the larger range (4°C < T < 200°C) are presented in F. L. Smith and A. H. Harvey, "Avoid Common Pitfalls When Using Henry's Law," *Chem. Eng. Prog.*, **103**(9), 2007. See also Y.-L. Huang, J. D. Olson, and G. E. Keller II, "Steam Stripping for Removal of Organic Pollutants from Water. 2. Vapor-Liquid Equilibrium Data," *Ind. Eng. Chem. Res.*, **31**, pp. 1759-1768, 1992. (Also see the Supplementary Material, which contains the databank of 404 compounds of environmental interest and other useful property data.)

The H in Tables 2-123 to 2-134 is the proportionality constant in Henry's law, $p = Hx$, where x is the mole fraction of the solute in the aqueous liquid phase; p is the partial pressure in atm of the solute in the gas phase; and H is a proportionality constant, generally referred to as Henry's constant. Values of H often have considerable uncertainty and are strong functions of temperature. To convert values of H at 25°C from atm to atm/(mol/m³), divide by the molar density of water at 25°C, which is 55,342 mol/m³. Henry's law is valid only for dilute solutions.

Additional values of Henry's constant can be found in "Environmental Simulation Program": OLI Systems, Inc., Morris Plains, N.J.; "Estimated Henry's Law Constant," EPA Online Tools for Site Assessment Calculation (<http://www.epa.gov/athens/learn2model/part-two/onsite/esthenry.htm>); "Compilation of Henry's Law Constants for Inorganic and Organic Species of Potential Importance in Environmental Chemistry," Rolf Sander, Air Chemistry Department, Max-Planck Institute of Chemistry, Mainz, Germany; "Modeling Atmospheric Chemistry: Interactions between Gas-Phase Species and Liquid Cloud/Aerosol Particles," Rolf Sander, *Surv. Geophys.* **20**:1-31, 1999 (<http://www.henrys-law.org>).

TABLE 2-125 Henry's Constant H for Various Compounds in Water at 25 °C from Infinite Dilution Activity Coefficients

Compound	CAS no.	Formula	$H = \gamma^*P_{vp}$, atm
Pentane	109660	C ₅ H ₁₂	63700
Hexane	1100543	C ₆ H ₁₄	84600
Heptane	142825	C ₇ H ₁₆	120000
Benzene	71432	C ₆ H ₆	309
Toluene	108883	C ₇ H ₈	344
<i>o</i> -Xylene	95476	C ₈ H ₁₀	267
Cumene	98,828	C ₉ H ₁₂	613
Styrene	100425	C ₈ H ₈	145
Formaldehyde	50000	CH ₂ O	14.3
Acetaldehyde	75070	C ₂ H ₄ O	4.54
Propanal	123386	C ₃ H ₆ O	5.45
Acetone	67641	C ₃ H ₆ O	2.13
Methyl ethyl ketone	78933	C ₄ H ₈ O	3.11
Methyl <i>n</i> -propyl ketone	107879	C ₅ H ₁₀ O	4.60
Formic acid	64186	C ₁ H ₂ O ₂	0.0404
Methyl acetate	79209	C ₃ H ₆ O ₂	6.38
Ethyl acetate	141786	C ₄ H ₈ O ₂	8.01
Butyl acetate	123864	C ₆ H ₁₂ O ₂	12.3
Chloroethane	75003	C ₂ H ₅ Cl	626
1-Chloropropane	74986	C ₃ H ₇ Cl	792
Chlorobenzene	108907	C ₆ H ₅ Cl	219
Methanol	67561	CH ₃ O	0.263
Ethanol	64175	C ₂ H ₅ O	0.293
Pyridine	110861	C ₅ H ₅ N	0.544
Diethyl ether	60297	C ₄ H ₁₀ O	48.7
Thiophene	110021	C ₄ H ₄ S	160

Henry's constant H at 25 °C is the vapor pressure at 25 °C times the infinite dilution activity coefficient, also at 25 °C. Infinite dilution activity coefficients are from Mitchell and Jurs, *J. Chem. Inf. Comput. Sci.* **38**: 200 (1998). Henry's constant is a strong nonlinear function of temperature. A single value measured at one temperature, if used for calculation at a different temperature, can lead to serious errors. Procedures for extrapolation of single-point values over the ambient temperature range (4°C < T < 50°C) are presented in Sec. 22, p. 22–49, under "Estimating Henry's law constants." Estimation procedures for the larger range (4°C < T < 200°C) are presented in F. L. Smith and A. H. Harvey, "Avoid Common Pitfalls When Using Henry's Law," *Chem. Eng. Prog.*, **103**(9), 2007. See also Y.-L. Huang, J. D. Olson, and G. E. Keller II, "Steam Stripping for Removal of Organic Pollutants from Water. 2. Vapor-Liquid Equilibrium Data," *Ind. Eng. Chem. Res.*, **31**, pp. 1759–1768, 1992. (Also see the Supplementary Material, which contains the databank of 404 compounds of environmental interest and other useful property data.)

TABLE 2-126 Air*

t , °C	0	5	10	15	20	25	30	35
$10^{-4} \times H^\dagger$	4.32	4.88	5.49	6.07	6.64	7.20	7.71	8.23
t , °C	40	45	50	60	70	80	90	100
$10^{-4} \times H^\dagger$	8.70	9.11	9.46	10.1	10.5	10.7	10.8	10.7

**International Critical Tables*, vol. 3, p. 257.

$\dagger H$ is calculated from the absorption coefficients of O₂ and N₂, taking into consideration the correction for constant argon content.

TABLE 2-127 Ammonia-Water at 10 and 20 °C*

Mass fraction NH ₃ in liquid	10 °C		20 °C	
	P , kPa	Mass fraction NH ₃ in vapor	P , kPa	Mass fraction NH ₃ in vapor
0.0	1.23	0.0	2.34	0.0
0.00467	1.37	0.1		
0.00495			2.60	0.1
0.1	7.07	0.84164	11.95	0.82096
0.2	20.07	0.95438	32.34	0.94541
0.3	47.37	0.98565	73.85	0.98199
0.4	99.84	0.99544	150.56	0.99393
0.5	184.44	0.99848	269.50	0.99783
0.6	292.15	0.99943	416.63	0.99913
0.7	399.03	0.99975	560.61	0.99960
0.8	486.44	0.99988	678.61	0.99980
0.9	554.33	0.99995	771.87	0.99991
1.0	615.05	1.0	857.48	1.0

*Selected values from R. Tillner-Roth and D. G. Friend, *J. Phys. Chem. Ref. Data* **27**:63 (1998). This reference lists solubilities for temperatures from –70 to 340°C. Densities, enthalpies, and entropies are listed for both the two-phase and single-phase regions for pressures up to 40 MPa.

TABLE 2-128 Carbon Dioxide (CO₂)*

Total pressure, atm	Liquid mol fraction CO ₂ × 10 ³							
	0 °C	10 °C	15 °C	20 °C	25 °C	35 °C	50 °C	100 °C
1	1.445	0.985	0.802	0.692	0.608	0.473	0.342	0.187
2	2.89	1.946	1.587	1.374	1.207	0.943	0.683	0.373
10	12.71	8.81	7.32	6.44	5.74	4.54	3.30	1.841
20	21.23	15.38	13.13	11.84	10.75	8.64	6.34	3.62
30	25.79	19.80	17.49	16.22	15.05	12.80	9.10	5.35
36		21.45	19.42	18.30	17.29	14.80	10.63	6.35

*Values selected from G. Houghton, A. M. McLean, and P. D. Ritchie, *Chem. Eng. Sci.* **6**:132–137, 1957.

TABLE 2-129 Carbonyl Sulfide (COS)*

t , °C	0	5	10	15	20	25	30
$10^{-3} \times H$	0.92	1.17	1.48	1.82	2.19	2.59	3.04

**International Critical Tables*, vol. 3, p. 261.

2-132 PHYSICAL AND CHEMICAL DATA

TABLE 2-130 Chlorine (Cl₂)

Partial pressure of Cl ₂ , mmHg	Solubility, g of Cl ₂ per liter					
	0 °C	10 °C	20 °C	30 °C	40 °C	50 °C
5	0.488	0.451	0.438	0.424	0.412	0.398
10	0.679	0.603	0.575	0.553	0.532	0.512
30	1.221	1.024	0.937	0.873	0.821	0.781
50	1.717	1.354	1.210	1.106	1.025	0.962
100	2.79	2.08	1.773	1.573	1.424	1.313
150	3.81	2.73	2.27	1.966	1.754	1.599
200	4.78	3.35	2.74	2.34	2.05	1.856
250	5.71	3.95	3.19	2.69	2.34	2.09
300		4.54	3.63	3.03	2.61	2.31
350		5.13	4.06	3.35	2.86	2.53
400		5.71	4.48	3.69	3.11	2.74
450		6.26	4.88	3.98	3.36	2.94
500		6.85	5.29	4.30	3.61	3.14
550		7.39	5.71	4.60	3.84	3.33
600		7.97	6.12	4.91	4.08	3.52
650		8.52	6.52	5.21	4.32	3.71
700		9.09	6.90	5.50	4.54	3.89
750		9.65	7.29	5.80	4.77	4.07
800		10.21	7.69	6.08	4.99	4.27
900			8.46	6.68	5.44	4.62
1000			9.27	7.27	5.89	4.97
1200	Cl ₂ ·8H ₂ O ₂ separates		10.84	8.42	6.81	5.67
1500			13.23	10.14	8.05	6.70
2000			17.07	13.02	10.22	8.38
2500			21.0	15.84	12.32	10.03
3000				18.73	14.47	11.70
3500				21.7	16.62	13.38
4000				24.7	18.84	15.04
4500				27.7	20.7	16.75
5000				30.8	23.3	18.46

Partial pressure of Cl ₂ , mmHg	Solubility, g of Cl ₂ per liter					
	60 °C	70 °C	80 °C	90 °C	100 °C	110 °C
5	0.383	0.369	0.351	0.339	0.326	0.316
10	0.492	0.470	0.447	0.431	0.415	0.402
30	0.743	0.704	0.671	0.642	0.627	0.598
50	0.912	0.863	0.815	0.781	0.747	0.722
100	1.228	1.149	1.085	1.034	0.987	0.950
150	1.482	1.382	1.294	1.227	1.174	1.137
200	1.706	1.580	1.479	1.396	1.333	1.276
250	1.914	1.764	1.642	1.553	1.480	1.413
300	2.10	1.932	1.793	1.700	1.610	1.542
350	2.28	2.10	1.940	1.831	1.736	1.661
400	2.47	2.25	2.08	1.965	1.854	1.773
450	2.64	2.41	2.22	2.09	1.972	1.880
500	2.80	2.55	2.35	2.21	2.08	1.986
550	2.97	2.69	2.47	2.32	2.19	2.09
600	3.13	2.83	2.59	2.43	2.29	2.19
650	3.29	2.97	2.72	2.55	2.41	2.28
700	3.44	3.10	2.84	2.66	2.50	2.37
750	3.59	3.23	2.96	2.76	2.60	2.47
800	3.75	3.37	3.08	2.87	2.69	2.56
900	4.04	3.63	3.30	3.08	2.89	2.74
1000	4.36	3.88	3.53	3.28	3.07	2.91
1200	4.92	4.37	3.95	3.67	3.43	3.25
1500	5.76	5.09	4.58	4.23	3.95	3.74
2000	7.14	6.26	5.63	5.17	4.78	4.49
2500	8.48	7.40	6.61	6.05	5.59	5.25
3000	9.83	8.52	7.54	6.92	6.38	5.97
3500	11.22	9.65	8.53	7.79	7.16	6.72
4000	12.54	10.76	9.52	8.65	7.94	7.42
4500	13.88	11.91	10.46	9.49	8.72	8.13
5000	15.26	13.01	11.42	10.35	9.48	8.84

TABLE 2-131 Chlorine Dioxide (ClO₂)

Vol % of ClO ₂ in gas phase	Weight of ClO ₂ , grams per liter of solution						
	0 °C	5 °C	10 °C	15 °C	20 °C	30 °C	40 °C
1	2.00	1.50	1.25	1.00	0.90	0.60	0.46
3	6.00	4.7	3.85	3.20	2.70	1.95	1.30
5	10.0	7.8	6.30	5.25	4.30	3.20	2.25
7	14.0	10.9	8.95	7.35	6.15	4.40	3.20
10	20.0	15.5	12.8	10.5	8.80	6.30	4.50
11		17.0	14.0	11.7	9.70	7.00	5.00
12		18.6	15.3	12.8	10.55	7.50	5.45
13		20.3	16.6	13.8	11.5	8.20	5.85
14			18.0	14.9	12.3	8.80	6.35
15			19.2	16.0	13.2	9.50	6.80
16			20.3	17.0	14.2	10.1	7.20

Ishi, *Chem. Eng. (Japan)*, **22**:153 (1958).

TABLE 2-132 Hydrogen Chloride (HCl)

Weights of HCl per 100 weights of H ₂ O	Partial pressure of HCl, mmHg			
	0 °C	10 °C	20 °C	30 °C
78.6	510	840		
66.7	130	233		
56.3	29.0	56.4	399	627
47.0	5.7	11.8	105.5	188
38.9	1.0	2.27	23.5	44.5
31.6	0.175	0.43	4.90	9.90
25.0	0.0316	0.084	1.00	2.17
19.05	0.0056	0.016	0.205	0.48
13.64	0.00099	0.00305	0.0428	0.106
8.70	0.000118	0.000583	0.0088	0.0234
4.17	0.000018	0.000069	0.00178	0.00515
2.04		0.0000117	0.00024	0.00077
			0.000044	0.000151

Weights of HCl per 100 weights of H ₂ O	Partial pressure of HCl, mm Hg		
	50 °C	80 °C	110 °C
78.6			
66.7			
56.3	535		
47.0	141	623	
38.9	35.7	188	760
31.6	8.9	54.5	253
25.0	2.21	15.6	83
19.05	0.55	4.66	28
13.64	0.136	1.34	9.3
8.70	0.0344	0.39	3.10
4.17	0.0064	0.095	0.93
2.04	0.00140	0.0245	0.280

Enthalpy and phase-equilibrium data for the binary system HCl-H₂O are given by Van Nuy, *Trans. Am. Inst. Chem. Engrs.*, **39**, 663 (1943).

TABLE 2-133 Hydrogen Sulfide (H₂S)

<i>t</i> , °C	0	5	10	15	20	25	30	35
10 ⁻² × <i>H</i>	2.68	3.15	3.67	4.23	4.83	5.45	6.09	6.76
<i>t</i> , °C	40	45	50	60	70	80	90	100
10 ⁻² × <i>H</i>	7.45	8.14	8.84	10.3	11.9	13.5	14.4	14.8

International Critical Tables, vol. 3, p. 259.

TABLE 2-134 Partial Vapor Pressure of Sulfur Dioxide over Water, mmHg

g SO ₂ / 100 g H ₂ O	Temperature, °C								
	0	10	20	30	40	50	60	90	120
0.01	0.02	0.04	0.07	0.12	0.19	0.29	0.43	1.21	2.82
0.05	0.38	0.66	1.07	1.68	2.53	3.69	5.24	12.9	27.0
0.10	1.15	1.91	3.03	4.62	6.80	9.71	13.5	31.7	63.9
0.15	2.10	3.44	5.37	8.07	11.7	16.5	22.7	52.2	104
0.20	3.17	5.13	7.93	11.8	17.0	23.8	32.6	73.7	145
0.25	4.34	6.93	10.6	15.7	22.5	31.4	42.8	95.8	186
0.30	5.57	8.84	13.5	19.8	28.2	39.2	53.3	118	229
0.40	8.17	12.8	19.4	28.3	40.1	55.3	74.7	164	316
0.50	10.9	17.0	25.6	37.1	52.3	72.0	96.8	211	404
1.00	25.8	39.5	58.4	83.7	117	159	212	454	856
2.00	58.6	88.5	129	183	253	342	453	955	
3.00	93.2	139	202	285	393	530	700		
4.00	129	192	277	389	535	720			
5.00	165	245	353	496	679				
6.00	202	299	430	602	824				
8.00	275	407	585	818					
10.00	351	517	741						
15.00	542	796							
20.00	735								

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THERMAL EXPANSION

UNITS CONVERSIONS

For this subsection, the following units conversion is applicable:

$$^{\circ}\text{F} = \% ^{\circ}\text{C} + 32$$

ADDITIONAL REFERENCES

The tables given under this subject are reprinted by permission from the *Smithsonian Tables*. For more detailed data on thermal expansion, see *International Critical Tables*: tabular index, vol. 3, p. 1; abrasives, vol. 2, p. 87; alloys, vol. 2, p. 463; building stones, vol. 2, p. 54; carbons, vol. 2, p. 303; elements, vol. 1, p. 102; enamels, vol. 2, p. 115; glass, vol.

2, p. 93; metals, vol. 2, p. 459; petroleum, vol. 2, p. 145; porcelains, vol. 2, pp. 70, 78; refractory materials, vol. 2, p. 83; solid insulators, vol. 2, p. 310.

THERMAL EXPANSION OF GASES

No tables of the coefficients of thermal expansion of gases are given in this edition. The coefficient at constant pressure, $1/\nu(\partial\nu/\partial T)_p$, for an ideal gas is merely the reciprocal of the absolute temperature. For a real gas or liquid, both it and the coefficient at constant volume, $1/p(\partial p/\partial T)_v$, should be calculated either from the equation of state or from tabulated PVT data.

2-134 PHYSICAL AND CHEMICAL DATA

TABLE 2-135 Linear Expansion of the Solid Elements*

C is the true expansion coefficient at the given temperature; M is the mean coefficient between given temperatures; where one temperature is given, the true coefficient at that temperature is indicated; α and β are coefficients in formula $l_t = l_0(1 + \alpha t + \beta t^2)$; l_0 is length at 0 °C (unless otherwise indicated, when, if x is the reference temperature, $l_t = l_x[1 + \alpha(t - t_x) + \beta(t - t_x)^2]$; l_t is length at t °C).

Element	Temp., °C	$C \times 10^4$	Temp. range, °C	$M \times 10^4$	Temp. range, °C	$\alpha \times 10^4$	$\beta \times 10^6$
Aluminum	20	0.224	100	0.235	0, 500	0.22	0.009
Aluminum	300	0.284	500	0.311			
Antimony	20	0.136	20	0.080⊥			
Arsenic	20	0.05					
Bismuth	20	0.014	20	0.103⊥			
Cadmium	0	0.54	-180, -140	0.59	20, 100	0.526	
Cadmium	0	0.20⊥	-180, -140	0.117⊥	20, 100	0.214⊥	
Carbon, diamond	50	0.012					
graphite	50	0.06					
Chromium			20, 100	0.068	20, 500	0.086	
Cobalt	20	0.123			6, 121	0.121	0.0064
Copper	20	0.162	100	0.166	0, 625	0.161	0.0040
Copper	200	0.170	300	0.175			
Gold	20	0.140	17, 100	0.143	0, 520	0.142	0.0022
Gold			-191, 17	0.132			
Indium	40	0.417					
Iodine			-190, 17	0.837			
Iridium	20	0.065			0, 80	0.0636	0.0032
Iridium					1070, 1720	0.0679	0.0011
Iron, soft	40	0.1210	0, 100	0.11			
cast	20	0.118			0, 750	0.1158	0.0053
wrought	20	0.119			0, 750	0.1170	0.0053
steel	20	0.114			0, 750	0.1118	0.0053
Lead (99.9)			20, 100	0.291	100, 240	0.269	0.011
	100	0.291	20, 200	0.300			
	280	0.343					
Magnesium	20	0.254	-100, +20	0.240	+20, 500	0.2480	0.0096
			20, 100	0.260			
Manganese	20	0.233	0, 100	0.228			
			-190, 0	0.159	20, 300	0.216	0.0121
Molybdenum†	20	0.053	0, 100	0.052	-142, 19	0.0515	0.0057
			25, 100	0.049	19, +305	0.0501	0.0014
			25, 500	0.055			
Nickel	20	0.126	0, 100	0.130	-190, +20	0.1308	0.0166
					+20, +300	0.1236	0.0066
					500, 1000	0.1346	0.0033
Osmium	40	0.066					
Palladium	20	0.1173			-190, +100	0.1152	0.00517
					0, 1000	0.1167	0.0022
Platinum	20	0.0887			-190, -100	0.0875	0.00314
	20	0.0893			0, +80	0.0890	0.00121
					0, 1000	0.0887	0.00132
Potassium			0, 50	0.83			
Rhodium	40	0.0850	6, 21	0.0876	-75, -112	0.0746	
Ruthenium	40	0.0963					
Selenium	0	0.439	0, 100	0.660			
Silicon	40	0.0763	-3, +18	0.0249	-75, -67	0.0182	
Silver	20	0.1846	0, 100	0.197	0, 875	0.1827	0.00479
	20	0.195			20, 500	0.1939	0.00295
Sodium			-190, -17	0.622	0, 50	0.72	
Steel, 36.4Ni			20, 260	0.031	260, 500	0.144	
			20, 340	0.055	340, 500	0.136	
Tantalum†	20	0.065	-78, 0	0.059	20, 400	0.0646	0.0009
			0, 100	0.0655			
Tellurium	20	0.016	0, 20	0.272⊥			
Thallium	40	0.302					
Tin	20	0.214			8, 95	0.2033	0.0263
	20	0.305	20	0.154⊥			
Tungsten†	27	0.0444	0, 100	0.045	-105, +502	0.0428	0.00058
Zinc	20†	0.643	-140, -100	0.656	+0, 400	0.354	0.010
	20†	0.125⊥	+20, 100	0.639			
	20	0.358	+20, 100	0.141⊥			

**Smithsonian Tables*. For more complete tabulations see Table 142, *Smithsonian Physical Tables*, 9th ed., 1954; *Handbook of Chemistry and Physics*, 40th ed., pp. 2239–2245. Chemical Rubber Publishing Co.; Goldsmith, and Waterman, WADC-TR-58-476, 1959; Johnson (ed.), WADD-TR-60-56, 1960, etc.

†Molybdenum, 300 to 2500 °C; $l_t = l_{300}[1 + 5.00 \times 10^{-6}(t - 300) + 10.5 \times 10^{-10}(t - 300)^2]$

Tantalum, 300 to 2800 °C; $l_t = l_{300}[1 + 6.60 \times 10^{-6}(t - 300) + 5.2 \times 10^{-10}(t - 300)^2]$

Tungsten, 300 to 2700 °C; $l_t = l_{300}[1 + 4.44 \times 10^{-6}(t - 300) + 4.5 \times 10^{-10}(t - 300)^2]$

Beryllium, 20 to 100 °C; 12.3×10^{-6} per °C.

Columbium, 0 to 100 °C; 7.2×10^{-6} per °C.

Tantalum, 20 to 100 °C; 6.6×10^{-6} per °C.

†These values for zinc were taken from Grüneisen and Goens, *Z. Physik.*, **29**:141 (1924).

TABLE 2-136 Linear Expansion of Miscellaneous Substances*

The coefficient of cubical expansion may be taken as three times the linear coefficient. In the following table, t is the temperature or range of temperature, and C , the coefficient of expansion.

Substance	t , °C	$C \times 10^4$	Substance	t , °C	$C \times 10^4$	Substance	t , °C	$C \times 10^4$
Amber	0–30	0.50	Jena thermometer 59 ^{III}	0–100	0.058	Topas:		
	0–09	0.61	Jena thermometer 59 ^{III}	–191–+16	0.424	Parallel to lesser horizontal axis	0–100	0.0832
Bakelite, bleached	20–60	0.22	Gutta percha	20	1.983	Parallel to greater horizontal axis	0–100	0.0836
Brass:			Ice	–20––1	0.51	Parallel to vertical axis	0–100	0.0472
Cast	0–100	0.1875	Iceland spar:			Tourmaline:		
Wire	0–100	0.1930	Parallel to axis	0–80	0.2631	Parallel to longitudinal axis	0–100	0.0937
Wire	0–100	0.1783–0.193	Perpendicular to axis	0–80	0.0544	Parallel to horizontal axis	0–100	0.0773
71.5 Cu + 27.7 Zn + 0.3 Sn + 0.5 Pb	40	0.1859	Lead tin (solder) 2 Pb + 1 Sn	0–100	0.2508	Type metal	16.6–254	0.1952
71 Cu + 29 Zn	0–100	0.1906	Limestone	25–100	0.09	Vulcanite	0–18	0.6360
Bronze:			Magnesium	12–39	0.238	Wedgwood ware	0–100	0.0890
3 Cu + 1 Sn	16.6–100	0.1844	Manganin		0.181	Wood:		
3 Cu + 1 Sn	16.6–350	0.2116	Marble	15–100	0.117	Parallel to fiber:		
3 Cu + 1 Sn	16.6–957	0.1737	Monel metal	25–100	0.14	Ash	0–100	0.0951
86.3 Cu + 9.7 Sn + 4 Zn	40	0.1782		25–600	0.16	Beech	2.34	0.0257
97.6 Cu + [hard	0–80	0.1713	Paraffin	0–16	1.0662	Chestnut	2.34	0.0649
2.2 Sn + [soft	0–80	0.1708	Paraffin	16–38	1.3030	Elm	2.34	0.0565
0.2 P			Paraffin	38–49	4.7707	Mahogany	2.34	0.0361
Caoutchouc		0.657–0.686	Platinum-iridium, 10 Pt + 1 Ir	40	0.0884	Maple	2.34	0.0638
Caoutchouc	16.7–25.3	0.770	Platinum-silver, 1 Pt + 2 Ag	0–100	0.1523	Oak	2.34	0.0492
Celluloid	20–70	1.00	Porcelain	20–790	0.0413	Pine	2.34	0.0541
Constantan	4–29	0.1523	Porcelain Bayeux	1000–1400	0.0553	Walnut	2.34	0.0658
Duralumin, 94Al	20–100	0.23	Quartz:			Across the fiber:		
	20–300	0.25	Parallel to axis	0–80	0.0797	Beech	2.34	0.614
Ebonite	25.3–35.4	0.842	Parallel to axis	–190 to + 16	0.0521	Chestnut	2.34	0.325
Fluorspar, CaF ₂	0–100	0.1950	Perpend. to axis	0–80	0.1337	Elm	2.34	0.443
German silver	0–100	0.1836	Quartz glass	–190 to + 16	–0.0026	Mahogany	2.34	0.404
Gold-platinum, 2 Au + 1 Pt	0–100	0.1523	Quartz glass	16 to 500	0.0057	Maple	2.34	0.484
Gold-copper, 2 Au + 1 Cu	0–100	0.1552	Quartz glass	16 to 1000	0.0058	Oak	2.34	0.544
Glass:			Rock salt	40	0.4040	Pine	2.34	0.341
Tube	0–100	0.0833	Rubber, hard	0	0.691	Walnut	2.34	0.484
Tube	0–100	0.0828	Rubber, hard	–160	0.300	Wax white	10–26	2.300
Plate	0–100	0.0891	Speculum metal	0–100	0.1933	Wax white	26–31	3.120
Crown (mean)	0–100	0.0897	Steel, 0.14 C, 34.5 Ni	25–100	0.037	Wax white	31–43	4.860
Crown (mean)	50–60	0.0954		25–600	0.136	Wax white	43–57	15.227
Flint	50–60	0.0788						
Jena thermometer 16 ^{III} normal	0–100	0.081						

* *Smithsonian Tables*. For a more complete tabulation see Tables 143, 144. *Smithsonian Physical Tables*. 9th ed., 1954, also reprinted in *American Institute of Physics Handbook*, McGraw-Hill, New York, 1957; *Handbook of Chemistry and Physics*, 40th ed., pp. 2239–2245, Chemical Rubber Publishing Co. For data on many solids prior to 1926, see Gruneisen, *Handbuch der Physik*, vol. 10, pp. 1–52, 1926, translation available as N.A.S.A. RE 2-18-59W, 1959. For eight plastic solids below 300 K, see Scott, *Cryogenic Engineering*, p. 331, Van Nostrand, Princeton, NJ, 1959. For 11 other materials to 300 K, see Scott, *loc. cit.*, p. 333. For quartz and silica, see Cook, *Brit. J. Appl. Phys.*, **7**, 285 (1956).

TABLE 2-137 Volume Expansion of Liquids*

If V_0 is the volume at 0° , then at t° the expansion formula is $V_t = V_0(1 + \alpha t + \beta t^2 + \gamma t^3)$. The table gives values of α , β , and γ , and of C , the true coefficient of volume expansion at 20° for some liquids and solutions. The temperature range of the observation is Δt . Values for the coefficient of volume expansion of liquids can be derived from the tables of specific volumes of the saturated liquid given as a function of temperature later in this section. $C = (dV/dt)/V_0$

Liquid	Range	$\alpha \times 10^3$	$\beta \times 10^6$	$\gamma \times 10^8$	$C \times 10^3$ at 20°
Acetic acid	16–107	1.0630	0.12636	1.0876	1.071
Acetone	0–54	1.3240	3.8090	–0.87983	1.487
Alcohol:					
Amyl	–15–80	0.9001	0.6573	1.18458	0.902
Ethyl, 30% by volume	18–39	0.2928	10.790	–11.87	
Ethyl, 50% by volume	0–39	0.7450	1.85	0.730	
Ethyl, 99.3% by volume					
27–46	1.012	2.20			1.12
Ethyl, 500 atm pressure					
0–40	0.866				
Ethyl, 3000 atm pressure					
0–40	0.524				
Methyl	0–61	1.1342	1.3635	0.8741	1.199
Benzene	11–81	1.17626	1.27776	0.80648	1.237
Bromine	0–59	1.06218	1.87714	–0.30854	1.132
Calcium chloride:					
5.8% solution	18–25	0.07878	4.2742		0.250
40.9% solution	17–24	0.42383	0.8571		0.458
Carbon disulfide	–34–60	1.13980	1.37065	1.91225	1.218
500 atm pressure	0–50	0.940			
3000 atm pressure	0–50	0.581			
Carbon tetrachloride	0–76	1.18384	0.89881	1.35135	1.236
Chloroform	0–63	1.10715	4.66473	–1.74328	1.273
Ether	–15–38	1.51324	2.35918	4.00512	1.656
Glycerin		0.4853	0.4895		0.505
Hydrochloric acid,					
33.2% solution	0–33	0.4460	0.215		0.455
Mercury	0–100	0.18182	0.0078		0.18186
Olive oil		0.6821	1.1405	–0.539	0.721
Pentane	0–33	1.4646	3.09319	1.6084	1.608
Potassium chloride,					
24.3% solution	16–25	0.2695	2.080		0.353
Phenol	36–157	0.8340	0.10732	0.4446	1.090
Petroleum, 0.8467 density	24–120	0.8994	1.396		0.955
Sodium chloride, 20.6% solution	0–29	0.3640	1.237		0.414
Sodium sulfate, 24% solution	11–40	0.3599	1.258		0.410
Sulfuric acid:					
10.9% solution	0–30	0.2835	2.580		0.387
100.0%	0–30	0.5758	–0.432		0.558
Turpentine	–9–106	0.9003	1.9595	–0.44998	0.973
Water	0–33	–0.06427	8.5053	–6.7900	0.207

* *Smithsonian Tables*, Table 269. For a detailed discussion of mercury data, see Cook, *Brit. J. Appl. Phys.*, **7**, 285 (1956). For data on nitrogen and argon, see Johnson (ed.), WADD-TR-60-56, 1960.

Bromoform¹ 7.7–50 °C.

$$V_t = 0.34204[1 + 0.00090411(t - 7.7) + 0.000006766(t - 7.7)^2]$$

0.34204 is the specific volume of bromoform at 7.7 °C.

Glycerin² –62 to 0 °C.

$$\bar{V}_t = V_0(1 + 4.83 \times 10^{-4}t - 0.49 \times 10^{-6}t^2)$$

0–80 °C.

$$V_t = V_0(1 + 4.83 \times 10^{-4}t + 0.49 \times 10^{-6}t^2)$$

Mercury³ 0–300 °C.

$$V_t - V_0[1 + 10^{-8}(18,153.8t + 0.7548t^2 + 0.001533t^3 + 0.00000536t^4)]$$

¹ Sherman and Sherman, *J. Am. Chem. Soc.*, **50**, 1119 (1928). (An obvious error in their equation has been corrected.)

² Samsoen, *Ann. phys.*, (10) **9**, 91 (1928).

³ Harlow, *Phil. Mag.*, (7) **7**, 674 (1929).

TABLE 2-138 Volume Expansion of Solids*

If v_2 and v_1 are the volumes at t_2 and t_1 , respectively, then $v_2 = v_1(1 + C\Delta t)$, C being the coefficient of cubical expansion and Δt the temperature interval. Where only a single temperature is stated, C represents the true coefficient of volume expansion at that temperature.

Substance	t or Δt	$C \times 10^4$
Antimony	0–100	0.3167
Beryl	0–100	0.0105
Bismuth	0–100	0.3948
Copper†	0–100	0.4998
Diamond	40	0.0354
Emerald	40	0.0168
Galena	0–100	0.558
Glass, common tube	0–100	0.276
hard	0–100	0.214
Jena, borosilicate 59 III	20–100	0.156
pure silica	0–80	0.0129
Gold	0–100	0.4411
Ice	–20 to –1	1.1250
Iron	0–100	0.3550
Lead†	0–100	0.8399
Paraffin	20	5.88
Platinum	0–100	0.265
Porcelain, Berlin	20	0.0814
chloride	0–100	1.094
nitrate	0–100	1.967
sulfate	20	1.0754
Quartz	0–100	0.3840
Rock salt	50–60	1.2120
Rubber	20	4.87
Silver	0–100	0.5831
Sodium	20	2.13
Stearic acid	33.8–45.4	8.1
Sulfur, native	13.2–50.3	2.23
Tin	0–100	0.6889
Zinc†	0–100	0.8928

* *Smithsonian Tables*, Table 268.

† See additional data below.

Aluminum¹ 100–530 °C.

$$V = V_0(1 + 2.16 \times 10^{-5}t + 0.95 \times 10^{-8}t^2)$$

Cadmium¹ 130–270 °C.

$$V = V_0(1 + 8.04 \times 10^{-5}t + 5.9 \times 10^{-8}t^2)$$

Copper¹ 110–300 °C.

$$V = V_0(1 + 1.62 \times 10^{-5}t + 0.20 \times 10^{-8}t^2)$$

Colophony² 0–34 °C.

$$V = V_0(1 + 2.21 \times 10^{-4}t + 0.31 \times 10^{-6}t^2)$$

34–150 °C.

$$V = V_{34}[1 + 7.40 \times 10^{-4}(t - 34) + 5.91 \times 10^{-6}(t - 34)^2]$$

Lead¹ 100–280 °C.

$$V = V_0(1 + 1.60 \times 10^{-5}t + 3.2 \times 10^{-8}t^2)$$

Shellac² 0–46 °C.

$$V = V_0(1 + 2.73 \times 10^{-4}t + 0.39 \times 10^{-6}t^2)$$

46–100 °C.

$$V = V_{46}[1 + 13.10 \times 10^{-4}(t - 46) + 0.62 \times 10^{-6}(t - 46)^2]$$

Silica (vitreous)³ 0–300 °C.

$$V_t = V_0[1 + 10^{-8}(93.6t + 0.7776t^2 - 0.003315t^3 + 0.000005244t^4)]$$

Sugar (cane, amorphous)² 0–67 °C.

$$V_t = V_0(1 + 2.34 \times 10^{-4}t + 0.14 \times 10^{-6}t^2)$$

67–160 °C.

$$V_t = V_{67}[1 + 5.02 \times 10^{-4}(t - 67) + 0.43 \times 10^{-6}(t - 67)^2]$$

Zinc¹ 120–360 °C.

$$V_t = V_0(1 + 8.50 \times 10^{-5}t + 3.9 \times 10^{-8}t^2)$$

¹ Uffelman, *Phil. Mag.*, (7) **10**, 633 (1930).

² Samsoen, *Ann. phys.*, (10) **9**, 83 (1928).

³ Harlow, *Phil. Mag.*, (7) **7**, 674 (1929).

JOULE-THOMSON EFFECT

UNITS CONVERSIONS

Joule-Thomson coefficients for substances listed in Table 2-184 are given in tables in the Thermodynamic Properties section.

For this subsection, the following units conversions are applicable:

To convert the Joule-Thomson coefficient, μ , in degrees Celsius per atmosphere to degrees Fahrenheit per atmosphere, multiply by 1.8.

$$^{\circ}\text{F} = \% ^{\circ}\text{C} + 32; ^{\circ}\text{R} = \% \text{K}$$

To convert bars to pounds-force per square inch, multiply by 14.504; to convert bars to kilopascals, multiply by 1×10^2 .

TABLE 2-139 Additional References Available for the Joule-Thomson Coefficient

Gas	Pressure range, atm				Temp. range, $^{\circ}\text{C}$			Other references
	0–10	10–50	50–200	>200	<0	0–300	>300	
Air	12, 15, 16 19, 35	12, 15, 19 35	15, 19, 35		19, 35	12, 15, 16 19, 35		3, 4, 18
Ammonia	28					28		2, 3
Argon	39	39	39		39	39		
Benzene	31	31	31			31	31	
Butane	26	26				26		
Carbon dioxide	7, 8, 28 37	7, 8, 37	7, 8, 37		7, 8, 37	7, 8, 9, 10 37		
Carbon monoxide	17	17			17	17		
Deuterium		22, 24, 25 1°	1, ° 22, 24 25		1, ° 22, 24, 25			
Dowtherm A	46	46				46	46	
Ethane	45	45				45		
Ethylene						9, 10		
Helium	1, 38	1, 38	38		1, 38	38		48
Hydrogen	24, 30	22, 24, 25 30	24, 30		22, 24, 25 30	24		
Methane		6	6			6		
Mixtures						9, 11		
Natural gas			33	33	33	33		
Nitrogen	13, 28, 40	13, 40	13, 40	13	13, 40	9, 10, 13 28, 40 9, 10	13	19
Nitrous oxide						26, 34, 44		
Pentane	26, 34, 44	34	34			43		
Propane	41	43				28, 29, 42		
Steam	28, 29, 42	29, 42, 47	42, 47			45	29, 42, 47	29, 47

*See also 14 (generalized chart); 18 (review, to 1919); 20–22; 23 (review, to 1948); 27 (review, to 1905); 32, 36, 41, 50.

REFERENCES: 1. Baehr, Z. *Elektrochem.*, **60**, 515 (1956). 2. Beattie, J. *Math. Phys.*, 9, 11 (1930). 3. Beattie, *Phys. Rev.*, **35**, 643 (1930). 4. Bradley and Hale, *Phys. Rev.*, **29**, 258 (1909). 5. Brown and Dean, *Bur. Stand. J. Res.*, **60**, 161 (1958). 6. Budenholzer, Sage, et al., *Ind. Eng. Chem.*, **29**, 658 (1937). 7. Burnett, *Phys. Rev.*, **22**, 590 (1923). 8. Burnett, Univ. Wisconsin Bull. 9(6), 1926. 9. Charnley, Ph.D. thesis, University of Manchester, 1952. 10. Charnley, Isles, et al., *Proc. R. Soc. (London)*, **A217**, 133 (1953). 11. Charnley, Rowlinson, et al., *Proc. R. Soc. (London)*, **A230**, 354 (1955). 12. Dalton, *Commun. Phys. Lab. Univ. Leiden*, no. 109c, 1909. 13. Deming and Deming, *Phys. Rev.*, **48**, 448 (1935). 14. Edmister, *Pet. Refiner*, **28**, 128 (1949). 15. Eucken, Clusius, et al., *Z. Tech. Phys.*, **13**, 267 (1932). 16. Eumorfopoulos and Rai, *Phil. Mag.*, **7**, 961 (1926). 17. Huang, Lin, et al., *Z. Phys.*, **100**, 594 (1936). 18. Hoxton, *Phys. Rev.*, **13**, 438 (1919). 19. Ishkin and Kaganov, *J. Tech. Phys. U.S.S.R.*, **26**, 2323 (1956). 20. Isles, Ph.D. thesis, Leeds University. 21. Jenkin and Pye, *Phil. Trans. R. Soc. (London)*, **A213**, 67 (1914); **A215**, 353 (1915). 22. Johnston, *J. Am. Chem. Soc.*, **68**, 2362 (1946). 23. Johnston, *Trans. Am. Soc. Mech. Eng.*, **70**, 651 (1948). 24. Johnston, Bezman, et al., *J. Am. Chem. Soc.*, **68**, 2367 (1946). 25. Johnston, Swanson, et al., *J. Am. Chem. Soc.*, **68**, 2373 (1946). 26. Kennedy, Sage, et al., *Ind. Eng. Chem.*, **28**, 718 (1936). 27. Kester, *Phys. Rev.*, **21**, 260 (1905). 28. Keyes and Collins, *Proc. Nat. Acad. Sci.*, **18**, 328 (1932). 29. Kleinschmidt, *Mech. Eng.*, **45**, 165 (1923); **48**, 155 (1926). 30. Koepppe, *Kältetechnik*, **8**, 275 (1956). 31. Lindsay and Brown, *Ind. Eng. Chem.*, **27**, 817 (1935). 32. Noell, dissertation, Munich, 1914, *Forschungsdienst*, 184, p. 1, 1916. 33. Palienko, *Tr. Inst. Ispol' z. Gaza, Akad. Nauk Ukr. SSR*, no. 4, p. 87, 1956. 34. Pattee and Brown, *Ind. Eng. Chem.*, **26**, 511, (1934). 35. Roebuck, *Proc. Am. Acad. Arts Sci.*, **60**, 537 (1925); **64**, 287 (1930). 36. Roebuck, see 49 below. 37. Roebuck and Murrell, *Phys. Rev.*, **55**, 240 (1939). 38. Roebuck and Osterberg, *Phys. Rev.*, **37**, 110 (1931); **43**, 60 (1933). 39. Roebuck and Osterberg, *Phys. Rev.*, **46**, 785 (1934). 40. Roebuck and Osterberg, *Phys. Rev.*, **48**, 450 (1935). 41. Roebuck, Murrell, et al., *J. Am. Chem. Soc.*, **64**, 400 (1942). 42. Sage, unpublished data, California Institute of Technology, 1959. 43. Sage and Lacy, *Ind. Eng. Chem.*, **27**, 1484 (1934). 44. Sage, Kennedy, et al., *Ind. Eng. Chem.*, **28**, 601 (1936). 45. Sage, Webster, et al., *Ind. Eng. Chem.*, **29**, 658 (1937). 46. Ullock, Gaffert, et al., *Trans. Am. Inst. Chem. Eng.*, **32**, 73 (1936). 47. Yang, *Ind. Eng. Chem.*, **45**, 786 (1953). 48. Zelmanov, *J. Phys. U.S.S.R.*, **3**, 43 (1940). 49. Roebuck, recalculated data. 50. Michels et al., van der Waals laboratory publications. Gunn, Cheuh, and Prausnitz, *Cryogenics*, **6**, 324 (1966), review equations relating the inversion temperatures and pressures. The ability of various equations of state to relate these was also discussed by Miller, *Ind. Eng. Chem. Fundam.*, **9**, 585 (1970); and Juris and Wenzel, *Am. Inst. Chem. Eng. J.*, **18**, 684 (1972). Perhaps the most detailed review is that of Hendricks, Peller, and Baron. NASA Tech. Note D 6807, 1972.

TABLE 2-140 Approximate Inversion-Curve Locus in Reduced Coordinates ($T_r = T/T_c$; $P_r = P/P_c$)^a

P_r	0	0.5	1	1.5	2	2.5	3	4
T_{rL}	0.782	0.800	0.818	0.838	0.859	0.880	0.903	0.953
T_{rU}	4.984	4.916	4.847	4.777	4.706	4.633	4.550	4.401
P_r	5	6	7	8	9	10	11	11.79
T_{rL}	1.01	1.08	1.16	1.25	1.35	1.50	1.73	2.24
T_{rU}	4.23	4.06	3.88	3.68	3.45	3.18	2.86	2.24

^aCalculated from the best three-constant equation recommended by Miller, *Ind. Eng. Chem. Fundam.*, **9**, 585 (1970). T_{rL} refers to the lower curve, and T_{rU} , to the upper curve.

CRITICAL CONSTANTS

ADDITIONAL REFERENCES

Other data and estimation techniques for the elements are contained in Gates and Thodos, *Am. Inst. Chem. Eng. J.*, **6** (1960):50-54; and Ohse and von Tippelskirch, *High Temperatures—High Pressures*, **9** (1977):367-385. For inorganic substances see Mathews, *Chem. Rev.*,

72 (1972):71-100; for organics see Kudchaker, Alani, and Zwolinski, *Chem. Rev.*, **68** (1968):659-735; and for fluorocarbons see *Advances in Fluorine Chemistry*, App. B, Butterworth, Washington, 1963, pp. 173-175. Pages 6-49 and 6-50 of the 84th edition of the *Handbook of Chemistry and Physics* provide an excellent list of references for critical properties.

TABLE 2-141 Critical Constants and Acentric Factors of Inorganic and Organic Compounds

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	T_c , K	P_c , MPa	V_c , m ³ /kmol	Z_c	Acentric factor
1	Acetaldehyde	C ₂ H ₄ O	75-07-0	44.053	466	5.55	0.154	0.221	0.2907
2	Acetamide	C ₂ H ₅ NO	60-35-5	59.067	761	6.6	0.215	0.224	0.4210
3	Acetic acid	C ₂ H ₄ O ₂	64-19-7	60.052	591.95	5.786	0.177	0.208	0.4665
4	Acetic anhydride	C ₄ H ₆ O ₃	108-24-7	102.089	606	4	0.29	0.23	0.4535
5	Acetone	C ₃ H ₆ O	67-64-1	58.079	508.2	4.701	0.209	0.233	0.3065
6	Acetonitrile	C ₂ H ₃ N	75-05-8	41.052	545.5	4.83	0.173	0.184	0.3379
7	Acetylene	C ₂ H ₂	74-86-2	26.037	308.3	6.138	0.112	0.268	0.1912
8	Acrolein	C ₃ H ₄ O	107-02-8	56.063	506	5	0.197	0.234	0.3198
9	Acrylic acid	C ₃ H ₄ O ₂	79-10-7	72.063	615	5.66	0.208	0.23	0.5383
10	Acrylonitrile	C ₃ H ₃ N	107-13-1	53.063	535	4.48	0.212	0.214	0.3498
11	Air	Mixture	132259-10-0	28.960	132.45	3.774	0.09147	0.313	
12	Ammonia	H ₃ N	7664-41-7	17.031	405.65	11.28	0.07247	0.242	0.2526
13	Anisole	C ₇ H ₈ O	100-66-3	108.138	645.6	4.25	0.337	0.267	0.3502
14	Argon	Ar	7440-37-1	39.948	150.86	4.898	0.07459	0.291	0.0000
15	Benzamide	C ₇ H ₇ NO	55-21-0	121.137	824	5.05	0.346	0.255	0.5585
16	Benzene	C ₆ H ₆	71-43-2	78.112	562.05	4.895	0.256	0.268	0.2103
17	Benzenethiol	C ₆ H ₆ S	108-98-5	110.177	689	4.74	0.315	0.261	0.2628
18	Benzoic acid	C ₇ H ₆ O ₂	65-85-0	122.121	751	4.47	0.344	0.246	0.6028
19	Benzonitrile	C ₇ H ₅ N	100-47-0	103.121	699.35	4.215	0.3132	0.227	0.3662
20	Benzophenone	C ₁₃ H ₁₀ O	119-61-9	182.218	830	3.352	0.5677	0.276	0.5019
21	Benzyl alcohol	C ₇ H ₈ O	100-51-6	108.138	720.15	4.374	0.382	0.279	0.3631
22	Benzyl ethyl ether	C ₉ H ₁₂ O	539-30-0	136.191	662	3.11	0.442	0.25	0.4332
23	Benzyl mercaptan	C ₇ H ₈ S	100-53-8	124.203	718	4.06	0.367	0.25	0.3126
24	Biphenyl	C ₁₂ H ₁₀	92-52-4	154.208	773	3.38	0.497	0.261	0.4029
25	Bromine	Br ₂	7726-95-6	159.808	584.15	10.3	0.135	0.286	0.1290
26	Bromobenzene	C ₆ H ₅ Br	108-86-1	157.008	670.15	4.519	0.324	0.263	0.2506
27	Bromoethane	C ₂ H ₅ Br	74-96-4	108.965	503.8	6.23	0.215	0.32	0.2548
28	Bromomethane	CH ₃ Br	74-83-9	94.939	467	8	0.156	0.321	0.1922
29	1,2-Butadiene	C ₄ H ₆	590-19-2	54.090	452	4.36	0.22	0.255	0.1659
30	1,3-Butadiene	C ₄ H ₆	106-99-0	54.090	425	4.32	0.221	0.27	0.1950
31	Butane	C ₄ H ₁₀	106-97-8	58.122	425.12	3.796	0.255	0.274	0.2002
32	1,2-Butanediol	C ₄ H ₁₀ O ₂	584-03-2	90.121	680	5.21	0.303	0.279	0.6305
33	1,3-Butanediol	C ₄ H ₁₀ O ₂	107-88-0	90.121	676	4.02	0.305	0.218	0.7043
34	1-Butanol	C ₄ H ₁₀ O	71-36-3	74.122	563.1	4.414	0.273	0.258	0.5883
35	2-Butanol	C ₄ H ₁₀ O	78-92-2	74.122	535.9	4.188	0.27	0.254	0.5692
36	1-Butene	C ₄ H ₈	106-98-9	56.106	419.5	4.02	0.241	0.278	0.1845
37	cis-2-Butene	C ₄ H ₈	590-18-1	56.106	435.5	4.21	0.234	0.272	0.2019
38	trans-2-Butene	C ₄ H ₈	624-64-6	56.106	428.6	4.1	0.238	0.274	0.2176
39	Butyl acetate	C ₆ H ₁₂ O ₂	123-86-4	116.158	575.4	3.09	0.389	0.251	0.4394
40	Butylbenzene	C ₁₀ H ₁₄	104-51-8	134.218	660.5	2.89	0.497	0.261	0.3941
41	Butyl mercaptan	C ₄ H ₁₀ S	109-79-5	90.187	570.1	3.97	0.307	0.257	0.2714
42	sec-Butyl mercaptan	C ₄ H ₁₀ S	513-53-1	90.187	554	4.06	0.307	0.271	0.2506
43	1-Butyne	C ₄ H ₆	107-00-6	54.090	440	4.6	0.208	0.262	0.2470
44	Butyraldehyde	C ₄ H ₈ O	123-72-8	72.106	537.2	4.32	0.258	0.25	0.2774
45	Butyric acid	C ₄ H ₈ O ₂	107-92-6	88.105	615.7	4.06	0.293	0.232	0.6805
46	Butyronitrile	C ₄ H ₇ N	109-74-0	69.105	582.25	3.79	0.278	0.218	0.3714
47	Carbon dioxide	CO ₂	124-38-9	44.010	304.21	7.383	0.094	0.274	0.2236
48	Carbon disulfide	CS ₂	75-15-0	76.141	552	7.9	0.16	0.275	0.1107
49	Carbon monoxide	CO	630-08-0	28.010	132.92	3.499	0.0944	0.299	0.0482
50	Carbon tetrachloride	CCl ₄	56-23-5	153.823	556.35	4.56	0.276	0.272	0.1926
51	Carbon tetrafluoride	CF ₄	75-73-0	88.004	227.51	3.745	0.143	0.283	0.1790
52	Chlorine	Cl ₂	7782-50-5	70.906	417.15	7.71	0.124	0.276	0.0688
53	Chlorobenzene	C ₆ H ₅ Cl	108-90-7	112.557	632.35	4.519	0.308	0.265	0.2499
54	Chloroethane	C ₂ H ₅ Cl	75-00-3	64.514	460.35	5.27	0.2	0.275	0.1902
55	Chloroform	CHCl ₃	67-66-3	119.378	536.4	5.472	0.239	0.293	0.2219
56	Chloromethane	CH ₃ Cl	74-87-3	50.488	416.25	6.68	0.143	0.276	0.1531
57	1-Chloropropane	C ₃ H ₇ Cl	540-54-5	78.541	503.15	4.58	0.247	0.27	0.2277
58	2-Chloropropane	C ₃ H ₇ Cl	75-29-6	78.541	489	4.54	0.247	0.276	0.1986
59	m-Cresol	C ₇ H ₈ O	108-39-4	108.138	705.85	4.56	0.312	0.242	0.4480
60	o-Cresol	C ₇ H ₈ O	95-48-7	108.138	697.55	5.01	0.282	0.244	0.4339
61	p-Cresol	C ₇ H ₈ O	106-44-5	108.138	704.65	5.15	0.277	0.244	0.5072
62	Cumene	C ₉ H ₁₂	98-82-8	120.192	631	3.209	0.434	0.265	0.3274
63	Cyanogen	C ₂ N ₂	460-19-5	52.035	400.15	5.98	0.195	0.351	0.2790
64	Cyclobutane	C ₄ H ₈	287-23-0	56.106	459.93	4.98	0.21	0.273	0.1847

TABLE 2-141 Critical Constants and Acentric Factors of Inorganic and Organic Compounds (Continued)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	T_c , K	P_c , MPa	V_c , m ³ /kmol	Z_c	Acentric factor
65	Cyclohexane	C ₆ H ₁₂	110-82-7	84.159	553.8	4.08	0.308	0.273	0.2081
66	Cyclohexanol	C ₆ H ₁₂ O	108-93-0	100.159	650.1	4.26	0.322	0.254	0.3690
67	Cyclohexanone	C ₆ H ₁₀ O	108-94-1	98.143	653	4	0.311	0.229	0.2990
68	Cyclohexene	C ₆ H ₁₀	110-83-8	82.144	560.4	4.35	0.291	0.272	0.2123
69	Cyclopentane	C ₅ H ₁₀	287-92-3	70.133	511.7	4.51	0.26	0.276	0.1949
70	Cyclopentene	C ₅ H ₈	142-29-0	68.117	507	4.8	0.245	0.279	0.1961
71	Cyclopropane	C ₃ H ₆	75-19-4	42.080	398	5.54	0.162	0.271	0.1278
72	Cyclohexyl mercaptan	C ₆ H ₁₂ S	1569-69-3	116.224	664	3.97	0.355	0.255	0.2641
73	Decanal	C ₁₀ H ₂₀ O	112-31-2	156.265	674.2	2.6	0.58	0.269	0.5820
74	Decane	C ₁₀ H ₂₂	124-18-5	142.282	617.7	2.11	0.617	0.254	0.4923
75	Decanoic acid	C ₁₀ H ₂₀ O ₂	334-48-5	172.265	722.1	2.28	0.639	0.243	0.8126
76	1-Decanol	C ₁₀ H ₂₂ O	112-30-1	158.281	688	2.308	0.645	0.26	0.6070
77	1-Decene	C ₁₀ H ₂₀	872-05-9	140.266	616.6	2.223	0.584	0.253	0.4805
78	Decyl mercaptan	C ₁₀ H ₂₂ S	143-10-2	174.347	696	2.13	0.624	0.23	0.5874
79	1-Decyne	C ₁₀ H ₁₈	764-93-2	138.250	619.85	2.37	0.552	0.254	0.5178
80	Deuterium	D ₂	7782-39-0	4.032	38.35	1.6617	0.060263	0.314	-0.1449
81	1,1-Dibromoethane	C ₂ H ₄ Br ₂	557-91-5	187.861	628	6.03	0.276	0.319	0.1250
82	1,2-Dibromoethane	C ₂ H ₄ Br ₂	106-93-4	187.861	650.15	5.477	0.2616	0.265	0.2067
83	Dibromomethane	CH ₂ Br ₂	74-95-3	173.835	611	7.17	0.223	0.315	0.2095
84	Dibutyl ether	C ₈ H ₁₈ O	142-96-1	130.228	584.1	2.46	0.487	0.247	0.4476
85	<i>m</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	541-73-1	147.002	683.95	4.07	0.351	0.251	0.2790
86	<i>o</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	95-50-1	147.002	705	4.07	0.351	0.244	0.2192
87	<i>p</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	106-46-7	147.002	684.75	4.07	0.351	0.251	0.2846
88	1,1-Dichloroethane	C ₂ H ₄ Cl ₂	75-34-3	98.959	523	5.07	0.24	0.28	0.2339
89	1,2-Dichloroethane	C ₂ H ₄ Cl ₂	107-06-2	98.959	561.6	5.37	0.22	0.253	0.2866
90	Dichloromethane	CH ₂ Cl ₂	75-09-2	84.933	510	6.08	0.185	0.265	0.1986
91	1,1-Dichloropropane	C ₃ H ₆ Cl ₂	78-99-9	112.986	560	4.24	0.291	0.265	0.2529
92	1,2-Dichloropropane	C ₃ H ₆ Cl ₂	78-87-5	112.986	572	4.24	0.291	0.259	0.2564
93	Diethanol amine	C ₄ H ₁₁ NO ₂	111-42-2	105.136	736.6	4.27	0.349	0.243	0.9529
94	Diethyl amine	C ₄ H ₁₁ N	109-89-7	73.137	496.6	3.71	0.301	0.27	0.3039
95	Diethyl ether	C ₄ H ₁₀ O	60-29-7	74.122	466.7	3.64	0.28	0.263	0.2811
96	Diethyl sulfide	C ₄ H ₁₀ S	352-93-2	90.187	557.15	3.96	0.318	0.272	0.2900
97	1,1-Difluoroethane	C ₂ H ₄ F ₂	75-37-6	66.050	386.44	4.52	0.179	0.252	0.2751
98	1,2-Difluoroethane	C ₂ H ₄ F ₂	624-72-6	66.050	445	4.34	0.195	0.229	0.2224
99	Difluoromethane	CH ₂ F ₂	75-10-5	52.023	351.255	5.784	0.123	0.244	0.2771
100	Di-isopropyl amine	C ₆ H ₁₅ N	108-18-9	101.190	523.1	3.2	0.418	0.308	0.3883
101	Di-isopropyl ether	C ₆ H ₁₄ O	108-20-3	102.175	500.05	2.88	0.386	0.267	0.3387
102	Di-isopropyl ketone	C ₇ H ₁₄ O	565-80-0	114.185	576	3.02	0.416	0.262	0.4044
103	1,1-Dimethoxyethane	C ₄ H ₁₀ O ₂	534-15-6	90.121	507.8	3.773	0.297	0.265	0.3277
104	1,2-Dimethoxypropane	C ₅ H ₁₂ O ₂	7778-85-0	104.148	543	3.446	0.35	0.267	0.3522
105	Dimethyl acetylene	C ₄ H ₆	503-17-3	54.090	473.2	4.87	0.221	0.274	0.2385
106	Dimethyl amine	C ₂ H ₇ N	124-40-3	45.084	437.2	5.34	0.18	0.264	0.2999
107	2,3-Dimethylbutane	C ₆ H ₁₄	79-29-8	86.175	500	3.15	0.361	0.274	0.2493
108	1,1-Dimethylcyclohexane	C ₈ H ₁₆	590-66-9	112.213	591.15	2.938	0.45	0.269	0.2326
109	<i>cis</i> -1,2-Dimethylcyclohexane	C ₈ H ₁₆	2207-01-4	112.213	606.15	2.938	0.46	0.268	0.2324
110	<i>trans</i> -1,2-Dimethylcyclohexane	C ₈ H ₁₆	6876-23-9	112.213	596.15	2.938	0.46	0.273	0.2379
111	Dimethyl disulfide	C ₂ H ₆ S ₂	624-92-0	94.199	615	5.36	0.252	0.264	0.2059
112	Dimethyl ether	C ₂ H ₆ O	115-10-6	46.068	400.1	5.37	0.17	0.2744	0.2002
113	<i>N,N</i> -Dimethyl formamide	C ₃ H ₇ NO	68-12-2	73.094	649.6	4.42	0.26199	0.214	0.3177
114	2,3-Dimethylpentane	C ₇ H ₁₆	565-59-3	100.202	537.3	2.91	0.393	0.256	0.2964
115	Dimethyl phthalate	C ₁₀ H ₁₀ O ₄	131-11-3	194.184	766	2.78	0.53	0.231	0.6568
116	Dimethylsilane	C ₂ H ₆ Si	1111-74-6	60.170	402	3.56	0.258	0.275	0.1300
117	Dimethyl sulfide	C ₂ H ₆ S	75-18-3	62.134	503.04	5.53	0.201	0.266	0.1943
118	Dimethyl sulfoxide	C ₂ H ₆ OS	67-68-5	78.133	729	5.65	0.227	0.212	0.2806
119	Dimethyl terephthalate	C ₁₀ H ₁₀ O ₄	120-61-6	194.184	772	2.78	0.529	0.229	0.6371
120	1,4-Dioxane	C ₄ H ₈ O ₂	123-91-1	88.105	587	5.208	0.238	0.254	0.2793
121	Diphenyl ether	C ₁₂ H ₁₀ O	101-84-8	170.207	766.8	3.08	0.503	0.243	0.4389
122	Dipropyl amine	C ₆ H ₁₅ N	142-84-7	101.190	550	3.14	0.402	0.276	0.4497
123	Dodecane	C ₁₂ H ₂₆	112-40-3	170.335	658	1.82	0.755	0.251	0.5764
124	Eicosane	C ₂₀ H ₄₂	112-95-8	282.547	768	1.16	1.34	0.243	0.9069
125	Ethane	C ₂ H ₆	74-84-0	30.069	305.32	4.872	0.1455	0.279	0.0995
126	Ethanol	C ₂ H ₆ O	64-17-5	46.068	514	6.137	0.168	0.241	0.6436
127	Ethyl acetate	C ₄ H ₈ O ₂	141-78-6	88.105	523.3	3.88	0.286	0.255	0.3664
128	Ethyl amine	C ₂ H ₇ N	75-04-7	45.084	456.15	5.62	0.207	0.307	0.2848
129	Ethylbenzene	C ₈ H ₁₀	100-41-4	106.165	617.15	3.609	0.374	0.263	0.3035
130	Ethyl benzoate	C ₉ H ₁₀ O ₂	93-89-0	150.175	698	3.18	0.489	0.268	0.4771
131	2-Ethyl butanoic acid	C ₆ H ₁₂ O ₂	88-09-5	116.158	655	3.41	0.389	0.244	0.6326
132	Ethyl butyrate	C ₆ H ₁₂ O ₂	105-54-4	116.158	571	2.95	0.403	0.25	0.4011
133	Ethylcyclohexane	C ₈ H ₁₆	1678-91-7	112.213	609.15	3.04	0.43	0.258	0.2455
134	Ethylcyclopentane	C ₇ H ₁₄	1640-89-7	98.186	569.5	3.4	0.375	0.269	0.2701
135	Ethylene	C ₂ H ₄	74-85-1	28.053	282.34	5.041	0.131	0.281	0.0862
136	Ethylenediamine	C ₂ H ₈ N ₂	107-15-3	60.098	593	6.29	0.264	0.337	0.4724
137	Ethylene glycol	C ₂ H ₆ O ₂	107-21-1	62.068	720	8.2	0.191	0.262	0.5068
138	Ethyleneimine	C ₂ H ₅ N	151-56-4	43.068	537	6.85	0.173	0.265	0.2007
139	Ethylene oxide	C ₂ H ₄ O	75-21-8	44.053	469.15	7.19	0.140296	0.25876	0.1974
140	Ethyl formate	C ₃ H ₆ O ₂	109-94-4	74.079	508.4	4.74	0.229	0.257	0.2847

TABLE 2-141 Critical Constants and Acentric Factors of Inorganic and Organic Compounds (Continued)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	T_c , K	P_c , MPa	V_c , $m^3/kmol$	Z_c	Acentric factor
141	2-Ethyl hexanoic acid	$C_8H_{16}O_2$	149-57-5	144.211	674.6	2.778	0.528	0.262	0.8067
142	Ethylhexyl ether	$C_8H_{18}O$	5756-43-4	130.228	583	2.46	0.487	0.247	0.4944
143	Ethylisopropyl ether	$C_5H_{12}O$	625-54-7	88.148	489	3.41	0.329	0.276	0.3056
144	Ethylisopropyl ketone	$C_6H_{12}O$	565-69-5	100.159	567	3.32	0.369	0.26	0.3891
145	Ethyl mercaptan	C_2H_6S	75-08-1	62.134	499.15	5.49	0.207	0.274	0.1878
146	Ethyl propionate	$C_5H_{10}O_2$	105-37-3	102.132	546	3.362	0.345	0.256	0.3944
147	Ethylpropyl ether	$C_5H_{12}O$	628-32-0	88.148	500.23	3.37	0.339	0.275	0.3473
148	Ethyltrichlorosilane	$C_2H_5Cl_3Si$	115-21-9	163.506	559.95	3.33	0.414	0.296	0.2691
149	Fluorine	F_2	7782-41-4	37.997	144.12	5.172	0.066547	0.287	0.0530
150	Fluorobenzene	C_6H_5F	462-06-6	96.102	560.09	4.551	0.269	0.263	0.2472
151	Fluoroethane	C_2H_5F	353-36-6	48.060	375.31	5.028	0.164	0.264	0.2200
152	Fluoromethane	CH_3F	593-53-3	34.033	317.42	5.875	0.113	0.252	0.1980
153	Formaldehyde	CH_2O	50-00-0	30.026	408	6.59	0.115	0.223	0.2818
154	Formamide	CH_3NO	75-12-7	45.041	771	7.8	0.163	0.198	0.4124
155	Formic acid	CH_2O_2	64-18-6	46.026	588	5.81	0.125	0.149	0.3173
156	Furan	C_4H_4O	110-00-9	68.074	490.15	5.5	0.218	0.294	0.2015
157	Helium-4	He	7440-59-7	4.003	5.2	0.2275	0.0573	0.302	-0.3900
158	Heptadecane	$C_{17}H_{36}$	629-78-7	240.468	736	1.34	1.11	0.244	0.7697
159	Heptanal	$C_7H_{14}O$	111-71-7	114.185	616.8	3.16	0.434	0.267	0.4279
160	Heptane	C_7H_{16}	142-82-5	100.202	540.2	2.74	0.428	0.261	0.3495
161	Heptanoic acid	$C_7H_{14}O_2$	111-14-8	130.185	677.3	3.043	0.466	0.252	0.7564
162	1-Heptanol	$C_7H_{16}O$	111-70-6	116.201	632.3	3.085	0.444	0.261	0.5621
163	2-Heptanol	$C_7H_{16}O$	543-49-7	116.201	608.3	3.001	0.447	0.265	0.5628
164	3-Heptanone	$C_7H_{14}O$	106-35-4	114.185	606.6	2.92	0.433	0.251	0.4076
165	2-Heptanone	$C_7H_{14}O$	110-43-0	114.185	611.4	2.94	0.434	0.251	0.4190
166	1-Heptene	C_7H_{14}	592-76-7	98.186	537.4	2.92	0.402	0.263	0.3432
167	Heptyl mercaptan	$C_7H_{16}S$	1639-09-4	132.267	645	2.77	0.465	0.24	0.4226
168	1-Heptyne	C_7H_{12}	628-71-7	96.170	547	3.21	0.387	0.273	0.3778
169	Hexadecane	$C_{16}H_{34}$	544-76-3	226.441	723	1.4	1.04	0.243	0.7174
170	Hexanal	$C_6H_{12}O$	66-25-1	100.159	591	3.46	0.369	0.26	0.3872
171	Hexane	C_6H_{14}	110-54-3	86.175	507.6	3.025	0.371	0.266	0.3013
172	Hexanoic acid	$C_6H_{12}O_2$	142-62-1	116.158	660.2	3.308	0.408	0.246	0.7299
173	1-Hexanol	$C_6H_{14}O$	111-27-3	102.175	611.3	3.446	0.382	0.259	0.5586
174	2-Hexanol	$C_6H_{14}O$	626-93-7	102.175	585.3	3.311	0.385	0.262	0.5574
175	2-Hexanone	$C_6H_{12}O$	591-78-6	100.159	587.61	3.287	0.378	0.254	0.3846
176	3-Hexanone	$C_6H_{12}O$	589-38-8	100.159	582.82	3.32	0.378	0.259	0.3801
177	1-Hexene	C_6H_{12}	592-41-6	84.159	504	3.21	0.348	0.267	0.2888
178	3-Hexyne	C_6H_{10}	928-49-4	82.144	544	3.53	0.331	0.258	0.2183
179	Hexyl mercaptan	$C_6H_{14}S$	111-31-9	118.240	623	3.08	0.412	0.245	0.3681
180	1-Hexyne	C_6H_{10}	693-02-7	82.144	516.2	3.62	0.322	0.272	0.3327
181	2-Hexyne	C_6H_{10}	764-35-2	82.144	549	3.53	0.331	0.256	0.2214
182	Hydrazine	H_4N_2	302-01-2	32.045	653.15	14.7	0.158	0.428	0.3143
183	Hydrogen	H_2	1333-74-0	2.016	33.19	1.313	0.064147	0.305	-0.2160
184	Hydrogen bromide	HBr	10035-10-6	80.912	363.15	8.552	0.1	0.283	0.0734
185	Hydrogen chloride	HCl	7647-01-0	36.461	324.65	8.31	0.081	0.249	0.1315
186	Hydrogen cyanide	CHN	74-90-8	27.025	456.65	5.39	0.139	0.197	0.4099
187	Hydrogen fluoride	HF	7664-39-3	20.006	461.15	6.48	0.069	0.117	0.3823
188	Hydrogen sulfide	H_2S	7783-06-4	34.081	373.53	8.963	0.0985	0.284	0.0942
189	Isobutyric acid	$C_4H_8O_2$	79-31-2	88.105	605	3.7	0.292	0.215	0.6141
190	Isopropyl amine	C_3H_9N	75-31-0	59.110	471.85	4.54	0.221	0.256	0.2759
191	Malonic acid	$C_3H_4O_4$	141-82-2	104.061	805	5.64	0.258	0.217	0.9418
192	Methacrylic acid	$C_4H_6O_2$	79-41-4	86.089	662	4.79	0.28	0.244	0.3318
193	Methane	CH_4	74-82-8	16.042	190.564	4.599	0.0986	0.286	0.0115
194	Methanol	CH_4O	67-56-1	32.042	512.5	8.084	0.117	0.222	0.5658
195	N-Methyl acetamide	C_3H_7NO	79-16-3	73.094	718	4.98	0.267	0.223	0.4351
196	Methyl acetate	$C_3H_6O_2$	79-20-9	74.079	506.55	4.75	0.228	0.257	0.3313
197	Methyl acetylene	C_3H_4	74-99-7	40.064	402.4	5.63	0.164	0.276	0.2115
198	Methyl acrylate	$C_4H_6O_2$	96-33-3	86.089	536	4.25	0.27	0.258	0.3423
199	Methyl amine	CH_5N	74-89-5	31.057	430.05	7.46	0.154	0.321	0.2814
200	Methyl benzoate	$C_8H_8O_2$	93-58-3	136.148	693	3.59	0.436	0.272	0.4205
201	3-Methyl-1,2-butadiene	C_5H_8	598-25-4	68.117	490	3.83	0.291	0.274	0.1874
202	2-Methylbutane	C_5H_{12}	78-78-4	72.149	460.4	3.38	0.306	0.27	0.2279
203	2-Methylbutanoic acid	$C_5H_{10}O_2$	116-53-0	102.132	643	3.89	0.347	0.252	0.5894
204	3-Methyl-1-butanol	$C_5H_{12}O$	123-51-3	88.148	577.2	3.93	0.329	0.269	0.5939
205	2-Methyl-1-butene	C_5H_{10}	563-46-2	70.133	465	3.447	0.292	0.26	0.2341
206	2-Methyl-2-butene	C_5H_{10}	513-35-9	70.133	470	3.42	0.292	0.256	0.2870
207	2-Methyl-1-butene-3-yne	C_5H_8	78-80-8	66.101	492	4.38	0.248	0.266	0.1370
208	Methylbutyl ether	$C_5H_{12}O$	628-28-4	88.148	512.74	3.371	0.329	0.26	0.3130
209	Methylbutyl sulfide	$C_5H_{12}S$	628-29-5	104.214	593	3.47	0.36	0.253	0.3229
210	3-Methyl-1-butyne	C_5H_8	598-23-2	68.117	463.2	4.2	0.275	0.3	0.3081
211	Methyl butyrate	$C_5H_{10}O_2$	623-42-7	102.132	554.5	3.473	0.34	0.256	0.3775
212	Methylchlorosilane	CH_3ClSi	993-00-0	80.589	442	4.17	0.246	0.279	0.2252
213	Methylcyclohexane	C_7H_{14}	108-87-2	98.186	572.1	3.48	0.369	0.27	0.2361
214	1-Methylcyclohexanol	$C_7H_{14}O$	590-67-0	114.185	686	4	0.374	0.262	0.2213
215	cis-2-Methylcyclohexanol	$C_7H_{14}O$	7443-70-1	114.185	614	3.79	0.374	0.278	0.6805
216	trans-2-Methylcyclohexanol	$C_7H_{14}O$	7443-52-9	114.185	617	3.79	0.374	0.276	0.6790

TABLE 2-141 Critical Constants and Acentric Factors of Inorganic and Organic Compounds (Continued)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	T_c , K	P_c , MPa	V_c , m ³ /kmol	Z_c	Acentric factor
217	Methylcyclopentane	C ₆ H ₁₂	96-37-7	84.159	532.7	3.79	0.319	0.273	0.2288
218	1-Methylcyclopentene	C ₆ H ₁₀	693-89-0	82.144	542	4.13	0.303	0.278	0.2318
219	3-Methylcyclopentene	C ₆ H ₁₀	1120-62-3	82.144	526	4.13	0.303	0.286	0.2296
220	Methyldichlorosilane	CH ₃ Cl ₂ Si	75-54-7	115.034	483	3.95	0.289	0.284	0.2758
221	Methylethyl ether	C ₃ H ₈ O	540-67-0	60.095	437.8	4.4	0.221	0.267	0.2314
222	Methylethyl ketone	C ₄ H ₈ O	78-93-3	72.106	535.5	4.15	0.267	0.249	0.3234
223	Methylethyl sulfide	C ₃ H ₈ S	624-89-5	76.161	533	4.26	0.254	0.244	0.2091
224	Methyl formate	C ₂ H ₄ O ₂	107-31-3	60.052	487.2	6	0.172	0.255	0.2556
225	Methylisobutyl ether	C ₅ H ₁₂ O	625-44-5	88.148	497	3.41	0.329	0.272	0.3078
226	Methylisobutyl ketone	C ₆ H ₁₂ O	108-10-1	100.159	574.6	3.27	0.369	0.253	0.3557
227	Methyl isocyanate	C ₂ H ₃ NO	624-83-9	57.051	488	5.48	0.202	0.273	0.3007
228	Methylisopropyl ether	C ₄ H ₁₀ O	598-53-8	74.122	464.48	3.762	0.276	0.269	0.2656
229	Methylisopropyl ketone	C ₅ H ₁₀ O	563-80-4	86.132	553.4	3.8	0.31	0.256	0.3208
230	Methylisopropyl sulfide	C ₄ H ₁₀ S	1551-21-9	90.187	553.1	4.021	0.328	0.28718	0.2461
231	Methyl mercaptan	CH ₃ S	74-93-1	48.107	469.95	7.23	0.145	0.268	0.1582
232	Methyl methacrylate	C ₅ H ₈ O ₂	80-62-6	100.116	566	3.68	0.323	0.253	0.2802
233	2-Methyloctanoic acid	C ₉ H ₁₈ O ₂	3004-93-1	158.238	694	2.54	0.572	0.252	0.7913
234	2-Methylpentane	C ₆ H ₁₄	107-83-5	86.175	497.7	3.04	0.368	0.27	0.2791
235	Methyl pentyl ether	C ₆ H ₁₄ O	628-80-8	102.175	546.49	3.042	0.38	0.254	0.3442
236	2-Methylpropane	C ₄ H ₁₀	75-28-5	58.122	407.8	3.64	0.259	0.278	0.1835
237	2-Methyl-2-propanol	C ₄ H ₁₀ O	75-65-0	74.122	506.2	3.972	0.275	0.26	0.6152
238	2-Methyl propene	C ₄ H ₈	115-11-7	56.106	417.9	4	0.239	0.275	0.1948
239	Methyl propionate	C ₄ H ₈ O ₂	554-12-1	88.105	530.6	4.004	0.282	0.256	0.3466
240	Methylpropyl ether	C ₄ H ₁₀ O	557-17-5	74.122	476.25	3.801	0.276	0.265	0.2770
241	Methylpropyl sulfide	C ₄ H ₁₀ S	3877-15-4	90.187	565	3.97	0.307	0.259	0.2737
242	Methylsilane	CH ₃ Si	992-94-9	46.144	352.5	4.7	0.205	0.329	0.1314
243	alpha-Methyl styrene	C ₉ H ₁₀	98-83-9	118.176	654	3.36	0.399	0.247	0.3230
244	Methyl <i>tert</i> -butyl ether	C ₅ H ₁₂ O	1634-04-4	88.148	497.1	3.287	0.314	0.25	0.2466
245	Methyl vinyl ether	C ₃ H ₆ O	107-25-5	58.079	437	4.67	0.21	0.27	0.2416
246	Naphthalene	C ₁₀ H ₈	91-20-3	128.171	748.4	4.05	0.407	0.265	0.3020
247	Neon	Ne	7440-01-9	20.180	44.4	2.653	0.0417	0.3	-0.0396
248	Nitroethane	C ₂ H ₅ NO ₂	79-24-3	75.067	593	5.16	0.236	0.247	0.3803
249	Nitrogen	N ₂	7727-37-9	28.013	126.2	3.4	0.08921	0.289	0.0377
250	Nitrogen trifluoride	F ₃ N	7783-54-2	71.002	234	4.461	0.11875	0.272	0.1200
251	Nitromethane	CH ₃ NO ₂	75-52-5	61.040	588.15	6.31	0.173	0.223	0.3480
252	Nitrous oxide	N ₂ O	10024-97-2	44.013	309.57	7.245	0.0974	0.274	0.1409
253	Nitric oxide	NO	10102-43-9	30.006	180.15	6.48	0.058	0.251	0.5829
254	Nonadecane	C ₁₉ H ₄₀	629-92-5	268.521	758	1.21	1.26	0.242	0.8522
255	Nonanal	C ₉ H ₁₈ O	124-19-6	142.239	658	2.73	0.527	0.263	0.5117
256	Nonane	C ₉ H ₂₀	111-84-2	128.255	594.6	2.29	0.551	0.255	0.4435
257	Nonanoic acid	C ₉ H ₁₈ O ₂	112-05-0	158.238	710.7	2.514	0.584	0.248	0.7724
258	1-Nonanol	C ₉ H ₂₀ O	143-08-8	144.255	670.9	2.527	0.576	0.261	0.5841
259	2-Nonanol	C ₉ H ₂₀ O	628-99-9	144.255	649.5	2.541	0.577	0.271	0.5911
260	1-Nonene	C ₉ H ₁₈	124-11-8	126.239	593.1	2.428	0.524	0.258	0.4367
261	Nonyl mercaptan	C ₉ H ₂₀ S	1455-21-6	160.320	681	2.31	0.571	0.233	0.5260
262	1-Nonyne	C ₉ H ₁₆	3452-09-3	124.223	598.05	2.61	0.497	0.261	0.4710
263	Octadecane	C ₁₈ H ₃₈	593-45-3	254.494	747	1.27	1.19	0.243	0.8114
264	Octanal	C ₈ H ₁₆ O	124-13-0	128.212	638.9	2.96	0.488	0.272	0.4636
265	Octane	C ₈ H ₁₈	111-65-9	114.229	568.7	2.49	0.486	0.256	0.3996
266	Octanoic acid	C ₈ H ₁₆ O ₂	124-07-2	144.211	694.26	2.779	0.523	0.252	0.7706
267	1-Octanol	C ₈ H ₁₈ O	111-87-5	130.228	652.3	2.783	0.509	0.261	0.5697
268	2-Octanol	C ₈ H ₁₈ O	123-96-6	130.228	629.8	2.749	0.512	0.269	0.5807
269	2-Octanone	C ₈ H ₁₆ O	111-13-7	128.212	632.7	2.64	0.497	0.249	0.4549
270	3-Octanone	C ₈ H ₁₆ O	106-68-3	128.212	627.7	2.704	0.497	0.257	0.4406
271	1-Octene	C ₈ H ₁₆	111-66-0	112.213	566.9	2.663	0.464	0.262	0.3921
272	Octyl mercaptan	C ₈ H ₁₈ S	111-88-6	146.294	667.3	2.52	0.518	0.235	0.4497
273	1-Octyne	C ₈ H ₁₄	629-05-0	110.197	574	2.88	0.442	0.267	0.4233
274	Oxalic acid	C ₂ H ₂ O ₄	144-62-7	90.035	804	7.02	0.205	0.215	0.9176
275	Oxygen	O ₂	7782-44-7	31.999	154.58	5.043	0.0734	0.288	0.0222
276	Ozone	O ₃	10028-15-6	47.998	261	5.57	0.089	0.228	0.2119
277	Pentadecane	C ₁₅ H ₃₂	629-62-9	212.415	708	1.48	0.969	0.244	0.6863
278	Pentanal	C ₅ H ₁₀ O	110-62-3	86.132	566.1	3.97	0.313	0.264	0.3472
279	Pentane	C ₅ H ₁₂	109-66-0	72.149	469.7	3.37	0.313	0.27	0.2515
280	Pentanoic acid	C ₅ H ₁₀ O ₂	109-52-4	102.132	639.16	3.63	0.35	0.239	0.7052
281	1-Pentanol	C ₅ H ₁₂ O	71-41-0	88.148	588.1	3.897	0.326	0.258	0.5748
282	2-Pentanol	C ₅ H ₁₂ O	6032-29-7	88.148	561	3.7	0.326	0.259	0.5549
283	3-Pentanone	C ₅ H ₁₀ O	107-87-9	86.132	561.08	3.694	0.301	0.238	0.3433
284	2-Pentanone	C ₅ H ₁₀ O	96-22-0	86.132	560.95	3.74	0.336	0.269	0.3448
285	1-Pentene	C ₅ H ₁₀	109-67-1	70.133	464.8	3.56	0.293	0.27	0.2372
286	2-Pentyl mercaptan	C ₅ H ₁₂ S	2084-19-7	104.214	584.3	3.536	0.385	0.28	0.2685
287	Pentyl mercaptan	C ₅ H ₁₂ S	110-66-7	104.214	598	3.47	0.359	0.251	0.3207
288	1-Pentyne	C ₅ H ₈	627-19-0	68.117	481.2	4.17	0.277	0.289	0.2899
289	2-Pentyne	C ₅ H ₈	627-21-4	68.117	519	4.03	0.276	0.258	0.1752
290	Phenanthrene	C ₁₄ H ₁₀	85-01-8	178.229	869	2.9	0.554	0.222	0.4707
291	Phenol	C ₆ H ₆ O	108-95-2	94.111	694.25	6.13	0.229	0.243	0.4435
292	Phenyl isocyanate	C ₇ H ₅ NO	103-71-9	119.121	653	4.06	0.37	0.277	0.4123

TABLE 2-141 Critical Constants and Acentric Factors of Inorganic and Organic Compounds (Concluded)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	T_c , K	P_c , MPa	V_c , m ³ /kmol	Z_c	Acentric factor
293	Phthalic anhydride	C ₈ H ₄ O ₃	85-44-9	148.116	791	4.72	0.421	0.302	0.7025
294	Propadiene	C ₃ H ₄	463-49-0	40.064	394	5.25	0.165	0.264	0.1041
295	Propane	C ₃ H ₈	74-98-6	44.096	369.83	4.248	0.2	0.276	0.1523
296	1-Propanol	C ₃ H ₈ O	71-23-8	60.095	536.8	5.169	0.219	0.254	0.6209
297	2-Propanol	C ₃ H ₈ O	67-63-0	60.095	508.3	4.765	0.222	0.25	0.6544
298	Propenylcyclohexene	C ₉ H ₁₄	13511-13-2	122.207	636	3.12	0.437	0.258	0.3420
299	Propionaldehyde	C ₃ H ₆ O	123-38-6	58.079	504.4	4.92	0.204	0.239	0.2559
300	Propionic acid	C ₃ H ₆ O ₂	79-09-4	74.079	600.81	4.668	0.235	0.22	0.5796
301	Propionitrile	C ₃ H ₅ N	107-12-0	55.079	564.4	4.18	0.229	0.204	0.3243
302	Propyl acetate	C ₅ H ₁₀ O ₂	109-60-4	102.132	549.73	3.36	0.345	0.254	0.3889
303	Propyl amine	C ₃ H ₉ N	107-10-8	59.110	496.95	4.74	0.26	0.298	0.2798
304	Propylbenzene	C ₉ H ₁₂	103-65-1	120.192	638.35	3.2	0.44	0.265	0.3444
305	Propylene	C ₃ H ₆	115-07-1	42.080	364.85	4.6	0.185	0.281	0.1376
306	Propyl formate	C ₄ H ₈ O ₂	110-74-7	88.105	538	4.02	0.285	0.256	0.3088
307	2-Propyl mercaptan	C ₃ H ₇ S	75-33-2	76.161	517	4.75	0.254	0.281	0.2138
308	Propyl mercaptan	C ₃ H ₇ S	107-03-9	76.161	536.6	4.63	0.254	0.264	0.2318
309	1,2-Propylene glycol	C ₃ H ₈ O ₂	57-55-6	76.094	626	6.1	0.239	0.28	1.1065
310	Quinone	C ₆ H ₄ O ₂	106-51-4	108.095	683	5.96	0.291	0.305	0.4945
311	Silicon tetrafluoride	SiF ₄	7783-61-1	104.079	259	3.72	0.202	0.349	0.3858
312	Styrene	C ₈ H ₈	100-42-5	104.149	636	3.84	0.352	0.256	0.2971
313	Succinic acid	C ₄ H ₆ O ₄	110-15-6	118.088	806	4.71	0.317	0.223	0.9922
314	Sulfur dioxide	SO ₂	7446-09-5	64.064	430.75	7.884	0.122	0.269	0.2454
315	Sulfur hexafluoride	SF ₆	2551-62-4	146.055	318.69	3.76	0.19852	0.282	0.2151
316	Sulfur trioxide	SO ₃	7446-11-9	80.063	490.85	8.21	0.127	0.255	0.4240
317	Terephthalic acid	C ₈ H ₆ O ₄	100-21-0	166.131	1113	3.95	0.424	0.181	1.0591
318	<i>o</i> -Terphenyl	C ₁₈ H ₁₄	84-15-1	230.304	857	2.99	0.731	0.307	0.5513
319	Tetradecane	C ₁₄ H ₃₀	629-59-4	198.388	693	1.57	0.897	0.244	0.6430
320	Tetrahydrofuran	C ₄ H ₈ O	109-99-9	72.106	540.15	5.19	0.224	0.259	0.2254
321	1,2,3,4-Tetrahydronaphthalene	C ₁₀ H ₁₂	119-64-2	132.202	720	3.65	0.408	0.249	0.3353
322	Tetrahydrothiophene	C ₄ H ₈ S	110-01-0	88.171	631.95	5.16	0.249	0.245	0.1996
323	2,2,3,3-Tetramethylbutane	C ₈ H ₁₈	594-82-1	114.229	568	2.87	0.461	0.28	0.2450
324	Thiophene	C ₄ H ₄ S	110-02-1	84.140	579.35	5.69	0.219	0.259	0.1970
325	Toluene	C ₇ H ₈	108-88-3	92.138	591.75	4.108	0.316	0.264	0.2640
326	1,1,2-Trichloroethane	C ₂ H ₃ Cl ₃	79-00-5	133.404	602	4.48	0.281	0.252	0.2591
327	Tridecane	C ₁₃ H ₂₈	629-50-5	184.361	675	1.68	0.826	0.247	0.6174
328	Triethyl amine	C ₆ H ₁₅ N	121-44-8	101.190	535.15	3.04	0.39	0.266	0.3162
329	Trimethyl amine	C ₃ H ₉ N	75-50-3	59.110	433.25	4.07	0.254	0.287	0.2062
330	1,2,3-Trimethylbenzene	C ₉ H ₁₂	526-73-8	120.192	664.5	3.454	0.414	0.259	0.3666
331	1,2,4-Trimethylbenzene	C ₉ H ₁₂	95-63-6	120.192	649.1	3.232	0.43	0.258	0.3787
332	2,2,4-Trimethylpentane	C ₈ H ₁₈	540-84-1	114.229	543.8	2.57	0.468	0.266	0.3035
333	2,3,3-Trimethylpentane	C ₈ H ₁₈	560-21-4	114.229	573.5	2.82	0.455	0.269	0.2903
334	1,3,5-Trinitrobenzene	C ₆ H ₃ N ₃ O ₆	99-35-4	213.105	846	3.39	0.479	0.231	0.8623
335	2,4,6-Trinitrotoluene	C ₇ H ₅ N ₃ O ₆	118-96-7	227.131	828	3.04	0.572	0.253	0.8972
336	Undecane	C ₁₁ H ₂₄	1120-21-4	156.308	639	1.95	0.685	0.252	0.5303
337	1-Undecanol	C ₁₁ H ₂₄ O	112-42-5	172.308	703.9	2.119	0.715	0.259	0.6236
338	Vinyl acetate	C ₄ H ₆ O ₂	108-05-4	86.089	519.13	3.958	0.27	0.248	0.3513
339	Vinyl acetylene	C ₄ H ₄	689-97-4	52.075	454	4.86	0.205	0.264	0.1069
340	Vinyl chloride	C ₂ H ₃ Cl	75-01-4	62.498	432	5.67	0.179	0.283	0.1001
341	Vinyl trichlorosilane	C ₂ H ₃ Cl ₃ Si	75-94-5	161.490	543.15	3.06	0.408	0.276	0.2815
342	Water	H ₂ O	7732-18-5	18.015	647.096	22.064	0.0559472	0.229	0.3449
343	<i>m</i> -Xylene	C ₈ H ₁₀	108-38-3	106.165	617	3.541	0.375	0.259	0.3265
344	<i>o</i> -Xylene	C ₈ H ₁₀	95-47-6	106.165	630.3	3.732	0.37	0.264	0.3101
345	<i>p</i> -Xylene	C ₈ H ₁₀	106-42-3	106.165	616.2	3.511	0.378	0.259	0.3218

All substances are listed by chemical family in Table 2-6 and by formula in Table 2-7.

Values in this table were taken from the Design Institute for Physical Properties (DIPPR) of the American Institute of Chemical Engineers (AIChE), copyright 2007 AIChE and reproduced with permission of AIChE and of the DIPPR Evaluated Process Design Data Project Steering Committee. Their source should be cited as R. L. Rowley, W. V. Wilding, J. L. Oscarson, Y. Yang, N. A. Zundel, T. E. Daubert, R. P. Danner, DIPPR® Data Compilation of Pure Chemical Properties, Design Institute for Physical Properties, AIChE, New York (2007).

The number of digits provided for the acentric factor was chosen for uniformity of appearance and formatting; these do not represent the uncertainties of the physical quantities, but are the result of calculations from the standard thermophysical property formulations within a fixed format.

COMPRESSIBILITIES

INTRODUCTION

The compressibility factor Z can be calculated by using the defining equation $Z = PV/(RT)$, where P is pressure, V is molar volume, R is the gas constant, and T is absolute temperature. Values of P , V , and T for substances listed in Table 2-184 are given in tables in the Thermodynamic Properties section. For the units used in these tables, R is 0.008314472 MPa·dm³/(mol·K). Values at temperatures and pressures other than those in the tables can be generated for many of the substances in Table 2-184 by going to <http://webbook.nist.gov> and selecting NIST Chemistry WebBook, then Thermo-

physical Properties of Fluid Systems High Accuracy Data. Results can be pasted into a spreadsheet to facilitate calculation of the compressibility factor.

UNITS CONVERSIONS

For this subsection, the following units conversions are applicable:

$$^{\circ}\text{R} = \% \text{ K}$$

To convert bars to pounds-force per cubic inch, multiply by 14.504.

To convert bars to kilopascals, multiply by 1×10^2 .

TABLE 2-142 Composition of Selected Refrigerant Mixtures

Mixture	Tables	Composition (mass percent)			
		R-32	R-125	R-134a	R-143a
R-410A	2-290	50	50		
R-404A	2-288		44	4	52
R-507A	2-296		50		50
R-407C (Klea 66)	2-263, 2-264, 2-289	23	25	52	
R-407A (Klea 60)	2-143, 2-261	20	40	40	
R-407B (Klea 61)	2-144, 2-262	10	70	20	

TABLE 2-143 Compressibility Factors for R 407A (Klea 60)

T, K	Pressure, bar								Z_{dew}	P_{dew}
	1	5	10	15	20	25	30			
250	0.9691	0.0163	0.0325	0.0487	0.0648	0.0809	0.0970	0.9340	2.05	
260	0.9737	0.0161	0.0321	0.0480	0.0640	0.0798	0.0957	0.9136	3.08	
270	0.9773	0.4268	0.0318	0.0476	0.0633	0.0790	0.0947	0.8895	4.46	
280	0.9803	0.8932	0.0316	0.0473	0.0630	0.0785	0.0940	0.8614	6.28	
290	0.9828	0.9080	0.0360	0.0473	0.0629	0.0784	0.0938	0.8290	8.60	
300	0.9848	0.9199	0.8253	0.0476	0.0632	0.0787	0.0942	0.7916	11.53	
310	0.9866	0.9298	0.8495	0.7518	0.0641	0.0797	0.0952	0.7485	15.15	
320	0.9881	0.9380	0.8689	0.7889	0.5737	0.0816	0.0972	0.6983	19.58	
330	0.9893	0.9449	0.8847	0.8173	0.7386	0.6279	0.1011	0.6386	24.96	
340	0.9904	0.9509	0.8980	0.8401	0.7752	0.6993	0.6016	0.5641	31.46	
350	0.9914	0.9560	0.9092	0.8588	0.8038	0.7425	0.6713	0.4564	39.42	
Z_{dew}	0.9593	0.8809	0.8107	0.7502	0.6936	0.6381	0.5813			
T_{dew}, K	234.3	273.3	295.1	309.6	320.9	330.1	337.9			

The values in this table were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties-REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1).

TABLE 2-144 Compressibility Factors for R 407B (Klea 61)

T, K	Pressure, bar								Z_{dew}	P_{dew}
	1	5	10	15	20	25	30			
250	0.9703	0.0180	0.0359	0.0538	0.0716	0.0894	0.1071	0.9251	2.40	
260	0.9745	0.0178	0.0355	0.0532	0.0708	0.0883	0.1058	0.9024	3.56	
270	0.9779	0.8785	0.0352	0.0527	0.0702	0.0875	0.1048	0.8757	5.10	
280	0.9808	0.8961	0.0351	0.0526	0.0699	0.0872	0.1043	0.8446	7.10	
290	0.9831	0.9101	0.5520	0.0527	0.0700	0.0872	0.1043	0.8085	9.64	
300	0.9851	0.9215	0.8290	0.0533	0.0707	0.0879	0.1050	0.7666	12.81	
310	0.9868	0.9310	0.8522	0.7569	0.0721	0.0895	0.1067	0.7178	16.71	
320	0.9883	0.9390	0.8709	0.7924	0.6949	0.0925	0.1098	0.6600	21.45	
330	0.9895	0.9457	0.8864	0.8200	0.7428	0.6453	0.1164	0.5890	27.18	
340	0.9906	0.9515	0.8993	0.8422	0.7784	0.7042	0.6108	0.4916	34.12	
350	0.9915	0.9565	0.9103	0.8606	0.8064	0.7462	0.6770			
Z_{dew}	0.9587	0.8774	0.8036	0.7389	0.6776	0.6164	0.5518			
T_{dew}, K	230.6	269.4	291.3	305.9	317.1	326.4	334.3			

The values in this table were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties-REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1).

2-144 PHYSICAL AND CHEMICAL DATA

TABLE 2-145 Compressibilities of Liquids*

At the constant temperature T , the compressibility $\beta = (1/\bar{V}_0)(dV/dP)$. In general as P increases, β decreases rapidly at first and then slowly; the change of β with T is large at low pressures but very small at pressures above 1000 to 2000 megabars. 1 megabar = 0.987 atm = 10^6 dynes/cm² based upon the older usage, 1 bar = 1 dyne/cm².

Substance	Temp., °C	Pres- sure, mega- bars	Compres- sibility per megabar $\beta \times 10^6$	Substance	Temp., °C	Pres- sure, mega- bars	Compres- sibility per megabar $\beta \times 10^6$	Substance	Temp., °C	Pres- sure, mega- bars	Compres- sibility per megabar $\beta \times 10^6$
Acetone	14	23	111	Ethyl acetate	20	400	75	Methyl alcohol	15	23	103
Acetone	20	500	61	alcohol	14	23	100	alcohol	20	200	95
Acetone	20	1,000	52	alcohol	20	500	63	alcohol	20	400	80
Acetone	40	12,000	9	alcohol	20	1,000	54	alcohol	20	500	65
Amyl alcohol	14	23	88	alcohol	20	12,000	8	alcohol	20	1,000	54
alcohol, iso.	20	200	84	bromide	20	200	100	alcohol	20	12,000	8
alcohol, iso.	20	400	70	bromide	20	400	82	Nitric acid	0	17	32
alcohol, n	20	500	61	bromide	20	500	70	Oils:			
alcohol, n	20	1,000	46	bromide	20	1,000	54	Almond	15	5	53
alcohol, n	20	12,000	8	bromide	20	12,000	8	Castor	15	5	46
alcohol, n	40	12,000	8	chloride	15	23	151	Linseed	15	5	51
Benzene	17	5	89	chloride	20	500	102	Olive	15	5	55
Benzene	20	200	77	chloride	20	1,000	66	Rapeseed	20		59
Benzene	20	400	67	chloride	20	12,000	8	Phosphorus trichloride	10	250	71
Bromine	20	200	56	ether	25	23	188	trichloride	20	500	63
Bromine	20	400	51	ether	20	500	84	trichloride	20	1,000	47
Butyl alcohol, iso	18	8	97	ether	20	1,000	61	trichloride	20	12,000	8
alcohol, iso	20	200	81	ether	20	12,000	10	Propyl alcohol (n)	20	200	77
alcohol, iso	20	400	64	iodide	20	200	81	alcohol (n)	20	400	67
alcohol, iso	20	500	56	iodide	20	400	69	alcohol (n ?)	20	500	65
alcohol, iso	20	1,000	46	iodide	20	500	64	alcohol (n ?)	20	1,000	47
alcohol, iso	20	12,000	8	iodide	20	1,000	50	alcohol (n ?)	20	12,000	7
Carbon bisulfide	16	21	86	iodide	20	12,000	8	Toluene	20	200	74
bisulfide	20	500	57	Gallium	30	300	3.97	Toluene	20	400	64
bisulfide	20	1,000	48	Glycerol	15	5	22	Turpentine	20		74
bisulfide	20	12,000	6	Hexane	20	200	117	Water	20	13	49
tetrachloride	20	200	86	Hexane	20	400	91	Water	20	200	43
tetrachloride	20	400	73	Kerosene	20	500	55	Water	20	400	41
Chloroform	20	200	83	Kerosene	20	1,000	45	Water	20	500	39
Chloroform	20	400	70	Kerosene	20	12,000	8	Water	40	500	38
Dichloroethylsulfide	32	1,000	34	Mercury	20	300	3.95	Water	40	1,000	33
Dichloroethylsulfide	32	2,000	24	Mercury	22	500	3.97	Water	40	12,000	9
Ethyl acetate	13	23	103	Mercury	22	1,000	3.91	Xylene, meta	20	200	69
acetate	20	200	90	Mercury	22	12,000	2.37	meta	20	400	60

* Smithsonian Tables, Table 106.

Scott (*Cryogenic Engineering*, Van Nostrand, Princeton, N.J., 1959) gives data for liquid nitrogen (p. 283), oxygen (p. 276), and hydrogen (p. 303). For a convenient index to the high-pressure work of Bridgman, see *American Institute of Physics Handbook*, p. 2-163, McGraw-Hill, New York, 1957.

TABLE 2-146 Compressibilities of Solids

Many data on the compressibility of solids obtained prior to 1926 are contained in Gruneisen, *Handbuch der Physik*, vol. 10, Springer, Berlin, 1926, pp. 1–52; also available as translation, NASA RE 2-18-59W, 1959. See also Tables 271, 273, 276, 278, and other material in *Smithsonian Physical Tables*, 9th ed., 1954. For a review of high-pressure work to 1946, see Bridgman, *Rev. Mod. Phys.*, **18**, 1 (1946).

LATENT HEATS

UNITS CONVERSIONS

For this subsection, the following units conversions are applicable:

$$^{\circ}\text{F} = \frac{9}{5}^{\circ}\text{C} + 32$$

To convert calories per gram-mole to British thermal units per

pound-mole, multiply by 1.799; to convert calories per gram to British thermal units per pound, multiply by 1.799.

To convert millimeters of mercury to pounds-force per square inch, multiply by 1.934×10^{-2} .

TABLE 2-147 Heats of Fusion and Vaporization of the Elements and Inorganic Compounds*

Unless stated otherwise, the values have been taken from the compilations by K. K. Kelley on "Heats of Fusion of Inorganic Compounds," U.S. Bur. Mines Bull. 393 (1936), and "The Free Energies of Vaporization and Vapor Pressures of Inorganic Substances," U.S. Bur. Mines Bull. 383 (1935).

Substance	mp, °C	Heat of fusion, ^{a,b} cal/mol	bp at 1 atm, °C	Heat of vaporization, ^{a,b} cal/mol	Substance	mp, °C	Heat of fusion, ^{a,b} cal/mol	bp at 1 atm, °C	Heat of vaporization, ^{a,b} cal/mol
Aluminum					Carbon (<i>Cont.</i>)				
Al	660.0	2,550	2057	61,020	CNF			-72.8	5,780 ^f
Al ₂ Br ₆	97.5	5,420	256.4	10,920	CNI			141	13,980 ^f
Al ₂ Cl ₆	192.5	16,960	180.2 ^e	26,750 ^e	CO	-205.0	200	-191.5	1,444
AlF ₃ ·3NaF	1000	16,380			CO ₂	-57.5	1,900	-78.4 ^e	6,030 ^{c,r}
Al ₂ I ₆	191.0	7,960	385.5	15,360	COS	-138.8	1,129 ^k	-50.2	4,423 ^k
Al ₂ O ₃	2045	(26,000)	3000		COCl ₂			8.0	5,990
Antimony					CS ₂	-112.0	1,049 ^j		
Sb	630.5	4,770	1440	46,670	Cerium				
SbBr ₃	97	3,510			Ce	775	2,120		
SbCl ₃	73.4	3,030	219	10,360	Cesium				
SbCl ₅	4	2,400	172 ^j	11,570	Cs	28.4	500	690	16,320
Sb ₂ O ₆	655	(27,000)	1425	17,820	CsBr			1300	35,990
Sb ₂ S ₆	546	11,200			CsCl	642	3,600	1300	35,690
Argon					CsF	715	(2,450)	1251	34,330
A	-189.3	290	-185.8	1,590	CsI			1280	35,930
Arsenic					CsNO ₃	407	3,250		
As	814	(6,620)	610 ^e	31,000 ^e	Chlorine				
AsBr ₃	31	2,810			Cl ₂	-101.0	1,531 ^m	-34.1	4,878 ^m
AsCl ₃	-16	2,420	122	7,570	ClF			-101	
AsF ₃	-80.7	2,800	-52.8	4,980	ClF ₃			11.3	5,890
As ₂ O ₆	313	8,000	457.2	14,300	Cl ₂ O			2.0	6,280
Barium					ClO ₂			10.9	7,100
Ba	704	(1,400) ^e	1638	35,670	Cl ₂ O ₇			79	8,480
BaBr ₂	847	6,000			Chromium				
BaCl ₂	960	5,370			Cr	1550	3,930	2475	
BaF ₂	1287	3,000			CrO ₂ Cl ₂			117	8,250
Ba(NO ₃) ₂	595	(5,980)			Cobalt				
Ba ₃ (PO ₄) ₂	1730	18,600			Co	1490	3,660		
BaSO ₄	1350	9,700			CoCl ₂	727	7,390	1050	27,170
Beryllium					Copper				
Be	1280	2,500 ^e			Cu	1083.0	3,110	2595	72,810
Bismuth					Cu ₂ Br ₂			1355	16,310
Bi	271.3	2,505	1420		Cu ₂ Cl ₂	430	4,890	1490	11,920
BiBr ₃			461	18,020	CuI			1336	15,940
BiCl ₃	224	2,600	441	17,350	Cu ₂ (CN) ₂	473	(5,400)		
Bi ₂ O ₃	817	6,800			Cu ₂ O	1230	(13,400)		
Bi ₂ S ₅	747	8,900			CuO	1447	2,820		
Boron					Cu ₂ S	1127	5,500		
BBr ₃			91.3	7,300	Fluorine				
BCl ₃			12.5	5,680	F ₂	-223		-188.2	1,640
BF ₃	-128	480	-100.9	4,620	F ₂ O			-144.8	2,650
B ₂ H ₆	-165.5		-92.4	3,685	Gallium				
B ₂ H ₁₀	-119.8		16	6,470	Ga	29.8	1,336	2071	
B ₂ H ₉	-46.9		58	7,700	Germanium				
B ₂ H ₁₁			67	8,500	Ge	959	(8,300)		
B ₁₀ H ₁₄	99.7	7,800	<i>f</i>	11,600	GeH ₄	-165		-89.1	3,580
B ₂ H ₅ Br	-104		16	6,230	Ge ₂ H ₆	-109		31.4	5,900
B ₃ N ₃ H ₆	-58		50.4	7,670	Ge ₃ H ₈	-105.6		110.6	7,550
Bromine					GeHCl ₃	-71		75 ^e	8,000
Br ₂	-7.2	2,580	58.0	7,420	GeBr ₄	26.1		189	8,560
BrF ₅	-61.3	1,355	40.4	7,470	GeCl ₄	-49.5		84	7,030
Cadmium					Ge(CH ₃) ₄	-88		44	6,460
Cd	320.9	1,460	765	23,870	Gold				
CdBr ₂	568	(5,000)			Au	1063.0	3,030	2966	81,800
CdCl ₂	568	5,300	967	29,860	Helium				
CdF ₂	1110	(5,400)			He	-271.4		-268.4	22
CdI ₂	387	3,660	796	25,400	Hydrogen				
CdO			1559 ^e	53,820 ^e	H ₂	-259.2	28	-252.7	216
CdSO ₄	1000	4,790			HBr	-86.9	575	-66.7	4,210
Calcium					HCl	-114.2	476	-85.0	3,860
Ca	851	2,230	1487	36,580	HCN	-13.2	2,009 ^e	25.7	6,027 ^j
CaBr ₂	730	4,180			HF	-83.0	1,094	33.3	7,460
CaCO ₃	1282	(12,700)			(HF) ₆			51.2	5,020
CaCl ₂	782	6,100			HI	-50.8	686		
CaF ₂	1392	4,100			H ₂ O	0.0	1,436	100.0	9,729 ^{k,q}
Ca(NO ₃) ₂	561	5,120			H ₂ ² O (= D ₂ O)	3.8	1,501 ^r	101.4	9,945 ^q
CaO	2707	(12,240)			H ₂ O ₂	-2	2,520 ^r	158	10,270
CaO·Al ₂ O ₃ ·2SiO ₂	1550	29,400			HNO ₃	-47	600		
CaO·MgO·2SiO ₂	1392	(18,200)			H ₃ PO ₃	17.4	2,310		
CaO·SiO ₂	1512	13,400			H ₃ PO ₄	74	3,070		
CaSO ₄	1297	6,700			H ₃ PO ₄	42.4	2,520		
Carbon					H ₄ P ₂ O ₆	55	8,300		
C (graphite)	3600	11,000 ^e			H ₂ S	-85.5	568 ^f	-60.3	4,463 ^f
CBR ₄	90	1,050			H ₂ S ₂	-87.6	1,805		
CCl ₄	-24.0	644	77	7,280	H ₂ SO ₄	10.5	2,360		
CF ₄			-127.9	3,110	H ₂ Se			-41.3	4,880
CH ₄	-182.5	224	-161.4	2,040	H ₂ SeO ₄	58	3,450		
C ₂ N ₂	-27.8	1,938 ^e	-21.1	5,576 ^e	H ₂ Te	-48.9	1,670	-2.2	5,650
CNBr	52			11,010 ^e	Indium				
CNCl	-5	2,240	13	6,300	In	156.4	781		

*See also subsection "Thermodynamic Properties."

TABLE 2-147 Heats of Fusion and Vaporization of the Elements and Inorganic Compounds (Continued)

Substance	mp, °C	Heat of fusion, ^{a,b} cal/mol	bp at 1 atm, °C	Heat of vaporization, ^{a,b} cal/mol	Substance	mp, °C	Heat of fusion, ^{a,b} cal/mol	bp at 1 atm, °C	Heat of vaporization, ^{a,b} cal/mol
Iodine					Palladium				
I ₂	113.0	3,650	183	10,390	Pd	1554	4,120		
ICl(α)	17.2	2,660			Phosphorus				
ICl(β)	13.9	2,270			P ₄ (yellow)	44.2	615	280	12,520
IF ₇			4°	7,460 ^c	P ₄ (violet)			417°	25,600 ^c
Iron					P ₄ (black)			453°	33,100
Fe	1530	3,560	2735	84,600	PCl ₃			74.2	7,280
FeCl ₂	677	7,800	1026	30,210	PH ₃	-133.8	270°	-87.7	3,489°
Fe ₂ Cl ₆	304	20,590	319	12,040	P ₂ O ₃	23.8	3,360	174	10,380
Fe(CO) ₅	-21	3,250	105	9,000	P ₄ O ₁₀ (α)	569	17,080	591	20,670
FeO	1380	(7,700)			P ₄ O ₁₀ (β)			358°	
FeS	1195	5,000			POCl ₃	1.1	3,110	105.1	8,380
Krypton					P ₂ S ₃			508	
Kr	-157	360°	152.9	2,310°	Platinum				
Lead					Pt	1773.5	4,700	(4400)	(107,000)
Pb	327.4	1,224	1744	42,060	Potassium				
PbBr ₂	488	4,290	914	27,700	K	63.5	574	776	18,920
PbCl ₂	498	5,650	954	29,600	KBO ₂	947	(5,700)		
PbF ₂	824	1,860	1293	38,300	KBr	742	5,000	1383	37,060
PbI ₂	412	5,970	872	24,850	KCl	770	6,410	1407	38,840
PbMoO ₄	1065	(25,800)			KCN	623	(3,500)		
PbO	890	2,820	1472	51,310	KCNS	179	2,250		
PbS	1114	4,150	1281	(50,000)	K ₂ CO ₃	897	7,800		
PbSO ₄	1087	9,600			K ₂ CrO ₄	984	6,920		
PbWO ₄	1123	(15,200)			K ₂ Cr ₂ O ₇	398	8,770		
Lithium					KF	857	6,500		
Li	179	1,100	1372	32,250	KI	682	4,100	1324	34,690
LiBO ₂	845	(5,570)			K ₂ MoO ₄	922	(4,000)		
LiBr	552	2,900	1310	35,420	KNO ₃	338	2,840		
LiCl	614	3,200	1382	35,960	KOH	360	(2,000)	1327	30,850
LiF	847	(2,360)	1681	50,970	KPO ₃	817	2,110		
LiI	440	(1,420)	1171	40,770	K ₃ PO ₄	1340	8,900		
LiOH	462	2,480			K ₄ P ₂ O ₇	1092	14,000		
Li ₂ MoO ₄	705	4,200			K ₂ SO ₄	1074	8,100		
LiNO ₃					K ₂ TiO ₃	810	(10,600)		
Li ₂ SiO ₃	1177	7,210			K ₂ WO ₄	927	(4,400)		
Li ₂ SiO ₄	1249	7,430			Praseodymium				
Li ₂ SO ₄	857	3,040			Pr	932	2,700		
Li ₂ WO ₄	742	(6,700)			Radon				
Magnesium					Rn	-71		-61.8	4,010
Mg	650	2,160	1107	32,520	Rhenium				
MgBr ₂	711	8,300			Re	(3000)			
MgCl ₂	712	8,100	1418	32,690	Re ₂ O ₇	296	15,340	362.4	18,060
MgF ₂	1221	5,900			Re ₂ O ₈	147	3,800		
MgO	2642	18,500			Rubidium				
Mg ₂ (PO ₄) ₂	1184	(11,300)			Rb	39.1	525	679	18,110
MgSiO ₃	1524	14,700			RbBr	677	3,700	1352	37,120
MgSO ₄	1127	3,500			RbCl	717	4,400	1381	36,920
MgZn ₂	589	(8,270)			RbF	833	4,130	1408	39,510
Manganese					RbI	638	2,990	1304	35,960
Mn	1220	3,450	2152	55,150	RbNO ₃	305	1,340		
MnCl ₂	650	7,340	1190	29,630	Selenium				
MnSiO ₃	1274	(8,200)			Se ₂	217	1,220	753	25,490
MnTiO ₃	1404	(7,960)			Se ₆			736	20,600
Mercury					SeF ₆			-45.8°	6,350°
Hg	-38.9	557	361	13,980	SeO ₂			317°	20,900
HgBr ₂	241	3,960	319	14,080	SeOCl ₂	10	1,010	168	
HgCl ₂	277	4,150	304	14,080	Silicon				
HgI ₂	250	4,500	354	14,260	Si	1427	9,470	2290	
HgSO ₄	850	(1,440)			SiCl ₄	-67.6	1,845	56.8	6,860
Molybdenum					Si ₂ Cl ₆	-1		139	
Mo	2622	(6,660)	(4800)	(128,000)	Si ₂ Cl ₈			211.4	12,340
MoF ₆	17	2,500	36	6,000	(SiCl ₃) ₂ O	-33		135.6	8,820
MoO ₃	745	(2,500)	1151		SiF ₄			-94.8°	6,130°
Neon					Si ₂ F ₆	-18.5	3,900	-18.9°	10,400°
Ne	-248.5	77	-246.0	440°	SiF ₃ Cl	-138		-70.1	4,460
Nickel					SiF ₂ Cl ₂	-144		-31.5	5,080
Ni	1455	4,200	2730	87,300	SiH ₄	-185		-111.6	2,960
NiCl ₂			987°	48,360°	Si ₂ H ₆	-132.5		-14.3	5,110
Ni(CO) ₄			42.5	7,000	Si ₃ H ₈	-117		53.1	6,780
Ni ₂ S	645	(2,980)			Si ₄ H ₁₀	-93.5		100	8,890
Ni ₃ S ₂	790	5,800			SiH ₃ Br	-93.8		2.4	5,650
Nitrogen					SiH ₂ Br ₂	-70.0		70.5	6,840
N ₂	-210.0	172	-195.8	1,336	SiHCl ₃	-126.5		31.8	6,360
NF ₃			-129.0	3,000	(SiH ₃) ₃ N	-105.6		48.7	6,850
NH ₃	-77.7	1,352°	-33.4	5,581°	(SiH ₃) ₂ O	-144		-15.4	5,350
NH ₄ CNS	146	(4,700)			SiO ₂ (quartz)	1470	3,400	2230	
NH ₄ NO ₃	169.6	1,460			SiO ₂ (cristobalite)	1700	2,100		
N ₂ O	-90.8	1,563	-88.5	3,950	Silver				
NO	-163.6	550	-151.7	3,307	Ag	960.5	2,700	2212	60,720
N ₂ O ₄	-13	5,540	30	7,040	AgBr	430	2,180		
N ₂ O ₅			32.4	13,800°	AgCl	455	3,155	1564	42,520
NOCl			-6.4	6,140	AgCN	350	2,750		
Osmium					AgI	557	2,250	1506	34,450
OsF ₈			47.4	6,840	AgNO ₃	209	2,755		
OsO ₄ (yellow)	56	4,060	130	9,450	Ag ₂ S	842	3,360		
OsO ₄ (white)	42	2,340			Ag ₂ SO ₄	657	(4,300)		
Oxygen					Sodium				
O ₂	-218.9	106	-183.0	1,629	Na	97.7	630	914	23,120
O ₃			-111	2,880	NaBO ₂	966	8,660		

TABLE 2-147 Heats of Fusion and Vaporization of the Elements and Inorganic Compounds (Concluded)

Substance	mp, °C	Heat of fusion, ^{a,b} cal/mol	bp at 1 atm, °C	Heat of vaporization, ^{a,b} cal/mol	Substance	mp, °C	Heat of fusion, ^{a,b} cal/mol	bp at 1 atm, °C	Heat of vaporization, ^{a,b} cal/mol
Sodium (<i>Cont.</i>)					Thallium				
NaBr	747	6,140	1392	37,950	Tl	302.5	1,030	1457	38,810
NaCl	800	7,220	1465	40,810	TlBr	460	5,990	819	23,800
NaClO ₃	255	5,290			TlCl	427	4,260	807	24,420
NaCN	562	(4,400)	1500	37,280	Tl ₂ CO ₃	273	4,400		
NaCNS	323	4,450			TlI	440	3,125	823	25,030
Na ₂ CO ₃	854	7,000			TlNO ₃	207	2,290		
NaF	992	7,000	1704	53,260	Tl ₂ S	449	3,000		
NaI	662	5,240			Tl ₂ SO ₄	632	5,500		
Na ₂ MoO ₄	687	3,600			Tin				
NaNO ₃	310	3,760			Sn	231.8	1,720	2270	68,000
NaOH	322	2,000	1378		SnBr ₂	232	(1,700)		
½Na ₂ O·½Al ₂ O ₃ ·3SiO ₂	1107	13,150			SnBr ₄	30	3,000		
NaPO ₃	988	(5,000)			SnCl ₂	247	3,050	623	20,740
Na ₄ P ₂ O ₇	970	(13,700)			SnCl ₄	-33.2	2,190	113	8,330
Na ₂ S	920	(1,200)			Sn(CH ₃) ₄			78.3	7,320
Na ₂ SiO ₃	1087	10,300			SnH ₄	-149.8		-52.3	4,420
Na ₂ Si ₂ O ₅	884	8,460			SnI ₄	143.5	(4,300)		
Na ₂ SO ₄	884	5,830			Titanium				
Na ₂ WO ₄	702	5,800			TiBr ₄	38.2	(2,060)		
Strontium					TiCl ₄	-23	2,240	136	8,350
Sr	757	2,190	1384	33,610	TiO ₂	1825	(11,400)		
SrBr ₂	643	4,780			Tungsten				
SrCl ₂	872	4,100			W	3390	(8,400)	(5900)	(176,000)
SrF ₂	1400	4,260			WF ₆	-0.4	1,800	17.3	6,350
Sr ₃ (PO ₄) ₂	1770	18,500			Uranium				
Sulfur					UF ₆			55.1 ^c	9,990 ^c
S (rhombic)	112.8		444.6	2,200	Xenon				
S (monoclinic)	119.2				Xe	-111.5	740	-108.0	3,110
S ₂ Cl ₂			138	8,720	Zinc				
SF ₆			-63.5 ^c	5,600 ^c	Zn	419.5	1,595	907	27,430
SO ₂	-75.5	1,769 ^d	-5.0	5,960 ^d	ZnCl ₂	283	(5,500)	732	28,710
SO ₃ (α)	17	2,060	44.8	10,190	Zn(C ₂ H ₅) ₂			118	8,960
SO ₃ (β)	32.4	2,890			ZnO	1975	4,470		
SO ₃ (γ)	62.2	6,310			ZnS	1645	(9,000)		
SOBr ₂			139.5	9,920	Zirconium				
SOCl ₂			75.4	7,600	ZrBr ₄			357 ^e	25,800 ^c
SO ₂ Cl ₂			69.2	7,760	ZrCl ₄			311 ^c	25,290 ^c
Tellurium					ZrI ₄			431 ^c	29,030 ^c
Te	453	3,230	1090		ZrO ₂	2715	20,800		
TeCl ₄			392	16,830					
TeF ₆			-38.6 ^c	6,700 ^c					

^a Values in parentheses are uncertain.^b For the freezing point or the normal boiling point unless otherwise stated.^c Sublimation.^d Decomposes at about 75 °C; value obtained by extrapolation.^e Bichowsky and Rossini, *Thermochemistry of the Chemical Substances*, Reinhold, New York (1936).^f Decomposes before the normal boiling point is reached.^g Decomposes at about 40 °C; value obtained by extrapolation.^h See also pp. 2-304 through 2-307 on steam table.ⁱ Giauque and Ruehrwein, *J. Am. Chem. Soc.*, **61** (1939): 2626.^j Giauque and Egan, *J. Chem. Phys.*, **5** (1937): 45.^k Kemp and Giauque, *J. Am. Chem. Soc.*, **59** (1937): 79.^l Brown and Manov, *J. Am. Chem. Soc.*, **59** (1937): 500.^m Giauque and Powell, *J. Am. Chem. Soc.*, **61** (1939): 1970.ⁿ Overstreet and Giauque, *J. Am. Chem. Soc.*, **59** (1937): 254.^o Stephenson and Giauque, *J. Chem. Phys.*, **5** (1937): 149.^p Giauque and Stephenson, *J. Am. Chem. Soc.*, **60** (1938): 1389.^q Osborne, Stimson, and Ginnings, *Bur. Standards J. Research*, **23**, 197 (1939): 261.^r Miles and Menzies, *J. Am. Chem. Soc.*, **58** (1936): 1067.^s Long and Kemp, *J. Am. Chem. Soc.*, **58** (1936): 1829.^t Giauque and Blue, *J. Am. Chem. Soc.*, **58** (1936): 831.^u Ruehrwein and Giauque, *J. Am. Chem. Soc.*, **61** (1939): 2940.

TABLE 2-148 Heats of Fusion of Miscellaneous Materials

Material	mp, °C	Heat of fusion, cal/g
Alloys		
30.5 Pb + 69.5 Sn	183	17
36.9 Pb + 63.1 Sn	179	15.5
63.7 Pb + 36.3 Sn	177.5	11.6
77.8 Pb + 22.2 Sn	176.5	9.54
1 Pb + 9 Sn	236	28
24 Pb + 27.3 Sn + 48.7 Bi	98.8	6.85
25.8 Pb + 14.7 Sn + 52.4 Bi + 7 Cd	75.5	8.4
Silicates		
Anorthite (CaAl ₂ Si ₂ O ₈)		100
Orthoclase (KAlSi ₃ O ₈)		100
Microcline (KAlSi ₃ O ₈)		83
Wollastonite (CaSiO ₃)		100
Malacolite (Ca ₈ MgSi ₄ O ₁₂)		94
Diopside (CaMgSi ₂ O ₆)		100
Olivine (Mg ₂ SiO ₄)		130
Fayalite (Fe ₂ SiO ₄)		85
Spermaceti	43.9	37.0
Wax (bees')	61.8	42.3

TABLE 2-149 Heats of Fusion of Organic Compounds

The values for the hydrocarbons are from the tables of the American Petroleum Institute Research Project 44 at the National Bureau of Standards, with some from Parks and Huffman, *Ind. Eng. Chem.*, **23**, 1138 (1931).

The values for the nonhydrocarbon compounds were recalculated from data in *International Critical Tables*, vol. 5.

Hydrocarbon compounds	Formula	mp, °C	Heat of fusion, cal/g	Hydrocarbon compounds	Formula	mp, °C	Heat of fusion, cal/g
Paraffins				Aromatics—(Cont.)			
Methane	CH ₄	-182.48	14.03	1-Methyl-3-ethylbenzene	C ₉ H ₁₂	-95.55	15.14
Ethane	C ₂ H ₆	-183.23	22.712	1-Methyl-4-ethylbenzene	C ₉ H ₁₂	-62.350	25.29
Propane	C ₃ H ₈	-187.65	19.100	1,2,3-Trimethylbenzene	C ₉ H ₁₂	-25.375	16.64
<i>n</i> -Butane	C ₄ H ₁₀	-138.33	19.167	1,2,4-Trimethylbenzene	C ₉ H ₁₂	-43.80	24.54
2-Methylpropane	C ₄ H ₁₀	-159.60	18.668	1,3,5-Trimethylbenzene	C ₉ H ₁₂	-44.720	18.97
<i>n</i> -Pentane	C ₅ H ₁₂	-129.723	27.874	Naphthalene	C ₁₀ H ₈	+80.0	36.0
2-Methylbutane	C ₅ H ₁₂	-159.890	17.076	Camphene	C ₁₀ H ₁₆	+51	57
2,2-Dimethylpropane	C ₅ H ₁₂	-16.6	10.786	Durene	C ₁₀ H ₁₄	+79.3	37.4
<i>n</i> -Hexane	C ₆ H ₁₄	-95.320	36.138	Isodurene	C ₁₀ H ₁₄	-24.0	23.0
2-Methylpentane	C ₆ H ₁₄	-153.680	17.407	Prehnitene	C ₁₀ H ₁₄	-7.7	20.0
2,2-Dimethylbutane	C ₆ H ₁₄	-99.73	1.607	<i>p</i> -Cymene	C ₁₀ H ₁₄	-68.9	17.1
2,3-Dimethylbutane	C ₆ H ₁₄	-128.41	2.251	<i>n</i> -Butyl benzene	C ₁₀ H ₁₄	-88.5	19.5
<i>n</i> -Heptane	C ₇ H ₁₆	-90.595	33.513	<i>tert</i> -Butyl benzene	C ₁₀ H ₁₄	-58.1	14.9
2-Methylhexane	C ₇ H ₁₆	-118.270	21.158	β -Methyl naphthalene	C ₁₁ H ₁₀	+34.1	20.1
3-Ethylpentane	C ₇ H ₁₆	-118.593	22.555	Diphenyl	C ₁₂ H ₁₀	+68.6	28.8
2,2-Dimethylpentane	C ₇ H ₁₆	-123.790	13.982	Hexamethyl benzene	C ₁₂ H ₁₈	+165.5	30.4
2,4-Dimethylpentane	C ₇ H ₁₆	-119.230	15.968	Diphenyl methane	C ₁₃ H ₁₂	+25.2	26.4
3,3-Dimethylpentane	C ₇ H ₁₆	-134.46	16.856	Anthracene	C ₁₄ H ₁₀	+216.5	38.7
2,2,3-Trimethylbutane	C ₇ H ₁₆	-24.96	5.250	Phenanthrene	C ₁₄ H ₁₀	+96.3	25.0
<i>n</i> -Octane	C ₈ H ₁₈	-56.798	43.169	Tolane	C ₁₄ H ₁₀	+60	28.7
2-Methylheptane	C ₈ H ₁₈	-109.04	21.458	Stilbene	C ₁₄ H ₁₂	+124	40.0
3-Methylheptane	C ₈ H ₁₈	-120.50	23.795	Dibenzil	C ₁₄ H ₁₄	+51.4	30.7
4-Methylheptane	C ₈ H ₁₈	-120.955	22.692	Triphenyl methane	C ₁₈ H ₁₆	+92.1	21.1
2,2-Dimethylhexane	C ₈ H ₁₈	-121.18	24.226	Alkyl cyclohexanes			
2,5-Dimethylhexane	C ₈ H ₁₈	-91.200	26.903	Cyclohexane	C ₆ H ₁₂	+6.67	7.569
3,3-Dimethylhexane	C ₈ H ₁₈	-126.10	14.9	Methylcyclohexane	C ₇ H ₁₄	-126.58	16.429
2-Methyl-3-ethylpentane	C ₈ H ₁₈	-114.960	23.690	Alkyl cyclopentanes			
3-Methyl-3-ethylpentane	C ₈ H ₁₈	-90.870	22.657	Cyclopentane	C ₅ H ₁₀	-93.80	2.068
2,2,3-Trimethylpentane	C ₈ H ₁₈	-112.27	18.061	Methylcyclopentane	C ₆ H ₁₂	-142.445	19.68
2,2,4-Trimethylpentane	C ₈ H ₁₈	-107.365	19.278	Ethylcyclopentane	C ₇ H ₁₄	-138.435	11.10
2,3,3-Trimethylpentane	C ₈ H ₁₈	-100.70	3.204	1,1-Dimethylcyclopentane	C ₇ H ₁₄	-69.73	3.36
2,3,4-Trimethylpentane	C ₈ H ₁₈	-109.210	19.392	<i>cis</i> -1,2-Dimethylcyclopentane	C ₇ H ₁₄	-53.85	3.87
2,2,3,3-Tetramethylbutane	C ₈ H ₁₈	+100.69	14.900	<i>trans</i> -1,2-Dimethylcyclopentane	C ₇ H ₁₄	-117.57	15.68
<i>n</i> -Nonane	C ₉ H ₂₀	-53.9	41.2	<i>trans</i> -1,3-Dimethylcyclopentane	C ₇ H ₁₄	-133.680	17.93
<i>n</i> -Decane	C ₁₀ H ₂₂	-30.0	48.3	Monoolefins			
<i>n</i> -Undecane	C ₁₁ H ₂₄	-25.9	34.1	Ethene (Ethylene)	C ₂ H ₄	-169.15	28.547
<i>n</i> -Dodecane	C ₁₂ H ₂₆	-9.6	51.3	Propene (Propylene)	C ₃ H ₆	-185.25	17.054
Eicosane	C ₂₀ H ₄₂	+36.4	52.0	1-Butene	C ₄ H ₈	-185.35	16.393
Pentacosane	C ₂₅ H ₅₂	+53.3	53.6	<i>cis</i> -2-Butene	C ₄ H ₈	-138.91	31.135
Tritriacontane	C ₃₃ H ₆₈	+71.1	54.0	<i>trans</i> -2-Butene	C ₄ H ₈	-105.55	41.564
Aromatics				2-Methylpropene (isobutene)	C ₄ H ₈	-140.35	25.265
Benzene	C ₆ H ₆	+5.533	30.100	1-Pentene	C ₅ H ₁₀	-165.27	16.82
Methylbenzene (Toluene)	C ₇ H ₈	-94.991	17.171	<i>cis</i> -2-pentene	C ₅ H ₁₀	-151.363	24.239
Ethylbenzene	C ₈ H ₁₀	-94.950	20.629	<i>trans</i> -2-pentene	C ₅ H ₁₀	-140.235	26.536
<i>o</i> -Xylene	C ₈ H ₁₀	-25.187	30.614	2-Methyl-1-butene	C ₅ H ₁₀	-137.560	26.879
<i>m</i> -Xylene	C ₈ H ₁₀	-47.872	26.045	3-Methyl-1-butene	C ₅ H ₁₀	-168.500	18.009
<i>p</i> -Xylene	C ₈ H ₁₀	+13.263	38.526	2-Methyl-2-butene	C ₅ H ₁₀	-133.780	25.738
<i>n</i> -Propylbenzene	C ₉ H ₁₂	-99.500	16.97	Acetylenes			
Isopropylbenzene	C ₉ H ₁₂	-96.028	19.22	Acetylene	C ₂ H ₂	-81.5	23.04
1-Methyl-2-ethylbenzene	C ₉ H ₁₂	-80.833	21.13	2-Butyne (dimethylacetylene)	C ₄ H ₆	-132.23	40.808
Nonhydrocarbon compounds	Formula	mp, °C	Heat of fusion, cal/g	Nonhydrocarbon compounds	Formula	mp, °C	Heat of fusion, cal/g
Acetic acid	C ₂ H ₄ O ₂	16.7	46.68	Butyl alcohol (<i>n</i> -)	C ₄ H ₁₀ O	-89.2	29.93
Acetone	C ₃ H ₆ O	-95.5	23.42	(<i>t</i> -)	C ₄ H ₁₀ O	25.4	21.88
Acrylic acid	C ₃ H ₄ O ₂	12.3	37.03	Butyric acid (<i>n</i> -)	C ₄ H ₈ O ₂	-5.7	30.04
Allo-cinnamic acid	C ₉ H ₈ O ₂	68	27.35	Capric acid (<i>n</i> -)	C ₁₀ H ₂₀ O ₂	31.99	38.87
Aminobenzoic acid (<i>o</i> -)	C ₇ H ₇ NO ₂	145	35.48	Caprylic acid (<i>n</i> -)	C ₈ H ₁₆ O ₂	16.3	35.40
(<i>m</i> -)	C ₇ H ₇ NO ₂	179.5	38.03	Carbazole	C ₁₂ H ₉ N	243	42.05
(<i>p</i> -)	C ₇ H ₇ NO ₂	188.5	36.46	Carbon tetrachloride	CCl ₄	-22.8	41.57
Amyl alcohol	C ₈ H ₁₈ O	-78.9	26.65	Carvoxime (<i>d</i> -)	C ₁₀ H ₁₅ NO	71.5	23.29
Anethole	C ₁₀ H ₁₂ O	22.5	25.80	(<i>l</i> -)	C ₁₀ H ₁₅ NO	71	23.41
Aniline	C ₆ H ₅ NH ₂	-6.3	27.09	(<i>dl</i> -)	C ₁₀ H ₁₅ NO	91	24.61
Anthraquinone	C ₁₄ H ₈ O ₂	284.8	37.48	Cetyl alcohol	C ₁₆ H ₃₄ O	49.27	33.80
Apiol	C ₁₂ H ₁₄ O ₄	29.5	25.80	Chloroacetic acid (α -)	C ₂ H ₃ ClO ₂	61.2	31.06
Azobenzene	C ₁₂ H ₁₀ N ₂	67.1	28.91	(β -)	C ₂ H ₃ ClO ₂	56	35.12
Azoxybenzene	C ₁₂ H ₁₀ N ₂ O	36	21.62	Chloral alcoholate	C ₂ H ₃ Cl ₂ O ₂	9	24.03
Benzil	C ₁₄ H ₁₀ O ₂	95.2	22.15	hydrate	C ₂ H ₃ Cl ₂ O ₂	47.4	33.18
Benzoic acid	C ₇ H ₆ O ₂	122.45	33.90	Chloroaniline (<i>p</i> -)	C ₆ H ₆ ClN	71	37.15
Benzophenone	C ₁₃ H ₁₀ O	47.85	23.53	Chlorobenzoic acid (<i>o</i> -)	C ₇ H ₅ ClO ₂	140.2	39.30
Benzylaniline	C ₁₅ H ₁₃ N	32.37	21.86	(<i>m</i> -)	C ₇ H ₅ ClO ₂	154.25	36.41
Bromocamphor	C ₁₀ H ₁₅ BrO	78	41.57	(<i>p</i> -)	C ₇ H ₅ ClO ₂	239.7	49.21
Bromochlorobenzene (<i>o</i> -)	C ₆ H ₄ BrCl	-12.6	15.41	Chloronitrobenzene (<i>m</i> -)	C ₆ H ₄ ClNO ₂	44.4	29.38
(<i>m</i> -)	C ₆ H ₄ BrCl	-21.2	15.29	(<i>p</i> -)	C ₆ H ₄ ClNO ₂	83.5	31.51
(<i>p</i> -)	C ₆ H ₄ BrCl	64.6	23.41	Cinnamic acid	C ₉ H ₈ O ₂	133	36.50
Bromiodobenzene (<i>o</i> -)	C ₆ H ₄ BrI	21	12.18	anhydride	C ₁₈ H ₁₄ O ₃	48	28.14
(<i>m</i> -)	C ₆ H ₄ BrI	9.3	10.27	Cresol (<i>p</i> -)	C ₇ H ₈ O	34.6	26.28
(<i>p</i> -)	C ₆ H ₄ BrI	90.1	16.60	Crotonic acid (α -)	C ₄ H ₆ O ₂	72	25.32
Bromol hydrate	C ₂ H ₃ Br ₂ O ₂	46	16.90	(<i>cis</i> -)	C ₄ H ₆ O ₂	71.2	34.90
Bromophenol (<i>p</i> -)	C ₆ H ₅ BrO	63.5	20.50	Cyanamide	CH ₃ N ₂	44	49.81
Bromotoluene (<i>p</i> -)	C ₇ H ₇ Br	28	20.86	Cyclohexanol	C ₆ H ₁₂ O	25.46	4.19

TABLE 2-149 Heats of Fusion of Organic Compounds (Concluded)

Nonhydrocarbon compounds	Formula	mp, °C	Heat of fusion, cal/g	Nonhydrocarbon compounds	Formula	mp, °C	Heat of fusion, cal/g
Dibromobenzene (<i>o</i> -)	C ₆ H ₄ Br ₂	1.8	12.78	Naphthol (α -)	C ₁₀ H ₈ O	95.0	38.94
(<i>m</i> -)	C ₆ H ₄ Br ₂	-6.9	13.38	(β -)	C ₁₀ H ₈ O	120.6	31.30
(<i>p</i> -)	C ₆ H ₄ Br ₂	86	20.55	Naphthylamine (α -)	C ₁₀ H ₉ N	50	22.34
Dibromophenol (2, 4-)	C ₆ H ₃ Br ₂ O	12	13.97	Nitroaniline (<i>o</i> -)	C ₆ H ₅ N ₂ O ₂	71.2	27.88
Dichloroacetic acid	C ₂ H ₂ Cl ₂ O ₂	-4(?)	14.21	(<i>m</i> -)	C ₆ H ₅ N ₂ O ₂	114.0	40.97
Dichlorobenzene (<i>o</i> -)	C ₆ H ₄ Cl ₂	-16.7	21.02	(<i>p</i> -)	C ₆ H ₅ N ₂ O ₂	147.3	36.46
(<i>m</i> -)	C ₆ H ₄ Cl ₂	-24.8	20.55	Nitrobenzene	C ₆ H ₅ NO ₂	5.85	22.52
(<i>p</i> -)	C ₆ H ₄ Cl ₂	53.13	29.67	Nitrobenzoic acid (<i>o</i> -)	C ₇ H ₅ NO ₄	145.8	40.06
Dihydroxybenzene (<i>o</i> -)	C ₆ H ₆ O ₂	104.3	49.40	(<i>m</i> -)	C ₇ H ₅ NO ₄	141.1	27.59
(<i>m</i> -)	C ₆ H ₆ O ₂	109.65	46.20	(<i>p</i> -)	C ₇ H ₅ NO ₄	239.2	52.80
(<i>p</i> -)	C ₆ H ₆ O ₂	172.3	58.77	Nitronaphthalene	C ₁₀ H ₇ NO ₂	56.7	25.44
Di-iodobenzene (<i>o</i> -)	C ₆ H ₄ I ₂	23.4	10.15	Nitrophenol (<i>o</i> -)	C ₆ H ₅ NO ₃	45.13	26.76
(<i>m</i> -)	C ₆ H ₄ I ₂	34.2	11.54				
(<i>p</i> -)	C ₆ H ₄ I ₂	129	16.20	Palmitic acid	C ₁₆ H ₃₂ O ₂	61.82	39.18
Dimethyl tartrate (<i>dl</i> -)	C ₈ H ₁₀ O ₆	87	35.12	Paraldehyde	C ₆ H ₁₂ O ₃	10.5	25.02
(<i>d</i> -)	C ₈ H ₁₀ O ₆	49	21.50	Pelargic acid (<i>n</i> -) (β -)	C ₉ H ₁₈ O ₂	70	39.04
pyrone	C ₅ H ₄ O ₂	132	56.14	Pelargonic acid (<i>n</i> -) (α -)	C ₉ H ₁₈ O ₂	12.35	30.63
Dinitrobenzene (<i>o</i> -)	C ₆ H ₄ N ₂ O ₄	116.93	32.25	Phenol	C ₆ H ₆ O	40.92	29.03
(<i>m</i> -)	C ₆ H ₄ N ₂ O ₄	89.7	24.70	Phenylacetic acid	C ₈ H ₈ O ₂	76.7	25.44
(<i>p</i> -)	C ₆ H ₄ N ₂ O ₄	173.5	39.99	Phenylhydrazine	C ₈ H ₈ N ₂	19.6	36.31
Dinitrotoluene (2, 4-)	C ₇ H ₆ N ₂ O ₄	70.14	26.40	Propyl ether (<i>n</i>)	C ₆ H ₁₄ O	-126.1	20.66
Dioxane	C ₆ H ₈ O ₂	11.0	34.85				
Diphenyl amine	C ₁₂ H ₁₁ N	52.98	25.23	Quinone	C ₆ H ₄ O ₂	115.7	40.85
				Stearic acid	C ₁₈ H ₃₆ O ₂	68.82	47.54
Elaidic acid	C ₁₈ H ₃₄ O ₂	44.4	52.08	Succinic anhydride	C ₆ H ₄ O ₃	119	48.74
Ethyl acetate	C ₄ H ₈ O ₂	83.8	28.43	Succinonitrile	C ₄ H ₄ N ₂	54.5	11.71
alcohol	C ₂ H ₆ O	-114.4	25.76				
Ethylene dibromide	C ₂ H ₄ Br ₂	10.012	13.52	Tetrachloroethylene (<i>o</i> -)	C ₂ Cl ₄	86	21.02
Ethyl ether	C ₄ H ₁₀ O	-116.3	23.54	(<i>p</i> -)	C ₂ Cl ₄	95	22.10
Formic acid	CH ₂ O ₂	8.40	58.89	Thiophene	C ₄ H ₄ S	-39.4	14.11
				Thiosinamine	C ₄ H ₄ N ₂ S	77	33.45
Glutaric acid	C ₆ H ₈ O ₄	97.5	37.39	Thymol	C ₁₀ H ₁₄ O	51.5	27.47
Glycerol	C ₃ H ₈ O ₃	18.07	47.49	Toluic acid (<i>o</i> -)	C ₈ H ₈ O ₂	103.7	35.40
Glycol, ethylene	C ₂ H ₆ O ₂	-11.5	43.26	(<i>m</i> -)	C ₈ H ₈ O ₂	108.75	27.59
				(<i>p</i> -)	C ₈ H ₈ O ₂	179.6	39.90
Hydrazo benzene	C ₁₂ H ₁₂ N ₂	134	22.89	Toluidine (<i>p</i> -)	C ₇ H ₉ N	43.3	39.90
Hydrocinnamic acid	C ₉ H ₁₀ O ₂	48	28.14	Tribromophenol (2, 4, 6-)	C ₆ H ₃ Br ₃ O	93	13.38
Hydroxyacetanilide	C ₈ H ₉ NO ₂	91.3	33.59	Trichloroacetic acid	C ₂ HCl ₃ O ₂	57.5	8.60
				Trinitroglycerol	C ₃ H ₅ N ₃ O ₉	12.3	23.02
Iodotoluene (<i>p</i> -)	C ₇ H ₇ I	34	18.75	Trinitrotoluene (2, 4, 6-)	C ₇ H ₅ N ₃ O ₆	80.83	22.34
Isopropyl alcohol	C ₃ H ₈ O	-88.5	21.08	Tristearin	C ₅₇ H ₁₁₆ O ₆	70.8, 54.5	45.63
ether	C ₆ H ₁₄ O	-86.8	25.79				
				Undecylic acid (α -) (<i>n</i> -)	C ₁₁ H ₂₂ O ₂	28.25	32.20
Lauric acid (<i>n</i> -)	C ₁₂ H ₂₄ O ₂	43.22	43.72	(β -) (<i>n</i> -)	C ₁₁ H ₂₂ O ₂		42.91
Levulinic acid	C ₅ H ₈ O ₃	33	18.97	Urethane	C ₃ H ₇ NO ₂	48.7	40.85
Menthol (<i>l</i> -) (α)	C ₁₀ H ₂₀ O	43.5	18.63	Veratrol	C ₈ H ₁₀ O ₂	22.5	27.45
Methyl alcohol	CH ₄ O	-97.8	23.7				
Myristic acid	C ₁₄ H ₂₈ O ₂	53.86	47.49	Xylene dibromide (<i>o</i> -)	C ₈ H ₈ Br ₂	95	24.25
Methyl cinnamate	C ₁₀ H ₁₀ O ₂	36	26.53	(<i>m</i> -)	C ₈ H ₈ Br ₂	77	21.45
fumarate	C ₈ H ₈ O ₄	102	57.93	dichloride (<i>o</i> -)	C ₈ H ₆ Cl ₂	55	29.03
oxalate	C ₄ H ₄ O ₄	54.35	42.64	(<i>m</i> -)	C ₈ H ₆ Cl ₂	34	26.64
phenylpropionate	C ₁₀ H ₈ O ₂	18	22.86	(<i>p</i> -)	C ₈ H ₆ Cl ₂	100	32.73
succinate	C ₈ H ₁₀ O ₄	19.5	35.72				

TABLE 2-150 Heats of Vaporization of Inorganic and Organic Liquids (J/kmol)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1 × 1E-07	C2	C3	C4	$T_{\min}$, K	ΔH_v at $T_{\min} \times 1E-07$	$T_{\max}$, K	ΔH_v at $T_{\max}$
1	Acetaldehyde	C ₂ H ₄ O	75-07-0	44.053	3.8366	0.40081			150.15	3.2828	466.00	0
2	Acetamide	C ₂ H ₅ NO	60-35-5	59.067	8.107	0.42			353.15	6.2386	761.00	0
3	Acetic acid	C ₂ H ₄ O ₂	64-19-7	60.052	4.0179	2.6037	-5.0031	2.7069	289.81	2.3412	591.95	0
4	Acetic anhydride	C ₄ H ₆ O ₃	108-24-7	102.089	6.352	0.3986			200.15	5.4139	606.00	0
5	Acetone	C ₃ H ₆ O	67-64-1	58.079	4.215	0.3397			178.45	3.6390	508.20	0
6	Acetonitrile	C ₂ H ₃ N	75-05-8	41.052	4.3511	0.34765			229.32	3.5996	545.50	0
7	Acetylene	C ₂ H ₂	74-86-2	26.037	2.3214	0.35938			192.40	1.6333	308.30	0
8	Acrolein	C ₃ H ₄ O	107-02-8	56.063	3.8736	0.29335			185.45	3.3881	506.00	0
9	Acrylic acid	C ₃ H ₄ O ₂	79-10-7	72.063	4.3756	2.2571	-4.5116	2.5738	286.15	2.7965	615.00	0
10	Acrylonitrile	C ₃ H ₃ N	107-13-1	53.063	4.155	0.2733			189.63	3.6866	535.00	0
11	Air	Mixture	132259-10-0	28.960	0.8474	0.3822			59.15	0.6759	132.45	0
12	Ammonia	H ₃ N	7664-41-7	17.031	3.1523	0.3914	-0.2289	0.2309	195.41	2.5298	405.65	0
13	Anisole	C ₇ H ₈ O	100-66-3	108.138	5.8662	0.37127			235.65	4.9560	645.60	0
14	Argon	Ar	7440-37-1	39.948	0.87308	0.3526			83.78	0.6561	150.86	0
15	Benzamide	C ₇ H ₇ NO	55-21-0	121.137	8.7809	0.1933	0.30877	-0.14162	403.00	7.1286	824.00	0
16	Benzene	C ₆ H ₆	71-43-2	78.112	4.5346	0.39053			278.68	3.4705	562.05	0
17	Benzenethiol	C ₆ H ₆ S	108-98-5	110.177	6.225	0.4412			258.27	5.0597	689.00	0
18	Benzoic acid	C ₇ H ₆ O ₂	65-85-0	122.121	10.19	0.478			395.45	7.1277	751.00	0
19	Benzonitrile	C ₇ H ₅ N	100-47-0	103.121	6.8077	0.63344	-0.27365		260.40	5.3147	699.35	0
20	Benzophenone	C ₁₃ H ₁₀ O	119-61-9	182.218	10.523	0.87091	-0.45568		321.35	7.4895	830.00	0
21	Benzyl alcohol	C ₇ H ₈ O	100-51-6	108.138	8.4762	0.35251	0.43853	-0.3026	257.85	6.8800	720.15	0
22	Benzyl ethyl ether	C ₉ H ₁₂ O	539-30-0	136.191	6.228	0.3411			275.65	5.1829	662.00	0
23	Benzyl mercaptan	C ₇ H ₈ S	100-53-8	124.203	6.9642	0.44354			243.95	5.7930	718.00	0
24	Biphenyl	C ₁₂ H ₁₀	92-52-4	154.208	7.635	0.39182			342.20	6.0719	773.00	0
25	Bromine	Br ₂	7726-95-6	159.808	4	0.351			265.85	3.2323	584.15	0
26	Bromobenzene	C ₆ H ₅ Br	108-86-1	157.008	5.552	0.37694			242.43	4.6875	670.15	0
27	Bromoethane	C ₂ H ₅ Br	74-96-4	108.965	3.9004	0.38012			154.55	3.3933	503.80	0
28	Bromomethane	CH ₃ Br	74-83-9	94.939	3.169	0.3015			179.47	2.7379	467.00	0
29	1,2-Butadiene	C ₄ H ₆	590-19-2	54.090	3.522	0.395			136.95	3.0540	452.00	0
30	1,3-Butadiene	C ₄ H ₆	106-99-0	54.090	3.2632	0.3701			164.25	2.7235	425.00	0
31	Butane	C ₄ H ₁₀	106-97-8	58.122	3.6238	0.8337	-0.82274	0.39613	134.86	2.8684	425.12	0
32	1,2-Butanediol	C ₄ H ₁₀ O ₂	584-03-2	90.121	8.9754	0.45316			220.00	7.5185	680.00	0
33	1,3-Butanediol	C ₄ H ₁₀ O ₂	107-88-0	90.121	9.2247	0.42442			196.15	7.9759	676.00	0
34	1-Butanol	C ₄ H ₁₀ O	71-36-3	74.122	7.1274	0.0483	0.8966	-0.5116	183.85	6.3643	563.10	0
35	2-Butanol	C ₄ H ₁₀ O	78-92-2	74.122	7.9227	0.58361	0.02016	-0.08654	158.45	6.4607	535.90	0
36	1-Butene	C ₄ H ₈	106-98-9	56.106	3.3774	0.5107	-0.17304	0.05181	87.80	3.0197	419.50	0
37	cis-2-Butene	C ₄ H ₈	590-18-1	56.106	3.4358	0.38004			134.26	2.9867	435.50	0
38	trans-2-Butene	C ₄ H ₈	624-64-6	56.106	3.3191	0.36968			167.62	2.7630	428.60	0
39	Butyl acetate	C ₆ H ₁₂ O ₂	123-86-4	116.158	5.8276	0.38854			199.65	4.9384	575.40	0
40	Butylbenzene	C ₁₀ H ₁₄	104-51-8	134.218	6.3487	0.38222			185.30	5.5979	660.50	0
41	Butyl mercaptan	C ₄ H ₁₀ S	109-79-5	90.187	4.9702	0.41199			157.46	4.3505	570.10	0
42	sec-Butyl mercaptan	C ₄ H ₁₀ S	513-53-1	90.187	4.6432	0.399			133.02	4.1614	554.00	0
43	1-Butyne	C ₄ H ₆	107-00-6	54.090	3.6972	0.39168			147.43	3.1511	440.00	0
44	Butyraldehyde	C ₄ H ₈ O	123-72-8	72.106	4.6403	0.3849			176.75	3.9797	537.20	0
45	Butyric acid	C ₄ H ₈ O ₂	107-92-6	88.105	6.1947	1.6524	-2.8505	1.6285	250.00	4.1619	615.70	0
46	Butyronitrile	C ₄ H ₇ N	109-74-0	69.105	5.22	0.165	0.6692	-0.539	161.25	4.7223	582.25	0
47	Carbon dioxide	CO ₂	124-38-9	44.010	2.173	0.382	-0.4339	0.42213	216.58	1.5202	304.21	0
48	Carbon disulfide	CS ₂	75-15-0	76.141	3.496	0.2986			161.11	3.1537	552.00	0
49	Carbon monoxide	CO	630-08-0	28.010	0.8585	0.4921	-0.326	0.2231	68.13	0.6517	132.50	915,280
50	Carbon tetrachloride	CCl ₄	56-23-5	153.823	4.3252	0.37688			250.33	3.4528	556.35	0
51	Carbon tetrafluoride	CF ₄	75-73-0	88.004	1.9311	0.94983	-1.0615	0.51894	89.56	1.4215	227.51	0
52	Chlorine	Cl ₂	7782-50-5	70.906	3.068	0.8458	-0.9001	0.453	172.12	2.2878	417.15	0
53	Chlorobenzene	C ₆ H ₅ Cl	108-90-7	112.557	5.148	0.36614			227.95	4.3707	632.35	0
54	Chloroethane	C ₂ H ₅ Cl	75-00-3	64.514	3.524	0.3652			134.80	3.1052	460.35	0
55	Chloroform	CHCl ₃	67-66-3	119.378	4.186	0.3584			209.63	3.5047	536.40	0
56	Chloromethane	CH ₃ Cl	74-87-3	50.488	2.9745	0.353			175.43	2.4520	416.25	0
57	1-Chloropropane	C ₃ H ₇ Cl	540-54-5	78.541	3.989	0.37956			150.35	3.4862	503.15	0
58	2-Chloropropane	C ₃ H ₇ Cl	75-29-6	78.541	3.8871	0.38043			155.97	3.3586	489.00	0
59	m-Cresol	C ₇ H ₈ O	108-39-4	108.138	8.0082	0.45314			285.39	6.3326	705.85	0

60	<i>o</i> -Cresol	C ₇ H ₈ O	95-48-7	108.138	7.1979	0.40317			304.19	5.7135	697.55	0
61	<i>p</i> -Cresol	C ₇ H ₈ O	106-44-5	108.138	8.4942	0.50234			307.93	6.3649	704.65	0
62	Cumene	C ₉ H ₁₂	98-82-8	120.192	5.766	0.38939			177.14	5.0717	631.00	0
63	Cyanogen	C ₂ N ₂	460-19-5	52.035	3.384	0.3707			245.25	2.3803	400.15	0
64	Cyclobutane	C ₄ H ₈	287-23-0	56.106	3.334	0.3395			182.48	2.8083	459.93	0
65	Cyclohexane	C ₆ H ₁₂	110-82-7	84.159	4.4902	0.39881			279.69	3.3920	553.80	0
66	Cyclohexanol	C ₆ H ₁₂ O	108-93-0	100.159	9.1791	0.6382			296.60	6.2221	650.10	0
67	Cyclohexanone	C ₆ H ₁₀ O	108-94-1	98.143	5.6772	0.37431			242.00	4.7739	653.00	0
68	Cyclohexene	C ₆ H ₁₀	110-83-8	82.144	4.4405	0.37479			169.67	3.8791	560.40	0
69	Cyclopentane	C ₅ H ₁₀	287-92-3	70.133	3.8911	0.36111			179.28	3.3299	511.70	0
70	Cyclopentene	C ₅ H ₈	142-29-0	68.117	3.8107	0.3543			138.13	3.4046	507.00	0
71	Cyclopropane	C ₃ H ₆	75-19-4	42.080	2.7672	0.35588			145.59	2.3532	398.00	0
72	Cyclohexyl mercaptan	C ₆ H ₁₂ S	1569-69-3	116.224	5.6067	0.38729			189.64	4.9220	664.00	0
73	Decanal	C ₁₀ H ₂₀ O	112-31-2	156.265	7.9073	0.4129			267.15	6.4201	674.20	0
74	Decane	C ₁₀ H ₂₂	124-18-5	142.282	6.6126	0.39797			243.51	5.4168	617.70	0
75	Decanoic acid	C ₁₀ H ₂₀ O ₂	334-48-5	172.265	13.107	1.0674	-0.97372	0.40491	304.55	8.7931	722.10	0
76	1-Decanol	C ₁₀ H ₂₂ O	112-30-1	158.281	7.9041	-1.36	4.0854	-2.3871	280.05	8.2959	688.00	0
77	1-Decene	C ₁₀ H ₂₀	872-05-9	140.266	6.6985	0.76944	-0.79975	0.42379	206.89	5.3524	616.60	0
78	Decyl mercaptan	C ₁₀ H ₂₂ S	143-10-2	174.347	8.0617	0.41045			247.56	6.7308	696.00	0
79	1-Decyne	C ₁₀ H ₁₈	764-93-2	138.250	6.9461	0.42109			229.15	5.7192	619.85	0
80	Deuterium	D ₂	7782-39-0	4.032	0.1657	0.352			18.73	0.1309	38.35	0
81	1,1-Dibromoethane	C ₂ H ₄ Br ₂	557-91-5	187.861	5.712	0.5255			210.15	4.6111	628.00	0
82	1,2-Dibromoethane	C ₂ H ₄ Br ₂	106-93-4	187.861	5.37	0.416			282.85	4.2346	650.15	0
83	Dibromomethane	CH ₂ Br ₂	74-95-3	173.835	4.82	0.3771			220.60	4.0709	611.00	0
84	Dibutyl ether	C ₈ H ₁₈ O	142-96-1	130.228	5.9616	0.38833			175.30	5.1902	584.10	0
85	<i>m</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	541-73-1	147.002	5.6899	0.35765			248.39	4.8419	683.95	0
86	<i>o</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	95-50-1	147.002	6.2117	0.42845			256.15	5.1191	705.00	0
87	<i>p</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	106-46-7	147.002	5.9765	0.38559			326.14	4.6573	684.75	0
88	1,1-Dichloroethane	C ₂ H ₄ Cl ₂	75-34-3	98.959	4.2117	0.36927			176.19	3.6189	523.00	0
89	1,2-Dichloroethane	C ₂ H ₄ Cl ₂	107-06-2	98.959	4.5507	0.34444			237.49	3.7657	561.60	0
90	Dichloromethane	CH ₂ Cl ₂	75-09-2	84.933	4.186	0.4092			178.01	3.5116	510.00	0
91	1,1-Dichloropropane	C ₃ H ₆ Cl ₂	78-99-9	112.986	4.774	0.39204			200.00	4.0147	560.00	0
92	1,2-Dichloropropane	C ₃ H ₆ Cl ₂	78-87-5	112.986	4.675	0.36529			172.71	4.0997	572.00	0
93	Diethanol amine	C ₄ H ₁₁ NO ₂	111-42-2	105.136	10.154	0.3403			301.15	8.4908	736.60	0
94	Diethyl amine	C ₄ H ₁₁ N	109-89-7	73.137	4.6133	0.42628			223.35	3.5761	496.60	0
95	Diethyl ether	C ₄ H ₁₀ O	60-29-7	74.122	4.06	0.3868			156.85	3.4651	466.70	0
96	Diethyl sulfide	C ₄ H ₁₀ S	352-93-2	90.187	4.7659	0.37987			169.20	4.1537	557.15	0
97	1,1-Difluoroethane	C ₂ H ₄ F ₂	75-37-6	66.050	3.2312	0.37653			154.56	2.6659	386.44	0
98	1,2-Difluoroethane	C ₂ H ₄ F ₂	624-72-6	66.050	3.4552	0.3499			215.00	2.7427	445.00	0
99	Difluoromethane	CH ₂ F ₂	75-10-5	52.023	2.8081	0.3364			136.95	2.3781	351.26	0
100	Di-isopropyl amine	C ₆ H ₁₅ N	108-18-9	101.190	5.007	0.4362			176.85	4.1823	523.10	0
101	Di-isopropyl ether	C ₆ H ₁₄ O	108-20-3	102.175	4.6117	0.4			187.65	3.8207	500.05	0
102	Di-isopropyl ketone	C ₇ H ₁₄ O	565-80-0	114.185	5.0256	0.29611			204.81	4.4125	576.00	0
103	1,1-Dimethoxyethane	C ₄ H ₁₀ O ₂	534-15-6	90.121	4.3872	0.56226	-0.60662	0.4202	159.95	3.7528	507.80	0
104	1,2-Dimethoxypropane	C ₅ H ₁₂ O ₂	7778-85-0	104.148	4.7999	0.30724	-0.024545	0.091361	226.10	4.0557	543.00	0
105	Dimethyl acetylene	C ₄ H ₆	503-17-3	54.090	3.856	0.3737			240.91	2.9557	473.20	0
106	Dimethyl amine	C ₃ H ₇ N	124-40-3	45.084	4.09	0.42005			180.96	3.2678	437.20	0
107	2,3-Dimethylbutane	C ₆ H ₁₄	79-29-8	86.175	4.1509	0.38383			145.19	3.6388	500.00	0
108	1,1-Dimethylcyclohexane	C ₈ H ₁₆	590-66-9	112.213	5.0402	0.4036			239.66	4.0862	591.15	0
109	<i>cis</i> -1,2-Dimethylcyclohexane	C ₈ H ₁₆	2207-01-4	112.213	5.2852	0.41607			223.16	4.3662	606.15	0
110	<i>trans</i> -1,2-Dimethylcyclohexane	C ₈ H ₁₆	6876-23-9	112.213	5.1194	0.405			184.99	4.4043	596.15	0
111	Dimethyl disulfide	C ₂ H ₆ S ₂	624-92-0	94.199	4.9825	0.3958			188.44	4.3108	615.00	0
112	Dimethyl ether	C ₂ H ₆ O	115-10-6	46.068	2.994	0.3505			131.65	2.6032	400.10	0
113	<i>N,N</i> -Dimethyl formamide	C ₃ H ₇ NO	68-12-2	73.094	5.9217	0.37996			212.72	5.0931	649.60	0
114	2,3-Dimethylpentane	C ₇ H ₁₆	565-59-3	100.202	4.6533	0.37577			160.00	4.0745	537.30	0
115	Dimethyl phthalate	C ₁₀ H ₁₀ O ₄	131-11-3	194.184	8.1578	0.29346			274.18	7.1632	766.00	0
116	Dimethylsilane	C ₂ H ₆ Si	1111-74-6	60.170	2.8365	0.35393			122.93	2.4928	402.00	0
117	Dimethyl sulfide	C ₂ H ₆ S	75-18-3	62.134	3.9022	0.37731			174.88	3.3213	503.04	0
118	Dimethyl sulfoxide	C ₂ H ₆ OS	67-68-5	78.133	6.629	0.4084			291.67	5.3804	729.00	0
119	Dimethyl terephthalate	C ₁₀ H ₁₀ O ₄	120-61-6	194.184	7.236	0.2424			413.80	6.0070	772.00	0
120	1,4-Dioxane	C ₄ H ₈ O ₂	123-91-1	88.105	5.051	0.3791			284.95	3.9263	587.00	0
121	Diphenyl ether	C ₁₂ H ₁₀ O	101-84-8	170.207	6.8243	0.30877			300.03	5.8546	766.80	0
122	Dipropyl amine	C ₆ H ₁₅ N	142-84-7	101.190	5.428	0.3665			210.15	4.5500	550.00	0

TABLE 2-150 Heats of Vaporization of Inorganic and Organic Liquids (J/kmol) (Continued)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1 × 1E-07	C2	C3	C4	$T_{\min}$, K	ΔH_v at $T_{\min} \times 1E-07$	$T_{\max}$, K	ΔH_v at $T_{\max}$
123	Dodecane	C ₁₂ H ₂₆	112-40-3	170.335	7.7337	0.40681			263.57	6.2802	658.00	0
124	Eicosane	C ₂₀ H ₄₂	112-95-8	282.547	12.86	0.50351	0.32986	-0.42184	309.58	9.5933	768.00	0
125	Ethane	C ₂ H ₆	74-84-0	30.069	2.1091	0.60646	-0.55492	0.32799	90.35	1.7879	305.32	0
126	Ethanol	C ₂ H ₆ O	64-17-5	46.068	5.5789	0.31245			159.05	4.9694	514.00	0
127	Ethyl acetate	C ₄ H ₈ O ₂	141-78-6	88.105	4.933	0.3847			189.60	4.1490	523.30	0
128	Ethyl amine	C ₂ H ₇ N	75-04-7	45.084	4.275	0.5857	-0.332	0.169	192.15	3.2955	456.15	0
129	Ethylbenzene	C ₈ H ₁₀	100-41-4	106.165	5.4805	0.39524			178.20	4.7900	617.15	0
130	Ethyl benzoate	C ₉ H ₁₀ O ₂	93-89-0	150.175	6.7093	0.33273			238.45	5.8382	698.00	0
131	2-Ethyl butanoic acid	C ₆ H ₁₂ O ₂	88-09-5	116.158	7.898	0.39445			258.15	6.4816	655.00	0
132	Ethyl butyrate	C ₆ H ₁₂ O ₂	105-54-4	116.158	5.6419	0.37985			175.15	4.9090	571.00	0
133	Ethylcyclohexane	C ₈ H ₁₆	1678-91-7	112.213	5.3832	0.41763			161.84	4.7318	609.15	0
134	Ethylcyclopentane	C ₇ H ₁₄	1640-89-7	98.186	4.8287	0.37804			134.71	4.3603	569.50	0
135	Ethylene	C ₂ H ₄	74-85-1	28.053	1.8844	0.36485			104.00	1.5936	282.34	0
136	Ethylenediamine	C ₂ H ₆ N ₂	107-15-3	60.098	5.7521	0.34513			284.29	4.5918	593.00	0
137	Ethylene glycol	C ₂ H ₆ O ₂	107-21-1	62.068	8.3518	0.42625			260.15	6.8989	720.00	0
138	Ethyleneimine	C ₂ H ₅ N	151-56-4	43.068	4.94	0.466			195.20	4.0022	537.00	0
139	Ethylene oxide	C ₂ H ₄ O	75-21-8	44.053	3.6652	0.37878			160.65	3.1271	469.15	0
140	Ethyl formate	C ₃ H ₆ O ₂	109-94-4	74.079	4.5909	0.4123			193.55	3.7679	508.40	0
141	2-Ethyl hexanoic acid	C ₈ H ₁₆ O ₂	149-57-5	144.211	11.184	0.86189	-0.47845	0.048646	235.00	8.2832	674.60	0
142	Ethylhexyl ether	C ₈ H ₁₈ O	5756-43-4	130.228	6.2786	0.39513			180.00	5.4262	583.00	0
143	Ethylisopropyl ether	C ₅ H ₁₂ O	625-54-7	88.148	4.258	0.37221			140.00	3.7556	489.00	0
144	Ethylisopropyl ketone	C ₆ H ₁₂ O	565-69-5	100.159	5.2207	0.34893			204.15	4.4677	567.00	0
145	Ethyl mercaptan	C ₂ H ₆ S	75-08-1	62.134	3.844	0.37534			125.26	3.4489	499.15	0
146	Ethyl propionate	C ₅ H ₁₀ O ₂	105-37-3	102.132	5.3325	0.401			199.25	4.4449	546.00	0
147	Ethylpropyl ether	C ₅ H ₁₂ O	628-32-0	88.148	5.438	0.60624			145.65	4.4140	500.23	0
148	Ethyltrichlorosilane	C ₂ H ₅ Cl ₃ Si	115-21-9	163.506	4.9482	0.39871			167.55	4.2942	559.95	0
149	Fluorine	F ₂	7782-41-4	37.997	0.88757	0.34072			53.48	0.7578	144.12	0
150	Fluorobenzene	C ₆ H ₅ F	462-06-6	96.102	4.582	0.3717			230.94	3.7605	560.09	0
151	Fluoroethane	C ₂ H ₅ F	353-36-6	48.060	2.7617	0.32162			129.95	2.4089	375.31	0
152	Fluoromethane	CH ₃ F	593-53-3	34.033	2.4708	0.37014			131.35	2.0276	317.42	0
153	Formaldehyde	CH ₂ O	50-00-0	30.026	3.076	0.2954			181.15	2.5863	408.00	0
154	Formamide	CH ₃ NO	75-12-7	45.041	7.358	0.3564			275.70	6.2844	771.00	0
155	Formic acid	CH ₂ O ₂	64-18-6	46.026	2.3195	1.9091	-5.0003	3.2641	250.00	1.8865	588.00	0
156	Furan	C ₄ H ₄ O	110-00-9	68.074	4.005	0.3995			196.29	3.2647	490.15	0
157	Helium-4	He	7440-59-7	4.003	0.012504	1.3038	-2.6954	1.7098	2.20	0.0097	5.20	0
158	Heptadecane	C ₁₇ H ₃₆	629-78-7	240.468	10.473	0.4374			295.13	8.3699	736.00	0
159	Heptanal	C ₇ H ₁₄ O	111-71-7	114.185	5.956	0.36474			229.80	5.0248	616.80	0
160	Heptane	C ₇ H ₁₆	142-82-5	100.202	5.0014	0.38795			182.57	4.2619	540.20	0
161	Heptanoic acid	C ₇ H ₁₄ O ₂	111-14-8	130.185	11.274	0.86047	-0.40661	-0.012644	265.83	7.9579	677.30	0
162	1-Heptanol	C ₇ H ₁₆ O	111-70-6	116.201	7.0236	-1.3652	3.987	-2.2545	239.15	7.6498	632.30	0
163	2-Heptanol	C ₇ H ₁₆ O	543-49-7	116.201	9.6433	0.783	-0.27273	0.038495	230.00	6.9638	608.30	0
164	3-Heptanone	C ₇ H ₁₄ O	106-35-4	114.185	6.3357	0.42167			234.15	5.1579	606.60	0
165	2-Heptanone	C ₇ H ₁₄ O	110-43-0	114.185	6.1425	0.39802			238.15	5.0471	611.40	0
166	1-Heptene	C ₇ H ₁₄	592-76-7	98.186	4.9437	0.35428	0.22149	-0.2353	154.12	4.3208	537.40	0
167	Heptyl mercaptan	C ₇ H ₁₆ S	1639-09-4	132.267	6.5473	0.40968			229.92	5.4656	645.00	0
168	1-Heptyne	C ₇ H ₁₂	628-71-7	96.170	4.8222	0.33858			192.22	4.1647	547.00	0
169	Hexadecane	C ₁₆ H ₃₄	544-76-3	226.441	10.156	0.45726			291.31	8.0225	723.00	0
170	Hexanal	C ₆ H ₁₂ O	66-25-1	100.159	5.6661	0.38533			217.15	4.7495	591.00	0
171	Hexane	C ₆ H ₁₄	110-54-3	86.175	4.4544	0.39002			177.83	3.7647	507.60	0
172	Hexanoic acid	C ₆ H ₁₂ O ₂	142-62-1	116.158	9.0746	0.8926	-0.75172	0.34378	269.25	6.4783	660.20	0
173	1-Hexanol	C ₆ H ₁₄ O	111-27-3	102.175	7.035	-0.9575	3.1431	-1.8066	228.55	7.1509	611.30	0
174	2-Hexanol	C ₆ H ₁₄ O	626-93-7	102.175	11.55	2.2877	-3.6724	2.1326	223.00	6.5014	585.30	0
175	2-Hexanone	C ₆ H ₁₂ O	591-78-6	100.159	5.6231	0.38207			217.35	4.7135	587.61	0
176	3-Hexanone	C ₆ H ₁₂ O	589-38-8	100.159	5.6232	0.39972			217.50	4.6655	582.82	0
177	1-Hexene	C ₆ H ₁₂	592-41-6	84.159	4.1429	0.49118	-0.44821	0.32105	133.39	3.6691	504.00	0
178	3-Hexyne	C ₆ H ₁₀	928-49-4	82.144	4.808	0.436			170.05	4.0831	544.00	0
179	Hexyl mercaptan	C ₆ H ₁₄ S	111-31-9	118.240	5.8422	0.38704			192.62	5.0630	623.00	0
180	1-Hexyne	C ₆ H ₁₀	693-02-7	82.144	4.574	0.3698			141.25	4.0640	516.20	0
181	2-Hexyne	C ₆ H ₁₀	764-35-2	82.144	4.911	0.4392			183.65	4.1067	549.00	0

182	Hydrazine	H ₂ N ₂	302-01-2	32.045	5.9794	0.9424	-1.398	0.8862	274.69	4.5238	653.15	0
183	Hydrogen	H ₂	1333-74-0	2.016	0.10127	0.698	-1.817	1.447	13.95	0.0913	33.19	0
184	Hydrogen bromide	HBr	10035-10-6	80.912	2.485	0.39			185.15	1.8817	363.15	0
185	Hydrogen chloride	HCl	7647-01-0	36.461	2.2093	0.3466			158.97	1.7498	324.65	0
186	Hydrogen cyanide	CHN	74-90-8	27.025	3.349	0.2053			259.83	2.8176	456.65	0
187	Hydrogen fluoride	HF	7664-39-3	20.006	13.451	13.36	-23.383	10.785	277.56	0.7104	461.15	0
188	Hydrogen sulfide	H ₂ S	7783-06-4	34.081	2.5676	0.37358			187.68	1.9782	373.53	0
189	Isobutyric acid	C ₄ H ₈ O ₂	79-31-2	88.105	4.0385	0.82698	-2.033	1.4769	227.15	3.5534	605.00	0
190	Isopropyl amine	C ₃ H ₇ N	75-31-0	59.110	4.4041	0.43325			177.95	3.5874	471.85	0
191	Malonic acid	C ₃ H ₄ O ₄	141-82-2	104.061	11.767	0.37877			407.95	9.0033	805.00	0
192	Methacrylic acid	C ₄ H ₆ O ₂	79-41-4	86.089	4.6095	0.23331			288.15	4.0342	662.00	0
193	Methane	CH ₄	74-82-8	16.042	1.0194	0.26087	-0.14694	0.22154	90.69	0.8724	190.56	0
194	Methanol	CH ₃ O	67-56-1	32.042	5.0451	0.33594			175.47	4.3825	512.50	0
195	N-Methyl acetamide	C ₃ H ₇ NO	79-16-3	73.094	7.3402	0.38974			301.15	5.9384	718.00	0
196	Methyl acetate	C ₃ H ₆ O ₂	79-20-9	74.079	4.492	0.3685			175.15	3.8418	506.55	0
197	Methyl acetylene	C ₃ H ₄	74-99-7	40.064	3.1889	0.37881			170.45	2.5882	402.40	0
198	Methyl acrylate	C ₄ H ₆ O ₂	96-33-3	86.089	4.68	0.349			196.32	3.9913	536.00	0
199	Methyl amine	CH ₃ N	74-89-5	31.057	3.858	0.404			179.69	3.1006	430.05	0
200	Methyl benzoate	C ₈ H ₈ O ₂	93-58-3	136.148	6.8504	0.38852			260.75	5.7026	693.00	0
201	3-Methyl-1,2-butadiene	C ₅ H ₈	598-25-4	68.117	4.1233	0.426			159.53	3.4864	490.00	0
202	2-Methylbutane	C ₅ H ₁₂	78-78-4	72.149	3.7593	0.39173			113.25	3.3657	460.40	0
203	2-Methylbutanoic acid	C ₅ H ₁₀ O ₂	116-53-0	102.132	7.48	0.3933			193.00	6.5004	643.00	0
204	3-Methyl-1-butanol	C ₅ H ₁₂ O	123-51-3	88.148	10.178	1.3211	-1.2234	0.44836	155.95	7.7338	577.20	0
205	2-Methyl-1-butene	C ₅ H ₁₀	563-46-2	70.133	3.9091	0.39866			135.58	3.4072	465.00	0
206	2-Methyl-2-butene	C ₅ H ₁₀	513-35-9	70.133	3.9248	0.36173			139.39	3.4558	470.00	0
207	2-Methyl-1-butene-3-yne	C ₅ H ₆	78-80-8	66.101	3.648	0.3863			160.15	3.1332	492.00	0
208	Methylbutyl ether	C ₅ H ₁₂ O	628-28-4	88.148	4.5302	0.37779			157.48	3.9438	512.74	0
209	Methylbutyl sulfide	C ₅ H ₁₂ S	628-29-5	104.214	5.3416	0.3835			175.30	4.6699	593.00	0
210	3-Methyl-1-butyne	C ₅ H ₈	598-23-2	68.117	3.792	0.3565			183.45	3.1681	463.20	0
211	Methyl butyrate	C ₅ H ₁₀ O ₂	623-42-7	102.132	5.3781	0.39523			187.35	4.5694	554.50	0
212	Methylchlorosilane	CH ₃ ClSi	993-00-0	80.589	3.2835	0.33116			139.05	2.8974	442.00	0
213	Methylcyclohexane	C ₇ H ₁₄	108-87-2	98.186	4.7528	0.39437			146.58	4.2291	572.10	0
214	1-Methylcyclohexanol	C ₇ H ₁₄ O	590-67-0	114.185	6.477	0.4853			299.15	4.9050	686.00	0
215	cis-2-Methylcyclohexanol	C ₇ H ₁₄ O	7443-70-1	114.185	7.8011	0.4172			280.15	6.0500	614.00	0
216	trans-2-Methylcyclohexanol	C ₇ H ₁₄ O	7443-52-9	114.185	7.8995	0.42479			269.15	6.1926	617.00	0
217	Methylcyclopentane	C ₆ H ₁₂	96-37-7	84.159	4.3595	0.38507			130.73	3.9115	532.70	0
218	1-Methylcyclopentene	C ₆ H ₁₀	693-89-0	82.144	4.3541	0.36805			146.62	3.8769	542.00	0
219	3-Methylcyclopentene	C ₆ H ₁₀	1120-62-3	82.144	4.209	0.36779			115.00	3.8439	526.00	0
220	Methyldichlorosilane	CH ₂ Cl ₂ Si	75-54-7	115.034	3.6756	0.31266			182.55	3.1686	483.00	0
221	Methylethyl ether	C ₃ H ₈ O	540-67-0	60.095	3.53	0.376			160.00	2.9751	437.80	0
222	Methylethyl ketone	C ₄ H ₈ O	78-93-3	72.106	4.622	0.355			186.48	3.9704	535.50	0
223	Methylethyl sulfide	C ₃ H ₈ S	624-89-5	76.161	4.4842	0.41151			167.23	3.8406	533.00	0
224	Methyl formate	C ₂ H ₄ O ₂	107-31-3	60.052	4.103	0.3825			174.15	3.4644	487.20	0
225	Methylisobutyl ether	C ₅ H ₁₂ O	625-44-5	88.148	4.2678	0.37995			150.00	3.7232	497.00	0
226	Methylisobutyl ketone	C ₆ H ₁₂ O	108-10-1	100.159	5.4687	0.40583			189.15	4.6507	574.60	0
227	Methyl Isocyanate	C ₂ H ₃ NO	624-83-9	57.051	4.2967	0.37922			256.15	3.2402	488.00	0
228	Methylisopropyl ether	C ₄ H ₁₀ O	598-53-8	74.122	3.8501	0.36453			127.93	3.4235	464.48	0
229	Methylisopropyl ketone	C ₅ H ₁₀ O	563-80-4	86.132	4.7075	0.33601			180.15	4.1240	553.40	0
230	Methylisopropyl sulfide	C ₄ H ₁₀ S	1551-21-9	90.187	4.5052	0.36493			171.64	3.9340	553.10	0
231	Methyl mercaptan	CH ₄ S	74-93-1	48.107	3.4448	0.37427			150.18	2.9825	469.95	0
232	Methyl methacrylate	C ₅ H ₈ O ₂	80-62-6	100.116	5.468	0.4472			224.95	4.3596	566.00	0
233	2-Methyloctanoic acid	C ₉ H ₁₈ O ₂	3004-93-1	158.238	10.53	0.7454	-0.39297	0.047214	240.00	8.1106	694.00	0
234	2-Methylpentane	C ₆ H ₁₄	107-83-5	86.175	4.2522	0.3807			119.55	3.8300	497.70	0
235	Methyl pentyl ether	C ₆ H ₁₄ O	628-80-8	102.175	5.0002	0.3781			176.00	4.3168	546.49	0
236	2-Methylpropane	C ₄ H ₁₀	75-28-5	58.122	3.188	0.39006			113.54	2.8070	407.80	0
237	2-Methyl-2-propanol	C ₄ H ₁₀ O	75-65-0	74.122	7.7646	0.56757			298.97	4.6771	506.20	0
238	2-Methyl propene	C ₄ H ₈	115-11-7	56.106	3.2614	0.38073			132.81	2.8195	417.90	0
239	Methyl propionate	C ₄ H ₈ O ₂	554-12-1	88.105	5.008	0.3959			185.65	4.2231	530.60	0
240	Methylpropyl ether	C ₄ H ₁₀ O	557-17-5	74.122	4.2719	0.43175			133.97	3.7041	476.25	0
241	Methylpropyl sulfide	C ₄ H ₁₀ S	3877-15-4	90.187	4.8253	0.38087			160.17	4.2499	565.00	0
242	Methylsilane	CH ₃ Si	992-94-9	46.144	2.2656	0.30269			116.34	2.0069	352.50	0
243	alpha-Methyl styrene	C ₉ H ₁₀	98-83-9	118.176	5.8071	0.37009			249.95	4.8591	654.00	0
244	Methyl tert-butyl ether	C ₅ H ₁₂ O	1634-04-4	88.148	3.872	0.044	0.448	-0.112	164.55	3.6017	497.10	0

TABLE 2-150 Heats of Vaporization of Inorganic and Organic Liquids (J/kmol) (Concluded)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1 × 1E-07	C2	C3	C4	$T_{\min}$, K	ΔH_v at $T_{\min} \times 1E-07$	$T_{\max}$, K	ΔH_v at $T_{\max}$
245	Methyl vinyl ether	C ₃ H ₆ O	107-25-5	58.079	3.587	0.3769			151.15	3.0567	437.00	0
246	Naphthalene	C ₁₀ H ₈	91-20-3	128.171	7.0911	0.46468			353.43	5.2691	748.40	0
247	Neon	Ne	7440-01-9	20.180	0.2389	0.3494			24.56	0.1803	44.40	0
248	Nitroethane	C ₂ H ₅ NO ₂	79-24-3	75.067	5.1459	0.33017			183.63	4.5533	593.00	0
249	Nitrogen	N ₂	7727-37-9	28.013	0.74905	0.40406	−0.317	0.27343	63.15	0.6024	126.20	0
250	Nitrogen trifluoride	F ₃ N	7783-54-2	71.002	1.6402	0.36494			66.46	1.4519	234.00	0
251	Nitromethane	CH ₃ NO ₂	75-52-5	61.040	4.7417	0.3062			244.60	4.0220	588.15	0
252	Nitrous oxide	N ₂ O	10024-97-2	44.013	2.3215	0.384			182.30	1.6502	309.57	0
253	Nitric oxide	NO	10102-43-9	30.006	2.131	0.4056			109.50	1.4578	180.15	0
254	Nonadecane	C ₁₉ H ₄₀	629-92-5	268.521	11.674	0.45865			305.04	9.2185	758.00	0
255	Nonanal	C ₉ H ₁₈ O	124-19-6	142.239	7.3363	0.41735			255.15	5.9779	658.00	0
256	Nonane	C ₉ H ₂₀	111-84-2	128.255	6.037	0.38522			219.66	5.0545	594.60	0
257	Nonanoic acid	C ₉ H ₁₈ O ₂	112-05-0	158.238	12.38	0.69869	0.097854	−0.35082	285.55	8.7232	710.70	0
258	1-Nonanol	C ₉ H ₂₀ O	143-08-8	144.255	7.5429	−1.5966	4.6489	−2.7229	268.15	8.2411	670.90	0
259	2-Nonanol	C ₉ H ₂₀ O	628-99-9	144.255	7.9797	−1.0341	3.553	−2.1149	238.15	8.0370	649.50	0
260	1-Nonene	C ₉ H ₁₈	124-11-8	126.239	5.9054	0.61039	−0.54533	0.30683	191.91	4.9218	593.10	0
261	Nonyl mercaptan	C ₉ H ₂₀ S	1455-21-6	160.320	7.5239	0.3991			253.05	6.2506	681.00	0
262	1-Nonyne	C ₉ H ₁₆	3452-09-3	124.223	6.3337	0.3975			223.15	5.2606	598.05	0
263	Octadecane	C ₁₈ H ₃₈	593-45-3	254.494	10.969	0.44327			301.31	8.7246	747.00	0
264	Octanal	C ₈ H ₁₆ O	124-13-0	128.212	6.7735	0.40607			246.00	5.5600	638.90	0
265	Octane	C ₈ H ₁₈	111-65-9	114.229	5.518	0.38467			216.38	4.5898	568.70	0
266	Octanoic acid	C ₈ H ₁₆ O ₂	124-07-2	144.211	12.23	0.69294	0.12287	−0.36132	289.65	8.4658	694.26	0
267	1-Octanol	C ₈ H ₁₈ O	111-87-5	130.228	7.2468	−1.2464	3.6797	−2.0665	257.65	7.6793	652.30	0
268	2-Octanol	C ₈ H ₁₈ O	123-96-6	130.228	7.6376	−0.7612	2.7875	−1.6033	241.55	7.3759	629.80	0
269	2-Octanone	C ₈ H ₁₆ O	111-13-7	128.212	6.5363	0.38718			252.85	5.3646	632.70	0
270	3-Octanone	C ₈ H ₁₆ O	106-68-3	128.212	6.6142	0.58562	−0.40512	0.22144	255.55	5.2076	627.70	0
271	1-Octene	C ₈ H ₁₆	111-66-0	112.213	5.4859	0.26207	0.50642	−0.43873	171.45	4.7927	566.90	0
272	Octyl mercaptan	C ₈ H ₁₈ S	111-88-6	146.294	6.8907	0.40017			223.95	5.8506	667.30	0
273	1-Octyne	C ₈ H ₁₄	629-05-0	110.197	5.4046	0.35299			193.55	4.6743	574.00	0
274	Oxalic acid	C ₂ H ₂ O ₄	144-62-7	90.035	11.473	0.37238			462.65	8.3393	804.00	0
275	Oxygen	O ₂	7782-44-7	31.999	0.9008	0.4542	−0.4096	0.3183	54.36	0.7742	154.58	0
276	Ozone	O ₃	10028-15-6	47.998	1.8587	0.30416			80.15	1.6625	261.00	0
277	Pentadecane	C ₁₅ H ₃₂	629-62-9	212.415	9.6741	0.45399			283.07	7.6728	708.00	0
278	Pentanal	C ₅ H ₁₀ O	110-62-3	86.132	5.1478	0.37541			182.00	4.4502	566.10	0
279	Pentane	C ₅ H ₁₂	109-66-0	72.149	3.9109	0.38681			143.42	3.3968	469.70	0
280	Pentanoic acid	C ₅ H ₁₀ O ₂	109-52-4	102.132	7.3197	1.2093	−1.9114	1.1591	239.15	5.3813	639.16	0
281	1-Pentanol	C ₅ H ₁₂ O	71-41-0	88.148	7.39	−0.1464	1.4751	−0.9208	195.56	6.7005	588.10	0
282	2-Pentanol	C ₅ H ₁₂ O	6032-29-7	88.148	11.111	1.8011	−2.1801	1.0641	200.00	6.6655	561.00	0
283	2-Pentanone	C ₅ H ₁₀ O	107-87-9	86.132	5.174	0.39422			196.29	4.3663	561.08	0
284	3-Pentanone	C ₅ H ₁₀ O	96-22-0	86.132	5.2359	0.40465			234.18	4.2075	560.95	0
285	1-Pentene	C ₅ H ₁₀	109-67-1	70.133	3.5027	0.3481	−0.19672	0.22394	108.02	3.2232	464.80	0
286	2-Pentyl mercaptan	C ₅ H ₁₂ S	2084-19-7	104.214	5.0573	0.45827	−0.22568	0.16393	160.75	4.4343	584.30	0
287	Pentyl mercaptan	C ₅ H ₁₂ S	110-66-7	104.214	5.4315	0.3972			197.45	4.6322	598.00	0
288	1-Pentyne	C ₅ H ₈	627-19-0	68.117	3.954	0.3512			167.45	3.4025	481.20	0
289	2-Pentyne	C ₅ H ₈	627-21-4	68.117	4.4158	0.44347			163.83	3.7321	519.00	0
290	Phenanthrene	C ₁₄ H ₁₀	85-01-8	178.229	8.3482	0.33172			372.38	6.9340	869.00	0
291	Phenol	C ₆ H ₆ O	108-95-2	94.111	7.306	0.4246			314.06	5.6577	694.25	0
292	Phenyl isocyanate	C ₇ H ₅ NO	103-71-9	119.121	5.5769	0.30346			243.15	4.8418	653.00	0
293	Phthalic anhydride	C ₈ H ₆ O ₃	85-44-9	148.116	6.916	0.1755			404.15	6.1001	791.00	0
294	Propadiene	C ₃ H ₄	463-49-0	40.064	2.9535	0.41367			136.87	2.4755	394.00	0
295	Propane	C ₃ H ₈	74-98-6	44.096	2.9209	0.78237	−0.77319	0.39246	85.47	2.4787	369.83	0
296	1-Propanol	C ₃ H ₈ O	71-23-8	60.095	6.8988	0.6458	−0.5384	0.3317	146.95	5.8356	536.80	0
297	2-Propanol	C ₃ H ₈ O	67-63-0	60.095	7.2542	0.79137	−0.66092	0.34223	185.26	5.5370	508.30	0
298	Propenylcyclohexene	C ₉ H ₁₄	13511-13-2	122.207	5.8866	0.38533			199.00	5.0941	636.00	0
299	Propionaldehyde	C ₃ H ₆ O	123-38-6	58.079	4.1492	0.36751			170.00	3.5675	504.40	0
300	Propionic acid	C ₃ H ₆ O ₂	79-09-4	74.079	4	1.3936	−2.9465	1.794	252.45	3.0922	600.81	0
301	Propionitrile	C ₃ H ₅ N	107-12-0	55.079	4.9348	0.41873			180.26	4.2005	564.40	0
302	Propyl acetate	C ₅ H ₁₀ O ₂	109-60-4	102.132	5.4327	0.407			178.15	4.6322	549.73	0

303	Propyl amine	C ₃ H ₉ N	107-10-8	59.110	4.4488	0.39494			188.36	3.6857	496.95	0
304	Propylbenzene	C ₉ H ₁₂	103-65-1	120.192	5.8887	0.38534			173.55	5.2110	638.35	0
305	Propylene	C ₃ H ₆	115-07-1	42.080	2.5216	0.33721	-0.18399	0.22377	87.89	2.3177	364.85	0
306	Propyl formate	C ₄ H ₈ O ₂	110-74-7	88.105	4.9687	0.4025			180.25	4.2162	538.00	0
307	2-Propyl mercaptan	C ₃ H ₇ S	75-33-2	76.161	4.2191	0.41161			142.61	3.6942	517.00	0
308	Propyl mercaptan	C ₃ H ₇ S	107-03-9	76.161	4.4782	0.41073			159.95	3.8723	536.60	0
309	1,2-Propylene glycol	C ₃ H ₈ O ₂	57-55-6	76.094	8.07	0.295			213.15	7.1374	626.00	0
310	Quinone	C ₆ H ₄ O ₂	106-51-4	108.095	6.49	0.3112			388.85	4.9933	683.00	0
311	Silicon tetrafluoride	F ₄ Si	7783-61-1	104.079	2.4105	0.37988			186.35	1.4873	259.00	0
312	Styrene	C ₈ H ₈	100-42-5	104.149	5.726	0.4055			242.54	4.7128	636.00	0
313	Succinic acid	C ₄ H ₆ O ₄	110-15-6	118.088	12.018	0.37149			460.65	8.7719	806.00	0
314	Sulfur dioxide	O ₃ S	7446-09-5	64.064	3.676	0.4			197.67	2.8753	430.75	0
315	Sulfur hexafluoride	F ₆ S	2551-62-4	146.055	2.571	0.383			223.15	1.6208	318.69	0
316	Sulfur trioxide	O ₃ S	7446-11-9	80.063	7.337	0.5647			289.95	4.4303	490.85	0
317	Terephthalic acid	C ₈ H ₆ O ₄	100-21-0	166.131	8.824				298.15	8.8240	298.15	88,240,000
318	<i>o</i> -Terphenyl	C ₁₈ H ₁₄	84-15-1	230.304	8.7165	0.3224			329.35	7.4548	857.00	0
319	Tetradecane	C ₁₄ H ₃₀	629-59-4	198.388	9.0539	0.44467			279.01	7.2002	693.00	0
320	Tetrahydrofuran	C ₄ H ₈ O	109-99-9	72.106	4.3021	0.36972			164.65	3.7610	540.15	0
321	1,2,3,4-Tetrahydronaphthalene	C ₁₀ H ₁₂	119-64-2	132.202	6.8086	0.43054			237.38	5.7314	720.00	0
322	Tetrahydrothiophene	C ₄ H ₆ S	110-01-0	88.171	5.0642	0.38904			176.99	4.4565	631.95	0
323	2,2,3,3-Tetramethylbutane	C ₈ H ₁₈	594-82-1	114.229	4.9055	0.40678			373.96	3.1691	568.00	0
324	Thiophene	C ₄ H ₄ S	110-02-1	84.140	4.5854	0.38756			234.94	3.7484	579.35	0
325	Toluene	C ₇ H ₈	108-88-3	92.138	4.9507	0.37742			178.18	4.3246	591.75	0
326	1,1,2-Trichloroethane	C ₂ H ₃ Cl ₃	79-00-5	133.404	5.0929	0.38013			236.50	4.2130	602.00	0
327	Tridecane	C ₁₃ H ₂₈	629-50-5	184.361	8.4339	0.4257			267.76	6.8015	675.00	0
328	Triethyl amine	C ₆ H ₁₅ N	121-44-8	101.190	4.664	0.3663			158.45	4.1011	535.15	0
329	Trimethyl amine	C ₃ H ₉ N	75-50-3	59.110	3.305	0.354			156.08	2.8216	433.25	0
330	1,2,3-Trimethylbenzene	C ₉ H ₁₂	526-73-8	120.192	5.9996	0.35578			247.79	5.0818	664.50	0
331	1,2,4-Trimethylbenzene	C ₉ H ₁₂	95-63-6	120.192	5.9254	0.35709			229.33	5.0713	649.10	0
332	2,2,4-Trimethylpentane	C ₈ H ₁₈	540-84-1	114.229	4.7711	0.37949			165.78	4.1561	543.80	0
333	2,3,3-Trimethylpentane	C ₈ H ₁₈	560-21-4	114.229	4.991	0.383			172.22	4.3530	573.50	0
334	1,3,5-Trinitrobenzene	C ₆ H ₃ N ₃ O ₆	99-35-4	213.105	10.687	0.38			398.40	8.3906	846.00	0
335	2,4,6-Trinitrotoluene	C ₇ H ₅ N ₃ O ₆	118-96-7	227.131	10.686	0.40074			354.00	8.5455	828.00	0
336	Undecane	C ₁₁ H ₂₄	1120-21-4	156.308	7.2284	0.40607			247.57	5.9240	639.00	0
337	1-Undecanol	C ₁₁ H ₂₄ O	112-42-5	172.308	8.7274	-1.5834	5.0913	-3.2171	288.45	8.9007	703.90	0
338	Vinyl acetate	C ₄ H ₆ O ₂	108-05-4	86.089	4.77	0.3765			180.35	4.0619	519.13	0
339	Vinyl acetylene	C ₄ H ₄	689-97-4	52.075	3.649	0.4	0.043		173.15	2.9876	454.00	0
340	Vinyl chloride	C ₂ H ₃ Cl	75-01-4	62.498	3.4125	0.4513			119.36	2.9491	432.00	0
341	Vinyl trichlorosilane	C ₂ H ₃ Cl ₃ Si	75-94-5	161.490	4.5659	0.36278			178.35	3.9520	543.15	0
342	Water	H ₂ O	7732-18-5	18.015	5.2053	0.3199	-0.212	0.25795	273.16	4.4733	647.10	0
343	<i>m</i> -Xylene	C ₈ H ₁₀	108-38-3	106.165	5.4626	0.37289			225.30	4.6112	617.00	0
344	<i>o</i> -Xylene	C ₈ H ₁₀	95-47-6	106.165	5.5395	0.37788			247.98	4.5859	630.30	0
345	<i>p</i> -Xylene	C ₈ H ₁₀	106-42-3	106.165	5.3819	0.36695			286.41	4.2787	616.20	0

The heat of vaporization ΔH_v is calculated by

$$\Delta H_v = C1(1 - T_r)^{C2+C3T_r+C4T_r^2+C5T_r^3}$$

where $T_r = T/T_c$, T_c is the critical temperature from Table 2-141, ΔH_v is in J/kmol, and T is in K. All substances are listed by chemical family in Table 2-6 and by formula in Table 2-7.

Values in this table were taken from the Design Institute for Physical Properties (DIPPR) of the American Institute of Chemical Engineers (AIChE), copyright 2007 AIChE and reproduced with permission of AIChE and of the DIPPR Evaluated Process Design Data Project Steering Committee. Their source should be cited as R. L. Rowley, W. V. Wilding, J. L. Oscarson, Y. Yang, N. A. Zundel, T. E. Daubert, R. P. Danner, DIPPR® Data Compilation of Pure Chemical Properties, Design Institute for Physical Properties, AIChE, New York (2007).

The number of digits provided for values at $T_{\min}$ and $T_{\max}$ was chosen for uniformity of appearance and formatting; these do not represent the uncertainties of the physical quantities, but are the result of calculations from the standard thermophysical property formulations within a fixed format.

SPECIFIC HEATS OF PURE COMPOUNDS

UNITS CONVERSIONS

For this subsection, the following units conversions are applicable:

$$^{\circ}\text{F} = \% ^{\circ}\text{C} + 32$$

$$^{\circ}\text{R} = 1.8 \text{ K}$$

To convert calories per gram-kelvin to British thermal units per pound-degree Rankine, multiply by 1.0; to convert calories per mole-kelvin to British thermal units per pound-mole-degree Rankine, multiply by 1.0.

To convert kilojoules per kilogram-kelvin to British thermal units per pound-degree Rankine, multiply by 0.2388.

ADDITIONAL REFERENCES

Additional data are contained in the subsection "Thermodynamic Properties." Data on water are also contained in that subsection. Additional tables for water are found in Eng. Sci. Data Item 68008, 251 Regent Street, London, England, which contains about 5000 values from 1 to 1000 bar, 0 to 1500 °C.

TABLE 2-151 Heat Capacities of the Elements and Inorganic Compounds*

Substance	State†	Heat capacity at constant pressure ($T = \text{K}$; $0^{\circ}\text{C} = 273.1 \text{ K}$), cal/(mol·K)	Range of temperature, K	Uncertainty, %
Aluminum ¹				
Al	<i>c</i>	$4.80 + 0.00322T$	273–931	1
	<i>l</i>	7.00	931–1273	5
AlBr ₃	<i>c</i>	$18.74 + 0.01866T$	273–370	3
	<i>l</i>	29.5	370–407	5
AlCl ₃	<i>c</i>	$13.25 + 0.02800T$	273–465	3
	<i>l</i>	31.2	465–504	3
AlCl ₃ ·6H ₂ O	<i>c</i>	76	288–327	?
AlF ₃	<i>c</i>	19.3	288–326	?
AlF ₃ ·3½H ₂ O	<i>c</i>	50.5	288–326	?
AlF ₃ ·3NaF	<i>c</i>	$38.63 + 0.04760T - 449200/T^2$	273–1273	2
	<i>l</i>	142	1273–1373	?
AlI ₃	<i>c</i>	$16.88 + 0.02266T$	273–464	3
	<i>l</i>	28.8	464–480	5
Al ₂ O ₃	<i>c</i>	$22.08 + 0.008971T - 522500/T^2$	273–1973	3
Al ₂ O ₃ ·SiO ₂	<i>c</i> , sillimanite	$40.79 + 0.004763T - 992800/T^2$	273–1573	3
	<i>c</i> , disthene	$41.81 + 0.005283T - 1211000/T^2$	273–1673	2
	<i>c</i> , andalusite	$43.96 + 0.001923T - 1086000/T^2$	273–1573	3
3Al ₂ O ₃ ·2SiO ₂	<i>c</i> , mullite	$59.65 + 0.0670T$	273–576	5
4Al ₂ O ₃ ·3SiO ₂	<i>c</i>	$113.2 + 0.0652T$	273–575	3
Al ₂ (SO ₄) ₃	<i>c</i>	63.5	273–373	?
Al ₂ (SO ₄) ₃ ·18H ₂ O	<i>c</i>	235	288–325	?
Antimony				
Sb	<i>c</i>	$5.51 + 0.00178T$	273–903	2
	<i>l</i>	7.15	903–1273	5
SbBr ₃	<i>c</i>	$17.2 + 0.0293T$	273–370	?
SbCl ₃	<i>c</i>	$10.3 + 0.0511T$	273–346	?
Sb ₂ O ₃	<i>c</i>	$19.1 + 0.0171T$	273–929	?
Sb ₂ O ₄	<i>c</i>	$22.6 + 0.0162T$	273–1198	?
Sb ₂ S ₃	<i>c</i>	$24.2 + 0.0132T$	273–821	?
Argon ²				
A	<i>g</i>	4.97	All	0
Arsenic				
As	<i>c</i>	$5.17 + 0.00234T$	273–1168	5
AsCl ₃	<i>l</i>	31.9	286–371	?
As ₂ O ₃	<i>c</i>	$8.37 + 0.0486T$	273–548	?
As ₂ S ₃	<i>c</i>	25.8	293–373	?
Barium				
BaCl ₂	<i>c</i>	$17.0 + 0.00334T$	273–1198	?
BaCl ₂ ·H ₂ O	<i>c</i>	28.2	273–307	?
BaCl ₂ ·2H ₂ O	<i>c</i>	37.3	273–307	?
Ba(ClO ₃) ₂ ·H ₂ O	<i>c</i>	51	289–320	?
BaCO ₃	<i>c</i> , α	$17.26 + 0.0131T$	273–1083	5
	<i>c</i> , β	30.0	1083–1255	15
BaMoO ₄	<i>c</i>	34	273–297	?
Ba(NO ₃) ₂	<i>c</i>	39.8	285–371	?
BaSO ₄	<i>c</i>	$21.35 + 0.0141T$	273–1323	5
Beryllium ^{3,4}				
Be	<i>c</i>	$4.698 + 0.001555T - 121000/T^2$	273–1173	1
BeO	<i>c</i>	$8.69 + 0.00365T - 313000/T^2$	273–1175	5
BeO·Al ₂ O ₃	<i>c</i>	25.4	273–373	?
BeSO ₄	<i>c</i>	20.8	273–373	?

*From Kelley, U.S. Bur. Mines Bull. 371, 1934. For a revision see Kelley, U.S. Bur. Mines Bull. 477, 1948. Data for many elements and compounds are given by Johnson (ed.), WADD-TR-60-56, 1960, for cryogenic temperatures. Tabulated data for gases can be obtained from many of the references cited in the "Thermodynamic Properties" subsection and other tables in this section. Tinh, Duran, et al., *Hydrocarbon Process.*, **50**, 98 (January 1971), review previous equation fits and give newer fits for 408 hydrocarbons and related compounds. Later publications include Duran, Tinh, et al., *Hydrocarbon Process.*, **55**, 153 (August 1976); Thompson, *J. Chem. Eng. Data*, **22**(4), 431 (1977); and Passut and Danner, *Ind. Eng. Chem. Process Des. Dev.*, **11**, 543 (1972); **13**, 193 (1974).

†The symbols in this column have the following meaning: *c*, crystal; *l*, liquid; *g*, gas; *gls*, glass.

TABLE 2-151 Heat Capacities of the Elements and Inorganic Compounds (Continued)

Substance	State†	Heat capacity at constant pressure ($T = K$; $0\text{ }^{\circ}\text{C} = 273.1\text{ K}$), cal/(mol·K)	Range of temperature, K	Uncertainty, %
Bismuth ⁴				
Bi	<i>c</i>	$5.38 + 0.00260T$	273–544	3
	<i>l</i>	7.60	544–1273	3
Bi ₂ O ₃	<i>c</i>	$23.27 + 0.01105T$	273–777	2
Bi ₂ S ₃	<i>c</i>	30.4	284–372	?
Boron				
B	<i>c</i>	$1.54 + 0.00440T$	273–1174	5
B ₂ O ₃	<i>gls</i>	$5.14 + 0.0320T$	273–513	3
	<i>gls</i>	30.4	513–623	3
BN	<i>c</i>	$1.61 + 0.00400T$	273–1173	5
Bromine				
Br ₂	<i>g</i>	9.00	300–2000	5
Cadmium				
Cd	<i>c</i>	$5.46 + 0.002466T$	273–594	1
	<i>l</i>	7.13	594–973	5
CdO	<i>c</i>	$9.65 + 0.00208T$	273–2086	?
CdS	<i>c</i>	$12.9 + 0.00090T$	273–1273	?
CdSO ₄ ·8/3H ₂ O	<i>c</i>	51.3	293	?
Calcium				
Ca	<i>c</i>	$5.31 + 0.00333T$	273–673	2
	<i>c</i>	$6.29 + 0.00140T$	673–873	2
CaCl ₂	<i>c</i>	$16.9 + 0.00386T$	273–1055	?
CaCO ₃	<i>c</i>	$19.68 + 0.01189T - 307600/T^2$	273–1033	3
CaF ₂	<i>c</i>	$14.7 + 0.00380T$	273–1651	?
CaMg(CO ₃) ₂	<i>c</i>	40.1	299–372	?
CaMoO ₄	<i>c</i>	33	273–297	?
CaO	<i>c</i>	$10.00 + 0.00484T - 108000/T^2$	273–1173	2
Ca(OH) ₂	<i>c</i>	21.4	276–373	?
CaO·Al ₂ O ₃ ·2SiO ₂	<i>c</i> , anorthite	$63.13 + 0.01500T - 1537000/T^2$	273–1673	1
	<i>gls</i>	$67.41 + 0.01048T - 1874000/T^2$	273–973	1
CaO·MgO·2SiO ₂	<i>c</i> , diopside	$54.46 + 0.005746T - 1500000/T^2$	273–1573	1
	<i>gls</i>	$51.68 + 0.009724T - 1308000/T^2$	273–973	1
CaO·SiO ₂	<i>c</i> , wollastonite	$27.95 + 0.002056T - 745600/T^2$	273–1573	1
	<i>c</i> , pseudowollastonite	$25.48 + 0.004132T - 488100/T^2$	273–1673	1
	<i>gls</i>	$23.16 + 0.009672T - 487100/T^2$	273–973	1
CaP ₂ O ₆	<i>c</i>	39.5	287–371	?
CaSO ₄	<i>c</i>	$18.52 + 0.02197T - 156800/T^2$	273–1373	5
CaSO ₄ ·2H ₂ O	<i>c</i>	46.8	282–373	?
CaWO ₄	<i>c</i>	27.9	292–322	?
Carbon ⁵				
C	<i>c</i> , graphite	$2.673 + 0.002617T - 116900/T^2$	273–1373	2
	<i>c</i> , diamond	$2.162 + 0.003059T - 130300/T^2$	273–1313	3
CH ₄	<i>g</i>	$5.34 + 0.0115T$	273–1200	2
CO ⁶	<i>g</i>	$6.60 + 0.00120T$	273–2500	1½
CO ₂	<i>g</i>	$10.34 + 0.00274T - 195500/T^2$	273–1200	1½
CS ₂	<i>l</i>	18.4	293	?
Cerium				
Ce	<i>c</i>	$5.88 + 0.00123T$	273–908	?
CeO ₂	<i>c</i>	15.1	273–373	?
Ce ₂ (MoO ₄) ₃	<i>c</i>	96	273–297	?
Ce ₂ (SO ₄) ₃	<i>c</i>	66.4	273–373	?
Ce ₂ (SO ₄) ₃ ·5H ₂ O	<i>c</i>	131.6	273–319	?
Cesium				
Cs	<i>c</i>	$1.96 + 0.0182T$	273–301	3
	<i>l</i>	8.00	302	3
	<i>g</i>	4.97	All	0
CsBr	<i>c</i>	$12.6 + 0.00259T$	273–909	?
CsCl	<i>c</i>	$11.7 + 0.00309T$	273–752	?
CsF	<i>c</i>	$11.3 + 0.00285T$	273–957	?
CsI	<i>c</i>	$11.6 + 0.00268T$	273–894	?
Chlorine				
Cl ₂	<i>g</i>	$8.28 + 0.00056T$	273–2000	1½
Chromium ⁴				
Cr	<i>c</i>	$4.84 + 0.00295T$	273–1823	5
	<i>l</i>	9.70	1823–1923	10
CrCl ₃	<i>c</i>	23	286–319	?
Cr ₂ O ₃	<i>c</i>	$26.0 + 0.00400T$	273–2263	?
CrSb	<i>c</i>	$12.3 + 0.00120T$	273–1383	?
CrSb ₂	<i>c</i>	$19.2 + 0.00184T$	273–949	?
Cr ₂ (SO ₄) ₃	<i>c</i>	67.4	273–373	?
Cobalt ⁴				
Co	<i>c</i>	$5.12 + 0.00333T$	273–1763	5
	<i>l</i>	8.40	1763–1873	5
CoAs ₂ ·CoS ₂	<i>c</i>	32.9	283–373	?
CoSb	<i>c</i>	$11.7 + 0.00156T$	273–1464	?
Co ₂ Sn	<i>c</i>	$15.83 + 0.00950T$	273–903	2
CoS	<i>c</i>	$10.6 + 0.00251T$	273–1373	?
CoSO ₄ ·7H ₂ O	<i>c</i>	96	286–303	?

TABLE 2-151 Heat Capacities of the Elements and Inorganic Compounds (Continued)

Substance	State†	Heat capacity at constant pressure ($T = K$; $0^\circ C = 273.1 K$), cal/(mol·K)	Range of temperature, K	Uncertainty, %
Copper ⁷				
Cu	<i>c</i>	$5.44 + 0.001462T$	273–1357	1
	<i>l</i>	7.50	1357–1573	3
CuAl	<i>c</i>	$9.88 + 0.00500T$	273–733	2
CuAl ₂	<i>c</i>	$16.78 + 0.00366T$	273–773	2
Cu ₃ Al	<i>c</i>	$19.61 + 0.01054T$	273–775	2
CuI	<i>c</i>	$12.1 + 0.00286T$	273–675	?
CuI ₂	<i>c</i>	20.1	274–328	?
CuO	<i>c</i>	$10.87 + 0.003576T - 150600/T^2$	273–810	2
CuO·SiO ₂ ·H ₂ O	<i>c</i>	29	293–323	?
CuS	<i>c</i>	$10.6 + 0.00264T$	273–1273	?
Cu ₂ S	<i>c, α</i>	$9.38 + 0.0312T$	273–376	3
	<i>c, β</i>	20.9	376–1173	2
CuS·FeS	<i>c</i>	24	292–321	?
Cu ₂ Sb	<i>c</i>	$13.73 + 0.01350T$	273–573	2
Cu ₂ Sb	<i>c</i>	$21.79 + 0.00900T$	273–693	2
Cu ₂ Se	<i>c, α</i>	20.85	273–383	5
	<i>c, β</i>	20.35	383–488	5
Cu ₃ Si	<i>c</i>	$20.3 + 0.00587T$	273–1135	?
CuSO ₄	<i>c</i>	24.1	282	?
CuSO ₄ ·H ₂ O	<i>c</i>	31.3	282	?
CuSO ₄ ·3H ₂ O	<i>c</i>	49.0	282	?
CuSO ₄ ·5H ₂ O	<i>c</i>	67.2	282	?
Fluorine ⁸				
F ₂	<i>g</i>	$6.50 + 0.00100T$	300–3000	5
Gallium				
Ga ₂ O ₃	<i>c</i>	$18.2 + 0.0252T$	273–923	?
Ga ₂ (SO ₄) ₃	<i>c</i>	62.4	273–373	?
Germanium ¹				
Ge	<i>c</i>			
Gold				
Au	<i>c</i>	$5.61 + 0.00144T$	273–1336	2
	<i>l</i>	7.00	1336–1573	5
AuSb ₂	<i>c, α</i>	$17.12 + 0.00465T$	273–628	1
	<i>c, βγ</i>	$11.47 + 0.01756T$	628–713	?
Helium ⁹				
He	<i>g</i>	4.97	All	0
Hydrogen ¹⁰				
H	<i>g</i>	4.97	All	0
H ₂	<i>g</i>	$6.62 + 0.00081T$	273–2500	2
HBr	<i>g</i>	$6.80 + 0.00084T$	273–2000	2
HCl	<i>g</i>	$6.70 + 0.00084T$	273–2000	1½
HI	<i>g</i>	$6.93 + 0.00083T$	273–2000	2
H ₂ O	<i>l</i>	See Tables 2-153 and 2-305		
	<i>g</i>	$8.22 + 0.00015T + 0.00000134T^2$	300–2500	?
H ₂ S	<i>g</i>	$7.20 + 0.00360T$	300–600	8
H ₂ S ₂ O ₇	<i>c</i>	27	281	?
	<i>l</i>	58	308	?
Indium				
In	<i>c</i>			
Iodine				
I ₂	<i>g</i>	9.00	300–2000	5
Iridium				
Ir	<i>c</i>	$5.50 + 0.00148T$	273–1873	1
Iron ⁴				
Fe	<i>c, α</i>	$4.13 + 0.00638T$	273–1041	3
	<i>c, β</i>	$6.12 + 0.00336T$	1041–1179	3
	<i>c, γ</i>	8.40	1179–1674	5
	<i>c, δ</i>	10.0	1674–1803	5
	<i>l</i>	8.15	1803–1873	5
FeAs ₂	<i>c</i>	17.8	283–373	?
Fe ₃ C	<i>c</i>	$25.17 + 0.00223T$	273–1173	10
FeCO ₃	<i>c</i>	22.7	293–368	?
FeO	<i>c</i>	$12.62 + 0.001492T - 76200/T^2$	273–1173	2
Fe ₂ O ₃	<i>c</i>	$24.72 + 0.01604T - 423400/T^2$	273–1097	2
Fe ₃ O ₄	<i>c</i>	$41.17 + 0.01882T - 979500/T^2$	273–1065	2
Fe ₂ O ₃ ·3H ₂ O	<i>c</i>	47.8	286–373	?
FeS	<i>c, α</i>	$2.03 + 0.0390T$	273–411	5
	<i>c, β</i>	$12.05 + 0.00273T$	411–1468	3
FeS ₂	<i>c</i>	$10.7 + 0.01336T$	273–773	?
FeSi	<i>c</i>	$10.54 + 0.00458T$	273–903	2
Fe ₂ SiO ₄	<i>c</i>	$33.57 + 0.01907T - 879700/T^2$	273–1161	2
FeSO ₄	<i>c</i>	22	293–373	?
Fe ₂ (SO ₄) ₃	<i>c</i>	66.2	273–373	?
FeSO ₄ ·4H ₂ O	<i>c</i>	63.6	282	?
FeSO ₄ ·7H ₂ O	<i>c</i>	96	291–319	?
Krypton				
Kr	<i>g</i>	4.97	All	0

TABLE 2-151 Heat Capacities of the Elements and Inorganic Compounds (Continued)

Substance	State†	Heat capacity at constant pressure ($T = K$; $0\text{ }^{\circ}\text{C} = 273.1\text{ K}$), $\text{cal}/(\text{mol}\cdot\text{K})$	Range of temperature, K	Uncertainty, %
Lanthanum				
La	<i>c</i>	$5.91 + 0.00100T$	273–1009	?
La_2O_3	<i>c</i>	$22.6 + 0.00544T$	273–2273	?
$\text{La}_2(\text{MoO}_4)_3$	<i>c</i>	86	273–307	?
$\text{La}_2(\text{SO}_4)_3$	<i>c</i>	66.9	273–373	?
$\text{La}_2(\text{SO}_4)_3\cdot 9\text{H}_2\text{O}$	<i>c</i>	152	273–319	?
Lead [‡]				
Pb	<i>c</i>	$5.77 + 0.00202T$	273–600	2
	<i>l</i>	6.8	600–1273	5
$\text{Pb}_3(\text{AsO}_4)_2$	<i>c</i>	65.5	286–370	?
PbB_2O_4	<i>c</i>	26.5	288–371	?
PbB_2O_7	<i>c</i>	41.4	289–371	?
PbBr_2	<i>c</i>	$18.13 + 0.00310T$	273–761	2
	<i>l</i>	27.4	761–860	10
PbCl_2	<i>c</i>	$15.88 + 0.00835T$	273–771	2
	<i>l</i>	27.2	771–851	10
$2\text{PbCl}_2\cdot\text{NH}_4\text{Cl}$	<i>c</i>	53.1	293	?
PbCO_3	<i>c</i>	21.1	286–320	?
PbCrO_4	<i>c</i>	29.1	292–323	?
PbF_2	<i>c</i>	$16.5 + 0.00412T$	273–1091	?
PbI_2	<i>c</i>	$18.66 + 0.00293T$	273–648	2
	<i>l</i>	32.3	648–776	20
PbMoO_4	<i>c</i>	30.4	292–322	?
$\text{Pb}(\text{NO}_3)_2$	<i>c</i>	36.4	286–320	?
PbO	<i>c</i>	$10.33 + 0.00318T$	273–544	2
PbO_2	<i>c</i>	$12.7 + 0.00780T$	273–?	?
$\text{Pb}_2\text{P}_2\text{O}_7$	<i>c</i>	48.3	284–371	?
PbS	<i>c</i>	$10.63 + 0.00401T$	273–873	3
PbSO_4	<i>c</i>	26.4	293–372	?
PbS_2O_3	<i>c</i>	29	293–373	?
PbWO_4	<i>c</i>	35	273–297	?
Lithium				
Li	<i>c</i>	$0.68 + 0.0180T$	273–459	10
	<i>g</i>	4.97	All	0
LiBr	<i>c</i>	$11.5 + 0.00302T$	273–825	?
$\text{LiBr}\cdot\text{H}_2\text{O}$	<i>c</i>	22.6	278–318	?
LiCl	<i>c</i>	$11.0 + 0.00339T$	273–887	?
$\text{LiCl}\cdot\text{H}_2\text{O}$	<i>c</i>	23.6	279–360	?
LiF	<i>c</i>	$8.20 + 0.00520T$	273–1117	?
LiI	<i>c</i>	$12.5 + 0.00208T$	273–723	?
$\text{LiI}\cdot\text{H}_2\text{O}$	<i>c</i>	23.6	277–359	?
$\text{LiI}\cdot 2\text{H}_2\text{O}$	<i>c</i>	32.9	277–345	?
$\text{LiI}\cdot 3\text{H}_2\text{O}$	<i>c</i>	43.2	277–347	?
LiNO_3	<i>c</i>	$9.17 + 0.0360T$	273–523	5
	<i>l</i>	26.8	523–575	5
Magnesium [‡]				
Mg	<i>c</i>	$6.20 + 0.00133T - 67800/T^2$	273–923	1
	<i>l</i>	7.4	923–1048	10
MgAg	<i>c</i>	$10.58 + 0.00412T$	273–905	2
Mg_7Al_3	<i>c</i>	$34.4 + 0.0198T$	273–736	?
MgAu	<i>c</i>	$11.3 + 0.00189T$	273–1433	?
Mg_{25}Au	<i>c</i>	$16.2 + 0.00451T$	273–1073	?
Mg_3Au	<i>c</i>	$21.2 + 0.00614T$	273–1103	?
MgCl_2	<i>c</i>	$17.3 + 0.00377T$	273–991	?
$\text{MgCl}_2\cdot 6\text{H}_2\text{O}$	<i>c</i>	77.1	292–342	?
MgCO_3	<i>c</i>	16.9	290	?
MgCu_2	<i>c</i>	$14.96 + 0.00776T$	273–903	3
Mg_2Cu	<i>c</i>	$15.5 + 0.00652T$	273–843	?
MgNi_2	<i>c</i>	$15.87 + 0.00692T$	273–903	2
MgO	<i>c</i>	$10.86 + 0.001197T - 208700/T^2$	273–2073	2
$\text{MgO}\cdot\text{Al}_2\text{O}_3$	<i>c</i>	28	288–319	?
$\text{MgO}\cdot\text{SiO}_2$	<i>c</i> , amphibole	$25.60 + 0.004380T - 674200/T^2$	273–1373	1
	<i>c</i> , pyroxene	$23.35 + 0.008062T - 558800/T^2$	273–773	1
	<i>gls</i>	$23.30 + 0.007734T - 542000/T^2$	273–973	1
$6\text{MgO}\cdot\text{MgCl}_2\cdot 8\text{B}_2\text{O}_3$	<i>c</i> , α	$58.7 + 0.408T$	273–538	5
	<i>c</i> , β	$107.2 + 0.2876T$	538–623	5
	<i>c</i>	18.2	292–323	?
$\text{Mg}(\text{OH})_2$	<i>c</i>	$28.2 + 0.00560T$	273–1234	?
Mg_3Sb_2	<i>c</i>	$15.4 + 0.00415T$	273–1343	?
Mg_2Si	<i>c</i>	26.7	296–372	?
MgSO_4	<i>c</i>	33	282	?
$\text{MgSO}_4\cdot\text{H}_2\text{O}$	<i>c</i>	80	282	?
$\text{MgSO}_4\cdot 6\text{H}_2\text{O}$	<i>c</i>	89	291–319	?
$\text{MgSO}_4\cdot 7\text{H}_2\text{O}$	<i>c</i>			?

TABLE 2-151 Heat Capacities of the Elements and Inorganic Compounds (Continued)

Substance	State†	Heat capacity at constant pressure ($T = K$; $0\text{ }^{\circ}\text{C} = 273.1\text{ K}$), cal/(mol·K)	Range of temperature, K	Uncertainty, %
Manganese				
Mn	<i>c</i> , α	$3.76 + 0.00747T$	273–1108	5
	<i>c</i> , β	$5.06 + 0.00395T$	1108–1317	5
	<i>c</i> , γ	$4.80 + 0.00422T$	1317–1493	5
	<i>l</i>	11.0	1493–1673	10
MnCl ₂	<i>c</i>	$16.2 + 0.00520T$	273–923	?
MnCO ₃	<i>c</i>	$7.79 + 0.0421T + 0.0000090T^2$	273–773	?
MnO	<i>c</i>	$7.43 + 0.01038T - 0.00000362T^2$	273–1923	?
Mn ₂ O ₃	<i>c</i>	$10.33 + 0.0530T - 0.0000257T^2$	273–1173	?
Mn ₃ O ₄	<i>c</i>	$19.25 + 0.0538T - 0.0000209T^2$	273–1773	?
MnO ₂	<i>c</i>	$1.92 + 0.0471T - 0.0000297T^2$	273–773	?
Mn ₂ O ₃ ·H ₂ O	<i>c</i>	31	291–322	?
MnS	<i>c</i>	$10.21 + 0.00656T - 0.00000242T^2$	273–1883	?
MnSO ₄	<i>c</i>	27.5	293–373	?
MnSO ₄ ·5H ₂ O	<i>c</i>	78	290–319	?
Mercury ¹¹				
Hg	<i>l</i>	6.61	273–630	1
	<i>g</i>	4.97	All	0
Hg ₂	<i>g</i>	9.00	300–2000	5
HgCl	<i>c</i>	$11.05 + 0.00370T$	273–798	?
HgCl ₂	<i>c</i>	$15.3 + 0.0103T$	273–553	?
Hg(CN) ₂	<i>c</i>	25	285–319	?
HgI	<i>c</i>	$11.4 + 0.00461T$	273–563	?
HgI ₂	<i>c</i> , α	$17.4 + 0.004001T$	273–403	3
	<i>c</i> , β	20.2	403–523	3
HgO	<i>c</i>	11.5	278–371	?
HgS	<i>c</i>	$10.9 + 0.00365T$	273–853	?
Hg ₂ SO ₄	<i>c</i>	31.0	273–307	?
Molybdenum				
Mo	<i>c</i>	$5.69 + 0.00188T - 50300/T^2$	273–1773	5
MoO ₃	<i>c</i>	$15.1 + 0.0121T$	273–1068	?
MoS ₂	<i>c</i>	$19.7 + 0.00315T$	273–729	?
Neon ¹²				
Ne	<i>g</i>	4.97	All	0
Nickel ⁴				
Ni	<i>c</i> , α	$4.26 + 0.00640T$	273–626	2
	<i>c</i> , β	$6.99 + 0.000905T$	626–1725	5
	<i>l</i>	8.55	1725–1903	10
NiO	<i>c</i>	$11.3 + 0.00215T$	273–1273	?
NiS	<i>c</i>	$9.25 + 0.00640T$	273–597	3
Ni ₂ Si	<i>c</i>	$15.8 + 0.00329T$	273–1582	?
NiSi	<i>c</i>	$10.0 + 0.00312T$	273–1273	?
Ni ₃ Sn	<i>c</i>	$20.78 + 0.0102T$	273–904	2
NiSO ₄	<i>c</i>	33.4	293–373	?
NiSO ₄ ·6H ₂ O	<i>c</i>	82	291–325	?
NiTe	<i>c</i>	$11.00 + 0.00433T$	273–700	2
Nitrogen ¹³				
N ₂	<i>g</i>	$6.50 + 0.00100T$	300–3000	3
NH ₃	<i>g</i>	$6.70 + 0.00630T$	300–800	1½
NH ₄ Br	<i>c</i>	22.8	274–328	?
NH ₄ Cl	<i>c</i> , α	$9.80 + 0.0368T$	273–457	5
	<i>c</i> , β	$5.0 + 0.0340T$	457–523	5
NH ₄ I	<i>c</i>	17.8	273–328	?
NH ₄ NO ₃	<i>c</i>	31.8	273–293	?
(NH ₄) ₂ SO ₄	<i>c</i>	51.6	275–328	?
NO	<i>g</i>	$8.05 + 0.000233T - 156300/T^2$	300–5000	2
Osmium				
Os	<i>c</i>	$5.686 + 0.000875T$	273–1877	1
Oxygen ¹⁴				
O ₂	<i>g</i>	$8.27 + 0.000258T - 187700/T^2$	300–5000	1
Palladium				
Pd	<i>c</i>	$5.41 + 0.00184T$	273–1822	2
Phosphorus				
P	<i>c</i> , yellow	5.50	273–317	5
	<i>c</i> , red	$0.21 + 0.0180T$	273–472	10
	<i>l</i>	6.6	317–373	10
PCl ₃	<i>l</i>	28.7	284–371	?
P ₄ O ₁₀	<i>c</i>	$15.72 + 0.1092T$	273–631	2
	<i>g</i>	73.6	631–1371	3
Platinum ⁴				
Pt	<i>c</i>	$5.92 + 0.00116T$	273–1873	1
Potassium				
K	<i>c</i>	$5.24 + 0.00555T$	273–336	5
	<i>l</i>	7.7	336–373	5

TABLE 2-151 Heat Capacities of the Elements and Inorganic Compounds (Continued)

Substance	State†	Heat capacity at constant pressure ($T = K$; $0\text{ }^{\circ}\text{C} = 273.1\text{ K}$), cal/(mol·K)	Range of temperature, K	Uncertainty, %
Potassium—(Cont.)				
K	<i>g</i>	4.97	All	0
K ₂	<i>gg</i>	9.00	300–2000	5
KAsO ₃	<i>c</i>	25.3	290–372	?
KBO ₂	<i>c</i>	12.6 + 0.0126 <i>T</i>	273–1220	?
K ₂ B ₄ O ₇	<i>c</i>	51.3	290–372	?
KBr	<i>c</i>	11.49 + 0.00360 <i>T</i>	273–543	2
KCl	<i>c</i>	10.93 + 0.00376 <i>T</i>	273–1043	2
KClO ₃	<i>c</i>	25.7	289–371	?
KClO ₄	<i>c</i>	26.3	287–318	?
2KCl·CuCl ₂ ·2H ₂ O	<i>c</i>	63	292–323	?
2KCl·PtCl ₄	<i>c</i>	55	286–319	?
2KCl·SnCl ₄	<i>c</i>	54.5	292–323	?
2KCl·ZnCl ₂	<i>c</i>	43.4	279–319	?
2KCN·Zn(CN) ₂	<i>c</i>	57.4	277–319	?
K ₂ CO ₃	<i>c</i>	29.9	296–372	?
K ₂ CrO ₄	<i>c</i>	35.9	289–371	?
K ₂ Cr ₂ O ₇	<i>c</i>	42.80 + 0.0410 <i>T</i>	273–671	5
	<i>l</i>	96.9	671–757	5
KF	<i>c</i>	10.8 + 0.00284 <i>T</i>	273–1129	?
K ₄ Fe(CN) ₆	<i>c</i>	80.1	273–319	?
K ₄ Fe(CN) ₆ ·3H ₂ O	<i>c</i>	114.5	273–310	?
KH ₂ AsO ₄	<i>c</i>	32	289–319	?
KH ₂ PO ₄	<i>c</i>	28.3	290–320	?
KHSO ₄	<i>c</i>	30	292–324	?
KMnO ₄	<i>c</i>	28	287–318	?
KNO ₃	<i>c</i>	6.42 + 0.0530 <i>T</i>	273–401	10
	<i>c</i>	28.8	401–611	5
	<i>l</i>	29.5	611–683	10
K ₂ O·Al ₂ O ₃ ·3SiO ₂	<i>c</i> , orthoclase	69.26 + 0.00821 <i>T</i> – 2331000/ <i>T</i> ²	273–1373	1½
	<i>gls</i> , orthoclase	69.81 + 0.01053 – 2403000/ <i>T</i> ²	273–1373	1½
	<i>c</i> , microcline	65.65 + 0.01102 <i>T</i> – 1748000/ <i>T</i> ²	273–1373	1½
	<i>gls</i> , microcline	64.83 + 0.01438 <i>T</i> – 1641000/ <i>T</i> ²	273–1373	1½
K ₄ P ₂ O ₇	<i>c</i>	63.1	290–371	?
K ₂ SO ₄	<i>c</i>	33.1	287–371	?
K ₂ S ₂ O ₃	<i>c</i>	37	293–373	?
K ₂ SO ₄ ·Al ₂ (SO ₄) ₃ ·24H ₂ O	<i>c</i>	352	292–322	?
K ₂ SO ₄ ·Cr ₂ (SO ₄) ₃ ·24H ₂ O	<i>c</i>	324	292–324	?
K ₂ SO ₄ ·MgSO ₄ ·6H ₂ O	<i>c</i>	106	292–323	?
K ₂ SO ₄ ·NiSO ₄ ·6H ₂ O	<i>c</i>	107	289–319	?
K ₂ SO ₄ ·ZnSO ₄ ·6H ₂ O	<i>c</i>	120	293–317	?
Promethium				
Pr	<i>c</i>			
Radon				
Rn	<i>g</i>	4.97	All	0
Rhenium				
Re	<i>c</i>	6.30 + 0.00053 <i>T</i>	273–2273	?
Rhodium				
Rh	<i>c</i>	5.40 + 0.00219 <i>T</i>	273–1877	2
Rubidium				
Rb	<i>c</i>	3.27 + 0.0131 <i>T</i>	273–312	2
	<i>l</i>	7.85	312–373	5
RbBr	<i>c</i>	11.6 + 0.00255 <i>T</i>	273–954	?
RbCl	<i>c</i>	11.5 + 0.00249 <i>T</i>	273–987	?
Rb ₂ CO ₃	<i>c</i>	28.4	291–320	?
RbF	<i>c</i>	11.3 + 0.00256 <i>T</i>	273–1048	?
RbI	<i>c</i>	11.6 + 0.00263 <i>T</i>	273–913	?
Scandium				
Sc ₂ O ₃	<i>c</i>	21.1	273–373	?
Sc ₂ (SO ₄) ₃	<i>c</i>	62.0	273–373	?
Selenium				
Se	<i>c</i>	4.53 + 0.00550 <i>T</i>	273–490	2
	<i>l</i>	8.35	490–570	3
Silicon				
Si	<i>c</i>	5.74 + 0.000617 <i>T</i> – 101000/ <i>T</i> ²	273–1174	2
SiC	<i>c</i>	8.89 + 0.00291 <i>T</i> – 284000/ <i>T</i> ²	273–1629	2
SiCl ₄	<i>l</i>	32.4	293–373	?
SiO ₂	<i>c</i> , quartz, α	10.87 + 0.008712 <i>T</i> – 241200/ <i>T</i> ²	273–848	1
	<i>c</i> , quartz, β	10.95 + 0.00550 <i>T</i>	848–1873	3½
	<i>c</i> , cristobalite, α	3.65 + 0.0240 <i>T</i>	273–523	2½
	<i>c</i> , cristobalite, β	17.09 + 0.000454 <i>T</i> – 897200/ <i>T</i> ²	523–1973	2
	<i>gls</i>	12.80 + 0.00447 <i>T</i> – 302000/ <i>T</i> ²	273–1973	3½
Silver ⁴				
Ag	<i>c</i>	5.60 + 0.00150 <i>T</i>	273–1234	1
	<i>l</i>	8.2	1234–1573	3

TABLE 2-151 Heat Capacities of the Elements and Inorganic Compounds (Continued)

Substance	State†	Heat capacity at constant pressure ($T = K$; $0\text{ }^{\circ}\text{C} = 273.1\text{ K}$), cal/(mol·K)	Range of temperature, K	Uncertainty, %
Silver—(Cont.)				
Ag ₃ Al	<i>c</i>	$22.56 + 0.00570T$	273–902	2
Ag ₅ Al	<i>c</i>	$16.85 + 0.00450T$	273–903	2
AgAl ₁₂	<i>c</i>	$58.62 + 0.0575T$	273–768	5
AgBr	<i>c</i>	$8.58 + 0.0141T$	273–703	6
	<i>l</i>	14.9	703–836	5
AgCl	<i>c</i>	$9.60 + 0.00929T$	273–728	2
	<i>l</i>	14.05	728–806	5
AgCNO	<i>c</i>	18.7	273–353	?
AgI	<i>c, α</i>	$8.58 + 0.0141T$	273–423	6
AgNO ₃	<i>c, α</i>	$18.83 + 0.0160T$	273–433	2
	<i>c, β</i>	25.7	433–482	5
	<i>l</i>	30.2	482–541	5
Ag ₃ PO ₄	<i>c</i>	37.5	293–325	?
Ag ₂ S	<i>c, α</i>	18.8	273–448	5
	<i>c, β</i>	21.8	448–597	5
Ag ₃ Sb	<i>c</i>	$19.53 + 0.0160T$	273–694	5
Ag ₂ Se	<i>c, α</i>	20.2	273–406	5
	<i>c, β</i>	20.4	406–460	5
Sodium ¹⁵				
Na	<i>c</i>	$5.01 + 0.00536T$	273–371	1½
	<i>l</i>	7.50	371–451	2
	<i>g</i>	4.97	All	0
NaBO ₂	<i>c</i>	$10.4 + 0.0199T$	273–1239	?
Na ₂ B ₄ O ₇	<i>c</i>	47.9	289–371	?
Na ₂ B ₄ O ₇ ·10H ₂ O	<i>c</i>	147	292–323	?
NaBr	<i>c</i>	$11.74 + 0.00233T$	273–543	2
NaCl	<i>c</i>	$10.79 + 0.00420T$	273–1074	2
	<i>l</i>	15.9	1073–1205	3
NaClO ₃	<i>c</i>	$9.48 + 0.0468T$	273–528	3
	<i>l</i>	31.8	528–572	5
NaCNO	<i>c</i>	13.1	273–353	?
Na ₂ CO ₃	<i>c</i>	28.9	288–371	?
NaF	<i>c</i>	$10.4 + 0.00289T$	273–1261	?
Na ₂ HPO ₄ ·7H ₂ O	<i>c</i>	86.6	275–307	?
Na ₂ HPO ₄ ·12H ₂ O	<i>c</i>	133.4	275–307	?
NaI	<i>c</i>	$12.5 + 0.00162T$	273–936	?
NaNO ₃	<i>c</i>	$4.56 + 0.0580T$	273–583	5
	<i>l</i>	37.2	583–703	10
Na ₂ O·Al ₂ O ₃ ·3SiO ₂	<i>c, albite</i>	$63.78 + 0.01171T - 1678000/T^2$	273–1373	1
	<i>gls</i>	$61.25 + 0.01768T - 1545000/T^2$	273–1173	1
NaPO ₃	<i>c</i>	22.1	290–319	?
Na ₄ P ₂ O ₇	<i>c</i>	60.7	290–371	?
Na ₂ SO ₄	<i>c</i>	32.8	289–371	?
Na ₂ S ₂ O ₃	<i>c</i>	34.9	273–307	?
Na ₂ S ₂ O ₃ ·5H ₂ O	<i>c</i>	86.2	273–307	?
Sodium-potassium alloys ¹⁵	<i>l</i>			
Strontium				
SrBr ₂	<i>c</i>	$18.1 + 0.00311T$	273–923	?
SrBr ₂ ·H ₂ O	<i>c</i>	28.9	277–370	?
SrBr ₂ ·6H ₂ O	<i>c</i>	82.1	276–327	?
SrCl ₂	<i>c</i>	$18.2 + 0.00244T$	273–1143	?
SrCl ₂ ·H ₂ O	<i>c</i>	28.7	276–365	?
SrCl ₂ ·2H ₂ O	<i>c</i>	38.3	277–366	?
SrCO ₃	<i>c</i>	21.8	281–371	?
SrI ₂	<i>c</i>	$18.6 + 0.00304T$	273–783	?
SrI ₂ ·H ₂ O	<i>c</i>	28.5	276–363	?
SrI ₂ ·2H ₂ O	<i>c</i>	39.1	275–336	?
SrI ₂ ·6H ₂ O	<i>c</i>	84.9	275–333	?
SrMoO ₄	<i>c</i>	37	273–297	?
Sr(NO ₃) ₂	<i>c</i>	38.3	290–320	?
SrSO ₄	<i>c</i>	26.2	293–369	?
Sulfur ¹⁶				
S	<i>c, rhombic</i>	$3.63 + 0.00640T$	273–368	3
	<i>c, monoclinic</i>	$4.38 + 0.00440T$	368–392	3
S ₂	<i>g</i>	$8.58 + 0.00030T$	300–2500	5
S ₂ Cl ₂	<i>l</i>	27.5	273–332	?
SO ₂	<i>g</i>	$7.70 + 0.00530T - 0.00000083T^2$	300–2500	2½
Tantalum				
Ta	<i>c</i>	$5.91 + 0.00099T$	273–1173	2
Tellurium				
Te	<i>c</i>	$5.19 + 0.00250T$	273–600	3
Thallium				
Tl	<i>c, α</i>	$5.32 + 0.00385T$	273–500	1
	<i>c, β</i>	8.12	500–576	1

TABLE 2-151 Heat Capacities of the Elements and Inorganic Compounds (Concluded)

Substance	State†	Heat capacity at constant pressure ($T = K$; $0\text{ }^{\circ}\text{C} = 273.1\text{ K}$), cal/(mol·K)	Range of temperature, K	Uncertainty, %
Thallium—(Cont.)				
Tl	<i>l</i>	7.12	576–773	3
TlBr	<i>c</i>	$12.53 + 0.00100T$	273–733	10
	<i>l</i>	16.0	733–800	10
TlCl	<i>c</i>	$12.56 + 0.00088T$	273–700	5
	<i>l</i>	14.2	700–803	10
Thorium				
Th	<i>c</i>	6.40	273–373	?
ThO ₂	<i>c</i>	$14.6 + 0.00507T$	273–1273	?
Th(SO ₄) ₂	<i>c</i>	41.2	273–373	?
Tin ⁴				
Sn	<i>c</i>	$5.05 + 0.00480T$	273–504	2
	<i>l</i>	6.6	504–1273	10
SnAu	<i>c</i>	$11.79 + 0.00233T$	273–581	1
SnCl ₂	<i>c</i>	$16.2 + 0.00926T$	273–520	?
SnCl ₄	<i>l</i>	38.4	286–371	?
SnO	<i>c</i>	$9.40 + 0.00362T$	273–1273	?
SnO ₂	<i>c</i>	$13.94 + 0.00565T - 252000/T^2$	273–1373	?
SnPt	<i>c</i>	$11.49 + 0.00190T$	273–1318	1
SnS	<i>c</i>	$12.1 + 0.00165T$	273–1153	?
SnS ₂	<i>c</i>	$20.5 + 0.00400T$	273–873	?
Titanium				
Ti	<i>c</i>	$8.91 + 0.00114T - 433000/T^2$	273–713	3
TiCl ₄	<i>l</i>	35.7	285–372	?
TiO ₂	<i>c</i>	$11.81 + 0.00754T - 41900/T^2$	273–713	3
Tungsten				
W	<i>c</i>	$5.65 + 0.00866$	273–2073	1
WO ₃	<i>c</i>	$16.0 + 0.00774T$	273–1550	?
Uranium				
U	<i>c</i>	6.64	273–372	?
U ₃ O ₈	<i>c</i>	59.8	276–314	?
Vanadium				
V	<i>c</i>	$5.57 + 0.00097T$	273–1993	?
Xenon				
Xe	<i>g</i>	4.97	All	0
Zinc ⁴				
Zn	<i>c</i>	$5.25 + 0.00270T$	273–692	1
	<i>l</i>	$7.59 + 0.00055T$	692–1122	3
ZnCl ₂	<i>c</i>	$15.9 + 0.00800T$	273–638	?
ZnO	<i>c</i>	$11.40 + 0.00145T - 182400/T^2$	273–1573	1
ZnS	<i>c</i>	$12.81 + 0.00095T - 194600/T^2$	273–1173	5
ZnSb	<i>c</i>	$11.5 + 0.00313T$	273–810	?
ZnSO ₄	<i>c</i>	28	293–373	?
ZnSO ₄ ·H ₂ O	<i>c</i>	34.7	282	?
ZnSO ₄ ·6H ₂ O	<i>c</i>	80.8	282	?
ZnSO ₄ ·7H ₂ O	<i>c</i>	100.2	273–307	?
Zirconium				
ZrO ₂	<i>c</i>	$11.62 + 0.01046T - 177700/T^2$	273–1673	5
ZrO ₂ ·SiO ₂	<i>c</i>	26.7	297–372	?

¹ See also Table 2-152. Data to 298 K are also given by Scott, *Cryogenic Engineering*, Van Nostrand, Princeton, N.J., 1959.

² For liquid and gas data, see Johnson (ed.), WADD-TR-60-56, 1960.

³ Stalder, NACA Tech. Note 4141, 1957 (Fig. 5), gives data from 400 to 2600°R.

⁴ See also Table 2-152.

⁵ For data from 400 to 5500 °R see Stalder, NACA Tech. Note 4141, 1975 (Fig. 4).

⁶ For solid, liquid, and gas data, see Johnson (ed.), WADD-TR-60-56, 1960.

⁷ For data from 400 to 2350 °R see Stalder, NACA Tech. Note 4141, 1957.

⁸ For solid, liquid, and gas data, see Johnson (ed.), WADD-TR-60-56, 1960.

⁹ For liquid and gas data, see Johnson (ed.), WADD-TR-60-56, 1960.

¹⁰ For solid, liquid, and gas data, see Johnson (ed.), WADD-TR-60-56, 1960.

¹¹ See also Table 2-152; Douglas, Ball, et al., *Bur. Stand. J. Res.*, **46** (1951): 334; Busey and Giaque, *J. Am. Chem. Soc.*, **75** (1953): 806; Sheldon, ASME Pap. 49-A-30, 1949.

¹² For solid, liquid, and gas data, see Johnson (ed.), WADD-TR-56-60, 1960.

¹³ For solid, liquid, and gas data, see Johnson (ed.), WADD-TR-56-60, 1960.

¹⁴ For solid, liquid, and gas data, see Johnson (ed.), WADD-TR-56-60, 1960. Ozone: For liquid see Brabets and Waterman, *J. Chem. Phys.*, **28** (1958): 1212.

¹⁵ For data on liquid Na-K alloys to 1500 °F and for liquid Na to 1460 °F, see Lubarsky and Kaufman, NACA Rep. 1270, 1956.

¹⁶ See also Evans and Wagman, *Bur. Stand. J. Res.*, **49** (1952): 141; Gratch, OTS PB 124957, 1950; Guthrie, Scott et al., *J. Am. Chem. Soc.*, **76** (1954): 1488.

2-164 PHYSICAL AND CHEMICAL DATA

TABLE 2-152 Specific Heat [kJ/(kg·K)] of Selected Elements

Symbol	Temperature, K														
	4	6	8	10	20	40	60	80	100	200	250	300	400	600	800
Al	0.00026	0.00050	0.00088	0.00140	0.0089	0.0775	0.214	0.357	0.481	0.797	0.859	0.902	0.949	1.042	1.134
Be	0.00008			0.00028	0.0014				0.195	1.109	1.537	1.840	2.191	2.605	2.823
Bi	0.00054	0.00220	0.00541	0.01040	0.0340	0.0729	0.092	0.102	0.109	0.120	0.121	0.122	0.123	0.142	0.136
Cr	0.00016	0.00029	0.00050	0.00081	0.0021	0.0107	0.059	0.127	0.190	0.382	0.424	0.450	0.501	0.565	0.611
Co	0.00036	0.00059	0.00085	0.00121	0.0048	0.0404	0.110	0.184	0.234	0.376	0.406	0.426	0.451	0.509	0.543
Cu	0.00011	0.00024	0.00048	0.00086	0.0076	0.059	0.137	0.203	0.254	0.357	0.377	0.386	0.396	0.431	0.448
Ge			0.00037	0.00081	0.0129	0.0619	0.108	0.153	0.192	0.286	0.305	0.323	0.343	0.364	0.377
Au	0.00018	0.00047	0.00126	0.00255	0.0163	0.0569	0.084	0.100	0.109	0.124	0.127	0.129	0.131	0.136	0.141
Ir				0.00032	0.0021				0.090	0.122	0.128	0.131	0.133	0.140	0.146
Fe	0.00038	0.00061	0.00090	0.00127	0.0039	0.0276	0.086	0.154	0.216	0.384	0.422	0.450	0.491	0.555	0.692
Pb	0.00075	0.00242	0.00747	0.01350	0.0531	0.0944	0.108	0.114	0.118	0.125	0.127	1.129	0.132	0.142	
Mg	0.00034	0.00080	0.00155	0.00172	0.0148	0.138	0.336	0.513	0.648	0.929	0.985	1.005	1.082	1.177	1.263
Hg	0.00417	0.01420	0.01820	0.02250	0.0515	0.0895	0.107	0.116	0.121	0.136	0.141	0.139	0.136	0.135	0.104
Mo	0.00011	0.00019	0.00032	0.00050	0.0029	0.0236	0.061	0.105	0.140	0.223	0.241	0.248	0.261	0.280	0.292
Ni	0.00054	0.00086	0.00121	0.00178	0.0058	0.0380	0.103	0.173	0.232	0.383	0.416	0.444	0.490	0.590	0.530
Pt	0.00019	0.00028	0.00067	0.00112	0.0077	0.0382	0.069	0.088	0.101	0.127	0.132	0.134	0.136	0.140	0.146
Ag	0.00016	0.00035	0.00093	0.00186	0.0159	0.0778	0.133	0.166	0.187	0.225	0.232	0.236	0.240	0.251	0.264
Sn	0.00024	0.00127	0.00423	0.00776	0.0400	0.108	0.149	0.173	0.189	0.214	0.220	0.222	0.245	0.257	0.257
Zn	0.00011	0.00029	0.00096	0.00250	0.0269	0.123	0.205	0.258	0.295	0.366	0.380	0.389	0.404	0.435	0.479

TABLE 2-153 Heat Capacities of Inorganic and Organic Liquids [J/(kmol·K)]

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1	C2	C3	C4	C5	$T_{\min}$, K	C_p at $T_{\min}$ × 1E-05	$T_{\max}$, K	C_p at $T_{\max}$ × 1E-05
1	Acetaldehyde	C ₂ H ₄ O	75-07-0	44.053	115,100	-433	1.425			150.15	0.8221	294.00	1.1097
2	Acetamide	C ₂ H ₅ NO	60-35-5	59.067	102,300	128.7				354.15	1.4788	571.00	1.7579
3	Acetic acid	C ₂ H ₄ O ₂	64-19-7	60.052	139,640	-320.8	0.8985			289.81	1.2213	391.05	1.5159
4	Acetic anhydride	C ₄ H ₆ O ₃	108-24-7	102.089	36,600	511				250.00	1.6435	350.00	2.1545
5	Acetone	C ₃ H ₆ O	67-64-1	58.079	135,600	-177	0.2837	0.000689		178.45	1.1696	329.44	1.3271
6	Acetonitrile	C ₂ H ₃ N	75-05-8	41.052	97,582	-122.2	0.34085			229.32	0.8748	354.75	0.9713
7	Acetylene	C ₂ H ₂	74-86-2	26.037	-122,020	3,082.7	-15.895	0.027732		192.40	0.8021	250.00	0.8853
8	Acrolein	C ₃ H ₄ O	107-02-8	56.063	103,090	-247.8	1.0343			253.00	1.0660	379.50	1.5801
9	Acrylic acid	C ₃ H ₄ O ₂	79-10-7	72.063	55,300	300				286.15	1.4114	375.00	1.6780
10	Acrylonitrile	C ₃ H ₃ N	107-13-1	53.063	109,900	-109.75	0.35441			189.63	1.0183	400.00	1.2271
11	Air	Mixture	132259-10-0	28.960	-214,460	9,185.1	-106.12	0.41616		75.00	0.5307	115.00	0.7132
12	Ammonia [use Eq. (2)]	H ₃ N	7664-41-7	17.031	61.289	80,925	799.4	-2,651		203.15	0.7575	401.15	4.1847
13	Anisole	C ₇ H ₈ O	100-66-3	108.138	150,940	93.455	0.23602			298.15	1.9978	484.20	2.5153
14	Argon	Ar	7440-37-1	39.948	134,390	-1,989.4	11.043			83.78	0.4523	135.00	0.6708
15	Benzamide	C ₇ H ₇ NO	55-21-0	121.137	161,440	260.66				403.00	2.6649	563.15	3.0823
16	Benzene	C ₆ H ₆	71-43-2	78.112	129,440	-169.5	0.64781			278.68	1.3251	353.24	1.5040
16	Benzene	C ₆ H ₆	71-43-2	78.112	162,940	-344.94	0.85562			278.68	1.3326	500.00	2.0437
17	Benzenethiol	C ₆ H ₆ S	108-98-5	110.177	119,780	180.34				258.27	1.6636	442.29	1.9954
18	Benzoic acid	C ₇ H ₆ O ₂	65-85-0	122.121	-5,480	647.12				395.45	2.5042	450.00	2.8572
19	Benzonitrile	C ₇ H ₅ N	100-47-0	103.121	93,383	242.61				260.40	1.5656	464.15	2.0599
20	Benzophenone	C ₁₃ H ₁₀ O	119-61-9	182.218	156,130	454.49				321.35	3.0218	640.00	4.4700
21	Benzyl alcohol	C ₇ H ₈ O	100-51-6	108.138	-334,997	3,644.21	-7.77514	0.00591102		257.85	1.8905	478.60	2.7617
22	Benzyl ethyl ether	C ₉ H ₁₀ O	539-30-0	136.191	87,500	480				275.65	2.1981	458.15	3.0741
23	Benzyl mercaptan	C ₇ H ₈ S	100-53-8	124.203	100,320	346.89				243.95	1.8494	472.03	2.6406
24	Biphenyl	C ₁₂ H ₁₀	92-52-4	154.208	121,770	429.3				342.20	2.6868	533.37	3.5075
25	Bromine	Br ₂	7726-95-6	159.808	179,400	-667.11	1.0701			265.90	0.7768	331.90	0.7587
26	Bromobenzene	C ₆ H ₅ Br	108-86-1	157.008	121,600	-9.45	0.358			293.15	1.4960	495.08	2.0467
27	Bromoethane	C ₂ H ₅ Br	74-96-4	108.965	94,364	-109.12	0.44032			160.00	0.8818	320.00	1.0453
28	Bromomethane	CH ₃ Br	74-83-9	94.939	129,730	-596.54	2.16	-0.0024234		184.45	0.7798	276.71	0.7870
29	1,2-Butadiene	C ₄ H ₆	590-19-2	54.090	135,150	-311.14	0.97007	-0.0001523		136.95	1.1034	290.00	1.2279
30	1,3-Butadiene	C ₄ H ₆	106-99-0	54.090	128,860	-323.1	1.015	0.000032		165.00	1.0333	350.00	1.4148
31	Butane	C ₄ H ₁₀	106-97-8	58.122	191,030	-1,675	12.5	-0.03874	4.6121E-05	134.86	1.1272	400.00	2.2237
32	1,2-Butanediol [use Eq. (2)]	C ₄ H ₁₀ O ₂	584-03-2	90.121	55,136	314,200	280.19	1,413.9		220.00	1.5590	670.00	5.2045
33	1,3-Butanediol [use Eq. (2)]	C ₄ H ₁₀ O ₂	107-88-0	90.121	42,152	324,580	517.35	1,449.5		196.15	0.6251	670.00	5.2437
34	1-Butanol	C ₄ H ₁₀ O	71-36-3	74.122	191,200	-730.4	2.2998			183.85	1.3465	391.90	2.5817
35	2-Butanol	C ₄ H ₁₀ O	78-92-2	74.122	426,790	-3,694.6	13.828	-0.0135		158.45	1.3485	372.90	2.7190
36	1-Butene	C ₄ H ₈	106-98-9	56.106	182,050	-1,611	11.963	-0.037454	4.5027E-05	87.80	1.1015	380.00	1.8103
37	cis-2-Butene	C ₄ H ₈	590-18-1	56.106	126,680	-65.47	-0.64	0.002912		134.26	1.1340	350.00	1.5022
38	trans-2-Butene	C ₄ H ₈	624-64-6	56.106	112,760	-104.7	0.5214			167.62	1.0986	274.03	1.2322
39	Butyl acetate	C ₆ H ₁₂ O ₂	123-86-4	116.158	111,850	384.52				298.15	2.2649	399.26	2.6537
40	Butylbenzene	C ₁₀ H ₁₄	104-51-8	134.218	182,470	-13,912	0.72897			185.30	2.0492	400.00	2.9354
41	Butyl mercaptan	C ₄ H ₁₀ S	109-79-5	90.187	232,190	-804.35	2.7063	-0.0023017		157.46	1.6365	390.00	1.9359
42	sec-Butyl mercaptan	C ₄ H ₁₀ S	513-53-1	90.187	197,890	-491.54	1.7219	-0.0012499		133.02	1.6003	370.00	1.8844
43	1-Butyne	C ₄ H ₆	107-00-6	54.090	136,340	-300.4	1.0216			147.43	1.1426	298.15	1.3759
44	Butyraldehyde	C ₄ H ₈ O	123-72-8	72.106	65,682	1,329.1	-7.1579	0.012755		176.75	1.4741	300.00	1.6459
45	Butyric acid	C ₄ H ₈ O ₂	107-92-6	88.105	237,700	-746.4	1.829			267.95	1.6902	436.42	2.6031
46	Butyronitrile	C ₄ H ₇ N	109-74-0	69.105	104,000	174				161.25	1.3206	390.75	1.7199
47	Carbon dioxide	CO ₂	124-38-9	44.010	-8,304,300	104,370	-433.33	0.60052		220.00	0.7827	290.00	1.6603
48	Carbon disulfide	CS ₂	75-15-0	76.141	85,600	-122	0.5605	-0.001452	2.008E-06	161.11	0.7577	552.00	1.3125
49	Carbon monoxide [use Eq. (2)]	CO	630-08-0	28.010	65,429	28,723	-847.39	1,959.6		68.15	0.5912	132.00	6.4799
50	Carbon tetrachloride	CCl ₄	56-23-5	153.823	-752,700	8,966.1	-30.394	0.034455		250.33	1.2763	388.71	1.6374
51	Carbon tetrafluoride	CF ₄	75-73-0	88.004	104,600	-500.6	2.2851			89.56	0.7810	145.10	0.8007
52	Chlorine	Cl ₂	7782-50-5	70.906	63,936	46.35	-0.1623			172.12	0.6711	239.12	0.6574
53	Chlorobenzene	C ₆ H ₅ Cl	108-90-7	112.557	-1,307,500	15,338	-53.974	0.063483		227.95	1.3617	360.00	1.8101
54	Chloroethane	C ₂ H ₅ Cl	75-00-3	64.514	127,900	-345.15	0.915			134.80	0.9800	340.00	1.1632
55	Chloroform	CHCl ₃	67-66-3	119.378	124,850	-166.34	0.43209			233.15	1.0956	366.48	1.2192
56	Chloromethane	CH ₃ Cl	74-87-3	50.488	96,910	-207.9	0.37456	0.000488		175.43	0.7460	373.15	0.9684
57	1-Chloropropane	C ₃ H ₇ Cl	540-54-5	78.541	132,280	-153.27	0.50836			150.35	1.2073	319.67	1.3523
58	2-Chloropropane	C ₃ H ₇ Cl	75-29-6	78.541	69,362	215.01				200.00	1.1236	308.85	1.3577

TABLE 2-153 Heat Capacities of Inorganic and Organic Liquids [J/(kmol·K)] (Continued)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1	C2	C3	C4	C5	$T_{\min}$, K	C_p at $T_{\min}$ $\times 1E-05$	$T_{\max}$, K	C_p at $T_{\max}$ $\times 1E-05$
59	<i>m</i> -Cresol	C ₇ H ₈ O	108-39-4	108.138	-246,700	3,256.8	-7.4202	0.0060467		285.39	2.1895	400.00	2.5578
60	<i>o</i> -Cresol	C ₇ H ₈ O	95-48-7	108.138	-185,150	3,148	-8.0367	0.007254		304.20	2.3297	400.00	2.5243
61	<i>p</i> -Cresol	C ₇ H ₈ O	106-44-5	108.138	259,980	-1,112.3	4.9427	-0.0054367		307.93	2.2740	400.00	2.5794
62	Cumene	C ₉ H ₁₂	98-82-8	120.192	61,723	494.81				177.14	1.4937	425.56	2.7229
63	Cyanogen	C ₂ N ₂	460-19-5	52.035	65,516	144.7	0.063229			200.08	0.9700	300.08	1.1463
64	Cyclobutane	C ₄ H ₈	287-23-0	56.106	101,920	-215.81	0.8103			190.00	0.9017	298.15	1.0961
65	Cyclohexane	C ₆ H ₁₂	110-82-7	84.159	-220,600	3,118.3	-9.4216	0.010687		279.69	1.4836	400.00	2.0323
66	Cyclohexanol	C ₆ H ₁₂ O	108-93-0	100.159	-40,000	853				296.60	2.1300	434.00	3.3020
67	Cyclohexanone	C ₆ H ₁₀ O	108-94-1	98.143	6,110.4	600.94				290.00	1.8038	489.75	3.0042
68	Cyclohexene	C ₆ H ₁₀	110-83-8	82.144	105,850	-60	0.68			169.67	1.1525	356.12	1.7072
69	Cyclopentane	C ₅ H ₁₀	287-92-3	70.133	122,530	-403.8	1.7344	-0.0010975		179.28	0.9956	322.40	1.3584
70	Cyclopentene	C ₅ H ₈	142-29-0	68.117	125,380	-349.7	1.143			138.13	0.9888	317.38	1.2953
71	Cyclopropane	C ₃ H ₆	75-19-4	42.080	89,952	-196.63	0.65237			150.00	0.7514	298.15	0.8932
72	Cyclohexyl mercaptan	C ₆ H ₁₂ S	1569-69-3	116.224	177,560	-179.12	0.76723			189.64	1.7118	431.95	2.4334
73	Decanal	C ₁₀ H ₂₀ O	112-31-2	156.265	150,460	586.63				267.15	3.0718	488.15	4.3682
74	Decane	C ₁₀ H ₂₂	124-18-5	142.282	278,620	-197.91	1.0737			243.51	2.9409	460.00	4.1478
75	Decanoic acid	C ₁₀ H ₂₀ O ₂	334-48-5	172.265	219,840	140.41	0.9968			304.75	3.5521	543.15	5.9017
76	1-Decanol	C ₁₀ H ₂₂ O	112-30-1	158.281	4,988,500	-52,898	216.35	-0.37538	0.00023674	280.05	3.5373	503.00	5.0169
77	1-Decene	C ₁₀ H ₂₀	872-05-9	140.266	417,440	-1,616.5	5.3948	-0.004348		206.89	2.7541	443.75	3.8250
78	Decyl mercaptan	C ₁₀ H ₂₂ S	143-10-2	174.347	314,570	-160.93	0.95561			247.56	3.3330	512.35	4.8297
79	1-Decyne	C ₁₀ H ₁₈	764-93-2	138.250	276,900	-371.23	1.5774			229.15	2.7466	447.15	4.2629
80	Deuterium	D ₂	7782-39-0	4.032									
81	1,1-Dibromoethane	C ₂ H ₄ Br ₂	557-91-5	187.861	149,400	-231.8	0.5946			210.15	1.2695	381.15	1.4743
82	1,2-Dibromoethane	C ₂ H ₄ Br ₂	106-93-4	187.861	200,560	-491.44	0.9187			282.85	1.3506	410.00	1.5350
83	Dibromomethane	CH ₂ Br ₂	74-95-3	173.835	202,580	-726.3	1.3377			240.00	1.0532	370.10	1.1701
84	Diethyl ether	C ₄ H ₁₀ O	142-96-1	130.228	270,720	-259.83	0.95427			175.30	2.5450	450.00	3.4704
85	<i>m</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	541-73-1	147.002	114,880	187.25				248.39	1.6139	400.00	1.8978
86	<i>o</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	95-50-1	147.002	93,093	183.97	0.2314			273.15	1.6061	528.75	2.5506
87	<i>p</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	106-46-7	147.002	133,950	-24.84	0.48191			326.14	1.7711	513.56	2.4829
88	1,1-Dichloroethane	C ₂ H ₄ Cl ₂	75-34-3	98.959	126,340	-94.63	0.32			176.19	1.1960	330.45	1.3001
89	1,2-Dichloroethane	C ₂ H ₄ Cl ₂	107-06-2	98.959	179,170	-444.74	0.93009			237.49	1.2601	356.59	1.3885
90	Dichloromethane	CH ₂ Cl ₂	75-09-2	84.933	98,968	-62.941	0.23265			180.00	0.9518	320.00	1.0265
91	1,1-Dichloropropane	C ₃ H ₆ Cl ₂	78-99-9	112.986	144,560	-53.605	0.30617			180.00	1.4483	361.25	1.6515
92	1,2-Dichloropropane	C ₃ H ₆ Cl ₂	78-87-5	112.986	111,560	149.44				275.00	1.5266	369.52	1.6678
93	Diethanol amine	C ₄ H ₁₁ NO ₂	111-42-2	105.136	184,200	286				301.15	2.7033	541.54	3.3908
94	Diethyl amine	C ₄ H ₁₁ N	109-89-7	73.137	101,330	243.18				223.35	1.5564	328.60	1.8124
95	Diethyl ether	C ₄ H ₁₀ O	60-29-7	74.122	44,400	1,301	-5.5	0.008763		156.92	1.4698	460.00	3.3202
96	Diethyl sulfide	C ₄ H ₁₀ S	352-93-2	90.187	238,520	-1,038.4	4.0587	-0.0044691		181.95	1.5703	322.08	1.7579
97	1,1-Difluoroethane [use Eq. (2)]	C ₂ H ₄ F ₂	75-37-6	66.050	67.155	105,580	310.21	-490.54		154.56	0.9915	359.98	1.6874
98	1,2-Difluoroethane	C ₂ H ₄ F ₂	624-72-6	66.050	82,577	109.85				215.00	1.0619	283.65	1.1374
99	Difluoromethane	CH ₂ F ₂	75-10-5	52.023	263,980	-1,791.1	4.3666			200.00	0.8042	250.00	0.8912
100	Di-isopropyl amine	C ₆ H ₁₅ N	108-18-9	101.190	98,434	429.04				275.00	2.1642	357.05	2.5162
101	Di-isopropyl ether	C ₆ H ₁₄ O	108-20-3	102.175	163,000	-4.5	0.62			187.65	1.8399	341.45	2.3375
102	Di-isopropyl ketone	C ₇ H ₁₄ O	565-80-0	114.185	179,270	28.37	0.5375			204.81	2.0763	410.00	2.8126
103	1,1-Dimethoxyethane	C ₄ H ₁₀ O ₂	534-15-6	90.121	187,790	-313.41	1.1023			159.95	1.6586	337.45	2.0755
104	1,2-Dimethoxypropane	C ₅ H ₁₂ O ₂	7778-85-0	104.148	199,930	-191.5	0.87664			226.10	2.0145	366.15	2.4734
105	Dimethyl acetylene	C ₄ H ₆	503-17-3	54.090	88,153	124.16				240.91	1.1806	300.13	1.2542
106	Dimethyl amine	C ₂ H ₇ N	124-40-3	45.084	-214,870	3,787.2	-13.781	0.016924		180.96	1.1947	298.15	1.3779
107	2,3-Dimethylbutane	C ₆ H ₁₄	79-29-8	86.175	129,450	18.5	0.608			145.19	1.4495	331.13	2.0224
108	1,1-Dimethylcyclohexane	C ₈ H ₁₆	590-66-9	112.213	134,500	8.765	0.81151			239.66	1.8321	392.70	2.6309
109	<i>cis</i> -1,2-Dimethylcyclohexane	C ₈ H ₁₆	2207-01-4	112.213	150,130	-62.38	0.8851			223.16	1.8029	402.94	2.6870
110	<i>trans</i> -1,2-Dimethylcyclohexane	C ₈ H ₁₆	6876-23-9	112.213	155,560	-145.26	1.0932			184.99	1.6610	396.58	2.6989
111	Dimethyl disulfide	C ₂ H ₆ S ₂	624-92-0	94.199	171,580	-256.67	0.5727			188.44	1.4355	360.00	1.5340
112	Dimethyl ether	C ₂ H ₆ O	115-10-6	46.068	110,100	-157.47	0.51853			131.65	0.9836	250.00	1.0314
113	<i>N,N</i> -Dimethyl formamide	C ₃ H ₇ NO	68-12-2	73.094	147,900	-106	0.384			273.82	1.4767	466.44	1.8200
114	2,3-Dimethylpentane	C ₇ H ₁₆	565-59-3	100.202	146,420	59.2	0.604			90.00	1.5664	380.00	2.5613
115	Dimethyl phthalate	C ₁₀ H ₁₀ O ₄	131-11-3	194.184	206,560	325.75				274.16	2.9587	360.00	3.2383
116	Dimethylsilane	C ₂ H ₈ Si	1111-74-6	60.170	131,810					298.15	1.3181	298.15	1.3181

117	Dimethyl sulfide	C ₂ H ₆ S	75-18-3	62.134	146,950	-380.06	1.2035	-0.00084787		174.88	1.1276	310.48	1.1959
118	Dimethyl sulfoxide	C ₂ H ₆ OS	67-68-5	78.133	240,300	-595	1.013			291.67	1.5293	422.15	1.6965
119	Dimethyl terephthalate	C ₁₀ H ₁₀ O ₄	120-61-6	194.184	190,020	431.04				423.40	3.7252	466.35	3.9104
120	1,4-Dioxane	C ₄ H ₈ O ₂	123-91-1	88.105	956,860	-5,559.9	9.6124			284.95	1.5306	374.47	2.2277
121	Diphenyl ether	C ₁₂ H ₁₀ O	101-84-8	170.207	134,160	447.67				300.03	2.6847	570.00	3.8933
122	Dipropyl amine	C ₆ H ₁₅ N	142-84-7	101.190	49,120	562.24				277.90	2.0537	407.90	2.7846
123	Dodecane	C ₁₂ H ₂₆	112-40-3	170.335	508,210	-1,368.7	3.1015			263.57	3.6292	330.00	3.9429
124	Eicosane	C ₂₀ H ₄₂	112-95-8	282.547	352,720	807.32	0.2122			309.58	6.2299	616.93	9.3154
125	Ethane [use Eq. (2)]	C ₂ H ₆	74-84-0	30.069	44.009	89,718	918.77	-1,886		92.00	0.6855	290.00	1.2444
126	Ethanol	C ₂ H ₆ O	64-17-5	46.068	102,640	-139.63	-0.030341	0.0020386		159.05	0.8787	390.00	1.6450
127	Ethyl acetate	C ₄ H ₈ O ₂	141-78-6	88.105	226,230	-624.8	1.472			189.60	1.6068	350.21	1.8796
128	Ethyl amine	C ₂ H ₇ N	75-04-7	45.084	121,700	38,993				192.15	1.2919	289.73	1.3300
129	Ethylbenzene	C ₈ H ₁₀	100-41-4	106.165	154,040	-142.29	0.80539			178.20	1.5426	409.35	2.3075
130	Ethyl benzoate	C ₉ H ₁₀ O ₂	93-89-0	150.175	124,500	370.6				238.45	2.1287	486.55	3.0482
131	2-Ethyl butanoic acid	C ₆ H ₁₂ O ₂	88-09-5	116.158	56,359	603.02				258.15	2.1203	466.95	3.3794
132	Ethyl butyrate	C ₆ H ₁₂ O ₂	105-54-4	116.158	82,434	422.45	0.20992			285.50	2.2015	428.25	3.0185
133	Ethylcyclohexane	C ₈ H ₁₆	1678-91-7	112.213	132,360	72.74	0.64738			161.84	1.6109	404.95	2.6798
134	Ethylcyclopentane	C ₇ H ₁₄	1640-89-7	98.186	178,520	-518.35	2.3255	-0.0016818		134.71	1.4678	301.82	1.8767
135	Ethylene	C ₂ H ₄	74-85-1	28.053	247,390	-4,428	40.936	-0.1697	0.00026816	104.00	0.7012	252.70	0.9758
136	Ethylenediamine	C ₂ H ₈ N ₂	107-15-3	60.098	184,440	-150.2	0.37044			284.29	1.7168	390.41	1.8226
137	Ethylene glycol	C ₂ H ₆ O ₂	107-21-1	62.068	35,540	436.78	-0.18486			260.15	1.3666	493.15	2.0598
138	Ethyleneimine	C ₂ H ₅ N	151-56-4	43.068	46,848	205.35				250.00	0.9819	329.00	1.1441
139	Ethylene oxide	C ₂ H ₄ O	75-21-8	44.053	144,710	-758.87	2.8261	-0.003064		160.65	0.8303	283.85	0.8693
140	Ethyl formate	C ₃ H ₆ O ₂	109-94-4	74.079	80,000	223.6				254.20	1.3684	374.20	1.6367
141	2-Ethyl hexanoic acid	C ₈ H ₁₆ O ₂	149-57-5	144.211	207,670	-17,907	1.0493			235.00	2.6141	510.10	4.7157
142	Ethylhexyl ether	C ₈ H ₁₈ O	5756-43-4	130.228	146,040	458.22				298.15	2.8266	417.15	3.3719
143	Ethylisopropyl ether	C ₅ H ₁₂ O	625-54-7	88.148	106,250	292.15				298.15	1.9335	326.15	2.0153
144	Ethylisopropyl ketone	C ₆ H ₁₂ O	565-69-5	100.159	229,250	-404.54	1.1382			204.15	1.9410	386.55	2.4295
145	Ethyl mercaptan	C ₂ H ₆ S	75-08-1	62.134	134,670	-234.39	0.59656			125.26	1.1467	315.25	1.2007
146	Ethyl propionate	C ₅ H ₁₀ O ₂	105-37-3	102.132	76,330	400.1				298.15	1.9562	410.00	2.4037
147	Ethylpropyl ether	C ₅ H ₁₂ O	628-32-0	88.148	103,680	726.3	-2.6047	0.0040957		145.65	1.6686	320.00	2.0358
148	Ethyltrichlorosilane	C ₂ H ₅ Cl ₃ Si	115-21-9	163.506	105,150	0.46693				167.55	1.3255	371.05	2.0109
149	Fluorine	F ₂	7782-41-4	37.997	-94,585	7,529.9	-139.6	1.1301	-0.0033241	58.00	0.5541	98.00	0.5966
149	Fluorine	F ₂	7782-41-4	37.997	1724,400	-59,924	537.85			53.48	0.5798	56.00	0.5535
150	Fluorobenzene	C ₆ H ₅ F	462-06-6	96.102	-991,200	11,734	-40.669	0.047333		239.99	1.3675	319.99	1.5018
151	Fluoroethane	C ₂ H ₅ F	353-36-6	48.060	85,663	-118.56	0.55459			140.00	0.7994	240.00	0.8915
152	Fluoromethane	CH ₃ F	593-53-3	34.033	74,746	-132.32	0.53772			140.00	0.6676	220.00	0.7166
153	Formaldehyde	CH ₂ O	50-00-0	30.026	61,900	28.3				204.00	0.6767	234.00	0.6852
154	Formamide	CH ₃ NO	75-12-7	45.041	63,400	150.6				292.00	1.0738	493.00	1.3765
155	Formic acid	CH ₂ O ₂	64-18-6	46.026	78,060	71.54				281.45	0.9820	380.00	1.0525
156	Furan	C ₄ H ₄ O	110-00-9	68.074	114,370	-215.69	0.72691			187.55	0.9949	304.50	1.1609
157	Helium-4	He	7440-59-7	4.003	387,220	-465,570	211,800	-42,494	3212.9	2.20	0.1087	4.60	0.2965
157	Helium-4	He	7440-59-7	4.003	410,430	-464,890	135,100			1.80	0.1135	2.10	0.2995
158	Heptadecane	C ₁₇ H ₃₆	629-78-7	240.468	376,970	347.82	0.57895			295.13	5.3005	575.30	7.6869
159	Heptanal	C ₇ H ₁₄ O	111-71-7	114.185	222,360	-105.17	0.65074			229.80	2.3256	381.25	2.7685
160	Heptane [use Eq. (2)]	C ₇ H ₁₆	142-82-5	100.202	61.26	314,410	1,824.6	-2,547.9		182.57	1.9989	520.00	4.0657
161	Heptanoic acid	C ₇ H ₁₄ O ₂	111-14-8	130.185	194,570	-23,206	0.88395			265.83	2.5087	496.15	4.0065
162	1-Heptanol	C ₇ H ₁₆ O	111-70-6	116.201	2,416,800	-26,105	110.03	-0.19172	0.00011968	239.15	2.3590	448.60	3.8766
163	2-Heptanol	C ₇ H ₁₆ O	543-49-7	116.201	283,127	-1,037.63	3.44064			230.00	2.2649	432.90	4.7873
164	3-Heptanone	C ₇ H ₁₄ O	106-35-4	114.185	270,730	-399.89	1.0601			234.15	2.3522	480.00	3.2303
165	2-Heptanone	C ₇ H ₁₄ O	110-43-0	114.185	265,040	-375.68	1.0024			238.15	2.3242	490.00	3.2163
166	1-Heptene	C ₇ H ₁₄	592-76-7	98.186	267,950	-1,315.9	6.5242	-0.011994	9.3808E-06	154.12	1.8150	366.79	2.4096
167	Heptyl mercaptan	C ₇ H ₁₆ S	1639-09-4	132.267	236,870	-158.01	0.78982			229.92	2.4229	460.00	3.3131
168	1-Heptyne	C ₇ H ₁₂	628-71-7	96.170	46,798	761.13	-0.62882			200.00	1.7387	372.93	2.4319
169	Hexadecane	C ₁₆ H ₃₄	544-76-3	226.441	370,350	231.47	0.68632			291.31	4.9602	560.01	7.1521
170	Hexanal	C ₆ H ₁₂ O	66-25-1	100.159	117,700	329.52				217.15	1.8926	401.45	2.4999
171	Hexane	C ₆ H ₁₄	110-54-3	86.175	172,120	-183.78	0.88734			177.83	1.6750	460.00	2.7534
172	Hexanoic acid	C ₆ H ₁₂ O ₂	142-62-1	116.158	161,980	44.116	0.709			269.25	2.2526	478.85	3.4568
173	1-Hexanol	C ₆ H ₁₄ O	111-27-3	102.175	1,638,600	-17,261	71.721	-0.12026	0.000071087	228.55	1.9821	460.00	3.5197
174	2-Hexanol	C ₆ H ₁₄ O	626-93-7	102.175	267,628	-1,033.06	3.35185			223.00	2.0394	585.30	8.1124
175	2-Hexanone	C ₆ H ₁₂ O	591-78-6	100.159	208,250	-107.47	0.2062	0.00070293		217.35	2.0185	460.00	2.7087
176	3-Hexanone	C ₆ H ₁₂ O	589-38-8	100.159	235,960	-345.94	0.94278			217.50	2.0532	460.00	2.7632
177	1-Hexene	C ₆ H ₁₂	592-41-6	84.159	164,640	-200.37	0.8784			133.39	1.5354	336.63	1.9673

TABLE 2-153 Heat Capacities of Inorganic and Organic Liquids [J/(kmol·K)] (Continued)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1	C2	C3	C4	C5	$T_{\min}$, K	C_p at $T_{\min}$ × 1E-05	$T_{\max}$, K	C_p at $T_{\max}$ × 1E-05
178	3-Hexyne	C ₆ H ₁₀	928-49-4	82.144	82,795	283.4				300.00	1.6781	354.35	1.8322
179	Hexyl mercaptan	C ₆ H ₁₄ S	111-31-9	118.240	303,320	-1,009	3.3885	-0.002762		192.62	2.1495	430.00	2.7639
180	1-Hexyne	C ₆ H ₁₀	693-02-7	82.144	93,000	326				200.00	1.5820	344.48	2.0530
181	2-Hexyne	C ₆ H ₁₀	764-35-2	82.144	94,860	254.15				300.00	1.7110	357.67	1.8576
182	Hydrazine	H ₄ N ₂	302-01-2	32.045	79,815	50.929	0.043379			274.69	0.9708	653.15	1.3158
183	Hydrogen [use Eq. (2)]	H ₂	1333-74-0	2.016	66,653	6,765.9	-123.63	478.27		13.95	0.1262	32.00	1.3122
184	Hydrogen bromide	HBr	10035-10-6	80.912	57,720	9.9				185.15	0.5955	206.45	0.5976
185	Hydrogen chloride	HCl	7647-01-0	36.461	47,300	90				165.00	0.6215	185.00	0.6395
186	Hydrogen cyanide	CHN	74-90-8	27.025	95,398	-197.52	0.3883			259.83	0.7029	298.85	0.7105
187	Hydrogen fluoride	HF	7664-39-3	20.006	62,520	-223.02	0.6297			189.79	0.4288	292.67	0.5119
188	Hydrogen sulfide [use Eq. (2)]	H ₂ S	7783-06-4	34.081	64,666	49,354	22.493	-1,623		187.68	0.6733	370.00	4.9183
189	Isobutyric acid	C ₄ H ₈ O ₂	79-31-2	88.105	127,540	-65.35	0.82867			270.00	1.7031	427.65	2.5114
190	Isopropyl amine	C ₃ H ₉ N	75-31-0	59.110	-32,469	1,977.1	-7.0145	0.0086913		177.95	1.4621	320.00	1.6671
191	Malonic acid	C ₃ H ₄ O ₄	141-82-2	104.061	157,850	-41.619	0.42817			407.95	2.1213	603.75	2.8880
192	Methacrylic acid	C ₄ H ₆ O ₂	79-41-4	86.089	146,290	-58.59	0.3582			288.15	1.5915	434.15	1.8837
193	Methane [use Eq. (2)]	CH ₄	74-82-8	16.042	65,708	38,883	-257.95	614.07		90.69	0.5361	190.00	14.9780
194	Methanol	CH ₃ O	67-56-1	32.042	105,800	-362.23	0.9379			175.47	0.7112	400.00	1.1097
195	N-Methyl acetamide	C ₃ H ₇ NO	79-16-3	73.094	62,600	243.4				359.00	1.4998	538.50	1.9367
196	Methyl acetate	C ₃ H ₆ O ₂	79-20-9	74.079	61,260	270.9				253.40	1.2991	373.40	1.6241
197	Methyl acetylene	C ₃ H ₄	74-99-7	40.064	79,791	89.49				200.00	0.9769	249.94	1.0216
198	Methyl acrylate	C ₄ H ₆ O ₂	96-33-3	86.089	275,500	-1,147	2.568			196.32	1.4930	353.35	1.9084
199	Methyl amine	CH ₃ N	74-89-5	31.057	92,520	37.45				179.69	0.9925	266.82	1.0251
200	Methyl benzoate	C ₈ H ₈ O ₂	93-58-3	136.148	125,630	279.75				260.75	1.9857	472.65	2.5785
201	3-Methyl-1,2-butadiene	C ₅ H ₈	598-25-4	68.117	135,370	-133.34	0.63868			159.53	1.3035	314.56	1.5662
202	2-Methylbutane	C ₅ H ₁₂	78-78-4	72.149	108,300	146	-0.292	0.00151		113.25	1.2328	310.00	1.7048
203	2-Methylbutanoic acid	C ₅ H ₁₀ O ₂	116-53-0	102.132	74,200	417.4				321.50	2.0839	481.50	2.7518
204	3-Methyl-1-butanol	C ₅ H ₁₂ O	123-51-3	88.148	247,870	-1,145	3.4223			155.95	1.5254	404.15	3.4411
205	2-Methyl-1-butene	C ₅ H ₁₀	563-46-2	70.133	149,510	-247.63	0.91849			135.58	1.3282	304.31	1.5921
206	2-Methyl-2-butene	C ₅ H ₁₀	513-35-9	70.133	151,600	-266.72	0.90847			139.39	1.3207	311.71	1.5673
207	2-Methyl-1-butene-3-yne	C ₆ H ₈	78-80-8	66.101	81,919	181.01				298.15	1.3589	305.40	1.3720
208	Methylbutyl ether	C ₅ H ₁₂ O	628-28-4	88.148	177,850	-171.57	0.74379			157.48	1.6928	343.31	2.0661
209	Methylbutyl sulfide	C ₅ H ₁₂ S	628-29-5	104.214	198,390	-220.35	0.76096			175.30	1.8315	510.00	2.8394
210	3-Methyl-1-butyne	C ₅ H ₈	598-23-2	68.117	105,200	191.1				200.00	1.4342	299.49	1.6243
211	Methyl butyrate	C ₅ H ₁₀ O ₂	623-42-7	102.132	102,930	129.1	0.62516			277.25	1.8678	415.87	2.6474
212	Methylchlorosilane	CH ₃ ClSi	993-00-0	80.589	47,726	338.4				250.00	1.3233	325.00	1.5771
213	Methylcyclohexane	C ₇ H ₁₄	108-87-2	98.186	131,340	-63.1	0.8125			146.58	1.3955	320.00	1.9435
214	1-Methylcyclohexanol	C ₇ H ₁₄ O	590-67-0	114.185	50,578	508.59				300.00	2.0315	441.15	2.7494
215	cis-2-Methylcyclohexanol	C ₇ H ₁₄ O	7443-70-1	114.185	118,600	447.07				300.00	2.5272	438.15	3.1448
216	trans-2-Methylcyclohexanol	C ₇ H ₁₄ O	7443-52-9	114.185	118,170	447.99				300.00	2.5257	440.15	3.1535
217	Methylcyclopentane	C ₆ H ₁₂	96-37-7	84.159	155,920	-490	2.1383	-0.0015585		130.73	1.2492	366.48	1.8682
218	1-Methylcyclopentene	C ₆ H ₁₀	693-89-0	82.144	53,271	327.92				200.00	1.1885	348.64	1.6760
219	3-Methylcyclopentene	C ₆ H ₁₀	1120-62-3	82.144	46,457	346.93				200.00	1.1584	338.05	1.6374
220	Methyldichlorosilane	CH ₂ Cl ₂ Si	75-54-7	115.034	27,030	413				250.00	1.3028	350.00	1.7158
221	Methylethyl ether	C ₃ H ₈ O	540-67-0	60.095	85,383	199.08	-0.061547			160.00	1.1566	280.50	1.3638
222	Methylethyl ketone	C ₄ H ₈ O	78-93-3	72.106	132,300	200.87	-0.9597	0.0019533		186.48	1.4905	373.15	1.7511
223	Methylethyl sulfide	C ₃ H ₈ S	624-89-5	76.161	161,240	-288.61	0.78179			167.23	1.3484	339.80	1.5344
224	Methyl formate	C ₂ H ₄ O ₂	107-31-3	60.052	130,200	-396	1.21			174.15	0.9793	304.90	1.2195
225	Methylisobutyl ether	C ₅ H ₁₂ O	625-44-5	88.148	92,919	324.43				298.15	1.8965	350.00	2.0647
226	Methylisobutyl ketone	C ₆ H ₁₂ O	108-10-1	100.159	183,650	-79.862	0.60769			189.15	1.9029	389.15	2.4460
227	Methyl Isocyanate	C ₂ H ₃ NO	624-83-9	57.051	149,770	-529.82	1.3499			256.15	1.0263	366.00	1.3668
228	Methylisopropyl ether	C ₄ H ₁₀ O	598-53-8	74.122	143,440	-154.07	0.7255			127.93	1.3560	310.00	1.6540
229	Methylisopropyl ketone	C ₅ H ₁₀ O	563-80-4	86.132	191,170	-331.04	0.98445			180.15	1.6348	440.00	2.3610
230	Methylisopropyl sulfide	C ₄ H ₁₀ S	1551-21-9	90.187	211,170	-661.97	2.4216	-0.0021383		171.64	1.5808	357.91	1.8641
231	Methyl mercaptan	CH ₃ S	74-93-1	48.107	115,300	-263.23	0.60412			150.18	0.8939	298.15	0.9052
232	Methyl methacrylate	C ₅ H ₈ O ₂	80-62-6	100.116	255,100	-938.4	2.413			224.95	1.6611	373.45	2.4118
233	2-Methyloctanoic acid	C ₈ H ₁₆ O ₂	3004-93-1	158.238	226,650	15,421	1.0578			240.00	2.9128	518.15	5.1864
234	2-Methylpentane	C ₆ H ₁₄	107-83-5	86.175	142,220	-47.83	0.739			119.55	1.4706	333.41	2.0842
235	Methyl pentyl ether	C ₆ H ₁₄ O	628-80-8	102.175	251,890	-468.32	1.2209			176.00	2.0728	372.00	2.4663
236	2-Methylpropane	C ₄ H ₁₀	75-28-5	58.122	172,370	-1,783.9	14.759	-0.047909	0.00005805	113.54	0.9961	380.00	2.0725

237	2-Methyl-2-propanol	C ₄ H ₁₀ O	75-65-0	74.122	-925,460	7,894.9	-17.661	0.013617		298.96	2.2016	460.00	2.9455
238	2-Methyl propene	C ₄ H ₈	115-11-7	56.106	87,680	217.1	-0.9153	0.002266		132.81	1.0568	343.15	1.4596
239	Methyl propionate	C ₅ H ₈ O ₂	554-12-1	88.105	71,140	335.5				300.00	1.7179	390.00	2.0198
240	Methylpropyl ether	C ₄ H ₁₀ O	557-17-5	74.122	144,110	-102.09	0.58113			133.97	1.4086	312.20	1.6888
241	Methylpropyl sulfide	C ₄ H ₁₀ S	3877-15-4	90.187	179,850	-264.1	0.79202			160.17	1.5787	368.69	1.9014
242	Methylsilane	CH ₃ Si	992-94-9	46.144	113,470					298.15	1.1347	298.15	1.1347
243	alpha-Methyl styrene	C ₉ H ₁₀	98-83-9	118.176	76,822	421.6				249.95	1.8220	438.65	2.6176
244	Methyl <i>tert</i> -butyl ether	C ₅ H ₁₂ O	1634-04-4	88.148	134,300	94.356	-0.0032	0.0009795		164.55	1.5410	328.18	1.9954
245	Methyl vinyl ether	C ₃ H ₆ O	107-25-5	58.079	73,600	184.7				151.15	1.0152	278.65	1.2507
246	Naphthalene	C ₁₀ H ₈	91-20-3	128.171	29,800	527.5				353.43	2.1623	491.14	2.8888
247	Neon	Ne	7440-01-9	20.180	1,034,100	-138,770	7,154	-162.55	1.3841	24.56	0.3666	40.00	0.6980
248	Nitroethane	C ₂ H ₅ NO ₂	79-24-3	75.067	187,740	-497.6	1.0691			183.63	1.3242	387.22	1.5536
249	Nitrogen	N ₂	7727-37-9	28.013	281,970	-12,281	248	-2.2182	0.0074902	63.15	0.5593	112.00	0.7960
250	Nitrogen trifluoride	F ₃ N	7783-54-2	71.002	101,400	-682.11	3.8912			117.00	0.7486	175.50	1.0154
251	Nitromethane	CH ₃ NO ₂	75-52-5	61.040	116,270	-135.3	0.345			244.60	1.0382	473.15	1.2949
252	Nitrous oxide	N ₂ O	10024-97-2	44.013	67,556	54.373				182.30	0.7747	200.00	0.7843
253	Nitric oxide	NO	10102-43-9	30.006	-2,979,600	76,602	-652.59	1.8879		109.50	0.6229	150.00	1.9909
254	Nonadecane	C ₁₉ H ₄₀	629-92-5	268.521	342,570	762.08	0.20481			305.04	5.9409	603.05	8.7663
255	Nonanal	C ₉ H ₁₈ O	124-19-6	142.239	136,820	531.29				255.15	2.7238	468.15	3.8554
256	Nonane	C ₉ H ₂₀	111-84-2	128.255	383,080	-1,139.8	2.7101			219.66	2.6348	325.00	2.9890
257	Nonanoic acid	C ₉ H ₁₈ O ₂	112-05-0	158.238	224,336	49.726	0.9813			285.55	3.1855	528.75	5.2498
258	1-Nonanol	C ₉ H ₂₀ O	143-08-8	144.255	10,483,000	-115,220	476.87	-0.85381	0.00056246	310.00	3.5059	460.00	4.6494
259	2-Nonanol	C ₉ H ₂₀ O	628-99-9	144.255	329,641	-1,046.78	3.61823			238.15	2.8555	649.50	11.7608
260	1-Nonene	C ₉ H ₁₈	124-11-8	126.239	254,490	-298.06	1.1707			191.91	2.4041	420.02	3.3583
261	Nonyl mercaptan	C ₉ H ₂₀ S	1455-21-6	160.320	265,350	-46.22	0.79154			253.05	3.0434	492.95	4.3491
262	1-Nonyne	C ₉ H ₁₆	3452-09-3	124.223	253,580	-366.3	1.4881			223.15	2.4594	423.85	3.6566
263	Octadecane	C ₁₈ H ₃₈	593-45-3	254.494	399,430	374.64	0.58156			301.31	5.6511	589.86	8.2276
264	Octanal	C ₈ H ₁₆ O	124-13-0	128.212	130,650	463.61				246.00	2.4470	447.15	3.3795
265	Octane	C ₈ H ₁₈	111-65-9	114.229	224,830	-186.63	0.95891			216.38	2.2934	460.00	3.4189
266	Octanoic acid	C ₈ H ₁₆ O ₂	124-07-2	144.211	205,260	44.392	0.8956			289.65	2.9326	512.85	4.6358
267	1-Octanol	C ₈ H ₁₈ O	111-87-5	130.228	571,370	-4,849	19.725	-0.021532		250.00	2.5550	467.10	4.1566
268	2-Octanol	C ₈ H ₁₈ O	123-96-6	130.228	319,198	-1,042.21	3.52943			241.55	2.7338	452.90	5.7113
269	2-Octanone	C ₈ H ₁₆ O	111-13-7	128.212	300,400	-426.2	1.1172			252.86	2.6406	500.00	3.6660
270	3-Octanone	C ₈ H ₁₆ O	106-68-3	128.212	289,980	-417.27	1.2218			255.55	2.6314	440.65	3.4335
271	1-Octene	C ₈ H ₁₆	111-66-0	112.213	509,420	-4,279.1	21.477	-0.044462	3.5028E-05	171.45	2.1327	394.41	2.8235
272	Octyl mercaptan	C ₈ H ₁₈ S	111-88-6	146.294	240,040	-33.198	0.67889			240.00	2.7118	472.19	3.7573
273	1-Octyne	C ₈ H ₁₄	629-05-0	110.197	42,642	886.67	-0.69315			200.00	1.9225	399.35	2.8619
274	Oxalic acid	C ₂ H ₂ O ₄	144-62-7	90.035	175,510	-381.36	0.64623			462.65	1.3740	603.00	1.8052
275	Oxygen	O ₂	7782-44-7	31.999	175,430	-6,152.3	113.92	-0.92382	0.0027963	54.36	0.5365	142.00	0.9066
276	Ozone	O ₃	10028-15-6	47.998	60,046	281.16				90.00	0.8535	150.00	1.0222
277	Pentadecane	C ₁₅ H ₃₂	629-62-9	212.415	346,910	219.54	0.65632			283.07	4.6165	543.84	6.6042
278	Pentanal	C ₅ H ₁₀ O	110-62-3	86.132	112,050	257.78				200.00	1.6361	376.15	2.0901
279	Pentane	C ₅ H ₁₂	109-66-0	72.149	159,080	-270.5	0.99537			143.42	1.4076	390.00	2.0498
280	Pentanoic acid	C ₅ H ₁₀ O ₂	109-52-4	102.132	145,050	28.344	0.6372			239.15	1.8827	458.95	2.9228
281	1-Pentanol	C ₅ H ₁₂ O	71-41-0	88.148	201,200	-651.3	2.275			200.14	1.6198	389.15	2.9227
282	2-Pentanol	C ₅ H ₁₂ O	6032-29-7	88.148	251,596	-1,028.49	3.26306			200.00	1.7642	561.00	7.0158
283	2-Pentanone	C ₅ H ₁₀ O	107-87-9	86.132	194,590	-263.86	0.76808			196.29	1.7239	375.46	2.0380
284	3-Pentanone	C ₅ H ₁₀ O	96-22-0	86.132	193,020	-176.43	0.5669			234.18	1.8279	375.14	2.0661
285	1-Pentene	C ₅ H ₁₀	109-67-1	70.133	156,100	-456.94	2.255	-0.003163	0.00000238	108.02	1.2939	350.00	1.7251
286	2-Pentyl mercaptan	C ₅ H ₁₂ S	2084-19-7	104.214	188,200	-140.84	0.63581			160.75	1.8199	385.15	2.2827
287	Pentyl mercaptan	C ₅ H ₁₂ S	110-66-7	104.214	213,760	-324.4	0.9472			197.45	1.8664	399.79	2.3546
288	1-Pentyne	C ₅ H ₈	627-19-0	68.117	86,200	256.6				200.00	1.3752	313.33	1.6660
289	2-Pentyne	C ₅ H ₈	627-21-4	68.117	68,671	246.66				200.00	1.1800	329.27	1.4989
290	Phenanthrene	C ₁₄ H ₁₀	85-01-8	178.229	103,370	527.03				372.39	2.9963	500.00	3.6688
291	Phenol	C ₆ H ₆ O	108-95-2	94.111	101,720	317.61				314.06	2.0147	425.00	2.3670
292	Phenyl isocyanate	C ₇ H ₅ NO	103-71-9	119.121	60,834	215.89	0.29552			243.15	1.3080	489.75	2.3745
293	Phthalic anhydride	C ₈ H ₄ O ₃	85-44-9	148.116	145,400	252.4				404.15	2.4741	557.65	2.8615
294	Propadiene	C ₃ H ₄	463-49-0	40.064	66,230	98.275				200.00	0.8589	238.65	0.8968
295	Propane [use Eq. (2)]	C ₃ H ₈	74-98-6	44.096	62,983	113,630	633.21	-873.46		85.47	0.8488	360.00	2.6079
296	1-Propanol	C ₃ H ₈ O	71-23-8	60.095	158,760	-635	1.969			146.95	1.0797	400.00	2.1980
297	2-Propanol	C ₃ H ₈ O	67-63-0	60.095	471,710	-4,172.1	14.745	-0.0144		185.26	1.1329	355.30	2.0487
298	Propenylcyclohexene	C ₉ H ₁₄	13511-13-2	122.207	201,400	-450.6	1.7053			199.00	1.7926	431.65	3.2463
299	Propionaldehyde	C ₃ H ₆ O	123-38-6	58.079	99,306	115.73				200.00	1.2245	328.75	1.3735

TABLE 2-153 Heat Capacities of Inorganic and Organic Liquids [J/(kmol·K)] (Concluded)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1	C2	C3	C4	C5	$T_{\min}$, K	C_p at $T_{\min}$ × 1E-05	$T_{\max}$, K	C_p at $T_{\max}$ × 1E-05
300	Propionic acid	C ₃ H ₆ O ₂	79-09-4	74.079	213,660	-702.7	1.6605			252.45	1.4209	414.32	2.0756
301	Propionitrile	C ₃ H ₅ N	107-12-0	55.079	118,190	-120.98	0.42075			180.26	1.1005	370.50	1.3112
302	Propyl acetate	C ₅ H ₁₀ O ₂	109-60-4	102.132	83,400	384.1				274.70	1.8891	404.70	2.3885
303	Propyl amine	C ₃ H ₉ N	107-10-8	59.110	139,530	78				188.36	1.5422	340.00	1.6605
304	Propylbenzene	C ₉ H ₁₂	103-65-1	120.192	174,380	-101.8	0.79			173.55	1.8051	432.39	2.7806
305	Propylene	C ₃ H ₆	115-07-1	42.080	114,140	-343.72	1.0905			87.89	0.9235	225.45	0.9208
306	Propyl formate	C ₄ H ₈ O ₂	110-74-7	88.105	75,700	326.1				298.15	1.7293	398.15	2.0554
307	2-Propyl mercaptan	C ₃ H ₇ S	75-33-2	76.161	135,390	-117.11	0.47059			142.61	1.3126	350.00	1.5505
308	Propyl mercaptan	C ₃ H ₇ S	107-03-9	76.161	167,330	-319.1	0.8127			159.95	1.3708	340.87	1.5299
309	1,2-Propylene glycol	C ₃ H ₈ O ₂	57-55-6	76.094	58,080	445.2				213.15	1.5297	460.75	2.6321
310	Quinone	C ₆ H ₄ O ₂	106-51-4	108.095	45,810	368.33				388.85	1.8904	683.00	2.9738
311	Silicon tetrafluoride	F ₄ Si	7783-61-1	104.079	829,380	-7,331.5	19.203			186.35	1.3000	253.15	2.0403
312	Styrene	C ₈ H ₈	100-42-5	104.149	113,340	290.2	-0.6051	0.0013567		242.54	1.6749	418.31	2.2816
313	Succinic acid	C ₄ H ₆ O ₄	110-15-6	118.088	244,770	-236.96	0.63148			460.65	2.6961	604.50	3.3228
314	Sulfur dioxide	O ₂ S	7446-09-5	64.064	85,743	5,7443				197.67	0.8688	350.00	0.8775
315	Sulfur hexafluoride	F ₆ S	2551-62-4	146.055	119,500					230.15	1.1950	230.15	1.1950
316	Sulfur trioxide	O ₃ S	7446-11-9	80.063	258,090					303.15	2.5809	303.15	2.5809
317	Terephthalic acid	C ₈ H ₆ O ₄	100-21-0	166.131									
318	<i>o</i> -Terphenyl	C ₁₈ H ₁₄	84-15-1	230.304	182,900	635.09				329.35	3.9207	609.15	5.6977
319	Tetradecane	C ₁₄ H ₃₀	629-59-4	198.388	353,140	29.13	0.86116			279.01	4.2831	526.73	6.0741
320	Tetrahydrofuran	C ₄ H ₈ O	109-99-9	72.106	171,730	-800.47	2.8934	-0.0025015		164.65	1.0721	339.12	1.3546
321	1,2,3,4-Tetrahydronaphthalene	C ₁₀ H ₁₂	119-64-2	132.202	81,760	455.38				237.38	1.8986	480.77	3.0069
322	Tetrahydrothiophene	C ₄ H ₈ S	110-01-0	88.171	123,300	-130.1	0.6229			176.98	1.1979	394.27	1.6883
323	2,2,3,3-Tetramethylbutane	C ₈ H ₁₈	594-82-1	114.229	43,326	630.73				375.41	2.8011	426.00	3.1202
324	Thiophene	C ₄ H ₄ S	110-02-1	84.140	84,864	91.725	0.13243			234.94	1.1372	357.31	1.3455
325	Toluene	C ₇ H ₈	108-88-3	92.138	140,140	-152.3	0.695			178.18	1.3507	500.00	2.3774
326	1,1,2-Trichloroethane	C ₂ H ₃ Cl ₃	79-00-5	133.404	103,350	159.3				236.50	1.4102	300.00	1.5114
327	Tridecane	C ₁₃ H ₂₈	629-50-5	184.361	350,180	-104.7	1.0022			267.76	3.9400	508.62	5.5619
328	Triethyl amine	C ₆ H ₁₅ N	121-44-8	101.190	111,480	368.13				200.00	1.8511	361.92	2.4471
329	Trimethyl amine	C ₃ H ₉ N	75-50-3	59.110	136,050	-288	0.9913			156.08	1.1525	276.02	1.3208
330	1,2,3-Trimethylbenzene	C ₉ H ₁₂	526-73-8	120.192	119,450	324.54				247.79	1.9987	449.27	2.6526
331	1,2,4-Trimethylbenzene	C ₉ H ₁₂	95-63-6	120.192	178,800	-128.47	0.83741			229.33	1.9338	350.00	2.3642
332	2,2,4-Trimethylpentane	C ₈ H ₁₈	540-84-1	114.229	95,275	696.7	-1.3765	0.0021734		165.78	1.8285	520.00	3.9095
333	2,3,3-Trimethylpentane	C ₈ H ₁₈	560-21-4	114.229	388,620	-1,439.5	3.2187			280.00	2.3791	320.00	2.5757
334	1,3,5-Trinitrobenzene	C ₆ H ₃ N ₃ O ₆	99-35-4	213.105	40,364	664.46				398.40	3.0508	475.47	3.5629
335	2,4,6-Trinitrotoluene	C ₇ H ₅ N ₃ O ₆	118-96-7	227.131	133,530	514.64				354.00	3.1571	475.00	3.7798
336	Undecane	C ₁₁ H ₂₄	1120-21-4	156.308	293,980	-114.98	0.96936			247.57	3.2493	433.42	4.2624
337	1-Undecanol	C ₁₁ H ₂₄ O	112-42-5	172.308	129,450	-3,039.5	27.927	-0.061847	4.3042E-05	289.05	3.9103	520.30	5.5127
338	Vinyl acetate	C ₄ H ₆ O ₂	108-05-4	86.089	136,300	-106.17	0.75175			259.56	1.5939	389.35	2.0892
339	Vinyl acetylene	C ₄ H ₄	689-97-4	52.075	68,720	135				200.00	0.9572	278.25	1.0628
340	Vinyl chloride	C ₂ H ₃ Cl	75-01-4	62.498	-10,320	322.8				200.00	0.5424	400.00	1.1880
341	Vinyl trichlorosilane	C ₂ H ₃ Cl ₃ Si	75-94-5	161.490	49,516	420.35				178.35	1.2449	363.85	2.0246
342	Water	H ₂ O	7732-18-5	18.015	276,370	-2,090.1	8.125	-0.014116	9.3701E-06	273.16	0.7615	533.15	0.8939
343	<i>m</i> -Xylene	C ₈ H ₁₀	108-38-3	106.165	133,860	7,8754	0.52265			217.00	1.6018	540.15	2.9060
344	<i>o</i> -Xylene	C ₈ H ₁₀	95-47-6	106.165	36,500	1,017.5	-2.63	0.00302		247.98	1.7314	417.58	2.2269
345	<i>p</i> -Xylene	C ₈ H ₁₀	106-42-3	106.165	-35,500	1,287.2	-2.599	0.002426		286.41	1.7697	600.00	3.2520

For the 11 substances, ammonia, 1,2-butanediol, 1,3-butanediol, carbon monoxide, 1,1-difluoroethane, ethane, heptane, hydrogen, hydrogen sulfide, methane, and propane, the liquid heat capacity C_{pL} is calculated with Eq. (2) below. For all other compounds, Eq. (1) is used. For benzene, fluorine, and helium, two sets of constants are given for Eq. (1) that cover different temperature ranges, as shown in the table.

$$(1) C_{pL} = C1 + C2T + C3T^2 + C4T^3 + C4T^4$$

$$(2) C_{pL} = \frac{C1^2}{t} + C2 - 2C1C3t - C1C4t^2 - \frac{C3^2t^3}{3} - \frac{C3C4t^4}{2} - \frac{C4^2t^5}{5}$$

where $t = 1 - T_r$, $T_r = T/T_c$, T_c is the critical temperature from Table 2-141, C_{pL} is in J/(kmol·K) and T is in K. All substances are listed by chemical family in Table 2-6 and by formula in Table 2-7. For temperatures less than the normal boiling point, the pressure is 1 atm. Above the normal boiling point, the pressure is the vapor pressure.

Values in this table were taken from the Design Institute for Physical Properties (DIPPR) of the American Institute of Chemical Engineers (AIChE), copyright 2007 AIChE and reproduced with permission of AIChE and of the DIPPR Evaluated Process Design Data Project Steering Committee. Their source should be cited as R. L. Rowley, W. V. Wilding, J. L. Oscarson, Y. Yang, N. A. Zundel, T. E. Daubert, R. P. Danner, DIPPR® Data Compilation of Pure Chemical Properties, Design Institute for Physical Properties, AIChE, New York (2007).

The number of digits provided for values at $T_{\min}$ and $T_{\max}$ was chosen for uniformity of appearance and formatting; these do not represent the uncertainties of the physical quantities, but are the result of calculations from the standard thermophysical property formulations within a fixed format.

TABLE 2-154 Specific Heats of Organic Solids

Recalculated from *International Critical Tables*, vol. 5, pp. 101-105

Compound	Formula	Temperature, °C	sp ht, cal/(g·°C)
Acetic acid	C ₂ H ₄ O ₂	-200 to +25	0.330 + 0.00080 <i>t</i>
Acetone	C ₃ H ₆ O	-210 to -80	0.540 + 0.0156 <i>t</i>
Aminobenzoic acid (<i>o</i> -)	C ₇ H ₇ NO ₂	85 to mp	0.254 + 0.00136 <i>t</i>
(<i>m</i> -)	C ₇ H ₇ NO ₂	120 to mp	0.253 + 0.00122 <i>t</i>
(<i>p</i> -)	C ₇ H ₇ NO ₂	128 to mp	0.287 + 0.00088 <i>t</i>
Aniline	C ₆ H ₇ N		0.741
Anthracene	C ₁₄ H ₁₀	50	0.308
		100	0.350
		150	0.382
Anthraquinone	C ₁₄ H ₈ O ₂	0 to 270	0.258 + 0.00069 <i>t</i>
Apiol	C ₁₂ H ₁₄ O ₄	10	0.299
Azobenzene	C ₁₂ H ₁₀ N ₂	28	0.330
Benzene	C ₆ H ₆	-250	0.0399
		-225	0.0908
		-200	0.124
		-150	0.170
		-100	0.227
		-50	0.299
		0	0.375
Benzoic acid	C ₇ H ₆ O ₂	20 to mp	0.287 + 0.00050 <i>t</i>
Benzophenone	C ₁₃ H ₁₀ O	-150	0.115
		-100	0.172
		-50	0.220
		0	0.275
		+20	0.303
Betol	C ₁₇ H ₁₂ O ₃	-150	0.129
		-100	0.167
		0	0.248
		+50	0.308
Bromiodobenzene (<i>o</i> -)	C ₆ H ₄ BrI	-50 to 0	0.143 + 0.00025 <i>t</i>
(<i>m</i> -)	C ₆ H ₄ BrI	-75 to -15	0.143
(<i>p</i> -)	C ₆ H ₄ BrI	-40 to 50	0.116 + 0.00032 <i>t</i>
Bromonaphthalene (β-)	C ₁₀ H ₇ Br	41	0.260
Bromophenol	C ₆ H ₅ BrO	32	0.263
Camphene	C ₁₀ H ₁₆	35	0.380
Capric acid	C ₁₀ H ₂₀ O ₂	8	0.695
Caprylic acid	C ₈ H ₁₆ O ₂	-2	0.628
Carbon tetrachloride	CCl ₄	-240	0.013
		-200	0.081
		-160	0.131
		-120	0.162
		-80	0.182
		-40	0.201
Cerotic acid	C ₂₇ H ₅₄ O ₂	15	0.387
Chloral alcoholate	C ₄ H ₇ Cl ₃ O ₂	78	0.509
hydrate	C ₂ H ₃ Cl ₃ O ₂	32	0.213
Chloroacetic acid	C ₂ H ₃ ClO ₂	60	0.363
Chlorobenzoic acid (<i>o</i> -)	C ₇ H ₅ ClO ₂	80 to mp	0.228 + 0.00084 <i>t</i>
(<i>m</i> -)	C ₇ H ₅ ClO ₂	94 to mp	0.232 + 0.00073 <i>t</i>
(<i>p</i> -)	C ₇ H ₅ ClO ₂	180 to mp	0.242 + 0.00055 <i>t</i>
Chlorobromobenzene (<i>o</i> -)	C ₆ H ₄ BrCl	-34	0.192
(<i>m</i> -)	C ₆ H ₄ BrCl	-52	0.150
(<i>p</i> -)	C ₆ H ₄ BrCl	-40	0.150
Crotonic acid	C ₄ H ₆ O ₂	38 to 70	0.520 + 0.00020 <i>t</i>
Cyamelide	C ₃ H ₃ N ₃ O ₃	40	0.263
Cyanamide	CH ₂ N ₂	20	0.547
Cyanuric acid	C ₃ H ₃ N ₃ O ₃	40	0.318
Dextrin	(C ₆ H ₁₀ O ₅) _x	0 to 90	0.291 + 0.00096 <i>t</i>
Dextrose	C ₆ H ₁₂ O ₆	-250	0.016
		-200	0.077
		-100	0.160
		0	0.277
		20	0.300
Dibenzyl	C ₁₄ H ₁₄	28	0.363
Dibromobenzene (<i>o</i> -)	C ₆ H ₄ Br ₂	-36	0.248
(<i>m</i> -)	C ₆ H ₄ Br ₂	-25	0.134
(<i>p</i> -)	C ₆ H ₄ Br ₂	-50 to +50	0.139 + 0.00038 <i>t</i>
Dichloroacetic acid	C ₂ H ₂ Cl ₂ O ₂		0.406
Dichlorobenzene (<i>o</i> -)	C ₆ H ₄ Cl ₂	-48.5	0.185
(<i>m</i> -)	C ₆ H ₄ Cl ₂	-52	0.186
(<i>p</i> -)	C ₆ H ₄ Cl ₂	-50 to +53	0.219 + 0.0021 <i>t</i>
Dicyandiamide	C ₂ H ₄ N ₄	0 to 204	0.456

TABLE 2-154 Specific Heats of Organic Solids (Continued)

Recalculated from *International Critical Tables*, vol. 5, pp. 101-105

Compound	Formula	Temperature, °C	sp. ht., cal/(g·°C)
Dihydroxybenzene (<i>o</i> -)	C ₆ H ₆ O ₂	-163 to mp	0.278 + 0.00098 <i>t</i>
(<i>m</i> -)	C ₆ H ₆ O ₂	-160 to mp	0.269 + 0.00118 <i>t</i>
(<i>p</i> -)	C ₆ H ₆ O ₂	-250	0.025
		-240	0.038
		-220	0.061
		-200	0.081
		-150 to mp	0.268 + 0.00093 <i>t</i>
Di-iodobenzene (<i>o</i> -)	C ₆ H ₄ I ₂	-50 to +15	0.109 + 0.00026 <i>t</i>
(<i>m</i> -)	C ₆ H ₄ I ₂	-52 to -42	0.100 + 0.00026 <i>t</i>
(<i>p</i> -)	C ₆ H ₄ I ₂	-50 to +80	0.101 + 0.00026 <i>t</i>
Dimethyl oxalate	C ₄ H ₆ O ₄	10 to 50	0.212 + 0.0044 <i>t</i>
Dimethylpyrene	C ₇ H ₈ O ₂	50	0.368
Dinitrobenzene (<i>o</i> -)	C ₆ H ₄ N ₂ O ₄	-160 to mp	0.252 + 0.00083 <i>t</i>
(<i>m</i> -)	C ₆ H ₄ N ₂ O ₄	-160 to mp	0.248 + 0.00077 <i>t</i>
(<i>p</i> -)	C ₆ H ₄ N ₂ O ₄	119 to mp	0.259 + 0.00057 <i>t</i>
Diphenyl	C ₁₂ H ₁₀	40	0.385
Diphenylamine	C ₁₂ H ₁₁ N	26	0.337
Dulcitol	C ₆ H ₁₄ O ₆	20	0.282
Erythritol	C ₄ H ₁₀ O ₄	60	0.351
Ethyl alcohol	C ₂ H ₆ O (crystalline)	-190	0.232
		-180	0.248
		-160	0.282
		-140	0.318
		-130	0.376
	(vitreous)	-190	0.260
		-180	0.296
		-175	0.380
		-170	0.399
Ethylene glycol	C ₂ H ₆ O ₂	-190 to -40	0.366 + 0.00110 <i>t</i>
Formic acid	CH ₂ O ₂	-22	0.387
		0	0.430
Glutaric acid	C ₅ H ₈ O ₄	20	0.299
Glycerol	C ₃ H ₈ O ₃	-265	0.009
		-260	0.022
		-250	0.047
		-220	0.085
		-200	0.115
		-100	0.217
		0	0.330
Hexachloroethane	C ₂ Cl ₆	25	0.174
Hexadecane	C ₁₆ H ₃₄		0.495
Hydroxyacetanilide	C ₈ H ₉ NO ₂	41 to mp	0.249 + 0.00154 <i>t</i>
Iodobenzene	C ₆ H ₅ I	40	0.191
Isopropyl alcohol	C ₃ H ₈ O	-200 to -160	0.051 + 0.00165 <i>t</i>
Lactose	C ₁₂ H ₂₂ O ₁₁	20	0.287
	C ₁₂ H ₂₂ O ₁₁ ·H ₂ O	20	0.299
Lauric acid	C ₁₂ H ₂₄ O ₂	-30 to +40	0.430 + 0.000027 <i>t</i>
Levoglucothane	C ₆ H ₁₀ O ₅	40	0.607
Levulose	C ₆ H ₁₂ O ₆	20	0.275
Malonic acid	C ₃ H ₄ O ₄	20	0.275
Maltose	C ₁₂ H ₂₂ O ₁₁	20	0.320
Mannitol	C ₆ H ₁₄ O ₆	0 to 100	0.313 + 0.00025 <i>t</i>
Melamine	C ₃ H ₆ N ₆	40	0.351
Myristic acid	C ₁₄ H ₂₈ O ₂	0 to 35	0.381 + 0.00545 <i>t</i>
Naphthalene	C ₁₀ H ₈	-130 to mp	0.281 + 0.00111 <i>t</i>
Naphthol (α -)	C ₁₀ H ₈ O	50 to mp	0.240 + 0.00147 <i>t</i>
(β -)	C ₁₀ H ₈ O	61 to mp	0.252 + 0.00128 <i>t</i>
Naphthylamine (α -)	C ₁₀ H ₉ N	0 to 50	0.270 + 0.0031 <i>t</i>
Nitroaniline (<i>o</i> -)	C ₆ H ₆ N ₂ O ₂	-160 to mp	0.269 + 0.000920 <i>t</i>
(<i>m</i> -)	C ₆ H ₆ N ₂ O ₂	-160 to mp	0.275 + 0.000946 <i>t</i>
(<i>p</i> -)	C ₆ H ₆ N ₂ O ₂	-160 to mp	0.276 + 0.001000 <i>t</i>
Nitrobenzoic acid (<i>o</i> -)	C ₇ H ₅ NO ₄	-163 to mp	0.256 + 0.00085 <i>t</i>
(<i>m</i> -)	C ₇ H ₅ NO ₄	66 to mp	0.258 + 0.00091 <i>t</i>
(<i>p</i> -)	C ₇ H ₅ NO ₄	-160 to mp	0.247 + 0.00077 <i>t</i>
Nitronaphthalene	C ₁₀ H ₇ NO ₂	0 to 55	0.236 + 0.00215 <i>t</i>

TABLE 2-154 Specific Heats of Organic Solids (Concluded)Recalculated from *International Critical Tables*, vol. 5, pp. 101-105

Compound	Formula	Temperature, °C	sp ht, cal/(g·°C)
Oxalic acid	$C_2H_2O_4$ $C_2H_2O_4 \cdot 2H_2O$	-200 to +50	0.259 + 0.00076 <i>t</i>
		-200	0.117
		-100	0.239
		0	0.338
		+50	0.385
		100	0.416
Palmitic acid	$C_{16}H_{32}O_2$	-180	0.167
		-140	0.208
		-100	0.251
		-50	0.306
		0	0.382
		+20	0.430
Phenol	C_6H_6O	14 to 26	0.561
Phthalic acid	$C_8H_6O_4$	20	0.232
Picric acid	$C_6H_3N_3O_7$	-100	0.165
		0	0.240
		+50	0.263
		100	0.297
		120	0.332
		-33	0.726
Propionic acid	$C_3H_6O_2$	-200	0.170
Propyl alcohol (<i>n</i> -)	C_3H_8O	-175	0.363
		-150	0.471
		-130	0.497
		20	0.301
Pyrotartaric acid	$C_6H_8O_4$	20	0.301
		-250	0.017
		-225	0.061
		-200	0.098
		-100	0.191
Quinhydrone	$C_{12}H_{10}O_4$	0	0.256
		-250	0.031
		-225	0.082
		-200	0.113
		-150 to mp	0.282 + 0.00083 <i>t</i>
Quinone	$C_6H_4O_2$	-250	0.031
		-225	0.082
		-200	0.113
		-150 to mp	0.282 + 0.00083 <i>t</i>
Salol	$C_{13}H_{10}O_3$	32	0.289
Stearic acid	$C_{18}H_{36}O_2$	15	0.399
Succinic acid	$C_4H_6O_4$	0 to 160	0.248 + 0.00153 <i>t</i>
Sucrose	$C_{12}H_{22}O_{11}$	20	0.299
Sugar (cane)	$C_{12}H_{22}O_{11}$	22 to 51	0.301
Tartaric acid	$C_4H_6O_6$	36	0.287
Tartaric acid	$C_4H_6O_6 \cdot H_2O$	-150	0.112
		-100	0.170
		-50	0.231
		0	0.308
		+50	0.366
		-40 to 0	0.198 + 0.00018 <i>t</i>
		-100	0.182
		-50	0.199
Tetrachloroethylene	C_2Cl_4	-40 to 0	0.198 + 0.00018 <i>t</i>
Tetryl	$C_7H_5N_5O_8$	-100	0.182
		-50	0.199
		0	0.212
		+100	0.236
		-100 to +100	0.253 + 0.00072 <i>t</i>
1 Tetryl + 1 picric acid	$C_{13}H_8N_8O_{15}$	-100	0.172
1 Tetryl + 2 TNT	$C_{21}H_{15}N_{11}O_{20}$	0	0.280
		+50	0.325
		-100 to +100	0.253 + 0.00072 <i>t</i>
Thymol	$C_{10}H_{14}O$	0 to 49	0.315 + 0.0031 <i>t</i>
Toluic acid (<i>o</i> -)	$C_8H_6O_2$	54 to mp	0.277 + 0.00120 <i>t</i>
		54 to mp	0.239 + 0.00195 <i>t</i>
		130 to mp	0.271 + 0.00106 <i>t</i>
		0	0.337
Toluidine (<i>p</i> -)	C_7H_9N	20	0.387
		40	0.440
		solid	0.459
		-4	0.559
Trichloroacetic acid	$C_2HCl_3O_2$	solid	0.459
Trimethyl carbinol	$C_4H_{10}O$	-4	0.559
Trinitrotoluene	$C_7H_5N_3O_6$	-100	0.170
		-50	0.253
		0	0.311
		+100	0.385
		-185 to +23	0.241
Trinitroxylenes	$C_8H_7N_3O_6$	20 to 50	0.423
Triphenylmethane	$C_{19}H_{16}$	0 to 91	0.189 + 0.0027 <i>t</i>
Urea	CH_4N_2O	20	0.320

TABLE 2-155 Heat Capacity at Constant Pressure of Inorganic and Organic Compounds in the Ideal Gas State Fit to a Polynomial C_p [J/(kmol·K)]

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1	C2	C3	C4 × 1E 05	C5 × 1E 10	$T_{\min}$, K	C_p at $T_{\min}$ × 1E-05	$T_{\max}$, K	C_p at $T_{\max}$ × 1E-05
1	Acetaldehyde	C ₂ H ₄ O	75-07-0	44.053	29,705	127.43	-0.21793			50	0.3553	200	0.4647
7	Acetylene	C ₂ H ₂	74-86-2	26.037	30,800	-53.08	0.384			50	0.2911	200	0.3554
8	Acrolein	C ₃ H ₄ O	107-02-8	56.063	30,702	80.95	0.191			50	0.3523	200	0.5453
14	Argon	Ar	7440-37-1	39.948	20,786	0	0			100	0.2079	1,500	0.2079
16	Benzene	C ₆ H ₆	71-43-2	78.112	35,978	-101.69	0.939			50	0.3324	200	0.5320
27	Bromoethane	C ₂ H ₅ Br	74-96-4	108.965	27,112	117.99	0			100	0.3891	200	0.5071
29	1,2-Butadiene	C ₄ H ₆	590-19-2	54.090	27,400	177.6	0			50	0.3628	200	0.6292
31	Butane	C ₄ H ₁₀	106-97-8	58.122	17,330	458.16	-0.816			50	0.3820	200	0.7632
34	1-Butanol	C ₄ H ₁₀ O	71-36-3	74.122	25,300	371.2	-0.461			50	0.4271	200	0.8110
37	cis-2-Butene	C ₄ H ₈	590-18-1	56.106	39,760	108.8	0			50	0.4520	200	0.6152
38	trans-2-Butene	C ₄ H ₈	624-64-6	56.106	20,908	324.73	-0.411			50	0.3612	200	0.6941
43	1-Butyne	C ₄ H ₆	107-00-6	54.090	25,300	183.2	0			50	0.3446	200	0.6194
59	m-Cresol	C ₇ H ₈ O	108-39-4	108.138	29,002	158.79	0.635			50	0.3853	200	0.8616
60	o-Cresol	C ₇ H ₈ O	95-48-7	108.138	16,192	469.81	-0.479			50	0.3849	200	0.9099
61	p-Cresol	C ₇ H ₈ O	106-44-5	108.138	29,090	166	0.616			50	0.3893	200	0.8693
64	Cyclobutane	C ₄ H ₈	287-23-0	56.106	31,863	37.226	0.23616			50	0.3432	200	0.4876
67	Cyclohexanone	C ₆ H ₁₀ O	108-94-1	98.143	32,182	116.87	0.547			50	0.3939	200	0.7744
81	1,1-Dibromoethane	C ₂ H ₄ Br ₂	557-91-5	187.861	20,560	285.2	-0.332			100	0.4576	200	0.6432
88	1,1-Dichloroethane	C ₂ H ₄ Cl ₂	75-34-3	98.959	19,560	249.01	-0.22187			100	0.4224	200	0.6049
95	Diethyl ether	C ₄ H ₁₀ O	60-29-7	74.122	26,040	388	-0.268			50	0.4477	200	0.9292
97	1,1-Difluoroethane	C ₂ H ₄ F ₂	75-37-6	66.050	29,736	72.364	0.228			50	0.3392	200	0.5333
98	1,2-Difluoroethane	C ₂ H ₄ F ₂	624-72-6	66.050	27,581	169.88	-0.1581			50	0.3568	200	0.5523
99	Difluoromethane	CH ₂ F ₂	75-10-5	52.023	33,851	-20.966	0.17584			50	0.3324	200	0.3669
112	Dimethyl ether	C ₂ H ₆ O	115-10-6	46.068	25,940	178.46	-0.186			50	0.3440	200	0.5419
120	1,4-Dioxane	C ₄ H ₈ O ₂	123-91-1	88.105	28,345	88.3	0.446			50	0.3388	200	0.6385
125	Ethane	C ₂ H ₆	74-84-0	30.069	31,742	26.567	0.12927			50	0.3339	200	0.4223
126	Ethanol	C ₂ H ₆ O	64-17-5	46.068	32,585	87.4	0.05			50	0.3708	200	0.5207
134	Ethylcyclopentane	C ₇ H ₁₄	1640-89-7	98.186	34,710	304.96	-0.084			50	0.4975	200	0.9234
145	Ethyl mercaptan	C ₂ H ₆ S	75-08-1	62.134	23,014	271.36	-0.4427			50	0.3548	200	0.5958
151	Fluoroethane	C ₂ H ₅ F	353-36-6	48.060	30,358	62.839	0.1067			50	0.3377	200	0.4719
156	Furan	C ₄ H ₄ O	110-00-9	68.074	40,860	-160.3	0.87			100	0.3353	200	0.4360
157	Helium-4	He	7440-59-7	4.003	20,786	0	0			100	0.2079	1,500	0.2079
182	Hydrazine	H ₂ N ₂	302-01-2	32.045	32,998	-5.2147	0.21379			50	0.3327	200	0.4051
183	Hydrogen	H ₂	1333-74-0	2.016	64,979	-788.17	5.8287	-1845.9	216400	50	0.3797	250	0.2834
190	Isopropyl amine	C ₃ H ₉ N	75-31-0	59.110	23,590	310.42	-0.274			50	0.3843	200	0.7471
194	Methanol	CH ₄ O	67-56-1	32.042	30,270	84.64	-0.188			50	0.3403	200	0.3968
197	Methyl acetylene	C ₃ H ₄	74-99-7	40.064	30,810	35.8	0.27			50	0.3328	200	0.4877
217	Methylcyclopentane	C ₆ H ₁₂	96-37-7	84.159	35,465	147.38	0.242			50	0.4344	200	0.7462
221	Methylethyl ether	C ₃ H ₈ O	540-67-0	60.095	23,337	309.03	-0.285			50	0.3808	200	0.7374
231	Methyl mercaptan	CH ₄ S	74-93-1	48.107	31,520	60.1	0			50	0.3453	200	0.4354
236	2-Methylpropane	C ₄ H ₁₀	75-28-5	58.122	21,380	271.2	-0.092			50	0.3471	200	0.7194

237	2-Methyl-2-propanol	C ₄ H ₁₀ O	75-65-0	74.122	17,080	381.7	-0.199			50	0.3567	200	0.8546
238	2-Methyl propene	C ₄ H ₈	115-11-7	56.106	24,970	211.8	0			50	0.3556	200	0.6733
243	alpha-Methyl styrene	C ₉ H ₁₀	98-83-9	118.176	37,735	112.94	0.846			50	0.4550	200	0.9416
246	Naphthalene	C ₁₀ H ₈	91-20-3	128.171	29,120	82.88	0.964			50	0.3567	200	0.8426
247	Neon	Ne	7440-01-9	20.180	20,786	0	0			100	0.2079	1,500	0.2079
248	Nitroethane	C ₂ H ₅ NO ₂	79-24-3	75.067	33,055	89.54	0.238			50	0.3813	200	0.6048
251	Nitromethane	CH ₃ NO ₂	75-52-5	61.040	38,782	-48.39	0.413			50	0.3740	200	0.4562
253	Nitric oxide	NO	10102-43-9	30.006	34,980	-35.32	0.07729	-5.7357	0.0014526	100	0.3217	1,500	0.3586
289	2-Pentyne	C ₅ H ₈	627-21-4	68.117	24,330	335.7	-0.37			50	0.4019	200	0.7667
290	Phenanthrene	C ₁₄ H ₁₀	85-01-8	178.229	27,700	210	1.24			50	0.4130	200	1.1930
294	Propadiene	C ₃ H ₄	463-49-0	40.064	31,690	17.1	0.282			50	0.3325	200	0.4639
295	Propane	C ₃ H ₈	74-98-6	44.096	26,675	147.04	0			50	0.3403	200	0.5608
296	1-Propanol	C ₃ H ₈ O	71-23-8	60.095	28,800	257	-0.35			50	0.4078	200	0.6620
304	Propylbenzene	C ₉ H ₁₂	103-65-1	120.192	22,880	538.46	-0.546			50	0.4844	200	1.0873
310	Quinone	C ₆ H ₄ O ₂	106-51-4	108.095	29,668	129.07	0.53105			50	0.3745	200	0.7672
320	Tetrahydrofuran	C ₄ H ₈ O	109-99-9	72.106	36,970	-12.28	0.444			50	0.3747	200	0.5227
321	1,2,3,4-Tetrahydronaphthalene	C ₁₀ H ₁₂	119-64-2	132.202	28,560	225.1	0.616			50	0.4136	200	0.9822
322	Tetrahydrothiophene	C ₄ H ₈ S	110-01-0	88.171	41,195	-88.3	0.942			50	0.3914	200	0.6122
324	Thiophene	C ₄ H ₄ S	110-02-1	84.140	36,765	-112.82	0.862			50	0.3328	200	0.4868
331	1,2,4-Trimethylbenzene	C ₉ H ₁₂	95-63-6	120.192	35,652	323.89	0.305			50	0.5261	200	1.1263

Constants in this table can be used in the following equation to calculate the ideal gas heat capacity C_p^0 .

$$C_p^0 = C_1 + C_2T + C_3T^2 + C_4T^3 + C_5T^4$$

where C_p^0 is in J/(kmol·K) and T is in K. All substances are listed by chemical family in Table 2-6 and by formula in Table 2-7.

Values in this table were taken from the Design Institute for Physical Properties (DIPPR) of the American Institute of Chemical Engineers (AIChE), copyright 2007 AIChE and reproduced with permission of AIChE and of the DIPPR Evaluated Process Design Data Project Steering Committee. Their source should be cited as R. L. Rowley, W. V. Wilding, J. L. Oscarson, Y. Yang, N. A. Zundel, T. E. Daubert, R. P. Danner, DIPPR® Data Compilation of Pure Chemical Properties, Design Institute for Physical Properties, AIChE, New York (2007).

The number of digits provided for values at $T_{\min}$ and $T_{\max}$ was chosen for uniformity of appearance and formatting; these do not represent the uncertainties of the physical quantities, but are the result of calculations from the standard thermophysical property formulations within a fixed format.

TABLE 2-156 Heat Capacity at Constant Pressure of Inorganic and Organic Compounds in the Ideal Gas State Fit to Hyperbolic Functions C_p [J/(kmol·K)]

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1 × 1E-05	C2 × 1E-05	C3 × 1E-03	C4 × 1E-05	C5	$T_{\min}$, K	C_p at $T_{\min}$ × 1E-05	$T_{\max}$, K	C_p at $T_{\max}$ × 1E-05
1	Acetaldehyde	C ₂ H ₄ O	75-07-0	44.053	0.4451	1.0687	1.6141	0.6135	737.8	200	0.4660	1500	1.2994
2	Acetamide	C ₂ H ₅ NO	60-35-5	59.067	0.342	1.294	1.075	0.64	502	100	0.3448	1500	1.4997
3	Acetic acid	C ₂ H ₄ O ₂	64-19-7	60.052	0.402	1.3675	1.262	0.7003	569.7	50	0.4020	1500	1.5756
4	Acetic anhydride	C ₄ H ₆ O ₃	108-24-7	102.089	0.713	2.222	1.6203	1.676	746.5	200	0.7665	1500	2.5675
5	Acetone	C ₃ H ₆ O	67-64-1	58.079	0.5704	1.632	1.607	0.968	731.5	200	0.6049	1500	1.8820
6	Acetonitrile	C ₂ H ₃ N	75-05-8	41.052	0.41914	0.8876	1.5818	0.5032	699.8	100	0.4192	1500	1.1285
7	Acetylene	C ₂ H ₂	74-86-2	26.037	0.3199	0.5424	1.594	0.4325	607.1	200	0.3566	1500	0.7575
8	Acrolein	C ₃ H ₄ O	107-02-8	56.063	0.48449	1.2546	1.3979	0.87243	633.26	200	0.5467	1500	1.5620
9	Acrylic acid	C ₃ H ₄ O ₂	79-10-7	72.063	0.6059	1.3703	1.6475	1.0446	751.49	250	0.6984	1500	1.7424
10	Acrylonitrile	C ₃ H ₃ N	107-13-1	53.063	0.4678	1.0366	1.3998	0.6536	629.35	200	0.5156	1500	1.3464
11	Air	Mixture	132259-10-0	28.960	0.28958	0.0939	3.012	0.0758	1484	50	0.2896	1500	0.3496
12	Ammonia	H ₃ N	7664-41-7	17.031	0.33427	0.4898	2.036	0.2256	882	100	0.3343	1500	0.6647
13	Anisole	C ₇ H ₈ O	100-66-3	108.138	0.7637	2.9377	1.6051	2.17	751.2	300	1.1302	1200	3.0226
15	Benzamide	C ₇ H ₇ NO	55-21-0	121.137	1.9581	1.7019	1.3257	-37.417	41.232	298.15	1.2745	1500	3.2501
16	Benzene	C ₆ H ₆	71-43-2	78.112	0.44767	2.3085	1.4792	1.6836	677.66	200	0.5358	1500	2.4157
17	Benzenethiol	C ₆ H ₆ S	108-98-5	110.177	0.6895	2.3275	1.512	1.7516	697.9	200	0.7689	1500	2.6739
18	Benzoic acid	C ₇ H ₆ O ₂	65-85-0	122.121	0.77594	2.6455	1.7925	2.2382	835.9	200	0.8126	1500	2.9712
19	Benzonitrile	C ₇ H ₅ N	100-47-0	103.121	0.7186	2.27	1.4669	1.693	680.77	200	0.8053	1500	2.6706
20	Benzophenone	C ₁₃ H ₁₀ O	119-61-9	182.218	1.0099	4.4898	1.311	2.8395	627.4	300	1.8001	1500	4.9311
21	Benzyl alcohol	C ₉ H ₈ O	100-51-6	108.138	0.84115	3.1428	1.9539	2.5743	850.06	298.15	1.1198	1500	3.2880
22	Benzyl ethyl ether	C ₉ H ₁₂ O	539-30-0	136.191	0.9521	2.8868	0.70207	1.6385	2002.6	300	1.5501	1500	4.3445
23	Benzyl mercaptan	C ₇ H ₈ S	100-53-8	124.203	0.99192	2.9633	1.5583	2.2116	719.16	300	1.4156	1200	3.2957
24	Biphenyl	C ₁₂ H ₁₀	92-52-4	154.208	1.0759	4.2105	1.9041	4.1785	828.81	200	1.1481	1500	4.5557
25	Bromine	Br ₂	7726-95-6	159.808	0.30113	0.08009	0.7514	0.1078	314.6	100	0.3090	1500	0.3794
26	Bromobenzene	C ₆ H ₅ Br	108-86-1	157.008	0.721	2.064	1.6504	1.687	765.3	200	0.7679	1500	2.4628
27	Bromoethane	C ₂ H ₅ Br	74-96-4	108.965	0.47191	1.2787	1.5957	0.85166	703.87	200	0.5089	1500	1.5121
28	Bromomethane	CH ₃ Br	74-83-9	94.939	0.3377	0.715	1.578	0.4175	691.4	100	0.3378	1500	0.9107
29	1,2-Butadiene	C ₄ H ₆	590-19-2	54.090	0.575	1.6476	1.527	0.99	677.3	200	0.6269	1500	1.9202
30	1,3-Butadiene	C ₄ H ₆	106-99-0	54.090	0.5095	1.705	1.5324	1.337	685.6	200	0.5756	1500	1.9555
31	Butane	C ₄ H ₁₀	106-97-8	58.122	0.7134	2.43	1.63	1.5033	730.42	200	0.7673	1500	2.6602
32	1,2-Butanediol	C ₄ H ₁₀ O ₂	584-03-2	90.121	1.0478	2.549	1.8776	1.875	833	298.15	1.2667	1500.1	3.0289
33	1,3-Butanediol	C ₄ H ₁₀ O ₂	107-88-0	90.121	1.066	2.575	1.967	1.951	860.5	298.15	1.2679	1500.15	3.0311
34	1-Butanol	C ₄ H ₁₀ O	71-36-3	74.122	0.7454	2.5907	1.6073	1.732	712.4	200	0.8162	1500	2.8509
35	2-Butanol	C ₄ H ₁₀ O	78-92-2	74.122	0.90878	2.5508	1.893	1.852	832.13	298.15	1.1257	1500	2.8730
36	1-Butene	C ₄ H ₈	106-98-9	56.106	0.64257	2.0618	1.6768	1.3324	757.06	250	0.7571	1500	2.2898
37	cis-2-Butene	C ₄ H ₈	590-18-1	56.106	0.5765	2.115	1.6299	1.2872	739.1	200	0.6199	1500	2.2715
38	trans-2-Butene	C ₄ H ₈	624-64-6	56.106	0.6592	2.07	1.6733	1.251	742.2	200	0.7004	1500	2.2904
39	Butyl acetate	C ₆ H ₁₂ O ₂	123-86-4	116.158	1.1684	3.769	1.956	2.818	811.2	300	1.5358	1200	3.6724
40	Butylbenzene	C ₁₀ H ₁₄	104-51-8	134.218	1.138	4.454	1.5507	3.0497	708.86	200	1.2659	1500	4.8435
41	Butyl mercaptan	C ₄ H ₁₀ S	109-79-5	90.187	0.92478	2.7795	1.6837	1.5974	758.68	200	0.9714	1500	3.1008
42	sec-Butyl mercaptan	C ₄ H ₁₀ S	513-53-1	90.187	0.92367	2.5166	1.6109	1.5641	739.2	200	0.9763	1500	2.9615
43	1-Butyne	C ₄ H ₆	107-00-6	54.090	0.5587	1.6694	1.5328	1.07	656	200	0.6238	1500	1.9209
44	Butyraldehyde	C ₄ H ₈ O	123-72-8	72.106	0.89657	2.3731	1.9754	1.5866	904.13	200	0.9119	1500	2.6775
45	Butyric acid	C ₄ H ₈ O ₂	107-92-6	88.105	1.488	1.3522	1.146	-678	6.98	298.15	1.1533	1500	2.5905
46	Butyronitrile	C ₄ H ₇ N	109-74-0	69.105	0.6906	1.9996	1.5494	1.3146	675	200	0.7607	1500	2.3273
47	Carbon dioxide	CO ₂	124-38-9	44.010	0.2937	0.3454	1.428	0.264	588	50	0.2937	5000	0.6335
48	Carbon disulfide	CS ₂	75-15-0	76.141	0.301	0.3338	0.896	0.2893	374.7	100	0.3100	1500	0.6148
49	Carbon monoxide	CO	630-08-0	28.010	0.29108	0.08773	3.0851	0.084553	1538.2	60	0.2911	1500	0.3521
50	Carbon tetrachloride	CCl ₄	56-23-5	153.823	0.37582	0.7054	0.5121	0.485	236.1	100	0.4730	1500	1.0662
51	Carbon tetrafluoride	CF ₄	75-73-0	88.004	0.92004	0.16446	1.0764	-5083.8	2.3486	298	0.6106	1500	1.0465
52	Chlorine	Cl ₂	7782-50-5	70.906	0.29142	0.09176	0.949	0.1003	425	50	0.2914	1500	0.3793
53	Chlorobenzene	C ₆ H ₅ Cl	108-90-7	112.557	0.8011	2.31	2.157	2.046	897.6	200	0.8219	1500	2.5327
54	Chloroethane	C ₂ H ₅ Cl	75-00-3	64.514	0.4568	1.2967	1.5992	0.859	708.8	100	0.4569	1500	1.5112
55	Chloroform	CHCl ₃	67-66-3	119.378	0.3942	0.6573	0.928	0.493	399.6	100	0.4048	1500	1.0063
56	Chloromethane	CH ₃ Cl	74-87-3	50.488	0.3409	0.7246	1.723	0.448	780.5	150	0.3424	1500	0.9097
57	1-Chloropropane	C ₃ H ₇ Cl	540-54-5	78.541	0.621	1.843	1.629	1.2337	724	200	0.6674	1500	2.1126
58	2-Chloropropane	C ₃ H ₇ Cl	75-29-6	78.541	0.61809	1.8023	1.5438	1.1893	685.93	200	0.6768	1500	2.1023
59	m-Cresol	C ₇ H ₈ O	108-39-4	108.138	0.7515	2.09	0.6666	1.212	2214	200	0.8701	1500	3.2075
60	o-Cresol	C ₇ H ₈ O	95-48-7	108.138	0.7988	2.853	1.4765	2.042	664.7	200	0.9158	1500	3.2163

61	<i>p</i> -Cresol	C ₇ H ₈ O	106-44-5	108.138	0.7384	2.908	1.4559	2.091	650.42	200	0.8707	1500	3.2102
62	Cumene	C ₉ H ₁₂	98-82-8	120.192	1.081	3.7932	1.7505	3.0027	794.8	200	1.1480	1500	4.1808
63	Cyanogen	C ₂ N ₂	460-19-5	52.035	0.3545	0.5015	1.057	0.452	396	100	0.3648	1500	0.8100
64	Cyclobutane	C ₄ H ₈	287-23-0	56.106	0.44004	2.3074	1.6283	1.5571	744.9	200	0.4903	1500	2.3234
65	Cyclohexane	C ₆ H ₁₂	110-82-7	84.159	0.432	3.735	1.192	1.635	530.1	100	0.4366	1500	3.6516
66	Cyclohexanol	C ₆ H ₁₂ O	108-93-0	100.159	0.9043	2.5771	0.7882	1.3068	1952.2	200	0.9648	1500	3.8251
67	Cyclohexanone	C ₆ H ₁₀ O	108-94-1	98.143	0.67384	3.2598	1.3955	2.0209	677.33	200	0.7802	1500	3.4743
68	Cyclohexene	C ₆ H ₁₀	110-83-8	82.144	0.58171	3.1717	1.5435	2.1273	701.62	150	0.5978	1500	3.2132
69	Cyclopentane	C ₅ H ₁₀	287-92-3	70.133	0.416	3.014	1.4617	1.8095	668.8	100	0.4165	1500	2.9298
70	Cyclopentene	C ₅ H ₈	142-29-0	68.117	0.48074	2.5159	1.5803	1.7454	718.37	150	0.4918	1500	2.5619
71	Cyclopropane	C ₃ H ₆	75-19-4	42.080	0.338	1.6894	1.6135	1.1768	722.8	100	0.3381	1500	1.7213
72	Cyclohexyl mercaptan	C ₆ H ₁₂ S	1569-69-3	116.224	0.54305	3.9962	1.3575	2.5623	618.54	300	1.2644	1200	3.7236
73	Decanal	C ₁₀ H ₂₀ O	112-31-2	156.265	1.9641	5.1412	1.8989	4.1278	862.51	200	2.0192	1500	6.0539
74	Decane	C ₁₀ H ₂₂	124-18-5	142.282	1.672	5.353	1.6141	3.782	742	200	1.7967	1500	6.0932
75	Decanoic acid	C ₁₀ H ₂₀ O ₂	334-48-5	172.265	0.24457	6.546	1.0899	4.8642	424	298.15	2.5232	1500	6.1099
76	1-Decanol	C ₁₀ H ₂₂ O	112-30-1	158.281	1.6984	5.392	1.568	3.938	720.5	200	1.8502	1500	6.2186
77	1-Decene	C ₁₀ H ₂₀	872-05-9	140.266	1.7101	5.2089	1.7265	3.5935	782.92	298.15	2.2304	1500	5.8745
78	Decyl mercaptan	C ₁₀ H ₂₂ S	143-10-2	174.347	1.931	5.4815	1.6085	3.74	754.75	200	2.0434	1500	6.4613
79	1-Decyne	C ₁₀ H ₁₈	764-93-2	138.250	1.5045	4.3794	1.3291	2.5557	632.01	298	2.1938	1500	5.2794
80	Deuterium	D ₂	7782-39-0	4.032	0.3029	0.0975	2.515	-0.0275	368	100	0.3020	1500	0.3425
81	1,1-Dibromoethane	C ₂ H ₄ Br ₂	557-91-5	187.861	0.5927	1.158	1.4931	0.8428	655.5	200	0.6442	1500	1.5673
82	1,2-Dibromoethane	C ₂ H ₄ Br ₂	106-93-4	187.861	0.74906	1.2725	1.981	0.9437	845.2	200	0.7635	1500	1.7041
83	Dibromomethane	CH ₂ Br ₂	74-95-3	173.835	0.391	0.648	1.194	0.42	501	100	0.3929	1500	0.9599
84	Dibutyl ether	C ₈ H ₁₈ O	142-96-1	130.228	1.6122	4.4777	1.6831	2.918	781.6	200	1.6841	1500	5.2145
85	<i>m</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	541-73-1	147.002	0.7	2.0746	1.3664	1.5983	620.16	200	0.8245	1500	2.5161
86	<i>o</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	95-50-1	147.002	0.6948	2.0804	1.3632	1.594	619.2	200	0.8198	1500	2.5161
87	<i>p</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	106-46-7	147.002	0.6978	2.078	1.3635	1.5965	619.37	200	0.8228	1500	2.5175
88	1,1-Dichloroethane	C ₂ H ₄ Cl ₂	75-34-3	98.959	0.5521	1.205	1.502	0.8719	653.5	200	0.6061	1500	1.5615
89	1,2-Dichloroethane	C ₂ H ₄ Cl ₂	107-06-2	98.959	0.65271	1.1254	1.7376	0.878	795.45	200	0.6722	1500	1.5743
90	Dichloromethane	CH ₂ Cl ₂	75-09-2	84.933	0.3628	0.6804	1.256	0.4275	548	100	0.3637	1500	0.9543
91	1,1-Dichloropropane	C ₃ H ₆ Cl ₂	78-99-9	112.986	0.7145	1.7344	1.524	1.223	674.2	150	0.7268	1500	2.1609
92	1,2-Dichloropropane	C ₃ H ₆ Cl ₂	78-87-5	112.986	0.78658	1.7429	1.7157	1.2627	765.1	200	0.8217	1500	2.1894
93	Diethanol amine	C ₄ H ₁₁ NO ₂	111-42-2	105.136	1.208	3.066	2.089	2.343	891	298.15	1.4197	1500.1	3.4674
94	Diethyl amine	C ₄ H ₁₁ N	109-89-7	73.137	0.9102	2.674	1.719	1.7926	794.94	200	0.9502	1500	3.0519
95	Diethyl ether	C ₄ H ₁₀ O	60-29-7	74.122	0.8621	2.551	1.5413	1.437	688.9	200	0.9316	1500	2.9244
96	Diethyl sulfide	C ₄ H ₁₀ S	352-93-2	90.187	0.91273	2.41	1.6686	1.652	771.08	200	0.9567	1500	2.8724
97	1,1-Difluoroethane	C ₂ H ₄ F ₂	75-37-6	66.050	0.49653	1.2546	1.5394	0.87561	694.17	200	0.5373	1500	1.5424
98	1,2-Difluoroethane	C ₂ H ₄ F ₂	624-72-6	66.050	0.51889	1.2431	1.5048	0.76269	697.51	200	0.5536	1500	1.5510
99	Difluoromethane	CH ₂ F ₂	75-10-5	52.023	0.35489	0.71002	1.5936	0.4622	762	200	0.3681	1500	0.9419
100	Di-isopropyl amine	C ₆ H ₁₅ N	108-18-9	101.190	1.1384	2.5747	0.7384	1.62	2143	300	1.5995	1500	4.1941
101	Di-isopropyl ether	C ₆ H ₁₄ O	108-20-3	102.175	1.093	3.683	1.6057	2.342	699	298.15	1.5669	1500	4.0535
102	Di-isopropyl ketone	C ₇ H ₁₄ O	565-80-0	114.185	1.0869	4.054	1.7802	2.9786	791.6	300	1.5102	1500	4.3093
103	1,1-Dimethoxyethane	C ₄ H ₁₀ O ₂	534-15-6	90.121	1.1556	1.8305	0.95919	0.99605	2826.3	298.15	1.2777	1500	3.0678
104	1,2-Dimethoxypropane	C ₅ H ₁₂ O ₂	7778-85-0	104.148	1.0113	3.2393	1.5611	2.1501	689.3	298.15	1.4638	1500	3.6669
105	Dimethyl acetylene	C ₄ H ₆	503-17-3	54.090	0.6534	1.6179	1.7837	1.0242	821.4	200	0.6721	1500	1.9148
106	Dimethyl amine	C ₂ H ₇ N	124-40-3	45.084	0.5565	1.6384	1.7341	1.0899	793.04	200	0.5812	1500	1.8585
107	2,3-Dimethylbutane	C ₆ H ₁₄	79-29-8	86.175	0.7772	4.032	1.544	2.508	649.95	200	0.9363	1500	4.0353
108	1,1-Dimethylcyclohexane	C ₈ H ₁₆	590-66-9	112.213	1.0776	4.6718	1.654	3.3397	792.5	200	1.1535	1500	4.9543
109	<i>cis</i> -1,2-Dimethylcyclohexane	C ₈ H ₁₆	2207-01-4	112.213	1.1039	4.6445	1.6943	3.3949	798.35	200	1.1777	1500	4.9243
110	<i>trans</i> -1,2-Dimethylcyclohexane	C ₈ H ₁₆	6876-23-9	112.213	1.0991	4.6401	1.6679	3.3736	781.97	200	1.1820	1500	4.9275
111	Dimethyl disulfide	C ₂ H ₆ S ₂	624-92-0	94.199	0.7843	1.4364	1.5836	0.871	730.65	200	0.8155	1500	1.9523
112	Dimethyl ether	C ₂ H ₆ O	115-10-6	46.068	0.5148	1.442	1.6034	0.7747	725.4	200	0.5436	1500	1.6581
113	<i>N,N</i> -Dimethyl formamide	C ₃ H ₇ NO	68-12-2	73.094	0.722	1.783	1.532	1.31	762	200	0.7594	1500	2.2596
114	2,3-Dimethylpentane	C ₇ H ₁₆	565-59-3	100.202	0.85438	4.5772	1.5181	2.974	641.01	200	1.0550	1500	4.5983
115	Dimethyl phthalate	C ₁₀ H ₁₀ O ₄	131-11-3	194.184	1.396	4.78	2.19	3.9705	900.6	300	1.7481	1200	4.4740
116	Dimethylsilane	C ₂ H ₆ Si	1111-74-6	60.170	0.61453	1.7438	1.3418	1.0102	592.09	200	0.7095	1500	2.0944
117	Dimethyl sulfide	C ₂ H ₆ S	75-18-3	62.134	0.6037	1.3747	1.641	0.7988	743.5	200	0.6298	1500	1.6949
118	Dimethyl sulfoxide	C ₂ H ₆ OS	67-68-5	78.133	0.6949	1.524	1.6514	1.0658	722.2	200	0.7355	1500	1.9255
119	Dimethyl terephthalate	C ₁₀ H ₁₀ O ₄	120-61-6	194.184	1.174	5.32	2.105	4.1	818	298.15	1.6816	1000.15	4.1139
120	1,4-Dioxane	C ₄ H ₈ O ₂	123-91-1	88.105	0.56184	2.7034	1.5171	1.7658	700.76	200	0.6403	1500	2.8174
121	Diphenyl ether	C ₁₂ H ₁₀ O	101-84-8	170.207	1.0985	4.3412	1.6222	3.6455	743.62	300	1.7298	1200	4.5143
122	Dipropyl amine	C ₆ H ₁₅ N	142-84-7	101.190	1.2114	2.6127	0.78956	1.6903	2394.4	300	1.5900	1500	4.2484
123	Dodecane	C ₁₂ H ₂₆	112-40-3	170.335	2.1295	6.633	1.7155	4.5161	777.5	200	2.2442	1500	7.4325

TABLE 2-156 Heat Capacity at Constant Pressure of Inorganic and Organic Compounds in the Ideal Gas State Fit to Hyperbolic Functions C_p [J/(kmol·K)] (Continued)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1 × 1E-05	C2 × 1E-05	C3 × 1E-03	C4 × 1E-05	C5	$T_{\min}$, K	C_p at $T_{\min}$ × 1E-05	$T_{\max}$, K	C_p at $T_{\max}$ × 1E-05
124	Eicosane	C ₂₀ H ₄₂	112-95-8	282.547	3.2481	11.09	1.636	7.45	726.27	200	3.5235	1500	12.2110
125	Ethane	C ₂ H ₆	74-84-0	30.069	0.40326	1.6555	0.73223		752.87	200	0.4256	1500	1.4562
126	Ethanol	C ₂ H ₆ O	64-17-5	46.068	0.492	1.4577	1.6628	0.939	744.7	200	0.5224	1500	1.6576
127	Ethyl acetate	C ₄ H ₈ O ₂	141-78-6	88.105	0.9981	2.0931	2.0226	1.803	928.05	200	1.0126	1500	2.6594
128	Ethyl amine	C ₂ H ₇ N	75-04-7	45.084	0.594	1.618	1.812	1.078	820	200	0.6139	1500	1.8528
129	Ethylbenzene	C ₈ H ₁₀	100-41-4	106.165	0.7844	3.399	1.559	2.426	702	200	0.8912	1500	3.6147
130	Ethyl benzoate	C ₉ H ₁₀ O ₂	93-89-0	150.175	1.0944	4.1794	0.88375	-1.609	1183.1	300	1.4598	1500	4.2540
131	2-Ethyl butanoic acid	C ₆ H ₁₂ O ₂	88-09-5	116.158	1.0455	2.3148	0.71	1.471	2061.6	300	1.5102	1200.15	3.6330
132	Ethyl butyrate	C ₈ H ₁₆ O ₂	105-54-4	116.158	1.115	3.391	1.6705	2.518	733.6	298	1.5583	1200	3.6213
133	Ethylcyclohexane	C ₈ H ₁₆	1678-91-7	112.213	1.1059	4.6306	1.6628	3.299	781.1	200	1.1875	1500	4.9184
134	Ethylcyclopentane	C ₇ H ₁₄	1640-89-7	98.186	0.82052	4.0342	1.567	2.6697	715.52	200	0.9272	1500	4.1472
135	Ethylene	C ₂ H ₄	74-85-1	28.053	0.3338	0.9479	1.596	0.551	740.8	60	0.3338	1500	1.0987
136	Ethylenediamine	C ₂ H ₈ N ₂	107-15-3	60.098	0.7286	1.8436	1.688	1.199	767.3	300	0.9178	1500	2.2016
137	Ethylene glycol	C ₂ H ₆ O ₂	107-21-1	62.068	0.63012	1.4584	1.673	0.97296	773.65	300	0.7800	1500	1.8095
138	Ethyleneimine	C ₂ H ₅ N	151-56-4	43.068	0.343	1.427	1.638	1.037	744.7	150	0.3480	1500	1.5178
139	Ethylene oxide	C ₂ H ₄ O	75-21-8	44.053	0.3346	1.2116	1.6084	0.8241	737.3	50	0.3346	1500	1.3297
140	Ethyl formate	C ₃ H ₆ O ₂	109-94-4	74.079	0.537	1.886	1.207	0.864	496	100	0.5412	1500	2.1485
141	2-Ethyl hexanoic acid	C ₈ H ₁₆ O ₂	149-57-5	144.211	1.5777	4.4017	1.7494	3.2378	792.34	298.15	2.0279	1500	5.1201
142	Ethylhexyl ether	C ₈ H ₁₈ O	5756-43-4	130.228	1.634	4.5119	1.7532	3.1032	809.75	298.15	2.0360	1200	4.8744
143	Ethylisopropyl ether	C ₅ H ₁₂ O	625-54-7	88.148	1.0953	3.0032	1.7988	2.1311	817.35	298.15	1.3620	1200	3.2289
144	Ethylisopropyl ketone	C ₆ H ₁₂ O	565-69-5	100.159	1.24	3.2	1.967	2.346	896	298.15	1.4479	1200	3.4234
145	Ethyl mercaptan	C ₂ H ₆ S	75-08-1	62.134	0.5576	1.3617	1.5221	0.8073	687.5	200	0.5970	1500	1.6729
146	Ethyl propionate	C ₅ H ₁₀ O ₂	105-37-3	102.132	0.937	2.829	1.648	2.155	724.7	300	1.3377	1200	3.0569
147	Ethylpropyl ether	C ₅ H ₁₂ O	628-32-0	88.148	1.132	2.94	1.827	2.055	852	298.15	1.3538	1500	3.4535
148	Ethyltrichlorosilane	C ₂ H ₅ Cl ₃ Si	115-21-9	163.506	0.85105	1.0378	0.59737	0.94745	2122.7	167	0.8926	1500	2.2349
149	Fluorine	F ₂	7782-41-4	37.997	0.29122	0.10132	1.453	0.094101	662.91	50	0.2912	1500	0.3812
150	Fluorobenzene	C ₆ H ₅ F	462-06-6	96.102	0.62653	2.1646	1.564	1.7278	724.29	200	0.6914	1500	2.4736
151	Fluoroethane	C ₂ H ₅ F	353-36-6	48.060	0.44373	1.3119	1.6422	0.85441	738.77	200	0.4726	1500	1.5008
152	Fluoromethane	CH ₃ F	593-53-3	34.033	0.33289	0.73989	1.8639	0.46079	891.16	50	0.3329	1500	0.9024
153	Formaldehyde	CH ₂ O	50-00-0	30.026	0.3327	0.49542	1.8666	0.28075	934.9	50	0.3327	1500	0.7113
154	Formamide	CH ₃ NO	75-12-7	45.041	0.3822	0.93	1.845	0.69	850	150	0.3833	1500	1.1203
155	Formic acid	CH ₂ O ₂	64-18-6	46.026	0.3381	0.7593	1.1925	0.318	550	50	0.3381	1500	0.9933
156	Furan	C ₄ H ₄ O	110-00-9	68.074	0.3727	1.6606	1.5112	1.3145	686	200	0.4376	1500	1.7940
158	Heptadecane	C ₁₇ H ₃₆	629-78-7	240.468	2.7878	9.5247	1.6935	6.6651	744.57	200	3.0034	1500	10.4160
159	Heptanal	C ₇ H ₁₄ O	111-71-7	114.185	1.404	2.5907	0.8315	1.312	2201	200	1.4479	1500	4.2863
160	Heptane	C ₇ H ₁₆	142-82-5	100.202	1.2015	4.001	1.6766	2.74	756.4	200	1.2828	1500	4.4283
161	Heptanoic acid	C ₇ H ₁₄ O ₂	111-14-8	130.185	1.3135	2.3317	0.67567	1.824	1846	300	1.8497	1500	4.2941
162	1-Heptanol	C ₇ H ₁₆ O	111-70-6	116.201	1.2215	3.991	1.58	2.835	717.7	200	1.3330	1500	4.5346
163	2-Heptanol	C ₇ H ₁₆ O	543-49-7	116.201	1.4569	2.8252	0.81695	1.766	2537.2	298.15	1.8136	1500	4.6604
164	3-Heptanone	C ₇ H ₁₄ O	106-35-4	114.185	1.2768	3.381	1.3831	1.888	650.3	200	1.3968	1500	4.1386
165	2-Heptanone	C ₇ H ₁₄ O	110-43-0	114.185	1.2507	2.148	0.6912	1.619	1759.3	150	1.2688	1200	3.8446
166	1-Heptene	C ₇ H ₁₄	592-76-7	98.186	1.1851	3.6362	1.7359	2.5048	785.73	298.15	1.5434	1500	4.0836
167	Heptyl mercaptan	C ₇ H ₁₆ S	1639-09-4	132.267	1.442	4.1603	1.6603	2.6572	759.39	200	1.5191	1500	4.7831
168	1-Heptyne	C ₇ H ₁₂	628-71-7	96.170	1.0712	3.0258	1.5273	2.0975	689.62	200	1.1721	1500	3.5985
169	Hexadecane	C ₁₆ H ₃₄	544-76-3	226.441	2.6283	8.9733	1.6912	6.264	744.41	200	2.8312	1500	9.8182
170	Hexanal	C ₆ H ₁₂ O	66-25-1	100.159	1.232	2.2146	0.84	1.219	2205	200	1.2672	1500	3.7314
171	Hexane	C ₆ H ₁₄	110-54-3	86.175	1.044	3.523	1.6946	2.369	761.6	200	1.1117	1500	3.8620
172	Hexanoic acid	C ₆ H ₁₂ O ₂	142-62-1	116.158	1.1622	2.0708	0.68661	1.5355	1932.5	298.15	1.6107	1500	3.7636
173	1-Hexanol	C ₆ H ₁₄ O	111-27-3	102.175	1.0625	3.521	1.5835	2.462	715.75	200	1.1607	1500	3.9726
174	2-Hexanol	C ₆ H ₁₄ O	626-93-7	102.175	1.2615	3.5964	1.8445	2.594	819.17	298.15	1.5829	1500	4.0672
175	2-Hexanone	C ₆ H ₁₂ O	591-78-6	100.159	1.094	1.807	0.689	1.474	1772	200	1.1815	1200	3.3207
176	3-Hexanone	C ₆ H ₁₂ O	589-38-8	100.159	1.1237	2.936	1.401	1.601	650.5	150	1.1443	1500	3.5874
177	1-Hexene	C ₆ H ₁₂	592-41-6	84.159	1.0434	3.0749	1.7459	2.0728	793.53	298	1.3301	1500	3.4819
178	3-Hexyne	C ₆ H ₁₀	928-49-4	82.144	0.9376	3.015	1.9057	1.986	817	300	1.1909	1500	3.1889
179	Hexyl mercaptan	C ₆ H ₁₄ S	111-31-9	118.240	1.2662	3.7294	1.6574	2.308	757.8	200	1.3340	1500	4.2483
180	1-Hexyne	C ₆ H ₁₀	693-02-7	82.144	0.9129	2.5577	1.529	1.737	683	200	1.0004	1500	3.0371
181	2-Hexyne	C ₆ H ₁₀	764-35-2	82.144	1.036	3.009	2.116	2.106	902.4	300	1.2215	1500	3.1894
182	Hydrazine	H ₄ N ₂	302-01-2	32.045	0.38711	0.8576	1.7228	0.56635	733.53	200	0.4070	1500	1.0571
183	Hydrogen	H ₂	1333-74-0	2.016	0.27617	0.0956	2.466	0.0376	567.6	250	0.2843	1500	0.3225

184	Hydrogen bromide	HBr	10035-10-6	80.912	0.2912	0.0953	2.142	0.0157	1400	50	0.2912	1500	0.3479
185	Hydrogen chloride	HCl	7647-01-0	36.461	0.29157	0.09048	2.0938	-0.00107	120	50	0.2914	1500	0.3406
186	Hydrogen cyanide	HCN	74-90-8	27.025	0.30125	0.3171	1.6102	0.2179	626	100	0.3014	1500	0.5522
187	Hydrogen fluoride	HF	7664-39-3	20.006	0.29134	0.093252	2.905	1.95E-03	1.33E+03	50	0.2913	1500	0.3224
188	Hydrogen sulfide	H ₂ S	7783-06-4	34.081	0.33288	0.26086	0.9134	-0.17979	949.4	100	0.3329	1500	0.5143
189	Isobutyric acid	C ₄ H ₈ O ₂	79-31-2	88.105	0.74694	2.4356	1.715	1.8484	757.75	298.15	1.0427	1200	2.5383
190	Isopropyl amine	C ₃ H ₉ N	75-31-0	59.110	0.68545	2.1876	1.5831	1.3855	691.76	200	0.7510	1500	2.4540
191	Malonic acid	C ₃ H ₄ O ₄	141-82-2	104.061	0.49522	1.8718	1.2958	1.4852	569.96	300	0.9790	1200	2.0517
192	Methacrylic acid	C ₄ H ₆ O ₂	79-41-4	86.089	0.7251	2.089	1.8516	1.6483	798.43	298.15	0.9475	1200.1	2.2057
193	Methane	CH ₄	74-82-8	16.042	0.33298	0.79933	2.0869	0.41602	991.96	50	0.3330	1500	0.8890
194	Methanol	CH ₃ O	67-56-1	32.042	0.39252	0.879	1.9165	0.53654	896.7	200	0.3980	1500	1.0533
195	N-Methyl acetamide	C ₃ H ₇ NO	79-16-3	73.094	0.6116	2.029	1.7683	1.3302	835.5	300	0.7698	1500	2.2209
196	Methyl acetate	C ₃ H ₆ O ₂	79-20-9	74.079	0.555	1.782	1.26	0.853	562	298	0.8489	1500	2.0754
197	Methyl acetylene	C ₃ H ₄	74-99-7	40.064	0.4478	1.0917	1.5508	0.675	658.2	200	0.4882	1500	1.3293
198	Methyl acrylate	C ₄ H ₆ O ₂	96-33-3	86.089	0.1206	2.3766	1.0543	1.8186	418.8	298.15	0.9908	1200.1	2.1663
199	Methyl amine	CH ₃ N	74-89-5	31.057	0.41	1.0578	1.708	0.6836	735	150	0.4136	1500	1.2388
200	Methyl benzoate	C ₈ H ₈ O ₂	93-58-3	136.148	0.9396	2.559	0.825	1.36	3000	300	1.2586	1200	3.3569
201	3-Methyl-1,2-butadiene	C ₅ H ₈	598-25-4	68.117	0.671	2.222	1.421	1.194	614.7	150	0.6931	1500	2.5028
202	2-Methylbutane	C ₅ H ₁₂	78-78-4	72.149	0.746	3.265	1.545	1.923	666.7	200	0.8546	1500	3.3792
203	2-Methylbutanoic acid	C ₆ H ₁₀ O ₂	116-53-0	102.132	1.8458	1.743	1.22	-56.11	31.2	300	1.2793	1500	3.2262
204	3-Methyl-1-butanol	C ₅ H ₁₂ O	123-51-3	88.148	0.92165	3.3371	1.8365	2.4645	757.99	298.15	1.3135	1500	3.4856
205	2-Methyl-1-butene	C ₅ H ₁₀	563-46-2	70.133	0.87026	2.5556	1.7757	1.7636	807.82	200	0.9060	1500	2.8923
206	2-Methyl-2-butene	C ₅ H ₁₀	513-35-9	70.133	0.81924	2.6038	1.7593	1.7195	800.93	200	0.8559	1500	2.8709
207	2-Methyl-1-butene-3-yne	C ₅ H ₈	78-80-8	66.101	0.7906	1.656	1.6926	1.2167	788.4	298.15	0.9632	1500.15	2.1502
208	Methylbutyl ether	C ₅ H ₁₂ O	628-28-4	88.148	0.82051	3.0869	1.3864	1.7886	613.87	300	1.3300	1200	3.1994
209	Methylbutyl sulfide	C ₅ H ₁₂ S	628-29-5	104.214	1.0785	2.7388	1.5885	1.9067	749.6	273.15	1.3173	1200	3.1687
210	3-Methyl-1-butyne	C ₅ H ₈	598-23-2	68.117	0.8274	2.1377	1.755	1.5149	782	200	0.8646	1500	2.5255
211	Methyl butyrate	C ₆ H ₁₀ O ₂	623-42-7	102.132	0.894	2.91	1.57	2.073	678.3	298	1.3461	1200	3.0766
212	Methylchlorosilane	CH ₃ ClSi	993-00-0	80.589	0.59895	1.1636	1.565	0.81581	690.39	200	0.6380	1500	1.5593
213	Methylcyclohexane	C ₇ H ₁₄	108-87-2	98.186	0.9227	4.115	1.6504	2.9006	779.48	200	0.9953	1500	4.3180
214	1-Methylcyclohexanol	C ₇ H ₁₄ O	590-67-0	114.185	0.7959	2.596	0.6213	2.288	1698.6	300	1.5302	1200	4.1359
215	cis-2-Methylcyclohexanol	C ₇ H ₁₄ O	7443-70-1	114.185	0.92279	2.6709	0.68784	1.9847	1732.4	300	1.5099	1200	4.1467
216	trans-2-Methylcyclohexanol	C ₇ H ₁₄ O	7443-52-9	114.185	0.92279	2.6709	0.68784	1.9847	1732.4	300	1.5099	1200	4.1467
217	Methylcyclopentane	C ₆ H ₁₂	96-37-7	84.159	0.66456	3.507	1.5892	2.3526	727.13	200	0.7510	1500	3.5495
218	1-Methylcyclopentene	C ₆ H ₁₀	693-89-0	82.144	0.69411	3.0209	1.6903	2.1209	781.56	200	0.7464	1500	3.1496
219	3-Methylcyclopentene	C ₆ H ₁₀	1120-62-3	82.144	0.6422	3.0711	1.6387	2.1298	750.25	200	0.7083	1500	3.1549
220	Methyldichlorosilane	CH ₂ Cl ₂ Si	75-54-7	115.034	0.7283	1.0307	1.5429	0.7811	668.94	200	0.7717	1500	1.5893
221	Methylethyl ether	C ₃ H ₈ O	540-67-0	60.095	0.68681	1.9959	1.5534	1.1168	692.04	200	0.7396	1500	2.2931
222	Methylethyl ketone	C ₄ H ₈ O	78-93-3	72.106	0.784	2.1032	1.5488	1.1855	693	200	0.8397	1500	2.4816
223	Methylethyl sulfide	C ₃ H ₈ S	624-89-5	76.161	0.75083	1.9577	1.6424	1.1949	749.19	273.16	0.9004	1500	2.3178
224	Methyl formate	C ₂ H ₄ O ₂	107-31-3	60.052	0.506	1.219	1.637	0.894	743	250	0.5888	1500	1.5109
225	Methylisobutyl ether	C ₆ H ₁₂ O	625-44-5	88.148	0.7284	3.1713	1.352	1.8948	585.14	300	1.3200	1200	3.1987
226	Methylisobutyl ketone	C ₆ H ₁₂ O	108-10-1	100.159	1.227	2.195	0.842	1.191	2460	298.15	1.4755	1500.15	3.6532
227	Methyl Isocyanate	C ₂ H ₃ NO	624-83-9	57.051	0.474	1.226	2.188	0.85983	1008.2	298.15	0.5195	1500	1.3595
228	Methylisopropyl ether	C ₄ H ₁₀ O	598-53-8	74.122	0.89232	2.4765	1.696	1.5598	791.4	200	0.9280	1500	2.8696
229	Methylisopropyl ketone	C ₅ H ₁₀ O	563-80-4	86.132	1.5914	1.764	1.2076	-407.4	10.503	300	1.1291	1500	2.9991
230	Methylisopropyl sulfide	C ₄ H ₁₀ S	1551-21-9	90.187	0.99247	2.7275	2.003	1.8974	849.64	273	1.1377	1500	2.9952
231	Methyl mercaptan	CH ₃ S	74-93-1	48.107	0.4146	0.8307	1.589	0.4612	716.7	200	0.4329	1500	1.0781
232	Methyl methacrylate	C ₅ H ₈ O ₂	80-62-6	100.116	0.864	1.811	0.7543	0.8	2160	298.15	1.1621	1500	2.8637
233	2-Methyloctanoic acid	C ₁₀ H ₁₈ O ₂	3004-93-1	158.238	1.7483	4.9288	1.7384	3.5897	788.01	298.15	2.2567	1500	5.7177
234	2-Methylpentane	C ₆ H ₁₄	107-83-5	86.175	0.903	3.801	1.602	2.453	691.6	200	1.0192	1500	3.9617
235	Methyl pentyl ether	C ₆ H ₁₄ O	628-80-8	102.175	0.94326	3.5965	1.3533	2.0569	599.92	300	1.5600	1200	3.7409
236	2-Methylpropane	C ₄ H ₁₀	75-28-5	58.122	0.6549	2.4776	1.587	1.575	706.99	200	0.7218	1500	2.6656
237	2-Methyl-2-propanol	C ₄ H ₁₀ O	75-65-0	74.122	0.7704	2.539	1.5502	1.669	679.3	200	0.8567	1500	2.8508
238	2-Methyl propene	C ₄ H ₈	115-11-7	56.106	0.6125	2.066	1.545	1.2057	676	200	0.6763	1500	2.2814
239	Methyl propionate	C ₄ H ₈ O ₂	554-12-1	88.105	0.7765	2.442	1.714	1.818	716	300	1.1242	1200	2.5276
240	Methylpropyl ether	C ₄ H ₁₀ O	557-17-5	74.122	0.92151	2.3943	1.6936	1.4896	797.79	298	1.1251	1200	2.6391
241	Methylpropyl sulfide	C ₄ H ₁₀ S	3877-15-4	90.187	0.93775	2.6178	1.7291	1.6236	783.23	298.15	1.1728	1500	2.9904
242	Methylsilane	CH ₃ Si	992-94-9	46.144	0.46149	1.2781	1.4565	0.79115	643.23	200	0.5141	1500	1.5253
243	alpha-Methyl styrene	C ₉ H ₁₀	98-83-9	118.176	0.78548	3.5969	1.4342	2.5336	651.69	200	0.9445	1500	3.8592
244	Methyl tert-butyl ether	C ₅ H ₁₂ O	1634-04-4	88.148	0.9779	3.091	1.643	2.099	731.191	298	1.3522	1500	3.4779
245	Methyl vinyl ether	C ₃ H ₆ O	107-25-5	58.079	0.60865	1.5965	1.619	0.93783	739.55	300	0.7748	1500	1.8871
246	Naphthalene	C ₁₀ H ₈	91-20-3	128.171	0.6805	3.5494	1.4262	2.5984	650.1	200	0.8454	1500	3.7359

TABLE 2-156 Heat Capacity at Constant Pressure of Inorganic and Organic Compounds in the Ideal Gas State Fit to Hyperbolic Functions C_p [J/(kmol·K)] (Concluded)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1 × 1E-05	C2 × 1E-05	C3 × 1E-03	C4 × 1E-05	C5	$T_{\min}$, K	C_p at $T_{\min}$ × 1E-05	$T_{\max}$, K	C_p at $T_{\max}$ × 1E-05
248	Nitroethane	C ₂ H ₅ NO ₂	79-24-3	75.067	0.54619	1.6492	1.4803	1.0635	666.94	200	0.6062	1500	1.9237
249	Nitrogen	N ₂	7727-37-9	28.013	0.29105	0.086149	1.7016	0.0010347	909.79	50	0.2911	1500	0.3484
250	Nitrogen trifluoride	F ₃ N	7783-54-2	71.002	0.33284	0.49837	0.7093	0.23264	372.91	100	0.3404	1500	0.8092
251	Nitromethane	CH ₃ NO ₂	75-52-5	61.040	0.42267	1.0842	1.4885	0.68603	683.57	200	0.4571	1500	1.3280
252	Nitrous oxide	N ₂ O	10024-97-2	44.013	0.29338	0.3236	1.1238	0.2177	479.4	100	0.2948	1500	0.5828
254	Nonadecane	C ₁₉ H ₄₀	629-92-5	268.521	3.1062	10.575	0.76791	-4.5661	912.03	200	3.3533	1500	11.6130
255	Nonanal	C ₉ H ₁₈ O	124-19-6	142.239	1.7347	4.5115	1.712	3.3256	810.96	200	1.8005	1500	5.4439
256	Nonane	C ₉ H ₂₀	111-84-2	128.255	1.5175	4.915	1.6448	3.47	749.6	200	1.6257	1500	5.5407
257	Nonanoic acid	C ₉ H ₁₈ O ₂	112-05-0	158.238	0.1266	6.011	1.0815	4.5946	418.2	298.15	2.2953	1500	5.5267
258	1-Nonanol	C ₉ H ₂₀ O	143-08-8	144.255	1.54	4.936	1.578	3.588	721.11	200	1.6777	1500	5.6606
259	2-Nonanol	C ₉ H ₂₀ O	628-99-9	144.255	1.8197	3.5542	0.81514	2.1974	2508.8	298.15	2.2720	1500	5.8526
260	1-Nonene	C ₉ H ₁₈	124-11-8	126.239	1.5352	4.6844	1.7288	3.2304	783.67	298.15	2.0014	1500	5.2776
261	Nonyl mercaptan	C ₉ H ₂₀ S	1455-21-6	160.320	1.7646	5.044	1.6182	3.3857	755.48	200	1.8658	1500	5.9082
262	1-Nonyne	C ₉ H ₁₆	3452-09-3	124.223	1.6289	3.9708	1.8928	3.2136	855.52	298.15	1.9693	1500	4.7924
263	Octadecane	C ₁₈ H ₃₈	593-45-3	254.494	2.9502	10.034	0.77107	-4.3012	916.73	200	3.1800	1500	11.0160
264	Octanal	C ₈ H ₁₆ O	124-13-0	128.212	1.6088	4.218	1.9126	3.278	869	200	1.6504	1500	4.9286
265	Octane	C ₈ H ₁₈	111-65-9	114.229	1.3554	4.431	1.6356	3.054	746.4	200	1.4529	1500	4.9764
266	Octanoic acid	C ₈ H ₁₆ O ₂	124-07-2	144.211	1.4082	4.3436	1.4662	2.7687	659.38	298.15	2.0652	1500	5.0411
267	1-Octanol	C ₈ H ₁₈ O	111-87-5	130.228	1.3805	4.459	1.5751	3.2016	718.8	200	1.5055	1500	5.0965
268	2-Octanol	C ₈ H ₁₈ O	123-96-6	130.228	1.6383	3.1897	0.81595	1.9814	2521.3	298.15	2.0428	1500	5.2565
269	2-Octanone	C ₈ H ₁₆ O	111-13-7	128.212	1.3901	3.806	1.3717	2.2573	660.96	150	1.4162	1500	4.6547
270	3-Octanone	C ₈ H ₁₆ O	106-68-3	128.212	1.4952	4.4103	0.80211	-2.0958	981.95	200	1.5775	1500	4.9067
271	1-Octene	C ₈ H ₁₆	111-66-0	112.213	1.3599	4.1605	1.7317	2.8675	784.47	298.15	1.7723	1500	4.6807
272	Octyl mercaptan	C ₈ H ₁₈ S	111-88-6	146.294	1.5981	4.6063	1.6295	3.0301	756.28	200	1.6881	1500	5.3549
273	1-Octyne	C ₈ H ₁₄	629-05-0	110.197	1.2307	3.4942	1.528	2.4617	694.81	200	1.3448	1500	4.1604
274	Oxalic acid	C ₂ H ₂ O ₄	144-62-7	90.035	0.25751	1.1734	2.7969	0.65788	878.91	298.15	0.3201	1000.1	0.6502
275	Oxygen	O ₂	7782-44-7	31.999	0.29103	0.1004	2.5265	0.09356	1153.8	50	0.2910	1500	0.3653
276	Ozone	O ₃	10028-15-6	47.998	0.33483	0.29577	1.5217	0.27151	680.35	100	0.3349	1500	0.5928
277	Pentadecane	C ₁₅ H ₃₂	629-62-9	212.415	2.4679	8.4212	1.6865	5.8537	743.6	200	2.6586	1500	9.2209
278	Pentanal	C ₅ H ₁₀ O	110-62-3	86.132	1.0743	2.8363	1.9549	2.0146	890.44	200	1.0960	1500	3.2404
279	Pentane	C ₅ H ₁₂	109-66-0	72.149	0.8805	3.011	1.6502	1.892	747.6	200	0.9404	1500	3.2927
280	Pentanoic acid	C ₅ H ₁₀ O ₂	109-52-4	102.132	2.836	1.08	2.107	-3.56	283	298.15	1.3824	1500	3.2952
281	1-Pentanol	C ₅ H ₁₂ O	71-41-0	88.148	0.906	3.062	1.6054	2.115	717.97	200	0.9890	1500	3.4133
282	2-Pentanol	C ₅ H ₁₂ O	6032-29-7	88.148	1.0853	3.0747	1.8672	2.2271	825.4	298.15	1.3539	1500	3.4701
283	2-Pentanone	C ₅ H ₁₀ O	107-87-9	86.132	0.90053	2.7085	1.6592	1.8012	743.96	200	0.9591	1500	3.0797
284	3-Pentanone	C ₅ H ₁₀ O	96-22-0	86.132	0.96896	2.4907	1.4177	1.301	646.7	200	1.0536	1500	3.0358
285	1-Pentene	C ₅ H ₁₀	109-67-1	70.133	0.82523	2.5943	1.7291	1.768	778.7	298.15	1.0856	1500	2.8897
286	2-Pentyl mercaptan	C ₅ H ₁₂ S	2084-19-7	104.214	1.1327	2.947	1.7418	2.0987	795.78	298	1.4202	1500	3.4994
287	Pentyl mercaptan	C ₅ H ₁₂ S	110-66-7	104.214	1.0974	3.2959	1.6761	1.9486	757.67	200	1.1547	1500	3.6956
288	1-Pentyne	C ₅ H ₈	627-19-0	68.117	0.753	2.0905	1.5307	1.378	672.8	200	0.8276	1500	2.4754
289	2-Pentyne	C ₅ H ₈	627-21-4	68.117	0.70737	2.2229	1.557	1.3125	690.78	200	0.7700	1500	2.5052
290	Phenanthrene	C ₁₄ H ₁₀	85-01-8	178.229	0.9374	4.758	1.382	3.485	627.4	200	1.1959	1500	5.0645
291	Phenol	C ₆ H ₆ O	108-95-2	94.111	0.434	2.445	1.152	1.512	507	100	0.4401	1500	2.6045
292	Phenyl isocyanate	C ₇ H ₅ NO	103-71-9	119.121	0.59683	2.5533	1.2397	1.5519	576.78	298.15	1.1054	1500	2.8390
293	Phthalic anhydride	C ₈ H ₄ O ₃	85-44-9	148.116	0.7364	2.544	1.0852	0.808	573	298.15	1.0745	1000.15	2.6737
294	Propadiene	C ₃ H ₄	463-49-0	40.064	0.426	1.1194	1.5772	0.7546	680.8	200	0.4646	1500	1.3376
295	Propane	C ₃ H ₈	74-98-6	44.096	0.5192	1.9245	1.6265	1.168	723.6	200	0.5632	1500	2.0556
296	1-Propanol	C ₃ H ₈ O	71-23-8	60.095	0.619	2.0213	1.6293	1.2956	727.4	200	0.6665	1500	2.2458
297	2-Propanol	C ₃ H ₈ O	67-63-0	60.095	0.73145	2.0313	1.9375	1.4815	843.37	298.15	0.8966	1500	2.2760
298	Propenylcyclohexene	C ₉ H ₁₄	13511-13-2	122.207	1.0563	4.3397	1.6098	3.181	729.66	300	1.6392	1500	4.6527
299	Propionaldehyde	C ₃ H ₆ O	123-38-6	58.079	0.7174	1.914	2.0144	1.1708	930.6	200	0.7266	1500	2.1149
300	Propionic acid	C ₃ H ₆ O ₂	79-09-4	74.079	0.6959	1.7778	1.7098	1.2654	763.78	298.15	0.8938	1500	2.1248
301	Propionitrile	C ₃ H ₅ N	107-12-0	55.079	0.5357	1.4617	1.553	0.91197	678.2	200	0.5832	1500	1.7235
302	Propyl acetate	C ₅ H ₁₀ O ₂	109-60-4	102.132	1.7994	1.753	1.196	-4.12	108.2	298.15	1.3594	1500	3.2024
303	Propyl amine	C ₃ H ₉ N	107-10-8	59.110	0.76078	2.1049	1.7256	1.3936	789.03	200	0.7933	1500	2.4353

304	Propylbenzene	C ₉ H ₁₂	103-65-1	120.192	0.96885	3.7954	1.5168	2.6618	694.3	200	1.0927	1500	4.1613
305	Propylene	C ₃ H ₆	115-07-1	42.080	0.43852	1.506	1.3988	0.74754	616.46	130	0.4436	1500	1.6817
306	Propyl formate	C ₄ H ₈ O ₂	110-74-7	88.105	0.871	2.447	1.9254	1.888	821.3	298.15	1.1022	1500	2.7484
307	2-Propyl mercaptan	C ₃ H ₇ S	75-33-2	76.161	0.73815	1.9529	1.5954	1.2356	730.5	200	0.7825	1500	2.3287
308	Propyl mercaptan	C ₃ H ₇ S	107-03-9	76.161	0.7474	1.9523	1.631	1.2112	750.92	200	0.7848	1500	2.3216
309	1,2-Propylene glycol	C ₃ H ₈ O ₂	57-55-6	76.094	2.0114	0.8082	1.8656	-2.4404	279.98	298.15	1.0218	1000.15	2.1175
310	Quinone	C ₆ H ₄ O ₂	106-51-4	108.095	0.6487	2.1227	1.3491	1.514	614.8	200	0.7711	1500	2.4969
311	Silicon tetrafluoride	F ₄ Si	7783-61-1	104.079	0.3681	0.71245	0.65201	0.46721	286.03	100	0.4182	1500	1.0537
312	Styrene	C ₈ H ₈	100-42-5	104.149	0.893	2.1503	0.772	0.999	2442	100	0.8931	1500	3.2416
313	Succinic acid	C ₄ H ₆ O ₄	110-15-6	118.088	0.71806	2.2669	1.2739	1.7342	537.65	300	1.3370	1200	2.5823
314	Sulfur dioxide	O ₂ S	7446-09-5	64.064	0.33375	0.25864	0.9328	0.1088	423.7	100	0.3354	1500	0.5695
315	Sulfur hexafluoride	F ₆ S	2551-62-4	146.055	0.35256	1.227	0.67938	0.78407	351.27	100	0.3872	1500	1.5397
316	Sulfur trioxide	O ₃ S	7446-11-9	80.063	0.33408	0.49677	0.87322	0.28563	393.74	100	0.3408	1500	0.7967
317	Terephthalic acid	C ₈ H ₆ O ₄	100-21-0	166.131	0.945	2.526	0.829	0.5	2010	298.15	1.2478	1500	3.4444
318	<i>o</i> -Terphenyl	C ₁₈ H ₁₄	84-15-1	230.304	2.0719	6.2668	2.4044	6.345	967.71	298.15	2.4763	1500	6.6947
319	Tetradecane	C ₁₄ H ₃₀	629-59-4	198.388	2.3082	7.8678	1.6823	5.4486	743.1	200	2.4864	1500	8.6225
320	Tetrahydrofuran	C ₄ H ₈ O	109-99-9	72.106	0.46905	2.5314	1.5998	1.7051	740.64	200	0.5259	1500	2.5538
321	1,2,3,4-Tetrahydronaphthalene	C ₁₀ H ₁₂	119-64-2	132.202	0.8145	4.395	1.471	3.065	666.4	200	0.9881	1500	4.5348
322	Tetrahydrothiophene	C ₄ H ₆ S	110-01-0	88.171	0.51848	2.4535	1.5018	1.6871	665.31	200	0.6147	1500	2.5679
323	2,2,3,3-Tetramethylbutane	C ₈ H ₁₈	594-82-1	114.229	1.1352	5.6331	1.6211	3.3829	681.9	200	1.3069	1500	5.5784
324	Thiophene	C ₄ H ₄ S	110-02-1	84.140	0.40399	1.627	1.4562	1.322	648.81	200	0.4886	1500	1.8098
325	Toluene	C ₇ H ₈	108-88-3	92.138	0.5814	2.863	1.4406	1.898	650.43	200	0.7016	1500	3.0029
326	1,1,2-Trichloroethane	C ₂ H ₃ Cl ₃	79-00-5	133.404	0.66554	1.1257	1.5454	0.97196	717.04	298.15	0.8496	1500	1.6433
327	Tridecane	C ₁₃ H ₂₈	629-50-5	184.361	2.1496	7.3045	1.6695	4.9998	741.02	200	2.3156	1500	8.0251
328	Triethyl amine	C ₆ H ₁₅ N	121-44-8	101.190	1.2766	2.5559	0.80937	1.4829	2231.7	200	1.3278	1500	4.2046
329	Trimethyl amine	C ₃ H ₉ N	75-50-3	59.110	0.7107	1.5051	0.79662	0.84537	2187.6	200	0.7439	1500	2.4322
330	1,2,3-Trimethylbenzene	C ₉ H ₁₂	526-73-8	120.192	1.052	3.79	1.4814	2.331	667.3	200	1.1832	1500	4.1983
331	1,2,4-Trimethylbenzene	C ₉ H ₁₂	95-63-6	120.192	1.0106	3.8314	1.501	2.395	678.3	200	1.1354	1500	4.1854
332	2,2,4-Trimethylpentane	C ₈ H ₁₈	540-84-1	114.229	1.139	5.286	1.594	3.351	677.94	200	1.3139	1500	5.3769
333	2,3,3-Trimethylpentane	C ₈ H ₁₈	560-21-4	114.229	0.982	5.402	1.531	3.493	639.9	200	1.2194	1500	5.3754
334	1,3,5-Trinitrobenzene	C ₆ H ₃ N ₃ O ₆	99-35-4	213.105	2.0367	1.8181	1.2089	0.79777	1060.8	298.15	2.1054	1500	3.7585
335	2,4,6-Trinitrotoluene	C ₇ H ₅ N ₃ O ₆	118-96-7	227.131	2.154	2.4432	1.1126	0.58651	950.59	298.15	2.2726	1500	4.3560
336	Undecane	C ₁₁ H ₂₄	1120-21-4	156.308	1.9529	6.0998	1.7087	4.1302	775.4	200	2.0594	1500	6.8342
337	1-Undecanol	C ₁₁ H ₂₄ O	112-42-5	172.308	1.859	5.869	1.5718	4.326	722.7	200	2.0232	1500	6.7834
338	Vinyl acetate	C ₄ H ₆ O ₂	108-05-4	86.089	0.536	2.119	1.198	1.147	510	100	0.5404	1500	2.3750
339	Vinyl acetylene	C ₄ H ₄	689-97-4	52.075	0.55978	1.2141	1.6102	0.89079	710.4	200	0.5967	1500	1.5590
340	Vinyl chloride	C ₂ H ₃ Cl	75-01-4	62.498	0.42364	0.8735	1.6492	0.6556	739.07	200	0.4457	1500	1.1423
341	Vinyl trichlorosilane	C ₂ H ₃ Cl ₃ Si	75-94-5	161.490	0.84894	1.1471	1.38	0.9	644.61	298.15	1.0788	1500	1.8595
342	Water	H ₂ O	7732-18-5	18.015	0.33363	0.2679	2.6105	0.08896	1169	100	0.3336	2273.15	0.5276
343	<i>m</i> -Xylene	C ₈ H ₁₀	108-38-3	106.165	0.7568	3.3924	1.496	2.247	675.9	200	0.8759	1500	3.5920
344	<i>o</i> -Xylene	C ₈ H ₁₀	95-47-6	106.165	0.8521	3.2954	1.4944	2.115	675.8	200	0.9643	1500	3.5965
345	<i>p</i> -Xylene	C ₈ H ₁₀	106-42-3	106.165	0.7512	3.397	1.4928	2.247	675.1	200	0.8710	1500	3.5923

Constants in this table can be used in the following equation to calculate the ideal gas heat capacity C_p^0 .

$$C_p^0 = C_1 + C_2 \left[\frac{C_3/T}{\sinh(C_3/T)} \right]^2 + C_4 \left[\frac{C_5/T}{\cosh(C_5/T)} \right]^2$$

where C_p^0 is in J/(kmol·K) and T is in K. All substances are listed by chemical family in Table 2-6 and by formula in Table 2-7.

Values in this table were taken from the Design Institute for Physical Properties (DIPPR) of the American Institute of Chemical Engineers (AIChE), copyright 2007 AIChE and reproduced with permission of AIChE and of the DIPPR Evaluated Process Design Data Project Steering Committee. Their source should be cited as R. L. Rowley, W. V. Wilding, J. L. Oscarson, Y. Yang, N. A. Zundel, T. E. Daubert, R. P. Danner, DIPPR® Data Compilation of Pure Chemical Properties, Design Institute for Physical Properties, AIChE, New York (2007).

The number of digits provided for values at $T_{\min}$ and $T_{\max}$ was chosen for uniformity of appearance and formatting; these do not represent the uncertainties of the physical quantities, but are the result of calculations from the standard thermophysical property formulations within a fixed format.

2-182 PHYSICAL AND CHEMICAL DATA

TABLE 2-157 C_p/C_v : Ratios of Specific Heats of Gases at 1 atm Pressure*

Compound	Formula	Temperature, °C	Ratio of specific heats, $(\gamma) = C_p/C_v$	Compound	Formula	Temperature, °C	Ratio of specific heats, $(\gamma) = C_p/C_v$
Acetaldehyde	C ₂ H ₄ O	30	1.14	Hydrogen (<i>Cont.</i>)			
Acetic acid	C ₂ H ₄ O ₂	136	1.15	iodide	HI	20–100	1.40
Acetylene	C ₂ H ₂	15	1.26	sulfide	H ₂ S	15	1.332
		–71	1.31			–45	1.350
Air		925	1.36			–57	1.356
		17	1.403				
		–78	1.408	Iodine	I ₂	185	1.30
		–118	1.415	Isobutane	C ₄ H ₁₀	15	1.110
Ammonia	NH ₃	15	1.320				
Argon	Ar	15	1.670	Krypton	Kr	19	1.672
		–180	1.715				
		0–100	1.67	Mercury	Hg	360	1.67
				Methane	CH ₄	600	1.113
Benzene	C ₆ H ₆	90	1.10			300	1.196
Bromine	Br ₂	20–350	1.32			15	1.310
						–80	1.339
Carbon dioxide	CO ₂	15	1.299			–115	1.347
		–75	1.37	Methyl acetate	C ₃ H ₆ O ₂	15	1.14
disulfide	CS ₂	100	1.21	alcohol	CH ₃ O	77	1.237
monoxide	CO	15	1.402	ether	C ₂ H ₆ O	6–30	1.11
		–180	1.433	Methylal	C ₃ H ₈ O ₂	13	1.06
Chlorine	Cl ₂	15	1.355			40	1.09
Chloroform	CHCl ₃	100	1.15				
Cyanogen	(CN) ₂	15	1.256	Neon	Ne	19	1.667
Cyclohexane	C ₆ H ₁₂	80	1.315	Nitric oxide	NO	15	1.400
						–45	1.39
Dichlorodifluoromethane	CCl ₂ F ₂	25	1.139			–80	1.38
						15	1.402
Ethane	C ₂ H ₆	100	1.157	Nitrogen	N ₂	–181	1.433
		15	1.200			100	1.28
		–82	1.28	Nitrous oxide	N ₂ O	15	1.303
Ethyl alcohol	C ₂ H ₆ O	90	1.13			–30	1.31
ether	C ₄ H ₁₀ O	35	1.08			–70	1.34
		80	1.086				
Ethylene	C ₂ H ₄	100	1.201	Oxygen	O ₂	15	1.398
		15	1.253			–76	1.405
		–91	1.345			–181	1.439
Helium	He	–180	1.667	Pentane (<i>n</i> -)	C ₅ H ₁₂	86	1.071
Hexane (<i>n</i> -)	C ₆ H ₁₄	80	1.066	Phosphorus	P	300	1.17
Hydrogen	H ₂	15	1.407	Potassium	K	850	1.77
		–76	1.441				
		–181	1.607	Sodium	Na	750–920	1.68
bromide	HBr	20	1.42	Sulfur dioxide	SO ₂	15	1.290
chloride	HCl	15	1.41				
		100	1.40	Xenon	Xe	19	1.678
cyanide	HCN	65	1.31				
		140	1.28				
		210	1.24				

*For compounds that appear in Table 2-184, values are from E. W. Lemmon, M. O. McLinden, and D. G. Friend, "Thermophysical Properties of Fluid Systems" in *NIST Chemistry WebBook*, **NIST Standard Reference Database Number 69**, Eds. P. J. Linstrom and W. G. Mallard, June 2005, National Institute of Standards and Technology, Gaithersburg, Md. (<http://webbook.nist.gov>). Values for other compounds are from *International Critical Tables*, vol. 5, pp. 80–82.

SPECIFIC HEATS OF AQUEOUS SOLUTIONS

UNITS CONVERSIONS

For this subsection, the following units conversions are applicable:

$$^{\circ}\text{F} = \% ^{\circ}\text{C} + 32$$

To convert calories per gram-degree Celsius to British thermal units per pound-degree Fahrenheit, multiply by 1.0.

TABLE 2-158 Acetic Acid (at 38 °C)

Mole % acetic acid Cal/(g·°C)	0	6.98	30.9	54.5	100
	1.0	0.911	0.73	0.631	0.535

TABLE 2-159 Ammonia

Mole % NH ₃	Specific heat, cal/(g·°C)			
	2.4 °C	20.6 °C	41 °C	61 °C
0	1.01	1.0	0.995	1.0
10.5	0.98	0.995	1.06	1.02
20.9	0.96	0.99	1.03	
31.2	0.956	1.0		
41.4	0.985			

TABLE 2-160 Aniline (at 20 °C)

Mol % aniline Cal/(g·°C)	100	95	90.5	82.3	75.2
	0.497	0.52	0.53	0.56	0.581

TABLE 2-161 Copper Sulfate

Composition	Temperature	Specific heat, cal/(g·°C)
CuSO ₄ + 50H ₂ O	12 to 15 °C	0.848
CuSO ₄ + 200H ₂ O	12 to 14 °C	0.951
CuSO ₄ + 400H ₂ O	13 to 17 °C	0.975

TABLE 2-162 Ethyl Alcohol

Mole % C ₂ H ₅ OH	Specific heat, cal/(g·°C)		
	3 °C	23 °C	41 °C
4.16	1.05	1.02	1.02
11.5	1.02	1.03	1.03
37.0	0.805	0.86	0.875
61.0	0.67	0.727	0.748
100.0	0.54	0.577	0.621

TABLE 2-163 Glycerol

Mole % C ₃ H ₅ (OH) ₃	Specific heat, cal/(g·°C)	
	15 °C	32 °C
2.12	0.961	0.960
4.66	0.929	0.924
11.5	0.851	0.841
22.7	0.765	0.758
43.9	0.67	0.672
100.0	0.555	0.576

TABLE 2-164 Hydrochloric Acid

Mole % HCl	Specific heat, cal/(g·°C)				
	0 °C	10 °C	20 °C	40 °C	60 °C
0.0	1.00				
9.09	0.72		0.74	0.75	0.78
16.7	0.61	0.605	0.631	0.645	0.67
20.0	0.58	0.575	0.591	0.615	0.638
25.9	0.55				0.61

ADDITIONAL REFERENCES

Most of the tables below are from *International Critical Tables*, vol. 5, pp. 115–116, 122–125.

TABLE 2-165 Methyl Alcohol

Mole % CH ₃ OH	Specific heat, cal/(g·°C)		
	5 °C	20 °C	40 °C
5.88	1.02	1.0	0.995
12.3	0.975	0.982	0.98
27.3	0.877	0.917	0.92
45.8	0.776	0.811	0.83
69.6	0.681	0.708	0.726
100	0.576	0.60	0.617

TABLE 2-166 Nitric Acid

% HNO ₃ by Weight	Specific heat at 20 °C, cal/(g·°C)
0	1.000
10	0.900
20	0.810
30	0.730
40	0.675
50	0.650
60	0.640
70	0.615
80	0.575
90	0.515

TABLE 2-167 Phosphoric Acid*

%H ₃ PO ₄	C _p at 21.3 °C cal/(g·°C)	%H ₃ PO ₄	C _p at 21.3 °C cal/(g·°C)
2.50	0.9903	50.00	0.6350
3.80	0.9970	52.19	0.6220
5.33	0.9669	53.72	0.6113
8.81	0.9389	56.04	0.5972
10.27	0.9293	58.06	0.5831
14.39	0.8958	60.23	0.5704
16.23	0.8796	62.10	0.5603
19.99	0.8489	64.14	0.5460
22.10	0.8300	66.13	0.5349
24.56	0.8125	68.14	0.5242
25.98	0.8004	69.97	0.5157
28.15	0.7856	69.50	0.5160
29.96	0.7735	71.88	0.5046
32.09	0.7590	73.71	0.4940
33.95	0.7432	75.79	0.4847
36.26	0.7270	77.69	0.4786
38.10	0.7160	79.54	0.4680
40.10	0.7024	80.00	0.4686
42.08	0.6877	82.00	0.4593
44.11	0.6748	84.00	0.4500
46.22	0.6607	85.98	0.4419
48.16	0.6475	88.01	0.4359
49.79	0.6370	89.72	0.4206

*Z. Physik. Chem., A167, 42 (1933).

2-184 PHYSICAL AND CHEMICAL DATA

TABLE 2-168 Potassium Chloride

Mole % KCl	Specific heat, cal/(g·°C)			
	6 °C	20 °C	33 °C	40 °C
0.99	0.945	0.947	0.947	0.947
3.85	0.828	0.831	0.835	0.837
5.66	0.77	0.775	0.778	0.775
7.41		0.727		

TABLE 2-169 Potassium Hydroxide (at 19 °C)

Mole % KOH	0	0.497	1.64	4.76	9.09
Cal/(g·°C)	1.0	0.975	0.93	0.814	0.75

TABLE 2-170 Normal Propyl Alcohol

Mole % C ₃ H ₇ OH	Specific heat, cal/(g·°C)		
	5 °C	20 °C	40 °C
1.55	1.03	1.02	1.01
5.03	1.07	1.06	1.03
11.4	1.035	1.032	0.99
23.1	0.877	0.90	0.91
41.2	0.75	0.78	0.815
73.0	0.612	0.645	0.708
100.0	0.534	0.57	0.621

TABLE 2-171 Sodium Carbonate*

% Na ₂ CO ₃ by weight	Temperature, °C			
	17.6	30.0	76.6	98.0
0.000	0.9992	0.9986	1.0098	1.0084
1.498	0.9807			
2.000		0.9786		
2.901	0.9597			
4.000		0.9594		
5.000	0.9428		0.9761	
6.000		0.9392		
8.000	0.9183			
10.000	0.9086		0.9452	
13.790	0.8924			
13.840		0.8881		
20.000		0.8631	0.8936	
25.000			0.8615	0.8911

*J. Chem. Soc. 3062–3079 (1931).

TABLE 2-172 Sodium Chloride

Mole % NaCl	Specific heat, cal/(g·°C)			
	6 °C	20 °C	33 °C	57 °C
0.249		0.99		
0.99	0.96	0.97	0.97	
2.44	0.91	0.915	0.915	0.923
9.09	0.805	0.81	0.81	0.82

TABLE 2-173 Sodium Hydroxide (at 20 °C)

Mole % NaOH	0	0.5	1.0	9.09	16.7	28.6	37.5
Cal/(g·°C)	1.0	0.985	0.97	0.835	0.80	0.784	0.782

TABLE 2-174 Sulfuric Acid*

%H ₂ SO ₄	C _p at 20 °C, cal/(g·°C)	%H ₂ SO ₄	C _p at 20 °C, cal/(g·°C)
0.34	0.9968	35.25	0.7238
0.68	0.9937	37.69	.7023
1.34	0.9877	40.49	.6770
2.65	0.9762	43.75	.6476
3.50	0.9688	47.57	.6153
5.16	0.9549	52.13	.5801
9.82	0.9177	57.65	.5420
15.36	0.8767	64.47	.5012
21.40	0.8339	73.13	.4628
22.27	0.8275	77.91	.4518
23.22	0.8205	81.33	.4481
24.25	0.8127	82.49	.4467
25.39	0.8041	84.48	.4408
26.63	0.7945	85.48	.4346
28.00	0.7837	89.36	.4016
29.52	0.7717	91.81	.3787
30.34	0.7647	94.82	.3554
31.20	0.7579	97.44	.3404
33.11	0.7422	100.00	.3352

*Vinal and Craig, *Bur. Standards J. Research*, **24**, 475 (1940).

TABLE 2-175 Zinc Sulfate

Composition	Temperature	Specific heat, cal/(g·°C)
ZnSO ₄ + 50H ₂ O	20 to 52 °C	0.842
ZnSO ₄ + 200H ₂ O	20 to 52 °C	0.952

SPECIFIC HEATS OF MISCELLANEOUS MATERIALS

TABLE 2-176 Specific Heats of Miscellaneous Liquids and Solids

Material	Specific heat, cal/(g·°C)
Alumina	0.2 (100 °C); 0.274 (1500 °C)
Alundum	0.186 (100 °C)
Asbestos	0.25
Asphalt	0.22
Bakelite	0.3 to 0.4
Brickwork	About 0.2
Carbon	0.168 (26 to 76 °C) 0.314 (40 to 892 °C) 0.387 (56 to 1450 °C)
(gas retort) (see under Graphite)	0.204
Cellulose	0.32
Cement, Portland Clinker	0.186
Charcoal (wood)	0.242
Chrome brick	0.17
Clay	0.224
Coal	0.26 to 0.37
tar oils	0.34 (15 to 90 °C)
Coal tars	0.35 (40 °C); 0.45 (200 °C)
Coke	0.265 (21 to 400 °C) 0.359 (21 to 800 °C) 0.403 (21 to 1300 °C)
Concrete	0.156 (70 to 312 °F); 0.219 (72 to 1472 °F)
Cryolite	0.253 (16 to 55 °C)
Diamond	0.147
Fireclay brick	0.198 (100 °C); 0.298 (1500 °C)
Fluorspar	0.21 (30 °C)
Gasoline	0.53
Glass (crown) (flint) (pyrex) (silicate)	0.16 to 0.20 0.117 0.20 0.188 to 0.204 (0 to 100 °C) 0.24 to 0.26 (0 to 700 °C)
wool	0.157
Granite	0.20 (20 to 100 °C)
Graphite	0.165 (26 to 76 °C); 0.390 (56 to 1450 °C)
Gypsum	0.259 (16 to 46 °C)
Kerosene	0.47
Limestone	0.217
Litharge	0.055
Magnesia	0.234 (100 °C); 0.188 (1500 °C)
Magnesite brick	0.222 (100 °C); 0.195 (1500 °C)
Marble	0.21 (18 °C)
Porcelain, fired Berlin	0.189 (60 °C)
Porcelain, green Berlin	0.185 (60 °C)
Porcelain, fired earthenware	0.186 (60 °C)
Porcelain, green earthenware	0.181 (60 °C)

TABLE 2-176 Specific Heats of Miscellaneous Liquids and Solids (Concluded)

Material	Specific heat, cal/(g·°C)
Pyrex glass	0.20
Pyrites (copper)	0.131 (30 °C)
Pyrites (iron)	0.136 (30 °C)
Pyroxylin plastics	0.34 to 0.38
Quartz	0.17 (0 °C); 0.28 (350 °C)
Rubber (vulcanized)	0.415
Sand	0.191
Silica	0.316
Silica brick	0.202 (100 °C); 0.195 (1500 °C)
Silicon carbide brick	0.202 (100 °C)
Silk	0.33
Steel	0.12
Stone	about 0.2
Stoneware (common)	0.188 (60 °C)
Turpentine	0.42 (18 °C)
Wood (Oak)	0.570
Woods, miscellaneous	0.45 to 0.65
Wool	0.325
Zirconium oxide	0.11 (100 °C); 0.179 (1500 °C)

TABLE 2-177 Oils (Animal, Vegetable, Mineral Oils)

$$C_p[\text{cal/(g·°C)}] = A/\sqrt{d^{15}} + B(t - 15)$$

where d = density, g/cm³.

°F = 1.8 °C + 32; to convert calories per gram-degree Celsius to British thermal units per pound-degree Fahrenheit, multiply by 1.0; to convert grams per cubic centimeter to pounds per cubic foot, multiply by 62.43.

Oils	A	B
Castor	0.500	0.0007
Citron	(0.438 at 54 °C)	
Fatty drying	0.440	0.0007
nondrying	0.450	0.0007
semidrying	0.445	0.0007
oils (except castor)	0.450	0.0007
Naphthene base	0.405	0.0009
Olive	(0.47 at 7 °C)	
Paraffin base	0.425	0.0009
Petroleum oils	0.415	0.0009

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°F = 1.8 °C + 32; to convert kilocalories per gram-mole to British thermal units per pound-mole, multiply by 1.799×10^{-3} .

TABLE 2-178 Heats and Free Energies of Formation of Inorganic Compounds

The values given in the following table for the heats and free energies of formation of inorganic compounds are derived from (a) Bichowsky and Rossini, "Thermochemistry of the Chemical Substances," Reinhold, New York, 1936; (b) Latimer, "Oxidation States of the Elements and Their Potentials in Aqueous Solution," Prentice-Hall, New York, 1938; (c) the tables of the American Petroleum Institute Research Project 44 at the National Bureau of Standards; and (d) the tables of Selected Values of Chemical Thermodynamic Properties of the National Bureau of Standards. The reader is referred to the preceding books and tables for additional details as to methods of calculation, standard states, and so on.

Compound	State†	Heat of formation‡ § ΔH (formation) at 25 °C, kcal/mol	Free energy of formation ¶ ΔF (formation) at 25 °C, kcal/mol	Compound	State†	Heat of formation‡ § ΔH (formation) at 25 °C, kcal/mol	Free energy of formation ¶ ΔF (formation) at 25 °C, kcal/mol
Aluminum				Barium (Cont.)			
Al	c	0.00	0.00	BaF ₂	c	-287.9	
AlBr ₃	c	-123.4			aq, 1600	-284.6	-265.3
	aq	-209.5	-189.2	BaH ₂	c	-40.8	-31.5
Al ₂ C ₃	c	-30.8	-29.0	Ba(HCO ₃) ₂	aq	-459	-414.4
AlCl ₃	c	-163.8		BaI ₂	c	-144.6	
	aq, 600	-243.9	-209.5		aq, 400	-155.17	-158.52
AlF ₃	c	-329		Ba(IO ₃) ₂	c	-264.5	
	aq	-360.8	-312.6		aq	-237.50	-198.35
AlI ₃	c	-72.8		BaMoO ₄	c	-370	
	aq	-163.4	-152.5	Ba ₃ N ₂	c	-90.7	
AlN	c	-57.7	-50.4	Ba(NO ₂) ₂	c	-184.5	
Al(NH ₄)(SO ₄) ₂	c	-561.19	-486.17		aq	-179.05	-150.75
Al(NH ₄)(SO ₄) ₂ ·12H ₂ O	c	-1419.36	-1179.26	Ba(NO ₃) ₂	c	-236.99	-189.94
Al(NO ₃) ₃ ·6H ₂ O	c	-680.89	-526.32		aq, 600	-227.74	
Al(NO ₃) ₃ ·9H ₂ O	c	-897.59		BaO	c	-133.0	
Al ₂ O ₃	c, corundum	-399.09	-376.87	Ba(OH) ₂	c	-225.9	
Al(OH) ₃	c	-304.8	-272.9		aq, 400	-237.76	-209.02
Al ₂ O ₃ ·SiO ₂	c, sillimanite	-648.7		BaO·SiO ₂	c	-363	
Al ₂ O ₃ ·SiO ₂	c, disthene	-642.4		Ba ₃ (PO ₄) ₂	c	-992	
Al ₂ O ₃ ·SiO ₂	c, andalusite	-642.0		BaPtCl ₆	c	-284.9	
3Al ₂ O ₃ ·2SiO ₂	c, mullite	-1874		BaS	c	-111.2	
Al ₂ S ₃	c	-121.6		BaSO ₃	c	-282.5	
Al ₂ (SO ₄) ₃	c	-820.99	-739.53	BaSO ₄	c	-340.2	-313.4
	aq	-893.9	-759.3	BaWO ₄	c	-402	
Al ₂ (SO ₄) ₃ ·6H ₂ O	c	-1268.15	-1103.39	Beryllium			
Al ₂ (SO ₄) ₃ ·18H ₂ O	c	-2120		Be	c	0.00	0.00
Antimony				BeBr ₂	c	-79.4	
Sb	c	0.00	0.00		aq	-142	-127.9
SbBr ₃	c	-59.9		BeCl ₂	c	-112.6	
SbCl ₃	c	-91.3	-77.8		aq	-163.9	-141.4
SbCl ₅	l	-104.8		BeI ₂	c	-39.4	
SbF ₃	c	-216.6			aq	-112	-103.4
SbI ₃	c	-22.8		Be ₃ N ₂	c	-134.5	-122.4
Sb ₂ O ₃	c, I, orthorhombic	-165.4	-146.0	BeO	c	-145.3	-138.3
	c, II, octahedral	-166.6		Be(OH) ₂	c	-215.6	
Sb ₂ O ₄	c	-213.0	-186.6	BeS	c	-56.1	
Sb ₂ O ₅	c	-230.0	-196.1	BeSO ₄	c	-281	
Sb ₂ S ₃	c, black	-38.2	-36.9		aq		-254.8
Arsenic				Bismuth			
As	c	0.00	0.00	Bi	c	0.00	0.00
AsBr ₃	c	-45.9		BiCl ₃	c	-90.5	-76.4
AsCl ₃	l	-80.2	-70.5		aq	-101.6	
AsF ₃	l	-223.76	-212.27	BiI ₃	c	-24	
AsH ₃	g	43.6	37.7		aq	-27	
AsI ₃	c	-13.6		BiO	c	-49.5	-43.2
As ₂ O ₃	c	-154.1	-134.8	Bi ₂ O ₃	c	-137.1	-117.9
As ₂ O ₅	c	-217.9	-183.9	Bi(OH) ₃	c	-171.1	
As ₂ S ₃	c	-20	-20	Bi ₂ S ₃	c	-43.9	-39.1
	amorphous	-34.76		Bi ₂ (SO ₄) ₃	c	-607.1	
Barium				Boron			
Ba	c	0.00	0.00	B	c	0.00	0.00
BaBr ₂	c	-180.38		BBr ₃	l	-52.7	
	aq, 400	-185.67	-183.0		g	-44.6	-50.9
BaCl ₂	c	-205.25		BCl ₃	g	-94.5	-90.8
	aq, 300	-207.92	-196.5	BF ₃	g	-265.2	-261.0
Ba(ClO ₃) ₂	c	-176.6		B ₂ H ₆	g	7.5	19.9
	aq, 1600	-170.0	-134.4	BN	c	-32.1	-27.2
Ba(ClO ₄) ₂	c	-210.2		B ₂ O ₃	c	-302.0	-282.9
	aq, 800		-155.3		gls	-297.6	-280.3
Ba(CN) ₂	c	-48		B(OH) ₃	c	-260.0	-229.4
Ba(CNO) ₂	c	-212.1		B ₂ S ₃	c	-56.6	
	aq		-180.7	Bromine			
BaCN ₂	c	-63.6		Br ₂	l	0.00	0.00
BaCO ₃	c, witherite	-284.2	-271.4		g	7.47	0.931
BaCrO ₄	c	-342.2		BrCl	g	3.06	-0.63

*For footnotes see end of table.

TABLE 2-178 Heats and Free Energies of Formation of Inorganic Compounds (Continued)

Compound	State†	Heat of formation‡§ ΔH (formation) at 25 °C, kcal/mol	Free energy of formation ¶ ΔF (formation) at 25 °C, kcal/mol	Compound	State†	Heat of formation‡§ ΔH (formation) at 25 °C, kcal/mol	Free energy of formation ¶ ΔF (formation) at 25 °C, kcal/mol
Cadmium				Cesium (Cont.)			
Cd	c	0.00	0.00	Cs ₂ CO ₃	c	-271.88	
CdBr ₂	c	-75.8	-70.7	CsF	c	-131.67	
	aq, 400	-76.6	-67.6		aq, 400	-140.48	-135.98
CdCl ₂	c	-92.149	-81.889	CsH	c	-12	-7.30
	aq, 400	-96.44	-81.2	CsHCO ₃	c	-230.6	
Cd(CN) ₂	c	36.2			aq, 2000	-226.6	-210.56
CdCO ₃	c	-178.2	-163.2	CsI	c	-83.91	
CdI ₂	c	-48.40			aq, 400	-75.74	-82.61
	aq, 400	-47.46	-43.22	CsNH ₂	c	-28.2	
Cd ₃ N ₂	c	39.8		CsNO ₃	c	-121.14	
Cd(NO ₃) ₂	aq, 400	-115.67	-71.05		aq, 400	-111.54	-96.53
CdO	c	-62.35	-55.28	Cs ₂ O	c	-82.1	
Cd(OH) ₂	c	-135.0	-113.7	CsOH	c	-100.2	
CdS	c	-34.5	-33.6		aq, 200	-117.0	-107.87
CdSO ₄	c	-222.23		Cs ₂ S	c	-87	
	aq, 400	-232.635	-194.65	Cs ₂ SO ₄	c	-344.86	
Calcium					aq	-340.12	-316.66
Ca	c	0.00	0.00	Chlorine			
CaBr ₂	c	-162.20		Cl ₂	g	0.00	0.00
	aq, 400	-187.19	-181.86	ClF	g	-25.7	
CaC ₂	c	-14.8	-16.0	ClO	g	33	
CaCl ₂	c	-190.6	-179.8	ClO ₂	g	24.7	29.5
	aq	-209.15	-195.36	ClO ₃	g	37	
CaCN ₂	c	-85		Cl ₂ O	g	18.20	22.40
Ca(CN) ₂	c	-43.3		Cl ₂ O ₇	g	63	
	aq		-54.0	Chromium			
CaCO ₃	c, calcite	-289.5	-270.8	Cr	c	0.00	0.00
	c, aragonite	-289.54	-270.57	CrBr ₃	aq		-122.7
CaCO ₃ ·MgCO ₃	c	-558.8		Cr ₃ C ₂	c	-21.008	-21.20
CaC ₂ O ₄	c	-332.2		Cr ₄ C	c	-16.378	-16.74
Ca(C ₂ H ₃ O ₂) ₂	c	-356.3		CrCl ₂	c	-103.1	-93.8
	aq	-364.1	-311.3		aq		-102.1
CaF ₂	c	-290.2		CrF ₂	c	-152	
	aq	-286.5	-264.1	CrF ₃	c	-231	
CaH ₂	c	-46	-35.7	CrI ₂	c	-63.7	
CaI ₂	c	-128.49			aq		-64.1
	aq, 400	-156.63	-157.37	CrO ₃	c	-139.3	
Ca ₃ N ₂	c	-103.2	-88.2	Cr ₂ O ₃	c	-268.8	-249.3
Ca(NO ₃) ₂	c	-224.05	-177.38	Cr ₂ (SO ₄) ₃	aq		-626.3
	aq, 400	-228.29		Cobalt			
Ca(NO ₃) ₂ ·2H ₂ O	c	-367.95	-293.57	Co	c	0.00	0.00
Ca(NO ₃) ₂ ·3H ₂ O	c	-439.05	-351.58	CoBr ₂	c	-55.0	
Ca(NO ₃) ₂ ·4H ₂ O	c	-509.43	-409.32		aq	-73.61	-61.96
CaO	c	-151.7	-144.3	Co ₃ C	c	9.49	7.08
Ca(OH) ₂	c	-235.58	-213.9	CoCl ₂	c	-76.9	-66.6
	aq, 800	-239.2	-207.9		aq, 400	-95.58	-75.46
CaO·SiO ₂	c, II, wollastonite	-377.9	-357.5	CoCO ₃	c	-172.39	-155.36
	c, I, pseudo-wollastonite	-376.6	-356.6	CoF ₂	aq	-172.98	-144.2
				CoI ₂	c	-24.2	
CaS	c	-114.3	-113.1		aq	-43.15	-37.4
CaSO ₄	c, insoluble form	-338.73	-311.9	Co(NO ₃) ₂	c	-102.8	
	c, soluble form α	-336.58	-309.8		aq	-114.9	-65.3
	c, soluble form β	-335.52	-308.8	CoO	c	-57.5	
CaSO ₄ · $\frac{1}{2}$ H ₂ O	c	-376.13		Co ₃ O ₄	c	-196.5	
CaSO ₄ ·2H ₂ O	c	-479.33	-425.47	Co(OH) ₂	c	-131.5	-108.9
CaWO ₄	c	-387		Co(OH) ₃	c	-177.0	-142.0
Carbon				CoS	c	-22.3	-19.8
C	c, graphite	0.00	0.00	Co ₂ S ₃	c	-40.0	
	c, diamond	0.453	0.685	CoSO ₄	c	-216.6	
CO	g	-26.416	-32.808		aq, 400		-188.9
CO ₂	g	-94.052	-94.260	Columbium			
Cerium				Cb	c		0.00
Ce	c	0.00	0.00	Cb ₂ O ₅	c	-462.96	
CeN	c	-78.2	-70.8	Copper			
Cesium				Cu	c	0.00	0.00
Cs	c	0.00	0.00	CuBr	c	-26.7	-23.8
CsBr	c	-97.64		CuBr ₂	c	-34.0	
	aq, 500	-91.39	-94.86		aq	-42.4	-33.25
CsCl	c	-106.31		CuCl	c	-31.4	-24.13
	aq, 400	-102.01	-101.61	CuCl ₂	c	-48.83	
					aq, 400	-64.7	

TABLE 2-178 Heats and Free Energies of Formation of Inorganic Compounds (Continued)

Compound	State†	Heat of formation‡§ ΔH (formation) at 25 °C, kcal/mol	Free energy of formation ¶ ΔF (formation) at 25 °C, kcal/mol	Compound	State†	Heat of formation‡§ ΔH (formation) at 25 °C, kcal/mol	Free energy of formation ¶ ΔF (formation) at 25 °C, kcal/mol
Copper (Cont.)				Hydrogen (Cont.)			
CuClO ₄	aq	-28.3	1.34	H ₂ CO ₃	aq	-167.19	-149.0
Cu(ClO ₃) ₂	aq, 400		15.4	HF	g	-64.2	-64.7
Cu(ClO ₄) ₂	aq		-5.5		aq, 200	-75.75	
CuI	c	-17.8	-16.66	HI	g	6.27	0.365
CuI ₂	c	-4.8			aq, 400	-13.47	-12.35
	aq	-11.9	-8.76	HIO	aq	-38	-23.33
Cu ₃ N	c	17.78		HIO ₃	c	-56.77	
Cu(NO ₃) ₂	c	-73.1			aq	-54.8	-32.25
	aq, 200	-83.6	-36.6	HN ₃	g	70.3	78.50
CuO	c	-38.5	-31.9	HNO ₃	g	-31.99	-17.57
Cu ₂ O	c	-43.00	-38.13		l	-41.35	-19.05
Cu(OH) ₂	c	-108.9	-85.5		aq, 400	-49.210	
CuS	c	-11.6	-11.69	HNO ₃ ·H ₂ O	l	-112.91	-78.36
Cu ₂ S	c	-18.97	-20.56	HNO ₃ ·3H ₂ O	l	-252.15	-193.70
CuSO ₄	c	-184.7	-158.3	H ₂ O	g	-57.7979	-54.6351
	aq, 800	-200.78	-160.19		l	-68.3174	-56.6899
Cu ₂ SO ₄	c	-179.6		H ₂ O ₂	l	-45.16	-28.23
	aq		-152.0		aq, 200	-45.80	-31.47
Erbium				H ₃ PO ₂	c	-145.5	
Er	c	0.00	0.00		aq	-145.6	-120.0
Er(OH) ₃	c	-326.8		H ₃ PO ₃	c	-232.2	
Fluorine					aq	-232.2	-204.0
F ₂	g	0.00	0.00	H ₃ PO ₄	c	-306.2	
F ₂ O	g	5.5	9.7		aq, 400	-309.32	-270.0
Gallium				H ₂ S	g	-4.77	-7.85
Ga	c	0.00	0.00		aq, 2000	-9.38	
GaBr ₃	c	-92.4		H ₂ S ₂	l	-3.6	
GaCl ₃	c	-125.4		H ₂ SO ₃	aq, 200	-146.88	-128.54
GaN	c	-26.2		H ₂ SO ₄	l	-193.69	
Ga ₂ O	c	-84.3			aq, 400	-212.03	
Ga ₂ O ₃	c	-259.9		H ₂ Se	g	20.5	17.0
Germanium					aq	18.1	18.4
Ge	c	0.00	0.00	H ₂ SeO ₃	c	-126.5	
Ge ₃ N ₄	c	-15.7			aq	-122.4	-101.36
GeO ₂	c	-128.6		H ₂ SeO ₄	c	-130.23	
Gold					aq, 400	-143.4	
Au	c	0.00	0.00	H ₂ SiO ₃	c	-267.8	-247.9
AuBr	c	-3.4		H ₄ SiO ₄	c	-340.6	
AuBr ₃	c	-14.5		H ₂ Te	g	36.9	33.1
	aq	-11.0	24.47	H ₂ TeO ₃	c	-145.0	-115.7
AuCl	c	-8.3			aq	-145.0	
AuCl ₃	c	-28.3		H ₂ TeO ₄	aq	-165.6	
	aq	-32.96	4.21	Indium			
AuI	c	0.2	-0.76	In	c	0.00	0.00
Au ₂ O ₃	c	11.0	18.71	InBr ₃	c	-97.2	
Au(OH) ₃	c	-100.6			aq	-112.9	-97.2
Hafnium				InCl ₃	c	-128.5	
Hf	c	0.00	0.00		aq	-145.6	-117.5
HfO ₂	c	-271.1	-258.2	InI ₃	c	-56.5	
Hydrogen					aq	-67.2	-60.5
H ₃ AsO ₃	aq	-175.6	-153.04	InN	c	-4.8	
H ₃ AsO ₄	c	-214.9		In ₂ O ₃	c	-222.47	
	aq	-214.8	-183.93	Iodine			
HBr	g	-8.66	-12.72	I ₂	c	0.00	0.00
	aq, 400	-28.80	-24.58		g	14.88	4.63
HBrO	aq	-25.4	-19.90	IBr	g	10.05	1.24
HBrO ₃	aq	-11.51	5.00	ICl	g	4.20	-1.32
HCl	g	-22.063	-22.778	ICl ₃	c	-21.8	-6.05
	aq, 400	-39.85	-31.330	I ₂ O ₅	c	-42.5	
HCN	g	31.1	27.94	Iridium			
	aq, 100	24.2	26.55	Ir	c	0.00	0.00
HClO	aq, 400	-28.18	-19.11	IrCl	c	-20.5	-16.9
HClO ₃	aq	-23.4	-0.25	IrCl ₂	c	-40.6	-32.0
HClO ₄	aq, 660	-31.4	-10.70	IrCl ₃	c	-60.5	-46.5
HC ₂ H ₃ O ₂	l	-116.2	-93.56	IrF ₆	l	-130	
	aq, 400	-116.74	-96.8	IrO ₂	c	-40.14	
H ₃ C ₂ O ₄	c	-196.7		Iron			
	aq, 300	-194.6	-165.64	Fe	c, α	0.00	0.00
HCOOH	l	-97.8	-82.7	FeBr ₂	c	-57.15	
	aq, 200	-98.0	-85.1		aq, 540	-78.7	-69.47

TABLE 2-178 Heats and Free Energies of Formation of Inorganic Compounds (Continued)

Compound	State†	Heat of formation‡§ ΔH (formation) at 25 °C, kcal/mol	Free energy of formation ¶ ΔF (formation) at 25 °C, kcal/mol	Compound	State†	Heat of formation‡§ ΔH (formation) at 25 °C, kcal/mol	Free energy of formation ¶ ΔF (formation) at 25 °C, kcal/mol
Iron (Cont.)				Lithium (Cont.)			
FeBr ₃	aq	-95.5	-76.26	LiC ₂ H ₃ O ₂	aq	-183.9	-160.00
Fe ₃ C	c	5.69	4.24	Li ₂ CO ₃	c	-289.7	-269.8
Fe(CO) ₅	l	-187.6			aq, 1900	-293.1	-267.58
FeCO ₃	c, siderite	-172.4	-154.8	LiCl	c	-97.63	
FeCl ₂	c	-81.9	-72.6		aq, 278	-106.45	-102.03
	aq	-100.0	-83.0	LiClO ₃	aq	-87.5	-70.95
FeCl ₃	c	-96.4		LiClO ₄	aq	-106.3	-81.4
	aq, 2000	-128.5	-96.5	LiF	c	-145.57	
FeF ₂	aq, 1200	-177.2	-151.7		aq, 400	-144.85	-136.40
FeI ₂	c	-24.2		LiH	c	-22.9	
	aq	-47.7	-45	LiHCO ₃	aq, 2000	-231.1	-210.98
FeI ₃	aq	-49.7	-39.5	LiI	c	-65.07	
Fe ₄ N	c	-2.55	0.862		aq, 400	-80.09	-83.03
Fe(NO ₃) ₂	aq	-118.9	-72.8	LiIO ₃	aq	-121.3	-102.95
Fe(NO ₃) ₃	aq, 800	-156.5	-81.3	Li ₃ N	c	-47.45	-37.33
FeO	c	-64.62	-59.38	LiNO ₃	c	-115.350	
Fe ₂ O ₃	c	-198.5	-179.1		aq, 400	-115.88	-96.95
Fe ₃ O ₄	c	-266.9	-242.3	Li ₂ O	c	-142.3	
Fe(OH) ₂	c	-135.9	-115.7	Li ₂ O ₂	c	-151.9	-138.0
Fe(OH) ₃	c	-197.3	-166.3		aq	-159	
FeO·SiO ₂	c	-273.5		LiOH	c	-116.58	-106.44
Fe ₂ P	c	-13			aq, 400	-121.47	-108.29
FeSi	c	-19.0		LiOH·H ₂ O	c	-188.92	
FeS	c	-22.64	-23.23	Li ₂ O·SiO ₂	gls	-374	
FeS ₂	c, pyrites	-38.62	-35.93	Li ₃ Se	c	-84.9	
	c, marcasite	-33.0			aq	-95.5	-105.64
FeSO ₄	c	-221.3	-195.5	Li ₂ SO ₄	c	-340.23	-314.66
	aq, 400	-236.2	-196.4		aq, 400	-347.02	
Fe ₂ (SO ₄) ₃	aq, 400	-653.3	-533.4	Li ₃ SO ₄ ·H ₂ O	c	-411.57	-375.07
FeTiO ₃	c, ilmenite	-295.51	-277.06	Magnesium			
Lanthanum				Mg	c	0.00	0.00
La	c	0.00	0.00	Mg(AsO ₄) ₂	c	-731.3	
LaCl ₃	c	-253.1			aq	-749	-630.14
	aq	-284.7		MgBr ₂	c	-123.9	
La ₃ H ₈	c	-160			aq, 400	-167.33	-156.94
LaN	c	-72.0	-64.6	Mg(CN) ₂	aq	-39.7	-29.08
La ₂ O ₃	c	-539		MgCN ₂	c	-61	
LaS ₂	c	-148.3		Mg(C ₂ H ₃ O ₂) ₂	aq	-344.6	-286.38
La ₂ S ₃	c	-351.4		MgCO ₃	c	-261.7	-241.7
La ₂ (SO ₄) ₃	aq	-972		MgCl ₂	c	-153.220	-143.77
Lead					aq, 400	-189.76	
Pb	c	0.00	0.00	MgCl ₂ ·H ₂ O	c	-230.970	-205.93
PbBr ₂	c	-66.24	-62.06	MgCl ₂ ·2H ₂ O	c	-305.810	-267.20
	aq	-56.4	-54.97	MgCl ₂ ·4H ₂ O	c	-453.820	-387.98
PbCO ₃	c, cerussite	-167.6	-150.0	MgCl ₂ ·6H ₂ O	c	-597.240	-505.45
Pb(C ₂ H ₃ O ₂) ₂	c	-232.6		MgF ₂	c	-263.8	
	aq, 400	-234.2	-184.40	MgI ₂	c	-86.8	
PbC ₂ O ₄	c	-205.3			aq, 400	-136.79	-132.45
PbCl ₂	c	-85.68	-75.04	MgMoO ₄	c	-329.9	
	aq	-82.5	-68.47	Mg ₃ N ₂	c	-115.2	-100.8
PbF ₂	c	-159.5	-148.1	Mg(NO ₃) ₂	c	-188.770	-140.66
PbI ₂	c	-41.77	-41.47		aq, 400	-209.927	-160.28
Pb(NO ₃) ₂	c	-106.88		Mg(NO ₃) ₂ ·2H ₂ O	c	-336.625	
	aq, 400	-99.46	-58.3	Mg(NO ₃) ₂ ·6H ₂ O	c	-624.48	-496.03
PbO	c, red	-51.72	-45.53	MgO	c	-143.84	-136.17
	c, yellow	-50.86	-43.88	MgO·SiO ₂	c	-347.5	-326.7
PbO ₂	c	-65.0	-52.0	Mg(OH) ₂	c, ppt.	-221.90	-200.17
Pb ₃ O ₄	c	-172.4	-142.2		c, brucite	-223.9	-193.3
Pb(OH) ₂	c	-123.0	-102.2	MgS	c	-84.2	
PbS	c	-22.38	-21.98		aq	-108	
PbSO ₄	c	-218.5	-192.9	MgSO ₄	c	-304.94	-277.7
Lithium					aq, 400	-325.4	-283.88
Li	c	0.00	0.00	MgTe	c	-25	
LiBr	c	-83.75		MgWO ₄	c	-345.2	
	aq, 400	-95.40	-95.28	Manganese			
LiBrO ₃	aq	-77.9	-65.70	Mn	c, α	0.00	0.00
Li ₂ C ₂	c	-13.0		MnBr ₂	c	-91	
LiCN	aq	-31.4	-31.35		aq	-106	-97.8
LiCNO	aq	-101.2	-94.12	Mn ₃ C	c	1.1	1.26

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TABLE 2-178 Heats and Free Energies of Formation of Inorganic Compounds (Continued)

Compound	State†	Heat of formation‡§ ΔH (formation) at 25 °C, kcal/mol	Free energy of formation ¶ ΔF (formation) at 25 °C, kcal/mol	Compound	State†	Heat of formation‡§ ΔH (formation) at 25 °C, kcal/mol	Free energy of formation ¶ ΔF (formation) at 25 °C, kcal/mol
Manganese (Cont.)				Nickel (Cont.)			
Mn(C ₂ H ₃ O ₂) ₂	c	-270.3		NiF ₂	aq, 400	-94.34	-74.19
	aq	-282.7	-227.2		c	-157.5	
MnCO ₃	c	-211	-192.5	NiI ₂	aq	-171.6	-142.9
MnC ₂ O ₄	c	-240.9			c	-22.4	
MnCl ₂	c	-112.0	-102.2	Ni(NO ₃) ₂	aq	-42.0	-36.2
	aq, 400	-128.9			c	-101.5	
MnF ₂	aq, 1200	-206.1	-180.0	NiO	aq, 200	-113.5	-64.0
MnI ₂	c	-49.8		Ni(OH) ₂	c	-58.4	-51.7
	aq	-76.2	-73.3	Ni(OH) ₃	c	-129.8	-105.6
Mn ₃ N ₂	c	-57.77	-46.49	NiS	c	-163.2	
Mn(NO ₃) ₂	c	-134.9			c	-20.4	
	aq, 400	-148.0	-101.1	NiSO ₄	c	-216	
Mn(NO ₃) ₂ ·6H ₂ O	c	-557.07	-441.2		aq, 200	-231.3	-187.6
MnO	c	-92.04	-86.77	Nitrogen			
MnO ₂	c	-124.58	-111.49	N ₂	g	0.00	0.00
Mn ₂ O ₃	c	-229.5	-209.9	NF ₃	g	-27	
Mn ₃ O ₄	c	-331.65	-306.22	NH ₃	g	-10.96	-3.903
MnO·SiO ₂	c	-301.3	-282.1		aq, 200	-19.27	
Mn(OH) ₂	c	-163.4	-143.1	NH ₄ Br	c	-64.57	
Mn(OH) ₃	c	-221	-190		aq	-60.27	-43.54
Mn ₃ (PO ₄) ₂	c	-736		NH ₄ C ₂ H ₃ O ₂	c	-148.1	
MnSe	c	-26.3	-27.5		aq, 400	-148.58	-108.26
MnS	c, green	-47.0	-48.0	NH ₄ CN	c	-0.7	
MnSO ₄	c	-254.18	-228.41		aq	3.6	20.4
	aq, 400	-265.2		NH ₄ CNS	c	-17.8	
Mn ₂ (SO ₄) ₃	c	-635			aq	-12.3	4.4
	aq	-657		(NH ₄) ₂ CO ₃	aq	-223.4	-164.1
Mercury				(NH ₄) ₂ C ₂ O ₄	c	-266.3	
Hg	l	0.00	0.00		aq	-260.6	-196.2
HgBr	g	23	18	NH ₄ Cl	c	-75.23	-48.59
HgBr ₂	c	-40.68	-38.8		aq, 400	-71.20	
	aq	-38.4	-9.74	NH ₄ ClO ₄	c	-69.4	
Hg(C ₂ H ₃ O ₂) ₂	c	-196.3			aq	-63.2	-21.1
	aq	-192.5	-139.2	(NH ₄) ₂ CrO ₄	c	-276.9	
HgCl ₂	c	-53.4	-42.2		aq	-271.3	-209.3
	aq	-50.3	-23.25	NH ₄ F	c	-111.6	
HgCl	g	19	14		aq	-110.2	-84.7
Hg ₂ Cl ₂	c	-63.13		NH ₄ I	c	-48.43	
Hg(CN) ₂	c	62.8			aq	-44.97	-31.3
	aq, 1110	66.25		NH ₄ NO ₃	c	-87.40	
HgC ₂ O ₄	c	-159.3			aq, 500	-80.89	
HgH	g	57.1	52.25	NH ₄ OH	aq	-87.59	
HgI ₂	c, red	-25.3	-24.0	(NH ₄) ₂ S	aq, 400	-55.21	-14.50
HgI	g	33	23	(NH ₄) ₂ SO ₄	c	-281.74	-215.06
Hg ₂ I ₂	c	-28.88	-26.53		aq, 400	-279.33	-214.02
Hg(NO ₃) ₂	aq	-56.8	-13.09	N ₂ H ₄	l	12.06	
Hg ₂ (NO ₃) ₂	aq	-58.5	-15.65	N ₂ H ₄ ·H ₂ O	l	-57.96	
HgO	c, red	-21.6	-13.94	N ₂ H ₄ ·H ₂ SO ₄	c	-232.2	
	c, yellow ppt.	-20.8		N ₂ O	g	19.55	24.82
Hg ₂ O	c	-21.6	-12.80	NO	g	21.600	20.719
HgS	c, black	-10.7	-8.80	NO ₂	g	7.96	12.26
HgSO ₄	c	-166.6		N ₂ O ₄	g	2.23	23.41
Hg ₂ SO ₄	c	-177.34	-149.12	N ₂ O ₅	c	-10.0	
Molybdenum				NOBr	l	11.6	19.26
Mo	c	0.00	0.00	NOCl	g	12.8	16.1
Mo ₂ C	c	4.36	2.91	Osmium			
Mo ₂ N	c	-8.3		Os	c	0.00	0.00
MoO ₂	c	-130	-118.0	OsO ₄	c	-93.6	-70.9
MoO ₃	c	-180.39	-162.01		g	-80.1	-68.1
MoS ₂	c	-56.27	-54.19	Oxygen			
MoS ₃	c	-61.48	-57.38	O ₂	g	0.00	0.00
Nickel				O ₃	g	33.88	38.86
Ni	c	0.00	0.00	Palladium			
NiBr ₂	c	-53.4		Pd	c	0.00	0.00
	aq	-72.6	-60.7	PdO	c	-20.40	
Ni ₃ C	c	9.2	8.88	Phosphorus			
Ni(C ₂ H ₃ O ₂) ₂	aq	-249.6	-190.1	P	c, white ("yellow")	0.00	0.00
Ni(CN) ₂	aq	230.9	66.3		c, red ("violet")	-4.22	-1.80
NiCl ₂	c	-75.0		P	g	150.35	141.88

TABLE 2-178 Heats and Free Energies of Formation of Inorganic Compounds (Continued)

Compound	State†	Heat of formation‡§ ΔH (formation) at 25 °C, kcal/mol	Free energy of formation ¶ ΔF (formation) at 25 °C, kcal/mol	Compound	State†	Heat of formation‡§ ΔH (formation) at 25 °C, kcal/mol	Free energy of formation ¶ ΔF (formation) at 25 °C, kcal/mol
Phosphorus (Cont.)				Potassium (Cont.)			
P ₂	c	33.82	24.60	KNH ₂	c	-28.25	
P ₄	c	13.2	5.89	KNO ₂	aq	-86.0	-75.9
PBr ₃	c	-45		KNO ₃	c	-118.08	-94.29
PBr ₅	c	-60.6			aq, 400	-109.79	-93.68
PCl ₃	c	-70.0	-65.2	K ₂ O	c	-86.2	
	c	-76.8	-63.3	K ₂ O·Al ₂ O ₃ ·SiO ₂	c, leucite	-1379.6	
PCl ₅	c	-91.0	-73.2		gls	-1368.2	
PH ₃	c	2.21	-1.45	K ₂ O·Al ₂ O ₃ ·SiO ₂	c, adularia	-1784.5	
PI ₃	c	-10.9			c, microcline	-1784.5	
P ₂ O ₃	c	-360.0			gls	-1747	
POCl ₃	g	-138.4	-127.2	KOH	c	-102.02	
Platinum					aq, 400	-114.96	-105.0
Pt	c	0.00	0.00	K ₃ PO ₃	aq	-397.5	
PtBr ₄	c	-40.6		K ₃ PO ₄	aq	-478.7	-443.3
	aq	-50.7		KH ₂ PO ₄	c	-362.7	-326.1
PtCl ₂	c	-34		K ₂ PtCl ₄	c	-254.7	
PtCl ₄	c	-62.6			aq	-242.6	-226.5
	aq	-82.3		K ₂ PtCl ₆	c	-299.5	-263.6
PtI ₄	c	-18			aq, 9400	-286.1	
Pt(OH) ₂	c	-87.5	-67.9	K ₂ Se	c	-74.4	
PtS	c	-20.18	-18.55		aq	-83.4	-99.10
PtS ₂	c	-26.64	-24.28	K ₂ SeO ₄	aq	-267.1	-240.0
Potassium				K ₂ S	c	-121.5	
K	c	0.00	0.00		aq, 400	-110.75	-111.44
K ₃ AsO ₃	aq	-323.0		K ₂ SO ₃	c	-267.7	
K ₃ AsO ₄	aq	-390.3	-355.7		aq	-269.7	-251.3
KH ₂ AsO ₄	c	-271.2	-236.7	K ₂ SO ₄	c	-342.65	-314.62
KBr	c	-94.06	-90.8		aq, 400	-336.48	-310.96
KBrO ₃	c	-89.19	-92.0	K ₂ SO ₄ ·Al ₂ (SO ₄) ₃	c	-1178.38	-1068.48
	aq, 400	-81.58	-60.30	K ₂ SO ₄ ·Al ₂ (SO ₄) ₃ ·24H ₂ O	c	-2895.44	-2455.68
KC ₂ H ₃ O ₂	c	-71.68		K ₂ S ₂ O ₆	c	-418.62	
	aq, 400	-173.80		Rhenium			
KCl	c	-177.38	-156.73	Re	c	0.00	0.00
	aq, 400	-104.348	-97.76	ReF ₆	g	-274	
KClO ₃	c	-93.5	-69.30	Rhodium			
	aq, 400	-81.34		Rh	c	0.00	0.00
KClO ₄	c	-103.8	-72.86	RhO	c	-21.7	
	aq, 400	-101.14		Rh ₂ O	c	-22.7	
KCN	c	-28.1		Rh ₂ O ₃	c	-68.3	
KCNO	c	-25.3	-28.08	Rubidium			
	aq	-99.6		Rb	c	0.00	0.00
KCNS	c	-94.5	-90.85	RbBr	c	-95.82	
	aq, 400	-47.0			g	-45.0	-52.50
K ₂ CO ₃	c	-41.07	-44.08		aq, 500	-90.54	-93.38
	aq, 400	-274.01		RbCN	aq	-25.9	
K ₂ C ₂ O ₄	c	-280.90	-264.04	Rb ₂ CO ₃	c	-273.22	
	aq, 400	-319.9			aq, 220	-282.61	-263.78
K ₂ CrO ₄	c	-315.5	-293.1	RbCl	c	-105.06	-98.48
	aq, 400	-333.4			g	-53.6	-57.9
K ₂ Cr ₂ O ₇	c	-328.2	-306.3		aq, ∞	-101.06	-100.13
	aq, 400	-488.5		RbF	c	-133.23	
KF	c	-472.1	-440.9		aq, 400	-139.31	-134.5
	aq, 180	-134.50		RbHCO ₃	c	-230.01	
K ₃ Fe(CN) ₆	c	-138.36	-133.13		aq, 2000	-225.59	-209.07
	aq	-48.4		RbI	c	-81.04	
K ₄ Fe(CN) ₆	c	-34.5			g	-31.2	-40.5
	aq	-131.8		RbNH ₂	aq, 400	-74.57	-81.13
KH	c	-119.9		RbNO ₃	c	-27.74	
KHCO ₃	c	-10	-5.3		c	-119.22	
	aq, 2000	-229.8			aq, 400	-110.52	-95.05
KI	c	-224.85	-207.71	Rb ₂ O	c	-82.9	
	aq, 500	-78.88	-77.37	Rb ₂ O ₂	c	-107	
KIO ₃	c	-73.95	-79.76	RbOH	c	-101.3	
	aq, 400	-121.69	-101.87		aq, 200	-115.8	-106.39
KIO ₄	c	-115.18	-99.68	Ruthenium			
KMnO ₄	c	-98.1		Ru	c	0.00	0.00
	aq, 400	-192.9	-169.1	RuS ₂	c	-46.99	-44.11
K ₂ MoO ₄	c	-182.5	-168.0	Selenium			
	aq, 880	-364.2	-342.9	Se	c, I, hexagonal	0.00	0.00

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TABLE 2-178 Heats and Free Energies of Formation of Inorganic Compounds (Continued)

Compound	State†	Heat of formation‡§ ΔH (formation) at 25 °C, kcal/mol	Free energy of formation ¶ ΔF (formation) at 25 °C, kcal/mol	Compound	State†	Heat of formation‡§ ΔH (formation) at 25 °C, kcal/mol	Free energy of formation ¶ ΔF (formation) at 25 °C, kcal/mol
Selenium (<i>Cont.</i>)				Sodium (<i>Cont.</i>)			
	c, II, red, mono-clinic	0.2			aq, 476	-97.66	-73.29
Se ₂ Cl ₂	l	-22.06	-13.73	Na ₂ CrO ₄	c	-319.8	
SeF ₆	g	-246	-222		aq, 800	-323.0	-296.58
SeO ₂	c	-56.33		Na ₂ Cr ₂ O ₇	aq, 1200	-465.9	-431.18
Silicon				NaF	c	-135.94	-129.0
Si	c	0.00	0.00		aq, 400	-135.711	-128.29
SiBr ₄	l	-93.0		NaH	c	-14	-9.30
SiC	c	-28	-27.4	NaHCO ₃	c	-226.0	-202.66
SiCl ₄	l	-150.0	-133.9		aq	-222.1	-202.87
	g	-142.5	-133.0	NaI	c	-69.28	
SiF ₄	g	-370	-360		aq, ∞	-71.10	-74.92
SiH ₄	g	-14.8	-9.4	NaIO ₃	aq, 400	-112.300	-94.84
SiI ₄	c	-29.8		Na ₂ MoO ₄	c	-364	
Si ₃ N ₄	c	-179.25	-154.74		aq	-358.7	-333.18
SiO ₂	c, cristobalite, 1600° form	-202.62		NaNO ₂	c	-86.6	
	c, cristobalite, 1100° form	-202.46			aq	-83.1	-71.04
	c, quartz	-203.35	-190.4	NaNO ₃	c	-111.71	-87.62
	c, tridymite	-203.23			aq, 400	-106.880	-88.84
Silver				Na ₂ O	c	-99.45	-90.06
Ag	c	0.00	0.00	Na ₂ O ₂	c	-119.2	-105.0
AgBr	c	-23.90	-23.02	Na ₂ O·SiO ₂	c	-383.91	-361.49
Ag ₂ C ₂	c	84.5		Na ₂ O·Al ₂ O ₃ ·3SiO ₂	c, natrolite	-1180	
AgC ₂ H ₃ O ₂	c	-95.9		Na ₂ O·Al ₂ O ₃ ·4SiO ₂	c	-1366	
	aq	-91.7	-70.86	NaOH	c	-101.96	-90.60
AgCN	c	33.8	38.70		aq, 400	-112.193	-100.18
Ag ₂ CO ₃	c	-119.5	-103.0	Na ₃ PO ₃	aq, 1000	-389.1	
Ag ₂ C ₂ O ₄	c	-158.7		Na ₃ PO ₄	c	-457	
AgCl	c	-30.11	-25.98		aq, 400	-471.9	-428.74
AgF	c	-48.7		Na ₂ PtCl ₄	c	-237.2	-216.78
	aq, 400	-53.1	-47.26	Na ₂ PtCl ₆	aq	-272.1	
AgI	c	-15.14	-16.17		aq	-280.9	
AgIO ₃	c	-42.02	-24.08	Na ₂ Se	c	-59.1	
AgNO ₂	c	-11.6	3.76		aq, 440	-78.1	-89.42
	aq	-2.9	9.99	Na ₂ SeO ₄	c	-254	
AgNO ₃	c	-29.4	-7.66		aq, 800	-261.5	-230.30
	aq, 6500	-24.02	-7.81	Na ₂ S	c	-89.8	
Ag ₂ O	c	-6.95	-2.23		aq, 400	-105.17	-101.76
Ag ₂ S	c	-5.5	-7.6	Na ₂ SO ₃	c	-261.2	-240.14
Ag ₂ SO ₄	c	-170.1	-146.8		aq, 800	-264.1	-241.58
	aq	-165.8	-139.22	Na ₂ SO ₄	c	-330.50	-302.38
Sodium					aq, 1100	-330.82	-301.28
Na	c	0.00	0.00	Na ₂ SO ₄ ·10H ₂ O	c	-1033.85	-870.52
Na ₃ AsO ₃	aq, 500	-314.61		Na ₂ WO ₄	c	-391	
Na ₃ AsO ₄	c	-366			aq	-381.5	-345.18
	aq, 500	-381.97	-341.17	Strontium			
NaBr	c	-86.72		Sr	c	0.00	0.00
	aq, 400	-86.33	-87.17	SrBr ₂	c	-171.0	
NaBrO	aq	-78.9			aq, 400	-187.24	-182.36
NaBrO ₃	aq, 400	-68.89	-57.59	Sr(C ₂ H ₃ O ₂) ₂	c	-358.0	
NaC ₂ H ₃ O ₂	c	-170.45			aq	-364.4	-311.80
	aq, 400	-175.450	-152.31	Sr(CN) ₂	aq	-59.5	-54.50
NaCN	c	-22.47		SrCO ₃	c	-290.9	-271.9
	aq, 200	-22.29	-23.24	SrCl ₂	c	-197.84	
NaCNO	c	-96.3			aq, 400	-209.20	-195.86
	aq	-91.7	-86.00	SrF ₂	c	-289.0	
NaCNS	c	-39.94		Sr(HCO ₃) ₂	aq	-459.1	-413.76
	aq, 400	-38.23	-39.24	SrI ₂	c	-136.1	
Na ₂ CO ₃	c	-269.46	-249.55		aq, 400	-156.70	-157.87
	aq, 1000	-275.13	-251.36	Sr ₃ N ₂	c	-91.4	-76.5
NaCO ₂ NH ₂	c	-142.17		Sr(NO ₃) ₂	c	-233.2	
Na ₂ C ₂ O ₄	c	-313.8			aq, 400	-228.73	-185.70
	aq, 600	-309.92	-283.42	SrO	c	-140.8	-133.7
NaCl	c	-98.321	-91.894	SrO·SiO ₂	gls	-364	
	aq, 400	-97.324	-93.92	SrO ₂	c	-153.3	-139.0
NaClO ₃	c	-83.59		Sr ₂ O	c	-153.6	
	aq, 400	-78.42	-62.84	Sr(OH) ₂	c	-228.7	
NaClO ₄	c	-101.12			aq, 800	-239.4	-208.27
				Sr ₃ (PO ₄) ₂	c	-980	
					aq	-985	-881.54
				SrS	c	-113.1	

TABLE 2-178 Heats and Free Energies of Formation of Inorganic Compounds (Continued)

Compound	State†	Heat of formation‡§ ΔH (formation) at 25 °C, kcal/mol	Free energy of formation ¶ ΔF (formation) at 25 °C, kcal/mol	Compound	State†	Heat of formation‡§ ΔH (formation) at 25 °C, kcal/mol	Free energy of formation ¶ ΔF (formation) at 25 °C, kcal/mol
Strontium (Cont.)				Tin			
SrSO ₄	aq	-120.4	-109.78	Sn	c, II, tetragonal	0.00	0.00
	c	-345.3			c, III, "gray," cubic	0.6	1.1
SrWO ₄	aq, 400	-345.0	-309.30	SnBr ₂	c	-61.4	
Sulfur	c	-393			aq	-60.0	-55.43
S	c, rhombic	0.00	0.00	SnBr ₄	c	-94.8	
	c, monoclinic	-0.071	0.023		aq	-110.6	-97.66
	l, λ	0.257	0.072	SnCl ₂	c	-83.6	
	l, $\lambda\mu$ equilibrium		0.071		aq	-81.7	-68.94
	g	53.25	43.57	SnCl ₄	l	-127.3	-110.4
S ₂		31.02	19.36		aq	-157.6	-124.67
S ₆		27.78	13.97	SnI ₂	c	-38.9	
S ₈		27.090	12.770		aq	-33.3	-30.95
S ₂ Br ₂		-4		SnO	c	-67.7	-60.75
SCl ₄	l	-13.7		SnO ₂	c	-138.1	-123.6
S ₂ Cl ₂	l	-14.2	-5.90	Sn(OH) ₂	c	-136.2	-115.95
S ₂ Cl ₄	l	-24.1		Sn(OH) ₄	c	-268.9	-226.00
SF ₆		-262	-237	SnS	c	-18.61	
SO		19.02	12.75	Titanium			
SO ₂		-70.94	-71.68	Ti	c	0.00	0.00
SO ₃		-94.39	-88.59	TiC	c	-110	-109.2
		-103.03	-88.28	TiCl ₄	l	-181.4	-165.5
	c, α	-105.09	-88.22	TiN	c	-80.0	-73.17
	c, β	-105.92	-88.34	TiO ₂	c, III, rutil	-225.0	-211.9
	c, γ	-109.34	-88.98		amorphous	-214.1	-201.4
SO ₂ Cl ₂		-82.04	-74.06	Tungsten			
	l	-89.80	-75.06	W	c	0.00	0.00
Tantalum				WO ₂	c	-130.5	-118.3
Ta	c	0.00	0.00	WO ₃	c	-195.7	-177.3
TaN	c	-51.2	-45.11	WS ₂	c	-84	
Ta ₂ O ₅	c	-486.0	-453.7	Uranium			
Tellurium				U	c	0.00	0.00
Te	c	0.00	0.00	UC ₂	c	-29	
TeBr ₄	c	-49.3		UCl ₃	c	-213	
TeCl ₄	c	-77.4	-57.4	UCl ₄	c	-251	
TeF ₆	g	-315	-292	U ₃ N ₄	c	-274	-249.6
TeO ₂	c	-77.56	-64.66	UO ₂	c	-256.6	-242.2
Thallium				UO ₂ (NO ₃) ₂ ·6H ₂ O	c	-756.8	-617.8
Tl	c	0.00	0.00	UO ₃	c	-291.6	
TlBr	c	-41.5	-39.43	U ₃ O ₈	c	-845.1	
	aq	-28.0	-32.34	Vanadium			
TlCl	c	-49.37	-44.46	V	c	0.00	0.00
	aq	-38.4	-39.09	VCl ₂	c	-147	
TlCl ₃	c	-82.4		VCl ₃	l	-187	
	aq	-91.0	-44.25	VCl ₄	l	-165	
TlF	aq	-77.6	-73.46	VN	c	-41.43	-35.08
TlI	c	-31.1	-31.3	V ₂ O ₂	c	-195	
	aq	-12.7	-20.09	V ₂ O ₃	c	-296	-277
TlNO ₃	c	-58.2	-36.32	V ₂ O ₄	c	-342	-316
	aq	-48.4	-34.01	V ₂ O ₅	c	-373	-342
Tl ₂ O	c	-43.18		Zinc			
Tl ₂ O ₃	c	-120		Zn	c	0.00	0.00
TlOH	c	-57.44	-45.54	ZnSb	c	-3.6	-3.88
	aq	-53.9	-45.35	ZnBr ₂	c	-77.0	-72.9
Tl ₂ S	c	-22			aq, 400	-93.6	
Tl ₂ SO ₄	c	-222.8	-197.79	Zn(C ₂ H ₃ O ₂) ₂	c	-259.4	
	aq, 800	-214.1	-191.62		aq, 400	-269.4	-214.4
Thorium				Zn(CN) ₂	c	17.06	
Th	c	0.00	0.00	ZnCO ₃	c	-192.9	-173.5
ThBr ₄	c	-281.5		ZnCl ₂	c	-99.9	-88.8
	aq	-352.0	-295.31		aq, 400	-115.44	
ThC ₂	c	-45.1		ZnF ₂	aq	-192.9	-166.6
ThCl ₄	c	-335		ZnI ₂	c	-50.50	-49.93
	aq	-392	-322.32		aq	-61.6	
ThI ₄	aq	-292.0	-246.33	Zn(NO ₃) ₂	aq, 400	-134.9	-87.7
Th ₃ N ₄	c	-309.0	-282.3	ZnO	c, hexagonal	-83.36	-76.19
ThO ₂	c	-291.6	-280.1	ZnO·SiO ₂	c	-282.6	
Th(OH) ₄	c, "soluble"	-336.1		Zn(OH) ₂	c, rhombic	-153.66	
Th(SO ₄) ₂	c	-632		ZnS	c, wurtzite	-45.3	-44.2
	aq	-668.1	-549.2	ZnSO ₄	c	-233.4	
					aq, 400	-252.12	-211.28

2-194 PHYSICAL AND CHEMICAL DATA

TABLE 2-178 Heats and Free Energies of Formation of Inorganic Compounds (Concluded)

Compound	State†	Heat of formation‡§ ΔH (formation) at 25 °C, kcal/mol	Free energy of formation ¶ ΔF (formation) at 25 °C, kcal/mol	Compound	State†	Heat of formation‡§ ΔH (formation) at 25 °C, kcal/mol	Free energy of formation ¶ ΔF (formation) at 25 °C, kcal/mol
Zirconium				Zirconium (Cont.)			
Zr	c	0.00	0.00	ZrO ₂	c, monoclinic	-258.5	-244.6
ZrC	c	-29.8	-34.6	Zr(OH) ₄	c	-411.0	
ZrCl ₄	c	-268.9		ZrO(OH) ₂	c	-337	-307.6
ZrN	c	-82.5	-75.9				

† The physical state is indicated as follows: *c*, crystal (solid); *l*, liquid; *g*, gas; *gls*, glass or solid supercooled liquid; *aq*, in aqueous solution. A number following the symbol *aq* applies only to the values of the heats of formation (not to those of free energies of formation); and indicates the number of moles of water per mole of solute; when no number is given, the solution is understood to be dilute. For the free energy of formation of a substance in aqueous solution, the concentration is always that of the hypothetical solution of unit molality.

‡ The increment in heat content, ΔH , is the reaction of forming the given substance from its elements in their standard states. When ΔH is negative, heat is evolved in the process, and, when positive, heat is absorbed.

§ The heat of solution in water of a given solid, liquid, or gaseous compound is given by the difference in the value for the heat of formation of the given compound in the solid, liquid, or gaseous state and its heat of formation in aqueous solution. The following two examples serve as an illustration of the procedure: (1) For NaCl(*c*) and NaCl(*aq*, 400H₂O), the values of ΔH (formation) are, respectively, -98.321 and -97.324 kcal/mol. Subtraction of the first value from the second gives $\Delta H = 0.998$ kcal/mol for the reaction of dissolving crystalline sodium chloride in 400 mol of water. When this process occurs at a constant pressure of 1 atm, 0.998 kcal of energy are absorbed. (2) For HCl(*g*) and HCl(*aq*, 400H₂O), the values for ΔH (formation) are, respectively, -22.06 and -39.85 kcal/mol. Subtraction of the first from the second gives $\Delta H = -17.79$ kcal/mol for the reaction of dissolving gaseous hydrogen chloride in 400 mol of water. At a constant pressure of 1 atm, 17.79 kcal of energy are evolved in this process.

|| The increment in the free energy, ΔF , is the reaction of forming the given substance in its standard state from its elements in their standard states. The standard states are: for a gas, fugacity (approximately equal to the pressure) of 1 atm; for a pure liquid or solid, the substance at a pressure of 1 atm; for a substance in aqueous solution, the hypothetical solution of unit molality, which has all the properties of the infinitely dilute solution except the property of concentration.

¶ The free energy of solution of a given substance from its normal standard state as a solid, liquid, or gas to the hypothetical one molal state in aqueous solution may be calculated in a manner similar to that described in footnote § for calculating the heat of solution.

TABLE 2-179 Enthalpies and Gibbs Energies of Formation, Entropies, and Net Enthalpies of Combustion of Inorganic and Organic Compounds at 298.15 K

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	Ideal gas enthalpy of formation, J/kmol × 1E-07	Ideal gas Gibbs energy of formation, J/kmol × 1E-07	Ideal gas entropy, J/(kmol·K) × 1E-05	Standard net enthalpy of combustion, J/kmol × 1E-09
1	Acetaldehyde	C ₂ H ₄ O	75-07-0	44.053	-16.64	-13.33	2.642	-1.1045
2	Acetamide	C ₂ H ₅ NO	60-35-5	59.067	-23.83	-15.96	2.722	-1.0741
3	Acetic acid	C ₂ H ₄ O ₂	64-19-7	60.052	-46.11	-40.3	2.825	-0.7866
4	Acetic anhydride	C ₄ H ₆ O ₃	108-24-7	102.089	-57.25	-47.34	3.899	-1.675
5	Acetone	C ₃ H ₆ O	67-64-1	58.079	-21.57	-15.13	2.954	-1.659
6	Acetonitrile	C ₂ H ₃ N	75-05-8	41.052	7.404	9.1868	2.4329	-1.19043
7	Acetylene	C ₂ H ₂	74-86-2	26.037	22.82	21.068	2.0081	-1.257
8	Acrolein	C ₃ H ₄ O	107-02-8	56.063	-8.18	-5.68	2.97	-1.5468
9	Acrylic acid	C ₃ H ₄ O ₂	79-10-7	72.063	-35.591	-30.6	3.15	-1.32717
10	Acrylonitrile	C ₃ H ₃ N	107-13-1	53.063	18.37	19.37	2.753	-1.69
11	Air	Mixture	132259-10-0	28.960	0	0	1.99	0
12	Ammonia	H ₃ N	7664-41-7	17.031	-4.5898	-1.64	1.9266	-0.31683
13	Anisole	C ₇ H ₈ O	100-66-3	108.138	-6.79	2.27	3.61	-3.6072
14	Argon	Ar	7440-37-1	39.948	0	0	1.54737	0
15	Benzamide	C ₇ H ₇ NO	55-21-0	121.137	-10.09	-0.211	3.641	-3.39877
16	Benzene	C ₆ H ₆	71-43-2	78.112	8.288	12.96	2.693	-3.136
17	Benzenethiol	C ₆ H ₆ S	108-98-5	110.177	11.15	14.76	3.369	-3.4474
18	Benzoic acid	C ₇ H ₆ O ₂	65-85-0	122.121	-29.41	-21.42	3.69	-3.0951
19	Benzonitrile	C ₇ H ₅ N	100-47-0	103.121	21.57	25.78	3.21	-3.5238
20	Benzophenone	C ₁₃ H ₁₀ O	119-61-9	182.218	5.68	17.3	4.4	-6.2876
21	Benzyl alcohol	C ₇ H ₈ O	100-51-6	108.138	-9.025	-0.254	3.713	-3.56
22	Benzyl ethyl ether	C ₉ H ₁₂ O	539-30-0	136.191	-11.5	3.37	4.39	-4.83
23	Benzyl mercaptan	C ₇ H ₈ S	100-53-8	124.203	9.33	16.3	3.607	-4.06
24	Biphenyl	C ₁₂ H ₁₀	92-52-4	154.208	17.849	27.63	3.9367	-6.248
25	Bromine	Br ₂	7726-95-6	159.808	3.091	0.314	2.4535	
26	Bromobenzene	C ₆ H ₅ Br	108-86-1	157.008	10.5018	13.8532	3.24386	-3.01917
27	Bromoethane	C ₂ H ₅ Br	74-96-4	108.965	-6.36	-2.582	2.873	-1.285
28	Bromomethane	CH ₃ Br	74-83-9	94.939	-3.77	-2.819	2.458	-0.70542
29	1,2-Butadiene	C ₄ H ₆	590-19-2	54.090	16.23	19.86	2.93	-2.4617
30	1,3-Butadiene	C ₄ H ₆	106-99-0	54.090	10.924	14.972	2.7889	-2.409
31	Butane	C ₄ H ₁₀	106-97-8	58.122	-12.579	-1.67	3.0991	-2.65732
32	1,2-Butanediol	C ₄ H ₁₀ O ₂	584-03-2	90.121	-44.58	-30.44	4.065	-2.2678
33	1,3-Butanediol	C ₄ H ₁₀ O ₂	107-88-0	90.121	-43.32	-29.18	4.065	-2.2824
34	1-Butanol	C ₄ H ₁₀ O	71-36-3	74.122	-27.51	-15.07	3.618	-2.454
35	2-Butanol	C ₄ H ₁₀ O	78-92-2	74.122	-29.29	-16.7	3.566	-2.446
36	1-Butene	C ₄ H ₈	106-98-9	56.106	-0.05	7.041	3.074	-2.5408
37	<i>cis</i> -2-Butene	C ₄ H ₈	590-18-1	56.106	-0.74	6.536	3.012	-2.5339
38	<i>trans</i> -2-Butene	C ₄ H ₈	624-64-6	56.106	-1.1	6.32	2.965	-2.53
39	Butyl acetate	C ₆ H ₁₂ O ₂	123-86-4	116.158	-48.56	-31.26	4.425	-3.28
40	Butylbenzene	C ₁₀ H ₁₄	104-51-8	134.218	-1.314	14.54	4.3949	-5.5644
41	Butyl mercaptan	C ₄ H ₁₀ S	109-79-5	90.187	-8.78	1.139	3.752	-2.9554
42	<i>sec</i> -Butyl mercaptan	C ₄ H ₁₀ S	513-53-1	90.187	-9.66	0.512	3.667	-2.949
43	1-Butyne	C ₄ H ₆	107-00-6	54.090	16.52	20.225	2.9039	-2.4647
44	Butyraldehyde	C ₄ H ₈ O	123-72-8	72.106	-20.7	-11.63	3.4365	-2.3035
45	Butyric acid	C ₄ H ₈ O ₂	107-92-6	88.105	-47.58	-36	3.601	-2.008
46	Butyronitrile	C ₄ H ₇ N	109-74-0	69.105	3.40578	10.8658	3.25432	-2.4148
47	Carbon dioxide	CO ₂	124-38-9	44.010	-39.351	-39.437	2.13677	
48	Carbon disulfide	CS ₂	75-15-0	76.141	11.69	6.68	2.379	-1.0769
49	Carbon monoxide	CO	630-08-0	28.010	-11.053	-13.715	1.97556	-0.283
50	Carbon tetrachloride	CCl ₄	56-23-5	153.823	-9.581	-5.354	3.0991	-0.2653
51	Carbon tetrafluoride	CF ₄	75-73-0	88.004	-92.21	-87.76	2.62	0.5286
52	Chlorine	Cl ₂	7782-50-5	70.906	0	0	2.22972	
53	Chlorobenzene	C ₆ H ₅ Cl	108-90-7	112.557	5.109	9.829	3.1403	-2.976
54	Chloroethane	C ₂ H ₅ Cl	75-00-3	64.514	-11.226	-6.0499	2.7578	-1.2849
55	Chloroform	CHCl ₃	67-66-3	119.378	-10.29	-7.01	2.956	-0.38
56	Chloromethane	CH ₃ Cl	74-87-3	50.488	-8.196	-5.844	2.3418	-0.67538
57	1-Chloropropane	C ₃ H ₇ Cl	540-54-5	78.541	-13.318	-5.261	3.1547	-1.867
58	2-Chloropropane	C ₃ H ₇ Cl	75-29-6	78.541	-14.477	-6.136	3.0594	-1.863

TABLE 2-179 Enthalpies and Gibbs Energies of Formation, Entropies, and Net Enthalpies of Combustion of Inorganic and Organic Compounds at 298.15 K (Continued)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	Ideal gas enthalpy of formation, J/kmol $\times 1\text{E-}07$	Ideal gas Gibbs energy of formation, J/kmol $\times 1\text{E-}07$	Ideal gas entropy, J/(kmol $\times 1\text{E-}05$)	Standard net enthalpy of combustion, J/kmol $\times 1\text{E-}09$
59	<i>m</i> -Cresol	C ₇ H ₈ O	108-39-4	108.138	-13.23	-4.019	3.5604	-3.52783
60	<i>o</i> -Cresol	C ₇ H ₈ O	95-48-7	108.138	-12.857	-3.543	3.5259	-3.528
61	<i>p</i> -Cresol	C ₇ H ₈ O	106-44-5	108.138	-12.535	-3.166	3.5075	-3.52256
62	Cumene	C ₉ H ₁₂	98-82-8	120.192	0.4	13.79	3.86	-4.951
63	Cyanogen	C ₂ N ₂	460-19-5	52.035	30.9072	29.7598	2.41463	-1.0961
64	Cyclobutane	C ₄ H ₈	287-23-0	56.106	2.85	11.22	2.64396	-2.5678
65	Cyclohexane	C ₆ H ₁₂	110-82-7	84.159	-12.33	3.191	2.97276	-3.656
66	Cyclohexanol	C ₆ H ₁₂ O	108-93-0	100.159	-28.62	-10.95	3.277	-3.4639
67	Cyclohexanone	C ₆ H ₁₀ O	108-94-1	98.143	-22.61	-9.028	3.3426	-3.299
68	Cyclohexene	C ₆ H ₁₀	110-83-8	82.144	-0.46	10.77	3.10518	-3.532
69	Cyclopentane	C ₅ H ₁₀	287-92-3	70.133	-7.703	3.885	2.929	-3.0709
70	Cyclopentene	C ₅ H ₈	142-29-0	68.117	3.23	11.05	2.91267	-2.9393
71	Cyclopropane	C ₃ H ₆	75-19-4	42.080	5.33	10.44	2.37378	-1.9593
72	Cyclohexyl mercaptan	C ₆ H ₁₂ S	1569-69-3	116.224	-9.602	4.886	3.646	-3.968
73	Decanal	C ₁₀ H ₂₀ O	112-31-2	156.265	-33.17	-6.739	5.7912	-5.959
74	Decane	C ₁₀ H ₂₂	124-18-5	142.282	-24.946	3.318	5.457	-6.29422
75	Decanoic acid	C ₁₀ H ₂₀ O ₂	334-48-5	172.265	-59.43	-30.5	5.99	-5.72
76	1-Decanol	C ₁₀ H ₂₂ O	112-30-1	158.281	-39.85	-10.02	5.971	-6.116
77	1-Decene	C ₁₀ H ₂₀	872-05-9	140.266	-12.21	12.27	5.433	-6.1809
78	Decyl mercaptan	C ₁₀ H ₂₂ S	143-10-2	174.347	-21.09	6.165	6.116	-6.6161
79	1-Decyne	C ₁₀ H ₁₈	764-93-2	138.250	4.1	25.16	5.263	-6.1037
80	Deuterium	D ₂	7782-39-0	4.032	0	0	1.4486	-0.24625
81	1,1-Dibromoethane	C ₂ H ₄ Br ₂	557-91-5	187.861	-4.08	-1.181	3.276	-1.16
82	1,2-Dibromoethane	C ₂ H ₄ Br ₂	106-93-4	187.861	-3.89	-1.054	3.297	-1.1769
83	Dibromomethane	CH ₂ Br ₂	74-95-3	173.835			2.92964	
84	Dibutyl ether	C ₈ H ₁₈ O	142-96-1	130.228	-33.34	-8.827	5.014	-4.94691
85	<i>m</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	541-73-1	147.002	2.57	7.79	3.4353	-2.825
86	<i>o</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	95-50-1	147.002	3.02	8.29	3.4185	-2.826
87	<i>p</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	106-46-7	147.002	2.25	7.67	3.3674	-2.802
88	1,1-Dichloroethane	C ₂ H ₄ Cl ₂	75-34-3	98.959	-12.941	-7.259	3.0501	-1.1104
89	1,2-Dichloroethane	C ₂ H ₄ Cl ₂	107-06-2	98.959	-12.979	-7.3945	3.0828	-1.105
90	Dichloromethane	CH ₂ Cl ₂	75-09-2	84.933	-9.552	-6.896	2.7018	-0.51388
91	1,1-Dichloropropane	C ₃ H ₆ Cl ₂	78-99-9	112.986	-15.08	-6.52	3.448	-1.72
92	1,2-Dichloropropane	C ₃ H ₆ Cl ₂	78-87-5	112.986	-16.28	-8.018	3.548	-1.707
93	Diethanol amine	C ₄ H ₁₁ NO ₂	111-42-2	105.136	-40.847	-22.574	4.29	-2.4105
94	Diethyl amine	C ₄ H ₁₁ N	109-89-7	73.137	-7.142	7.308	3.522	-2.8003
95	Diethyl ether	C ₄ H ₁₀ O	60-29-7	74.122	-25.21	-12.21	3.423	-2.5035
96	Diethyl sulfide	C ₄ H ₁₀ S	352-93-2	90.187	-8.356	1.774	3.681	-2.9607
97	1,1-Difluoroethane	C ₂ H ₄ F ₂	75-37-6	66.050	-49.7	-43.9485	2.824	-0.773662
98	1,2-Difluoroethane	C ₂ H ₄ F ₂	624-72-6	66.050	-44.77	-39.19	2.88194	-0.823
99	Difluoromethane	CH ₂ F ₂	75-10-5	52.023	-45.23	-42.4747	2.4658	-0.183031
100	Di-isopropyl amine	C ₆ H ₁₅ N	108-18-9	101.190	-14.38	6.42	4.12	-3.99
101	Di-isopropyl ether	C ₆ H ₁₄ O	108-20-3	102.175	-31.92	-12.48	3.989	-3.70261
102	Di-isopropyl ketone	C ₇ H ₁₄ O	565-80-0	114.185	-31.14	-12.37	4.27	-4.095
103	1,1-Dimethoxyethane	C ₄ H ₁₀ O ₂	534-15-6	90.121	-38.97	-23.8	3.726	-2.394
104	1,2-Dimethoxypropane	C ₅ H ₁₂ O ₂	7778-85-0	104.148	-38.42	-20.11	4.038	-2.996
105	Dimethyl acetylene	C ₄ H ₆	503-17-3	54.090	14.57	18.49	2.833	-2.4189
106	Dimethyl amine	C ₂ H ₇ N	124-40-3	45.084	-1.845	6.839	2.7296	-1.6146
107	2,3-Dimethylbutane	C ₆ H ₁₄	79-29-8	86.175	-17.68	-0.3125	3.6592	-3.84761
108	1,1-Dimethylcyclohexane	C ₈ H ₁₆	590-66-9	112.213	-18.1	3.52293	3.65012	-4.8639
109	<i>cis</i> -1,2-Dimethylcyclohexane	C ₈ H ₁₆	2207-01-4	112.213	-17.2172	4.12124	3.7451	-4.87084
110	<i>trans</i> -1,2-Dimethylcyclohexane	C ₈ H ₁₆	6876-23-9	112.213	-17.9996	3.44761	3.70912	-4.86436
111	Dimethyl disulfide	C ₂ H ₆ S ₂	624-92-0	94.199	-2.42	1.516	3.35291	-2.0441
112	Dimethyl ether	C ₂ H ₆ O	115-10-6	46.068	-18.41	-11.28	2.667	-1.3284
113	<i>N,N</i> -Dimethyl formamide	C ₃ H ₇ NO	68-12-2	73.094	-19.17	-8.84	3.26	-1.78871
114	2,3-Dimethylpentane	C ₇ H ₁₆	565-59-3	100.202	-19.41	0.5717	4.1455	-4.46075
115	Dimethyl phthalate	C ₁₀ H ₁₀ O ₄	131-11-3	194.184	-60.5	-46.7749	6.6	-4.4662
116	Dimethylsilane	C ₂ H ₆ Si	1111-74-6	60.170	-9.47	-1.925	2.9953	-2.569

117	Dimethyl sulfide	C ₂ H ₆ S	75-18-3	62.134	-3.724	0.7302	2.8585	-1.7443
118	Dimethyl sulfoxide	C ₂ H ₆ OS	67-68-5	78.133	-15.046	-8.1441	3.0627	-1.6054
119	Dimethyl terephthalate	C ₁₀ H ₁₀ O ₄	120-61-6	194.184	-64.4	-47.4	5.5	-4.4115
120	1,4-Dioxane	C ₄ H ₈ O ₂	123-91-1	88.105	-31.58	-18.16	3.0012	-2.1863
121	Diphenyl ether	C ₁₂ H ₁₀ O	101-84-8	170.207	5.2	17.5	4.13	-5.8939
122	Dipropyl amine	C ₆ H ₁₅ N	142-84-7	101.190	-11.6	8.68	4.29	-4.0189
123	Dodecane	C ₁₂ H ₂₆	112-40-3	170.335	-29.072	4.981	6.2415	-7.51368
124	Eicosane	C ₂₀ H ₄₂	112-95-8	282.547	-45.646	11.57	9.3787	-12.3908
125	Ethane	C ₂ H ₆	74-84-0	30.069	-8.382	-3.192	2.2912	-1.42864
126	Ethanol	C ₂ H ₆ O	64-17-5	46.068	-23.495	-16.785	2.8064	-1.235
127	Ethyl acetate	C ₄ H ₈ O ₂	141-78-6	88.105	-44.45	-32.8	3.597	-2.061
128	Ethyl amine	C ₂ H ₇ N	75-04-7	45.084	-4.715	3.616	2.848	-1.5874
129	Ethylbenzene	C ₈ H ₁₀	100-41-4	106.165	2.992	13.073	3.6063	-4.3448
130	Ethyl benzoate	C ₉ H ₁₀ O ₂	93-89-0	150.175	-32.6	-19.05	4.55	-4.41
131	2-Ethyl butanoic acid	C ₆ H ₁₂ O ₂	88-09-5	116.158	-53.78	-35.9	4.23	-3.21203
132	Ethyl butyrate	C ₆ H ₁₂ O ₂	105-54-4	116.158	-48.55	-31.22	4.417	-3.284
133	Ethylcyclohexane	C ₈ H ₁₆	1678-91-7	112.213	-17.15	3.955	3.826	-4.87051
134	Ethylcyclopentane	C ₇ H ₁₄	1640-89-7	98.186	-12.69	4.48	3.783	-4.2839
135	Ethylene	C ₂ H ₄	74-85-1	28.053	5.251	6.844	2.192	-1.323
136	Ethylenediamine	C ₂ H ₆ N ₂	107-15-3	60.098	-1.73	10.3	3.21833	-1.691
137	Ethylene glycol	C ₂ H ₆ O ₂	107-21-1	62.068	-39.22	-30.18	3.04891	-1.0527
138	Ethyleneimine	C ₂ H ₅ N	151-56-4	43.068	12.3428	17.7987	2.5062	-1.481
139	Ethylene oxide	C ₂ H ₄ O	75-21-8	44.053	-5.263	-1.323	2.4299	-1.218
140	Ethyl formate	C ₃ H ₆ O ₂	109-94-4	74.079	-38.83	-30.31	3.282	-1.50696
141	2-Ethyl hexanoic acid	C ₈ H ₁₆ O ₂	149-57-5	144.211	-55.95	-32.5	5.1	-4.448
142	Ethylhexyl ether	C ₈ H ₁₈ O	5756-43-4	130.228	-33.37	-9.042	5.076	-4.943
143	Ethylisopropyl ether	C ₅ H ₁₂ O	625-54-7	88.148	-28.58	-12.64	3.8	-3.103
144	Ethylisopropyl ketone	C ₆ H ₁₂ O	565-69-5	100.159	-28.61	-13.3	4.069	-3.4863
145	Ethyl mercaptan	C ₂ H ₆ S	75-08-1	62.134	-4.63	-0.4814	2.961	-1.7366
146	Ethyl propionate	C ₅ H ₁₀ O ₂	105-37-3	102.132	-46.36	-31.93	4.025	-2.674
147	Ethylpropyl ether	C ₅ H ₁₂ O	628-32-0	88.148	-27.22	-11.52	3.881	-3.12
148	Ethyltrichlorosilane	C ₂ H ₅ Cl ₃ Si	115-21-9	163.506	-59.54	-51.01	4.07	-1.671
149	Fluorine	F ₂	7782-41-4	37.997	0	0	2.02682	
150	Fluorobenzene	C ₆ H ₅ F	462-06-6	96.102	-11.6566	-6.9036	3.02629	-2.81451
151	Fluoroethane	C ₂ H ₅ F	353-36-6	48.060	-26.44	-21.23	2.644	-1.127
152	Fluoromethane	CH ₃ F	593-53-3	34.033	-23.43	-21.04	2.22734	-0.5219
153	Formaldehyde	CH ₂ O	50-00-0	30.026	-10.86	-10.26	2.1866	-0.5268
154	Formamide	CH ₃ NO	75-12-7	45.041	-19.22	-14.71	2.4857	-0.5021
155	Formic acid	CH ₂ O ₂	64-18-6	46.026	-40.55	-37.78	2.487	-0.2115
156	Furan	C ₄ H ₄ O	110-00-9	68.074	-3.48	0.08225	2.6714	-1.9959
157	Helium-4	He	7440-59-7	4.003	0	0	1.26044	0
158	Heptadecane	C ₁₇ H ₃₆	629-78-7	240.468	-39.445	9.083	8.2023	-10.5618
159	Heptanal	C ₇ H ₁₄ O	111-71-7	114.185	-26.94	-9.191	4.6138	-4.136
160	Heptane	C ₇ H ₁₆	142-82-5	100.202	-18.765	0.8165	4.2798	-4.46473
161	Heptanoic acid	C ₇ H ₁₄ O ₂	111-14-8	130.185	-53.62	-33.4	4.8	-3.839
162	1-Heptanol	C ₇ H ₁₆ O	111-70-6	116.201	-33.68	-12.55	4.795	-4.285
163	2-Heptanol	C ₇ H ₁₆ O	543-49-7	116.201	-35.54	-14.25	4.74	-4.282
164	3-Heptanone	C ₇ H ₁₄ O	106-35-4	114.185	-30.1	-12.25	4.58	-4.098
165	2-Heptanone	C ₇ H ₁₄ O	110-43-0	114.185	-30.0453	-11.96	4.486	-4.09952
166	1-Heptene	C ₇ H ₁₄	592-76-7	98.186	-6.289	9.482	4.252	-4.3499
167	Heptyl mercaptan	C ₇ H ₁₆ S	1639-09-4	132.267	-14.95	3.622	4.939	-4.7865
168	1-Heptyne	C ₇ H ₁₂	628-71-7	96.170	10.3	22.7	4.085	-4.2717
169	Hexadecane	C ₁₆ H ₃₄	544-76-3	226.441	-37.417	8.216	7.8102	-9.95145
170	Hexanal	C ₆ H ₁₂ O	66-25-1	100.159	-24.86	-10.005	4.2214	-3.52
171	Hexane	C ₆ H ₁₄	110-54-3	86.175	-16.694	-0.006634	3.8874	-3.8551
172	Hexanoic acid	C ₆ H ₁₂ O ₂	142-62-1	116.158	-51.19	-33.8	4.41	-3.23
173	1-Hexanol	C ₆ H ₁₄ O	111-27-3	102.175	-31.62	-13.39	4.402	-3.675
174	2-Hexanol	C ₆ H ₁₄ O	626-93-7	102.175	-33.46	-15.06	4.349	-3.67
175	2-Hexanone	C ₆ H ₁₂ O	591-78-6	100.159	-27.9826	-13.0081	4.17856	-3.49
176	3-Hexanone	C ₆ H ₁₂ O	589-38-8	100.159	-27.76	-12.6	4.092	-3.492
177	1-Hexene	C ₆ H ₁₂	592-41-6	84.159	-4.167	8.7	3.863	-3.7397
178	3-Hexyne	C ₆ H ₁₀	928-49-4	82.144	10.6	19.9	3.76	-3.64
179	Hexyl mercaptan	C ₆ H ₁₄ S	111-31-9	118.240	-12.92	2.759	4.546	-4.1762

TABLE 2-179 Enthalpies and Gibbs Energies of Formation, Entropies, and Net Enthalpies of Combustion of Inorganic and Organic Compounds at 298.15 K (Continued)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	Ideal gas enthalpy of formation, J/kmol × 1E-07	Ideal gas Gibbs energy of formation, J/kmol × 1E-07	Ideal gas entropy, J/(kmol·K) × 1E-05	Standard net enthalpy of combustion, J/kmol × 1E-09
180	1-Hexyne	C ₆ H ₁₀	693-02-7	82.144	12.37	21.85	3.694	-3.661
181	2-Hexyne	C ₆ H ₁₀	764-35-2	82.144	10.5	19.9	3.72	-3.64
182	Hydrazine	H ₄ N ₂	302-01-2	32.045	9.5353	15.917	2.3861	-0.5342
183	Hydrogen	H ₂	1333-74-0	2.016	0	0	1.30571	-0.24182
184	Hydrogen bromide	HBr	10035-10-6	80.912	-3.629	-5.334	1.98591	-0.06904
185	Hydrogen chloride	HCl	7647-01-0	36.461	-9.231	-9.53	1.86786	-0.0286
186	Hydrogen cyanide	CHN	74-90-8	27.025	13.5143	12.4725	2.01719	-0.62329
187	Hydrogen fluoride	HF	7664-39-3	20.006	-27.33	-27.54	1.7367	0.1524
188	Hydrogen sulfide	H ₂ S	7783-06-4	34.081	-2.063	-3.344	2.056	-0.518
189	Isobutyric acid	C ₄ H ₈ O ₂	79-31-2	88.105	-48.41	-36.21	3.412	-2.0004
190	Isopropyl amine	C ₃ H ₉ N	75-31-0	59.110	-8.38	3.192	3.124	-2.1566
191	Malonic acid	C ₃ H ₄ O ₄	141-82-2	104.061	-76.68	-67	3.7	-0.7732
192	Methacrylic acid	C ₄ H ₆ O ₂	79-41-4	86.089	-36.8	-28.8	3.5	-1.93
193	Methane	CH ₄	74-82-8	16.042	-7.452	-5.049	1.8627	-0.80262
194	Methanol	CH ₃ O	67-56-1	32.042	-20.094	-16.232	2.3988	-0.6382
195	N-Methyl acetamide	C ₃ H ₇ NO	79-16-3	73.094	-24	-13.5	3.2	-1.71
196	Methyl acetate	C ₃ H ₆ O ₂	79-20-9	74.079	-41.19	-32.42	3.198	-1.461
197	Methyl acetylene	C ₃ H ₄	74-99-7	40.064	18.49	19.384	2.4836	-1.8487
198	Methyl acrylate	C ₄ H ₆ O ₂	96-33-3	86.089	-33.3	-25.7	3.66	-1.9303
199	Methyl amine	CH ₃ N	74-89-5	31.057	-2.297	3.207	2.433	-0.97508
200	Methyl benzoate	C ₈ H ₈ O ₂	93-58-3	136.148	-28.79	-18.1	4.14	-3.772
201	3-Methyl-1,2-butadiene	C ₅ H ₈	598-25-4	68.117	12.908	19.75	3.2151	-3.032
202	2-Methylbutane	C ₅ H ₁₂	78-78-4	72.149	-15.37	-1.405	3.4374	-3.23954
203	2-Methylbutanoic acid	C ₅ H ₁₀ O ₂	116-53-0	102.132	-49.8	-34.99	3.9	-2.622
204	3-Methyl-1-butanol	C ₅ H ₁₂ O	123-51-3	88.148	-30.3	-14.1	3.869	-3.062
205	2-Methyl-1-butene	C ₅ H ₁₀	563-46-2	70.133	-3.53	6.668	3.395	-3.1159
206	2-Methyl-2-butene	C ₅ H ₁₀	513-35-9	70.133	-4.18	6.045	3.386	-3.1088
207	2-Methyl -1-butene-3-yne	C ₅ H ₈	78-80-8	66.101	26	30.25	2.78	-2.93
208	Methylbutyl ether	C ₅ H ₁₂ O	628-28-4	88.148	-25.81	-10.17	3.901	-3.12818
209	Methylbutyl sulfide	C ₅ H ₁₂ S	628-29-5	104.214	-10.2	2.691	4.118	-3.5723
210	3-Methyl-1-butene	C ₅ H ₈	598-23-2	68.117	13.8	20.72	3.189	-3.046
211	Methyl butyrate	C ₅ H ₁₀ O ₂	623-42-7	102.132	-45.07	-30.53	3.988	-2.686
212	Methylchlorosilane	CH ₃ ClSi	993-00-0	80.589	-21.5	-16.61	2.98277	-1.693
213	Methylcyclohexane	C ₇ H ₁₄	108-87-2	98.186	-15.48	2.733	3.433	-4.25714
214	1-Methylcyclohexanol	C ₇ H ₁₄ O	590-67-0	114.185	-33.2	-12.9	3.75	-4.058
215	cis-2-Methylcyclohexanol	C ₇ H ₁₄ O	7443-70-1	114.185	-32.7	-12.68	3.853	-4.0574
216	trans-2-Methylcyclohexanol	C ₇ H ₁₄ O	7443-52-9	114.185	-35.26	-15.24	3.853	-4.0318
217	Methylcyclopentane	C ₆ H ₁₂	96-37-7	84.159	-10.62	3.63	3.399	-3.6741
218	1-Methylcyclopentene	C ₆ H ₁₀	693-89-0	82.144	-0.38	10.38	3.264	-3.534
219	3-Methylcyclopentene	C ₆ H ₁₀	1120-62-3	82.144	0.74	11.38	3.305	-3.5464
220	Methyldichlorosilane	CH ₂ Cl ₂ Si	75-54-7	115.034	-40.2	-34.83	3.287	-1.357
221	Methylethyl ether	C ₃ H ₈ O	540-67-0	60.095	-21.64	-11.71	3.0881	-1.9314
222	Methylethyl ketone	C ₄ H ₈ O	78-93-3	72.106	-23.9	-14.7	3.394	-2.268
223	Methylethyl sulfide	C ₃ H ₈ S	624-89-5	76.161	-5.96	1.147	3.332	-2.354
224	Methyl formate	C ₂ H ₄ O ₂	107-31-3	60.052	-35.24	-29.5	2.852	-0.8924
225	Methylisobutyl ether	C ₅ H ₁₂ O	625-44-5	88.148	-26.6	-10.7	3.81	-3.122
226	Methylisobutyl ketone	C ₆ H ₁₂ O	108-10-1	100.159	-28.64	-13.51	4.129	-3.4762
227	Methyl Isocyanate	C ₂ H ₃ NO	624-83-9	57.051	-6.24	0.0244	1.955	-1.06
228	Methylisopropyl ether	C ₄ H ₁₀ O	598-53-8	74.122	-25.2	-12.18	3.416	-2.5311
229	Methylisopropyl ketone	C ₅ H ₁₀ O	563-80-4	86.132	-26.26	-13.93	3.699	-2.877
230	Methylisopropyl sulfide	C ₄ H ₁₀ S	1551-21-9	90.187	-8.96	1.4509	3.59	-2.957
231	Methyl mercaptan	CH ₄ S	74-93-1	48.107	-2.29	-0.98	2.55	-1.1517
232	Methyl methacrylate	C ₅ H ₈ O ₂	80-62-6	100.116	-36	-25.4	4.01	-2.54
233	2-Methyloctanoic acid	C ₉ H ₁₈ O ₂	3004-93-1	158.238	-57.95	-31.8	5.533	-5.056
234	2-Methylpentane	C ₆ H ₁₄	107-83-5	86.175	-17.455	-0.5338	3.8089	-3.84915
235	Methyl pentyl ether	C ₆ H ₁₄ O	628-80-8	102.175	-27.8	-9.35	4.32	-3.739
236	2-Methylpropane	C ₄ H ₁₀	75-28-5	58.122	-13.499	-2.144	2.955	-2.64895
237	2-Methyl-2-propanol	C ₄ H ₁₀ O	75-65-0	74.122	-31.24	-17.76	3.263	-2.4239

238	2-Methyl propene	C ₄ H ₈	115-11-7	56.106	-1.71	5.808	2.9309	-2.5242
239	Methyl propionate	C ₄ H ₈ O ₂	554-12-1	88.105	-42.75	-31.1	3.596	-2.078
240	Methylpropyl ether	C ₄ H ₁₀ O	557-17-5	74.122	-23.82	-11.1	3.52	-2.51739
241	Methylpropyl sulfide	C ₄ H ₁₀ S	3877-15-4	90.187	-8.23	1.793	3.717	-2.962
242	Methylsilane	CH ₃ Si	992-94-9	46.144	-2.91	1.853	2.565	-1.999
243	alpha-Methyl styrene	C ₉ H ₁₀	98-83-9	118.176	11.83	21.73	3.725	-4.8214
244	Methyl <i>tert</i> -butyl ether	C ₅ H ₁₂ O	1634-04-4	88.148	-28.32	-11.7	3.578	-3.105
245	Methyl vinyl ether	C ₃ H ₆ O	107-25-5	58.079	-10.8	-4.73	3.08	-1.77431
246	Naphthalene	C ₁₀ H ₈	91-20-3	128.171	15.058	22.408	3.3315	-4.9809
247	Neon	Ne	7440-01-9	20.180	0	0	1.46219	0
248	Nitroethane	C ₂ H ₅ NO ₂	79-24-3	75.067	-10.21	-0.6125	3.168	-1.25
249	Nitrogen	N ₂	7727-37-9	28.013	0	0	1.915	
250	Nitrogen trifluoride	F ₃ N	7783-54-2	71.002	-13.2089	-9.06	2.6062	
251	Nitromethane	CH ₃ NO ₂	75-52-5	61.040	-7.47	-0.6934	2.751	-0.6432
252	Nitrous oxide	N ₂ O	10024-97-2	44.013	8.205	10.416	2.1985	-0.0820482
253	Nitric oxide	NO	10102-43-9	30.006	9.025	8.657	2.106	-0.0902489
254	Nonadecane	C ₁₉ H ₄₀	629-92-5	268.521	-43.579	10.74	8.9866	-11.7812
255	Nonanal	C ₉ H ₁₈ O	124-19-6	142.239	-31.09	-7.553	5.3988	-5.35
256	Nonane	C ₉ H ₂₀	111-84-2	128.255	-22.874	2.498	5.064	-5.68455
257	Nonanoic acid	C ₉ H ₁₈ O ₂	112-05-0	158.238	-57.73	-31.7	5.59	-5.061
258	1-Nonanol	C ₉ H ₂₀ O	143-08-8	144.255	-37.79	-10.86	5.579	-5.506
259	2-Nonanol	C ₉ H ₂₀ O	628-99-9	144.255	-39.71	-12.61	5.523	-5.506
260	1-Nonene	C ₉ H ₁₈	124-11-8	126.239	-10.35	11.23	5.041	-5.5716
261	Nonyl mercaptan	C ₉ H ₂₀ S	1455-21-6	160.320	-19.08	5.28	5.724	-6.006
262	1-Nonyne	C ₉ H ₁₆	3452-09-3	124.223	6.17	24.34	4.8699	-5.493
263	Octadecane	C ₁₈ H ₃₈	593-45-3	254.494	-41.512	9.91	8.5945	-11.1715
264	Octanal	C ₈ H ₁₆ O	124-13-0	128.212	-29.02	-8.377	5.0063	-4.74
265	Octane	C ₈ H ₁₈	111-65-9	114.229	-20.875	1.6	4.6723	-5.07415
266	Octanoic acid	C ₈ H ₁₆ O ₂	124-07-2	144.211	-55.6	-32.5	5.2	-4.448
267	1-Octanol	C ₈ H ₁₈ O	111-87-5	130.228	-35.73	-11.7	5.187	-4.895
268	2-Octanol	C ₈ H ₁₈ O	123-96-6	130.228	-37.62	-13.43	5.132	-4.894
269	2-Octanone	C ₈ H ₁₆ O	111-13-7	128.212	-32.16	-11.38	4.962	-4.6984
270	3-Octanone	C ₈ H ₁₆ O	106-68-3	128.212	-33.9	-12.81	4.879	-4.711
271	1-Octene	C ₈ H ₁₆	111-66-0	112.213	-8.194	10.57	4.637	-4.961
272	Octyl mercaptan	C ₈ H ₁₈ S	111-88-6	146.294	-17.01	4.457	5.331	-5.3962
273	1-Octyne	C ₈ H ₁₄	629-05-0	110.197	8.23	23.5	4.478	-4.88145
274	Oxalic acid	C ₂ H ₂ O ₄	144-62-7	90.035	-72.37	-66.14	3.433	-0.1989
275	Oxygen	O ₂	7782-44-7	31.999	0	0	2.05043	0
276	Ozone	O ₃	10028-15-6	47.998	14.2671	16.3164	2.38823	-0.142671
277	Pentadecane	C ₁₅ H ₃₂	629-62-9	212.415	-35.311	7.426	7.4181	-9.34237
278	Pentanal	C ₅ H ₁₀ O	110-62-3	86.132	-22.78	-10.71	3.8289	-2.91
279	Pentane	C ₅ H ₁₂	109-66-0	72.149	-14.676	-0.8813	3.4945	-3.24494
280	Pentanoic acid	C ₅ H ₁₀ O ₂	109-52-4	102.132	-49.13	-34.7	4.02	-2.617
281	1-Pentanol	C ₅ H ₁₂ O	71-41-0	88.148	-29.57	-14.23	4.01	-3.064
282	2-Pentanol	C ₅ H ₁₂ O	6032-29-7	88.148	-31.37	-15.88	3.958	-3.058
283	2-Pentanone	C ₅ H ₁₀ O	107-87-9	86.132	-25.92	-13.83	3.786	-2.87956
284	3-Pentanone	C ₅ H ₁₀ O	96-22-0	86.132	-25.79	-13.44	3.7	-2.8804
285	1-Pentene	C ₅ H ₁₀	109-67-1	70.133	-2.162	7.837	3.462	-3.13037
286	2-Pentyl mercaptan	C ₅ H ₁₂ S	2084-19-7	104.214	-11.3	1.814	4.05	-3.564
287	Pentyl mercaptan	C ₅ H ₁₂ S	110-66-7	104.214	-10.84	1.94408	4.154	-3.5641
288	1-Pentyne	C ₅ H ₈	627-19-0	68.117	14.44	21.03	3.298	-3.051
289	2-Pentyne	C ₅ H ₈	627-21-4	68.117	12.89	19.45	3.3084	-3.0291
290	Phenanthrene	C ₁₄ H ₁₀	85-01-8	178.229	20.12	30.219	3.945	-6.8282
291	Phenol	C ₆ H ₆ O	108-95-2	94.111	-9.6399	-3.2637	3.1481	-2.921
292	Phenyl isocyanate	C ₇ H ₅ NO	103-71-9	119.121	-1.454	4.87212	3.527	-3.298
293	Phthalic anhydride	C ₈ H ₄ O ₃	85-44-9	148.116	-37.14	-30.7001	3.995	-3.1715
294	Propadiene	C ₃ H ₄	463-49-0	40.064	19.05	20.08	2.439	-1.8563
295	Propane	C ₃ H ₈	74-98-6	44.096	-10.468	-2.439	2.702	-2.04311
296	1-Propanol	C ₃ H ₈ O	71-23-8	60.095	-25.46	-15.99	3.226	-1.844
297	2-Propanol	C ₃ H ₈ O	67-63-0	60.095	-27.21	-17.52	3.175	-1.834
298	Propenylcyclohexene	C ₉ H ₁₄	13511-13-2	122.207	4.677	20.85	4.233	-5.232
299	Propionaldehyde	C ₃ H ₆ O	123-38-6	58.079	-18.63	-12.46	3.044	-1.6857
300	Propionic acid	C ₃ H ₆ O ₂	79-09-4	74.079	-47.99	-38.5	2.949	-1.395

TABLE 2-179 Enthalpies and Gibbs Energies of Formation, Entropies, and Net Enthalpies of Combustion of Inorganic and Organic Compounds at 298.15 K (Concluded)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	Ideal gas enthalpy of formation, J/kmol × 1E-07	Ideal gas Gibbs energy of formation, J/kmol × 1E-07	Ideal gas entropy, J/(kmol·K) × 1E-05	Standard net enthalpy of combustion, J/kmol × 1E-09
301	Propionitrile	C ₃ H ₅ N	107-12-0	55.079	5.18	9.74949	2.8614	-1.8007
302	Propyl acetate	C ₅ H ₁₀ O ₂	109-60-4	102.132	-46.48	-32.04	4.023	-2.672
303	Propyl amine	C ₃ H ₇ N	107-10-8	59.110	-7.05	4.17	3.242	-2.165
304	Propylbenzene	C ₉ H ₁₂	103-65-1	120.192	0.79	13.76	4.0014	-4.95415
305	Propylene	C ₃ H ₆	115-07-1	42.080	2.023	6.264	2.67	-1.9262
306	Propyl formate	C ₄ H ₈ O ₂	110-74-7	88.105	-40.76	-29.36	3.678	-2.041
307	2-Propyl mercaptan	C ₃ H ₆ S	75-33-2	76.161	-7.59	-0.218	3.243	-2.3398
308	Propyl mercaptan	C ₃ H ₇ S	107-03-9	76.161	-6.75	0.2583	3.365	-2.3458
309	1,2-Propylene glycol	C ₃ H ₈ O ₂	57-55-6	76.094	-42.15	-30.4	3.52	-1.6476
310	Quinone	C ₆ H ₄ O ₂	106-51-4	108.095	-12.29	-6.92	3.205	-2.658
311	Silicon tetrafluoride	F ₄ Si	7783-61-1	104.079	-161.494	-157.27	2.82651	0.7055
312	Styrene	C ₈ H ₈	100-42-5	104.149	14.74	21.39	3.451	-4.219
313	Succinic acid	C ₄ H ₆ O ₄	110-15-6	118.088	-82.29	-69.73	4.034	-1.3591
314	Sulfur dioxide	O ₂ S	7446-09-5	64.064	-29.684	-30.012	2.481	
315	Sulfur hexafluoride	F ₆ S	2551-62-4	146.055	-122.047	-111.653	2.91625	0.924
316	Sulfur trioxide	O ₃ S	7446-11-9	80.063	-39.572	-37.095	2.5651	0.1422
317	Terephthalic acid	C ₈ H ₆ O ₄	100-21-0	166.131	-71.79	-59.9	4.48	-3.0576
318	<i>o</i> -Terphenyl	C ₁₈ H ₁₄	84-15-1	230.304	27.66	42.3	5.263	-9.053
319	Tetradecane	C ₁₄ H ₃₀	629-59-4	198.388	-33.244	6.599	7.0259	-8.73282
320	Tetrahydrofuran	C ₄ H ₈ O	109-99-9	72.106	-18.418	-7.969	2.9729	-2.325
321	1,2,3,4-Tetrahydronaphthalene	C ₁₀ H ₁₂	119-64-2	132.202	2.661	16.71	3.6964	-5.3575
322	Tetrahydrothiophene	C ₄ H ₆ S	110-01-0	88.171	-3.376	4.59	3.1	-2.76549
323	2,2,3,3-Tetramethylbutane	C ₈ H ₁₈	594-82-1	114.229	-22.56	2.239	3.893	-5.0639
324	Thiophene	C ₄ H ₄ S	110-02-1	84.140	11.544	12.67	2.784	-2.4352
325	Toluene	C ₇ H ₈	108-88-3	92.138	5.017	12.22	3.2099	-3.734
326	1,1,2-Trichloroethane	C ₂ H ₂ Cl ₃	79-00-5	133.404	-14.2	-8.097	3.371	-0.9685
327	Tridecane	C ₁₃ H ₂₈	629-50-5	184.361	-31.177	5.771	6.6337	-8.1229
328	Triethyl amine	C ₆ H ₁₅ N	121-44-8	101.190	-9.58	11.41	4.054	-4.0405
329	Trimethyl amine	C ₃ H ₉ N	75-50-3	59.110	-2.431	9.899	2.87	-2.2449
330	1,2,3-Trimethylbenzene	C ₉ H ₁₂	526-73-8	120.192	-0.95	12.61	3.805	-4.934
331	1,2,4-Trimethylbenzene	C ₉ H ₁₂	95-63-6	120.192	-1.38	11.71	3.961	-4.9307
332	2,2,4-Trimethylpentane	C ₈ H ₁₈	540-84-1	114.229	-22.401	1.394	4.2296	-5.06528
333	2,3,3-Trimethylpentane	C ₈ H ₁₈	560-21-4	114.229	-21.845	1.828	4.2702	-5.06876
334	1,3,5-Trinitrobenzene	C ₆ H ₃ N ₃ O ₆	99-35-4	213.105	6.24	26.79	4.435	-2.6867
335	2,4,6-Trinitrotoluene	C ₇ H ₅ N ₃ O ₆	118-96-7	227.131	4.34	28.44	4.607	-3.2959
336	Undecane	C ₁₁ H ₂₄	1120-21-4	156.308	-27.043	4.116	5.8493	-6.9036
337	1-Undecanol	C ₁₁ H ₂₄ O	112-42-5	172.308	-41.9	-9.177	6.363	-6.726
338	Vinyl acetate	C ₄ H ₆ O ₂	108-05-4	86.089	-31.49	-22.79	3.28	-1.95
339	Vinyl acetylene	C ₄ H ₄	689-97-4	52.075	30.46	30.6	2.794	-2.362
340	Vinyl chloride	C ₂ H ₃ Cl	75-01-4	62.498	2.845	4.195	2.7354	-1.178
341	Vinyl trichlorosilane	C ₂ H ₃ Cl ₃ Si	75-94-5	161.490	-48.116	-42.5514	3.73966	-1.544
342	Water	H ₂ O	7732-18-5	18.015	-24.1814	-22.859	1.88724	
343	<i>m</i> -Xylene	C ₈ H ₁₀	108-38-3	106.165	1.732	11.876	3.5854	-4.3318
344	<i>o</i> -Xylene	C ₈ H ₁₀	95-47-6	106.165	1.908	12.2	3.5383	-4.333
345	<i>p</i> -Xylene	C ₈ H ₁₀	106-42-3	106.165	1.803	12.14	3.52165	-4.333

The compounds are considered to be formed from the elements in their standard states at 298.15 K and 101,325 Pa. These include C (graphite) and S (rhombic). Enthalpy of combustion is the net value for the compound in its standard state at 298.15 K and 101,325 Pa. Products of combustion are taken to be CO₂ (gas), H₂O (gas), Cl₂ (gas), Br₂ (gas), I₂ (gas), SO₂ (gas), N₂ (gas), P₄O₁₀ (crystalline), SiO₂ (cristobalite), and Al₂O₃ (crystal, alpha).

Values in this table were taken from the Design Institute for Physical Properties (DIPPR) of the American Institute of Chemical Engineers (AIChE), copyright 2007 AIChE and reproduced with permission of AIChE and of the DIPPR Evaluated Process Design Data Project Steering Committee. Their source should be cited as R. L. Rowley, W. V. Wilding, J. L. Oscarson, Y. Yang, N. A. Zundel, T. E. Daubert, R. P. Danner, DIPPR® Data Compilation of Pure Chemical Properties, Design Institute for Physical Properties, AIChE, New York (2007).

TABLE 2-180 Ideal Gas Sensible Enthalpies, $h_T - h_{298}$ (kJ/kmol), of Combustion Products

Temperature, K	CO	CO ₂	H	OH	H ₂	N	NO	NO ₂	N ₂	N ₂ O	O	O ₂	SO ₂	H ₂ O
200	-2858	-3414	-2040	-2976	-2774	-2040	-2951	-3495	-2857	-3553	-2186	-2868	-3736	-3282
240	-1692	-2079	-1209	-1756	-1656	-1209	-1743	-2104	-1692	-2164	-1285	-1703	-2258	-1948
260	-1110	-1383	-793	-1150	-1091	-793	-1142	-1392	-1110	-1438	-840	-1118	-1496	-1279
280	-529	-665	-377	-546	-522	-378	-543	-672	-528	-692	-398	-533	-718	-609
298.15	0	0	0	0	0	0	0	0	0	0	0	0	0	0
300	54	69	38	55	53	38	55	68	54	72	41	54	74	62
320	638	823	454	654	630	454	652	816	636	854	478	643	881	735
340	1221	1594	870	1251	1209	870	1248	1571	1219	1654	913	1234	1702	1410
360	1805	2382	1285	1847	1791	1286	1845	2347	1802	2470	1346	1828	2538	2088
380	2389	3184	1701	2442	2373	1701	2442	3130	2386	3302	1777	2425	3387	2769
400	2975	4003	2117	3035	2959	2117	3040	3927	2971	4149	2207	3025	4250	3452
420	3563	4835	2532	3627	3544	2533	3638	4735	3557	5010	2635	3629	5126	4139
440	4153	5683	2948	4219	4131	2949	4240	5557	4143	5584	3063	4236	6015	4829
460	4643	6544	3364	4810	4715	3364	4844	6392	4731	6771	3490	4847	6917	5523
480	5335	7416	3779	5401	5298	3780	5450	7239	5320	7670	3918	5463	7831	6222
500	5931	8305	4196	5992	5882	4196	6059	8099	5911	8580	4343	6084	8758	6925
550	7428	10572	5235	7385	6760	5235	7592	10340	7395	10897	5402	7653	11123	8699
600	8942	12907	6274	8943	8811	6274	9144	12555	8894	13295	6462	9244	13544	10501
650	10477	15303	7314	10423	10278	7314	10716	14882	10407	15744	7515	10859	16022	12321
700	12023	17754	8353	11902	11749	8353	12307	17250	11937	18243	8570	12499	18548	14192
750	13592	20260	9392	13391	13223	9329	13919	19671	13481	20791	9620	14158	21117	16082
800	15177	22806	10431	14880	14702	10431	15548	22136	15046	23383	10671	15835	23721	18002
850	16781	25398	11471	16384	16186	11471	17195	24641	16624	26014	11718	17531	26369	19954
900	18401	28030	12510	17888	17676	12510	18858	27179	18223	28681	12767	19241	29023	21938
950	20031	30689	13550	19412	19175	13550	20537	29749	19834	31381	13812	20965	31714	23954
1000	21690	33397	14589	20935	20680	14589	22229	32344	21463	34110	14860	22703	34428	26000
1100	25035	38884	16667	24024	23719	16667	25653	37605	24760	39647	16950	26212	39914	30191
1200	28430	44473	18746	27160	26797	18746	29120	42946	28109	45274	19039	29761	45464	34506
1300	31868	50148	20824	30342	29918	20824	32626	48351	31503	50976	21126	33344	51069	38942
1400	35343	55896	22903	33569	33082	22903	36164	53808	34936	56740	23212	36957	56718	43493
1500	38850	61705	24982	36839	36290	24982	39729	59309	38405	62557	25296	40599	62404	48151
1600	42385	67569	27060	40151	39541	27060	43319	64846	41904	68420	27381	44266	68123	52908
1700	45945	73480	29139	43502	42835	29139	46929	70414	45429	74320	29464	47958	73870	57758
1800	49526	79431	31217	46889	46169	31218	50557	76007	48978	80254	31547	51673	79642	62693
1900	53126	85419	33296	50310	49541	33296	54201	81624	52548	86216	33630	55413	85436	67706
2000	56744	91439	35375	53762	52951	35375	57859	87259	56137	92203	35713	59175	91250	72790
2100	60376	97488	37453	57243	56397	37454	61530	92911	59742	98212	37796	62961	97081	77941
2200	64021	103562	39532	60752	59876	39534	65212	98577	63361	104240	39878	66769	102929	83153
2300	67683	109660	41610	64285	63387	41614	68904	104257	66995	110284	41962	70600	108792	88421
2400	71324	115779	43689	67841	66928	43695	72606	109947	70640	116344	44045	74453	114669	93741
2500	74985	121917	45768	71419	70498	45777	76316	115648	74296	122417	46130	78328	120559	99108
2600	78673	128073	47846	75017	74096	47860	80034	121357	77963	128501	48216	82224	126462	104520
2700	82369	134246	49925	78633	77720	49945	83759	127075	81639	134596	50303	86141	132376	109973
2800	86074	140433	52004	82267	81369	52033	87491	132799	85323	140701	52391	90079	138302	115464
2900	89786	146636	54082	85918	85043	54124	91229	138530	89015	146814	54481	94036	144238	120990
3000	93504	152852	56161	89584	88740	56218	94973	144267	92715	152935	56574	98013	150184	126549
3500	112185	184109	66554	108119	107555	66769	113768	173020	111306	183636	67079	118165	180057	154768
4000	130989	215622	75947	126939	126874	77532	132671	201859	130027	214453	77675	188705	210145	183552
4500	149895	247354	87340	145991	146660	88614	151662	230756	148850	245348	88386	159572	240427	212764
5000	168890	279283	97733	165246	166876	100111	170730	259692	167763	276299	99222	180749	270893	242313

Converted and usually rounded off from JANAF Thermochemical Tables, NSRDS-NBS-37, 1971 (1141 pp.).

2-202 PHYSICAL AND CHEMICAL DATA

TABLE 2-181 Ideal Gas Entropies s° , kJ/(kmol·K), of Combustion Products

Temperature, K	CO	CO ₂	H	OH	H ₂	N	NO	NO ₂	N ₂	N ₂ O	O	O ₂	SO ₂	H ₂ O
200	186.0	200.0	106.4	171.6	119.4	145.0	198.7	225.9	180.0	205.6	152.2	193.5	233.0	175.5
240	191.3	206.0	110.1	177.1	124.5	148.7	204.1	232.2	185.2	211.9	156.2	198.7	239.9	181.4
260	193.7	208.8	111.8	179.5	126.8	150.4	206.6	235.0	187.6	214.8	158.0	201.1	242.8	184.1
280	195.3	211.5	113.3	181.8	129.2	151.9	208.8	237.7	189.8	217.5	159.7	203.3	245.8	186.6
298.15	197.7	213.8	114.7	183.7	130.7	153.3	210.8	240.0	191.6	220.0	161.1	205.1	248.2	188.8
300	197.8	214.0	114.8	183.9	130.9	153.4	210.9	240.3	191.8	220.2	161.2	205.3	248.5	189.0
320	199.7	216.5	116.2	185.9	132.8	154.8	212.9	242.7	193.7	222.7	162.6	207.2	251.1	191.2
340	201.5	218.8	117.4	187.7	134.5	156.0	214.7	245.0	195.5	225.2	163.9	209.0	253.6	193.3
360	203.2	221.0	118.6	189.4	136.2	157.2	216.4	247.2	197.2	227.5	165.2	210.7	256.0	195.2
380	204.7	223.2	119.7	191.0	137.7	158.3	218.0	249.3	198.7	229.7	166.3	212.5	258.2	197.1
400	206.2	225.3	120.8	192.5	139.2	159.4	219.5	251.3	200.2	231.9	167.4	213.8	260.4	198.8
420	207.7	227.3	121.8	194.0	140.6	160.4	221.0	253.2	201.5	234.0	168.4	215.3	262.5	200.5
440	209.0	229.3	122.8	195.3	141.9	161.4	222.3	255.1	202.9	236.0	169.4	216.7	264.6	202.0
460	210.4	231.2	123.7	196.6	143.2	162.3	223.7	257.0	204.2	238.0	170.4	218.0	266.6	203.6
480	211.6	233.1	124.6	197.9	144.5	163.1	225.0	258.8	205.5	239.9	171.3	219.4	268.5	205.1
500	212.8	234.9	125.5	199.1	145.7	164.0	226.3	260.6	206.7	241.8	172.2	220.7	270.5	206.5
550	215.7	239.2	127.5	201.8	148.6	166.0	229.1	264.7	209.4	246.2	174.2	223.7	274.9	210.5
600	218.3	243.3	129.3	204.4	151.1	167.8	231.9	268.8	212.2	250.4	176.1	226.5	279.2	213.1
650	220.8	247.1	131.0	206.8	153.4	169.4	234.4	272.6	214.6	254.3	177.7	229.1	283.1	215.9
700	223.1	250.8	132.5	209.0	155.6	171.0	236.8	276.0	216.9	258.0	179.3	231.5	286.9	218.7
750	225.2	255.4	133.9	211.1	157.6	172.5	239.0	279.3	219.0	261.5	180.7	233.7	290.4	221.3
800	227.3	257.5	135.2	213.0	159.5	173.8	241.1	282.5	221.0	264.8	182.1	235.9	293.8	223.8
850	229.2	260.6	136.4	214.8	161.4	175.1	243.0	285.5	223.0	268.0	183.4	237.9	297.0	226.2
900	231.1	263.6	137.7	216.5	163.1	176.3	245.0	288.4	224.8	271.1	184.6	239.9	300.1	228.5
950	232.8	266.5	138.8	218.1	164.7	177.4	246.8	291.3	226.5	274.0	185.7	241.8	303.0	230.6
1000	234.5	269.3	139.9	219.7	166.2	178.5	248.4	293.9	228.2	276.8	186.8	243.6	305.8	232.7
1100	237.7	274.5	141.9	222.7	169.1	180.4	251.8	298.9	231.3	282.1	188.8	246.9	311.0	236.7
1200	240.7	279.4	143.7	225.4	171.8	182.2	254.8	303.6	234.2	287.0	190.6	250.0	315.8	240.5
1300	243.4	283.9	145.3	228.0	174.3	183.9	257.6	307.9	236.9	291.5	192.3	252.9	320.3	244.0
1400	246.0	288.2	146.9	230.3	176.6	185.4	260.2	311.9	239.5	295.8	193.8	255.6	324.5	247.4
1500	248.4	292.2	148.3	232.6	178.8	186.9	262.7	315.7	241.9	299.8	195.3	258.1	328.4	250.6
1600	250.7	296.0	149.6	234.7	180.9	188.2	265.0	319.3	244.1	303.6	196.6	260.4	332.1	253.7
1700	252.9	299.6	150.9	236.8	182.9	189.5	267.2	322.7	246.3	307.2	197.9	262.7	335.6	256.6
1800	254.9	303.0	152.1	238.7	184.8	190.7	269.3	325.9	248.3	310.6	199.1	264.8	338.9	259.5
1900	256.8	306.2	153.2	240.6	186.7	191.8	271.3	328.9	250.2	313.8	200.2	266.8	342.0	262.2
2000	258.7	309.3	154.3	242.3	188.4	192.9	273.1	331.8	252.1	316.9	201.3	268.7	345.0	264.8
2100	260.5	312.2	155.3	244.0	190.1	193.9	274.9	334.5	253.8	319.8	202.3	270.6	347.9	267.3
2200	262.2	315.1	156.3	245.7	191.7	194.8	276.6	337.2	255.5	322.6	203.2	272.4	350.6	269.7
2300	263.8	317.8	157.2	247.2	193.3	195.8	278.3	339.7	257.1	325.3	204.2	274.1	353.2	272.0
2400	265.4	320.4	158.1	248.7	194.8	196.7	279.8	342.1	258.7	327.9	205.0	275.7	355.7	274.3
2500	266.9	322.9	158.9	250.2	196.2	197.5	281.4	344.5	260.2	330.4	205.9	277.3	358.1	276.5
2600	268.3	325.3	159.7	251.6	197.7	198.3	282.8	346.7	261.6	332.7	206.7	278.8	360.4	278.6
2700	269.7	327.6	160.5	253.0	199.0	199.1	284.2	348.9	263.0	335.0	207.5	280.3	362.6	280.7
2800	271.0	329.9	161.3	254.3	200.3	199.9	285.6	350.9	264.3	337.3	208.3	281.7	364.8	282.7
2900	272.3	332.1	162.0	255.6	201.6	200.6	286.9	352.9	265.6	339.4	209.0	283.1	366.9	284.6
3000	273.6	334.2	162.7	256.8	202.9	201.3	288.2	354.9	266.9	341.5	209.7	284.4	368.9	286.5
3500	279.4	343.8	165.9	262.5	208.7	204.6	294.0	363.8	272.6	350.9	212.9	290.7	378.1	295.2
4000	284.4	352.2	168.7	267.6	213.8	207.4	299.0	371.5	277.6	359.2	215.8	296.2	386.1	302.9
4500	288.8	359.7	171.1	272.1	218.5	210.1	303.5	378.3	282.1	366.5	218.3	301.1	393.3	309.8
5000	292.8	366.4	173.3	276.1	222.8	212.5	307.5	384.4	286.0	373.0	220.6	305.5	399.7	316.0

Usually rounded off from JANAF Thermochemical Tables, NSRDS-NBS-37, 1971 (1141 pp.). Equilibrium constants can be calculated by combining Δh_f° values from Table 2-179, $h_T - h_{298}$ from Table 2-180, and s° values from the above, using the formula $\ln K_p = -\Delta G/(RT)$, where $\Delta G = \Delta h_f^\circ + (h_T - h_{298}) - T^\circ s^\circ$.

HEATS OF SOLUTION

TABLE 2-182 Heats of Solution of Inorganic Compounds in Water

Heat evolved, in kilocalories per gram formula weight, on solution in water at 18 °C. Computed from data in Bichowsky and Rossini, *Thermochemistry of Chemical Substances*, Reinhold, New York, 1936.

Substance	Dilution*	Formula	Heat, kcal/mol	Substance	Dilution*	Formula	Heat, kcal/mol
Aluminum bromide	aq	AlBr ₃	+85.3	Calcium—(Cont.)			
chloride	600	AlCl ₃	+77.9	bromide	∞	CaBr ₂	+24.86
	600	AlCl ₃ ·6H ₂ O	+13.2		∞	CaBr ₂ ·6H ₂ O	−0.9
fluoride	aq	AlF ₃	+31	chloride	∞	CaCl ₂	+4.9
	aq	AlF ₃ ·½H ₂ O	+19.0		∞	CaCl ₂ ·H ₂ O	+12.3
	aq	AlF ₃ ·3½H ₂ O	−1.7		∞	CaCl ₂ ·2H ₂ O	+12.5
iodide	aq	AlI ₃	+89.0		∞	CaCl ₂ ·4H ₂ O	+2.4
sulfate	aq	Al ₂ (SO ₄) ₃	+126		∞	CaCl ₂ ·6H ₂ O	−4.11
	aq	Al ₂ (SO ₄) ₃ ·6H ₂ O	+56.2	formate	400	Ca(CHO ₂) ₂	+0.7
	aq	Al ₂ (SO ₄) ₃ ·18H ₂ O	+6.7	iodide	∞	CaI ₂	+28.0
Ammonium bromide	aq	NH ₄ Br	−4.45		∞	CaI ₂ ·8H ₂ O	+1.8
chloride	∞	NH ₄ Cl	−3.82	nitrate	∞	Ca(NO ₃) ₂	+4.1
chromate	aq	(NH ₄) ₂ CrO ₄	−5.82		∞	Ca(NO ₃) ₂ ·H ₂ O	+0.7
dichromate	600	(NH ₄) ₂ Cr ₂ O ₇	−12.9		∞	Ca(NO ₃) ₂ ·2H ₂ O	−3.2
iodide	aq	NH ₄ I	−3.56		∞	Ca(NO ₃) ₂ ·3H ₂ O	−4.2
nitrate	∞	NH ₄ NO ₃	−6.47		∞	Ca(NO ₃) ₂ ·4H ₂ O	−7.99
perborate	aq	NH ₄ BO ₃ ·H ₂ O	−9.0	phosphate, mono-	aq	Ca(H ₂ PO ₄) ₂ ·H ₂ O	−0.6
sulfate	∞	(NH ₄) ₂ SO ₄	−2.75	dibasic	aq	CaHPO ₄ ·2H ₂ O	−1
sulfate, acid	800	NH ₄ HSO ₄	+0.56	sulfate	∞	CaSO ₄	+5.1
sulfite	aq	(NH ₄) ₂ SO ₃	−1.2		∞	CaSO ₄ ·½H ₂ O	+3.6
	aq	(NH ₄) ₂ SO ₃ ·H ₂ O	−4.13		∞	CaSO ₄ ·2H ₂ O	−0.18
Antimony fluoride	aq	SbF ₃	−1.7	Chromous chloride	aq	CrCl ₂	+18.6
iodide	aq	SbI ₃	−0.8			CrCl ₂ ·3H ₂ O	+5.3
Arsenic acid	aq	H ₃ AsO ₄	−0.4			CrCl ₂ ·4H ₂ O	+2.0
				iodide	aq	CrI ₂	+5.7
Barium bromate	∞	Ba(BrO ₃) ₂ ·H ₂ O	−15.9	Cobaltous bromide	aq	CoBr ₂	+18.4
bromide	∞	BaBr ₂	+5.3		aq	CoBr ₂ ·6H ₂ O	−1.25
	∞	BaBr ₂ ·H ₂ O	−0.8	chloride	400	CoCl ₂	+18.5
	∞	BaBr ₂ ·2H ₂ O	−3.87		400	CoCl ₂ ·2H ₂ O	+9.8
chlorate	∞	Ba(ClO ₃) ₂	−6.7		400	CoCl ₂ ·6H ₂ O	−2.9
	∞	Ba(ClO ₃) ₂ ·H ₂ O	−10.6	iodide	aq	CoI ₂	+18.8
chloride	∞	BaCl ₂	+2.4	sulfate	400	CoSO ₄	+15.0
	∞	BaCl ₂ ·H ₂ O	−2.17		400	CoSO ₄ ·6H ₂ O	−1.4
	∞	BaCl ₂ ·2H ₂ O	−4.5		400	CoSO ₄ ·7H ₂ O	−3.6
cyanide	aq	Ba(CN) ₂	+1.5	Cupric acetate	aq	Cu(C ₂ H ₃ O ₂) ₂	+2.4
	aq	Ba(CN) ₂ ·H ₂ O	−2.4	formate	aq	Cu(CHO ₂) ₂	+0.5
	aq	Ba(CN) ₂ ·2H ₂ O	−4.9	nitrate	200	Cu(NO ₃) ₂	+10.3
iodate	∞	Ba(IO ₃) ₂	−9.1		200	Cu(NO ₃) ₂ ·3H ₂ O	−2.6
	∞	Ba(IO ₃) ₂ ·H ₂ O	−11.3		200	Cu(NO ₃) ₂ ·6H ₂ O	−10.7
iodide	∞	BaI ₂	+10.5	sulfate	800	CuSO ₄	+15.9
	∞	BaI ₂ ·H ₂ O	+2.7			CuSO ₄ ·H ₂ O	+9.3
	∞	BaI ₂ ·2H ₂ O	+0.14			CuSO ₄ ·3H ₂ O	+3.65
	∞	BaI ₂ ·2½H ₂ O	−0.58			CuSO ₄ ·5H ₂ O	−2.85
	∞	BaI ₂ ·7H ₂ O	−6.61	Cuprous sulfate	aq	Cu ₂ SO ₄	+11.6
nitrate	∞	Ba(NO ₃) ₂	−10.2				
perchlorate	∞	Ba(ClO ₄) ₂	−2.8	Ferric chloride	1000	FeCl ₃	+31.7
	∞	Ba(ClO ₄) ₂ ·3H ₂ O	−10.5		1000	FeCl ₃ ·2½H ₂ O	+21.0
sulfide	∞	BaS	+7.2		1000	FeCl ₃ ·6H ₂ O	+5.6
Beryllium bromide	aq	BeBr ₂	+62.6	nitrate	800	Fe(NO ₃) ₃ ·9H ₂ O	−9.1
chloride	aq	BeCl ₂	+51.1	Ferrous bromide	aq	FeBr ₂	+18.0
iodide	aq	BeI ₂	+72.6	chloride	400	FeCl ₂	+17.9
sulfate	aq	BeSO ₄	+18.1		400	FeCl ₂ ·2H ₂ O	+8.7
	aq	BeSO ₄ ·H ₂ O	+13.5		400	FeCl ₂ ·4H ₂ O	+2.7
	aq	BeSO ₄ ·2H ₂ O	+7.9	iodide	aq	FeI ₂	+23.3
	aq	BeSO ₄ ·4H ₂ O	+1.1	sulfate	400	FeSO ₄	+14.7
Bismuth iodide	aq	BiI ₃	+3		400	FeSO ₄ ·H ₂ O	+7.35
Boric acid	aq	H ₃ BO ₃	−5.4		400	FeSO ₄ ·4H ₂ O	+1.4
					400	FeSO ₄ ·7H ₂ O	−4.4
Cadmium bromide	400	CdBr ₂	+0.4	Lead acetate	400	Pb(C ₂ H ₃ O ₂) ₂	+1.4
	400	CdBr ₂ ·4H ₂ O	−7.3		400	Pb(C ₂ H ₃ O ₂) ₂ ·3H ₂ O	−5.9
chloride	400	CdCl ₂	+3.1	bromide	aq	PbBr ₂	−10.1
	400	CdCl ₂ ·H ₂ O	+0.6	chloride	aq	PbCl ₂	−3.4
	400	CdCl ₂ ·2½H ₂ O	−3.00	formate	aq	Pb(CHO ₂) ₂	−6.9
nitrate	400	Cd(NO ₃) ₂ ·H ₂ O	+4.17	nitrate	400	Pb(NO ₃) ₂	−7.61
	400	Cd(NO ₃) ₂ ·4H ₂ O	−5.08	Lithium bromide	∞	LiBr	+11.54
sulfate	400	CdSO ₄	+10.69		∞	LiBr·H ₂ O	+5.30
	400	CdSO ₄ ·H ₂ O	+6.05		∞	LiBr·2H ₂ O	+2.05
	400	CdSO ₄ ·2½H ₂ O	+2.51		∞	LiBr·3H ₂ O	−1.59
Calcium acetate	∞	Ca(C ₂ H ₃ O ₂) ₂	+7.6	chloride	∞	LiCl	+8.66
	∞	Ca(C ₂ H ₃ O ₂) ₂ ·H ₂ O	+6.5				

*The numbers represent moles of water used to dissolve 1 g formula weight of substance; ∞ means "infinite dilution"; and aq means "aqueous solution of unspecified dilution."

TABLE 2-182 Heats of Solution of Inorganic Compounds in Water (Continued)

Substance	Dilution°	Formula	Heat, kcal/mol	Substance	Dilution°	Formula	Heat, kcal/mol
Lithium—(Cont.)				Phosphoric acid, ortho-	400	H ₃ PO ₄	+2.79
	∞	LiCl·H ₂ O	+4.45		400	H ₃ PO ₄ ·½H ₂ O	−0.1
	∞	LiCl·2H ₂ O	+1.07	pyro-	aq	H ₄ P ₂ O ₇	+25.9
fluoride	∞	LiCl·3H ₂ O	−1.98		aq	H ₄ P ₂ O ₇ ·1½H ₂ O	+4.65
hydroxide	∞	LiF	−0.74	Potassium acetate	∞	KC ₂ H ₃ O ₂	+3.55
	∞	LiOH	+4.74	aluminum sulfate	600	KAl(SO ₄) ₂	+48.5
	∞	LiOH·½H ₂ O	+4.39		600	KAl(SO ₄) ₂ ·3H ₂ O	+26.6
	∞	LiOH·H ₂ O	+9.6			KAl(SO ₄) ₂ ·12H ₂ O	−10.1
iodide	∞	LiI	+14.92	bicarbonate	2000	KHCO ₃	−5.1
	∞	LiI·½H ₂ O	+10.08	bromate	∞	KBrO ₃	−10.13
	∞	LiI·H ₂ O	+6.93	bromide	∞	KBr	−5.13
	∞	LiI·2H ₂ O	+3.43	carbonate	∞	K ₂ CO ₃	+6.58
	∞	LiI·3H ₂ O	−0.17			K ₂ CO ₃ ·½H ₂ O	+4.25
nitrate	∞	LiNO ₃	+0.466			K ₂ CO ₃ ·1½H ₂ O	−0.43
	∞	LiNO ₃ ·3H ₂ O	−7.87	chlorate	∞	KClO ₃	−10.31
sulfate	∞	Li ₂ SO ₄	+6.71	chloride	∞	KCl	−4.404
	∞	Li ₂ SO ₄ ·H ₂ O	+3.77	chromate	2185	K ₂ CrO ₄	−4.9
Magnesium bromide	∞	MgBr ₂	+43.7	chrome sulfate	600	KCr(SO ₄) ₂	+55
	∞	MgBr ₂ ·H ₂ O	+35.9			KCr(SO ₄) ₂ ·H ₂ O	+42
	∞	MgBr ₂ ·6H ₂ O	+19.8			KCr(SO ₄) ₂ ·2H ₂ O	+33
chloride	∞	MgCl ₂	+36.3			KCr(SO ₄) ₂ ·6H ₂ O	+7
	∞	MgCl ₂ ·2H ₂ O	+20.8			KCr(SO ₄) ₂ ·12H ₂ O	−9.5
	∞	MgCl ₂ ·4H ₂ O	+10.5	cyanide	200	KCN	−3.0
	∞	MgCl ₂ ·6H ₂ O	+3.4	dichromate	1600	K ₂ Cr ₂ O ₇	−17.8
iodide	∞	MgI ₂	+50.2	fluoride	∞	KF	+3.96
nitrate	∞	Mg(NO ₃) ₂ ·6H ₂ O	−3.7		∞	KF·2H ₂ O	−1.85
phosphate	aq	Mg ₃ (PO ₄) ₂	+10.2	hydrosulfide	∞	KHS	+0.86
sulfate	∞	MgSO ₄	+21.1		∞	KHS·½H ₂ O	+1.21
	∞	MgSO ₄ ·H ₂ O	+14.0	hydroxide	∞	KOH	+12.91
	∞	MgSO ₄ ·2H ₂ O	+11.7		∞	KOH·¾H ₂ O	+4.27
	∞	MgSO ₄ ·4H ₂ O	+4.9		∞	KOH·H ₂ O	+3.48
	∞	MgSO ₄ ·6H ₂ O	+0.55		∞	KOH·7H ₂ O	+0.86
	∞	MgSO ₄ ·7H ₂ O	−3.18	iodate	∞	KIO ₃	−6.93
sulfide	aq	MgS	+25.8	iodide	∞	KI	−5.23
Manganic nitrate	400	Mn(NO ₃) ₂	+12.9	nitrate	∞	KNO ₃	−8.633
	400	Mn(NO ₃) ₂ ·3H ₂ O	−3.9	oxalate	400	K ₂ C ₂ O ₄	−4.6
	400	Mn(NO ₃) ₂ ·6H ₂ O	−6.2			K ₂ C ₂ O ₄ ·H ₂ O	−7.5
sulfate	aq	Mn ₂ (SO ₄) ₃	+22	perchlorate	∞	KClO ₄	−12.94
Manganous acetate	aq	Mn(C ₂ H ₃ O ₂) ₂	+12.2	permanganate	400	KMnO ₄	−10.4
	aq	Mn(C ₂ H ₃ O ₂) ₂ ·4H ₂ O	+1.6	phosphate, dihydrogen	aq	KH ₂ PO ₄	+4.7
bromide	aq	MnBr ₂	+15	pyrosulfite	aq	K ₂ S ₂ O ₅	−11.0
	aq	MnBr ₂ ·H ₂ O	+14.4		aq	K ₂ S ₂ O ₅ ·½H ₂ O	−10.22
	aq	MnBr ₂ ·4H ₂ O	+16.1	sulfate	∞	K ₂ SO ₄	−6.32
chloride	400	MnCl ₂	+16.0	sulfate, acid	800	KHSO ₄	−3.10
	400	MnCl ₂ ·2H ₂ O	+8.2	sulfide	∞	K ₂ S	−11.0
	400	MnCl ₂ ·4H ₂ O	+1.5	sulfite	aq	K ₂ SO ₃	+1.8
formate	aq	Mn(CHO ₂) ₂	+4.3		aq	K ₂ SO ₃ ·H ₂ O	+1.37
	aq	Mn(CHO ₂) ₂ ·2H ₂ O	−2.9	thiocyanate	∞	KCNS	−6.08
iodide	aq	MnI ₂	+26.2	thionate, di-	aq	K ₂ S ₂ O ₆	−13.0
	aq	MnI ₂ ·H ₂ O	+24.1	thiosulfate	∞	K ₂ S ₂ O ₃	−4.5
	aq	MnI ₂ ·2H ₂ O	+22.7				
	aq	MnI ₂ ·4H ₂ O	+19.9	Silver acetate	aq	AgC ₂ H ₃ O ₂	−5.4
	aq	MnI ₂ ·6H ₂ O	+21.2	nitrate	200	AgNO ₃	−4.4
sulfate	400	MnSO ₄	+13.8	Sodium acetate	∞	NaC ₂ H ₃ O ₂	+4.085
	400	MnSO ₄ ·H ₂ O	+11.9		∞	NaC ₂ H ₃ O ₂ ·3H ₂ O	−4.665
	400	MnSO ₄ ·7H ₂ O	−1.7	arsenate	500	Na ₃ AsO ₄	+15.6
Mercuric acetate	aq	Hg(C ₂ H ₃ O ₂) ₂	−4.0		500	Na ₃ AsO ₄ ·12H ₂ O	−12.61
bromide	aq	HgBr ₂	−2.4	bicarbonate	1800	NaHCO ₃	−4.1
chloride	aq	HgCl ₂	−3.3	borate, tetra-	900	Na ₂ B ₄ O ₇	+10.0
nitrate	aq	Hg(NO ₃) ₂ ·½H ₂ O	−0.7		900	Na ₂ B ₄ O ₇ ·10H ₂ O	−16.8
Mercurous nitrate	aq	Hg ₂ (NO ₃) ₂ ·2H ₂ O	−11.5	bromide	∞	NaBr	−0.58
					∞	NaBr·2H ₂ O	−4.57
Nickel bromide	aq	NiBr ₂	+19.0	carbonate	∞	Na ₂ CO ₃	+5.57
	aq	NiBr ₂ ·3H ₂ O	+0.2		∞	Na ₂ CO ₃ ·H ₂ O	+2.19
Nickel chloride	800	NiCl ₂	+19.23		∞	Na ₂ CO ₃ ·7H ₂ O	−10.81
	800	NiCl ₂ ·2H ₂ O	+10.4		∞	Na ₂ CO ₃ ·10H ₂ O	−16.22
	800	NiCl ₂ ·4H ₂ O	+4.2	chlorate	∞	NaClO ₃	−5.37
	800	NiCl ₂ ·6H ₂ O	−1.15	chloride	∞	NaCl	−1.164
iodide	aq	NiI ₂	+19.4	chromate	800	Na ₂ CrO ₄	+2.50
nitrate	200	Ni(NO ₃) ₂	+11.8		800	Na ₂ CrO ₄ ·4H ₂ O	−7.52
	200	Ni(NO ₃) ₂ ·6H ₂ O	−7.5		800	Na ₂ CrO ₄ ·10H ₂ O	−16.0
sulfate	200	NiSO ₄	+15.1	cyanide	200	NaCN	−0.37
	200	NiSO ₄ ·7H ₂ O	−4.2		200	NaCN·½H ₂ O	−0.92

TABLE 2-182 Heats of Solution of Inorganic Compounds in Water (Concluded)

Substance	Dilution°	Formula	Heat, kcal/mol	Substance	Dilution°	Formula	Heat, kcal/mol
Sodium—(Cont.)	200	NaCN·2H ₂ O	−4.41	Sodium—(Cont.)			
fluoride	∞	NaF	−0.27	thionate, di-	aq	Na ₂ S ₂ O ₆	−5.80
hydrosulfide	∞	NaHS	+4.62		aq	Na ₂ S ₂ O ₆ ·2H ₂ O	−11.86
	∞	NaHS·2H ₂ O	−1.49	Sodium thiosulfate	aq	Na ₂ S ₂ O ₃	+2.0
Sodium hydroxide	∞	NaOH	+10.18		aq	Na ₂ S ₂ O ₃ ·5H ₂ O	−11.30
	∞	NaOH·½H ₂ O	+5.17	Stannic bromide	aq	SnBr ₄	+15.5
	∞	NaOH·¾H ₂ O	+7.08	Stannous bromide	aq	SnBr ₂	−1.6
	∞	NaOH·¾H ₂ O	+6.48	iodide	aq	SnI ₂	−5.8
	∞	NaOH·H ₂ O	+5.17	Strontium acetate	∞	Sr(C ₂ H ₃ O ₂) ₂	+6.2
iodide	∞	NaI	+1.57		∞	Sr(C ₂ H ₃ O ₂) ₂ ·½H ₂ O	+5.9
	∞	NaI·2H ₂ O	−3.89	bromide	∞	SrBr ₂	+16.4
metaphosphate	600	NaPO ₃	+3.97		∞	SrBr ₂ ·H ₂ O	+9.25
nitrate	∞	NaNO ₃	−5.05		∞	SrBr ₂ ·2H ₂ O	+6.5
nitrite	aq	NaNO ₂	−3.6		∞	SrBr ₂ ·4H ₂ O	+0.4
perchlorate	∞	NaClO ₄	−4.15		∞	SrBr ₂ ·6H ₂ O	−6.1
phosphate di-	1600	Na ₂ HPO ₄	+5.21	chloride	∞	SrCl ₂	+11.54
tri-	1600	Na ₃ PO ₄	+13		∞	SrCl ₂ ·H ₂ O	+6.4
phosphate di-	1600	Na ₃ PO ₄ ·12H ₂ O	−15.3		∞	SrCl ₂ ·2H ₂ O	+2.95
	1600	Na ₂ HPO ₄ ·2H ₂ O	−0.82	iodide	∞	SrCl ₂ ·6H ₂ O	−7.1
	1600	Na ₂ HPO ₄ ·7H ₂ O	−12.04		∞	SrI ₂	+20.7
	1600	Na ₂ HPO ₄ ·12H ₂ O	−23.18		∞	SrI ₂ ·H ₂ O	+12.65
phosphite, mono-	600	NaH ₂ PO ₃	+0.90		∞	SrI ₂ ·2H ₂ O	+10.4
	600	NaH ₂ PO ₃ ·2½H ₂ O	−5.29	nitrate	∞	SrI ₂ ·6H ₂ O	−4.5
di-	800	Na ₂ HPO ₃	+9.30		∞	Sr(NO ₃) ₂	−4.8
	800	Na ₂ HPO ₃ ·5H ₂ O	−4.54	sulfate	∞	Sr(NO ₃) ₂ ·4H ₂ O	−12.4
pyrophosphate	1600	Na ₄ P ₂ O ₇	+11.9		∞	SrSO ₄	+0.5
	1600	Na ₄ P ₂ O ₇ ·10H ₂ O	−11.7	Sulfuric acid, pyro-	∞	H ₂ S ₂ O ₇	−18.08
di-	1200	Na ₂ H ₂ P ₂ O ₇	−2.2	Zinc acetate	400	Zn(C ₂ H ₃ O ₂) ₂	+9.8
	1200	Na ₂ H ₂ P ₂ O ₇ ·6H ₂ O	−14.0		400	Zn(C ₂ H ₃ O ₂) ₂ ·H ₂ O	+7.0
sulfate	∞	Na ₂ SO ₄	+0.28		400	Zn(C ₂ H ₃ O ₂) ₂ ·2H ₂ O	+3.9
	∞	Na ₂ SO ₄ ·10H ₂ O	−18.74	bromide	400	ZnBr ₂	+15.0
sulfate, acid	800	NaHSO ₄	+1.74	chloride	400	ZnCl ₂	+15.72
	800	NaHSO ₄ ·H ₂ O	+0.15	iodide	aq	ZnI ₂	+11.6
sulfide	∞	Na ₂ S	+15.2	nitrate	400	Zn(NO ₃) ₂ ·3H ₂ O	−5
	∞	Na ₂ S·4½H ₂ O	+0.09		400	Zn(NO ₃) ₂ ·6H ₂ O	−6.0
	∞	Na ₂ S·5H ₂ O	−6.54	sulfate	400	ZnSO ₄	+18.5
	∞	Na ₂ S·9H ₂ O	−16.65		400	ZnSO ₄ ·H ₂ O	+10.0
sulfite	∞	Na ₂ SO ₃	+2.8		400	ZnSO ₄ ·6H ₂ O	−0.8
	∞	Na ₂ SO ₃ ·7H ₂ O	−11.1		400	ZnSO ₄ ·7H ₂ O	−4.3
thiocyanate	∞	NaCNS	−1.83				

NOTE: To convert kilocalories per mole to British thermal units per pound-mole, multiply by 1.799×10^{-3} .

TABLE 2-183 Heats of Solution of Organic Compounds in Water (at Infinite Dilution and Approximately Room Temperature)Recalculated and rearranged from *International Critical Tables*, vol. 5, pp. 148–150. cal/mol = Btu/(lb·mol) × 1.799.

Solute	Heat of solution, cal/mol solute*	Solute	Heat of solution, cal/mol solute*
Acetic acid (solid), C ₂ H ₄ O ₂	-2,251	Oxalic acid, C ₂ H ₂ O ₄	-2,290
Acetylacetone, C ₈ H ₈ O ₂	-641	(2H ₂ O)	-8,485
Acetylurea, C ₃ H ₆ N ₂ O ₂	-6,812	Phenol (solid), C ₆ H ₆ O	-2,605
Aconitic acid, C ₆ H ₆ O ₆	-4,206	Phthalic acid, C ₈ H ₆ O ₄	-4,871
Ammonium benzoate, C ₇ H ₅ NO ₂	-2,700	Picric acid, C ₆ H ₃ N ₃ O ₇	-7,098
picrate	-8,700	Piperic acid, C ₁₂ H ₁₀ O ₄	-10,492
succinate (<i>n</i> -)	-3,489	Piperonylic acid, C ₈ H ₆ O ₄	-9,106
Aniline, hydrochloride, C ₆ H ₅ ClN	-2,732	Potassium benzoate	-1,506
Barium picrate	-4,708	citrate	2,820
Benzoic acid, C ₇ H ₆ O ₂	-6,501	tartrate (<i>n</i> -) (0.5 H ₂ O)	-5,562
Camphoric acid, C ₁₀ H ₁₆ O ₄	-502	Pyrogallol, C ₆ H ₆ O ₃	-3,705
Citric acid, C ₆ H ₈ O ₇	-5,401	Pyrotartaric acid	-5,019
Dextrin, C ₁₂ H ₂₀ O ₁₀	268	Quinone	-3,991
Fumaric acid, C ₄ H ₄ O ₄	-5,903	Raffinose, C ₁₈ H ₃₂ O ₁₆ (5H ₂ O)	-9,703
Hexamethylenetetramine, C ₆ H ₁₂ N ₄	4,780	Resorcinol, C ₆ H ₆ O ₂	-3,960
Hydroxybenzamide (<i>m</i> -), C ₇ H ₇ NO ₂	-4,161	Silver malonate (<i>n</i> -)	-9,799
(<i>m</i> -), (HCl)	-7,003	Sodium citrate (tri-)	5,270
(<i>o</i> -), C ₇ H ₇ NO ₂	-4,340	picrate	-6,441
(<i>p</i> -)	-5,392	potassium tartrate	-1,817
Hydroxybenzoic acid (<i>o</i> -), C ₇ H ₆ O ₃	-6,350	succinate (<i>n</i> -) (4H ₂ O)	-12,342
(<i>p</i> -), C ₇ H ₆ O ₃	-5,781	tartrate (<i>n</i> -) (6H ₂ O)	2,390
Hydroxybenzyl alcohol (<i>o</i> -), C ₇ H ₈ O ₂	-3,203	tartrate (<i>n</i> -) (2H ₂ O)	-10,994
Inulin, C ₃₆ H ₆₂ O ₃₁	-96	Strontium picrate	-1,121
Isosuccinic acid, C ₄ H ₆ O ₄	-3,420	(6H ₂ O)	-5,882
Itaconic acid, C ₅ H ₆ O ₄	-5,922	Succinic acid, C ₄ H ₆ O ₄	7,887
Lactose, C ₁₂ H ₂₂ O ₁₁ ·H ₂ O	-3,705	Succinimide, C ₄ H ₅ NO ₂	-14,412
Lead picrate	-7,098	Sucrose, C ₁₂ H ₂₂ O ₁₁	-6,405
(2H ₂ O)	-13,193	Tartaric acid (<i>d</i> -)	-4,302
Magnesium picrate	14,699	Thiourea, CH ₄ N ₂ S	-1,319
(8H ₂ O)	-15,894	Urea, CH ₄ N ₂ O	-3,451
Maleic acid, C ₄ H ₄ O ₄	-4,441	acetate	-5,330
Malic acid, C ₄ H ₆ O ₅	-3,150	formate	-3,609
Malonic acid, C ₃ H ₄ O ₄	-4,493	nitrate	-8,795
Mandelic acid, C ₈ H ₈ O ₃	-3,090	oxalate	-7,194
Mannitol, C ₆ H ₁₄ O ₆	-5,260	Vanillic acid	-10,803
Menthol, C ₁₀ H ₂₀ O	0	Vanillin	-17,806
Nicotine dihydrochloride, C ₁₀ H ₁₆ Cl ₂ N ₂	6,561	Zinc picrate	-5,160
Nitrobenzoic acid (<i>m</i> -), C ₇ H ₅ NO ₄	-5,593	(8H ₂ O)	-5,210
(<i>o</i> -), C ₇ H ₅ NO ₄	-5,306		-11,496
(<i>p</i> -), C ₇ H ₅ NO ₄	-8,891		-15,894
Nitrophenol (<i>m</i> -), C ₆ H ₅ NO ₃	-5,210		
(<i>o</i> -), C ₆ H ₅ NO ₃	-6,310		
(<i>p</i> -), C ₆ H ₅ NO ₃	-4,493		

* + denotes heat evolved, and - denotes heat absorbed. The data in the *International Critical Tables* were calculated by E. Anderson.

THERMODYNAMIC PROPERTIES

EXPLANATION OF TABLES

The following subsection presents information on the thermodynamic properties of a number of fluids. In some cases transport properties are also included.

Properties for the compounds listed in Table 2-184 were generated by using the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). Megan Friend's help in generating these tables is acknowledged and gratefully appreciated. The number of digits provided in these tables was chosen for uniformity of appearance and formatting; these do not represent the uncertainties of the physical quantities, but are the result of calculations from the standard thermophysical property formulations within a fixed format.

Properties for many of these compounds can also be generated by going to <http://webbook.nist.gov> and selecting NIST Chemistry WebBook, then Thermophysical Properties of Fluid Systems High Accuracy Data. This site allows the user to generate tables of thermodynamic properties. The user can select the units as well as the temperatures and/or pressure increments for which properties are to be generated. The resulting table can be copied into a spreadsheet. Because of this capability, properties for the compounds listed in Table 2-184 are not tabulated at as many temperatures and pressures as might otherwise be the case.

Notation

- c_p = isobaric specific heat
- c_v = isochoric specific heat
- e = specific internal energy
- h = enthalpy
- k = thermal conductivity
- p = pressure
- s = specific entropy
- t = temperature
- T = absolute temperature
- u = specific internal energy
- μ = viscosity
- v = specific volume
- f = subscript denoting saturated liquid
- g = subscript denoting saturated vapor

UNITS CONVERSIONS

For this subsection, the following units conversions are applicable:

- c_p , specific heat: To convert kilojoules per kilogram-kelvin to British thermal units per pound-degree Fahrenheit, multiply by 0.23885.
- e , internal energy: To convert kilojoules per kilogram to British thermal units per pound, multiply by 0.42992.
- g , gravity acceleration: To convert meters per second squared to feet per second squared, multiply by 3.2808.

h , enthalpy: To convert kilojoules per kilogram to British thermal units per pound, multiply by 0.42992.

k , thermal conductivity: To convert watts per meter-kelvin to British thermal unit-foot per hour-square foot-degree Fahrenheit, multiply by 0.57779.

p , pressure: To convert bars to kilopascals, multiply by 1×10^5 ; to convert bars to pounds-force per square inch, multiply by 14.504; and to convert millimeters of mercury to pounds-force per square inch, multiply by 0.01934.

s , entropy: to convert kilojoules per kilogram-kelvin to British thermal units per pound-degree Rankine, multiply by 0.23885.

t , temperature: $^{\circ}\text{F} = \% ^{\circ}\text{C} + 32$.

T , absolute temperature: $^{\circ}\text{R} = \% \text{K}$.

u , internal energy: to convert kilojoules per kilogram to British thermal units per pound, multiply by 0.42992.

μ , viscosity: to convert pascal-seconds to pound-force-seconds per square foot, multiply by 0.020885; to convert pascal-seconds to c_p , multiply by 1000.

v , specific volume: to convert cubic meters per kilogram to cubic feet per pound, multiply by 16.018.

ρ , density: to convert kilograms per cubic meter to pounds per cubic foot, multiply by 0.062428.

ADDITIONAL REFERENCES

Bretsznajder, *Prediction of Transport and Other Physical Properties of Fluids*, Pergamon, New York, 1971. D'Ans and Lax, *Handbook for Chemists and Physicists* (in German), 3 vols., Springer-Verlag, Berlin. *Engineering Data Book*, Natural Gas Processors Suppliers Association, Tulsa, Okla. Ganic, Hartnett, and Rohsenow, *Handbook of Heat Transfer*, 2d ed., McGraw-Hill, New York, 1984. Gray, *American Institute of Physics Handbook*, 3d ed., McGraw-Hill, New York, 1972. Kay and Laby, *Tables of Physical and Chemical Constants*, Longman, London, various editions and dates. Landolt-Börnstein *Tables*, many volumes and dates, Springer-Verlag, Berlin. Lange, *Handbook of Chemistry*, McGraw-Hill, New York, various editions and dates. Partington, *Advanced Treatise on Physical Chemistry*, 5 vols., Longman, London, 1950. Raznjevic, *Handbook of Thermodynamic Tables and Charts*, McGraw-Hill, New York, 1976 and other editions. Reynolds, *Thermodynamic Properties in SI*, Department of Mechanical Engineering, Stanford University, 1979. Stephan and Lucas, *Viscosity of Dense Fluids*, Plenum, New York and London, 1979. *Selected Values of Properties of Chemical Compounds and Selected Values of the Properties of Hydrocarbons and Related Compounds*, Thermodynamics Research Center, NIST, Boulder, Colo., looseleaf, periodic publication. Vargaftik, *Tables of the Thermophysical Properties of Gases and Liquids*, Wiley, New York, 1975. Vargaftik, Filippov, Tarzimanov, and Totiskiy, *Thermal Conductivity of Liquids and Gases* (in Russian), Standartov, Moscow, 1978. Weast, *Handbook of Chemistry and Physics*, Chemical Rubber Co., Boca Raton, FL, annually.

TABLE 2-184 List of Substances for Which Thermodynamic Property Tables Were Generated from NIST Standard Reference Database 23

Table no.	Name	Chemical formula	Alternate name
2-185	Acetone	C ₃ H ₆ O	
2-187	Air		
2-189	Ammonia	NH ₃	
2-190	Argon	Ar	
2-193	Benzene	C ₆ H ₆	
2-195	Butane	C ₄ H ₁₀	
2-196	1-Butene	C ₄ H ₈	
2-197	<i>cis</i> -2-Butene	C ₄ H ₈	
2-198	<i>trans</i> -2-butene	C ₄ H ₈	
2-199	Carbon dioxide	CO ₂	
2-200	Carbon monoxide	CO	
2-203	Carbonyl sulfide	COS	
2-207	Cyclohexane	C ₆ H ₁₂	
2-208	Decane	C ₁₀ H ₂₂	
2-209	Deuterium oxide	D ₂ O	Heavy water
2-210	2,2-Dimethylpropane	C ₅ H ₁₂	Neopentane
2-212	Dodecane	C ₁₂ H ₂₆	
2-213	Ethane	C ₂ H ₆	
2-214	Ethanol	C ₂ H ₆ O	
2-215	Ethylene	C ₂ H ₄	
2-216	Fluorine	F ₂	
2-219	Helium	He	
2-220	Heptane	C ₇ H ₁₆	
2-221	Hexane	C ₆ H ₁₄	
2-223	Normal hydrogen	H ₂	
2-224	<i>para</i> -Hydrogen	H ₂	
2-226	Hydrogen sulfide	H ₂ S	
2-227	Isobutane	C ₄ H ₁₀	
2-228	Isobutene	C ₄ H ₈	2-methyl propene
2-229	Krypton	Kr	
2-233	Methane	CH ₄	
2-234	Methanol	CH ₄ O	
2-235	2-Methylbutane	C ₅ H ₁₂	Isopentane
2-236	2-Methylpentane	C ₆ H ₁₄	Isohexane
2-238	Neon	Ne	
2-239	Nitrogen	N ₂	
2-241	Nitrogen trifluoride	NF ₃	
2-242	Nitrous oxide	N ₂ O	
2-243	Nonane	C ₉ H ₂₀	
2-244	Octane	C ₈ H ₁₈	
2-245	Oxygen	O ₂	
2-246	Pentane	C ₅ H ₁₂	
2-248	Propane	C ₃ H ₈	
2-249	Propylene	C ₃ H ₆	
2-250	R-11	CCl ₃ F	Trichlorofluoromethane
2-251	R-12	CCl ₂ F ₂	Dichlorodifluoromethane
2-252	R-13	CClF ₃	Chlorotrifluoromethane
2-255	R-22	CHClF ₂	Chlorodifluoromethane
2-256	R-23	CHF ₃	Trifluoromethane
2-257	R-32	CH ₂ F ₂	Difluoromethane
2-258	R-41	CH ₃ F	Fluoromethane
2-271	R-113	C ₂ Cl ₃ F ₃	1,1,2-Trichlorotrifluoroethane
2-272	R-114	C ₂ Cl ₂ F ₄	1,2-Dichlorotetrafluoroethane
2-274	R-116	C ₂ F ₆	Hexafluoroethane
2-275	R-123	C ₂ HF ₅	2,2-Dichloro-1,1,1-trifluoroethane
2-276	R-124	C ₂ HClF ₄	1-Chloro-1,2,2,2-tetrafluoroethane
2-277	R-125	C ₂ HF ₅	Pentafluoroethane
2-278	R-134a	C ₂ H ₂ F ₄	1,1,1,2-Tetrafluoroethane
2-279	R-141b	C ₂ H ₃ ClF ₃	1,1-Dichloro-1-fluoroethane
2-280	R-142b	C ₂ H ₃ ClF ₂	1-Chloro-1,1-difluoroethane
2-281	R-143a	C ₂ H ₃ F ₃	1,1,1-Trifluoroethane
2-282	R-152a	C ₂ H ₄ F ₂	1,1-Difluoroethane
2-284	R-218	C ₃ F ₈	Octafluoropropane
2-285	R227ea	C ₃ HF ₇	1,1,1,2,3,3,3-Heptafluoropropane
2-288	R-404A		
2-289	R-407C		
2-290	R-410A		
2-296	R-507A		
2-300	Sulfur dioxide	SO ₂	
2-301	Sulfur hexafluoride	SF ₆	
2-303	Toluene	C ₇ H ₈	
2-305	Water	H ₂ O	
2-307	Xenon	Xe	

TABLE 2-185 Thermodynamic Properties of Acetone

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule- Thomson K/MPa
Saturated Properties										
178.50	2.3265E-06	15.723	0.063601	0.47366	0.47366	0.0080825	0.082500	0.11544	1765.7	-0.43351
180.00	2.8743E-06	15.695	0.063715	0.64687	0.64687	0.0090488	0.082598	0.11550	1757.0	-0.43308
195.00	1.9454E-05	15.416	0.064868	2.3835	2.3835	0.018316	0.083407	0.11604	1672.3	-0.42849
210.00	9.6588E-05	15.141	0.066048	4.1282	4.1282	0.026935	0.084076	0.11660	1591.8	-0.42274
225.00	0.00037556	14.867	0.067264	5.8823	5.8823	0.035003	0.084758	0.11731	1514.4	-0.41520
240.00	0.0012008	14.593	0.068525	7.6487	7.6488	0.042602	0.085541	0.11825	1439.4	-0.40545
255.00	0.0032765	14.319	0.069840	9.4311	9.4314	0.049806	0.086468	0.11946	1366.3	-0.39322
270.00	0.0078514	14.041	0.071218	11.234	11.234	0.056674	0.087553	0.12094	1294.8	-0.37827
285.00	0.016899	13.760	0.072673	13.060	13.062	0.063259	0.088794	0.12270	1224.5	-0.36033
300.00	0.033259	13.474	0.074217	14.915	14.918	0.069601	0.090180	0.12474	1155.2	-0.33907
315.00	0.060720	13.181	0.075867	16.802	16.807	0.075739	0.091697	0.12704	1086.7	-0.31399
330.00	0.10404	12.880	0.077643	18.725	18.733	0.081702	0.093329	0.12962	1018.8	-0.28437
345.00	0.16891	12.568	0.079569	20.687	20.701	0.087517	0.095063	0.13249	951.24	-0.24915
360.00	0.26188	12.243	0.081677	22.693	22.714	0.093209	0.096886	0.13568	883.84	-0.20678
375.00	0.39033	11.904	0.084008	24.746	24.779	0.098798	0.098794	0.13924	816.36	-0.15495
390.00	0.56235	11.545	0.086616	26.852	26.900	0.10431	0.10078	0.14328	748.57	-0.090162
405.00	0.78681	11.163	0.089578	29.015	29.085	0.10975	0.10286	0.14794	680.21	-0.0069455
420.00	1.0733	10.753	0.093001	31.243	31.343	0.11516	0.10504	0.15350	610.99	0.10371
435.00	1.4324	10.304	0.097051	33.546	33.685	0.12056	0.10736	0.16042	540.51	0.25760
450.00	1.8759	9.8043	0.10200	35.938	36.130	0.12599	0.10986	0.16967	468.19	0.48516
465.00	2.4172	9.2319	0.10832	38.445	38.707	0.13150	0.11265	0.18350	392.99	0.85357
480.00	3.0725	8.5423	0.11706	41.117	41.476	0.13720	0.11600	0.20893	312.66	1.5474
495.00	3.8632	7.6072	0.13145	44.096	44.604	0.14341	0.12077	0.28551	221.66	3.3240
508.10	4.6924	4.7000	0.21277	49.249	50.247	0.15437			0	14.310
178.50	2.3265E-06	1.5677E-06	637.900	36.689	38.173	0.21928	0.050120	0.058440	172.60	3845.4
180.00	2.8743E-06	1.9207E-06	520.660	36.764	38.260	0.21801	0.050280	0.058600	173.29	3637.4
195.00	1.9454E-05	1.2001E-05	83.324	37.528	39.149	0.20686	0.051928	0.060265	179.95	2139.7
210.00	9.6588E-05	5.5355E-05	18.065	38.314	40.059	0.19803	0.053740	0.062119	186.29	1312.0
225.00	0.00037556	0.00020108	4.973.1	39.121	40.989	0.19103	0.055800	0.064267	192.29	834.10
240.00	0.0012008	0.00060385	1.656.0	39.947	41.936	0.18546	0.058169	0.066795	197.94	547.82
255.00	0.0032765	0.0015555	642.89	40.790	42.897	0.18104	0.060883	0.069763	203.19	370.79
270.00	0.0078514	0.0035368	282.74	41.649	43.869	0.17754	0.063945	0.073198	207.99	258.27
285.00	0.016899	0.0072603	137.74	42.522	44.849	0.17479	0.067329	0.077094	212.26	184.97
300.00	0.033259	0.013699	72.996	43.406	45.834	0.17266	0.070988	0.081429	215.93	136.14
315.00	0.060720	0.024107	41.482	44.302	46.821	0.17102	0.074863	0.086172	218.90	102.93
330.00	0.10404	0.040034	24.979	45.207	47.806	0.16980	0.078895	0.091302	221.08	79.878
345.00	0.16891	0.063362	15.782	46.119	48.784	0.16892	0.083030	0.096822	222.35	63.590
360.00	0.26188	0.096367	10.377	47.033	49.751	0.16831	0.087227	0.10277	222.60	51.884
375.00	0.39033	0.14184	7.0503	47.946	50.698	0.16791	0.091459	0.10927	221.70	43.343
390.00	0.56235	0.20329	4.9192	48.849	51.615	0.16768	0.095718	0.11649	219.53	37.032
405.00	0.78681	0.28530	3.5050	49.733	52.490	0.16754	0.10001	0.12481	215.94	32.325
420.00	1.0733	0.39420	2.5368	50.582	53.305	0.16745	0.10438	0.13483	210.76	28.797
435.00	1.4324	0.53918	1.8547	51.376	54.033	0.16734	0.10887	0.14772	203.80	26.154
450.00	1.8759	0.73472	1.3611	52.083	54.636	0.16711	0.11357	0.16583	194.82	24.184
465.00	2.4172	1.0061	0.99393	52.648	55.050	0.16664	0.11865	0.19480	183.50	22.717
480.00	3.0725	1.4051	0.71168	52.968	55.154	0.16569	0.12436	0.25197	169.39	21.551
495.00	3.8632	2.0767	0.48154	52.771	54.631	0.16367	0.13126	0.42947	151.36	20.240
508.10	4.6924	4.7000	0.21277	49.249	50.247	0.15437			0	14.310
Single-Phase Properties										
200.00	0.10000	15.325	0.065254	2.9626	2.9691	0.021248	0.083638	0.11621	1645.6	-0.42678
250.00	0.10000	14.411	0.069389	8.8328	8.8397	0.047436	0.086143	0.11902	1391.1	-0.39768
300.00	0.10000	13.475	0.074210	14.913	14.921	0.069594	0.090180	0.12473	1155.7	-0.33922
328.84	0.10000	12.903	0.077500	18.575	18.583	0.081247	0.093199	0.12941	1024.0	-0.28685
328.84	0.10000	0.038565	25.930	45.137	47.730	0.16988	0.078579	0.090892	220.94	81.384
350.00	0.10000	0.035712	28.002	46.843	49.643	0.17552	0.079533	0.090386	229.44	58.339
400.00	0.10000	0.030709	32.563	50.998	54.255	0.18783	0.085418	0.094849	246.85	30.192
450.00	0.10000	0.027083	36.923	55.474	59.166	0.19939	0.092823	0.10175	262.23	18.173
500.00	0.10000	0.024272	41.200	60.316	64.436	0.21049	0.10033	0.10903	276.40	12.201
550.00	0.10000	0.022008	45.437	65.522	70.066	0.22122	0.10753	0.11612	289.72	8.8355
200.00	1.0000	15.333	0.065220	2.9486	3.0138	0.021178	0.083649	0.11619	1649.7	-0.42708
250.00	1.0000	14.423	0.069336	8.8130	8.8824	0.047357	0.086152	0.11896	1396.0	-0.39848
300.00	1.0000	13.491	0.074123	14.885	14.959	0.069499	0.090182	0.12460	1162.0	-0.34115
350.00	1.0000	12.483	0.080107	21.312	21.392	0.089316	0.095644	0.13326	936.35	-0.24033
400.00	1.0000	11.308	0.088431	28.263	28.351	0.10788	0.10213	0.14605	707.25	-0.042437
416.48	1.0000	10.852	0.092149	30.714	30.806	0.11389	0.10452	0.15210	627.32	0.074613
416.48	1.0000	0.36582	2.7336	50.387	53.120	0.16747	0.10335	0.13228	212.13	29.536
450.00	1.0000	0.31254	3.1996	54.081	57.281	0.17709	0.10087	0.11921	233.76	20.211
500.00	1.0000	0.26538	3.7681	59.388	63.156	0.18947	0.10402	0.11743	256.99	12.984
550.00	1.0000	0.23391	4.2751	64.832	69.107	0.20081	0.10950	0.12100	275.55	9.1542

TABLE 2-185 Thermodynamic Properties of Acetone (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_p kJ/(mol·K)	C_v kJ/(mol·K)	Sound speed m/s	Joule- Thomson K/MPa
Single-Phase Properties										
200.00	5.0000	15.367	0.065073	2.8871	3.2125	0.020868	0.083704	0.11609	1667.9	-0.42837
250.00	5.0000	14.471	0.069106	8.7271	9.0726	0.047011	0.086197	0.11871	1417.7	-0.40187
300.00	5.0000	13.560	0.073747	14.762	15.130	0.069085	0.090197	0.12408	1189.0	-0.34909
350.00	5.0000	12.588	0.079439	21.128	21.525	0.088784	0.095584	0.13214	972.15	-0.25988
400.00	5.0000	11.490	0.087035	27.958	28.393	0.10711	0.10186	0.14320	759.27	-0.10136
450.00	5.0000	10.123	0.098782	35.450	35.944	0.12488	0.10898	0.16059	538.79	0.26123
500.00	5.0000	7.8139	0.12798	44.435	45.075	0.14406	0.11961	0.23343	262.33	2.3418
550.00	5.0000	1.7344	0.57657	60.563	63.446	0.17943	0.12191	0.17820	205.69	10.650
200.00	10.000	15.410	0.064894	2.8125	3.4614	0.020488	0.083781	0.11598	1689.9	-0.42983
250.00	10.000	14.528	0.068831	8.6237	9.3120	0.046589	0.086264	0.11843	1443.6	-0.40569
300.00	10.000	13.641	0.073307	14.616	15.349	0.068589	0.090234	0.12351	1220.9	-0.35775
350.00	10.000	12.709	0.078687	20.916	21.703	0.088163	0.095554	0.13100	1013.1	-0.27983
400.00	10.000	11.683	0.085592	27.629	28.485	0.10626	0.10166	0.14066	815.03	-0.15336
450.00	10.000	10.491	0.095320	34.864	35.818	0.12352	0.10827	0.15332	622.74	0.080235
500.00	10.000	8.9733	0.11144	42.815	43.930	0.14060	0.11552	0.17314	433.48	0.63674
550.00	10.000	6.6600	0.15015	52.079	53.581	0.15896	0.12442	0.22174	255.34	2.7218
250.00	100.00	15.320	0.065276	7.2620	13.790	0.040421	0.088285	0.11631	1791.8	-0.43634
300.00	100.00	14.657	0.068228	12.852	19.675	0.061873	0.092127	0.11946	1616.6	-0.42000
350.00	100.00	14.023	0.071312	18.631	25.763	0.080632	0.097243	0.12424	1466.4	-0.39555
400.00	100.00	13.409	0.074574	24.654	32.112	0.097579	0.10299	0.12980	1337.4	-0.36734
450.00	100.00	12.813	0.078044	30.941	38.745	0.11320	0.10892	0.13553	1226.9	-0.33807
500.00	100.00	12.234	0.081739	37.489	45.663	0.12777	0.11478	0.14112	1133.0	-0.30922
550.00	100.00	11.674	0.085664	44.286	52.852	0.14147	0.12045	0.14639	1053.8	-0.28171
450.00	500.00	15.616	0.064037	27.237	59.256	0.097266	0.11562	0.13393	2201.1	-0.39010
500.00	500.00	15.306	0.065335	33.413	66.081	0.11164	0.12123	0.13909	2129.8	-0.37710
550.00	500.00	15.012	0.066615	39.856	73.163	0.12514	0.12669	0.14416	2067.5	-0.36510

The values in this table were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., and Span, R., "Short Fundamental Equations of State for 20 Industrial Fluids," *J. Chem. Eng. Data*, **51**(3):785–850, 2006. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in the equation of state are 0.1% in the saturated liquid density between 280 and 310 K, 0.5% in density in the liquid phase below 380 K, and 1% in density elsewhere, including all states at pressures above 100 MPa. The uncertainties in vapor pressure are 0.5% above 270 K (0.25% between 290 and 390 K), and the uncertainties in heat capacities and speeds of sound are 1%. These uncertainties (in caloric properties and sound speeds) may be higher at pressures above the saturation pressure and at temperatures above 320 K in the liquid phase and at supercritical conditions.

TABLE 2-186 Saturated Acetylene*

Temperature, K	Pressure, bar	v_{cond} m ³ /kg	v_{g} m ³ /kg	h_{cond} kJ/kg	h_{g} kJ/kg	s_{cond} kJ/(kg·K)	s_{g} kJ/(kg·K)
162.0	0.101		5.081	158	983	2.967	8.062
169.3	0.203		2.644	173	994	3.039	7.889
173.9	0.304		1.805	182	999	3.095	7.797
180.0	0.507		1.116	194	1007	3.161	7.672
184.3	0.709		0.810	203	1011	3.216	7.596
189.1	1.013		0.5780	214	1015	3.272	7.511
192.4 ^a	1.283		0.4617	221	1018	3.312	7.455
192.4 ^a	1.283	0.00164	0.4617	378	1018	4.127	7.455
200.9	2.027	0.00165	0.3011	411	1027	4.296	7.362
209.4	3.040	0.00169	0.2074	445	1035	4.461	7.280
221.5	5.066	0.00174	0.1264	493	1046	4.684	7.180
230.4	7.093	0.00179	0.0907	528	1052	4.837	7.111
240.7	10.13	0.00186	0.0635	565	1058	4.990	7.037
253.2	15.20	0.00195	0.0420	602	1061	5.133	6.947
263.0	20.27	0.00204	0.0309	628	1061	5.231	6.878
271.6	25.33	0.00213	0.0240	654	1060	5.326	6.822
278.9	30.40	0.00223	0.0193	680	1057	5.414	6.767
284.9	35.46	0.00232	0.0159	704	1051	5.494	6.716
290.4	40.53	0.00242	0.0133	727	1041	5.576	6.658
300.0	50.66	0.00270	0.0093	778	1017	5.737	6.534
307.8	60.80	0.00335	0.0061	850	968	5.965	6.351
308.7 ^a	62.47	0.00434	0.0043	908	908	6.158	6.158

*Values recalculated into SI units from those of Din, *Thermodynamic Functions of Gases*, vol. 2, Butterworth, London, 1956. Above the solid line the condensed phase is solid; below the line it is liquid. t = triple point; c = critical point.

TABLE 2-187 Thermodynamic Properties of Air

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule- Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
59.75	0.005265	33.067	0.030242	-1.0619	-1.0617	-0.01536	0.034011	0.055064	1030.3	-0.40785	171.43	376.64
60	0.005546	33.031	0.030275	-1.0481	-1.0480	-0.01513	0.033955	0.055062	1028.3	-0.40743	171.02	371.92
61	0.006797	32.888	0.030406	-0.99308	-0.99287	-0.01422	0.033731	0.055060	1020.3	-0.40565	169.40	353.83
62	0.008270	32.745	0.030539	-0.93803	-0.93778	-0.01333	0.033512	0.055062	1012.2	-0.40375	167.78	336.91
63	0.009994	32.601	0.030674	-0.88298	-0.88267	-0.01245	0.033298	0.055069	1004.0	-0.40173	166.16	321.09
64	0.012000	32.457	0.030810	-0.82792	-0.82755	-0.01158	0.033089	0.055081	995.77	-0.39958	164.53	306.27
65	0.014320	32.312	0.030949	-0.77286	-0.77241	-0.01073	0.032884	0.055098	987.48	-0.39729	162.91	292.39
66	0.016988	32.166	0.031089	-0.71777	-0.71725	-0.00989	0.032683	0.055120	979.13	-0.39485	161.28	279.38
67	0.020042	32.020	0.031231	-0.66267	-0.66205	-0.00906	0.032486	0.055148	970.72	-0.39227	159.65	267.17
68	0.023520	31.873	0.031375	-0.60755	-0.60681	-0.00824	0.032294	0.055181	962.24	-0.38952	158.01	255.71
69	0.027461	31.725	0.031521	-0.55239	-0.55152	-0.00744	0.032105	0.055220	953.70	-0.38660	156.37	244.94
70	0.031908	31.576	0.031669	-0.49720	-0.49619	-0.00664	0.031920	0.055266	945.10	-0.38352	154.73	234.81
71	0.036905	31.427	0.031820	-0.44196	-0.44079	-0.00586	0.031739	0.055317	936.43	-0.38024	153.09	225.28
72	0.042498	31.277	0.031972	-0.38669	-0.38533	-0.00508	0.031562	0.055376	927.70	-0.37677	151.44	216.31
73	0.048733	31.126	0.032127	-0.33135	-0.32979	-0.00432	0.031388	0.055441	918.90	-0.37310	149.79	207.85
74	0.055659	30.974	0.032285	-0.27597	-0.27417	-0.00357	0.031217	0.055514	910.04	-0.36922	148.14	199.88
75	0.063326	30.821	0.032445	-0.22051	-0.21846	-0.00282	0.031050	0.055594	901.11	-0.36511	146.49	192.35
76	0.071786	30.668	0.032608	-0.16499	-0.16265	-0.00209	0.030886	0.055682	892.11	-0.36076	144.83	185.23
77	0.081091	30.513	0.032773	-0.10939	-0.10673	-0.00136	0.030725	0.055779	883.05	-0.35616	143.16	178.51
78	0.091294	30.357	0.032941	-0.05371	-0.05070	-0.00064	0.030568	0.055884	873.91	-0.35130	141.50	172.14
79	0.10245	30.200	0.033112	0.002063	0.005456	6.86E-05	0.030413	0.055988	864.71	-0.34616	139.83	166.11
80	0.11462	30.042	0.033287	0.057934	0.061749	0.000772	0.030262	0.056122	855.44	-0.34074	138.15	160.39
81	0.12785	29.883	0.033464	0.11391	0.11819	0.001467	0.030113	0.056256	846.09	-0.33500	136.48	154.96
82	0.14221	29.722	0.033645	0.17000	0.17478	0.002156	0.029968	0.056400	836.67	-0.32894	134.80	149.80
83	0.15775	29.560	0.033829	0.22621	0.23155	0.002838	0.029826	0.056556	827.18	-0.32254	133.11	144.90
84	0.17453	29.397	0.034017	0.28255	0.28849	0.003513	0.029686	0.056723	817.61	-0.31577	131.42	140.23
85	0.19262	29.232	0.034209	0.33903	0.34562	0.004181	0.029550	0.056902	807.96	-0.30862	129.78	135.78
86	0.21207	29.066	0.034404	0.39566	0.40296	0.004844	0.029417	0.057094	798.24	-0.30107	128.11	131.54
87	0.23295	28.898	0.034604	0.45245	0.46051	0.005501	0.029286	0.057300	788.44	-0.29308	126.44	127.50
88	0.25531	28.729	0.034808	0.50940	0.51829	0.006153	0.029158	0.057521	778.56	-0.28464	124.76	123.63
89	0.27922	28.558	0.035017	0.56653	0.57631	0.006799	0.029033	0.057757	768.59	-0.27572	123.07	119.93
90	0.30475	28.385	0.035230	0.62386	0.63459	0.007440	0.028911	0.058009	758.55	-0.26628	121.38	116.38
91	0.33196	28.210	0.035449	0.68138	0.69315	0.008077	0.028792	0.058278	748.42	-0.25629	119.69	112.98
92	0.36091	28.033	0.035672	0.73912	0.75199	0.008708	0.028676	0.058566	738.20	-0.24573	118.00	109.72
93	0.39166	27.854	0.035901	0.79709	0.81115	0.009336	0.028563	0.058874	727.90	-0.23455	116.30	106.59
94	0.42429	27.673	0.036137	0.85529	0.87062	0.009960	0.028453	0.059202	717.51	-0.22270	114.61	103.58
95	0.45886	27.489	0.036378	0.91375	0.93044	0.010579	0.028346	0.059553	707.03	-0.21016	112.91	100.68
96	0.49543	27.304	0.036625	0.97248	0.99063	0.011195	0.028241	0.059928	696.46	-0.19686	111.21	97.879
97	0.53408	27.115	0.036880	1.0315	1.0512	0.011808	0.028140	0.060329	685.80	-0.18275	109.51	95.179
98	0.57486	26.924	0.037142	1.0908	1.1122	0.012418	0.028042	0.060757	675.05	-0.16779	107.81	92.571
99	0.61786	26.730	0.037411	1.1505	1.1736	0.013025	0.027948	0.061216	664.20	-0.15189	106.11	90.048
100	0.66313	26.533	0.037688	1.2104	1.2354	0.013630	0.027856	0.061707	653.26	-0.13501	104.41	87.605
101	0.71074	26.333	0.037975	1.2708	1.2978	0.014232	0.027768	0.062232	642.22	-0.11705	102.71	85.236
102	0.76077	26.130	0.038270	1.3315	1.3606	0.014833	0.027684	0.062796	631.08	-0.09794	101.01	82.937
103	0.81329	25.923	0.038575	1.3926	1.4240	0.015431	0.027603	0.063401	619.84	-0.07758	99.316	80.703
104	0.86836	25.713	0.038891	1.4542	1.4880	0.016029	0.027525	0.064052	608.50	-0.05588	97.623	78.529
105	0.92606	25.499	0.039217	1.5162	1.5525	0.016625	0.027452	0.064753	597.06	-0.03271	95.933	76.412
106	0.98645	25.281	0.039556	1.5787	1.6177	0.017221	0.027383	0.065508	585.51	-0.00795	94.247	74.347
107	1.0496	25.058	0.039908	1.6417	1.6836	0.017816	0.027317	0.066323	573.85	0.018543	92.565	72.331
108	1.1156	24.831	0.040273	1.7053	1.7502	0.018411	0.027256	0.067206	562.09	0.046927	90.888	70.361
109	1.1845	24.598	0.040653	1.7695	1.8176	0.019006	0.027200	0.068163	550.21	0.077386	89.216	68.432
110	1.2564	24.361	0.041050	1.8343	1.8858	0.019602	0.027149	0.069205	538.21	0.11012	87.551	66.542
111	1.3314	24.118	0.041464	1.8997	1.9549	0.020200	0.027103	0.070341	526.10	0.14538	85.893	64.688
112	1.4095	23.868	0.041896	1.9659	2.0250	0.020799	0.027062	0.071585	513.86	0.18342	84.242	62.867
113	1.4908	23.613	0.042350	2.0329	2.0960	0.021400	0.027028	0.072951	501.48	0.22456	82.599	61.075
114	1.5753	23.350	0.042826	2.1007	2.1682	0.022004	0.027000	0.074459	488.97	0.26917	80.965	59.311

TABLE 2-187 Thermodynamic Properties of Air (Continued)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule- Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
115	1.6633	23.080	0.043328	2.1695	2.2415	0.022611	0.026979	0.076131	476.31	0.31767	79.340	57.571
116	1.7546	22.801	0.043857	2.2392	2.3161	0.023223	0.026965	0.077996	463.48	0.37057	77.724	55.852
117	1.8495	22.514	0.044417	2.3100	2.3922	0.023840	0.026961	0.080090	450.49	0.42848	76.119	54.152
118	1.9479	22.217	0.045011	2.3821	2.4697	0.024462	0.026966	0.082459	437.29	0.49214	74.523	52.467
119	2.0499	21.908	0.045645	2.4554	2.5490	0.025092	0.026982	0.085163	423.88	0.56243	72.938	50.794
120	2.1557	21.588	0.046323	2.5303	2.6302	0.025731	0.027010	0.088280	410.23	0.64047	71.363	49.130
121	2.2653	21.253	0.047052	2.6069	2.7135	0.026380	0.027053	0.091919	396.30	0.72765	69.798	47.469
122	2.3787	20.903	0.047841	2.6854	2.7992	0.027041	0.027113	0.096227	382.04	0.82574	68.243	45.809
123	2.4960	20.534	0.048700	2.7662	2.8878	0.027717	0.027194	0.10142	367.40	0.93703	66.700	44.141
124	2.6173	20.144	0.049643	2.8496	2.9796	0.028412	0.027300	0.10781	352.31	1.0646	65.170	42.460
125	2.7427	19.727	0.050691	2.9363	3.0753	0.029131	0.027438	0.11589	336.67	1.2125	63.658	40.755
126	2.8721	19.278	0.051871	3.0269	3.1759	0.029880	0.027618	0.12645	320.36	1.3865	62.176	39.013
127	3.0055	18.788	0.053225	3.1227	3.2827	0.030668	0.027855	0.14089	303.21	1.5951	60.751	37.215
128	3.1431	18.242	0.054818	3.2253	3.3976	0.031512	0.028171	0.16186	285.00	1.8510	59.445	35.332
129	3.2845	17.616	0.056765	3.3379	3.5243	0.032436	0.028607	0.19519	265.37	2.1752	58.409	33.316
130	3.4295	16.863	0.059300	3.4661	3.6695	0.033492	0.029242	0.25624	243.75	2.6058	58.054	31.072
131	3.5770	15.869	0.063015	3.6243	3.8497	0.034804	0.030266	0.40151	219.07	3.2246	59.591	28.384
132	3.7228	14.198	0.070432	3.8680	4.1302	0.036863	0.032343	1.0148	189.12	4.2808	67.802	24.467
132.63	3.7858	10.448	0.095715	4.4004	4.7627	0.041603			0	6.3978		
59.75	0.002432	0.004907	203.80	4.8774	5.3730	0.096708	0.020805	0.029217	154.83	58.283	5.2938	4.2197
60	0.002584	0.005192	192.59	4.8825	5.3800	0.096323	0.020809	0.029225	155.14	57.634	5.3199	4.2382
61	0.003274	0.006475	154.45	4.9025	5.4081	0.094825	0.020825	0.029261	156.38	55.151	5.4244	4.3119
62	0.004111	0.008005	124.93	4.9225	5.4361	0.093392	0.020843	0.029302	157.60	52.832	5.5291	4.3855
63	0.005120	0.009817	101.86	4.9424	5.4639	0.092020	0.020864	0.029348	158.81	50.666	5.6340	4.4590
64	0.006325	0.011948	83.693	4.9621	5.4915	0.090705	0.020886	0.029399	159.99	48.640	5.7391	4.5324
65	0.007756	0.014438	69.263	4.9817	5.5189	0.089445	0.020911	0.029455	161.16	46.742	5.8444	4.6057
66	0.009442	0.017326	57.715	5.0012	5.5461	0.088235	0.020938	0.029518	162.30	44.963	5.9500	4.6788
67	0.011416	0.020659	48.406	5.0205	5.5731	0.087074	0.020968	0.029587	163.42	43.293	6.0559	4.7519
68	0.013713	0.024481	40.849	5.0397	5.5998	0.085959	0.021000	0.029663	164.53	41.724	6.1621	4.8248
69	0.016372	0.028841	34.673	5.0587	5.6263	0.084887	0.021035	0.029746	165.60	40.248	6.2688	4.8976
70	0.019431	0.033789	29.595	5.0774	5.6525	0.083855	0.021072	0.029836	166.66	38.858	6.3759	4.9703
71	0.022933	0.039379	25.394	5.0960	5.6784	0.082862	0.021113	0.029934	167.69	37.548	6.4835	5.0429
72	0.026921	0.045664	21.899	5.1144	5.7040	0.081906	0.021156	0.030040	168.70	36.313	6.5917	5.1154
73	0.031443	0.052702	18.975	5.1326	5.7292	0.080983	0.021201	0.030155	169.69	35.146	6.7005	5.1878
74	0.036547	0.060550	16.515	5.1505	5.7541	0.080094	0.021250	0.030278	170.65	34.043	6.8099	5.2602
75	0.042282	0.069268	14.437	5.1682	5.7786	0.079235	0.021302	0.030410	171.58	32.999	6.9202	5.3325
76	0.048702	0.078918	12.671	5.1856	5.8027	0.078406	0.021356	0.030552	172.49	32.010	7.0312	5.4048
77	0.055859	0.089564	11.165	5.2028	5.8264	0.077604	0.021414	0.030703	173.37	31.072	7.1431	5.4771
78	0.063810	0.10127	9.8746	5.2196	5.8497	0.076828	0.021474	0.030865	174.23	30.183	7.2560	5.5494
79	0.072611	0.11410	8.7639	5.2362	5.8726	0.076076	0.021538	0.031037	175.05	29.337	7.3700	5.6217
80	0.082321	0.12813	7.8043	5.2525	5.8949	0.075348	0.021605	0.031220	175.85	28.534	7.4851	5.6940
81	0.093001	0.14343	6.9721	5.2684	5.9169	0.074643	0.021674	0.031415	176.62	27.769	7.6014	5.7664
82	0.10471	0.16006	6.2475	5.2841	5.9383	0.073957	0.021747	0.031621	177.36	27.041	7.7192	5.8389
83	0.11751	0.17811	5.6145	5.2994	5.9591	0.073292	0.021822	0.031840	178.07	26.346	7.8384	5.9116
84	0.13147	0.19765	5.0595	5.3143	5.9795	0.072645	0.021901	0.032072	178.75	25.684	7.9591	5.9844
85	0.14665	0.21875	4.5715	5.3289	5.9993	0.072016	0.021983	0.032317	179.40	25.051	8.0817	6.0574
86	0.16312	0.24150	4.1408	5.3431	6.0185	0.071403	0.022068	0.032577	180.02	24.447	8.2060	6.1307
87	0.18094	0.26598	3.7597	5.3569	6.0372	0.070806	0.022155	0.032851	180.61	23.869	8.3324	6.2043
88	0.20018	0.29228	3.4214	5.3703	6.0552	0.070224	0.022246	0.033141	181.17	23.316	8.4610	6.2781
89	0.22091	0.32048	3.1203	5.3832	6.0726	0.069655	0.022340	0.033447	181.69	22.786	8.5919	6.3524
90	0.24320	0.35068	2.8516	5.3958	6.0893	0.069099	0.022436	0.033770	182.19	22.278	8.7254	6.4272
91	0.26712	0.38298	2.6111	5.4079	6.1054	0.068556	0.022535	0.034111	182.65	21.791	8.8616	6.5024
92	0.29273	0.41747	2.3954	5.4195	6.1207	0.068024	0.022638	0.034472	183.08	21.324	9.0008	6.5782
93	0.32011	0.45426	2.2014	5.4307	6.1354	0.067503	0.022744	0.034853	183.47	20.876	9.1433	6.6547
94	0.34934	0.49345	2.0265	5.4413	6.1492	0.066991	0.022852	0.035256	183.84	20.445	9.2893	6.7318
95	0.38047	0.53517	1.8686	5.4514	6.1624	0.066489	0.022964	0.035681	184.17	20.031	9.4390	6.8098
96	0.41359	0.57953	1.7255	5.4610	6.1747	0.065995	0.023078	0.036132	184.46	19.632	9.5929	6.8887

97	0.44878	0.62667	1.5957	5.4701	6.1862	0.065510	0.023196	0.036610	184.72	19.249	9.7513	6.9686
98	0.48609	0.67671	1.4777	5.4785	6.1968	0.065031	0.023317	0.037116	184.95	18.879	9.9145	7.0495
99	0.52562	0.72980	1.3702	5.4864	6.2066	0.064560	0.023441	0.037654	185.14	18.523	10.083	7.1317
100	0.56742	0.78609	1.2721	5.4936	6.2154	0.064094	0.023568	0.038225	185.30	18.180	10.257	7.2153
101	0.61159	0.84575	1.1824	5.5002	6.2233	0.063633	0.023698	0.038834	185.42	17.848	10.438	7.3003
102	0.65820	0.90895	1.1002	5.5060	6.2302	0.063177	0.023833	0.039483	185.51	17.528	10.626	7.3870
103	0.70732	0.97587	1.0247	5.5112	6.2360	0.062726	0.023970	0.040176	185.55	17.218	10.821	7.4755
104	0.75903	1.0467	0.95535	5.5156	6.2408	0.062277	0.024112	0.040918	185.57	16.918	11.024	7.5659
105	0.81341	1.1217	0.89147	5.5193	6.2444	0.061832	0.024258	0.041714	185.54	16.628	11.237	7.6586
106	0.87055	1.2011	0.83254	5.5221	6.2469	0.061389	0.024400	0.042570	185.48	16.346	11.459	7.7537
107	0.93052	1.2852	0.77810	5.5240	6.2481	0.060947	0.024563	0.043492	185.38	16.072	11.693	7.8514
108	0.9934	1.3742	0.72772	5.5250	6.2480	0.060506	0.024722	0.044490	185.24	15.805	11.939	7.9521
109	1.0593	1.4684	0.68102	5.5251	6.2465	0.060065	0.024887	0.045573	185.07	15.546	12.198	8.0560
110	1.1282	1.5682	0.63767	5.5241	6.2436	0.059623	0.025058	0.046751	184.85	15.292	12.473	8.1634
111	1.2004	1.6740	0.59737	5.5221	6.2391	0.059180	0.025234	0.048038	184.60	15.044	12.764	8.2749
112	1.2757	1.7862	0.55985	5.5188	6.2330	0.058735	0.025418	0.049450	184.30	14.800	13.074	8.3907
113	1.3545	1.9053	0.52486	5.5143	6.2252	0.058286	0.025608	0.051005	183.97	14.561	13.406	8.5114
114	1.4366	2.0318	0.49217	5.5085	6.2156	0.057833	0.025807	0.052727	183.59	14.324	13.762	8.6375
115	1.5223	2.1664	0.46160	5.5012	6.2039	0.057375	0.026015	0.054644	183.17	14.090	14.145	8.7696
116	1.6115	2.3097	0.43296	5.4924	6.1901	0.056910	0.026232	0.056790	182.71	13.856	14.559	8.9086
117	1.7045	2.4625	0.40608	5.4819	6.1740	0.056437	0.026461	0.059209	182.21	13.623	15.008	9.0552
118	1.8013	2.6259	0.38082	5.4695	6.1554	0.055955	0.026701	0.061956	181.66	13.388	15.499	9.2104
119	1.9020	2.8009	0.35702	5.4550	6.1341	0.055461	0.026956	0.065102	181.08	13.151	16.039	9.3755
120	2.0067	2.9889	0.33457	5.4383	6.1097	0.054954	0.027226	0.068738	180.45	12.909	16.635	9.5518
121	2.1156	3.1913	0.31335	5.4190	6.0819	0.054432	0.027514	0.072988	179.78	12.661	17.298	9.7412
122	2.2287	3.4103	0.29323	5.3969	6.0504	0.053890	0.027823	0.078015	179.06	12.405	18.042	9.9456
123	2.3462	3.6481	0.27412	5.3715	6.0147	0.053326	0.028155	0.084052	178.31	12.137	18.884	10.168
124	2.4682	3.9078	0.25590	5.3424	5.9740	0.052735	0.028516	0.091426	177.51	11.854	19.849	10.411
125	2.5949	4.1934	0.23847	5.3089	5.9277	0.052112	0.028910	0.10063	176.68	11.553	20.968	10.681
126	2.7266	4.5101	0.22173	5.2701	5.8746	0.051448	0.029344	0.11241	175.81	11.229	22.288	10.982
127	2.8633	4.8653	0.20554	5.2248	5.8133	0.050732	0.029827	0.12801	174.91	10.874	23.877	11.324
128	3.0055	5.2697	0.18976	5.1713	5.7417	0.049950	0.030371	0.14959	173.96	10.480	25.841	11.720
129	3.1536	5.7405	0.17420	5.1069	5.6563	0.049076	0.030994	0.18134	172.98	10.033	28.367	12.191
130	3.3084	6.3074	0.15854	5.0268	5.5513	0.048067	0.031726	0.23261	171.93	9.5119	31.807	12.775
131	3.4712	7.0343	0.14216	4.9209	5.4143	0.046830	0.032619	0.32992	170.79	8.8740	37.001	13.553
132	3.6462	8.1273	0.12304	4.7566	5.2053	0.045064	0.033814	0.59804	169.40	7.9854	46.996	14.798
132.63	3.7858	10.448	0.095715	4.4004	4.7627	0.041603			0	6.3978		

Single-Phase Properties

100	0.1	0.12283	8.1414	5.6800	6.4941	0.080463	0.021087	0.030116	198.24	17.423	9.4692	7.1068
300	0.1	0.040103	24.936	9.8544	12.348	0.11269	0.020796	0.029149	347.36	2.2510	26.384	18.537
500	0.1	0.024046	41.586	14.072	18.231	0.12770	0.021504	0.029830	446.40	0.50305	39.944	27.090
700	0.1	0.017175	58.223	18.500	24.323	0.13794	0.022817	0.031137	523.89	-0.12430	51.755	34.176
900	0.1	0.013359	74.855	23.201	30.686	0.14593	0.024150	0.032467	589.60	-0.41124	62.543	40.394
1100	0.1	0.010931	91.486	28.145	37.293	0.15255	0.025246	0.033562	648.15	-0.56194	72.680	46.051
1300	0.1	0.009249	108.12	33.282	44.094	0.15823	0.026091	0.034406	701.76	-0.64963	82.381	51.325
1500	0.1	0.008016	124.75	38.568	51.042	0.16320	0.026734	0.035049	751.59	-0.70457	91.781	56.325
1700	0.1	0.007073	141.38	43.966	58.104	0.16762	0.027229	0.035544	798.38	-0.74078	100.97	61.127
1900	0.1	0.006329	158.00	49.453	65.253	0.17160	0.027619	0.035934	842.62	-0.76547	110.01	65.783
100	1	26.593	0.037604	1.2007	1.2383	0.013532	0.027868	0.061355	658.25	-0.14308	104.97	88.326
106.22	1	25.232	0.039632	1.5924	1.6321	0.017351	0.027368	0.065680	582.97	-0.00232	93.879	73.903
108.1	1	1.3836	0.72278	5.5251	6.2479	0.060461	0.024739	0.044597	185.23	15.779	11.965	7.9625
300	1	0.40205	2.4873	9.8022	12.289	0.093372	0.020859	0.029563	348.45	2.1789	26.684	18.672
500	1	0.23974	4.1711	14.046	18.218	0.10851	0.021526	0.029954	448.46	0.47425	40.110	27.179
700	1	0.17119	5.8415	18.485	24.326	0.11877	0.022830	0.031194	525.96	-0.13809	51.868	34.242
900	1	0.13319	7.5079	23.190	30.698	0.12677	0.024159	0.032498	591.54	-0.41899	62.628	40.446
1100	1	0.10902	9.1727	28.138	37.311	0.13340	0.025253	0.033582	649.96	-0.56686	72.748	46.094
1300	1	0.092279	10.837	33.278	44.114	0.13908	0.026096	0.034419	703.44	-0.65304	82.438	51.361
1500	1	0.079999	12.500	38.565	51.065	0.14405	0.026738	0.035057	753.17	-0.70711	91.830	56.357
1700	1	0.070604	14.163	43.964	58.128	0.14847	0.027233	0.035550	799.86	-0.74278	101.01	61.155
1900	1	0.063185	15.827	49.451	65.278	0.15245	0.027622	0.035939	844.02	-0.76711	110.05	65.808
100	5	27.222	0.036735	1.0983	1.2820	0.012483	0.028034	0.058181	710.56	-0.21837	111.13	96.436
300	5	2.0232	0.49426	9.5710	12.042	0.079244	0.021131	0.031423	355.63	1.8817	28.389	19.420

TABLE 2-187 Thermodynamic Properties of Air (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _e kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule- Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Single Phase Properties												
500	5	1.1814	0.84642	13.935	18.167	0.094907	0.021621	0.030478	458.30	0.36370	40.969	27.606
700	5	0.84321	1.1859	18.417	24.347	0.102529	0.022885	0.031434	535.45	-0.19118	52.433	34.545
900	5	0.65711	1.5218	23.146	30.755	0.11334	0.024197	0.032632	600.34	-0.44905	63.045	40.682
1100	5	0.53874	1.8562	28.107	37.388	0.11999	0.025282	0.033664	658.10	-0.58606	73.076	46.287
1300	5	0.45667	2.1898	33.256	44.205	0.12568	0.026119	0.034473	711.01	-0.66646	82.707	51.523
1500	5	0.39636	2.5229	38.550	51.165	0.13066	0.026757	0.035095	760.23	-0.71716	92.057	56.497
1700	5	0.35015	2.8559	43.954	58.234	0.13509	0.027249	0.035577	806.49	-0.75073	101.21	61.275
1900	5	0.31361	3.1887	49.445	65.389	0.13907	0.027636	0.035958	850.28	-0.77366	110.22	65.917
100	10	27.863	0.035889	0.99444	1.3533	0.011382	0.028284	0.055716	763.47	-0.27969	117.77	105.78
300	10	4.0370	0.24771	9.2885	11.766	0.072612	0.021441	0.033664	369.50	1.5212	31.116	20.637
500	10	2.3157	0.43183	13.802	18.120	0.088894	0.021733	0.031078	471.81	0.25100	42.260	28.194
700	10	1.6542	0.60452	18.336	24.382	0.099422	0.022952	0.031710	547.83	-0.24405	53.257	34.944
900	10	1.2922	0.77388	23.092	30.831	0.10752	0.024243	0.032786	611.64	-0.47890	63.641	40.985
1100	10	1.0618	0.94184	28.070	37.489	0.11420	0.025317	0.033760	668.47	-0.60517	73.538	46.531
1300	10	0.90165	1.1091	33.231	44.321	0.11990	0.026146	0.034537	720.60	-0.67990	83.082	51.728
1500	10	0.78374	1.2759	38.532	51.292	0.12489	0.026780	0.035139	769.17	-0.72730	92.372	56.673
1700	10	0.69321	1.4426	43.943	58.368	0.12932	0.027268	0.035608	814.87	-0.75881	101.48	61.432
1900	10	0.62149	1.6090	49.438	65.528	0.13330	0.027653	0.035981	858.18	-0.78039	110.45	66.054
100	100	33.161	0.030156	0.24746	3.2631	0.001378	0.031980	0.048218	1192.4	-0.47290	179.20	252.46
300	100	21.138	0.047309	7.0356	11.767	0.049067	0.023981	0.038366	818.47	-0.49747	86.312	53.642
500	100	15.089	0.066273	12.371	18.999	0.067619	0.023117	0.034686	772.41	-0.55640	71.549	42.159
700	100	11.803	0.084722	17.367	25.840	0.079134	0.023855	0.034011	790.14	-0.62591	73.572	43.339
900	100	9.7481	0.10258	22.408	32.667	0.087711	0.024903	0.034331	821.78	-0.67702	79.057	46.948
1100	100	8.3307	0.12004	27.580	39.584	0.09465	0.025831	0.034845	857.40	-0.71435	85.797	51.158
1300	100	7.2877	0.13722	32.880	46.602	0.10051	0.026565	0.035323	894.00	-0.74281	93.151	55.511
1500	100	6.4847	0.15421	38.287	53.708	0.10559	0.027131	0.035723	930.40	-0.76506	100.84	59.875
1700	100	5.8456	0.17107	43.779	60.886	0.11009	0.027569	0.036049	966.13	-0.78264	108.75	64.208
1900	100	5.3239	0.18783	49.340	68.123	0.11411	0.027915	0.036317	1001.0	-0.79653	116.78	68.504
300	500	34.106	0.029320	6.2145	20.875	0.033155	0.028875	0.039265	1678.8	-0.57656	208.23	181.12
500	500	29.826	0.033528	11.583	28.348	0.052311	0.026614	0.036111	1573.6	-0.65015	178.50	120.62
700	500	26.714	0.037433	16.768	35.484	0.064323	0.026496	0.035494	1514.8	-0.67879	161.67	97.470
900	500	24.283	0.041180	22.008	42.598	0.073261	0.026991	0.035702	1482.8	-0.68796	151.95	86.531
1100	500	22.305	0.044833	27.358	49.775	0.080460	0.027539	0.036073	1468.3	-0.69130	146.88	81.387
1300	500	20.651	0.048423	32.814	57.025	0.086515	0.028000	0.036415	1465.1	-0.69354	144.95	79.411
1500	500	19.243	0.051966	38.354	64.337	0.091746	0.02836	0.036693	1469.3	-0.69594	145.84	79.312
1700	500	18.027	0.055473	43.961	71.698	0.096353	0.02864	0.036911	1478.5	-0.69875	148.48	80.393
1900	500	16.963	0.058952	49.623	79.098	0.10047	0.02886	0.037085	1491.1	-0.70188	152.39	82.251
300	1000	40.130	0.024919	6.8286	31.747	0.024761	0.032271	0.041510	2208.5	-0.50493	274.96	337.76
500	1000	36.567	0.027347	12.271	39.618	0.044944	0.029334	0.037843	2104.7	-0.57316	247.30	219.41
700	1000	33.895	0.029503	17.554	47.057	0.057468	0.028754	0.036801	2033.9	-0.60504	230.60	174.51
900	1000	31.736	0.031510	22.890	54.399	0.066695	0.028917	0.036702	1984.7	-0.61882	219.72	149.43
1100	1000	29.916	0.033427	28.327	61.754	0.074073	0.029215	0.036558	1951.3	-0.62560	212.46	133.76
1300	1000	28.338	0.035288	33.857	69.145	0.080246	0.029476	0.037051	1929.3	-0.62968	207.70	123.58
1500	1000	26.946	0.037111	39.461	76.573	0.085561	0.029675	0.037224	1915.7	-0.63251	204.81	116.94
1700	1000	25.701	0.038909	45.123	84.032	0.090229	0.029821	0.037369	1908.3	-0.63465	203.41	112.74
1900	1000	24.577	0.040688	50.830	91.519	0.094392	0.029928	0.037491	1905.8	-0.63632	203.25	110.27

This table was generated for a standard three-component dry air containing mole fractions 0.7812 nitrogen, 0.2096 oxygen, and 0.0092 argon. The values in this table were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., Jacobsen, R. T., Penoncello, S. G., and Friend, D. G., "Thermodynamic Properties of Air and Mixtures of Nitrogen, Argon, and Oxygen from 60 to 2000 K at Pressures to 2000 MPa," *J. Phys. Chem. Ref. Data* **29**(3):331–385, 2000. The source for viscosity and thermal conductivity is Lemmon, E. W., and Jacobsen, R. T., "Viscosity and Thermal Conductivity Equations for Nitrogen, Oxygen, Argon, and Air," *Int. J. Thermophys.* **25**:21–69, 2004.

Properties at the freezing point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

In the range from the solidification point to 873 K at pressures to 70 MPa, the estimated uncertainty of density values calculated with the equation of state is 0.1%. The estimated uncertainty of calculated speed of sound values is 0.2% and that for calculated heat capacities is 1%. At temperatures above 873 K and 70 MPa, the estimated uncertainty of calculated density values is 0.5%, increasing to 1.0% at 2000 K and 2000 MPa. For viscosity, the uncertainty is 1% in the dilute gas. The uncertainty is around 2% between 270 and 300 K and increases to 5% outside of this region. There are very few measurements between 130 and 270 K for air to validate this claim, and the uncertainties may be even higher in this supercritical region. For thermal conductivity, the uncertainty for the dilute gas is 2% with increasing uncertainties near the triple points. The uncertainties range from 3% between 140 and 300 K to 5% at the triple point and at high temperatures. The uncertainties above 100 MPa are not known due to a lack of experimental data.

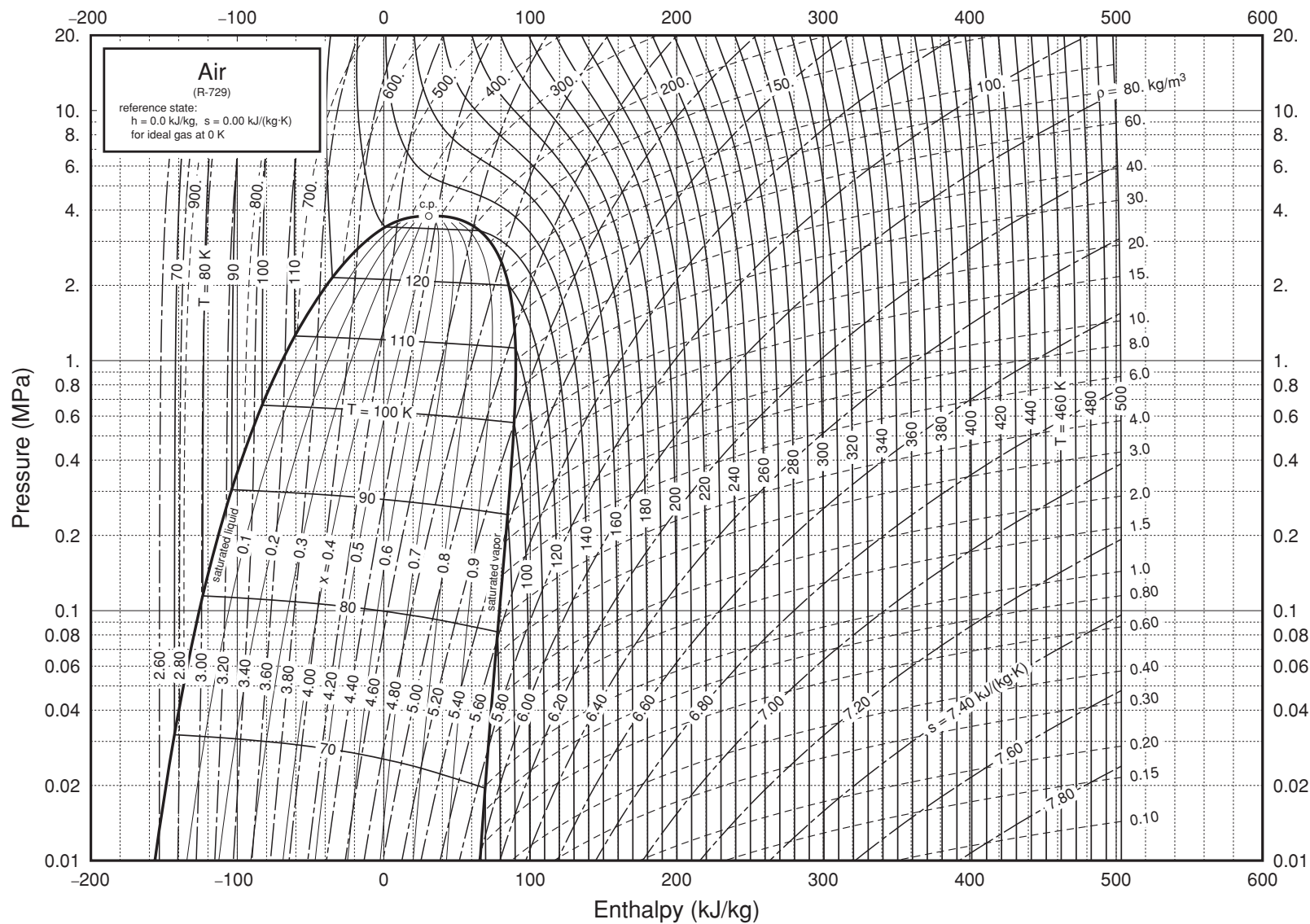


FIG. 2-5 Pressure-enthalpy diagram for dry air. Properties computed with the NIST REFPROP Database, Version 7.0 (Lemmon, E. W., McLinden, M. O., and Huber, M. L., 2002, NIST Standard Reference Database 23, NIST Reference Fluid Thermodynamic and Transport Properties—REFPROP, Version 7.0, Standard Reference Data Program, National Institute of Standards and Technology), based on the equation of state of Lemmon, E. W., Jacobsen, R. T., Penoncello, S. G., and Friend, D. G., "Thermodynamic Properties of Air and Mixtures of Nitrogen, Argon, and Oxygen from 60 to 2000 K at Pressures to 2000 MPa," *J. Phys. Chem. Ref. Data* **29**:331–355, 2000.

TABLE 2-188 Air

Other tables include Stewart, R. B., S. G. Penoncello, et al., University of Idaho CATS report, 85-5, 1985 (0.1–700 bar, 85–750 K), and Lemmon, E. W., Jacobsen, R. T., Penoncello, S. G., and Friend, D. G., *Thermodynamic Properties of Air and Mixtures of Nitrogen, Argon, and Oxygen from 60 to 2000 K at Pressures to 2000 MPa*, J. Phys. Chem. Ref. Data, **29**(3): 331–385, 2000. Tables including reactions with hydrocarbons include Gordon, S., NASA Techn. Paper 1907, 4 vols., 1982. See also Gupta, R. N., K-P. Lee, et al., NASA RP 1232, 1990 (89 pp.) and RP 1260, 1991 (75 pp.). Analytic expressions for high temperatures were given by Matsuzaki, R., *Jap. J. Appl. Phys.*, **21**, 7 (1982): 1009–1013 and Japanese National Aerospace Laboratory report NAL TR 671, 1981 (45 pp.). Functions from 1500 to 15,000 K were tabulated by Hilsenrath, J. and M. Klein, AEDC-TR-65-58 = AD 612 301, 1965 (333 pp.). Tables from 10000 to 10,000,000 K were authored by Gilmore, F. R., Lockheed rept. 3-27-67-1, vol 1., 1967 (340 pp.), also published as *Radiative Properties of Air*, IFI/Plenum, New York, 1969 (648 pp.). Saturation and superheat tables and a chart to 7000 psia, 660°R appear in Stewart, R. B., R. T. Jacobsen, et al., *Thermodynamic Properties of Refrigerants*, ASHRAE, Atlanta, Ga, 1986 (521 pp.). For specific heat, thermal conductivity, and viscosity see *Thermophysical Properties of Refrigerants*, ASHRAE, 1993.

AIR, MOIST

An ASHRAE publication, *Thermodynamic Properties of Dry Air and Water and S. I. Psychrometric Charts*, 1983 (360 pp.), extensively reviews moist air properties. Candiduson, P., *Chem. Eng.*, Oct. 29, 1984 gives on page 118 a nomograph from 50 to 120°F, while equations in SI units were given by Nelson, B., *Chem. Eng. Progr.* **76**, 5 (May 1980): 83–85. Liley, P. E., *2000 Solved Problems in M.E. Thermodynamics*, McGraw-Hill, New York, 1989, gives four simple equations with which most calculations can be made. Devres, Y.O., *Appl. Energy* **48** (1994): 1–18 gives equations with which three known properties can be used to determine four others. Klapert, M. T. and G. F. Schilling, Rand RM-4244-PR = AD 604 856, 1984 (40 pp.) gives tables from 100 to 270 K, while programs from –60 to 2°F are given by Sando, F. A., *ASHRAE Trans.*, **96**, 2 (1990): 299–308.

Viscosity references include Kestin, J. and J. H. Whitelaw, *Int. J. Ht. Mass Transf.* **7**, 11 (1964): 1245–1255; Studnokov, E. L., *Inz.-Fiz. Zhur.* **19**, 2 (1970): 338–340; Hochrainer, D. and F. Munczak, *Setzb. Ost. Acad. Wiss II* **175**, 10 (1966): 540–550. For thermal conductivity see, for instance, Mason, E. A. and L. Monchick, *Humidity and Moisture Control in Science and Industry*, Reinhold, New York, 1965 (257–272).

TABLE 2-189 Thermodynamic Properties of Ammonia

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
195.50	0.0060912	43.035	0.023237	0.00000	0.00014154	0.00000	0.049972	0.071565	2124.2	-0.23362	818.99	559.57
200.00	0.0086509	42.754	0.023389	0.32333	0.32353	0.0016351	0.049837	0.071988	2080.2	-0.22917	803.14	507.28
210.00	0.017739	42.111	0.023747	1.0480	1.0484	0.0051707	0.049521	0.072971	1992.7	-0.21883	768.02	414.98
220.00	0.033790	41.442	0.024130	1.7825	1.7833	0.0085874	0.049207	0.073950	1913.7	-0.20813	733.17	346.68
230.00	0.060407	40.748	0.024541	2.5265	2.5279	0.011894	0.048906	0.074883	1839.2	-0.19712	698.80	294.94
240.00	0.10223	40.032	0.024980	3.2793	3.2818	0.015098	0.048613	0.075764	1766.9	-0.18561	665.09	254.85
250.00	0.16494	39.293	0.025450	4.0403	4.0445	0.018205	0.048327	0.076608	1695.6	-0.17327	632.16	223.08
260.00	0.25531	38.533	0.025952	4.8093	4.8160	0.021222	0.048047	0.077448	1624.5	-0.15963	600.07	197.34
270.00	0.38107	37.748	0.026491	5.5862	5.5963	0.024154	0.047774	0.078328	1553.1	-0.14414	568.85	176.06
280.00	0.55092	36.939	0.027072	6.3712	6.3861	0.027010	0.047511	0.079296	1481.0	-0.12612	538.50	158.12
290.00	0.77436	36.101	0.027700	7.1651	7.1866	0.029797	0.047266	0.080412	1407.8	-0.10470	508.99	142.74
300.00	1.0617	35.230	0.028385	7.9691	7.9993	0.032525	0.047044	0.081747	1333.2	-0.078790	480.25	129.33
310.00	1.4240	34.320	0.029138	8.7850	8.8265	0.035203	0.046856	0.083390	1256.7	-0.046923	452.23	117.49
320.00	1.8728	33.363	0.029973	9.6153	9.6714	0.037843	0.046715	0.085465	1177.9	-0.0070718	424.83	106.91
330.00	2.4205	32.350	0.030912	10.463	10.538	0.040458	0.046636	0.088145	1096.5	0.043673	397.96	97.325
340.00	3.0802	31.264	0.031986	11.333	11.432	0.043065	0.046642	0.091701	1011.8	0.10967	371.51	88.555
350.00	3.8660	30.087	0.033237	12.232	12.361	0.045682	0.046767	0.096576	923.38	0.19774	345.32	80.430
360.00	4.7929	28.788	0.034737	13.169	13.335	0.048339	0.047064	0.10357	830.62	0.31928	319.25	72.796
370.00	5.8778	27.321	0.036602	14.158	14.373	0.051075	0.047619	0.11435	732.78	0.49497	293.07	65.493
380.00	7.1402	25.606	0.039054	15.224	15.503	0.053961	0.048589	0.13314	628.75	0.76738	266.57	58.315
390.00	8.6045	23.465	0.042616	16.424	16.790	0.057149	0.050319	0.17550	515.88	1.2455	239.65	50.877
400.00	10.305	20.232	0.049426	17.969	18.478	0.061223	0.054109	0.38707	384.58	2.3557	216.00	41.802
405.40	11.339	13.212	0.075690	20.640	21.499	0.068559			0	5.0513		
195.50	0.0060912	0.0037635	265.71	23.661	25.279	0.12931	0.026510	0.035130	354.12	171.13	19.636	6.8396
200.00	0.0086509	0.0052305	191.19	23.770	25.424	0.12714	0.026650	0.035345	357.91	152.55	19.684	6.9515
210.00	0.017739	0.010249	97.573	24.006	25.737	0.12273	0.027053	0.035961	365.94	120.01	19.860	7.2115
220.00	0.033790	0.018721	53.415	24.233	26.038	0.11884	0.027583	0.036783	373.38	96.215	20.132	7.4846
230.00	0.060407	0.032214	31.043	24.450	26.325	0.11536	0.028245	0.037836	380.19	78.430	20.503	7.7679
240.00	0.10223	0.052667	18.987	24.655	26.596	0.11224	0.029043	0.039142	386.30	64.852	20.978	8.0587
250.00	0.16494	0.082417	12.133	24.846	26.847	0.10942	0.029978	0.040728	391.66	54.280	21.560	8.3552
260.00	0.25531	0.12421	8.0506	25.021	27.077	0.10684	0.031050	0.042623	396.20	45.905	22.258	8.6558
270.00	0.38107	0.18126	5.5168	25.179	27.281	0.10447	0.032253	0.044859	399.86	39.175	23.079	8.9595
280.00	0.55092	0.25729	3.8867	25.317	27.459	0.10227	0.033581	0.047476	402.59	33.701	24.034	9.2664
290.00	0.77436	0.35664	2.8040	25.435	27.606	0.10021	0.035028	0.050530	404.30	29.207	25.138	9.5771
300.00	1.0617	0.48448	2.0641	25.528	27.720	0.098259	0.036584	0.054099	404.95	25.489	26.408	9.8938
310.00	1.4240	0.64702	1.5455	25.595	27.796	0.096395	0.038244	0.058302	404.45	22.391	27.872	10.220
320.00	1.8728	0.85202	1.1737	25.632	27.830	0.094589	0.040004	0.063320	402.70	19.794	29.568	10.561
330.00	2.4205	1.1094	0.90139	25.634	27.816	0.092817	0.041868	0.069443	399.61	17.599	31.559	10.927
340.00	3.0802	1.4325	0.69810	25.595	27.746	0.091046	0.043844	0.077150	395.05	15.728	33.945	11.330
350.00	3.8660	1.8399	0.54350	25.505	27.606	0.089242	0.045954	0.087280	388.86	14.112	36.900	11.792
360.00	4.7929	2.3598	0.42377	25.350	27.381	0.087355	0.048233	0.10141	380.83	12.690	40.752	12.346
370.00	5.8778	3.0375	0.32922	25.107	27.042	0.085316	0.050744	0.12286	370.69	11.400	46.149	13.053
380.00	7.1402	3.9558	0.25279	24.734	26.539	0.083003	0.053589	0.16000	357.96	10.172	54.556	14.025
390.00	8.6045	5.2979	0.18875	24.144	25.768	0.080169	0.056957	0.24170	341.67	8.9038	70.114	15.527
400.00	10.305	7.6973	0.12992	23.047	24.386	0.075992	0.061281	0.59477	318.22	7.3513	113.54	18.529
405.40	11.339	13.212	0.075690	20.640	21.499	0.068559			0	5.0513		

TABLE 2-189 Thermodynamic Properties of Ammonia (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _p kJ/(mol·K)	C _v kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Single-Phase Properties												
200.00	0.10000	42.756	0.023388	0.32270	0.32504	0.0016320	0.049842	0.071983	2080.3	-0.22921	803.24	507.47
239.56	0.10000	40.064	0.024960	3.2461	3.2486	0.014960	0.048626	0.075726	1770.0	-0.18613	666.56	256.42
239.56	0.10000	0.051595	19.382	24.646	26.584	0.11237	0.029005	0.039079	386.05	65.377	20.955	8.0459
300.00	0.10000	0.040502	24.690	26.378	28.847	0.12080	0.028021	0.036849	434.39	27.493	25.100	10.161
400.00	0.10000	0.030171	33.144	29.297	32.612	0.13162	0.030417	0.038883	497.93	10.681	37.215	13.971
500.00	0.10000	0.024091	41.509	32.514	36.665	0.14065	0.033897	0.042280	550.96	5.5276	53.119	17.863
600.00	0.10000	0.020060	49.849	36.096	41.081	0.14869	0.037731	0.046083	597.69	3.2702	68.607	21.682
700.00	0.10000	0.017188	58.179	40.068	45.885	0.15609	0.041678	0.050015	640.16	2.0841	78.312	25.391
200.00	1.0000	42.774	0.023379	0.31651	0.33989	0.0016010	0.049890	0.071938	2081.5	-0.22959	804.23	509.28
298.05	1.0000	35.403	0.028246	7.8111	7.8393	0.031996	0.047085	0.081465	1347.9	-0.084271	485.81	131.82
298.05	1.0000	0.45697	2.1883	25.512	27.700	0.098633	0.036271	0.053356	404.91	26.163	26.145	9.8313
300.00	1.0000	0.45215	2.2117	25.592	27.804	0.098979	0.035866	0.052493	407.16	25.620	26.308	9.9115
400.00	1.0000	0.31157	3.2095	29.019	32.229	0.11177	0.031641	0.041627	488.94	10.494	38.087	13.927
500.00	1.0000	0.24426	4.0940	32.359	36.453	0.12119	0.034312	0.043338	546.79	5.4884	53.750	17.877
600.00	1.0000	0.20197	4.9513	35.994	40.945	0.12937	0.037928	0.046628	595.60	3.2544	69.123	21.717
700.00	1.0000	0.17248	5.7977	39.994	45.792	0.13684	0.041791	0.050341	639.15	2.0746	78.751	25.434
200.00	5.0000	42.852	0.023336	0.28942	0.40611	0.0014649	0.050097	0.071739	2086.8	-0.23126	808.60	517.30
300.00	5.0000	35.450	0.028209	7.8852	8.0263	0.032243	0.047090	0.080899	1361.2	-0.089577	487.57	132.49
362.03	5.0000	28.505	0.035081	13.365	13.540	0.048887	0.047152	0.10538	811.17	0.34968	313.94	71.291
362.03	5.0000	2.4828	0.40277	25.309	27.323	0.086956	0.048722	0.10501	378.95	12.419	41.693	12.475
400.00	5.0000	1.8706	0.53459	27.540	30.213	0.094581	0.038466	0.061581	441.81	9.6373	45.730	14.036
500.00	5.0000	1.3046	0.76650	31.630	35.462	0.10634	0.036193	0.048779	528.14	5.2830	57.294	18.073
600.00	5.0000	1.0412	0.96040	35.527	40.329	0.11521	0.038798	0.049210	586.79	3.1693	71.791	21.941
700.00	5.0000	0.87563	1.1420	39.662	45.373	0.12298	0.042289	0.051836	635.17	2.0254	80.941	25.662
200.00	10.000	42.947	0.023284	0.25644	0.48928	0.0012980	0.050342	0.071495	2093.5	-0.23328	814.02	527.29
300.00	10.000	35.714	0.028000	7.7848	8.0648	0.031903	0.047164	0.079960	1394.2	-0.10159	496.50	136.36
398.32	10.000	20.945	0.047744	17.655	18.132	0.060394	0.053149	0.30653	409.04	2.0704	218.73	43.632
398.32	10.000	7.1390	0.14008	23.303	24.704	0.076892	0.060447	0.46915	323.12	7.6606	101.04	17.793
400.00	10.000	6.5455	0.15278	23.801	25.329	0.078458	0.057611	0.30552	336.28	7.7633	95.455	17.230
500.00	10.000	2.8656	0.34897	30.616	34.106	0.098525	0.038603	0.057806	505.64	4.9335	63.922	18.722
600.00	10.000	2.1650	0.46190	34.920	39.539	0.10844	0.039862	0.052806	577.38	3.0278	76.053	22.393
700.00	10.000	1.7835	0.56069	39.241	44.848	0.11663	0.042896	0.053796	631.50	1.9491	84.235	26.035
300.00	100.00	38.995	0.025644	6.5830	9.1474	0.027511	0.048894	0.072740	1774.7	-0.19551	622.86	193.71
400.00	100.00	33.105	0.030207	13.432	16.453	0.048523	0.046636	0.073557	1378.2	-0.11309	431.98	96.237
500.00	100.00	27.067	0.036945	20.212	23.907	0.065147	0.045999	0.075495	1081.8	0.049919	305.65	60.386
600.00	100.00	21.518	0.046473	26.825	31.472	0.078942	0.046723	0.075193	918.11	0.23722	234.79	46.188
700.00	100.00	17.303	0.057794	33.074	38.854	0.090326	0.048331	0.072317	861.52	0.32753	196.04	41.237
300.00	500.00	45.670	0.021896	4.7114	15.660	0.018023	0.052877	0.067831	2597.1	-0.25055	989.00	376.31
400.00	500.00	42.416	0.023576	10.633	22.421	0.037482	0.051527	0.066802	2353.2	-0.25064	804.05	188.46
500.00	500.00	39.515	0.025307	16.367	29.021	0.052215	0.050431	0.065418	2176.9	-0.25260	674.00	120.77
600.00	500.00	36.909	0.027094	22.007	35.554	0.064127	0.050614	0.065476	2044.6	-0.24722	582.63	91.251
700.00	500.00	34.550	0.028943	27.680	42.152	0.074295	0.051816	0.066615	1943.8	-0.23682	511.57	77.538
300.00	1000.0	49.944	0.020022	4.1818	24.204	0.011750	0.055176	0.065784	3230.2	-0.25989	1324.0	554.62
400.00	1000.0	47.551	0.021030	9.8612	30.891	0.030984	0.054864	0.066677	2997.6	-0.25431	1138.9	274.91
500.00	1000.0	45.362	0.022045	15.432	37.477	0.045686	0.053323	0.065150	2842.8	-0.26084	996.49	174.11
600.00	1000.0	43.378	0.023053	20.911	43.964	0.057514	0.052940	0.064819	2728.6	-0.26235	887.73	129.02
700.00	1000.0	41.556	0.024064	26.418	50.481	0.067559	0.053649	0.065697	2639.0	-0.25820	797.25	107.14

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Tillner-Roth, R., Harms-Watzenberg, F., and Baehr, H. D., "Eine neue Fundamentalgleichung fuer Ammoniak," *DKV-Tagungsbericht*, **20**:167–181, 1993. The source for viscosity is Fenghour, A., Wakeham, W. A., Vesovic, V., Watson, J. T. R., Millat, J., and Vogel, E., "The Viscosity of Ammonia," *J. Phys. Chem. Ref. Data* **24**:1649–1667, 1995. The source for thermal conductivity is Tufeu, R., Ivanov, D. Y., Garrabos, Y., and Le Neindre, B., "Thermal Conductivity of Ammonia in a Large Temperature and Pressure Range Including the Critical Region," *Ber. Bunsenges. Phys. Chem.* **88**:422–427, 1984.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties of the equation of state are 0.2% in density, 2% in heat capacity, and 2% in the speed of sound, except in the critical region. The uncertainty in vapor pressure is 0.2%. The uncertainty varies from 0.5% for the viscosity of the dilute gas phase at moderate temperatures to about 5% for the viscosity at high pressures and temperatures. The uncertainty in thermal conductivity is 2%.

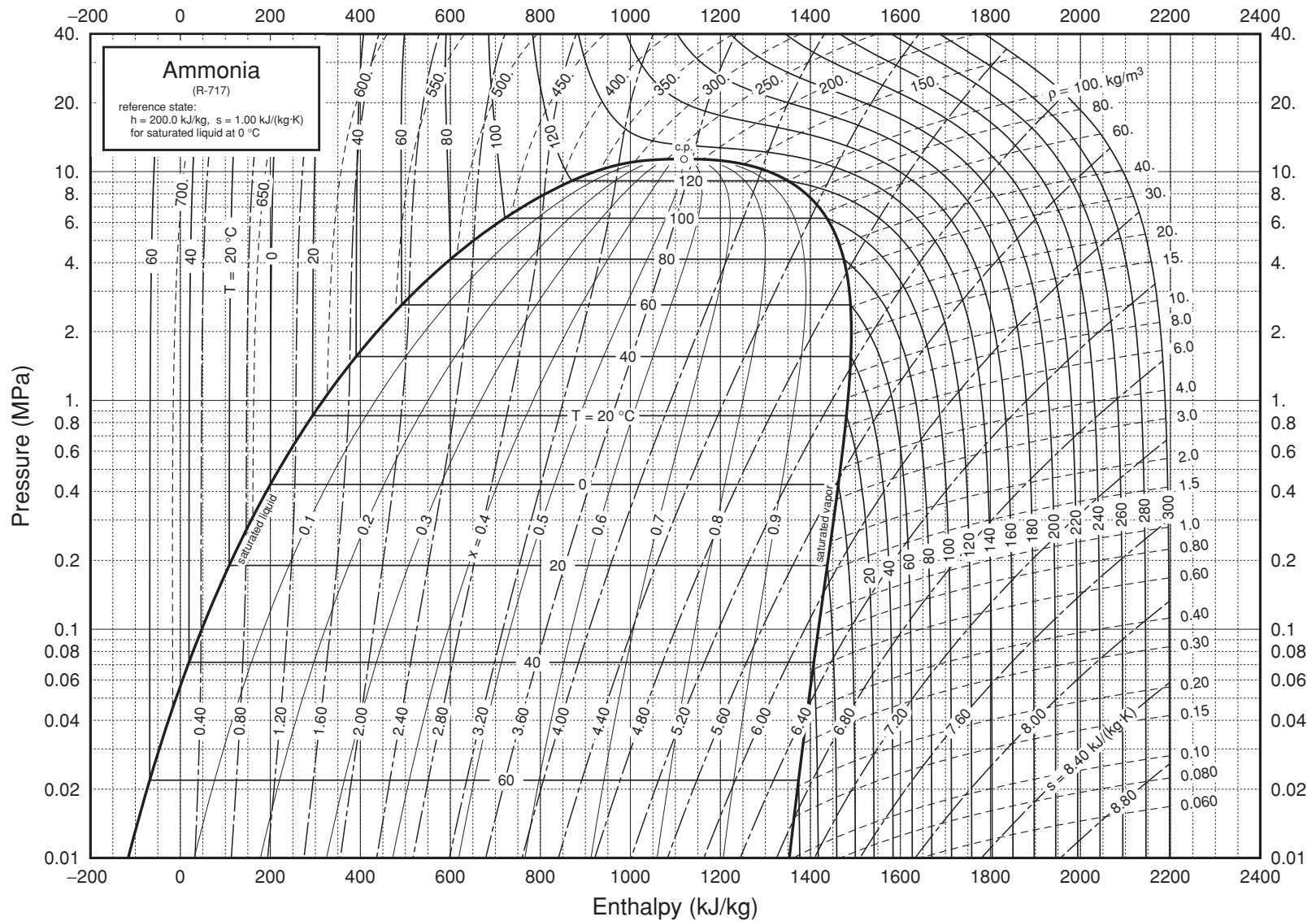


FIG. 2-6 Pressure-enthalpy diagram for ammonia. Properties computed with the NIST REFPROP Database, Version 7.0 (Lemmon, E. W., McLinden, M. O., and Huber, M. L., 2002, NIST Standard Reference Database 23, NIST Reference Fluid Thermodynamic and Transport Properties—REFPROP, Version 7.0, Standard Reference Data Program, National Institute of Standards and Technology), based on the equation of state of Tillner-Roth, R., Harms-Watzenberg, F., and Baehr, H. D., "Eine neue Fundamentalgleichung für Ammoniak, *DKV-Tagungsbericht* 20(II):167–181, 1993.

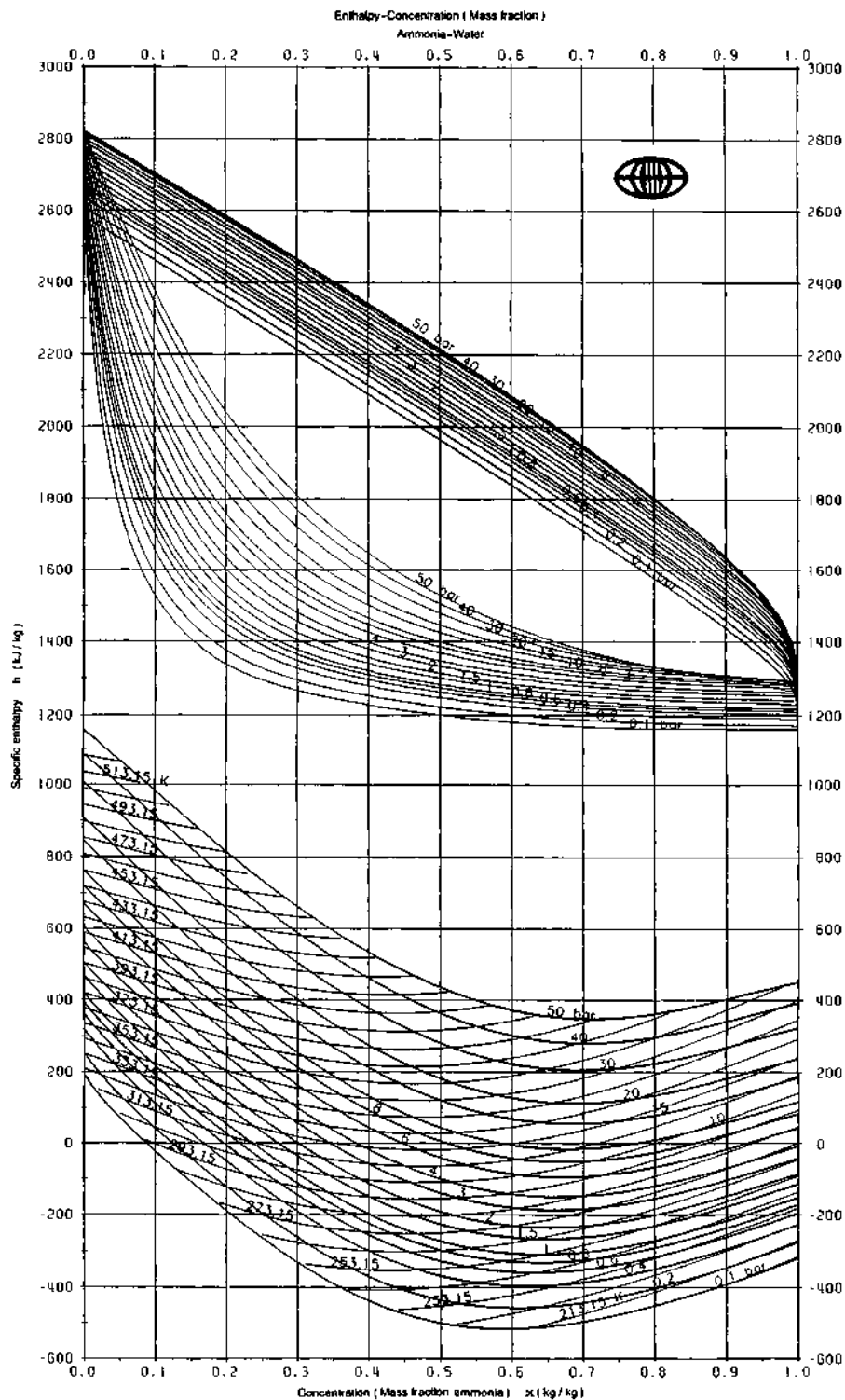


FIG. 2-7 Enthalpy-concentration diagram for aqueous ammonia. From *Thermodynamic and Physical Properties* $\text{NH}_3\text{-H}_2\text{O}$, Int. Inst. Refrigeration, Paris, France, 1994 (88 pp.). Reproduced by permission. In order to determine equilibrium compositions, draw a vertical from any liquid composition on any boiling line (the lowest plots) to intersect the appropriate auxiliary curve (the intermediate curves). A horizontal then drawn from this point to the appropriate dew line (the upper curves) will establish the vapor composition. The Int. Inst. Refrigeration publication also gives extensive P - x tables from -50 to 316°C . Other sources include Park, Y. M. and Sonntag, R. E., *ASHRAE Trans.*, **96**, 1 (1990): 150–159 (x , h , s , tables, 360 to 640 K); Ibrahim, O. M. and S. A. Klein, *ASHRAE Trans.*, **99**, 1 (1993): 1495–1502 (Eqs., 0.2 to 110 bar, 293 to 413 K); Smolen, T. M., D. B. Manley, et al., *J. Chem. Eng. Data*, **36** (1991): 202–208 (p - x correlation, 0.9 to 450 psia, 293–413 K); Ruiter, J. P., *Int. J. Refrig.*, **13** (1990): 223–236 gives ten sub-routines for computer calculations.

TABLE 2-190 Thermodynamic Properties of Argon

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
83.806	0.068891	35.465	0.028197	-4.8531	-4.8512	0.053110	0.021956	0.044570	862.43	-0.40500	133.63	290.17
85.000	0.078897	35.284	0.028342	-4.8000	-4.7977	0.053740	0.021764	0.044571	854.24	-0.40120	131.86	279.46
87.000	0.098131	34.977	0.028590	-4.7110	-4.7082	0.054774	0.021461	0.044620	840.43	-0.39386	128.91	262.70
89.000	0.12078	34.667	0.028846	-4.6220	-4.6185	0.055787	0.021178	0.044720	826.48	-0.38530	125.97	247.29
91.000	0.14723	34.353	0.029109	-4.5328	-4.5285	0.056778	0.020912	0.044869	812.39	-0.37551	123.06	233.09
93.000	0.17785	34.035	0.029381	-4.4433	-4.4380	0.057752	0.020662	0.045063	798.14	-0.36447	120.16	219.98
95.000	0.21305	33.713	0.029662	-4.3534	-4.3471	0.058708	0.020424	0.045301	783.72	-0.35214	117.28	207.87
97.000	0.25323	33.386	0.029953	-4.2632	-4.2556	0.059649	0.020198	0.045583	769.14	-0.33847	114.44	196.65
99.000	0.29882	33.054	0.030254	-4.1725	-4.1634	0.060575	0.019982	0.045909	754.37	-0.32336	111.63	186.24
101.00	0.35023	32.715	0.030567	-4.0812	-4.0705	0.061489	0.019777	0.046282	739.40	-0.30671	108.82	176.58
103.00	0.40789	32.371	0.030892	-3.9893	-3.9767	0.062391	0.019580	0.046705	724.21	-0.28840	106.03	167.58
105.00	0.47224	32.020	0.031230	-3.8968	-3.8820	0.063283	0.019393	0.047181	708.81	-0.26827	103.26	159.18
107.00	0.54371	31.662	0.031584	-3.8034	-3.7862	0.064165	0.019214	0.047715	693.16	-0.24614	100.50	151.32
109.00	0.62276	31.296	0.031953	-3.7092	-3.6893	0.065039	0.019044	0.048313	677.26	-0.22181	97.768	143.96
111.00	0.70982	30.921	0.032340	-3.6141	-3.5912	0.065906	0.018883	0.048984	661.08	-0.19503	95.055	137.04
113.00	0.80535	30.538	0.032747	-3.5180	-3.4916	0.066767	0.018731	0.049737	644.60	-0.16549	92.363	130.52
115.00	0.90981	30.144	0.033174	-3.4207	-3.3905	0.067623	0.018589	0.050584	627.79	-0.13284	89.695	124.35
117.00	1.0237	29.739	0.033626	-3.3222	-3.2878	0.068477	0.018458	0.051541	610.64	-0.096667	87.049	118.50
119.00	1.1473	29.322	0.034104	-3.2223	-3.1831	0.069328	0.018337	0.052626	593.11	-0.056465	84.428	112.93
121.00	1.2814	28.891	0.034613	-3.1208	-3.0764	0.070179	0.018229	0.053865	575.16	-0.011619	81.832	107.61
123.00	1.4262	28.446	0.035155	-3.0175	-2.9674	0.071031	0.018134	0.055288	556.75	0.038624	79.260	102.49
125.00	1.5823	27.983	0.035735	-2.9123	-2.8558	0.071886	0.018056	0.056938	537.83	0.095199	76.711	97.565
127.00	1.7503	27.502	0.036361	-2.8049	-2.7413	0.072747	0.017995	0.058870	518.34	0.15928	74.185	92.791
129.00	1.9305	26.999	0.037039	-2.6949	-2.6234	0.073616	0.017955	0.061161	498.21	0.23236	71.679	88.143
131.00	2.1237	26.471	0.037778	-2.5820	-2.5018	0.074496	0.017941	0.063916	477.36	0.31637	69.190	83.595
133.00	2.3303	25.913	0.038591	-2.4655	-2.3756	0.075392	0.017957	0.067291	455.69	0.41386	66.713	79.119
135.00	2.5509	25.320	0.039495	-2.3449	-2.2442	0.076309	0.018011	0.071519	433.10	0.52823	64.243	74.689
137.00	2.7862	24.683	0.040513	-2.2192	-2.1063	0.077253	0.018109	0.076964	409.47	0.66427	61.773	70.276
139.00	3.0369	23.993	0.041678	-2.0871	-1.9605	0.078235	0.018264	0.084237	384.62	0.82889	59.297	65.850
141.00	3.3037	23.233	0.043041	-1.9468	-1.8046	0.079268	0.018501	0.094443	358.17	1.0326	56.816	61.374
143.00	3.5876	22.379	0.044686	-1.7953	-1.6350	0.080375	0.018882	0.10982	329.41	1.2928	54.352	56.802
145.00	3.8896	21.385	0.046762	-1.6277	-1.4458	0.081593	0.019546	0.13580	297.06	1.6405	52.009	52.060
147.00	4.2111	20.163	0.049597	-1.4335	-1.2247	0.083001	0.020842	0.18993	258.79	2.1416	50.224	46.982
149.00	4.5541	18.446	0.054213	-1.1822	-0.93533	0.084835	0.023975	0.37941	209.33	2.9836	51.152	41.002
150.69	4.8630	13.407	0.074586	-0.53575	-0.17304	0.089789			0	5.2503		
83.806	0.068891	0.10150	9.8526	1.0103	1.6891	0.13115	0.012972	0.022172	168.12	35.712	5.3590	6.8558
85.000	0.078897	0.11492	8.7019	1.0212	1.7078	0.13028	0.013015	0.022310	169.08	34.706	5.4485	6.9622
87.000	0.098131	0.14035	7.1253	1.0388	1.7380	0.12887	0.013092	0.022563	170.64	33.127	5.6006	7.1414
89.000	0.12078	0.16979	5.8895	1.0554	1.7668	0.12753	0.013176	0.022845	172.12	31.668	5.7559	7.3219
91.000	0.14723	0.20365	4.9104	1.0710	1.7939	0.12626	0.013267	0.023159	173.52	30.315	5.9147	7.5039
93.000	0.17785	0.24233	4.1266	1.0855	1.8194	0.12504	0.013364	0.023506	174.85	29.056	6.0776	7.6877
95.000	0.21305	0.28625	3.4935	1.0988	1.8431	0.12387	0.013469	0.023892	176.11	27.883	6.2452	7.8736
97.000	0.25323	0.33587	2.9774	1.1108	1.8648	0.12275	0.013580	0.024318	177.29	26.785	6.4180	8.0617
99.000	0.29882	0.39165	2.5533	1.1216	1.8845	0.12167	0.013700	0.024789	178.39	25.755	6.5969	8.2524
101.00	0.35023	0.45412	2.2021	1.1309	1.9021	0.12062	0.013828	0.025310	179.41	24.786	6.7825	8.4461
103.00	0.40789	0.52379	1.9092	1.1387	1.9174	0.11962	0.013964	0.025887	180.36	23.873	6.9760	8.6432
105.00	0.47224	0.60126	1.6632	1.1449	1.9303	0.11864	0.014108	0.026526	181.22	23.009	7.1783	8.8440
107.00	0.54371	0.68714	1.4553	1.1494	1.9406	0.11769	0.014263	0.027235	182.01	22.189	7.3908	9.0492
109.00	0.62276	0.78213	1.2786	1.1521	1.9483	0.11676	0.014427	0.028024	182.71	21.410	7.6149	9.2592
111.00	0.70982	0.88696	1.1274	1.1528	1.9531	0.11585	0.014602	0.028904	183.33	20.667	7.8524	9.4749
113.00	0.80535	1.0025	0.99752	1.1515	1.9549	0.11497	0.014789	0.029890	183.87	19.955	8.1054	9.6969
115.00	0.90981	1.1296	0.88525	1.1480	1.9534	0.11409	0.014989	0.030998	184.33	19.272	8.3764	9.9262
117.00	1.0237	1.2694	0.78776	1.1421	1.9485	0.11323	0.015203	0.032251	184.70	18.615	8.6685	10.164
119.00	1.1473	1.4231	0.70269	1.1336	1.9398	0.11238	0.015432	0.033674	184.98	17.979	8.9853	10.411
121.00	1.2814	1.5920	0.62813	1.1223	1.9271	0.11153	0.015679	0.035305	185.17	17.361	9.3315	10.670
123.00	1.4262	1.7778	0.56248	1.1079	1.9101	0.11069	0.015945	0.037186	185.27	16.758	9.7128	10.942
125.00	1.5823	1.9824	0.50443	1.0901	1.8883	0.10984	0.016234	0.039377	185.28	16.167	10.136	11.228

TABLE 2-190 Thermodynamic Properties of Argon (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
127.00	1.7503	2.2081	0.45288	1.0685	1.8612	0.10899	0.016549	0.041958	185.18	15.585	10.611	11.533
129.00	1.9305	2.4576	0.40690	1.0428	1.8283	0.10813	0.016892	0.045036	184.98	15.007	11.150	11.860
131.00	2.1237	2.7345	0.36570	1.0122	1.7888	0.10725	0.017269	0.048764	184.68	14.430	11.768	12.212
133.00	2.3303	3.0431	0.32861	0.97615	1.7419	0.10635	0.017684	0.053370	184.27	13.851	12.487	12.596
135.00	2.5509	3.3893	0.29505	0.93360	1.6862	0.10542	0.018152	0.059209	183.74	13.262	13.337	13.018
137.00	2.7862	3.7806	0.26451	0.88331	1.6203	0.10445	0.018697	0.066854	183.03	12.655	14.363	13.489
139.00	3.0369	4.2280	0.23652	0.82353	1.5418	0.10343	0.019352	0.077279	182.09	12.019	15.633	14.022
141.00	3.3037	4.7471	0.21066	0.75170	1.4476	0.10233	0.020167	0.092257	180.81	11.341	17.255	14.638
143.00	3.5876	5.3629	0.18647	0.66386	1.3328	0.10113	0.021213	0.11542	179.05	10.606	19.424	15.370
145.00	3.8896	6.1190	0.16342	0.55308	1.1887	0.099762	0.022619	0.15563	176.57	9.7892	22.530	16.274
147.00	4.2111	7.1057	0.14073	0.40516	0.99779	0.098120	0.024694	0.24158	172.74	8.8407	27.572	17.475
149.00	4.5541	8.5869	0.11646	0.17894	0.70930	0.095873	0.028532	0.54500	165.39	7.6233	38.688	19.366
150.69	4.8630	13.407	0.074586	-0.53575	-0.17304	0.089789			133.87	0		
Single-Phase Properties												
100.00	0.10000	0.12304	8.1275	1.2122	2.0249	0.13179	0.012807	0.021852	184.16	25.048	6.4504	8.2341
200.00	0.10000	0.060310	16.581	2.4833	4.1414	0.14649	0.012497	0.020918	263.21	7.4099	12.539	15.998
300.00	0.10000	0.040115	24.928	3.7351	6.2279	0.15495	0.012479	0.020834	322.67	3.6153	17.837	22.741
400.00	0.10000	0.030069	33.257	4.9842	8.3099	0.16094	0.012475	0.020810	372.65	1.9970	22.516	28.704
500.00	0.10000	0.024050	41.580	6.2324	10.390	0.16559	0.012474	0.020801	416.64	1.1105	26.726	34.077
600.00	0.10000	0.020041	49.899	7.4803	12.470	0.16938	0.012473	0.020796	456.40	0.55603	30.573	38.997
700.00	0.10000	0.017177	58.217	8.7279	14.550	0.17258	0.012473	0.020793	492.95	0.17948	34.134	43.556
100.00	1.0000	32.955	0.030345	-4.1375	-4.1072	0.060927	0.019909	0.045870	751.59	-0.32095	110.85	183.08
116.60	1.0000	29.821	0.033533	-3.3421	-3.3086	0.068305	0.018483	0.051339	614.12	-0.10424	87.579	119.65
116.60	1.0000	1.2403	0.80628	1.1435	1.9497	0.11340	0.015159	0.031986	184.63	18.745	8.6079	10.115
200.00	1.0000	0.61937	1.6145	2.3849	3.9995	0.12686	0.012725	0.022195	261.59	7.2269	12.995	16.256
300.00	1.0000	0.40331	2.4795	3.6807	6.1603	0.13563	0.012545	0.021266	323.44	3.5287	18.142	22.901
400.00	1.0000	0.30073	3.3252	4.9476	8.2728	0.14171	0.012507	0.021025	374.06	1.9489	22.749	28.817
500.00	1.0000	0.24013	4.1644	6.2054	10.370	0.14639	0.012493	0.020928	418.25	1.0808	26.915	34.163
600.00	1.0000	0.19997	5.0007	7.4592	12.460	0.15020	0.012487	0.020879	458.06	0.53628	30.733	39.065
700.00	1.0000	0.17137	5.8354	8.7110	14.546	0.15342	0.012483	0.020851	494.61	0.16568	34.272	43.612
100.00	5.0000	33.343	0.029992	-4.1962	-4.0463	0.060330	0.020096	0.044750	777.34	-0.35058	114.41	193.32
200.00	5.0000	3.5342	0.28295	1.8892	3.3039	0.11098	0.013847	0.030481	257.20	6.1700	16.132	18.205
300.00	5.0000	2.0596	0.48554	3.4362	5.8640	0.12145	0.012825	0.023284	328.37	3.1276	19.721	23.850
400.00	5.0000	1.5022	0.66567	4.7860	8.1144	0.12793	0.012643	0.021972	381.06	1.7370	23.882	29.440
500.00	5.0000	1.1914	0.83938	6.0868	10.284	0.13277	0.012578	0.021480	425.81	0.95180	27.811	34.619
600.00	5.0000	0.98986	1.0102	7.3671	12.418	0.13667	0.012547	0.021239	465.68	0.45113	31.477	39.420
700.00	5.0000	0.84774	1.1796	8.6370	14.535	0.13993	0.012530	0.021104	502.09	0.10636	34.910	43.899
100.00	10.000	33.779	0.029604	-4.2615	-3.9654	0.059648	0.020331	0.043670	805.91	-0.37991	118.60	205.82
200.00	10.000	8.4545	0.11828	1.1163	2.2991	0.10142	0.015240	0.048544	267.70	4.1966	23.573	23.429
300.00	10.000	4.1955	0.23835	3.1284	5.5119	0.11472	0.013145	0.025892	338.42	2.6066	22.192	25.583
400.00	10.000	2.9896	0.33449	4.5884	7.9333	0.12170	0.012803	0.023107	391.51	1.4795	25.505	30.479
500.00	10.000	2.3554	0.42455	5.9428	10.188	0.12674	0.012681	0.022129	436.14	0.79870	29.044	35.348
600.00	10.000	1.9532	0.51198	7.2556	12.375	0.13073	0.012621	0.021662	475.69	0.35092	32.479	39.971
700.00	10.000	1.6722	0.59801	8.5473	14.527	0.13404	0.012587	0.021400	511.73	0.036761	35.757	44.338

200.00	100.00	30.368	0.032930	-1.9234	1.3696	0.077319	0.017598	0.034137	854.48	-0.51219	99.994	120.12
300.00	100.00	24.152	0.041404	0.48428	4.6247	0.090561	0.015534	0.031005	733.39	-0.45856	70.273	75.589
400.00	100.00	19.710	0.050736	2.5103	7.5839	0.099092	0.014579	0.028287	690.42	-0.45296	59.077	61.649
500.00	100.00	16.607	0.060216	4.2850	10.307	0.10518	0.014048	0.026288	683.08	-0.49513	54.952	57.655
600.00	100.00	14.372	0.069578	5.9046	12.862	0.10984	0.013716	0.024915	690.94	-0.55858	53.635	57.185
700.00	100.00	12.695	0.078769	7.4268	15.304	0.11361	0.013492	0.023969	705.52	-0.62622	53.660	58.260
200.00	500.00	41.639	0.024016	-2.6867	9.3214	0.063085	0.022923	0.032604	1519.6	-0.60932	218.05	504.85
300.00	500.00	38.382	0.026054	-0.59281	12.434	0.075743	0.019980	0.029872	1443.8	-0.67083	184.82	259.42
400.00	500.00	35.672	0.028033	1.3125	15.329	0.084083	0.018316	0.028142	1393.0	-0.71809	161.66	183.29
500.00	500.00	33.368	0.029969	3.0950	18.079	0.090226	0.017262	0.026929	1358.5	-0.75700	145.02	151.69
600.00	500.00	31.379	0.031868	4.7903	20.724	0.095050	0.016535	0.026007	1334.9	-0.79113	132.94	135.48
700.00	500.00	29.643	0.033734	6.4198	23.287	0.099002	0.016000	0.025273	1318.8	-0.82207	124.11	125.84
300.00	1000.0	44.836	0.022304	-0.13952	22.164	0.068356	0.022820	0.030815	1866.0	-0.61332	267.67	666.74
400.00	1000.0	42.715	0.023411	1.7386	25.150	0.076958	0.020840	0.029020	1823.7	-0.65733	243.17	410.53
500.00	1000.0	40.856	0.024476	3.5106	27.987	0.083293	0.019528	0.027790	1791.9	-0.69220	223.98	297.24
600.00	1000.0	39.201	0.025510	5.2086	30.718	0.088276	0.018596	0.026881	1767.8	-0.72129	208.59	239.91
700.00	1000.0	37.711	0.026518	6.8515	33.369	0.092364	0.017896	0.026165	1749.3	-0.74683	196.06	207.86

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Tegeler, Ch., Span, R., and Wagner, W., “A New Equation of State for Argon Covering the Fluid Region for Temperatures from the Melting Line to 700 K at Pressures up to 1000 MPa,” *J. Phys. Chem. Ref. Data* **28**(3):779–850, 1999. The source for viscosity and thermal conductivity is Lemmon, E. W., and Jacobsen, R. T., “Viscosity and Thermal Conductivity Equations for Nitrogen, Oxygen, Argon, and Air,” *Int. J. Thermophys.* **25**:21–69, 2004.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The estimated uncertainty in density is less than 0.02% for pressures up to 12 MPa and temperatures up to 340 K with the exception of the critical region and less than 0.03% for pressures up to 30 MPa and temperatures between 235 and 520 K. Elsewhere, the uncertainty in density is generally within 0.2%. In the region with densities up to one-half the critical density and for temperatures between 90 and 450 K, the estimated uncertainty of calculated speeds of sound is in general less than 0.02%. In the liquid and supercritical regions, the uncertainty is less than 1%. The uncertainty in heat capacities is within 0.3% for the vapor and 2% for the liquid. The formulation gives reasonable extrapolation behavior up to very high pressures (50 GPa) and temperatures (17,000 K).

For viscosity, the uncertainty is 0.5% in the dilute gas. Away from the dilute gas (pressures greater than 1 MPa and in the liquid), the uncertainties are as low as 1% between 270 and 300 K at pressures less than 100 MPa, and increase outside that range. The uncertainties are around 2% at temperatures of 180 K and higher. Below this and away from the critical region, the uncertainties steadily increase to around 5% at the triple points of the fluids. The uncertainties in the critical region are higher.

For thermal conductivity, the uncertainty for the dilute gas is 2% with increasing uncertainties near the triple point. For the nondilute gas, the uncertainty is 2% for temperatures greater than 170 K. The uncertainty is 3% at temperatures less than the critical point and 5% in the critical region, except for states very near the critical point.

TABLE 2-191 Liquid-Vapor Equilibrium Data for the Argon-Nitrogen-Oxygen System

Liquid mole fraction				Vapor mole fraction			Temperature K	Relative volatility			p^0 (kPa)			Activity coefficient			Enthalpy kJ/kmol		Isobaric specific heat, kJ/(kmol·K)	
$N_2/(N_2+O_2)$	Ar	N_2	O_2	Ar	N_2	O_2		N_2/Ar	N_2/O_2	Ar/O_2	Ar	N_2	O_2	Ar	N_2	O_2	Liquid	Vapor	Liquid	Vapor
Pressure, 101.325 kPa (1 atm)																				
0.00	0.00	0.00	1.00	0.00000	0.0000	1.0000	90.188	—	—	1.606	136.010	366.29	101.33	1.196	—	1.000	-4268	2550	54.378	31.061
0.00	0.01	0.00	0.99	0.01590	0.0000	0.9841	90.128	—	—	1.600	135.210	364.42	100.69	1.192	—	1.000	-4271	2536	54.269	30.929
0.00	0.02	0.00	0.98	0.03151	0.0000	0.9685	90.070	—	—	1.594	134.430	362.62	100.08	1.188	—	1.001	-4275	2523	54.160	30.801
0.00	0.03	0.00	0.97	0.04684	0.0000	0.9532	90.013	—	—	1.589	133.680	360.86	99.49	1.183	—	1.001	-4278	2510	54.052	30.674
0.00	0.04	0.00	0.96	0.06189	0.0000	0.9381	89.956	—	—	1.583	132.930	359.10	98.89	1.179	—	1.001	-4282	2497	53.944	30.553
0.00	0.05	0.00	0.95	0.07667	0.0000	0.9233	89.901	—	—	1.578	132.200	357.41	98.32	1.175	—	1.002	-4285	2484	53.837	30.429
0.00	0.07	0.00	0.93	0.10548	0.0000	0.8945	89.794	—	—	1.567	130.810	354.14	97.22	1.167	—	1.002	-4292	2460	53.622	30.192
0.00	0.10	0.00	0.90	0.14692	0.0000	0.8531	89.640	—	—	1.550	128.810	349.47	95.65	1.156	—	1.004	-4302	2425	53.302	29.851
0.00	0.20	0.00	0.80	0.27211	0.0000	0.7279	89.183	—	—	1.495	123.040	335.88	91.10	1.120	—	1.012	-4336	2320	52.253	28.822
0.00	0.40	0.00	0.60	0.48104	0.0000	0.5190	88.470	—	—	1.390	114.430	315.46	84.35	1.065	—	1.039	-4409	2147	50.224	27.089
0.00	0.60	0.00	0.40	0.65969	0.0000	0.3403	87.948	—	—	1.292	108.420	301.10	79.66	1.028	—	1.082	-4491	2004	48.281	25.578
0.00	0.80	0.00	0.20	0.82774	0.0000	0.1723	87.566	—	—	1.201	104.180	290.90	76.35	1.006	—	1.143	-4586	1873	46.417	24.123
0.00	0.90	0.00	0.10	0.91247	0.0000	0.0875	87.420	—	—	1.158	102.590	287.07	75.12	1.001	—	1.181	-4638	1808	45.515	23.378
0.10	0.00	0.10	0.90	0.0000	0.3150	0.6850	87.546	2.573	4.138	1.608	103.960	290.37	76.18	1.193	1.099	1.012	-4235	2471	54.388	30.764
0.10	0.01	0.10	0.89	0.0122	0.3108	0.6771	87.529	2.579	4.131	1.602	103.780	289.93	76.04	1.188	1.097	1.013	-4238	2461	54.281	30.669
0.10	0.02	0.10	0.88	0.0242	0.3066	0.6692	87.512	2.584	4.124	1.596	103.590	289.48	75.89	1.184	1.095	1.013	-4241	2452	54.175	30.576
0.10	0.03	0.10	0.87	0.0361	0.3025	0.6614	87.495	2.589	4.117	1.590	103.410	289.03	75.75	1.180	1.093	1.013	-4244	2443	54.068	30.484
0.10	0.04	0.10	0.86	0.0479	0.2985	0.6536	87.479	2.595	4.110	1.584	103.230	288.61	75.61	1.176	1.092	1.014	-4248	2434	53.962	30.392
0.10	0.05	0.10	0.86	0.0596	0.2945	0.6459	87.463	2.600	4.103	1.578	103.060	288.19	75.48	1.172	1.090	1.014	-4251	2425	53.856	30.302
0.10	0.07	0.09	0.84	0.0826	0.2866	0.6308	87.432	2.611	4.089	1.566	102.720	287.38	75.22	1.164	1.086	1.015	-4257	2408	53.645	30.123
0.10	0.10	0.09	0.81	0.1164	0.2751	0.6086	87.389	2.627	4.068	1.549	102.260	286.26	74.86	1.153	1.082	1.017	-4266	2382	53.330	29.861
0.10	0.20	0.08	0.72	0.2230	0.2390	0.5380	87.270	2.680	3.999	1.492	100.980	283.17	73.87	1.119	1.069	1.025	-4300	2302	52.291	29.029
0.10	0.40	0.06	0.54	0.4179	0.1749	0.4072	87.121	2.790	3.866	1.386	99.401	279.34	72.64	1.065	1.057	1.052	-4378	2157	50.269	27.486
0.10	0.60	0.04	0.36	0.6027	0.1167	0.2807	87.077	2.904	3.742	1.288	98.938	278.22	72.28	1.029	1.062	1.093	-4469	2023	48.319	25.986
0.10	0.80	0.02	0.18	0.7916	0.0598	0.1486	87.132	3.023	3.625	1.199	99.517	279.63	72.73	1.008	1.084	1.150	-4575	1889	46.440	24.404
0.10	0.90	0.01	0.09	0.8923	0.0306	0.0771	87.201	3.085	3.569	1.157	100.250	281.40	73.00	1.002	1.101	1.185	-4633	1818	45.527	23.542
0.20	0.00	0.20	0.80	0.0000	0.5104	0.4896	85.511	2.606	4.170	1.600	83.506	240.35	60.37	1.188	1.076	1.027	-4176	2409	54.481	30.819
0.20	0.01	0.20	0.79	0.0098	0.5050	0.4853	85.519	2.611	4.162	1.594	83.580	240.53	60.42	1.184	1.074	1.027	-4179	2403	54.374	30.742
0.20	0.02	0.20	0.78	0.0195	0.4995	0.4810	85.526	2.616	4.154	1.588	83.645	240.69	60.47	1.180	1.073	1.028	-4183	2396	54.267	30.665
0.20	0.03	0.19	0.78	0.0292	0.4941	0.4767	85.534	2.621	4.146	1.582	83.719	240.88	60.53	1.176	1.071	1.028	-4186	2389	54.161	30.588
0.20	0.04	0.19	0.77	0.0388	0.4888	0.4724	85.542	2.626	4.139	1.576	83.793	241.06	60.59	1.172	1.070	1.029	-4190	2383	54.054	30.512
0.20	0.05	0.19	0.76	0.0484	0.4835	0.4682	85.551	2.631	4.131	1.570	83.876	241.27	60.65	1.169	1.069	1.029	-4193	2376	53.948	30.436
0.20	0.07	0.19	0.74	0.0674	0.4729	0.4597	85.568	2.641	4.115	1.558	84.033	241.66	60.77	1.161	1.066	1.030	-4200	2363	53.736	30.285
0.20	0.10	0.18	0.72	0.0957	0.4573	0.4470	85.594	2.656	4.092	1.541	84.274	242.25	60.96	1.150	1.063	1.032	-4211	2344	53.420	30.061
0.20	0.20	0.16	0.64	0.1879	0.4069	0.4052	85.695	2.707	4.017	1.484	85.216	244.59	61.68	1.117	1.054	1.040	-4250	2282	52.376	29.328
0.20	0.40	0.12	0.48	0.3685	0.3108	0.3206	85.947	2.812	3.878	1.379	87.602	250.48	63.52	1.066	1.048	1.066	-4340	2161	50.337	27.881
0.20	0.60	0.08	0.32	0.5541	0.2158	0.2302	86.280	2.921	3.750	1.284	90.832	258.43	66.01	1.030	1.057	1.104	-4444	2038	48.366	26.371
0.20	0.80	0.04	0.16	0.7583	0.1150	0.1267	86.717	3.033	3.630	1.197	95.211	269.15	69.39	1.009	1.082	1.157	-4563	1903	46.464	24.673
0.20	0.90	0.02	0.08	0.8729	0.0599	0.0671	86.987	3.090	3.572	1.156	97.996	275.93	71.55	1.003	1.101	1.188	-4628	1827	45.539	23.701
0.40	0.00	0.40	0.60	0.0000	0.7339	0.2661	82.557	2.639	4.138	1.568	—	179.50	42.12	—	1.036	1.067	-4012	2320	54.860	31.014
0.40	0.01	0.40	0.59	0.0070	0.7285	0.2646	82.590	2.644	4.131	1.562	—	180.11	42.29	—	1.035	1.067	-4017	2316	54.749	30.957
0.40	0.02	0.39	0.59	0.0139	0.7230	0.2630	82.623	2.649	4.123	1.557	—	180.72	42.47	—	1.034	1.067	-4022	2313	54.637	30.899
0.40	0.03	0.39	0.58	0.0209	0.7176	0.2615	82.656	2.654	4.116	1.551	—	181.33	42.65	—	1.033	1.068	-4028	2309	54.526	30.841
0.40	0.04	0.38	0.58	0.0279	0.7121	0.2600	82.689	2.659	4.108	1.545	—	181.95	42.82	—	1.033	1.068	-4033	2305	54.415	30.783
0.40	0.05	0.38	0.57	0.0349	0.7067	0.2585	82.722	2.664	4.101	1.540	—	182.56	43.00	—	1.032	1.068	-4038	2301	54.304	30.724
0.40	0.07	0.37	0.56	0.0490	0.6957	0.2553	82.789	2.673	4.087	1.529	—	183.81	43.37	—	1.031	1.069	-4049	2294	54.083	30.607
0.40	0.10	0.36	0.54	0.0702	0.6792	0.2506	82.890	2.688	4.066	1.513	—	185.71	43.92	—	1.029	1.071	-4066	2282	53.752	30.431
0.40	0.20	0.32	0.48	0.1424	0.6237	0.2340	83.238	2.738	3.998	1.460	—	192.37	45.87	—	1.027	1.077	-4123	2243	52.662	29.830
0.40	0.40	0.24	0.36																	

0.60	0.04	0.58	0.38	0.0218	0.8392	0.1390	80.619	2.676	4.024	1.504	—	146.40	32.74	—	1.008	1.121	-3851	2247	54.973	31.002
0.60	0.05	0.57	0.38	0.0273	0.8343	0.1384	80.664	2.681	4.018	1.499	—	147.12	32.94	—	1.008	1.121	-3859	2245	54.853	30.954
0.60	0.07	0.56	0.37	0.0384	0.8244	0.1372	80.755	2.690	4.007	1.489	—	148.56	33.34	—	1.008	1.121	-3874	2240	54.614	30.857
0.60	0.10	0.54	0.36	0.0554	0.8094	0.1352	80.893	2.705	3.991	1.475	—	150.78	33.95	—	1.007	1.121	-3898	2233	54.257	30.709
0.60	0.20	0.48	0.32	0.1145	0.7573	0.1282	81.368	2.756	3.938	1.429	—	158.60	36.14	—	1.008	1.123	-3978	2207	53.083	30.196
0.60	0.40	0.36	0.24	0.2487	0.6402	0.1112	82.415	2.860	3.839	1.342	—	176.90	41.36	—	1.019	1.135	-4148	2147	50.811	29.035
0.60	0.60	0.24	0.16	0.4170	0.4949	0.0881	83.658	2.967	3.743	1.262	—	200.64	48.32	—	1.041	1.155	-4327	2066	48.644	27.586
0.60	0.80	0.12	0.08	0.6475	0.2979	0.0546	85.216	3.068	3.640	1.186	80.82	233.67	58.30	1.015	1.077	1.186	-4512	1945	46.584	25.615
0.60	0.90	0.06	0.04	0.8025	0.1665	0.0310	86.175	3.111	3.581	1.151	89.80	255.91	65.21	1.006	1.099	1.204	-4604	1858	45.595	24.292
0.80	0.00	0.80	0.20	0.0000	0.9404	0.0596	78.768	2.673	3.942	1.475	—	119.30	25.42	—	0.998	1.189	-3620	2205	56.237	31.347
0.80	0.01	0.79	0.20	0.0044	0.9362	0.0594	78.820	2.677	3.938	1.471	—	120.01	25.60	—	0.998	1.188	-3630	2204	56.103	31.306
0.80	0.02	0.78	0.20	0.0089	0.9319	0.0592	78.871	2.682	3.934	1.467	—	120.70	25.79	—	0.998	1.187	-3640	2202	55.970	31.266
0.80	0.03	0.78	0.19	0.0133	0.9276	0.0590	78.923	2.687	3.930	1.463	—	121.41	25.97	—	0.998	1.187	-3650	2201	55.836	31.225
0.80	0.04	0.77	0.19	0.0179	0.9233	0.0588	78.974	2.691	3.926	1.459	—	122.12	26.16	—	0.998	1.186	-3660	2200	55.704	31.183
0.80	0.05	0.76	0.19	0.0224	0.9190	0.0586	79.027	2.696	3.921	1.455	—	122.85	26.35	—	0.997	1.186	-3670	2198	55.571	31.142
0.80	0.07	0.74	0.19	0.0317	0.9102	0.0581	79.132	2.705	3.913	1.447	—	124.31	26.74	—	0.997	1.185	-3691	2195	55.307	31.058
0.80	0.10	0.72	0.18	0.0458	0.8967	0.0575	79.292	2.720	3.901	1.434	—	126.55	27.34	—	0.997	1.183	-3721	2191	54.914	30.929
0.80	0.20	0.64	0.16	0.0958	0.8492	0.0550	79.848	2.769	3.863	1.395	—	134.60	29.51	—	0.999	1.179	-3825	2175	53.625	30.476
0.80	0.40	0.48	0.12	0.2139	0.7375	0.0486	81.090	2.874	3.793	1.320	—	153.99	34.85	—	1.011	1.178	-4039	2133	51.154	29.417
0.80	0.60	0.32	0.08	0.3708	0.5897	0.0396	82.596	2.982	3.723	1.249	—	180.22	42.32	—	1.036	1.185	-4261	2069	48.829	28.030
0.80	0.80	0.16	0.04	0.6029	0.3716	0.0255	84.549	3.082	3.639	1.181	74.99	219.07	53.85	1.018	1.074	1.201	-4483	1960	46.656	26.011
0.80	0.90	0.08	0.02	0.7712	0.2139	0.0149	85.793	3.121	3.584	1.148	86.14	246.87	62.39	1.008	1.098	1.212	-4591	1872	45.625	24.563
0.90	0.00	0.90	0.10	0.0000	0.9722	0.0278	78.036	2.683	3.891	1.450	—	109.71	22.91	—	0.998	1.228	-3519.2	2182.4	56.7	31.418
0.90	0.01	0.89	0.10	0.0040	0.9683	0.0277	78.090	2.687	3.888	1.447	—	110.4	23.088	—	0.997	1.227	-3530.4	2181.5	56.6	31.380
0.90	0.02	0.88	0.10	0.0081	0.9643	0.0276	78.144	2.692	3.884	1.443	—	111.09	23.267	—	0.997	1.226	-3541.6	2180.5	56.4	31.342
0.90	0.03	0.87	0.10	0.0122	0.9603	0.0275	78.199	2.696	3.881	1.439	—	111.79	23.45	—	0.997	1.225	-3552.8	2179.5	56.3	31.303
0.90	0.04	0.86	0.10	0.0164	0.9562	0.0274	78.254	2.701	3.877	1.436	—	112.5	23.634	—	0.997	1.224	-3564.1	2178.5	56.1	31.264
0.90	0.05	0.86	0.10	0.0206	0.9521	0.0273	78.309	2.705	3.874	1.432	—	113.22	23.82	—	0.997	1.223	-3575.3	2177.5	56.0	31.225
0.90	0.07	0.84	0.09	0.0291	0.9438	0.0271	78.420	2.714	3.867	1.425	—	114.67	24.198	—	0.996	1.221	-3598	2175.3	55.7	31.146
0.90	0.10	0.81	0.09	0.0421	0.9310	0.0268	78.589	2.728	3.857	1.414	—	116.9	24.783	—	0.996	1.219	-3632.1	2172	55.3	31.024
0.90	0.20	0.72	0.08	0.0886	0.8857	0.0257	79.177	2.776	3.825	1.378	—	124.93	26.908	—	0.998	1.211	-3747.3	2159.4	53.9	30.594
0.90	0.40	0.54	0.06	0.1999	0.7772	0.0229	80.497	2.880	3.768	1.308	—	144.49	32.209	—	1.009	1.202	-3983.9	2124.8	51.3	29.578
0.90	0.60	0.36	0.04	0.3513	0.6299	0.0189	82.110	2.988	3.711	1.242	—	171.42	39.784	—	1.034	1.201	-4226.4	2068.5	48.9	28.223
0.90	0.80	0.18	0.02	0.5828	0.4049	0.0124	84.233	3.088	3.638	1.178	72.35	212.4	51.833	1.020	1.073	1.209	-4468.2	1965.6	46.7	26.192
0.90	0.90	0.09	0.01	0.7563	0.2364	0.0073	85.608	3.125	3.585	1.147	84.404	242.58	61.055	1.009	1.097	1.216	-4584.8	1877.8	45.6	24.692
0.97	0.00	0.97	0.03	0.0000	0.9921	0.0080	77.555	2.691	3.857	1.433	—	103.74	21.375	—	0.999	1.257	-3448.5	2167.6	57.0	31.465
0.97	0.01	0.96	0.03	0.0038	0.9883	0.0079	77.610	2.695	3.854	1.430	—	104.41	21.546	—	0.999	1.256	-3460.5	2166.9	56.9	31.429
0.97	0.02	0.95	0.03	0.0077	0.9844	0.0079	77.666	2.700	3.851	1.426	—	105.09	21.722	—	0.998	1.254	-3472.5	2166.1	56.7	31.392
0.97	0.03	0.94	0.03	0.0116	0.9806	0.0079	77.722	2.704	3.848	1.423	—	105.78	21.898	—	0.998	1.253	-3484.6	2165.3	56.6	31.355
0.97	0.04	0.93	0.03	0.0155	0.9767	0.0079	77.779	2.708	3.844	1.420	—	106.49	22.079	—	0.998	1.252	-3496.6	2164.5	56.5	31.318
0.97	0.05	0.92	0.03	0.0195	0.9727	0.0078	77.836	2.712	3.841	1.416	—	107.2	22.261	—	0.998	1.251	-3508.7	2163.7	56.3	31.280
0.97	0.07	0.90	0.03	0.0275	0.9647	0.0078	77.950	2.721	3.835	1.409	—	108.62	22.629	—	0.998	1.249	-3533	2162.1	56.0	31.203
0.97	0.10	0.87	0.03	0.0399	0.9524	0.0077	78.125	2.735	3.826	1.399	—	110.85	23.204	—	0.997	1.245	-3569.5	2159.4	55.6	31.086
0.97	0.20	0.78	0.02	0.0842	0.9084	0.0074	78.734	2.781	3.798	1.366	—	118.85	25.294	—	0.998	1.235	-3692.7	2149.1	54.2	30.670
0.97	0.40	0.58	0.02	0.1912	0.8022	0.0066	80.104	2.884	3.750	1.300	—	138.44	30.551	—	1.009	1.219	-3944.8	2119.2	51.5	29.681
0.97	0.60	0.39	0.01	0.3389	0.6557	0.0055	81.786	2.992	3.702	1.237	—	165.73	38.159	—	1.033	1.212	-4202.1	2068.1	49.0	28.347
0.97	0.80	0.19	0.01	0.5694	0.4270	0.0036	84.018	3.092	3.636	1.176	70.593	207.94	50.496	1.022	1.072	1.215	-4457.7	1969.5	46.7	26.313
0.97	0.90	0.10	0.00	0.7462	0.2516	0.0022	85.481	3.128	3.585	1.146	83.23	239.66	60.153	1.009	1.097	1.219	-4580.2	1881.9	45.7	24.780
1.00	0.00	1.00	0.00	0.0000	1.0000	0.0000	77.355	2.695	—	—	—	101.33	20.762	—	1.000	—	-3418.2	2161.5	57.2	31.485
1.00	0.01	0.99	0.00	0.0037	0.9963	0.0000	77.411	2.699	—	—	—	102	20.932	—	1.000	—	-3430.5	2160.8	57.0	31.449
1.00	0.02	0.98	0.00	0.0075	0.9925	0.0000	77.468	2.703	—	—	—	102.68	21.106	—	0.999	—	-3442.9	2160.1	56.9	31.413
1.00	0.03	0.97	0.00	0.0113	0.9887	0.0000	77.525	2.708	—	—	—	103.37	21.282	—	0.999	—	-3455.3	2159.4	56.7	31.377
1.00	0.04	0.96	0.00	0.0151	0.9849	0.0000	77.582	2.712	—	—	—	104.06	21.459	—	0.999	—	-3467.7	2158.7	56.6	31.340
1.00	0.05	0.95	0.00	0.0190	0.9810	0.0000	77.640	2.716	—	—	—	104.77	21.64	—	0.999	—	-3480.2	2158	56.4	31.303
1.00	0.07	0.93	0.00	0.0269	0.9731	0.0000	77.756	2.725	—	—	—	106.2	22.006	—	0.998	—	-3505.1	2156.5	56.2	31.227
1.00	0.10	0.90	0.00	0.0390	0.9610	0.0000	77.933	2.738	—	—	—	108.41	22.574	—	0.998	—	-3542.7	2154.1	55.7	31.112
1.00	0.20	0.80	0.00	0.0824	0.9176	0.0000	78.550	2.784	—	—	—	116.38	24.647	—	0.999	—	-3669.3	2144.7	54.3	30.701
1.00	0.40	0.60	0.00	0.1877	0.8123	0.0000	79.941	2.885	—	—	—	135.99	29.883	—	1.009	—	-3927.9	2116.8	51.6	29.723
1.00	0.60	0.40	0.00	0.3338	0.6662	0.0000	81.650	2.994	—	—	—	163.38	37.493	—	1.033	—	-4191.7	2067.7	49.0	28.398
1.00	0.80	0.2																		

TABLE 2-191 Liquid-Vapor Equilibrium Data for the Argon-Nitrogen-Oxygen System (Concluded)

Liquid mole fraction				Vapor mole fraction			Temperature K	Relative volatility			p ⁰ (kPa)			Activity coefficient			Enthalpy kJ/kmol		Isobaric specific heat, kJ/(kmol·K)	
N ₂ /(N ₂ +O ₂)	Ar	N ₂	O ₂	Ar	N ₂	O ₂		N ₂ /Ar	N ₂ /O ₂	Ar/O ₂	Ar	N ₂	O ₂	Ar	N ₂	O ₂	Liquid	Vapor	Liquid	Vapor
Pressure, 405.3 kPa (4 atm)																				
0.00	0.02	0.00	0.98	0.0275	0.0000	0.9725	105.790	—	—	1.385	499.6	1138.3	401.72	1.115	—	1.001	−3402.6	2829.5	56.5	33.534
0.00	0.03	0.00	0.97	0.0410	0.0000	0.9590	105.740	—	—	1.381	497.83	1134.8	400.22	1.112	—	1.001	−3407.8	2816.1	56.4	33.431
0.00	0.04	0.00	0.96	0.0543	0.0000	0.9457	105.680	—	—	1.378	495.72	1130.5	398.43	1.110	—	1.002	−3412.9	2802.8	56.3	33.329
0.00	0.05	0.00	0.95	0.0675	0.0000	0.9326	105.630	—	—	1.374	493.96	1127	396.94	1.107	—	1.002	−3418	2789.7	56.2	33.229
0.00	0.07	0.00	0.93	0.0933	0.0000	0.9067	105.520	—	—	1.367	490.12	1119.3	393.68	1.102	—	1.004	−3428.3	2764	56.0	33.031
0.00	0.10	0.00	0.90	0.1310	0.0000	0.8690	105.370	—	—	1.357	484.91	1108.8	389.27	1.095	—	1.005	−3443.6	2726.7	55.7	32.744
0.00	0.20	0.00	0.80	0.2484	0.0000	0.7516	104.900	—	—	1.322	468.86	1076.5	375.67	1.074	—	1.014	−3495.1	2611.3	54.6	31.846
0.00	0.40	0.00	0.60	0.4556	0.0000	0.5445	104.150	—	—	1.255	444.04	1026.3	354.7	1.040	—	1.037	−3601.7	2411.9	52.5	30.249
0.00	0.60	0.00	0.40	0.6412	0.0000	0.3588	103.590	—	—	1.191	426.16	989.9	339.63	1.016	—	1.071	−3716.7	2238.3	50.5	28.788
0.00	0.80	0.00	0.20	0.8190	0.0000	0.1810	103.190	—	—	1.131	413.71	964.49	329.15	1.003	—	1.115	−3842.6	2076.7	48.6	27.354
0.00	0.90	0.00	0.10	0.9084	0.0000	0.0916	103.030	—	—	1.102	408.8	954.46	325.03	1.001	—	1.142	−3910.1	1997.2	47.6	26.620
0.10	0.00	0.10	0.90	0.0000	0.2399	0.7601	103.430	2.045	2.840	1.389	421.14	979.68	335.41	1.129	0.992	1.021	−3351.1	2779.8	56.8	33.931
0.10	0.01	0.10	0.89	0.0117	0.2369	0.7515	103.400	2.048	2.837	1.385	420.21	977.77	334.62	1.127	0.992	1.022	−3356.1	2769.2	56.7	33.841
0.10	0.02	0.10	0.88	0.0233	0.2339	0.7429	103.380	2.051	2.833	1.381	419.59	976.5	334.1	1.124	0.990	1.022	−3361.1	2758.7	56.5	33.752
0.10	0.03	0.10	0.87	0.0348	0.2309	0.7343	103.360	2.054	2.830	1.378	418.97	975.23	333.57	1.121	0.989	1.022	−3366.1	2748.2	56.4	33.663
0.10	0.04	0.10	0.86	0.0462	0.2280	0.7259	103.340	2.057	2.827	1.374	418.34	973.97	333.05	1.118	0.988	1.022	−3371.1	2737.9	56.3	33.575
0.10	0.05	0.10	0.86	0.0575	0.2251	0.7174	103.320	2.060	2.823	1.370	417.72	972.7	332.53	1.116	0.987	1.023	−3376.2	2727.6	56.2	33.488
0.10	0.07	0.09	0.84	0.0799	0.2193	0.7008	103.280	2.066	2.817	1.363	416.49	970.17	331.49	1.111	0.985	1.024	−3386.3	2707.4	56.0	33.315
0.10	0.10	0.09	0.81	0.1129	0.2109	0.6762	103.220	2.075	2.807	1.352	414.63	966.38	329.93	1.104	0.983	1.026	−3401.7	2677.6	55.7	33.059
0.10	0.20	0.08	0.72	0.2186	0.1841	0.5973	103.060	2.105	2.774	1.317	409.72	956.33	325.8	1.081	0.975	1.032	−3454.5	2583	54.6	32.238
0.10	0.40	0.06	0.54	0.4159	0.1352	0.4489	102.840	2.166	2.709	1.251	403.04	942.64	320.19	1.046	0.968	1.052	−3568	2409.5	52.6	30.684
0.10	0.60	0.04	0.36	0.6047	0.0899	0.3054	102.740	2.229	2.648	1.188	400.02	936.47	317.66	1.021	0.972	1.082	−3693.2	2247.1	50.6	29.163
0.10	0.80	0.02	0.18	0.7958	0.0456	0.1586	102.760	2.294	2.591	1.129	400.62	937.7	318.16	1.006	0.986	1.122	−3830.8	2086.1	48.6	27.585
0.10	0.90	0.01	0.09	0.8955	0.0232	0.0813	102.820	2.328	2.563	1.101	402.43	941.41	319.68	1.002	0.997	1.146	−3904.2	2003.1	47.6	26.748
0.20	0.00	0.20	0.80	0.0000	0.4159	0.5841	101.340	2.061	2.848	1.382	359.58	853.02	283.81	1.138	0.988	1.043	−3293.1	2714.9	56.9	34.186
0.20	0.01	0.20	0.79	0.0101	0.4114	0.5785	101.340	2.064	2.845	1.378	359.58	853.02	283.81	1.135	0.987	1.043	−3298.4	2706.6	56.8	34.106
0.20	0.02	0.20	0.78	0.0201	0.4069	0.5730	101.340	2.067	2.841	1.375	359.58	853.02	283.81	1.132	0.986	1.044	−3303.7	2698.3	56.6	34.026
0.20	0.03	0.19	0.78	0.0301	0.4025	0.5674	101.350	2.070	2.837	1.371	359.85	853.59	284.04	1.129	0.985	1.043	−3309.1	2690.1	56.5	33.946
0.20	0.04	0.19	0.77	0.0400	0.3981	0.5619	101.350	2.072	2.834	1.367	359.85	853.59	284.04	1.127	0.984	1.044	−3314.5	2681.9	56.4	33.867
0.20	0.05	0.19	0.76	0.0499	0.3937	0.5564	101.350	2.075	2.830	1.364	359.85	853.59	284.04	1.124	0.984	1.045	−3319.9	2673.7	56.3	33.787
0.20	0.07	0.19	0.74	0.0696	0.3849	0.5455	101.360	2.081	2.823	1.356	360.13	854.17	284.27	1.119	0.982	1.045	−3330.8	2657.6	56.1	33.630
0.20	0.10	0.18	0.72	0.0989	0.3720	0.5291	101.380	2.089	2.812	1.346	360.69	855.32	284.74	1.111	0.979	1.046	−3347.5	2633.6	55.8	33.395
0.20	0.20	0.16	0.64	0.1948	0.3300	0.4753	101.440	2.118	2.777	1.311	362.36	858.79	286.13	1.089	0.973	1.052	−3405.2	2555.4	54.7	32.622
0.20	0.40	0.12	0.48	0.3822	0.2496	0.3682	101.640	2.176	2.711	1.246	367.98	870.43	290.82	1.052	0.968	1.069	−3530.2	2404.5	52.6	31.093
0.20	0.60	0.08	0.32	0.5719	0.1706	0.2575	101.940	2.237	2.649	1.185	376.52	888.1	297.97	1.026	0.973	1.095	−3668.1	2253.6	50.6	29.517
0.20	0.80	0.04	0.16	0.7738	0.0889	0.1373	102.350	2.299	2.592	1.127	388.44	912.66	307.94	1.009	0.987	1.129	−3818.6	2094.6	48.6	27.809
0.20	0.90	0.02	0.08	0.8829	0.0457	0.0713	102.610	2.330	2.564	1.100	396.13	928.48	314.4	1.004	0.998	1.150	−3898.3	2008.8	47.6	26.873
0.40	0.00	0.40	0.60	0.0000	0.6538	0.3462	97.997	2.082	2.832	1.360	275.25	675.50	213.91	1.156	0.981	1.093	−3143.7	2610.3	57.346	34.697
0.40	0.01	0.40	0.59	0.0079	0.6484	0.3438	98.032	2.084	2.829	1.357	276.05	677.21	214.57	1.153	0.980	1.093	−3150.5	2605.1	57.232	34.629
0.40	0.02	0.39	0.59	0.0157	0.6429	0.3414	98.067	2.087	2.825	1.354	276.85	678.92	215.23	1.150	0.979	1.093	−3157.4	2599.8	57.118	34.560
0.40	0.03	0.39	0.58	0.0236	0.6375	0.3389	98.101	2.090	2.822	1.350	277.63	680.58	215.87	1.148	0.978	1.093	−3164.4	2594.5	57.004	34.492
0.40	0.04	0.38	0.58	0.0315	0.6321	0.3364	98.136	2.092	2.818	1.347	278.43	682.30	216.53	1.145	0.978	1.093	−3171.3	2589.2	56.891	34.423
0.40	0.05	0.38	0.57	0.0394	0.6267	0.3340	98.172	2.095	2.815	1.344	279.26	684.07	217.21	1.142	0.977	1.093	−3178.3	2584.0	56.778	34.354
0.40	0.07	0.37	0.56	0.0552	0.6159	0.3290	98.242	2.100	2.808	1.337	280.88	687.51	218.54	1.137	0.976	1.093	−3192.4	2573.4	56.551	34.217
0.40	0.10	0.36	0.54	0.0790	0.5996	0.3214	98.350	2.108	2.798	1.327	283.38	692.86	220.61	1.130	0.974	1.094	−3213.9	2557.4	56.212	34.009
0.40	0.20	0.32	0.48	0.1595	0.5449	0.2956	98.721	2.136	2.765	1.295	292.12	711.44	227.81	1.106	0.970	1.096	−3287.5	2503.3	55.091	33.304
0.40	0.40	0.24	0.36	0.3285	0.4320	0.2396	99.531	2.192	2.704	1.234	311.88	753.27	244.15	1.067	0.968	1.105	−3			

0.60	0.07	0.56	0.37	0.0457	0.7694	0.1850	95.771	2.113	2.773	1.312	227.93	573.19	175.17	1.160	0.975	1.150	-3032.6	2503.5	57.243	34.742
0.60	0.10	0.54	0.36	0.0657	0.7528	0.1815	95.934	2.121	2.765	1.304	231.17	580.27	177.81	1.152	0.974	1.149	-3060.2	2492.8	56.870	34.550
0.60	0.20	0.48	0.32	0.1350	0.6957	0.1694	96.496	2.148	2.738	1.275	242.61	605.18	187.15	1.127	0.971	1.146	-3154.0	2455.3	55.639	33.887
0.60	0.40	0.36	0.24	0.2878	0.5707	0.1415	97.721	2.203	2.688	1.220	269.02	662.15	208.79	1.084	0.970	1.145	-3349.9	2369.9	53.233	32.433
0.60	0.60	0.24	0.16	0.4690	0.4239	0.1070	99.133	2.260	2.641	1.169	302.05	732.50	236.01	1.049	0.977	1.149	-3554.9	2263.2	50.933	30.730
0.60	0.80	0.12	0.08	0.6962	0.2417	0.0622	100.820	2.314	2.592	1.120	345.35	823.43	271.95	1.021	0.991	1.158	-3766.5	2121.1	48.741	28.628
0.60	0.90	0.06	0.04	0.8358	0.1303	0.0339	101.810	2.339	2.565	1.096	372.80	880.41	294.86	1.010	1.000	1.164	-3873.6	2029.2	47.694	27.351
0.80	0.00	0.80	0.20	0.0000	0.9165	0.0835	93.251	2.106	2.744	1.303	182.01	471.46	138.01	1.212	0.985	1.226	-2783.0	2458.2	59.157	35.652
0.80	0.01	0.79	0.20	0.0055	0.9114	0.0831	93.315	2.108	2.742	1.301	183.08	473.86	138.88	1.209	0.984	1.225	-2794.3	2456.1	59.014	35.593
0.80	0.02	0.78	0.20	0.0110	0.9063	0.0827	93.380	2.111	2.740	1.298	184.17	476.32	139.75	1.205	0.984	1.224	-2805.5	2454.0	58.871	35.533
0.80	0.03	0.78	0.19	0.0165	0.9012	0.0823	93.445	2.113	2.738	1.296	185.27	478.78	140.64	1.202	0.983	1.223	-2816.8	2451.8	58.729	35.474
0.80	0.04	0.77	0.19	0.0221	0.8961	0.0819	93.510	2.116	2.736	1.293	186.37	481.25	141.53	1.199	0.983	1.221	-2828.1	2449.7	58.586	35.414
0.80	0.05	0.76	0.19	0.0277	0.8909	0.0815	93.576	2.118	2.734	1.291	187.50	483.77	142.43	1.196	0.982	1.220	-2839.4	2447.5	58.444	35.354
0.80	0.07	0.74	0.19	0.0390	0.8804	0.0806	93.708	2.123	2.729	1.286	189.76	488.84	144.26	1.190	0.981	1.218	-2862.1	2443.0	58.161	35.232
0.80	0.10	0.72	0.18	0.0563	0.8643	0.0793	93.910	2.131	2.723	1.278	193.27	496.68	147.09	1.181	0.980	1.215	-2896.3	2436.0	57.739	35.047
0.80	0.20	0.64	0.16	0.1171	0.8082	0.0747	94.605	2.157	2.704	1.254	205.71	524.32	157.14	1.153	0.976	1.205	-3011.9	2410.8	56.353	34.405
0.80	0.40	0.48	0.12	0.2562	0.6801	0.0637	96.135	2.212	2.667	1.206	235.21	589.09	181.11	1.104	0.975	1.189	-3249.2	2348.3	53.683	32.970
0.80	0.60	0.32	0.08	0.4302	0.5204	0.0494	97.925	2.268	2.631	1.160	273.62	672.00	212.57	1.062	0.981	1.178	-3492.7	2261.3	51.165	31.234
0.80	0.80	0.16	0.04	0.6627	0.3076	0.0297	100.110	2.321	2.590	1.116	326.60	784.23	256.36	1.028	0.994	1.174	-3758.9	2130.7	48.820	28.996
0.80	0.90	0.08	0.02	0.8139	0.1696	0.0165	101.420	2.343	2.565	1.095	361.80	857.63	285.67	1.013	1.002	1.172	-3860.8	2038.1	47.724	27.577
0.90	0.00	0.90	0.10	0.0000	0.9607	0.0393	92.292	2.111	2.719	1.288	166.50	436.43	125.58	1.231	0.991	1.267	-2687.7	2426.8	59.774	35.885
0.90	0.01	0.89	0.10	0.0051	0.9558	0.0391	92.361	2.114	2.717	1.286	167.59	438.89	126.45	1.227	0.991	1.265	-2700.1	2425.2	59.619	35.827
0.90	0.02	0.88	0.10	0.0102	0.9509	0.0389	92.431	2.116	2.716	1.283	168.69	441.39	127.33	1.224	0.990	1.264	-2712.5	2423.6	59.466	35.768
0.90	0.03	0.87	0.10	0.0153	0.9459	0.0387	92.501	2.119	2.714	1.281	169.79	443.90	128.22	1.221	0.989	1.262	-2724.9	2422.0	59.312	35.709
0.90	0.04	0.86	0.10	0.0205	0.9409	0.0385	92.571	2.121	2.712	1.279	170.91	446.42	129.11	1.218	0.989	1.261	-2737.3	2420.3	59.159	35.650
0.90	0.05	0.86	0.10	0.0258	0.9359	0.0384	92.642	2.123	2.710	1.276	172.04	448.99	130.02	1.214	0.988	1.259	-2749.8	2418.6	59.007	35.590
0.90	0.07	0.84	0.09	0.0364	0.9256	0.0380	92.784	2.128	2.707	1.272	174.33	454.16	131.85	1.208	0.987	1.256	-2774.7	2415.2	58.703	35.469
0.90	0.10	0.81	0.09	0.0526	0.9100	0.0374	93.001	2.136	2.702	1.265	177.86	462.14	134.69	1.199	0.985	1.251	-2812.4	2409.7	58.250	35.286
0.90	0.20	0.72	0.08	0.1099	0.8548	0.0354	93.751	2.161	2.686	1.242	190.51	490.50	144.86	1.169	0.981	1.237	-2939.0	2389.6	56.769	34.648
0.90	0.40	0.54	0.06	0.2429	0.7267	0.0304	95.408	2.216	2.655	1.198	220.83	557.64	169.40	1.115	0.978	1.212	-3197.3	2337.1	53.939	33.216
0.90	0.60	0.36	0.04	0.4131	0.5631	0.0238	97.360	2.272	2.625	1.156	261.03	644.98	202.23	1.069	0.983	1.194	-3460.7	2259.1	51.297	31.467
0.90	0.80	0.18	0.02	0.6471	0.3384	0.0145	99.771	2.324	2.589	1.114	317.92	765.99	249.16	1.031	0.995	1.181	-3724.5	2134.7	48.865	29.170
0.90	0.90	0.09	0.01	0.8034	0.1884	0.0082	101.230	2.346	2.565	1.094	356.53	846.69	281.27	1.015	1.002	1.176	-3854.3	2042.2	47.740	27.687
0.97	0.00	0.97	0.03	0.0000	0.9887	0.0113	91.656	2.116	2.702	1.277	156.79	414.28	117.83	1.245	0.997	1.298	-2620.4	2405.9	60.245	36.049
0.97	0.01	0.96	0.03	0.0048	0.9839	0.0113	91.728	2.118	2.700	1.275	157.87	416.75	118.68	1.242	0.996	1.296	-2633.6	2404.6	60.082	35.990
0.97	0.02	0.95	0.03	0.0097	0.9791	0.0112	91.801	2.120	2.699	1.273	158.97	419.26	119.56	1.238	0.996	1.294	-2646.8	2403.3	59.920	35.932
0.97	0.03	0.94	0.03	0.0146	0.9742	0.0112	91.874	2.123	2.697	1.271	160.07	421.78	120.44	1.235	0.995	1.292	-2660.0	2402.0	59.758	35.873
0.97	0.04	0.93	0.03	0.0196	0.9693	0.0111	91.948	2.125	2.696	1.268	161.20	424.35	121.34	1.231	0.994	1.290	-2673.3	2400.6	59.596	35.813
0.97	0.05	0.92	0.03	0.0246	0.9643	0.0111	92.022	2.128	2.694	1.266	162.33	426.93	122.24	1.228	0.993	1.288	-2686.5	2399.3	59.436	35.754
0.97	0.07	0.90	0.03	0.0347	0.9543	0.0110	92.171	2.132	2.691	1.262	164.62	432.16	124.08	1.221	0.992	1.284	-2713.1	2396.5	59.115	35.634
0.97	0.10	0.87	0.03	0.0503	0.9389	0.0108	92.397	2.140	2.687	1.256	168.15	440.17	126.90	1.212	0.990	1.279	-2753.1	2392.0	58.639	35.450
0.97	0.20	0.78	0.02	0.1053	0.8845	0.0102	93.182	2.165	2.673	1.235	180.86	468.87	137.09	1.180	0.985	1.261	-2887.4	2375.1	57.084	34.814
0.97	0.40	0.58	0.02	0.2344	0.7567	0.0088	94.921	2.219	2.647	1.193	211.56	537.25	161.88	1.123	0.981	1.230	-3160.7	2329.1	54.131	33.381
0.97	0.60	0.39	0.01	0.4019	0.5911	0.0070	96.979	2.274	2.621	1.152	252.78	627.20	195.47	1.074	0.984	1.205	-3437.9	2257.3	51.395	31.622
0.97	0.80	0.19	0.01	0.6366	0.3591	0.0043	99.538	2.326	2.588	1.112	312.06	753.64	244.29	1.034	0.995	1.187	-3714.4	2137.3	48.897	29.289
0.97	0.90	0.10	0.00	0.7962	0.2014	0.0024	101.100	2.347	2.565	1.093	352.96	839.27	278.29	1.016	1.003	1.179	-3849.7	2045.0	47.752	27.763
1.00	0.00	1.00	0.00	0.0000	1.0000	0.0000	91.391	2.118	—	—	152.87	405.30	114.70	1.252	1.000	—	-2591.4	2397.1	60.458	36.119
1.00	0.01	0.99	0.00	0.0047	0.9953	0.0000	91.465	2.120	—	—	153.96	407.79	115.57	1.248	0.999	—	-2605.0	2396.0	60.291	36.060
1.00	0.02	0.98	0.00	0.0095	0.9905	0.0000	91.539	2.122	—	—	155.05	410.30	116.44	1.245	0.998	—	-2618.5	2394.8	60.124	36.002
1.00	0.03	0.97	0.00	0.0143	0.9857	0.0000	91.613	2.125	—	—	156.15	412.81	117.31	1.241	0.998	—	-2632.1	2393.6	59.959	35.943
1.00	0.04	0.96	0.00	0.0192	0.9808	0.0000	91.688	2.127	—	—	157.27	415.38	118.21	1.238	0.997	—	-2645.7	2392.4	59.793	35.883
1.00	0.05	0.95	0.00	0.0241	0.9759	0.0000	91.763	2.129	—	—	158.39	417.95	119.10	1.234	0.996	—	-2659.3	2391.2	59.629	35.824
1.00	0.07	0.93	0.00	0.0341	0.9659	0.0000	91.915	2.134	—	—	160.69	423.20	120.94	1.228	0.995	—	-2686.6	2388.6	59.301	35.704
1.00	0.10	0.90	0.00	0.0493	0.9507	0.0000	92.146	2.141	—	—	164.24	431.27	123.77	1.217	0.993	—	-2727.6	2384.6	58.814	35.520
1.00	0.20	0.80	0.00	0.1035	0.8965	0.0000	92.945	2.166	—	—	176.95	460.07	133.95	1.185	0.987	—	-2865.3	2369.0	57.225	34.883
1.00	0.40	0.60	0.00	0.2310	0.7690	0.0000	94.718	2.220	—	—	207.79	528.92	158.82	1.126	0.982	—	-3144.8	2325.6	54.217	33.450
1.00	0.60	0.40	0.00																	

TABLE 2-192 Thermodynamic Properties of the International Standard Atmosphere*

Z, m	T, K	P, bar	ρ , kg/m ³	g, m/s ²	M	a, m/s	μ , Pa·s	k, W/(m·K)	λ , m	H, m
0	288.15	1.01325	1.2250	9.80665	28.964	340.29	1.79-5	2.54-5	6.63-8	0
1,000	281.65	0.89876	1.1117	9.8036	28.964	336.43	1.76-5	2.49-5	7.31-8	1,000
2,000	275.15	0.79501	1.0066	9.8005	28.964	332.53	1.73-5	2.43-5	8.07-8	2,999
3,000	268.66	0.70121	0.90925	9.7974	28.964	328.58	1.69-5	2.38-5	8.94-8	2,999
4,000	262.17	0.61660	0.81935	9.7943	28.964	324.59	1.66-5	2.33-5	9.92-8	3,997
5,000	255.68	0.54048	0.73643	9.7912	28.964	320.55	1.63-5	2.28-5	1.10-7	4,996
6,000	249.19	0.47217	0.66011	9.7882	28.964	316.45	1.59-5	2.22-5	1.23-7	5,994
7,000	242.70	0.41105	0.59002	9.7851	28.964	312.31	1.56-5	2.17-5	1.38-7	6,992
8,000	236.22	0.35651	0.52579	9.7820	28.964	308.11	1.53-5	2.12-5	1.55-7	7,990
9,000	229.73	0.30800	0.46706	9.7789	28.964	303.85	1.49-5	2.06-5	1.74-7	8,987
10,000	223.25	0.26499	0.41351	9.7759	28.964	299.53	1.46-5	2.01-5	1.97-7	9,984
15,000	216.65	0.12111	0.19476	9.7605	28.964	295.07	1.42-5	1.95-5	4.17-7	14,965
20,000	216.65	0.05529	0.08891	9.7452	28.964	295.07	1.42-5	1.95-5	9.14-7	19,937
25,000	221.55	0.02549	0.04008	9.7300	28.964	298.39	1.45-5	1.99-5	2.03-6	24,902
30,000	226.51	0.01197	0.01841	9.7147	28.964	301.71	1.48-5	2.04-5	4.42-6	29,859
40,000	250.35	2.87-3	4.00-3	9.6844	28.964	317.19	1.60-5	2.23-5	2.03-5	39,750
50,000	270.65	8.00-4	1.03-3	9.6542	28.964	329.80	1.70-5	2.40-5	7.91-5	49,610
60,000	247.02	2.20-4	3.10-4	9.6241	28.964	315.07	1.58-5	2.21-5	2.62-4	59,439
70,000	219.59	5.22-5	8.28-5	9.5942	28.964	297.06	1.44-5	1.98-5	9.81-4	69,238
80,000	198.64	1.05-5	1.85-5	9.5644	28.964	282.54	1.32-5	1.80-5	4.40-3	79,006
90,000	186.87	1.84-6	3.43-6	9.5348	28.95				2.37-2	88,744
100,000	195.08	3.20-7	5.60-7	9.5052	28.40				0.142	98,451
150,000	634.39	4.54-9	2.08-9	9.3597	24.10				33	146,542
200,000	854.56	8.47-10	2.54-10	9.2175	21.30				240	193,899
250,000	941.33	2.48-10	6.07-11	9.0785	19.19				890	240,540
300,000	976.01	8.77-11	1.92-11	8.9427	17.73				2600	286,480
400,000	995.83	1.45-11	2.80-12	8.6799	15.98				1.6+4	376,320
500,000	999.24	3.02-12	5.22-13	8.4286	14.33				7.7+4	463,540
600,000	999.85	8.21-13	1.14-13	8.1880	11.51				2.8+5	548,252
800,000	999.99	1.70-13	1.14-14	7.7368	5.54				1.4+6	710,574
1,000,000	1000.00	7.51-14	3.56-15	7.3218	3.94				3.1+6	864,071

*Extracted from *U.S. Standard Atmosphere, 1976, National Oceanic and Atmospheric Administration*, National Aeronautics and Space Administration and the U.S. Air Force, Washington, 1976. Z = geometric altitude, T = temperature, P = pressure, g = acceleration of gravity, M = molecular weight, a = velocity of sound, μ = viscosity, k = thermal conductivity, λ = mean free path, ρ = density, and H = geopotential altitude. The notation 1.79-5 signifies 1.79×10^{-5} .

TABLE 2-193 Thermodynamic Properties of Benzene

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
278.7	0.0047988	11.441	0.087404	-10.278	-10.278	-0.03257	0.095741	0.13301	1369.2	-0.45084
280	0.0051472	11.424	0.087532	-10.105	-10.105	-0.03196	0.095210	0.13274	1366.0	-0.45037
295	0.010954	11.227	0.089073	-8.1218	-8.1208	-0.02506	0.092757	0.13249	1316.2	-0.43580
310	0.021374	11.024	0.090715	-6.1160	-6.1140	-0.01842	0.094410	0.13531	1251.1	-0.41192
325	0.038777	10.816	0.092455	-4.0576	-4.0540	-0.01194	0.097865	0.13934	1180.0	-0.38479
340	0.066128	10.605	0.094297	-1.9360	-1.9297	-0.00556	0.10197	0.14373	1108.1	-0.35639
355	0.10696	10.389	0.096251	0.25114	0.26143	0.000737	0.10619	0.14815	1037.7	-0.32675
370	0.16531	10.170	0.098331	2.5029	2.5192	0.006950	0.11029	0.15251	969.46	-0.29499
385	0.24565	9.9444	0.10056	4.8180	4.8427	0.013084	0.11418	0.15684	903.49	-0.25976
400	0.35284	9.7123	0.10296	7.1959	7.2322	0.019144	0.11788	0.16124	839.45	-0.21937
415	0.49204	9.4716	0.10558	9.6372	9.6892	0.025138	0.12142	0.16581	776.89	-0.17166
430	0.66869	9.2204	0.10845	12.144	12.217	0.031076	0.12487	0.17070	715.33	-0.11372
445	0.88851	8.9560	0.11166	14.721	14.820	0.036970	0.12827	0.17609	654.26	-0.04137
460	1.1575	8.6747	0.11528	17.372	17.506	0.042839	0.13170	0.18223	593.15	0.05173
475	1.4820	8.3719	0.11945	20.108	20.285	0.048703	0.13524	0.18948	531.41	0.17604
490	1.8689	8.0406	0.12437	22.940	23.172	0.054589	0.13896	0.19847	468.41	0.35018
505	2.3256	7.6701	0.13038	25.887	26.190	0.060537	0.14300	0.21044	403.32	0.61096
520	2.8608	7.2419	0.13809	28.980	29.375	0.066611	0.14755	0.22839	335.01	1.0427
535	3.4853	6.7170	0.14888	32.284	32.802	0.072938	0.15298	0.26197	261.60	1.8878
550	4.2145	5.9774	0.16730	35.991	36.696	0.079902	0.16038	0.37221	179.15	4.2186
562.05	4.9012	3.9561	0.25278	41.513	42.752	0.090526			0	15.682
278.7	0.0047988	0.0020778	481.28	22.307	24.617	0.092630	0.067199	0.075641	182.14	69.790
280	0.0051472	0.0022187	450.72	22.394	24.714	0.092396	0.067633	0.076082	182.47	68.760
295	0.010954	0.0044939	222.52	23.427	25.864	0.090149	0.072601	0.081164	186.19	58.188
310	0.021374	0.0083797	119.34	24.522	27.073	0.088632	0.077506	0.086243	189.61	49.702
325	0.038777	0.014587	68.556	25.676	28.334	0.087716	0.082363	0.091352	192.67	42.885
340	0.066128	0.023968	41.723	26.881	29.640	0.087296	0.087187	0.096528	195.28	37.396
355	0.10696	0.037514	26.657	28.135	30.986	0.087285	0.091992	0.10181	197.37	32.960
370	0.16531	0.056356	17.744	29.431	32.365	0.087613	0.096792	0.10724	198.86	29.366
385	0.24565	0.081780	12.228	30.765	33.769	0.088218	0.10160	0.11287	199.67	26.447
400	0.35284	0.11526	8.6760	32.132	35.194	0.089048	0.10643	0.11877	199.73	24.076
415	0.49204	0.15852	6.3082	33.526	36.630	0.090056	0.11129	0.12502	198.94	22.158
430	0.66869	0.21364	4.6807	34.942	38.072	0.091203	0.11620	0.13176	197.20	20.623
445	0.88851	0.28321	3.5310	36.370	39.508	0.092449	0.12117	0.13918	194.42	19.421
460	1.1575	0.37058	2.6984	37.803	40.927	0.093754	0.12624	0.14759	190.45	18.522
475	1.4820	0.48038	2.0817	39.228	42.313	0.095076	0.13144	0.15753	185.11	17.915
490	1.8689	0.61931	1.6147	40.625	43.642	0.096364	0.13684	0.17002	178.19	17.610
505	2.3256	0.79794	1.2532	41.966	44.880	0.097548	0.14252	0.18715	169.36	17.642
520	2.8608	1.0349	0.96626	43.200	45.965	0.098514	0.14867	0.21415	158.20	18.090
535	3.4853	1.3697	0.73008	44.221	46.766	0.099038	0.15562	0.26827	144.00	19.073
550	4.2145	1.9242	0.51970	44.709	46.899	0.098452	0.16433	0.45840	125.56	20.568
562.05	4.9012	3.9561	0.25278	41.513	42.752	0.090526			0	15.682
Single-Phase Properties										
300.00	0.10000	11.161	0.089602	-7.4606	-7.4516	-0.022832	0.092996	0.13320	1296.2	-0.42861
350.00	0.10000	10.462	0.095585	-0.48554	-0.47598	-0.0013533	0.10479	0.14668	1061.0	-0.33684
352.81	0.10000	10.421	0.095958	-0.072586	-0.062991	-0.00017807	0.10558	0.14751	1047.8	-0.33119
352.81	0.10000	0.035232	28.383	27.949	30.787	0.087264	0.091290	0.10103	197.10	33.552
400.00	0.10000	0.030711	32.562	32.595	35.851	0.10072	0.10432	0.11349	210.66	20.922
450.00	0.10000	0.027110	36.887	38.153	41.842	0.11482	0.11715	0.12601	223.76	13.727
500.00	0.10000	0.024299	41.154	44.322	48.437	0.12870	0.12898	0.13766	235.93	9.5972
550.00	0.10000	0.022032	45.388	51.054	55.592	0.14233	0.13981	0.14838	247.41	7.0666
600.00	0.10000	0.020161	49.600	58.300	63.260	0.15567	0.14967	0.15817	258.31	5.4319

TABLE 2-193 Thermodynamic Properties of Benzene (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Single-Phase Properties										
300.00	1.0000	11.170	0.089523	-7.4897	-7.4002	-0.022929	0.092978	0.13308	1301.8	-0.43004
350.00	1.0000	10.476	0.095456	-0.52680	-0.43135	-0.0014714	0.10480	0.14651	1068.0	-0.33971
400.00	1.0000	9.7282	0.10279	7.1526	7.2554	0.019036	0.11785	0.16097	846.37	-0.22423
450.00	1.0000	8.8655	0.11280	15.593	15.706	0.038922	0.12940	0.17801	634.37	-0.013693
451.58	1.0000	8.8350	0.11319	15.873	15.987	0.039545	0.12977	0.17867	627.51	-0.0036091
451.58	1.0000	0.31908	3.1340	36.999	40.133	0.093016	0.12338	0.14272	192.83	18.991
500.00	1.0000	0.27026	3.7002	43.395	47.096	0.10766	0.13222	0.14607	214.43	11.641
550.00	1.0000	0.23728	4.2144	50.362	54.577	0.12192	0.14175	0.15344	231.44	7.9458
600.00	1.0000	0.21298	4.6952	57.755	62.450	0.13561	0.15089	0.16150	245.85	5.8511
300.00	5.0000	11.213	0.089185	-7.6160	-7.1700	-0.023353	0.092913	0.13259	1325.8	-0.43605
350.00	5.0000	10.537	0.094904	-0.70378	-0.22926	-0.0019818	0.10485	0.14581	1097.9	-0.35136
400.00	5.0000	9.8210	0.10182	6.8987	7.4078	0.018394	0.11775	0.15956	886.30	-0.25035
450.00	5.0000	9.0240	0.11082	15.189	15.743	0.038012	0.12877	0.17438	692.84	-0.088322
500.00	5.0000	8.0418	0.12435	24.331	24.953	0.057399	0.13990	0.19602	496.64	0.27193
550.00	5.0000	6.4075	0.15607	35.199	35.979	0.078369	0.15609	0.27120	246.43	2.1959
600.00	5.0000	1.5540	0.64348	53.945	57.162	0.11554	0.16178	0.21373	176.29	9.3898
300.00	10.000	11.263	0.088783	-7.7671	-6.8793	-0.023867	0.092869	0.13203	1354.1	-0.44280
350.00	10.000	10.608	0.094266	-0.91178	0.030879	-0.0025897	0.10497	0.14508	1132.3	-0.36378
400.00	10.000	9.9256	0.10075	6.6097	7.6172	0.017651	0.11774	0.15816	930.56	-0.27601
450.00	10.000	9.1886	0.10883	14.760	15.848	0.037025	0.12832	0.17132	752.83	-0.15003
500.00	10.000	8.3418	0.11988	23.600	24.799	0.055872	0.13831	0.18753	586.00	0.073722
550.00	10.000	7.2764	0.13743	33.351	34.726	0.074776	0.14918	0.21149	420.33	0.56941
600.00	10.000	5.7109	0.17510	44.567	46.318	0.094920	0.16240	0.25868	257.52	2.1761
350.00	50.000	11.061	0.090412	-2.2210	2.2996	-0.0066440	0.10685	0.14195	1337.7	-0.41928
400.00	50.000	10.527	0.094991	4.9365	9.6861	0.013066	0.11931	0.15301	1173.3	-0.37184
450.00	50.000	10.001	0.099985	12.570	17.570	0.031627	0.12871	0.16219	1041.8	-0.32474
500.00	50.000	9.4723	0.10557	20.624	25.902	0.049177	0.13669	0.17116	932.63	-0.26918
550.00	50.000	8.9322	0.11196	29.094	34.691	0.065924	0.14419	0.18047	839.96	-0.20274
600.00	50.000	8.3797	0.11934	37.984	43.951	0.082032	0.15154	0.18989	761.25	-0.12687
350.00	75.000	11.280	0.088652	-2.8344	3.8145	-0.0087087	0.10831	0.14116	1431.4	-0.43585
400.00	75.000	10.798	0.092613	4.2071	11.153	0.010874	0.12070	0.15181	1277.8	-0.39637
450.00	75.000	10.331	0.096793	11.699	18.958	0.029250	0.12977	0.16021	1157.1	-0.36122
500.00	75.000	9.8716	0.10130	19.570	27.167	0.046542	0.13730	0.16819	1058.5	-0.32227
550.00	75.000	9.4118	0.10625	27.813	35.782	0.062956	0.14435	0.17644	975.53	-0.27775
600.00	75.000	8.9496	0.11174	36.435	44.815	0.078670	0.15124	0.18489	905.02	-0.22901

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Polt, A., Platzler, B., and Maurer, G., "Parameter der thermischen Zustandsgleichung von Bender fuer 14 mehratomige reine Stoffe," *Chem. Tech. (Leipzig)*, **44**(6):216–224, 1992. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in density for benzene are 1% in the vapor phase, 0.3% in the liquid phase up to 400 K (with lower uncertainties at lower temperatures), 1% in the liquid phase between 400 and 500 K, and 2% and rising at temperatures above 500 K. Near the saturation line at temperatures below 350 K, the liquid-phase uncertainty decreases to 0.05%. The uncertainties in vapor pressures are 0.15% at temperatures below 380 K, and 0.5% at higher temperatures. The uncertainties in heat capacities and sound speeds are 2% in the vapor phase and 5% in the liquid phase.

TABLE 2-194 Saturated Bromine*

T, K	P, bar	$v_f, \text{m}^3/\text{kg}$	$v_g, \text{m}^3/\text{kg}$	$h_f, \text{kJ/kg}$	$h_g, \text{kJ/kg}$	$s_f, \text{kJ}/(\text{kg}\cdot\text{K})$	$s_g, \text{kJ}/(\text{kg}\cdot\text{K})$	$c_{pf}, \text{kJ}/(\text{kg}\cdot\text{K})$	$\mu_f, 10^{-4} \text{Pa}\cdot\text{s}$	$k_f, \text{W}/(\text{m}\cdot\text{K})$
260	0.042	3.106-4	3.195	-147.2	51.8	0.903	1.669	0.486	13.4	0.131
280	0.124	3.168-4	1.169	-138.9	56.2	0.933	1.629	0.479	11.5	0.127
300	0.310	3.232-4	0.5002	-131.6	60.6	0.956	1.597	0.475	9.3	0.122
320	0.680	3.311-4	0.2425	-124.2	64.8	0.978	1.570	0.473	7.8	0.118
340	1.330	3.385-4	0.1309	-112.3	71.1	1.004	1.539	0.471	6.7	0.114
360	2.384	3.464-4	0.0767	-108.6	73.1	1.026	1.531	0.470	5.7	0.109
380	4.010	3.550-4	0.0477	-100.6	76.9	1.048	1.515	0.471	5.0	0.104
400	6.390	3.647-4	0.0311	-93.4	80.6	1.063	1.501	0.475	4.5	0.099
420	9.730	3.752-4	0.0211	-85.8	84.0	1.084	1.488	0.480	4.0	0.094
440	14.25	3.885-4	0.0148	-77.7	87.1	1.103	1.477	0.489	3.7	0.089
460	20.17	4.023-4	0.0107	-69.0	89.9	1.122	1.467	0.503	3.3	0.084
480	27.75	4.179-4	0.00786	-59.7	92.2	1.142	1.457	0.527	3.1	0.079
500	37.21	4.378-4	0.00589	-49.3	94.0	1.161	1.448	0.595	2.8	0.073
520	48.81	4.623-4	0.00445	-37.7	95.0	1.183	1.438	0.710	2.6	0.066
540	62.80	4.938-4	0.00337	-24.0	94.8	1.207	1.428	0.860	2.5	0.059
560	79.41	5.368-4	0.00251	-7.1	92.5	1.237	1.414	1.063	2.3	0.050
580	98.90	6.250-4	0.00167	18.8	82.5	1.280	1.390	2.31	2.2	0.035
584.2 ^c	103.4	8.475-4	0.00085	64.8	64.8	1.356	1.356	∞	2.1	∞

*Reproduced or converted from a tabulation by Seshadri, Viswanath, and Kuloor, *Ind. J. Technol.*, 6 (1970): 191-198. c = critical point.

TABLE 2-195 Thermodynamic Properties of Butane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
134.9	6.6566E-07	12.645	0.079082	-5.2207	-5.2207	-0.02703	0.083783	0.11467	1826.8	-0.57143	176.56	2304.3
140	1.6922E-06	12.563	0.079598	-4.6347	-4.6347	-0.02277	0.083802	0.11492	1793.1	-0.56883	174.84	1887.3
155	0.00001767	12.323	0.081151	-2.9051	-2.9051	-0.01103	0.083864	0.11573	1699.3	-0.55989	169.24	1195.3
170	0.00011616	12.082	0.082767	-1.1617	-1.1617	-0.00030	0.084118	0.11677	1610.1	-0.54850	163.00	844.68
185	0.00054113	11.840	0.084458	0.59963	0.59967	0.009630	0.084658	0.11813	1523.4	-0.53413	156.32	634.79
200	0.0019390	11.596	0.086236	2.3840	2.3842	0.018903	0.085627	0.11986	1438.2	-0.51637	149.36	496.50
215	0.0056671	11.349	0.088115	4.1972	4.1977	0.027645	0.086949	0.12199	1354.1	-0.49482	142.26	399.50
230	0.014106	11.097	0.090113	6.0455	6.0467	0.035954	0.088641	0.12455	1270.8	-0.46909	135.13	328.38
245	0.030885	10.840	0.092253	7.9351	7.9379	0.043912	0.090679	0.12755	1188.2	-0.43868	128.09	274.41
260	0.060978	10.575	0.094562	9.8722	9.8780	0.051585	0.093026	0.13098	1106.4	-0.40288	121.19	232.32
275	0.11065	10.301	0.097075	11.863	11.874	0.059030	0.095645	0.13488	1025.2	-0.36055	114.50	198.71
290	0.18734	10.016	0.099837	13.914	13.933	0.066292	0.098497	0.13925	944.51	-0.30993	108.06	171.28
305	0.29946	9.7171	0.10291	16.031	16.062	0.073412	0.10155	0.14418	864.09	-0.24825	101.92	148.46
320	0.45624	9.4002	0.10638	18.222	18.271	0.080426	0.10478	0.14976	783.62	-0.17103	96.103	129.09
335	0.66761	9.0607	0.11037	20.494	20.568	0.087371	0.10817	0.15620	702.63	-0.07074	90.627	112.34
350	0.94418	8.6916	0.11505	22.858	22.967	0.094284	0.11173	0.16389	620.37	0.066051	85.498	97.559
365	1.2974	8.2822	0.12074	25.328	25.485	0.10121	0.11552	0.17360	535.75	0.26565	80.706	84.220
380	1.7399	7.8141	0.12797	27.928	28.150	0.10822	0.11966	0.18719	447.13	0.58704	76.219	71.859
395	2.2868	7.2504	0.13792	30.703	31.018	0.11543	0.12445	0.21028	352.08	1.1936	71.976	59.958
410	2.9578	6.4885	0.15412	33.782	34.237	0.12318	0.13061	0.27186	246.50	2.7451	67.979	47.596
425.13	3.796	3.9228	0.25492	39.364	40.331	0.13735			0	15.851		
134.9	6.6566E-07	5.935E-07	1684900	22.482	23.604	0.18665	0.055993	0.064308	148.87	361.89	4.8545	3.3208
140	1.6922E-06	1.4538E-06	687860	22.771	23.935	0.18130	0.057048	0.065363	151.48	315.59	5.0913	3.4522
155	0.00001767	1.3712E-05	72930	23.648	24.937	0.16859	0.059941	0.068258	158.89	218.48	5.8353	3.8372
170	0.00011616	8.2203E-05	12165	24.567	25.980	0.15936	0.062670	0.070995	165.94	157.83	6.6513	4.2205
185	0.00054113	0.00035209	2840.2	25.525	27.062	0.15267	0.065393	0.073744	172.61	117.96	7.5389	4.6014
200	0.0019390	0.0011686	855.70	26.519	28.179	0.14788	0.068227	0.076640	178.87	90.683	8.4974	4.9792
215	0.0056671	0.0031862	313.85	27.550	29.328	0.14453	0.071254	0.079787	184.64	71.432	9.5258	5.3531
230	0.014106	0.0074503	134.22	28.614	30.507	0.14230	0.074521	0.083261	189.81	57.513	10.624	5.7226
245	0.030885	0.015434	64.792	29.709	31.711	0.14094	0.078047	0.087120	194.27	47.254	11.792	6.0877
260	0.060978	0.029042	34.433	30.834	32.934	0.14026	0.081835	0.091404	197.86	39.578	13.034	6.4502
275	0.11065	0.050611	19.759	31.984	34.170	0.14011	0.085872	0.096154	200.46	33.766	14.357	6.8133
290	0.18734	0.082943	12.056	33.156	35.414	0.14037	0.090138	0.10142	201.90	29.330	15.775	7.1826
305	0.29946	0.12941	7.7271	34.343	36.657	0.14094	0.094611	0.10730	202.04	25.936	17.311	7.5668
320	0.45624	0.19420	5.1494	35.541	37.890	0.14174	0.099274	0.11394	200.69	23.353	18.999	7.9782
335	0.66761	0.28270	3.5374	36.739	39.100	0.14269	0.10412	0.12161	197.62	21.423	20.889	8.4346
350	0.94418	0.40241	2.4850	37.923	40.270	0.14372	0.10910	0.13080	192.54	20.061	23.058	8.9610
365	1.2974	0.56471	1.7708	39.073	41.370	0.14473	0.11417	0.14259	185.07	19.268	25.626	9.5964
380	1.7399	0.78888	1.2676	40.147	42.353	0.14559	0.11962	0.16009	174.66	19.058	28.813	10.407
395	2.2868	1.1133	0.89820	41.076	43.130	0.14609	0.12600	0.19220	160.51	19.446	33.129	11.532
410	2.9578	1.6409	0.60943	41.673	43.475	0.14572	0.13401	0.28133	141.36	20.469	40.394	13.373
425.13	3.796	3.9228	0.25492	39.364	40.331	0.13735			0	15.851		
Single-Phase Properties												
150.00	0.10000	12.404	0.080622	-3.4846	-3.4765	-0.014834	0.083837	0.11544	1730.2	-0.56316	171.22	1370.9
200.00	0.10000	11.597	0.086227	2.3816	2.3902	0.018891	0.085632	0.11985	1438.8	-0.51649	149.40	496.90
250.00	0.10000	10.753	0.092993	8.5729	8.5822	0.046489	0.091432	0.12863	1161.4	-0.42760	125.81	259.42
272.31	0.10000	10.351	0.096608	11.502	11.512	0.057711	0.095158	0.13415	1039.7	-0.36868	115.68	204.22
272.31	0.10000	0.046045	21.718	31.776	33.948	0.14010	0.085131	0.095267	200.07	34.694	14.113	6.7480
300.00	0.10000	0.041289	24.220	34.241	36.663	0.14959	0.091498	0.10103	211.28	24.802	16.747	7.4522
350.00	0.10000	0.034963	28.601	39.149	42.009	0.16605	0.10399	0.11299	229.21	14.912	22.115	8.6907
400.00	0.10000	0.030400	32.894	44.681	47.970	0.18195	0.11669	0.12545	245.29	9.8194	28.275	9.8987
450.00	0.10000	0.026924	37.142	50.834	54.548	0.19743	0.12893	0.13755	260.14	6.9019	35.230	11.083
500.00	0.10000	0.024176	41.363	57.578	61.714	0.21252	0.14043	0.14897	274.06	5.0916	42.982	12.248
550.00	0.10000	0.021945	45.569	64.875	69.432	0.22723	0.15112	0.15960	287.25	3.8945	51.533	13.395
150.00	1.0000	12.411	0.080576	-3.4986	-3.4180	-0.014927	0.083887	0.11540	1733.9	-0.56354	171.48	1381.3
200.00	1.0000	11.608	0.086150	2.3598	2.4460	0.018782	0.085677	0.11977	1443.9	-0.51758	149.79	500.58

250.00	1.0000	10.770	0.092855	8.5389	8.6318	0.046353	0.091477	0.12845	1168.5	-0.43053	126.34	261.81
300.00	1.0000	9.8412	0.10161	15.271	15.373	0.070896	0.10054	0.14209	899.57	-0.27749	104.51	157.29
350.00	1.0000	8.6953	0.11500	22.851	22.966	0.094264	0.11173	0.16378	621.53	0.063603	85.560	97.696
352.62	1.0000	8.6234	0.11596	23.281	23.397	0.095491	0.11237	0.16540	605.80	0.095319	84.637	95.139
352.62	1.0000	0.42728	2.3404	38.127	40.468	0.14390	0.10997	0.13262	191.42	19.881	23.473	9.0627
400.00	1.0000	0.34181	2.9256	43.808	46.734	0.16057	0.11928	0.13410	221.49	11.545	29.230	10.156
450.00	1.0000	0.28958	3.4533	50.185	53.638	0.17683	0.13036	0.14250	244.24	7.5723	36.235	11.323
500.00	1.0000	0.25369	3.9418	57.063	61.005	0.19234	0.14132	0.15221	262.94	5.3868	44.067	12.475
550.00	1.0000	0.22679	4.4094	64.449	68.858	0.20730	0.15171	0.16189	279.30	4.0325	52.702	13.611
150.00	5.0000	12.442	0.080374	-3.5595	-3.1577	-0.015338	0.084107	0.11525	1749.9	-0.56513	172.63	1428.2
200.00	5.0000	11.653	0.085815	2.2656	2.6947	0.018306	0.085873	0.11943	1466.4	-0.52214	151.52	516.90
250.00	5.0000	10.838	0.092266	8.3942	8.8555	0.045767	0.091674	0.12773	1199.3	-0.44237	128.64	272.30
300.00	5.0000	9.9548	0.10045	15.037	15.539	0.070104	0.10072	0.14034	943.54	-0.31083	107.50	165.96
350.00	5.0000	8.9238	0.11206	22.411	22.971	0.092982	0.11170	0.15798	694.05	-0.067667	89.620	106.76
400.00	5.0000	7.5216	0.13295	30.873	31.537	0.11581	0.12435	0.18921	430.26	0.64313	75.035	65.426
450.00	5.0000	3.1209	0.32042	44.259	45.861	0.14921	0.14234	0.35080	147.05	11.475	55.862	19.322
500.00	5.0000	1.7051	0.58647	54.030	56.962	0.17272	0.14611	0.18483	208.77	6.6929	51.142	14.775
550.00	5.0000	1.3443	0.74389	62.232	65.951	0.18986	0.15448	0.17804	244.93	4.4692	57.838	14.928
150.00	10.000	12.480	0.080129	-3.6330	-2.8317	-0.015840	0.084373	0.11508	1769.4	-0.56694	174.03	1488.3
200.00	10.000	11.707	0.085415	2.1534	3.0075	0.017730	0.086111	0.11906	1493.4	-0.52718	153.61	537.30
250.00	10.000	10.919	0.091586	8.2254	9.1412	0.045071	0.091916	0.12697	1235.4	-0.45489	131.40	285.19
300.00	10.000	10.082	0.099189	14.776	15.768	0.069202	0.10096	0.13868	993.02	-0.34276	110.99	176.33
350.00	10.000	9.1484	0.10931	21.971	23.064	0.091668	0.11182	0.15374	766.31	-0.16435	94.017	116.79
400.00	10.000	8.0346	0.12446	29.964	31.209	0.11339	0.12357	0.17293	554.16	0.17567	81.081	77.782
450.00	10.000	6.5680	0.15225	38.976	40.499	0.13524	0.13606	0.20093	360.83	1.0509	71.907	48.326
500.00	10.000	4.6089	0.21697	49.153	51.323	0.15803	0.14784	0.22343	243.50	2.9827	65.910	27.075
550.00	10.000	3.2339	0.30923	58.940	62.032	0.17846	0.15662	0.20486	242.94	3.2916	65.615	19.386
150.00	30.000	12.622	0.079224	-3.9002	-1.5235	-0.017740	0.085345	0.11453	1842.9	-0.57242	179.28	1748.0
200.00	30.000	11.905	0.083996	1.7557	4.2756	0.015602	0.086999	0.11796	1591.7	-0.54200	161.37	619.22
250.00	30.000	11.197	0.089313	7.6534	10.333	0.042606	0.092823	0.12491	1361.4	-0.48867	141.44	335.06
300.00	30.000	10.483	0.095388	13.956	16.817	0.066223	0.10187	0.13486	1153.4	-0.41613	123.09	214.53
350.00	30.000	9.7548	0.10251	20.772	23.848	0.087875	0.11265	0.14654	971.05	-0.32840	108.07	150.45
400.00	30.000	9.0036	0.11107	28.147	31.479	0.10824	0.12399	0.15873	816.42	-0.22551	97.051	110.91
450.00	30.000	8.2322	0.12147	36.070	39.714	0.12762	0.13519	0.17053	691.11	-0.10687	90.031	83.811
500.00	30.000	7.4563	0.13411	44.490	48.514	0.14616	0.14590	0.18124	595.43	0.023732	86.541	64.099
550.00	30.000	6.7039	0.14917	53.337	57.812	0.16387	0.15594	0.19043	527.27	0.15444	85.903	49.561
150.00	65.000	12.844	0.077859	-4.2847	0.77614	-0.020732	0.086776	0.11392	1957.6	-0.57756	187.43	2300.4
200.00	65.000	12.196	0.081993	1.2046	6.5342	0.012378	0.088354	0.11690	1737.3	-0.55581	173.22	766.46
250.00	65.000	11.576	0.086384	6.9077	12.523	0.039078	0.094214	0.12318	1537.0	-0.51712	156.34	419.12
300.00	65.000	10.979	0.091087	12.978	18.898	0.062302	0.10329	0.13220	1359.9	-0.46758	140.34	275.14
350.00	65.000	10.399	0.096160	19.516	25.767	0.083456	0.11406	0.14267	1208.7	-0.41511	126.97	199.77
400.00	65.000	9.8373	0.10165	26.564	33.172	0.10322	0.12536	0.15351	1082.5	-0.36425	116.89	154.25
450.00	65.000	9.2935	0.10760	34.118	41.112	0.12191	0.13648	0.16402	979.42	-0.31694	110.18	123.77
500.00	65.000	8.7710	0.11401	42.152	49.563	0.13971	0.14709	0.17389	896.76	-0.27399	106.55	101.78
550.00	65.000	8.2737	0.12087	50.632	58.488	0.15671	0.15706	0.18299	831.73	-0.23593	105.59	85.075

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, NIST Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Buecker, D., and Wagner, W., "Reference Equations of State for the Thermodynamic Properties of Fluid Phase *n*-Butane and Isobutane," *J. Phys. Chem. Ref. Data* **35**(2): 929–1019, 2006. The source for viscosity is Vogel, E., Kuechenmeister, C., and Bich, E., "Viscosity for *n*-Butane in the Fluid Region," *High Temp.—High Pressures* **31**(2):173–186, 1999. The source for thermal conductivity is Perkins, R. A., Ramires, M. L. V., Nieto de Castro, C. A., and Cusco, L., "Measurement and Correlation of the Thermal Conductivity of Butane from 135 K to 600 K at Pressures to 70 MPa," *J. Chem. Eng. Data* **47**(5):1263–1271, 2002.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in density are 0.02% at temperatures below 340 K and pressures below 12 MPa (both liquid and vapor states), 0.1% at temperatures below 270 K and pressures above 12 MPa, 0.2% between 340 and 515 K at pressures less than 0.6 MPa, and 0.4% elsewhere. In the critical region, deviations in pressure are 0.5%. At temperatures above 500 K, the uncertainties in density increase up to 1%. Uncertainties in heat capacities are typically 1%, rising to 5% in the critical region and at pressures above 30 MPa. Uncertainties in the speed of sound are typically 0.5%, rising to 1% at temperatures below 200 K and to 4% in a large area around the critical point.

The uncertainty in viscosity varies from 0.4% in the dilute gas between room temperature and 600 K, to 3.0% over the rest of the fluid surface.

Uncertainty in thermal conductivity is 3%, except in the critical region and dilute gas which have an uncertainty of 5%.

TABLE 2-196 Thermodynamic Properties of 1-Butene

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_v kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
87.800	5.95E-13	14.582	0.068578	-19.753	-19.753	-0.12145	0.080229	0.10913	2086.5	-0.55371
150.00	1.2730E-05	13.379	0.074746	-13.108	-13.108	-0.064173	0.071962	0.10631	1744.9	-0.55469
165.00	9.3335E-05	13.097	0.076351	-11.509	-11.509	-0.054011	0.072351	0.10701	1649.6	-0.54558
180.00	0.00046966	12.818	0.078017	-9.8961	-9.8961	-0.044655	0.073193	0.10813	1555.6	-0.53276
195.00	0.0017787	12.538	0.079756	-8.2633	-8.2632	-0.035943	0.074385	0.10964	1463.4	-0.51623
210.00	0.0054125	12.257	0.081585	-6.6054	-6.6050	-0.027753	0.075854	0.11149	1373.4	-0.49587
225.00	0.013876	11.973	0.083519	-4.9174	-4.9162	-0.019990	0.077551	0.11367	1285.6	-0.47138
240.00	0.031052	11.685	0.085582	-3.1944	-3.1918	-0.012577	0.079443	0.11619	1199.6	-0.44220
255.00	0.062301	11.390	0.087798	-1.4319	-1.4264	-0.0054542	0.081507	0.11904	1115.4	-0.40748
270.00	0.11440	11.086	0.090204	0.37503	0.38535	0.0014312	0.083732	0.12227	1032.5	-0.36587
285.00	0.19537	10.771	0.092841	2.2316	2.2497	0.0081238	0.086107	0.12593	950.65	-0.31542
300.00	0.31425	10.442	0.095769	4.1435	4.1736	0.014663	0.088628	0.13010	869.48	-0.25308
315.00	0.48093	10.094	0.099067	6.1173	6.1650	0.021087	0.091289	0.13492	788.58	-0.17406
330.00	0.70602	9.7227	0.10285	8.1613	8.2339	0.027432	0.094094	0.14061	707.49	-0.070421
345.00	1.0009	9.3197	0.10730	10.286	10.393	0.033738	0.097052	0.14758	625.58	0.071832
360.00	1.3777	8.8735	0.11269	12.507	12.662	0.040056	0.10019	0.15666	541.98	0.27973
375.00	1.8505	8.3641	0.11956	14.848	15.069	0.046456	0.10358	0.16976	455.28	0.61278
390.00	2.4356	7.7513	0.12901	17.357	17.672	0.053070	0.10741	0.19258	362.75	1.2326
405.00	3.1556	6.9234	0.14444	20.167	20.622	0.060243	0.11236	0.25490	258.01	2.7886
419.29	4.0057	4.2400	0.23585	25.194	26.139	0.073219			0	14.832
87.800	5.95E-13	8.1441E-13	1.2279E+12	10.763	11.493	0.23442	0.035901	0.044215	126.59	6453.8
150.00	1.2730E-05	1.0208E-05	97966.	13.339	14.586	0.12046	0.046819	0.055138	161.79	519.37
165.00	9.3335E-05	6.8053E-05	14694.	14.060	15.432	0.10927	0.049589	0.057921	168.95	334.05
180.00	0.00046966	0.00031411	3183.5	14.822	16.317	0.10097	0.052497	0.060867	175.70	224.29
195.00	0.0017787	0.0010999	909.21	15.621	17.238	0.094833	0.055552	0.064008	182.01	156.42
210.00	0.0054125	0.0031172	320.80	16.456	18.192	0.090329	0.058761	0.067378	187.84	112.91
225.00	0.013876	0.0074985	133.36	17.323	19.174	0.087076	0.062133	0.071013	193.09	84.169
240.00	0.031052	0.015861	63.049	18.219	20.177	0.084791	0.065668	0.074946	197.63	64.687
255.00	0.062301	0.030305	32.998	19.139	21.195	0.083257	0.069362	0.079208	201.32	51.184
270.00	0.11440	0.053404	18.725	20.081	22.223	0.082311	0.073208	0.083840	204.02	41.642
285.00	0.19537	0.088232	11.334	21.039	23.253	0.081820	0.077198	0.088901	205.56	34.789
300.00	0.31425	0.13847	7.2216	22.009	24.279	0.081681	0.081324	0.094492	205.77	29.808
315.00	0.48093	0.20868	4.7920	22.985	25.290	0.081801	0.085583	0.10078	204.47	26.172
330.00	0.70602	0.30478	3.2811	23.957	26.274	0.082098	0.089980	0.10808	201.43	23.538
345.00	1.0009	0.43505	2.2986	24.912	27.212	0.082489	0.094533	0.11696	196.37	21.690
360.00	1.3777	0.61215	1.6336	25.827	28.078	0.082878	0.099287	0.12863	188.91	20.504
375.00	1.8505	0.85778	1.1658	26.665	28.823	0.083133	0.10434	0.14599	178.55	19.927
390.00	2.4356	1.2160	0.82239	27.351	29.354	0.083025	0.10992	0.17800	164.58	19.957
405.00	3.1556	1.8085	0.55294	27.680	29.425	0.081979	0.11666	0.27163	145.93	20.477
419.29	4.0057	4.2400	0.23585	25.194	26.139	0.073219			0	14.832

Single-Phase Properties

100.00	0.10000	14.342	0.069727	-18.432	-18.425	-0.10737	0.076348	0.10761	2041.3	-0.55898
200.00	0.10000	12.446	0.080347	-7.7161	-7.7081	-0.033172	0.074849	0.11021	1433.8	-0.51000
266.51	0.10000	11.158	0.089625	-0.050006	-0.041044	-0.00015346	0.083200	0.12149	1051.7	-0.37627
266.51	0.10000	0.047098	21.232	19.860	21.983	0.082486	0.072300	0.082725	203.49	43.586
300.00	0.10000	0.041199	24.273	22.407	24.834	0.092558	0.078346	0.087850	217.19	27.387
400.00	0.10000	0.030377	32.920	31.321	34.613	0.12055	0.099614	0.10831	251.29	10.192
500.00	0.10000	0.024172	41.370	42.371	46.508	0.14700	0.12081	0.12931	280.25	5.2895
100.00	1.0000	14.347	0.069701	-18.441	-18.371	-0.10745	0.076343	0.10760	2045.4	-0.55906
200.00	1.0000	12.457	0.080273	-7.7377	-7.6574	-0.033280	0.074873	0.11013	1439.7	-0.51117
300.00	1.0000	10.465	0.095554	4.1009	4.1965	0.014521	0.088626	0.12974	878.06	-0.26069
344.96	1.0000	9.3208	0.10729	10.280	10.388	0.033722	0.097044	0.14756	625.79	0.071399
344.96	1.0000	0.43467	2.3006	24.909	27.210	0.082488	0.094521	0.11694	196.39	21.694
400.00	1.0000	0.33780	2.9603	30.561	33.521	0.099463	0.10210	0.11588	229.62	11.604
500.00	1.0000	0.25311	3.9508	41.920	45.871	0.12695	0.12152	0.13206	269.53	5.5503
100.00	5.0000	14.371	0.069585	-18.478	-18.131	-0.10783	0.076326	0.10752	2063.2	-0.55939
200.00	5.0000	12.507	0.079955	-7.8313	-7.4315	-0.033753	0.074982	0.10983	1464.8	-0.51602
300.00	5.0000	10.593	0.094404	3.8689	4.3409	0.013736	0.088648	0.12795	924.82	-0.29848
400.00	5.0000	7.7924	0.12833	18.376	19.018	0.055643	0.10867	0.17944	398.89	0.92138
500.00	5.0000	1.6432	0.60858	39.334	42.377	0.10830	0.12539	0.15726	221.46	6.5598
100.00	10.000	14.400	0.069444	-18.524	-17.830	-0.10830	0.076316	0.10743	2084.7	-0.55976
200.00	10.000	12.566	0.079577	-7.9428	-7.1471	-0.034325	0.075127	0.10949	1494.9	-0.52136
300.00	10.000	10.734	0.093162	3.6108	4.5424	0.012845	0.088724	0.12625	976.91	-0.33426
400.00	10.000	8.4254	0.11869	17.389	18.576	0.053002	0.10763	0.15809	531.56	0.27589
500.00	10.000	4.4457	0.22494	35.004	37.254	0.094450	0.12805	0.20133	235.89	3.6822
100.00	30.000	14.511	0.068913	-18.695	-16.628	-0.11011	0.076383	0.10716	2164.7	-0.56070
200.00	30.000	12.781	0.078241	-8.3397	-5.9925	-0.036441	0.075766	0.10853	1602.5	-0.53694
300.00	30.000	11.176	0.089478	2.8019	5.4862	0.0099119	0.089276	0.12244	1142.6	-0.41541
400.00	30.000	9.5263	0.10497	15.547	18.696	0.047777	0.10751	0.14231	802.89	-0.21113
500.00	30.000	7.8006	0.12820	30.094	33.940	0.081710	0.12649	0.16189	578.80	0.077488
200.00	70.000	13.133	0.076142	-8.9642	-3.6342	-0.040078	0.077144	0.10763	1777.6	-0.55246
300.00	70.000	11.778	0.084903	1.7245	7.6677	0.0055841	0.090645	0.11961	1376.9	-0.47674
400.00	70.000	10.537	0.094902	13.811	20.454	0.042252	0.10876	0.13655	1099.3	-0.37969
500.00	70.000	9.3918	0.10648	27.536	34.989	0.074613	0.12760	0.15389	914.99	-0.28997

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., and Ihmel, E. C., "Thermodynamic Properties of the Butenes. Part II. Short Fundamental Equations of State," *Fluid Phase Equilibria* **228–229C**:173–187, 2005. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties of densities calculated by the equation of state (based on a coverage factor of 2) are 0.1% in the liquid phase at temperatures above 270 K (rising to 0.5% in density at temperatures below 200 K), 0.2% at temperatures above the critical temperature and at pressures above 10 MPa, and 0.5% in the vapor phase, including supercritical conditions below 10 MPa. The uncertainty in vapor pressure is 0.25% above 200 K. The uncertainty in heat capacities is 0.5% at saturated liquid conditions, rising to 5% at much higher pressures and at temperatures above 350 K.

TABLE 2-197 Thermodynamic Properties of *cis*-2-Butene

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. Energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
134.30	2.6365E-07	14.084	0.071002	-16.205	-16.205	-0.081920	0.079059	0.11421	1931.8	-0.50947
140.00	8.0149E-07	13.976	0.071549	-15.556	-15.556	-0.077189	0.078138	0.11344	1897.3	-0.51181
155.00	9.6316E-06	13.696	0.073014	-13.866	-13.866	-0.065723	0.076346	0.11195	1805.9	-0.51506
170.00	7.0502E-05	13.419	0.074523	-12.194	-12.194	-0.055424	0.075319	0.11114	1714.3	-0.51402
185.00	0.00035720	13.143	0.076087	-10.529	-10.529	-0.046038	0.074926	0.11096	1623.4	-0.50857
200.00	0.0013683	12.868	0.077715	-8.8626	-8.8625	-0.037376	0.075072	0.11135	1533.6	-0.49860
215.00	0.0042205	12.591	0.079421	-7.1860	-7.1857	-0.029294	0.075681	0.11228	1445.0	-0.48403
230.00	0.010977	12.312	0.081218	-5.4919	-5.4910	-0.021677	0.076690	0.11371	1357.7	-0.46470
245.00	0.024918	12.030	0.083127	-3.7731	-3.7710	-0.014438	0.078045	0.11560	1271.7	-0.44035
260.00	0.050686	11.742	0.085168	-2.0229	-2.0186	-0.0075050	0.079699	0.11795	1186.9	-0.41046
275.00	0.094280	11.446	0.087369	-0.23513	-0.22689	-0.00082003	0.081613	0.12075	1103.2	-0.37418
290.00	0.16294	11.140	0.089765	1.5965	1.6111	0.0056653	0.083753	0.12400	1020.4	-0.33016
305.00	0.26496	10.822	0.092403	3.4781	3.5026	0.011993	0.086090	0.12777	938.24	-0.27626
320.00	0.40952	10.488	0.095344	5.4164	5.4554	0.018199	0.088602	0.13211	856.41	-0.20906
335.00	0.60651	10.134	0.098675	7.4188	7.4786	0.024318	0.091273	0.13720	774.55	-0.12307
350.00	0.86653	9.7540	0.10252	9.4943	9.5831	0.030386	0.094094	0.14328	692.22	-0.0089736
365.00	1.2009	9.3387	0.10708	11.655	11.783	0.036443	0.097073	0.15087	608.84	0.14999
380.00	1.6221	8.8746	0.11268	13.918	14.101	0.042540	0.10024	0.16100	523.58	0.38688
395.00	2.1441	8.3381	0.11993	16.313	16.570	0.048756	0.10367	0.17620	435.01	0.77697
410.00	2.7840	7.6790	0.13023	18.901	19.263	0.055246	0.10763	0.20480	340.18	1.5369
425.00	3.5654	6.7389	0.14839	21.868	22.397	0.062490	0.11310	0.30217	231.16	3.6646
435.75	4.2360	4.2440	0.23563	26.307	27.305	0.073584			0	14.802
134.30	2.6365E-07	2.3611E-07	4235300.	13.340	14.457	0.14638	0.045782	0.054097	153.35	1022.9
140.00	8.0149E-07	6.8856E-07	1452300.	13.603	14.767	0.13940	0.046445	0.054760	156.40	846.64
155.00	9.6316E-06	7.4740E-06	133800.	14.312	15.601	0.12439	0.048209	0.056526	164.10	534.19
170.00	7.0502E-05	4.9890E-05	20044.	15.048	16.461	0.11314	0.050082	0.058410	171.37	352.76
185.00	0.00035720	0.00023239	4303.0	15.811	17.348	0.10465	0.052140	0.060499	178.22	241.97
200.00	0.0013683	0.00082449	1212.9	16.601	18.261	0.098239	0.054433	0.062862	184.64	171.51
215.00	0.0042205	0.0023717	421.65	17.417	19.197	0.093417	0.056990	0.065554	190.57	125.19
230.00	0.010977	0.0057914	172.67	18.259	20.154	0.089825	0.059826	0.068614	195.95	93.915
245.00	0.024918	0.012427	80.467	19.124	21.129	0.087196	0.062937	0.072067	200.67	72.325
260.00	0.050686	0.024061	41.562	20.011	22.117	0.085325	0.066305	0.075928	204.61	57.138
275.00	0.094280	0.042902	23.309	20.917	23.114	0.084057	0.069907	0.080211	207.65	46.280
290.00	0.16294	0.071600	13.966	21.839	24.115	0.083264	0.073716	0.084945	209.64	38.405
305.00	0.26496	0.11331	8.8256	22.775	25.113	0.082846	0.077706	0.090186	210.44	32.629
320.00	0.40952	0.17184	5.8195	23.718	26.101	0.082717	0.081857	0.096045	209.88	28.363
335.00	0.60651	0.25199	3.9684	24.662	27.069	0.082798	0.086156	0.10272	207.78	25.214
350.00	0.86653	0.36018	2.7764	25.597	28.003	0.083015	0.090605	0.11059	203.91	22.923
365.00	1.2009	0.50560	1.9779	26.506	28.882	0.083288	0.095223	0.12039	197.97	21.324
380.00	1.6221	0.70265	1.4232	27.364	29.673	0.083518	0.10007	0.13368	189.62	20.319
395.00	2.1441	0.97689	1.0237	28.124	30.319	0.083563	0.10526	0.15445	178.35	19.863
410.00	2.7840	1.3825	0.72331	28.692	30.706	0.083155	0.11109	0.19596	163.51	19.920
425.00	3.5654	2.0863	0.47931	28.785	30.494	0.081542	0.11843	0.34488	144.08	20.169
435.75	4.2360	4.2440	0.23563	26.307	27.305	0.073584			0	14.802

Single-Phase Properties

150.00	0.10000	13.790	0.072517	-14.429	-14.421	-0.069410	0.076853	0.11236	1836.9	-0.51448
250.00	0.10000	11.936	0.083783	-3.1958	-3.1874	-0.012105	0.078567	0.11633	1243.8	-0.43122
276.53	0.10000	11.415	0.087603	-0.050523	-0.041763	-0.00015051	0.081821	0.12106	1094.7	-0.37008
276.53	0.10000	0.045328	22.062	21.010	23.216	0.083956	0.070286	0.080672	207.90	45.358
350.00	0.10000	0.034963	28.602	26.707	29.567	0.10426	0.084207	0.093226	235.49	18.349
450.00	0.10000	0.026927	37.137	36.211	39.925	0.13019	0.10536	0.11396	266.57	8.0313
150.00	1.0000	13.797	0.072480	-14.442	-14.369	-0.069498	0.076879	0.11234	1840.9	-0.51475
250.00	1.0000	11.951	0.083676	-3.2258	-3.1421	-0.012226	0.078590	0.11619	1250.7	-0.43363
350.00	1.0000	9.7615	0.10244	9.4809	9.5833	0.030348	0.094087	0.14310	694.58	-0.013023
356.43	1.0000	9.5810	0.10437	10.409	10.513	0.032980	0.095350	0.14630	656.67	0.052242
356.43	1.0000	0.41725	2.3967	25.991	28.388	0.083130	0.092560	0.11449	201.64	22.160
450.00	1.0000	0.28985	3.4501	35.598	39.048	0.10966	0.10690	0.11882	250.49	8.7484
150.00	5.0000	13.827	0.072321	-14.500	-14.138	-0.069886	0.076995	0.11223	1858.7	-0.51590
250.00	5.0000	12.016	0.083219	-3.3549	-2.9388	-0.012747	0.078700	0.11561	1280.3	-0.44345
350.00	5.0000	9.9638	0.10036	9.1178	9.6196	0.029293	0.093949	0.13889	758.54	-0.10935
450.00	5.0000	3.4526	0.28964	29.287	30.736	0.080884	0.12364	0.49664	144.63	13.395
150.00	10.000	13.864	0.072127	-14.569	-13.848	-0.070362	0.077144	0.11212	1880.2	-0.51720
250.00	10.000	12.094	0.082686	-3.5069	-2.6800	-0.013371	0.078849	0.11500	1315.2	-0.45403
350.00	10.000	10.171	0.098318	8.7400	9.7232	0.028171	0.093912	0.13550	825.19	-0.18764
450.00	10.000	7.3847	0.13542	23.767	25.121	0.066644	0.11375	0.17964	387.53	1.0233
150.00	20.000	13.936	0.071757	-14.703	-13.268	-0.071287	0.077453	0.11192	1921.2	-0.51938
250.00	20.000	12.236	0.081725	-3.7846	-2.1501	-0.014539	0.079171	0.11403	1379.1	-0.47093
350.00	20.000	10.498	0.095255	8.1362	10.041	0.026317	0.094047	0.13149	932.72	-0.28055
450.00	20.000	8.4814	0.11791	22.018	24.376	0.062213	0.11264	0.15577	586.98	0.13495
250.00	50.000	12.592	0.079414	-4.4655	-0.49483	-0.017580	0.080215	0.11231	1538.7	-0.50151
350.00	50.000	11.161	0.089596	6.9055	11.385	0.022259	0.094937	0.12655	1160.5	-0.39521
450.00	50.000	9.7684	0.10237	19.807	24.926	0.056195	0.11311	0.14428	893.45	-0.26115

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., and Ihmels, E. C., "Thermodynamic Properties of the Butenes. Part II. Short Fundamental Equations of State," *Fluid Phase Equilibria* **228–229C**:173–187, 2005. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in densities calculated using the equation of state are 0.1% in the liquid phase at temperatures above 270 K (rising to 0.5% at temperatures below 200 K), 0.2% at temperatures above the critical temperature and at pressures above 10 MPa, and 0.5% in the vapor phase, including supercritical conditions below 10 MPa. The uncertainty in the vapor phase may be higher than 0.5% in some regions. The uncertainty in vapor pressure is 0.2% between 220 and 310 K and 0.5% above 310 K, and the uncertainty in heat capacities is 0.5% at saturated liquid conditions, rising to 5% at much higher pressures and at temperatures above 300 K.

TABLE 2-198 Thermodynamic Properties of *trans*-2-Butene

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_v kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
167.60	0.000074817	13.141	0.076099	-12.322	-12.322	-0.056582	0.075471	0.10906	1653.9	-0.53710
170.00	0.000098940	13.097	0.076351	-12.060	-12.060	-0.055030	0.075651	0.10931	1639.3	-0.53489
185.00	0.00047141	12.827	0.077960	-10.409	-10.409	-0.045722	0.076779	0.11086	1549.4	-0.52040
200.00	0.0017229	12.556	0.079642	-8.7337	-8.7336	-0.037018	0.077989	0.11250	1461.5	-0.50389
215.00	0.0051236	12.284	0.081408	-7.0328	-7.0324	-0.028818	0.079349	0.11434	1375.6	-0.48446
230.00	0.012948	12.008	0.083277	-5.3024	-5.3013	-0.021038	0.080901	0.11646	1291.2	-0.46138
245.00	0.028726	11.728	0.085267	-3.5381	-3.5357	-0.013608	0.082660	0.11890	1208.2	-0.43391
260.00	0.057367	11.441	0.087404	-1.7351	-1.7301	-0.0064657	0.084626	0.12171	1126.4	-0.40117
275.00	0.10513	11.146	0.089718	0.11181	0.12124	0.00044042	0.086786	0.12491	1045.4	-0.36199
290.00	0.17951	10.840	0.092251	2.0081	2.0247	0.0071551	0.089124	0.12854	965.18	-0.31463
305.00	0.28905	10.520	0.095056	3.9596	3.9871	0.013718	0.091618	0.13267	885.29	-0.25645
320.00	0.44317	10.183	0.098206	5.9728	6.0163	0.020164	0.094251	0.13740	805.47	-0.18323
335.00	0.65207	9.8227	0.10181	8.0553	8.1216	0.026529	0.097011	0.14294	725.30	-0.088064
350.00	0.92670	9.4330	0.10601	10.217	10.315	0.032849	0.099897	0.14962	644.30	0.041045
365.00	1.2790	9.0030	0.11107	12.471	12.613	0.039171	0.10292	0.15815	561.78	0.22658
380.00	1.7221	8.5154	0.11743	14.840	15.042	0.045556	0.10614	0.17004	476.68	0.51577
395.00	2.2718	7.9376	0.12598	17.363	17.650	0.052111	0.10969	0.18946	387.01	1.0273
410.00	2.9483	7.1898	0.13909	20.137	20.547	0.059086	0.11399	0.23383	288.25	2.1709
425.00	3.7851	5.8610	0.17062	23.657	24.303	0.067765	0.12174	0.63363	165.30	7.0491
428.61	4.0191	4.2130	0.23736	26.217	27.171	0.074373			0	14.836
167.60	0.000074817	0.000053703	18.621.	13.917	15.310	0.10829	0.056121	0.064451	168.85	333.88
170.00	0.000098940	0.000070020	14.282.	14.051	15.465	0.10688	0.056517	0.064850	169.97	313.10
185.00	0.00047141	0.00030677	3259.8	14.913	16.450	0.099458	0.058961	0.067333	176.77	214.55
200.00	0.0017229	0.0010387	962.76	15.805	17.464	0.093969	0.061448	0.069906	183.16	152.47
215.00	0.0051236	0.0028818	347.00	16.724	18.502	0.089946	0.064084	0.072698	189.08	111.85
230.00	0.012948	0.0068419	146.16	17.667	19.560	0.087053	0.066934	0.075804	194.41	84.452
245.00	0.028726	0.014361	69.632	18.632	20.632	0.085037	0.070029	0.079283	199.05	65.512
260.00	0.057367	0.027327	36.594	19.616	21.716	0.083710	0.073369	0.083170	202.86	52.156
275.00	0.10513	0.048069	20.803	20.617	22.804	0.082924	0.076939	0.087492	205.70	42.578
290.00	0.17951	0.079385	12.597	21.632	23.893	0.082564	0.080711	0.092287	207.43	35.618
305.00	0.28905	0.12463	8.0236	22.656	24.976	0.082533	0.084658	0.097621	207.89	30.512
320.00	0.44317	0.18794	5.3209	23.685	26.043	0.082747	0.088756	0.10363	206.90	26.754
335.00	0.65207	0.27462	3.6414	24.709	27.083	0.083130	0.092992	0.11056	204.25	24.009
350.00	0.92670	0.39199	2.5511	25.716	28.080	0.083605	0.097368	0.11889	199.66	22.059
365.00	1.2790	0.55094	1.8151	26.685	29.006	0.084083	0.10191	0.12962	192.81	20.773
380.00	1.7221	0.76954	1.2995	27.583	29.821	0.084447	0.10670	0.14503	183.21	20.089
395.00	2.2718	1.0825	0.92378	28.347	30.445	0.084505	0.11191	0.17158	170.22	19.999
410.00	2.9483	1.5757	0.63463	28.822	30.693	0.083831	0.11800	0.23687	152.90	20.460
425.00	3.7851	2.6760	0.37369	28.287	29.702	0.080468	0.12690	0.83276	129.63	20.057
428.61	4.0191	4.2130	0.23736	26.217	27.171	0.074373			0	14.836

Single-Phase Properties

200.00	0.10000	12.557	0.079634	-8.7360	-8.7280	-0.037029	0.077991	0.11249	1462.2	-0.50400
273.69	0.10000	11.172	0.089507	-0.051921	-0.042970	-0.00015647	0.086590	0.12461	1052.5	-0.36572
273.69	0.10000	0.045873	21.799	20.529	22.709	0.082975	0.076618	0.087096	205.49	43.299
300.00	0.10000	0.041314	24.205	22.623	25.044	0.091118	0.080917	0.090604	216.41	30.137
400.00	0.10000	0.030400	32.895	31.727	35.016	0.11968	0.10071	0.10946	251.03	11.036
500.00	0.10000	0.024180	41.357	42.796	46.932	0.14618	0.12004	0.12856	280.24	5.7119
200.00	1.0000	12.568	0.079566	-8.7565	-8.6769	-0.037132	0.078010	0.11242	1467.8	-0.50499
300.00	1.0000	10.652	0.093883	3.2597	3.3536	0.011401	0.090766	0.13089	920.75	-0.28393
353.43	1.0000	9.3386	0.10708	10.724	10.831	0.034294	0.10058	0.15138	625.59	0.077370
353.43	1.0000	0.42419	2.3574	25.942	28.299	0.083718	0.098392	0.12109	198.31	21.710
400.00	1.0000	0.34144	2.9288	30.880	33.808	0.098365	0.10374	0.11833	227.44	12.735
500.00	1.0000	0.25402	3.9367	42.308	46.245	0.12606	0.12089	0.13164	268.79	6.0235
200.00	5.0000	12.615	0.079271	-8.8455	-8.4492	-0.037581	0.078096	0.11215	1492.3	-0.50911
300.00	5.0000	10.766	0.092881	3.0452	3.5096	0.010676	0.090769	0.12936	964.95	-0.31451
400.00	5.0000	8.1724	0.12236	17.547	18.158	0.052539	0.10973	0.17328	451.39	0.61704
500.00	5.0000	1.7031	0.58716	39.420	42.355	0.10676	0.12565	0.16251	215.66	7.3063
200.00	10.000	12.671	0.078919	-8.9520	-8.1628	-0.038127	0.078213	0.11184	1521.6	-0.51368
300.00	10.000	10.895	0.091783	2.8039	3.7218	0.0098449	0.090820	0.12787	1014.7	-0.34416
400.00	10.000	8.6867	0.11512	16.708	17.859	0.050312	0.10892	0.15802	568.97	0.18397
500.00	10.000	4.7944	0.20858	34.462	36.547	0.091756	0.12831	0.21038	241.07	3.5123
200.00	25.000	12.827	0.077961	-9.2436	-7.2946	-0.039668	0.078610	0.11112	1602.1	-0.52436
300.00	25.000	11.215	0.089166	2.2040	4.4332	0.0076959	0.091136	0.12501	1139.2	-0.40118
400.00	25.000	9.5005	0.10526	15.305	17.937	0.046401	0.10862	0.14563	778.83	-0.17082
500.00	25.000	7.5935	0.13169	30.258	33.551	0.081157	0.12604	0.16587	531.13	0.22774
200.00	50.000	13.054	0.076608	-9.6567	-5.8263	-0.041984	0.079350	0.11038	1718.0	-0.53554
300.00	50.000	11.620	0.086061	1.4532	5.7562	0.0048127	0.091871	0.12263	1299.8	-0.44882
400.00	50.000	10.240	0.097652	13.991	18.873	0.042428	0.10919	0.13997	994.88	-0.32579
500.00	50.000	8.9015	0.11234	28.100	33.717	0.075482	0.12638	0.15648	788.67	-0.19336

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., and Ihmel, E. C., "Thermodynamic Properties of the Butenes. Part II. Short Fundamental Equations of State," *Fluid Phase Equilibria* **228–229C**:173–187, 2005. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in densities calculated using the equation of state are 0.1% in the liquid phase at temperatures above 270 K (rising to 0.5% at temperatures below 200 K), 0.2% at temperatures above the critical temperature and at pressures above 10 MPa, and 0.5% in the vapor phase, including supercritical conditions below 10 MPa. The uncertainty in the vapor phase may be higher than 0.5% in some regions. The uncertainty in vapor pressure is 0.3% above 200 K, and the uncertainty in heat capacities is 0.5% at saturated liquid conditions, rising to 5% at much higher pressures and at temperatures above 250 K.

TABLE 2-199 Thermodynamic Properties of Carbon Dioxide

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
216.59	0.51796	26.777	0.037345	3.5030	3.5223	0.022943	0.042895	0.085960	975.85	-0.14430	180.63	256.70
220.00	0.59913	26.497	0.037740	3.7943	3.8169	0.024279	0.042682	0.086338	951.21	-0.13180	176.15	242.01
225.00	0.73509	26.078	0.038347	4.2235	4.2517	0.026209	0.042383	0.087024	915.16	-0.11104	169.67	222.19
230.00	0.89291	25.646	0.038992	4.6554	4.6902	0.028110	0.042103	0.087886	879.09	-0.086994	163.28	204.23
235.00	1.0747	25.201	0.039680	5.0908	5.1334	0.029986	0.041843	0.088954	842.88	-0.059053	156.98	187.88
240.00	1.2825	24.742	0.040418	5.5303	5.5821	0.031840	0.041605	0.090263	806.38	-0.026454	150.75	172.96
245.00	1.5185	24.264	0.041213	5.9749	6.0375	0.033678	0.041393	0.091866	769.44	0.011808	144.58	159.30
250.00	1.7850	23.767	0.042075	6.4256	6.5007	0.035505	0.041212	0.093831	731.78	0.057087	138.47	146.74
255.00	2.0843	23.246	0.043018	6.8836	6.9733	0.037326	0.041079	0.096251	693.01	0.11121	132.40	135.14
260.00	2.4188	22.697	0.044059	7.3505	7.4571	0.039148	0.041029	0.099258	652.58	0.17663	126.35	124.40
265.00	2.7909	22.114	0.045219	7.8282	7.9544	0.040979	0.041109	0.10306	610.07	0.25672	120.31	114.40
270.00	3.2033	21.491	0.046531	8.3190	8.4681	0.042829	0.041351	0.10798	565.46	0.35639	114.25	105.02
275.00	3.6589	20.817	0.048037	8.8266	9.0024	0.044711	0.041750	0.11457	519.14	0.48324	108.17	96.174
280.00	4.1607	20.077	0.049808	9.3560	9.5633	0.046643	0.042270	0.12385	471.54	0.64959	102.03	87.731
285.00	4.7123	19.247	0.051957	9.9154	10.160	0.048657	0.042900	0.13790	422.75	0.87650	95.810	79.548
290.00	5.3177	18.284	0.054693	10.519	10.810	0.050805	0.043734	0.16176	371.95	1.2037	89.546	71.409
295.00	5.9822	17.100	0.058480	11.197	11.547	0.053196	0.045175	0.21098	315.91	1.7218	83.558	62.936
300.00	6.7131	15.434	0.064793	12.036	12.471	0.056151	0.049288	0.38279	245.67	2.7258	80.593	53.107
304.13	7.3773	10.625	0.094118	13.928	14.622	0.063094			0	5.8665		
216.59	0.51796	0.31268	3.1982	17.286	18.943	0.094138	0.027691	0.039992	222.78	26.174	11.014	10.951
220.00	0.59913	0.35941	2.7824	17.329	18.996	0.093276	0.028120	0.040943	223.15	25.084	11.301	11.135
225.00	0.73509	0.43766	2.2849	17.387	19.067	0.092055	0.028782	0.042489	223.49	23.617	11.745	11.409
230.00	0.89291	0.52878	1.8912	17.438	19.127	0.090878	0.029488	0.044244	223.57	22.288	12.221	11.689
235.00	1.0747	0.63442	1.5762	17.481	19.175	0.089736	0.030241	0.046248	223.40	21.077	12.736	11.976
240.00	1.2825	0.75654	1.3218	17.515	19.210	0.088622	0.031042	0.048555	222.96	19.969	13.297	12.272
245.00	1.5185	0.89743	1.1143	17.538	19.230	0.087526	0.031899	0.051242	222.24	18.950	13.917	12.579
250.00	1.7850	1.0599	0.94353	17.550	19.234	0.086439	0.032827	0.054421	221.22	18.005	14.610	12.902
255.00	2.0843	1.2472	0.80180	17.549	19.220	0.085352	0.033844	0.058244	219.87	17.117	15.396	13.245
260.00	2.4188	1.4637	0.68320	17.532	19.185	0.084254	0.034955	0.062912	218.19	16.277	16.306	13.614
265.00	2.7909	1.7149	0.58314	17.498	19.125	0.083133	0.036164	0.068721	216.15	15.476	17.381	14.017
270.00	3.2033	2.0080	0.49800	17.441	19.037	0.081972	0.037482	0.076168	213.75	14.704	18.687	14.469
275.00	3.6589	2.3535	0.42490	17.359	18.913	0.080750	0.038949	0.086123	210.96	13.947	20.325	14.987
280.00	4.1607	2.7663	0.36150	17.241	18.746	0.079437	0.040628	0.10020	207.72	13.185	22.468	15.601
285.00	4.7123	3.2702	0.30579	17.078	18.519	0.077987	0.042629	0.12177	203.94	12.387	25.424	16.361
290.00	5.3177	3.9074	0.25593	16.848	18.209	0.076319	0.045155	0.15906	199.45	11.509	29.821	17.357
295.00	5.9822	4.7654	0.20985	16.509	17.764	0.074270	0.048677	0.23904	193.84	10.459	37.215	18.792
300.00	6.7131	6.1028	0.16386	15.935	17.035	0.071364	0.054908	0.52463	185.33	9.0093	53.689	21.306
304.13	7.3773	10.625	0.094118	13.928	14.622	0.063094			0	5.8665		

Single-Phase Properties

250.00	0.10000	0.048542	20.601	18.448	20.509	0.11415	0.026766	0.035428	247.79	17.399	12.950	12.565
450.00	0.10000	0.026758	37.372	24.664	28.401	0.13712	0.034775	0.043148	324.41	4.0212	29.346	21.901
650.00	0.10000	0.018506	54.037	32.199	37.602	0.15397	0.040192	0.048529	385.01	1.6551	45.466	29.873
850.00	0.10000	0.014148	70.683	40.636	47.705	0.16750	0.043944	0.052271	437.11	0.78058	60.295	36.707
1050.0	0.10000	0.011452	87.321	49.704	58.436	0.17883	0.046573	0.054895	483.65	0.34646	73.843	42.692
250.00	1.0000	0.53250	1.8779	18.023	19.901	0.093263	0.029361	0.042504	235.08	17.606	13.584	12.691
450.00	1.0000	0.27038	3.6985	24.546	28.244	0.11771	0.034954	0.043866	322.89	3.9880	29.620	21.954
650.00	1.0000	0.18527	5.3976	32.133	37.530	0.13473	0.040239	0.048779	385.36	1.6311	45.651	29.907
850.00	1.0000	0.14131	7.0767	40.591	47.668	0.14830	0.043965	0.052397	438.06	0.76632	60.435	36.732
1050.0	1.0000	0.11430	8.7487	49.671	58.419	0.15965	0.046585	0.054970	484.84	0.33777	73.956	42.712
250.00	5.0000	24.060	0.041563	6.2824	6.4902	0.034925	0.041321	0.090937	762.21	0.015208	142.22	153.15
287.43	5.0000	18.798	0.053196	10.202	10.468	0.049681	0.043268	0.14775	398.39	1.0195	92.760	75.598
287.43	5.0000	3.5600	0.28090	16.977	18.381	0.077209	0.043774	0.13705	201.86	11.974	27.323	16.808
450.00	5.0000	1.4155	0.70647	24.000	27.533	0.10313	0.035769	0.047478	317.50	3.8034	31.164	22.429
650.00	5.0000	0.92982	1.0755	31.842	37.219	0.12091	0.040445	0.049898	387.59	1.5263	46.589	30.157
850.00	5.0000	0.70241	1.4237	40.395	47.513	0.13469	0.044055	0.052945	442.64	0.70611	61.117	36.899
1050.0	5.0000	0.56658	1.7650	49.524	58.349	0.14613	0.046637	0.055297	490.31	0.30129	74.494	42.836
250.00	10.000	24.459	0.040885	6.0862	6.4950	0.034120	0.041488	0.087624	804.05	-0.034849	147.52	162.47
450.00	10.000	2.9910	0.33433	23.276	26.619	0.095787	0.036785	0.052935	314.60	3.4705	33.917	23.679
650.00	10.000	1.8632	0.53671	31.482	36.849	0.11461	0.040693	0.051293	391.91	1.3965	48.005	30.687
850.00	10.000	1.3930	0.71790	40.155	47.334	0.12866	0.044164	0.053603	449.04	0.63635	62.093	37.224
1050.0	10.000	1.1205	0.89248	49.347	58.271	0.14021	0.046701	0.055685	497.48	0.25964	75.242	43.066
250.00	100.00	28.075	0.035619	4.3002	7.8621	0.026023	0.043569	0.073521	1227.6	-0.27302	206.28	287.05
450.00	100.00	19.246	0.051959	16.560	21.756	0.067062	0.040841	0.066107	753.30	-0.11128	106.65	83.996
650.00	100.00	13.677	0.073117	27.132	34.444	0.090445	0.043108	0.061252	646.36	-0.054084	86.093	58.868
850.00	100.00	10.636	0.094022	37.076	46.478	0.10660	0.045620	0.059534	646.61	-0.13292	87.259	54.445
1050.0	100.00	8.7929	0.11373	46.995	58.368	0.11916	0.047676	0.059512	668.90	-0.21482	94.022	55.058
450.00	500.00	28.922	0.034576	13.014	30.302	0.050604	0.047702	0.063434	1576.4	-0.38514	239.59	303.64
650.00	500.00	25.661	0.038969	23.302	42.786	0.073576	0.048419	0.061885	1404.7	-0.40369	197.25	191.14
850.00	500.00	23.144	0.043208	33.551	55.155	0.090166	0.049676	0.061912	1320.1	-0.41098	177.31	145.07
1050.0	500.00	21.126	0.047334	43.903	67.570	0.10328	0.050818	0.062247	1278.7	-0.41674	168.50	123.33

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Span, R., and Wagner, W., "A New Equation of State for Carbon Dioxide Covering the Fluid Region from the Triple-Point Temperature to 1100 K at Pressures up to 800 MPa," *J. Phys. Chem. Ref. Data* **25**(6):1509–1596, 1996. The source for viscosity is Fenghour, A., Wakeham, W. A., and Vesovic, V., "The Viscosity of Carbon Dioxide," *J. Phys. Chem. Ref. Data* **27**:31–44, 1998. The source for thermal conductivity is Vesovic, V., Wakeham, W. A., Olchowy, G. A., Sengers, J. V., Watson, J. T. R., and Millat, J., "The Transport Properties of Carbon Dioxide," *J. Phys. Chem. Ref. Data* **19**:763–808, 1990.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

At pressures up to 30 MPa and temperatures up to 523 K, the estimated uncertainty ranges from 0.03% to 0.05% in density, 0.03% (in the vapor) to 1% in the speed of sound (0.5% in the liquid), and 0.15% (in the vapor) to 1.5% (in the liquid) in heat capacity. Special interest has been focused on the description of the critical region and the extrapolation behavior of the formulation (to the limits of chemical stability). The uncertainty in viscosity ranges from 0.3% in the dilute gas near room temperature to 5% at the highest pressures. The uncertainty in thermal conductivity is less than 5%.

TABLE 2-200 Thermodynamic Properties of Carbon Monoxide

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
68.160	0.015537	30.330	0.032971	-0.81158	-0.81106	-0.010820	0.035351	0.060430	998.20	-0.36906	180.28	274.18
70.000	0.021053	30.064	0.033262	-0.70065	-0.69995	-0.0092140	0.034805	0.060226	980.50	-0.36553	175.49	252.15
72.000	0.028718	29.773	0.033588	-0.58046	-0.57950	-0.0075210	0.034248	0.060064	961.22	-0.36074	170.45	232.14
74.000	0.038447	29.478	0.033924	-0.46058	-0.45927	-0.0058785	0.033724	0.059961	941.89	-0.35489	165.55	215.32
76.000	0.050599	29.180	0.034270	-0.34088	-0.33915	-0.0042823	0.033232	0.059917	922.49	-0.34794	160.76	201.01
78.000	0.065559	28.878	0.034628	-0.22127	-0.21900	-0.0027285	0.032768	0.059930	903.01	-0.33981	156.06	188.69
80.000	0.083738	28.573	0.034999	-0.10165	-0.098716	-0.0012138	0.032329	0.060002	883.44	-0.33041	151.45	177.96
82.000	0.10556	28.262	0.035383	0.018099	0.021834	0.00026503	0.031915	0.060132	863.76	-0.31966	146.89	168.52
84.000	0.13148	27.947	0.035782	0.13806	0.14277	0.0017110	0.031522	0.060324	843.95	-0.30742	142.40	160.13
86.000	0.16196	27.626	0.036197	0.25835	0.26421	0.0031269	0.031150	0.060578	824.00	-0.29356	137.96	152.60
88.000	0.19748	27.300	0.036630	0.37906	0.38629	0.0045153	0.030798	0.060899	803.89	-0.27794	133.57	145.77
90.000	0.23852	26.967	0.037082	0.50030	0.50915	0.0058787	0.030463	0.061291	783.60	-0.26034	129.23	139.52
92.000	0.28559	26.627	0.037556	0.62218	0.63291	0.0072195	0.030146	0.061760	763.12	-0.24056	124.94	133.75
94.000	0.33919	26.280	0.038052	0.74482	0.75773	0.0085399	0.029846	0.062314	742.41	-0.21834	120.69	128.38
96.000	0.39983	25.924	0.038574	0.86835	0.88377	0.0098422	0.029562	0.062962	721.45	-0.19335	116.51	123.34
98.000	0.46805	25.559	0.039125	0.99289	1.0112	0.011129	0.029294	0.063716	700.22	-0.16523	112.38	118.57
100.00	0.54438	25.184	0.039708	1.1186	1.1402	0.012402	0.029043	0.064590	678.68	-0.13353	108.31	114.02
102.00	0.62934	24.798	0.040326	1.2457	1.2710	0.013663	0.028809	0.065604	656.78	-0.097704	104.30	109.66
104.00	0.72348	24.399	0.040985	1.3742	1.4039	0.014916	0.028592	0.066781	634.50	-0.057078	100.36	105.45
106.00	0.82736	23.987	0.041689	1.5045	1.5390	0.016162	0.028395	0.068153	611.77	-0.010824	96.482	101.35
108.00	0.94154	23.560	0.042446	1.6368	1.6768	0.017404	0.028218	0.069759	588.54	0.042104	92.679	97.342
110.00	1.0666	23.114	0.043263	1.7713	1.8175	0.018646	0.028066	0.071656	564.73	0.10304	88.948	93.404
112.00	1.2031	22.649	0.044151	1.9085	1.9616	0.019891	0.027941	0.073916	540.25	0.17371	85.290	89.510
114.00	1.3517	22.161	0.045124	2.0487	2.1097	0.021142	0.027850	0.076648	515.01	0.25641	81.702	85.641
116.00	1.5130	21.646	0.046197	2.1925	2.2624	0.022406	0.027800	0.080005	488.86	0.35427	78.180	81.774
118.00	1.6877	21.099	0.047395	2.3405	2.4205	0.023688	0.027803	0.084225	461.63	0.47167	74.716	77.888
120.00	1.8765	20.513	0.048749	2.4938	2.5853	0.024996	0.027874	0.089692	433.11	0.61495	71.296	73.954
122.00	2.0802	19.878	0.050307	2.6536	2.7583	0.026343	0.028038	0.097070	403.00	0.79382	67.896	69.940
124.00	2.2997	19.179	0.052141	2.8221	2.9420	0.027744	0.028333	0.10762	370.88	1.0239	64.476	65.797
126.00	2.5360	18.390	0.054377	3.0024	3.1403	0.029230	0.028826	0.12411	336.15	1.3325	60.972	61.448
128.00	2.7904	17.464	0.057259	3.2010	3.3608	0.030854	0.029646	0.15392	297.82	1.7728	57.261	56.748
130.00	3.0647	16.288	0.061393	3.4328	3.6210	0.032745	0.031097	0.22603	254.03	2.4703	53.107	51.348
132.86	3.4982	10.850	0.092166	4.2912	4.6137	0.040039			0	6.1475		
68.160	0.015537	0.027707	36.091	5.1252	5.6859	0.084499	0.021089	0.029785	167.25	40.804	6.6865	4.6366
70.000	0.021053	0.036656	27.281	5.1600	5.7343	0.082704	0.021155	0.029947	169.22	38.426	6.8845	4.7768
72.000	0.028718	0.048780	20.500	5.1971	5.7859	0.080887	0.021238	0.030153	171.27	36.126	7.1009	4.9329
74.000	0.038447	0.063796	15.675	5.2334	5.8361	0.079194	0.021333	0.030394	173.22	34.080	7.3188	5.0934
76.000	0.050599	0.082130	12.176	5.2688	5.8849	0.077613	0.021441	0.030672	175.07	32.250	7.5382	5.2589
78.000	0.065559	0.10424	9.5935	5.3031	5.9320	0.076131	0.021563	0.030993	176.80	30.604	7.7592	5.4300
80.000	0.083738	0.13059	7.6573	5.3363	5.9775	0.074739	0.021699	0.031360	178.42	29.116	7.9820	5.6076
82.000	0.10556	0.16171	6.1841	5.3682	6.0210	0.073426	0.021850	0.031777	179.92	27.763	8.2067	5.7922
84.000	0.13148	0.19810	5.0478	5.3988	6.0625	0.072185	0.022017	0.032250	181.29	26.527	8.4335	5.9847
86.000	0.16196	0.24036	4.1605	5.4280	6.1019	0.071007	0.022199	0.032783	182.54	25.392	8.6627	6.1860
88.000	0.19748	0.28906	3.4595	5.4556	6.1388	0.069885	0.022397	0.033383	183.66	24.345	8.8944	6.3968
90.000	0.23852	0.34486	2.8997	5.4816	6.1733	0.068813	0.022611	0.034057	184.65	23.377	9.1291	6.6182
92.000	0.28559	0.40845	2.4483	5.5058	6.2050	0.067785	0.022842	0.034813	185.51	22.477	9.3672	6.8512
94.000	0.33919	0.48058	2.0808	5.5280	6.2338	0.066796	0.023089	0.035661	186.22	21.638	9.6091	7.0969
96.000	0.39983	0.56209	1.7791	5.5482	6.2595	0.065840	0.023352	0.036615	186.80	20.853	9.8555	7.3566
98.000	0.46805	0.65388	1.5293	5.5661	6.2819	0.064912	0.023633	0.037690	187.23	20.118	10.107	7.6317
100.00	0.54438	0.75700	1.3210	5.5816	6.3007	0.064007	0.023931	0.038906	187.52	19.426	10.366	7.9239
102.00	0.62934	0.87260	1.1460	5.5945	6.3157	0.063120	0.024248	0.040288	187.67	18.773	10.632	8.2350
104.00	0.72348	1.0020	0.99799	5.6044	6.3265	0.062248	0.024586	0.041869	187.66	18.154	10.909	8.5675
106.00	0.82736	1.1468	0.87198	5.6112	6.3327	0.061385	0.024945	0.043694	187.51	17.565	11.198	8.9238
108.00	0.94154	1.3088	0.76404	5.6145	6.3339	0.060526	0.025329	0.045820	187.20	17.001	11.502	9.3073
110.00	1.0666	1.4903	0.67102	5.6138	6.3295	0.059665	0.025741	0.048326	186.73	16.458	11.828	9.7221
112.00	1.2031	1.6938	0.59039	5.6088	6.3191	0.058797	0.026186	0.051322	186.11	15.930	12.181	10.173

114.00	1.3517	1.9228	0.52008	5.5986	6.3016	0.057914	0.026671	0.054966	185.33	15.411	12.569	10.667
116.00	1.5130	2.1815	0.45841	5.5827	6.2762	0.057008	0.027203	0.059493	184.38	14.894	13.005	11.213
118.00	1.6877	2.4754	0.40397	5.5598	6.2416	0.056070	0.027794	0.065263	183.27	14.372	13.507	11.821
120.00	1.8765	2.8123	0.35558	5.5286	6.1959	0.055084	0.028462	0.072864	181.99	13.833	14.101	12.509
122.00	2.0802	3.2027	0.31224	5.4872	6.1367	0.054034	0.029229	0.083320	180.52	13.265	14.826	13.301
124.00	2.2997	3.6629	0.27301	5.4324	6.0602	0.052892	0.030133	0.098585	178.84	12.648	15.747	14.234
126.00	2.5360	4.2194	0.23700	5.3595	5.9605	0.051613	0.031233	0.12291	176.93	11.956	16.981	15.373
128.00	2.7904	4.9212	0.20320	5.2594	5.8264	0.050117	0.032636	0.16759	174.68	11.140	18.777	16.840
130.00	3.0647	5.8832	0.16998	5.1113	5.6322	0.048216	0.034579	0.27599	171.86	10.100	21.845	18.936
132.86	3.4982	10.850	0.092166	4.2912	4.6137	0.040039		0		6.1475		

Single-Phase Properties

100.00	0.10000	0.12298	8.1315	5.7653	6.5785	0.080014	0.021118	0.030153	201.29	17.820	10.075	6.9147
200.00	0.10000	0.060293	16.586	7.8674	9.5259	0.10048	0.020812	0.029239	288.05	5.3111	19.227	12.897
300.00	0.10000	0.040104	24.935	9.9522	12.446	0.11231	0.020833	0.029191	353.12	2.5186	26.605	17.731
400.00	0.10000	0.030062	33.265	12.045	15.371	0.12073	0.021028	0.029364	407.29	1.2653	33.106	21.870
500.00	0.10000	0.024045	41.588	14.169	18.328	0.12733	0.021479	0.029807	454.00	0.56244	39.272	25.540
100.00	1.0000	25.261	0.039586	1.1047	1.1443	0.012262	0.029062	0.064114	685.44	-0.14414	112.87	113.83
108.96	1.0000	23.349	0.042829	1.7009	1.7437	0.017998	0.028142	0.070627	577.22	0.070176	90.884	95.450
108.96	1.0000	1.3931	0.71782	5.6147	6.3325	0.060114	0.025522	0.046966	186.99	16.739	11.655	9.5017
200.00	1.0000	0.61727	1.6200	7.7647	9.3847	0.080819	0.020996	0.030510	286.20	5.1924	19.474	13.192
300.00	1.0000	0.40214	2.4867	9.8936	12.380	0.092976	0.020895	0.029646	354.42	2.4256	26.760	17.918
400.00	1.0000	0.29999	3.3334	12.005	15.338	0.10149	0.021064	0.029598	409.43	1.2088	33.222	22.024
500.00	1.0000	0.23962	4.1732	14.140	18.313	0.10812	0.021505	0.029948	456.39	0.52786	39.364	25.676
100.00	5.0000	25.864	0.038663	0.99666	1.1900	0.011154	0.029263	0.060925	737.92	-0.21740	152.30	112.19
200.00	5.0000	3.4130	0.29299	7.2656	8.7305	0.064994	0.021878	0.038000	285.27	4.3757	22.190	15.094
300.00	5.0000	2.0232	0.49426	9.6364	12.108	0.078767	0.021174	0.031673	362.95	2.0288	27.812	18.716
400.00	5.0000	1.4824	0.67458	11.834	15.207	0.087691	0.021224	0.030585	420.15	0.98413	33.871	22.588
500.00	5.0000	1.1786	0.84845	14.015	18.258	0.094498	0.021618	0.030535	467.56	0.39254	39.837	26.139
100.00	10.000	26.482	0.037761	0.88669	1.2643	0.0099878	0.029539	0.058409	792.04	-0.27800	200.46	110.33
200.00	10.000	7.4298	0.13459	6.5960	7.9419	0.056188	0.022832	0.048831	307.84	2.8854	34.772	19.114
300.00	10.000	4.0263	0.24837	9.3290	11.813	0.072068	0.021511	0.034036	379.01	1.5731	30.972	19.862
400.00	10.000	2.9079	0.34389	11.634	15.073	0.081462	0.021420	0.031689	435.68	0.74980	35.414	23.298
500.00	10.000	2.3052	0.43381	13.870	18.208	0.088461	0.021755	0.031188	482.48	0.25486	40.797	26.682
100.00	50.000	29.422	0.033988	0.39153	2.0910	0.0040257	0.031398	0.052094	1066.8	-0.43831	567.46	99.463
200.00	50.000	20.591	0.048566	4.5424	6.9707	0.038097	0.025036	0.045541	706.01	-0.28689	256.88	50.929
300.00	50.000	14.766	0.067725	7.7949	11.181	0.055259	0.023212	0.039083	609.73	-0.21242	139.18	34.086
400.00	50.000	11.439	0.087418	10.518	14.889	0.065951	0.022620	0.035519	604.51	-0.27153	94.476	31.319
500.00	50.000	9.3865	0.10654	13.024	18.350	0.073681	0.022659	0.033937	622.30	-0.36911	76.899	32.167
100.00	100.00	31.474	0.031772	0.095937	3.2732	-0.00053857	0.033037	0.050530	1282.4	-0.47725	1005.7	90.560
200.00	100.00	24.888	0.040181	3.9022	7.9203	0.031951	0.026437	0.043352	987.69	-0.50923	536.84	73.380
300.00	100.00	20.200	0.049505	7.0625	12.013	0.048608	0.024358	0.038799	866.60	-0.54516	331.69	54.648
400.00	100.00	16.970	0.058928	9.8487	15.741	0.059353	0.023555	0.036051	822.33	-0.59581	229.18	45.748
500.00	100.00	14.662	0.068206	12.449	19.269	0.067230	0.023434	0.034683	808.91	-0.64123	173.58	42.504

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., and Span, R., "Short Fundamental Equations of State for 20 Industrial Fluids," *J. Chem. Eng. Data*, **51**(3):785–850, 2006. The source for viscosity and thermal conductivity is Version 9.08 of the NIST14 database.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The equation of state is valid from the triple point to 500 K with pressures to 100 MPa. At higher pressures, the deviations from the equation increase rapidly, and it is not recommended to use the equation above 100 MPa. The uncertainties in the equation are 0.3% in density (approaching 1% near the critical point), 0.2% in vapor pressure, and 2% in heat capacities. For viscosity, estimated uncertainty is 2%. For thermal conductivity, estimated uncertainty, except near the critical region, is 4–6%.

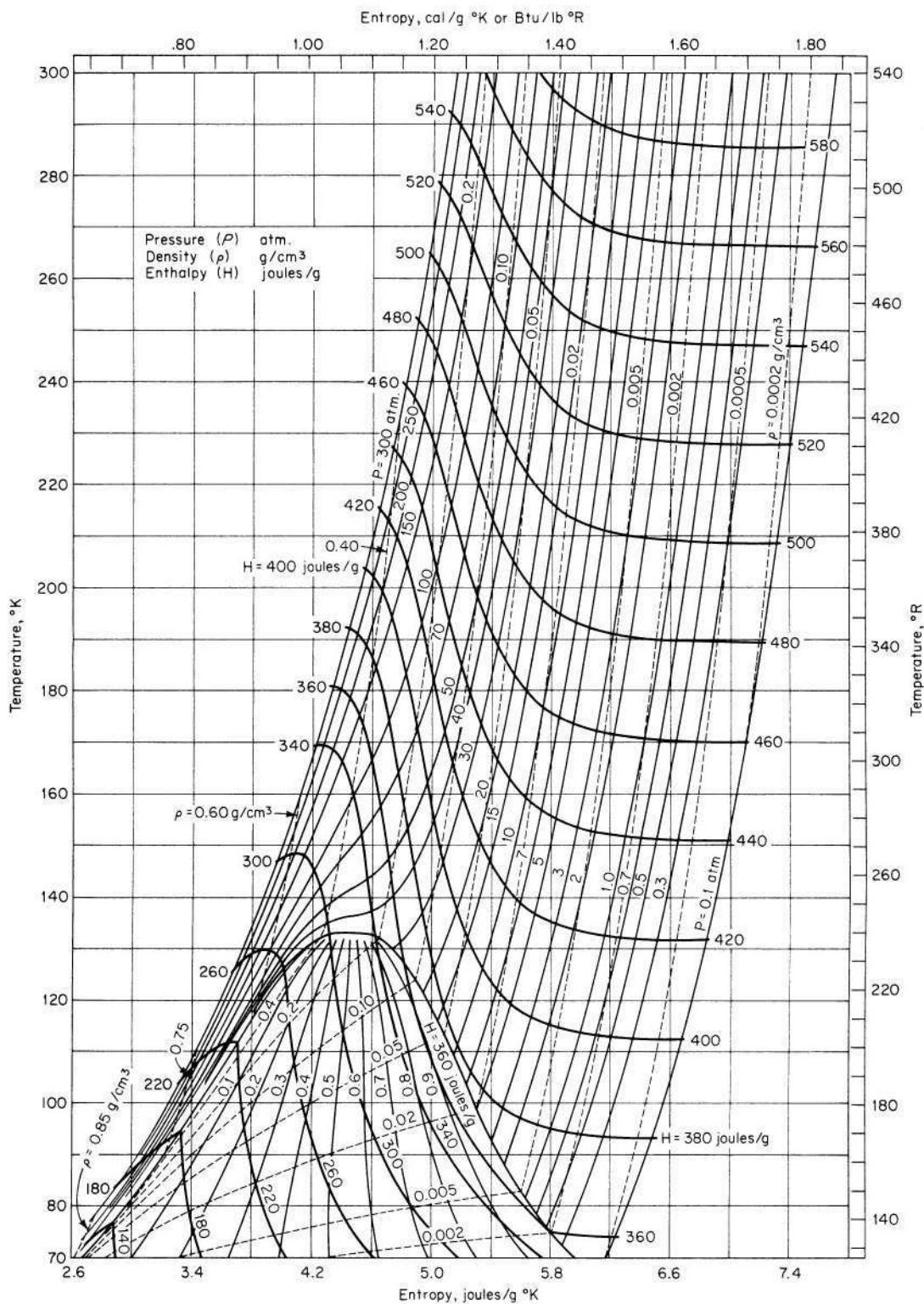


FIG. 2-8 Temperature-entropy diagram for carbon monoxide. Pressure P , in atmospheres; density ρ , in grams per cubic centimeter; enthalpy H , in joules per gram. (From Hust and Stewart, NBS Tech. Note 202, 1963.)

TABLE 2-201 Thermophysical Properties of Saturated Carbon Tetrachloride

T, K	P, bar	v_f , m ³ /kg	v_g , m ³ /kg	h_f , kJ/kg	h_g , kJ/kg	s_f , kJ/(kg·K)	s_g , kJ/(kg·K)	c_{pf} , kJ/(kg·K)	μ_f , 10 ⁻⁶ Pa·s	k_f , W/(m·K)	Pr
280	0.064	0.000 619	2.414	205.5	420.7	1.018	1.787	0.835	1042	0.1043	8.34
290	0.105	0.000 625	1.495	212.9	425.7	1.042	1.775	0.844	892	0.1020	7.38
300	0.165	0.000 633	0.971	220.9	430.9	1.068	1.768	0.853	774	0.0998	6.62
310	0.251	0.000 641	0.669	228.8	436.1	1.095	1.764	0.863	679	0.0975	6.01
320	0.370	0.000 649	0.463	236.9	441.3	1.121	1.760	0.874	603	0.0952	5.54
330	0.531	0.000 657	0.3306	246.0	446.4	1.149	1.756	0.885	539	0.0930	5.13
340	0.743	0.000 666	0.2407	254.5	451.5	1.174	1.754	0.897	486	0.0907	4.81
350	1.017	0.000 674	0.1802	263.1	456.6	1.199	1.752	0.910	441	0.0884	4.54
360	1.361	0.000 684	0.1370	271.8	461.7	1.224	1.751	0.924	402	0.0861	4.31
370	1.795	0.000 694	0.1053	280.8	466.6	1.248	1.751	0.939	368	0.0839	4.12
380	2.327	0.000 704	0.0820	289.7	471.5	1.272	1.750	0.954	338	0.0816	3.95
390	2.970	0.000 715	0.0651	298.1	475.8	1.295	1.751	0.970	311	0.0794	3.80
400	3.735	0.000 727	0.0525	307.9	481.2	1.319	1.752	0.987	287	0.0771	3.67
410	4.642	0.000 739	0.0426	317.1	485.8	1.341	1.753	1.010	265	0.0749	3.57
420	5.700	0.000 753	0.0350	326.0	490.4	1.363	1.754	1.034	246	0.0726	3.50
430	6.927	0.000 766	0.02899	335.2	494.9	1.384	1.756	1.060	227	0.0704	3.42
440	8.342	0.000 780	0.02413	344.3	499.2	1.405	1.757	1.094	211	0.0682	3.38
450	9.958	0.000 796	0.02020	353.6	503.4	1.426	1.759	1.141	195	0.0660	3.37
460	11.792	0.000 801	0.01692	363.1	507.3	1.446	1.760	1.207	180	0.0638	3.36
470	13.869	0.000 834	0.01425	372.8	511.1	1.467	1.761	1.240	167	0.0666	3.36
480	16.21	0.000 856	0.01205	382.6	514.6	1.487	1.762	1.278	156	0.0594	3.36
490	18.83	0.000 880	0.01011	392.0	517.5	1.507	1.763	1.320	145	0.0511	3.35
500	21.77	0.000 858	0.00858	402.5	520.2	1.526	1.762	1.375	133	0.0549	3.35
510	25.02	0.000 945	0.00722	412.9	522.6	1.546	1.761	1.44			
520	28.68	0.000 987	0.00607	424.3	524.2	1.568	1.760	1.52			
530	32.71	0.001 041	0.00500	436.4	524.5	1.590	1.756				
540	37.18	0.001 121	0.00400	448.3	522.7	1.614	1.749				
550	44.12	0.001 248	0.00309	463.4	518.2	1.638	1.738				
556.4 ^c	45.60	0.001 792	0.00179	494.4	494.4	1.692	1.692				

c = critical point. Base points: $h_f = 200$ at 273.15 K = 0°C = $h_A - 300$ kJ/kg; $s_f = 1.000$ at 273.15 K = 0°C = $s_A - 4.000$ kJ/(kg·K). Values mostly rounded and converted from Altunin, V. V., V. Z. Geller, et al., *Thermophysical Properties of Freons*, vol. 9, Hemisphere, Washington, DC, 1987 (243 pp.). Some irregularities exist in these data.

TABLE 2-202 Saturated Carbon Tetrafluoride (R14)*

T, K	P, bar	v_f , m ³ /kg	v_g , m ³ /kg	h_f , kJ/kg	h_g , kJ/kg	s_f , kJ/(kg·K)	s_g , kJ/(kg·K)	c_{pf} , kJ/(kg·K)	μ_f , 10 ⁻⁴ Pa·s	k_f , W/(m·K)
100	0.0089	5.370.-4	10.77	495.8	648.4	5.487	7.003	0.887		0.136
110	0.0286	5.515.-4	3.648	502.7	652.9	5.556	6.919	0.887		0.128
120	0.0924	5.668.-4	1.228	510.4	657.1	5.624	6.847	0.890		0.119
130	0.2986	5.834.-4	0.4051	518.8	661.1	5.691	6.786	0.896		0.111
140	0.6901	6.018.-4	0.1855	527.7	664.8	5.757	6.736	0.904	3.56	0.104
150	1.4074	6.225.-4	0.0951	537.2	668.3	5.822	6.696	0.922	3.28	0.097
160	2.598	6.460.-4	0.0532	549.4	671.4	5.885	6.662	0.975	3.03	0.089
170	4.426	6.733.-4	0.0318	557.6	674.0	5.947	6.629	1.031	2.80	0.081
180	7.067	7.055.-4	0.0200	568.2	676.1	6.007	6.607	1.104	2.59	0.072
190	10.702	7.449.-4	0.0131	579.3	677.4	6.066	6.583	1.203	2.39	0.064
200	15.531	7.957.-4	0.0087	591.0	677.8	6.124	6.558	1.334	2.19	0.057
210	21.794	8.674.-4	0.0058	603.5	676.4	6.182	6.536	1.506	2.01	0.049
220	29.269	9.931.-4	0.0036	618.5	671.4	6.233	6.490	1.73	1.85	0.042
227.5 ^c	37.45	1.598.-3	0.0016	646.9	646.9	6.371	6.371	∞		∞

* P , v , h , and s values interpolated, extrapolated, and converted from Oguchi, *Reito*, 52 (1977): 869-889. c = critical point. The notation 5.370.-4 signifies 5.370×10^{-4} .

Equations and constants approximated to ASHRAE tables are given by Mecaryk, K. and M. Masaryk, *Heat Recovery Systems and CHP*, 11, 2-3 (1991). The 1993 ASHRAE *Handbook—Fundamentals* (S.I. ed.) contains a saturation table from -140 to -45.65 °C and an enthalpy-log-pressure diagram from 0.1 to 300 bar, -140 to 300 °C. For properties to 1000 bar from 90 to 420 K, see Rublo, R. G., J. A. Zollweg, et al., *J. Chem. Eng. Data*, 36 (1991): 171-184. Saturation and superheat tables and a diagram to 80 bar, 600 K are given by Reynolds, W. C., *Thermodynamic Properties in S.I.*, Stanford Univ. publ., 1979 (173 pp.). Chari, Ph.D. thesis, University of Michigan, 1960, presents saturation-temperature tables in fps units for 1°F increments from -270 to -51°F. Thermodynamic and transport properties, equations, and computer code and tables at constant entropy from 89 to 845 K are given by Hunt, J. L. and Boney, L. R., NASA TN D-7181, 1973 (105 pp.), largely based upon the Chari data.

TABLE 2-203 Thermodynamic Properties of Carbonyl Sulfide

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
134.30	0.000064435	22.518	0.044409	-6.2965	-6.2965	-0.036049	0.050295	0.074835	1449.7	-0.47647
140.00	0.00014316	22.330	0.044783	-5.8733	-5.8733	-0.032962	0.049071	0.073691	1423.0	-0.48236
150.00	0.00049244	22.002	0.045451	-5.1446	-5.1446	-0.027934	0.047317	0.072140	1376.1	-0.48929
160.00	0.0014232	21.674	0.046138	-4.4289	-4.4289	-0.023315	0.045963	0.071067	1329.2	-0.49217
170.00	0.0035714	21.346	0.046847	-3.7220	-3.7219	-0.019029	0.044919	0.070377	1282.3	-0.49125
180.00	0.0079826	21.016	0.047583	-3.0204	-3.0201	-0.015019	0.044118	0.070000	1235.4	-0.48674
190.00	0.016210	20.683	0.048349	-2.3213	-2.3206	-0.011239	0.043507	0.069885	1188.5	-0.47875
200.00	0.030380	20.345	0.049151	-1.6224	-1.6209	-0.0076537	0.043047	0.069996	1141.6	-0.46727
210.00	0.053219	20.002	0.049994	-0.92144	-0.91878	-0.0042337	0.042708	0.070307	1094.7	-0.45221
220.00	0.088035	19.653	0.050884	-0.21676	-0.21228	-0.00095536	0.042464	0.070804	1047.8	-0.43335
230.00	0.13868	19.295	0.051827	0.49332	0.50050	0.0022014	0.042299	0.071479	1000.8	-0.41029
240.00	0.20947	18.927	0.052834	1.2104	1.2214	0.0052538	0.042198	0.072335	953.72	-0.38249
250.00	0.30513	18.548	0.053914	1.9359	1.9524	0.0082168	0.042150	0.073381	906.38	-0.34913
260.00	0.43069	18.155	0.055080	2.6716	2.6954	0.011104	0.042147	0.074640	858.76	-0.30911
270.00	0.59148	17.747	0.056348	3.4192	3.4525	0.013927	0.042186	0.076145	810.74	-0.26092
280.00	0.79300	17.320	0.057737	4.1805	4.2263	0.016699	0.042262	0.077949	762.21	-0.20244
290.00	1.0409	16.870	0.059276	4.9579	5.0196	0.019432	0.042376	0.080132	713.03	-0.13069
300.00	1.3412	16.394	0.060998	5.7542	5.8360	0.022138	0.042533	0.082812	663.05	-0.041371
310.00	1.6998	15.885	0.062953	6.5728	6.6798	0.024832	0.042738	0.086180	612.06	0.071920
320.00	2.1232	15.335	0.065210	7.4185	7.5570	0.027531	0.043007	0.090546	559.82	0.21913
330.00	2.6180	14.733	0.067877	8.2981	8.4758	0.030256	0.043365	0.096471	505.95	0.41667
340.00	3.1915	14.060	0.071123	9.2217	9.4487	0.033041	0.043857	0.10507	449.91	0.69379
350.00	3.8523	13.287	0.075261	10.207	10.497	0.035938	0.044569	0.11899	390.71	1.1092
360.00	4.6109	12.352	0.080962	11.289	11.662	0.039054	0.045699	0.14652	326.44	1.8033
370.00	5.4827	11.072	0.090320	12.572	13.067	0.042698	0.047851	0.23509	252.51	3.2373
378.77	6.3688	7.4100	0.13495	15.239	16.099	0.050522			0	8.9233
134.30	0.000064435	0.000057710	17.328	14.600	15.716	0.12786	0.022770	0.031089	159.29	230.42
140.00	0.00014316	0.00012301	8.129.4	14.730	15.894	0.12252	0.023105	0.031429	162.31	199.85
150.00	0.00049244	0.00039506	2.531.3	14.963	16.209	0.11442	0.023740	0.032079	167.40	158.07
160.00	0.0014232	0.0010711	933.62	15.201	16.529	0.10767	0.024427	0.032795	172.22	127.26
170.00	0.0035714	0.0025330	394.79	15.443	16.853	0.10200	0.025159	0.033580	176.77	104.13
180.00	0.0079826	0.0053579	186.64	15.689	17.179	0.097198	0.025933	0.034438	181.06	86.482
190.00	0.016210	0.010340	96.715	15.938	17.505	0.093108	0.026745	0.035375	185.07	72.846
200.00	0.030380	0.018492	54.078	16.187	17.830	0.089601	0.027592	0.036399	188.78	62.184
210.00	0.053219	0.031038	32.218	16.436	18.151	0.086575	0.028471	0.037521	192.15	53.756
220.00	0.088035	0.049400	20.243	16.684	18.466	0.083945	0.029382	0.038751	195.15	47.024
230.00	0.13868	0.075184	13.301	16.927	18.772	0.081643	0.030323	0.040105	197.76	41.595
240.00	0.20947	0.11018	9.0759	17.166	19.067	0.079610	0.031293	0.041602	199.92	37.175
250.00	0.30513	0.15639	6.3942	17.397	19.348	0.077800	0.032292	0.043268	201.62	33.545
260.00	0.43069	0.21605	4.6285	17.619	19.613	0.076171	0.033320	0.045137	202.81	30.541
270.00	0.59148	0.29173	3.4278	17.830	19.858	0.074687	0.034376	0.047258	203.46	28.040
280.00	0.79300	0.38642	2.5879	18.027	20.079	0.073316	0.035463	0.049700	203.53	25.945
290.00	1.0409	0.50374	1.9852	18.207	20.273	0.072030	0.036585	0.052570	202.98	24.185
300.00	1.3412	0.64823	1.5427	18.365	20.434	0.070799	0.037748	0.056029	201.77	22.698
310.00	1.6998	0.82581	1.2109	18.497	20.556	0.069593	0.038964	0.060340	199.84	21.437
320.00	2.1232	1.0446	0.95730	18.596	20.629	0.068379	0.040251	0.065951	197.14	20.357
330.00	2.6180	1.3165	0.75961	18.651	20.640	0.067117	0.041639	0.073691	193.61	19.412
340.00	3.1915	1.6600	0.60241	18.647	20.570	0.065750	0.043181	0.085260	189.17	18.546
350.00	3.8523	2.1078	0.47442	18.555	20.383	0.064185	0.044969	0.10478	183.76	17.674
360.00	4.6109	2.7278	0.36660	18.321	20.011	0.062245	0.047203	0.14532	177.26	16.625
370.00	5.4827	3.7183	0.26894	17.782	19.256	0.059425	0.050441	0.28170	169.47	14.934
378.77	6.3688	7.4100	0.13495	15.239	16.099	0.050522			0	8.9233

Single-Phase Properties

150.00	0.10000	22.003	0.045448	-5.1456	-5.1411	-0.027941	0.047320	0.072138	1376.4	-0.48934
200.00	0.10000	20.347	0.049148	-1.6235	-1.6186	-0.0076594	0.043049	0.069990	1141.9	-0.46739
222.70	0.10000	19.557	0.051133	-0.025288	-0.020175	-0.000090227	0.042413	0.070969	1035.1	-0.42755
222.70	0.10000	0.055571	17.995	16.750	18.550	0.083293	0.029634	0.039104	195.90	45.441
250.00	0.10000	0.049051	20.387	17.590	19.628	0.087860	0.030973	0.040014	207.32	32.301
300.00	0.10000	0.040528	24.674	19.215	21.683	0.095346	0.033493	0.042188	226.21	19.802
350.00	0.10000	0.034596	28.905	20.954	23.845	0.10201	0.035716	0.044263	243.37	13.574
150.00	1.0000	22.015	0.045423	-5.1547	-5.1093	-0.028002	0.047344	0.072112	1379.4	-0.48974
200.00	1.0000	20.365	0.049103	-1.6382	-1.5891	-0.0077331	0.043071	0.069917	1145.9	-0.46887
250.00	1.0000	18.572	0.053843	1.9164	1.9703	0.0081385	0.042163	0.073223	910.91	-0.35287
288.48	1.0000	16.940	0.059031	4.8385	4.8975	0.019018	0.042357	0.079771	720.56	-0.14261
288.48	1.0000	0.48426	2.0650	18.180	20.245	0.072221	0.036412	0.052100	203.11	24.434
300.00	1.0000	0.45571	2.1944	18.640	20.834	0.074222	0.036112	0.050249	209.62	21.627
350.00	1.0000	0.36985	2.7038	20.566	23.270	0.081735	0.036823	0.048032	233.01	14.214
150.00	5.0000	22.068	0.045315	-5.1945	-4.9679	-0.028269	0.047452	0.072006	1392.3	-0.49145
200.00	5.0000	20.446	0.048909	-1.7019	-1.4574	-0.0080546	0.043169	0.069609	1163.4	-0.47508
250.00	5.0000	18.707	0.053456	1.8087	2.0760	0.0077029	0.042245	0.072395	935.94	-0.37256
300.00	5.0000	16.644	0.060080	5.5592	5.8596	0.021479	0.042516	0.080087	701.33	-0.11429
350.00	5.0000	13.583	0.073622	10.000	10.369	0.035328	0.044139	0.10909	422.84	0.86866
150.00	10.000	22.132	0.045183	-5.2426	-4.7907	-0.028596	0.047587	0.071886	1408.1	-0.49341
200.00	10.000	20.543	0.048678	-1.7780	-1.2912	-0.0084436	0.043292	0.069265	1184.3	-0.48206
250.00	10.000	18.864	0.053012	1.6837	2.2138	0.0071898	0.042351	0.071522	965.13	-0.39347
300.00	10.000	16.938	0.059038	5.3293	5.9197	0.020687	0.042551	0.077434	746.52	-0.18740
350.00	10.000	14.430	0.069300	9.3887	10.082	0.033492	0.043442	0.091456	517.30	0.36632
150.00	20.000	22.256	0.044932	-5.3340	-4.4353	-0.029231	0.047854	0.071679	1438.4	-0.49677
200.00	20.000	20.726	0.048249	-1.9196	-0.95458	-0.0091833	0.043536	0.068683	1223.7	-0.49387
250.00	20.000	19.146	0.052230	1.4600	2.5046	0.0062486	0.042574	0.070163	1018.0	-0.42634
300.00	20.000	17.418	0.057411	4.9523	6.1005	0.019350	0.042705	0.074036	821.57	-0.28388
350.00	20.000	15.414	0.064877	8.6515	9.9491	0.031203	0.043268	0.080401	634.40	-0.00046299
200.00	50.000	21.201	0.047167	-2.2782	0.080117	-0.011163	0.044246	0.067514	1327.4	-0.51767
250.00	50.000	19.828	0.050433	0.93091	3.4526	0.0038870	0.043245	0.067745	1148.2	-0.48534
300.00	50.000	18.430	0.054259	4.1637	6.8767	0.016368	0.043321	0.069343	986.75	-0.42082
350.00	50.000	16.985	0.058876	7.4493	10.393	0.027205	0.043748	0.071325	844.44	-0.32515

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., and Span, R., "Short Fundamental Equations of State for 20 Industrial Fluids," *J. Chem. Eng. Data*, **51**(3):785–850, 2006. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The resulting equation has uncertainties of 0.1% in density in the liquid below 450 K, 1% in density at temperatures between 450 and 500 K, 3% in density at temperatures above 500 K, 0.5% in density in the vapor phase and at supercritical conditions below 10 MPa and 450 K, 0.5% in vapor pressure, and 2% in isobaric heat capacity.

TABLE 2-204 Saturated Cesium*

T , K	P , bar	v_f , m ³ /kg	v_g , m ³ /kg	h_f , kJ/kg	h_g , kJ/kg	s_f , kJ/(kg·K)	s_g , kJ/(kg·K)	c_{pf} , kJ/(kg·K)
301.6 ^m	2.66.-9	5.444.-4	7.01.+7	74.6	637.6	0.696	2.563	0.245
400	3.83.-6	5.615.-4	6.54.+4	98.5	651.9	0.765	2.148	0.240
500	3.11.-4	5.800.-4	1001	122.0	666.1	0.817	1.905	0.232
600	5.65.-3	5.999.-4	65.63	144.9	678.4	0.859	1.748	0.224
700	0.0440	6.215.-4	9.671	167.0	688.9	0.893	1.638	0.219
800	0.2029	6.443.-4	2.353	188.7	698.3	0.922	1.559	0.217
900	0.6620	6.689.-4	0.796	210.6	707.3	0.975	1.500	0.222
1000	1.693	6.954.-4	0.335	233.2	716.4	0.972	1.455	0.231
1200	6.790	7.628.-4	0.097	281.1	736.1	1.015	1.394	0.248
1500	27.6	8.84.-4	0.029	358.8	772.2	1.072	1.345	0.275

*Converted from tables in Vargaftik, *Tables of the Thermophysical Properties of Liquids and Gases*, Nauka, Moscow, 1972, and Hemisphere, Washington, 1975. m = melting point. The notation 2.66.-9 signifies 2.66×10^{-9} .

Many of the Vargaftik values also appear in Ohse, R. W., *Handbook of Thermodynamic and Transport Properties of Alkali Metals*, Blackwell Sci. Pubs., Oxford, 1985 (1020 pp.). This source contains superheat data.

Saturation and superheat tables and a diagram to 30 bar, 1550 K are given by Reynolds, W. C., *Thermodynamic Properties in S.I.*, Stanford Univ. publ., 1979 (173 pp.).

For a Mollier diagram from 0.1 to 327 psia, 1300–2700 °R, see Weatherford, W. D., J. C. Tyler, et al., WADD-TR-61-96, 1961.

An extensive review of properties of the solid and the saturated liquid was given by Alcock, C. B., M. W. Chase, et al., *J. Phys. Chem. Ref. Data*, **23**, 3 (1994): 385–497.

TABLE 2-205 Thermophysical Properties of Saturated Chlorine

$T, ^\circ\text{C}$	P, bar	$v_f, \text{m}^3/\text{kg}$	$v_g, \text{m}^3/\text{kg}$	$h_f, \text{kJ/kg}$	$h_g, \text{kJ/kg}$	$s_f, \text{kJ/(kg}\cdot\text{K)}$	$s_g, \text{kJ/(kg}\cdot\text{K)}$	$c_{pf}, \text{kJ/(kg}\cdot\text{K)}$	$c_{pg}, \text{kJ/(kg}\cdot\text{K)}$	$\mu_f, 10^{-6} \text{Pa}\cdot\text{s}$	$\mu_g, 10^{-6} \text{Pa}\cdot\text{s}$	$k_f, \text{W/(m}\cdot\text{K)}$	$k_g, \text{W/(m}\cdot\text{K)}$	Pr_f	Pr_g
-50	0.475	0.000 623	0.5448	221.5	518.2	1.7650	3.0946	0.9454	0.476	565	10.3	0.1684	0.0061	3.17	0.809
-40	0.773	0.000 634	0.3481	231.0	522.2	1.8074	3.0562	0.9474	0.484	520	10.8	0.1650	0.0065	2.99	0.815
-30	1.203	0.000 645	0.2314	240.6	526.1	1.8480	3.0223	0.9496	0.497	483	11.4	0.1613	0.0069	2.85	0.820
-20	1.802	0.000 656	0.1593	250.3	529.9	1.8869	2.9921	0.9520	0.513	452	11.9	0.1573	0.0074	2.74	0.826
-10	2.608	0.000 668	0.1134	260.0	533.9	1.9243	2.9649	0.9547	0.532	422	12.4	0.1527	0.0078	2.64	0.841
0	3.664	0.000 681	0.0829	269.7	537.4	1.9604	2.9402	0.9579	0.554	393	13.0	0.1478	0.0083	2.55	0.864
10	5.014	0.000 695	0.0619	279.4	540.5	1.9953	2.9177	0.9618	0.579	368	13.5	0.1427	0.0088	2.48	0.888
20	6.702	0.000 710	0.0471	289.2	543.3	2.0291	2.8924	0.9667	0.607	348	14.1	0.1378	0.0093	2.45	0.918
30	8.774	0.000 726	0.0364	299.0	545.7	2.0622	2.8777	0.9728	0.638	333	14.7	0.1327	0.0099	2.44	0.950
40	11.27	0.000 744	0.0286	308.8	548.0	2.0946	2.8593	0.9816	0.674	318	15.2	0.1282	0.0104	2.43	0.985
50	14.25	0.000 763	0.02276	318.6	549.8	2.1264	2.8417	0.9968	0.720	304	15.8	0.1230	0.0110	2.46	1.034
60	17.76	0.000 784	0.01827	329.1	551.2	2.1578	2.8245	1.022	0.786	290	16.4	0.1171	0.0117	2.53	1.107
70	21.85	0.000 808	0.01481	340.0	552.1	2.1892	2.8074	1.054	0.885	278	17.1	0.1122	0.0126	2.61	1.201
80	26.65	0.000 834	0.01202	351.4	552.5	2.2207	2.7900	1.124	1.017	267	17.9	0.1050	0.0137	2.85	1.331
90	32.17	0.000 865	0.00972	364.1	552.4	2.2528	2.7714	1.253	1.205	256	18.7	0.0986	0.0149	3.26	1.510
100	38.44	0.000 901	0.00789	377.8	551.0	2.2860	2.7502	1.418	1.434	247	19.5	0.0916	0.0163	3.82	1.700
110	45.54	0.000 956	0.00639	391.3	548.8	2.3207	2.7317	1.632	1.696	238	20.6	0.0850	0.0178	4.57	1.96
120	53.57	0.001 016	0.00508	407.1	543.7	2.3590	2.7064	1.891	1.960	230	22.2	0.0775	0.0195	5.61	2.23
130	62.68	0.001 121	0.00392	426.1	535.0	2.4032	2.6733								
140	72.84	0.001 335	0.00282	451.1	517.3	2.4595	2.6198								
144 ^c	77.10	0.001 77	0.00177	483.1	483.1	2.5365	2.5365								

^c = critical point.

Values interpolated and converted from Martin, J. J., 1977 (private communication), and from *Heat Exchanger Design Handbook*, vol. 5, Hemisphere, Washington, DC, 1983. Values of Ziegler, *Chem.-Ing.-Tech.*, **22** (1950): 229, apparently were also used in Landolt-Bornstein, **IVa**, (1967): 238–239, and in Ullmans *Enzyklopädie der technische Chemie*, 9, Verlag Chemie, Weinheim, 1975 (317–372).

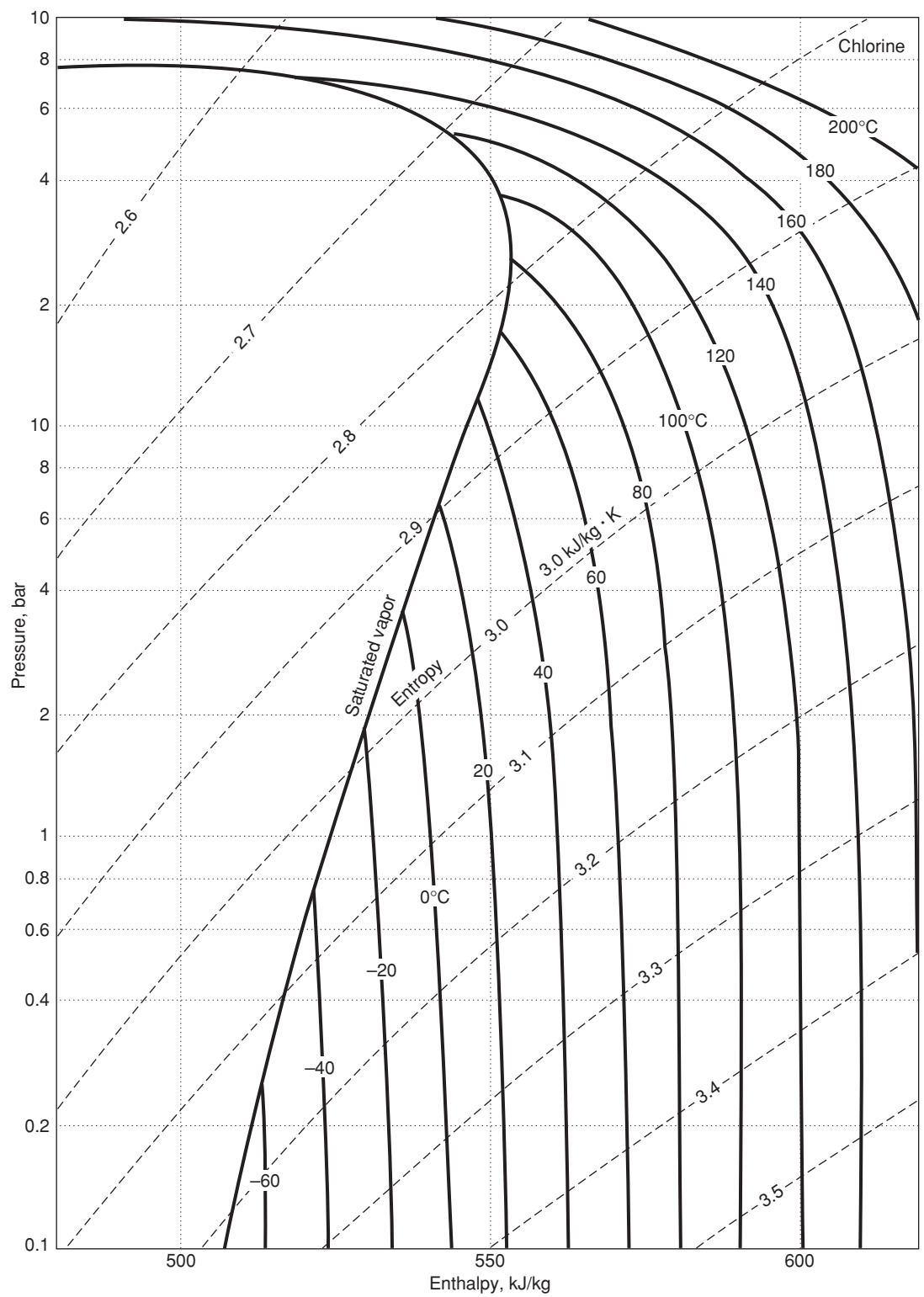


FIG. 2-9 Enthalpy-log-pressure diagram for chlorine.

TABLE 2-206 Saturated Chloroform (R20)

T , K	P , bar	v_f , m ³ /kg	v_g , m ³ /kg	h_f , kJ/kg	h_g , kJ/kg	s_f , kJ/(kg·K)	s_g , kJ/(kg·K)	c_{pf} , kJ/(kg·K)	μ_f , 10 ⁻⁶ Pa·s	k_f , W/(m·K)	Pr_f
280	0.115	0.000 660	1.689	-46.0	219.5	-0.165	0.798		748	0.120	
300	0.293	0.000 678	0.714	-32.6	230.6	-0.105	0.773		587	0.114	
320	0.620	0.000 695	0.358	-13.4	241.1	-0.041	0.754		468	0.109	
340	1.224	0.000 715	0.190	5.2	252.1	0.015	0.741		381	0.103	
360	2.255	0.000 739	0.107	23.3	263.0	0.065	0.731	1.03	319	0.095	3.35
380	3.830	0.000 765	0.0653	41.7	273.7	0.114	0.725	1.07	273	0.0921	3.17
400	6.039	0.000 795	0.0425	61.4	284.2	0.165	0.722	1.11	237	0.0863	3.04
420	9.058	0.000 822	0.0288	82.8	294.2	0.217	0.721	1.15	206	0.0808	2.93
440	13.39	0.000 871	0.0195	106.1	303.6	0.270	0.719	1.21	177	0.0750	2.86
460	18.80	0.000 921	0.0137	131.6	311.2	0.325	0.716	1.32	155	0.0694	2.95
480	26.00	0.000 980	0.00962	157.4	316.5	0.380	0.711	1.43	129.6	0.0641	2.89
500	34.66	0.001 059	0.00673	186.2	320.8	0.436	0.706	1.59	105.5	0.0584	2.87
520	44.68	0.001 193	0.00467	219.6	321.3	0.499	0.694		81.2	0.0518	
530	50.44	0.001 328	0.00359	242.7	315.7	0.540	0.678		67.7	0.0461	
536.6°	54.72	0.002 00	0.00200	284.1	284.1	0.602	0.602				

c = critical point. $h_f = s_f = 0$ at n.b.p., 334.5 K.

P , v , h , and s interpolated from Altunin, V. V., V. Z. Geller, et al., *Thermophysical Properties of Freons*, U.S.S.R. N.S.R.D.S. series, vol. 9., Hemisphere, 1987.

TABLE 2-207 Thermodynamic Properties of Cyclohexane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_p kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
279.47	0.0052538	9.4045	0.10633	-12.076	-12.075	-0.038084	0.094354	0.13918	1403.6	-0.50892
290.00	0.0089097	9.2862	0.10769	-10.567	-10.566	-0.032787	0.10358	0.14714	1314.7	-0.47482
300.00	0.014139	9.1736	0.10901	-9.0626	-9.0611	-0.027686	0.11089	0.15372	1245.3	-0.44719
310.00	0.021670	9.0604	0.11037	-7.4959	-7.4935	-0.022550	0.11715	0.15959	1185.7	-0.42239
320.00	0.032188	8.9462	0.11178	-5.8733	-5.8697	-0.017398	0.12261	0.16495	1132.8	-0.39928
330.00	0.046481	8.8309	0.11324	-4.1995	-4.1942	-0.012248	0.12749	0.16991	1084.4	-0.37708
340.00	0.065433	8.7142	0.11476	-2.4777	-2.4702	-0.0071081	0.13193	0.17461	1039.4	-0.35516
350.00	0.090023	8.5958	0.11634	-0.71047	-0.70000	-0.0019852	0.13604	0.17911	996.64	-0.33304
360.00	0.12131	8.4756	0.11799	1.1006	1.1149	0.0031169	0.13992	0.18349	955.45	-0.31028
370.00	0.16044	8.3533	0.11971	2.9544	2.9736	0.0081966	0.14362	0.18781	915.28	-0.28642
380.00	0.20862	8.2285	0.12153	4.8503	4.8756	0.013253	0.14720	0.19210	875.71	-0.26103
390.00	0.26711	8.1009	0.12344	6.7880	6.8210	0.018287	0.15069	0.19642	836.40	-0.23356
400.00	0.33727	7.9701	0.12547	8.7677	8.8100	0.023301	0.15412	0.20079	797.11	-0.20340
410.00	0.42046	7.8357	0.12762	10.790	10.843	0.028296	0.15751	0.20528	757.60	-0.16978
420.00	0.51814	7.6970	0.12992	12.855	12.922	0.033275	0.16088	0.20990	717.72	-0.13174
430.00	0.63180	7.5533	0.13239	14.964	15.048	0.038241	0.16425	0.21472	677.33	-0.087993
440.00	0.76300	7.4037	0.13507	17.119	17.222	0.043199	0.16762	0.21980	636.33	-0.036876
450.00	0.91333	7.2473	0.13798	19.321	19.447	0.048153	0.17101	0.22522	594.66	0.023882
460.00	1.0845	7.0828	0.14119	21.572	21.726	0.053108	0.17442	0.23106	552.32	0.097405
470.00	1.2781	6.9085	0.14475	23.876	24.061	0.058071	0.17788	0.23747	509.35	0.18807
480.00	1.4961	6.7225	0.14875	26.236	26.458	0.063050	0.18140	0.24464	465.88	0.30221
490.00	1.7402	6.5223	0.15332	28.655	28.922	0.068054	0.18499	0.25284	422.08	0.44929
500.00	2.0124	6.3048	0.15861	31.140	31.459	0.073094	0.18868	0.26259	378.12	0.64479
510.00	2.3150	6.0655	0.16487	33.699	34.081	0.078189	0.19249	0.27496	333.83	0.91793
520.00	2.6505	5.7961	0.17253	36.350	36.807	0.083372	0.19650	0.29284	287.99	1.3377
530.00	3.0222	5.4765	0.18260	39.135	39.687	0.088731	0.20083	0.32631	236.86	2.1175
540.00	3.4352	5.0307	0.19878	42.215	42.898	0.094586	0.20611	0.43566	171.37	4.2415
550.00	3.8928	4.1555	0.24065	46.176	47.113	0.10213	0.21549	0.97375	108.39	10.542
553.64	4.0750	3.2438	0.30828	48.978	50.235	0.10770			0	19.224
279.47	0.0052538	0.0022692	440.68	19.476	21.791	0.083095	0.088889	0.097282	173.20	37.820
290.00	0.0089097	0.0037161	269.10	20.430	22.828	0.082366	0.093915	0.10238	175.72	39.495
300.00	0.014139	0.0057156	174.96	21.380	23.854	0.082030	0.098579	0.10715	177.99	39.306
310.00	0.021670	0.0085063	117.56	22.370	24.917	0.082001	0.10318	0.11190	180.09	38.189
320.00	0.032188	0.012293	81.348	23.397	26.015	0.082241	0.10775	0.11667	182.00	36.613
330.00	0.046481	0.017306	57.785	24.459	27.145	0.082719	0.11233	0.12149	183.70	34.853
340.00	0.065433	0.023800	42.017	25.555	28.305	0.083406	0.11692	0.12641	185.14	33.070
350.00	0.090023	0.032056	31.195	26.684	29.492	0.084277	0.12155	0.13145	186.30	31.353
360.00	0.12131	0.042386	23.593	27.842	30.704	0.085309	0.12623	0.13663	187.14	29.750
370.00	0.16044	0.055130	18.139	29.029	31.939	0.086481	0.13097	0.14199	187.65	28.282
380.00	0.20862	0.070668	14.151	30.242	33.194	0.087775	0.13577	0.14756	187.78	26.956
390.00	0.26711	0.089425	11.183	31.480	34.467	0.089174	0.14063	0.15338	187.51	25.772
400.00	0.33727	0.11188	8.9379	32.739	35.754	0.090660	0.14557	0.15949	186.80	24.723
410.00	0.42046	0.13860	7.2152	34.019	37.053	0.092220	0.15059	0.16596	185.62	23.803
420.00	0.51814	0.17021	5.8751	35.315	38.359	0.093839	0.15569	0.17286	183.94	23.004
430.00	0.63180	0.20749	4.8195	36.625	39.670	0.095502	0.16088	0.18030	181.73	22.319
440.00	0.76300	0.25135	3.9785	37.945	40.980	0.097195	0.16615	0.18840	178.96	21.739
450.00	0.91333	0.30293	3.3011	39.270	42.285	0.098905	0.17153	0.19737	175.58	21.259
460.00	1.0845	0.36362	2.7501	40.597	43.579	0.10062	0.17701	0.20746	171.56	20.872
470.00	1.2781	0.43522	2.2977	41.918	44.855	0.10231	0.18259	0.21903	166.87	20.572
480.00	1.4961	0.52006	1.9228	43.228	46.105	0.10398	0.18828	0.23261	161.44	20.351
490.00	1.7402	0.62133	1.6095	44.519	47.319	0.10560	0.19406	0.24895	155.22	20.204
500.00	2.0124	0.74350	1.3450	45.779	48.486	0.10715	0.19990	0.26922	148.10	20.128
510.00	2.3150	0.89342	1.1193	46.995	49.586	0.10859	0.20573	0.29534	139.91	20.128
520.00	2.6505	1.0826	0.92368	48.145	50.593	0.10988	0.21142	0.33088	130.29	20.237
530.00	3.0222	1.3338	0.74972	49.186	51.452	0.11093	0.21669	0.38411	118.65	20.538
540.00	3.4352	1.7017	0.58765	50.020	52.039	0.11151	0.22086	0.48310	104.10	21.201
550.00	3.8928	2.3597	0.42378	50.334	51.984	0.11099	0.22221	0.96007	87.229	22.656
553.64	4.0750	3.2438	0.30828	48.978	50.235	0.10770			0	19.224

Single-Phase Properties

300.00	0.10000	9.1745	0.10900	-9.0661	-9.0552	-0.027698	0.11091	0.15371	1245.8	-0.44737
353.45	0.10000	8.5546	0.11690	-0.090837	-0.079147	-0.00022335	0.13740	0.18063	982.29	-0.32529
353.45	0.10000	0.035368	28.274	27.080	29.907	0.084616	0.12316	0.13322	186.62	30.786
400.00	0.10000	0.030817	32.450	33.301	36.546	0.10224	0.14262	0.15196	200.08	18.560
500.00	0.10000	0.024334	41.095	49.598	53.708	0.14037	0.18205	0.19081	224.91	8.2156
600.00	0.10000	0.020172	49.575	69.610	74.568	0.17831	0.21697	0.22554	246.67	4.7321
700.00	0.10000	0.017244	57.992	92.840	98.639	0.21535	0.24653	0.25502	266.50	3.1661
300.00	1.0000	9.1840	0.10888	-9.1021	-8.9932	-0.027818	0.11106	0.15360	1250.6	-0.44930
400.00	1.0000	7.9861	0.12522	8.7122	8.8374	0.023162	0.15416	0.20043	803.90	-0.20919
455.22	1.0000	7.1625	0.13962	20.490	20.630	0.050740	0.17279	0.22821	572.63	0.060445
455.22	1.0000	0.33339	2.9995	39.963	42.963	0.099799	0.17438	0.20248	173.56	21.046
500.00	1.0000	0.27733	3.6058	48.369	51.975	0.11868	0.18654	0.20332	198.76	11.678
600.00	1.0000	0.21435	4.6653	68.879	73.544	0.15793	0.21804	0.22955	233.28	5.2822
700.00	1.0000	0.17821	5.6112	92.287	97.898	0.19542	0.24682	0.25703	258.70	3.2649
300.00	5.0000	9.2252	0.10840	-9.2570	-8.7150	-0.028339	0.11167	0.15315	1272.4	-0.45730
400.00	5.0000	8.0768	0.12351	8.3977	9.0168	0.022365	0.15438	0.19856	843.04	-0.23916
500.00	5.0000	6.5774	0.15204	30.362	31.122	0.071493	0.18830	0.24756	459.37	0.29542
600.00	5.0000	1.8230	0.54855	63.154	65.897	0.13428	0.23018	0.32421	141.27	9.9456
700.00	5.0000	1.0620	0.94165	89.416	94.125	0.17787	0.24921	0.27380	220.29	3.9618
300.00	10.000	9.2742	0.10783	-9.4403	-8.3620	-0.028964	0.11229	0.15264	1300.3	-0.46622
400.00	10.000	8.1782	0.12228	8.0451	9.2679	0.021455	0.15456	0.19678	888.19	-0.26775
500.00	10.000	6.8741	0.14547	29.472	30.927	0.069619	0.18805	0.23749	554.89	0.059727
600.00	10.000	4.9059	0.20384	55.388	57.426	0.11776	0.22275	0.30052	265.77	1.5290
700.00	10.000	2.6733	0.37407	84.812	88.552	0.16576	0.25026	0.30095	217.61	2.8239
300.00	25.000	9.4073	0.10630	-9.9312	-7.2736	-0.030689	0.11356	0.15139	1386.2	-0.48739
400.00	25.000	8.4285	0.11864	7.1716	10.138	0.019116	0.15480	0.19342	1006.6	-0.32157
500.00	25.000	7.3968	0.13519	27.819	31.199	0.065971	0.18805	0.22750	739.78	-0.16655
600.00	25.000	6.2973	0.15880	51.613	55.583	0.11034	0.21982	0.25993	540.01	0.046982
700.00	25.000	5.1531	0.19406	78.272	83.123	0.15273	0.24808	0.28977	414.69	0.35809
400.00	75.000	8.9762	0.11141	5.2440	13.599	0.013443	0.15529	0.18906	1308.3	-0.39015
500.00	75.000	8.2410	0.12134	25.023	34.124	0.059111	0.18956	0.22091	1084.3	-0.31328
600.00	75.000	7.5637	0.13221	47.767	57.683	0.10198	0.22144	0.24976	931.19	-0.25806
700.00	75.000	6.9477	0.14393	73.174	83.969	0.14245	0.24950	0.27541	824.46	-0.21348

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Penoncello, S. G., Goodwin, A. R. H., and Jacobsen, R. T., "A Thermodynamic Property Formulation for Cyclohexane," *Int. J. Thermophys.* **16**(2):519–531, 1995. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties of the equation of state are 0.1% in density, 2% in heat capacity, and 1% in the speed of sound, except in the critical region.

TABLE 2-208 Thermodynamic Properties of Decane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _e kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
243.50	1.4042E-06	5.4064	0.18497	-69.252	-69.252	-0.20307	0.22393	0.28635	1468.5	-0.48339	144.08	2433.6
250.00	2.8673E-06	5.3702	0.18621	-67.383	-67.383	-0.19549	0.22704	0.28898	1438.4	-0.47764	142.30	2066.6
270.00	1.9993E-05	5.2597	0.19012	-61.516	-61.516	-0.17292	0.23712	0.29787	1349.4	-0.45826	136.88	1350.0
290.00	0.00010141	5.1502	0.19417	-55.461	-55.461	-0.15129	0.24780	0.30773	1265.6	-0.43680	131.57	957.69
310.00	0.00039985	5.0409	0.19838	-49.202	-49.202	-0.13043	0.25893	0.31832	1186.0	-0.41360	126.38	718.75
330.00	0.0012883	4.9312	0.20279	-42.725	-42.725	-0.11018	0.27034	0.32946	1109.9	-0.38881	121.32	561.90
350.00	0.0035240	4.8206	0.20744	-36.021	-36.020	-0.090462	0.28191	0.34100	1036.6	-0.36228	116.40	452.88
370.00	0.0084305	4.7082	0.21239	-29.084	-29.082	-0.071189	0.29354	0.35282	965.68	-0.33365	111.63	373.57
390.00	0.018060	4.5936	0.21770	-21.908	-21.905	-0.052306	0.30512	0.36483	896.48	-0.30225	107.04	313.65
410.00	0.035300	4.4757	0.22343	-14.493	-14.485	-0.033764	0.31659	0.37700	828.54	-0.26699	102.62	266.90
430.00	0.063919	4.3538	0.22968	-6.8339	-6.8192	-0.015527	0.32790	0.38931	761.40	-0.22626	98.409	229.40
450.00	0.10855	4.2266	0.23659	1.0694	1.0950	0.0024370	0.33901	0.40181	694.57	-0.17752	94.401	198.56
470.00	0.17465	4.0927	0.24434	9.2198	9.2624	0.020158	0.34991	0.41460	627.57	-0.11677	90.612	172.63
490.00	0.26846	3.9499	0.25317	17.622	17.690	0.037667	0.36059	0.42793	559.85	-0.037250	87.051	150.36
510.00	0.39696	3.7952	0.26349	26.283	26.388	0.054997	0.37108	0.44218	490.80	0.073195	83.726	130.83
530.00	0.56801	3.6240	0.27594	35.219	35.375	0.072193	0.38143	0.45818	419.70	0.23865	80.640	113.31
550.00	0.79054	3.4286	0.29166	44.457	44.687	0.089321	0.39176	0.47770	345.68	0.51353	77.794	97.183
570.00	1.0751	3.1943	0.31306	54.055	54.391	0.10649	0.40231	0.50540	267.65	1.0476	75.192	81.781
590.00	1.4353	2.8870	0.34638	64.149	64.646	0.12397	0.41372	0.55950	184.05	2.4223	72.934	66.154
610.00	1.8918	2.3754	0.42098	75.300	76.096	0.14276	0.42875	0.87382	91.074	9.9501	73.097	47.430
617.70	2.1014	1.6400	0.60976	82.386	83.668	0.15491			0	34.686		
243.50	1.4042E-06	6.9358E-07	1,441,800.	-15.410	-13.386	0.026360	0.19210	0.20042	121.84	320.14	6.4788	4.3408
250.00	2.8673E-06	1.3795E-06	724,920.	-14.150	-12.071	0.025751	0.19573	0.20405	123.41	287.01	6.7500	4.4520
270.00	1.9993E-05	8.9067E-06	112,280.	-10.121	-7.8760	0.025744	0.20739	0.21571	128.09	208.26	7.6856	4.7935
290.00	0.00010141	4.2071E-05	23,769.	-5.8536	-3.4432	0.028079	0.21963	0.22797	132.58	154.49	8.7681	5.1337
310.00	0.00039985	0.00015529	6,439.4	-1.3417	1.2331	0.032268	0.23228	0.24065	136.85	116.99	9.9889	5.4722
330.00	0.0012883	0.00047076	2,124.2	3.4167	6.1533	0.037934	0.24518	0.25363	140.87	90.369	11.339	5.8086
350.00	0.0035240	0.0012178	821.18	8.4178	11.312	0.044772	0.25823	0.26684	144.56	71.168	12.808	6.1419
370.00	0.0084305	0.0027701	361.00	13.654	16.697	0.052538	0.27134	0.28022	147.82	57.142	14.385	6.4713
390.00	0.018060	0.0056755	176.20	19.113	22.295	0.061027	0.28446	0.29376	150.52	46.790	16.056	6.7963
410.00	0.035300	0.010678	93.649	24.781	28.087	0.070070	0.29754	0.30748	152.52	39.094	17.808	7.1172
430.00	0.063919	0.018742	53.355	30.642	34.053	0.079524	0.31053	0.32142	153.66	33.355	19.629	7.4356
450.00	0.10855	0.031091	32.164	36.677	40.168	0.089267	0.32343	0.33567	153.78	29.089	21.506	7.7555
470.00	0.17465	0.049284	20.290	42.864	46.408	0.099191	0.33621	0.35042	152.66	25.969	23.438	8.0842
490.00	0.26846	0.075371	13.268	49.176	52.738	0.10919	0.34890	0.36600	150.07	23.781	25.431	8.4344
510.00	0.39696	0.11219	8.9134	55.580	59.118	0.11917	0.36154	0.38308	145.68	22.413	27.522	8.8280
530.00	0.56801	0.16399	6.0979	62.026	65.490	0.12901	0.37421	0.40300	139.04	21.855	29.799	9.3033
550.00	0.79054	0.23784	4.2045	68.438	71.762	0.13855	0.38707	0.42896	129.53	22.255	32.477	9.9327
570.00	1.0751	0.34741	2.8784	74.676	77.770	0.14751	0.40046	0.47018	116.14	24.065	36.104	10.874
590.00	1.4353	0.52650	1.8993	80.422	83.148	0.15533	0.41513	0.56612	97.344	28.492	42.347	12.566
610.00	1.8918	0.92143	1.0853	84.493	86.546	0.15989	0.43359	1.2127	71.032	37.614	59.648	17.240
617.70	2.1014	1.6400	0.60976	82.386	83.668	0.15491			0	34.686		

Single-Phase Properties

250.00	0.10000	5.3706	0.18620	-67.388	-67.369	-0.19551	0.22705	0.28898	1438.9	-0.47770	142.33	2069.3
350.00	0.10000	4.8213	0.20741	-36.029	-36.008	-0.090485	0.28192	0.34098	1037.4	-0.36251	116.44	453.43
446.75	0.10000	4.2477	0.23542	-0.23039	-0.20685	-0.00046204	0.33722	0.39976	705.42	-0.18612	95.037	203.20
446.75	0.10000	0.028735	34.801	35.686	39.166	0.087670	0.32134	0.33333	153.83	29.697	21.198	7.7032
450.00	0.10000	0.028470	35.125	36.739	40.251	0.090090	0.32313	0.33498	154.68	28.767	21.529	7.7604
550.00	0.10000	0.022477	44.491	71.828	76.277	0.16222	0.37514	0.38484	176.60	12.764	32.572	9.4815
650.00	0.10000	0.018766	53.287	111.67	117.00	0.23014	0.41928	0.42830	194.21	7.0015	44.613	11.158
250.00	1.0000	5.3745	0.18606	-67.431	-67.245	-0.19569	0.22712	0.28892	1443.8	-0.47822	142.62	2093.8
350.00	1.0000	4.8284	0.20711	-36.104	-35.897	-0.090699	0.28198	0.34075	1044.7	-0.36461	116.88	458.58
450.00	1.0000	4.2417	0.23575	0.92464	1.1604	0.0021144	0.33902	0.40097	706.99	-0.18764	95.062	201.71
550.00	1.0000	3.4433	0.29042	44.348	44.638	0.089120	0.39165	0.47560	353.85	0.47267	78.107	98.236
565.17	1.0000	3.2558	0.30715	51.697	52.004	0.10233	0.39972	0.49744	286.94	0.87812	75.797	85.478
565.17	1.0000	0.31644	3.1602	73.196	76.356	0.14542	0.39714	0.45776	119.81	23.453	35.085	10.603
650.00	1.0000	0.21839	4.5790	109.33	113.91	0.20734	0.42517	0.44558	167.63	8.8875	44.104	11.626
250.00	5.0000	5.3914	0.18548	-67.618	-66.691	-0.19644	0.22744	0.28869	1465.1	-0.48040	143.89	2204.8
350.00	5.0000	4.8586	0.20582	-36.424	-35.395	-0.091624	0.28223	0.33985	1075.7	-0.37303	118.77	481.47
450.00	5.0000	4.3034	0.23237	0.32894	1.4908	0.00076852	0.33912	0.39795	757.95	-0.22390	97.851	215.38
550.00	5.0000	3.6402	0.27471	42.808	44.181	0.086242	0.39052	0.45616	467.92	0.098466	82.872	114.20
650.00	5.0000	2.6391	0.37891	91.156	93.050	0.16772	0.43712	0.52513	213.89	1.6230	75.400	55.304
350.00	10.000	4.8938	0.20434	-36.797	-34.754	-0.092723	0.28255	0.33892	1112.2	-0.38180	121.04	510.17
450.00	10.000	4.3700	0.22883	-0.32038	1.9680	-0.00073280	0.33929	0.39532	813.41	-0.25552	101.03	231.67
550.00	10.000	3.7932	0.26363	41.527	44.163	0.083766	0.39008	0.44754	562.29	-0.060225	87.304	129.37
650.00	10.000	3.1242	0.32009	88.002	91.203	0.16225	0.43362	0.49145	369.51	0.31317	81.597	76.394
350.00	100.00	5.3085	0.18838	-41.033	-22.195	-0.10704	0.28785	0.33463	1562.8	-0.43121	152.54	1088.0
450.00	100.00	4.9747	0.20102	-6.2415	13.860	-0.016697	0.34377	0.38610	1360.6	-0.37226	139.48	498.37
550.00	100.00	4.6758	0.21387	33.447	54.834	0.065371	0.39337	0.43213	1216.3	-0.32986	130.26	308.65
650.00	100.00	4.4066	0.22693	77.332	100.03	0.14078	0.43497	0.47043	1111.9	-0.30079	124.76	219.90
550.00	300.00	5.3265	0.18774	28.160	84.482	0.047128	0.40016	0.43412	1858.1	-0.34462	192.00	698.31
650.00	300.00	5.1406	0.19453	71.521	129.88	0.12288	0.44112	0.47265	1766.6	-0.32003	188.95	480.95

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., and Span, R., "Short Fundamental Equations of State for 20 Industrial Fluids," *J. Chem. Eng. Data*, **51**(3):785–850, 2006. The source for viscosity is Huber, M. L., Laesecke, A., and Xiang, H. W., "Viscosity Correlations for Minor Constituent Fluids in Natural Gas: *n*-Octane, *n*-Nonane and *n*-decane," *Fluid Phase Equilibria* **224**:263–270, 2004. The source for thermal conductivity is Huber, M. L., and Perkins, R. A., "Thermal Conductivity Correlations for Minor Constituent Fluids in Natural Gas: *n*-Octane, *n*-Nonane and *n*-Decane," *Fluid Phase Equilibria* **227**:47–55, 2004.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in density are 0.05% in the saturated liquid density between 290 and 320 K, 0.2% in the liquid phase at temperatures to 400 K (with somewhat higher uncertainties above 100 MPa, up to 0.5%), 1% in the liquid phase up to 500 MPa, and 2% at higher temperatures as well as in the vapor phase. Vapor pressures have an uncertainty of 0.2%, and the uncertainties in liquid heat capacities and liquid sound speeds are 1%. The uncertainty in heat capacities may be higher at pressures above 10 MPa. The estimated uncertainty in viscosity is 1% along the saturated liquid line, 2% in compressed liquid to 200 MPa, 5% in vapor and supercritical regions. Uncertainty in thermal conductivity is 3%, except in the supercritical region and dilute gas which have an uncertainty of 5%.

TABLE 2-209 Thermodynamic Properties of Deuterium Oxide (Heavy Water)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
276.97	0.00066103	55.198	0.018117	-8.2294	-8.2294	-0.025460	0.084185	0.084334	1324.3	-0.22218	564.56	2085.6
280.00	0.00082243	55.214	0.018111	-7.9735	-7.9735	-0.024541	0.084523	0.084574	1332.9	-0.21843	569.58	1868.6
300.00	0.0030641	55.126	0.018140	-6.2759	-6.2758	-0.018685	0.084432	0.084957	1394.9	-0.19991	596.96	1046.6
320.00	0.0094511	54.780	0.018255	-4.5798	-4.5796	-0.013212	0.082466	0.084598	1436.7	-0.18770	615.88	688.90
340.00	0.025012	54.254	0.018432	-2.8930	-2.8926	-0.0080990	0.079894	0.084075	1455.5	-0.17806	628.04	497.65
360.00	0.058391	53.593	0.018659	-1.2168	-1.2157	-0.0033082	0.077173	0.083580	1455.6	-0.16908	634.49	381.94
380.00	0.12292	52.817	0.018933	0.45077	0.45310	0.0011999	0.074503	0.083248	1440.4	-0.15944	635.99	306.06
400.00	0.23743	51.937	0.019254	2.1139	2.1185	0.0054654	0.071990	0.083196	1413.3	-0.14815	633.11	253.36
420.00	0.42676	50.957	0.019624	3.7786	3.7870	0.0095268	0.069689	0.083502	1376.7	-0.13444	626.31	215.11
440.00	0.72190	49.875	0.020050	5.4521	5.4666	0.013420	0.067622	0.084212	1331.9	-0.11766	616.01	186.33
460.00	1.1598	48.687	0.020539	7.1417	7.1655	0.017176	0.065790	0.085360	1279.7	-0.097022	602.53	164.01
480.00	1.7833	47.386	0.021103	8.8551	8.8927	0.020824	0.064181	0.087004	1219.9	-0.071339	586.08	146.23
500.00	2.6406	45.961	0.021758	10.601	10.658	0.024390	0.062780	0.089266	1152.3	-0.038642	566.82	131.71
520.00	3.7850	44.390	0.022528	12.390	12.475	0.027902	0.061580	0.092393	1076.5	0.0043364	544.81	119.53
540.00	5.2758	42.641	0.023452	14.236	14.360	0.031394	0.060585	0.096848	991.90	0.063056	520.11	109.01
560.00	7.1787	40.660	0.024594	16.162	16.339	0.034909	0.059829	0.10351	897.91	0.14707	492.75	99.548
580.00	9.5679	38.357	0.026071	18.203	18.453	0.038511	0.059403	0.11428	793.19	0.27477	462.70	90.597
600.00	12.530	35.563	0.028119	20.425	20.777	0.042313	0.059524	0.13431	674.99	0.48803	430.22	81.477
620.00	16.171	31.867	0.031381	22.980	23.487	0.046577	0.060792	0.18616	537.16	0.91327	396.40	70.992
640.00	20.654	24.976	0.040039	26.745	27.572	0.052801	0.066781	0.82671	359.72	2.2860	420.92	53.746
643.89	21.660	17.875	0.055943	29.926	31.137	0.058280			0	3.7063		
276.97	0.00066103	0.00028721	3481.8	36.006	38.307	0.14256	0.025895	0.034265	389.85	394.78	16.529	9.6017
280.00	0.00082242	0.00035350	2828.9	36.083	38.409	0.14111	0.025952	0.034329	391.88	363.66	16.753	9.6779
300.00	0.0030641	0.0012303	812.82	36.589	39.079	0.13250	0.026375	0.034822	404.89	220.57	18.277	10.228
320.00	0.0094511	0.0035636	280.61	37.091	39.743	0.12530	0.026887	0.035451	417.19	143.76	19.890	10.840
340.00	0.025012	0.0089021	112.33	37.585	40.394	0.11922	0.027487	0.036239	428.74	100.20	21.616	11.492
360.00	0.058391	0.019720	50.709	38.064	41.025	0.11403	0.028182	0.037225	439.47	74.063	23.487	12.169
380.00	0.12292	0.039608	25.247	38.521	41.624	0.10955	0.029002	0.038484	449.26	57.443	25.539	12.856
400.00	0.23743	0.073418	13.621	38.946	42.180	0.10562	0.030006	0.040139	457.95	46.206	27.812	13.544
420.00	0.42676	0.12741	7.8485	39.329	42.679	0.10213	0.031275	0.042357	465.33	38.108	30.356	14.226
440.00	0.72190	0.20947	4.7740	39.660	43.106	0.098964	0.032903	0.045350	471.15	31.903	33.231	14.897
460.00	1.1598	0.32939	3.0359	39.925	43.447	0.096048	0.034981	0.049371	475.13	26.904	36.529	15.558
480.00	1.7833	0.49953	2.0019	40.116	43.686	0.093309	0.037576	0.054720	476.97	22.747	40.398	16.213
500.00	2.6406	0.73576	1.3591	40.219	43.808	0.090688	0.040723	0.061780	476.36	19.244	45.081	16.871
520.00	3.7850	1.0594	0.94392	40.219	43.792	0.088127	0.044416	0.071116	472.96	16.294	50.974	17.547
540.00	5.2758	1.5008	0.66632	40.096	43.612	0.085564	0.048619	0.083703	466.44	13.819	58.649	18.267
560.00	7.1787	2.1062	0.47479	39.815	43.224	0.082917	0.053289	0.10149	456.44	11.743	68.218	19.071
580.00	9.5679	2.9537	0.33856	39.318	42.557	0.080069	0.058406	0.12892	442.43	9.9759	79.570	20.029
600.00	12.530	4.1942	0.23842	38.492	41.479	0.076817	0.064014	0.17822	423.35	8.4154	98.667	21.290
620.00	16.171	6.2033	0.16120	37.081	39.688	0.072707	0.070293	0.29956	395.96	6.9363	133.97	23.259
640.00	20.654	11.096	0.090120	33.747	35.608	0.065358	0.077762	1.5264	343.38	5.1241	285.74	28.595
643.89	21.660	17.875	0.055943	29.926	31.137	0.058280			0	3.7063		

Single-Phase Properties

300.00	0.10000	55.129	0.018139	-6.2760	-6.2742	-0.018686	0.084425	0.084951	1395.1	-0.19991	597.01	1046.5
374.20	0.10000	53.053	0.018849	-0.032221	-0.030336	-0.000080962	0.075265	0.083320	1446.2	-0.16237	636.03	325.15
374.20	0.10000	0.032646	30.631	38.391	41.454	0.11078	0.028748	0.038084	446.53	61.573	24.923	12.656
400.00	0.10000	0.030411	32.883	39.139	42.427	0.11330	0.028403	0.037410	462.35	46.430	27.024	13.658
500.00	0.10000	0.024157	41.395	42.023	46.162	0.12163	0.029147	0.037685	515.85	19.445	36.485	17.747
600.00	0.10000	0.020087	49.784	45.017	49.996	0.12861	0.030636	0.039048	562.30	10.077	47.629	21.980
700.00	0.10000	0.017202	58.134	48.166	53.980	0.13475	0.032285	0.040652	604.21	6.0901	60.182	26.224
800.00	0.10000	0.015045	66.469	51.481	58.128	0.14029	0.033959	0.042306	642.79	4.1486	73.869	30.405
300.00	1.0000	55.152	0.018132	-6.2770	-6.2589	-0.018689	0.084361	0.084894	1397.1	-0.19988	597.49	1046.3
400.00	1.0000	51.959	0.019246	2.1086	2.1279	0.0054522	0.071968	0.083157	1415.4	-0.14835	633.54	253.63
453.53	1.0000	49.083	0.020374	6.5927	6.6131	0.015974	0.066357	0.084938	1297.4	-0.10418	607.22	170.66
453.53	1.0000	0.28575	3.4996	39.847	43.347	0.096969	0.034254	0.047940	474.06	28.417	35.408	15.345
500.00	1.0000	0.25193	3.9693	41.482	45.451	0.10139	0.032240	0.043385	503.79	19.634	38.239	17.445
600.00	1.0000	0.20474	4.8843	44.748	49.632	0.10902	0.031708	0.041087	556.10	10.106	48.345	21.886
700.00	1.0000	0.17382	5.7529	48.001	53.754	0.11537	0.032710	0.041565	600.64	6.1060	60.651	26.233
800.00	1.0000	0.15139	6.6052	51.363	57.969	0.12100	0.034141	0.042787	640.67	4.1567	74.254	30.468
300.00	5.0000	55.254	0.018098	-6.2816	-6.1911	-0.018705	0.084679	0.084647	1405.7	-0.19974	599.61	1045.2
400.00	5.0000	52.073	0.019204	2.0813	2.1773	0.0053836	0.071858	0.082958	1426.2	-0.14937	635.76	255.02
500.00	5.0000	46.085	0.021699	10.558	10.667	0.024304	0.062722	0.085880	1163.3	-0.042258	568.91	132.48
536.66	5.0000	42.947	0.023284	13.923	14.039	0.030810	0.060736	0.095981	1006.7	0.051819	524.42	110.68
536.66	5.0000	1.4172	0.70563	40.127	43.655	0.085995	0.047884	0.081310	467.76	14.202	57.219	18.142
600.00	5.0000	1.1342	0.88170	43.329	47.738	0.093213	0.038071	0.054713	524.33	9.9010	53.862	21.508
700.00	5.0000	0.91389	1.0942	47.210	52.681	0.10085	0.034830	0.046443	584.32	6.1037	63.474	26.310
800.00	5.0000	0.77903	1.2836	50.818	57.237	0.10693	0.034981	0.045153	631.66	4.1656	76.389	30.769
300.00	10.000	55.380	0.018057	-6.2874	-6.1068	-0.018725	0.083735	0.084346	1415.9	-0.19956	602.23	1043.9
400.00	10.000	52.214	0.019152	2.0479	2.2394	0.0052992	0.071726	0.082718	1439.3	-0.15059	638.50	256.74
500.00	10.000	46.341	0.021579	10.471	10.687	0.024128	0.062611	0.088119	1185.7	-0.049352	573.22	134.06
583.19	10.000	37.950	0.026351	18.544	18.807	0.039101	0.059377	0.11663	775.32	0.30140	457.66	89.173
583.19	10.000	3.1198	0.32053	39.212	42.417	0.079584	0.059266	0.13479	439.77	9.7157	81.935	20.204
600.00	10.000	2.7732	0.36059	40.719	44.325	0.082812	0.051641	0.097980	468.72	9.0626	72.595	21.210
700.00	10.000	1.9660	0.50864	46.072	51.158	0.093421	0.038049	0.054970	562.64	5.9486	69.164	26.519
800.00	10.000	1.6183	0.61793	50.085	56.265	0.10025	0.036102	0.048671	621.09	4.1215	80.100	31.206
300.00	50.000	56.338	0.017750	-6.3323	-5.4448	-0.018905	0.081303	0.082295	1484.2	-0.19786	622.24	1038.3
400.00	50.000	53.254	0.018778	1.8073	2.7462	0.0046700	0.070872	0.081058	1527.2	-0.15849	659.16	269.56
500.00	50.000	48.076	0.020801	9.8899	10.930	0.022920	0.062103	0.083801	1327.9	-0.088396	603.54	145.01
600.00	50.000	40.498	0.024693	18.559	19.794	0.039048	0.057208	0.096053	997.60	0.10313	493.46	98.391
700.00	50.000	26.703	0.037449	29.678	31.551	0.057059	0.055751	0.15684	588.37	1.0407	331.10	62.248
800.00	50.000	12.014	0.083233	42.333	46.495	0.077124	0.045204	0.10482	569.23	2.4839	169.04	40.739
300.00	100.00	57.439	0.017410	-6.3859	-4.6449	-0.019168	0.078994	0.080433	1546.7	-0.19571	645.56	1041.5
700.00	100.00	35.855	0.027890	25.888	28.677	0.050728	0.052858	0.095839	944.96	0.21724	435.02	84.790
800.00	100.00	26.322	0.037991	34.906	38.706	0.064110	0.045160	0.10162	785.38	0.69848	323.29	66.016

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Hill, P. G., MacMillan, R. D. C., and Lee, V., "A Fundamental Equation of State for Heavy Water," *J. Phys. Chem. Ref. Data*, **11**(1):1-14, 1982. The source for viscosity is International Association for the Properties of Water and Steam, "Viscosity and Thermal Conductivity of Heavy Water Substance," Physical Chemistry of Aqueous Systems: Proceedings of the 12th International Conference on the Properties of Water and Steam, Orlando, Fla., Sept. 11-16, 1994, A107-A138. The source for thermal conductivity is International Association for the Properties of Water and Steam, "Viscosity and Thermal Conductivity of Heavy Water Substance," Physical Chemistry of Aqueous Systems: Proceedings of the 12th International Conference on the Properties of Water and Steam, Orlando, Fla., Sept. 11-16, 1994, A107-A138.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

For a discussion of the uncertainties associated with the equation of state and thermal conductivity entries of this table, please see the source references given above. The uncertainty in viscosity is 1% in the liquid below 474 K, 2% in the liquid at higher temperatures and in the vapor, and 5% between 623 and 723 K at pressures between 16 and 50 MPa. The uncertainty in viscosity is 2% in the liquid below 623 K and in the vapor below 573 K, 5% elsewhere in the liquid and vapor, and 10% in the critical region (623 to 723 K and 21.66 to 50 MPa).

TABLE 2-210 Thermodynamic Properties of 2,2-Dimethylpropane (Neopentane)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
256.60	0.035401	8.7017	0.11492	-4.0448	-4.0407	-0.014961	0.11106	0.14987	1061.0	-0.45488
260.00	0.041178	8.6547	0.11554	-3.5333	-3.5286	-0.012981	0.11212	0.15116	1043.4	-0.44532
270.00	0.062611	8.5146	0.11745	-2.0036	-1.9963	-0.0072081	0.11525	0.15504	992.29	-0.41587
280.00	0.091973	8.3711	0.11946	-0.43520	-0.42422	-0.0015039	0.11839	0.15903	941.76	-0.38403
290.00	0.13106	8.2236	0.12160	1.1729	1.1889	0.0041396	0.12153	0.16315	891.68	-0.34919
300.00	0.18184	8.0715	0.12389	2.8218	2.8444	0.0097305	0.12467	0.16742	841.93	-0.31054
310.00	0.24636	7.9140	0.12636	4.5127	4.5439	0.015276	0.12781	0.17188	792.35	-0.26700
320.00	0.32680	7.7503	0.12903	6.2470	6.2892	0.020784	0.13096	0.17655	742.82	-0.21713
330.00	0.42545	7.5792	0.13194	8.0264	8.0826	0.026263	0.13411	0.18151	693.20	-0.15892
340.00	0.54466	7.3993	0.13515	9.8530	9.9266	0.031720	0.13728	0.18683	643.34	-0.089543
350.00	0.68688	7.2090	0.13872	11.729	11.825	0.037165	0.14047	0.19263	593.09	-0.0048767
360.00	0.85466	7.0060	0.14273	13.659	13.781	0.042610	0.14369	0.19910	542.29	0.10132
370.00	1.0507	6.7873	0.14733	15.647	15.802	0.048067	0.14696	0.20651	490.74	0.23893
380.00	1.2778	6.5488	0.15270	17.699	17.894	0.053555	0.15032	0.21536	438.21	0.42456
390.00	1.5390	6.2842	0.15913	19.824	20.069	0.059099	0.15380	0.22655	384.34	0.68847
400.00	1.8380	5.9837	0.16712	22.038	22.345	0.064738	0.15752	0.24199	328.55	1.0924
410.00	2.1788	5.6295	0.17764	24.368	24.755	0.070543	0.16164	0.26680	269.76	1.7855
420.00	2.5668	5.1798	0.19306	26.878	27.373	0.076679	0.16662	0.32099	205.53	3.2489
430.00	3.0111	4.4559	0.22442	29.850	30.525	0.083875	0.17440	0.65083	129.02	8.3551
433.74	3.1963	3.2700	0.30581	32.506	33.484	0.090610			0	19.128
256.60	0.035401	0.016951	58.993	18.019	20.107	0.079147	0.097392	0.10663	176.08	42.543
260.00	0.041178	0.019508	51.262	18.336	20.447	0.079232	0.098838	0.10818	176.77	40.673
270.00	0.062611	0.028804	34.717	19.288	21.461	0.079672	0.10307	0.11278	178.55	35.912
280.00	0.091973	0.041222	24.259	20.267	22.498	0.080361	0.10727	0.11744	179.94	32.069
290.00	0.13106	0.057423	17.415	21.271	23.554	0.081260	0.11146	0.12220	180.88	28.950
300.00	0.18184	0.078145	12.797	22.298	24.625	0.082334	0.11562	0.12709	181.34	26.412
310.00	0.24636	0.10423	9.5944	23.346	25.710	0.083553	0.11978	0.13214	181.26	24.346
320.00	0.32680	0.13664	7.3187	24.411	26.803	0.084890	0.12394	0.13741	180.58	22.672
330.00	0.42545	0.17651	5.6655	25.490	27.901	0.086318	0.12810	0.14299	179.24	21.330
340.00	0.54466	0.22521	4.4402	26.580	28.999	0.087815	0.13228	0.14898	177.18	20.277
350.00	0.68688	0.28448	3.5152	27.676	30.091	0.089355	0.13648	0.15554	174.30	19.485
360.00	0.85466	0.35651	2.8049	28.773	31.170	0.090912	0.14073	0.16293	170.51	18.940
370.00	1.0507	0.44433	2.2506	29.862	32.226	0.092458	0.14505	0.17158	165.68	18.642
380.00	1.2778	0.55219	1.8110	30.932	33.246	0.093957	0.14948	0.18223	159.65	18.607
390.00	1.5390	0.68654	1.4566	31.969	34.211	0.095360	0.15407	0.19633	152.23	18.876
400.00	1.8380	0.85798	1.1655	32.946	35.088	0.096596	0.15892	0.21707	143.15	19.520
410.00	2.1788	1.0863	0.92056	33.817	35.822	0.097537	0.16422	0.25302	132.06	20.655
420.00	2.5668	1.4175	0.70547	34.476	36.287	0.097901	0.17039	0.33783	118.40	22.431
430.00	3.0111	2.0429	0.48949	34.498	35.972	0.096542	0.17895	0.86590	100.88	24.389
433.74	3.1963	3.2700	0.30581	32.506	33.484	0.090610			0	19.128

Single-Phase Properties

300.00	0.10000	0.041573	24.054	22.482	24.888	0.087934	0.11446	0.12428	186.71	24.744
350.00	0.10000	0.035109	28.482	28.668	31.516	0.10834	0.13169	0.14083	203.23	14.420
400.00	0.10000	0.030488	32.800	35.683	38.963	0.12820	0.14811	0.15694	217.95	9.4234
450.00	0.10000	0.026981	37.063	43.493	47.199	0.14758	0.16371	0.17238	231.47	6.6348
500.00	0.10000	0.024215	41.296	52.060	56.190	0.16652	0.17858	0.18715	244.10	4.9220
550.00	0.10000	0.021973	45.510	61.352	65.903	0.18502	0.19273	0.20124	256.02	3.7932
300.00	1.0000	8.0927	0.12357	2.7638	2.8874	0.0095365	0.12470	0.16698	852.01	-0.31857
350.00	1.0000	7.2257	0.13839	11.687	11.825	0.037043	0.14045	0.19204	599.54	-0.017539
367.56	1.0000	6.8424	0.14615	15.155	15.301	0.046731	0.14616	0.20459	503.42	0.20165
367.56	1.0000	0.42120	2.3741	29.597	31.971	0.092083	0.14399	0.16932	166.96	18.691
400.00	1.0000	0.35635	2.8062	34.611	37.418	0.10628	0.15145	0.16863	188.60	12.099
450.00	1.0000	0.29713	3.3655	42.712	46.077	0.12667	0.16541	0.17859	211.98	7.6415
500.00	1.0000	0.25832	3.8712	51.444	55.315	0.14613	0.17958	0.19109	230.27	5.3700
550.00	1.0000	0.22988	4.3501	60.842	65.192	0.16494	0.19337	0.20397	245.89	4.0038
300.00	5.0000	8.1890	0.12211	2.4999	3.1104	0.0086423	0.12485	0.16516	897.98	-0.35167
350.00	5.0000	7.4104	0.13495	11.213	11.887	0.035660	0.14039	0.18654	671.51	-0.13585
400.00	5.0000	6.4044	0.15614	21.099	21.879	0.062302	0.15611	0.21533	444.29	0.37314
450.00	5.0000	4.6105	0.21690	33.057	34.142	0.091100	0.17461	0.30156	198.33	3.5214
500.00	5.0000	1.9914	0.50215	47.248	49.759	0.12411	0.18590	0.24909	166.16	7.2775
550.00	5.0000	1.4527	0.68839	57.986	61.428	0.14637	0.19664	0.22704	205.52	4.5988
300.00	10.000	8.2958	0.12054	2.2072	3.4126	0.0076277	0.12506	0.16345	949.31	-0.38267
350.00	10.000	7.5920	0.13172	10.739	12.056	0.034237	0.14048	0.18250	743.85	-0.22237
400.00	10.000	6.7750	0.14760	20.216	21.692	0.059940	0.15568	0.20328	555.11	0.041740
450.00	10.000	5.7764	0.17312	30.691	32.422	0.085190	0.17074	0.22624	391.52	0.55851
500.00	10.000	4.5715	0.21875	42.114	44.301	0.11020	0.18530	0.24770	276.41	1.5460
550.00	10.000	3.4249	0.29198	53.968	56.887	0.13419	0.19832	0.25208	234.55	2.4058
400.00	100.00	8.5990	0.11629	15.633	27.262	0.045388	0.15862	0.18487	1257.1	-0.42302
450.00	100.00	8.2554	0.12113	24.735	36.848	0.067954	0.17287	0.19848	1172.6	-0.38739
500.00	100.00	7.9290	0.12612	34.489	47.101	0.089546	0.18665	0.21152	1103.1	-0.35738
550.00	100.00	7.6203	0.13123	44.867	57.990	0.11029	0.19993	0.22395	1046.3	-0.33233
550.00	200.00	8.6161	0.11606	42.742	65.955	0.10250	0.20230	0.22337	1444.9	-0.36818

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., and Span, R., "Short Fundamental Equations of State for 20 Industrial Fluids," *J. Chem. Eng. Data*, **51**(3):785–850, 2006. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in density in the equation of state range from 0.2% in the liquid phase at pressures less than 10 MPa to 1% in the liquid phase at higher pressures (up to 200 MPa) and at temperatures above the critical point (up to 550 K). The uncertainty in density in the vapor phase is 0.5%. Uncertainties in other properties are 0.1% for the vapor pressure, 2% in liquid-phase heat capacities, 0.5% in vapor-phase heat capacities, 1% for liquid-phase sound speeds, and 0.02% for vapor-phase sound speeds.

TABLE 2-211 Saturated Diphenyl*

T , K	P , bar	v_f , m ³ /kg	v_g , m ³ /kg	h_f , kJ/kg	h_g , kJ/kg	s_f , kJ/(kg·K)	s_g , kJ/(kg·K)	c_{pf} , kJ/(kg·K)	μ_f , 10 ⁻⁴ Pa·s	k_f , W/(m·K)
343	0.0010	1.010.-3	252.5	0.0	444.2	0.000	1.298	1.760	15.0	0.139
350	0.0016	1.014.-3	156.1	13.0	444.2	0.036	1.266	1.782	13.5	0.138
360	0.0029	1.021.-3	85.0	30.0	446.7	0.084	1.236	1.813	11.7	0.136
370	0.0049	1.030.-3	49.9	47.2	449.7	0.130	1.213	1.844	10.3	0.135
380	0.0064	1.037.-3	29.9	65.0	454.5	0.178	1.200	1.875	9.1	0.133
390	0.0129	1.046.-3	18.3	82.7	462.7	0.224	1.194	1.906	8.1	0.132
400	0.0200	1.054.-3	11.7	99.3	461.2	0.273	1.202	1.936	7.3	0.130
420	0.0432	1.072.-3	5.84	139.9	499.0	0.358	1.228	1.998	6.0	0.127
440	0.0879	1.092.-3	3.021	180.3	532.4	0.451	1.267	2.060	5.0	0.125
460	0.1694	1.112.-3	1.652	222.7	569.7	0.545	1.378	2.122	4.3	0.122
480	0.3112	1.132.-3	0.9594	267.6	611.6	0.652	1.367	2.184	3.7	0.119
500	0.5218	1.154.-3	0.4452	314.9	651.8	0.746	1.424	2.246	3.3	0.116
520	0.8375	1.177.-3	0.3652	361.5	687.8	0.824	1.477	2.308	2.7	0.113
540	1.290	1.204.-3	0.2261	404.5	723.8	0.915	1.529	2.370	2.4	0.110
560	1.941	1.230.-3	0.1447	457.2	762.7	1.032	1.582	2.432	2.2	0.107
580	2.818	1.258.-3	0.0977	522.3	801.7	1.125	1.635	2.494	1.90	0.105
600	3.926	1.291.-3	0.0685	563.7	842.4	1.223	1.688	2.556	1.71	0.102
620	5.408	1.326.-3	0.0504	630.4	886.4	1.316	1.740	2.618	1.54	0.099
640	7.328	1.366.-3	0.0381	689.1	930.9	1.375	1.748	2.680	1.39	0.096
660	9.572	1.412.-3	0.0301	745.9	977.1	1.457	1.791	2.741	1.24	0.093
680	12.05	1.465.-3	0.0236	802.8	1024.9	1.585	1.856	2.803	1.10	0.090
700	15.21	1.529.-3	0.0186	860.1	1073.1	1.663	1.951	2.865	0.97	0.087
720	19.14	1.56.-3	0.0147	917.5	1116.7	1.746	2.003	2.93		
740	23.93	1.70.-3	0.0113	975.2	1152.8	1.822	2.058	3.00		
760	28.71	1.95.-3	0.0085	1033.1	1182.5	1.901	2.099			
780	34.83	2.16.-3	0.0058	1091.2	1163.0	1.977	2.107			
800	42.46	3.18.-3	0.0032	1148.4	1148.4	2.047	2.047			

*Interpolated by P. E. Liley from the Landolt-Börnstein band IVa, p. 557, 1967 tables based on *Technical Data on Fuel*, British National Committee, World Energy Conference, London.

TABLE 2-212 Thermodynamic Properties of Dodecane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_p kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
263.60	6.2621E-07	4.5291	0.22079	-96.621	-96.621	-0.26003	0.28923	0.35985	1427.2	-0.45503	143.93	2892.5
270.00	1.2617E-06	4.5008	0.22218	-94.309	-94.309	-0.25136	0.29275	0.36265	1398.7	-0.45038	142.27	2447.1
290.00	8.8545E-06	4.4131	0.22660	-86.960	-86.960	-0.22511	0.30440	0.37246	1313.5	-0.43398	137.24	1578.5
310.00	4.5799E-05	4.3261	0.23115	-79.402	-79.402	-0.19991	0.31679	0.38355	1233.4	-0.41526	132.45	1112.9
330.00	0.00018556	4.2393	0.23589	-71.612	-71.612	-0.17556	0.32968	0.39559	1157.5	-0.39464	127.86	832.98
350.00	0.00061679	4.1522	0.24084	-63.574	-63.574	-0.15192	0.34285	0.40831	1085.1	-0.37230	123.43	650.53
370.00	0.0017432	4.0641	0.24605	-55.276	-55.276	-0.12887	0.35616	0.42150	1015.6	-0.34817	119.12	524.02
390.00	0.0043088	3.9747	0.25159	-46.712	-46.711	-0.10633	0.36947	0.43503	948.42	-0.32197	114.92	431.83
410.00	0.0095271	3.8833	0.25751	-37.875	-37.872	-0.084231	0.38269	0.44878	883.07	-0.29316	110.81	361.82
430.00	0.019190	3.7893	0.26390	-28.762	-28.757	-0.062532	0.39574	0.46268	819.18	-0.26089	106.77	306.78
450.00	0.035739	3.6920	0.27085	-19.371	-19.361	-0.041187	0.40859	0.47672	756.37	-0.22391	102.79	262.21
470.00	0.062299	3.5907	0.27850	-9.6999	-9.6825	-0.020162	0.42118	0.49091	694.32	-0.18039	98.863	225.20
490.00	0.10268	3.4842	0.28701	0.25270	0.28217	0.00057539	0.43351	0.50530	632.70	-0.12755	94.976	193.79
510.00	0.16136	3.3713	0.29662	10.490	10.538	0.021053	0.44557	0.52003	571.20	-0.061022	91.129	166.63
530.00	0.24347	3.2503	0.30766	21.016	21.091	0.041300	0.45738	0.53534	509.51	0.026306	87.324	142.76
550.00	0.35486	3.1189	0.32063	31.839	31.953	0.061352	0.46897	0.55164	447.33	0.14688	83.568	121.46
570.00	0.50211	2.9736	0.33629	42.975	43.144	0.081249	0.48040	0.56974	384.30	0.32430	79.876	102.19
590.00	0.69279	2.8091	0.35598	54.448	54.695	0.10105	0.49179	0.59128	320.04	0.60884	76.283	84.516
610.00	0.93585	2.6161	0.38225	66.310	66.667	0.12086	0.50334	0.62041	253.82	1.1288	72.856	68.050
630.00	1.2426	2.3747	0.42111	78.676	79.199	0.14087	0.51558	0.67253	183.58	2.3418	69.764	52.302
650.00	1.6298	2.0078	0.49805	92.046	92.858	0.16193	0.53081	0.91317	100.02	8.0937	68.073	35.652
658.10	1.8176	1.3300	0.75188	101.36	102.72	0.17683			0	40.219		
263.60	6.2621E-07	2.8572E-07	3,499,900.	-33.699	-31.507	-0.013011	0.24443	0.25274	115.35	330.02	6.7106	3.8271
270.00	1.2617E-06	5.6205E-07	1,779,200.	-32.120	-29.875	-0.012718	0.24903	0.25734	116.70	296.37	7.0413	3.9297
290.00	8.8545E-06	3.6724E-06	272,300.	-26.993	-24.582	-0.010011	0.26382	0.27214	120.83	215.32	8.1425	4.2511
310.00	4.5799E-05	1.7772E-05	56,267.	-21.566	-18.989	-0.0050291	0.27906	0.28739	124.81	160.20	9.3434	4.5733
330.00	0.00018556	6.7670E-05	14,778.	-15.834	-13.092	0.0017708	0.29454	0.30288	128.63	121.85	10.640	4.8958
350.00	0.00061679	0.00021228	4,710.8	-9.7983	-6.8927	0.010028	0.31008	0.31848	132.26	94.623	12.030	5.2183
370.00	0.0017432	0.00056862	1,758.6	-3.4632	-0.39757	0.019454	0.32559	0.33408	135.66	74.946	13.512	5.5399
390.00	0.0043088	0.0013382	747.29	3.1615	6.3814	0.029808	0.34098	0.34965	138.74	60.510	15.090	5.8599
410.00	0.0095271	0.0028309	353.24	10.063	13.428	0.040891	0.35619	0.36515	141.40	49.785	16.769	6.1775
430.00	0.019190	0.0054855	182.30	17.224	20.723	0.052536	0.37119	0.38060	143.51	41.738	18.561	6.4920
450.00	0.035739	0.0098881	101.13	24.628	28.242	0.064598	0.38596	0.39603	144.95	35.662	20.479	6.8034
470.00	0.062299	0.016797	59.534	32.254	35.962	0.076955	0.40048	0.41151	145.55	31.061	22.543	7.1122
490.00	0.10268	0.027184	36.786	40.078	43.855	0.089500	0.41475	0.42715	145.16	27.591	24.775	7.4202
510.00	0.16136	0.042311	23.635	48.075	51.889	0.10213	0.42878	0.44316	143.58	25.017	27.206	7.7302
530.00	0.24347	0.063865	15.658	56.215	60.027	0.11477	0.44259	0.45987	140.61	23.184	29.871	8.0477
550.00	0.35486	0.094219	10.614	64.460	68.226	0.12730	0.45625	0.47792	135.98	22.013	32.819	8.3808
570.00	0.50211	0.13692	7.3034	72.758	76.425	0.13964	0.46984	0.49854	129.36	21.503	36.118	8.7440
590.00	0.69279	0.19777	5.0563	81.036	84.539	0.15163	0.48353	0.52443	120.31	21.771	39.870	9.1635
610.00	0.93585	0.28746	3.4787	89.170	92.425	0.16308	0.49764	0.56276	108.12	23.181	44.272	9.6956
630.00	1.2426	0.43012	2.3249	96.902	99.791	0.17355	0.51290	0.64061	91.557	26.817	49.828	10.493
650.00	1.6298	0.71569	1.3973	103.34	105.62	0.18156	0.53205	1.0334	67.007	37.545	60.081	12.293
658.10	1.8176	1.3300	0.75188	101.36	102.72	0.17683			49.653	0		

TABLE 2-212 Thermodynamic Properties of Dodecane (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_v kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Single-Phase Properties												
300.00	0.10000	4.3700	0.22883	-83.216	-83.193	-0.21241	0.31053	0.37786	1273.5	-0.42497	134.86	1315.3
400.00	0.10000	3.9300	0.25445	-42.339	-42.313	-0.095254	0.37610	0.44184	916.43	-0.30827	112.91	395.13
488.89	0.10000	3.4902	0.28651	-0.30755	-0.27889	-0.00056937	0.43283	0.50449	636.11	-0.13079	95.191	195.41
488.89	0.10000	0.026496	37.741	39.638	43.413	0.088800	0.41397	0.42628	145.21	27.759	24.647	7.4030
500.00	0.10000	0.025735	38.857	44.299	48.185	0.098451	0.42081	0.43268	147.79	25.222	25.670	7.5935
600.00	0.10000	0.020711	48.284	89.425	94.254	0.18229	0.47769	0.48757	167.29	12.534	35.525	9.2766
700.00	0.10000	0.017494	57.163	139.71	145.42	0.26108	0.52517	0.53435	183.12	7.4958	46.039	10.929
300.00	1.0000	4.3739	0.22863	-83.277	-83.048	-0.21262	0.31061	0.37775	1279.2	-0.42580	135.18	1330.4
400.00	1.0000	3.9374	0.25398	-42.444	-42.190	-0.095518	0.37615	0.44147	924.90	-0.31138	113.40	400.40
500.00	1.0000	3.4447	0.29030	5.1293	5.4196	0.010431	0.43951	0.51125	616.23	-0.11178	93.839	183.76
600.00	1.0000	2.7332	0.36588	60.165	60.531	0.11067	0.49726	0.60093	296.10	0.75711	74.943	77.601
614.58	1.0000	2.5662	0.38968	69.090	69.480	0.12541	0.50605	0.62919	238.23	1.3130	72.109	64.404
614.58	1.0000	0.31406	3.1841	90.990	94.175	0.16559	0.50099	0.57515	104.78	23.748	45.410	9.8451
700.00	1.0000	0.21164	4.7249	136.66	141.39	0.23754	0.53113	0.55438	152.31	9.1496	51.144	11.188
300.00	5.0000	4.3910	0.22774	-83.541	-82.402	-0.21351	0.31099	0.37733	1303.7	-0.42924	136.59	1398.2
400.00	5.0000	3.9683	0.25200	-42.890	-41.630	-0.096648	0.37639	0.44002	960.72	-0.32355	115.50	423.55
500.00	5.0000	3.5099	0.28491	4.2808	5.7054	0.0087025	0.43938	0.50646	674.60	-0.16525	97.177	201.12
600.00	5.0000	2.9472	0.33930	57.896	59.593	0.10677	0.49475	0.57114	420.49	0.19333	81.179	99.888
700.00	5.0000	2.1498	0.46517	117.69	120.02	0.19979	0.54281	0.63863	215.10	1.5765	70.846	42.370
300.00	10.000	4.4115	0.22668	-83.856	-81.590	-0.21459	0.31147	0.37689	1333.1	-0.43298	138.29	1484.7
400.00	10.000	4.0039	0.24976	-43.403	-40.906	-0.097972	0.37674	0.43856	1001.9	-0.33581	117.98	452.04
500.00	10.000	3.5776	0.27952	3.3852	6.1804	0.0068314	0.43944	0.50252	735.90	-0.20845	100.85	221.12
600.00	10.000	3.0995	0.32263	56.144	59.370	0.10365	0.49390	0.55974	517.52	-0.0053841	86.682	119.96
700.00	10.000	2.5593	0.39074	113.82	117.72	0.19351	0.53945	0.60482	359.64	0.33979	77.506	65.788
400.00	100.00	4.3951	0.22753	-48.840	-26.088	-0.11422	0.38400	0.43214	1472.8	-0.39899	149.79	967.99
500.00	100.00	4.1426	0.24140	-4.0985	20.041	-0.011523	0.44564	0.48959	1297.7	-0.35002	140.67	510.30
600.00	100.00	3.9126	0.25559	46.019	71.578	0.082301	0.49884	0.53969	1169.1	-0.31375	134.18	334.45
700.00	100.00	3.7019	0.27013	100.67	127.69	0.16871	0.54304	0.58105	1073.9	-0.28792	129.56	242.80
600.00	300.00	4.4838	0.22302	40.123	107.03	0.062648	0.50752	0.54106	1807.8	-0.33126	190.55	737.48
700.00	300.00	4.3441	0.23020	94.208	163.27	0.14926	0.55080	0.58241	1725.5	-0.31036	190.65	531.88

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., and Huber, M. L., "Thermodynamic Properties of *n*-Dodecane," *Energy & Fuels*, **18**:960–967, 2004. The source for viscosity and thermal conductivity is Huber, M. L., Laesecke, A., and Perkins, R. A., "Transport Properties of *n*-Dodecane," *Energy & Fuels* **18**: 968–975, 2004.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties (where the uncertainties can be considered as estimates of a combined expanded uncertainty with a coverage factor of 2) of density values calculated using the equation of state in the liquid phase (including at saturation) are 0.2% for pressures less than 200 MPa and 0.5% for higher pressures. The uncertainty for heat capacities is 1%, and that for sound speeds is 0.5%. The estimated uncertainties of vapor pressures calculated using the Maxwell criterion are 0.2% for temperatures above 350 K and approach 5% as the temperature decreases to the triple point temperature. These estimated uncertainties for calculated properties are consistent with the experimental accuracies of the various available experimental data. The estimated uncertainty in viscosity is 0.5% along the saturated-liquid line, 2% in compressed liquid to 200 MPa, 5% in vapor and supercritical regions. Uncertainty in thermal conductivity is 3%, except in the supercritical region and dilute gas which have an uncertainty of 5%.

TABLE 2-213 Thermodynamic Properties of Ethane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
90.368	1.1421E-06	21.668	0.046151	-6.5907	-6.5907	-0.04975	0.048264	0.069935	2008.7	-0.55899	255.62	1280.8
100	1.1081E-05	21.316	0.046913	-5.9245	-5.9245	-0.04275	0.046324	0.068639	1938.4	-0.56656	247.83	873.22
110	7.4287E-05	20.951	0.04773	-5.2401	-5.2401	-0.03623	0.045172	0.068351	1866.4	-0.56429	239.12	633.38
120	0.0003523	20.584	0.048581	-4.5559	-4.5559	-0.03027	0.044453	0.068544	1794.4	-0.55668	229.91	485.78
130	0.0012839	20.214	0.049469	-3.8686	-3.8685	-0.02477	0.043962	0.068959	1722.0	-0.54585	220.35	388.17
140	0.0038136	19.840	0.050402	-3.1764	-3.1762	-0.01964	0.043607	0.069499	1649.1	-0.53246	210.57	319.79
150	0.009638	19.461	0.051385	-2.4784	-2.4779	-0.01483	0.043351	0.070139	1575.5	-0.51640	200.71	269.55
160	0.021405	19.074	0.052426	-1.7736	-1.7724	-0.01028	0.043183	0.070888	1501.3	-0.49723	190.84	231.16
170	0.042819	18.680	0.053534	-1.0608	-1.0585	-0.00596	0.043100	0.071768	1426.3	-0.47424	181.03	200.83
180	0.078638	18.275	0.054720	-0.3389	-0.33459	-0.00183	0.043108	0.072812	1350.5	-0.44655	171.34	176.18
190	0.13459	17.858	0.055998	0.3938	0.40133	0.002133	0.043210	0.074056	1273.7	-0.41304	161.84	155.68
200	0.21723	17.426	0.057386	1.1391	1.1515	0.005957	0.043411	0.075548	1196.0	-0.37225	152.56	138.27
210	0.33380	16.976	0.058907	1.8991	1.9188	0.009667	0.043717	0.077345	1117.3	-0.32220	143.47	123.23
220	0.49205	16.504	0.060590	2.6766	2.7064	0.013287	0.044131	0.079529	1037.3	-0.26011	134.60	110.05
230	0.70018	16.006	0.062476	3.4747	3.5184	0.016839	0.044661	0.082221	956.09	-0.18185	125.95	98.339
240	0.96679	15.475	0.064622	4.2973	4.3598	0.020347	0.045314	0.085612	873.25	-0.08101	117.51	87.799
250	1.3008	14.901	0.067112	5.1501	5.2374	0.023839	0.046109	0.090024	788.33	0.05295	109.29	78.190
260	1.7118	14.270	0.070079	6.0406	6.1606	0.027349	0.047076	0.096059	700.52	0.23862	101.24	69.305
270	2.2100	13.559	0.073750	6.9809	7.1438	0.030924	0.048269	0.10496	608.92	0.51141	93.309	60.936
280	2.8067	12.728	0.078565	7.9919	8.2124	0.034644	0.049743	0.11989	512.38	0.94945	85.434	52.839
290	3.5159	11.684	0.085590	9.1194	9.4204	0.038677	0.051927	0.15219	405.70	1.7713	77.574	44.603
300	4.3573	10.094	0.099071	10.525	10.957	0.043620	0.057488	0.30137	274.91	3.9421	71.489	34.970
305.32	4.8722	6.8569	0.14584	12.490	13.200	0.050820			0	9.6957		
90.368	1.1421E-06	1.5201E-06	657.870	10.542	11.293	0.14815	0.026809	0.035124	180.93	249.88	2.9082	3.0427
100	1.1081E-05	1.3327E-05	75.033	10.803	11.634	0.13284	0.027384	0.035699	189.86	199.02	3.4563	3.3157
110	7.4287E-05	8.1234E-05	12.310	11.080	11.994	0.12045	0.027997	0.036318	198.61	177.19	4.0447	3.6034
120	0.0003523	0.00035326	2,830.8	11.362	12.359	0.11069	0.028651	0.036992	206.89	158.49	4.6568	3.8961
130	0.0012839	0.0011893	840.82	11.648	12.728	0.10289	0.029380	0.037770	214.69	129.38	5.2959	4.1937
140	0.0038136	0.0032857	304.35	11.938	13.099	0.096610	0.030154	0.038623	222.01	96.433	5.9661	4.4956
150	0.0096380	0.0077732	128.65	12.231	13.471	0.091501	0.030876	0.039449	228.84	70.230	6.6725	4.8011
160	0.021405	0.016263	61.489	12.525	13.841	0.087308	0.031513	0.040235	235.12	54.075	7.4216	5.1100
170	0.042819	0.030843	32.422	12.817	14.205	0.083831	0.032173	0.041138	240.73	45.339	8.2209	5.4223
180	0.078638	0.054053	18.500	13.104	14.559	0.080912	0.033015	0.042375	245.54	40.255	9.0790	5.7388
190	0.13459	0.088865	11.253	13.384	14.898	0.078433	0.034120	0.044070	249.41	36.410	10.005	6.0609
200	0.21723	0.13870	7.2100	13.654	15.221	0.076302	0.035457	0.046220	252.26	32.914	11.011	6.3908
210	0.33380	0.20749	4.8195	13.914	15.523	0.074447	0.036937	0.048771	254.02	29.687	12.109	6.7319
220	0.49205	0.29989	3.3346	14.160	15.801	0.072806	0.038486	0.051718	254.65	26.853	13.318	7.0889
230	0.70018	0.42157	2.3721	14.389	16.050	0.071324	0.040077	0.055176	254.05	24.485	14.662	7.4687
240	0.96679	0.57983	1.7246	14.597	16.264	0.069948	0.041742	0.059419	252.14	22.563	16.179	7.8814
250	1.3008	0.78456	1.2746	14.776	16.434	0.068624	0.043548	0.064956	248.79	21.017	17.930	8.3424
260	1.7118	1.0502	0.9522	14.915	16.545	0.067290	0.045586	0.072712	243.81	19.757	20.023	8.8764
270	2.2100	1.3998	0.71441	14.999	16.578	0.065865	0.047971	0.084634	237.02	18.700	22.667	9.5257
280	2.8067	1.8748	0.53338	14.998	16.495	0.064224	0.050986	0.10591	228.10	17.721	26.327	10.373
290	3.5159	2.5679	0.38943	14.846	16.215	0.062107	0.055181	0.15523	216.50	16.582	32.319	11.617
300	4.3573	3.8079	0.26261	14.314	15.458	0.058625	0.062820	0.39989	200.51	14.497	47.465	14.023
305.32	4.8722	6.8569	0.14584	12.490	13.200	0.050820			0	9.6957		

TABLE 2-213 Thermodynamic Properties of Ethane (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_v kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Single-Phase Properties												
100.00	0.10000	21.317	0.046910	-5.9253	-5.9206	-0.042756	0.046328	0.068636	1938.8	-0.56660	247.88	873.92
200.00	0.10000	0.061716	16.203	13.768	15.388	0.083336	0.034553	0.043810	257.90	30.845	10.723	6.3683
300.00	0.10000	0.040388	24.760	17.702	20.178	0.10263	0.044475	0.053048	312.23	10.722	21.218	9.4081
400.00	0.10000	0.030155	33.162	22.785	26.102	0.11958	0.057153	0.065586	355.24	5.0759	35.965	12.213
500.00	0.10000	0.024084	41.521	29.137	33.289	0.13557	0.069620	0.078001	393.08	2.7932	53.769	14.778
600.00	0.10000	0.020056	49.861	36.675	41.661	0.15080	0.080870	0.089227	427.63	1.6883	73.338	17.135
100.00	1.0000	21.329	0.046885	-5.9325	-5.8856	-0.042828	0.046362	0.068612	1942.4	-0.56691	248.30	880.24
200.00	1.0000	17.457	0.057283	1.1164	1.1737	0.0058431	0.043441	0.075331	1204.0	-0.37737	153.30	139.33
241.10	1.0000	15.414	0.064876	4.3894	4.4543	0.020731	0.045394	0.086038	864.04	-0.068172	116.60	86.702
241.10	1.0000	0.59982	1.6672	14.618	16.285	0.069802	0.041932	0.059951	251.85	22.377	16.358	7.9293
300.00	1.0000	0.43455	2.3012	17.331	19.632	0.082230	0.045482	0.057061	296.92	11.381	22.177	9.6518
400.00	1.0000	0.30970	3.2290	22.568	25.797	0.099896	0.057476	0.067069	348.76	5.1250	36.459	12.417
500.00	1.0000	0.24355	4.1060	28.986	33.092	0.11613	0.069783	0.078784	390.24	2.7821	54.084	14.954
600.00	1.0000	0.20145	4.9639	36.562	41.526	0.13147	0.080969	0.089708	426.64	1.6715	73.562	17.289
100.00	5.0000	21.381	0.046771	-5.9638	-5.7300	-0.043144	0.046517	0.068510	1957.7	-0.56823	250.17	909.15
200.00	5.0000	17.610	0.056786	1.0063	1.2902	0.0052852	0.043595	0.074345	1242.5	-0.40076	156.99	144.70
300.00	5.0000	10.907	0.091680	10.087	10.546	0.042046	0.053743	0.17141	359.63	2.3430	73.067	39.648
400.00	5.0000	1.7653	0.56647	21.484	24.316	0.083790	0.059017	0.076481	322.27	5.2035	39.974	13.905
500.00	5.0000	1.2769	0.78316	28.293	32.209	0.10138	0.070476	0.082676	380.40	2.6757	55.976	15.995
600.00	5.0000	1.0245	0.97605	36.055	40.936	0.11726	0.081384	0.091919	424.37	1.5738	74.801	18.124
100.00	10.000	21.444	0.046634	-6.0016	-5.5352	-0.043532	0.046712	0.068393	1976.2	-0.56975	252.48	947.29
200.00	10.000	17.786	0.056225	0.88047	1.4427	0.0046352	0.043786	0.073343	1286.7	-0.42471	161.36	151.27
300.00	10.000	12.680	0.078863	9.0258	9.8145	0.038217	0.050799	0.10280	579.32	0.58968	86.388	52.867
400.00	10.000	4.1977	0.23822	19.826	22.208	0.073976	0.060880	0.096368	310.38	4.4129	48.699	17.942
500.00	10.000	2.6794	0.37322	27.390	31.122	0.093896	0.071226	0.088180	377.77	2.3866	59.523	17.954
600.00	10.000	2.0757	0.48177	35.421	40.239	0.11050	0.081836	0.094716	427.41	1.4041	76.885	19.504
100.00	30.000	21.682	0.046121	-6.1384	-4.7548	-0.045002	0.047466	0.068003	2044.8	-0.57447	261.51	1127.3
200.00	30.000	18.371	0.054434	0.46960	2.1026	0.0024068	0.044528	0.070795	1433.6	-0.48623	177.27	177.04
300.00	30.000	14.784	0.067640	7.6392	9.6684	0.032933	0.050618	0.081850	909.38	-0.18133	112.64	77.610
400.00	30.000	10.812	0.092490	15.788	18.563	0.058427	0.061395	0.094994	585.71	0.40872	81.974	42.216
500.00	30.000	7.6781	0.13024	24.381	28.288	0.080115	0.072527	0.098504	497.87	0.71195	78.720	30.510
600.00	30.000	5.8933	0.16968	33.201	38.291	0.098343	0.082911	0.10204	505.75	0.59060	88.635	27.539
200.00	70.000	19.230	0.052001	0.098974	3.5411	0.0010269	0.045892	0.068554	1648.3	-0.54018	204.45	230.70
300.00	70.000	16.537	0.060469	6.4673	10.700	0.027905	0.051766	0.075512	1238.5	-0.42067	145.70	113.52
400.00	70.000	14.033	0.071259	13.756	18.744	0.050977	0.062135	0.085472	970.70	-0.27243	115.84	71.766
500.00	70.000	11.869	0.084254	21.871	27.769	0.071077	0.073299	0.094785	827.80	-0.15903	107.39	53.331
600.00	70.000	10.144	0.098582	30.754	37.654	0.089078	0.083730	0.10274	764.56	-0.10326	111.32	44.741

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Buecker, D., and Wagner, W., "A Reference Equation of State for the Thermodynamic Properties of Ethane for Temperatures from the Melting Line to 675 K and Pressures up to 900 MPa," *J. Phys. Chem. Ref. Data*, **35**(1):205–206, 2006. The source for viscosity and thermal conductivity is Friend, D. G., Ingham, H., and Ely, J. F., "Thermophysical Properties of Ethane," *J. Phys. Chem. Ref. Data*, **35**(1):205–266, 2006.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in the equation of state are 0.02% to 0.04% in density from the melting line up to temperatures of 520 K and pressures of 30 MPa. The uncertainties increase to 0.3% at higher temperatures and to 1% at higher pressures. The uncertainty in speed of sound ranges from 0.02% in the gaseous phase to 0.15% in the liquid phase. Above 450 K, the uncertainties increase to 0.3% at lower pressures and to 1% at higher pressures. At pressures above 40 MPa at all temperatures, the uncertainties are 1% up to 100 MPa, and 5% at higher pressures. The uncertainties in heat capacities range from 2% in the vapor and liquid regions below 450 K and 30 MPa to 5% at high pressures. The uncertainties in vapor pressure are 0.01% above 170 K and 10 MPa below 170 K. The uncertainty in viscosity is 2%. The uncertainty in thermal conductivity is 2%.

TABLE 2-214 Thermodynamic Properties of Ethanol

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
250.00	0.00027007	17.911	0.055831	6.9274	6.9275	0.037330	0.076657	0.093612	1325.0	-0.44553	178.12	3140.9
265.00	0.00089527	17.642	0.056681	8.3792	8.3793	0.042968	0.083798	0.10028	1260.8	-0.41423	173.58	2182.0
280.00	0.0025823	17.376	0.057551	9.9424	9.9426	0.048704	0.091653	0.10829	1202.8	-0.37872	169.56	1564.4
295.00	0.0066146	17.106	0.058460	11.630	11.631	0.054574	0.099433	0.11678	1149.2	-0.34323	165.87	1152.5
310.00	0.015298	16.828	0.059426	13.445	13.446	0.060574	0.10670	0.12524	1098.1	-0.30910	162.38	869.40
325.00	0.032394	16.537	0.060469	15.385	15.387	0.066684	0.11322	0.13340	1048.0	-0.27615	159.01	669.49
340.00	0.063544	16.231	0.061610	17.444	17.448	0.072875	0.11893	0.14115	997.94	-0.24356	155.69	524.87
355.00	0.11663	15.905	0.062872	19.615	19.622	0.079123	0.12381	0.14847	947.31	-0.21011	152.39	417.88
370.00	0.20205	15.557	0.064281	21.892	21.905	0.085403	0.12792	0.15543	895.56	-0.17428	149.09	337.04
385.00	0.33279	15.181	0.065871	24.268	24.290	0.091699	0.13130	0.16215	842.31	-0.13410	145.78	274.76
400.00	0.52446	14.774	0.067684	26.740	26.775	0.098000	0.13405	0.16883	787.16	-0.086812	142.47	225.91
415.00	0.79509	14.331	0.069779	29.307	29.362	0.10430	0.13625	0.17576	729.67	-0.028333	139.18	186.93
430.00	1.1649	13.843	0.072241	31.970	32.054	0.11061	0.13798	0.18341	669.25	0.047976	135.93	155.35
445.00	1.6559	13.298	0.075202	34.737	34.862	0.11695	0.13934	0.19262	605.07	0.15384	132.78	129.37
460.00	2.2916	12.676	0.078889	37.629	37.810	0.12335	0.14041	0.20504	535.80	0.31228	129.85	107.62
475.00	3.0963	11.941	0.083745	40.684	40.943	0.12991	0.14134	0.22469	459.19	0.57597	127.41	88.972
490.00	4.0954	11.007	0.090848	44.002	44.374	0.13684	0.14234	0.26508	371.03	1.0976	126.33	72.213
505.00	5.3159	9.5842	0.10434	47.926	48.480	0.14485	0.14382	0.41790	264.74	2.5369	129.43	55.104
513.90	6.1480	5.9910	0.16692	53.880	54.906	0.15723			0	8.6373		
250.00	0.00027007	0.00012998	7693.7	49.039	51.116	0.21409	0.058885	0.067215	226.86	149.30	14.936	7.2715
265.00	0.00089527	0.00040670	2458.8	49.932	52.134	0.20808	0.060795	0.069146	233.03	111.11	15.737	7.7433
280.00	0.0025823	0.0011115	899.69	50.851	53.174	0.20310	0.062753	0.071149	238.89	87.283	16.612	8.2114
295.00	0.0066146	0.0027080	369.28	51.792	54.234	0.19899	0.064753	0.073238	244.41	71.858	17.566	8.6756
310.00	0.015298	0.0059814	167.18	52.749	55.307	0.19561	0.066816	0.075464	249.49	61.180	18.602	9.1353
325.00	0.032394	0.012150	82.305	53.717	56.383	0.19282	0.068988	0.077921	254.02	53.164	19.731	9.5902
340.00	0.063544	0.022975	43.525	54.684	57.450	0.19053	0.071336	0.080736	257.88	46.697	20.969	10.040
355.00	0.11663	0.040873	24.466	55.640	58.494	0.18862	0.073932	0.084059	260.92	41.226	22.341	10.486
370.00	0.20205	0.069039	14.485	56.573	59.500	0.18701	0.076838	0.088058	263.02	36.486	23.886	10.929
385.00	0.33279	0.11160	8.9606	57.469	60.451	0.18562	0.080106	0.092930	264.03	32.353	25.659	11.372
400.00	0.52446	0.17385	5.7521	58.312	61.329	0.18438	0.083774	0.098936	263.82	28.756	27.741	11.820
415.00	0.79509	0.26261	3.8080	59.087	62.115	0.18322	0.087876	0.10646	262.27	25.644	30.251	12.283
430.00	1.1649	0.38683	2.5851	59.774	62.785	0.18208	0.092450	0.11610	259.21	22.967	33.369	12.774
445.00	1.6559	0.55876	1.7897	60.348	63.312	0.18088	0.097544	0.12898	254.44	20.681	37.377	13.318
460.00	2.2916	0.79629	1.2558	60.777	63.654	0.17954	0.10323	0.14727	247.61	18.747	42.735	13.961
475.00	3.0963	1.1286	0.88602	61.004	63.747	0.17792	0.10966	0.17612	238.10	17.136	50.248	14.786
490.00	4.0954	1.6143	0.61945	60.916	63.453	0.17578	0.11709	0.23200	224.59	15.831	61.578	15.982
505.00	5.3159	2.4339	0.41086	60.144	62.328	0.17228	0.12644	0.42053	203.70	14.728	82.512	18.148
513.90	6.1480	5.9910	0.16692	53.880	54.906	0.15723			0	8.6373		

TABLE 2-214 Thermodynamic Properties of Ethanol (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_p kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Single-Phase Properties												
300.00	0.10000	17.016	0.058768	12.219	12.225	0.056554	0.10193	0.11962	1132.5	-0.33179	164.74	1047.2
351.05	0.10000	15.993	0.062527	19.033	19.040	0.077475	0.12261	0.14658	960.72	-0.21908	153.26	443.11
351.05	0.10000	0.035314	28.317	55.390	58.222	0.18909	0.073221	0.083127	260.21	42.587	21.965	10.369
400.00	0.10000	0.030577	32.704	59.207	62.477	0.20043	0.080640	0.089997	279.09	28.685	26.374	11.853
500.00	0.10000	0.024191	41.338	67.925	72.058	0.22176	0.092910	0.10162	312.39	11.830	37.865	14.768
600.00	0.10000	0.020086	49.786	77.796	82.775	0.24127	0.10403	0.11252	341.55	5.6356	52.622	17.543
300.00	1.0000	17.034	0.058707	12.202	12.261	0.056497	0.10191	0.11954	1137.9	-0.33273	165.24	1053.2
400.00	1.0000	14.795	0.067589	26.715	26.783	0.097937	0.13400	0.16857	791.85	-0.089821	142.87	227.32
423.85	1.0000	14.049	0.071181	30.866	30.938	0.10802	0.13732	0.18015	694.41	0.013963	137.25	167.52
423.85	1.0000	0.33095	3.0216	59.504	62.526	0.18255	0.090516	0.11184	260.65	24.014	32.003	12.568
500.00	1.0000	0.25567	3.9114	67.014	70.925	0.20078	0.096953	0.11008	300.64	12.007	39.539	14.859
600.00	1.0000	0.20473	4.8846	77.311	82.195	0.22131	0.10581	0.11605	337.33	5.6301	53.583	17.678
300.00	5.0000	17.111	0.058443	12.129	12.421	0.056249	0.10185	0.11922	1161.4	-0.33665	167.43	1079.6
400.00	5.0000	14.961	0.066842	26.516	26.851	0.097435	0.13359	0.16658	829.44	-0.11211	146.07	238.82
500.00	5.0000	10.220	0.097846	46.419	46.908	0.14179	0.14311	0.32410	308.88	1.7596	127.42	61.882
501.39	5.0000	10.013	0.099875	46.876	47.375	0.14272	0.14340	0.35152	292.31	2.0063	128.00	59.510
501.39	5.0000	2.1809	0.45852	60.445	62.737	0.17336	0.12389	0.34099	209.80	15.000	75.676	17.454
600.00	5.0000	1.1372	0.87939	74.966	79.363	0.20395	0.11419	0.13659	314.06	5.6703	61.725	18.972
300.00	10.000	17.203	0.058131	12.041	12.623	0.055950	0.10179	0.11885	1189.5	-0.34096	170.07	1111.8
400.00	10.000	15.147	0.066020	26.293	26.953	0.096860	0.13313	0.16456	872.36	-0.13414	149.80	252.40
500.00	10.000	11.521	0.086800	44.752	45.620	0.13830	0.14031	0.22204	464.50	0.60618	130.15	80.680
600.00	10.000	2.8001	0.35713	71.266	74.837	0.19172	0.12599	0.18744	273.66	5.6926	84.190	23.411
300.00	100.00	18.389	0.054380	10.984	16.422	0.051802	0.10149	0.11571	1558.1	-0.37198	207.54	1611.3
400.00	100.00	17.030	0.058722	24.075	29.947	0.090466	0.12901	0.15081	1348.2	-0.25352	195.29	435.09
500.00	100.00	15.408	0.064899	39.356	45.846	0.12589	0.13221	0.16433	1166.1	-0.17199	188.35	192.15
600.00	100.00	13.601	0.073523	55.055	62.407	0.15608	0.12575	0.16553	1015.1	-0.082822	187.31	109.49
300.00	200.00	19.244	0.051963	10.349	20.742	0.048505	0.10196	0.11495	1830.4	-0.37578	238.67	2085.4
400.00	200.00	18.138	0.055134	22.905	33.931	0.086238	0.12678	0.14539	1660.3	-0.28090	228.57	591.02
500.00	200.00	16.878	0.059250	37.295	49.145	0.12014	0.12868	0.15623	1525.6	-0.22946	224.49	269.30
600.00	200.00	15.505	0.064494	51.902	64.801	0.14869	0.12066	0.15566	1422.7	-0.19099	226.40	148.43

The values in these tables were generated from the NIST REFPROP software (Lemmon, E.W., McLinden, M.O., and Huber, M.L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Dillon, H.E., and Penoncello, S.G., "A Fundamental Equation for Calculation of the Thermodynamic Properties of Ethanol," *Int. J. Thermophys.*, **25**(2):321–335, 2004. The source for viscosity is Kiselev, S. B., Ely, J. F., Abdulagatov, I. M., and Huber, M. L., "Generalized SAFT-DFT/DMT Model for the Thermodynamic, Interfacial, and Transport Properties of Associating Fluids: Application for *n*-Alkanols," *Ind. Eng. Chem. Res.*, **44**:6916–6927, 2005. The source for thermal conductivity is unpublished, 2004; however, the fit uses functional form found in Marsh, K., Perkins, R., and Ramirez, M. L.V., "Measurement and Correlation of the Thermal Conductivity of Propane from 86 to 600 K at Pressures to 70 MPa," *J. Chem. Eng. Data*, **47**(4):932–940, 2002.

Properties at the critical point temperature are given in the last entry of the saturation tables. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in the equation of state are 0.2% in density, 3% in heat capacities, 1% in speed of sound, and 0.5% in vapor pressure and saturation densities. The estimated uncertainty in the liquid phase along the saturation boundary is approximately 3%, increasing to 10% at pressures to 100 MPa, and is estimated at 10% in the vapor phase. The estimated uncertainty in the liquid phase is approximately 5% and is estimated as 10% in the vapor phase.

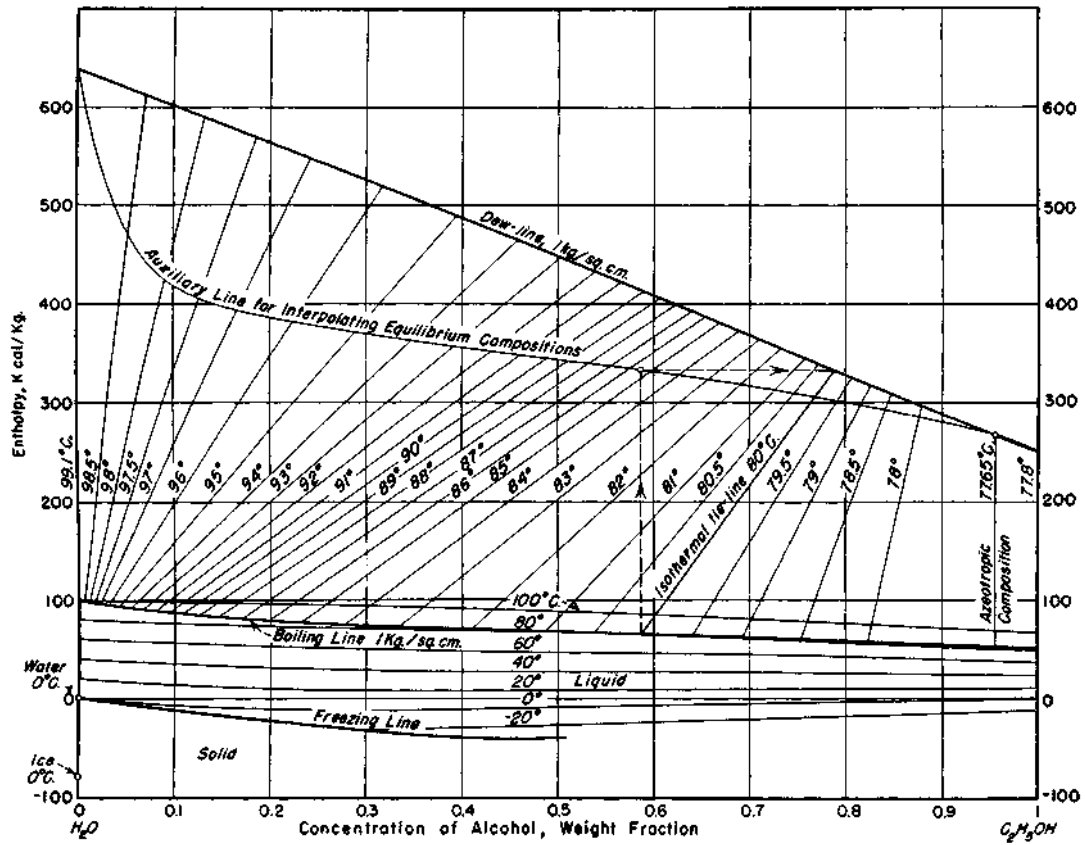


FIG. 2-10 Enthalpy-concentration diagram for aqueous ethyl alcohol. Reference states: Enthalpies of liquid water and ethyl alcohol at 0°C are zero. NOTE: In order to interpolate equilibrium compositions, a vertical may be erected from any liquid composition on the boiling line and its intersection with the auxiliary line determined. A horizontal from this intersection will establish the equilibrium vapor composition on the dew line. (Bosnjakovic, Technische Thermodynamik, T. Steinkopff, Leipzig, 1935.)

TABLE 2-215 Thermodynamic Properties of Ethylene

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_v kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
103.99	0.00012196	23.334	0.042856	-4.4352	-4.4352	-0.033074	0.045503	0.068155	1766.6	-0.50257	270.65	685.73
105.00	0.00014568	23.288	0.042940	-4.3661	-4.3661	-0.032412	0.045394	0.068183	1760.2	-0.50171	269.25	662.48
115.00	0.00069745	22.834	0.043793	-3.6839	-3.6838	-0.026206	0.044202	0.068194	1694.4	-0.49455	255.56	488.07
125.00	0.0025267	22.375	0.044692	-3.0030	-3.0029	-0.020529	0.042929	0.067951	1626.0	-0.48779	242.07	378.67
135.00	0.0073921	21.909	0.045643	-2.3250	-2.3247	-0.015311	0.041693	0.067669	1556.2	-0.47928	228.83	305.49
145.00	0.018309	21.435	0.046652	-1.6496	-1.6488	-0.010484	0.040562	0.067475	1485.3	-0.46747	215.95	253.84
155.00	0.039755	20.952	0.047729	-0.97554	-0.97364	-0.0059884	0.039568	0.067448	1413.2	-0.45105	203.50	215.68
165.00	0.077693	20.456	0.048885	-0.30104	-0.29724	-0.0017710	0.038729	0.067649	1339.9	-0.42866	191.54	186.36
175.00	0.13944	19.945	0.050137	0.37612	0.38311	0.0022141	0.038051	0.068127	1265.0	-0.39877	180.09	163.03
185.00	0.23344	19.417	0.051502	1.0585	1.0706	0.0060075	0.037536	0.068937	1188.5	-0.35944	169.21	143.88
195.00	0.36901	18.865	0.053008	1.7492	1.7688	0.0096456	0.037184	0.070151	1110.0	-0.30804	158.78	127.73
205.00	0.55614	18.286	0.054687	2.4516	2.4820	0.013162	0.036995	0.071867	1029.3	-0.24081	148.78	113.76
215.00	0.80534	17.671	0.056589	3.1699	3.2155	0.016589	0.036970	0.074242	946.13	-0.15205	139.15	101.40
225.00	1.1276	17.011	0.058784	3.9095	3.9758	0.019960	0.037115	0.077537	860.08	-0.032706	129.81	90.217
235.00	1.5342	16.291	0.061382	4.6774	4.7715	0.023314	0.037446	0.082229	770.62	0.13270	120.64	79.891
245.00	2.0376	15.488	0.064566	5.4842	5.6157	0.026699	0.037996	0.089292	676.97	0.37286	111.48	70.142
255.00	2.6509	14.650	0.068682	6.3476	6.5297	0.030191	0.038851	0.10110	577.38	0.74844	102.11	60.685
265.00	3.3898	13.419	0.074519	7.3042	7.5568	0.033937	0.040322	0.12549	467.22	1.4173	92.171	51.112
275.00	4.2752	11.793	0.084795	8.4637	8.8262	0.038375	0.043922	0.21287	334.73	2.9732	81.576	40.397
282.35	5.0417	7.6368	0.13095	10.545	11.206	0.046609			0	8.7988		
103.99	0.00012196	0.00014109	7087.6	10.621	11.486	0.12003	0.024972	0.033295	202.67	154.90	6.8012	0.77270
105.00	0.00014568	0.00016690	5991.6	10.647	11.520	0.11888	0.024975	0.033300	203.65	150.43	6.7393	1.0011
115.00	0.00069745	0.00073000	1369.9	10.895	11.850	0.10888	0.025028	0.033376	213.03	114.92	6.6223	2.7342
125.00	0.0025267	0.0024363	410.46	11.142	12.179	0.10093	0.025131	0.033533	221.86	90.667	7.0116	3.8262
135.00	0.0073921	0.0066178	151.11	11.385	12.502	0.094519	0.025307	0.033814	230.09	73.502	7.5850	4.5578
145.00	0.018309	0.015333	65.217	11.624	12.818	0.089286	0.025577	0.034263	237.62	60.977	8.2013	5.0886
155.00	0.039755	0.031379	31.869	11.855	13.122	0.084954	0.025956	0.034922	244.36	51.588	8.8125	5.5112
165.00	0.077693	0.058231	17.173	12.078	13.412	0.081316	0.026454	0.035833	250.18	44.379	9.4170	5.8799
175.00	0.13944	0.099997	10.000	12.289	13.683	0.078216	0.027081	0.037045	254.99	38.733	10.036	6.2279
185.00	0.23344	0.16142	6.1950	12.486	13.933	0.075532	0.027844	0.038617	258.69	34.236	10.700	6.5765
195.00	0.36901	0.24802	4.0319	12.667	14.155	0.073165	0.028751	0.040637	261.18	30.602	11.446	6.9410
205.00	0.55614	0.36638	2.7294	12.829	14.346	0.071037	0.029816	0.043243	262.38	27.626	12.313	7.3341
215.00	0.80534	0.52471	1.9058	12.966	14.501	0.069079	0.031060	0.046651	262.18	25.155	13.346	7.7689
225.00	1.1276	0.73386	1.3627	13.074	14.610	0.067225	0.032511	0.051237	260.50	23.074	14.607	8.2612
235.00	1.5342	1.0092	0.99086	13.144	14.664	0.065409	0.034212	0.057696	257.19	21.287	16.186	8.8342
245.00	2.0376	1.3747	0.72745	13.162	14.645	0.063551	0.036243	0.067503	252.09	19.706	18.250	9.5251
255.00	2.6509	1.8719	0.53422	13.105	14.522	0.061532	0.038789	0.084393	244.94	18.213	21.166	10.404
265.00	3.3898	2.5886	0.38631	12.923	14.232	0.059128	0.042250	0.12086	235.28	16.604	26.045	11.632
275.00	4.2752	3.7947	0.26352	12.460	13.587	0.055687	0.048136	0.26071	221.57	14.315	39.168	13.765
282.35	5.0417	7.6368	0.13095	10.545	11.206	0.046609			0	8.7988		

Single-Phase Properties

150.00	0.10000	21.197	0.047176	-1.3137	-1.3090	-0.0082061	0.040048	0.067428	1449.9	-0.46008	209.74	233.39
169.16	0.10000	20.246	0.049393	-0.020110	-0.015171	-0.000089238	0.038427	0.067811	1309.0	-0.41725	186.72	176.06
169.16	0.10000	0.073519	13.602	12.167	13.527	0.079970	0.026699	0.036297	252.31	41.871	9.6709	6.0256
225.00	0.10000	0.054182	18.456	13.722	15.568	0.090386	0.028730	0.037508	291.07	20.438	12.903	7.8384
300.00	0.10000	0.040315	24.805	16.100	18.580	0.10190	0.034782	0.043294	330.82	10.015	20.558	10.381
375.00	0.10000	0.032159	31.096	18.990	22.099	0.11234	0.042202	0.050626	364.16	5.6845	31.002	12.827
450.00	0.10000	0.026764	37.364	22.436	26.173	0.12222	0.049547	0.057930	394.34	3.5485	42.200	15.092
150.00	1.0000	21.223	0.047119	-1.3281	-1.2810	-0.0083026	0.040075	0.067329	1456.3	-0.46205	210.67	232.98
221.33	1.0000	17.260	0.057938	3.6350	3.6929	0.018726	0.037041	0.067196	892.05	-0.080890	133.22	94.208
221.33	1.0000	0.65028	1.5378	13.038	14.576	0.067898	0.031952	0.049381	261.30	23.800	14.112	8.0724
225.00	1.0000	0.63206	1.5821	13.173	14.755	0.068701	0.031799	0.048297	265.10	22.637	14.323	8.1765
300.00	1.0000	0.42535	2.3510	15.822	18.173	0.081821	0.035459	0.046096	320.09	10.271	21.362	10.589
375.00	1.0000	0.32964	3.0336	18.803	21.836	0.092696	0.042478	0.051961	358.72	5.7053	31.533	13.017
450.00	1.0000	0.27097	3.6904	22.297	25.987	0.10277	0.049700	0.058724	391.57	3.5291	42.581	15.284
150.00	5.0000	21.333	0.046876	-1.3902	-1.1558	-0.0087214	0.040199	0.066921	1483.4	-0.47016	214.70	231.18
225.00	5.0000	17.364	0.057591	3.7130	4.0009	0.019071	0.037097	0.073970	926.49	-0.13327	135.08	93.288
281.98	5.0000	9.0494	0.11050	9.9242	10.477	0.040404	0.061633	0.076196	189.76	7.3487	145.71	27.667
281.98	5.0000	6.2408	0.16024	11.247	12.048	0.049617	0.066988	5.9539	191.76	10.151	182.63	19.004
300.00	5.0000	3.1277	0.31972	13.977	15.576	0.061893	0.040822	0.089599	263.71	10.910	30.862	13.344
375.00	5.0000	1.8590	0.53792	17.877	20.567	0.076837	0.043817	0.060244	337.12	5.6244	35.453	14.287
450.00	5.0000	1.4297	0.69947	21.650	25.147	0.087965	0.050366	0.062753	381.96	3.3730	45.106	16.277
150.00	10.000	21.464	0.046589	-1.4634	-0.99754	-0.0092238	0.040364	0.066475	1515.1	-0.47905	219.60	229.03
225.00	10.000	17.739	0.056372	3.5022	4.0659	0.018094	0.037190	0.070996	996.58	-0.22081	140.93	96.609
300.00	10.000	11.510	0.086881	9.6517	10.521	0.042563	0.041889	0.11956	420.01	1.9372	79.635	37.621
375.00	10.000	4.3583	0.22945	16.483	18.778	0.067434	0.045508	0.077265	328.58	4.7029	45.198	17.402
450.00	10.000	3.0293	0.33011	20.798	24.099	0.080391	0.051131	0.068674	379.55	2.9790	50.075	18.014
150.00	100.00	23.148	0.043200	-2.3062	2.0139	-0.015964	0.043582	0.063328	1891.7	-0.54092	294.42	199.89
225.00	100.00	20.862	0.047935	1.8396	6.6331	0.0090426	0.040468	0.060888	1581.0	-0.54008	200.24	126.05
300.00	100.00	18.775	0.053261	5.9492	11.275	0.026831	0.043293	0.063490	1328.5	-0.48350	154.23	98.312
375.00	100.00	16.889	0.059210	10.287	16.208	0.041488	0.048765	0.068211	1143.7	-0.41039	131.89	81.449
450.00	100.00	15.219	0.065708	14.942	21.513	0.054370	0.054851	0.073224	1020.9	-0.34600	121.24	70.072
150.00	300.00	25.298	0.039530	-2.9624	8.8964	-0.024942	0.048504	0.062644	2343.7	-0.54689	432.13	161.49
225.00	300.00	23.718	0.042162	0.79998	13.448	-0.00029268	0.044752	0.059805	2151.3	-0.57336	275.32	155.50
300.00	300.00	22.336	0.044772	4.5667	17.998	0.017143	0.047124	0.062135	1984.6	-0.55352	209.21	157.18
375.00	300.00	21.120	0.047350	8.6213	22.826	0.031488	0.052222	0.066814	1847.1	-0.51701	178.33	156.74
450.00	300.00	20.042	0.049894	13.065	28.033	0.044131	0.057988	0.072043	1740.6	-0.48199	163.32	153.81

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Smukala, J., Span, R., and Wagner, W., "A New Equation of State for Ethylene Covering the Fluid Region for Temperatures from the Melting Line to 450 K at Pressures up to 300 MPa," *J. Phys. Chem. Ref. Data*, **29**(5):1053–1122, 2000. The source for viscosity is Holland, P. M., Eaton, B. E., and Hanley, H. J. M., "A Correlation of the Viscosity and Thermal Conductivity Data of Gaseous and Liquid Ethylene," *J. Phys. Chem. Ref. Data*, **12**(4):917–932, 1983. The source for thermal conductivity is Holland, P. M., Eaton, B. E., and Hanley, H. J. M., "A Correlation of the Viscosity and Thermal Conductivity Data of Gaseous and Liquid Ethylene," *J. Phys. Chem. Ref. Data*, **12**(4):917–932, 1983.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in density of the equation of state range from 0.02% in the liquid and most of the vapor phase to 0.1% for supercritical states. At $p > 100$ MPa, the uncertainty in density is 0.5%. The uncertainty in heat capacity is 3% in the liquid phase, 0.2% in the vapor phase, and as high as 5% in the supercritical region at higher pressures. For the speed of sound, the uncertainty is 0.05 to 0.1% in the vapor phase, rising to 3% in the liquid phase. The uncertainty in vapor pressure is less than 0.05% above 140 K. The uncertainty in viscosity is 5%, increasing to 10% in the dense liquid. The uncertainty in thermal conductivity is 5%, increasing to 10% in the dense liquid.

TABLE 2-216 Thermodynamic Properties of Fluorine

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
53.481	0.00023881	44.917	0.022263	-1.7783	-1.7783	-0.026073	0.036802	0.057963	1041.6	-0.30422
55.000	0.00038159	44.657	0.022393	-1.6923	-1.6923	-0.024487	0.035214	0.055732	1040.8	-0.31810
60.000	0.0014872	43.829	0.022816	-1.4151	-1.4151	-0.019663	0.034254	0.055680	1043.5	-0.31768
65.000	0.0046126	43.005	0.023253	-1.1355	-1.1354	-0.015187	0.033385	0.056098	1019.2	-0.31059
70.000	0.011981	42.169	0.023714	-0.85459	-0.85430	-0.011024	0.032745	0.056274	973.15	-0.30322
75.000	0.027062	41.315	0.024204	-0.57283	-0.57217	-0.0071357	0.032241	0.056496	921.41	-0.29375
80.000	0.054668	40.437	0.024730	-0.28975	-0.28840	-0.0034817	0.031737	0.056860	871.16	-0.28123
85.000	0.10090	39.531	0.025296	-0.0046707	-0.0021183	-0.000024789	0.031196	0.057394	823.90	-0.26509
90.000	0.17296	38.590	0.025913	0.28316	0.28764	0.0032663	0.030634	0.058117	779.02	-0.24474
95.000	0.27894	37.609	0.026589	0.57448	0.58190	0.0064180	0.030073	0.059051	735.47	-0.21935
100.00	0.42751	36.582	0.027336	0.87013	0.88182	0.0094534	0.029532	0.060243	692.27	-0.18762
105.00	0.62778	35.499	0.028170	1.1711	1.1887	0.012394	0.029024	0.061773	648.60	-0.14746
110.00	0.88912	34.351	0.029112	1.4785	1.5044	0.015261	0.028561	0.063791	603.67	-0.095557
115.00	1.2212	33.119	0.030194	1.7943	1.8312	0.018078	0.028159	0.066559	556.65	-0.026510
120.00	1.6342	31.777	0.031470	2.1213	2.1728	0.020877	0.027847	0.070566	506.54	0.068825
125.00	2.1389	30.280	0.033025	2.4642	2.5349	0.023700	0.027681	0.076800	452.05	0.20691
130.00	2.7475	28.549	0.035027	2.8314	2.9276	0.026617	0.027791	0.087576	391.41	0.42042
135.00	3.4739	26.416	0.037855	3.2397	3.3712	0.029765	0.028512	0.10995	321.84	0.78572
140.00	4.3357	23.427	0.042685	3.7349	3.9200	0.033502	0.030963	0.18235	238.35	1.5459
144.41	5.2394	15.603	0.064090	4.7226	5.0584	0.041160			0	4.3227
53.481	0.00023881	0.00053725	1861.3	5.3412	5.7857	0.11536	0.020800	0.029129	127.97	76.144
55.000	0.00038159	0.00083488	1197.8	5.3726	5.8296	0.11228	0.020805	0.029139	129.76	71.404
60.000	0.0014872	0.0029857	334.92	5.4753	5.9734	0.10348	0.020831	0.029204	135.46	58.461
65.000	0.0046126	0.0085667	116.73	5.5765	6.1149	0.096356	0.020886	0.029341	140.83	48.604
70.000	0.011981	0.020743	48.208	5.6750	6.2526	0.090504	0.020984	0.029589	145.85	40.910
75.000	0.027062	0.044003	22.726	5.7697	6.3847	0.085622	0.021139	0.029984	150.47	34.772
80.000	0.054668	0.084093	11.892	5.8592	6.5093	0.081489	0.021361	0.030562	154.65	29.788
85.000	0.10090	0.14789	6.7617	5.9423	6.6246	0.077936	0.021653	0.031349	158.34	25.696
90.000	0.17296	0.24329	4.1103	6.0181	6.7290	0.074837	0.022013	0.032368	161.52	22.322
95.000	0.27894	0.37919	2.6372	6.0855	6.8212	0.072094	0.022432	0.033644	164.12	19.545
100.00	0.42751	0.56568	1.7678	6.1438	6.8995	0.069630	0.022900	0.035219	166.12	17.279
105.00	0.62778	0.81460	1.2276	6.1916	6.9623	0.067380	0.023406	0.037174	167.46	15.449
110.00	0.88912	1.1406	0.87675	6.2274	7.0070	0.065284	0.023951	0.039668	168.12	13.988
115.00	1.2212	1.5630	0.63979	6.2487	7.0300	0.063285	0.024547	0.043016	168.05	12.827
120.00	1.6342	2.1098	0.47398	6.2511	7.0256	0.061318	0.025234	0.047844	167.26	11.886
125.00	2.1389	2.8245	0.35404	6.2277	6.9850	0.059301	0.026096	0.055492	165.81	11.069
130.00	2.7475	3.7825	0.26438	6.1664	6.8928	0.057119	0.027303	0.069276	163.76	10.250
135.00	3.4739	5.1364	0.19469	6.0431	6.7195	0.054567	0.029246	0.099896	161.03	9.2577
140.00	4.3357	7.3237	0.13654	5.7921	6.3841	0.051103	0.033084	0.21077	156.46	7.8082
144.41	5.2394	15.603	0.064090	4.7226	5.0584	0.041160			0	4.3227
Single-Phase Properties										
84.922	0.10000	39.546	0.025287	-0.0091317	-0.0066030	-0.000077307	0.031205	0.057385	824.62	-0.26537
84.922	0.10000	0.14668	6.8177	5.9411	6.6228	0.077988	0.021648	0.031335	158.29	25.754
100.00	0.10000	0.12275	8.1468	6.2721	7.0868	0.083019	0.021268	0.030345	173.11	17.864
150.00	0.10000	0.080628	12.403	7.3358	8.5761	0.095107	0.021022	0.029539	213.56	7.4423
200.00	0.10000	0.060267	16.593	8.3977	10.057	0.10363	0.021401	0.029803	246.33	4.1219
250.00	0.10000	0.048155	20.766	9.4869	11.564	0.11035	0.022142	0.030505	274.27	2.6600
300.00	0.10000	0.040106	24.934	10.617	13.110	0.11598	0.023021	0.031368	298.96	1.8799
100.00	1.0000	36.645	0.027289	0.86105	0.88834	0.0093623	0.029558	0.060025	696.10	-0.19109
111.80	1.0000	33.918	0.029483	1.5911	1.6206	0.016279	0.028409	0.064682	587.03	-0.073051
111.80	1.0000	1.2803	0.78106	6.2369	7.0180	0.064557	0.024158	0.040751	168.18	13.540
150.00	1.0000	0.85139	1.1746	7.1946	8.3692	0.075007	0.021931	0.032606	207.71	7.3873

200.00	1.0000	0.61498	1.6261	8.3181	9.9441	0.084080	0.021681	0.030928	244.27	4.1390
250.00	1.0000	0.48577	2.0586	9.4313	11.490	0.090978	0.022225	0.031046	273.76	2.6581
300.00	1.0000	0.40251	2.4844	10.573	13.057	0.096691	0.023032	0.031678	299.28	1.8692
100.00	5.0000	37.065	0.026980	0.80142	0.93632	0.0087568	0.029727	0.058693	721.70	-0.21256
143.33	5.0000	19.879	0.050304	4.2229	4.4744	0.037196	0.036294	0.63033	169.23	2.8620
143.33	5.0000	10.490	0.095333	5.3751	5.8518	0.046806	0.039553	1.0196	148.40	6.0573
150.00	5.0000	6.5097	0.15362	6.2260	6.9941	0.054653	0.027354	0.084441	177.58	6.7256
200.00	5.0000	3.3899	0.29499	7.9204	9.3953	0.068706	0.022993	0.037593	237.67	3.8562
250.00	5.0000	2.5221	0.39650	9.1743	11.157	0.076582	0.022698	0.033813	273.06	2.4478
300.00	5.0000	2.0427	0.48954	10.377	12.824	0.082665	0.023148	0.033134	301.36	1.6957
100.00	10.000	37.536	0.026641	0.73479	1.0012	0.0080653	0.029857	0.057385	752.43	-0.23402
150.00	10.000	24.141	0.041423	3.9349	4.3491	0.034822	0.027090	0.094016	324.06	0.86131
200.00	10.000	7.6450	0.13080	7.3273	8.6354	0.060060	0.024489	0.050206	240.08	3.1116
250.00	10.000	5.2188	0.19162	8.8378	10.754	0.069563	0.023418	0.037864	278.39	2.0656
300.00	10.000	4.1242	0.24247	10.136	12.561	0.076160	0.023411	0.035025	307.87	1.4058
100.00	15.000	37.961	0.026343	0.67509	1.0702	0.0074311	0.029870	0.056345	783.34	-0.25114
150.00	15.000	27.008	0.037026	3.5725	4.1279	0.032052	0.026615	0.070563	415.88	0.33099
200.00	15.000	12.412	0.080569	6.7026	7.9111	0.053898	0.025220	0.060759	263.46	2.0529
250.00	15.000	7.9757	0.12538	8.4961	10.377	0.064977	0.024073	0.042125	289.64	1.7085
300.00	15.000	6.1790	0.16184	9.9020	12.330	0.072112	0.023725	0.036972	318.16	1.1770
100.00	20.000	38.348	0.026077	0.62089	1.1424	0.0068429	0.029763	0.055495	815.20	-0.26507
150.00	20.000	28.645	0.034910	3.3586	4.0568	0.030382	0.026518	0.062828	480.50	0.11826
200.00	20.000	16.458	0.060762	6.1944	7.4096	0.049659	0.025474	0.062710	304.19	1.2436
250.00	20.000	10.648	0.093913	8.1664	10.045	0.061490	0.024527	0.045421	307.96	1.3200
300.00	20.000	8.1626	0.12251	9.6749	12.125	0.069094	0.024015	0.038840	331.68	0.98352

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is de Reuck, K. M., “International Thermodynamic Tables of the Fluid State—11 Fluorine,” *International Union of Pure and Applied Chemistry*, Pergamon Press, Oxford, 1990. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties of the equation of state are 0.2% in density, 2% in heat capacity, and 1.5% in the speed of sound, except in the critical region.

TABLE 2-217 Flutec

Proprietary name for a series of fluorocarbons produced by the Imperial Smelting Corp., Avonmouth, Bristol, UK. Bulletins of thermodynamic properties include PP1 (C₆F₁₄), PP2 (C₇F₁₄), PP3 (C₈F₁₆), PP5 (C₁₀F₁₈), PP9 (C₁₁F₂₀), and PP50, usually for 0.1–100 kg/m³, 0–500 °C. See also Green, S. W., *Chem. & Ind.* (1969): 63–67.

TABLE 2-218 Halon

A series of fire-extinguishing fluids. Halon 1211 is produced by ICI, and Halon 1301, by duPont, the latter issuing a bulletin with thermodynamic properties and a diagram for the range 0.6–600 psia, –160–460 °F.

TABLE 2-219 Thermodynamic Properties of Helium

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_v kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
2.1768	0.0048565	36.537	0.027370	-0.030158	-0.030025	-0.0087120	0.025164	0.025289	216.83	-1.0411	13.522	3.5963
2.2000	0.0051477	36.523	0.027380	-0.029596	-0.029455	-0.0084552	0.023062	0.023216	216.59	-1.1290	13.631	3.6153
2.3500	0.0073079	36.396	0.027475	-0.026858	-0.026658	-0.0072478	0.014101	0.014502	215.68	-1.7580	14.294	3.7039
2.5000	0.010001	36.217	0.027611	-0.025011	-0.024734	-0.0064839	0.010092	0.010808	216.03	-2.2958	14.894	3.7456
2.6500	0.013298	35.992	0.027784	-0.023514	-0.023145	-0.0059016	0.0085050	0.0095913	216.74	-2.5162	15.442	3.7541
2.8000	0.017270	35.727	0.027990	-0.022106	-0.021623	-0.0053835	0.0080743	0.0095802	216.57	-2.4447	15.946	3.7391
2.9500	0.021983	35.423	0.028230	-0.020660	-0.020040	-0.0048790	0.0081637	0.010142	215.08	-2.2326	16.408	3.7074
3.1000	0.027502	35.082	0.028504	-0.019120	-0.018336	-0.0043676	0.0084575	0.010969	212.50	-1.9844	16.829	3.6637
3.2500	0.033890	34.705	0.028814	-0.017461	-0.016485	-0.0038422	0.0088040	0.011921	209.15	-1.7406	17.209	3.6113
3.4000	0.041209	34.291	0.029162	-0.015675	-0.014473	-0.0033010	0.0091348	0.012944	205.24	-1.5095	17.548	3.5525
3.5500	0.049518	33.838	0.029553	-0.013757	-0.012294	-0.0027443	0.0094238	0.014037	200.91	-1.2873	17.844	3.4890
3.7000	0.058879	33.342	0.029993	-0.011706	-0.0099402	-0.0021719	0.0096652	0.015223	196.21	-1.0669	18.098	3.4218
3.8500	0.069351	32.799	0.030489	-0.0095155	-0.0074010	-0.0015832	0.0098629	0.016553	191.15	-0.84045	18.307	3.3516
4.0000	0.080998	32.203	0.031053	-0.0071756	-0.0046603	-0.00097628	0.010024	0.018104	185.70	-0.59987	18.474	3.2785
4.1500	0.093886	31.545	0.031700	-0.0046702	-0.0016940	-0.00034763	0.010158	0.019993	179.82	-0.33639	18.599	3.2025
4.3000	0.10809	30.813	0.032454	-0.0019735	0.0015343	0.00030858	0.010273	0.022416	173.44	-0.03948	18.688	3.1231
4.4500	0.12368	29.986	0.033349	0.0005551	0.0050800	0.0010017	0.010378	0.025720	166.48	0.30472	18.753	3.0392
4.6000	0.14075	29.032	0.034445	0.0041842	0.0090322	0.0017471	0.010481	0.030604	158.80	0.71637	18.811	2.9490
4.7500	0.15942	27.895	0.035849	0.0078334	0.013548	0.0025725	0.010592	0.038735	150.22	1.2278	18.892	2.8488
4.9000	0.17983	26.456	0.037798	0.012155	0.018953	0.0035365	0.010726	0.055286	140.44	1.8978	19.051	2.7312
5.0500	0.20225	24.384	0.041010	0.017863	0.026157	0.0048066	0.010920	0.10769	128.80	2.8635	19.490	2.5752
5.1953	0.22637	17.399	0.057475	0.034500	0.047510	0.0087388			0	5.6142		
2.1768	0.0048565	0.28619	3.4942	0.045977	0.062946	0.033998	0.013890	0.024259	83.225	27.717	3.9767	0.53761
2.2000	0.0051477	0.30076	3.3249	0.046200	0.063316	0.033714	0.013900	0.024319	83.564	27.045	4.0381	0.54487
2.3500	0.0073079	0.40515	2.4682	0.047628	0.065665	0.032039	0.013945	0.024692	85.668	23.375	4.4135	0.59162
2.5000	0.010001	0.52873	1.8913	0.049016	0.067931	0.030582	0.013954	0.025049	87.618	20.607	4.7651	0.63825
2.6500	0.013298	0.67363	1.4845	0.050355	0.070096	0.029283	0.013937	0.025413	89.415	18.480	5.1055	0.68502
2.8000	0.017270	0.84193	1.1878	0.051635	0.072147	0.028106	0.013897	0.025800	91.063	16.819	5.4423	0.73220
2.9500	0.021983	1.0357	0.96551	0.052848	0.074072	0.027023	0.013840	0.026230	92.568	15.504	5.7797	0.77999
3.1000	0.027502	1.2573	0.79536	0.053985	0.075859	0.026018	0.013767	0.026720	93.934	14.450	6.1203	0.82863
3.2500	0.033890	1.5091	0.66264	0.055037	0.077494	0.025074	0.013682	0.027292	95.167	13.597	6.4659	0.87831
3.4000	0.041209	1.7941	0.55738	0.055993	0.078962	0.024180	0.013588	0.027973	96.271	12.902	6.8180	0.92926
3.5500	0.049518	2.1157	0.47265	0.056842	0.080247	0.023324	0.013485	0.028799	97.250	12.329	7.1823	0.98171
3.7000	0.058879	2.4783	0.40351	0.057570	0.081328	0.022495	0.013376	0.029817	98.108	11.856	7.5561	1.0359
3.8500	0.069351	2.8871	0.34637	0.058158	0.082179	0.021684	0.013262	0.031095	98.847	11.460	7.9453	1.0922
4.0000	0.080998	3.3494	0.29856	0.058586	0.082769	0.020881	0.013144	0.032736	99.469	11.127	8.3558	1.1509
4.1500	0.093886	3.8747	0.25808	0.058825	0.083055	0.020074	0.013022	0.034897	99.978	10.841	8.7965	1.2125
4.3000	0.10809	4.4768	0.22337	0.058834	0.082977	0.019249	0.012898	0.037844	100.38	10.587	9.2815	1.2776
4.4500	0.12368	5.1759	0.19320	0.058554	0.082449	0.018388	0.012770	0.042056	100.67	10.349	9.8336	1.3473
4.6000	0.14075	6.0046	0.16654	0.057896	0.081335	0.017465	0.012637	0.048501	100.88	10.104	10.490	1.4230
4.7500	0.15942	7.0212	0.14243	0.056698	0.079403	0.016437	0.012496	0.059461	101.05	9.8116	11.319	1.5075
4.9000	0.17983	8.3467	0.11981	0.054637	0.076183	0.015216	0.012341	0.081908	101.28	9.3953	12.465	1.6065
5.0500	0.20225	10.326	0.096844	0.050816	0.070403	0.013568	0.012146	0.15172	101.95	8.6477	14.402	1.7364
5.1953	0.22637	17.399	0.057475	0.034500	0.047510	0.0087388			0	5.6142		

Single-Phase Properties

100.00	0.10000	0.12010	8.3265	1.2679	2.1006	0.089342	0.012475	0.020791	589.24	-0.54790	73.713	9.7778
350.00	0.10000	0.034351	29.111	4.3860	7.2972	0.11538	0.012472	0.020785	1101.2	-0.61403	173.53	22.154
600.00	0.10000	0.020042	49.896	7.5039	12.494	0.12659	0.012472	0.020785	1441.5	-0.57336	252.40	32.215
850.00	0.10000	0.014148	70.681	10.622	17.690	0.13383	0.012472	0.020786	1715.7	-0.54223	321.87	41.153
1100.0	0.10000	0.010933	91.467	13.740	22.886	0.13919	0.012472	0.020786	1951.7	-0.51882	385.47	49.382
1350.0	0.10000	0.0089085	112.25	16.857	28.083	0.14344	0.012472	0.020786	2162.0	-0.50057	444.91	57.105
100.00	1.0000	1.1854	0.84359	1.2672	2.1108	0.070191	0.012506	0.020837	596.88	-0.54470	74.600	9.8922
350.00	1.0000	0.34233	2.9212	4.3875	7.3086	0.096243	0.012477	0.020782	1104.7	-0.61441	174.19	22.180
600.00	1.0000	0.20005	4.9988	7.5054	12.504	0.10744	0.012474	0.020783	1444.0	-0.57410	253.00	32.231
850.00	1.0000	0.14131	7.0768	10.623	17.700	0.11468	0.012473	0.020784	1717.6	-0.54296	322.43	41.164
1100.0	1.0000	0.10923	9.1550	13.741	22.896	0.12004	0.012473	0.020784	1953.3	-0.51949	386.00	49.391
1350.0	1.0000	0.089021	11.233	16.859	28.092	0.12430	0.012472	0.020784	2163.5	-0.50117	445.41	57.112
100.00	5.0000	5.6058	0.17839	1.2641	2.1560	0.056777	0.012635	0.021029	630.65	-0.53640	79.465	10.374
350.00	5.0000	1.6858	0.59320	4.3938	7.3598	0.082880	0.012500	0.020769	1120.3	-0.61554	176.72	22.297
600.00	5.0000	0.99204	1.0080	7.5120	12.552	0.094074	0.012484	0.020771	1454.9	-0.57711	255.23	32.299
850.00	5.0000	0.70266	1.4232	10.630	17.745	0.10131	0.012479	0.020775	1726.3	-0.54605	324.52	41.212
1100.0	5.0000	0.54393	1.8385	13.747	22.939	0.10667	0.012477	0.020777	1960.6	-0.52237	387.99	49.428
1350.0	5.0000	0.44368	2.2539	16.864	28.134	0.11092	0.012475	0.020779	2169.9	-0.50380	447.32	57.143
100.00	10.000	10.505	0.095194	1.2606	2.2125	0.050975	0.012775	0.021226	672.30	-0.53355	85.417	10.926
350.00	10.000	3.3086	0.30224	4.4013	7.4237	0.077139	0.012527	0.020755	1140.1	-0.61584	179.43	22.435
600.00	10.000	1.9638	0.50921	7.5200	12.612	0.088326	0.012497	0.020756	1468.8	-0.58023	257.42	32.383
850.00	10.000	1.3957	0.71649	10.637	17.802	0.095556	0.012487	0.020763	1737.3	-0.54953	326.55	41.272
1100.0	10.000	1.0823	0.92394	13.754	22.994	0.10091	0.012482	0.020769	1969.9	-0.52572	389.94	49.475
1350.0	10.000	0.88379	1.1315	16.871	28.186	0.10516	0.012479	0.020772	2178.0	-0.50691	449.21	57.181
100.00	50.000	35.384	0.028261	1.2602	2.6732	0.037457	0.013484	0.021607	962.59	-0.53310	127.36	14.393
350.00	50.000	14.387	0.069507	4.4519	7.9272	0.063904	0.012710	0.020725	1299.0	-0.59388	197.39	23.353
600.00	50.000	9.0593	0.11038	7.5781	13.097	0.075052	0.012587	0.020666	1584.0	-0.58658	268.59	32.979
850.00	50.000	6.6048	0.15140	10.695	18.265	0.082253	0.012543	0.020682	1829.4	-0.56519	334.90	41.713
1100.0	50.000	5.1933	0.19256	13.811	23.438	0.087587	0.012521	0.020701	2047.7	-0.54419	397.07	49.825
1350.0	50.000	4.2772	0.23380	16.926	28.615	0.091828	0.012508	0.020716	2245.9	-0.52586	455.79	57.471
100.00	100.00	51.724	0.019333	1.3032	3.2366	0.031667	0.013965	0.021327	1228.1	-0.51530	197.79	18.001
350.00	100.00	24.886	0.040183	4.5034	8.5217	0.058239	0.012884	0.020817	1482.2	-0.55155	217.91	24.221
600.00	100.00	16.512	0.060563	7.6393	13.696	0.069398	0.012683	0.020630	1727.9	-0.57040	281.89	33.590
850.00	100.00	12.359	0.080910	10.759	18.850	0.076579	0.012605	0.020615	1948.4	-0.56473	342.93	42.188
1350.0	100.00	8.2113	0.12178	16.988	29.166	0.086124	0.012542	0.020653	2335.4	-0.53766	458.27	57.800

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is McCarty, R. D., and Arp, V. D., “A New Wide Range Equation of State for Helium,” *Adv. Cryo. Eng.* **35**:1465–1475, 1990. The source for viscosity is Arp, V. D., McCarty, R. D., and Friend, D. G., “Thermophysical Properties of Helium-4 from 0.8 to 1500 K with Pressures to 2000 MPa,” *NIST Technical Note* 1334, Boulder, Colo., 1998. The source for thermal conductivity is Hands, B. A., and Arp, V. D., “A Correlation of Thermal Conductivity Data for Helium,” *Cryogenics*, **21**(12):697–703, 1981.

Properties at the triple point and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties of the equation of state range from 1% at low temperatures (<20 K) to 0.1% at temperatures between 200 and 400 K, and from 3% in the speed of sound in the liquid phase to 0.1% in the speed of sound between 100 and 500 K. The uncertainty in heat capacities is about 5%. The uncertainty in viscosity is 10%. The uncertainty in thermal conductivity is 5%, except at low temperatures where it increases to 10%.

TABLE 2-220 Thermodynamic Properties of Heptane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
182.55	1.7549E-07	7.7457	0.12910	-41.621	-41.621	-0.15398	0.14886	0.19662	1669.5	-0.52328	155.74	3857.1
185.00	2.5788E-07	7.7246	0.12946	-41.139	-41.139	-0.15136	0.14924	0.19687	1656.1	-0.52236	155.56	3523.8
205.00	4.0249E-06	7.5549	0.13236	-37.178	-37.178	-0.13103	0.15264	0.19936	1550.9	-0.51267	153.03	1889.1
225.00	3.5822E-05	7.3882	0.13535	-33.158	-33.158	-0.11232	0.15666	0.20287	1452.3	-0.49878	149.21	1175.0
245.00	0.00021055	7.2229	0.13845	-29.056	-29.056	-0.094861	0.16146	0.20746	1358.8	-0.48063	144.65	805.42
265.00	0.00090411	7.0579	0.14169	-24.852	-24.852	-0.078369	0.16706	0.21311	1269.3	-0.45837	139.70	589.93
285.00	0.0030469	6.8917	0.14510	-20.526	-20.525	-0.062631	0.17337	0.21972	1183.0	-0.43224	134.59	452.87
305.00	0.0084921	6.7233	0.14874	-16.058	-16.057	-0.047486	0.18029	0.22715	1099.5	-0.40232	129.47	359.75
325.00	0.020357	6.5515	0.15264	-11.436	-11.433	-0.032809	0.18765	0.23527	1018.0	-0.36834	124.41	293.16
345.00	0.043242	6.3747	0.15687	-6.6463	-6.6396	-0.018509	0.19534	0.24397	938.09	-0.32951	119.45	243.54
365.00	0.083287	6.1914	0.16151	-1.6791	-1.6656	-0.0045150	0.20321	0.25320	859.20	-0.28429	114.61	205.28
385.00	0.14809	5.9995	0.16668	3.4744	3.4991	0.0092301	0.21119	0.26293	780.76	-0.22999	109.90	174.90
405.00	0.24655	5.7965	0.17252	8.8226	8.8651	0.022774	0.21920	0.27327	702.10	-0.16200	105.28	150.12
425.00	0.38874	5.5785	0.17926	14.375	14.445	0.036160	0.22720	0.28444	622.51	-0.072112	100.73	129.39
445.00	0.58581	5.3401	0.18726	20.147	20.256	0.049436	0.23517	0.29693	541.08	0.055234	96.175	111.58
465.00	0.85019	5.0723	0.19715	26.159	26.327	0.062666	0.24315	0.31186	456.64	0.25318	91.522	95.804
485.00	1.1961	4.7587	0.21014	32.455	32.706	0.075947	0.25123	0.33203	367.46	0.60537	86.595	81.246
505.00	1.6406	4.3628	0.22921	39.129	39.506	0.089485	0.25971	0.36700	270.69	1.3922	81.058	66.925
525.00	2.2070	3.7651	0.26559	46.501	47.087	0.10393	0.26971	0.48246	161.32	4.2679	74.208	50.757
540.13	2.7311	2.3153	0.43191	55.157	56.336	0.12095			0	26.539		
182.55	1.7549E-07	1.1562E-07	8,649,000.	1.0688	2.5867	0.088185	0.11339	0.12171	127.51	478.91	4.8076	0.93944
185.00	2.5788E-07	1.6766E-07	5,964,600.	1.3477	2.8859	0.086612	0.11424	0.12255	128.33	454.07	4.9177	1.6197
205.00	4.0249E-06	2.3614E-06	423,480.	3.7004	5.4048	0.076690	0.12103	0.12935	134.83	301.84	5.8582	3.8170
225.00	3.5822E-05	1.9151E-05	52,218.	6.1891	8.0596	0.070867	0.12796	0.13628	140.99	208.67	6.8855	4.4411
245.00	0.00021055	0.00010342	9,669.6	8.8178	10.854	0.068037	0.13529	0.14363	146.83	148.90	8.0159	4.8725
265.00	0.00090411	0.00041102	2,433.0	11.592	13.792	0.067457	0.14313	0.15153	152.32	109.17	9.2615	5.2813
285.00	0.0030469	0.0012912	774.44	14.516	16.875	0.068598	0.15149	0.16002	157.38	82.064	10.631	5.6934
305.00	0.0084921	0.0033793	295.92	17.589	20.102	0.071069	0.16033	0.16910	161.89	63.190	12.128	6.1135
325.00	0.020357	0.0076646	130.47	20.809	23.465	0.074569	0.16957	0.17874	165.69	49.846	13.757	6.5434
345.00	0.043242	0.015529	64.395	24.169	26.954	0.078862	0.17912	0.18891	168.60	40.301	15.521	6.9849
365.00	0.083287	0.028777	34.750	27.660	30.555	0.083760	0.18889	0.19961	170.42	33.422	17.424	7.4417
385.00	0.14809	0.049694	20.123	31.271	34.251	0.089105	0.19880	0.21088	170.92	28.458	19.473	7.9194
405.00	0.24655	0.081188	12.317	34.984	38.021	0.094763	0.20878	0.22285	169.84	24.914	21.682	8.4272
425.00	0.38874	0.12711	7.8674	38.780	41.838	0.10061	0.21881	0.23587	166.88	22.470	24.071	8.9798
445.00	0.58581	0.19291	5.1838	42.627	45.663	0.10653	0.22889	0.25064	161.58	20.951	26.681	9.6020
465.00	0.85019	0.28714	3.4827	46.478	49.439	0.11237	0.23906	0.26877	153.35	20.312	29.592	10.339
485.00	1.1961	0.42516	2.3521	50.254	53.067	0.11793	0.24945	0.29447	141.27	20.689	32.980	11.280
505.00	1.6406	0.64076	1.5607	53.786	56.346	0.12283	0.26036	0.34231	123.97	22.542	37.355	12.649
525.00	2.2070	1.0396	0.96195	56.608	58.731	0.12611	0.27248	0.50359	99.655	26.764	45.104	15.245
540.13	2.7311	2.3153	0.43191	55.157	56.336	0.12095			0	26.539		

Single-Phase Properties

200.00	0.10000	7.5975	0.13162	-38.176	-38.163	-0.13596	0.15175	0.19864	1576.9	-0.51551	153.84	2174.6
300.00	0.10000	6.7666	0.14778	-17.194	-17.180	-0.051241	0.17852	0.22520	1120.8	-0.41037	130.79	380.51
371.09	0.10000	6.1340	0.16303	-0.13101	-0.11470	-0.00030834	0.20564	0.25610	835.31	-0.26890	113.16	195.32
371.09	0.10000	0.034210	29.231	28.747	31.670	0.085345	0.19189	0.20297	170.72	31.736	18.032	7.5845
400.00	0.10000	0.031296	31.953	34.506	37.702	0.10099	0.20404	0.21430	179.24	23.441	20.824	8.1460
500.00	0.10000	0.024471	40.864	57.027	61.114	0.15307	0.24438	0.25347	203.88	10.262	31.544	10.052
600.00	0.10000	0.020221	49.453	83.290	88.236	0.20242	0.27936	0.28808	224.61	5.6176	43.213	11.883
200.00	1.0000	7.6022	0.13154	-38.202	-38.071	-0.13609	0.15181	0.19861	1581.3	-0.51586	154.03	2203.0
300.00	1.0000	6.7756	0.14759	-17.244	-17.096	-0.051406	0.17857	0.22503	1127.5	-0.41229	131.16	385.02
400.00	1.0000	5.8668	0.17045	7.3729	7.5433	0.019169	0.21722	0.26994	732.61	-0.19065	107.01	158.14
474.30	1.0000	4.9336	0.20269	29.049	29.251	0.068829	0.24689	0.32029	415.88	0.38883	89.279	88.930
474.30	1.0000	0.34452	2.9026	48.251	51.154	0.11501	0.24385	0.27933	148.28	20.343	31.091	10.743
500.00	1.0000	0.30146	3.1372	54.943	58.260	0.12960	0.25108	0.27563	166.14	14.530	33.576	11.093
600.00	1.0000	0.22064	4.5322	82.131	86.663	0.18133	0.28189	0.29560	207.02	6.3945	44.625	12.599
200.00	5.0000	7.6229	0.13118	-38.316	-37.661	-0.13667	0.15208	0.19848	1600.2	-0.51732	154.89	2331.7
300.00	5.0000	6.8141	0.14675	-17.456	-16.722	-0.052121	0.17879	0.22435	1156.4	-0.42010	132.76	405.07
400.00	5.0000	5.9526	0.16799	6.9316	7.7715	0.018047	0.21734	0.26714	783.30	-0.23205	109.78	169.17
500.00	5.0000	4.8210	0.20743	35.998	37.035	0.083125	0.25622	0.32159	423.74	0.32892	89.492	84.960
600.00	5.0000	2.2376	0.44691	72.724	74.958	0.15188	0.29509	0.43674	134.50	7.5010	63.671	27.947
200.00	10.000	7.6479	0.13075	-38.454	-37.147	-0.13737	0.15241	0.19835	1623.3	-0.51893	155.94	2498.6
300.00	10.000	6.8595	0.14578	-17.705	-16.247	-0.052975	0.17908	0.22363	1190.5	-0.42839	134.68	430.15
400.00	10.000	6.0454	0.16542	6.4514	8.1056	0.016799	0.21753	0.26465	838.61	-0.26850	112.88	182.05
500.00	10.000	5.0885	0.19652	34.818	36.783	0.080607	0.25576	0.30908	532.87	0.040826	95.612	98.261
600.00	10.000	3.8337	0.26084	67.259	69.867	0.14081	0.29039	0.35093	303.32	0.87871	82.499	55.622
300.00	50.000	7.1493	0.13987	-19.254	-12.260	-0.058701	0.18121	0.22068	1412.7	-0.46360	147.62	635.00
400.00	50.000	6.5292	0.15316	3.9425	11.600	0.0096885	0.21935	0.25730	1139.8	-0.37509	130.89	269.35
500.00	50.000	5.9397	0.16836	30.741	39.159	0.071034	0.25677	0.29309	940.60	-0.29668	119.90	158.19
600.00	50.000	5.3846	0.18572	60.741	70.026	0.12723	0.28945	0.32327	800.70	-0.23327	114.54	109.31
300.00	100.00	7.4106	0.13494	-20.560	-7.0660	-0.064263	0.18354	0.21976	1622.2	-0.47764	160.40	909.65
400.00	100.00	6.8927	0.14508	2.1360	16.644	0.0037009	0.22143	0.25527	1385.0	-0.40643	146.71	364.81
500.00	100.00	6.4248	0.15565	28.397	43.961	0.064512	0.25865	0.29041	1216.1	-0.35005	137.75	209.24
600.00	100.00	6.0005	0.16665	57.894	74.559	0.12021	0.29113	0.32068	1095.6	-0.31028	133.53	143.14

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Span, R., and Wagner, W., "Equations of State for Technical Applications. II. Results for Nonpolar Fluids," *Int. J. Thermophys.*, **24**(1):41–109, 2003. The source for viscosity is NIST14, Version 9.08. The source for thermal conductivity is NIST14, Version 9.08.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties of the equation of state are approximately 0.2% (to 0.5% at high pressures) in density, 1% (in the vapor phase) to 2% in heat capacity, 1% (in the vapor phase) to 2% in the speed of sound, and 0.2% in vapor pressure, except in the critical region. For viscosity, estimated uncertainty is 2%. For thermal conductivity, estimated uncertainty, except near the critical region, is 4–6%.

TABLE 2-221 Thermodynamic Properties of Hexane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_v kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
177.83	1.2771E-06	8.8394	0.11313	-30.053	-30.053	-0.11799	0.11891	0.16166	1638.9	-0.55189	155.43	2392.6
180.00	1.7416E-06	8.8166	0.11342	-29.702	-29.702	-0.11602	0.11944	0.16212	1626.3	-0.54996	155.64	2235.0
195.00	1.2006E-05	8.6607	0.11546	-27.247	-27.247	-0.10292	0.12299	0.16527	1542.5	-0.53626	155.51	1474.2
210.00	6.0531E-05	8.5066	0.11756	-24.744	-24.744	-0.090556	0.12642	0.16850	1463.6	-0.52146	153.36	1048.0
225.00	0.00023840	8.3535	0.11971	-22.190	-22.190	-0.078815	0.12992	0.17196	1388.4	-0.50504	149.89	786.13
240.00	0.00077035	8.2006	0.12194	-19.583	-19.583	-0.067597	0.13361	0.17575	1316.2	-0.48667	145.59	613.59
255.00	0.0021200	8.0472	0.12427	-16.916	-16.916	-0.056819	0.13753	0.17992	1246.3	-0.46617	140.79	493.58
270.00	0.0051148	7.8927	0.12670	-14.184	-14.183	-0.046408	0.14172	0.18448	1178.5	-0.44339	135.74	406.45
285.00	0.011066	7.7363	0.12926	-11.380	-11.379	-0.036304	0.14616	0.18943	1112.3	-0.41819	130.60	340.96
300.00	0.021865	7.5774	0.13197	-8.4998	-8.4969	-0.026456	0.15084	0.19474	1047.6	-0.39031	125.50	290.31
315.00	0.040031	7.4151	0.13486	-5.5378	-5.5324	-0.016823	0.15571	0.20040	984.03	-0.35939	120.53	250.18
330.00	0.068721	7.2487	0.13796	-2.4895	-2.4800	-0.0073691	0.16074	0.20638	921.37	-0.32478	115.73	217.72
345.00	0.11169	7.0772	0.14130	0.64962	0.66541	0.0019333	0.16590	0.21269	859.40	-0.28553	111.15	190.99
360.00	0.17326	6.8995	0.14494	3.8834	3.9085	0.011109	0.17114	0.21932	797.86	-0.24020	106.82	168.61
375.00	0.25821	6.7141	0.14894	7.2160	7.2544	0.020180	0.17645	0.22633	736.47	-0.18658	102.74	149.58
390.00	0.37176	6.5194	0.15339	10.652	10.709	0.029166	0.18178	0.23378	674.93	-0.12130	98.910	133.16
405.00	0.51954	6.3129	0.15840	14.195	14.278	0.038087	0.18713	0.24184	612.85	-0.038945	95.319	118.78
420.00	0.70759	6.0916	0.16416	17.854	17.970	0.046965	0.19251	0.25078	549.78	0.069646	91.941	105.97
435.00	0.94243	5.8506	0.17092	21.638	21.799	0.055828	0.19791	0.26113	485.05	0.22117	88.736	94.348
450.00	1.2313	5.5825	0.17913	25.562	25.782	0.064717	0.20338	0.27400	417.67	0.44954	85.638	83.551
465.00	1.5827	5.2736	0.18963	29.656	29.956	0.073697	0.20902	0.29208	346.06	0.83549	82.537	73.177
480.00	2.0071	4.8943	0.20432	33.983	34.393	0.082910	0.21509	0.32426	267.23	1.6275	79.234	62.632
495.00	2.5196	4.3496	0.22990	38.752	39.331	0.092809	0.22251	0.42766	174.59	4.0963	75.402	50.415
507.82	3.0429	2.7059	0.36957	45.698	46.822	0.10743			0	21.950		
177.83	1.2771E-06	8.6377E-07	1,157,700.	6.5712	8.0497	0.096281	0.095572	0.10389	136.57	381.55	5.2567	3.0564
180.00	1.7416E-06	1.1637E-06	859,330.	6.7793	8.2759	0.094966	0.096248	0.10456	137.36	363.60	5.3651	3.2844
195.00	1.2006E-05	7.4054E-06	135,040.	8.2570	9.8783	0.087463	0.10079	0.10910	142.71	265.13	6.1371	4.1334
210.00	6.0531E-05	3.4674E-05	28,840.	9.8013	11.547	0.082256	0.10523	0.11356	147.84	198.42	6.9555	4.5617
225.00	0.00023840	0.00012750	7,842.9	11.411	13.281	0.078835	0.10975	0.11809	152.75	151.75	7.8286	4.9114
240.00	0.00077035	0.00038658	2,586.8	13.086	15.079	0.076827	0.11444	0.12282	157.43	118.24	8.7644	5.2472
255.00	0.0021200	0.0010029	997.06	14.827	16.941	0.075951	0.11936	0.12783	161.83	93.691	9.7697	5.5832
270.00	0.0051148	0.0022917	436.35	16.633	18.865	0.075993	0.12457	0.13317	165.91	75.403	10.851	5.9228
285.00	0.011066	0.0047185	211.93	18.505	20.850	0.076780	0.13005	0.13886	169.59	61.606	12.012	6.2669
300.00	0.021865	0.0089152	112.17	20.440	22.893	0.078177	0.13579	0.14492	172.79	51.090	13.258	6.6161
315.00	0.040031	0.015688	63.742	22.438	24.989	0.080072	0.14178	0.15136	175.43	43.011	14.592	6.9712
330.00	0.068721	0.026023	38.427	24.494	27.135	0.082372	0.14798	0.15817	177.42	36.767	16.018	7.3333
345.00	0.11169	0.041099	24.332	26.606	29.324	0.085000	0.15435	0.16537	178.64	31.926	17.540	7.7043
360.00	0.17326	0.062314	16.048	28.769	31.549	0.087889	0.16085	0.17297	179.01	28.172	19.162	8.0869
375.00	0.25821	0.091349	10.947	30.977	33.804	0.090978	0.16746	0.18106	178.39	25.277	20.892	8.4853
390.00	0.37176	0.13027	7.6763	33.223	36.077	0.094213	0.17415	0.18976	176.64	23.080	22.739	8.9054
405.00	0.51954	0.18173	5.5026	35.497	38.356	0.097538	0.18092	0.19931	173.59	21.471	24.719	9.3563
420.00	0.70759	0.24933	4.0108	37.783	40.621	0.10090	0.18775	0.21014	169.01	20.386	26.858	9.8524
435.00	0.94243	0.33826	2.9563	40.061	42.848	0.10422	0.19467	0.22312	162.60	19.809	29.203	10.417
450.00	1.2313	0.45679	2.1892	42.299	44.994	0.10741	0.20173	0.24003	153.94	19.784	31.843	11.090
465.00	1.5827	0.61964	1.6138	44.438	46.992	0.11033	0.20903	0.26537	142.42	20.448	34.979	11.951
480.00	2.0071	0.85826	1.1651	46.363	48.702	0.11272	0.21678	0.31367	127.17	22.098	39.160	13.186
495.00	2.5196	1.2664	0.78962	47.752	49.742	0.11384	0.22558	0.46871	106.85	25.172	46.755	15.416
507.82	3.0429	2.7059	0.36957	45.698	46.822	0.10743			0	21.950		

Single-Phase Properties

200.00	0.10000	8.6098	0.11615	-26.420	-26.409	-0.098737	0.12414	0.16632	1516.3	-0.53154	155.01	1308.5
300.00	0.10000	7.5784	0.13195	-8.5042	-8.4910	-0.026471	0.15084	0.19472	1048.3	-0.39056	125.55	290.66
341.45	0.10000	7.1184	0.14048	-0.10255	-0.088504	-0.00025850	0.16467	0.21117	874.04	-0.29532	112.22	196.89
341.45	0.10000	0.037028	27.006	26.101	28.801	0.084352	0.15282	0.16363	178.43	32.965	17.170	7.6155
400.00	0.10000	0.030895	32.368	35.725	38.962	0.11177	0.17421	0.18376	196.30	17.891	23.256	8.9061
500.00	0.10000	0.024344	41.078	54.980	59.088	0.15654	0.20940	0.21824	221.55	8.1885	35.216	11.054
600.00	0.10000	0.020168	49.583	77.506	82.464	0.19908	0.23990	0.24850	243.38	4.6052	48.247	13.104
200.00	1.0000	8.6159	0.11606	-26.445	-26.329	-0.098862	0.12419	0.16628	1521.0	-0.53204	155.38	1327.2
300.00	1.0000	7.5906	0.13174	-8.5541	-8.4223	-0.026637	0.15088	0.19451	1055.7	-0.39338	126.12	294.62
400.00	1.0000	6.4028	0.15618	12.928	13.084	0.034934	0.18534	0.23836	642.57	-0.081596	96.993	124.96
438.24	1.0000	5.7953	0.17255	22.473	22.645	0.057745	0.19908	0.26364	470.76	0.26228	88.060	91.958
438.24	1.0000	0.36099	2.7701	40.550	43.320	0.10492	0.19618	0.22634	160.93	19.754	29.744	10.551
500.00	1.0000	0.27702	3.6098	53.661	57.271	0.13470	0.21315	0.22970	196.29	10.036	36.713	11.711
600.00	1.0000	0.21383	4.6765	76.704	81.381	0.17859	0.24143	0.25320	230.76	5.0002	49.365	13.575
200.00	5.0000	8.6424	0.11571	-26.553	-25.975	-0.099407	0.12440	0.16610	1541.5	-0.53416	157.01	1411.9
300.00	5.0000	7.6426	0.13085	-8.7667	-8.1124	-0.027355	0.15106	0.19368	1087.5	-0.40464	128.58	312.23
400.00	5.0000	6.5335	0.15306	12.434	13.199	0.033675	0.18536	0.23424	702.66	-0.15548	100.50	136.08
500.00	5.0000	4.8882	0.20457	38.358	39.381	0.091842	0.22066	0.30042	311.48	1.0544	81.341	64.368
600.00	5.0000	1.5349	0.65149	71.648	74.906	0.15663	0.24956	0.30605	173.73	6.3809	59.195	20.153
200.00	10.000	8.6744	0.11528	-26.683	-25.530	-0.10007	0.12466	0.16591	1566.5	-0.53649	159.01	1521.9
300.00	10.000	7.7032	0.12982	-9.0138	-7.7156	-0.028204	0.15128	0.19280	1124.9	-0.41638	131.56	334.21
400.00	10.000	6.6684	0.14996	11.917	13.416	0.032326	0.18546	0.23093	765.76	-0.21470	104.41	148.74
500.00	10.000	5.3956	0.18533	36.755	38.608	0.088363	0.21933	0.27365	455.87	0.25439	88.518	80.764
600.00	10.000	3.6515	0.27386	65.462	68.201	0.14220	0.24998	0.31536	248.16	1.7834	79.582	43.840
300.00	50.000	8.0787	0.12378	-10.511	-4.3222	-0.033766	0.15301	0.18931	1362.7	-0.46412	152.57	513.27
400.00	50.000	7.3187	0.13664	9.3999	16.232	0.025137	0.18689	0.22225	1090.0	-0.36559	128.45	231.13
500.00	50.000	6.5956	0.15162	32.487	40.068	0.078194	0.21979	0.25371	895.54	-0.27952	115.10	140.55
600.00	50.000	5.9215	0.16888	58.351	66.794	0.12685	0.24844	0.27993	763.82	-0.21072	111.13	100.20
300.00	100.00	8.4063	0.11896	-11.737	0.15835	-0.039032	0.15494	0.18811	1582.2	-0.48273	175.12	753.68
400.00	100.00	7.7789	0.12855	7.6799	20.535	0.019368	0.18864	0.21997	1345.0	-0.40550	152.26	319.21
500.00	100.00	7.2120	0.13866	30.244	44.110	0.071845	0.22138	0.25089	1178.8	-0.34615	138.03	188.34
600.00	100.00	6.7006	0.14924	55.635	70.559	0.11999	0.24989	0.27732	1062.8	-0.30501	132.17	132.10

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Span, R., and Wagner, W., “Equations of State for Technical Applications. II. Results for Nonpolar Fluids,” *Int. J. Thermophys.* **24**(1):41–109, 2003. The source for viscosity and thermal conductivity is NIST14, Version 9.08.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties of the equation of state are approximately 0.2% (to 0.5% at high pressures) in density, 1% (in the vapor phase) to 2% in heat capacity, 1% (in the vapor phase) to 2% in the speed of sound, and 0.2% in vapor pressure, except in the critical region. For viscosity, estimated uncertainty is 2%. For thermal conductivity, estimated uncertainty, except near the critical region, is 4–6 %.

2-278 PHYSICAL AND CHEMICAL DATA

TABLE 2-222 Saturated Hydrazine

Temperature, K	Pressure, bar	v_f , m ³ /kg	v_g , m ³ /kg	h_f , kJ/kg	h_g , kJ/kg	s_f , kJ/(kg·K)	s_g , kJ/(kg·K)
386.6	1.013	0.001 053	0.9833	-105.9	65.4	0.5994	1.0426
390	1.135	0.001 060	0.8850	-104.8	66.0	0.6029	1.0409
400	1.560	0.001 081	0.6579	-101.4	68.2	0.6120	1.0360
410	2.102	0.001 104	0.4994	-97.6	70.6	0.6211	1.0314
420	2.786	0.001 127	0.3850	-93.9	73.0	0.6300	1.0275
440	4.732	0.001 178	0.2355	-86.1	77.6	0.6492	1.0212
460	7.610	0.001 235	0.1500	-76.9	82.1	0.6707	1.0163
480	11.76	0.001 299	0.1005	-67.1	86.6	0.6916	1.0118
500	17.42	0.001 374	0.0690	-57.3	90.8	0.7124	1.0086
520	29.59	0.001 460	0.0407	-47.8	94.6	0.7320	1.0058
540	34.75	0.001 563	0.0353	-36.0	97.7	0.7566	1.0042
560	47.09	0.001 681	0.0263	-25.2	101.2	0.7762	1.0020
580	62.44	0.001 835	0.0196	-12.4	103.6	0.8002	1.0002
600	81.17	0.002 045	0.0142	5.2	104.2	0.8335	0.9988
620	102.7	0.002 320	0.0106	23.2	103.6	0.8671	0.9967
640	128.1	0.002 86	0.0074	45.9	98.1	0.9035	0.9906
653 ^c	146.9	0.004 33	0.0043	83.7	83.7	0.9715	0.9715

Converted from E. F. Fricke, Republic Aviation Co. rept. F-5028-101. c = critical point.

TABLE 2-223 Thermodynamic Properties of Normal Hydrogen

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
13.957	0.0077031	38.148	0.026214	-0.10434	-0.10414	-0.0059480	0.011064	0.015654	1361.1	-1.4137	76.293	25.463
14.000	0.0078936	38.129	0.026226	-0.10367	-0.10346	-0.0059000	0.010957	0.015547	1359.6	-1.4230	76.650	25.310
15.000	0.013436	37.701	0.026524	-0.088896	-0.088539	-0.0048799	0.0096961	0.014420	1318.5	-1.5204	84.106	22.215
16.000	0.021534	37.261	0.026838	-0.074446	-0.073868	-0.0039471	0.0096482	0.014709	1271.6	-1.4695	90.079	19.784
17.000	0.032848	36.802	0.027172	-0.059414	-0.058521	-0.0030355	0.010003	0.015557	1226.6	-1.3623	94.784	17.815
18.000	0.048078	36.321	0.027533	-0.043440	-0.042116	-0.0021219	0.010462	0.016642	1185.5	-1.2409	98.405	16.182
19.000	0.067960	35.812	0.027923	-0.026375	-0.024477	-0.0011983	0.010915	0.017842	1147.3	-1.1194	101.10	14.799
20.000	0.093249	35.274	0.028350	-0.0081516	-0.0055080	-0.00026211	0.011323	0.019120	1110.7	-1.0003	103.01	13.607
21.000	0.12472	34.702	0.028817	0.011274	0.014868	0.00068790	0.011677	0.020476	1074.7	-0.88232	104.24	12.565
22.000	0.16314	34.092	0.029333	0.031947	0.036732	0.0016528	0.011978	0.021935	1038.2	-0.76268	104.87	11.641
23.000	0.20932	33.439	0.029905	0.053929	0.060188	0.0026344	0.012235	0.023539	1000.5	-0.63795	104.98	10.811
24.000	0.26406	32.738	0.030546	0.077308	0.085375	0.0036357	0.012457	0.025351	960.99	-0.50414	104.60	10.057
25.000	0.32818	31.979	0.031271	0.10222	0.11248	0.0046610	0.012655	0.027465	919.10	-0.35648	103.79	9.3625
26.000	0.40250	31.152	0.032101	0.12884	0.14176	0.0057167	0.012840	0.030024	874.29	-0.18882	102.53	8.7151
27.000	0.48788	30.242	0.033067	0.15744	0.17357	0.0068122	0.013025	0.033265	826.00	0.0073109	100.83	8.1034
28.000	0.58524	29.225	0.034217	0.18843	0.20846	0.0079614	0.013224	0.037610	773.58	0.24446	98.654	7.5160
29.000	0.69554	28.067	0.035629	0.22245	0.24723	0.0091865	0.013460	0.043909	716.22	0.54283	95.935	6.9409
30.000	0.81989	26.706	0.037444	0.26061	0.29132	0.010527	0.013764	0.054194	652.73	0.93827	92.547	6.3620
31.000	0.95964	25.017	0.039973	0.30524	0.34360	0.012063	0.014198	0.074872	581.16	1.5038	88.221	5.7518
32.000	1.1168	22.637	0.044175	0.36302	0.41236	0.014035	0.014926	0.14185	497.24	2.4292	82.176	5.0391
33.190	1.3301	14.940	0.066934	0.53004	0.61907	0.020012			0	5.3208		
13.957	0.0077031	0.067540	14.806	0.68715	0.80120	0.058918	0.013157	0.021964	304.61	31.943	10.375	0.66345
14.000	0.0078936	0.069018	14.489	0.68764	0.80201	0.058777	0.013129	0.021944	305.17	31.808	10.431	0.66695
15.000	0.013436	0.11050	9.0494	0.69864	0.82024	0.055705	0.012872	0.021898	316.15	28.572	11.624	0.74268
16.000	0.021534	0.16764	5.9651	0.70899	0.83745	0.053010	0.012907	0.022199	325.05	25.724	12.681	0.81064
17.000	0.032848	0.24349	4.1069	0.71875	0.85365	0.050622	0.012992	0.022618	333.00	23.407	13.681	0.87421
18.000	0.048078	0.34126	2.9303	0.72783	0.86871	0.048480	0.013083	0.023121	340.22	21.522	14.669	0.93555
19.000	0.067960	0.46437	2.1535	0.73614	0.88249	0.046537	0.013178	0.023724	346.75	19.961	15.675	0.99611
20.000	0.093249	0.61652	1.6220	0.74359	0.89484	0.044755	0.013280	0.024449	352.59	18.642	16.716	1.0569
21.000	0.12472	0.80187	1.2471	0.75005	0.90558	0.043103	0.013392	0.025329	357.75	17.507	17.806	1.1186
22.000	0.16314	1.0251	0.97549	0.75541	0.91455	0.041554	0.013514	0.026401	362.25	16.513	18.956	1.1819
23.000	0.20932	1.2919	0.77406	0.75951	0.92154	0.040085	0.013650	0.027724	366.11	15.629	20.180	1.2472
24.000	0.26406	1.6089	0.62153	0.76218	0.92630	0.038674	0.013802	0.029376	369.34	14.829	21.493	1.3151
25.000	0.32818	1.9848	0.50383	0.76318	0.92853	0.037303	0.013973	0.031482	371.95	14.091	22.916	1.3863
26.000	0.40250	2.4307	0.41141	0.76224	0.92783	0.035950	0.014167	0.034234	373.96	13.396	24.477	1.4619
27.000	0.48788	2.9618	0.33763	0.75895	0.92368	0.034594	0.014392	0.037960	375.38	12.726	26.218	1.5433
28.000	0.58524	3.6003	0.27775	0.75276	0.91531	0.033206	0.014655	0.043253	376.19	12.061	28.202	1.6331
29.000	0.69554	4.3810	0.22826	0.74277	0.90154	0.031749	0.014971	0.051322	376.39	11.374	30.535	1.7362
30.000	0.81989	5.3643	0.18642	0.72747	0.88031	0.030160	0.015358	0.065054	375.97	10.624	33.407	1.8628
31.000	0.95964	6.6763	0.14978	0.70374	0.84748	0.028317	0.015854	0.093486	374.91	9.7362	37.226	2.0375
32.000	1.1168	8.6823	0.11518	0.66274	0.79136	0.025879	0.016535	0.18606	373.31	8.5059	43.200	2.3378
33.190	1.3301	14.940	0.066934	0.53004	0.61907	0.020012			0	5.3208		

TABLE 2-223 Thermodynamic Properties of Normal Hydrogen (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Single-Phase Properties												
25.000	0.10000	0.50823	1.9676	0.81207	1.0088	0.049309	0.012734	0.022519	403.66	12.894	20.761	1.3142
100.00	0.10000	0.12030	8.3127	1.7949	2.6261	0.079163	0.014263	0.022637	808.92	1.4058	68.334	4.1896
175.00	0.10000	0.068680	14.560	3.0224	4.4785	0.092882	0.018150	0.026480	1026.9	0.13575	117.11	6.1845
250.00	0.10000	0.048077	20.800	4.4642	6.5442	0.10269	0.020003	0.028323	1209.1	-0.22980	160.59	7.9025
325.00	0.10000	0.036986	27.037	5.9947	8.6984	0.11022	0.020681	0.028998	1371.7	-0.38965	197.72	9.4561
400.00	0.10000	0.030054	33.273	7.5545	10.882	0.11626	0.020865	0.029180	1519.7	-0.47650	234.06	10.892
25.000	1.0000	32.746	0.030538	0.089693	0.12023	0.0041410	0.012580	0.025394	985.14	-0.51115	106.80	9.9923
31.268	1.0000	24.474	0.040861	0.31894	0.35980	0.012531	0.014353	0.084709	560.14	1.7031	86.829	5.5759
31.268	1.0000	7.1182	0.14049	0.69511	0.83559	0.027747	0.016014	0.10713	374.52	9.4548	38.524	2.0997
100.00	1.0000	1.2044	0.83027	1.7679	2.5982	0.059751	0.014331	0.023244	817.03	1.3036	70.413	4.2550
175.00	1.0000	0.68243	1.4653	3.0101	4.4754	0.073667	0.018190	0.026659	1035.8	0.11718	118.31	6.2213
250.00	1.0000	0.47788	2.0926	4.4576	6.5501	0.083515	0.020028	0.028401	1217.3	-0.23428	161.46	7.9283
325.00	1.0000	0.36797	2.7176	5.9910	8.7086	0.091062	0.020699	0.029039	1379.1	-0.39039	198.43	9.4759
400.00	1.0000	0.29924	3.3418	7.5525	10.894	0.097113	0.020878	0.029204	1526.4	-0.47605	234.65	10.908
25.000	5.0000	35.661	0.028042	0.046611	0.18682	0.0021443	0.012376	0.020610	1223.4	-0.90198	119.36	13.101
100.00	5.0000	5.9683	0.16755	1.6549	2.4927	0.045314	0.014583	0.025613	865.94	0.86369	80.395	4.5875
175.00	5.0000	3.3132	0.30183	2.9582	4.4673	0.059998	0.018352	0.027370	1077.3	0.032971	123.58	6.3871
250.00	5.0000	2.3268	0.42978	4.4292	6.5781	0.070022	0.020136	0.028723	1254.0	-0.25678	165.19	8.0420
325.00	5.0000	1.7990	0.55587	5.9750	8.7543	0.077631	0.020776	0.029211	1412.0	-0.39563	201.36	9.5631
400.00	5.0000	1.4680	0.68120	7.5440	10.950	0.083710	0.020937	0.029304	1556.2	-0.47537	237.09	10.979
25.000	10.000	37.930	0.026364	0.020221	0.28386	0.00059913	0.012222	0.018499	1402.1	-1.0762	131.12	16.625
100.00	10.000	11.417	0.087588	1.5346	2.4105	0.038585	0.014838	0.027423	955.43	0.39679	94.196	5.1692
175.00	10.000	6.3697	0.15699	2.9000	4.4699	0.053931	0.018532	0.028063	1133.3	-0.068718	130.39	6.6199
250.00	10.000	4.5028	0.22209	4.3966	6.6175	0.064133	0.020260	0.029065	1300.6	-0.28786	169.92	8.1898
325.00	10.000	3.5006	0.28567	5.9563	8.8130	0.071811	0.020867	0.029402	1453.0	-0.40519	205.05	9.6733
400.00	10.000	2.8687	0.34859	7.5339	11.020	0.077921	0.021006	0.029419	1593.1	-0.47687	240.15	11.067
100.00	50.000	31.993	0.031257	1.1768	2.7397	0.023964	0.016349	0.026254	1710.1	-0.57213	192.47	10.534
175.00	50.000	22.700	0.044053	2.6415	4.8442	0.039587	0.019545	0.029321	1632.6	-0.46415	189.52	9.1377
250.00	50.000	17.524	0.057066	4.2297	7.0830	0.050225	0.020999	0.030163	1690.7	-0.46214	211.67	9.7772
325.00	50.000	14.304	0.069911	5.8539	9.3494	0.058153	0.021434	0.030202	1784.8	-0.48443	237.63	10.827
400.00	50.000	12.107	0.082595	7.4773	11.607	0.064404	0.021458	0.029985	1887.2	-0.51016	267.11	11.965
175.00	100.00	33.019	0.030286	2.5589	5.5875	0.033643	0.020316	0.029140	2128.4	-0.52750	282.17	13.079
250.00	100.00	27.257	0.036688	4.1604	7.8292	0.044291	0.021603	0.030346	2125.4	-0.50698	303.19	12.218
325.00	100.00	23.228	0.043051	5.8083	10.113	0.052281	0.021923	0.030469	2170.6	-0.51215	327.83	12.546
400.00	100.00	20.261	0.049356	7.4555	12.391	0.058588	0.021864	0.030246	2235.8	-0.52481	356.83	13.289

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Younglove, B. A., "Thermophysical Properties of Fluids. I. Argon, Ethylene, Parahydrogen, Nitrogen, Nitrogen Trifluoride, and Oxygen," *J. Phys. Chem. Ref. Data*, Suppl. 1, **11**: 1–11, 1982. The source for viscosity is McCarty, R. D., and Weber, L. A., "Thermophysical Properties of Parahydrogen from the Freezing Liquid Line to 5000 R for Pressures to 10,000 psia," *N.B.S. Tech. Note* 617, 1972. The source for thermal conductivity is McCarty, R. D., and Weber, L. A., "Thermophysical Properties of Parahydrogen from the Freezing Liquid Line to 5000 R for Pressures to 10,000 psia," *N.B.S. Tech. Note* 617, 1972.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in density are 0.1% in the liquid phase, 0.25% in the vapor phase, and 0.2% in the supercritical region. The uncertainty in heat capacity is 3%, and the uncertainty in speed of sound is 2% in the liquid phase and 1% elsewhere. The uncertainty in viscosity ranges from 4% to 15%. The uncertainty in thermal conductivity below 100 K is estimated to be 3% below 150 atm and up to 10% below 700 atm. For temperatures around 100 K at low densities, the uncertainty is about 1%. Above 100 K, the uncertainty is estimated to be on the order of 10%.

TABLE 2-224 Thermodynamic Properties of *para*-Hydrogen

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _p kJ/(mol·K)	C _v kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
13.800	0.0070373	38.214	0.026168	−0.10640	−0.10621	−0.0060986	0.010937	0.015530	1376.3	−1.4263	75.249	26.035
14.000	0.0078936	38.129	0.026226	−0.10334	−0.10313	−0.0058783	0.010548	0.015138	1367.4	−1.4615	76.869	25.310
15.000	0.013436	37.701	0.026524	−0.088758	−0.088401	−0.0048719	0.0096208	0.014345	1320.2	−1.5284	84.107	22.215
16.000	0.021534	37.261	0.026838	−0.074351	−0.073773	−0.0039418	0.0096210	0.014682	1272.2	−1.4722	90.003	19.784
17.000	0.032848	36.802	0.027172	−0.059344	−0.058452	−0.0030317	0.0099778	0.015532	1227.2	−1.3645	94.698	17.815
18.000	0.048078	36.321	0.027533	−0.043395	−0.042071	−0.0021196	0.010438	0.016618	1186.0	−1.2427	98.339	16.182
19.000	0.067960	35.812	0.027923	−0.026352	−0.024454	−0.0011972	0.010895	0.017822	1147.7	−1.1207	101.06	14.799
20.000	0.093249	35.274	0.028350	−0.0081471	−0.0055035	−0.00026188	0.011306	0.019103	1111.1	−1.0012	103.00	13.607
21.000	0.12472	34.702	0.028817	0.011263	0.014857	0.00068738	0.011663	0.020462	1075.0	−0.88292	104.25	12.565
22.000	0.16314	34.092	0.029333	0.031923	0.036709	0.0016517	0.011966	0.021923	1038.4	−0.76309	104.89	11.641
23.000	0.20932	33.439	0.029905	0.053894	0.060154	0.0026329	0.012226	0.023529	1000.7	−0.63820	105.01	10.811
24.000	0.26406	32.738	0.030546	0.077266	0.085332	0.0036338	0.012450	0.025344	961.13	−0.50428	104.64	10.057
25.000	0.32818	31.979	0.031271	0.10217	0.11243	0.0046588	0.012652	0.027461	919.17	−0.35652	103.82	9.3625
26.000	0.40250	31.152	0.032101	0.12879	0.14171	0.0057145	0.012841	0.030025	874.28	−0.18881	102.56	8.7151
27.000	0.48788	30.242	0.033067	0.15739	0.17353	0.0068101	0.013030	0.033270	825.90	0.0073097	100.85	8.1034
28.000	0.58524	29.225	0.034217	0.18839	0.20842	0.0079596	0.013235	0.037620	773.39	0.24439	98.669	7.5160
29.000	0.69554	28.067	0.035629	0.22242	0.24721	0.0091852	0.013476	0.043924	715.93	0.54264	95.941	6.9409
30.000	0.81989	26.706	0.037444	0.26061	0.29131	0.010526	0.013785	0.054215	652.36	0.93791	92.543	6.3620
31.000	0.95964	25.017	0.039973	0.30526	0.34362	0.012063	0.014225	0.074899	580.72	1.5033	88.209	5.7518
32.000	1.1168	22.637	0.044175	0.36307	0.41240	0.014036	0.014957	0.14188	496.76	2.4286	82.155	5.0391
32.938	1.2838	15.556	0.064284	0.51279	0.59532	0.019385			0	5.2019		
13.800	0.0070373	0.062340	16.041	0.68576	0.79865	0.059471	0.012705	0.021484	305.17	33.288	10.454	0.65046
14.000	0.0078936	0.069018	14.489	0.68798	0.80235	0.058799	0.012720	0.021535	307.13	32.412	10.649	0.66695
15.000	0.013436	0.11050	9.0494	0.69878	0.82037	0.055713	0.012797	0.021823	316.53	28.670	11.625	0.74268
16.000	0.021534	0.16764	5.9651	0.70909	0.83754	0.053015	0.012879	0.022172	325.20	25.755	12.604	0.81064
17.000	0.032848	0.24349	4.1069	0.71881	0.85372	0.050625	0.012967	0.022593	333.13	23.433	13.595	0.87421
18.000	0.048078	0.34126	2.9303	0.72787	0.86876	0.048482	0.013059	0.023097	340.35	21.544	14.603	0.93555
19.000	0.067960	0.46437	2.1535	0.73616	0.88251	0.046538	0.013158	0.023703	346.87	19.978	15.637	0.99611
20.000	0.093249	0.61652	1.6220	0.74359	0.89484	0.044755	0.013263	0.024433	352.69	18.654	16.705	1.0569
21.000	0.12472	0.80187	1.2471	0.75004	0.90557	0.043102	0.013378	0.025315	357.84	17.516	17.816	1.1186
22.000	0.16314	1.0251	0.97549	0.75539	0.91453	0.041553	0.013503	0.026390	362.33	16.521	18.981	1.1819
23.000	0.20932	1.2919	0.77406	0.75948	0.92151	0.040083	0.013641	0.027714	366.17	15.634	20.214	1.2472
24.000	0.26406	1.6089	0.62153	0.76213	0.92626	0.038672	0.013795	0.029369	369.39	14.832	21.529	1.3151
25.000	0.32818	1.9848	0.50383	0.76314	0.92848	0.037301	0.013969	0.031478	371.98	14.092	22.951	1.3863
26.000	0.40250	2.4307	0.41141	0.76219	0.92778	0.035948	0.014168	0.034235	373.96	13.396	24.507	1.4619
27.000	0.48788	2.9618	0.33763	0.75891	0.92363	0.034592	0.014397	0.037965	375.33	12.725	26.241	1.5433
28.000	0.58524	3.6003	0.27775	0.75272	0.91527	0.033204	0.014666	0.043264	376.10	12.058	28.217	1.6331
29.000	0.69554	4.3810	0.22826	0.74275	0.90151	0.031747	0.014986	0.051337	376.25	11.370	30.541	1.7362
30.000	0.81989	5.3643	0.18642	0.72747	0.88031	0.030159	0.015380	0.065075	375.77	10.621	33.404	1.8628
31.000	0.95964	6.6763	0.14978	0.70375	0.84749	0.028317	0.015881	0.093513	374.65	9.7335	37.214	2.0375
32.000	1.1168	8.6823	0.11518	0.66278	0.79141	0.025880	0.016566	0.18609	372.98	8.5044	43.179	2.3378
32.938	1.2838	15.556	0.064284	0.51279	0.59532	0.019385			0	5.2019		
Single-Phase Properties												
50.000	0.10000	0.24255	4.1229	1.1351	1.5474	0.064284	0.012710	0.021301	583.04	4.8322	38.576	2.4717
125.00	0.10000	0.096180	10.397	2.4062	3.4459	0.086764	0.022470	0.030820	841.25	0.59482	106.18	4.9004
200.00	0.10000	0.060094	16.641	4.2197	5.8837	0.10202	0.024087	0.032413	1054.3	−0.023305	155.64	6.7798
275.00	0.10000	0.043708	22.879	5.9434	8.2313	0.11200	0.022011	0.030330	1250.9	−0.27924	183.10	8.4358
350.00	0.10000	0.034345	29.116	7.5556	10.467	0.11919	0.021169	0.029486	1418.7	−0.41834	212.22	9.9461
50.000	1.0000	2.6201	0.38166	1.0703	1.4520	0.043845	0.012853	0.024464	575.43	4.4711	42.291	2.6080
125.00	1.0000	0.95806	1.0438	2.3861	3.4298	0.067460	0.022526	0.031195	848.73	0.55405	107.83	4.9523
200.00	1.0000	0.59708	1.6748	4.2097	5.8845	0.082821	0.024121	0.032545	1062.6	−0.032981	156.70	6.8120
275.00	1.0000	0.43458	2.3010	5.9379	8.2390	0.092837	0.022033	0.030392	1258.8	−0.28185	183.91	8.4592
350.00	1.0000	0.34179	2.9257	7.5525	10.478	0.10004	0.021186	0.029519	1425.9	−0.41854	212.88	9.9645

TABLE 2-224 Thermodynamic Properties of para-Hydrogen (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_v kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Single-Phase Properties												
50.000	5.0000	16.715	0.059826	0.73484	1.0340	0.024551	0.013420	0.041376	692.39	1.6074	79.882	4.3382
125.00	5.0000	4.6789	0.21372	2.3021	3.3707	0.053441	0.022747	0.032653	888.35	0.37288	115.33	5.1978
200.00	5.0000	2.9000	0.34482	4.1677	5.8918	0.069234	0.024261	0.033076	1100.2	-0.078236	161.31	6.9555
275.00	5.0000	2.1189	0.47194	5.9145	8.2742	0.079371	0.022129	0.030649	1293.6	-0.29566	187.32	8.5624
350.00	5.0000	1.6731	0.59770	7.5394	10.528	0.086623	0.021255	0.029661	1457.5	-0.42110	215.63	10.045
50.000	10.000	26.141	0.038254	0.53828	0.92082	0.017729	0.013420	0.031369	1020.9	-0.00090690	109.43	6.6673
125.00	10.000	8.9624	0.11158	2.2107	3.3264	0.047053	0.022979	0.033916	952.54	0.16557	125.40	5.5904
200.00	10.000	5.5879	0.17896	4.1201	5.9097	0.063249	0.024419	0.033613	1149.8	-0.13543	167.23	7.1499
275.00	10.000	4.1087	0.24339	5.8874	8.3212	0.073512	0.022240	0.030928	1337.5	-0.31595	191.64	8.6948
350.00	10.000	3.2606	0.30669	7.5240	10.591	0.080815	0.021338	0.029822	1497.0	-0.42707	219.08	10.147
50.000	50.000	41.635	0.024018	0.33890	1.5398	0.0075571	0.014831	0.021072	1903.0	-0.85893	205.72	18.078
125.00	50.000	28.231	0.035422	1.8792	3.6503	0.032377	0.024288	0.034391	1571.6	-0.41268	207.03	9.5718
200.00	50.000	20.660	0.048403	3.8955	6.3157	0.049075	0.025325	0.034876	1590.0	-0.39075	219.10	9.2598
275.00	50.000	16.295	0.061368	5.7445	8.8129	0.059704	0.022913	0.031923	1701.1	-0.44355	229.75	10.106
350.00	50.000	13.485	0.074155	7.4386	11.146	0.067214	0.021862	0.030536	1813.9	-0.48674	249.51	11.203
125.00	100.00	38.364	0.026066	1.8103	4.4169	0.026596	0.025170	0.033554	2093.0	-0.46785	308.69	15.278
200.00	100.00	30.845	0.032420	3.8148	7.0568	0.043119	0.026034	0.034871	2056.2	-0.43974	320.11	12.564
275.00	100.00	25.763	0.038815	5.6828	9.5643	0.053790	0.023474	0.032151	2114.9	-0.48043	324.21	12.252
350.00	100.00	22.144	0.045159	7.4012	11.917	0.061362	0.022320	0.030806	2185.5	-0.50940	340.22	12.765

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Younglove, B. A., "Thermophysical Properties of Fluids. I. Argon, Ethylene, Parahydrogen, Nitrogen, Nitrogen Trifluoride, and Oxygen," *J. Phys. Chem. Ref. Data*, Suppl. 1, **11**: 1–11, 1982. The source for viscosity is McCarty, R. D., and Weber, L. A., "Thermophysical Properties of Parahydrogen from the Freezing Liquid Line to 5000 R for Pressures to 10,000 psia," *N.B.S. Tech. Note* 617, 1972. The source for thermal conductivity is McCarty, R. D., and Weber, L. A., "Thermophysical Properties of Parahydrogen from the Freezing Liquid Line to 5000 R for Pressures to 10,000 psia," *N.B.S. Tech. Note* 617, 1972.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in density are 0.1% in the liquid phase, 0.25% in the vapor phase, and 0.2% in the supercritical region. The uncertainty in heat capacity is 3%, and the uncertainty in speed of sound is 2% in the liquid phase and 1% elsewhere. The uncertainty in viscosity ranges from 4% to 15%. The uncertainty in thermal conductivity below 100 K is estimated to be 3% below 150 atm and up to 10% below 700 atm. For temperatures around 100 K at low densities, the uncertainty is about 1%. Above 100 K, the uncertainty is estimated to be on the order of 10%.

TABLE 2-225 Saturated Hydrogen Peroxide*

T , K	P , bar	v_f , m ³ /kg	v_g , m ³ /kg	h_f , kJ/kg	h_g , kJ/kg	s_f , kJ/(kg·K)	s_g , kJ/(kg·K)	c_{pf} , kJ/(kg·K)	μ_f , 10 ⁻⁴ Pa·s	k_f , W/(m·K)
273	0.0004	0.00068	1672	-5577	-4027	2.990	8.662	1.45	18.0	0.483
300	0.0031	0.00069	235	-5510	-3995	3.224	8.269	1.48	11.3	0.481
350	0.0564	0.00072	15.1	-5376	-3933	3.631	7.758	1.54	4.3	0.474
400	0.4521	0.00076	2.12	-5238	-3878	4.032	7.440	1.61	2.2	0.464
450	2.143	0.00081	0.487	-5091	-3820	4.346	7.172	1.68	1.3	0.453
500	7.126	0.00088	0.155	-4945	-3777	4.656	6.992	1.75	0.89	0.443
550	18.56	0.00095	0.0605	-4794	-3745	4.941	6.846	1.82	0.65	0.431
600	40.75	0.00107	0.0268	-4635	-3731	5.209	6.720	1.90	0.50	0.416
650	79.27	0.00125	0.0125	-4463	-3746	5.485	6.582			
700	141.7	0.00171	0.0048	-4195	-3860	5.682	6.339			
708.5°	155.3	0.00284	0.0028	-4012	-4012	5.732	5.732			

*Values reproduced or converted from a tabulation by Tsykalo and Tabachnikov in V. A. Rabinovich (ed.), *Thermophysical Properties of Gases and Liquids*, Standartov, Moscow, 1968; NBS-NSF transl. TT 69-55091, 1970. The reader may be reminded that very pure hydrogen peroxide is very difficult to obtain owing to its decomposition or instability. c = critical point. The FMC Corp., Philadelphia, PA tech. bull. 67, 1969 (100 pp.) contains an enthalpy-pressure diagram to 3000 psia, 1100 K.

TABLE 2-226 Thermodynamic Properties of Hydrogen Sulfide

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
187.70	0.023259	29.116	0.034345	-1.7210	-1.7202	-0.0085877	0.044390	0.068835	1437.8	-0.34039	254.24	439.13
190.00	0.027106	29.003	0.034479	-1.5628	-1.5619	-0.0077504	0.044124	0.068707	1425.8	-0.33923	251.74	428.67
200.00	0.050340	28.505	0.035082	-0.87841	-0.87664	-0.0042394	0.043042	0.068273	1373.4	-0.33253	240.93	385.68
210.00	0.087474	27.998	0.035717	-0.19759	-0.19446	-0.00091743	0.042067	0.068029	1321.0	-0.32287	230.26	346.75
220.00	0.14366	27.480	0.036390	0.48138	0.48661	0.0022415	0.041188	0.067975	1268.4	-0.30988	219.81	311.74
230.00	0.22485	26.949	0.037107	1.1602	1.1686	0.0052596	0.040393	0.068115	1215.6	-0.29305	209.52	280.37
240.00	0.33767	26.403	0.037875	1.8406	1.8534	0.0081564	0.039677	0.068461	1162.3	-0.27174	199.43	252.29
250.00	0.48934	25.838	0.038702	2.5245	2.5434	0.010949	0.039030	0.069032	1108.5	-0.24509	189.56	227.14
260.00	0.68751	25.253	0.039599	3.2135	3.2408	0.013654	0.038449	0.069859	1053.9	-0.21197	179.91	204.60
270.00	0.94022	24.642	0.040580	3.9100	3.9481	0.016285	0.037930	0.070989	998.54	-0.17084	170.48	184.32
280.00	1.2558	24.002	0.041662	4.6161	4.6685	0.018857	0.037470	0.072490	942.08	-0.11959	161.26	166.02
290.00	1.6429	23.327	0.042868	5.3348	5.4053	0.021385	0.037070	0.074466	884.34	-0.055218	152.24	149.44
300.00	2.1103	22.609	0.044230	6.0696	6.1629	0.023885	0.036732	0.077082	825.04	0.026636	143.40	134.32
310.00	2.6672	21.838	0.045791	6.8248	6.9469	0.026373	0.036462	0.080603	763.84	0.13260	134.71	120.43
320.00	3.3233	21.000	0.047618	7.6068	7.7650	0.028873	0.036273	0.085498	700.23	0.27324	126.16	107.58
330.00	4.0889	20.073	0.049818	8.4246	8.6283	0.031414	0.036191	0.092666	633.51	0.46655	117.71	95.533
340.00	4.9755	19.021	0.052573	9.2932	9.5548	0.034044	0.036265	0.10410	562.59	0.74618	109.36	84.050
350.00	5.9969	17.776	0.056256	10.241	10.578	0.036848	0.036600	0.12534	485.59	1.1841	101.19	72.784
360.00	7.1713	16.172	0.061837	11.335	11.779	0.040035	0.037471	0.17963	398.86	1.9714	93.864	61.060
370.00	8.5294	13.436	0.074429	12.903	13.538	0.044599	0.040079	0.63367	292.76	3.9324	92.754	46.102
373.10	8.9987	10.190	0.098135	14.470	15.353	0.049374			0	6.3885		
187.70	0.023259	0.015024	66.559	16.328	17.876	0.095815	0.025347	0.034000	245.84	55.730	10.628	8.0025
190.00	0.027106	0.017314	57.758	16.382	17.947	0.094930	0.025386	0.034078	247.20	53.868	10.775	8.1053
200.00	0.050340	0.030704	32.569	16.611	18.250	0.091395	0.025586	0.034487	252.82	46.796	11.429	8.5566
210.00	0.087474	0.051165	19.545	16.832	18.541	0.088301	0.025837	0.035021	257.96	41.090	12.107	9.0159
220.00	0.14366	0.080932	12.356	17.043	18.818	0.085567	0.026142	0.035698	262.58	36.435	12.816	9.4844
230.00	0.22485	0.12253	8.1613	17.244	19.079	0.083129	0.026502	0.036537	266.64	32.601	13.566	9.9634
240.00	0.33767	0.17879	5.5933	17.431	19.320	0.080933	0.026917	0.037563	270.10	29.412	14.365	10.455
250.00	0.48934	0.25286	3.9547	17.604	19.539	0.078934	0.027388	0.038807	272.91	26.737	15.227	10.961
260.00	0.68751	0.34834	2.8707	17.761	19.735	0.077092	0.027914	0.040312	275.05	24.476	16.166	11.485
270.00	0.94022	0.46937	2.1305	17.899	19.902	0.075375	0.028496	0.042139	276.47	22.550	17.202	12.031
280.00	1.2558	0.62086	1.6107	18.016	20.039	0.073752	0.029136	0.044378	277.15	20.897	18.360	12.604
290.00	1.6429	0.80887	1.2363	18.108	20.139	0.072193	0.029838	0.047166	277.05	19.466	19.675	13.213
300.00	2.1103	1.0411	0.96050	18.171	20.198	0.070669	0.030608	0.050723	276.12	18.212	21.197	13.867
310.00	2.6672	1.3280	0.75300	18.199	20.207	0.069149	0.031458	0.055410	274.34	17.097	22.997	14.582
320.00	3.3233	1.6843	0.59373	18.183	20.156	0.067594	0.032407	0.061879	271.64	16.081	25.187	15.380
330.00	4.0889	2.1323	0.46898	18.109	20.027	0.065956	0.033485	0.071400	268.00	15.116	27.946	16.300
340.00	4.9755	2.7096	0.36906	17.957	19.793	0.064157	0.034746	0.086837	263.35	14.142	31.600	17.405
350.00	5.9969	3.4881	0.28669	17.684	19.403	0.062064	0.036293	0.11617	257.65	13.053	36.820	18.833
360.00	7.1713	4.6442	0.21532	17.192	18.736	0.059360	0.038364	0.19265	250.84	11.629	45.513	20.940
370.00	8.5294	6.9933	0.14299	16.046	17.266	0.054674	0.041755	0.80649	242.80	9.0701	70.939	25.604
373.10	8.9987	10.190	0.098135	14.470	15.353	0.049374			0	6.3885		

TABLE 2-226 Thermodynamic Properties of Hydrogen Sulfide (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _e kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Single-Phase Properties												
200.00	0.10000	28.506	0.035080	-0.87902	-0.87551	-0.0042425	0.043042	0.068269	1373.6	-0.33258	240.95	385.79
212.60	0.10000	27.865	0.035888	-0.021243	-0.017654	-0.000082766	0.041830	0.067997	1307.4	-0.31984	227.54	337.30
212.60	0.10000	0.057900	17.271	16.888	18.615	0.087559	0.025911	0.035183	259.21	39.791	12.288	9.1366
300.00	0.10000	0.040389	24.759	19.164	21.640	0.099486	0.025979	0.034563	309.73	16.968	17.999	12.954
400.00	0.10000	0.030157	33.160	21.830	25.146	0.10956	0.027268	0.035693	356.35	8.9467	24.990	17.172
500.00	0.10000	0.024088	41.515	24.642	28.794	0.11770	0.028923	0.037297	396.06	5.5432	32.218	21.094
600.00	0.10000	0.020059	49.853	27.626	32.611	0.12465	0.030708	0.039059	431.20	3.7250	39.592	24.714
700.00	0.10000	0.017187	58.182	30.789	36.607	0.13081	0.032534	0.040873	463.05	2.6185	47.091	28.082
200.00	1.0000	28.528	0.035053	-0.89011	-0.85506	-0.0042980	0.043058	0.068210	1377.6	-0.33351	241.31	387.91
272.07	1.0000	24.513	0.040795	4.0550	4.0958	0.016821	0.037830	0.071266	986.97	-0.16116	168.56	180.39
272.07	1.0000	0.49800	2.0080	17.925	19.933	0.075033	0.028623	0.042564	276.67	22.188	17.430	12.147
300.00	1.0000	0.43539	2.2968	18.775	21.072	0.079019	0.027626	0.039427	296.60	17.369	19.015	13.337
400.00	1.0000	0.31003	3.2255	21.626	24.852	0.089907	0.027708	0.037234	351.09	9.0111	25.609	17.465
500.00	1.0000	0.24394	4.0993	24.507	28.606	0.098281	0.029100	0.038036	393.73	5.5366	32.691	21.319
600.00	1.0000	0.20183	4.9547	27.525	32.480	0.10534	0.030799	0.039492	430.30	3.7023	39.980	24.893
700.00	1.0000	0.17237	5.8015	30.710	36.511	0.11155	0.032589	0.041157	462.94	2.5947	47.423	28.227
200.00	5.0000	28.625	0.034935	-0.93837	-0.76369	-0.0045411	0.043127	0.067957	1394.7	-0.33745	242.90	397.26
300.00	5.0000	22.858	0.043749	5.9433	6.1620	0.023458	0.036740	0.075047	855.61	-0.017492	146.15	139.57
340.26	5.0000	18.992	0.052654	9.3164	9.5797	0.034113	0.036269	0.10449	560.70	0.75500	109.14	83.760
340.26	5.0000	2.7266	0.36675	17.952	19.786	0.064108	0.034782	0.087361	263.22	14.116	31.710	17.437
400.00	5.0000	1.8047	0.55412	20.549	23.319	0.073747	0.030043	0.048271	325.86	9.1749	29.786	19.070
500.00	5.0000	1.2939	0.77288	23.863	27.728	0.083609	0.029940	0.041996	384.33	5.4471	35.063	22.459
600.00	5.0000	1.0365	0.96476	27.065	31.889	0.091196	0.031216	0.041590	427.30	3.5816	41.773	25.773
700.00	5.0000	0.87211	1.1466	30.353	36.086	0.097665	0.032838	0.042472	463.26	2.4840	48.924	28.935
200.00	10.000	28.741	0.034793	-0.99643	-0.64850	-0.0048367	0.043212	0.067668	1415.5	-0.34197	244.82	408.90
300.00	10.000	23.238	0.043033	5.7496	6.1800	0.022795	0.036779	0.072377	902.78	-0.077399	150.49	148.08
400.00	10.000	5.0473	0.19812	18.370	20.351	0.062081	0.034651	0.10189	291.29	8.2243	44.719	23.639
500.00	10.000	2.8037	0.35667	22.959	26.526	0.076030	0.031080	0.048875	375.68	5.1487	38.963	24.438
600.00	10.000	2.1399	0.46730	26.466	31.139	0.084452	0.031755	0.044597	426.01	3.3812	44.210	27.165
700.00	10.000	1.7663	0.56617	29.903	35.564	0.091275	0.033155	0.044212	465.45	2.3347	50.878	30.013
300.00	75.000	26.050	0.038388	4.3332	7.2123	0.017506	0.037754	0.061705	1276.9	-0.33612	187.18	232.59
400.00	75.000	21.973	0.045510	9.9713	13.384	0.035260	0.035381	0.061962	983.51	-0.18247	134.39	124.22
500.00	75.000	17.947	0.055720	15.404	19.583	0.049092	0.034762	0.061649	786.35	0.057516	103.90	81.074
600.00	75.000	14.519	0.068874	20.474	25.640	0.060142	0.034994	0.059227	688.11	0.27314	87.712	62.531
700.00	75.000	11.974	0.083511	25.148	31.412	0.069045	0.035721	0.056291	654.55	0.36585	82.684	54.563
300.00	150.00	27.794	0.035979	3.5100	8.9069	0.013888	0.038777	0.058983	1538.4	-0.40376	214.24	311.25
400.00	150.00	24.751	0.040403	8.6429	14.703	0.030575	0.036402	0.057226	1302.6	-0.37006	165.64	174.85
500.00	150.00	21.937	0.045585	13.539	20.377	0.043238	0.035779	0.056273	1132.1	-0.31802	135.75	119.57
600.00	150.00	19.449	0.051416	18.248	25.960	0.053420	0.036044	0.055409	1019.1	-0.26874	118.38	93.323
700.00	150.00	17.335	0.057687	22.811	31.464	0.061906	0.036804	0.054711	949.06	-0.23292	110.03	79.581

The values in these tables were generated from the NIST REFPROP software (Lemmon, E.W., McLinden, M.O., and Huber, M.L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., and Span, R., "Short Fundamental Equations of State for 20 Industrial Fluids," *J. Chem. Eng. Data* **51**(3): 785–850, 2006. The source for viscosity and thermal conductivity is NIST14, Version 9.08.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in density are 0.1% in the liquid phase below the critical temperature, 0.4% in the vapor phase, 1% at supercritical temperatures up to 500 K, and 2.5% at higher temperatures. Uncertainties will be higher near the critical point, and may be lower than 0.5% between 400 and 500 K. The uncertainty in vapor pressure is 0.25%, and the uncertainty in heat capacities is estimated to be 1%. For viscosity, estimated uncertainty is 2%. For thermal conductivity, estimated uncertainty, except near the critical region, is 4–6%.

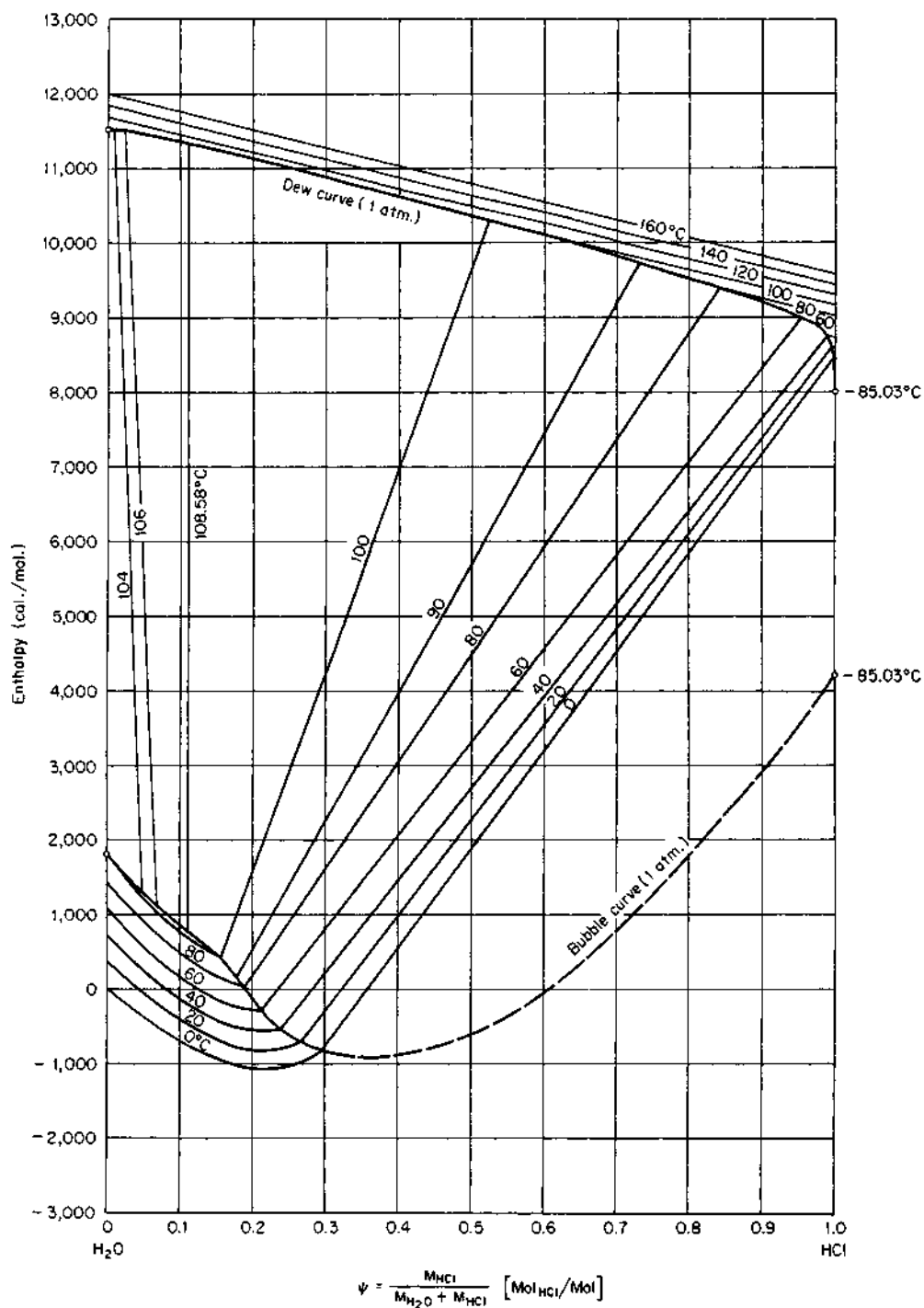


FIG. 2-11 Enthalpy-concentration diagram for aqueous hydrogen chloride at 1 atm. Reference states: enthalpy of liquid water at 0°C is zero; enthalpy of pure saturated HCl vapor at 1 atm (−85.03°C) is 8000 kcal/mol. NOTE: It should be observed that the weight basis includes the vapor, which is particularly important in the two-phase region. Saturation values may be read at the ends of the tie lines. [Van Nuys, Trans. Am. Inst. Chem. Eng., 39, 663 (1943).]

TABLE 2-227 Thermodynamic Properties of Isobutane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_v kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
113.73	2.2891E-08	12.738	0.078508	-6.5316	-6.5316	-0.03951	0.068243	0.098159	1999.8	-0.68375	157.92	8767.2
115	3.1711E-08	12.717	0.078635	-6.4068	-6.4068	-0.03842	0.068392	0.098388	1988.6	-0.68180	157.70	8996.1
130	8.8064E-07	12.473	0.080171	-4.9101	-4.9101	-0.02619	0.070255	0.10118	1863.4	-0.65814	154.39	3698.4
145	1.1510E-05	12.229	0.081772	-3.3714	-3.3714	-0.01499	0.072110	0.10397	1750.1	-0.63434	150.06	2060.1
160	8.8176E-05	11.984	0.083447	-1.7912	-1.7911	-0.00463	0.073940	0.10673	1645.3	-0.61032	144.95	1302.4
175	0.00045673	11.736	0.085209	-0.16946	-0.16942	0.005061	0.075811	0.10951	1546.1	-0.58545	139.28	897.76
190	0.0017628	11.485	0.087069	1.4945	1.4946	0.014182	0.077792	0.11237	1450.9	-0.55901	133.23	657.25
205	0.0054294	11.231	0.089043	3.2024	3.2029	0.022832	0.079933	0.11540	1358.6	-0.53024	126.94	502.16
220	0.014023	10.971	0.091150	4.9572	4.9585	0.031092	0.082269	0.11865	1268.6	-0.49839	120.56	395.73
235	0.031511	10.705	0.093415	6.7622	6.7652	0.039028	0.084817	0.12216	1180.3	-0.46255	114.21	319.15
250	0.06335	10.431	0.095866	8.6216	8.6277	0.046697	0.087581	0.12600	1093.4	-0.42158	107.94	262.04
265	0.11640	10.148	0.098544	10.540	10.551	0.054149	0.090556	0.13021	1007.7	-0.37389	101.85	218.23
280	0.19876	9.8523	0.10150	12.522	12.543	0.061427	0.093728	0.13486	922.77	-0.31712	95.986	183.83
295	0.31952	9.5416	0.10480	14.575	14.608	0.068569	0.097085	0.14006	838.36	-0.24757	90.398	156.24
310	0.48858	9.2114	0.10856	16.705	16.758	0.075615	0.10062	0.14598	753.97	-0.15928	85.118	133.64
325	0.71658	8.8560	0.11292	18.920	19.001	0.082601	0.10432	0.15290	669.03	-0.04193	80.169	114.69
340	1.0148	8.4662	0.11812	21.234	21.354	0.089574	0.10821	0.16139	582.74	0.12359	75.561	98.394
355	1.3957	8.0276	0.12457	23.665	23.839	0.096591	0.11234	0.17260	494.00	0.37690	71.293	83.919
370	1.8727	7.5139	0.13309	26.245	26.494	0.10375	0.11686	0.18958	401.01	0.81512	67.356	70.526
385	2.4620	6.8650	0.14567	29.045	29.404	0.11123	0.12220	0.22324	300.29	1.7502	63.794	57.352
400	3.1856	5.8674	0.17043	32.319	32.862	0.11975	0.13078	0.36901	184.38	4.9259	61.898	42.498
407.81	3.629	3.8798	0.25775	35.910	36.846	0.12937			0	16.282		
113.73	2.2891E-08	2.4207E-08	41,310,000	20.462	21.408	0.20615	0.042812	0.051127	139.39	577.37	2.2717	2.8476
115	3.1711E-08	3.3165E-08	30,152,000	20.517	21.473	0.20401	0.043157	0.051472	140.07	553.27	2.3426	2.8811
130	8.8064E-07	8.1474E-07	1,227,400	21.194	22.275	0.18292	0.047087	0.055402	147.92	348.08	3.2114	3.2763
145	1.1510E-05	9.5478E-06	104,740	21.928	23.133	0.16780	0.050787	0.059103	155.36	233.14	4.1386	3.6690
160	8.8176E-05	6.6294E-05	15,084	22.716	24.046	0.15686	0.054332	0.062655	162.43	164.07	5.1239	4.0588
175	0.00045673	0.0003141	3,183.6	23.555	25.009	0.14894	0.057806	0.066151	169.14	120.18	6.1666	4.4452
190	0.0017628	0.0011181	894.34	24.442	26.018	0.14325	0.061291	0.06969	175.44	91.024	7.2652	4.8275
205	0.0054294	0.0032007	312.43	25.374	27.070	0.13926	0.064859	0.07337	181.26	70.957	8.4173	5.2047
220	0.014023	0.0077418	129.17	26.348	28.159	0.13655	0.068567	0.077281	186.48	56.748	9.6201	5.5763
235	0.031511	0.016419	60.904	27.360	29.279	0.13483	0.072455	0.081501	190.96	46.450	10.872	5.9424
250	0.06335	0.031405	31.842	28.406	30.423	0.13388	0.076548	0.086099	194.52	38.843	12.174	6.3045
265	0.11640	0.055358	18.064	29.481	31.584	0.13352	0.080855	0.091145	197.02	33.140	13.532	6.6660
280	0.19876	0.091469	10.933	30.581	32.754	0.13361	0.085375	0.096722	198.29	28.823	14.961	7.0328
295	0.31952	0.14360	6.9638	31.700	33.925	0.13405	0.090101	0.10295	198.13	25.545	16.487	7.4136
310	0.48858	0.21659	4.6171	32.830	35.086	0.13474	0.095003	0.11003	196.37	23.074	18.151	7.8217
325	0.71658	0.31684	3.1561	33.960	36.222	0.13559	0.10017	0.11832	192.73	21.254	20.022	8.2772
340	1.0148	0.45354	2.2049	35.075	37.313	0.13651	0.10541	0.12842	186.90	20.032	22.207	8.8129
355	1.3957	0.64121	1.5596	36.146	38.323	0.13739	0.11078	0.14204	178.45	19.459	24.900	9.4859
370	1.8727	0.90588	1.1039	37.123	39.190	0.13806	0.11703	0.16455	166.77	19.526	28.498	10.409
385	2.4620	1.3046	0.76649	37.906	39.793	0.13822	0.12502	0.21348	150.88	20.215	34.050	11.844
400	3.1856	2.0369	0.49095	38.148	39.712	0.13688	0.13683	0.43911	128.90	21.293	46.452	14.761
407.81	3.629	3.8798	0.25775	35.910	36.846	0.12937			0	16.282		

Single-Phase Properties

200.00	0.10000	11.317	0.088361	2.6255	2.6343	0.019983	0.079206	0.11436	1389.7	-0.54030	129.10	547.68
261.07	0.10000	10.223	0.097817	10.031	10.040	0.052213	0.089756	0.12906	1030.1	-0.38717	103.43	228.69
261.07	0.10000	0.048038	20.817	29.197	31.278	0.13356	0.079705	0.089774	196.48	34.485	13.170	6.5710
300.00	0.10000	0.041143	24.305	32.513	34.944	0.14664	0.089474	0.098798	212.04	21.436	17.082	7.5436
400.00	0.10000	0.030372	32.925	42.834	46.127	0.17863	0.11631	0.12501	245.50	8.6878	28.880	9.9294
500.00	0.10000	0.024169	41.375	55.743	59.881	0.20922	0.14101	0.14952	274.09	4.5740	43.232	12.195
200.00	1.0000	11.329	0.088272	2.6017	2.6900	0.019864	0.079268	0.11426	1395.0	-0.54183	129.47	552.65
300.00	1.0000	9.4575	0.10574	15.229	15.335	0.070771	0.098270	0.14150	819.00	-0.23012	89.092	149.88
339.34	1.0000	8.4843	0.11786	21.129	21.247	0.089265	0.10803	0.16097	586.60	0.11482	75.758	99.071
339.34	1.0000	0.44655	2.2394	35.027	37.266	0.13647	0.10518	0.12792	187.21	20.073	22.102	8.7869
400.00	1.0000	0.33698	2.9675	42.086	45.054	0.15757	0.11833	0.13200	9.9934	9.9934	29.559	10.129
500.00	1.0000	0.25270	3.9573	55.288	59.246	0.18916	0.14174	0.15229	263.78	4.7852	44.232	12.391
200.00	5.0000	11.379	0.087884	2.4992	2.9387	0.019346	0.079538	0.11385	1417.9	-0.54814	131.11	574.91
300.00	5.0000	9.5950	0.10422	14.958	15.479	0.069851	0.098458	0.13915	870.18	-0.28002	92.042	160.04
400.00	5.0000	6.7797	0.14750	31.029	31.767	0.11631	0.12520	0.20584	328.20	1.3689	64.364	56.352
500.00	5.0000	1.6118	0.62041	52.732	55.834	0.17060	0.14556	0.17644	219.06	5.5914	50.270	15.871
200.00	10.000	11.438	0.087424	2.3778	3.2521	0.018721	0.079860	0.11339	1445.4	-0.55506	133.10	603.11
300.00	10.000	9.7443	0.10262	14.662	15.689	0.068827	0.098710	0.13705	926.11	-0.32494	95.422	172.11
400.00	10.000	7.5550	0.13236	29.780	31.103	0.11292	0.12349	0.17456	486.23	0.34096	70.562	72.797
500.00	10.000	4.0999	0.24391	48.735	51.174	0.15751	0.14786	0.21137	239.25	3.1085	61.481	28.154
200.00	15.000	11.496	0.086989	2.2631	3.5679	0.018121	0.080166	0.11300	1471.7	-0.56110	135.03	631.76
300.00	15.000	9.8755	0.10126	14.403	15.922	0.067905	0.098964	0.13552	975.70	-0.35793	98.548	183.72
400.00	15.000	7.9692	0.12548	29.065	30.948	0.11092	0.12324	0.16619	583.15	0.062447	75.048	84.081
500.00	15.000	5.5655	0.17968	46.521	49.217	0.15156	0.14693	0.19670	343.01	1.1026	67.712	41.020
200.00	30.000	11.654	0.085807	1.9531	4.5274	0.016439	0.081000	0.11206	1544.8	-0.57512	140.47	720.69
300.00	30.000	10.198	0.098056	13.768	16.709	0.065552	0.099691	0.13266	1100.2	-0.41943	106.90	216.98
400.00	30.000	8.6989	0.11496	27.757	31.205	0.10708	0.12354	0.15745	775.70	-0.21676	85.299	109.96
500.00	30.000	7.1701	0.13947	43.945	48.129	0.14475	0.14649	0.18019	572.71	0.033041	78.479	66.356

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Buecker, D., and Wagner, W., "Reference Equations of State for the Thermodynamic Properties of Fluid Phase *n*-Butane and Isobutane," *J. Phys. Chem. Ref. Data* **35**(2):929–1019, 2006. The source for viscosity is Vogel, E., Kuechenmeister, C., and Bich, E., "Viscosity Correlation for Isobutane over Wide Ranges of the Fluid Region," *Int. J. Thermophys.* **21**(2):343–356, 2000. The source for thermal conductivity is Perkins, R. A., "Measurement and Correlation of the Thermal Conductivity of Isobutane from 114 K to 600 K at Pressures to 70 MPa," *J. Chem. Eng. Data* **47**(5):1272–1279, 2002.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in density are 0.02% at temperatures below 340 K and pressures below 12 MPa (both liquid and vapor states), 0.3% at temperatures below 300 K and pressures above 12 MPa, 0.1% in the vapor phase between 340 and 450 K, and 0.5% elsewhere. In the critical region, deviations in pressure are 0.5%. Uncertainties in heat capacities are typically 1–2%, rising to 5% in the critical region and at temperatures below 200 K. Uncertainties in the speed of sound are typically 1–2%, rising to 5% at temperatures below 200 K and in the critical region. The uncertainty in viscosity varies from 0.4% in the dilute gas between room temperature and 600 K to 3.0% over the rest of the fluid surface. Uncertainty in thermal conductivity is 3%, except in the critical region and dilute gas which have an uncertainty of 5%.

TABLE 2-228 Thermodynamic Properties of Isobutene (2-Methyl 1-Propene)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
132.40	6.7619E-07	13.667	0.073170	-15.196	-15.196	-0.078577	0.074156	0.10623	1842.9	-0.56574
135.00	1.1116E-06	13.619	0.073428	-14.919	-14.919	-0.076509	0.074179	0.10642	1827.0	-0.56403
150.00	1.3438E-05	13.343	0.074943	-13.315	-13.315	-0.065239	0.074543	0.10755	1734.4	-0.55349
165.00	9.7764E-05	13.070	0.076514	-11.692	-11.692	-0.054931	0.075233	0.10881	1641.6	-0.54130
180.00	0.00048920	12.796	0.078149	-10.049	-10.049	-0.045402	0.076215	0.11028	1549.8	-0.52677
195.00	0.0018451	12.522	0.079860	-8.3822	-8.3821	-0.036508	0.077475	0.11203	1459.4	-0.50932
210.00	0.0055971	12.246	0.081662	-6.6870	-6.6865	-0.028133	0.079001	0.11407	1370.6	-0.48847
225.00	0.014316	11.966	0.083571	-4.9587	-4.9575	-0.020185	0.080776	0.11645	1283.4	-0.46371
240.00	0.031980	11.681	0.085610	-3.1926	-3.1899	-0.012587	0.082776	0.11917	1197.6	-0.43442
255.00	0.064075	11.389	0.087805	-1.3838	-1.3782	-0.0052767	0.084977	0.12224	1113.1	-0.39974
270.00	0.11754	11.088	0.090191	0.47283	0.48343	0.0017979	0.087351	0.12570	1029.6	-0.35836
285.00	0.20057	10.774	0.092812	2.3824	2.4010	0.0086815	0.089877	0.12958	946.77	-0.30827
300.00	0.32243	10.446	0.095729	4.3506	4.3814	0.015413	0.092533	0.13397	864.37	-0.24634
315.00	0.49324	10.098	0.099027	6.3837	6.4326	0.022030	0.095306	0.13900	781.98	-0.16750
330.00	0.72386	9.7251	0.10283	8.4899	8.5643	0.028568	0.098189	0.14491	699.15	-0.063276
345.00	1.0259	9.3181	0.10732	10.680	10.790	0.035068	0.10119	0.15215	615.32	0.081482
360.00	1.4121	8.8642	0.11281	12.969	13.128	0.041581	0.10433	0.16163	529.74	0.29633
375.00	1.8963	8.3412	0.11989	15.383	15.611	0.048183	0.10770	0.17548	441.23	0.64712
390.00	2.4955	7.7041	0.12980	17.975	18.299	0.055015	0.11149	0.20028	347.41	1.3162
405.00	3.2319	6.8249	0.14652	20.892	21.365	0.062470	0.11647	0.27285	242.00	3.0714
418.09	4.0157	4.1700	0.23981	25.759	26.722	0.075104			0	15.166
132.40	6.7619E-07	6.1426E-07	1,628,000.	12.185	13.286	0.13654	0.045424	0.053739	152.35	932.58
135.00	1.1116E-06	9.9037E-07	1,009,700.	12.304	13.427	0.13346	0.046004	0.054319	153.69	849.42
150.00	1.3438E-05	1.0775E-05	92,803.	13.018	14.265	0.11863	0.049224	0.057543	161.19	514.47
165.00	9.7764E-05	7.1284E-05	14,028.	13.778	15.150	0.10775	0.052312	0.060645	168.31	329.24
180.00	0.00048920	0.00032720	3,056.2	14.581	16.076	0.099740	0.055381	0.063755	175.06	220.54
195.00	0.0018451	0.0011411	876.38	15.424	17.041	0.093867	0.058529	0.066994	181.39	153.60
210.00	0.0055971	0.0032245	310.13	16.303	18.039	0.089606	0.061825	0.070461	187.22	110.74
225.00	0.014316	0.0077400	129.20	17.214	19.064	0.086578	0.065309	0.074224	192.45	82.434
240.00	0.031980	0.016347	61.174	18.155	20.112	0.084503	0.068992	0.078326	196.96	63.248
255.00	0.064075	0.031202	32.049	19.122	21.176	0.083170	0.072865	0.082797	200.59	49.962
270.00	0.11754	0.054953	18.197	20.111	22.250	0.082415	0.076909	0.087668	203.21	40.590
285.00	0.20057	0.090771	11.017	21.118	23.328	0.082110	0.081099	0.092989	204.65	33.877
300.00	0.32243	0.14247	7.0191	22.138	24.401	0.082147	0.085414	0.098854	204.73	29.016
315.00	0.49324	0.21479	4.6557	23.164	25.460	0.082436	0.089840	0.10544	203.27	25.486
330.00	0.72386	0.31394	3.1853	24.185	26.491	0.082890	0.094375	0.11307	200.03	22.947
345.00	1.0259	0.44866	2.2289	25.186	27.473	0.083425	0.099036	0.12238	194.71	21.185
360.00	1.4121	0.63242	1.5812	26.144	28.377	0.083938	0.10387	0.13473	186.95	20.076
375.00	1.8963	0.88850	1.1255	27.018	29.152	0.084292	0.10900	0.15340	176.23	19.567
390.00	2.4955	1.2646	0.79073	27.725	29.699	0.084244	0.11468	0.18881	161.89	19.640
405.00	3.2319	1.8955	0.52758	28.046	29.751	0.083176	0.12162	0.29879	142.88	20.140
418.09	4.0157	4.1700	0.23981	25.759	26.722	0.075104			0	15.166

Single-Phase Properties

150.00	0.10000	13.344	0.074938	-13.316	-13.309	-0.065249	0.074545	0.10754	1734.9	-0.55354
250.00	0.10000	11.488	0.087047	-1.9936	-1.9849	-0.0076921	0.084223	0.12117	1141.5	-0.41214
265.81	0.10000	11.173	0.089503	-0.050895	-0.041944	-0.00015725	0.086672	0.12469	1052.8	-0.37070
265.81	0.10000	0.047251	21.164	19.833	21.949	0.082576	0.075763	0.086265	202.59	42.897
350.00	0.10000	0.034911	28.645	26.949	29.814	0.10818	0.092639	0.10158	234.72	15.195
450.00	0.10000	0.026911	37.160	37.274	40.990	0.13616	0.11313	0.12171	266.02	6.9323
550.00	0.10000	0.021943	45.572	49.523	54.081	0.16237	0.13123	0.13969	293.54	4.0208
150.00	1.0000	13.352	0.074895	-13.330	-13.255	-0.065342	0.074562	0.10751	1739.5	-0.55391
250.00	1.0000	11.506	0.086914	-2.0267	-1.9398	-0.0078249	0.084239	0.12099	1149.4	-0.41526
343.85	1.0000	9.3507	0.10694	10.509	10.616	0.034571	0.10096	0.15153	621.79	0.068431
343.85	1.0000	0.43682	2.2893	25.111	27.400	0.083383	0.098674	0.12159	195.20	21.295
350.00	1.0000	0.42121	2.3741	25.767	28.142	0.085521	0.098979	0.11981	199.99	19.480
450.00	1.0000	0.28769	3.4759	36.712	40.188	0.11575	0.11442	0.12599	251.31	7.4906
550.00	1.0000	0.22650	4.4150	49.150	53.565	0.14255	0.13168	0.14162	285.82	4.1331
150.00	5.0000	13.386	0.074707	-13.390	-13.017	-0.065747	0.074643	0.10737	1759.5	-0.55549
250.00	5.0000	11.581	0.086351	-2.1681	-1.7364	-0.0083971	0.084319	0.12027	1182.8	-0.42781
350.00	5.0000	9.4333	0.10601	10.992	11.522	0.035956	0.10188	0.14839	663.68	-0.014643
450.00	5.0000	2.4761	0.40387	32.462	34.481	0.092351	0.12422	0.22025	169.25	10.828
550.00	5.0000	1.3207	0.75720	47.230	51.016	0.12568	0.13385	0.15463	255.06	4.4151
150.00	10.000	13.426	0.074480	-13.463	-12.718	-0.066244	0.074751	0.10722	1783.6	-0.55725
250.00	10.000	11.668	0.085703	-2.3329	-1.4758	-0.0090755	0.084439	0.11952	1221.8	-0.44099
350.00	10.000	9.6916	0.10318	10.547	11.579	0.034624	0.10172	0.14363	741.89	-0.13006
450.00	10.000	6.6358	0.15070	26.495	28.002	0.075649	0.12143	0.19240	325.60	1.5013
550.00	10.000	3.1171	0.32081	44.413	47.621	0.11510	0.13595	0.17757	248.73	3.4960
150.00	20.000	13.504	0.074050	-13.601	-12.120	-0.067205	0.074987	0.10695	1829.3	-0.56021
250.00	20.000	11.827	0.084554	-2.6298	-0.93871	-0.010332	0.084720	0.11836	1292.1	-0.46137
350.00	20.000	10.075	0.099256	9.8719	11.857	0.032532	0.10175	0.13862	862.17	-0.25114
450.00	20.000	8.0205	0.12468	24.414	26.907	0.070229	0.11990	0.16238	539.83	0.19834
550.00	20.000	5.7872	0.17279	40.594	44.050	0.10458	0.13615	0.17749	377.20	0.90256
250.00	50.000	12.214	0.081874	-3.3402	0.75353	-0.013539	0.085706	0.11640	1463.7	-0.49631
350.00	50.000	10.805	0.092548	8.5711	13.199	0.028177	0.10258	0.13309	1105.6	-0.38284
450.00	50.000	9.4377	0.10596	22.086	27.384	0.063736	0.12031	0.15031	858.20	-0.25457
550.00	50.000	8.1524	0.12266	37.030	43.163	0.095351	0.13651	0.16471	704.72	-0.13437

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W. and Ihmels, E. C., “Thermodynamic Properties of the Butenes. Part II. Short Fundamental Equations of State,” *Fluid Phase Equilibria*, **228–229C**:173–187, 2005. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in densities calculated using the equation of state are 0.1% in the liquid phase at temperatures above 270 K (rising to 0.5% at temperatures below 200 K), 0.2% at temperatures above the critical temperature and at pressures above 10 MPa, and 0.5% in the vapor phase, including supercritical conditions below 10 MPa. The uncertainty in the vapor phase may be higher than 0.5% in some regions. The uncertainty in vapor pressure is 0.5% above 250 K, and the uncertainty in heat capacities is 0.5% at saturated liquid conditions, rising to 5% at much higher pressures and at temperatures above 250 K.

TABLE 2-229 Thermodynamic Properties of Krypton

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_v kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
115.77	0.073503	29.197	0.034250	-0.17537	-0.17285	-0.00146	0.021225	0.043334	698.97	-0.50800
120	0.10344	28.814	0.034705	0.008003	0.011593	9.61E-05	0.020889	0.043577	681.44	-0.49215
125	0.15007	28.353	0.035270	0.22606	0.23136	0.001877	0.020519	0.043953	660.52	-0.47002
130	0.21125	27.880	0.035867	0.44593	0.45351	0.003602	0.020175	0.044431	639.35	-0.44379
135	0.28957	27.396	0.036502	0.66802	0.67859	0.005280	0.019855	0.045022	617.89	-0.41290
140	0.38778	26.896	0.037180	0.89279	0.90720	0.006916	0.019558	0.045743	596.06	-0.37662
145	0.50867	26.381	0.037906	1.1207	1.1400	0.008518	0.019283	0.046614	573.82	-0.33403
150	0.65513	25.847	0.038689	1.3524	1.3778	0.010092	0.019030	0.047664	551.09	-0.28396
155	0.83007	25.291	0.039539	1.5885	1.6213	0.011644	0.018799	0.048934	527.76	-0.22485
160	1.0365	24.710	0.040469	1.8297	1.8716	0.013181	0.018594	0.050480	503.74	-0.15466
165	1.2774	24.100	0.041494	2.0770	2.1300	0.014711	0.018417	0.052386	478.86	-0.07060
170	1.5560	23.455	0.042636	2.3317	2.398	0.016241	0.018276	0.054776	452.95	0.03121
175	1.8754	22.766	0.043925	2.5952	2.6775	0.017781	0.018179	0.057846	425.75	0.15636
180	2.2390	22.024	0.045404	2.8697	2.9713	0.019345	0.018142	0.061924	396.95	0.31329
185	2.6505	21.213	0.047140	3.1584	3.2833	0.020950	0.018193	0.067602	366.08	0.51538
190	3.1140	20.309	0.049240	3.4661	3.6194	0.022623	0.018377	0.076087	332.48	0.78550
195	3.6340	19.267	0.051903	3.8015	3.9901	0.024412	0.018781	0.090283	295.17	1.1669
200	4.2163	17.995	0.055571	4.1824	4.4167	0.026413	0.019602	0.11946	252.47	1.7554
205	4.8690	16.222	0.061645	4.6620	4.9622	0.028917	0.021432	0.21813	200.85	2.8313
209.48	5.5255	10.850	0.092166	5.8827	6.3920	0.035568			0	6.4431
115.77	0.073503	0.078418	12.752	7.9735	8.9108	0.077003	0.013270	0.022596	136.12	49.758
120	0.10344	0.10724	9.3246	8.0103	8.9748	0.074790	0.013453	0.023062	138.00	45.644
125	0.15007	0.15089	6.6272	8.0498	9.0444	0.072381	0.013699	0.023713	140.03	41.448
130	0.21125	0.20670	4.8380	8.0846	9.1066	0.070164	0.013976	0.024485	141.85	37.834
135	0.28957	0.27668	3.6142	8.1140	9.1606	0.068109	0.014283	0.025395	143.45	34.693
140	0.38778	0.36308	2.7542	8.1375	9.2055	0.066190	0.014618	0.026463	144.82	31.940
145	0.50867	0.46836	2.1351	8.1544	9.2405	0.064384	0.014982	0.027716	145.98	29.510
150	0.65513	0.59537	1.6796	8.1640	9.2644	0.062670	0.015372	0.029193	146.92	27.351
155	0.83007	0.74740	1.3380	8.1655	9.2761	0.061030	0.015789	0.030948	147.64	25.418
160	1.0365	0.92843	1.0771	8.1578	9.2742	0.059447	0.016235	0.033060	148.13	23.675
165	1.2774	1.1433	0.87464	8.1395	9.2567	0.057903	0.016713	0.035646	148.39	22.088
170	1.5560	1.3983	0.71513	8.1089	9.2216	0.056379	0.017230	0.038887	148.42	20.627
175	1.8754	1.7017	0.58764	8.0637	9.1657	0.054856	0.017796	0.043072	148.21	19.260
180	2.2390	2.0649	0.48428	8.0007	9.0850	0.053310	0.018426	0.048693	147.75	17.955
185	2.6505	2.5046	0.39926	7.9154	8.9736	0.051708	0.019147	0.056647	147.01	16.679
190	3.1140	3.0471	0.32818	7.8004	8.8224	0.050007	0.019998	0.068771	145.97	15.388
195	3.6340	3.7380	0.26752	7.6434	8.6156	0.048132	0.021054	0.089467	144.56	14.023
200	4.2163	4.6709	0.21409	7.4192	8.3219	0.045939	0.022465	0.13256	142.62	12.486
205	4.8690	6.1126	0.16360	7.0576	7.8541	0.043024	0.024624	0.27498	139.66	10.545
209.48	5.5255	10.850	0.092166	5.8827	6.3920	0.035568			0	6.4431

Single-Phase Properties

200.00	0.10000	0.060557	16.513	9.0557	10.707	0.086165	0.012553	0.021091	181.32	15.987
300.00	0.10000	0.040171	24.893	10.313	12.803	0.094664	0.012488	0.020880	222.65	7.9817
400.00	0.10000	0.030088	33.236	11.564	14.888	0.10066	0.012478	0.020832	257.24	4.8486
500.00	0.10000	0.024059	41.565	12.813	16.970	0.10531	0.012475	0.020813	287.64	3.1571
600.00	0.10000	0.020044	49.889	14.062	19.051	0.10910	0.012474	0.020804	315.11	2.0967
700.00	0.10000	0.017179	58.210	15.310	21.131	0.11231	0.012474	0.020799	340.36	1.3720
200.00	1.0000	0.64884	1.5412	8.8431	10.384	0.065940	0.013359	0.024440	176.18	15.667
300.00	1.0000	0.40911	2.4443	10.207	12.651	0.075164	0.012634	0.021770	221.94	7.8257
400.00	1.0000	0.30268	3.3039	11.493	14.797	0.081341	0.012536	0.021250	257.69	4.7410
500.00	1.0000	0.24094	4.1505	12.761	16.911	0.086060	0.012509	0.021058	288.54	3.0825
600.00	1.0000	0.20035	4.9913	14.020	19.012	0.089890	0.012498	0.020965	316.19	2.0440
700.00	1.0000	0.17155	5.8291	15.276	21.105	0.093117	0.012492	0.020912	341.52	1.3340
200.00	5.0000	18.695	0.053490	4.0257	4.2932	0.025582	0.018694	0.092820	286.26	1.3105
205.94	5.0000	15.764	0.063436	4.7766	5.0938	0.029518	0.022035	0.27057	189.55	3.1528
205.94	5.0000	6.5051	0.15373	6.9575	7.7261	0.042300	0.025196	0.34754	138.88	10.086
300.00	5.0000	2.2268	0.44908	9.6909	11.936	0.060067	0.013335	0.026859	220.60	6.9488
400.00	5.0000	1.5501	0.64511	11.172	14.397	0.067171	0.012800	0.023237	260.80	4.2506
500.00	5.0000	1.2103	0.82624	12.528	16.659	0.072221	0.012660	0.022154	293.14	2.7636
600.00	5.0000	0.99847	1.0015	13.839	18.847	0.076212	0.012603	0.021667	321.36	1.8241
700.00	5.0000	0.85191	1.1738	15.130	20.999	0.079530	0.012572	0.021402	346.89	1.1768
200.00	10.000	20.764	0.048160	3.5349	4.0165	0.022941	0.017485	0.059683	386.03	0.41868
300.00	10.000	4.9580	0.20169	8.9533	10.970	0.051900	0.014221	0.035983	225.22	5.4452
400.00	10.000	3.1738	0.31508	10.764	13.915	0.060436	0.013132	0.025901	267.23	3.6190
500.00	10.000	2.4249	0.41238	12.241	16.365	0.065913	0.012848	0.023506	300.18	2.3913
600.00	10.000	1.9842	0.50399	13.620	18.660	0.070099	0.012733	0.022506	328.56	1.5769
700.00	10.000	1.6870	0.59277	14.954	20.881	0.073525	0.012672	0.021979	354.04	1.0028
200.00	100.00	27.903	0.035839	1.8128	5.3966	0.012020	0.019196	0.036607	779.27	-0.62535
300.00	100.00	23.193	0.043116	4.6453	8.9569	0.026484	0.016847	0.034479	643.59	-0.55209
400.00	100.00	19.351	0.051678	7.1023	12.270	0.036035	0.015591	0.031754	570.49	-0.49661
500.00	100.00	16.428	0.060871	9.2300	15.317	0.042844	0.014841	0.029280	537.49	-0.48427
600.00	100.00	14.234	0.070256	11.121	18.146	0.048008	0.014355	0.027397	525.89	-0.50576
700.00	100.00	12.558	0.079631	12.850	20.814	0.052122	0.014021	0.026021	525.42	-0.54812
200.00	200.00	30.673	0.032602	1.2743	7.7946	0.0070011	0.020666	0.035096	976.34	-0.69885
300.00	200.00	27.185	0.036785	3.8223	11.179	0.020757	0.018195	0.032708	867.80	-0.72858
400.00	200.00	24.274	0.041196	6.1068	14.346	0.029882	0.016813	0.030662	798.01	-0.75899
500.00	200.00	21.877	0.045709	8.1787	17.321	0.036526	0.015938	0.028881	754.77	-0.79823
600.00	200.00	19.905	0.050239	10.086	20.134	0.041659	0.015341	0.027435	728.59	-0.84213
700.00	200.00	18.266	0.054748	11.868	22.818	0.045799	0.014910	0.026300	713.29	-0.88605

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W. and Span, R., "Short Fundamental Equations of State for 20 Industrial Fluids," *J. Chem. Eng. Data* **51**(3):785–850, 2006. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The equation of state is valid from the triple point to 750 K with pressures to 200 MPa, although the uncertainties increase substantially above 100 MPa. The uncertainties in density are typically 0.2% below 100 MPa, increasing to 1% at pressures up to 200 MPa. The uncertainty in vapor pressure is 0.2%, and the uncertainties in speed of sound are 0.01% in the vapor phase (including supercritical conditions) at low pressures, 1% below 20 MPa in the liquid phase, and 3% below 100 MPa at other state points. The limited amount of heat capacity data shows that the uncertainty is 1% near the triple point, and uncertainties in heat capacities at other states are probably within 2%, at least at pressures up to 20 MPa.

TABLE 2-230 Saturated Lithium*

T, K	P, bar	$v_f, \text{m}^3/\text{kg}$	$v_g, \text{m}^3/\text{kg}$	$h_f, \text{kJ/kg}$	$h_g, \text{kJ/kg}$	$s_f, \text{kJ}/(\text{kg}\cdot\text{K})$	$s_g, \text{kJ}/(\text{kg}\cdot\text{K})$	$c_{pf}, \text{kJ}/(\text{kg}\cdot\text{K})$
453.7 ^m	1.78.-13	1.912.-3		1703	24259	6.776	56.492	4.30
500	8.21.-12	1.946.-3		1905	24390	7.199	52.169	4.34
600	4.18.-9	1.988.-3		2334	24674	7.983	45.216	4.23
700	3.51.-7	2.028.-3	2.40.+7	2697	24869	8.633	40.307	4.19
800	9.57.-6	2.070.-3	9.94.+5	3174	25162	9.192	36.678	4.17
900	1.24.-4	2.114.-3	8.55.+4	3590	25341	9.682	33.850	4.16
1000	9.60.-4	2.160.-3	1.22.+4	4006	25477	10.120	31.591	4.16
1200	0.0204	2.262.-3	669.3	4835	25654	10.876	28.225	4.14
1400	0.1794	2.370.-3	86.06	5668	25778	11.518	25.882	4.19
1500	0.4269	2.433.-3	38.17	6088	25845	11.808	24.979	4.20

*Converted from tables in Vargaftik, *Tables of the Thermophysical Properties of Liquids and Gases*, Nauka, Moscow, 1972, and Hemisphere, Washington, 1975. m = melting point. The notation 1.78.-13 signifies 1.78×10^{-13} .

Many of the Vargaftik values also appear in Ohse, R. W., *Handbook of Thermodynamic and Transport Properties of Alkali Metals*, Blackwell Sci. Pubs., Oxford, 1985 (1020 pp.). This source contains superheat data.

Saturation and superheat tables and a diagram to 14 bar, 2200 K are given by Reynolds, W. C., *Thermodynamic properties in S.I.*, Stanford Univ. publ., 1979 (173 pp.). For a Mollier diagram from 0.1 to 140 psia, 2100–3600 °R, see Weatherford, P. M., J. C. Tyler, et al., WADD-TR-61-96, 1961.

An extensive review of properties of the solid and the saturated liquid was given by Alcock, C. B., M. W. Chase, et al., *J. Phys. Chem. Ref. Data*, **23**, 3 (1994): 385–497.

TABLE 2-231 Lithium Bromide—Water Solutions

Ruiter, J. P., *Rev. Int. Froid = Int. J. Refrig.*, **13** (1990): 223–236 gives subroutines for computer calculations. See also ASHRAE *Handbook—Fundamentals*.

TABLE 2-232 Saturated Mercury*

T, K	P, bar	$v_f \times 10^5, \text{m}^3/\text{kg}$	$v_g, \text{m}^3/\text{kg}$	$h_f, \text{kJ/kg}$	$h_g, \text{kJ/kg}$	$h_{fg}, \text{kJ/kg}$	$s_f, \text{kJ/(kg}\cdot\text{K)}$	$s_g, \text{kJ/(kg}\cdot\text{K)}$
203.15	$2.298 \cdot 10^{-11}$	7.26239	$3.665 \cdot 10^9$	33.131	342.637	309.506	0.32434	1.84787
213.15	$1.288 \cdot 10^{-10}$	7.27570	$6.862 \cdot 10^8$	34.567	343.674	309.107	0.33124	1.78142
223.15	$6.169 \cdot 10^{-10}$	7.28900	$1.499 \cdot 10^8$	35.997	344.710	308.713	0.33780	1.72123
233.15	$2.580 \cdot 10^{-9}$	7.30231	$3.746 \cdot 10^7$	37.422	345.746	308.324	0.34404	1.66647
243.15	$9.573 \cdot 10^{-9}$	7.31563	$1.053 \cdot 10^7$	38.842	346.782	307.940	0.35001	1.61647
253.15	$3.198 \cdot 10^{-8}$	7.32896	$3.281 \cdot 10^6$	40.258	347.819	307.561	0.35571	1.57065
263.15	$9.736 \cdot 10^{-8}$	7.34229	$1.120 \cdot 10^6$	41.668	348.855	307.187	0.36118	1.52852
273.15	$2.728 \cdot 10^{-7}$	7.35563	$4.150 \cdot 10^5$	43.074	349.891	306.817	0.36642	1.48967
283.15	$7.101 \cdot 10^{-7}$	7.36898	$1.653 \cdot 10^5$	44.476	350.927	306.451	0.37146	1.45375
293.15	$1.729 \cdot 10^{-6}$	7.38234	$7.026 \cdot 10^4$	45.874	351.964	306.090	0.37631	1.42045
303.15	$3.968 \cdot 10^{-6}$	7.39572	$3.167 \cdot 10^4$	47.268	353.000	305.732	0.38099	1.38951
313.15	$8.626 \cdot 10^{-6}$	7.40911	$1.505 \cdot 10^4$	48.659	354.036	305.377	0.38550	1.36068
323.15	$1.786 \cdot 10^{-5}$	7.42252	$7.501 \cdot 10^3$	50.046	355.072	305.026	0.38986	1.33378
333.15	$3.356 \cdot 10^{-5}$	7.43594	$3.905 \cdot 10^3$	51.430	356.108	304.678	0.39408	1.30862
343.15	$6.724 \cdot 10^{-5}$	7.44938	$2.115 \cdot 10^3$	52.810	357.145	304.335	0.39816	1.28505
353.15	$1.232 \cdot 10^{-4}$	7.46285	$1.188 \cdot 10^3$	54.188	358.181	303.993	0.40212	1.26292
363.15	$2.182 \cdot 10^{-4}$	7.47633	$6.899 \cdot 10^2$	55.563	359.217	303.654	0.40596	1.24213
373.15	$3.745 \cdot 10^{-4}$	7.48984	413.0	56.936	360.253	303.317	0.40969	1.22255
383.15	$6.247 \cdot 10^{-4}$	7.50337	254.2	58.306	361.289	302.983	0.41331	1.20408
393.15	$1.015 \cdot 10^{-3}$	7.51693	153.6	59.674	362.326	302.652	0.41684	1.18665
403.15	$1.608 \cdot 10^{-3}$	7.53052	103.9	61.039	363.362	302.323	0.42027	1.17017
413.15	$2.491 \cdot 10^{-3}$	7.55415	68.75	62.403	364.397	301.994	0.42361	1.15456
423.15	$3.778 \cdot 10^{-3}$	7.55780	46.43	63.765	365.433	301.668	0.42687	1.13978
433.15	$5.618 \cdot 10^{-3}$	7.57148	31.96	65.125	366.469	301.344	0.43004	1.12575
443.15	$8.204 \cdot 10^{-3}$	7.58520	22.39	66.484	367.504	301.020	0.43314	1.11242
453.15	$1.178 \cdot 10^{-2}$	7.59897	15.95	67.842	368.539	300.697	0.43617	1.09975
463.15	$1.664 \cdot 10^{-2}$	7.61277	11.54	69.198	369.574	300.376	0.43913	1.08768
473.15	$2.315 \cdot 10^{-2}$	7.62662	8.469	70.553	370.609	300.056	0.44203	1.07619
483.15	$3.177 \cdot 10^{-2}$	7.64051	6.301	71.908	371.642	299.734	0.44486	1.06524
493.15	$4.304 \cdot 10^{-2}$	7.65444	4.748	73.261	372.676	299.415	0.44763	1.05478
503.15	$5.758 \cdot 10^{-2}$	7.66843	3.621	74.614	373.708	299.094	0.45035	1.04479
513.15	$7.614 \cdot 10^{-2}$	7.68247	2.793	75.967	374.740	298.773	0.45301	1.03525
523.15	$9.959 \cdot 10^{-2}$	7.69656	2.176	77.319	375.771	298.452	0.45562	1.02611
533.15	0.12892	7.71071	1.7132	78.671	376.800	298.129	0.45818	1.01737
543.15	0.16527	7.72491	1.3613	80.023	377.829	297.806	0.46069	1.00899
553.15	0.20993	7.73918	1.0912	81.375	378.855	297.480	0.46316	1.00095
563.15	0.26435	7.75351	0.88213	82.728	379.880	297.152	0.46558	0.99324
573.15	0.33015	7.7679	0.71874	84.080	380.904	296.824	0.46796	0.98554
583.15	0.40910	7.7823	0.59002	85.434	381.925	296.491	0.47030	0.97893
593.15	0.50320	7.7969	0.48779	86.788	382.944	296.156	0.47260	0.97190
603.15	0.61460	7.8115	0.40600	88.143	383.960	295.817	0.47487	0.96532
613.15	0.74567	7.8262	0.34008	89.499	384.973	295.474	0.47709	0.95899
623.15	0.89896	7.8409	0.28660	90.856	385.984	295.128	0.47929	0.95289
633.15	1.0772	7.8558	0.24291	92.215	386.991	294.776	0.48145	0.94702
643.15	1.2834	7.8707	0.20702	93.575	387.994	294.419	0.48358	0.94135
653.15	1.5207	7.8858	0.17735	94.937	388.994	294.057	0.48568	0.93589
663.15	1.9725	7.9008	0.15269	96.300	389.989	293.689	0.48774	0.93061
673.15	2.1024	7.9160	0.13207	97.666	390.980	293.314	0.48978	0.92552
683.15	2.454	7.9313	0.11476	99.033	391.966	292.933	0.49180	0.92059
693.15	2.852	7.9467	0.10014	100.403	392.947	292.544	0.49378	0.91583
703.15	3.299	7.9622	0.08775	101.775	393.923	292.148	0.49574	0.91123
713.15	3.801	7.9778	0.07719	103.150	394.893	291.743	0.49768	0.90677
723.15	4.362	7.9935	0.06815	104.528	395.858	291.330	0.49959	0.90245
733.15	4.986	8.0094	0.06039	105.908	396.816	290.908	0.50148	0.89827
743.15	5.679	8.0252	0.05369	107.292	397.767	290.475	0.50335	0.89422
753.15	6.446	8.0413	0.04789	108.679	398.711	290.032	0.50519	0.89029
763.15	7.292	8.0574	0.04285	110.069	399.649	289.580	0.50702	0.88647
773.15	8.222	8.074	0.03846	111.463	400.579	289.116	0.50882	0.88277
783.15	9.242	8.090	0.03462	112.861	401.501	288.640	0.51061	0.87917
793.15	10.358	8.106	0.03124	114.262	402.415	288.153	0.51238	0.87568
803.15	11.576	8.123	0.02827	115.668	403.321	287.653	0.51412	0.87228
813.15	12.901	8.140	0.02565	117.078	404.218	287.140	0.51586	0.86898
823.15	14.340	8.157	0.02333	118.492	405.106	286.614	0.51757	0.86576
833.15	15.899	8.174	0.02126	119.911	405.985	286.074	0.51927	0.86263

*From Vukalovich, Ivanov, Fokin, and Yakovlev, *Thermophysical Properties of Mercury*, Standartov, Moscow, 1971. For the saturated liquid the specific volume at 203.15 K is $7.26239 \times 10^{-5} \text{ m}^3/\text{kg}$, etc. All the tabular values for 203.15 K, 213.15 K, 223.15 K, and 233.15 K represent a metastable equilibrium between the subcooled liquid and the saturated vapor.

Saturation and superheat tables and a diagram to 100 bar, 1600 K are given by Reynolds, W. C., *Thermodynamic properties in S.I.*, Stanford Univ. publ., 1979 (173 pp.). For a Mollier diagram from 1 to 8200 psia and 2700°R, see Weatherford, W. D., J. C. Tyler, et al., WADD-TR-61-96, 1961.

2-294 PHYSICAL AND CHEMICAL DATA

TABLE 2-232 Saturated Mercury* (Concluded)

T, K	P, bar	$v_f \times 10^5, \text{m}^3/\text{kg}$	$v_g, \text{m}^3/\text{kg}$	$h_f, \text{kJ/kg}$	$h_g, \text{kJ/kg}$	$h_{fg}, \text{kJ/kg}$	$s_f, \text{kJ}/(\text{kg}\cdot\text{K})$	$s_g, \text{kJ}/(\text{kg}\cdot\text{K})$
843.15	17.584	8.191	0.019426	121.335	406.855	285.520	0.52095	0.85959
853.15	19.403	8.209	0.017785	122.763	407.715	284.952	0.52262	0.85662
863.15	21.36	8.226	0.016317	124.197	408.565	284.368	0.52427	0.85372
873.15	23.46	8.244	0.015000	125.636	409.405	283.769	0.52591	0.85090
883.15	25.72	8.262	0.013815	127.080	410.235	283.155	0.52753	0.84815
893.15	28.14	8.280	0.012748	128.530	411.054	282.524	0.52914	0.84546
903.15	30.72	8.298	0.011784	129.986	411.861	281.875	0.53074	0.84284
913.15	33.47	8.316	0.010911	131.448	412.658	281.210	0.53232	0.84028
923.15	36.41	8.335	0.010120	132.915	413.444	280.529	0.53389	0.83777
933.15	39.53	8.353	0.009401	134.389	414.218	279.829	0.53545	0.83533
943.15	42.85	8.372	0.008746	135.869	414.980	279.111	0.53700	0.83294
953.15	46.36	8.391	0.008150	137.356	415.731	278.375	0.53854	0.83060
963.15	50.09	8.410	0.007604	138.850	416.469	277.619	0.54006	0.82831
973.15	54.03	8.430	0.007105	140.350	417.195	276.845	0.54158	0.82606
983.15	58.20	8.450	0.006648	141.858	417.909	276.051	0.54308	0.82387
993.15	62.59	8.468	0.006228	143.372	418.610	275.238	0.54458	0.82172
1003.15	67.22	8.488	0.005842	144.894	419.298	274.404	0.54607	0.81961
1013.15	72.10	8.508	0.005487	146.424	419.974	273.550	0.54754	0.81754
1023.15	77.22	8.529	0.005159	147.961	420.636	272.675	0.54901	0.81552
1033.15	82.60	8.550	0.004856	149.506	421.286	271.780	0.55047	0.81353
1043.15	88.25	8.570	0.004576	151.059	421.923	270.864	0.55192	0.81158
1053.15	94.17	8.590	0.004317	152.619	422.546	269.927	0.55336	0.80966
1063.15	100.37	8.612	0.004077	154.188	423.156	268.968	0.55479	0.80778
1073.15	106.85	8.632	0.003854	155.766	423.752	267.986	0.55621	0.80593

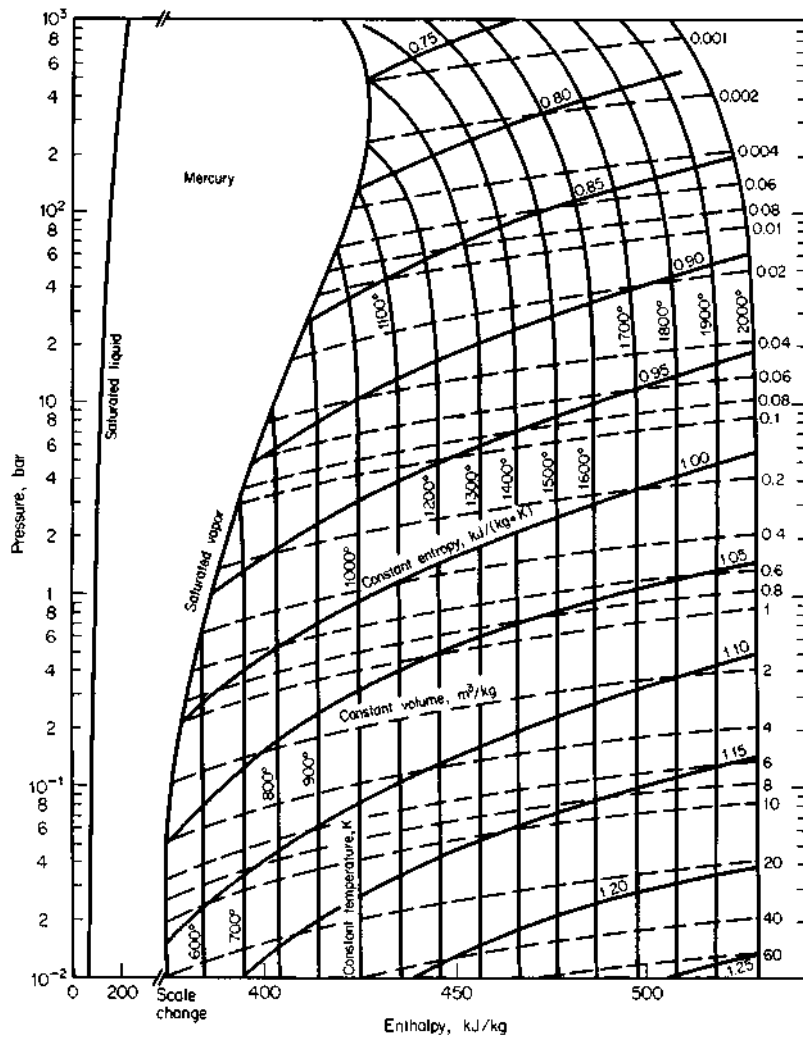


FIG. 2-12 Enthalpy-log-pressure diagram for mercury. (Drawn from tabular data in footnote reference to Table 2-232.)

TABLE 2-233 Thermodynamic Properties of Methane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_v kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
90.694	0.011696	28.142	0.035534	-1.1526	-1.1522	-0.011389	0.034776	0.054029	1538.6	-0.48191	211.24	204.52
100.00	0.034376	27.357	0.036554	-0.64728	-0.64602	-0.0060856	0.033908	0.054681	1452.0	-0.45812	199.67	155.78
105.00	0.056377	26.923	0.037143	-0.37306	-0.37097	-0.0034096	0.033500	0.055135	1403.9	-0.44202	193.03	136.86
110.00	0.088130	26.478	0.037768	-0.096585	-0.093257	-0.00083691	0.033115	0.055656	1354.7	-0.42328	186.18	121.34
115.00	0.13221	26.021	0.038431	0.18242	0.18750	0.0016441	0.032749	0.056253	1304.6	-0.40145	179.21	108.39
120.00	0.19143	25.551	0.039138	0.46425	0.47174	0.0040439	0.032400	0.056941	1253.5	-0.37589	172.15	97.432
125.00	0.26876	25.065	0.039896	0.74927	0.75999	0.0063722	0.032069	0.057741	1201.3	-0.34578	165.04	88.031
130.00	0.36732	24.562	0.040714	1.0379	1.0529	0.0086383	0.031757	0.058684	1148.1	-0.31006	157.91	79.868
135.00	0.49035	24.038	0.041600	1.3307	1.3511	0.010851	0.031469	0.059809	1093.6	-0.26735	150.78	72.699
140.00	0.64118	23.491	0.042569	1.6284	1.6557	0.013020	0.031206	0.061169	1037.7	-0.21579	143.65	66.333
145.00	0.82322	22.917	0.043636	1.9317	1.9676	0.015154	0.030974	0.062840	980.17	-0.15286	136.54	60.620
150.00	1.0400	22.309	0.044825	2.2418	2.2884	0.017264	0.030780	0.064932	920.85	-0.075032	129.43	55.437
155.00	1.2950	21.661	0.046165	2.5602	2.6199	0.019362	0.030631	0.067613	859.39	0.022798	122.32	50.682
160.00	1.5921	20.964	0.047702	2.8887	2.9647	0.021462	0.030541	0.071156	795.43	0.14836	115.19	46.266
165.00	1.9351	20.202	0.049500	3.2304	3.3262	0.023584	0.030531	0.076044	728.42	0.31398	108.01	42.105
170.00	2.3283	19.355	0.051667	3.5895	3.7098	0.025755	0.030634	0.083218	657.52	0.54087	100.73	38.115
175.00	2.7765	18.384	0.054394	3.9734	4.1244	0.028021	0.030920	0.094816	581.27	0.86918	93.324	34.196
180.00	3.2852	17.218	0.058078	4.3965	4.5873	0.030467	0.031554	0.11699	497.01	1.3866	85.799	30.193
185.00	3.8617	15.668	0.063825	4.8955	5.1420	0.033313	0.033085	0.17822	398.59	2.3397	78.733	25.773
190.00	4.5186	12.515	0.079902	5.7074	6.0685	0.038000	0.041746	1.5082	250.31	5.2488	96.970	18.982
190.56	4.5992	10.139	0.098628	6.2136	6.6672	0.041109			0	6.8877		
90.694	0.011696	0.015630	63.981	6.8310	7.5793	0.084885	0.025243	0.033851	249.13	47.921	8.8517	3.6388
100.00	0.034376	0.042048	23.782	7.0469	7.8644	0.079019	0.025487	0.034425	260.09	37.826	10.015	3.9976
105.00	0.056377	0.066154	15.116	7.1582	8.0104	0.076413	0.025652	0.034853	265.31	33.883	10.669	4.1951
110.00	0.088130	0.099622	10.038	7.2654	8.1501	0.074103	0.025842	0.035378	270.01	30.662	11.350	4.3964
115.00	0.13221	0.14457	6.9171	7.3680	8.2825	0.072036	0.026056	0.036016	274.17	28.004	12.062	4.6019
120.00	0.19143	0.20332	4.9183	7.4652	8.4067	0.070168	0.026295	0.036786	277.76	25.790	12.811	4.8123
125.00	0.26876	0.27844	3.5915	7.5562	8.5215	0.068464	0.026560	0.037714	280.76	23.928	13.604	5.0285
130.00	0.36732	0.37278	2.6825	7.6403	8.6257	0.066891	0.026854	0.038836	283.13	22.347	14.449	5.2517
135.00	0.49035	0.48962	2.0424	7.7165	8.7180	0.065421	0.027182	0.040203	284.86	20.993	15.355	5.4833
140.00	0.64118	0.63279	1.5803	7.7837	8.7970	0.064029	0.027549	0.041885	285.93	19.819	16.334	5.7254
145.00	0.82322	0.80691	1.2393	7.8406	8.8608	0.062694	0.027965	0.043985	286.31	18.789	17.402	5.9806
150.00	1.0400	1.0177	0.98256	7.8856	8.9074	0.061391	0.028439	0.046657	285.97	17.870	18.581	6.2526
155.00	1.2950	1.2728	0.78568	7.9166	8.9340	0.060098	0.028989	0.050144	284.88	17.035	19.904	6.5462
160.00	1.5921	1.5821	0.63206	7.9306	8.9369	0.058789	0.029636	0.054849	283.01	16.255	21.423	6.8688
165.00	1.9351	1.9603	0.51014	7.9238	8.9109	0.057431	0.030412	0.061496	280.30	15.500	23.225	7.2313
170.00	2.3283	2.4294	0.41163	7.8898	8.8482	0.055982	0.031374	0.071527	276.66	14.732	25.477	7.6515
175.00	2.7765	3.0268	0.33038	7.8184	8.7357	0.054371	0.032615	0.088273	271.99	13.896	28.545	8.1609
180.00	3.2852	3.8257	0.26139	7.6893	8.5480	0.052471	0.034338	0.12151	266.04	12.892	33.392	8.8251
185.00	3.8617	5.0137	0.19945	7.4515	8.2217	0.049961	0.037087	0.21701	258.03	11.492	43.706	9.8238
190.00	4.5186	7.8027	0.12816	6.7850	7.3641	0.044819	0.045796	2.2590	238.55	8.4951	119.40	12.455
190.56	4.5992	10.139	0.098628	6.2136	6.6672	0.041109			0	6.8877		

Single-Phase Properties

100.00	0.10000	27.360	0.036549	-0.64803	-0.64438	-0.0060931	0.033911	0.054672	1452.6	-0.45829	199.74	155.91
111.51	0.10000	26.341	0.037963	-0.012738	-0.0089413	-0.000079677	0.033003	0.055828	1339.7	-0.41705	184.09	117.20
111.51	0.10000	0.11186	8.9395	7.2969	8.1908	0.073456	0.025904	0.035558	271.33	29.808	11.561	4.4579
200.00	0.10000	0.060518	16.524	9.5570	11.209	0.093427	0.025259	0.033784	369.98	9.2893	21.941	7.8096
300.00	0.10000	0.040158	24.901	12.175	14.665	0.10741	0.027479	0.035869	449.74	4.3216	34.552	11.245
400.00	0.10000	0.030082	33.243	15.151	18.475	0.11834	0.032300	0.040652	510.56	2.2395	50.127	14.272
500.00	0.10000	0.024055	41.572	18.673	22.831	0.12803	0.038196	0.046533	561.86	1.2245	68.564	16.976
600.00	0.10000	0.020042	49.895	22.795	27.784	0.13705	0.044179	0.052509	608.04	0.68722	88.921	19.431
100.00	1.0000	27.403	0.036493	-0.65829	-0.62179	-0.0061960	0.033950	0.054562	1459.6	-0.46060	200.62	157.63
149.14	1.0000	22.416	0.044610	2.1878	2.2325	0.016902	0.030810	0.064535	931.21	-0.089695	130.66	56.297
149.14	1.0000	0.97852	1.0220	7.8788	8.9007	0.061614	0.028353	0.046147	286.08	18.022	18.368	6.2043
200.00	1.0000	0.64363	1.5537	9.3582	10.912	0.073276	0.025879	0.036730	357.81	9.5001	23.028	8.0145
300.00	1.0000	0.40776	2.4524	12.072	14.524	0.087922	0.027621	0.036721	447.04	4.2699	35.152	11.367
400.00	1.0000	0.30205	3.3108	15.083	18.393	0.099023	0.032360	0.041056	510.57	2.2001	50.558	14.357
500.00	1.0000	0.24058	4.1567	18.623	22.780	0.10879	0.038227	0.046766	562.99	1.1998	68.902	17.040
600.00	1.0000	0.20012	4.9971	22.755	27.752	0.11784	0.044198	0.052659	609.73	0.67124	89.200	19.483
100.00	5.0000	27.586	0.036250	-0.70190	-0.52065	-0.0066393	0.034116	0.054117	1490.0	-0.46993	204.45	165.28
200.00	5.0000	5.4706	0.18279	7.8197	8.7337	0.051495	0.032029	0.11667	291.29	8.9784	40.612	10.828
300.00	5.0000	2.1799	0.45874	11.590	13.884	0.072954	0.028262	0.041234	439.25	3.9428	38.480	12.194
400.00	5.0000	1.5333	0.65221	14.779	18.040	0.084897	0.032614	0.042903	513.11	2.0089	52.693	14.872
500.00	5.0000	1.2013	0.83240	18.401	22.563	0.094971	0.038361	0.047789	569.49	1.0870	70.509	17.410
600.00	5.0000	0.99281	1.0072	22.581	27.617	0.10417	0.044277	0.053309	618.13	0.60013	90.498	19.768
100.00	10.000	27.802	0.035969	-0.75239	-0.39270	-0.0071652	0.034314	0.053642	1525.7	-0.47979	209.07	174.83
200.00	10.000	16.593	0.060268	5.1551	5.7578	0.034542	0.030129	0.085085	567.92	1.0266	84.234	29.399
300.00	10.000	4.6859	0.21340	10.942	13.077	0.065137	0.028995	0.048165	444.53	3.2606	44.730	13.896
400.00	10.000	3.1002	0.32256	14.401	17.627	0.078246	0.032902	0.045220	522.58	1.7355	55.941	15.766
500.00	10.000	2.3887	0.41863	18.132	22.318	0.088698	0.038516	0.049007	580.99	0.94125	72.781	18.011
600.00	10.000	1.9619	0.50971	22.371	27.468	0.098073	0.044371	0.054070	630.63	0.51200	92.268	20.217
200.00	100.00	25.496	0.039222	3.0510	6.9732	0.020596	0.032058	0.048512	1541.0	-0.51619	188.05	80.392
300.00	100.00	21.266	0.047024	7.0865	11.789	0.040126	0.031823	0.048281	1267.5	-0.44889	137.68	47.835
400.00	100.00	17.881	0.055926	11.121	16.713	0.054276	0.035273	0.050523	1115.8	-0.37484	120.38	37.584
500.00	100.00	15.305	0.065340	15.405	21.939	0.065922	0.040312	0.054139	1044.8	-0.32811	120.87	33.590
600.00	100.00	13.357	0.074869	20.074	27.561	0.076160	0.045724	0.058364	1018.4	-0.30439	130.36	32.111
200.00	500.00	33.003	0.030301	2.3322	17.482	0.0061671	0.037832	0.047821	2664.2	-0.53926	429.60	205.24
300.00	500.00	30.786	0.032482	5.9505	22.192	0.025271	0.037006	0.047114	2500.0	-0.55416	358.93	106.90
400.00	500.00	28.929	0.034567	9.7401	27.024	0.039152	0.039890	0.049933	2360.3	-0.52806	312.36	78.768
500.00	500.00	27.331	0.036588	13.934	32.228	0.050747	0.044407	0.054280	2250.3	-0.49035	285.41	66.669
600.00	500.00	25.934	0.038559	18.612	37.892	0.061061	0.049344	0.059017	2168.1	-0.45514	272.14	60.413

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Setzmann, U., and Wagner, W., "A New Equation of State and Tables of Thermodynamic Properties for Methane Covering the Range from the Melting Line to 625 K at Pressures up to 1000 MPa," *J. Phys. Chem. Ref. Data* **20**(6):1061–1151, 1991. The source for viscosity is Younglove, B. A., and Ely, J. F., "Thermophysical Properties of Fluids. II. Methane, Ethane, Propane, Isobutane and Normal Butane," *J. Phys. Chem. Ref. Data* **16**:577–798, 1987. The source for thermal conductivity is Friend, D. G., Ely, J. F., and Ingham, H., "Tables for the Thermophysical Properties of Methane," *NIST Tech. Note* 1325, 1989.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in density are 0.03% for pressures below 12 MPa and temperatures below 350 K and up to 0.07% for pressures less than 50 MPa. For the speed of sound, the uncertainty ranges from 0.03% (in the vapor phase) to 0.3% depending on temperature and pressure. Heat capacities may be generally calculated within an uncertainty of 1%. The uncertainty in viscosity is 2%, except in the critical region which is 5%. The uncertainty in thermal conductivity of the dilute gas between 130 and 625 K is 2.5%. For temperatures below 130 K, the uncertainty is less than 10%. Excluding the dilute gas, the uncertainty is 2% between 110 and 725 K at pressures up to 70 MPa, except near the critical point which has an uncertainty of 5% or greater. For the vapor at lower temperatures and the dense liquid near the triple point, an uncertainty of 10% is possible.

TABLE 2-234 Thermodynamic Properties of Methanol

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
175.61	1.8635E-07	28.230	0.035423	-12.440	-12.440	-0.049524	0.056728	0.070390	1625.1	-0.40884
180.00	3.7619E-07	28.096	0.035592	-12.130	-12.130	-0.047781	0.056689	0.070750	1590.2	-0.40373
195.00	3.2175E-06	27.629	0.036194	-11.067	-11.067	-0.042108	0.056604	0.070897	1496.4	-0.39791
210.00	1.9841E-05	27.163	0.036815	-10.001	-10.001	-0.036846	0.057072	0.071215	1425.3	-0.39361
225.00	9.4330E-05	26.703	0.037449	-8.9277	-8.9277	-0.031908	0.057992	0.072004	1363.2	-0.38674
240.00	0.00036348	26.250	0.038096	-7.8395	-7.8395	-0.027226	0.059275	0.073141	1304.6	-0.37793
255.00	0.0011791	25.802	0.038756	-6.7318	-6.7318	-0.022750	0.060916	0.074617	1248.2	-0.36733
270.00	0.0033166	25.360	0.039432	-5.5991	-5.5990	-0.018434	0.062917	0.076487	1194.1	-0.35457
285.00	0.0082787	24.922	0.040125	-4.4351	-4.4347	-0.014239	0.065250	0.078803	1142.6	-0.33915
300.00	0.018682	24.484	0.040844	-3.2329	-3.2322	-0.010129	0.067864	0.081584	1093.5	-0.32073
315.00	0.038692	24.041	0.041595	-1.9858	-1.9842	-0.0060725	0.070693	0.084823	1046.3	-0.29904
330.00	0.074453	23.590	0.042390	-0.68700	-0.68385	-0.0020451	0.073674	0.088505	1000.2	-0.27385
345.00	0.13447	23.124	0.043244	0.66961	0.67543	0.0019747	0.076752	0.092616	954.32	-0.24479
360.00	0.22992	22.638	0.044174	2.0901	2.1003	0.0060049	0.079881	0.097164	907.64	-0.21121
375.00	0.37483	22.123	0.045203	3.5806	3.5975	0.010061	0.083033	0.10219	859.31	-0.17199
390.00	0.58617	21.571	0.046358	5.1475	5.1746	0.014159	0.086189	0.10776	808.48	-0.12529
405.00	0.88399	20.973	0.047681	6.7983	6.8404	0.018314	0.089346	0.11405	754.41	-0.068163
420.00	1.2914	20.315	0.049226	8.5423	8.6058	0.022546	0.092514	0.12133	696.47	0.0042669
435.00	1.8349	19.579	0.051075	10.392	10.485	0.026879	0.095722	0.13007	634.17	0.10021
450.00	2.5433	18.741	0.053360	12.364	12.500	0.031347	0.099032	0.14120	567.09	0.23444
465.00	3.4456	17.759	0.056310	14.488	14.682	0.036009	0.10257	0.15679	494.36	0.43762
480.00	4.5713	16.553	0.060411	16.820	17.096	0.040977	0.10666	0.18345	412.12	0.79465
495.00	5.9794	14.880	0.067203	19.521	19.923	0.046590	0.11250	0.25717	308.94	1.6506
510.00	7.7496	11.689	0.085547	23.297	23.960	0.054351	0.12653	1.1088	192.83	4.6061
513.38	8.2159	8.7852	0.11383	25.917	26.852	0.059911			0	6.7425
Subcooled Properties										
175.61	1.8635E-07	1.2764E-07	7,834,400.	28.219	29.679	0.19032	0.031874	0.040287	239.95	1187400.
180.00	3.7619E-07	2.5140E-07	3,977,700.	28.353	29.850	0.18544	0.032397	0.040854	242.62	857090.
195.00	3.2175E-06	1.9855E-06	503,660.	28.810	30.430	0.17069	0.035224	0.043954	251.06	293110.
210.00	1.9841E-05	1.1378E-05	87,892.	29.259	31.003	0.15841	0.040104	0.049389	258.49	105090.
225.00	9.4330E-05	5.0556E-05	19,780.	29.698	31.564	0.14806	0.047248	0.057480	265.30	39363.
240.00	0.00036348	0.00018304	5,463.4	30.123	32.109	0.13923	0.056324	0.067973	271.89	15552.
255.00	0.0011791	0.00056065	1,783.7	30.534	32.637	0.13164	0.066572	0.080135	278.43	6557.5
270.00	0.0033166	0.0014959	668.48	30.932	33.149	0.12508	0.077055	0.093000	284.90	2971.8
285.00	0.0082787	0.0035581	281.05	31.321	33.648	0.11938	0.086920	0.10564	291.19	1449.9
300.00	0.018682	0.0076845	130.13	31.703	34.134	0.11442	0.095581	0.11740	297.15	759.96
315.00	0.038692	0.015300	65.359	32.077	34.606	0.11009	0.10279	0.12798	302.63	426.34
330.00	0.074453	0.028438	35.164	32.442	35.060	0.10627	0.10860	0.13749	307.47	254.84
345.00	0.13447	0.049870	20.052	32.789	35.485	0.10287	0.11331	0.14644	311.48	161.53
360.00	0.22992	0.083267	12.009	33.108	35.869	0.099808	0.11736	0.15559	314.48	108.02
375.00	0.37483	0.13344	7.4940	33.385	36.194	0.096984	0.12125	0.16605	316.20	75.802
390.00	0.58617	0.20674	4.8370	33.601	36.436	0.094317	0.12546	0.17917	316.34	55.467
405.00	0.88399	0.31179	3.2073	33.736	36.571	0.091723	0.13033	0.19663	314.53	42.002
420.00	1.2914	0.46071	2.1706	33.767	36.570	0.089128	0.13587	0.21986	310.36	32.587
435.00	1.8349	0.67055	1.4913	33.687	36.423	0.086505	0.14101	0.24709	303.71	25.532
450.00	2.5433	0.96219	1.0393	33.541	36.184	0.083978	0.14238	0.26502	295.26	19.860
465.00	3.4456	1.3555	0.73775	33.439	35.981	0.081813	0.13589	0.25879	285.25	15.568
480.00	4.5713	1.9102	0.52352	33.258	35.652	0.079634	0.12618	0.27959	267.83	13.904
495.00	5.9794	2.9050	0.34423	32.267	34.325	0.075685	0.12608	0.42448	247.46	12.099
510.00	7.7496	5.1706	0.19340	29.688	31.187	0.068520	0.13259	1.9096	212.65	9.5115
513.38	8.2159	8.7852	0.11383	25.917	26.852	0.059911			0	6.7425

Single-Phase Properties

200.00	0.10000	27.474	0.036398	-10.713	-10.709	-0.040317	0.056702	0.070943	1471.5	-0.39677
300.00	0.10000	24.486	0.040839	-3.2341	-3.2300	-0.010133	0.067862	0.081580	1094.1	-0.32081
337.30	0.10000	23.366	0.042798	-0.034546	-0.030266	-0.000089518	0.075163	0.090451	977.93	-0.26023
337.30	0.10000	0.037626	26.577	32.613	35.271	0.10457	0.11100	0.14187	309.54	202.71
400.00	0.10000	0.030452	32.839	36.075	39.359	0.11581	0.044972	0.054208	349.19	40.941
500.00	0.10000	0.024157	41.396	40.921	45.060	0.12851	0.051823	0.060380	387.15	12.933
600.00	0.10000	0.020089	49.779	46.476	51.454	0.14014	0.059065	0.067441	420.71	4.3382
200.00	1.0000	27.491	0.036376	-10.720	-10.684	-0.040354	0.056724	0.070932	1475.1	-0.39705
300.00	1.0000	24.514	0.040793	-3.2472	-3.2064	-0.010176	0.067848	0.081541	1100.0	-0.32177
400.00	1.0000	21.193	0.047185	6.2298	6.2770	0.016901	0.088257	0.11177	775.46	-0.090229
409.75	1.0000	20.772	0.048143	7.3401	7.3883	0.019645	0.090347	0.11623	736.51	-0.047179
409.75	1.0000	0.35352	2.8287	33.758	36.586	0.090904	0.13203	0.20333	313.48	38.678
500.00	1.0000	0.25202	3.9680	40.335	44.303	0.10818	0.056676	0.068069	376.08	13.330
600.00	1.0000	0.20501	4.8778	46.300	51.178	0.12070	0.061344	0.070369	413.09	4.5635
200.00	5.0000	27.561	0.036283	-10.752	-10.571	-0.040517	0.056820	0.070883	1490.9	-0.39825
300.00	5.0000	24.635	0.040592	-3.3039	-3.1010	-0.010367	0.067795	0.081377	1125.5	-0.32568
400.00	5.0000	21.441	0.046640	6.0896	6.3228	0.016546	0.087676	0.11029	818.58	-0.11504
484.95	5.0000	16.076	0.062203	17.655	17.966	0.042725	0.10826	0.19836	381.15	0.98684
484.95	5.0000	2.1711	0.46060	33.047	35.350	0.078574	0.12499	0.32263	260.06	13.826
500.00	5.0000	1.7679	0.56566	35.907	38.735	0.085457	0.098975	0.17315	301.43	12.155
600.00	5.0000	1.1389	0.87808	45.247	49.638	0.10553	0.072927	0.087489	379.21	5.1009
200.00	10.000	27.648	0.036169	-10.791	-10.430	-0.040716	0.056935	0.070820	1509.9	-0.39966
300.00	10.000	24.779	0.040357	-3.3713	-2.9677	-0.010598	0.067746	0.081196	1155.3	-0.32990
400.00	10.000	21.717	0.046048	5.9321	6.3925	0.016141	0.087087	0.10880	865.91	-0.13884
500.00	10.000	15.932	0.062765	19.374	20.002	0.046226	0.10760	0.18959	424.68	0.83939
600.00	10.000	2.6640	0.37537	43.262	47.015	0.096406	0.088868	0.12122	343.60	5.0965
200.00	100.00	28.911	0.034588	-11.305	-7.8460	-0.043691	0.057827	0.068992	1772.1	-0.41976
300.00	100.00	26.630	0.037552	-4.2043	-0.44914	-0.013840	0.067889	0.079818	1515.8	-0.35799
400.00	100.00	24.493	0.040827	4.3449	8.4277	0.011565	0.082823	0.098951	1334.7	-0.26099
500.00	100.00	22.020	0.045413	14.917	19.458	0.036085	0.095694	0.12152	1164.1	-0.14407
600.00	100.00	19.139	0.052250	27.300	32.525	0.059862	0.10406	0.13787	977.50	-0.023905
300.00	500.00	30.547	0.032736	-5.4195	10.949	-0.022106	0.070293	0.080627	2316.0	-0.34762
400.00	500.00	29.154	0.034300	2.2795	19.430	0.0022308	0.077541	0.089761	2194.8	-0.30897
500.00	500.00	27.670	0.036140	11.020	29.089	0.023726	0.084883	0.10419	2123.2	-0.24751
600.00	500.00	26.003	0.038457	21.094	40.322	0.044161	0.092281	0.12017	2074.1	-0.19279

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is de Reuck, K. M., and Craven, R. J. B., "Methanol, International Thermodynamic Tables of the Fluid State—12," *IUPAC*, Blackwell Scientific Publications, London, 1993. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties of the equation of state are generally 0.1% in density and 2% in the speed of sound, except in the critical region and high pressures.

TABLE 2-235 Thermodynamic Properties of 2-Methyl Butane (Isopentane)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. Cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
112.65	8.9347E-11	10.936	0.091442	-26.072	-26.072	-0.13287	0.088268	0.12675	2024.6	-0.61677	177.46	18091
115	1.9071E-10	10.903	0.091719	-25.774	-25.774	-0.13026	0.087968	0.12625	2007.2	-0.61938	177.30	15987
130	1.1761E-08	10.698	0.093478	-23.896	-23.896	-0.11491	0.087279	0.12456	1899.0	-0.62810	174.59	7701.7
145	2.8119E-07	10.500	0.095242	-22.028	-22.028	-0.10131	0.088069	0.12473	1796.0	-0.62623	169.97	4126.6
160	3.4585E-06	10.306	0.097028	-20.148	-20.148	-0.08897	0.089805	0.12613	1698.3	-0.61691	164.30	2438.0
175	2.6222E-05	10.116	0.09885	-18.240	-18.240	-0.07757	0.092192	0.12840	1605.5	-0.60215	158.07	1564.5
190	0.00013839	9.9281	0.10072	-16.293	-16.293	-0.06690	0.095056	0.13133	1517.0	-0.58328	151.59	1074.5
205	0.00055325	9.7405	0.10266	-14.298	-14.298	-0.05679	0.098282	0.13478	1432.2	-0.56116	145.02	779.51
220	0.0017804	9.5525	0.10468	-12.247	-12.247	-0.04714	0.10179	0.13865	1350.7	-0.53631	138.51	591.10
235	0.0048228	9.3629	0.10680	-10.136	-10.136	-0.03786	0.10553	0.14289	1271.9	-0.50892	132.11	464.45
250	0.011375	9.1708	0.10904	-7.9595	-7.9583	-0.02888	0.10944	0.14744	1195.4	-0.47898	125.89	375.48
265	0.023974	8.9750	0.11142	-5.7124	-5.7097	-0.02016	0.11351	0.15230	1120.7	-0.44620	119.87	310.55
280	0.046069	8.7745	0.11397	-3.3908	-3.3856	-0.01163	0.11771	0.15744	1047.6	-0.41003	114.07	261.53
295	0.082015	8.5679	0.11671	-0.99085	-0.98128	-0.00329	0.12200	0.16288	975.50	-0.36957	108.49	223.37
310	0.13700	8.3537	0.11971	1.4914	1.5078	0.004922	0.12638	0.16864	904.21	-0.32346	103.13	192.84
325	0.21693	8.1302	0.12300	4.0602	4.0869	0.013015	0.13084	0.17476	833.33	-0.26964	98.008	167.77
340	0.32835	7.8950	0.12666	6.7200	6.7616	0.021017	0.13535	0.18133	762.50	-0.20494	93.105	146.69
355	0.47831	7.6451	0.13080	9.4764	9.5389	0.028954	0.13992	0.18849	691.33	-0.12436	88.417	128.56
370	0.67436	7.3765	0.13557	12.336	12.428	0.036852	0.14455	0.19649	619.38	-0.01964	83.934	112.60
385	0.92461	7.0835	0.14117	15.310	15.440	0.044740	0.14926	0.20576	546.13	0.12385	79.645	98.258
400	1.2378	6.7572	0.14799	18.410	18.593	0.052658	0.15409	0.21716	470.92	0.33435	75.537	85.057
415	1.6238	6.3830	0.15667	21.662	21.916	0.060668	0.15915	0.23260	392.75	0.67407	71.596	72.595
430	2.0941	5.9327	0.16856	25.110	25.463	0.068880	0.16468	0.25764	309.77	1.3120	67.829	60.422
445	2.6640	5.3333	0.18750	28.871	29.370	0.077578	0.17145	0.31907	217.58	2.9280	64.403	47.740
460.35	3.3782	3.2710	0.30572	35.298	36.331	0.092549			0	18.924		
112.65	8.9527E-11	9.5585E-11	1.05E+10	7.8069	8.7435	0.17618	0.044856	0.053170	124.05		2.3120	2.775
115	1.9071E-10	1.9975E-10	5.01E+09	7.9134	8.8695	0.17099	0.045769	0.054083	125.14	1474.9	2.3996	2.827
130	1.1761E-08	1.0881E-08	91,905,000	8.6432	9.7241	0.14371	0.051515	0.059829	131.91	867.62	3.0011	3.1624
145	2.8119E-07	2.3324E-07	4,287,500	9.4580	10.664	0.12415	0.057098	0.065412	138.36	543.86	3.6746	3.5046
160	3.4585E-06	2.5998E-06	384,650	10.355	11.686	0.10999	0.062541	0.070856	144.53	358.94	4.4196	3.853
175	2.6222E-05	1.8023E-05	55,485	11.333	12.788	0.099732	0.067888	0.076206	150.45	247.17	5.2372	4.2074
190	0.00013839	8.7629E-05	11,412	12.390	13.969	0.092376	0.073180	0.081510	156.12	176.38	6.1289	4.5669
205	0.00055325	0.00032489	3,077.9	13.524	15.227	0.087227	0.078451	0.086811	161.53	129.80	7.0964	4.931
220	0.0017804	0.00097556	1,025.1	14.732	16.557	0.083784	0.083733	0.092156	166.66	98.138	8.1416	5.2989
235	0.0048228	0.0024804	403.15	16.011	17.955	0.081674	0.089051	0.09759	171.43	76.055	9.2673	5.6698
250	0.011375	0.0055234	181.05	17.357	19.416	0.080615	0.094425	0.10316	175.77	60.317	10.477	6.0432
265	0.023974	0.011056	90.449	18.766	20.935	0.080389	0.099871	0.10890	179.56	48.903	11.776	6.4187
280	0.046069	0.020302	49.256	20.234	22.503	0.080825	0.10539	0.11485	182.71	40.504	13.169	6.7963
295	0.082015	0.034759	28.769	21.756	24.115	0.081787	0.11100	0.12105	185.07	34.252	14.664	7.1769
310	0.13700	0.056212	17.790	23.326	25.763	0.083164	0.11669	0.12757	186.54	29.560	16.270	7.5619
325	0.21693	0.086783	11.523	24.939	27.438	0.084865	0.12246	0.13447	186.97	26.024	17.997	7.9543
340	0.32835	0.12904	7.7495	26.587	29.132	0.086812	0.12831	0.14187	186.20	23.372	19.860	8.3586
355	0.47831	0.18620	5.3705	28.263	30.831	0.088933	0.13425	0.14999	184.05	21.417	21.877	8.7821
370	0.67436	0.26252	3.8092	29.953	32.522	0.091161	0.14029	0.15916	180.31	20.043	24.077	9.2355
385	0.92461	0.36398	2.7474	31.643	34.183	0.093423	0.14648	0.17005	174.70	19.189	26.507	9.7357
400	1.2378	0.49975	2.0010	33.305	35.782	0.095630	0.15286	0.18403	166.85	18.851	29.248	10.311
415	1.6238	0.68534	1.4591	34.899	37.268	0.097660	0.15958	0.20431	156.26	19.092	32.469	11.011
430	2.0941	0.95152	1.0510	36.342	38.543	0.099299	0.16690	0.24064	142.23	20.076	36.596	11.947
445	2.6640	1.3789	0.72521	37.430	39.362	0.10003	0.17558	0.34061	123.63	22.072	43.254	13.438
460.35	3.3782	3.2710	0.30572	35.298	36.331	0.092549			0	18.924		

Single-Phase Properties

150.00	0.10000	10.435	0.095828	-21.406	-21.396	-0.097086	0.088566	0.12508	1763.3	-0.62386	168.19	3428.5
225.00	0.10000	9.4905	0.10537	-11.554	-11.543	-0.044025	0.10302	0.14001	1324.8	-0.52760	136.40	543.85
300.00	0.10000	8.4975	0.11768	-0.17289	-0.16112	-0.00053568	0.12345	0.16476	951.69	-0.35493	106.68	212.48
300.60	0.10000	8.4890	0.11780	-0.074609	-0.062829	-0.00020836	0.12363	0.16499	948.83	-0.35312	106.46	211.22
300.60	0.10000	0.041839	23.901	22.336	24.726	0.082257	0.11311	0.12344	185.73	32.344	15.250	7.3199
375.00	0.10000	0.032734	30.549	31.687	34.742	0.11194	0.13702	0.14615	210.31	14.520	23.271	9.1730
450.00	0.10000	0.027025	37.003	42.874	46.575	0.14064	0.16045	0.16919	231.26	8.0746	32.656	10.981
150.00	1.0000	10.441	0.095776	-21.422	-21.326	-0.097192	0.088617	0.12507	1767.4	-0.62411	168.39	3436.2
225.00	1.0000	9.5002	0.10526	-11.582	-11.477	-0.044150	0.10306	0.13992	1330.6	-0.52893	136.76	547.08
300.00	1.0000	8.5157	0.11743	-0.22558	-0.10815	-0.00071178	0.12349	0.16444	960.79	-0.36015	107.23	214.82
375.00	1.0000	7.2951	0.13708	13.279	13.417	0.039387	0.14609	0.19894	600.15	0.013112	82.749	108.44
388.92	1.0000	7.0019	0.14282	16.107	16.250	0.046804	0.15050	0.20848	526.70	0.17073	78.555	94.715
388.92	1.0000	0.39567	2.5273	32.081	34.609	0.094009	0.14812	0.17332	172.88	19.050	27.189	9.8772
450.00	1.0000	0.30367	3.2931	41.914	45.207	0.11931	0.16299	0.17764	207.91	9.5995	34.291	11.376
150.00	5.0000	10.466	0.095550	-21.491	-21.013	-0.097661	0.088840	0.12500	1785.1	-0.62511	169.25	3470.1
225.00	5.0000	9.5418	0.10480	-11.704	-11.180	-0.044697	0.10325	0.13954	1355.9	-0.53441	138.31	561.26
300.00	5.0000	8.5922	0.11638	-0.44721	0.13471	-0.0014609	0.12363	0.16321	999.20	-0.38047	109.58	224.97
375.00	5.0000	7.4791	0.13371	12.787	13.456	0.038048	0.14600	0.19336	670.90	-0.10299	86.612	119.86
450.00	5.0000	5.7922	0.17265	28.764	29.627	0.077218	0.16985	0.24769	327.62	1.0932	68.533	59.364
150.00	10.000	10.496	0.095275	-21.575	-20.622	-0.098235	0.089117	0.12494	1806.7	-0.62619	170.30	3511.8
225.00	10.000	9.5915	0.10426	-11.848	-10.805	-0.045356	0.10348	0.13913	1386.1	-0.54037	140.18	578.68
300.00	10.000	8.6795	0.11521	-0.69963	0.45251	-0.0023313	0.12382	0.16197	1043.2	-0.40072	112.31	237.18
375.00	10.000	7.6607	0.13054	12.293	13.598	0.036668	0.14603	0.18923	742.26	-0.18850	90.691	132.45
450.00	10.000	6.3848	0.15662	27.389	28.955	0.073910	0.16843	0.22130	471.81	0.27965	75.593	78.440
225.00	100.00	10.235	0.097700	-13.576	-3.8061	-0.054491	0.10727	0.13669	1787.0	-0.57796	165.84	860.73
300.00	100.00	9.6137	0.10402	-3.2651	7.1368	-0.012648	0.12720	0.15576	1539.5	-0.50239	145.78	420.70
375.00	100.00	9.0393	0.11063	8.5444	19.607	0.024366	0.14895	0.17686	1352.4	-0.43356	130.89	290.41
450.00	100.00	8.5029	0.11761	21.892	33.653	0.058454	0.17040	0.19748	1210.1	-0.37778	121.48	234.19
375.00	500.00	10.851	0.092161	5.2760	51.356	0.0033611	0.15705	0.17792	2411.2	-0.44269	211.59	864.12
450.00	500.00	10.550	0.094785	18.067	65.459	0.037589	0.17779	0.19805	2302.8	-0.40062	205.82	747.58

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., and Span, R., "Short Fundamental Equations of State for 20 Industrial Fluids," *J. Chem. Eng. Data* **51**(3):785–850, 2006. The source for viscosity and thermal conductivity is NIST14, Version 9.08.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties are approximately 0.2% in density at temperatures up to 320 K, 0.5% in density at higher temperatures, 2% in heat capacity above 250 K, 4% in heat capacity at lower temperatures, 0.1% in the vapor phase speed of sound, 3% in the liquid-phase speed of sound, and 0.4% in vapor pressure at temperatures above 200 K. For viscosity, estimated uncertainty is 2%. For thermal conductivity, estimated uncertainty, except near the critical region, is 4–6%.

TABLE 2-236 Thermodynamic Properties of 2-Methyl Pentane (Isohexane)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. Energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
119.60	7.3432E-12	9.3708	0.10671	-36.470	-36.470	-0.17002	0.10838	0.15299	2019.9	-0.59545
120.00	8.6393E-12	9.3662	0.10677	-36.409	-36.409	-0.16951	0.10835	0.15291	2016.9	-0.59577
140.00	2.6628E-09	9.1438	0.10936	-33.376	-33.376	-0.14612	0.10816	0.15094	1877.8	-0.60440
160.00	1.6516E-07	8.9315	0.11196	-30.354	-30.354	-0.12595	0.10991	0.15162	1750.1	-0.60052
180.00	3.6662E-06	8.7257	0.11460	-27.297	-27.297	-0.10795	0.11321	0.15435	1631.2	-0.58671
200.00	4.0345E-05	8.5239	0.11732	-24.169	-24.169	-0.091478	0.11773	0.15866	1519.7	-0.56534
220.00	0.00026966	8.3237	0.12014	-20.943	-20.943	-0.076105	0.12318	0.16418	1414.3	-0.53846
240.00	0.0012504	8.1236	0.12310	-17.596	-17.596	-0.061550	0.12928	0.17061	1314.2	-0.50751
260.00	0.0044049	7.9216	0.12624	-14.114	-14.113	-0.047617	0.13582	0.17772	1218.4	-0.47322
280.00	0.012572	7.7163	0.12960	-10.484	-10.483	-0.034172	0.14266	0.18535	1126.1	-0.43565
300.00	0.030459	7.5058	0.13323	-6.6984	-6.6944	-0.021115	0.14969	0.19342	1036.6	-0.39411
320.00	0.064855	7.2881	0.13721	-2.7483	-2.7394	-0.0083705	0.15683	0.20190	948.94	-0.34721
340.00	0.12457	7.0608	0.14163	1.3732	1.3908	0.0041212	0.16404	0.21081	862.49	-0.29249
360.00	0.22023	6.8210	0.14661	5.6736	5.7059	0.016411	0.17128	0.22027	776.48	-0.22585
380.00	0.36396	6.5643	0.15234	10.162	10.218	0.028548	0.17854	0.23047	690.10	-0.14022
400.00	0.56922	6.2850	0.15911	14.852	14.942	0.040580	0.18581	0.24183	602.46	-0.022615
420.00	0.85081	5.9735	0.16740	19.761	19.903	0.052569	0.19312	0.25521	512.43	0.15352
440.00	1.2253	5.6136	0.17814	24.925	25.143	0.064603	0.20057	0.27266	418.45	0.45116
460.00	1.7125	5.1716	0.19337	30.414	30.745	0.076849	0.20838	0.30051	317.90	1.0612
480.00	2.3384	4.5506	0.21975	36.430	36.944	0.089761	0.21745	0.37420	204.95	2.9251
497.70	3.0426	2.7150	0.36832	44.813	45.934	0.10771			0	22.418
119.60	7.3432E-12	7.7171E-12	1.2958E+11	3.2114	4.2058	0.17008	0.062593	0.070908	114.33	1878.9
120.00	8.6393E-12	8.8145E-12	1.1345E+11	3.2365	4.2342	0.16919	0.062751	0.071065	114.51	1852.6
140.00	2.6628E-09	2.2870E-09	43.7250,000.	4.5677	5.7317	0.13321	0.070285	0.078600	122.90	971.14
160.00	1.6516E-07	1.2415E-07	8.054,900.	6.0465	7.3769	0.10987	0.077610	0.085925	130.73	557.20
180.00	3.6662E-06	2.4497E-06	4.08,210.	7.6734	9.1700	0.094644	0.085136	0.093451	138.07	341.59
200.00	4.0345E-05	2.4265E-05	41.212.	9.4535	11.116	0.084949	0.092992	0.10131	144.98	220.67
220.00	0.00026966	0.00014751	6,779.4	11.391	13.220	0.079177	0.10116	0.10950	151.49	148.98
240.00	0.0012504	0.00062783	1,592.8	13.489	15.480	0.076268	0.10956	0.11797	157.59	104.56
260.00	0.0044049	0.0020480	488.27	15.742	17.893	0.075483	0.11814	0.12670	163.19	76.030
280.00	0.012572	0.0054602	183.14	18.144	20.447	0.076291	0.12686	0.13570	168.11	57.168
300.00	0.030459	0.012471	80.187	20.686	23.128	0.078294	0.13569	0.14499	172.17	44.401
320.00	0.064855	0.025273	39.568	23.355	25.921	0.081194	0.14461	0.15463	175.13	35.601
340.00	0.12457	0.046680	21.422	26.139	28.807	0.084758	0.15360	0.16471	176.74	29.461
360.00	0.22023	0.080240	12.463	29.022	31.767	0.088803	0.16265	0.17537	176.73	25.167
380.00	0.36396	0.13054	7.6606	31.987	34.775	0.093173	0.17176	0.18690	174.77	22.214
400.00	0.56922	0.20388	4.9048	35.009	37.801	0.097727	0.18095	0.19986	170.42	20.301
420.00	0.85081	0.30983	3.2276	38.050	40.796	0.10231	0.19027	0.21550	163.11	19.294
440.00	1.2253	0.46493	2.1509	41.048	43.683	0.10674	0.19983	0.23692	151.97	19.227
460.00	1.7125	0.70383	1.4208	43.875	46.308	0.11068	0.20995	0.27434	135.67	20.404
480.00	2.3384	1.1281	0.88647	46.190	48.263	0.11334	0.22144	0.38625	112.14	23.602
497.70	3.0426	2.7150	0.36832	44.813	45.934	0.10771			0	22.418

Single-Phase Properties

150.00	0.10000	9.0371	0.11065	-31.868	-31.857	-0.13572	0.10883	0.15099	1813.1	-0.60387
250.00	0.10000	8.0239	0.12463	-15.876	-15.864	-0.054530	0.13251	0.17408	1266.4	-0.49091
332.95	0.10000	7.1422	0.14001	-0.099839	-0.085838	-0.00025709	0.16149	0.20761	892.88	-0.31289
332.95	0.10000	0.037947	26.353	25.145	27.780	0.083438	0.15042	0.16110	176.34	31.381
350.00	0.10000	0.035789	27.941	27.784	30.578	0.091632	0.15693	0.16712	181.93	25.747
450.00	0.10000	0.027168	36.808	45.429	49.110	0.13801	0.19438	0.20338	209.65	10.406
550.00	0.10000	0.022045	45.361	66.610	71.146	0.18212	0.22795	0.23660	232.80	5.5017
150.00	1.0000	9.0415	0.11060	-31.886	-31.775	-0.13584	0.10889	0.15098	1816.8	-0.60402
250.00	1.0000	8.0324	0.12450	-15.912	-15.787	-0.054671	0.13256	0.17397	1272.3	-0.49228
350.00	1.0000	6.9610	0.14366	3.4284	3.5720	0.010082	0.16769	0.21498	829.36	-0.26842
428.63	1.0000	5.8255	0.17166	21.956	22.127	0.057750	0.19631	0.26204	472.47	0.26084
428.63	1.0000	0.36934	2.7075	39.354	42.062	0.10426	0.19435	0.22372	158.83	19.140
450.00	1.0000	0.33044	3.0263	43.784	46.810	0.11507	0.19990	0.22188	173.64	14.396
550.00	1.0000	0.23887	4.1863	65.703	69.889	0.16131	0.22981	0.24260	216.16	6.1837
150.00	5.0000	9.0608	0.11037	-31.962	-31.410	-0.13635	0.10915	0.15096	1833.1	-0.60459
250.00	5.0000	8.0692	0.12393	-16.063	-15.443	-0.055282	0.13278	0.17352	1297.7	-0.49791
350.00	5.0000	7.0426	0.14199	3.1052	3.8152	0.0091444	0.16784	0.21301	873.72	-0.29846
450.00	5.0000	5.7124	0.17506	26.615	27.490	0.068380	0.20366	0.26396	476.76	0.22550
550.00	5.0000	2.3237	0.43035	58.312	60.464	0.13398	0.24134	0.37655	137.16	8.5257
150.00	10.000	9.0844	0.11008	-32.055	-30.954	-0.13698	0.10949	0.15093	1853.0	-0.60519
250.00	10.000	8.1131	0.12326	-16.241	-15.009	-0.056017	0.13305	0.17304	1328.1	-0.50401
350.00	10.000	7.1333	0.14019	2.7454	4.1473	0.0080780	0.16805	0.21116	923.39	-0.32667
450.00	10.000	5.9777	0.16729	25.695	27.368	0.066210	0.20340	0.25395	576.50	-0.0027791
550.00	10.000	4.3857	0.22801	52.778	55.059	0.12162	0.23673	0.29917	305.87	1.0063
250.00	100.00	8.6765	0.11525	-18.375	-6.8492	-0.066142	0.13735	0.17026	1729.6	-0.54229
350.00	100.00	8.0271	0.12458	-0.67131	11.787	-0.0037487	0.17182	0.20310	1445.5	-0.44895
450.00	100.00	7.4403	0.13440	20.331	33.771	0.051327	0.20635	0.23620	1247.4	-0.37634
550.00	100.00	6.9067	0.14479	44.426	58.905	0.10166	0.23764	0.26573	1108.2	-0.32474
450.00	500.00	9.0895	0.11002	16.069	71.078	0.028505	0.21492	0.23767	2315.0	-0.39042
550.00	500.00	8.7910	0.11375	39.474	96.350	0.079120	0.24537	0.26717	2204.0	-0.35059

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M.O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., and Span, R., "Short Fundamental Equations of State for 20 Industrial Fluids," *J. Chem. Eng. Data* **51**(3):785–850, 2006. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in the equation of state are 0.2% in density in the liquid phase, 0.5% in density in the vapor phase, 1% in density above the critical temperature, 0.2% in vapor pressure between 280 and 350 K, 0.5% in vapor pressure at higher temperatures, 2% in heat capacities, and 1% in the speed of sound.

2-304 PHYSICAL AND CHEMICAL DATA

TABLE 2-237 Saturated Methyl Chloride*

T, K	P, bar	$v_f, \text{m}^3/\text{kg}$	$v_g, \text{m}^3/\text{kg}$	$h_f, \text{kJ/kg}$	$h_g, \text{kJ/kg}$	$s_f, \text{kJ}/(\text{kg}\cdot\text{K})$	$s_g, \text{kJ}/(\text{kg}\cdot\text{K})$	$c_{pf}, \text{kJ}/(\text{kg}\cdot\text{K})$	$\mu_f, 10^{-4} \text{ Pa}\cdot\text{s}$	$k_f, \text{W}/(\text{m}\cdot\text{K})$
175	0.0117	8.84-4	27.90	274.5	764.3	3.529	6.328	1.469		
180	0.0165	8.91-4	19.85	280.9	767.7	3.570	6.274	1.472		
185	0.0233	8.97-4	14.12	287.5	771.0	3.603	6.222	1.475		
190	0.0327	9.04-4	10.12	294.5	774.3	3.647	6.172	1.477		
195	0.0462	9.10-4	7.208	301.7	777.5	3.684	6.124	1.480		
200	0.0653	9.17-4	5.137	309.0	780.7	3.722	6.080	1.483	4.44	0.241
205	0.0919	9.25-4	3.835	316.3	783.9	3.756	6.038	1.486	4.27	0.236
210	0.1315	9.33-4	2.656	323.7	787.0	3.791	5.998	1.489	4.11	0.232
215	0.181	9.40-4	1.975	331.0	790.1	3.825	5.961	1.492	3.96	0.228
220	0.243	9.48-4	1.505	338.4	793.2	3.859	5.928	1.496	3.82	0.224
225	0.319	9.56-4	1.168	345.7	796.3	3.892	5.896	1.500	3.69	0.219
230	0.417	9.65-4	0.911	353.1	799.3	3.925	5.866	1.504	3.57	0.215
235	0.539	9.73-4	0.718	360.5	802.3	3.957	5.845	1.508	3.46	0.211
240	0.688	9.81-4	0.572	368.0	805.3	3.988	5.822	1.513	3.35	0.207
245	0.866	9.89-4	0.462	375.6	808.2	4.019	5.786	1.518	3.25	0.202
250	1.076	9.98-4	0.377	383.2	811.1	4.050	5.762	1.523	3.16	0.198
255	1.328	10.08-4	0.311	390.7	814.0	4.080	5.740	1.528	3.08	0.194
260	1.627	10.18-4	0.257	398.3	816.8	4.110	5.720	1.533	3.00	0.190
265	1.970	10.27-4	0.215	406.0	819.4	4.139	5.699	1.539	2.92	0.186
270	2.364	10.36-4	0.1807	413.7	822.0	4.168	5.680	1.546	2.85	0.182
275	2.830	10.46-4	0.1524	421.5	824.4	4.197	5.662	1.554	2.78	0.177
280	3.347	10.57-4	0.1301	429.4	826.8	4.225	5.644	1.565	2.72	0.173
285	3.936	10.68-4	0.1115	437.3	829.0	4.253	5.628	1.574	2.66	0.169
290	4.612	10.79-4	0.0960	445.2	831.2	4.280	5.612	1.583	2.61	0.165
295	5.361	10.91-4	0.0830	453.2	833.2	4.308	5.597	1.594	2.56	0.160
300	6.189	11.03-4	0.0723	461.2	835.2	4.334	5.581	1.605	2.51	0.156
305	7.110	11.15-4	0.0632	469.3	837.0	4.361	5.567	1.617	2.46	0.152
310	8.111	11.27-4	0.0556	477.4	838.8	4.388	5.553	1.631	2.42	0.148
315	9.243	11.40-4	0.0489	485.6	840.5	4.414	5.540	1.644	2.37	0.143
320	10.47	11.55-4	0.0433	493.8	841.9	4.440	5.527	1.658	2.33	0.139
325	11.78	11.70-4	0.0386	502.1	843.3	4.465	5.516		2.30	0.135
330	13.27	11.86-4	0.0343	510.4	844.5	4.491	5.504		2.27	0.131
340	16.52	12.17-4	0.0282	518.8	846.4	4.542	5.481		2.12	0.124
350	20.53	12.54-4	0.0228	538.3	847.5	4.592	5.457		1.99	0.117
360	25.29	12.97-4	0.0186	562.9	847.6	4.643	5.434		1.87	0.110
370	30.74	13.47-4	0.0151	581.6	845.9	4.694	5.398		1.77	0.103
380	36.99	14.11-4	0.0117	602.8	842.6	4.747	5.382		1.67	0.095
390	44.05	14.67-4	0.0096	622.9	837.4	4.805	5.358		1.59	0.086
400	52.29	15.66-4	0.0075	643.6	826.4	4.870	5.323		1.51	0.075
405	56.6	16.48-4	0.0063	663.2	819.1	4.904	5.289			
410	61.5	17.97-4	0.0052	677.3	807.1	4.954	5.256			
415	67.4	21.10-4	0.0038	714.1	778.6	5.025	5.200			
416 ^c	69.0	27.40-4	0.0027	749.3	749.3	5.116	5.116			

*Interpolated by P. E. Liley from the Landolt-Börnstein band IVa, p. 677, 1967 tables by Steinle/Dienemann. c = critical point. The notation 8.84-4 signifies 8.84×10^{-4} .

TABLE 2-238 Thermodynamic Properties of Neon

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
24.562	0.043464	62.059	0.016114	-0.099873	-0.099172	-0.0038092	0.018679	0.043081	674.66	-0.22893
25.000	0.050920	61.645	0.016222	-0.081689	-0.080863	-0.0030751	0.018439	0.040416	652.08	-0.25101
26.000	0.071611	60.761	0.016458	-0.042983	-0.041805	-0.0015561	0.018193	0.037847	617.51	-0.27322
27.000	0.098173	59.903	0.016694	-0.0055930	-0.0039541	-0.00014422	0.018006	0.037540	596.00	-0.27053
28.000	0.13159	59.032	0.016940	0.031967	0.034196	0.0012227	0.017803	0.038200	579.65	-0.25492
29.000	0.17287	58.129	0.017203	0.070340	0.073314	0.0025706	0.017589	0.039303	564.56	-0.23297
30.000	0.22307	57.186	0.017487	0.10979	0.11370	0.0039099	0.017371	0.040595	548.99	-0.20788
31.000	0.28324	56.198	0.017794	0.15041	0.15545	0.0052441	0.017156	0.041956	532.26	-0.18090
32.000	0.35446	55.162	0.018128	0.19218	0.19860	0.0065735	0.016952	0.043347	514.19	-0.15212
33.000	0.43782	54.078	0.018492	0.23507	0.24317	0.0078977	0.016768	0.044788	494.83	-0.12096
34.000	0.53443	52.941	0.018889	0.27908	0.28917	0.0092172	0.016614	0.046343	474.28	-0.086349
35.000	0.64543	51.749	0.019324	0.32426	0.33673	0.010534	0.016498	0.048115	452.65	-0.046782
36.000	0.77202	50.493	0.019805	0.37073	0.38602	0.011853	0.016426	0.050251	429.96	-0.00027396
37.000	0.91543	49.162	0.020341	0.41874	0.43736	0.013181	0.016404	0.052967	406.17	0.055871
38.000	1.0770	47.736	0.020948	0.46869	0.49125	0.014529	0.016438	0.056597	381.08	0.12549
39.000	1.2581	46.187	0.021651	0.52120	0.54844	0.015914	0.016536	0.061719	354.35	0.21441
40.000	1.4603	44.462	0.022491	0.57729	0.61014	0.017363	0.016715	0.069475	325.39	0.33222
41.000	1.6855	42.471	0.023545	0.63879	0.67848	0.018922	0.017014	0.082509	293.27	0.49632
42.000	1.9355	40.021	0.024987	0.70940	0.75776	0.020686	0.017522	0.10852	256.54	0.74180
43.000	2.2121	36.640	0.027292	0.79805	0.85843	0.022883	0.018471	0.17680	213.78	1.1433
44.000	2.5168	31.327	0.031921	0.92402	1.0044	0.026030	0.020210	0.44810	170.57	1.8283
44.492	2.6786	23.882	0.041873	1.0877	1.1999	0.030313			0	2.9448
24.562	0.043464	0.22026	4.5402	1.4817	1.6790	0.068586	0.035186	0.047919	113.01	27.054
25.000	0.050920	0.25406	3.9361	1.4898	1.6902	0.067767	0.029819	0.041791	115.48	23.404
26.000	0.071611	0.34598	2.8903	1.5046	1.7116	0.065882	0.022043	0.033098	121.12	18.114
27.000	0.098173	0.46111	2.1687	1.5159	1.7288	0.064031	0.018031	0.028839	126.17	15.446
28.000	0.13159	0.60299	1.6584	1.5247	1.7430	0.062250	0.016001	0.026943	130.23	14.023
29.000	0.17287	0.77543	1.2896	1.5318	1.7547	0.060549	0.015021	0.026349	133.28	13.110
30.000	0.22307	0.98260	1.0177	1.5372	1.7642	0.058928	0.014604	0.026517	135.55	12.393
31.000	0.28324	1.2292	0.81356	1.5411	1.7716	0.057377	0.014497	0.027183	137.25	11.758
32.000	0.35446	1.5204	0.65772	1.5435	1.7767	0.055888	0.014566	0.028231	138.54	11.166
33.000	0.43782	1.8625	0.53690	1.5443	1.7794	0.054449	0.014745	0.029633	139.52	10.606
34.000	0.53443	2.2630	0.44190	1.5432	1.7794	0.053047	0.015000	0.031418	140.26	10.073
35.000	0.64543	2.7308	0.36619	1.5401	1.7765	0.051670	0.015316	0.033669	140.78	9.5602
36.000	0.77202	3.2778	0.30508	1.5347	1.7702	0.050303	0.015688	0.036531	141.11	9.0641
37.000	0.91543	3.9193	0.25515	1.5265	1.7601	0.048930	0.016118	0.040243	141.24	8.5780
38.000	1.0770	4.6762	0.21385	1.5150	1.7453	0.047531	0.016614	0.045207	141.18	8.0944
39.000	1.2581	5.5788	0.17925	1.4994	1.7249	0.046079	0.017189	0.052136	140.93	7.6033
40.000	1.4603	6.6734	0.14985	1.4784	1.6972	0.044539	0.017863	0.062405	140.49	7.0917
41.000	1.6855	8.0363	0.12443	1.4500	1.6597	0.042855	0.018664	0.079029	139.83	6.5404
42.000	1.9355	9.8091	0.10195	1.4107	1.6080	0.040930	0.019634	0.10997	138.95	5.9193
43.000	2.2121	12.303	0.081282	1.3530	1.5328	0.038566	0.020823	0.18428	137.87	5.1760
44.000	2.5168	16.541	0.060455	1.2537	1.4058	0.035154	0.022208	0.55080	136.83	4.1789
44.492	2.6786	23.882	0.041873	1.0877	1.1999	0.030313			0	2.9448

TABLE 2-238 Thermodynamic Properties of Neon (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Single-Phase Properties										
100.00	0.10000	0.12034	8.3098	2.4474	3.2784	0.091975	0.012479	0.020851	262.24	1.6033
200.00	0.10000	0.060106	16.637	3.6964	5.3601	0.10641	0.012475	0.020800	370.87	0.12447
300.00	0.10000	0.040072	24.955	4.9441	7.4396	0.11484	0.012473	0.020791	454.13	-0.30667
400.00	0.10000	0.030057	33.270	6.1915	9.5185	0.12082	0.012473	0.020789	524.31	-0.49467
500.00	0.10000	0.024047	41.585	7.4388	11.597	0.12546	0.012473	0.020787	586.15	-0.59137
600.00	0.10000	0.020040	49.899	8.6861	13.676	0.12925	0.012472	0.020787	642.06	-0.64538
700.00	0.10000	0.017178	58.214	9.9333	15.755	0.13245	0.012472	0.020786	693.47	-0.67670
100.00	1.0000	1.2090	0.82716	2.4215	3.2486	0.072572	0.012547	0.021444	264.06	1.5285
200.00	1.0000	0.59828	1.6715	3.6864	5.3579	0.087212	0.012502	0.020926	373.32	0.11185
300.00	1.0000	0.39903	2.5061	4.9392	7.4453	0.095677	0.012491	0.020841	456.29	-0.30952
400.00	1.0000	0.29952	3.3387	6.1891	9.5278	0.10167	0.012485	0.020813	526.19	-0.49518
500.00	1.0000	0.23978	4.1705	7.4379	11.608	0.10631	0.012482	0.020800	587.81	-0.59127
600.00	1.0000	0.19992	5.0020	8.6861	13.688	0.11010	0.012480	0.020794	643.54	-0.64516
700.00	1.0000	0.17143	5.8334	9.9339	15.767	0.11331	0.012478	0.020791	694.81	-0.67651
100.00	5.0000	6.1145	0.16355	2.3075	3.1252	0.058098	0.012814	0.024056	275.27	1.1885
200.00	5.0000	2.9287	0.34145	3.6436	5.3508	0.073620	0.012619	0.021453	384.50	0.054603
300.00	5.0000	1.9580	0.51074	4.9181	7.4718	0.082226	0.012565	0.021051	465.95	-0.32325
400.00	5.0000	1.4747	0.67810	6.1787	9.5693	0.088261	0.012539	0.020917	534.57	-0.49810
500.00	5.0000	1.1837	0.84478	7.4338	11.658	0.092921	0.012523	0.020858	595.18	-0.59116
600.00	5.0000	0.98897	1.0112	8.6860	13.742	0.096721	0.012513	0.020828	650.12	-0.64435
700.00	5.0000	0.84934	1.1774	9.9366	15.824	0.099930	0.012506	0.020810	700.76	-0.67575
100.00	10.000	12.113	0.082559	2.1736	2.9992	0.051158	0.013095	0.026878	296.54	0.80676
200.00	10.000	5.6967	0.17554	3.5933	5.3487	0.067620	0.012752	0.022026	399.17	-0.014628
300.00	10.000	3.8255	0.26141	4.8930	7.5070	0.076381	0.012653	0.021291	478.16	-0.34148
400.00	10.000	2.8938	0.34556	6.1661	9.6217	0.082467	0.012603	0.021040	545.08	-0.50266
500.00	10.000	2.3306	0.42908	7.4286	11.719	0.087148	0.012573	0.020927	604.43	-0.59158
600.00	10.000	1.9519	0.51231	8.6858	13.809	0.090958	0.012553	0.020868	658.38	-0.64363
700.00	10.000	1.6795	0.59540	9.9399	15.894	0.094172	0.012539	0.020835	708.23	-0.67492
100.00	100.00	50.564	0.019777	1.4006	3.3783	0.028294	0.016249	0.028809	728.18	-0.37331
200.00	100.00	34.632	0.028875	3.1327	6.0202	0.046773	0.014308	0.024694	673.52	-0.43751
300.00	100.00	26.457	0.037796	4.6205	8.4001	0.056445	0.013708	0.023102	694.50	-0.49312
400.00	100.00	21.512	0.046486	6.0160	10.665	0.062965	0.013422	0.022262	729.70	-0.54535
500.00	100.00	18.185	0.054990	7.3645	12.863	0.067874	0.013245	0.021756	767.75	-0.58985
600.00	100.00	15.781	0.063366	8.6852	15.022	0.071810	0.013121	0.021432	805.72	-0.62513
700.00	100.00	13.955	0.071657	9.9877	17.153	0.075096	0.013028	0.021216	842.83	-0.65210
300.00	500.00	59.131	0.016912	4.6368	13.093	0.042072	0.015948	0.024091	1306.5	-0.47221
400.00	500.00	53.366	0.018738	6.0941	15.463	0.048897	0.015200	0.023386	1293.9	-0.49185
500.00	500.00	48.708	0.020530	7.5120	17.777	0.054063	0.014770	0.022915	1294.1	-0.50859
600.00	500.00	44.864	0.022290	8.9046	20.049	0.058207	0.014481	0.022544	1301.3	-0.52455
700.00	500.00	41.633	0.024020	10.278	22.288	0.061658	0.014265	0.022235	1312.5	-0.53993

The values in these tables were generated from the NIST REFPROP software (Lenmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Katti, R. S., Jacobsen, R. T., Stewart, R. B., and Jahangiri, M., “Thermodynamic Properties for Neon for Temperatures from the Triple Point to 700 K at Pressures to 700 MPa,” *Adv. Cryo. Eng.* **31**:1189–1197, 1986. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties of the equation of state are 0.1% in density, 2% in heat capacity, and 1% in the speed of sound, except in the critical region. The uncertainty in vapor pressure is 0.2%.

TABLE 2-239 Thermodynamic Properties of Nitrogen

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
63.151	0.012520	30.957	0.032303	-4.2230	-4.2226	0.067951	0.032951	0.056033	995.28	-0.40419	173.24	311.59
65.000	0.017404	30.685	0.032589	-4.1194	-4.1188	0.069569	0.032591	0.056121	976.36	-0.39833	169.51	282.07
67.000	0.024300	30.387	0.032909	-4.0071	-4.0063	0.071270	0.032207	0.056231	956.04	-0.39135	165.49	254.55
69.000	0.033213	30.085	0.033239	-3.8946	-3.8935	0.072924	0.031831	0.056360	935.83	-0.38364	161.47	230.85
71.000	0.044527	29.779	0.033581	-3.7819	-3.7804	0.074535	0.031463	0.056512	915.66	-0.37508	157.47	210.32
73.000	0.058656	29.468	0.033935	-3.6689	-3.6669	0.076105	0.031106	0.056690	895.49	-0.36560	153.46	192.43
75.000	0.076043	29.153	0.034302	-3.5556	-3.5530	0.077637	0.030760	0.056899	875.28	-0.35506	149.47	176.75
77.000	0.097152	28.832	0.034683	-3.4419	-3.4385	0.079133	0.030427	0.057142	855.00	-0.34334	145.48	162.94
79.000	0.12247	28.506	0.035080	-3.3278	-3.3235	0.080597	0.030105	0.057425	834.61	-0.33029	141.50	150.71
81.000	0.15251	28.175	0.035493	-3.2132	-3.2078	0.082030	0.029795	0.057752	814.07	-0.31574	137.55	139.82
83.000	0.18780	27.837	0.035924	-3.0980	-3.0913	0.083436	0.029499	0.058130	793.36	-0.29951	133.61	130.07
85.000	0.22886	27.492	0.036375	-2.9822	-2.9739	0.084815	0.029215	0.058566	772.44	-0.28135	129.66	121.31
87.000	0.27626	27.139	0.036847	-2.8657	-2.8555	0.086172	0.028944	0.059068	751.28	-0.26099	125.72	113.38
89.000	0.33055	26.779	0.037343	-2.7483	-2.7360	0.087507	0.028687	0.059647	729.84	-0.23813	121.77	106.18
91.000	0.39230	26.409	0.037865	-2.6301	-2.6152	0.088823	0.028444	0.060315	708.09	-0.21237	117.83	99.602
93.000	0.46210	26.030	0.038417	-2.5107	-2.4930	0.090123	0.028215	0.061088	685.99	-0.18326	113.89	93.568
95.000	0.54052	25.640	0.039002	-2.3902	-2.3691	0.091408	0.028001	0.061983	663.50	-0.15025	109.95	88.004
97.000	0.62817	25.238	0.039623	-2.2683	-2.2434	0.092682	0.027804	0.063026	640.57	-0.11264	106.02	82.847
99.000	0.72566	24.822	0.040288	-2.1449	-2.1156	0.093946	0.027624	0.064246	617.14	-0.069613	102.08	78.042
101.00	0.83358	24.390	0.041000	-2.0196	-1.9854	0.095204	0.027464	0.065684	593.17	-0.020100	98.144	73.543
103.00	0.95259	23.941	0.041769	-1.8923	-1.8525	0.096459	0.027327	0.067392	568.58	0.037239	94.208	69.306
105.00	1.0833	23.471	0.042605	-1.7625	-1.7163	0.097715	0.027214	0.069443	543.30	0.10414	90.272	65.292
107.00	1.2264	22.978	0.043520	-1.6298	-1.5765	0.098977	0.027133	0.071937	517.24	0.18288	86.337	61.464
109.00	1.3826	22.457	0.044530	-1.4938	-1.4323	0.10025	0.027088	0.075021	490.29	0.27654	82.404	57.786
111.00	1.5526	21.902	0.045658	-1.3537	-1.2828	0.10154	0.027089	0.078914	462.32	0.38936	78.472	54.224
113.00	1.7371	21.306	0.046935	-1.2086	-1.1271	0.10285	0.027149	0.083966	433.19	0.52741	74.544	50.740
115.00	1.9370	20.658	0.048407	-1.0571	-0.96336	0.10420	0.027290	0.090771	402.67	0.69974	70.626	47.290
117.00	2.1533	19.943	0.050144	-0.89741	-0.78944	0.10561	0.027545	0.10044	370.43	0.92076	66.728	43.824
119.00	2.3869	19.134	0.052262	-0.72635	-0.60161	0.10710	0.027981	0.11531	335.85	1.2154	62.883	40.270
121.00	2.6391	18.187	0.054985	-0.53833	-0.39322	0.10873	0.028755	0.14140	297.68	1.6317	59.196	36.509
123.00	2.9116	16.997	0.058834	-0.32093	-0.14962	0.11059	0.030317	0.20028	253.32	2.2811	56.121	32.310
125.00	3.2069	15.210	0.065747	-0.031475	0.17937	0.11310	0.034680	0.46831	195.48	3.5308	56.435	26.935
126.19	3.3958	11.184	0.089414	0.51527	0.81891	0.11807			0	6.0831		
Subcooled Properties												
63.151	0.012520	0.024070	41.546	1.2945	1.8147	0.16355	0.021007	0.029647	161.11	40.718	5.6209	4.3763
65.000	0.017404	0.032594	30.680	1.3299	1.8639	0.16161	0.021059	0.029788	163.20	38.268	5.8164	4.5123
67.000	0.024300	0.044300	22.573	1.3675	1.9160	0.15966	0.021123	0.029969	165.37	35.907	6.0298	4.6601
69.000	0.033213	0.059031	16.940	1.4042	1.9668	0.15786	0.021196	0.030180	167.43	33.803	6.2457	4.8088
71.000	0.044527	0.077273	12.941	1.4400	2.0162	0.15618	0.021278	0.030427	169.39	31.922	6.4645	4.9585
73.000	0.058656	0.099542	10.046	1.4747	2.0639	0.15461	0.021370	0.030712	171.23	30.231	6.6870	5.1096
75.000	0.076043	0.12638	7.9124	1.5082	2.1099	0.15314	0.021472	0.031039	172.95	28.707	6.9138	5.2621
77.000	0.097152	0.15838	6.3140	1.5404	2.1539	0.15176	0.021585	0.031413	174.55	27.328	7.1458	5.4164
79.000	0.12247	0.19613	5.0986	1.5713	2.1957	0.15046	0.021709	0.031839	176.03	26.074	7.3839	5.5727
81.000	0.15251	0.24030	4.1614	1.6007	2.2353	0.14923	0.021845	0.032323	177.38	24.931	7.6295	5.7313
83.000	0.18780	0.29157	3.4297	1.6284	2.2725	0.14806	0.021994	0.032873	178.60	23.884	7.8837	5.8924
85.000	0.22886	0.35069	2.8515	1.6544	2.3070	0.14694	0.022157	0.033496	179.68	22.923	8.1483	6.0565
87.000	0.27626	0.41846	2.3897	1.6784	2.3386	0.14587	0.022334	0.034204	180.63	22.035	8.4251	6.2238
89.000	0.33055	0.49576	2.0171	1.7005	2.3672	0.14485	0.022528	0.035008	181.43	21.212	8.7163	6.3948
91.000	0.39230	0.58355	1.7137	1.7203	2.3925	0.14385	0.022738	0.035925	182.10	20.446	9.0247	6.5700
93.000	0.46210	0.68291	1.4643	1.7377	2.4143	0.14289	0.022967	0.036973	182.62	19.730	9.3533	6.7499
95.000	0.54052	0.79504	1.2578	1.7525	2.4324	0.14195	0.023217	0.038177	182.99	19.057	9.7060	6.9353
97.000	0.62817	0.92134	1.0854	1.7645	2.4463	0.14103	0.023489	0.039568	183.21	18.421	10.087	7.1270
99.000	0.72566	1.0634	0.94038	1.7733	2.4557	0.14012	0.023787	0.041185	183.28	17.815	10.503	7.3260
101.00	0.83358	1.2231	0.81759	1.7788	2.4603	0.13922	0.024113	0.043081	183.18	17.236	10.960	7.5334
103.00	0.95259	1.4027	0.71291	1.7804	2.4595	0.13832	0.024471	0.045326	182.93	16.676	11.467	7.7509
105.00	1.0833	1.6049	0.62309	1.7778	2.4528	0.13742	0.024860	0.048012	182.51	16.132	12.035	7.9804
107.00	1.2264	1.8331	0.54553	1.7703	2.4394	0.13651	0.025284	0.051276	181.93	15.600	12.679	8.2245

TABLE 2-239 Thermodynamic Properties of Nitrogen (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
109.00	1.3826	2.0916	0.47811	1.7573	2.4183	0.13557	0.025750	0.055332	181.19	15.075	13.419	8.4867
111.00	1.5526	2.3860	0.41911	1.7377	2.3884	0.13461	0.026284	0.060528	180.28	14.546	14.284	8.7716
113.00	1.7371	2.7240	0.36711	1.7102	2.3479	0.13360	0.026924	0.067435	179.15	13.996	15.315	9.0860
115.00	1.9370	3.1162	0.32091	1.6730	2.2946	0.13253	0.027721	0.077010	177.75	13.409	16.580	9.4395
117.00	2.1533	3.5786	0.27944	1.6234	2.2251	0.13138	0.028723	0.091003	176.01	12.767	18.186	9.8474
119.00	2.3869	4.1370	0.24172	1.5572	2.1341	0.13009	0.029997	0.11312	173.87	12.045	20.329	10.336
121.00	2.6391	4.8380	0.20670	1.4665	2.0119	0.12860	0.031683	0.15295	171.17	11.203	23.424	10.953
123.00	2.9116	5.7846	0.17287	1.3343	1.8376	0.12675	0.034185	0.24490	167.43	10.148	28.604	11.813
125.00	3.2069	7.3244	0.13653	1.1039	1.5417	0.12400	0.039278	0.66512	160.26	8.6030	41.535	13.326
126.19	3.3958	11.184	0.089414	0.51527	0.81891	0.11807			0	6.0831		
Single-Phase Properties												
100.00	0.10000	0.12268	8.1514	2.0396	2.8547	0.15950	0.021049	0.030012	201.64	16.082	9.3806	6.9581
600.00	0.10000	0.020037	49.908	12.573	17.564	0.21217	0.021796	0.030118	496.27	0.021483	44.840	29.577
1100.0	0.10000	0.010930	91.489	24.284	33.433	0.23131	0.024932	0.033248	660.05	-0.65654	70.075	44.199
1600.0	0.10000	0.0075152	133.06	37.272	50.579	0.24414	0.026815	0.035130	788.94	-0.81543	92.344	56.398
100.00	1.0000	24.658	0.040554	-2.0907	-2.0501	0.094493	0.027546	0.064564	609.42	-0.054514	100.58	76.255
103.75	1.0000	23.768	0.042073	-1.8441	-1.8020	0.096928	0.027281	0.065113	559.22	0.060996	92.738	67.783
103.75	1.0000	1.4754	0.67778	1.7800	2.4577	0.13799	0.024612	0.046272	182.79	16.471	11.671	7.8351
600.00	1.0000	0.19960	5.0099	12.554	17.564	0.19300	0.021812	0.030198	498.66	0.0061465	44.992	29.626
1100.0	1.0000	0.10899	9.1755	24.277	33.452	0.21216	0.024938	0.033267	662.07	-0.65940	70.155	44.221
1600.0	1.0000	0.074993	13.335	37.270	50.605	0.22499	0.026820	0.035138	790.64	-0.81612	92.399	56.411
100.00	5.0000	25.436	0.039314	-2.2176	-2.0210	0.093188	0.027713	0.059868	673.24	-0.17096	108.13	84.510
600.00	5.0000	0.98084	1.0195	12.469	17.567	0.17948	0.021881	0.030539	509.60	-0.057679	45.797	29.882
1100.0	5.0000	0.53797	1.8588	24.247	33.541	0.19875	0.024969	0.033350	671.08	-0.67112	70.555	44.330
1600.0	5.0000	0.37146	2.6921	37.259	50.720	0.21161	0.026839	0.035170	798.18	-0.81886	92.663	56.476
100.00	10.000	26.188	0.038186	-2.3398	-1.9580	0.091882	0.028004	0.056646	734.22	-0.25658	115.90	93.648
600.00	10.000	1.9183	0.52130	12.368	17.581	0.17355	0.021965	0.030926	523.87	-0.12928	46.995	30.284
1100.0	10.000	1.0590	0.94433	24.211	33.654	0.19296	0.025006	0.033447	682.37	-0.68394	71.130	44.493
1600.0	10.000	0.73435	1.3618	37.246	50.864	0.20583	0.026863	0.035209	807.57	-0.82170	93.033	56.570
600.00	500.00	27.434	0.036451	10.778	29.003	0.13791	0.026493	0.035336	1574.4	-0.70223	177.40	103.10
1100.0	500.00	21.868	0.045729	23.840	46.705	0.15935	0.027586	0.035848	1501.4	-0.72394	149.37	79.801
1600.0	500.00	18.335	0.054541	37.584	64.855	0.17295	0.028647	0.036665	1506.6	-0.73166	147.47	79.226
600.00	1000.0	34.270	0.029180	11.714	40.894	0.13093	0.029169	0.036905	2107.2	-0.61888	278.97	208.00
1100.0	1000.0	29.362	0.034057	25.065	59.122	0.15303	0.029373	0.036577	1985.1	-0.65271	232.49	129.86
1600.0	1000.0	25.920	0.038580	38.999	77.579	0.16685	0.029977	0.037212	1942.0	-0.65439	214.36	110.35

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Span, R., Lemmon, E. W., Jacobsen, R. T., Wagner, W., and Yokozeki, A., "A Reference Quality Thermodynamic Property Formulation for Nitrogen," *J. Phys. Chem. Ref. Data* **29**(6):1361–1433, 2000. See also *Int. J. Thermophys.* **14**(4):1121–1132, 1998. The source for viscosity is Lemmon, E. W., and Jacobsen, R. T., "Viscosity and Thermal Conductivity Equations for Nitrogen, Oxygen, Argon, and Air," *Int. J. Thermophys.* **25**:21–69, 2004. The source for thermal conductivity is Lemmon, E. W., and Jacobsen, R. T., "Viscosity and Thermal Conductivity Equations for Nitrogen, Oxygen, Argon, and Air," *Int. J. Thermophys.* **25**:21–69, 2004. Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainty in density of the equation of state is 0.02% from the triple point up to temperatures of 523 K and pressures up to 12 MPa and from temperatures of 240 to 523 K at pressures less than 30 MPa. In the range from 270 to 350 K at pressures less than 12 MPa, the uncertainty in density is 0.01%. The uncertainty at very high pressures (>1 GPa) is 0.6% in density. The uncertainty in pressure in the critical region is estimated to be 0.02%. In the gaseous and supercritical region, the speed of sound can be calculated with a typical uncertainty of 0.005% to 0.1%. At liquid states and at high pressures, the uncertainty increases to 0.5% to 1.5%. For pressures up to 30 MPa, the estimated uncertainty for heat capacities ranges from 0.3% at gaseous and gaslike supercritical states up to 0.8% at liquid states and at certain gaseous and supercritical states at low temperatures. The uncertainty is 2% for pressures up to 200 MPa and larger at higher pressures. The estimated uncertainties of vapor pressure, saturated-liquid density, and saturated-vapor density are in general 0.02% for each property. The formulation yields a reasonable extrapolation behavior up to the limits of chemical stability of nitrogen.

For viscosity, the uncertainty is 0.5% in dilute gas. Away from the dilute gas (pressures greater than 1 MPa and in the liquid), the uncertainties are as low as 1% between 270 and 300 K at pressures less than 100 MPa, and increase outside that range. The uncertainties are around 2% at temperatures of 180 K and higher. Below this and away from the critical region, the uncertainties steadily increase to around 5% at the triple points of the fluids. The uncertainties in the critical region are higher.

For thermal conductivity, the uncertainty for the dilute gas is 2% with increasing uncertainties near the triple point. For the nondilute gas, the uncertainty is 2% for temperatures greater than 150 K. The uncertainty is 3% at temperatures less than the critical point and 5% in the critical region, except for states very near the critical point.

TABLE 2-240 Saturated Nitrogen Tetroxide

Pressure, bar	Temperature, K	v_f , m ³ /kg	v_g , m ³ /kg	M_f	M_g
1.0133	299.32	0.000 694	0.2996	91.857	79.157
2	309.57	0.000 711	0.1630	91.886	76.503
4	326.66	0.000 733	0.0876	91.766	73.538
6	337.43	0.000 749	0.0608	91.625	71.748
8	345.45	0.000 762	0.0469	91.488	70.480
10	351.88	0.000 774	0.0382	91.346	69.483
15	364.09	0.000 800	0.0262	90.979	67.742
20	373.17	0.000 822	0.0199	90.601	66.547
30	386.57	0.000 863	0.0133	89.823	64.997
40	396.52	0.000 903	0.0098	89.018	64.099
50	404.50	0.000 945	0.00761	88.191	63.532
60	411.20	0.000 993	0.00607	87.344	63.181
80	422.07	0.001 129	0.00394	85.602	62.959
100	430.76	0.001 577	0.00209	83.817	63.366

Condensed from McCarty, R. D., H-U. Steurer, et al., NBS IR 86 - 3054, 1986 (106 pp.). M = mol wt for the reaction $\text{N}_2\text{O}_4 \rightarrow 2\text{NO}_2 \rightarrow 2\text{NO} + \text{O}_2$. No derived thermodynamic functions were tabulated due to unduly large differences in literature values, but 92 references are given.

TABLE 2-241 Thermodynamic Properties of Nitrogen Trifluoride

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
90.000	0.00015612	25.006	0.039990	-3.8551	-3.8551	-0.033490	0.045791	0.070957	1081.4	-0.44772
95.000	0.00040547	24.719	0.040455	-3.4999	-3.4999	-0.029648	0.045221	0.071116	1049.3	-0.44233
100.00	0.00094625	24.427	0.040938	-3.1442	-3.1442	-0.026000	0.044514	0.071131	1018.1	-0.43718
105.00	0.0020164	24.132	0.041439	-2.7887	-2.7887	-0.022531	0.043768	0.071067	986.86	-0.43184
110.00	0.0039755	23.831	0.041962	-2.4336	-2.4335	-0.019227	0.043048	0.070980	955.57	-0.42594
115.00	0.0073310	23.526	0.042506	-2.0790	-2.0786	-0.016074	0.042394	0.070913	924.52	-0.41921
120.00	0.012759	23.217	0.043073	-1.7245	-1.7240	-0.013057	0.041825	0.070898	894.02	-0.41143
125.00	0.021119	22.902	0.043664	-1.3701	-1.3691	-0.010163	0.041346	0.070958	864.29	-0.40241
130.00	0.033457	22.583	0.044281	-1.0151	-1.0137	-0.0073784	0.040957	0.071113	835.40	-0.39193
135.00	0.051008	22.259	0.044925	-0.65928	-0.65699	-0.0046923	0.040653	0.071379	807.28	-0.37972
140.00	0.075184	21.931	0.045599	-0.30195	-0.29853	-0.0020929	0.040428	0.071771	779.79	-0.36549
145.00	0.10757	21.596	0.046304	0.057452	0.062433	0.00042986	0.040274	0.072303	752.68	-0.34889
150.00	0.14989	21.256	0.047046	0.41959	0.42664	0.0028558	0.040186	0.072988	725.72	-0.32951
155.00	0.20401	20.908	0.047829	0.78514	0.79490	0.0052840	0.040159	0.073841	698.67	-0.30688
160.00	0.27193	20.552	0.048658	1.1549	1.1681	0.0076328	0.040187	0.074876	671.28	-0.28044
165.00	0.35573	20.186	0.049539	1.5295	1.5472	0.0099402	0.040269	0.076114	643.36	-0.24950
170.00	0.45760	19.809	0.050483	1.9100	1.9331	0.012214	0.040404	0.077577	614.73	-0.21319
175.00	0.57979	19.418	0.051499	2.2972	2.3270	0.014461	0.040591	0.079300	585.25	-0.17042
180.00	0.72462	19.011	0.052602	2.6920	2.7301	0.016690	0.040834	0.081326	554.78	-0.11971
185.00	0.89450	18.584	0.053808	3.0957	3.1438	0.018907	0.041136	0.083719	523.23	-0.059078
190.00	1.0918	18.135	0.055141	3.5095	3.5697	0.021121	0.041505	0.086571	490.50	0.014249
195.00	1.3192	17.659	0.056630	3.9350	4.0097	0.023341	0.041953	0.090023	456.52	0.10427
200.00	1.5791	17.148	0.058316	4.3742	4.4663	0.025577	0.042494	0.094298	421.22	0.21696
205.00	1.8742	16.595	0.060260	4.8297	4.9426	0.027843	0.043150	0.099779	384.52	0.36170
210.00	2.2073	15.987	0.062552	5.3052	5.4433	0.030157	0.043953	0.10717	346.33	0.55413
215.00	2.5815	15.305	0.065340	5.8067	5.9754	0.032548	0.044950	0.11795	306.49	0.82239
220.00	3.0003	14.514	0.068899	6.3449	6.5516	0.035068	0.046220	0.13575	264.69	1.2231
225.00	3.4682	13.541	0.073850	6.9434	7.1995	0.037830	0.047920	0.17267	220.19	1.8903
230.00	3.9918	12.148	0.082320	7.6794	8.0080	0.041203	0.050506	0.30655	171.03	3.2539
234.00	4.4607	7.9200	0.12626	9.3687	9.9320	0.049280			0	8.0982
90.000	0.00015612	0.00020869	4791.8	9.0527	9.8008	0.11824	0.025384	0.033709	118.26	125.93
95.000	0.00040547	0.00051363	1946.9	9.1794	9.9688	0.11213	0.025553	0.033889	121.38	109.58
100.00	0.00094625	0.0011394	877.68	9.3064	10.137	0.10681	0.025759	0.034117	124.38	96.037
105.00	0.0020164	0.0023144	432.07	9.4337	10.305	0.10217	0.026007	0.034400	127.25	84.718
110.00	0.0039755	0.0043618	229.26	9.5609	10.472	0.098099	0.026301	0.034747	129.99	75.177
115.00	0.0073310	0.0077094	129.71	9.6879	10.639	0.094513	0.026641	0.035166	132.58	67.076
120.00	0.012759	0.012896	77.545	9.8143	10.804	0.091341	0.027032	0.035666	135.01	60.154
125.00	0.021119	0.020570	48.615	9.9397	10.966	0.088521	0.027474	0.036255	137.27	54.206
130.00	0.033457	0.031490	31.756	10.064	11.126	0.086004	0.027968	0.036942	139.35	49.068
135.00	0.051008	0.046520	21.496	10.186	11.282	0.083746	0.028515	0.037735	141.23	44.607
140.00	0.075184	0.066627	15.009	10.305	11.434	0.081709	0.029116	0.038644	142.91	40.717
145.00	0.10757	0.092885	10.766	10.422	11.580	0.079863	0.029770	0.039680	144.37	37.310
150.00	0.14989	0.12648	7.9063	10.536	11.721	0.078179	0.030477	0.040854	145.59	34.315
155.00	0.20401	0.16872	5.9269	10.645	11.854	0.076634	0.031236	0.042183	146.58	31.671
160.00	0.27193	0.22107	4.5236	10.750	11.980	0.075207	0.032049	0.043684	147.30	29.330
165.00	0.35573	0.28514	3.5071	10.850	12.097	0.073880	0.032913	0.045384	147.77	27.250
170.00	0.45760	0.36279	2.7564	10.943	12.205	0.072636	0.033831	0.047313	147.95	25.397
175.00	0.57979	0.45613	2.1923	11.031	12.302	0.071459	0.034801	0.049515	147.85	23.743
180.00	0.72462	0.56766	1.7616	11.110	12.386	0.070337	0.035825	0.052050	147.44	22.263
185.00	0.89450	0.70033	1.4279	11.181	12.458	0.069254	0.036905	0.055001	146.73	20.936
190.00	1.0918	0.85776	1.1658	11.241	12.514	0.068196	0.038044	0.058491	145.68	19.746
195.00	1.3192	1.0445	0.95740	11.289	12.552	0.067150	0.039245	0.062704	144.29	18.678
200.00	1.5791	1.2665	0.78960	11.323	12.570	0.066097	0.040516	0.067933	142.53	17.718
205.00	1.8742	1.5317	0.65287	11.340	12.563	0.065016	0.041867	0.074677	140.38	16.854
210.00	2.2073	1.8518	0.54001	11.333	12.525	0.063879	0.043315	0.083847	137.79	16.072
215.00	2.5815	2.2450	0.44543	11.296	12.446	0.062642	0.044890	0.097325	134.71	15.354

TABLE 2-241 Thermodynamic Properties of Nitrogen Trifluoride (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
220.00	3.0003	2.7432	0.36453	11.213	12.307	0.061229	0.046643	0.11971	131.07	14.662
225.00	3.4682	3.4146	0.29286	11.055	12.071	0.059479	0.048690	0.16599	126.78	13.900
230.00	3.9918	4.4712	0.22365	10.725	11.618	0.056898	0.051363	0.32814	121.80	12.709
234.00	4.4607	7.9200	0.12626	9.3687	9.9320	0.049280			0	8.0982
Single-Phase Properties										
100.00	0.10000	24.430	0.040934	-3.1452	-3.1411	-0.026009	0.044502	0.071124	1018.9	-0.43728
143.95	0.10000	21.667	0.046153	-0.018242	-0.013626	-0.000094174	0.040301	0.072179	758.35	-0.35259
143.95	0.10000	0.086803	11.520	10.398	11.550	0.080236	0.029628	0.039451	144.08	37.990
200.00	0.10000	0.060926	16.413	12.215	13.856	0.093729	0.034723	0.043508	169.06	15.650
300.00	0.10000	0.040232	24.856	16.237	18.723	0.11332	0.045284	0.053741	203.45	5.5743
400.00	0.10000	0.030102	33.220	21.208	24.530	0.12997	0.053558	0.061943	232.46	2.8668
500.00	0.10000	0.024060	41.562	26.869	31.025	0.14444	0.059251	0.067609	258.38	1.7661
100.00	1.0000	24.449	0.040901	-3.1540	-3.1131	-0.026098	0.044398	0.071055	1026.3	-0.43816
187.76	1.0000	18.339	0.054527	3.3230	3.3775	0.020131	0.041331	0.085231	505.29	-0.020346
187.76	1.0000	0.78400	1.2755	11.215	12.491	0.068667	0.037527	0.056853	146.19	20.263
200.00	1.0000	0.70199	1.4245	11.738	13.162	0.072132	0.037614	0.053386	154.62	16.645
300.00	1.0000	0.41548	2.4068	16.042	18.449	0.093526	0.045687	0.055540	199.37	5.5586
400.00	1.0000	0.30407	3.2887	21.082	24.371	0.11051	0.053632	0.062664	231.33	2.8289
500.00	1.0000	0.24110	4.1476	26.771	30.918	0.12510	0.059244	0.067998	258.62	1.7349
100.00	5.0000	24.532	0.040763	-3.1921	-2.9883	-0.026483	0.043955	0.070771	1057.9	-0.44174
200.00	5.0000	17.659	0.056627	4.1435	4.4266	0.024397	0.041665	0.087072	477.43	0.053822
300.00	5.0000	2.4334	0.41095	15.064	17.119	0.076882	0.047559	0.067409	183.97	5.3277
400.00	5.0000	1.5833	0.63161	20.512	23.670	0.095734	0.053907	0.066129	228.50	2.5958
500.00	5.0000	1.2132	0.82425	26.341	30.462	0.11087	0.059206	0.069735	260.85	1.5696
100.00	10.000	24.630	0.040600	-3.2377	-2.8317	-0.026951	0.043446	0.070455	1094.6	-0.44552
200.00	10.000	18.214	0.054902	3.8853	4.4343	0.023042	0.041251	0.081506	536.12	-0.078962
300.00	10.000	5.9501	0.16806	13.534	15.214	0.066202	0.049590	0.093780	185.51	3.9475
400.00	10.000	3.2770	0.30515	19.798	22.849	0.088300	0.054171	0.070777	231.76	2.1918
500.00	10.000	2.4256	0.41227	25.826	29.949	0.10414	0.059167	0.071809	266.86	1.3296
100.00	25.000	24.895	0.040168	-3.3631	-2.3589	-0.028280	0.042128	0.069766	1191.1	-0.45304
200.00	25.000	19.339	0.051709	3.3539	4.6466	0.020118	0.041959	0.074322	642.55	-0.26070
300.00	25.000	12.709	0.078686	10.815	12.782	0.052928	0.049110	0.086464	338.34	0.50765
400.00	25.000	7.8701	0.12706	17.964	21.141	0.077032	0.054897	0.079284	285.01	0.90188
500.00	25.000	5.7388	0.17425	24.506	28.863	0.094275	0.059344	0.076236	303.16	0.66132
100.00	50.000	25.271	0.039572	-3.5465	-1.5679	-0.030335	0.040341	0.069490	1322.6	-0.45498
200.00	50.000	20.560	0.048639	2.8054	5.2373	0.016816	0.044958	0.070043	734.96	-0.38097
300.00	50.000	15.995	0.062520	9.4909	12.617	0.046643	0.052084	0.076746	484.09	-0.15530
400.00	50.000	12.237	0.081722	16.326	20.412	0.069057	0.057087	0.078504	397.17	0.026532
500.00	50.000	9.6861	0.10324	23.086	28.248	0.086544	0.060630	0.078201	380.22	0.066751

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Younglove, B. A., "Thermophysical Properties of Fluids. I. Argon, Ethylene, Parahydrogen, Nitrogen, Nitrogen Trifluoride, and Oxygen," *J. Phys. Chem. Ref. Data Suppl.* 1, **11**: 1–11, 1982. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the critical point temperature are given in the last entry of the saturation tables. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in density are 0.25% in the liquid phase and 0.3% in the vapor and supercritical regions. The uncertainty in speed of sound and heat capacity is 5%.

TABLE 2-242 Thermodynamic Properties of Nitrous Oxide

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
182.33	0.087837	28.113	0.035570	-0.18142	-0.17830	-0.00096900	0.042295	0.075603	1149.6	-0.26723
185.00	0.10325	27.935	0.035797	0.020258	0.023954	0.00012922	0.042134	0.075669	1132.4	-0.26287
190.00	0.13782	27.599	0.036234	0.39840	0.40340	0.0021464	0.041852	0.075851	1100.2	-0.25377
195.00	0.18085	27.257	0.036688	0.77745	0.78408	0.0041159	0.041596	0.076111	1068.2	-0.24334
200.00	0.23367	26.910	0.037161	1.1577	1.1664	0.0060420	0.041365	0.076452	1036.3	-0.23146
205.00	0.29767	26.557	0.037656	1.5396	1.5508	0.0079284	0.041155	0.076875	1004.5	-0.21798
210.00	0.37431	26.197	0.038173	1.9234	1.9377	0.0097789	0.040967	0.077388	972.70	-0.20272
215.00	0.46509	25.829	0.038716	2.3094	2.3274	0.011597	0.040797	0.077995	940.94	-0.18545
220.00	0.57160	25.453	0.039288	2.6982	2.7206	0.013385	0.040645	0.078706	909.16	-0.16592
225.00	0.69544	25.069	0.039891	3.0899	3.1177	0.015148	0.040510	0.079533	877.35	-0.14379
230.00	0.83828	24.674	0.040529	3.4852	3.5192	0.016888	0.040392	0.080489	845.46	-0.11869
235.00	1.0018	24.268	0.041207	3.8844	3.9257	0.018607	0.040290	0.081595	813.47	-0.090113
240.00	1.1878	23.849	0.041930	4.2881	4.3379	0.020311	0.040203	0.082875	781.31	-0.057462
245.00	1.3979	23.417	0.042705	4.6969	4.7566	0.022000	0.040132	0.084362	748.95	-0.019968
250.00	1.6341	22.968	0.043538	5.1115	5.1826	0.023681	0.040078	0.086100	716.31	0.023355
255.00	1.8982	22.502	0.044440	5.5326	5.6170	0.025355	0.040041	0.088150	683.32	0.073794
260.00	2.1920	22.015	0.045423	5.9614	6.0609	0.027028	0.040025	0.090597	649.86	0.13307
265.00	2.5177	21.505	0.046501	6.3990	6.5161	0.028704	0.040030	0.093564	615.81	0.20351
270.00	2.8772	20.966	0.047697	6.8471	6.9843	0.030391	0.040064	0.097233	580.97	0.28843
275.00	3.2728	20.393	0.049037	7.3078	7.4682	0.032097	0.040134	0.10189	545.11	0.39261
280.00	3.7068	19.777	0.050564	7.7841	7.9715	0.033833	0.040251	0.10803	507.90	0.52333
285.00	4.1820	19.106	0.052339	8.2804	8.4993	0.035615	0.040437	0.11651	468.86	0.69225
290.00	4.7012	18.361	0.054464	8.8039	9.0600	0.037468	0.040727	0.12910	427.31	0.91943
295.00	5.2681	17.505	0.057125	9.3670	9.6680	0.039439	0.041190	0.14997	382.17	1.2430
300.00	5.8874	16.466	0.060732	9.9960	10.354	0.041620	0.041979	0.19222	331.57	1.7475
305.00	6.5663	15.022	0.066569	10.768	11.205	0.044293	0.043564	0.32907	271.24	2.6810
309.52	7.2447	10.270	0.097371	12.745	13.450	0.051414			0	6.2530
182.33	0.087837	0.059336	16.853	14.931	16.411	0.090017	0.025545	0.034993	211.93	44.590
185.00	0.10325	0.068951	14.503	14.984	16.481	0.089089	0.025844	0.035439	212.91	42.793
190.00	0.13782	0.090180	11.089	15.082	16.610	0.087443	0.026427	0.036334	214.62	39.645
195.00	0.18085	0.11615	8.6098	15.176	16.733	0.085908	0.027038	0.037309	216.14	36.768
200.00	0.23367	0.14754	6.7780	15.268	16.852	0.084469	0.027674	0.038365	217.48	34.146
205.00	0.29767	0.18508	5.4030	15.356	16.965	0.083117	0.028331	0.039506	218.62	31.765
210.00	0.37431	0.22958	4.3558	15.440	17.071	0.081842	0.029006	0.040736	219.56	29.606
215.00	0.46509	0.28190	3.5474	15.521	17.170	0.080634	0.029694	0.042062	220.30	27.653
220.00	0.57160	0.34298	2.9156	15.596	17.262	0.079485	0.030395	0.043494	220.82	25.888
225.00	0.69544	0.41387	2.4162	15.666	17.346	0.078387	0.031105	0.045048	221.13	24.295
230.00	0.83828	0.49575	2.0171	15.730	17.421	0.077332	0.031824	0.046741	221.20	22.858
235.00	1.0018	0.58993	1.6951	15.789	17.487	0.076314	0.032552	0.048600	221.04	21.560
240.00	1.1878	0.69791	1.4329	15.840	17.541	0.075325	0.033289	0.050660	220.64	20.387
245.00	1.3979	0.82145	1.2174	15.883	17.584	0.074359	0.034037	0.052967	219.98	19.325
250.00	1.6341	0.96261	1.0388	15.917	17.614	0.073408	0.034798	0.055585	219.07	18.361
255.00	1.8982	1.1239	0.88979	15.941	17.630	0.072464	0.035577	0.058600	217.88	17.483
260.00	2.1920	1.3082	0.76438	15.953	17.629	0.071519	0.036377	0.062136	216.41	16.679
265.00	2.5177	1.5196	0.65808	15.952	17.609	0.070564	0.037205	0.066370	214.64	15.939
270.00	2.8772	1.7627	0.56730	15.935	17.567	0.069588	0.038072	0.071572	212.56	15.251
275.00	3.2728	2.0443	0.48917	15.899	17.500	0.068575	0.038988	0.078164	210.15	14.603
280.00	3.7068	2.3734	0.42135	15.839	17.401	0.067508	0.039971	0.086856	207.39	13.981
285.00	4.1820	2.7632	0.36189	15.749	17.262	0.066361	0.041046	0.098934	204.24	13.366
290.00	4.7012	3.2349	0.30912	15.618	17.071	0.065093	0.042252	0.11700	200.67	12.734
295.00	5.2681	3.8252	0.26142	15.429	16.806	0.063636	0.043653	0.14723	196.63	12.043
300.00	5.8874	4.6102	0.21691	15.145	16.422	0.061848	0.045376	0.20883	192.02	11.210
305.00	6.5663	5.8106	0.17210	14.661	15.791	0.059328	0.047733	0.40691	186.66	10.014
309.52	7.2447	10.270	0.097371	12.745	13.450	0.051414			0	6.2530

TABLE 2-242 Thermodynamic Properties of Nitrous Oxide (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_v kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Single-Phase Properties										
200.00	0.10000	0.061319	16.308	15.382	17.013	0.092110	0.025918	0.035025	221.52	31.472
300.00	0.10000	0.040304	24.811	18.214	20.695	0.10698	0.030437	0.038943	267.85	10.733
400.00	0.10000	0.030129	33.190	21.467	24.786	0.11873	0.034363	0.042764	306.03	5.6159
500.00	0.10000	0.024075	41.537	25.072	29.226	0.12862	0.037563	0.045926	339.54	3.4059
200.00	1.0000	26.940	0.037120	1.1429	1.1800	0.0059677	0.041381	0.076317	1040.5	-0.23345
234.95	1.0000	24.272	0.041200	3.8803	3.9215	0.018590	0.040291	0.081583	813.80	-0.090429
234.95	1.0000	0.58888	1.6981	15.788	17.486	0.076324	0.032545	0.048580	221.04	21.573
300.00	1.0000	0.42412	2.3578	17.944	20.302	0.086931	0.031268	0.041833	260.01	10.924
400.00	1.0000	0.30696	3.2578	21.310	24.568	0.099190	0.034610	0.043833	303.02	5.5924
500.00	1.0000	0.24260	4.1220	24.963	29.085	0.10926	0.037680	0.046488	338.49	3.3701
200.00	5.0000	27.091	0.036913	1.0679	1.2525	0.0055897	0.041467	0.075665	1061.6	-0.24314
292.69	5.0000	17.918	0.055811	9.1008	9.3799	0.038510	0.040948	0.13883	403.56	1.0778
292.69	5.0000	3.5344	0.28294	15.525	16.940	0.064339	0.042976	0.13106	198.56	12.373
300.00	5.0000	3.1047	0.32209	16.114	17.725	0.066990	0.039518	0.091593	212.60	11.486
400.00	5.0000	1.6773	0.59621	20.558	23.539	0.083920	0.035787	0.049876	290.66	5.4199
500.00	5.0000	1.2538	0.79755	24.469	28.457	0.094897	0.038199	0.049186	334.89	3.1966
200.00	10.000	27.269	0.036672	0.97935	1.3461	0.0051379	0.041579	0.074951	1086.5	-0.25384
300.00	10.000	18.407	0.054327	9.1891	9.7324	0.038771	0.040304	0.11033	462.21	0.72534
400.00	10.000	3.7981	0.26329	19.470	22.102	0.075432	0.037387	0.061591	279.96	4.9171
500.00	10.000	2.6054	0.38381	23.835	27.673	0.087896	0.038831	0.052973	333.21	2.9341
200.00	25.000	27.751	0.036035	0.74199	1.6429	0.0038961	0.041933	0.073298	1154.2	-0.27902
300.00	25.000	20.902	0.047843	8.0747	9.2708	0.034710	0.039937	0.082004	654.64	0.10041
400.00	25.000	11.546	0.086610	16.070	18.235	0.060433	0.040060	0.087395	352.41	1.7332
500.00	25.000	6.9433	0.14402	21.962	25.563	0.076883	0.040346	0.063761	357.12	1.8422
300.00	50.000	22.869	0.043727	7.1665	9.3528	0.031190	0.040453	0.072893	828.55	-0.14288
400.00	50.000	17.051	0.058649	13.791	16.723	0.052389	0.040588	0.073379	568.05	0.23352
500.00	50.000	12.463	0.080236	19.764	23.776	0.068153	0.041531	0.067192	477.36	0.52378

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., and Span, R., "Short Fundamental Equations of State for 20 Industrial Fluids," *J. Chem. Eng. Data* **51**(3):785–850, 2006. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in the equation of state are 0.1% in density in the liquid and vapor phases between 220 and 300 K, 0.25% at temperatures above 300 K and at temperatures below 220 K, and 0.5% in the critical region, except very close to the critical point. The uncertainty in vapor pressure is 0.2%, that for heat capacities is 3%, and that for the speed of sound in the vapor phase is 0.05% above 220 K. The uncertainty in the liquid phase is not known but is estimated to be within 5%.

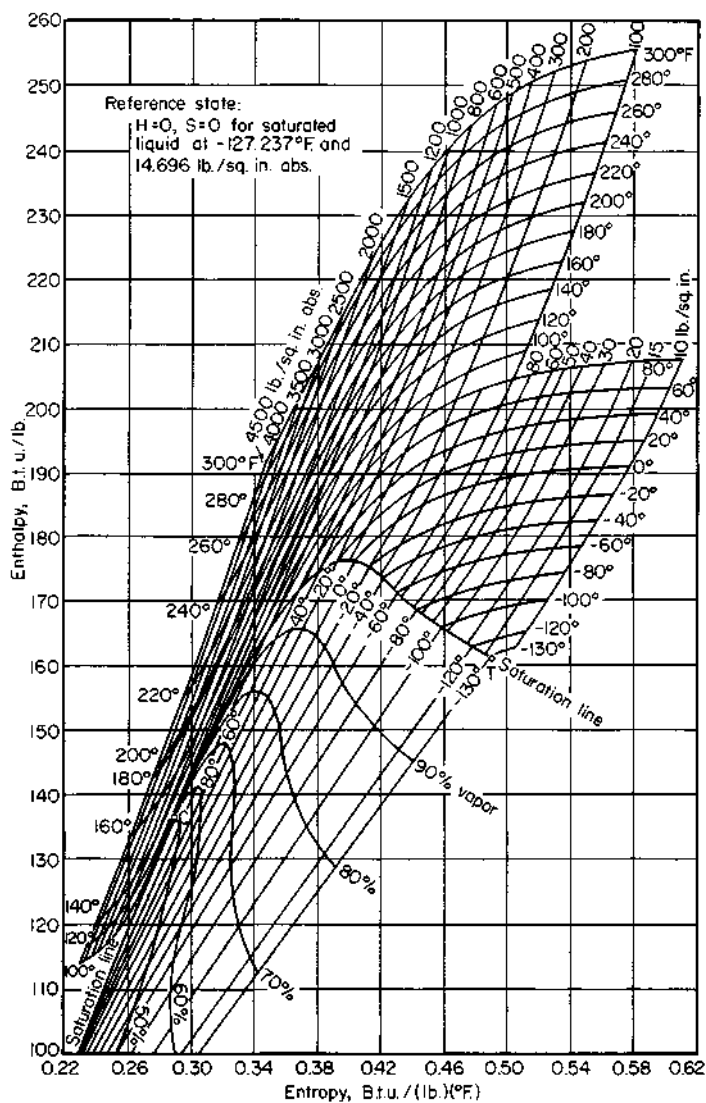


FIG. 2-14 Mollier diagram for nitrous oxide. (Fig. 9, Univ. Texas Rep., Cont. DAI-23-072-ORD-685, June 1, 1956, by Couch and Kobe. Reproduced by permission.) Some irregularity in the compressibility factors from 80 to 160 atm, 50 to 100 °C exists (Couch, private communication, 1967). See Couch et al., *J. Chem. Eng. Data*, **6**, (1961) for PVT data.

TABLE 2-243 Thermodynamic Properties of Nonane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_v kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
219.70	4.4449E-07	6.0520	0.16524	-60.675	-60.675	-0.19174	0.19748	0.25433	1544.5	-0.50071	153.02	4037.3
220.00	4.6229E-07	6.0501	0.16529	-60.599	-60.599	-0.19139	0.19757	0.25440	1543.0	-0.50053	152.90	3978.4
240.00	4.8933E-06	5.9249	0.16878	-55.459	-55.459	-0.16904	0.20402	0.25984	1449.8	-0.48613	145.29	1914.0
260.00	3.3858E-05	5.8013	0.17237	-50.195	-50.195	-0.14797	0.21164	0.26681	1361.7	-0.46790	138.35	1194.3
280.00	0.00016888	5.6785	0.17610	-44.779	-44.779	-0.12791	0.22016	0.27497	1277.9	-0.44671	131.98	842.44
300.00	0.00065182	5.5557	0.18000	-39.190	-39.190	-0.10863	0.22932	0.28402	1197.9	-0.42315	126.11	638.13
320.00	0.0020515	5.4321	0.18409	-33.413	-33.413	-0.089996	0.23894	0.29376	1121.1	-0.39747	120.68	505.83
340.00	0.0054791	5.3071	0.18843	-27.437	-27.436	-0.071883	0.24886	0.30400	1046.8	-0.36960	115.63	413.37
360.00	0.012806	5.1799	0.19305	-21.252	-21.249	-0.054209	0.25894	0.31464	974.60	-0.33918	110.92	344.96
380.00	0.026830	5.0496	0.19803	-14.851	-14.846	-0.036909	0.26908	0.32557	903.98	-0.30546	106.50	292.08
400.00	0.051367	4.9154	0.20344	-8.2313	-8.2208	-0.019932	0.27921	0.33677	834.45	-0.26723	102.33	249.76
420.00	0.091247	4.7760	0.20938	-1.3870	-1.3679	-0.0032367	0.28927	0.34824	765.54	-0.22257	98.374	214.92
440.00	0.15227	4.6299	0.21599	5.6854	5.7183	0.013213	0.29920	0.36002	696.76	-0.16849	94.594	185.56
460.00	0.24110	4.4754	0.22345	12.991	13.045	0.029452	0.30899	0.37228	627.57	-0.10004	90.947	160.30
480.00	0.36527	4.3096	0.23204	20.537	20.622	0.045513	0.31863	0.38530	557.35	-0.0087139	87.384	138.18
500.00	0.53313	4.1286	0.24221	28.335	28.464	0.061438	0.32814	0.39965	485.36	0.12153	83.848	118.43
520.00	0.75413	3.9259	0.25472	36.408	36.600	0.077283	0.33756	0.41650	410.60	0.32444	80.262	100.46
540.00	1.0392	3.6898	0.27102	44.798	45.080	0.093141	0.34703	0.43881	331.60	0.68436	76.518	83.674
560.00	1.4023	3.3948	0.29457	53.602	54.015	0.10920	0.35685	0.47629	245.76	1.4810	72.450	67.309
594.55	2.2820	1.8100	0.55249	73.819	75.080	0.144498	—	—	—	31.599	—	—
219.70	4.4449E-07	2.4333E-07	4,109,700.	-9.8808	-8.0542	0.047770	0.16161	0.16993	122.37	363.94	6.2016	4.0513
220.00	4.6229E-07	2.5273E-07	3,956,800.	-9.8323	-8.0032	0.047676	0.16174	0.17006	122.45	361.95	6.2133	4.0567
240.00	4.8933E-06	2.4522E-06	407,790.	-6.5064	-4.5110	0.043246	0.17103	0.17935	127.73	254.36	7.0659	4.4125
260.00	3.3858E-05	1.5664E-05	63,840.	-2.9862	-0.82471	0.041911	0.18119	0.18951	132.76	183.36	8.0682	4.7673
280.00	0.00016888	7.2578E-05	13,778.	0.74180	3.0688	0.042975	0.19199	0.20033	137.56	135.32	9.2127	5.1206
300.00	0.00065182	0.00026168	3,821.5	4.6851	7.1760	0.045920	0.20323	0.21161	142.11	102.10	10.493	5.4722
320.00	0.0020515	0.00077364	1,292.6	8.8457	11.497	0.050349	0.21478	0.22326	146.36	78.695	11.902	5.8211
340.00	0.0054791	0.0019519	512.31	13.220	16.027	0.055950	0.22652	0.23519	150.21	61.932	13.436	6.1663
360.00	0.012806	0.0043356	230.65	17.802	20.756	0.062471	0.23839	0.24737	153.55	49.767	15.091	6.5069
380.00	0.026830	0.0086887	115.09	22.580	25.668	0.069707	0.25032	0.25979	156.23	40.846	16.864	6.8425
400.00	0.051367	0.016023	62.412	27.541	30.747	0.077488	0.26226	0.27245	158.10	34.258	18.757	7.1737
420.00	0.091247	0.027625	36.200	32.671	35.974	0.085673	0.27416	0.28541	158.99	29.384	20.773	7.5025
440.00	0.15227	0.045117	22.165	37.952	41.327	0.094142	0.28601	0.29878	158.70	25.803	22.918	7.8332
460.00	0.24110	0.070581	14.168	43.363	46.779	0.10279	0.29779	0.31276	157.00	23.233	25.205	8.1731
480.00	0.36527	0.10681	9.3627	48.876	52.296	0.11150	0.30950	0.32778	153.58	21.504	27.655	8.5341
500.00	0.53313	0.15781	6.3369	54.456	57.834	0.12018	0.32118	0.34467	148.05	20.545	30.306	8.9361
520.00	0.75413	0.22994	4.3490	60.045	63.325	0.12868	0.33293	0.36530	139.82	20.405	33.233	9.4142
540.00	1.0392	0.33476	2.9872	65.548	68.652	0.13679	0.34492	0.39454	127.99	21.346	36.613	10.039
560.00	1.4023	0.49815	2.0074	70.762	73.577	0.14413	0.35757	0.44973	111.07	24.126	40.989	10.990
594.55	2.2820	1.8100	0.55249	73.819	75.080	0.144498	—	—	—	31.599	—	—

Single-Phase Properties

250.00	0.10000	5.8635	0.17055	-52.849	-52.832	-0.15838	0.20771	0.26315	1405.7	-0.47750	141.78	1483.4
300.00	0.10000	5.5564	0.17997	-39.196	-39.178	-0.10865	0.22933	0.28401	1198.6	-0.42329	126.15	638.82
350.00	0.10000	5.2447	0.19067	-24.378	-24.359	-0.063016	0.25389	0.30925	1011.2	-0.35500	113.28	377.18
400.00	0.10000	4.9160	0.20342	-8.2368	-8.2164	-0.019946	0.27921	0.33675	834.97	-0.26752	102.36	249.91
423.42	0.10000	4.7515	0.21046	-0.19510	-0.17406	-0.00041018	0.29097	0.35022	753.80	-0.21411	97.717	209.56
423.42	0.10000	0.030145	33.173	33.563	36.880	0.087103	0.27619	0.28766	159.03	28.690	21.130	7.5587
450.00	0.10000	0.027993	35.724	41.125	44.697	0.10500	0.28978	0.30044	165.87	22.301	23.838	8.0448
500.00	0.10000	0.024798	40.326	56.284	60.317	0.13790	0.31440	0.32419	177.26	14.792	29.272	8.9428
250.00	1.0000	5.8679	0.17042	-52.889	-52.719	-0.15854	0.20778	0.26309	1410.5	-0.47809	142.11	1501.9
300.00	1.0000	5.5624	0.17978	-39.250	-39.070	-0.10883	0.22940	0.28387	1204.4	-0.42451	126.55	645.13
350.00	1.0000	5.2530	0.19037	-24.450	-24.260	-0.063223	0.25395	0.30900	1018.5	-0.35745	113.76	380.90
400.00	1.0000	4.9280	0.20292	-8.3375	-8.1346	-0.020199	0.27927	0.33627	844.45	-0.27280	102.96	252.79
450.00	1.0000	4.5708	0.21878	9.1743	9.3930	0.021057	0.30414	0.36516	674.11	-0.14850	93.482	174.91
500.00	1.0000	4.1469	0.24114	28.203	28.444	0.061172	0.32810	0.39811	496.26	0.096651	84.482	120.06
250.00	5.0000	5.8875	0.16985	-53.064	-52.214	-0.15925	0.20809	0.26282	1431.5	-0.48055	143.55	1587.0
300.00	5.0000	5.5884	0.17894	-39.480	-38.585	-0.10961	0.22969	0.28333	1229.8	-0.42952	128.25	673.50
350.00	5.0000	5.2884	0.18909	-24.758	-23.813	-0.064113	0.25422	0.30802	1049.7	-0.36725	115.82	397.43
400.00	5.0000	4.9781	0.20088	-8.7590	-7.7546	-0.021267	0.27950	0.33447	884.19	-0.29292	105.52	265.35
450.00	5.0000	4.6466	0.21521	8.5689	9.6450	0.019689	0.30429	0.36159	727.45	-0.19528	96.800	186.34
500.00	5.0000	4.2772	0.23380	27.246	28.415	0.059215	0.32797	0.38941	574.69	-0.043276	89.174	132.45
250.00	10.000	5.9111	0.16917	-53.273	-51.582	-0.16011	0.20848	0.26253	1456.9	-0.48327	145.29	1699.9
300.00	10.000	5.6194	0.17796	-39.753	-37.973	-0.11054	0.23005	0.28275	1260.0	-0.43494	130.31	709.84
350.00	10.000	5.3297	0.18763	-25.116	-23.240	-0.065168	0.25456	0.30700	1086.3	-0.37743	118.27	418.13
400.00	10.000	5.0347	0.19862	-9.2354	-7.2492	-0.022500	0.27981	0.33273	929.33	-0.31235	108.48	280.67
450.00	10.000	4.7272	0.21154	7.9189	10.034	0.018184	0.30452	0.35854	784.82	-0.23486	100.47	199.70
500.00	10.000	4.3995	0.22730	26.321	28.594	0.057270	0.32803	0.38371	650.34	-0.13338	93.894	145.66
300.00	100.00	6.0143	0.16627	-43.037	-26.410	-0.12342	0.23562	0.27987	1662.5	-0.46881	158.86	1638.6
350.00	100.00	5.8100	0.17212	-29.086	-11.875	-0.078654	0.25980	0.30194	1534.1	-0.43419	149.98	851.44
400.00	100.00	5.6178	0.17800	-13.998	3.8029	-0.036820	0.28476	0.32520	1426.4	-0.40174	143.57	557.39
450.00	100.00	5.4358	0.18396	2.2421	20.639	0.0028129	0.30915	0.34808	1335.5	-0.37331	139.19	406.92
500.00	100.00	5.2628	0.19001	19.589	38.590	0.040621	0.33224	0.36974	1258.2	-0.34915	136.49	316.22
500.00	300.00	5.9601	0.16778	15.017	65.352	0.023367	0.33895	0.37096	1885.2	-0.36365	184.31	733.45

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., and Span, R., "Short Fundamental Equations of State for 20 Industrial Fluids," *J. Chem. Eng. Data* **51**(3):785–850, 2006. The source for viscosity is Huber, M. L., Laesecke, A., and Xiang, H. W., "Viscosity Correlations for Minor Constituent Fluids in Natural Gas: *n*-Octane, *n*-Nonane and *n*-Decane," *Fluid Phase Equilibria* **224**:263–270, 2004. The source for thermal conductivity is Huber, M. L., and Perkins, R. A., "Thermal Conductivity Correlations for Minor Constituent Fluids in Natural Gas: *n*-Octane, *n*-Nonane and *n*-Decane," *Fluid Phase Equilibria* **227**:47–55, 2004.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in the equation are 0.05% in the saturated-liquid density between 280 and 335 K and 0.2% in density in the liquid phase below 430 K and 10 MPa. The uncertainty increases to 0.3% up to 100 MPa and 0.5% up to 800 MPa. In the vapor phase and at supercritical state points, the uncertainty in density is 1%, whereas in the liquid phase between 430 K and the critical point it is 0.5% in density. Other uncertainties are 0.2% in vapor pressure between 300 and 430 K, 0.5% in vapor pressure at higher temperatures, 2% in heat capacities below 550 K, 5% at higher temperatures, and 1% in the liquid-phase speed of sound below 430 K. The estimated uncertainty in viscosity is 1.0% along the saturated-liquid line, 5% elsewhere. Uncertainty in thermal conductivity is 3%, except in the supercritical region and dilute gas which have an uncertainty of 5%.

TABLE 2-244 Thermodynamic Properties of Octane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_v kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
216.37	1.9889E-06	6.6864	0.14956	-47.586	-47.586	-0.15718	0.18017	0.22965	1496.9	-0.50088	153.28	2275.3
220.00	3.0719E-06	6.6606	0.15014	-46.751	-46.751	-0.15335	0.18100	0.23031	1479.7	-0.49876	151.81	2055.6
240.00	2.5565E-05	6.5193	0.15339	-42.103	-42.103	-0.13313	0.18613	0.23473	1388.4	-0.48455	144.08	1275.2
260.00	0.00014507	6.3793	0.15676	-37.353	-37.353	-0.11412	0.19222	0.24044	1301.6	-0.46633	136.89	875.51
280.00	0.00061336	6.2396	0.16027	-32.477	-32.477	-0.096062	0.19917	0.24730	1218.5	-0.44452	130.16	644.21
300.00	0.0020600	6.0991	0.16396	-27.455	-27.454	-0.078739	0.20684	0.25511	1138.6	-0.41949	123.83	498.03
320.00	0.0057644	5.9569	0.16787	-22.268	-22.267	-0.062005	0.21505	0.26370	1061.2	-0.39136	117.86	399.06
340.00	0.013932	5.8120	0.17206	-16.904	-16.901	-0.045746	0.22363	0.27290	985.87	-0.35994	112.19	328.21
360.00	0.029907	5.6633	0.17657	-11.351	-11.345	-0.029879	0.23245	0.28260	912.09	-0.32455	106.78	275.06
380.00	0.058269	5.5097	0.18150	-5.6010	-5.5904	-0.014338	0.24139	0.29271	839.36	-0.28395	101.59	233.59
400.00	0.10483	5.3495	0.18693	0.35204	0.37164	0.00092888	0.25035	0.30321	767.20	-0.23602	96.614	200.14
420.00	0.17652	5.1809	0.19302	6.5150	6.5490	0.015963	0.25926	0.31416	695.05	-0.17723	91.804	172.37
440.00	0.28132	5.0012	0.19995	12.895	12.951	0.030805	0.26810	0.32572	622.34	-0.10167	87.139	148.75
460.00	0.42812	4.8068	0.20804	19.502	19.591	0.045494	0.27685	0.33823	548.39	0.0013050	82.591	128.19
480.00	0.62676	4.5922	0.21776	26.352	26.489	0.060080	0.28552	0.35240	472.36	0.15251	78.136	109.92
500.00	0.88820	4.3477	0.23000	33.473	33.677	0.074631	0.29419	0.36979	393.17	0.39829	73.755	93.322
520.00	1.2251	4.0556	0.24657	40.921	41.223	0.089269	0.30300	0.39457	309.23	0.86399	69.457	77.803
540.00	1.6533	3.6731	0.27225	48.832	49.282	0.10426	0.31238	0.44291	217.99	2.0228	65.431	62.572
560.00	2.1958	3.0341	0.32959	57.752	58.476	0.12068	0.32413	0.68855	115.40	7.6023	64.209	45.346
569.32	2.4978	2.0564	0.48629	64.527	65.741	0.13332			0	28.218		
216.37	1.9889E-06	1.1056E-06	904.510	-1.8491	-0.050104	0.062519	0.14278	0.15110	129.10	314.66	6.1594	4.2034
220.00	3.0719E-06	1.6794E-06	595.460	-1.3282	0.50093	0.061430	0.14422	0.15253	130.14	294.38	6.3194	4.2717
240.00	2.5565E-05	1.2813E-05	78.047	1.6370	3.6323	0.057431	0.15246	0.16078	135.72	207.62	7.2936	4.6468
260.00	0.00014507	6.7137E-05	14.895	4.7709	6.9318	0.056201	0.16128	0.16962	141.02	150.42	8.4195	5.0206
280.00	0.00061336	0.00026380	3,790.7	8.0822	10.407	0.057097	0.17068	0.17906	146.04	111.63	9.6907	5.3924
300.00	0.0020600	0.00082859	1,206.9	11.576	14.062	0.059649	0.18058	0.18906	150.71	84.719	11.102	5.7614
320.00	0.0057644	0.0021820	458.30	15.253	17.894	0.063500	0.19087	0.19955	154.94	65.728	12.651	6.1263
340.00	0.013932	0.0049961	200.15	19.108	21.897	0.068366	0.20147	0.21047	158.59	52.133	14.338	6.4865
360.00	0.029907	0.010232	97.731	23.136	26.058	0.074020	0.21227	0.22178	161.53	42.292	16.164	6.8419
380.00	0.058269	0.019165	52.177	27.323	30.363	0.080278	0.22318	0.23348	163.56	35.112	18.138	7.1948
400.00	0.10483	0.033420	29.922	31.658	34.794	0.086985	0.23415	0.24557	164.51	29.858	20.268	7.5505
420.00	0.17652	0.055042	18.168	36.123	39.330	0.094013	0.24512	0.25818	164.17	26.034	22.571	7.9191
440.00	0.28132	0.086662	11.539	40.698	43.945	0.10124	0.25607	0.27154	162.27	23.316	25.072	8.3171
460.00	0.42812	0.13183	7.5854	45.358	48.605	0.10857	0.26700	0.28610	158.49	21.503	27.811	8.7715
480.00	0.62676	0.19572	5.1093	50.064	53.266	0.11587	0.27795	0.30284	152.37	20.514	30.863	9.3258
500.00	0.88820	0.28672	3.4877	54.757	57.855	0.12299	0.28899	0.32404	143.24	20.401	34.377	10.056
520.00	1.2251	0.42068	2.3771	59.333	62.245	0.12970	0.30032	0.35598	130.05	21.459	38.702	11.113
540.00	1.6533	0.63573	1.5730	63.555	66.156	0.13551	0.31238	0.42387	111.06	24.546	44.855	12.885
560.00	2.1958	1.0851	0.92161	66.584	68.608	0.13877	0.32642	0.78365	83.717	31.412	57.602	17.102
569.32	2.4978	2.0564	0.48629	64.527	65.741	0.13332			0	28.218		

Single-Phase Properties

300.00	0.10000	6.0999	0.16394	-27.460	-27.444	-0.078758	0.20684	0.25509	1139.3	-0.41966	123.88	498.56
398.30	0.10000	5.3634	0.18645	-0.16168	-0.14303	-0.00035827	0.24959	0.30230	773.32	-0.24045	97.030	202.73
398.30	0.10000	0.031955	31.294	31.284	34.413	0.086401	0.23322	0.24453	164.48	30.243	20.080	7.5200
400.00	0.10000	0.031787	31.459	31.684	34.830	0.087444	0.23402	0.24527	164.97	29.707	20.249	7.5528
500.00	0.10000	0.024611	40.633	57.485	61.548	0.14688	0.27938	0.28878	189.53	12.431	31.224	9.4433
600.00	0.10000	0.020274	49.325	87.490	92.422	0.20307	0.31885	0.32771	209.46	6.6034	43.584	11.290
300.00	1.0000	6.1073	0.16374	-27.511	-27.347	-0.078927	0.20690	0.25495	1145.7	-0.42119	124.30	503.48
400.00	1.0000	5.3654	0.18638	0.25030	0.43668	0.00067378	0.25038	0.30262	778.09	-0.24368	97.295	202.65
500.00	1.0000	4.3556	0.22959	33.432	33.661	0.074548	0.29416	0.36900	396.85	0.38286	73.948	93.722
507.20	1.0000	4.2495	0.23532	36.111	36.346	0.079880	0.29733	0.37746	363.62	0.52889	72.196	87.646
507.20	1.0000	0.32874	3.0419	56.425	59.467	0.12547	0.29302	0.33370	139.04	20.620	35.810	10.386
600.00	1.0000	0.22849	4.3766	85.870	90.247	0.18117	0.32280	0.33940	186.79	8.0078	45.491	11.637
300.00	5.0000	6.1393	0.16288	-27.730	-26.915	-0.079664	0.20714	0.25435	1173.4	-0.42743	126.15	525.55
400.00	5.0000	5.4311	0.18413	-0.17211	0.74852	-0.00039889	0.25054	0.30044	823.37	-0.27213	100.17	213.73
500.00	5.0000	4.5686	0.21888	32.275	33.369	0.072174	0.29359	0.35339	499.49	0.076618	79.631	105.97
600.00	5.0000	3.0593	0.32688	70.832	72.466	0.14322	0.33497	0.44331	188.23	2.6621	64.841	42.657
300.00	10.000	6.1772	0.16188	-27.987	-26.368	-0.080546	0.20745	0.25373	1206.2	-0.43408	128.36	553.60
400.00	10.000	5.5037	0.18170	-0.64047	1.1765	-0.0016150	0.25077	0.29842	873.81	-0.29834	103.45	227.34
500.00	10.000	4.7443	0.21078	31.275	33.383	0.070056	0.29343	0.34556	588.81	-0.072942	84.985	118.55
600.00	10.000	3.7971	0.26336	67.577	70.211	0.13708	0.33156	0.38999	358.22	0.45738	73.121	61.952
300.00	50.000	6.4220	0.15572	-29.605	-21.819	-0.086527	0.20976	0.25121	1422.7	-0.46254	143.51	798.49
400.00	50.000	5.8987	0.16953	-3.1776	5.2989	-0.0087968	0.25276	0.29206	1159.5	-0.38084	123.35	335.22
500.00	50.000	5.4043	0.18504	27.307	36.559	0.060791	0.29481	0.33234	965.74	-0.30867	110.66	198.43
600.00	50.000	4.9374	0.20254	61.436	71.563	0.12452	0.33146	0.36666	826.07	-0.24994	103.74	134.83
300.00	100.00	6.6452	0.15048	-30.984	-15.936	-0.092407	0.21232	0.25061	1629.4	-0.47349	158.61	1163.7
400.00	100.00	6.2050	0.16116	-5.0534	11.063	-0.015013	0.25503	0.29032	1399.1	-0.40628	141.31	478.53
500.00	100.00	5.8089	0.17215	24.893	42.108	0.054099	0.29686	0.32988	1233.7	-0.35233	131.06	291.34
600.00	100.00	5.4489	0.18352	58.505	76.857	0.11736	0.33331	0.36412	1113.8	-0.31375	125.82	208.31

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Span, R., and Wagner, W., "Equations of State for Technical Applications. II. Results for Nonpolar Fluids," *Int. J. Thermophys.* **24**(1):41–109, 2003. The source for viscosity is Huber, M. L., Laesecke, A., and Xiang, H. W., "Viscosity Correlations for Minor Constituent Fluids in Natural Gas: *n*-Octane, *n*-Nonane and *n*-Decane," *Fluid Phase Equilibria* **224**:263–270, 2004. The source for thermal conductivity is Huber, M. L., and Perkins, R. A., "Thermal Conductivity Correlations for Minor Constituent Fluids in Natural Gas: *n*-Octane, *n*-Nonane and *n*-Decane," *Fluid Phase Equilibria* **227**:47–55, 2004.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties of the equation of state are approximately 0.2% (to 0.5% at high pressures) in density, 1% (in the vapor phase) to 2% in heat capacity, 1% (in the vapor phase) to 2% in the speed of sound, and 0.2% in vapor pressure, except in the critical region. The estimated uncertainty in viscosity is 0.5% along the saturated-liquid line, 2% in compressed liquid to 200 MPa, 5% in vapor and supercritical regions. Uncertainty in thermal conductivity is 3%, except in the supercritical region and dilute gas which have an uncertainty of 5%.

TABLE 2-245 Thermodynamic Properties of Oxygen

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_v kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
54.361	0.00014628	40.816	0.024500	-6.1954	-6.1954	0.066946	0.038252	0.053541	1123.4	-0.37992	201.92	773.62
55.000	0.00017857	40.734	0.024549	-6.1613	-6.1612	0.067571	0.037651	0.053489	1126.9	-0.37886	201.02	747.53
60.000	0.00072582	40.064	0.024960	-5.8938	-5.8938	0.072225	0.034835	0.053548	1127.4	-0.37011	193.94	578.07
65.000	0.0023349	39.367	0.025402	-5.6258	-5.6257	0.076516	0.033469	0.053668	1101.7	-0.36312	186.82	457.94
70.000	0.0062623	38.656	0.025869	-5.3573	-5.3572	0.080495	0.032532	0.053697	1066.3	-0.35686	179.70	371.79
75.000	0.014547	37.936	0.026360	-5.0889	-5.0885	0.084199	0.031745	0.053719	1027.5	-0.34972	172.58	308.66
80.000	0.030123	37.203	0.026879	-4.8202	-4.8194	0.087667	0.031030	0.053808	987.43	-0.34056	165.44	261.22
85.000	0.056831	36.457	0.027430	-4.5510	-4.5495	0.090931	0.030365	0.054012	946.87	-0.32856	158.27	224.62
90.000	0.099350	35.692	0.028017	-4.2806	-4.2778	0.094023	0.029745	0.054361	905.90	-0.31302	151.05	195.64
95.000	0.16308	34.905	0.028649	-4.0084	-4.0038	0.096967	0.029169	0.054880	864.40	-0.29316	143.81	172.12
100.00	0.25400	34.092	0.029333	-3.7337	-3.7263	0.099787	0.028636	0.055599	822.19	-0.26804	136.55	152.56
105.00	0.37853	33.245	0.030079	-3.4556	-3.4442	0.10250	0.028146	0.056557	779.06	-0.23637	129.25	135.93
110.00	0.54340	32.360	0.030903	-3.1732	-3.1564	0.10513	0.027703	0.057816	734.77	-0.19639	121.92	121.52
115.00	0.75559	31.426	0.031820	-2.8853	-2.8612	0.10770	0.027311	0.059469	689.03	-0.14551	114.57	108.81
120.00	1.0223	30.434	0.032858	-2.5904	-2.5568	0.11022	0.026976	0.061666	641.52	-0.079899	107.23	97.426
125.00	1.3509	29.367	0.034051	-2.2867	-2.2407	0.11271	0.026712	0.064659	591.86	0.0063780	99.912	87.086
130.00	1.7491	28.203	0.035457	-1.9711	-1.9091	0.11520	0.026536	0.068905	539.50	0.12309	92.634	77.571
135.00	2.2250	26.907	0.037165	-1.6394	-1.5567	0.11773	0.026485	0.075327	483.69	0.28750	85.404	68.687
140.00	2.7878	25.415	0.039347	-1.2839	-1.1742	0.12035	0.026634	0.086099	423.10	0.53357	78.217	60.223
145.00	3.4477	23.599	0.042375	-0.88908	-0.74298	0.12319	0.027189	0.10778	355.20	0.93865	71.056	51.869
150.00	4.2186	21.110	0.047372	-0.41330	-0.21346	0.12654	0.028982	0.17484	273.80	1.7389	64.190	42.900
154.58	5.0428	13.630	0.073368	0.66752	1.0375	0.13442			0	5.0628		
54.361	0.00014628	0.00032370	3089.2	1.1195	1.5714	0.20982	0.021241	0.029631	140.32	507.90	4.4204	4.0962
55.000	0.00017857	0.00039060	2560.2	1.1327	1.5898	0.20850	0.021297	0.029698	141.11	480.26	4.4842	4.1481
60.000	0.00072582	0.0014561	686.75	1.2355	1.7339	0.19935	0.021815	0.030320	147.03	284.62	4.9840	4.5528
65.000	0.0023349	0.0043291	230.99	1.3377	1.8770	0.19194	0.022310	0.030934	152.65	156.71	5.4863	4.9555
70.000	0.0062623	0.010804	92.556	1.4393	2.0189	0.18587	0.022565	0.031294	158.07	87.254	5.9925	5.3557
75.000	0.014547	0.023509	42.536	1.5397	2.1584	0.18083	0.022513	0.031336	163.33	52.570	6.5051	5.7533
80.000	0.030123	0.045891	21.791	1.6377	2.2941	0.17659	0.022239	0.031177	168.36	35.817	7.0277	6.1486
85.000	0.056831	0.082138	12.175	1.7320	2.4239	0.17297	0.021896	0.031019	173.06	27.728	7.5654	6.5423
90.000	0.099350	0.13710	7.2938	1.8209	2.5455	0.16984	0.021624	0.031053	177.30	23.649	8.1241	6.9355
95.000	0.16308	0.21627	4.6239	1.9031	2.6571	0.16708	0.021515	0.031420	180.99	21.338	8.7113	7.3301
100.00	0.25400	0.32579	3.0695	1.9772	2.7569	0.16462	0.021605	0.032204	184.06	19.753	9.3362	7.7281
105.00	0.37853	0.47267	2.1156	2.0421	2.8430	0.16238	0.021894	0.033461	186.44	18.446	10.010	8.1324
110.00	0.54340	0.66506	1.5036	2.0966	2.9136	0.16032	0.022361	0.035245	188.14	17.250	10.748	8.5467
115.00	0.75559	0.91283	1.0955	2.1391	2.9668	0.15838	0.022978	0.037647	189.13	16.118	11.571	8.9760
120.00	1.0223	1.2284	0.81405	2.1678	3.0000	0.15652	0.023726	0.040839	189.41	15.045	12.509	9.4273
125.00	1.3509	1.6285	0.61407	2.1801	3.0097	0.15471	0.024597	0.045146	188.96	14.029	13.607	9.9112
130.00	1.7491	2.1366	0.46803	2.1722	2.9908	0.15289	0.025604	0.051204	187.75	13.062	14.940	10.445
135.00	2.2250	2.7893	0.35852	2.1380	2.9357	0.15100	0.026794	0.060349	185.74	12.120	16.641	11.061
140.00	2.7878	3.6487	0.27407	2.0670	2.8311	0.14896	0.028269	0.075824	182.82	11.155	18.977	11.823
145.00	3.4477	4.8412	0.20656	1.9383	2.6505	0.14659	0.030276	0.10781	178.78	10.071	22.582	12.881
150.00	4.2186	6.7170	0.14888	1.6938	2.3219	0.14345	0.033574	0.21201	172.82	8.6358	29.666	14.721
154.58	5.0428	13.630	0.073368	0.66752	1.0375	0.13442			0	5.0628		

Single-Phase Properties

100.00	0.10000	0.12316	8.1192	2.0355	2.8474	0.17297	0.020885	0.029925	188.37	18.479	9.0852	7.7121
300.00	0.10000	0.040116	24.928	6.2338	8.7265	0.20531	0.021078	0.029435	329.72	2.6530	26.485	20.652
500.00	0.10000	0.024050	41.579	10.604	14.762	0.22069	0.022781	0.031108	421.27	0.75388	41.046	30.486
700.00	0.10000	0.017177	58.216	15.357	21.179	0.23147	0.024672	0.032992	493.31	0.10517	53.966	38.653
900.00	0.10000	0.013360	74.849	20.438	27.923	0.23994	0.026045	0.034363	555.60	-0.18735	65.867	45.806
100.00	1.0000	34.158	0.029276	-3.7444	-3.7151	0.099680	0.028683	0.055399	826.85	-0.27181	137.23	153.89
119.62	1.0000	30.512	0.032774	-2.6131	-2.5803	0.11003	0.027000	0.061476	645.19	-0.085501	107.79	98.249
119.62	1.0000	1.2018	0.83209	2.1662	2.9983	0.15666	0.023665	0.040564	189.41	15.124	12.433	9.3921
300.00	1.0000	0.40337	2.4791	6.1772	8.6563	0.18598	0.021148	0.029887	329.90	2.6066	26.894	20.846
500.00	1.0000	0.24010	4.1649	10.576	14.741	0.20149	0.022802	0.031240	422.68	0.73726	41.288	30.630
700.00	1.0000	0.17135	5.8360	15.340	21.176	0.21230	0.024682	0.033052	494.87	0.098376	54.139	38.766
900.00	1.0000	0.13328	7.5029	20.426	27.929	0.22078	0.026051	0.034395	557.14	-0.19062	66.001	45.899
100.00	5.0000	34.497	0.028988	-3.7983	-3.6533	0.099132	0.028935	0.054458	850.39	-0.28978	140.71	160.92
154.36	5.0000	16.011	0.062457	0.35374	0.66602	0.13204	0.038878	3.5718	163.89	4.2044	75.954	29.668
154.36	5.0000	11.160	0.089610	1.0294	1.4774	0.13729	0.041906	4.2513	158.85	6.0016	72.313	20.574
300.00	5.0000	2.0616	0.48505	5.9227	8.3480	0.17177	0.021448	0.032003	332.25	2.3730	28.797	21.766
500.00	5.0000	1.1908	0.83975	10.454	14.653	0.18787	0.022894	0.031815	429.36	0.66261	42.362	31.267
700.00	5.0000	0.84728	1.1802	15.264	21.165	0.19881	0.024726	0.033309	501.98	0.068114	54.901	39.261
900.00	5.0000	0.65931	1.5167	20.373	27.956	0.20734	0.026076	0.034537	564.07	-0.20519	66.593	46.305
100.00	10.000	34.885	0.028665	-3.8593	-3.5726	0.098498	0.029235	0.053516	877.07	-0.30803	144.82	169.49
300.00	10.000	4.2056	0.23778	5.6024	7.9802	0.16499	0.021790	0.034749	339.35	2.0332	31.466	23.153
500.00	10.000	2.3538	0.42484	10.306	14.554	0.18182	0.022999	0.032491	438.67	0.56900	43.708	32.074
700.00	10.000	1.6705	0.59861	15.171	21.157	0.19292	0.024776	0.033613	511.24	0.030534	55.839	39.873
900.00	10.000	1.3010	0.76866	20.307	27.993	0.20150	0.026104	0.034706	572.92	-0.22339	67.321	46.804
100.00	25.000	35.884	0.027867	-4.0109	-3.3142	0.096845	0.030037	0.051627	945.24	-0.34532	155.97	194.38
300.00	25.000	10.393	0.096215	4.7194	7.1247	0.15490	0.022521	0.040917	390.80	1.0167	41.851	29.605
500.00	25.000	5.6243	0.17780	9.8920	14.337	0.17346	0.023256	0.034167	472.62	0.30658	47.943	34.705
700.00	25.000	3.9923	0.25048	14.907	21.169	0.18495	0.024901	0.034397	541.32	-0.076019	58.651	41.714
900.00	25.000	3.1222	0.32028	20.117	28.124	0.19369	0.026174	0.035155	600.66	-0.27597	69.464	48.271
100.00	75.000	38.263	0.026135	-4.3340	-2.3739	0.092788	0.031906	0.049123	1115.1	-0.39472	184.96	274.96
300.00	75.000	21.603	0.046289	3.1884	6.6601	0.14315	0.023601	0.041272	645.54	-0.18640	75.261	53.378
500.00	75.000	13.760	0.072675	8.8798	14.330	0.16284	0.023725	0.036534	619.75	-0.20732	64.149	45.084
700.00	75.000	10.201	0.098029	14.192	21.544	0.17498	0.025126	0.035903	657.04	-0.31840	68.835	48.269
900.00	75.000	8.1749	0.12233	19.571	28.745	0.18403	0.026293	0.036153	701.72	-0.40609	76.863	53.163

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Schmidt, R., and Wagner, W., "A New Form of the Equation of State for Pure Substances and Its Application to Oxygen," *Fluid Phase Equilibria*, **19**:175–200, 1985. The source for viscosity and thermal conductivity is Lemmon, E. W., and Jacobsen, R. T., "Viscosity and Thermal Conductivity Equations for Nitrogen, Oxygen, Argon, and Air," *Int. J. Thermophys.* **25**:21–69, 2004.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties of the equation of state are 0.1% in density, 2% in heat capacity, and 1% in the speed of sound, except in the critical region. For viscosity, the uncertainty is 1% in the dilute gas at temperatures above 200 K, and 5% in the dilute gas at lower temperatures. The uncertainty is around 2% between 270 and 300 K, and increases to 5% outside of this region. The uncertainty may be higher in the liquid near the triple point. The uncertainty for the dilute gas is 2% with increasing uncertainties near the triple point. For thermal conductivity, the uncertainties range from 3% between 270 and 300 K to 5% elsewhere. The uncertainties above 100 MPa are not known due to a lack of experimental data.

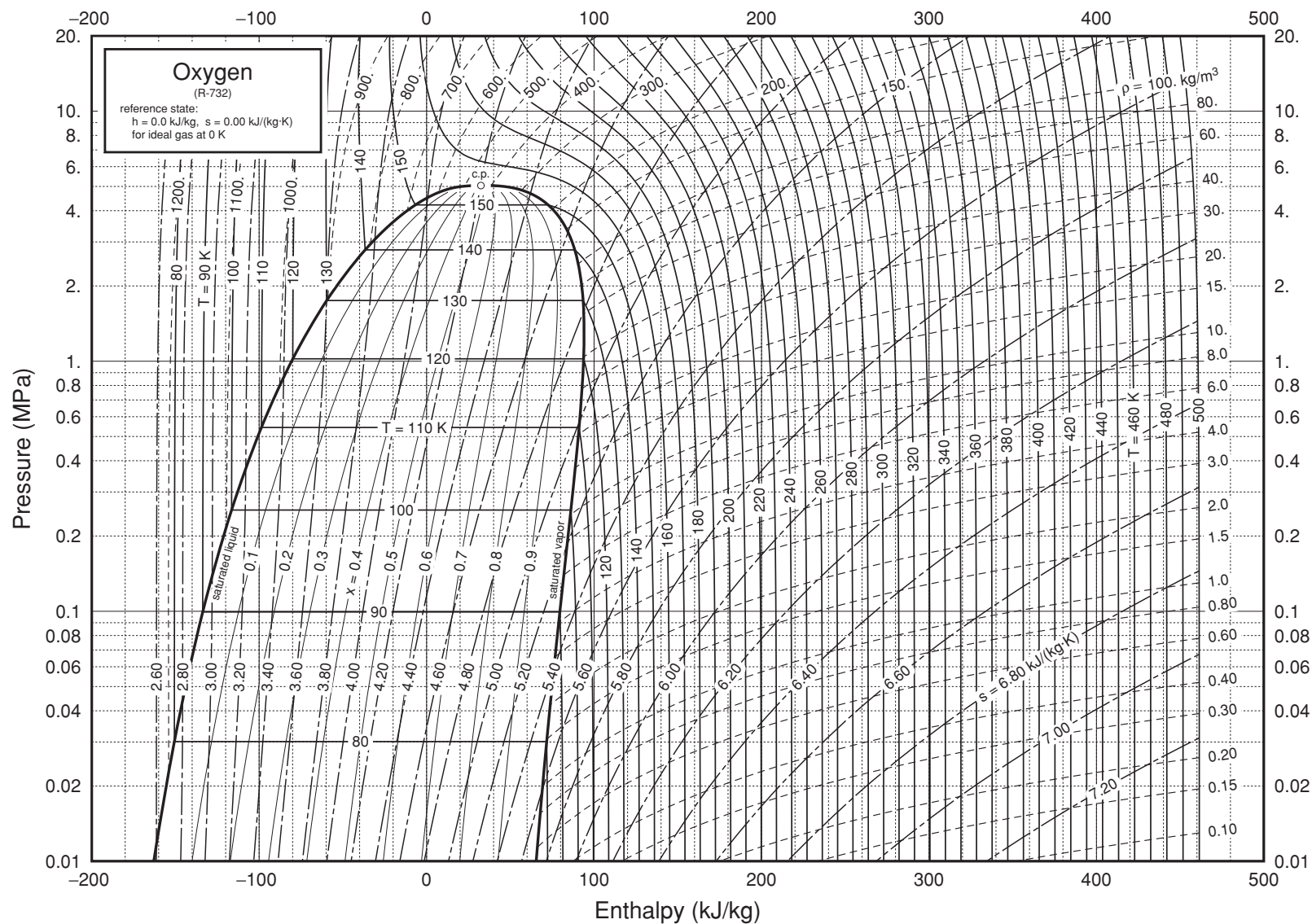


FIG. 2-15 Pressure-enthalpy diagram for oxygen. Properties computed with the NIST REFPROP Database, Version 7.0 (Lemmon, E. W., McLinden, M. O. and Huber, M. L., 2002, NIST Standard Reference Database 23, NIST Reference Fluid Thermodynamic and Transport Properties—REFPROP, Version 7.0, Standard Reference Data Program, National Institute of Standards and Technology), based on the equation of state of Schmidt, R., and Wagner, W., "A New Form of the Equation of State for Pure Substances and Its Application to Oxygen," *Fluid Phase Equilibria* **19**:175–200, 1985.

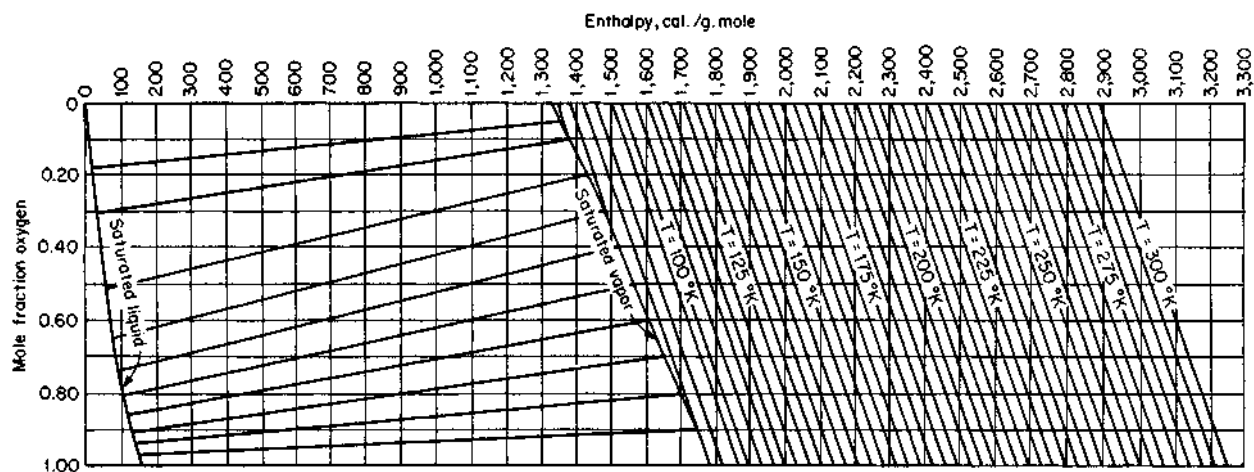


FIG. 2-16 Enthalpy-concentration diagram for oxygen-nitrogen mixture at 1 atm. Reference states: Enthalpies of liquid oxygen and liquid nitrogen at the normal boiling point of nitrogen are zero. (*Dodge, Chemical Engineering Thermodynamics, McGraw-Hill, New York, 1944.*) Wilson, Silverberg, and Zellner, AFAPL TDR 64-64 (AD 603151), 1964, p. 314, present extensive vapor-liquid equilibrium data for the three-component system argon-nitrogen-oxygen as well as for binary systems including oxygen-nitrogen. Calculations for this mixture are also available with the NIST REFPROP software.

TABLE 2-246 Thermodynamic Properties of Pentane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
143.47	7.6322E-08	10.566	0.094640	-25.092	-25.092	-0.11487	0.10392	0.14205	1829.9	-0.54905	175.03	3709.1
150.00	2.6809E-07	10.482	0.095399	-24.164	-24.164	-0.10855	0.10408	0.14201	1788.3	-0.54859	172.92	2750.8
165.00	3.1471E-06	10.292	0.097163	-22.032	-22.032	-0.095008	0.10469	0.14231	1696.9	-0.54514	167.37	1587.7
180.00	2.3256E-05	10.105	0.098964	-19.892	-19.892	-0.082594	0.10558	0.14312	1610.3	-0.53850	161.20	1048.4
195.00	0.00012116	9.9190	0.10082	-17.737	-17.737	-0.071091	0.10680	0.14441	1527.4	-0.52866	154.70	757.19
210.00	0.00048191	9.7339	0.10273	-15.558	-15.557	-0.060326	0.10837	0.14621	1447.4	-0.51549	148.08	581.46
225.00	0.0015504	9.5483	0.10473	-13.348	-13.347	-0.050163	0.11032	0.14853	1369.7	-0.49891	141.47	466.22
240.00	0.0042112	9.3613	0.10682	-11.099	-11.099	-0.040489	0.11267	0.15137	1293.8	-0.47884	134.95	385.57
255.00	0.0099767	9.1719	0.10903	-8.8042	-8.8031	-0.031215	0.11539	0.15473	1219.4	-0.45516	128.57	326.15
270.00	0.021139	8.9791	0.11137	-6.4555	-6.4532	-0.022266	0.11847	0.15857	1146.3	-0.42770	122.40	280.49
285.00	0.040858	8.7818	0.11387	-4.0461	-4.0414	-0.013582	0.12186	0.16289	1074.0	-0.39609	116.43	244.17
300.00	0.073168	8.5788	0.11657	-1.5694	-1.5608	-0.0051134	0.12552	0.16766	1002.5	-0.35971	110.69	214.42
315.00	0.12293	8.3688	0.11949	0.98088	0.99557	0.0031816	0.12940	0.17288	931.46	-0.31748	105.17	189.45
330.00	0.19575	8.1499	0.12270	3.6106	3.6346	0.011338	0.13348	0.17857	860.59	-0.26769	99.882	168.04
345.00	0.29786	7.9200	0.12626	6.3259	6.3635	0.019386	0.13770	0.18478	789.61	-0.20766	94.813	149.32
360.00	0.43606	7.6765	0.13027	9.1332	9.1900	0.027354	0.14204	0.19161	718.18	-0.13303	89.958	132.67
375.00	0.61766	7.4155	0.13485	12.040	12.123	0.035271	0.14649	0.19928	645.89	-0.036610	85.304	117.61
390.00	0.85052	7.1319	0.14022	15.057	15.176	0.043168	0.15103	0.20814	572.27	0.094152	80.839	103.78
405.00	1.1432	6.8175	0.14668	18.197	18.364	0.051083	0.15570	0.21894	496.72	0.28299	76.544	90.837
420.00	1.5050	6.4595	0.15481	21.482	21.715	0.059073	0.16056	0.23325	418.39	0.58010	72.402	78.483
435.00	1.9472	6.0340	0.16573	24.952	25.275	0.067234	0.16578	0.25525	335.90	1.1123	68.402	66.354
450.00	2.4836	5.4866	0.18226	28.698	29.150	0.075780	0.17187	0.30145	246.17	2.3196	64.610	53.842
465.00	3.1355	4.5754	0.21856	33.125	33.811	0.085678	0.18139	0.60959	138.75	7.5108	63.369	38.530
469.70	3.3710	3.2156	0.31099	36.504	37.552	0.093548			0	19.135		
143.47	7.6322E-08	6.3981E-08	15,630,000.	8.3431	9.5360	0.12648	0.070619	0.078934	135.94	542.15	4.0211	3.2806
150.00	2.6809E-07	2.1496E-07	4,652,000.	8.8107	10.058	0.11959	0.072591	0.080906	138.80	453.68	4.3338	3.4236
165.00	3.1471E-06	2.2940E-06	435,920.	9.9311	11.303	0.10702	0.076717	0.085033	145.17	311.80	5.0759	3.7558
180.00	2.3256E-05	1.5540E-05	64,349.	11.110	12.607	0.097955	0.080473	0.088791	151.28	222.93	5.8544	4.0934
195.00	0.00012116	7.4749E-05	13,378.	12.343	13.964	0.091475	0.084066	0.092395	157.11	164.55	6.6764	4.4362
210.00	0.00048191	0.00027623	3,620.1	13.627	15.372	0.086957	0.087665	0.096021	162.67	124.69	7.5505	4.7837
225.00	0.0015504	0.00083048	1,204.1	14.962	16.829	0.083956	0.091392	0.099805	167.92	96.614	8.4862	5.1356
240.00	0.0042112	0.0021198	471.75	16.347	18.333	0.082144	0.095331	0.10385	172.81	76.365	9.4926	5.4918
255.00	0.0099767	0.0047454	210.73	17.780	19.882	0.081276	0.099526	0.10823	177.25	61.487	10.578	5.8525
270.00	0.021139	0.0095553	104.65	19.260	21.472	0.081162	0.10400	0.11297	181.16	50.401	11.752	6.2179
285.00	0.040858	0.017654	56.645	20.787	23.101	0.081654	0.10874	0.11811	184.42	42.052	13.022	6.5890
300.00	0.073168	0.030405	32.889	22.357	24.763	0.082633	0.11373	0.12367	186.93	35.715	14.394	6.9672
315.00	0.12293	0.049446	20.224	23.968	26.454	0.084003	0.11894	0.12966	188.56	30.879	15.879	7.3547
330.00	0.19575	0.076727	13.033	25.617	28.168	0.085681	0.12435	0.13613	189.17	27.184	17.484	7.7547
345.00	0.29786	0.11460	8.7259	27.297	29.896	0.087596	0.12993	0.14317	188.63	24.375	19.221	8.1719
360.00	0.43606	0.16601	6.0239	29.002	31.629	0.089684	0.13567	0.15092	186.76	22.275	21.106	8.6130
375.00	0.61766	0.23477	4.2594	30.721	33.352	0.091880	0.14155	0.15968	183.35	20.767	23.164	9.0883
390.00	0.85052	0.32623	3.0654	32.439	35.046	0.094117	0.14759	0.17000	178.13	19.785	25.433	9.6135
405.00	1.1432	0.44834	2.2305	34.133	36.682	0.096313	0.15383	0.18300	170.76	19.316	27.980	10.215
420.00	1.5050	0.61427	1.6279	35.764	38.214	0.098358	0.16037	0.20125	160.78	19.406	30.941	10.939
435.00	1.9472	0.84898	1.1779	37.264	39.557	0.10007	0.16742	0.23193	147.55	20.186	34.634	11.885
450.00	2.4836	1.2118	0.82522	38.471	40.520	0.10105	0.17552	0.30440	130.14	21.884	40.108	13.312
465.00	3.1355	1.9618	0.50974	38.730	40.328	0.099694	0.18671	0.82173	106.55	24.088	55.658	16.549
469.70	3.3710	3.2156	0.31099	36.504	37.552	0.093548			0	19.135		

Single-Phase Properties

200.00	0.10000	9.8581	0.10144	-17.016	-17.006	-0.067441	0.10728	0.14495	1501.0	-0.52472	152.53	690.14
300.00	0.10000	8.5793	0.11656	-1.5709	-1.5592	-0.0051184	0.12552	0.16765	1002.8	-0.35984	110.71	214.50
308.83	0.10000	8.4562	0.11826	-0.077672	-0.065846	-0.00021257	0.12778	0.17068	960.65	-0.33565	107.41	199.24
308.83	0.10000	0.040733	24.550	23.300	25.755	0.083398	0.11677	0.12714	188.00	32.712	15.254	7.1940
400.00	0.10000	0.030611	32.668	35.260	38.527	0.11949	0.14472	0.15376	217.34	12.986	24.749	9.3018
500.00	0.10000	0.024257	41.224	51.273	55.396	0.15700	0.17449	0.18314	243.86	6.4310	37.084	11.487
600.00	0.10000	0.020135	49.665	70.070	75.037	0.19274	0.20055	0.20906	267.28	3.8205	50.416	13.525
200.00	1.0000	9.8656	0.10136	-17.038	-16.937	-0.067555	0.10733	0.14489	1505.8	-0.52539	152.82	696.76
300.00	1.0000	8.5958	0.11634	-1.6210	-1.5046	-0.0052858	0.12555	0.16737	1011.2	-0.36413	111.23	217.09
398.07	1.0000	6.9671	0.14353	16.730	16.874	0.047422	0.15353	0.21364	531.90	0.18628	78.507	96.722
398.07	1.0000	0.38746	2.5809	33.355	35.936	0.095310	0.15092	0.17655	174.46	19.468	26.763	9.9253
400.00	1.0000	0.38317	2.6098	33.666	36.276	0.096162	0.15126	0.17612	175.97	18.915	26.951	9.9649
500.00	1.0000	0.26375	3.7914	50.465	54.257	0.13622	0.17624	0.18896	226.09	7.2197	38.498	12.017
600.00	1.0000	0.20980	4.7663	69.531	74.297	0.17270	0.20129	0.21181	257.70	3.9909	51.523	13.960
200.00	5.0000	9.8980	0.10103	-17.137	-16.632	-0.068053	0.10753	0.14467	1527.0	-0.52817	154.08	726.41
300.00	5.0000	8.6652	0.11540	-1.8329	-1.2559	-0.0060014	0.12571	0.16627	1047.2	-0.38106	113.46	228.42
400.00	5.0000	7.1612	0.13964	16.501	17.199	0.046820	0.15388	0.20601	606.55	0.017312	82.387	107.19
500.00	5.0000	3.2553	0.30719	42.207	43.743	0.10528	0.19073	0.42935	135.39	9.4810	59.214	26.017
600.00	5.0000	1.2834	0.77916	66.610	70.506	0.15448	0.20500	0.23291	220.81	4.4347	57.890	17.494
200.00	10.000	9.9373	0.10063	-17.255	-16.249	-0.068660	0.10778	0.14442	1552.6	-0.53127	155.61	763.99
300.00	10.000	8.7452	0.11435	-2.0763	-0.93280	-0.0068390	0.12592	0.16515	1088.8	-0.39826	116.08	242.27
400.00	10.000	7.3792	0.13552	15.896	17.251	0.045232	0.15383	0.20020	687.58	-0.10747	86.892	120.04
500.00	10.000	5.4581	0.18321	37.622	39.455	0.094589	0.18299	0.24567	349.25	0.88491	69.546	57.792
600.00	10.000	3.0774	0.32495	62.405	65.655	0.14230	0.20754	0.26294	229.05	2.8865	67.434	28.573
200.00	50.000	10.209	0.097953	-18.043	-13.146	-0.072988	0.10968	0.14331	1731.1	-0.54573	166.40	1086.9
300.00	50.000	9.2247	0.10840	-3.5039	1.9164	-0.012155	0.12755	0.16079	1343.9	-0.46478	132.95	346.81
400.00	50.000	8.2838	0.12072	13.310	19.346	0.037795	0.15501	0.18829	1056.0	-0.35370	109.81	195.02
500.00	50.000	7.3733	0.13562	32.748	39.529	0.082724	0.18248	0.21473	853.23	-0.25198	97.812	130.14
600.00	50.000	6.5265	0.15322	54.478	62.139	0.12389	0.20695	0.23672	723.60	-0.17131	94.951	95.489
200.00	100.00	10.479	0.095431	-18.760	-9.2168	-0.077502	0.11180	0.14305	1911.9	-0.55097	177.42	1550.8
300.00	100.00	9.6305	0.10384	-4.6330	5.7508	-0.017033	0.12937	0.15917	1570.5	-0.49017	148.83	473.34
400.00	100.00	8.8688	0.11275	11.680	22.955	0.032278	0.15665	0.18547	1325.1	-0.40837	128.55	270.23
500.00	100.00	8.1734	0.12235	30.587	42.822	0.076502	0.18394	0.21138	1153.2	-0.34418	117.64	187.06
600.00	100.00	7.5451	0.13254	51.860	65.113	0.11708	0.20826	0.23383	1035.8	-0.29936	114.07	142.50

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Span, R., and Wagner, W., "Equations of State for Technical Applications. II. Results for Nonpolar Fluids," *Int. J. Thermophys.* **24**(1):41–109, 2003. The source for viscosity and thermal conductivity is NIST14, Version 9.08.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties of the equation of state are approximately 0.2% (to 0.5% at high pressures) in density, 1% (in the vapor phase) to 2% in heat capacity, 1% (in the vapor phase) to 2% in the speed of sound, and 0.2% in vapor pressure, except in the critical region. For viscosity, estimated uncertainty is 2%. For thermal conductivity, estimated uncertainty, except near the critical region, is 4–6%.

TABLE 2-247 Saturated Potassium*

T, K	$P, \text{ bar}$	$v_f, \text{ m}^3/\text{kg}$	$v_g, \text{ m}^3/\text{kg}$	$h_f, \text{ kJ/kg}$	$h_g, \text{ kJ/kg}$	$s_f, \text{ kJ/(kg}\cdot\text{K)}$	$s_g, \text{ kJ/(kg}\cdot\text{K)}$	$c_{pf}, \text{ kJ/(kg}\cdot\text{K)}$
336.4 ^m	1.37-9	0.001208		93.8	2327	1.928	8.567	0.822
400	1.84-7	0.001229	4.64+6	145.5	2342	2.068	7.559	0.805
500	3.13-5	0.001266	3.39+4	225.1	2390	2.246	6.576	0.785
600	9.26-4	0.001304	3164	302.7	2433	2.388	5.937	0.771
700	0.01022	0.001346	142.3	379.4	2468	2.506	5.490	0.762
800	0.06116	0.001389	26.75	455.5	2498	2.608	5.161	0.761
1000	0.7322	0.001488	2.691	609.7	2552	2.780	4.722	0.792
1200	3.913	0.001605	0.584	773.5	2610	2.929	4.459	0.846
1400	12.44	0.001742	0.207	948.0	2679	3.063	4.299	0.899
1500	20.0	0.001816	0.132	1040.0	2718	3.123	4.209	0.924

*Converted from tables in Vargaftik, *Tables of the Thermophysical Properties of Liquids and Gases*, Nauka, Moscow, 1972; and Hemisphere, Washington, 1975. m = melting point. The notation 1.37-9 signifies 1.37×10^{-9} .

Many of the Vargaftik values also appear in Ohse, R. W., *Handbook of Thermodynamic and Transport Properties of Alkali Metals*, Blackwell Sci. Pubs., Oxford, 1985 (1020 pp.). This source contains superheat data. Saturation and superheat tables and a diagram to 30 bar, 1650 K are given by Reynolds, W. C., *Thermodynamic Properties in S.I.*, Stanford Univ. publ., 1979 (173 pp.). For a Mollier diagram from 0.1 to 250 psia, 1300 to 2700°R, see Weatherford, W. D., J. C. Tyler, et al., WADD-TR-61-96, 1961. An extensive review of properties of the solid and the saturated liquid is given by Alcock, C. B., M. W. Chase, et al., *J. Phys. Chem. Ref. Data*, 23, 3 (1994):385-497.

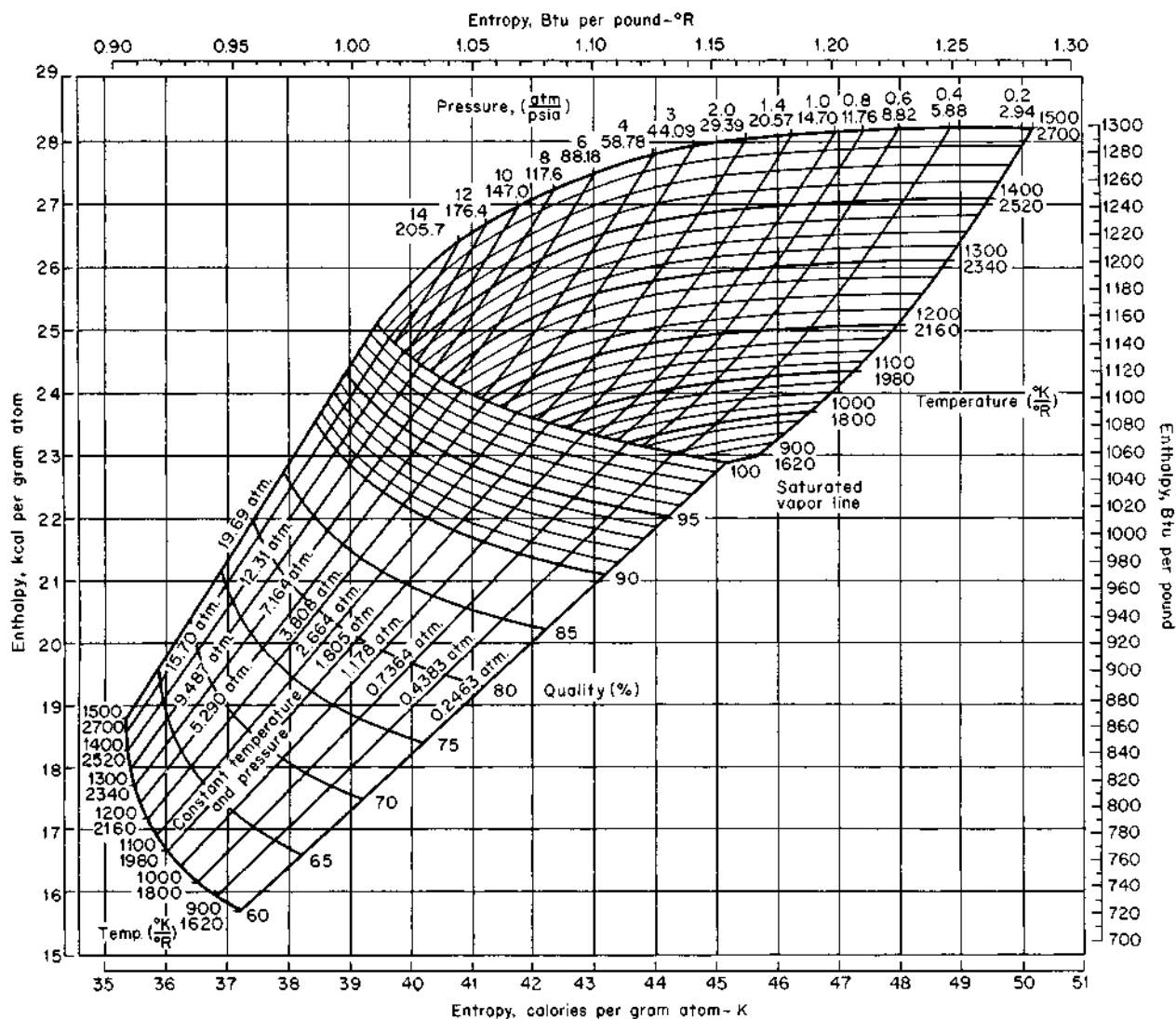


FIG. 2-17 Mollier diagram for potassium. Basis: enthalpy = 0.0 cal/g atom at 298 K; entropy = 15.8 cal/(g atom-K) at 298 K. (Aerojet-General Rep. AGN8194, vol. 2, 1967. Reproduced by permission.)

TABLE 2-248 Thermodynamic Properties of Propane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
100.00	2.5330E-08	16.287	0.061397	-7.4323	-7.4323	-0.048110	0.058376	0.083833	2030.4	-0.62852	203.23	3780.3
115.00	1.0677E-06	15.941	0.062730	-6.1638	-6.1638	-0.036293	0.058304	0.085262	1930.7	-0.61354	196.90	1822.7
130.00	1.7600E-05	15.596	0.064117	-4.8753	-4.8753	-0.025763	0.058442	0.086525	1827.2	-0.59904	189.40	1080.5
145.00	0.00015328	15.251	0.065568	-3.5682	-3.5682	-0.016249	0.058710	0.087764	1724.5	-0.58338	181.11	729.42
160.00	0.00084980	14.904	0.067094	-2.2418	-2.2418	-0.0075455	0.059141	0.089109	1623.4	-0.56521	172.30	534.62
175.00	0.0033874	14.554	0.068710	-0.89395	-0.89372	0.00050603	0.059774	0.090658	1523.6	-0.54342	163.19	412.89
190.00	0.010547	14.198	0.070434	0.47907	0.47981	0.0080327	0.060636	0.092484	1424.7	-0.51702	153.97	329.94
205.00	0.027195	13.834	0.072288	1.8816	1.8836	0.015137	0.061740	0.094646	1326.6	-0.48497	144.76	269.81
220.00	0.060583	13.459	0.074301	3.3189	3.3234	0.021903	0.063087	0.097193	1229.1	-0.44601	135.72	224.28
235.00	0.12030	13.070	0.076511	4.7964	4.8056	0.028399	0.064672	0.10018	1131.9	-0.39841	126.91	188.66
250.00	0.21798	12.663	0.078968	6.3203	6.3375	0.034686	0.066484	0.10368	1034.8	-0.33958	118.40	160.10
265.00	0.36693	12.234	0.081740	7.8975	7.9275	0.040815	0.068514	0.10780	937.63	-0.26538	110.26	136.69
280.00	0.58173	11.775	0.084925	9.5360	9.5854	0.046833	0.070755	0.11272	839.90	-0.16892	102.52	117.11
295.00	0.87805	11.277	0.088673	11.246	11.324	0.052790	0.073209	0.11880	741.08	-0.037850	95.209	100.38
310.00	1.2726	10.726	0.093229	13.041	13.159	0.058740	0.075898	0.12672	640.30	0.15178	88.324	85.735
325.00	1.7837	10.097	0.099036	14.943	15.120	0.064759	0.078882	0.13817	536.11	0.45297	81.819	72.510
340.00	2.4320	9.3403	0.10706	16.996	17.257	0.070985	0.082328	0.15818	425.33	1.0100	75.581	60.014
355.00	3.2432	8.3162	0.12025	19.317	19.707	0.077769	0.087215	0.21184	300.04	2.3857	69.491	47.137
369.83	4.2477	5.0000	0.20000	23.653	24.503	0.090537			0	12.890		
100.00	2.5330E-08	3.0465E-08	32,825,000.	15.887	16.718	0.19339	0.032981	0.041296	153.65	420.53	2.4171	2.9792
115.00	1.0677E-06	1.1166E-06	895,550.	16.399	17.356	0.16822	0.035381	0.043696	163.64	267.23	3.2182	3.3431
130.00	1.7600E-05	1.6284E-05	61,410.	16.947	18.028	0.15042	0.037630	0.045946	172.99	181.79	4.0849	3.7167
145.00	0.00015328	0.00012717	7,863.4	17.527	18.732	0.13755	0.039780	0.048104	181.78	130.32	5.0168	4.0976
160.00	0.00084980	0.00063940	1,564.0	18.137	19.466	0.12813	0.041911	0.050262	190.03	97.395	6.0131	4.4832
175.00	0.0033874	0.0023346	428.34	18.775	20.226	0.12119	0.044098	0.052519	197.68	75.331	7.0721	4.8708
190.00	0.010547	0.0067219	148.77	19.437	21.007	0.11607	0.046406	0.054975	204.61	59.995	8.1917	5.2578
205.00	0.027195	0.016180	61.804	20.120	21.801	0.11230	0.048888	0.057727	210.64	49.008	9.3706	5.6429
220.00	0.060583	0.033974	29.435	20.819	22.602	0.10953	0.051579	0.060867	215.59	40.931	10.611	6.0262
235.00	0.12030	0.064224	15.570	21.528	23.401	0.10753	0.054504	0.064488	219.25	34.865	11.922	6.4107
250.00	0.21798	0.11195	8.9325	22.242	24.189	0.10609	0.057677	0.068703	221.39	30.235	13.323	6.8027
265.00	0.36693	0.18328	5.4561	22.954	24.956	0.10507	0.061104	0.073667	221.82	26.669	14.845	7.2123
280.00	0.58173	0.28598	3.4968	23.656	25.690	0.10435	0.064794	0.079635	220.30	23.923	16.544	7.6551
295.00	0.87805	0.43050	2.3229	24.336	26.376	0.10381	0.068763	0.087064	216.53	21.832	18.503	8.1548
310.00	1.2726	0.63231	1.5815	24.977	26.989	0.10335	0.072895	0.096756	210.16	20.354	20.863	8.7494
325.00	1.7837	0.91726	1.0902	25.543	27.488	0.10282	0.077315	0.11131	200.74	19.511	23.881	9.5073
340.00	2.4320	1.3363	0.74835	25.971	27.791	0.10197	0.083072	0.13963	187.44	19.120	28.187	10.574
355.00	3.2432	2.0253	0.49375	26.098	27.699	0.10028	0.091334	0.22264	168.95	18.854	36.060	12.373
369.83	4.2477	5.0000	0.20000	23.653	24.503	0.090537			0	12.890		
Single-Phase Properties												
100.00	0.10000	16.288	0.061394	-7.4332	-7.4270	-0.048119	0.058377	0.083831	2030.8	-0.62854	203.26	3784.5
200.00	0.10000	13.958	0.071646	1.4085	1.4157	0.012800	0.061348	0.093874	1359.8	-0.49655	147.87	288.12
230.74	0.10000	13.182	0.075860	4.3719	4.3795	0.026576	0.064197	0.099283	1159.5	-0.41295	129.38	197.97
230.74	0.10000	0.054083	18.490	21.326	23.175	0.10803	0.053648	0.063403	218.35	36.423	11.542	6.3010
300.00	0.10000	0.040726	24.554	25.482	27.937	0.12602	0.065918	0.074796	249.37	16.357	18.513	8.1962
400.00	0.10000	0.030257	33.050	33.084	36.389	0.15020	0.085775	0.094327	286.19	7.1977	30.995	10.819
500.00	0.10000	0.024125	41.452	42.616	46.761	0.17327	0.10431	0.11276	318.30	3.9113	46.363	13.290
100.00	1.0000	16.295	0.061368	-7.4410	-7.3796	-0.048197	0.058391	0.083810	2034.6	-0.62871	203.49	3822.7
200.00	1.0000	13.974	0.071560	1.3862	1.4577	0.012688	0.061383	0.093756	1366.3	-0.49871	148.40	290.36
300.00	1.0000	11.101	0.090085	11.833	11.924	0.054770	0.074077	0.12118	707.81	0.017030	92.871	95.307
300.09	1.0000	11.097	0.090112	11.844	11.934	0.054806	0.074092	0.12123	707.18	0.018141	92.828	95.215

TABLE 2-248 Thermodynamic Properties of Propane (Continued)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Single-Phase Properties												
300.09	1.0000	0.49153	2.0344	24.559	26.593	0.10366	0.070157	0.090035	214.68	21.259	19.249	8.3433
400.00	1.0000	0.32145	3.1109	32.638	35.749	0.12993	0.086666	0.097825	272.57	7.6356	31.915	10.996
500.00	1.0000	0.24776	4.0362	42.322	46.359	0.15353	0.10468	0.11438	311.60	3.9684	47.555	13.477
100.00	5.0000	16.326	0.061251	-7.4751	-7.1688	-0.048542	0.058457	0.083721	2051.3	-0.62944	204.51	3995.6
200.00	5.0000	14.047	0.071191	1.2900	1.6459	0.012202	0.061542	0.093272	1394.2	-0.50758	150.70	300.33
300.00	5.0000	11.363	0.088002	11.506	11.946	0.053658	0.074005	0.11643	779.30	-0.10225	97.139	103.08
400.00	5.0000	2.5448	0.39296	29.499	31.463	0.10822	0.093912	0.16341	197.49	10.349	42.438	14.948
500.00	5.0000	1.4059	0.71129	40.868	44.425	0.13725	0.10638	0.12446	284.88	4.1092	52.411	15.207
100.00	10.000	16.365	0.061108	-7.5164	-6.9053	-0.048966	0.058554	0.083620	2071.2	-0.63025	205.77	4219.8
200.00	10.000	14.133	0.070758	1.1767	1.8843	0.011619	0.061744	0.092746	1427.0	-0.51720	153.49	312.82
300.00	10.000	11.627	0.086007	11.173	12.033	0.052499	0.074065	0.11280	851.72	-0.19531	101.80	111.73
400.00	10.000	7.5861	0.13182	24.108	25.426	0.090675	0.093138	0.16714	338.94	1.7653	68.250	41.091
500.00	10.000	3.2486	0.30783	38.754	41.832	0.12746	0.10818	0.14277	275.78	3.4683	58.851	19.828
100.00	50.000	16.644	0.060081	-7.8016	-4.7976	-0.052119	0.059685	0.083087	2203.4	-0.63391	214.99	6389.5
200.00	50.000	14.696	0.068047	0.46479	3.8672	0.0076714	0.063347	0.090345	1638.0	-0.56103	173.15	416.61
300.00	50.000	12.850	0.077822	9.6226	13.514	0.046601	0.075522	0.10376	1203.5	-0.42987	129.30	167.38
400.00	50.000	11.043	0.090552	20.208	24.735	0.078767	0.092492	0.12069	901.34	-0.26933	103.15	95.902
500.00	50.000	9.3527	0.10692	32.246	37.592	0.10739	0.10930	0.13596	721.97	-0.11844	94.269	63.347
100.00	100.00	16.946	0.059010	-8.0684	-2.1674	-0.055583	0.061388	0.082816	2328.3	-0.63418	225.01	10396.
200.00	100.00	15.221	0.065700	-0.13496	6.4350	0.0038087	0.065107	0.089250	1833.0	-0.58053	193.78	561.59
300.00	100.00	13.700	0.072994	8.6070	15.906	0.042047	0.077244	0.10131	1467.7	-0.49303	155.37	229.66
400.00	100.00	12.331	0.081095	18.712	26.822	0.073340	0.094108	0.11714	1214.6	-0.40159	129.74	138.80
500.00	100.00	11.114	0.089974	30.308	39.306	0.10113	0.11078	0.13224	1052.7	-0.33205	118.57	98.032

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Buecker, D., and Wagner, W., "Reference Equations of State for the Thermodynamic Properties of Fluid Phase *n*-Butane and Isobutane," *J. Phys. Chem. Ref. Data* **35**(2):929–1019, 2006. The source for viscosity is Vogel, E., Kuechenmeister, C., Bich, E., and Laesecke, A., "Reference Correlation of the Viscosity of Propane," *J. Phys. Chem. Ref. Data* **27**(5):947–970, 1998. The source for thermal conductivity is Marsh, K., Perkins, R., and Ramires, M. L. V., "Measurement and Correlation of the Thermal Conductivity of Propane from 86 to 600 K at Pressures to 70 MPa," *J. Chem. Eng. Data* **47**(4):932–940, 2002.

Properties at the critical point temperature are given in the last entry of the saturation tables. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

Typical uncertainties in density are 0.02% in the liquid phase, 0.05% in the vapor phase and at supercritical temperatures, and 0.1% in the critical region, except very near the critical point, where the uncertainty in pressure is 0.1%. For vapor pressures, the uncertainty is 0.02% above 180 K, 0.05% above 1 Pa (115 K), and dropping to 0.001 mPa at the triple point. The uncertainty in heat capacity (isobaric, isochoric, and saturated) is 0.5% at temperatures above 125 K and 2% at temperatures below 125 K for the liquid, and is 0.5% for all vapor states. The uncertainty in the liquid-phase speed of sound is 0.5%, and that for the vapor phase is 0.05%. The uncertainties are higher for all properties very near the critical point except pressure (saturated vapor/liquid and single phase). The uncertainty in viscosity varies from 0.4% in the dilute gas between room temperature and 600 K, to about 2.5% from 100 to 475 K up to about 30 MPa, and to about 4% outside this range. Uncertainty in thermal conductivity is 3%, except in the critical region and dilute gas which have an uncertainty of 5%.

TABLE 2-249 Thermodynamic Properties of Propylene

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
105.00	1.8242E-07	17.799	0.056183	-6.2877	-6.2877	-0.039412	0.047146	0.074152	1984.2	-0.63335
120.00	4.8828E-06	17.383	0.057526	-5.1660	-5.1660	-0.029431	0.048479	0.076350	1860.6	-0.61019
135.00	5.9371E-05	16.973	0.058917	-3.9917	-3.9917	-0.020215	0.051603	0.080066	1740.6	-0.57672
150.00	0.00041697	16.567	0.060361	-2.7699	-2.7699	-0.011635	0.053564	0.082659	1637.2	-0.55180
165.00	0.0019723	16.162	0.061873	-1.5160	-1.5159	-0.0036693	0.054606	0.084443	1542.5	-0.53067
180.00	0.0069671	15.755	0.063470	-0.23811	-0.23766	0.0037429	0.055253	0.085952	1450.8	-0.50839
195.00	0.019747	15.343	0.065175	1.0625	1.0638	0.010683	0.055851	0.087538	1359.1	-0.48173
210.00	0.047272	14.922	0.067016	2.3884	2.3916	0.017233	0.056582	0.089404	1266.3	-0.44843
225.00	0.099223	14.487	0.069027	3.7442	3.7510	0.023468	0.057524	0.091663	1171.8	-0.40645
240.00	0.18775	14.035	0.071251	5.1356	5.1490	0.029455	0.058697	0.094401	1075.9	-0.35329
255.00	0.32701	13.559	0.073750	6.5692	6.5933	0.035251	0.060098	0.097711	978.69	-0.28530
270.00	0.53269	13.054	0.076607	8.0523	8.0931	0.040906	0.061715	0.10174	880.57	-0.19657
285.00	0.82165	12.509	0.079945	9.5938	9.6595	0.046469	0.063540	0.10676	781.63	-0.077005
300.00	1.2118	11.911	0.083956	11.205	11.307	0.051993	0.065577	0.11327	681.71	0.092061
315.00	1.7225	11.240	0.088971	12.904	13.057	0.057541	0.067855	0.12246	580.13	0.34911
330.00	2.3751	10.455	0.095644	14.722	14.949	0.063218	0.070450	0.13754	475.08	0.78805
345.00	3.1956	9.4631	0.10567	16.734	17.071	0.069261	0.073575	0.17154	362.18	1.7115
360.00	4.2202	7.8236	0.12782	19.290	19.829	0.076741	0.078178	0.39594	229.03	4.9073
365.57	4.6646	5.3086	0.18837	21.862	22.741	0.084569			0	12.042
105.00	1.8242E-07	2.0896E-07	4,785,700.	16.041	16.914	0.18156	0.031371	0.039685	162.00	260.32
120.00	4.8828E-06	4.8940E-06	204,330.	16.525	17.522	0.15964	0.033039	0.041354	172.27	186.64
135.00	5.9371E-05	5.2900E-05	18,904.	17.032	18.154	0.14383	0.034573	0.042891	181.89	139.37
150.00	0.00041697	0.00033450	2,989.5	17.560	18.807	0.13221	0.036090	0.044423	190.90	107.24
165.00	0.0019723	0.0014403	694.32	18.109	19.479	0.12357	0.037686	0.046064	199.27	84.470
180.00	0.0069671	0.0046776	213.78	18.676	20.166	0.11710	0.039423	0.047910	206.90	67.837
195.00	0.019747	0.012310	81.237	19.258	20.862	0.11221	0.041343	0.050046	213.67	55.441
210.00	0.047272	0.027633	36.188	19.850	21.560	0.10851	0.043468	0.052548	219.38	46.072
225.00	0.099223	0.054946	18.200	20.446	22.252	0.10570	0.045811	0.055492	223.85	38.921
240.00	0.18775	0.099531	10.047	21.042	22.928	0.10354	0.048376	0.058970	226.88	33.424
255.00	0.32701	0.16779	5.9598	21.630	23.579	0.10186	0.051168	0.063112	228.25	29.182
270.00	0.53269	0.26768	3.7358	22.202	24.192	0.10053	0.054191	0.068131	227.73	25.909
285.00	0.82165	0.40967	2.4410	22.749	24.754	0.099433	0.057457	0.074408	225.06	23.398
300.00	1.2118	0.60872	1.6428	23.254	25.245	0.098452	0.060989	0.082685	219.92	21.502
315.00	1.7225	0.88863	1.1253	23.693	25.632	0.097460	0.064833	0.094593	211.93	20.107
330.00	2.3751	1.2929	0.77343	24.022	25.859	0.096280	0.069077	0.11448	200.59	19.116
345.00	3.1956	1.9214	0.52045	24.137	25.801	0.094563	0.073899	0.15900	185.21	18.381
360.00	4.2202	3.1801	0.31446	23.631	24.958	0.090988	0.079791	0.42332	164.25	17.115
365.57	4.6646	5.3086	0.18837	21.862	22.741	0.084569			0	12.042
Single-Phase Properties										
100.00	0.10000	17.939	0.055745	-6.6629	-6.6573	-0.043075	0.049534	0.076050	2009.3	-0.61954
200.00	0.10000	15.206	0.065766	1.4997	1.5063	0.012896	0.056075	0.088113	1328.9	-0.47165
225.17	0.10000	14.482	0.069051	3.7598	3.7667	0.023538	0.057536	0.091692	1170.7	-0.40591
225.17	0.10000	0.055346	18.068	20.453	22.260	0.10567	0.045839	0.055529	223.89	38.850
300.00	0.10000	0.040648	24.601	24.309	26.769	0.12289	0.056666	0.065495	258.13	16.866
400.00	0.10000	0.030232	33.078	30.772	34.080	0.14382	0.072240	0.080757	295.63	7.2207
500.00	0.10000	0.024118	41.463	38.745	42.891	0.16342	0.086809	0.095223	328.33	3.7462
600.00	0.10000	0.020074	49.815	48.088	53.070	0.18194	0.099688	0.10806	357.96	2.2036
100.00	1.0000	17.947	0.055720	-6.6707	-6.6149	-0.043152	0.049534	0.076021	2014.7	-0.61988
200.00	1.0000	15.224	0.065684	1.4781	1.5437	0.012788	0.056080	0.087997	1335.9	-0.47384
292.39	1.0000	12.222	0.081820	10.378	10.460	0.049191	0.064516	0.10973	732.56	-0.0016360
292.39	1.0000	0.49938	2.0025	23.004	25.006	0.098942	0.059161	0.078164	222.86	22.395
300.00	1.0000	0.47670	2.0978	23.500	25.598	0.10094	0.059932	0.077330	228.89	20.287

TABLE 2-249 Thermodynamic Properties of Propylene (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Single-Phase Properties										
400.00	1.0000	0.31858	3.1389	30.390	33.529	0.12371	0.073331	0.084055	283.73	7.6465
500.00	1.0000	0.24704	4.0480	38.516	42.564	0.14381	0.087329	0.096722	322.07	3.8350
600.00	1.0000	0.20332	4.9183	47.935	52.854	0.16254	0.099980	0.10888	354.28	2.2229
200.00	5.0000	15.306	0.065334	1.3850	1.7117	0.012317	0.056114	0.087522	1365.9	-0.48279
300.00	5.0000	12.206	0.081924	10.889	11.299	0.050918	0.065369	0.10822	752.67	-0.041773
400.00	5.0000	2.2829	0.43803	27.896	30.087	0.10377	0.080080	0.12610	221.38	9.5490
500.00	5.0000	1.3832	0.72298	37.351	40.966	0.12811	0.089729	0.10566	297.48	4.0173
600.00	5.0000	1.0713	0.93348	47.197	51.865	0.14796	0.10124	0.11304	342.05	2.2088
200.00	10.000	15.402	0.064925	1.2752	1.9245	0.011753	0.056174	0.087009	1401.2	-0.49245
300.00	10.000	12.515	0.079905	10.553	11.352	0.049747	0.065258	0.10422	828.84	-0.15061
400.00	10.000	7.5283	0.13283	22.740	24.068	0.085918	0.081711	0.16868	302.55	2.6128
500.00	10.000	3.1462	0.31784	35.603	38.781	0.11902	0.092261	0.12166	287.51	3.5544
600.00	10.000	2.2419	0.44605	46.180	50.641	0.14065	0.10253	0.11891	340.40	1.9833
200.00	100.00	16.616	0.060184	-0.021002	5.9974	0.0040990	0.058338	0.083715	1815.4	-0.55416
300.00	100.00	14.839	0.067390	8.0040	14.743	0.039443	0.067573	0.092245	1457.4	-0.48492
400.00	100.00	13.274	0.075333	17.017	24.550	0.067574	0.081108	0.10402	1192.7	-0.40335
500.00	100.00	11.892	0.084087	27.127	35.536	0.092040	0.094320	0.11555	1020.4	-0.33132
600.00	100.00	10.680	0.093635	38.256	47.620	0.11404	0.10601	0.12591	913.65	-0.27115
200.00	200.00	17.488	0.057182	-0.77627	10.660	-0.0018754	0.060831	0.083362	2084.9	-0.55991
300.00	200.00	16.028	0.062389	6.8870	19.365	0.033304	0.070484	0.091727	1796.1	-0.51205
400.00	200.00	14.825	0.067452	15.619	29.110	0.061259	0.084735	0.10329	1568.7	-0.45996
500.00	200.00	13.819	0.072363	25.539	40.011	0.085539	0.098618	0.11458	1403.8	-0.42037
600.00	200.00	12.963	0.077142	36.558	51.987	0.10734	0.11085	0.12471	1282.8	-0.39117

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Angus, S., Armstrong, B., and de Reuck, K. M., "International Thermodynamic Tables of the Fluid State—7 Propylene," *International Union of Pure and Applied Chemistry*, Pergamon Press, Oxford, 1980. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties of the equation of state are generally 0.1% in density (except in the critical region), 1% in the heat capacity in the vapor phase, and 2–5% in the heat capacity in the liquid phase.

TABLE 2-250 Thermodynamic Properties of R-11, Trichlorofluoromethane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
162.68	6.5101E-06	12.874	0.077673	15.035	15.035	0.079300	0.074722	0.10506	1224.2	-0.60355
165.00	9.0030E-06	12.841	0.077878	15.280	15.280	0.080791	0.074630	0.10557	1221.1	-0.59915
180.00	5.8433E-05	12.618	0.079249	16.886	16.886	0.090108	0.075084	0.10851	1178.8	-0.57445
195.00	0.00027433	12.392	0.080694	18.531	18.531	0.098885	0.076137	0.11069	1119.8	-0.55514
210.00	0.0010018	12.164	0.082210	20.204	20.204	0.10715	0.077185	0.11231	1057.5	-0.53850
225.00	0.0030014	11.933	0.083801	21.899	21.899	0.11494	0.078127	0.11364	996.42	-0.52214
240.00	0.0076770	11.699	0.085475	23.613	23.614	0.12232	0.078986	0.11489	937.75	-0.50429
255.00	0.017281	11.462	0.087246	25.346	25.347	0.12932	0.079799	0.11621	881.67	-0.48363
270.00	0.035048	11.219	0.089131	27.099	27.102	0.13600	0.080594	0.11767	827.91	-0.45904
285.00	0.065240	10.971	0.091150	28.875	28.881	0.14240	0.081391	0.11933	776.09	-0.42946
300.00	0.11311	10.715	0.093329	30.677	30.687	0.14856	0.082199	0.12124	725.77	-0.39364
315.00	0.18478	10.449	0.095702	32.508	32.525	0.15452	0.083025	0.12343	676.56	-0.34999
330.00	0.28718	10.172	0.098308	34.371	34.399	0.16030	0.083873	0.12595	628.04	-0.29629
345.00	0.42787	9.8812	0.10120	36.271	36.315	0.16594	0.084749	0.12888	579.81	-0.22926
360.00	0.61496	9.5733	0.10446	38.213	38.277	0.17145	0.085663	0.13232	531.47	-0.14389
375.00	0.85703	9.2442	0.10818	40.203	40.295	0.17687	0.086633	0.13648	482.57	-0.032121
390.00	1.1631	8.8882	0.11251	42.247	42.378	0.18223	0.087686	0.14171	432.58	0.11986
405.00	1.5427	8.4966	0.11769	44.359	44.541	0.18756	0.088870	0.14871	380.86	0.33760
420.00	2.0061	8.0545	0.12415	46.558	46.808	0.19292	0.090270	0.15899	326.51	0.67402
435.00	2.5648	7.5340	0.13273	48.882	49.223	0.19840	0.092058	0.17674	268.20	1.2582
450.00	3.2329	6.8663	0.14564	51.423	51.894	0.20422	0.094658	0.21832	203.77	2.4973
465.00	4.0318	5.7930	0.17262	54.535	55.231	0.21124	0.099494	0.43941	132.04	6.3140
471.11	4.4076	4.0330	0.24796	57.789	58.881	0.21886			0	15.427
162.68	6.5101E-06	4.8133E-06	207.760	44.924	46.276	0.27134	0.049536	0.057858	107.23	1885.7
165.00	9.0030E-06	6.5631E-06	152.370	45.039	46.411	0.26946	0.050011	0.058335	107.92	1701.5
180.00	5.8433E-05	3.9056E-05	25.605	45.809	47.305	0.25910	0.052998	0.061344	112.26	905.28
195.00	0.00027433	0.00016934	5.905.3	46.618	48.238	0.25123	0.055869	0.064262	116.42	508.88
210.00	0.0010018	0.00057483	1,739.7	47.463	49.206	0.24525	0.058662	0.067140	120.39	301.60
225.00	0.0030014	0.0016103	621.01	48.339	50.203	0.24074	0.061395	0.070004	124.15	188.62
240.00	0.0076770	0.0038729	258.21	49.242	51.224	0.23736	0.064063	0.072855	127.67	124.75
255.00	0.017281	0.0082428	121.32	50.168	52.265	0.23488	0.066646	0.075684	130.89	87.418
270.00	0.035048	0.015895	62.911	51.115	53.320	0.23310	0.069122	0.078487	133.75	64.864
285.00	0.065240	0.028297	35.340	52.075	54.381	0.23188	0.071480	0.081281	136.17	50.774
300.00	0.11311	0.047202	21.186	53.045	55.441	0.23108	0.073723	0.084113	138.07	41.663
315.00	0.18478	0.074674	13.391	54.015	56.490	0.23060	0.075868	0.087058	139.37	35.564
330.00	0.28718	0.11315	8.8379	54.979	57.517	0.23036	0.077940	0.090222	140.01	31.344
345.00	0.42787	0.16555	6.0405	55.928	58.512	0.23028	0.079972	0.093753	139.89	28.342
360.00	0.61496	0.23549	4.2464	56.851	59.462	0.23030	0.081998	0.097861	138.95	26.165
375.00	0.85703	0.32766	3.0519	57.736	60.352	0.23035	0.084057	0.10286	137.09	24.577
390.00	1.1631	0.44843	2.2300	58.569	61.163	0.23039	0.086191	0.10929	134.19	23.443
405.00	1.5427	0.60714	1.6471	59.331	61.872	0.23035	0.088457	0.11813	130.10	22.687
420.00	2.0061	0.81886	1.2212	59.992	62.442	0.23014	0.090937	0.13148	124.58	22.277
435.00	2.5648	1.1115	0.89966	60.498	62.805	0.22962	0.093775	0.15496	117.29	22.202
450.00	3.2329	1.5503	0.64504	60.727	62.813	0.22849	0.097282	0.20995	107.74	22.379
465.00	4.0318	2.3952	0.41751	60.210	61.894	0.22557	0.10241	0.51972	95.245	21.779
471.11	4.4076	4.0330	0.24796	57.789	58.881	0.21886			0	15.427
Single-Phase Properties										
200.00	0.10000	12.317	0.081186	19.084	19.092	0.10168	0.076484	0.11128	1099.8	-0.54951
296.49	0.10000	10.776	0.092803	30.252	30.262	0.14714	0.082009	0.12077	737.44	-0.40266
296.49	0.10000	0.042088	23.760	52.817	55.193	0.23123	0.073207	0.083443	137.67	43.472
300.00	0.10000	0.041523	24.083	53.078	55.486	0.23221	0.073294	0.083384	138.61	41.094
400.00	0.10000	0.030462	32.827	60.756	64.039	0.25678	0.079843	0.088628	161.80	15.373
500.00	0.10000	0.024206	41.312	69.066	73.197	0.27719	0.085650	0.094219	181.31	9.4764
600.00	0.10000	0.020110	49.728	77.853	82.825	0.29474	0.089636	0.098111	198.74	6.5190

TABLE 2-250 Thermodynamic Properties of R-11, Trichlorofluoromethane (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_v kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Single-Phase Properties										
200.00	1.0000	12.325	0.081134	19.066	19.147	0.10159	0.076374	0.11122	1106.2	-0.55026
300.00	1.0000	10.733	0.093174	30.637	30.730	0.14843	0.082161	0.12101	731.85	-0.39806
382.43	1.0000	9.0718	0.11023	41.208	41.318	0.17953	0.087142	0.13891	457.99	0.036901
382.43	1.0000	0.38339	2.6083	58.156	60.765	0.23038	0.085101	0.10582	135.79	23.965
400.00	1.0000	0.35322	2.8311	59.743	62.574	0.23501	0.083835	0.10080	143.19	19.717
500.00	1.0000	0.25738	3.8853	68.469	72.354	0.25684	0.086291	0.097662	172.67	10.081
600.00	1.0000	0.20723	4.8256	77.413	82.239	0.27486	0.090004	0.10006	194.29	6.6365
200.00	5.0000	12.359	0.080912	18.988	19.393	0.10120	0.075954	0.11098	1133.5	-0.55346
300.00	5.0000	10.810	0.092509	30.464	30.926	0.14785	0.082048	0.12006	757.72	-0.41608
400.00	5.0000	8.8674	0.11277	43.167	43.731	0.18456	0.087727	0.13909	448.07	0.070413
500.00	5.0000	2.1555	0.46393	64.141	66.460	0.23416	0.094709	0.18205	120.07	14.043
600.00	5.0000	1.2189	0.82040	75.225	79.327	0.25776	0.091726	0.11326	173.08	7.0186
200.00	10.000	12.399	0.080649	18.894	19.700	0.10072	0.075562	0.11072	1164.8	-0.55716
300.00	10.000	10.899	0.091755	30.263	31.181	0.14716	0.082018	0.11907	787.03	-0.43502
400.00	10.000	9.1187	0.10966	42.641	43.737	0.18319	0.087335	0.13358	500.70	-0.076836
500.00	10.000	6.3465	0.15757	56.993	58.568	0.21613	0.093373	0.17100	231.94	1.9364
600.00	10.000	3.0124	0.33196	71.945	75.265	0.24670	0.093817	0.13897	167.77	5.3068
200.00	20.000	12.475	0.080163	18.716	20.320	0.099796	0.075119	0.11027	1220.3	-0.56373
300.00	20.000	11.057	0.090438	29.905	31.714	0.14590	0.082196	0.11753	838.26	-0.46461
400.00	20.000	9.4912	0.10536	41.844	43.951	0.18104	0.087334	0.12773	578.74	-0.23372
500.00	20.000	7.6205	0.13123	54.651	57.275	0.21072	0.091707	0.13823	378.80	0.31683
600.00	20.000	5.5967	0.17868	67.668	71.241	0.23619	0.094708	0.13794	265.18	1.2664
200.00	30.000	12.543	0.079723	18.552	20.943	0.098918	0.075012	0.10990	1268.1	-0.56938
300.00	30.000	11.197	0.089311	29.591	32.270	0.14476	0.082559	0.11638	882.50	-0.48678
400.00	30.000	9.7739	0.10231	41.235	44.304	0.17932	0.087676	0.12450	638.80	-0.31979
500.00	30.000	8.2190	0.12167	53.466	57.116	0.20788	0.091885	0.13097	464.35	-0.028029
600.00	30.000	6.6706	0.14991	65.781	70.278	0.23188	0.095040	0.13106	357.26	0.34752

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Jacobsen, R. T., Penoncello, S. G., and Lemmon, E. W., "A Fundamental Equation for Trichlorofluoromethane (R-11)," *Fluid Phase Equilibria* **80**:45–56, 1992. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties of the equation of state are 0.1% in density for the liquid, and 0.25% for the vapor, 2% in heat capacity, and 1% in the speed of sound, except in the critical region. The uncertainty in vapor pressure is 0.2%.

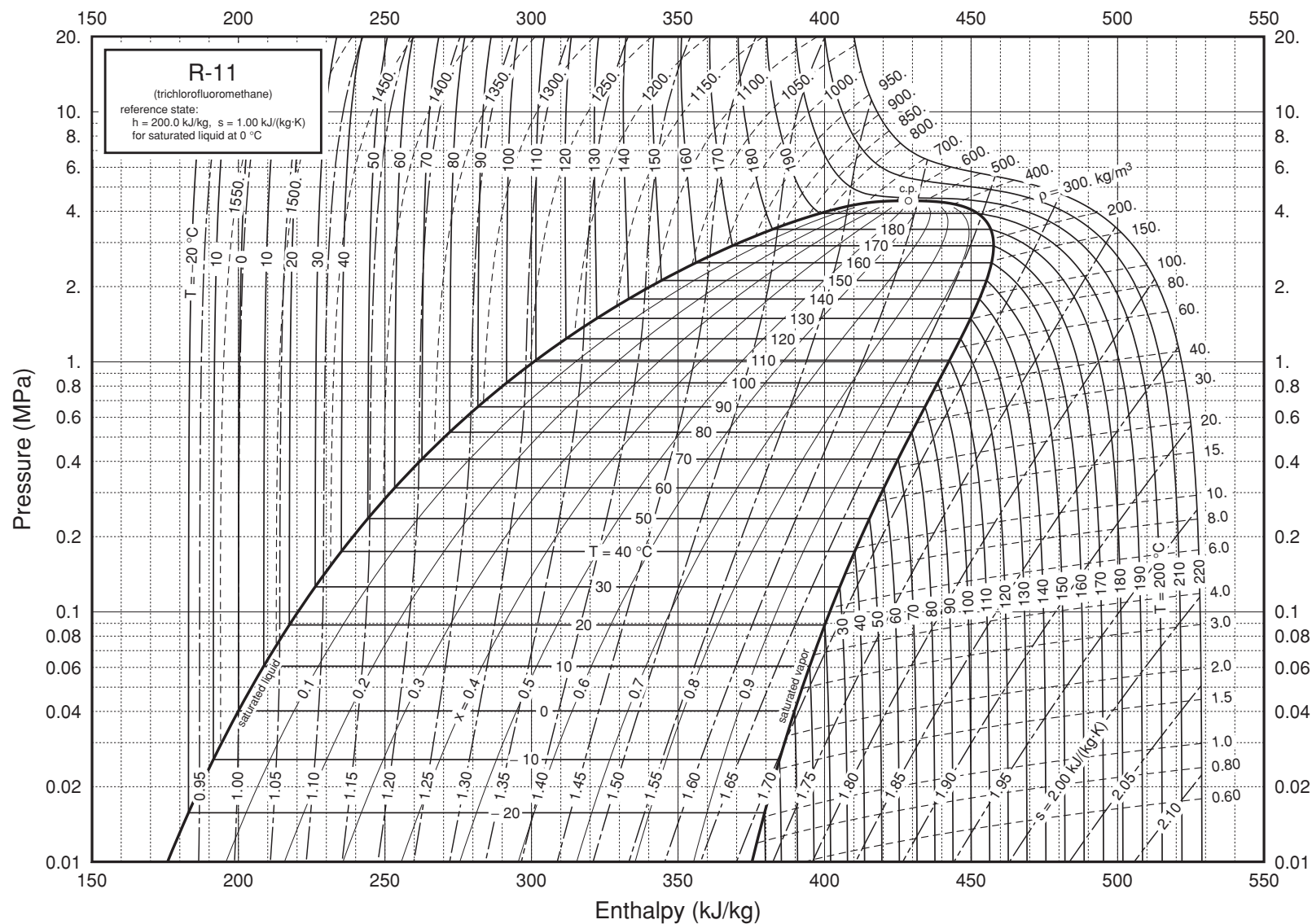


FIG. 2-18 Pressure-enthalpy diagram for Refrigerant 11. Properties computed with the NIST REFPROP Database, Version 7.0 (Lemmon, E. W., McLinden, M. O., and Huber, M. L., 2002, NIST Standard Reference Database 23, NIST Reference Fluid Thermodynamic and Transport Properties—REFPROP, Version 7.0, Standard Reference Data Program, National Institute of Standards and Technology), based on the equation of state of Jacobsen, R. T., Penoncello, S. G., and Lemmon, E. W., "A Fundamental Equation for Trichlorofluoromethane (R-11)," *Fluid Phase Equilibria* **80**:45–56, 1992.

TABLE 2-251 Thermodynamic Properties of R-12, Dichlorodifluoromethane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
116.10	2.4255E-07	15.125	0.066114	8.0204	8.0204	0.033619	0.069219	0.10352	1310.0	-0.53048
120.00	5.8432E-07	15.039	0.066492	8.4220	8.4220	0.037021	0.068095	0.10239	1290.2	-0.53554
135.00	1.0092E-05	14.712	0.067972	9.9342	9.9342	0.048899	0.065219	0.099608	1215.7	-0.54646
150.00	9.1829E-05	14.387	0.069505	11.419	11.419	0.059328	0.063937	0.098564	1143.1	-0.54643
165.00	0.00053030	14.064	0.071106	12.897	12.897	0.068717	0.063633	0.098622	1072.5	-0.53813
180.00	0.0021945	13.738	0.072790	14.381	14.381	0.077328	0.063942	0.099406	1003.9	-0.52327
195.00	0.0070692	13.409	0.074574	15.881	15.882	0.085332	0.064638	0.10070	937.07	-0.50270
210.00	0.018793	13.075	0.076482	17.403	17.405	0.092852	0.065582	0.10237	872.00	-0.47654
225.00	0.043015	12.733	0.078538	18.953	18.956	0.099978	0.066683	0.10436	808.46	-0.44424
240.00	0.087479	12.380	0.080776	20.533	20.540	0.10678	0.067882	0.10664	746.23	-0.40455
255.00	0.16186	12.013	0.083241	22.149	22.162	0.11331	0.069143	0.10925	685.03	-0.35528
270.00	0.27747	11.629	0.085990	23.803	23.827	0.11961	0.070443	0.11226	624.56	-0.29292
285.00	0.44693	11.223	0.089104	25.502	25.541	0.12574	0.071775	0.11579	564.41	-0.21173
300.00	0.68394	10.787	0.092703	27.250	27.314	0.13172	0.073152	0.12012	504.07	-0.10203
315.00	1.0032	10.312	0.096973	29.059	29.157	0.13762	0.074611	0.12572	442.82	0.053692
330.00	1.4203	9.7818	0.10223	30.943	31.088	0.14348	0.076246	0.13360	379.59	0.29048
345.00	1.9528	9.1671	0.10909	32.930	33.143	0.14940	0.078279	0.14625	312.77	0.68965
360.00	2.6209	8.4050	0.11898	35.078	35.390	0.15556	0.081286	0.17199	239.96	1.4874
375.00	3.4527	7.3032	0.13693	37.570	38.042	0.16248	0.087150	0.26475	158.26	3.7100
385.12	4.1362	4.6728	0.21401	41.164	42.049	0.17270			0	13.369
116.10	2.4255E-07	2.5127E-07	3,979,800.	33.279	34.244	0.25949	0.034581	0.042896	99.513	532.61
120.00	5.8432E-07	5.8564E-07	1,707,500.	33.415	34.413	0.25361	0.035365	0.043680	100.95	464.92
135.00	1.0092E-05	8.9910E-06	111,220.	33.968	35.091	0.23524	0.038409	0.046725	106.27	288.39
150.00	9.1829E-05	7.3643E-05	13,579.	34.567	35.814	0.22196	0.041427	0.049750	111.28	190.47
165.00	0.00053030	0.00038683	2,585.1	35.208	36.579	0.21224	0.044365	0.052712	116.02	132.65
180.00	0.0021945	0.0014695	680.49	35.888	37.381	0.20511	0.047211	0.055621	120.49	96.659
195.00	0.0070692	0.0043835	228.13	36.602	38.214	0.19986	0.049977	0.058515	124.63	73.304
210.00	0.018793	0.010883	91.884	37.342	39.069	0.19601	0.052681	0.061451	128.34	57.635
225.00	0.043015	0.023468	42.611	38.102	39.935	0.19322	0.055348	0.064497	131.52	46.836
240.00	0.087479	0.045368	22.042	38.873	40.801	0.19120	0.058002	0.067735	134.05	39.210
255.00	0.16186	0.080547	12.415	39.647	41.657	0.18976	0.060669	0.071271	135.80	33.704
270.00	0.27747	0.13381	7.4735	40.415	42.488	0.18873	0.063373	0.075247	136.64	29.653
285.00	0.44693	0.21108	4.7376	41.165	43.283	0.18799	0.066138	0.079875	136.46	26.640
300.00	0.68394	0.32006	3.1244	41.887	44.024	0.18742	0.068991	0.085504	135.10	24.405
315.00	1.0032	0.47149	2.1209	42.562	44.689	0.18693	0.071972	0.092789	132.40	22.790
330.00	1.4203	0.68182	1.4667	43.166	45.249	0.18639	0.075158	0.10314	128.14	21.703
345.00	1.9528	0.97985	1.0206	43.655	45.648	0.18565	0.078715	0.12016	122.00	21.085
360.00	2.6209	1.4275	0.70055	43.941	45.777	0.18441	0.083065	0.15653	113.54	20.831
375.00	3.4527	2.2257	0.44930	43.738	45.289	0.18181	0.089754	0.30373	102.14	20.225
385.12	4.1362	4.6728	0.21401	41.164	42.049	0.17270			0	13.369

Single-Phase Properties

125.00	0.10000	14.931	0.066977	8.9296	8.9363	0.041166	0.066915	0.10121	1265.3	-0.54064
225.00	0.10000	12.734	0.078531	18.951	18.959	0.099970	0.066686	0.10435	808.74	-0.44443
243.09	0.10000	12.306	0.081264	20.863	20.871	0.10814	0.068138	0.10715	733.55	-0.39528
243.09	0.10000	0.051383	19.462	39.033	40.979	0.19086	0.058549	0.068435	134.48	37.929
325.00	0.10000	0.037543	26.636	44.257	46.920	0.21187	0.067874	0.076721	156.65	16.166
425.00	0.10000	0.028469	35.126	51.527	55.039	0.23360	0.076661	0.085212	179.16	8.3282
525.00	0.10000	0.022973	43.529	59.527	63.880	0.25225	0.082795	0.091242	198.90	5.1283
125.00	1.0000	14.938	0.066945	8.9186	8.9856	0.041078	0.066988	0.10119	1267.0	-0.54091
225.00	1.0000	12.752	0.078422	18.922	19.001	0.099842	0.066739	0.10418	813.10	-0.44741
314.87	1.0000	10.316	0.096933	29.043	29.140	0.13757	0.074598	0.12566	443.35	0.052069
314.87	1.0000	0.46996	2.1279	42.556	44.684	0.18693	0.071945	0.092716	132.43	22.802
325.00	1.0000	0.44273	2.2587	43.348	45.606	0.18982	0.071678	0.089701	137.48	19.919
425.00	1.0000	0.30150	3.1617	51.055	54.372	0.21333	0.077527	0.088660	171.39	8.7630
525.00	1.0000	0.23568	4.2430	59.210	63.453	0.23250	0.083185	0.092926	195.20	5.1839
125.00	5.0000	14.968	0.066807	8.8706	9.2046	0.040691	0.067308	0.10110	1274.3	-0.54205
225.00	5.0000	12.827	0.077958	18.799	19.189	0.099289	0.066963	0.10352	831.96	-0.45954
325.00	5.0000	10.252	0.097546	29.898	30.385	0.14025	0.075544	0.12368	450.15	0.017671
425.00	5.0000	2.2637	0.44176	47.937	50.146	0.19225	0.083408	0.14054	131.42	10.771
525.00	5.0000	1.3295	0.75217	57.642	61.402	0.21615	0.085038	0.10337	180.93	5.2339
125.00	10.000	15.006	0.066639	8.8124	9.4788	0.040215	0.067691	0.10099	1283.4	-0.54334
225.00	10.000	12.917	0.077418	18.655	19.429	0.098631	0.067222	0.10281	854.41	-0.47253
325.00	10.000	10.544	0.094842	29.472	30.420	0.13887	0.075613	0.11858	501.05	-0.12043
425.00	10.000	6.6383	0.15064	42.586	44.093	0.17528	0.084440	0.16445	204.25	2.4031
525.00	10.000	3.0541	0.32743	55.399	58.673	0.20629	0.087008	0.12146	177.27	4.2532
225.00	100.00	13.992	0.071470	17.017	24.164	0.090059	0.069779	0.098455	1155.1	-0.55347
325.00	100.00	12.599	0.079369	26.406	34.343	0.12742	0.077370	0.10504	937.26	-0.49945
425.00	100.00	11.358	0.088042	36.302	45.106	0.15626	0.084005	0.10984	788.21	-0.45110
525.00	100.00	10.262	0.097444	46.498	56.243	0.17978	0.088827	0.11260	690.80	-0.41225
225.00	200.00	14.702	0.068017	16.054	29.657	0.083541	0.070631	0.097995	1397.8	-0.56210
325.00	200.00	13.559	0.073751	25.016	39.766	0.12065	0.077764	0.10414	1220.9	-0.52944
425.00	200.00	12.582	0.079477	34.535	50.431	0.14923	0.084238	0.10881	1091.3	-0.50685
525.00	200.00	11.738	0.085195	44.429	61.468	0.17254	0.088994	0.11168	999.41	-0.49445

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Marx, V., Pruss, A., and Wagner, W., "Neue Zustandsgleichungen fuer R 12, R 22, R 11 und R 113. Beschreibung des thermodynamischen Zustandsverhaltens bei Temperaturen bis 525 K und Druecken bis 200 MPa," Duesseldorf: *VDI Verlag*, Series 19 (Waermetechnik/Kaelte-technik), No. 57, 1992. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in density are 0.2% below the critical point temperature and increase to 1% in and above the critical region. The uncertainties for vapor pressures are 0.2% above 200 K and greater than 1% below 200 K. The uncertainties in heat capacities and sound speeds are 1% each.

FIG. 2-19 Pressure-enthalpy diagram for Refrigerant 12. Properties computed with the NIST REFPROP Database, Version 7.0 (Lemmon, E. W., McLinden, M. O., and Huber, M. L., 2002, NIST Standard Reference Database 23, NIST Reference Fluid Thermodynamic and Transport Properties—REFPROP, Version 7.0, Standard Reference Data Program, National Institute of Standards and Technology), based on the equation of state of Marx, V., Pruß, A., and Wagner, W., “Neue Zustandsgleichungen für R 12, R 22, R 11 und R 113. Beschreibung des thermodynamischen Zustandsverhaltens bei Temperaturen bis 525 K und Drücken bis 200 MPa,” VDI-Fortschritt-Ber. Series 19, No. 57, Düsseldorf: VDI Verlag, 1992.

TABLE 2-252 Thermodynamic Properties of R-13, Chlorotrifluoromethane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
92.000	3.2889E-07	17.840	0.056054	3.4760	3.4760	0.0038039	0.059797	0.077526	918.05	-0.61098
100.00	2.6891E-06	17.596	0.056830	4.1217	4.1217	0.010530	0.059177	0.083231	1032.1	-0.56374
110.00	2.3101E-05	17.288	0.057844	4.9721	4.9721	0.018633	0.057632	0.086329	1078.3	-0.53784
120.00	0.00013272	16.976	0.058907	5.8413	5.8413	0.026196	0.055848	0.087308	1072.1	-0.52512
130.00	0.00056195	16.660	0.060023	6.7158	6.7159	0.033195	0.054335	0.087538	1039.3	-0.51544
140.00	0.0018797	16.339	0.061202	7.5918	7.5919	0.039687	0.053311	0.087670	992.15	-0.50447
150.00	0.0052264	16.012	0.062451	8.4698	8.4701	0.045744	0.052818	0.087987	937.71	-0.49023
160.00	0.012543	15.679	0.063780	9.3523	9.3531	0.051439	0.052815	0.088595	879.99	-0.47186
170.00	0.026741	15.337	0.065200	10.242	10.244	0.056835	0.053227	0.089527	821.49	-0.44883
180.00	0.051763	14.987	0.066725	11.143	11.146	0.061982	0.053972	0.090788	763.71	-0.42058
190.00	0.092557	14.626	0.068370	12.057	12.063	0.066925	0.054972	0.092386	707.45	-0.38628
200.00	0.15496	14.254	0.070158	12.988	12.999	0.071701	0.056159	0.094338	653.03	-0.34462
210.00	0.24553	13.866	0.072118	13.939	13.956	0.076341	0.057477	0.096688	600.48	-0.29355
220.00	0.37145	13.461	0.074289	14.913	14.940	0.080874	0.058882	0.099513	549.57	-0.22995
230.00	0.54033	13.033	0.076727	15.914	15.955	0.085328	0.060340	0.10294	499.97	-0.14899
240.00	0.76025	12.577	0.079511	16.947	17.007	0.089732	0.061829	0.10719	451.26	-0.043013
250.00	1.0397	12.083	0.082762	18.018	18.104	0.094118	0.063338	0.11266	402.91	0.10085
260.00	1.3878	11.537	0.086674	19.137	19.258	0.098526	0.064870	0.12009	354.31	0.30583
270.00	1.8145	10.919	0.091587	20.319	20.485	0.10301	0.066451	0.13115	304.57	0.61895
280.00	2.3316	10.183	0.098198	21.591	21.820	0.10769	0.068161	0.15049	252.29	1.1529
290.00	2.9538	9.2260	0.10839	23.021	23.341	0.11280	0.070241	0.19837	194.59	2.2680
300.00	3.7065	7.4440	0.13434	25.005	25.503	0.11981	0.074099	0.74301	122.38	6.3031
302.00	3.8790	5.5800	0.17921	26.462	27.158	0.12522			0	11.123
92.000	3.2889E-07	4.2996E-07	2,325,800.	22.976	23.741	0.22408	0.026152	0.034466	98.239	846.97
100.00	2.6891E-06	3.2343E-06	309,190.	23.192	24.023	0.20954	0.027668	0.035983	101.74	607.63
110.00	2.3101E-05	2.5261E-05	39,587.	23.477	24.392	0.19518	0.029536	0.037856	105.92	414.42
120.00	0.00013272	0.00013307	7,514.8	23.781	24.778	0.18400	0.031384	0.039718	109.91	291.57
130.00	0.00056195	0.00052045	1,921.4	24.101	25.181	0.17523	0.033228	0.041601	113.70	210.73
140.00	0.0018797	0.0016191	617.64	24.435	25.596	0.16829	0.035092	0.043546	117.29	155.99
150.00	0.0052264	0.0042135	237.33	24.782	26.022	0.16276	0.037002	0.045603	120.64	118.01
160.00	0.012543	0.0095248	104.99	25.137	26.454	0.15832	0.038978	0.047816	123.73	91.120
170.00	0.026741	0.019245	51.961	25.499	26.888	0.15474	0.041033	0.050218	126.48	71.760
180.00	0.051763	0.035532	28.144	25.864	27.321	0.15184	0.043168	0.052835	128.85	57.628
190.00	0.092557	0.060985	16.398	26.231	27.749	0.14948	0.045375	0.055688	130.77	47.196
200.00	0.15496	0.098644	10.137	26.597	28.168	0.14755	0.047639	0.058800	132.18	39.425
210.00	0.24553	0.15204	6.5771	26.959	28.574	0.14595	0.049946	0.062214	133.00	33.599
220.00	0.37145	0.22536	4.4373	27.315	28.963	0.14462	0.052282	0.066006	133.16	29.214
230.00	0.54033	0.32372	3.0891	27.659	29.329	0.14347	0.054636	0.070318	132.58	25.919
240.00	0.76025	0.45374	2.2039	27.987	29.663	0.14246	0.057009	0.075407	131.16	23.463
250.00	1.0397	0.62459	1.6010	28.289	29.954	0.14152	0.059411	0.081759	128.80	21.667
260.00	1.3878	0.84999	1.1765	28.552	30.185	0.14055	0.061872	0.090357	125.37	20.399
270.00	1.8145	1.1527	0.86755	28.753	30.327	0.13947	0.064450	0.10345	120.73	19.555
280.00	2.3316	1.5762	0.63445	28.850	30.329	0.13808	0.067265	0.12745	114.75	19.007
290.00	2.9538	2.2289	0.44866	28.740	30.066	0.13599	0.070591	0.19031	107.22	18.455
300.00	3.7065	3.7166	0.26906	27.886	28.884	0.13108	0.075537	0.91719	98.019	15.921
302.00	3.8790	5.5800	0.17921	26.462	27.158	0.12522			0	11.123
Single-Phase Properties										
100.00	0.10000	17.598	0.056826	4.1207	4.1264	0.010520	0.059127	0.083227	1033.2	-0.56370
175.00	0.10000	15.165	0.065942	10.689	10.696	0.059427	0.053557	0.090103	792.98	-0.43564
191.43	0.10000	14.574	0.068616	12.189	12.196	0.067617	0.055131	0.092642	699.56	-0.38082
191.43	0.10000	0.065539	15.258	26.284	27.809	0.14918	0.045695	0.056115	131.00	45.943
250.00	0.10000	0.048881	20.458	29.189	31.235	0.16476	0.052574	0.061514	150.16	17.864
325.00	0.10000	0.037259	26.839	33.487	36.171	0.18198	0.061422	0.069986	170.52	8.3711
400.00	0.10000	0.030170	33.145	38.384	41.699	0.19726	0.068694	0.077154	188.46	5.2121

TABLE 2-252 Thermodynamic Properties of R-13, Chlorotrifluoromethane (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_v kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Single-Phase Properties										
100.00	1.0000	17.609	0.056790	4.1118	4.1686	0.010431	0.058689	0.083187	1042.6	-0.56328
175.00	1.0000	15.188	0.065843	10.666	10.731	0.059291	0.053474	0.089928	800.56	-0.43907
248.71	1.0000	12.149	0.082311	17.878	17.960	0.093552	0.063142	0.11187	409.14	0.079502
248.71	1.0000	0.59985	1.6671	28.252	29.919	0.14164	0.059099	0.080842	129.16	21.866
250.00	1.0000	0.59355	1.6848	28.338	30.023	0.14205	0.059028	0.080216	129.97	21.418
325.00	1.0000	0.39827	2.5108	33.099	35.610	0.16162	0.062521	0.073958	162.24	8.9819
400.00	1.0000	0.31126	3.2128	38.116	41.329	0.17744	0.068870	0.078791	184.36	5.3350
100.00	5.0000	17.658	0.056633	4.0724	4.3555	0.010032	0.056929	0.083080	1081.6	-0.56120
175.00	5.0000	15.284	0.065427	10.564	10.891	0.058704	0.053220	0.089221	831.21	-0.45278
250.00	5.0000	12.436	0.080411	17.696	18.098	0.092801	0.062047	0.10644	461.89	-0.057256
325.00	5.0000	3.5611	0.28081	29.869	31.273	0.13759	0.071896	0.18862	120.24	10.597
400.00	5.0000	1.8042	0.55425	36.726	39.497	0.16061	0.070112	0.090875	170.26	5.4004
100.00	10.000	17.716	0.056446	4.0235	4.5880	0.0095294	0.055111	0.083086	1124.6	-0.55799
175.00	10.000	15.396	0.064954	10.446	11.096	0.058009	0.053119	0.088480	863.64	-0.46709
250.00	10.000	12.772	0.078295	17.377	18.160	0.091462	0.061300	0.10203	518.33	-0.17104
325.00	10.000	8.8242	0.11332	25.758	26.891	0.12181	0.069005	0.13617	240.23	1.3924
400.00	10.000	4.2213	0.23689	34.655	37.023	0.14997	0.071938	0.11434	173.36	3.9992
100.00	15.000	17.772	0.056268	3.9751	4.8192	0.0090235	0.053642	0.083232	1162.0	-0.55407
175.00	15.000	15.499	0.064521	10.336	11.304	0.057352	0.053200	0.087860	890.77	-0.47907
250.00	15.000	13.041	0.076680	17.115	18.265	0.090334	0.061058	0.099258	562.82	-0.24338
325.00	15.000	9.9265	0.10074	24.846	26.357	0.11854	0.068084	0.11665	322.91	0.49723
400.00	15.000	6.3827	0.15667	32.962	35.312	0.14333	0.072431	0.11694	213.68	1.9494
175.00	35.000	15.857	0.063065	9.9615	12.169	0.055005	0.054657	0.086150	962.40	-0.51269
250.00	35.000	13.811	0.072404	16.358	18.892	0.086896	0.062146	0.093881	679.17	-0.38660
325.00	35.000	11.718	0.085341	23.258	26.245	0.11257	0.069359	0.10153	492.75	-0.18758
400.00	35.000	9.7082	0.10301	30.364	33.969	0.13395	0.074599	0.10349	383.00	0.022671

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Magee, J. W., Outcalt, S. L., and Ely, J. F., "Molar Heat Capacity $C(v)$, Vapor Pressure, and (p, Rho, T) Measurements from 92 to 350 K at Pressures to 35 MPa and a New Equation of State for Chlorotrifluoromethane (R13)," *Int. J. Thermophys.* **21**(5):1097–1121, 2000. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties of the equation of state are 0.15% in density and 2% in heat capacity, except in the critical region. The uncertainty in vapor pressure is 0.1%.

TABLE 2-253 Saturated Refrigerant 13B1, Bromotrifluoromethane*

T, K	P, bar	$v_f, \text{m}^3/\text{kg}$	$v_g, \text{m}^3/\text{kg}$	$h_f, \text{kJ/kg}$	$h_g, \text{kJ/kg}$	$s_f, \text{kJ}/(\text{kg}\cdot\text{K})$	$s_g, \text{kJ}/(\text{kg}\cdot\text{K})$	$c_{pf}, \text{kJ}/(\text{kg}\cdot\text{K})$	$\mu_f, 10^{-4} \text{ Pa}\cdot\text{s}$	$k_f, \text{W}/(\text{m}\cdot\text{K})$
170	0.059	4.594-4	1.6015	-40.90	90.95	-0.2033	0.5723	0.597	9.54	0.101
180	0.127	4.677-4	0.7840	-34.75	94.37	-0.1682	0.5491	0.618	7.60	0.096
190	0.250	4.765-4	0.4190	-28.51	97.83	-0.1345	0.5305	0.634	6.20	0.091
200	0.455	4.860-4	0.2407	-22.17	101.32	-0.1020	0.5154	0.648	5.13	0.086
210	0.777	4.961-4	0.1467	-15.68	104.82	-0.0704	0.5033	0.663	4.33	0.082
215.4	1.013	5.020-4	0.1147	-12.09	106.70	-0.0536	0.4978	0.670	3.97	0.079
220	1.254	5.071-4	0.0940	-9.02	108.28	-0.0396	0.4936	0.676	3.71	0.077
230	1.933	5.190-4	0.0628	-2.19	111.68	-0.0094	0.4857	0.690	3.22	0.073
240	2.863	5.321-4	0.0433	4.83	114.99	0.0202	0.4793	0.703	2.83	0.068
250	4.096	5.466-4	0.0308	12.03	118.16	0.0494	0.4739	0.721	2.51	0.063
260	5.690	5.627-4	0.0224	19.44	121.16	0.0781	0.4693	0.742	2.25	0.059
270	7.703	5.809-4	0.0166	27.06	123.93	0.1064	0.4652	0.767	2.04	0.054
280	10.20	6.018-4	0.0124	34.94	126.41	0.1345	0.4612	0.800	1.84	0.049
290	13.25	6.264-4	0.0094	43.11	128.51	0.1625	0.4570	0.842	1.69	0.045
300	16.91	6.562-4	0.0072	51.68	130.09	0.1908	0.4522	0.891	1.57	0.040
310	21.28	6.940-4	0.0055	60.81	130.97	0.2197	0.4460	0.951	1.45	0.035
320	26.44	7.458-4	0.0041	70.80	130.76	0.2503	0.4376	1.09	1.26	0.030
330	32.48	8.295-4	0.0030	82.42	128.59	0.2845	0.4245	1.29	0.99	0.026
340.2°	39.64	1.344-3	0.0013	108.70	108.70	0.3605	0.3605	∞	0.35	∞

*Values reproduced or converted from Table 4, p. 17.83, *ASHRAE Handbook, 1981: Fundamentals*, American Society of Heating, Refrigerating and Air-Conditioning Engineers, Atlanta, 1981. Copyright material. Reproduced by permission of the copyright owner. c = critical point. The notation 4.594-4 signifies 4.594×10^{-4} .

The 1993 *ASHRAE Handbook—Fundamentals* (SI ed.) contains a table at closer temperature increments and also an enthalpy-log-pressure diagram from 0.1 to 35 bar, -80 to 220 °C. For tables and a chart to 500 psia, 480 °F, see Stewart, R. B., R. T. Jacobsen, et al., *Thermodynamic Properties of Refrigerants*, ASHRAE, Atlanta, GA, 1986 (521 pp.). For specific heat, thermal conductivity, and viscosity, see *Thermophysical Properties of Refrigerants*, ASHRAE, 1993.

Refrigerant 14 (tetrafluoromethane) See Carbon Tetrafluoride (Table 2-202).

Refrigerant 20 See Chloroform (Table 2-206).

TABLE 2-254 Saturated Refrigerant 21, Dichlorofluoromethane

Temperature, K	Pressure, bar	$v_f, \text{m}^3/\text{kg}$	$v_g, \text{m}^3/\text{kg}$	$h_f, \text{kJ/kg}$	$h_g, \text{kJ/kg}$	$s_f, \text{kJ}/(\text{kg}\cdot\text{K})$	$s_g, \text{kJ}/(\text{kg}\cdot\text{K})$
250	0.2415	0.000 677	0.8292	16.6	274.8	0.0687	1.1015
260	0.3953	0.000 687	0.5247	26.5	279.9	0.1076	1.0820
270	0.6200	0.000 698	0.3455	36.6	284.9	0.1454	1.0653
280	0.9364	0.000 709	0.2355	46.7	290.0	0.1824	1.0511
290	1.3682	0.000 722	0.1654	57.1	295.0	0.2186	1.0389
300	1.9417	0.000 735	0.1192	67.7	300.0	0.2543	1.0286
310	2.6849	0.000 748	0.0879	78.4	304.8	0.2894	1.0196
320	3.6279	0.000 763	0.0661	89.5	309.5	0.3242	1.0119
330	4.8022	0.000 778	0.0505	100.7	314.1	0.3586	1.0051
340	6.2409	0.000 794	0.0391	112.3	318.4	0.3927	0.9989
350	7.978	0.000 812	0.0307	124.1	322.4	0.4266	0.9932
360	10.049	0.000 830	0.0243	136.2	326.1	0.4602	0.9877
370	12.489	0.000 850	0.0194	148.6	329.3	0.4935	0.9820
380	15.337	0.000 870	0.0155	161.2	331.9	0.5264	0.9758
390	18.630	0.000 893	0.0125	173.9	333.8	0.5587	0.9688
400	22.41	0.000 918	0.01011	186.4	334.8	0.5896	0.9605
410	26.72	0.000 944	0.00820	198.3	334.7	0.6180	0.9506
420	31.60	0.000 972	0.00672	208.7	333.7	0.6418	0.9394
430	37.10	0.001 002	0.00564	216.4	332.4	0.6587	0.9286
440	43.26	0.001 034	0.00491	221.1	332.3	0.6682	0.9208

Reproduced and rounded from unpublished Center for Applied Thermodynamic Studies, Moscow ID report, 1981. For a thermodynamic diagram to 350 bar, 370 °C, see Rombusch, U. K., *Allgem. Warme.*, **11**, 3 (1962).

TABLE 2-255 Thermodynamic Properties of R-22, Chlorodifluoromethane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
115.73	3.7947E-07	19.907	0.050235	2.5595	2.5595	0.0065813	0.061918	0.092976	1410.9	-0.44463
120.00	9.9588E-07	19.777	0.050564	2.9559	2.9559	0.0099451	0.061567	0.092700	1388.4	-0.44526
135.00	1.7187E-05	19.325	0.051747	4.3400	4.3400	0.020814	0.060123	0.091960	1312.3	-0.44448
150.00	0.00015627	18.873	0.052985	5.7179	5.7179	0.030493	0.059099	0.091824	1239.0	-0.43872
165.00	0.00089946	18.420	0.054289	7.0951	7.0951	0.039243	0.058356	0.091789	1166.5	-0.43084
180.00	0.0037009	17.963	0.055670	8.4715	8.4717	0.047227	0.057679	0.091751	1095.0	-0.42063
195.00	0.011835	17.500	0.057141	9.8483	9.8490	0.054574	0.057097	0.091898	1024.5	-0.40608
210.00	0.031218	17.028	0.058726	11.230	11.232	0.061399	0.056707	0.092431	954.55	-0.38505
225.00	0.070909	16.542	0.060453	12.622	12.627	0.067804	0.056579	0.093482	884.79	-0.35555
240.00	0.14319	16.036	0.062359	14.034	14.043	0.073877	0.056737	0.095120	814.96	-0.31561
255.00	0.26329	15.506	0.064493	15.472	15.489	0.079692	0.057171	0.097391	744.97	-0.26263
270.00	0.44888	14.944	0.066919	16.945	16.975	0.085308	0.057856	0.10037	674.69	-0.19248
285.00	0.71966	14.341	0.069730	18.462	18.513	0.090782	0.058767	0.10424	603.91	-0.097827
300.00	1.0970	13.686	0.073070	20.034	20.114	0.096166	0.059887	0.10941	532.11	0.035333
315.00	1.6039	12.956	0.077181	21.676	21.800	0.10152	0.061218	0.11688	458.37	0.23608
330.00	2.2661	12.114	0.082547	23.418	23.605	0.10696	0.062814	0.12929	381.15	0.57205
345.00	3.1130	11.069	0.090340	25.325	25.606	0.11267	0.064858	0.15550	298.16	1.2334
360.00	4.1837	9.5229	0.10501	27.613	28.053	0.11931	0.068488	0.25950	201.90	3.0745
369.30	4.9900	6.0582	0.16506	30.901	31.725	0.12907			0	10.366
115.73	3.7947E-07	3.9436E-07	2,535,700.	27.807	28.769	0.23305	0.028465	0.036779	119.91	398.80
120.00	9.9588E-07	9.9813E-07	1,001,900.	27.929	28.927	0.22637	0.028872	0.037186	121.91	367.18
135.00	1.7187E-05	1.5313E-05	65,305.	28.373	29.495	0.20715	0.030386	0.038703	128.58	269.83
150.00	0.00015627	0.00012533	7,979.0	28.840	30.087	0.19295	0.031990	0.040316	134.79	197.57
165.00	0.00089946	0.00065634	1,523.6	29.329	30.699	0.18230	0.033655	0.042018	140.59	146.30
180.00	0.0037009	0.0024808	403.09	29.836	31.328	0.17421	0.035388	0.043849	145.98	110.28
195.00	0.011835	0.0073561	135.94	30.357	31.966	0.16800	0.037218	0.045881	150.87	84.902
210.00	0.031218	0.018161	55.062	30.885	32.603	0.16317	0.039177	0.048204	155.15	66.865
225.00	0.070909	0.038991	25.647	31.411	33.229	0.15937	0.041300	0.050920	158.70	53.872
240.00	0.14319	0.075182	13.301	31.928	33.833	0.15634	0.043615	0.054144	161.35	44.348
255.00	0.26329	0.13342	7.4950	32.428	34.401	0.15386	0.046138	0.058023	162.96	37.240
270.00	0.44888	0.22208	4.5029	32.900	34.922	0.15178	0.048881	0.062763	163.38	31.852
285.00	0.71966	0.35201	2.8409	33.335	35.380	0.14996	0.051851	0.068713	162.45	27.725
300.00	1.0970	0.53822	1.8580	33.717	35.755	0.14830	0.055064	0.076534	159.98	24.549
315.00	1.6039	0.80363	1.2443	34.021	36.017	0.14666	0.058574	0.087659	155.73	22.102
330.00	2.2661	1.1882	0.84159	34.207	36.114	0.14486	0.062516	0.10579	149.36	20.201
345.00	3.1130	1.7777	0.56253	34.184	35.935	0.14261	0.067254	0.14389	140.39	18.613
360.00	4.1837	2.8529	0.35052	33.661	35.127	0.13896	0.074178	0.29995	127.92	16.641
369.30	4.9900	6.0582	0.16506	30.901	31.725	0.12907			0	10.366

Single-Phase Properties

150.00	0.10000	18.874	0.052982	5.7167	5.7220	0.030484	0.059102	0.091820	1239.3	-0.43876
232.06	0.10000	16.306	0.061325	13.284	13.290	0.070699	0.056618	0.094177	851.94	-0.33819
232.06	0.10000	0.053734	18.610	31.656	33.517	0.15786	0.042364	0.052366	160.06	49.032
250.00	0.10000	0.049441	20.226	32.442	34.465	0.16180	0.043860	0.053391	166.39	38.154
350.00	0.10000	0.034652	28.859	37.325	40.211	0.18105	0.053191	0.061845	196.16	13.439
450.00	0.10000	0.026819	37.287	43.093	46.822	0.19762	0.061672	0.070141	221.08	6.7638
550.00	0.10000	0.021901	45.660	49.620	54.186	0.21238	0.068475	0.076877	243.30	4.0604
150.00	1.0000	18.885	0.052952	5.7053	5.7582	0.030408	0.059134	0.091780	1241.8	-0.43921
250.00	1.0000	15.711	0.063648	14.961	15.024	0.077665	0.057020	0.096318	773.25	-0.28642
296.57	1.0000	13.841	0.072248	19.669	19.741	0.094938	0.059612	0.10807	548.68	0.00029448
296.57	1.0000	0.48960	2.0425	33.635	35.677	0.14867	0.054305	0.074523	160.70	25.205
350.00	1.0000	0.37703	2.6523	36.754	39.407	0.16025	0.055369	0.068138	184.85	14.183
450.00	1.0000	0.27693	3.6110	42.774	46.385	0.17776	0.062289	0.072300	216.22	6.8582
550.00	1.0000	0.22209	4.5027	49.400	53.903	0.19283	0.068737	0.077971	241.16	4.0624
150.00	5.0000	18.931	0.052823	5.6555	5.9197	0.030074	0.059277	0.091614	1253.0	-0.44109
250.00	5.0000	15.837	0.063142	14.822	15.138	0.077105	0.057135	0.095206	797.37	-0.30700
350.00	5.0000	11.141	0.089759	25.585	26.034	0.11341	0.064765	0.14356	317.33	1.0314
450.00	5.0000	1.6422	0.60893	41.131	44.175	0.16067	0.065284	0.087278	194.56	7.0507
550.00	5.0000	1.1832	0.84520	48.377	52.603	0.17760	0.069855	0.083655	232.93	3.9600
150.00	10.000	18.987	0.052667	5.5953	6.1220	0.029665	0.059460	0.091420	1266.7	-0.44331
250.00	10.000	15.982	0.062569	14.663	15.289	0.076450	0.057274	0.094054	825.41	-0.32844
350.00	10.000	12.008	0.083275	24.782	25.615	0.11098	0.063647	0.11843	412.24	0.39255
450.00	10.000	4.2433	0.23566	38.423	40.780	0.14893	0.068870	0.12461	184.82	5.5220
550.00	10.000	2.5432	0.39321	47.019	50.951	0.16944	0.071068	0.092204	229.21	3.5059
150.00	30.000	19.198	0.052089	5.3741	6.9367	0.028113	0.060251	0.090769	1318.0	-0.45090
250.00	30.000	16.469	0.060719	14.132	15.953	0.074181	0.057799	0.091018	921.00	-0.38542
350.00	30.000	13.518	0.073976	23.279	25.499	0.10621	0.063170	0.10053	603.47	-0.13764
450.00	30.000	10.241	0.097648	33.086	36.016	0.13259	0.069234	0.10864	392.35	0.40072
550.00	30.000	7.3810	0.13548	42.738	46.802	0.15425	0.073417	0.10533	317.51	0.89994
150.00	60.000	19.480	0.051336	5.0916	8.1717	0.026007	0.061560	0.089983	1384.9	-0.45992
250.00	60.000	17.029	0.058724	13.533	17.056	0.071434	0.058519	0.088686	1034.6	-0.42947
350.00	60.000	14.650	0.068258	22.111	26.206	0.10216	0.063912	0.094730	764.59	-0.31532
450.00	60.000	12.381	0.080770	31.083	35.929	0.12657	0.070220	0.099099	589.21	-0.17799
550.00	60.000	10.405	0.096108	40.152	45.919	0.14662	0.075020	0.10030	492.11	-0.055349

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Kamei, A., Beyerlein, S. W., and Jacobsen, R. T., "Application of Nonlinear Regression in the Development of a Wide Range Formulation for HCFC-22," *Int. J. Thermophys.* **16**:1155–1164, 1995. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties of the equation of state are 0.1% in density, 1% in heat capacity, and 0.3% in the speed of sound, except in the critical region. The uncertainty in vapor pressure is 0.2%.

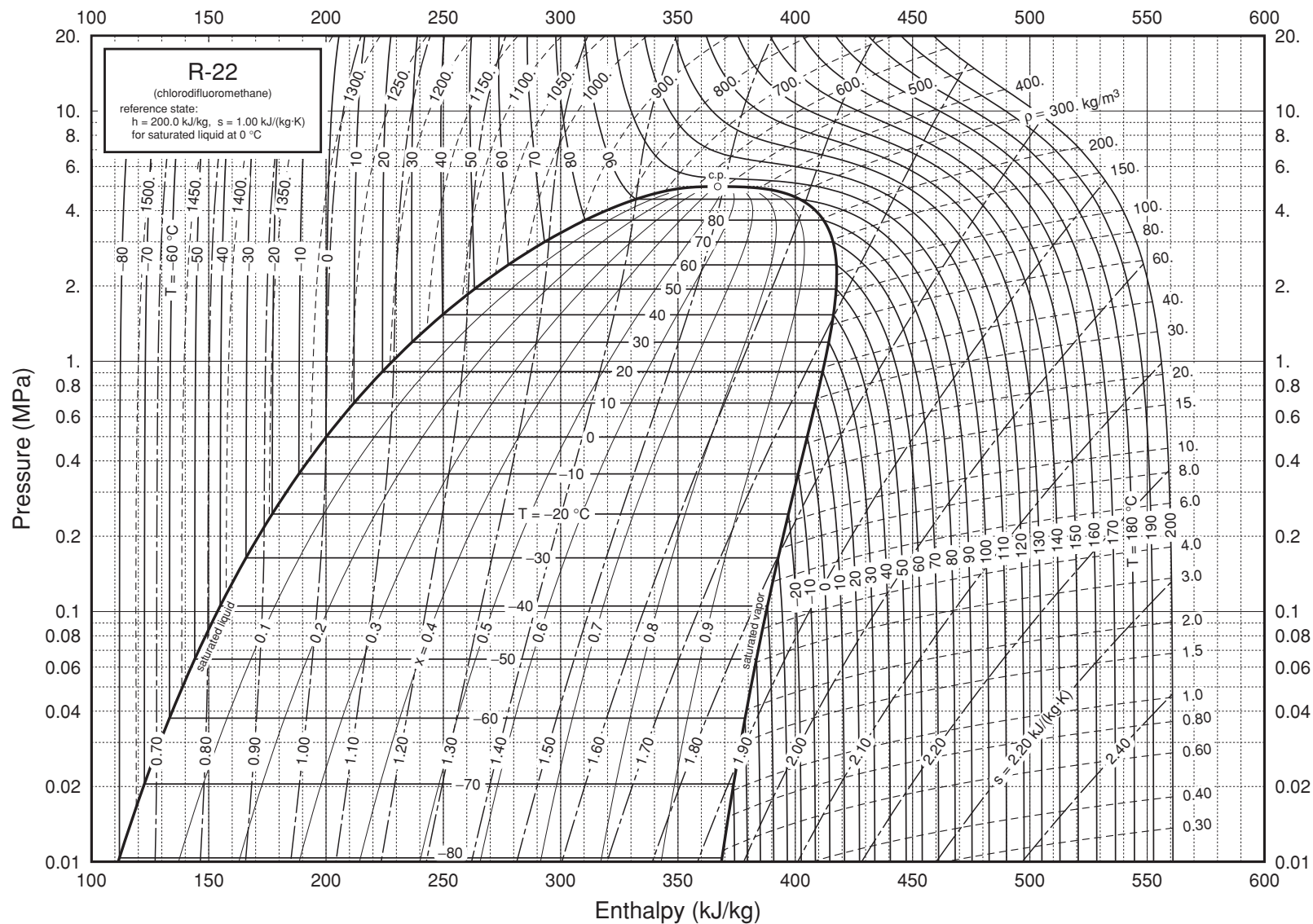


FIG. 2-20 Pressure-enthalpy diagram for Refrigerant 22. Properties computed with the NIST REFPROP Database, Version 7.0 (Lemmon, E. W., McLinden, M. O., and Huber, M. L., 2002, NIST Standard Reference Database 23, NIST Reference Fluid Thermodynamic and Transport Properties—REFPROP, Version 7.0, Standard Reference Data Program, National Institute of Standards and Technology), based on the equation of state of Kamei, A., Beyerlein, S. W., and Jacobsen, R. T., "Application of Nonlinear Regression in the Development of a Wide Range Formulation for HCFC-22," *Int. J. Thermophysics* **16**:1155–1164, 1995.

TABLE 2-256 Thermodynamic Properties of R-23, Trifluoromethane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
118.02	5.8041E-05	24.308	0.041139	-0.25719	-0.25719	-0.0049376	0.057095	0.085477	1211.2	-0.37494	268.59	2055.0
120.00	8.1883E-05	24.217	0.041293	-0.088512	-0.088509	-0.0035202	0.055755	0.084936	1208.8	-0.37515	261.17	1889.2
130.00	0.00038808	23.746	0.042112	0.75355	0.75356	0.0032204	0.052298	0.083783	1168.8	-0.37102	229.32	1285.4
140.00	0.0014276	23.263	0.042987	1.5903	1.5904	0.0094218	0.051128	0.083657	1111.2	-0.36298	204.76	927.74
150.00	0.0043019	22.771	0.043915	2.4276	2.4278	0.015198	0.050605	0.083836	1051.8	-0.35278	185.36	702.09
160.00	0.011055	22.272	0.044900	3.2675	3.2680	0.020619	0.050302	0.084188	994.03	-0.34018	169.64	551.48
170.00	0.024979	21.763	0.045950	4.1116	4.1127	0.025736	0.050114	0.084721	938.04	-0.32445	156.60	446.01
180.00	0.050823	21.243	0.047074	4.9618	4.9642	0.030596	0.050024	0.085477	883.23	-0.30465	145.55	369.03
190.00	0.094862	20.709	0.048288	5.8204	5.8250	0.035238	0.050032	0.086504	829.00	-0.27966	135.98	310.79
200.00	0.16485	20.158	0.049608	6.6899	6.6981	0.039699	0.050140	0.087863	774.87	-0.24798	127.54	265.32
210.00	0.26991	19.584	0.051061	7.5735	7.5873	0.044011	0.050350	0.089628	720.45	-0.20758	119.92	228.80
220.00	0.42035	18.982	0.052680	8.4748	8.4969	0.048206	0.050666	0.091906	665.40	-0.15554	112.91	198.71
230.00	0.62755	18.345	0.054512	9.3983	9.4325	0.052315	0.051093	0.094862	609.38	-0.087415	106.33	173.29
240.00	0.90390	17.660	0.056625	10.350	10.401	0.056370	0.051645	0.098763	552.02	0.0038879	100.01	151.31
250.00	1.2628	16.913	0.059125	11.336	11.411	0.060409	0.052347	0.10409	492.89	0.13049	93.771	131.85
260.00	1.7190	16.081	0.062186	12.370	12.477	0.064481	0.053248	0.11180	431.39	0.31494	87.445	114.17
270.00	2.2887	15.123	0.066126	13.469	13.621	0.068658	0.054447	0.12412	366.67	0.60438	80.778	97.624
280.00	2.9913	13.958	0.071643	14.670	14.885	0.073078	0.056175	0.14787	297.30	1.1166	73.344	81.444
290.00	3.8516	12.367	0.080863	16.074	16.386	0.078112	0.059136	0.21909	220.12	2.2579	64.064	64.222
299.29	4.8317	7.5200	0.13298	19.029	19.672	0.088893			0	8.4094		
118.02	5.8041E-05	5.9176E-05	16.899	19.268	20.249	0.16881	0.026617	0.034991	135.67	1947.8	3.7962	5.3460
120.00	8.1883E-05	8.2116E-05	12.178	19.320	20.317	0.16652	0.026770	0.035155	136.72	1684.9	3.9025	5.4612
130.00	0.00038808	0.00035955	2.781.2	19.581	20.661	0.15635	0.027678	0.036154	141.80	854.63	4.4400	6.0411
140.00	0.0014276	0.0012302	812.86	19.845	21.005	0.14810	0.028813	0.037439	146.52	472.27	4.9794	6.6166
150.00	0.0043019	0.0034701	288.17	20.109	21.348	0.14134	0.030153	0.039008	150.89	282.89	5.5230	7.1868
160.00	0.011055	0.0083994	119.06	20.372	21.688	0.13574	0.031671	0.040857	154.87	182.40	6.0746	7.7511
170.00	0.024979	0.017990	55.587	20.632	22.020	0.13107	0.033344	0.042992	158.43	125.40	6.6400	8.3091
180.00	0.050823	0.034921	28.636	20.886	22.342	0.12714	0.035155	0.045436	161.51	90.972	7.2268	8.8611
190.00	0.094862	0.062608	15.972	21.133	22.648	0.12378	0.037094	0.048225	164.02	68.962	7.8452	9.4081
200.00	0.16485	0.10526	9.5007	21.369	22.935	0.12088	0.039149	0.051416	165.89	54.182	8.5073	9.9521
210.00	0.26991	0.16798	5.9532	21.589	23.196	0.11834	0.041312	0.055095	167.03	43.846	9.2284	10.497
220.00	0.42035	0.25707	3.8900	21.791	23.426	0.11607	0.043578	0.059399	167.35	36.378	10.028	11.048
230.00	0.62755	0.38050	2.6281	21.968	23.617	0.11399	0.045950	0.064561	166.76	30.839	10.931	11.616
240.00	0.90390	0.54888	1.8219	22.112	23.759	0.11203	0.048442	0.070995	165.16	26.639	11.970	12.216
250.00	1.2628	0.77721	1.2866	22.214	23.839	0.11012	0.051088	0.079472	162.43	23.387	13.197	12.873
260.00	1.7190	1.0885	0.91866	22.256	23.835	0.10816	0.053955	0.091578	158.44	20.801	14.687	13.633
270.00	2.2887	1.5223	0.65691	22.209	23.712	0.10603	0.057163	0.11107	153.01	18.652	16.578	14.579
280.00	2.9913	2.1579	0.46341	22.017	23.403	0.10350	0.060947	0.14938	145.86	16.699	19.150	15.906
290.00	3.8516	3.2161	0.31093	21.529	22.727	0.099978	0.065857	0.26628	136.34	14.524	23.215	18.223
299.29	4.8317	7.5200	0.13298	19.029	19.672	0.088893			0	8.4094		
Single-Phase Properties												
150.00	0.10000	22.773	0.043911	2.4262	2.4306	0.015189	0.050608	0.083827	1052.1	-0.35289	185.41	702.64
190.90	0.10000	20.660	0.048402	5.8986	5.9034	0.035649	0.050037	0.086612	824.11	-0.27709	135.18	306.22
190.90	0.10000	0.065782	15.202	21.155	22.675	0.12350	0.037275	0.048496	164.22	67.381	7.9031	9.4574
250.00	0.10000	0.048799	20.492	23.373	25.422	0.13604	0.038440	0.047375	188.56	22.325	11.046	12.547
350.00	0.10000	0.034513	28.974	27.705	30.603	0.15338	0.048023	0.056526	220.23	8.0459	16.386	17.318
450.00	0.10000	0.026771	37.355	32.978	36.713	0.16869	0.057058	0.065464	247.21	4.0690	21.735	21.621
150.00	1.0000	22.792	0.043874	2.4134	2.4573	0.015103	0.050638	0.083752	1055.5	-0.35386	185.86	707.90
242.93	1.0000	17.448	0.057313	10.635	10.692	0.057555	0.051834	0.10015	534.88	0.036629	98.175	145.37
242.93	1.0000	0.60881	1.6425	22.147	23.790	0.11147	0.049200	0.073220	164.48	25.601	12.308	12.401
250.00	1.0000	0.57474	1.7399	22.544	24.284	0.11347	0.046933	0.067137	169.72	23.136	12.612	12.737
350.00	1.0000	0.35949	2.7818	27.401	30.182	0.13337	0.048882	0.059316	214.59	8.0837	17.536	17.376
450.00	1.0000	0.27164	3.6813	32.791	36.472	0.14914	0.057361	0.066629	245.15	4.0345	22.720	21.629

TABLE 2-256 Thermodynamic Properties of R-23, Trifluoromethane (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_v kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Single-Phase Properties												
150.00	5.0000	22.876	0.043714	2.3577	2.5763	0.014729	0.050778	0.083431	1070.0	−0.35795	187.83	731.40
250.00	5.0000	17.302	0.057797	11.094	11.383	0.059422	0.051903	0.098833	539.89	0.027079	97.307	141.75
350.00	5.0000	2.2512	0.44420	25.765	27.986	0.11523	0.053415	0.081317	190.19	7.7729	23.888	19.400
450.00	5.0000	1.4480	0.69062	31.931	35.384	0.13386	0.058685	0.072566	237.91	3.7699	27.113	22.593
150.00	10.000	22.976	0.043523	2.2910	2.7262	0.014275	0.050969	0.083065	1086.9	−0.36259	190.23	761.17
250.00	10.000	17.719	0.056437	10.828	11.392	0.058319	0.051625	0.094482	590.19	−0.061719	101.24	153.45
350.00	10.000	6.4822	0.15427	22.835	24.378	0.10116	0.058836	0.14411	186.97	4.7798	39.852	30.808
450.00	10.000	3.1053	0.32203	30.818	34.039	0.12572	0.060182	0.081111	235.46	3.2093	33.390	25.446
150.00	50.000	23.670	0.042247	1.8477	3.9600	0.011072	0.052531	0.080990	1193.2	−0.38760	207.85	1020.9
250.00	50.000	19.633	0.050935	9.5803	12.127	0.052740	0.052051	0.083522	826.45	−0.29649	121.46	226.71
350.00	50.000	15.604	0.064086	17.587	20.792	0.081836	0.057678	0.089445	563.57	−0.085583	92.030	113.25
450.00	50.000	12.055	0.082953	25.724	29.871	0.10464	0.064073	0.091553	437.72	0.13382	77.100	73.803
250.00	100.00	20.957	0.047717	8.7360	13.508	0.048428	0.053244	0.080401	1002.1	−0.36685	137.85	307.24
350.00	100.00	17.954	0.055698	16.172	21.741	0.076094	0.059172	0.084434	778.97	−0.30018	111.99	167.69
450.00	100.00	15.405	0.064915	23.879	30.370	0.097761	0.065748	0.087934	652.70	−0.24430	100.34	119.14

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Penoncello, S. G., Lemmon, E. W., Jacobsen, R. T., and Shan, Z., “A Fundamental Equation for Trifluoromethane (R-23),” *J. Phys. Chem. Ref. Data*, **32**(4):1473–1499, 2003. The source for viscosity and thermal conductivity is Shan, Z., Penoncello, S. G., and Jacobsen, R. T., “A Generalized Model for Viscosity and Thermal Conductivity of Trifluoromethane (R-23),” *ASHRAE Trans.* **106**:1–11, 2000.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties of the equation of state are 0.1% in density, 0.5% in heat capacities and speed of sound, and 0.2% in vapor pressures. Uncertainties in the critical region will be higher. The uncertainty in viscosity is 1%. The uncertainty in thermal conductivity is 2%.

TABLE 2-257 Thermodynamic Properties of R-32, Difluoromethane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)
Saturated Properties											
136.34	4.8000E-05	27.473	0.036399	-0.99220	-0.99220	-0.0054608	0.055447	0.082847	1414.4	-0.33760	242.91
140.00	8.3535E-05	27.302	0.036627	-0.68946	-0.68946	-0.0032696	0.054980	0.082588	1395.1	-0.33728	241.74
150.00	0.00032474	26.835	0.037265	0.13324	0.13325	0.0024067	0.053793	0.081975	1342.3	-0.33542	237.64
160.00	0.0010410	26.364	0.037930	0.95053	0.95057	0.0076815	0.052740	0.081513	1289.9	-0.33191	232.45
170.00	0.0028536	25.889	0.038626	1.7640	1.7641	0.012613	0.051818	0.081215	1237.6	-0.32650	226.39
180.00	0.0068782	25.409	0.039357	2.5753	2.5756	0.017251	0.051021	0.081087	1185.7	-0.31891	219.64
190.00	0.014904	24.921	0.040127	3.3862	3.3868	0.021635	0.050345	0.081137	1133.8	-0.30886	212.37
200.00	0.029545	24.424	0.040944	4.1983	4.1995	0.025800	0.049783	0.081373	1082.1	-0.29600	204.70
210.00	0.054344	23.916	0.041814	5.0135	5.0158	0.029778	0.049333	0.081803	1030.4	-0.27988	196.75
220.00	0.093819	23.394	0.042745	5.8337	5.8377	0.033594	0.048988	0.082443	978.59	-0.25996	188.62
230.00	0.15345	22.858	0.043749	6.6608	6.6675	0.037271	0.048747	0.083313	926.62	-0.23550	180.39
240.00	0.23965	22.303	0.044838	7.4969	7.5077	0.040830	0.048604	0.084442	874.35	-0.20551	172.14
250.00	0.35967	21.726	0.046028	8.3443	8.3609	0.044291	0.048559	0.085874	821.65	-0.16864	163.92
260.00	0.52157	21.124	0.047340	9.2056	9.2303	0.047671	0.048610	0.087676	768.33	-0.12292	155.78
270.00	0.73415	20.491	0.048802	10.084	10.120	0.050989	0.048761	0.089947	714.18	-0.065518	147.75
280.00	1.0069	19.820	0.050454	10.983	11.034	0.054264	0.049019	0.092852	658.88	0.0079177	139.86
290.00	1.3501	19.102	0.052350	11.908	11.979	0.057517	0.049399	0.096659	602.05	0.10428	132.11
300.00	1.7749	18.323	0.054577	12.866	12.963	0.060775	0.049934	0.10185	543.11	0.23517	124.48
310.00	2.2934	17.460	0.057273	13.867	13.998	0.064076	0.050685	0.10938	481.27	0.42163	116.94
320.00	2.9194	16.477	0.060691	14.930	15.107	0.067477	0.051776	0.12140	415.41	0.70602	109.42
330.00	3.6686	15.299	0.065364	16.088	16.328	0.071088	0.053487	0.14404	343.84	1.1876	101.80
340.00	4.5614	13.740	0.072779	17.428	17.760	0.075179	0.056594	0.20457	263.77	2.1660	94.166
350.00	5.6311	10.732	0.093180	19.453	19.977	0.081343	0.066340	1.2085	163.70	5.4955	97.067
351.26	5.7826	8.1501	0.12270	20.836	21.546	0.085769			0	8.0731	
136.34	4.8000E-05	4.2353E-05	23.611	21.981	23.115	0.17135	0.025987	0.034319	169.60	881.12	6.9492
140.00	8.3535E-05	7.1788E-05	13.930	22.076	23.239	0.16765	0.026110	0.034451	171.76	769.01	6.9554
150.00	0.00032474	0.00026061	3.837.2	22.335	23.581	0.15872	0.026507	0.034889	177.47	541.12	7.0006
160.00	0.0010410	0.00078411	1,275.3	22.593	23.921	0.15125	0.027014	0.035477	182.88	391.73	7.0875
170.00	0.0028536	0.0020270	493.35	22.850	24.258	0.14493	0.027667	0.036272	187.97	291.02	7.2166
180.00	0.0068782	0.0046295	216.01	23.103	24.589	0.13955	0.028505	0.037336	192.69	221.02	7.3887
190.00	0.014904	0.0095503	104.71	23.350	24.910	0.13492	0.029560	0.038728	197.00	170.81	7.6049
200.00	0.029545	0.018112	55.213	23.588	25.219	0.13090	0.030843	0.040483	200.85	133.79	7.8668
210.00	0.054344	0.032028	31.223	23.816	25.513	0.12738	0.032341	0.042613	204.20	106.00	8.1765
220.00	0.093819	0.053428	18.717	24.032	25.788	0.12428	0.034016	0.045105	206.99	84.924	8.5374
230.00	0.15345	0.084890	11.780	24.234	26.042	0.12151	0.035821	0.047943	209.19	68.870	8.9546
240.00	0.23965	0.12949	7.7224	24.421	26.272	0.11901	0.037709	0.051127	210.73	56.605	9.4365
250.00	0.35967	0.19093	5.2375	24.590	26.474	0.11674	0.039648	0.054696	211.57	47.194	9.9965
260.00	0.52157	0.27370	3.6537	24.738	26.643	0.11464	0.041621	0.058741	211.65	39.922	10.656
270.00	0.73415	0.38340	2.6083	24.860	26.775	0.11268	0.043631	0.063434	210.90	34.246	11.449
280.00	1.0069	0.52726	1.8966	24.952	26.862	0.11079	0.045693	0.069063	209.26	29.758	12.431
290.00	1.3501	0.71503	1.3985	25.006	26.894	0.10895	0.047840	0.076110	206.64	26.148	13.691
300.00	1.7749	0.96054	1.0411	25.011	26.858	0.10709	0.050119	0.085424	202.95	23.187	15.376
310.00	2.2934	1.2848	0.77830	24.950	26.735	0.10516	0.052598	0.098649	198.02	20.693	17.748
320.00	2.9194	1.7233	0.58029	24.797	26.491	0.10305	0.055390	0.11948	191.66	18.510	21.309
330.00	3.6686	2.3442	0.42658	24.503	26.068	0.10060	0.058707	0.15836	183.49	16.477	27.173
340.00	4.5614	3.3211	0.30111	23.943	25.316	0.097403	0.063103	0.26199	172.68	14.312	38.601
350.00	5.6311	5.7166	0.17493	22.380	23.365	0.091024	0.071998	1.9028	154.59	10.637	87.141
351.26	5.7826	8.1501	0.12270	20.836	21.546	0.085769			0	8.0731	
Single-Phase Properties											
150.00	0.10000	26.836	0.037263	0.13226	0.13599	0.0024002	0.053795	0.081971	1342.7	-0.33547	237.67
221.24	0.10000	23.329	0.042866	5.9359	5.9402	0.034057	0.048953	0.082538	972.15	-0.25719	187.60
221.24	0.10000	0.056727	17.628	24.058	25.821	0.12392	0.034234	0.045439	207.30	82.688	8.5859
225.00	0.10000	0.055592	17.988	24.191	25.989	0.12467	0.033681	0.044513	209.39	76.114	8.7199
300.00	0.10000	0.040576	24.645	26.760	29.224	0.13708	0.035327	0.044188	241.95	25.024	12.643
375.00	0.10000	0.032244	31.013	29.631	32.733	0.14749	0.041082	0.049630	267.64	12.117	18.907

TABLE 2-257 Thermodynamic Properties of R-32, Difluoromethane (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)
Single-Phase Properties											
150.00	1.0000	26.851	0.037243	0.12350	0.16075	0.0023417	0.053807	0.081935	1345.7	−0.33584	237.89
225.00	1.0000	23.159	0.043179	6.2265	6.2697	0.035360	0.048867	0.082690	957.47	−0.25084	185.10
279.77	1.0000	19.836	0.050414	10.962	11.013	0.054190	0.049012	0.092777	660.15	0.0060372	140.04
279.77	1.0000	0.52357	1.9100	24.951	26.861	0.11084	0.045646	0.068922	209.31	29.848	12.406
300.00	1.0000	0.46100	2.1692	25.950	28.120	0.11518	0.041298	0.057916	223.83	23.889	13.334
375.00	1.0000	0.33917	2.9484	29.227	32.175	0.12727	0.042549	0.053538	259.50	11.929	19.059
150.00	5.0000	26.915	0.037154	0.085198	0.27097	0.0020846	0.053863	0.081784	1358.9	−0.33741	238.83
225.00	5.0000	23.298	0.042923	6.1394	6.3541	0.034970	0.048922	0.082027	978.81	−0.26135	187.71
300.00	5.0000	18.710	0.053448	12.637	12.905	0.060001	0.049679	0.096911	586.17	0.13431	128.72
344.33	5.0000	12.808	0.078078	18.130	18.520	0.077304	0.059015	0.28240	224.84	2.9944	91.412
344.33	5.0000	3.9887	0.25071	23.524	24.777	0.095475	0.065786	0.39574	166.61	13.148	48.235
375.00	5.0000	2.3228	0.43051	26.928	29.080	0.10755	0.050835	0.090625	218.44	10.800	26.239
150.00	10.000	26.993	0.037047	0.038702	0.40917	0.0017692	0.053932	0.081608	1375.0	−0.33924	239.97
225.00	10.000	23.461	0.042625	6.0369	6.4632	0.034504	0.048996	0.081303	1004.1	−0.27287	190.81
300.00	10.000	19.196	0.052094	12.346	12.867	0.058997	0.049538	0.092105	639.69	0.031716	134.49
375.00	10.000	10.448	0.095714	21.112	22.069	0.085980	0.057265	0.21222	231.14	3.5190	77.284
150.00	30.000	27.285	0.036650	−0.13357	0.96593	0.00056834	0.054209	0.081027	1436.1	−0.34524	244.02
225.00	30.000	24.030	0.041614	5.6813	6.9297	0.032836	0.049307	0.079197	1094.1	−0.30661	201.89
300.00	30.000	20.517	0.048739	11.542	13.004	0.056105	0.049670	0.083697	789.57	−0.15824	152.78
375.00	30.000	16.472	0.060708	17.786	19.607	0.075712	0.052941	0.093069	536.98	0.21392	112.94
225.00	70.000	24.916	0.040135	5.1400	7.9495	0.030109	0.049916	0.076915	1240.7	−0.34341	219.54
300.00	70.000	22.090	0.045270	10.583	13.752	0.052355	0.050281	0.078370	986.17	−0.28333	179.29
375.00	70.000	19.309	0.051788	16.127	19.752	0.070195	0.053544	0.081767	788.81	−0.18583	147.00

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Tillner-Roth, R., and Yokozeki, A., "An International Standard Equation of State for Difluoromethane (R-32) for Temperatures from the Triple Point at 136.34 K to 435 K and Pressures up to 70 MPa," *J. Phys. Chem. Ref. Data* **26**(6):1273–1328, 1997. Validated equations for the viscosity are not currently available for this fluid. The source for thermal conductivity is unpublished; however, the fit uses the functional form found in Marsh, K., Perkins, R., and Ramires, M. L. V., "Measurement and Correlation of the Thermal Conductivity of Propane from 86 to 600 K at Pressures to 70 MPa," *J. Chem. Eng. Data* **47**(4):932–940, 2002.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

For the equation of state, typical uncertainties are 0.05% for density, 0.02% for the vapor pressure, and 0.5% to 1% for the heat capacity and speed of sound in the liquid phase. In the vapor phase, the uncertainty in the speed of sound is 0.02%. For thermal conductivity, the estimated uncertainty of the correlation is 5%, except for the dilute gas and points approaching critical where the uncertainty rises to 10%.

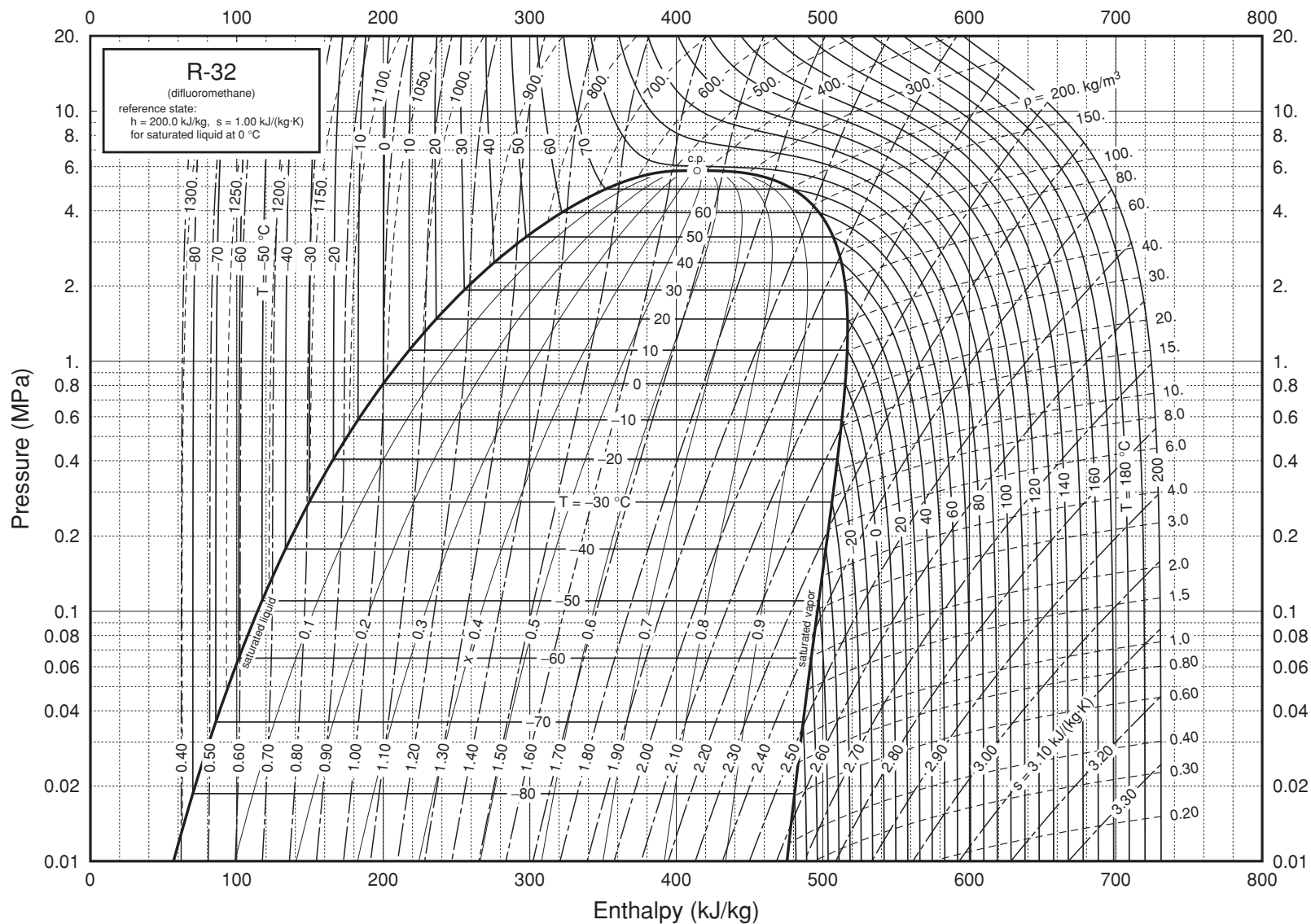


FIG. 2-21 Pressure-enthalpy diagram for Refrigerant 32. Properties computed with the NIST REFPROP Database, Version 7.0 (Lemmon, E. W., McLinden, M. O., and Huber, M. L., 2002, NIST Standard Reference Database 23, NIST Reference Fluid Thermodynamic and Transport Properties—REFPROP, Version 7.0, Standard Reference Data Program, National Institute of Standards and Technology), based on the equation of state of Tillner-Roth, R., and Yokozeki, A., “An International Standard Equation of State for Difluoromethane (R-32) for Temperatures from the Triple Point at 136.34 K to 435 K and Pressures up to 70 MPa,” *J. Phys. Chem. Ref. Data* **26**(6):1273–1328, 1997.

TABLE 2-258 Thermodynamic Properties of R-41, Fluoromethane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
129.82	0.00034504	29.650	0.033727	-3.7550	-3.7549	-0.019920	0.050428	0.072206	1413.8	-0.35314
130.00	0.00035370	29.640	0.033738	-3.7420	-3.7420	-0.019820	0.050396	0.072196	1413.1	-0.35309
140.00	0.0012519	29.080	0.034388	-3.0231	-3.0230	-0.014492	0.048727	0.071584	1371.7	-0.35022
150.00	0.0036677	28.514	0.035070	-2.3102	-2.3100	-0.0095735	0.047265	0.071018	1325.5	-0.34606
160.00	0.0092373	27.942	0.035788	-1.6024	-1.6021	-0.0050057	0.045989	0.070568	1276.1	-0.33979
170.00	0.020585	27.362	0.036548	-0.89848	-0.89772	-0.00073794	0.044881	0.070278	1224.2	-0.33071
180.00	0.041505	26.770	0.037355	-0.19668	-0.19513	0.0032736	0.043923	0.070180	1170.6	-0.31808
190.00	0.077054	26.165	0.038218	0.50493	0.50787	0.0070672	0.043101	0.070303	1115.6	-0.30109
200.00	0.13355	25.544	0.039148	1.2085	1.2137	0.010676	0.042407	0.070678	1059.3	-0.27874
210.00	0.21852	24.901	0.040159	1.9164	1.9251	0.014131	0.041832	0.071342	1001.8	-0.24979
220.00	0.34056	24.233	0.041266	2.6312	2.6453	0.017458	0.041372	0.072345	943.19	-0.21256
230.00	0.50928	23.534	0.042492	3.3561	3.3778	0.020682	0.041022	0.073761	883.36	-0.16475
240.00	0.73518	22.796	0.043868	4.0946	4.1268	0.023828	0.040783	0.075695	822.19	-0.10306
250.00	1.0296	22.010	0.045434	4.8508	4.8976	0.026921	0.040657	0.078315	759.47	-0.022535
260.00	1.4048	21.164	0.047251	5.6304	5.6968	0.029987	0.040652	0.081902	694.85	0.084654
270.00	1.8740	20.239	0.049410	6.4408	6.5334	0.033058	0.040786	0.086969	627.81	0.23174
280.00	2.4519	19.207	0.052065	7.2930	7.4207	0.036178	0.041098	0.094564	557.39	0.44318
290.00	3.1548	18.018	0.055501	8.2059	8.3810	0.039414	0.041677	0.10725	481.85	0.77036
300.00	4.0027	16.564	0.060372	9.2189	9.4606	0.042906	0.042763	0.13357	397.63	1.3453
310.00	5.0232	14.501	0.068961	10.456	10.802	0.047088	0.045267	0.23071	295.99	2.6641
317.28	5.9062	9.3000	0.10753	12.765	13.400	0.055113			0	7.4905
129.82	0.00034504	0.00031992	3125.8	14.466	15.544	0.12874	0.025226	0.033592	205.35	365.81
130.00	0.00035370	0.00032749	3053.5	14.470	15.550	0.12858	0.025228	0.033595	205.49	363.97
140.00	0.0012519	0.0010777	927.93	14.716	15.878	0.12051	0.025399	0.033848	213.06	278.51
150.00	0.0036677	0.0029538	338.55	14.957	16.199	0.11382	0.025715	0.034322	220.17	217.09
160.00	0.0092373	0.0070026	142.80	15.190	16.509	0.10819	0.026225	0.035097	226.74	170.82
170.00	0.020585	0.014779	67.662	15.411	16.804	0.10339	0.026955	0.036230	232.71	135.12
180.00	0.041505	0.028398	35.214	15.620	17.081	0.099254	0.027900	0.037744	238.00	107.41
190.00	0.077054	0.050558	19.779	15.813	17.337	0.095640	0.029030	0.039636	242.53	85.993
200.00	0.13355	0.084573	11.824	15.988	17.567	0.092444	0.030304	0.041897	246.25	69.518
210.00	0.21852	0.13444	7.4384	16.144	17.769	0.089580	0.031684	0.044538	249.08	56.880
220.00	0.34056	0.20496	4.8791	16.278	17.940	0.086979	0.033143	0.047606	250.96	47.167
230.00	0.50928	0.30200	3.3112	16.388	18.075	0.084582	0.034668	0.051200	251.81	39.654
240.00	0.73518	0.43294	2.3098	16.470	18.168	0.082334	0.036262	0.055502	251.54	33.780
250.00	1.0296	0.60737	1.6464	16.519	18.214	0.080186	0.037939	0.060820	250.07	29.121
260.00	1.4048	0.83844	1.1927	16.527	18.203	0.078086	0.039727	0.067681	247.30	25.360
270.00	1.8740	1.1452	0.87319	16.484	18.121	0.075974	0.041668	0.077052	243.07	22.257
280.00	2.4519	1.5578	0.64192	16.374	17.948	0.073776	0.043822	0.090909	237.22	19.624
290.00	3.1548	2.1293	0.46963	16.167	17.649	0.071373	0.046290	0.11407	229.42	17.292
300.00	4.0027	2.9728	0.33638	15.803	17.149	0.068535	0.049264	0.16236	219.09	15.070
310.00	5.0232	4.4455	0.22495	15.090	16.220	0.064565	0.053237	0.33830	204.65	12.533
317.28	5.9062	9.3000	0.10753	12.765	13.400	0.055113			0	7.4905

Single-Phase Properties

150.00	0.10000	28.517	0.035067	-2.3112	-2.3077	-0.0095803	0.047264	0.071013	1326.0	-0.34613
194.60	0.10000	25.882	0.038637	0.82800	0.83187	0.0087476	0.042767	0.070442	1089.8	-0.29155
194.60	0.10000	0.064499	15.504	15.896	17.446	0.094125	0.029600	0.040629	244.34	77.876
225.00	0.10000	0.054701	18.281	16.776	18.604	0.099660	0.027396	0.036814	265.51	44.413
300.00	0.10000	0.040421	24.739	18.906	21.380	0.11030	0.029496	0.038140	305.33	18.192
375.00	0.10000	0.032197	31.059	21.292	24.398	0.11926	0.034142	0.042610	336.84	9.5450
150.00	1.0000	28.539	0.035039	-2.3206	-2.2855	-0.0096430	0.047252	0.070965	1330.7	-0.34675
225.00	1.0000	23.924	0.041800	2.9761	3.0179	0.019009	0.041178	0.072808	918.60	-0.19408
249.10	1.0000	22.083	0.045284	4.7819	4.8272	0.026644	0.040664	0.078045	765.18	-0.030727
249.10	1.0000	0.58959	1.6961	16.516	18.212	0.080377	0.037784	0.060289	250.26	29.499
300.00	1.0000	0.43831	2.2815	18.437	20.718	0.089572	0.032042	0.044524	291.25	17.553
375.00	1.0000	0.33366	2.9971	21.029	24.026	0.099411	0.034956	0.044943	329.85	9.3564
150.00	5.0000	28.637	0.034919	-2.3615	-2.1869	-0.0099181	0.047205	0.070762	1351.1	-0.34938
225.00	5.0000	24.159	0.041392	2.8706	3.0775	0.018535	0.041150	0.071655	952.55	-0.21829
300.00	5.0000	17.057	0.058628	9.0202	9.3133	0.042218	0.042199	0.11693	436.81	1.0494
309.79	5.0000	14.557	0.068696	10.426	10.769	0.046985	0.045181	0.22600	298.41	2.6188
309.79	5.0000	4.4014	0.22720	15.113	16.249	0.064673	0.053137	0.32991	205.02	12.597
375.00	5.0000	2.0056	0.49861	19.699	22.192	0.082450	0.038898	0.060151	299.64	8.3579
150.00	10.000	28.756	0.034776	-2.4107	-2.0629	-0.010253	0.047153	0.070531	1375.8	-0.35235
225.00	10.000	24.428	0.040937	2.7503	3.1596	0.017985	0.041136	0.070486	991.68	-0.24309
300.00	10.000	18.494	0.054071	8.4188	8.9595	0.040105	0.041153	0.090910	556.97	0.45639
375.00	10.000	5.4445	0.18367	17.496	19.333	0.070759	0.043934	0.10163	274.07	6.1896
150.00	30.000	29.189	0.034259	-2.5895	-1.5617	-0.011513	0.047004	0.069806	1467.3	-0.36156
225.00	30.000	25.306	0.039516	2.3574	3.5429	0.016117	0.041195	0.067503	1123.5	-0.30724
300.00	30.000	20.991	0.047639	7.3266	8.7557	0.036076	0.040629	0.072733	794.05	-0.067863
375.00	30.000	15.862	0.063045	12.699	14.590	0.053392	0.043123	0.082994	538.73	0.53694
225.00	70.000	26.558	0.037653	1.8062	4.4419	0.013266	0.041529	0.064822	1323.8	-0.36575
300.00	70.000	23.298	0.042923	6.3044	9.3090	0.031929	0.041001	0.065644	1056.2	-0.28880
375.00	70.000	20.079	0.049804	10.864	14.350	0.046913	0.043364	0.068927	851.76	-0.15494

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., and Span, R., "Short Fundamental Equations of State for 20 Industrial Fluids," *J. Chem. Eng. Data*, **51**(3):785–850, 2006. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in the equation of state are 0.1% in density (except near the critical point), 0.25% in vapor-pressure, 1% in heat capacities, 0.2% in the vapor-phase speed of sound, and 3% in the liquid speed of sound. The liquid speed of sound uncertainty is an estimate and cannot be verified without experimental information. The uncertainties above 290 K in vapor pressure may be as high as 0.5%.

2-350 PHYSICAL AND CHEMICAL DATA

TABLE 2-259 Saturated R-401A (SUVA MP 39)

Temp., °C	P_g , bar	P_g , bar	v_f , m ³ /kg	v_g , m ³ /kg	h_f , kJ/kg	h_g , kJ/kg	s_f , kJ/(kg·K)	s_g , kJ/(kg·K)	c_{pf} , kJ/(kg·K)	μ_f , 10 ⁻⁶ Pa·s	k_f , W/(m·K)	Pr_f
-40	0.733	0.533	0.000 712	0.3778	154.0	385.0	0.8188	1.8244	1.078	351	0.1209	3.13
-30	1.155	0.871	0.000 728	0.2391	164.9	390.6	0.8647	1.8059	1.109	323	0.1154	3.06
-20	1.748	1.361	0.000 744	0.1576	176.2	396.3	0.9099	1.7907	1.137	291	0.1107	2.99
-10	2.553	2.043	0.000 762	0.1075	188.6	401.8	0.9577	1.7781	1.165	266	0.1057	2.93
0	3.615	2.965	0.000 781	0.0755	200.0	407.3	1.0000	1.7675	1.197	241	0.1012	2.85
10	4.984	4.177	0.000 803	0.0544	212.7	412.6	1.0454	1.7587	1.233	221	0.0967	2.82
20	6.712	5.733	0.000 826	0.0399	225.3	417.6	1.0884	1.7510	1.277	202	0.0922	2.80
30	8.857	7.697	0.000 851	0.0298	238.3	422.2	1.1316	1.7439	1.329	186	0.0877	2.83
40	11.475	10.133	0.000 878	0.0225	252.0	426.5	1.1752	1.7372	1.392	170	0.0830	2.85
50	14.628	13.112	0.000 909	0.0172	266.4	430.1	1.2194	1.7304	1.468	157	0.0781	2.95
60	18.378	16.711	0.000 944	0.01313	281.6	433.0	1.2647	1.7228	1.564	143	0.0737	3.04
70	22.79	21.01	0.000 988	0.01005	297.9	434.9	1.3118	1.7138	1.652	131	0.0684	3.16
80	27.92	26.12	0.001 028	0.00764	315.9	435.4	1.3616	1.7022	1.802	122	0.0631	3.48
90	33.83	32.13	0.001 084	0.00570	336.2	433.5	1.4163	1.6858	1.958	115	0.0577	3.90
100	40.53	39.22	0.001 140	0.00403	361.4	426.9	1.4820	1.6584	2.16	110	0.0533	4.46
108.0°	46.04	46.04	0.001 96	0.00196	397	397						

c = critical point. SUVA MP 39 = R401A = CHClF₂ (R22) 53% wt + CH₃CHF₂ (R 152a) 13% wt + CHClFCF₃ (R124) 34% wt, near-azeotropic blend. Some values read from charts are approximate. Material used by permission of DuPont Fluoroproducts.

TABLE 2-260 R-401A (SUVA MP 39) at Atmospheric Pressure

Temp., °C	-27.01	-20	0	20	40	60	80	100	120	140
v (m ³ /kg)	0.2102	0.2167	0.2351	0.2534	0.2715	0.2896	0.3076	0.3256	0.3435	0.3613
h (kJ/kg)	351.7	396.9	410.4	424.5	439.2	454.4	470.3	486.6	503.5	521.2
s (kJ/kg·K)	1.8009	1.8193	1.8706	1.9204	1.9689	2.0161	2.0623	2.1073	2.1513	2.1943
c_p (kJ/kg·K)	0.648	0.669	0.698	0.727	0.757	0.787	0.811	0.836	0.859	0.883
μ (10 ⁻⁶ Pa·s)	10.17	10.43	11.18	11.93	12.68	13.42	14.17	14.89	15.61	16.32
k (W/m·K)	0.00878	0.00921	0.01041	0.01161	0.01282	0.01404	0.01536	0.01668	0.01796	0.01929
Pr	0.750	0.758	0.750	0.749	0.749	0.748	0.748	0.748	0.747	0.747
Z	0.9829	0.9852	0.9906	0.9949	0.9979	1.0005	1.0025	1.0043	1.0056	1.0060

For composition see footnote to Table 2-259. Some values read from charts are approximate. Material used by permission of DuPont Fluoroproducts.

TABLE 2-261 Thermodynamic Properties of Saturated R-407A (Klea 60)

Pressure, bar	T_f , K	T_g , K	v_f , m ³ /kg	v_g , m ³ /kg	h_f , kJ/kg	h_g , kJ/kg	s_f , kJ/(kg·K)	s_g , kJ/(kg·K)
1	227.3	234.0	0.000 7118	0.2097	-7.80	229.64		0.9965
1.5	236.1	242.5	0.000 7263	0.1433	3.92	235.02		0.9833
2	242.8	249.1	0.000 7381	0.1093	12.89	239.07		0.9744
2.5	248.3	254.5	0.000 7483	0.0885	20.27	242.35		0.9679
3	253.0	259.1	0.000 7573	0.0744	26.57	245.08		0.9629
4	260.7	266.8	0.000 7735	0.0564	37.23	249.54		0.9552
5	267.3	273.1	0.000 7880	0.0442	46.12	253.07		0.9496
6	272.9	278.5	0.000 8012	0.0384	53.84	255.24		0.9450
8	282.1	287.5	0.000 8254	0.0286	67.02	260.70		0.9378
10	289.8	295.0	0.000 8480	0.0228	78.23	263.86		0.9318
12.5	297.9	302.8	0.000 8750	0.01802	90.50	266.95		0.9257
15	304.8	309.5	0.000 9017	0.01481	101.51	269.12		0.9190
17.5	311.0	315.4	0.000 9290	0.01247	111.64	270.58		0.9128
20	316.5	320.7	0.000 9613	0.01069	121.18	271.46		0.9065
22.5	321.4	325.5	0.000 9884	0.00928	130.31	271.79		0.8999
25	326.1	329.8	0.001 023	0.00828	139.17	271.63		0.8927
27.5	330.4	333.9	0.001 063	0.00717	147.89	270.97		0.8850
30	334.5	337.6	0.001 115	0.00635	156.58	269.81		0.8765

$h_f = s_f = 0$ at 233.15 K = -40°C. Converted and interpolated from *Thermodynamic Properties of Klea 60* (British units, 20 pp.), copyright ICI Chemicals and Polymers Limited, 1993. Reproduced by permission. T_f = bubble point temperature; T_g = dew point temperature.

TABLE 2-262 Thermodynamic Properties of Saturated R-407B (Klea 61)

Pressure, bar	T_f , K	T_g , K	v_f , m ³ /kg	v_g , m ³ /kg	h_f , kJ/kg	h_g , kJ/kg	s_f , kJ/(kg·K)	s_g , kJ/(kg·K)
1	225.6	230.0	0.000 6852	0.1800	-9.45	191.64		0.8433
1.5	234.3	238.5	0.000 6994	0.1230	2.52	196.90		0.8341
2	241.8	245.0	0.000 7110	0.0937	9.72	200.88		0.8282
2.5	246.4	250.4	0.000 7211	0.0758	16.59	204.10		0.8245
3	251.1	254.9	0.000 7301	0.0637	22.47	206.80		0.8215
4	258.9	262.6	0.000 7463	0.04831	32.43	211.22		0.8172
5	265.4	269.0	0.000 7607	0.03888	40.76	214.74		0.8141
6	270.9	274.4	0.000 7740	0.03249	48.00	217.65		0.8123
8	280.2	283.4	0.000 7985	0.02435	59.82	222.21		0.8080
10	287.8	290.9	0.000 8214	0.01936	70.98	225.63		0.8048
12.5	295.8	298.7	0.000 8491	0.01528	82.59	228.80		0.8010
15	302.8	305.5	0.000 8768	0.01251	93.02	231.08		0.7971
17.5	308.8	311.4	0.000 9053	0.01049	102.67	232.64		0.7929
20	314.3	316.7	0.000 9353	0.00896	111.79	233.60		0.7882
22.5	319.3	321.5	0.000 9680	0.00774	120.55	233.99		0.7829
25	323.9	325.9	0.001 005	0.00674	129.11	233.85		0.7769
27.5	328.1	330.0	0.001 048	0.00590	137.62	233.16		0.7700
30	332.1	333.7	0.001 102	0.00518	146.21	231.84		0.7619

Converted and interpolated from *Thermodynamic Properties of Klea 61* (British units, 20 pp.), copyright ICI Chemicals and Polymers Limited, 1993. Reproduced by permission. T_f = bubble-point temperature; T_g = dew-point temperature. $h_f = s_f = 0$ at 233.15 K = -40°C.

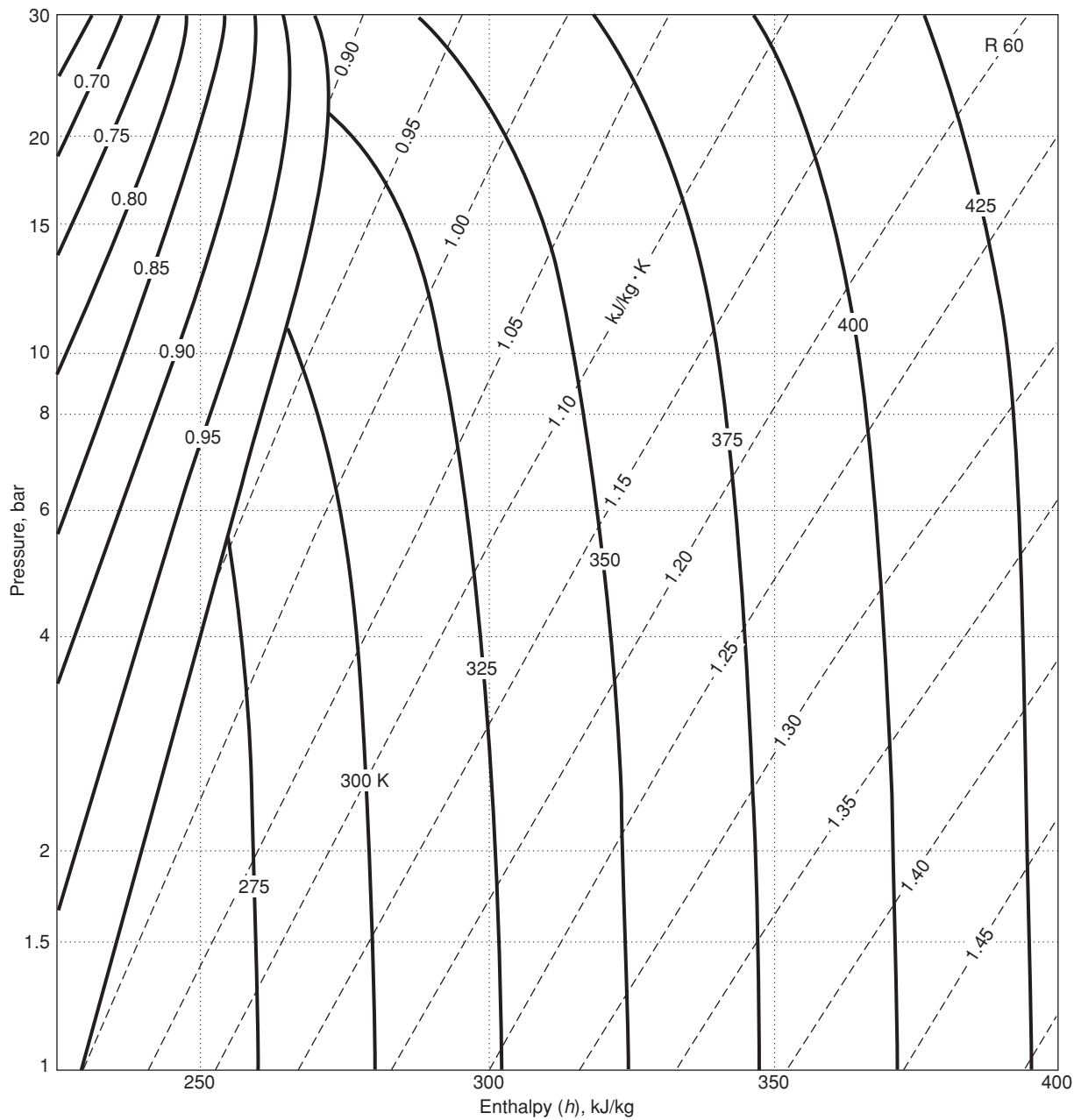


FIG. 2-22 Enthalpy–log–pressure diagram for R-407A (Klea 60).

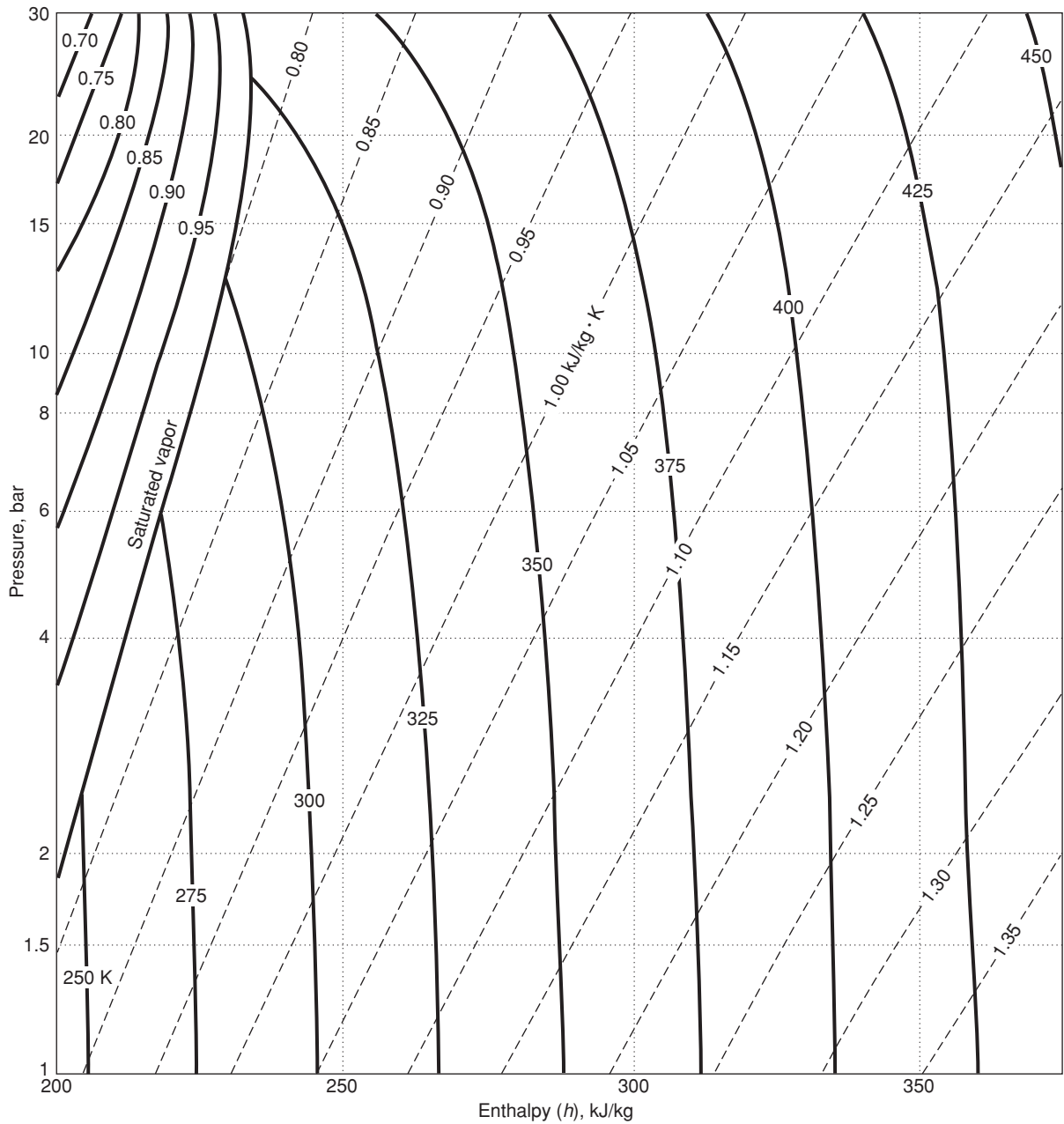


FIG. 2-23 Enthalpy-log-pressure diagram for R-407B (Klea 61).

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TABLE 2-263 Saturated R-404A (SUVA HP 62)

Temp., °C	P_f , bar	P_g , bar	v_f , m ³ /kg	v_g , m ³ /kg	h_f , kJ/kg	h_g , kJ/kg	s_f , kJ/(kg·K)	s_g , kJ/(kg·K)	c_{pf} , kJ/(kg·K)	μ_f , 10 ⁻⁶ Pa·s	k_f , W/(m·K)	Pr_f
-50	0.852	0.821	0.000 761	0.2244	133.1	337.3	0.7318	1.6487		370	0.0970	
-40	1.367	1.325	0.000 779	0.1434	145.6	343.8	0.7862	1.6380		318		
-30	2.095	2.041	0.000 799	0.0953	159.9	350.3	0.8460	1.6301	1.220	276	0.0868	3.88
-20	3.087	3.018	0.000 820	0.0656	172.8	356.5	0.8975	1.6245	1.260	238	0.0834	3.60
-10	4.404	4.321	0.000 843	0.0463	186.1	362.6	0.9487	1.6202	1.302	207	0.0801	3.37
0	6.111	6.013	0.000 868	0.03338	200.0	368.3	1.0000	1.6188	1.351	181	0.0767	3.19
10	8.278	8.165	0.000 898	0.02444	214.5	373.6	1.0515	1.6138	1.412	158	0.0733	3.04
20	10.977	10.851	0.000 933	0.01809	229.9	378.3	1.1038	1.6106	1.489	138	0.0698	2.94
30	14.287	14.150	0.000 977	0.01348	246.2	382.2	1.1574	1.6065	1.592	122	0.0663	2.93
40	18.292	18.148	0.001 037	0.01003	263.8	385.0	1.2130	1.6005	1.753	106	0.0624	2.98
50	23.08	22.94	0.001 122	0.00739	283.2	386.1	1.2723	1.5910	2.09	91	0.0583	3.26
60	28.75	28.63	0.001 261	0.00527	305.8	384.2	1.3389	1.5742		76	0.0535	
70	35.58		0.00285	0.00285	339.8	375.9				61		
72.1 ^c	37.32	37.32	0.002 06	0.00206	361	361						

c = critical point. SUVA HP 62 = CHF₂CF₃ (R125) 44% wt + CH₃CF₃ (R143a) 52% wt + CH₂FCF₃ (R134a) 4% wt, near-azeotropic blend. Material used by permission of DuPont Fluoroproducts. Some values read from charts may be approximate.

TABLE 2-264 R-404A (SUVA HP 62) at Atmospheric Pressure

Temp., °C	-45.63	-40	-20	0	20	40	60	80	100	120
v (m ³ /kg)	0.1866	0.1921	0.2100	0.2278	0.2455	0.2630	0.2805	0.2980	0.3153	0.3325
h (kJ/kg)	336.0	344.4	359.9	376.2	393.1	410.9	429.3	448.4	468.2	488.7
s (kJ/kg·K)	1.6599	1.6636	1.7274	1.7891	1.8491	1.9076	1.9646	2.0203	2.0747	2.1278
c_p (kJ/kg·K)	0.732	0.738	0.781	0.821	0.860	0.897	0.933	0.967	1.000	1.032
μ (10 ⁻⁶ Pa·s)	9.47	9.68	10.45	11.22	11.99	12.76	13.53	14.30	15.07	15.84
k (W/m·K)	0.00860	0.00932	0.01059	0.01186	0.01313	0.01440	0.01568	0.01695	0.01827	0.01949
Pr	0.806	0.767	0.771	0.777	0.785	0.795	0.805	0.816	0.827	0.839
Z	0.9755	0.9800	0.9867	0.9919	0.9961	0.9989	1.0014	1.0037	1.0050	1.0060

v , h , and s from DuPont bull. T—HP62—SI, June 1993 (17 pp.). c_p and k from DuPont bull. ART 18, June 1993 (37 pp.). Some values read from charts may be approximate. Material used by permission of DuPont Fluoroproducts.

TABLE 2-265 Saturated R-401B (SUVA MP 66)

Temp., °C	P_f , bar	P_g , bar	v_f , m ³ /kg	v_g , m ³ /kg	h_f , kJ/kg	h_g , kJ/kg	s_f , kJ/(kg·K)	s_g , kJ/(kg·K)	c_{pf} , kJ/(kg·K)	μ_f , 10 ⁻⁶ Pa·s	k_f , W/(m·K)	Pr_f
-40	0.788	0.585	0.000 710	0.3498	153.8	386.0	0.8184	1.8291	1.078	349	0.1209	3.11
-30	1.239	0.952	0.000 725	0.2224	164.8	391.6	0.8643	1.8100	1.109	313	0.1154	3.01
-20	1.872	1.479	0.000 740	0.1471	176.0	397.1	0.9095	1.7940	1.137	282	0.1106	2.90
-10	2.726	2.212	0.000 758	0.1008	188.6	402.6	0.9577	1.7807	1.165	257	0.1057	2.83
0	3.850	3.198	0.000 778	0.0710	200.0	407.8	1.0000	1.7694	1.197	236	0.1012	2.79
10	5.297	4.491	0.000 801	0.05124	212.6	412.9	1.0450	1.7598	1.233	217	0.0967	2.77
20	7.120	6.146	0.000 827	0.03771	225.1	417.7	1.0879	1.7512	1.277	198	0.0922	2.74
30	9.379	8.229	0.000 858	0.02818	238.2	422.1	1.1311	1.7433	1.329	181	0.0877	2.74
40	12.133	10.808	0.000 895	0.02131	251.9	426.1	1.1747	1.7357	1.392	168	0.0830	2.82
50	15.444	13.955	0.000 939	0.01625	266.3	429.4	1.2190	1.7278	1.468	151	0.0781	2.84
60	19.378	17.750	0.000 994	0.01244	281.6	431.9	1.2645	1.7191	1.564	139	0.0737	2.95
70	24.00	22.28	0.001 066	0.00951	298.1	433.4	1.3120	1.7088	1.652	127	0.0684	3.07
80	29.37	27.64	0.001 164	0.00721	316.3	433.2	1.3625	1.6956	1.802	116	0.0631	3.31
90	35.55	33.96	0.001 313	0.00534	337.2	430.4	1.4187	1.6768			0.0577	
100	42.30										0.0533	
106.1 ^c	46.82	46.82	0.001 95	0.00195	389	389						

c = critical point. SUVA MP 66 = R-401B = CHClF₂ (R22) 61% wt + CH₃CHF₂ (R152a) 11% wt + CHClFCF₃ (R124) 28% wt, near-azeotropic blend. Material used by permission of DuPont Fluoroproducts. Some values read from charts are approximate.

TABLE 2-266 R-401B (SUVA MP 66) at Atmospheric Pressure

Temp., °C	-28.63 ^b	-20	0	20	40	60	80	100	120	140
v (m ³ /kg)	0.2086	0.2177	0.2362	0.2545	0.2727	0.2908	0.3089	0.3269	0.3449	0.3629
h (kJ/kg)	392.2	397.9	411.2	425.1	439.6	454.6	470.1	486.2	502.7	519.4
s (kJ/kg·K)	1.8081	1.8299	1.8804	1.9295	1.9772	2.0237	2.0690	2.1132	2.1564	2.1986
c_p (kJ/kg·K)	0.641	0.652	0.688	0.716	0.744	0.771	0.796	0.822	0.844	0.866
μ (10 ⁻⁶ Pa·s)	9.78	10.43	11.18	11.93	12.68	13.42	14.17	14.89	15.61	16.32
k (W/m·K)	0.00817	0.00921	0.01041	0.01161	0.01282	0.01404	0.01536	0.01668	0.01796	0.01929
Pr	0.767	0.738	0.737	0.736	0.735	0.735	0.734	0.734	0.733	0.733
Z	0.9652	0.9730	0.9783	0.9822	0.9852	0.9876	0.9896	0.9912	0.9925	0.9937

v , h , and s from DuPont bull. T—MP 66—SI, Jan. 1993 (17 pp.). c_p , μ , and k from DuPont bull. ART 10, Jan. 1993 (27 pp.). Some values read from charts may be approximate. Material used by permission of DuPont Fluoroproducts. b = normal boiling point.

TABLE 2-267 Saturated R-402A (SUVA HP 80)

Temp., °C	P_f , bar	P_g , bar	v_f , m ³ /kg	v_g , m ³ /kg	h_f , kJ/kg	h_g , kJ/kg	s_f , kJ/(kg·K)	s_g , kJ/(kg·K)	c_{pf} , kJ/(kg·K)	μ_f , 10 ⁻⁶ Pa·s	k_f , W/(m·K)	Pr_f
-50	0.962	0.872	0.000 679	0.2033	139.6	334.1	0.7578	1.6327		377	0.0970	
-40	1.520	1.403	0.000 695	0.1303	150.8	339.9	0.8070	1.6206		317		
-30	2.305	2.156	0.000 713	0.0869	163.1	345.6	0.8584	1.6110	1.193	283	0.0880	3.84
-20	3.370	3.188	0.000 733	0.0598	174.9	351.1	0.9053	1.6034	1.217	247	0.0849	3.54
-10	4.776	4.560	0.000 757	0.0423	187.6	356.4	0.9541	1.5972	1.236	215	0.0813	3.27
0	6.588	6.336	0.000 785	0.03060	200.0	361.3	1.0000	1.5919	1.253	188	0.0778	3.03
10	8.877	8.592	0.000 819	0.02248	213.0	365.9	1.0461	1.5870	1.286	165	0.0743	2.86
20	11.720	11.404	0.000 860	0.01671	226.7	369.8	1.0927	1.5820	1.340	146	0.0708	2.76
30	15.195	14.855	0.000 911	0.01250	241.2	373.1	1.1403	1.5762	1.412	128	0.0672	2.69
40	19.388	19.034	0.000 977	0.00936	256.8	375.4	1.1897	1.5690	1.512	113	0.0634	2.70
50	24.39	24.04	0.001 070	0.00696	273.9	376.2	1.2420	1.5589	1.64	98	0.0593	2.71
60	30.30	29.97	0.001 212	0.00505	293.6	374.6	1.2998	1.5433	1.81	83	0.0551	2.79
70										68		
75.5 ^c	41.35	41.35	0.001 850	0.00185	340	340						

c = critical point. SUVA HP 80 = R-402A = CHF₂CF₃ (R125) 60% wt + CH₃CH₂CH₃ (R290) 2% wt + CHClF₂ (R22) 38% wt, near-azeotropic blend. Material used by permission of DuPont Fluoroproducts. Some values, read from charts, may be approximate.

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TABLE 2-268 R-402A (SUVA HP 80) at Atmospheric Pressure

Temp., °C	-46.95 ^b	-40	-20	0	20	40	60	80	100	120
<i>v</i> (m ³ /kg)	0.1768	0.1827	0.1996	0.2164	0.2331	0.2497	0.2663	0.2828	0.2992	0.3155
<i>h</i> (kJ/kg)	335.9	340.5	354.3	368.6	383.5	398.7	414.9	431.4	448.5	466.1
<i>s</i> (kJ/kg·K)	1.6286	1.6490	1.7055	1.7599	1.8124	1.8633	1.9128	1.9610	2.0081	2.0541
<i>c_p</i> (kJ/kg·K)	0.648	0.654	0.687	0.721	0.749	0.779	0.807	0.836	0.863	0.890
<i>μ</i> (10 ⁻⁶ Pa·s)	9.42	9.69	10.45	11.22	11.99	12.75	13.52	14.29	15.06	15.82
<i>k</i> (W/m·K)	0.00888	0.00932	0.01059	0.01186	0.01313	0.01440	0.01568	0.01695	0.01822	0.01949
<i>Pr</i>	0.687	0.680	0.678	0.681	0.685	0.690	0.696	0.703	0.713	0.722
<i>Z</i>	0.9673	0.9697	0.9758	0.9804	0.9840	0.9868	0.9892	0.9910	0.9923	0.9932

b = normal boiling pt. *v*, *h*, and *s* from DuPont bull. T—HP 80—SI, Jan. 1993 (17 pp.). *c_p*, *μ*, and *k* from DuPont bull. ART 18, June 1993 (37 pp.). Some values read from charts may be approximate. Material used by permission of DuPont Fluoroproducts.

TABLE 2-269 Saturated R-402B (SUVA HP 81)

Temp., °C	<i>P_f</i> bar	<i>P_g</i> bar	<i>v_f</i> m ³ /kg	<i>v_g</i> m ³ /kg	<i>h_f</i> kJ/kg	<i>h_g</i> kJ/kg	<i>s_f</i> kJ/(kg·K)	<i>s_g</i> kJ/(kg·K)	<i>c_{pf}</i> kJ/(kg·K)	<i>μ_f</i> 10 ⁻⁶ Pa·s	<i>k_f</i> W/(m·K)	<i>Pr_f</i>
-50	0.883	0.787	0.000 687	0.2425	140.3	351.7	0.7606	1.7122		383	0.1031	
-40	1.403	1.273	0.000 702	0.1548	151.4	357.2	0.8092	1.6957		333	0.0983	
-30	2.135	1.967	0.000 719	0.1028	163.3	362.7	0.8589	1.6820	1.178	290	0.0941	3.63
-20	3.132	2.923	0.000 739	0.0707	174.9	368.0	0.9054	1.6706	1.191	253	0.0900	3.35
-10	4.451	4.198	0.000 761	0.0499	187.8	373.0	0.9550	1.6611	1.204	223	0.0863	3.11
0	6.153	5.852	0.000 787	0.03610	200.0	377.8	1.0000	1.6528	1.221	195	0.0818	2.91
10	8.307	7.959	0.000 817	0.02656	212.7	382.2	1.0450	1.6451	1.288	173	0.0790	2.82
20	10.984	10.591	0.000 854	0.01980	226.0	386.0	1.0905	1.6376	1.313	151	0.0753	2.63
30	14.261	13.827	0.000 899	0.01490	240.1	389.3	1.1367	1.6299	1.37	137	0.0715	2.49
40	18.216	17.750	0.000 955	0.01125	255.1	391.5	1.1842	1.6211	1.75	122	0.0676	3.16
50	22.93	22.45	0.001 030	0.00848	271.4	392.8	1.2339	1.6104	2.07	106	0.0633	3.47
60	28.50	28.03	0.001 136	0.00632	289.5	392.2	1.2873	1.5961		91	0.0586	
70	35.01	34.60	0.001 307	0.00456	299.6	390.9	1.3164	1.5866		75	0.0544	
80												
82.6°	44.45	44.45	0.001 88	0.00188	351	351						

c = critical point. SUVA HP 81 = R402B (38/2/60) = CHF₂CF₃ (R125) 38% wt + CH₃CH₂CH₃ (R290) 2% wt + CHClF₂ (R22) 60% wt, near-azeotropic blend. Material used by permission of DuPont Fluoroproducts. Some values read from charts may be approximate.

TABLE 2-270 R-402B (SUVA HP 81) at Atmospheric Pressure

Temp., °C	-44.87 ^b	-40	-20	0	20	40	60	80	100	120
<i>v</i> (m ³ /kg)	0.1903	0.1960	0.2142	0.2322	0.2500	0.2678	0.2856	0.3032	0.3209	0.3386
<i>h</i> (kJ/kg)	354.7	357.7	370.8	384.6	398.8	413.6	428.9	444.7	461.0	477.7
<i>s</i> (kJ/kg·K)	1.7032	1.7169	1.7711	1.8232	1.8735	1.9222	1.9696	2.0158	2.0607	2.1047
<i>c_p</i> (kJ/kg·K)	1.187	1.177	1.169	1.159	1.149	1.143	1.134	1.128	1.124	1.120
<i>μ</i> (10 ⁻⁶ Pa·s)	10.16	10.33	11.10	11.86	12.62	13.39	14.15	14.78	15.54	16.30
<i>k</i> (W/m·K)	0.00739	0.00768	0.00902	0.01036	0.01170	0.01304	0.01438	0.01572	0.01706	0.01840
<i>Pr</i>	1.632	1.583	1.439	1.327	1.239	1.174	1.124	1.061	1.024	0.992
<i>Z</i>	0.9622	0.9703	0.9766	0.9811	0.9843	0.9870	0.9894	0.9909	0.9926	0.9940

b = normal boiling point. *v*, *h*, and *s* from DuPont bull. T—HP 81—SI, Jan. 1993 (17 pp.). *c_p*, *μ*, and *k* from DuPont bull. ART 18, June 1993 (37 pp.). Some values, read from charts, may be approximate. Material used by permission of DuPont Fluoroproducts.

TABLE 2-271 Thermodynamic Properties of R-113, 1,1,2-Trichlorotrifluoroethane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
236.93	0.0018714	9.0987	0.10991	31.483	31.484	0.16386	0.11672	0.16341	907.79	-0.45956
240.00	0.0022999	9.0614	0.11036	31.985	31.986	0.16596	0.11694	0.16366	897.63	-0.45692
250.00	0.0043168	8.9395	0.11186	33.627	33.627	0.17266	0.11797	0.16467	863.61	-0.44732
260.00	0.0076484	8.8172	0.11342	35.279	35.280	0.17914	0.11931	0.16593	828.94	-0.43622
270.00	0.012885	8.6941	0.11502	36.945	36.947	0.18543	0.12084	0.16735	794.26	-0.42364
280.00	0.020766	8.5702	0.11668	38.626	38.629	0.19154	0.12245	0.16891	759.92	-0.40951
290.00	0.032185	8.4450	0.11841	40.323	40.327	0.19750	0.12410	0.17056	726.09	-0.39367
300.00	0.048190	8.3183	0.12022	42.036	42.042	0.20331	0.12576	0.17231	692.84	-0.37590
310.00	0.069977	8.1897	0.12211	43.767	43.776	0.20898	0.12740	0.17414	660.17	-0.35588
320.00	0.098880	8.0588	0.12409	45.516	45.528	0.21453	0.12901	0.17606	628.04	-0.33321
330.00	0.13636	7.9252	0.12618	47.284	47.301	0.21998	0.13060	0.17807	596.38	-0.30739
340.00	0.18400	7.7884	0.12840	49.071	49.095	0.22531	0.13215	0.18020	565.13	-0.27774
350.00	0.24349	7.6479	0.13076	50.879	50.911	0.23055	0.13367	0.18247	534.20	-0.24340
360.00	0.31660	7.5029	0.13328	52.708	52.751	0.23571	0.13517	0.18491	503.50	-0.20323
370.00	0.40521	7.3529	0.13600	54.560	54.616	0.24079	0.13665	0.18755	472.94	-0.15569
380.00	0.51127	7.1968	0.13895	56.437	56.508	0.24579	0.13813	0.19047	442.41	-0.098686
390.00	0.63682	7.0336	0.14217	58.340	58.430	0.25074	0.13961	0.19374	411.80	-0.029233
400.00	0.78398	6.8619	0.14573	60.271	60.386	0.25564	0.14112	0.19749	380.97	0.057005
410.00	0.95499	6.6797	0.14971	62.235	62.378	0.26050	0.14266	0.20191	349.77	0.16660
420.00	1.1522	6.4845	0.15421	64.236	64.414	0.26533	0.14428	0.20730	317.99	0.31000
430.00	1.3781	6.2728	0.15942	66.280	66.500	0.27015	0.14601	0.21416	285.39	0.50481
440.00	1.6354	6.0390	0.16559	68.379	68.650	0.27500	0.14794	0.22344	251.65	0.78295
450.00	1.9271	5.7742	0.17318	70.548	70.881	0.27990	0.15016	0.23712	216.41	1.2078
460.00	2.2568	5.4624	0.18307	72.817	73.230	0.28494	0.15288	0.25998	179.46	1.9196
470.00	2.6288	5.0712	0.19719	75.247	75.765	0.29024	0.15638	0.30706	141.14	3.2778
480.00	3.0498	4.5104	0.22171	78.016	78.692	0.29621	0.16109	0.47179	101.80	6.5781
487.21	3.3923	2.9887	0.33460	82.401	83.536	0.30602			0	19.849
236.93	0.0018714	0.00095199	1050.4	61.277	63.243	0.29790	0.098915	0.10733	106.58	82.183
240.00	0.0022999	0.0011554	865.51	61.580	63.570	0.29756	0.099759	0.10819	107.21	77.881
250.00	0.0043168	0.0020850	479.61	62.580	64.650	0.29675	0.10244	0.11094	109.17	66.150
260.00	0.0076484	0.0035597	280.92	63.602	65.750	0.29634	0.10502	0.11361	111.03	57.141
270.00	0.012885	0.0057917	172.66	64.643	66.868	0.29625	0.10751	0.11623	112.77	50.111
280.00	0.020766	0.0090358	110.67	65.701	67.999	0.29644	0.10994	0.11883	114.38	44.541
290.00	0.032185	0.013590	73.586	66.774	69.143	0.29686	0.11231	0.12142	115.82	40.065
300.00	0.048190	0.019793	50.523	67.860	70.295	0.29748	0.11463	0.12402	117.08	36.421
310.00	0.069977	0.028029	35.677	68.957	71.453	0.29826	0.11691	0.12666	118.14	33.419
320.00	0.098880	0.038727	25.822	70.061	72.615	0.29918	0.11917	0.12935	118.97	30.923
330.00	0.13636	0.052364	19.097	71.172	73.776	0.30020	0.12140	0.13214	119.57	28.833
340.00	0.18400	0.069475	14.394	72.286	74.934	0.30131	0.12362	0.13503	119.89	27.074
350.00	0.24349	0.090662	11.030	73.400	76.086	0.30248	0.12583	0.13807	119.93	25.594
360.00	0.31660	0.11662	8.5752	74.513	77.228	0.30370	0.12804	0.14130	119.66	24.353
370.00	0.40521	0.14813	6.7508	75.621	78.356	0.30495	0.13024	0.14477	119.05	23.322
380.00	0.51127	0.18615	5.3719	76.720	79.467	0.30621	0.13246	0.14857	118.07	22.480
390.00	0.63682	0.23182	4.3136	77.808	80.555	0.30747	0.13469	0.15281	116.69	21.816
400.00	0.78398	0.28656	3.4897	78.878	81.614	0.30871	0.13695	0.15765	114.87	21.322
410.00	0.95499	0.35216	2.8396	79.926	82.638	0.30991	0.13926	0.16333	112.57	21.000
420.00	1.1522	0.43104	2.3200	80.944	83.617	0.31105	0.14165	0.17026	109.72	20.858
430.00	1.3781	0.52654	1.8992	81.922	84.539	0.31210	0.14414	0.17911	106.25	20.915
440.00	1.6354	0.64356	1.5539	82.844	85.386	0.31304	0.14679	0.19114	102.05	21.204
450.00	1.9271	0.78990	1.2660	83.689	86.129	0.31379	0.14969	0.20894	97.003	21.779
460.00	2.2568	0.97939	1.0210	84.417	86.721	0.31426	0.15299	0.23897	90.902	22.716
470.00	2.6288	1.2415	0.80548	84.942	87.059	0.31427	0.15703	0.30270	83.480	24.091
480.00	3.0498	1.6684	0.59937	85.005	86.833	0.31317	0.16277	0.54016	74.333	25.627
487.21	3.3923	2.9887	0.33460	82.401	83.536	0.30602			0	19.849

TABLE 2-271 Thermodynamic Properties of R-113, 1,1,2-Trichlorotrifluoroethane (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Single-Phase Properties										
250.00	0.10000	8.9405	0.11185	33.623	33.634	0.17265	0.11797	0.16466	864.02	-0.44747
320.34	0.10000	8.0543	0.12416	45.576	45.588	0.21472	0.12907	0.17612	626.96	-0.33239
320.34	0.10000	0.039138	25.551	70.099	72.654	0.29921	0.11925	0.12945	119.00	30.846
325.00	0.10000	0.038492	25.979	70.661	73.259	0.30109	0.11993	0.12998	120.05	29.068
400.00	0.10000	0.030653	32.623	80.120	83.382	0.32908	0.13100	0.14005	135.10	14.858
475.00	0.10000	0.025598	39.065	90.324	94.230	0.35391	0.14017	0.14891	148.01	9.5374
250.00	1.0000	8.9494	0.11174	33.589	33.701	0.17251	0.11802	0.16456	867.87	-0.44882
325.00	1.0000	8.0093	0.12485	46.338	46.463	0.21708	0.12985	0.17671	618.26	-0.32646
400.00	1.0000	6.8729	0.14550	60.238	60.383	0.25555	0.14107	0.19705	384.09	0.047064
412.41	1.0000	6.6340	0.15074	62.713	62.864	0.26166	0.14304	0.20311	342.18	0.19754
412.41	1.0000	0.36984	2.7039	80.174	82.878	0.31019	0.13983	0.16487	111.93	20.948
475.00	1.0000	0.28704	3.4839	89.325	92.809	0.33262	0.14241	0.15696	133.62	11.300
250.00	5.0000	8.9881	0.11126	33.441	33.998	0.17191	0.11826	0.16413	884.22	-0.45442
325.00	5.0000	8.0819	0.12373	46.082	46.701	0.21629	0.13008	0.17538	644.07	-0.34900
400.00	5.0000	7.0485	0.14187	59.694	60.403	0.25417	0.14058	0.19113	433.49	-0.086367
475.00	5.0000	5.5199	0.18116	74.959	75.865	0.28950	0.15167	0.23166	217.18	1.2219
250.00	10.000	9.0343	0.11069	33.266	34.373	0.17119	0.11861	0.16368	903.13	-0.46053
325.00	10.000	8.1644	0.12248	45.791	47.016	0.21536	0.13042	0.17407	672.87	-0.37126
400.00	10.000	7.2197	0.13851	59.149	60.534	0.25274	0.14040	0.18683	481.58	-0.18229
475.00	10.000	6.0609	0.16499	73.507	75.157	0.28621	0.14885	0.20446	313.71	0.32498
325.00	100.00	9.0570	0.11041	42.816	53.857	0.20444	0.13658	0.16811	970.17	-0.48114
400.00	100.00	8.5204	0.11737	54.955	66.691	0.23995	0.14351	0.17396	867.10	-0.45624
475.00	100.00	8.0191	0.12470	67.454	79.924	0.27026	0.14864	0.17872	788.68	-0.42986
475.00	200.00	8.7846	0.11384	65.196	87.963	0.26222	0.14928	0.17693	1035.9	-0.46436

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Marx, V., Pruss, A., and Wagner, W., "Neue Zustandsgleichungen fuer R 12, R 22, R 11 und R 113. Beschreibung des thermodynamischen Zustandsverhaltens bei Temperaturen bis 525 K und Druecken bis 200 MPa," Duesseldorf: *VDI Verlag*, Series 19 (Waermetechnik/Kaeltechnik), No. 57, 1992. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainty in density is 0.2%, that for vapor pressure is 0.3%, and that for the isobaric heat capacity is 2%. The uncertainties are higher in and above the critical region.

TABLE 2-272 Thermodynamic Properties of R-114, 1,2-Dichlorotetrafluoroethane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
275.00	0.094764	8.9110	0.11222	34.479	34.490	0.17204	0.11622	0.16411	636.17	-0.32927
280.00	0.11455	8.8268	0.11329	35.301	35.314	0.17500	0.11691	0.16513	617.82	-0.31514
285.00	0.13740	8.7415	0.11440	36.128	36.143	0.17792	0.11763	0.16623	599.54	-0.29975
290.00	0.16362	8.6548	0.11554	36.960	36.979	0.18082	0.11838	0.16741	581.32	-0.28301
295.00	0.19353	8.5667	0.11673	37.798	37.820	0.18368	0.11917	0.16866	563.14	-0.26478
300.00	0.22746	8.4771	0.11796	38.641	38.668	0.18652	0.11997	0.16998	545.01	-0.24491
305.00	0.26574	8.3858	0.11925	39.491	39.523	0.18933	0.12079	0.17138	526.89	-0.22318
310.00	0.30872	8.2927	0.12059	40.348	40.385	0.19212	0.12162	0.17286	508.80	-0.19936
315.00	0.35677	8.1976	0.12199	41.211	41.255	0.19488	0.12247	0.17441	490.70	-0.17315
320.00	0.41024	8.1004	0.12345	42.082	42.133	0.19763	0.12333	0.17606	472.58	-0.14418
325.00	0.46953	8.0007	0.12499	42.960	43.019	0.20035	0.12419	0.17781	454.45	-0.11201
330.00	0.53501	7.8984	0.12661	43.846	43.913	0.20306	0.12507	0.17967	436.26	-0.076088
335.00	0.60708	7.7932	0.12832	44.739	44.817	0.20575	0.12595	0.18167	418.03	-0.035744
340.00	0.68615	7.6847	0.13013	45.642	45.731	0.20843	0.12684	0.18383	399.72	0.0098734
345.00	0.77265	7.5726	0.13205	46.553	46.655	0.21109	0.12775	0.18617	381.32	0.061838
350.00	0.86701	7.4565	0.13411	47.475	47.591	0.21375	0.12866	0.18875	362.81	0.12152
355.00	0.96967	7.3358	0.13632	48.407	48.539	0.21640	0.12959	0.19162	344.17	0.19072
360.00	1.0811	7.2098	0.13870	49.350	49.500	0.21904	0.13054	0.19485	325.38	0.27180
365.00	1.2018	7.0779	0.14129	50.306	50.476	0.22169	0.13152	0.19854	306.40	0.36794
370.00	1.3323	6.9389	0.14412	51.276	51.468	0.22434	0.13252	0.20284	287.21	0.48354
375.00	1.4730	6.7916	0.14724	52.261	52.478	0.22700	0.13356	0.20795	267.77	0.62482
380.00	1.6246	6.6345	0.15073	53.265	53.510	0.22967	0.13465	0.21417	248.03	0.80081
385.00	1.7878	6.4651	0.15468	54.291	54.567	0.23237	0.13581	0.22201	227.95	1.0252
390.00	1.9630	6.2804	0.15922	55.343	55.655	0.23511	0.13705	0.23227	207.45	1.3196
395.00	2.1511	6.0759	0.16459	56.428	56.782	0.23790	0.13842	0.24644	186.45	1.7200
400.00	2.3530	5.8442	0.17111	57.557	57.960	0.24078	0.13997	0.26750	164.86	2.2906
405.00	2.5694	5.5731	0.17943	58.749	59.211	0.24379	0.14180	0.30245	142.55	3.1570
410.00	2.8015	5.2381	0.19091	60.042	60.577	0.24704	0.14408	0.37230	119.37	4.5962
415.00	3.0507	4.7774	0.20932	61.532	62.171	0.25078	0.14726	0.57876	95.263	7.3420
418.83	3.2516	3.3932	0.29471	64.391	65.349	0.25828			0	18.568
275.00	0.094764	0.043032	23.239	55.626	57.828	0.25690	0.10567	0.11538	116.26	24.935
280.00	0.11455	0.051392	19.458	56.123	58.352	0.25728	0.10683	0.11680	116.64	24.301
285.00	0.13740	0.060964	16.403	56.622	58.875	0.25769	0.10799	0.11826	116.94	23.725
290.00	0.16362	0.071869	13.914	57.121	59.397	0.25812	0.10916	0.11977	117.15	23.203
295.00	0.19353	0.084238	11.871	57.620	59.917	0.25859	0.11032	0.12133	117.27	22.730
300.00	0.22746	0.098213	10.182	58.119	60.435	0.25908	0.11150	0.12295	117.29	22.303
305.00	0.26574	0.11395	8.7760	58.617	60.949	0.25958	0.11268	0.12465	117.20	21.919
310.00	0.30872	0.13161	7.5984	59.113	61.459	0.26010	0.11387	0.12643	117.00	21.576
315.00	0.35677	0.15138	6.6059	59.608	61.965	0.26063	0.11508	0.12831	116.68	21.272
320.00	0.41024	0.17347	5.7648	60.100	62.465	0.26117	0.11630	0.13031	116.25	21.005
325.00	0.46953	0.19810	5.0480	60.589	62.959	0.26171	0.11754	0.13245	115.69	20.776
330.00	0.53501	0.22553	4.4340	61.074	63.446	0.26225	0.11880	0.13475	115.00	20.582
335.00	0.60708	0.25605	3.9055	61.554	63.925	0.26279	0.12008	0.13725	114.17	20.423
340.00	0.68615	0.28999	3.4484	62.029	64.395	0.26332	0.12139	0.13998	113.19	20.301
345.00	0.77265	0.32774	3.0512	62.497	64.855	0.26384	0.12274	0.14299	112.05	20.216
350.00	0.86701	0.36975	2.7045	62.957	65.302	0.26435	0.12412	0.14634	110.75	20.169
355.00	0.96967	0.41655	2.4007	63.408	65.736	0.26484	0.12555	0.15013	109.28	20.160
360.00	1.0811	0.46878	2.1332	63.849	66.155	0.26531	0.12703	0.15444	107.63	20.193
365.00	1.2018	0.52722	1.8967	64.277	66.557	0.26575	0.12856	0.15945	105.77	20.269
370.00	1.3323	0.59285	1.6868	64.690	66.938	0.26615	0.13016	0.16534	103.71	20.392
375.00	1.4730	0.66690	1.4995	65.086	67.295	0.26651	0.13185	0.17244	101.42	20.565
380.00	1.6246	0.75097	1.3316	65.461	67.624	0.26681	0.13363	0.18118	98.885	20.790
385.00	1.7878	0.84722	1.1803	65.810	67.920	0.26705	0.13552	0.19228	96.087	21.072
390.00	1.9630	0.95861	1.0432	66.126	68.174	0.26720	0.13755	0.20692	92.999	21.411
395.00	2.1511	1.0895	0.91784	66.401	68.376	0.26725	0.13975	0.22720	89.593	21.805
400.00	2.3530	1.2467	0.80213	66.621	68.509	0.26715	0.14217	0.25728	85.834	22.243

TABLE 2-272 Thermodynamic Properties of R-114, 1,2-Dichlorotetrafluoroethane (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
405.00	2.5694	1.4416	0.69369	66.763	68.545	0.26684	0.14487	0.30666	81.678	22.687
410.00	2.8015	1.6966	0.58941	66.782	68.433	0.26620	0.14792	0.40278	77.066	23.035
415.00	3.0507	2.0686	0.48342	66.570	68.045	0.26493	0.15144	0.67085	71.914	22.963
418.83	3.2516	3.3932	0.29471	64.391	65.349	0.25828			0	18.568
Single-Phase Properties										
300.00	0.10000	0.041312	24.206	58.335	60.756	0.26666	0.11051	0.11990	122.03	19.386
350.00	0.10000	0.035003	28.569	64.107	66.964	0.28578	0.11931	0.12828	132.80	12.959
400.00	0.10000	0.030432	32.860	70.284	73.570	0.30341	0.12703	0.13580	142.49	9.4836
450.00	0.10000	0.026945	37.113	76.817	80.529	0.31980	0.13376	0.14241	151.42	7.3132
500.00	0.10000	0.024188	41.342	83.661	87.796	0.33510	0.13956	0.14813	159.78	5.8366
300.00	1.0000	8.4984	0.11767	38.583	38.701	0.18633	0.11994	0.16950	551.87	-0.25320
350.00	1.0000	7.4651	0.13396	47.454	47.588	0.21369	0.12863	0.18840	364.95	0.11384
356.40	1.0000	7.3010	0.13697	48.670	48.807	0.21714	0.12986	0.19249	338.92	0.21213
356.40	1.0000	0.43061	2.3223	63.533	65.855	0.26498	0.12596	0.15128	108.84	20.165
400.00	1.0000	0.34682	2.8833	69.357	72.240	0.28188	0.12912	0.14474	126.86	11.867
450.00	1.0000	0.29218	3.4226	76.104	79.526	0.29904	0.13466	0.14726	141.14	8.1166
500.00	1.0000	0.25510	3.9201	83.069	86.989	0.31477	0.14003	0.15132	152.73	6.1412
300.00	5.0000	8.6001	0.11628	38.303	38.884	0.18538	0.11983	0.16738	584.86	-0.28910
350.00	5.0000	7.6832	0.13015	46.919	47.570	0.21213	0.12793	0.18096	420.00	-0.048046
400.00	5.0000	6.4329	0.15545	56.400	57.177	0.23775	0.13659	0.20814	251.68	0.75207
450.00	5.0000	3.3217	0.30105	69.251	70.756	0.26953	0.14905	0.34888	94.479	9.2719
500.00	5.0000	1.7533	0.57034	79.658	82.509	0.29444	0.14423	0.18709	121.74	7.0501
300.00	10.000	8.7118	0.11479	37.991	39.139	0.18430	0.11975	0.16542	621.41	-0.32232
350.00	10.000	7.8877	0.12678	46.398	47.666	0.21057	0.12746	0.17597	473.47	-0.15701
400.00	10.000	6.9197	0.14452	55.329	56.774	0.23488	0.13488	0.18894	338.62	0.18601
450.00	10.000	5.6865	0.17586	64.892	66.650	0.25812	0.14158	0.20710	225.09	1.0459
500.00	10.000	4.2030	0.23793	74.962	77.341	0.28064	0.14594	0.21449	163.22	2.5802
300.00	15.000	8.8106	0.11350	37.713	39.415	0.18332	0.11973	0.16394	654.14	-0.34729
350.00	15.000	8.0508	0.12421	45.970	47.834	0.20926	0.12720	0.17290	517.56	-0.22334
400.00	15.000	7.2160	0.13858	54.628	56.707	0.23294	0.13421	0.18200	398.49	-0.017272
450.00	15.000	6.2819	0.15919	63.643	66.030	0.25489	0.14025	0.19084	301.49	0.34360
500.00	15.000	5.2683	0.18981	72.897	75.744	0.27536	0.14487	0.19667	235.39	0.89448
300.00	20.000	8.8998	0.11236	37.460	39.707	0.18241	0.11973	0.16278	683.99	-0.36679
350.00	20.000	8.1881	0.12213	45.603	48.046	0.20811	0.12704	0.17079	555.75	-0.26854
400.00	20.000	7.4361	0.13448	54.086	56.775	0.23141	0.13385	0.17821	446.47	-0.12632
450.00	20.000	6.6421	0.15056	62.835	65.846	0.25277	0.13972	0.18438	358.35	0.081032
500.00	20.000	5.8257	0.17165	71.746	75.179	0.27243	0.14442	0.18846	294.31	0.35805

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Platzer, B., Polt, A., and Maurer, G., *Thermophysical Properties of Refrigerants*, Springer-Verlag, Berlin, 1990. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the critical point temperature are given in the last entry of the saturation tables. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainty in density is 0.2% up to 400 K and 1% at higher temperatures. The vapor pressure uncertainty is 1.5%. In the liquid phase, the uncertainty in isobaric heat capacity is 3%.

TABLE 2-273 Saturated Refrigerant 115, Chloropentafluoroethane*

Temp., °F	Pressure, lb/in ² abs.	Volume, ft ³ /lb		Enthalpy, Btu/lb		Entropy, Btu/(lb)(°F)	
		Liquid	Vapor	Liquid	Vapor	Liquid	Vapor
-100	2.327	0.00966	10.57	-13.07	45.83	-0.0335	0.1302
-80	4.573	0.00986	5.624	-8.78	48.39	-0.0219	0.1286
-60	8.306	0.01009	3.218	-4.43	50.96	-0.0108	0.1278
-40	14.13	0.01033	1.953	0.00	53.53	0.0000	0.1275
-20	22.74	0.01060	1.245	4.50	56.07	0.0104	0.1277
0	34.94	0.01090	0.8257	9.09	58.56	0.0206	0.1282
20	51.59	0.01123	0.5657	13.76	61.00	0.0305	0.1290
40	73.65	0.01161	0.3979	18.54	63.35	0.0401	0.1298
60	102.1	0.01204	0.2857	23.45	65.60	0.0496	0.1308
80	138.1	0.01255	0.2081	28.54	67.71	0.0591	0.1317
100	182.7	0.01316	0.1530	33.85	69.63	0.0686	0.1325
120	237.3	0.01393	0.1125	39.50	71.24	0.0782	0.1330
140	303.2	0.01496	0.0817	45.67	72.36	0.0884	0.1329
160	382.0	0.01664	0.0567	52.76	72.42	0.0996	0.1314
170	427.0	0.01838	0.0444	56.56	71.33	0.1055	0.1290
175.89°	457.6	0.0261	0.0261	64.30	64.30	0.1175	0.1175

*Unpublished data of General Chemicals Division, Allied Chemical Company. Used by permission. *c* = critical temperature.

No material in SI units appears in the 1993 ASHRAE *Handbook—Fundamentals* (SI ed.). Tables and a chart to 50 ata, 200 °C are given by Mathias, H. and H. J. Löffler, Techn. Univ. Berlin rept., 1966 (42 pp.). A chart to 1500 psia, 500 °F was given by Mears, W. H., E. Rosenthal, et al., *J. Chem. Eng. Data*, **11**, 3 (1966): 338–343.

TABLE 2-274 Thermodynamic Properties of R-116, Hexafluoroethane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_p kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
173.10	0.026084	12.304	0.081274	13.031	13.033	0.073325	0.083104	0.12322	653.29	-0.38383
175.00	0.029802	12.247	0.081649	13.266	13.268	0.074672	0.083413	0.12375	645.09	-0.37807
180.00	0.041661	12.097	0.082664	13.887	13.891	0.078175	0.084236	0.12518	623.59	-0.36202
185.00	0.057021	11.945	0.083720	14.516	14.521	0.081622	0.085077	0.12670	602.22	-0.34452
190.00	0.076558	11.789	0.084824	15.153	15.159	0.085018	0.085933	0.12829	580.95	-0.32540
195.00	0.10101	11.631	0.085979	15.797	15.806	0.088366	0.086803	0.12996	559.75	-0.30443
200.00	0.13116	11.469	0.087192	16.450	16.461	0.091671	0.087686	0.13173	538.61	-0.28132
205.00	0.16784	11.303	0.088469	17.111	17.126	0.094937	0.088580	0.13361	517.48	-0.25573
210.00	0.21194	11.134	0.089818	17.781	17.800	0.098169	0.089485	0.13560	496.36	-0.22722
215.00	0.26438	10.959	0.091248	18.461	18.485	0.10137	0.090402	0.13773	475.20	-0.19525
220.00	0.32611	10.779	0.092769	19.151	19.181	0.10454	0.091330	0.14002	453.97	-0.15913
225.00	0.39814	10.594	0.094395	19.851	19.889	0.10769	0.092270	0.14249	432.64	-0.11798
230.00	0.48150	10.401	0.096140	20.563	20.609	0.11082	0.093224	0.14520	411.18	-0.070666
235.00	0.57725	10.201	0.098026	21.286	21.343	0.11394	0.094193	0.14818	389.53	-0.015673
240.00	0.68650	9.9926	0.10007	22.023	22.092	0.11705	0.095179	0.15152	367.64	0.049034
245.00	0.81042	9.7735	0.10232	22.774	22.857	0.12015	0.096188	0.15531	345.46	0.12626
250.00	0.95020	9.5423	0.10480	23.541	23.640	0.12326	0.097226	0.15970	322.91	0.21998
255.00	1.1071	9.2968	0.10756	24.325	24.444	0.12638	0.098301	0.16491	299.89	0.33604
260.00	1.2826	9.0336	0.11070	25.130	25.272	0.12952	0.099429	0.17129	276.28	0.48338
265.00	1.4781	8.7482	0.11431	25.959	26.128	0.13269	0.10063	0.17942	251.91	0.67638
270.00	1.6952	8.4341	0.11857	26.818	27.019	0.13593	0.10195	0.19040	226.54	0.93994
275.00	1.9360	8.0803	0.12376	27.715	27.955	0.13926	0.10345	0.20651	199.81	1.3211
280.00	2.2027	7.6671	0.13043	28.667	28.954	0.14274	0.10525	0.23354	171.13	1.9215
285.00	2.4983	7.1502	0.13986	29.710	30.059	0.14651	0.10768	0.29195	139.42	3.0109
290.00	2.8276	6.3727	0.15692	30.973	31.417	0.15106	0.11183	0.54643	102.12	5.6531
293.03	3.0477	4.4440	0.22502	33.029	33.715	0.15879			0	13.224
173.10	0.026084	0.018437	54.238	28.993	30.408	0.17370	0.067328	0.076383	106.90	43.932
175.00	0.029802	0.020874	47.907	29.113	30.541	0.17338	0.067932	0.077064	107.29	42.424
180.00	0.041661	0.028519	35.064	29.433	30.893	0.17263	0.069527	0.078887	108.24	38.839
185.00	0.057021	0.038215	26.168	29.755	31.247	0.17203	0.071130	0.080765	109.07	35.743
190.00	0.076558	0.050323	19.872	30.079	31.601	0.17155	0.072742	0.082706	109.78	33.062
195.00	0.10101	0.065236	15.329	30.406	31.954	0.17118	0.074365	0.084719	110.35	30.736
200.00	0.13116	0.083387	11.992	30.733	32.306	0.17090	0.076000	0.086816	110.78	28.716
205.00	0.16784	0.10525	9.5011	31.061	32.656	0.17069	0.077648	0.089011	111.05	26.960
210.00	0.21194	0.13135	7.6132	31.389	33.003	0.17056	0.079312	0.091322	111.15	25.435
215.00	0.26438	0.16228	6.1624	31.716	33.345	0.17049	0.080992	0.093772	111.07	24.112
220.00	0.32611	0.19868	5.0331	32.041	33.683	0.17046	0.082691	0.096389	110.80	22.967
225.00	0.39814	0.24133	4.1437	32.364	34.014	0.17047	0.084412	0.099211	110.33	21.983
230.00	0.48150	0.29110	3.4353	32.682	34.336	0.17051	0.086158	0.10229	109.63	21.143
235.00	0.57725	0.34902	2.8652	32.995	34.649	0.17056	0.087932	0.10569	108.69	20.435
240.00	0.68650	0.41636	2.4018	33.302	34.950	0.17063	0.089742	0.10951	107.50	19.850
245.00	0.81042	0.49465	2.0216	33.599	35.237	0.17068	0.091594	0.11388	106.02	19.381
250.00	0.95020	0.58587	1.7069	33.885	35.507	0.17072	0.093497	0.11901	104.24	19.024
255.00	1.1071	0.69256	1.4439	34.157	35.755	0.17073	0.095465	0.12520	102.13	18.777
260.00	1.2826	0.81820	1.2222	34.409	35.977	0.17069	0.097518	0.13296	99.661	18.637
265.00	1.4781	0.96766	1.0334	34.636	36.164	0.17056	0.099682	0.14316	96.789	18.605
270.00	1.6952	1.1482	0.87090	34.830	36.306	0.17033	0.10200	0.15741	93.477	18.677
275.00	1.9360	1.3717	0.72904	34.975	36.387	0.16992	0.10454	0.17918	89.681	18.837
280.00	2.2027	1.6592	0.60271	35.047	36.375	0.16924	0.10743	0.21720	85.354	19.040
285.00	2.4983	2.0572	0.48610	34.992	36.207	0.16807	0.11094	0.30201	80.445	19.127
290.00	2.8276	2.7232	0.36721	34.631	35.669	0.16572	0.11584	0.66697	74.869	18.438
293.03	3.0477	4.4440	0.22502	33.029	33.715	0.15879			0	13.224

Single-Phase Properties

175.00	0.10000	12.249	0.081637	13.263	13.271	0.074658	0.083417	0.12373	645.46	-0.37835
194.81	0.10000	11.637	0.085935	15.773	15.782	0.088242	0.086771	0.12990	560.54	-0.30525
194.81	0.10000	0.064625	15.474	30.394	31.941	0.17119	0.074304	0.084642	110.33	30.817
250.00	0.10000	0.049033	20.394	34.892	36.932	0.19368	0.087457	0.096533	126.48	13.742
325.00	0.10000	0.037293	26.815	42.103	44.784	0.22106	0.10389	0.11253	144.51	6.5149
400.00	0.10000	0.030174	33.141	50.430	53.744	0.24582	0.11740	0.12590	160.19	3.7890
175.00	1.0000	12.272	0.081484	13.232	13.314	0.074480	0.083472	0.12349	650.25	-0.38185
250.00	1.0000	9.5480	0.10473	23.534	23.638	0.12323	0.097224	0.15951	323.69	0.21614
251.65	1.0000	9.4631	0.10567	23.797	23.903	0.12428	0.097576	0.16132	315.37	0.25543
251.65	1.0000	0.61920	1.6150	33.976	35.591	0.17073	0.094138	0.12091	103.59	18.931
325.00	1.0000	0.40237	2.4853	41.594	44.080	0.20033	0.10529	0.11782	135.84	7.1157
400.00	1.0000	0.31155	3.2097	50.096	53.305	0.22584	0.11796	0.12829	156.26	3.8772
250.00	5.0000	9.9221	0.10078	23.073	23.577	0.12134	0.097206	0.14935	376.03	0.0088070
325.00	5.0000	3.7219	0.26868	37.500	38.843	0.17386	0.11497	0.24037	101.35	7.5484
400.00	5.0000	1.7961	0.55676	48.386	51.170	0.20825	0.12060	0.14444	144.06	3.8425
250.00	10.000	10.261	0.097459	22.648	23.622	0.11954	0.097376	0.14319	425.10	-0.12130
325.00	10.000	7.1151	0.14055	34.117	35.522	0.16098	0.11174	0.17544	207.64	1.0712
400.00	10.000	3.9471	0.25335	46.132	48.666	0.19742	0.12255	0.16440	156.43	2.4329
250.00	20.000	10.747	0.093053	22.031	23.892	0.11682	0.097852	0.13725	498.98	-0.24812
325.00	20.000	8.5892	0.11643	32.498	34.826	0.15497	0.11132	0.15339	328.94	0.093523
400.00	20.000	6.4948	0.15397	43.572	46.651	0.18769	0.12291	0.16013	246.48	0.50074
250.00	50.000	11.642	0.085893	20.902	25.197	0.11135	0.099289	0.13120	647.29	-0.37511
325.00	50.000	10.197	0.098067	30.607	35.511	0.14736	0.11250	0.14332	516.00	-0.28102
400.00	50.000	8.8968	0.11240	40.984	46.604	0.17805	0.12396	0.15192	437.11	-0.20643

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., and Span, R., “Short Fundamental Equations of State for 20 Industrial Fluids,” *J. Chem. Eng. Data* **51**(3):785–850, 2006. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in the equation are 0.5% in density for liquid and vapor states and 1% in density or pressure for supercritical states. For vapor pressure, the uncertainty is 0.3%, that for vapor-phase speed of sounds is 0.2%, and the uncertainty for heat capacities is 5%.

TABLE 2-275 Thermodynamic Properties of R-123, 2,2-Dichloro-1,1,1-Trifluoroethane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_v kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
166.00	4.2021E-06	11.580	0.086355	15.111	15.111	0.081219	0.096271	0.14206	1243.8	-0.47555	115.89	4953.8
170.00	7.5115E-06	11.520	0.086807	15.678	15.678	0.084599	0.096537	0.14181	1227.9	-0.47661	114.18	4180.7
185.00	5.1172E-05	11.298	0.088511	17.800	17.800	0.096559	0.096881	0.14120	1167.0	-0.47672	109.51	2571.4
200.00	0.00024954	11.079	0.090257	19.919	19.919	0.10757	0.097283	0.14151	1104.3	-0.47027	105.59	1795.0
215.00	0.00093965	10.861	0.092076	22.050	22.050	0.11785	0.098162	0.14272	1040.9	-0.45811	101.48	1337.4
230.00	0.0028868	10.640	0.093989	24.204	24.204	0.12753	0.099479	0.14455	977.54	-0.44149	97.035	1036.8
245.00	0.0075380	10.415	0.096017	26.388	26.389	0.13673	0.10110	0.14676	914.89	-0.42124	92.395	825.41
260.00	0.017260	10.185	0.098182	28.607	28.609	0.14552	0.10291	0.14919	853.42	-0.39765	87.716	669.88
275.00	0.035500	9.9497	0.10051	30.863	30.867	0.15395	0.10483	0.15176	793.34	-0.37048	83.129	551.59
290.00	0.066848	9.7073	0.10302	33.158	33.165	0.16208	0.10680	0.15443	734.70	-0.33894	78.719	459.38
305.00	0.11700	9.4567	0.10575	35.492	35.504	0.16993	0.10879	0.15722	677.42	-0.30164	74.533	386.07
320.00	0.19264	9.1962	0.10874	37.867	37.888	0.17753	0.11080	0.16020	621.30	-0.25640	70.585	326.86
335.00	0.30136	8.9238	0.11206	40.286	40.319	0.18492	0.11283	0.16350	566.09	-0.19988	66.863	278.37
350.00	0.45147	8.6365	0.11579	42.751	42.804	0.19212	0.11491	0.16729	511.47	-0.12681	63.340	238.12
365.00	0.65201	8.3301	0.12005	45.270	45.349	0.19917	0.11704	0.17186	457.07	-0.028618	59.972	204.30
380.00	0.91268	7.9984	0.12502	47.853	47.967	0.20612	0.11927	0.17772	402.48	0.10963	56.709	175.46
395.00	1.2440	7.6316	0.13103	50.514	50.677	0.21300	0.12167	0.18579	347.23	0.31613	53.486	150.45
410.00	1.6577	7.2132	0.13863	53.281	53.510	0.21990	0.12433	0.19804	290.69	0.65103	50.222	128.23
425.00	2.1672	6.7113	0.14900	56.204	56.527	0.22695	0.12747	0.21999	231.85	1.2687	46.810	107.74
440.00	2.7898	6.0457	0.16541	59.412	59.873	0.23446	0.13159	0.27613	168.40	2.7240	43.102	87.306
455.00	3.5547	4.6936	0.21305	63.738	64.496	0.24446	0.13975	1.2865	90.823	10.053	39.324	58.622
456.83	3.6619	3.5964	0.27805	65.873	66.891	0.24965			0	16.566		
166.00	4.2021E-06	3.0446E-06	328,450.	47.940	49.320	0.28730	0.064141	0.072457	100.97	335.67	1.6736	5.5300
170.00	7.5115E-06	5.3144E-06	188,170.	48.198	49.612	0.28421	0.065216	0.073532	102.08	305.67	1.9014	5.7017
185.00	5.1172E-05	3.3272E-05	30,055.	49.206	50.744	0.27463	0.069152	0.077473	106.14	219.47	2.7552	6.3386
200.00	0.00024954	0.00015013	6,660.7	50.269	51.931	0.26763	0.072951	0.081290	110.02	162.12	3.6080	6.9646
215.00	0.00093965	0.00052637	1,899.8	51.386	53.171	0.26259	0.076634	0.085017	113.72	122.83	4.4590	7.5791
230.00	0.0028868	0.0015145	660.29	52.550	54.456	0.25906	0.080226	0.088699	117.20	95.252	5.3073	8.1817
245.00	0.0075380	0.0037253	268.44	53.755	55.779	0.25669	0.083755	0.092390	120.41	75.497	6.1533	8.7712
260.00	0.017260	0.0080830	123.72	54.996	57.131	0.25522	0.087245	0.096144	123.27	61.102	6.9987	9.3465
275.00	0.035500	0.015852	63.083	56.264	58.504	0.25445	0.090715	0.10001	125.72	50.463	7.8477	9.9065
290.00	0.066848	0.028647	34.907	57.553	59.887	0.25422	0.094177	0.10405	127.66	42.504	8.7073	10.450
305.00	0.11700	0.048445	20.642	58.855	61.270	0.25440	0.097639	0.10832	129.01	36.493	9.5872	10.979
320.00	0.19264	0.077628	12.882	60.163	62.645	0.25489	0.10111	0.11290	129.68	31.923	10.500	11.494
335.00	0.30136	0.11198	8.3975	61.468	63.999	0.25560	0.10459	0.11790	129.58	28.442	11.463	12.002
350.00	0.45147	0.17640	5.6690	62.760	65.319	0.25645	0.10810	0.12353	128.62	25.804	12.497	12.515
365.00	0.65201	0.25423	3.9334	64.026	66.591	0.25737	0.11165	0.13011	126.67	23.846	13.627	13.054
380.00	0.91268	0.35900	2.7855	65.252	67.794	0.25829	0.11529	0.13821	123.57	22.466	14.889	13.656
395.00	1.2440	0.50026	1.9990	66.412	68.899	0.25913	0.11908	0.14902	119.13	21.617	16.332	14.390
410.00	1.6577	0.69385	1.4412	67.468	69.857	0.25977	0.12313	0.16522	113.03	21.309	18.039	15.384
425.00	2.1672	0.97042	1.0305	68.348	70.581	0.26002	0.12767	0.19466	104.85	21.605	20.168	16.913
440.00	2.7898	1.4081	0.71016	68.866	70.847	0.25940	0.13325	0.27310	93.850	22.561	23.176	19.744
455.00	3.5547	2.5178	0.39718	67.881	69.293	0.25500	0.14227	1.6671	78.287	21.950	30.870	29.346
456.83	3.6619	3.5964	0.27805	65.873	66.891	0.24965			0	16.566		

Single-Phase Properties

200.00	0.10000	11.080	0.090251	19.917	19.926	0.10756	0.097300	0.14151	1104.5	-0.47033	105.61	1797.1
300.00	0.10000	9.5412	0.10481	34.709	34.720	0.16734	0.10812	0.15627	696.38	-0.31484	75.903	408.75
300.61	0.10000	9.5309	0.10492	34.805	34.815	0.16766	0.10820	0.15639	694.05	-0.31326	75.734	405.88
300.61	0.10000	0.041795	23.926	58.473	60.866	0.25432	0.096626	0.10704	128.68	38.084	9.3269	10.826
400.00	0.10000	0.030530	32.754	68.902	72.178	0.28673	0.11170	0.12066	150.93	14.728	15.085	14.262
500.00	0.10000	0.024219	41.290	80.750	84.879	0.31502	0.12437	0.13300	169.33	7.9263	20.780	17.224
600.00	0.10000	0.020111	49.724	93.743	98.716	0.34022	0.13502	0.14352	185.60	5.0504	26.460	19.696
200.00	1.0000	11.087	0.090194	19.896	19.986	0.10745	0.097447	0.14145	1105.5	-0.47094	105.80	1815.8
300.00	1.0000	9.5586	0.10462	34.660	34.764	0.16717	0.10822	0.15597	701.57	-0.31949	76.273	413.78
384.30	1.0000	7.8974	0.12662	48.607	48.733	0.20809	0.11994	0.17975	386.73	0.16009	55.784	167.94
384.30	1.0000	0.39525	2.5301	65.592	68.122	0.25855	0.11636	0.14096	122.45	22.169	15.282	13.849
400.00	1.0000	0.36390	2.7480	67.553	70.301	0.26410	0.11698	0.13715	129.70	18.389	16.135	14.368
500.00	1.0000	0.25890	3.8625	80.024	83.887	0.29441	0.12571	0.13778	160.11	8.3865	21.585	17.431
600.00	1.0000	0.20738	4.8220	93.232	98.054	0.32022	0.13546	0.14581	181.06	5.1181	27.069	19.994
200.00	5.0000	11.118	0.089944	19.803	20.253	0.10699	0.098069	0.14121	1110.5	-0.47345	106.62	1902.1
300.00	5.0000	9.6324	0.10382	34.450	34.969	0.16646	0.10865	0.15478	723.64	-0.33792	77.874	436.12
400.00	5.0000	7.7754	0.12861	50.791	51.434	0.21368	0.12180	0.17810	388.06	0.14142	55.285	160.58
500.00	5.0000	2.1822	0.45826	74.934	77.225	0.27035	0.13698	0.23468	111.01	11.061	30.182	28.291
600.00	5.0000	1.2117	0.82529	90.757	94.883	0.30270	0.13749	0.16051	161.94	5.1664	30.009	24.445
200.00	10.000	11.156	0.089639	19.691	20.588	0.10642	0.098779	0.14095	1117.0	-0.47622	107.63	2017.3
300.00	10.000	9.7177	0.10290	34.208	35.237	0.16564	0.10912	0.15357	749.33	-0.35684	79.793	464.14
400.00	10.000	8.0432	0.12433	50.152	51.395	0.21200	0.12141	0.17093	447.83	-0.033773	58.469	181.18
500.00	10.000	5.5005	0.18180	68.419	70.237	0.25388	0.13402	0.21202	205.61	1.7914	41.785	78.103
600.00	10.000	2.8283	0.35357	87.359	90.895	0.29164	0.13939	0.18482	163.33	3.7808	35.942	39.777
200.00	20.000	11.229	0.089054	19.480	21.261	0.10532	0.099998	0.14055	1130.7	-0.48062	109.63	2276.1
300.00	20.000	9.8704	0.10131	33.778	35.804	0.16412	0.10993	0.15175	795.84	-0.38528	83.416	521.04
400.00	20.000	8.4163	0.11882	49.229	51.605	0.20949	0.12134	0.16430	534.29	-0.19523	63.653	217.09
500.00	20.000	6.7515	0.14812	65.791	68.753	0.24770	0.13081	0.17889	348.69	0.27303	49.919	115.74
600.00	20.000	4.9944	0.20022	83.034	87.039	0.28102	0.13822	0.18342	257.14	0.96765	41.677	74.025
200.00	40.000	11.368	0.087968	19.099	22.618	0.10325	0.10179	0.14011	1160.1	-0.48571	113.48	2947.0
300.00	40.000	10.125	0.098764	33.072	37.023	0.16152	0.11120	0.14952	875.49	-0.42067	90.039	640.89
400.00	40.000	8.9071	0.11227	47.981	52.472	0.20590	0.12199	0.15901	653.37	-0.32210	72.037	282.73
500.00	40.000	7.7025	0.12983	63.586	68.779	0.24225	0.13053	0.16697	503.02	-0.17413	60.777	165.05
600.00	40.000	6.5412	0.15288	79.686	85.801	0.27327	0.13791	0.17293	409.64	0.011715	52.389	115.87

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Younglove, B. A., and McLinden, M. O., “An International Standard Equation of State for the Thermodynamic Properties of Refrigerant 123 (2,2-Dichloro-1,1,1-Trifluoroethane),” *J. Phys. Chem. Ref. Data* **23**:731–779, 1994. The source for viscosity is Tanaka, Y. and Sotani, T., “Transport Properties (Thermal Conductivity and Viscosity),” in McLinden, M. O., Ed., *R123—Thermodynamic and Physical Properties*, Paris: International Institute of Refrigeration, 1995. See also *Int. J. Thermophys.* **17**(2):293–328, 1996. The source for thermal conductivity is Laesecke, A., Perkins, R. A., and Howley, J. B., “An Improved Correlation for the Thermal Conductivity of HCFC123 (2,2-Dichloro-1,1,1-Trifluoroethane),” *Int. J. Refrigeration* **19**:231–238, 1996.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties of the equation of state are 0.1% in density, 1.5% in heat capacity, and 2% in the speed of sound, except in the critical region. The uncertainty in vapor pressure is 0.1%. Uncertainties are greater below 180 K. The uncertainty in viscosity is 5%. The uncertainty in thermal conductivity is 2%.

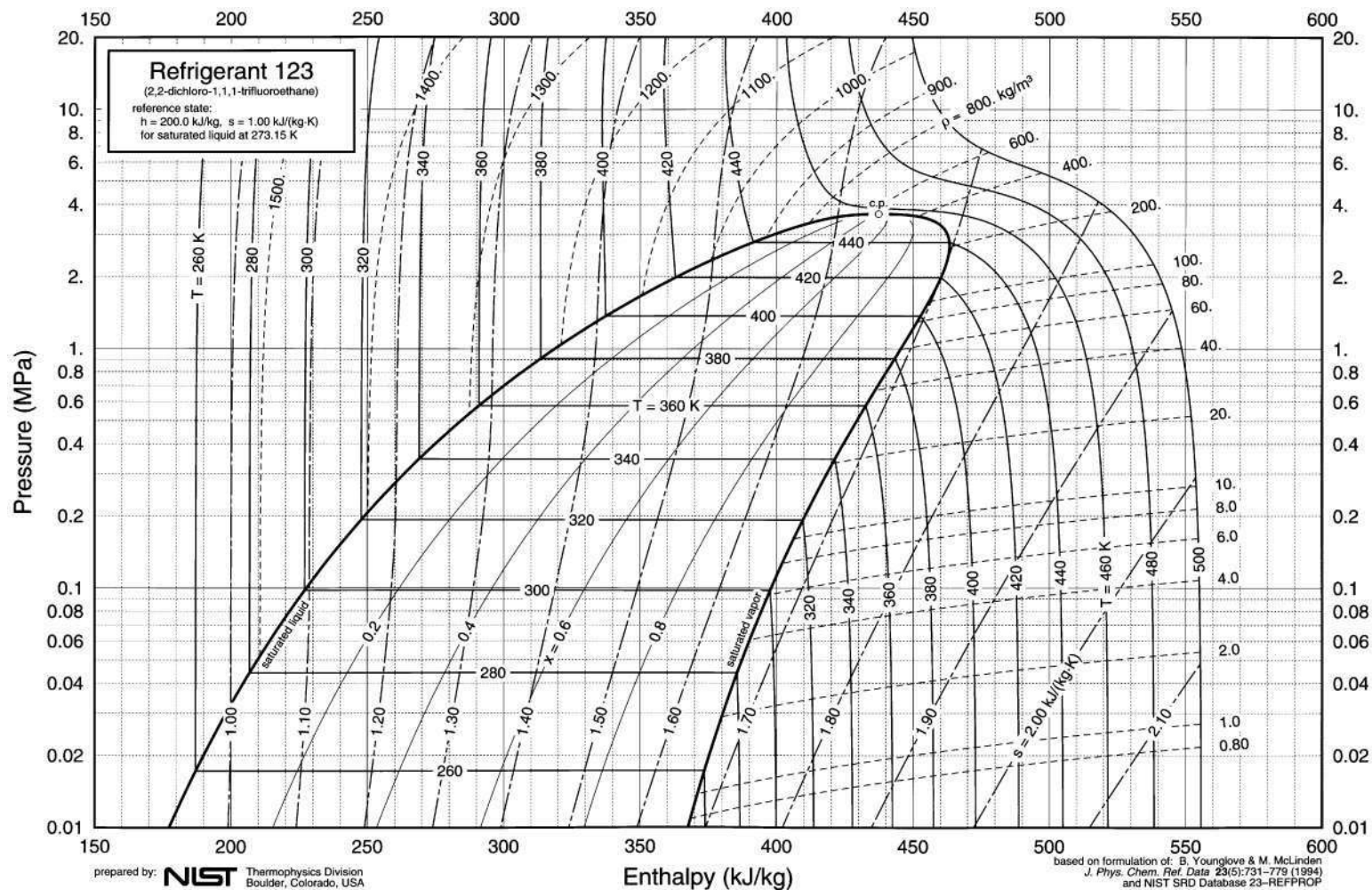


FIG. 2-24 Enthalpy-log-pressure diagram for Refrigerant 123.

TABLE 2-276 Thermodynamic Properties of R-124, 2-Chloro-1,1,1,2-Tetrafluoroethane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _p kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
120.00	2.6739E-08	13.576	0.073661	6.7015	6.7015	0.027106	0.088123	0.12662	1293.6	-0.48257
135.00	8.0309E-07	13.288	0.075255	8.6014	8.6014	0.042024	0.087495	0.12684	1225.8	-0.47832
150.00	1.1223E-05	13.003	0.076906	10.510	10.510	0.055431	0.087696	0.12778	1157.4	-0.47012
165.00	9.0881E-05	12.718	0.078626	12.437	12.437	0.067673	0.088451	0.12919	1089.0	-0.45872
180.00	0.00049265	12.433	0.080432	14.388	14.388	0.078987	0.089589	0.13095	1021.3	-0.44440
195.00	0.0019732	12.145	0.082339	16.367	16.367	0.089547	0.091000	0.13299	954.41	-0.42707
210.00	0.0062626	11.853	0.084369	18.378	18.379	0.099484	0.092610	0.13527	888.65	-0.40640
225.00	0.016573	11.554	0.086547	20.426	20.427	0.10890	0.094368	0.13778	824.09	-0.38173
240.00	0.037986	11.248	0.088904	22.512	22.515	0.11787	0.096239	0.14055	760.75	-0.35205
255.00	0.077612	10.931	0.091481	24.641	24.648	0.12648	0.098199	0.14361	698.57	-0.31583
270.00	0.14453	10.601	0.094332	26.816	26.830	0.13477	0.10024	0.14702	637.46	-0.27079
285.00	0.24955	10.253	0.097528	29.043	29.067	0.14280	0.10235	0.15090	577.23	-0.21337
300.00	0.40500	9.8843	0.10117	31.327	31.368	0.15061	0.10454	0.15545	517.54	-0.13780
315.00	0.62447	9.4868	0.10541	33.677	33.743	0.15826	0.10685	0.16102	457.87	-0.034029
330.00	0.92279	9.0506	0.11049	36.106	36.208	0.16580	0.10933	0.16825	397.44	0.11699
345.00	1.3163	8.5590	0.11684	38.634	38.788	0.17332	0.11213	0.17860	335.05	0.35590
360.00	1.8234	7.9793	0.12532	41.302	41.531	0.18092	0.11553	0.19587	269.18	0.78492
375.00	2.4663	7.2348	0.13822	44.202	44.543	0.18889	0.12005	0.23450	198.17	1.7419
390.00	3.2770	6.0066	0.16648	47.714	48.259	0.19827	0.12812	0.46802	115.48	5.4469
395.43	3.6243	4.1033	0.24371	50.813	51.696	0.20684			0	14.607
120.00	2.6739E-08	2.6799E-08	37,314,000.	36.727	37.725	0.28563	0.051678	0.059992	92.125	997.50
135.00	8.0309E-07	7.1548E-07	1,397,700.	37.530	38.653	0.26462	0.055405	0.063720	97.256	558.90
150.00	1.1223E-05	8.9989E-06	111,120.	38.388	39.636	0.24960	0.059044	0.067361	102.10	338.42
165.00	9.0881E-05	6.6260E-05	15,092.	39.300	40.671	0.23879	0.062605	0.070933	106.70	218.53
180.00	0.00049265	0.00032946	3,035.3	40.262	41.757	0.23104	0.066113	0.074474	111.05	149.08
195.00	0.0019732	0.0012200	819.64	41.269	42.887	0.22555	0.069600	0.078043	115.13	106.83
210.00	0.0062626	0.0036077	277.19	42.318	44.054	0.22175	0.073111	0.081721	118.89	80.088
225.00	0.016573	0.0089652	111.54	43.400	45.248	0.21922	0.076698	0.085612	122.21	62.512
240.00	0.037986	0.019455	51.400	44.507	46.459	0.21764	0.080420	0.089837	124.99	50.458
255.00	0.077612	0.037965	26.340	45.631	47.675	0.21678	0.084321	0.094522	127.10	41.824
270.00	0.14453	0.068150	14.674	46.762	48.883	0.21645	0.088420	0.099789	128.40	35.421
285.00	0.24955	0.11457	8.7285	47.892	50.070	0.21649	0.092708	0.10578	128.75	30.583
300.00	0.40500	0.18300	5.4645	49.009	51.222	0.21679	0.097166	0.11271	128.00	26.921
315.00	0.62447	0.28114	3.5569	50.100	52.321	0.21724	0.10179	0.12102	125.97	24.195
330.00	0.92279	0.42003	2.3808	51.145	53.342	0.21772	0.10663	0.13162	122.47	22.252
345.00	1.3163	0.61719	1.6202	52.111	54.244	0.21811	0.11180	0.14664	117.17	21.001
360.00	1.8234	0.90480	1.1052	52.940	54.955	0.21821	0.11757	0.17189	109.67	20.405
375.00	2.4663	1.3577	0.73653	53.497	55.314	0.21761	0.12466	0.23071	99.240	20.431
390.00	3.2770	2.2903	0.43662	53.233	54.664	0.21470	0.13608	0.61311	84.410	20.196
395.43	3.6243	4.1033	0.24371	50.813	51.696	0.20684			0	14.607

TABLE 2-276 Thermodynamic Properties of R-124, 2-Chloro-1,1,1,2-Tetrafluoroethane (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Single-Phase Properties										
150.00	0.10000	13.004	0.076901	10.509	10.516	0.055420	0.087698	0.12777	1157.7	-0.47016
225.00	0.10000	11.556	0.086537	20.423	20.431	0.10889	0.094369	0.13777	824.54	-0.38192
260.87	0.10000	10.804	0.092562	25.487	25.496	0.12976	0.098988	0.14489	674.53	-0.29941
260.87	0.10000	0.048164	20.763	46.073	48.150	0.21660	0.085902	0.096509	127.71	39.096
300.00	0.10000	0.041129	24.314	49.579	52.011	0.23038	0.092039	0.10148	138.33	22.359
375.00	0.10000	0.032447	30.820	57.006	60.088	0.25436	0.10524	0.11403	155.52	10.936
450.00	0.10000	0.026891	37.187	65.395	69.114	0.27627	0.11804	0.12661	170.43	6.3680
150.00	1.0000	13.011	0.076859	10.494	10.570	0.055319	0.087712	0.12773	1160.8	-0.47053
225.00	1.0000	11.570	0.086432	20.392	20.479	0.10875	0.094382	0.13761	829.31	-0.38397
300.00	1.0000	9.9082	0.10093	31.280	31.381	0.15045	0.10452	0.15494	523.28	-0.14686
333.28	1.0000	8.9488	0.11175	36.649	36.760	0.16744	0.10991	0.17017	384.03	0.15954
333.28	1.0000	0.45731	2.1867	51.364	53.551	0.21782	0.10772	0.13441	121.47	21.922
375.00	1.0000	0.36697	2.7250	56.110	58.835	0.23277	0.10870	0.12428	140.23	12.660
450.00	1.0000	0.28511	3.5074	64.848	68.356	0.25590	0.11949	0.13091	162.32	6.7329
150.00	5.0000	13.042	0.076674	10.428	10.811	0.054877	0.087777	0.12756	1174.3	-0.47207
225.00	5.0000	11.630	0.085982	20.262	20.692	0.10817	0.094440	0.13696	849.95	-0.39231
300.00	5.0000	10.055	0.099451	30.991	31.488	0.14947	0.10448	0.15210	558.86	-0.19712
375.00	5.0000	7.8072	0.12809	43.322	43.963	0.18645	0.11692	0.19006	276.48	0.71939
450.00	5.0000	2.0764	0.48161	61.347	63.755	0.23451	0.12778	0.18398	124.73	8.2236
150.00	10.000	13.080	0.076452	10.348	11.113	0.054334	0.087856	0.12736	1190.9	-0.47382
225.00	10.000	11.702	0.085459	20.109	20.963	0.10747	0.094513	0.13626	874.54	-0.40129
300.00	10.000	10.213	0.097912	30.675	31.654	0.14838	0.10449	0.14957	597.87	-0.24225
375.00	10.000	8.3628	0.11958	42.388	43.584	0.18379	0.11566	0.17088	359.58	0.22172
450.00	10.000	5.4895	0.18217	56.064	57.886	0.21844	0.12834	0.21040	179.08	2.3542
150.00	20.000	13.153	0.076029	10.197	11.717	0.053282	0.088011	0.12702	1222.7	-0.47677
225.00	20.000	11.832	0.084516	19.827	21.518	0.10615	0.094657	0.13514	920.41	-0.41563
300.00	20.000	10.473	0.095482	30.150	32.059	0.14651	0.10459	0.14630	664.60	-0.30098
375.00	20.000	8.9796	0.11136	41.285	43.513	0.18054	0.11536	0.15949	462.43	-0.080024
450.00	20.000	7.2897	0.13718	53.276	56.020	0.21090	0.12652	0.17382	319.21	0.35516
150.00	40.000	13.287	0.075260	9.9211	12.932	0.051293	0.088302	0.12653	1282.4	-0.48099
225.00	40.000	12.057	0.082941	19.345	22.662	0.10380	0.094922	0.13366	1002.2	-0.43482
300.00	40.000	10.865	0.092042	29.347	33.028	0.14350	0.10483	0.14289	772.84	-0.36270
375.00	40.000	9.6799	0.10331	39.971	44.103	0.17641	0.11550	0.15243	600.05	-0.26440
450.00	40.000	8.5132	0.11746	51.192	55.890	0.20504	0.12642	0.16186	479.46	-0.14030

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is de Vries, B., Tillner-Roth, R., and Baehr, H. D., "Thermodynamic Properties of HCFC 124," *19th Int. Congress of Refrigeration*, The Hague, The Netherlands, International Institute of Refrigeration, IVa:582–589, 1995. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties of the equation of state are 0.05% in density, 1% in heat capacity, and 1% in the speed of sound, except in the critical region. The uncertainty in vapor pressure is 0.1%.

TABLE 2-277 Thermodynamic Properties of R-125, Pentafluoroethane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
172.52	0.0029140	14.086	0.070990	10.457	10.457	0.058837	0.081329	0.12417	932.57	-0.38374	116.02	1152.4
180.00	0.0056285	13.885	0.072020	11.389	11.389	0.064124	0.082012	0.12500	893.63	-0.37406	112.52	957.54
190.00	0.012328	13.613	0.073461	12.646	12.647	0.070919	0.083102	0.12647	843.11	-0.35818	107.79	768.40
200.00	0.024602	13.336	0.074988	13.919	13.921	0.077448	0.084327	0.12825	793.91	-0.33901	103.06	631.00
210.00	0.045417	13.052	0.076615	15.210	15.214	0.083750	0.085644	0.13028	745.67	-0.31627	98.331	527.00
220.00	0.078505	12.762	0.078360	16.523	16.529	0.089856	0.087029	0.13254	698.07	-0.28935	93.653	445.76
230.00	0.12833	12.461	0.080247	17.858	17.869	0.095792	0.088472	0.13505	650.89	-0.25723	89.019	380.67
240.00	0.20004	12.150	0.082305	19.219	19.235	0.10158	0.089971	0.13785	603.92	-0.21839	84.443	327.41
250.00	0.29934	11.824	0.084572	20.607	20.632	0.10725	0.091529	0.14102	556.99	-0.17062	79.940	283.01
260.00	0.43250	11.481	0.087098	22.025	22.063	0.11282	0.093153	0.14468	510.02	-0.11056	75.520	245.39
270.00	0.60624	11.117	0.089954	23.478	23.532	0.11830	0.094843	0.14903	462.91	-0.032921	71.187	213.02
280.00	0.82782	10.724	0.093245	24.971	25.048	0.12374	0.096594	0.15440	415.46	0.070972	66.940	184.73
290.00	1.1050	10.295	0.097130	26.511	26.619	0.12916	0.098430	0.16135	367.36	0.21606	62.772	159.60
300.00	1.4463	9.8162	0.10187	28.112	28.259	0.13461	0.10043	0.17099	318.17	0.43036	58.667	136.86
310.00	1.8610	9.2637	0.10795	29.793	29.994	0.14015	0.10274	0.18593	267.31	0.77370	54.597	115.81
320.00	2.3600	8.5923	0.11638	31.595	31.869	0.14593	0.10571	0.21395	213.55	1.4029	50.534	95.602
330.00	2.9579	7.6744	0.13030	33.632	34.017	0.15231	0.11043	0.29625	153.34	2.9184	46.661	74.602
339.17	3.6179	4.7790	0.20925	37.417	38.174	0.16438			0	12.361		
172.52	0.0029140	0.0020381	490.65	31.863	33.293	0.19120	0.059815	0.068285	116.43	90.257	5.2349	7.4339
180.00	0.0056285	0.0037809	264.49	32.307	33.795	0.18860	0.061648	0.070217	118.54	77.516	5.7185	7.7624
190.00	0.012328	0.0078784	126.93	32.913	34.477	0.18582	0.064126	0.072893	121.15	64.018	6.3724	8.1999
200.00	0.024602	0.015031	66.529	33.530	35.167	0.18368	0.066646	0.075712	123.47	53.589	7.0353	8.6344
210.00	0.045417	0.026661	37.508	34.157	35.860	0.18207	0.069223	0.078713	125.44	45.456	7.7081	9.0657
220.00	0.078505	0.044514	22.465	34.788	36.552	0.18087	0.071864	0.081939	127.01	39.066	8.3929	9.4944
230.00	0.12833	0.070679	14.148	35.421	37.237	0.18000	0.074575	0.085437	128.11	34.014	9.0929	9.9221
240.00	0.20004	0.10763	9.2907	36.052	37.911	0.17940	0.077362	0.089271	128.68	29.998	9.8136	10.353
250.00	0.29934	0.15835	6.3153	36.678	38.568	0.17900	0.080230	0.093526	128.64	26.787	10.563	10.791
260.00	0.43250	0.22645	4.4159	37.292	39.202	0.17874	0.083141	0.098283	127.93	24.232	11.356	11.246
270.00	0.60624	0.31661	3.1585	37.890	39.805	0.17857	0.086003	0.10368	126.44	22.293	12.213	11.732
280.00	0.82782	0.43510	2.2983	38.460	40.363	0.17844	0.088869	0.11025	124.10	20.938	13.169	12.266
290.00	1.1050	0.59084	1.6925	38.988	40.858	0.17826	0.092050	0.11918	120.81	20.043	14.286	12.884
300.00	1.4463	0.79742	1.2540	39.453	41.267	0.17797	0.095933	0.13255	116.42	19.438	15.680	13.638
310.00	1.8610	1.0777	0.92787	39.828	41.554	0.17744	0.10077	0.15449	110.71	19.016	17.586	14.635
320.00	2.3600	1.4778	0.67670	40.054	41.651	0.17649	0.10679	0.19725	103.29	18.738	20.574	16.104
330.00	2.9579	2.1269	0.47016	39.964	41.355	0.17454	0.11481	0.32843	93.550	18.404	26.607	18.766
339.17	3.6179	4.7790	0.20925	37.417	38.174	0.16438			0	12.361		

TABLE 2-277 Thermodynamic Properties of R-125, Pentafluoroethane (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Single-Phase Properties												
200.00	0.10000	13.337	0.074979	13.916	13.924	0.077436	0.084327	0.12823	794.34	-0.33920	103.09	631.60
224.79	0.10000	12.619	0.079245	17.160	17.168	0.092721	0.087714	0.13371	675.42	-0.27468	91.425	412.85
224.79	0.10000	0.055877	17.897	35.092	36.881	0.18042	0.073155	0.083578	127.60	36.498	8.7263	9.6994
300.00	0.10000	0.040689	24.576	41.155	43.613	0.20616	0.086960	0.095910	149.16	13.603	14.156	13.041
400.00	0.10000	0.030228	33.082	50.746	54.054	0.23608	0.10398	0.11252	172.25	5.6807	22.115	17.070
500.00	0.10000	0.024108	41.479	61.864	66.012	0.26271	0.11764	0.12607	192.24	3.1093	30.917	20.691
200.00	1.0000	13.355	0.074878	13.888	13.963	0.077295	0.084329	0.12805	799.42	-0.34145	103.50	638.78
286.46	1.0000	10.452	0.095673	25.960	26.055	0.12724	0.097767	0.15865	384.49	0.15862	64.240	168.19
286.46	1.0000	0.53072	1.8842	38.807	40.691	0.17833	0.090862	0.11564	122.09	20.318	13.866	12.653
300.00	1.0000	0.48261	2.0720	40.159	42.231	0.18359	0.092286	0.11224	129.90	16.539	14.732	13.270
400.00	1.0000	0.31780	3.1466	50.310	53.456	0.21585	0.10520	0.11626	165.47	5.9354	22.513	17.404
500.00	1.0000	0.24593	4.0661	61.591	65.657	0.24302	0.11806	0.12764	189.33	3.1109	31.336	21.014
200.00	5.0000	13.432	0.074450	13.768	14.140	0.076686	0.084364	0.12732	820.94	-0.35061	105.29	671.00
300.00	5.0000	10.214	0.097901	27.606	28.095	0.13288	0.099404	0.15790	379.39	0.16755	62.727	155.81
400.00	5.0000	2.1222	0.47120	47.739	50.095	0.19593	0.11164	0.15240	136.55	6.6581	27.340	22.244
500.00	5.0000	1.3333	0.75004	60.288	64.038	0.22710	0.12001	0.13643	180.53	2.9952	33.551	23.530
200.00	10.000	13.522	0.073953	13.626	14.366	0.075959	0.084459	0.12655	845.71	-0.36040	107.42	712.07
300.00	10.000	10.597	0.094369	27.096	28.039	0.13109	0.098728	0.14971	441.85	-0.0032571	67.164	177.37
400.00	10.000	5.5436	0.18039	43.724	45.528	0.18103	0.11417	0.19389	165.42	2.8474	40.583	46.568
500.00	10.000	2.8438	0.35165	58.555	62.072	0.21813	0.12198	0.14925	183.47	2.4316	37.529	29.745
200.00	30.000	13.835	0.072281	13.138	15.306	0.073355	0.085197	0.12439	927.73	-0.38765	115.19	889.29
300.00	30.000	11.489	0.087039	25.838	28.449	0.12645	0.098551	0.13883	597.35	-0.23171	79.855	245.90
400.00	30.000	9.0272	0.11078	39.705	43.028	0.16830	0.11217	0.15193	391.05	0.060756	59.824	112.78
500.00	30.000	6.8239	0.14654	54.141	58.538	0.20288	0.12339	0.15662	307.14	0.34144	53.649	67.694
300.00	60.000	12.259	0.081575	24.732	29.626	0.12196	0.10001	0.13426	735.32	-0.32748	93.938	338.04
400.00	60.000	10.465	0.095559	37.854	43.587	0.16206	0.11339	0.14453	563.68	-0.23451	75.022	173.26
500.00	60.000	8.9121	0.11221	51.703	58.436	0.19516	0.12470	0.15193	471.23	-0.15582	67.810	113.08

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., and Jacobsen, R. T., "A New Functional Form and New Fitting Techniques for Equations of State with Application to Pentafluoroethane (HFC-125)," *J. Phys. Chem. Ref. Data* **34**(1):69–108, 2005. The source for viscosity is Huber, M. L., and Laesecke, A., "Correlation for the Viscosity of Pentafluoroethane (R125) from the Triple Point to 500 K at Pressures up to 60 MPa," *Ind. Eng. Chem. Res.*, **45**(12):4447–4453, 2006. The source for thermal conductivity is Perkins, R., and Huber, M. L., "Measurement and Correlation of the Thermal Conductivity of Pentafluoroethane (R125) from 190 K to 512 K at Pressures to 70 MPa," *J. Chem. Eng. Data* **51**(3):898–904, 2006.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainty in density is 0.1% at temperatures from the triple point to 400 K at pressures up to 60 MPa, except in the critical region, where an uncertainty of 0.2% in pressure is generally attained. In the limited region between 340 and 400 K and at pressures from 4 to 10 MPa, as well as for all states above 400 K, the uncertainty in density increases to 0.5%. At temperatures below 330 K and pressures below 30 MPa, the uncertainty in density in the liquid phase may be as low as 0.04%. In the vapor and supercritical region, speed of sound data are represented within 0.05% at pressures below 1 MPa. The estimated uncertainty for heat capacities is 0.5%, and the estimated uncertainty for the speed of sound in the liquid phase is 0.5% for $T > 250$ K. The estimated uncertainties of vapor pressures and saturated liquid densities calculated using the Maxwell criterion are 0.1% for each property, and the estimated uncertainty for saturated vapor densities is 0.2%. The uncertainty in density increases as the critical point is approached, while the accompanying uncertainty in calculated pressures is 0.2%. The viscosity correlation has an estimated uncertainty of 3.0% along the saturation boundary in the liquid phase, and 0.8% in the vapor. For thermal conductivity, the estimated uncertainty of the correlation is 3%, except for the dilute gas and points approaching critical, where the uncertainty rises to 5%.

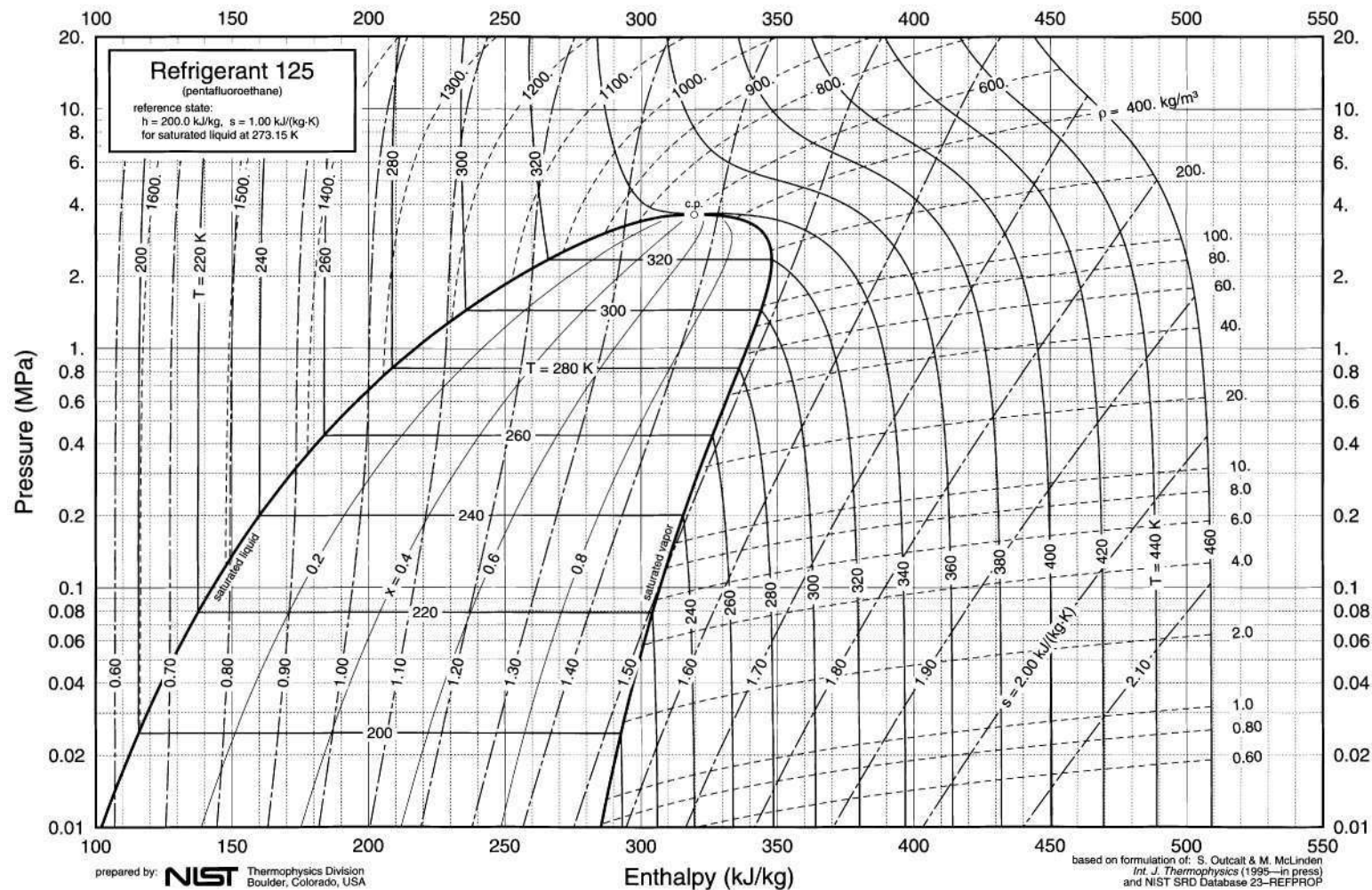


FIG. 2-25 Enthalpy–log-pressure diagram for Refrigerant 125.

TABLE 2-278 Thermodynamic Properties of R-134a, 1,1,1,2-Tetrafluoroethane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _e kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
169.85	0.00038956	15.594	0.064126	7.2907	7.2907	0.042100	0.080831	0.12079	1120.0	-0.38145	145.24	2153.6
170.00	0.00039617	15.590	0.064142	7.3088	7.3088	0.042207	0.080824	0.12079	1119.2	-0.38136	145.15	2139.7
180.00	0.0011275	15.331	0.065228	8.5179	8.5179	0.049117	0.080732	0.12112	1068.3	-0.37370	139.12	1479.1
190.00	0.0028170	15.069	0.066362	9.7328	9.7330	0.055686	0.081114	0.12193	1017.7	-0.36352	133.32	1106.2
200.00	0.0063130	14.804	0.067550	10.957	10.958	0.061966	0.081784	0.12303	967.61	-0.35119	127.74	867.31
210.00	0.012910	14.535	0.068798	12.194	12.195	0.067999	0.082633	0.12434	918.33	-0.33678	122.36	702.27
220.00	0.024433	14.262	0.070116	13.444	13.446	0.073815	0.083595	0.12582	869.85	-0.32011	117.17	582.15
230.00	0.043287	13.984	0.071512	14.710	14.713	0.079441	0.084636	0.12746	822.11	-0.30082	112.14	491.22
240.00	0.072481	13.699	0.072999	15.992	15.997	0.084899	0.085734	0.12927	775.00	-0.27839	107.27	420.20
250.00	0.11561	13.406	0.074593	17.293	17.301	0.090209	0.086879	0.13126	728.39	-0.25204	102.53	363.25
260.00	0.17684	13.104	0.076311	18.613	18.627	0.095389	0.088067	0.13348	682.14	-0.22073	97.922	316.57
270.00	0.26082	12.791	0.078179	19.956	19.976	0.10046	0.089298	0.13597	636.12	-0.18299	93.414	277.54
280.00	0.37271	12.465	0.080227	21.322	21.352	0.10543	0.090576	0.13883	590.17	-0.13675	88.995	244.34
290.00	0.51805	12.121	0.082499	22.716	22.759	0.11032	0.091908	0.14216	544.15	-0.079015	84.644	215.64
300.00	0.70282	11.758	0.085050	24.141	24.201	0.11516	0.093303	0.14615	497.89	-0.0052732	80.341	190.46
310.00	0.93340	11.368	0.087965	25.603	25.685	0.11996	0.094777	0.15108	451.23	0.091533	76.063	168.04
320.00	1.2166	10.945	0.091364	27.108	27.219	0.12475	0.096352	0.15740	404.00	0.22306	71.781	147.78
330.00	1.5599	10.478	0.095439	28.667	28.816	0.12956	0.098067	0.16598	355.90	0.41006	67.464	129.20
340.00	1.9715	9.9483	0.10052	30.297	30.495	0.13446	0.10001	0.17863	306.37	0.69376	63.075	111.81
350.00	2.4611	9.3237	0.10725	32.029	32.293	0.13952	0.10241	0.20012	254.06	1.1714	58.581	95.095
360.00	3.0405	8.5279	0.11726	33.932	34.289	0.14496	0.10601	0.24863	196.05	2.1419	54.062	78.146
370.00	3.7278	7.2558	0.13782	36.283	36.797	0.15159	0.11372	0.52085	127.23	5.1434	51.767	57.956
374.21	4.0591	5.0171	0.19932	38.947	39.756	0.15938			0	11.931		
169.85	0.00038956	0.00027611	3621.7	32.764	34.175	0.20038	0.051318	0.059719	126.79	373.57	3.0801	6.8294
170.00	0.00039617	0.00028055	3564.4	32.772	34.184	0.20029	0.051354	0.059756	126.84	370.78	3.0921	6.8353
180.00	0.0011275	0.00075481	1324.8	33.287	34.781	0.19502	0.053742	0.062208	130.05	234.43	3.8934	7.2319
190.00	0.0028170	0.0017896	558.79	33.821	35.395	0.19075	0.056118	0.064682	133.11	160.10	4.6952	7.6253
200.00	0.0063130	0.0038201	261.77	34.371	36.023	0.18729	0.058489	0.067201	135.98	116.94	5.4978	8.0147
210.00	0.012910	0.0074704	133.86	34.934	36.662	0.18451	0.060874	0.069802	138.63	90.215	6.3018	8.3993
220.00	0.024433	0.013574	73.669	35.508	37.308	0.18228	0.063296	0.072534	141.01	72.584	7.1080	8.7786
230.00	0.043287	0.023188	43.125	36.090	37.956	0.18050	0.065783	0.075455	143.06	60.236	7.9176	9.1524
240.00	0.072481	0.037603	26.593	36.675	38.602	0.17909	0.068357	0.078618	144.73	51.130	8.7324	9.5209
250.00	0.11561	0.058360	17.135	37.261	39.242	0.17797	0.071031	0.082078	145.98	44.137	9.5551	9.8553
260.00	0.17684	0.087278	11.458	37.844	39.870	0.17709	0.073812	0.085888	146.75	38.613	10.389	10.247
270.00	0.26082	0.12651	7.9043	38.420	40.482	0.17640	0.076698	0.090115	146.99	34.169	11.241	10.611
280.00	0.37271	0.17865	5.5976	38.986	41.073	0.17586	0.079686	0.094850	146.63	30.561	12.118	10.980
290.00	0.51805	0.24685	4.0511	39.538	41.636	0.17542	0.082776	0.10023	145.61	27.621	13.035	11.363
300.00	0.70282	0.33512	2.9840	40.069	42.166	0.17504	0.085974	0.10650	143.88	25.230	14.011	11.771
310.00	0.93340	0.44874	2.2285	40.573	42.653	0.17469	0.089297	0.11404	141.33	23.301	15.081	12.219
320.00	1.2166	0.59505	1.6805	41.038	43.083	0.17432	0.092780	0.12355	137.86	21.768	16.303	12.735
330.00	1.5599	0.78498	1.2739	41.451	43.438	0.17387	0.096484	0.13638	133.33	20.578	17.780	13.358
340.00	1.9715	1.0363	0.96498	41.785	43.687	0.17326	0.10052	0.15548	127.57	19.687	19.711	14.164
350.00	2.4611	1.3818	0.72368	41.994	43.775	0.17232	0.10510	0.18870	120.33	19.033	22.525	15.300
360.00	3.0405	1.8973	0.52707	41.973	43.576	0.17075	0.11074	0.26594	111.25	18.448	27.365	17.140
370.00	3.7278	2.8805	0.34717	41.323	42.617	0.16731	0.11928	0.70016	99.370	17.050	40.137	21.336
374.21	4.0591	5.0171	0.19932	38.947	39.756	0.15938			0	11.931		

Single-Phase Properties

200.00	0.10000	14.805	0.067543	10.955	10.962	0.061955	0.081787	0.12301	968.03	-0.35132	127.78	868.18
246.79	0.10000	13.501	0.074068	16.873	16.880	0.088519	0.086506	0.13060	743.31	-0.26099	104.04	380.27
246.79	0.10000	0.050898	19.647	37.073	39.037	0.17830	0.070161	0.080931	145.63	46.198	9.2899	9.7687
275.00	0.10000	0.044972	22.236	39.124	41.348	0.18716	0.073781	0.083445	154.76	29.047	11.540	10.906
350.00	0.10000	0.034753	28.775	45.154	48.032	0.20860	0.086248	0.095065	175.31	12.552	17.537	13.823
425.00	0.10000	0.028455	35.143	52.096	55.610	0.22818	0.098386	0.10695	192.97	6.7852	23.539	16.650
200.00	1.0000	14.819	0.067479	10.933	11.001	0.061846	0.081812	0.12291	972.08	-0.35256	128.11	876.60
275.00	1.0000	12.657	0.079009	20.597	20.676	0.10281	0.089915	0.13695	619.10	-0.16746	91.627	262.84
312.54	1.0000	11.264	0.088775	25.980	26.069	0.12117	0.095165	0.15252	439.31	0.12101	74.978	162.71
312.54	1.0000	0.48242	2.0729	40.695	42.768	0.17460	0.090164	0.11623	140.54	22.877	15.374	12.343
350.00	1.0000	0.39132	2.5555	44.290	46.846	0.18694	0.090315	0.10603	159.63	13.885	17.989	13.936
425.00	1.0000	0.29983	3.3352	51.597	54.933	0.20785	0.099891	0.11116	185.14	7.0297	23.806	16.917
200.00	5.0000	14.880	0.067202	10.839	11.175	0.061371	0.081929	0.12246	989.55	-0.35768	129.56	915.11
275.00	5.0000	12.804	0.078103	20.385	20.776	0.10203	0.089864	0.13495	651.41	-0.19924	94.015	277.35
350.00	5.0000	9.8674	0.10134	31.397	31.904	0.13765	0.10066	0.17178	320.01	0.59952	63.012	109.32
425.00	5.0000	2.0736	0.48225	48.647	51.058	0.18734	0.10791	0.15381	148.25	7.9048	28.574	20.974
200.00	10.000	14.954	0.066874	10.727	11.395	0.060796	0.082085	0.12196	1010.3	-0.36339	131.31	966.05
275.00	10.000	12.967	0.077121	20.149	20.920	0.10115	0.089868	0.13304	687.36	-0.22964	96.744	295.02
350.00	10.000	10.478	0.095440	30.642	31.597	0.13537	0.099573	0.15486	400.60	0.21924	68.919	128.79
425.00	10.000	6.1370	0.16295	43.563	45.193	0.17038	0.11141	0.20870	177.89	3.0434	44.888	46.711
200.00	30.000	15.216	0.065720	10.326	12.298	0.058683	0.082769	0.12047	1084.1	-0.38053	137.79	1211.0
275.00	30.000	13.479	0.074190	19.398	21.624	0.098210	0.090220	0.12865	801.47	-0.30014	105.87	364.87
350.00	30.000	11.662	0.085750	29.071	31.644	0.13038	0.098885	0.13885	582.52	-0.15240	82.955	183.26
425.00	30.000	9.7202	0.10288	39.385	42.471	0.15838	0.10808	0.14967	425.63	0.10364	67.154	107.03
275.00	70.000	14.181	0.070517	18.373	23.310	0.093839	0.091314	0.12519	962.41	-0.35619	119.84	521.91
350.00	70.000	12.797	0.078141	27.492	32.961	0.12484	0.099542	0.13226	787.10	-0.30093	99.868	277.09
425.00	70.000	11.494	0.087004	37.066	43.157	0.15121	0.10829	0.13963	661.39	-0.23655	86.640	181.77

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Tillner-Roth, R., and Baehr, H. D., "An International Standard Formulation of the Thermodynamic Properties of 1,1,1,2-Tetrafluoroethane (HFC-134a) for temperatures from 170 K to 455 K at Pressures up to 70 MPa," *J. Phys. Chem. Ref. Data* **23**:657–729, 1994. The source for viscosity is Huber, M. L., Laesecke, A., and Perkins, R. A., "Model for the Viscosity and Thermal Conductivity of Refrigerants, Including a New Correlation for the Viscosity of R134a," *Ind. Eng. Chem. Res.* **42**:3163–3178, 2003. The source for thermal conductivity is Perkins, R. A., Laesecke, A., Howley, J., Ramires, M. L. V., Gurova, A. N., and Cusco, L., "Experimental Thermal Conductivity Values for the IUPAC Round-Robin Sample of 1,1,1,2-Tetrafluoroethane (R134a)," NISTIR, 2000.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

Typical uncertainties are 0.05% for density, 0.02% for vapor pressure, 0.5% to 1% for heat capacity, 0.05% for vapor speed of sound, and 1% for liquid speed of sound, except in the critical region. The uncertainty in viscosity is 1.5% along the saturated-liquid line, 3% in the liquid phase, 0.5% in the dilute gas, 3% to 5% in the vapor phase, and 5% in the supercritical region, rising to 8% at pressures above 40 MPa. Below 200 K, the uncertainty is 8%. The uncertainty in thermal conductivity is 5%.

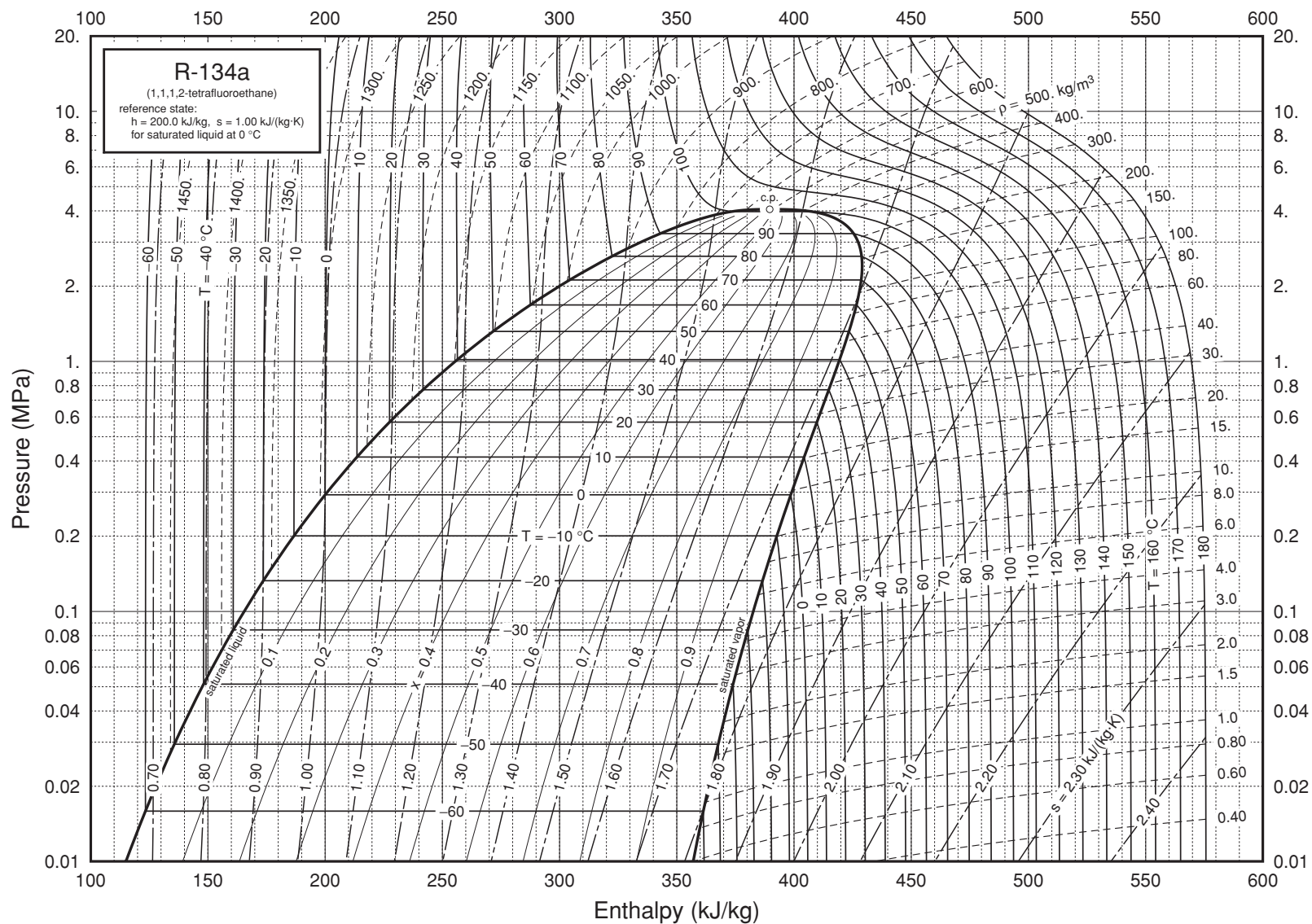


FIG. 2-26 Pressure-enthalpy diagram for Refrigerant 134a. Properties computed with the NIST REFPROP Database, Version 7.0 (Lemmon, E. W., McLinden, M. O., and Huber, M. L., 2002, NIST Standard Reference Database 23, NIST Reference Fluid Thermodynamic and Transport Properties—REFPROP, Version 7.0, Standard Reference Data Program, National Institute of Standards and Technology), based on the equation of state of Tillner-Roth, R., and Baehr, H. D., "An International Standard Formulation of the Thermodynamic Properties of 1,1,1,2-Tetrafluoroethane (HFC-134a) Covering Temperatures from 170 K to 455 K at Pressures up to 70 MPa," *J. Phys. Chem. Ref. Data* **23**(5):657–729, 1994.

TABLE 2-279 Thermodynamic Properties of R-141b, 1,1-Dichloro-1-Fluoroethane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
169.68	6.4927E-06	12.560	0.079620	9.9164	9.9164	0.054895	0.094555	0.13580	1362.2	-0.46011
175.00	1.3392E-05	12.475	0.080158	10.635	10.635	0.059062	0.093422	0.13422	1336.2	-0.46486
190.00	7.9884E-05	12.241	0.081696	12.622	12.622	0.069959	0.091221	0.13103	1264.8	-0.47325
205.00	0.00035111	12.009	0.083271	14.573	14.573	0.079842	0.090109	0.12928	1196.3	-0.47511
220.00	0.0012182	11.779	0.084897	16.505	16.506	0.088942	0.089760	0.12859	1130.5	-0.47131
235.00	0.0035074	11.549	0.086588	18.434	18.434	0.097423	0.089963	0.12869	1067.1	-0.46244
250.00	0.0086986	11.318	0.088358	20.369	20.370	0.10540	0.090575	0.12943	1005.9	-0.44884
265.00	0.019117	11.083	0.090224	22.319	22.321	0.11298	0.091495	0.13069	946.56	-0.43061
280.00	0.038053	10.845	0.092205	24.291	24.294	0.12022	0.092648	0.13238	888.79	-0.40759
295.00	0.069799	10.602	0.094324	26.290	26.297	0.12717	0.093975	0.13446	832.34	-0.37931
310.00	0.11960	10.351	0.096609	28.323	28.334	0.13389	0.095432	0.13691	776.95	-0.34494
325.00	0.19353	10.092	0.099093	30.393	30.412	0.14041	0.096986	0.13972	722.38	-0.30314
340.00	0.29839	9.8211	0.10182	32.505	32.535	0.14677	0.098610	0.14292	668.39	-0.25184
355.00	0.44157	9.5371	0.10485	34.664	34.710	0.15299	0.10029	0.14659	614.74	-0.18789
370.00	0.63091	9.2365	0.10827	36.876	36.945	0.15909	0.10201	0.15085	561.17	-0.10633
385.00	0.87470	8.9148	0.11217	39.148	39.246	0.16512	0.10377	0.15592	507.38	0.00079658
400.00	1.1817	8.5660	0.11674	41.489	41.627	0.17110	0.10557	0.16222	452.97	0.14720
415.00	1.5611	8.1809	0.12224	43.912	44.103	0.17706	0.10745	0.17053	397.38	0.35834
430.00	2.0234	7.7444	0.12913	46.442	46.703	0.18308	0.10945	0.18264	339.70	0.68768
445.00	2.5805	7.2267	0.13838	49.122	49.480	0.18925	0.11171	0.20353	278.17	1.2701
460.00	3.2487	6.5517	0.15263	52.069	52.565	0.19586	0.11464	0.25500	208.92	2.5762
475.00	4.0574	5.2532	0.19036	55.987	56.760	0.20453	0.12111	0.92342	118.23	8.4848
477.50	4.2117	3.9210	0.25504	58.424	59.498	0.21020			0	15.683
169.68	6.4927E-06	4.6024E-06	217.280.	42.292	43.703	0.25401	0.054464	0.062782	117.92	881.00
175.00	1.3392E-05	9.2047E-06	108.640.	42.585	44.040	0.24995	0.055738	0.064059	119.57	725.60
190.00	7.9884E-05	5.0581E-05	19.770.	43.445	45.025	0.24050	0.059180	0.067515	124.11	439.35
205.00	0.00035111	0.00020615	4.850.8	44.353	46.056	0.23342	0.062459	0.070830	128.46	282.90
220.00	0.0012182	0.00066719	1.498.8	45.303	47.129	0.22814	0.065653	0.074095	132.61	192.43
235.00	0.0035074	0.0018021	554.92	46.291	48.237	0.22424	0.068824	0.077393	136.53	137.32
250.00	0.0086986	0.0042156	237.22	47.312	49.376	0.22143	0.072021	0.080793	140.16	102.08
265.00	0.019117	0.0087866	113.81	48.362	50.538	0.21946	0.075271	0.084345	143.44	78.530
280.00	0.038053	0.016681	59.949	49.437	51.718	0.21816	0.078582	0.088081	146.30	62.188
295.00	0.069799	0.029348	34.074	50.531	52.910	0.21739	0.081944	0.092019	148.66	50.500
310.00	0.11960	0.048516	20.612	51.641	54.106	0.21703	0.085336	0.096171	150.46	41.951
325.00	0.19353	0.076209	13.122	52.763	55.302	0.21700	0.088731	0.10056	151.61	35.598
340.00	0.29839	0.11480	8.7112	53.890	56.489	0.21722	0.092106	0.10521	152.02	30.835
355.00	0.44157	0.16710	5.9846	55.016	57.659	0.21763	0.095442	0.11023	151.61	27.254
370.00	0.63091	0.23659	4.2267	56.134	58.801	0.21817	0.098733	0.11578	150.27	24.582
385.00	0.87470	0.32780	3.0507	57.234	59.902	0.21877	0.10198	0.12213	147.85	22.633
400.00	1.1817	0.44696	2.2373	58.300	60.944	0.21939	0.10522	0.12983	144.19	21.289
415.00	1.5611	0.60344	1.6572	59.311	61.898	0.21994	0.10848	0.13996	139.04	20.490
430.00	2.0234	0.81289	1.2302	60.231	62.720	0.22033	0.11185	0.15495	132.06	20.229
445.00	2.5805	1.1059	0.90427	60.993	63.326	0.22037	0.11548	0.18176	122.74	20.562
460.00	3.2487	1.5609	0.64064	61.427	63.508	0.21965	0.11977	0.25134	110.25	21.544
475.00	4.0574	2.6472	0.37776	60.578	62.110	0.21579	0.12640	1.1603	93.316	21.336
477.50	4.2117	3.9210	0.25504	58.424	59.498	0.21020			0	15.683

TABLE 2-279 Thermodynamic Properties of R-141b, 1,1-Dichloro-1-Fluoroethane (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_v kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Single-Phase Properties										
200.00	0.10000	12.087	0.082735	13.923	13.931	0.076634	0.090387	0.12973	1219.1	-0.47521
300.00	0.10000	10.519	0.095063	26.963	26.973	0.12943	0.094448	0.13523	813.87	-0.36865
304.82	0.10000	10.438	0.095799	27.617	27.627	0.13160	0.094917	0.13602	795.96	-0.35757
304.82	0.10000	0.041044	24.364	51.257	53.693	0.21711	0.084163	0.094713	149.91	44.615
400.00	0.10000	0.030497	32.790	59.875	63.154	0.24405	0.096356	0.10524	173.74	15.111
500.00	0.10000	0.024216	41.296	70.152	74.282	0.26884	0.10833	0.11691	194.56	8.0045
200.00	1.0000	12.094	0.082685	13.904	13.987	0.076539	0.090431	0.12969	1222.5	-0.47573
300.00	1.0000	10.536	0.094914	26.923	27.018	0.12930	0.094476	0.13502	819.45	-0.37223
391.53	1.0000	8.7668	0.11407	40.158	40.272	0.16773	0.10455	0.15848	483.80	0.058567
391.53	1.0000	0.37577	2.6612	57.703	60.364	0.21904	0.10339	0.12527	146.42	21.978
400.00	1.0000	0.36041	2.7746	58.638	61.413	0.22169	0.10298	0.12252	150.55	19.770
500.00	1.0000	0.25865	3.8663	69.520	73.387	0.24841	0.10961	0.12123	184.02	8.7395
200.00	5.0000	12.127	0.082464	13.822	14.234	0.076124	0.090629	0.12954	1237.0	-0.47789
300.00	5.0000	10.606	0.094284	26.751	27.222	0.12872	0.094606	0.13416	843.29	-0.38665
400.00	5.0000	8.7814	0.11388	41.014	41.584	0.16989	0.10538	0.15604	502.40	0.0060632
500.00	5.0000	2.5394	0.39380	63.911	65.880	0.22278	0.12220	0.29767	113.94	14.372
200.00	10.000	12.166	0.082198	13.722	14.544	0.075616	0.090880	0.12937	1254.5	-0.48033
300.00	10.000	10.688	0.093559	26.550	27.486	0.12803	0.094784	0.13326	871.09	-0.40188
400.00	10.000	9.0042	0.11106	40.515	41.625	0.16859	0.10531	0.15122	554.81	-0.10586
500.00	10.000	6.4723	0.15451	56.831	58.376	0.20581	0.11637	0.19130	265.53	1.3996
300.00	100.00	11.645	0.085873	24.317	32.904	0.11935	0.098509	0.12788	1200.2	-0.49651
400.00	100.00	10.676	0.093670	36.738	46.105	0.15726	0.10834	0.13645	995.82	-0.44654
500.00	100.00	9.7763	0.10229	49.945	60.174	0.18863	0.11752	0.14460	852.98	-0.39437
400.00	400.00	12.505	0.079970	33.644	65.632	0.14209	0.11569	0.13564	1606.9	-0.48581
500.00	400.00	11.986	0.083429	46.201	79.572	0.17316	0.12401	0.14302	1498.0	-0.46416

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., and Span, R., "Short Fundamental Equations of State for 20 Industrial Fluids," *J. Chem. Eng. Data* **51**(3):785–850, 2006. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The equation has uncertainties of 0.2% in density between 180 and 400 K at pressures to 100 MPa, and 0.5% in density at higher pressures. The uncertainty in density may be higher as temperatures approach 400 K. Vapor pressures are represented with an uncertainty of 0.2% from 270 to 400 K. The uncertainty in speed of sound is 0.01% in the vapor phase and 0.5% in the liquid phase. There are no heat capacity data to verify the equation of state; however, the uncertainties are estimated to be within 2%.

TABLE 2-280 Thermodynamic Properties of R-142b, 1-Chloro-1,1-Difluoroethane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
142.72	3.6327E-06	14.439	0.069257	5.0138	5.0138	0.026013	0.067327	0.11000	1457.9	-0.49370
150.00	1.1727E-05	14.281	0.070022	5.8147	5.8147	0.031486	0.068270	0.11005	1403.7	-0.49255
165.00	9.0545E-05	13.963	0.071619	7.4687	7.4687	0.041995	0.070168	0.11057	1301.7	-0.48684
180.00	0.00047371	13.650	0.073262	9.1343	9.1343	0.051656	0.072051	0.11158	1209.7	-0.47689
195.00	0.0018484	13.339	0.074968	10.818	10.818	0.060640	0.073955	0.11299	1125.1	-0.46285
210.00	0.0057557	13.028	0.076756	12.526	12.526	0.069077	0.075895	0.11477	1046.0	-0.44466
225.00	0.015025	12.715	0.078649	14.263	14.264	0.077064	0.077877	0.11689	971.28	-0.42204
240.00	0.034108	12.396	0.080672	16.033	16.036	0.084681	0.079897	0.11931	899.90	-0.39443
255.00	0.069220	12.069	0.082856	17.841	17.847	0.091990	0.081949	0.12205	831.11	-0.36085
270.00	0.12829	11.732	0.085239	19.692	19.703	0.099042	0.084027	0.12513	764.28	-0.31977
285.00	0.22078	11.380	0.087873	21.590	21.609	0.10588	0.086128	0.12859	698.87	-0.26876
300.00	0.35741	11.010	0.090824	23.538	23.571	0.11255	0.088252	0.13253	634.41	-0.20400
315.00	0.54995	10.617	0.094190	25.545	25.597	0.11908	0.090406	0.13714	570.43	-0.11928
330.00	0.81110	10.193	0.098109	27.617	27.696	0.12551	0.092605	0.14271	506.46	-0.0039877
345.00	1.1545	9.7272	0.10280	29.765	29.884	0.13189	0.094876	0.14984	441.95	0.16142
360.00	1.5950	9.2033	0.10866	32.009	32.182	0.13828	0.097271	0.15984	376.09	0.41716
375.00	2.1496	8.5896	0.11642	34.381	34.632	0.14477	0.099910	0.17613	307.44	0.86137
390.00	2.8397	7.8118	0.12801	36.963	37.327	0.15160	0.10313	0.21217	232.74	1.8155
405.00	3.6991	6.5381	0.15295	40.117	40.683	0.15973	0.10871	0.44141	141.73	5.3582
410.26	4.0548	4.4380	0.22533	43.151	44.065	0.16786			0	14.009
142.72	3.6327E-06	3.0614E-06	326.650.	32.685	33.872	0.22821	0.043630	0.051946	118.57	826.84
150.00	1.1727E-05	9.4031E-06	106.350.	33.008	34.256	0.22109	0.045288	0.053606	121.19	618.59
165.00	9.0545E-05	6.6018E-05	15.147.	33.711	35.083	0.20935	0.048631	0.056964	126.42	363.33
180.00	0.00047371	0.00031681	3,156.4	34.461	35.956	0.20067	0.051923	0.060296	131.39	230.77
195.00	0.0018484	0.0011428	875.03	35.253	36.870	0.19424	0.055225	0.063688	136.07	156.62
210.00	0.0057557	0.0033148	301.68	36.082	37.819	0.18952	0.058589	0.067227	140.40	112.18
225.00	0.015025	0.0081206	123.14	36.944	38.794	0.18609	0.062056	0.070991	144.30	83.865
240.00	0.034108	0.017432	57.365	37.831	39.787	0.18365	0.065643	0.075042	147.66	64.873
255.00	0.069220	0.033721	29.655	38.737	40.790	0.18196	0.069344	0.079422	150.35	51.637
270.00	0.12829	0.060064	16.649	39.657	41.793	0.18086	0.073133	0.084167	152.25	42.170
285.00	0.22078	0.10020	9.9797	40.582	42.786	0.18019	0.076978	0.089331	153.23	35.289
300.00	0.35741	0.15871	6.3006	41.506	43.758	0.17984	0.080851	0.095026	153.15	30.249
315.00	0.54995	0.24138	4.1428	42.419	44.697	0.17972	0.084737	0.10147	151.86	26.559
330.00	0.81110	0.35595	2.8094	43.306	45.585	0.17972	0.088646	0.10911	149.19	23.894
345.00	1.1545	0.51364	1.9469	44.149	46.397	0.17975	0.092615	0.11883	144.89	22.043
360.00	1.5950	0.73250	1.3652	44.916	47.094	0.17970	0.096730	0.13261	138.64	20.877
375.00	2.1496	1.0462	0.95585	45.551	47.605	0.17937	0.10116	0.15586	129.97	20.328
390.00	2.8397	1.5347	0.65159	45.920	47.771	0.17838	0.10632	0.21004	118.18	20.306
405.00	3.6991	2.5436	0.39314	45.483	46.938	0.17518	0.11368	0.55965	102.00	19.783
410.26	4.0548	4.4380	0.22533	43.151	44.065	0.16786			0	14.009

TABLE 2-280 Thermodynamic Properties of R-142b, 1-Chloro-1,1-Difluoroethane (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/mol·K	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Single-Phase Properties										
150.00	0.10000	14.282	0.070018	5.8131	5.8201	0.031476	0.068275	0.11005	1404.1	-0.49259
225.00	0.10000	12.716	0.078641	14.260	14.268	0.077053	0.077880	0.11688	971.75	-0.42221
263.70	0.10000	11.875	0.084211	18.910	18.918	0.096110	0.083152	0.12379	792.14	-0.33806
263.70	0.10000	0.047556	21.028	39.269	41.372	0.18126	0.071534	0.082128	151.56	45.775
300.00	0.10000	0.041113	24.323	41.979	44.411	0.19205	0.076432	0.085864	162.77	26.250
375.00	0.10000	0.032446	30.820	48.175	51.257	0.21237	0.087964	0.096734	182.58	12.432
450.00	0.10000	0.026895	37.182	55.179	58.897	0.23092	0.098211	0.10678	199.94	7.4502
150.00	1.0000	14.289	0.069982	5.7989	5.8689	0.031381	0.068317	0.11003	1407.4	-0.49289
225.00	1.0000	12.730	0.078553	14.234	14.312	0.076937	0.077905	0.11676	976.69	-0.42404
300.00	1.0000	11.034	0.090630	23.498	23.589	0.11241	0.088238	0.13215	640.59	-0.21142
338.73	1.0000	9.9277	0.10073	28.857	28.957	0.12923	0.093915	0.14661	469.03	0.084297
338.73	1.0000	0.44153	2.2648	43.804	46.069	0.17974	0.090944	0.11443	146.90	22.728
375.00	1.0000	0.36746	2.7214	47.311	50.033	0.19087	0.091549	0.10695	164.80	14.567
450.00	1.0000	0.28570	3.5002	54.646	58.146	0.21058	0.099383	0.11074	190.29	7.9246
150.00	5.0000	14.321	0.069826	5.7370	6.0861	0.030965	0.068506	0.10997	1421.7	-0.49417
225.00	5.0000	12.791	0.078178	14.121	14.512	0.076430	0.078025	0.11629	997.91	-0.43157
300.00	5.0000	11.170	0.089523	23.263	23.710	0.11162	0.088194	0.13015	676.41	-0.25069
375.00	5.0000	8.9978	0.11114	33.787	34.343	0.14314	0.098897	0.15862	373.38	0.42121
450.00	5.0000	2.2231	0.44982	50.992	53.241	0.18867	0.10800	0.17517	139.23	10.314
150.00	10.000	14.360	0.069636	5.6618	6.3581	0.030454	0.068749	0.10991	1439.0	-0.49558
225.00	10.000	12.864	0.077739	13.987	14.765	0.075821	0.078190	0.11578	1023.0	-0.43976
300.00	10.000	11.321	0.088335	23.001	23.885	0.11072	0.088215	0.12826	716.03	-0.28769
375.00	10.000	9.4491	0.10583	33.097	34.155	0.14120	0.098194	0.14723	451.41	0.11527
450.00	10.000	6.5247	0.15326	45.004	46.536	0.17118	0.10853	0.18939	220.16	2.1126
225.00	30.000	13.119	0.076223	13.519	15.805	0.073606	0.078954	0.11435	1110.6	-0.46323
300.00	30.000	11.787	0.084836	22.183	24.728	0.10777	0.088691	0.12401	840.46	-0.37159
375.00	30.000	10.406	0.096097	31.547	34.430	0.13659	0.097974	0.13464	635.15	-0.22643
450.00	30.000	8.9530	0.11169	41.538	44.889	0.16199	0.10623	0.14384	484.32	-0.010974
225.00	60.000	13.434	0.074437	12.957	17.423	0.070756	0.080223	0.11326	1217.3	-0.48274
300.00	60.000	12.284	0.081409	21.322	26.207	0.10439	0.089774	0.12130	975.81	-0.42589
375.00	60.000	11.180	0.089443	30.259	35.626	0.13238	0.098881	0.12975	801.25	-0.35484
450.00	60.000	10.119	0.098828	39.706	45.636	0.15670	0.10699	0.13689	675.38	-0.27949

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., and Span, R., "Short Fundamental Equations of State for 20 Industrial Fluids," *J. Chem. Eng. Data* **51**(3):785–850, 2006. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in density are 0.3% in the liquid phase below 370 K, 1% at higher temperatures in the liquid and supercritical regions, and 0.5% in the vapor phase. Uncertainties for other properties are 0.5% for vapor pressure, 2% for heat capacities and liquid sound speeds, and 0.2% for vapor sound speeds.

TABLE 2-281 Thermodynamic Properties of R-143a, 1,1,1-Trifluoroethane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
161.34	0.0010749	15.832	0.063163	4.4138	4.4138	0.026403	0.068393	0.10179	1058.1	-0.43936
170.00	0.0025084	15.583	0.064174	5.2969	5.2971	0.031735	0.068179	0.10225	1016.7	-0.42914
180.00	0.0059324	15.291	0.065399	6.3240	6.3244	0.037606	0.068405	0.10325	969.14	-0.41402
190.00	0.012629	14.994	0.066693	7.3629	7.3637	0.043223	0.068990	0.10460	921.61	-0.39585
200.00	0.024624	14.692	0.068062	8.4164	8.4181	0.048626	0.069825	0.10621	874.04	-0.37472
210.00	0.044602	14.384	0.069519	9.4869	9.4900	0.053849	0.070836	0.10803	826.46	-0.35034
220.00	0.075908	14.069	0.071077	10.576	10.581	0.058915	0.071969	0.11005	778.88	-0.32211
230.00	0.12252	13.745	0.072753	11.685	11.694	0.063848	0.073190	0.11227	731.28	-0.28906
240.00	0.18902	13.410	0.074570	12.817	12.831	0.068665	0.074475	0.11474	683.65	-0.24979
250.00	0.28049	13.062	0.076556	13.973	13.995	0.073385	0.075809	0.11750	635.90	-0.20231
260.00	0.40251	12.698	0.078751	15.156	15.188	0.078027	0.077186	0.12066	587.94	-0.14368
270.00	0.56112	12.314	0.081208	16.368	16.414	0.082607	0.078605	0.12435	539.61	-0.069548
280.00	0.76276	11.904	0.084004	17.615	17.680	0.087149	0.080078	0.12879	490.70	0.026895
290.00	1.0144	11.461	0.087249	18.903	18.991	0.091675	0.081625	0.13438	440.95	0.15683
300.00	1.3234	10.975	0.091119	20.239	20.360	0.096221	0.083293	0.14180	389.93	0.34002
310.00	1.6983	10.426	0.095914	21.638	21.801	0.10083	0.085172	0.15244	337.04	0.61491
320.00	2.1483	9.7846	0.10220	23.125	23.344	0.10559	0.087455	0.16980	281.21	1.0681
330.00	2.6850	8.9829	0.11132	24.750	25.048	0.11065	0.090641	0.20591	220.39	1.9469
340.00	3.3250	7.7913	0.12835	26.688	27.114	0.11659	0.096654	0.35704	149.32	4.3897
345.86	3.7618	5.1285	0.19499	29.429	30.163	0.12527			0	12.397
161.34	0.0010749	0.00080362	1244.4	25.521	26.859	0.16552	0.044397	0.052938	137.57	385.09
170.00	0.0025084	0.0017832	560.78	25.895	27.302	0.16118	0.046371	0.055037	140.62	262.77
180.00	0.0059324	0.0039967	250.20	26.340	27.824	0.15705	0.048691	0.057550	143.91	176.43
190.00	0.012629	0.0081006	123.45	26.796	28.355	0.15370	0.051040	0.060156	146.92	124.70
200.00	0.024624	0.015110	66.180	27.262	28.892	0.15100	0.053424	0.062886	149.60	92.835
210.00	0.044602	0.026311	38.007	27.736	29.431	0.14881	0.055867	0.065796	151.89	72.442
220.00	0.075908	0.043269	23.111	28.213	29.968	0.14704	0.058395	0.068954	153.71	58.764
230.00	0.12252	0.067850	14.738	28.693	30.498	0.14560	0.061026	0.072428	155.02	49.133
240.00	0.18902	0.10227	9.7783	29.170	31.018	0.14444	0.063766	0.076287	155.74	42.049
250.00	0.28049	0.14916	6.7041	29.641	31.521	0.14349	0.066612	0.080611	155.81	36.657
260.00	0.40251	0.21175	4.7225	30.102	32.003	0.14270	0.069560	0.085515	155.15	32.456
270.00	0.56112	0.29409	3.4004	30.548	32.456	0.14202	0.072606	0.091184	153.70	29.136
280.00	0.76276	0.40144	2.4910	30.972	32.872	0.14141	0.075756	0.097932	151.36	26.498
290.00	1.0144	0.54109	1.8481	31.366	33.240	0.14081	0.079035	0.10632	148.04	24.406
300.00	1.3234	0.72367	1.3818	31.715	33.544	0.14017	0.082489	0.11743	143.59	22.765
310.00	1.6983	0.96617	1.0350	32.000	33.758	0.13940	0.086214	0.13359	137.85	21.499
320.00	2.1483	1.2991	0.76974	32.183	33.837	0.13838	0.090400	0.16076	130.56	20.526
330.00	2.6850	1.7898	0.55871	32.184	33.684	0.13682	0.095488	0.22018	121.34	19.682
340.00	3.3250	2.6696	0.37458	31.732	32.977	0.13384	0.10298	0.47999	109.29	18.259
345.86	3.7618	5.1285	0.19499	29.429	30.163	0.12527			0	12.397

TABLE 2-281 Thermodynamic Properties of R-143a, 1,1,1-Trifluoroethane (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Single-Phase Properties										
200.00	0.10000	14.694	0.068054	8.4143	8.4211	0.048616	0.069829	0.10620	874.45	-0.37493
225.63	0.10000	13.888	0.072006	11.198	11.205	0.061709	0.072648	0.11128	752.07	-0.30416
225.63	0.10000	0.056043	17.843	28.483	30.268	0.14619	0.059863	0.070868	154.52	52.958
300.00	0.10000	0.040759	24.534	33.380	35.833	0.16745	0.070758	0.079793	179.93	18.414
400.00	0.10000	0.030247	33.061	41.260	44.566	0.19247	0.086121	0.094695	207.36	7.5341
500.00	0.10000	0.024114	41.469	50.551	54.698	0.21503	0.099065	0.10751	231.13	4.1010
600.00	0.10000	0.020066	49.836	61.004	65.987	0.23558	0.10950	0.11790	252.55	2.5191
200.00	1.0000	14.715	0.067956	8.3891	8.4570	0.048489	0.069866	0.10604	879.27	-0.37736
289.48	1.0000	11.485	0.087067	18.835	18.922	0.091441	0.081543	0.13406	443.55	0.14902
289.48	1.0000	0.53292	1.8765	31.346	33.223	0.14084	0.078861	0.10583	148.24	24.503
300.00	1.0000	0.49298	2.0285	32.269	34.298	0.14449	0.077834	0.099455	155.53	20.787
400.00	1.0000	0.32001	3.1249	40.778	43.903	0.17212	0.087343	0.098681	198.53	7.7199
500.00	1.0000	0.24665	4.0544	50.244	54.298	0.19527	0.099519	0.10927	227.30	4.0976
600.00	1.0000	0.20246	4.9391	60.781	65.720	0.21607	0.10975	0.11891	251.11	2.4913
200.00	5.0000	14.806	0.067539	8.2811	8.6188	0.047943	0.070021	0.10541	900.01	-0.38724
300.00	5.0000	11.380	0.087876	19.812	20.251	0.094764	0.082741	0.13222	452.11	0.11636
400.00	5.0000	2.2504	0.44436	38.004	40.225	0.15165	0.093496	0.13815	159.25	8.1026
500.00	5.0000	1.3639	0.73318	48.802	52.468	0.17903	0.10135	0.11872	213.49	3.8903
600.00	5.0000	1.0491	0.95318	59.789	64.554	0.20106	0.11075	0.12364	246.86	2.2996
200.00	10.000	14.913	0.067054	8.1545	8.8250	0.047292	0.070195	0.10473	924.61	-0.39776
300.00	10.000	11.776	0.084916	19.382	20.231	0.093258	0.082583	0.12585	514.06	-0.037998
400.00	10.000	6.3531	0.15740	33.679	35.253	0.13608	0.095653	0.17697	196.98	2.9315
500.00	10.000	3.0122	0.33199	46.933	50.252	0.16976	0.10303	0.13206	211.16	3.0876
600.00	10.000	2.1596	0.46305	58.580	63.211	0.19339	0.11178	0.12945	248.28	1.9304
200.00	50.000	15.598	0.064113	7.3673	10.573	0.042937	0.070934	0.10170	1089.7	-0.44295
300.00	50.000	13.358	0.074862	17.612	21.355	0.086486	0.083360	0.11389	794.12	-0.33796
400.00	50.000	11.268	0.088747	28.859	33.296	0.12077	0.096003	0.12454	590.09	-0.20571
500.00	50.000	9.4018	0.10636	40.839	46.157	0.14943	0.10668	0.13213	478.93	-0.077157
600.00	50.000	7.8978	0.12662	53.306	59.637	0.17399	0.11548	0.13720	431.69	-0.0052131
300.00	100.00	14.343	0.069721	16.500	23.472	0.081539	0.084059	0.11166	1008.9	-0.40055
400.00	100.00	12.767	0.078330	27.298	35.131	0.11502	0.097186	0.12126	833.69	-0.35302
500.00	100.00	11.435	0.087453	38.923	47.668	0.14296	0.10821	0.12921	723.22	-0.31534
600.00	100.00	10.314	0.096952	51.227	60.923	0.16711	0.11716	0.13566	656.57	-0.28902

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., and Jacobsen, R. T., "An International Standard Formulation for the Thermodynamic Properties of 1,1,1-Trifluoroethane (HFC-143a) for Temperatures from 161 to 450 K and Pressures to 50 MPa," *J. Phys. Chem. Ref. Data* **29**(4):521–552, 2000. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The estimated uncertainties of properties calculated using the equation of state are 0.1% in density, 0.5% in heat capacities, 0.02% in the speed of sound for the vapor at pressures less than 1 MPa, 0.5% in speed of sound elsewhere, and 0.1% in vapor pressure, except in the critical region.

TABLE 2-282 Thermodynamic Properties of R-152a, 1,1-Difluoroethane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
154.56	6.4139E-05	18.061	0.055369	0.91117	0.91117	0.0073929	0.065711	0.097583	1400.9	-0.43262
160.00	0.00012983	17.910	0.055836	1.4460	1.4460	0.010794	0.066998	0.098932	1359.2	-0.42405
170.00	0.00041463	17.630	0.056720	2.4420	2.4421	0.016832	0.067926	0.10007	1293.7	-0.41381
180.00	0.0011411	17.350	0.057638	3.4453	3.4454	0.022566	0.068069	0.10053	1236.1	-0.40568
190.00	0.0027751	17.067	0.058594	4.4524	4.4525	0.028011	0.068043	0.10090	1182.6	-0.39686
200.00	0.0060859	16.781	0.059592	5.4638	5.4641	0.033199	0.068097	0.10143	1131.3	-0.38620
210.00	0.012233	16.491	0.060638	6.4814	6.4822	0.038164	0.068309	0.10218	1081.2	-0.37317
220.00	0.022836	16.198	0.061737	7.5077	7.5091	0.042938	0.068689	0.10317	1031.8	-0.35753
230.00	0.040024	15.899	0.062897	8.5448	8.5474	0.047548	0.069217	0.10439	982.80	-0.33902
240.00	0.066451	15.594	0.064128	9.5949	9.5992	0.052017	0.069868	0.10583	933.98	-0.31735
250.00	0.10530	15.281	0.065440	10.660	10.667	0.056365	0.070618	0.10747	885.23	-0.29203
260.00	0.16024	14.960	0.066846	11.741	11.752	0.060607	0.071447	0.10932	836.44	-0.26235
270.00	0.23541	14.628	0.068363	12.841	12.857	0.064759	0.072342	0.11140	787.50	-0.22731
280.00	0.33537	14.283	0.070011	13.962	13.985	0.068835	0.073294	0.11374	738.29	-0.18547
290.00	0.46506	13.924	0.071819	15.105	15.138	0.072848	0.074300	0.11641	688.71	-0.13478
300.00	0.62978	13.546	0.073821	16.273	16.319	0.076812	0.075363	0.11951	638.61	-0.072262
310.00	0.83519	13.147	0.076065	17.470	17.533	0.080741	0.076490	0.12318	587.86	0.0064939
320.00	1.0873	12.719	0.078621	18.699	18.785	0.084652	0.077694	0.12765	536.30	0.10833
330.00	1.3925	12.257	0.081587	19.968	20.081	0.088567	0.078997	0.13333	483.73	0.24433
340.00	1.7577	11.748	0.085118	21.284	21.434	0.092513	0.080433	0.14093	429.89	0.43380
350.00	2.1907	11.176	0.089475	22.662	22.858	0.096532	0.082059	0.15192	374.37	0.71349
360.00	2.7000	10.510	0.095147	24.126	24.383	0.10070	0.083973	0.16999	316.40	1.1636
370.00	3.2967	9.6829	0.10328	25.730	26.070	0.10515	0.086395	0.20774	254.38	2.0014
380.00	3.9966	8.4686	0.11808	27.655	28.127	0.11043	0.090038	0.35969	183.93	4.1151
386.41	4.5168	5.5715	0.17949	30.732	31.543	0.11914			0	11.292
154.56	6.4139E-05	4.9919E-05	20,032.	26.412	27.697	0.18069	0.037824	0.046148	154.04	336.54
160.00	1.2983E-04	9.7623E-05	10,244.	26.619	27.949	0.17644	0.038592	0.046923	156.44	294.63
170.00	0.00041463	0.00029357	3,406.3	27.009	28.422	0.16965	0.040032	0.048389	160.70	233.22
180.00	0.0011411	0.00076374	1,309.3	27.412	28.906	0.16401	0.041518	0.049922	164.78	186.99
190.00	0.0027751	0.0017625	567.37	27.825	29.399	0.15931	0.043061	0.051550	168.64	151.71
200.00	0.0060859	0.0036817	271.61	28.247	29.900	0.15538	0.044675	0.053297	172.27	124.45
210.00	0.012233	0.0070756	141.33	28.676	30.405	0.15208	0.046371	0.055192	175.63	103.17
220.00	0.022836	0.012679	78.873	29.110	30.911	0.14931	0.048158	0.057260	178.66	86.396
230.00	0.040024	0.021415	46.696	29.546	31.415	0.14697	0.050042	0.059528	181.32	73.068
240.00	0.066451	0.034408	29.063	29.984	31.915	0.14500	0.052027	0.062017	183.57	62.400
250.00	0.10530	0.052990	18.871	30.419	32.406	0.14332	0.054111	0.064754	185.35	53.805
260.00	0.16024	0.078721	12.703	30.851	32.887	0.14189	0.056291	0.067769	186.60	46.838
270.00	0.23541	0.11343	8.8162	31.277	33.352	0.14067	0.058564	0.071103	187.28	41.163
280.00	0.33537	0.15926	6.2791	31.694	33.800	0.13960	0.060926	0.074811	187.33	36.517
290.00	0.46506	0.21879	4.5705	32.099	34.224	0.13866	0.063373	0.078977	186.70	32.701
300.00	0.62978	0.29518	3.3877	32.488	34.622	0.13782	0.065904	0.083731	185.34	29.557
310.00	0.83519	0.39242	2.5483	32.857	34.985	0.13704	0.068522	0.089270	183.16	26.966
320.00	1.0873	0.51573	1.9390	33.199	35.308	0.13629	0.071233	0.095924	180.10	24.832
330.00	1.3925	0.67231	1.4874	33.506	35.577	0.13552	0.074054	0.10426	176.06	23.083
340.00	1.7577	0.87269	1.1459	33.764	35.779	0.13470	0.077013	0.11534	170.91	21.662
350.00	2.1907	1.1336	0.88218	33.953	35.886	0.13375	0.080160	0.13140	164.49	20.515
360.00	2.7000	1.4847	0.67352	34.036	35.855	0.13256	0.083590	0.15806	156.55	19.584
370.00	3.2967	1.9912	0.50220	33.933	35.589	0.13088	0.087501	0.21461	146.71	18.738
380.00	3.9966	2.8726	0.34812	33.381	34.772	0.12792	0.092461	0.44168	134.10	17.420
386.41	4.5168	5.5715	0.17949	30.732	31.543	0.11914			0	11.292

TABLE 2-282 Thermodynamic Properties of R-152a, 1,1-Difluoroethane (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _c kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Single-Phase Properties										
200.00	0.10000	16.782	0.059586	5.4618	5.4678	0.033189	0.068102	0.10142	1131.7	-0.38632
248.83	0.10000	15.318	0.065282	10.535	10.541	0.055862	0.070526	0.10726	890.93	-0.29520
248.83	0.10000	0.050481	19.809	30.369	32.350	0.14351	0.053862	0.064421	185.16	54.718
300.00	0.10000	0.040939	24.427	33.323	35.766	0.15598	0.060341	0.069611	204.37	27.531
400.00	0.10000	0.030290	33.014	40.103	43.405	0.17786	0.074770	0.083404	235.25	10.509
500.00	0.10000	0.024131	41.441	48.286	52.430	0.19794	0.088343	0.096818	261.80	5.4858
200.00	1.0000	16.798	0.059530	5.4436	5.5031	0.033097	0.068150	0.10133	1135.5	-0.38750
300.00	1.0000	13.568	0.073703	16.249	16.323	0.076732	0.075359	0.11920	642.81	-0.078450
316.76	1.0000	12.861	0.077752	18.296	18.374	0.083384	0.077294	0.12609	553.13	0.072197
316.76	1.0000	0.47247	2.1165	33.092	35.208	0.13653	0.070342	0.093615	181.20	25.479
400.00	1.0000	0.32539	3.0733	39.505	42.579	0.15720	0.076729	0.088940	222.89	10.836
500.00	1.0000	0.24847	4.0246	47.922	51.947	0.17807	0.088892	0.098930	256.19	5.4940
200.00	5.0000	16.867	0.059288	5.3645	5.6609	0.032698	0.068345	0.10098	1152.1	-0.39242
300.00	5.0000	13.783	0.072554	16.011	16.373	0.075926	0.075345	0.11645	684.64	-0.13417
400.00	5.0000	3.1526	0.31720	34.823	36.409	0.13126	0.092011	0.26333	148.13	12.808
500.00	5.0000	1.4336	0.69753	46.169	49.657	0.16117	0.091345	0.11166	232.93	5.3800
200.00	10.000	16.949	0.059000	5.2700	5.8600	0.032215	0.068554	0.10058	1172.7	-0.39792
300.00	10.000	14.014	0.071358	15.752	16.466	0.075036	0.075376	0.11401	729.99	-0.18420
400.00	10.000	9.5224	0.10502	28.480	29.530	0.11233	0.087211	0.16027	310.79	1.2458
500.00	10.000	3.4436	0.29039	43.674	46.578	0.15056	0.094041	0.13591	219.92	4.4385
200.00	30.000	17.248	0.057977	4.9338	6.6732	0.030434	0.069111	0.099355	1253.5	-0.41462
300.00	30.000	14.709	0.067986	14.971	17.010	0.072215	0.075740	0.10883	869.51	-0.29079
400.00	30.000	11.934	0.083794	26.048	28.561	0.10535	0.085262	0.12249	576.62	-0.0089214
500.00	30.000	8.9569	0.11165	38.069	41.418	0.13400	0.094199	0.13295	407.30	0.48690
200.00	60.000	17.625	0.056738	4.5216	7.9258	0.028097	0.069524	0.098187	1367.5	-0.42991
300.00	60.000	15.424	0.064836	14.174	18.064	0.069098	0.076558	0.10570	1020.4	-0.35644
400.00	60.000	13.309	0.075135	24.599	29.107	0.10080	0.086520	0.11507	770.60	-0.24663
500.00	60.000	11.312	0.088399	35.736	41.039	0.12739	0.095954	0.12324	613.84	-0.11541

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Outcalt, S. L., and McLinden, M. O., "A Modified Benedict-Webb-Rubin Equation of State for the Thermodynamic Properties of R152a (1,1-Difluoroethane)," *J. Phys. Chem. Ref. Data* **25**(2):605–636, 1996. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties of the equation of state are 0.1% in density, 2% in heat capacity, and 0.05% in the vapor speed of sound, except in the critical region. The uncertainty in vapor pressure is 0.1%.

TABLE 2-283 Saturated Refrigerant 216a, 1,3-Dichloro-1,1,2,2,3,3-Hexafluoropropane

Temp., °F	Pressure, lb/in ² abs.	Volume, ft ³ /lb		Enthalpy, Btu/lb		Entropy, Btu/(lb)(°F)	
		Liquid	Vapor	Liquid	Vapor	Liquid	Vapor
-40	0.339	0.00927	59.957	0.000	62.415	0.0000	0.1487
-20	0.713	0.00942	29.749	4.778	65.276	0.0111	0.1487
0	1.382	0.00958	15.986	9.541	68.208	0.0217	0.1493
20	2.497	0.00974	9.184	14.298	71.199	0.0318	0.1504
40	4.247	0.00992	5.582	19.056	74.239	0.0415	0.1520
60	6.862	0.01010	3.558	23.821	77.319	0.0509	0.1538
80	10.612	0.01030	2.361	28.598	80.429	0.0599	0.1559
100	15.797	0.01050	1.6215	33.391	83.559	0.0686	0.1582
120	22.753	0.01073	1.1462	38.205	86.701	0.0770	0.1607
140	31.845	0.01097	0.8304	43.049	89.845	0.0852	0.1632
160	43.468	0.01124	0.6142	47.930	92.981	0.0931	0.1658
180	58.046	0.01153	0.4623	52.861	96.099	0.1009	0.1685
200	76.033	0.01186	0.3529	57.857	99.186	0.1085	0.1712
220	97.913	0.01223	0.2725	62.939	102.225	0.1161	0.1739
240	124.21	0.01266	0.2121	68.132	105.196	0.1235	0.1765
260	155.50	0.01317	0.1660	73.474	108.066	0.1309	0.1790
280	192.40	0.01378	0.1300	79.015	110.789	0.1384	0.1813
300	235.63	0.01458	0.1013	84.835	113.282	0.1460	0.1834
320	286.03	0.01570	0.0776	91.089	115.373	0.1539	0.1851
340	344.81	0.01764	0.0565	98.234	116.538	0.1628	0.1856
355.98 ^c	399.45	0.02771	0.0277	110.248	110.248	0.1773	0.1773

^cFrom published data, Chemicals Division, Union Carbide Corporation. Used by permission. The paper describing these data is by Shank, *ASHRAE J.*, **1**(1965): 94-101. *c* = critical temperature.

The 1993 ASHRAE *Handbook—Fundamentals* (SI ed.) gives material for integral degrees Celsius with temperatures on the ITS 90 scale from -118.59 to 113.26 °C. The thermodynamic diagram from 0.1 to 30 bar extends to 180 °C. For tables and a diagram to 400 psia, 360 °F, see Stewart, R. B., R. T. Jacobsen, et al., *Thermodynamic Properties of Refrigerants*, ASHRAE, Atlanta, GA, 1986 (521 pp.). For specific heat and viscosity, see *Thermophysical Properties of Refrigerants*, ASHRAE, 1993. Thermal conductivity data as a function of pressure and temperature are reported by Krauss, R. and K. Stephan, *Proc. 12th Symp. Thermophys. Props.*, Boulder, CO, 1994.

TABLE 2-284 Thermodynamic Properties of R-218, Octafluoropropane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
125.45	2.0186E-06	10.687	0.093572	12.346	12.346	0.057736	0.086723	0.14866	1157.7	-0.48474
130.00	4.8985E-06	10.598	0.094353	13.024	13.024	0.063044	0.088773	0.14939	1119.0	-0.48168
140.00	2.7215E-05	10.407	0.096088	14.528	14.528	0.074187	0.093214	0.15145	1041.4	-0.47276
150.00	0.00011605	10.219	0.097856	16.054	16.054	0.084719	0.097514	0.15392	972.09	-0.46146
160.00	0.00040065	10.033	0.099670	17.607	17.607	0.094739	0.10165	0.15668	909.32	-0.44819
170.00	0.0011659	9.8482	0.10154	19.189	19.189	0.10433	0.10561	0.15964	851.65	-0.43313
180.00	0.0029499	9.6632	0.10349	20.800	20.801	0.11354	0.10942	0.16276	797.99	-0.41623
190.00	0.0066482	9.4771	0.10552	22.444	22.445	0.12242	0.11310	0.16603	747.52	-0.39731
200.00	0.013606	9.2890	0.10765	24.121	24.122	0.13102	0.11667	0.16944	699.61	-0.37604
210.00	0.025681	9.0979	0.10992	25.832	25.835	0.13937	0.12014	0.17301	653.72	-0.35194
220.00	0.045264	8.9027	0.11233	27.580	27.585	0.14750	0.12355	0.17675	609.45	-0.32433
230.00	0.075274	8.7022	0.11491	29.365	29.374	0.15544	0.12690	0.18069	566.44	-0.29229
240.00	0.11911	8.4950	0.11772	31.190	31.204	0.16320	0.13022	0.18489	524.41	-0.25451
250.00	0.18060	8.2795	0.12078	33.056	33.077	0.17082	0.13351	0.18940	483.09	-0.20911
260.00	0.26395	8.0537	0.12417	34.966	34.998	0.17832	0.13680	0.19432	442.24	-0.15332
270.00	0.37369	7.8151	0.12796	36.923	36.971	0.18571	0.14010	0.19980	401.64	-0.082915
280.00	0.51466	7.5602	0.13227	38.933	39.001	0.19302	0.14343	0.20607	361.06	0.0089029
290.00	0.69206	7.2843	0.13728	41.000	41.095	0.20029	0.14683	0.21355	320.24	0.13369
300.00	0.91150	6.9802	0.14326	43.134	43.265	0.20754	0.15035	0.22299	278.86	0.31279
310.00	1.1793	6.6366	0.15068	45.351	45.528	0.21483	0.15405	0.23603	236.46	0.59020
320.00	1.5030	6.2324	0.16045	47.677	47.918	0.22226	0.15812	0.25690	192.26	1.0733
330.00	1.8924	5.7189	0.17486	50.177	50.508	0.23002	0.16294	0.30134	144.68	2.1079
340.00	2.3634	4.9092	0.20370	53.102	53.583	0.23893	0.17021	0.52335	89.750	5.6878
345.02	2.6402	3.3400	0.29940	56.252	57.042	0.24883			0	17.096
125.45	2.0186E-06	1.9353E-06	516,720.	37.874	38.917	0.26954	0.072287	0.080604	78.648	854.14
130.00	4.8985E-06	4.5321E-06	220,650.	38.208	39.289	0.26508	0.074826	0.083144	79.920	682.22
140.00	2.7215E-05	2.3384E-05	42,765.	38.983	40.147	0.25718	0.080190	0.088516	82.654	432.76
150.00	1.1605E-04	9.3093E-05	10,742.	39.808	41.055	0.25139	0.085274	0.093623	85.299	288.12
160.00	0.00040065	0.00030151	3,316.6	40.680	42.009	0.24725	0.090119	0.098514	87.846	200.28
170.00	0.0011659	0.00082688	1,209.4	41.595	43.005	0.24442	0.094776	0.10326	90.278	144.70
180.00	0.0029499	0.0019804	504.96	42.548	44.038	0.24263	0.099300	0.10792	92.569	108.17
190.00	0.0066482	0.0042437	235.65	43.536	45.102	0.24167	0.10374	0.11258	94.686	83.339
200.00	0.013606	0.0082953	120.55	44.553	46.193	0.24138	0.10814	0.11730	96.589	65.928
210.00	0.025681	0.015024	66.559	45.596	47.305	0.24161	0.11253	0.12213	98.233	53.394
220.00	0.045264	0.025536	39.161	46.661	48.433	0.24227	0.11693	0.12712	99.570	44.174
230.00	0.075274	0.041158	24.297	47.744	49.573	0.24326	0.12135	0.13232	100.55	37.278
240.00	0.11911	0.063461	15.758	48.843	50.720	0.24452	0.12577	0.13776	101.12	32.063
250.00	0.18060	0.094301	10.604	49.952	51.867	0.24598	0.13020	0.14351	101.22	28.095
260.00	0.26395	0.13591	7.3580	51.067	53.009	0.24759	0.13463	0.14970	100.79	25.083
270.00	0.37369	0.19105	5.2343	52.183	54.139	0.24929	0.13906	0.15649	99.737	22.826
280.00	0.51466	0.26332	3.7977	53.291	55.246	0.25104	0.14352	0.16420	97.968	21.188
290.00	0.69206	0.35768	2.7958	54.381	56.316	0.25277	0.14802	0.17340	95.356	20.088
300.00	0.91150	0.48153	2.0767	55.437	57.330	0.25442	0.15262	0.18521	91.730	19.493
310.00	1.1793	0.64702	1.5456	56.433	58.256	0.25589	0.15742	0.20207	86.859	19.423
320.00	1.5030	0.87704	1.1402	57.322	59.035	0.25700	0.16259	0.23055	80.422	19.958
330.00	1.8924	1.2252	0.81620	57.996	59.540	0.25739	0.16855	0.29532	72.007	21.201
340.00	2.3634	1.8924	0.52844	58.081	59.330	0.25584	0.17647	0.62247	61.241	22.532
345.02	2.6402	3.3400	0.29940	56.252	57.042	0.24883			0	17.096

Single-Phase Properties

150.00	0.10000	10.220	0.097848	16.052	16.061	0.084701	0.097521	0.15391	972.51	-0.46154
225.00	0.10000	8.8041	0.11358	28.465	28.477	0.15148	0.12523	0.17867	588.11	-0.30918
236.07	0.10000	8.5773	0.11659	30.468	30.479	0.16017	0.12892	0.18321	540.83	-0.27015
236.07	0.10000	0.053789	18.591	48.410	50.269	0.24400	0.12403	0.13558	100.95	33.943
300.00	0.10000	0.040933	24.430	56.919	59.362	0.27801	0.14104	0.15028	116.42	12.416
375.00	0.10000	0.032372	30.891	68.272	71.361	0.31363	0.16055	0.16927	131.00	6.1732
150.00	1.0000	10.228	0.097772	16.028	16.125	0.084540	0.097582	0.15385	976.23	-0.46230
225.00	1.0000	8.8236	0.11333	28.413	28.527	0.15125	0.12522	0.17828	594.72	-0.31433
300.00	1.0000	6.9891	0.14308	43.116	43.259	0.20748	0.15030	0.22248	280.68	0.30296
303.52	1.0000	6.8645	0.14568	43.905	44.051	0.21010	0.15163	0.22705	264.07	0.39586
303.52	1.0000	0.53424	1.8718	55.797	57.669	0.25497	0.15428	0.19038	90.170	19.406
375.00	1.0000	0.35613	2.8080	67.521	70.329	0.29246	0.16284	0.17689	120.35	7.1384
150.00	5.0000	10.262	0.097443	15.923	16.411	0.083839	0.097864	0.15361	992.31	-0.46544
225.00	5.0000	8.9050	0.11230	28.196	28.757	0.15027	0.12525	0.17677	622.21	-0.33410
300.00	5.0000	7.2996	0.13699	42.444	43.129	0.20518	0.14888	0.20895	346.02	0.039275
375.00	5.0000	4.0113	0.24930	60.519	61.765	0.26018	0.17483	0.31884	101.99	4.7668
150.00	10.000	10.304	0.097050	15.799	16.769	0.082988	0.098234	0.15336	1011.4	-0.46885
225.00	10.000	8.9969	0.11115	27.949	29.060	0.14913	0.12538	0.17531	653.07	-0.35341
300.00	10.000	7.5606	0.13227	41.847	43.170	0.20307	0.14819	0.20166	403.66	-0.10223
375.00	10.000	5.6952	0.17559	57.659	59.415	0.25128	0.16841	0.23156	219.93	0.67650
150.00	20.000	10.382	0.096321	15.565	17.492	0.081360	0.099023	0.15301	1046.6	-0.47434
225.00	20.000	9.1565	0.10921	27.517	29.701	0.14709	0.12580	0.17326	706.41	-0.38080
300.00	20.000	7.9189	0.12628	40.989	43.515	0.19992	0.14793	0.19500	486.73	-0.22897
375.00	20.000	6.5981	0.15156	55.836	58.867	0.24552	0.16668	0.21331	339.02	0.012472
150.00	30.000	10.454	0.095654	15.351	18.220	0.079818	0.099855	0.15280	1078.7	-0.47844
225.00	30.000	9.2933	0.10760	27.147	30.375	0.14527	0.12636	0.17191	751.98	-0.39925
300.00	30.000	8.1774	0.12229	40.353	44.022	0.19747	0.14819	0.19173	549.34	-0.28972
375.00	30.000	7.0690	0.14146	54.800	59.044	0.24210	0.16662	0.20796	416.45	-0.15627

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., and Span, R., "Short Fundamental Equations of State for 20 Industrial Fluids," *J. Chem. Eng. Data* **51**(3):785–850, 2006. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainty in density is 0.2% for the liquid phase and 0.5% for the vapor phase. Above the critical temperature, the uncertainties are estimated to be 1% in density and 0.5% in pressure. Calculated vapor pressures have an uncertainty of 0.5%. The uncertainties for heat capacities and sound speeds are 1%.

TABLE 2-285 Thermodynamic Properties of R-227ea, 1,1,1,2,3,3,3-Heptafluoropropane

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
146.35	7.0312E-06	11.086	0.090205	12.227	12.227	0.063811	0.11121	0.16308	1068.8	-0.42071
150.00	1.2811E-05	11.020	0.090746	12.821	12.821	0.067824	0.11136	0.16271	1048.3	-0.42085
160.00	5.6086E-05	10.841	0.092245	14.446	14.446	0.078309	0.11222	0.16238	994.14	-0.41876
170.00	0.00019958	10.663	0.093780	16.072	16.072	0.088164	0.11357	0.16286	942.81	-0.41349
180.00	0.00059967	10.487	0.095359	17.705	17.705	0.097501	0.11528	0.16396	893.95	-0.40546
190.00	0.0015672	10.310	0.096992	19.352	19.352	0.10641	0.11723	0.16554	847.26	-0.39497
200.00	0.0036466	10.133	0.098689	21.017	21.017	0.11495	0.11934	0.16750	802.44	-0.38217
210.00	0.0076967	9.9539	0.10046	22.703	22.704	0.12317	0.12157	0.16976	759.23	-0.36707
220.00	0.014961	9.7727	0.10233	24.413	24.414	0.13112	0.12386	0.17228	717.37	-0.34954
230.00	0.027114	9.5883	0.10429	26.148	26.151	0.13884	0.12618	0.17502	676.63	-0.32930
240.00	0.046286	9.3999	0.10638	27.912	27.917	0.14635	0.12851	0.17797	636.81	-0.30590
250.00	0.075058	9.2065	0.10862	29.706	29.714	0.15367	0.13085	0.18114	597.69	-0.27867
260.00	0.11644	9.0068	0.11103	31.531	31.544	0.16083	0.13318	0.18454	559.10	-0.24663
270.00	0.17384	8.7997	0.11364	33.390	33.410	0.16784	0.13549	0.18822	520.85	-0.20842
280.00	0.25102	8.5834	0.11650	35.286	35.315	0.17474	0.13781	0.19225	482.77	-0.16205
290.00	0.35208	8.3560	0.11967	37.220	37.262	0.18153	0.14011	0.19673	444.69	-0.10460
300.00	0.48145	8.1148	0.12323	39.197	39.256	0.18824	0.14243	0.20183	406.42	-0.031613
310.00	0.64388	7.8563	0.12729	41.221	41.303	0.19488	0.14477	0.20782	367.76	0.064014
320.00	0.84450	7.5756	0.13200	43.299	43.411	0.20149	0.14716	0.21515	328.49	0.19426
330.00	1.0889	7.2652	0.13764	45.442	45.592	0.20810	0.14964	0.22467	288.32	0.38093
340.00	1.3836	6.9135	0.14465	47.666	47.866	0.21476	0.15227	0.23817	246.84	0.66802
350.00	1.7358	6.4988	0.15387	50.002	50.269	0.22158	0.15517	0.26018	203.37	1.1593
360.00	2.1552	5.9733	0.16741	52.517	52.877	0.22873	0.15867	0.30696	156.53	2.1701
370.00	2.6553	5.1649	0.19361	55.458	55.972	0.23696	0.16399	0.51614	102.78	5.2930
375.95	2.9991	3.4100	0.29326	59.193	60.073	0.24773			0	16.489
146.35	7.0312E-06	5.7786E-06	173.050	41.158	42.374	0.26981	0.072002	0.080321	89.345	630.83
150.00	1.2811E-05	1.0273E-05	97.345	41.423	42.670	0.26682	0.073562	0.081883	90.352	540.51
160.00	5.6086E-05	4.2170E-05	23.714	42.179	43.509	0.25995	0.077861	0.086194	93.045	363.36
170.00	0.00019958	0.00014128	7.078.0	42.976	44.389	0.25474	0.082177	0.090535	95.643	253.26
180.00	0.00059967	0.00040123	2.492.3	43.814	45.308	0.25085	0.086488	0.094896	98.141	182.56
190.00	0.0015672	0.00099473	1.005.3	44.689	46.265	0.24805	0.090782	0.099275	100.53	135.75
200.00	0.0036466	0.0022037	453.78	45.600	47.255	0.24613	0.095053	0.10368	102.77	103.86
210.00	0.0076967	0.0044452	224.96	46.543	48.275	0.24494	0.099298	0.10813	104.85	81.549
220.00	0.014961	0.0082891	120.64	47.515	49.320	0.24433	0.10352	0.11263	106.73	65.553
230.00	0.027114	0.014469	69.111	48.513	50.387	0.24421	0.10771	0.11721	108.37	53.830
240.00	0.046286	0.023890	41.858	49.533	51.470	0.24448	0.11188	0.12189	109.71	45.075
250.00	0.075058	0.037635	26.571	50.570	52.565	0.24507	0.11601	0.12670	110.73	38.434
260.00	0.11644	0.056982	17.550	51.622	53.666	0.24591	0.12010	0.13166	111.36	33.337
270.00	0.17384	0.083437	11.985	52.684	54.768	0.24695	0.12416	0.13684	111.55	29.395
280.00	0.25102	0.11880	8.4176	53.752	55.864	0.24813	0.12818	0.14232	111.26	26.339
290.00	0.35208	0.16526	6.0510	54.818	56.949	0.24942	0.13216	0.14822	110.42	23.978
300.00	0.48145	0.22559	4.4328	55.878	58.012	0.25076	0.13612	0.15475	108.94	22.183
310.00	0.64388	0.30344	3.2956	56.922	59.044	0.25211	0.14009	0.16227	106.75	20.864
320.00	0.84450	0.40387	2.4761	57.938	60.029	0.25342	0.14409	0.17136	103.73	19.969
330.00	1.0889	0.53442	1.8712	58.909	60.946	0.25463	0.14817	0.18320	99.740	19.477
340.00	1.3836	0.70732	1.4138	59.808	61.764	0.25564	0.15244	0.20027	94.584	19.399
350.00	1.7358	0.94497	1.0582	60.587	62.424	0.25630	0.15704	0.22897	88.022	19.775
360.00	2.1552	1.2970	0.77102	61.146	62.807	0.25632	0.16228	0.29185	79.757	20.625
370.00	2.6553	1.9293	0.51831	61.164	62.540	0.25471	0.16893	0.56287	69.452	21.487
375.95	2.9991	3.4100	0.29326	59.193	60.073	0.24773			0	16.489

Single-Phase Properties

150.00	0.10000	11.021	0.090739	12.819	12.828	0.067809	0.11137	0.16271	1048.6	-0.42089
225.00	0.10000	9.6823	0.10328	25.274	25.284	0.13499	0.12502	0.17360	697.31	-0.34002
256.43	0.10000	9.0790	0.11014	30.875	30.886	0.15829	0.13235	0.18330	572.84	-0.25870
256.43	0.10000	0.049336	20.269	51.245	53.272	0.24559	0.11864	0.12987	111.18	35.009
300.00	0.10000	0.041177	24.286	56.684	59.112	0.26660	0.12925	0.13881	122.16	17.772
375.00	0.10000	0.032454	30.813	67.081	70.162	0.29941	0.14667	0.15550	137.79	8.4581
450.00	0.10000	0.026888	37.191	78.645	82.364	0.32903	0.16079	0.16939	151.34	5.1086
150.00	1.0000	11.028	0.090682	12.799	12.890	0.067676	0.11146	0.16267	1051.5	-0.42131
225.00	1.0000	9.6973	0.10312	25.234	25.337	0.13482	0.12507	0.17337	702.21	-0.34268
300.00	1.0000	8.1405	0.12284	39.137	39.260	0.18804	0.14240	0.20096	412.50	-0.045381
326.58	1.0000	7.3754	0.13559	44.701	44.837	0.20584	0.14878	0.22108	302.19	0.30867
326.58	1.0000	0.48574	2.0587	58.583	60.642	0.25423	0.14676	0.17873	101.23	19.600
375.00	1.0000	0.36801	2.7173	66.118	68.835	0.27764	0.15025	0.16647	123.32	10.030
450.00	1.0000	0.28468	3.5127	78.039	81.552	0.30854	0.16198	0.17375	144.14	5.4173
150.00	5.0000	11.058	0.090430	12.712	13.164	0.067092	0.11182	0.16253	1064.0	-0.42305
225.00	5.0000	9.7613	0.10245	25.066	25.578	0.13406	0.12533	0.17247	723.03	-0.35330
300.00	5.0000	8.3150	0.12026	38.729	39.330	0.18665	0.14230	0.19592	454.01	-0.12586
375.00	5.0000	6.0063	0.16649	54.845	55.677	0.23506	0.15968	0.25736	188.73	1.3729
450.00	5.0000	1.9568	0.51103	74.454	77.009	0.28716	0.16853	0.21820	117.53	5.8791
225.00	10.000	9.8358	0.10167	24.870	25.887	0.13316	0.12567	0.17153	747.22	-0.36439
300.00	10.000	8.4921	0.11776	38.308	39.485	0.18518	0.14237	0.19196	496.71	-0.18923
375.00	10.000	6.7736	0.14763	53.368	54.845	0.23077	0.15755	0.21921	286.63	0.34861
450.00	10.000	4.4095	0.22679	70.066	72.334	0.27322	0.17015	0.23938	162.15	2.0189
225.00	30.000	10.091	0.099103	24.209	27.182	0.13000	0.12709	0.16914	829.42	-0.39369
300.00	30.000	8.9874	0.11127	37.111	40.449	0.18079	0.14335	0.18477	619.36	-0.30379
375.00	30.000	7.8569	0.12728	51.038	54.856	0.22359	0.15736	0.19887	467.67	-0.17451
450.00	30.000	6.7284	0.14862	65.707	70.165	0.26078	0.16895	0.20846	368.25	-0.017558
225.00	60.000	10.391	0.096240	23.457	29.231	0.12610	0.12922	0.16752	926.46	-0.41551
300.00	60.000	9.4675	0.10562	35.965	42.302	0.17614	0.14519	0.18109	743.76	-0.36285
375.00	60.000	8.5978	0.11631	49.374	56.353	0.21789	0.15896	0.19315	617.08	-0.30676
450.00	60.000	7.7868	0.12842	63.495	71.201	0.25396	0.17036	0.20230	530.91	-0.25596

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is an interim equation from Lemmon, E. W., personal communication (NIST, Boulder, CO, 2006). Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in the equation of state are 0.2% in density, except in a small region close to the critical point, 0.2% in vapor pressure between 250 and 360 K, 0.4% in vapor pressure outside this region, 1% in heat capacities (with increasing uncertainties in the critical region and at higher temperatures), 0.1% in the vapor-phase speed of sound, and 3% in the liquid-phase speed of sound.

TABLE 2-286 Saturated Refrigerant 245cb 1,1,1,2,2-Pentafluoropropane*

T, K	P, bar	v_f , m ³ /kg	v_g , m ³ /kg	h_f , kJ/kg	h_g , kJ/kg	s_f , kJ/(kg·K)	s_g , kJ/(kg·K)
172	0.0034	6.46.-4	31.49	-63.4	133.8	-0.3131	0.8327
180	0.0076	6.57.-4	14.63	-55.9	138.7	-0.2707	0.8099
190	0.0190	6.70.-4	6.20	-46.2	145.1	-0.2182	0.7885
200	0.0425	6.83.-4	2.91	-36.0	151.7	-0.1666	0.7725
210	0.0870	6.97.-4	1.48	-25.7	158.5	-0.1157	0.7612
220	0.1654	7.11.-4	0.822	-14.8	165.4	-0.0654	0.7539
230	0.2946	7.25.-4	0.475	-3.6	172.5	-0.0156	0.7500
240	0.4958	7.40.-4	0.292	8.0	179.6	0.0337	0.7487
250	0.7946	7.55.-4	0.192	19.9	186.8	0.0824	0.7497
260	1.2204	7.72.-4	0.125	32.3	194.0	0.1305	0.7525
270	1.806	7.89.-4	0.0862	44.9	201.1	0.1781	0.7567
280	2.584	8.08.-4	0.0611	57.9	208.3	0.2249	0.7621
290	3.600	8.30.-4	0.0443	71.1	215.3	0.2711	0.7683
300	4.888	8.53.-4	0.0327	84.6	222.2	0.3161	0.7751
310	6.491	8.80.-4	0.0246	98.4	228.9	0.3614	0.7822
320	8.456	9.11.-4	0.0186	112.6	235.3	0.4057	0.7893
330	10.83	9.48.-4	0.0143	127.1	241.4	0.4497	0.7960
340	13.67	9.93.-4	0.0111	142.1	246.9	0.4937	0.8018
350	17.04	0.00105	0.0084	157.2	251.5	0.5382	0.8060
360	21.02	0.00113	0.0063	174.7	254.8	0.5844	0.8071
370	25.71	0.00125	0.0045	193.6	255.2	0.6349	0.8013
375	28.46	0.00137	0.0036	205.2	252.5	0.6649	0.7953
380.1 ^c	31.37	0.00204	0.0020	231.8	231.8	0.7341	0.7341

*Values converted from tables of Shank, *Thermodynamic Properties of UCON 245 Refrigerant*, Union Carbide Corporation, New York, 1966. See also Shank, *J. Chem. Eng. Data*, **12**, 474-480 (1967). *c* = critical point. The notation 6.46.-4 signifies 6.46×10^{-4} .

TABLE 2-287 Refrigerant RC 318, Octafluorocyclobutane*

T, K	P, bar	v_f , m ³ /kg	v_g , m ³ /kg	h_f , kJ/kg	h_g , kJ/kg	s_f , kJ/(kg·K)	s_g , kJ/(kg·K)	c_{pf} , kJ/(kg·K)	μ_f , 10 ⁻⁴ Pa·s	k_f , W/(m·K)
200	0.0216	5.507.-4	3.810	353.5	498.0	3.909	4.560			
210	0.0449	5.593.-4	1.931	361.0	500.1	3.947	4.564			
220	0.0875	5.683.-4	1.038	369.2	502.2	3.984	4.569			
230	0.1608	5.778.-4	0.588	377.6	504.4	4.022	4.574	0.98	11.7	0.088
240	0.2810	5.879.-4	0.349	386.4	510.9	4.060	4.578	1.00	9.55	0.085
250	0.466	5.988.-4	0.2166	395.6	517.4	4.097	4.584	1.02	7.90	0.082
260	0.741	6.106.-4	0.1401	405.2	524.0	4.133	4.592	1.03	6.63	0.078
270	1.133	6.234.-4	0.0938	415.1	530.7	4.172	4.599	1.05	5.64	0.075
280	1.672	6.375.-4	0.0647	425.8	537.3	4.210	4.609	1.07	4.85	0.071
290	2.392	6.529.-4	0.0458	436.2	543.9	4.247	4.618	1.09	4.22	0.068
300	3.325	6.694.-4	0.0332	447.3	550.4	4.284	4.626	1.12	3.70	0.065
310	4.522	6.893.-4	0.0245	458.7	556.9	4.322	4.638	1.15	3.20	0.061
320	6.007	7.115.-4	0.0184	470.5	563.3	4.359	4.648	1.18	2.94	0.058
330	7.826	7.365.-4	0.0139	482.7	569.4	4.396	4.659	1.23	2.66	0.054
340	10.018	7.666.-4	0.0106	495.2	575.4	4.433	4.669	1.27	2.33	0.051
350	12.632	8.034.-4	0.0082	508.1	581.0	4.469	4.678	1.32	2.00	0.048
360	15.71	8.508.-4	0.0062	521.5	585.8	4.507	4.685	1.39		
370	19.33	9.172.-4	0.0047	535.6	589.9	4.544	4.691			
380	23.59	1.031.-3	0.0033	551.4	591.5	4.585	4.691			
388.5 ^c	27.83	1.613.-3	0.0016	577.2	577.2	4.651	4.651			

*Values of *P*, *v*, *h*, and *s* interpolated, extrapolated, and converted from tables of Oguchi, *Reito*, **52** (1977): 869-889. Values of *c_p*, *μ*, and *k* interpolated and converted from tables in *Thermophysical Properties of Refrigerants*, American Society of Heating, Refrigerating and Air-Conditioning Engineers, New York, 1976. *c* = critical point. Saturation and superheat tables and a diagram to 80 bar, 580 K are given by Reynolds, W. C., *Thermodynamic Properties in S.I.*, Stanford Univ. publ., 1979 (173 pp.). For equations, see Cipollone, R., *ASHRAE Trans.*, **97**, 2 (1991): 262-267.

TABLE 2-288 Thermodynamic Properties of R-404A

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
200.00	0.022649	14.209	0.070377	10.353	10.355	0.058867	0.076939	0.11881	859.89	-0.34161
205.00	0.030989	14.059	0.071131	10.948	10.950	0.061803	0.077522	0.11915	831.56	-0.33384
210.00	0.041658	13.907	0.071905	11.544	11.547	0.064678	0.078116	0.11965	804.42	-0.32460
215.00	0.055101	13.755	0.072703	12.144	12.148	0.067498	0.078719	0.12028	778.20	-0.31394
220.00	0.071804	13.600	0.073527	12.746	12.751	0.070269	0.079326	0.12103	752.69	-0.30185
225.00	0.092293	13.444	0.074380	13.353	13.359	0.072995	0.079940	0.12188	727.72	-0.28830
230.00	0.11713	13.286	0.075265	13.963	13.972	0.075679	0.080558	0.12282	703.16	-0.27318
235.00	0.14693	13.126	0.076185	14.578	14.590	0.078326	0.081183	0.12386	678.91	-0.25636
240.00	0.18232	12.963	0.077145	15.199	15.213	0.080939	0.081815	0.12499	654.88	-0.23766
245.00	0.22397	12.796	0.078147	15.824	15.842	0.083520	0.082456	0.12621	631.01	-0.21683
250.00	0.27258	12.627	0.079197	16.456	16.478	0.086073	0.083107	0.12754	607.23	-0.19356
255.00	0.32888	12.453	0.080301	17.094	17.120	0.088600	0.083770	0.12899	583.50	-0.16749
260.00	0.39363	12.275	0.081465	17.738	17.770	0.091104	0.084446	0.13057	559.77	-0.13814
265.00	0.46763	12.092	0.082697	18.390	18.429	0.093589	0.085137	0.13229	535.99	-0.10493
270.00	0.55168	11.904	0.084006	19.049	19.096	0.096056	0.085846	0.13419	512.13	-0.067104
275.00	0.64664	11.709	0.085402	19.717	19.772	0.098510	0.086574	0.13630	488.15	-0.023728
280.00	0.75338	11.508	0.086899	20.394	20.460	0.10095	0.087326	0.13866	464.02	0.026418
285.00	0.87280	11.298	0.088513	21.081	21.159	0.10339	0.088104	0.14133	439.69	0.084928
290.00	1.0059	11.078	0.090266	21.780	21.870	0.10583	0.088914	0.14438	415.13	0.15392
295.00	1.1536	10.848	0.092182	22.490	22.597	0.10826	0.089761	0.14793	390.28	0.23626
300.00	1.3169	10.605	0.094295	23.215	23.339	0.11071	0.090653	0.15211	365.11	0.33594
305.00	1.4970	10.346	0.096652	23.956	24.101	0.11317	0.091603	0.15715	339.56	0.45865
310.00	1.6950	10.069	0.099314	24.717	24.885	0.11566	0.092625	0.16339	313.56	0.61280
315.00	1.9122	9.7686	0.10237	25.500	25.695	0.11818	0.093744	0.17139	287.00	0.81141
320.00	2.1499	9.4384	0.10595	26.310	26.538	0.12075	0.094998	0.18212	259.73	1.0757
325.00	2.4096	9.0688	0.11027	27.157	27.423	0.12341	0.096455	0.19751	231.53	1.4433
330.00	2.6932	8.6431	0.11570	28.054	28.365	0.12619	0.098241	0.22197	201.95	1.9881
335.00	3.0027	8.1285	0.12302	29.026	29.396	0.12918	0.10064	0.26871	170.20	2.8807
340.00	3.3414	7.4362	0.13448	30.147	30.597	0.13261	0.10445	0.40392	134.59	4.6353
345.00	3.7150	5.7429	0.17413	32.108	32.755	0.13874	0.11650	8.2559	89.976	10.564
345.27	3.7348	4.9400	0.20243	32.875	33.631	0.14126			0	12.409
200.00	0.021264	0.013010	76.866	29.920	31.555	0.16521	0.058696	0.068032	138.13	88.073
205.00	0.029285	0.017550	56.979	30.185	31.853	0.16408	0.059968	0.069509	139.28	77.215
210.00	0.039592	0.023271	42.971	30.451	32.152	0.16307	0.061245	0.071021	140.34	68.305
215.00	0.052629	0.030378	32.919	30.718	32.451	0.16217	0.062526	0.072573	141.29	60.948
220.00	0.068883	0.039095	25.579	30.987	32.749	0.16138	0.063815	0.074172	142.12	54.835
225.00	0.088879	0.049667	20.134	31.256	33.046	0.16068	0.065113	0.075826	142.84	49.721
230.00	0.11318	0.062359	16.036	31.525	33.340	0.16006	0.066424	0.077546	143.43	45.413
235.00	0.14240	0.077463	12.909	31.795	33.633	0.15952	0.067750	0.079344	143.88	41.761
240.00	0.17718	0.095292	10.494	32.063	33.922	0.15903	0.069095	0.081232	144.18	38.642
245.00	0.21817	0.11619	8.6063	32.330	34.208	0.15861	0.070463	0.083226	144.33	35.962
250.00	0.26610	0.14055	7.1149	32.596	34.489	0.15823	0.071855	0.085343	144.32	33.645
255.00	0.32169	0.16879	5.9247	32.859	34.765	0.15789	0.073276	0.087604	144.14	31.630
260.00	0.38571	0.20137	4.9659	33.119	35.034	0.15759	0.074728	0.090033	143.77	29.871
265.00	0.45896	0.23885	4.1867	33.375	35.296	0.15732	0.076214	0.092660	143.22	28.327
270.00	0.54225	0.28183	3.5483	33.626	35.550	0.15707	0.077737	0.095524	142.46	26.970
275.00	0.63645	0.33102	3.0210	33.872	35.794	0.15684	0.079300	0.098671	141.49	25.773
280.00	0.74245	0.38725	2.5823	34.111	36.028	0.15661	0.080909	0.10217	140.29	24.718
285.00	0.86115	0.45152	2.2148	34.342	36.249	0.15639	0.082567	0.10609	138.85	23.789
290.00	0.99353	0.52501	1.9047	34.563	36.455	0.15616	0.084282	0.11056	137.16	22.972
295.00	1.1406	0.60922	1.6414	34.773	36.645	0.15592	0.086062	0.11574	135.19	22.256
300.00	1.3034	0.70599	1.4165	34.968	36.814	0.15566	0.087917	0.12186	132.92	21.633
305.00	1.4830	0.81772	1.2229	35.147	36.960	0.15536	0.089863	0.12929	130.34	21.094

TABLE 2-288 Thermodynamic Properties of R-404A (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _e kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties (Concluded)										
310.00	1.6806	0.94761	1.0553	35.304	37.078	0.15501	0.091922	0.13856	127.41	20.630
315.00	1.8975	1.1001	0.90903	35.435	37.159	0.15459	0.094123	0.15062	124.10	20.234
320.00	2.1351	1.2815	0.78032	35.530	37.196	0.15408	0.096513	0.16713	120.38	19.889
325.00	2.3950	1.5019	0.66583	35.578	37.173	0.15343	0.099162	0.19141	116.21	19.571
330.00	2.6789	1.7781	0.56239	35.558	37.065	0.15257	0.10220	0.23111	111.51	19.233
335.00	2.9893	2.1438	0.46645	35.429	36.824	0.15136	0.10585	0.30867	106.19	18.763
340.00	3.3299	2.6882	0.37199	35.084	36.323	0.14946	0.11074	0.53035	100.03	17.851
345.00	3.7109	4.2113	0.23746	33.615	34.496	0.14379	0.12022	8.6291	90.307	14.130
345.27	3.7348	4.9400	0.20243	32.875	33.631	0.14126			0	12.409
Single-Phase Properties										
226.65	0.10000	13.392	0.074669	13.554	13.561	0.073887	0.080143	0.12218	719.55	-0.28348
227.41	0.10000	0.055492	18.021	31.386	33.188	0.16038	0.065742	0.076645	143.14	47.558
300.00	0.10000	0.040750	24.540	36.596	39.050	0.18269	0.076875	0.085907	166.24	16.998
400.00	0.10000	0.030243	33.066	45.121	48.428	0.20956	0.092844	0.10141	191.81	6.9229
500.00	0.10000	0.024113	41.471	55.103	59.250	0.23365	0.10612	0.11456	213.92	3.7455
289.79	1.0000	11.088	0.090189	21.750	21.840	0.10572	0.088879	0.14425	416.16	0.15078
290.23	1.0000	0.52866	1.8916	34.573	36.464	0.15615	0.084363	0.11078	137.07	22.936
300.00	1.0000	0.49205	2.0323	35.486	37.518	0.15972	0.083854	0.10554	143.42	19.634
400.00	1.0000	0.31957	3.1292	44.644	47.773	0.18922	0.094142	0.10545	183.69	7.1401
500.00	1.0000	0.24648	4.0571	54.804	58.861	0.21391	0.10659	0.11631	210.41	3.7496
300.00	5.0000	10.994	0.090955	22.770	23.225	0.10919	0.089725	0.14193	427.56	0.11428
400.00	5.0000	2.2256	0.44932	41.867	44.113	0.16875	0.10069	0.14547	148.00	7.6582
500.00	5.0000	1.3561	0.73741	53.389	57.076	0.19774	0.10859	0.12579	198.35	3.5826
300.00	10.000	11.371	0.087944	22.323	23.203	0.10763	0.089307	0.13525	489.01	-0.035312
400.00	10.000	6.1241	0.16329	37.557	39.190	0.15323	0.10301	0.18569	184.17	2.8522
500.00	10.000	2.9622	0.33758	51.539	54.915	0.18852	0.11046	0.13925	198.11	2.8527
300.00	25.000	12.107	0.082594	21.419	23.484	0.10432	0.089393	0.12713	614.12	-0.22002
400.00	25.000	9.2730	0.10784	34.286	36.982	0.14304	0.10166	0.14227	381.56	0.20064
500.00	25.000	6.6326	0.15077	47.758	51.527	0.17548	0.11215	0.14633	292.53	0.65319
300.00	50.000	12.867	0.077719	20.472	24.358	0.10057	0.090168	0.12262	753.19	-0.32391
400.00	50.000	10.834	0.092300	32.538	37.153	0.13731	0.10251	0.13302	559.24	-0.19415
500.00	50.000	9.0225	0.11083	45.317	50.859	0.16786	0.11339	0.14051	455.37	-0.070101

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., "Pseudo Pure-Fluid Equations of State for the Refrigerant Blends R-410A, R-404A, R-507A, and R-407C," *Int. J. Thermophys.* **24**(4):991–1006, 2003. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the critical point temperature are given in the last entry of the saturation tables. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The estimated uncertainty of density values calculated with the equation of state is 0.1%. The estimated uncertainty of calculated heat capacities and speed of sound values is 0.5%. Uncertainties of bubble and dew point pressures are 0.5%.

TABLE 2-289 Thermodynamic Properties of R-407C

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_v kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
200.00	0.019158	17.036	0.058698	8.8272	8.8283	0.050593	0.070988	0.11073	956.60	-0.31996
210.00	0.035795	16.697	0.059892	9.9359	9.9380	0.056002	0.071320	0.11118	903.06	-0.30662
220.00	0.062640	16.352	0.061156	11.051	11.055	0.061189	0.071817	0.11203	851.40	-0.28934
230.00	0.10366	15.999	0.062503	12.175	12.182	0.066188	0.072410	0.11319	801.12	-0.26790
240.00	0.16353	15.637	0.063949	13.312	13.323	0.071026	0.073074	0.11465	751.77	-0.24161
250.00	0.24755	15.264	0.065512	14.464	14.480	0.075728	0.073803	0.11641	703.01	-0.20937
260.00	0.36157	14.877	0.067218	15.632	15.657	0.080314	0.074597	0.11853	654.53	-0.16951
270.00	0.51193	14.472	0.069099	16.822	16.857	0.084805	0.075463	0.12111	606.09	-0.11964
280.00	0.70540	14.045	0.071198	18.035	18.085	0.089223	0.076412	0.12427	557.44	-0.056172
290.00	0.94916	13.591	0.073576	19.278	19.348	0.093590	0.077458	0.12822	508.32	0.026372
300.00	1.2507	13.102	0.076322	20.555	20.651	0.097931	0.078624	0.13331	458.46	0.13683
310.00	1.6182	12.567	0.079573	21.877	22.006	0.10228	0.079950	0.14013	407.51	0.29038
320.00	2.0599	11.969	0.083552	23.255	23.427	0.10668	0.081509	0.14989	354.98	0.51547
330.00	2.5851	11.278	0.088671	24.711	24.940	0.11119	0.083453	0.16541	300.07	0.87274
340.00	3.2038	10.435	0.095832	26.287	26.594	0.11596	0.086157	0.19551	241.20	1.5203
350.00	3.9255	9.2661	0.10792	28.110	28.534	0.12137	0.090943	0.28993	174.57	3.0499
359.35	4.6317	5.2600	0.19011	32.145	33.025	0.13372			0	10.947
200.00	0.011312	0.0068643	145.68	30.051	31.699	0.16726	0.048920	0.057805	149.59	109.12
210.00	0.022624	0.013151	76.041	30.504	32.224	0.16412	0.050967	0.060200	152.36	88.122
220.00	0.041929	0.023450	42.644	30.957	32.745	0.16151	0.053143	0.062854	154.78	72.006
230.00	0.072846	0.039384	25.391	31.409	33.259	0.15933	0.055439	0.065784	156.79	59.560
240.00	0.11979	0.062913	15.895	31.857	33.761	0.15749	0.057839	0.069010	158.33	49.904
250.00	0.18793	0.096374	10.376	32.298	34.248	0.15593	0.060328	0.072563	159.36	42.378
260.00	0.28317	0.14256	7.0147	32.728	34.715	0.15460	0.062897	0.076500	159.80	36.483
270.00	0.41203	0.20484	4.8820	33.145	35.157	0.15343	0.065542	0.080917	159.60	31.840
280.00	0.58173	0.28739	3.4796	33.544	35.568	0.15240	0.068266	0.085971	158.69	28.163
290.00	0.80008	0.39560	2.5278	33.918	35.940	0.15144	0.071085	0.091920	156.99	25.236
300.00	1.0757	0.53670	1.8632	34.259	36.263	0.15051	0.074027	0.099203	154.41	22.898
310.00	1.4179	0.72101	1.3869	34.556	36.523	0.14956	0.077140	0.10861	150.83	21.024
320.00	1.8375	0.96439	1.0369	34.790	36.696	0.14852	0.080507	0.12170	146.11	19.516
330.00	2.3470	1.2939	0.77283	34.931	36.745	0.14727	0.084283	0.14208	140.03	18.284
340.00	2.9627	1.7642	0.56682	34.916	36.595	0.14560	0.088501	0.18030	132.24	17.206
350.00	3.7100	2.5260	0.39588	34.578	36.047	0.14299	0.095065	0.28700	122.09	15.951
359.35	4.6317	5.2600	0.19011	32.145	33.025	0.13372			0	10.947

TABLE 2-289 Thermodynamic Properties of R-407C (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_v kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Single-Phase Properties										
200.00	0.10000	17.038	0.058692	8.8253	8.8312	0.050583	0.070990	0.11072	956.99	-0.32010
229.25	0.10000	16.026	0.062399	12.091	12.097	0.065819	0.072363	0.11310	804.85	-0.26966
236.25	0.10000	0.053062	18.846	31.690	33.574	0.15814	0.056928	0.067764	157.81	53.242
300.00	0.10000	0.040722	24.557	35.535	37.991	0.17467	0.063341	0.072378	179.00	20.041
400.00	0.10000	0.030231	33.079	42.554	45.862	0.19722	0.076588	0.085147	205.99	7.7925
500.00	0.10000	0.024109	41.479	50.849	54.997	0.21756	0.088895	0.097330	229.27	4.0471
291.84	1.0000	13.504	0.074050	19.510	19.584	0.094388	0.077662	0.12906	499.23	0.044233
297.47	1.0000	0.49738	2.0105	34.177	36.187	0.15075	0.073268	0.097199	155.15	23.442
300.00	1.0000	0.48865	2.0465	34.384	36.431	0.15156	0.072744	0.095419	156.77	22.566
400.00	1.0000	0.31821	3.1425	42.101	45.244	0.17694	0.078067	0.089213	198.26	7.9701
500.00	1.0000	0.24608	4.0637	50.576	54.639	0.19786	0.089416	0.099001	225.88	4.0390
300.00	5.0000	13.412	0.074559	20.240	20.613	0.096862	0.078093	0.12762	507.10	0.027202
400.00	5.0000	2.1880	0.45703	39.458	41.743	0.15675	0.086188	0.12964	161.10	8.5257
500.00	5.0000	1.3504	0.74050	49.289	52.992	0.18193	0.091753	0.10811	213.00	3.9036
300.00	10.000	13.740	0.072780	19.898	20.626	0.095679	0.077756	0.12301	558.94	-0.063246
400.00	10.000	7.1029	0.14079	34.426	35.834	0.13888	0.090433	0.20031	184.71	3.3408
500.00	10.000	2.9957	0.33381	47.547	50.885	0.17282	0.094263	0.12254	207.67	3.3327
300.00	25.000	14.443	0.069238	19.146	20.877	0.092972	0.077634	0.11624	672.43	-0.19834
400.00	25.000	10.899	0.091752	30.990	33.284	0.12855	0.087056	0.13223	399.92	0.28513
500.00	25.000	7.3363	0.13631	43.479	46.886	0.15889	0.096319	0.13592	289.22	0.97116
300.00	50.000	15.220	0.065703	18.302	21.587	0.089730	0.078160	0.11176	802.78	-0.28843
400.00	50.000	12.648	0.079064	29.255	33.209	0.12310	0.087179	0.12077	579.57	-0.12975
500.00	50.000	10.260	0.097468	40.787	45.660	0.15086	0.096753	0.12761	457.54	0.047330

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., "Pseudo Pure-Fluid Equations of State for the Refrigerant Blends R-410A, R-404A, R-507A, and R-407C," *Int. J. Thermophys.* **24**(4):991–1006, 2003. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the critical point temperature are given in the last entry of the saturation tables. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The estimated uncertainty of density values calculated with the equation of state is 0.1%. The estimated uncertainty of calculated heat capacities and speed of sound values is 0.5%. Uncertainties of bubble and dew point pressures are 0.5%.

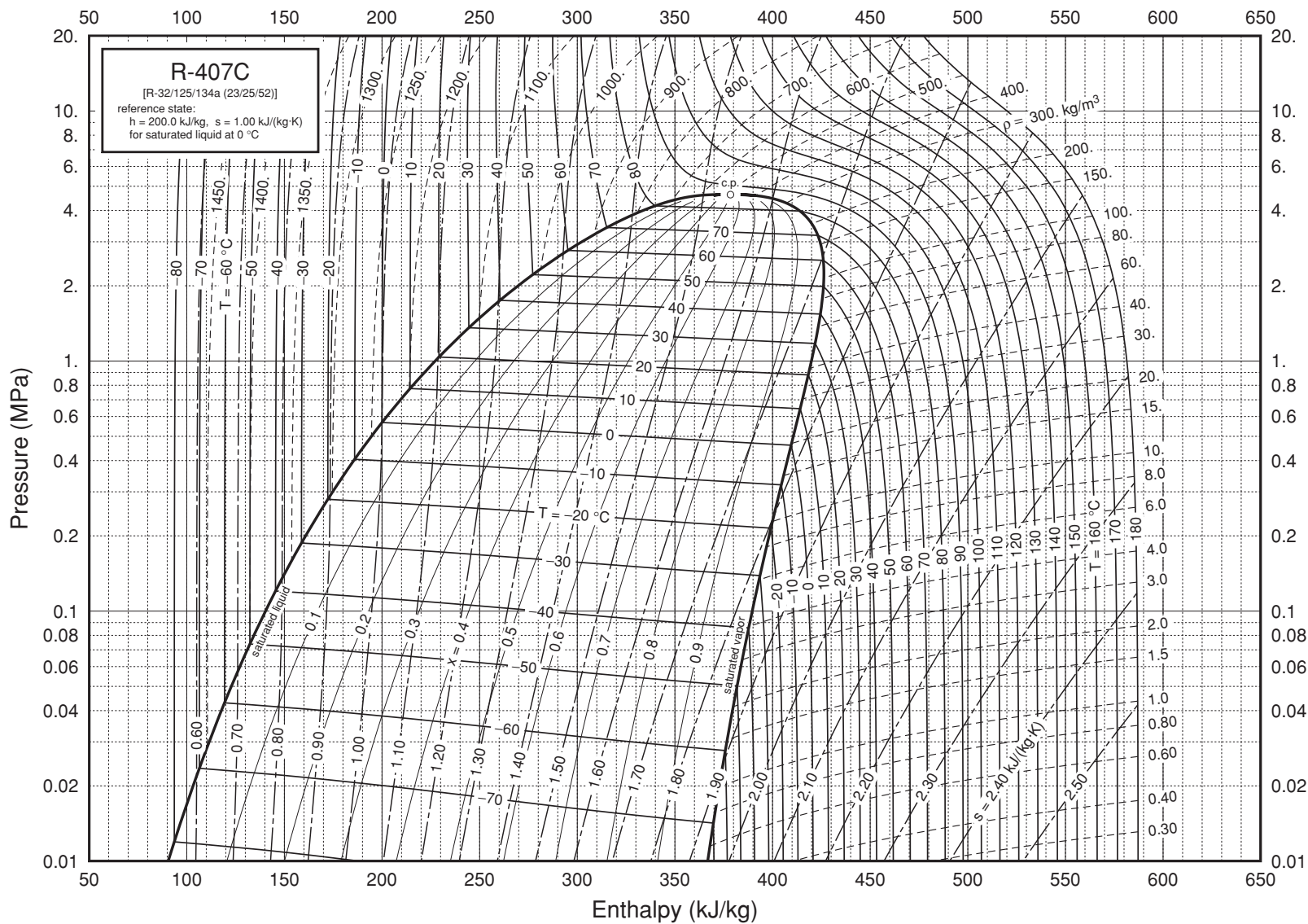


FIG. 2-27 Pressure-enthalpy diagram for Refrigerant 407C. Properties computed with the NIST REFPROP Database, Version 7.0 (Lemmon, E. W., McLinden, M. O., and Huber, M. L., 2002, NIST Standard Reference Database 23, NIST Reference Fluid Thermodynamic and Transport Properties—REFPROP, Version 7.0, Standard Reference Data Program, National Institute of Standards and Technology), based on the mixture model of Lemmon, E. W., and Jacobsen, R. T., "Equations of State for Mixtures of R-32, R-125, R-134a, R-143a, and R-152a," *J. Phys. Chem. Ref. Data* **33**:593–620, 2004.

TABLE 2-290 Thermodynamic Properties of R-410A

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
200.00	0.029160	19.510	0.051256	7.0380	7.0395	0.040995	0.062260	0.097942	929.01	-0.30179
210.00	0.053727	19.093	0.052375	8.0188	8.0217	0.045781	0.062050	0.098396	879.84	-0.28524
215.00	0.071143	18.881	0.052962	8.5112	8.5149	0.048098	0.062014	0.098729	855.20	-0.27544
220.00	0.092819	18.667	0.053571	9.0052	9.0101	0.050370	0.062020	0.099138	830.52	-0.26446
225.00	0.11946	18.449	0.054202	9.5012	9.5077	0.052600	0.062066	0.099628	805.81	-0.25217
230.00	0.15182	18.229	0.054858	9.9997	10.008	0.054791	0.062151	0.10020	781.06	-0.23841
235.00	0.19070	18.005	0.055542	10.501	10.512	0.056948	0.062271	0.10088	756.26	-0.22300
240.00	0.23697	17.776	0.056255	11.006	11.019	0.059073	0.062426	0.10165	731.41	-0.20574
245.00	0.29152	17.543	0.057002	11.514	11.530	0.061169	0.062615	0.10253	706.48	-0.18637
250.00	0.35531	17.305	0.057786	12.026	12.047	0.063240	0.062837	0.10353	681.45	-0.16459
255.00	0.42933	17.062	0.058611	12.543	12.568	0.065289	0.063092	0.10466	656.31	-0.14006
260.00	0.51461	16.812	0.059482	13.065	13.096	0.067318	0.063380	0.10594	631.02	-0.11232
265.00	0.61223	16.555	0.060406	13.593	13.630	0.069331	0.063701	0.10738	605.55	-0.080861
270.00	0.72330	16.290	0.061388	14.127	14.172	0.071331	0.064057	0.10902	579.88	-0.045000
275.00	0.84899	16.016	0.062439	14.669	14.722	0.073321	0.064451	0.11088	553.95	-0.0039006
280.00	0.99048	15.732	0.063567	15.218	15.281	0.075304	0.064884	0.11300	527.72	0.043515
285.00	1.1490	15.436	0.064785	15.776	15.851	0.077284	0.065363	0.11543	501.14	0.098651
290.00	1.3260	15.127	0.066109	16.344	16.432	0.079266	0.065893	0.11825	474.14	0.16337
295.00	1.5226	14.802	0.067559	16.924	17.026	0.081254	0.066483	0.12156	446.66	0.24022
300.00	1.7404	14.459	0.069160	17.516	17.636	0.083253	0.067147	0.12550	418.60	0.33275
305.00	1.9809	14.095	0.070948	18.123	18.263	0.085270	0.067901	0.13029	389.87	0.44607
310.00	2.2456	13.704	0.072969	18.747	18.911	0.087314	0.068773	0.13630	360.33	0.58788
315.00	2.5364	13.282	0.075293	19.392	19.583	0.089398	0.069800	0.14413	329.82	0.77028
320.00	2.8550	12.816	0.078025	20.064	20.287	0.091537	0.071046	0.15493	298.10	1.0135
325.00	3.2037	12.294	0.081343	20.772	21.032	0.093762	0.072616	0.17109	264.83	1.3544
330.00	3.5848	11.685	0.085578	21.531	21.837	0.096123	0.074717	0.19853	229.46	1.8665
335.00	4.0009	10.930	0.091491	22.376	22.742	0.098732	0.077843	0.25685	190.98	2.7232
340.00	4.4556	9.8413	0.10161	23.414	23.867	0.10194	0.083650	0.46832	147.49	4.4554
344.49	4.9012	6.3240	0.15813	25.988	26.763	0.11022			0	9.7623
200.00	0.029010	0.017797	56.190	26.495	28.125	0.14644	0.042482	0.052236	164.41	113.67
210.00	0.053489	0.031567	31.678	26.835	28.530	0.14345	0.044604	0.055055	167.03	90.100
215.00	0.070844	0.041089	24.338	27.002	28.726	0.14212	0.045719	0.056590	168.16	80.508
220.00	0.092447	0.052763	18.953	27.167	28.919	0.14087	0.046862	0.058205	169.16	72.155
225.00	0.11900	0.066925	14.942	27.329	29.107	0.13972	0.048026	0.059899	170.03	64.889
230.00	0.15125	0.083936	11.914	27.488	29.290	0.13864	0.049206	0.061674	170.75	58.571
235.00	0.19000	0.10420	9.5972	27.645	29.468	0.13762	0.050400	0.063533	171.32	53.077
240.00	0.23611	0.12814	7.8039	27.798	29.640	0.13667	0.051603	0.065483	171.73	48.299
245.00	0.29049	0.15625	6.4000	27.947	29.806	0.13577	0.052814	0.067535	171.97	44.137
250.00	0.35407	0.18905	5.2895	28.092	29.965	0.13492	0.054033	0.069705	172.04	40.508
255.00	0.42786	0.22714	4.4026	28.232	30.116	0.13411	0.055260	0.072011	171.93	37.336
260.00	0.51287	0.27117	3.6877	28.367	30.258	0.13333	0.056497	0.074481	171.62	34.560
265.00	0.61019	0.32190	3.1066	28.496	30.392	0.13259	0.057747	0.077148	171.11	32.123
270.00	0.72092	0.38018	2.6303	28.619	30.515	0.13187	0.059014	0.080057	170.39	29.979
275.00	0.84622	0.44702	2.2371	28.733	30.626	0.13116	0.060302	0.083265	169.45	28.087
280.00	0.98729	0.52357	1.9100	28.839	30.725	0.13047	0.061618	0.086846	168.27	26.412
285.00	1.1454	0.61123	1.6360	28.935	30.809	0.12978	0.062969	0.090901	166.84	24.924
290.00	1.3218	0.71170	1.4051	29.019	30.876	0.12908	0.064364	0.095568	165.16	23.598
295.00	1.5179	0.82707	1.2091	29.090	30.925	0.12837	0.065814	0.10104	163.20	22.410
300.00	1.7351	0.95997	1.0417	29.144	30.951	0.12764	0.067335	0.10760	160.94	21.338

305.00	1.9749	1.1138	0.89779	29.178	30.951	0.12688	0.068945	0.11566	158.36	20.365
310.00	2.2390	1.2933	0.77322	29.189	30.920	0.12606	0.070671	0.12589	155.44	19.470
315.00	2.5291	1.5048	0.66456	29.170	30.850	0.12517	0.072551	0.13943	152.13	18.632
320.00	2.8472	1.7576	0.56894	29.112	30.732	0.12418	0.074643	0.15831	148.40	17.826
325.00	3.1955	2.0668	0.48384	29.002	30.548	0.12305	0.077041	0.18670	144.16	17.018
330.00	3.5766	2.4582	0.40681	28.817	30.272	0.12169	0.079915	0.23464	139.30	16.153
335.00	3.9935	2.9848	0.33503	28.510	29.848	0.11995	0.083629	0.33370	133.59	15.123
340.00	4.4504	3.7974	0.26334	27.951	29.123	0.11740	0.089197	0.65947	126.39	13.641
344.49	4.9012	6.3240	0.15813	25.988	26.763	0.11022			0	9.7623

Single-Phase Properties

221.45	0.10000	18.604	0.053751	9.1488	9.1541	0.051020	0.062030	0.099271	823.36	−0.26104
221.53	0.10000	0.056810	17.603	27.217	28.977	0.14051	0.047215	0.058714	169.44	69.827
300.00	0.10000	0.040605	24.628	31.028	33.491	0.15794	0.050980	0.059877	198.35	19.643
400.00	0.10000	0.030202	33.111	36.670	39.981	0.17654	0.061554	0.070067	227.37	7.7467
500.00	0.10000	0.024099	41.495	43.331	47.480	0.19323	0.071249	0.079663	252.59	4.0758
280.32	1.0000	15.713	0.063641	15.253	15.317	0.075429	0.064913	0.11314	526.05	0.046760
280.42	1.0000	0.53054	1.8849	28.848	30.733	0.13041	0.061731	0.087169	168.16	26.279
300.00	1.0000	0.46599	2.1460	30.151	32.297	0.13580	0.057548	0.075210	180.65	20.254
400.00	1.0000	0.31478	3.1769	36.304	39.481	0.15648	0.062713	0.073258	220.92	7.7768
500.00	1.0000	0.24505	4.0808	43.106	47.187	0.17364	0.071665	0.081012	249.79	4.0343
300.00	5.0000	14.870	0.067248	17.202	17.539	0.082188	0.066139	0.11773	472.56	0.17344
400.00	5.0000	1.9755	0.50621	34.349	36.880	0.13813	0.068570	0.097588	192.34	7.6786
500.00	5.0000	1.3185	0.75845	42.072	45.864	0.15821	0.073521	0.087959	239.51	3.7957
300.00	10.000	15.342	0.065180	16.830	17.482	0.080897	0.065435	0.11125	533.86	0.036775
400.00	10.000	5.7949	0.17257	30.845	32.570	0.12363	0.074099	0.16518	182.45	5.0121
500.00	10.000	2.8642	0.34914	40.710	44.202	0.14982	0.075667	0.098492	233.93	3.3106
300.00	25.000	16.289	0.061392	16.058	17.592	0.078110	0.065000	0.10273	658.09	−0.14678
400.00	25.000	11.685	0.085582	26.530	28.670	0.10987	0.072157	0.11830	379.59	0.50709
500.00	25.000	7.4115	0.13493	37.197	40.570	0.13644	0.078574	0.11515	291.31	1.2724
300.00	50.000	17.287	0.057845	15.231	18.123	0.074923	0.065499	0.097459	792.80	−0.26163
400.00	50.000	14.049	0.071182	24.722	28.281	0.10409	0.072480	0.10533	566.13	−0.063564
500.00	50.000	11.128	0.089864	34.526	39.019	0.12804	0.079657	0.10880	449.76	0.13566

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., “Pseudo Pure-Fluid Equations of State for the Refrigerant Blends R-410A, R-404A, R-507A, and R-407C,” *Int. J. Thermophys.* **24**(4):991–1006, 2003. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the critical point temperature are given in the last entry of the saturation tables. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The estimated uncertainty of density values calculated with the equation of state is 0.1%. The estimated uncertainty of calculated heat capacities and speed of sound values is 0.5%. Uncertainties of bubble and dew point pressures are 0.5%.

TABLE 2-291 Saturated Refrigerant 500*

T, K	P, bar	v_f , m ³ /kg	v_g , m ³ /kg	h_f , kJ/kg	h_g , kJ/kg	s_f , kJ/(kg·K)	s_g , kJ/(kg·K)	c_{pf} , kJ/(kg·K)	μ_f , 10 ⁻⁴ Pa·s	k_f , W/(m·K)
200	0.1219	6.966-4	1.360	-29.56	185.87	-0.1363	0.9408	1.044	6.11	0.113
210	0.2258	7.090-4	0.766	-21.03	191.25	-0.0948	0.9161	1.018	5.15	0.109
220	0.3936	7.222-4	0.457	-12.17	196.63	-0.0536	0.8955	0.997	4.42	0.106
230	0.6511	7.361-4	0.286	-2.97	201.96	-0.0130	0.8782	0.987	3.85	0.102
240	1.0291	7.509-4	0.187	6.58	207.23	0.0277	0.8638	0.987	3.42	0.098
250	1.5632	7.668-4	0.1261	16.50	212.40	0.0680	0.8517	0.997	3.04	0.094
260	2.2932	7.839-4	0.0879	26.78	217.45	0.1082	0.8415	1.017	2.74	0.090
270	3.2624	8.024-4	0.0628	37.44	222.35	0.1481	0.8329	1.048	2.48	0.086
280	4.5172	8.226-4	0.0459	48.48	227.06	0.1878	0.8257	1.089	2.26	0.082
290	6.1064	8.450-4	0.0342	59.91	231.56	0.2275	0.8194	1.140	2.08	0.078
300	8.0809	8.699-4	0.0259	71.76	235.79	0.2671	0.8139	1.201	1.92	0.074
310	10.49	8.981-4	0.0198	84.05	239.69	0.3067	0.8088	1.273	1.77	0.070
320	13.40	9.306-4	0.0154	96.83	243.19	0.3464	0.8038	1.355	1.63	0.066
330	16.86	9.690-4	0.0119	110.17	246.14	0.3864	0.7985	1.447	1.48	0.062
340	20.93	1.016-3	0.0093	124.20	248.36	0.4271	0.7922	1.550	1.34	0.058
350	25.70	1.077-3	0.0072	139.18	249.47	0.4689	0.7841	1.663		
360	31.25	1.162-3	0.0055	155.66	248.71	0.5135	0.7721	1.919		
370	37.72	1.307-3	0.0040	175.59	244.26	0.5650	0.7509	2.07		
378.6°	44.26	2.012-3	0.0020	219.50	219.50	0.6729	0.6729	∞		

*Values reproduced and converted from Table 12, p. 17.99, *ASHRAE Handbook, 1981: Fundamentals*, American Society of Heating, Refrigerating and Air-Conditioning Engineers, Atlanta, 1981. Copyright material. Reproduced by permission of the copyright owner. c = critical point. The notation 6.966-4 signifies 6.966×10^{-4} .

The 1993 *ASHRAE Handbook—Fundamentals* (SI ed.) gives material for integral degrees Celsius with temperatures on the IPTS 68 scale from -70 to 105.60 °C. The thermodynamic diagram from 0.1 to 240 °C. Equations and constants approximated to the 1985 ASHRAE tables were given by Mecaryk, K. and M. Masaryk, *Heat Recovery Systems and CHP*, 11, 2/3 (1991): 193-197. Saturation and superheat tables and a diagram to 80 bar, 560 K are given by Reynolds, W. C., *Thermodynamic Properties in S.I.*, Stanford Univ. publ., 1979 (173 pp.). Tables and a chart to 1000 psia, 480 °F are given by Stewart, R. B., R. T. Jacobsen, et al., *Thermodynamic Properties of Refrigerants*, ASHRAE, Atlanta, GA, 1986 (521 pp.). Specific heat and viscosity appear in *Thermophysical Properties of Refrigerants*, ASHRAE, 1993.

TABLE 2-292 Saturated Refrigerant 502*

T, K	P, bar	v_f , m ³ /kg	v_g , m ³ /kg	h_f , kJ/kg	h_g , kJ/kg	s_f , kJ/(kg·K)	s_g , kJ/(kg·K)	c_{pf} , kJ/(kg·K)	μ_f , 10 ⁻⁴ Pa·s	k_f , W/(m·K)
200	0.2274	6.381-4	0.646	-29.04	153.34	-0.1337	0.7782	1.018	5.72	0.103
210	0.4098	6.507-4	0.374	-20.83	158.42	-0.0937	0.7599	1.036	4.88	0.099
220	0.6965	6.640-4	0.228	-12.15	163.49	-0.0534	0.7449	1.055	4.23	0.095
230	1.1251	6.783-4	0.146	-2.99	168.50	-0.0128	0.7328	1.075	3.71	0.091
240	1.7392	6.938-4	0.0969	6.66	173.42	0.0280	0.7228	1.097	3.28	0.087
250	2.5867	7.105-4	0.0665	16.78	178.20	0.0691	0.7148	1.120	2.94	0.083
260	3.7188	7.289-4	0.0470	27.36	182.81	0.1102	0.7082	1.144	2.65	0.079
270	5.1893	7.492-4	0.0340	38.36	187.21	0.1514	0.7027	1.170	2.41	0.075
280	7.0530	7.720-4	0.0251	49.77	191.35	0.1923	0.6980	1.197	2.18	0.072
290	9.3660	7.979-4	0.0188	61.55	195.16	0.2330	0.6937	1.225	1.99	0.068
300	12.19	8.280-4	0.0143	73.68	198.56	0.2734	0.6896	1.254	1.79	0.064
310	15.57	8.637-4	0.0109	86.17	201.43	0.3134	0.6852	1.285	1.59	0.060
320	19.60	9.081-4	0.0084	99.06	203.57	0.3532	0.6798	1.317	1.40	0.056
330	24.35	9.666-4	0.0064	112.53	204.62	0.3933	0.6723	1.351	1.23	0.052
340	29.95	1.053-3	0.0048	127.13	203.71	0.4351	0.6604	1.386	1.07	0.048
350	36.62	1.220-3	0.0033	145.44	197.82	0.4859	0.6355	1.422	0.93	0.044
355.3°	40.75	1.786-3	0.0018	174.00	174.00	0.5634	0.5634			

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The 1993 *ASHRAE Handbook—Fundamentals* (SI ed.) gives material for integral degrees Celsius with temperatures on the IPTS 68 scale from -70 to 82.2 °C. The thermodynamic diagram from 0.1 to 80 bar extends to 180 °C. Equations and constants approximated to 1985 ASHRAE tables are given by Mecaryk, K., and M. Masaryk, *Heat Recovery Systems and CHP*, 11, 2/3 (1991): 193-197. Saturation and superheat tables and a diagram to 20 bar, 515 K are given by Reynolds, W. C., *Thermodynamic Properties in S.I.*, Stanford Univ. publ., 1979 (173 pp.). Tables and a chart to 1000 psia, 400 °F appear in Stewart, R. B., R. T. Jacobsen, et al., *Thermodynamic Properties of Refrigerants*, ASHRAE, Atlanta, GA, 1986 (521 pp.). For specific heat and viscosity, see *Thermophysical Properties of Refrigerants*, ASHRAE, 1993.

TABLE 2-293 Saturated Refrigerant 503*

T, K	P, bar	v_f , m ³ /kg	v_g , m ³ /kg	h_f , kJ/kg	h_g , kJ/kg	s_f , kJ/(kg·K)	s_g , kJ/(kg·K)	c_{pf} , kJ/(kg·K)	μ_f , 10 ⁻⁴ Pa·s	k_f , W/(m·K)
150	0.0750	6.384-4	1.894	-89.60	111.02	-0.4694	0.8681	0.482	6.12	0.128
160	0.1798	6.478-4	0.837	-79.73	115.40	-0.4057	0.8139	0.554	5.05	0.123
170	0.3828	6.585-4	0.414	-69.55	119.70	-0.3441	0.7691	0.620	4.16	0.116
180	0.7395	6.700-4	0.224	-59.08	123.84	-0.2844	0.7318	0.682	3.43	0.111
190	1.3187	6.850-4	0.130	-48.36	127.77	-0.2267	0.7003	0.747	2.94	0.105
200	2.1999	7.014-4	0.0803	-37.45	131.45	-0.1710	0.6735	0.817	2.56	0.099
210	3.4713	7.204-4	0.0520	-26.36	134.84	-0.1173	0.6503	0.896	2.25	0.094
220	5.2281	7.426-4	0.0350	-15.10	137.87	-0.0656	0.6298	0.988	1.98	0.088
230	7.5713	7.687-4	0.0242	-3.65	140.49	-0.0155	0.6112	1.017	1.73	0.082
240	10.61	8.001-4	0.0172	8.07	142.58	0.0334	0.5939	1.227	1.52	0.076
250	14.46	8.386-4	0.0124	20.22	143.98	0.0817	0.5767	1.382	1.33	0.070
260	19.25	8.874-4	0.0090	33.10	144.38	0.1305	0.5585	1.57	1.17	0.065
270	25.13	9.526-4	0.0064	47.22	143.23	0.1816	0.5373	1.79	1.03	0.059
280	32.27	1.050-3	0.0045	63.64	139.25	0.2354	0.5085	2.03	0.91	0.054
290	40.87	1.264-3	0.0028	86.41	127.51	0.3131	0.4548	2.35		
292.6°	43.57	1.773-3	0.0018	110.20	110.20	0.3864	0.3864	∞		∞

* P , v , h , and s values reproduced and converted from Table 14, p. 17.103, ASHRAE Handbook, 1981: Fundamentals, American Society of Heating, Refrigerating and Air-Conditioning Engineers, Atlanta, 1981. Copyright material. Reproduced by permission of the copyright owner. c_p , μ , and k values interpolated and converted from *Thermophysical Properties of Refrigerants*, American Society of Heating, Refrigerating and Air-Conditioning Engineers, New York, 1976. c = critical point. The notation 6.384-4 signifies 6.384×10^{-4} .

Saturation and superheat tables and a diagram to 80 bar, 600 K are given by Reynolds, W. C., *Thermodynamic Properties in S.I.*, Stanford Univ. publ., 1979 (173 pp.). Tables and a chart to 1000 psia, 460 °F are given by Stewart, R. B., R. T. Jacobsen, et al., *Thermodynamic Properties of Refrigerants*, ASHRAE, Atlanta, GA, 1986 (521 pp.). For specific heat and viscosity see *Thermophysical Properties of Refrigerants*, ASHRAE, 1993. The 1993 ASHRAE Handbook—Fundamentals (SI ed.) gives material for integral degrees Celsius with temperatures on the IPTS 68 scale for saturation conditions from -125 to 19.50 °C. The thermodynamic diagram from 0.1 to 80 bar extends to 220 °C.

TABLE 2-294 Saturated Refrigerant 504*

Temp., °F	Pressure, lb/in ² abs.	Volume, ft ³ /lb		Enthalpy, Btu/lb		Entropy, Btu/(lb)(°F)	
		Liquid	Vapor	Liquid	Vapor	Liquid	Vapor
-120	2.964	0.01095	15.31	-21.48	86.69	-0.0565	0.2609
-100	6.042	0.01119	7.874	-16.39	89.31	-0.0420	0.2519
-80	11.34	0.01146	4.372	-11.12	91.84	-0.0277	0.2435
-60	19.85	0.01175	2.585	-5.65	94.25	-0.0137	0.2362
-40	32.76	0.01206	1.609	0.00	96.50	0.0000	0.2299
-20	51.44	0.01242	1.045	5.85	98.58	0.0135	0.2244
0	77.41	0.01282	0.7029	11.91	100.45	0.0269	0.2195
20	112.3	0.01328	0.4859	18.22	102.09	0.0401	0.2150
40	158.0	0.01379	0.3431	24.81	103.44	0.0533	0.2107
60	216.2	0.01443	0.2458	31.78	104.41	0.0667	0.2065
80	289.2	0.01522	0.1773	39.25	104.85	0.0804	0.2020
100	379.1	0.01629	0.1274	47.43	104.49	0.0948	0.1968
120	488.3	0.01783	0.0893	56.78	102.72	0.1107	0.1899
140	618.1	0.02083	0.0578	69.97	97.70	0.1322	0.1784
150	692.2	0.02597	0.0394	76.96	89.76	0.1432	0.1642

*Unpublished data of Allied Chemical Company, 1970. Used by permission.

TABLE 2-295 Thermodynamic Properties of Refrigerant 507*

Temp., K	Pressure, bar	v_f , m ³ /kg	v_g , m ³ /kg	h_f , kJ/kg	h_g , kJ/kg	s_f , kJ/(kg·K)	s_g , kJ/(kg·K)
230.5	1.013	0.000 574	0.1280	-3.1	143.3	-0.015	0.620
240	1.59	0.000 602	0.0826	10.3	150.2	0.042	0.623
250	2.42	0.000 627	0.0546	22.6	154.5	0.092	0.619
260	3.54	0.000 658	0.0377	37.6	159.0	0.149	0.617
270	4.95	0.000 695	0.0270	51.6	163.8	0.202	0.618
280	6.70	0.000 738	0.0198	64.7	169.0	0.250	0.620
290	8.85	0.000 787	0.0148	77.2	174.6	0.295	0.634
300	11.52	0.000 839	0.0112	89.4	180.3	0.336	0.640
310	14.74	0.000 903	0.0084	101.6	185.4	0.378	0.648
320	18.76	0.001 006	0.0062	115.7	188.6	0.422	0.649
330	23.65	0.001 221	0.0042	135.5	189.3	0.481	0.641
340	29.57	0.001 618	0.0025	161.7	179.9	0.557	0.611
341.5°	32.67	0.001 97	0.0020	172.7	172.7	0.590	0.590

*Azeotropic mixture of R152a and R218. $h_f = s_f = 0$ at 233.15 K = -40 °C. Interpolated, extrapolated and converted from Lavrenchenko, G. K., M. G. Khmelniuk, et al., *Int. J. Refrig.*, 17, 7 (1994): 461. Some values are tentative. This source also gives a $\ln P$ - h diagram from 0.6 to 30 bar, -50 to 70 °C. Differences exist between the published diagram and tables. c = critical point.

TABLE 2-296 Thermodynamic Properties of R-507A

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
200.00	0.023233	14.130	0.070772	10.553	10.555	0.059915	0.077250	0.11931	851.53	-0.34068
210.00	0.042731	13.828	0.072318	11.749	11.752	0.065751	0.078412	0.12015	796.42	-0.32336
215.00	0.056515	13.675	0.073126	12.351	12.355	0.068583	0.079028	0.12081	770.28	-0.31246
220.00	0.073637	13.521	0.073961	12.956	12.962	0.071366	0.079658	0.12158	744.80	-0.30008
225.00	0.094634	13.365	0.074825	13.566	13.573	0.074104	0.080295	0.12247	719.84	-0.28620
230.00	0.12008	13.206	0.075722	14.179	14.188	0.076802	0.080939	0.12345	695.27	-0.27072
235.00	0.15060	13.045	0.076655	14.798	14.809	0.079463	0.081589	0.12453	671.01	-0.25349
240.00	0.18683	12.882	0.077628	15.421	15.436	0.082089	0.082245	0.12569	646.97	-0.23433
245.00	0.22945	12.715	0.078646	16.051	16.069	0.084685	0.082907	0.12696	623.07	-0.21297
250.00	0.27919	12.545	0.079714	16.686	16.708	0.087253	0.083579	0.12833	599.27	-0.18910
255.00	0.33679	12.371	0.080837	17.328	17.355	0.089796	0.084260	0.12982	575.51	-0.16232
260.00	0.40302	12.192	0.082022	17.976	18.009	0.092316	0.084954	0.13144	551.75	-0.13214
265.00	0.47868	12.008	0.083277	18.632	18.672	0.094817	0.085661	0.13321	527.94	-0.097937
270.00	0.56461	11.819	0.084612	19.296	19.344	0.097301	0.086385	0.13517	504.05	-0.058916
275.00	0.66165	11.623	0.086039	19.968	20.025	0.099772	0.087128	0.13735	480.04	-0.014071
280.00	0.77071	11.419	0.087571	20.650	20.718	0.10223	0.087895	0.13978	455.86	0.037902
285.00	0.89272	11.208	0.089225	21.343	21.422	0.10469	0.088688	0.14254	431.49	0.098716
290.00	1.0286	10.986	0.091024	22.046	22.140	0.10714	0.089513	0.14571	406.87	0.17067
295.00	1.1794	10.753	0.092996	22.763	22.873	0.10960	0.090377	0.14941	381.97	0.25689
300.00	1.3462	10.507	0.095177	23.495	23.623	0.11207	0.091290	0.15379	356.72	0.36179
305.00	1.5300	10.244	0.097618	24.243	24.392	0.11455	0.092263	0.15912	331.07	0.49171
310.00	1.7322	9.9616	0.10039	25.011	25.185	0.11707	0.093316	0.16577	304.91	0.65619
315.00	1.9539	9.6544	0.10358	25.803	26.006	0.11962	0.094477	0.17441	278.14	0.87027
320.00	2.1967	9.3151	0.10735	26.626	26.862	0.12224	0.095794	0.18622	250.55	1.1591
325.00	2.4622	8.9322	0.11195	27.487	27.763	0.12494	0.097351	0.20363	221.86	1.5687
330.00	2.7523	8.4856	0.11785	28.405	28.729	0.12779	0.099317	0.23262	191.54	2.1936
335.00	3.0697	7.9323	0.12607	29.412	29.799	0.13089	0.10208	0.29323	158.62	3.2672
340.00	3.4178	7.1361	0.14013	30.620	31.099	0.13460	0.10695	0.52176	121.00	5.5872
343.77	3.7049	4.9640	0.20145	32.908	33.654	0.14193			0	12.382
200.00	0.023222	0.014226	70.296	29.992	31.624	0.16527	0.059362	0.068766	137.04	85.105
210.00	0.042726	0.025154	39.755	30.526	32.225	0.16325	0.061910	0.071764	139.19	65.962
215.00	0.056512	0.032678	30.602	30.796	32.525	0.16240	0.063188	0.073318	140.11	58.876
220.00	0.073634	0.041875	23.881	31.067	32.825	0.16166	0.064473	0.074919	140.92	53.008
225.00	0.094628	0.052996	18.869	31.338	33.124	0.16100	0.065766	0.076576	141.60	48.113
230.00	0.12007	0.066313	15.080	31.610	33.420	0.16043	0.067072	0.078300	142.16	44.001
235.00	0.15057	0.082125	12.177	31.881	33.715	0.15992	0.068393	0.080105	142.57	40.521
240.00	0.18678	0.10075	9.9251	32.152	34.005	0.15947	0.069735	0.082005	142.84	37.557
245.00	0.22938	0.12256	8.1593	32.421	34.292	0.15907	0.071100	0.084017	142.95	35.013
250.00	0.27908	0.14794	6.7597	32.688	34.575	0.15872	0.072492	0.086159	142.90	32.816
255.00	0.33664	0.17732	5.6396	32.953	34.851	0.15841	0.073915	0.088453	142.67	30.909
260.00	0.40281	0.21120	4.7348	33.214	35.121	0.15814	0.075372	0.090927	142.26	29.244
265.00	0.47840	0.25014	3.9977	33.471	35.384	0.15788	0.076865	0.093613	141.66	27.785
270.00	0.56424	0.29478	3.3924	33.724	35.638	0.15765	0.078399	0.096551	140.85	26.502
275.00	0.66119	0.34585	2.8914	33.970	35.882	0.15744	0.079978	0.099795	139.83	25.372
280.00	0.77015	0.40424	2.4738	34.209	36.115	0.15723	0.081605	0.10341	138.57	24.376
285.00	0.89202	0.47097	2.1233	34.440	36.334	0.15702	0.083287	0.10750	137.08	23.500
290.00	1.0278	0.54731	1.8271	34.661	36.539	0.15680	0.085030	0.11218	135.32	22.730
295.00	1.1784	0.63483	1.5752	34.869	36.726	0.15656	0.086845	0.11764	133.29	22.057
300.00	1.3450	0.73552	1.3596	35.063	36.892	0.15630	0.088742	0.12415	130.95	21.472
305.00	1.5287	0.85197	1.1737	35.239	37.033	0.15600	0.090740	0.13211	128.30	20.967

310.00	1.7307	0.98768	1.0125	35.391	37.144	0.15565	0.092861	0.14217	125.29	20.534
315.00	1.9522	1.1475	0.87142	35.515	37.217	0.15521	0.095142	0.15545	121.90	20.163
320.00	2.1948	1.3389	0.74689	35.601	37.241	0.15467	0.097636	0.17401	118.07	19.838
325.00	2.4602	1.5733	0.63559	35.634	37.198	0.15397	0.10043	0.20216	113.77	19.531
330.00	2.7503	1.8717	0.53427	35.589	37.058	0.15303	0.10368	0.25052	108.90	19.179
335.00	3.0678	2.2784	0.43891	35.411	36.758	0.15166	0.10772	0.35438	103.35	18.634
340.00	3.4166	2.9318	0.34109	34.936	36.101	0.14932	0.11346	0.74186	96.734	17.407
343.77	3.7049	4.9640	0.20145	32.908	33.654	0.14193			0	12.382

Single-Phase Properties

226.14	0.10000	13.329	0.075025	13.705	13.712	0.074720	0.080441	0.12268	714.23	-0.28283
226.14	0.10000	0.055821	17.915	31.400	33.191	0.16087	0.066061	0.076961	141.74	47.115
300.00	0.10000	0.040743	24.544	36.736	39.190	0.18376	0.077477	0.086497	165.13	16.625
400.00	0.10000	0.030241	33.067	45.323	48.629	0.21080	0.093474	0.10204	190.54	6.8071
500.00	0.10000	0.024113	41.472	55.366	59.514	0.23504	0.10672	0.11516	212.51	3.6966
288.99	1.0000	11.032	0.090648	21.903	21.994	0.10665	0.089344	0.14503	411.87	0.15510
289.02	1.0000	0.53150	1.8815	34.618	36.500	0.15684	0.084683	0.11121	135.69	22.873
300.00	1.0000	0.49044	2.0390	35.647	37.686	0.16087	0.084205	0.10551	142.77	19.202
400.00	1.0000	0.31937	3.1312	44.851	47.983	0.19047	0.094741	0.10601	182.55	7.0158
500.00	1.0000	0.24643	4.0580	55.070	59.128	0.21530	0.10718	0.11688	209.06	3.6962
300.00	5.0000	10.905	0.091702	23.040	23.498	0.11052	0.090271	0.14306	420.12	0.12774
400.00	5.0000	2.2078	0.45293	42.120	44.385	0.17014	0.10115	0.14483	147.77	7.5193
500.00	5.0000	1.3533	0.73892	53.669	57.364	0.19916	0.10913	0.12621	197.37	3.5221
300.00	10.000	11.289	0.088585	22.583	23.469	0.10892	0.089813	0.13611	482.39	-0.028276
400.00	10.000	6.0317	0.16579	37.890	39.548	0.15478	0.10349	0.18529	182.13	2.8871
500.00	10.000	2.9478	0.33924	51.842	55.234	0.18999	0.11096	0.13942	197.45	2.8080
300.00	25.000	12.033	0.083106	21.667	23.744	0.10556	0.089890	0.12779	608.04	-0.21781
400.00	25.000	9.2040	0.10865	34.588	37.304	0.14446	0.10218	0.14278	377.87	0.20389
500.00	25.000	6.5873	0.15181	48.091	51.887	0.17699	0.11266	0.14661	290.93	0.64689
300.00	50.000	12.796	0.078149	20.711	24.618	0.10177	0.090684	0.12322	746.86	-0.32316
400.00	50.000	10.770	0.092847	32.830	37.472	0.13868	0.10306	0.13357	554.88	-0.19407
500.00	50.000	8.9711	0.11147	45.655	51.229	0.16935	0.11396	0.14097	452.46	-0.072291

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., "Pseudo Pure-Fluid Equations of State for the Refrigerant Blends R-410A, R-404A, R-507A, and R-407C," *Int. J. Thermophys.* **24**(4):991–1006, 2003. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the critical point temperature are given in the last entry of the saturation tables. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The estimated uncertainty of density values calculated with the equation of state is 0.1%. The estimated uncertainty of calculated heat capacities and speed of sound values is 0.5%. Uncertainties of bubble and dew point pressures are 0.5%.

TABLE 2-297 Saturated Rubidium*

T, K	P, bar	v_f , m ³ /kg	v_g , m ³ /kg	h_f , kJ/kg	h_g , kJ/kg	s_f , kJ/(kg·K)	s_g , kJ/(kg·K)	c_{pf} , kJ/(kg·K)
312.7 ^m	2.46.-9	6.75.-4		118.7	1036	0.998	3.932	0.379
400	1.69.-6	6.98.-4	2.3.+5	151.6	1057	1.091	3.355	0.375
500	1.73.-4	7.22.-4	2790	188.8	1078	1.174	2.953	0.369
600	0.0037	7.46.-4	156.6	225.4	1096	1.241	2.692	0.362
700	0.0317	7.73.-4	20.75	261.3	1111	1.296	2.511	0.357
800	0.1584	8.10.-4	4.662	296.8	1124	1.343	2.378	0.353
1000	1.467	8.65.-4	0.605	367.6	1150	1.422	2.205	0.360
1200	6.466	9.40.-4	0.159	440.1	1179	1.490	2.104	0.385
1400	18.6	1.03.-3						
1500	28.5	1.08.-3						

*Converted from tables in Vargaftik, *Tables of the Thermophysical Properties of Liquids and Gases*, Nauka, Moscow, 1972, and Hemisphere, Washington, 1975. m = melting point. The notation 2.46.-9 signifies 2.46×10^{-9} .

Many of the Vargaftik values also appear in Ohse, R. W., *Handbook of Thermodynamic and Transport Properties of Alkali Metals*, Blackwell Sci. Pubs., Oxford, 1985 (1020 pp.). This source contains superheat data.

Saturation and superheat tables and a diagram to 40 bar, 1600 K are given by Reynolds, W. C., *Thermodynamic Properties in S.I.*, Stanford Univ. publ., 1979 (173 pp.).

For a Mollier diagram from 0.1 to 320 psia, 1200 to 2700 °R, see Weatherford, W. D., J. C. Tyler, et al., WADD-TR-61-96, 1961. An extensive review of properties of the solid and the saturated liquid was given by Alcock, C. B., M. W. Chase, et al., *J. Phys. Chem. Ref. Data*, **23**, 3 (1994): 385-497.

TABLE 2-298 Thermophysical Properties of Saturated Seawater

Temp., °C	Pressure, bar	v , (m ³ /kg)10 ³	c_p , kJ/(kg·K)	μ , Ns/m ²	k , W/(m·K)	N_p	10 ⁵ κ , 1/bar
0	0.005993	1.000158	4.000	0.001884	0.560	13.46	5.06
1	0.006438	1.000099	4.000	0.001827	0.563	12.98	5.02
2	0.006916	1.000057	4.000	0.001772	0.565	12.55	4.98
3	0.007427	1.000033	4.000	0.001720	0.567	12.13	4.95
4	0.007970	1.000025	4.001	0.001669	0.569	11.74	4.92
5	0.008548	1.000033	4.001	0.001620	0.571	11.35	4.89
6	0.009163	1.000057	4.001	0.001574	0.574	10.97	4.86
7	0.009816	1.000096	4.002	0.001529	0.576	10.62	4.83
8	0.010511	1.000149	4.002	0.001486	0.578	10.29	4.80
9	0.011248	1.000261	4.002	0.001445	0.580	9.97	4.78
10	0.01203	1.000298	4.003	0.001405	0.582	9.70	4.76
11	0.01286	1.000392	4.003	0.001367	0.584	9.37	4.74
12	0.01374	1.000500	4.003	0.001330	0.586	9.09	4.72
13	0.01467	1.000620	4.004	0.001294	0.588	8.81	4.70
14	0.01566	1.000727	4.004	0.001259	0.590	8.54	4.68
15	0.01671	1.000899	4.005	0.001226	0.592	8.29	4.66
16	0.01781	1.001055	4.005	0.001195	0.594	8.06	4.65
17	0.01898	1.001224	4.006	0.001165	0.595	7.82	4.63
18	0.02022	1.001404	4.006	0.001136	0.597	7.62	4.62
19	0.02153	1.001595	4.007	0.001107	0.599	7.41	4.60
20	0.02291	1.001796	4.007	0.001080	0.600	7.21	4.59
21	0.02437	1.002009	4.007	0.001054	0.602	7.02	4.57
22	0.02591	1.002232	4.008	0.001029	0.604	6.82	4.56
23	0.02753	1.002465	4.008	0.001005	0.605	6.66	4.55
24	0.02924	1.002708	4.009	0.000981	0.607	6.48	4.54
25	0.03104	1.002961	4.009	0.000958	0.608	6.31	4.53
26	0.03294	1.003224	4.009	0.000936	0.609	6.16	4.52
27	0.03494	1.003496	4.010	0.000915	0.611	6.01	4.51
28	0.03705	1.003778	4.010	0.000895	0.612	5.86	4.50
29	0.03926	1.004069	4.011	0.000875	0.614	5.72	4.49
30	0.04159	1.004369	4.011	0.000855	0.615	5.58	4.48

$\kappa = (-1/V)(\partial v/\partial p)_T \cdot 10^5$. Thus, at 0 °C, the compressibility is 5.06×10^{-5} /bar.

For further information see, for instance, Bromley, LeR. A., *J. Chem. Eng. Data*, **12**, 2 (1967): 202-206; **13**, 1 (1968): 60-62 and **13**, 3: 399-402; **15**, 2 (1970): 246-253; and *A.I.Ch.E.J.*, **20**, 2 (1974): 326-335.

Thermal conductivity data sources include Castelli, V. J., E. M. Stanley, et al., *Deep Sea Res.*, **211** (1974): 311-318; Levy, F. L., *Int. J. Refrig.*, **5**, 3 (1982): 155-159.

For velocity of sound, see, for instance, U.S. Naval Oceanographic Office SP 58, 1962 (50 pp.). More recent information is contained in UNESCO technical papers. See *Marine Science* No. 38, 1981 (6 pp.) and No. 44, 1983 (53 pp.).

For sea ice properties, see Fukusako, S., *Int. J. Thermophys.*, **11**, 2 (1990): 353-372.

TABLE 2-299 Saturated Sodium

Temp., K	Pressure, bar	v_f , m ³ /kg	v_g , m ³ /kg	h_f , kJ/kg	h_g , kJ/kg	s_f , kJ/(kg·K)	s_g , kJ/(kg·K)	c_{pf} , kJ/(kg·K)	c_{pg} , kJ/(kg·K)	μ_f , 10 ⁻⁶ Pa·s	μ_g , 10 ⁻⁶ Pa·s	k_f , W/(m·K)	k_g , W/(m·K)	Pr_f	Pr_g
371	1.59.-10	0.001 078	8.54.+9	207	4739	2.259	14.475	1.383		688		89.4		0.0106	
400	1.80.-9	0.001 088	8.08.+8	247	4757	2.920	14.195	1.372	0.86	599		87.2		0.0094	
500	8.99.-7	0.001 115	1.99.+6	382	4817	3.222	12.092	1.334	1.25	415		80.1		0.0069	
600	5.57.-5	0.001 144	38022	514	4872	3.462	10.745	1.301	1.80	321		73.7		0.0057	
700	0.00105	0.001 174	2320	642	4921	3.661	10.631	1.277	2.28	264		68.0		0.0050	
800	0.00941	0.001 208	291.5	769	4966	3.830	9.076	1.260	2.59	227	19.6	62.9	0.0343	0.0045	1.48
900	0.05147	0.001 242	58.8	895	5007	3.978	8.547	1.252	2.72	201	20.6	58.3	0.0406	0.0043	1.38
1000	0.1995	0.001 280	16.6	1020	5044	4.110	8.134	1.252	2.70	181	23.0	54.2	0.0455	0.0042	1.36
1100	0.6016	0.001 323	5.95	1146	5079	4.230	7.805	1.261	2.62	166	25.3	50.5	0.0492	0.0042	1.35
1154.7	1.013	0.001 347	3.89	1215	5097	4.290	7.652	1.271	2.56	159	26.5	48.7	0.0522	0.0041	1.30
1200	1.50	0.001 366	2.54	1273	5111	4.340	7.538	1.279	2.51	153	27.5	47.2	0.0547	0.0041	1.26
1300	3.26	0.001 416	1.24	1402	5140	4.444	7.319	1.305	2.43	143	29.9	44.0	0.0570	0.0042	1.27
1400	6.30	0.001 471	0.676	1534	5168	4.542	7.138	1.340	2.39	135	32.2	41.1	0.0592	0.0044	1.30
1500	11.13	0.001 531	0.400	1671	5193	4.636	6.984	1.384	2.36	128	34.6	38.2		0.0046	
1600	18.28	0.001 597	0.253	1812	5217	4.727	6.855	1.437	2.34	122	37.1	35.4		0.0050	
1700	28.28	0.001 675	0.168	1959	5238	4.816	6.745	1.500	2.41	117		32.6		0.0054	
1800	41.61	0.001 761	0.117	2113	5256	4.904	6.650	1.574	2.46	112		29.7		0.0059	
1900	58.70	0.001 862	0.084	2274	5268	4.992	6.568	1.661	2.53	108		26.6		0.0067	
2000	79.91	0.001 984	0.063	2444	5273	5.079	6.494	1.764	2.66	104		23.2		0.0079	
2100	105.5	0.002 174	0.0472	2625	5265			1.926	2.91						
2200	135.7	0.002 320	0.0361	2822	5241			2.190	3.40						
2300	170.6	0.002 584	0.0275	3047	5188			2.690	4.47						
2400	210.3	0.002 985	0.0203	3331	5078			4.012	8.03						
2500	254.7	0.004 19	0.0098	3965	4617			39.3	417.						
2503.7 ^c	256.4	0.004 57	0.0046	4294	4294										

^c = critical point.

s_f values converted from Cordfunke, E. H. P. and R. J. M. Konings, *Thermochemical Data for Reactor Materials and Fission Products*, North Holland Elsevier, NY, 1990. s_g determined as $s_f + (h_g - h_f)/T$. μ_g and k_g values estimated by P. E. Liley. All other values are from Fink, J. K. and L. Leibowitz, Argonne Nat. Lab Rept. ANL/RE-95-2, 1995. The Fink and Leibowitz work also appeared in *High Temp. Materials Sci.*, **35**, 65–103, 1996. Saturation and superheat tables and a diagram to 14 bar, 1700 K are given by Reynolds, W. C., *Thermodynamic Properties in S.I.*, Stanford Univ. publ., 1979 (173 pp.). For a Mollier diagram for 0.1–150 psia, 1500–2700 °R, see Weatherford, P. M., J. C. Tyler, et al., WADD-TR-61-96, 1961.

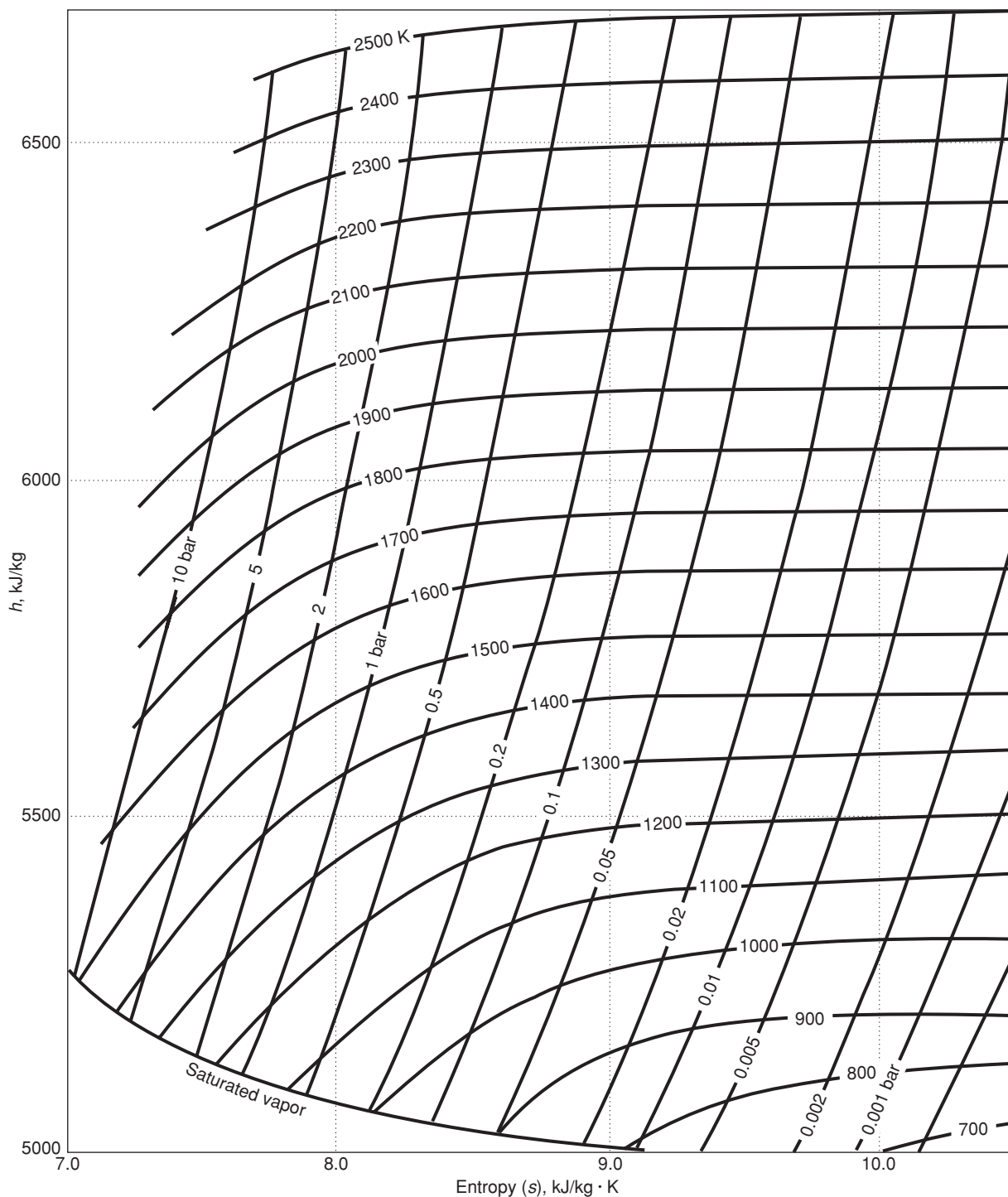


FIG. 2-28 Mollier Diagram for Sodium. Drawn from the Vargaftik et al. values in Ohse, R. W., *Handbook of Thermodynamic and Transport Properties of Alkali Metals*, Blackwell Sci. Pubs., Oxford, UK, 1985. These values are identical with those of Vargaftik, N. B., *Handbook of Thermophysical Properties of Gases and Liquids*, Moscow, 1972, and the Hemisphere translation, p. 19. An apparent discontinuity exists between the superheat values and the saturation values, not reproduced here. For a Mollier diagram in f.p.s. units from 0.1 to 150 psia, 1500 to 2700°R, see Fig. 3-36, p. 3-232 of the 6th edition of this handbook. An extensive review of properties of the solid and the saturated liquid was given by Alcock, C. B., Chase, M. W. et al., *J. Phys. Chem. Ref. Data*, **23**(3), 385–497, 1994.

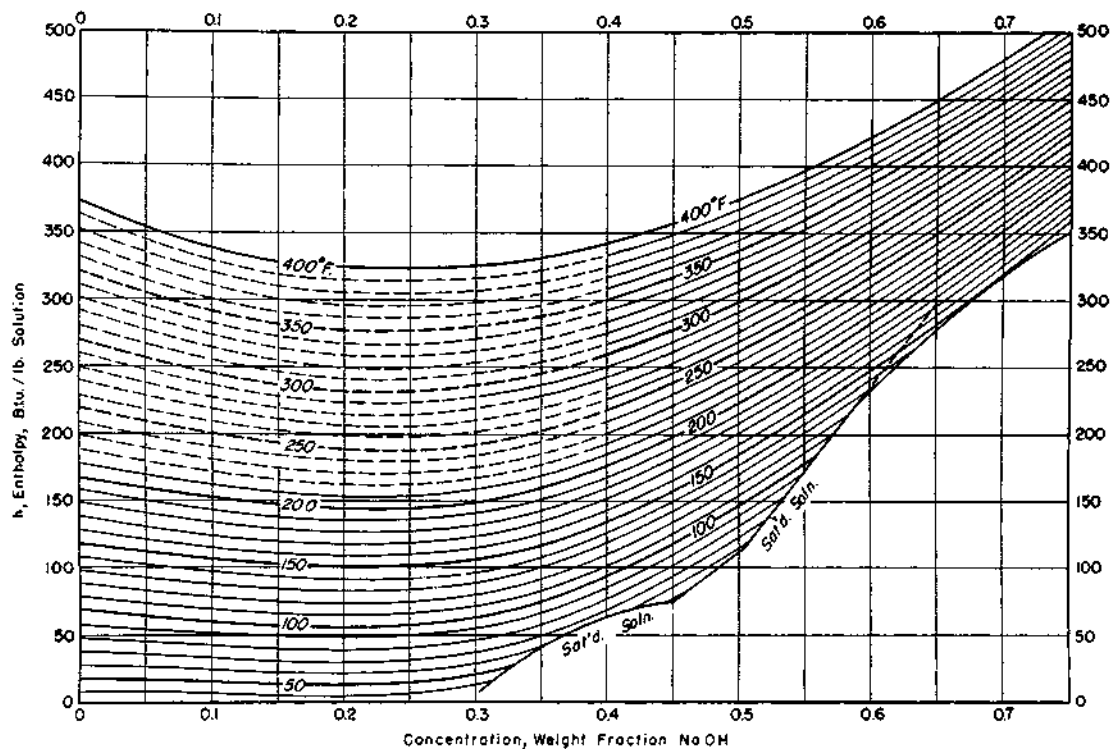


FIG. 2-29 Enthalpy-concentration diagram for aqueous sodium hydroxide at 1 atm. Reference states: enthalpy of liquid water at 32 °F and vapor pressure is zero; partial molal enthalpy of infinitely dilute NaOH solution at 64 °F and 1 atm is zero. [McCabe, Trans. Am. Inst. Chem. Eng., 31, 129 (1935).]

TABLE 2-300 Thermodynamic Properties of Sulfur Dioxide

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_v kJ/(mol·K)	C_p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
197.70	0.0016602	25.290	0.039541	-5.7212	-5.7211	-0.024986	0.056222	0.088123	1361.6	-0.31925
200.00	0.0020260	25.205	0.039674	-5.5186	-5.5185	-0.023967	0.056094	0.088031	1350.1	-0.31864
210.00	0.0045390	24.835	0.040265	-4.6402	-4.6400	-0.019681	0.055527	0.087673	1301.1	-0.31528
220.00	0.0093340	24.463	0.040878	-3.7650	-3.7646	-0.015610	0.054955	0.087396	1253.3	-0.31064
230.00	0.017835	24.088	0.041514	-2.8921	-2.8914	-0.011730	0.054393	0.087214	1206.4	-0.30450
240.00	0.031988	23.709	0.042179	-2.0207	-2.0194	-0.0080208	0.053849	0.087137	1160.3	-0.29665
250.00	0.054309	23.324	0.042875	-1.1497	-1.1473	-0.0044649	0.053331	0.087177	1114.8	-0.28683
260.00	0.087910	22.932	0.043608	-0.27785	-0.27402	-0.0010454	0.052843	0.087344	1069.7	-0.27473
270.00	0.13649	22.532	0.044382	0.59591	0.60197	0.0022525	0.052388	0.087651	1024.9	-0.26000
280.00	0.20431	22.122	0.045204	1.4729	1.4821	0.0054424	0.051969	0.088112	980.26	-0.24217
290.00	0.29614	21.701	0.046081	2.3545	2.3682	0.0085368	0.051587	0.088748	935.65	-0.22070
300.00	0.41725	21.267	0.047021	3.2423	3.2619	0.011547	0.051243	0.089582	890.99	-0.19485
310.00	0.57327	20.818	0.048035	4.1378	4.1653	0.014485	0.050938	0.090647	846.15	-0.16370
320.00	0.77025	20.352	0.049136	5.0430	5.0809	0.017362	0.050675	0.091988	801.04	-0.12603
330.00	1.0145	19.865	0.050340	5.9601	6.0112	0.020187	0.050455	0.093668	755.54	-0.080167
340.00	1.3128	19.354	0.051670	6.8916	6.9594	0.022972	0.050281	0.095777	709.53	-0.023832
350.00	1.6720	18.814	0.053153	7.8408	7.9296	0.025730	0.050157	0.098448	662.87	0.046220
360.00	2.0994	18.239	0.054828	8.8116	8.9267	0.028473	0.050089	0.10189	615.37	0.13476
370.00	2.6028	17.621	0.056750	9.8095	9.9572	0.031220	0.050089	0.10644	566.77	0.24913
380.00	3.1904	16.949	0.059001	10.842	11.030	0.033991	0.050173	0.11271	516.72	0.40128
390.00	3.8713	16.204	0.061713	11.921	12.160	0.036818	0.050373	0.12189	464.59	0.61221
400.00	4.6557	15.355	0.065124	13.066	13.369	0.039753	0.050753	0.13673	409.36	0.92296
410.00	5.5562	14.340	0.069734	14.317	14.704	0.042899	0.051457	0.16539	349.05	1.4276
420.00	6.5903	12.987	0.077001	15.782	16.290	0.046536	0.052928	0.24772	279.41	2.4089
430.00	7.7908	9.8261	0.10177	18.415	19.207	0.053143	0.058931	3.7896	182.70	5.8716
430.64	7.8753	8.1950	0.12203	19.585	20.546	0.056233			0	7.4962
197.70	0.0016602	0.0010120	988.16	21.102	22.743	0.11899	0.028340	0.036808	182.19	299.66
200.00	0.0020260	0.0012211	818.93	21.164	22.824	0.11774	0.028464	0.036952	183.16	280.77
210.00	0.0045390	0.0026096	383.20	21.435	23.174	0.11277	0.029049	0.037650	187.23	214.32
220.00	0.0093340	0.0051346	194.76	21.702	23.520	0.10841	0.029713	0.038475	191.08	166.88
230.00	0.017835	0.0094158	106.20	21.967	23.861	0.10458	0.030462	0.039438	194.70	132.33
240.00	0.031988	0.016257	61.512	22.226	24.194	0.10120	0.031293	0.040551	198.07	106.70
250.00	0.054309	0.026653	37.519	22.480	24.518	0.098195	0.032204	0.041819	201.15	87.372
260.00	0.087910	0.041793	23.928	22.726	24.830	0.095507	0.033186	0.043246	203.92	72.572
270.00	0.13649	0.063061	15.858	22.964	25.128	0.093090	0.034230	0.044838	206.35	61.089
280.00	0.20431	0.092046	10.864	23.191	25.411	0.090902	0.035327	0.046602	208.42	52.073
290.00	0.29614	0.13055	7.6600	23.407	25.676	0.088908	0.036465	0.048552	210.09	44.917
300.00	0.41725	0.18062	5.5366	23.611	25.921	0.087077	0.037638	0.050709	211.33	39.185
310.00	0.57327	0.24460	4.0883	23.799	26.142	0.085379	0.038838	0.053111	212.12	34.550
320.00	0.77025	0.32522	3.0748	23.969	26.338	0.083789	0.040062	0.055813	212.43	30.772
330.00	1.0145	0.42570	2.3491	24.120	26.503	0.082284	0.041310	0.058901	212.22	27.666
340.00	1.3128	0.54995	1.8184	24.247	26.634	0.080838	0.042587	0.062502	211.47	25.092
350.00	1.6720	0.70287	1.4227	24.345	26.724	0.079427	0.043900	0.066814	210.13	22.939
360.00	2.0994	0.89084	1.1225	24.409	26.765	0.078025	0.045264	0.072150	208.17	21.118
370.00	2.6028	1.1225	0.89086	24.430	26.748	0.076601	0.046699	0.079031	205.54	19.558
380.00	3.1904	1.4103	0.70908	24.396	26.658	0.075116	0.048236	0.088395	202.18	18.192
390.00	3.8713	1.7733	0.56391	24.289	26.472	0.073516	0.049922	0.10210	198.02	16.958
400.00	4.6557	2.2444	0.44556	24.079	26.153	0.071714	0.051835	0.12442	192.95	15.777
410.00	5.5562	2.8887	0.34618	23.705	25.628	0.069543	0.054121	0.16795	186.81	14.529
420.00	6.5903	3.8804	0.25770	23.009	24.707	0.066577	0.057117	0.29279	179.31	12.934
430.00	7.7908	6.6270	0.15090	20.801	21.976	0.059582	0.062123	4.6423	168.96	9.1885
430.64	7.8753	8.1950	0.12203	19.585	20.546	0.056233			0	7.4962

Single-Phase Properties

200.00	0.10000	25.206	0.039672	-5.5197	-5.5158	-0.023973	0.056096	0.088028	1350.4	-0.31868
262.84	0.10000	22.819	0.043823	-0.030118	-0.025736	-0.000097639	0.052710	0.087416	1057.0	-0.27084
262.84	0.10000	0.047141	21.213	22.794	24.916	0.094795	0.033476	0.043681	204.65	69.019
300.00	0.10000	0.040764	24.532	24.043	26.497	0.10042	0.032826	0.041994	219.46	39.646
400.00	0.10000	0.030240	33.068	27.458	30.765	0.11269	0.035338	0.043891	252.48	15.140
500.00	0.10000	0.024117	41.465	31.151	35.297	0.12279	0.038292	0.046720	280.65	8.3690
200.00	1.0000	25.219	0.039653	-5.5302	-5.4905	-0.024025	0.056110	0.087995	1353.1	-0.31905
300.00	1.0000	21.287	0.046976	3.2251	3.2721	0.011490	0.051245	0.089444	894.23	-0.19703
329.46	1.0000	19.892	0.050272	5.9101	5.9604	0.020035	0.050466	0.093567	758.01	-0.082891
329.46	1.0000	0.41969	2.3827	24.112	26.495	0.082363	0.041242	0.058722	212.25	27.819
400.00	1.0000	0.31968	3.1281	27.007	30.135	0.092405	0.037220	0.048421	244.07	15.238
500.00	1.0000	0.24701	4.0483	30.891	34.939	0.10312	0.038821	0.048357	276.82	8.3664
200.00	5.0000	25.271	0.039570	-5.5759	-5.3780	-0.024255	0.056174	0.087853	1365.0	-0.32064
300.00	5.0000	21.420	0.046686	3.1113	3.3448	0.011108	0.051267	0.088566	915.79	-0.21095
400.00	5.0000	15.439	0.064770	13.003	13.327	0.039592	0.050649	0.13367	417.04	0.87444
403.99	5.0000	14.976	0.066773	13.548	13.882	0.040973	0.050981	0.14563	386.08	1.0922
403.99	5.0000	2.4748	0.40407	23.955	25.975	0.070906	0.052690	0.13790	190.64	15.297
500.00	5.0000	1.3950	0.71686	29.591	33.176	0.087109	0.041506	0.058653	259.84	8.2323
300.00	10.000	21.575	0.046350	2.9776	3.4411	0.010654	0.051308	0.087615	941.12	-0.22613
400.00	10.000	16.326	0.061251	12.327	12.940	0.037839	0.049804	0.11122	501.11	0.46626
500.00	10.000	3.4161	0.29273	27.477	30.404	0.076991	0.045669	0.085639	241.12	7.4245
300.00	20.000	21.859	0.045748	2.7336	3.6486	0.0098108	0.051422	0.086075	987.39	-0.25095
400.00	20.000	17.379	0.057542	11.494	12.644	0.035620	0.049297	0.096839	608.17	0.15187
500.00	20.000	9.7192	0.10289	22.208	24.266	0.061366	0.049635	0.13354	292.48	2.7261
300.00	35.000	22.232	0.044980	2.4136	3.9878	0.0086740	0.051645	0.084373	1048.5	-0.27869
400.00	35.000	18.377	0.054417	10.681	12.586	0.033379	0.049231	0.088711	717.32	-0.043488
500.00	35.000	13.607	0.073492	19.322	21.894	0.054113	0.048429	0.097089	462.83	0.67011

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., and Span, R., *J. Chem. Eng. Data*, **51**(3):785–850, 2006. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainty in density of the equation of state ranges from 0.1% at low temperatures in the liquid and vapor to 0.5% at the highest temperatures. The uncertainty in heat capacities is 2%, and the uncertainty in vapor pressure is 0.4% at temperatures above 270 K. The uncertainty in vapor pressure increases at lower temperatures due to the lack of experimental data. In the critical region, the uncertainties are higher for all properties except vapor pressure.

TABLE 2-301 Thermodynamic Properties of Sulfur Hexafluoride

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
222.38	0.22436	12.677	0.078884	22.560	22.578	0.11968	0.063933	0.10620	578.22	-0.32279
225.00	0.25012	12.592	0.079415	22.842	22.862	0.12094	0.066189	0.11001	565.05	-0.29395
230.00	0.30561	12.424	0.080489	23.405	23.430	0.12342	0.069989	0.11631	539.86	-0.24573
235.00	0.37011	12.249	0.081642	23.996	24.026	0.12596	0.073230	0.12154	514.71	-0.20297
240.00	0.44448	12.066	0.082875	24.610	24.647	0.12855	0.076028	0.12596	489.70	-0.16256
245.00	0.52962	11.878	0.084192	25.244	25.288	0.13117	0.078482	0.12981	464.88	-0.12208
250.00	0.62644	11.682	0.085600	25.894	25.948	0.13380	0.080672	0.13330	440.25	-0.079442
255.00	0.73591	11.480	0.087108	26.559	26.623	0.13643	0.082667	0.13661	415.80	-0.032633
260.00	0.85899	11.270	0.088731	27.238	27.315	0.13908	0.084519	0.13994	391.49	0.020496
265.00	0.99671	11.051	0.090487	27.932	28.022	0.14172	0.086274	0.14344	367.25	0.082465
270.00	1.1502	10.822	0.092400	28.640	28.746	0.14438	0.087968	0.14730	343.00	0.15646
275.00	1.3204	10.581	0.094507	29.363	29.488	0.14704	0.089631	0.15173	318.65	0.24678
280.00	1.5088	10.325	0.096852	30.105	30.251	0.14973	0.091292	0.15702	294.10	0.35955
285.00	1.7164	10.050	0.099503	30.867	31.037	0.15244	0.092978	0.16356	269.23	0.50392
290.00	1.9447	9.7505	0.10256	31.653	31.852	0.15519	0.094719	0.17202	243.92	0.69427
295.00	2.1952	9.4189	0.10617	32.470	32.703	0.15801	0.096558	0.18353	218.04	0.95460
300.00	2.4696	9.0428	0.11059	33.327	33.600	0.16093	0.098567	0.20035	191.41	1.3279
305.00	2.7698	8.6013	0.11626	34.240	34.562	0.16399	0.10090	0.22773	163.77	1.8996
310.00	3.0984	8.0537	0.12417	35.238	35.623	0.16732	0.10399	0.28254	134.40	2.8784
318.73	3.7539	5.0926	0.19636	38.575	39.312	0.17871			0	11.978
222.38	0.22436	0.13009	7.6872	36.790	38.515	0.19135	0.069340	0.080626	112.78	24.151
225.00	0.25012	0.14419	6.9355	36.943	38.678	0.19124	0.070316	0.081915	112.82	23.609
230.00	0.30561	0.17448	5.7314	37.235	38.987	0.19106	0.072196	0.084476	112.77	22.653
235.00	0.37011	0.20966	4.7696	37.528	39.293	0.19093	0.074101	0.087187	112.56	21.791
240.00	0.44448	0.25032	3.9949	37.820	39.595	0.19083	0.076031	0.090076	112.19	21.015
245.00	0.52962	0.29713	3.3655	38.110	39.892	0.19077	0.077991	0.093176	111.65	20.317
250.00	0.62644	0.35086	2.8502	38.397	40.183	0.19074	0.079983	0.096531	110.92	19.693
255.00	0.73591	0.41240	2.4248	38.681	40.465	0.19072	0.082010	0.10020	109.99	19.139
260.00	0.85899	0.48285	2.0711	38.960	40.739	0.19071	0.084077	0.10426	108.86	18.652
265.00	0.99671	0.56348	1.7747	39.232	41.001	0.19070	0.086191	0.10882	107.50	18.230
270.00	1.1502	0.65592	1.5246	39.496	41.249	0.19069	0.088360	0.11403	105.91	17.872
275.00	1.3204	0.76219	1.3120	39.749	41.481	0.19066	0.090597	0.12013	104.06	17.578
280.00	1.5088	0.88492	1.1301	39.988	41.693	0.19059	0.092917	0.12746	101.93	17.349
285.00	1.7164	1.0276	0.97310	40.209	41.879	0.19048	0.095345	0.13657	99.489	17.185
290.00	1.9447	1.1953	0.83661	40.407	42.034	0.19030	0.097913	0.14840	96.707	17.086
295.00	2.1952	1.3951	0.71677	40.573	42.147	0.19003	0.10068	0.16473	93.543	17.049
300.00	2.4696	1.6387	0.61023	40.695	42.202	0.18960	0.10372	0.18917	89.944	17.065
305.00	2.7698	1.9464	0.51377	40.751	42.174	0.18895	0.10722	0.23074	85.845	17.098
310.00	3.0984	2.3611	0.42353	40.693	42.006	0.18791	0.11155	0.31922	81.148	17.036
318.73	3.7539	5.0926	0.19636	38.575	39.312	0.17871			0	11.978

Single-Phase Properties

225.00	0.10000	0.054962	18.194	37.124	38.943	0.19969	0.069072	0.078403	117.21	20.832
300.00	0.10000	0.040537	24.669	43.122	45.589	0.22504	0.089337	0.098060	135.40	8.9939
375.00	0.10000	0.032243	31.015	50.424	53.525	0.24859	0.10425	0.11277	151.16	4.9674
450.00	0.10000	0.026801	37.312	58.668	62.399	0.27014	0.11482	0.12326	165.37	3.1241
525.00	0.10000	0.022943	43.585	67.581	71.940	0.28973	0.12235	0.13075	178.44	2.1242
225.00	1.0000	12.618	0.079249	22.807	22.886	0.12079	0.066092	0.10946	570.46	-0.30187
265.11	1.0000	11.046	0.090528	27.948	28.038	0.14178	0.086313	0.14352	366.71	0.083989
265.11	1.0000	0.56543	1.7686	39.238	41.007	0.19070	0.086239	0.10893	107.47	18.222
300.00	1.0000	0.45567	2.1946	42.504	44.699	0.20379	0.091458	0.10595	122.90	10.582
375.00	1.0000	0.33888	2.9509	50.051	53.002	0.22845	0.10508	0.11588	145.14	5.2070
450.00	1.0000	0.27473	3.6399	58.408	62.048	0.25042	0.11528	0.12496	162.15	3.1619
525.00	1.0000	0.23243	4.3024	67.387	71.689	0.27022	0.12264	0.13181	176.71	2.1142
300.00	5.0000	9.6705	0.10341	32.693	33.210	0.15873	0.096509	0.16433	257.24	0.56212
375.00	5.0000	2.2583	0.44281	47.826	50.040	0.20902	0.10970	0.14796	119.18	6.1481
450.00	5.0000	1.5252	0.65566	57.112	60.390	0.23421	0.11717	0.13497	151.89	3.1819
525.00	5.0000	1.2155	0.82269	66.464	70.578	0.25515	0.12378	0.13716	173.02	2.0030
300.00	10.000	10.303	0.097057	31.996	32.966	0.15625	0.095358	0.14728	332.30	0.14584
375.00	10.000	5.9487	0.16810	44.159	45.840	0.19435	0.11121	0.18759	145.44	2.6476
450.00	10.000	3.3384	0.29955	55.295	58.291	0.22473	0.11863	0.15056	154.01	2.5753
525.00	10.000	2.4910	0.40144	65.252	69.266	0.24729	0.12474	0.14433	177.25	1.6978
300.00	25.000	11.266	0.088766	30.853	33.072	0.15199	0.095576	0.13478	455.74	-0.17165
375.00	25.000	9.1060	0.10982	41.023	43.768	0.18375	0.10890	0.14958	314.62	0.14966
450.00	25.000	7.0706	0.14143	51.697	55.233	0.21161	0.11870	0.15437	248.92	0.47538
525.00	25.000	5.5874	0.17897	62.312	66.786	0.23536	0.12549	0.15348	234.29	0.57992
300.00	50.000	12.151	0.082298	29.792	33.906	0.14767	0.098484	0.12950	565.06	-0.30996
375.00	50.000	10.625	0.094118	39.323	44.029	0.17775	0.10946	0.14045	454.58	-0.19915
450.00	50.000	9.2093	0.10859	49.451	54.881	0.20410	0.11846	0.14799	394.01	-0.10589
525.00	50.000	8.0206	0.12468	59.888	66.122	0.22720	0.12538	0.15134	361.13	-0.058213

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is de Reuck, K. M., Craven, R. J. B., and Cole, W. A., "Report on the Development of an Equation of State for Sulphur Hexafluoride," *IUPAC Thermodynamic Tables Project Centre*, London, 1991. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties of the equation of state are 0.1% in density, 2% in heat capacity, and 5% in the speed of sound, except in the critical region.

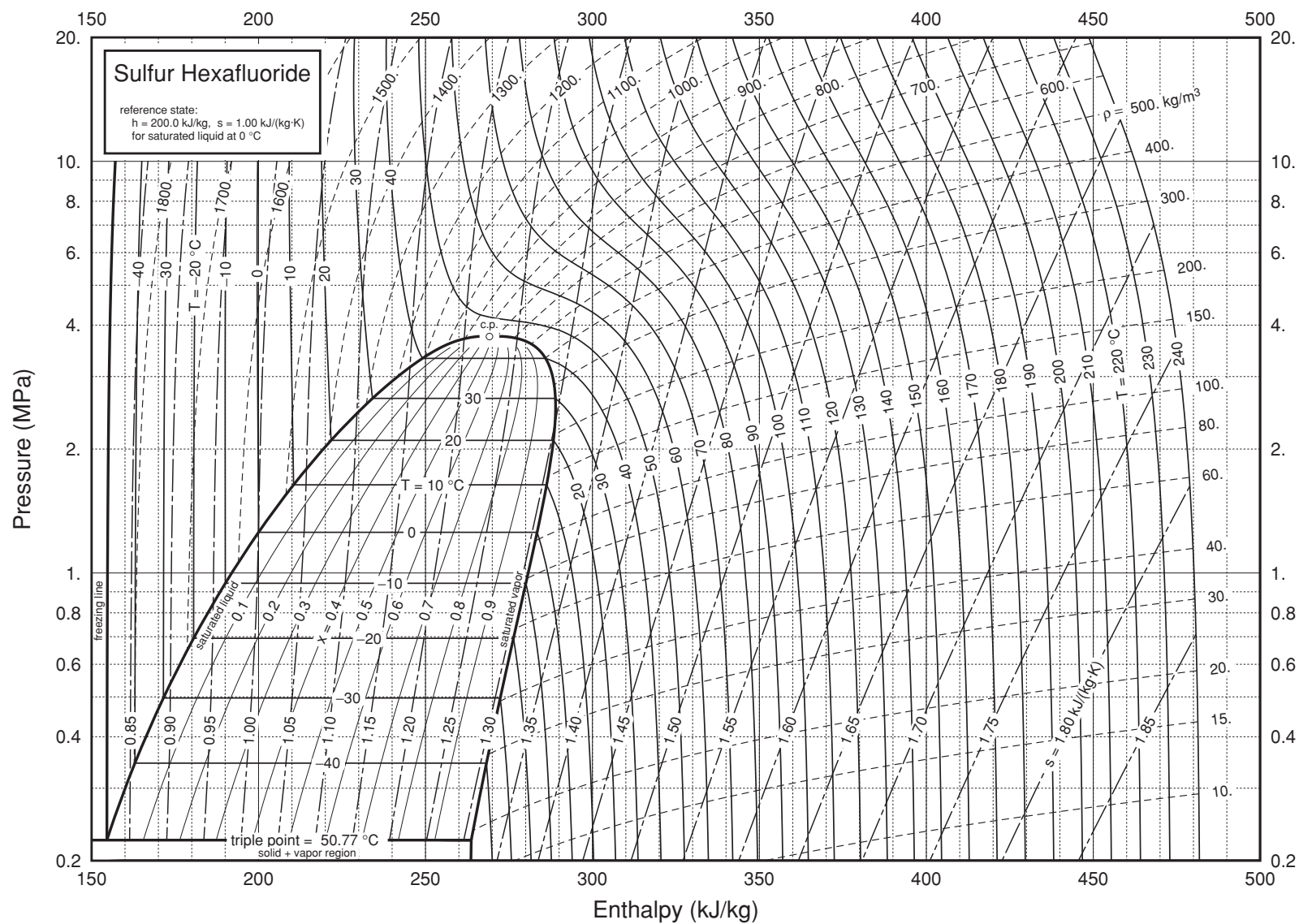


FIG. 2-30 Pressure-enthalpy diagram for sulfur hexafluoride (SF_6). Properties computed with the NIST REFPROP Database, Version 7.0 (Lemmon, E. W., McLinden, M. O., and Huber, M. L., 2002, NIST Standard Reference Database 23, NIST Reference Fluid Thermodynamic and Transport Properties—REFPROP, Version 7.0, Standard Reference Data Program, National Institute of Standards and Technology), based on the equation of state of de Reuck, K. M., Craven, R. J. B., and Cole, W. A., "Report on the Development of an Equation of State for Sulphur Hexafluoride," IUPAC Thermodynamic Tables Project Centre, London, 1991.

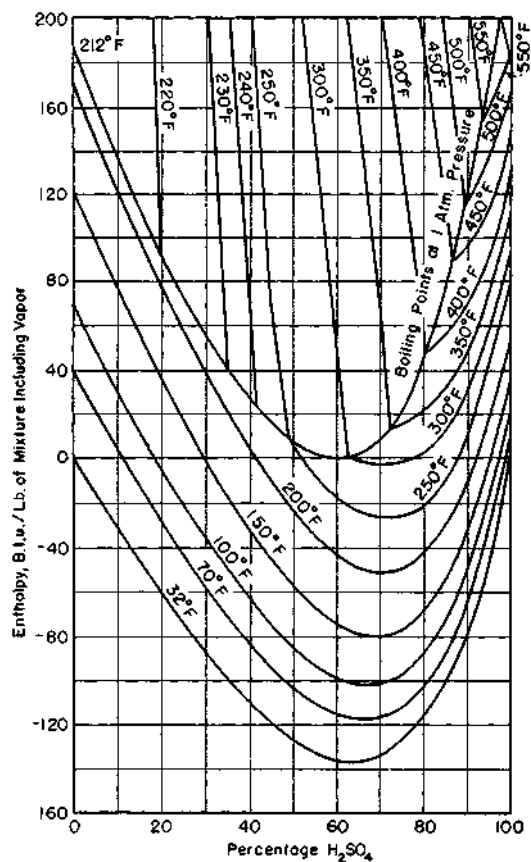


FIG. 2-31 Enthalpy-concentration diagram for aqueous sulfuric acid at 1 atm. Reference states: enthalpies of pure-liquid components at 32° F and vapor pressures are zero. NOTE: It should be observed that the weight basis includes the vapor, which is particularly important in the two-phase region. The upper ends of the tie lines in this region are assumed to be pure water. (*Hougen and Watson, Chemical Process Principles, part I, Wiley, New York, 1943.*)

TABLE 2-302 Saturated SUVA AC 9000

DuPont bulletin T-AC-9000-SI, 1994 (16 pp.) gives tables and a chart to 100 bar, 235°C. With a stated composition of 23% wt CH_2F_2 (R23), 25% wt CHF_2CF_3 (R125), and 52% wt CH_2FCH_3 (R134a) this is apparently identical to KLEA 66, to which the reader is referred.

TABLE 2-303 Thermodynamic Properties of Toluene

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C_p kJ/(mol·K)	C_v kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa
Saturated Properties										
178.00	3.9393E-08	10.580	0.094517	-31.779	-31.779	-0.11617	0.094332	0.13565	1887.6	-0.57170
180.00	5.5336E-08	10.559	0.094708	-31.508	-31.508	-0.11466	0.094377	0.13561	1876.5	-0.57178
200.00	1.0833E-06	10.349	0.096627	-28.792	-28.792	-0.10035	0.095732	0.13627	1768.3	-0.56718
220.00	1.1479E-05	10.144	0.098582	-26.046	-26.046	-0.087266	0.098418	0.13855	1664.8	-0.55455
240.00	7.7542E-05	9.9416	0.10059	-23.242	-23.242	-0.075068	0.10209	0.14205	1565.9	-0.53604
260.00	0.00037312	9.7408	0.10266	-20.358	-20.358	-0.063529	0.10651	0.14648	1471.5	-0.51329
280.00	0.0013829	9.5402	0.10482	-17.378	-17.378	-0.052490	0.11147	0.15162	1381.2	-0.48747
300.00	0.0041774	9.3385	0.10708	-14.290	-14.289	-0.041839	0.11682	0.15729	1294.5	-0.45928
320.00	0.010727	9.1347	0.10947	-11.084	-11.083	-0.031496	0.12243	0.16337	1210.9	-0.42904
340.00	0.024170	8.9275	0.11201	-7.7538	-7.7511	-0.021403	0.12819	0.16976	1130.0	-0.39664
360.00	0.048980	8.7157	0.11474	-4.2939	-4.2882	-0.011516	0.13403	0.17640	1051.3	-0.36162
380.00	0.090988	8.4979	0.11768	-0.70017	-0.68946	-0.0018023	0.13988	0.18325	974.11	-0.32305
400.00	0.15731	8.2722	0.12089	3.0306	3.0496	0.0077653	0.14571	0.19032	898.08	-0.27947
420.00	0.25622	8.0367	0.12443	6.9017	6.9336	0.017209	0.15148	0.19764	822.67	-0.22860
440.00	0.39698	7.7887	0.12839	10.917	10.968	0.026551	0.15718	0.20531	747.37	-0.16688
460.00	0.58974	7.5246	0.13290	15.082	15.160	0.035811	0.16280	0.21350	671.66	-0.088509
480.00	0.84559	7.2394	0.13813	19.404	19.521	0.045015	0.16836	0.22253	594.94	0.016497
500.00	1.1766	6.9258	0.14439	23.896	24.066	0.054196	0.17388	0.23298	516.53	0.16681
520.00	1.5964	6.5719	0.15216	28.580	28.823	0.063400	0.17943	0.24610	435.57	0.40139
540.00	2.1207	6.1562	0.16244	33.494	33.839	0.072709	0.18518	0.26495	350.82	0.81616
560.00	2.7691	5.6334	0.17751	38.729	39.220	0.082291	0.19153	0.30025	260.06	1.7226
580.00	3.5688	4.8513	0.20613	44.585	45.321	0.092723	0.20008	0.44025	157.29	4.9610
591.75	4.1264	3.1690	0.31556	50.827	52.129	0.10409			0	19.734
178.00	3.9393E-08	2.6618E-08	37,569,000.	12.076	13.556	0.13852	0.054735	0.063049	136.02	821.69
180.00	5.5336E-08	3.6974E-08	27,046,000.	12.186	13.683	0.13640	0.055322	0.063636	136.69	783.08
200.00	1.0833E-06	6.5146E-07	1,535,000.	13.353	15.016	0.11869	0.061435	0.069750	143.14	496.50
220.00	1.1479E-05	6.2760E-06	159,340.	14.646	16.475	0.10601	0.067930	0.076246	149.27	328.33
240.00	7.7542E-05	3.8866E-05	25,729.	16.071	18.066	0.097048	0.074733	0.083057	155.12	225.22
260.00	0.00037312	0.00017271	5,790.1	17.632	19.793	0.090897	0.081779	0.090124	160.70	159.61
280.00	0.0013829	0.00059503	1,680.6	19.331	21.655	0.086914	0.089005	0.097403	166.00	116.50
300.00	0.0041774	0.0016815	594.71	21.165	23.649	0.084624	0.096358	0.10486	170.96	87.391
320.00	0.010727	0.0040644	246.04	23.130	25.769	0.083666	0.10379	0.11248	175.47	67.283
340.00	0.024170	0.0086767	115.25	25.218	28.003	0.083758	0.11127	0.12026	179.42	53.119
360.00	0.048980	0.016770	59.631	27.421	30.342	0.084678	0.11875	0.12819	182.67	42.979
380.00	0.090988	0.029922	33.420	29.730	32.770	0.086250	0.12622	0.13630	185.07	35.624
400.00	0.15731	0.050060	19.976	32.133	35.275	0.088329	0.13364	0.14463	186.44	30.237
420.00	0.25622	0.079525	12.575	34.618	37.840	0.090796	0.14100	0.15326	186.62	26.277
440.00	0.39698	0.12122	8.2496	37.172	40.447	0.093548	0.14832	0.16234	185.42	23.381
460.00	0.58974	0.17889	5.5901	39.776	43.073	0.096491	0.15558	0.17214	182.60	21.313
480.00	0.84559	0.25767	3.8810	42.408	45.690	0.099533	0.16283	0.18316	177.87	19.928
500.00	1.1766	0.36515	2.7386	45.035	48.257	0.10258	0.17013	0.19639	170.89	19.156
520.00	1.5964	0.51366	1.9468	47.606	50.713	0.10550	0.17758	0.21403	161.17	19.007
540.00	2.1207	0.72590	1.3776	50.033	52.955	0.10811	0.18540	0.24190	148.11	19.581
560.00	2.7691	1.0524	0.95020	52.136	54.768	0.11005	0.19407	0.30118	130.92	21.067
580.00	3.5688	1.6605	0.60222	53.371	55.520	0.11031	0.20499	0.55687	108.57	23.353
591.75	4.1264	3.1690	0.31556	50.827	52.129	0.10409			0	19.734

Single-Phase Properties

200.00	0.10000	10.350	0.096623	-28.794	-28.784	-0.10036	0.095737	0.13627	1768.7	-0.56720
325.00	0.10000	9.0842	0.11008	-10.267	-10.256	-0.028962	0.12386	0.16493	1191.0	-0.42130
383.28	0.10000	8.4614	0.11818	-0.097277	-0.085459	-0.00022242	0.14084	0.18440	961.57	-0.31629
383.28	0.10000	0.032692	30.589	30.118	33.177	0.086560	0.12744	0.13765	185.37	34.622
450.00	0.10000	0.027344	36.572	39.369	43.026	0.11021	0.14828	0.15755	202.98	18.656
575.00	0.10000	0.021125	47.338	60.145	64.879	0.15289	0.18229	0.19099	230.86	8.1724
700.00	0.10000	0.017270	57.904	84.697	90.487	0.19311	0.20915	0.21767	255.08	4.5991
200.00	1.0000	10.354	0.096581	-28.811	-28.715	-0.10045	0.095778	0.13625	1771.8	-0.56742
325.00	1.0000	9.0933	0.10997	-10.303	-10.193	-0.029074	0.12389	0.16482	1196.3	-0.42281
450.00	1.0000	7.6732	0.13032	12.928	13.058	0.031073	0.16000	0.20893	715.87	-0.13702
489.95	1.0000	7.0876	0.14109	21.616	21.757	0.049581	0.17110	0.22749	556.21	0.083851
489.95	1.0000	0.30691	3.2583	43.717	46.976	0.10105	0.16645	0.18937	174.71	19.469
575.00	1.0000	0.23372	4.2786	59.065	63.344	0.13184	0.18481	0.19848	210.35	9.4517
700.00	1.0000	0.18116	5.5199	84.035	89.555	0.17301	0.21011	0.22077	244.28	4.8519
200.00	5.0000	10.373	0.096400	-28.887	-28.405	-0.10083	0.095961	0.13619	1785.8	-0.56831
325.00	5.0000	9.1328	0.10950	-10.460	-9.9129	-0.029562	0.12404	0.16434	1219.1	-0.42907
450.00	5.0000	7.7765	0.12859	12.548	13.191	0.030217	0.16002	0.20635	761.15	-0.18092
575.00	5.0000	5.6051	0.17841	41.699	42.591	0.087531	0.19430	0.28257	285.95	1.3449
700.00	5.0000	1.1698	0.85487	80.250	84.524	0.15419	0.21547	0.24819	197.87	5.8706
200.00	10.000	10.397	0.096178	-28.980	-28.018	-0.10130	0.096187	0.13612	1802.9	-0.56932
325.00	10.000	9.1800	0.10893	-10.647	-9.5580	-0.030150	0.12422	0.16382	1246.4	-0.43597
450.00	10.000	7.8890	0.12676	12.131	13.398	0.029259	0.16009	0.20400	810.93	-0.22080
575.00	10.000	6.1856	0.16166	40.014	41.630	0.084396	0.19242	0.24975	425.25	0.38997
700.00	10.000	3.2697	0.30584	73.814	76.873	0.13965	0.21978	0.30359	189.44	3.9797
325.00	100.00	9.7930	0.10211	-12.967	-2.7551	-0.038333	0.12727	0.16019	1609.1	-0.48598
450.00	100.00	8.9423	0.11183	8.2140	19.397	0.019038	0.16256	0.19397	1308.6	-0.38806
575.00	100.00	8.1649	0.12248	33.274	45.521	0.070122	0.19323	0.22291	1104.9	-0.31962
700.00	100.00	7.4598	0.13405	61.439	74.845	0.11620	0.21782	0.24523	967.84	-0.27316
450.00	500.00	10.545	0.094835	3.6375	51.055	-0.00069710	0.17022	0.19432	2238.9	-0.40813
575.00	500.00	10.097	0.099035	27.662	77.179	0.050390	0.20000	0.22274	2087.2	-0.35992
700.00	500.00	9.7036	0.10305	54.974	106.50	0.096461	0.22392	0.24550	1977.5	-0.32978

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., and Span, R., “Short Fundamental Equations of State for 20 Industrial Fluids,” *J. Chem. Eng. Data* **51**(3):785–850, 2006. Validated equations for the viscosity and thermal conductivity are not currently available for this fluid.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in density in the equation of state are 0.05% in the liquid phase up to 540 K, 0.5% up to the critical temperature, 1% at higher temperatures, 0.5% at pressures from 100 to 500 MPa, and 0.2% in the vapor phase. The uncertainty for the saturated-liquid density (and densities near atmospheric pressure) approaches 0.01% around 300 K. The uncertainties in vapor pressure are 0.3% from 270 to 305 K, 0.05% from 305 to 425 K, 0.1% up to 555 K, and 0.15% up to the critical temperature. The uncertainty in heat capacities is 0.5% and rises to 3% in the critical region. The uncertainty in the speed of sound is 1% up to 500 K and 100 MPa and rises to 2% at higher pressures and higher temperatures.

TABLE 2-304 Saturated Solid/Vapor Water*

Temp., °F	Pressure, lb/in ² abs.	Volume, ft ³ /lb		Enthalpy, Btu/lb		Entropy, Btu/(lb)(°F)	
		Solid	Vapor	Solid	Vapor	Solid	Vapor
-160	4.949.-8	0.01722	3.607.+9	-222.05	990.38	-0.4907	3.5549
-150	1.620.-7	0.01723	1.139.+9	-218.82	994.80	-0.4801	3.4387
-140	4.928.-7	0.01724	3.864.+8	-215.49	999.21	-0.4695	3.3301
-130	1.403.-6	0.01725	1.400.+8	-212.08	1003.63	-0.4590	3.2284
-120	3.757.-6	0.01726	5.386.+7	-208.58	1008.05	-0.4485	3.1330
-110	9.517.-6	0.01728	2.189.+7	-204.98	1012.47	-0.4381	3.0434
-100	2.291.-5	0.01729	9.352.+6	-201.28	1016.89	-0.4277	2.9591
-90	5.260.-5	0.01730	4.186.+6	-197.49	1021.31	-0.4173	2.8796
-80	1.157.-4	0.01731	1.955.+6	-193.60	1025.73	-0.4069	2.8045
-70	2.443.-4	0.01732	9.501.+5	-189.61	1030.15	-0.3965	2.7336
-60	4.972.-4	0.01734	4.788.+5	-185.52	1034.58	-0.3862	2.6664
-50	9.776.-4	0.01735	2.496.+5	-181.34	1039.00	-0.3758	2.6028
-45	1.354.-3	0.01736	1.824.+5	-179.21	1041.21	-0.3707	2.5723
-40	1.861.-3	0.01737	1.343.+5	-177.06	1043.42	-0.3655	2.5425
-35	2.540.-3	0.01737	9.961.+4	-174.88	1045.63	-0.3604	2.5135
-30	3.440.-3	0.01738	7.441.+4	-172.68	1047.84	-0.3552	2.4853
-25	4.627.-3	0.01739	5.596.+4	-170.46	1050.05	-0.3501	2.4577
-20	6.181.-3	0.01739	4.237.+4	-168.21	1052.26	-0.3449	2.4308
-15	8.204.-3	0.01740	3.228.+4	-165.94	1054.47	-0.3398	2.4046
-10	1.082.-2	0.01741	2.475.+4	-163.65	1056.67	-0.3347	2.3791
-5	1.419.-2	0.01741	1.909.+4	-161.33	1058.88	-0.3295	2.3541
0	1.849.-2	0.01742	1.481.+4	-158.98	1061.09	-0.3244	2.3297
5	2.396.-2	0.01743	1.155.+4	-156.61	1063.29	-0.3193	2.3039
10	3.087.-2	0.01744	9.060.+3	-154.22	1065.50	-0.3142	2.2827
15	3.957.-2	0.01744	7.144.+3	-151.80	1067.70	-0.3090	2.2600
16	4.156.-2	0.01745	6.817.+3	-151.32	1068.14	-0.3080	2.2555
18	4.581.-2	0.01745	6.210.+3	-150.34	1069.02	-0.3060	2.2466
20	5.045.-2	0.01745	5.662.+3	-149.36	1069.90	-0.3039	2.2378
22	5.552.-2	0.01746	5.166.+3	-148.38	1070.38	-0.3019	2.2291
24	6.105.-2	0.01746	4.717.+3	-147.39	1071.66	-0.2998	2.2205
26	6.708.-2	0.01746	4.311.+3	-146.40	1072.53	-0.2978	2.2119
28	7.365.-2	0.01746	3.943.+3	-145.40	1073.41	-0.2957	2.2034
30	8.080.-2	0.01747	3.608.+3	-144.40	1074.29	-0.2937	2.1950
31	8.461.-2	0.01747	3.453.+3	-143.90	1074.73	-0.2927	2.1908
32	8.858.-2	0.01747	3.305.+3	-143.40	1075.16	-0.2916	2.1867

*Condensed from *Fundamentals*, American Society of Heating, Refrigerating and Air-Conditioning Engineers, 1967 and 1972. Reproduced by permission. The validity of many standard reference tables has been critically reviewed by Janco, Pupezin, and van Hook, *J. Phys. Chem.*, **74** (1970):2984. Current information on the properties of solid, vapor, and liquid water properties can be found at <http://www.iapws.org>. The notation 4.949.-8, 3.607.+9, etc., means 4.949×10^{-8} , 3.607×10^9 , etc.

TABLE 2-305 Thermodynamic Properties of Water

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
273.16	0.000612	55.497	0.018019	0	1.1E-05	0	0.075978	0.076023	1402.3	-0.24142	561.04	1791.2
280	0.000992	55.501	0.018018	0.51875	0.51877	0.001876	0.075669	0.075668	1434.1	-0.23515	574.04	1433.7
290	0.001920	55.440	0.018038	1.2742	1.2742	0.004527	0.075095	0.075429	1472.1	-0.22720	592.73	1084.0
300	0.003537	55.315	0.018078	2.0278	2.0279	0.007082	0.074412	0.075320	1501.4	-0.22024	610.28	853.84
310	0.006231	55.139	0.018136	2.7808	2.7810	0.009551	0.073645	0.075294	1523.2	-0.21393	626.05	693.54
320	0.010546	54.919	0.018209	3.5339	3.5340	0.011941	0.072811	0.075317	1538.7	-0.20804	639.71	577.02
330	0.017213	54.662	0.018294	4.2873	4.2876	0.014260	0.071927	0.075373	1548.7	-0.20241	651.18	489.49
340	0.027188	54.371	0.018392	5.0414	5.0419	0.016511	0.071008	0.075456	1553.9	-0.19690	660.55	421.97
350	0.041682	54.049	0.018502	5.7964	5.7972	0.018700	0.070070	0.075567	1554.8	-0.19140	668.00	368.77
360	0.062194	53.698	0.018623	6.5526	6.5538	0.020830	0.069124	0.075708	1552.0	-0.18581	673.76	326.10
370	0.090535	53.321	0.018754	7.3104	7.3121	0.022906	0.068180	0.075883	1545.8	-0.18005	678.02	291.36
380	0.12885	52.918	0.018897	8.0701	8.0725	0.024932	0.067247	0.076098	1536.5	-0.17404	681.00	262.69
390	0.17964	52.490	0.019051	8.8320	8.8354	0.026911	0.066331	0.076357	1524.3	-0.16769	682.83	238.77
400	0.24577	52.038	0.019217	9.5966	9.6013	0.028847	0.065438	0.076664	1509.5	-0.16092	683.64	218.60
410	0.33045	51.563	0.019394	10.364	10.371	0.030743	0.064570	0.077026	1492.2	-0.15366	683.52	201.43
420	0.43730	51.064	0.019583	11.136	11.144	0.032602	0.063731	0.077447	1472.5	-0.14581	682.53	186.68
430	0.57026	50.541	0.019786	11.911	11.923	0.034427	0.062920	0.077934	1450.6	-0.13728	680.70	173.91
440	0.73367	49.994	0.020003	12.692	12.706	0.036222	0.062140	0.078495	1426.5	-0.12794	678.05	162.77
450	0.9322	49.421	0.020234	13.477	13.496	0.037988	0.061390	0.079136	1400.4	-0.11767	674.59	152.98
460	1.1709	48.824	0.020482	14.269	14.293	0.039729	0.060671	0.079869	1372.2	-0.10631	670.28	144.31
470	1.4551	48.199	0.020748	15.068	15.098	0.041448	0.059984	0.080706	1342.0	-0.09369	665.12	136.58
480	1.7905	47.545	0.021033	15.875	15.913	0.043147	0.059327	0.081662	1309.8	-0.07959	659.07	129.64
490	2.1831	46.861	0.021340	16.690	16.737	0.044830	0.058702	0.082757	1275.7	-0.06372	652.06	123.37
500	2.6392	46.145	0.021671	17.515	17.573	0.046498	0.058109	0.084013	1239.6	-0.04578	644.05	117.66
510	3.1655	45.393	0.022030	18.352	18.421	0.048156	0.057548	0.085464	1201.5	-0.02534	634.95	112.42
520	3.7690	44.603	0.022420	19.200	19.285	0.049807	0.057023	0.087149	1161.3	-0.00189	624.68	107.57
530	4.4569	43.770	0.022847	20.064	20.165	0.051454	0.056536	0.089124	1119.1	0.025264	613.15	103.05
540	5.2369	42.889	0.023316	20.943	21.065	0.053102	0.056089	0.091464	1074.6	0.057002	600.26	98.792
550	6.1172	41.954	0.023836	21.841	21.987	0.054756	0.055690	0.094275	1027.9	0.094527	585.95	94.746
560	7.1062	40.956	0.024417	22.762	22.935	0.056422	0.055347	0.097713	978.54	0.13949	570.21	90.857
570	8.2132	39.885	0.025072	23.709	23.915	0.058106	0.055071	0.10201	926.44	0.19425	553.08	87.074
580	9.448	38.725	0.025823	24.688	24.932	0.059821	0.054881	0.10754	871.23	0.26220	534.74	83.342
590	10.821	37.456	0.026698	25.707	25.996	0.061577	0.054808	0.11491	812.49	0.34857	515.43	79.600
600	12.345	36.048	0.027741	26.777	27.119	0.063396	0.054902	0.12526	749.57	0.46172	495.46	75.773
610	14.033	34.451	0.029026	27.917	28.324	0.065309	0.055258	0.14100	681.27	0.61660	475.03	71.759
620	15.901	32.577	0.030697	29.160	29.648	0.067371	0.056100	0.16852	604.73	0.84473	454.10	67.382
630	17.969	30.210	0.033101	30.585	31.180	0.069715	0.058152	0.23108	513.19	1.2251	432.51	62.244
640	20.265	26.729	0.037413	32.422	33.180	0.072737	0.064521	0.46736	400.66	1.9542	414.93	55.247
647.1	22.064	17.874	0.055948	36.314	37.548	0.079393			0	3.7410		
273.16	0.000612	0.000269	3711.0	42.785	45.055	0.16494	0.025553	0.033947	409.00	592.65	17.071	9.2163
280	0.000992	0.000426	2345.4	42.954	45.280	0.16174	0.025657	0.034073	413.92	477.26	17.442	9.3815
290	0.001920	0.000797	1254.3	43.201	45.609	0.15741	0.025816	0.034270	420.99	351.65	18.031	9.6414
300	0.003537	0.001420	704.01	43.446	45.936	0.15344	0.025982	0.034483	427.89	264.35	18.673	9.9195
310	0.006231	0.002424	412.60	43.690	46.261	0.14981	0.026158	0.034716	434.63	203.74	19.369	10.213
320	0.010546	0.003978	251.39	43.931	46.582	0.14647	0.026350	0.034980	441.18	161.25	20.117	10.518
330	0.017213	0.006304	158.62	44.169	46.900	0.14339	0.026568	0.035287	447.54	130.92	20.922	10.833
340	0.027188	0.009681	103.30	44.404	47.212	0.14054	0.026821	0.035653	453.68	108.77	21.784	11.157
350	0.041682	0.014448	69.213	44.634	47.519	0.13791	0.027118	0.036091	459.58	92.178	22.707	11.487
360	0.062194	0.021014	47.586	44.860	47.819	0.13546	0.027469	0.036617	465.22	79.440	23.695	11.823
370	0.090535	0.029859	33.491	45.079	48.111	0.13317	0.027883	0.037249	470.57	69.427	24.750	12.162
380	0.12885	0.041537	24.075	45.291	48.393	0.13104	0.028372	0.038004	475.61	61.373	25.875	12.504
390	0.17964	0.056683	17.642	45.496	48.665	0.12904	0.028944	0.038903	480.32	54.749	27.074	12.848
400	0.24577	0.076014	13.156	45.691	48.924	0.12715	0.029608	0.039963	484.67	49.181	28.347	13.192
410	0.33045	0.10034	9.9666	45.876	49.170	0.12537	0.030369	0.041203	488.65	44.405	29.699	13.538
420	0.43730	0.13055	7.6601	46.050	49.400	0.12369	0.031230	0.042634	492.22	40.237	31.128	13.883
430	0.57026	0.16765	5.9649	46.211	49.613	0.12208	0.032187	0.044269	495.39	36.550	32.638	14.228
440	0.73367	0.21276	4.7002	46.359	49.807	0.12054	0.033234	0.046114	498.12	33.259	34.230	14.573

TABLE 2-305 Thermodynamic Properties of Water (*Continued*)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
450	0.93220	0.26711	3.7438	46.492	49.982	0.11907	0.034362	0.048177	500.41	30.307	35.904	14.917
460	1.1709	0.33209	3.0113	46.609	50.134	0.11764	0.035561	0.050469	502.24	27.653	37.663	15.261
470	1.4551	0.40925	2.4435	46.708	50.263	0.11627	0.036821	0.053005	503.60	25.265	39.512	15.606
480	1.7905	0.50035	1.9986	46.788	50.367	0.11493	0.038137	0.055809	504.45	23.118	41.455	15.952
490	2.1831	0.60738	1.6464	46.848	50.442	0.11362	0.039503	0.058919	504.78	21.187	43.502	16.300
500	2.6392	0.73265	1.3649	46.885	50.487	0.11233	0.040920	0.062388	504.55	19.450	45.666	16.653
510	3.1655	0.87884	1.1379	46.898	50.500	0.11105	0.042391	0.066289	503.71	17.886	47.969	17.011
520	3.7690	1.0491	0.95318	46.883	50.475	0.10979	0.043920	0.070723	502.23	16.475	50.442	17.377
530	4.4569	1.2473	0.80174	46.838	50.411	0.10852	0.045519	0.075827	500.05	15.197	53.130	17.755
540	5.2369	1.4780	0.67659	46.758	50.302	0.10724	0.047197	0.081789	497.10	14.035	56.102	18.149
550	6.1172	1.7471	0.57238	46.641	50.142	0.10595	0.048968	0.088873	493.31	12.973	59.456	18.563
560	7.1062	2.0620	0.48497	46.478	49.925	0.10462	0.050848	0.097461	488.58	11.997	63.341	19.007
570	8.2132	2.4325	0.41110	46.264	49.641	0.10324	0.052856	0.10813	482.79	11.093	67.981	19.489
580	9.4480	2.8720	0.34819	45.988	49.278	0.10180	0.055017	0.12178	475.80	10.248	73.721	20.024
590	10.821	3.3994	0.29417	45.636	48.819	0.10026	0.057361	0.13994	467.41	9.4499	81.108	20.634
600	12.345	4.0434	0.24732	45.188	48.242	0.098600	0.059939	0.16540	457.33	8.6837	91.052	21.350
610	14.033	4.8497	0.20620	44.613	47.506	0.096755	0.062831	0.20384	445.11	7.9329	105.17	22.229
620	15.901	5.9009	0.16946	43.855	46.550	0.094631	0.066197	0.26923	429.99	7.1743	126.66	23.374
630	17.969	7.3737	0.13562	42.801	45.238	0.092029	0.070465	0.40819	410.21	6.3669	163.44	25.018
640	20.265	9.8331	0.10170	41.095	43.156	0.088324	0.077576	0.94736	379.64	5.3854	250.01	27.938
647.1	22.064	17.874	0.055948	36.314	37.548	0.079393			0	3.7410		
Single-Phase Properties												
300	0.1	55.317	0.018078	2.0277	2.0295	0.007081	0.074406	0.075315	1501.5	-0.22024	610.32	853.83
372.76	0.1	53.212	0.018793	7.5196	7.5214	0.02347	0.067921	0.075938	1543.5	-0.17843	678.97	282.91
372.76	0.1	0.032769	30.517	45.138	48.190	0.13257	0.02801	0.037444	471.99	67.038	25.053	12.256
400	0.1	0.030397	32.898	45.900	49.189	0.13516	0.02717	0.036170	490.31	47.254	27.008	13.285
500	0.1	0.024154	41.401	48.619	52.759	0.14313	0.02717	0.035693	548.31	19.298	35.861	17.270
600	0.1	0.020086	49.786	51.387	56.365	0.14970	0.028103	0.036513	598.61	10.567	46.367	21.407
700	0.1	0.017201	58.136	54.256	60.069	0.15541	0.029225	0.037592	643.92	6.6444	57.964	25.564
800	0.1	0.015044	66.471	57.240	63.887	0.16050	0.030431	0.038778	685.47	4.5167	70.385	29.669
900	0.1	0.013369	74.799	60.347	67.827	0.16514	0.031687	0.040024	724.03	3.2280	83.466	33.685
1000	0.1	0.012030	83.123	63.581	71.893	0.16943	0.032963	0.041293	760.17	2.3885	97.085	37.592
1100	0.1	0.010936	91.444	66.941	76.085	0.17342	0.034228	0.042554	794.33	1.8122	111.15	41.382
1200	0.1	0.010024	99.763	70.426	80.402	0.17718	0.035458	0.043781	826.85	1.4006	125.58	45.054
300	1	55.340	0.018070	2.0263	2.0444	0.007077	0.074353	0.075270	1503.0	-0.22022	610.73	853.67
400	1	52.060	0.019209	9.5914	9.6106	0.028834	0.065422	0.076628	1511.3	-0.16113	684.10	218.80
453.03	1	49.243	0.020307	13.717	13.737	0.038518	0.061169	0.079348	1392.0	-0.11435	673.37	150.24
453.03	1	0.28559	3.5015	46.529	50.030	0.11863	0.034718	0.048846	501.02	29.473	36.427	15.021
500	1	0.25158	3.9749	48.111	52.086	0.12295	0.030084	0.041065	535.74	19.741	38.799	17.051
600	1	0.20466	4.8861	51.123	56.009	0.13011	0.029002	0.038358	592.58	10.615	47.636	21.329
700	1	0.17377	5.7547	54.087	59.842	0.13602	0.029629	0.038495	640.55	6.6387	58.735	25.550
800	1	0.15134	6.6074	57.121	63.729	0.14121	0.030651	0.039301	683.48	4.5077	70.983	29.687
900	1	0.13418	7.4524	60.258	67.710	0.14590	0.031821	0.040358	722.85	3.2212	84.000	33.718
1000	1	0.12058	8.2932	63.511	71.804	0.15021	0.033051	0.041522	759.50	2.3837	97.573	37.630
1100	1	0.10951	9.1313	66.885	76.016	0.15422	0.034290	0.042719	794.01	1.8089	111.57	41.420
1200	1	0.10032	9.9677	70.380	80.347	0.15799	0.035504	0.043905	826.77	1.3982	125.89	45.088
300	5	55.439	0.018038	2.0204	2.1106	0.007057	0.074119	0.075070	1509.8	-0.22012	612.54	853.00
400	5	52.173	0.019167	9.5643	9.6601	0.028766	0.065337	0.076438	1520.9	-0.16222	686.54	219.84
500	5	46.267	0.021614	17.474	17.582	0.046415	0.058082	0.083643	1250.0	-0.04945	646.52	118.27
537.09	5	43.151	0.023175	20.685	20.801	0.052622	0.056215	0.090740	1087.8	0.047232	604.15	100.01
537.09	5	1.4072	0.71063	46.785	50.338	0.10762	0.046699	0.079952	498.04	14.362	55.203	18.032
600	5	1.1320	0.88340	49.734	54.151	0.11436	0.034611	0.051045	561.07	10.407	54.653	21.062
700	5	0.91269	1.0957	53.286	58.765	0.12148	0.031678	0.043318	624.59	6.5536	62.680	25.547

800	5	0.77805	1.2853	56.576	63.002	0.12714	0.031683	0.041848	674.39	4.4532	73.950	29.806
900	5	0.68224	1.4658	59.855	67.183	0.13207	0.032430	0.041922	717.57	3.1856	86.626	33.891
1000	5	0.60918	1.6416	63.197	71.405	0.13652	0.033447	0.042571	756.57	2.3599	99.971	37.821
1100	5	0.55109	1.8146	66.632	75.705	0.14061	0.034565	0.043465	792.63	1.7924	113.64	41.606
1200	5	0.50355	1.9859	70.172	80.101	0.14444	0.035704	0.044458	826.45	1.3865	127.51	45.257
300	10	55.561	0.017998	2.0131	2.1931	0.007031	0.073834	0.074829	1518.2	-0.21999	614.81	852.28
400	10	52.312	0.019116	9.5311	9.7222	0.028682	0.065233	0.076208	1532.7	-0.16351	689.57	221.13
500	10	46.517	0.021497	17.389	17.604	0.046244	0.058028	0.082910	1271.3	-0.05669	651.64	119.55
584.15	10	38.213	0.026169	25.105	25.367	0.060543	0.054835	0.11032	847.33	0.29540	526.83	81.795
584.15	10	3.0787	0.32482	45.852	49.100	0.10117	0.055964	0.128640	472.51	9.9124	76.543	20.267
600	10	2.7628	0.36195	47.183	50.802	0.10405	0.047271	0.092535	503.34	9.4382	71.110	21.036
700	10	1.9625	0.50956	52.145	57.241	0.11405	0.034838	0.051779	602.20	6.3228	69.301	25.704
800	10	1.6157	0.61893	55.851	62.040	0.12046	0.033089	0.045603	662.61	4.3529	78.476	30.054
900	10	1.3945	0.71709	59.334	66.505	0.12572	0.033219	0.044062	710.98	3.1289	90.516	34.176
1000	10	1.2345	0.81002	62.798	70.898	0.13035	0.033947	0.043952	753.03	2.3241	103.50	38.111
1100	10	1.1111	0.90002	66.314	75.314	0.13456	0.034908	0.044427	791.02	1.7683	116.73	41.882
1200	10	1.0119	0.98820	69.910	79.792	0.13846	0.035954	0.045164	826.16	1.3695	130.00	45.506
300	100	57.573	0.017369	1.8921	3.6290	0.006516	0.069812	0.071696	1667.9	-0.21618	654.50	856.88
400	100	54.500	0.018349	9.0423	10.877	0.027360	0.063582	0.073086	1717.3	-0.17905	741.80	243.50
500	100	49.914	0.020034	16.289	18.292	0.043895	0.057324	0.075607	1555.7	-0.12564	730.42	138.92
600	100	43.935	0.022761	23.820	26.097	0.058109	0.052776	0.081104	1300.4	-0.02079	645.83	101.51
700	100	36.179	0.027640	31.916	34.680	0.071320	0.049610	0.091576	1020.0	0.21155	510.14	79.363
800	100	26.768	0.037359	40.700	44.435	0.084331	0.047143	0.10108	813.97	0.65939	351.46	62.042
900	100	19.073	0.052429	48.805	54.048	0.095669	0.043932	0.088057	765.30	1.0399	257.03	53.250
1000	100	14.734	0.067868	55.188	61.975	0.10404	0.041345	0.071678	792.50	1.0944	232.07	51.518
1100	100	12.246	0.081656	60.470	68.635	0.11039	0.040131	0.062539	832.67	0.98401	223.70	52.497
1200	100	10.631	0.094062	65.222	74.628	0.11561	0.039810	0.057826	872.28	0.83544	219.07	54.415
300	500	63.750	0.015686	1.5247	9.3678	0.003746	0.063403	0.068296	2228.6	-0.19915	763.82	1089.4
400	500	60.862	0.016431	7.9635	16.179	0.023347	0.059634	0.067603	2258.7	-0.19486	929.09	320.18
500	500	57.695	0.017332	14.264	22.930	0.038412	0.055769	0.067522	2200.7	-0.18339	1096.6	189.08
600	500	54.316	0.018411	20.481	29.687	0.050731	0.052734	0.067584	2093.8	-0.16883	1097.9	141.83
700	500	50.847	0.019667	26.606	36.439	0.061141	0.050315	0.067436	1970.5	-0.15188	935.15	118.47
800	500	47.385	0.021104	32.615	43.167	0.070124	0.048442	0.067080	1850.1	-0.13256	738.72	104.70
900	500	44.018	0.022718	38.492	49.851	0.077998	0.047068	0.066596	1743.4	-0.11124	572.49	95.388
1000	500	40.814	0.024501	44.233	56.484	0.084987	0.046126	0.066041	1655.7	-0.08910	445.17	88.418
1100	500	37.834	0.026432	49.839	63.055	0.091251	0.045537	0.065356	1589.3	-0.06907	350.97	83.021
1200	500	35.124	0.028470	55.312	69.547	0.096900	0.045218	0.064451	1543.9	-0.05511	282.78	78.952
400	1000	65.942	0.015165	7.4792	22.644	0.019833	0.057934	0.065743	2718.6	-0.19303	1172.7	329.93
500	1000	63.253	0.015810	13.357	29.167	0.034391	0.055063	0.064967	2677.2	-0.19158	2199.5	190.55
600	1000	60.572	0.016509	19.141	35.650	0.046212	0.053055	0.064676	2602.3	-0.18789	3250.5	137.73
700	1000	57.937	0.017260	24.836	42.096	0.056150	0.051393	0.064219	2513.7	-0.18439	3202.2	108.98
800	1000	55.384	0.018056	30.435	48.491	0.064689	0.050059	0.063663	2423.7	-0.18105	2408.7	91.430
900	1000	52.937	0.018890	35.938	54.828	0.072155	0.049062	0.063101	2338.7	-0.17779	1610.7	80.198
1000	1000	50.611	0.019759	41.354	61.113	0.078776	0.048373	0.062594	2261.5	-0.17459	1052.9	72.716
1100	1000	48.415	0.020655	46.695	67.350	0.084722	0.047942	0.062176	2193.2	-0.17139	703.41	67.520
1200	1000	46.349	0.021575	51.976	73.551	0.090117	0.047713	0.061861	2133.6	-0.16808	487.61	63.774

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Wagner, W., and Pruss, A., “The IAPWS Formulation 1995 for the Thermodynamic Properties of Ordinary Water Substance for General and Scientific Use,” *J. Phys. Chem. Ref. Data* **31**(2):387–535, 2002. The source for viscosity is International Association for the Properties of Water and Steam, *Revised Release on the IAPS Formulation 1985 for the Viscosity of Ordinary Water Substance*, IAPWS, 1997. The source for thermal conductivity is the International Association for the Properties of Water and Steam, *Revised Release on the IAPS Formulation 1985 for the Thermal Conductivity of Ordinary Water Substance*, IAPWS, 1998.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainty in density of the equation of state is 0.0001% at 1 atm in the liquid phase, and 0.001% at other liquid states at pressures up to 10 MPa and temperatures to 423 K. In the vapor phase, the uncertainty is 0.05% or less. The uncertainties rise at higher temperatures and/or pressures, but are generally less than 0.1% in density except at extreme conditions. The uncertainty in pressure in the critical region is 0.1%. The uncertainty of the speed of sound is 0.15% in the vapor and 0.1% or less in the liquid, and increases near the critical region and at high temperatures and pressures. The uncertainty in isobaric heat capacity is 0.2% in the vapor and 0.1% in the liquid, with increasing values in the critical region and at high pressures. The uncertainties of saturation conditions are 0.025% in vapor pressure, 0.0025% in saturated-liquid density, and 0.1% in saturated-vapor density. The uncertainties in the saturated densities increase substantially as the critical region is approached. For the uncertainties in the viscosity and thermal conductivity, see the IAPWS Release.

TABLE 2-306 Thermodynamic Properties of Water Substance along the Melting Line

P , bar	T , °C	$10^3 v_f$, m ³ /kg	h_f , kJ/kg	s_f , kJ/kg·K	c_{pf} , kJ/kg·K	c_{melt} , kJ/kg·K	$10^6 \alpha_f$, K ⁻¹	$10^6 K_{f,T}$ bar ⁻¹
6.117×10^{-3}	0.0100	1.00021	0	0	4.219	3.969	-67.42	50.90
1.01325	0.0026	1.00016	0.0719	-0.0001	4.218	3.970	-67.17	50.88
50	-0.3618	0.99770	3.5140	-0.0054	4.196	3.997	-54.92	50.30
100	-0.7410	0.99523	6.9794	-0.0110	4.174	4.023	-42.52	49.73
150	-1.1249	0.99278	10.3964	-0.0167	4.152	4.047	-30.24	49.17
200	-1.5166	0.99037	13.7648	-0.0225	4.132	4.070	-18.05	48.63
250	-1.9151	0.98798	17.0843	-0.0285	4.112	4.092	-5.93	48.11
300	-2.3206	0.98562	20.3547	-0.0347	4.092	4.113	6.12	47.59
400	-3.1532	0.98098	26.7472	-0.0474	4.056	4.150	30.09	46.61
500	-4.0156	0.97643	32.9403	-0.0607	4.022	4.184	53.97	45.68
600	-4.909	0.97196	38.932	-0.0747	3.992	4.215	77.87	44.80
800	-6.790	0.96326	50.300	-0.1046	3.937	4.270	126.18	43.19
1000	-8.803	0.95493	60.836	-0.1371	3.893	4.320	175.98	41.74

Condensed from U. Grigull, Private communication, January 18, 1995.

Materials prepared at Technical University München, Germany by U. Grigull and S. Marek. For a table as a function of temperature, see Grigull, U. and S. Marek, *Wärme u. Stoff.*, **30** (1994): 1–8.

t = the triple point (at 6.117×10^{-3} bar, 0.01 °C); $v_f = 0.0010021$ m³/kg; $\alpha_f = -67.42 \times 10^{-6}$ /K.

Other equations for properties are given by Jones, F. E. and G. L. Harris, *J. Res. N.I.S.T.*, **97**, 3 (1992): 335–340, and by Wagner, W. and A. Pruss, *J. Phys. Chem. Ref. Data*, **22**, 3 (1993): 783–787. Steam tables include Walker, W. A., U.S. Naval Ordn. Lab. rept. NOLTR-66-217 = AD 651105 (0–1000 bar, 0–150°C), 1967 (72 pp.); Grigull, U., J. Straub, et al., *Steam Tables in S.I. Units* (0.01–1000 bar, 0–1000 °C), Springer-Verlag, Berlin, 1990 (133 pp.); Tseng, C. M., T. A. Hamp, et al., Atomic Energy of Canada rept. (30 props, sat liq & vap., 1–220 bar), AECL-5910 1977 (90 pp.). For dissociation, see e.g., Knonicek, V., *Rozpr. Česko Acad Ved., Rada techn ved* (0.01–100 bar, 1000–5000 K), **77**, 1 (1967). The proceedings of the 10th international conference on the properties of steam were edited by Sytchev, V. V. and A. A. Aleksandrov, Plenum, NY, 1984; and for the 11th conference by Pichal, M. and O. Sifner, Hemisphere, 1989 (550 pp.). Current information on the properties of solid, vapor, and liquid water properties can be found at <http://www.iapws.org>.

For electrical conductivity, see e.g., Marshall, W. L., *J. Chem. Eng. Data*, **32** (1987): 221–226.

TABLE 2-307 Thermodynamic Properties of Xenon

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Saturated Properties												
161.40	0.081748	22.592	0.044263	−0.16663	−0.16301	−0.00099337	0.022064	0.044446	653.38	−0.63704	72.081	516.89
170.00	0.13343	22.155	0.045137	0.21574	0.22176	0.0013152	0.021486	0.044730	629.12	−0.60760	68.920	457.99
175.00	0.17325	21.895	0.045673	0.43915	0.44706	0.0026109	0.021176	0.044972	614.85	−0.58689	67.037	427.78
180.00	0.22153	21.630	0.046232	0.66366	0.67390	0.0038765	0.020884	0.045274	600.43	−0.56326	65.137	400.07
185.00	0.27933	21.361	0.046815	0.88953	0.90261	0.0051149	0.020609	0.045640	585.85	−0.53642	63.229	374.57
190.00	0.34774	21.085	0.047426	1.1170	1.1335	0.0063292	0.020349	0.046075	571.10	−0.50604	61.320	351.01
195.00	0.42789	20.804	0.048068	1.3464	1.3669	0.0075220	0.020104	0.046585	556.14	−0.47173	59.415	329.16
200.00	0.52091	20.516	0.048743	1.5779	1.6033	0.0086959	0.019873	0.047179	540.97	−0.43300	57.519	308.84
205.00	0.62797	20.220	0.049455	1.8119	1.8429	0.0098535	0.019656	0.047868	525.55	−0.38927	55.633	289.88
210.00	0.75025	19.916	0.050210	2.0487	2.0864	0.010997	0.019453	0.048666	509.85	−0.33987	53.759	272.13
215.00	0.88893	19.603	0.051013	2.2887	2.3341	0.012130	0.019264	0.049590	493.84	−0.28392	51.897	255.47
220.00	1.0452	19.279	0.051869	2.5324	2.5866	0.013254	0.019091	0.050663	477.47	−0.22039	50.048	239.78
225.00	1.2203	18.944	0.052787	2.7802	2.8446	0.014373	0.018933	0.051914	460.71	−0.14796	48.209	224.96
230.00	1.4155	18.596	0.053776	3.0328	3.1089	0.015488	0.018794	0.053385	443.49	−0.064982	46.379	210.92
235.00	1.6321	18.232	0.054847	3.2907	3.3802	0.016605	0.018675	0.055128	425.74	0.030670	44.554	197.57
240.00	1.8712	17.852	0.056017	3.5550	3.6598	0.017726	0.018580	0.057218	407.38	0.14178	42.730	184.83
245.00	2.1344	17.451	0.057304	3.8265	3.9488	0.018857	0.018515	0.059766	388.30	0.27205	40.903	172.63
250.00	2.4229	17.026	0.058735	4.1068	4.2491	0.020002	0.018487	0.062932	368.36	0.42662	39.071	160.90
255.00	2.7382	16.572	0.060344	4.3976	4.5628	0.021170	0.018507	0.066972	347.41	0.61272	37.238	149.54
260.00	3.0820	16.082	0.062183	4.7014	4.8930	0.022371	0.018591	0.072312	325.20	0.84107	35.418	138.48
265.00	3.4561	15.545	0.064330	5.0218	5.2441	0.023618	0.018767	0.079722	301.44	1.1284	33.653	127.61
270.00	3.8623	14.945	0.066910	5.3646	5.6230	0.024935	0.019077	0.090763	275.68	1.5023	32.016	116.78
275.00	4.3032	14.254	0.070156	5.7397	6.0416	0.026360	0.019604	0.10917	247.28	2.0135	30.622	105.75
280.00	4.7818	13.411	0.074566	6.1681	6.5247	0.027976	0.020521	0.14668	215.12	2.7688	29.654	94.067
285.00	5.3025	12.242	0.081686	6.7103	7.1435	0.030022	0.022329	0.26914	176.78	4.0593	29.742	80.469
289.73	5.8420	8.4000	0.11905	8.2423	8.9377	0.036074			0	8.3700		
161.40	0.081748	0.062613	15.971	11.189	12.495	0.077430	0.013321	0.022698	128.33	64.799	3.1160	13.386
170.00	0.13343	0.098131	10.190	11.261	12.621	0.074250	0.013615	0.023442	130.83	57.103	3.3040	14.014
175.00	0.17325	0.12476	8.0157	11.298	12.687	0.072554	0.013809	0.023957	132.15	53.274	3.4175	14.392
180.00	0.22153	0.15646	6.3913	11.332	12.748	0.070955	0.014020	0.024539	133.36	49.835	3.5348	14.781
185.00	0.27933	0.19385	5.1587	11.362	12.803	0.069442	0.014247	0.025194	134.47	46.729	3.6563	15.180
190.00	0.34774	0.23755	4.2096	11.388	12.852	0.068004	0.014491	0.025929	135.46	43.913	3.7826	15.592
195.00	0.42789	0.28826	3.4691	11.409	12.893	0.066633	0.014750	0.026755	136.35	41.350	3.9143	16.018
200.00	0.52091	0.34671	2.8842	11.426	12.928	0.065319	0.015025	0.027681	137.13	39.009	4.0523	16.460
205.00	0.62797	0.41374	2.4170	11.437	12.954	0.064056	0.015314	0.028722	137.79	36.862	4.1975	16.920
210.00	0.75025	0.49026	2.0397	11.442	12.973	0.062836	0.015618	0.029898	138.35	34.888	4.3511	17.399
215.00	0.88893	0.57731	1.7322	11.442	12.981	0.061653	0.015938	0.031232	138.80	33.066	4.5144	17.901
220.00	1.0452	0.67606	1.4791	11.434	12.980	0.060498	0.016273	0.032757	139.13	31.378	4.6894	18.429
225.00	1.2203	0.78791	1.2692	11.420	12.968	0.059367	0.016625	0.034515	139.35	29.805	4.8787	18.987
230.00	1.4155	0.91450	1.0935	11.397	12.945	0.058252	0.016996	0.036563	139.45	28.333	5.0855	19.579
235.00	1.6321	1.0578	0.94537	11.365	12.908	0.057147	0.017389	0.038982	139.43	26.946	5.3151	20.212
240.00	1.8712	1.2202	0.81952	11.322	12.856	0.056042	0.017808	0.041885	139.29	25.629	5.5745	20.892
245.00	2.1344	1.4049	0.71181	11.268	12.787	0.054930	0.018258	0.045436	139.02	24.367	5.8750	21.630
250.00	2.4229	1.6157	0.61892	11.199	12.699	0.053801	0.018748	0.049883	138.62	23.142	6.2336	22.437
255.00	2.7382	1.8581	0.53818	11.114	12.588	0.052641	0.019288	0.055621	138.07	21.938	6.6778	23.333
260.00	3.0820	2.1394	0.46742	11.009	12.450	0.051434	0.019894	0.063311	137.36	20.732	7.2524	24.341
265.00	3.4561	2.4705	0.40477	10.878	12.277	0.050157	0.020590	0.074154	136.47	19.500	8.0324	25.501
270.00	3.8623	2.8685	0.34861	10.713	12.059	0.048773	0.021413	0.090577	135.36	18.207	9.1495	26.874
275.00	4.3032	3.3629	0.29736	10.500	11.779	0.047224	0.022428	0.11833	133.95	16.800	10.852	28.575
280.00	4.7818	4.0140	0.24913	10.209	11.400	0.045388	0.023764	0.17512	132.11	15.185	13.688	30.845
285.00	5.3025	4.9911	0.20036	9.7610	10.823	0.042934	0.025741	0.35495	129.38	13.131	19.481	34.410
289.73	5.8420	8.4000	0.11905	8.2423	8.9377	0.036074			0	8.3700		
Single-Phase Properties												
200.00	0.10000	0.061174	16.347	11.693	13.328	0.080422	0.012825	0.021719	143.95	39.340	3.8225	16.051
300.00	0.10000	0.040297	24.816	12.971	15.453	0.089046	0.012529	0.021011	177.58	18.027	5.5664	23.114
400.00	0.10000	0.030132	33.188	14.227	17.546	0.095069	0.012489	0.020884	205.37	10.987	7.2216	30.061

TABLE 2-307 Thermodynamic Properties of Xenon (Concluded)

Temperature K	Pressure MPa	Density mol/dm ³	Volume dm ³ /mol	Int. energy kJ/mol	Enthalpy kJ/mol	Entropy kJ/(mol·K)	C _v kJ/(mol·K)	C _p kJ/(mol·K)	Sound speed m/s	Joule-Thomson K/MPa	Therm. cond. mW/(m·K)	Viscosity μPa·s
Single-Phase Properties (Concluded)												
500.00	0.10000	0.024077	41.533	15.479	19.632	0.099724	0.012480	0.020841	229.73	7.4876	8.7865	36.679
600.00	0.10000	0.020054	49.866	16.728	21.715	0.10352	0.012477	0.020822	251.71	5.3775	10.259	42.919
700.00	0.10000	0.017184	58.193	17.977	23.796	0.10673	0.012475	0.020811	271.90	3.9611	11.644	48.797
200.00	1.0000	20.545	0.048673	1.5644	1.6131	0.0086285	0.019883	0.047017	543.44	-0.43967	57.759	310.61
218.61	1.0000	19.370	0.051625	2.4644	2.5160	0.012943	0.019137	0.050348	482.05	-0.23885	50.560	244.05
218.61	1.0000	0.64740	1.5446	11.437	12.982	0.060816	0.016178	0.032313	139.05	31.834	4.6396	18.280
300.00	1.0000	0.42338	2.3619	12.735	15.097	0.069105	0.013071	0.023342	174.22	17.710	5.9066	23.825
400.00	1.0000	0.30719	3.2553	14.081	17.337	0.075560	0.012653	0.021812	204.53	10.799	7.4612	30.617
500.00	1.0000	0.24284	4.1179	15.373	19.491	0.080369	0.012555	0.021347	229.84	7.3466	8.9749	37.129
600.00	1.0000	0.20128	4.9682	16.646	21.614	0.084241	0.012521	0.021143	252.27	5.2698	10.414	43.291
700.00	1.0000	0.17205	5.8123	17.910	23.723	0.087491	0.012506	0.021033	272.69	3.8786	11.777	49.110
200.00	5.0000	20.777	0.048130	1.4588	1.6995	0.0080923	0.019975	0.045835	562.91	-0.48921	59.693	324.93
282.15	5.0000	12.969	0.077107	6.3800	6.7656	0.028774	0.021130	0.17857	199.64	3.2255	29.468	88.606
282.15	5.0000	4.3744	0.22860	10.045	11.188	0.044447	0.024500	0.22292	131.10	14.382	15.582	32.133
300.00	5.0000	3.0178	0.33137	11.189	12.846	0.050182	0.017212	0.057127	155.40	14.736	9.7191	29.671
400.00	5.0000	1.6846	0.59361	13.369	16.337	0.060390	0.013445	0.027264	202.15	9.6973	8.7332	33.661
500.00	5.0000	1.2596	0.79390	14.889	18.589	0.066031	0.012902	0.023853	231.38	6.6609	9.8856	39.357
600.00	5.0000	1.0212	0.97922	16.278	21.175	0.070256	0.012724	0.022627	255.44	4.7873	11.137	45.049
700.00	5.0000	0.86365	1.1579	17.615	23.404	0.073694	0.012645	0.022028	276.67	3.5219	12.377	50.547
200.00	10.000	21.038	0.047533	1.3406	1.8159	0.0074790	0.020097	0.044678	584.79	-0.53901	61.959	341.91
300.00	10.000	13.284	0.075281	6.6026	7.3554	0.029493	0.018200	0.088497	258.88	2.0533	29.649	94.113
400.00	10.000	3.8169	0.26199	12.321	14.941	0.052029	0.014472	0.037766	204.59	7.6552	11.158	39.566
500.00	10.000	2.6239	0.38111	14.257	18.068	0.059048	0.013339	0.027456	235.98	5.6862	11.280	42.848
600.00	10.000	2.0706	0.48294	15.817	20.646	0.063756	0.012977	0.024556	260.98	4.1777	12.172	47.602
700.00	10.000	1.7304	0.57792	17.251	23.030	0.067433	0.012819	0.023262	282.62	3.0965	13.210	52.559
200.00	100.00	23.707	0.042181	0.22766	4.4458	0.00066778	0.022077	0.038788	819.84	-0.80697	91.685	582.16
300.00	100.00	20.755	0.048181	3.3951	8.2132	0.015969	0.018781	0.036811	690.23	-0.77003	62.741	293.11
400.00	100.00	18.043	0.055422	6.2728	11.815	0.026343	0.017019	0.035156	597.21	-0.68464	46.987	188.96
500.00	100.00	15.689	0.063739	8.8555	15.229	0.033970	0.015937	0.033084	538.16	-0.60849	38.256	141.62
600.00	100.00	13.750	0.072728	11.158	18.431	0.039814	0.015222	0.030987	504.61	-0.56578	33.282	118.14
700.00	100.00	12.188	0.082045	13.232	21.437	0.044451	0.014724	0.029184	487.44	-0.55679	30.399	106.02
400.00	500.00	25.007	0.039989	4.1551	24.150	0.012220	0.020738	0.032768	1164.9	-0.96827	96.278	544.58
500.00	500.00	23.787	0.042040	6.3322	27.352	0.019372	0.019369	0.031337	1121.3	-1.0167	80.779	424.61
600.00	500.00	22.695	0.044062	8.3949	30.426	0.024980	0.018396	0.030178	1086.3	-1.0606	71.145	352.34
700.00	500.00	21.712	0.046058	10.364	33.393	0.029555	0.017664	0.029185	1058.1	-1.1030	64.734	304.58

The values in these tables were generated from the NIST REFPROP software (Lemmon, E. W., McLinden, M. O., and Huber, M. L., NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties—REFPROP, National Institute of Standards and Technology, Standard Reference Data Program, Gaithersburg, Md., 2002, Version 7.1). The primary source for the thermodynamic properties is Lemmon, E. W., and Span, R., "Short Fundamental Equations of State for 20 Industrial Fluids," *J. Chem. Eng. Data* **51**(3):785–850, 2006. The source for viscosity and thermal conductivity is McCarty, R. D., *Correlations for the Thermophysical Properties of Xenon*, National Institute of Standards and Technology, Boulder, Colo., 1989.

Properties at the triple point temperature and the critical point temperature are given in the first and last entries of the saturation tables, respectively. In the single-phase table, when the temperature range for a given isobar includes a vapor-liquid phase boundary, the temperature of phase equilibrium is noted, and properties for both the saturated liquid and saturated vapor are given (with liquid properties given in the upper line). Lines are omitted from the temperature-pressure grid of the single-phase table, when the system would be in the solid phase or if there are potential problems with the source property surface.

The uncertainties in the equation of state are 0.2% in density up to 100 MPa, rising to 1% at higher pressures, 0.2% in vapor pressure, 1% in the speed of sound, and 2% in heat capacities. For viscosity, estimated uncertainty is less than 5%. For thermal conductivity, estimated uncertainty is less than 6%.

TABLE 2-308 Surface Tension (N/m) of Saturated Liquid Refrigerants*

R no.	Temperature, °C								
	-50	-25	0	25	50	75	100	125	150
11	0.0279	0.0244	0.0210	0.0178	0.0146	0.0116	0.0087	0.0060	0.0036
12	0.0188	0.0152	0.0118	0.0085	0.0055	0.0029	0.0007	—	—
13	0.0092	0.0056	0.0025	0.0002	—	—	—	—	—
22	0.0197	0.0156	0.0117	0.0081	0.0047	0.0018	—	—	—
23	0.0115	0.0065	0.0025	—	—	—	—	—	—
32	—	—	—	0.0069	0.0032	0.0002	—	—	—
113	—	0.0231	0.0201	0.0172	0.0144	0.0118	0.0092	0.0067	0.0045
114	—	—	0.0138	0.0109	0.0082	0.0056	0.0033	0.0012	—
115	—	—	0.0075	0.0047	0.0022	—	—	—	—
134a	0.0192	0.0154	0.0117	0.0082	0.0050	0.0021	0.0000	—	—
142b	0.0213	0.0178	0.0145	0.0113	0.0083	0.0055	0.0029	—	—
152a	0.0201	0.0166	0.0132	0.0100	0.0068	0.0038	0.0011	—	—
170	0.0100	0.0051	0.0032	0.0005	—	—	—	—	—
290	—	—	0.0101	0.0082	0.0041	0.0016	—	—	—
C318	—	0.0143	0.0113	0.0085	0.0048	0.0033	0.0011	—	—
502	0.0159	0.0121	0.0086	0.0054	0.0026	—	—	—	—
503	0.0094	0.0053	0.0018	—	—	—	—	—	—
600	—	0.0180	0.0150	0.0122	0.0094	0.0068	0.0043	0.0020	0.0001
600a	—	—	0.0132	0.0101	0.0073	0.0047	0.0024	0.0005	—
718	—	—	0.0755	0.0720	0.0680	0.0636	0.0590	0.0540	0.0488
744	—	0.0096	0.0044	0.0005	—	—	—	—	—
1150	0.0100	0.0055	0.0013	—	—	—	—	—	—
1270	0.0171	0.0136	0.0102	0.0070	0.0041	0.0014	—	—	—

*Dashes indicate inaccessible states; blanks indicate no available data.

Values and equations were given by Srinivasan, K., *Can. J. Chem. Eng.* (27 liquids), **68** (1990): 493; Lielmezs, J. and T. A. Herrick, *Chem. Eng. J.* (34 liquids), **32** (1986): 165–169; Somayajulu, G. R., *Int. J. Thermophys.* (64 liquids), **9**, 4 (1988): 559–566; Ibrahim, N. and S. Murad, *Chem. Eng. Commun.* (29 polar liquids), **79** (1979): 165–174; Yaws, C. L.; Morachevsky, A. G. and I. B. Sladkov, *Physico-Chemical Properties of Molecular Inorganic Compounds* (200 compounds), Khimiya, Leningrad, 1987; Jasper, J., *J. Phys. Chem. Ref. Data* (2200 compounds), **1**, (1972): 841–1009; and Vargaftik, N. B., B. N. Volkov, et al., *J. Phys. Chem. Ref. Data* (water), **12**, 3 (1983): 817–820. See also Escobedo, J. and Mansoori, G. R., *AIChE J.*, **42**(5), May 1996: 1425–1433.

TABLE 2-309 Surface Tension σ (dyn/cm) of Various Liquids

Compound	T, K	σ	Compound	T, K	σ	Compound	T, K	σ
Acetic acid	293	27.59	<i>p</i> -Cresol	313	34.88	Isobutyric acid	293	25.04
	333	23.62		373	29.32		313	23.2
Acetone	298	24.02	Cyclohexane	293	25.24		333	21.36
	308	22.34		313	22.87	Methyl formate	363	18.6
	318	21.22		333	20.49		293	24.62
	293	42.67	Cyclopentane	293	22.61		323	20.05
Aniline	313	40.5		313	19.68		373	12.9
	333	38.33	Diethyl ether	288	17.56		423	6.3
	353	36.15		303	16.2		473	0.87
	293	28.88	2,3-Dimethylbutane	293	17.38	Methyl alcohol	293	22.56
Benzene	313	26.25		313	15.38		313	20.96
	333	23.67	Ethyl acetate	293	23.97		333	19.41
	353	21.2		313	21.65	Phenol	313	39.27
	293	39.37		333	19.32		333	37.13
Benzonitrile	323	35.89		353	17		373	32.96
	363	31.26		373	14.68	<i>n</i> -Propyl alcohol	293	23.71
Bromobenzene	293	35.82		293	35.04		313	22.15
	323	32.34	Ethyl benzoate	313	32.92		333	20.6
	373	26.54		333	30.81		363	18.27
	203	23.31	Ethyl bromide	283	25.36	<i>n</i> -Propyl benzene	293	29.98
	233	19.69		303	23.04		313	26.83
	293	12.46	Ethyl mercaptan	288	23.87		333	24.68
Carbon disulfide	293	32.32		303	22.68		353	22.53
	313	29.35	Formamide	298	57.02		373	20.38
Carbon tetrachloride	288	27.65		338	53.66	Pyridine	293	37.21
	308	25.21		373	50.71		313	34.6
	328	22.76		293	20.14		333	31.98
	348	20.31	<i>n</i> -Heptane	313	18.18			
	368	17.86		333	16.22			
	293	33.59		353	14.26			
Chlorobenzene	323	30.01						
	373	24.06						

Methyl formate values from D. B. Macleod, *Trans. Faraday Soc.* **19**:38, 1923. All others from J. J. Jasper, *J. Phys. Chem. Ref. Data* **1**:841, 1972.

TABLE 2-310 Velocity of Sound (m/s) in Gaseous Refrigerants at Atmospheric Pressure*

R. no.	Temperature, °C								
	-50	-25	0	25	50	75	100	125	150
14	158	166	173	180	187	194	200	206	212
170	272	286	299	311	323	334	344	355	364
290	—	227	238	249	258	268	277	286	294
600	—	—	200	210	220	228	237	245	252
600a	—	—	201	211	221	229	237	246	253
718	—	—	—	—	—	—	473	490	505
744	—	248	258	269	279	288	297	307	316
1150	290	305	318	330	341	352	363	373	384
1270	—	235	246	257	267	277	286	295	303

*Dashes indicate inaccessible states; blanks indicate no available data. Values for the velocity of sound for all compounds listed in Table 2-184 are given in their respective tables earlier in this section.

TABLE 2-311 Velocity of Sound (m/s) in Saturated Liquid Refrigerants*

R. no.	Temperature, °C								
	-50	-25	0	25	50	75	100	125	150
14	182	—	—	—	—	—	—	—	—
290	1210	982	884	719	551	367	—	—	—
600	1290	1163	1031	896	759	609	477	325	142
600a	1205	1078	947	812	661	528	378	208	—
718	—	—	1402	1495	1542	1554	1543	1514	1468
744	—	751	525	272	—	—	—	—	—
1150	874	644	372	—	—	—	—	—	—
1270	1184	1022	859	694	524	335	—	—	—

*Dashes indicate inaccessible states; blanks indicate no available data. Values for the velocity of sound for all compounds listed in Table 2-184 are given in their respective tables earlier in this section.

TRANSPORT PROPERTIES

INTRODUCTION

Extensive tables of the viscosity and thermal conductivity of air and of water or steam for various pressures and temperatures are given with the thermodynamic-property tables. The thermal conductivity and the viscosity for the saturated-liquid state are also tabulated for many fluids along with the thermodynamic-property tables earlier in this section.

UNITS CONVERSIONS

For this subsection the following units conversions are applicable:

Diffusivity: to convert square centimeters per second to square feet per hour, multiply by 3.8750; to convert square meters per second to square feet per hour, multiply by 38,750.

Pressure: to convert bars to pounds-force per square inch, multiply by 14.504.

Temperature: °F = % °C + 32; °R = % K.

Thermal conductivity: to convert watts per meter-kelvin to British thermal unit–feet per hour–square foot–degree Fahrenheit, multiply by 0.57779; and to convert British thermal unit–feet per hour–square foot–degree Fahrenheit to watts per meter-kelvin, multiply by 1.7307.

Viscosity: to convert pascal-seconds to centipoises, multiply by 1000.

ADDITIONAL REFERENCES

An extensive coverage of the general pressure and temperature variation of thermal conductivity is given in the monograph by Vargaftik, Filippov, Tarzimanov, and Totitskiy, *Thermal Conductivity of Liquids and Gases* (in Russian), Standartov, Moscow, 1978, now published in English translation by CRC Press, Miami, FL.

For a similar work on viscosity, see Stephan and Lucas, *Viscosity of Dense Fluids*, Plenum, New York and London, 1979. Tables and

polynomial fits for refrigerants in both the gaseous and the liquid state are contained in *ASHRAE Handbook—Fundamentals*, SI ed., ASHRAE, Atlanta, 2005. Other sources for viscosity include Fischer & Porter Co. catalog 10-A-94, "Fluid densities and viscosities," 1953 (200 industrial fluids in 48 pp.) and van Velzen, D., R. L. Cardozo et al., EURATOM Ispra, Italy rept. 4735 e, 1972 (160 pp.). Liquid viscosity, 314 cpds, is summarized in *I&EC Fundtls.*, 11 (1972): 20–26. Five hundred forty-nine binary and ternary systems are discussed in Skubla, P., *Coll. Czech. Chem. Commun.*, **46** (1981): 303–339.

See also Duhne, C. R., *Chem. Eng.* (NY), **86**, 15 (July 16, 1979): 83–91 (equations and 326 liquids); and Rao, K. V. K., *Chem. Eng.* (NY), **90**, 11 (May 30, 1983): 90–91 (nomograph, 87 liquids). For rheology, non-Newtonian behavior, and the like, see, for instance, Barnes, H., *The Chem. Engr.* (UK), (June 24, 1993): 17–23; Hyman, W. A., *I&EC Fundtls.*, **16** (1976): 215–218; and Ferguson, J. and Z. Kemblowski, *Applied Fluid Rheology*, Elsevier, 1991 (325 pp.). Other sources for thermal conductivity include Ho, C. Y., R. W. Powell et al., *J. Phys. Chem. Ref. Data*, **1** (1972) and **3**, suppl. 1 (1974); Childs, Ericks et al., N.B.S. Monogr. 131, 1973; Jamieson, D. T., J. B. Irving et al., *Liquid Thermal Conductivity*, H.M.S.O., Edinburgh, Scotland, 1975 (220 pp.).

Other references include Poling, Prausnitz, and O'Connell, *The Properties of Gases and Liquids*, 5th ed., McGraw-Hill, New York, 2000; Vargaftik, Vinogradov, and Yargin, *Handbook of Physical Properties of Liquids and Gases*, Begell House, New York, 1996; Yaws, *Chemical Properties Handbook: Physical, Thermodynamics, Environmental Transport, Safety & Health Related Properties for Organic & Inorganic Chemicals*, McGraw-Hill, New York, 1998; and Riazi, *Characterization and Properties of Petroleum Fractions*, ASTM, West Conshohocken, Pa., 2005. Free web resources include the NIST Webbook at <http://webbook.nist.gov> and the KDB (Korea thermo-physical properties) database at <http://www.cheric.org/research/kdb/>.

TABLE 2-312 Vapor Viscosity of Inorganic and Organic Substances (Pa·s)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1	C2	C3	C4	$T_{\min}$, K	Viscosity at $T_{\min}$	$T_{\max}$, K	Viscosity at $T_{\max}$
1	Acetaldehyde	C ₂ H ₄ O	75-07-0	44.053	1.222E-07	0.787	77		150.15	4.171E-06	1000	2.605E-05
2	Acetamide	C ₂ H ₅ NO	60-35-5	59.067	1.423E-07	0.7574	272.14		353.33	6.842E-06	1000	2.094E-05
3	Acetic acid	C ₂ H ₄ O ₂	64-19-7	60.052	2.7449E-08	1.0123	7.4948		289.81	8.315E-06	1000	2.966E-05
4	Acetic anhydride	C ₄ H ₆ O ₃	108-24-7	102.089	1.20E-07	0.7915	105.3		200.15	5.213E-06	1000	2.572E-05
5	Acetone	C ₃ H ₆ O	67-64-1	58.079	3.1005E-08	0.9762	23.139		178.45	4.329E-06	1000	2.571E-05
6	Acetonitrile	C ₂ H ₃ N	75-05-8	41.052	5.1964E-08	0.8857	38.805		229.32	5.476E-06	1000	2.271E-05
7	Acetylene	C ₂ H ₂	74-86-2	26.037	1.2025E-06	0.4952	291.4		192.40	6.468E-06	600	1.923E-05
8	Acrolein	C ₃ H ₄ O	107-02-8	56.063	6.523E-07	0.579	410.8		185.45	4.174E-06	1000	2.523E-05
9	Acrylic acid	C ₃ H ₄ O ₂	79-10-7	72.063	1.7154E-07	0.7418	138.4		286.15	7.680E-06	1000	2.532E-05
10	Acrylonitrile	C ₃ H ₃ N	107-13-1	53.063	4.302E-08	0.9114	54.3		189.63	3.985E-06	1000	2.213E-05
11	Air	Mixture	132259-10-0	28.960	1.425E-06	0.5039	108.3		80.00	5.508E-06	2000	6.227E-05
12	Ammonia	H ₃ N	7664-41-7	17.031	4.1855E-08	0.9806	30.8		195.41	6.378E-06	1000	3.551E-05
13	Anisole	C ₇ H ₈ O	100-66-3	108.138	1.7531E-07	0.72	176.17		235.65	5.122E-06	1000	2.155E-05
14	Argon	Ar	7440-37-1	39.948	9.2121E-07	0.60529	83.24		83.78	6.742E-06	3273.1	1.001E-11
15	Benzamide	C ₇ H ₇ NO	55-21-0	121.137	2.5082E-08	0.96663			403.00	8.274E-06	1000	1.992E-05
16	Benzene	C ₆ H ₆	71-43-2	78.112	3.134E-08	0.9676	7.9		278.68	7.077E-06	1000	2.486E-05
17	Benzenethiol	C ₆ H ₆ S	108-98-5	110.177	1.1184E-07	0.8002	152.43		442.29	1.089E-05	1000	2.441E-05
18	Benzoic acid	C ₇ H ₆ O ₂	65-85-0	122.121	7.4266E-08	0.8289	91.197		395.45	8.578E-06	1000	2.087E-05
19	Benzonitrile	C ₇ H ₅ N	100-47-0	103.121	4.2137E-08	0.92271	45.387		260.40	6.079E-06	1000	2.363E-05
20	Benzophenone	C ₁₃ H ₁₀ O	119-61-9	182.218	3.779E-07	0.6005	409		321.35	5.324E-06	1000	1.698E-05
21	Benzyl alcohol	C ₇ H ₈ O	100-51-6	108.138	6.9022E-08	0.84014	74.746		257.85	5.680E-06	1000	2.129E-05
22	Benzyl ethyl ether	C ₉ H ₁₂ O	539-30-0	136.191	1.56E-07	0.7181	180		458.15	9.122E-06	1000	1.886E-05
23	Benzyl mercaptan	C ₇ H ₈ S	100-53-8	124.203	4.0138E-08	0.90735	34.714		243.95	5.151E-06	1000	2.045E-05
24	Biphenyl	C ₁₂ H ₁₀	92-52-4	154.208	1.3874E-06	0.4434	678.22		342.20	6.186E-06	1000	1.768E-05
25	Bromine	Br ₂	7726-95-6	159.808	7.3534E-08	0.93798			265.85	1.383E-05	600	2.967E-05
26	Bromobenzene	C ₆ H ₅ Br	108-86-1	157.008	2.232E-07	0.7146	184.9		429.24	1.187E-05	1000	2.623E-05
27	Bromoethane	C ₂ H ₅ Br	74-96-4	108.965	5.29E-07	0.632	226		154.55	5.195E-06	1000	3.396E-05
28	Bromomethane	CH ₃ Br	74-83-9	94.939	7.8796E-08	0.90476			179.47	8.626E-06	1000	4.081E-05
29	1,2-Butadiene	C ₄ H ₆	590-19-2	54.090	6.0259E-07	0.5309	199.64		136.95	3.340E-06	1000	1.966E-05
30	1,3-Butadiene	C ₄ H ₆	106-99-0	54.090	2.696E-07	0.6715	134.7		164.25	4.553E-06	1000	2.457E-05
31	Butane	C ₄ H ₁₀	106-97-8	58.122	3.4387E-08	0.94604			134.86	3.559E-06	1000	2.369E-05
32	1,2-Butanediol	C ₄ H ₁₀ O ₂	584-03-2	90.121	7.5626E-08	0.83521	71.798		220.00	5.157E-06	1000	2.260E-05
33	1,3-Butanediol	C ₄ H ₁₀ O ₂	107-88-0	90.121	7.0728E-08	0.84383	64.391		196.15	4.580E-06	1000	2.259E-05
34	1-Butanol	C ₄ H ₁₀ O	71-36-3	74.122	1.4031E-06	0.4611	537		183.85	3.961E-06	1000	2.207E-05
35	2-Butanol	C ₄ H ₁₀ O	78-92-2	74.122	1.2114E-07	0.76972	92.661		158.45	3.772E-06	1000	2.259E-05
36	1-Butene	C ₄ H ₈	106-98-9	56.106	6.9744E-07	0.5462	305.25		87.80	1.795E-06	1000	2.325E-05
37	cis-2-Butene	C ₄ H ₈	590-18-1	56.106	4.2898E-08	0.91349			134.26	3.770E-06	1000	2.360E-05
38	trans-2-Butene	C ₄ H ₈	624-64-6	56.106	1.05E-06	0.4867	358.7		167.62	4.044E-06	1000	2.229E-05
39	Butyl acetate	C ₆ H ₁₂ O ₂	123-86-4	116.158	1.006E-07	0.77881	95.108		199.65	4.216E-06	1000	1.993E-05
40	Butylbenzene	C ₁₀ H ₁₄	104-51-8	134.218	3.4205E-07	0.59764	234.21		185.30	3.425E-06	1000	1.720E-05
41	Butyl mercaptan	C ₄ H ₁₀ S	109-79-5	90.187	5.4539E-08	0.88896	43.687		157.46	3.833E-06	1000	2.427E-05
42	sec-Butyl mercaptan	C ₄ H ₁₀ S	513-53-1	90.187	3.1378E-08	0.96513			133.02	3.520E-06	1000	2.466E-05
43	1-Butyne	C ₄ H ₆	107-00-6	54.090	2.7856E-06	0.377	663.14		147.43	3.329E-06	800	1.893E-05
44	Butyraldehyde	C ₄ H ₈ O	123-72-8	72.106	8.0079E-08	0.8178	65.855		176.75	4.017E-06	1000	2.134E-05
45	Butyric acid	C ₄ H ₈ O ₂	107-92-6	88.105	2.128E-06	0.4273	886	-41,400	267.95	6.220E-06	1000	2.208E-05
46	Butyronitrile	C ₄ H ₇ N	109-74-0	69.105	8.0223E-08	0.80561	75.207		161.25	3.284E-06	1000	1.948E-05
47	Carbon dioxide	CO ₂	124-38-9	44.010	2.148E-06	0.46	290		194.67	9.749E-06	1500	5.203E-05
48	Carbon disulfide	CS ₂	75-15-0	76.141	5.8204E-08	0.9262	44.581		161.11	5.048E-06	800	2.693E-05
49	Carbon monoxide	CO	630-08-0	28.010	1.1127E-06	0.5338	94.7		68.15	4.434E-06	1250	4.654E-05
50	Carbon tetrachloride	CCl ₄	56-23-5	153.823	3.137E-06	0.3742	491.5		250.33	8.361E-06	1000	2.789E-05
51	Carbon tetrafluoride	CF ₄	75-73-0	88.004	2.1709E-06	0.45853	208		89.56	5.132E-06	1000	4.267E-05
52	Chlorine	Cl ₂	7782-50-5	70.906	2.60E-07	0.7423	98.3		200.00	8.900E-06	1000	3.992E-05
53	Chlorobenzene	C ₆ H ₅ Cl	108-90-7	112.557	1.065E-07	0.7942	94.7		227.95	5.611E-06	1000	2.348E-05
54	Chloroethane	C ₂ H ₅ Cl	75-00-3	64.514	3.12E-07	0.6711	139		134.80	4.127E-06	1000	2.824E-05
55	Chloroform	CHCl ₃	67-66-3	119.378	1.696E-07	0.7693	96.6		209.63	7.091E-06	1000	3.143E-05
56	Chloromethane	CH ₃ Cl	74-87-3	50.488	8.60E-08	0.8706	35.8		230.00	8.468E-06	700	2.454E-05
57	1-Chloropropane	C ₃ H ₇ Cl	540-54-5	78.541	3.638E-07	0.6417	208.3		150.35	3.805E-06	1000	2.534E-05
58	2-Chloropropane	C ₃ H ₇ Cl	75-29-6	78.541	3.8802E-07	0.6367	205.08		155.97	4.175E-06	1000	2.618E-05
59	m-Cresol	C ₇ H ₈ O	108-39-4	108.138	1.4427E-07	0.7438	166.15		285.39	6.113E-06	1000	2.108E-05

TABLE 2-312 Vapor Viscosity of Inorganic and Organic Substances (Pa·s) (Continued)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1	C2	C3	C4	$T_{\min}$, K	Viscosity at $T_{\min}$	$T_{\max}$, K	Viscosity at $T_{\max}$
60	<i>o</i> -Cresol	C ₇ H ₈ O	95-48-7	108.138	8.7371E-08	0.80775	98.538		304.19	6.688E-06	1000	2.108E-05
61	<i>p</i> -Cresol	C ₇ H ₈ O	106-44-5	108.138	1.4305E-07	0.7451	159.8		307.93	6.731E-06	1000	2.120E-05
62	Cumene	C ₉ H ₁₂	98-82-8	120.192	3.3699E-07	0.60751	221.17		177.14	3.480E-06	1000	1.834E-05
63	Cyanogen	C ₂ N ₂	460-19-5	52.035	4.0854E-08	0.97182	6.6762		245.25	8.353E-06	600	2.024E-05
64	Cyclobutane	C ₄ H ₈	287-23-0	56.106	1.0881E-06	0.48359	330.86		182.48	4.797E-06	1000	2.308E-05
65	Cyclohexane	C ₆ H ₁₂	110-82-7	84.159	6.77E-08	0.8367	36.7		279.69	6.671E-06	900	1.928E-05
66	Cyclohexanol	C ₆ H ₁₂ O	108-93-0	100.159	7.9581E-08	0.8376	104.97		296.60	6.917E-06	1000	2.346E-05
67	Cyclohexanone	C ₆ H ₁₀ O	108-94-1	98.143	5.2312E-08	0.89422	58.008		242.00	5.714E-06	1000	2.381E-05
68	Cyclohexene	C ₆ H ₁₀	110-83-8	82.144	1.3326E-06	0.4537	445		169.67	3.778E-06	1000	2.118E-05
69	Cyclopentane	C ₅ H ₁₀	287-92-3	70.133	2.3619E-07	0.67465	139		179.28	4.409E-06	1000	2.191E-05
70	Cyclopentene	C ₅ H ₈	142-29-0	68.117	3.026E-07	0.64991	167.14		138.13	3.369E-06	1000	2.309E-05
71	Cyclopropane	C ₃ H ₆	75-19-4	42.080	1.7578E-06	0.4265	370.34		145.59	4.150E-06	1000	2.441E-05
72	Cyclohexyl mercaptan	C ₆ H ₁₂ S	1569-69-3	116.224	3.915E-08	0.91427	22.264		189.64	4.238E-06	1000	2.118E-05
73	Decanal	C ₁₀ H ₂₀ O	112-31-2	156.265	1.2638E-07	0.7248	176.88		267.15	4.365E-06	1000	1.605E-05
74	Decane	C ₁₀ H ₂₂	124-18-5	142.282	2.64E-08	0.9487	71		243.51	3.755E-06	1000	1.730E-05
75	Decanoic acid	C ₁₀ H ₂₀ O ₂	334-48-5	172.265	7.1748E-08	0.7982	109.38		304.55	5.070E-06	1000	1.605E-05
76	1-Decanol	C ₁₀ H ₂₂ O	112-30-1	158.281	5.5065E-08	0.8341	79.56		280.05	4.715E-06	1000	1.622E-05
77	1-Decene	C ₁₀ H ₂₀	872-05-9	140.266	6.1192E-08	0.82546	77.434		206.89	3.632E-06	1000	1.701E-05
78	Decyl mercaptan	C ₁₀ H ₂₂ S	143-10-2	174.347	3.272E-08	0.9302	39.13		247.56	4.761E-06	1000	1.944E-05
79	1-Decyne	C ₁₀ H ₁₈	764-93-2	138.250	5.6914E-07	0.50744	273.3		229.15	4.091E-06	1000	1.488E-05
80	Deuterium	D ₂	7782-39-0	4.032	2.4999E-07	0.6878	0.5962		60.00	4.137E-06	480	1.744E-05
81	1,1-Dibromoethane	C ₂ H ₄ Br ₂	557-91-5	187.861	1.4125E-07	0.8097	83.243		210.15	7.685E-06	1000	3.502E-05
82	1,2-Dibromoethane	C ₂ H ₄ Br ₂	106-93-4	187.861	1.1379E-07	0.8502	93.816		282.85	1.038E-05	1000	3.696E-05
83	Dibromomethane	CH ₂ Br ₂	74-95-3	173.835	2.9444E-07	0.728	154.74		370.10	1.538E-05	1000	3.895E-05
84	Dibutyl ether	C ₈ H ₁₈ O	142-96-1	130.228	7.7147E-08	0.79906	80.765		175.30	3.278E-06	1000	1.781E-05
85	<i>m</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	541-73-1	147.002	2.334E-07	0.714	260		248.39	5.850E-06	1000	2.569E-05
86	<i>o</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	95-50-1	147.002	1.603E-07	0.763	205		256.15	6.127E-06	1000	2.588E-05
87	<i>p</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	106-46-7	147.002	1.5913E-07	0.7639	193.14		326.14	8.313E-06	1000	2.611E-05
88	1,1-Dichloroethane	C ₂ H ₄ Cl ₂	75-34-3	98.959	2.0135E-07	0.73421	111.98		176.19	5.487E-06	1000	2.887E-05
89	1,2-Dichloroethane	C ₂ H ₄ Cl ₂	107-06-2	98.959	1.4321E-07	0.7785	98.159		237.49	7.164E-06	1000	2.824E-05
90	Dichloromethane	CH ₂ Cl ₂	75-09-2	84.933	7.6787E-07	0.5741	276.16		178.01	5.895E-06	1000	3.175E-05
91	1,1-Dichloropropane	C ₃ H ₆ Cl ₂	78-99-9	112.986	1.4906E-07	0.7617	105.9		200.00	5.515E-06	1000	2.599E-05
92	1,2-Dichloropropane	C ₃ H ₆ Cl ₂	78-87-5	112.986	1.1989E-07	0.79108	84.37		172.71	4.742E-06	1000	2.611E-05
93	Diethanol amine	C ₄ H ₁₁ NO ₂	111-42-2	105.136	3.3628E-08	0.9426	39.587		301.15	6.450E-06	1000	2.176E-05
94	Diethyl amine	C ₄ H ₁₁ N	109-89-7	73.137	4.3184E-07	0.6035	247		223.35	5.364E-06	1000	2.239E-05
95	Diethyl ether	C ₄ H ₁₀ O	60-29-7	74.122	1.948E-06	0.41	495.8		156.85	3.720E-06	1000	2.212E-05
96	Diethyl sulfide	C ₄ H ₁₀ S	352-93-2	90.187	6.5492E-08	0.86232	59.455		169.20	4.046E-06	1000	2.388E-05
97	1,1-Difluoroethane	C ₂ H ₄ F ₂	75-37-6	66.050	2.7228E-06	0.39531	445.07		154.56	5.148E-06	1000	2.891E-05
98	1,2-Difluoroethane	C ₂ H ₄ F ₂	624-72-6	66.050	4.3934E-07	0.64867	169.64		215.00	8.001E-06	1000	3.317E-05
99	Difluoromethane	CH ₂ F ₂	75-10-5	52.023	7.7484E-07	0.57978	198.7		136.95	5.478E-06	1000	1.172E-12
100	Di-isopropyl amine	C ₆ H ₁₅ N	108-18-9	101.190	4.138E-07	0.5999	269.5		357.05	8.016E-06	1000	2.055E-05
101	Di-isopropyl ether	C ₆ H ₁₄ O	108-20-3	102.175	1.691E-07	0.7114	124		187.65	4.218E-06	1000	2.049E-05
102	Di-isopropyl ketone	C ₇ H ₁₄ O	565-80-0	114.185	9.2797E-08	0.7819	93.399		204.81	4.089E-06	1000	1.881E-05
103	1,1-Dimethoxyethane	C ₄ H ₁₀ O ₂	534-15-6	90.121	4.4172E-08	0.91098			159.95	4.497E-06	1000	2.388E-05
104	1,2-Dimethoxypropane	C ₅ H ₁₂ O ₂	7778-85-0	104.148	3.9833E-08	0.91566			226.10	5.701E-06	1000	2.225E-05
105	Dimethyl acetylene	C ₄ H ₆	503-17-3	54.090	1.9377E-06	0.4093	492.69		240.91	6.006E-06	1000	2.194E-05
106	Dimethyl amine	C ₂ H ₇ N	124-40-3	45.084	2.757E-07	0.6841	133.2		180.96	5.563E-06	1000	2.744E-05
107	2,3-Dimethylbutane	C ₆ H ₁₄	79-29-8	86.175	6.8567E-07	0.52542	278.82		145.19	3.211E-06	1000	2.021E-05
108	1,1-Dimethylcyclohexane	C ₈ H ₁₆	590-66-9	112.213	7.822E-07	0.4994	371.6		392.70	7.936E-06	1000	1.796E-05
109	<i>cis</i> -1,2-Dimethylcyclohexane	C ₈ H ₁₆	2207-01-4	112.213	8.4576E-07	0.487	398		402.94	7.900E-06	1000	1.749E-05
110	<i>trans</i> -1,2-Dimethylcyclohexane	C ₈ H ₁₆	6876-23-9	112.213	9.9104E-07	0.4723	436.89		396.58	7.957E-06	1000	1.801E-05
111	Dimethyl disulfide	C ₂ H ₆ S ₂	624-92-0	94.199	3.2282E-08	0.97742			188.44	5.405E-06	1000	2.762E-05
112	Dimethyl ether	C ₂ H ₆ O	115-10-6	46.068	0.00000268	0.3975	534		131.65	3.688E-06	1000	2.722E-05
113	<i>N,N</i> -Dimethyl formamide	C ₃ H ₇ NO	68-12-2	73.094	3.5538E-06	0.3766	1176.1		212.72	4.097E-06	1000	2.202E-05
114	2,3-Dimethylpentane	C ₇ H ₁₆	565-59-3	100.202	5.0372E-07	0.54462	227.44		160.00	3.300E-06	1000	1.766E-05
115	Dimethyl phthalate	C ₁₀ H ₁₀ O ₄	131-11-3	194.184	5.2195E-08	0.85584	69.036		274.18	5.089E-06	1000	1.804E-05
116	Dimethylsilane	C ₂ H ₆ Si	1111-74-6	60.170	4.7238E-08	0.90849			122.93	3.739E-06	1000	2.511E-05
117	Dimethyl sulfide	C ₂ H ₆ S	75-18-3	62.134	5.2854E-07	0.6112	302.85		174.88	4.544E-06	1000	2.766E-05
118	Dimethyl sulfoxide	C ₂ H ₆ OS	67-68-5	78.133	8.6101E-08	0.8345	167.86		291.67	6.231E-06	1000	2.350E-05

119	Dimethyl terephthalate	C ₁₀ H ₁₀ O ₄	120-61-6	194.184	4.8739E-08	0.8749	51.885		413.80	8.433E-06	1000	1.953E-05
120	1,4-Dioxane	C ₄ H ₈ O ₂	123-91-1	88.105	2.7334E-07	0.7393	129.93		284.95	1.226E-05	1000	3.995E-05
121	Diphenyl ether	C ₁₂ H ₁₀ O	101-84-8	170.207	2.8451E-08	0.93622			300.03	5.933E-06	1000	1.831E-05
122	Dipropyl amine	C ₆ H ₁₅ N	142-84-7	101.190	1.29E-07	0.744	117.03		210.15	4.429E-06	1000	1.970E-05
123	Dodecane	C ₁₂ H ₂₆	112-40-3	170.335	6.344E-08	0.8287	219.5		263.57	3.511E-06	1000	1.593E-05
124	Eicosane	C ₂₀ H ₄₂	112-95-8	282.547	2.9236E-07	0.62458	702.84		309.58	3.214E-06	1000	1.284E-05
125	Ethane	C ₂ H ₆	74-84-0	30.069	2.5906E-07	0.67988	98.902		90.35	2.643E-06	1000	2.583E-05
126	Ethanol	C ₂ H ₆ O	64-17-5	46.068	1.0613E-07	0.8066	52.7		200.00	6.029E-06	1000	2.651E-05
127	Ethyl acetate	C ₄ H ₈ O ₂	141-78-6	88.105	3.214E-06	0.3572	667		189.60	4.632E-06	1000	2.274E-05
128	Ethyl amine	C ₂ H ₇ N	75-04-7	45.084	4.934E-07	0.5924	239.17		192.15	4.953E-06	1000	2.384E-05
129	Ethylbenzene	C ₈ H ₁₀	100-41-4	106.165	4.2231E-07	0.58154	239.21		178.20	3.673E-06	1000	1.893E-05
130	Ethyl benzoate	C ₉ H ₁₀ O ₂	93-89-0	150.175	6.3441E-08	0.8369	73.63		238.45	4.733E-06	1000	1.915E-05
131	2-Ethyl butanoic acid	C ₆ H ₁₂ O ₂	88-09-5	116.158	9.2371E-08	0.7908	102.32		258.15	5.344E-06	1000	1.975E-05
132	Ethyl butyrate	C ₆ H ₁₂ O ₂	105-54-4	116.158	1.6175E-07	0.7163	142.27	3,590	175.15	3.392E-06	1000	1.989E-05
133	Ethylcyclohexane	C ₈ H ₁₆	1678-91-7	112.213	4.107E-07	0.57143	230.06		161.84	3.103E-06	1000	1.729E-05
134	Ethylcyclopentane	C ₇ H ₁₄	1640-89-7	98.186	2.1696E-06	0.3812	577.77		134.71	2.659E-06	1000	1.914E-05
135	Ethylene	C ₂ H ₄	74-85-1	28.053	2.0789E-06	0.4163	352.7		169.41	5.714E-06	1000	2.726E-05
136	Ethylenediamine	C ₂ H ₈ N ₂	107-15-3	60.098	1.3744E-07	0.7557	122.8		284.29	6.863E-06	1000	2.264E-05
137	Ethylene glycol	C ₂ H ₆ O ₂	107-21-1	62.068	8.6706E-08	0.83923	75.512		260.15	7.150E-06	1000	2.655E-05
138	Ethyleneimine	C ₂ H ₅ N	151-56-4	43.068	2.8132E-07	0.6792	238.46		329.00	8.359E-06	1000	2.477E-05
139	Ethylene oxide	C ₂ H ₄ O	75-21-8	44.053	4.3403E-08	0.94806			160.65	5.356E-06	1000	3.032E-05
140	Ethyl formate	C ₃ H ₆ O ₂	109-94-4	74.079	6.761E-07	0.5804	354.9		193.55	5.069E-06	1000	2.750E-05
141	2-Ethyl hexanoic acid	C ₈ H ₁₆ O ₂	149-57-5	144.211	2.5704E-08	0.94738			235.00	4.532E-06	1000	1.787E-05
142	Ethylhexyl ether	C ₈ H ₁₈ O	5756-43-4	130.228	7.9129E-08	0.79565	83.193		180.00	3.371E-06	1000	1.781E-05
143	Ethylisopropyl ether	C ₅ H ₁₂ O	625-54-7	88.148	1.3974E-07	0.74266	98.58		140.00	3.219E-06	1000	2.150E-05
144	Ethylisopropyl ketone	C ₆ H ₁₂ O	565-69-5	100.159	1.0498E-07	0.76988	100.41		204.15	4.224E-06	1000	1.946E-05
145	Ethyl mercaptan	C ₂ H ₆ S	75-08-1	62.134	8.5992E-08	0.8427	58.148		125.26	3.441E-06	1000	2.742E-05
146	Ethyl propionate	C ₅ H ₁₀ O ₂	105-37-3	102.132	5.53E-07	0.6061	273.66		199.25	5.768E-06	1000	2.857E-05
147	Ethylpropyl ether	C ₅ H ₁₂ O	628-32-0	88.148	5.1539E-07	0.5726	288.76		145.65	2.994E-06	1000	2.088E-05
148	Ethyltrichlorosilane	C ₂ H ₅ Cl ₃ Si	115-21-9	163.506	3.2769E-08	0.9729			167.55	4.779E-06	1000	2.718E-05
149	Fluorine	F ₂	7782-41-4	37.997	6.36E-07	0.6638	61.6		53.48	4.148E-06	1000	5.874E-05
150	Fluorobenzene	C ₆ H ₅ F	462-06-6	96.102	2.1174E-07	0.7087	157.42		357.88	9.491E-06	1000	2.446E-05
151	Fluoroethane	C ₂ H ₅ F	353-36-6	48.060	8.8742E-07	0.5404	251.82		129.95	4.192E-06	1000	2.963E-05
152	Fluoromethane	CH ₃ F	593-53-3	34.033	1.2269E-07	0.82167			131.35	6.752E-06	1000	3.579E-05
153	Formaldehyde	CH ₂ O	50-00-0	30.026	4.758E-07	0.6405	161.7		181.15	7.025E-06	1000	3.419E-05
154	Formamide	CH ₃ NO	75-12-7	45.041	6.829E-08	0.8774	54.864		275.60	7.882E-06	1000	2.776E-05
155	Formic acid	CH ₂ O ₂	64-18-6	46.026	2.8608E-07	0.6958	184.25		281.45	8.751E-06	1000	2.954E-05
156	Furan	C ₄ H ₄ O	110-00-9	68.074	6.432E-07	0.5854	325.3		187.55	5.037E-06	1000	2.768E-05
157	Helium-4	He	7440-59-7	4.003	3.253E-07	0.7162	-9.6	107	20.00	3.531E-06	2000	7.561E-05
158	Heptadecane	C ₁₇ H ₃₆	629-78-7	240.468	3.1338E-07	0.6238	692.2		295.13	3.254E-06	1000	1.377E-05
159	Heptanal	C ₇ H ₁₄ O	111-71-7	114.185	8.4633E-08	0.79185	94.487		229.80	4.444E-06	1000	1.836E-05
160	Heptane	C ₇ H ₁₆	142-82-5	100.202	6.672E-08	0.82837	85.752		182.57	3.391E-06	1000	1.878E-05
161	Heptanoic acid	C ₇ H ₁₄ O ₂	111-14-8	130.185	7.2834E-08	0.81279	89.874		265.83	5.088E-06	1000	1.834E-05
162	1-Heptanol	C ₇ H ₁₆ O	111-70-6	116.201	2.572E-07	0.6502	248.6		239.15	4.440E-06	1000	1.838E-05
163	2-Heptanol	C ₇ H ₁₆ O	543-49-7	116.201	9.1526E-08	0.78346	100.28		230.00	4.516E-06	1000	1.864E-05
164	3-Heptanone	C ₇ H ₁₄ O	106-35-4	114.185	8.9656E-08	0.78236	100.14		234.15	4.485E-06	1000	1.812E-05
165	2-Heptanone	C ₇ H ₁₄ O	110-43-0	114.185	8.8629E-08	0.78376	100.18		238.15	4.550E-06	1000	1.809E-05
166	1-Heptene	C ₇ H ₁₄	592-76-7	98.186	7.7509E-08	0.81089	69.927		154.12	3.169E-06	1000	1.962E-05
167	Heptyl mercaptan	C ₇ H ₁₆ S	1639-09-4	132.267	4.697E-08	0.8932	57.6		229.92	4.832E-06	1000	2.124E-05
168	1-Heptyne	C ₇ H ₁₂	628-71-7	96.170	5.9501E-07	0.52758	274.02		192.22	3.932E-06	1000	1.787E-05
169	Hexadecane	C ₁₆ H ₃₄	544-76-3	226.441	1.2463E-07	0.7322	395	6,000	291.31	3.274E-06	1000	1.399E-05
170	Hexanal	C ₆ H ₁₂ O	66-25-1	100.159	9.6724E-08	0.78044	97.798		217.15	4.444E-06	1000	1.933E-05
171	Hexane	C ₆ H ₁₄	110-54-3	86.175	1.7514E-07	0.70737	157.14		177.83	3.631E-06	1000	2.005E-05
172	Hexanoic acid	C ₆ H ₁₂ O ₂	142-62-1	116.158	8.4032E-08	0.80073	96.779		269.25	5.457E-06	1000	1.934E-05
173	1-Hexanol	C ₆ H ₁₄ O	111-27-3	102.175	1.5773E-07	0.7189	163.3		228.55	4.567E-06	1000	1.945E-05
174	2-Hexanol	C ₆ H ₁₄ O	626-93-7	102.175	1.0652E-07	0.77022	105.85		223.00	4.650E-06	1000	1.970E-05
175	2-Hexanone	C ₆ H ₁₂ O	591-78-6	100.159	9.782E-08	0.7772	99.53		217.35	4.397E-06	1000	1.909E-05
176	3-Hexanone	C ₆ H ₁₂ O	589-38-8	100.159	9.8882E-08	0.7755	99.825		217.50	4.403E-06	1000	1.907E-05
177	1-Hexene	C ₆ H ₁₂	592-41-6	84.159	8.006E-08	0.81293	65.274		133.39	2.871E-06	1000	2.064E-05
178	3-Hexyne	C ₆ H ₁₀	928-49-4	82.144	5.2127E-07	0.5444	237.01		170.05	3.567E-06	1000	1.811E-05
179	Hexyl mercaptan	C ₆ H ₁₄ S	111-31-9	118.240	4.3636E-08	0.90747	42.32		192.62	4.235E-06	1000	2.209E-05
180	1-Hexyne	C ₆ H ₁₀	693-02-7	82.144	2.9986E-07	0.62647	178.17		141.25	2.947E-06	1000	3.758E-13
181	2-Hexyne	C ₆ H ₁₀	764-35-2	82.144	5.5562E-07	0.5337	244.38		183.65	3.851E-06	1000	1.782E-05

TABLE 2-312 Vapor Viscosity of Inorganic and Organic Substances (Pa·s) (Continued)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1	C2	C3	C4	$T_{\min}$, K	Viscosity at $T_{\min}$	$T_{\max}$, K	Viscosity at $T_{\max}$
182	Hydrazine	H ₄ N ₂	302-01-2	32.045	2.3489E-07	0.7151	205.05		274.69	7.460E-06	1673.15	4.225E-05
183	Hydrogen	H ₂	1333-74-0	2.016	1.797E-07	0.685	-0.59	140	13.95	6.517E-07	3000	4.330E-05
184	Hydrogen bromide	HBr	10035-10-6	80.912	9.17E-08	0.9273			206.45	1.285E-05	800	4.512E-05
185	Hydrogen chloride	HCl	7647-01-0	36.461	4.924E-07	0.6702	157.7		200.00	9.594E-06	1000	4.358E-05
186	Hydrogen cyanide	CHN	74-90-8	27.025	1.278E-08	1.0631	340		300.00	2.576E-06	425	4.421E-06
187	Hydrogen fluoride	HF	7664-39-3	20.006	4.5101E-14	3.0005	-521.83	76,111	285.50	9.931E-06	472.68	2.019E-05
188	Hydrogen sulfide	H ₂ S	7783-06-4	34.081	3.9314E-08	1.0134			250.00	1.058E-05	480	2.050E-05
189	Isobutyric acid	C ₄ H ₈ O ₂	79-31-2	88.105	1.1202E-07	0.7822	100.3		227.15	5.415E-06	1000	2.261E-05
190	Isopropyl amine	C ₃ H ₉ N	75-31-0	59.110	5.2542E-08	0.88063			177.95	5.037E-06	1000	2.304E-05
191	Malonic acid	C ₃ H ₄ O ₄	141-82-2	104.061	8.4213E-08	0.82573	102.08		407.95	9.639E-06	1000	2.293E-05
192	Methacrylic acid	C ₄ H ₆ O ₂	79-41-4	86.089	9.113E-08	0.8222	93.57		288.15	7.242E-06	1000	2.440E-05
193	Methane	CH ₄	74-82-8	16.042	5.2546E-07	0.59006	105.67		90.69	3.468E-06	1000	2.800E-05
194	Methanol	CH ₃ O	67-56-1	32.042	3.0663E-07	0.69655	205		240.00	7.523E-06	1000	3.128E-05
195	N-Methyl acetamide	C ₃ H ₇ NO	79-16-3	73.094	8.0599E-08	0.8392	77.332		301.15	7.714E-06	1000	2.464E-05
196	Methyl acetate	C ₃ H ₆ O ₂	79-20-9	74.079	1.3226E-06	0.4885	504.3		250.00	6.505E-06	800	2.125E-05
197	Methyl acetylene	C ₃ H ₄	74-99-7	40.064	1.163E-06	0.4787	316		170.45	4.769E-06	800	2.045E-05
198	Methyl acrylate	C ₄ H ₆ O ₂	96-33-3	86.089	1.648E-06	0.4444	510.66		196.32	4.781E-06	1000	2.350E-05
199	Methyl amine	CH ₃ N	74-89-5	31.057	5.6409E-07	0.5863	231.9		179.69	5.167E-06	1000	2.628E-05
200	Methyl benzoate	C ₈ H ₈ O ₂	93-58-3	136.148	7.4106E-08	0.82436	83.086		260.75	5.515E-06	1000	2.034E-05
201	3-Methyl-1,2-butadiene	C ₅ H ₈	598-25-4	68.117	4.0824E-07	0.5923	208.22		159.53	3.572E-06	1000	2.022E-05
202	2-Methylbutane	C ₅ H ₁₂	78-78-4	72.149	2.4344E-08	0.97376	-91.597	18,720	150.00	2.621E-06	1000	2.191E-05
203	2-Methylbutanoic acid	C ₅ H ₁₀ O ₂	116-53-0	102.132	1.869E-07	0.7096	192		450.15	1.000E-05	1000	2.109E-05
204	3-Methyl-1-butanol	C ₅ H ₁₂ O	123-51-3	88.148	8.8783E-08	0.80279	77.075		155.95	3.423E-06	1000	2.111E-05
205	2-Methyl-1-butene	C ₅ H ₁₀	563-46-2	70.133	5.0602E-07	0.55258	199.82		135.58	3.083E-06	1000	1.918E-05
206	2-Methyl-2-butene	C ₅ H ₁₀	513-35-9	70.133	8.5423E-07	0.47389	239.34		139.39	3.263E-06	1000	1.820E-05
207	2-Methyl-1-butene-3-yne	C ₅ H ₆	78-80-8	66.101	5.6844E-07	0.553	227.18		160.15	3.893E-06	1000	2.112E-05
208	Methylbutyl ether	C ₅ H ₁₂ O	628-28-4	88.148	3.9342E-08	0.91086			157.48	3.947E-06	1000	2.125E-05
209	Methylbutyl sulfide	C ₅ H ₁₂ S	628-29-5	104.214	4.995E-08	0.89479	44.662		175.30	4.052E-06	1000	2.312E-05
210	3-Methyl-1-butyne	C ₅ H ₈	598-23-2	68.117	4.0748E-08	0.92709			183.45	5.112E-06	1000	2.463E-05
211	Methyl butyrate	C ₅ H ₁₀ O ₂	623-42-7	102.132	3.733E-07	0.6177	256.5		187.35	3.993E-06	1000	2.118E-05
212	Methylchlorosilane	CH ₃ ClSi	993-00-0	80.589	4.8806E-08	0.92549			139.05	4.698E-06	1000	2.917E-05
213	Methylcyclohexane	C ₇ H ₁₄	108-87-2	98.186	6.5281E-07	0.5294	310.59		146.58	2.934E-06	1000	1.930E-05
214	1-Methylcyclohexanol	C ₇ H ₁₄ O	590-67-0	114.185	8.5736E-08	0.80277	100.77		299.15	6.232E-06	1000	1.994E-05
215	cis-2-Methylcyclohexanol	C ₇ H ₁₄ O	7443-70-1	114.185	2.40E-07	0.68	210		280.15	6.331E-06	1000	2.175E-05
216	trans-2-Methylcyclohexanol	C ₇ H ₁₄ O	7443-52-9	114.185	2.00E-07	0.704	187		269.15	6.062E-06	1000	2.181E-05
217	Methylcyclopentane	C ₆ H ₁₂	96-37-7	84.159	9.0798E-07	0.495	355.89		130.73	2.722E-06	1000	2.046E-05
218	1-Methylcyclopentene	C ₆ H ₁₀	693-89-0	82.144	3.7026E-08	0.92849			146.62	3.800E-06	1000	2.259E-05
219	3-Methylcyclopentene	C ₆ H ₁₀	1120-62-3	82.144	3.9771E-08	0.92242			115.00	3.165E-06	1000	2.327E-05
220	Methyldichlorosilane	CH ₂ Cl ₂ Si	75-54-7	115.034	1.977E-07	0.7453	131.22		182.55	5.574E-06	1000	3.009E-05
221	Methylethyl ether	C ₃ H ₈ O	540-67-0	60.095	2.6098E-07	0.68276	133.4		160.00	4.552E-06	1000	2.573E-05
222	Methylethyl ketone	C ₄ H ₈ O	78-93-3	72.106	2.6552E-08	0.98316			186.48	4.534E-06	1000	2.364E-05
223	Methylethyl sulfide	C ₃ H ₈ S	624-89-5	76.161	8.6219E-08	0.83591	72.564		167.23	4.341E-06	1000	2.588E-05
224	Methyl formate	C ₂ H ₄ O ₂	107-31-3	60.052	6.9755E-06	0.3154	1034.5		174.15	5.117E-06	1000	3.029E-05
225	Methylisobutyl ether	C ₅ H ₁₂ O	625-44-5	88.148	1.5035E-07	0.7338	108.5		150.00	3.448E-06	1000	2.157E-05
226	Methylisobutyl ketone	C ₆ H ₁₂ O	108-10-1	100.159	9.4257E-08	0.7845	90.183		189.15	3.901E-06	1000	1.951E-05
227	Methyl Isocyanate	C ₂ H ₃ NO	624-83-9	57.051	3.1573E-07	0.66404	173.59		256.15	7.481E-06	1000	2.642E-05
228	Methylisopropyl ether	C ₄ H ₁₀ O	598-53-8	74.122	1.925E-07	0.7091	109		127.93	3.242E-06	1000	2.327E-05
229	Methylisopropyl ketone	C ₅ H ₁₀ O	563-80-4	86.132	1.0826E-07	0.77382	93.349		180.15	3.968E-06	1000	2.076E-05
230	Methylisopropyl sulfide	C ₄ H ₁₀ S	1551-21-9	90.187	8.6077E-08	0.81669	71.294		171.64	4.065E-06	1000	2.265E-05
231	Methyl mercaptan	CH ₃ S	74-93-1	48.107	1.637E-07	0.76706	107.97		150.18	4.450E-06	1000	2.956E-05
232	Methyl methacrylate	C ₅ H ₈ O ₂	80-62-6	100.116	4.889E-07	0.6096	342.23		224.95	5.265E-06	1000	2.456E-05
233	2-Methyloctanoic acid	C ₉ H ₁₈ O ₂	3004-93-1	158.238	7.2131E-08	0.80319	99.437		240.00	4.162E-06	1000	1.685E-05
234	2-Methylpentane	C ₆ H ₁₄	107-83-5	86.175	1.1164E-06	0.4537	374.74		119.55	2.366E-06	1000	1.865E-05
235	Methyl pentyl ether	C ₆ H ₁₄ O	628-80-8	102.175	1.0546E-07	0.77106	93.745		176.00	3.707E-06	1000	1.983E-05
236	2-Methylpropane	C ₄ H ₁₀	75-28-5	58.122	1.0871E-07	0.78135	70.639		150.00	3.707E-06	1000	2.242E-05
237	2-Methyl-2-propanol	C ₄ H ₁₀ O	75-65-0	74.122	9.605E-07	0.4856	381		298.97	6.727E-06	600	1.312E-05
238	2-Methyl propene	C ₄ H ₈	115-11-7	56.106	9.0981E-07	0.49288	260.08		132.81	3.423E-06	1000	2.174E-05
239	Methyl propionate	C ₄ H ₈ O ₂	554-12-1	88.105	3.5642E-07	0.6327	232.2		185.65	4.316E-06	1000	2.288E-05
240	Methylpropyl ether	C ₄ H ₁₀ O	557-17-5	74.122	4.4941E-08	0.90199			133.97	3.725E-06	1000	2.284E-05

241	Methylpropyl sulfide	C ₄ H ₁₀ S	3877-15-4	90.187	5.8223E-08	0.88057	48.298	160.17	3.908E-06	1000	2.434E-05
242	Methylsilane	CH ₃ Si	992-94-9	46.144	3.8926E-07	0.63159	169.45	116.34	3.196E-06	1000	5.762E-13
243	alpha-Methyl styrene	C ₉ H ₁₀	98-83-9	118.176	7.1455E-07	0.49832	303.31	249.95	5.057E-06	1000	1.714E-05
244	Methyl <i>tert</i> -butyl ether	C ₅ H ₁₂ O	1634-04-4	88.148	1.571E-07	0.733	111.578	164.55	3.944E-06	1000	2.235E-05
245	Methyl vinyl ether	C ₃ H ₆ O	107-25-5	58.079	7.646E-07	0.5476	284	278.65	8.264E-06	1000	2.616E-05
246	Naphthalene	C ₁₀ H ₈	91-20-3	128.171	6.4318E-07	0.5389	400.16	353.43	7.125E-06	1000	1.900E-05
247	Neon	Ne	7440-01-9	20.180	7.19E-07	0.6659	5.3	30.00	5.884E-06	3273.1	1.573E-04
248	Nitroethane	C ₂ H ₅ NO ₂	79-24-3	75.067	2.4391E-07	0.702	280	183.63	3.752E-06	1000	2.432E-05
249	Nitrogen	N ₂	7727-37-9	28.013	6.5592E-07	0.6081	54.714	63.15	4.372E-06	1970	6.432E-05
250	Nitrogen trifluoride	F ₃ N	7783-54-2	71.002	8.2005E-07	0.61423	114.58	66.46	3.964E-06	1000	1.975E-12
251	Nitromethane	CH ₃ NO ₂	75-52-5	61.040	4.070E-07	0.6485	367.5	244.60	5.756E-06	1000	2.625E-05
252	Nitrous oxide	N ₂ O	10024-97-2	44.013	2.115E-06	0.4642	305.7	182.30	8.854E-06	1000	4.000E-05
253	Nitric oxide	NO	10102-43-9	30.006	1.467E-06	0.5123	125.4	110.00	7.618E-06	1500	5.737E-05
254	Nonadecane	C ₁₉ H ₄₀	629-92-5	268.521	3.0465E-07	0.62218	705.34	305.04	3.231E-06	1000	1.314E-05
255	Nonanal	C ₉ H ₁₈ O	124-19-6	142.239	7.1902E-08	0.8013	92.051	255.15	4.483E-06	1000	1.669E-05
256	Nonane	C ₉ H ₂₀	111-84-2	128.255	1.0344E-07	0.77301	220.47	219.66	3.335E-06	1000	1.767E-05
257	Nonanoic acid	C ₉ H ₁₈ O ₂	112-05-0	158.238	7.0165E-08	0.8062	100.36	285.65	4.957E-06	1000	1.672E-05
258	1-Nonanol	C ₉ H ₂₀ O	143-08-8	144.255	1.20E-07	0.74	180	268.15	4.499E-06	1000	1.688E-05
259	2-Nonanol	C ₉ H ₂₀ O	628-99-9	144.255	7.0111E-08	0.80701	89.582	238.15	4.219E-06	1000	1.697E-05
260	1-Nonene	C ₉ H ₁₈	124-11-8	126.239	6.6329E-08	0.82027	76.204	191.91	3.542E-06	1000	1.781E-05
261	Nonyl mercaptan	C ₉ H ₂₀ S	1455-21-6	160.320	3.8673E-08	0.91142	50.646	253.05	4.995E-06	1000	1.996E-05
262	1-Nonyne	C ₉ H ₁₆	3452-09-3	124.223	6.1447E-07	0.50705	287.19	223.15	4.170E-06	1000	1.585E-05
263	Octadecane	C ₁₈ H ₃₈	593-45-3	254.494	3.2095E-07	0.61839	709.09	301.31	3.266E-06	1000	1.345E-05
264	Octanal	C ₈ H ₁₆ O	124-13-0	128.212	1.0321E-07	0.7589	121.26	246.00	4.510E-06	1000	1.741E-05
265	Octane	C ₈ H ₁₈	111-65-9	114.229	3.1191E-08	0.92925	55.092	216.38	3.677E-06	1000	1.813E-05
266	Octanoic acid	C ₈ H ₁₆ O ₂	124-07-2	144.211	7.9611E-08	0.7948	106.6	289.65	5.267E-06	1000	1.743E-05
267	1-Octanol	C ₈ H ₁₈ O	111-87-5	130.228	1.752E-07	0.6941	206.8	257.65	4.583E-06	1000	1.755E-05
268	2-Octanol	C ₈ H ₁₈ O	123-96-6	130.228	8.1701E-08	0.79241	97.709	241.55	4.498E-06	1000	1.774E-05
269	2-Octanone	C ₈ H ₁₆ O	111-13-7	128.212	8.0901E-08	0.79062	99.338	252.85	4.611E-06	1000	1.733E-05
270	3-Octanone	C ₈ H ₁₆ O	106-68-3	128.212	6.1515E-11	1.8808		255.55	2.075E-06	1000	2.700E-05
271	1-Octene	C ₈ H ₁₆	111-66-0	112.213	5.0324E-05	0.077611	3604.6	171.45	3.406E-06	1000	1.868E-05
272	Octyl mercaptan	C ₈ H ₁₈ S	111-88-6	146.294	3.3253E-08	0.9351	32.426	223.95	4.579E-06	1000	2.057E-05
273	1-Octyne	C ₈ H ₁₄	629-05-0	110.197	5.7084E-07	0.52446	271.76	193.55	3.758E-06	1000	1.681E-05
274	Oxalic acid	C ₂ H ₂ O ₄	144-62-7	90.035	1.1453E-07	0.7968	126.34	462.65	1.196E-05	1000	2.498E-05
275	Oxygen	O ₂	7782-44-7	31.999	1.101E-06	0.5634	96.3	54.35	3.773E-06	1500	6.371E-05
276	Ozone	O ₃	10028-15-6	47.998	1.196E-07	0.84797		80.15	4.922E-06	1000	4.184E-05
277	Pentadecane	C ₁₅ H ₃₂	629-62-9	212.415	4.0828E-08	0.8766	212.68	283.07	3.288E-06	1000	1.436E-05
278	Pentanal	C ₅ H ₁₀ O	110-62-3	86.132	2.27E-07	0.6767	191.74	182.00	3.740E-06	1000	2.042E-05
279	Pentane	C ₅ H ₁₂	109-66-0	72.149	6.3412E-08	0.84758	41.718	143.42	3.305E-06	1000	2.124E-05
280	Pentanoic acid	C ₅ H ₁₀ O ₂	109-52-4	102.132	9.4314E-08	0.7932	98.279	239.15	5.150E-06	1000	2.058E-05
281	1-Pentanol	C ₅ H ₁₂ O	71-41-0	88.148	1.8903E-07	0.7031	175.9	410.95	9.111E-06	1000	2.068E-05
282	2-Pentanol	C ₅ H ₁₂ O	6032-29-7	88.148	1.1749E-07	0.7649	103.78	200.00	4.452E-06	1000	2.098E-05
283	2-Pentanone	C ₅ H ₁₀ O	107-87-9	86.132	2.463E-07	0.6653	208.7	196.29	4.003E-06	1000	2.019E-05
284	3-Pentanone	C ₅ H ₁₀ O	96-22-0	86.132	1.164E-07	0.7615	107.94	234.18	5.079E-06	1000	2.023E-05
285	1-Pentene	C ₅ H ₁₀	109-67-1	70.133	1.6378E-06	0.44337	636.11	108.02	2.813E-06	1000	2.176E-05
286	2-Pentyl mercaptan	C ₅ H ₁₂ S	2084-19-7	104.214	8.8646E-08	0.81492	85.198	160.75	3.638E-06	1000	2.275E-05
287	Pentyl mercaptan	C ₅ H ₁₂ S	110-66-7	104.214	2.7467E-08	0.97555		197.45	4.766E-06	1000	2.320E-05
288	1-Pentyne	C ₅ H ₈	627-19-0	68.117	4.1022E-08	0.90585		167.45	4.242E-06	1000	2.141E-05
289	2-Pentyne	C ₅ H ₈	627-21-4	68.117	5.765E-07	0.53498	235.2	163.83	3.621E-06	1000	1.879E-05
290	Phenanthrene	C ₁₄ H ₁₀	85-01-8	178.229	4.3478E-07	0.5272	238.27	372.38	6.010E-06	1000	1.340E-05
291	Phenol	C ₆ H ₆ O	108-95-2	94.111	1.0094E-07	0.799	103.1	314.06	7.514E-06	1000	2.283E-05
292	Phenyl isocyanate	C ₇ H ₅ NO	103-71-9	119.121	8.536E-08	0.80872	88.273	243.15	5.324E-06	1000	2.093E-05
293	Phthalic anhydride	C ₈ H ₄ O ₃	85-44-9	148.116	4.3511E-08	0.908	102.73	404.15	8.072E-06	1000	2.090E-05
294	Propadiene	C ₃ H ₄	463-49-0	40.064	6.0758E-07	0.53845	173.45	136.87	3.788E-06	1000	2.135E-05
295	Propane	C ₃ H ₈	74-98-6	44.096	4.9054E-08	0.90125		85.47	2.702E-06	1000	2.480E-05
296	1-Propanol	C ₃ H ₈ O	71-23-8	60.095	7.942E-07	0.5491	415.8	200.00	4.732E-06	1000	2.490E-05
297	2-Propanol	C ₃ H ₈ O	67-63-0	60.095	1.2003E-06	0.494	479.78	187.35	4.471E-06	1000	2.461E-05
298	Propenylcyclohexene	C ₉ H ₁₄	13511-13-2	122.207	5.4749E-07	0.53893	283.52	199.00	3.914E-06	1000	1.765E-05
299	Propionaldehyde	C ₃ H ₆ O	123-38-6	58.079	1.7526E-07	0.72691	119.93	170.00	4.297E-06	1000	2.373E-05
300	Propionic acid	C ₃ H ₆ O ₂	79-09-4	74.079	1.61E-07	0.7457	159.3	252.45	6.105E-06	1000	2.397E-05
301	Propionitrile	C ₃ H ₅ N	107-12-0	55.079	1.0111E-07	0.7821	89.5	180.26	3.927E-06	1000	2.060E-05
302	Propyl acetate	C ₅ H ₁₀ O ₂	109-60-4	102.132	2.1372E-07	0.6894	178.57	178.15	3.802E-06	1000	2.122E-05
303	Propyl amine	C ₃ H ₉ N	107-10-8	59.110	1.62E-07	0.7285	117	188.36	4.540E-06	1000	2.223E-05

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TABLE 2-312 Vapor Viscosity of Inorganic and Organic Substances (Pa·s) (Concluded)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1	C2	C3	C4	$T_{\min}$, K	Viscosity at $T_{\min}$	$T_{\max}$, K	Viscosity at $T_{\max}$
304	Propylbenzene	C ₉ H ₁₂	103-65-1	120.192	3.0387E-07	0.61945	210.35		173.55	3.351E-06	1000	1.812E-05
305	Propylene	C ₃ H ₆	115-07-1	42.080	7.3919E-07	0.5423	263.73		87.89	2.093E-06	1000	2.477E-05
306	Propyl formate	C ₄ H ₈ O ₂	110-74-7	88.105	6.0741E-07	0.5863	367.29		180.25	4.203E-06	1000	2.550E-05
307	2-Propyl mercaptan	C ₃ H ₈ S	75-33-2	76.161	3.5532E-08	0.95654			142.61	4.085E-06	1000	2.632E-05
308	Propyl mercaptan	C ₃ H ₈ S	107-03-9	76.161	7.9457E-08	0.84656	65.878		159.95	4.132E-06	1000	2.583E-05
309	1,2-Propylene glycol	C ₃ H ₈ O ₂	57-55-6	76.094	4.543E-08	0.9173	61		213.15	4.832E-06	1000	2.418E-05
310	Quinone	C ₆ H ₄ O ₂	106-51-4	108.095	1.1085E-07	0.8008	152.51		388.85	9.439E-06	1000	2.429E-05
311	Silicon tetrafluoride	F ₄ Si	7783-61-1	104.079	2.1671E-07	0.76757	16.28		250.00	1.410E-05	500	2.475E-05
312	Styrene	C ₈ H ₈	100-42-5	104.149	6.3863E-07	0.5254	295.1		242.54	5.158E-06	1000	1.858E-05
313	Succinic acid	C ₄ H ₆ O ₄	110-15-6	118.088	2.273E-07	0.6845	229.8		460.65	1.009E-05	1000	2.091E-05
314	Sulfur dioxide	O ₂ S	7446-09-5	64.064	6.863E-07	0.6112	217		197.67	8.280E-06	1000	3.844E-05
315	Sulfur hexafluoride	F ₆ S	2551-62-4	146.055	5.3986E-07	0.6349	34.5	19,000	205.15	9.790E-06	5000	1.195E-04
316	Sulfur trioxide	O ₃ S	7446-11-9	80.063	3.9067E-06	0.3845	470.1		297.93	1.355E-05	694.19	2.883E-05
317	Terephthalic acid	C ₈ H ₆ O ₄	100-21-0	166.131	2.2452E-08	0.97631			700.15	1.346E-05	1000	1.906E-05
318	<i>o</i> -Terphenyl	C ₁₈ H ₁₄	84-15-1	230.304	7.0859E-07	0.51971	652.24		329.35	4.837E-06	1000	1.554E-05
319	Tetradecane	C ₁₄ H ₃₀	629-59-4	198.388	5.1567E-09	1.1561			279.01	3.465E-06	1000	1.516E-05
320	Tetrahydrofuran	C ₄ H ₈ O	109-99-9	72.106	3.778E-07	0.6533	271.01		164.65	4.006E-06	1000	2.710E-05
321	1,2,3,4-Tetrahydronaphthalene	C ₁₀ H ₁₂	119-64-2	132.202	5.0784E-07	0.5614	328.55		237.38	4.592E-06	1000	1.847E-05
322	Tetrahydrothiophene	C ₄ H ₈ S	110-01-0	88.171	8.5988E-08	0.82841	68.172		176.99	4.520E-06	1000	2.461E-05
323	2,2,3,3-Tetramethylbutane	C ₈ H ₁₈	594-82-1	114.229	8.1458E-07	0.50257	380.29		373.96	7.930E-06	1000	1.900E-05
324	Thiophene	C ₄ H ₄ S	110-02-1	84.140	1.03E-06	0.5497	569.4		234.94	6.049E-06	1000	2.926E-05
325	Toluene	C ₇ H ₈	108-88-3	92.138	8.7268E-07	0.49397	323.79		178.18	4.008E-06	1000	2.000E-05
326	1,1,2-Trichloroethane	C ₂ H ₃ Cl ₃	79-00-5	133.404	2.7081E-07	0.6955	187.93		236.50	6.756E-06	1000	2.782E-05
327	Tridecane	C ₁₃ H ₂₈	629-50-5	184.361	3.5585E-08	0.8987	165.3		267.76	3.344E-06	1000	1.517E-05
328	Triethyl amine	C ₆ H ₁₅ N	121-44-8	101.190	2.411E-07	0.6845	223		158.45	3.210E-06	1000	2.230E-05
329	Trimethyl amine	C ₃ H ₉ N	75-50-3	59.110	1.2434E-06	0.4832	447.7		156.08	3.689E-06	1000	2.418E-05
330	1,2,3-Trimethylbenzene	C ₉ H ₁₂	526-73-8	120.192	7.8498E-07	0.49855	362.79		247.79	4.975E-06	1000	1.803E-05
331	1,2,4-Trimethylbenzene	C ₉ H ₁₂	95-63-6	120.192	6.8812E-07	0.51063	330.88		229.33	4.520E-06	1000	1.760E-05
332	2,2,4-Trimethylpentane	C ₈ H ₁₈	540-84-1	114.229	1.107E-07	0.746	72.4		165.78	3.488E-06	1000	1.786E-05
333	2,3,3-Trimethylpentane	C ₈ H ₁₈	560-21-4	114.229	8.2418E-07	0.4931	371.44		387.91	7.958E-06	1000	1.812E-05
334	1,3,5-Trinitrobenzene	C ₆ H ₃ N ₃ O ₆	99-35-4	213.105	3.4066E-08	0.95252	43.528		398.40	9.208E-06	1000	2.352E-05
335	2,4,6-Trinitrotoluene	C ₇ H ₅ N ₃ O ₆	118-96-7	227.131	2.8471E-08	0.96571	30.83		354.00	7.581E-06	1000	2.179E-05
336	Undecane	C ₁₁ H ₂₄	1120-21-4	156.308	3.594E-08	0.9052	125		247.57	3.506E-06	1000	1.660E-05
337	1-Undecanol	C ₁₁ H ₂₄ O	112-42-5	172.308	5.9537E-08	0.81842	90.245		288.45	4.677E-06	1000	1.558E-05
338	Vinyl acetate	C ₄ H ₆ O ₂	108-05-4	86.089	1.388E-07	0.7599	98		180.35	4.660E-06	1000	2.407E-05
339	Vinyl acetylene	C ₄ H ₄	689-97-4	52.075	6.7484E-07	0.5304	230.17		173.15	4.459E-06	1000	2.140E-05
340	Vinyl chloride	C ₂ H ₃ Cl	75-01-4	62.498	2.379E-07	0.71517	102.84		119.36	3.907E-06	1000	1.263E-12
341	Vinyl trichlorosilane	C ₂ H ₃ Cl ₃ Si	75-94-5	161.490	3.6429E-08	0.95924			178.35	5.260E-06	1000	2.749E-05
342	Water	H ₂ O	7732-18-5	18.015	1.7096E-08	1.1146			273.16	8.882E-06	1073.15	4.082E-05
343	<i>m</i> -Xylene	C ₈ H ₁₀	108-38-3	106.165	6.8293E-07	0.52199	324.17		225.30	4.735E-06	1000	1.899E-05
344	<i>o</i> -Xylene	C ₈ H ₁₀	95-47-6	106.165	8.3436E-07	0.49713	365.86		247.98	5.225E-06	1000	1.894E-05
345	<i>p</i> -Xylene	C ₈ H ₁₀	106-42-3	106.165	9.3485E-07	0.47683	371.96		286.41	6.037E-06	1000	1.836E-05

The vapor viscosity is calculated by

$$\mu = \frac{C1T^{C2}}{1 + C3T + C4T^2}$$

where μ is the viscosity in Pa·s and T is the temperature in K. Viscosities are at either 1 atm or the vapor pressure, whichever is lower. All substances are listed by chemical family in Table 2-6 and by formula in Table 2-7.

Values in this table were taken from the Design Institute for Physical Properties (DIPPR) of the American Institute of Chemical Engineers (AIChE), copyright 2007 AIChE and reproduced with permission of AIChE and of the DIPPR Evaluated Process Design Data Project Steering Committee. Their source should be cited as R. L. Rowley, W. V. Wilding, J. L. Oscarson, Y. Yang, N. A. Zundel, T. E. Daubert, R. P. Danner, DIPPR® Data Compilation of Pure Chemical Properties, Design Institute for Physical Properties, AIChE, New York (2007).

The number of digits provided for values at $T_{\min}$ and $T_{\max}$ was chosen for uniformity of appearance and formatting; these do not represent the uncertainties of the physical quantities, but are the result of calculations from the standard thermophysical property formulations within a fixed format.

TABLE 2-313 Viscosity of Inorganic and Organic Liquids (Pa-s)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1	C2	C3	C4	C5	T_{min} , K	Viscosity at T_{min}	T_{max} , K	Viscosity at T_{max}
1	Acetaldehyde	C ₂ H ₄ O	75-07-0	44.053	-5.895	668.21	-0.84323			150.15	3.446E-03	294.00	2.216E-04
2	Acetamide	C ₂ H ₅ NO	60-35-5	59.067	1.5525	1376.4	-2.0126			353.33	1.728E-03	494.30	2.895E-04
3	Acetic acid	C ₂ H ₄ O ₂	64-19-7	60.052	-9.03	1212.3	-0.322			289.81	1.265E-03	391.05	3.890E-04
4	Acetic anhydride	C ₄ H ₆ O ₃	108-24-7	102.089	-14.164	1350.3	0.4492			210.00	4.834E-03	412.70	2.783E-04
5	Acetone	C ₃ H ₆ O	67-64-1	58.079	-14.918	1023.4	0.5961			190.00	1.655E-03	329.44	2.351E-04
6	Acetonitrile	C ₂ H ₃ N	75-05-8	41.052	-10.906	872.02				229.32	8.220E-04	436.40	1.350E-04
7	Acetylene	C ₂ H ₂	74-86-2	26.037	6.224	-151.8	-2.6554			193.15	1.958E-04	273.15	9.819E-05
8	Acrolein	C ₃ H ₄ O	107-02-8	56.063	-12.032	867.34	0.19534			185.45	1.773E-03	353.22	2.181E-04
9	Acrylic acid	C ₃ H ₄ O ₂	79-10-7	72.063	-28.12	2280.2	2.3956			286.15	1.359E-03	460.00	2.086E-04
10	Acrylonitrile	C ₃ H _{3.5} N	107-13-1	53.063	2.019	239.7	-1.8975			220.00	8.040E-04	350.50	2.215E-04
11	Air	Mixture	132259-10-0	28.960	-20.077	285.15	1.784	-6.2382E-22	10	59.15	3.430E-04	130.00	4.276E-05
12	Ammonia	H ₃ N	7664-41-7	17.031	-6.743	598.3	-0.7341	-3.69E-27	10	195.41	5.240E-04	393.15	4.858E-05
13	Anisole	C ₇ H ₈ O	100-66-3	108.138	-15.407	1518.7	0.60172			235.65	3.429E-03	426.73	2.736E-04
14	Argon	Ar	7440-37-1	39.948	-8.8685	204.29	-0.38305	-1.2937E-22	10	83.78	2.950E-04	150.00	3.823E-05
15	Benzamide	C ₇ H ₇ NO	55-21-0	121.137	-12.632	2668.2				403.00	2.451E-03	563.15	3.730E-04
16	Benzene	C ₆ H ₆	71-43-2	78.112	7.5117	294.68	-2.794			278.68	7.761E-04	545.00	7.106E-05
17	Benzenethiol	C ₆ H ₆ S	108-98-5	110.177	-8.4562	1024.4	-0.30635			258.27	2.047E-03	442.29	3.333E-04
18	Benzoic acid	C ₇ H ₆ O ₂	65-85-0	122.121	-12.947	2557.9	395.52			395.52	1.530E-03	600.80	1.680E-04
19	Benzonitrile	C ₇ H ₅ N	100-47-0	103.121	-20.236	1737.4	1.3531			270.00	1.977E-03	450.00	3.009E-04
20	Benzophenone	C ₁₃ H ₁₀ O	119-61-9	182.218	13.354	-232.91	-3.2685	1.7488E+20	-8.052	321.35	6.100E-03	664.00	2.660E-04
21	Benzyl alcohol	C ₇ H ₈ O	100-51-6	108.138	-14.152	2652				257.85	2.092E-02	478.60	1.821E-04
22	Benzyl ethyl ether	C ₉ H ₁₂ O	539-30-0	136.191	-11.46	1497	-0.043397			275.65	1.887E-03	458.15	2.121E-04
23	Benzyl mercaptan	C ₇ H ₈ S	100-53-8	124.203	-11.459	1334.4	0.00049694			243.95	2.513E-03	472.03	1.788E-04
24	Biphenyl	C ₁₂ H ₁₀	92-52-4	154.208	-9.9265	1576.3	-0.21119			342.20	1.427E-03	723.15	1.076E-04
25	Bromine	Br ₂	7726-95-6	159.808	16.775	-314	-3.9763			265.85	1.353E-03	350.00	6.021E-04
26	Bromobenzene	C ₆ H ₅ Br	108-86-1	157.008	-20.611	1656.5	1.4415			242.43	2.842E-03	429.24	3.310E-04
27	Bromoethane	C ₂ H ₅ Br	74-96-4	108.965	-10.015	823.43	-0.11122			154.55	5.260E-03	311.50	3.321E-04
28	Bromomethane	CH ₃ Br	74-83-9	94.939	-8.103	570.8	-0.32958			179.47	1.316E-03	276.71	3.732E-04
29	1,2-Butadiene	C ₄ H ₆	590-19-2	54.090	-10.143	472.79	-0.028241			136.95	1.081E-03	284.00	1.773E-04
30	1,3-Butadiene	C ₄ H ₆	106-99-0	54.090	17.844	-310.2	-4.5058			250.00	2.547E-04	400.00	4.880E-05
31	Butane	C ₄ H ₁₀	106-97-8	58.122	-7.2471	534.82	-0.57469	-4.6625E-27	10	134.86	2.243E-03	420.00	3.566E-05
32	1,2-Butanediol	C ₄ H ₁₀ O ₂	584-03-2	90.121	-393.86	19042	59.978	-0.049479	1	220.00	2.020E+02	544.00	3.440E-04
33	1,3-Butanediol	C ₄ H ₁₀ O ₂	107-88-0	90.121	-390.03	18609	60.014	-0.055844	1	196.15	4.410E+04	540.80	2.890E-04
34	1-Butanol	C ₄ H ₁₀ O	71-36-3	74.122	0.87669	1602.9	-2.1475	3.3866E+22	-9.9231	190.00	3.237E-01	391.90	3.877E-04
35	2-Butanol	C ₄ H ₁₀ O	78-92-2	74.122	-16.323	3141.7				158.45	3.327E+01	372.90	3.715E-04
36	1-Butene	C ₄ H ₈	106-98-9	56.106	-10.773	591.61				87.80	1.770E-02	335.60	1.220E-04
37	cis-2-Butene	C ₄ H ₈	590-18-1	56.106	-10.346	522.3	-0.011847			134.26	1.483E-03	276.87	1.982E-04
38	trans-2-Butene	C ₄ H ₈	624-64-6	56.106	-10.335	521.39	-0.013184			167.62	6.810E-04	274.03	2.022E-04
39	Butyl acetate	C ₆ H ₁₂ O ₂	123-86-4	116.158	-17.488	1478.2	0.91828			250.00	1.496E-03	399.26	2.521E-04
40	Butylbenzene	C ₁₀ H ₁₄	104-51-8	134.218	-23.802	1887.2	1.8479			200.00	1.030E-02	456.46	2.359E-04
41	Butyl mercaptan	C ₄ H ₁₀ S	109-79-5	90.187	-10.807	966.74	-0.014851			157.46	8.717E-03	373.15	2.475E-04
42	sec-Butyl mercaptan	C ₄ H ₁₀ S	513-53-1	90.187	-10.903	932.82	0.023034			133.02	2.288E-02	358.13	2.851E-04
43	1-Butyne	C ₄ H ₆	107-00-6	54.090	-3.4644	334.5	-1.0811			147.43	1.369E-03	373.15	1.271E-04
44	Butyraldehyde	C ₄ H ₈ O	123-72-8	72.106	-10.057	903.73	-0.13186			176.75	3.602E-03	348.05	2.660E-04
45	Butyric acid	C ₄ H ₈ O ₂	107-92-6	88.105	-9.817	1388	-0.238			267.95	2.561E-03	436.42	3.087E-04
46	Butyronitrile	C ₄ H ₇ N	109-74-0	69.105	-10.136	1006.4	-0.1337			161.25	1.031E-02	390.75	2.344E-04
47	Carbon dioxide	CO ₂	124-38-9	44.010	18.775	-402.92	-4.6854	-6.9171E-26	10	216.58	2.488E-04	303.15	5.652E-05
48	Carbon disulfide	CS ₂	75-15-0	76.141	-10.306	703.01				161.58	2.590E-03	441.60	1.640E-04
49	Carbon monoxide	CO	630-08-0	28.010	-4.9735	97.67	-1.1088			68.15	2.689E-04	131.37	6.515E-05
50	Carbon tetrachloride	CCl ₄	56-23-5	153.823	-8.0738	1121.1	-0.4726			250.00	2.032E-03	455.00	2.030E-04
51	Carbon tetrafluoride	CF ₄	75-73-0	88.004	-9.9212	300.5				89.56	1.408E-03	145.10	3.897E-04
52	Chlorine	Cl ₂	7782-50-5	70.906	-9.5412	456.62				172.12	1.020E-03	333.72	2.820E-04
53	Chlorobenzene	C ₆ H ₅ Cl	108-90-7	112.557	0.15772	540.5	-1.6075			250.00	1.422E-03	540.00	1.291E-04
54	Chloroethane	C ₂ H ₅ Cl	75-00-3	64.514	-10.216	702	-0.072			150.00	2.748E-03	373.15	1.567E-04
55	Chloroform	CHCl ₃	67-66-3	119.378	-14.109	1049.2	0.5377			209.63	1.970E-03	353.20	3.410E-04
56	Chloromethane	CH ₃ Cl	74-87-3	50.488	-25.132	1381.9	2.0811	-4.4999E-27	10	175.43	1.501E-03	403.15	5.951E-05
57	1-Chloropropane	C ₃ H ₇ Cl	540-54-5	78.541	-13.994	949.4	0.50223	-6.163E-17	6	150.35	5.728E-03	423.15	1.154E-04
58	2-Chloropropane	C ₃ H ₇ Cl	75-29-6	78.541	-15.458	1086	0.654			250.00	5.515E-04	308.85	2.767E-04
59	m-Cresol	C ₇ H ₈ O	108-39-4	108.138	59.686	-3517.9	-9.838	9.0312E+12	-5	273.15	8.670E-02	564.68	1.630E-04

TABLE 2-313 Viscosity of Inorganic and Organic Liquids (Pa-s) (Continued)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1	C2	C3	C4	C5	$T_{\min}$ K	Viscosity at $T_{\min}$	$T_{\max}$ K	Viscosity at $T_{\max}$
60	<i>o</i> -Cresol	C ₇ H ₈ O	95-48-7	108.138	-0.033937	390.77	-1.4547	5.0187E+12	-5	293.15	9.600E-03	558.04	2.160E-04
61	<i>p</i> -Cresol	C ₇ H ₈ O	106-44-5	108.138	-1.6355	1052.9	-1.3891	6.6844E+17	-7	273.15	9.770E-02	563.72	1.940E-04
62	Cumene	C ₉ H ₁₂	98-82-8	120.192	-24.988	1807.9	2.0556			200.00	6.363E-03	400.00	2.881E-04
63	Cyanogen	C ₂ N ₂	460-19-5	52.035	-12.086	994.23				245.25	3.250E-04	320.12	1.260E-04
64	Cyclobutane	C ₄ H ₈	287-23-0	56.106	-3.4968	397.94	-1.1087			182.48	8.345E-04	367.94	1.278E-04
65	Cyclohexane	C ₆ H ₁₂	110-82-7	84.159	-33.763	2497.2	3.2236			279.69	1.260E-03	443.04	2.070E-04
66	Cyclohexanol	C ₆ H ₁₂ O	108-93-0	100.159	280.87	-31869	-38.837	3,994,500	-2.002	296.60	6.330E-02	520.08	1.650E-04
67	Cyclohexanone	C ₆ H ₁₀ O	108-94-1	98.143	-44.877	3227.7	4.887			242.00	8.960E-03	428.58	4.402E-04
68	Cyclohexene	C ₆ H ₁₀	110-83-8	82.144	-11.641	1154.3	0.066511			200.00	4.018E-03	373.15	2.877E-04
69	Cyclopentane	C ₅ H ₁₀	287-92-3	70.133	-3.2612	614.16	-1.156			225.00	1.122E-03	325.00	3.167E-04
70	Cyclopentene	C ₅ H ₈	142-29-0	68.117	-4.1508	599.77	-1.0308			138.13	7.531E-03	405.60	1.416E-04
71	Cyclopropane	C ₃ H ₆	75-19-4	42.080	-3.524	342.54	-1.1599			145.59	9.601E-04	318.40	1.080E-04
72	Cyclohexyl mercaptan	C ₆ H ₁₂ S	1569-69-3	116.224	-11.338	1304.1	0.000092396			189.64	1.155E-02	431.95	2.440E-04
73	Decanal	C ₁₀ H ₂₀ O	112-31-2	156.265	-10.115	1111.9	-0.015659			267.15	2.381E-03	488.15	3.583E-04
74	Decane	C ₁₀ H ₂₂	124-18-5	142.282	-9.6489	1181.1	-0.24367	9.0522E+34	-15	240.05	2.780E-03	494.16	1.550E-04
75	Decanoic acid	C ₁₀ H ₂₀ O ₂	334-48-5	172.265	-12.305	2324.1	-0.055494			304.55	6.798E-03	543.15	2.304E-04
76	1-Decanol	C ₁₀ H ₂₂ O	112-30-1	158.281	-69.985	5818.8	8.0715			285.00	1.937E-02	503.00	2.727E-04
77	1-Decene	C ₁₀ H ₂₀	872-05-9	140.266	-15.868	1434.8	0.68071			206.89	4.975E-03	443.75	2.064E-04
78	Decyl mercaptan	C ₁₀ H ₂₂ S	143-10-2	174.347	-11.464	1510.1	-0.012754			247.56	4.364E-03	512.35	1.848E-04
79	1-Decyne	C ₁₀ H ₁₈	764-93-2	138.250	-2.3633	791.93	-1.2272			229.15	3.786E-03	505.60	2.167E-04
80	Deuterium	D ₂	7782-39-0	4.032	0.000001348					20.35	1.348E-06	20.35	1.348E-06
81	1,1-Dibromoethane	C ₂ H ₄ Br ₂	557-91-5	187.861	-10.457	1101.1	-0.0031354			210.15	5.331E-03	381.15	5.071E-04
82	1,2-Dibromoethane	C ₂ H ₄ Br ₂	106-93-4	187.861	-17.582	1635.4	0.9932			282.85	2.042E-03	404.51	5.120E-04
83	Dibromomethane	CH ₂ Br ₂	74-95-3	173.835	-10.013	921.31				220.60	2.920E-03	488.80	2.950E-04
84	Dibutyl ether	C ₈ H ₁₈ O	142-96-1	130.228	10.027	206	-3.1607			175.30	5.931E-03	414.15	1.989E-04
85	<i>m</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	541-73-1	147.002	-1.9265	387.67	-1.1335	1.5037E+14	-6	248.39	2.540E-03	547.16	2.340E-04
86	<i>o</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	95-50-1	147.002	-30.6	2153.4	2.9371			256.15	2.727E-03	453.57	3.761E-04
87	<i>p</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	106-46-7	147.002	31.63	-1080	-6.114			326.14	8.543E-04	447.21	3.039E-04
88	1,1-Dichloroethane	C ₂ H ₄ Cl ₂	75-34-3	98.959	-8.991	870.2	-0.2805			176.19	4.076E-03	330.45	3.407E-04
89	1,2-Dichloroethane	C ₂ H ₄ Cl ₂	107-06-2	98.959	15.312	-41.12	-3.919			237.49	1.839E-03	400.00	2.557E-04
90	Dichloromethane	CH ₂ Cl ₂	75-09-2	84.933	-13.071	940.03	0.3733			208.38	1.407E-03	373.93	2.374E-04
91	1,1-Dichloropropane	C ₃ H ₆ Cl ₂	78-99-9	112.986	-10.872	1033.1	-0.00067435			200.00	3.312E-03	361.25	3.301E-04
92	1,2-Dichloropropane	C ₃ H ₆ Cl ₂	78-87-5	112.986	-11.269	1195.3	0.012736			172.71	1.381E-02	369.52	3.95E-04
93	Diethanol amine	C ₄ H ₁₁ NO ₂	111-42-2	105.136	-375.21	171.77	66.66	-3.6367	0.5	293.15	8.130E-01	589.28	1.090E-04
94	Diethyl amine	C ₄ H ₁₁ N	109-89-7	73.137	-17.57	1385.7	0.85647			223.35	1.191E-03	329.10	2.260E-04
95	Diethyl ether	C ₄ H ₁₀ O	60-29-7	74.122	10.197	-63.8	-3.226			200.00	7.359E-04	373.15	1.141E-04
96	Diethyl sulfide	C ₄ H ₁₀ S	352-93-2	90.187	-5.135	667.5	-0.8553			225.00	1.113E-03	365.25	2.354E-04
97	1,1-Difluoroethane	C ₂ H ₄ F ₂	75-37-6	66.050	10.501	-52.181	-3.3459			154.56	1.229E-03	343.15	1.026E-04
98	1,2-Difluoroethane	C ₂ H ₄ F ₂	624-72-6	66.050	-10.072	710.48	-0.14677			215.00	5.231E-04	283.65	2.257E-04
99	Difluoromethane	CH ₂ F ₂	75-10-5	52.023	-17.723	850.2	1.0601	-1.1719E-18	7	137.00	1.830E-03	343.15	6.050E-05
100	Di-isopropyl amine	C ₆ H ₁₅ N	108-18-9	101.190	-1.7366	599.8	-1.4237			250.00	7.479E-04	357.05	2.193E-04
101	Di-isopropyl ether	C ₆ H ₁₄ O	108-20-3	102.175	-11.5	993	0.022			187.65	2.259E-03	341.45	2.110E-04
102	Di-isopropyl ketone	C ₇ H ₁₄ O	565-80-0	114.185	-15.097	1426.9	0.51512			204.81	4.569E-03	397.55	2.194E-04
103	1,1-Dimethoxyethane	C ₄ H ₁₀ O ₂	534-15-6	90.121	-10.968	885.49				159.95	4.375E-03	337.45	2.378E-04
104	1,2-Dimethoxypropane	C ₅ H ₁₂ O ₂	7778-85-0	104.148	-10.631	1086.4				226.10	2.950E-03	366.15	4.695E-04
105	Dimethyl acetylene	C ₄ H ₆	503-17-3	54.090	0.10842	300.2	-1.6831			240.91	3.796E-04	371.00	1.186E-04
106	Dimethyl amine	C ₂ H ₇ N	124-40-3	45.084	-10.93	699.5				200.00	5.917E-04	308.15	1.734E-04
107	2,3-Dimethylbutane	C ₆ H ₁₄	79-29-8	86.175	7.2565	221.4	-2.7946			220.00	1.103E-03	331.13	2.509E-04
108	1,1-Dimethylcyclohexane	C ₈ H ₁₆	590-66-9	112.213	-10.716	1140.5	-0.047736			239.66	1.992E-03	392.70	3.045E-04
109	<i>cis</i> -1,2-Dimethylcyclohexane	C ₈ H ₁₆	2207-01-4	112.213	-11.796	1463.5				223.16	5.310E-03	484.92	1.540E-04
110	<i>trans</i> -1,2-Dimethylcyclohexane	C ₈ H ₁₆	6876-23-9	112.213	-11.344	1168.9	0.04513			184.99	8.315E-03	396.58	2.956E-04
111	Dimethyl disulfide	C ₂ H ₆ S ₂	624-92-0	94.199	-10.577	1172.6	-0.14244			188.44	6.093E-03	382.90	2.336E-04
112	Dimethyl ether	C ₂ H ₆ O	115-10-6	46.068	-10.62	448.99	0.000083967			131.65	7.398E-04	248.31	1.490E-04
113	<i>N,N</i> -Dimethyl formamide	C ₃ H ₇ NO	68-12-2	73.094	-20.425	1515.5	1.4444			240.00	2.041E-03	425.15	2.981E-04
114	2,3-Dimethylpentane	C ₇ H ₁₆	565-59-3	100.202	-12.08	1112.2	0.09654			160.00	9.669E-03	362.93	2.147E-04
115	Dimethyl phthalate	C ₁₀ H ₁₀ O ₄	131-11-3	194.184	16.961	-423.16	-3.8178	1.362E+15	-6	274.18	6.030E-02	612.80	2.730E-04
116	Dimethylsilane	C ₂ H ₆ Si	1111-74-6	60.170									
117	Dimethyl sulfide	C ₂ H ₆ S	75-18-3	62.134	-17.641	1067.5	1.0317			225.00	6.696E-04	310.48	2.528E-04
118	Dimethyl sulfoxide	C ₂ H ₆ OS	67-68-5	78.133	-37.347	2835	3.7937			291.67	2.253E-03	464.00	3.547E-04

119	Dimethyl terephthalate	C ₁₀ H ₁₀ O ₄	120-61-6	194.184	-11.488	1922.6						413.80	1.068E-03	561.15	3.154E-04
120	1,4-Dioxane	C ₄ H ₈ O ₂	123-91-1	88.105	-46.166	3086.2						284.95	1.525E-03	374.65	4.610E-04
121	Diphenyl ether	C ₁₂ H ₁₀ O	101-84-8	170.207	-12.373	2017.5						293.15	4.120E-03	613.44	1.130E-04
122	Dipropyl amine	C ₆ H ₁₅ N	142-84-7	101.190	-15.404	1390						260.00	9.454E-04	382.35	2.118E-04
123	Dodecane	C ₁₂ H ₂₆	112-40-3	170.335	-7.8244	1191.9						262.15	3.020E-03	526.40	1.680E-04
124	Eicosane	C ₂₀ H ₄₂	112-95-8	282.547	-18.315	2283.5						309.58	4.243E-03	616.93	2.078E-04
125	Ethane	C ₂ H ₆	74-84-0	30.069	-7.0046	276.38						90.35	1.247E-03	300.00	3.587E-05
126	Ethanol	C ₂ H ₆ O	64-17-5	46.068	7.875	781.98						200.00	1.315E-02	440.00	1.416E-04
127	Ethyl acetate	C ₄ H ₈ O ₂	141-78-6	88.105	14.354	-154.6						220.00	1.132E-03	473.15	9.061E-05
128	Ethyl amine	C ₂ H ₇ N	75-04-7	45.084	19.822	-0.12598						192.15	1.727E-03	289.73	2.236E-04
129	Ethylbenzene	C ₈ H ₁₀	100-41-4	106.165	-13.563	1208.6						178.20	8.012E-03	413.10	2.326E-04
130	Ethyl benzoate	C ₉ H ₁₀ O ₂	93-89-0	150.175	-40.706	3035						250.00	6.643E-03	486.55	3.109E-04
131	2-Ethyl butanoic acid	C ₆ H ₁₂ O ₂	88-09-5	116.158	-12.24	1836.4						258.15	6.705E-03	466.95	2.822E-04
132	Ethyl butyrate	C ₆ H ₁₂ O ₂	105-54-4	116.158	-15.485	1325.6						250.00	1.319E-03	394.65	2.533E-04
133	Ethylcyclohexane	C ₈ H ₁₆	1678-91-7	112.213	-22.11	1673						200.00	6.406E-03	404.94	2.956E-04
134	Ethylcyclopentane	C ₇ H ₁₄	1640-89-7	98.186	-6.894	818.6						253.15	9.605E-04	378.15	2.599E-04
135	Ethylene	C ₂ H ₄	74-85-1	28.053	1.8878	78.865						104.00	6.334E-04	250.00	6.143E-05
136	Ethylenediamine	C ₂ H ₈ N ₂	107-15-3	60.098	-53.908	4030.8						284.29	2.487E-03	483.15	1.723E-04
137	Ethylene glycol	C ₂ H ₆ O ₂	107-21-1	62.068	-20.515	2468.5						260.15	1.340E-01	576.00	2.520E-04
138	Ethyleneimine	C ₂ H ₅ N	151-56-4	43.068	-11.012	967.4						250.00	7.909E-04	329.00	3.123E-04
139	Ethylene oxide	C ₂ H ₄ O	75-21-8	44.053	-8.521	634.2						160.65	1.918E-03	283.85	2.863E-04
140	Ethyl formate	C ₃ H ₆ O ₂	109-94-4	74.079	-9.8417	876.4						245.00	7.435E-04	345.00	2.486E-04
141	2-Ethyl hexanoic acid	C ₈ H ₁₆ O ₂	149-57-5	144.211	-13.037	2346						235.00	4.717E-02	510.10	2.165E-04
142	Ethylhexyl ether	C ₈ H ₁₈ O	5756-43-4	130.228	-11.311	1337.2						180.00	1.765E-02	417.15	2.522E-04
143	Ethylisopropyl ether	C ₅ H ₁₂ O	625-54-7	88.148	-11.331	908.46						140.00	7.908E-03	326.15	1.949E-04
144	Ethylisopropyl ketone	C ₆ H ₁₂ O	565-69-5	100.159	-11.452	1172.7						204.15	3.319E-03	386.55	2.207E-04
145	Ethyl mercaptan	C ₂ H ₆ S	75-08-1	62.134	-9.7574	729.43						125.26	9.520E-03	308.15	2.626E-04
146	Ethyl propionate	C ₅ H ₁₀ O ₂	105-37-3	102.132	-8.9215	950.8						250.00	9.848E-04	372.25	2.480E-04
147	Ethylpropyl ether	C ₅ H ₁₂ O	628-32-0	88.148	0.7109	386.51						200.00	1.156E-03	337.01	2.086E-04
148	Ethyltrichlorosilane	C ₂ H ₅ Cl ₃ Si	115-21-9	163.506	7.8744	-106.34						167.55	1.010E-02	447.96	1.550E-04
149	Fluorine	F ₂	7782-41-4	37.997	8.18	-75.6						53.48	7.317E-04	140.00	5.954E-05
150	Fluorobenzene	C ₆ H ₅ F	462-06-6	96.102	-10.064	1058.7						232.15	1.599E-03	453.15	1.542E-04
151	Fluoroethane	C ₂ H ₅ F	353-36-6	48.060	-10.758	558.81						129.95	1.448E-03	235.45	2.087E-04
152	Fluoromethane	CH ₃ F	593-53-3	34.033	-10.501	427.78						131.35	7.450E-04	194.82	2.587E-04
153	Formaldehyde	CH ₂ O	50-00-0	30.026	-11.24	751.69						181.15	7.331E-04	254.05	2.210E-04
154	Formamide	CH ₃ NO	75-12-7	45.041	40.153	-912.39						273.15	7.408E-03	493.00	1.821E-04
155	Formic acid	CH ₂ O ₂	64-18-6	46.026	-48.529	3394.7						281.45	2.319E-03	373.71	5.444E-04
156	Furan	C ₄ H ₄ O	110-00-9	68.074	-10.923	894.63						200.00	1.575E-03	304.50	3.392E-04
157	Helium-4	He	7440-59-7	4.003	-9.6312	-3.841						2.20	3.628E-06	5.10	2.532E-06
158	Heptadecane	C ₁₇ H ₃₆	629-78-7	240.468	-19.991	2245.1						295.13	3.814E-03	575.30	2.088E-04
159	Heptanal	C ₇ H ₁₄ O	111-71-7	114.185	-10.443	1063.2						229.80	2.510E-03	425.95	2.924E-04
160	Heptane	C ₇ H ₁₆	142-82-5	100.202	-9.4622	877.07						180.15	4.420E-03	432.16	1.430E-04
161	Heptanoic acid	C ₇ H ₁₄ O ₂	111-14-8	130.185	-40.543	3328.3						265.83	9.242E-03	496.15	3.754E-04
162	1-Heptanol	C ₇ H ₁₆ O	111-70-6	116.201	-66.654	5325.8						239.15	8.805E-02	448.60	3.190E-04
163	2-Heptanol	C ₇ H ₁₆ O	543-49-7	116.201	11.225	25.319						230.00	4.036E-01	432.90	2.723E-04
164	3-Heptanone	C ₇ H ₁₄ O	106-35-4	114.185	-9.3874	1204.9						234.15	2.427E-03	421.15	2.040E-04
165	2-Heptanone	C ₇ H ₁₄ O	110-43-0	114.185	-13.929	1321.9						250.00	1.642E-03	424.18	2.318E-04
166	1-Heptene	C ₇ H ₁₄	592-76-7	98.186	-10.819	841.33						154.12	4.700E-03	429.92	1.420E-04
167	Heptyl mercaptan	C ₇ H ₁₆ S	1639-09-4	132.267	-11.812	1291.9						229.92	3.097E-03	450.09	2.087E-04
168	1-Heptyne	C ₇ H ₁₂	628-71-7	96.170	-2.7947	563.86						192.22	2.528E-03	447.20	1.777E-04
169	Hexadecane	C ₁₆ H ₃₄	544-76-3	226.441	-20.182	2203.5						291.31	3.536E-03	564.15	2.054E-04
170	Hexanal	C ₆ H ₁₂ O	66-25-1	100.159	-10.745	1021.4						217.15	2.378E-03	401.45	2.744E-04
171	Hexane	C ₆ H ₁₄	110-54-3	86.175	-6.3276	640						174.65	2.400E-03	406.08	1.340E-04
172	Hexanoic acid	C ₆ H ₁₂ O ₂	142-62-1	116.158	-46.402	3448.6						269.25	5.854E-03	478.85	4.019E-04
173	1-Hexanol	C ₆ H ₁₄ O	111-27-3	102.175	-39.324	3841						250.00	2.822E-02	429.90	3.343E-04
174	2-Hexanol	C ₆ H ₁₄ O	626-93-7	102.175	-82.705	7404.9						223.00	4.919E-01	412.40	3.274E-04
175	2-Hexanone	C ₆ H ₁₂ O	591-78-6	100.159	-11.445	1187.2						217.35	2.561E-03	400.70	2.108E-04
176	3-Hexanone	C ₆ H ₁₂ O	589-38-8	100.159	-13.684	1283.4						217.50	2.563E-03	396.65	2.185E-04
177	1-Hexene	C ₆ H ₁₂	592-41-6	84.159	-10.36	775.85						133.39	7.108E-03	336.63	1.966E-04
178	3-Hexyne	C ₆ H ₁₀	928-49-4	82.144	-4.2684	647.6						170.05	3.550E-03	432.00	1.377E-04
179	Hexyl mercaptan	C ₆ H ₁₄ S	111-31-9	118.240	-10.073	1123.3						192.62	6.035E-03	425.81	2.172E-04
180	1-Hexyne	C ₆ H ₁₀	693-02-7	82.144	-4.7263	594.43						141.25	8.332E-03	412.00	2.083E-04
181	2-Hexyne	C ₆ H ₁₀	764-35-2	82.144	-3.7464	624.2						183.65	2.483E-03	435.00	1.368E-04

TABLE 2-313 Viscosity of Inorganic and Organic Liquids (Pa-s) (Continued)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1	C2	C3	C4	C5	$T_{\min}$, K	Viscosity at $T_{\min}$	$T_{\max}$, K	Viscosity at $T_{\max}$
182	Hydrazine	H ₄ N ₂	302-01-2	32.045	-75.781	4175.4	9.6508	-7.27E-09	3	274.69	1.450E-03	522.52	2.190E-04
183	Hydrogen	H ₂	1333-74-0	2.016	-11.661	24.7	-0.261	-4.1E-16	10	13.95	2.546E-05	33.00	3.906E-06
184	Hydrogen bromide	HBr	10035-10-6	80.912	-11.633	316.38	0.56191			185.15	9.207E-04	206.45	8.206E-04
185	Hydrogen chloride	HCl	7647-01-0	36.461	-116.34	3834.6	16.864	-2.5875E-10	4	158.97	1.000E-03	318.15	5.780E-05
186	Hydrogen cyanide	HCN	74-90-8	27.025	-21.927	1266.5	1.5927			259.83	2.754E-04	298.85	1.821E-04
187	Hydrogen fluoride	HF	7664-39-3	20.006	353.99	13928	-41.717	-2,962	-0.5	189.79	1.550E-03	368.92	1.190E-04
188	Hydrogen sulfide	H ₂ S	7783-06-4	34.081	-10.905	762.11	-0.11863			187.68	5.726E-04	350.00	8.089E-05
189	Isobutyric acid	C ₄ H ₈ O ₂	79-31-2	88.105	-11.497	1365.7	0.036966			250.00	2.938E-03	450.00	2.649E-04
190	Isopropyl amine	C ₃ H ₉ N	75-31-0	59.110	-31.157	1926	2.925			250.00	6.737E-04	453.15	1.214E-04
191	Malonic acid	C ₃ H ₄ O ₄	141-82-2	104.061	-19.834	2784.5	1.1161			404.15	1.939E-03	580.00	3.593E-04
192	Methacrylic acid	C ₄ H ₆ O ₂	79-41-4	86.089	-14.527	1497.7	0.51747			288.15	1.664E-03	434.15	3.582E-04
193	Methane	CH ₄	74-82-8	16.042	-6.1572	178.15	-0.95239	-9.0606E-24	10	90.69	2.063E-04	188.00	2.263E-05
194	Methanol	CH ₃ O	67-56-1	32.042	-25.317	1789.2	2.069			175.47	1.193E-02	337.85	3.442E-04
195	N-Methyl acetamide	C ₃ H ₇ NO	79-16-3	73.094	-4.648	1832	-1.2191			301.15	3.995E-03	478.15	2.392E-04
196	Methyl acetate	C ₃ H ₆ O ₂	79-20-9	74.079	13.557	-187.3	-3.6592			250.00	6.135E-04	425.00	1.198E-04
197	Methyl acetylene	C ₃ H ₄	74-99-7	40.064	-2.8737	301.35	-1.2271			170.45	6.045E-04	373.15	8.846E-05
198	Methyl acrylate	C ₄ H ₆ O ₂	96-33-3	86.089	10.848	75	-3.297			275.00	6.126E-04	400.00	1.636E-04
199	Methyl amine	CH ₅ N	74-89-5	31.057	-17.044	1074	0.84203			179.69	1.236E-03	273.15	2.275E-04
200	Methyl benzoate	C ₈ H ₈ O ₂	93-58-3	136.148	-21.971	2267.4	1.4173			288.15	2.299E-03	472.65	2.149E-04
201	3-Methyl-1,2-butadiene	C ₅ H ₈	598-25-4	68.117	-10.481	648.37	-0.041947			159.53	1.321E-03	314.00	1.739E-04
202	2-Methylbutane	C ₅ H ₁₂	78-78-4	72.149	-12.596	889.11	0.20469			150.00	3.542E-03	310.00	1.928E-04
203	2-Methylbutanoic acid	C ₅ H ₁₀ O ₂	116-53-0	102.132	-1.035	1048.5	-1.5474			298.15	1.774E-03	450.15	2.859E-04
204	3-Methyl-1-butanol	C ₅ H ₁₂ O	123-51-3	88.148	-25.882	3359.4	1.5787			155.95	3.776E+01	404.15	3.051E-04
205	2-Methyl-1-butene	C ₅ H ₁₀	563-46-2	70.133	-10.755	705.48	-0.011113			135.58	3.675E-03	304.30	2.034E-04
206	2-Methyl-2-butene	C ₅ H ₁₀	513-35-9	70.133	-8.4453	639.21	-0.38409			139.39	3.164E-03	311.70	1.841E-04
207	2-Methyl-1-butene-3-yne	C ₅ H ₆	78-80-8	66.101	-3.6585	441.1	-1.0547			160.15	1.915E-03	390.15	1.476E-04
208	Methylbutyl ether	C ₅ H ₁₂ O	628-28-4	88.148	-11.278	949.12	-0.00012343			157.48	5.239E-03	343.31	2.006E-04
209	Methylbutyl sulfide	C ₅ H ₁₂ S	628-29-5	104.214	-10.97	1067.3	-0.017484			175.30	6.930E-03	396.58	2.286E-04
210	3-Methyl-1-butyne	C ₅ H ₈	598-23-2	68.117	-1.8842	433.58	-1.3238			183.45	1.628E-03	364.00	2.035E-04
211	Methyl butyrate	C ₅ H ₁₀ O ₂	623-42-7	102.132	-12.206	1141.7	0.15014			200.00	3.339E-03	375.90	2.539E-04
212	Methylchlorosilane	CH ₃ ClSi	993-00-0	80.589	-12.002	1009.7				139.05	8.730E-03	353.60	1.070E-04
213	Methylcyclohexane	C ₇ H ₁₄	108-87-2	98.186	-11.358	1213.1				146.58	4.590E-02	457.68	1.650E-04
214	1-Methylcyclohexanol	C ₇ H ₁₄ O	590-67-0	114.185	-6.1534	3219	-1.4494			299.15	2.584E-02	548.80	8.026E-05
215	cis-2-Methylcyclohexanol	C ₇ H ₁₄ O	7443-70-1	114.185	-6.6904	3150.5	-1.392			280.15	3.729E-02	491.20	1.360E-04
216	trans-2-Methylcyclohexanol	C ₇ H ₁₄ O	7443-52-9	114.185	-6.6915	3173.2	-1.3046			269.15	1.107E-01	493.60	2.356E-04
217	Methylcyclopentane	C ₆ H ₁₂	96-37-7	84.159	-1.8553	612.62	-1.3774			248.15	9.288E-04	353.15	2.742E-04
218	1-Methylcyclopentene	C ₆ H ₁₀	693-89-0	82.144	-4.8515	679.07	-0.93238			146.62	7.669E-03	433.60	1.301E-04
219	3-Methylcyclopentene	C ₆ H ₁₀	1120-62-3	82.144	-6.7424	788.86	-0.69862			115.00	4.086E-02	420.80	1.129E-04
220	Methyldichlorosilane	CH ₂ Cl ₂ Si	75-54-7	115.034	-10.517	745.32				275.00	4.070E-04	314.70	2.891E-04
221	Methylethyl ether	C ₃ H ₈ O	540-67-0	60.095	-11.104	627.18	0.036581			160.00	9.133E-04	280.50	1.731E-04
222	Methylethyl ketone	C ₄ H ₈ O	78-93-3	72.106	-1.0598	520.68	-1.4961			186.48	2.266E-03	535.50	7.577E-05
223	Methylethyl sulfide	C ₃ H ₆ S	624-89-5	76.161	-10.842	863.65	-0.00074603			167.23	3.409E-03	339.80	2.474E-04
224	Methyl formate	C ₂ H ₄ O ₂	107-31-3	60.052	-39.641	2113.3	4.308			250.00	6.104E-04	304.90	3.134E-04
225	Methylisobutyl ether	C ₅ H ₁₂ O	625-44-5	88.148	-11.27	888.42	0.024736			150.00	5.390E-03	331.70	2.143E-04
226	Methylisobutyl ketone	C ₆ H ₁₂ O	108-10-1	100.159	-11.394	1168.7	-0.007539			189.15	5.222E-03	389.15	2.170E-04
227	Methyl Isocyanate	C ₂ H ₃ NO	624-83-9	57.051									
228	Methylisopropyl ether	C ₄ H ₁₀ O	598-53-8	74.122	-11.216	737.75	0.019308			127.93	4.722E-03	303.92	1.703E-04
229	Methylisopropyl ketone	C ₅ H ₁₀ O	563-80-4	86.132	-11.272	1048.9	0.00030493			180.15	4.305E-03	367.55	2.212E-04
230	Methylisopropyl sulfide	C ₄ H ₁₀ S	1551-21-9	90.187	-11.075	990.72				171.64	4.977E-03	553.10	9.292E-05
231	Methyl mercaptan	CH ₄ S	74-93-1	48.107	-10.628	645	0.025885			150.18	2.023E-03	279.11	2.827E-04
232	Methyl methacrylate	C ₅ H ₈ O ₂	80-62-6	100.116	-0.099	496	-1.5939			260.00	8.635E-04	400.00	2.229E-04
233	2-Methyloctanoic acid	C ₉ H ₁₈ O ₂	3004-93-1	158.238	-12.579	2224.2				240.00	3.646E-02	518.15	2.519E-04
234	2-Methylpentane	C ₆ H ₁₄	107-83-5	86.175	-12.86	946.91	0.26191			119.55	2.506E-02	333.41	2.038E-04
235	Methyl pentyl ether	C ₆ H ₁₄ O	628-80-8	102.175	-11.391	1090.8	1.0752E-07			176.00	5.554E-03	372.00	2.120E-04
236	2-Methylpropane	C ₄ H ₁₀	75-28-5	58.122	-13.912	797.09	0.45308			110.00	1.072E-02	310.95	1.588E-04
237	2-Methyl-2-propanol	C ₄ H ₁₀ O	75-65-0	74.122	51.356	-1249.5	-9.4593	3.694E+24	-9.8759	295.56	5.440E-03	451.21	1.010E-04
238	2-Methyl propene	C ₄ H ₈	115-11-7	56.106	-10.385	599.59	-0.046088			132.81	2.253E-03	266.25	2.270E-04
239	Methyl propionate	C ₄ H ₈ O ₂	554-12-1	88.105	-4.841	696.7	-0.9194			250.00	8.002E-04	352.60	2.593E-04
240	Methylpropyl ether	C ₄ H ₁₀ O	557-17-5	74.122	-10.705	788.94	-0.048383			133.97	6.390E-03	312.20	2.127E-04

241	Methylpropyl sulfide	C ₄ H ₁₀ S	3877-15-4	90.187	-10.569	952.38	-0.063873			160.17	7.103E-03	368.69	2.333E-04
242	Methylsilane	CH ₆ Si	992-94-9	46.144									
243	alpha-Methyl styrene	C ₉ H ₁₀	98-83-9	118.176	-11.632	1251.6	0.071692			249.95	1.972E-03	438.65	2.382E-04
244	Methyl <i>tert</i> -butyl ether	C ₅ H ₁₂ O	1634-04-4	88.148	-6.921	790.773	-0.654			164.55	4.284E-03	450.00	1.052E-04
245	Methyl vinyl ether	C ₃ H ₆ O	107-25-5	58.079	-10.34	519.61	-0.013899			151.15	9.377E-04	278.65	1.929E-04
246	Naphthalene	C ₁₀ H ₈	91-20-3	128.171	-19.308	1822.5	1.218			353.43	9.077E-04	633.15	1.892E-04
247	Neon	Ne	7440-01-9	20.180	-17.945	115.57	1.428	-2.14E-17	10	25.09	1.602E-04	44.13	2.706E-05
248	Nitroethane	C ₂ H ₅ NO ₂	79-24-3	75.067	-4.438	746.5	-0.9385			200.00	3.421E-03	387.22	3.027E-04
249	Nitrogen	N ₂	7727-37-9	28.013	16.004	-181.61	-5.1551			63.15	2.633E-04	124.00	3.331E-05
250	Nitrogen trifluoride	F ₃ N	7783-54-2	71.002									
251	Nitromethane	CH ₃ NO ₂	75-52-5	61.040	-9.5556	981.64	-0.19453			244.60	1.344E-03	374.35	3.078E-04
252	Nitrous oxide	N ₂ O	10024-97-2	44.013	19.329	-381.68	-4.8618			210.00	2.065E-04	283.09	7.730E-05
253	Nitric oxide	NO	10102-43-9	30.006	-246.65	3150.3	49.98	-0.22541	1	109.50	3.858E-04	180.05	3.791E-05
254	Nonadecane	C ₁₉ H ₄₀	629-92-5	268.521	-16.403	2119.5	0.6881			305.04	4.012E-03	603.15	2.068E-04
255	Nonanal	C ₉ H ₁₈ O	124-19-6	142.239	-12.94	1257.6	0.37191			255.15	2.606E-03	468.15	3.468E-04
256	Nonane	C ₉ H ₂₀	111-84-2	128.255	-68.54	3165.3	9.0919	-0.000013519	2	218.15	3.310E-03	593.15	5.000E-05
257	Nonanoic acid	C ₉ H ₁₈ O ₂	112-05-0	158.238	-48.851	4095	5.294			285.55	1.030E-02	528.75	3.670E-04
258	1-Nonanol	C ₉ H ₂₀ O	143-08-8	144.255	-39.863	4089	3.7631			280.00	1.733E-02	485.20	2.852E-04
259	2-Nonanol	C ₉ H ₂₀ O	628-99-9	144.255	-98.854	7183.8	12.283			238.15	2.310E-01	471.70	3.334E-04
260	1-Nonene	C ₉ H ₁₈	124-11-8	126.239	-21.921	1603.9	1.5971			191.91	5.699E-03	420.02	2.127E-04
261	Nonyl mercaptan	C ₉ H ₂₀ S	1455-21-6	160.320	-11.319	1428	-0.022545			253.05	3.026E-03	492.95	1.913E-04
262	1-Nonyne	C ₉ H ₁₆	3452-09-3	124.223	-2.3409	715.52	-1.222			223.15	3.206E-03	487.20	2.172E-04
263	Octadecane	C ₁₈ H ₃₈	593-45-3	254.494	-22.688	2466	1.5703			301.31	3.926E-03	589.86	2.057E-04
264	Octanal	C ₈ H ₁₆ O	124-13-0	128.212	-10.191	1072.4	-0.030553			246.00	2.479E-03	447.15	3.425E-04
265	Octane	C ₈ H ₁₈	111-65-9	114.229	-7.556	881.09	-0.52502	4.6342E+22	-10	211.15	2.660E-03	454.96	1.460E-04
266	Octanoic acid	C ₈ H ₁₆ O ₂	124-07-2	144.211	-60.795	4617.8	7.028			289.65	6.652E-03	512.85	3.576E-04
267	1-Octanol	C ₈ H ₁₈ O	111-87-5	130.228	-19.907	2791.7	0.94296	2.3041E+24	-10.09	280.00	1.569E-02	468.35	2.901E-04
268	2-Octanol	C ₈ H ₁₈ O	123-96-6	130.228	16.792	1353.6	-4.6357	2.6663E+31	-13.039	241.55	4.576E-01	452.90	1.899E-04
269	2-Octanone	C ₈ H ₁₆ O	111-13-7	128.212	-11.736	1415.2	0.0003618			252.85	2.161E-03	446.15	1.913E-04
270	3-Octanone	C ₈ H ₁₆ O	106-68-3	128.212	-20.804	1834.6	1.3403			255.55	2.039E-03	440.65	2.075E-04
271	1-Octene	C ₈ H ₁₆	111-66-0	112.213	-11.19	1057.4				171.45	6.590E-03	453.52	1.420E-04
272	Octyl mercaptan	C ₈ H ₁₆ S	111-88-6	146.294	-11.498	1362.1	0.015575			223.95	4.837E-03	472.19	2.000E-04
273	1-Octyne	C ₈ H ₁₄	629-05-0	110.197	-3.8552	684.22	-1.0071			193.55	3.614E-03	468.00	1.868E-04
274	Oxalic acid	C ₂ H ₂ O ₄	144-62-7	90.035									
275	Oxygen	O ₂	7782-44-7	31.999	-4.1476	94.04	-1.207			54.36	7.170E-04	150.00	6.990E-05
276	Ozone	O ₃	10028-15-6	47.998	-10.94	415.96				77.55	3.790E-03	208.80	1.300E-04
277	Pentadecane	C ₁₅ H ₃₂	629-62-9	212.415	-19.299	2088.6	1.1091			283.07	3.486E-03	543.84	2.091E-04
278	Pentanal	C ₅ H ₁₀ O	110-62-3	86.132	-10.846	980.01	-0.0054565			182.00	4.129E-03	376.15	2.553E-04
279	Pentane	C ₅ H ₁₂	109-66-0	72.149	-53.509	1836.6	7.1409	-0.000019627	2	143.42	3.529E-03	465.15	4.796E-05
280	Pentanoic acid	C ₅ H ₁₀ O ₂	109-52-4	102.132	-37.067	2856.7	3.7344			270.00	3.773E-03	458.65	3.516E-04
281	1-Pentanol	C ₅ H ₁₂ O	71-41-0	88.148	-36.561	3542.2	3.3364	-8.0487E-37	12.84	253.15	1.649E-02	410.90	3.844E-04
282	2-Pentanol	C ₅ H ₁₂ O	6032-29-7	88.148	-410.49	18371	61.985	-0.0095612	1.2201	200.00	3.823E+01	392.20	5.828E-04
283	2-Pentanone	C ₅ H ₁₀ O	107-87-9	86.132	-11.055	1005.3	0.0039301			250.00	9.009E-04	375.46	2.354E-04
284	3-Pentanone	C ₅ H ₁₀ O	96-22-0	86.132	-2.8695	596.32	-1.2025			234.18	1.024E-03	375.14	2.232E-04
285	1-Pentene	C ₅ H ₁₀	109-67-1	70.133	-10.667	659.56				108.02	1.045E-02	303.22	2.051E-04
286	2-Pentyl mercaptan	C ₅ H ₁₂ S	2084-19-7	104.214	-6.9168	818.76	-0.59628			220.00	1.643E-03	385.15	2.385E-04
287	Pentyl mercaptan	C ₅ H ₁₂ S	110-66-7	104.214	-11.677	1091.2	0.10658			197.45	3.746E-03	399.79	2.463E-04
288	1-Pentyne	C ₅ H ₈	627-19-0	68.117	-1.7273	424.34	-1.342			167.45	2.323E-03	378.00	1.898E-04
289	2-Pentyne	C ₅ H ₈	627-21-4	68.117	-3.7241	516.54	-1.1167			163.83	1.902E-03	415.20	9.980E-05
290	Phenanthrene	C ₁₄ H ₁₀	85-01-8	178.229	-22.472	2566.9	1.5749			372.38	1.920E-03	610.03	2.849E-04
291	Phenol	C ₆ H ₆ O	108-95-2	94.111	-43.335	3881.7	4.3983	3.0548E+24	-10	291.45	1.270E-02	555.40	1.940E-04
292	Phenyl isocyanate	C ₇ H ₅ NO	103-71-9	119.121	-11.31	1280				243.15	2.370E-03	522.40	1.420E-04
293	Phthalic anhydride	C ₈ H ₆ O ₃	85-44-9	148.116	195.25	-11072	-29.084			404.15	1.229E-03	557.65	1.986E-04
294	Propadiene	C ₃ H ₄	463-49-0	40.064	-6.3528	240.85	-0.58229			136.87	5.772E-04	298.15	1.416E-04
295	Propane	C ₃ H ₈	74-98-6	44.096	-17.156	646.25	1.1101	-7.3439E-11	4	85.47	9.458E-03	360.00	4.275E-05
296	1-Propanol	C ₃ H ₈ O	71-23-8	60.095	23.467	116.07	-5.3372	2.8801E+09	-4.0267	146.95	2.069E+01	370.35	4.735E-04
297	2-Propanol	C ₃ H ₈ O	67-63-0	60.095	-8.8918	2357.6	-0.91376			185.26	3.917E-01	355.30	4.892E-04
298	Propenylcyclohexene	C ₉ H ₁₄	13511-13-2	122.207	-11.208	1079.8				199.00	3.080E-03	508.80	1.130E-04
299	Propionaldehyde	C ₃ H ₆ O	123-38-6	58.079	-9.9177	839.53	-0.16735			170.00	2.912E-03	321.15	2.562E-04
300	Propionic acid	C ₃ H ₆ O ₂	79-09-4	74.079	-23.931	1834.6	1.9124			252.45	2.275E-03	414.32	3.430E-04
301	Propionitrile	C ₃ H ₅ N	107-12-0	55.079	-5.7136	703.62	-0.78123			250.00	7.372E-04	370.50	2.171E-04
302	Propyl acetate	C ₅ H ₁₀ O ₂	109-60-4	102.132	17.797	-252.43	-4.291			250.00	1.002E-03	473.15	1.045E-04
303	Propyl amine	C ₃ H ₉ N	107-10-8	59.110	-9.8074	1010.4	-0.25697			188.36	3.060E-03	321.00	2.908E-04

TABLE 2-313 Viscosity of Inorganic and Organic Liquids (Pa·s) (Concluded)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1	C2	C3	C4	C5	$T_{\min}$, K	Viscosity at $T_{\min}$	$T_{\max}$, K	Viscosity at $T_{\max}$
304	Propylbenzene	C ₉ H ₁₂	103-65-1	120.192	-18.282	1549.7	1.0454			200.00	6.774E-03	432.39	2.357E-04
305	Propylene	C ₃ H ₆	115-07-1	42.080	-92.082	1907.3	15.639	-0.043098	1	87.90	1.550E-02	333.15	5.150E-05
306	Propyl formate	C ₄ H ₈ O ₂	110-74-7	88.105	-73.735	2668.2	10.993	-0.018364	1	180.25	5.852E-03	353.97	2.810E-04
307	2-Propyl mercaptan	C ₃ H ₈ S	75-33-2	76.161	-5.7244	638.2	-0.76415			142.61	6.477E-03	325.71	2.784E-04
308	Propyl mercaptan	C ₃ H ₈ S	107-03-9	76.161	-10.153	840.71	-0.093763			159.95	4.641E-03	340.87	2.656E-04
309	1,2-Propylene glycol	C ₃ H ₈ O ₂	57-55-6	76.094	-804.54	30487	130.79	-0.15449	1	213.15	9.500E+02	500.80	3.310E-04
310	Quinone	C ₆ H ₄ O ₂	106-51-4	108.095	-14.846	1829.4	0.3729			388.85	3.643E-04	454.00	1.965E-04
311	Silicon tetrafluoride	F ₄ Si	7783-61-1	104.079									
312	Styrene	C ₈ H ₈	100-42-5	104.149	-22.675	1758	1.6701			242.54	1.919E-03	418.31	2.268E-04
313	Succinic acid	C ₄ H ₆ O ₄	110-15-6	118.088	-13.422	3431.8				460.65	2.550E-03	644.80	3.040E-04
314	Sulfur dioxide	O ₂ S	7446-09-5	64.064	46.223	-1378	-8.7475			225.00	6.900E-04	400.00	6.557E-05
315	Sulfur hexafluoride	F ₆ S	2551-62-4	146.055	3.8305	41.21	-2.1342			223.15	5.388E-04	318.69	2.383E-04
316	Sulfur trioxide	O ₃ S	7446-11-9	80.063	-88.793	6400.7	10.709			289.95	2.477E-03	318.15	9.456E-04
317	Terephthalic acid	C ₈ H ₆ O ₄	100-21-0	166.131									
318	<i>o</i> -Terphenyl	C ₁₈ H ₁₄	84-15-1	230.304	-215.09	11612	31.849	-0.026882	1	329.35	1.736E-02	723.15	1.522E-04
319	Tetradecane	C ₁₄ H ₃₀	629-59-4	198.388	-14.493	1710.8	0.4417	3.0895E+28	-12	277.65	3.350E-03	554.40	1.810E-04
320	Tetrahydrofuran	C ₄ H ₈ O	109-99-9	72.106	-10.321	900.92	-0.069128			164.65	5.505E-03	373.15	2.446E-04
321	1,2,3,4-Tetrahydronaphthalene	C ₁₀ H ₁₂	119-64-2	132.202	-11.167	1193.2	0.096226	9.5986E+11	-5	237.40	1.300E-02	576.00	2.100E-04
322	Tetrahydrothiophene	C ₄ H ₈ S	110-01-0	88.171	-10.843	1165.2				293.15	1.040E-03	303.15	9.125E-04
323	2,2,3,3-Tetramethylbutane	C ₈ H ₁₈	594-82-1	114.229	5.5351	632.38	-2.6576			373.96	1.999E-04	454.00	8.859E-05
324	Thiophene	C ₄ H ₄ S	110-02-1	84.140	-16.671	1342.5	0.8388			250.00	1.269E-03	393.15	2.625E-04
325	Toluene	C ₇ H ₈	108-88-3	92.138	-226.08	6805.7	37.542	-0.060853	1	178.18	1.569E-02	383.78	2.428E-04
326	1,1,2-Trichloroethane	C ₂ H ₃ Cl ₃	79-00-5	133.404	0.388	736.5	-1.7063			236.50	2.955E-03	387.00	3.798E-04
327	Tridecane	C ₁₃ H ₂₈	629-50-5	184.361	-4.1103	1005.3	-1.0188	1.0017E+19	-8	267.67	3.450E-03	540.00	1.740E-04
328	Triethyl amine	C ₆ H ₁₅ N	121-44-8	101.190	-3.7067	585.78	-1.0926			250.00	6.135E-04	359.05	2.028E-04
329	Trimethyl amine	C ₃ H ₉ N	75-50-3	59.110	10.142	-130.41	-3.2199			200.00	5.156E-04	308.15	1.612E-04
330	1,2,3-Trimethylbenzene	C ₉ H ₁₂	526-73-8	120.192	-11.756	1483.1	-0.040387			247.79	2.495E-03	449.27	1.663E-04
331	1,2,4-Trimethylbenzene	C ₉ H ₁₂	95-63-6	120.192	-9.6461	1281.2	-0.29478			229.33	3.477E-03	442.53	1.942E-04
332	2,2,4-Trimethylpentane	C ₈ H ₁₈	540-84-1	114.229	-12.928	1137.5	0.25725	-3.6929E-28	10	165.78	8.636E-03	541.15	4.530E-05
333	2,3,3-Trimethylpentane	C ₈ H ₁₈	560-21-4	114.229	-4.0309	990.76	-1.1771			172.22	1.305E-02	387.91	2.049E-04
334	1,3,5-Trinitrobenzene	C ₆ H ₃ N ₃ O ₆	99-35-4	213.105	-10.707	1818.5				398.40	2.150E-03	676.80	3.290E-04
335	2,4,6-Trinitrotoluene	C ₇ H ₅ N ₃ O ₆	118-96-7	227.131	-11.504	3301	-0.39102			353.15	1.167E-02	625.00	1.601E-04
336	Undecane	C ₁₁ H ₂₄	1120-21-4	156.308	52.176	-4951.9	-8.5676	570,980	-2	247.57	3.240E-03	511.20	1.570E-04
337	1-Undecanol	C ₁₁ H ₂₄ O	112-42-5	172.308	-69.778	5905.2	8.0214			288.45	2.089E-02	590.15	1.856E-04
338	Vinyl acetate	C ₄ H ₆ O ₂	108-05-4	86.089	-22.407	1462.8	1.7006			225.00	1.237E-03	345.65	2.654E-04
339	Vinyl acetylene	C ₄ H ₄	689-97-4	52.075	-2.2333	320.37	-1.2915			173.15	8.764E-04	364.00	1.273E-04
340	Vinyl chloride	C ₂ H ₃ Cl	75-01-4	62.498	0.26297	276.55	-1.7282			130.00	2.425E-03	400.00	8.272E-05
341	Vinyl trichlorosilane	C ₂ H ₃ Cl ₃ Si	75-94-5	161.490	-10.37	823.31				178.35	3.170E-03	434.52	2.090E-04
342	Water	H ₂ O	7732-18-5	18.015	-52.843	3703.6	5.866	-5.879E-29	10	273.16	1.702E-03	646.15	5.028E-05
343	<i>m</i> -Xylene	C ₈ H ₁₀	108-38-3	106.165	-11.91	1094.9	0.13825			225.30	1.834E-03	413.10	2.189E-04
344	<i>o</i> -Xylene	C ₈ H ₁₀	95-47-6	106.165	-15.489	1393.5	0.63711			247.98	1.735E-03	418.10	2.459E-04
345	<i>p</i> -Xylene	C ₈ H ₁₀	106-42-3	106.165	-7.381	911.7	-0.54152			286.41	7.021E-04	413.10	2.169E-04

The liquid viscosity is calculated by

$$\mu = \exp(C1 + C2/T + C3 \ln T + C4T^{C5})$$

where μ is the viscosity in Pa·s and T is the temperature in K. Viscosities are at either 1 atm or the vapor pressure, whichever is higher. All substances are listed by chemical family in Table 2-6 and by formula in Table 2-7.

Values in this table were taken from the Design Institute for Physical Properties (DIPPR) of the American Institute of Chemical Engineers (AIChE), copyright 2007 AIChE and reproduced with permission of AIChE and of the DIPPR Evaluated Process Design Data Project Steering Committee. Their source should be cited as R. L. Rowley, W. V. Wilding, J. L. Oscarson, Y. Yang, N. A. Zundel, T. E. Daubert, R. P. Danner, DIPPR® Data Compilation of Pure Chemical Properties, Design Institute for Physical Properties, AIChE, New York (2007).

The number of digits provided for values at $T_{\min}$ and $T_{\max}$ was chosen for uniformity of appearance and formatting; these do not represent the uncertainties of the physical quantities, but are the result of calculations from the standard thermophysical property formulations within a fixed format.

TABLE 2-314 Vapor Thermal Conductivity of Inorganic and Organic Substances [W/(m·K)]

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1	C2	C3	C4	$T_{\min}$, K	Thermal cond. at $T_{\min}$	$T_{\max}$, K	Thermal cond. at $T_{\max}$
1	Acetaldehyde	C ₂ H ₄ O	75-07-0	44.05	3.7272E-07	1.8129			273.15	0.00973	994	0.10124
2	Acetamide	C ₂ H ₅ NO	60-35-5	59.07	0.00013195	0.97	728.3		494.30	0.02189	1000	0.06206
3	Acetic acid	C ₂ H ₄ O ₂	64-19-7	60.05	0.000001691	1.6692		-95,400	294.70	0.01049	686.88	0.05236
4	Acetic anhydride	C ₄ H ₆ O ₃	108-24-7	102.09	0.00042004	0.8066	658	142,620	412.70	0.01864	1000	0.06981
5	Acetone	C ₃ H ₆ O	67-64-1	58.08	-26.8	0.9098	439.37		329.44	0.01363	1000	0.11362
6	Acetonitrile	C ₂ H ₃ N	75-05-8	41.05	4.901E-08	2.1091	-126500000		354.75	0.01170	994.75	0.10298
7	Acetylene	C ₂ H ₂	74-86-2	26.04	0.000075782	1.0327	-36.227	31,432	189.35	0.01011	1000	0.09545
8	Acrolein	C ₃ H ₄ O	107-02-8	56.06	0.024098	0.3285	1325.3	577,830	325.84	0.01534	1000	0.08028
9	Acrylic acid	C ₃ H ₄ O ₂	79-10-7	72.06	0.0009265	0.7035	627.58	112,460	414.15	0.02027	1000	0.06867
10	Acrylonitrile	C ₃ H ₃ N	107-13-1	53.06	0.0013784	1.2093	50594		350.50	0.01133	990.5	0.11107
11	Air	Mixture	132259-10-0	28.96	0.00031417	0.7786	-0.7116	2,121.7	70.00	0.00603	2000	0.11675
12	Ammonia	H ₃ N	7664-41-7	17.03	9.6608E-06	1.3799			200.00	0.01446	900	0.11523
13	Anisole	C ₇ H ₈ O	100-66-3	108.14	0.00059858	0.7527	354.04	241,830	426.73	0.01809	1000	0.06796
14	Argon	Ar	7440-37-1	39.95	0.000633	0.6221	70		90.00	0.00585	3273.1	0.09525
15	Benzamide	C ₇ H ₇ NO	55-21-0	121.14	0.025389	0.28547	1018.3	1,228,600	563.15	0.02317	1000	0.05618
16	Benzene	C ₆ H ₆	71-43-2	78.11	0.00001652	1.3117	491		339.15	0.01407	1000	0.09542
17	Benzenethiol	C ₆ H ₆ S	108-98-5	110.18	0.00047951	0.7818	463.4	189,410	442.29	0.01861	1000	0.06427
18	Benzoic acid	C ₇ H ₆ O ₂	65-85-0	122.12	0.0001163	0.9705	740		522.40	0.02090	1000	0.05452
19	Benzonitrile	C ₇ H ₅ N	100-47-0	103.12	0.00015674	0.95503	711.32		464.15	0.02180	1000	0.06713
20	Benzophenone	C ₁₃ H ₁₀ O	119-61-9	182.22	0.0001235	0.9495	778.7		579.24	0.02213	1000	0.04899
21	Benzyl alcohol	C ₇ H ₈ O	100-51-6	108.14	0.00023476	0.8639	187.8	193,840	478.60	0.02167	1000	0.06636
22	Benzyl ethyl ether	C ₉ H ₁₂ O	539-30-0	136.19	0.00096451	0.69225	519.99	278,930	458.15	0.01936	1000	0.06398
23	Benzyl mercaptan	C ₇ H ₈ S	100-53-8	124.20	0.00015525	0.9446	715.78		472.03	0.02071	1000	0.06171
24	Biphenyl	C ₁₂ H ₁₀	92-52-4	154.21	2.8646E-06	1.4098	-391.35	156,820	373.15	0.01123	1000	0.06347
25	Bromine	Br ₂	7726-95-6	159.81	1.0404E-06	1.4685			300.00	0.00452	500	0.00956
26	Bromobenzene	C ₆ H ₅ Br	108-86-1	157.01	0.00027085	0.7932	278.33	165,880	429.24	0.01302	1000	0.04495
27	Bromoethane	C ₂ H ₅ Br	74-96-4	108.97	0.00019745	0.8824	647		311.50	0.01018	1000	0.05321
28	Bromomethane	CH ₃ Br	74-83-9	94.94	0.000038314	1.0484	287.38		273.00	0.00669	1000	0.04158
29	1,2-Butadiene	C ₄ H ₆	590-19-2	54.09	0.000088221	1.0273	75.316	99,063	284.00	0.01172	1000	0.09071
30	1,3-Butadiene	C ₄ H ₆	106-99-0	54.09	-20,890	0.9593	-9.382E+10		268.74	0.01281	1000	0.16809
31	Butane	C ₄ H ₁₀	106-97-8	58.12	0.051094	0.45253	5455.5	1,979,800	272.65	0.01357	1000	0.13799
32	1,2-Butanediol	C ₄ H ₁₀ O ₂	584-03-2	90.12	-295.44	-0.21463	91602	-7.6032E+08	469.57	0.02426	1000	0.10046
33	1,3-Butanediol	C ₄ H ₁₀ O ₂	107-88-0	90.12	-918.39	-0.21199	334420	-2.8842E+09	481.38	0.02110	1000	0.08332
34	1-Butanol	C ₄ H ₁₀ O	71-36-3	74.12	0.00111484	0.87647	3253.7		370.70	0.02097	712.94	0.06536
35	2-Butanol	C ₄ H ₁₀ O	78-92-2	74.12	4.5894E-06	1.4484			372.90	0.02435	1000	0.10161
36	1-Butene	C ₄ H ₈	106-98-9	56.11	0.000096809	1.1153	781.82		266.91	0.01252	1000	0.12049
37	cis-2-Butene	C ₄ H ₈	590-18-1	56.11	0.000067737	1.0709	-65.881	129,390	273.15	0.01105	1273.15	0.13926
38	trans-2-Butene	C ₄ H ₈	624-64-6	56.11	0.000078576	1.0565	14.63	105,920	274.03	0.01200	1257	0.13704
39	Butyl acetate	C ₈ H ₁₂ O ₂	123-86-4	116.16	5.86E-09	2.376	-401.32	69,280	273.00	0.00783	800	0.07634
40	Butylbenzene	C ₁₀ H ₁₄	104-51-8	134.22	0.1807	0.0082225	-129.42	1,691,500	456.46	0.02151	1000	0.07465
41	Butyl mercaptan	C ₈ H ₁₀ S	109-79-5	90.19	0.00097826	0.78643	1531.5	67,115	371.61	0.01832	1000	0.08610
42	sec-Butyl mercaptan	C ₈ H ₁₀ S	513-53-1	90.19	0.9719	-0.111	1167.2	3,163,200	358.13	0.01749	1000	0.08470
43	1-Butyne	C ₄ H ₆	107-00-6	54.09	0.000037269	1.1427	-43.844	79,421	281.22	0.01268	1000	0.09644
44	Butyraldehyde	C ₄ H ₈ O	123-72-8	72.11	1138.6	0.95596	7.5086E+09		347.95	0.01419	1000	0.11186
45	Butyric acid	C ₄ H ₈ O ₂	107-92-6	88.11	0.002751	0.2734	-314.55		436.40	0.05192	526.32	0.03792
46	Butyronitrile	C ₄ H ₇ N	109-74-0	69.11	5.2879E-07	1.6715	-381.9	95,650	273.00	0.00706	1000	0.07660
47	Carbon dioxide	CO ₂	124-38-9	44.01	3.69	-0.3838	964	1,860,000	194.67	0.00887	1500	0.09025
48	Carbon disulfide	CS ₂	75-15-0	76.14	0.0003467	0.7345	479		273.15	0.00776	1000	0.03745
49	Carbon monoxide	CO	630-08-0	28.01	0.00059882	0.6863	57.13	501.92	70.00	0.00576	1500	0.08724
50	Carbon tetrachloride	CCl ₄	56-23-5	153.82	0.00016599	0.94375	1449.6		349.79	0.00812	1000	0.04595
51	Carbon tetrafluoride	CF ₄	75-73-0	88.00	0.000092004	1.0164	270.83		145.10	0.00505	1000	0.08108
52	Chlorine	Cl ₂	7782-50-5	70.91	0.0009993	0.5472	458.6		200.00	0.00551	1000	0.03002
53	Chlorobenzene	C ₆ H ₅ Cl	108-90-7	112.56	0.0004783	0.8994	1845.5	163,000	400.00	0.01579	1000	0.07935
54	Chloroethane	C ₂ H ₅ Cl	75-00-3	64.51	-19.283	0.20238	-715050	-271,300,000	273.15	0.00960	1000	0.07920
55	Chloroform	CHCl ₃	67-66-3	119.38	0.00043073	0.83878	1874.5		334.33	0.00854	1000	0.04920
56	Chloromethane	CH ₃ Cl	74-87-3	50.49	-22136	0.7666	-4.8749E+10		213.15	0.00590	750	0.05448
57	1-Chloropropane	C ₃ H ₇ Cl	540-54-5	78.54	0.00004861	1.1407	593		319.67	0.01225	1000	0.08065
58	2-Chloropropane	C ₃ H ₇ Cl	75-29-6	78.54	0.00009154	1.0681	746.6		308.85	0.01222	1000	0.08389

TABLE 2-314 Vapor Thermal Conductivity of Inorganic and Organic Substances [W/(m·K)] (Continued)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1	C2	C3	C4	$T_{\min}$, K	Thermal cond. at $T_{\min}$	$T_{\max}$, K	Thermal cond. at $T_{\max}$
59	<i>m</i> -Cresol	C ₇ H ₈ O	108-39-4	108.14	0.00019307	0.9248	710		475.43	0.02316	1000	0.06716
60	<i>o</i> -Cresol	C ₇ H ₈ O	95-48-7	108.14	0.00018648	0.9302	709.37		464.15	0.02230	1000	0.06736
61	<i>p</i> -Cresol	C ₇ H ₈ O	106-44-5	108.14	0.00019063	0.9282	716.91		475.13	0.02319	1000	0.06762
62	Cumene	C ₉ H ₁₂	98-82-8	120.19	1.6743E-07	1.8369	-449.46	112,760	380.00	0.01534	1000	0.08181
63	Cyanogen	C ₂ N ₂	460-19-5	52.03	0.000026933	1.137	28.119		252.00	0.01302	1000	0.06749
64	Cyclobutane	C ₄ H ₈	287-23-0	56.11	-449.910	0.27364	-1.0001E+10	-9.8654E+12	285.66	0.01356	1000	0.14994
65	Cyclohexane	C ₆ H ₁₂	110-82-7	84.16	0.000000859	1.7709	243		325.00	0.01380	1000	0.14198
66	Cyclohexanol	C ₆ H ₁₂ O	108-93-0	100.16	0.0032207	0.5991	608.69	509,290	434.00	0.02399	1000	0.09535
67	Cyclohexanone	C ₆ H ₁₀ O	108-94-1	98.14	-1.095.5	-0.023408	498,780	-7.8355E+09	428.58	0.02291	1000	0.12704
68	Cyclohexene	C ₆ H ₁₀	110-83-8	82.14	0.0000901	1.0897	655		356.12	0.01914	1000	0.10116
69	Cyclopentane	C ₅ H ₁₀	287-92-3	70.13	9.5461E-06	1.4641	632.62		273.00	0.01061	1000	0.14429
70	Cyclopentene	C ₅ H ₈	142-29-0	68.12	0.0010949	0.71644	175.55	346,040	317.38	0.01360	1000	0.10148
71	Cyclopropane	C ₃ H ₆	75-19-4	42.08	-91.383	0.89718	-283,310,000		240.37	0.01061	1000	0.15854
72	Cyclohexyl mercaptan	C ₆ H ₁₂ S	1569-69-3	116.22	0.0000813	1.0674	697.6		431.95	0.02022	1000	0.07629
73	Decanal	C ₁₀ H ₂₀ O	112-31-2	156.27	-4.9825	0.04928	-1107	-67,349,000	488.15	0.02381	1000	0.10382
74	Decane	C ₁₀ H ₂₂	124-18-5	142.28	-668.4	0.9323	-4.071E+09		447.30	0.02173	1000	0.10286
75	Decanoic acid	C ₁₀ H ₂₀ O ₂	334-48-5	172.27	0.00017047	0.9313	757.67		543.15	0.02508	1000	0.06034
76	1-Decanol	C ₁₀ H ₂₂ O	112-30-1	158.28	-0.3072	0.489	-67,500	-29,400,000	504.07	0.02591	1000	0.09389
77	1-Decene	C ₁₀ H ₂₀	872-05-9	140.27	0.000027232	1.257	751.7		443.75	0.02149	1000	0.09175
78	Decyl mercaptan	C ₁₀ H ₂₂ S	143-10-2	174.35	0.00012058	1.0111	740		512.35	0.02709	1000	0.07482
79	1-Decyne	C ₁₀ H ₁₈	764-93-2	138.25	0.000016707	1.2128	-206.08	153,850	447.15	0.02092	1000	0.07667
80	Deuterium	D ₂	7782-39-0	4.03	0.00028527	0.9874	-200.51	21,807	233.15	0.11474	1500	0.44547
81	1,1-Dibromoethane	C ₂ H ₄ Br ₂	557-91-5	187.86	0.00021231	0.8052	649.51		381.15	0.00940	1000	0.03351
82	1,2-Dibromoethane	C ₂ H ₄ Br ₂	106-93-4	187.86	0.00015878	0.8636	659.5		404.51	0.01077	1000	0.03729
83	Dibromomethane	CH ₂ Br ₂	74-95-3	173.83	0.00021302	0.8719	1,620		370.10	0.00687	1000	0.03356
84	Dibutyl ether	C ₈ H ₁₈ O	142-96-1	130.23	0.0032694	0.58633	1259.9	300,890	323.15	0.01244	1000	0.07330
85	<i>m</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	541-73-1	147.00	-1.067.8	0.754	-3.0361E+09		446.23	0.01561	1000	0.06430
86	<i>o</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	95-50-1	147.00	-1.420	0.7614	-4.5040E+09		453.57	0.01507	1000	0.06066
87	<i>p</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	106-46-7	147.00	-1.520.8	0.754	-4.3328E+09		447.21	0.01564	1000	0.06417
88	1,1-Dichloroethane	C ₂ H ₄ Cl ₂	75-34-3	98.96	0.0001315	1.0113	1,023.8		330.45	0.01132	1000	0.07025
89	1,2-Dichloroethane	C ₂ H ₄ Cl ₂	107-06-2	98.96	0.00021054	0.9574	1,414		356.59	0.01177	1000	0.06498
90	Dichloromethane	CH ₂ Cl ₂	75-09-2	84.93	0.0014796	0.69531	2,657.4		312.90	0.00847	1000	0.04931
91	1,1-Dichloropropane	C ₃ H ₆ Cl ₂	78-99-9	112.99	0.000057603	1.1148	849.98		361.25	0.01220	1000	0.06881
92	1,2-Dichloropropane	C ₃ H ₆ Cl ₂	78-87-5	112.99	0.000062435	1.103	913.43		369.52	0.01222	1000	0.06647
93	Diethanol amine	C ₄ H ₁₁ NO ₂	111-42-2	105.14	-11.633	0.4621	-3.7939E+09		541.54	0.03044	1000	0.07463
94	Diethyl amine	C ₄ H ₁₁ N	109-89-7	73.14	0.00001706	1.248	-112.8	77,960	273.15	0.01148	1000	0.09804
95	Diethyl ether	C ₄ H ₁₀ O	60-29-7	74.12	-0.0044894	0.6155	-3266.3		200.00	0.00764	600	0.05181
96	Diethyl sulfide	C ₄ H ₁₀ S	352-93-2	90.19	0.0018097	0.67406	1179.7	174,850	365.25	0.01743	1000	0.08089
97	1,1-Difluoroethane	C ₂ H ₄ F ₂	75-37-6	66.05	0.000059249	1.0713	101.84	45,974	248.95	0.01016	1000	0.08447
98	1,2-Difluoroethane	C ₂ H ₄ F ₂	624-72-6	66.05	2.4194E-06	1.4456			303.65	0.00938	993.65	0.05206
99	Difluoromethane	CH ₂ F ₂	75-10-5	52.02	0.000013015	1.1897			221.50	0.00803	1000	0.04826
100	Di-isopropyl amine	C ₆ H ₁₅ N	108-18-9	101.19	0.00051305	0.8076	360.19	154,510	357.05	0.01836	1000	0.08967
101	Di-isopropyl ether	C ₆ H ₁₄ O	108-20-3	102.17	0.00019879	0.9423	306.8	106,230	328.05	0.01598	1000	0.09444
102	Di-isopropyl ketone	C ₇ H ₁₄ O	565-80-0	114.19	-8.5357	-0.0056423	1882.1	-65,622,000	397.55	0.02015	1000	0.13085
103	1,1-Dimethoxyethane	C ₄ H ₁₀ O ₂	534-15-6	90.12	0.00046265	0.81968	539.34	104,530	337.45	0.01554	1000	0.08099
104	1,2-Dimethoxypropane	C ₅ H ₁₂ O ₂	7778-85-0	104.15	3.7962E-06	1.4462			366.15	0.01936	1000	0.08279
105	Dimethyl acetylene	C ₄ H ₆	503-17-3	54.09	0.00021761	0.9187	217	132,070	300.13	0.01288	1000	0.09199
106	Dimethyl amine	C ₂ H ₇ N	124-40-3	45.08	1.6085	-0.1103	2160.3	2,989,300	280.03	0.01845	1000	0.12209
107	2,3-Dimethylbutane	C ₆ H ₁₄	79-29-8	86.18	0.000034741	1.1646	-99.956	130,820	331.13	0.01581	1000	0.10506
108	1,1-Dimethylcyclohexane	C ₈ H ₁₆	590-66-9	112.21	0.008856	0.4215	-50.645	764,580	392.70	0.01884	1000	0.09500
109	<i>cis</i> -1,2-Dimethylcyclohexane	C ₈ H ₁₆	2207-01-4	112.21	0.013298	0.3692	0.1027	852,540	402.94	0.01948	1000	0.09196
110	<i>trans</i> -1,2-Dimethylcyclohexane	C ₈ H ₁₆	6876-23-9	112.21	0.012144	0.3854	52.191	803,590	396.58	0.01952	1000	0.09376
111	Dimethyl disulfide	C ₂ H ₆ S ₂	624-92-0	94.20	0.00022578	0.892	697		382.90	0.01613	1000	0.06310
112	Dimethyl ether	C ₂ H ₆ O	115-10-6	46.07	0.059975	0.2667	1,018.6	1,098,800	248.31	0.01139	1500	0.19458
113	<i>N,N</i> -Dimethyl formamide	C ₃ H ₇ NO	68-12-2	73.09	0.014449	0.3612	595.22	728,130	425.15	0.02001	1000	0.07539
114	2,3-Dimethylpentane	C ₇ H ₁₆	565-59-3	100.20	0.000022421	1.2137	-146.91	131,830	362.93	0.01797	1000	0.09962
115	Dimethyl phthalate	C ₁₀ H ₁₀ O ₄	131-11-3	194.18	0.00012822	0.9324	752.5		556.85	0.01981	1000	0.04587
116	Dimethylsilane	C ₂ H ₆ Si	1111-74-6	60.17	0.0011808	0.742	1131	6,400	253.55	0.01291	1000	0.09296

117	Dimethyl sulfide	C ₂ H ₆ S	75-18-3	62.13	0.00023614	0.9204	638		310.48	0.01520	1000	0.08319
118	Dimethyl sulfoxide	C ₂ H ₆ OS	67-68-5	78.13	0.00064761	0.7716	1013.3		462.15	0.02059	1000	0.06379
119	Dimethyl terephthalate	C ₁₀ H ₁₀ O ₄	120-61-6	194.18	-25,190	0.3639	-6,869,000,000		561.15	0.02060	1000	0.04529
120	1,4-Dioxane	C ₄ H ₈ O ₂	123-91-1	88.11	6.4032E-07	1.7194			337.85	0.01427	768.01	0.05855
121	Diphenyl ether	C ₁₂ H ₁₀ O	101-84-8	170.21	0.00014629	0.9377	745.89		531.46	0.02188	1000	0.05449
122	Dipropyl amine	C ₆ H ₁₅ N	142-84-7	101.19	0.0001123	0.9958	183.2	98,000	279.65	0.01055	1000	0.08515
123	Dodecane	C ₁₂ H ₂₆	112-40-3	170.33	0.000005719	1.4699	579.4		489.47	0.02354	1000	0.09301
124	Eicosane	C ₂₀ H ₄₂	112-95-8	282.55	-375.32	1.0708	-8,783,600,000		616.93	0.02563	1000	0.06968
125	Ethane	C ₂ H ₆	74-84-0	30.07	0.000073869	1.1689	500.73		184.55	0.00886	1000	0.15807
126	Ethanol	C ₂ H ₆ O	64-17-5	46.07	-0.010109	0.6475	-7,332	-268,000	293.15	0.01475	1000	0.13417
127	Ethyl acetate	C ₄ H ₈ O ₂	141-78-6	88.11	1.3575E-07	1.9681			273.15	0.00847	990.21	0.10682
128	Ethyl amine	C ₂ H ₇ N	75-04-7	45.08	0.3935	0.0131	1,380	1,710,000	289.73	0.01622	1000	0.10532
129	Ethylbenzene	C ₈ H ₁₀	100-41-4	106.17	0.000017537	1.3144	560.65		409.35	0.02007	1000	0.09859
130	Ethyl benzoate	C ₉ H ₁₀ O ₂	93-89-0	150.17	0.00002012	1.1513	-89.583	125,410	486.55	0.01855	1000	0.05524
131	2-Ethyl butanoic acid	C ₆ H ₁₂ O ₂	88-09-5	116.16	0.00017727	0.9428	712.4		466.95	0.02306	1000	0.06973
132	Ethyl butyrate	C ₆ H ₁₂ O ₂	105-54-4	116.16	829.29	1.0156	8,955,300,000		394.65	0.01583	1000	0.10314
133	Ethylcyclohexane	C ₈ H ₁₆	1678-91-7	112.21	0.0000748	1.1103	686		404.95	0.02180	1000	0.09505
134	Ethylcyclopentane	C ₇ H ₁₄	1640-89-7	98.19	0.0043244	0.5429	333.67	570,470	376.62	0.01832	1000	0.09659
135	Ethylene	C ₂ H ₄	74-85-1	28.05	8.6806E-06	1.4559	299.72	-29,403	170.00	0.00879	590.92	0.06613
136	Ethylenediamine	C ₂ H ₈ N ₂	107-15-3	60.10	0.1655	0.1798	3,827.9	1,600,000	390.41	0.02272	1000	0.08915
137	Ethylene glycol	C ₂ H ₆ O ₂	107-21-1	62.07	-8,145,800	-0.30502	1,832,500,000	-1.1842E+13	470.45	0.02513	1000	0.09896
138	Ethyleneimine	C ₂ H ₅ N	151-56-4	43.07	0.00077079	0.7713	446.16	197,930	329.00	0.01610	1000	0.09659
139	Ethylene oxide	C ₂ H ₄ O	75-21-8	44.05	-0.0003788	1.115	-5,641		273.15	0.01004	1000	0.18063
140	Ethyl formate	C ₃ H ₆ O ₂	109-94-4	74.08	508	0.9023	2,170,000,000		327.46	0.01426	1000	0.11921
141	2-Ethyl hexanoic acid	C ₈ H ₁₆ O ₂	149-57-5	144.21	2.5804E-06	1.4669			500.66	0.02353	1000	0.06492
142	Ethylhexyl ether	C ₈ H ₁₈ O	5756-43-4	130.23	0.0052833	0.52982	1,415.7	378,180	417.15	0.01967	1000	0.07348
143	Ethylisopropyl ether	C ₅ H ₁₂ O	625-54-7	88.15	0.00021652	0.94192	632.16		326.15	0.01717	1000	0.08882
144	Ethylisopropyl ketone	C ₆ H ₁₂ O	565-69-5	100.16	-152,400	-0.049106	80,955,000	-9.3122E+11	386.55	0.01889	1000	0.12768
145	Ethyl mercaptan	C ₂ H ₆ S	75-08-1	62.13	0.0015251	0.70243	1,347.5	35,085	308.15	0.01487	1000	0.08195
146	Ethyl propionate	C ₅ H ₁₀ O ₂	105-37-3	102.13	1.0507E-07	1.9854			400.00	0.01540	1000	0.09499
147	Ethylpropyl ether	C ₅ H ₁₂ O	628-32-0	88.15	5.8174E-08	2.0116	-372.68	57,690	273.15	0.01133	550	0.03690
148	Ethyltrichlorosilane	C ₂ H ₅ Cl ₃ Si	115-21-9	163.51	2.9354E-06	1.4153			373.15	0.01281	573.15	0.02352
149	Fluorine	F ₂	7782-41-4	38.00	0.00012144	0.93831			70.00	0.00654	700	0.05675
150	Fluorobenzene	C ₆ H ₅ F	462-06-6	96.10	0.000053432	1.1576	760.75		357.88	0.01546	600	0.03874
151	Fluoroethane	C ₂ H ₅ F	353-36-6	48.06	0.0004104	0.8333	723		235.45	0.00955	1000	0.07531
152	Fluoromethane	CH ₃ F	593-53-3	34.03	0.003959	0.4834	997.4		194.82	0.00827	1000	0.05589
153	Formaldehyde	CH ₂ O	50-00-0	30.03	44.847	-0.7096	-3,493.5	5,353,200	254.05	0.01256	994.05	0.11532
154	Formamide	CH ₃ NO	75-12-7	45.04	0.00025893	0.9083	723.6		493.00	0.02930	1000	0.07973
155	Formic acid	CH ₂ O ₂	64-18-6	46.03	0.0003754	0.8459	674.4		373.71	0.02008	1000	0.07733
156	Furan	C ₄ H ₄ O	110-00-9	68.07	-644,950	0.2862	-1.6794E+10	-1.7372E+13	304.50	0.01367	1000	0.13631
157	Helium-4	He	7440-59-7	4.00	0.00226	0.7305	-18.63	440	30.00	0.03124	2000	0.58820
158	Heptadecane	C ₁₇ H ₃₆	629-78-7	240.47	-114.41	1.0566	-2,211,400,000		575.30	0.02454	1000	0.07649
159	Heptanal	C ₇ H ₁₄ O	111-71-7	114.19	1,556.7	1.0284	17,049,000,000		425.95	0.01967	1000	0.11110
160	Heptane	C ₇ H ₁₆	142-82-5	100.20	-0.070028	0.38068	-7,049.9	-2,400,500	339.15	0.01583	1000	0.11493
161	Heptanoic acid	C ₇ H ₁₄ O ₂	111-14-8	130.19	0.00019376	0.92434	739.56		496.15	0.02413	1000	0.06605
162	1-Heptanol	C ₇ H ₁₆ O	111-70-6	116.20	-0.061993	0.2792	-3,336	-1,642,000	449.45	0.02345	1000	0.10722
163	2-Heptanol	C ₇ H ₁₆ O	543-49-7	116.20	0.00017569	0.97218	686.56		432.90	0.02484	1000	0.08596
164	3-Heptanone	C ₇ H ₁₄ O	106-35-4	114.19	1,348.6	1.0313	14,832,000,000		420.55	0.01943	1000	0.11287
165	2-Heptanone	C ₇ H ₁₄ O	110-43-0	114.19	2,049.3	1.0323	22,983,000,000		424.18	0.01951	1000	0.11145
166	1-Heptene	C ₇ H ₁₄	592-76-7	98.19	0.00002133	1.2885	487.8		366.79	0.01845	1000	0.10518
167	Heptyl mercaptan	C ₇ H ₁₆ S	1639-09-4	132.27	0.0083145	0.51862	2,253	532,590	450.09	0.02289	1000	0.07899
168	1-Heptyne	C ₇ H ₁₂	628-71-7	96.17	0.000060732	1.0586	-102.79	143,140	372.93	0.01827	1000	0.08751
169	Hexadecane	C ₁₆ H ₃₄	544-76-3	226.44	0.000004438	1.4949	682		560.01	0.02568	1000	0.08055
170	Hexanal	C ₆ H ₁₂ O	66-25-1	100.16	-7,157,100	-0.05819	4,089,000,000	-4.5826E+13	401.45	0.01842	1000	0.11472
171	Hexane	C ₆ H ₁₄	110-54-3	86.18	-650.5	0.8053	-1,412,100,000		339.09	0.01704	1000	0.12003
172	Hexanoic acid	C ₆ H ₁₂ O ₂	142-62-1	116.16	0.00021014	0.91616	727.64		478.85	0.02381	1000	0.06816
173	1-Hexanol	C ₆ H ₁₄ O	111-27-3	102.17	-4,935,500	-0.1653	1,563,100,000	-1.5752E+13	429.90	0.02220	1000	0.11104
174	2-Hexanol	C ₆ H ₁₄ O	626-93-7	102.18	0.00018361	0.97199	677.05		412.40	0.02421	1000	0.09022
175	2-Hexanone	C ₆ H ₁₂ O	591-78-6	100.16	-1.2158	0.026637	-1,711.6	-13,176,000	273.00	0.00775	1000	0.10523
176	3-Hexanone	C ₆ H ₁₂ O	589-38-8	100.16	-0.33262	0.12054	-2,472.6	-5,493,400	273.00	0.00800	1000	0.10980
177	1-Hexene	C ₆ H ₁₂	592-41-6	84.16	0.000064256	1.1355	445.15	64,810	336.63	0.01644	1000	0.10850
178	3-Hexyne	C ₆ H ₁₀	928-49-4	82.14	6.9682E-06	1.347	-214.35	110,480	354.35	0.01485	1000	0.08546
179	Hexyl mercaptan	C ₆ H ₁₄ S	111-31-9	118.24	0.074318	0.30035	4,470.1	1,775,800	425.81	0.02151	1000	0.08167

TABLE 2-314 Vapor Thermal Conductivity of Inorganic and Organic Substances [W/(m·K)] (Continued)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1	C2	C3	C4	$T_{\min}$, K	Thermal cond. at $T_{\min}$	$T_{\max}$, K	Thermal cond. at $T_{\max}$
180	1-Hexyne	C ₆ H ₁₀	693-02-7	82.14	0.000058116	1.0724	-77.165	123,900	344.48	0.01679	1000	0.09155
181	2-Hexyne	C ₆ H ₁₀	764-35-2	82.14	0.000011631	1.2753	-202.84	122,990	357.67	0.01506	1000	0.08466
182	Hydrazine	H ₄ N ₂	302-01-2	32.05	0.00043196	0.86603	641.48		386.65	0.02828	1000	0.10430
183	Hydrogen	H ₂	1333-74-0	2.02	0.002653	0.7452	12		22.00	0.01718	1600	0.64299
184	Hydrogen bromide	HBr	10035-10-6	80.91	0.00049725	0.63088	331.62		206.45	0.00551	600	0.01812
185	Hydrogen chloride	HCl	7647-01-0	36.46	0.001865	0.49755	358		190.00	0.00880	700	0.03213
186	Hydrogen cyanide	CHN	74-90-8	27.03	4.6496E-06	1.3669	-210.76	58,295	273.15	0.00985	673.15	0.04185
187	Hydrogen fluoride	HF	7664-39-3	20.01	0.000034629	1.1224	18.744		350.00	0.02356	450	0.03160
188	Hydrogen sulfide	H ₂ S	7783-06-4	34.08	1.381E-07	1.8379	-352.09	46,041	212.80	0.00724	600	0.03258
189	Isobutyric acid	C ₄ H ₈ O ₂	79-31-2	88.11	0.000214	0.9248	698		427.85	0.02206	1000	0.07497
190	Isopropyl amine	C ₃ H ₉ N	75-31-0	59.11	0.00028183	0.92094	619.17		304.92	0.01804	1000	0.10081
191	Malonic acid	C ₃ H ₄ O ₄	141-82-2	104.06	0.00033075	0.81895	777.75		580.00	0.02590	1000	0.05327
192	Methacrylic acid	C ₄ H ₆ O ₂	79-41-4	86.09	0.00019847	0.9284	678.69		434.15	0.02176	1000	0.07210
193	Methane	CH ₄	74-82-8	16.04	8.3983E-06	1.4268	-49.654		111.63	0.01263	600	0.08425
194	Methanol	CH ₃ O	67-56-1	32.04	5.7992E-07	1.7862			273.00	0.01303	684.37	0.06726
195	N-Methyl acetamide	C ₃ H ₇ NO	79-16-3	73.09	0.034177	0.3312	2,070	1,195,600	478.15	0.02498	1000	0.07895
196	Methyl acetate	C ₃ H ₆ O ₂	79-20-9	74.08	-25.343	-0.1934	11,164,000	-67,259,000,000	330.09	0.01415	1000	0.11878
197	Methyl acetylene	C ₃ H ₄	74-99-7	40.06	0.00026544	0.8921	222.19	79,869	249.94	0.01154	1000	0.09675
198	Methyl acrylate	C ₄ H ₆ O ₂	96-33-3	86.09	0.4734	-0.1111	533.57	1,649,600	353.35	0.01569	1000	0.06904
199	Methyl amine	CH ₃ N	74-89-5	31.06	-55.13	1.065	-448,200,000		266.82	0.01259	650	0.07917
200	Methyl benzoate	C ₈ H ₈ O ₂	93-58-3	136.15	0.000023963	1.1308	-67.272	125,720	472.65	0.01784	1000	0.05588
201	3-Methyl-1,2-butadiene	C ₅ H ₈	598-25-4	68.12	0.0002509	0.899	253.4	149,500	314.00	0.01326	1000	0.08902
202	2-Methylbutane	C ₅ H ₁₂	78-78-4	72.15	0.0008968	0.7742	456	230,640	273.15	0.01198	1000	0.11176
203	2-Methylbutanoic acid	C ₅ H ₁₀ O ₂	116-53-0	102.13	0.0001799	0.9457	704.6		450.15	0.02266	1000	0.07253
204	3-Methyl-1-butanol	C ₅ H ₁₂ O	123-51-3	88.15	2,053.4	0.90109	8,755,500,000		404.15	0.02116	1000	0.11843
205	2-Methyl-1-butene	C ₅ H ₁₀	563-46-2	70.13	0.00019098	0.9341	84.07	155,720	304.30	0.01348	1000	0.09771
206	2-Methyl-2-butene	C ₅ H ₁₀	513-35-9	70.13	0.00021736	0.9171	112.3	177,690	311.71	0.01320	1000	0.09504
207	2-Methyl-1-butene-3-yne	C ₅ H ₆	78-80-8	66.10	0.00015498	0.9364	15.366	137,400	305.40	0.01304	1000	0.08664
208	Methylbutyl ether	C ₆ H ₁₄ O	628-28-4	88.15	0.000023993	1.1976	58.59	35,667	273.15	0.01173	1000	0.08586
209	Methylbutyl sulfide	C ₆ H ₁₂ S	628-29-5	104.21	0.079414	0.23442	2,671.9	1,366,100	396.58	0.01966	1000	0.07960
210	3-Methyl-1-butene	C ₅ H ₁₀	598-23-2	68.12	0.000065855	1.072	-36.369	106,430	302.15	0.01468	1000	0.10120
211	Methyl butyrate	C ₆ H ₁₀ O ₂	623-42-7	102.13	1,333.1	0.9962	12,317,000,000		375.90	0.01495	1000	0.10543
212	Methylchlorosilane	CH ₃ ClSi	993-00-0	80.59	0.00037057	0.81367	609.17		281.85	0.01155	1000	0.06357
213	Methylcyclohexane	C ₇ H ₁₄	108-87-2	98.19	0.0000719	1.1274	667		374.08	0.02056	1000	0.10399
214	1-Methylcyclohexanol	C ₇ H ₁₄ O	590-67-0	114.19	0.00011359	1.0311	709.27		441.15	0.02322	1000	0.08238
215	cis-2-Methylcyclohexanol	C ₇ H ₁₄ O	7443-70-1	114.19	0.069565	0.1633	208.7	1,209,500	438.15	0.02415	1000	0.08888
216	trans-2-Methylcyclohexanol	C ₇ H ₁₄ O	7443-52-9	114.19	0.075448	0.155	218.44	1,252,500	440.15	0.02435	1000	0.08908
217	Methylcyclopentane	C ₆ H ₁₂	96-37-7	84.16	0.0024385	0.61774	223.01	477,570	344.96	0.01592	1000	0.10227
218	1-Methylcyclopentene	C ₆ H ₁₀	693-89-0	82.14	0.0040082	0.54462	242.12	559,040	348.64	0.01544	1000	0.09578
219	3-Methylcyclopentene	C ₆ H ₁₀	1120-62-3	82.14	0.0019845	0.6393	227.11	434,120	338.05	0.01501	1000	0.09888
220	Methyldichlorosilane	CH ₂ Cl ₂ Si	75-54-7	115.03	0.00041077	0.75688	591.5		314.70	0.01109	1000	0.04813
221	Methylethyl ether	C ₃ H ₈ O	540-67-0	60.10	0.00024036	0.93177	588.14		273.00	0.01419	1000	0.09447
222	Methylethyl ketone	C ₄ H ₈ O	78-93-3	72.11	-4,202,700	-0.1524	2,084,600,000	-1.4577E+13	352.79	0.01546	1000	0.11740
223	Methylethyl sulfide	C ₄ H ₈ S	624-89-5	76.16	0.0034805	0.61906	1,810.8	166,290	339.80	0.01653	1000	0.08415
224	Methyl formate	C ₂ H ₄ O ₂	107-31-3	60.05	-800,040	-0.2285	248,100,000	-1.5034E+12	300.00	0.01369	1000	0.13148
225	Methylisobutyl ether	C ₅ H ₁₂ O	625-44-5	88.15	0.00020053	0.95381	644.42		331.70	0.01729	1000	0.08863
226	Methylisobutyl ketone	C ₆ H ₁₂ O	108-10-1	100.16	-2,483,300	-0.046517	1,313,100,000	-1.5798E+13	389.65	0.01869	1000	0.12433
227	Methyl Isocyanate	C ₂ H ₃ NO	624-83-9	57.05	0.0026136	0.62	1,631.7	126,720	312.00	0.01221	1000	0.06864
228	Methylisopropyl ether	C ₄ H ₁₀ O	598-53-8	74.12	2.1191	-0.19015	1,453.4	3,575,500	303.92	0.01606	1000	0.09451
229	Methylisopropyl ketone	C ₅ H ₁₀ O	563-80-4	86.13	-5,935,000	-0.089497	3,098,800,000	-2.7994E+13	367.55	0.01760	1000	0.12847
230	Methylisopropyl sulfide	C ₅ H ₁₀ S	1551-21-9	90.19	0.0071536	0.53907	2,700.7	241,730	171.64	0.00459	1000	0.07516
231	Methyl mercaptan	CH ₃ S	74-93-1	48.11	0.00002653	1.1631	29.996	32,519	273.15	0.01171	1000	0.07704
232	Methyl methacrylate	C ₅ H ₈ O ₂	80-62-6	100.12	0.00072502	0.7395	365.68	204,360	373.45	0.01680	1000	0.07637
233	2-Methyloctanoic acid	C ₈ H ₁₆ O ₂	3004-93-1	158.24	0.0001813	0.92912	793.45		518.15	0.02383	1000	0.06195
234	2-Methylpentane	C ₆ H ₁₄	107-83-5	86.18	0.000061119	1.0861	-59.592	141,260	333.41	0.01606	1000	0.10242
235	Methyl pentyl ether	C ₆ H ₁₄ O	628-80-8	102.17	0.93312	-0.1172	1,154.3	2,961,700	372.00	0.01828	1000	0.08117
236	2-Methylpropane	C ₄ H ₁₀	75-28-5	58.12	0.089772	0.18501	639.23	1,114,700	261.43	0.01273	1000	0.11701
237	2-Methyl-2-propanol	C ₄ H ₁₀ O	75-65-0	74.12	1.1776E-06	1.6618			333.82	0.01839	766.87	0.07325

238	2-Methyl propene	C ₄ H ₈	115-11-7	56.11	-488.1	0.8877	-1,448,500,000	266.25	0.01276	1000	0.15513
239	Methyl propionate	C ₅ H ₈ O ₂	554-12-1	88.11	-200.9	-0.1321	104,000	350.00	0.01402	1000	0.10886
240	Methylpropyl ether	C ₅ H ₁₀ O	557-17-5	74.12	0.011136	0.4831	2,170.3	312.20	0.01648	1000	0.09079
241	Methylpropyl sulfide	C ₅ H ₁₀ S	3877-15-4	90.19	0.0023574	0.67434	1,804.1	368.69	0.01802	1000	0.08398
242	Methylsilane	CH ₃ Si	992-94-9	46.14	12.248	-0.5611	-1,067	216.25	0.01108	1000	0.09590
243	alpha-Methyl styrene	C ₉ H ₁₀	98-83-9	118.18	0.21276	-0.022299	-194.68	438.65	0.01969	1000	0.07255
244	Methyl tert-butyl ether	C ₅ H ₁₂ O	1634-04-4	88.15	0.0002084	0.93034	364.832	273.00	0.01161	1000	0.08958
245	Methyl vinyl ether	C ₃ H ₆ O	107-25-5	58.08	0.00032359	0.8892	623.22	278.65	0.01493	1000	0.09273
246	Naphthalene	C ₁₀ H ₈	91-20-3	128.17	0.000091828	1.0345	731.78	491.14	0.02243	1000	0.06730
247	Neon	Ne	7440-01-9	20.18	0.0011385	0.6646	8.7	30.00	0.00846	3273.1	0.24616
248	Nitroethane	C ₂ H ₅ NO ₂	79-24-3	75.07	0.0011282	0.6895	679.11	387.22	0.01580	1000	0.06887
249	Nitrogen	N ₂	7727-37-9	28.01	0.00033143	0.7722	16.323	63.15	0.00602	2000	0.11638
250	Nitrogen trifluoride	F ₃ N	7783-54-2	71.00	2.1443	-0.30545	1,860.3	144.09	0.00648	1000	0.06377
251	Nitromethane	CH ₃ NO ₂	75-52-5	61.04	0.00003135	1.1119	-91.6	374.35	0.01365	1000	0.06553
252	Nitrous oxide	N ₂ O	10024-97-2	44.01	0.001096	0.667	540	182.30	0.00891	1000	0.07133
253	Nitric oxide	NO	10102-43-9	30.01	0.0004096	0.7509	45.6	121.38	0.01094	750	0.05567
254	Nonadecane	C ₁₉ H ₄₀	629-92-5	268.52	0.000049571	1.2652	3,332.3	603.05	0.02502	1000	0.07147
255	Nonanal	C ₉ H ₁₈ O	124-19-6	142.24	-11.621	0.025653	2,248.3	468.15	0.02228	1000	0.10522
256	Nonane	C ₉ H ₂₀	111-84-2	128.26	-0.065771	0.27198	-3,482.3	423.97	0.02130	1000	0.10597
257	Nonanoic acid	C ₉ H ₁₈ O ₂	112-05-0	158.24	0.000178	0.9288	753	528.75	0.02484	1000	0.06209
258	1-Nonanol	C ₉ H ₂₀ O	143-08-8	144.25	-30.715	-0.1075	8,107	485.20	0.02436	1000	0.09895
259	2-Nonanol	C ₉ H ₂₀ O	628-99-9	144.26	0.00016337	0.97256	709.74	471.70	0.02599	1000	0.07905
260	1-Nonene	C ₉ H ₁₈	124-11-8	126.24	0.000021269	1.2943	662.21	420.02	0.02051	1000	0.09771
261	Nonyl mercaptan	C ₉ H ₂₀ S	1455-21-6	160.32	0.047041	0.29733	2,460.6	492.95	0.02559	1000	0.07598
262	1-Nonyne	C ₉ H ₁₆	3452-09-3	124.22	0.000016681	1.218	-199.41	423.85	0.01981	1000	0.07956
263	Octadecane	C ₁₈ H ₃₈	593-45-3	254.49	-291.08	1.0615	-6,019,900,000	589.86	0.02491	1000	0.07395
264	Octanal	C ₈ H ₁₆ O	124-13-0	128.21	-110.89	-0.000042988	51,384	447.15	0.02117	1000	0.10893
265	Octane	C ₈ H ₁₈	111-65-9	114.23	-8,758	0.8448	-2,7121E+10	339.00	0.01503	1000	0.11053
266	Octanoic acid	C ₈ H ₁₆ O ₂	124-07-2	144.21	0.00018263	0.9283	741.3	513.05	0.02450	1000	0.06391
267	1-Octanol	C ₈ H ₁₈ O	111-87-5	130.23	-0.0030238	0.8745	-13,352	468.35	0.02380	1000	0.10288
268	2-Octanol	C ₈ H ₁₈ O	123-96-6	130.23	0.00016915	0.97238	698.55	452.90	0.02545	1000	0.08229
269	2-Octanone	C ₈ H ₁₆ O	111-13-7	128.21	-0.0020184	1.0027	-20,406	446.15	0.02046	1000	0.10597
270	3-Octanone	C ₈ H ₁₆ O	106-68-3	128.21	8.1833E-08	2.0418		440.65	0.02050	1000	0.10923
271	1-Octene	C ₈ H ₁₆	111-66-0	112.21	0.0000133	1.3554	504.59	394.41	0.01926	1000	0.10295
272	Octyl mercaptan	C ₈ H ₁₆ S	111-88-6	146.29	-3,965.5	0.5213	-1,851,900,000	472.19	0.02505	1000	0.07845
273	1-Octyne	C ₈ H ₁₄	629-05-0	110.20	0.000060734	1.0516	-124.91	399.35	0.01967	1000	0.08394
274	Oxalic acid	C ₂ H ₂ O ₄	144-62-7	90.03	0.05881	0.278	14,815	569.00	0.01269	1000	0.02537
275	Oxygen	O ₂	7782-44-7	32.00	0.00044994	0.7456	56.699	80.00	0.00691	2000	0.12655
276	Ozone	O ₃	10028-15-6	48.00	0.0043147	0.47999	700.09	161.85	0.00931	1000	0.06990
277	Pentadecane	C ₁₅ H ₃₂	629-62-9	212.41	4.7796E-06	1.4851	643.13	543.84	0.02529	1000	0.08299
278	Pentanal	C ₅ H ₁₀ O	110-62-3	86.13	-4,918,700	-0.10297	2,691,100,000	376.15	0.01705	1000	0.11788
279	Pentane	C ₅ H ₁₂	109-66-0	72.15	-684.4	0.764	-1,055,000,000	273.15	0.01288	1000	0.12707
280	Pentanoic acid	C ₅ H ₁₀ O ₂	109-52-4	102.13	0.00024601	0.8946	696.42	458.65	0.02349	1000	0.07002
281	1-Pentanol	C ₅ H ₁₂ O	71-41-0	88.15	2,896	0.8985	12,735,000,000	410.90	0.02084	990.95	0.11087
282	2-Pentanol	C ₅ H ₁₂ O	6032-29-7	88.15	0.00019575	0.9692	664.04	392.20	0.02372	1000	0.09509
283	2-Pentanone	C ₅ H ₁₀ O	107-87-9	86.13	-0.01719	0.4832	-3,798	273.00	0.00877	1000	0.12002
284	3-Pentanone	C ₅ H ₁₀ O	96-22-0	86.13	22.775	1.0019	191,000,000	273.00	0.00898	1000	0.12082
285	1-Pentene	C ₅ H ₁₀	109-67-1	70.13	2.7081E-06	1.5493	41,075	303.22	0.01546	1000	0.11472
286	2-Pentyl mercaptan	C ₅ H ₁₂ S	2084-19-7	104.21	0.00022307	0.93358	794.16	385.15	0.01890	1000	0.07858
287	Pentyl mercaptan	C ₅ H ₁₂ S	110-66-7	104.21	0.00011261	1.034	693.05	399.79	0.02019	1000	0.08412
288	1-Pentyne	C ₅ H ₈	627-19-0	68.12	0.000052415	1.0948	-51.09	313.33	0.01517	1000	0.09608
289	2-Pentyne	C ₅ H ₈	627-21-4	68.12	0.00025623	1.0073	1,423.7	329.27	0.01653	1000	0.11119
290	Phenanthrene	C ₁₄ H ₁₀	85-01-8	178.23	0.00010167	0.988	797	610.03	0.02490	1000	0.05208
291	Phenol	C ₆ H ₆ O	108-95-2	94.11	0.038846	0.2392	985.81	454.99	0.02183	1000	0.06936
292	Phenyl isocyanate	C ₇ H ₅ NO	103-71-9	119.12	0.00016675	0.91777	730.1	439.43	0.01669	1000	0.05461
293	Phthalic anhydride	C ₈ H ₄ O ₃	85-44-9	148.12	0.0000593	1.046	765.5	557.65	0.01864	1000	0.04615
294	Propadiene	C ₃ H ₄	463-49-0	40.06	0.000061629	1.0731	1,8579	238.65	0.00980	1000	0.09526
295	Propane	C ₃ H ₈	74-98-6	44.10	-1.12	0.10972	-9,834.6	231.11	0.01114	1000	0.14599
296	1-Propanol	C ₃ H ₈ O	71-23-8	60.10	-613.84	0.7927	-1,157,400,000	370.35	0.02135	720.25	0.07034
297	2-Propanol	C ₃ H ₈ O	67-63-0	60.10	7.3907E-07	1.7419		355.30	0.02049	1000	0.12428
298	Propenylcyclohexene	C ₉ H ₁₄	13511-13-2	122.21	0.00010242	1.0486	701.56	431.65	0.02262	1000	0.08421
299	Propionaldehyde	C ₃ H ₆ O	123-38-6	58.08	1,165.1	0.90419	5,472,900,000	321.15	0.01263	1000	0.10983
300	Propionic acid	C ₃ H ₆ O ₂	79-09-4	74.08	0.00022286	0.91704	678.21	414.32	0.02124	1000	0.07487

TABLE 2-314 Vapor Thermal Conductivity of Inorganic and Organic Substances [W/(m·K)] (Concluded)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1	C2	C3	C4	$T_{\min}$, K	Thermal cond. at $T_{\min}$	$T_{\max}$, K	Thermal cond. at $T_{\max}$
301	Propionitrile	C ₃ H ₅ N	107-12-0	55.08	0.001321	1.2202	51,822		370.50	0.01278	990.5	0.11209
302	Propyl acetate	C ₅ H ₁₀ O ₂	109-60-4	102.13	1,325.3	1	12,235,000,000		374.65	0.01520	1000	0.10832
303	Propyl amine	C ₃ H ₉ N	107-10-8	59.11	0.2833	0.055046	1,325.9	1,817,600	321.00	0.01709	1000	0.10000
304	Propylbenzene	C ₉ H ₁₂	103-65-1	120.19	0.16992	0.021258	-54.484	1,624,800	432.39	0.02022	1000	0.07658
305	Propylene	C ₃ H ₆	115-07-1	42.08	0.0000449	1.2018	421		225.45	0.01054	1000	0.12737
306	Propyl formate	C ₄ H ₈ O ₂	110-74-7	88.11	740.1	0.9732	5,646,000,000		353.97	0.01403	1000	0.10893
307	2-Propyl mercaptan	C ₃ H ₈ S	75-33-2	76.16	0.00018367	0.9627	646.01		325.71	0.01616	1000	0.08624
308	Propyl mercaptan	C ₃ H ₈ S	107-03-9	76.16	0.0087425	0.51733	2,358.1	334,590	340.87	0.01654	1000	0.08439
309	1,2-Propylene glycol	C ₃ H ₈ O ₂	57-55-6	76.09	0.0001666	0.9765	706		460.75	0.02624	1000	0.08302
310	Quinone	C ₆ H ₄ O ₂	106-51-4	108.09	-5,678,600	-0.045252	2,615,700,000	-3.5415E+13	454.00	0.02593	1000	0.12665
311	Silicon tetrafluoride	F ₄ Si	7783-61-1	104.08	0.0000955	0.928	63.6		333.55	0.01761	702.45	0.03837
312	Styrene	C ₈ H ₈	100-42-5	104.15	0.010048	0.4033	553.74	685,570	418.31	0.01837	1000	0.07276
313	Succinic acid	C ₄ H ₆ O ₄	110-15-6	118.09	0.00032875	0.8172	740.97		591.00	0.02685	1000	0.05342
314	Sulfur dioxide	O ₂ S	7446-09-5	64.06	10.527	-0.7732	-1,333	1,506,400	250.00	0.00745	900	0.03969
315	Sulfur hexafluoride	F ₆ S	2551-62-4	146.06	0.00048883	0.6518	-117.08	78,863	273.15	0.01163	1000	0.04587
316	Sulfur trioxide	O ₃ S	7446-11-9	80.06	1.0702	-0.2348	2,010.4	1,277,000	317.90	0.01386	1000	0.04930
317	Terephthalic acid	C ₈ H ₆ O ₄	100-21-0	166.13	0.00017531	0.8901	909.56		832.00	0.03328	1000	0.04297
318	<i>o</i> -Terphenyl	C ₁₈ H ₁₄	84-15-1	230.30	0.000078652	0.95174	-282.82	289,490	373.15	0.00950	1000	0.05598
319	Tetradecane	C ₁₄ H ₃₀	629-59-4	198.39	-163.62	0.9193	-1.0876E+09		526.73	0.02517	1000	0.08615
320	Tetrahydrofuran	C ₄ H ₈ O	109-99-9	72.11	9.5521E-06	1.4561	662.22		339.12	0.01564	1000	0.13419
321	1,2,3,4-Tetrahydronaphthalene	C ₁₀ H ₁₂	119-64-2	132.20	0.00007754	1.0778	729		480.77	0.02395	1000	0.07676
322	Tetrahydrothiophene	C ₄ H ₈ S	110-01-0	88.17	0.00085604	0.7297	531.99	213,840	394.27	0.01801	1000	0.07579
323	2,2,3,3-Tetramethylbutane	C ₈ H ₁₈	594-82-1	114.23	0.000015235	1.2816	-111.88	124,120	379.44	0.01964	1000	0.10528
324	Thiophene	C ₄ H ₄ S	110-02-1	84.14	0.00013384	0.98115	645.95		357.31	0.01525	1000	0.07139
325	Toluene	C ₇ H ₈	108-88-3	92.14	0.00002392	1.2694	537		383.78	0.01901	1000	0.10007
326	1,1,2-Trichloroethane	C ₂ H ₃ Cl ₃	79-00-5	133.40	0.0000952	1.0423	1,243.3		387.00	0.01125	1000	0.05684
327	Tridecane	C ₁₃ H ₂₈	629-50-5	184.36	5.3701E-06	1.4751	599.09		508.62	0.02422	1000	0.08942
328	Triethyl amine	C ₆ H ₁₅ N	121-44-8	101.19	0.000106	1.0161	91	132,900	273.15	0.01018	1000	0.09680
329	Trimethyl amine	C ₃ H ₉ N	75-50-3	59.11	0.00027648	0.901	167.68	132,200	273.15	0.01280	1000	0.10734
330	1,2,3-Trimethylbenzene	C ₉ H ₁₂	526-73-8	120.19	0.000098408	1.0452	720.49		449.27	0.02238	1000	0.07816
331	1,2,4-Trimethylbenzene	C ₉ H ₁₂	95-63-6	120.19	0.00008498	1.061	708		442.53	0.02098	1000	0.07583
332	2,2,4-Trimethylpentane	C ₈ H ₁₈	540-84-1	114.23	0.00001758	1.3114	392.9		355.15	0.01846	1000	0.10847
333	2,3,3-Trimethylpentane	C ₈ H ₁₈	560-21-4	114.23	0.000020248	1.2284	-174.72	147,800	387.91	0.02001	1000	0.10079
334	1,3,5-Trinitrobenzene	C ₆ H ₃ N ₃ O ₆	99-35-4	213.10	0.00020544	0.87137	807.3		629.60	0.02474	1000	0.04675
335	2,4,6-Trinitrotoluene	C ₇ H ₅ N ₃ O ₆	118-96-7	227.13	0.00018189	0.88744	803.39		625.00	0.02410	1000	0.04635
336	Undecane	C ₁₁ H ₂₄	1120-21-4	156.31	0.038012	0.68615	34,663	8,721,900	469.08	0.02259	1000	0.09798
337	1-Undecanol	C ₁₁ H ₂₄ O	112-42-5	172.31	2,498.8	0.95209	2,0167E+10		520.30	0.02486	1000	0.08899
338	Vinyl acetate	C ₄ H ₆ O ₂	108-05-4	86.09	-3,279,500	-0.12941	1.7104E+09	-1.2727E+13	345.65	0.01515	1000	0.12177
339	Vinyl acetylene	C ₄ H ₄	689-97-4	52.07	0.000054197	1.0632	-70.589	9,0617	278.25	0.01123	1000	0.08222
340	Vinyl chloride	C ₂ H ₃ Cl	75-01-4	62.50	-229.41	0.59582	-169,430,000		259.25	0.00963	1000	0.08300
341	Vinyl trichlorosilane	C ₂ H ₃ Cl ₃ Si	75-94-5	161.49	3510.8	0.225	401,720,000		363.85	0.01198	1000	0.04135
342	Water	H ₂ O	7732-18-5	18.02	6.2041E-06	1.3973			273.16	0.01574	1073.15	0.10652
343	<i>m</i> -Xylene	C ₈ H ₁₀	108-38-3	106.17	3.0593E-09	2.4182	-569.28	121,060	320	0.00867	1000	0.09965
344	<i>o</i> -Xylene	C ₈ H ₁₀	95-47-6	106.17	4.9707E-06	1.3787	-225.64	66,786	320	0.01492	1000	0.08084
345	<i>p</i> -Xylene	C ₈ H ₁₀	106-42-3	106.17	9.9305E-08	1.9229	-469.93	113,460	320	0.01019	1000	0.09060

The vapor thermal conductivity is calculated by

$$k = \frac{C1T^{C2}}{1 + C3/T + C4/T^2}$$

where k is the thermal conductivity in W/(m·K) and T is the temperature in K. Thermal conductivities are at either 1 atm or the vapor pressure, whichever is lower. All substances are listed by chemical family in Table 2-6 and by formula in Table 2-7.

Values in this table were taken from the Design Institute for Physical Properties (DIPPR) of the American Institute of Chemical Engineers (AIChE), copyright 2007 AIChE and reproduced with permission of AIChE and of the DIPPR Evaluated Process Design Data Project Steering Committee. Their source should be cited as R. L. Rowley, W. V. Wilding, J. L. Oscarson, Y. Yang, N. A. Zundel, T. E. Daubert, R. P. Danner, DIPPR® Data Compilation of Pure Chemical Properties, Design Institute for Physical Properties, AIChE, New York (2007).

The number of digits provided for values at $T_{\min}$ and $T_{\max}$ was chosen for uniformity of appearance and formatting; these do not represent the uncertainties of the physical quantities, but are the result of calculations from the standard thermophysical property formulations within a fixed format.

TABLE 2-315 Thermal Conductivity of Inorganic and Organic Liquids [W/(m·K)]

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1	C2	C3	C4	C5	$T_{\min}$, K	Thermal cond. at $T_{\min}$	$T_{\max}$, K	Thermal cond. at $T_{\max}$
1	Acetaldehyde	C ₂ H ₄ O	75-07-0	44.05	0.311	-0.000436				150.15	0.2455	294.00	0.1828
2	Acetamide	C ₂ H ₅ NO	60-35-5	59.07	0.39363	-0.00037053				353.33	0.2627	494.30	0.2105
3	Acetic acid	C ₂ H ₄ O ₂	64-19-7	60.05	0.214	-0.0001834				289.81	0.1609	391.05	0.1423
4	Acetic anhydride	C ₄ H ₆ O ₃	108-24-7	102.09	0.23638	-0.00024263				200.15	0.1878	412.70	0.1363
5	Acetone	C ₃ H ₆ O	67-64-1	58.08	0.2878	-0.000427				178.45	0.2116	343.15	0.1413
6	Acetonitrile	C ₂ H ₃ N	75-05-8	41.05	0.33192	-0.00043243				229.32	0.2328	349.32	0.1809
7	Acetylene	C ₂ H ₂	74-86-2	26.04	0.33363	-0.00083655				192.40	0.1727	250.00	0.1245
8	Acrolein	C ₃ H ₄ O	107-02-8	56.06	0.2703	-0.0003764				185.45	0.2005	325.84	0.1477
9	Acrylic acid	C ₃ H ₄ O ₂	79-10-7	72.06	0.2441	-0.0002904				286.15	0.1610	484.50	0.1034
10	Acrylonitrile	C ₃ H _{3.5} N	107-13-1	53.06	0.28941	-0.00041691				189.63	0.2104	350.50	0.1433
11	Air	Mixture	132259-10-0	28.96	0.28472	-0.0017393				75.00	0.1543	125.00	0.0673
12	Ammonia	H ₃ N	7664-41-7	17.03	1.169	-0.002314				195.41	0.1768	400.05	0.2433
13	Anisole	C ₇ H ₈ O	100-66-3	108.14	0.23494	-0.00026477				235.65	0.1726	512.50	0.0992
14	Argon	Ar	7440-37-1	39.95	0.1819	-0.0003176	-0.00000411			83.78	0.1264	150.00	0.0418
15	Benzamide	C ₇ H ₇ NO	55-21-0	121.14	0.28485	-0.00025225				403.00	0.1832	563.15	0.1428
16	Benzene	C ₆ H ₆	71-43-2	78.11	0.23444	-0.00030572				278.68	0.1492	413.10	0.1082
17	Benzenethiol	C ₆ H ₆ S	108-98-5	110.18	0.20996	-0.0002146				258.27	0.1545	442.29	0.1150
18	Benzoic acid	C ₇ H ₆ O ₂	65-85-0	122.12	0.2391	-0.0002325				395.45	0.1472	596.00	0.1005
19	Benzonitrile	C ₇ H ₅ N	100-47-0	103.12	0.21284	-0.00021587				260.40	0.1566	464.15	0.1126
20	Benzophenone	C ₁₃ H ₁₀ O	119-61-9	182.22	0.25867	-0.00022516				321.35	0.1863	664.00	0.1092
21	Benzyl alcohol	C ₇ H ₈ O	100-51-6	108.14	0.17847	-0.000065843				257.85	0.1615	478.60	0.1470
22	Benzyl ethyl ether	C ₉ H ₁₂ O	539-30-0	136.19	0.2029	-0.0002226				275.65	0.1415	528.60	0.0852
23	Benzyl mercaptan	C ₇ H ₈ S	100-53-8	124.20	0.20316	-0.00019912				243.95	0.1546	472.03	0.1092
24	Biphenyl	C ₁₂ H ₁₀	92-52-4	154.21	0.19053	-0.00015145				342.20	0.1387	723.15	0.0810
25	Bromine	Br ₂	7726-95-6	159.81	-0.2185	0.0042143	-0.000017753	3.1041E-08	-2.0108E-11	266.00	0.1299	584.00	0.0316
26	Bromobenzene	C ₆ H ₅ Br	108-86-1	157.01	0.16983	-0.0001981				242.43	0.1218	429.24	0.0848
27	Bromoethane	C ₂ H ₅ Br	74-96-4	108.97	0.1799	-0.000262				154.55	0.1394	414.14	0.0714
28	Bromomethane	CH ₃ Br	74-83-9	94.94	0.1912	-0.000299				179.47	0.1375	370.10	0.0805
29	1,2-Butadiene	C ₄ H ₆	590-19-2	54.09	0.21966	-0.0003436				136.95	0.1726	284.00	0.1221
30	1,3-Butadiene	C ₄ H ₆	106-99-0	54.09	0.22231	-0.0003664				164.25	0.1621	268.74	0.1238
31	Butane	C ₄ H ₁₀	106-97-8	58.12	0.27349	-0.00071267	5.1555E-07			134.86	0.1868	400.00	0.0709
32	1,2-Butanediol	C ₄ H ₁₀ O ₂	584-03-2	90.12	0.064621	0.00067625	-1.0491E-06			220.00	0.1626	469.57	0.1509
33	1,3-Butanediol	C ₄ H ₁₀ O ₂	107-88-0	90.12	-0.0032865	0.0011463	-1.5525E-06			196.15	0.1618	481.38	0.1888
34	1-Butanol	C ₄ H ₁₀ O	71-36-3	74.12	0.2136	-0.0002034				183.85	0.1762	391.90	0.1339
35	2-Butanol	C ₄ H ₁₀ O	78-92-2	74.12	0.22787	-0.00030727				158.45	0.1792	372.90	0.1133
36	1-Butene	C ₄ H ₈	106-98-9	56.11	0.22153	-0.00035023				87.80	0.1908	266.91	0.1281
37	cis-2-Butene	C ₄ H ₈	590-18-1	56.11	0.21378	-0.00035445				134.26	0.1662	276.87	0.1156
38	trans-2-Butene	C ₄ H ₈	624-64-6	56.11	0.21153	-0.00035056				167.62	0.1528	274.03	0.1155
39	Butyl acetate	C ₆ H ₁₂ O ₂	123-86-4	116.16	0.21721	-0.00026563				199.65	0.1642	453.75	0.0967
40	Butylbenzene	C ₁₀ H ₁₄	104-51-8	134.22	0.18707	-0.00020037				185.30	0.1499	473.15	0.0923
41	Butyl mercaptan	C ₄ H ₁₀ S	109-79-5	90.19	0.21143	-0.000258				157.46	0.1708	371.61	0.1156
42	sec-Butyl mercaptan	C ₄ H ₁₀ S	513-53-1	90.19	0.2069	-0.0002568				133.02	0.1727	358.13	0.1149
43	1-Butyne	C ₄ H ₆	107-00-6	54.09	0.22334	-0.0003515				147.43	0.1715	281.22	0.1245
44	Butyraldehyde	C ₄ H ₈ O	123-72-8	72.11	0.21915	-0.00024846				176.75	0.1752	382.15	0.1242
45	Butyric acid	C ₄ H ₈ O ₂	107-92-6	88.11	0.1967	-0.000168				267.95	0.1517	573.15	0.1004
46	Butyronitrile	C ₄ H ₇ N	109-74-0	69.11	0.2597	-0.00031				161.25	0.2097	390.75	0.1386
47	Carbon dioxide	CO ₂	124-38-9	44.01	0.4406	-0.0012175				216.58	0.1769	300.00	0.0754
48	Carbon disulfide	CS ₂	75-15-0	76.14	0.2333	-0.000275				161.11	0.1890	319.37	0.1455
49	Carbon monoxide	CO	630-08-0	28.01	0.2855	-0.001784				68.15	0.1639	125.00	0.0625
50	Carbon tetrachloride	CCl ₄	56-23-5	153.82	0.1589	-0.0001987				250.33	0.1092	349.79	0.0894
51	Carbon tetrafluoride	CF ₄	75-73-0	88.00	0.20771	-0.00078883				89.56	0.1371	145.10	0.0933
52	Chlorine	Cl ₂	7782-50-5	70.91	0.2246	-0.000064	-0.000000788			172.12	0.1902	410.00	0.0659
53	Chlorobenzene	C ₆ H ₅ Cl	108-90-7	112.56	0.1841	-0.0001917				227.95	0.1404	404.87	0.1065
54	Chloroethane	C ₂ H ₅ Cl	75-00-3	64.51	0.2438	-0.000419				134.80	0.1873	373.15	0.0875
55	Chloroform	CHCl ₃	67-66-3	119.38	0.1778	-0.0002023				209.63	0.1354	400.00	0.0969
56	Chloromethane	CH ₃ Cl	74-87-3	50.49	0.41067	-0.0008478				175.43	0.2619	350.00	0.1139
57	1-Chloropropane	C ₃ H ₇ Cl	540-54-5	78.54	0.20143	-0.00028925				150.35	0.1579	400.95	0.0855
58	2-Chloropropane	C ₃ H ₇ Cl	75-29-6	78.54	0.21232	-0.0003149				155.97	0.1632	386.70	0.0905

TABLE 2-315 Thermal Conductivity of Inorganic and Organic Liquids [W/(m·K)] (Continued)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1	C2	C3	C4	C5	$T_{\min}$, K	Thermal cond. at $T_{\min}$	$T_{\max}$, K	Thermal cond. at $T_{\max}$
59	<i>m</i> -Cresol	C ₇ H ₈ O	108-39-4	108.14	0.18241	-0.00011109				285.39	0.1507	475.43	0.1296
60	<i>o</i> -Cresol	C ₇ H ₈ O	95-48-7	108.14	0.19186	-0.0001303				304.19	0.1522	464.15	0.1314
61	<i>p</i> -Cresol	C ₇ H ₈ O	106-44-5	108.14	0.17971	-0.00012037				307.93	0.1426	475.13	0.1225
62	Cumene	C ₉ H ₁₂	98-82-8	120.19	0.1855	-0.00020895				177.14	0.1485	413.15	0.0992
63	Cyanogen	C ₂ N ₂	460-19-5	52.03	0.4685	-0.00086594				245.25	0.2561	252.00	0.2503
64	Cyclobutane	C ₄ H ₈	287-23-0	56.11	0.22262	-0.00034082				182.48	0.1604	285.66	0.1253
65	Cyclohexane	C ₆ H ₁₂	110-82-7	84.16	0.19813	-0.0002505				279.69	0.1281	353.87	0.1095
66	Cyclohexanol	C ₆ H ₁₂ O	108-93-0	100.16	0.1715	-0.0001255				296.60	0.1343	563.15	0.1008
67	Cyclohexanone	C ₆ H ₁₀ O	108-94-1	98.14	0.17557	-0.00012392				242.00	0.1456	428.58	0.1225
68	Cyclohexene	C ₆ H ₁₀	110-83-8	82.14	0.20926	-0.00026037				169.67	0.1651	356.12	0.1165
69	Cyclopentane	C ₅ H ₁₀	287-92-3	70.13	0.2066	-0.0002696				179.28	0.1583	322.40	0.1197
70	Cyclopentene	C ₅ H ₈	142-29-0	68.12	0.21776	-0.00027783				138.13	0.1794	333.15	0.1252
71	Cyclopropane	C ₃ H ₆	75-19-4	42.08	0.24348	-0.00042568				145.59	0.1815	240.37	0.1412
72	Cyclohexyl mercaptan	C ₆ H ₁₂ S	1569-69-3	116.22	0.18374	-0.0001925				189.64	0.1472	431.95	0.1006
73	Decanal	C ₁₀ H ₂₀ O	112-31-2	156.27	0.20383	-0.0002				267.15	0.1504	488.15	0.1062
74	Decane	C ₁₀ H ₂₂	124-18-5	142.28	0.2063	-0.00025				243.51	0.1454	447.30	0.0945
75	Decanoic acid	C ₁₀ H ₂₀ O ₂	334-48-5	172.27	0.206	-0.0002				304.75	0.1451	543.15	0.0974
76	1-Decanol	C ₁₀ H ₂₂ O	112-30-1	158.28	0.228	-0.000223				280.05	0.1656	503.00	0.1158
77	1-Decene	C ₁₀ H ₂₀	872-05-9	140.27	0.20237	-0.00024187				206.89	0.1523	443.75	0.0950
78	Decyl mercaptan	C ₁₀ H ₂₂ S	143-10-2	174.35	0.20134	-0.00020826				247.56	0.1498	512.35	0.0946
79	1-Decyne	C ₁₀ H ₁₈	764-93-2	138.25	0.20839	-0.00023622				229.15	0.1543	447.15	0.1028
80	Deuterium	D ₂	7782-39-0	4.03	1.264					20.40	1.2640	20.40	1.2640
81	1,1-Dibromoethane	C ₂ H ₄ Br ₂	557-91-5	187.86	0.1426	-0.00016402				210.15	0.1081	498.40	0.0609
82	1,2-Dibromoethane	C ₂ H ₄ Br ₂	106-93-4	187.86	0.13622	-0.0001179				282.85	0.1029	404.51	0.0885
83	Dibromomethane	CH ₂ Br ₂	74-95-3	173.83	0.17558	-0.00022499				220.60	0.1260	370.10	0.0923
84	Dibutyl ether	C ₈ H ₁₈ O	142-96-1	130.23	0.19418	-0.00022246				175.30	0.1552	523.15	0.0778
85	<i>m</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	541-73-1	147.00	0.16694	-0.0001667				248.39	0.1255	446.23	0.0926
86	<i>o</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	95-50-1	147.00	0.16994	-0.0001637				262.87	0.1269	351.71	0.1124
87	<i>p</i> -Dichlorobenzene	C ₆ H ₄ Cl ₂	106-46-7	147.00	0.16977	-0.0001799				326.14	0.1111	548.00	0.0712
88	1,1-Dichloroethane	C ₂ H ₄ Cl ₂	75-34-3	98.96	0.18881	-0.00026083				176.19	0.1429	416.90	0.0801
89	1,2-Dichloroethane	C ₂ H ₄ Cl ₂	107-06-2	98.96	0.214	-0.000266				253.15	0.1467	356.59	0.1192
90	Dichloromethane	CH ₂ Cl ₂	75-09-2	84.93	0.23847	-0.00033366				178.01	0.1791	325.00	0.1300
91	1,1-Dichloropropane	C ₃ H ₆ Cl ₂	78-99-9	112.99	0.18	-0.00023144				200.00	0.1337	438.00	0.0786
92	1,2-Dichloropropane	C ₃ H ₆ Cl ₂	78-87-5	112.99	0.19653	-0.00025012				172.71	0.1533	457.60	0.0821
93	Diethanol amine	C ₄ H ₁₁ NO ₂	111-42-2	105.14	0.0218	0.0010315	-0.000001355			301.15	0.2096	673.15	0.1022
94	Diethyl amine	C ₄ H ₁₁ N	109-89-7	73.14	0.2587	-0.00054343	4.2097E-07			223.35	0.1583	453.15	0.0989
95	Diethyl ether	C ₄ H ₁₀ O	60-29-7	74.12	0.2495	-0.000407				156.85	0.1857	433.15	0.0732
96	Diethyl sulfide	C ₄ H ₁₀ S	352-93-2	90.19	0.21065	-0.0002623				169.20	0.1663	365.25	0.1148
97	1,1-Difluoroethane	C ₂ H ₄ F ₂	75-37-6	66.05	0.27019	-0.000661	3.443E-07			154.56	0.1763	363.15	0.0756
98	1,2-Difluoroethane	C ₂ H ₄ F ₂	624-72-6	66.05	0.23171	-0.00038503				215.00	0.1489	372.80	0.0882
99	Diffuoromethane	CH ₂ F ₂	75-10-5	52.02	0.37296	-0.00088707	2.5762E-07			136.95	0.2563	302.56	0.1282
100	Di-isopropyl amine	C ₆ H ₁₅ N	108-18-9	101.19	0.1844	-0.000239				176.85	0.1421	357.05	0.0991
101	Di-isopropyl ether	C ₆ H ₁₄ O	108-20-3	102.17	0.19162	-0.0002762				187.65	0.1398	400.10	0.0811
102	Di-isopropyl ketone	C ₇ H ₁₄ O	565-80-0	114.19	0.22076	-0.00027624				204.81	0.1642	460.00	0.0937
103	1,1-Dimethoxyethane	C ₄ H ₁₀ O ₂	534-15-6	90.12	0.22078	-0.00031271				159.95	0.1708	337.45	0.1153
104	1,2-Dimethoxypropane	C ₅ H ₁₂ O ₂	7778-85-0	104.15	0.22998	-0.00030372				226.10	0.1613	366.15	0.1188
105	Dimethyl acetylene	C ₄ H ₆	503-17-3	54.09	0.22773	-0.00034804				240.91	0.1439	300.13	0.1233
106	Dimethyl amine	C ₂ H ₇ N	124-40-3	45.08	0.2454	-0.000338				180.96	0.1842	403.15	0.1091
107	2,3-Dimethylbutane	C ₆ H ₁₄	79-29-8	86.18	0.1774	-0.0002436				145.19	0.1420	331.15	0.0967
108	1,1-Dimethylcyclohexane	C ₈ H ₁₆	590-66-9	112.21	0.1807	-0.0002177				239.66	0.1285	392.70	0.0952
109	<i>cis</i> -1,2-Dimethylcyclohexane	C ₈ H ₁₆	2207-01-4	112.21	0.18092	-0.0002108				223.16	0.1339	402.94	0.0960
110	<i>trans</i> -1,2-Dimethylcyclohexane	C ₈ H ₁₆	6876-23-9	112.21	0.17675	-0.0002077				184.99	0.1383	596.15	0.0529
111	Dimethyl disulfide	C ₂ H ₆ S ₂	624-92-0	94.20	0.21373	-0.0002447				188.44	0.1676	382.90	0.1200
112	Dimethyl ether	C ₂ H ₆ O	115-10-6	46.07	0.31174	-0.0005638				131.65	0.2375	320.03	0.1313
113	<i>N,N</i> -Dimethyl formamide	C ₃ H ₇ NO	68-12-2	73.09	0.26	-0.000255				250.00	0.1963	425.15	0.1516
114	2,3-Dimethylpentane	C ₇ H ₁₆	565-59-3	100.20	0.17964	-0.000246				160.00	0.1403	362.93	0.0904
115	Dimethyl phthalate	C ₁₀ H ₁₀ O ₄	131-11-3	194.18	0.13905	0.0001509	-3.978E-07			273.15	0.1506	556.85	0.0997

116	Dimethylsilane	C ₂ H ₆ Si	1111-74-6	60.17	0.25547	-0.0004411			122.93	0.2013	253.55	0.1436
117	Dimethyl sulfide	C ₂ H ₆ S	75-18-3	62.13	0.23942	-0.0003311			174.88	0.1815	310.48	0.1366
118	Dimethyl sulfoxide	C ₂ H ₆ OS	67-68-5	78.13	0.3142	-0.00030809			291.67	0.2243	464.00	0.1713
119	Dimethyl terephthalate	C ₁₀ H ₁₀ O ₄	120-61-6	194.18	0.21593	-0.00020805			413.80	0.1298	561.15	0.0992
120	1,4-Dioxane	C ₄ H ₈ O ₂	123-91-1	88.11	0.3027	-0.0004827			284.95	0.1652	374.47	0.1219
121	Diphenyl ether	C ₁₂ H ₁₀ O	101-84-8	170.21	0.18686	-0.00014953			300.03	0.1420	531.46	0.1074
122	Dipropyl amine	C ₆ H ₁₅ N	142-84-7	101.19	0.2224	-0.000314			210.15	0.1564	382.00	0.1025
123	Dodecane	C ₁₂ H ₂₆	112-40-3	170.33	0.2047	-0.0002326			263.57	0.1434	489.47	0.0908
124	Eicosane	C ₂₀ H ₄₂	112-95-8	282.55	0.2178	-0.0002233			309.58	0.1487	616.93	0.0800
125	Ethane	C ₂ H ₆	74-84-0	30.07	0.35758	-0.0011458	6.1866E-07		90.35	0.2591	300.00	0.0695
126	Ethanol	C ₂ H ₆ O	64-17-5	46.07	0.2468	-0.000264			159.05	0.2048	353.15	0.1536
127	Ethyl acetate	C ₄ H ₈ O ₂	141-78-6	88.11	0.2501	-0.0003563			189.60	0.1826	350.21	0.1253
128	Ethyl amine	C ₂ H ₇ N	75-04-7	45.08	0.30059	-0.000581	6.602E-07		192.15	0.2133	293.15	0.1870
129	Ethylbenzene	C ₈ H ₁₀	100-41-4	106.17	0.1999	-0.00023823			178.20	0.1575	413.10	0.1015
130	Ethyl benzoate	C ₉ H ₁₀ O ₂	93-89-0	150.17	0.20771	-0.00021265			238.45	0.1570	549.40	0.0909
131	2-Ethyl butanoic acid	C ₆ H ₁₂ O ₂	88-09-5	116.16	0.2175	-0.0002407			258.15	0.1554	516.50	0.0932
132	Ethyl butyrate	C ₈ H ₁₆ O ₂	105-54-4	116.16	0.21043	-0.00024903			175.15	0.1668	453.15	0.0976
133	Ethylcyclohexane	C ₈ H ₁₆	1678-91-7	112.21	0.17662	-0.0002014			161.84	0.1440	404.94	0.0951
134	Ethylcyclopentane	C ₇ H ₁₄	1640-89-7	98.19	0.18334	-0.0002228			134.71	0.1533	376.62	0.0994
135	Ethylene	C ₂ H ₄	74-85-1	28.05	0.4194	-0.001591	0.000001306		104.00	0.2681	280.00	0.0763
136	Ethylenediamine	C ₂ H ₈ N ₂	107-15-3	60.10	0.36434	-0.0004433			284.29	0.2383	390.41	0.1913
137	Ethylene glycol	C ₂ H ₆ O ₂	107-21-1	62.07	0.088067	0.00094712	-1.3114E-06		260.15	0.2457	470.45	0.2434
138	Ethyleneimine	C ₂ H ₅ N	151-56-4	43.07	0.3097	-0.0004023			195.20	0.2312	329.00	0.1773
139	Ethylene oxide	C ₂ H ₄ O	75-21-8	44.05	0.26957	-0.0003984			160.65	0.2056	283.85	0.1565
140	Ethyl formate	C ₃ H ₆ O ₂	109-94-4	74.08	0.2587	-0.00033			193.55	0.1948	433.15	0.1158
141	2-Ethyl hexanoic acid	C ₈ H ₁₆ O ₂	149-57-5	144.21	0.20954	-0.00022251			235.00	0.1573	500.66	0.0981
142	Ethylhexyl ether	C ₈ H ₁₈ O	5756-43-4	130.23	0.19356	-0.00024102			180.00	0.1502	466.40	0.0811
143	Ethylisopropyl ether	C ₆ H ₁₄ O	625-54-7	88.15	0.21928	-0.00032568			140.00	0.1737	391.20	0.0919
144	Ethylisopropyl ketone	C ₆ H ₁₂ O	565-69-5	100.16	0.22873	-0.0002913			204.15	0.1693	450.10	0.0976
145	Ethyl mercaptan	C ₂ H ₆ S	75-08-1	62.13	0.23392	-0.0003206			125.26	0.1938	308.15	0.1351
146	Ethyl propionate	C ₅ H ₁₀ O ₂	105-37-3	102.13	0.2137	-0.0002515			199.25	0.1636	495.00	0.0892
147	Ethylpropyl ether	C ₅ H ₁₂ O	628-32-0	88.15	0.22717	-0.0003298			145.65	0.1791	400.07	0.0952
148	Ethyltrichlorosilane	C ₂ H ₅ Cl ₃ Si	115-21-9	163.51	0.19769	-0.00017713	-1.5448E-07		167.55	0.1637	371.05	0.1107
149	Fluorine	F ₂	7782-41-4	38.00	0.2758	-0.0016297			53.48	0.1886	130.00	0.0639
150	Fluorobenzene	C ₆ H ₅ F	462-06-6	96.10	0.20962	-0.00028034			238.15	0.1429	353.15	0.1106
151	Fluoroethane	C ₂ H ₅ F	353-36-6	48.06	0.2595	-0.0005008			129.95	0.1944	292.59	0.1130
152	Fluoromethane	CH ₃ F	593-53-3	34.03	0.445	-0.001023			131.35	0.3106	283.15	0.1553
153	Formaldehyde	CH ₂ O	50-00-0	30.03	0.37329	-0.00065			204.00	0.2407	234.00	0.2212
154	Formamide	CH ₃ NO	75-12-7	45.04	0.3847	-0.0001065			275.70	0.3553	493.00	0.3322
155	Formic acid	CH ₂ O ₂	64-18-6	46.03	0.302	-0.000108			281.45	0.2716	373.71	0.2616
156	Furan	C ₄ H ₄ O	110-00-9	68.07	0.2198	-0.00031405			187.55	0.1609	304.50	0.1242
157	Helium-4	He	7440-59-7	4.00	-0.013833	0.022913	-0.0054872	0.0004585	2.20	0.0149	4.80	0.0204
158	Heptadecane	C ₁₇ H ₃₆	629-78-7	240.47	0.20926	-0.0002215			295.13	0.1439	575.30	0.0818
159	Heptanal	C ₇ H ₁₄ O	111-71-7	114.19	0.21816	-0.0003015	1.033E-07		229.80	0.1543	553.15	0.0830
160	Heptane	C ₇ H ₁₆	142-82-5	100.20	0.215	-0.000303			182.57	0.1597	371.58	0.1024
161	Heptanoic acid	C ₇ H ₁₄ O ₂	111-14-8	130.19	0.202	-0.0002			265.83	0.1488	496.15	0.1028
162	1-Heptanol	C ₇ H ₁₆ O	111-70-6	116.20	0.2239	-0.000226			239.15	0.1699	573.15	0.0944
163	2-Heptanol	C ₇ H ₁₆ O	543-49-7	116.20	0.21134	-0.00024776			230.00	0.1544	432.90	0.1041
164	3-Heptanone	C ₇ H ₁₄ O	106-35-4	114.19	0.2026	-0.0002234			234.15	0.1503	553.15	0.0790
165	2-Heptanone	C ₇ H ₁₄ O	110-43-0	114.19	0.2108	-0.000246			238.15	0.1522	424.05	0.1065
166	1-Heptene	C ₇ H ₁₄	592-76-7	98.19	0.19664	-0.00016623	-2.5241E-07		154.12	0.1650	366.79	0.1017
167	Heptyl mercaptan	C ₇ H ₁₆ S	1639-09-4	132.27	0.2037	-0.0002252			229.92	0.1519	450.09	0.1023
168	1-Heptyne	C ₇ H ₁₂	628-71-7	96.17	0.21098	-0.00026652			192.22	0.1598	372.93	0.1116
169	Hexadecane	C ₁₆ H ₃₄	544-76-3	226.44	0.20749	-0.00021917			291.31	0.1436	560.01	0.0848
170	Hexanal	C ₆ H ₁₂ O	66-25-1	100.16	0.22196	-0.00032053	1.1554E-07		217.15	0.1578	533.15	0.0839
171	Hexane	C ₆ H ₁₄	110-54-3	86.18	0.22492	-0.0003533			177.83	0.1621	370.00	0.0942
172	Hexanoic acid	C ₆ H ₁₂ O ₂	142-62-1	116.16	0.1855	-0.000146			269.25	0.1462	603.15	0.0974
173	1-Hexanol	C ₆ H ₁₄ O	111-27-3	102.17	0.2193	-0.00022			228.55	0.1690	575.00	0.0928
174	2-Hexanol	C ₆ H ₁₄ O	626-93-7	102.18	0.21391	-0.00026042			223.00	0.1558	412.40	0.1065
175	2-Hexanone	C ₆ H ₁₂ O	591-78-6	100.16	0.21076	-0.00024			217.35	0.1586	400.85	0.1146
176	3-Hexanone	C ₆ H ₁₂ O	589-38-8	100.16	0.23493	-0.0002912			217.50	0.1716	466.00	0.0992
177	1-Hexene	C ₆ H ₁₂	592-41-6	84.16	0.19112	-0.00083519	-5.1407E-07		133.39	0.1708	336.63	0.1048
178	3-Hexyne	C ₆ H ₁₀	928-49-4	82.14	0.20996	-0.0002692			170.05	0.1642	354.35	0.1146

TABLE 2-315 Thermal Conductivity of Inorganic and Organic Liquids [W/(m·K)] (Continued)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1	C2	C3	C4	C5	$T_{\min}$, K	Thermal cond. at $T_{\min}$	$T_{\max}$, K	Thermal cond. at $T_{\max}$
179	Hexyl mercaptan	C ₆ H ₁₄ S	111-31-9	118.24	0.2058	-0.0002324	-0.000382	-3.3324E-06	1.0266E-07	192.62	0.1610	425.81	0.1068
180	1-Hexyne	C ₆ H ₁₀	693-02-7	82.14	0.21492	-0.0002899				141.25	0.1740	344.48	0.1151
181	2-Hexyne	C ₆ H ₁₀	764-35-2	82.14	0.2119	-0.00027048				183.65	0.1622	357.67	0.1152
182	Hydrazine	H ₄ N ₂	302-01-2	32.05	1.3675	-0.0015895				274.69	0.9309	623.15	0.3770
183	Hydrogen	H ₂	1333-74-0	2.02	-0.0917	0.017678				13.95	0.0754	31.00	0.0847
184	Hydrogen bromide	HBr	10035-10-6	80.91	0.234	-0.0004636	3.5588E-06	8.033E-07		185.15	0.1482	290.62	0.0993
185	Hydrogen chloride	HCl	7647-01-0	36.46	0.8045	-0.002102				273.15	0.2303	323.15	0.1252
186	Hydrogen cyanide	CHN	74-90-8	27.03	0.43454	-0.0007008				259.83	0.2525	298.85	0.2251
187	Hydrogen fluoride	HF	7664-39-3	20.01	0.7516	-0.0010874				189.79	0.5452	394.45	0.3227
188	Hydrogen sulfide	H ₂ S	7783-06-4	34.08	0.4842	-0.001184				193.15	0.2555	292.42	0.1380
189	Isobutyric acid	C ₄ H ₈ O ₂	79-31-2	88.11	0.21668	-0.0002556	8.033E-07			227.15	0.1586	482.75	0.0933
190	Isopropyl amine	C ₃ H ₉ N	75-31-0	59.11	0.237	-0.000332				177.95	0.1779	305.55	0.1356
191	Malonic acid	C ₃ H ₄ O ₄	141-82-2	104.06	0.28918	-0.0002614				407.95	0.1825	644.00	0.1208
192	Methacrylic acid	C ₄ H ₆ O ₂	79-41-4	86.09	0.2306	-0.00025201				288.15	0.1580	530.00	0.0970
193	Methane	CH ₄	74-82-8	16.04	0.41768	-0.0024528				90.69	0.2245	180.00	0.0915
194	Methanol	CH ₃ O	67-56-1	32.04	0.2837	-0.000281	8.033E-07			175.47	0.2344	337.85	0.1888
195	N-Methyl acetamide	C ₃ H ₇ NO	79-16-3	73.09	0.23743	-0.0002362				301.15	0.1663	478.15	0.1245
196	Methyl acetate	C ₃ H ₆ O ₂	79-20-9	74.08	0.2777	-0.000417				175.15	0.2047	386.15	0.1167
197	Methyl acetylene	C ₃ H ₄	74-99-7	40.06	0.23648	-0.00041639				170.45	0.1655	249.94	0.1324
198	Methyl acrylate	C ₄ H ₆ O ₂	96-33-3	86.09	0.26082	-0.0003506				196.32	0.1920	421.00	0.1132
199	Methyl amine	CH ₃ N	74-89-5	31.06	0.33446	-0.00067427	8.033E-07			179.69	0.2392	283.15	0.2079
200	Methyl benzoate	C ₈ H ₈ O ₂	93-58-3	136.15	0.22142	-0.00022759				260.75	0.1621	547.90	0.0967
201	3-Methyl-1,2-butadiene	C ₅ H ₈	598-25-4	68.12	0.1983	-0.0002822				159.53	0.1533	314.00	0.1097
202	2-Methylbutane	C ₅ H ₁₂	78-78-4	72.15	0.21246	-0.00033581				113.25	0.1744	368.13	0.0888
203	2-Methylbutanoic acid	C ₅ H ₁₀ O ₂	116-53-0	102.13	0.22284	-0.0002516				357.15	0.1330	480.90	0.1019
204	3-Methyl-1-butanol	C ₅ H ₁₂ O	123-51-3	88.15	0.17471	-0.0001256	8.033E-07			155.95	0.1551	404.15	0.1239
205	2-Methyl-1-butene	C ₅ H ₁₀	563-46-2	70.13	0.19447	-0.0002901				135.58	0.1551	304.30	0.1062
206	2-Methyl-2-butene	C ₅ H ₁₀	513-35-9	70.13	0.19636	-0.000291				139.39	0.1558	311.70	0.1057
207	2-Methyl-1-butene-3-yne	C ₅ H ₆	78-80-8	66.10	0.20385	-0.0002874				160.15	0.1578	305.40	0.1161
208	Methylbutyl ether	C ₅ H ₁₂ O	628-28-4	88.15	0.22235	-0.0003044				157.48	0.1744	463.15	0.0814
209	Methylbutyl sulfide	C ₅ H ₁₂ S	628-29-5	104.21	0.20698	-0.00024439	8.033E-07			175.30	0.1641	396.58	0.1101
210	3-Methyl-1-butyne	C ₅ H ₈	598-23-2	68.12	0.20348	-0.0003106				183.45	0.1465	302.15	0.1096
211	Methyl butyrate	C ₅ H ₁₀ O ₂	623-42-7	102.13	0.21748	-0.00025913				187.35	0.1689	493.15	0.0897
212	Methylchlorosilane	CH ₃ ClSi	993-00-0	80.59	0.24683	-0.00038854				139.05	0.1928	281.85	0.1373
213	Methylcyclohexane	C ₇ H ₁₄	108-87-2	98.19	0.1791	-0.0002291				273.15	0.1165	374.08	0.0934
214	1-Methylcyclohexanol	C ₇ H ₁₄ O	590-67-0	114.19	0.21558	-0.00022728	8.033E-07			299.15	0.1476	548.80	0.0908
215	cis-2-Methylcyclohexanol	C ₇ H ₁₄ O	7443-70-1	114.19	0.21839	-0.00025776				280.15	0.1462	484.20	0.0936
216	trans-2-Methylcyclohexanol	C ₇ H ₁₄ O	7443-52-9	114.19	0.21828	-0.0002557				269.15	0.1495	484.80	0.0943
217	Methylcyclopentane	C ₆ H ₁₂	96-37-7	84.16	0.1929	-0.0002492				130.73	0.1603	344.95	0.1069
218	1-Methylcyclopentene	C ₆ H ₁₀	693-89-0	82.14	0.20023	-0.00025581				146.62	0.1627	348.64	0.1110
219	3-Methylcyclopentene	C ₆ H ₁₀	1120-62-3	82.14	0.1994	-0.00026149	8.033E-07			115.00	0.1693	338.05	0.1110
220	Methyldichlorosilane	CH ₂ Cl ₂ Si	75-54-7	115.03	0.21956	-0.00032153				182.55	0.1609	314.70	0.1184
221	Methylethyl ether	C ₃ H ₈ O	540-67-0	60.10	0.27304	-0.0004518				160.00	0.2008	341.34	0.1188
222	Methylethyl ketone	C ₄ H ₈ O	78-93-3	72.11	0.2197	-0.0002505				186.48	0.1730	352.79	0.1313
223	Methylethyl sulfide	C ₃ H ₈ S	624-89-5	76.16	0.22136	-0.00028938				167.23	0.1730	339.80	0.1230
224	Methyl formate	C ₂ H ₄ O ₂	107-31-3	60.05	0.3246	-0.000468	8.033E-07			174.15	0.2431	373.15	0.1500
225	Methylisobutyl ether	C ₅ H ₁₂ O	625-44-5	88.15	0.222	-0.00032217				150.00	0.1737	390.00	0.0964
226	Methylisobutyl ketone	C ₆ H ₁₂ O	108-10-1	100.16	0.2301	-0.00028899				189.15	0.1754	451.42	0.0996
227	Methyl isocyanate	C ₂ H ₃ NO	624-83-9	57.05	0.2822	-0.00042037				256.15	0.1745	312.00	0.1510
228	Methylisopropyl ether	C ₄ H ₁₀ O	598-53-8	74.12	0.24154	-0.0003774				127.93	0.1933	370.00	0.1019
229	Methylisopropyl ketone	C ₅ H ₁₀ O	563-80-4	86.13	0.2332	-0.0003044	8.033E-07			180.15	0.1784	435.90	0.1005
230	Methylisopropyl sulfide	C ₄ H ₁₀ S	1551-21-9	90.19	0.20978	-0.00026468				171.64	0.1644	357.91	0.1151
231	Methyl mercaptan	CH ₃ S	74-93-1	48.11	0.26119	-0.00038345				150.18	0.2036	279.11	0.1542
232	Methyl methacrylate	C ₅ H ₈ O ₂	80-62-6	100.12	0.2583	-0.000379				290.15	0.1483	363.45	0.1206
233	2-Methyloctanoic acid	C ₉ H ₁₈ O ₂	3004-93-1	158.24	0.20911	-0.00021852	8.033E-07			208.20	0.1636	555.20	0.0878
234	2-Methylpentane	C ₆ H ₁₄	107-83-5	86.18	0.19334	-0.00028038				119.55	0.1598	389.25	0.0842
235	Methyl pentyl ether	C ₆ H ₁₄ O	628-80-8	102.17	0.21698	-0.00028998				176.00	0.1659	432.30	0.0916
236	2-Methylpropane	C ₄ H ₁₀	75-28-5	58.12	0.20455	-0.00036589				113.54	0.1630	400.00	0.0582

237	2-Methyl-2-propanol	C ₄ H ₁₀ O	75-65-0	74.12	0.21258	-0.00029864				298.97	0.1233	404.96	0.0916
238	2-Methyl propene	C ₄ H ₈	115-11-7	56.11	0.2802	-0.000786	6.516E-07			132.81	0.1873	395.20	0.0713
239	Methyl propionate	C ₄ H ₈ O ₂	554-12-1	88.11	0.22534	-0.0002683				185.65	0.1755	475.00	0.0979
240	Methylpropyl ether	C ₄ H ₁₀ O	557-17-5	74.12	0.24817	-0.0003774				133.97	0.1976	373.00	0.1074
241	Methylpropyl sulfide	C ₄ H ₁₀ S	3877-15-4	90.19	0.21103	-0.00025985				160.17	0.1694	368.69	0.1152
242	Methylsilane	CH ₅ Si	992-94-9	46.14	0.2774	-0.00054608				116.34	0.2139	216.25	0.1593
243	α-Methyl styrene	C ₉ H ₁₀	98-83-9	118.18	0.19657	-0.0002118				249.95	0.1436	438.65	0.1037
244	Methyl <i>tert</i> -butyl ether	C ₅ H ₁₂ O	1634-04-4	88.15	0.2253	-0.00037273	1.1728E-07			164.55	0.1671	328.18	0.1156
245	Methyl vinyl ether	C ₃ H ₆ O	107-25-5	58.08	0.28035	-0.0004646				151.15	0.2101	341.10	0.1219
246	Naphthalene	C ₁₀ H ₈	91-20-3	128.17	0.17096	-0.00010059				353.43	0.1354	646.97	0.1059
247	Neon	Ne	7440-01-9	20.18	0.2971	-0.017356	0.0005911	-0.000007421		25.00	0.1167	44.00	0.0457
248	Nitroethane	C ₂ H ₅ NO ₂	79-24-3	75.07	0.247	-0.0002814				183.63	0.1953	387.22	0.1380
249	Nitrogen	N ₂	7727-37-9	28.01	0.2654	-0.001677				63.15	0.1595	124.00	0.0575
250	Nitrogen trifluoride	F ₃ N	7783-54-2	71.00									
251	Nitromethane	CH ₃ NO ₂	75-52-5	61.04	0.3276	-0.000405				244.60	0.2285	374.35	0.1760
252	Nitrous oxide	N ₂ O	10024-97-2	44.01	0.10112					277.59	0.1011	277.59	0.1011
253	Nitric oxide	NO	10102-43-9	30.01	0.1878	0.0010293	-0.00000943			110.00	0.1869	176.40	0.0759
254	Nonadecane	C ₁₉ H ₄₀	629-92-5	268.52	0.21229	-0.00022				305.04	0.1452	603.05	0.0796
255	Nonanal	C ₉ H ₁₈ O	124-19-6	142.24	0.21523	-0.0002799	9.572E-08			255.15	0.1501	593.15	0.0829
256	Nonane	C ₉ H ₂₀	111-84-2	128.26	0.209	-0.000264				219.66	0.1510	423.97	0.0971
257	Nonanoic acid	C ₉ H ₁₈ O ₂	112-05-0	158.24	0.204	-0.0002				285.55	0.1469	528.75	0.0983
258	1-Nonanol	C ₉ H ₂₀ O	143-08-8	144.25	0.2292	-0.00023				268.15	0.1675	578.65	0.0961
259	2-Nonanol	C ₉ H ₂₀ O	628-99-9	144.26	0.20829	-0.00022922				238.15	0.1537	471.70	0.1002
260	1-Nonene	C ₉ H ₁₈	124-11-8	126.24	0.20468	-0.00025738				191.91	0.1553	420.02	0.0966
261	Nonyl mercaptan	C ₉ H ₂₀ S	1455-21-6	160.32	0.20244	-0.00021343				253.05	0.1484	492.95	0.0972
262	1-Nonyne	C ₉ H ₁₆	3452-09-3	124.22	0.20954	-0.00024588				223.15	0.1547	423.85	0.1053
263	Octadecane	C ₁₈ H ₃₈	593-45-3	254.49	0.2137	-0.0002252				301.31	0.1458	589.86	0.0809
264	Octanal	C ₈ H ₁₆ O	124-13-0	128.21	0.20143	-0.00021102				246.00	0.1495	573.15	0.0805
265	Octane	C ₈ H ₁₈	111-65-9	114.23	0.2156	-0.00029483				216.38	0.1518	398.83	0.0980
266	Octanoic acid	C ₈ H ₁₆ O ₂	124-07-2	144.21	0.203	-0.0002				289.65	0.1451	512.85	0.1004
267	1-Octanol	C ₈ H ₁₈ O	111-87-5	130.23	0.2316	-0.0002407				257.65	0.1696	570.15	0.0944
268	2-Octanol	C ₈ H ₁₈ O	123-96-6	130.23	0.20955	-0.00023733				241.55	0.1522	452.90	0.1021
269	2-Octanone	C ₈ H ₁₆ O	111-13-7	128.21	0.2132	-0.0002494				252.85	0.1501	499.00	0.0887
270	3-Octanone	C ₈ H ₁₆ O	106-68-3	128.21	0.21732	-0.00024969				255.55	0.1535	440.65	0.1073
271	1-Octene	C ₈ H ₁₆	111-66-0	112.21	0.20467	-0.0002675				171.45	0.1588	394.41	0.0992
272	Octyl mercaptan	C ₈ H ₁₈ S	111-88-6	146.29	0.2012	-0.0002142				223.95	0.1532	472.19	0.1001
273	1-Octyne	C ₈ H ₁₄	629-05-0	110.20	0.2095	-0.00025334				193.55	0.1605	399.35	0.1083
274	Oxalic acid	C ₂ H ₂ O ₄	144-62-7	90.03	0.3074	-0.00028101				462.65	0.1774	643.20	0.1267
275	Oxygen	O ₂	7782-44-7	32.00	0.2741	-0.00138				60.00	0.1913	150.00	0.0671
276	Ozone	O ₃	10028-15-6	48.00	0.17483	0.00075288	-2.5228E-06			77.35	0.2180	161.85	0.2306
277	Pentadecane	C ₁₅ H ₃₂	629-62-9	212.41	0.20649	-0.00021911				283.07	0.1445	543.84	0.0873
278	Pentanal	C ₅ H ₁₀ O	110-62-3	86.13	0.22697	-0.00033227	1.177E-07			182.00	0.1704	513.15	0.0875
279	Pentane	C ₅ H ₁₂	109-66-0	72.15	0.2537	-0.000576	0.000000344			143.42	0.1782	445.00	0.0655
280	Pentanoic acid	C ₅ H ₁₀ O ₂	109-52-4	102.13	0.1848	-0.0001434				239.15	0.1505	458.65	0.1190
281	1-Pentanol	C ₅ H ₁₂ O	71-41-0	88.15	0.2006	-0.0001603				273.15	0.1568	353.15	0.1440
282	2-Pentanol	C ₅ H ₁₂ O	6032-29-7	88.15	0.21875	-0.00027849				200.00	0.1631	392.20	0.1095
283	2-Pentanone	C ₅ H ₁₀ O	107-87-9	86.13	0.2161	-0.00024866				196.29	0.1673	375.46	0.1227
284	3-Pentanone	C ₅ H ₁₀ O	96-22-0	86.13	0.21569	-0.00024081				234.18	0.1593	375.14	0.1254
285	1-Pentene	C ₅ H ₁₀	109-67-1	70.13	0.21361	-0.00030777				108.02	0.1804	303.22	0.1203
286	2-Pentyl mercaptan	C ₅ H ₁₂ S	2084-19-7	104.21	0.20597	-0.00024518				160.75	0.1666	385.15	0.1115
287	Pentyl mercaptan	C ₅ H ₁₂ S	110-66-7	104.21	0.2086	-0.00024536				197.45	0.1602	399.79	0.1105
288	1-Pentyne	C ₅ H ₈	627-19-0	68.12	0.22102	-0.000322				167.45	0.1671	313.33	0.1201
289	2-Pentyne	C ₅ H ₈	627-21-4	68.12	0.21282	-0.0002856				163.83	0.1660	329.27	0.1188
290	Phenanthrene	C ₁₄ H ₁₀	85-01-8	178.23	0.13753	-0.00025247				372.38	0.1281	610.03	0.1221
291	Phenol	C ₆ H ₆ O	108-95-2	94.11	0.18831	-0.0001				314.06	0.1569	454.99	0.1428
292	Phenyl isocyanate	C ₇ H ₅ NO	103-71-9	119.12	0.16326	-0.00017777				243.15	0.1200	439.43	0.0851
293	Phthalic anhydride	C ₈ H ₄ O ₃	85-44-9	148.12	0.22946	-0.00021345				404.15	0.1432	557.65	0.1104
294	Propadiene	C ₃ H ₄	463-49-0	40.06	0.23081	-0.0004078				136.87	0.1750	238.65	0.1335
295	Propane	C ₃ H ₈	74-98-6	44.10	0.26755	-0.00066457	2.774E-07			85.47	0.2128	350.00	0.0689
296	1-Propanol	C ₃ H ₈ O	71-23-8	60.10	0.2203	-0.0002155				200.00	0.1772	370.35	0.1405
297	2-Propanol	C ₃ H ₈ O	67-63-0	60.10	0.20161	-0.00021529				185.26	0.1617	425.00	0.1101
298	Propenylcyclohexene	C ₉ H ₁₄	13511-13-2	122.21	0.1831	-0.00020275				199.00	0.1428	431.65	0.0956
299	Propionaldehyde	C ₃ H ₆ O	123-38-6	58.08	0.2498	-0.00030075				170.00	0.1987	453.15	0.1135

TABLE 2-315 Thermal Conductivity of Inorganic and Organic Liquids [W/(m·K)] (Concluded)

Cmpd. no.	Name	Formula	CAS no.	Mol. wt.	C1	C2	C3	C4	C5	$T_{\min}$, K	Thermal cond. at $T_{\min}$	$T_{\max}$, K	Thermal cond. at $T_{\max}$
300	Propionic acid	C ₃ H ₆ O ₂	79-09-4	74.08	0.1954	-0.000164	0.000000412			252.45	0.1540	543.15	0.1063
301	Propionitrile	C ₃ H ₅ N	107-12-0	55.08	0.26626	-0.0003307				180.26	0.2067	370.50	0.1437
302	Propyl acetate	C ₅ H ₁₀ O ₂	109-60-4	102.13	0.2332	-0.0003096				178.15	0.1780	434.82	0.0986
303	Propyl amine	C ₃ H ₉ N	107-10-8	59.11	0.2632	-0.0004278				188.36	0.1972	333.15	0.1664
304	Propylbenzene	C ₉ H ₁₂	103-65-1	120.19	0.18707	-0.00019846				173.55	0.1526	583.15	0.0713
305	Propylene	C ₃ H ₆	115-07-1	42.08	0.24719	-0.00048824				87.89	0.2043	340.49	0.0809
306	Propyl formate	C ₄ H ₈ O ₂	110-74-7	88.11	0.2247	-0.000264				180.25	0.1771	483.15	0.0971
307	2-Propyl mercaptan	C ₃ H ₈ S	75-33-2	76.16	0.21706	-0.00028952				142.61	0.1758	325.71	0.1228
308	Propyl mercaptan	C ₃ H ₈ S	107-03-9	76.16	0.2202	-0.00028535				159.95	0.1746	340.87	0.1229
309	1,2-Propylene glycol	C ₃ H ₈ O ₂	57-55-6	76.09	0.2152	-0.0000497				213.15	0.2046	460.75	0.1923
310	Quinone	C ₆ H ₄ O ₂	106-51-4	108.09	0.26524	-0.00028676	0.00000266			388.85	0.1537	545.00	0.1090
311	Silicon tetrafluoride	F ₄ Si	7783-61-1	104.08									
312	Styrene	C ₈ H ₈	100-42-5	104.15	0.20215	-0.0002201				242.54	0.1488	418.31	0.1101
313	Succinic acid	C ₄ H ₆ O ₄	110-15-6	118.09	0.28215	-0.0002585				460.65	0.1631	642.00	0.1162
314	Sulfur dioxide	O ₂ S	7446-09-5	64.06	0.38218	-0.0006254				197.67	0.2586	400.00	0.1320
315	Sulfur hexafluoride	F ₆ S	2551-62-4	146.06	0.2544	-0.0006595				223.15	0.1072	318.69	0.0442
316	Sulfur trioxide	O ₃ S	7446-11-9	80.06	0.92882	-0.0030803				289.95	0.2593	481.47	0.0624
317	Terephthalic acid	C ₈ H ₆ O ₄	100-21-0	166.13									
318	<i>o</i> -Terphenyl	C ₁₈ H ₁₄	84-15-1	230.30	0.16853	-0.00010817				329.35	0.1329	723.15	0.0903
319	Tetradecane	C ₁₄ H ₃₀	629-59-4	198.39	0.20293	-0.00021798				279.01	0.1421	526.73	0.0881
320	Tetrahydrofuran	C ₄ H ₈ O	109-99-9	72.11	0.19428	-0.000249	-0.000010066			164.65	0.1533	339.12	0.1098
321	1,2,3,4-Tetrahydronaphthalene	C ₁₀ H ₁₂	119-64-2	132.20	0.14563	-0.0000536				237.38	0.1329	480.77	0.1199
322	Tetrahydrothiophene	C ₄ H ₈ S	110-01-0	88.17	0.20414	-0.00021217				176.98	0.1666	394.27	0.1205
323	2,2,3,3-Tetramethylbutane	C ₈ H ₁₈	594-82-1	114.23	0.17835	-0.00023704				373.96	0.0897	426.00	0.0774
324	Thiophene	C ₄ H ₄ S	110-02-1	84.14	0.20571	-0.00020028				234.94	0.1587	357.31	0.1342
325	Toluene	C ₇ H ₈	108-88-3	92.14	0.20463	-0.00024252				178.18	0.1614	474.85	0.0895
326	1,1,2-Trichloroethane	C ₂ H ₃ Cl ₃	79-00-5	133.40	0.20731	-0.00024997				236.50	0.1482	482.00	0.0868
327	Tridecane	C ₁₃ H ₂₈	629-50-5	184.36	0.20447	-0.00022612				267.76	0.1439	508.62	0.0895
328	Triethyl amine	C ₆ H ₁₅ N	121-44-8	101.19	0.1918	-0.0002453				158.45	0.1529	483.15	0.0733
329	Trimethyl amine	C ₃ H ₉ N	75-50-3	59.11	0.23813	-0.00038397				156.08	0.1782	276.02	0.1322
330	1,2,3-Trimethylbenzene	C ₉ H ₁₂	526-73-8	120.19	0.18854	-0.0001963	-0.000008078			247.79	0.1399	449.27	0.1004
331	1,2,4-Trimethylbenzene	C ₉ H ₁₂	95-63-6	120.19	0.19216	-0.0002105				229.33	0.1439	442.53	0.0990
332	2,2,4-Trimethylpentane	C ₈ H ₁₈	540-84-1	114.23	0.1659	-0.00022686				165.78	0.1283	372.39	0.0814
333	2,3,3-Trimethylpentane	C ₈ H ₁₈	560-21-4	114.23	0.16815	-0.00020535				172.22	0.1328	387.91	0.0885
334	1,3,5-Trinitrobenzene	C ₆ H ₃ N ₃ O ₆	99-35-4	213.10	0.18421	-0.00016097				398.40	0.1201	629.60	0.0829
335	2,4,6-Trinitrotoluene	C ₇ H ₅ N ₃ O ₆	118-96-7	227.13	-1.5128	0.0079553				357.20	0.0445	393.20	0.0590
336	Undecane	C ₁₁ H ₂₄	1120-21-4	156.31	0.20515	-0.00023933				247.57	0.1459	469.08	0.0929
337	1-Undecanol	C ₁₁ H ₂₄ O	112-42-5	172.31	0.21211	-0.00021815				288.45	0.1492	561.20	0.0897
338	Vinyl acetate	C ₄ H ₆ O ₂	108-05-4	86.09	0.256	-0.0003542				180.35	0.1921	410.00	0.1108
339	Vinyl acetylene	C ₄ H ₄	689-97-4	52.07	0.22838	-0.00035173	1.861E-09			173.15	0.1675	278.25	0.1305
340	Vinyl chloride	C ₂ H ₃ Cl	75-01-4	62.50	0.2333	-0.00039223				119.36	0.1865	345.60	0.0977
341	Vinyl trichlorosilane	C ₂ H ₃ Cl ₃ Si	75-94-5	161.49	0.21831	-0.00029122				178.35	0.1664	434.52	0.0918
342	Water	H ₂ O	7732-18-5	18.02	-0.432	0.0057255				273.16	0.5672	633.15	0.4272
343	<i>m</i> -Xylene	C ₈ H ₁₀	108-38-3	106.17	0.20044	-0.00023544				225.30	0.1474	413.10	0.1032
344	<i>o</i> -Xylene	C ₈ H ₁₀	95-47-6	106.17	0.19989	-0.0002299				247.98	0.1429	417.58	0.1039
345	<i>p</i> -Xylene	C ₈ H ₁₀	106-42-3	106.17	0.20003	-0.00023573				286.41	0.1325	413.10	0.1027

The liquid thermal conductivity is calculated by

$$k = C1 + C2T + C3T^2 + C4T^3 + C5T^4.$$

where k is the thermal conductivity in W/(m·K) and T is the temperature in K. Thermal conductivities are at either 1 atm or the vapor pressure, whichever is higher. All substances are listed by chemical family in Table 2-6 and by formula in Table 2-7.

Values in this table were taken from the Design Institute for Physical Properties (DIPPR) of the American Institute of Chemical Engineers (AIChE), copyright 2007 AIChE and reproduced with permission of AIChE and of the DIPPR Evaluated Process Design Data Project Steering Committee. Their source should be cited as R. L. Rowley, W. V. Wilding, J. L. Oscarson, Y. Yang, N. A. Zundel, T. E. Daubert, R. P. Danner, DIPPR® Data Compilation of Pure Chemical Properties, Design Institute for Physical Properties, AIChE, New York (2007).

The number of digits provided for values at $T_{\min}$ and $T_{\max}$ was chosen for uniformity of appearance and formatting; these do not represent the uncertainties of the physical quantities, but are the result of calculations from the standard thermophysical property formulations within a fixed format.

TABLE 2-316 Transport Properties of Selected Gases at Atmospheric Pressure*

Substance	Thermal conductivity, W/(m·K) Temperature, K					Viscosity, 10 ⁻⁴ Pa·s Temperature, K					Prandtl number, dimensionless Temperature, K			
	250	300	400	500	600	250	300	400	500	600	250	300	400	500
Acetone	0.0080	0.0115	0.0201	0.0310			0.077	0.101	0.128	0.156				
Acetylene	0.0162	0.0213	0.0332	0.0452	0.0561		0.104	0.135	0.164					
Benzene	0.0077	0.0104	0.0195	0.0335	0.0524		0.076	0.101	0.127	0.154				
Bromine	0.0038	0.0048	0.0067					0.203	0.260	0.291				
CCl ₄	0.0053	0.0067	0.0099	0.0126			0.101	0.131	0.162	0.191				
Chlorine	0.0071	0.0089	0.0124	0.0156	0.0190		0.136	0.178	0.218	0.259				
Deuterium	0.122	0.141	0.176			0.111	0.126	0.153	0.178	0.201				
Propylene	0.0114	0.0168	0.0226	0.0430	0.0580	0.073	0.087	0.115	0.141		0.860	0.797	0.762	
R 11		0.0078	0.0119			0.094	0.110	0.144				0.814	0.761	
R 12	0.0072	0.0097	0.0151	0.0208		0.108	0.126	0.162			0.827	0.781	0.745	0.708
R 13	0.0091	0.0121	0.0185	0.0248		0.123	0.145	0.190			0.796	0.766	0.759	0.757
R 21		0.0088	0.0135	0.0181		0.100	0.115	0.154				0.779	0.773	
R 22	0.0080	0.0109	0.0170	0.0230	0.0290	0.109	0.129	0.168			0.820	0.771	0.760	
SO ₂	0.0078	0.0096	0.0143	0.0200	0.0256		0.129	0.175	0.217	0.256				

*An approximate interpolation scheme is to plot the logarithm of the viscosity or the thermal conductivity versus the logarithm of the absolute temperature. At 250 K the viscosity of gaseous argon is to be read as 1.95×10^{-5} Pa·s = 0.0000195 N·s/m².

TABLE 2-317 Lower and Upper Flammability Limits, Flash Point, and Autoignition Temperature for Selected Hydrocarbons

Compound	CAS no.	Formula	LFL	LFL rating	UFL	UFL rating	Flash point °C	Flash point rating	Autoignition T °C	Autoignition rating
<i>Paraffin hydrocarbons</i>										
Methane	74828	CH ₄	5.00	12	15.00	12	-187.15	4	536.85	11
Ethane	74840	C ₂ H ₆	2.90	12	13.00	12	-130.15	4	471.85	11
Propane	74986	C ₃ H ₈	2.00	12	9.50	12	-104.00	4	449.85	11
Butane	106978	C ₄ H ₁₀	1.50	12	9.00	12	-69.00	6	287.55	11
Isobutane	75285	C ₄ H ₁₀	1.80	12	8.40	12	-82.59	5	460.00	11
Pentane	109660	C ₅ H ₁₂	1.30	14	8.00	14	-40.00	11	242.85	11
Isopentane	78784	C ₅ H ₁₂	1.30	12	8.00	12	-56.00	11	420.00	11
2,2-Dimethylpropane	463821	C ₅ H ₁₂	1.40	11	7.50	11	-72.15	4	450.00	11
Hexane	1100543	C ₆ H ₁₄	1.05	13	7.68	13	-21.65	11	224.85	11
Heptane	142825	C ₇ H ₁₆	1.00	14	7.00	14	-4.15	11	203.85	11
2,3-Dimethylpentane	565593	C ₇ H ₁₆	1.12	4	6.75	4	-6.00	4	335.00	11
Octane	111659	C ₈ H ₁₈	0.80	12	6.50	12	12.85	11	205.85	11
Isooctane	540841	C ₈ H ₁₈	0.95	11	6.00	11	-12.22	11	411.00	11
Nonane	111842	C ₉ H ₂₀	0.70	12	5.60	12	30.85	11	204.85	11
Decane	124185	C ₁₀ H ₂₂	0.70	12	5.40	12	45.85	11	200.85	11
<i>Olefins</i>										
Ethylene	74851	C ₂ H ₄	2.70	11	36.00	11	-146.15	4	450.00	11
Propylene	115071	C ₃ H ₆	2.00	11	11.00	11	-108.15	11	455.00	11
1-Butene	106989	C ₄ H ₈	1.60	11	9.30	11	-79.81	5	383.85	11
2-Butene	107017	C ₄ H ₈	1.70	8	9.70	8	<-34.00	1	325.00	8
1-Pentene	109671	C ₅ H ₁₀	1.50	11	8.70	11	-18.15	11	272.85	11
<i>Acetylenes</i>										
Acetylene	74862	C ₂ H ₂	2.50	12	80.00	12	-18.15	11	305.00	11
Vinylacetylene	689974	C ₄ H ₄	1.80	6	100.00	4	-66.15	4	310.00	6
Methylacetylene	613878	C ₃ H ₄	1.70	11	16.80	11	-86.15	4	340.00	11
<i>Aromatics</i>										
Benzene	71432	C ₆ H ₆	1.40	11	7.10	11	-11.15	11	561.85	11
Toluene	108583	C ₇ H ₈	1.20	11	7.10	11	4.85	11	535.85	11
o-Xylene	95476	C ₈ H ₁₀	1.00	11	6.00	11	16.85	11	463.85	11
Ethylbenzene	100414	C ₈ H ₁₀	1.00	11	6.70	11	15.00	11	431.85	11
Isopropylbenzene (Cumene)	98828	C ₉ H ₁₂	0.88	12	6.50	12	31.00	12	423.85	11
Anthracene	120127	C ₁₄ H ₁₀	0.65	8	5.46	4	120.85	4	540.00	8
<i>Cyclic hydrocarbons</i>										
Cyclopropane	75194	C ₃ H ₆	2.40	12	10.40	12	92.00	4	497.85	11
Furan	110009	C ₄ H ₄ O	2.30	11	14.30	11	-36.15	11	510.00	6
Cyclopentadiene	542927	C ₅ H ₆	1.62	4	7.70	4	-28.15	4	429.85	4
Cyclohexane	110827	C ₆ H ₁₂	1.30	8	7.80	8	-18.15	4	245.00	11
Methylcyclohexane	108872	C ₇ H ₁₄	1.10	11	6.70	11	-6.00	11	250.00	11
Phenol	108952	C ₆ H ₆ O	1.52	6	8.76	6	79.85	11	714.85	11
Dicyclopentadiene	77736	C ₁₀ H ₁₂	1.00	4	6.76	4	26.00	4	505.00	5
<i>Alcohols</i>										
Methyl alcohol	67561	CH ₃ O	7.30	11	36.00	11	10.85	11	463.85	11
Ethyl alcohol	64175	C ₂ H ₅ O	3.30	8	19.00	8	12.85	4	356.00	8
2-propen-1-ol	107186	C ₃ H ₆ O	2.50	11	18.00	11	20.85	11	377.85	11
n-Propyl alcohol	71238	C ₃ H ₇ O	2.10	8	13.50	8	24.63	5	440.00	8
Isopropyl alcohol	67630	C ₃ H ₇ O	2.20	8	12.00	8	21.85	4	399.00	5
n-Butyl alcohol	71363	C ₄ H ₉ O	1.40	8	11.20	8	37.19	5	359.00	5
2-Butanol	78922	C ₄ H ₁₀ O	1.68	4	9.80	4	23.85	4	405.00	8
2-Methyl-1-propanol	78831	C ₄ H ₁₀ O	1.70	11	10.90	11	27.85	11	408.00	11
2-Methyl-2-propanol	75650	C ₄ H ₁₀ O	1.90	4	9.00	4	10.85	4	480.00	8
Cyclohexanol	108930	C ₆ H ₁₂ O	1.20	11	9.30	10	67.85	11	300.00	11
<i>Aldehydes</i>										
Formaldehyde	50000	CH ₂ O	7.00	11	73.00	11	-52.96	3	430.00	11
Acetaldehyde	75070	C ₂ H ₄ O	4.00	11	60.00	11	-38.15	11	175.00	8
Acrolein	107028	C ₃ H ₄ O	2.80	11	31.00	11	-26.00	11	233.85	11
Propionaldehyde	123386	C ₃ H ₆ O	2.60	11	16.10	11	-30.00	11	206.85	11
Crotonic aldehyde	123739	C ₄ H ₆ O	2.10	11	15.50	11	8.00	11	231.85	11
Crotonaldehyde	4170303	C ₄ H ₆ O	2.10	8	15.50	8	13.00	6	232.00	5
Isobutyraldehyde	78842	C ₄ H ₈ O	1.60	4	10.60	4	-18.15	4	204.85	11
Butyraldehyde	123728	C ₄ H ₈ O	2.50	8	12.50	4	-7.15	4	215.76	5
2-Furancarboxaldehyde	98011	C ₅ H ₄ O ₂	2.10	8	19.30	4	72.85	4	316.00	5
<i>Ethers</i>										
Dimethyl ether	115106	C ₂ H ₆ O	3.30	11	27.30	11	-41.15	11	350.00	11
Vinyl methyl ether	107255	C ₃ H ₆ O	2.60	11	39.00	11	-56.00	11	287.00	11
Diethyl ether	60297	C ₄ H ₁₀ O	1.90	11	48.00	11	-45.00	11	160.00	11
Diphenyl ether	101848	C ₁₂ H ₁₀ O	0.80	11	1.50	11	95.85	11	618.00	11
<i>Ketones</i>										
Acetone	67641	C ₃ H ₆ O	2.60	8	13.00	8	-18.15	4	465.00	8
Methylethyl ketone	78933	C ₄ H ₈ O	1.80	11	10.00	11	-6.15	11	515.85	11
Methyl phenyl ketone	98862	C ₉ H ₈ O	1.10	8	6.89	4	82.00	8	571.11	11
<i>Acids</i>										
Acetic acid	64197	C ₂ H ₄ O ₂	4.05	8	16.00	4	42.85	11	426.85	11
Hydrocyanic acid	74908	HCN	6.00	11	41.00	11	-18.00	6	537.85	11
Formic acid	64186	CH ₂ O ₂	15.70	10	38.00	11	48.00	12	480.00	11
<i>Esters</i>										
Methyl formate	107313	C ₂ H ₄ O ₂	5.90	11	20.00	11	-19.15	11	455.85	11
Ethyl formate	109944	C ₃ H ₆ O ₂	2.80	8	16.00	8	-19.50	6	455.00	8

Methyl acetate	79209	C ₄ H ₈ O ₂	3.20	8	16.00	8	-15.50	6	476.85	14
Vinyl acetate	108054	C ₄ H ₆ O ₂	2.60	11	13.40	11	-8.15	11	426.85	11
Ethyl acetate	141786	C ₄ H ₈ O ₂	2.20	11	11.40	11	-4.15	11	426.85	11
Propyl acetate	109604	C ₅ H ₁₀ O ₂	2.05	5	8.00	4	-	-	450.00	4
Isopropyl acetate	108214	C ₅ H ₁₀ O ₂	2.00	5	7.80	4	4.50	6	460.00	4
Butyl acetate	123864	C ₆ H ₁₂ O ₂	1.70	8	7.60	8	21.85	4	425.00	8
Isobutyl acetate	110190	C ₆ H ₁₂ O ₂	1.30	11	10.50	11	17.85	11	422.85	11
Amyl acetate	628637	C ₇ H ₁₄ O ₂	1.10	11	7.50	11	23.00	11	360.00	11
<i>Inorganic</i>										
Hydrogen	1333740	H ₂	4.00	11	75.00	11	-260.15	4	400.00	11
Ammonia	7664417	NH ₃	16.00	11	25.00	11	Gas	3	650.85	11
Cyanogen	460195	C ₂ N ₂	6.00	11	32.00	11	-62.15	4	661.85	4
<i>Oxides</i>										
Carbon monoxide	630380	CO	12.50	11	74.00	11	-206.15	4	608.85	11
Ethylene oxide	75218	C ₂ H ₄ O	3.00	11	100.00	8	-19.81	5	428.85	11
Propylene oxide	75569	C ₃ H ₆ O	2.10	11	21.50	11	-37.15	11	465.00	11
Dioxan	123911	C ₈ H ₈ O ₂	2.00	11	22.00	11	11.85	11	180.00	11
Mesityl oxide	141797	C ₈ H ₁₀ O	1.30	4	7.20	4	30.00	4	344.85	4
Styrene oxide	100425	C ₈ H ₈ O	1.10	11	-	-	31.85	11	463.00	11
<i>Peroxides</i>										
Di- <i>tert</i> -butyl peroxide	110054	C ₈ H ₁₈ O ₂	1.00	6	7.05	4	1.00	-	79.85	4
<i>Sulfur containing</i>										
Carbon disulfide	75150	CS ₂	1.30	11	50.00	11	-30.00	-	90.00	11
Hydrogen sulfide	7783064	H ₂ S	4.30	14	45.50	14	-107.15	-	260.00	11
Carbon oxysulfide	463581	COS	12.00	11	29.00	11	-88.15	-	250.00	6
Dimethyl sulfide	75183	C ₂ H ₆ S	2.20	8	19.70	8	-36.00	-	205.00	11
<i>Chlorine containing</i>										
Methyl chloride	74873	CH ₃ Cl	8.10	11	17.20	11	-70.15	4	631.85	11
Ethyl chloride	75003	C ₂ H ₅ Cl	3.80	11	15.40	11	-50.00	11	518.85	11
Isopropyl chloride	75296	C ₃ H ₇ Cl	2.50	11	10.70	11	-35.00	11	592.85	11
Ethylene dichloride	107062	C ₂ H ₄ Cl ₂	6.20	11	16.00	6	12.85	11	412.85	11
Propylene dichloride	75875	C ₃ H ₄ Cl ₂	3.40	11	14.50	11	13.00	11	556.85	11
Dichloromethane	75092	CH ₂ Cl ₂	14.00	6	22.00	4	-9.15	4	615.00	11
2-Chloroethanol	107073	C ₂ H ₅ ClO	4.90	11	15.90	11	40.55	12	425.00	11
Trichloroethylene	79016	C ₂ HCl ₃	8.00	11	40.00	5	32.22	11	410.00	11
Hexachlorobutadiene	87683	C ₄ Cl ₆	2.90	1	15.70	1	121.85	10	610.00	11
Vinyl chloride	75014	C ₂ H ₃ Cl	3.60	11	33.00	11	-78.15	11	471.85	11
Chlorobenzene	108907	C ₆ H ₅ Cl	1.30	11	7.10	11	31.85	11	637.85	11
Benzyl chloride	100447	C ₇ H ₇ Cl	1.10	11	8.00	4	60.00	11	585.00	11
<i>Bromides</i>										
Methyl bromide	74839	CH ₃ Br	10.00	11	15.00	11	-48.15	4	537.22	11
<i>Glycols</i>										
Ethylene glycol	107211	C ₂ H ₆ O ₂	3.20	11	33.00	10	110.85	11	395.85	11
Diethylene glycol	111466	C ₄ H ₁₀ O ₃	2.00	10	17.10	10	123.85	11	223.85	11
Triethylene glycol	112276	C ₆ H ₁₄ O ₄	0.90	11	9.20	11	156.00	11	370.85	11
<i>Amines</i>										
Methyl amine	74895	CH ₃ N	4.90	11	20.70	11	-62.00	7	430.00	11
Ethyl amine	75047	C ₂ H ₅ N	3.50	8	14.00	8	-39.00	6	383.85	7
Dimethyl amine	124403	C ₂ H ₇ N	2.80	11	14.40	11	-57.00	7	400.00	11
Isopropylamine	75310	C ₃ H ₇ N	2.00	11	10.40	11	-37.00	11	400.00	12
Trimethylamine	75503	C ₃ H ₉ N	2.00	11	11.60	11	-7.15	11	190.00	11
Isopropylamine	75310	C ₃ H ₉ N	2.00	11	10.40	11	-37.00	11	400.00	12
Allylamine	107119	C ₃ H ₇ N	2.20	11	22.00	11	-28.59	11	373.89	11
Diethyl amine	109807	C ₄ H ₁₁ N	1.70	12	10.10	12	-39.00	11	310.00	12
<i>tert</i> -Butylamine	75349	C ₄ H ₁₁ N	1.70	11	8.90	11	-9.00	11	380.00	5
Triethyl amine	121448	C ₆ H ₁₅ N	1.20	11	8.00	11	-12.15	12	231.85	10
Cyclohexylamine	108918	C ₆ H ₁₃ N	0.66	11	8.42	4	31.00	13	293.00	11
Monoethanolamine	141435	C ₂ H ₇ NO	3.02	4	24.06	4	85.00	11	410.00	5
Diethanolamine	111422	C ₄ H ₁₁ NO ₂	1.69	4	16.25	4	151.85	11	661.85	11
Dimethylethanolamine	108010	C ₄ H ₁₁ NO	1.67	4	11.36	4	41.00	11	295.00	11
<i>Miscellaneous</i>										
Acrylonitrile	107131	C ₃ H ₃ N	2.42	11	17.34	13	0.00	11	480.85	11
Aniline	62533	C ₆ H ₅ N	1.30	11	11.00	11	70.00	11	616.85	11
Diborane	19287457	B ₂ H ₆	0.90	11	98.00	11	-148.15	4	51.85	11
Methyl methacrylate	80626	C ₅ H ₈ O ₂	2.10	11	12.50	11	11.00	11	435.85	11
Naphtha	8030366	-	0.50	8	5.00	8	-6.00	4	290.00	4
Styrene	100425	C ₈ H ₈	1.10	11	6.10	11	31.85	11	490.00	11
Biphenyl	92524	C ₁₂ H ₁₀	0.60	11	5.80	11	113.00	11	540.00	11
Methylacrylate	96333	C ₅ H ₈ O ₂	2.80	4	25.00	4	5.85	4	468.00	5
Phthalic anhydride	85449	C ₈ H ₄ O ₃	1.70	8	10.50	4	151.85	6	583.85	11

LFL and UFL are lower and upper flammability limits respectively in volume percent fuel in air. Values in this table were taken from the Design Institute for Physical Properties (DIPPR) of the American Institute of Chemical Engineers (AIChE), copyright 2007 AIChE and reproduced with permission of AIChE, the DIPPR Environmental Safety Property Data and Estimations Steering Committee and of the DIPPR Evaluated Process Design Data Project Steering Committee. Their source should be cited as T. N. Rogers, D. A. Zei, R. L. Rowley, W. V. Wilding, J. L. Oscarson, Y. Yang, N. A. Zundel, T. E. Daubert, R. P. Danner, DIPPR® Data Compilation of Pure Chemical Properties, Design Institute for Physical Properties, AIChE, New York (2007). Flammability limits are from B. Lewis and G. Von Elbe, *Combustion, Flames and Explosions of Gases*, New York: Harcourt Brace Jovanovich (1987). Flash point data are from N. I. Sax, *Dangerous Properties of Industrial Materials*, 6th ed., New York: Van Nostrand Reinhold (1984). Autoignition data are from I. Glassman, *Combustion*, 3d ed., New York: Academic Press (1996). The ratings reflect DIPPR® ESP's effort to provide a critical evaluation and quality assessment of each data point with 15 being the highest score possible. The rating is not directly correlated with the estimated experimental uncertainty.

TABLE 2-318 Viscosities of Liquids: Coordinates for Use with Fig. 2-32

Liquid	X	Y	Liquid	X	Y
Acetaldehyde	15.2	4.8	Freon-113	12.5	11.4
Acetic acid, 100%	12.1	14.2	Glycerol, 100%	2.0	30.0
Acetic acid, 70%	9.5	17.0	Glycerol, 50%	6.9	19.6
Acetic anhydride	12.7	12.8	Heptane	14.1	8.4
Acetone, 100%	14.5	7.2	Hexane	14.7	7.0
Acetone, 35%	7.9	15.0	Hydrochloric acid, 31.5%	13.0	16.6
Acetonitrile	14.4	7.4	Iodobenzene	12.8	15.9
Acrylic acid	12.3	13.9	Isobutyl alcohol	7.1	18.0
Allyl alcohol	10.2	14.3	Isobutyric acid	12.2	14.4
Allyl bromide	14.4	9.6	Isopropyl alcohol	8.2	16.0
Allyl iodide	14.0	11.7	Isopropyl bromide	14.1	9.2
Ammonia, 100%	12.6	2.0	Isopropyl chloride	13.9	7.1
Ammonia, 26%	10.1	13.9	Isopropyl iodide	13.7	11.2
Amyl acetate	11.8	12.5	Kerosene	10.2	16.9
Amyl alcohol	7.5	18.4	Linseed oil, raw	7.5	27.2
Aniline	8.1	18.7	Mercury	18.4	16.4
Anisole	12.3	13.5	Methanol, 100%	12.4	10.5
Arsenic trichloride	13.9	14.5	Methanol, 90%	12.3	11.8
Benzene	12.5	10.9	Methanol, 40%	7.8	15.5
Brine, CaCl ₂ , 25%	6.6	15.9	Methyl acetate	14.2	8.2
Brine, NaCl, 25%	10.2	16.6	Methyl acrylate	13.0	9.5
Bromine	14.2	13.2	Methyl <i>i</i> -butyrate	12.3	9.7
Bromotoluene	20.0	15.9	Methyl <i>n</i> -butyrate	13.2	10.3
Butyl acetate	12.3	11.0	Methyl chloride	15.0	3.8
Butyl acrylate	11.5	12.6	Methyl ethyl ketone	13.9	8.6
Butyl alcohol	8.6	17.2	Methyl formate	14.2	7.5
Butyric acid	12.1	15.3	Methyl iodide	14.3	9.3
Carbon dioxide	11.6	0.3	Methyl propionate	13.5	9.0
Carbon disulfide	16.1	7.5	Methyl propyl ketone	14.3	9.5
Carbon tetrachloride	12.7	13.1	Methyl sulfide	15.3	6.4
Chlorobenzene	12.3	12.4	Napthalene	7.9	18.1
Chloroform	14.4	10.2	Nitric acid, 95%	12.8	13.8
Chlorosulfonic acid	11.2	18.1	Nitric acid, 60%	10.8	17.0
Chlorotoluene, ortho	13.0	13.3	Nitrobenzene	10.6	16.2
Chlorotoluene, meta	13.3	12.5	Nitrogen dioxide	12.9	8.6
Chlorotoluene, para	13.3	12.5	Nitrotoluene	11.0	17.0
Cresol, meta	2.5	20.8	Octane	13.7	10.0
Cyclohexanol	2.9	24.3	Octyl alcohol	6.6	21.1
Cyclohexane	9.8	12.9	Pentachloroethane	10.9	17.3
Dibromomethane	12.7	15.8	Pentane	14.9	5.2
Dichloroethane	13.2	12.2	Phenol	6.9	20.8
Dichloromethane	14.6	8.9	Phosphorus tribromide	13.8	16.7
Diethyl ketone	13.5	9.2	Phosphorus trichloride	16.2	10.9
Diethyl oxalate	11.0	16.4	Propionic acid	12.8	13.8
Diethylene glycol	5.0	24.7	Propyl acetate	13.1	10.3
Diphenyl	12.0	18.3	Propyl alcohol	9.1	16.5
Dipropyl ether	13.2	8.6	Propyl bromide	14.5	9.6
Dipropyl oxalate	10.3	17.7	Propyl chloride	14.4	7.5
Ethyl acetate	13.7	9.1	Propyl formate	13.1	9.7
Ethyl acrylate	12.7	10.4	Propyl iodide	14.1	11.6
Ethyl alcohol, 100%	10.5	13.8	Sodium	16.4	13.9
Ethyl alcohol, 95%	9.8	14.3	Sodium hydroxide, 50%	3.2	25.8
Ethyl alcohol, 40%	6.5	16.6	Stannic chloride	13.5	12.8
Ethyl benzene	13.2	11.5	Succinonitrile	10.1	20.8
Ethyl bromide	14.5	8.1	Sulfur dioxide	15.2	7.1
2-Ethyl butyl acrylate	11.2	14.0	Sulfuric acid, 110%	7.2	27.4
Ethyl chloride	14.8	6.0	Sulfuric acid, 100%	8.0	25.1
Ethyl ether	14.5	5.3	Sulfuric acid, 98%	7.0	24.8
Ethyl formate	14.2	8.4	Sulfuric acid, 60%	10.2	21.3
2-Ethyl hexyl acrylate	9.0	15.0	Sulfuryl chloride	15.2	12.4
Ethyl iodide	14.7	10.3	Tetrachloroethane	11.9	15.7
Ethyl propionate	13.2	9.9	Thiophene	13.2	11.0
Ethyl propyl ether	14.0	7.0	Titanium tetrachloride	14.4	12.3
Ethyl sulfide	13.8	8.9	Toluene	13.7	10.4
Ethylene bromide	11.9	15.7	Trichloroethylene	14.8	10.5
Ethylene chloride	12.7	12.2	Triethylene glycol	4.7	24.8
Ethylene glycol	6.0	23.6	Turpentine	11.5	14.9
Ethylidene chloride	14.1	8.7	Vinyl acetate	14.0	8.8
Fluorobenzene	13.7	10.4	Vinyl toluene	13.4	12.0
Formic acid	10.7	15.8	Water	10.2	13.0
Freon-11	14.4	9.0	Xylene, ortho	13.5	12.1
Freon-12	16.8	5.6	Xylene, meta	13.9	10.6
Freon-21	15.7	7.5	Xylene, para	13.9	10.9
Freon-22	17.2	4.7			

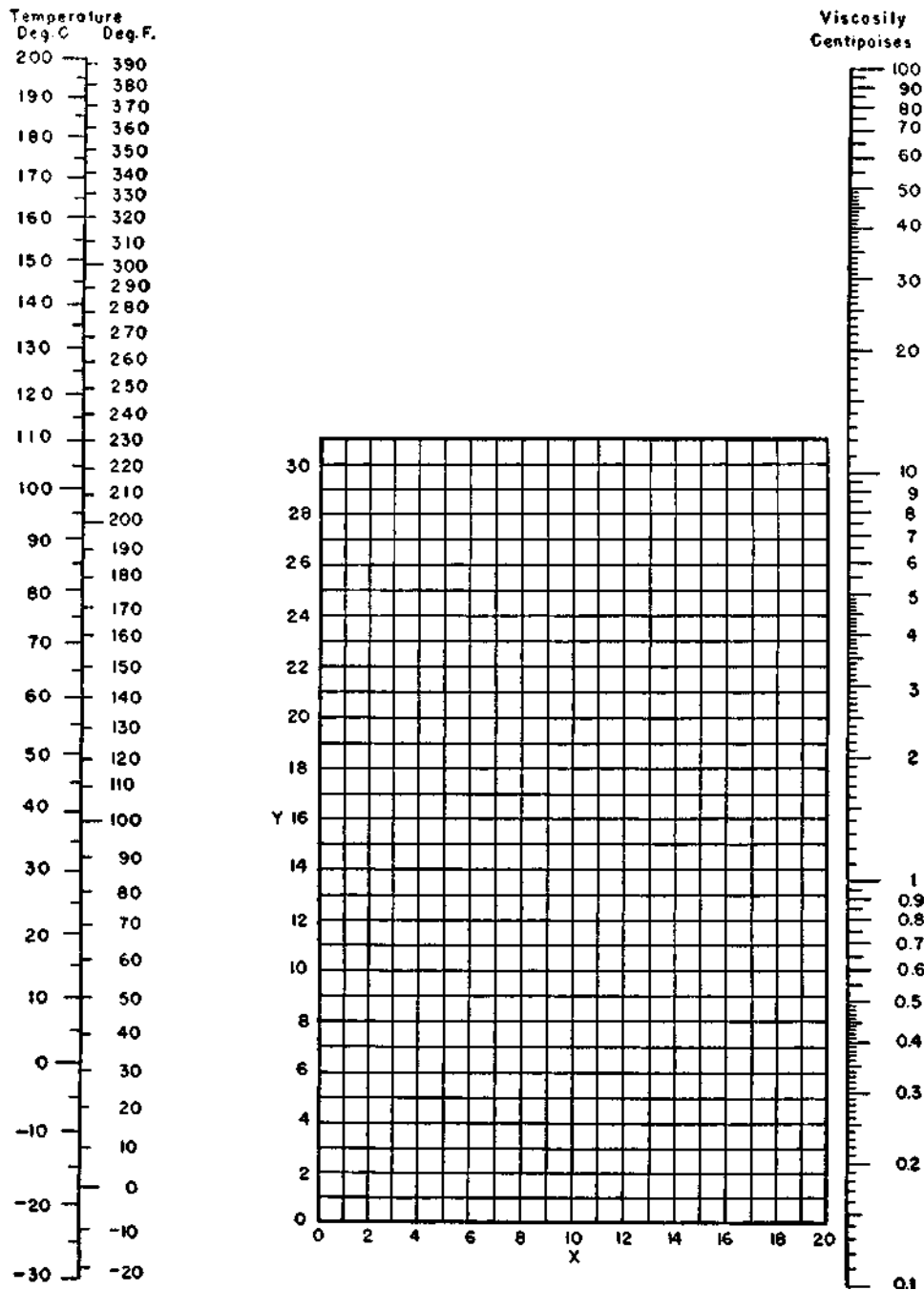


FIG. 2-32 Nomograph for viscosities of liquids at 1 atm. For coordinates see Table 2-318. To convert centipoises to pascal-seconds, multiply by 0.001.

TABLE 2-320

No.	Compound	Range, °C	Exp pt	% Avg abs devi
31	Acetaldehyde	0 - 30	2	0.4
40	Acetic acid	18 - 79	5	0.1
29	Acetone	0 - 40	3	0.4
20	Aniline	0 - 93	6	1.3
2	Benzaldehyde	16 - 68	4	0.7
13	Benzene	20 - 116	5	0.6
47	n-Butane	-21 - 0	2	0.2
16	n-Butanol	0 - 102	7	0.4
48	i-Butanol	0 - 20	2	0.6
49	s-Butanol	0 - 65	3	0.2
18	t-Butanol	20 - 77	3	2.5
28	Butyl acetate	0 - 38	4	1.1
23	Carbon tetrachloride	-20 - 25	4	1.1
25	Chlorobenzene	-40 - 80	6	0.5
42	Chloroform	0 - 40	3	1.3
19	m-Cresol	20 - 80	2	0.2
11	Cyclohexane	20 - 38	2	0.4
7	n-Decane	20 - 76	5	1.0
30	Diethyl ether	0 - 25	3	0.7
9	2,3-Dimethylbutane	32 - 49	2	0.3
38	n-Dodecane	0 - 63	6	1.1
15	Ethanol	17 - 77	7	0.5
27	Ethyl acetate	6 - 160	9	1.6
14	Ethylbenzene	0 - 80	3	0.7
24	Ethyl bromide	0 - 30	3	0.3
4	n-Heptane	0 - 60	3	0.3
3	n-Hexane	17 - 40	5	1.4
26	Iodobenzene	-20 - 80	4	0.7
41	Methanol	20 - 62	6	0.2
39	Methyl acetate	4 - 21	2	0.4
12	Methylcyclopentane	20 - 38	2	0.1
22	Methylene chloride	-20 - 20	3	0.9
8	2-Methylpentane	32 - 49	2	0.5
6	n-Nonane	16 - 77	5	2.0
5	n-Octane	0 - 77	3	0.3
10	i-Octane	20 - 77	4	1.6
17	n-Octanol	20 - 212	3	1.3
43	n-Pentanol	0 - 38	3	0.4
1	Propane	-60 - 50	4	1.6
32	n-Propanol	16 - 86	6	0.6
21	Propionic acid	12 - 30	3	1.7
44	n-Propyl acetate	0 - 38	2	0.1
33	i-Propylbenzene	0 - 38	4	0.2
45	Refrigerant-11, CFCl ₃	0 - 20	4	0.5
46	Refrigerant-12, CF ₂ Cl ₂	-42 - 70	4	1.8
50	Refrigerant-13, CF ₃ Cl	-20 - 20	4	1.7
51	Refrigerant-22, CHF ₂ Cl	-65 - 60	7	2.5
52	Refrigerant-113, C ₂ F ₃ Cl ₃	0 - 20	4	0.7
53	Refrigerant-114, C ₂ F ₄ Cl ₂	-25 - 20	4	0.8
54	Refrigerant-142, C ₂ H ₃ F ₂ Cl	0 - 90	2	0.9
34	Toluene	-70 - 200	5	1.8
55	n-Tridecane	20 - 228	5	2.4
36	m-Xylene	0 - 208	5	1.0
35	o-Xylene	0 - 208	5	0.7
37	p-Xylene	0 - 251	5	1.4

TABLE 2-319 Viscosity of Sucrose Solutions*

Viscosity in centipoises

Temp., °C	Percentage sucrose by weight			Temp., °C	Percentage sucrose by weight		
	20	40	60		20	40	60
0	3.818	14.82		50	0.974	2.506	14.06
5	3.166	11.60		55	0.887	2.227	11.71
10	2.662	9.830	113.9	60	0.811	1.989	9.87
15	2.275	7.496	74.9	65	0.745	1.785	8.37
20	1.967	6.223	56.7	70	0.688	1.614	7.18
25	1.710	5.206	44.02	75	0.637	1.467	6.22
30	1.510	4.398	34.01	80	0.592	1.339	5.42
35	1.336	3.776	26.62	85	0.552	1.226	4.75
40	1.197	3.261	21.30	90		1.127	4.17
45	1.074	2.858	17.24	95		1.041	3.73

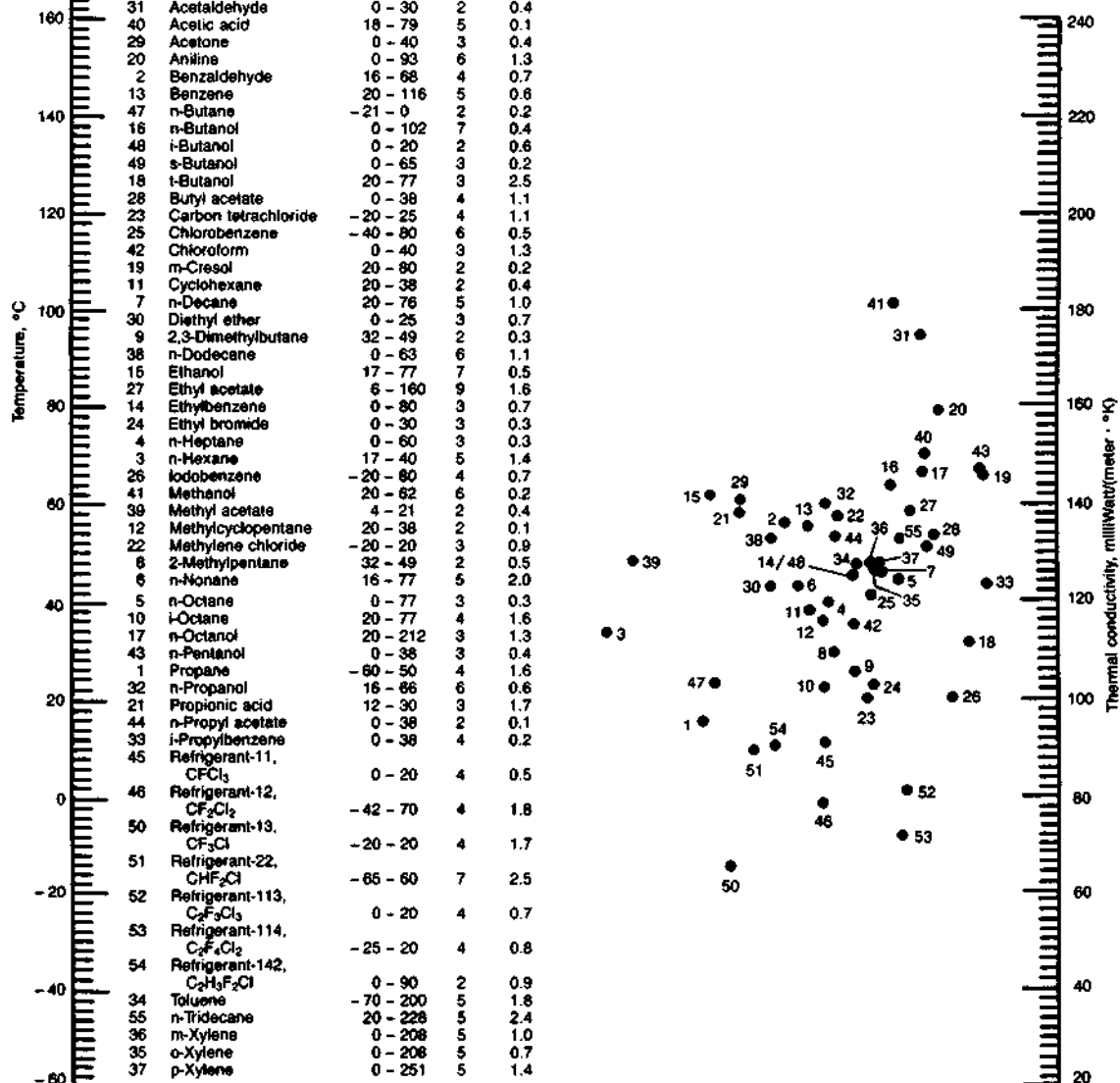
*International Critical Tables, vol. 5, p. 23. Bingham and Jackson, *Bur. Standards Bull.* 14 (1919): 59.FIG. 2-33 and TABLE 2-320 Nomograph (right) for thermal conductivity of organic liquids. (From Mallu and Rao, *Hydroc. Proc.* 78, 1988.)

TABLE 2-321 Prandtl Number of Air*

Temperature, K	Pressure, bar											
	1	5	10	20	30	40	50	60	70	80	90	100
80	mix	2.31	2.32	2.35	2.37	2.40	2.42	2.45	2.48	2.51	2.54	2.57
90	0.796	1.76	1.77	1.78	1.79	1.81	1.82	1.83	1.85	1.87	1.89	1.91
100	0.786	0.872	1.54	1.53	1.53	1.53	1.53	1.53	1.53	1.54	1.54	1.55
120	0.773	0.813	0.89	1.44	1.65	1.54	1.48	1.43	1.40	1.38	1.36	1.34
140	0.763	0.782	0.82	0.94	1.20	1.59	2.14	2.43	2.07	1.78	1.62	1.52
160	0.754	0.765	0.78	0.84	0.92	1.03	1.13	1.25	1.37	1.65	1.83	1.72
180	0.745	0.754	0.763	0.792	0.830	0.876	0.932	1.00	1.07	1.14	1.20	1.25
200	0.738	0.743	0.749	0.766	0.788	0.812	0.841	0.87	0.90	0.95	0.97	1.00
240	0.724	0.727	0.729	0.737	0.746	0.756	0.767	0.78	0.80	0.81	0.81	0.82
280	0.710	0.711	0.713	0.717	0.721	0.726	0.731	0.737	0.742	0.75	0.75	0.76
300	0.705	0.707	0.708	0.712	0.715	0.717	0.721	0.725	0.728	0.732	0.737	0.742
350	0.699	0.699	0.699	0.701	0.703	0.705	0.707	0.709	0.711	0.712	0.714	0.716
400	0.694	0.694	0.694	0.695	0.696	0.697	0.698	0.699	0.700	0.701	0.703	0.704
450	0.691	0.691	0.691	0.691	0.692	0.692	0.693	0.693	0.694	0.695	0.695	0.696
500	0.689	0.689	0.689	0.689	0.689	0.690	0.690	0.690	0.690	0.691	0.691	0.691
600	0.690	0.690	0.690	0.689	0.689	0.689	0.689	0.689	0.689	0.690	0.690	0.690
700	0.696	0.696	0.695	0.695	0.695	0.695	0.695	0.695	0.695	0.695	0.695	0.695
800	0.705	0.704	0.704	0.704	0.704	0.703	0.703	0.703	0.703	0.702	0.702	0.702
900	0.709	0.709	0.708	0.708	0.708	0.708	0.708	0.708	0.708	0.708	0.708	0.708
1000	0.711	0.711	0.711	0.711	0.711	0.710	0.710	0.710	0.710	0.709	0.709	0.709

*Compiled by P. E. Liley from tables of specific heat at constant pressure, thermal conductivity, and viscosity given in SI units for integral kelvin temperatures and pressures in bars by Vasserman. *Thermophysical Properties of Air and Its Components* and *Thermophysical Properties of Liquid Air and Its Components*. Nauka, Moscow, and in translated form by the National Bureau of Standards, Washington. The number of significant figures given above reflects the similar numbers appearing for the constituent properties in the source references. While reasonable agreement occurs for atmospheric pressure with some other works, the fragmentary data available for the saturated, etc., states show large deviations.

TABLE 2-322 Prandtl Number of Liquid Refrigerants*

Refrigerant	No.	Temperature, K										
		180	200	220	240	260	280	300	320	340	360	380
Trichlorofluoromethane	11		11.9	8.64	6.73	5.33	4.74	4.18				
Dichlorodifluoromethane	12	7.00	5.25	4.27	3.65	3.27	3.08	3.04	3.19	3.44	4.00	
Chlorotrifluoromethane	13		2.96	2.67	2.69	3.05	3.57			—	—	—
Bromotrifluoromethane	13B1	4.80	3.75	3.27	2.94	2.83	3.03	3.61	4.52			
Dichlorofluoromethane	21			5.72	4.50	3.87	3.48	3.25	3.16	3.17		
Chlorodifluoromethane	22	4.68	3.76	3.23	2.93	2.79	2.77	2.87	3.18	3.54		—
Methyl chloride	40			2.53	2.42	2.40	2.45	2.60	2.85			
Trichlorotrifluoroethane	113	—	—	—	—	—	—	—	7.04	6.23	5.61	5.18
Dichlorotetrafluoroethane	114	25.7	15.13	11.18	8.59	6.94	5.77	5.06	4.78	4.82		
Chloropentafluoroethane	115	—	7.85	6.16	5.21	4.67	4.40	4.46	4.90		—	—
Ethane	170	2.55	2.29	2.22	2.40	2.70			—	—	—	—
Propane	290	5.28	4.46	3.88	3.44	3.16	3.02	3.16				
Octafluorocyclobutane	C318	—	—	—	11.2	8.74	7.35	6.37	5.87	5.96		—
Dichlorodifluoromethane/difluoroethane	500		5.78	4.23	3.40	3.13	3.01	3.13	3.35	3.72		—
Chlorodifluoromethane/chloropentafluoroethane	502		5.73	4.71	4.13	3.81					—	—
Trifluoromethane/chlorotrifluoromethane	503	2.10	2.09	2.24	2.43	2.89		—	—	—	—	—
Methylene fluoride/chloropentafluoroethane	504		4.90	3.60	3.04	2.79	2.69	2.85	3.30	—	—	—
Butane	600	8.35	6.19	5.20	4.44	3.83	3.44	3.22	3.07	3.02		
Isobutane (2-methyl propane)	600a		8.26	6.36	5.18	4.49	3.93	3.66	3.53	3.53	3.77	4.68
Ammonia	717	—			1.97	1.76	1.54	1.40	1.29	1.24	1.25	1.34
Water	718	—	—	—	—	—	10.3	5.69	3.65	2.60	1.99	1.59
Ethylene	1150	1.85	1.74	1.78	2.07	2.70	4.4	—	—	—	—	—
Propylene	1270	3.80	2.24	1.88	1.71	1.71	1.88	2.24	3.91	4.73		

*Dashes indicate inaccessible states. Average uncertainty is about 20 percent. Values derived from formulations for thermal conductivity, specific heat at constant pressure, and viscosity contained in *Thermophysical Properties of Refrigerants*. American Society of Heating, Refrigerating and Air-Conditioning Engineers, New York, 1976. For further details see M. W. Johnson, M.S.M.E. thesis, Purdue University, West Lafayette, Ind., 1976.

TABLE 2-323 Thermophysical Properties of Miscellaneous Saturated Liquids

Substance	Property	Temperature, °C															
		−50	−40	−30	−20	−10	0	10	20	30	40	50	60	70	80	90	100
Acetaldehyde	ρ (kg/m³)	863	852	840	828	816	804	794	783								
	c _p (kJ/kg·K)	2.05	2.08	2.11	2.14	2.17	2.20	2.24	2.28								
	μ (10 ^{−6} Pa·s)	460	404	358	321	290	263	241	222								
	k (W/m·K)	0.211	0.206	0.200	0.195	0.189	0.184	0.182	0.180								
	Pr	4.47	4.08	3.78	3.52	3.33	3.14	2.97	2.81								
Acetic acid	ρ (kg/m³)								1049	1039	1028	1018	1006	995	984	972	960
	c _p (kJ/kg·K)								2.031								
	μ (10 ^{−6} Pa·s)								1210	1102	1010	795	600				
	k (W/m·K)								0.173	0.170	0.168	0.167	0.165	0.163	0.161		
	Pr								14.2								
Aniline	ρ (kg/m³)	—	—	—	—	—	1039	1030	1022	1013	1005	996	987	978	969	960	951
	c _p (kJ/kg·K)	—	—	—	—	—	2.024	2.047	2.071	2.093	2.113	2.132	2.17	2.20	2.23	2.27	2.32
	μ (10 ^{−6} Pa·s)	—	—	—	—	—	10200	6500	4400	3160	2370	1850	1510	1270	1090	935	825
	k (W/m·K)	—	—	—	—	—	0.186	0.184	0.182	0.180	0.177	0.174	0.171	0.169	0.168	0.167	0.167
	Pr	—	—	—	—	—	111	72	50	36.7	28.3	22.7	19.2	16.5	14.5	12.7	11.5
Butanol	ρ (kg/m³)	845	841	837	833	829	825	817	810	803	797	791	784	776	768	760	753
	c _p (kJ/kg·K)	1.947	1.996	2.046	2.100	2.153	2.202	2.262	2.345	2.437	2.524	2.621					
	μ (10 ^{−6} Pa·s)	34700	22400	14700	10300	7400	5190	3870	2950	2300	1780	1410	1140	930	760	630	535
	k (W/m·K)	0.175	0.174	0.173	0.172	0.171	0.170	0.168	0.167	0.166	0.165	0.164	0.163	0.162	0.161	0.160	0.159
	Pr	3860	2570	1740	1260	930	670	120	41	33.8	27.2	22.5					
Carbon disulfide	ρ (kg/m³)	1362	1348	1334	1320	1306	1292	1278	1263								
	c _p (kJ/kg·K)	0.988	0.989	0.990	0.991	0.993	0.996	1.004	1.017								
	μ (10 ^{−6} Pa·s)	630	580	535	496	463	435	405	375	350	330						
	k (W/m·K)	0.194	0.190	0.186	0.182	0.178	0.174	0.170	0.166	0.161	0.158	0.156	0.154	0.152	0.150		
	Pr	3.21	3.02	2.85	2.70	2.58	2.49	2.39	2.30								
Cyclohexane	ρ (kg/m³)	—	—	—	—	—	789	779	779	769	759	750	740	731	721		
	c _p (kJ/kg·K)	—	—	—	—	—	2.068	2.081	2.094	2.106	2.119						
	μ (10 ^{−6} Pa·s)	—	—	—	—	—	1175	980	820	710	605	540					
	k (W/m·K)	—	—	—	—	—	0.122	0.120	0.119	0.118	0.117	0.116	0.114	0.112			
	Pr	—	—	—	—	—	19.9	17.0	14.4	12.7	11.0						
Ethanol	ρ (kg/m³)						806	798	789	781	776	763	754	745	735	725	716
	c _p (kJ/kg·K)						2.27	2.35	2.43	2.52	2.62	2.73	2.83	2.93	3.03	3.19	3.30
	μ (10 ^{−6} Pa·s)	6400	4790	3650	2825	2220	1770	1470	1200	1000	835	700	590	500	435	370	314
	k (W/m·K)	0.188	0.186	0.184	0.181	0.179	0.177	0.175	0.173	0.171	0.168	0.165	0.162	0.159	0.156	0.153	0.151
	Pr	68.4	52.5	41.3	33.2	27.2	22.7	19.7	16.9	14.7	13.0	11.6	10.3	9.2	8.4	7.7	6.9
Ethyl acetate	ρ (kg/m³)					947	935	924	912	901	888	876	863	851	838	825	811
	c _p (kJ/kg·K)									2.01							
	μ (10 ^{−6} Pa·s)	1090					580	510	455	400	370	345	310	280	250	230	220
	k (W/m·K)								0.145	0.142	0.139	0.136	0.133	0.130	0.127	0.123	0.119
	Pr								6.3								
Ethylamine	ρ (kg/m³)	761	750	739	729	718	707	695	683	671	658	646	633	620	607		
	c _p (kJ/kg·K)	2.95	2.97	2.98	3.00	3.01	3.03										
	μ (10 ^{−6} Pa·s)	580	500	435	390	350	320										
	k (W/m·K)	0.204	0.201	0.199	0.196	0.194	0.191										
	Pr	8.39	7.39	6.51	5.97	5.43	5.08										
Ethyl ether	ρ (kg/m³)	790	780	769	758	747	736	725	714	702	689	676	666	653	640	625	611
	c _p (kJ/kg·K)	2.135	2.156	2.179	2.205	2.233	2.265	2.299	2.332	2.36	2.39	2.43	2.47	2.51			
	μ (10 ^{−6} Pa·s)	550	470	410	365	330	290	265	233	214	197	181	166	153	140	129	118
	k (W/m·K)	0.159	0.155	0.151	0.147	0.144	0.140	0.139	0.134	0.129	0.125	0.120	0.116	0.112	0.112		
	Pr	7.39	6.54	5.92	5.48	5.12	4.69	4.38	4.05	3.92	3.77	3.67	3.54	3.43			
Ethyl iodide	ρ (kg/m³)																
	c _p (kJ/kg·K)																
	μ (10 ^{−6} Pa·s)			0.656	0.663	0.670	0.677	0.684	0.691	0.698	0.705	0.712	0.718	0.724			
	k (W/m·K)						730	655	590	539	495	455	420	390			
	Pr						0.092	0.090	0.088	0.086	0.085	0.083	0.081	0.080			
Ethylene glycol	ρ (kg/m³)						1127	1120	1113	1106	1099	1092	1085	1077	1070	1063	1056
	c _p (kJ/kg·K)						2.272	2.327	2.381	2.431	2.484	2.536	2.586	2.636	2.685	2.734	2.779
	μ (10 ^{−6} Pa·s)						57000	33300	20200	13400	9100	7070	4000	3450	3000	2440	2000
	k (W/m·K)						0.254	0.255	0.256	0.258	0.259	0.260					
	Pr						510	305	190	126	87.3	69.0					
Formic acid	ρ (kg/m³)						1241	1231	1220	1209	1196	1184	1170	1156	1140	1124	1108
	c _p (kJ/kg·K)																
	μ (10 ^{−6} Pa·s)							2260	1800	1470	1220	1030	890	780	680	615	550
	k (W/m·K)						0.265	0.261	0.257	0.257	0.253	0.250	0.246	0.243	0.240	0.236	0.232
	Pr																

TABLE 2-323 Thermophysical Properties of Miscellaneous Saturated Liquids (Concluded)

Substance	Property	Temperature, °C															
		−50	−40	−30	−20	−10	0	10	20	30	40	50	60	70	80	90	100
Gasoline	ρ (kg/m ³)				784	775	767	759	751	743	735	721	717	708	699	690	681
	c_p (kJ/kg·K)				1.88	1.92	1.97	2.02	2.06	2.11	2.15	2.20	2.25	2.30	2.35	2.41	2.46
	μ (10 ^{−6} Pa·s)	1710	1400	1170	990	850	735	645	530	464	410	367	330	298	270	246	225
	k (W/m·K)	0.131	0.128	0.125	0.123	0.121	0.120	0.118	0.116	0.114	0.112	0.110	0.108	0.106	0.104	0.102	0.100
	Pr				15.1	13.5	12.1	11.0	9.41	8.59	7.87	7.34	6.88	6.47	6.10	5.81	5.54
Glycerol	ρ (kg/m ³)	—	—	—	—	—	1276	1270	1260	1254	1248	1242					
	c_p (kJ/kg·K)								2.393	2.406	2.457	2.504	2.548	2.588	2.625	2.657	2.686
	μ (10 ^{−6} Pa·s)						1.2.+7	4.0.+6	1.5.+6								
	k (W/m·K)								0.284	0.285	0.287	0.288	0.289	0.291	0.293	0.294	0.295
	Pr								12650								
Kerosene	ρ (kg/m ³)						781	774	767	760	754	748	742				
	c_p (kJ/kg·K)						1.91	1.96	2.02	2.07	2.13	2.18	2.23	2.28	2.32	2.35	2.38
	μ (10 ^{−6} Pa·s)	1150	725	500	360	275	215	173	149	126	108	95	83	73	66	60	55
	k (W/m·K)						0.140	0.139	0.139	0.138	0.138	0.137	0.137				
	Pr						2.93	2.44	2.17	1.89	1.67	1.51	1.35				
Methanol	ρ (kg/m ³)									783	774	766	756	746	736	725	711
	c_p (kJ/kg·K)	2.30	2.32	2.35	2.37	2.40	2.42	2.45	2.47	2.49	2.52	2.55	2.65	2.78	2.94	3.13	3.30
	μ (10 ^{−6} Pa·s)	2305	1800	1410	1170	975	820	692	590	510	455	400	355	315	271	240	218
	k (W/m·K)	0.225	0.222	0.219	0.216	0.212	0.209	0.206	0.203	0.199	0.195	0.192	0.189	0.187	0.184	0.182	0.180
	Pr	23.6	18.8	15.1	12.9	11.0	9.53	8.23	7.18	6.38	5.88	5.31	4.98	4.68	4.34	4.13	3.99
Methyl formate	ρ (kg/m ³)	1069	1056	1043	1030	1017	1003	989	975	960	944	929	913	897	880	863	845
	c_p (kJ/kg·K)	1.84	1.86	1.88	1.90	1.92	1.95	1.99	2.03	2.08							
	μ (10 ^{−6} Pa·s)	830	711	618	544	481	430	380	345	315							
	k (W/m·K)	0.217	0.213	0.209	0.205	0.200	0.195	0.191	0.186	0.180							
	Pr	7.04	6.21	5.56	5.04	4.62	4.30	3.96	3.77	3.64							
Oil, castor	ρ (kg/m ³)																
	c_p (kJ/kg·K)																
	μ (10 ^{−6} Pa·s)							2,420,000	986,000	451,000	231,000	125,000	74,000	43,000			
	k (W/m·K)							0.182	0.181	0.180	0.179	0.178	0.177	0.176	0.175	0.174	0.17
	Pr																
Oil, olive	ρ (kg/m ³)								914								
	c_p (kJ/kg·K)								1.633								
	μ (10 ^{−6} Pa·s)							138,000	84,000	52,000	36,300	24,500	17,000	12,400			
	k (W/m·K)							0.170	0.169	0.168	0.167	0.166	0.166	0.165	0.165	0.164	0.164
	Pr								810								
Pentane	ρ (kg/m ³)	693	684	674	665	656	646	636	626	616	606	596	585	574	562	550	538
	c_p (kJ/kg·K)	2.060	2.084	2.110	2.137	2.167	2.206	2.239	2.273								
	μ (10 ^{−6} Pa·s)	489	428	379	339	307	279	254	234	209	190	175	161	148	137	124	113
	k (W/m·K)	0.142	0.139	0.136	0.132	0.128	0.125	0.122	0.119	0.115	0.112	0.108	0.105	0.101	0.098	0.095	0.091
	Pr	7.14	6.42	5.88	5.49	5.20	4.92	4.66	4.47								
Propanol	ρ (kg/m ³)	849					819	811	814	796	788	779	770	761	752	747	743
	c_p (kJ/kg·K)	1.955					2.219										
	μ (10 ^{−6} Pa·s)	20,200	13,500	9500	6900	5110	3900	2900	2245	1720	1400	1130	921	760	630	508	447
	k (W/m·K)	0.167	0.166	0.165						0.171	0.169	0.168	0.167	0.165	0.164	0.163	0.162
	Pr	236															
Sulfuric acid	ρ (kg/m ³)								1834								
	c_p (kJ/kg·K)								1.382								
	μ (10 ^{−6} Pa·s)						48,400	35,200	25,400	15,700	11,500	8820	7220	6090	5190		
	k (W/m·K)						0.314										
	Pr																
Toluene	ρ (kg/m ³)	932	923	913	904	895	886	876	867	858	848	839	829	820	810	800	790
	c_p (kJ/kg·K)	1.514	1.535	1.556	1.579	1.602	1.633	1.652	1.675	1.701	1.73	1.76	1.80	1.83	1.87	1.92	1.97
	μ (10 ^{−6} Pa·s)	2120	1670	1345	1100	915	770	670	590	520	470	420	380	355	325	295	270
	k (W/m·K)	0.152	0.149	0.147	0.144	0.142	0.139	0.137	0.134	0.132	0.129	0.126	0.124	0.122	0.119	0.117	0.114
	Pr	21.1	17.8	14.2	12.1	10.3	9.0	8.1	7.4	6.7	6.3	5.9	5.5	5.3	5.1	4.8	4.7
Turpentine	ρ (kg/m ³)																
	c_p (kJ/kg·K)						1.72	1.76	1.80			1.93					
	μ (10 ^{−6} Pa·s)						2250	1780	1490	1270	1070	925	820	730	675		
	k (W/m·K)						0.130	0.129	0.128	0.127	0.126	0.125					
	Pr						29.8	24.3	20.9	18.4	16.1	14.3					

2-454 PHYSICAL AND CHEMICAL DATA

TABLE 2-324 Diffusivities of Pairs of Gases and Vapors (1 atm)

 D_p in cm^2/s [illegible]

TABLE 2-324 Diffusivities of Pairs of Gases and Vapors (1 atm) (Concluded)
 D_e in cm^2/s

Substance	Temp., °C	Air	A	H ₂	O ₂	N ₂	CO ₂	N ₂ O	CH ₄	C ₂ H ₆	C ₂ H ₄	<i>n</i> -C ₄ H ₁₀	<i>i</i> -C ₄ H ₁₀	Ref.
Hydrogen cyanide	0	0.173												10
Hydrogen peroxide	60	.188												11
Iodine	0	.07				0.070								8, 12, 14
Mercury	0	.112		0.53		.13								8, 12, 13
Mesitylene	0	.056												8
Methane	500				1.1									18
Methyl acetate	0	.084		.333			0.0567							8
Methyl alcohol	0	.132		.506			.0879							8
Methyl butyrate	0	.0633		.242			.0446							8
Methyl <i>i</i> -butyrate	0	.0639		.257			.0451							8
Methyl cyclopentane	15		0.0731	.318	0.0742	0.0758								3
Methyl formate	0	.0872												8
Methyl propionate	0	.0735		.295			.0528							8
Methyl valerate	0	0.0569												8
Naphthalene	0	.0513												8
Nitrogen	0				0.181									8
	25						0.165			0.148	0.163	0.0960	0.0908	2
Nitrous oxide	0			0.535			.096							8
<i>n</i> -Octane	0	.0505												8
	30		0.0642	.271	0.0705	0.0710								3
Oxygen	0	.178		.697		0.181	.139							8
Phosgene	0	.095												10
Propionic acid	0	.0829		.330			.0588							8
Propyl acetate	0	.067												8
<i>n</i> -Propyl alcohol	0	.085		.315			.0577							8
<i>i</i> -Propyl alcohol	0	.0818												8
	30	.101												5
<i>n</i> -Propyl benzene	0	.0481												8
<i>i</i> -Propyl benzene	0	.0489												8
<i>n</i> -Propyl bromide	0	.085												8
<i>i</i> -Propyl bromide	0	.0902												8
Propyl butyrate	0	.0530		.206			.0364							8
Propyl formate	0	.0712		.281			.0490							8
<i>n</i> -Propyl iodide	0	.079												8
<i>i</i> -Propyl iodide	0	.0802												8
<i>n</i> -Propyl isobutyrate	0	.0549		.212			.0388							8
<i>i</i> -Propyl isobutyrate	0	.059												8
Propyl propionate	0	.057		.212			.0395							8
Propyl valerate	0	.0466		.189			.0341							8
Safrol	0	.0434												8
<i>i</i> -Safrol	0	.0455												8
Sulfur hexafluoride	25			.418										2
Toluene	0	.076	0.071											4, 8
	30	.088												5
Trimethyl carbinol	0	.087												8
2,2,4-Trimethyl pentane	30		0.0618	.288	0.0688	0.0705								3
2,2,3-Trimethyl heptane	90			.270		0.0684								3
<i>n</i> -Valeric acid	0	0.050												8
<i>i</i> -Valeric acid	0	0.0544		.212			.0376							8
Water	0	0.220		.75			.138							8, 20
	450				1.3									18

° 320 mmHg.

† 40 atm.

‡ Also at other temperatures.

§ Strong function of concentration.

References¹ Amdur, Irvine, Mason, and Ross, *J. Chem. Phys.*, **20**, 436 (1952).² Boyd, Stein, Steingrimsen, and Rumpel, *J. Chem. Phys.*, **19**, 548 (1951).³ Cummings and Ubbelohde, *J. Chem. Soc. (London)*, 1953, p. 3751.⁴ Fairbanks and Wilke, *Ind. Eng. Chem.*, **42**, 471 (1950).⁵ Gilliland, *Ind. Eng. Chem.*, **26**, 681 (1934).⁶ Gorynova and Kuvskinskii, *Zhur. Tekh. Fiz.*, **18**, 1421 (1948).⁷ Hansen, Dissertation, Jena, 1907.⁸ *International Critical Tables*, vol. 5, p. 62.⁹ Jeffries and Drickamer, *J. Chem. Phys.*, **22**, 436 (1954).¹⁰ Klotz and Miller, *J. Am. Chem. Soc.*, **69**, 2557 (1947).¹¹ McMurtrie and Keyes, *J. Am. Chem. Soc.*, **70**, 3755 (1948).¹² Mullaly and Jacques, *Phil. Mag.*, **48**, 6, 1105 (1924).¹³ Spier, *Physica*, 6 (1939): 453; **7**, 381 (1940).¹⁴ Topley and Whytlaw-Gray, *Phil. Mag.*, **4**, 873 (1927).¹⁵ Trautz and Ludvig, *Ann. Physik*, **5**, 5, 887 (1930).¹⁶ Trautz and Muller, *Ann. Physik*, **22**, 353 (1935).¹⁷ Trautz and Ries, *Ann. Physik*, **8**, 163 (1931).¹⁸ Walker and Westenberg, *J. Chem. Phys.*, **32**, 136 (1960).¹⁹ Westenberg and Walker, *J. Chem. Phys.*, **26**, 1753 (1957).²⁰ Winkelmann, *Wied. Ann.*, **22**, 152 (1884); **23**, 203 (1884); **26**, 105 (1885); **33**, 445 (1888); **36**, 92 (1889).

In this table are a representative selection of diffusion coefficients. The subsection "Prediction and Correlation of Physical Properties" should be consulted for estimation techniques. As general references, the works by Hirschfelder, Curtiss, and Bird, *Molecular Theory of Gases and Liquids*, Wiley, New York, 1964; Chapman and Cowling, *The Mathematical Theory of Non-Uniform Gases*, Cambridge, New York, 1970; Reid and Sherwood, *The Properties of Gases and Liquids*,

McGraw-Hill, New York, 1964; and Bretsznajder, *Prediction of Transport and Other Physical Properties of Fluids*, Pergamon, New York, 1971, may be found useful. The most exhaustive recent compilation for gases is by Mason and Marrero, *J. Phys. Chem. Ref. Data*, 1 (1972). Unfortunately, the Mason and Marrero work cites only equations and equation constants and not direct tabulations. For these, the Landolt-Börnstein series is suggested.

TABLE 2-325 Diffusivities in Liquids (25 °C)

Dilute solutions and 1 atm unless otherwise noted; use $D_L\mu/T$ = constant to estimate effect of temperature; * indicates that reference gives effect of concentration.

Solute	Solvent	$D_L \times 10^5$, sq cm/sec	Estimated possible, error, $\pm$ %1	Ref.
Acetal*	Ethanol	1.25	5	11
Acetamide*	Ethanol	0.68	5	11
Acetamide*	Water	1.19	3	11
Acetic acid	Acetone	3.31		4
Acetic acid	Benzene	2.11		1, 4
Acetic acid	Carbon tetrachloride	1.49		4
Acetic acid	Ethylene glycol	0.13		4
Acetic acid	Toluene	2.26		4
Acetic acid*	Water	1.24	3	11
Acetonitrile	Water	1.66	5	11
Acetylene	Water	1.78, 2.11		1, 24
Allyl alcohol*	Ethanol	1.06	5	11
Allyl alcohol	Water	1.19	6	11
Ammonia*	Water	1.7, 2.0, 2.3		1, 11
<i>i</i> -Amyl alcohol*	Ethanol	0.87	5	11
<i>i</i> -Amyl alcohol	Water	1.0	8	11, 25
Benzene	Carbon tetrachloride	1.53		7
Benzene (50 mole %)	<i>n</i> -Decane	1.72		26
Benzene (50 mole %)	2,4-Dimethyl pentane	2.49		26
Benzene (50 mole %)	<i>n</i> -Dodecane	1.40		26
Benzene (50 mole %)	<i>n</i> -Heptane	2.47		26
Benzene (50 mole %)	<i>n</i> -Hexadecane	0.96		26
Benzene (50 mole %)	<i>n</i> -Octadecane	0.86		26
Benzoic acid	Acetone	2.62		4
Benzoic acid	Benzene	1.38		4
Benzoic acid	Carbon tetrachloride	0.91		4
Benzoic acid	Ethylene glycol	0.043		4
Benzoic acid	Toluene	1.49		4
Bromine	Benzene	2.7		11
Bromine	Carbon disulfide	4.1		11
Bromine	Water	1.3		11
Bromobenzene	Benzene	2.30		25
Bromoform*	Acetone	2.90		11
Bromoform	<i>i</i> -Amyl alcohol	0.53		11
Bromoform	Ethanol	1.08	5	11
Bromoform*	Ethyl ether	3.62		11
Bromoform	Methanol	2.20		23
Bromoform	<i>n</i> -Propanol	0.94		11
<i>n</i> -Butanol	Water	0.96	5	1, 11, 18, 25
Caffeine	Water	0.63	6	11
Carbon dioxide	Ethanol	4.0	6	11
Carbon dioxide	Water	1.96	1	1, 3, 5, 20, 24, 28
Carbon disulfide (50 mole %, 200 atm.)	<i>n</i> -Butanol	3.57		14
Carbon disulfide (50 mole %, 200 atm.)	<i>i</i> -Butanol	2.42		14
Carbon disulfide (50 mole %, 218 atm.)	Chlorobenzene	3.00		14
Carbon disulfide (50 mole %, 200 atm.)	2,4-Dimethyl pentane	3.63		14
Carbon disulfide (50 mole %, 100 atm.)	<i>n</i> -Heptane	3.0		14
Carbon disulfide (50 mole %, 50 atm.)	Methyl cyclohexane	3.5		14
Carbon disulfide (50 mole %, 200 atm.)	<i>n</i> -Octane	3.10		14
Carbon disulfide (50 mole %)	Toluene	2.06		14
Carbon tetrachloride	Benzene	2.04	3	7, 9
Carbon tetrachloride*	Cyclohexane	1.49	2	9, 10*
Carbon tetrachloride	Decalin	0.776	2	9
Carbon tetrachloride	Dioxane	1.02	2	9
Carbon tetrachloride*	Ethanol	1.50	2	9, 10*
Carbon tetrachloride	<i>n</i> -Heptane	3.17	2	9
Carbon tetrachloride	Kerosene	0.961	2	9
Carbon tetrachloride	Methanol	2.30	2	9
Carbon tetrachloride	<i>i</i> -Octane	2.57	2	9
Carbon tetrachloride	Tetralin	0.735	2	9
Chloral*	Ethanol	0.68	5	11
Chloral hydrate	Water	0.77	7	11

TABLE 2-325 Diffusivities in Liquids (25 °C) (Continued)

Dilute solutions and 1 atm unless otherwise noted; use $D_L\mu/T$ = constant to estimate effect of temperature; ° indicates that reference gives effect of concentration.

Solute	Solvent	$D_L \times 10^5$, sq cm/sec	Estimated possible, error, $\pm$ %1	Ref.
Chlorine	Water	1.44	4	1, 28
Chlorobenzene	Benzene	2.66		25
Chloroform	Benzene	2.50	6	1, 25
Chloroform	Ethanol	1.38	3	11
Cinnamic acid	Acetone	2.41		4
Cinnamic acid	Benzene	1.12		4
Cinnamic acid	Carbon tetrachloride	0.76		4
Cinnamic acid	Toluene	2.41		4
1,1'-Dichloropropanol	Water	1.0	6	11
Dicyanodiamide°	Water	1.18	4	11
Diethyl ether	Benzene	2.73		25
Diethyl ether	Water	0.85		2
2,4-Dimethyl pentane (50 mole %)	<i>n</i> -Dodecane	1.44		26
2,4-Dimethyl pentane (50 mole %)	<i>n</i> -Hexadecane	0.88		26
Ethanol°	Water	1.28	4	1, 7, 9, ° 11, ° 22
Ethyl acetate	Ethyl benzoate	0.94		6
Ethylene dichloride	Benzene	2.8		1, 25
Formic acid	Acetone	3.77		4
Formic acid	Benzene	2.28		4
Formic acid	Carbon tetrachloride	1.89		4
Formic acid	Ethylene glycol	0.094		4
Formic acid	Toluene	2.65		
Formic acid	Water	1.37	10	11
Glucose	Water	0.69	6	11
Glycerol	<i>i</i> -Amyl alcohol	0.12		11
Glycerol	Ethanol	0.56		11
Glycerol°	Water	0.94	6	1, 11°
<i>n</i> -Heptane (50 mole %)	<i>n</i> -Dodecane	1.58		26
<i>n</i> -Heptane (50 mole %)	<i>n</i> -Hexadecane	1.00		26
<i>n</i> -Heptane (50 mole %)	<i>n</i> -Octadecane	0.92		26
<i>n</i> -Heptane (50 mole %)	<i>n</i> -Tetradecane	1.29		26
Hexamethylene tetramine	Water	0.67		11
Hydrogen chloride°	Water	3.10	3	4, 11, ° 12°
Hydrogen	Water	5.85 (4.4)		1, 11, 24(?)
Hydrogen sulfide	Water	1.61		1
Hydroquinone°	Ethanol	0.53	5	11
Hydroquinone°	Water	0.88, 1.12		2, 11°
Iodine	Acetic acid	1.13		11
Iodine	Anisole	1.25		11
Iodine	Benzene	1.98		9, 19, 23
Iodine	Bromobenzene	1.25	10	4, 11, 19
Iodine	Carbon disulfide	3.2		11, 19, 23
Iodine	Carbon tetrachloride	1.45	8	9, 11, 19
Iodine	Chloroform	2.30	3	11, 23
Iodine	Cyclohexane	1.80		4
Iodine	Dioxane	1.07		9
Iodine°	Ethanol	1.30		4, 11°
Iodine	Ethyl acetate	2.2		11, 19
Iodine	Ethyl ether	3.61		11
Iodine	Ethylene bromide	0.93		11
Iodine	<i>n</i> -Heptane	3.4, 2.5		9, 11, 19
Iodine	<i>n</i> -Hexane	4.15		4, 9
Iodine	Mesitylene	1.49		9
Iodine	Methanol	1.74		19
Iodine	Methyl cyclohexane	2.1		4
Iodine	<i>n</i> -Octane	2.76		4
Iodine	Tetrabromoethane	2.0		11
Iodine	<i>n</i> -Tetradecane	0.96		4
Iodine	Toluene	2.1		11
Iodine	<i>m</i> -Xylene	1.82		9, 11
Iodobenzene	Ethanol	1.09	3	11
Lactose°	Water	0.49	5	11
Maltose°	Water	0.48	5	11
Mannitol°	Water	0.65	5	11
Methanol	Water	1.6		1, 7, 11
Nicotine°	Water	0.60	8	11
Nitric acid°	Water	2.98	2	11
Nitrobenzene	Carbon tetrachloride	1.00		7
Nitrogen	Water	1.9		1, 24
Nitrous oxide	Water	1.8		1, 11
Oxalic acid°	Water	1.61	2	11

TABLE 2-325 Diffusivities in Liquids (25 °C) (Concluded)

Dilute solutions and 1 atm unless otherwise noted; use $D_L\mu/T$ = constant to estimate effect of temperature; ° indicates that reference gives effect of concentration.

Solute	Solvent	$D_L \times 10^5$, sq cm/sec	Estimated possible, error, $\pm$ %1	Ref.
Oxygen	Glycerol°-water (106 poise)	0.24		13
Oxygen	Sucrose°-water (125 poise)	0.25		13
Oxygen	Water	2.5	20	1, 3, 15, 21, 24
Pentaerythritol°	Water	0.77	4	11
Phenol	<i>i</i> -Amyl alcohol	0.2		11
Phenol	Benzene	1.68		1
Phenol	Carbon disulfide	3.7		11
Phenol	Chloroform	2.0		11
Phenol	Ethanol	0.89		11
Phenol	Ethyl ether	3.9		11
<i>n</i> -Propanol	Water	1.1		1, 7, 11
Pyridine°	Ethanol	1.24	3	11
Pyridine	Water	0.76	7	11
Pyrogallol	Water	0.74	7	11
Raffinose°	Water	0.41	4	11
Resorcinol°	Ethanol	0.46	5	11
Resorcinol°	Water	0.87	4	11
Saccharose°	Water	0.49	4	11
Stearic acid°	Ethanol	0.65	5	11
Succinic acid°	Water	0.94		11
Sucrose	Water	0.56	6	2, 27
Sulfur dioxide	Water	1.7		15, 17
Sulfuric acid°	Water	1.97	3	11
Tartaric acid°	Water	0.80	10	11
1,1,2,2-Tetrabromoethane	1,1,2,2-Tetra- chloroethane	0.61	4	11
Toluene	<i>n</i> -Decane	2.09		4
Toluene	<i>n</i> -Dodecane	1.38		4
Toluene	<i>n</i> -Heptane	3.72		4
Toluene	<i>n</i> -Hexane	4.21		4
Toluene	<i>n</i> -Tetradecane	1.02		4
Urea	Ethanol	0.73		11
Urea	Water	1.37	2	8, 11
Urethane	Water	1.06		11, 25
Water	Glycerol	0.021		16

References

- ¹Arnold, *J. Am. Chem. Soc.*, **52**, 3937 (1930).
- ²Calvet, *J. Chim. Phys.*, **44**, 47 (1947).
- ³Carlson, *J. Am. Chem. Soc.*, **33**, 1027 (1911).
- ⁴Chang and Wilke, *J. Phys. Chem.*, **59**, 592 (1955).
- ⁵Davidson and Cullen, *Trans. Inst. Chem. Eng.*, **35**, 51 (1957).
- ⁶Dummer, *Z. Anorg. Chem.*, **109**, 31 (1949).
- ⁷Gerlach, *Ann. Phys. (Leipzig)*, **10**, 437 (1931).
- ⁸Gosting and Akeley, *J. Am. Chem. Soc.*, **74**, 2058 (1952).
- ⁹Hammond and Stokes, *Trans. Faraday Soc.*, **49**, 890 (1953); **49**, 886 (1953).
- ¹⁰Hammond and Stokes, *Trans. Faraday Soc.*, **52**, 781 (1956).
- ¹¹*International Critical Tables*, vol. 5, p. 63.
- ¹²James, Hollingshead, and Gordon, *J. Chem. Phys.*, **7**, 89 (1939); **7**, 836 (1939).
- ¹³Jordon, Ackermann, and Berger, *J. Am. Chem. Soc.*, **78**, 2979 (1956).
- ¹⁴Koeller and Drickamer, *J. Chem. Phys.*, **21**, 575 (1953).
- ¹⁵Kolthoff and Miller, *J. Am. Chem. Soc.*, **63**, 1013 (1941).

TABLE 2-326 Thermal Conductivities of Some Building and Insulating Materials*

 $k = \text{Btu}/(\text{h}\cdot\text{ft}^2)(^\circ\text{F}/\text{ft})$

Material	Apparent density ρ , lb/ft ³ at room temperature	t , °C	k	Material	Apparent density ρ , lb/ft ³ at room temperature	t , °C	k
Aerogel, silica, opacified	8.5	120	0.013	Cotton wool	5	30	0.024
Asbestos-cement boards	120	290	.026	Cork board	10	30	.025
Asbestos sheets	55.5	20	.43	Cork (regranulated)	8.1	30	.026
Asbestos slate	112	51	.096	(ground)	9.4	30	.025
Asbestos	112	0	.087	Diatomaceous earth powder, coarse	20.0	38	.036
	112	60	.114	(Note 2)	20.0	871	.082
	29.3	-200	.043	fine (Note 2)	17.2	204	.040
	29.3	0	.090		17.2	871	.074
	36	0	.087	molded pipe covering (Note 2)	26.0	204	.051
	36	100	.111		26.0	871	.088
	36	200	.120	4 vol. calcined earth and 1 vol. cement, poured and fired (Note 2)	61.8	204	.16
	36	400	.129		61.8	871	.23
	43.5	-200	.090	Dolomite	167	50	1.0
	43.5	0	.135	Ebonite			0.10
Aluminum foil (7 air spaces per 2.5 in.)	0.2	38	.025	Enamel, silicate	38		0.5-0.75
		177	.038	Felt, wool	20.6	30	0.03
Ashes, wood		0-100	.041	Fiber insulating board	14.8	21	.028
Asphalt	132	20	.43	Fiber, red	80.5	20	.27
Boiler scale (Note 1)				(with binder, baked)		20-97	.097
Bricks:				Gas carbon		0-100	2.0
Alumina (92-99% Al ₂ O ₃ by wt.) fused		427	1.8	Glass			0.2-0.73
Alumina (64-65% Al ₂ O ₃ by wt.)		1315	2.7	Borosilicate type	139	30-75	0.63
(See also Bricks, fire clay)	115	800	0.62	Window glass			0.3-0.61
	115	1100	.63	Soda glass			0.3-0.44
Building brick work		20	.4	Granite			1.0-2.3
Carbon	96.7	3.0	3.0	Graphite, longitudinal		20	95
Chrome brick (32% Cr ₂ O ₃ by wt.)	200	200	.67	powdered, through 100 mesh	30	40	0.104
	200	650	.85	Gypsum (molded and dry)	78	20	.25
	200	1315	1.0	Hair felt (perpendicular to fibers)	17	30	.021
Diatomaceous earth, natural, across strata				Ice	57.5	0	1.3
(Note 2)	27.7	204	0.051	Infusorial earth, see diatomaceous earth			
	27.7	871	.077	Kapok	0.88	20	0.020
Diatomaceous, natural, parallel to strata				Lampblack	10	40	.038
(Note 2)	27.7	204	.081	Lava			.49
	27.7	871	.106	Leather, sole	62.4		.092
Diatomaceous earth, molded and fired	38	204	.14	Limestone (15.3 vol. % H ₂ O)	103	24	.54
(Note 2)	38	871	.18	Linen		30	.05
Diatomaceous earth and clay, molded and				Magnesia (powdered)	49.7	47	.35
fired (Note 2)	42.3	204	.14	Magnesia (light carbonate)	13	21	0.034
	42.3	871	.19	Magnesium oxide (compressed)	49.9	20	.32
Diatomaceous earth, high burn, large pores				Marble			1.2-1.7
(Note 3)	37	200	.13	Mica (perpendicular to planes)		50	0.25
	37	1000	.34	Mill shavings			0.033-0.05
Fire clay (Missouri)		200	.58	Mineral wool	9.4	30	0.0225
		600	.85		19.7	30	.024
		1000	.95	Paper			.075
		1400	1.02	Paraffin wax		0	.14
Kaolin insulating brick (Note 3)	27	500	0.15	Petroleum coke		100	3.4
	27	1150	.26			500	2.9
Kaolin insulating firebrick (Note 4)	19	200	.050	Porcelain		200	0.88
	19	760	.113	Portland cement, see concrete		90	.17
Magnesite (86.8% MgO, 6.3% Fe ₂ O ₃ , 3% CaO, 2.6% SiO ₂ by wt.)	158	204	2.2	Pumice stone		21-66	.14
	158	650	1.6	Pyroxylin plastics			.075
	158	1200	1.1	Rubber (hard)	74.8	0	.087
Silicon carbide brick, recrystallized				(para)		21	.109
(Note 3)	129	600	10.7	(soft)		21	0.075-0.092
	129	800	9.2	Sand (dry)	94.6	20	0.19
	129	1000	8.0	Sandstone	140	40	1.06
	129	1200	7.0	Sawdust	12	21	0.03
	129	1400	6.3	Scale (Note 1)			
Calcium carbonate, natural	162	30	1.3	Silk	6.3		.026
White marble			1.7	varnished		38	.096
Chalk	96	0.4	0.4	Slag, blast furnace		24-127	.064
Calcium sulfate (4H ₂ O), artificial	84.6	40	.22	Slag wool	12	30	.022
plaster (artificial)	132	75	.43	Slate		94	.86
(building)	77.9	25	.25	Snow	34.7	0	.27
Cambric (varnished)		38	.091	Sulfur (monoclinic)		100	0.09-0.097
Carbon, gas		0-100	2.0	(rhombic)		21	0.16
Carbon stock	94	-184	0.55	Wall board, insulating type	14.8	21	.028
		0	3.6	Wall board, stiff paste board	43	30	.04
Cardboard, corrugated			0.037	Wood shavings	8.8	30	.034

TABLE 2-326 Thermal Conductivities of Some Building and Insulating Materials* (Concluded) $k = \text{Btu}/(\text{h}\cdot\text{ft}^2)(^\circ\text{F}/\text{ft})$

Material	Apparent density ρ , lb/ft ³ at room temperature	t , $^\circ\text{C}$	k	Material	Apparent density ρ , lb/ft ³ at room temperature	t , $^\circ\text{C}$	k
Celluloid	87.3	30	.12	Wood (across grain):			
Charcoal flakes	11.9	80	.043	Balsa	7–8	30	0.025–0.03
	15	80	.051	Oak	51.5	15	0.12
Clinker (granular)		0–700	.27	Maple	44.7	50	.11
Coke, petroleum		100	3.4	Pine, white	34.0	15	.087
		500	2.9	Teak	40.0	15	.10
Coke, petroleum (20–100 mesh)	62	400	0.55	White fir	28.1	60	.062
Coke (powdered)		0–100	.11	Wood (parallel to grain):			
Concrete (cinder)			.20	Pine	34.4	21	.20
(stone)			.54	Wool, animal	6.9	30	.021
(1:4 dry)			.44				

*Marks, *Mechanical Engineers' Handbook*, 4th ed., McGraw-Hill, New York, 1941. *International Critical Tables*, McGraw-Hill, 1929, and other sources.

Note 1: B. Kamp [*Z. tech. Physik*, **12**, 30 (1931)] shows the effect of increased porosity in decreasing thermal conductivity of boiler scale. Partridge [University of Michigan, *Eng. Research Bull.*, **15**, 1930] has published a 170-page treatise on Formation and Properties of Boiler Scale.

Note 2: Townshend and Williams, *Chem. & Met.*, **39**, 219 (1932).

Note 3: Norton, *Refractories*, 2d ed., McGraw-Hill, New York, 1942.

Note 4: Norton, private communication.

TABLE 2-327 Thermal-Conductivity-Temperature Table for Metals*

Thermal conductivities tabulated in watts per meter-kelvin

Substance	Temperature, K														
	10	20	40	60	80	100	200	300	400	500	600	800	1000	1200	1400
Alumina	7	32	121	174	160	125	55	36	26	20	16	10	8	7	6
Aluminum	38,000	13,500	2,300	850	380	300	237	273	240	237	232	220	93	99	105
Antimony	470	230	110	80	60	48	32	26	22	20					
Beryllium oxide	47	196	810	1,400	1,650	1,490	480	272	196	146	111	70	47	33	25
Bismuth	240	100	45	31	24	22	18	16	14	12					
Boron	165	305	400	327	230	170	45	25	15	12					
Cadmium	900	250	150	120	110	110	105	104	101	99					
Chromium	400	570	450	250	180	158	111	90	87	85	81	71	65	62	61
Cobalt	250	450	380	250	190	160	120	100	85	70					
Constantan	4	9	16	18	19	20	23	25	27	30					
Copper	19,000	10,700	2,100	850	570	483	413	398	392	388	383	371	357	342	
Gallium	2,200	640	250	200	170	140	100	85							
Gold	2,800	1,500	520	380	350	345	327	315	312	309	304	292	278	262	
Graphite†	27	108	135	81	54	39	15	10	7	5	4	3	3	2	2
Graphite‡	81	420	1,630	2,980	4,290	4,980	3,250	2,000	1,460	1,140	930	680	530	440	370
Hastelloy	1	3	4	5	6	7	9	10	11	13					
Inconel	2	4	8	10	11	11	14	15							
Iridium	1,300	1,900	750	360	230	172	147	145	143	140					
Iron	710	1,000	560	270	170	132	94	80	69	61	55	43	33	28	31
Lead	175	57	43	42	41	40	37	35	34	33	31	19	22	24	26
Magnesium	1,200	1,300	620	290	190	169	159	156	153	151	149	146	84	98	112
Magnesium oxide	1,100	3,100	2,200	950	460	260	75	48	36	27	21	13	10	8	7
Manganese	2	2	4	5	5	6	7	8	9	9					
Manganin	2	4	9	11	13	13	17	22	28	34	40				
Mercury	54	40	35	33	32	32	32	8	10	11	12	13	14		
Molybdenum	150	280	350	250	210	179	143	138	134	130	126	118	112	105	100
Nickel	2,600	1,700	570	290	200	158	106	91	80	72	66	67	72	76	80
Nylon	0.04	0.10	0.17	0.20	0.23	0.25	0.28	0.30							
Palladium	1,200	610	160	100	88	80	78	78	78	80					
Platinum	1,200	490	130	92	82	79	75	73	72	72	72	73	78	78	81
PTFE§	0.94	1.43	1.94	2.1	2.15	2.16	2.20	2.25	2.3	2.5					
Pyrex	0.12	0.20	0.33	0.42	0.51	0.57	0.88	1.1	1.6	2.1					
Quartz	1,200	480	82	40	30										
Rhodium	2,900	3,900	1,000	370	250	190	160	150	145	140					
Rubber			0.13	0.15	0.16	0.17	0.20	0.22	0.24	0.25					
Selenium (axis)	140	57	25	15	10	8	6	4	3	2					
Silica								1.34	1.52	1.70	1.87	2.22	2.60		
Silver	16,500	5,200	1,100	630	500	430	425	424	420	413	405	389	374	358	
Tantalum	108	146	88	68	62	59	58	57	58	58	59	59	60	61	62
Tellurium	300	93	29	17	13	11	6	4	3	3					
Tin		320	130	101	90	84	72	67	62	60					
Titanium	14	28	39	37	33	31	26	21	20	20	19				
Tungsten			880	330	310	280	190	180	170	150	140				
Uranium				20	22	23	26	28	30	32					
Zinc				150	135	130	123	120	116	110	110				
Zirconium	100	110	59	42	38	34	25	23	22	21	21				

* Especially at low temperatures, the thermal conductivity can often be markedly reduced by even small traces of impurities. This table, for the highest-purity specimens available, should thus be used with caution in applications with commercial materials. From Perry, *Engineering Manual*, 3d ed., McGraw-Hill, New York, 1976. A more detailed table appears as Section 5.5.6 in the *Heat Exchanger Design Handbook*, Hemisphere Pub. Corp., Washington, DC, 1983.

† Parallel to basal plane.

‡ Perpendicular to basal plane.

§ Also known as Teflon, etc.

TABLE 2-328 Thermal Conductivity of Chromium Alloys*

$$k = \text{Btu}/(\text{h}\cdot\text{ft}^2)(^\circ\text{F}/\text{ft})$$

American Iron and Steel Institute Type No.	k at 212 °F	k at 932 °F
301, 302, 302B, 303, 304, 316†	9.4	12.4
308	8.8	12.5
309, 310	8.0	10.8
321, 347	9.3	12.8
403, 406, 410, 414, 416†	14.4	16.6
430, 430F†	15.1	15.2
442	12.5	14.2
501, 502†	21.2	19.5

*Table 2-328 is based on information from manufacturers.

† Shelton and Swanger (National Bureau of Standards), *Trans. Am. Soc. Steel Treat.*, 21, 1061–1078 (1933).

TABLE 2-329 Thermal Conductivity of Some Alloys at High Temperature*

Thermal conductivity, Btu/(ft)(hr)(°R)						
°R	Kovar	Advance	Monel	Hastelloy A	Inconel	Nichrome V
500	7.8		9.0	5.6	6.0	5.5
600	8.3	11.4	10.2	6.2	6.5	6.1
700	8.6	12.6	11.2	6.8	7.0	6.7
800	8.7	13.9	12.3	7.3	7.6	7.3
900	8.7	15.1	13.4	7.8	8.1	7.8
1000	8.9	16.4	14.4	8.4	8.6	8.4
1100	9.2	17.6	15.4	9.0	9.1	9.0
1200	9.5	18.8	16.5	9.5	9.7	9.5
1300	9.8	20.0	17.6	10.1	10.2	10.1
1400	10.2	21.2	18.7	10.7	10.8	10.7
1500	10.5	22.5	19.8	11.3	11.3	11.3
1600	10.8	23.8	20.8	11.8	11.8	11.9
1700	11.1	25.0	21.9	12.3	12.4	12.4
1800	11.3	26.2	23.0	12.9	13.0	13.0
1900	11.5	27.4	24.0	13.4	13.6	13.5
2000	11.8	28.7	25.1	14.0	14.0	14.1
2100	12.1	30.0	26.1	14.6	14.5	14.7
2200	12.3		27.2	15.1	15.0	15.3

*Silverman, *J. Metals*, 5, 631 (1953). Copyright American Institute of Mining, Metallurgical and Petroleum Engineers, Inc.

TABLE 2-330 Thermal Conductivities of Some Materials for Refrigeration and Building Insulation*

$$k = \text{Btu}/(\text{h}\cdot\text{ft}^2)(^\circ\text{F}/\text{ft}) \text{ at approximately room temperature}$$

Material	Apparent density, lb/cu ft room temp.	k
Soft flexible materials in sheet form:		
Chemically treated wood fiber	2.2	0.023
Eel grass between paper	3.4–4.6	0.021–0.022
Felted cattle hair	11–13	0.022
Flax fibers between paper	4.9	.023
Hair and asbestos fibers, felted	7.8	.023
Insulating hair, and jute	6.1–6.3	0.022–0.023
Jute and asbestos fibers, felted	10.0	0.031
Loose materials:		
Cork, regranulated, fine particles	8–9	.025
Charcoal, 6 mesh	15.2	.031
Diatomaceous earth, powdered	10.6	.026
Glass wool, curled	4–10	.024
Gypsum in powdered form	26–34	0.043–0.05
Mineral wool, fibrous	6	0.0217
	10	.0225
	14	.0233
	18	.0242
Sawdust	12	.034
Wood shavings, from planer	8.8	.034
Semiflexible materials in sheet form:		
Flax fiber	13.0	.026
Semirigid materials in board form:		
Corkboard	7.0	.0225
	10.6	.025
Mineral wool, block, with binder	16.7	.031
Stiff fibrous materials in sheet form:		
Wood pulp	16.2–16.9	.028
Sugar-cane fiber	13.2–14.8	.028
Cellular gypsum	8	.029
	12	.037
	18	.049
	24	.064
	30	.083

*Abstracted from *U.S. Bur. Standards Letter Circ.* 227, Apr. 19, 1927.

TABLE 2-331 Thermal Conductivities of Insulating Materials at High Temperatures*

$$k = \text{Btu}/(\text{h}\cdot\text{ft}^2)(^\circ\text{F}/\text{ft})$$

Material	For tem- peratures, °F up to	Mean temperatures, °F									
		100	200	300	400	500	600	800	1000	1500	2000
Laminated asbestos felt (approx. 40 laminations per in)	700	0.033	0.037	0.040	0.044	0.048					
Laminated asbestos felt (approx. 20 laminations per in)	500	.045	.050	.055	.060	.065					
Corrugated asbestos (4 plies per in)	300	.050	.058	.069							
85% magnesia (density, 13 lb/ft³)	600	.034	.036	.038	.040						
Diatomaceous earth, asbestos and bonding material	1600	.045	.047	.049	.050	.053	0.055	0.060	0.065		
Diatomaceous earth brick	1600	.054	.056	.058	.060	.063	.065	.069	.073		
Diatomaceous earth brick	2000	.127	.130	.133	.137	.140	.143	.150	.158	0.176	
Diatomaceous earth brick	2500	.128	.131	.135	.139	.143	.148	.155	.163	.183	0.203
Diatomaceous earth powder (density, 18 lb/ft³)		.039	.042	.044	.048	.051	.054	.061	.068		
Rock wool		.030	.034	.039	.044	.050	.057				

Asbestos cement, 1.2; 85% magnesia cement, 0.05; asbestos and rock wool cement, 0.075 approx.

*Marks, *Mechanical Engineers' Handbook*, 4th ed., McGraw-Hill, New York, 1941.

TABLE 2-332 Thermal Conductivities of Insulating Materials at Moderate Temperatures (Nusselt)*
 $k = \text{Btu}/(\text{h}\cdot\text{ft}^2)(^\circ\text{F}/\text{ft})$

Material	Weight, lb/cu ft	Temperatures, °F						
		32	100	200	300	400	600	800
Asbestos	36.0	0.087	0.097	0.110	0.117	0.121	0.125	0.130
Burned infusorial earth for pipe coverings	12.5	.043	.046	.052	.057	.062	.073	.085
Insulating composition (loose)	25.0	.040	.046	.050	.053	.055		
Cotton	5.0	.032	.035	.039				
Silk hair	9.1	.026	.030	.034				
Silk	6.3	.025	.028	.034				
Wool	8.5	.022	.027	.033				
Pulverized cork	10.0	.021	.026	.032				
Infusorial earth (loose)	22.0	.035	.039	.045	.047	.050	.053	

*Marks, *Mechanical Engineers' Handbook*, 4th ed., McGraw-Hill, New York, 1941.

TABLE 2-333 Thermal Conductivities of Insulating Materials at Low Temperatures (Gröber)*
 $k = \text{Btu}/(\text{h}\cdot\text{ft}^2)(^\circ\text{F}/\text{ft})$

Material	Weight, lb/cu ft	Temperatures, °F				
		32	−50	−100	−200	−300
Asbestos	44.0	0.135	0.132	0.130	0.125	0.100
Asbestos	29.0	.0894	.0860	.0820	.0720	.0545
Cotton	5.0	.0325	.0302	.0276	.0235	.0198
Silk	6.3	.0290	.0256	.0235	.0196	.0155

*Marks, *Mechanical Engineers' Handbook*, 4th ed., McGraw-Hill, New York, 1941.

TABLE 2-334 Thermal Diffusivity (m²/s) of Selected Elements*

Element	Temperature, K									
	20	40	60	80	100	200	400	600	800	1000
Aluminum	0.50	0.012	0.0014	4.4. − 4	2.3. − 4	1.1. − 4	9.4. − 5	8.4. − 5	7.4. − 5	6.6. − 5
Beryllium					0.0036	1.5. − 4	4.0. − 5	2.6. − 5	2.1. − 5	1.7. − 5
Chromium	0.038	0.0037	5.9. − 4	2.0. − 4	1.2. − 4	4.1. − 5	2.6. − 5	2.0. − 5	1.7. − 5	1.4. − 5
Copper	0.16	0.0040	6.9. − 4	3.1. − 4	2.2. − 4	1.3. − 4	1.1. − 4	1.0. − 4	9.0. − 5	9.0. − 5
Gold	0.005	4.5. − 4	2.3. − 4	1.8. − 4	1.5. − 4	1.3. − 4	1.2. − 4	1.2. − 4	1.1. − 4	9.8. − 5
Iridium	0.046				8.4. − 5	5.6. − 5	4.8. − 5	4.4. − 5	4.1. − 5	3.5. − 5
Iron	0.043	3.2. − 3	4.9. − 4	1.6. − 4	8.2. − 5	3.1. − 5	1.8. − 5	1.3. − 5	1.1. − 5	1.0. − 5
Lead	9.3. − 5	3.9. − 5	3.3. − 5	3.1. − 5	2.9. − 5	2.6. − 5	2.3. − 5	2.0. − 5	1.3. − 5	1.5. − 5
Molybdenum	0.0095	0.0014	4.0. − 4	2.0. − 4	1.3. − 4	6.3. − 5	5.1. − 5	4.5. − 5	4.2. − 5	3.8. − 5
Nickel	0.033	0.0017	3.1. − 4	1.3. − 4	8.0. − 5	3.1. − 5	1.9. − 5	1.3. − 5	1.4. − 5	1.5. − 5
Platinum	0.0029	1.6. − 4	6.3. − 5	4.3. − 5	3.6. − 5	2.7. − 5	2.5. − 5	2.5. − 5	2.5. − 5	2.5. − 5
Silver	0.031	0.0013	4.5. − 4	2.8. − 4	2.3. − 4	1.8. − 4	1.7. − 4	1.6. − 4	1.5. − 4	1.4. − 4
Zinc	0.0046	3.1. − 4	1.0. − 4	7.0. − 5	5.5. − 5	4.7. − 5	3.9. − 5	3.4. − 5	1.8. − 5	2.2. − 5

*Tables for up to 24 temperatures for 47 elements appear in the *Handbook of Heat Transfer*, 2d ed., McGraw-Hill, New York, 1984. The notation 3.2. − 4 signifies 2.3×10^{-4} .

TABLE 2-335 Thermophysical Properties of Selected Nonmetallic Solid Substances

Material	Density, kg/m ³	Emissivity	Specific heat, kJ/(kg·K)	Thermal conductivity, W/(m·K)	Thermal diffusivity, m ² /s × 10 ⁶
Alumina	3975		0.765	36	11.9
Asphalt	2110		0.920	0.06	0.03
Bakelite	1300		1.465	1.4	0.74
Beryllia	3000	0.82	1.030	270	88
Brick	1925	0.93	0.835	0.72	0.45
Brick, fireclay	2640	0.93	0.960	1.0	0.39
Carbon, amorphous	1950	0.86	0.724	1.6	1.13
Clay	1460	0.91	0.880	1.3	1.01
Coal	1350	0.80	1.26	0.26	0.15
Cotton	80		1.30	0.06	0.58
Diamond	3500		0.509	2300	1290
Granite	2630		0.775	2.79	1.37
Hardboard	1000		1.38	0.15	0.11
Magnesite	3025	0.38	1.13	4.0	1.2
Magnesia	3635	0.72	0.943	48	14
Oak	770	0.90	2.38	0.18	0.10
Paper	930	0.83	1.34	0.011	0.01
Pine	525	0.84	2.75	0.12	0.54
Plaster board	800	0.91		0.17	
Plywood	540		1.22	0.12	0.18
Pyrex	2250	0.92	0.835	1.4	0.74
Rubber	1150	0.92	2.00	0.2	0.09
Rubber, foam	70	0.90		0.03	
Salt		0.34	0.854	7.1	
Sandstone	2150	0.59	0.745	2.9	1.8
Silica		0.79	0.743	1.3	
Sapphire	3975	0.48	0.765	46	15
Silicon carbide	3160	0.86	0.675	110	230
Soil	2050	0.38	1.84	0.52	0.14
Teflon	2200	0.92	0.35	0.26	0.34
Thoria	4160	0.28	0.71	14	4.7
Urethane foam	70		1.05	0.03	0.36
Vermiculite	120		0.84	0.06	0.60

NOTE: Difficulties of accurately characterizing many of the specimens mean that many of the values presented here must be regarded as being of order of magnitude only. For some materials, actual measurement may be the only way to obtain data of the required accuracy. To convert kilograms per cubic meter to pounds per cubic foot, multiply by 0.062428; to convert kilojoules per kilogram-kelvin to British thermal units per pound-degree Fahrenheit, multiply by 0.23885.

PREDICTION AND CORRELATION OF PHYSICAL PROPERTIES*

INTRODUCTION

Physical property values, sufficiently accurate for many engineering applications, can be estimated in the absence of reliable experimental data. The purpose of this subsection is to provide a set of recommended prediction methods for general engineering use; it is not a comprehensive review, and many alternative methods are available in the literature. Methods recommended were selected on the basis of accuracy, generality, and, in most cases, simplicity. They generally correspond to the methods tested and given priority in the DIPPR® 801 database project.†

Properties included in this subsection are divided into 10 categories: (1) physical constants: critical properties, normal melting and boiling points, acentric factor, radius of gyration, and dipole moment; (2) vapor

pressure: liquid and solid; (3) thermal properties: enthalpy and Gibbs' energy of formation and ideal gas entropy; (4) latent enthalpy: vaporization, fusion, and sublimation; (5) heat capacity: ideal gas, gas, liquid, and solid; (6) density: gas, liquid, and solid; (7) viscosity: gas and liquid; (8) thermal conductivity: gas and liquid; (9) surface tension; and (10) flammability properties: flash point, flammability limits, and autoignition temperature. Each of the 10 subsections provides a definition of the relevant properties and a description of one or more recommended prediction methods. Each description lists the type of method, its uncertainty, its limitations, and the expected uncertainty of the predicted value. A numerical example is also given to illustrate use of the method. For brevity, symbols used for physical properties and for variables and constants in the equations are defined under Nomenclature and are not generally defined after their use except where doing so clarifies usage. A list of equation and table numbers in which variables appear is included in the Nomenclature section for quick cross-referencing. Although emphasis is on pure-component properties, some mixture estimation techniques have been included for physical constants, density, viscosity, thermal conductivity, surface tension, and flammability. Correlation and estimation of properties that are inherently multicomponent (e.g., diffusion coefficients, mixture excess properties, activity coefficients, etc.) are treated elsewhere in the Handbook.

*Some material in this subsection has been retained from the corresponding subsection in the Seventh Edition, which was coauthored by Thomas E. Daubert and Evan Buck.

†The Design Institute for Physical Properties (DIPPR®) is an industrial consortium under the auspices of AIChE; Project 801, Evaluated Process Design Data, is a pure-component database of industrially important compounds.

UNITS

The International Metric System (SI) of units has been used throughout this subsection. Where possible, the estimation equations are set up in dimensionless groups. This makes transparent any conversion factors that should be applied to obtain the property in a desired set of units and eliminates the requirement of specific units for variables. For example, rather than use P_c as a variable with defined units, the dimensionless group P_c/Pa is used. When a value for P_c expressed in

any units (say, $P_c = 6.53 \text{ MPa}$) is inserted into this group, the result is dimensionless with an explicit indication of conversion factors that must be included:

$$\frac{P_c}{\text{Pa}} = \frac{6.53 \text{ MPa}}{\text{Pa}} = \left(\frac{6.53 \text{ MPa}}{\text{Pa}} \right) \left(\frac{10^6 \text{ Pa}}{\text{MPa}} \right) = 6.53 \times 10^6$$

Section 1 of this handbook should be used for appropriate unit conversion factors.

Nomenclature

Physical constants	Definition	Value
h	Planck's constant	$6.626 \times 10^{-34} \text{ J}\cdot\text{s}$
k	Boltzmann's constant	$1.3806 \times 10^{-23} \text{ J}/(\text{molecule}\cdot\text{K})$
N_A	Avogadro's number	$6.022 \times 10^{26} \text{ molecule/kmol}$
R	Gas constant	$8.3143 \text{ Pa}\cdot\text{m}^3/(\text{kmol}\cdot\text{K})$
Properties	Definition	Typical units
A, B, C	Molecular principal moments of inertia	$\text{kg}\cdot\text{m}^2$
$B, B(T)$	Second virial coefficient	m^3/kmol
B_m	Second virial coefficient for a mixture	m^3/kmol
C_p	Constant-pressure molar heat capacity	$\text{J}/(\text{kmol}\cdot\text{K})$
C_p^o	Ideal gas constant-pressure molar heat capacity	$\text{J}/(\text{kmol}\cdot\text{K})$
C_v	Constant-volume molar heat capacity	$\text{J}/(\text{kmol}\cdot\text{K})$
H_i	Enthalpy of compound i	J/kmol
k	Thermal conductivity	$\text{W}/(\text{m}\cdot\text{K})$
k_b	Thermal conductivity at T_b	$\text{W}/(\text{m}\cdot\text{K})$
LFL	Lower flammability limit	%
M	Molecular weight	kg/kmol
P	Pressure	Pa
$\mathbf{P}$	Parachor	unitless
P_c	Critical pressure	Pa
P_r	Reduced pressure; $P_r = P/P_c$	unitless
P^*	Vapor pressure	Pa
P^*_{meas}	Measured vapor pressure value	Pa
P_r^*	Reduced vapor pressure; $P_r^* = P^*/P_c$	unitless
P_t^*	Vapor pressure at triple point	Pa
R_g	Radius of gyration	m
S^o	Ideal gas entropy	$\text{J}/(\text{kmol}\cdot\text{K})$
S^o	Standard state entropy	$\text{J}/(\text{kmol}\cdot\text{K})$
S_r	Rotational contribution to entropy	$\text{J}/(\text{kmol}\cdot\text{K})$
S_{vib}	Vibrational contribution to entropy	$\text{J}/(\text{kmol}\cdot\text{K})$
T	Temperature	K
T_b	Normal boiling point temperature	K
T_{br}	Reduced temperature at T_b ; $T_{br} = T_b/T_c$	unitless
T_c	Critical temperature	K
T_m	Melting temperature	K
T_{meas}	T at which a dependent property was measured	K
T_r	Reduced temperature; $T_r = T/T_c$	unitless
T_t	Triple point temperature	K
UFL	Upper flammability limit	%
V	Molar volume	m^3/kmol
V_m	Mixture molar volume	m^3/kmol
V_r	Reduced volume; $V_r = ZT_r/P_r$	unitless
w_i	Mass fraction of component i	unitless
x_i	Mole fraction of component i	unitless
y_i	Mole fraction of component i in vapor phase	unitless
Z	Compressibility factor; $Z = PV/RT$	unitless
Z_c	Critical compressibility factor; $Z_c = P_c V_c/RT_c$	unitless
Z_i	Compressibility factor of reference fluid i	unitless
ΔG_f^o	Ideal gas standard Gibbs energy of formation	J/kmol
ΔG_f^o	Standard state Gibbs energy of formation	J/kmol
ΔH_f^o	Ideal gas standard enthalpy of formation	J/kmol
ΔH_f^o	Standard state enthalpy of formation	J/kmol
ΔH_{fus}	Enthalpy of fusion	J/kmol
ΔH_{rxn}	Enthalpy change per mole of reaction as written	J/kmol
ΔH_{sub}	Enthalpy of sublimation	J/kmol
ΔH_v	Enthalpy of vaporization	J/kmol
ΔS_f^o	Standard state entropy of formation	$\text{J}/(\text{kmol}\cdot\text{K})$
ΔS_f^o	Ideal gas entropy of formation	$\text{J}/(\text{kmol}\cdot\text{K})$
ΔS_{fus}	Latent entropy of fusion	$\text{J}/(\text{kmol}\cdot\text{K})$
ΔZ_v	Change in compressibility factor upon vaporization	unitless
ρ	Molar density; $\rho = V^{-1}$	kmol/m^3
ρ_c	Critical molar density; $\rho_c = V_c^{-1}$	kmol/m^3

Nomenclature (Continued)

Properties	Definition	Typical units
ρ_r	Reduced molar density; $\rho_r = \rho/\rho_c$	unitless
ω	Acentric factor	unitless
η	Viscosity	Pa·s
η^0	Viscosity at low pressure	Pa·s
σ	Surface tension	mN/m
σ_m	Surface tension of mixture	mN/m
τ	Complementary reduced temperature ($= 1 - T_r$)	unitless
τ_b	Complementary reduced normal boiling temperature ($= 1 - T_{br}$)	unitless
μ	Dipole moment	D
μ_r	Reduced dipole moment [defined in Eq. (2-62)]	unitless
Eq. variables	Definition	(Equations), [Tables]
a	EoS constant	(2-66), [2-354]
$a, b, c, \dots$	GC values for C_p and η	(2-51), (2-52), (2-53), (2-97), [2-356]
a, b, c	Correlation coefficients	(2-21), (2-23)
a_i	GC values	(2-43), (2-97), [2-346, 2-356]
a, b	Terms in second virial correlation	(2-61)
a, b	Chickos correlation parameters	(2-39), (2-40), (2-41)
a_b, b_b, d_i	GC values for liquid C_p	(2-51), [2-348]
$\bar{a}, \bar{\alpha}$	EoS constant for mixture	(2-76)
$A, B, C, \dots$	Correlation constants/parameters	(2-2), (2-20), (2-22), (2-34), (2-36), (2-50), (2-65), (2-68), (2-83), (2-85), (2-87), (2-88), (2-95), (2-96), (2-101), above (2-114)
A	Factor in liquid k correlation	(2-108), [2-358]
A_i	Constants in C_p^0 correlation	(2-45), (2-46)
b	EoS constant	(2-66), [2-354]
$b_b, c_b, \dots$	Reference EoS constants	(2-65), [2-353]
b_i	GC value for AIT	(2-124), [2-363, 2-364]
$\bar{b}$	EoS constant for mixture	(2-75)
$B^{(i)}$	Second virial expansion terms	(2-58), (2-59), (2-60), (2-61)
C	Number of components in mixture	(2-55), (2-72), (2-73), (2-74), (2-75), (2-76), (2-77), (2-80), (2-81), (2-82), (2-98), (2-113), (2-119)
C_i	GC values for T_b or η	(2-15), (2-87), [2-341, 2-355]
$C_{i,int}$	Sum of intramolecular group-group interactions	(2-16)
C_{ij}	Group-group intramolecular interaction pair	(2-16), [2-342]
$(C_p^0)_i$	GC values for ideal gas heat capacity	(2-49), [2-347]
C_{sj}	Chickos: GC value for C—H group	(2-41), [2-344]
C_{tj}	Chickos: GC value for functional group	(2-41), [2-345]
f_i	Halogen correction for ΔH_{sub} correlation	(2-43), [2-346]
$f^{(i)}$	Vapor pressure deviation function	(2-25)
F	Factor in surface tension equation	(2-116), (2-117)
G_{ij}	Adjustable mixture viscosity parameter	(2-98)
g_{sc}^E	UNIFAC combinatorial excess Gibbs energy	(2-99)
g_r^E	UNIFAC residual excess Gibbs energy	(2-99)
h	Parameter in Riedel vapor pressure equation	following (2-24)
K	Parameter in Riedel vapor pressure equation	following (2-24)
LFL_i	GC contribution	(2-122), [2-361, 2-362]
n	Number of nonhydrogen atoms	(2-15)
n_A	Number of atoms in molecule	(2-1), (2-30), (2-31), (2-48)
n_E	Number of occurrences of element E in compound	(2-54)
n_i	Number of occurrences of group i	(2-27), (2-43), (2-49), (2-51), (2-53), (2-87), (2-109), (2-115), (2-122), (2-123), (2-124)
n_f	Chickos: no. of different functional groups	(2-41)
n_s	Chickos: no. of different nonring or aromatic C—H groups bonded to functional groups	(2-41)
n_x	Total no. of halogen and H atoms attached to C and Si atoms for ΔH_{sub} correlation	(2-43)
N	Total number of groups in molecule	(2-12), (2-15) (2-16), (2-43), (2-49), (2-51), (2-53), (2-54), (2-87), (2-97), (2-109), (2-115)
N_C	Number of C atoms	(2-121)
N_f^i	Chickos: no. of functional groups of type i	(2-41)
N_{gi}^i	Chickos: no. of C—H groups of type i bonded to other C atoms	(2-41)
N_H	Number of H atoms	(2-121)

Nomenclature (*Continued*)

Eq. variables	Definition	(Equations), [Tables]
N_{CR}	Chickos: no. of CH_2 groups in nonaromatic ring to form cyclic paraffin of same ring size	(2-40)
N_O	Chickos: Number of O atoms	(2-121)
N_R	Chickos: no. of nonaromatic rings	(2-40)
N_S	Chickos: Number of S atoms	(2-121)
N_{S_i}	Chickos: no. of C—H groups of type i bonded to at least one functional group	(2-41)
N_X	Number of halogen atoms	(2-121)
$\bar{P}_c$	Pseudocritical pressure for mixture	(2-73)
$P_{c,ij}$	Cross term in mixing rule	(2-79)
q	Rackett equation power for Z_c	(2-69), (2-70), (2-81)
q_i	UNIFAC molecular surface area	following (2-100)
Q_k	UNIFAC group surface area	following (2-100)
r_i	UNIFAC molecular volume	following (2-100)
r°	Dimensionless separation distance	(2-4)
R_k	UNIFAC group volume	following (2-100)
$(S^\circ)_i$	GC value for entropy	(2-27), [2-343]
t	Chickos: total no. of functional groups	(2-41)
$t_{m1,i}$	First-order GC contribution for T_m	(2-13), [2-339]
$t_{m2,i}$	Second-order GC contribution for T_m	(2-13), [2-340]
$\bar{T}_c$	Pseudocritical temperature for mixture	(2-72), (2-73), (2-80)
$T_{c,ij}$	Cross term in mixing rule	(2-78), (2-80)
x_P	Term in Pailhes method = $\log(1 \text{ atm}/P)$	(2-14)
U°	Dimensionless intermolecular potential	(2-4)
UFL_i	GC contribution	(2-123), [2-361, 2-362]
$V_{c,ij}$	Cross term in mixing rule	(2-78), (2-79)
$Z^{(0)}$	Compressibility factor of simple fluid	(2-63), (2-64), [2-351]
$Z^{(1)}$	Acentric deviation term for Z	(2-63), (2-64), [2-352]
$Z_{c,ij}$	Cross term in mixing rule	(2-78), (2-79)
Z_{RA}	Modified Rackett correlation parameter	(2-70)
$\bar{Z}_{RA}$	Modified Rackett parameter for mixture	(2-82)
$\alpha, \beta, \gamma, \dots$	Correlation parameters for k	(2-106), (2-107), (2-108), (2-109), [2-358]
$\alpha(T_r)$	EoS temperature-dependent function	(2-66), [2-354]
α_c	Parameter in Riedel vapor pressure equation	following (2-24)
α_{mn}	Viscosity group-group interactions	(2-100), [2-357]
β	Reference EoS constant	(2-65), [2-353]
β	Stoichiometric coefficient in FP correlation	(2-120), (2-121)
β_i	Nonlinear correction term in correlation	(2-43), (2-53), [2-346, 2-349]
γ	Reference EoS constant	(2-65), [2-353]
δ	= 0 for nonlinear molecules; = 1 for linear	(2-1), (2-48)
δ	EoS parameter	(2-66), [2-354]
Δ_E	Contribution of element E to heat capacity	(2-54), [2-350]
Δ_P	GC contribution to P_c	(2-7), (2-10), [2-336, 2-337]
Δ_T	GC contribution to T_c	(2-6), (2-9), [2-336, 2-337]
Δ_V	GC contribution to V_c	(2-8), (2-11), [2-336, 2-337]
$(\Delta H_f^\circ)_i$	GC value for enthalpy of formation	(2-27), [2-343]
Δk_i	GC for thermal conductivity at T_b	(2-109), [2-359]
ΔP_i	GC for parachor	(2-115), [2-360]
ΔS_i	Chickos: GC value for group i	(2-41), [2-344, 2-345]
ε	Lennard-Jones well depth parameter	following (2-4)
ε	EoS parameter	(2-66), [2-354]
ϕ	UNIFAC molecular volume fraction	following (2-100)
ϕ_i	Volume fraction of component i	(2-80), following (2-100), (2-113)
ν_i	Stoichiometric coefficient (+ for product and – for reactant) for compound i in reaction	(2-28), (2-29), (2-30)
ν_j	Frequency of vibrational mode j	(2-47)
θ	UNIFAC molecular surface fraction	following (2-100)
Θ	UNIFAC group surface fraction	following (2-100)
$\Theta_A, \Theta_B, \Theta_C$	Characteristic rotational T of molecule	before and following (2-31)
Θ_j	Characteristic vibrational T of mode j	(2-1), (2-31), (2-48)
σ	Lennard-Jones size parameter	(2-4)
σ	Rotational external symmetry number	following (2-31)
μ_r°	Modified reduced dipole moment	(2-85), (2-86)
ψ	Parameter in Riedel vapor pressure equation	following (2-24)
Ψ	Parameter in correlation of k for gases	(2-106)
Ψ_{mn}	UNIFAC interaction factor	(2-100)
ξ	Viscosity dedimensionalizing factor	(2-89), (2-90), (2-91), (2-92), (2-93), (2-94)
ω	Pseudoacentric factor for mixture	(2-74)
ω_{ij}	Cross term in mixing rule	(2-79)

Acronyms and abbreviations

Definition

CC	Computational chemistry
CS	Corresponding states
DIPPR®	Design Institute for Physical Properties
EoS	Equation of state

Nomenclature (Concluded)

Acronyms and abbreviations	Definition
GC	Group contributions
LJ	Lennard-Jones
MC	Monte Carlo
MD	Molecular dynamics
QSPR	Quantitative structure-property relationships
TRC	Thermodynamics Research Center

GENERAL REFERENCES

- (a) *Prediction methods:*
 [PGL4] Reid, R. C., J. M. Prausnitz, and B. E. Poling, *The Properties of Gases and Liquids*, 4th ed., McGraw-Hill, New York, 1987.
 [PGL5] Poling, B. E., J. M. Prausnitz, and J. P. O'Connell, *The Properties of Gases and Liquids*, 5th ed., McGraw-Hill, New York, 2001.
 (b) *Property databases:*
 [DIPPR] Rowley, R. L., et al., *DIPPR® Data Compilation of Pure Chemicals Properties*, Design Institute for Physical Properties, AIChE, New York, 2007.
 [TRC] *TRC Thermodynamic Tables—Non-Hydrocarbons*, Thermodynamics Research Center, The Texas A&M University System, College Station, Tex., extant 2004; *TRC Thermodynamic Tables—Hydrocarbons*, Thermodynamics Research Center, The Texas A&M University System, College Station, Tex., extant 2004.
 [JANAF] Chase, M. W., Jr., et al., "JANAF Thermochemical Tables," *J. Phys. Chem. Ref. Data*, **14**, suppl. 1, 1985.
 [SWS] Stull, D. R., F. F. Westrum, Jr., and G. C. Sinke, *The Chemical Thermodynamics of Organic Compounds*, John Wiley & Sons, New York, 1969.

CLASSIFICATION OF ESTIMATION METHODS

Physical property estimation methods may be classified into six general areas: (1) theory and empirical extension of theory, (2) corresponding states, (3) group contributions, (4) computational chemistry, (5) empirical and quantitative structure property relations (QSPR) correlations, and (6) molecular simulation. A quick overview of each class is given below to provide context for the methods and to define the general assumptions, accuracies, and limitations inherent in each.

Theory and Empirical Extension of Theory Methods based on theory generally provide better extrapolation capability than empirical fits of experimental data. Assumptions required to simplify the theory to a manageable equation suggest accuracy limitations and possible improvements, if necessary. For example, the ideal gas isobaric heat capacity, rigorously obtained from statistical mechanics under the assumption of independent harmonic vibrational modes, is [Rowley, R. L., *Statistical Mechanics for Thermophysical Property Calculations*, Prentice-Hall, Englewood Cliffs, N.J., 1994]

$$\frac{C_p}{R} = \frac{8 - \delta}{2} + \sum_{j=1}^{3n_A - 6 + \delta} \left(\frac{\Theta_j}{T} \right)^2 \frac{e^{\Theta_j/T}}{(e^{\Theta_j/T} - 1)^2}$$

$$\delta = \begin{cases} 0 & \text{nonlinear molecules} \\ 1 & \text{linear molecules} \end{cases} \quad (2-1)$$

where Θ_j is the characteristic temperature for the j th vibrational frequency in a molecule of n_A atoms. The temperature dependence of this equation is exact to the extent that the frequencies are harmonic. Corrections for anharmonicity can be applied (albeit with difficulty) where warranted.

Extension of theory often requires introduction of empirical models and parameters in lieu of terms that cannot be rigorously calculated. Good accuracy is expected in the region where the model parameters were fitted to experimental data, but only limited accuracy when an empirical model is extrapolated to other conditions. For example, a simplified theory suggests that vapor pressure should have the form

$$\ln P^\circ = A - \frac{B}{T} \quad (2-2)$$

where the empirical parameter B is given by

$$B = \frac{\Delta H_v}{R\Delta Z_v} \quad (2-3)$$

and ΔH_v and ΔZ_v are differences between the vapor and liquid enthalpies and compressibility factors, respectively. Vapor pressures over a narrow temperature range can be effectively correlated using Eq. (2-2), but this equation should not be used to extrapolate vapor pressures over a wide range of temperatures.

Corresponding States (CS) The principle of CS applies to conforal fluids [Leland, T. L., Jr., and P. S. Chapple, *Ind. Eng. Chem.*, **60** (1968): 15]. Two fluids are conformal if their intermolecular interactions are equivalent when scaled in dimensionless form. For example, the Lennard-Jones (LJ) intermolecular pair potential energy U can be written in dimensionless form as

$$U^\circ = 4[(r^\circ)^{-12} - (r^\circ)^{-6}] \quad (2-4)$$

where $r^\circ = r/\sigma$, $U^\circ = U/\epsilon$, σ is the LJ size parameter, and ϵ is the LJ attractive well depth parameter. At equivalent scaled temperatures kT/ϵ (k is Boltzmann's constant) and pressures $P\sigma^3/\epsilon$, all LJ fluids will have identical dimensionless properties because the molecules interact through the identical scaled intermolecular potential given by Eq. (2-4). Generalization of this scaling principle is commonly done using critical temperature T_c and critical pressure P_c as scaling factors. At the same reduced coordinates ($T_r = T/T_c$ and $P_r = P/P_c$) all conformal fluids will have the same dimensionless properties; for example, $Z = Z(T_r, P_r)$ where the compressibility factor is defined as $Z = PV/RT$. A correlation of experimental data for one fluid can be used as the reference for the properties of all conformal fluids. Nonconformality is the main accuracy limitation. For instance, interactions between non-spherical or polar molecules may not be adequately represented by Eq. (2-4), and so the scaled properties of these fluids will not conform to those of a fluid with interactions well represented by Eq. (2-4). A correction for nonconformality is usually made by the addition of one or more reference fluids whose deviations from the first reference fluid are used to characterize the effect of nonconformality. For example, in the Lee-Kesler method [Lee, B. I., and M. G. Kesler, *AIChE J.*, **21** (1975): 510] n -octane is used as a second, nonspherical reference fluid, and deviations of n -octane scaled properties from those of the spherical reference fluid at equivalent reduced conditions are assumed to be a linear function of the acentric factor. An extended Lee-Kesler method [Wilding, W. V., and R. L. Rowley, *Int. J. Thermophys.*, **7** (1986): 525] uses three reference fluids: n -octane to correct for size-shape nonconformality relative to methane and water to correct for polar effects. Some CS methods [e.g., Teja, A., S. I. Sandler, and N. C. Patel, *Chem. Eng. J.*, **21** (1981): 21] utilize different reference fluids for different classes of fluids to maintain close conformality between the fluid whose properties are to be estimated and the reference fluids.

Group Contributions (GC) Chemical and physical properties generally correlate well with molecular structure. GC methods assume a summative behavior of the structural groups of the constituent molecules. For example, ethanol ($\text{CH}_3\text{—CH}_2\text{—OH}$) properties would be obtained as the sum of contributions from the —CH_3 , —CH_2 , and —OH groups. The contribution of each group is obtained by regression of experimental data that include as many different compounds containing that group as possible. Structural groups must be used exactly as defined in the original correlation. A general principle in deciding how to make the groupings is to give the more specific group priority. For example, although the structural piece —COOCH_3 is ambiguous in a methyl ester, a more specific group value available for an ester —COO— would take precedence over a combination the two smaller groups —(C=O)— and

—O— whose values were most likely regressed only from ketone and ether data, respectively. Excellent accuracy can usually be expected from GC methods when group values were regressed from large quantities of experimental data. However, if the ratio of number of groups to regressed experimental data is large, significant errors can result when the method is applied to new compounds (extrapolation). Such excessive specificity in the group definitions leads to poor extrapolation capabilities even though the fit of the regressed data may be excellent.

Computational Chemistry (CC) Commercial software is available that solves the Schrödinger equation for approximate forms of the wave function. Various levels of sophistication (termed *model chemistry*) for the wave function can be chosen at the expense of computational time. Results include structural information (bond lengths, bond angles, dihedral angles, etc.), electron/charge distribution information, internal vibrational modes (for ideal gas properties), and energy of the molecule, valid for the chosen model chemistry. Because calculations are performed on individual molecules, they are primarily suited for ideal gas properties. Relative energies for the same model chemistry are more accurately obtained than absolute energies, so enthalpies and entropies of reaction are common industrial uses of CC predictions in addition to individual structural and ideal gas properties.

Empirical QSPR Correlations In quantitative structure property relationship (QSPR) methods, physical properties are correlated with molecular descriptors that characterize the molecular and electronic structure of the molecule. Large amounts of experimental data are used to statistically determine the most significant descriptors to be used in the correlation and their contributions. The resultant correlations are simple to apply if the descriptors are available. Descriptors must generally be generated by the user with computational chemistry software, although the DIPPR® 801 database now contains a table of molecular descriptors for most of the compounds in it. QSPR methods are often very accurate for specific families of compounds for which the correlation was developed, but extrapolation problems are even more of an issue than with GC methods.

Molecular Simulations Molecular simulations are useful for predicting properties of bulk fluids and solids. Molecular dynamics (MD) simulations solve Newton's equations of motion for a small number (on the order of 10^3) of molecules to obtain the time evolution of the system. MD methods can be used for equilibrium and transport properties. Monte Carlo (MC) simulations use a model for the potential energy between molecules to simulate configurations of the molecules in proportion to their probability of occurrence. Statistical averages of MC configurations are useful for equilibrium properties, particularly for saturated densities, vapor pressures, etc. Property estimations using molecular simulation techniques are not illustrated in the remainder of this section as commercial software implementations are not generally available at this time.

PHYSICAL CONSTANTS

Critical Properties The critical temperature T_c , pressure P_c , and volume V_c of a compound are important, widely used constants. They are important in determining the phase boundaries of a compound and (particularly T_c and P_c) are required input parameters for most thermal and volumetric property calculations of the equilibrium phases using CS or analytical equations of state. Most estimation methods employ weighted group, atom, or bond contributions.

The critical temperature of a compound is the temperature above which a liquid phase cannot be formed, no matter the pressure of the system. The critical pressure is the vapor pressure of the compound at the critical temperature. The critical volume is the volume occupied by a set amount of a compound (typically 1 mol) at its critical temperature and pressure.

The critical compressibility factor Z_c is determined from the experimental or predicted values of the critical properties by the definition

$$Z_c = \frac{P_c V_c}{RT_c} \quad (2-5)$$

Recommended Methods

Organic molecules: The Ambrose method is recommended for all three critical properties of hydrocarbons and *n*-alcohols. The Joback method is recommended for T_c and P_c of all other organic molecules. Fedors' method is recommended for V_c of these compounds, but the Joback method can also be used.

Inorganic molecules: The simple correlation $T_c = 1.64T_b$ is recommended if the normal boiling point is known. Critical pressure P_c is best obtained by extrapolating vapor pressure data to T_c , and V_c is best obtained from a correlation of liquid density extrapolated to T_c .

Other recent methods including the Wilson-Jasperson (Wilson, G. M., and L.V. Jasperson, *Critical Constants T_c , P_c , Estimation Based on Zero, First and Second Order Methods*, AIChE Spring Meeting, New Orleans, La., 1996) and Marrero-Pardillo [Marrero-Morejon, J., and E. Pardillo-Fontdevila, *AIChE J.*, **45** (1999): 615] methods have proved to be as good as or better for some classes of compounds than the methods presented here; however, their application is more difficult.

Method: Ambrose method.

Reference: Ambrose, D., Correlation and Estimation of Vapour-Liquid Critical Properties. I. Critical Temperatures of Organic Compounds, *Natl. Phys. Lab. Report Chem.* **92** (1978); Correlation and Estimation of Vapour-Liquid Critical Properties. II. Critical Pressures and Volumes of Organic Compounds, *Natl. Phys. Lab Report Chem.* **98** (1979).

Classification: Group contributions.

Expected uncertainty: ~6 K for T_c (about 1 percent), ~2 bar for P_c (about 5 percent), ~8 cm³/mol for V_c (about 3 percent).

Applicability: Organic compounds.

Input data: T_b , M , group contributions Δ_T , Δ_P , and Δ_V from Table 2-336.

Description: A GC method with first-order contributions and corrections (delta Platt number) for branched alkanes. Variables T_c , P_c , and V_c are given by the following relations:

$$T_c = T_b \left[1 + \left(1.242 + \sum \Delta_T \right)^{-1} \right] \quad (2-6)$$

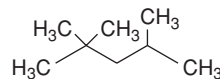
$$\frac{P_c}{\text{bar}} = \frac{M}{\text{kg/kmol}} \left(0.339 + \sum \Delta_P \right)^{-2} \quad (2-7)$$

$$\frac{V_c}{\text{cm}^3/\text{mol}} = 40 + \sum \Delta_V \quad (2-8)$$

Example Use the Ambrose method to estimate the critical constants of 2,2,4-trimethylpentane.

Required data: From the DIPPR® 801 database, $T_b = 372.39$ K and $M = 114.229$ kg/kmol.

Structure:



Group contributions from Table 2-336:

Group	n_i	Δ_T	Δ_P	Δ_V
Alkyl carbons	8	0.138	0.226	55.1
>CH— (correction)	1	−0.043	−0.006	−8
>C< (correction)	1	−0.120	−0.030	−17
Delta Platt no.	0	−0.023	−0.026	—

Calculations using Eqs. (2-6), (2-7), and (2-8):

$$\sum \Delta_T = (8)(0.138) + (1)(-0.043) + (1)(-0.120) = 0.941$$

$$T_c = T_b(1.4581) = (372.39 \text{ K})(1.4581) = 543.0 \text{ K}$$

$$\sum \Delta_P = (8)(0.226) + (1)(-0.006) + (1)(-0.030) = 1.772$$

$$\frac{P_c}{\text{bar}} = \frac{M}{\text{kg/kmol}} \left(0.339 + \sum \Delta_P \right)^{-2} = \frac{114.229}{(0.339 + 1.772)^2} = 25.63$$

$P_c = 25.63 \text{ bar}$

TABLE 2-336 Ambrose Group^a Contributions for Critical Constants

Group	Δ_T	Δ_P	Δ_V
Carbon atoms in alkyl groups	0.138	0.226	55.1
Corrections			
>CH— (each)	-0.043	-0.006	-8
>C< (each)	-0.120	-0.030	-17
Double bonds (nonaromatic)	-0.050	-0.065	-20
Triple bonds	-0.200	-0.170	-40
Delta Platt number, ^b multiply by	-0.023	-0.026	—
Aliphatic functional groups:			
—O—	0.138	0.160	20
>CO	0.220	0.282	60
—CHO	0.220	0.220	55
—COOH	0.578	0.450	80
—CO—O—OC—	1.156	0.900	160
—CO—O—	0.330	0.470	80
—NO ₂ —	0.370	0.420	78
—NH ₂	0.208	0.095	30
—NH—	0.208	0.135	30
>N—	0.088	0.170	30
—CN	0.423	0.360	80
—S—	0.105	0.270	55
—SH	0.090	0.270	55
—SiH ₃	0.200	0.460	119
—O—Si(CH ₃) ₂	0.496	—	—
—F	0.055	0.223	14
—Cl	0.055	0.318	45
—Br	0.055	0.500	67
—I	0.055	—	90
Halogen correction in aliphatic compounds:			
F is present	0.125		
F is absent, but Cl, Br, I are present	0.055		
Aliphatic alcohols ^c	<i>d</i>	<i>e</i>	15
Ring compound increments (listed only when different from aliphatic values):			
—CH ₂ —, >CH—, >C<	0.090	0.182	44.5
>CH— in fused ring	0.030	0.182	44.5
Double bond	-0.030	—	-15
—O—	0.090	—	10
—NH—	0.090	—	—
—S—	0.090	—	30
Aromatic compounds:			<i>f</i>
Benzene	0.448	0.924	
Pyridine	0.448	0.850	
C ₆ H ₄ (fused as in naphthalene)	0.220	0.515	
—F	0.080	0.183	
—Cl	0.080	0.318	
—Br	0.080	0.600	
—I	0.080	0.850	
—OH	0.198	-0.025	
Corrections for nonhalogenated substitutions:			
First	0.010	0	
Each subsequent	0.030	0.020	
Ortho pairs containing —OH	-0.080	-0.050	
Ortho pairs with no —OH	-0.040	-0.050	
Highly fluorinated aliphatic compounds:			
—CF ₃ , —CF ₂ —, >CF—	0.200	0.550	
—CF ₂ —, >CF— (ring)	0.140	0.420	
>CF— (in fused ring)	0.030	—	
—H (monosubstitution)	-0.050	-0.350	
Double bond (nonring)	-0.150	-0.500	
Double bond (ring)	-0.030	—	
(other increments as in nonfluorinated compounds)			

^aAmbrose, D., Correlation and Estimation of Vapour-Liquid Critical Properties. I. Critical Temperatures of Organic Compounds, *Natl. Phys. Lab Report Chem.* **92** (1978); Correlation and Estimation of Vapour-Liquid Critical Properties. II. Critical Pressures and Volumes of Organic Compounds, *Natl. Phys. Lab Report Chem.* **98** (1979).

^bThe delta Platt number is defined as the Platt number of the isomer minus the Platt number of the corresponding alkane. (For *n*-alkanes the Platt number is $n - 3$.) The Platt number is the total number of groups of four carbon atoms three bonds apart [Platt, J. R., *J. Chem. Phys.*, **15**(1947): 419; **56**(1952): 328]. This correction is used only for branched alkanes.

^cIncludes naphthenic alcohols and glycols but not aromatic alcohols such as xylenol.

^dFirst determine the hydrocarbon homomorph, i.e., substitute —CH₃ for each —OH and calculate $\sum \Delta_T$ for this compound. Subtract 0.138 from $\sum \Delta_T$ for each —OH substituted. Next, add $0.87 - 0.11n + 0.003n^2$ where $n = [T_b/K(\text{alcohol}) - 314]/19.2$. Exceptions include methanol ($\sum \Delta_T = 0$), ethanol ($\sum \Delta_T = 0.939$), and any alcohol whose value of *n* exceeds 10.

^eDetermine the hydrocarbon homomorph as in footnote *d*. Calculate $\sum \Delta_P$ and subtract 0.226 for each —OH substituted. Add $0.100 - 0.013n$, where *n* is computed as in footnote *d*.

^fWhen estimating the critical volumes of aromatic substances, use ring compound values, if available, and correct for double bonds.

$$\sum \Delta_V = (8)(55.1) + (1)(-8) + (1)(-17) = 415.8$$

$$V_c = (40 + 415.8) \text{ cm}^3/\text{mol} = 455.8 \text{ cm}^3/\text{mol}$$

Results:

Property	DIPPR® recommended value	Ambrose estimation	% Difference
T_c/K	543.8	543.0	-0.15
P_c/bar	25.70	25.63	0.27
$V_c/(\text{cm}^3/\text{mol})$	468.0	455.8	-2.6

Method: Joback method.

Reference: Joback, K. G., M.S. Thesis in Chemical Engineering, Massachusetts Institute of Technology, Cambridge, Mass., June 1984.

Classification: Group contributions.

Expected uncertainty: 7 K (~1 percent) for T_c ; 2 bar (~5 percent) for P_c .

Applicability: Organic compounds.

Input data: T_b , group contributions Δ_T , Δ_P , Δ_V from Table 2-337, and the number of atoms in the molecule n_A .

Description: A GC method with first-order contributions. Variables T_c , P_c , and V_c are given by the following relations:

$$T_c = T_b \left[0.584 + 0.965 \sum \Delta_T - \left(\sum \Delta_T \right)^2 \right]^{-1} \quad (2-9)$$

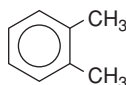
$$\frac{P_c}{\text{bar}} = \left(0.113 + 0.0032 n_A - \sum \Delta_P \right)^{-2} \quad (2-10)$$

$$\frac{V_c}{\text{cm}^3/\text{mol}} = 17.5 + \sum \Delta_V \quad (2-11)$$

where Δ_T , Δ_P , Δ_V are group contributions from Table 2-337 and n_A is the number of atoms in the molecule.

Example Estimate the critical constants of *o*-xylene by using the Joback method.

Structure:



Required input data: From the DIPPR® 801 database, $T_b = 417.58 \text{ K}$. From Table 2-337:

Group	n_i	Δ_T	Δ_P	Δ_V
$\text{CH}(\text{ring})$	4	0.0082	0.0011	41
$\text{C}(\text{ring})$	2	0.0143	0.0008	32
CH_3	2	0.0141	-0.0012	65

From Eqs. (2-9), (2-10), and (2-11):

$$\sum \Delta_T = (4)(0.0082) + (2)(0.0143) + (2)(0.0141) = 0.0896$$

$$T_c = T_b [0.584 + 0.965 (0.896) - (0.896)^2]^{-1}$$

$$T_c = T_b (1.5096) = (417.58 \text{ K})(1.5096) = 630.37 \text{ K}$$

$$n_A = 18 \quad \sum \Delta_P = (4)(0.0011) + (2)(0.0008) + (2)(-0.0012) = 0.0036$$

$$\frac{P_c}{\text{bar}} = \left(0.113 + 0.0032 n_A - \sum \Delta_P \right)^{-2} = [0.113 + (0.0032)(18) - 0.0036]^{-2} = 35.86$$

$$P_c = 35.86 \text{ bar}$$

$$\sum \Delta_V = (4)(41) + (2)(32) + (2)(65) = 358$$

$$V_c = (17.5 + 358) \text{ cm}^3/\text{mol} = 375.5 \text{ cm}^3/\text{mol}$$

TABLE 2-337 Joback * Group Contributions for Critical Constants

Group	Δ_T	Δ_P	Δ_V
Nonring increments:			
CH_3	0.0141	-0.0012	65
CH_2	0.0189	0	56
$\text{CH}(\text{ring})$	0.0164	0.0020	41
$\text{C}(\text{ring})$	0.0067	0.0043	27
CH_2	0.0113	-0.0028	56
$\text{CH}(\text{ring})$	0.0129	-0.0006	46
$\text{C}(\text{ring})$	0.0117	0.0011	38
$\text{C}(\text{ring})$	0.0026	0.0028	36
$\text{CH}(\text{ring})$	0.0027	-0.0008	46
$\text{C}(\text{ring})$	0.0020	0.0016	37
Ring increments:			
CH_2	0.0100	0.0025	48
$\text{CH}(\text{ring})$	0.0122	0.0004	38
$\text{C}(\text{ring})$	0.0042	0.0061	27
$\text{CH}(\text{ring})$	0.0082	0.0011	41
$\text{C}(\text{ring})$	0.0143	0.0008	32
Halogen increments:			
F	0.0111	-0.0057	27
Cl	0.0105	-0.0049	58
Br	0.0133	0.0057	71
I	0.0068	-0.0034	97
Oxygen increments:			
OH (alcohol)	0.0741	0.0112	28
OH (phenol)	0.0240	0.0184	-25
O (nonring)	0.0168	0.0015	18
O (ring)	0.0098	0.0048	13
$\text{C}=\text{O}$ (nonring)	0.0380	0.0031	62
$\text{C}=\text{O}$ (ring)	0.0284	0.0028	55
$\text{CH}=\text{O}$ (aldehyde)	0.0379	0.0030	82
COOH (acid)	0.0791	0.0077	89
COO (ester)	0.0481	0.0005	82
O (except as above)	0.0143	0.0101	36
Nitrogen increments:			
NH_2	0.0243	0.0109	38
NH (nonring)	0.0295	0.0077	35
NH (ring)	0.0130	0.0114	29
N (nonring)	0.0169	0.0074	9
$\text{N}(\text{nonring})$	0.0255	-0.0099	—
$\text{N}(\text{ring})$	0.0085	0.0076	34
CN	0.0496	-0.0101	91
NO_2	0.0437	0.0064	91
Sulfur increments:			
SH	0.0031	0.0084	63
S (nonring)	0.0119	0.0049	54
S (ring)	0.0019	0.0051	38

*Joback, K. G., M.S. thesis in chemical engineering, Massachusetts Institute of Technology, Cambridge, Mass., June 1984.

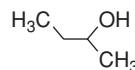
Results:

Property	DIPPR® 801 recommendation	Joback estimation	% Difference
T_c/K	630.3	630.37	0.00
P_c/bar	37.32	35.86	-3.92
$V_c/(\text{cm}^3/\text{mol})$	370	375.5	1.49

Example Estimate the critical constants of *sec*-butanol by using the Joback method.

Required input data: From DIPPR® 801 database, $T_b = 372.7 \text{ K}$, $M = 74.1216 \text{ kg/kmol}$.

Structure:



Group contributions from Table 2-337:

Group	n_i	Δ_T	Δ_P	Δ_V
CH_3	2	0.0141	-0.0012	65
CH_2	1	0.0189	0	56
$\text{CH}(\text{ring})$	1	0.0164	0.0020	41
OH (alcohol)	1	0.0741	0.0112	28

From Eqs. (2-9), (2-10), and (2-11):

TABLE 2-338 Fedors* Method Atomic and Structural Contributions

Atomic increments		Structural increments	
Atom	Δ_V	Feature	Δ_V
C	34.426	Three-member ring	-15.824
H	9.172	Four-member ring	-17.247
O	20.291	Five-member ring	-39.126
O (alcohols)	18.000	Six-member ring	-39.508
N	48.855	Double bond	5.028
N (amines)	47.422	Triple bond	0.797
F	22.242	Ring attached to another	35.524
Cl	52.801		
Br	71.774		
I	96.402		
S	50.866		

*Fedors, R. F., *AIChE J.*, **25** (1979): 202.

$$\sum \Delta_r = (2)(0.0141) + (1)(0.0189) + (1)(0.0164) + (1)(0.0741) = 0.1376$$

$$T_c = T_b(1.4330) = (372.7 \text{ K})(1.4330) = 534.1 \text{ K}$$

$$\sum \Delta_r = (2)(-0.0012) + (1)(0.0020) + (1)(0.0112) = 0.0108 \quad n_A = 15$$

$$\frac{P_c}{\text{bar}} = (0.1022 + 0.0032n_A)^{-2} = \left[\frac{1}{0.1022 + (0.0032)(15)} \right]^2 = 44.33$$

$$P_c = 44.33 \text{ bar}$$

$$\sum \Delta_V = (2)(65) + (1)(56) + (1)(41) + (1)(28) = 255$$

$$V_c = (17.5 + 255) \text{ cm}^3/\text{mol} = 272.5 \text{ cm}^3/\text{mol}$$

Results:

Property	DIPPR® 801 recommendation	Joback estimation	% Difference
T_c/K	536.2	534.1	-0.39
P_c/bar	42.02	44.33	5.50
$V_c/(\text{cm}^3/\text{mol})$	269	272.5	1.30

Method: Fedor method.

Reference: Fedors, R. F., *AIChE J.*, **25** (1979): 202.

Classification: Atom/structure contributions.

Expected uncertainty: ~8 cm³/mol (3 percent).

Applicability: Organic compounds.

Input data: Molecular structure, atom and structural increments, Δ_V in Table 2-238.

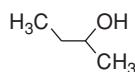
Description: An atom contribution method with structural increments for V_c , given by

$$\frac{V_c}{\text{cm}^3/\text{mol}} = 26.6 + \sum_{i=1}^N n_i \Delta_V \quad (2-12)$$

where Δ_V are structural contributions from Table 2-238.

Example Use Fedors' method to estimate the critical volume of *sec*-butanol.

Group contributions from Table 2-238:



Group	n_i	Δ_V
C	4	34.426
H	10	9.172
O (alcohol)	1	18.000

Calculation using Eq. (2-12):

$$\sum \Delta_V = (4)(34.426) + (10)(9.172) + (1)(18.000) = 247.4$$

$$V_c = (26.6 + 247.4) \text{ cm}^3/\text{mol} = 274.0 \text{ cm}^3/\text{mol}$$

This value differs from the DIPPR® 801 recommended value of 269 cm³/mol by 1.9 percent.

Mixtures Application of CS methods and analytical equations of state (EoS) to mixtures typically requires pseudocritical temperatures and pressures. These values are obtained from the pure-component critical properties by applying mixing rules specific to the method under consideration. While pseudocritical values should be used for mixture CS and EoS applications, the values are usually quite different from the mixture's true critical properties. A variety of methods are available for estimating true critical properties for mixtures (see PGL4), but only modest accuracy can be expected.

Normal Melting Point The *normal melting point* is defined as the temperature at which melting occurs at atmospheric pressure. Methods to estimate the melting point have not been particularly effective because the melting point depends strongly on solid crystal structure and that structure is not effectively correlated with standard GC or CS methods. If the triple point temperature is known, then the melting point is best estimated as being equal to the triple point temperature. However, rarely is the triple point temperature available if the melting point has not also been determined.

Recommended Method The method of Constantinou and Gani is recommended with caution.

Reference: Constantinou, L., and R. Gani, *AIChE J.*, **40** (1994): 1697.

Classification: Group contributions.

Expected uncertainty: 25 percent.

Applicability: Organic compounds.

Input data: First-order and second-order group contributions from molecular structure.

Description: A group contribution method given by

$$T_m = (102.425 \text{ K}) \cdot \ln \left(\sum_i N_i t_{m1,i} + \sum_j N_j t_{m2,j} \right) \quad (2-13)$$

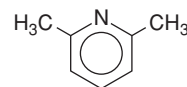
where N_i , N_j = number of first- and second-order groups, respectively

$t_{m1,i}$ = first-order group contributions from Table 2-339

$t_{m2,j}$ = second-order group contributions from Table 2-340

Example Estimate the melting point of 2,6-dimethylpyridine.

Structure and group contributions:



Group	N_i	$t_{m1,i}$	$t_{m2,i}$
-CH ₃	2	0.4640	
-C ₅ H ₃ (N)-	1	12.6275	
Six-member ring	1		1.5656

Calculation using Eq. (2-13):

$$T_m = (102.425 \text{ K}) \ln [(2)(0.4640) + 12.6275 + 1.5656] = 278 \text{ K}$$

The predicted value is 4 percent higher than the recommended experimental value of 267 K in the DIPPR® 801 database.

Normal Boiling Point The normal boiling temperature T_b is the temperature at which the vapor pressure of the liquid equals 101.325 kPa (1.0 atm). In some (usually older) literature sources, T_b values were reported at ambient pressures rather than at 101.325 kPa. If two or more such values are available, they can be used to obtain T_b by using Eq. (2-2) to linearly interpolate $\ln P^*$ vs. $1/T$ values.

If there are sufficient vapor pressure data available, then T_b may be found from a regression of the data using an appropriate vapor pressure equation [e.g., Eqs. (2-21)–(2-26)]. If one or a few vapor pressure data at low pressure are available, a common occurrence, then the method of Pailhes can be used to estimate T_b . The most accurate method for prediction of normal boiling temperatures without experimental data is the Nannoolal method.

TABLE 2-339 First-Order Groups and Their Contributions for Melting Point*

Group	$t_{m1,i}$	Group	$t_{m1,i}$	Group	$t_{m1,i}$
—CH ₃	0.4640	—COOCH ₂ —	3.5572	—CCl ₃	10.2337
>CH ₂	0.9246	—OOCH	4.2250	>ACCl	2.7336
>CH—	0.3557	—OCH ₃	2.9248	—CH ₂ NO ₂	5.5424
>C<	1.6479	—OCH ₂ —	2.0695	>CHNO ₂	4.9738
—CH=CH ₂	1.6472	—OCH<	4.0352	>ACNO ₂	8.4724
—CH=CH—	1.6322	—OCH ₂ F	4.5047	—CH ₂ SH	3.0044
>C=CH ₂	1.7899	—CH ₂ NH ₂	6.7684	—I	4.6089
>C=CH—	2.0018	>CHNH ₂	4.1187	—Br	3.7442
>C=C<	5.1175	—NHCH ₃	4.5341	—C≡CH	3.9106
—CH=C=CH ₂	3.3439	—CH ₂ NH—	6.0609	—C≡C—	9.5793
>ACH	1.4669	>CHNH—	3.4100	>C=CCl—	1.5598
>AC—	0.2098	>NCH ₃	4.0580	>ACF	2.5015
>ACCH ₃	1.8635	—NCH ₂ —	0.9544	—CF ₃	3.2411
>ACCH ₂ —	0.4177	>ACNH ₂	10.1031	—COO—	3.4448
>ACCH<	-1.7567	—C ₂ H ₅ (N)—	12.6275	—CCl ₃ F	7.4756
—OH	3.5979	—CH ₃ CN	4.1859	—CClF ₂	2.7523
>ACOH	13.7349	—COOH	11.5630	—F (other)	1.9623
—COCH ₃	4.8776	—CH ₂ Cl	3.3376	—CONH ₂	31.2786
—COCH ₂ —	5.6622	>CHCl	2.9933	—CON(CH ₃) ₂	11.3770
—CHO	4.2927	>CCL—	9.8409	—CH ₃ S	5.0506
—COOCH ₃	4.0823	—CHCl ₂	5.1638	>CH ₂ S	3.1468

*Constantinou, L., and R. Gani, *AIChE J.*, **40** (1994): 1697.**Recommended Method** Pailhes method.**Classification:** Group contributions.**Expected uncertainty:** 3 K (~1 to 2 percent).**Applicability:** Organic compounds.**Reference:** Pailhes, F., *Fluid Phase Equilib.*, **41** (1988): 97.

Input data: Molecular structure and one measured vapor pressure value P_{meas} , (often at a low pressure). The method requires estimation of T_c and P_c using a group contribution method. The original Pailhes method used the Lydersen method [Lydersen, A. L., *AIChE J.*, **21** (1975): 510], but the Joback method is illustrated here for consistency with the previous examples on critical constant estimation.

Description: A simple group contribution method given by

$$T_b = T_{\text{meas}} \frac{\log(P_c/\text{bar}) + (1 - T_{br})x_p}{\log(P_c/\text{bar})} - 3x_p - 1.49x_p^2 \quad (2-14)$$

where T_b = estimate of normal boiling point P_c = critical pressure estimated from group contributions T_{br} = reduced normal boiling point estimated from Eq. (2-9) $x_p = \log(1 \text{ atm}/P_{\text{meas}})$ T_{meas} = temperature at which experimental vapor pressure is known

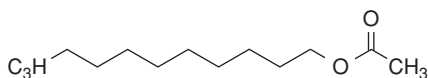
TABLE 2-340 Second-Order Groups and Their Contributions for Melting Point*

Group	$t_{m2,i}$	Group	$t_{m2,i}$
—CH(CH ₃) ₂	0.0381	CHCOOH; CCOOH	-3.1034
—C(CH ₃) ₃	-0.2355	ACCOOH	28.4324
—CH(CH ₃)CH(CH ₃)—	0.4401	CH ₃ COOCH; CH ₃ COOC	0.4838
—CH(CH ₃)C(CH ₃) ₂ —	-0.4923	COCH ₂ COO or COCHCOO or COCCOO	0.0127
—C(CH ₃) ₂ C(CH ₃) ₂ —	6.0650	CO—O—CO	-2.3598
Three-member ring	1.3772	ACCOO	-2.0198
Five-member ring	0.6824	CHOH	-0.5480
Six-member ring	1.5656	COH	0.3189
Seven-member ring	6.9709	CH _m (OH)CH _n (OH) [$m, n = 0, 1, 2$]	0.9124
CH _n =CH _m —CH _p =CH _k	1.9913	CH _{m cyclic} —OH [$m = 0, 1$]	9.5209
[$k, n, m, p = 0, 1, 2$]			
CH ₃ CH _m =CH _n [$m, n = 0, 1, 2$]	0.2476	CH _m (OH)CH _n (NH _p)	2.7826
CH ₂ CH _m =CH _n [$m, n = 0, 1, 2$]	-0.5870	[$m, n, p = 0, 1, 2, 3$]	
CHCH _m =CH _n or CCH _m =CH _n	-0.2361	CH _m (NH ₂)CH _n (NH ₂)	2.5114
[$m, n = 0, 1, 2$]		[$m, n = 0, 1, 2$]	
Alicyclic side chain: C _{cyclic} C _m	-2.8298	CH _{m cyclic} —NH _p —CH _{n cyclic}	1.0729
[$m > 1$]		[$m, n, p = 0, 1, 2$]	
CH ₃ CH ₃	1.4880	CH _m —O—CH _n =CH _p	0.2476
CHCHO; CCHO	2.0547	[$m, n, p = 0, 1, 2$]	
CH ₃ COCH ₂	-0.2951	AC—O—CH _m	0.1175
CH ₃ COCH; CH ₃ COC	-0.2986	[$m = 0, 1, 2, 3$]	
C _{cyclic} (=O)	0.7143	CH _{m cyclic} —S—CH _{n cyclic}	-0.2914
ACCHO	-0.6697	[$m, n = 0, 1, 2$]	
		CH _m =CH _n —F	-0.0514
		[$m, n = 0, 1, 2$]	
		CH _m =CH _n —Br	-1.6425
		[$m, n = 0, 1, 2$]	
		ACBr	2.5832
		ACl	-1.5511

*Constantinou, L., and R. Gani, *AIChE J.*, **40** (1994): 1697.

Example The vapor pressure of *n*-decylacetate at 348.65 K is 106.66 Pa. Estimate the normal boiling point of this compound.

Structure and group contributions from Table 2-337:



Group	n_i	Δ_r	Δ_p
—CH ₃	2	0.0141	-0.0012
>CH ₂	9	0.0189	0
—COO— (ester)	1	0.0481	0.0005

Group contribution calculations using Eqs. (2-9) and (2-10):

$$\sum \Delta_r = (2)(0.0141) + (9)(0.0189) + (1)(0.0481) = 0.2464$$

$$T_{br} = 0.584 + 0.965(0.2464) - (0.2464)^2 = 0.76106$$

$$\sum \Delta_p = (2)(-0.0012) + (1)(0.0005) = -0.0019$$

$$n_A = 38 \quad P_c = (0.113 + 0.0032n_A + 0.0019)^{-2} \text{ bar} = 17.88 \text{ bar} = 17.65 \text{ atm}$$

Calculation of auxiliary quantities:

$$x_p = \log \frac{1 \text{ atm}}{P_{\text{meas}}} = \log \frac{101325 \text{ Pa}}{106.66 \text{ Pa}} = 2.9777$$

Calculation of normal boiling point using Eq. (2-14):

$$\frac{T_b}{K} = 348.65 \frac{\log 17.88 + (1 - 0.76016)(2.9777)}{\log 17.88} - 3(2.9777) - 1.49(2.9777)^2$$

$$T_b = 525.3 \text{ K}$$

The estimated value is 1.6 percent higher than the DIPPR® 801 recommended value of 517.15 K.

Recommended Method Nannoolal method.

Reference: Nannoolal, Y., et al., *Fluid Phase Equilib.*, **226** (2004): 45.

Classification: Group contributions.

Expected uncertainty: 7 K (on the order of 2 percent).

Applicability: Organic compounds for which group values are available.

Input data: GC values C_i in Table 2-341; intramolecular group-group interactions C_{ij} in Table 2-342.

Description: A GC method that includes second-order corrections for steric effects and intramolecular interactions. Variable T_b is found from

$$\frac{T_b}{K} = \frac{\sum_{i=1}^N n_i \cdot C_i}{n^{0.6583} + 1.6865} + 84.3395 \quad (2-15)$$

where n = number of nonhydrogen atoms

n_i = number of occurrences of group i

N = number of groups

C_i = group contribution from Table 2-341 or Eq. (2-16)

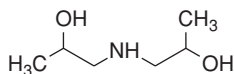
Corrections for intramolecular group-group interactions $C_{i,\text{int}}$ are made by dividing the sum of all unique group pairs within the molecule by n . This can be written as

$$C_{i,\text{int}} = \frac{1}{n} \sum_{i=1}^N \sum_{j>i}^N C_{ij} \quad (2-16)$$

or thought of as the sum of terms in the upper half (above and to the right of the diagonal) of the square matrix formed by listing each group in the molecule along the row and column. The values for the interactions are shown in this format in Table 2-342.

Example Estimate the normal boiling point of di-isopropanolamine by using the Nannoolal method.

Structure:



Group contributions and values:

Group	n_i	C_i	Group total
—CH ₃	2	177.3066	354.6132
>C(c)<(e)	4	266.8769	1067.508
—OH sec	2	390.2446	780.4892
—NH—	1	223.0992	223.0992
Corrections			
—OH::—OH	1/9	291.7985	32.42206
—OH::—NH—	2/9	286.9698	63.77107
Total			2521.902

Note that there are three interacting groups (—OH, —OH, —NH—) in the molecule which gives two —OH::—NH— interactions and one —OH::—OH interaction which are then divided by n ($=9$) as in Eq. (2-16) to give the frequency.

Calculation using Eq. (2-15):

$$\frac{T_b}{K} = \frac{2521.902}{9^{0.6583} + 1.6865} + 84.3395 = 509.3 \quad T_b = 509.3 \text{ K}$$

The calculated value differs by -2.4 percent from the DIPPR® 801 recommended value of 521.9 K.

Characterizing and Correlating Constants

Acentric Factor The acentric factor of a compound ω is defined in terms of the reduced vapor pressure evaluated at a reduced temperature of 0.7 as

$$\omega = -\log P_r \Big|_{T_r=0.7} - 1.0000 \quad (2-17)$$

It is primarily used as a third parameter (beyond T_c and P_c) in CS predictions as a measure of deviations from nonspherical molecular shape, hence the name, suggesting molecular interactions that are not between centers of molecules. However, as defined in Eq. (2-17), ω also contains polarity information, and ω increases slightly with increasing polarity for molecules of similar size and shape. The value of ω is close to zero for small, spherically shaped, nonpolar molecules (argon, methane, etc.). It increases in value with larger deviations of molecular shape from spherical (longer chain lengths, less chain branching, etc.) and with increasing molecular polarity. When possible, ω should be obtained from experimental vapor pressure correlations by using Eq. (2-17), but an accurate estimation of ω can be made by using the critical constants and a single vapor pressure point by application of CS vapor pressure equations.

Recommended Method Ambrose-Walton modification of Lee-Kesler vapor pressure equations.

References: Ambrose, D., and J. Walton, *Pure & Appl. Chem.*, **61** (1989): 1395; Lee, B. I., and M. G. Kesler, *AIChE J.*, **21** (1975): 510.

Classification: Corresponding states.

Expected uncertainty: Generally within 5 percent.

Applicability: Most organic compounds.

Input data: T_c , P_c , and a single vapor pressure point (e.g., the normal boiling point).

Description: See Eq. (2-25) for the equations used in this method. The vapor pressure equation is inverted to obtain the acentric factor from a single experimental vapor pressure point.

Example Calculate the acentric factor of chlorobenzene with a known value for T_b .

Input information: From the DIPPR® 801 database, $T_b = 404.87 \text{ K}$, $T_c = 632.35 \text{ K}$, and $P_c = 45.1911 \text{ bar}$.

Calculation of auxiliary quantities:

$$T_{br} = \frac{T_b}{T_c} = \frac{404.87}{632.35} = 0.64 \quad \tau = 1 - 0.64 = 0.36$$

$$f^{(0)} = \frac{(-5.97616)(0.36) + (1.29874)(0.36)^{1.5} - (0.60394)(0.36)^{2.5} - (1.06841)(0.36)^5}{0.64}$$

$$= -3.0034$$

$$f^{(1)} = \frac{(-5.03365)(0.36) + (1.11505)(0.36)^{1.5} - (5.41217)(0.36)^{2.5} - (7.46628)(0.36)^5}{0.64}$$

$$= -3.1788$$

2-474 PHYSICAL AND CHEMICAL DATA

TABLE 2-341 Group Contributions for the Nannoolal* Method for Normal Boiling Point

Table-specific nomenclature: (e) = connected to N, O, F, Cl; (ne) = not connected to N, O, F, Cl; (r) = in a ring; (c) = in a chain; (a) = aromatic, not necessarily carbon; (Ca) = aromatic carbon; b = any nonhydrogen atom

ID	Group	Description	Value
1	CH ₃ —(ne)	CH ₃ — not connected to N, O, F, or Cl	177.3066
2	CH ₃ —(e)	CH ₃ — connected to N, O, F, or Cl	251.8338
3	CH ₃ —(a)	CH ₃ — connected to an aromatic atom (not necessarily C)	157.9527
4	—C(c)H ₂ —	—CH ₂ — in a chain	239.4531
5	>C(c)H—	>CH— in a chain	240.6785
6	>C(c)<	>C< in a chain	249.5809
7	>C(c)<(e)	>C< in a chain connected to at least one F, Cl, N, or O	266.8769
8	>C(c)<(Ca)	>C< in a chain connected to at least one aromatic carbon	201.0115
9	—C(r)H ₂ —	—CH ₂ — in a ring	239.4957
10	>C(r)H—	>CH— in a ring	222.1163
11	>C(r)<	>C< in a ring	209.9749
12	>C(r)<(e, c)	>C< in a ring; connected to at least one N, O, Cl, or F which are not part of the ring	250.9584
13	>C(r)<(e, r)	>C< in a ring connected to at least one N or O which are part of the ring	291.2291
14	>C(r)<(Ca)	>C< in a ring connected to at least one aromatic carbon	244.3581
15	==C(a)H—	aromatic ==CH—	235.3462
16	==C(a)<(ne)	aromatic ==C< not connected to O, N, Cl, or F	315.4128
17	==C(a)<(e)	aromatic ==C< connected to O, N, Cl, or F	348.2779
18	(a) ==C(a)<2(a)	aromatic ==C< with three aromatic neighbors	367.9649
19	F—(C, Si)	F— connected to C or Si	106.5492
20	—CF==C<	F— on a C=C (vinylfluoride)	49.2701
21	F—(C, Si)(F)(2b)	F— connected to C or Si already substituted with at least one F and two other atoms	53.1871
22	F—(C, Si)([F, Cl])(b)	F— connected to a C or Si already substituted with one F or Cl and one other atom	78.7578
23	F—(C, Si)([F, Cl]2)	F— connected to C or Si already substituted with two F or Cl atoms	103.5672
24	F—(Ca)	F— connected to an aromatic carbon	-19.5575
25	Cl—(C, Si)	Cl— connected to C or Si not already substituted with F or Cl	330.9117
26	Cl—(C, Si)([F, Cl])	Cl— connected to C or Si already substituted with one F or Cl	287.1863
27	Cl—(C, Si)([F, Cl]2)	Cl— connected to C or Si already substituted with at least two F or Cl	267.4170
28	Cl—(Ca)	Cl— connected to aromatic C	205.7363
29	—CCl=C<	Cl— on a C=C (vinylchloride)	292.5816
30	Br—(C, Si)	Br— connected to a nonaromatic C or Si	419.4959
31	Br—(Ca)	Br— connected to an aromatic C	377.6775
32	I—(C, Si)	I— connected to C or Si	556.3944
33	—OH <i>tert</i>	—OH connected to tertiary carbon	349.9409
34	HO—(C, Si) <i>sec</i>	—OH connected to secondary C or Si	390.2446
35	HO—(C, Si) <i>long</i>	—OH connected to primary C or Si; chain > 4 C or Si	443.8712
36	HO—(C, Si) <i>short</i>	—OH connected to primary C or Si; chain < 5 C or Si	488.0819
37	—OH (Ca)	—OH connected to an aromatic C (phenols)	361.4775
38	(C, Si)—O—(C, Si)	ether —O— connected to two C or Si	146.4836
39	>(OC ₂)<	>(OC ₂)< (epoxide)	820.7118
40	NH ₂ —(C, Si)	NH ₂ — connected to either C or Si	321.1759
41	NH ₂ —(Ca)	NH ₂ — connected to an aromatic C	441.4388
42	(C, Si)—NH—(C, Si)	—NH— connected to two C or Si (secondary amine)	223.0992
43	(C, Si)2>N—(C, Si)	>N— connected to three C or Si (tertiary amine)	126.2952
44	COOH—(C)	—COOH connected to C	1080.3139
45	(C)—COO—(C)	—COO— connected to two C (ester)	636.2020
46	HCOO—(C)	HCOO— connected to C (formic acid ester)	642.0427
47	—C(c)OO—	—COO— in ring, C is connected to C (lactone)	1142.6119
48	—CON<	—CON< disubstituted amide	1052.6072
49	—CONH—	—CONH— (monosubstituted amide)	1364.5333
50	—CONH ₂	—CONH ₂ (amide)	1487.4109
51	O=C<(Can)2	—CO— connected to two nonaromatic C (ketones)	618.9782
52	CHO—(Can)	CHO— connected to nonaromatic C (aldehydes)	553.8090
53	SH—(C)	—SH connected to C (thioles)	434.0811
54	(C)—S—(C)	—S— connected to two C	461.5784
55	(C)—S—S—(C)	—S—S— (disulfide) connected to two C	864.5074
56	—S(a)—	—S— in an aromatic ring	304.3321
57	(C)—C≡N	—C≡N (cyanide) connected to C	719.2462
58	>C(c)=C(c)<	>C=C< (both C have at least one non-H neighbor)	475.7958
59	>C(c)=C(c)<(Ca)	noncyclic >C=C< connected to at least one aromatic C	586.1413
60	—(e)C(c)=C(c)<	noncyclic >C=C< with at least one F, Cl, N, or O	500.2434
61	H ₂ C(c)=C<	H ₂ C=C< (1-ene)	412.6276
62	>C(r)=C(r)<	cyclic >C=C<	475.9623
63	—C≡C—	—C≡C—	512.2893
64	HC≡C—	HC≡C— (1-yne)	422.2307
65	(Ca)—O(a)—(Ca)	—O— in an aromatic ring with aromatic C neighbors	37.1936
66	==N(a)—(r5)	aromatic ==N— in a five-member ring, free electron pair	453.3397
67	==N(a)—(r6)	aromatic ==N— in a six-member ring	306.7139
68	NO ₂ —(C)	NO ₂ — connected to aliphatic C	866.5843
69	NO ₂ —(Ca)	NO ₂ — connected to aromatic C	821.4141
70	>Si<	>Si<	282.0181
71	>Si<(O)	>Si< connected to at least one O	207.9312
72	NO ₃ —	nitrate (esters of nitric acid)	920.3617
73	PO ₄ —	phosphates	1153.1344
74	O=N—O—(C)	nitrites (esters of nitrous acid)	494.2668

TABLE 2-341 Group Contributions for the Nannoolal* Method for Normal Boiling Point (Concluded)

Table-specific nomenclature: (e) = connected to N, O, F, Cl; (ne) = not connected to N, O, F, Cl; (r) = in a ring; (c) = in a chain; (a) = aromatic, not necessarily carbon; (Ca) = aromatic carbon; b = any nonhydrogen atom

ID	Group	Description	Value
75	ONC—	ONC— (oxime)	1041.0851
76	—C=O—O—C=O—	anhydride connected to two C	1251.2675
77	COCl—	COCl— connected to C (acid chloride)	778.9151
78	>Si<(F,Cl)	>Si< connected to at least one F or Cl	540.0895
79	O=C(—O—) ₂	noncyclic carbonate	879.7062
80	OCN—	OCN— connected to C or Si (cyanate)	660.4645
81	SCN—(C)	SCN— (thiocyanate) connected to C	1018.4865
82	(C)—SO ₂ —(C)	noncyclic sulfone connected to two C (sulfones)	1559.9840
83	(C) ₂ >Sn<(C) ₂	>Sn< connected to four carbons	510.4223
84	AsCl ₂ —	AsCl ₂ connected to C	1149.9670
85	GeCl ₃ —	GeCl ₃ — connected to carbons	1209.2972
86	(C) ₂ >Ge<(C) ₂	>Ge< connected to four carbons	347.7717
87	>C=C=C<	cumulated double bond	664.0903
88	>C=C—C=C<(r)	conjugated double bond in a ring	957.6388
89	>C=C—C=C<(c)	conjugated double bond in a chain	928.9954
90	CHO—(Ca)	CHO— connected to aromatic C (aldehydes)	560.1024
91	(C,Si)=N—	double-bonded amine connected to at least one C or Si	229.2288
92	(O=C<(C) ₂) _a	—CO— connected to two C with at least one aromatic C (ketones)	606.1797
94	—O—O—	peroxide	273.1755
95	—C≡C—C≡C—	conjugated triple bond	1218.1878
96	(—C=O—O—C=O—) _r	cyclic anhydride connected to two C	2082.3288
97	(C,Si) _a —NH—(Ca,Si) _a	—NH— connected to two C or Si with at least one aromatic (secondary amines)	201.3224
99	—OCON<	—CO connected to O and N (carbamate)	886.7613
100	>N—(C=O)—N<	—CO connected to two N (urea)	1045.0343
101	(C,Si) ₂ >N<(C,Si) ₂	Quaternary amine connected to four C or Si	-109.6269
102	F—(C,Si)(Cl)(b) ₂	F— connected to C or Si already substituted with at least one Cl and two other atoms	111.0590
103	—OCOO—	—CO connected to two O (carbonates)	1573.3769
104	>SO ₄	S(=O) ₂ connected to two O (sulfates)	1483.1289
105	—SO ₂ N<	—S(=O) ₂ connected to N	1506.8136
106	...=CNC=NC=...	imidazole	484.6371
107	>S=O	sulfoxide	1379.4485
108	(S)—C≡N	—C≡N (cyanide) connected to S	659.7336
109	>N(C=O)	—CO connected to N	492.0707
111	(N)—C≡N	—C≡N (cyanide) connected to N	971.0365
113	>P<	phosphorus connected to at least 1 C or S (phosphine)	428.8911
115	—ON=(C,Si)	—ON= connected to C or Si (isozole)	612.9506
116	>Se<	>Se< connected to at least one C or Si	562.1791
117	>Al<	>Al< connected to at least one C or Si	761.6006
Corrections			
118	C=C—C=O	—C=O connected to sp ₂ carbon	40.4205
119	(C=O)—C([F,Cl] ₂ ,3)	carbonyl connected to C with two or more halogens	-82.2328
120	(C=O)—C([F,Cl] ₂ ,3) ₂	carbonyl connected to two C, each with at least two halogens	-247.8893
121	C—[F,Cl] ₃	carbon with three halogens	-20.3996
122	(C) ₂ —C—[F,Cl] ₂	secondary carbon with two halogens	15.4720
123	No hydrogen	component has no hydrogen	-172.4201
124	One-hydrogen	component has one hydrogen	-99.8035
125	(3,4) ring	a three- or four-member nonaromatic ring	-62.3740
126	5-ring	a five-member nonaromatic ring	-40.0058
127	Ortho pair(s)	Ortho- position counted only once and only if there are no meta or para pairs	-27.2705
128	Meta pair(s)	Meta- position counted only once and only if there are no para or ortho pairs	-3.5075
129	Para pair(s)	Para- position counted only once and only if there are no meta or ortho pairs	16.1061
130	((C=)/(C/C)—CC3)	carbon with four carbon neighbors and one double-bonded carbon neighbor	25.8348
131	C2C—CC2	carbon with four carbon neighbors, two on each side	35.8330
132	C3C—CC2	carbon with five carbon neighbors	51.9098
133	C3C—CC3	carbon with six carbon neighbors	111.8372

*Nannoolal, Y., et al., *Fluid Phase Equilib.* **226** (2004): 45.

$$f^{(2)} = \frac{(-0.64771)(0.36) + (2.41539)(0.36)^{1.5} - (4.26979)(0.36)^{2.5} + (3.25259)(0.36)^5}{0.64}$$

$$= -0.037$$

Calculation using Eq. (2-25) at the normal boiling point:

$$\ln \frac{1.01325}{45.1911} = -3.798 = f^{(0)} + \omega f^{(1)} + \omega f^{(2)} = 3.0034 + 3.1788\omega + 0.037\omega^2$$

$$\omega = 0.249$$

The value obtained from the Ambrose-Walton-Lee-Kesler method compares favorably with the value of 0.2499 recommended in the DIPPR® 801 database (obtained from the vapor pressure correlation).

Radius of Gyration The radius of gyration R_g is a measure of the mass distribution about the center of mass of a molecule. Radius R_g increases with molecular size. It is useful in CS applications to separate molecular size and shape effects from polar effects. It is defined in terms of the principal moments of inertia of a molecule (A, B, and C) as

$$R_g = \sqrt{\frac{(AB)^{1/2}N_A}{M}} \quad (2-18)$$

for planar molecules and as

$$R_g = \sqrt{\frac{2\pi(ABC)^{1/3}N_A}{M}} \quad (2-19)$$

TABLE 2-342 Intermolecular Interaction Corrections for the Nannoolal et al.* Method for Normal Boiling Point

	—OH	—OH(a)	—COOH	—O—	>(OC ₂)<	—COOC—	—CO—	—CHO
—OH	291.7985	0	146.7286	135.3991	226.4980	211.6814	46.3754	0
—OH(a)		288.6155	-1477.9671	130.3742	0	-1184.9784	0	43.9722
—COOH			117.2044	612.8821	0	-183.2986	-55.9871	0
—O—				91.4997	178.7845	322.5671	15.6980	17.0400
>(OC ₂)<					1006.388	0	22.5208	163.5475
—COOC—						431.0990	22.5208	0
—CO—							-303.9653	-391.3690
—CHO								582.1763

	—O(a)—	—S(na)—	—S(a)—	—SH	—NH ₂	>NH	—OCN	—CN
—OH	435.0923	-74.0193	0	38.6974	314.6126	286.9698	0	306.3979
—OH(a)	0	0	0	0	797.4327	0	0	0
—COOH	0	0	0	0	0	0	0	0
—O—	329.0050	394.5505	0	0	124.3549	101.8475	0	293.5974
>(OC ₂)<	0	0	0	0	0	0	0	0
—COOC—	707.9404	0	0	0	182.6291	317.0200	0	517.0677
—CO—	176.5481	0	381.0107	0	0	-215.3532	0	-574.2230
—CHO	674.6858	0	397.575	0	0	0	0	0
—O(a)—	0	0	0	0	395.4093	0	0	0
—S(na)—		-11.9406	0	0	-562.306	0	0	0
—S(a)—			0	0	0	0	0	-101.232
—SH				217.6360	0	0	0	0
—NH ₂					174.0258	510.3473	0	0
>NH						239.8076	0	0
—OCN							-356.5017	0
—CN								0

	Nitrate	==N(a)—(r5)	==N(a)—(r6)
—OH	0	0	1334.6747
—OH(a)	-1048.124	0	-614.3624
—COOH	0	0	0
—O—	963.6518	0	0
>(OC ₂)<	0	0	0
—COOC—	-205.6165	0	0
—CO—	-3628.903	0	124.1943
—CHO	140.9644	0	0
—O(a)—	0	-888.612	0
—S(na)—	0	0	0
—S(a)—	0	-348.740	0
—SH	0	0	0
—NH ₂	663.8009	0	27.2735
>NH	0	0	758.9855
—OCN	-263.0807	0	0
—CN	0	0	-370.9729
Nitrate	65.1432	0	0
==N(a)—(r5)		0	0
==N(a)—(r6)			-271.9449

*Nannoolal, Y., et al., *Fluid Phase Equilib.*, **226** (2004): 45.

for nonplanar molecules. Radii of gyration can be calculated from these defining equations and principal moments of inertia obtained from spectral data or from computational chemistry software.

Recommended Method Principal moments of inertia.

Classification: Computational chemistry.

Expected uncertainty: Less than 5 percent.

Applicability: All molecules.

Input data: *M* and molecular structure.

Description: Computational chemistry software is used to optimize the geometry of the molecule and obtain the principal moments of inertia to be used in Eqs. (2-18) and (2-19).

Example Calculate the radius of gyration for hydrazine.

Input information: From the DIPPR® 801 database, *M* = 32.0452 kg/kmol. The structure of hydrazine is



Calculation of the principal moments of inertia: Optimizing hydrazine with HF/6-31G model chemistry gives the following principal moments of inertia:

$$A = 12.24050 \text{ amu} \cdot \text{Bohr}^2 \quad B = 72.41081 \text{ amu} \cdot \text{Bohr}^2$$

$$C = 79.16893 \text{ amu} \cdot \text{Bohr}^2$$

Conversion from atomic units to SI gives

$$A = (12.24050 \text{ amu} \cdot \text{Bohr}^2) \left(\frac{5.29177 \times 10^{-11} \text{ m}}{\text{Bohr}} \right)^2 \left(\frac{1.66054 \times 10^{-27} \text{ kg}}{\text{amu}} \right) = 5.692 \times 10^{-47} \text{ kg} \cdot \text{m}^2$$

$$B = (72.41081 \text{ amu} \cdot \text{Bohr}^2) \left(\frac{4.65010^{-48} \text{ kg} \cdot \text{m}^2}{\text{amu} \cdot \text{Bohr}^2} \right) = 3.367 \times 10^{-46} \text{ kg} \cdot \text{m}^2$$

$$C = (79.16893 \text{ amu} \cdot \text{Bohr}^2) \left(\frac{4.65010^{-48} \text{ kg} \cdot \text{m}^2}{\text{amu} \cdot \text{Bohr}^2} \right) = 3.681 \times 10^{-46} \text{ kg} \cdot \text{m}^2$$

Calculation using Eq. (2-19):

$$(ABC)^{1/3} = [(5.692 \times 10^{-47})(3.367 \times 10^{-46})(3.681 \times 10^{-46})]^{1/3} \text{ kg} \cdot \text{m}^2 = 1.918 \times 10^{-46} \text{ kg} \cdot \text{m}^2$$

$$R_g = \sqrt{\frac{2\pi(1.918 \times 10^{-46} \text{ kg}\cdot\text{m}^2)(6.022 \cdot 10^{26} \text{ kmol}^{-1})}{32.0452 \text{ kg/kmol}}} = 1.505 \times 10^{-10} \text{ m}$$

This is 3.8 percent below the DIPPR® 801 database value of 1.564×10^{-10} m which was obtained from spectral principal moments of inertia.

Dipole Moment The dipole moment of a molecule is the first moment of the electric charge density expansion. All normal paraffins have a value of zero. Charge separation within the molecule due to electronegativity differences between bonded atoms increases the dipole moment. Computational chemistry software uses the electron density distribution of the optimized molecule to calculate dipole moments.

Recommended Method Electron density distribution.

Classification: Computational chemistry.

Expected uncertainty: Uncertainty varies depending upon the model chemistry chosen, but it can be as large as 60 percent.

Applicability: All molecules.

Input data: Molecular structure.

Example Calculate the dipole moment for methanol.

Draw structure and optimize molecule using computational chemistry software: The dipole moment obtained from a geometry optimized with the HF/6-31G model chemistry for methanol is 2.288 D. This value is 35 percent larger than the experimental gas-phase value of 1.700 D in the DIPPR® 801 database.

VAPOR PRESSURE

Liquids Vapor pressure is the most important of the basic thermodynamic properties of fluids. It is the pressure of equilibrium, coexisting liquid and vapor phases at a specified temperature. The vapor pressure curve is a monotonic function of temperature from its minimum value (the triple point pressure) at the triple point temperature T_t to its maximum value (the critical pressure) at T_c .

Liquid vapor pressure data over a limited temperature range can be correlated with the Antoine [Antoine, C. C.R., **107** (1888): 681, 836] equation

$$\ln \frac{P^*}{\text{Pa}} = A - \frac{B}{T/K + C} \quad (2-20)$$

Data from the triple point to the critical point can be correlated with either a modified form of the Wagner equation [Wagner, W., "A New Correlation Method for Thermodynamic Data Applied to the Vapor-Pressure Curve of Argon, Nitrogen, and Water," J.T.R. Watson (trans. and ed.), IUPAC Thermodynamic Tables Project Centre, London, 1977; Ambrose, D., *J. Chem. Thermodyn.*, **18** (1986): 45; Ambrose, D., and N. B. Giassee, *J. Chem. Thermodyn.*, **19** (1987): 903, 911]

$$\ln P_r^* = \frac{a\tau + b\tau^{1.5} + c\tau^{2.5} + d\tau^5}{1 - \tau} \quad \text{where } \tau \equiv 1 - T_r \quad (2-21)$$

or the Riedel [Riedel, L., *Chem. Ing. Tech.*, **26** (1954): 679] equation

$$\ln \frac{P^*}{\text{Pa}} = A + \frac{B}{T/K} + C \ln \frac{T}{K} + D \left(\frac{T}{K} \right)^E \quad (2-22)$$

Generally, E in Eq. (2-22) is assigned a value of 6, but values of 2 or 1 have also been used, particularly when correlating low-temperature data. While the Wagner equation can be used to correlate most fluids over the whole liquid range, a fifth term is often required for alcohols [Poling, B. E., *Fluid Phase Equil.*, **116** (1996): 102]:

$$\ln P_r^* = \frac{a\tau + b\tau^{1.5} + c\tau^{2.5} + d\tau^5 + e\tau^6}{1 - \tau} \quad (\text{for alcohols}) \quad (2-23)$$

Correlation of experimental data within a few tenths of a percent over the entire fluid range can usually be obtained with either the Wagner or Riedel equations.

Two prediction methods are recommended for liquid vapor pressure. The first method is based on the Riedel equation; the second is a CS method. Both methods require T_c and P_c as input, but these can

be estimated by the methods shown earlier if experimental values are unavailable.

Recommended Method 1 Riedel method.

Reference: Riedel, L., *Chem. Ing. Tech.*, **26** (1954): 679

Classification: Empirical extension of theory and corresponding states.

Expected uncertainty: Varies strongly depending upon relative T , but 1 percent or less above T_b is typical with uncertainties of 5 to 30 percent near the triple point.

Applicability: Most organic compounds.

Input data: T_b , T_c , P_c .

Description: Equation (2-22) in reduced form

$$\ln P_r = A + \frac{B}{T_r} + C \ln T_r + DT_r^6 \quad (2-24)$$

Constants for this equation are determined from the following set of relationships:

$$\psi = -35 + \frac{36}{T_{br}} + 42 \ln T_{br} - T_{br}^6 \quad \alpha_c = \frac{3.758K\psi + \ln(P_c/1.01325 \text{ bar})}{K\psi - \ln T_{br}}$$

$$h = T_{br} \frac{\ln(P_c/1.01325 \text{ bar})}{1 - T_{br}} \quad D = K(\alpha_c - 3.758)$$

$$C = \alpha_c - 42D \quad B = -36D \quad A = 35D$$

Values of the constant K [Vetere, A., *Ind. Eng. Chem. Res.*, **30** (1991): 2487]:

Class	Value
Acids	$K = -0.120 + 0.025h$
Alcohols	$K = 0.373 - 0.030h$
All other organic compounds	$K = 0.0838$

Example Estimate the vapor pressure of chlorobenzene at 50 K intervals from 300 to 600 K.

Input information: From the DIPPR® 801 database, $T_b = 404.87$ K, $T_c = 632.35$ K, and $P_c = 45.1911$ bar.

Auxiliary Quantities:

$$K = 0.0838 \quad T_{br} = 404.87/632.35 = 0.640$$

$$\psi = -35 + \frac{36}{0.640} + 42 \ln 0.640 - (0.640)^6 = 2.431$$

$$\alpha_c = \frac{(3.758)(0.0838)(2.431) + \ln(45.191/1.01325)}{(0.0838)(2.431) - \ln(0.640)} = 7.0248$$

$$D = (0.0838)(7.0248 - 3.758) = 0.2738 \quad C = 7.0248 - (42)(0.2738) = -4.4729$$

$$B = -(36)(0.2738) = -9.8552 \quad A = -(35)(0.2738) = 9.5814$$

Calculation using Eq. (2-24) at each T (detailed calculation shown for $T = 500$ K):

$$T_r = 500/632.35 = 0.7907$$

$$\ln P_r = 9.5814 - \frac{9.8552}{0.7907} - 4.4729 \ln 0.7907 + (0.2738)(0.7907)^6 = -1.7651$$

$$P_r = \exp(-1.7651) = 0.1712 \quad P = P_r P_c = (0.1712)(45.1911 \text{ bar}) = 7.74 \text{ bar}$$

T/K	T_r	$\ln P_r$	P/bar	$P_{\text{DIPPR}}/\text{bar}$	% Error
300	0.4744	-7.8532	0.0176	0.0175	0.3
350	0.5535	-5.5704	0.172	0.172	0.1
400	0.6326	-3.9323	0.886	0.880	0.6
450	0.7116	-2.7101	3.01	2.98	0.9
500	0.7907	-1.7651	7.74	7.67	0.9
550	0.8698	-1.0067	16.51	16.39	0.8
600	0.9488	-0.3705	31.20	31.11	0.3

Recommended Method 2 Ambrose-Walton method.

References: Ambrose, D., and J. Walton, *Pure & Appl. Chem.*, **61** (1989): 1395; Lee, B. I., and M. G. Kesler, *AIChE J.*, **21** (1975): 510.

Classification: Corresponding states.

Expected uncertainty: Varies strongly with relative T , but less than 1 percent is typical above T_b if the acentric factor is known.

Applicability: Most organic compounds.

Input data: T_b , T_c , P_c , and ω .

Description: The acentric factor is used to linearly interpolate within the simple-fluid and deviation terms for the $\ln P^*$ values of the reference fluids, which themselves have been correlated with the Wagner vapor pressure equation.

$$\ln P_r^* = f^{(0)} + \omega f^{(1)} + \omega^2 f^{(2)}$$

$$\begin{aligned} f^{(0)} &= \frac{-5.97616\tau + 1.29874\tau^{1.5} - 0.60394\tau^{2.5} - 1.06841\tau^5}{1 - \tau} \\ f^{(1)} &= \frac{-5.03365\tau + 1.11505\tau^{1.5} - 5.41217\tau^{2.5} - 7.46628\tau^5}{1 - \tau} \\ f^{(2)} &= \frac{-0.64771\tau + 2.41539\tau^{1.5} - 4.26979\tau^{2.5} + 3.25259\tau^5}{1 - \tau} \end{aligned} \quad (2-25)$$

where $\tau = 1 - T_r$.

Example Repeat the calculation of the liquid vapor pressure of chlorobenzene at 50 K intervals from 300 to 600 K.

Input information: From the DIPPR® 801 database, $T_c = 632.35$ K, $P_c = 45.1911$ bar, and $\omega = 0.249857$.

Auxiliary quantities:

$$T_r = 500/632.35 = 0.7907 \quad \tau = 1 - 0.7907 = 0.2093$$

Simple-fluid and deviation vapor pressure terms at each T (shown for $T = 500$ K):

$$\begin{aligned} f^{(0)} &= \frac{(-5.97616)(0.2093) + (1.29874)(0.2093)^{1.5} - (0.60394)(0.2093)^{2.5} - (1.06841)(0.2093)^5}{0.7907} = -1.4405 \\ f^{(1)} &= \frac{(-5.03365)(0.2093) + (1.11505)(0.2093)^{1.5} - (5.41217)(0.2093)^{2.5} - (7.46628)(0.2093)^5}{0.7907} = -1.3383 \\ f^{(2)} &= \frac{(-0.64771)(0.2093) + (2.41539)(0.2093)^{1.5} - (4.26979)(0.2093)^{2.5} + (3.25259)(0.2093)^5}{0.7907} = 0.0145 \end{aligned}$$

Calculation using Eq. (2-25):

$$\ln P_r^* = -1.4405 + (0.249857)(-1.3383) + (0.249857)^2(0.0145) = -1.774$$

$$P^* = (45.1911 \text{ bar})[\exp(-1.774)] = 7.667 \text{ bar}$$

T	τ	$f^{(0)}$	$f^{(1)}$	$f^{(2)}$	$\ln P_r^*$	P^*/bar	$P^*_{\text{DIPPR}}/\text{bar}$	% Error
300	0.5256	-5.9228	-7.5966	-0.3050	-7.840	0.0178	0.0175	1.4
350	0.4465	-4.3006	-5.0017	-0.1439	-5.559	0.174	0.172	1.5
400	0.3674	-3.1036	-3.3106	-0.0437	-3.933	0.885	0.880	0.5
450	0.2884	-2.1800	-2.1576	0.0043	-2.719	2.98	2.98	0.0
500	0.2093	-1.4405	-1.3383	0.0145	-1.774	7.67	7.67	0.0
550	0.1302	-0.8289	-0.7318	0.0036	-1.012	16.43	16.39	0.3
600	0.0512	-0.3068	-0.2612	-0.0081	-0.373	31.14	31.11	0.1

Solids Below the triple point, the pressure at which the solid and vapor phases of a pure component are in equilibrium at any given temperature is the vapor pressure of the solid. It is a monotonic function of temperature with a maximum at the triple point. Solid vapor pressures can be correlated with the same equations used for liquids. Estimation of solid vapor pressure can be made from the integrated form of the Clausius-Clapeyron equation

$$\ln \frac{P^*}{P_t^*} = \frac{\Delta H_{\text{sub}}}{RT_t} \left(1 - \frac{T_t}{T} \right) \quad (2-26)$$

where T_t = triple point temperature

P_t^* = triple point pressure

ΔH_{sub} = enthalpy of sublimation

The liquid and solid vapor pressures are identical at the triple point. A good vapor pressure correlation that is valid at the triple point may be used to obtain the triple point pressure. Estimating solid vapor pressures by using Eq. (2-26) generally requires an estimation of ΔH_{sub} , and so the illustrative example is combined with the example on enthalpy of sublimation in the section on latent enthalpy.

THERMAL PROPERTIES

Enthalpy of Formation The standard enthalpy (heat) of formation is the enthalpy change upon formation of 1 mol of the compound in its standard state from its constituent elements in their standard states. Two different standard enthalpies of formation are commonly defined based on the chosen standard state. The standard state enthalpy of formation ΔH_f° uses the naturally occurring phase at 298.15 K and 1 bar as the standard state; the ideal gas standard enthalpy (heat) of formation ΔH_f° uses the compound in the ideal gas state at 298.15 K and 1 bar as the standard state. In both cases, the standard state for the elements is their naturally occurring state of aggregation at 298.15 K and 1 atm. Sources for data include DIPPR®, TRC, SWS, JANAF, and Daubert, T. E., and R. P. Danner, *Technical Data Book—Petroleum Refining*, 5th ed., American Petroleum Institute, Washington, extant 1994. The Domalski-Hearing method is the most accurate general method for estimating either ΔH_f° or ΔH_f° if the appropriate GC values are available, but a CC method is also as accurate for estimating ΔH_f° if an isodesmic reaction can be formulated and used. The Domalski-Hearing method also applies to entropies, and the entropy predictive equations are listed in this section for convenience because they are equivalent in form to the enthalpy equations. However, discussion and illustration of the estimation methods for entropy are delayed to the next subsection.

Recommended Method Domalski-Hearing method.

Reference: Domalski, E. S., and E. D. Hearing, *J. Phys. Chem. Ref. Data*, **22** (1993): 805.

Classification: Group contributions.

Expected uncertainty: 3 percent.

Applicability: Organic compounds for which group contributions have been regressed.

Input data: Molecular structure.

Description: GC values from Table 2-343 are directly additive for both enthalpy of formation and absolute third-law entropies:

$$\frac{\Delta H_f^\circ}{\text{kJ/mol}} = \sum_{i=1}^N n_i (\Delta H_f^\circ)_i \quad \frac{S^\circ}{\text{J} \cdot \text{mol}^{-1} \text{K}^{-1}} = \sum_{i=1}^N n_i (S^\circ)_i \quad (2-27)$$

where $(\Delta H_f^\circ)_i$ = enthalpy of formation GC value from Table 2-343 and $(S^\circ)_i$ = entropy GC value from Table 2-343.

Group values in Table 2-343 are defined by the central, nonhydrogen group and the atoms bonded to that group. Thus, C—(2H)(2C) represents a C atom to which 2 H and 2 C atoms are bonded. For example, propane (CH₃—CH₂—CH₃) is composed of three groups: two C—(3H)(C) and one C—(2H)(2C).

Example Estimate the standard and ideal gas enthalpies of formation of *o*-toluidine.

Input information: The melting point (256.8 K) and boiling point (473.49 K) given for *o*-toluidine in the DIPPR® 801 database bracket 298.15 K, and so the standard state phase at 298.15 K and 1 bar must be liquid.

Structure:

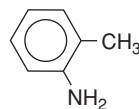


TABLE 2-343 Domalski-Hearing* Group Contribution Values for Standard State Thermal Properties

This table is a partial listing of GC values available from the original Domalski-Hearing tables. Table-specific nomenclature: Cd = carbon with double bond; Ct = carbon with triple bond; Cb = carbon in benzene ring; Ca = allenic carbon; corr = correction term; Cbf = fused benzene ring; N_A = azo nitrogen; N_I = imino nitrogen.

Group	ΔH_f°	S°	ΔH_f° liq.	S° liq.	ΔH_f° solid	S° solid
CH Groups						
C—(3H)(C)	-42.26	127.32	-47.61	83.30	-46.74	56.69
C—(2H)(2C)	-20.63	39.16	-25.73	32.38	-29.41	23.01
C—(H)(3C)	-1.17	-53.60	-4.77	-23.89	-5.98	-16.89
—CH ₃ corr (tertiary)	-2.26	0.00	-2.18	0.00	-2.34	0.00
C—(4C)	19.20	-149.49	17.99	-98.65	12.47	-33.19
—CH ₃ corr (quaternary)	-4.56	0.00	-4.39	0.00	-4.35	0.00
—CH ₃ corr (tert/quat)	-1.80	0.00	-1.77	0.00	-2.70	0.00
—CH ₃ corr (quat/quat)	-0.64	0.00	-0.64	0.00	-2.24	0.00
Cd—(2H)	-26.32	115.52	21.75	86.19	22.43	
Cd—(H)(C)	36.32	33.05	31.05	28.58	25.48	
Cd—(2C)	44.14	-50.84	39.16	-29.83	32.97	
Cd—(H)(Cd)	28.28	27.74	22.18	13.30	17.53	21.75
Cd—(C)(Cd)	36.78	-61.33	30.42	-41.92	27.91	
Cd—(Cd)(Cb)					56.07	
Cd—(H)(Cb)	28.28	27.74	22.18	13.30	17.53	21.75
Cd—(C)(Cb)	37.95	-51.97	38.58			
Cd—(H)(Ct)	28.28	27.74	22.18	13.30	17.53	21.75
C—(4H), Methane	-74.48	206.92				
Cd—(2Cb)	32.88		30.83		49.91	
C—(2H)(C)(Cd)	-20.88	38.20	-25.73	31.67	-24.35	
C—(H)(2C)(Cd)	-1.63	-50.38	-5.02	-28.07	-6.49	
—CH ₃ corr (tertiary)	-2.26	0.00	-2.18	0.00	-2.34	0.00
C—(3C)(Cd)	22.13	-150.23	20.79	-108.20	12.51	
—CH ₃ corr (quaternary)	-4.56	0.00	-4.39	0.00	-4.35	0.00
C—(H)(C)(2Cd)	-1.17	-53.60	-4.77	-23.89	-5.98	-16.89
C—(2H)(2Cd)	-18.92	42.08	-24.43	19.32	-21.60	
C—(2H)(Cd)(Cb)			-24.73			
C—(H)(C)(Cd)(Cb)			-6.90			
cis (unsat) corr	4.85	5.06	5.27	0.00	5.73	0.00
tert—Butyl cis corr	17.24	0.00	17.48	0.00	17.57	0.00
Ct—(H)	113.50	101.96	104.47	67.57	110.34	
Ct—(C)	115.10	26.32	107.15	14.25	101.66	
Ct—(Cd)	121.42	39.92	114.77			
Ct—(Cb)	120.76	17.77	119.00		103.28	
Ct—(Ct)	120.76	25.94	104.80		103.28	
C—(2H)(C)(Ct)	-19.70	42.80	-22.13	32.36	-29.41	
C—(H)(2C)(Ct)	-3.16	-45.69				
—CH ₃ corr (tertiary)	-2.26	0.00	-2.18	0.00	-2.34	0.00
C—(3C)(Ct)			22.83		26.38	
—CH ₃ corr (quaternary)	-4.56	0.00	-4.39	0.00	-4.35	0.00
C—(2H)(2Ct)	-41.14		-39.08			
C—(2C)(2Ct)			20.67			
Ca	142.67	26.28	134.68	14.39	131.08	
Cb—(H)(2Cb)	13.81	48.31	8.16	28.87	6.53	22.75
Cb—(C)(2Cb)	23.64	-35.61	19.16	-19.50	13.90	-5.50
Cb—(Cd)(2Cb)	24.17	-33.85	19.12	-9.04	20.27	-10.00
Cb—(Ct)(2Cb)	24.17	-33.85	19.12	-9.04	20.07	-10.00
Cb—(3Cb)	21.66	-36.57	17.21		17.03	-6.00
C—(2C)(2Cb)					52.81	
C—(2H)(C)(Cb)	-21.34	42.59	-24.81	47.40	-22.10	26.90
C—(H)(2C)(Cb)	-4.52	-48.00	-5.82	-13.90	-3.50	
C—(Cb)(3C)	18.28	-147.19	18.70	-96.10	21.57	
C—(2H)(2Cb)	-46.43		-26.50	51.97	-21.44	22.85
C—(H)(C)(2Cb)			-21.47	28.12	16.40	
C—(H)(3Cb)	-6.86				34.48	-12.62
C—(3Cb)(C)					116.25	
C—(4Cb)	27.04				64.89	
Cbf—(Cb)(2Cb)	20.10	0.00	15.83	-5.54	14.10	-6.00
Cbf—(Cb)(2Cb)	16.00		11.50		12.00	2.00
Cbf—(3Cb)	3.59		-0.90		1.94	7.00
Cb—(2Cb)(Cb)					-8.77	
Cb—(Cb)(2Cb)	22.46				47.93	
ortho corr hydrocarbons	1.26	-2.50	3.26	0.00	5.00	0.00
meta corr hydrocarbons	-0.63	0.00	0.00	0.00	2.00	0.00
Cyclopropane rsc (unsub)	115.15	134.86	111.58			
Cyclobutane rsc	110.89	126.04	106.64	51.48	114.43	
Cyclopentane rsc (unsub)	26.75	116.22	22.84	42.24	34.00	
Cyclohexane rsc (unsub)	0.68	78.18	-1.77	10.07	10.94	
Cycloheptane rsc	26.34	73.97	23.50	15.89		
Cyclooctane rsc	40.65	70.78	38.10	2.96		
Cyclononane rsc	52.91		50.40			
Cyclodecane rsc	51.99		50.61			

TABLE 2-343 Domalski-Hearing* Group Contribution Values for Standard State Thermal Properties (Continued)

This table is a partial listing of GC values available from the original Domalski-Hearing tables. Table-specific nomenclature: Cd = carbon with double bond; Ct = carbon with triple bond; Cb = carbon in benzene ring; Ca = allenic carbon; corr = correction term; Cbf = fused benzene ring; N_A = azo nitrogen; N_I = imino nitrogen.

Group	ΔH_f°	S°	ΔH_f° liq.	S° liq.	ΔH_f° solid	S° solid
CHO Groups						
CO—(2H), formaldehyde	-108.60	224.54				
CO—(C)(CO)	-121.29		-135.04		-140.75	
CO—(H)(CO)	-105.98					
CO—(CO)(Cb)	-112.30				-117.75	
CO—(O)(CO)	-123.75		-123.30		-120.81	
CO—(Cd)(O)	-136.73	62.59	-155.56		-134.10	32.90
CO—(C)(O)	-137.24	62.59	-149.37	32.72	-153.60	32.13
CO—(H)(O)	-124.39	147.03	-142.42	94.68		
CO—(2O)	-111.88		-122.00		-123.00	-42.92
CO—(H)(Cd)	-126.96		-153.05			
CO—(2Cb)	-110.00		-119.00		-116.00	
CO—(C)(Cb)	-148.82		-145.22		-143.70	23.72
CO—(H)(Cb)	-121.35		-138.12		-160.18	
CO—(O)(Cb)	-125.00		-140.00		-145.00	32.13
CO—(2C)	-132.67	64.31	-152.76	33.81	-157.95	
CO—(H)(C)	-124.39	147.03	-142.42	93.55		
CO—(C)(Cd)						
O—(2CO), aliphatic	-214.50	34.16	-230.50		-235.00	
O—(2CO), aromatic	-238.30		-220.90		-207.00	
O—(Cd)(CO)	-198.03		-201.42			
O—(C)(CO)	-188.87	36.03	-196.02	38.28	-210.60	12.09
O—(H)(CO)	-254.30	101.71	-285.64	38.28	-282.15	21.78
O—(Cb)(CO)	-167.00		-165.50		-170.00	45.32
O—(C)(O)	-20.75		-23.50		-30.20	
O—(H)(O)	-72.26		-101.75		-105.30	
O—(2Cd)	-139.29		-137.32			
O—(H)(Cd)						
O—(C)(Cd)	-129.33		-133.72			
O—(2Cb)	-77.66		-85.27	23.31	-96.20	3.14
O—(C)(Cb)	-92.55		-104.85		-122.87	
O—(H)(Cb)	-160.30	121.50	-191.75	43.89	-199.25	28.62
O—(2C)	-101.42	29.33	-110.83	26.78	-119.00	
O—(H)(C)	-159.33	121.50	-191.50	43.89	-199.66	28.62
Cd—(H)(CO)	32.30	35.19	26.61		7.82	27.53
Cd—(C)(CO)						
Cd—(O)(Cd)	36.78	-61.34	30.42	-41.92	27.91	
Cd—(O)(C)	44.14	-50.84	39.08	-29.83	32.97	
Cd—(O)(H)	36.32	33.05	31.05	28.58	25.48	
Ct—(CO)					144.52	
Cb—(CO)(2Cb)	15.50		10.50		8.15	0.08
Cb—(O)(2Cb)	-4.75	-43.72	-5.61	-10.59	1.00	1.59
C—(2H)(2CO)	-30.74		-23.06		-19.10	
C—(CO)(3C)	23.93		26.15	-85.98	24.02	
C—(H)(CO)(2C)	-0.25		-3.89	-24.52	-9.83	
C—(2H)(CO)(C)	-21.84	39.58	-24.14	39.87	-27.90	24.73
C—(3H)(CO)	-42.26	127.32	-47.61	83.30	-46.74	56.69
C—(2H)(CO)(Cd)	-16.95		-19.62			
C—(2H)(CO)(Ct)	-25.48		-26.61			
C—(2H)(CO)(Cb)	-16.20		-11.67			
C—(H)(CO)(C)(Cb)					14.81	
C—(H)(O)(CO)(C)	126.63		123.43	-46.71	-14.39	8.08
C—(4O)	-152.46		-133.34			
C—(H)(3O)	-113.97		-107.74			
C—(3O)(C)	-114.39		-99.54			
C—(2O)(2C)	-53.56		-41.30			
C—(H)(2O)(C)	-57.78		-51.42			
C—(2H)(2O)	-62.22		-62.89	23.85		
C—(2H)(O)(Cb)	-33.76		-29.17			
C—(2H)(O)(Cd)	-27.49	37.49	-28.62			
C—(H)(CO)(C)(Cb)					-14.39	
C—(H)(CO)(2Cb)					3.72	
C—(O)(3Cb)					60.46	
C—(O)(3C) (ethers, esters)	9.50	-141.92	0.79	-94.68	-0.50	
C—(H)(O)(2C) (ethers, esters)	-19.46	-52.80	-21.00	-25.31	-20.08	
C—(O)(3C) (alcohols, peroxides)	-13.50	-144.60	-11.13	-122.48	-12.25	-14.77
C—(H)(O)(2C) (alcohols, peroxides)	-26.10	-43.05	-27.60	-29.83	-29.08	6.95
C—(2H)(O)(C)	-32.90	43.43	-35.80	32.59	-33.00	24.73
C—(3H)(O)	-42.26	127.32	-47.61	83.30	-46.74	56.69
O—(CO)(O)	-88.00		-90.00		-80.50	
C—(2C)(O)(Cb)	15.30		25.80		29.30	
C—(H)(C)(2O)					-52.50	

TABLE 2-343 Domalski-Hearing* Group Contribution Values for Standard State Thermal Properties (Continued)

This table is a partial listing of GC values available from the original Domalski-Hearing tables. Table-specific nomenclature: Cd = carbon with double bond; Ct = carbon with triple bond; Cb = carbon in benzene ring; Ca = allenic carbon; corr = correction term; Cbf = fused benzene ring; N_A = azo nitrogen; N_I = imino nitrogen.

Group	ΔH_f°	S°	ΔH_f° liq.	S° liq.	ΔH_f° solid	S° solid
CHN and CHNO Groups						
C—(3H)(N)	-42.26	127.32	-47.61	83.30	-46.74	56.69
C—(2H)(C)(N)	-28.30	42.26	-30.80	32.38	-34.00	23.01
C—(H)(2C)(N)	-16.70	-63.55	-14.65	-20.00	-13.90	
—CH ₃ corr (tertiary)	-2.26	0.00	-2.18	0.00	-2.34	0.00
C—(3C)(N)	0.29	-152.59	5.10	-87.99	1.00	
—CH ₃ corr (quaternary)	-4.56	0.00	-4.39	0.00	-4.35	0.00
C—(2H)(2N)	-30.00				-26.00	
C—(2H)(Cb)(N)	-24.14		-26.09		-33.31	
N—(2H)(C) (first, amino acids)	19.25	124.40	0.33	71.71	-6.30	39.00
N—(2H)(C) (second, amino acids)	19.25	126.90	0.33	71.71	-46.00	48.75
N—(H)(2C)	67.55	33.96	51.50	32.09	47.80	
N—(3C)	116.50	-61.71	112.00	-38.62	101.00	
N—(2H)(N)	47.70	122.18	25.30	60.58	18.97	
N—(H)(C)(N)	89.16		75.00	22.05		
N—(2C)(N)	120.71		119.00	-26.94		
N—(2Cb)(N)					137.35	
N—(H)(Cb)(N)	87.50		73.40		66.90	
N—(2CO)(N)					73.62	
N—(H)(2Cd)	83.55		50.50		45.40	
N—(C)(2Cd)	120.64		97.38		88.92	
N—(2H)(Cb)	19.25	126.90	-11.00	71.71	-21.60	70.00
N—(H)(C)(Cb)	59.00		26.25		36.55	
N—(2C)(Cb)	126.40		109.40		96.50	
N—(C)(2Cb)	120.44		97.38		89.30	
N—(H)(2Cb)	83.55		50.50		45.40	
N—(3Cb)	123.15		121.80		107.50	
N _I —(C)	81.46		73.68			
N _I —(Cb)	69.00	47.01	54.50	36.40	57.00	
N _A —(C)	109.50		104.85		103.00	
N _A —(Cb)	109.50		104.85		103.00	
N _A —(oxide)(C)	40.80		22.65			
C—(2H)(C)(N _A)	-20.70		-25.70		-29.41	
C—(H)(2C)(N _A)	-2.66		-5.42			
C—(3C)(N _A)	11.50		15.50		10.50	
Cd—(H)(N)	-16.00		-15.50		-13.00	
Cd—(C)(N)	-5.74		-5.62		-3.95	
Cb—(N)(2Cb)	-1.30	-43.53	1.50	-24.43	9.75	-37.57
Cb—(NO)(2Cb)	21.50				23.00	
Cb—(NO ₂)(2Cb)	-1.45		-28.30	79.95	-32.50	110.46
Cb—(CNO)(2Cb)	-177.63				155.69	
Cb—(CN)(2Cb)	151.00	85.25	122.38	64.75	121.20	50.45
Cb—(N _A)(2Cb)	22.55		20.08		18.65	
Cb—(H)(2N _I)	6.30				0.25	
CO—(H)(N)	-124.39	147.03	-188.00	93.55		
CO—(C)(N)	-133.26	56.70	-185.00		-194.60	40.00
CO—(Cb)(N) (amides)					-177.75	
CO—(Cb)(N) (amino acids)					-177.75	
CO—(Cd)(N)	-171.80					
CO—(2N)	-111.00	96.00	-190.50		-203.10	69.00
N—(2H)(CO) (amides, ureas)	-63.00	88.25	-63.90		-65.25	18.00
N—(2H)(CO) (amino acids)	-63.00		-63.90		-59.75	33.03
N—(H)(C)(CO) (amides, ureas)	-16.28		-17.10		-9.80	
N—(H)(C)(CO) (amino acids)	-16.28		-17.10		5.50	
N—(2C)(CO)	45.00		62.00		55.00	
N—(H)(Cb)(CO)	-20.84				-3.50	
N—(H)(2CO)	-91.00				-30.80	
N—(C)(2CO)	-11.64		56.20		64.00	
N—(Cb)(2CO)	9.12					
N—(2Cb)(CO)					60.85	
N—(C)(Cb)(CO)					72.00	
C—(3H)(CN), acetonitrile	74.04	252.60	40.56	149.62		
C—(2H)(C)(CN)	94.52	167.25	66.07	106.02	69.85	96.15
C—(H)(2C)(CN)	113.50	67.86	81.50		69.00	
C—(3C)(CN)	137.96		116.20	-17.91	102.07	
C—(2C)(2CN)						74.57
C—(2H)(Cd)(CN)	95.31		66.40			
Cd—(H)(CN)	146.65	158.41	117.28	92.72		
Ct—(CN)	264.60		250.20			
C—(3H)(NO ₂), nitromethane	-74.86	284.14	-112.60	171.75		
C—(2H)(2NO ₂), dinitromethane	-58.90		-104.90			
C—(H)(3NO ₂), trinitromethane	-0.30				-48.00	
C—(4NO ₂), tetranitromethane	82.30		38.30			
C—(2H)(C)(NO ₂)	-60.50	203.60	-93.50		-99.00	

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TABLE 2-343 Domalski-Hearing* Group Contribution Values for Standard State Thermal Properties (Continued)

This table is a partial listing of GC values available from the original Domalski-Hearing tables. Table-specific nomenclature: Cd = carbon with double bond; Ct = carbon with triple bond; Cb = carbon in benzene ring; Ca = allenic carbon; corr = correction term; Cbf = fused benzene ring; N_A = azo nitrogen; N_I = imino nitrogen.

Group	ΔH_f°	S°	ΔH_f° liq.	S° liq.	ΔH_f° solid	S° solid
CHN and CHNO Groups						
C—(H)(2C)(NO ₂)	-53.00	115.32	-82.50		-89.00	
C—(3C)(NO ₂)	-36.65		-61.20		-76.55	
C—(2H)(Cb)(NO ₂)	-62.00		-82.76		-81.00	
C—(H)(C)(2NO ₂)	-36.80		-88.80		-91.50	
C—(2C)(2NO ₂)	-28.50		-77.20		-90.30	
C—(H)(C)(CO)(N)	-18.70				-11.65	-4.00
C—(2H)(CO)(N)	-3.10				-30.95	24.00
C—(H)(Cb)(CO)(N)						
O—(C)(NO)	-24.23	166.11	-46.50			
O—(C)(NO ₂)	-79.71	191.92	-108.96	127.50	-124.00	
N—(H)(C)(NO ₂)					16.50	
N—(H)(Cb)(NO ₂)						
N—(H)(CO)(NO ₂)					-14.00	
N—(C)(2NO ₂)	100.30		53.50			
N—(C)(Cb)(NO ₂)	183.00		167.00		150.50	
N—(2C)(NO)	90.00		59.00		55.00	
N—(2C)(NO ₂)	88.00		50.00		40.00	
C—(2H)(C)(N ₃)			321.70			
C—(H)(2C)(N ₃)	274.00		255.00			
C—(2H)(Cb)(N ₃)	347.00		327.40			
C—(3Cb)(N ₃)	328.60				346.50	
Cb—(N ₃)(2Cb)	320.00		303.50			
CHS and CHSO Groups						
C—(3H)(S)	-42.26	127.32	-47.61	83.30	-46.74	56.69
C—(2H)(C)(S)	-23.17	41.87	-26.77	41.09		
C—(H)(2C)(S)	-5.88	-47.36	-6.07	-16.61		
—CH ₃ corr (tertiary)	-2.26	0.00	-2.18	0.00	-2.34	0.00
C—(3C)(S)	13.52	-145.38	16.69	-86.86		
—CH ₃ corr (quaternary)	-4.56	0.00	-4.39	0.00	-4.35	0.00
—CH ₃ corr (tert/quat)	-1.80	0.00	-1.77	0.00	-2.70	0.00
—CH ₃ corr (quat/quat)	-0.64	0.00	-0.64	0.00	-2.24	0.00
C—(2H)(Cb)(S)	-18.53		-23.82			
C—(2H)(Cd)(S)	-25.93		-32.44			
C—(2H)(2S)	-25.10					
Cb—(S)(2Cb)	-4.75	43.72	-5.61	-10.59	1.00	1.59
Cd—(H)(S)	36.32	33.05	31.05	28.58	25.48	
Cd—(C)(S)	45.73	-51.92				
S—(C)(H)	18.64	137.67	0.06	85.95		
S—(Cb)(H)	48.10	57.34	28.51	89.04		
S—(2C)	46.99	55.19	29.82	29.80		
S—(H)(Cd)	25.52					
S—(C)(Cd)	54.39					
S—(2Cd)	102.60	68.59				
S—(Cb)(C)	76.21		58.20	35.44	42.00	
S—(C)(S)	27.62	50.50	14.36	30.84		
S—(Cb)(S)	57.45				40.60	
S—(2S)	12.59	56.07				
S—(2Cb)	102.60	68.59	93.02			
S—(H)(S)	7.95					
S—(H)(CO)	-5.90	130.54				
CO—(C)(S)	-132.67	64.31	-152.76	33.81		
C—(3H)(SO)	-42.26	127.32	-47.61	83.30	-46.74	56.69
C—(2H)(C)(SO)	-29.16		-36.88			
C—(H)(2C)(SO)						
—CH ₃ corr (tertiary)	-2.26	0.00	-2.18	0.00	-2.34	0.00
C—(3C)(SO)	4.56		0.97			
—CH ₃ corr (quaternary)	-4.56	0.00	-4.39	0.00	-4.35	0.00
C—(2H)(Cd)(SO)	-27.56		-32.63			
cis correction	4.11	5.06	5.27	0.00	5.73	0.00
Cb—(SO)(2Cb)	15.48		25.44		7.55	0.08
O—(SO)(H)	-158.60					
O—(C)(SO)	-92.60					
SO—(2C)	-66.78	75.73	-108.98	22.18		
SO—(2Cb)	-62.26					
SO—(2O)	-213.00					
SO—(C)(Cb)	-72.00					
C—(3H)(SO ₂)	-42.26	127.32	-47.61	83.30	-46.74	56.69
C—(2H)(C)(SO ₂)	-27.03		-33.76		-35.96	
C—(H)(2C)(SO ₂)	-14.00					
—CH ₃ corr (tertiary)	-2.26	0.00	-2.18	0.00	-2.34	0.00
C—(3C)(SO ₂)	1.52		2.00		3.78	
—CH ₃ corr (quaternary)	-4.56	0.00	-4.39	0.00	-4.35	0.00

TABLE 2-343 Domalski-Hearing* Group Contribution Values for Standard State Thermal Properties (Continued)

This table is a partial listing of GC values available from the original Domalski-Hearing tables. Table-specific nomenclature: Cd = carbon with double bond; Ct = carbon with triple bond; Cb = carbon in benzene ring; Ca = allenic carbon; corr = correction term; Cbf = fused benzene ring; N_A = azo nitrogen; N_I = imino nitrogen.

Group	ΔH_f°	S°	ΔH_f° liq.	S° liq.	ΔH_f° solid	S° solid
CHN and CHNO Groups						
—CH ₃ _{corr} (quat/quat)	−0.64		−0.64		−2.24	
C—(2H)(Cd)(SO ₂)	−29.49		−49.05			
C—(H)(C)(Cd)(SO ₂)	−71.99					
C—(2H)(Cb)(SO ₂)	−29.80					
C—(2H)(Ct)(SO ₂)	16.36					
Cb—(SO ₂)(2Cb)	15.48		25.44		7.55	0.08
Cd—(H)(SO ₂)	51.58					
Cd—(C)(SO ₂)	64.01					
Ct—(SO ₂)	177.10					
SO ₂ —(Cd)(Cb)	−291.55					
SO ₂ —(2Cd)	−306.70					
SO ₂ —(2C)	−288.58	87.37	−341.14		−356.62	32.10
SO ₂ —(C)(Cb)	−289.10					
SO ₂ —(2Cb)	−287.76				−305.40	
SO ₂ —(SO ₂)(Cb)	−325.18				−361.75	
SO ₂ —(2O)	−417.30					
SO ₂ —(C)(Cd)	−316.80					
SO ₂ —(Ct)(Cb)	−296.30					
O—(SO ₂)(H)	−158.60					
O—(C)(SO ₂)	−91.40					
CHX and CHXO Groups						
C—(3H)(F), methyl fluoride	−247.00	231.93				
C—(3H)(Cl), methyl chloride	−81.90	243.60				
C—(3H)(Br), methyl bromide	−37.66	254.94	−61.10			
C—(3H)(I), methyl iodide	14.30	263.14	−11.70			
C—(C)(3F)	−673.81	178.22	−709.07	135.56		
C—(2H)(C)(F)	−221.12	146.80				
C—(H)(2C)(F)	−204.46	55.76				
C—(3C)(F)	−202.92					
C—(H)(C)(2F)	−454.74	164.32	−487.23			
C—(2C)(2F)	−411.39	74.48	−400.37		−428.77	
C—(C)(Cl)(2F)	−462.70	169.45	−466.00	138.31		
C—(H)(C)(Cl)(F)	−271.14					
C—(C)(3Cl)	−81.98	202.14	−112.93	145.91		
C—(H)(C)(2Cl)	−79.10	183.28	−102.60	128.45		
C—(2H)(C)(Cl)	−69.45	159.24	−86.90	104.27	−85.65	
C—(2C)(2Cl)	−79.56	95.41	−101.80			
C—(H)(2C)(Cl)	−55.61	71.34	−71.17			
C—(3C)(Cl)	−43.70	−24.26	−56.78			
C—(C)(3Br)		233.05				
C—(H)(C)(2Br)						
C—(2H)(C)(Br)	−21.78	173.31	−42.65	113.00		
C—(2C)(2Br)						
C—(H)(2C)(Br)	−10.75	84.69	−27.31			
C—(3C)(Br)	7.26	−13.46	−7.40			
C—(C)(3I)						
C—(H)(C)(2I)	108.78	228.45				
C—(2H)(C)(I)	33.54	177.78	4.14		3.65	
C—(2C)(2I)						
C—(H)(2C)(I)	48.74	88.10	24.78			
C—(3C)(I)	68.46	−3.21	48.60			
C—(H)(C)(Br)(Cl)	−18.45	191.21				
N—(C)(2F)	−32.64					
C—(H)(C)(Cl)(O)	−90.37	66.53				
C—(2H)(I)(O)	15.90	170.29				
C—(C)(2Cl)(F)	−322.54		−343.87	141.71		
C—(C)(Br)(2F)	−394.55			149.70		
C—(C)(2Br)(F)						
C—(Br)(Cl)(F)						
Cd—(H)(F)	−165.12	137.24				
Cd—(H)(Cl)	4.37	147.85	−12.67			
Cd—(H)(Br)	50.94	159.91				
Cd—(H)(I)	102.36	169.45				
Cd—(C)(Cl)	−5.06	62.76	−2.23			
Cd—(2F)	−329.90	155.63				
Cd—(2Cl)	−11.51	175.41	−32.08	115.35		
Cd—(2Br)		199.16				
Cd—(2I)						
Cd—(Cl)(F)	−235.10	175.61				
Cd—(Br)(F)		177.82				
Cd—(Cl)(Br)		188.70				
Ct—(F)						

TABLE 2-343 Domalski-Hearing* Group Contribution Values for Standard State Thermal Properties (Concluded)

This table is a partial listing of GC values available from the original Domalski-Hearing tables. Table-specific nomenclature: Cd = carbon with double bond; Ct = carbon with triple bond; Cb = carbon in benzene ring; Ca = allenic carbon; corr = correction term; Cbf = fused benzene ring; N_A = azo nitrogen; N_I = imino nitrogen.

Group	ΔH_f°	S°	ΔH_f° liq.	S° liq.	ΔH_f° solid	S° solid
CHX and CHXO Groups						
Ct—(Cl)		140.00				
Ct—(Br)		151.30				
Ct—I	35.53					
Cb—(F)(2Cb)	-181.26	67.52	-191.20	54.19	-194.00	39.79
Cb—(Cl)(2Cb)	-17.03	77.08	-32.20	55.47	-32.00	43.37
Cb—(Br)(2Cb)	36.35	88.60	19.90	74.85	13.50	54.45
Cb—I(2Cb)	94.50	98.26	73.70	61.08	70.40	
cis _{corr} —(I)(I)	3.00	0.00	0.00	0.00	0.00	0.00
C—(2H)(CO)(Cl)	-44.26		-58.41		-74.75	
C—(H)(CO)(2Cl)	-40.40		-55.11			
CO—(C)(F)	-379.84		-419.59			
C—(Cb)(3F)	-691.79	179.08	-696.66			
C—(2H)(Cb)(Br)	-29.49		-44.06			
C—(2H)(Cb)(I)	7.31		-7.24			
C—(2H)(Cb)(Cl)	-73.79		-92.56			
CO—(C)(Cl)	-200.54	176.66	-225.29			
CO—(Cb)(Cl)			-216.67		-212.99	
CO—(C)(Br)	-148.54		-175.49			
CO—(C)(I)	-83.94		-117.09			
C—(H)(C)(CO)(Cl)	-39.88		-35.46			
C—(C)(CO)(2Cl)						
ortho _{corr} —(I)(I)	7.56	0.00	6.96	0.00	5.50	0.00
ortho _{corr} —(F)(F)	20.90	0.00	25.00	0.00	25.50	0.00
ortho _{corr} —(Cl)(Cl)	9.50	0.00	14.00	0.00	8.50	0.00
ortho _{corr} —(alkyl)(X)	2.51	0.00	6.30	0.00	0.00	0.00
cis _{corr} —(Cl)(Cl)	-4.00	0.00	0.00	0.00	0.00	0.00
cis _{corr} —(CH ₃)(Br)	-4.00	0.00	0.00	0.00	0.00	0.00
ortho _{corr} —(F)(Cl)	13.50	0.00	18.50	0.00	19.50	0.00
ortho _{corr} —(F)(Br)	37.25	0.00	40.60	0.00	42.50	0.00
ortho _{corr} —(F)(I)	85.40	0.00	83.55	0.00	85.20	0.00
meta _{corr} —(I)(I)	0.00	0.00	0.00	0.00	20.08	0.00
meta _{corr} —(COCl)(COCl)	0.00	0.00	0.00	0.00	16.06	0.00
ortho _{corr} —(COCl)(COCl)	0.00	0.00	0.00	0.00		0.00
ortho _{corr} —(F)(CF ₃)	111.00	0.00	112.00	0.00	0.00	0.00
meta _{corr} —(F)(CF ₃)	2.00	0.00	6.00	0.00	0.00	0.00
ortho _{corr} —(F)(CH ₃)	-3.30	0.00	-6.00	0.00	0.00	0.00
ortho _{corr} —(F)(F')	8.00	0.00	8.00	0.00	8.00	0.00
ortho _{corr} —(Cl)(Cl')	8.00	0.00	8.00	0.00	8.00	0.00
meta _{corr} —(F)(F)	0.00	0.00	6.00	0.00	8.50	0.00
meta _{corr} —(Cl)(Cl)	-5.00	0.00	10.00	0.00	4.00	0.00
ortho _{corr} —(Cl)(CHO)	-6.75	0.00	8.50	0.00	0.00	0.00
ortho _{corr} —(F)(COOH)	20.00	0.00	0.00	0.00	20.00	0.00
ortho _{corr} —(Cl)(COCl)	0.00	0.00	34.43	0.00	0.00	0.00
ortho _{corr} —(F)(OH)	25.50	0.00	23.00	0.00	20.00	0.00
ortho _{corr} —(Cl)(COOH)	0.00	0.00	0.00	0.00	20.00	0.00
ortho _{corr} —(Br)(COOH)	0.00	0.00	0.00	0.00	20.00	0.00
ortho _{corr} —(I)(COOH)	0.00	0.00	0.00	0.00	20.00	0.00
ortho _{corr} —(NH ₂)(NH ₂)	-10.00	0.00	0.00	0.00	0.00	0.00
meta _{corr} —(NH ₂)(NH ₂)	0.00	0.00	0.00	0.00	14.00	0.00
ortho _{corr} —(OH)(Cl)	7.50	0.00	0.00	0.00	11.00	0.00
cis _{corr} —(CH ₃)(I)	-4.00	0.00	0.00	0.00	0.00	0.00

*Domalski, E. S. and E. D. Hearing, *J. Phys. Chem. Ref. Data*, **22** (1993): 805

Group contributions:

Group	n_i	ΔH_f° gas	ΔH_f° liq.	S° gas	S° liq.
Cb—(H)(2Cb)	4	13.81	8.16	48.31	28.87
Cb—(C)(2Cb)	1	23.64	19.16	-35.61	-19.50
Cb—(N)(2Cb)	1	-1.30	1.50	-43.53	-24.43
C—(3H)(C)	1	-42.26	-47.61	127.32	83.30
N—(2H)(Cb)	1	19.25	-11.00	126.90	71.71
Total		54.57	-5.31	368.32	226.56

Calculation from Eq. (2-27):

$$\frac{\Delta H_f^\circ}{\text{kJ/mol}} = 54.57 \quad \frac{\Delta H_f^\circ}{\text{kJ/mol}} = -5.31$$

$$\frac{S^\circ}{\text{J/(mol}\cdot\text{K)}} = 368.32 \quad \frac{S^\circ}{\text{J/(mol}\cdot\text{K)}} = 226.56$$

The recommended DIPPR® 801 standard enthalpies of formation are $\Delta H_f^\circ = 53.20$ kJ/mol and $\Delta H_f^\circ = -4.72$ kJ/mol; the estimated values are higher than the recommended values by 2.6 and 12.5 percent, respectively. The recommended DIPPR® 801 standard entropies are $S^\circ = 355.8$ J/(mol·K) and $S^\circ = 231.2$ J/(mol·K). The estimated values differ from these by 3.5 and -2.0 percent, respectively.

Recommended Method Isodesmic reaction.

Reference: Foresman, J. B., and A. Frisch, *Exploring Chemistry with Electronic Structure Methods*, 2d ed., Gaussian Inc., Pittsburgh, Pa., 1996.

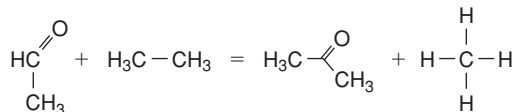
Classification: Computational chemistry.

Expected uncertainty: 5 to 10 percent depending upon the level of theory and basis set size used.

Applicability: Compounds for which an isodesmic reaction can be formulated.

Input data: Experimental ΔH_f° values for all other participants in the isodesmic reaction.

Description: While *ab initio* calculations of absolute enthalpies are not currently as accurate as GC methods, relative enthalpies of molecules calculated with the same level of theory and basis set can be very accurate, as in the case of isodesmic reactions. An isodesmic reaction is one in which the number and type of bonds are preserved during the reaction. For example, the reaction of acetaldehyde with ethane to form acetone and methane



is isodesmic with 12 single bonds and 1 double bond in both reactants and products. To use this method, one devises an isodesmic reaction involving the compound for which ΔH_f° is to be determined and other compounds for which experimental ΔH_f° values are available. *Ab initio* calculations are performed on all the participating compounds, all at the same level of theory and basis set size, to obtain the enthalpy for each at 298.15 K. The enthalpy of reaction is then calculated from

$$\Delta H_{\text{rxn}} = \sum v_i H_i \quad (2-28)$$

where v_i = stoichiometric coefficient of i (+ for products, - for reactants). The enthalpy of reaction is also related to ΔH_f° by

$$\Delta H_{\text{rxn}} = \sum v_i (\Delta H_f^\circ)_i \quad (2-29)$$

With experimental values available for all ΔH_f° except the desired compound, its value can be back-calculated from Eq. (2-29).

Example Estimate the standard ideal gas enthalpy of formation of acetaldehyde.

Input information: The isodesmic reaction shown above will be used. The recommended ΔH_f° values from DIPPR® 801 for the other three compounds are

Acetone	Methane	Ethane
-215.70 kJ/mol	-74.52 kJ/mol	-83.82 kJ/mol

Ab initio calculations of enthalpy: With structures optimized using HF/6-31G(d) model chemistry and energies calculated with B3LYP/6-311 + G(3df,2p), the following enthalpies are obtained (including the zero point energy):

Acetone	Methane	Ethane	Acetaldehyde
-5.071×10^5 kJ/mol	-1.063×10^5 kJ/mol	-2.095×10^5 kJ/mol	-4.039×10^5 kJ/mol

Calculation using Eq. (2-28):

$$\Delta H_{\text{rxn}} = (-1.063 - 5.071 + 2.095 + 4.039) \times 10^5 \text{ kJ/mol} = -41.67 \text{ kJ/mol}$$

Calculation using Eq. (2-29):

$$\Delta H_{f,\text{acetaldehyde}}^\circ = \Delta H_{f,\text{acetone}}^\circ + \Delta H_{f,\text{methane}}^\circ - \Delta H_{f,\text{ethane}}^\circ - \Delta H_{\text{rxn}}$$

$$\Delta H_{f,\text{acetaldehyde}}^\circ = (-215.70 - 74.52 + 83.82 + 41.67) \frac{\text{kJ}}{\text{mol}} = -164.73 \frac{\text{kJ}}{\text{mol}}$$

The estimated value is 1.0 percent above the DIPPR® 801 recommended value of -166.40 kJ/mol.

Entropy Absolute or third-law entropies (relative to a perfectly ordered crystal at 0 K) of a compound in its standard state S° or of an

ideal gas S° at 298.15 K and 1 bar can be found in various literature sources (DIPPR, JANAF, TRC, SWS, and Daubert, T. E., and R. P. Danner, *Technical Data Book-Petroleum Refining*, 5th ed., American Petroleum Institute, Washington, extant 1994). Very good estimates for S° or S° can be obtained by using the Domalski-Hearing method. Excellent S° values can also be obtained from statistical mechanics by using experimental vibrational frequencies or values of the frequencies generated from computational chemistry.

The standard state ΔS_f° and standard ideal gas ΔS_f° entropies of formation at 298.15 K and 1 bar are related to the standard entropies by

$$\Delta S_f^\circ = S_{\text{compound}}^\circ - \sum_{i=1}^{n_A} v_i S_{\text{element},i}^\circ \quad \Delta S_f^\circ = S_{\text{compound}}^\circ - \sum_{i=1}^{n_A} v_i S_{\text{element},i}^\circ \quad (2-30)$$

where $S_{\text{element},i}^\circ$ is the absolute entropy of element i in its standard state at 298.15 K and 1 bar.

Recommended Method Domalski-Hearing method.

Reference: Domalski, E. S., and E. D. Hearing, *J. Phys. Chem. Ref. Data*, **22** (1993): 805.

Classification: Group contributions.

Expected uncertainty: 3 percent.

Applicability: Organic compounds for which group contributions have been regressed.

Input data: Molecular structure.

Description: See description given under Enthalpy of Formation above.

Example Estimate the standard and ideal gas entropies of formation of *o*-toluidine.

Standard state entropies: Estimation of S° and S° using the Domalski-Hearing method was illustrated above in the Enthalpy of Formation section. The standard entropies of formation can be obtained from the values determined in that example.

Formula: $\text{C}_7\text{H}_9\text{N}$. The standard state entropies of the elements from the DIPPR® 801 database are as follows:

Compound:	N_2	H_2	C, graphite
v_i :	1/2	9/2	7
$S^\circ/[\text{J}(\text{kmol} \cdot \text{K})]$:	1.9151×10^5	1.3057×10^5	5740

Entropies of formation can be calculated from these values by using Eq. (2-30):

$$\Delta S_f^\circ = \left[0.22656 - \left(\frac{1}{2} \right) (1.9151) - \left(\frac{9}{2} \right) (1.3057) - (7) (0.0574) \right] \frac{10^5 \text{ J}}{\text{kmol} \cdot \text{K}}$$

$$= -7.008 \cdot 10^5 \frac{\text{J}}{\text{kmol} \cdot \text{K}}$$

$$\Delta S_f^\circ = \left[0.36832 - \left(\frac{1}{2} \right) (1.9151) - \left(\frac{9}{2} \right) (1.3057) - (7) (0.0574) \right] \frac{10^5 \text{ J}}{\text{kmol} \cdot \text{K}}$$

$$= -6.867 \cdot 10^5 \frac{\text{J}}{\text{kmol} \cdot \text{K}}$$

Recommended Method Statistical mechanics.

Classification: Theory and computational chemistry.

Expected uncertainty: 0.2 percent if vibrational frequencies (or their characteristic temperatures) are experimentally available; uncertainty depends upon model chemistry if frequencies are determined from computational chemistry, but generally within about 5 percent.

Applicability: Ideal gases.

Input data: M ; σ (external symmetry number); characteristic rotational temperature(s) (Θ_A for linear molecules; Θ_A , Θ_B , and Θ_C for nonlinear molecules); and $3n_A - 6 + \delta$ characteristic vibrational temperatures Θ_i .

Description: For harmonic frequencies, the rigorous temperature dependence of S° is given by

$$\frac{S^\circ}{R} = \frac{3}{2} \ln \left(6175 \frac{M}{\text{kg/kmol}} \right) + \frac{S_e}{R} + \frac{3n_A - 6 + \delta}{\sum_{j=1}^J} \left[\left(\frac{\Theta_j}{T} \right) (e^{\Theta_j/T} - 1)^{-1} - \ln (1 - e^{-\Theta_j/T}) \right] \quad (2-31)$$

where $\delta = \begin{cases} 0 & \text{nonlinear} \\ 1 & \text{linear} \end{cases}$

$$\frac{S_r}{R} = \begin{cases} \ln \left[\frac{1}{\sigma} \left(\frac{\pi T^3 e^3}{\Theta_A \Theta_B \Theta_C} \right)^{1/2} \right] & \text{nonlinear} \\ \ln \left[\left(\frac{T e}{\sigma \Theta_A} \right)^{1/2} \right] & \text{linear} \end{cases}$$

$$T = 298.15 \text{ K}$$

Example Calculate S° for ammonia.

Structure: NH_3 .

Input data: $M = 17 \text{ kg/kmol}$. McQuarrie [McQuarrie, D. A., *Statistical Mechanics*, Harper & Row, New York, 1976] gives the following $3m - 6 + \delta = 12 - 6 + 0 = 6$ characteristic vibrational temperatures (K): 1360, 2330, 2330, 4800, 4880, 4880. The characteristic rotational temperatures given by McQuarrie are $\Theta_A = 13.6 \text{ K}$, $\Theta_B = 13.6 \text{ K}$, and $\Theta_C = 8.92 \text{ K}$. For NH_3 , $\sigma = 3$.

Vibrational contribution: The table below shows a spreadsheet calculation of the vibrational terms inside the summation sign in Eq. (2-31).

Θ_j/K	Θ_j/T	S_{vib}
1207.91	4.051	0.08929
1850.16	6.205	0.01457
1850.16	6.205	0.01457
3688.19	12.370	0.00006
3821.36	12.817	0.00004
3821.36	12.817	0.00004
		Sum 0.1186

Rotational contribution:

$$\frac{S_r}{R} = \ln \left\{ \frac{1}{3} \cdot \left[\frac{(298.15 \text{ K})^3 \pi c^3}{(13.6 \text{ K})(13.6 \text{ K})(8.92 \text{ K})} \right]^{1/2} \right\} = 5.81593$$

Calculation using Eq. (2-31):

$$\frac{S_{298}^\circ}{R} = \frac{3}{2} \ln(6175.17) + 5.81593 + 0.1186 = 23.277 \quad \Delta H_{\text{res}} = \sum \nu_i (\Delta H_f^\circ)_i$$

$$S_{298}^\circ = 1.935 \times 10^5 \frac{\text{J}}{\text{kmol} \cdot \text{K}}$$

The calculated value differs from the DIPPR® 801 recommended value of $1.9266 \times 10^5 \text{ J/(kmol} \cdot \text{K)}$ by 0.5 percent.

Gibbs' Energy of Formation The standard Gibbs energy of formation is the Gibbs energy change upon formation of 1 mol of the compound in its standard state from its constituent elements in their standard states. The standard state Gibbs energy of formation ΔG_f° uses the naturally occurring phase at 298.15 K and 1 bar as the standard state, while the ideal gas Gibbs energy of formation ΔG_f° uses the compound in the ideal gas state at 298.15 K and 1 bar as the standard state. In both cases, the standard state for the elements is their naturally occurring state of aggregation at 298.15 K and 1 atm. Sources for data include DIPPR, TRC, JANAF, and Daubert and Danner, (Daubert, T. E., and R. P. Danner *Technical Data Book—Petroleum Refining*, 5th ed., American Petroleum Institute, Washington, extant 1994). The Gibbs energies of formation are related to the corresponding enthalpies and entropies of formation by

$$\Delta G_f^\circ = \Delta H_f^\circ - T \Delta S_f^\circ \quad \text{and} \quad \Delta G_f^\circ = \Delta H_f^\circ - T \Delta S_f^\circ \quad (2-32)$$

and predicted values of ΔG_f° and ΔG_f° are obtained from Eq. (2-32) by estimating the enthalpies and entropies of formation as shown above.

LATENT ENTHALPY

Enthalpy of Vaporization The enthalpy (heat) of vaporization ΔH_v is the difference between the molar enthalpies of the saturated

vapor and saturated liquid at a temperature between the triple point and critical point (at the corresponding vapor pressure). Variable ΔH_v is related to the vapor pressure P^* by the thermodynamically exact Clapeyron equation

$$\Delta H_v = -R \Delta Z_v \frac{d \ln P^*}{d(1/T)} = RT^2 \Delta Z_v \frac{d \ln P^*}{dT} \quad (2-33)$$

where $\Delta Z_v = Z_G - Z_L$
 $Z_G = Z$ of saturated vapor
 $Z_L = Z$ of saturated liquid

Experimental heats of vaporization can be effectively correlated with

$$\Delta H_v = A(1 - T_r)^{B+CT_r+DT_r^2+ET_r^3} \quad (2-34)$$

A simple method for obtaining ΔH_v at one temperature from a known value at a reference temperature, say, at the normal boiling point, is to truncate Eq. (2-34) after the B term, set $B = 0.38$, and take a ratio of the ΔH_v values at the two conditions to give the Watson [Thek, R. E., and L. I. Stiel, *AIChE J.*, **12** (1966): 599; 13(1967): 626] correlation

$$\Delta H_v = \Delta H_{v,\text{ref}} \left(\frac{1 - T_r}{1 - T_{r,\text{ref}}} \right)^{0.38} \quad (2-35)$$

If an accurate correlation for P^* and accurate values for Z_G and Z_L are available, Eq. (2-33) is the preferred method for obtaining enthalpies of vaporization. Otherwise, the CS methods shown below should be used.

Recommended Method 1 Vapor pressure correlation.

Classification: Extension of theory.

Expected uncertainty: Varies significantly with temperature and with the quality and temperature range of the vapor pressure data used in the correlation.

Applicability: Organic compounds for which group contributions have been regressed.

Input data: Correlations for P^* , Z_G , and Z_L . If $T_r < 0.8$, then Z_G can be set to 1.

Description: An expression for ΔH_v can be obtained from Eq. (2-33) by using an appropriate vapor pressure correlation. If one differentiates the Riedel vapor pressure correlation, Eq. (2-22), in accordance with Eq. (2-33), one obtains the heat of vaporization as

$$\Delta H_v = R \Delta Z_v (-B + CT + DET^{E+1}) \quad (2-36)$$

The Z_G and Z_L values can be evaluated by using the methods given in the subsection on densities below.

Example Calculate ΔH_v for anisole at 452 K.

Input data: The vapor pressure coefficients in the DIPPR® 801 database, based on Eq. (2-22), are

$$A = 128.06 \quad B = -9307.7 \quad C = -16.693 \quad D = 0.014919 \quad E = 1$$

The vapor pressure at 452 K is

$$\ln \frac{P^*}{\text{Pa}} = 128.06 - \frac{9307.7}{452} - 16.693 \ln 452 + 0.014919(452)^1 = 12.155$$

$$P^* = \exp(12.155) \cdot \text{Pa} = 1.901 \times 10^5 \text{ Pa}$$

Determine ΔZ : Required data from the DIPPR® 801 database for this calculation are $T_c = 645.6 \text{ K}$, $P_c = 4.25 \text{ MPa}$, and $\omega = 0.35017$. These values are used to determine the reduced conditions and the values of Z_G and Z_L from the Lee-Kesler corresponding states method as discussed in the subsection on density.

$$T_r = \frac{452}{645.6} = 0.7 \quad P_r = \frac{0.1901}{4.25} = 0.045$$

Interpolation of the P_r values in Tables 2-351 and 2-352 at a T_r of 0.7 gives

$$Z_G^{(0)} = 0.9904 + \frac{0.045 - 0.010}{0.050 - 0.010} (0.9504 - 0.9904) = 0.9554$$

$$Z_G^{(1)} = -0.0064 + \frac{0.045 - 0.010}{0.050 - 0.010} (-0.0507 + 0.0064) = -0.0452$$

$$Z_G = Z_G^{(0)} + \omega Z_G^{(1)} = 0.9554 + (0.35017)(-0.0452) = 0.94$$

At this low pressure, Z_L is very small compared to Z_G and may be neglected; so

$$\Delta Z_V = Z_G - Z_L = 0.94$$

Calculation using Eq. (2-36):

$$\begin{aligned} \Delta H_v &= \left(8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}} \right) (0.94) [9307.7 - (16.693)(452) + (0.014919)(1)(452)^2] \\ &= 37.59 \frac{\text{kJ}}{\text{mol} \cdot \text{K}} \end{aligned}$$

This value is 0.2 percent higher than the value of 37.51 kJ/(mol·K) obtained from the DIPPR® 801 database.

Recommended Method 2 Corresponding states correlation.

Reference: PGL5, p. 7.18.

Classification: Corresponding states.

Expected uncertainty: Less than about 6 percent.

Applicability: Organic compounds.

Input data: T_c , P_c , and ω .

Description: The following correlation is used:

$$\frac{\Delta H_v}{RT_c} = 7.08\tau^{0.354} + 10.95\omega\tau^{0.456} \quad \text{where } \tau = 1 - T_r \quad (2-37)$$

Example Repeat the above calculation for anisole's ΔH_v at 452 K.

Input data: $T_c = 645.6$ K, $P_c = 4.25$ MPa, and $\omega = 0.35017$.

Auxiliary quantities: From the previous example, the reduced temperature is

$$T_r = 0.7 \quad \tau = 1 - 0.7 = 0.3$$

Calculation using Eq. (2-37):

$$\frac{\Delta H_v}{RT_c} = 7.08(0.3)^{0.354} + 10.95(0.35017)(0.3)^{0.456} = 6.838$$

$$\Delta H_v = (6.838) \left(8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}} \right) (645.6 \text{ K}) = 36.70 \frac{\text{kJ}}{\text{mol} \cdot \text{K}}$$

This value is 2.2 percent below the DIPPR® 801 recommended value of 37.51 kJ/(mol·K).

Recommended Method 3 Vetere method for ΔH_v at T_b .

Reference: Vetere, A., *Fluid Phase Equilib.*, **106** (1995): 1.

Classification: Corresponding states.

Expected uncertainty: About 4 percent.

Applicability: Valid only at the normal boiling point.

Input data: T_c , P_c , and T_b .

Description: The following correlation is used:

$$\frac{\Delta H_v}{RT_b} = \frac{\tau_b^{0.38} \left[\ln(P_c/\text{bar}) - 0.513 + \frac{0.5066 \cdot \text{bar}}{P_c T_b^2} \right]}{\tau_b + F(1 - \tau_b^{0.38}) \ln T_{br}} \quad (2-38)$$

where $\tau_b = 1 - T_{br}$

For most compounds, $F = 1$; compounds that dimerize (e.g., SO_2 , NO , NO_2) and alcohols with more than two C atoms are assigned $F = 1.05$.

Example Calculate ΔH_v at the normal boiling point for acetaldehyde.

Input data: Recommended values from the DIPPR® 801 database are $T_c = 466.0$ K, $P_c = 5.55$ MPa, and $T_b = 294.0$ K.

Auxiliary quantities: From the previous example, the reduced temperature is

$$T_{br} = \frac{294}{466} = 0.631 \quad \tau_b = 1 - T_{br} = 0.369 \quad F = 1$$

Calculation using Eq. (2-38):

$$\frac{\Delta H_v}{RT_b} = \frac{(0.369)^{0.38} \left[\ln 55.5 - 0.513 + \frac{0.5066/(55.5)(0.631)^2}{(0.369) + [1 - (0.369)^{0.38}] \ln 0.631} \right]}{(0.369) + [1 - (0.369)^{0.38}] \ln 0.631} = 10.785$$

$$\Delta H_v = (10.785) \left(8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}} \right) (294 \text{ K}) = 26.36 \frac{\text{kJ}}{\text{mol} \cdot \text{K}}$$

This value is 2.4 percent above the DIPPR® 801 recommended value of 25.73 kJ/(mol·K).

Enthalpy of Fusion The enthalpy (heat) of fusion ΔH_{fus} is the difference between the molar enthalpies of the equilibrium liquid and solid at the melting temperature and 1.0 atm pressure. There is no generally applicable, high-accuracy estimation method for ΔH_{fus} , but the GC method of Chickos can be used to obtain approximate results if the melting temperature is known.

Recommended Method Chickos method.

Reference: Chickos, J. S., et al., *J. Org. Chem.*, **56** (1991): 927.

Classification: QSPR and group contributions.

Expected uncertainty: Considerable variation but generally less than 50 percent.

Applicability: Only valid at the melting temperature. The method is based on the ΔS_{fus} between a solid at 0 K and the liquid at the T_m so no solid-solid transitions are taken into account. Values of ΔH_{fus} will be overestimated if there are solid-solid transitions for the actual material.

Input data: T_m and molecular structure.

Description:

$$\frac{\Delta H_{\text{fus}}}{\text{J/mol}} = \frac{\Delta S_{\text{fus}}}{\text{J/(mol} \cdot \text{K)}} \left(\frac{T_m}{\text{K}} \right) = (T_m/\text{K})(a + b) \quad (2-39)$$

$$a = \begin{cases} 0 & \text{no nonaromatic rings} \\ 35.19N_R + 4.289(N_{CR} - 3N_R) & \text{nonaromatic rings} \end{cases} \quad (2-40)$$

$$b = \sum_{i=1}^{n_g} N g_i \Delta s_i + \sum_{j=1}^{n_s} N s_j C s_j \Delta s_j + \sum_{k=1}^{n_f} N f_k C t_k \Delta s_k \quad (2-41)$$

where $N g_i$ = number of C—H groups of type i bonded to other carbon atoms

n_g = number of different nonring or aromatic C—H groups bonded to other carbon atoms

$N s_j$ = number of C—H groups of type j bonded to at least one functional group or atom

n_s = number of different nonring or aromatic C—H groups bonded to at least one functional group or atom

$N f_k$ = number of functional groups of type k

n_f = number of different functional groups or atoms

i = total number of functional groups or atoms with the exception that F atoms count as one regardless of number of occurrences

$C s_j$ = value from Table 2-344 for C—H group j bonded to at least one functional group or atom

$C t_k$ = value from Table 2-345 for functional group k

N_R = number of nonaromatic rings

N_{CR} = number of —CH₂— groups in nonaromatic ring(s) required to form cyclic paraffin of same ring size(s)

Δs_i = contribution from Table 2-344 for group i

Δs_k = contribution from Table 2-345 for group k

TABLE 2-344 Cs (C—H) Group Values for Chickos Estimation* of ΔH_{fus}

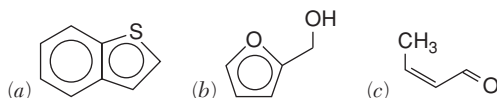
Group	Description	Cs	Δ_s	Group	Description	Cs	Δ_s
—CH ₃	methyl	1.0	18.33	—CH _{Ar}	aromatic C	1.0	6.44
>CH ₂	methylene	1.0	9.41	—C _{Ar} —	ar. C bonded to paraffinic C	1.0	-10.33
>CH—	secondary C	0.69	-16.91	—C _{Ar} —	ar. C bonded to olefinic C or non-C group	1.0	-4.27
>C<	tertiary C	0.67	-38.70	—C _{Ar} —	ar. C bonded to acetylinic C	1.0	-2.51
CH ₂ =	terminal alkene	1.0	14.56	>C _r H—	ring structure	0.76	-15.98
—CH=	alkene	3.23	4.85	>C _r <	ring structure	1.0	-32.97
>C=	subst. alkene	1.0	-11.38	—C _r H=	ring structure	0.62	-4.35
≡CH	term. alkyne	1.0	10.88	>C _r =	ring structure	0.86	-11.72
≡C—	alkyne	1.0	2.18	≡C _r — or =C _r =	ring structure	1.0	-5.36

*Chickos, J. S., et al., *J. Org. Chem.*, **56** (1991): 927.

Note that nonaromatic ring —CH₂ groups are accounted for in the *a* term and are *not* included in the *b* term.

Example Calculate ΔH_{fus} at the melting point for (a) benzothiophene, (b) furfuryl alcohol, and (c) *cis*-crotonaldehyde.

Structures:



(a) $t = 1$ (1 total “functional group”), so the C_1 column in Table 2-345 is used.

$$N_R = 1 \quad N_{CR} = 5 \quad a = 35.19 + (5 - 3)(4.289) = 43.77$$

Group	Description	N	C	Δ_s	Total
≡CH—	aromatic (Ng type)	4	1	6.44	25.76
≡C—	ring (Ng type)	1	1	-11.72	-11.72
≡C—	ring (Ns type)	1	0.86	-11.72	-10.08
≡CH—	ring (Ng type)	1	1	-4.35	-4.35
≡CH—	ring (Ns type)	1	0.62	-4.35	-2.70
—S—	ring	1	1	2.18	2.18
				Total	-0.91

TABLE 2-345 Ct (Functional) Group Values for Chickos Estimation* of ΔH_{fus}

Group	Description	C ₁	C ₂	C ₃	C ₄	Δ_s
—OH	alcohol	1.0	12.6	18.9	26.4	1.13
—OH	phenol	1.0	1.0	1.0	1.0	16.57
—O—	nonring ether	1.0	1.0	1.0	1.0	1.09
—O—	ring ether	1.0	1.0	1.0	1.0	1.34
>C=O	nonring ketone	1.0	1.0			3.14
>C=O	ring ketone	1.0	1.0			-1.88
—CHO	aldehyde	1.0	1.0			19.66
—COOH	acid	1.0	1.83	1.88	1.72	14.90
—COO—	ester	1.0	1.0	1.0	1.0	3.68
—NH ₂	aliphatic	1.0	1.0			16.23
—NH ₂	aromatic	1.0	1.0			15.48
>NH	nonring	1.0	1.0			-2.18
>NH	ring	1.0	1.0			1.84
>N—	nonring	1.0	1.0			-15.90
>N—	ring	1.0	1.0			-17.07
≡N—	ring	1.0	1.0			1.67
≡N—	aromatic	1.0	1.0	1.0		7.32
—CN	nitrile	1.0	1.4			9.62
—NO ₂	nitro	1.0	1.0	1.0		17.36
—CONH ₂	primary amide	1.0	1.0			26.19
—CONH—	secondary amide	1.0	1.0			-0.42
—SH		1.0	1.0			17.99
—S—	nonring	1.0	1.0		0.36	7.20
—S—	ring	1.0	1.0			2.18
—SO ₂	nonring	1.0	1.0			3.26
—F	on aliph. C	1.0	1.0	1.0	1.0	14.73
—F	on olefinic C	1.0	1.0	1.0	1.0	13.01
—F	on ring C	1.0	1.0	1.0	1.0	15.90
—Cl		1.0	2.0	2.0	1.93	8.37
—Br		1.0	1.0	1.0	0.82	17.95
—I		1.0	1.0			16.95

*Chickos, J. S., et al., *J. Org. Chem.*, **56** (1991): 927.

$$T_m = 304.5 \text{ K} \quad \text{from DIPPR}^{\circ} 801 \text{ database}$$

$$\Delta H_{\text{fus}} = (T_m/K)(a + b) \text{ J/mol} = (304.5)(43.77 - 0.91) \text{ J/mol} = 13.05 \text{ kJ/mol}$$

This value is 10 percent higher than the DIPPR[®] 801 recommended value of 11.83 kJ/mol.

(b) $t = 2$ (2 total “functional groups”), so the C_2 column in Table 2-345 is used.

$$N_R = 1 \quad N_{CR} = 5 \quad a = 35.19 + (5 - 3)(4.289) = 43.77$$

Group	Description	N	C	Δ_s	Total
≡CH—	ring (Ng type)	2	1	-4.35	-8.70
≡CH—	ring (Ns type)	1	0.62	-4.35	-2.70
≡C<	ring (Ns type)	1	0.86	-11.72	-10.08
—O—	ring ether	1	1	1.34	1.34
—CH ₂ —	Ns type	1	1	9.41	9.41
—OH	alcohol	1	12.6	1.13	14.24
				Total	3.51

$$T_m = 258.52 \text{ K} \quad \text{from DIPPR}^{\circ} 801 \text{ database}$$

$$\Delta H_{\text{fus}} = (T_m/K)(a + b) \text{ J/mol} = (258.52)(43.77 + 3.51) \text{ J/mol} = 12.22 \text{ kJ/mol}$$

This value is 7 percent lower than the DIPPR[®] 801 recommended value of 13.13 kJ/mol.

(c) $t = 1 \quad N_R = 0 \quad a = 0$

Group	Description	N	C	Δ_s	Total
—CH ₃	nonring (Ng type)	1	1	18.33	18.33
≡CH—	nonring (Ng type)	1	1	4.85	4.85
≡CH—	nonring (Ns type)	1	3.23	4.85	15.67
—CHO	aldehyde	1	1	19.66	19.66
				Total	58.51

$$T_m = 158.38 \text{ K} \quad \text{from DIPPR}^{\circ} 801 \text{ database}$$

$$\Delta H_{\text{fus}} = (T_m/K)(a + b) \text{ J/mol} = (158.38)(0 + 58.51) \text{ J/mol} = 9.27 \text{ kJ/mol}$$

This value is 5 percent higher than the DIPPR[®] 801 recommended value of 8.86 kJ/mol.

Enthalpy of Sublimation The enthalpy (heat) of sublimation ΔH_{fus} is the difference between the molar enthalpies of the equilibrium vapor and solid along the sublimation curve below the triple point. The effects of pressure on ΔH_{fus} and melting temperature are very small so that T_i and the normal melting point are nearly equal and

$$\Delta H_{\text{sub}}(T_i) = \Delta H_v(T_i) + \Delta H_{\text{fus}}(T_i) \quad (2-42)$$

Equation (2-42) can be used to estimate ΔH_{sub} at the triple point if ΔH_v is accurately known at T_i . Because ΔH_v is usually obtained from Eq. (2-33), $\Delta H_v(T)$ correlations may be less accurate near T_i where $P^{\circ}(T_i)$ is very small and difficult to measure. In this case, it is better to estimate ΔH_{sub} directly by using the following recommended

method. Although Eqs. (2-33) and (2-34) apply to T_i , ΔH_{sub} is only a weak function of temperature and can generally be taken as a constant from the triple point temperature down to the first solid-solid phase transition.

Recommended Method Goodman method.

Reference: Goodman, B., et al., *Int. J. Thermophys.* **25** (2004): 337.

Classification: QSPR and group contributions.

Expected uncertainty: 6 percent.

Applicability: Organic compounds for which group contributions have been regressed.

Input data: Molecular structure and radius of gyration R_G .

Description:

$$\frac{\Delta H_{\text{sub}}(T_i)}{R K} = 698.04 + 3.83798 \times 10^{12} \left(\frac{R_G}{\text{m}} \right) + \sum_{i=1}^N n_i a_i + \sum_{i=1}^N n_i^2 \beta_i + \sum_{i=1}^N \frac{n_i}{n_x} f_i \quad (2-43)$$

where a_i = GC values from Table 2-346

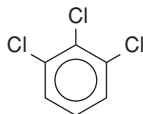
β_i = nonlinear corrections for $>\text{CH}_2$ and $\text{Ar}-\text{CH}$ = groups

f_i = halogen corrections

n_x = total number of all halogen and hydrogen atoms attached to C and Si atoms

Example Calculate ΔH_{sub} and the solid vapor pressure for 1,2,3-trichlorobenzene at 301.15 K.

Structure:



Group contributions:

Linear groups			Nonlinear and correction terms			
Group	n_i	a_i	Group	n_i	β_i	f_i
Ar—CH=	3	626.7621	Ar—CH=	3	-2.21614	
Ar >C=	3	348.8092	—Cl	3		-1543.66
—Cl	3	1243.445	n_x	6		
		$\sum_i n_i a_i$				
		6657.049				

Input data: The value of R_G from the DIPPR® 801 database is 4.455×10^{-10} m. **Calculation using Eq. (2-43):**

$$\frac{\Delta H_{\text{sub}}(T_i)}{R K} = 698.04 + (3.83798 \times 10^{12})(4.455 \times 10^{-10}) + 6657.049 + (3^2)(-2.21614) + \left(\frac{3}{6} \right)(-1543.66)$$

$$\Delta H_{\text{sub}}(T_i) = (8273.134 \text{ K}) \left(0.008314 \frac{\text{kJ}}{\text{mol} \cdot \text{K}} \right) = 68.78 \frac{\text{kJ}}{\text{mol}}$$

The estimated value is 5.6 percent above the DIPPR® 801 recommended value of 65.11 kJ/mol.

Estimate the solid vapor pressure from Eq. (2-26): The solid vapor pressure at 301.15 K can be calculated from Eq. (2-26) by using the estimated ΔH_{sub} . Recommended values for T_i and P_i^* from the DIPPR® 801 database are 325.65 K and 182.957 Pa, respectively.

$$\ln \frac{P^*}{182.957 \text{ Pa}} = \frac{68.78 \text{ kJ/mol}}{[0.008314 \text{ kJ/(mol} \cdot \text{K)}](325.65 \text{ K})} \left(1 - \frac{325.65}{301.15} \right) = -2.067$$

$$P^* = (182.957 \text{ Pa})[\exp(-2.067)] = 23.16 \text{ Pa}$$

The estimated value is 0.3 percent above the DIPPR® 801 recommended value of 23.09 Pa.

HEAT CAPACITY

Heat capacity C_p is defined as the energy required to change the temperature of a unit mass (specific heat) or mole (molar heat capacity) of the material by one degree. Typical units are J/(kg·K).

Gases The heat capacity of a gas is related rigorously to the ideal gas heat capacity C_p^0 by

$$C_p = C_p^0 - T \int_0^P \left(\frac{\partial^2 V}{\partial T^2} \right)_P dP \quad (2-44)$$

The second term, giving the deviation of the real fluid heat capacity from the ideal gas value, can be neglected at low to moderate pressures, or it can be calculated directly from an appropriate EoS.

Ideal gas heat capacities are available from several sources (DIPPR, JANAF, TRC, and SWS). Two common correlating equations for C_p^0 are the Aly-Lee [Aly, F. A., and L. L. Lee, *Fluid Phase Equilib.*, **6**

TABLE 2-346 Group Contributions and Corrections* for ΔH_{sub}

Group	Description	a_i	Group	Description	a_i
—CH ₃	methyl	736.5889	>C=O	ketone	1816.093
>CH ₂	methylene	561.3543	—COO—	ester	2674.525
>CH—	secondary C	111.0344	—COOH	acid	5006.188
>C<	tertiary C	-800.517	—NH ₂	primary amine	2219.148
CH ₂ =	terminal alkene	572.6245	—NH—	sec. amine	1561.222
—CH=	alkene	541.2918	>N—	tertiary amine	325.9442
>C=	substituted alkene	117.9504	—NO ₂	nitro	3661.233
Ar—CH=	aromatic C	626.7621	—SH	thiol/mercaptan	1921.097
Ar >C=	subst. aromatic C	348.8092	—S—	sulfide	1930.84
Ar—O—	furan O	763.284	—SS—	disulfide	2782.054
Ar—N=	pyridine N	1317.056	—F	fluoride	626.4494
Ar—S—	thiophene S	911.2903	—Cl	chloride	1243.445
—O—	ether	970.4474	—Br	bromide	669.9302
—OH	alcohol	3278.446	>Si<	silane	-83.7034
—COH	aldehyde	2402.093	>Si(O—)—	siloxane	-16.0597
Nonlinear terms		β_i	Halogen correction terms		f_i
>CH ₂	methylene	9.5553	—F	F fraction	-1397.4
Ar—CH=	aromatic C	-2.21614	—Cl	Cl fraction	-1543.66
			—Br	Br fraction	5812.49

*Goodman, B., et al., *Int. J. Thermophys.*, **25** (2004): 337.

(1981): 169] equation

$$C_P^o = A_o + A_1 \left[\frac{A_2/T}{\sinh(A_2/T)} \right]^2 + A_3 \left[\frac{A_4/T}{\cosh(A_4/T)} \right]^2 \quad (2-45)$$

and a polynomial form (generally fourth-order)

$$C_P^o = \sum_{i=0}^4 A_i T^i \quad (2-46)$$

Ideal gas heat capacities may also be estimated from several techniques, of which two of the most accurate and commonly used are recommended here.

Recommended Method 1 Statistical mechanics.

Reference: Rowley, R. L., *Statistical Mechanics for Thermophysical Property Calculations*, Prentice-Hall, Englewood Cliffs, N.J., 1994.

Classification: Theory and computational chemistry.

Expected uncertainty: 0.2 percent if vibrational frequencies (or their characteristic temperatures) are experimentally available; accuracy depends upon model chemistry if frequencies are determined from computational chemistry, but generally within 3 percent.

Applicability: Ideal gases.

Input data: $3n_A - 6 + \delta$ vibrational frequencies ν_j , or the corresponding characteristic vibrational temperatures Θ_j . The two are related by

$$\Theta_j = h\nu_j/k \quad (2-47)$$

where n_A = number of atoms in molecule and $\delta = 1$ for linear molecules and 0 for nonlinear molecules.

Description: For harmonic frequencies, the rigorous temperature dependence of C_P^o is given by

$$\frac{C_P^o}{R} = \frac{8-\delta}{2} + \sum_{j=1}^{3n_A-6+\delta} \left(\frac{\Theta_j}{T} \right)^2 \left[\frac{e^{\Theta_j/T}}{(e^{\Theta_j/T} - 1)^2} \right] \quad \delta = \begin{cases} 0 & \text{nonlinear} \\ 1 & \text{linear} \end{cases} \quad (2-48)$$

Example Calculate the ideal gas heat capacity of ammonia at 300 K.

Structure:



Input data: McQuarrie (McQuarrie, D. A., *Statistical Mechanics*, Harper & Row, New York, 1976) gives the following $3m - 6 + \delta = 12 - 6 + 0 = 6$ characteristic vibrational temperatures (K): 1360, 2330, 2330, 4800, 4880, and 4880. Alternatively, a computational chemistry package gives the following *scaled* frequencies for HF/6-31G+ model chemistry (10^{13} Hz): 3.24, 4.97, 4.97, 9.90, 10.26, and 10.26.

Calculation: The table on the left uses the experimental Θ values to determine the individual terms in the summation of Eq. (2-48); the table on the right uses the scaled frequencies from computational chemistry software and Eq. (2-47) to obtain Θ values and the individual terms in Eq. (2-48).

Experimental frequencies		
Θ/K	$\Theta/(300 \text{ K})$	Term
1360	4.533	0.2256
2330	7.767	0.0256
2330	7.767	0.0256
4800	16.000	0.0000
4880	16.267	0.0000
4880	16.267	0.0000
Sum		0.2769

From experimental frequencies:

$$C_P^o = \left(\frac{8}{2} + 0.2769 \right) R = (4.2769) \left(8.3143 \frac{\text{J}}{\text{mol}\cdot\text{K}} \right) = 35.56 \frac{\text{J}}{\text{mol}\cdot\text{K}}$$

From computational chemistry frequencies:

$$C_P^o = \left(\frac{8}{2} + 0.197 \right) R = (4.197) \left(8.3143 \frac{\text{J}}{\text{mol}\cdot\text{K}} \right) = 34.90 \frac{\text{J}}{\text{mol}\cdot\text{K}}$$

The value calculated from experimental frequencies is 0.1 percent lower than the DIPPR® 801 recommended value of 35.61 J/(mol·K); the value calculated from frequencies generated from computational chemistry software is 2.0 percent lower than the DIPPR® 801 value.

Recommended Method 2 Benson method as implemented in CHETAH program.

References: Benson, S. W., et al., *Chem. Rev.*, **69** (1969): 279; CHETAH Version 8.0: *The ASTM Computer Program for Chemical Thermodynamic and Energy Release Evaluation (NIST Special Database 16)*.

Classification: Group contributions.

Expected uncertainty: 4 percent.

Applicability: Ideal gases of organic compounds.

Input data: Table 2-347 group values at the seven specified temperatures.

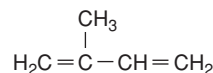
Description: Groups are summed at each individual temperature:

$$C_P^o = \sum_{i=1}^N n_i \cdot (C_P^o)_i \quad (2-49)$$

where n_i = number of occurrences of group i and $(C_P^o)_i$ = individual group contribution. Either Eq. (2-45) or Eq. (2-46) can be used to interpolate between the discrete temperatures.

Example Calculate the ideal gas heat capacity of isoprene (2-methyl-1,3-butadiene) at 400 K.

Structure:



Group identification and values:

Group	No.	Value, J/(mol·K)	Contribution, J/(mol·K)
$=\text{CH}_2$	2	26.62	53.24
$=\text{C}-(2\text{C})$	1	19.3	19.3
$-\text{CH}_2-(=\text{C})$	1	32.82	32.82
$=\text{CH}-(\text{C})$	1	21.05	21.05
Total			126.41

The value of 126.4 J/(mol·K) is 3.1 percent below the DIPPR® 801 recommended value of 130.4 J/(mol·K).

Liquids Liquid heat capacity increases with increasing temperature, although a minimum occurs near the triple point for many compounds. Usually liquid heat capacity is correlated as a function of temperature with a polynomial equation. A cubic equation is usually adequate.

Estimation of liquid heat capacity can be done by using a number of methods [Ruzicka, V., and E. S. Domalski, *J. Phys. Chem. Ref. Data*, **22** (1993): 597, 619; Chueh, C. F., and A. C. Swanson, *Chem. Eng.*

HF/6-31G + scaled frequencies			
$\nu_{\text{scaled}}/\text{Hz}$	Θ/K	$\Theta/(300 \text{ K})$	Term
3.24	1555.0	5.183	0.1524
4.97	2385.3	7.951	0.0223
4.97	2385.3	7.951	0.0223
9.90	4751.4	15.838	0.0000
10.26	4924.2	16.414	0.0000
10.26	4924.2	16.414	0.0000
Sum			0.1970

NOTE: Empirical scaling factors have been developed for each model chemistry to help correct theoretical frequencies for anharmonic effects [Scott, A. P., and L. Radom, *J. Phys. Chem.*, **100** (1996): 16502].

TABLE 2-347 Benson* and CHETAH[†] Group Contributions for Ideal Gas Heat Capacity

Table-specific nomenclature: Cb = carbon in benzene ring; Ct = carbon with a triple bond, ($\equiv$ C) = carbon with a double bond; Cp = carbon in fused ring; Naz = azide; Nim = imino.

Group	298 K	400 K	500 K	600 K	800 K	1000 K	1500 K
CH ₃ Groups							
CH ₃ —(Cb)	25.91	32.82	39.35	45.17	54.5	61.83	73.59
CH ₃ —(CO)	25.91	32.82	39.35	45.17	54.5	61.83	73.59
CH ₃ —(Ct)	25.91	32.82	39.35	45.17	54.5	61.83	73.59
CH ₃ —(C)	25.91	32.82	39.35	45.17	54.5	61.83	73.59
CH ₃ —(N)	25.95	32.65	39.35	45.21	54.42	61.95	73.67
CH ₃ —(O)	25.91	32.82	39.35	45.17	54.54	61.83	73.59
CH ₃ —(PO)	25.91	32.82	39.35	45.17	54.54	61.83	73.59
CH ₃ —(P)	25.91	32.82	39.35	45.17	54.54	61.83	73.59
CH ₃ —(P $\equiv$ N)	25.91	32.82	39.35	45.17	54.54	61.83	73.59
CH ₃ —(Si)	25.91	32.82	39.35	45.17	54.5	61.83	73.59
CH ₃ —(SO ₂)	25.91	32.82	39.35	45.17	54.5	61.83	
CH ₃ —(SO)	25.91	32.82	39.35	45.17	54.5	61.83	
CH ₃ —(S)	25.91	32.82	39.35	45.17	54.5	61.83	
CH ₃ —($\equiv$ C)	25.91	32.82	39.35	45.17	54.5	61.83	73.59
Ct Groups							
Ct—(Cb)	10.76	14.82	14.65	20.59	22.35	23.02	24.28
Ct—(Ct)	14.82	16.99	18.42	19.42	20.93	21.89	23.32
Ct—(C)	13.1	14.57	15.95	17.12	19.25	20.59	26.58
Ct—($\equiv$ C)	10.76	14.82	14.65	20.59	22.35	23.02	24.28
CtBr	34.74	36.42	37.67	38.51	39.77	40.6	
CtCl	33.07	35.16	36.42	37.67	39.35	40.18	
CtF	28.55	31.65	33.99	35.79	38.3	39.85	41.77
CtH	22.06	25.07	27.17	28.76	31.27	33.32	37.04
CtI	35.16	36.84	38.09	38.93	40.18	41.02	
Ct(CN)	43.11	47.3	50.65	53.16	56.93	59.86	64.04
CH ₂ Groups							
CH ₂ —(2CO)	16.03	26.66	32.15	37.8	45.46	51.74	
CH ₂ —(2C)	23.02	29.09	34.53	39.14	46.34	51.65	59.65
CH ₂ —(2O)	11.85	21.18	31.48	38.17	43.2	47.26	
CH ₂ —(2 $\equiv$ C)	19.67	28.46	35.16	40.18	47.3	52.74	60.28
CH ₂ —(Cb,O)	15.53	26.26	34.66	40.98	49.35	55.25	
CH ₂ —(Cb,SO ₂)	15.53	27.5	34.66	40.98	49.77	55.25	
CH ₂ —(Cb,S)	38.09	49.02	57.43	63.71	72.58	78.82	
CH ₂ —(Cb, $\equiv$ C)	19.67	28.46	35.16	40.18	47.3	52.74	60.28
CH ₂ —(C,Cb)	24.45	31.85	37.59	41.9	48.1	52.49	57.6
CH ₂ —(C,CO)	25.95	32.23	36.42	39.77	46.46	51.07	
CH ₂ —(C,Ct)	20.72	27.46	33.19	38.01	45.46	51.03	59.44
CH ₂ —(C,N)	21.77	28.88	34.74	39.35	46.46	51.49	
CH ₂ —(C,O)	20.89	28.67	34.74	39.47	46.5	51.61	61.11
CH ₂ —(C,SO ₂)	17.12	24.99	31.56	36.84	44.58	49.94	
CH ₂ —(C,SO)	19.05	26.87	33.28	38.34	45.84	51.15	
CH ₂ —(C,S)	22.52	29.64	36	41.73	51.32	59.23	
CH ₂ —(C, $\equiv$ C)	21.43	28.71	34.83	39.72	46.97	52.24	60.11
CH ₂ —($\equiv$ C,O)	19.51	29.18	36.21	41.36	48.3	53.29	
CH ₂ —($\equiv$ C,SO ₂)	20.34	28.51	34.95	40.1	47.17	52.49	
CH ₂ —($\equiv$ C,SO)	18.42	26.62	29.05	38.72	45.92	51.28	
CH ₂ —($\equiv$ C,S)	22.23	28.59	34.45	40.85	50.98	59.48	
CH Groups							
CH—(2C,Cb)	20.43	27.88	33.07	36.63	40.73	42.9	44.7
CH—(2C,CO)	18.96	25.87	30.89	35.12	41.11	43.99	
CH—(2C,Ct)	16.7	23.48	28.67	32.57	38.09	41.44	46.55
CH—(2C,N)	19.67	26.37	31.81	35.16	40.18	42.7	
CH—(2C,O)	20.09	27.79	33.91	36.54	41.06	43.53	
CH—(2C,SO ₂)	18.5	26.16	31.65	35.5	40.35	43.11	
CH—(2C,S)	20.3	27.25	32.57	36.38	41.44	44.24	
CH—(2C, $\equiv$ C)	17.41	24.74	30.72	34.28	39.6	42.65	47.22
CH—(3C)	19	25.12	30.01	33.7	38.97	42.07	46.76
CH—(C,2O)	22.02	23.06	27.67	31.77	35.41	38.97	
C Groups							
C—(2C,2O)	19.25	19.25	23.02	25.53	27.63	28.46	
C—(3C,Cb)	19.72	28.42	33.86	36.75	38.47	37.51	31.94
C—(3C,CO)	9.71	18.33	23.86	27.17	30.43	31.69	
C—(3C,Ct)	0.33	7.33	14.36	19.97	25.2	26.71	
C—(3C,N)	18.42	25.95	30.56	33.07	35.58	35.58	
C—(3C,O)	18.12	25.91	30.35	32.23	34.32	34.49	
C—(3C,SO ₂)	9.71	18.33	23.86	27.17	30.43	31.23	
C—(3C,SO)	12.81	19.17	20.26	27.63	31.56	33.32	
C—(3C,S)	19.13	26.25	31.18	34.11	36.5	33.91	
C—(3C, $\equiv$ C)	16.7	25.28	31.1	34.58	37.34	37.51	34.45
C—(4C)	18.29	25.66	30.81	33.99	36.71	36.67	33.99

2-492 PHYSICAL AND CHEMICAL DATA

TABLE 2-347 Benson* and CHETAH† Group Contributions for Ideal Gas Heat Capacity (Continued)

Table-specific nomenclature: Cb = carbon in benzene ring; Ct = carbon with a triple bond, ($\equiv$ C) = carbon with a double bond; Cp = carbon in fused ring; Naz = azide; Nim = imino.

Group	298 K	400 K	500 K	600 K	800 K	1000 K	1500 K
Aromatic (Cb and Cp Groups)							
Cb—(Cb)	13.94	17.66	20.47	22.06	24.11	24.91	25.32
Cb—(CO)	11.18	13.14	15.4	17.37	20.76	22.77	
Cb—(Ct)	15.03	16.62	18.33	19.76	22.1	23.48	24.07
Cb—(C)	11.18	13.14	15.4	17.37	20.76	22.77	25.03
Cb—(N)	16.53	21.81	24.86	26.45	27.33	27.46	
Cb—(O)	16.32	22.19	25.95	27.63	28.88	28.88	
Cb—(Si)	11.18	13.14	15.4	17.37	20.76	22.77	25.03
Cb—(SO ₂)	11.18	13.14	15.4	17.37	20.76	22.77	
Cb—(SO)	11.18	13.14	15.4	17.37	20.76	22.77	
Cb—(S)	16.32	22.19	25.95	27.63	28.88	28.88	
Cb—($\equiv$ C)	15.03	16.62	18.33	19.76	22.1	23.48	24.07
Cb—($\equiv$ Nim)	16.53	21.81	24.86	26.45	27.33	27.46	
CbBr	32.65	36.42	39.35	41.44	43.11	43.95	
CbCl	30.98	35.16	38.51	40.6	42.7	43.53	
CbF	26.37	31.81	35.58	38.09	41.02	42.7	
CbH	13.56	18.59	22.85	26.37	31.56	35.2	40.73
CbI	33.49	37.25	40.18	41.44	43.11	43.95	
Cb(CHN ₂)	47.3						
Cb(CN)	41.86	48.14	52.74	55.67	59.86	62.79	
Cb(N ₃)	34.74						
Cb(NCO)	55.25	64.04	70.32	74.51	79.95	82.88	85.81
Cb(NCS)	32.23						
Cb(NO ₂)	38.93	50.23	59.44	66.56	76.18	80.37	
Cb(SO ₃ OH)	65.42	79.49	84.51	97.61	109.25	113.31	
Cp—(2Cb,Cp)	12.56	15.49	17.58	19.25	21.77	23.02	
Cp—(3Cp)	8.37	12.14	14.65	16.74	19.67	21.35	
Cp—(Cb,2Cp)	12.56	15.49	17.58	19.25	21.77	23.02	
$\equiv$ C=, $\equiv$ C—, $\equiv$ CH— Groups							
$\equiv$ C—(2C)	17.16	19.3	20.89	22.02	24.28	25.45	26.62
$\equiv$ C—(CO,O)	23.4	29.3	31.31	32.44	33.57	34.03	
$\equiv$ C—(C,Cb)	18.42	22.48	24.82	25.87	27.21	27.71	28.13
$\equiv$ C—(C,CO)	22.94	29.22	31.02	31.98	33.53	34.32	
$\equiv$ C—(C,O)	17.16	19.3	20.89	22.02	24.28	25.45	
$\equiv$ C—(C,SO ₂)	15.49	26.04	33.32	38.51	44.62	47.47	
$\equiv$ C—(C,S)	14.65	14.94	16.03	17.12	18.46	20.93	
$\equiv$ C—(C, $\equiv$ C)	18.42	22.48	24.82	25.87	27.21	27.71	28.13
$\equiv$ CC—($\equiv$ C,O)	18.42	22.9	24.82	26.29	27.21	27.71	
$\equiv$ CH—(Cb)	18.67	24.24	28.25	31.06	34.95	37.63	41.77
$\equiv$ CH—(CO)	31.73	37.04	38.8	40.31	43.45	46.21	
$\equiv$ CH—(Ct)	18.67	24.24	28.25	31.06	34.95	37.63	41.77
$\equiv$ CH—(C)	17.41	21.05	24.32	27.21	32.02	35.37	40.27
$\equiv$ CH—(O)	17.41	21.05	24.32	27.21	32.02	35.37	40.27
$\equiv$ CH—(SO ₂)	12.72	19.55	24.82	28.63	32.94	36.29	
$\equiv$ CH—(S)	17.41	21.05	24.32	27.21	32.02	35.37	
$\equiv$ CH—($\equiv$ C)	18.67	24.24	28.25	31.06	34.95	37.63	41.77
$\equiv$ CH ₂	21.35	26.62	31.44	35.58	42.15	47.17	55.21
$\equiv$ C=	16.32	18.42	19.67	20.93	22.19	23.02	23.86
Oxygen Groups							
O—(2C)	14.23	15.49	15.49	15.91	18.42	19.25	
O—(2O)	15.49	15.49	15.49	15.49	17.58	17.58	20.09
O—(2 $\equiv$ C)	14.02	16.32	17.58	18.84	21.35	22.6	
O—(Cb,CO)	8.62	11.3	13.02	14.32	16.24	17.5	
O—(CO,O)	1.51	6.28	9.63	11.89	15.28	17.33	
O—(C,Cb)	2.6	3.01	4.94	7.45	11.89	14.99	
O—(C,CO)	11.64	15.86	18.33	19.8	20.55	21.05	
O—(C,O)	15.49	15.49	15.49	15.49	17.58	17.58	20.09
O—(C, $\equiv$ C)	12.72	13.9	14.65	15.49	17.54	18.96	
O—($\equiv$ C,CO)	6.03	12.47	16.66	18.79	20.8	21.77	
OH—(Cb)	18	18.84	20.09	21.77	25.12	27.63	
OH—(CO)	15.95	20.85	24.28	26.54	30.01	32.44	37.34
OH—(C)	18.12	18.63	20.18	21.89	25.2	27.67	33.65
OH—(O)	21.64	24.24	26.29	27.88	29.93	31.44	34.2
O(CN)—(Cb)	34.74						
O(CN)—(C)	41.86						
O(CN)—($\equiv$ C)	54.42						
O(NO ₂)—(C)	39.93	48.3	55.5	65.3	68.61	72.75	
O(NO)—(C)	38.09	43.11	46.88	50.23	55.67	58.18	60.69
(CO)Cl—(C)	42.28	46.04	49.39	51.9	55.67	57.76	
(CO)H—(Cb)	33.53	44.2	48.77	59.48	68.56	74.01	
(CO)H—(CO)	28.13	32.78	37.25	41.4	47.84	50.73	

TABLE 2-347 Benson* and CHETAH[†] Group Contributions for Ideal Gas Heat Capacity (Continued)

Table-specific nomenclature: Cb = carbon in benzene ring; Ct = carbon with a triple bond, ($\equiv$ C) = carbon with a double bond; Cp = carbon in fused ring; Naz = azide; Nim = imino.

Group	298 K	400 K	500 K	600 K	800 K	1000 K	1500 K
Oxygen Groups							
(CO)H—(C)	29.43	32.94	36.92	40.52	46.71	51.07	
(CO)H—(N)	29.43	32.94	36.92	40.52	46.71	51.07	
(CO)H—(O)	29.43	32.94	36.92	40.52	46.71	51.07	
(CO)H—($\equiv$ C)	24.32	30.22	39.77	48.77	63.12	74.68	
CO—(Cb)(O)	9.12	11.51	16.65	21.05	26.32	29.54	
Halide Groups							
CBr—(3C)	39.35	47.72	52.74	55.25	56.93	56.09	
CBr ₃ —(C)	72.12	78.65	82.92	85.64	88.66	89.66	
CCl—(3C)	36.96	43.87	47.72	49.52	52.07	53.12	
CCl ₂ —(2C)	51.07	62.29	66.76	68.98	70.99	71.24	
CCl ₃ —(C)	68.23	75.35	79.95	82.88	86.23	87.9	
CClF ₃ —(C)	57.35	67.39	73.25	77.86	82.88	85.39	
CF—(3C)	28.46	37.09	42.7	46.71	52.03	53.24	
CF ₂ —(2C)	39.01	46.97	53.24	57.85	63.46	65.84	
CF ₃ —(Cb)	52.32	64.04	72	77.44	84.14	87.9	
CF ₃ —(C)	53.16	62.79	68.65	74.93	80.79	83.72	
CF ₃ —(S)	41.36	54.46	62.08	68.52	76.06	79.99	
CH ₂ Br—(Cb)	30.51	46.46	52.2	57.3	65.26	69.95	
CH ₂ Br—(C)	38.09	46.04	52.74	57.35	64.88	70.32	
CH ₂ Br—($\equiv$ C)	40.6	47.72	54.42	59.86	67.81	73.67	
CH ₂ Cl—(C)	37.25	44.79	51.49	56.09	64.04	69.9	
CH ₂ F—(C)	33.91	41.86	50.23	54.42	63.62	69.49	
CH ₂ I—(Cb)	33.91	45.17	53.7	59.9	68.15	73.8	
CH ₂ I—(C)	38.51	46.04	54	58.18	66.14	72	
CH ₂ I—(O)	34.41	43.91	51.19	56.72	64.25	69.36	
CHBr—(2C)	37.38	44.62	50.06	53.75	58.81	61.62	
CHBrCl—(C)	51.9	58.6	63.3	68.23	74.93	79.53	
CHCl—(2C)	35.45	42.7	48.89	53.41	59.82	64.38	
CHCl—(C,O)	37.67	41.44	43.95	46.88			
CHCl ₂ —(C)	50.65	58.6	64.46	69.07	74.93	78.28	
CHF—(2C)	30.56	37.84	43.83	48.39	54.83	58.64	
CHF ₂ —(C)	41.44	50.23	57.35	63.21	69.9	74.51	
CHI—(2C)	38.64	45.67	50.9	54.42	59.31	61.95	
CHI ₂ —(C)	56.93	63.42	69.61	74.17	79.7	81.58	
CI—(3C)	41.15	49.18	54.08	56.3	57.72	56.93	
$\equiv$ CBr ₂	51.49	55.25	58.18	59.86	62.37	63.62	
$\equiv$ CBrCl	50.65	53.16	56.51	59.02	61.53	62.79	
$\equiv$ CBrF	45.21	50.23	53.58	56.51	59.86	61.53	
$\equiv$ CCl ₂	47.72	52.32	55.67	58.18	61.11	62.79	
$\equiv$ CClF	43.11	48.97	52.74	55.67	59.44	61.53	
$\equiv$ CF ₂	40.6	46.04	50.23	53.16	57.76	60.69	
$\equiv$ CHBr	33.91	39.77	44.37	47.72	51.9	55.25	
$\equiv$ CHCl	33.07	38.51	43.11	46.88	51.49	54.83	
$\equiv$ CHF	28.46	35.16	39.77	43.95	49.39	53.16	
$\equiv$ CHI	36.84	41.86	45.63	48.56	52.74	55.67	
Nitrogen Groups							
CH ₂ (N ₃)—(C)	64.46						
$\equiv$ CH(N ₃)	54.42						
N—(2C,Cb)	2.6	8.46	13.69	17.29	21.89	23.4	
N—(2C,CO)	13.02	19.17	23.52	26.16	28.42	28.76	
N—(2C,SO ₂)	25.2	26.58	31.56	34.45	37.8	38.47	
N—(2C,SO)	17.58	24.61	25.62	27.33	28.59	34.91	
N—(2C,S)	15.99	21.64	25.99	29.05	30.93	38.68	
N—(3C)	14.57	19.09	22.73	24.99	27.46	27.92	27.21
N—(Cb,2CO)	4.1	12.81	17.71	20.3	22.1	22.14	
N—(C,2CO)	4.48	12.99	18.04	20.93	22.94	27.08	
Nb pyrid—N	10.88	13.48	15.95	17.66	20.05	21.43	
NF ₂ —(C)	26.5	34.58	40.9	45.63	50.9	53.54	
NH—(2Cb)	9.04	13.06	17.29	21.35	28.3	32.98	
NH—(2CO)	15.03	23.19	28.05	30.93	33.28	34.28	
NH—(2C)	17.58	21.81	25.66	28.59	33.07	36.21	39.97
NH—(Cb,CO)	2.39	6.32	9.96	13.94	16.91	18.21	
NH—(C,Cb)	15.99	20.47	23.9	26.29	30.1	32.36	
NH—(C,CO)	2.76	6.49	10.3	14.57	17.75	18.96	
NH—(C,N)	20.09	24.28	27.21	29.3	32.65	34.74	37.67
NH ₂ —(Cb)	23.94	27.25	30.64	33.78	39.39	43.83	51.4
NH ₂ —(CO)	17.04	24.03	29.85	34.7	41.69	46.97	
NH ₂ —(C)	23.94	27.25	30.64	33.78	39.39	43.83	51.4
NH ₂ —N	25.53	30.98	35.16	38.93	43.95	48.14	55.25
$\equiv$ Naz—(C)	11.3	17.16	20.59	22.35	23.82	23.9	
$\equiv$ Naz—(N)	8.87	17.5	23.06	28.34	28.71	29.51	

2-494 PHYSICAL AND CHEMICAL DATA

TABLE 2-347 Benson* and CHETAH† Group Contributions for Ideal Gas Heat Capacity (Continued)

Table-specific nomenclature: Cb = carbon in benzene ring; Ct = carbon with a triple bond, ($\equiv$ C) = carbon with a double bond; Cp = carbon in fused ring; Naz = azide; Nim = imino.

Group	298 K	400 K	500 K	600 K	800 K	1000 K	1500 K
Nitrogen Groups							
$\equiv$ NazH	18.33	20.47	22.77	24.86	28.34	31.06	35.33
$\equiv$ Nim—(Cb)	12.56						
$\equiv$ Nim—(C)	10.38	13.98	16.53	17.96	19.21	19.25	
$\equiv$ NimH	12.35	19.17	27	32.27	38.22	41.52	
Sulfur Groups							
S—(2Cb)	8.37	8.41	9.38	11.47	15.91	19.72	
S—(2C)	20.89	20.76	21.01	21.22	22.65	23.98	
S—(2S)	19.67	20.93	21.35	21.77	22.19	22.6	
S—(2 $\equiv$ C)	20.05	23.36	23.15	26.33	33.24	40.73	
S—(Cb,S)	12.1	14.19	15.57	17.37	20.01	21.35	
S—(C,Cb)	12.64	14.19	15.53	16.91	19.34	20.93	
S—(C,S)	21.89	22.69	23.06	23.06	22.52	21.43	
S—(C, $\equiv$ C)	17.66	21.26	23.27	24.15	24.57	24.57	
SH—(Cb)	21.43	22.02	23.32	25.24	29.26	32.82	
SH—(CO)	31.94	33.86	33.99	34.2	35.58	34.49	
SH—(C)	24.53	25.95	27.25	28.38	30.56	32.27	
SO—(2Cb)	23.94	38.05	40.6	47.93	47.97	47.09	
SO—(2C)	37.17	41.98	43.95	45.17	45.96	46.76	
SO ₂ —(2Cb)	34.99	46.17	56.72	62.54	66.39	66.81	
SO ₂ —(2C)	48.22	50.1	55.88	59.77	64.38	66.47	
SO ₂ —(2 $\equiv$ C)	48.22	50.1	55.88	59.77	64.38	66.47	
SO ₂ —(Cb,SO ₂)	41.06	48.14	56.59	61.66	65.76	67.1	
SO ₂ —(Cb, $\equiv$ C)	41.4	48.14	55.88	61.16	65.8	66.64	
SO ₂ —(C,Cb)	41.61	48.14	56.3	60.74	65.38	66.64	
S(CN)—(Cb)	39.77						
S(CN)—(C)	46.88						
S(CN)—($\equiv$ C)	59.44						
Boron and Silicon Groups							
Si—(4C)	113.23	134.95	154.5	171.2	198.62	219.72	252.91
SiH ₃ —(C)	-39.64						
Monovalent Ligands							
CH ₂ (CN)—(C)	47.72	56.93	64.04	70.74	80.79	85.81	
CH ₂ (NCS)—(C)	61.95						
CH ₂ (NO ₂)—(C)	52.7	66.22	77.52	86.48	99.58	108.41	
CH(CN)—(2C)	45.21	54	60.69	66.14	72	79.11	
CH(NO ₂)—(2C)	50.19	63.67	74.17	82.08	92.84	99.2	
CH(NO ₂) ₂ —(C)	80.79	101.3	117.2	129.76	146.09	156.13	
C(CN)—(3C)	36.21	46.71	53.96	58.81	64.92	67.77	
C(CN) ₂ —(2C)	61.62	74.47	83.72	90.46	99.54	104.48	
C(NO ₂)—(3C)	41.4	55.84	66.39	73.75	82.92	87.32	
$\equiv$ CH(CHN ₂)	72.42						
$\equiv$ CH(CN)	43.11	50.23	56.09	61.11	68.65	73.67	
$\equiv$ CH(NCS)	51.90						
$\equiv$ CH(NO ₂)	51.49	63.21	72.83	80.37	90.41	97.11	105.9
$\equiv$ C(CN) ₂	56.93	69.28	78.19	84.76	93.51	98.74	
3,4 Member Ring Corrections							
cyclobutane ring	-19.3	-16.28	-13.14	-11.05	-7.87	-5.78	-2.8
cyclobutene ring	-10.59	-9.17	-7.91	-7.03	-6.2	-5.57	-5.11
cyclopropane ring	-12.77	-10.59	-8.79	-7.95	-7.41	-6.78	-6.36
ethylene oxide ring	-8.37	-11.72	-12.56	-10.88	-9.63	-8.63	
ethylene sulfide ring	-11.93	-10.84	-11.13	-12.64	18.09	24.35	
thietane ring	-19.21	-17.5	-16.37	-16.37	-19.25	-23.86	
trimethylene oxide ring	-19.25	-20.93	-17.58	-14.56	-10.88	0.84	
5,6 Member Ring Corrections							
1,4 dioxane ring	-19.21	-20.8	-15.91	-10.97	-6.4	-1.8	
cyclohexane ring	-24.28	-17.16	-12.14	-5.44	4.6	9.21	13.81
cyclohexene ring	-17.92	-12.72	-8.29	-5.99	-1.21	0.33	3.39
cyclopentadiene ring	-14.44	-11.85	-8.96	-6.91	-5.36	-4.35	
cyclopentane ring	-27.21	-23.02	-18.84	-15.91	-11.72	-8.08	-1.55
cyclopentene ring	-25.03	-22.39	-20.47	-17.33	-12.26	-9.46	-4.52
furan ring	-20.51	-18	-15.07	-12.56	-10.88	-10.05	
piperidine ring	-24.7	-19.67	-12.14	-3.77	9.21	17.58	
pyrrolidine ring	-25.83	-23.36	-20.09	-16.74	-12.01	-9.08	
tetrahydrofuran ring	-25.12	-24.28	-20.09	-15.91	-11.3	-7.53	
thiacyclohexane ring	-26.04	-17.83	-9.38	-2.89	3.6	5.4	
thiolane ring	-20.51	-19.55	-15.4	-15.32	-18.46	-23.32	
thiophene ring	-20.51	-19.55	-15.4	-15.32	-18.46	-23.32	

TABLE 2-347 Benson* and CHETAH† Group Contributions for Ideal Gas Heat Capacity (Concluded)

Table-specific nomenclature: Cb = carbon in benzene ring; Ct = carbon with a triple bond, (≡C) = carbon with a double bond; Cp = carbon in fused ring; Naz = azide; Nim = imino.

Group	298 K	400 K	500 K	600 K	800 K	1000 K	1500 K
7 and 8 Member Ring Corrections							
cycloheptane ring	-38.01						
cyclooctane ring	-44.16						
Gauche and 1,5 Repulsion Corrections							
but-2-ene structure C—C=C—C	-5.61	-4.56	-3.39	-2.55	-1.63	-1.09	
but-3-ene structure C—C—C=C	-5.61	-4.56	-3.39	-2.55	-1.63	-1.09	
cis- between 2 t-butyl groups	-5.61	-4.56	-3.39	-2.55	-1.63	-1.09	
cis- involving 1 t-butyl group	-5.61	-4.56	-3.39	-2.55	-1.63	-1.09	
cis- (not with t-butyl group)	-5.61	-4.56	-3.39	-2.55	-1.63	-1.09	
ortho- between Cl atoms	-2.09	5.02	2.09	-2.51	-1.26		
ortho- between F atoms		-0.84	-0.42	1.26	2.93		
other ortho- (nonpolar-nonpolar)	4.69	5.65	5.44	4.9	3.68	2.76	-0.21

*Benson, S. W., et al., *Chem. Rev.*, **69** (1969): 279.

†CHETAH Version 8.0: *The ASTM Computer Program for Chemical Thermodynamic and Energy Release Evaluation (NIST Special Database 16)*.

Prog., **69**, 7 (1973): 83; Lee, B. I., and M. G. Kesler, *AIChE J.*, **21** (1975): 510; Tarakad, R. R., and R. P. Danner, *AIChE J.*, **23** (1977): 944] and thermodynamic differentiation. The most accurate and generally applicable method is that by Ruzicka and Domalski.

Recommended Method Ruzicka-Domalski.

References: Ruzicka, V., and E. S. Domalski, *J. Phys. Chem. Ref. Data*, **22** (1993): 597, 619.

Classification: Group contributions.

Expected Uncertainty: 4 percent.

Applicability: Organic compounds for which group values are available.

Input data: Molecular structure and Table 2-348 values.

Description: Groups are summed to find the temperature coefficients for a cubic polynomial correlation:

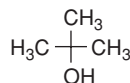
$$\frac{C_p}{R} = A + B\left(\frac{T}{100 \text{ K}}\right) + D\left(\frac{T}{100 \text{ K}}\right)^2 \quad (2-50)$$

$$A = \sum_{i=1}^N n_i a_i \quad B = \sum_{i=1}^N n_i b_i \quad D = \sum_{i=1}^N n_i d_i \quad (2-51)$$

where n_i = number of occurrences of group i and a_i , b_i , d_i = individual group contributions.

Example Estimate the liquid heat capacity for 2-methyl-2-propanol at 340 K.

Structure:



Group contributions:

Group	n_i	a_i	b_i	d_i
C—(3C,O) (alcohol)	1	-44.690	31.769	-4.8791
O—(H)(C)	1	12.952	-10.145	2.6261
C—(3H)(C)	3	3.8452	-0.33997	0.19489
Sum		-20.202	20.604	-1.668

$$\begin{aligned} C_p &= \left(8.3143 \frac{\text{J}}{\text{mol} \cdot \text{K}} \right) \left[-20.202 + 20.604 \left(\frac{340}{100} \right) - 1.668 \left(\frac{340}{100} \right)^2 \right] \\ &= 254.16 \frac{\text{J}}{\text{mol} \cdot \text{K}} \end{aligned}$$

This value is 0.7 percent higher than the DIPPR® 801 recommended value of 252.40 J/(mol·K).

Solids Solid heat capacity increases with increasing temperature and is proportional to T^3 near absolute zero. The heat capacity at a solid-solid phase transition becomes large, and there can be a substantial dif-

ference in the heat capacity of the two equilibrium solid phases that exist on either side of the transition temperature. The heat capacity generally rises steeply with increasing temperature near the triple point.

For a quick estimation of solid heat capacity specifically at 298.15 K, the very simple modification of Kopp's rule [Kopp, H., *Ann. Chem. Pharm. (Liebig)*, **126** (1863): 362] by Hurst and Harrison [Hurst, J. E., and B. K. Harrison, *Chem. Eng. Comm.*, **112** (1992): 21] can be used. At other temperatures and to obtain the temperature dependence of the solid heat capacity, the method given below by Goodman et al. should be used.

Recommended Method 1 Goodman method.

Reference: Goodman, B. T., et al., *J. Chem. Eng. Data*, **49** (2004): 24.

Classification: Group contributions.

Expected uncertainty: 10 percent.

Applicability: Organic compounds for which group values are available.

Input data: Molecular structure and Table 2-349 group values.

Description:

$$\frac{C_p}{\text{J}/(\text{mol} \cdot \text{K})} = \frac{A}{1000} \left(\frac{T}{\text{K}} \right)^{0.79267} \quad (2-52)$$

$$A = \exp \left(6.7796 + \sum_{i=1}^N n_i a_i + \sum_{i=1}^N n_i^2 \beta_i \right) \quad (2-53)$$

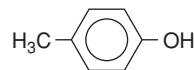
where n_i = number of occurrences of group i

a_i = individual group contribution

β_i = nonlinear correction terms for chain and aromatic carbons

Example Estimate the solid heat capacity for *p*-cresol at 307.93 K.

Structure:



Group contributions:

Group	n_i	a_i	β_i
—CH ₃	1	0.20184	0
Ar—CH≡	4	0.082478	-0.00033
Ar >C≡	2	0.012958	0
—OH	1	0.10341	0

From Eq. (2-53):

$$\begin{aligned} A &= \exp [6.7796 + 0.20184 + (4)(0.082478) + (2)(0.012958) + 0.10341 \\ &\quad + (4)^2(-0.00033)] = 1694.9 \end{aligned}$$

TABLE 2-348 Liquid Heat Capacity Group Parameters for Ruzicka-Domalski Method*

Table-specific nomenclature: Ct refers to a carbon atom with a triple bond; Cb refers to a carbon atom in benzene ring; =C refers to a carbon atom with a double bond; Cp refers to a carbon atom in a fused benzene ring; =C= refers to an allenic carbon atom.

Group Definition	<i>a</i>	<i>b</i>	<i>d</i>	<i>T</i> range (K)	Group Definition	<i>a</i>	<i>b</i>	<i>d</i>	<i>T</i> range (K)
Hydrocarbon Groups					Nitrogen Groups				
C—(3H,C)	3.8452	-0.33997	0.19489	80–490	N—(H,2C)	-0.10987	0.73024	0.89325	170–400
C—(2H,2C)	2.7972	-0.054967	0.10679	80–490	N—(3C)	4.5942	-2.2134	0.55316	160–360
C—(H,3C)	-0.42867	0.93805	0.0029498	85–385	N—(H,C,Cb)	0.49631	3.4617	-0.57161	240–380
C—(4C)	-2.9353	1.4255	-0.085271	145–395	N—(2C,Cb)	-0.23640	16.260	-2.5258	285–390
=C—(2H)	4.1763	-0.47392	0.099928	90–355	N—(C,2Cb)	4.5942	-2.2134	0.55316	160–360
=C—(H,C)	4.0749	-1.0735	0.21413	90–355	Cb—(N)	-0.78169	1.5059	-0.25287	240–455
=C—(2C)	1.9570	-0.31938	0.11911	140–315	N—(2H,N)	6.8050	-0.72563	0.15634	215–465
=C—(H,=C)	3.6968	-1.6037	0.55022	130–305	N—(H,C,N)	1.1411	3.5981	-0.69350	205–300
=C—(C,=C)	1.0679	-0.50952	0.33607	130–305	N—(2C,N)	-1.0570	4.0038	-0.71494	205–300
C—(3H,=C)	3.8452	-0.33997	0.19489	80–490	N—(H,Cb,N)	-0.74531	3.6258	-0.53306	295–385
C—(2H,C,=C)	2.0268	-0.20137	0.11624	90–355	C—(2H,C,CN)	11.976	-2.4886	0.52358	185–345
C—(H,2C,=C)	-0.87558	0.82109	0.18415	110–300	C—(3C,CN)	2.5774	3.5218	-0.58466	295–345
C—(3C,=C)	-4.8006	2.6004	-0.040688	165–295	=C—(H,CN)	9.0789	-0.86929	0.32986	195–345
C—(2H,2=C)	1.4973	-0.46017	0.52861	130–300	Cb—(CN)	1.9389	3.0269	-0.47276	265–480
Ct—(H)	9.1633	-4.6695	1.1400	150–275	C—(2H,C,NO ₂)	18.520	-5.4568	1.05080	190–300
Ct—(C)	1.4822	1.0770	-0.19489	150–285	O—(C,NO ₂)	-2.0181	10.505	-1.83980	180–350
=C=	3.0880	-0.62917	0.25779	140–315	Cb—(NO ₂)	15.277	-4.4049	0.71161	280–415
Ct—(Cb)	12.377	-7.5742	1.3760	230–550	N—(H,2Cb) (pyrrole)	-7.3662	6.3622	-0.68137	255–450
Cb—(H)	2.2609	-0.2500	0.12592	180–670	Nb—(2Cb)	0.84237	1.25560	-0.20336	210–395
Cb—(C)	1.5070	-0.13366	0.011799	180–670	Oxygen Groups				
Cb—(=C)	-5.7020	5.8271	-1.2013	230–550	O—(H,C)	12.952	-10.145	2.6261	155–505
Cb—(Cb)	5.8685	-0.86054	-0.063611	295–670	O—(H,C) (diol)	5.2302	-1.5124	0.54075	195–475
C—(2H,C,Ct)	2.0268	-0.20137	0.11624	90–355	O—(H,Cb) (diol)	5.2302	-1.5124	0.54075	195–475
C—(3H,Ct)	3.8452	-0.33997	0.19489	80–490	O—(H,Cb)	-7.9768	8.10450	-0.87263	285–400
C—(3H,Cb)	3.8452	-0.33997	0.19489	80–490	C—(3H,O)	3.8452	-0.33997	0.19489	80–490
C—(2H,C,Cb)	1.4142	0.56919	0.0053465	180–470	C—(2H,C,O)	1.4596	1.4657	-0.27140	135–505
C—(H,2C,Cb)	-0.10495	1.0141	-0.071918	180–670	C—(2H,Cb,O)	-35.127	28.409	-4.9593	260–460
C—(3C,Cb)	1.2367	-1.3997	0.41385	220–295	C—(2H,=C,O)	-35.127	28.409	-4.9593	260–460
C—(2H,2Cb)	-18.583	11.344	-1.4108	300–420	C—(H,2C,O) (alcohol)	2.2209	-1.4350	0.69508	185–460
C—(H,3Cb)	-46.611	24.987	-3.0249	375–595	C—(H,2C,O) (ether, ester)	0.98790	0.39403	-0.016124	130–170
=C—(H,Cb)	3.6968	-1.6037	0.55022	130–305	C—(3C,O) (alcohol)	-44.690	31.769	-4.8791	200–355
=C—(C,Cb)	1.0679	-0.50952	0.33607	130–305	C—(3C,O) (ether, ester)	-3.3182	2.6317	-0.44354	170–310
Cp—(Cp,2Cb)	-3.5572	2.8308	-0.39125	250–510	O—(2C)	5.0312	-1.5718	0.37860	130–350
Cp—(2Cp,Cb)	-11.635	6.4068	-0.78182	370–510	O—(C,Cb)	-22.5240	13.1150	-1.44210	320–350
Cp—(3Cp)	26.164	-11.353	1.2756	385–480	O—(2Cb)	-4.5788	0.94150	0.31655	300–535
Halogen Groups					C—(2H,2O)	1.0852	1.5402	-0.31693	170–310
C—(C,3F)	15.423	-9.2464	2.8647	125–345	C—(2C,2O)	-12.955	9.10270	-1.53670	275–335
C—(2C,2F)	-8.9527	10.550	-1.9986	125–345	Cb—(O)	-1.0686	3.52210	-0.79259	285–530
C—(C,3Cl)	8.5430	2.6966	-0.42564	245–310	C—(3H,CO)	3.8452	-0.33997	0.19489	80–490
C—(H,C,2Cl)	10.880	-0.35391	0.08488	180–355	C—(2H,C,CO)	6.6782	-2.44730	0.47121	180–465
C—(2H,C,Cl)	9.6663	-1.8601	0.41360	140–360	C—(H,2C,CO)	3.92380	-2.12100	0.49646	185–375
C—(2H,=C,Cl)	9.6663	-1.8601	0.41360	140–360	C—(3C,CO)	-2.2681	1.75580	-0.25674	225–360
C—(H,2C,Cl)	-2.0600	5.3281	-0.82721	275–360	CO—(H,C)	-3.82680	7.67190	-1.27110	180–430
C—(2H,C,Br)	6.3944	-0.10298	0.19403	168–360	CO—(H,=C)	-8.00240	3.63790	-0.15377	220–430
C—(H,2C,Br)	10.784	-2.4754	0.33288	190–420	CO—(H,Cb)	-8.00240	3.63790	-0.15377	220–430
C—(2H,C,I)	0.037620	5.6204	-0.92054	245–340	CO—(2C)	5.4375	0.72091	-0.18312	185–380
C—(C,2Cl,F)	13.532	-3.2794	0.80145	240–420	CO—(C,=C)	41.507	-32.632	6.0326	275–355
C—(C,Cl,2F)	7.2295	0.41759	0.15892	180–420	CO—(C,Cb)	-47.21100	24.36800	-2.82740	300–465
C—(C,Br,2F)	8.7956	-0.19165	0.24596	165–415	CO—(H,O)	13.11800	16.12000	-5.12730	280–340
=C—(H,Cl)	7.1564	-0.84442	0.27199	120–300	CO—(C,O)	29.24600	3.42610	-2.89620	180–445
=C—(2F)	7.6646	-2.0750	0.82003	120–240	CO—(=C,O)	41.61500	-12.78900	0.53631	195–350
=C—(2Cl)	9.3249	-1.2478	0.44241	155–300	CO—(O,CO)	23.99000	6.25730	-3.24270	320–345
=C—(Cl,F)	7.8204	-0.69005	0.19165	120–240	O—(C,O)	-21.43400	-4.01640	3.05310	175–440
Cb—(F)	3.0794	0.46959	-0.0055745	210–365	O—(H,CO)	-27.58700	-0.16485	2.74830	230–500
Cb—(Cl)	4.5479	0.22250	-0.0097873	230–460	=C—(H,CO)	-0.01080	15.14800	-3.04360	195–355
Cb—(Br)	2.2857	2.2573	-0.40942	245–370	=C—(C,CO)	-12.81800	15.99700	-3.05670	195–430
Cb—(I)	2.9033	2.9763	-0.62960	250–320	Cb—(CO)	12.15100	-1.67050	-0.12758	175–500
C—(Cb,3F)	7.4477	-0.92230	0.39346	210–365	CO—(Cb,O)	16.58600	5.44910	-2.68490	175–500
C—(2H,Cb,Cl)	16.752	-6.7938	1.2520	245–345	Sulfur Groups				
Nitrogen Groups					C—(3H,S)	3.84520	-0.33997	0.19489	80–490
C—(3H,N)	3.8452	-0.33997	0.19489	80–490	C—(2H,C,S)	1.54560	0.88228	-0.08349	130–390
C—(2H,C,N)	2.4555	1.0431	-0.24054	190–375	C—(H,2C,S)	-1.64300	2.30700	-0.31234	150–390
C—(2H,Cb,N)	2.4555	1.0431	-0.24054	190–375	C—(3C,S)	-5.38250	4.50230	-0.72356	190–365
C—(H,2C,N)	2.6322	-2.0135	0.45109	240–370	Cb—(S)	-4.45070	4.43240	-0.75674	260–375
C—(3C,N)	1.9630	-1.7235	0.31086	255–375	S—(H,C)	10.99400	-3.21130	0.47368	130–380
N—(2H,C)	8.2758	-0.18365	0.035272	185–455	S—(H,Cb)	10.99400	-3.21130	0.47368	130–380
N—(2H,Cb)	8.2758	-0.18365	0.035272	185–455	S—(2C)	9.23060	-3.00870	0.45625	165–390
					S—(2Cb)	9.23060	-3.00870	0.45625	165–390
					S—(C,S)	6.65900	-1.35570	0.17938	170–350
					S—(Cb,S)	9.23060	-3.00870	0.45625	165–390
					S—(2Cb) (thiophene)	3.84610	0.36718	-0.06131	205–345

TABLE 2-348 Liquid Heat Capacity Group Parameters for Ruzicka-Domalski Method* (Concluded)

Table-specific nomenclature: Ct refers to a carbon atom with a triple bond; Cb refers to a carbon atom in benzene ring; =C refers to a carbon atom with a double bond; Cp refers to a carbon atom in a fused benzene ring; =C= refers to an allenic carbon atom.

Group Definition	<i>a</i>	<i>b</i>	<i>d</i>	<i>T</i> range (K)	Group Definition	<i>a</i>	<i>b</i>	<i>d</i>	<i>T</i> range (K)
Ring Strain Contributions					decahydronaphthalene	-6.8984	0.66846	-0.070012	235–485
Hydrocarbons (ring strain)					hexahydroindan	-3.9271	-0.29239	0.048561	210–425
cyclopropane	4.4297	-4.3392	1.0222	155–240	dodecahydrofluorene	-19.687	8.8265	-1.4031	315–485
cyclobutane	1.2313	-2.8988	0.75099	140–300	tetradecahydrophenanthrene	-0.67632	-1.4753	-0.13087	315–485
cyclopentane (unsub)	-0.33642	-2.8663	0.70123	180–300	hexadecahydropyrene	61.213	-30.927	3.2269	310–485
cyclopentane (sub)	0.21983	-1.5118	0.28172	135–365	Nitrogen compounds				
cyclohexane	-2.0097	-0.72656	0.14758	145–485	ethyleneimine	15.281	-2.3360	-0.13720	195–330
cycloheptane	-11.460	4.9507	-0.74754	270–300	pyrrolidine	12.703	1.3109	-1.18130	170–400
cyclooctane	-4.1696	0.52991	-0.018423	295–320	piperidine	25.681	-7.0966	0.14304	265–370
spiropentane	5.9700	-3.7965	0.74612	175–310	Oxygen compounds				
cyclopentene	0.21433	-2.5214	0.63136	140–300	ethylene oxide	6.8459	-5.8759	1.2408	135–325
cyclohexene	-1.2086	-1.5041	0.42863	160–320	trimethylene oxide	-7.0148	7.3764	-2.1901	185–300
cycloheptene	-5.6817	1.5073	-0.19810	220–300	1, 3-dioxolane	-2.3985	-0.48585	0.10253	175–300
cyclooctene	-14.885	7.4878	-1.0879	260–330	furan	9.6704	-2.8138	0.11376	190–305
cyclohexadiene	-8.9683	6.4959	-1.5272	170–300	tetrahydrofuran	3.2842	-5.8260	1.2681	160–320
cyclooctadiene	-7.2890	3.1119	-0.43040	205–320	tetrahydropyran	-13.017	3.7416	-0.15622	295–325
cycloheptatriene	-8.7885	8.2530	-2.4573	200–310	Sulfur compounds				
cyclooctatetraene	-12.914	13.583	-4.0230	275–330	thiacyclobutane	-0.73127	-1.3426	0.40114	200–320
indan	-6.1414	3.5709	-0.48620	170–395	thiacyclopentane	-3.2899	0.38399	0.089358	170–390
1H-indene	-3.6501	2.4707	-0.60531	280–375	thiacyclohexane	-12.766	5.2886	-0.59558	295–340
tetrahydronaphthalene	-6.3861	2.6257	-0.19578	250–320					

*Ruzicka, V., and E. S. Domalski, *J. Phys. Chem. Ref. Data*, **22** (1993): 597, 619.

From Eq. (2-52):

$$C_p = \frac{1694.9}{1000} (307.93)^{0.79267} \frac{\text{J}}{\text{mol} \cdot \text{K}} = 159.1 \frac{\text{J}}{\text{mol} \cdot \text{K}}$$

This value is 2.5 percent higher than the DIPPR® 801 recommended value of 155.2 J/(mol·K).

Recommended Method 2 Modified Kopp's rule.

Reference: Kopp, H., *Ann. Chem. Pharm. (Liebig)*, **126** (1863): 362; Hurst, J. E., and B. K. Harrison, *Chem. Eng. Comm.*, **112** (1992): 21.

Classification: Group contributions.

Expected uncertainty: 10 percent.

Applicability: 298.15 K; organic compounds that are solids at 298.15 K.

Input data: Compound chemical formula and element contributions of Table 2-350.

Description:

$$\frac{C_p}{\text{J/(mol} \cdot \text{K)}} = \sum_{E=1}^N n_E \Delta_E \quad (2-54)$$

where N = number of different elements in compound
 n_E = number of occurrences of element E in compound
 Δ_E = contribution of element E from Table 2-350

Example Estimate the solid heat capacity at 298.15 K for dibenzothio-
 phene.

Structure: $\text{C}_{12}\text{H}_8\text{S}$.

Group values from Table 2-350:

$$\Delta_C = 10.89 \quad \Delta_H = 7.56 \quad \Delta_S = 12.36$$

Calculation using Eq. (2-54):

$$C_p = (12)(10.89) + (8)(7.56) + (1)(12.36) = 203.52 \text{ J/(mol} \cdot \text{K)}$$

This value is 2.5 percent higher than the DIPPR® 801 recommended value of 198.45 J/(mol·K).

Mixtures The heat capacity of liquid and vapor mixtures can be estimated as mole fraction averages of the pure-component values

$$C_{p,m} = \sum_{i=1}^C x_i C_{p,i} \quad (2-55)$$

This neglects the excess heat capacity, which if available can be added to the mole fraction average to improve the estimated value.

DENSITY

Density is defined as the mass of a substance per unit volume. Density is given in kg/m³ in SI units, but lb_m/ft³ and g/cm³ are common AES and cgs units, respectively. Other commonly used forms of density include *molar density* (density divided by molecular weight) in kmol/m³, *relative density* (density relative to water at 15°C), and the older term *specific gravity* (density relative to water at 60°F). Often the inverse of density, *specific volume*, and the inverse of molar density, *molar volume*, are correlated and used to convey equivalent information.

Gases Gases/vapors are compressible and their densities are strong functions of both temperature and pressure. Equations of state (EoS) are commonly used to correlate molar densities or molar volumes. The most accurate EoS are those developed for specific fluids with parameters regressed from all available data for that fluid. Super EoS are available for some of the most industrially important gases and may contain 50 or more constants specific to that chemical. Different predictive methods may be used for gas densities depending upon the conditions:

1. At *very low densities* (high temperatures, generally above the critical, and very low pressures, generally below a few bar), the ideal gas EoS

$$Z \equiv \frac{PV}{RT} = 1 \quad (2-56)$$

may be applied.

2. At *moderate densities* (below 40 percent of the critical density), the virial equation truncated after the second virial coefficient

$$Z = 1 + \frac{B(T)}{V} \quad (2-57)$$

may be used. Second virial coefficients $B(T)$ are available in the DIPPR® 801 database for many chemicals and can be estimated for others by using the Tsonopoulos method.

TABLE 2-349 Group Values and Nonlinear Correction Terms for Estimation of Solid Heat Capacity with the Goodman et al.* Method

Group	Description	a_i	Group	Description	a_i
—CH ₃	methyl	0.20184	—CO ₃ —	carbonate	0.2517
>CH ₂	methylene	0.11644	—NH ₂	primary amine	0.056138
>CH—	secondary C	0.030492	>NH	secondary amine	−0.00717
>C<	tertiary C	−0.04064	>N—	tertiary amine	−0.01661
CH ₂ ==	terminal alkene	0.18511	==NH	double-bond NH	0.17689
—CH==	alkene	0.11224	#N	nitrile	0.015355
>C==	subst. alkene	0.028794	—N==N—	diazide	0.3687
==C==	allene	0.053464	—NO ₂	nitro	0.23327
#CH	terminal alkyne	−0.02914	—N≡C≡O	isocyanate	0.2698
#C—	alkyne	0.13298	—SH	thiol/mercaptan	0.21123
Ar—CH==	arom. C	0.082478	—S—	sulfide	0.14232
Ar>C==	subst. arom. C	0.012958	—SS—	disulfide	0.31457
Ar—O—	furan O	0.066027	==S	sulfur double bond	0.13753
Ar—N==	pyridine N	0.056641	>S=O	sulfoxide	0.040002
Ar>N—	subst. pyrrole N	0.008938	—F	fluoride	0.15511
Ar—NH—	pyrrole N	−0.05246	—Cl	chloride	0.16995
Ar—S—	thiophene S	0.090926	—Br	bromide	0.19112
—O—	ether	0.064068	—I	iodide	0.11318
—OH	alcohol	0.10341	>Si<	silane	0.12213
—COH	aldehyde	0.15699	>Si(O—)—	linear siloxane	0.10125
>C=O	ketone	0.12939	cyc >Si(O—)—	cyclic siloxane	0.063438
—COO—	ester	0.13686	P(=O)(O—) ₃	phosphate	0.15016
—COOH	acid	0.21019	>P—	phosphine	0.069602
—COOCO—	anhydride	0.33091	>P(=O)—	phosphine oxide	0.21875

Nonlinear Terms		
Groups	β_i	Usage
>CH ₂	−0.00188	Methylene
Ar==CH—	−0.00033	Aromatic carbon

*Goodman, B. T., W. V. Wilding, J. L. Oscarson, and R. L. Rowley, *J. Chem. Eng. Data*, **49** (2004): 24**Recommended Method** Tsonopoulos method.Reference: Tsonopoulos, C., *AIChE J.*, **20** (1974): 263; **21** (1975): 827; **24** (1978): 1112.

Classification: Corresponding states.

Expected uncertainty: 8 percent for $B(T)$.

Applicability: Nonpolar organic compounds and some classes of polar compounds.

Input data: Class of fluid, ω , P_c , T_c , and μ .

Description:

$$\frac{BP_c}{RT_c} = B^{(0)} + \omega B^{(1)} + B^{(2)} \quad (2-58)$$

where ω = acentric factor P_c = critical pressure T_c = critical temperature

$$B^{(0)} = 0.1445 - \frac{0.330}{T_r} - \frac{0.1385}{T_r^2} - \frac{0.0121}{T_r^3} - \frac{0.000607}{T_r^8} \quad (2-59)$$

TABLE 2-350 Element Contributions to Solid Heat Capacity for the Modified Kopp's Rule*†

Element	Δ_E	Element	Δ_E	Element	Δ_E
C	10.89	Ba	32.37	Mo	29.44
H	7.56	Be	12.47	Na	26.19
O	13.42	Ca	28.25	Ni	25.46
N	18.74	Co	25.71	Pb	31.60
S	12.36	Cu	26.92	Si	17.00
F	26.16	Fe	29.08	Sr	28.41
Cl	24.69	Hg	27.87	Ti	27.24
Br	25.36	K	28.78	V	29.36
I	25.29	Li	23.25	W	30.87
Al	18.07	Mg	22.69	Zr	26.82
B	10.10	Mn	28.06	All others	26.63

*Kopp, H., *Ann. Chem. Pharm. (Liebig)*, **126** (1863): 362.†Hurst, J. E., and B. K. Harrison, *Chem. Eng. Comm.*, **112** (1992): 21.

$$B^{(1)} = 0.0637 + \frac{0.331}{T_r^2} - \frac{0.423}{T_r^3} - \frac{0.008}{T_r^8} \quad (2-60)$$

$$B^{(2)} = \frac{a}{T_r^6} - \frac{b}{T_r^8} \quad (2-61)$$

$$\mu_r = \left(\frac{\mu}{D} \right)^2 \left(\frac{P_c}{\text{bar}} \right) \left(\frac{T_c}{K} \right)^{-2} \quad (2-62)$$

where μ = dipole moment. The values of a and b used in Eq. (2-61) depend upon the class of fluid, as given in the table below:

Class	a	b
Nonpolar fluids	0	0
Ketones, aldehydes, nitriles, ethers, esters, NH ₃ , H ₂ S, HCN	−21.4 μ_r − 4.308 × 10 ¹⁰ μ_r^8	0
Monoalkylhalides, mercaptans, sulfides	−2.188 × 10 ¹⁶ μ_r^4 − 7.831 × 10 ¹⁰ μ_r^8	0
1-Alcohols except methanol	0.0878	0.00908 + 69.57 μ_r
Methanol	0.0878	0.0525

Example Estimate the molar volume of ammonia at 430 K and 2.82 MPa.Input properties: Recommended values from the DIPPR® 801 database are $T_c = 405.65$ K, $P_c = 11.28$ MPa, $\mu = 1.469$ D, and $\omega = 0.252608$.

Reduced conditions:

$$T_r = (430 \text{ K}) / (405.65 \text{ K}) = 1.06$$

$$P_r = (2.82 \text{ MPa}) / (11.28 \text{ MPa}) = 0.25$$

$$\mu_r = (1.469)^2 (112.8) / (405.65)^2 = 0.0014793$$

Second virial coefficient from Eqs. (2-59) to (2-61):

$$B^{(0)} = 0.1445 - 0.330/1.06 - 0.1385/(1.06)^2 - 0.0121/(1.06)^3$$

$$-0.000607/(1.06)^8 = -0.301$$

$$B^{(1)} = 0.0637 + 0.331/(1.06)^2 - 0.423/(1.06)^3 - 0.008/(1.06)^8 = -0.00189$$

$$a = (-21.4)(0.0014793) - (4.308 \times 10^{10})(0.0014793)^8 = -0.033$$

$$b = 0$$

$$B^{(2)} = (-0.033)/(1.06)^6 = -0.023$$

From Eq. (2-58):

$$BP_c/(RT_c) = -0.301 - (0.252608)(0.00189) - 0.023 = -0.324$$

$$B = (-0.324)[0.008314 \text{ m}^3 \cdot \text{MPa}/(\text{kmol} \cdot \text{K})](405.65 \text{ K})/(11.28 \text{ MPa}) \\ = -0.097 \text{ m}^3/\text{kmol}$$

Molar volume from Eq. (2-56):

$$V = \frac{RT}{P} \left(1 + \frac{B}{V} \right) = \frac{(0.0083143 \text{ m}^3 \cdot \text{MPa}/(\text{kmol} \cdot \text{K}))(430 \text{ K})}{2.82 \text{ MPa}} \left(1 + \frac{-0.097 \text{ m}^3/\text{kmol}}{V} \right) \\ = 1.162 \text{ m}^3/\text{kmol}$$

Note that the ideal gas value, $1.268 \text{ m}^3/\text{kmol}$, deviates by 9.1 percent from this more accurate value. The truncated virial EoS should be valid for this density since $\rho = V^{-1} = 0.86 \text{ kmol}/\text{m}^3$ is much less than 40 percent of the critical density (the DIPPR[®] 801 recommended value for the critical density is $13.8 \text{ kmol}/\text{m}^3$).

3. For higher gas densities, the Lee-Kesler method described below provides excellent predictions for nonpolar and slightly polar fluids. Extended four-parameter corresponding-states methods are available for polar and slightly associating compounds.

Recommended Method Lee-Kesler method.

Reference: Lee, B. I., and M. G. Kesler, *AIChE J.*, **21** (1975): 510.

Classification: Corresponding states.

Expected uncertainty: 1 percent except near the critical point where errors can be up to 30 percent.

Applicability: Nonpolar and moderately polar compounds. An extended Lee-Kesler method, not described here, may be used for polar and slightly associating compounds [Wilding, W. V., and R. L. Rowley, *Int. J. Thermophys.*, **8** (1986): 525].

Input data: T_c , P_c , ω , $Z^{(0)}$, $Z^{(1)}$.

Description:

$$Z = Z^{(0)} + \omega Z^{(1)} \quad (2-63)$$

where Z = compressibility factor

$Z^{(0)}$ = compressibility factor of simple fluid obtained from Table 2-351.

$Z^{(1)}$ = deviation from simple fluid obtained from Table 2-352.

Analytical expressions for $Z^{(0)}$ and $Z^{(1)}$ can also be generated by using

$$Z^{(0)} = Z_0 \quad Z^{(1)} = \frac{Z_1 - Z_0}{0.3978} \quad (2-64)$$

where Z_0 and Z_1 are determined from

$$Z_1 = \frac{PV_r}{T_r} = 1 + \frac{B}{V_r} + \frac{C}{V_r^2} + \frac{D}{V_r^3} + \frac{c_4}{T_r^3 V_r^2} \left(\beta + \frac{\gamma}{V_r^2} \right) \exp \left(\frac{-\gamma}{V_r^2} \right) \quad (2-65)$$

$$B = b_1 - \frac{b_2}{T_r} - \frac{b_3}{T_r^2} - \frac{b_4}{T_r^3}$$

$$C = c_1 - \frac{c_2}{T_r} + \frac{c_3}{T_r^2}$$

$$D = d_1 + \frac{d_2}{T_r}$$

as applied to the simple reference fluid and to the acentric reference fluid (*n*-octane), respectively. The constants for Eq. (2-65) for the two reference fluids are given in Table 2-353.

Example Estimate the molar volume of saturated decane vapor at 540.5 K. Input properties: Recommended values from the DIPPR[®] 801 database are $T_c = 617.7 \text{ K}$, $P_c = 2.11 \text{ MPa}$, $P^*(540.5 \text{ K}) = 0.6799 \text{ MPa}$ (vapor pressure), and $\omega = 0.492328$.

Reduced conditions:

$$T_r = (540.5 \text{ K})/(617.7 \text{ K}) = 0.875$$

$$P_r = (0.6799 \text{ MPa})/(2.11 \text{ MPa}) = 0.322$$

LK compressibility factor: Since vapor phase values are needed, the appropriate values from Tables 2-351 and 2-352 that can be used to double-interpolate are

$Z^{(0)}$		
$T_r \backslash P_r$	0.2	0.4
0.85	0.8810	(0.7222)
0.90	0.9015	0.7800

$Z^{(1)}$		
$T_r \backslash P_r$	0.2	0.4
0.85	-0.0715	(-0.1503)
0.90	-0.0442	-0.1118

Double linear interpolation within these values gives $Z^{(0)} = 0.8058$ and $Z^{(1)} = -0.1025$.

From Eq. (2-63):

$$Z = 0.8058 + (0.492328)(-0.1025) = 0.7553$$

Note: If the analytical form available in Eq. (2-65) is used, the following more accurate values are obtained: $Z^{(0)} = 0.8131$, $Z^{(1)} = -0.1067$, and $Z = 0.7606$.

Molar volume:

$$V = \frac{ZRT}{P} = \frac{(0.7553) \left(0.0083143 \frac{\text{m}^3 \cdot \text{MPa}}{\text{kmol} \cdot \text{K}} \right) (540.5 \text{ K})}{0.6799 \text{ MPa}} = 4.992 \frac{\text{m}^3}{\text{kmol}}$$

4. Cubic EoS can be used to obtain both vapor and liquid densities as an alternative method to those mentioned above.

Recommended Method Cubic EoS.

Classification: Empirical extension of theory.

Expected uncertainty: Varies depending upon compound and conditions, but a general expectation is perhaps 10 to 20 percent.

Applicability: Nonpolar and moderately polar compounds.

Input data: T_c , P_c , ω .

Description: The more common cubic EoS can be written in the form

$$Z = \frac{V}{V-b} - \frac{V}{V^2 + \delta V + \epsilon} \frac{a \alpha(T_r)}{RT} \quad (2-66)$$

where a , b , δ , and ϵ are constants that depend upon the model EoS chosen, as does the temperature dependence of the function $\alpha(T_r)$. Definitions of these constants and $\alpha(T_r)$ for some of the more commonly used EoS models are shown in Table 2-354. The corresponding relations for many other EoS models in this same form are available [Soave, G., *Chem. Eng. Sci.*, **27** (1972): 1197]. The independent parameters a and b in these models can be regressed from experimental data to correlate densities or obtained from known critical constants to predict density data. In the latter case, the relationships between a and b and the critical constants shown in Table 2-354 were obtained from the critical point requirements

$$\left(\frac{\partial P}{\partial V} \right)_{T=T_c} = 0 = \left(\frac{\partial^2 P}{\partial V^2} \right)_{T=T_c} \quad (2-67)$$

TABLE 2-351 Simple Fluid Compressibility Factors $Z^{(0)}$ Values in parentheses are for the opposite phase and may be used to interpolate to or near the phase boundary [PGL4; Wilding, W. V., J. K. Johnson, and R. L. Rowley, *Int. J. Thermophys.*, **8**(1987):717].

$Tr \backslash Pr$	0.010	0.050	0.100	0.200	0.400	0.600	0.800	1.000	1.200	1.500	2.000	3.000	5.000	7.000	10.000
0.30	0.0029	0.0145	0.0290	0.0579	0.1158	0.1737	0.2315	0.2892	0.3470	0.4335	0.5775	0.8648	1.4366	2.0048	2.8507
0.35	0.0026	0.0130	0.0261	0.0522	0.1043	0.1564	0.2084	0.2604	0.3123	0.3901	0.5195	0.7775	1.2902	1.7987	2.5539
0.40	0.0024	0.0119	0.0239	0.0477	0.0953	0.1429	0.1904	0.2379	0.2853	0.3563	0.4744	0.7095	1.1758	1.6373	2.3211
0.45	0.0022	0.0110	0.0221	0.0442	0.0882	0.1322	0.1762	0.2200	0.2638	0.3294	0.4384	0.6551	1.0841	1.5077	2.1338
	(0.9648)														
0.50	0.0021	0.0103	0.0207	0.0413	0.0825	0.1236	0.1647	0.2056	0.2465	0.3077	0.4092	0.6110	1.0094	1.4017	1.9801
	(0.9741)	(0.8699)													
0.55	0.9804	0.0098	0.0195	0.0390	0.0778	0.1166	0.1553	0.1939	0.2323	0.2899	0.3853	0.5747	0.9475	1.3137	1.8520
	(0.0020)	(0.9000)	(0.7995)												
0.60	0.9849	0.0093	0.0186	0.0371	0.0741	0.1109	0.1476	0.1842	0.2207	0.2753	0.3657	0.5446	0.8959	1.2398	1.7440
	(0.0019)	(0.9211)	(0.8405)												
0.65	0.9881	0.9377	0.0178	0.0356	0.0710	0.1063	0.1415	0.1765	0.2113	0.2634	0.3495	0.5197	0.8526	1.1773	1.6519
	(0.0018)	(0.0089)	(0.8707)	(0.7367)											
0.70	0.9904	0.9504	0.8958	0.0344	0.0687	0.1027	0.1366	0.1703	0.2038	0.2538	0.3364	0.4991	0.8161	1.1241	1.5729
		(0.0086)	(0.0172)	(0.7805)											
0.75	0.9922	0.9598	0.9165	0.0336	0.0670	0.1001	0.1330	0.1656	0.1981	0.2464	0.3260	0.4823	0.7854	1.0787	1.5047
		(0.0085)	(0.0169)	(0.8181)	(0.6122)										
0.80	0.9935	0.9669	0.9319	0.8539	0.0661	0.0985	0.1307	0.1626	0.1942	0.2411	0.3182	0.4690	0.7598	1.0400	1.4456
			(0.0168)	(0.0332)	(0.6659)	(0.4746)									
0.85	0.9946	0.9725	0.9436	0.8810	0.0661	0.0983	0.1301	0.1614	0.1924	0.2382	0.3132	0.4591	0.7388	1.0071	1.3943
			(0.0336)	(0.7322)	(0.5346)										
0.90	0.9954	0.9768	0.9528	0.9015	0.7800	0.1006	0.1321	0.1630	0.1935	0.2383	0.3114	0.4527	0.7220	0.9793	1.3496
			(0.0364)	(0.0685)	(0.6040)	(0.4034)									
0.93	0.9959	0.9790	0.9573	0.9115	0.8059	0.6635	0.1359	0.1664	0.1963	0.2405	0.3122	0.4507	0.7138	0.9648	1.3257
					(0.7350)	(0.1047)	(0.4499)								
0.95	0.9961	0.9803	0.9600	0.9174	0.8206	0.6967	0.1410	0.1705	0.1998	0.2432	0.3138	0.4501	0.7092	0.9561	1.3108
					(0.0822)	(0.1116)	(0.4853)								
0.97	0.9963	0.9815	0.9625	0.9227	0.8338	0.7240	0.5580	0.1779	0.2055	0.2474	0.3164	0.4504	0.7052	0.9480	1.2968
						(0.1312)	(0.1532)								
0.98	0.9965	0.9821	0.9637	0.9253	0.8398	0.7360	0.5887	0.1844	0.2097	0.2503	0.3182	0.4508	0.7035	0.9442	1.2901
							(0.1703)								
0.99	0.9966	0.9826	0.9648	0.9277	0.8455	0.7471	0.6138	0.1959	0.2154	0.2538	0.3204	0.4514	0.7018	0.9406	1.2835
							(0.2324)								
1.00	0.9967	0.9832	0.9659	0.9300	0.8509	0.7574	0.6353	0.2901	0.2237	0.2583	0.3229	0.4522	0.7004	0.9372	1.2772
1.01	0.9968	0.9837	0.9669	0.9322	0.8561	0.7671	0.6542	0.4648	0.2370	0.2640	0.3260	0.4533	0.6991	0.9339	1.2710
1.02	0.9969	0.9842	0.9679	0.9343	0.8610	0.7761	0.6710	0.5146	0.2629	0.2715	0.3297	0.4547	0.6980	0.9307	1.2650
1.05	0.9971	0.9855	0.9707	0.9401	0.8743	0.8002	0.7130	0.6026	0.4437	0.3131	0.3452	0.4604	0.6956	0.9222	1.2481
1.10	0.9975	0.9874	0.9747	0.9485	0.8930	0.8323	0.7649	0.6880	0.5984	0.4580	0.3953	0.4770	0.6950	0.9110	1.2232
1.15	0.9978	0.9891	0.9780	0.9554	0.9081	0.8576	0.8032	0.7443	0.6803	0.5798	0.4760	0.5042	0.6987	0.9033	1.2021
1.20	0.9981	0.9904	0.9808	0.9611	0.9205	0.8779	0.8330	0.7858	0.7363	0.6605	0.5605	0.5425	0.7069	0.8990	1.1844
1.30	0.9985	0.9926	0.9852	0.9702	0.9396	0.9083	0.8764	0.8438	0.8111	0.7624	0.6908	0.6344	0.7358	0.8998	1.1580
1.40	0.9988	0.9942	0.9884	0.9768	0.9534	0.9298	0.9062	0.8827	0.8595	0.8256	0.7753	0.7202	0.7761	0.9112	1.1419
1.50	0.9991	0.9954	0.9909	0.9818	0.9636	0.9456	0.9278	0.9103	0.8933	0.8689	0.8328	0.7887	0.8200	0.9297	1.1339
1.60	0.9993	0.9964	0.9928	0.9856	0.9714	0.9575	0.9439	0.9308	0.9180	0.9000	0.8738	0.8410	0.8617	0.9518	1.1320
1.70	0.9994	0.9971	0.9943	0.9886	0.9775	0.9667	0.9563	0.9463	0.9367	0.9234	0.9043	0.8809	0.8984	0.9745	1.1343
1.80	0.9995	0.9977	0.9955	0.9910	0.9823	0.9739	0.9659	0.9583	0.9511	0.9413	0.9275	0.9118	0.9297	0.9961	1.1391
1.90	0.9996	0.9982	0.9964	0.9929	0.9861	0.9796	0.9735	0.9678	0.9624	0.9552	0.9456	0.9359	0.9557	1.0157	1.1452
2.00	0.9997	0.9986	0.9972	0.9944	0.9892	0.9842	0.9796	0.9754	0.9715	0.9664	0.9599	0.9550	0.9772	1.0328	1.1516
2.20	0.9998	0.9992	0.9983	0.9967	0.9937	0.9910	0.9886	0.9865	0.9847	0.9826	0.9806	0.9827	1.0094	1.0600	1.1635
2.40	0.9999	0.9996	0.9991	0.9983	0.9969	0.9957	0.9948	0.9941	0.9936	0.9935	0.9945	1.0011	1.0313	1.0793	1.1728
2.60	1.0000	0.9998	0.9997	0.9994	0.9991	0.9990	0.9990	0.9993	0.9998	1.0010	1.0040	1.0137	1.0463	1.0926	1.1792
2.80	1.0000	1.0000	1.0001	1.0002	1.0007	1.0013	1.0021	1.0031	1.0042	1.0063	1.0106	1.0223	1.0565	1.1016	1.1830
3.00	1.0000	1.0002	1.0004	1.0008	1.0018	1.0030	1.0043	1.0057	1.0074	1.0101	1.0153	1.0284	1.0635	1.1075	1.1848
3.50	1.0001	1.0004	1.0008	1.0017	1.0035	1.0055	1.0075	1.0097	1.0120	1.0156	1.0221	1.0368	1.0723	1.1138	1.1834
4.00	1.0001	1.0005	1.0010	1.0021	1.0043	1.0066	1.0090	1.0115	1.0140	1.0179	1.0249	1.0401	1.0741	1.1136	1.1773

Table 2-352 Acentric Deviations $Z^{(1)}$ from the Simple Fluid Compressibility FactorValues in parentheses are for the opposite phase and may be used to interpolate to or near the phase boundary [PGL4; Wilding, W. V., J. K. Johnson, and R. L. Rowley, *Int. J. Thermophys.*, 8(1987):717].

$TrPr$	0.010	0.050	0.100	0.200	0.400	0.600	0.800	1.000	1.200	1.500	2.000	3.000	5.000	7.000	10.000
0.30	-0.0008	-0.0040	-0.0081	-0.0161	-0.0323	-0.0484	-0.0645	-0.0806	-0.0966	-0.1207	-0.1608	-0.2407	-0.3996	-0.5572	-0.7915
0.35	-0.0009	-0.0046	-0.0093	-0.0185	-0.0370	-0.0554	-0.0738	-0.0921	-0.1105	-0.1379	-0.1834	-0.2738	-0.4523	-0.6279	-0.8863
0.40	-0.0010	-0.0048	-0.0095	-0.0190	-0.0380	-0.0570	-0.0758	-0.0946	-0.1134	-0.1414	-0.1879	-0.2799	-0.4603	-0.6365	-0.8936
0.45	-0.0009 (-0.0740)	-0.0047	-0.0094	-0.0187	-0.0374	-0.0560	-0.0745	-0.0929	-0.1113	-0.1387	-0.1840	-0.2734	-0.4475	-0.6162	-0.8606
0.50	-0.0009 (-0.0457)	-0.0045 (-0.2270)	-0.0090	-0.0181	-0.0360	-0.0539	-0.0716	-0.0893	-0.1069	-0.1330	-0.1762	-0.2611	-0.4253	-0.5831	-0.8099
0.55	-0.0314 (-0.0009)	-0.0043 (-0.1438)	-0.0086 (-0.2864)	-0.0172	-0.0343	-0.0513	-0.0682	-0.0849	-0.1015	-0.1263	-0.1669	-0.2465	-0.3991	-0.5446	-0.7521
0.60	-0.0205 (0.0005)	-0.0041 (0.0949)	-0.0082 (-0.1857)	-0.0164	-0.0326	-0.0487	-0.0646	-0.0803	-0.0960	-0.1192	-0.1572	-0.2312	-0.3718	-0.5047	-0.6928
0.65	-0.0137 (-0.0005)	-0.0772 (0.0039)	-0.0078 (-0.1262)	-0.0156 (-0.2424)	-0.0309	-0.0461	-0.0611	-0.0759	-0.0906	-0.1122	-0.1476	-0.2160	-0.3447	-0.4653	-0.6346
0.70	-0.0093	-0.0507 (-0.0035)	-0.0161 (-0.0075)	-0.0148 (-0.1685)	-0.0294	-0.0438	-0.0579	-0.0718	-0.0855	-0.1057	-0.1385	-0.2013	-0.3184	-0.4270	-0.5785
0.75	-0.0064	-0.0339 (-0.0037)	-0.0744 (-0.0072)	-0.0143 (-0.1295)	-0.0282 (-0.2203)	-0.0417	-0.0550	-0.0681	-0.0808	-0.0996	-0.1298	-0.1872	-0.2929	-0.3901	-0.5250
0.80	-0.0044	-0.0228	-0.0487 (-0.0073)	-0.1160 (-0.0139)	-0.0272 (-0.1682)	-0.0401 (-0.2185)	-0.0526	-0.0648	-0.0767	-0.0940	-0.1217	-0.1736	-0.2682	-0.3545	-0.4740
0.85	-0.0029	-0.0152	-0.0319	-0.0715 (-0.0144)	-0.0268 (-0.1503)	-0.0391 (-0.1692)	-0.0509	-0.0622	-0.0731	-0.0888	-0.1138	-0.1602	-0.2439	-0.3201	-0.4254
0.90	-0.0019	-0.0099	-0.0205	-0.1118 (-0.0179)	-0.0396 (-0.0286)	-0.0503 (-0.1580)	-0.0503	-0.0604	-0.0701	-0.0840	-0.1059	-0.1463	-0.2195	-0.2862	-0.3788
0.93	-0.0015	-0.0075	-0.0154	-0.0326	-0.0763 (-0.0340)	-0.0514 (-0.0424)	-0.0514 (-0.1418)	-0.0602	-0.0687	-0.0810	-0.1007	-0.1374	-0.2045	-0.2661	-0.3516
0.95	-0.0012	-0.0062	-0.0126	-0.0262	-0.0589 (-0.0444)	-0.1110 (-0.0490)	-0.0540 (-0.1532)	-0.0607	-0.0678	-0.0788	-0.0967	-0.1310	-0.1943	-0.2526	-0.3339
0.97	-0.0010	-0.0050	-0.0101	-0.0208	-0.0450	-0.0770 (-0.0714)	-0.1647 (-0.0643)	-0.0623	-0.0669	-0.0759	-0.0921	-0.1240	-0.1837	-0.2391	-0.3163
0.98	-0.0009	-0.0044	-0.0090	-0.0184	-0.0390	-0.0641	-0.1100 (-0.0828)	-0.0641	-0.0661	-0.0740	-0.0893	-0.1202	-0.1783	-0.2322	-0.3075
0.99	-0.0008	-0.0039	-0.0079	-0.0161	-0.0335	-0.0531	-0.0796 (-0.1621)	-0.0680	-0.0646	-0.0715	-0.0861	-0.1162	-0.1728	-0.2254	-0.2989
1.00	-0.0007	-0.0034	-0.0069	-0.0140	-0.0285	-0.0435	-0.0588	-0.0879	-0.0609	-0.0678	-0.0824	-0.1118	-0.1672	-0.2185	-0.2902
1.01	-0.0006	-0.0030	-0.0060	-0.0120	-0.0240	-0.0351	-0.0429	-0.0223	-0.0473	-0.0621	-0.0778	-0.1072	-0.1615	-0.2116	-0.2816
1.02	-0.0005	-0.0026	-0.0051	-0.0102	-0.0198	-0.0277	-0.0303	-0.0062	0.0227	-0.0524	-0.0722	-0.1021	-0.1556	-0.2047	-0.2731
1.05	-0.0003	-0.0015	-0.0029	-0.0054	-0.0092	-0.0097	-0.0032	0.0220	0.1059	0.0451	-0.0432	-0.0838	-0.1370	-0.1835	-0.2476
1.10	0.0000	0.0000	0.0001	0.0007	0.0038	0.0106	0.0236	0.0476	0.0897	0.1630	0.0698	-0.0373	-0.1021	-0.1469	-0.2056
1.15	0.0002	0.0011	0.0023	0.0052	0.0127	0.0237	0.0396	0.0625	0.0943	0.1548	0.1667	0.0332	-0.0611	-0.1084	-0.1642
1.20	0.0004	0.0019	0.0039	0.0084	0.0190	0.0326	0.0499	0.0719	0.0991	0.1477	0.1990	0.1095	-0.0141	-0.0678	-0.1231
1.30	0.0006	0.0030	0.0061	0.0125	0.0267	0.0429	0.0612	0.0819	0.1048	0.1420	0.1991	0.2079	0.0875	0.0176	-0.0423
1.40	0.0007	0.0036	0.0072	0.0147	0.0306	0.0477	0.0661	0.0857	0.1063	0.1383	0.1894	0.2397	0.1737	0.1008	0.0350
1.50	0.0008	0.0039	0.0078	0.0158	0.0323	0.0497	0.0677	0.0864	0.1055	0.1345	0.1806	0.2433	0.2309	0.1717	0.1058
1.60	0.0008	0.0040	0.0080	0.0162	0.0330	0.0501	0.0677	0.0855	0.1035	0.1303	0.1729	0.2381	0.2631	0.2255	0.1673
1.70	0.0008	0.0040	0.0081	0.0163	0.0329	0.0497	0.0667	0.0838	0.1008	0.1259	0.1658	0.2305	0.2788	0.2628	0.2179
1.80	0.0008	0.0040	0.0081	0.0162	0.0325	0.0488	0.0652	0.0816	0.0978	0.1216	0.1593	0.2224	0.2846	0.2871	0.2576
1.90	0.0008	0.0040	0.0079	0.0159	0.0318	0.0477	0.0635	0.0792	0.0947	0.1173	0.1532	0.2144	0.2848	0.3017	0.2876
2.00	0.0008	0.0039	0.0078	0.0155	0.0310	0.0464	0.0617	0.0767	0.0916	0.1133	0.1476	0.2069	0.2819	0.3097	0.3096
2.20	0.0007	0.0037	0.0074	0.0147	0.0293	0.0437	0.0579	0.0719	0.0857	0.1057	0.1374	0.1932	0.2720	0.3135	0.3355
2.40	0.0007	0.0035	0.0070	0.0139	0.0276	0.0411	0.0544	0.0675	0.0803	0.0989	0.1285	0.1812	0.2602	0.3089	0.3459
2.60	0.0007	0.0033	0.0066	0.0131	0.0260	0.0387	0.0512	0.0634	0.0754	0.0929	0.1207	0.1706	0.2484	0.3009	0.3475
2.80	0.0006	0.0031	0.0062	0.0124	0.0245	0.0365	0.0483	0.0598	0.0711	0.0876	0.1138	0.1613	0.2372	0.2915	0.3443
3.00	0.0006	0.0029	0.0059	0.0117	0.0232	0.0345	0.0456	0.0565	0.0672	0.0828	0.1076	0.1529	0.2268	0.2817	0.3385
3.50	0.0005	0.0026	0.0052	0.0103	0.0204	0.0303	0.0401	0.0497	0.0591	0.0728	0.0949	0.1356	0.2042	0.2584	0.3194
4.00	0.0005	0.0023	0.0046	0.0091	0.0182	0.0270	0.0357	0.0443	0.0527	0.0651	0.0849	0.1219	0.1857	0.2378	0.2994

TABLE 2-353 Constants for the Two Reference Fluids Used in Lee-Kesler Method*

Constant	Simple reference fluid	Acentric reference fluid
b_1	0.1181193	0.2026579
b_2	0.265728	0.331511
b_3	0.154790	0.027655
b_4	0.030323	0.203488
c_1	0.0236744	0.0313385
c_2	0.0186984	0.0503618
c_3	0.0	0.016901
c_4	0.042724	0.041577
$d_1 \times 10^4$	0.155488	0.48736
$d_2 \times 10^4$	0.623689	0.0740336
β	0.65392	1.226
γ	0.060167	0.03754

*Lee, B. I., and M. G. Kesler, *AIChE J.*, **21** (1975): 510.

Of the cubic EoS given in Table 2-354, the Soave and Peng-Robinson are the most accurate, but there is no general rule for which EoS produces the best estimated volumes for specific fluids or conditions. The Peng-Robinson equation has been better tuned to liquid densities, while the Soave equation has been better tuned to vapor-liquid equilibrium and vapor densities. In solving the cubic equation for volume, a convenient initial guess to find the vapor root is the ideal gas value, while an initial value of $1.05b$ is convenient to locate the liquid root.

Example Estimate the molar density of liquid and vapor saturated ammonia at 353.15 K, using the Soave and Peng-Robinson EoS.

Required properties: Recommended values in the DIPPR® 801 database are

$$T_c = 405.65 \text{ K} \quad P_c = 112.8 \text{ bar} \quad \omega = 0.252608$$

$$P^* (353.15 \text{ K}) = 41.352 \text{ bar (vapor pressure at 353.15 K)}$$

EoS parameters (shown for Soave EoS):

$$a = \frac{0.42748(RT_c)^2}{P_c} = \frac{0.42748 \left[\left(83.145 \frac{\text{bar}\cdot\text{cm}^3}{\text{mol}\cdot\text{K}} \right) (405.65 \text{ K}) \right]^2}{112.8 \text{ bar}}$$

$$= 4.311 \times 10^6 \frac{\text{cm}^6\cdot\text{bar}}{\text{mol}^2}$$

$$b = \frac{0.08664(RT_c)}{P_c} = \frac{0.08664 \left(83.145 \frac{\text{bar}\cdot\text{cm}^3}{\text{mol}\cdot\text{K}} \right) (405.65 \text{ K})}{112.8 \text{ bar}}$$

$$= 25.906 \frac{\text{cm}^3}{\text{mol}}$$

$$T_r = (353.15 \text{ K})/(405.65 \text{ K}) = 0.871$$

$$\alpha = [1 + [0.48 + (1.574)(0.252608) - (0.176)(0.252608)^2][1 - (0.871)^{0.5}]]^2 = 1.119$$

Rearrange and solve Eq. (2-66) for V :

$$P = \frac{RT}{V-b} - \frac{a\alpha}{V(V+b)} \quad \text{or} \quad PV^3 - RTV^2 + (a\alpha - bRT - Pb^2)V - ab\alpha = 0$$

$$41.352 \left(\frac{V}{\text{m}^3/\text{mol}} \right)^3 - \left(0.029 \frac{\text{m}^3}{\text{mol}} \right) \left(\frac{V}{\text{m}^3/\text{mol}} \right)^2 + \left(4.037 \times 10^{-6} \frac{\text{m}^6}{\text{mol}^2} \right) \times \left(\frac{V}{\text{m}^3/\text{mol}} \right) - 1.25 \times 10^{-10} = 0$$

Vapor root (initial guess of $V = 7.1 \times 10^{-7} \text{ m}^3/\text{mol}$ from ideal gas equation):

$$V_{\text{vap}} = 5.395 \times 10^{-4} \text{ m}^3/\text{mol} \quad \text{and} \quad \rho_{\text{vap}} = 1/V_{\text{vap}} = 1.854 \text{ kmol/m}^3$$

Liquid root (initial guess of $V = 2.72 \times 10^{-5} \text{ m}^3/\text{mol}$ from $1.05b$):

$$V_{\text{liq}} = 4.441 \times 10^{-5} \text{ m}^3/\text{mol} \quad \text{and} \quad \rho_{\text{liq}} = 1/V_{\text{liq}} = 22.516 \text{ kmol/m}^3$$

The corresponding values and equation for the Peng-Robinson EoS are

$$a = 4.611 \times 10^6 \text{ cm}^6\cdot\text{bar/mol}^2 \quad b = 23.262 \text{ cm}^3/\text{mol}$$

$$\alpha = 1.103$$

$$P = \frac{RT}{V-b} - \frac{a\alpha}{V^2 + 2bV - b^2}$$

or

$$PV^3 + (bP - RT)V^2 + (a\alpha - 2bRT - 3Pb^2)V + (bP^3 + RTb^2 - ab\alpha) = 0$$

$$41.352 \left(\frac{V}{\text{m}^3/\text{mol}} \right)^3 - \left(0.0284 \frac{\text{m}^3}{\text{mol}} \right) \left(\frac{V}{\text{m}^3/\text{mol}} \right)^2 + \left(3.651 \times 10^{-6} \frac{\text{m}^6}{\text{mol}^2} \right) \times \left(\frac{V}{\text{m}^3/\text{mol}} \right) - 1.018 \times 10^{-10} = 0$$

$$V_{\text{vap}} = 5.286 \times 10^{-4} \text{ m}^3/\text{mol} \quad \text{and} \quad \rho_{\text{vap}} = 1.892 \text{ kmol/m}^3$$

$$V_{\text{liq}} = 3.914 \times 10^{-5} \text{ m}^3/\text{mol} \quad \text{and} \quad \rho_{\text{liq}} = 25.55 \text{ kmol/m}^3$$

The liquid density calculated from the Soave EoS is 24.2 percent below the DIPPR® 801 recommended value of 29.69 kmol/m³, while that calculated from the Peng-Robinson EoS is 13.9 percent below the recommended value.

TABLE 2-354 Relationships for Eq. (2-66) for Common Cubic EoS

EoS	δ	ϵ	$\alpha(T_r)$	$aP_c/(RT_c)^2$	$bP_c/(RT_c)$
van der Waals*	0	0	1	0.42188	0.125
Relich-Kwong†	0	0	$T_r^{-0.5}$	0.42748	0.08664
Soave‡	b	0	$[1 + (0.48 + 1.574\omega - 0.176\omega^2)(1 - T_r^{0.5})]^2$	0.42748	0.08664
Peng-Robinson§	$2b$	$-b^2$	$[1 + (0.37464 + 1.54226\omega - 0.2699\omega^2)(1 - T_r^{0.5})]^2$	0.45724	0.0778

*van der Waal, J. H., *Z. Phys. Chem.*, **5** (1890): 133.

†Redlich, O., and J. N. S. Kwong, *Chem. Rev.*, **44** (1949): 233.

‡Soave, G., *Chem. Eng. Sci.*, **27** (1972): 1197.

§Peng, D. Y., and D. B. Robinson, *Ind. Eng. Chem. Fundam.*, **15** (1976): 59.

Liquids For most liquids, the saturated molar liquid density ρ can be effectively correlated with

$$\rho = \frac{A}{B^{[1+(1-T/C)^D]}} \quad (2-68)$$

adapted from the Rackett prediction equation [Rackett, H. G., *J. Chem. Eng. Data*, **15** (1970): 514]. The regression constants A , B , and D are determined from the nonlinear regression of available data, while C is usually taken as the critical temperature. The liquid density decreases approximately linearly from the triple point to the normal boiling point and then nonlinearly to the critical density (the reciprocal of the critical volume). A few compounds such as water cannot be fit with this equation over the entire range of temperature.

The recommended method for estimation of saturated liquid density for pure organic compounds is the Rackett prediction method.

Recommended Method Rackett method.

Reference: Rackett, H. G., *J. Chem. Eng. Data*, **15** (1970): 514.

Classification: Corresponding states.

Expected uncertainty: 8 percent as purely predictive equation; 2 percent if Z_{RA} (see Description below) or some liquid density data are available.

Applicability: Saturated liquid densities of organic compounds.

Input data: T_c , P_c , and Z_c (or, equivalently, V_c).

Description: A predictive form of the equation is given by

$$\frac{1}{\rho} = V = \left(\frac{RT_c}{P_c} \right) Z_c^q \quad \text{where } q = 1.0 + (1.0 - T_r)^{2.7} \quad (2-69)$$

A modification of the Rackett method by Spencer and Danner [Spencer, C. F., and R. P. Danner, *J. Chem. Eng. Data*, **17** (1972): 236] replaces Z_c with an adjustable parameter Z_{RA}

$$\frac{1}{\rho} = V = \left(\frac{RT_c}{P_c} \right) Z_{RA}^q \quad (2-70)$$

to provide better estimations of liquid density away from the critical point [Eq. (2-70) gives the correct critical density only when $Z_{RA} = Z_c$]. An alternative to this modification when several liquid density data points are available is to replace the 2/7 power in q of Eq. (2-69) with an adjustable parameter. This generally provides good agreement with the experimental values and permits accurate extrapolation of the densities all the way to the critical point.

Example Estimate the saturated liquid density of acetonitrile at 376.69 K.

Required properties: The recommended values from the DIPPR® 801 database are

$$T_c = 545.5 \text{ K} \quad P_c = 4.83 \text{ MPa} \quad Z_c = 0.184$$

Calculate supporting quantities:

$$T_r = (376.69 \text{ K})/(545.5 \text{ K}) = 0.691$$

$$q = 1 + (1 - 0.691)^{2.7} = 1.715$$

Calculate saturated liquid density from Eq. (2-69):

$$\rho = \left[\frac{4.83 \times 10^6 \text{ Pa}}{\left(8.314 \frac{\text{Pa} \cdot \text{m}^3}{\text{mol} \cdot \text{K}} \right) (545.5 \text{ K})} \right] (0.184)^{-1.715} = 19.42 \frac{\text{kmol}}{\text{m}^3}$$

This estimated value is 16.1 percent above the DIPPR® 801 recommended value of 16.726 kmol/m³.

Calculate ρ_{sat} from Eq. (2-70): Kratzke [Kratzke, H., and S. Muller, *J. Chem. Thermo.*, **17** (1985): 151] reported an experimental density of 18.919 kmol/m³ at 298.08 K. Use of this experimental value in Eq. (2-70) to calculate Z_{RA} gives

$$T_r = (298.08 \text{ K})/(545.5 \text{ K}) = 0.546 \quad q = 1 + (1 - 0.546)^{2.7} = 1.798$$

$$Z_{RA} = \left[\frac{4.83 \times 10^6 \text{ Pa}}{\left(8.314 \frac{\text{Pa} \cdot \text{m}^3}{\text{mol} \cdot \text{K}} \right) (545.5 \text{ K}) \left(18.919 \frac{\text{kmol}}{\text{m}^3} \right)} \right]^{1/1.798} = 0.202$$

$$\rho = \left[\frac{4.83 \times 10^6 \text{ Pa}}{\left(8.314 \frac{\text{Pa} \cdot \text{m}^3}{\text{mol} \cdot \text{K}} \right) (545.5 \text{ K})} \right] (0.202)^{-1.715} = 16.577 \frac{\text{kmol}}{\text{m}^3}$$

The value obtained by the modified Rackett method is 0.9 percent below the DIPPR® 801 recommended value. Note, however, that with $Z_{RA} = 0.202$, Eq. (2-70) gives $\rho_c = 5.28 \text{ kmol/m}^3$ as opposed to the DIPPR® 801 recommended value of 5.79 kmol/m³. If the power is regressed from the Kratzke density, one obtains $q_1 = 0.452$ and $\rho = 15.68 \text{ kmol/m}^3$ (4 percent below the experimental value), while still retaining $\rho_c = 5.79 \text{ kmol/m}^3$.

Solids Solid density data are sparse and usually available only within a narrow temperature range. For most solids, density decreases approximately linearly with increasing temperature. Prediction of solid densities is an inexact science, but reasonable correlation has been found between the density of the liquid phase at the triple point and the solid that is stable at the triple point conditions.

Recommended Method Goodman method.

Reference: Goodman, B. T., et al., *J. Chem. Eng. Data*, **49** (2004): 1512.

Classification: Empirical correlation.

Expected uncertainty: 6 percent.

Applicability: Organic compounds; applicable to the stable solid phase at the triple point temperature T_i , to either the next solid-phase transition temperature or to approximately 0.3 T_i .

Input data: Liquid density at the triple point.

Description: The density for the solid phase that is stable at the triple point has been correlated as a function of temperature and the liquid density at T_i :

$$\rho_s = \left(1.28 - 0.16 \frac{T}{T_i} \right) \rho_L(T_i) \quad (2-71)$$

Example Estimate the density of solid naphthalene at 281.46 K.

Required properties: The recommended values from the DIPPR® 801 database for T_i and the liquid density at T_i are

$$T_i = 353.43 \text{ K} \quad \rho_L(T_i) = 7.6326 \text{ kmol/m}^3$$

From Eq. (2-71):

$$\rho_s = \left(1.28 - 0.16 \frac{281.46 \text{ K}}{353.43 \text{ K}} \right) \left(7.6326 \frac{\text{kmol}}{\text{m}^3} \right) = 8.797 \frac{\text{kmol}}{\text{m}^3}$$

The estimated value is 4.3 percent lower than the DIPPR® 801 recommended value of 9.1905 kmol/m³.

Mixtures Both liquid and vapor densities can be estimated using pure-component CS and EoS methods by treating the fluid as a pseudo-pure component with effective parameters calculated from the pure-component parameters and using ad hoc mixing rules.

To apply the Lee-Kesler CS method to mixtures, pseudo-pure fluid constants are required. One of the simplest set of mixing rules for these quantities is [Prausnitz, J. M., and R. D. Gunn, *AIChE J.*, **4** (1958): 430, 494; Joffe, J., *Ind. Eng. Chem. Fundam.*, **10** (1971): 532]:

$$\bar{T}_c = \sum_{i=1}^C x_i T_{c,i} \quad (2-72)$$

$$\bar{P}_c = \frac{\sum_{i=1}^C x_i Z_{c,i}}{\sum_{i=1}^C x_i V_{c,i}} \bar{R} \bar{T}_c \quad (2-73)$$

$$\bar{\omega} = \sum_{i=1}^C x_i \omega_i \quad (2-74)$$

The procedures are identical to those for pure components with the replacement of T_c , P_c , and ω with the effective mixture values calculated by using these equations.

To use a cubic EoS for a mixture, mixing rules are used to calculate effective mixture parameters in terms of the pure-component values. Although there are more complex mixing rules available that may improve prediction accuracy, the simplest forms are recommended here for their simplicity and reasonable accuracy without adjustable parameters:

$$\bar{b} = \sum_{i=1}^C x_i b_i \quad (2-75)$$

$$\bar{a}\alpha = \left[\sum_{i=1}^C x_i (a_i \alpha_i)^{1/2} \right]^2 \quad (2-76)$$

Mixture calculations are then identical to the pure-component calculations using these effective mixture parameters for the pure-component $a\alpha$ and b values.

The actual mixture second virial coefficient B_m is related to the pure-component values by

$$B_m = \sum_{i=1}^C \sum_{j=1}^C x_i x_j B_{ij} \quad \text{where} \quad B_{ii} = B_i \quad (2-77)$$

This requires calculation of all possible binary pair interaction virials (B_{ij} , $i \neq j$) for the mixture. Again the pure-component methods can be used to provide estimates of these values by using the following combining rules:

$$T_{c,ij} = \sqrt{T_{c,i} T_{c,j}} \quad V_{c,ij} = \left(\frac{V_{c,i}^{1/3} + V_{c,j}^{1/3}}{2} \right)^3 \quad Z_{c,ij} = \frac{Z_{c,i} + Z_{c,j}}{2} \quad (2-78)$$

$$\omega_{ij} = \frac{\omega_i + \omega_j}{2} \quad P_{c,ij} = \frac{Z_{c,ij} \bar{R} T_{c,ij}}{V_{c,ij}} \quad (2-79)$$

These interaction parameters are used in place of the corresponding pure-component parameters to determine the B_{ij} values.

The modified Rackett method has also been extended to liquid mixtures [Spencer, C. F., and R. P. Danner, *J. Chem. Eng. Data*, **17** (1972): 236; Li, C. C., *Can. J. Chem. Eng.*, **19** (1971): 709]:

$$T_{c,ij} = \sqrt{T_{c,i} T_{c,j}} \quad \phi_i = \frac{x_i V_{c,i}}{\sum_{j=1}^C x_j V_{c,j}} \quad \bar{T}_c = \sum_{i=1}^C \sum_{j=1}^C \phi_i \phi_j T_{c,ij} \quad (2-80)$$

Recommended Method Spencer-Danner-Li mixing rules with Rackett equation.

References: Spencer, C. F., and R. P. Danner, *J. Chem. Eng. Data*, **17** (1972): 236; Li, C. C., *Can. J. Chem. Eng.*, **19** (1971): 709.

Classification: Corresponding states.

Expected uncertainty: About 7 percent on average; higher near the T_c of any of the components.

Applicability: Saturated (at the bubble point) liquid mixtures.

Input data: T_c , V_c , and x_i .

Description: The predictive form of the equation is given by

$$\frac{1}{\rho} = V = \bar{R} \left(\sum_{i=1}^C \frac{x_i T_{c,i}}{P_{c,i}} \right) \bar{Z}_{RA}^q \quad q = 1.0 + (1.0 - T_r)^{2/7} \quad (2-81)$$

where

$$\bar{Z}_{RA} = 0.29056 - 0.08775 \sum_{i=1}^C x_i \omega_i \quad \text{and} \quad T_r = \frac{T}{T_c} \quad (2-82)$$

Example Estimate the saturated liquid density of a liquid mixture of 50 mol % ethane(1) and 50 mol % *n*-decane(2) at 377.6 K.

Required properties: The recommended values from the DIPPR® 801 database for the required properties are as follows:

	T_c /K	V_c /(m ³ ·kmol ⁻¹)	P_c /bar	ω
Ethane	305.32	0.1455	48.72	0.0995
Decane	617.7	0.617	21.1	0.4923

Auxiliary quantities from Eq. (2-80):

$$\phi_1 = \frac{(0.5)(0.1455)}{(0.5)(0.1455) + (0.5)(0.617)} = 0.191; \quad \phi_2 = 0.809$$

$$T_{c,12} = \sqrt{(305.32 \text{ K})(617.7 \text{ K})} = 434.3 \text{ K}$$

$$\begin{aligned} \bar{T}_c &= \phi_1^2 T_{c,1} + 2\phi_1 \phi_2 T_{c,12} + \phi_2^2 T_{c,2} \\ &= (0.191)^2 (305.32) + (2)(0.191)(0.809)(434.3) + (0.809)^2 (617.7) \end{aligned}$$

$$\bar{T}_c = 549.68 \text{ K}$$

Calculations from Eqs. (2-81) and (2-82):

$$T_r = (377.6 \text{ K})/(549.68 \text{ K}) = 0.687$$

$$q = 1 + (1 - 0.687)^{2/7} = 1.718$$

$$\bar{Z}_{RA} = 0.29056 - 0.08775[(0.5)(0.0995) + (0.5)(0.4923)] = 0.2646$$

$$\begin{aligned} V &= \left(0.08314 \frac{\text{m}^3 \cdot \text{bar}}{\text{K} \cdot \text{kmol}} \right) \left[\frac{(0.5)(305.32 \text{ K})}{48.72 \text{ bar}} + \frac{(0.5)(617.7 \text{ K})}{21.1 \text{ bar}} \right] (0.2646)^{1.718} \\ &= 0.151 \frac{\text{m}^3}{\text{kmol}} \end{aligned}$$

The experimental value [Reamer, H. H., and B. H. Sage, *J. Chem. Eng. Data*, **7** (1962): 161] is 0.149 m³/kmol, and the error in the estimated value is 1.3 percent.

VISCOSITY

Viscosity is defined as the shear stress per unit area at any point in a confined fluid, divided by the velocity gradient in the direction perpendicular to the direction of flow. The *absolute viscosity* η is the shear stress at a point, divided by the velocity gradient at that point. The SI unit of viscosity is Pa·s [1 kg/(m·s)], but the cgs unit of poise (P) [1 g/(cm·s)] is also commonly used. Because many common fluids have viscosities on the order of 0.01 P, the unit of centipoise (cP) is also frequently used (1 cP = 1 mPa·s). The *kinematic viscosity* ν is defined as the ratio of the absolute viscosity to density at the same temperature and pressure. The SI unit for ν is m²/s, but again cgs units are very common and ν is often given in stokes (St) (1 cm²/s) or centistokes (cSt) (0.01 cm²/s).

Gases Experimental data for gases and vapors at low density are often correlated with

$$\eta^o = \frac{AT^B}{1 + C/T + D/T^2} \quad (2-83)$$

Over smaller temperature ranges, parameters C and D may not be necessary as $\ln \eta$ is often reasonably linear with $\ln T$. Care should be taken in extrapolating using Eq. (2-83) as there can be unintended mathematical poles where the denominator approaches zero.

Numerous methods have been developed for estimation of vapor viscosity. For nonpolar vapors, the Yoon-Thodos CS method works well, but for polar fluids the Reichenberg method is preferred. Both methods are illustrated below.

Recommended Method Yoon-Thodos method.

Reference: Yoon, P., and G. Thodos, *AIChE J.*, **16** (1970): 300.

Classification: Corresponding states.

Expected uncertainty: 5 percent.

Applicability: Nonpolar and slightly polar organic vapors.

Input data: T_c , P_c , and M .

Description: The correlation for viscosity as a function of reduced temperature is

$$\frac{\eta^\circ}{\text{Pa} \cdot \text{s}} = \frac{46.1T_r^{0.618} - 20.4 \exp(-0.449T_r) + 19.4 \exp(-4.058T_r) + 1}{2.173424 \times 10^{11}(T_c/K)^{1/6}(M/\text{g} \cdot \text{mol})^{-1}(P_c/\text{Pa})^{-2/3}} \quad (2-84)$$

Example Estimate the low-pressure vapor viscosity of propane at 353 K.

Required constants: The DIPPR® 801 database recommends the following values:

$$T_c = 369.83 \text{ K} \quad P_c = 4.248 \text{ MPa} \quad M = 44.0956 \text{ g/mol}$$

Reduced temperature:

$$T_r = (353 \text{ K})/(369.83 \text{ K}) = 0.9545$$

Calculation using Eq. (2-84):

$$\begin{aligned} \frac{\eta^\circ}{\text{Pa} \cdot \text{s}} &= \frac{(46.1)(0.9545)^{0.618} - 20.4 \exp[-0.449(0.9545)] + 19.4 \exp[-4.058(0.9545)] + 1}{(2.173424 \times 10^{11})(369.83)^{-1/2}(44.0956)^{-1/2}(4.248 \times 10^6)^{-2/3}} \\ &= 9.84 \times 10^{-6} \end{aligned}$$

The estimated value is 1.5 percent higher than the DIPPR® 801 recommended value of 9.70×10^{-6} Pa·s.

Recommended Method Reichenberg method.

Reference: Reichenberg, D., *AIChE J.*, **21** (1975): 181.

Classification: Group contributions and corresponding states.

Expected uncertainty: 5 percent.

Applicability: Nonpolar and polar organic and inorganic vapors.

Input data: T_c , P_c , M , μ , and molecular structure.

Description: The temperature dependence of the viscosity is given by

$$\frac{\eta^\circ}{\text{Pa} \cdot \text{s}} = \frac{AT_r^2}{[1 + 0.36T_r(T_r - 1)]^{1/6}} \left[\frac{1 + 270(\mu_r^*)^4}{T_r + 270(\mu_r^*)^4} \right] \quad (2-85)$$

where the parameter A is determined from group contributions and the modified reduced dipole μ_r^* is found from

$$\mu_r^* = 52.46\mu_r \quad (2-86)$$

and Eq. (2-62).

For organic compounds, A is found from the group values C_i , listed in Table 2-355, using

$$A = 10^{-7} \frac{\left[\frac{M}{(\text{kg} / \text{kmol})} \right]^{1/2} (T_c / \text{K})}{\sum_{i=1}^N n_i C_i} \quad (2-87)$$

For inorganic gases, A is obtained from

$$A = 1.6104 \times 10^{-10} \left[\left(\frac{M}{\text{g/mol}} \right)^{1/2} \left(\frac{P_c}{\text{Pa}} \right)^{2/3} \left(\frac{T_c}{\text{K}} \right)^{-1/6} \right] \quad (2-88)$$

TABLE 2-355 Reichenberg* Group Contribution Values

Group	C_i	Group	C_i
—CH ₃	9.04	—F	4.46
>CH ₂	6.47	—Cl	10.06
>CH—	2.67	—Br	12.83
>C<	−1.53	—OH alcohol	7.96
≡CH ₂	7.68	>O	3.59
≡CH—	5.53	>C=O	12.02
>C=	1.78	—CHO	14.02
≡CH	7.41	—COOH	18.65
≡C—	5.24	—COO— or HCOO—	13.41
>CH ₂ ring	6.91	—NH ₂	9.71
>CH— ring	1.16	>NH	3.68
>C< ring	0.23	—N— ring	4.97
≡CH— ring	5.90	—CN	18.13
>C= ring	3.59	>S ring	8.86

*Reichenberg, D., *AIChE J.*, **21** (1975): 181.

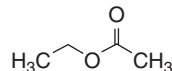
Example Estimate the low-pressure vapor viscosity of ethyl acetate at 401.25 K.

Required constants: The DIPPR® 801 database recommends the following values:

$$M = 88.1051 \text{ g/mol} \quad T_c = 523.3 \text{ K} \quad P_c = 3.88 \text{ MPa} \quad \mu = 1.78 \text{ D}$$

Supporting quantities:

Structural groups:



Group	n_i	C_i	Contribution
—CH ₃	2	9.04	18.08
>CH ₂	1	6.47	6.47
—COO—	1	13.41	13.41
Total			37.96

$$T_r = (401.25 \text{ K})/(523.3 \text{ K}) = 0.767$$

From Eqs. (2-62) and (2-86):

$$\mu_r^* = 52.46 \frac{(1.78)^2(38.8)}{(523.3)^2} = 0.024$$

From Eq. (2-87):

$$A = 10^{-7} \frac{(88.1051)^{1/2}(523.3)}{37.96} = 1.294 \times 10^{-5}$$

Calculation using Eq. (2-84):

$$\frac{\eta^\circ}{\text{Pa} \cdot \text{s}} = \frac{(1.294 \times 10^{-5})(0.767)^2}{[1 + (0.36)(0.767)(0.767 - 1)]^{1/6}} \frac{1 + (270)(0.024)^4}{0.767 + (270)(0.024)^4} = 1.003 \times 10^{-5}$$

The estimated value is 1.5 percent lower than the DIPPR® 801 recommended value of 1.018×10^{-5} Pa·s.

The dependence of viscosity upon pressure is principally a density effect. Estimation of vapor viscosity at elevated pressures is commonly done by correlating density deviations from the low-pressure values, which are in turn estimated by using the procedures mentioned above. Several methods are available, but the method developed by Jossi et al. and extended to polar fluids by Stiel and Thodos is relatively accurate and easy to apply.

Recommended Method Jossi-Stiel-Thodos Method.

References: Stiel, L. I., and G. Thodos, *AIChE J.*, **10** (1964): 26; Jossi, J. A., L. I. Stiel, and G. Thodos, *AIChE J.*, **8** (1962): 59.

Classification: Empirical correlation and corresponding states.

Expected uncertainty: 9 percent—often less for nonpolar gases, larger for polar gases.

Applicability: Nonassociating gases; $p_r < 2.6$.

Input data: M , T_c , P_c , Z_c , μ , η° (low-pressure viscosity at same T may be estimated by using methods given above), and ρ (may be calculated from T and P by using density methods given above).

2-506 PHYSICAL AND CHEMICAL DATA

Description: Deviation of η from the low-pressure value η^0 is given by one of the following correlations depending upon its polarity and reduced density range:

For nonpolar gases, $0.1 < p_r < 3.0$:

$$\left[\left(\frac{\eta - \eta^0}{\text{mPa}\cdot\text{s}} \right) \xi + 1 \right]^{1/4} = 1.0230 + 0.23364p_r + 0.58533p_r^2 - 0.40758p_r^3 + 0.093324p_r^4 \quad (2-89)$$

For polar gases, $p_r \leq 0.1$:

$$\left(\frac{\eta - \eta^0}{\text{mPa}\cdot\text{s}} \right) \xi = 1.656p_r^{1.111} \quad (2-90)$$

For polar gases, $0.1 < p_r \leq 0.9$:

$$\left(\frac{\eta - \eta^0}{\text{mPa}\cdot\text{s}} \right) \xi = 0.0607(9.045p_r + 0.63)^{1.739} \quad (2-91)$$

For polar gases, $0.9 < p_r \leq 2.2$:

$$\log \left\{ 4 - \log \left[\left(\frac{\eta - \eta^0}{\text{mPa}\cdot\text{s}} \right) \xi \right] \right\} = 0.6439 - 0.1005p_r \quad (2-92)$$

For polar gases, $2.2 < p_r \leq 2.6$:

$$\log \left\{ 4 - \log \left[\left(\frac{\eta - \eta^0}{\text{mPa}\cdot\text{s}} \right) \xi \right] \right\} = 0.6439 - 0.1005p_r - 0.000475(p_r^3 - 10.65)^2 \quad (2-93)$$

where $p_r = P_c/(Z_cRT_c)$ and

$$\xi = 2173.4 \left(\frac{T_c}{\text{K}} \right)^{1/6} \left(\frac{M}{\text{kg/kmol}} \right)^{-1/2} \left(\frac{P_c}{\text{MPa}} \right)^{-2/3} \quad (2-94)$$

Example Estimate the vapor viscosity of CO_2 at 350 K and 20 MPa if $\eta^0 = 0.0174 \text{ mPa}\cdot\text{s}$ and $Z = 0.4983$ (estimated from Lee-Kesler method, see section on density).

Required properties: From the DIPPR® 801 database,

$$M = 44.01 \text{ kg/kmol} \quad T_c = 304.21 \text{ K} \quad P_c = 7.383 \text{ MPa} \\ Z_c = 0.274 \quad \mu = 0 \text{ D (nonpolar)}$$

Auxiliary quantities:

$$\xi = 2173.4(304.21)^{1/6}(44.01)^{-1/2}(7.383)^{-2/3} = 224.1$$

$$p_r = \frac{7.383 \text{ MPa}}{0.274[8.314 \text{ m}^3\text{Pa}/(\text{K}\cdot\text{mol})](304.21 \text{ K})} = 10.654 \frac{\text{kmol}}{\text{m}^3}$$

$$p_r = \frac{p}{p_c} = \frac{P}{ZRTp_c} = \frac{20 \text{ MPa}}{0.4983[8.314 \text{ m}^3\text{Pa}/(\text{K}\cdot\text{mol})](350 \text{ K})(10.654 \text{ m}^3\cdot\text{kmol})} = 1.295$$

Calculation using Eq. (2-89) for nonpolar fluids:

$$\left[224.1 \left(\frac{\eta - \eta^0}{\text{mPa}\cdot\text{s}} \right) + 1 \right]^{1/4} = 1.0230 + 0.23364(1.295) + 0.58533(1.295)^2 - 0.40758(1.295)^3 + 0.093324(1.295)^4 = 1.684 \\ \eta = \frac{1.684^4 - 1}{224.1} \text{ mPa}\cdot\text{s} + 0.0174 \text{ mPa}\cdot\text{s} = 0.0489 \text{ mPa}\cdot\text{s}$$

This differs from the experimental value of $0.0473 \text{ mPa}\cdot\text{s}$ by 3.4 percent.

Liquids Liquid viscosity can be correlated as a function of temperature for low pressures. Usually the correlation is based on the Andrade equation [Andrade, E. N. da C., *Nature*, **125** (1930): 309]

$$\ln \eta = A + \frac{B}{T} \quad (2-95)$$

or an extension of it. For example, the DIPPR® 801 database uses the equation

$$\ln \eta = A + \frac{B}{T} + C \ln T + DT^E \quad (2-96)$$

which is analogous to the Riedel [Riedel, L., *Chem. Ing. Tech.*, **26** (1954): 83] vapor pressure equation.

Currently the most accurate method for predicting pure liquid viscosity is the following GC method.

Recommended Method Hsu method.

Reference: Hsu, H.-C., Y.-W. Sheu, and C.-H. Tu, *Chem. Eng. J.*, **88** (2002): 27.

Classification: Group contributions.

Expected uncertainty: 20 percent.

Applicability: Organic liquids; $T_r < 0.75$.

Input data: P_c and molecular structure.

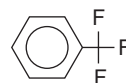
Description: The temperature dependence of the liquid viscosity is given by

$$\ln \left(\frac{\eta}{\text{mPa}\cdot\text{s}} \right) = \sum_{i=1}^N a_i + T \sum_{i=1}^N b_i + \frac{\sum_{i=1}^N c_i}{T^2} + \left(\sum_{i=1}^N d_i \right) \ln \left(\frac{P_c}{\text{bar}} \right) \quad (2-97)$$

where P_c is critical pressure and the coefficients a , b , c , and d are the sum of the group contributions obtained from Table 2-356.

Example Estimate the liquid viscosity of benzotrifluoride at 303.15 K.

Structural information:



Group	Number	a	$100b$	$0.0001c$	d
>C<	1	1.0031	-0.3677	-6.0316	1.1972
(=CH—) _A	5	-0.8570	-0.0098	2.4376	0.1311
(=C<) _A	1	0.7896	-0.0231	-0.9222	0.1928
(—F) ₃	1	1.5394	0.8465	17.8121	-2.9915
Total		-0.9529	0.4067	23.0463	-0.9460

Supporting values:

$$P_c = 32.1 \text{ MPa}$$

Calculation using Eq. (2-97):

$$\frac{\eta}{\text{mPa}\cdot\text{s}} = \exp \left[-0.9529 + (0.004067)(303.15) + \frac{230463}{(303.15)^2} - 0.9460 \ln 32.1 \right] = 0.610$$

The estimated value is 20 percent higher than the DIPPR® 801 recommended value of $0.509 \text{ mPa}\cdot\text{s}$.

Liquid Mixtures Almost all methods for estimating liquid mixture viscosity interpolate between the pure-component values at the same temperature. The Grunberg-Nissan [Grunberg, L., and A. H. Nissan, *Nature*, **164** (1949): 799] equation

$$\ln \eta = \sum_i^C x_i \ln \eta_i + \frac{1}{2} \sum_i^C \sum_{j=1}^C x_i x_j G_{ij} \quad (2-98)$$

is commonly used for nonaqueous mixtures. The parameter G_{ij} generally must be regressed from an experimental mixture viscosity. However, in the case of hydrocarbon mixtures G_{ij} can be set to zero with expected errors in the mixture viscosity of about 15 percent.

TABLE 2-356 Group Contributions for the Hsu et al. Method*

Table-specific nomenclature: R = in nonaromatic ring, A = in aromatic ring, RC = attached to nonaromatic ring, AC = attached to aromatic ring, X = halogen, $(-X)_n = n$ X atoms attached to same C atom

Group	<i>a</i>	100 <i>b</i>	0.0001 <i>c</i>	<i>d</i>
C, H Groups				
CH ₄	-1.7296	-1.0563	0.8928	-0.0019
-CH ₃	0.0570	-0.2382	0.7556	-0.1765
-CH ₂ -	-0.1497	0.0060	1.4157	0.0751
>CH-	-2.2942	0.4028	4.5094	0.6679
>C<	1.0031	-0.3677	-6.0316	1.1972
=CH ₂	0.9256	-0.2656	0.9860	-0.4417
=CH-	1.3365	0.1612	1.9408	0.2507
=C<	-3.5020	0.4305	3.1287	1.0465
≡CH	87.6040	-0.1106	4.4245	-24.1836
≡C-	-91.6154	-0.0111	0.3265	25.0542
(-CH ₂ -) _R	6.0416	-0.1778	0.8437	-1.5184
(>CH-) _R	-33.8745	0.7637	7.2433	8.5951
(=CH-) _R cycloalkene	1.2028	-0.0120	2.0143	-0.3677
(>C<) _R spirocyclane	-56.2158	1.7694	19.0452	13.3885
(=CH-) _A	-0.8570	-0.0098	2.4376	0.1311
(=C<) _A cycloalkene	0.7896	-0.0231	-0.9222	0.1928
(=C<) _A bi/terphenyl	2.0973	0.0444	8.1690	-0.4351
(=C<) _A naphthalene	0.4392	0.0683	8.8426	-0.1685
(=C<) _A turpentine	27.3350	1.2165	34.2857	-11.6500
(=C<) _A tetralin	14.2586	-0.8665	-14.7474	-2.7574
O, S Groups				
-OH primary for C<3	5.7852	-0.5310	9.5499	-1.0300
-OH primary for C>2	1.4351	-1.0010	13.8366	0.3418
-OH secondary	-2.6895	-0.3645	29.8404	0.4246
-OH tertiary	-18.5630	2.4275	78.5417	0.9650
(-OH) _{RC}	16.7808	0.8509	77.1759	-6.9285
-OH polyhydric	-0.0125	-0.3634	23.2329	-0.0172
(-OH) _{AC}	-2.0856	0.6362	50.0840	-1.0539
-OH alkoxyalcohol	-2.6991	-0.4377	17.2243	0.7139
-O-	-0.7185	0.0985	2.9405	0.1149
(-O-) _R	-29.8045	-0.2847	-4.3145	8.3131
(-O-) _{AC}	-2.3454	0.0872	6.4296	0.5389
-CHO	-0.8288	-0.2612	3.7241	0.2386
>CO	-2.6622	0.1142	6.7008	0.7348
(>CO) _R	45.9143	-0.2405	3.8828	-12.4994
HCOOH	-2.7291	0.0413	27.4079	0.0002
-COOH for C<7	-4.0451	-0.1841	12.6878	1.1139
-COOH for C>6	-0.6721	-0.1693	20.0309	0.0279
HCOO-	-3.3731	-0.0113	9.4694	0.6071
-COO- for C<8	-0.0635	-0.2162	1.9325	0.4686
-COO- for C>7	-2.5390	0.0006	5.4231	0.8717
>CHO-	-5.4872	1.5834	34.5474	-0.4244
-(CO)-O-(CO)- anhydride	-11.8236	0.0111	7.2831	3.6587
-O-(CO)-O- carbonate	-8.0314	0.2848	9.3746	2.1486
(>NO) _R	-16.9531	1.0614	49.1049	2.8583
-NO ₂	-13.0333	0.1801	12.9392	2.8987
=CHNO ₂	-1.9653	0.1322	15.8672	-0.0701
(-NO ₂) _{AC}	-1.2954	0.0427	12.1837	-0.0948
-S-	-3.2767	0.0779	4.4123	0.9549
-SH primary	-2.1030	-0.0965	6.0066	0.3464
-SH secondary	-0.2481	-0.3285	1.9387	0.1148
-SH tertiary	-12.3498	1.2621	23.1473	1.3950
-CSO- for C<13	-15.2678	0.5248	14.2694	3.7646
-CSO- for C>12	3.7475	-1.2592	-23.9353	0.8329
>SO	-32.8607	0.6232	27.5184	7.7525
N, X Groups				
-NH ₂	-1.1345	-0.2126	7.0544	0.1336
-NH-	-6.9489	-0.1723	5.7804	1.6467
-N<	-2.1403	0.4842	6.1893	0.4718
(-NH ₂) _{AC}	-6.3646	-0.0180	23.2752	1.0653
(-NH-) _{AC}	-1.7592	0.2208	14.9707	0.1171
(-N<) _{AC}	-1.2982	0.5975	14.0415	-0.0031
HCONH ₂	-1.5435	-0.2774	31.8007	0.0001
HCONH-	-8.1097	0.0432	20.9135	1.8795
HCON<	-122.3280	26.4615	394.1670	0.3530
-CONH ₂	-6.7363	0.1316	45.5193	1.2172
-CONH-	8.9977	1.5664	60.8742	-4.6399
-COONH ₂	17.8400	-4.5188	-62.0987	1.2353
-COONH-	-10.1316	0.6712	37.9465	1.9199
(>NH) _R	-0.1589	0.1910	12.0578	-0.0276

TABLE 2-356 Group Contributions for the Hsu et al. Method* (Concluded)

Table-specific nomenclature: R = in nonaromatic ring, A = in aromatic ring, RC = attached to nonaromatic ring, AC = attached to aromatic ring, X = halogen, $(-X)_n = n$ X atoms attached to same C atom

Group	<i>a</i>	100 <i>b</i>	0.0001 <i>c</i>	<i>d</i>
N, X Groups				
(=N→) _R	-4.7601	0.1120	6.98437	0.9719
—C≡N	-2.7194	-0.1324	7.7955	0.6293
(—C≡N) _{AC}	0.9435	-0.0086	8.6310	-0.6443
—Cl primary	-1.7997	-0.3851	3.0118	0.5524
=CHCl	1.5851	-0.1934	3.7798	-0.4748
(—Cl) ₂	-3.0561	-1.0770	0.1882	1.2223
(—Cl) ₃	-1.3357	-0.3220	8.8683	0.1702
(—Cl) ₄	4.2070	-0.4130	13.3194	-1.1972
(—Cl) _{AC}	-0.3083	-0.0623	4.1382	-0.2644
—F primary	-9.4982	0.2607	11.3406	1.8461
(—F) ₂	-10.3980	-1.1189	1.3134	2.6681
(—F) ₃	1.5394	0.8465	17.8121	-2.9915
(—F) _{AC}	0.4079	-0.2352	-0.1505	-0.2893
(—F)(—Cl)	-0.8565	-0.3682	4.6451	-0.0751
(—F)(—Cl) ₂	-3.4552	-0.5629	3.6831	0.3613
(—F) ₂ (—Cl)	54.2824	0.0109	5.9474	-14.5771
(—F) ₂ (—Cl) ₂	-2.1710	0.1403	10.3743	-1.1972
—Br primary	-0.7586	-0.6623	-2.4228	0.7385
—Br secondary	-279.0030	-0.3420	1.4253	73.6293
(—Br) _{AC}	-8.1919	-0.1635	3.0150	0.0621
—I primary	-1.4672	-0.2787	4.3362	0.5635
(—I) _{AC}	70.9918	-0.0245	7.2061	-18.9106
—(CO)—Cl	-2.3300	-0.0470	8.2815	0.4485

*Hsu, H.-C., Y.-W. Sheu, and C.-H. Tu, *Chem. Eng. J.*, **88** (2002): 27

Estimation of liquid mixture viscosity without any mixture data is difficult because the viscosity is strongly affected by large molecular size differences and strong cross interactions between the different types of molecules. Viscosity-composition plots for aqueous mixtures can have maxima or minima, and viscosities for these mixtures are particularly difficult to estimate. The UNIFAC-VISCO method described below can be used to predict liquid viscosity of organic mixtures without any mixture data. It is relatively successful even for large differences in molecular size, but it is currently limited in scope by the small number of group contributions available.

Recommended Method UNIFAC-VISCO method.

Reference: Chevalier, J. L., P. Petrino, and Y. Gaston-Bonhomme, *Chem. Eng. Sci.*, **43** (1988): 1303; Gaston-Bonhomme, Y., P. Petrino, and J. L. Chevalier, *Chem. Eng. Sci.*, **49** (1994): 1799.

Classification: Group contributions.

Expected uncertainty: 20 percent.

Applicability: Organic liquids.

Input data: Molecular structure; pure-component molar volumes and viscosities at the mixture temperature.

Description: The equations for this method are

$$\ln\left(\frac{\eta}{\text{mPa}\cdot\text{s}}\right) = \sum_{i=1}^{\text{comp}} x_i \ln\left(\frac{\eta_i}{\text{mPa}\cdot\text{s}} \cdot \frac{V_i}{V_m}\right) + \frac{g_c^E}{RT} - \frac{g_r^E}{RT} \quad (2-99)$$

where V_m is the mixture molar volume and V_i is the pure-component molar volume of component i . The combinatorial and residual excess Gibbs energies are calculated exactly as in the UNIFAC method (see Sec. 13) with the exception that the group interactions Ψ_{mn} are calculated by using

$$\Psi_{mn} = \exp\left(-\frac{\alpha_{mn}}{298.15}\right) \quad (2-100)$$

The interaction parameters α_{mn} are obtained from Table 2-357.

Example Estimate the viscosity of a mixture of 51.13 mol % ethanol(1) and 48.87 mol % benzene(2) at 298.15 K.

Required input: Values from the DIPPR® 801 database for the pure components at 298.15 K are $\eta_1 = 1.0774$ mPa·s, $\eta_2 = 0.5997$ mPa·s, $V_1 = 0.05862$ m³/kmol, and $V_2 = 0.08948$ m³/kmol.

Groups, area fractions, volume fractions:

Group	<i>R</i>	<i>Q</i>	<i>N</i> ₁	<i>N</i> ₂
CH ₃	0.9011	0.8480	1	0
CH ₂	0.6744	0.5400	1	0
CH _{ar}	0.5313	0.4000	0	6
OH	1.0000	1.2000	1	0

Group	<i>r</i> ₁	<i>q</i> ₁	<i>r</i> ₂	<i>q</i> ₂
CH ₃	0.9011	0.848	0	0
CH ₂	0.6744	0.54	0	0
CH _{ar}	0	0	3.1878	2.4
OH	1	1.2	0	0
Total	2.5755	2.588	3.1878	2.4

where in the above table

$$q_i = \sum_{k=1}^N N_{ik} Q_k \quad \text{and} \quad r_i = \sum_{k=1}^N N_{ik} R_k$$

$$\theta_1 = \frac{x_1 q_1}{\sum_{i=1}^4 x_i q_i} = \frac{(0.5113)(2.588)}{(0.5113)(2.588) + (0.4887)(2.4)} = 0.53 \quad \theta_2 = 0.47$$

$$\phi_1 = \frac{x_1 r_1}{\sum_{i=1}^4 x_i r_i} = \frac{(0.5113)(2.5755)}{(0.5113)(2.5755) + (0.4887)(3.1878)} = 0.458 \quad \phi_2 = 0.542$$

UNIFAC combinatorial term:

$$\frac{g_c^E}{RT} = \sum_{i=1}^2 x_i \ln \frac{\phi_i}{x_i} + 5 \sum_{i=1}^2 x_i q_i \ln \frac{\theta_i}{\phi_i} = 0.124$$

TABLE 2-357 UNIFAC-VISCO* Group Interaction Parameters α_{mn}

m/n	CH ₂	CH ₃	CH _{2cy}	CH _{ar}	Cl	CO	COO	OH	CH ₃ OH
CH ₂	0	66.53	224.9	406.7	60.30	859.5	1172.0	498.6	-219.7
CH ₃	-709.5	0	-130.7	-119.5	82.41	11.86	-172.4	594.4	-228.7
CH _{2cy}	-538.1	187.3	0	8.958	251.4	-125.4	-165.7	694.4	-381.53
CH _{ar}	-623.7	237.2	50.89	0	177.2	128.4	-49.85	419.3	-88.81
Cl	-710.3	375.3	-163.3	-139.8	0	-404.3	-525.4	960.2	-165.4
CO	586.2	-21.56	740.6	-117.9	-4.145	0	29.20	221.5	55.52
COO	541.6	-44.25	416.2	-36.17	240.5	22.92	0	186.8	69.62
OH	-634.5	1209.0	-138	197.7	195.7	664.1	68.35	0	416.4
CH ₃ OH	-526.1	653.1	751.3	51.31	-140.9	-22.59	-286.2	-23.91	0

*Chevalier, J. L., P. Petrino, and Y. Gaston-Bonhomme, *Chem. Eng. Sci.*, **43** (1988): 1303; Gaston-Bonhomme, Y., P. Petrino, and J. L. Chevalier, *Chem. Eng. Sci.*, **49** (1994): 1799.

Group interactions:

m/n	α_{mn}			
group	CH ₃	CH ₂	CH _{ar}	OH
CH ₃	0	-709.5	-119.5	594.4
CH ₂	66.53	0	406.7	498.6
CH _{ar}	237.2	-623.7	0	419.3
OH	1209	-634.5	197.7	0

m/n	Ψ_{mn}			
group	CH ₃	CH ₂	CH _{ar}	OH
CH ₃	1.000	10.801	1.493	0.136
CH ₂	0.800	1.000	0.256	0.188
CH _{ar}	0.451	8.100	1.000	0.245
OH	0.017	8.399	0.515	1.000

The α_{mn} values were obtained from Table 2-357, and Ψ_{mn} values were calculated from Eq. (2-100).

Group fractions in the mixture:

Group	N	X	XQ	Θ	$\ln \gamma$
CH ₃	0.5113	0.1145	0.097083	0.17370	0.293
CH ₂	0.5113	0.1145	0.061822	0.11061	-0.873
CH _{ar}	2.9322	0.6565	0.262618	0.46988	0.066
OH	0.5113	0.1145	0.137382	0.24581	1.077
Sum	4.4661		0.558905		

$$\text{Here } \Theta_m = \frac{X_m Q_m}{\sum_{i=1}^4 X_i Q_i}$$

and

$$\ln \gamma_m = Q_m \left[1 - \ln \left(\sum_{i=1}^4 \Theta_i \Psi_{i,m} \right) - \sum_{j=1}^4 \frac{\Theta_j \Psi_{m,j}}{\sum_{j=1}^4 \Theta_j \Psi_{j,j}} \right]$$

Group fractions in pure components:

Ethanol					
Group	N	X	XQ	Θ	$\ln \gamma$
CH ₃	1	0.3333	0.283	0.3277	0.5306
CH ₂	1	0.3333	0.180	0.2087	-0.9405
CH _{ar}	0	0.0000	0.000	0.0000	0.2095
OH	1	0.3333	0.400	0.4637	0.6179
Sum	3		0.863		

Benzene					
Group	N	X	XQ	Θ	$\ln \gamma$
CH ₃	0	0	0	0	0.257
CH ₂	0	0	0	0	-0.728
CH _{ar}	6	1	0.4	1	0.000
OH	0	0	0	0	2.270
Sum	6		0.4		

Here Θ and $\ln \gamma$ are defined by the same equations shown above for the mixture groups.

UNIFAC residual term:

$$\frac{g^E}{RT} = \sum_{i=1}^2 x_i \left[\sum_{m=1}^4 N_{m,i} (\ln \gamma_m - \ln \gamma_{m,i}) \right] = 0.3425$$

where N_m and $\ln \gamma_m$ refer to the mixture and $N_{m,i}$ and $\ln \gamma_{m,i}$ refer to the pure-component values.

Mixture volume:

$$V_m = \sum_{i=1}^2 x_i V_i = 0.5113 \left(0.05862 \frac{\text{m}^3}{\text{kmol}} \right) + 0.4887 \left(0.08948 \frac{\text{m}^3}{\text{kmol}} \right)$$

$$= 0.07370 \frac{\text{m}^3}{\text{kmol}}$$

Using Eq. (2-99):

$$\ln \left(\frac{\eta}{\text{mPa}\cdot\text{s}} \right) = 0.5113 \ln \left[1.0774 \left(\frac{0.05862}{0.07370} \right) \right] + 0.4887$$

$$\ln \left[0.5997 \left(\frac{0.08948}{0.07370} \right) \right] + 0.124 - 0.3425 = -0.4523$$

$$\eta = \exp (-0.4523) \text{ mPa}\cdot\text{s} = 0.636 \text{ mPa}\cdot\text{s}$$

The estimated value is 6.6 percent below the reported experimental value of 0.681 mPa·s [Kouris, S., and C. Panayiotou, *J. Chem. Eng. Data*, **34** (1989): 200].

THERMAL CONDUCTIVITY

Thermal conductivity k is a measure of the rate at which heat conducts through the material and is defined as the proportionality constant in Fourier's law of heat conduction that relates the gradient of temperature to the heat flux or flow per unit area. In SI units, it has the units of W/(m·K). The conduction mechanism in low-density media, gases,

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is primarily via molecular collisions, and k increases with increasing temperature (increasing molecular velocity). In dense media such as liquids, the energy transfer by conduction occurs more efficiently through the attractive intermolecular potential wells between the molecules than through collisions. As a result, liquid thermal conductivity generally decreases with increasing temperature (except for water, aqueous solutions, and a few multihydroxy and multiamine compounds), corresponding to the decrease in density with increased temperature. Below or near the normal boiling point, the temperature dependence of liquid thermal conductivity is nearly linear for modest temperature ranges and can be represented by

$$k = A - BT \quad (2-101)$$

where B is generally in the range of 1×10^{-4} to 3×10^{-4} W/(m·K²).

Gases Methods for estimating low-pressure gas thermal conductivities are based on kinetic theory and generally correlate the dimensionless group $kM/\eta C_p$ (M = molecular weight, η = viscosity, C_p = isochoric heat capacity) known as the Eucken factor. The method of Stiel and Thodos is recommended for pure nonpolar compounds, and the method of Chung is recommended for pure polar compounds.

Recommended Method Stiel-Thodos method.

Reference: Stiel, L. I., and G. Thodos, *AIChE J.*, **10** (1964): 26.

Classification: Empirical extension of theory.

Expected uncertainty: 15 percent.

Applicability: Pure nonpolar gases at low pressure.

Input data: M , T_c , η , and C_v .

Description: The following equations may be used depending upon the molecular shape:

$$\frac{kM}{\eta C_v} = 2.5 \quad \text{monatomic} \quad (2-102)$$

$$\frac{kM}{\eta C_v} = 1.30 + \left(\frac{R}{C_v}\right) \left(1.7614 - \frac{0.3523}{T_r}\right) \quad \text{linear molecules} \quad (2-103)$$

$$\frac{kM}{\eta C_v} = 1.15 + 2.033 \left(\frac{R}{C_v}\right) \quad \text{nonlinear molecules} \quad (2-104)$$

where η = viscosity at same conditions as desired for k . *Note:* Because this method is only applicable at low pressures, C_v may be calculated as $C_p - R$, where C_p is the ideal gas isobaric heat capacity.

Example Estimate the low-pressure thermal conductivity of toluene vapor at 500 K.

Required properties from the DIPPR® 801 database:

$$T_c = 591.75 \text{ K} \quad M = 92.138 \text{ g/mol} \quad \eta(500 \text{ K}) = 1.1408 \times 10^{-5} \text{ Pa}\cdot\text{s}$$

$$C_v = C_p - R = (170.78 - 8.314) \text{ J/(mol}\cdot\text{K)} = 162.47 \text{ J/(mol}\cdot\text{K)}$$

Auxiliary quantities:

$$T_r = 500/591.75 = 0.845 \quad R/C_v = (8.314)/(162.47) = 0.0512$$

From Eq. (2-104):

$$k = [1.15 + (2.033)(0.0512)] \left[\frac{(1.1408 \times 10^{-5} \text{ Pa}\cdot\text{s}) \left(162.47 \frac{\text{J}}{\text{mol}\cdot\text{K}}\right)}{92.138 \frac{\text{g}}{\text{mol}}} \right] = 25.2 \frac{\text{mW}}{\text{m}\cdot\text{K}}$$

The estimated value is 18 percent below the DIPPR® 801 recommended value of 30.76 mW/(m·K).

Recommended Method Chung-Lee-Starling method.

Reference: Chung, T.-H., L. L. Lee, and K. E. Starling, *Ind. Eng. Chem. Fundam.*, **23** (1984): 8.

Classification: Corresponding states.

Expected uncertainty: 15 percent.

Applicability: Pure organic gases at low pressure.

Input data: C_v , ω , T_c , M , and η .

Description: The following equations apply:

$$\frac{kM}{\eta C_v} = 3.75 \Psi \left(\frac{R}{C_v}\right) \quad (2-105)$$

$$\Psi = 1 + \alpha \left(\frac{0.215 + 0.28288\alpha - 1.061\beta + 0.26665\gamma}{0.6366 + \beta\gamma + 1.061\alpha\beta} \right) \quad (2-106)$$

$$\alpha = \frac{C_v}{R} - 1.5 \quad \beta = 0.7862 - 0.7109\omega + 1.3168\omega^2 \quad (2-107)$$

$$\gamma = 2.0 + 10.5T_r^2$$

Example Estimate the low-pressure thermal conductivity of naphthalene vapor at 500 K.

Required properties from the DIPPR® 801 database:

$$T_c = 748.4 \text{ K} \quad M = 128.17 \text{ g/mol} \quad \omega = 0.30203$$

$$\eta(500 \text{ K}) = 1.0173 \times 10^{-5} \text{ Pa}\cdot\text{s}$$

$$C_v = C_p - R = (219.82 - 8.314) \text{ J/(mol}\cdot\text{K)} = 211.51 \text{ J/(mol}\cdot\text{K)}$$

Auxiliary quantities [Eqs. (2-106) and (2-107)]:

$$T_r = 500/748.4 = 0.6681 \quad R/C_v = (8.314)/(211.51) = 0.0393$$

$$\gamma = 2.0 + (10.5)(0.6681)^2 = 6.6866 \quad \alpha = (0.0393)^{-1} - 1.5 = 23.9388$$

$$\beta = 0.7862 - (0.7109)(0.30203) + (1.3168)(0.30203)^2 = 0.6916$$

$$\Psi = 1 + 23.9388$$

$$\times \left[\frac{0.215 + 0.28288(23.9388) - 1.061(0.6916) + 0.26665(6.6866)}{0.6366 + 0.6916(6.6866) + 1.061(23.9388)(0.6916)} \right] = 9.4273$$

From Eq. (2-105):

$$k = (3.75)(9.4273) \left[\frac{(1.1408 \times 10^{-5} \text{ Pa}\cdot\text{s}) \left(8.314 \frac{\text{J}}{\text{mol}\cdot\text{K}}\right)}{128.17 \frac{\text{g}}{\text{mol}}} \right] = 23.33 \frac{\text{mW}}{\text{m}\cdot\text{K}}$$

The estimated value is 1.0 percent above the DIPPR® 801 recommended value of 23.09 mW/(m·K).

Liquids The Baroncini method provides accurate liquid thermal conductivity estimates if the compound clearly belongs to one of the specific families designated by the correlation. The Sastri-Rao method is recommended for other specific families as listed in the discussion of the method given below. As a general method for estimating thermal conductivity of pure liquids at ambient pressure for all other compounds, the Missenard method is recommended.

Recommended Method Baroncini method.

Reference: Baroncini, C., et al., *Int. J. Thermophys.*, **2** (1981): 21.

Classification: Empirical correlation.

Expected uncertainty: 10 percent.

Applicability: Particularly accurate for the following families: acetates, aliphatic ethers, halogenated compounds, dicarboxylic acids, ketones, aliphatic alcohols, aliphatic acids, propionates and butyrates, and unsaturated aliphatic esters.

Input data: M , T_b (normal boiling point), T_c .

Description:

$$\frac{k}{W/(\text{m}\cdot\text{K})} = A \left(\frac{T_b}{K}\right)^\alpha \left(\frac{M}{\text{g/mol}}\right)^\beta \left(\frac{T_c}{K}\right)^{-\gamma} \frac{(1 - T_r)^{0.38}}{T_r^{1/6}} \quad (2-108)$$

where A , α , β , and γ are obtained from Table 2-358.

TABLE 2-358 Correlation Parameters for Baroncini et al. Method^a for Estimation of Thermal Conductivity

Family	A	α	β	γ
Saturated hydrocarbons	0.00350	1.2	0.5	0.167
Olefins	0.0361	1.2	1	0.167
Cycloparaffins	0.0310	1.2	1	0.167
Aromatics	0.0346	1.2	1	0.167
Alcohols	0.00339	1.2	0.5	0.167
Organic acids	0.00319	1.2	0.5	0.167
Ketones	0.00383	1.2	0.5	0.167
Esters	0.0415	1.2	1	0.167
Ethers	0.0385	1.2	1	0.167
Refrigerants				
R20, R21, R22, R23	0.562	0	0.5	-0.167
Others	0.494	0	0.5	-0.167

^aBaroncini, C., et al., *Int. J. Thermophys.*, **2** (1981): 21.

Example Estimate the thermal conductivity of liquid *p*-cresol at 400 K. Required properties from DIPPR[®] 801 database:

$$M = 108.1378 \text{ g/mol} \quad T_c = 704.65 \text{ K} \quad T_b = 475.133 \text{ K}$$

Auxiliary properties:

$$T_r = T/T_c = (400 \text{ K})/(704.65 \text{ K}) = 0.5677$$

From Table 2-358 for alcohols:

$$A = 0.00339 \quad \alpha = 1.2 \quad \beta = \frac{1}{2} \quad \gamma = 0.167$$

From Eq. (2-108):

$$\frac{k}{W/(m \cdot K)} = (0.00339)(475.13)^{1.2}(108.1378)^{-1/2}(704.65)^{-0.167}$$

$$\frac{(1 - 0.5677)^{0.38}}{0.5677^{1/6}} = 0.142$$

The estimated value is 7.6 percent higher than the DIPPR[®] 801 recommended value of 0.132 W/(m·K).

Recommended Method Sastri-Rao method.

Reference: Sastri, S. R. S., and K. K. Rao, *Chem. Eng. J.*, **74** (1999): 161.

Classification: Group contributions and corresponding states.

Expected uncertainty: 10 percent.

Applicability: Particularly accurate for the following families: alkanes, alkenes, aromatic alcohols, cycloalkanes, epoxides, aliphatic acids, condensed rings.

Input data: T_c , T_b (normal boiling point), and molecular structure.

Description: The thermal conductivity at the normal boiling point k_b is calculated from the sum of the group contributions given in Table 2-359. This value is then scaled to the desired reduced temperature by using a factor α^β in which the power β is related to the fractional reduced temperature distance between the normal boiling point and the critical point as shown in the following equations:

$$k_b = \sum_{i=1}^N n_i \Delta k_i \quad \beta = 1 - \left(\frac{1 - T_r}{1 - T_{br}} \right)^\gamma \quad k = k_b \alpha^\beta \quad (2-109)$$

where $\gamma = 1.23$ for alcohols and 0.2 for all other compounds and $\alpha = 0.856$ for alcohols and 0.16 for all other compounds.

TABLE 2-359 Sastri-Rao^a Group Contributions for Liquid Thermal Conductivity at the Normal Boiling Point

Group	Δk_i (W/m·K)	Group	Δk_i (W/m·K)
Hydrocarbons		—COOH	0.0650
—CH ₃	0.0545	—NH ₂	0.0880
—CH ₂ —	-0.0008	—NH—	0.0065
>CH—	-0.0600	—NH— ring	0.0450
>C<	-0.1230	>N—	-0.0605
==CH ₂	0.0545	>N— ring	0.0135
==CH—	0.0020	—CN	0.0645
==C<	-0.0630	—NO ₂	0.0700
==C=	0.1200	—S	0.0100
ring ^c	0.1130	—F perfluoro	0.0568
Nonhydrocarbons		—F other	0.0510
—O—	0.0100	—Cl	0.0550
—OH primary ^b	0.0830	—Br	0.0415
—OH other	0.0680	—I	0.0245
>CO	0.0175	—H ^c	0.0675
>CHO	0.0730	3-ring	0.1500
—COO—	0.0070	other ring ^d	0.1100

Corrections for multigroup interactions

Hydrocarbons with 4 or fewer carbon atoms	^e 0.0150/(5 - nC)
Single CH ₃ group + nonhydrocarbon groups other than COOH, Br, I ^f	0.0600
Two hydrocarbon + nonhydrocarbon groups other than COOH, Br, I ^f	0.0285
Unsaturated aliphatic compounds with three hydrocarbon groups	0.0285
Special groups Cl(CH ₂) _n Cl	0.0350
More than one nonhydrocarbon group with hydrocarbon groups	0.0095
Nonhydrocarbon groups but no hydrocarbon groups	0.1165

^aAll rings are treated as separate rings in polycyclic compounds.

^bUsed only for aliphatic primary alcohols and phenols having no branch chains.

^cUsed in methane, formic acid, formates, etc.

^dUsed for all rings in polycyclic compounds with at least one nonhydrocarbon ring.

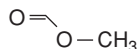
^eThe number of carbon atoms is nC.

^fAliphatic nonhydrocarbon liquids such as methylformate, acetanhydride, and ethylformate, having more than one type of nonhydrocarbon group and (1) one or two methyl groups or (2) one ethyl group only, require two correction factors. One is due to the hydrocarbon groups, and one is due to the presence of more than one type of nonhydrocarbon group.

^gSOURCE: Sastri, S. R. S., and K. K. Rao, *Chem. Eng. J.*, **74** (1999): 161.

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Example Estimate the thermal conductivity of liquid methyl formate at 323 K.
Molecular structure:



Group contributions from Table 2-359:

Group	n_i	Δk_i	$n_i \Delta k_i$
—CH ₃	1	0.0545	0.0545
—COO—	1	0.0070	0.0070
—H	1	0.0675	0.0675
Corrections for multigroup interactions			
Single CH ₃ group + nonhydrocarbon groups			0.0600
More than one nonhydrocarbon with hydrocarbon groups			0.0095
Total			0.1985

Required input properties from DIPPR® 801 recommended values:

$$T_c = 487.2 \text{ K} \quad T_b = 304.9 \text{ K}$$

$$T_r = T/T_c = 323/487.2 = 0.6630 \quad T_{br} = 304.9/487.2 = 0.6258$$

From Eq. (2-109):

$$\alpha = 0.16 \quad \gamma = 0.2 \quad (\text{nonalcohol values})$$

$$\beta = 1 - \left(\frac{1 - 0.633}{1 - 0.6257} \right)^{0.2} = 0.0207 \quad k_b = 0.1985 \text{ W/(m}\cdot\text{K)}$$

$$k = [0.1985 \text{ W/(m}\cdot\text{K)}] (0.16)^{0.0207} = 0.1911 \text{ W/(m}\cdot\text{K)}$$

The estimated value is 10.2 percent above the DIPPR® 801 recommended value of 0.1734 W/(m·K).

Recommended Method Missenard method.

Reference: Missenard, A., *Comptes Rendus*, **260** (1965): 5521.

Classification: Corresponding states.

Expected uncertainty: 20 percent.

Applicability: Organic compounds; nonassociating.

Input data: T_c , m (number of atoms in molecule), ρ_{273} (liquid density at 273.15 K), T_b , M , $C_{p,273}$ (liquid heat capacity at 273.15 K).

Description:

$$\frac{k_{273}}{\text{mW/(m}\cdot\text{K)}} = \frac{8.4}{m^{1/4}} \cdot \left(\frac{T_b}{\text{K}} \right)^{1/2} \left(\frac{\rho_{273}}{\text{g/m}^3} \right)^{1/2} \left(\frac{M}{\text{g/mol}} \right)^{-1/2} \times \left[\frac{C_{p,273}}{\text{J/(mol}\cdot\text{K)}} \right] \quad (2-110)$$

$$k = \frac{k_{273}[3 + 20(1 - T_r)^{2/3}]}{3 + 20(1 - T_{r,273})^{2/3}} \quad (2-111)$$

where $T_{r,273} = (273 \text{ K})/T_c$.

Example Estimate the thermal conductivity of *m*-xylene at 350 K.

Required properties from DIPPR® 801 database:

$$T_c = 617 \text{ K} \quad m = 18 \quad \rho_{273} = 7.6812 \text{ kmol/m}^3$$

$$T_b = 412.27 \text{ K} \quad M = 106.165 \text{ kg/kmol} \quad C_{p,273} = 200.64 \text{ kJ/(kmol}\cdot\text{K)}$$

Auxiliary properties:

$$T_r = 350/617 = 0.5673 \quad T_{br} = 412.27/617 = 0.6682$$

$$T_{r,273} = 273/617 = 0.4425$$

From Eq. (2-110):

$$\left(\frac{k_{273}}{\text{mW/(m}\cdot\text{K)}} \right) = (8.4)(412.27)^{1/2}(0.007681)^{1/2}(106.165)^{-1/2}(200.64)(18)^{-0.25} = 141.3$$

From Eq. (2-111):

$$k = \frac{k_{273}[3 + 20(1 - T_r)^{2/3}]}{3 + 20(1 - T_{r,273})^{2/3}} = \frac{\left(141.3 \frac{\text{mW}}{\text{m}\cdot\text{K}} \right) [3 + 20(1 - 0.5673)^{2/3}]}{3 + 20(1 - 0.4425)^{2/3}} = 123.3 \frac{\text{mW}}{\text{m}\cdot\text{K}}$$

The estimated value is 4.5 percent above the DIPPR® 801 recommended value of 118.0 mW/(m·K).

Liquid Mixtures The thermal conductivity of liquid mixtures generally shows a modest negative deviation from a linear mass-fraction-weighted average of the pure-component values. Although more complex methods with some improved accuracy are available, two simple methods are recommended here that require very little additional information. The first method applies only to binary mixtures while the second can be used for multiple components.

Recommended Method Filippov correlation.

References: Filippov, L. P., *Vest. Mosk. Univ., Ser. Fiz. Mat. Estestv. Nauk*, **10** (1955): 67; Filippov, L. P., and N. S. Novoselova, *Sugden, Vest. Mosk. Univ., Ser. Fiz. Mat. Estestv. Nauk*, **10** (1955): 37.

Classification: Empirical correlation.

Expected uncertainty: 4 to 8 percent.

Applicability: Binary liquid mixtures.

Input data: Pure-component thermal conductivities k_i at mixture conditions; w_i .

Description: The mixture thermal conductivity is calculated from the pure-component values using

$$k = w_1 k_1 + w_2 k_2 - 0.72 w_1 w_2 |k_2 - k_1| \quad (2-112)$$

where w_i is the mass fraction of pure fluid i .

Recommended Method Li correlation.

References: Li, C. C., *AIChE J.*, **22** (1976): 927.

Classification: Empirical correlation.

Expected uncertainty: 4 to 8 percent.

Applicability: Liquid mixtures.

Input data: Pure-component thermal conductivities k_i at mixture conditions; $\rho_{L,i}$

Description: The mixture thermal conductivity is correlated as a function of the mixture volume fractions ϕ_i :

$$k = \sum_{i=1}^C \sum_{j=1}^C \phi_i \phi_j \frac{2k_i k_j}{k_i + k_j} \quad (2-113)$$

$$\text{where } \phi_i = \frac{x_i \rho_{L,i}^{-1}}{\sum_{j=1}^C x_j \rho_{L,j}^{-1}}$$

Example Estimate the thermal conductivity of a mixture containing 30.2 mol % diethyl ether(1) and 69.8 mol % methanol(2) at 273.15 K and 0.1 MPa, using the Filippov and Li correlations.

Auxiliary data: The pure-component thermal conductivities and molar densities at 273.15 K recommended in the DIPPR® 801 database are

$$k_1 = 0.1383 \text{ W/(m}\cdot\text{K)} \quad \rho_1 = 9.9335 \text{ kmol/m}^3 \quad M_1 = 74.1216 \text{ kg/kmol}$$

$$k_2 = 0.2069 \text{ W/(m}\cdot\text{K)} \quad \rho_2 = 25.371 \text{ kmol/m}^3 \quad M_2 = 32.0419 \text{ kg/kmol}$$

The mass fractions corresponding to the mole fractions given above are

$$w_1 = 0.5 \quad w_2 = 0.5$$

The volume fractions are

$$\phi_1 = \frac{(0.302)(9.9335)^{-1}}{(0.302)(9.9335)^{-1} + (0.698)(25.371)^{-1}} = 0.525 \quad \phi_2 = 0.475$$

Calculation using Eq. (2-112):

$$k = [(0.5)(0.1383) + (0.5)(0.2069) - (0.72)(0.5)(0.5)(0.2069 - 0.1383)] \frac{W}{m \cdot K}$$

$$= 0.160 \text{ W/(m} \cdot \text{K)}$$

Calculation using Eq. (2-113):

$$k = \left[(0.525)^2(0.1383) + 2 \cdot \frac{(0.525)(0.475)(2)(0.1383)(0.2069)}{0.1383 + 0.2069} + (0.475)^2(0.2069) \right] \frac{W}{m \cdot K}$$

$$= 0.167 \text{ W/(m} \cdot \text{K)}$$

The Filippov value is 7.5 percent lower than the experimental value of 0.173 W/(m·K) [Jamieson, D. T., and B. K. Hastings, *Thermal Conductivity, Proceedings of the Eighth Conference*, C. Y. Ho and R. E. Taylor, eds., Plenum Press, New York, 1969]; the Li value is 3.5 percent lower than the experimental value.

Solids There is no reliable method for estimating solid thermal conductivity at this time.

SURFACE TENSION

The surface layer at a vapor-liquid interface is in tension and will contract to minimize the surface area. Qualitatively, the surface tension is due to the larger attractive forces that molecules at the interface experience from molecules in the dense liquid phase than from those in the low-density gas phase. Quantitatively, *surface tension* is defined as the force in the surface plane per unit length. Jasper [Jasper, J. J., *J. Phys. Chem. Ref. Data*, **1** (1972): 841] has made a critical evaluation of experimental surface tension data for approximately 2200 pure chemicals and correlated surface tension σ (mN/m = dyn/cm) with temperature as

$$\sigma = A - BT$$

Jasper's evaluation also includes values of A and B for most of the tabulated chemicals. Surface tension decreases with increasing temperature and increasing pressure.

Pure Liquids An approach suggested by Macleod [Macleod, D. B., *Trans. Faraday Soc.*, **19** (1923): 38] and modified by Sugden [Sugden, S. J., *Chem. Soc.*, **125** (1924): 32] relates σ to the liquid and vapor molar densities and a temperature-independent parameter called the parachor P

$$\frac{\sigma}{\text{mN/m}} = \left[P \cdot \left(\frac{\rho_L - \rho_V}{10^3 \text{ kmol/m}^3} \right) \right]^4 \quad (2-114)$$

where ρ_L and ρ_V are the saturated molar liquid and vapor densities, respectively. At low temperatures, where $\rho_L \gg \rho_V$, the vapor density can be neglected, but at higher temperatures the density of both phases must be calculated. At the critical point the surface tension is zero as $\rho_L = \rho_V$. Quayle [Quayle, O. R., *Chem. Rev.*, **53** (1953): 439] proposed a group contribution method for estimating P that has been improved in recent years by Knotts et al. [Knotts, T. A., et al., *J. Chem. Eng. Data*, **46** (2001): 1007]. This method using P is recommended when groups are available; otherwise, the Brock-Bird [Brock, J. R., and R. B. Bird, *AIChE J.*, **1** (1955): 174] corresponding-states method as modified by Miller [Miller, D. G., *Ind. Eng. Chem. Fundam.*, **2** (1963): 78] may be used to estimate surface tension for compounds that are not strongly polar or associating.

Recommended Method Parachor method.

References: Macleod, D. B., *Trans. Faraday Soc.*, **19** (1923): 38; Sugden, S. J., *Chem. Soc.*, **125** (1924): 32; Knotts, T. A., et al., *J. Chem. Eng. Data*, **46** (2001): 1007.

Classification: Group contributions and QSPR.

Expected uncertainty: 4 percent.

Applicability: Organic compounds for which group values are available.

Input Data: ρ_L , molecular structure, and Table 2-360.

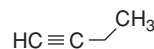
Description: Equation (2-114) is used with P calculated from

$$P = \sum_{i=1}^N n_i \Delta P_i \quad (2-115)$$

Group values for the parachor are given in Table 2-360.

Example Estimate the surface tension of ethylacetylene at 237.45 K.

Structure:



Group	n_i	ΔP_i	$n_i \Delta P_i$
$\equiv \text{CH}$	1	43.64	43.64
$\equiv \text{C}-$	1	28.64	28.64
$>\text{CH}_2$ ($n = 1-11$)	1	39.92	39.92
CH_3	1	55.25	55.25
Total			167.45

Required properties: The DIPPR® 801 database gives $\rho_L = 13.2573 \text{ kmol/m}^3$ at 237.45 K.

Calculation using Eq. (2-114):

$$\sigma = \left[(167.45) \left(\frac{13.2573}{1000} \right) \right]^4 \frac{\text{mN}}{\text{m}} = 0.02429 \frac{\text{N}}{\text{m}}$$

The estimated value is 0.9 percent above the DIPPR® 801 recommended value of 0.02407 N/m.

Recommended Method Brock-Bird method.

Reference: Brock, J. R., and R. B. Bird, *AIChE J.*, **1** (1955): 174; Miller, D. G., *Ind. Eng. Chem. Fundam.*, **2** (1963): 78.

Classification: Corresponding states.

Expected uncertainty: 5 percent.

Applicability: Nonpolar and moderately polar organic compounds.

Input data: T_c , P_c , and T_b .

Description:

$$\frac{\sigma}{\text{mN/m}} = (5.553 \times 10^{-5}) \left(\frac{P_c}{\text{Pa}} \right)^{2/3} \left(\frac{T_c}{\text{K}} \right)^{1/3} F(1 - T_r)^{11/9} \quad (2-116)$$

where

$$F = \frac{T_{br} [\ln(P_c/\text{Pa}) - 11.5261]}{1 - T_{br}} - 1.3281 \quad (2-117)$$

Example Estimate the surface tension for ethyl mercaptan at 303.15 K.

Required properties from DIPPR® 801:

$$T_c = 499.15 \text{ K} \quad P_c = 5.49 \times 10^6 \text{ Pa} \quad T_b = 308.15 \text{ K}$$

Supporting quantities:

$$T_r = (303.15 \text{ K}) / (499.15 \text{ K}) = 0.6073$$

$$T_{br} = (308.15 \text{ K}) / (499.15 \text{ K}) = 0.6173$$

$$F = [0.6173 [\ln(5.49 \times 10^6) - 11.5261] / (1 - 0.6173)] - 1.3281 = 5.113$$

[from Eq. (2-117)]

From Eq. (2-116):

$$\sigma = (5.553 \times 10^{-5}) (5.49 \times 10^6)^{2/3} (499.15)^{1/3} (5.113) (1 - 0.6073)^{11/9} \text{ mN/m}$$

$$= 22.36 \text{ mN/m}$$

The estimated value is 1.4 percent lower than the DIPPR® 801 recommended value of 22.68 mN/m.

TABLE 2-360 Knotts* Group Contributions for the Parachor in Estimating Surface Tension

Group	ΔP_i	Group	ΔP_i
(a) Nonring C		(e) Nitrogen groups	
—CH ₃	55.25	R—NH ₂ (primary R)	44.98
>CH ₂ (n = 1–11)	39.92	R—NH ₂ (sec R)	44.63
>CH ₂ (n = 12–20)	40.11	R—NH ₂ (tert R)	46.44
>CH ₂ (n > 20)	40.51	A—NH ₂ (attached to arom ring)	46.53
>CH—	28.90	>NH (nonring)	29.04
>C<	15.76	>NH (ring)	31.97
==CH ₂	49.76	>NH (in arom ring)	33.92
==CH—	34.57	>N- (nonring)	10.77
==C<	24.50	>N- (ring)	15.71
==C=	24.76	—N= (nonring)	23.24
==CH	43.64	>N (aromatic)	26.49
==C—	28.64	HC≡N (hyd cyanide)	80.94
Branch corrections		—C≡N	65.23
Per branch	−6.02	—C≡N (aromatic)	67.54
sec-sec adjacency	−2.73	(f) Nitrogen and oxygen groups	
sec-tert adjacency	−3.61	—C=ONH ₂ (amides)	93.43
tert-tert adjacency	−6.10	—C=ONH- (amides)	73.64
(b) Nonaromatic ring C		—C=ON< (amides)	57.05
—CH ₂ —	39.21	—NHCHO	91.69
>CH—	23.94	>NCHO	77.12
>C<	7.19	—N=O	64.32
==CH—	34.07	—NO ₂	73.86
==C<	18.85	—NO ₂ (aromatic)	75.05
>CH— (fused ring)	22.05	(g) Sulfur groups	
Ring corrections		R-SH (primary R)	66.89
Three-member ring	12.67	R-SH (sec R)	63.34
Four-member ring	15.76	R-SH (tert R)	65.33
Five-member ring	7.04	—SH (aromatic)	68.30
Six-member ring	5.19	—S— (nonring)	51.37
Seven-member ring	3.00	—S— (ring)	51.75
(c) Aromatic ring C		—S— (aromatic)	51.47
>CH	34.36	>S=O (nonring)	72.21
>C—	16.07	>SO ₂ (nonring)	93.20
—C— (fused arom/arom)	19.73	>SO ₂ (ring)	90.13
—C— (fused arom/aliph)	14.41	(h) Halogen groups	
Arom ring corr		—F	21.81
ortho	−0.60	—Cl	26.24
para	3.40	—Br	51.16
meta	2.24	—I	54.56
subst. naphthalene corr	−7.07	—F (aromatic)	66.30
(d) Oxygen groups		—Cl (aromatic)	70.39
—OH (alc, primary)	31.42	—Br (aromatic)	90.84
—OH (alc, sec)	22.68	—I (aromatic)	92.04
—OH (alc, tertiary)	20.66	(i) Si groups	
—OH (phenol)	30.32	SiH ₄	105.11
—O— (nonring)	20.61	>SiH—	54.50
—O— (ring)	21.67	>Si<	44.93
—O— (aromatic)	23.54	>Si< (ring)	28.64
>C=O (nonring)	47.02	(j) Other inorganic groups	
>C=O (ring)	50.04	—PO ₄	115.59
O=CH— (aldehyde)	66.06	>P—	48.84
CHOOH (formic)	94.01	>B—	22.65
—COOH (acid)	74.57	>Al—	25.06
—OCHO (formate)	82.29	—ClO ₃	106.03
—COO— (ester)	64.97		
—COOCO— (acid anhyd)	115.07		
—OC(=O)O— (ring)	84.05		

*Knotts, T. A., et al., *J. Chem. Eng. Data*, **46** (2001): 1007.

Liquid Mixtures Compositions at the liquid-vapor interface are not the same as in the bulk liquid, and so simple (bulk) composition-weighted averages of the pure-fluid values do not provide quantitative estimates of the surface tension at the vapor-liquid interface of a mixture. The behavior of aqueous mixtures is more difficult to correlate and estimate than that of nonpolar mixtures because small amounts of organic material can have a pronounced effect upon the surface concentrations and the resultant surface tension. These effects are usually modeled with thermodynamic methods that account for the activity coefficients. For example, a UNIFAC method [Suarez, J. T. C. Torres-Marchal, and P. Rasmussen, *Chem. Eng. Sci.*, **44** (1989): 782] is recommended and illustrated in PGL5. For nonaqueous systems the extension of the parachor method, used above for pure fluids, is a simple and reasonably effective method for estimating σ for mixtures.

Recommended Method Parachor correlation.

Reference: Hugill, J. A., and A. J. van Welsenes, *Fluid Phase Equil.*, **29** (1986): 383; Macleod, D. B., *Trans. Faraday Soc.*, **19** (1923): 38; Sugden, S. J., *Chem. Soc.*, **125** (1924).

Classification: Corresponding states.

Expected uncertainty: 3 to 10 percent.

Applicability: Nonaqueous mixtures.

Input data: Liquid and vapor ρ at mixture T ; parachors of pure components; x_i .

Description:

$$\frac{\sigma_m}{\text{mN/m}} = \left(P_{L,m} \frac{\rho_{L,m}}{10^3 \text{ kmol/m}^3} - P_{V,m} \frac{\rho_{V,m}}{10^3 \text{ kmol/m}^3} \right)^4 \quad (2-118)$$

where σ_m = surface tension of the mixture

$P_{L,m}$ = parachor of liquid mixture

$P_{V,m}$ = parachor of vapor mixture

$\rho_{L,m}$ = liquid mixture molar density

$\rho_{V,m}$ = vapor mixture molar density

The following definitions are used for the liquid and vapor mixture parachors:

$$P_{L,m} = \frac{1}{2} \sum_{i=1}^C \sum_{j=1}^C x_i x_j (P_i + P_j) \quad P_{V,m} = \frac{1}{2} \sum_{i=1}^C \sum_{j=1}^C y_i y_j (P_i + P_j) \quad (2-119)$$

where x_i is the mole fraction of component i in the liquid and y_i is the mole fraction of component i in the vapor.

Note that ρ_V is generally very small compared to ρ_L at temperatures substantially lower than T_c and can often be neglected.

Example Estimate the surface tension for a 16.06 mol % n -pentane(1) + 83.94 mol % dichloromethane(2) mixture at 298.15 K.

Required properties from DIPPR® 801:

	P	$\rho_L/(\text{kmol}\cdot\text{m}^{-3})$ at 298.15 K
n -Pentane	231.1	8.6173
Dichloromethane	146.6	15.5211

Mixture parachor from Eq. (2-119) and mixture density:

$$P_{L,m} = (0.1606)^2(231.1) + (0.1606)(0.8394)(231.1 + 146.6) + (0.8394)^2(146.6) = 160.17$$

$$\rho_{L,m} = \left(\sum_{i=1}^C \frac{x_i}{\rho_i} \right)^{-1} = \left(\frac{0.1606}{8.6173} + \frac{0.8394}{15.5211} \right)^{-1} \frac{\text{kmol}}{\text{m}^3} = 13.752 \frac{\text{kmol}}{\text{m}^3}$$

Calculation using Eq. (2-118): Because the temperature is low, the density of the vapor can be neglected, and

$$\frac{\sigma_m}{\text{mN/m}} = [(160.17)(0.013752)]^4 = 23.54 \frac{\text{mN}}{\text{m}}$$

The estimated value is 2.9 percent below the experimental value of 24.24 mN/m reported by De Soria et al. [De Soria, M. L. G., et al., *J. Colloid Interface Sci.*, **103** (1985): 354].

FLAMMABILITY PROPERTIES

Flash points, lower and upper flammability limits, and autoignition temperature are important properties for determining safe operating limits when processing organic compounds. As with any property, experimental values are preferable to predicted values, and prediction techniques for these properties are only modestly accurate.

The flash point is the lowest temperature at which a liquid gives off sufficient vapor to form an ignitable mixture with air near the surface of the liquid or within the vessel used. ASTM test methods include procedures using a closed-cup apparatus (ASTM D 56, ASTM D 93, and ASTM D 3828), which is preferred, and an open-cup apparatus (ASTM D 92 and ASTM D 1310). Closed-cup values are typically lower than open-cup values. When several values are available, the lowest reasonable temperature is usually accepted in order to ensure safe operations.

The lower and upper flammability limits are the boundary-line equilibrium mixtures of vapor or gas with air, which if ignited will just propagate a flame away from the ignition source. Each of these limits has a temperature at which the flammability limits are reached. The lower flammability limit temperature corresponds approximately to the flash point, but since the flash point is determined with downward flame propagation and nonuniform mixtures and the lower flammability temperature is determined with upward flame propagation and uniform vapor mixtures, the measured lower flammability temperature is often somewhat lower than the flash point.

The autoignition temperature is the minimum temperature for a substance to initiate self-combustion in air in the absence of an ignition source.

Recommended Methods Flash point: Thornton method.

Reference: As described by N. Y. Shebeko, A. V. Ivanov, and E. N. Alekhina, "Calculation of Flash Points and Ignition Temperatures of Organic Compounds," *Soviet Chem. Ind.* **16**(1984):1371.

Classification: Atomic contributions.

Expected uncertainty: 5 K.

Applicability: Organic compounds.

Input data: Number of carbon, hydrogen, sulfur, halogen, and oxygen atoms; vapor pressure correlation.

Description: A simple atom contribution method is given by

$$P^* - \frac{P_{\text{sys}}}{1 + 4.76(2\beta - 1)} = 0 \quad (2-120)$$

where P^* = vapor pressure at flash point temperature

P_{sys} = total system pressure, typically 1.01325×10^5 Pa

β = stoichiometric coefficient, defined by

$$\beta = N_C + N_S + \frac{N_H - N_X}{4} - \frac{N_O}{2} \quad (2-121)$$

where N_C = number of carbon atoms in compound

N_S = number of sulfur atoms in compound

N_H = number of hydrogen atoms in compound

N_X = number of halogen atoms in compound

N_O = number of oxygen atoms in compound

Procedure:

Step 1. Determine the number of carbon, sulfur, hydrogen, halogen, and oxygen atoms in the compound.

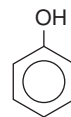
Step 2. Calculate β from Eq. (2-121).

Step 3. Substitute a temperature-dependent expression for vapor pressure in Eq. (2-120) for P^* .

Step 4. Solve Eq. (2-120) for temperature. This temperature is the flash point estimate.

Example Estimate the flash point of phenol.

Structure:



Atomic contributions:

Atom type	Number
C	6
H	6
O	1

$$\beta = 6 + \frac{6}{4} - \frac{1}{2} = 7$$

The DIPPR® 801 correlation for the vapor pressure of phenol is

$$P^* = \exp(95.444 - \frac{10,113}{T} - 10.09 \ln T + 6.7603 \times 10^{-18} T^6)$$

When this expression is used in Eq. (2-120) and solved for temperature, one obtains $T_{\text{FP}} = 348.7$ K, which is 1.2 percent below the DIPPR recommended value of 353 K.

Recommended Method Flammability limits: Pintar method.

Reference: Pintar, A. J., Technical Support Document DIPPR Project 912, Michigan Technological University, Houghton, 1996.

Classification: Group contributions.

Expected uncertainty: 25 percent.

TABLE 2-361 Group Contributions for Pintar* Flammability Limits Method for Organic Compounds

Group	LFL _i	UFL _i	Group	LFL _i	UFL _i
—CH ₃	17.2750	3.8461	>NH	3.2709	-1.9112
>CH ₂	13.7022	1.4959	—N=	-7.2149	-2.3309
>CH—	10.0000	0.2183	—CN	8.0990	3.6918
>C<	5.5291	-0.8422	=C=N—	2.5963	-1.1463
—H	2.7250	2.8206	=N—NH ₂	-3.5071	-0.29897
—OH	2.1797	0.5856	>N—NH ₂	0.5861	2.4811
—O—	-3.0156	-2.2427	—NO ₂	-3.1507	0.8011
—O—O—	-6.0312	-4.4854	—SH	7.9424	0.5344
=C=O	4.6752	0.6009	—S—	11.0079	-1.9832
—CHO	10.3801	-1.2148	—SO—	3.9115	-4.1834
—COOH	4.8890	1.6121	—SO ₂ —	5.5400	—
—COO—	1.2955	-1.1840	—SO ₃ —	2.8600	—
—CO—O—CO—	4.6740	2.4751	—SO ₄ —	0.1800	—
—C ₆ H ₅	73.8338	9.6661	—CO ₃ —	2.4103	-3.4894
<i>m</i> -C ₆ H ₄	57.4447	7.6126	—OPO ₂ —	7.1419	—
<i>o</i> -C ₆ H ₄	57.4447	7.2450	—P=	47.6909	—
<i>p</i> -C ₆ H ₄	57.4447	7.9291	—PO—	7.1515	—
Arom. ring	45.0633	-5.9925	—PO ₄ =	-11.5096	-6.0260
=	4.2821	2.0269	Si—C†	-2.2855	-3.0576
≡	17.5470	0.7842	Si—O†	2.5034	1.4282
—Cl	-2.9697	1.4008	Si—H†	8.3130	-24.4160
—Cl ₂	-5.9764	3.1943	Si—Cl†	4.1010	9.7131
—Cl ₃	-8.0982	4.2024	Si—N†	15.8960	1.6577
—F	-1.2615	0.3984	Si—Si	—	—
—F ₂	-2.1224	0.6847	Al	—	—
—F ₃	-5.1300	1.1952	B	47.3806	—
—Br	-8.0405	4.0018	Cr	—	—
—Br ₂	-16.0809	8.0036	Na	—	—
—Br ₃	-21.9000	12.0054	<i>cis</i>	-6.8350	1.8040
—I	-22.0000	11.4300	<i>trans</i>	0.5821	0.9183
—I ₂	-44.0000	22.8600	Nonarom. ring	2.9082	3.7760
—I ₃	-60.0000	34.2900	Add'l. ring	14.2712	3.1127
—NH ₂	3.7078	1.8802			

*Pintar, A. J., Technical Support Document DIPPR Project 912, Michigan Technological University, Houghton, 1996.

†Does not include contribution of atoms attached to silicon.

TABLE 2-362 Group Contributions for Pintar* Flammability Limits Method for Inorganic Compounds

Group	LFL _i	UFL _i	Group	LFL _i	UFL _i
B	24.5190	-1.3818	>N—NH ₂	0.5861	-0.2990
Br	—	—	O	-24.3242	2.2990
C	32.0745	-0.6259	P	24.8302	—
Cl	14.2658	-1.4231	S	26.6776	1.3171
F	—	—	Si—C†	-2.2855	-3.0576
Fe	—	—	Si—O†	2.5034	1.4282
H	10.3452	0.6500	Si—H†	8.3130	-24.4160
N	-24.5487	1.8453	Si—Cl†	4.1010	9.7131
Na	—	—	Si—N†	15.8960	1.6577
Ni	—	—	Si—Si	—	—

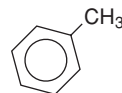
*Pintar, A. J., Technical Support Document DIPPR Project 912, Michigan Technological University, Houghton, 1996.

†Does not include contribution of atoms attached to silicon.

Applicability: Organic and inorganic compounds.**Input data:** Group contributions from Tables 2-361 and 2-362.**Description:** A simple GC method with first-order contributions with lower flammability limit (LFL) and upper flammability limit (UFL) in volume % given by

$$\text{LFL} = \frac{100\%}{\sum n_i \text{LFL}_i} \quad (2-122)$$

$$\text{UFL} = \frac{100\%}{\sum n_i \text{UFL}_i} \quad (2-123)$$

where n_i = number of groups of type i in moleculeLFL_{*i*} = contribution of group i to LFLUFL_{*i*} = contribution of group i to UFL**Example** Estimate the lower and upper flammability limits of toluene.**Structure:****Group contributions:**

Group	LFL	UFL
—CH ₃	17.2750	3.8461
—C ₆ H ₅	73.8338	9.6661

Calculations using Eqs. (2-122) and (2-123):

$$\text{LFL} = \frac{100\%}{17.2750 + 73.8338} = 1.10\%$$

$$\text{UFL} = \frac{100\%}{3.8461 + 9.6661} = 7.40\%$$

TABLE 2-363 Group Contributions for Pintar*Autoignition Temperature Method for Organic Compounds

Group	b_i	Group	b_i	Group	b_i
—CH ₃	301.91	—Cl ₃	1073.47	—SO ₃ —	—
>CH ₂	-10.86	—F	360.60	—SO ₄ —	-31.71
>CH—	-275.17	—F ₂	755.54	—CO ₃ —	442.26
>C<	-570.43	—F ₃	1082.00	—P=	-334.91
—H	391.48	—Br	420.96	—PO—	-549.59
—OH	324.10	—Br ₂	607.69	—OPO ₂ —	—
—O—	-18.60	—Br ₃	1260.00	—PO ₄ =	-329.45
—O—O—	-397.61	—I	310.53	Si—C†	-147.69
=C=O	57.65	—I ₂	—	Si—O†	-136.99
—CHO	195.20	—I ₃	—	Si—H†	-310.52
—COOH	370.75	—NH ₂	354.11	Si—Cl†	-200.88
—CO—	43.90	>NH	9.88	Si—N†	—
—CO—O—CO—	46.11	—N=	-249.91	Si—Si	—
—C ₆ H ₅	380.27	—CN	469.67	Al	—
<i>m</i> —C ₆ H ₄	153.15	=C=N—	-273.70	B	—
<i>o</i> —C ₆ H ₄	77.48	=N—NH ₂	378.27	Cr	—
<i>p</i> —C ₆ H ₄	99.87	>N—NH ₂	-215.02	Na	534.29
Aromatic ring	-1339.65	—NO ₂	292.57	<i>cis</i>	-29.19
=	578.72	—SH	273.84	<i>trans</i>	-38.31
=	1116.50	—S—	-60.75	Nonarom. ring	605.97
—Cl	347.39	—SO—	-91.10	Add'l. ring	565.11
—Cl ₂	726.03	—SO ₂ —	—	Zn	349.02

*Pintar, A. J., *Estimation of Autoignition Temperature*, Technical Support Document DIPPR Project 912, Michigan Technological University, Houghton, 1996.

†Does not include contribution of atoms attached to silicon.

TABLE 2-364 Group Contributions for Pintar*Autoignition Temperature Method for Inorganic Compounds

Group	b_i	Group	b_i	Group	b_i
B	-457.14	N	0.71	Si—C†	-147.69
Br	—	Na	—	Si—O†	-136.99
C	489.19	Ni	-1595.10	Si—H†	-310.52
Cl	395.42	>N—NH ₂	-215.02	Si—Cl†	-200.88
F	—	O	-13.39	Si—N†	—
Fe	-2050.90	P	108.45	Si—Si	—
H	204.55	S	-3.57		

*Pintar, A. J., *Estimation of Autoignition Temperature*, Technical Support Document DIPPR Project 912, Michigan Technological University, Houghton, 1996.

†Does not include contribution of atoms attached to silicon.

The values recommended in the DIPPR® 801 database are 1.2 and 7.1 percent, respectively.

Flammability temperatures are found by determining the temperature at which the vapor pressure equals the partial pressure corresponding to the LFL or UFL.

Recommended Methods Autoignition temperature: Pintar method.

Reference: Pintar, A. J., *Estimation of Autoignition Temperature*, Technical Support Document DIPPR Project 912, Michigan Technological University, Houghton, 1996.

Classification: Group contributions.

Expected uncertainty: 25 percent.

Applicability: Organic and inorganic compounds.

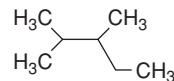
Input data: Group contributions from Tables 2-363 and 2-364.

Description: A simple GC method with first-order contributions is given by

$$\text{AIT} = \sum n_i b_i \quad (2-124)$$

where n_i is the number of groups of type i in the molecule and b_i is the contribution of group i to the autoignition temperature. A more accurate but somewhat more complicated logarithmic GC method was also developed by Pintar in the same reference cited here.

Example Estimate the autoignition temperature of 2,3-dimethylpentane. *Structure and group information:*



Group	n_i	b_i
—CH ₃	4	301.91
>CH ₂	1	-10.86
>CH—	2	-275.17

Calculation using Eq. (2-124):

$$\text{AIT} = 4(301.91) - 10.86 + 2(-275.17) = 646.4 \text{ K}$$

The estimated value is 6.3 percent above the DIPPR® 801 recommended value of 608.15 K.

SECTION 3

Mathematics

PERRY'S CHEMICAL ENGINEERS' HANDBOOK

8TH EDITION



BRUCE A. FINLAYSON
LORENZ T. BIEGLER

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Mathematics

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GENERAL REFERENCES: Abramowitz, M., and I. A. Stegun, *Handbook of Mathematical Functions*, National Bureau of Standards, Washington, D.C. (1972); Finlayson, B.A., *Nonlinear Analysis in Chemical Engineering*, McGraw-Hill, New York (1980), Ravenna Park, Seattle (2003); Jeffrey, A., *Mathematics for Engineers and Scientists*, Chapman & Hall/CRC, New York (2004); Jeffrey, A., *Essentials of Engineering Mathematics*, 2d ed., Chapman &

Hall/CRC, New York (2004); Weisstein, E. W., *CRC Concise Encyclopedia of Mathematics*, 2d ed., CRC Press, New York (2002); Wrede, R. C., and Murray R. Spiegel, *Schaum's Outline of Theory and Problems of Advanced Calculus*, 2d ed., McGraw-Hill, New York (2006); Zwillinger, D., *CRC Standard Mathematical Tables and Formulae*, 1st ed., CRC Press, New York (2002); <http://eqworld.ipmnet.ru/>.

MATHEMATICS

GENERAL

The basic problems of the sciences and engineering fall broadly into three categories:

1. *Steady state problems.* In such problems the configuration of the system is to be determined. This solution does not change with time but continues indefinitely in the same pattern, hence the name "steady state." Typical chemical engineering examples include steady temperature distributions in heat conduction, equilibrium in chemical reactions, and steady diffusion problems.

2. *Eigenvalue problems.* These are extensions of equilibrium problems in which critical values of certain parameters are to be determined in addition to the corresponding steady-state configurations. The determination of eigenvalues may also arise in propagation problems and stability problems. Typical chemical engineering problems include those in heat transfer and resonance in which certain boundary conditions are prescribed.

3. *Propagation problems.* These problems are concerned with predicting the subsequent behavior of a system from a knowledge of the initial state. For this reason they are often called the transient (time-varying) or unsteady-state phenomena. Chemical engineering examples include the transient state of chemical reactions (kinetics), the propagation of pressure waves in a fluid, transient behavior of an adsorption column, and the rate of approach to equilibrium of a packed distillation column.

The mathematical treatment of engineering problems involves four basic steps:

1. *Formulation.* The expression of the problem in mathematical language. That translation is based on the appropriate physical laws governing the process.

2. *Solution.* Appropriate mathematical and numerical operations are accomplished so that logical deductions may be drawn from the mathematical model.

3. *Interpretation.* Development of relations between the mathematical results and their meaning in the physical world.

4. *Refinement.* The recycling of the procedure to obtain better predictions as indicated by experimental checks.

Steps 1 and 2 are of primary interest here. The actual details are left to the various subsections, and only general approaches will be discussed.

The formulation step may result in algebraic equations, difference equations, differential equations, integral equations, or combinations of these. In any event these mathematical models usually arise from statements of physical laws such as the laws of mass and energy conservation in the form

$$\text{Input of } x - \text{output of } x + \text{production of } x = \text{accumulation of } x$$

or

$$\text{Rate of input of } x - \text{rate of output of } x + \text{rate of production of } x = \text{rate of accumulation of } x$$

where x = mass, energy, etc. These statements may be abbreviated by the statement

$$\text{Input} - \text{output} + \text{production} = \text{accumulation}$$

Many general laws of the physical universe are expressible by differential equations. Specific phenomena are then singled out from the infinity of solutions of these equations by assigning the individual initial or boundary conditions which characterize the given problem. For steady state or boundary-value problems (Fig. 3-1) the solution must

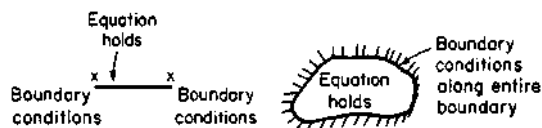


FIG. 3-1 Boundary conditions.

satisfy the differential equation inside the region and the prescribed conditions on the boundary.

In mathematical language, the propagation problem is known as an initial-value problem (Fig. 3-2). Schematically, the problem is characterized by a differential equation plus an open region in which the equation holds. The solution of the differential equation must satisfy the initial conditions plus any "side" boundary conditions.

The description of phenomena in a "continuous" medium such as a gas or a fluid often leads to partial differential equations. In particular, phenomena of "wave" propagation are described by a class of partial differential equations called "hyperbolic," and these are essentially different in their properties from other classes such as those that describe equilibrium ("elliptic") or diffusion and heat transfer ("parabolic"). Prototypes are:

1. *Elliptic.* Laplace's equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Poisson's equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = g(x, y)$$

These do not contain the variable t (time) explicitly; accordingly, their solutions represent equilibrium configurations. Laplace's equation corresponds to a "natural" equilibrium, while Poisson's equation corresponds to an equilibrium under the influence of $g(x, y)$. Steady heat-transfer and mass-transfer problems are elliptic.

2. *Parabolic.* The heat equation

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$$

describes unsteady or propagation states of diffusion as well as heat transfer.

3. *Hyperbolic.* The wave equation

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$$

describes wave propagation of all types when the assumption is made that the wave amplitude is small and that interactions are linear.

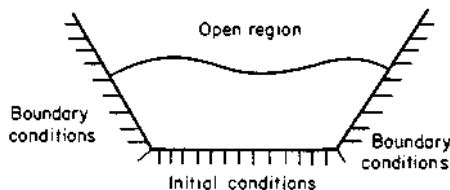


FIG. 3-2 Propagation problem.

3-4 MATHEMATICS

The solution phase has been characterized in the past by a concentration on methods to obtain analytic solutions to the mathematical equations. These efforts have been most fruitful in the area of the linear equations such as those just given. However, many natural phenomena are nonlinear. While there are a few nonlinear problems that can be solved analytically, most cannot. In those cases, numerical methods are used. Due to the widespread availability of software for computers, the engineer has quite good tools available.

Numerical methods almost never fail to provide an answer to any particular situation, but they can never furnish a general solution of any problem.

The mathematical details outlined here include both analytic and numerical techniques useful in obtaining solutions to problems.

Our discussion to this point has been confined to those areas in which the governing laws are well known. However, in many areas, information on the governing laws is lacking and statistical methods are reused. Broadly speaking, statistical methods may be of use whenever conclusions are to be drawn or decisions made on the basis of experimental evidence. Since statistics could be defined as the technology of the scientific method, it is primarily concerned with the first two aspects of the method, namely, the performance of experiments and the drawing of conclusions from experiments. Traditionally the field is divided into two areas:

1. *Design of experiments.* When conclusions are to be drawn or decisions made on the basis of experimental evidence, statistical techniques are most useful when experimental data are subject to errors. The design of experiments may then often be carried out in such a fashion as to avoid some of the sources of experimental error and make the necessary allowances for that portion which is unavoidable. Second, the results can be presented in terms of probability statements which express the reliability of the results. Third, a statistical approach frequently forces a more thorough evaluation of the experimental aims and leads to a more definitive experiment than would otherwise have been performed.

2. *Statistical inference.* The broad problem of statistical inference is to provide measures of the uncertainty of conclusions drawn from experimental data. This area uses the theory of probability, enabling scientists to assess the reliability of their conclusions in terms of probability statements.

Both of these areas, the mathematical and the statistical, are intimately intertwined when applied to any given situation. The methods of one are often combined with the other. And both in order to be successfully used must result in the numerical answer to a problem—that is, they constitute the means to an end. Increasingly the numerical answer is being obtained from the mathematics with the aid of computers. The mathematical notation is given in Table 3-1.

MISCELLANEOUS MATHEMATICAL CONSTANTS

Numerical values of the constants that follow are approximate to the number of significant digits given.

$\pi = 3.1415926536$	Pi
$e = 2.7182818285$	Napierian (natural) logarithm base
$\gamma = 0.5772156649$	Euler's constant
$\ln \pi = 1.1447298858$	Napierian (natural) logarithm of pi, base e
$\log \pi = 0.4971498727$	Briggsian (common logarithm of pi, base 10)
Radian = 57.2957795131°	
Degree = 0.0174532925 rad	
Minute = 0.0002908882 rad	
Second = 0.0000048481 rad	
$\gamma = \lim_{n \rightarrow \infty} \left(\sum_{m=1}^n \frac{1}{m} - \ln n \right) = 0.577215$	

THE REAL-NUMBER SYSTEM

The natural numbers, or counting numbers, are the positive integers: 1, 2, 3, 4, 5, The negative integers are -1, -2, -3,

A number in the form a/b , where a and b are integers, $b \neq 0$, is a rational number. A real number that cannot be written as the quotient of two integers is called an irrational number, e.g., $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, π , e , $\sqrt[3]{2}$.

TABLE 3-1 Mathematical Signs, Symbols, and Abbreviations

$\pm$ ($\mp$)	plus or minus (minus or plus)
:	divided by, ratio sign
::	proportional sign
<	less than
$\leftarrow$	not less than
>	greater than
$\rightarrow$	not greater than
$\cong$	approximately equals, congruent
$\sim$	similar to
$\approx$	equivalent to
$\neq$	not equal to
$\doteq$	approaches, is approximately equal to
∞	varies as
∞	infinity
$\therefore$	therefore
$\sqrt{\quad}$	square root
$\sqrt[3]{\quad}$	cube root
$\sqrt[n]{\quad}$	n th root
$\angle$	angle
$\perp$	perpendicular to
$\parallel$	parallel to
$ x $	numerical value of x
$\log$ or $\log_{10}$	common logarithm or Briggsian logarithm
$\log_e$ or $\ln$	natural logarithm or hyperbolic logarithm or Napierian logarithm
e	base (2.718) of natural system of logarithms
a°	an angle a degrees
a'	prime, an angle a minutes
a''	double prime, an angle a seconds, a second
$\sin$	sine
$\cos$	cosine
$\tan$	tangent
ctn or cot	cotangent
sec	secant
csc	cosecant
vers	versed sine
covers	covered sine
exsec	exsecant
$\sin^{-1}$	anti sine or angle whose sine is
$\sinh$	hyperbolic sine
$\cosh$	hyperbolic cosine
$\tanh$	hyperbolic tangent
$\sinh^{-1}$	anti hyperbolic sine or angle whose hyperbolic sine is
$f(x)$ or $\phi(x)$	function of x
Δx	increment of x
Σ	summation of
dx	differential of x
dy/dx or y'	derivative of y with respect to x
d^2y/dx^2 or y''	second derivative of y with respect to x
$d^n y/dx^n$	n th derivative of y with respect to x
$\partial y/\partial x$	partial derivative of y with respect to x
$\partial^n y/\partial x^n$	n th partial derivative of y with respect to x
$\frac{\partial^2 z}{\partial x \partial y}$	n th partial derivative with respect to x and y
$\int$	integral of
$\int_a^b$	integral between the limits a and b
$\dot{y}$	first derivative of y with respect to time
$\ddot{y}$	second derivative of y with respect to time
Δ or ∇^2	the "Laplacian"
$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right)$	
δ	sign of a variation
$\oint$	sign for integration around a closed path

There is a one-to-one correspondence between the set of real numbers and the set of points on an infinite line (coordinate line).

Order among Real Numbers; Inequalities

$a > b$ means that $a - b$ is a positive real number.

If $a < b$ and $b < c$, then $a < c$.

If $a < b$, then $a \pm c < b \pm c$ for any real number c .

If $a < b$ and $c > 0$, then $ac < bc$.

If $a < b$ and $c < 0$, then $ac > bc$.

If $a < b$ and $c < d$, then $a + c < b + d$.

If $0 < a < b$ and $0 < c < d$, then $ac < bd$.

If $a < b$ and $ab > 0$, then $1/a > 1/b$.

If $a < b$ and $ab < 0$, then $1/a < 1/b$.

Absolute Value For any real number x , $|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$

Properties

If $|x| = a$, where $a > 0$, then $x = a$ or $x = -a$.

$|x| = |-x|$; $-|x| \leq x \leq |x|$; $|xy| = |x||y|$.

If $|x| < c$, then $-c < x < c$, where $c > 0$.

$||x| - |y|| \leq |x + y| \leq |x| + |y|$.

$\sqrt{x^2} = |x|$.

Proportions If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a+b}{b} = \frac{c+d}{d}$, $\frac{a-b}{b} = \frac{c-d}{d}$,

$$\frac{a-b}{a+b} = \frac{c-d}{c+d}.$$

Indeterminants

Form	Example	
$(\infty)(0)$	xe^{-x}	$x \rightarrow \infty$
0^0	x^x	$x \rightarrow 0^+$
∞^0	$(\tan x)^{\cos x}$	$x \rightarrow \frac{1}{2}\pi^-$
1^∞	$(1+x)^{1/x}$	$x \rightarrow 0^+$
$\infty - \infty$	$\sqrt{x+1} - \sqrt{x-1}$	$x \rightarrow \infty$
$\frac{0}{0}$	$\frac{\sin x}{x}$	$x \rightarrow 0$
$\frac{\infty}{\infty}$	$\frac{e^x}{x}$	$x \rightarrow \infty$

Integral Exponents (Powers and Roots) If m and n are positive integers and a, b are numbers or functions, then the following properties hold:

$$a^{-n} = 1/a^n \quad a \neq 0$$

$$(ab)^n = a^n b^n$$

$$(a^n)^m = a^{nm}, \quad a^n a^m = a^{n+m}$$

$$\sqrt[n]{a} = a^{1/n} \quad \text{if } a > 0$$

$$\sqrt[n]{\sqrt[n]{a}} = \sqrt[n]{a}, \quad a > 0$$

$$a^{m/n} = (a^m)^{1/n} = \sqrt[n]{a^m}, \quad a > 0$$

$$a^0 = 1 \quad (a \neq 0)$$

$$0^a = 0 \quad (a \neq 0)$$

Logarithms $\log ab = \log a + \log b$, $a > 0$, $b > 0$

$$\log a^n = n \log a$$

$$\log(a/b) = \log a - \log b$$

$$\log \sqrt[n]{a} = (1/n) \log a$$

The common logarithm (base 10) is denoted $\log a$ or $\log_{10} a$. The natural logarithm (base e) is denoted $\ln a$ (or in some texts $\log_e a$). If the text is ambiguous (perhaps using $\log x$ for $\ln x$), test the formula by evaluating it.

Roots If a is a real number, n is a positive integer, then x is called the n th root of a if $x^n = a$. The number of n th roots is n , but not all of them are necessarily real. The principal n th root means the following: (1) if $a > 0$ the principal n th root is the unique positive root, (2) if

$a < 0$, and n odd, it is the unique negative root, and (3) if $a < 0$ and n even, it is any of the complex roots. In cases (1) and (2), the root can be found on a calculator by taking $y = \ln a/n$ and then $x = e^y$. In case (3), see the section on complex variables.

ALGEBRAIC INEQUALITIES

Arithmetic-Geometric Inequality Let A_n and G_n denote respectively the arithmetic and the geometric means of a set of positive numbers $a_1, a_2, \dots, a_n$. The $A_n \geq G_n$, i.e.,

$$\frac{a_1 + a_2 + \dots + a_n}{n} \geq (a_1 a_2 \dots a_n)^{1/n}$$

The equality holds only if all of the numbers a_i are equal.

Carleman's Inequality The arithmetic and geometric means just defined satisfy the inequality

$$\sum_{r=1}^n (a_1 a_2 \dots a_r)^{1/r} \leq n e A_n$$

where e is the best possible constant in this inequality.

Cauchy-Schwarz Inequality Let $a = (a_1, a_2, \dots, a_n)$, $b = (b_1, b_2, \dots, b_n)$, where the a_i 's and b_i 's are real or complex numbers. Then

$$\left| \sum_{k=1}^n a_k \bar{b}_k \right|^2 \leq \left(\sum_{k=1}^n |a_k|^2 \right) \left(\sum_{k=1}^n |b_k|^2 \right)$$

The equality holds if, and only if, the vectors a, b are linearly dependent (i.e., one vector is scalar times the other vector).

Minkowski's Inequality Let $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ be any two sets of complex numbers. Then for any real number $p > 1$,

$$\left(\sum_{k=1}^n |a_k + b_k|^p \right)^{1/p} \leq \left(\sum_{k=1}^n |a_k|^p \right)^{1/p} + \left(\sum_{k=1}^n |b_k|^p \right)^{1/p}$$

Hölder's Inequality Let $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ be any two sets of complex numbers, and let p and q be positive numbers with $1/p + 1/q = 1$. Then

$$\left| \sum_{k=1}^n a_k \bar{b}_k \right| \leq \left(\sum_{k=1}^n |a_k|^p \right)^{1/p} \left(\sum_{k=1}^n |b_k|^q \right)^{1/q}$$

The equality holds if, and only if, the sequences $|a_1|^p, |a_2|^p, \dots, |a_n|^p$ and $|b_1|^q, |b_2|^q, \dots, |b_n|^q$ are proportional and the argument (angle) of the complex numbers $a_k \bar{b}_k$ is independent of k . This last condition is of course automatically satisfied if $a_1, \dots, a_n$ and $b_1, \dots, b_n$ are positive numbers.

Lagrange's Inequality Let $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ be real numbers. Then

$$\left(\sum_{k=1}^n a_k b_k \right)^2 \leq \left(\sum_{k=1}^n a_k^2 \right) \left(\sum_{k=1}^n b_k^2 \right) - \sum_{1 \leq k < j \leq n} (a_k b_j - a_j b_k)^2$$

Example Two chemical engineers, John and Mary, purchase stock in the same company at times $t_1, t_2, \dots, t_n$ when the price per share is respectively $p_1, p_2, \dots, p_n$. Their methods of investment are different, however: John purchases x shares each time, whereas Mary invests P dollars each time (fractional shares can be purchased). Who is doing better?

While one can argue intuitively that the average cost per share for Mary does not exceed that for John, we illustrate a mathematical proof using inequalities. The average cost per share for John is equal to

$$\frac{\text{Total money invested}}{\text{Number of shares purchased}} = \frac{x \sum_{i=1}^n p_i}{nx} = \frac{1}{n} \sum_{i=1}^n p_i$$

The average cost per share for Mary is

$$\frac{nP}{\sum_{i=1}^n \frac{P}{p_i}} = \frac{n}{\sum_{i=1}^n \frac{1}{p_i}}$$

3-6 MATHEMATICS

Thus the average cost per share for John is the arithmetic mean of $p_1, p_2, \dots, p_n$, whereas that for Mary is the harmonic mean of these n numbers. Since the harmonic mean is less than or equal to the arithmetic mean for any set of positive numbers and the two means are equal only if $p_1 = p_2 = \dots = p_n$, we conclude that the average cost per share for Mary is less than that for John if two of the prices p_i are distinct. One can also give a proof based on the Cauchy-Schwarz inequality. To this end, define the vectors

$$a = (p_1^{-1/2}, p_2^{-1/2}, \dots, p_n^{-1/2}) \quad b = (p_1^{1/2}, p_2^{1/2}, \dots, p_n^{1/2})$$

Then $a \cdot b = 1 + \dots + 1 = n$, and so by the Cauchy-Schwarz inequality

$$(a \cdot b)^2 = n^2 \leq \sum_{i=1}^n \frac{1}{p_i} \sum_{i=1}^n p_i$$

with the equality holding only if $p_1 = p_2 = \dots = p_n$. Therefore

$$\frac{n}{\sum_{i=1}^n \frac{1}{p_i}} \leq \frac{\sum_{i=1}^n p_i}{n}$$

MENSURATION FORMULAS

REFERENCES: Liu, J., *Mathematical Handbook of Formulas and Tables*, McGraw-Hill, New York (1999); <http://mathworld.wolfram.com/SphericalSector.html>, etc.

Let A denote areas and V volumes in the following.

PLANE GEOMETRIC FIGURES WITH STRAIGHT BOUNDARIES

Triangles (see also "Plane Trigonometry") $A = \frac{1}{2}bh$ where b = base, h = altitude.

Rectangle $A = ab$ where a and b are the lengths of the sides.

Parallelogram (opposite sides parallel) $A = ah = ab \sin \alpha$ where a, b are the lengths of the sides, h the height, and α the angle between the sides. See Fig. 3-3.

Rhombus (equilateral parallelogram) $A = \frac{1}{2}ab$ where a, b are the lengths of the diagonals.

Trapezoid (four sides, two parallel) $A = \frac{1}{2}(a + b)h$ where the lengths of the parallel sides are a and b , and h = height.

Quadrilateral (four-sided) $A = \frac{1}{2}ab \sin \theta$ where a, b are the lengths of the diagonals and the acute angle between them is θ .

Regular Polygon of n Sides See Fig. 3-4.

$$A = \frac{1}{4}nl^2 \cot \frac{180^\circ}{n} \quad \text{where } l = \text{length of each side}$$

$$R = \frac{l}{2} \csc \frac{180^\circ}{n} \quad \text{where } R \text{ is the radius of the circumscribed circle}$$

$$r = \frac{l}{2} \cot \frac{180^\circ}{n} \quad \text{where } r \text{ is the radius of the inscribed circle}$$

Radius r of Circle Inscribed in Triangle with Sides a, b, c

$$r = \sqrt{\frac{(s-a)(s-b)(s-c)}{s}} \quad \text{where } s = \frac{1}{2}(a + b + c)$$

Radius R of Circumscribed Circle

$$R = \frac{abc}{4\sqrt{s(s-a)(s-b)(s-c)}}$$

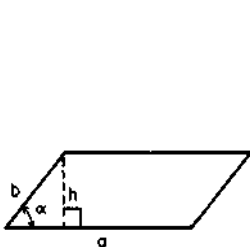


FIG. 3-3 Parallelogram.

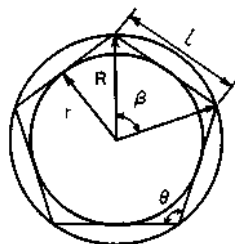


FIG. 3-4 Regular polygon.

Area of Regular Polygon of n Sides Inscribed in a Circle of Radius r

$$A = (nr^2/2) \sin (360^\circ/n)$$

Perimeter of Inscribed Regular Polygon

$$P = 2nr \sin (180^\circ/n)$$

Area of Regular Polygon Circumscribed about a Circle of Radius r

$$A = nr^2 \tan (180^\circ/n)$$

Perimeter of Circumscribed Regular Polygon

$$P = 2nr \tan \frac{180^\circ}{n}$$

PLANE GEOMETRIC FIGURES WITH CURVED BOUNDARIES

Circle (Fig. 3-5) Let

C = circumference

r = radius

D = diameter

A = area

S = arc length subtended by θ

l = chord length subtended by θ

H = maximum rise of arc above chord, $r - H = d$

θ = central angle (rad) subtended by arc S

$$C = 2\pi r = \pi D \quad (\pi = 3.14159 \dots)$$

$$S = r\theta = \frac{1}{2}D\theta$$

$$l = 2\sqrt{r^2 - d^2} = 2r \sin (\theta/2) = 2d \tan (\theta/2)$$

$$d = \frac{1}{2} \sqrt{4r^2 - l^2} = \frac{1}{2} l \cot \frac{\theta}{2}$$

$$\theta = \frac{S}{r} = 2 \cos^{-1} \frac{d}{r} = 2 \sin^{-1} \frac{l}{D}$$

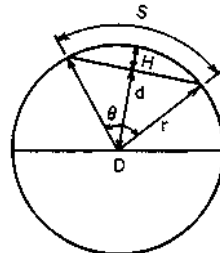


FIG. 3-5 Circle.

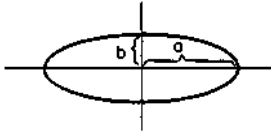


FIG. 3-6 Ellipse.

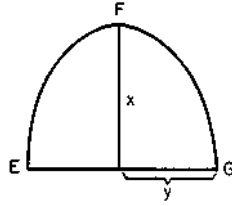


FIG. 3-7 Parabola.

$$A (\text{circle}) = \pi r^2 = \frac{1}{4}\pi D^2$$

$$A (\text{sector}) = \frac{1}{2}rS = \frac{1}{2}r^2\theta$$

$$A (\text{segment}) = A (\text{sector}) - A (\text{triangle}) = \frac{1}{2}r^2(\theta - \sin \theta)$$

Ring (area between two circles of radii r_1 and r_2) The circles need not be concentric, but one of the circles must enclose the other.

$$A = \pi(r_1 + r_2)(r_1 - r_2) \quad r_1 > r_2$$

Ellipse (Fig. 3-6) Let the semiaxes of the ellipse be a and b

$$A = \pi ab$$

$$C = 4aE(e)$$

where $e^2 = 1 - b^2/a^2$ and $E(e)$ is the complete elliptic integral of the second kind,

$$E(e) = \frac{\pi}{2} \left[1 - \left(\frac{1}{2} \right)^2 e^2 + \dots \right]$$

[an approximation for the circumference $C = 2\pi \sqrt{(a^2 + b^2)/2}$].

Parabola (Fig. 3-7)

$$\text{Length of arc } EFG = \sqrt{4x^2 + y^2} + \frac{y^2}{2x} \ln \frac{2x + \sqrt{4x^2 + y^2}}{y}$$

$$\text{Area of section } EFG = \frac{4}{3}xy$$

Catenary (the curve formed by a cord of uniform weight suspended freely between two points A, B ; Fig. 3-8)

$$y = a \cosh (x/a)$$

Length of arc between points A and B is equal to $2a \sinh (L/a)$. Sag of the cord is $D = a \cosh (L/a) - a$.

SOLID GEOMETRIC FIGURES WITH PLANE BOUNDARIES

Cube Volume = a^3 ; total surface area = $6a^2$; diagonal = $a\sqrt{3}$, where a = length of one side of the cube.

Rectangular Parallelepiped Volume = abc ; surface area = $2(ab + ac + bc)$; diagonal = $\sqrt{a^2 + b^2 + c^2}$, where a, b, c are the lengths of the sides.

Prism Volume = (area of base) $\times$ (altitude); lateral surface area = (perimeter of right section) $\times$ (lateral edge).

Pyramid Volume = $\frac{1}{3}$ (area of base) $\times$ (altitude); lateral area of regular pyramid = $\frac{1}{2}$ (perimeter of base) $\times$ (slant height) = $\frac{1}{2}$ (number of sides) (length of one side) (slant height).

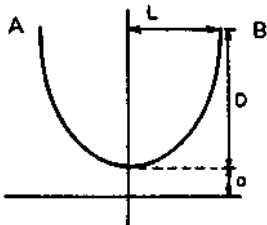


FIG. 3-8 Catenary.

Frustum of Pyramid (formed from the pyramid by cutting off the top with a plane

$$V = \frac{1}{3} (A_1 + A_2 + \sqrt{A_1 \cdot A_2})h$$

where h = altitude and A_1, A_2 are the areas of the base; lateral area of a regular figure = $\frac{1}{2}$ (sum of the perimeters of base) $\times$ (slant height).

Volume and Surface Area of Regular Polyhedra with Edge l

Type of surface	Name	Volume	Surface area
4 equilateral triangles	Tetrahedron	$0.1179 l^3$	$1.7321 l^2$
6 squares	Hexahedron (cube)	$1.0000 l^3$	$6.0000 l^2$
8 equilateral triangles	Octahedron	$0.4714 l^3$	$3.4641 l^2$
12 pentagons	Dodecahedron	$7.6631 l^3$	$20.6458 l^2$
20 equilateral triangles	Icosahedron	$2.1817 l^3$	$8.6603 l^2$

SOLIDS BOUNDED BY CURVED SURFACES

Cylinders (Fig. 3-9) V = (area of base) $\times$ (altitude); lateral surface area = (perimeter of right section) $\times$ (lateral edge).

Right Circular Cylinder $V = \pi$ (radius) $^2 \times$ (altitude); lateral surface area = 2π (radius) $\times$ (altitude).

Truncated Right Circular Cylinder

$$V = \pi r^2 h; \text{ lateral area} = 2\pi r h$$

$$h = \frac{1}{2} (h_1 + h_2)$$

Hollow Cylinders Volume = $\pi h(R^2 - r^2)$, where r and R are the internal and external radii and h is the height of the cylinder.

Sphere (Fig. 3-10)

$$V (\text{sphere}) = \frac{4}{3}\pi R^3, \frac{1}{6}\pi D^3$$

V (spherical sector) = $\frac{2}{3}\pi R^2 h_i = 2$ (open spherical sector), $i = 1$ (spherical cone)

$$V (\text{spherical segment of one base}) = \frac{1}{6}\pi h_1(3r_1^2 + h_1^2)$$

$$V (\text{spherical segment of two bases}) = \frac{1}{6}\pi h_2(3r_1^2 + 3r_2^2 + h_2^2)$$

$$A (\text{sphere}) = 4\pi R^2 = \pi D^2$$

$$A (\text{zone}) = 2\pi R h = \pi D h$$

A (lune on the surface included between two great circles, the inclination of which is θ radians) = $2R^2\theta$.

Cone $V = \frac{1}{3}$ (area of base) $\times$ (altitude).

Right Circular Cone $V = (\pi/3) r^2 h$, where h is the altitude and r is the radius of the base; curved surface area = $\pi r \sqrt{r^2 + h^2}$, curved surface of the frustum of a right cone = $\pi(r_1 + r_2) \sqrt{h^2 + (r_1 - r_2)^2}$, where r_1, r_2 are the radii of the base and top, respectively, and h is the altitude; volume of the frustum of a right cone = $\pi(h/3)(r_1^2 + r_1 r_2 + r_2^2) = h/3(A_1 + A_2 + \sqrt{A_1 A_2})$, where A_1 = area of base and A_2 = area of top.

Ellipsoid $V = (\frac{4}{3})\pi abc$, where a, b, c are the lengths of the semi-axes.

Torus (obtained by rotating a circle of radius r about a line whose distance is $R > r$ from the center of the circle)

$$V = 2\pi^2 R r^2 \quad \text{Surface area} = 4\pi^2 R r$$

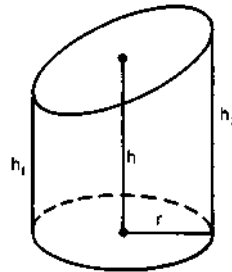


FIG. 3-9 Cylinder.

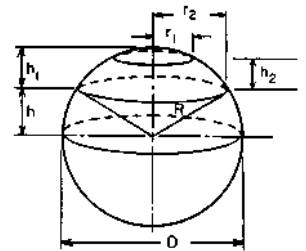


FIG. 3-10 Sphere.

3-8 MATHEMATICS

Prolate Spheroid (formed by rotating an ellipse about its major axis [$2a$])

$$\text{Surface area} = 2\pi b^2 + 2\pi(ab/e) \sin^{-1} e \quad V = \frac{4}{3}\pi ab^2$$

where a , b are the major and minor axes and e = eccentricity ($e < 1$).

Oblate Spheroid (formed by the rotation of an ellipse about its minor axis [$2b$]) Data as given previously.

$$\text{Surface area} = 2\pi a^2 + \pi \frac{b^2}{e} \ln \frac{1+e}{1-e} \quad V = \frac{4}{3}\pi a^2 b$$

For process vessels, the formulas reduce to the following:

Hemisphere

$$V = \frac{\pi}{12} D^3, \quad A = \frac{\pi}{2} D^2$$

For a hemisphere (concave up) partially filled to a depth h_1 , use the formulas for spherical segment with one base, which simplify to

$$V = \pi h_1^3 (R - h_1/3) = \pi h_1^3 (D/2 - h_1/3) \\ A = 2\pi R h_1 = \pi D h_1$$

For a hemisphere (concave down) partially filled from the bottom, use the formulas for a spherical segment of two bases, one of which is a plane through the center, where h = distance from the center plane to the surface of the partially filled hemisphere.

$$V = \pi h (R^2 - h^2/3) = \pi h (D^2/4 - h^2/3) \\ A = 2\pi R h = \pi D h$$

Cone For a cone partially filled, use the same formulas as for right circular cones, but use r and h for the region filled.

Ellipsoid If the base of a vessel is one-half of an oblate spheroid (the cross section fitting to a cylinder is a circle with radius of $D/2$ and the minor axis is smaller), then use the formulas for one-half of an oblate spheroid.

$$V = 0.1745 D^3, \quad S = 1.236 D^2, \quad \text{minor axis} = D/3 \\ V = 0.1309 D^3, \quad S = 1.084 D^2, \quad \text{minor axis} = D/4$$

MISCELLANEOUS FORMULAS

See also "Differential and Integral Calculus."

Volume of a Solid Revolution (the solid generated by rotating a plane area about the x axis)

$$V = \pi \int_a^b [f(x)]^2 dx$$

where $y = f(x)$ is the equation of the plane curve and $a \leq x \leq b$.

Area of a Surface of Revolution

$$S = 2\pi \int_a^b y ds$$

where $ds = \sqrt{1 + (dy/dx)^2} dx$ and $y = f(x)$ is the equation of the plane curve rotated about the x axis to generate the surface.

Area Bounded by $f(x)$, the x Axis, and the Lines $x = a$, $x = b$

$$A = \int_a^b f(x) dx \quad [f(x) \geq 0]$$

Length of Arc of a Plane Curve

If $y = f(x)$,

$$\text{Length of arc } s = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

If $x = g(y)$,

$$\text{Length of arc } s = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

If $x = f(t)$, $y = g(t)$,

$$\text{Length of arc } s = \int_{t_0}^{t_1} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

In general, $(ds)^2 = (dx)^2 + (dy)^2$.

IRREGULAR AREAS AND VOLUMES

Irregular Areas Let $y_0, y_1, \dots, y_n$ be the lengths of a series of equally spaced parallel chords and h be their distance apart (Fig. 3-11). The area of the figure is given approximately by any of the following:

$$A_T = (h/2)[(y_0 + y_n) + 2(y_1 + y_2 + \dots + y_{n-1})] \quad (\text{trapezoidal rule})$$

$$A_S = (h/3)[(y_0 + y_n) + 4(y_1 + y_3 + y_5 + \dots + y_{n-1}) \\ + 2(y_2 + y_4 + \dots + y_{n-2})] \quad (n \text{ even, Simpson's rule})$$

The greater the value of n , the greater the accuracy of approximation.

Irregular Volumes To find the volume, replace the y 's by cross-sectional areas A_j and use the results in the preceding equations.

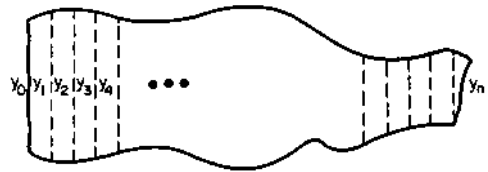


FIG. 3-11 Irregular area.

ELEMENTARY ALGEBRA

REFERENCES: Stillwell, J. C., *Elements of Algebra*, CRC Press, New York (1994); Rich, B., and P. Schmidt, *Schaum's Outline of Elementary Algebra*, McGraw-Hill, New York (2004).

OPERATIONS ON ALGEBRAIC EXPRESSIONS

An algebraic expression will here be denoted as a combination of letters and numbers such as

$$3ax - 3xy + 7x^2 + 7x^{3/2} - 2.8xy$$

Addition and Subtraction Only like terms can be added or subtracted in two algebraic expressions.

Example $(3x + 4xy - x^2) + (3x^2 + 2x - 8xy) = 5x - 4xy + 2x^2$.

Example $(2^x + 3xy - 4x^{1/2}) + (3^x + 6x - 8xy) = 2^x + 3^x + 6x - 5xy - 4x^{1/2}$.

Multiplication Multiplication of algebraic expressions is term by term, and corresponding terms are combined.

Example $(2x + 3y - 2xy)(3 + 3y) = 6x + 9y + 9y^2 - 6xy^2$.

Division This operation is analogous to that in arithmetic.

Example Divide $3e^{2x} + e^x + 1$ by $e^x + 1$.

$$\begin{array}{r}
 \text{Dividend} \\
 \text{Divisor } e^x + 1 \mid 3e^{2x} + e^x + 1 \quad \text{quotient} \\
 \underline{3e^{2x} + 3e^x} \\
 -2e^x + 1 \\
 \underline{-2e^x - 2} \\
 +3 \text{ (remainder)}
 \end{array}$$

Therefore, $3e^{2x} + e^x + 1 = (e^x + 1)(3e^x - 2) + 3$.

Operations with Zero All numerical computations (except division) can be done with zero: $a + 0 = 0 + a = a$; $a - 0 = a$; $0 - a = -a$; $(a)(0) = 0$; $a^0 = 1$ if $a \neq 0$; $0/a = 0$, $a \neq 0$. $a/0$ and $0/0$ have no meaning.

Fractional Operations

$$-\frac{x}{y} = -\left(\frac{-x}{-y}\right) = \frac{x}{-y} = \frac{-x}{y}; \quad \frac{x}{y} = \frac{-x}{-y}; \quad \frac{x}{y} = \frac{ax}{ay}, \text{ if } a \neq 0.$$

$$\frac{x}{y} \pm \frac{z}{y} = \frac{x \pm z}{y}; \quad \left(\frac{x}{y}\right)\left(\frac{z}{t}\right) = \frac{xz}{yt}; \quad \frac{x/y}{z/t} = \left(\frac{x}{y}\right)\left(\frac{t}{z}\right) = \frac{xt}{yz}$$

Factoring That process of analysis consisting of reducing a given expression into the product of two or more simpler expressions called *factors*. Some of the more common expressions are factored here:

- (1) $(x^2 - y^2) = (x - y)(x + y)$
- (2) $x^2 + 2xy + y^2 = (x + y)^2$
- (3) $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$
- (4) $(x^3 + y^3) = (x + y)(x^2 - xy + y^2)$
- (5) $(x^4 - y^4) = (x - y)(x + y)(x^2 + y^2)$
- (6) $x^5 + y^5 = (x + y)(x^4 - x^3y + x^2y^2 - xy^3 + y^4)$
- (7) $x^n - y^n = (x - y)(x^{n-1} + x^{n-2}y + x^{n-3}y^2 + \cdots + y^{n-1})$

Laws of Exponents

$(a^n)^m = a^{nm}$; $a^{n+m} = a^n \cdot a^m$; $a^{n/m} = (a^n)^{1/m}$; $a^{-n} = a^n/a^n$; $a^{1/m} = \sqrt[m]{a}$; $a^{1/2} = \sqrt{a}$; $\sqrt{x^2} = |x|$ (absolute value of x). For $x > 0$, $y > 0$, $\sqrt{xy} = \sqrt{x}\sqrt{y}$; for $x > 0$, $\sqrt[n]{x^m} = x^{m/n}$; $\sqrt[n]{1/x} = 1/\sqrt[n]{x}$

THE BINOMIAL THEOREM

If n is a positive integer,

$$\begin{aligned}
 (a + b)^n &= a^n + na^{n-1}b + \frac{n(n-1)}{2!}a^{n-2}b^2 \\
 &\quad + \frac{n(n-1)(n-2)}{3!}a^{n-3}b^3 + \cdots + b^n = \sum_{j=0}^n \binom{n}{j} a^{n-j} b^j
 \end{aligned}$$

where $\binom{n}{j} = \frac{n!}{j!(n-j)!}$ = number of combinations of n things taken j at

a time. $n! = 1 \cdot 2 \cdot 3 \cdot 4 \cdots n$, $0! = 1$.

Example Find the sixth term of $(x + 2y)^{12}$. The sixth term is obtained by setting $j = 5$. It is

$$\binom{12}{5} x^{12-5} (2y)^5 = 792x^7(2y)^5$$

Example $\sum_{j=0}^{14} \binom{n}{j} = (1 + 1)^{14} = 2^{14}$.

If n is not a positive integer, the sum formula no longer applies and an infinite series results for $(a + b)^n$. The coefficients are obtained from the first formulas in this case.

Example $(1 + x)^{1/2} = 1 + \frac{1}{2}x - \frac{1}{2} \cdot \frac{1}{4}x^2 + \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{3}{6}x^3 \cdots$ (convergent for $x^2 < 1$).

Additional discussion is under "Infinite Series."

PROGRESSIONS

An arithmetic progression is a succession of terms such that each term, except the first, is derivable from the preceding by the addition of a quantity d called the common difference. All arithmetic progressions have the form $a, a + d, a + 2d, a + 3d, \dots$. With a = first term, l = last term, d = common difference, n = number of terms, and s = sum of the terms, the following relations hold:

$$l = a + (n - 1)d = \frac{s}{n} + \frac{(n - 1)}{2}d$$

$$s = \frac{n}{2} [2a + (n - 1)d] = \frac{n}{2} (a + l) = \frac{n}{2} [2l - (n - 1)d]$$

$$a = l - (n - 1)d = \frac{s}{n} - \frac{(n - 1)d}{2} = \frac{2s}{n} - l$$

$$d = \frac{l - a}{n - 1} = \frac{2(s - an)}{n(n - 1)} = \frac{2(nl - s)}{n(n - 1)}$$

$$n = \frac{l - a}{d} + 1 = \frac{2s}{l + a}$$

The **arithmetic mean or average** of two numbers a, b is $(a + b)/2$; of n numbers $a_1, \dots, a_n$ is $(a_1 + a_2 + \cdots + a_n)/n$.

A **geometric progression** is a succession of terms such that each term, except the first, is derivable from the preceding by the multiplication of a quantity r called the common ratio. All such progressions have the form $a, ar, ar^2, \dots, ar^{n-1}$. With a = first term, l = last term, r = ratio, n = number of terms, s = sum of the terms, the following relations hold:

$$l = ar^{n-1} = \frac{[a + (r - 1)s]}{r} = \frac{(r - 1)s r^{n-1}}{r^n - 1}$$

$$s = \frac{a(r^n - 1)}{r - 1} = \frac{a(1 - r^n)}{1 - r} = \frac{rl - a}{r - 1} = \frac{lr^n - l}{r^n - r^{n-1}}$$

$$a = \frac{l}{r^{n-1}} = \frac{(r - 1)s}{r^n - 1}, \quad r = \frac{s - a}{s - l}, \quad \log r = \frac{\log l - \log a}{n - 1}$$

$$n = \frac{\log l - \log a}{\log r} + 1 = \frac{\log[a + (r - 1)s] - \log a}{\log r}$$

The geometric mean of two nonnegative numbers a, b is $\sqrt{ab}$; of n numbers is $(a_1 a_2 \cdots a_n)^{1/n}$. The geometric mean of a set of positive numbers is less than or equal to the arithmetic mean.

Example Find the sum of $1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{64}$. Here $a = 1$, $r = \frac{1}{2}$, $n = 7$. Thus

$$s = \frac{\frac{1}{2}(\frac{1}{64}) - 1}{\frac{1}{2} - 1} = 127/64$$

$$s = a + ar + ar^2 + \cdots + ar^{n-1} = \frac{a}{1 - r} - \frac{ar^n}{1 - r}$$

If $|r| < 1$, then $\lim_{n \rightarrow \infty} s = \frac{a}{1 - r}$

which is called the sum of the infinite geometric progression.

Example The present worth (PW) of a series of cash flows C_k at the end of year k is

$$PW = \sum_{k=1}^n \frac{C_k}{(1 + i)^k}$$

where i is an assumed interest rate. (Thus the present worth always requires specification of an interest rate.) If all the payments are the same, $C_k = R$, the present worth is

$$PW = R \sum_{k=1}^n \frac{1}{(1 + i)^k}$$

This can be rewritten as

$$PW = \frac{R}{1 + i} \sum_{k=1}^n \frac{1}{(1 + i)^{k-1}} = \frac{R}{1 + i} \sum_{j=0}^{n-1} \frac{1}{(1 + i)^j}$$

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This is a geometric series with $r = 1/(1+i)$ and $a = R/(1+i)$. The formulas above give

$$PW(=s) = \frac{R}{i} \frac{(1+i)^n - 1}{(1+i)^n}$$

The same formula applies to the value of an annuity (PW) now, to provide for equal payments R at the end of each of n years, with interest rate i .

A progression of the form $a, (a+d)r, (a+2d)r^2, (a+3d)r^3$, etc., is a combined arithmetic and geometric progression. The sum of n such terms is

$$s = \frac{a - [a + (n-1)d]r^n}{1-r} + \frac{rd(1-r^{n-1})}{(1-r)^2}$$

If $|r| < 1$, $\lim_{n \rightarrow \infty} s = \frac{a}{1-r} + \frac{rd}{(1-r)^2}$.

The non-zero numbers a, b, c , etc., form a harmonic progression if their reciprocals $1/a, 1/b, 1/c$, etc., form an arithmetic progression.

Example The progression $1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \dots, \frac{1}{31}$ is harmonic since $1, 3, 5, 7, \dots, 31$ form an arithmetic progression.

The **harmonic mean** of two numbers a, b is $2ab/(a+b)$.

PERMUTATIONS, COMBINATIONS, AND PROBABILITY

Each separate arrangement of all or a part of a set of things is called a **permutation**. The number of permutations of n things taken r at a time, written

$$P(n, r) = \frac{n!}{(n-r)!} = n(n-1)(n-2) \cdots (n-r+1)$$

Each separate selection of objects that is possible irrespective of the order in which they are arranged is called a combination. The number of combinations of n things taken r at a time, written $C(n, r) = n!/[r!(n-r)!]$.

An important relation is $r! C(n, r) = P(n, r)$.

If an event can occur in p ways and fail to occur in q ways, all ways being equally likely, the *probability* of its occurrence is $p/(p+q)$, and that of its failure $q/(p+q)$.

Example Two dice may be thrown in 36 separate ways. What is the probability of throwing such that their sum is 7? Seven may arise in 6 ways: 1 and 6, 2 and 5, 3 and 4, 4 and 3, 5 and 2, 6 and 1. The probability of shooting 7 is $\frac{1}{6}$.

THEORY OF EQUATIONS

Linear Equations A linear equation is one of the first degree (i.e., only the first powers of the variables are involved), and the process of obtaining definite values for the unknown is called solving the equation. Every linear equation in one variable is written $Ax + B = 0$ or $x = -B/A$. Linear equations in n variables have the form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

The solution of the system may then be found by elimination or matrix methods if a solution exists (see "Matrix Algebra and Matrix Computations").

Quadratic Equations Every quadratic equation in one variable is expressible in the form $ax^2 + bx + c = 0$, $a \neq 0$. This equation has two solutions, say, x_1, x_2 , given by

$$\left. \begin{aligned} x_1 \\ x_2 \end{aligned} \right\} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

If a, b, c are real, the discriminant $b^2 - 4ac$ gives the character of the roots. If $b^2 - 4ac > 0$, the roots are real and unequal. If $b^2 - 4ac < 0$, the roots are complex conjugates. If $b^2 - 4ac = 0$ the roots are **real and equal**. Two quadratic equations in two variables can in general be solved only by numerical methods (see "Numerical Analysis and Approximate Methods").

Cubic Equations A cubic equation, in one variable, has the form $x^3 + bx^2 + cx + d = 0$. Every cubic equation having complex coefficients has three complex roots. If the coefficients are real numbers, then at least one of the roots must be real. The cubic equation $x^3 + bx^2 + cx + d = 0$ may be reduced by the substitution $x = y - (b/3)$ to the form $y^3 + py + q = 0$, where $p = \frac{1}{3}(3c - b^2)$, $q = \frac{1}{27}(27d - 9bc + 2b^3)$. This equation has the solutions $y_1 = A + B$, $y_2 = -\frac{1}{2}(A + B) + (i\sqrt{3}/2)(A - B)$, $y_3 = -\frac{1}{2}(A + B) - (i\sqrt{3}/2)(A - B)$, where $i^2 = -1$, $A = \sqrt[3]{-q/2 + \sqrt{R}}$, $B = \sqrt[3]{-q/2 - \sqrt{R}}$, and $R = (p/3)^3 + (q/2)^2$. If b, c, d are all real and if $R > 0$, there are one real root and two conjugate complex roots; if $R = 0$, there are three real roots, of which at least two are equal; if $R < 0$, there are three real unequal roots. If $R < 0$, these formulas are impractical. In this case, the roots are given by $y_k = \sqrt[3]{2} \sqrt{-p/3} \cos[(\phi/3) + 120k]$, $k = 0, 1, 2$ where

$$\phi = \cos^{-1} \sqrt{\frac{q^2/4}{-p^3/27}}$$

and the upper sign applies if $q > 0$, the lower if $q < 0$.

Example $y^3 - 7y + 7 = 0$, $p = -7$, $q = 7$, $R < 0$. Hence

$$y_k = -\sqrt{\frac{28}{3}} \cos\left(\frac{\phi}{3} + 120k\right)$$

where $\phi = \cos^{-1} \sqrt{\frac{27}{28}}$, $\frac{\phi}{3} = 3.6311315 \text{ rad} = 3^\circ 37' 52''$

The roots are approximately -3.048917 , 1.692021 , and 1.356896 .

Example Many equations of state involve solving cubic equations for the compressibility factor Z . For example, the Redlich-Kwong-Soave equation of state requires solving

$$Z^3 - Z^2 + cZ + d = 0, \quad d < 0$$

where c and d depend on critical constants of the chemical species. In this case, only positive solutions, $Z > 0$, are desired.

Quartic Equations See Abramowitz and Stegun (1972, p. 17).

General Polynomials of the n th Degree Denote the general polynomial equation of degree n by

$$P(x) = a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n = 0$$

If $n > 4$, there is no formula which gives the roots of the general equation. For fourth and higher order (even third order), the roots can be found numerically (see "Numerical Analysis and Approximate Methods"). However, there are some general theorems that may prove useful.

Remainder Theorems When $P(x)$ is a polynomial and $P(x)$ is divided by $x - a$ until a remainder independent of x is obtained, this remainder is equal to $P(a)$.

Example $P(x) = 2x^4 - 3x^2 + 7x - 2$ when divided by $x + 1$ (here $a = -1$) results in $P(x) = (x + 1)(2x^3 - 2x^2 - x + 8) - 10$ where -10 is the remainder. It is easy to see that $P(-1) = -10$.

Factor Theorem If $P(a)$ is zero, the polynomial $P(x)$ has the factor $x - a$. In other words, if a is a root of $P(x) = 0$, then $x - a$ is a factor of $P(x)$.

If a number a is found to be a root of $P(x) = 0$, the division of $P(x)$ by $(x - a)$ leaves a polynomial of degree one less than that of the original equation, i.e., $P(x) = Q(x)(x - a)$. Roots of $Q(x) = 0$ are clearly roots of $P(x) = 0$.

Example $P(x) = x^3 - 6x^2 + 11x - 6 = 0$ has the root $+3$. Then $P(x) = (x - 3)(x^2 - 3x + 2)$. The roots of $x^2 - 3x + 2 = 0$ are 1 and 2. The roots of $P(x)$ are therefore 1, 2, 3.

Fundamental Theorem of Algebra Every polynomial of degree n has exactly n real or complex roots, counting multiplicities.

Every polynomial equation $a_0x^n + a_1x^{n-1} + \cdots + a_n = 0$ with *rational coefficients* may be rewritten as a polynomial, of the same degree, with *integral coefficients* by multiplying each coefficient by the least common multiple of the denominators of the coefficients.

Example The coefficients of $\frac{1}{2}x^4 + \frac{7}{3}x^3 - \frac{5}{6}x^2 + 2x - \frac{1}{6} = 0$ are rational numbers. The least common multiple of the denominators is $2 \times 3 \times 6 = 6$. Therefore, the equation is equivalent to $9x^4 + 14x^3 - 5x^2 + 12x - 1 = 0$.

Determinants Consider the system of two linear equations

$$a_{11}x_1 + a_{12}x_2 = b_1$$

$$a_{21}x_1 + a_{22}x_2 = b_2$$

If the first equation is multiplied by a_{22} and the second by $-a_{12}$ and the results added, we obtain

$$(a_{11}a_{22} - a_{21}a_{12})x_1 = b_1a_{22} - b_2a_{12}$$

The expression $a_{11}a_{22} - a_{21}a_{12}$ may be represented by the symbol

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12}$$

This symbol is called a determinant of second order. The value of the square array of n^2 quantities a_{ij} , where $i = 1, \dots, n$ is the row index, $j = 1, \dots, n$ the column index, written in the form

$$|A| = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

is called a determinant. The n^2 quantities a_{ij} are called the elements of the determinant. In the determinant $|A|$ let the i th row and j th column be deleted and a new determinant be formed having $n - 1$ rows and columns. This new determinant is called the minor of a_{ij} denoted M_{ij} .

Example $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ The minor of a_{23} is $M_{23} = \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}$

The cofactor A_{ij} of the element a_{ij} is the signed minor of a_{ij} determined by the rule $A_{ij} = (-1)^{i+j}M_{ij}$. The value of $|A|$ is obtained by forming any of the equivalent expressions $\sum_{j=1}^n a_{ij}A_{ij}$, $\sum_{i=1}^n a_{ij}A_{ij}$, where the elements a_{ij} must be taken from a single row or a single column of A .

Example

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{31}A_{31} + a_{32}A_{32} + a_{33}A_{33}$$

$$= a_{31} \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} - a_{32} \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} + a_{33} \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

In general, A_{ij} will be determinants of order $n - 1$, but they may in turn be expanded by the rule. Also,

$$\sum_{j=1}^n a_{ji}A_{jk} = \begin{cases} |A| & i = k \\ 0 & i \neq k \end{cases}$$

Fundamental Properties of Determinants

1. The value of a determinant $|A|$ is not changed if the rows and columns are interchanged.
2. If the elements of one row (or one column) of a determinant are all zero, the value of $|A|$ is zero.
3. If the elements of one row (or column) of a determinant are multiplied by the same constant factor, the value of the determinant is multiplied by this factor.
4. If one determinant is obtained from another by interchanging any two rows (or columns), the value of either is the negative of the value of the other.
5. If two rows (or columns) of a determinant are identical, the value of the determinant is zero.
6. If two determinants are identical except for one row (or column), the sum of their values is given by a single determinant obtained by adding corresponding elements of dissimilar rows (or columns) and leaving unchanged the remaining elements.
7. The value of a determinant is not changed if one row (or column) is multiplied by a constant and added to another row (or column).

ANALYTIC GEOMETRY

REFERENCES: Fuller, G., *Analytic Geometry*, 7th ed., Addison Wesley Longman (1994); Larson, R., R. P. Hostetler, and B. H. Edwards, *Calculus with Analytic Geometry*, 7th ed., Houghton Mifflin (2001); Riddle, D. F., *Analytic Geometry*, 6th ed., Thompson Learning (1996); Spiegel, M. R., and J. Liu, *Mathematical Handbook of Formulas and Tables*, 2d ed., McGraw-Hill (1999); Thomas, G. B., Jr., and R. L. Finney, *Calculus and Analytic Geometry*, 9th ed., Addison-Wesley (1996).

Analytic geometry uses algebraic equations and methods to study geometric problems. It also permits one to visualize algebraic equations in terms of geometric curves, which frequently clarifies abstract concepts.

PLANE ANALYTIC GEOMETRY

Coordinate Systems The basic concept of analytic geometry is the establishment of a one-to-one correspondence between the points of the plane and number pairs (x, y) . This correspondence may be done in a number of ways. The rectangular or cartesian coordinate system consists of two straight lines intersecting at right angles (Fig. 3-12). A point is designated by (x, y) , where x (the abscissa) is the distance of the point from the y axis measured parallel to the x axis,

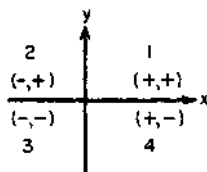


FIG. 3-12 Rectangular coordinates.

positive if to the right, negative to the left. y (ordinate) is the distance of the point from the x axis, measured parallel to the y axis, positive if above, negative if below the x axis. The **quadrants** are labeled 1, 2, 3, 4 in the drawing, the coordinates of points in the various quadrants having the depicted signs. Another common coordinate system is the polar coordinate system (Fig. 3-13). In this system the position of a point is designated by the pair (r, θ) , $r = \sqrt{x^2 + y^2}$ being the distance to the origin $O(0,0)$ and θ being the angle the line r makes with the positive x axis (polar axis). To change from polar to rectangular coordinates, use $x = r \cos \theta$ and $y = r \sin \theta$. To change from rectangular to polar coordinates, use $r = \sqrt{x^2 + y^2}$ and $\theta = \tan^{-1}(y/x)$ if $x \neq 0$; $\theta = \pi/2$ if $x = 0$. The distance between two points (x_1, y_1) , (x_2, y_2) is defined by $d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$ in rectangular coordinates or by $d = \sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos(\theta_1 - \theta_2)}$ in polar coordinates. Other coordinate systems are sometimes used. For example, on the surface of a sphere latitude and longitude prove useful.

The Straight Line (Fig. 3-14) The slope m of a straight line is the tangent of the inclination angle θ made with the positive x axis. If

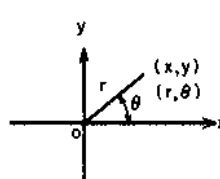


FIG. 3-13 Polar coordinates.

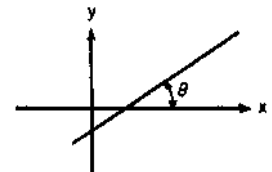


FIG. 3-14 Straight line.

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(x_1, y_1) and (x_2, y_2) are any two points on the line, slope $= m = (y_2 - y_1)/(x_2 - x_1)$. The slope of a line parallel to the x axis is zero; parallel to the y axis, it is undefined. Two lines are parallel if and only if they have the same slope. Two lines are perpendicular if and only if the product of their slopes is -1 (the exception being that case when the lines are parallel to the coordinate axes). Every equation of the type $Ax + By + C = 0$ represents a straight line, and every straight line has an equation of this form. A straight line is determined by a variety of conditions:

Given conditions	Equation of line
(1) Parallel to x axis	$y = \text{constant}$
(2) Parallel y axis	$x = \text{constant}$
(3) Point (x_1, y_1) and slope m	$y - y_1 = m(x - x_1)$
(4) Intercept on y axis $(0, b)$, m	$y = mx + b$
(5) Intercept on x axis $(a, 0)$, m	$y = m(x - a)$
(6) Two points (x_1, y_1) , (x_2, y_2)	$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$
(7) Two intercepts $(a, 0)$, $(0, b)$	$x/a + y/b = 1$

The angle β a line with slope m_1 makes with a line having slope m_2 is given by $\tan \beta = (m_2 - m_1)/(m_1 m_2 + 1)$. A line is determined if the length and direction of the perpendicular to it (the normal) from the origin are given (see Fig. 3-15). Let p = length of the perpendicular and α the angle that the perpendicular makes with the positive x axis. The equation of the line is $x \cos \alpha + y \sin \alpha = p$. The equation of a line perpendicular to a given line of slope m and passing through a point (x_1, y_1) is $y - y_1 = -(1/m)(x - x_1)$. The distance from a point (x_1, y_1) to a line with equation $Ax + By + C = 0$ is

$$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

Occasionally some nonlinear algebraic equations can be reduced to linear equations under suitable substitutions or changes of variables. In other words, certain curves become the graphs of lines if the scales or coordinate axes are appropriately transformed.

Example Consider $y = bx^a$. $B = \log b$. Taking logarithms $\log y = n \log x + \log b$. Let $Y = \log y$, $X = \log x$, $B = \log b$. The equation then has the form $Y = nX + B$, which is a linear equation. Consider $k = k_0 \exp(-E/RT)$, taking logarithms $\log k = \log k_0 - E/(RT)$. Let $Y = \log k$, $B = \log k_0$, and $m = -E/R$, $X = 1/T$, and the result is $Y = mX + B$. Next consider $y = a + bx^n$. If the substitution $t = x^n$ is made, then the graph of y is a straight line versus t .

Asymptotes The limiting position of the tangent to a curve as the point of contact tends to an infinite distance from the origin is called an **asymptote**. If the equation of a given curve can be expanded in a Laurent power series such that

$$f(x) = \sum_{k=0}^n a_k x^k + \sum_{k=1}^n \frac{b_k}{x^k}$$

and

$$\lim_{x \rightarrow \infty} f(x) = \sum_{k=0}^n a_k x^k$$

then the equation of the asymptote is $y = \sum_{k=0}^n a_k x^k$. If $n = 1$, then the asymptote is (in general oblique) a line. In this case, the equation of the asymptote may be written as

$$y = mx + b \quad m = \lim_{x \rightarrow \infty} f'(x)$$

$$b = \lim_{x \rightarrow \infty} [f(x) - xf'(x)]$$

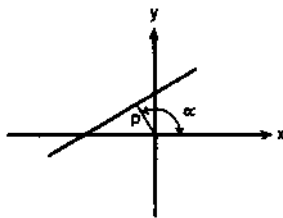


FIG. 3-15 Determination of line.

Geometric Properties of a Curve When the Equation Is Given The analysis of the properties of an equation is facilitated by the investigation of the equation by using the following techniques:

1. *Points of maximum, minimum, and inflection.* These may be investigated by means of the calculus.
2. *Symmetry.* Let $F(x, y) = 0$ be the equation of the curve.

Condition on $F(x, y)$	Symmetry
$F(x, y) = F(-x, y)$	With respect to y axis
$F(x, y) = F(x, -y)$	With respect to x axis
$F(x, y) = F(-x, -y)$	With respect to origin
$F(x, y) = F(y, x)$	With respect to the line $y = x$

3. *Extent.* Only real values of x and y are considered in obtaining the points (x, y) whose coordinates satisfy the equation. The extent of them may be limited by the condition that negative numbers do not have real square roots.

4. *Intercepts.* Find those points where the curves of the function cross the coordinate axes.

5. *Asymptotes.* See preceding discussion.

6. *Direction at a point.* This may be found from the derivative of the function at a point. This concept is useful for distinguishing among a family of similar curves.

Example $y^2 = (x^2 + 1)/(x^2 - 1)$ is symmetric with respect to the x and y axis, the origin, and the line $y = x$. It has the vertical asymptotes $x = \pm 1$. When $x = 0$, $y^2 = -1$; so there are no y intercepts. If $y = 0$, $(x^2 + 1)/(x^2 - 1) = 0$; so there are no x intercepts. If $|x| < 1$, y^2 is negative; so $|x| > 1$. From $x^2 = (y^2 + 1)/(y^2 - 1)$, $y = \pm 1$ are horizontal asymptotes and $|y| > 1$. As $x \rightarrow 1^+$, $y \rightarrow +\infty$; as $x \rightarrow +\infty$, $y \rightarrow +1$. The graph is given in Fig. 3-16.

Conic Sections The curves included in this group are obtained from plane sections of the cone. They include the circle, ellipse, parabola, hyperbola, and degeneratively the point and straight line. A **conic** is the locus of a point whose distance from a fixed point called the **focus** is in a constant ratio to its distance from a fixed line, called the **directrix**. This ratio is the eccentricity e . If $e = 0$, the conic is a circle; if $0 < e < 1$, the conic is an ellipse; if $e = 1$, the conic is a parabola; if $e > 1$, the conic is a hyperbola. Every conic section is representable by an equation of second degree. Conversely, every equation of second degree in two variables represents a conic. The general equation of the second degree is $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$. Let Δ be defined as the determinant

$$\Delta = \begin{vmatrix} 2A & B & D \\ B & 2C & E \\ D & E & 2F \end{vmatrix}$$

The table characterizes the curve represented by the equation.

	$B^2 - 4AC < 0$	$B^2 - 4AC = 0$	$B^2 - 4AC > 0$
$\Delta \neq 0$	$A\Delta < 0$ $A \neq C$, an ellipse $A\Delta < 0$ $A = C$, a circle $A\Delta > 0$, no locus	Parabola	Hyperbola
$\Delta = 0$	Point	2 parallel lines if $Q = D^2 + E^2 - 4(A + C)F > 0$ 1 straight line if $Q = 0$, no locus if $Q < 0$	2 intersecting straight lines

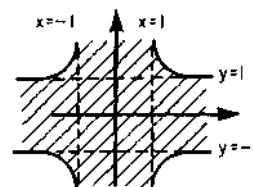


FIG. 3-16 Graph of $y^2 = (x^2 + 1)/(x^2 - 1)$.

Some common equations in parametric form are given below.

(1) $(x-h)^2 + (y-k)^2 = a^2$	$x = h + a \cos \theta$ $y = k + a \sin \theta$	Circle (Fig. 3-23) Parameter is angle θ .
(2) $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$	$x = h + a \cos \phi$ $y = k + a \sin \phi$	Ellipse (Fig. 3-20) Parameter is angle ϕ .
(3) $x^2 + y^2 = a^2$	$x = \frac{-at}{\sqrt{t^2 + 1}}$ $y = \frac{a}{\sqrt{t^2 + 1}}$	Circle Parameter is $t = \frac{dy}{dx}$ = slope of tangent at (x, y) .
(4) $x^2 = y + k$		Parabola (Fig. 3-22)
(5) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$		Hyperbola with the origin at the center (Fig. 3-21)
(6) $y = a \cosh \frac{x}{a}$	$x = a \sinh^{-1} \frac{s}{a}$ $y^2 = a^2 + s^2$	Catenary (such as hanging cable under gravity) Parameter s = arc length from $(0, a)$ to (x, y) .
(7) Cycloid	$x = a(\phi - \sin \phi)$ $y = a(1 - \cos \phi)$	Fig. 3.24

Example $3x^2 + 4xy - 2y^2 + 3x - 2y + 7 = 0$.

$$\Delta = \begin{vmatrix} 6 & 4 & 3 \\ 4 & -4 & -2 \\ 3 & -2 & 14 \end{vmatrix} = -596 \neq 0, B^2 - 4AC = 40 > 0$$

The curve is therefore a hyperbola.

The following tabulation gives the form of the more common equations.

Polar equation	Type of curve
(1) $r = a$	Circle, Fig. 3-17
(2) $r = 2a \cos \theta$	Circle, Fig. 3-18
(3) $r = 2a \sin \theta$	Circle, Fig. 3-19
(4) $r^2 - 2br \cos(\theta - \beta) + b^2 - a^2 = 0$	Circle at (b, β) , radius a
(5) $r = \frac{ke}{1 - e \cos \theta}$	$e = 1$ parabola, Fig. 3-22 $0 < e < 1$ ellipse, Fig. 3-20 $e > 1$ hyperbola, Fig. 3-21

Parametric Equations It is frequently useful to write the equations of a curve in terms of an auxiliary variable called a parameter. For example, a circle of radius a , center at $(0, 0)$, can be written in the equivalent form $x = a \cos \phi$, $y = a \sin \phi$ where ϕ is the parameter.

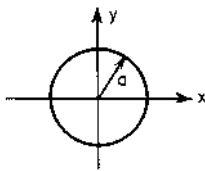


FIG. 3-17 Circle center $(0,0)$ $r = a$.

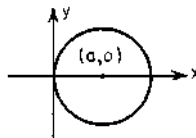


FIG. 3-18 Circle center $(a,0)$ $r = 2a \cos \theta$.

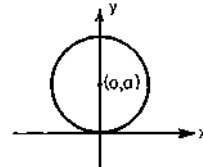


FIG. 3-19 Circle center $(0,a)$ $r = 2a \sin \theta$.

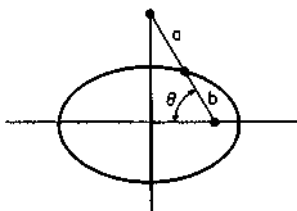


FIG. 3-20 Ellipse, $0 < e < 1$.

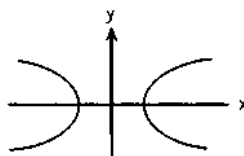


FIG. 3-21 Hyperbola, $e > 1$.

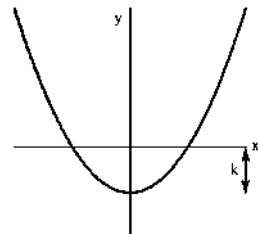


FIG. 3-22 Parabola, $e = 1$.

Similarly, $x = a \cos \phi$, $y = b \sin \phi$ are the parametric equations of the ellipse $x^2/a^2 + y^2/b^2 = 1$ with parameter ϕ .

SOLID ANALYTIC GEOMETRY

Coordinate Systems The commonly used coordinate systems are three in number. Others may be used in specific problems [see Morse, P. M., and H. Feshbach, *Methods of Theoretical Physics*, vols. I and II, McGraw-Hill, New York (1953)]. The **rectangular** (cartesian) system (Fig. 3-25) consists of mutually orthogonal axes x , y , z . A triple of numbers (x, y, z) is used to represent each point. The **cylindrical** coordinate system (r, θ, z) ; Fig. 3-26) is frequently used to locate a point in space. These are essentially the polar coordinates (r, θ) coupled with the z coordinate. As before, $x = r \cos \theta$, $y = r \sin \theta$, $z = z$ and $r^2 = x^2 + y^2$, $y/x = \tan \theta$. If r is held constant and θ and z are allowed to vary, the locus of (r, θ, z) is a right circular cylinder of radius r along the z axis. The locus of $r = C$ is a circle, and $\theta = \text{constant}$ is a plane containing the z axis and making an angle θ with the xz plane. Cylindrical coordinates are convenient to use when the problem has an axis of symmetry.

The **spherical** coordinate system is convenient if there is a point of symmetry in the system. This point is taken as the origin and the coordinates (ρ, ϕ, θ) illustrated in Fig. 3-27. The relations are $x =$

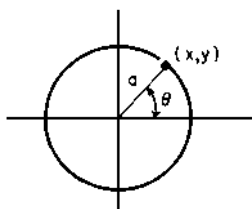


FIG. 3-23 Circle.

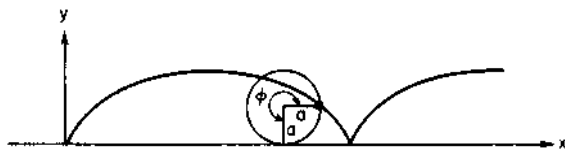


FIG. 3-24 Cycloid.

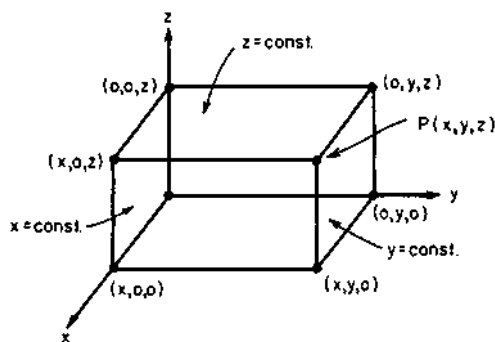


FIG. 3-25 Cartesian coordinates.

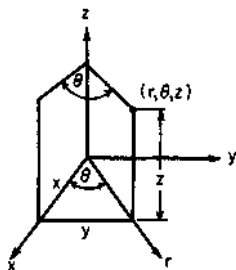


FIG. 3-26 Cylindrical coordinates.

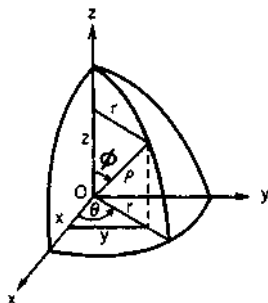


FIG. 3-27 Spherical coordinates.

$\rho \sin \phi \cos \theta$, $y = \rho \sin \phi \sin \theta$, $z = \rho \cos \phi$, and $r = \rho \sin \phi$. $\theta = \text{constant}$ is a plane containing the z axis and making an angle θ with the xz plane. $\phi = \text{constant}$ is a cone with vertex at 0. $\rho = \text{constant}$ is the surface of a sphere of radius ρ , center at the origin 0. Every point in the space may be given spherical coordinates restricted to the ranges $0 \leq \phi \leq \pi$, $\rho \geq 0$, $0 \leq \theta < 2\pi$.

Lines and Planes The distance between two points (x_1, y_1, z_1) , (x_2, y_2, z_2) is $d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$. There is nothing in the geometry of three dimensions quite analogous to the slope of a line in the plane case. Instead of specifying the direction of a line by a trigonometric function evaluated for one angle, a trigonometric function evaluated for three angles is used. The angles α , β , γ that a line segment makes with the positive x , y , and z axes, respectively, are called the **direction angles** of the line, and $\cos \alpha$, $\cos \beta$, $\cos \gamma$ are called the **direction cosines**. Let (x_1, y_1, z_1) , (x_2, y_2, z_2) be on the line. Then $\cos \alpha = (x_2 - x_1)/d$, $\cos \beta = (y_2 - y_1)/d$, $\cos \gamma = (z_2 - z_1)/d$, where d is the distance between the two points. Clearly $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$. If two lines are specified by the direction cosines $(\cos \alpha_1, \cos \beta_1, \cos \gamma_1)$, $(\cos \alpha_2, \cos \beta_2, \cos \gamma_2)$, then the angle θ between the lines is $\cos \theta = \cos \alpha_1 \cos \alpha_2 + \cos \beta_1 \cos \beta_2 + \cos \gamma_1 \cos \gamma_2$. Thus the lines are perpendicular if and only if $\theta = 90^\circ$ or $\cos \alpha_1 \cos \alpha_2 + \cos \beta_1 \cos \beta_2 + \cos \gamma_1 \cos \gamma_2 = 0$. The equation of a line with direction cosines $(\cos \alpha, \cos \beta, \cos \gamma)$ passing through (x_1, y_1, z_1) is $(x - x_1)/\cos \alpha = (y - y_1)/\cos \beta = (z - z_1)/\cos \gamma$.

The equation of every plane is of the form $Ax + By + Cz + D = 0$. The numbers

$$\frac{A}{\sqrt{A^2 + B^2 + C^2}}, \frac{B}{\sqrt{A^2 + B^2 + C^2}}, \frac{C}{\sqrt{A^2 + B^2 + C^2}}$$

are direction cosines of the normal lines to the plane. The plane through the point (x_1, y_1, z_1) whose normals have these as direction cosines is $A(x - x_1) + B(y - y_1) + C(z - z_1) = 0$.

Example Find the equation of the plane through $(1, 5, -2)$ perpendicular to the line $(x + 9)/7 = (y - 3)/-1 = z/8$. The numbers $(7, -1, 8)$ are called **direction numbers**. They are a constant multiple of the direction cosines. $\cos \alpha = 7/114$, $\cos \beta = -1/114$, $\cos \gamma = 8/114$. The plane has the equation $7(x - 1) - 1(y - 5) + 8(z + 2) = 0$ or $7x - y + 8z + 14 = 0$.

The distance from the point (x_1, y_1, z_1) to the plane $Ax + By + Cz + D = 0$ is

$$d = \frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}}$$

Space Curves Space curves are usually specified as the set of points whose coordinates are given parametrically by a system of equations $x = f(t)$, $y = g(t)$, $z = h(t)$ in the parameter t .

Example The equation of a straight line in space is $(x - x_1)/a = (y - y_1)/b = (z - z_1)/c$. Since all these quantities must be equal (say, to t), we may write $x = x_1 + at$, $y = y_1 + bt$, $z = z_1 + ct$, which represent the parametric equations of the line.

Example The equations $z = a \cos \beta t$, $y = a \sin \beta t$, $x = bt$, a, β, b positive constants, represent a circular helix.

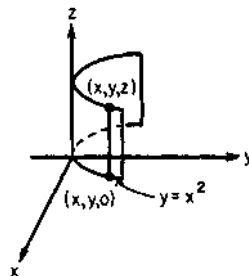


FIG. 3-28 Parabolic cylinder.

Surfaces The locus of points (x, y, z) satisfying $f(x, y, z) = 0$, broadly speaking, may be interpreted as a surface. The simplest surface is the **plane**. The next simplest is a **cylinder**, which is a surface generated by a straight line moving parallel to a given line and passing through a given curve.

Example The parabolic cylinder $y = x^2$ (Fig. 3-28) is generated by a straight line parallel to the z axis passing through $y = x^2$ in the plane $z = 0$.

A surface whose equation is a quadratic in the variables x, y , and z is called a **quadric surface**. Some of the more common such surfaces are tabulated and pictured in Figs. 3-28 to 3-36.

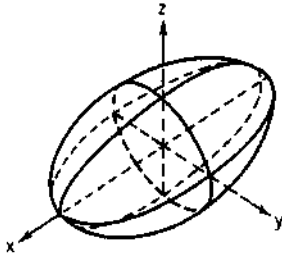


FIG. 3-29 Ellipsoid. $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ (sphere if $a = b = c$)

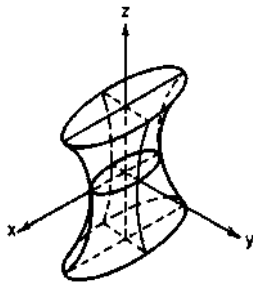


FIG. 3-30 Hyperboloid of one sheet. $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$

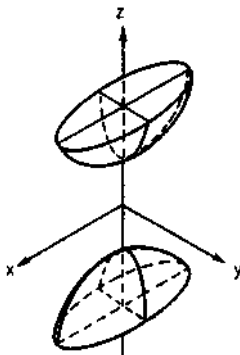


FIG. 3-31 Hyperboloid of two sheets. $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1$

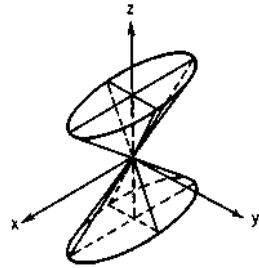


FIG. 3-32 Cone. $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 0$

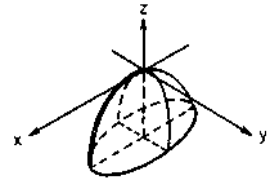


FIG. 3-33 Elliptic paraboloid.

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + cz = 0$$

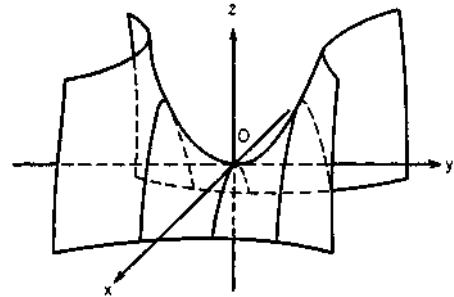


FIG. 3-34 Hyperbolic paraboloid. $\frac{x^2}{a^2} - \frac{y^2}{b^2} + cz = 0$

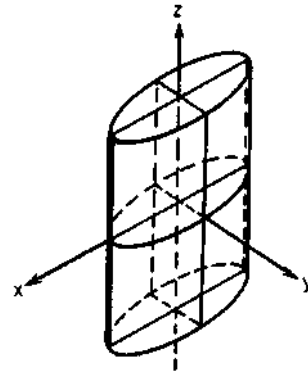


FIG. 3-35 Elliptic cylinder. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

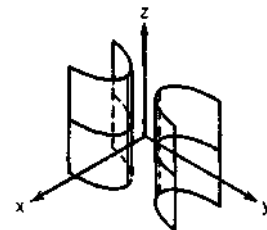


FIG. 3-36 Hyperbolic cylinder.

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

PLANE TRIGONOMETRY

REFERENCES: Gelfand, I. M., and M. Saul, *Trigonometry*, Birkhäuser, Boston (2001); Heineman, E. Richard, and J. Dalton Tarwater, *Plane Trigonometry*, 7th ed., McGraw-Hill (1993).

ANGLES

An angle is generated by the rotation of a line about a fixed center from some initial position to some terminal position. If the rotation is clockwise, the angle is negative; if it is counterclockwise, the angle is positive. Angle size is unlimited. If α, β are two angles such that $\alpha + \beta = 90^\circ$, they are complementary; they are supplementary if $\alpha + \beta = 180^\circ$. Angles are most commonly measured in the sexagesimal system or by radian measure. In the first system there are 360 degrees in one complete revolution; one degree = $\frac{1}{90}$ of a right angle. The degree is subdivided into 60 minutes; the minute is subdivided into 60 seconds. In the radian system one radian is the angle at the center of a circle subtended by an arc whose length is equal to the radius of the circle. Thus $2\pi \text{ rad} = 360^\circ$; $1 \text{ rad} = 57.29578^\circ$; $1^\circ = 0.01745 \text{ rad}$; $1 \text{ min} = 0.00029089 \text{ rad}$. The advantage of radian measure is that it is *dimensionless*. The quadrants are conventionally labeled as Fig. 3-37 shows.

FUNCTIONS OF CIRCULAR TRIGONOMETRY

The trigonometric functions of angles are the ratios between the various sides of the reference triangles shown in Fig. 3-38 for the various quadrants. Clearly $r = \sqrt{x^2 + y^2} \geq 0$. The fundamental functions (see Figs. 3-39, 3-40, 3-41) are

Plane Trigonometry

- Sine of $\theta = \sin \theta = y/r$
- Cosine of $\theta = \cos \theta = x/r$
- Tangent of $\theta = \tan \theta = y/x$
- Secant of $\theta = \sec \theta = r/x$
- Cosecant of $\theta = \csc \theta = r/y$
- Cotangent of $\theta = \cot \theta = x/y$

Values of the Trigonometric Functions for Common Angles

θ°	θ, rad	$\sin \theta$	$\cos \theta$	$\tan \theta$
0	0	0	1	0
30	$\pi/6$	$1/2$	$\sqrt{3}/2$	$\sqrt{3}/3$
45	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1
60	$\pi/3$	$\sqrt{3}/2$	$1/2$	$\sqrt{3}$
90	$\pi/2$	1	0	$+\infty$

If $90^\circ \leq \theta \leq 180^\circ$, $\sin \theta = \sin (180^\circ - \theta)$; $\cos \theta = -\cos (180^\circ - \theta)$; $\tan \theta = -\tan (180^\circ - \theta)$. If $180^\circ \leq \theta \leq 270^\circ$, $\sin \theta = -\sin (270^\circ - \theta)$; $\cos \theta = -\cos (270^\circ - \theta)$; $\tan \theta = \tan (270^\circ - \theta)$. If $270^\circ \leq \theta \leq 360^\circ$, $\sin \theta = -\sin (360^\circ - \theta)$; $\cos \theta = \cos (360^\circ - \theta)$; $\tan \theta = -\tan (360^\circ - \theta)$. The reciprocal properties may be used to find the values of the other functions.

If it is desired to find the angle when a function of it is given, the procedure is as follows: There will in general be two angles between 0° and 360° corresponding to the given value of the function.

Given ($a > 0$)	Find an acute angle θ_0 such that	Required angles are
$\sin \theta = +a$	$\sin \theta_0 = a$	θ_0 and $(180^\circ - \theta_0)$
$\cos \theta = +a$	$\cos \theta_0 = a$	θ_0 and $(360^\circ - \theta_0)$
$\tan \theta = +a$	$\tan \theta_0 = a$	θ_0 and $(180^\circ + \theta_0)$
$\sin \theta = -a$	$\sin \theta_0 = a$	$180^\circ + \theta_0$ and $360^\circ - \theta_0$
$\cos \theta = -a$	$\cos \theta_0 = a$	$180^\circ - \theta_0$ and $180^\circ + \theta_0$
$\tan \theta = -a$	$\tan \theta_0 = a$	$180^\circ - \theta_0$ and $360^\circ - \theta_0$

Relations between Functions of a Single Angle $\sec \theta = 1/\cos \theta$; $\csc \theta = 1/\sin \theta$, $\tan \theta = \sin \theta/\cos \theta = \sec \theta/\csc \theta = 1/\cot \theta$; $\sin^2 \theta + \cos^2 \theta = 1$; $1 + \tan^2 \theta = \sec^2 \theta$; $1 + \cot^2 \theta = \csc^2 \theta$. For $0 \leq \theta \leq 90^\circ$ the following results hold:

$$\sin \theta = 2 \sin \left(\frac{\theta}{2} \right) \cos \left(\frac{\theta}{2} \right)$$

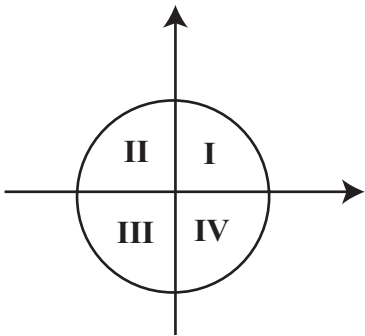


FIG. 3-37 Quadrants.

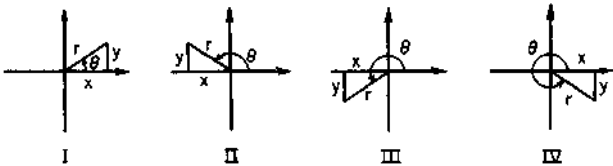


FIG. 3-38 Triangles.

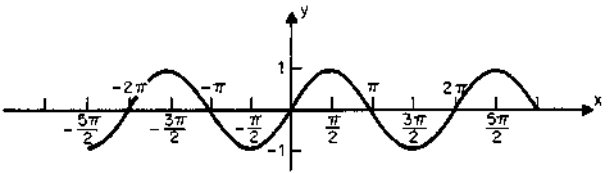


FIG. 3-39 Graph of $y = \sin x$.

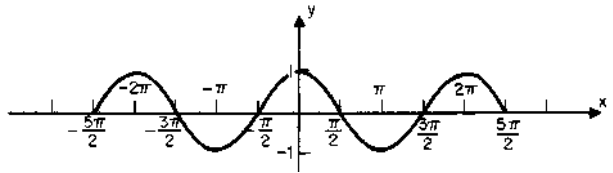


FIG. 3-40 Graph of $y = \cos x$.

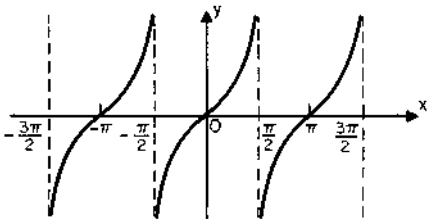


FIG. 3-41 Graph of $y = \tan x$.

$$\text{and } \cos \theta = \cos^2 \left(\frac{\theta}{2} \right) - \sin^2 \left(\frac{\theta}{2} \right)$$

The cofunction property is very important. $\cos \theta = \sin (90^\circ - \theta)$, $\sin \theta = \cos (90^\circ - \theta)$, $\tan \theta = \cot (90^\circ - \theta)$, $\cot \theta = \tan (90^\circ - \theta)$, etc.

Functions of Negative Angles $\sin (-\theta) = -\sin \theta$, $\cos (-\theta) = \cos \theta$, $\tan (-\theta) = -\tan \theta$, $\sec (-\theta) = \sec \theta$, $\csc (-\theta) = -\csc \theta$, $\cot (-\theta) = -\cot \theta$.

Identities

Sum and Difference Formulas Let x, y be two angles. $\sin (x \pm y) = \sin x \cos y \pm \cos x \sin y$; $\cos (x \pm y) = \cos x \cos y \mp \sin x \sin y$; $\tan (x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$; $\sin x \pm \sin y = 2 \sin \frac{1}{2}(x \pm y) \cos \frac{1}{2}(x \mp y)$; $\cos x + \cos y = 2 \cos \frac{1}{2}(x + y) \cos \frac{1}{2}(x - y)$; $\cos x - \cos y = -2 \sin \frac{1}{2}(x + y) \sin \frac{1}{2}(x - y)$; $\tan x \pm \tan y = \frac{\sin (x \pm y)}{\cos x \cos y}$; $\sin^2 x - \sin^2 y = \cos^2 y - \cos^2 x = \sin (x + y) \sin (x - y)$; $\cos^2 x - \sin^2 y = \cos^2 y - \sin^2 x = \cos (x + y) \cos (x - y)$; $\sin (45^\circ + x) = \cos (45^\circ - x)$; $\sin (45^\circ - x) = \cos (45^\circ + x)$; $\tan (45^\circ \pm x) = \cot (45^\circ \mp x)$.

Multiple and Half Angle Identities Let $x = \text{angle}$, $\sin 2x = 2 \sin x \cos x$; $\sin x = 2 \sin \frac{1}{2}x \cos \frac{1}{2}x$; $\cos 2x = \cos^2 x - \sin^2 x = 1 - 2 \sin^2 x = 2 \cos^2 x - 1$; $\tan 2x = (2 \tan x)/(1 - \tan^2 x)$; $\sin 3x = 3 \sin x - 4 \sin^3 x$; $\cos 3x = 4 \cos^3 x - 3 \cos x$; $\tan 3x = (3 \tan x - \tan^3 x)/(1 - 3 \tan^2 x)$; $\sin 4x = 4 \sin x \cos x - 8 \sin^3 x \cos x$; $\cos 4x = 8 \cos^4 x - 8 \cos^2 x + 1$.

$$\sin \left(\frac{x}{2} \right) = \sqrt{\frac{1 - \cos x}{2}}$$

$$\cos \left(\frac{x}{2} \right) = \sqrt{\frac{1 + \cos x}{2}}$$

$$\tan \left(\frac{x}{2} \right) = \sqrt{\frac{1 - \cos x}{1 + \cos x}} = \frac{\sin x}{1 + \cos x} = \frac{1 - \cos x}{\sin x}$$

INVERSE TRIGONOMETRIC FUNCTIONS

$y = \sin^{-1} x = \arcsin x$ is the angle y whose sine is x .

Example $y = \sin^{-1} \frac{1}{2}$, y is 30° .

The complete solution of the equation $x = \sin y$ is $y = (-1)^n \sin^{-1} x + n(180^\circ)$, $-\pi/2 \leq \sin^{-1} x \leq \pi/2$ where $\sin^{-1} x$ is the principal value of the angle whose sine is x . The range of principal values of the $\cos^{-1} x$ is $0 \leq \cos^{-1} x \leq \pi$ and $-\pi/2 \leq \tan^{-1} x \leq \pi/2$. If these restrictions are allowed to hold, the following formulas result:

$$\sin^{-1} x = \cos^{-1} \sqrt{1 - x^2} = \tan^{-1} \frac{x}{\sqrt{1 - x^2}} = \cot^{-1} \frac{\sqrt{1 - x^2}}{x}$$

$$= \sec^{-1} \frac{1}{\sqrt{1 - x^2}} = \csc^{-1} \frac{1}{x} = \frac{\pi}{2} - \cos^{-1} x$$

$$\cos^{-1} x = \sin^{-1} \sqrt{1 - x^2} = \tan^{-1} \frac{\sqrt{1 - x^2}}{x}$$

$$= \cot^{-1} \frac{x}{\sqrt{1 - x^2}} = \sec^{-1} \frac{1}{x}$$

$$= \csc^{-1} \frac{1}{\sqrt{1 - x^2}} = \frac{\pi}{2} - \sin^{-1} x$$

$$\tan^{-1} x = \sin^{-1} \frac{x}{\sqrt{1 + x^2}} = \cos^{-1} \frac{1}{\sqrt{1 + x^2}}$$

$$= \cot^{-1} \frac{1}{x} = \sec^{-1} \sqrt{1 + x^2} = \csc^{-1} \frac{\sqrt{1 + x^2}}{x}$$

RELATIONS BETWEEN ANGLES AND SIDES OF TRIANGLES

Solutions of Triangles (Fig. 3-42) Let a, b, c denote the sides and α, β, γ the angles opposite the sides in the triangle. Let $2s = a +$

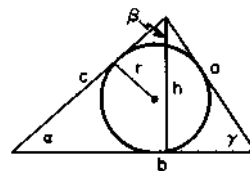


FIG. 3-42 Triangle.

$b + c$, $A = \text{area}$, $r = \text{radius of the inscribed circle}$, $R = \text{radius of the circumscribed circle}$, and $h = \text{altitude}$. In any triangle $\alpha + \beta + \gamma = 180^\circ$.

Law of Sines $\sin \alpha/a = \sin \beta/b = \sin \gamma/c = 1/(2R)$.

Law of Tangents

$$\frac{a+b}{a-b} = \frac{\tan \frac{1}{2}(\alpha+\beta)}{\tan \frac{1}{2}(\alpha-\beta)}; \frac{b+c}{b-c} = \frac{\tan \frac{1}{2}(\beta+\gamma)}{\tan \frac{1}{2}(\beta-\gamma)}; \frac{a+c}{a-c} = \frac{\tan \frac{1}{2}(\alpha+\gamma)}{\tan \frac{1}{2}(\alpha-\gamma)}$$

Law of Cosines $a^2 = b^2 + c^2 - 2bc \cos \alpha$; $b^2 = a^2 + c^2 - 2ac \cos \beta$; $c^2 = a^2 + b^2 - 2ab \cos \gamma$.

Other Relations In this subsection, where appropriate, two more formulas can be generated by replacing a by b , b by c , c by a , α by β , β by γ , and γ by α . $\cos \alpha = (b^2 + c^2 - a^2)/2bc$; $a = b \cos \gamma + c \cos \beta$; $\sin \alpha = (2/bc) \sqrt{s(s-a)(s-b)(s-c)}$;

$$\sin \left(\frac{\alpha}{2} \right) = \sqrt{\frac{(s-b)(s-c)}{bc}}; \cos \left(\frac{\alpha}{2} \right) = \sqrt{\frac{s(s-a)}{bc}};$$

$$A = \frac{1}{2} bh = \frac{1}{2} ab \sin \gamma = \frac{a^2 \sin \beta \sin \gamma}{2 \sin \alpha} = \sqrt{s(s-a)(s-b)(s-c)} = rs$$

$$\text{where } r = \frac{\sqrt{(s-a)(s-b)(s-c)}}{s}$$

$$R = a/(2 \sin \alpha) = abc/4A; h = c \sin a = a \sin \gamma = 2rs/b.$$

Right Triangle (Fig. 3-43) Given one side and any acute angle α or any two sides, the remaining parts can be obtained from the following formulas:

$$a = \sqrt{(c+b)(c-b)} = c \sin \alpha = b \tan \alpha$$

$$b = \sqrt{(c+a)(c-a)} = c \cos \alpha = a \cot \alpha$$

$$c = \sqrt{a^2 + b^2}, \sin \alpha = \frac{a}{c}, \cos \alpha = \frac{b}{c}, \tan \alpha = \frac{a}{b}, \beta = 90^\circ - \alpha$$

$$A = \frac{1}{2} ab = \frac{a^2}{2 \tan \alpha} = \frac{b^2 \tan \alpha}{2} = \frac{c^2 \sin 2\alpha}{4}$$

Oblique Triangles (Fig. 3-44) There are four possible cases.

1. Given b, c and the included angles α ,

$$\frac{1}{2} (\beta + \gamma) = 90^\circ - \frac{1}{2} \alpha; \tan \frac{1}{2} (\beta - \gamma) = \frac{b - c}{b + c} \tan \frac{1}{2} (\beta + \gamma)$$

$$\beta = \frac{1}{2} (\beta + \gamma) + \frac{1}{2} (\beta - \gamma); \gamma = \frac{1}{2} (\beta + \gamma) - \frac{1}{2} (\beta - \gamma); a = \frac{b \sin \alpha}{\sin \beta}$$

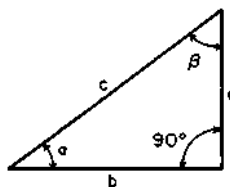


FIG. 3-43 Right triangle.

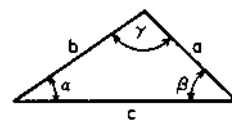


FIG. 3-44 Oblique triangle.

3-18 MATHEMATICS

2. Given the three sides a, b, c , $s = \frac{1}{2}(a + b + c)$;

$$r = \sqrt{\frac{(s-a)(s-b)(s-c)}{s}}$$

$$\tan \frac{1}{2} \alpha = \frac{r}{s-a}; \tan \frac{1}{2} \beta = \frac{r}{s-b}; \tan \frac{1}{2} \gamma = \frac{r}{s-c}$$

3. Given any two sides a, c and an angle opposite one of them α , $\sin \gamma = (c \sin \alpha)/a$; $\beta = 180^\circ - \alpha - \gamma$; $b = (a \sin \beta)/(\sin \alpha)$. There may be two solutions here. γ may have two values γ_1, γ_2 ; $\gamma_1 < 90^\circ$, $\gamma_2 = 180^\circ - \gamma_1 > 90^\circ$. If $\alpha + \gamma_2 > 180^\circ$, use only γ_1 . This case may be *impossible* if $\sin \gamma > 1$.

4. Given any side c and two angles α and β , $\gamma = 180^\circ - \alpha - \beta$; $a = (c \sin \alpha)/(\sin \gamma)$; $b = (c \sin \beta)/(\sin \gamma)$.

HYPERBOLIC TRIGONOMETRY

The hyperbolic functions are certain combinations of exponentials e^x and e^{-x} .

$$\cosh x = \frac{e^x + e^{-x}}{2}; \sinh x = \frac{e^x - e^{-x}}{2}; \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$\coth x = \frac{e^x + e^{-x}}{e^x - e^{-x}} = \frac{1}{\tanh x} = \frac{\cosh x}{\sinh x}; \operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}};$$

$$\operatorname{csch} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}$$

Fundamental Relationships $\sinh x + \cosh x = e^x$; $\cosh x - \sinh x = e^{-x}$; $\cosh^2 x - \sinh^2 x = 1$; $\operatorname{sech}^2 x + \tanh^2 x = 1$; $\coth^2 x - \operatorname{csch}^2 x = 1$; $\sinh 2x = 2 \sinh x \cosh x$; $\cosh 2x = \cosh^2 x + \sinh^2 x = 1 + 2 \sinh^2 x = 2 \cosh^2 x - 1$. $\tanh 2x = (2 \tanh x)/(1 + \tanh^2 x)$; $\sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y$; $\cosh(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y$;

$\cosh(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y$; $2 \sinh^2 x/2 = \cosh x - 1$; $2 \cosh^2 x/2 = \cosh x + 1$; $\sinh(-x) = -\sinh x$; $\cosh(-x) = \cosh x$; $\tanh(-x) = -\tanh x$.

When $u = a \cosh x$, $v = a \sinh x$, then $u^2 - v^2 = a^2$; which is the equation for a hyperbola. In other words, the hyperbolic functions in the parametric equations $u = a \cosh x$, $v = a \sinh x$ have the same relation to the hyperbola $u^2 - v^2 = a^2$ that the equations $u = a \cos \theta$, $v = a \sin \theta$ have to the circle $u^2 + v^2 = a^2$.

Inverse Hyperbolic Functions If $x = \sinh y$, then y is the inverse hyperbolic sine of x written $y = \sinh^{-1} x$ or $\operatorname{arsinh} x$. $\sinh^{-1} x = \log_e(x + \sqrt{x^2 + 1})$

$$\cosh^{-1} x = \log_e(x + \sqrt{x^2 - 1}); \tanh^{-1} x = \frac{1}{2} \log_e \frac{1+x}{1-x};$$

$$\coth^{-1} x = \frac{1}{2} \log_e \frac{x+1}{x-1}; \operatorname{sech}^{-1} x = \log_e \left(\frac{1 + \sqrt{1-x^2}}{x} \right);$$

$$\operatorname{csch}^{-1} x = \log_e \left(\frac{1 + \sqrt{1+x^2}}{x} \right)$$

Magnitude of the Hyperbolic Functions $\cosh x \geq 1$ with equality only for $x = 0$; $-\infty < \sinh x < \infty$; $-1 < \tanh x < 1$. $\cosh x \sim e^x/2$ as $x \rightarrow \infty$; $\sinh x \rightarrow e^x/2$ as $x \rightarrow \infty$.

APPROXIMATIONS FOR TRIGONOMETRIC FUNCTIONS

For small values of θ (θ measured in radians) $\sin \theta \approx \theta$, $\tan \theta \approx \theta$; $\cos \theta \approx 1 - (\theta^2/2)$. The behavior ratio of the functions as $\theta \rightarrow 0$ is given by the following:

$$\lim_{\theta \rightarrow 0} \sin \theta / \theta = 1; \sin \theta / \tan \theta = 1.$$

DIFFERENTIAL AND INTEGRAL CALCULUS

REFERENCES: Char, B. W., et al., *Maple V Language Reference Manual*, Springer-Verlag, New York (1991); Wolfram, S., *The Mathematics Book*, 5th ed., Wolfram Media (2003).

DIFFERENTIAL CALCULUS

An Example of Functional Notation Suppose that a storage warehouse of 16,000 ft³ is required. The construction costs per square foot are \$10, \$3, and \$2 for walls, roof, and floor respectively. What are the minimum cost dimensions? Thus, with h = height, x = width, and y = length, the respective costs are

$$\text{Walls} = 2 \times 10hy + 2 \times 10hx = 20h(y + x)$$

$$\text{Roof} = 3xy$$

$$\text{Floor} = 2xy$$

$$\text{Total cost} = 2xy + 3xy + 20h(x + y) = 5xy + 20h(x + y) \quad (3-1)$$

and the restriction

$$\text{Total volume} = xyh \quad (3-2)$$

Solving for h from Eq. (3-2),

$$h = \text{volume}/xy = 16,000/xy \quad (3-3)$$

$$\text{Cost} = 5xy + \frac{320,000}{xy} (y + x) = 5xy + 320,000 \left(\frac{1}{x} + \frac{1}{y} \right) \quad (3-4)$$

In this form it can be shown that the minimum cost will occur for $x = y$; therefore

$$\text{Cost} = 5x^2 + 640,000 (1/x)$$

By evaluation, the smallest cost will occur when $x = 40$.

$$\text{Cost} = 5(1600) + 640,000/40 = \$24,000$$

The dimensions are then $x = 40$ ft, $y = 40$ ft, $h = 16,000/(40 \times 40) = 10$ ft. Symbolically, the original cost relationship is written

$$\text{Cost} = f(x, y, h) = 5xy + 20h(y + x)$$

and the volume relation

$$\text{Volume} = g(x, y, h) = xyh = 16,000$$

In terms of the derived general relationships (3-1) and (3-2), x, y , and h are **independent variables**—cost and volume, **dependent variables**. That is, the cost and volume become fixed with the specification of dimensions. However, corresponding to the given restriction of the problem, relative to volume, the function $g(x, y, z) = xyh$ becomes a constraint function. In place of three independent and two dependent variables the problem reduces to two independent (volume has been constrained) and two dependent as in functions (3-3) and (3-4). Further, the requirement of minimum cost reduces the problem to three dependent variables (x, y, h) and no **degrees of freedom**, that is, freedom of independent selection.

Limits The limit of function $f(x)$ as x approaches a (a is finite or else x is said to increase without bound) is the number N .

$$\lim_{x \rightarrow a} f(x) = N$$

This states that $f(x)$ can be calculated as close to N as desirable by making x sufficiently close to a . This does not put any restriction on $f(x)$ when $x = a$. Alternatively, for any given positive number ϵ , a number δ can be found such that $0 < |a - x| < \delta$ implies that $|N - f(x)| < \epsilon$.

The following operations with limits (when they exist) are valid:

$$\lim_{x \rightarrow a} bf(x) = b \lim_{x \rightarrow a} f(x)$$

$$\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

$$\lim_{x \rightarrow a} [f(x)g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$$

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} \quad \text{if } \lim_{x \rightarrow a} g(x) \neq 0$$

See “Indeterminant Forms” below when $g(a) = 0$.

Continuity A function $f(x)$ is continuous at the point $x = a$ if

$$\lim_{h \rightarrow 0} [f(a+h) - f(a)] = 0$$

Rigorously, it is stated $f(x)$ is continuous at $x = a$ if for any positive ε there exists a $\delta > 0$ such that $|f(a+h) - f(a)| < \varepsilon$ for all x with $|x - a| < \delta$. For example, the function $(\sin x)/x$ is not continuous at $x = 0$ and therefore is said to be discontinuous. Discontinuities are classified into three types:

1. Removable $y = \sin x/x$ at $x = 0$
2. Infinite $y = 1/x$ at $x = 0$
3. Jump $y = 10/(1 + e^{1/x})$ at $x = 0^+$ $y = 0^+$
 $x = 0$ $y = 0$
 $x = 0^-$ $y = 10$

Derivative The function $f(x)$ has a derivative at $x = a$, which can be denoted as $f'(a)$, if

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

exists. This implies continuity at $x = a$. Conversely, a function may be continuous but not have a derivative. The derivative function is

$$f'(x) = \frac{df}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Differentiation Define $\Delta y = f(x + \Delta x) - f(x)$. Then dividing by Δx

$$\frac{\Delta y}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Call $\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx}$

then $\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$

Example Find the derivative of $y = \sin x$.

$$\begin{aligned} \frac{dy}{dx} &= \lim_{\Delta x \rightarrow 0} \frac{\sin(x + \Delta x) - \sin x}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{\sin x \cos \Delta x + \cos x \sin \Delta x - \sin x}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{\sin x (\cos \Delta x - 1)}{\Delta x} + \lim_{\Delta x \rightarrow 0} \frac{\sin \Delta x \cos x}{\Delta x} \\ &= \cos x \text{ since } \lim_{\Delta x \rightarrow 0} \frac{\sin \Delta x}{\Delta x} = 1 \end{aligned}$$

Differential Operations The following differential operations are valid: $f, g, \dots$ are differentiable functions of x , c and n are constants; e is the base of the natural logarithms.

$$\frac{dc}{dx} = 0 \quad (3-5)$$

$$\frac{dx}{dx} = 1 \quad (3-6)$$

$$\frac{d}{dx} (f + g) = \frac{df}{dx} + \frac{dg}{dx} \quad (3-7)$$

$$\frac{d}{dx} (f \times g) = f \frac{dg}{dx} + g \frac{df}{dx} \quad (3-8)$$

$$\frac{dy}{dx} = \frac{1}{dx/dy} \quad \text{if } \frac{dx}{dy} \neq 0 \quad (3-9)$$

$$\frac{d}{dx} f^n = n f^{n-1} \frac{df}{dx} \quad (3-10)$$

$$\frac{d}{dx} \left(\frac{f}{g} \right) = \frac{g(df/dx) - f(dg/dx)}{g^2} \quad (3-11)$$

$$\frac{df}{dx} = \frac{df}{dv} \times \frac{dv}{dx} \quad (\text{chain rule}) \quad (3-12)$$

$$\frac{df^g}{dx} = g f^{g-1} \frac{df}{dx} + f^g \ln f \frac{dg}{dx} \quad (3-13)$$

$$\frac{da^x}{dx} = (\ln a) a^x \quad (3-14)$$

Example Derive dy/dx for $x^2 + y^3 = x + xy + A$.

Here $\frac{d}{dx} x^2 + \frac{d}{dx} y^3 = \frac{d}{dx} x + \frac{d}{dx} xy + \frac{d}{dx} A$

$$2x + 3y^2 \frac{dy}{dx} = 1 + y + x \frac{dy}{dx} + 0$$

by rules (3-10), (3-10), (3-6), (3-8), and (3-5) respectively.

Thus $\frac{dy}{dx} = \frac{2x - 1 - y}{x - 3y^2}$

Differentials

$$de^x = e^x dx \quad (3-15a)$$

$$d(a^x) = a^x \log a dx \quad (3-15b)$$

$$d \ln x = (1/x) dx \quad (3-16)$$

$$d \log x = (\log e/x) dx \quad (3-17)$$

$$d \sin x = \cos x dx \quad (3-18)$$

$$d \cos x = -\sin x dx \quad (3-19)$$

$$d \tan x = \sec^2 x dx \quad (3-20)$$

$$d \cot x = -\csc^2 x dx \quad (3-21)$$

$$d \sec x = \tan x \sec x dx \quad (3-22)$$

$$d \csc x = -\cot x \csc x dx \quad (3-23)$$

$$d \sin^{-1} x = (1 - x^2)^{-1/2} dx \quad (3-24)$$

$$d \cos^{-1} x = -(1 - x^2)^{-1/2} dx \quad (3-25)$$

$$d \tan^{-1} x = (1 + x^2)^{-1} dx \quad (3-26)$$

$$d \cot^{-1} x = -(1 + x^2)^{-1} dx \quad (3-27)$$

$$d \sec^{-1} x = x^{-1} (x^2 - 1)^{-1/2} dx \quad (3-28)$$

$$d \csc^{-1} x = -x^{-1} (x^2 - 1)^{-1/2} dx \quad (3-29)$$

$$d \sinh x = \cosh x dx \quad (3-30)$$

$$d \cosh x = \sinh x dx \quad (3-31)$$

$$d \tanh x = \text{sech}^2 x dx \quad (3-32)$$

$$d \coth x = -\text{csch}^2 x dx \quad (3-33)$$

$$d \text{sech } x = -\text{sech } x \tanh x dx \quad (3-34)$$

$$d \text{csch } x = -\text{csch } x \coth x dx \quad (3-35)$$

$$d \sinh^{-1} x = (x^2 + 1)^{-1/2} dx \quad (3-36)$$

$$d \cosh^{-1} x = (x^2 - 1)^{-1/2} dx \quad (3-37)$$

$$d \tanh^{-1} x = (1 - x^2)^{-1} dx \quad (3-38)$$

$$d \coth^{-1} x = -(x^2 - 1)^{-1} dx \quad (3-39)$$

$$d \text{sech}^{-1} x = -(1/x)(1 - x^2)^{-1/2} dx \quad (3-40)$$

$$d \text{csch}^{-1} x = -x^{-1}(x^2 + 1)^{-1/2} dx \quad (3-41)$$

Example Find dy/dx for $y = \sqrt{x} \cos(1 - x^2)$.

Using

$$\frac{dy}{dx} = \sqrt{x} \frac{d}{dx} \cos(1 - x^2) + \cos(1 - x^2) \frac{d}{dx} \sqrt{x} \quad (3-8)$$

$$\frac{d}{dx} \cos(1 - x^2) = -\sin(1 - x^2) \frac{d}{dx} (1 - x^2) \quad (3-19)$$

$$= -\sin(1 - x^2)(0 - 2x) \quad (3-5), (3-10)$$

$$\frac{d\sqrt{x}}{dx} = \frac{1}{2} x^{-1/2} \quad (3-10)$$

$$\frac{dy}{dx} = 2x^{3/2} \sin(1 - x^2) + \frac{1}{2} x^{-1/2} \cos(1 - x^2)$$

Example Find the derivative of $\tan x$ with respect to $\sin x$.

$$v = \sin x$$

$$y = \tan x$$

Using

$$\frac{d \tan x}{d \sin x} = \frac{dy}{dv} = \frac{dy}{dx} \frac{dx}{dv} \quad (3-12)$$

$$= \frac{d \tan x}{dx} \frac{1}{\frac{d \sin x}{dx}} \quad (3-9)$$

$$= \sec^2 x / \cos x \quad (3-18), (3-20)$$

Very often in experimental sciences and engineering functions and their derivatives are available only through their numerical values. In particular, through measurements we may know the values of a function and its derivative only at certain points. In such cases the preceding operational rules for derivatives, including the chain rule, can be applied numerically.

Example Given the following table of values for differentiable functions f and g ; evaluate the following quantities:

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	3	1	4	-4
3	0	2	4	7
4	-2	10	3	6

$$\frac{d}{dx} [f(x) + g(x)]_{x=4} = f'(4) + g'(4) = 10 + 6 = 16$$

$$\left(\frac{f}{g}\right)'(1) = \frac{f'(1)g(1) - f(1)g'(1)}{[g(1)]^2} = \frac{1 \cdot 4 - 3(-4)}{(4)^2} = \frac{16}{16} = 1$$

Higher Differentials The first derivative of $f(x)$ with respect to x is denoted by f' or df/dx . The derivative of the first derivative is called the second derivative of $f(x)$ with respect to x and is denoted by f'' , $f^{(2)}$, or d^2f/dx^2 ; and similarly for the higher-order derivatives.

Example Given $f(x) = 3x^3 + 2x + 1$, calculate all derivative values at $x = 3$.

$$\frac{df(x)}{dx} = 9x^2 + 2 \quad x = 3, f'(3) = 9(9) + 2 = 83$$

$$\frac{d^2f(x)}{dx^2} = 18x \quad x = 3, f''(3) = 18(3) = 54$$

$$\frac{d^3f(x)}{dx^3} = 18 \quad x = 3, f'''(3) = 18$$

$$\frac{d^nf(x)}{dx^n} = 0 \quad \text{for } n \geq 4$$

If $f'(x) > 0$ on (a, b) , then f is increasing on (a, b) . If $f'(x) < 0$ on (a, b) , then f is decreasing on (a, b) .

The graph of a function $y = f(x)$ is concave up if f' is increasing on (a, b) ; it is concave down if f' is decreasing on (a, b) .

If $f''(x)$ exists on (a, b) and if $f''(x) > 0$, then f is concave up on (a, b) . If $f''(x) < 0$, then f is concave down on (a, b) .

An inflection point is a point at which a function changes the direction of its concavity.

Indeterminate Forms: L'Hospital's Theorem Forms of the type $0/0$, ∞/∞ , $0 \times \infty$, etc., are called indeterminates. To find the limiting values that the corresponding functions approach, L'Hospital's theorem is useful: If two functions $f(x)$ and $g(x)$ both become zero at $x = a$, then the limit of their quotient is equal to the limit of the quotient of their separate derivatives, if the limit exists, or is $+\infty$ or $-\infty$.

Example Find $\lim_{x \rightarrow 0} \frac{\sin x}{x}$.

$$\text{Here} \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{d \sin x}{dx} = \lim_{x \rightarrow 0} \frac{\cos x}{1} = 1$$

Example Find $\lim_{x \rightarrow \infty} \frac{(1.1)^x}{x^{1000}}$.

$$\lim_{x \rightarrow \infty} \frac{(1.1)^x}{x^{1000}} = \lim_{x \rightarrow \infty} \frac{d(1.1)^x}{dx^{1000}} = \lim_{x \rightarrow \infty} \frac{(\ln 1.1)(1.1)^x}{1000x^{999}}$$

Obviously $\lim_{x \rightarrow \infty} \frac{1.1^x}{x^{1000}} = \infty$ since repeated application of the rule will reduce the denominator to a finite number $1000!$ while the numerator remains infinitely large.

Example Find $\lim_{x \rightarrow \infty} x^3 e^{-x}$.

$$\lim_{x \rightarrow \infty} x^3 e^{-x} = \lim_{x \rightarrow \infty} \frac{x^3}{e^x} = \lim_{x \rightarrow \infty} \frac{6}{e^x} = 0$$

Example Find $\lim_{x \rightarrow 0} (1 - x)^{1/x}$.

Let

$$y = (1 - x)^{1/x}$$

$$\ln y = (1/x) \ln(1 - x)$$

$$\lim_{x \rightarrow 0} (\ln y) = \lim_{x \rightarrow 0} \frac{\ln(1 - x)}{x} = -1$$

Therefore,

$$\lim_{x \rightarrow 0} y = e^{-1}$$

Partial Derivative The abbreviation $z = f(x, y)$ means that z is a function of the two variables x and y . The derivative of z with respect to x , treating y as a constant, is called the partial derivative with respect to x and is usually denoted as $\partial z / \partial x$ or $\partial f(x, y) / \partial x$ or simply f_x . Partial differentiation, like full differentiation, is quite simple to apply. Conversely, the solution of partial differential equations is appreciably more difficult than that of differential equations.

Example Find $\partial z / \partial x$ and $\partial z / \partial y$ for $z = ye^{x^2} + xe^y$.

$$\begin{aligned} \frac{\partial z}{\partial x} &= y \frac{\partial e^{x^2}}{\partial x} + e^y \frac{\partial x}{\partial x} & \frac{\partial z}{\partial y} &= e^{x^2} \frac{\partial y}{\partial y} + x \frac{\partial e^y}{\partial y} \\ &= 2xye^{x^2} + e^y & &= e^{x^2} + xe^y \end{aligned}$$

Order of Differentiation It is generally true that the order of differentiation is immaterial for any number of differentiations or variables provided the function and the appropriate derivatives are continuous. For $z = f(x, y)$ it follows:

$$\frac{\partial^2 f}{\partial y^2 \partial x} = \frac{\partial^2 f}{\partial y \partial x \partial y} = \frac{\partial^2 f}{\partial x \partial y^2}$$

General Form for Partial Differentiation

1. Given $f(x, y) = 0$ and $x = g(t)$, $y = h(t)$.

$$\text{Then} \quad \frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

$$\begin{aligned} \frac{d^2f}{dt^2} &= \frac{\partial^2 f}{\partial x^2} \left(\frac{dx}{dt}\right)^2 + 2 \frac{\partial^2 f}{\partial x \partial y} \frac{dx}{dt} \frac{dy}{dt} + \frac{\partial^2 f}{\partial y^2} \left(\frac{dy}{dt}\right)^2 + \frac{\partial f}{\partial x} \frac{d^2x}{dt^2} \\ &\quad + \frac{\partial f}{\partial y} \frac{d^2y}{dt^2} \end{aligned}$$

Example Find df/dt for $f = xy$, $x = \rho \sin t$, $y = \rho \cos t$.

$$\begin{aligned}\frac{df}{dt} &= \frac{\partial(xy)}{\partial x} \left(\frac{d\rho \sin t}{dt} \right) + \frac{\partial(xy)}{\partial y} \left(\frac{d\rho \cos t}{dt} \right) \\ &= y(\rho \cos t) + x(-\rho \sin t) \\ &= \rho^2 \cos^2 t - \rho^2 \sin^2 t\end{aligned}$$

2. Given $f(x, y) = 0$ and $x = g(t, s)$, $y = h(t, s)$.

Then

$$\begin{aligned}\frac{\partial f}{\partial t} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial t} \\ \frac{\partial f}{\partial s} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial s}\end{aligned}$$

Differentiation of Composite Function

Rule 1. Given $f(x, y) = 0$, then $\frac{dy}{dx} = -\frac{\partial f/\partial x}{\partial f/\partial y} \left(\frac{\partial f}{\partial y} \neq 0 \right)$.

Rule 2. Given $f(u) = 0$ where $u = g(x)$, then

$$\begin{aligned}\frac{df}{dx} &= f'(u) \frac{du}{dx} \\ \frac{d^2 f}{dx^2} &= f''(u) \left(\frac{du}{dx} \right)^2 + f'(u) \frac{d^2 u}{dx^2}\end{aligned}$$

Example Find df/dx for $f = \sin^2 u$ and $u = \sqrt{1-x^2}$

$$\begin{aligned}\frac{df}{dx} &= \frac{d \sin^2 u}{du} \frac{d \sqrt{1-x^2}}{dx} \\ &= 2 \sin u \cos u \left(\frac{1}{2} \right) (-2x)(1-x^2)^{-1/2} \\ &= -2 \frac{\sqrt{1-u^2}}{u} \sin u \cos u\end{aligned}$$

Rule 3. Given $f(u) = 0$ where $u = g(x, y)$, then

$$\begin{aligned}\frac{\partial f}{\partial x} &= f'(u) \frac{\partial u}{\partial x} \quad \frac{\partial f}{\partial y} = f'(u) \frac{\partial u}{\partial y} \\ \frac{\partial^2 f}{\partial x^2} &= f'' \left(\frac{\partial u}{\partial x} \right)^2 + f' \frac{\partial^2 u}{\partial x^2} \\ \frac{\partial^2 f}{\partial x \partial y} &= f'' \frac{\partial u}{\partial x} \frac{\partial u}{\partial y} + f' \frac{\partial^2 u}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y^2} &= f'' \left(\frac{\partial u}{\partial y} \right)^2 + f' \frac{\partial^2 u}{\partial y^2}\end{aligned}$$

MULTIVARIABLE CALCULUS APPLIED TO THERMODYNAMICS

Many of the functional relationships needed in thermodynamics are direct applications of the rules of multivariable calculus. This section reviews those rules in the context of the needs of thermodynamics. These ideas were expounded in one of the classic books on chemical engineering thermodynamics [see Hougou, O. A., et al., Part II, "Thermodynamics," in *Chemical Process Principles*, 2d ed., Wiley, New York (1959)].

State Functions State functions depend only on the state of the system, not on past history or how one got there. If z is a function of two variables, x and y , then $z(x, y)$ is a state function, since z is known once x and y are specified. The differential of z is

$$dz = M dx + N dy$$

The line integral

$$\int_C (M dx + N dy)$$

is independent of the path in x - y space if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad (3-42)$$

The total differential can be written as

$$dz = \left(\frac{\partial z}{\partial x} \right)_y dx + \left(\frac{\partial z}{\partial y} \right)_x dy \quad (3-43)$$

and the following condition guarantees path independence.

$$\frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right)_y = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right)_x$$

or

$$\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial^2 z}{\partial x \partial y} \quad (3-44)$$

Example Suppose z is constant and apply Eq. (3-43).

$$0 = \left(\frac{\partial z}{\partial x} \right)_y dx + \left(\frac{\partial z}{\partial y} \right)_x dy$$

Rearrangement gives

$$\left(\frac{\partial z}{\partial x} \right)_y = - \left(\frac{\partial y}{\partial x} \right)_z \left(\frac{\partial z}{\partial y} \right)_x = - \frac{(\partial y/\partial x)_z}{(\partial y/\partial z)_x} \quad (3-45)$$

Alternatively, divide Eq. (3-43) by dy when holding some other variable w constant to obtain

$$\left(\frac{\partial z}{\partial y} \right)_w = \left(\frac{\partial z}{\partial x} \right)_y \left(\frac{\partial x}{\partial y} \right)_w + \left(\frac{\partial z}{\partial y} \right)_x \quad (3-46)$$

Also divide both numerator and denominator of a partial derivative by dw while holding a variable y constant to get

$$\left(\frac{\partial z}{\partial x} \right)_y = \frac{(\partial z/\partial w)_y}{(\partial x/\partial w)_y} = \left(\frac{\partial z}{\partial w} \right)_y \left(\frac{\partial w}{\partial x} \right)_y \quad (3-47)$$

Thermodynamic State Functions In thermodynamics, the state functions include the internal energy, U ; enthalpy, H ; and Helmholtz and Gibbs free energies, A and G , respectively, defined as follows:

$$H = U + pV$$

$$A = U - TS$$

$$G = H - TS = U + pV - TS = A + pV$$

S is the entropy, T the absolute temperature, p the pressure, and V the volume. These are also state functions, in that the entropy is specified once two variables (like T and p) are specified, for example. Likewise, V is specified once T and p are specified; it is therefore a state function.

All applications are for closed systems with constant mass. If a process is reversible and only p - V work is done, the first law and differentials can be expressed as follows.

$$dU = T dS - p dV$$

$$dH = T dS + V dp$$

$$dA = -S dT - p dV$$

$$dG = -S dT + V dp$$

Alternatively, if the internal energy is considered a function of S and V , then the differential is:

$$dU = \left(\frac{\partial U}{\partial S} \right)_V dS + \left(\frac{\partial U}{\partial V} \right)_S dV$$

This is the equivalent of Eq. (3-43) and gives the following definitions.

$$T = \left(\frac{\partial U}{\partial S} \right)_V, \quad p = - \left(\frac{\partial U}{\partial V} \right)_S$$

Since the internal energy is a state function, then Eq. (3-44) must be satisfied.

$$\frac{\partial^2 U}{\partial V \partial S} = \frac{\partial^2 U}{\partial S \partial V}$$

This is

$$\left(\frac{\partial T}{\partial V} \right)_S = - \left(\frac{\partial p}{\partial S} \right)_V$$

This is one of the Maxwell relations, and the other Maxwell relations can be derived in a similar fashion by applying Eq. (3-44).

See Sec. 4, Thermodynamics, "Constant-Composition Systems."

Partial Derivatives of All Thermodynamic Functions The various partial derivatives of the thermodynamic functions can be classified into six groups. In the general formulas below, the variables U, H, A, G or S are denoted by Greek letters, while the variables V, T , or p are denoted by Latin letters.

Type I (3 possibilities plus reciprocals)

$$\text{General: } \left(\frac{\partial a}{\partial b} \right)_c; \text{ Specific: } \left(\frac{\partial p}{\partial T} \right)_V$$

Eq. (3-45) gives

$$\left(\frac{\partial p}{\partial T} \right)_V = - \left(\frac{\partial V}{\partial T} \right)_p \left(\frac{\partial p}{\partial V} \right)_T = - \frac{(\partial V / \partial T)_p}{(\partial V / \partial p)_T}$$

Type II (30 possibilities plus reciprocals)

$$\text{General: } \left(\frac{\partial \alpha}{\partial b} \right)_c; \text{ Specific: } \left(\frac{\partial G}{\partial T} \right)_V$$

The differential for G gives

$$\left(\frac{\partial G}{\partial T} \right)_V = -S + V \left(\frac{\partial p}{\partial T} \right)_V$$

Using the other equations for U, H, A , or S gives the other possibilities.

Type III (15 possibilities plus reciprocals)

$$\text{General: } \left(\frac{\partial a}{\partial b} \right)_\alpha; \text{ Specific: } \left(\frac{\partial V}{\partial T} \right)_S$$

First expand the derivative using Eq. (3-45).

$$\left(\frac{\partial V}{\partial T} \right)_S = - \left(\frac{\partial T}{\partial S} \right)_V \left(\frac{\partial V}{\partial T} \right)_T = - \frac{(\partial S / \partial T)_V}{(\partial S / \partial V)_T}$$

Then evaluate the numerator and denominator as type II derivatives.

$$\left(\frac{\partial V}{\partial T} \right)_S = - \frac{\frac{C_V}{T}}{- \left(\frac{\partial V}{\partial T} \right)_p \left(\frac{\partial p}{\partial V} \right)_T} = \frac{C_V}{T} \left(\frac{\partial p}{\partial T} \right)_T$$

These derivatives are of importance for reversible, adiabatic processes (such as in an ideal turbine or compressor), since then the entropy is constant. An example is the Joule-Thomson coefficient for constant H .

$$\left(\frac{\partial T}{\partial p} \right)_H = \frac{1}{C_p} \left[-V + T \left(\frac{\partial V}{\partial T} \right)_p \right]$$

Type IV (30 possibilities plus reciprocals)

$$\text{General: } \left(\frac{\partial \alpha}{\partial \beta} \right)_c; \text{ Specific: } \left(\frac{\partial G}{\partial A} \right)_p$$

Use Eq. (3-47) to introduce a new variable.

$$\left(\frac{\partial G}{\partial A} \right)_p = \left(\frac{\partial G}{\partial T} \right)_p \left(\frac{\partial T}{\partial A} \right)_p = \frac{(\partial G / \partial T)_p}{(\partial A / \partial T)_p}$$

This operation has created two type II derivatives; by substitution we obtain

$$\left(\frac{\partial G}{\partial A} \right)_p = \frac{S}{S + p (\partial V / \partial T)_p}$$

Type V (60 possibilities plus reciprocals)

$$\text{General: } \left(\frac{\partial \alpha}{\partial b} \right)_\beta; \text{ Specific: } \left(\frac{\partial G}{\partial p} \right)_A$$

Start from the differential for dG . Then we get

$$\left(\frac{\partial G}{\partial p} \right)_A = -S \left(\frac{\partial T}{\partial p} \right)_A + V$$

The derivative is type III and can be evaluated by using Eq. (3-45).

$$\left(\frac{\partial G}{\partial p} \right)_A = S \frac{(\partial A / \partial p)_T}{(\partial A / \partial T)_p} + V$$

The two type II derivatives are then evaluated.

$$\left(\frac{\partial G}{\partial p} \right)_A = \frac{Sp (\partial V / \partial p)_T}{S + p (\partial V / \partial T)_p} + V$$

These derivatives are also of interest for free expansions or isentropic changes.

Type VI (30 possibilities plus reciprocals)

$$\text{General: } \left(\frac{\partial \alpha}{\partial \beta} \right)_\gamma; \text{ Specific: } \left(\frac{\partial G}{\partial A} \right)_H$$

We use Eq. (3-47) to obtain two type V derivatives.

$$\left(\frac{\partial G}{\partial A} \right)_H = \frac{(\partial G / \partial T)_H}{(\partial A / \partial T)_H}$$

These can then be evaluated using the procedures for Type V derivatives.

INTEGRAL CALCULUS

Indefinite Integral If $f'(x)$ is the derivative of $f(x)$, an antiderivative of $f'(x)$ is $f(x)$. Symbolically, the indefinite integral of $f'(x)$ is

$$\int f'(x) dx = f(x) + c$$

where c is an arbitrary constant to be determined by the problem. By virtue of the known formulas for differentiation the following relationships hold (a is a constant):

$$\int (du + dv + dw) = \int du + \int dv + \int dw \quad (3-48)$$

$$\int a dv = a \int dv \quad (3-49)$$

$$\int v^n dv = \frac{v^{n+1}}{n+1} + c \quad (n \neq -1) \quad (3-50)$$

$$\int \frac{dv}{v} = \ln |v| + c \quad (3-51)$$

$$\int a^v dv = \frac{a^v}{\ln a} + c \quad (3-52)$$

$$\int e^v dv = e^v + c \quad (3-53)$$

$$\int \sin v dv = -\cos v + c \quad (3-54)$$

$$\int \cos v dv = \sin v + c \quad (3-55)$$

$$\int \sec^2 v dv = \tan v + c \quad (3-56)$$

$$\int \csc^2 v dv = -\cot v + c \quad (3-57)$$

$$\int \sec v \tan v dv = \sec v + c \quad (3-58)$$

$$\int \csc v \cot v dv = -\csc v + c \quad (3-59)$$

$$\int \frac{dv}{v^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{v}{a} + c \quad (3-60)$$

$$\int \frac{dv}{\sqrt{a^2 - v^2}} = \sin^{-1} \frac{v}{a} + c \quad (3-61)$$

$$\int \frac{dv}{v^2 - a^2} = \frac{1}{2a} \ln \left| \frac{v - a}{v + a} \right| + c \quad (3-62)$$

$$\int \frac{dv}{\sqrt{v^2 \pm a^2}} = \ln |v + \sqrt{v^2 \pm a^2}| + c \quad (3-63)$$

$$\int \sec v dv = \ln (\sec v + \tan v) + c \quad (3-64)$$

$$\int \csc v \, dv = \ln (\csc v - \cot v) + c \quad (3-65)$$

Example Find $\int (3x^2 + e^x - 10) \, dx$ using Eq. (3-48). $\int (3x^2 + e^x - 10) \, dx = 3 \int x^2 \, dx + \int e^x \, dx - 10 \int 1 \, dx = x^3 + e^x - 10x + c$ (by Eqs. 3-50, 3-53).

Example Find $\int \frac{7x \, dx}{2 - 3x^2}$. Let $v = 2 - 3x^2$; $dv = -6x \, dx$

$$\begin{aligned} \text{Thus } \int \frac{7x \, dx}{2 - 3x^2} &= 7 \int \frac{x \, dx}{2 - 3x^2} = -\frac{7}{6} \int \frac{-6x \, dx}{2 - 3x^2} \\ &= -\frac{7}{6} \int \frac{dv}{v}, \text{ with } v = 2 - 3x^2 \text{ and } dv = -6x \, dx \\ &= -\frac{7}{6} \ln |v| + c = -\frac{7}{6} \ln |2 - 3x^2| + c \end{aligned}$$

Example—Constant of Integration By definition the derivative of x^3 is $3x^2$, and x^3 is therefore the integral of $3x^2$. However, if $f = x^3 + 10$, it follows that $f' = 3x^2$, and $x^3 + 10$ is therefore also the integral of $3x^2$. For this reason the constant c in $\int 3x^2 \, dx = x^3 + c$ must be determined by the problem conditions, i.e., the value of f for a specified x .

Methods of Integration In practice it is rare when generally encountered functions can be directly integrated. For example, the integrand in $\int \sqrt{\sin x} \, dx$ which appears quite simple has no elementary function whose derivative is $\sqrt{\sin x}$. In general, there is no explicit way of determining whether a particular function can be integrated into an elementary form. As a whole, integration is a trial-and-error proposition which depends on the effort and ingenuity of the practitioner. The following are general procedures which can be used to find the elementary forms of the integral when they exist. When they do not exist or cannot be found either from tabled integration formulas or directly, the only recourse is series expansion as illustrated later. Indefinite integrals cannot be solved numerically unless they are redefined as definite integrals (see "Definite Integral"), i.e., $F(x) = \int f(x) \, dx$, indefinite, whereas $F(x) = \int_a^x f(t) \, dt$, definite.

Direct Formula Many integrals can be solved by transformation in the integrand to one of the forms given previously.

Example Find $\int x^2 \sqrt{3x^3 + 10} \, dx$. Let $v = 3x^3 + 10$ for which $dv = 9x^2 \, dx$. Thus

$$\begin{aligned} \int x^2 \sqrt{3x^3 + 10} \, dx &= \int (3x^3 + 10)^{1/2} (x^2 \, dx) \\ &= \frac{1}{9} \int (3x^3 + 10)^{1/2} (9x^2 \, dx) = \frac{1}{9} \int v^{1/2} \, dv \\ &= \frac{1}{9} \frac{v^{3/2}}{3/2} + c \quad [\text{by Eq. (3-50)}] \\ &= \frac{2}{27} (3x^3 + 10)^{3/2} + c \end{aligned}$$

Trigonometric Substitution This technique is particularly well adapted to integrands in the form of radicals. For these the function is transformed into a trigonometric form. In the latter form they may be more easily recognizable relative to the identity formulas. These functions and their transformations are

$$\begin{aligned} \sqrt{x^2 - a^2} \quad &\text{Let } x = a \sec \theta \\ \sqrt{x^2 + a^2} \quad &\text{Let } x = a \tan \theta \\ \sqrt{a^2 - x^2} \quad &\text{Let } x = a \sin \theta \end{aligned}$$

Example Find $\int \frac{\sqrt{4-9x^2}}{x^2} \, dx$. Let $x = \frac{2}{3} \sin \theta$; then $dx = \frac{2}{3} \cos \theta \, d\theta$.

$$3 \int \frac{\sqrt{(2/3)^2 - x^2}}{x^2} \, dx = 3 \int \frac{2/3 \sqrt{1 - \sin^2 \theta}}{(2/3)^2 \sin^2 \theta} \left(\frac{2}{3} \cos \theta \, d\theta \right)$$

$$\begin{aligned} &= 3 \int \frac{\cos^2 \theta}{\sin^2 \theta} \, d\theta = 3 \int \cot^2 \theta \, d\theta \\ &= -3 \cot \theta - 3\theta + c \text{ by trigonometric transform} \\ &= -\frac{\sqrt{4-9x^2}}{x} - 3 \sin^{-1} \frac{3}{2} x + c \text{ in terms of } x \end{aligned}$$

Algebraic Substitution Functions containing elements of the type $(a + bx)^{1/n}$ are best handled by the algebraic transformation $y^n = a + bx$.

Example Find $\int \frac{x \, dx}{(3 + 4x)^{1/4}}$. Let $3 + 4x = y^4$; then $4dx = 4y^3 \, dy$ and

$$\begin{aligned} \int \frac{x \, dx}{(3 + 4x)^{1/4}} &= \int \frac{\frac{y^4 - 3}{4} y^3 \, dy}{y} = \frac{1}{4} \int y^2 (y^4 - 3) \, dy \\ &= \frac{1}{4} \frac{y^7}{7} - \frac{3}{4} \frac{y^3}{3} + c = \frac{1}{28} (3 + 4x)^{7/4} - \frac{1}{4} (3 + 4x)^{3/4} + c \end{aligned}$$

General The number of possible transformations one might use are unlimited. No specific overall rules can be given. Success in handling integration problems depends primarily upon experience and ingenuity. The following example illustrates the extent to which alternative approaches are possible.

Example Find $\int \frac{dx}{e^x - 1}$. Let $e^x = y$; then $e^x \, dx = dy$ or $dx = 1/y \, dy$.

$$\int \frac{dx}{e^x - 1} = \int \frac{(1/y) \, dy}{y - 1} = \int \frac{dy}{y^2 - y} = \ln \frac{y - 1}{y} = \ln \frac{e^x - 1}{e^x}$$

Partial Fractions Rational functions are of the type $f(x)/g(x)$ where $f(x)$ and $g(x)$ are polynomial expressions of degrees m and n respectively. If the degree of f is higher than g , perform the algebraic division—the remainder will then be at least one degree less than the denominator. Consider the following types:

Type 1 Reducible denominator to linear unequal factors. For example,

$$\begin{aligned} \frac{1}{x^3 - x^2 - 4x + 4} &= \frac{1}{(x + 2)(x - 2)(x - 1)} \\ &= \frac{A}{x + 2} + \frac{B}{x - 2} + \frac{C}{x - 1} \\ &= \frac{A(x - 2)(x - 1) + B(x + 2)(x - 1) + C(x + 2)(x - 2)}{(x + 2)(x - 2)(x - 1)} \\ &= \frac{x^2(A + B + C) + x(-3A + B) + (2A - 2B - 4C)}{(x + 2)(x - 2)(x - 1)} \end{aligned}$$

Equate coefficients and solve for A , B , and C .

$$A + B + C = 0$$

$$-3A + B = 0$$

$$2A - 2B - 4C = 1$$

$$A = 1/2, B = 1/4, C = -1/4$$

$$\frac{1}{x^3 - x^2 - 4x + 4} = \frac{1}{12(x + 2)} + \frac{1}{4(x - 2)} - \frac{1}{3(x - 1)}$$

Hence

$$\int \frac{dx}{x^3 - x^2 - 4x + 4} = \int \frac{dx}{12(x + 2)} + \int \frac{dx}{4(x - 2)} - \int \frac{dx}{3(x - 1)}$$

Parts An extremely useful formula for integration is the relation

$$d(uv) = u \, dv + v \, du$$

and

$$uv = \int u \, dv + \int v \, du$$

or

$$\int u \, dv = uv - \int v \, du$$

No general rule for breaking an integrand can be given. Experience alone limits the use of this technique. It is particularly useful for trigonometric and exponential functions.

Example Find $\int xe^x dx$. Let

$$u = x \quad \text{and} \quad dv = e^x dx$$

$$du = dx \quad \quad v = e^x$$

Therefore

$$\int xe^x dx = xe^x - \int e^x dx$$

$$= xe^x - e^x + c$$

Example Find $\int e^x \sin x dx$. Let

$$u = e^x \quad \quad dv = \sin x dx$$

$$du = e^x dx \quad \quad v = -\cos x$$

$$\int e^x \sin x dx = -e^x \cos x + \int e^x \cos x dx$$

Again

$$u = e^x \quad \quad dv = \cos x dx$$

$$du = e^x dx \quad \quad v = \sin x$$

$$\int e^x \sin x dx = -e^x \cos x + e^x \sin x - \int e^x \sin x dx + c$$

$$= (e^x/2)(\sin x - \cos x) + \frac{c}{2}$$

Series Expansion When an explicit function cannot be found, the integration can sometimes be carried out by a series expansion.

Example Find $\int e^{-x^2} dx$. Since

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \cdots$$

$$\int e^{-x^2} dx = \int dx - \int x^2 dx + \int \frac{x^4}{2!} dx - \int \frac{x^6}{3!} dx + \cdots$$

$$= x - \frac{x^3}{3} + \frac{x^5}{5 \cdot 2!} - \frac{x^7}{7 \cdot 3!} + \cdots \quad \text{for all } x$$

Definite Integral The concept and derivation of the definite integral are completely different from those for the indefinite integral. These are by definition different types of operations. However, the formal operation $\int$ as it turns out treats the integrand in the same way for both.

Consider the function $f(x) = 10 - 10e^{-2x}$. Define $x_1 = a$ and $x_n = b$, and suppose it is desirable to compute the area between the curve and the coordinate axis $y = 0$ and bounded by $x_1 = a$, $x_n = b$. Obviously, by a sufficiently large number of rectangles this area could be approximated as closely as desired by the formula

$$\sum_{i=1}^{n-1} f(\xi_i)(x_{i+1} - x_i) = f(\xi_1)(x_2 - a) + f(\xi_2)(x_3 - x_2)$$

$$+ \cdots + f(\xi_{n-1})(b - x_{n-1}) \quad x_{i-1} \leq \xi_{i-1} \leq x_i$$

The definite integral of $f(x)$ is defined as

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i)(x_{i+1} - x_i)$$

where the points $x_1, x_2, \dots, x_n$ are equally spaced.

Thus, the value of a definite integral depends on the limits a, b , and any selected variable coefficients in the function but not on the dummy variable of integration x . Symbolically

$$F(x) = \int f(x) dx \quad \text{indefinite integral where } dF/dx = f(x)$$

$$\text{or } F(a, b) = \int_a^b f(x) dx \quad \text{definite integral}$$

$$F(\alpha) = \int_a^b f(x, \alpha) dx$$

There are certain restrictions of the integration definition. "The function $f(x)$ must be continuous in the finite interval (a, b) with at most a finite number of finite discontinuities," which must be observed before integration formulas can be generally applied. Relaxing two of

these restrictions gives rise to so-called **improper integrals** and requires special handling. These occur when

1. The limits of integration are not both finite, i.e., $\int_0^\infty e^{-x} dx$.
2. The function becomes infinite within the interval of integration, i.e.,

$$\int_0^1 \frac{1}{\sqrt{x}} dx$$

Techniques for determining when integration is valid under these conditions are available in the references.

Properties The fundamental theorem of calculus states

$$\int_a^b f(x) dx = F(b) - F(a)$$

where

$$dF(x)/dx = f(x)$$

Other properties of the definite integral are

$$\int_a^b c[f(x)] dx = c \int_a^b f(x) dx$$

$$\int_a^b [f_1(x) + f_2(x)] dx = \int_a^b f_1(x) dx + \int_a^b f_2(x) dx$$

$$\int_a^b f(x) dx = -\int_b^a f(x) dx$$

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

$$\int_a^b f(x) dx = (b-a)f(\xi) \text{ for some } \xi \text{ in } (a, b)$$

$$\frac{\partial}{\partial b} \int_a^b f(x) dx = f(b)$$

$$\frac{\partial}{\partial a} \int_a^b f(x) dx = -f(a)$$

$$\frac{dF(\alpha)}{d\alpha} = \int_a^b \frac{\partial f(x, \alpha)}{\partial \alpha} dx \text{ if } a \text{ and } b \text{ are constant}$$

$$\int_a^b dx \int_c^d f(x, \alpha) d\alpha = \int_c^d d\alpha \int_a^b f(x, \alpha) dx \quad (3-66)$$

When $F(x) = \int_{a(x)}^{b(x)} f(x, y) dy$, the Leibniz rule gives

$$\frac{dF}{dx} = \frac{db}{dx} f[x, b(x)] - \frac{da}{dx} f[x, a(x)] + \int_{a(x)}^{b(x)} \frac{\partial f}{\partial x} dy$$

Example Find $\int_0^2 \frac{dx}{(x-1)^2}$. Direct application of the formula would yield the incorrect value

$$\int_0^2 \frac{dx}{(x-1)^2} = \left[-\frac{1}{x-1} \right]_0^2 = -2$$

It should be noted that $f(x) = 1/(x-1)^2$ becomes unbounded as $x \rightarrow 1$ and by Rule 2 the integral diverges and hence is said not to exist.

Methods of Integration All the methods of integration available for the indefinite integral can be used for definite integrals. In addition, several others are available for the latter integrals and are indicated below.

Change of Variable This substitution is basically the same as previously indicated for indefinite integrals. However, for definite integrals, the limits of integration must also be changed: i.e., for $x = \phi(t)$,

$$\int_a^b f(x) dx = \int_{t_0}^{t_1} f[\phi(t)] \phi'(t) dt$$

where $t = t_0$ when $x = a$
 $t = t_1$ when $x = b$

Example Find $\int_0^4 \sqrt{16-x^2} dx$. Let
 $x = 4 \sin \theta$ ($x=0, \theta=0$)
 $dx = 4 \cos \theta d\theta$ ($x=4, \theta=\pi/2$)

$$\text{Then } \int_0^4 \sqrt{16-x^2} dx = 16 \int_0^{\pi/2} \cos^2 \theta d\theta = 16[\frac{1}{2}\theta + \frac{1}{4}\sin 2\theta]_0^{\pi/2} = 4\pi$$

Integration It is sometimes useful to generate a double integral to solve a problem. By this approach, the fundamental theorem indicated by Eq. (3-66) can be used.

Example Find $\int_0^1 \frac{x^b - x^a}{\ln x} dx$

$$\text{Consider } \int_0^1 x^\alpha dx = \frac{1}{\alpha+1} \quad (\alpha > -1)$$

Then multiplying both sides by $d\alpha$ and integrating between a and b ,

$$\int_a^b d\alpha \int_0^1 x^\alpha dx = \int_a^b \frac{d\alpha}{\alpha+1} = \ln \left| \frac{b+1}{a+1} \right|$$

But also

$$\int_a^b d\alpha \int_0^1 x^\alpha dx = \int_0^1 dx \int_a^b x^\alpha d\alpha = \int_0^1 \frac{x^b - x^a}{\ln x} dx$$

$$\text{Therefore } \int_0^1 \frac{x^b - x^a}{\ln x} dx = \ln \left| \frac{b+1}{a+1} \right|$$

INFINITE SERIES

REFERENCES: de Bruijn, N. G., *Asymptotic Methods in Analysis*, Dover, New York (1981); Folland, G. B., *Advanced Calculus*, Prentice-Hall, Saddle River, N.J. (2002); Gradshteyn, I. S., and I. M. Ryzhik, *Tables of Integrals, Series, and Products*, Academic, New York (2000); Kaplan, W., *Advanced Calculus*, 5th ed., Addison-Wesley, Redwood City, Calif. (2003).

DEFINITIONS

A succession of numbers or terms that are formed according to some definite rule is called a sequence. The indicated sum of the terms of a sequence is called a series. A series of the form $a_0 + a_1(x-c) + a_2(x-c)^2 + \cdots + a_n(x-c)^n + \cdots$ is called a power series.

Consider the sum of a finite number of terms in the geometric series (a special case of a power series).

$$S_n = a + ar + ar^2 + ar^3 + \cdots + ar^{n-1} \quad (3-67)$$

For any number of terms n , the sum equals

$$S_n = a \frac{1-r^n}{1-r}$$

In this form, the geometric series is assumed finite.

In the form of Eq. (3-67), it can further be defined that the terms in the series be nonending and therefore an infinite series.

$$S = a + ar + ar^2 + \cdots + ar^n + \cdots \quad (3-68)$$

However, the defined sum of the terms [Eq. (3-67)]

$$S_n = a \frac{1-r^n}{1-r} \quad r \neq 1$$

while valid for any finite value of r and n now takes on a different interpretation. In this sense it is necessary to consider the limit of S_n as n increases indefinitely:

$$\begin{aligned} S &= \lim_{n \rightarrow \infty} S_n \\ &= a \lim_{n \rightarrow \infty} \frac{1-r^n}{1-r} \end{aligned}$$

For this, it is stated the infinite series converges if the limit of S_n approaches a fixed finite value as n approaches infinity. Otherwise, the series is **divergent**.

On this basis an analysis of

$$S = a \lim_{n \rightarrow \infty} \frac{1-r^n}{1-r}$$

shows that if r is less than 1 but greater than -1 , the infinite series is convergent. For values outside of the range $-1 < r < 1$, the series is divergent because the sum is not defined. The range $-1 < r < 1$ is called the **region of convergence**. (We assume $a \neq 0$.)

There are also two types of convergent series. Consider the new series

$$S = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + (-1)^{n+1} \frac{1}{n} + \cdots \quad (3-69)$$

It can be shown that the series (3-69) does converge to the value $S = \log 2$. However, if each term is replaced by its absolute value, the series becomes unbounded and therefore divergent (unbounded divergent):

$$S = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \cdots \quad (3-70)$$

In this case the series (3-69) is defined as a conditionally convergent series. If the replacement series of absolute values also converges, the series is defined to converge absolutely.

Series (3-69) is further defined as an alternating series, while series (3-70) is referred to as a positive series.

OPERATIONS WITH INFINITE SERIES

1. The convergence or divergence of an infinite series is unaffected by the removal of a finite number of finite terms. This is a trivial theorem but useful to remember, especially when using the comparison test to be described in the subsection "Tests for Convergence and Divergence."

2. If a series is conditionally convergent, its sums can be made to have any arbitrary value by a suitable rearrangement of the series; it can in fact be made divergent or oscillatory (Riemann's theorem).

3. A series of positive terms, if convergent, has a sum independent of the order of its terms; but if divergent, it remains divergent however its terms are rearranged.

4. An oscillatory series can always be made to converge by grouping terms.

5. A power series can be inverted, provided the first-degree term is not zero. Given

$$y = b_1x + b_2x^2 + b_3x^3 + b_4x^4 + b_5x^5 + b_6x^6 + b_7x^7 + \cdots$$

$$\text{then } x = B_1y + B_2y^2 + B_3y^3 + B_4y^4 + B_5y^5 + B_6y^6 + B_7y^7 + \cdots$$

where $B_1 = 1/b_1$

$$B_2 = -b_2/b_1^3$$

$$B_3 = (1/b_1^5) (2b_2^2 - b_1b_3)$$

$$B_4 = (1/b_1^7) (5b_1b_2b_3 - b_1^2b_4 - 5b_2^3)$$

Additional coefficients are available in the references.

6. Two series may be added or subtracted term by term provided each is a convergent series. The joint sum is equal to the sum (or difference) of the individuals.

7. The sum of two divergent series can be convergent. Similarly, the sum of a convergent series and a divergent series must be divergent.

8. A power series may be integrated term by term to represent the integral of the function within an interval of the region of convergence. If $f(x) = a_0 + a_1x + a_2x^2 + \cdots$, then

$$\int_{x_1}^{x_2} f(x) dx = \int_{x_1}^{x_2} a_0 dx + \int_{x_1}^{x_2} a_1x dx + \int_{x_1}^{x_2} a_2x^2 dx + \cdots$$

9. A power series may be differentiated term by term and represents the function $df(x)/dx$ within the same region of convergence as $f(x)$.

TESTS FOR CONVERGENCE AND DIVERGENCE

In general, the problem of determining whether a given series will converge or not can require a great deal of ingenuity and resourcefulness. There is no all-inclusive test which can be applied to all series. As the only alternative, it is necessary to apply one or more of the developed theorems in an attempt to ascertain the convergence or divergence of the series under study. The following defined tests are given in relative order of effectiveness. For examples, see references on advanced calculus.

1. **Comparison Test.** A series will converge if the absolute value of each term (with or without a finite number of terms) is less than the corresponding term of a known convergent series. Similarly, a positive series is divergent if it is termwise larger than a known divergent series of positive terms.

2. **n th-Term Test.** A series is divergent if the n th term of the series does not approach zero as n becomes increasingly large.

3. **Ratio Test.** If the absolute ratio of the $(n+1)$ term divided by the n th term as n becomes unbounded approaches

- a. A number less than 1, the series is absolutely convergent
- b. A number greater than 1, the series is divergent
- c. A number equal to 1, the test is inconclusive

4. **Alternating-Series Leibniz Test.** If the terms of a series are alternately positive and negative and never increase in value, the absolute series will converge, provided that the terms tend to zero as a limit.

5. **Cauchy's Root Test.** If the n th root of the absolute value of the n th term, as n becomes unbounded, approaches

- a. A number less than 1, the series is absolutely convergent
- b. A number greater than 1, the series is divergent
- c. A number equal to 1, the test is inconclusive

6. **Maclaurin's Integral Test.** Suppose $\sum a_n$ is a series of positive terms and f is a continuous decreasing function such that $f(x) \geq 0$ for $1 \leq x < \infty$ and $f(n) = a_n$. Then the series and the improper integral $\int_1^\infty f(x) dx$ either both converge or both diverge.

SERIES SUMMATION AND IDENTITIES

Sums for the First n Numbers to Integer Powers

$$\sum_{j=1}^n j = \frac{n(n+1)}{2} = 1 + 2 + 3 + 4 + \cdots + n$$

$$\sum_{j=1}^n j^2 = \frac{n(n+1)(2n+1)}{6} = 1^2 + 2^2 + 3^2 + 4^2 + \cdots + n^2$$

$$\sum_{j=1}^n j^3 = \frac{n^2(n+1)^2}{4} = 1^3 + 2^3 + 3^3 + \cdots + n^3$$

Arithmetic Progression

$$\begin{aligned} \sum_{k=1}^n [a + (k-1)d] &= a + (a+d) + (a+2d) \\ &\quad + (a+3d) + \cdots + [a + (n-1)d] \\ &= na + \frac{1}{2}n(n-1)d \end{aligned}$$

Geometric Progression

$$\sum_{j=1}^n ar^{j-1} = a + ar + ar^2 + ar^3 + \cdots + ar^{n-1} = a \frac{1-r^n}{1-r} \quad r \neq 1$$

Harmonic Progression

$$\sum_{k=0}^n \frac{1}{a+kd} = \frac{1}{a} + \frac{1}{a+d} + \frac{1}{a+2d} + \frac{1}{a+3d} + \frac{1}{a+4d} + \cdots + \frac{1}{a+nd}$$

The reciprocals of the terms of the arithmetic-progression series are called harmonic progression. No general summation formulas are available for this series.

Binomial Series (see also Elementary Algebra)

$$(1 \pm x)^n = 1 \pm nx + \frac{n(n-1)}{2!}x^2 \pm \frac{n(n-1)(n-2)}{3!}x^3 + \cdots \quad (x^2 < 1)$$

Taylor's Series

$$f(h+x) = f(h) + xf'(h) + \frac{x^2}{2!}f''(h) + \frac{x^3}{3!}f'''(h) + \cdots$$

$$\text{or } f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \frac{f'''(x_0)}{3!}(x-x_0)^3 + \cdots$$

Example Find a series expansion for $f(x) = \ln(1+x)$ about $x_0 = 0$.

$$f'(x) = (1+x)^{-1}, \quad f''(x) = -(1+x)^{-2}, \quad f'''(x) = 2(1+x)^{-3}, \text{ etc.}$$

thus $f(0) = 0, \quad f'(0) = 1, \quad f''(0) = -1, \quad f'''(0) = 2, \text{ etc.}$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots + (-1)^{n+1} \frac{x^n}{n} + \cdots$$

which converges for $-1 < x \leq 1$.

Maclaurin's Series

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots$$

This is simply a special case of Taylor's series when h is set to zero.

Exponential Series

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \cdots \quad -\infty < x < \infty$$

Logarithmic Series

$$\ln x = \frac{x-1}{x} + \frac{1}{2} \left(\frac{x-1}{x} \right)^2 + \frac{1}{3} \left(\frac{x-1}{x} \right)^3 + \cdots \quad (x > 1/2)$$

$$\ln x = 2 \left[\left(\frac{x-1}{x+1} \right) + \frac{1}{3} \left(\frac{x-1}{x+1} \right)^3 + \cdots \right] \quad (x > 0)$$

Trigonometric Series*

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \quad -\infty < x < \infty$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \quad -\infty < x < \infty$$

$$\sin^{-1} x = x + \frac{x^3}{6} + \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{x^5}{5} + \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdot \frac{x^7}{7} + \cdots \quad (x^2 < 1)$$

$$\tan^{-1} x = x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \frac{1}{7}x^7 + \cdots \quad (x^2 < 1)$$

Taylor Series The Taylor series for a function of two variables, expanded about the point (x_0, y_0) , is

$$^* \tan x \text{ series has awkward coefficients and should be computed as } \left[(\text{sign}) \frac{\sin x}{\sqrt{1-\sin^2 x}} \right].$$

$$f(x, y) = f(x_0, y_0) + \left. \frac{\partial f}{\partial x} \right|_{x_0, y_0} (x - x_0) + \left. \frac{\partial f}{\partial y} \right|_{x_0, y_0} (y - y_0) + \frac{1}{2!} \left[\left. \frac{\partial^2 f}{\partial x^2} \right|_{x_0, y_0} (x - x_0)^2 + 2 \left. \frac{\partial^2 f}{\partial x \partial y} \right|_{x_0, y_0} (x - x_0)(y - y_0) + \left. \frac{\partial^2 f}{\partial y^2} \right|_{x_0, y_0} (y - y_0)^2 \right] + \dots$$

Partial Sums of Infinite Series, and How They Grow Calculus textbooks devote much space to tests for convergence and divergence of series that are of little practical value, since a convergent series either converges rapidly, in which case almost any test (among those presented in the preceding subsections) will do; or it converges slowly, in which case it is not going to be of much use unless there is some way to get at its sum without adding up an unreasonable number of terms. To find out, as accurately as possible, how fast a convergent series converges and how fast a divergent series diverges, see Boas, R. P., Jr., *Am. Math. Mon.* **84**: 237–258 (1977).

COMPLEX VARIABLES

REFERENCES: Ablowitz, M. J., and A. S. Fokas, *Complex Variables: Introduction and Applications*, Cambridge University Press, New York (2003); Asmar, N., and G. C. Jones, *Applied Complex Analysis with Partial Differential Equations*, Prentice-Hall, Upper Saddle River, N.J. (2002); Brown, J. W., and R. V. Churchill, *Complex Variables and Applications*, 7th ed., McGraw-Hill, New York (2003); Kaplan, W., *Advanced Calculus*, 5th ed., Addison-Wesley, Redwood City, Calif. (2003); Kwok, Y. K., *Applied Complex Variables for Scientists and Engineers*, Cambridge University Press, New York (2002); McGehee, O. C., *An Introduction to Complex Analysis*, Wiley, New York (2000); Priestley, H. A., *Introduction to Complex Analysis*, Oxford University Press, New York (2003).

Numbers of the form $z = x + iy$, where x and y are real, $i^2 = -1$, are called complex numbers. The numbers $z = x + iy$ are representable in the plane as shown in Fig. 3-45. The following definitions and terminology are used:

1. Distance $OP = r =$ modulus of z written $|z|$. $|z| = \sqrt{x^2 + y^2}$.
2. x is the real part of z .
3. y is the imaginary part of z .
4. The angle θ , $0 \leq \theta < 2\pi$, measured counterclockwise from the positive x axis to OP is the argument of z . $\theta = \arctan y/x = \arcsin y/r = \arccos x/r$ if $x \neq 0$, $\theta = \pi/2$ if $x = 0$ and $y > 0$.
5. The numbers r , θ are the polar coordinates of z .
6. $\bar{z} = x - iy$ is the complex conjugate of z .

ALGEBRA

Let $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$.

Equality $z_1 = z_2$ if and only if $x_1 = x_2$ and $y_1 = y_2$.

Addition $z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$.

Subtraction $z_1 - z_2 = (x_1 - x_2) + i(y_1 - y_2)$.

Multiplication $z_1 \cdot z_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1)$.

Division $z_1/z_2 = \frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2} + i \frac{x_2 y_1 - x_1 y_2}{x_2^2 + y_2^2}$, $z_2 \neq 0$.

SPECIAL OPERATIONS

$z\bar{z} = x^2 + y^2 = |z|^2$; $\overline{z_1 \pm z_2} = \bar{z}_1 \pm \bar{z}_2$; $\overline{\bar{z}_1} = z_1$; $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$; $|z_1 \cdot z_2| = |z_1| \cdot |z_2|$; $\arg(z_1 \cdot z_2) = \arg z_1 + \arg z_2$; $\arg(z_1/z_2) = \arg z_1 - \arg z_2$; $i^{4n} = 1$ for n any integer; $i^{2n} = -1$ where n is any odd integer; $z + \bar{z} = 2x$; $z - \bar{z} = 2iy$.

Every complex quantity can be expressed in the form $x + iy$.

TRIGONOMETRIC REPRESENTATION

By referring to Fig. 3-45, there results $x = r \cos \theta$, $y = r \sin \theta$ so that $z = x + iy = r(\cos \theta + i \sin \theta)$, which is called the polar form of the

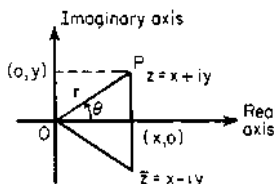


FIG. 3-45 Complex plane.

complex number. $\cos \theta + i \sin \theta = e^{i\theta}$. Hence $z = x + iy = re^{i\theta}$. $\bar{z} = x - iy = re^{-i\theta}$. Two important results from this are $\cos \theta = (e^{i\theta} + e^{-i\theta})/2$ and $\sin \theta = (e^{i\theta} - e^{-i\theta})/2i$. Let $z_1 = r_1 e^{i\theta_1}$, $z_2 = r_2 e^{i\theta_2}$. This form is convenient for multiplication for $z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}$ and for division for $z_1/z_2 = (r_1/r_2)e^{i(\theta_1 - \theta_2)}$, $z_2 \neq 0$.

POWERS AND ROOTS

If n is a positive integer, $z^n = (re^{i\theta})^n = r^n e^{in\theta} = r^n(\cos n\theta + i \sin n\theta)$.

If n is a positive integer,

$$z^{1/n} = r^{1/n} e^{i[(\theta + 2k\pi)/n]} = r^{1/n} \left[\cos \left(\frac{\theta + 2k\pi}{n} \right) + i \sin \left(\frac{\theta + 2k\pi}{n} \right) \right]$$

and selecting values of $k = 0, 1, 2, 3, \dots, n-1$ give the n distinct values of $z^{1/n}$. The n roots of a complex quantity are uniformly spaced around a circle, with radius $r^{1/n}$, in the complex plane in a symmetric fashion.

Example Find the three cube roots of -8 . Here $r = 8$, $\theta = \pi$. The roots are $z_0 = 2(\cos \pi/3 + i \sin \pi/3) = 1 + i\sqrt{3}$, $z_1 = 2(\cos \pi + i \sin \pi) = -2$, $z_2 = 2(\cos 5\pi/3 + i \sin 5\pi/3) = 1 - i\sqrt{3}$.

ELEMENTARY COMPLEX FUNCTIONS

Polynomials A polynomial in z , $a_n z^n + a_{n-1} z^{n-1} + \dots + a_0$, where n is a positive integer, is simply a sum of complex numbers times integral powers of z which have already been defined. Every polynomial of degree n has precisely n complex roots provided each multiple root of multiplicity m is counted m times.

Exponential Functions The exponential function e^z is defined by the equation $e^z = e^{x+iy} = e^x \cdot e^{iy} = e^x(\cos y + i \sin y)$. Properties: $e^0 = 1$; $e^{z_1} \cdot e^{z_2} = e^{z_1+z_2}$; $e^{z_1/z_2} = e^{z_1 - z_2}$; $e^{z+2k\pi i} = e^z$.

Trigonometric Functions $\sin z = (e^{iz} - e^{-iz})/2i$; $\cos z = (e^{iz} + e^{-iz})/2$; $\tan z = \sin z/\cos z$; $\cot z = \cos z/\sin z$; $\sec z = 1/\cos z$; $\csc z = 1/\sin z$. Fundamental identities for these functions are the same as their real counterparts. Thus $\cos^2 z + \sin^2 z = 1$, $\cos(z_1 \pm z_2) = \cos z_1 \cos z_2 \mp \sin z_1 \sin z_2$, $\sin(z_1 \pm z_2) = \sin z_1 \cos z_2 \pm \cos z_1 \sin z_2$. The sine and cosine of z are periodic functions of period 2π ; thus $\sin(z + 2\pi) = \sin z$. For computation purposes $\sin z = \sin(x + iy) = \sin x \cosh y + i \cos x \sinh y$, where $\sin x$, $\cosh y$, etc., are the real trigonometric and hyperbolic functions. Similarly, $\cos z = \cos x \cosh y - i \sin x \sinh y$. If $x = 0$ in the results given, $\cos iy = \cosh y$, $\sin iy = i \sinh y$.

Example Find all solutions of $\sin z = 3$. From previous data $\sin z = \sin x \cosh y + i \cos x \sinh y = 3$. Equating real and imaginary parts $\sin x \cosh y = 3$, $\cos x \sinh y = 0$. The second equation can hold for $y = 0$ or for $x = \pi/2, 3\pi/2, \dots$. If $y = 0$, $\cosh 0 = 1$ and $\sin x = 3$ is impossible for real x . Therefore, $x = \pm \pi/2, \pm 3\pi/2, \dots, \pm(2n+1)\pi/2$, $n = 0, \pm 1, \pm 2, \dots$. However, $\sin 3\pi/2 = -1$ and $\cosh y \geq 1$. Hence $x = \pi/2, 5\pi/2, \dots$. The solution is $z = [(4n+1)\pi/2 + i \cosh^{-1} 3, n = 0, 1, 2, 3, \dots]$.

Example Find all solutions of $e^z = -i$. $e^z = e^x(\cos y + i \sin y) = -i$. Equating real and imaginary parts gives $e^x \cos y = 0$, $e^x \sin y = -1$. From the first $y = \pm \pi/2, \pm 3\pi/2, \dots$. But $e^x > 0$. Therefore, $y = 3\pi/2, 7\pi/2, -\pi/2, \dots$. Then $x = 0$. The solution is $z = i[(4n+3)\pi/2]$.

Two important facets of these functions should be recognized. First, the $\sin z$ is *unbounded*; and, second, e^z takes *all* complex values *except* 0.

Hyperbolic Functions $\sinh z = (e^z - e^{-z})/2$; $\cosh z = (e^z + e^{-z})/2$; $\tanh z = \sinh z / \cosh z$; $\coth z = \cosh z / \sinh z$; $\operatorname{csch} z = 1 / \sinh z$; $\operatorname{sech} z = 1 / \cosh z$. Identities are: $\cosh^2 z - \sinh^2 z = 1$; $\sinh(z_1 + z_2) = \sinh z_1 \cosh z_2 + \cosh z_1 \sinh z_2$; $\cosh(z_1 + z_2) = \cosh z_1 \cosh z_2 + \sinh z_1 \sinh z_2$; $\cosh z + \sinh z = e^z$; $\cosh z - \sinh z = e^{-z}$. The hyperbolic sine and hyperbolic cosine are periodic functions with the imaginary period $2\pi i$. That is, $\sinh(z + 2\pi i) = \sinh z$.

Logarithms The logarithm of z , $\log z = \log |z| + i(\theta + 2n\pi)$, where $\log |z|$ is taken to the base e and θ is the principal argument of z , that is, the particular argument lying in the interval $0 \leq \theta < 2\pi$. The logarithm of z is infinitely many valued. If $n = 0$, the resulting logarithm is called the principal value. The familiar laws $\log z_1 z_2 = \log z_1 + \log z_2$, $\log z_1 / z_2 = \log z_1 - \log z_2$, $\log z^n = n \log z$ hold for the principal value.

General powers of z are defined by $z^\alpha = e^{\alpha \log z}$. Since $\log z$ is infinitely many valued, so too is z^α unless α is a rational number.

DeMoivre's formula can be derived from properties of e^z .

$$z^n = r^n (\cos \theta + i \sin \theta)^n = r^n (\cos n\theta + i \sin n\theta)$$

Thus $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$

COMPLEX FUNCTIONS (ANALYTIC)

In the real-number system a greater than b ($a > b$) and b less than c ($b < c$) define an order relation. These relations have no meaning for complex numbers. The absolute value is used for ordering. Some important relations follow: $|z| \geq x$; $|z| \geq y$; $|z_1 \pm z_2| \leq |z_1| + |z_2|$; $|z_1 - z_2| \geq ||z_1| - |z_2||$; $|z| \geq (|x| + |y|)/\sqrt{2}$. Parts of the complex plane, commonly called **regions** or **domains**, are described by using inequalities.

Example $|z - 3| \leq 5$. This is equivalent to $\sqrt{(x-3)^2 + y^2} \leq 5$, which is the set of all points within and on the circle, centered at $x = 3$, $y = 0$ of radius 5.

Example $|z - 1| \leq x$ represents the set of all points inside and on the parabola $2x = y^2 + 1$ or, equivalently, $2x \geq y^2 + 1$.

Functions of a Complex Variable If $z = x + iy$, $w = u + iv$ and if for each value of z in some region of the complex plane one or more values of w are defined, then w is said to be a function of z , $w = f(z)$. Some of these functions have already been discussed, e.g., $\sin z$, $\log z$. All functions are reducible to the form $w = u(x, y) + iv(x, y)$, where u , v are real functions of the real variables x and y .

Example $z^3 = (x + iy)^3 = x^3 + 3x^2(iy) + 3x(iy)^2 + (iy)^3 = (x^3 - 3xy^2) + i(3x^2y - y^3)$.

Differentiation The derivative of $w = f(z)$ is

$$\frac{dw}{dz} = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

and for the derivative to exist the limit must be the same no matter how Δz approaches zero. If w_1, w_2 are differentiable functions of z , the following rules apply:

$$\frac{d(w_1 \pm w_2)}{dz} = \frac{dw_1}{dz} \pm \frac{dw_2}{dz} \quad \frac{d(w_1 w_2)}{dz} = w_2 \frac{dw_1}{dz} + w_1 \frac{dw_2}{dz}$$

$$\frac{d(w_1/w_2)}{dz} = \frac{w_2(dw_1/dz) - w_1(dw_2/dz)}{w_2^2}$$

$$\text{and} \quad \frac{dw_1^n}{dz} = n w_1^{n-1} \frac{dw_1}{dz}$$

For $w = f(z)$ to be differentiable, it is necessary that $\partial u / \partial x = \partial v / \partial y$ and $\partial v / \partial x = -\partial u / \partial y$. The last two equations are called the Cauchy-Riemann equations. The derivative

$$\frac{dw}{dz} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}$$

If $f(z)$ possesses a derivative at z_0 and at every point in some neighborhood of z_0 , then $f(z)$ is said to be analytic or homomorphic at z_0 . If the

Cauchy-Riemann equations are satisfied and

$$u, v, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$$

are continuous in a region of the complex plane, then $f(z)$ is analytic in that region.

Example $w = z\bar{z} = x^2 + y^2$. Here $u = x^2 + y^2$, $v = 0$. $\partial u / \partial x = 2x$, $\partial u / \partial y = 2y$, $\partial v / \partial x = \partial v / \partial y = 0$. These are continuous everywhere, but the Cauchy-Riemann equations hold only at the origin. Therefore, w is nowhere analytic, but it is differentiable at $z = 0$ only.

Example $w = e^z = e^x \cos y + i e^x \sin y$. $u = e^x \cos y$, $v = e^x \sin y$. $\partial u / \partial x = e^x \cos y$, $\partial u / \partial y = -e^x \sin y$, $\partial v / \partial x = e^x \sin y$, $\partial v / \partial y = e^x \cos y$. The continuity and Cauchy-Riemann requirements are satisfied for all finite z . Hence e^z is analytic (except at ∞) and $dw/dz = \partial u / \partial x + i(\partial v / \partial x) = e^z$.

$$\text{Example} \quad w = \frac{1}{z} = \frac{x - iy}{x^2 + y^2} = \frac{x}{x^2 + y^2} - i \frac{y}{x^2 + y^2}$$

It is easy to see that dw/dz exists except at $z = 0$. Thus $1/z$ is analytic except at $z = 0$.

Singular Points If $f(z)$ is analytic in a region except at certain points, those points are called singular points.

Example $1/z$ has a singular point at zero.

Example $\tan z$ has singular points at $z = \pm(2n + 1)(\pi/2)$, $n = 0, 1, 2, \dots$

The derivatives of the common functions, given earlier, are the same as their real counterparts.

Example $(d/dz)(\log z) = 1/z$, $(d/dz)(\sin z) = \cos z$.

Harmonic Functions Both the *real* and the *imaginary* parts of any analytic function $f = u + iv$ satisfy Laplace's equation $\partial^2 \phi / \partial x^2 + \partial^2 \phi / \partial y^2 = 0$. A function which possesses continuous second partial derivatives and satisfies Laplace's equation is called a harmonic function.

Example $e^z = e^x \cos y + i e^x \sin y$. $u = e^x \cos y$, $\partial u / \partial x = e^x \cos y$, $\partial^2 u / \partial x^2 = e^x \cos y$, $\partial u / \partial y = -e^x \sin y$, $\partial^2 u / \partial y^2 = -e^x \cos y$. Clearly $\partial^2 u / \partial x^2 + \partial^2 u / \partial y^2 = 0$. Similarly, $v = e^x \sin y$ is also harmonic.

If $w = u + iv$ is analytic, the curves $u(x, y) = c$ and $v(x, y) = k$ intersect at right angles, if $w'(z) \neq 0$.

Integration In much of the work with complex variables a simple extension of integration called line or curvilinear integration is of fundamental importance. Since any complex line integral can be expressed in terms of real line integrals, we define only real line integrals. Let $F(x, y)$ be a real, continuous function of x and y and c be any continuous curve of finite length joining the points A and B (Fig. 3-46). $F(x, y)$ is not related to the curve c . Divide c up into n segments, Δs_i , whose projection on the x axis is Δx_i and on the y axis is Δy_i . Let (ϵ_i, η_i) be the coordinates of an arbitrary point on Δs_i . The limits of the sums

$$\lim_{\Delta s_i \rightarrow 0} \sum_{i=1}^n F(\epsilon_i, \eta_i) \Delta s_i = \int_c F(x, y) ds$$

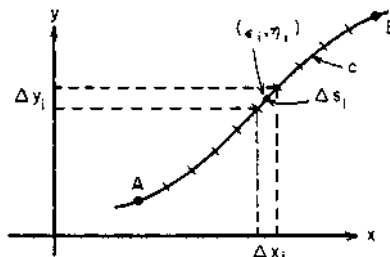


FIG. 3-46 Line integral.

$$\lim_{\Delta x_i \rightarrow 0} \sum_{i=1}^n F(\xi_i, \eta_i) \Delta x_i = \int_c F(x, y) dx$$

$$\lim_{\Delta y_i \rightarrow 0} \sum_{i=1}^n F(\xi_i, \eta_i) \Delta y_i = \int_c F(x, y) dy$$

are known as line integrals. Much of the initial strangeness of these integrals will vanish if it be observed that the ordinary definite integral $\int_a^b f(x) dx$ is just a line integral in which the curve c is a line segment on the x axis and $F(x, y)$ is a function of x alone. The evaluation of line integrals can be reduced to evaluation of ordinary integrals.

Example $\int_c y(1+x) dy$, where $c: y = 1 - x^2$ from $(-1, 0)$ to $(1, 0)$. Clearly $y = 1 - x^2$, $dy = -2x dx$. Thus $\int_c y(1+x) dy = -2 \int_{-1}^1 (1-x^2)(1+x) dx = -7/15$.

Example $\int_c x^2 y ds$, c is the square whose vertices are $(0, 0)$, $(1, 0)$, $(1, 1)$, $(0, 1)$. $ds = \sqrt{dx^2 + dy^2}$. When $dx = 0$, $ds = dy$. From $(0, 0)$ to $(1, 0)$, $y = 0$, $dy = 0$. Similar arguments for the other sides give

$$\int_c x^2 y ds = \int_0^1 0 \cdot x^2 dx + \int_0^1 y dy + \int_1^0 x^2 dx + \int_1^0 0 \cdot y dy = 1/2 - 1/6 = 1/6$$

Let $f(z)$ be any function of z , analytic or not, and c any curve as above. The complex integral is calculated as $\int_c f(z) dz = \int_c (u dx - v dy) + i \int_c (v dx + u dy)$, where $f(z) = u(x, y) + iv(x, y)$. Properties of line integrals are the same as those for ordinary integrals. That is, $\int_c [f(z) \pm g(z)] dz = \int_c f(z) dz \pm \int_c g(z) dz$; $\int_c kf(z) dz = k \int_c f(z) dz$ for any constant k , etc.

Example $\int_c (x^2 + iy) dz$ along $c: y = x$, 0 to $1 + i$. This becomes

$$\begin{aligned} \int_c (x^2 + iy) dz &= \int_c (x^2 dx - y dy) + i \int_c (y dx + x^2 dy) \\ &= \int_0^1 x^2 dx - \int_0^1 x dx + i \int_0^1 x dx + i \int_0^1 x^2 dx = -1/6 + 5i/6 \end{aligned}$$

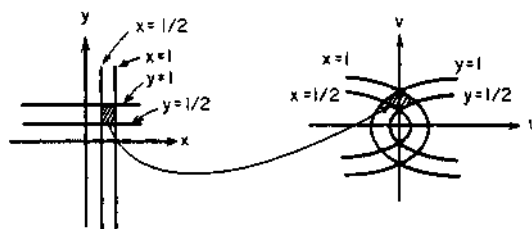


FIG. 3-47 Conformal transformation.

Conformal Mapping Every function of a complex variable $w = f(z) = u(x, y) + iv(x, y)$ transforms the x, y plane into the u, v plane in some manner. A conformal transformation is one in which angles between curves are preserved in *magnitude* and *sense*. Every analytic function, except at those points where $f'(z) = 0$, is a conformal transformation. See Fig. 3-47.

Example $w = z^2$. $u + iv = (x^2 - y^2) + 2ixy$ or $u = x^2 - y^2$, $v = 2xy$. These are the transformation equations between the (x, y) and (u, v) planes. Lines parallel to the x axis, $y = c_1$ map into curves in the u, v plane with parametric equations $u = x^2 - c_1^2$, $v = 2c_1x$. Eliminating x , $u = (v^2/4c_1^2) - c_1^2$, which represents a family of parabolas with the origin of the w plane as focus, the line $v = 0$ as axis and opening to the right. Similar arguments apply to $x = c_2$.

The principles of complex variables are useful in the solution of a variety of applied problems, including Laplace transforms and process control (Sec. 8).

DIFFERENTIAL EQUATIONS

REFERENCES: Ames, W. F., *Nonlinear Partial Differential Equations in Engineering*, Academic Press, New York (1965); Aris, R., and N. R. Amundson, *Mathematical Methods in Chemical Engineering*, vol. 2, *First-Order Partial Differential Equations with Applications*, Prentice-Hall, Englewood Cliffs, N.J. (1973); Asmar, N., and G. C. Jones, *Applied Complex Analysis with Partial Differential Equations*, Prentice-Hall, Upper Saddle River, N.J. (2002); Boyce, W. E., and R. C. Di Prima, *Elementary Differential Equations and Boundary Value Problems*, 7th ed., Wiley, New York (2004); Braun, M., *Differential Equations and Their Applications: An Introduction to Applied Mathematics*, 4th ed., Springer-Verlag, New York (1993); Bronson, R., and G. Costa, *Schaum's Outline of Differential Equations*, 3d ed., McGraw-Hill, New York (2007); Brown, J. W., and R. V. Churchill, *Fourier Series and Boundary Value Problems*, 6th ed., McGraw-Hill, New York (2000); Courant, R., and D. Hilbert, *Methods of Mathematical Physics*, vols. I and II, Interscience, New York (1953, 1962); Duffy, D., *Green's Functions with Applications*, Chapman and Hall/CRC (2001); Kreyszig, E., *Advanced Engineering Mathematics*, 8th ed., Wiley, New York (1999); Morse, P. M., and H. Feshbach, *Methods of Theoretical Physics*, vols. I and II, McGraw-Hill, New York (1953); Polyanin, A. D., *Handbook of Linear Partial Differential Equations for Engineers and Scientists*, Chapman and Hall/CRC (2002); Polyanin, A. D., and V. F. Zaitsev, *Handbook of Exact Solutions for Ordinary Differential Equations*, 2d ed., Chapman and Hall/CRC (2002); Ramkrishna, D., and N. R. Amundson, *Linear Operator Methods in Chemical Engineering with Applications to Transport and Chemical Reaction Systems*, Prentice-Hall, Englewood Cliffs, N.J. (1985).

The natural laws in any scientific or technological field are not regarded as precise and definitive until they have been expressed in mathematical form. Such a form, often an equation, is a relation between the quantity of interest, say, product yield, and independent variables such as time and temperature upon which yield depends. When it happens that this equation involves, besides the function itself, one or more of its derivatives it is called a differential equation.

Example The rate of the homogeneous bimolecular reaction $A + B \xrightarrow{k} C$ is characterized by the differential equation $dx/dt = k(a-x)(b-x)$, where a = initial

concentration of A , b = initial concentration of B , and $x = x(t)$ = concentration of C as a function of time t .

Example The differential equation of heat conduction in a moving fluid with velocity components v_x, v_y is

$$\frac{\partial u}{\partial t} + v_x \frac{\partial u}{\partial x} + v_y \frac{\partial u}{\partial y} = \frac{K}{\rho c_p} \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

where $u = u(x, y, t)$ = temperature, K = thermal conductivity, ρ = density, and c_p = specific heat at constant pressure.

ORDINARY DIFFERENTIAL EQUATIONS

When the function involved in the equation depends upon only one variable, its derivatives are ordinary derivatives and the differential equation is called an ordinary differential equation. When the function depends upon several independent variables, then the equation is called a partial differential equation. The theories of ordinary and partial differential equations are quite different. In almost every respect the latter is more difficult.

Whichever the type, a differential equation is said to be of n th order if it involves derivatives of order n but no higher. The equation in the first example is of first order and that in the second example of second order. The degree of a differential equation is the power to which the derivative of the highest order is raised after the equation has been cleared of fractions and radicals in the dependent variable and its derivatives.

A relation between the variables, involving no derivatives, is called a solution of the differential equation if this relation, when substituted in the equation, satisfies the equation. A solution of an ordinary differential equation which includes the maximum possible number of "arbitrary" constants is called the **general solution**. The maximum number of "arbitrary" constants is exactly equal to the order of the

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differential equation. If any set of specific values of the constants is chosen, the result is called a **particular solution**.

Example The general solution of $(d^2x/dt^2) + k^2x = 0$ is $x = A \cos kt + B \sin kt$, where A, B are arbitrary constants. A particular solution is $x = \frac{1}{2} \cos kt + 3 \sin kt$.

In the case of some equations still other solutions exist called singular solutions. A **singular solution** is any solution of the differential equation which is not included in the general solution.

Example $y = x(dy/dx) - \frac{1}{4}(dy/dx)^2$ has the general solution $y = cx - \frac{1}{4}c^2$, where c is an arbitrary constant; $y = x^2$ is a singular solution, as is easily verified.

ORDINARY DIFFERENTIAL EQUATIONS OF THE FIRST ORDER

Equations with Separable Variables Every differential equation of the first order and of the first degree can be written in the form $M(x, y) dx + N(x, y) dy = 0$. If the equation can be transformed so that M does not involve y and N does not involve x , then the variables are said to be separated. The solution can then be obtained by **quadrature**, which means that $y = \int f(x) dx + c$, which may or may not be expressible in simpler form.

Example Two liquids A and B are boiling together in a vessel. Experimentally it is found that the ratio of the rates at which A and B are evaporating at any time is proportional to the ratio of the amount of A (say, x) to the amount of B (say, y) still in the liquid state. This physical law is expressible as $(dy/dt)/(dx/dt) = ky/x$ or $dy/dx = ky/x$, where k is a proportionality constant. This equation may be written $dy/y = k(dx/x)$, in which the variables are separated. The solution is $\ln y = k \ln x + \ln c$ or $y = cx^k$.

Exact Equations The equation $M(x, y) dx + N(x, y) dy = 0$ is exact if and only if $\partial M/\partial y = \partial N/\partial x$. In this case there exists a function $w = f(x, y)$ such that $\partial f/\partial x = M$, $\partial f/\partial y = N$, and $f(x, y) = C$ is the required solution. $f(x, y)$ is found as follows: treat y as though it were constant and evaluate $\int M(x, y) dx$. Then treat x as though it were constant and evaluate $\int N(x, y) dy$. The sum of all unlike terms in these two integrals (including no repetitions) is $f(x, y)$.

Example $(2xy - \cos x) dx + (x^2 - 1) dy = 0$ is exact for $\partial M/\partial y = 2x$, $\partial N/\partial x = 2x$. $\int M dx = \int (2xy - \cos x) dx = x^2y - \sin x$, $\int N dy = \int (x^2 - 1) dy = x^2y - y$. The solution is $x^2y - \sin x - y = C$, as may easily be verified.

Linear Equations A differential equation is said to be linear when it is of first degree in the dependent variable and its derivatives. The general linear first-order differential equation has the form $dy/dx + P(x)y = Q(x)$. Its general solution is

$$y = e^{-\int P dx} \left[\int Q e^{\int P dx} dx + C \right]$$

Example A tank initially holds 200 gal of a salt solution in which 100 lb is dissolved. Six gallons of brine containing 4 lb of salt run into the tank per minute. If mixing is perfect and the output rate is 4 gal/min, what is the amount A of salt in the tank at time t ? The differential equation of A is $dA/dt = [2/(100 + t)]A = 4$. Its general solution is $A = (4/3)(100 + t) + C/(100 + t)^2$. At $t = 0$, $A = 100$; so the particular solution is $A = (4/3)(100 + t) - (1/3)10^6/(100 + t)^2$.

ORDINARY DIFFERENTIAL EQUATIONS OF HIGHER ORDER

The higher-order differential equations, especially those of order 2, are of great importance because of physical situations describable by them.

Equation $y^{(n)} = f(x)$ Such a differential equation can be solved by n integrations. The solution will contain n arbitrary constants.

Linear Differential Equations with Constant Coefficients and Right-Hand Member Zero (Homogeneous) The solution of $y'' + ay' + by = 0$ depends upon the nature of the roots of the characteristic equation $m^2 + am + b = 0$ obtained by substituting the trial solution $y = e^{mx}$ in the equation.

Distinct Real Roots If the roots of the characteristic equation are distinct real roots, r_1 and r_2 , say, the solution is $y = Ae^{r_1x} + Be^{r_2x}$, where A and B are arbitrary constants.

Example $y'' + 4y' + 3 = 0$. The characteristic equation is $m^2 + 4m + 3 = 0$. The roots are -3 and -1 , and the general solution is $y = Ae^{-3x} + Be^{-x}$.

Multiple Real Roots If $r_1 = r_2$, the solution of the differential equation is $y = e^{r_1x}(A + Bx)$.

Example $y'' + 4y' + 4 = 0$. The characteristic equation is $m^2 + 4m + 4 = 0$ with roots -2 and -2 . The solution is $y = e^{-2x}(A + Bx)$.

Complex Roots If the characteristic roots are $p \pm iq$, then the solution is $y = e^{px}(A \cos qx + B \sin qx)$.

Example The differential equation $My'' + Ay' + ky = 0$ represents the vibration of a linear system of mass M , spring constant k , and damping constant A . If $A < 2\sqrt{kM}$, the roots of the characteristic equation

$$Mm^2 + Am + k = 0 \text{ are complex } -\frac{A}{2M} \pm i \sqrt{\frac{k}{M} - \left(\frac{A}{2M}\right)^2}$$

and the solution is

$$y = e^{-(A/2M)t}$$

$$\left\{ c_1 \cos \left(\sqrt{\frac{k}{M} - \left(\frac{A}{2M}\right)^2} t \right) + ic_2 \sin \left(\sqrt{\frac{k}{M} - \left(\frac{A}{2M}\right)^2} t \right) \right\}$$

This solution is oscillatory, representing undercritical damping.

All these results generalize to homogeneous linear differential equations with constant coefficients of order higher than 2. These equations (especially of order 2) have been much used because of the ease of solution. Oscillations, electric circuits, diffusion processes, and heat-flow problems are a few examples for which such equations are useful.

Second-Order Equations: Dependent Variable Missing Such an equation is of the form

$$F \left(x, \frac{dy}{dx}, \frac{d^2y}{dx^2} \right) = 0$$

It can be reduced to a first-order equation by substituting $p = dy/dx$ and $dp/dx = d^2y/dx^2$.

Second-Order Equations: Independent Variable Missing Such an equation is of the form

$$F \left(y, \frac{dy}{dx}, \frac{d^2y}{dx^2} \right) = 0$$

Set

$$\frac{dy}{dx} = p, \quad \frac{d^2y}{dx^2} = p \frac{dp}{dy}$$

The result is a first-order equation in p ,

$$F \left(y, p, p \frac{dp}{dy} \right) = 0$$

Example The capillary curve for one vertical plate is given by

$$\frac{d^2y}{dx^2} = \frac{4y}{c^2} \left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2}$$

Its solution by this technique is

$$x + \sqrt{c^2 - y^2} - \sqrt{c^2 - h_0^2} = \frac{c}{2} \left(\cosh^{-1} \frac{c}{y} - \cosh^{-1} \frac{c}{h_0} \right)$$

where c, h_0 are physical constants.

Example The equation governing chemical reaction in a porous catalyst in plane geometry of thickness L is

$$D \frac{d^2c}{dx^2} = k f(c), \quad \frac{dc}{dx}(0) = 0, \quad c(L) = c_0$$

where D is a diffusion coefficient, k is a reaction rate parameter, c is the concentration, $k f(c)$ is the rate of reaction, and c_0 is the concentration at the

*The superscript (n) means n derivatives.

boundary. Making the substitution $p = \frac{dc}{ds}$ gives (Finlayson, 1980, p. 92)

$$p \frac{dp}{dc} = \frac{k}{D} f(c)$$

Integrating gives

$$\frac{p^2}{2} = \frac{k}{D} \int_{c(0)}^c f(c) dc$$

If the reaction is very fast, $c(0) = 0$ and the average reaction rate is related to $p(L)$. This variable is given by

$$p(L) = \left[\frac{2k}{D} \int_0^{c_0} f(c) dc \right]^{1/2}$$

Thus, the average reaction rate can be calculated without solving the complete problem.

Linear Nonhomogeneous Differential Equations

Linear Differential Equations Right-Hand Member $f(x) \neq 0$ Again the specific remarks for $y'' + ay' + by = f(x)$ apply to differential equations of similar type but higher order. We shall discuss two general methods.

Method of Undetermined Coefficients Use of this method is limited to equations exhibiting both constant coefficients and particular forms of the function $f(x)$. In most cases $f(x)$ will be a sum or product of functions of the type constant, x^n (n a positive integer), e^{mx} , $\cos kx$, $\sin kx$. When this is the case, the solution of the equation is $y = H(x) + P(x)$, where $H(x)$ is a solution of the homogeneous equations found by the method of the preceding subsection and $P(x)$ is a particular integral found by using the following table subject to these conditions: (1) When $f(x)$ consists of the sum of several terms, the appropriate form of $P(x)$ is the sum of the particular integrals corresponding to these terms individually. (2) When a term in any of the trial integrals listed is already a part of the homogeneous solution, the indicated form of the particular integral is multiplied by x .

Form of Particular Integral

If $f(x)$ is	Then $P(x)$ is
a (constant)	A (constant)
ax^n	$A_n x^n + A_{n-1} x^{n-1} + \cdots + A_1 x + A_0$
ae^{rx}	Be^{rx}
$\left. \begin{matrix} c \cos kx \\ d \sin kx \end{matrix} \right\}$	$A \cos kx + B \sin kx$
$\left. \begin{matrix} g e^{rx} \cos kx \\ h x^n e^{rx} \sin kx \end{matrix} \right\}$	$(A_n x^n + \cdots + A_0) e^{rx} \cos kx + (B_n x^n + \cdots + B_0) e^{rx} \sin kx$

Since the form of the particular integral is known, the constants may be evaluated by substitution in the differential equation.

Example $y'' + 2y' + y = 3e^{2x} - \cos x + x^3$. The characteristic equation is $(m+1)^2 = 0$ so that the homogeneous solution is $y = (c_1 + c_2 x)e^{-x}$. To find a particular solution we use the trial solution from the table, $y = a_1 e^{2x} + a_2 \cos x + a_3 \sin x + a_4 x^3 + a_5 x^2 + a_6 x + a_7$. Substituting this in the differential equation collecting and equating like terms, there results $a_1 = 1/3$, $a_2 = 0$, $a_3 = -1/2$, $a_4 = 1$, $a_5 = -6$, $a_6 = 18$, and $a_7 = -24$. The solution is $y = (c_1 + c_2 x)e^{-x} + 1/3 e^{2x} - 1/2 \sin x + x^3 - 6x^2 + 18x - 24$.

Method of Variation of Parameters This method is applicable to any linear equation. The technique is developed for a second-order equation but immediately extends to higher order. Let the equation be $y'' + a(x)y' + b(x)y = R(x)$ and let the solution of the homogeneous equation, found by some method, be $y = c_1 f_1(x) + c_2 f_2(x)$. It is now assumed that a particular integral of the differential equation is of the form $P(x) = u f_1 + v f_2$ where u, v are functions of x to be determined by two equations. One equation results from the requirement that $u f_1 + v f_2$ satisfy the differential equation, and the other is a degree of freedom open to the analyst. The best choice proves to be

$$u' f_1 + v' f_2 = 0 \quad \text{and} \quad u' f_1' + v' f_2' = R(x)$$

$$\text{Then} \quad u' = \frac{du}{dx} = -\frac{f_2}{f_1 f_2' - f_2' f_1} R(x)$$

$$v' = \frac{dv}{dx} = \frac{f_1}{f_1 f_2' - f_2' f_1} R(x)$$

and since f_1, f_2 , and R are known u, v may be found by direct integration.

Perturbation Methods If the ordinary differential equation has a parameter that is small and is not multiplying the highest derivative, perturbation methods can give solutions for small values of the parameter.

Example Consider the differential equation for reaction and diffusion in a catalyst; the reaction is second order: $c'' = ac^2$, $c'(0) = 0$, $c(1) = 1$. The solution is expanded in the following Taylor series in a .

$$c(x, a) = c_0(x) + ac_1(x) + a^2 c_2(x) + \cdots$$

The goal is to find equations governing the functions $\{c_i(x)\}$ and solve them. Substitution into the equations gives the following equations:

$$c_0''(x) + a c_1'(x) + a^2 c_2''(x) + \cdots = a[c_0(x) + ac_1(x) + a^2 c_2(x) + \cdots]^2$$

$$c_0'(0) + ac_1'(0) + a^2 c_2'(0) + \cdots = 0$$

$$c_0(1) + ac_1(1) + a^2 c_2(1) + \cdots = 1$$

Like terms in powers of a are collected to form the individual problems.

$$c_0'' = 0, \quad c_0'(0) = 0, \quad c_0(1) = 1$$

$$c_1'' = c_0^2, \quad c_1'(0) = 0, \quad c_1(1) = 0$$

$$c_2'' = 2c_0 c_1, \quad c_2'(0) = 0, \quad c_2(1) = 0$$

The solution proceeds in turn.

$$c_0(x) = 1, \quad c_1(x) = \frac{(x^2 - 1)}{2}, \quad c_2(x) = \frac{5 - 6x^2 + x^4}{12}$$

SPECIAL DIFFERENTIAL EQUATIONS [SEE ABRAMOWITZ AND STEGUN (1972)]

Euler's Equation The linear equation $x^n y^{(n)} + a_1 x^{n-1} y^{(n-1)} + \cdots + a_{n-1} x y' + a_n y = R(x)$ can be reduced to a linear equation with constant coefficients by the change of variable $x = e^r$. To solve the homogeneous equation substitute $y = x^r$ into it, cancel the powers of x , which are the same for all terms, and solve the resulting polynomial for r . In case of multiple or complex roots there results the form $y = x^r (\log x)^r$ and $y = x^{i\alpha} [\cos(\beta \log x) + i \sin(\beta \log x)]$.

Bessel's Equation The linear equation $x^2(d^2 y/dx^2) + (1 - 2\alpha)x(dy/dx) + [\beta^2 \gamma^2 x^{2\gamma} + (\alpha^2 - p^2 \gamma^2)]y = 0$ is the general Bessel equation. By series methods, not to be discussed here, this equation can be shown to have the solution

$$y = A x^\alpha J_p(\beta x^\gamma) + B x^\alpha J_{-p}(\beta x^\gamma) \quad p \text{ not an integer or zero}$$

$$y = A x^\alpha J_p(\beta x^\gamma) + B x^\alpha Y_p(\beta x^\gamma) \quad p \text{ an integer}$$

$$\text{where} \quad J_p(x) = \left(\frac{x}{2}\right)^p \sum_{k=0}^{\infty} \frac{(-1)^k (x/2)^{2k}}{k! \Gamma(p+k+1)}$$

$$J_{-p}(x) = \left(\frac{x}{2}\right)^{-p} \sum_{k=0}^{\infty} \frac{(-1)^k (x/2)^{2k}}{k! \Gamma(k+1-p)} \quad p \text{ not an integer}$$

$$\Gamma(n) = \int_0^{\infty} x^{n-1} e^{-x} dx \quad n > 0$$

is the gamma function. For p an integer

$$J_p(x) = \left(\frac{x}{2}\right)^p \sum_{k=0}^{\infty} \frac{(-1)^k (x/2)^{2k}}{k! (p+k)!}$$

(Bessel function of the first kind of order p)

$$Y_p(x) = \frac{[J_p(x) \cos(p\pi) - J_{-p}(x)]}{\sin(p\pi)}$$

(replace right-hand side by limiting value if p is an integer or zero).

The series converge for all x . Much of the importance of Bessel's equation and Bessel functions lies in the fact that the solutions of numerous linear differential equations can be expressed in terms of them.

Legendre's Equation The Legendre equation $(1 - x^2)y'' - 2xy' + n(n+1)y = 0$, $n \geq 0$, has the solution P_n for n an integer.

The polynomials P_n are the so-called Legendre polynomials, $P_0(x) = 1$, $P_1(x) = x$, $P_2(x) = \frac{1}{2}(3x^2 - 1)$, $P_3(x) = \frac{1}{2}(5x^3 - 3x)$, $\dots$

For n positive and not an integer, see Abramowitz and Stegun (1972).

Laguerre's Equation The Laguerre equation $x(d^2y/dx^2) + (c - x)(dy/dx) - ay = 0$ is satisfied by the confluent hypergeometric function. See Abramowitz and Stegun (1972) and Kreszig (1999).

Hermite's Equation The Hermite equation $y'' - 2xy' + 2ny = 0$ is satisfied by the Hermite polynomial of degree n , $y = AH_n(x)$ if n is a positive integer or zero. $H_0(x) = 1$, $H_1(x) = 2x$, $H_2(x) = 4x^2 - 2$, $H_3(x) = 8x^3 - 12x$, $H_4(x) = 16x^4 - 48x^2 + 12$, $H_{r+1}(x) = 2xH_r(x) - 2rH_{r-1}(x)$.

Chebyshev's Equation The equation $(1 - x^2)y'' - xy' + n^2y = 0$ for n a positive integer or zero is satisfied by the n th Chebyshev polynomial $y = AT_n(x)$. $T_0(x) = 1$, $T_1(x) = x$, $T_2(x) = 2x^2 - 1$, $T_3(x) = 4x^3 - 3x$, $T_4(x) = 8x^4 - 8x^2 + 1$; $T_{r+1}(x) = 2xT_r(x) - T_{r-1}(x)$.

PARTIAL DIFFERENTIAL EQUATIONS

The analysis of situations involving two or more independent variables frequently results in a partial differential equation.

Example The equation $\partial T/\partial t = K(\partial^2 T/\partial x^2)$ represents the unsteady one-dimensional conduction of heat.

Example The equation for the unsteady transverse motion of a uniform beam clamped at the ends is

$$\frac{\partial^4 y}{\partial x^4} + \frac{\rho}{EI} \frac{\partial^2 y}{\partial t^2} = 0$$

Example The expansion of a gas behind a piston is characterized by the simultaneous equations

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + \frac{c^2}{\rho} \frac{\partial \rho}{\partial x} = 0 \quad \text{and} \quad \frac{\partial \rho}{\partial t} + u \frac{\partial \rho}{\partial x} + \rho \frac{\partial u}{\partial x} = 0$$

The partial differential equation $\partial^2 f/\partial x \partial y = 0$ can be solved by two integrations yielding the solution $f = g(x) + h(y)$, where $g(x)$ and $h(y)$ are arbitrary differentiable functions. This result is an example of the fact that the general solution of partial differential equations involves arbitrary functions in contrast to the solution of ordinary differential equations, which involve only arbitrary constants. A number of methods are available for finding the general solution of a partial differential equation. In most applications of partial differential equations the general solution is of limited use. In such applications the solution of a partial differential equation must satisfy both the equation and certain auxiliary conditions called **initial** and/or **boundary** conditions, which are dictated by the problem. Examples of these include those in which the wall temperature is a fixed constant $T(x_0) = T_0$, there is no diffusion across a nonpermeable wall, and the like. In ordinary differential equations these auxiliary conditions allow definite numbers to be assigned to the constants of integration.

Partial Differential Equations of Second and Higher Order Many of the applications to scientific problems fall naturally into partial differential equations of second order, although there are important exceptions in elasticity, vibration theory, and elsewhere.

A second-order differential equation can be written as

$$a \frac{\partial^2 u}{\partial x^2} + b \frac{\partial^2 u}{\partial x \partial y} + c \frac{\partial^2 u}{\partial y^2} = f$$

where a , b , c , and f depend upon x , y , u , $\partial u/\partial x$, and $\partial u/\partial y$. This equation is hyperbolic, parabolic, or elliptic, depending on whether the discriminant $b^2 - 4ac$ is >0 , $=0$, or <0 , respectively. Since a , b , c , and f depend on the solution, the type of equation can be different at different x and y locations. If the equation is hyperbolic, discontinuities can be propagated. See Courant and Hilbert (1953, 1962) and LeVeque, R. J., *Numerical Methods for Conservation Laws*, Birkhäuser, Basel (1992).

Phenomena of **propagation** such as vibrations are characterized by equations of "hyperbolic" type which are essentially different in their

properties from other classes such as those which describe equilibrium (elliptic) or unsteady diffusion and heat transfer (parabolic). Prototypes are as follows:

Elliptic Laplace's equation $\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2 = 0$ and Poisson's equation $\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2 = g(x, y)$ do not contain the variable time explicitly and consequently represent equilibrium configurations. Laplace's equation is satisfied by static electric or magnetic potential at points free from electric charges or magnetic poles. Other important functions satisfying Laplace's equation are the velocity potential of the irrotational motion of an incompressible fluid, used in hydrodynamics; the steady temperature at points in a homogeneous solid, and the steady state of diffusion through a homogeneous body.

Parabolic The heat equation $\partial T/\partial t = \partial^2 T/\partial x^2 + \partial^2 T/\partial y^2$ represents nonequilibrium or unsteady states of heat conduction and diffusion.

Hyperbolic The wave equation $\partial^2 u/\partial t^2 = c^2(\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2)$ represents wave propagation of many varied types.

Quasilinear first-order differential equations are like

$$a \frac{\partial u}{\partial x} + b \frac{\partial u}{\partial y} = f$$

where a , b , and f depend on x , y , and u , with $a^2 + b^2 \neq 0$. This equation can be solved using the method of characteristics, which writes the solution in terms of a parameter s , which defines a path for the characteristic.

$$\frac{dx}{ds} = a, \quad \frac{dy}{ds} = b, \quad \frac{du}{ds} = f$$

These equations are integrated from some initial conditions. For a specified value of s , the value of x and y shows the location where the solution is u . The equation is semilinear if a and b depend just on x and y (and not u), and the equation is linear if a , b , and f all depend on x and y , but not u . Such equations give rise to shock propagation, and conditions have been derived to deduce the presence of shocks. Courant and Hilbert (1953, 1962); Rhee, H. K., R. Aris, and N. R. Amundson, *First-Order Partial Differential Equations*, vol. I, *Theory and Applications of Single Equations*, Prentice-Hall, Englewood Cliffs, N.J. (1986); and LeVeque (1992), *ibid*.

An example of a linear hyperbolic equation is the advection equation for flow of contaminants when the x and y velocity components are u and v , respectively.

$$\frac{\partial c}{\partial t} + u \frac{\partial c}{\partial x} + v \frac{\partial c}{\partial y} = 0$$

The equations for flow and adsorption in a packed bed or chromatography column give a quasilinear equation.

$$\phi \frac{\partial c}{\partial t} + \phi u \frac{\partial c}{\partial x} + (1 - \phi) \frac{df}{dc} \frac{\partial c}{\partial t} = 0$$

Here $n = f(c)$ is the relation between concentration on the adsorbent and fluid concentration.

The solution of problems involving partial differential equations often revolves about an attempt to reduce the partial differential equation to one or more ordinary differential equations. The solutions of the ordinary differential equations are then combined (if possible) so that the boundary conditions as well as the original partial differential equation are simultaneously satisfied. Three of these techniques are illustrated.

Similarity Variables The physical meaning of the term "similarity" relates to internal similitude, or self-similitude. Thus, similar solutions in boundary-layer flow over a horizontal flat plate are those for which the horizontal component of velocity u has the property that two velocity profiles located at different coordinates x differ only by a scale factor. The mathematical interpretation of the term similarity is a transformation of variables carried out so that a reduction in the number of independent variables is achieved. There are essentially two methods for finding similarity variables, "separation of variables" (not the classical concept) and the use of "continuous transformation groups." The basic theory is available in Ames (1965).

Example The equation $\partial\theta/\partial x = (A/y)(\partial^2\theta/\partial y^2)$ with the boundary conditions $\theta = 0$ at $x = 0$, $y > 0$; $\theta = 0$ at $y = \infty$, $x > 0$; $\theta = 1$ at $y = 0$, $x > 0$ represents the nondimensional temperature θ of a fluid moving past an infinitely wide flat plate immersed in the fluid. Turbulent transfer is neglected, as is molecular transport except in the y direction. It is now assumed that the equation and the boundary conditions can be satisfied by a solution of the form $\theta = f(y/x^n) = f(u)$, where $\theta = 0$ at $u = \infty$ and $\theta = 1$ at $u = 0$. The purpose here is to replace the independent variables x and y by the single variable u when it is hoped that a value of n exists which will allow x and y to be completely eliminated in the equation. In this case since $u = y/x^n$, there results after some calculation $\partial\theta/\partial x = -(nu/x)(d\theta/du)$, $\partial^2\theta/\partial y^2 = (1/x^{2n})(d^2\theta/du^2)$, and when these are substituted in the equation, $-(1/x)nu(d\theta/du) = (1/x^{2n})(A/u)(d^2\theta/du^2)$. For this to be a function of u only, choose $n = 1/2$. There results $(d^2\theta/du^2) + (u^2/3A)(d\theta/du) = 0$. Two integrations and use of the boundary conditions for this ordinary differential equation give the solution

$$\theta = \int_u^\infty \exp(-u^2/9A) du / \int_0^\infty \exp(-u^2/9A) du$$

Group Method The type of transformation can be deduced using group theory. For a complete exposition, see Ames (1965) and Hill, J. M., *Differential Equations and Group Methods for Scientists and Engineers*, CRC Press, New York (1992); a shortened version is in Finlayson (1980, 2003). Basically, a similarity transformation should be considered when one of the independent variables has no physical scale (perhaps it goes to infinity). The boundary conditions must also simplify (and combine) since each transformation leads to a differential equation with one fewer independent variable.

Example A similarity variable is found for the problem

$$\frac{\partial c}{\partial t} = \frac{\partial}{\partial x} \left(\frac{D(c)}{D} \frac{\partial c}{\partial x} \right), \quad c(0, t) = 1, \quad c(\infty, t) = 0, \quad c(x, 0) = 0$$

Note that the length dimension goes to infinity, so that there is no length scale in the problem statement; this is a clue to try a similarity transformation. The transformation examined here is

$$\bar{t} = a^\alpha t, \quad \bar{x} = a^\beta x, \quad \bar{c} = a^\gamma c$$

With this substitution, the equation becomes

$$a^{\alpha-\gamma} \frac{\partial \bar{c}}{\partial \bar{t}} = a^{2\beta-\gamma} \frac{\partial}{\partial \bar{x}} \left[D(a^{-\gamma} \bar{c}) \frac{\partial \bar{c}}{\partial \bar{x}} \right]$$

Group theory says a system is conformally invariant if it has the same form in the new variables; here, that is

$$\gamma = 0, \quad \alpha - \gamma = 2\beta - \gamma, \quad \text{or } \alpha = 2\beta$$

The invariants are

$$\eta = \frac{x}{t^\beta}, \quad \delta = \frac{\beta}{\alpha}$$

and the solution is

$$c(x, t) = f(\eta)t^{\gamma/\alpha}$$

We can take $\gamma = 0$ and $\delta = \beta/\alpha = 1/2$. Note that the boundary conditions combine because the point $x = \infty$ and $t = 0$ give the same value of η and the conditions on c at $x = \infty$ and $t = 0$ are the same. We thus make the transformation

$$\eta = \frac{x}{\sqrt{4D_0 t}}, \quad c(x, t) = f(\eta)$$

The use of the 4 and D_0 makes the analysis below simpler. The result is

$$\frac{d}{d\eta} \left[\frac{D(c)}{D_0} \frac{df}{d\eta} \right] + 2\eta \frac{df}{d\eta} = 0, \quad f(0) = 1, \quad f(\infty) = 0$$

Thus, we solve a two-point boundary value problem instead of a partial differential equation. When the diffusivity is constant, the solution is the error function, a tabulated function.

$$c(x, t) = 1 - \operatorname{erf} \eta = \operatorname{erfc} \eta$$

$$\operatorname{erf} \eta = \int_0^\eta e^{-\xi^2} d\xi / \int_0^\infty e^{-\xi^2} d\xi$$

Separation of Variables This is a powerful, well-utilized method which is applicable in certain circumstances. It consists of assuming that the solution for a partial differential equation has the form $U = f(x)g(y)$. If it is then possible to obtain an ordinary differential equation on one side of the equation depending only on x and on the other side only on y , the partial differential equation is said to be

separable in the variables x, y . If this is the case, one side of the equation is a function of x alone and the other of y alone. The two can be equal only if each is a constant, say λ . Thus the problem has again been reduced to the solution of ordinary differential equations.

Example Laplace's equation $\partial^2 V/\partial x^2 + \partial^2 V/\partial y^2 = 0$ plus the boundary conditions $V(0, y) = 0$, $V(l, y) = 0$, $V(x, \infty) = 0$, $V(x, 0) = f(x)$ represents the steady-state potential in a thin plate (in z direction) of infinite extent in the y direction and of width l in the x direction. A potential $f(x)$ is impressed (at $y = 0$) from $x = 0$ to $x = l$, and the sides are grounded. To obtain a solution of this boundary-value problem assume $V(x, y) = f(x)g(y)$. Substitution in the differential equation yields $f''(x)g(y) + f(x)g''(y) = 0$, or $g''(y)/g(y) = -f''(x)/f(x) = \lambda^2$ (say). This system becomes $g''(y) - \lambda^2 g(y) = 0$ and $f''(x) + \lambda^2 f(x) = 0$. The solutions of these ordinary differential equations are respectively $g(y) = Ae^{\lambda y} + Be^{-\lambda y}$, $f(x) = C \sin \lambda x + D \cos \lambda x$. Then $f(x)g(y) = (Ae^{\lambda y} + Be^{-\lambda y})(C \sin \lambda x + D \cos \lambda x)$. Now $V(0, y) = 0$ so that $f(0)g(y) = (Ae^{\lambda y} + Be^{-\lambda y})D = 0$ for all y . Hence $D = 0$. The solution then has the form $\sin \lambda x (Ae^{\lambda y} + Be^{-\lambda y})$ where the multiplicative constant C has been eliminated. Since $V(l, y) = 0$, $\sin \lambda l (Ae^{\lambda y} + Be^{-\lambda y}) = 0$. Clearly the bracketed function of y is not zero, for the solution would then be the identically zero solution. Hence $\sin \lambda l = 0$ or $\lambda_n = n\pi/l$, $n = 1, 2, \dots$ where $\lambda_n = n\pi/l$ is the n th eigenvalue.

The solution now has the form $\sin(n\pi x/l)(Ae^{n\pi y/l} + Be^{-n\pi y/l})$. Since $V(x, \infty) = 0$, A must be taken to be zero because e^y becomes arbitrarily large as $y \rightarrow \infty$. The solution then reads $B_n \sin(n\pi x/l)e^{-n\pi y/l}$, where B_n is the multiplicative constant. The differential equation is linear and homogeneous so that $\sum_{n=1}^\infty B_n e^{-n\pi y/l} \sin(n\pi x/l)$ is also a solution. Satisfaction of the last boundary condition is ensured by taking

$$B_n = \frac{2}{l} \int_0^l f(x) \sin(n\pi x/l) dx = \text{Fourier sine coefficients of } f(x)$$

Further, convergence and differentiability of this series are established quite easily. Thus the solution is

$$V(x, y) = \sum_{n=1}^\infty B_n e^{-n\pi y/l} \sin \frac{n\pi x}{l}$$

Example The diffusion problem in a slab of thickness L

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2}, \quad c(0, t) = 1, \quad c(L, t) = 0, \quad c(x, 0) = 0$$

can be solved by separation of variables. First transform the problem so that the boundary conditions are homogeneous (having zeros on the right-hand side). Let

$$c(x, t) = 1 - \frac{x}{L} + u(x, t)$$

Then $u(x, t)$ satisfies

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2}, \quad u(x, 0) = \frac{x}{L} - 1, \quad u(0, t) = 0, \quad u(L, t) = 0$$

Assume a solution of the form $u(x, t) = X(x)T(t)$, which gives

$$\frac{1}{DT} \frac{dT}{dt} = \frac{1}{X} \frac{d^2 X}{dx^2}$$

Since both sides are constant, this gives the following ordinary differential equations to solve.

$$\frac{1}{DT} \frac{dT}{dt} = -\lambda, \quad \frac{1}{X} \frac{d^2 X}{dx^2} = -\lambda$$

The solution of these is

$$T = A e^{-\lambda D t}, \quad X = B \cos \sqrt{\lambda} x + E \sin \sqrt{\lambda} x$$

The combined solution for $u(x, t)$ is

$$u = A (B \cos \sqrt{\lambda} x + E \sin \sqrt{\lambda} x) e^{-\lambda D t}$$

Apply the boundary condition that $u(0, t) = 0$ to give $B = 0$. Then the solution is

$$u = A (\sin \sqrt{\lambda} x) e^{-\lambda D t}$$

where the multiplicative constant E has been eliminated. Apply the boundary condition at $x = L$.

$$0 = A (\sin \sqrt{\lambda} L) e^{-\lambda D t}$$

This can be satisfied by choosing $A = 0$, which gives no solution. However, it can also be satisfied by choosing λ such that

$$\sin \sqrt{\lambda} L = 0, \quad \sqrt{\lambda} L = n\pi$$

Thus

$$\lambda = \frac{n^2 \pi^2}{L^2}$$

The combined solution can now be written as

$$u = A \left(\frac{\sin n\pi x}{L} \right) e^{-n^2\pi^2 D t / L^2}$$

Since the initial condition must be satisfied, we use an infinite series of these functions.

$$u = \sum_{n=1}^{\infty} A_n \left(\frac{\sin n\pi x}{L} \right) e^{-n^2\pi^2 D t / L^2}$$

At $t = 0$, we satisfy the initial condition.

$$\frac{x}{L} - 1 = \sum_{n=1}^{\infty} A_n \left(\frac{\sin n\pi x}{L} \right)$$

This is done by multiplying the equation by

$$\frac{\sin m\pi x}{L}$$

and integrating over x : $0 \rightarrow L$. (This is the same as minimizing the mean-square error of the initial condition.) This gives

$$\frac{A_m L}{2} = \int_0^L \left(\frac{x}{L} - 1 \right) \sin \frac{m\pi x}{L} dx$$

which completes the solution.

Integral-Transform Method A number of integral transforms are used in the solution of differential equations. Only one, the Laplace transform, will be discussed here [for others, see “Integral Transforms (Operational Methods)”. The one-sided Laplace transform indicated by $L[f(t)]$ is defined by the equation $L[f(t)] = \int_0^\infty f(t)e^{-st} dt$. It has numerous important properties. The ones of interest here are $L[f'(t)] = sL[f(t)] - f(0)$; $L[f''(t)] = s^2L[f(t)] - sf(0) - f'(0)$; $L[f^{(n)}(t)] = s^nL[f(t)] - s^{n-1}f(0) - s^{n-2}f'(0) - \dots - f^{(n-1)}(0)$ for ordinary derivatives. For partial derivatives an indication of which variable is being transformed avoids confusion. Thus, if

$$y = y(x, t), \quad L_t \left[\frac{\partial y}{\partial t} \right] = sL[y(x, t)] - y(x, 0)$$

REFERENCES: Elaydi, Saber, and S. N. Elaydi, *An Introduction to Difference Equations*, Springer-Verlag, New York (1999); Fulford, G., P. Forrester, and A. Jones, *Modelling with Differential and Difference Equations*, Cambridge University Press, New York (1997); Goldberg, S., *Introduction to Difference Equations*, Dover (1986); Kelley, W. G., and A. C. Peterson, *Difference Equations: An Introduction with Applications*, 2d ed., Harcourt/Academic Press, San Diego (2001).

Certain situations are such that the independent variable does not vary continuously but has meaning only for discrete values. Typical illustrations occur in the stagewise processes found in chemical engineering such as distillation, staged extraction systems, and absorption columns. In each of these the operation is characterized by a finite between-stage change of the dependent variable in which the independent variable is the integral number of the stage. The importance of difference equations is twofold: (1) to analyze problems of the type described and (2) to obtain approximate solutions of problems which lead, in their formulation, to differential equations. In this subsection only problems of analysis are considered; the application to approximate solutions is considered under “Numerical Analysis and Approximate Methods.”

ELEMENTS OF THE CALCULUS OF FINITE DIFFERENCES

Let $y = f(x)$ be defined for discrete equidistant values of x , which will be denoted by x_n . The corresponding value of y will be written $y_n = f(x_n)$. The first forward difference of $f(x)$ denoted by $\Delta f(x) = f(x+h) - f(x)$ where $h = x_n - x_{n-1}$ = interval length.

Example Let $f(x) = x^2$. Then $\Delta f(x) = (x+h)^2 - x^2 = 2hx + h^2$.

whereas

$$L_t \left[\frac{\partial y}{\partial x} \right] = \frac{dL_x[y(x, t)]}{dx}$$

since $L[y(x, t)]$ is “really” only a function of x . Otherwise the results are similar. These facts coupled with the linearity of the transform, i.e., $L[af(t) + bg(t)] = aL[f(t)] + bL[g(t)]$, make it a useful device in solving some linear differential equations. Its use reduces the solution of ordinary differential equations to the solution of algebraic equations for $L[y]$. The inverse transform must be obtained either from tables or by use of complex inversion methods.

Example The equation $\partial c / \partial t = D(\partial^2 c / \partial x^2)$ represents the diffusion in a semi-infinite medium, $x \geq 0$. Under the boundary conditions $c(0, t) = c_0$, $c(x, 0) = 0$ find a solution of the diffusion equation. By taking the Laplace transform of both sides with respect to t ,

$$\int_0^\infty e^{-st} \frac{\partial^2 c}{\partial x^2} dt = \frac{1}{D} \int_0^\infty e^{-st} \frac{\partial c}{\partial t} dt$$

$$\text{or} \quad \frac{d^2 F}{dx^2} = (1/D)sF - c(x, 0) = \frac{sF}{D}$$

where $F(x, s) = L_t[c(x, t)]$. Hence

$$\frac{d^2 F}{dx^2} - \left(\frac{s}{D} \right) F = 0$$

The other boundary condition transforms into $F(0, s) = c_0/s$. Finally the solution of the ordinary differential equation for F subject to $F(0, s) = c_0/s$ and F remains finite as $x \rightarrow \infty$ is $F(x, s) = (c_0/s)e^{-\sqrt{s/D}x}$. Reference to a table shows that the function having this as its Laplace transform is

$$c(x, t) = c_0 \left[1 - \frac{2}{\sqrt{\pi}} \int_0^{\sqrt{s/2D}x} e^{-u^2} du \right] = c_0 \operatorname{erfc} \left(\frac{x}{\sqrt{4Dt}} \right)$$

Matched-Asymptotic Expansions Sometimes the coefficient in front of the highest derivative is a small number. Special perturbation techniques can then be used, provided the proper scaling laws are found. See Kevorkian, J., and J. D. Cole, *Perturbation Methods in Applied Mathematics*, Springer-Verlag, New York (1981); and Lagerstrom, P. A., *Matched Asymptotic Expansions: Ideas and Techniques*, Springer-Verlag, New York (1988).

DIFFERENCE EQUATIONS

The second forward difference is obtained by taking the difference of the first; thus $\Delta^2 f(x) = \Delta^2 f(x) = \Delta f(x+h) - \Delta f(x) = f(x+2h) - 2f(x+h) + f(x)$.

Example $f(x) = x^2$, $\Delta^2 f(x) = \Delta[\Delta f(x)] = \Delta 2hx + \Delta h^2 = 2h(x+h) - 2hx + h^2 = h^2$.

Similarly the n th forward difference is defined by the relation $\Delta^n f(x) = \Delta[\Delta^{n-1} f(x)]$. Other difference relations are also quite useful. Some of these are $\nabla f(x) = f(x) - f(x-h)$, which is called the backward difference, and $\delta f(x) = f[x+(h/2)] - f[x-(h/2)]$, called the central difference. Some properties of the operator Δ are quite important. If C is any constant, $\Delta C = 0$; if $f(x)$ is any function of period h , $\Delta f(x) = 0$ (in fact, periodic functions of period h play the same role here as constants do in the differential calculus); $\Delta[f(x) + g(x)] = \Delta f(x) + \Delta g(x)$; $\Delta^m[\Delta^n f(x)] = \Delta^{m+n} f(x)$; $\Delta[f(x)g(x)] = f(x)\Delta g(x) + g(x+h)\Delta f(x)$

$$\Delta \left[\frac{f(x)}{g(x)} \right] = \frac{g(x)\Delta f(x) - f(x)\Delta g(x)}{g(x)g(x+h)}$$

Example $\Delta(x \sin x) = x\Delta \sin x + \sin(x+h)\Delta x = 2x \sin(h/2) \cos[x+(h/2)] + h \sin(x+h)$.

DIFFERENCE EQUATIONS

A difference equation is a relation between the differences and the independent variable, $\phi(\Delta^n y, \Delta^{n-1} y, \dots, \Delta y, y, x) = 0$, where ϕ is some

given function. The general case in which the interval between the successive points is any real number h , instead of 1, can be reduced to that with interval size 1 by the substitution $x = hx'$. Hence all further difference-equation work will assume the interval size between successive points is 1.

Example $f(x+1) - (\alpha+1)f(x) + \alpha f(x-1) = 0$. Common notation usually is $y_x = f(x)$. This equation is then written $y_{x+1} - (\alpha+1)y_x + \alpha y_{x-1} = 0$.

Example $y_{x+2} + 2y_x y_{x+1} + y_x = x^2$.

Example $y_{x+1} - y_x = 2^x$.

The order of the difference equation is the difference between the largest and smallest arguments when written in the form of the second example. The first and second examples are both of order 2, while the third example is of order 1. A linear difference equation involves no products or other nonlinear functions of the dependent variable and its differences. The first and third examples are linear, while the second example is nonlinear.

A solution of a difference equation is a relation between the variables which satisfies the equation. If the difference equation is of order n , the general solution involves n arbitrary constants. The techniques for solving difference equations resemble techniques used for differential equations.

Equation $\Delta^n y = a$ The solution of $\Delta^n y = a$, where a is a constant, is a polynomial of degree n plus an arbitrary periodic function of period 1. That is, $y = (a^n/n!) + c_1 x^{n-1} + c_2 x^{n-2} + \dots + c_n + f(x)$, where $f(x+1) = f(x)$.

Example $\Delta^3 y = 6$. The solution is $y = x^3 + c_1 x^2 + c_2 x + c_3 + f(x)$; c_1, c_2, c_3 are arbitrary constants, and $f(x)$ is an arbitrary periodic function of period 1.

Equation $y_{x+1} - y_x = \phi(x)$ This equation states that the first difference of the unknown function is equal to the given function $\phi(x)$. The solution by analogy with solving the differential equation $dy/dx = \phi(x)$ by integration is obtained by "finite integration" or summation. When there are only a finite number of data points, this is easily accomplished by writing $y_x = y_0 + \sum_{t=1}^x \phi(t-1)$, where the data points are numbered from 1 to x . This is the only situation considered here.

Examples If $\phi(x) = 1$, $y_x = x$. If $\phi(x) = x$, $y_x = [x(x-1)]/2$. If $\phi(x) = a^x$, $a \neq 0$, $y_x = a^x/(a-1)$. In all cases $y_0 = 0$.

Other examples may be evaluated by using summation, that is, $y_2 = y_1 + \phi(1)$, $y_3 = y_2 + \phi(2) = y_1 + \phi(1) + \phi(2)$, $y_4 = y_3 + \phi(3) = y_1 + \phi(1) + \phi(2) + \phi(3)$, $\dots$, $y_x = y_1 + \sum_{t=1}^{x-1} \phi(t)$.

Example $y_{x+1} - ry_x = 1$, r constant, $x > 0$ and $y_0 = 1$. $y_1 = 1 + r$; $y_2 = 1 + r + r^2$, $\dots$, $y_x = 1 + r + \dots + r^x = (1 - r^{x+1})/(1 - r)$ for $r \neq 1$ and $y_x = 1 + x$ for $r = 1$.

Linear Difference Equations The linear difference equation of order n has the form $P_n y_{x+n} + P_{n-1} y_{x+n-1} + \dots + P_1 y_{x+1} + P_0 y_x = Q(x)$ with $P_n \neq 0$ and $P_0 \neq 0$ and $P_j, j = 0, \dots, n$ are functions of x .

Constant Coefficient and $Q(x) = 0$ (Homogeneous) The solution is obtained by trying a solution of the form $y_x = c\beta^x$. When this trial solution is substituted in the difference equation, a polynomial of degree n results for β . If the solutions of this polynomial are denoted by $\beta_1, \beta_2, \dots, \beta_n$ then the following cases result: (1) if all the β_j 's are real and unequal, the solution is $y_x = \sum_{j=1}^n c_j \beta_j^x$, where the $c_1, \dots, c_n$ are arbitrary constants; (2) if the roots are real and repeated, say, β_1 has multiplicity m , then the partial solution corresponding to β_1 is $\beta_1^x(c_1 + c_2 x + \dots + c_m x^{m-1})$; (3) if the roots are complex conjugates, say, $a + ib = pe^{i\theta}$ and $a - ib = pe^{-i\theta}$, the partial solution corresponding to this pair is $p^x(c_1 \cos \theta x + ic_2 \sin \theta x)$; and (4) if the roots are multiple complex conjugates, say, $a + ib = pe^{i\theta}$ and $a - ib = pe^{-i\theta}$ are m -fold, then the partial solution corresponding to these is $p^x[(c_1 + c_2 x + \dots + c_m x^{m-1}) \cos \theta x + i(d_1 + d_2 x + \dots + d_m x^{m-1}) \sin \theta x]$.

Example The equation $y_{x+1} - (\alpha+1)y_x + \alpha y_{x-1} = 0$, $y_0 = c_0$ and $y_{m+1} = x_{m+1}/k$ represents the steady-state composition of transferable material in the raffinate stream of a staged countercurrent liquid-liquid extraction system.

Clearly y is a function of the stage number x . α is a combination of system constants. By using the trial solution $y_x = c\beta^x$, there results $\beta^2 - (\alpha+1)\beta + \alpha = 0$, so that $\beta_1 = 1$, $\beta_2 = \alpha$. The general solution is $y_x = c_1 + c_2 \alpha^x$. By using the side conditions, $c_1 = c_0 - c_2$, $c_2 = (y_{m+1} - c_0)/(\alpha^{m+1} - 1)$. The desired solution is $(y_x - c_0)/(\alpha^{x+1} - c_0) = (\alpha^x - 1)/(\alpha^{m+1} - 1)$.

Example $y_{x+3} - 3y_{x+2} + 4y_x = 0$. By setting $y_x = c\beta^x$, there results $\beta^3 - 3\beta^2 + 4 = 0$ or $\beta_1 = -1$, $\beta_2 = 2$, $\beta_3 = 2$. The general solution is $y_x = c_1(-1)^x + 2^x(c_2 + c_3 x)$.

Example $y_{x+1} - 2y_x + 2y_{x-1} = 0$, $\beta_1 = 1 + i$, $\beta_2 = 1 - i$, $p = \sqrt{1+1} = \sqrt{2}$, $\theta = \pi/4$. The solution is $y_x = 2^{x/2}[c_1 \cos(x\pi/4) + ic_2 \sin(x\pi/4)]$.

Constant Coefficients and $Q(x) \neq 0$ (Nonhomogeneous) In this case the general solution is found by first obtaining the homogeneous solution, say, y_x^h and adding to it any particular solution with $Q(x) \neq 0$, say, y_x^p . There are several means of obtaining the particular solution.

Method of Undetermined Coefficients If $Q(x)$ is a product or linear combination of products of the functions e^{bx} , a^x , x^p (p a positive integer or zero) $\cos cx$ and $\sin cx$, this method may be used. The "families" $[a^x]$, $[e^{bx}]$, $[\sin cx, \cos cx]$ and $[x^p, x^{p-1}, \dots, x, 1]$ are defined for each of the above functions in the following way: The family of a term f_x is the set of all functions of which f_x and all operations of the form a^{x+y} , $\cos c(x+y)$, $\sin c(x+y)$, $(x+y)^p$ on f_x and their linear combinations result in. The technique involves the following steps: (1) Solve the homogeneous system. (2) Construct the family of each term. (3) If the family has no representative in the homogeneous solution, assume y_x^p is a linear combination of the families of each term and determine the constants so that the equation is satisfied. (4) If a family has a representative in the homogeneous solution, multiply each member of the family by the smallest integral power of x for which all such representatives are removed and revert to step 3.

Example $y_{x+1} - 3y_x + 2y_{x-1} = 1 + a^x$, $a \neq 0$. The homogeneous solution is $y_x^h = c_1 + c_2 2^x$. The family of 1 is 1 and of a^x is a^x . However, 1 is a solution of the homogeneous system. Therefore, try $y_x^p = Ax + Ba^x$. Substituting in the equation there results

$$y_x = c_1 + c_2 2^x - x + \frac{a}{(a-1)(a-2)} a^x, a \neq 1, a \neq 2$$

If $a = 1$, $y_x = c_1 + c_2 2^x - 2x$. If $a = 2$, $y_x = c_1 + c_2 2^x - x + x 2^x$.

Example The family of $x^2 3^x$ is $[x^2 3^x, x 3^x, 3^x]$.

Method of Variation of Parameters This technique is applicable to general linear difference equations. It is illustrated for the second-order system $y_{x+2} + Ay_{x+1} + By_x = \phi(x)$. Assume that the homogeneous solution has been found by some technique and write $y_x^h = c_1 u_x + c_2 v_x$. Assume that a particular solution $y_x^p = D_x u_x + E_x v_x$. E_x and D_x can be found by solving the equations:

$$E_{x+1} - E_x = \frac{u_{x+1}\phi(x)}{u_{x+1}v_{x+2} - u_{x+2}v_{x+1}}$$

$$D_{x+1} - D_x = \frac{v_{x+1}\phi(x)}{v_{x+1}u_{x+2} - v_{x+2}u_{x+1}}$$

by summation. The general solution is then $y_x = y_x^p + y_x^h$.

Variable Coefficients The method of variation of parameters applies equally well to the linear difference equation with variable coefficients. Techniques are therefore needed to solve the homogeneous system with variable coefficients.

Equation $y_{x+1} - a_x y_x = 0$ By assuming that this equation is valid for $x \geq 0$ and $y_0 = c$, the solution is $y_x = c \prod_{n=1}^x a_{n-1}$.

Example $y_{x+1} + \frac{x+2}{x+1} y_x = 0$. The solution is

$$y_x = c \prod_{n=1}^x \left(-\frac{n+1}{n}\right) = c(-1)^x \frac{2}{1} \cdot \frac{3}{2} \cdot \dots \cdot \frac{x+1}{x} = (-1)^x c(x+1)$$

Example $y_{x+1} - xy_x = 0$. The solution is $y_x = c(x-1)!$

Reduction of Order If one homogeneous solution, say, u_x , can be found by inspection or otherwise, an equation of lower order can be obtained by the substitution $v_x = y_x/u_x$. The resultant equation must be satisfied by $v_x = \text{constant}$ or $\Delta v_x = 0$. Thus the equation will be of reduced order if the new variable $U_x = \Delta(y_x/u_x)$ is introduced.

Example $(x+2)y_{x+2} - (x+3)y_{x+1} + y_x = 0$. By observation $u_x = 1$ is a solution. Set $U_x = \Delta y_x = y_{x+1} - y_x$. There results $(x+2)U_{x+1} - U_x = 0$, which is of degree one lower than the original equation. The complete solution for y_x is finally

$$y_x = c_0 \sum_{n=0}^x \frac{1}{n!} + c_1$$

Factorization If the difference equation can be factored, then the general solution can be obtained by solving two or more successive equations of lower order. Consider $y_{x+2} + A_x y_{x+1} + B_x y_x = \phi(x)$. If there exists a_x, b_x such that $a_x + b_x = -A_x$ and $a_x b_x = B_x$, then the difference equation may be written $y_{x+2} - (a_x + b_x)y_{x+1} + a_x b_x y_x = \phi(x)$. First solve $U_{x+1} - b_x U_x = \phi(x)$ and then $y_{x+1} - a_x y_x = U_x$.

Example $y_{x+2} - (2x+1)y_{x+1} + (x^2+x)y_x = 0$. Set $a_x = x, b_x = x+1$. Solve $u_{x+1} - (x+1)u_x = 0$ and then $y_{x+1} - xy_x = u_x$.

Substitution If it is possible to rearrange a difference equation so that it takes the form $af_{x+2}y_{x+2} + bf_{x+1}y_{x+1} + cf_x y_x = \phi(x)$ with a, b, c constants, then the substitution $u_x = f_x y_x$ reduces the equation to one with constant coefficients.

Example $(x+2)^2 y_{x+2} - 3(x+1)^2 y_{x+1} + 2x^2 y_x = 0$. Set $u_x = x^2 y_x$. The equation becomes $u_{x+2} - 3u_{x+1} + 2u_x = 0$, which is linear and easily solved by previous methods.

The substitution $u_x = y_x/f_x$ reduces $af_{x+2}y_{x+2} + bf_{x+1}y_{x+1} + cf_x y_x = \phi(x)$ to an equation with constant coefficients.

Example $x(x+1)y_{x+2} + 3x(x+2)y_{x+1} - 4(x+1)(x+2)y_x = x$. Set $u_x = y_x/f_x = y_x/x$. Then $y_x = xu_x, y_{x+1} = (x+1)u_{x+1}$ and $y_{x+2} = (x+2)u_{x+2}$. Substitution in the equation yields $x(x+1)(x+2)u_{x+2} + 3x(x+2)(x+1)u_{x+1} - 4x(x+1)(x+2)u_x = x$ or $u_{x+2} + 3u_{x+1} - 4u_x = 1/(x+1)(x+2)$, which is a linear equation with constant coefficients.

Nonlinear Difference Equations: Riccati Difference Equation The Riccati equation $y_{x+1}y_x + ay_{x+1} + by_x + c = 0$ is a nonlinear difference equation which can be solved by reduction to linear form. Set $y = z + h$. The equation becomes $z_{x+1}z_x + (h+a)z_{x+1} + (h+b)z_x + h^2 + (a+b)h + c = 0$. If h is selected as a root of $h^2 + (a+b)h + c = 0$ and the equation is divided by $z_{x+1}z_x$ there results $[(h+b)/z_{x+1}] + [(h+a)/z_x] + 1 = 0$. This is a linear equation with constant coefficients. The solution is

$$y_x = h + \frac{1}{c \left[-\frac{a+h}{b+h} \right]^x - \frac{1}{(a+h) + (b+h)}}$$

Example This equation is obtained in distillation problems, among others, in which the number of theoretical plates is required. If the relative volatility is assumed to be constant, the plates are theoretically perfect, and the molar liquid and vapor rates are constant, then a material balance around the n th plate of the enriching section yields a Riccati difference equation.

INTEGRAL EQUATIONS

REFERENCES: Courant, R., and D. Hilbert, *Methods of Mathematical Physics*, vol. I, Interscience, New York (1953); Linz, P., *Analytical and Numerical Methods for Volterra Equations*, SIAM Publications, Philadelphia (1985); Porter, D., and D. S. G. Stirling, *Integral Equations: A Practical Treatment from Spectral Theory to Applications*, Cambridge University Press (1990); Statgold, I., *Green's Functions and Boundary Value Problems*, 2d ed., Interscience, New York (1997).

An integral equation is any equation in which the unknown function appears under the sign of integration and possibly outside the sign of integration. If derivatives of the dependent variable appear elsewhere in the equation, the equation is said to be integrodifferential.

CLASSIFICATION OF INTEGRAL EQUATIONS

Volterra integral equations have an integral with a variable limit. The Volterra equation of the second kind is

$$u(x) = f(x) + \lambda \int_a^x K(x, t)u(t) dt$$

whereas a Volterra equation of the first kind is

$$u(x) = \lambda \int_a^x K(x, t)u(t) dt$$

Equations of the first kind are very sensitive to solution errors so that they present severe numerical problems. Volterra equations are similar to initial value problems.

A Fredholm equation of the second kind is

$$u(x) = f(x) + \lambda \int_a^b K(x, t)u(t) dt$$

whereas a Fredholm equation of the first kind is

$$u(x) = \int_a^b K(x, t)u(t) dt$$

The limits of integration are fixed, and these problems are analogous to boundary value problems.

An eigenvalue problem is a homogeneous equation of the second kind, and solutions exist only for certain λ .

$$u(x) = \lambda \int_a^b K(x, t)u(t) dt$$

See Linz (1985) for further information and existence proofs.

If the unknown function u appears in the equation in any way except to the first power, the integral equation is said to be nonlinear. The equation $u(x) = f(x) + \int_a^b K(x, t)[u(t)]^{3/2} dt$ is nonlinear. The differential equation $du/dx = g(x, u)$ is equivalent to the nonlinear integral equation $u(x) = c + \int_a^x g[t, u(t)] dt$.

An integral equation is said to be singular when either one or both of the limits of integration become infinite or if $K(x, t)$ becomes infinite for one or more points of the interval under discussion.

Example $u(x) = x + \int_0^\infty \cos(xt)u(t) dt$ and $f(x) = \int_0^x \frac{u(t)}{x-t} dt$ are both singular. The kernel of the first equation is $\cos(xt)$, and that of the second is $(x-t)^{-1}$.

RELATION TO DIFFERENTIAL EQUATIONS

The Leibniz rule (see "Integral Calculus") can be used to show the equivalence of the initial-value problem consisting of the second-order differential equation $d^2y/dx^2 + A(x)(dy/dx) + B(x)y = f(x)$ together with the prescribed initial conditions $y(a) = y_0, y'(a) = y'_0$ to the integral equation.

$$y(x) = \int_a^x K(x, t)y(t) dt + F(x)$$

where $K(x, t) = (t-x)[B(t) - A'(t)] - A(t)$

$$\text{and} \quad F(x) = \int_a^x (x-t)f(t) dt + [A(a)y_0 + y_0'](x-a) + y_0$$

This integral equation is a **Volterra equation** of the second kind. Thus the initial-value problem is equivalent to a Volterra integral equation of the second kind.

Example $d^2y/dx^2 + x^2(dy/dx) + xy = x$, $y(0) = 1$, $y'(0) = 0$. Here $A(x) = x^2$, $B(x) = x$, $f(x) = x$. The equivalent integral equation is $y(x) = \int_0^x K(x, t)y(t) dt + F(x)$ where $K(x, t) = t(x-t) - t^2$ and $F(x) = \int_0^x (x-t)t dt + 1 = x^3/6 + 1$. Combining these $y(x) = \int_0^x t[x-2t]y(t) dt + x^3/6 + 1$.

Eigenvalue problems can also be related. For example, the problem $(d^2y/dx^2) + \lambda y = 0$ with $y(0) = 0$, $y(a) = 0$ is equivalent to the integral equation $y(x) = \lambda \int_0^a K(x, t)y(t) dt$, where $K(x, t) = (t/a)(a-x)$ when $t < x$ and $K(x, t) = (x/a)(a-t)$ when $t > x$. The differential equation may be recovered from the integral equation by differentiating the integral equation by using the Leibniz rule.

METHODS OF SOLUTION

In general, the solution of integral equations is not easy, and a few exact and approximate methods are given here. Often numerical methods must be employed, as discussed in "Numerical Solution of Integral Equations."

Equations of Convolution Type The equation $u(x) = f(x) + \lambda \int_0^x K(x-t)u(t) dt$ is a special case of the linear integral equation of the second kind of Volterra type. The integral part is the convolution integral discussed under "Integral Transforms (Operational Methods)"; so the solution can be accomplished by Laplace transforms; $L[u(x)] = L[f(x)] + \lambda L[u(x)]L[K(x)]$ or

$$L[u(x)] = \frac{L[f(x)]}{1 - \lambda L[K(x)]}, \quad u(x) = L^{-1} \left[\frac{L[f(x)]}{1 - \lambda L[K(x)]} \right]$$

Equations of the type considered here occur quite frequently in practice in what can be called "cause-and-effect" systems.

Example In a certain linear system, the effect $E(t)$ due to a cause $C = \lambda E$ at time τ is a function only of the elapsed time $t - \tau$. If the system has the activity level 1 at time $t < 0$, the cause λE and effect (E) relation is given by the integral equation $E(t) = 1 + \lambda \int_0^t K(t-\tau)E(\tau) d\tau$. Let $K(t-\tau) = t-\tau$. Then $E(t) = 1 + \lambda \int_0^t (t-\tau)E(\tau) d\tau$. By using the transform method

$$E(t) = L^{-1} \left[\frac{L[1]}{1 - \lambda L[K(t)]} \right] = L^{-1} \left[\frac{1/p}{1 - \lambda/p^2} \right] = L^{-1} \left[\frac{p}{p^2 - \lambda} \right] = \cosh \sqrt{\lambda} t$$

Method of Successive Approximations Consider the equation $y(x) = f(x) + \lambda \int_a^b K(x, t)y(t) dt$. In this method a unique solution is obtained in sequence form as follows: Substitute in the right-hand member of the equation $y_0(t)$ for $y(t)$. Upon integration there results $y_1(t) = f(x) + \lambda \int_a^b K(x, t)y_0(t) dt$. Continue in like manner by replacing y_0 by y_1 , y_1 by y_2 , etc. A series of functions $y_0(x)$, $y_1(x)$, $y_2(x)$, ... are obtained which satisfy the equations

$$y_n(x) = f(x) + \lambda \int_a^b K(x, t)y_{n-1}(t) dt$$

Then $y_n(x) = f(x) + \lambda \int_a^b K(x, t)f(t) dt + \lambda^2 \int_a^b K(x, t) \int_a^b K(t, t_1)f(t_1) dt_1 dt + \lambda^3 \int_a^b K(x, t) \int_a^b K(t, t_1) \int_a^b K(t_1, t_2)f(t_2) dt_2 dt_1 dt + \dots + R_n$, where R_n is the remainder, and

$$|R_n| \leq |\lambda^n| \left(\max_{a \leq x \leq b} y_0 \right) M^n (b-a)^n$$

where M = maximum value of $|K|$ in the rectangle $a \leq t \leq b$, $a \leq x \leq b$. If $|\lambda|M(b-a) < 1$, $\lim_{n \rightarrow \infty} R_n = 0$. Then $y_n(x) \rightarrow y(x)$, which is the unique solution.

INTEGRAL TRANSFORMS (OPERATIONAL METHODS)

REFERENCES: Brown, J. W., and R. V. Churchill, *Fourier Series and Boundary Value Problems*, 6th ed., McGraw-Hill, New York (2000); Churchill, R. V., *Operational Mathematics*, 3d ed., McGraw-Hill, New York (1972); Davies, B., *Integral Transforms and Their Applications*, 3d ed., Springer (2002); Duffy, D. G., *Transform Methods for Solving Partial Differential Equations*, Chapman & Hall/CRC, New York (2004); Varma, A., and M. Morbidelli, *Mathematical Methods in Chemical Engineering*, Oxford, New York (1997).

The term "operational method" implies a procedure of solving differential and difference equations by which the boundary or initial conditions are automatically satisfied in the course of the solution. The technique offers a very powerful tool in the applications of mathematics, but it is limited to linear problems.

Most integral transforms are special cases of the equation $g(s) = \int_a^b f(t)K(s, t) dt$ in which $g(s)$ is said to be the transform of $f(t)$ and $K(s, t)$ is called the kernel of the transform. A tabulation of the more important kernels and the interval (a, b) of applicability follows. The first three transforms are considered here.

Name of transform	(a, b)	$K(s, t)$
Laplace	$(0, \infty)$	e^{-st}
Fourier	$(-\infty, \infty)$	$\frac{1}{\sqrt{2\pi}} e^{-ist}$
Fourier cosine	$(0, \infty)$	$\sqrt{\frac{2}{\pi}} \cos st$
Fourier sine	$(0, \infty)$	$\sqrt{\frac{2}{\pi}} \sin st$
Mellin	$(0, \infty)$	t^{s-1}
Hankel	$(0, \infty)$	$tJ_\nu(st)$, $\nu \geq -1/2$

LAPLACE TRANSFORM

The Laplace transform of a function $f(t)$ is defined by $F(s) = L\{f(t)\} = \int_0^\infty e^{-st}f(t) dt$, where s is a complex variable. Note that the transform is an improper integral and therefore may not exist for all continuous functions and all values of s . We restrict consideration to those values of s and those functions f for which this improper integral converges. The Laplace transform is used in process control (see Sec. 8).

The function $L\{f(t)\} = g(s)$ is called the direct transform, and $L^{-1}\{g(s)\} = f(t)$ is called the inverse transform. Both the direct and the inverse transforms are tabulated for many often-occurring functions. In general,

$$L^{-1}\{g(s)\} = \frac{1}{2\pi i} \int_{\alpha-i\infty}^{\alpha+i\infty} e^{st}g(s) ds$$

and to evaluate this integral requires a knowledge of complex variables, the theory of residues, and contour integration.

A function is said to be piecewise continuous on an interval if it has only a finite number of finite (or jump) discontinuities. A function f on $0 < t < \infty$ is said to be of exponential growth at infinity if there exist constants M and α such that $|f(t)| \leq Me^{\alpha t}$ for sufficiently large t .

Sufficient Conditions for the Existence of Laplace Transform Suppose f is a function which is (1) piecewise continuous on every finite interval $0 < t < T$, (2) of exponential growth at infinity, and (3) $\int_0^\delta |f(t)| dt$ exist (finite) for every finite $\delta > 0$. Then the Laplace transform of f exists for all complex numbers s with sufficiently large real part.

Note that condition 3 is automatically satisfied if f is assumed to be piecewise continuous on every finite interval $0 \leq t < T$. The function $f(t) = t^{-1/2}$ is not piecewise continuous on $0 \leq t \leq T$ but satisfies conditions 1 to 3.

Let Λ denote the class of all functions on $0 < t < \infty$ which satisfy conditions 1 to 3.

Example Let $f(t)$ be the Heaviside step function at $t = t_0$; i.e., $f(t) = 0$ for $t \leq t_0$, and $f(t) = 1$ for $t > t_0$. Then

$$L\{f(t)\} = \int_0^\infty e^{-st} dt = \lim_{T \rightarrow \infty} \int_0^T e^{-st} dt = \lim_{T \rightarrow \infty} \frac{1}{s} (e^{-st_0} - e^{-sT}) = \frac{e^{-st_0}}{s}$$

provided $s > 0$.

Example Let $f(t) = e^{at}$, $t \geq 0$, where a is a real number. Then $L\{e^{at}\} = \int_0^\infty e^{-(s-a)t} dt = 1/(s-a)$, provided $\text{Re } s > a$.

Properties of the Laplace Transform

1. The Laplace transform is a linear operator: $L\{af(t) + bg(t)\} = aL\{f(t)\} + bL\{g(t)\}$ for any constants a, b and any two functions f and g whose Laplace transforms exist.

2. The Laplace transform of a real-valued function is real for real s . If $f(t)$ is a complex-valued function, $f(t) = u(t) + iv(t)$, where u and v are real, then $L\{f(t)\} = L\{u(t)\} + iL\{v(t)\}$. Thus $L\{u(t)\}$ is the real part of $L\{f(t)\}$, and $L\{v(t)\}$ is the imaginary part of $L\{f(t)\}$.

3. The Laplace transform of a function in the class Λ has derivatives of all orders, and $L\{t^k f(t)\} = (-1)^k d^k F(s)/ds^k$, $k = 1, 2, 3, \dots$

Example $\int_0^\infty e^{-st} \sin at \, dt = \frac{a}{s^2 + a^2}$, $s > 0$. By property 3, $\frac{2as}{(s^2 + a^2)^2} = \int_0^\infty e^{-st} t \sin at \, dt = L\{t \sin at\}$.

Example By applying property 3 with $f(t) = 1$ and using the preceding results, we obtain

$$L\{t^k\} = (-1)^k \frac{d^k}{ds^k} \left(\frac{1}{s} \right) = \frac{k!}{s^{k+1}}$$

provided $\text{Re } s > 0$; $k = 1, 2, \dots$. Similarly, we obtain

$$L\{t^k e^{at}\} = (-1)^k \frac{d^k}{ds^k} \left(\frac{1}{s-a} \right) = \frac{k!}{(s-a)^{k+1}}$$

4. Frequency-shift property (or, equivalently, the transform of an exponentially modulated function). If $F(s)$ is the Laplace transform of a function $f(t)$ in the class Λ , then for any constant a , $L\{e^{at}f(t)\} = F(s-a)$.

Example $L\{te^{-at}\} = \frac{1}{(s+a)^2}$, $s > 0$.

5. Time-shift property. Let $u(t-a)$ be the unit step function at $t = a$. Then $L\{f(t-a)u(t-a)\} = e^{-as}F(s)$.

6. Transform of a derivative. Let f be a differentiable function such that both f and f' belong to the class Λ . Then $L\{f'(t)\} = sF(s) - f(0)$.

7. Transform of a higher-order derivative. Let f be a function which has continuous derivatives up to order n on $(0, \infty)$, and suppose that f and its derivatives up to order n belong to the class Λ . Then $L\{f^{(j)}(t)\} = s^j F(s) - s^{j-1}f(0) - s^{j-2}f'(0) - \dots - sf^{(j-2)}(0) - f^{(j-1)}(0)$ for $j = 1, 2, \dots, k$.

Example $L\{f''(t)\} = s^2 L\{f(t)\} - sf(0) - f'(0)$

$$L\{f'''(t)\} = s^3 L\{f(t)\} - s^2 f(0) - sf'(0) - f''(0)$$

Example Solve $y'' + y = 2e^t$, $y(0) = y'(0) = 2$. $L[y''] = -y'(0) - sy(0) + s^2 L[y] = -2 - 2s + s^2 L[y]$. Thus

$$-2 - 2s + s^2 L[y] + L[y] = 2L\{e^t\} = \frac{2}{s-1}$$

$$L[y] = \frac{2s^2}{(s-1)(s^2+1)} = \frac{1}{s-1} + \frac{s}{s^2+1} + \frac{1}{s^2+1}$$

Hence $y = e^t + \cos t + \sin t$.

A short table (Table 3-2) of very common Laplace transforms and inverse transforms follows. The references include more detailed tables. NOTE: $\Gamma(n+1) = \int_0^\infty x^n e^{-x} dx$ (gamma function); $J_n(t)$ = Bessel function of the first kind of order n .

$$8. L\left[\int_a^t f(t) \, dt\right] = \frac{1}{s} L\{f(t)\} + \frac{1}{s} \int_a^0 f(t) \, dt$$

TABLE 3-2 Laplace Transforms

$f(t)$	$g(s)$	$f(t)$	$g(s)$
1	$1/s$	$e^{-at}(1-at)$	$\frac{s}{(s+a)^2}$
t^n , ($n, a + \text{integer}$)	$\frac{n!}{s^{n+1}}$	$\frac{t \sin at}{2a}$	$\frac{s}{(s^2+a^2)^2}$
t^n , $n \neq + \text{integer}$	$\frac{\Gamma(n+1)}{s^{n+1}}$	$\frac{1}{2a^2} \sin at \sinh at$	$\frac{s}{s^4+4a^4}$
$\cos at$	$\frac{s}{s^2+a^2}$	$\cos at \cosh at$	$\frac{s^2}{s^4+4a^4}$
$\sin at$	$\frac{a}{s^2+a^2}$	$\frac{1}{2a} (\sinh at + \sin at)$	$\frac{s^2}{s^4-a^4}$
$\cosh at$	$\frac{s}{s^2-a^2}$	$\frac{1}{2} (\cosh at + \cos at)$	$\frac{s^3}{s^4-a^4}$
$\sinh at$	$\frac{a}{s^2-a^2}$	$\frac{\sin at}{t}$	$\tan^{-1} \frac{a}{s}$
e^{-at}	$\frac{1}{s+a}$	$J_0(at)$	$\frac{1}{\sqrt{s^2+a^2}}$
$e^{-bt} \cos at$	$\frac{s+b}{(s+b)^2+a^2}$	$na^n \frac{J_n(at)}{t}$	$(\sqrt{s^2+a^2}-s)^n$ ($n > 0$)
$e^{-bt} \sin at$	$\frac{a}{(s+b)^2+a^2}$	$J_0(2\sqrt{at})$	$\frac{1}{s} e^{-a/s}$
$\text{erfc} \frac{k}{2\sqrt{t}}$	$\frac{1}{s} e^{-k\sqrt{s}}$		

Example Find $f(t)$ if $L\{f(t)\} = \frac{1}{s^2} \left[\frac{1}{s^2-a^2} \right]$. $L\left[\frac{1}{a} \sinh at\right] = \frac{1}{s^2-a^2}$.

$$\text{Therefore } f(t) = \int_0^t \left[\int_0^t \frac{1}{a} \sinh at \, dt \right] dt = \frac{1}{a^2} \left[\frac{\sinh at}{a} - t \right].$$

$$9. L\left[\frac{f(t)}{t}\right] = \int_s^\infty g(s) \, ds \quad L\left[\frac{f(t)}{t^k}\right] = \underbrace{\int_s^\infty \dots \int_s^\infty}_{k \text{ integrals}} g(s) (ds)^k$$

Example $L\left[\frac{\sin at}{t}\right] = \int_s^\infty L[\sin at] \, ds = \int_s^\infty \frac{a \, ds}{s^2+a^2} = \cot^{-1} \frac{s}{a}$

10. The unit step function

$$u(t-a) = \begin{cases} 0 & t < a \\ 1 & t > a \end{cases} \quad L[u(t-a)] = e^{-as}/s$$

11. The unit impulse function is

$$\delta(t) = u'(t-a) = \begin{cases} \infty & \text{at } t = a \\ 0 & \text{elsewhere} \end{cases} \quad L[u'(t-a)] = e^{-as}$$

12. $L^{-1}[e^{-as}g(s)] = f(t-a)u(t-a)$ (second shift theorem).

13. If $f(t)$ is periodic of period b , i.e., $f(t+b) = f(t)$, then

$$L\{f(t)\} = \left[\frac{1}{1-e^{-bs}} \right] \int_0^b e^{-st} f(t) \, dt$$

Example The partial differential equations relating gas composition to position and time in a gas chromatograph are $\partial y/\partial n + \partial x/\partial \theta = 0$, $\partial y/\partial n = x - y$, where $x = mx'$, $n = (k_G a P/G_m)h$, $\theta = (mk_G a P/\rho_b)t$ and G_m = molar velocity, y = mole fraction of the component in the gas phase, ρ_b = bulk density, h = distance from the entrance, P = pressure, k_G = mass-transfer coefficient, and m = slope of the equilibrium line. These equations are equivalent to $\partial^2 y/\partial n \partial \theta + \partial y/\partial n + \partial y/\partial \theta = 0$, where the boundary conditions considered here are $y(0, \theta) = 0$ and $x(n, 0) = y(n, 0) + (\partial y/\partial n)(n, 0) = \delta(0)$ (see property 11). The problem is conveniently

solved by using the Laplace transform of y with respect to n ; write $g(s, \theta) = \int_0^\infty e^{-s\theta} y(n, \theta) dn$. Operating on the partial differential equation gives $s(dg/d\theta) - (\partial y/\partial \theta)(0, \theta) + sg - y(0, \theta) + dg/d\theta = 0$ or $(s+1)(dg/d\theta) + sg = (\partial y/\partial \theta)(0, \theta) + y(0, \theta) = 0$. The second boundary condition gives $g(s, 0) + sg(s, 0) - y(0, 0) = 1$ or $g(s, 0) + sg(s, 0) = 1$ ($L[\delta(0)] = 1$). A solution of the ordinary differential equation for g consistent with this second condition is

$$g(s, \theta) = \frac{1}{s+1} e^{-\theta/(s+1)}$$

Inversion of this transform gives the solution $y(n, \theta) = e^{-(n+\theta)} I_0(2\sqrt{n\theta})$ where I_0 = zero-order Bessel function of an imaginary argument. For large u , $I_n(u) \sim e^u/\sqrt{2\pi u}$. Hence for large n ,

$$y(n, \theta) \sim \frac{\exp[-(\sqrt{\theta} - \sqrt{n})^2]}{2\pi^{1/2}(n\theta)^{1/4}}$$

or for sufficiently large n , the peak concentration occurs near $\theta = n$.

Other applications of Laplace transforms are given under "Differential Equations."

CONVOLUTION INTEGRAL

The convolution integral (faltung) of two functions $f(t)$, $r(t)$ is $x(t) = f(t) \circ r(t) = \int_0^t f(\tau)r(t-\tau) d\tau$.

Example $t \circ \sin t = \int_0^t \tau \sin(t-\tau) d\tau = t - \sin t$

$$L[f(t)]L[h(t)] = L[f(t) \circ h(t)]$$

z-TRANSFORM

See Ogunnaike, Babatunde A., and W. Harmon Ray, *Process Dynamics, Modeling, and Control*, Oxford University Press (1994); Seborg, D., T. F. Edgar, and D. A. Mellichamp, *Process Dynamics and Control*, 2d ed., Wiley, New York (2003). The z -transform is useful when data is available at only discrete points. Let

$$f^\circ(t) = f(t_k)$$

be the value of f at the sample points

$$t_k = k \Delta t, \quad k = 0, 1, 2, \dots$$

Then the function $f^\circ(t)$ is

$$f^\circ(t) = \sum_{k=0}^{\infty} f(t_k) \delta(t - t_k)$$

Take the Laplace transform of this.

$$g^\circ(s) = L[f^\circ(t)] = \sum_{k=0}^{\infty} f(t_k) e^{-st_k} = \sum_{k=0}^{\infty} f(t_k) e^{-s\Delta t k}$$

For convenience, replace $e^{s\Delta t}$ by z and call $g^\circ(z)$ the z -transform of $f^\circ(t)$.

$$g^\circ(z) = \sum_{k=0}^{\infty} f(t_k) z^{-k}$$

The z -transform is used in process control when the signals are at intervals of Δt . A brief table (Table 3-3) is provided here.

The z -transform can also be used to solve difference equations, just like the Laplace transform can be used to solve differential equations.

Example The difference equation for $y(k)$ is

$$y(k) + a_1 y(k-1) + a_2 y(k-2) = b_1 u(k)$$

Take the z -transform

$$(1 + a_1 z^{-1} + a_2 z^{-2}) y^\circ(z) = b_1 u^\circ(z)$$

Then

$$y^\circ(z) = \frac{u^\circ(z)}{1 + a_1 z^{-1} + a_2 z^{-2}}$$

The inverse transform must be found, usually from a table of inverse transforms.

FOURIER TRANSFORM

REFERENCES: Bateman, H., *Tables of Integral Transforms*, vol. I, McGraw-Hill, New York (1954); Varma, A., and M. Morbidelli, *Mathematical Methods in Chemical Engineering*, Oxford, New York (1997).

TABLE 3-3 z -Transforms

$f(k)$	$g^\circ(z)$
$1(k)$	$\frac{1}{1-z^{-1}}$
$k \Delta t$	$\frac{\Delta t z^{-1}}{(1-z^{-1})^2}$
$(k \Delta t)^{n-1}$	$\lim_{a \rightarrow 0} (-1)^{n-1} \frac{\partial^{n-1}}{\partial a^{n-1}} \left(\frac{1}{1-e^{-a\Delta t} z^{-1}} \right)$
$\sin a k \Delta t$	$\frac{z^{-1} \sin a \Delta t}{(1-2z^{-1} \cos a \Delta t + z^{-2})}$
$\cos a k \Delta t$	$\frac{1-z^{-1} \cos a \Delta t}{(1-2z^{-1} \cos a \Delta t + z^{-2})}$
$e^{-ak\Delta t}$	$\frac{1}{1-e^{-a\Delta t} z^{-1}}$
$e^{-bk\Delta t} \cos a k \Delta t$	$\frac{1-z^{-1} e^{-b\Delta t} \cos a \Delta t}{1-2z^{-1} e^{-b\Delta t} \cos a \Delta t + z^{-2} e^{-2b\Delta t}}$
$\frac{1}{b} e^{-bk\Delta t} \sin a k \Delta t$	$\frac{1}{b} \frac{z^{-1} e^{-b\Delta t} \sin a \Delta t}{1-2z^{-1} e^{-b\Delta t} \cos a \Delta t + z^{-2} e^{-2b\Delta t}}$

The Fourier transform is given by

$$F[f(t)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-ist} dt = g(s)$$

and its inverse by

$$F^{-1}[g(s)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(s) e^{ist} dt = f(t)$$

In brief, the condition for the Fourier transform to exist is that $\int_{-\infty}^{\infty} |f(t)| dt < \infty$, although certain functions may have a Fourier transform even if this is violated.

Example The function $f(t) = \begin{cases} 1-a \leq t \leq a \\ 0 \text{ elsewhere} \end{cases}$ has $F[f(t)] = \int_{-a}^a e^{-ist} dt =$

$$\int_0^a e^{ist} dt + \int_0^a e^{-ist} dt = 2 \int_0^a \cos st dt = \frac{2 \sin sa}{s}$$

Properties of the Fourier Transform Let $F[f(t)] = g(s)$;
 $F^{-1}[g(s)] = f(t)$.

- $F[f^{(n)}(t)] = (is)^n F[f(t)]$.
- $F[af(t) + bh(t)] = aF[f(t)] + bF[h(t)]$.
- $F[f(-t)] = g(-s)$.
- $F[f(at)] = \frac{1}{a} g\left(\frac{s}{a}\right), a > 0$.
- $F[e^{-iat} f(t)] = g(s+a)$.
- $F[f(t+t_1)] = e^{ist_1} g(s)$.
- $F[f(t)] = G(is) + G(-is)$ if $f(t) = f(-t)$ (f even)
 $F[f(t)] = G(is) - G(-is)$ if $f(t) = -f(-t)$ (f odd)

where $G(s) = L[f(t)]$. This result allows the use of the Laplace-transform tables to obtain the Fourier transforms.

Example Find $F[e^{-a|t|}]$ by property 7. $e^{-a|t|}$ is even. So $L[e^{-at}] = 1/(s+a)$. Therefore, $F[e^{-a|t|}] = 1/(is+a) + 1/(-is+a) = 2a/(s^2+a^2)$.

FOURIER COSINE TRANSFORM

The Fourier cosine transform is given by

$$F_c[f(t)] = g(s) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(t) \cos st dt$$

and its inverse by

$$F_c^{-1}[g(s)] = f(t) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} g(s) \cos st \, ds$$

The Fourier sine transform F_s is obtainable by replacing the cosine by the sine in these integrals. They can be used to solve linear differential equations; see the transform references.

MATRIX ALGEBRA AND MATRIX COMPUTATIONS

REFERENCES: Anton, H., and C. Rorres, *Elementary Linear Algebra with Applications*, 9th ed., Wiley (2004); Bernstein, D. S., *Matrix Mathematics: Theory, Facts, and Formulas with Application to Linear Systems Theory*, Princeton University Press, Princeton, N.J. (2005); Kolman, B., and D. R. Hill, *Introductory Linear Algebra: An Applied First Course*, 8th ed., Prentice-Hall, Englewood Cliffs, N.J. (2004); Lay, D. C., *Linear Algebra and Its Applications*, 3d ed., Addison Wesley (2002); Lipschutz, S., and M. Lipson, *Schaum's Outline of Linear Algebra*, McGraw-Hill, New York (2000); Noble, B., and J. W. Daniel, *Applied Linear Algebra*, 3d ed., Prentice-Hall, Englewood Cliffs, N.J. (1987); Press, W. H., et al., *Numerical Recipes*, Cambridge University Press, Cambridge (1986).

MATRIX ALGEBRA

Matrices A rectangular array of mn quantities, arranged in m rows and n columns

$$A = (a_{ij}) = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}$$

is called a matrix. The elements a_{ij} may be real or complex. The notation a_{ij} means the element in the i th row and j th column, i is called the row index, j the column index. If $m = n$ the matrix is said to be square and of order n . A matrix, even if it is square, does not have a numerical value, as a determinant does. However, if the matrix A is square, a determinant can be formed which has the same elements as the matrix A . This is called the determinant of the matrix and is written $\det(A)$ or $|A|$. If A is square and $\det(A) \neq 0$, A is said to be nonsingular; if $\det(A) = 0$, A is said to be singular. A matrix A has rank r if and only if it has a nonvanishing determinant of order r and no nonvanishing determinant of order $> r$.

Equality of Matrices Let $A = (a_{ij})$, $B = (b_{ij})$. Two matrices A and B are equal ($=$) if and only if they are identical; that is, they have the same number of rows and the same number of columns and equal corresponding elements ($a_{ij} = b_{ij}$ for all i and j).

Addition and Subtraction The operations of addition (+) and subtraction (−) of two or more matrices are possible if and only if they have the same number of rows and columns. Thus $A \pm B = (a_{ij} \pm b_{ij})$; i.e., addition and subtraction are of corresponding elements.

Transposition The matrix obtained from A by interchanging the rows and columns of A is called the transpose of A , written A^T or A^T .

Example $A = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 1 & 6 \end{bmatrix}$, $A^T = \begin{bmatrix} 1 & 2 \\ 3 & 1 \\ 4 & 6 \end{bmatrix}$

Note that $(A^T)^T = A$.

Multiplication Let $A = (a_{ij})$, $i = 1, \dots, m_1$; $j = 1, \dots, m_2$. $B = (b_{ij})$, $i = 1, \dots, n_1$, $j = 1, \dots, n_2$. The product AB is defined if and only if the number of columns of A (m_2) equals the number of rows of B (n_1), i.e., $n_1 = m_2$. For two such matrices the product $P = AB$ is defined by summing the element by element products of a row of A by a column of B . This is the row by column rule. Thus

$$p_{ij} = \sum_{k=1}^{n_1} a_{ik} b_{kj}$$

The resulting matrix has m_1 rows and n_2 columns.

Example $\begin{bmatrix} 3 & 2 \\ 1 & 1 \\ 5 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 & 5 & 6 \\ -2 & 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} -4 & 3 & 17 & 24 \\ -2 & 1 & 6 & 9 \\ -8 & 5 & 29 & 42 \end{bmatrix}$

It is helpful to remember that the element p_{ij} is formed from the i th row of the first matrix and the j th column of the second matrix. The matrix product is not commutative. That is, $AB \neq BA$ in general.

Inverse of a Matrix A square matrix A is said to have an inverse if there exists a matrix B such that $AB = BA = I$, where I is the identity matrix of order n .

$$\begin{bmatrix} 1 & 0 & \cdots & 0 \\ & 1 & & \\ & & \ddots & \\ 0 & & & 1 \end{bmatrix}$$

The inverse B is a square matrix of the order of A , designated by A^{-1} . Thus $AA^{-1} = A^{-1}A = I$. A square matrix A has an inverse if and only if A is nonsingular.

Certain relations are important:

- (1) $(AB)^{-1} = B^{-1}A^{-1}$
- (2) $(AB)^T = B^T A^T$
- (3) $(A^{-1})^T = (A^T)^{-1}$
- (4) $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$

Scalar Multiplication Let c be any real or complex number. Then $cA = (ca_{ij})$.

Adjugate Matrix of a Matrix Let A_{ij} denote the cofactor of the element a_{ij} in the determinant of the matrix A . The matrix B^T where $B = (A_{ij})$ is called the adjugate matrix of A written $\text{adj } A = B^T$. The elements b_{ij} are calculated by taking the matrix A , deleting the i th row and j th column, and calculating the determinant of the remaining matrix times $(-1)^{i+j}$. Then $A^{-1} = \text{adj } A / |A|$. This definition may be used to calculate A^{-1} . However, it is very laborious and the inversion is usually accomplished by numerical techniques shown under "Numerical Analysis and Approximate Methods."

Linear Equations in Matrix Form Every set of n nonhomogeneous linear equations in n unknowns

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ \vdots & \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned}$$

can be written in matrix form as $AX = B$, where $A = (a_{ij})$, $X^T = [x_1 \cdots x_n]$, and $B^T = [b_1 \cdots b_n]$. The solution for the unknowns is $X = A^{-1}B$.

Special Square Matrices

1. A triangular matrix is a matrix all of whose elements above or below the main diagonal (set of elements $a_{11}, \dots, a_{nn}$) are zero.

If A is triangular, $\det(A) = a_{11} \cdot a_{22} \cdots a_{nn}$.

2. A diagonal matrix is one such that all elements both above and below the main diagonal are zero (i.e., $a_{ij} = 0$ for all $i \neq j$). If all diagonal elements are equal, the matrix is called scalar. If A is diagonal, $A = (a_{ij})$, $A^{-1} = (1/a_{ij})$.

3. If $a_{ij} = a_{ji}$ for all i and j (i.e., $A = A^T$), the matrix is symmetric.

4. If $a_{ij} = -a_{ji}$ for $i \neq j$ but the a_{ij} are not all zero, the matrix is skew.

5. If $a_{ij} = -a_{ji}$ for all i and j (i.e., $a_{ii} = 0$), the matrix is skew symmetric.

6. If $A^T = A^{-1}$, the matrix A is orthogonal.

7. If the matrix $A^* = (\bar{a}_{ij})^T$, $\bar{a}_{ij}$ = complex conjugate of a_{ij} , A^* is the hermitian transpose of A .

8. If $A = A^{-1}$, A is involutory.

9. If $A = A^*$, A is hermitian.

10. If $A = -A^*$, A is skew hermitian.

11. If $A^{-1} = A^*$, A is unitary.

If A is any matrix, then AA^T and $A^T A$ are square symmetric matrices, usually of different order.

Example Let $A = \begin{bmatrix} 5 & 1 & 3 & 0 \\ 3 & 4 & 1 & 5 \\ 2 & -2 & 0 & 1 \end{bmatrix}$, $A^T = \begin{bmatrix} 5 & 3 & 2 \\ 1 & 4 & -2 \\ 3 & 1 & 0 \\ 0 & 5 & 1 \end{bmatrix}$

$$AA^T = \begin{bmatrix} 35 & 22 & 8 \\ 22 & 51 & 3 \\ 8 & 3 & 9 \end{bmatrix}, \quad A^T A = \begin{bmatrix} 38 & 13 & 18 & 17 \\ 13 & 21 & 7 & 18 \\ 18 & 7 & 10 & 5 \\ 17 & 18 & 5 & 26 \end{bmatrix}$$

Using a program such as MATLAB, these are easily calculated.

Matrix Calculus

Differentiation Let the elements of $A = [a_{ij}(t)]$ be differentiable functions of t . Then $\frac{dA}{dt} = \left[\frac{da_{ij}(t)}{dt} \right]$.

Example $A = \begin{bmatrix} \sin t & \cos t \\ -\cos t & \sin t \end{bmatrix}$, $\frac{dA}{dt} = \begin{bmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{bmatrix}$.

Integration The integral $\int A dt = [\int a_{ij}(t) dt]$.

Example $A = \begin{bmatrix} t & 2 \\ t^2 & e^t \end{bmatrix}$, $\int A dt = \begin{bmatrix} t^2/2 & 2t \\ t^3/3 & e^t \end{bmatrix}$.

The matrix $B = A - \lambda I$ is called the characteristic (eigen) matrix of A . Here A is square of order n , λ is a scalar parameter, and I is the $n \times n$ identity. $\det B = \det(A - \lambda I) = 0$ is the characteristic (eigen) equation for A . The characteristic equation is always of the same degree as the order of A . The roots of the characteristic equation are called the eigenvalues of A .

Example $A = \begin{bmatrix} 1 & 2 \\ 3 & 8 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 3 & 8 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 1-\lambda & 2 \\ 3 & 8-\lambda \end{bmatrix}$

is the characteristic matrix and $f(\lambda) = \det(B) = \det(A - \lambda I) = (1 - \lambda)(8 - \lambda) - 6 = 2 - 9\lambda + \lambda^2 = 0$ is the characteristic equation. The eigenvalues of A are the roots of $\lambda^2 - 9\lambda + 2 = 0$, which are $(9 \pm \sqrt{73})/2$.

A nonzero matrix X_i , which has one column and n rows, called a column vector satisfying the equation

$$(A - \lambda I)X_i = 0$$

and associated with the i th characteristic root λ_i is called an eigenvector.

Vector and Matrix Norms To carry out error analysis for approximate and iterative methods for the solutions of linear systems, one needs notions for vectors in R^n and for matrices that are analogous to the notion of length of a geometric vector. Let R^n denote the set of all vectors with n components, $x = (x_1, \dots, x_n)$. In dealing with matrices it is convenient to treat vectors in R^n as columns, and so $x = (x_1, \dots, x_n)^T$; however, we shall here write them simply as row vectors. A norm on R^n is a real-valued function f defined on R^n with the following properties:

1. $f(x) \geq 0$ for all $x \in R^n$.
2. $f(x) = 0$ if and only if $x = (0, 0, \dots, 0)$.
3. $f(ax) = |a|f(x)$ for all real numbers a and $x \in R^n$.
4. $f(x + y) \leq f(x) + f(y)$ for all $x, y \in R^n$.

The usual notation for a norm is $f(x) = \|x\|$.

The norm of a matrix is

$$\kappa(A) \equiv \|A\| \|A^{-1}\|$$

$$\|A\| = \sup_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \max_k \sum_{j=1}^n |a_{jk}|$$

The norm is useful when doing numerical calculations. If the computer's floating-point precision is 10^{-6} , then $\kappa = 10^6$ indicates an ill-conditioned matrix. If the floating-point precision is 10^{-12} (double precision), then a matrix with $\kappa = 10^{12}$ may be ill-conditioned. Two other measures are useful and are more easily calculated:

$$\text{Ratio} = \frac{\max_k |a_{kk}^{(k)}|}{\min_k |a_{kk}^{(k)}|}, \quad V = \frac{|\det A|}{\alpha_1 \alpha_2 \dots \alpha_n}, \quad \alpha_i = (a_{i1}^2 + a_{i2}^2 + \dots + a_{in}^2)^{1/2}$$

where $a_{kk}^{(k)}$ are the diagonal elements of the LU decomposition.

MATRIX COMPUTATIONS

The principal topics in linear algebra involve systems of linear equations, matrices, vector spaces, linear transformations, eigenvalues and eigenvectors, and least-squares problems. The calculations are routinely done on a computer.

LU Factorization of a Matrix To every $m \times n$ matrix A there exists a permutation matrix P , a lower triangular matrix L with unit diagonal elements, and an $m \times n$ (upper triangular) echelon matrix U such that $PA = LU$. The Gauss elimination is in essence an algorithm to determine U , P , and L . The permutation matrix P may be needed since it may be necessary in carrying out the Gauss elimination to interchange two rows of A to produce a (nonzero) pivot, such as if we start with

$$A = \begin{bmatrix} 0 & 2 \\ 1 & 6 \end{bmatrix}$$

If A is a square matrix and if principal submatrices of A are all nonsingular, then we may choose P as the identity in the preceding factorization and obtain $A = LU$. This factorization is unique if L is normalized (as assumed previously), so that it has unit elements on the main diagonal.

Solution of $Ax = b$ by Using LU Factorization Suppose that the indicated system is compatible and that $A = LU$ (the case $PA = LU$ is similarly handled and amounts to rearranging the equations). Let $z = Ux$. Then $Ax = LUx = b$ implies that $Lz = b$. Thus to solve $Ax = b$ we first solve $Lz = b$ for z and then solve $Ux = z$ for x . This procedure does not require that A be invertible and can be used to determine all solutions of a compatible system $Ax = b$. Note that the systems $Lz = b$ and $Ux = z$ are both in triangular forms and thus can be easily solved.

The LU decomposition is essentially a Gaussian elimination, arranged for maximum efficiency. The chief reason for doing an LU decomposition is that it takes fewer multiplications than would be needed to find an inverse. Also, once the LU decomposition has been found, it is possible to solve for multiple right-hand sides with little increase in work. The multiplication count for an $n \times n$ matrix and m right-hand sides is

$$\text{operation count} = \frac{1}{3} n^3 - \frac{1}{3} n + mn^2$$

If an inverse is desired, it can be calculated by solving for the LU decomposition and then solving n problems with right-hand sides consisting of all zeros except one entry. Thus $4n^2/3 - n/3$ multiplications are required for the inverse. The determinant is given by

$$\det A = \prod_{i=1}^n a_{ii}^{(i)}$$

where $a_{ii}^{(i)}$ are the diagonal elements obtained in the LU decomposition.

A tridiagonal matrix is one in which the only nonzero entries lie on the main diagonal and the diagonal just above and just below the main diagonal. The set of equations can be written as

$$a_i x_{i-1} + b_i x_i + c_i x_{i+1} = d_i$$

The LU decomposition is

$$b_1 = b_1$$

for $k=2, n$ do

$$a'_k = \frac{a_k}{b'_{k-1}}, \quad b'_k = b_k - \frac{a_k}{b'_{k-1}} c_{k-1}$$

enddo

$$d'_1 = d_1$$

for $k=2, n$ do

$$d'_k = d_k - a'_k d'_{k-1}$$

enddo

$$x_n = d'_n / b'_n$$

for $k=n-1, 1$ do

$$x_k = \frac{d'_k - c_k x_{k+1}}{b'_k}$$

enddo

The operation count for an $n \times n$ matrix with m right-hand sides is

$$2(n-1) + m(3n-2)$$

If $|b_i| > |a_i| + |c_i|$

no pivoting is necessary, and this is true for many boundary-value problems and partial-differential equations.

Sparse matrices are ones in which the majority of the elements are zero. If the structure of the matrix is exploited, the solution time on a computer is greatly reduced. See Duff, I. S., J. K. Reid, and A. M. Erisman (eds.), *Direct Methods for Sparse Matrices*, Clarendon Press, Oxford (1986); Saad, Y., *Iterative Methods for Sparse Linear Systems*, 2d ed., Society for Industrial and Applied Mathematics, Philadelphia (2003). The conjugate gradient method is one method for solving sparse matrix problems, since it only involves multiplication of a matrix times a vector. Thus the sparseness of the matrix is easy to exploit. The conjugate gradient method is an iterative method that converges for sure in n iterations where the matrix is an $n \times n$ matrix.

Matrix methods, in particular finding the rank of the matrix, can be used to find the number of independent reactions in a reaction set. If the stoichiometric numbers for the reactions and molecules are put in the form of a matrix, the rank of the matrix gives the number of independent reactions. See Amundson, N. R., *Mathematical Methods in Chemical Engineering*, Prentice-Hall, Englewood Cliffs, N.J. (1966, p. 50).

QR Factorization of a Matrix If A is an $m \times n$ matrix with $m \geq n$, there exists an $m \times m$ unitary matrix $Q = [q_1, q_2, \dots, q_m]$ and an $m \times n$ right triangular matrix R such that $A = QR$. The QR factorization is frequently used in the actual computations when the other transformations are unstable.

Singular-Value Decomposition If A is an $m \times n$ matrix with $m \geq n$ and rank $k \leq n$, consider the two following matrices.

$$AA^* \quad \text{and} \quad A^*A$$

An $m \times m$ unitary matrix U is formed from the eigenvectors u_i of the first matrix.

$$U = [u_1, u_2, \dots, u_m]$$

An $n \times n$ unitary matrix V is formed from the eigenvectors v_i of the second matrix.

$$V = [v_1, v_2, \dots, v_n]$$

Then matrix A can be decomposed into

$$A = U\Sigma V^*$$

where Σ is a $k \times k$ diagonal matrix with diagonal elements $d_{ii} = \sigma_i > 0$ for $1 \leq i \leq k$. The eigenvalues of $\Sigma^* \Sigma$ are σ_i^2 . The vectors u_i for $k+1 \leq i \leq m$ and v_i for $k+1 \leq i \leq n$ are eigenvectors associated with the eigenvalue zero; the eigenvalues for $1 \leq i \leq k$ are σ_i^2 . The values of σ_i are called the singular values of matrix A . If A is real, then U and V are real and hence orthogonal matrices. The value of the singular-value decomposition comes when a process is represented by a linear transformation and the elements of A , a_{ij} , are the contribution to an output i for a particular variable as input variable j . The input may be the size of a disturbance, and the output is the gain [Seborg, D. E., T. F. Edgar, and D. A. Mellichamp, *Process Dynamics and Control*, 2d ed., Wiley, New York (2004)]. If the rank is less than n , not all the variables are independent and they cannot all be controlled. Furthermore, if the singular values are widely separated, the process is sensitive to small changes in the elements of the matrix, and the process will be difficult to control.

Example Consider the following example from Noble and Daniel [*Applied Linear Algebra*, Prentice-Hall (1987)] with the MATLAB commands to do the analysis. Define the following real matrix with $m = 3$ and $n = 2$ (whose rank $k = 1$).

```
>> a = [ 1 1
        2 2
        2 2]
```

```
>> a1 = a'*a
a1 = 9    9    %(n x n or 2 x 2)
     9    9
>> a2 = a*a'
a2 = 2    4    4    %(m x m or 3 x 3)
     4    8    8
     4    8    8
>> [v,d1]=eig(a1)
v = -0.7071    0.7071    %(n x n or 2 x 2)
     0.7071    0.7071
d1 = 0    0
     0    18
>> [u,d2]=eig(a2)
u = 0.8944    0.2981    0.3333    %(m x m or 3 x 3)
    -0.4472    0.5963    0.6667
     0    -0.7454    0.6667
d2 = 0    0    0
     0    0    0
     0    0    18
```

Thus, $\sigma_1^2 = 18$ and the eigenfunctions are the rows of v and u . The second column of v is associated with the eigenvalue $\sigma_1^2 = 18$, and the third column of u is associated with the eigenvalue $\sigma_1^2 = 18$.

If A is square and nonsingular, the vector x that minimizes

$$\|Ax - b\| \quad (3-71)$$

is obtained by solving the linear equations

$$x = A^{-1}b$$

When A is not square, then the solution to

$$Ax = b$$

is

$$x = Vy$$

where $y_i = b'_i/\sigma_i$ for $i = 1, \dots, k$, $b' = U^*b$, and $y_{k+1}, y_{k+2}, \dots, y_m$ are arbitrary. The matrices U and V are those obtained in the singular-value decomposition. The solution which minimizes the norm, Eq. (3-71), is x with $y_{k+1}, y_{k+2}, \dots, y_m$ zero. These techniques can be used to monitor process variables. See Montgomery, D. C., *Introduction to Statistical Quality Control*, 4th ed., Wiley, New York (2001); Piovos, M. J., and K. A. Hoo, "Multivariate Statistics for Process Control," *IEEE Control Systems* 22(5):8 (2002).

Principal Component Analysis (PCA) PCA is used to recognize patterns in data and reduce the dimensionality of the problem. Let the matrix A now represent data with the columns of A representing different samples and the rows representing different variables. The covariance matrix is defined as

$$\text{cov}(A) = \frac{A^T A}{m-1}$$

This is just the same matrix discussed with singular value decomposition. For data analysis, though, it is necessary to adjust the columns to have zero mean by subtracting from each entry in the column the average of the column entries. Once this is done, the *loadings* are the v_i and satisfy

$$\text{cov}(A) v_i = \sigma_i^2 v_i$$

and the *score* vector u_i is given by

$$A v_i = \sigma_i u_i$$

In process analysis, the columns of A represent different measurement techniques (temperatures, pressures, etc.) and the rows represent the measurement output at different times. In that case

the columns of A are adjusted to have a zero mean and a variance of 1.0 (by dividing each entry in the column by the variance of the column). The goal is to represent the essential variation of the process with as few variables as possible. The $\mathbf{u}_i$, $\mathbf{v}_i$ pairs are arranged in descending order according to the associated σ_i . The σ_i can be thought of as the variance, and the $\mathbf{u}_i$, $\mathbf{v}_i$ pair captures the greatest amount of variation in the data. Instead of having to deal with n variables, one can capture most of the variation of the

data by using only the first few pairs. An excellent example of this is given by Wise, B. M., and B. R. Kowalski, "Process Chemometrics," Chap. 8 in *Process Analytical Chemistry*, F. McLennan and B. Kowalski (eds.), Blackie Academic & Professional, London (1995). When modeling a slurry-fed ceramic melter, they were able to capture 97 percent of the variation by using only four eigenvalues and eigenvectors, even though there were 16 variables (columns) measured.

NUMERICAL APPROXIMATIONS TO SOME EXPRESSIONS

APPROXIMATION IDENTITIES

For the following relationships the sign $\approx$ means approximately equal to, when X is small. These equations are derived by using a Taylor's series (see "Series Summation and Identities").

Approximation	Approximation
$\frac{1}{1 \pm X} \approx 1 \mp X$	$\sqrt{1 \pm X} \approx 1 \pm \frac{X}{2}$

Approximation	Approximation
$(1 \pm X)^n \approx 1 \pm nX$	$(1 \pm X)^{-n} \approx 1 \mp nX$
$(a \pm X)^2 \approx a^2 \pm 2aX$	$e^x \approx 1 + X$
$\sin X \approx X(X \text{ rad})$	$\tan X \approx X$
$\sqrt{Y(Y+X)} \approx \frac{2Y+X}{2}$	$\sqrt{Y^2+X^2} \approx Y + \frac{X^2}{2Y} \left(\frac{X}{Y} \text{ small} \right)$

NUMERICAL ANALYSIS AND APPROXIMATE METHODS

REFERENCES: Buchanan, G. R., *Schaum's Outline of Finite Element Analysis*, McGraw-Hill, New York (1995); Burden, R. L., J. D. Faires, and A. C. Reynolds, *Numerical Analysis*, 8th ed., Brooks Cole (2004); Chapra, S. C., and R. P. Canal, *Numerical Methods for Engineers*, 5th ed., McGraw-Hill, New York (2006); Finlayson, B. A., *Nonlinear Analysis in Chemical Engineering*, McGraw-Hill (1980), Ravenna Park (2003); Finlayson, B. A., and L. T. Biegler, "Mathematics in Chemical Engineering," *Ullmann's Encyclopedia of Industrial Chemistry*, vol. 20, VCH, Weinheim (2006); Gunzburger, M. D., *Finite Element Methods for Viscous Incompressible Flows*, Academic Press (1989); Kardestuncer, H., and D. H. Norrie (eds.), *Finite Element Handbook*, McGraw-Hill, New York (1987); Lau, H. T., *A Numerical Library in C for Scientists and Engineers*, CRC Press (1994); Lau, H. T., *A Numerical Library in Java for Scientists and Engineers*, CRC Press (2004); Morton, K. W., and D. F. Mayers, *Numerical Solution of Partial Differential Equations*, Cambridge University Press (1994); Press, W. H., et al., *Numerical Recipes*, Cambridge University Press, Cambridge (1986); Quarteroni, A., and A. Valli, *Numerical Approximation of Partial Differential Equations*, Springer (1997); Reddy, J. N., and D. K. Gartling, *The Finite Element Method in Heat Transfer and Fluid Dynamics*, 2d ed., CRC Press (2000); Scheid, F., *Schaum's Outline of Numerical Analysis*, 2d ed., McGraw-Hill, New York (1989); Schiesser, W. E., *The Numerical Method of Lines*, Academic Press (1991); Shampine, L., *Numerical Solution of Ordinary Differential Equations*, Chapman & Hall (1994); Zienkiewicz, O. C., R. L. Taylor, and J. Z. Zhu, *The Finite Element Method: Its Basis and Fundamentals*, vol. 1, 6th ed., Elsevier Butterworth-Heinemann (2005); Zienkiewicz, O. C., and R. L. Taylor, *The Finite Element Method: Solid Mechanics*, vol. 2, 5th ed., Butterworth-Heinemann (2000); Zienkiewicz, O. C., and R. L. Taylor, *The Finite Element Method: Fluid Mechanics*, vol. 2, 5th ed., Butterworth-Heinemann (2000).

INTRODUCTION

The goal of approximate and numerical methods is to provide convenient techniques for obtaining useful information from mathematical formulations of physical problems. Often this mathematical statement is not solvable by analytical means. Or perhaps analytic solutions are available but in a form that is inconvenient for direct interpretation. In the first case it is necessary either to attempt to approximate the problem satisfactorily by one which will be amenable to analysis, to obtain an approximate solution to the original problem by numerical means, or to use the two techniques in combination.

Numerical techniques therefore do not yield exact results in the sense of the mathematician. Since most numerical calculations are inexact, the concept of error is an important feature. The error associated with an approximate value is defined as

$$\text{True value} = \text{approximate value} + \text{error}$$

The four sources of error are as follows:

1. *Gross errors.* These result from unpredictable human, mechanical, or electrical mistakes.

2. *Round-off errors.* These are the consequence of using a number specified by m correct digits to approximate a number which requires more than m digits for its exact specification. For example, approximate the irrational number $\sqrt{2}$ by 1.414. Such errors are often present in experimental data, in which case they may be called inherent errors, due either to empiricism or to the fact that the computer dictates the number of digits. Such errors may be especially damaging in areas such as matrix inversion or the numerical solution of partial differential equations when the number of algebraic operations is extremely large.

3. *Truncation errors.* These errors arise from the substitution of a finite number of steps for an infinite sequence of steps which would yield the exact result. To illustrate this error consider the infinite series for e^{-x} : $e^{-x} = 1 - x + x^2/2 - x^3/6 + E_T(x)$, where E_T is the truncation error, $E_T = (1/24)e^{-\epsilon}x^4$, $0 < \epsilon < x$. If x is positive, ϵ is also positive. Hence $e^{-\epsilon} < 1$. The approximation $e^{-x} \approx 1 - x + x^2/2 - x^3/6$ is in error by a positive amount smaller than $(1/24)x^4$.

4. *Inherited errors.* These arise as a result of errors occurring in the previous steps of the computational algorithm.

The study of errors in a computation is related to the theory of probability. In what follows a relation for the error will be given in certain instances.

A variety of general-purpose computer programs are available commercially. Mathematica (<http://www.wolfram.com/>), Maple (<http://www.maplesoft.com/>), and Mathcad (<http://www.mathcad.com/>) all have the capability of doing symbolic manipulation so that algebraic solutions can be obtained. For example, Mathematica can solve some ordinary and partial differential equations analytically; Maple can make simple graphs and do linear algebra and simple computations; and Mathcad can do simple calculations. In this section, examples are given for the use of Matlab (<http://www.mathworks.com/>), which is a package of numerical analysis tools, some of which are accessed by simple commands and others of which are accessed by writing programs in C. Spreadsheets can also be used to solve certain problems, and these are described below too. A popular program used in chemical engineering education is Polymath (<http://www.polymath-software.com/>), which can numerically solve sets of linear or nonlinear equations, ordinary differential equations as initial-value problems, and perform data analysis and regression.

NUMERICAL SOLUTION OF LINEAR EQUATIONS

See the section entitled "Matrix Algebra and Matrix Computation."

NUMERICAL SOLUTION OF NONLINEAR EQUATIONS IN ONE VARIABLE

Special Methods for Polynomials Consider a polynomial equation of degree n :

$$P(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n = 0 \quad (3-72)$$

with real coefficients. $P(x)$ has exactly n roots, which may be real or complex. If the roots are complex, they occur in pairs with their complex conjugates.

One can obtain an upper and lower bound for the real roots by the following device: If $a_0 > 0$ in Eq. (3-72) and if in Eq. (3-72) the first negative coefficient is preceded by k coefficients which are positive or zero, and if G is the greatest of the absolute values of the negative coefficients, then each real root is less than $1 + \sqrt[k]{G/a_0}$.

Example $P(x) = x^5 + 3x^4 - 7x^3 - 40x + 2 = 0$. Here $a_0 = 1$, $G = 40$, and $k = 3$ since we must supply 0 as the coefficient for x^2 . Thus $1 + \sqrt[3]{40} \approx 4.42$ is an upper bound for the real roots. The largest real root is 2.19.

A lower bound to the real roots may be found by applying the criterion to the equation $P(-x)$.

Example $P(-x) = -x^5 + 3x^4 - 7x^3 + 40x + 2 = 0$, which is equivalent to $x^5 - 3x^4 + 7x^3 - 40x - 2 = 0$ since a_0 must be +. Then $a_0 = 1$, $G = 40$, and $k = 1$. Hence $-(1 + 40) = -41$ is a lower bound. The smallest real root is -3.41 . Thus all real roots are between -41 and 4.42 .

One last result is helpful in getting an estimate of how many positive and negative real roots there are.

Descartes Rule The number of positive real roots of a polynomial with real coefficients is either equal to the number of changes in sign v or is less than v by a positive even integer. The number of negative roots of $P(x)$ is either equal to the number of variations of sign of $P(-x)$ or is less than this by a positive even integer.

Example $f(x) = x^4 - 13x^2 + 4x - 2 = 0$ has three changes in sign; therefore, there are either three or one positive root. $f(-x) = x^4 - 13x^2 - 4x - 2$ has one change in sign. Therefore, there is one negative root. Using MATLAB, one defines the vector

$$C(i) = a_{i-1}$$

and uses the command `roots(c)` to find all the roots.

Methods for Nonlinear Equations in One Variable

Successive Substitutions Let $f(x) = 0$ be the nonlinear equation to be solved. If this is rewritten as $x = F(x)$, then an iterative scheme can be set up in the form $x_{k+1} = F(x_k)$. To start the iteration an initial guess must be obtained graphically or otherwise. The convergence or divergence of the procedure depends upon the method of writing $x = F(x)$, of which there will usually be several forms. However, if a is a root of $f(x) = 0$, and if $|F'(a)| < 1$, then for any initial approximation sufficiently close to a , the method converges to a . This process is called first order because the error in x_{k+1} is proportional to the first power of the error in x_k for large k .

One way of writing the equation is $x_{k+1} = x_k + \beta f(x_k)$. The choice of β is made such that $|1 + \beta df/dx(a)| < 1$. Convergence is guaranteed by the theorem given for simultaneous equations.

Methods of Perturbation Let $f(x) = 0$ be the equation. In general, the iterative relation is

$$x_{k+1} = x_k - [f(x_k)/\alpha_k]$$

where the iteration begins with x_0 as an initial approximation and α_k as some functional, derived below.

Newton-Raphson Procedure This variant chooses $\alpha_k = f'(x_k)$ where $f' = df/dx$ and geometrically consists of replacing the graph of

$f(x)$ by the tangent line at $x = x_k$ in each successive step. If $f'(x)$ and $f''(x)$ have the same sign throughout an interval $a \leq x \leq b$ containing the solution, with $f(a)$, $f(b)$ of opposite signs, then the process converges starting from any x_0 in the interval $a \leq x \leq b$. The process is second order.

Example $f(x) = x - 1 + \frac{(0.5)^x - 0.5}{0.3}$

$$f'(x) = 1 - 2.3105[0.5]^x$$

An approximate root (obtained graphically) is 2.

Step k	x_k	$f(x_k)$	$f'(x_k)$
0	2	0.1667	0.4224
1	1.6054	0.0342	0.2407
2	1.4632	0.0055	0.1620

Method of False Position This variant is commenced by finding x_0 and x_1 such that $f(x_0)$, $f(x_1)$ are of opposite signs. Then $\alpha_1 =$ slope of secant line joining $[x_0, f(x_0)]$ and $[x_1, f(x_1)]$ so that

$$x_2 = x_1 - \frac{x_1 - x_0}{f(x_1) - f(x_0)} f(x_1)$$

In each of the following steps α_k is the slope of the line joining $[x_k, f(x_k)]$ to the most recently determined point where $f(x)$ has the opposite sign from that of $f(x_k)$. This method is of first order. If one uses the most recently determined point (regardless of sign), the method is a second method.

Method of Wegstein This is a variant of the method of successive substitutions which forces and/or accelerates convergence. The iterative procedure $x_{k+1} = F(x_k)$ is revised by setting $\hat{x}_{k+1} = F(x_k)$ and then taking $x_{k+1} = qx_k + (1 - q)\hat{x}_{k+1}$, where q is a suitably chosen number which may be taken as constant throughout or may be adjusted at each step. Wegstein found that suitable q 's are:

Behavior of successive substitution process		Range of optimum q
Oscillatory convergence		$0 < q < 1/2$
Oscillatory divergence	without Wegstein	$1/2 < q < 1$
Monotonic convergence		$q < 0$
Monotonic divergence	without Wegstein	$1 < q$

At each step q may be calculated to give a locally optimum value by setting

$$q = \frac{\hat{x}_{k+1} - \hat{x}_k}{\hat{x}_{k+1} - 2\hat{x}_k + \hat{x}_{k-1}}$$

The Wegstein method is a secant method applied to $g(x) \equiv x - F(x)$. In Microsoft Excel, roots are found by using Goal Seek or Solver. Assign one cell to be x , put the equation for $f(x)$ in another cell, and let Goal Seek or Solver find the value of x that makes the equation cell zero. In MATLAB, the process is similar except that a function (m-file) is defined and the command `fzero('f',x0)` provides the solution x , starting from the initial guess x_0 .

METHODS FOR MULTIPLE NONLINEAR EQUATIONS

Method of Successive Substitutions Write a system of equations as

$$\alpha_i = f_i(\alpha), \quad \text{or } \alpha = f(\alpha)$$

The following theorem guarantees convergence. Let α be the solution to $\alpha = f(\alpha)$. Assume that given $h > 0$, there exists a number $0 < \mu < 1$ such that

$$\sum_{j=1}^n \left| \frac{\partial f_i}{\partial x_j} \right| \leq \mu \quad \text{for } |x_i - \alpha_i| < h, \quad i = 1, \dots, n$$

$$x_i^{k+1} = f_i(x_i^k)$$

Then $x_i^k \rightarrow \alpha_i$

as k increases [see Finlayson (2003)].

Newton-Raphson Method To solve the set of equations

$$F_i(x_1, x_2, \dots, x_n) = 0, \text{ or } F_i(\{x_j\}) = 0, \text{ or } F_i(\mathbf{x}) = 0$$

one uses a truncated Taylor series to give

$$0 = F_i(\{x^k\}) + \sum_{j=1}^n \frac{\partial F_i}{\partial x_j} \bigg|_{x^k} (x_j^{k+1} - x_j^k)$$

Thus one solves iteratively from one point to another.

$$\sum_{j=1}^n A_{ij}^k (x_j^{k+1} - x_j^k) = -F_i(\{x^k\})$$

where

$$A_{ij}^k = \frac{\partial F_i}{\partial x_j} \bigg|_{x^k}$$

This method requires solution of sets of linear equations until the functions are zero to some tolerance or the changes of the solution between iterations is small enough. Convergence is guaranteed provided the norm of the matrix $\mathbf{A}$ is bounded, $\mathbf{F}(\mathbf{x})$ is bounded for the initial guess, and the second derivative of $\mathbf{F}(\mathbf{x})$ with respect to all variables is bounded. See Finlayson (2003).

Method of Continuity (Homotopy) In the case of n equations in n unknowns, when n is large, determining the approximate solution may involve considerable effort. In such a case the method of continuity is admirably suited for use on digital computers. It consists basically of the introduction of an extra variable into the n equations

$$f_i(x_1, x_2, \dots, x_n) = 0 \quad i = 1, \dots, n \quad (3-73)$$

and replacing them by

$$f_i(x_1, x_2, \dots, x_n, \lambda) = 0 \quad i = 1, \dots, n \quad (3-74)$$

where λ is introduced in such a way that the functions (3-74) depend in a simple way upon λ and reduce to an easily solvable system for $\lambda = 0$ and to the original equations (3-73) for $\lambda = 1$. A system of ordinary differential equations, with independent variable λ , is then constructed by differentiating Eqs. (3-74) with respect to λ . There results

$$\sum_{j=1}^n \frac{\partial f_i}{\partial x_j} \frac{dx_j}{d\lambda} + \frac{\partial f_i}{\partial \lambda} = 0 \quad (3-75)$$

where $x_1, \dots, x_n$ are considered as functions of λ . Equations (3-75) are integrated, with initial conditions obtained from Eqs. (3-74) with $\lambda = 0$, from $\lambda = 0$ to $\lambda = 1$. If the solution can be continued to $\lambda = 1$, the values of $x_1, \dots, x_n$ for $\lambda = 1$ will be a solution of the original equations. If the integration becomes infinite, the parameter λ must be introduced in a different fashion. Integration of the differential equations (which are usually nonlinear in λ) may be accomplished by using techniques described under "Numerical Solution of Ordinary Differential Equations."

Other Methods Other methods can be found in the literature. See Chan, T. F. C., and H. B. Keller, *SIAM J. Sci. Stat. Comput.* 3:173-194 (1982); Seader, J. D., "Computer Modeling of Chemical Processes," *AIChE Monograph Series* 81(15) (1986).

INTERPOLATION AND FINITE DIFFERENCES

Linear Interpolation If a function $f(x)$ is approximately linear in a certain range, then the ratio

$$\frac{f(x_1) - f(x_0)}{x_1 - x_0} = f[x_0, x_1]$$

is approximately independent of x_0, x_1 in the range. The linear approximation to the function $f(x)$, $x_0 < x < x_1$ then leads to the interpolation formula

$$\begin{aligned} f(x) &\approx f(x_0) + (x - x_0)f[x_0, x_1] \\ &\approx f(x_0) + \frac{x - x_0}{x_1 - x_0} [f(x_1) - f(x_0)] \\ &\approx \frac{1}{x_1 - x_0} [(x_1 - x)f(x_0) - (x_0 - x)f(x_1)] \end{aligned}$$

Divided Differences of Higher Order and Higher-Order Interpolation The first-order divided difference $f[x_0, x_1]$ was defined previously. Divided differences of second and higher order are defined iteratively by

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$$

⋮

$$f[x_0, x_1, \dots, x_k] = \frac{f[x_1, \dots, x_k] - f[x_0, x_1, \dots, x_{k-1}]}{x_k - x_0}$$

and a convenient form for computational purposes is

$$f[x_0, x_1, \dots, x_k] = \sum_{j=0}^k \frac{f(x_j)}{(x_j - x_0)(x_j - x_1) \cdots (x_j - x_{k-1})}$$

for any $k \geq 0$, where the ' means that the term $(x_j - x_j)$ is omitted in the denominator. For example,

$$f[x_0, x_1, x_2] = \frac{f(x_0)}{(x_0 - x_1)(x_0 - x_2)} + \frac{f(x_1)}{(x_1 - x_0)(x_1 - x_2)} + \frac{f(x_2)}{(x_2 - x_0)(x_2 - x_1)}$$

If the accuracy afforded by a linear approximation is inadequate, a generally more accurate result may be based upon the assumption that $f(x)$ may be approximated by a polynomial of degree 2 or higher over certain ranges. This assumption leads to Newton's fundamental interpolation formula with divided differences

$$\begin{aligned} f(x) &\approx f(x_0) + (x - x_0)f[x_0, x_1] + (x - x_0)(x - x_1)f[x_0, x_1, x_2] \\ &\quad + \cdots + (x - x_0)(x - x_1) \cdots (x - x_{n-1})f[x_0, x_1, \dots, x_n] + E_n(x) \end{aligned}$$

where

$$E_n(x) = \text{error} = \frac{1}{(n+1)!} f^{n+1}(\epsilon)\pi(x)$$

where minimum $(x_0, \dots, x) < \epsilon < \text{maximum } (x_0, x_1, \dots, x_n, x)$ and $\pi(x) = (x - x_0)(x - x_1) \cdots (x - x_n)$. In order to use the previous equation most effectively one may first form a divided-difference table. For example, for third-order interpolation the difference table is

$$\begin{array}{l|l} x_0 & f(x_0) \\ x_1 & f(x_1) \\ x_2 & f(x_2) \\ x_3 & f(x_3) \end{array} \begin{array}{l} \nearrow f[x_0, x_1] \\ \nearrow f[x_1, x_2] \\ \nearrow f[x_2, x_3] \end{array} \begin{array}{l} \nearrow f[x_0, x_1, x_2] \\ \nearrow f[x_1, x_2, x_3] \end{array} \nearrow f[x_0, x_1, x_2, x_3]$$

where each entry is given by taking the difference between diagonally adjacent entries to the left, divided by the abscissas corresponding to the ordinates intercepted by the diagonals passing through the calculated entry.

Equally Spaced Forward Differences If the ordinates are equally spaced, i.e., $x_j - x_{j-1} = \Delta x$ for all j , then the first differences are denoted by $\Delta f(x_0) = f(x_1) - f(x_0)$ or $\Delta y_0 = y_1 - y_0$, where $y = f(x)$. The differences of these first differences, called second differences, are denoted by $\Delta^2 y_0, \Delta^2 y_1, \dots, \Delta^2 y_n$. Thus

$$\Delta^2 y_0 = \Delta y_1 - \Delta y_0 = y_2 - y_1 - y_1 + y_0 = y_2 - 2y_1 + y_0$$

And in general

$$\Delta^j y_0 = \sum_{n=0}^j (-1)^n \binom{j}{n} y_{j-n}$$

where $\binom{j}{n} = \frac{j!}{n!(j-n)!}$ = binomial coefficients.

If the ordinates are equally spaced,

$$\begin{aligned} x_{n+1} - x_n &= \Delta x \\ y_n &= y(x_n) \end{aligned}$$

then the first and second differences are denoted by

$$\begin{aligned} \Delta y_n &= y_{n+1} - y_n \\ \Delta^2 y_n &= \Delta y_{n+1} - \Delta y_n = y_{n+2} - 2y_{n+1} + y_n \end{aligned}$$

A new variable is defined

$$\alpha = \frac{x - x_0}{\Delta x}$$

and the finite interpolation formula through the points $y_0, y_1, \dots, y_n$ is written as follows:

$$y_\alpha = y_0 + \alpha \Delta y_0 + \frac{\alpha(\alpha-1)}{2!} \Delta^2 y_0 + \dots + \frac{\alpha(\alpha-1) \dots (\alpha-n+1)}{n!} \Delta^n y_0 \quad (3-76)$$

Keeping only the first two terms gives a straight line through (x_0, y_0) – (x_1, y_1) ; keeping the first three terms gives a quadratic function of position going through those points plus (x_2, y_2) . The value $\alpha = 0$ gives $x = x_0$; $\alpha = 1$ gives $x = x_1$, and so on.

Equally Spaced Backward Differences Backward differences are defined by

$$\begin{aligned} \nabla y_n &= y_n - y_{n-1} \\ \nabla^2 y_n &= \nabla y_n - \nabla y_{n-1} = y_n - 2y_{n-1} + y_{n-2} \end{aligned}$$

The interpolation polynomial of order n through the points $y_0, y_{-1}, \dots, y_{-n}$ is

$$y_\alpha = y_0 + \alpha \nabla y_0 + \frac{\alpha(\alpha+1)}{2!} \nabla^2 y_0 + \dots + \frac{\alpha(\alpha+1) \dots (\alpha+n-1)}{n!} \nabla^n y_0$$

The value of $\alpha = 0$ gives $x = x_0$; $\alpha = -1$ gives $x = x_{-1}$, and so on. Alternatively, the interpolation polynomial of order n through the points $y_1, y_0, y_{-1}, \dots, y_{-n}$ is

$$\begin{aligned} y_\alpha &= y_1 + (\alpha-1) \nabla y_1 + \frac{\alpha(\alpha-1)}{2!} \nabla^2 y_1 \\ &\quad + \dots + \frac{(\alpha-1) \alpha (\alpha+1) \dots (\alpha+n-2)}{n!} \nabla^n y_1 \end{aligned} \quad (3-77)$$

Now $\alpha = 1$ gives $x = x_1$; $\alpha = 0$ gives $x = x_0$.

Central Differences The central difference denoted by

$$\begin{aligned} \delta f(x) &= f\left(x + \frac{h}{2}\right) - f\left(x - \frac{h}{2}\right) \\ \delta^2 f(x) &= \delta f\left(x + \frac{h}{2}\right) - \delta f\left(x - \frac{h}{2}\right) = f(x+h) - 2f(x) + f(x-h) \\ \delta^n f(x) &= \delta^{n-1} f\left(x + \frac{h}{2}\right) - \delta^{n-1} f\left(x - \frac{h}{2}\right) \end{aligned}$$

is useful for calculating at the interior points of tabulated data.

Lagrange Interpolation Formulas A global polynomial is defined over the entire region of space

$$P_m(x) = \sum_{j=0}^m c_j x^j$$

This polynomial is of degree m (highest power is x^m) and order $m+1$ ($m+1$ parameters $\{c_j\}$). If we are given a set of $m+1$ points

$$y_1 = f(x_1), y_2 = f(x_2), \dots, y_{m+1} = f(x_{m+1})$$

then Lagrange's formula gives a polynomial of degree m that goes through the $m+1$ points:

$$\begin{aligned} P_m(x) &= \frac{(x-x_2)(x-x_3) \dots (x-x_{m+1})}{(x_1-x_2)(x_1-x_3) \dots (x_1-x_{m+1})} y_1 \\ &\quad + \frac{(x-x_1)(x-x_3) \dots (x-x_{m+1})}{(x_2-x_1)(x_2-x_3) \dots (x_2-x_{m+1})} y_2 + \dots \\ &\quad + \frac{(x-x_1)(x-x_2) \dots (x-x_{m+1})}{(x_{m+1}-x_1)(x_{m+1}-x_2) \dots (x_{m+1}-x_m)} y_{m+1} \end{aligned}$$

Note that each coefficient of y_j is a polynomial of degree m that vanishes at the points $\{x_j\}$ (except for one value of j) and takes the value of

1.0 at that point:

$$P_m(x_j) = y_j, \quad j = 1, 2, \dots, m+1$$

If the function $f(x)$ is known, the error in the approximation is, per Anderson, E., et al., LAPACK Users' Guide, SIAM(1992),

$$|\text{error}(x)| \leq \frac{|x_{m+1} - x_1|^{m+1}}{(m+2)!} \max_{x_1 \leq x \leq x_{m+1}} |f^{(m+2)}(x)|$$

The evaluation of $P_m(x)$ at a point other than at the defining points can be made with Neville's algorithm [Press et al. (1986)]. Let P_1 be the value at x of the unique function passing through the point (x_1, y_1) ; or $P_1 = y_1$. Let P_{12} be the value at x of the unique polynomial passing through the points x_1 and x_2 . Likewise, $P_{ijk \dots r}$ is the unique polynomial passing through the points $x_i, x_j, x_k, \dots, x_r$. Then use the table

x_1	$y_1 = P_1$			
		P_{12}		
x_2	$y_2 = P_2$		P_{123}	
		P_{23}		P_{1234}
x_3	$y_3 = P_3$		P_{234}	
		P_{34}		
x_4	$y_4 = P_4$			

These entries are defined using

$$P_{i(i+1) \dots (i+m)} = \frac{(x - x_{i+m}) P_{i(i+1) \dots (i+m-1)} + (x_i - x) P_{(i+1)(i+2) \dots (i+m)}}{x_i - x_{i+m}}$$

For example, consider P_{1234} . The terms on the right-hand side involve P_{123} and P_{234} . The "parents," P_{123} and P_{234} , already agree at points 2 and 3. Here $i = 1$, $m = 3$; thus, the parents agree at $x_{i+1}, \dots, x_{i+m-1}$ already. The formula makes $P_{i(i+1) \dots (i+m)}$ agree with the function at the additional points x_{i+m} and x_i . Thus, $P_{i(i+1) \dots (i+m)}$ agrees with the function at all the points $\{x_i, x_{i+1}, \dots, x_{i+m}\}$.

Spline Functions Splines are functions that match given values at the points $x_1, \dots, x_{NT}$ and have continuous derivatives up to some order at the knots, or the points $x_2, \dots, x_{NT-1}$. Cubic splines are most common; see de Boor, C., *A Practical Guide to Splines*, Springer (1978). The function is represented by a cubic polynomial within each interval (x_i, x_{i+1}) and has continuous first and second derivatives at the knots. Two more conditions can be specified arbitrarily. These are usually the second derivatives at the two end points, which are commonly taken as zero; this gives the natural cubic splines.

Take $y_i = y(x_i)$ at each of the points x_i , and let $\Delta x_i = x_{i+1} - x_i$. Then, in the interval (x_i, x_{i+1}) , the function is represented as a cubic polynomial.

$$C_i(x) = a_{0i} + a_{1i}x + a_{2i}x^2 + a_{3i}x^3$$

The interpolating function takes on specified values at the knots and has continuous first and second derivatives at the knots. Within the i th interval, the function is

$$\begin{aligned} C_i(x) &= C_i(x_i) + C'_i(x_i)(x - x_i) + C''_i(x_i) \frac{(x - x_i)^2}{2\Delta x_i} \\ &\quad + [C''_i(x_{i+1}) - C''_i(x_i)] \frac{(x - x_i)^3}{6\Delta x_i} \end{aligned}$$

where $C_i(x_i) = y_i$. The second derivative $C''_i(x_i) = y''_i$ is found by solving the following tridiagonal system of equations:

$$y''_{i-1} \Delta x_{i-1} + y''_i 2(\Delta x_{i-1} + \Delta x_i) + y''_{i+1} \Delta x_i = -6 \left(\frac{y_i - y_{i-1}}{\Delta x_{i-1}} - \frac{y_{i+1} - y_i}{\Delta x_i} \right)$$

Since the continuity conditions apply only for $i = 2, \dots, NT-1$, we have only $NT-2$ conditions for the NT values of y''_i . Two additional conditions are needed, and these are usually taken as the value of the second derivative at each end of the domain, y''_1, y''_{NT} . If these values are zero, we get the natural cubic splines; they can also be set to achieve some other purpose, such as making the first derivative match some desired condition at the two ends. With these values taken as zero in the natural cubic spline, we have a $NT-2$ system of tridiagonal equations, which is easily solved. Once the second derivatives are

known at each of the knots, the first derivatives are given by

$$y'_i = \frac{y_{i+1} - y_i}{\Delta x_i} - y''_i \frac{\Delta x_i}{3} - y''_{i+1} \frac{\Delta x_i}{6}$$

The function itself is then known within each element.

NUMERICAL DIFFERENTIATION

Numerical differentiation should be avoided whenever possible, particularly when data are empirical and subject to appreciable observation errors. Errors in data can affect numerical derivatives quite strongly; i.e., differentiation is a roughening process. When such a calculation must be made, it is usually desirable first to smooth the data to a certain extent.

Use of Interpolation Formula If the data are given over equidistant values of the independent variable x , an interpolation formula such as the Newton formula [Eq. (3-76) or (3-77)] may be used and the resulting formula differentiated analytically. If the independent variable is not at equidistant values, then Lagrange's formulas must be used. By differentiating three-point Lagrange interpolation formulas the following differentiation formulas result for equally spaced tabular points:

Three-Point Formulas Let x_0, x_1, x_2 be the three points.

$$f'(x_0) = \frac{1}{2h} [-3f(x_0) + 4f(x_1) - f(x_2)] + \frac{h^2}{3} f'''(\epsilon)$$

$$f'(x_1) = \frac{1}{2h} [-f(x_0) + f(x_2)] - \frac{h^2}{6} f'''(\epsilon)$$

$$f'(x_2) = \frac{1}{2h} [f(x_0) - 4f(x_1) + 3f(x_2)] + \frac{h^2}{3} f'''(\epsilon)$$

where the last term is an error term $\min_j x_j < \epsilon < \max_j x_j$.

Smoothing Techniques These techniques involve the approximation of the tabular data by a least-squares fit of the data by using some known functional form, usually a polynomial (for the concept of least squares see "Statistics"). In place of approximating $f(x)$ by a single least-squares polynomial of degree n over the entire range of the tabulation, it is often desirable to replace each tabulated value by the value taken on by a least-squares polynomial of degree n relevant to a subrange of $2M + 1$ points centered, when possible, at the point for which the entry is to be modified. Thus each smoothed value replaces a tabulated value. Let $f_j = f(x_j)$ be the tabular points and $y_j =$ smoothed values.

First-Degree Least Squares with Three Points

$$y_0 = 1/6[5f_0 + 2f_1 - f_2]$$

$$y_1 = 1/3[f_0 + f_1 + f_2]$$

$$y_2 = 1/6[-f_0 + 2f_1 + 5f_2]$$

The derivatives at all the points are

$$f'_0 = f'_1 = f'_2 = \frac{1}{2}h[y_2 - y_1]$$

Second-Degree Least Squares with Five Points For five evenly spaced points $x_{-2}, x_{-1}, x_0, x_1,$ and x_2 (separated by distance h) and their ordinates $f_{-2}, f_{-1}, f_0, f_1,$ and f_2 , assume a parabola is fit by least squares. Then the derivative at the center point is

$$f'_0 = 1/10h[-2f_{-2} - f_{-1} + f_1 + 2f_2]$$

The derivatives at the other points are

$$f'_{-2} = 1/70h[-54f_{-2} + 13f_{-1} + 40f_0 + 27f_1 - 26f_2]$$

$$f'_{-1} = 1/70h[-34f_{-2} + 3f_{-1} + 20f_0 + 17f_1 - 6f_2]$$

$$f'_1 = 1/70h[6f_{-2} - 17f_{-1} - 20f_0 - 3f_1 + 34f_2]$$

$$f'_2 = 1/70h[26f_{-2} - 27f_{-1} - 40f_0 - 13f_1 + 54f_2]$$

Numerical Derivatives The results given above can be used to obtain numerical derivatives when solving problems on the computer,

in particular for the Newton-Raphson method and homotopy methods. Suppose one has a program, subroutine, or other function evaluation device that will calculate f given x . One can estimate the value of the first derivative at x_0 using

$$\left. \frac{df}{dx} \right|_{x_0} \approx \frac{f[x_0(1 + \epsilon)] - f[x_0]}{\epsilon x_0}$$

(a first-order formula) or

$$\left. \frac{df}{dx} \right|_{x_0} \approx \frac{f[x_0(1 + \epsilon)] - f[x_0(1 - \epsilon)]}{2\epsilon x_0}$$

(a second-order formula). The value of ϵ is important; a value of 10^{-6} is typical, but smaller or larger values may be necessary depending on the computer precision and the application. One must also be sure that the value of x_0 is not zero and use a different increment in that case.

NUMERICAL INTEGRATION (QUADRATURE)

A multitude of formulas have been developed to accomplish numerical integration, which consists of computing the value of a definite integral from a set of numerical values of the integrand.

Newton-Cotes Integration Formulas (Equally Spaced Ordinates) for Functions of One Variable The definite integral $\int_a^b f(x) dx$ is to be evaluated.

Trapezoidal Rule This formula consists of subdividing the interval $a \leq x \leq b$ into n subintervals a to $a + h$, $a + h$ to $a + 2h$, $\dots$ and replacing the graph of $f(x)$ by the result of joining the ends of adjacent ordinates by line segments. If $f_j = f(x_j) = f(a + jh)$, $f_0 = f(a)$, $f_n = f(b)$, the integration formula is

$$\int_a^b f(x) dx = \frac{h}{2} [f_0 + 2f_1 + 2f_2 + \dots + 2f_{n-1} + f_n] + E_n$$

$$\text{where } |E_n| = \frac{nh^3}{12} |f'''(\epsilon)| = \frac{(b-a)^3}{12n^2} |f'''(\epsilon)| \quad a < \epsilon < b$$

This procedure is not of high accuracy. However, if $f'''(x)$ is continuous in $a < x < b$, the error goes to zero as $1/n^2$, $n \rightarrow \infty$.

Parabolic Rule (Simpson's Rule) This procedure consists of subdividing the interval $a < x < b$ into $n/2$ subintervals, each of length $2h$, where n is an even integer. By using the notation as above the integration formula is

$$\int_a^b f(x) dx = \frac{h}{3} [f_0 + 4f_1 + 2f_2 + 4f_3 + \dots + 4f_{n-3} + 2f_{n-2} + 4f_{n-1} + f_n] + E_n$$

$$\text{where } |E_n| = \frac{nh^5}{180} |f^{(IV)}(\epsilon)| = \frac{(b-a)^5}{180n^4} |f^{(IV)}(\epsilon)| \quad a < \epsilon < b$$

This method approximates $f(x)$ by a parabola on each subinterval. This rule is generally more accurate than the trapezoidal rule. It is the most widely used integration formula.

Gaussian Quadrature Gaussian quadrature provides a highly accurate formula based on irregularly spaced points, but the integral needs to be transformed onto the interval 0 to 1.

$$x = a + (b-a)u, \quad dx = (b-a)du$$

$$\int_a^b f(x) dx = (b-a) \int_0^1 f(u) du$$

$$\int_0^1 f(u) du = \sum_{i=1}^m W_i f(u_i)$$

The quadrature is exact when f is a polynomial of degree $2m - 1$ in x . Because there are m weights and m Gauss points, we have $2m$ parameters that are chosen to exactly represent a polynomial of degree

$2m - 1$, which has $2m$ parameters. The Gauss points and weights are given in the table.

Gaussian Quadrature Points and Weights

m	u_i	W_i
2	0.21132 48654	0.50000 00000
	0.78867 51346	0.50000 00000
3	0.11270 16654	0.27777 77778
	0.50000 00000	0.44444 44445
	0.88729 83346	0.27777 77778
4	0.06943 18442	0.17392 74226
	0.33000 94783	0.32607 25774
	0.66999 05218	0.32607 25774
	0.93056 81558	0.17392 74226
5	0.04691 00771	0.11846 34425
	0.23076 53450	0.23931 43353
	0.50000 00000	0.28444 44444
	0.76923 46551	0.23931 43353
	0.95308 99230	0.11846 34425

Example Calculate the value of the following integral.

$$I = \int_0^1 e^{-x} \sin x \, dx \quad (3-78)$$

Using the Gaussian quadrature formulas gives the following values for various values of m . Clearly, three internal points, requiring evaluation of the integrand at only three points, gives excellent results.

m	I
2	0.24609643062368
3	0.24583487736505
4	0.24583700444293
5	0.24583700700700

Romberg's Method Romberg's method uses extrapolation techniques to improve the answer [Press et al. (1986)]. If we let I_1 be the value of the integral obtained using interval size $h = \Delta x$, and I_2 be the value of I obtained when using interval size $h/2$, and I_0 the true value of I , then the error in a method is approximately h^m , or

$$I_1 \approx I_0 + ch^m$$

$$I_2 \approx I_0 + c\left(\frac{h}{2}\right)^m$$

Replacing the $\approx$ by an equality (an approximation) and solving for c and I_0 gives

$$I_0 = \frac{2^m I_2 - I_1}{2^m - 1}$$

This process can also be used to obtain $I_1, I_2, \dots$, by halving h each time, and then calculating new estimates from each pair, calling them $J_1, J_2, \dots$; that is, in the formula above, replace I_0 with J_1 . The formulas are reapplied for each pair of J to obtain $K_1, K_2, \dots$. The process continues until the required tolerance is obtained.

$$\begin{array}{cccc} I_1 & I_2 & I_3 & I_4 \\ J_1 & J_2 & J_3 & \\ & K_1 & K_2 & \\ & & L_1 & \end{array}$$

Romberg's method is most useful for a low-order method (small m) because significant improvement is then possible.

Example Evaluate the same integral (3-78) by using the trapezoid rule and then apply the Romberg method. Use 11, 21, 41, and 81 points with $m = 2$. To achieve six-digit accuracy, any result from J_2 through L_1 is suitable, even though the base results (I_1 through I_4) are not that accurate.

$I_1 = 0.24491148225216$	$I_2 = 0.24560560017077$	$I_3 = 0.24577915369183$	$I_4 = 0.24582254357310$
$J_1 = 0.24583697281030$	$J_2 = 0.24583700486552$	$J_3 = 0.24583700686685$	
	$K_1 = 0.24583701555059$	$K_2 = 0.24583700753396$	
		$L_1 = 0.24583700486175$	

Computer Methods These methods are easily programmed in a spreadsheet program such as Microsoft Excel. In MATLAB, the trapezoid rule can be calculated by using the command `trapz(x,y)`, where x is a vector of x values x_i and y is a vector of values $y(x_i)$. Alternatively, use the commands

$$F = @(x) \exp(-x) .* \sin(x)$$

$$Q = \text{quad}(F, 0, 1)$$

$I_1 = 0.24491148225216$	$I_2 = 0.24560560017077$	$I_3 = 0.24577915369183$	$I_4 = 0.24582254357310$
$J_1 = 0.24583697281030$	$J_2 = 0.24583700486552$	$J_3 = 0.24583700686685$	
	$K_1 = 0.24583701555059$	$K_2 = 0.24583700753396$	
		$L_1 = 0.24583700486175$	

Singularities When the integrand has singularities, a variety of techniques can be tried. The integral may be divided into one part that can be integrated analytically near the singularity and another part that is integrated numerically. Sometimes a change of argument allows analytical integration. Series expansion might be helpful, too. When the domain is infinite, it is possible to use Gauss-Legendre or Gauss-Hermite quadrature. Also a transformation can be made. For example, let $u = 1/x$ and then

$$\int_a^b f(x) \, dx = \int_{1/b}^{1/a} \frac{1}{u^2} f\left(\frac{1}{u}\right) du \quad ab > 0$$

Two-Dimensional Formula Two-dimensional integrals can be calculated by breaking down the integral into one-dimensional integrals.

$$\int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dx \, dy = \int_a^b G(x) \, dx$$

$$G(x) = \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy$$

Gaussian quadrature can also be used in two dimensions, provided the integration is on a square or can be transformed to one. (Domain transformations might be used to convert the domain to a square.)

$$\int_0^1 \int_0^1 f(x, y) \, dx \, dy = \sum_{i=1}^{mx} W_{xi} \sum_{j=1}^{my} W_{yj} f(x_i, y_j)$$

NUMERICAL SOLUTION OF ORDINARY DIFFERENTIAL EQUATIONS AS INITIAL VALUE PROBLEMS

A differential equation for a function that depends on only one variable, often time, is called an ordinary differential equation. The general solution to the differential equation includes many possibilities; the boundary or initial conditions are needed to specify which of those are desired. If all conditions are at one point, then the problem is an initial value problem and can be integrated from that point on. If some of the conditions are available at one point and others at another point, then the ordinary differential equations become two-point boundary value problems, which are treated in the next section. Initial value problems as ordinary differential equations arise in control of lumped parameter models, transient models of stirred tank reactors, and in all models where there are no spatial gradients in the unknowns.

A higher-order differential equation

$$y^{(n)} + F(y^{(n-1)}, y^{(n-2)}, \dots, y', y) = 0$$

with initial conditions

$$G_i(y^{(n-1)}(0), y^{(n-2)}(0), \dots, y(0), y'(0)) = 0, \quad i = 1, \dots, n$$

can be converted into a set of first-order equations using

$$y_i \equiv y^{(i-1)} = \frac{d^{(i-1)}y}{dt^{(i-1)}} = \frac{d}{dt} y^{(i-2)} = \frac{dy_{i-1}}{dt}$$

The higher-order equation can be written as a set of first-order equations.

$$\frac{dy_1}{dt} = y_2$$

$$\begin{aligned}\frac{dy_2}{dt} &= y_3 \\ \frac{dy_3}{dt} &= y_4 \\ &\vdots \\ \frac{dy_n}{dt} &= -F(y_{n-1}, y_{n-2}, \dots, y_2, y_1)\end{aligned}$$

The initial conditions would have to be specified for variables $y_1(0), \dots, y_n(0)$, or equivalently $y(0), \dots, y^{(n-1)}(0)$. The set of equations is then written as

$$\frac{dy}{dt} = \mathbf{f}(\mathbf{y}, t)$$

All the methods in this section are described for a single equation; the methods apply to multiple equations.

Euler's method is first-order.

$$y^{n+1} = y^n + \Delta t f(y^n)$$

and errors are proportional to Δt . The second-order Adams-Bashforth method is

$$y^{n+1} = y^n + \frac{\Delta t}{2} [3f(y^n) - f(y^{n-1})]$$

Errors are proportional to Δt^2 , and high-order methods are available. Notice that the higher-order explicit methods require knowing the solution (or the right-hand side) evaluated at times in the past. Since these were calculated to get to the current time, this presents no problem except for starting the problem. Then it may be necessary to use Euler's method with a very small step size for several steps in order to generate starting values at a succession of time points. The error terms, order of the method, function evaluations per step, and stability limitations are listed in Finlayson (2003). The advantage of the high-order Adams-Bashforth method is that it uses only one function evaluation per step yet achieves high-order accuracy. The disadvantage is the necessity of using another method to start.

Runge-Kutta methods are explicit methods that use several function evaluations for each time step. Runge-Kutta methods are traditionally written for $f(t, y)$. The first-order Runge-Kutta method is Euler's method. A second-order Runge-Kutta method is

$$y^{n+1} = y^n + \frac{\Delta t}{2} [f^n + f(t^n + \Delta t, y^n + \Delta t f^n)]$$

while the midpoint scheme is also a second-order Runge-Kutta method.

$$y^{n+1} = y^n + \Delta t f\left(t^n + \frac{\Delta t}{2}, y^n + \frac{\Delta t}{2} f^n\right)$$

A popular fourth-order Runge-Kutta method is the Runge-Kutta-Feldberg formulas, which have the property that the method is fourth-order but achieves fifth-order accuracy. The popular integration package RKF45 is based on this method.

$$\begin{aligned}k_1 &= \Delta t f(t^n, y^n) \\ k_2 &= \Delta t f\left(t^n + \frac{\Delta t}{4}, y^n + \frac{k_1}{4}\right) \\ k_3 &= \Delta t f\left(t^n + \frac{3}{8}\Delta t, y^n + \frac{3}{32}k_1 + \frac{9}{32}k_2\right) \\ k_4 &= \Delta t f\left(t^n + \frac{12}{13}\Delta t, y^n + \frac{1932}{2197}k_1 - \frac{7200}{2197}k_2 + \frac{7296}{2197}k_3\right) \\ k_5 &= \Delta t f\left(t^n + \Delta t, y^n + \frac{439}{216}k_1 - 8k_2 + \frac{3680}{513}k_3 - \frac{845}{4104}k_4\right) \\ k_6 &= \Delta t f\left(t^n + \frac{\Delta t}{2}, y^n - \frac{8}{27}k_1 + 2k_2 - \frac{3544}{2565}k_3 + \frac{1859}{4104}k_4 - \frac{11}{40}k_5\right)\end{aligned}$$

$$\begin{aligned}y^{n+1} &= y^n + \frac{25}{216}k_1 + \frac{1408}{2565}k_2 + \frac{2197}{4104}k_3 - \frac{1}{5}k_5 \\ z^{n+1} &= y^n + \frac{16}{135}k_1 + \frac{6656}{12825}k_2 + \frac{28561}{56430}k_3 - \frac{9}{50}k_5 + \frac{2}{55}k_6\end{aligned}$$

The value of $y^{n+1} - z^{n+1}$ is an estimate of the error in y^{n+1} and can be used in step-size control schemes.

Usually one would use a high-order method to achieve high accuracy. The Runge-Kutta-Feldberg method is popular because it is high order and does not require a starting method (as does an Adams-Bashforth method). However, it does require four function evaluations per time step, or four times as many as a fourth-order Adams-Bashforth method. For problems in which the function evaluations are a significant portion of the calculation time, this might be important. Given the speed and availability of desktop computers, the efficiency of the methods is most important only for very large problems that are going to be solved many times. For other problems, the most important criterion for choosing a method is probably the time the user spends setting up the problem.

The stability limits for the explicit methods are based on the largest eigenvalue of the linearized system of equations

$$\frac{dy_i}{dt} = \sum_{j=1}^n A_{ij} y_j, \quad A_{ij} = \left. \frac{\partial f_i}{\partial y_j} \right|_y$$

For linear problems, the eigenvalues do not change, so that the stability and oscillation limits must be satisfied for *every* eigenvalue of the matrix $\mathbf{A}$. When solving nonlinear problems, the equations are linearized about the solution at the local time, and the analysis applies for small changes in time, after which a new analysis about the new solution must be made. Thus, for nonlinear problems, the eigenvalues keep changing, and the largest stable time step changes, too. The stability limits are:

Euler method, $\lambda \Delta t \leq 2$

Runge-Kutta, 2nd order, $\lambda \Delta t < 2$

Runge-Kutta-Feldberg, $\lambda \Delta t < 3.0$

Richardson extrapolation can be used to improve the accuracy of a method. Suppose we step forward one step Δt with a p th-order method. Then redo the problem, this time stepping forward from the same initial point, but in two steps of length $\Delta t/2$, thus ending at the same point. Call the solution of the one-step calculation y_1 and the solution of the two-step calculation y_2 . Then an improved solution at the new time is given by

$$y = \frac{2^p y_2 - y_1}{2^p - 1}$$

This gives a good estimate provided Δt is small enough that the method is truly convergent with order p . This process can also be repeated in the same way Romberg's method was used for quadrature.

The error term in the various methods can be used to deduce a step size that will give a user-specified accuracy. Most packages today are based on a user-specified tolerance; the step-size is changed during the calculation to achieve that accuracy. The accuracy itself is not guaranteed, but it improves as the tolerance is decreased.

Implicit Methods By using different interpolation formulas involving y^{n+1} , it is possible to derive implicit integration methods. Implicit methods result in a nonlinear equation to be solved for y^{n+1} so that iterative methods must be used. The backward Euler method is a first-order method.

$$y^{n+1} = y^n + \Delta t f(y^{n+1})$$

Errors are proportional to Δt for small Δt . The trapezoid rule is a second-order method.

$$y^{n+1} = y^n + \frac{\Delta t}{2} [f(y^n) + f(y^{n+1})]$$

Errors are proportional to Δt^2 for small Δt . When the trapezoid rule is used with the finite difference method for solving partial differential equations, it is called the Crank-Nicolson method. The implicit methods are stable for any step size but do require the solution of a set of nonlinear equations, which must be solved iteratively. The set of equations can be solved using the successive substitution method

or Newton-Raphson method. See Bogacki, M. B., K. Alejski, and J. Szymanski, *Comp. Chem. Eng.* **13**: 1081–1085 (1989) for an application to dynamic distillation problems.

The best packages for stiff equations (see below) use Gear's backward difference formulas. The formulas of various orders are [Gear, G. W., *Numerical Initial Value Problems in Ordinary Differential Equations*, Prentice-Hall, Englewood Cliffs, N.J. (1971)]

$$\begin{aligned}(1) \quad y^{n+1} &= y^n + \Delta t f(y^{n+1}) \\(2) \quad y^{n+1} &= \frac{4}{3} y^n - \frac{1}{3} y^{n-1} + \frac{2}{3} \Delta t f(y^{n+1}) \\(3) \quad y^{n+1} &= \frac{18}{11} y^n - \frac{9}{11} y^{n-1} + \frac{2}{11} y^{n-2} + \frac{6}{11} \Delta t f(y^{n+1}) \\(4) \quad y^{n+1} &= \frac{48}{25} y^n - \frac{36}{25} y^{n-1} + \frac{16}{25} y^{n-2} - \frac{3}{25} y^{n-3} + \frac{12}{25} \Delta t f(y^{n+1}) \\(5) \quad y^{n+1} &= \frac{300}{137} y^n - \frac{300}{137} y^{n-1} + \frac{200}{137} y^{n-2} - \frac{75}{137} y^{n-3} + \frac{12}{137} y^{n-4} \\&\quad + \frac{60}{137} \Delta t f(y^{n+1})\end{aligned}$$

Stiffness The concept of stiffness is described for a system of linear equations.

$$\frac{dy}{dt} = \mathbf{A}y$$

Let λ_i be the eigenvalues of the matrix $\mathbf{A}$. The stiffness ratio is defined as

$$\text{SR} = \frac{\max_i |\text{Re}(\lambda_i)|}{\min_i |\text{Re}(\lambda_i)|} \quad (3-79)$$

SR = 20 is not stiff, SR = 10^3 is stiff, and SR = 10^6 is very stiff. If the problem is nonlinear, then the solution is expanded about the current state.

$$\frac{dy_i}{dt} = f_i[y(t^n)] + \sum_{j=1}^n \frac{\partial f_i}{\partial y_j} [y_j - y_j(t^n)]$$

The question of stiffness then depends on the solution at the current time. Consequently nonlinear problems can be stiff during one time period and not stiff during another. While the chemical engineer may not actually calculate the eigenvalues, it is useful to know that they determine the stability and accuracy of the numerical scheme and the step size used.

Problems are stiff when the time constants for different phenomena have very different magnitudes. Consider flow through a packed bed reactor. The time constants for different phenomena are:

1. Time for device flow-through

$$t_{\text{flow}} = \frac{L}{u} = \frac{\phi AL}{Q}$$

where Q is the volumetric flow rate, A is the cross sectional area, L is the length of the packed bed, and ϕ is the void fraction;

2. Time for reaction

$$t_{r \times n} = \frac{1}{k}$$

where k is a rate constant (time^{-1});

3. Time for diffusion inside the catalyst

$$t_{\text{internal diffusion}} = \frac{\epsilon R^2}{D_e}$$

where ϵ is the porosity of the catalyst, R is the catalyst radius, and D_e is the effective diffusion coefficient inside the catalyst;

4. Time for heat transfer is

$$t_{\text{internal heat transfer}} = \frac{R^2}{\alpha} = \frac{\rho_s C_s R^2}{k_e}$$

where ρ_s is the catalyst density, C_s is the catalyst heat capacity per unit mass, k_e is the effective thermal conductivity of the catalyst, and

α is the thermal diffusivity. For example, in the model of a catalytic converter for an automobile [Ferguson, N. B., and B. A. Finlayson, *AIChE J.* **20**:539–550 (1974)], the time constants for internal diffusion was 0.3 seconds; internal heat transfer, 21 seconds; and device flow-through, 0.003 seconds. The device flow-through is so fast that it might as well be instantaneous. The stiffness is approximately 7000. Implicit methods must be used to integrate the equations. Alternatively, a quasistate model can be developed [Ramirez, W. F., *Computational Methods for Process Simulations*, 2d ed., Butterworth-Heinemann, Boston (1997)].

Differential-Algebraic Systems Sometimes models involve ordinary differential equations subject to some algebraic constraints. For example, the equations governing one equilibrium stage (as in a distillation column) are

$$\begin{aligned}M \frac{dx^n}{dt} &= V^{n+1} y^{n+1} - L^n x^n - V^n y^n + L^{n-1} x^{n-1} \\x^{n-1} - x^n &= E^n (x^{n-1} - x^{*,n}) \\ \sum_{i=1}^N x_i &= 1\end{aligned}$$

where x and y are the mole fraction in the liquid and vapor, respectively; L and V are liquid and vapor flow rates, respectively; M is the holdup; and the superscript is the stage number. The efficiency is E , and the concentration in equilibrium with the vapor is x^* . The first equation is an ordinary differential equation for the mass of one component on the stage, while the third equation represents a constraint that the mass fractions add to one. This is a differential-algebraic system of equations.

Differential-algebraic equations can be written in the general notation

$$F\left(t, y, \frac{dy}{dt}\right) = 0$$

To solve the general problem using the backward Euler method, replace the nonlinear differential equation with the nonlinear algebraic equation for one step.

$$F\left(t, y^{n+1}, \frac{y^{n+1} - y^n}{\Delta t}\right) = 0$$

This equation must be solved for y^{n+1} . The Newton-Raphson method can be used, and if convergence is not achieved within a few iterations, the time step can be reduced and the step repeated. In actuality, the higher-order backward-difference Gear methods are used in DASSL [Ascher, U. M., and L. R. Petzold, *Computer Methods for Ordinary Differential Equations and Differential-Algebraic Equations*, SIAM, Philadelphia (1998); and Brenan, K. E., S. L. Campbell, and L. R. Petzold, *Numerical Solution of Initial-Value Problems in Differential-Algebraic Equations*, North Holland: Elsevier (1989)].

Differential-algebraic systems are more complicated than differential systems because the solution may not always be defined. Pontelides et al. [*Comp. Chem. Eng.* **12**: 449–454 (1988)] introduced the term *index* to identify the possible problems. The index is defined as the minimum number of times the equations need to be differentiated with respect to time to convert the system to a set of ordinary differential equations. These higher derivatives may not exist, and the process places limits on which variables can be given initial values. Sometimes the initial values must be constrained by the algebraic equations. For a differential-algebraic system modeling a distillation tower, Pontelides et al. show that the index depends on the specification of pressure for the column. Byrne and Ponzi [*Comp. Chem. Eng.* **12**: 377–382 (1988)]; also Chan, T. F. C., and H. B. Keller, *SIAM J. Sci. Stat. Comput.* **3**: 173–194 (1982)] also list several chemical engineering examples of differential-algebraic systems and solve one involving two-phase flow.

Computer Software Efficient computer packages are available for solving ordinary differential equations as initial value problems. The packages are widely available and good enough that most chemical engineers use them and do not write their own. Here we discuss three of them: RKF45, LSODE, and EPISODE. In each of the packages, the user specifies the differential equation to be solved and a

desired error criterion. The package then integrates in time and adjusts the step size to achieve the error criterion within the limitations imposed by stability.

A popular explicit, Runge-Kutta package is RKF45. Notice there that an estimate of the truncation error at each step is available. Then the step size can be reduced until this estimate is below the user-specified tolerance. The method is thus automatic, and the user is assured of the results. Note, however, that the tolerance is set on the local truncation error, namely from one step to another, whereas the user is usually interested in the global truncation error, or the error after several steps. The global error is generally made smaller by making the tolerance smaller, but the absolute accuracy is not the same as the tolerance. If the problem is stiff, then very small step sizes are used; the computation becomes very lengthy. The RKF45 code discovers this and returns control to the user with a message indicating the problem is too hard to solve with RKF45.

A popular implicit package is LSODE, a version of Gear's method (Gear, *ibid.*) written by Alan Hindmarsh at Lawrence Livermore Laboratory. In this package, the user specifies the differential equation to be solved and the tolerance desired. Now the method is implicit and therefore stable for any step size. The accuracy may not be acceptable, however, and sets of nonlinear equations must be solved. Thus, in practice the step size is limited but not nearly so much as in the Runge-Kutta methods. In these packages, both the step size and order of the method are adjusted by the package. Suppose we are calculating with a k th order method. The truncation error is determined by the $(k+1)$ th order derivative. This is estimated using difference formulas and the values of the right-hand sides at previous times. An estimate is also made for the k th and $(k+2)$ th derivative. Then it is possible to estimate the error in a $(k-1)$ th order method, a k th order method, and a $(k+1)$ th order method. Furthermore, the step size needed to satisfy the tolerance with each of these methods can be determined. Then we can choose the method and step size for the next step that achieves the biggest step, with appropriate adjustments due to the different work required for each order. The package generally starts with a very small step size and a first-order method, the backward Euler method. Then it integrates along, adjusting the order up (and later down) depending on the error estimates. The user is thus assured that the local truncation error meets the tolerance. There is a further difficulty, since the set of nonlinear equations must be solved. Usually a good guess of the solution is available, since the solution is evolving in time and past history can be extrapolated. Thus, the Newton-Raphson method will usually converge. The package protects itself, though, by only doing a few iterations. If convergence is not reached within this many iterations, then the step size is reduced and the calculation is redone for that time step. The convergence theorem for the Newton-Raphson method (p. 3-50) indicates that the method will converge if the step size is small enough. Thus the method is guaranteed to work. Further economies are possible. The Jacobian needed in the Newton-Raphson method can be fixed over several time steps. Then, if the iteration does not converge, the Jacobian can be reevaluated at the current time-step. If the iteration still does not converge, then the step-size is reduced and a new Jacobian is evaluated. Also the successive substitution method can be used, which is even faster, except that it may not converge. However, it, too, will converge if the time step is small enough.

The Runge-Kutta methods give extremely good accuracy, especially when the step size is kept small for stability reasons. If the problem is stiff, though, backward difference implicit methods must be used. Many chemical reactor problems are stiff, necessitating the use of implicit methods. In the MATLAB suite of ODE solvers, the ode45 uses a revision of the RKF45 program, while the ode15s program uses an improved backward difference method. Shampine and Reichelt [SIAM J. Sci. Comp. 18:1-22 (1997)] give details of the programs in MATLAB. Fortunately, many packages are available. On the NIST web page <http://gams.nist.gov/>, choose "problem decision tree" and then "differential and integral equations" to find packages that can be downloaded. On the Netlib web site <http://www.netlib.org/>, choose "ode" to find packages that can be downloaded. Using Microsoft Excel to solve ordinary differential equations is cumbersome, except for the simplest problems.

Stability, Bifurcations, Limit Cycles Some aspects of this subject involve the solution of nonlinear equations; other aspects involve

the integration of ordinary differential equations; applications include chaos and fractals as well as unusual operation of some chemical engineering equipment. Kubicek, M., and M. Marek, *Computational Methods in Bifurcation Theory and Dissipative Structures*, Springer-Verlag, Berlin (1983), give an excellent introduction to the subject and the details needed to apply the methods. Chan, T. F. C., and H. B. Keller, *SIAM J. Sci. Stat. Comput.* 3:173-194 (1982), give more details of the algorithms. A concise survey with some chemical engineering examples is given in Doherty, M. F., and J. M. Ottino, *Chem. Eng. Sci.* 43:139-183 (1988). Bifurcation results are closely connected with stability of the steady states, which is essentially a transient phenomenon.

Sensitivity Analysis When solving differential equations, it is frequently necessary to know the solution as well as the sensitivity of the solution to the value of a parameter. Such information is useful when doing parameter estimation (to find the best set of parameters for a model) and for deciding if a parameter needs to be measured accurately. See Finlayson, et al. (2006).

Molecular Dynamics Special integration methods have been developed for molecular dynamics calculations due to the structure of the equations. A very large number of equations are to be integrated, with the following form based on molecular interactions between molecules.

$$m_i \frac{d^2 \mathbf{r}_i}{dt^2} = \mathbf{F}_i(\{\mathbf{r}\}) \quad \mathbf{F}_i(\{\mathbf{r}\}) = -\nabla V$$

The symbol m_i is the mass of the i th particle, $\mathbf{r}_i$ is the position of the i th particle, $\mathbf{F}_i$ is the force acting on the i th particle, and V is the potential energy that depends upon the location of all the particles (but not their velocities). Since the major part of the calculation lies in the evaluation of the forces, or potentials, a method must be used that minimizes the number of times the forces are calculated to move from one time to another time. Rewrite this equation in the form of an acceleration as

$$\frac{d^2 \mathbf{r}_i}{dt^2} = \frac{1}{m_i} \mathbf{F}_i(\{\mathbf{r}\}) \equiv \mathbf{a}_i$$

In the Verlet method, this equation is written by using central finite differences (see "Interpolation and Finite Differences"). Note that the accelerations do not depend upon the velocities.

$$\mathbf{r}_i(t + \Delta t) = 2\mathbf{r}_i(t) - \mathbf{r}_i(t - \Delta t) + \mathbf{a}_i(t)\Delta t^2$$

The calculations are straightforward, and no explicit velocity is needed. The storage requirement is modest, and the precision is modest (it is a second-order method). Note that one must start the calculation with values of $\{\mathbf{r}\}$ at times t and $t - \Delta t$.

In the Verlet velocity method, an equation is written for the velocity, too.

$$\frac{d\mathbf{v}_i}{dt} = \mathbf{a}_i$$

The trapezoid rule [see "Numerical Integration (Quadrature)"] is applied to obtain

$$\mathbf{v}_i(t + \Delta t) = \mathbf{v}_i(t) + \frac{1}{2} [\mathbf{a}_i(t) + \mathbf{a}_i(t + \Delta t)] \Delta t$$

The position of the particles is expanded in a Taylor series.

$$\mathbf{r}_i(t + \Delta t) = \mathbf{r}_i(t) + \mathbf{v}_i \Delta t + \frac{1}{2} \mathbf{a}_i(t) \Delta t^2$$

Beginning with values of $\{\mathbf{r}\}$ and $\{\mathbf{v}\}$ at time 0, one calculates the new positions and then the new velocities. This method is second-order in Δt , too. For additional details, see Allen, M. P., and D. J. Tildesley, *Computer Simulation of Liquids*, Clarendon Press, Oxford (1989); Frenkel, D., and B. Smit, *Understanding Molecular Simulation*, Academic Press (2002); Haile, J. M., *Molecular Dynamics Simulation*, Wiley (1992); Leach, A. R., *Molecular Modelling: Principles and Applications*, Prentice-Hall (2001); Schlick, T., *Molecular Modeling and Simulations*, Springer, New York (2002).

ORDINARY DIFFERENTIAL EQUATIONS-BOUNDARY VALUE PROBLEMS

Diffusion problems in one dimension lead to boundary value problems. The boundary conditions are applied at two different spatial

locations: at one side the concentration may be fixed and at the other side the flux may be fixed. Because the conditions are specified at two different locations, the problems are not initial value in character. It is not possible to begin at one position and integrate directly because at least one of the conditions is specified somewhere else and there are not enough conditions to begin the calculation. Thus, methods have been developed especially for boundary value problems.

Boundary value methods provide a description of the solution either by providing values at specific locations or by an expansion in a series of functions. Thus, the key issues are the method of representing the solution, the number of points or terms in the series, and how the approximation converges to the exact answer, i.e., how the error changes with the number of points or number of terms in the series. These issues are discussed for each of the methods: finite difference, orthogonal collocation, and Galerkin finite element methods.

Finite Difference Method To apply the finite difference method, we first spread grid points through the domain. Figure 3-48 shows a uniform mesh of n points (nonuniform meshes are possible, too). The unknown, here $c(x)$, at a grid point x_i is assigned the symbol $c_i = c(x_i)$. The finite difference method can be derived easily by using a Taylor expansion of the solution about this point. Expressions for the derivatives are:

$$\left. \frac{dc}{dx} \right|_i = \frac{c_{i+1} - c_i}{\Delta x} - \frac{d^2c}{dx^2} \Big|_i \frac{\Delta x}{2} + \dots, \quad \left. \frac{dc}{dx} \right|_i = \frac{c_i - c_{i-1}}{\Delta x} + \frac{d^2c}{dx^2} \Big|_i \frac{\Delta x}{2} + \dots$$

$$\left. \frac{d^2c}{dx^2} \right|_i = \frac{c_{i+1} - 2c_i + c_{i-1}}{2\Delta x^2} - \frac{d^3c}{dx^3} \Big|_i \frac{\Delta x^2}{3!} + \dots$$

The truncation error in the first two expressions is proportional to Δx , and the methods are said to be first-order. The truncation error in the third expression is proportional to Δx^2 , and the method is said to be second-order. Usually the last equation is used to insure the best accuracy. The finite difference representation of the second derivative is:

$$\frac{d^2c}{dx^2} \Big|_i = \frac{c_{i+1} - 2c_i + c_{i-1}}{\Delta x^2} - \frac{d^4c}{dx^4} \Big|_i \frac{2\Delta x^2}{4!} + \dots$$

The truncation error is proportional to Δx^2 . To solve a differential equation, it is evaluated at a point i and then these expressions are inserted for the derivatives.

Example Consider the equation for convection, diffusion, and reaction in a tubular reactor.

$$\frac{1}{Pe} \frac{d^2c}{dx^2} - \frac{dc}{dx} = Da R(c)$$

The finite difference representation is

$$\frac{1}{Pe} \frac{c_{i+1} - 2c_i + c_{i-1}}{\Delta x^2} - \frac{c_{i+1} - c_{i-1}}{2\Delta x} = Da R(c_i)$$

This equation is written for $i = 2$ to $n - 1$, or the internal points. The equations would then be coupled but would also involve the values of c_1 and c_n , as well. These are determined from the boundary conditions.

If the boundary condition involves a derivative, it is important that the derivatives be evaluated using points that exist. Three possibilities exist:

$$\left. \frac{dc}{dx} \right|_1 = \frac{c_2 - c_1}{\Delta x}$$

$$\left. \frac{dc}{dx} \right|_1 = \frac{-3c_1 + 4c_2 - c_3}{2\Delta x}$$

The third alternative is to add a false point, outside the domain, as $c_0 = c(x = -\Delta x)$.

$$\left. \frac{dc}{dx} \right|_1 = \frac{c_2 - c_0}{2\Delta x}$$

Since this equation introduces a new variable, c_0 , another equation is needed and is obtained by writing the finite difference equation for $i = 1$, too.

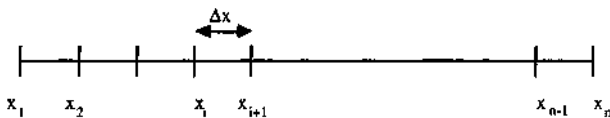


FIG. 3-48 Finite difference mesh; Δx uniform.

The sets of equations can be solved using the Newton-Raphson method. The first form of the derivative gives a tridiagonal system of equations, and the standard routines for solving tridiagonal equations suffice. For the other two options, some manipulation is necessary to put them into a tridiagonal form.

Frequently, the transport coefficients, such as diffusion coefficient or thermal conductivity, depend on the dependent variable, concentration, or temperature, respectively. Then the differential equation might look like

$$\frac{d}{dx} \left(D(c) \frac{dc}{dx} \right) = 0$$

This could be written as two equations.

$$-\frac{dJ}{dx} = 0 \quad J = -D(c) \frac{dc}{dx}$$

Because the coefficient depends on c , the equations are more complicated. A finite difference method can be written in terms of the fluxes at the midpoints, $i + 1/2$.

$$-\frac{J_{i+1/2} - J_{i-1/2}}{\Delta x} = 0 \quad J_{i+1/2} = -D(c_{i+1/2}) \frac{c_{i+1} - c_i}{\Delta x}$$

These are combined to give the complete equation.

$$\frac{D(c_{i+1/2}) (c_{i+1} - c_i) - D(c_{i-1/2}) (c_i - c_{i-1})}{\Delta x^2} = 0$$

This represents a set of nonlinear algebraic equations that can be solved with the Newton-Raphson method. However, in this case, a viable iterative strategy is to evaluate the transport coefficients at the last value and then solve

$$\frac{D(c_{i+1/2}^k) (c_{i+1}^{k+1} - c_i^{k+1}) - D(c_{i-1/2}^k) (c_i^{k+1} - c_{i-1}^{k+1})}{\Delta x^2} = 0$$

The advantage of this approach is that it is easier to program than a full Newton-Raphson method. If the transport coefficients do not vary radically, then the method converges. If the method does not converge, then it may be necessary to use the full Newton-Raphson method.

There are two common ways to evaluate the transport coefficient at the midpoint: Use the average value of the solution on each side to evaluate the diffusivity, or use the average value of the diffusivity on each side. Both methods have truncation error Δx^2 (Finlayson, 2003). The spacing of the grid points need not be uniform; see Finlayson (2003) and Finlayson et al. (2006) for the formulas in that case.

Example A reaction diffusion problem is solved with the finite difference method.

$$\frac{d^2c}{dx^2} = \phi^2 c, \quad \frac{dc}{dx}(0) = 0, \quad c(1) = 1$$

The solution is derived for $\phi = 2$. It is solved several times, first with two intervals and three points (at $x = 0, 0.5, 1$), then with four intervals, then with eight intervals. The reason is that when an exact solution is not known, one must use several Δx and see that the solution converges as Δx approaches zero. With two intervals, the equations are as follows. The points are $x_1 = 0$, $x_2 = 0.5$, and $x_3 = 1.0$; and the solution at those points are c_1 , c_2 , and c_3 , respectively. A false boundary is used at $x_0 = -0.5$.

$$\frac{c_0 - c_2}{2\Delta x} = 0, \quad \frac{c_0 - 2c_1 + c_2}{\Delta x^2} - \phi^2 c_1 = 0, \quad \frac{c_1 - 2c_2 + c_3}{\Delta x^2} - \phi^2 c_2 = 0, \quad c_3 = 1$$

The solution is $c_1 = 0.2857$, $c_2 = 0.4286$, and $c_3 = 1.0$. The problem is solved again with four and then eight intervals. The value of concentration at $x = 0$ takes the following values for different Δx . These values are extrapolated using the Richardson extrapolation technique to give $c(0) = 0.265718$. Using this value as the best estimate of the exact solution, the errors in the solution are tabulated versus Δx . Clearly the errors go as Δx^2 (decreasing by a factor of 4 when Δx decreases by a factor of 2), thus validating the solution. The exact solution is 0.265802.

$n - 1$	Δx	$c(0)$
2	0.5	0.285714
4	0.25	0.271043
8	0.125	0.267131

$n - 1$	Δx	Error in $c(0)$
2	0.5	0.02000
4	0.25	0.00532
8	0.125	0.00141

Finite Difference Methods Solved with Spreadsheets A convenient way to solve the finite difference equations for simple problems is to use a computer spreadsheet. The equations for the problem solved in the example can be cast into the following form

$$c_1 = \frac{2c_2}{2 + \phi^2 \Delta x^2}$$

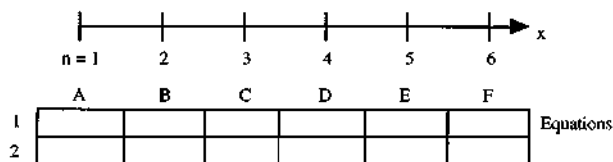


FIG. 3-49 Finite difference method using spreadsheets.

$$c_i = \frac{c_{i+1} + c_{i-1}}{2 + \phi^2 \Delta x^2}$$

$$c_{n+1} = 1$$

Let us solve the problem using 6 nodes, or 5 intervals. Then the connection between the cell in the spreadsheet and the nodal value is shown in Fig. 3-49. The following equations are placed into the various cells.

$$\begin{aligned} A1 &= 2*B1/(2.+(phi*dx)**2) \\ B1 &= (A1 + C1)/(2.+(phi*dx)**2) \\ F1 &= 1. \end{aligned}$$

The equation in cell B1 is copied into cells C1 through E1. Then turn on the iteration scheme in the spreadsheet and watch the solution converge. Whether or not convergence is achieved can depend on how you write the equations, so some experimentation may be necessary. Theorems for convergence of the successive substitution method are useful in this regard.

Orthogonal Collocation The orthogonal collocation method has found widespread application in chemical engineering, particularly for chemical reaction engineering. In the collocation method, the dependent variable is expanded in a series of orthogonal polynomials. See "Interpolation and Finite Differences: Lagrange Interpolation Formulas."

$$c(x) = \sum_{m=0}^N a_m P_m(x)$$

The differential equation is evaluated at certain collocation points. The collocation points are the roots to an orthogonal polynomial, as first used by Lanczos [Lanczos, C., *J. Math. Phys.* **17**:123–199 (1938); and Lanczos, C., *Applied Analysis*, Prentice-Hall (1956)]. A major improvement was proposed by Villadsen and Stewart [Villadsen, J. V., and W. E. Stewart, *Chem. Eng. Sci.* **22**:1483–1501 (1967)], who proposed that the entire solution process be done in terms of the solution at the collocation points rather than the coefficients in the expansion. This method is especially useful for reaction-diffusion problems that frequently arise when modeling chemical reactors. It is highly efficient when the solution is smooth, but the finite difference method is preferred when the solution changes steeply in some region of space. The error decreases very rapidly as N is increased since it is proportional to $[1/(1-N)]^{N-1}$. See Finlayson (2003) and Villadsen, J. V., and M. Michelsen, *Solution of Differential Equation Models by Polynomial Approximations*, Prentice-Hall (1978).

Galerkin Finite Element Method In the finite element method, the domain is divided into elements and an expansion is made for the solution on each finite element. In the Galerkin finite element method an additional idea is introduced: the Galerkin method is used to solve the equation. The Galerkin method is explained before the finite element basis set is introduced, using the equations for reaction and diffusion in a porous catalyst pellet.

$$\frac{d^2 c}{dx^2} = \phi^2 R(c)$$

$$\frac{dc}{dx}(0) = 0, \quad c(1) = 1$$

The unknown solution is expanded in a series of known functions $\{b_i(x)\}$ with unknown coefficients $\{a_i\}$.

$$c(x) = \sum_{i=1}^{NT} a_i b_i(x)$$

The trial solution is substituted into the differential equation to obtain

the residual.

$$\text{Residual} = \sum_{i=1}^{NT} a_i \frac{d^2 b_i}{dx^2} - \phi^2 R \left[\sum_{i=1}^{NT} a_i b_i(x) \right]$$

The residual is then made orthogonal to the set of basis functions.

$$\int_0^1 b_j(x) \left\{ \sum_{i=1}^{NT} a_i \frac{d^2 b_i}{dx^2} - \phi^2 R \left[\sum_{i=1}^{NT} a_i b_i(x) \right] \right\} dx = 0 \quad j = 1, \dots, NT$$

This is the process that makes the method a Galerkin method. The basis for the orthogonality condition is that a function that is made orthogonal to each member of a complete set is then zero. The residual is being made orthogonal, and if the basis functions are complete and you use infinitely many of them, then the residual is zero. Once the residual is zero, the problem is solved.

This equation is integrated by parts to give the following equation

$$-\sum_{i=1}^{NT} \int_0^1 \frac{db_i}{dx} \frac{db_i}{dx} dx a_i = \phi^2 \int_0^1 b_j(x) R \left[\sum_{i=1}^{NT} a_i b_i(x) \right] dx$$

$$j = 1, \dots, NT - 1 \quad (3-80)$$

This equation defines the Galerkin method and a solution that satisfies this equation (for all $j = 1, \dots, \infty$) is called a weak solution. For an approximate solution, the equation is written once for each member of the trial function, $j = 1, \dots, NT - 1$, and the boundary condition is applied.

$$\sum_{i=1}^{NT} a_i b_i(1) = c_B$$

The Galerkin finite element method results when the Galerkin method is combined with a finite element trial function. The domain is divided into elements separated by nodes, as in the finite difference method. The solution is approximated by a linear (or sometimes quadratic) function of position within the element. These approximations are substituted into Eq. (3-80) to provide the Galerkin finite element equations. For example, with the grid shown in Fig. 3-48, a linear interpolation would be used between points x_i and x_{i+1} .

$$c(x) = c_i(1-u) + c_{i+1}u \quad u = \frac{x-x_i}{x_{i+1}-x_i}$$

A finite element method based on these functions would have an error proportional to Δx^2 . The finite element representations for the first derivative and second derivative are the same as in the finite difference method, but this is not true for other functions or derivatives. With quadratic finite elements, take the region from x_{i-1} and x_{i+1} as one element. Then the interpolation would be

$$c(x) = c_{i-1}N_1(u) + c_iN_2(u) + c_{i+1}N_3(u)$$

$$N_1(u) = 2(u-1)\left(u - \frac{1}{2}\right) \quad N_2(u) = 4u(1-u)$$

$$N_3(u) = 2u\left(u - \frac{1}{2}\right)$$

A finite element method based on these functions would have an error proportional to Δx^3 . Thus, it would converge faster than one based on linear interpolation. A variety of other finite element functions can be used as well, including B splines (see "Interpolation and Finite Differences: Spline Functions").

Adaptive Meshes In many two-point boundary value problems, the difficulty in the problem is the formation of a boundary layer region, or a region in which the solution changes very dramatically. In such cases, it is prudent to use small mesh spacing there, either with the finite difference method or the finite element method. If the region is known *a priori*, small mesh spacings can be assumed at the boundary layer. If the region is not known, though, other techniques must be used. These techniques are known as adaptive mesh techniques. The mesh size is made small where some property of the solution is large. For example, if the truncation error of the method is n th order, then the n th-order derivative of the solution is evaluated and a small mesh is used where it is large. Alternatively, the residual (the differential equation with the numerical

solution substituted into it) can be used as a criterion. It is also possible to define the error that is expected from a method one order higher and one order lower. Then a decision about whether to increase or decrease the order of the method can be made, taking into account the relative work of the different orders. This provides a method of adjusting both the mesh spacing (Δx , or sometimes called h) and the degree of polynomial (p). Such methods are called h - p methods. Many finite element programs have the capability to do this mesh refinement automatically.

Singular Problems and Infinite Domains If the solution being sought has a singularity, it may be difficult to find a good numerical solution. Sometimes even the location of the singularity may not be known. One method of solving such problems is to refine the mesh near the singularity, relying on the better approximation due to a smaller Δx . Another approach is to incorporate the singular trial function into the approximation. Thus, if the solution approaches $f(x)$ as x goes to zero and $f(x)$ becomes infinite, one may define a new variable $u(x) = y(x) - f(x)$ and derive an equation for u . The differential equation is more complicated, but the solution is better near the singularity. Press et al. (1986).

Sometimes the domain is semi-infinite, as in boundary layer flow. The domain can be transformed from the x domain ($0 - \infty$) to the η domain ($1 - 0$) using the transformation $\eta = \exp(-x)$. Another approach is to use a variable mesh, perhaps with the same transformation. For example, use $\eta = \exp(-\beta x)$ and a constant mesh size in η ; the value of β is found experimentally. Still another approach is to solve on a finite mesh in which the last point is far enough away that its location does not influence the solution. A location that is far enough away must be found by trial and error.

Packages to solve boundary value problems are available on the Internet. On the NIST web page <http://gams.nist.gov/>, choose "problem decision tree" and then "differential and integral equations" and then "ordinary differential equations" and "multipoint boundary value problems." On the Netlib web site <http://www.netlib.org/>, search on "boundary value problem." Any spreadsheet that has an iteration capability can be used with the finite difference method. Some packages for partial differential equations also have a capability for solving one-dimensional boundary value problems [e.g. Comsol Multiphysics (formerly FEMLAB)].

NUMERICAL SOLUTION OF INTEGRAL EQUATIONS

In this subsection is considered a method of solving numerically the Fredholm integral equation of the second kind:

$$u(x) = f(x) + \lambda \int_a^b k(x, t)u(t) dt \quad \text{for } u(x) \quad (3-81)$$

The method discussed arises because a definite integral can be closely approximated by any of several numerical integration formulas (each of which arises by approximating the function by some polynomial over an interval). Thus the definite integral in Eq. (3-81) can be replaced by an integration formula, and Eq. (3-81) may be written

$$u(x) = f(x) + \lambda(b-a) \left[\sum_{i=1}^n c_i k(x, t_i) u(t_i) \right] \quad (3-82)$$

where $t_1, \dots, t_n$ are points of subdivision of the t axis, $a \leq t \leq b$, and the c 's are coefficients whose values depend upon the type of numerical integration formula used. Now Eq. (3-82) must hold for all values of x , $a \leq x \leq b$; so it must hold for $x = t_1, x = t_2, \dots, x = t_n$. Substituting for x successively $t_1, t_2, \dots, t_n$ and setting $u(t_i) = u_i, f(t_i) = f_i$, we get n linear algebraic equations for the n unknowns $u_1, \dots, u_n$. That is,

$$u_i = f_i + \lambda(b-a) [c_1 k(t_i, t_1) u_1 + c_2 k(t_i, t_2) u_2 + \dots + c_n k(t_i, t_n) u_n] \quad i = 1, 2, \dots, n$$

These u_i may be solved for by the methods under "Numerical Solution of Linear Equations and Associated Problems" and substituted into Eq. (3-82) to yield an approximate solution for Eq. (3-81).

Because of the work involved in solving large systems of simultaneous linear equations it is desirable that only a small number of u 's be computed. Thus the gaussian integration formulas are useful because of the economy they offer.

Solutions for Volterra equations are done in a similar fashion, except that the solution can proceed point by point, or in small groups of points depending on the quadrature scheme. See Linz, P., *Analytical and Numerical Methods for Volterra Equations*, SIAM, Philadelphia (1985). There are methods that are analogous to the usual methods for

integrating differential equations (Runge-Kutta, predictor-corrector, Adams methods, etc.). Explicit methods are fast and efficient until the time step is very small to meet the stability requirements. Then implicit methods are used, even though sets of simultaneous algebraic equations must be solved. The major part of the calculation is the evaluation of integrals, however, so that the added time to solve the algebraic equations is not excessive. Thus, implicit methods tend to be preferred. Volterra equations of the first kind are not well posed, and small errors in the solution can have disastrous consequences. The boundary element method uses Green's functions and integral equations to solve differential equations. See Brebbia, C. A., and J. Dominguez, *Boundary Elements—An Introductory Course*, 2d ed., Computational Mechanics Publications, Southampton (1992); and Mackerle, J., and C. A. Brebbia (eds.), *Boundary Element Reference Book*, Springer-Verlag (1988).

MONTE CARLO SIMULATIONS

Some physical problems, such as those involving interaction of molecules, are usually formulated as integral equations. Monte Carlo methods are especially well-suited to their solution. This section cannot give a comprehensive treatment of such methods, but their use in calculating the value of an integral will be illustrated. Suppose we wish to calculate the integral

$$G = \int_{\Omega_0} g(x)f(x) dx$$

where the distribution function $f(x)$ satisfies:

$$f(x) \geq 0, \quad \int_{\Omega_0} f(x) dx = 1$$

The distribution function $f(x)$ can be taken as constant; for example, $1/\Omega_0$. We choose variables $x_1, x_2, \dots, x_N$ randomly from $f(x)$ and form the arithmetic mean

$$G_N = \frac{1}{N} \sum_i g(x_i)$$

The quantity G_N is an estimation of G , and the fundamental theorem of Monte Carlo guarantees that the expected value of G_N is G , if G exists [Kalos, M. H., and P. A. Whitlock, *Monte Carlo Methods*, vol. 1, Wiley, New York (1986)]. The error in the calculation is given by

$$\epsilon = \frac{\sigma_1}{N^{1/2}}$$

where σ_1^2 is calculated from

$$\sigma_1^2 = \int_{\Omega_0} g^2(x)f(x) dx - G^2$$

Thus the number of terms needed to achieve a specified accuracy ϵ can be calculated once an estimate of σ_1^2 is known.

$$N = \frac{\sigma_1^2}{\epsilon^2}$$

Various methods, such as influence sampling, can be used to reduce the number of calculations needed. See also Lapeyre, B., *Introduction to Monte-Carlo Methods for Transport and Diffusion Equations*, Oxford University Press (2003), and Liu, J. S., *Monte Carlo Strategies in Scientific Computing*, Springer (2001). Some computer programs are available that perform simple Monte Carlo calculations using Microsoft Excel.

NUMERICAL SOLUTION OF PARTIAL DIFFERENTIAL EQUATIONS

The numerical methods for partial differential equations can be classified according to the type of equation (see "Partial Differential Equations"); parabolic, elliptic, and hyperbolic. This section uses the finite difference method to illustrate the ideas, and these results can be programmed for simple problems. For more complicated problems, though, it is common to rely on computer packages. Thus, some discussion is given to the issues that arise when using computer packages.

Parabolic Equations in One Dimension By combining the techniques applied to initial value problems and boundary value problems it

is possible to easily solve parabolic equations in one dimension. The method is often called the method of lines. It is illustrated here using the finite difference method, but the Galerkin finite element method and the orthogonal collocation method can also be combined with initial value methods in similar ways. The analysis is done by example. The finite volume method is described under Hyperbolic Equations.

Example Consider the diffusion equation, with boundary and initial conditions.

$$\begin{aligned}\frac{\partial c}{\partial t} &= D \frac{\partial^2 c}{\partial x^2} \\ c(x, 0) &= 0 \\ c(0, t) &= 1, \quad c(1, t) = 0\end{aligned}$$

We denote by c_i the value of $c(x_i, t)$ at any time. Thus, c_i is a function of time, and differential equations in c_i are ordinary differential equations. By evaluating the diffusion equation at the i th node and replacing the derivative with a finite difference equation, the following working equation is derived for each node $i, i = 2, \dots, n$ (see Fig. 3-50).

$$\frac{dc_i}{dt} = D \frac{c_{i+1} - 2c_i + c_{i-1}}{\Delta x^2}$$

This can be written in the general form of a set of ordinary differential equations by defining the matrix $\mathbf{A}\mathbf{A}$.

$$\frac{dc}{dt} = \mathbf{A}\mathbf{A}c$$

This set of ordinary differential equations can be solved using any of the standard methods, and the stability of the integration of these equations is governed by the largest eigenvalue of $\mathbf{A}\mathbf{A}$. When Euler's method is used to integrate in time, the equations become

$$\frac{c_i^{n+1} - c_i^n}{\Delta t} = D \frac{c_{i+1}^n - 2c_i^n + c_{i-1}^n}{\Delta x^2}$$

where $c_i^n = c(x_i, t^n)$. Notice that if the solution is known at every point at one time n , then it is a straightforward calculation to find the solution at every point at the new time $n+1$.

If Euler's method is used for integration, the time step is limited by

$$\Delta t \leq \frac{2}{|\lambda|_{\max}}$$

whereas if the Runge-Kutta-Feldberg method is used, the 2 in the numerator is replaced by 3.0. The largest eigenvalue of $\mathbf{A}\mathbf{A}$ is bounded by Gerschgorin's Theorem.

$$|\lambda|_{\max} \leq \max_{2 \leq j \leq n} \sum_{i=2}^n |\mathbf{A}\mathbf{A}_{ji}| = \frac{4D}{\Delta x^2}$$

This gives the well-known stability limit

$$\Delta t \frac{D}{\Delta x^2} \leq \frac{1}{2}$$

The smallest eigenvalue is independent of Δx (it is $D\pi^2/L^2$) so that the ratio of largest to smallest eigenvalue is proportional to $1/\Delta x^2$. Thus, the problem becomes stiff as Δx approaches zero. See Eq. (3-79).

The effect of the increased stiffness is that a smaller and smaller time step (Δt) must be taken as the mesh is refined ($\Delta x^2 \rightarrow 0$). At the same time, the number of points is increasing, so that the computation becomes very lengthy. Implicit methods are used to overcome this problem.

Write a finite difference form for the time derivative and average the right-hand sides, evaluated at the old and new time.

$$\frac{c_i^{n+1} - c_i^n}{\Delta t} = D(1 - \theta) \frac{c_{i+1}^n - 2c_i^n + c_{i-1}^n}{\Delta x^2} + D\theta \frac{c_{i+1}^{n+1} - 2c_i^{n+1} + c_{i-1}^{n+1}}{\Delta x^2}$$

Now the equations are of the form

$$\begin{aligned}-\frac{D\Delta t\theta}{\Delta x^2} c_{i+1}^{n+1} + \left[1 + 2\frac{D\Delta t\theta}{\Delta x^2}\right] c_i^{n+1} - \frac{D\Delta t\theta}{\Delta x^2} c_{i-1}^{n+1} \\ = c_i^n + \frac{D\Delta t(1 - \theta)}{\Delta x^2} (c_{i+1}^n - 2c_i^n + c_{i-1}^n)\end{aligned}$$

and require solving a set of simultaneous equations, which have a tridiagonal structure. Using $\theta = 0$ gives the Euler method (as above),

$$\left. \frac{du}{dx} \right|_{i,j} = \frac{1}{2h} \left\{ \begin{array}{c} (-1) \text{---} (0) \text{---} (1) \\ i-1,j \quad i,j \quad i+1,j \end{array} \right\} + O(h^2)$$

$$\left. \frac{du}{dy} \right|_{i,j} = \frac{1}{2k} \left\{ \begin{array}{c} (1) \\ i,j+1 \\ (0) \\ i,j \\ (-1) \\ i,j-1 \end{array} \right\} + O(k^2)$$

$$\left. \frac{d^2u}{dx^2} \right|_{i,j} = \frac{1}{h^2} \left\{ \begin{array}{c} (1) \text{---} (-2) \text{---} (1) \\ i-1,j \quad i,j \quad i+1,j \end{array} \right\} + O(h^2)$$

$$= \frac{1}{4h^2} \left\{ \begin{array}{ccc} (-1) & (0) & (1) \\ | & | & | \\ (0) & (0) & (0) \\ | & | & | \\ (1) & (0) & (-1) \end{array} \right\} + O(h^2)$$

$$\left. \nabla^2 u \right|_{i,j} = \frac{1}{h^2} \left\{ \begin{array}{c} (1) \\ | \\ (1) \text{---} (-4) \text{---} (1) \\ | \\ (1) \end{array} \right\} + O(h^2)$$

$$\left. \nabla^4 u \right|_{i,j} = \frac{1}{h^4} \left\{ \begin{array}{ccccc} & & (1) & & \\ & (2) & (-8) & (2) & \\ (1) & (-8) & (20) & (-8) & (1) \\ & (2) & (-8) & (2) & \\ & & (1) & & \end{array} \right\} + O(h^2)$$

$$\int_{\Omega} u \, d\Omega = \frac{h^2}{9} \left\{ \begin{array}{ccc} (1) & (4) & (1) \\ | & | & | \\ (4) & (16) & (4) \\ | & | & | \\ (1) & (4) & (1) \end{array} \right\} + O(h^6)$$

FIG. 3-50 Computational molecules. $h = \Delta x = \Delta y$.

$\theta = 0.5$ gives the Crank-Nicolson method, and $\theta = 1$ gives the backward Euler method. The Crank-Nicolson method is also the same as applying the trapezoid rule to do the integration. The stability limit is given by

$$\frac{D\Delta t}{\Delta x^2} \leq \frac{0.5}{1 - 2\theta}$$

The price of using implicit methods is that one now has a system of equations to solve at each time step, and the solution methods are more complicated (particularly for nonlinear problems) than the straightforward explicit methods. Phenomena that happen quickly can also be obliterated or smoothed over by using a large time step, so implicit methods are not suitable in all cases. The engineer must decide if he or she wants to track those fast phenomena, and choose an appropriate method that handles the time scales that are important in the problem.

Other methods can be used in space, such as the finite element method, the orthogonal collocation method, or the method of orthogonal collocation on finite elements. One simply combines the methods for ordinary differential equations (see “Ordinary Differential Equations—Boundary Value Problems”) with the methods for initial-value problems (see “Numerical Solution of Ordinary Differential Equations as Initial Value Problems”). Fast Fourier transforms can also be used on regular grids (see “Fast Fourier Transform”).

Elliptic Equations Elliptic equations can be solved with both finite difference and finite element methods. One-dimensional elliptic problems are two-point boundary value problems. Two- and three-dimensional elliptic problems are often solved with iterative methods when the finite difference method is used and direct methods when the finite element method is used. So there are two aspects to consider: how the equations are discretized to form sets of algebraic equations and how the algebraic equations are then solved.

The prototype elliptic problem is steady-state heat conduction or diffusion,

$$k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) = Q$$

possibly with a heat generation term per unit volume, Q . The boundary conditions taken here are $T = f(x, y)$ on the boundary (S) with f a known function. Illustrations are given for constant thermal conductivity k while Q is a known function of position. The finite difference formulation is given using the following nomenclature:

$$T_{i,j} = T(i\Delta x, j\Delta y)$$

The finite difference formulation is then (see Fig. 3-50)

$$\frac{T_{i+1,j} - 2T_{i,j} + T_{i-1,j}}{\Delta x^2} + \frac{T_{i,j+1} - 2T_{i,j} + T_{i,j-1}}{\Delta y^2} = Q_{i,j} \quad (3-83)$$

$$T_{i,j} = f(x_i, y_j) \text{ on } S$$

If the boundary is parallel to a coordinate axis any derivative is evaluated as in the section on boundary value problems, using either a one-sided, centered difference or a false boundary. If the boundary is more irregular and not parallel to a coordinate line then more complicated expressions are needed and the finite element method may be the better method.

Equation (3-83) provides a set of linear equations that must be solved. These equations and their boundary conditions may be written in matrix form as

$$\mathbf{A}\mathbf{t} = \mathbf{f}$$

where $\mathbf{t}$ is the set of temperatures at all the points, $\mathbf{f}$ is the set of heat generation terms at all points, and $\mathbf{A}$ is formed from the coefficients of T_{ij} in Eq. (3-83). The solution can be obtained simply by solving the set of linear equations. For three-dimensional problems, the matrix $\mathbf{A}$ is sparse, and iterative methods are used. These include Gauss-Seidel, alternating direction, overrelaxation methods, conjugate gradient, and multigrid methods. In Gauss-Seidel methods, one writes the equation for T_{ij} in terms of the other temperatures and cycles through all the points over and over. In the alternating direction method, one solves along one line (that is, $x = \text{constant}$), keeping the side values fixed, and then repeats this for all lines, and then repeats the process. Multigrid methods solve the problem on successively refined grids, which has advantages for both convergence and error estimation. Conjugate gradient methods frequently use a preconditioned matrix. The equation is multiplied by another matrix, which is chosen so that the resulting problem is easier to solve than the original one. Finding such matrices

is an art, but it can speed convergence. The generalized minimal residual method is described in <http://mathworld.wolfram.com/GeneralizedMinimalResidualMethod.html>. Additional resources can be found at http://www.netlib.org/linalg/html_templates/Templates.html. When the problem is nonlinear, the iterative methods may not converge, or the mesh may have to be refined before they converge, so some experimentation is sometimes necessary.

Spreadsheets can be used to solve two-dimensional problems on rectangular grids. The equation for T_{ij} is obtained by rearranging Eq. (3-83).

$$2 \left(1 + \frac{\Delta x^2}{\Delta y^2} \right) T_{i,j} = T_{i+1,j} + T_{i-1,j} + \frac{\Delta x^2}{\Delta y^2} (T_{i,j+1} + T_{i,j-1}) - \Delta x^2 \frac{Q_{i,j}}{k}$$

This equation is inserted into a cell and copied throughout the space represented by all the cells; when the iteration feature is turned on, the solution is obtained.

The Galerkin finite element method (FEM) is useful for solving elliptic problems and is particularly effective when the domain or geometry is irregular. As an example, cover the domain with triangles and define a trial function on each triangle. The trial function takes the value 1.0 at one corner and 0.0 at the other corners and is linear in between. See Fig. 3-51. These trial functions on each triangle are pieced together to give a trial function on the whole domain. General treatments of the finite element method are available (see references). The steps in the solution method are similar to those described for boundary value problems, except now the problems are much bigger so that the numerical analysis must be done very carefully to be efficient. Most engineers, though, just use a finite element program without generating it. There are three major caveats that must be addressed. The first one is that the solution is dependent on the mesh laid down, and the only way to assess the accuracy of the solution is to solve the problem with a more refined mesh. The second concern is that the solution obeys the shape of the trial function inside the element. Thus, if linear functions are used on triangles, a three-dimensional view of the solution, plotting the solution versus x and y , consists of a series of triangular planes joined together at the edges, as in a geodesic dome. The third caveat is that the Galerkin finite element method is applied to both the differential equations and the boundary conditions. Computer programs are usually quite general and may allow the user to specify boundary conditions that are not realistic. Also, natural boundary conditions are satisfied if no other boundary condition (ones involving derivatives) is set at a node. Thus, the user of finite element codes must be very clear what boundary conditions and differential equations are built into the computer code. When the problem is nonlinear, the Newton-Raphson method is used to iterate from an initial guess. Nonlinear problems lead to complicated integrals to evaluate, and they are usually evaluated using Gaussian quadrature.

One nice feature of the finite element method is the use of natural boundary conditions. It may be possible to solve the problem on a domain that is shorter than needed to reach some limiting condition (such as at an outflow boundary). The externally applied flux is still applied at the shorter domain, and the solution *inside* the truncated domain is still valid. Examples are given in Chang, M. W., and B. A. Finlayson, *Int. J. Num. Methods Eng.* **15**, 935–942 (1980), and Finlayson, B. A. (1992). The effect of this is to allow solutions in domains that are smaller, thus saving computation time and permitting the solution in semi-infinite domains.

The trial functions in the finite element method are not limited to linear ones. Quadratic functions and even higher-order functions are frequently used. The same considerations hold as for boundary value problems: The higher-order trial functions converge faster, but require more work. It is possible to refine both the mesh h and the power of polynomial in the trial function p in an hp method. Some problems have constraints on some of the variables. For flow problems, the pressure must usually be approximated by using a trial function that is one order lower than the polynomial used to approximate the velocity.

Hyperbolic Equations The most common situation yielding hyperbolic equations involves unsteady phenomena with convection. Two typical equations are the convective diffusive equation

$$\frac{\partial c}{\partial t} + u \frac{\partial c}{\partial x} = D \frac{\partial^2 c}{\partial x^2}$$

and the chromatography equation.

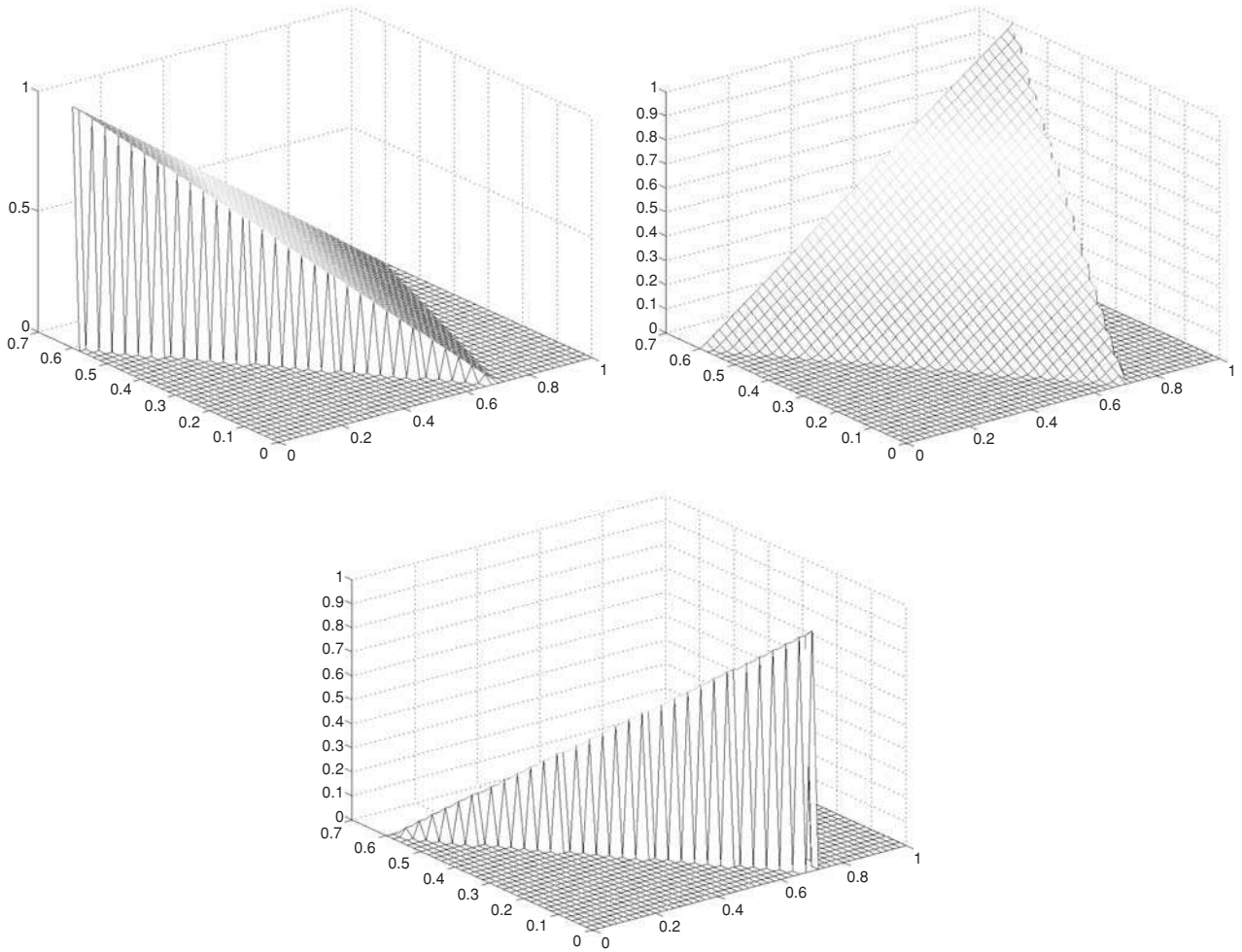


FIG. 3-51 Trial functions for Galerkin finite element method: linear polynomial on triangle.

(See “Partial Differential Equations.”) If the diffusion coefficient is zero, the convective diffusion equation is hyperbolic. If D is small, the phenomenon may be essentially hyperbolic, even though the equations are parabolic. Thus the numerical methods for hyperbolic equations may be useful even for parabolic equations.

Equations for several methods are given here, as taken from the book by Finlayson [Finlayson, B. A. (1992)]. If the convective term is treated with a centered difference expression, the solution exhibits oscillations from node to node, and these only go away if a very fine grid is used. The simplest way to avoid the oscillations with a hyperbolic equation is to use upstream derivatives. If the flow is from left to right, this would give

$$\frac{dc_i}{dt} + u \frac{c_i - c_{i-1}}{\Delta x} = D \frac{c_{i+1} - 2c_i + c_{i-1}}{\Delta x^2}$$

The effect of using upstream derivatives is to add artificial or numerical diffusion to the model. This can be ascertained by rearranging the finite difference form of the convective diffusion equation

$$\frac{dc_i}{dt} + u \frac{c_{i+1} - c_{i-1}}{2\Delta x} = \left(D + \frac{u\Delta x}{2} \right) \frac{c_{i+1} - 2c_i + c_{i-1}}{\Delta x^2}$$

Thus the diffusion coefficient has been changed from

$$D \text{ to } D + \frac{u\Delta x}{2}$$

Alternatively, the diffusion coefficient has been multiplied by the factor

$$D' = D \left(1 + \frac{u\Delta x}{2D} \right) = D \left(1 + \frac{\text{Pe}_{\text{cell}}}{2} \right)$$

where $\text{Pe}_{\text{cell}} = \frac{u\Delta x}{D} = \frac{uL}{D} \frac{\Delta x}{L} = \text{Pe} \frac{\Delta x}{L}$ is called the cell Peclet number.

When the diffusion coefficient is very small (or diffusion is slow compared with convection), the Peclet number will be large. In that case, extraneous diffusion will be included in the solution unless the mesh size (denoted by Δx) is small compared with the characteristic length of the problem. To avoid this problem (by keeping the factor small), very fine meshes must be used, and the smaller the diffusion coefficient, the smaller the required mesh size.

A variety of other methods are used to obtain a good solution without using extremely fine meshes. The flux correction methods keep track of the flux of material into and out of a cell (from one node to another) and put limits on the flux to make sure that no more material

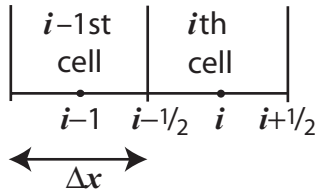


FIG. 3-52 Nomenclature for finite volume method.

leaves the cell than is there originally plus the input amount. See Finlayson, *ibid.*, for many examples.

All the methods have a limit to the time step that is set by the convection term. Essentially, the time step should not be so big as to take the material farther than it can go at its velocity. This is usually expressed as a Courant number limitation.

$$Co = \frac{u\Delta t}{\Delta x} \leq 1$$

Some methods require a smaller limit, depending upon the amount of diffusion present (see Finlayson, *ibid.*, Appendix).

In the finite element method, Petrov-Galerkin methods are used to minimize the unphysical oscillations. The Petrov-Galerkin method essentially adds a small amount of diffusion in the flow direction to smooth the unphysical oscillations. The amount of diffusion is usually proportional to Δx so that it becomes negligible as the mesh size is reduced. The value of the Petrov-Galerkin method lies in being able to obtain a smooth solution when the mesh size is large, so that the computation is feasible. This is not so crucial in one-dimensional problems, but it is essential in two- and three-dimensional problems and purely hyperbolic problems.

Finite Volume Methods Finite volume methods are utilized extensively in computational fluid dynamics. In this method, a mass balance is made over a cell, accounting for the change in what is in the cell, and the flow in and out. Figure 3-52 illustrates the geometry of the i th cell. A mass balance made on this cell (with area A perpendicular to the paper) is

$$A\Delta x (c_i^{n+1} - c_i^n) = \Delta t A (J_{i-1/2} - J_{i+1/2})$$

where J is the flux due to convection and diffusion, positive in the $+x$ direction.

$$J = uc - D \frac{\partial c}{\partial x}, \quad J_{i-1/2} = u_{i-1/2} c_{i-1/2} - D \frac{c_i - c_{i-1/2}}{\Delta x}$$

The concentration at the edge of the cell is taken as

$$c_{i-1/2} = \frac{1}{2}(c_i + c_{i-1/2})$$

Rearrangement for the case when the velocity u is the same for all nodes gives

$$\frac{c_i^{n+1} - c_i^n}{\Delta t} + \frac{u(c_{i+1} - c_{i-1})}{2\Delta x} = \frac{D}{\Delta x^2} (c_{i+1} - 2c_i + c_{i-1})$$

This is the same equation obtained by using the finite difference method. This isn't always true, and the finite volume equations are easy to derive. In two and three dimensions, the mesh need not be rectangular, as long as it is possible to compute the velocity normal to an edge of the cell. The finite volume method is useful for applications involving filling, such as injection molding, when only part of the cell is filled with fluid. Such applications do involve some approximations, since the interface is not tracked precisely, but they are useful engineering approximations.

Parabolic Equations in Two or Three Dimensions Computations become much more lengthy when there are two or more spatial dimensions. For example, we may have the unsteady heat conduction equation

$$\rho C_p \frac{\partial T}{\partial t} = k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) - Q$$

Most engineers use computer packages to solve such problems. If there is both convection and diffusion in the problem, the same considerations apply: A fine mesh is needed when the Peclet number is large. The upstream weighting and Petrov-Galerkin methods can be used, but it is important to apply the smoothing only in the direction of flow, since smoothing in the direction transverse to the flow direction would be incorrect. Some transverse smoothing is unavoidable, but the engineer needs to be sure that the smoothing is just enough to allow a good solution without creating large errors.

Computer Software When you are choosing computer software to solve your problem, there are a number of important considerations. The first decision is whether to use an approximate, engineering flow model, developed from correlations, or to solve the partial differential equations that govern the problem. Correlations are quick and easy to apply, but they may not be appropriate to your problem or give the needed detail. When you are using a computer package to solve partial differential equations, the first task is always to generate a mesh covering the problem domain. This is not a trivial task, and special methods have been developed to permit importation of a geometry from a computer-aided design (CAD) program. Then the mesh must be created automatically. If the boundary is irregular, the finite element method is especially well suited, although special embedding techniques can be used in finite difference methods (which are designed to be solved on rectangular meshes). Another capability to consider is the ability to track free surfaces that move during the computation. This phenomenon introduces the same complexity that occurs in problems with a large Peclet number, with the added difficulty that the free surface moves between mesh points and improper representation can lead to unphysical oscillations. The method used to solve the equations is important, and both explicit and implicit methods (as described above) can be used. Implicit methods may introduce unacceptable extra diffusion, so the engineer needs to examine the solution carefully. The methods used to smooth unphysical oscillations from node to node are also important, and the engineer needs to verify that the added diffusion or smoothing does not give inaccurate solutions. Since current-day problems are mostly nonlinear, convergence is always an issue since the problems are solved iteratively. Robust programs provide several methods for convergence, each of which is best in some circumstance or other. It is wise to have a program that includes many iterative methods. If the iterative solver is not very robust, the only recourse to solving a steady-state problem may be to integrate the time-dependent problem to steady state. The solution time may be long, and the final result may be further from convergence than would be the case if a robust iterative solver were used.

A variety of computer programs are available on the Internet, some free. First consider general-purpose programs. On the NIST web page <http://gams.nist.gov/>, choose "problem decision tree" then "differential and integral equations" and then "partial differential equations." The programs are organized by type of problem (elliptic, parabolic, and hyperbolic) and by the number of spatial dimensions (one or more than one). On the Netlib web site <http://www.netlib.org/>, search on "partial differential equation." Lau (1994, 2004) provides many programs in C++ (also see <http://www.nr.com/>). The multiphysics program Comsol Multiphysics (formerly FEMLAB) (<http://www.comsol.com/>), and ANSYS (<http://www.ansys.com/>). Of these, Comsol Multiphysics is particularly useful because it has a convenient graphical-user interface, permits easy mesh generation and refinement (including adaptive mesh refinement), allows the user to add phenomena and equations easily, permits solution by continuation methods (thus enhancing

Computational fluid dynamics (CFD) programs are more specialized, and most have been designed to solve sets of equations that are appropriate to specific industries. They can then include approximations and correlations for some features that would be difficult to solve for directly. Four major packages widely used are Fluent (<http://www.fluent.com/>), CFX (now part of ANSYS), Comsol Multiphysics (formerly FEMLAB) (<http://www.comsol.com/>), and ANSYS (<http://www.ansys.com/>). Of these, Comsol Multiphysics is particularly useful because it has a convenient graphical-user interface, permits easy mesh generation and refinement (including adaptive mesh refinement), allows the user to add phenomena and equations easily, permits solution by continuation methods (thus enhancing

convergence), and has extensive graphical output capabilities. Other packages are also available (see <http://cfd-online.com/>), and these may contain features and correlations specific to the engineer's industry. One important point to note is that for turbulent flow, all the programs contain approximations, using the k -epsilon models of turbulence, or large eddy simulations; the direct numerical simulation of turbulence is too slow to apply to very big problems, although it does give insight (independent of any approximations) that is useful for interpreting turbulent phenomena. Thus, the method used to include those turbulent correlations is important, and the method also may affect convergence or accuracy.

FAST FOURIER TRANSFORM

The Fourier transform and inverse transform are

$$Y(\omega) = \int_{-\infty}^{\infty} y(t)e^{i\omega t} dt$$

$$y(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} Y(\omega)e^{-i\omega t} d\omega$$

Suppose a signal $y(t)$ is sampled at equal intervals

$$y_n = y(n\Delta), \quad n = \dots, -2, -1, 0, 1, 2, \dots$$

Δ = sampling interval (e.g., time between samples)

The Nyquist critical frequency or critical angular frequency is

$$f_c = \frac{1}{2\Delta}, \quad \omega_c = \frac{\pi}{\Delta}$$

If a function $y(t)$ is bandwidth-limited to frequencies smaller than f_c , such as

$$Y(\omega) = 0 \quad \text{for } \omega > \omega_c$$

then the function is completely determined by its samples y_n . Thus, the entire information content of a signal can be recorded by sampling at a rate $\Delta^{-1} = 2f_c$. If the function is *not* bandwidth-limited, then aliasing occurs. Once a sample rate Δ is chosen, information corresponding to frequencies greater than f_c is simply aliased into that range. The way to detect this in a Fourier transform is to see if the transform approaches zero at $\pm f_c$; if not, aliasing has occurred, and a higher sampling rate is needed.

Next, suppose we have N samples, where N is even

$$y_k = y(t_k) \quad t_k = k\Delta \quad k = 0, 1, 2, \dots, N-1$$

and the sampling interval is Δ . With only N values $\{y_k\}$, it is not possible to determine the complete Fourier transform $Y(\omega)$. We calculate the value $Y(\omega_n)$ at the discrete points

$$\omega_n = \frac{2\pi n}{N\Delta}, \quad n = -\frac{N}{2}, \dots, 0, \dots, \frac{N}{2}$$

$$Y_n = \sum_{k=0}^{N-1} y_k e^{2\pi i k n / N}$$

$$Y(\omega_n) = \Delta Y_n$$

The discrete inverse Fourier transform is

$$y_k = \frac{1}{N} \sum_{n=0}^{N-1} Y_n e^{-2\pi i k n / N}$$

The fast Fourier transform (FFT) is used to calculate the Fourier transform as well as the inverse Fourier transform. A discrete Fourier transform of length N can be written as the sum of two discrete Fourier transforms, each of length $N/2$.

$$Y_k = Y_k^e + W^k Y_k^o$$

Here Y_k is the k th component of the Fourier transform of y , and Y_k^e is the k th component of the Fourier transform of the even components

of $\{y_j\}$ and is of length $N/2$. Similarly, Y_k^o is the k th component of the Fourier transform of the odd components of $\{y_j\}$ and is of length $N/2$. W is a constant, which is taken to the k th power.

$$W = e^{2\pi i / N}$$

Since Y_k has N components, while Y_k^e and Y_k^o have $N/2$ components, Y_k^e and Y_k^o are repeated once to give N components in the calculation of Y_k . This decomposition can be used recursively. Thus, Y_k is split into even and odd terms of length $N/4$.

$$Y_k^e = Y_k^{ee} + W^k Y_k^{eo}$$

$$Y_k^o = Y_k^{oe} + W^k Y_k^{oo}$$

This process is continued until there is only one component. For this reason, the number N is taken as a power of 2. The vector $\{y_j\}$ is filled with zeroes, if need be, to make $N = 2^p$ for some p . The standard Fourier transform takes N^2 operations to calculate, whereas the fast Fourier transform takes only $N \log_2 N$. For large N , the difference is significant; at $N = 100$ it is a factor of 15, but for $N = 1000$ it is a factor of 100.

The discrete Fourier transform can also be used for differentiating a function, and this is used in the spectral method for solving differential equations [Gottlieb, D., and S. A. Orszag, *Numerical Analysis of Spectral Methods: Theory and Applications*, SIAM, Philadelphia (1977); Trefethen, L. N., *Spectral Methods in Matlab*, SIAM, Philadelphia (2000)]. Suppose we have a grid of equidistant points

$$x_n = n\Delta x, \quad n = 0, 1, 2, \dots, 2N-1, \quad \Delta x = \frac{L}{2N}$$

The solution is known at each of these grid points $\{y(x_n)\}$. First the discrete Fourier transform is taken:

$$Y_k = \frac{1}{2N} \sum_{n=0}^{2N-1} y(x_n) e^{-2\pi i k x_n / L}, \quad k = -N, -N+1, \dots, 0, \dots, N-1, N$$

The inverse transformation is

$$y(x) = \frac{1}{L} \sum_{k=-N}^N Y_k e^{2\pi i k x / L}$$

Differentiate this to get

$$\frac{dy}{dx} = \frac{1}{L} \sum_{k=-N}^N Y_k \frac{2\pi i k}{L} e^{2\pi i k x / L}$$

Thus at the grid points

$$\left. \frac{dy}{dx} \right|_n = \sum_{k=-N}^N Y_k \frac{2\pi i k}{L} e^{2\pi i k x_n / L}$$

The process works as follows. From the solution at all grid points the Fourier transform is obtained using FFT, $\{Y_k\}$. Then this is multiplied by $2\pi i k / L$ to obtain the Fourier transform of the derivative.

$$Y'_k = Y_k \frac{2\pi i k}{L}$$

Then the inverse Fourier transform is taken using FFT, giving the value of the derivative at each of the grid points.

$$\left. \frac{dy}{dx} \right|_n = \frac{1}{L} \sum_{k=-N}^N Y'_k e^{2\pi i k x_n / L}$$

The spectral method is used for direct numerical simulation (DNS) of turbulence. The Fourier transform is taken of the differential equation, and the resulting equation is solved. Then the inverse transformation gives the solution. When there are nonlinear terms, they are calculated at each node in physical space, and the Fourier transform is taken of the result. This technique is especially suited to time-dependent problems, and the major computational effort is in the fast Fourier transform.

OPTIMIZATION

REFERENCES: General references include the following textbooks. For nonlinear programming, Fletcher, R., *Practical Methods of Optimization*, Wiley (1987); Nocedal, J., and S. J. Wright, *Numerical Optimization*, Springer, New York (1999); Conn, A. R., N. Gould, and P. Toint, *Trust Region Methods*, SIAM, Philadelphia (2000); Edgar, T. F., D. M. Himmelblau, and L. S. Lasdon, *Optimization of Chemical Processes*, McGraw-Hill (2002). For linear programming, Dantzig, G. B., *Linear Programming and Extensions*, Princeton, N.J.: Princeton University Press, (1963); Hillier, F., and G. J. Lieberman, *Introduction to Operations Research*, Holden-Day, San Francisco (1974). For mixed integer programming, Biegler, L. T., I. E. Grossmann, and A. W. Westerberg, *Systematic Methods for Chemical Process Design*, Prentice-Hall, Englewood Cliffs, N.J. (1997); Nemhauser, G. L., and L. A. Wolsey, *Integer and Combinatorial Optimization*, Wiley-Interscience, New York (1988). For global optimization, Floudas, C. A., *Deterministic Global Optimization: Theory, Algorithms and Applications*, Kluwer Academic Publishers (2000); Horst, R., and H. Tuy, *Global Optimization: Deterministic Approaches*, Springer-Verlag, Berlin (1993); Tawarmalani, M., and N. Sahinidis, *Convexification and Global Optimization in Continuous and Mixed-Integer Nonlinear Programming: Theory, Algorithms, Software, and Applications*, Kluwer Academic Publishers (2002). Many useful resources including descriptions, trial software, and examples can be found on the NEOS server (<http://www-neos.mcs.anl.gov>) maintained at Argonne National Laboratory. Background material for this section includes the two previous sections on matrix algebra and numerical analysis.

INTRODUCTION

Optimization is a key enabling tool for decision making in chemical engineering. It has evolved from a methodology of academic interest into a technology that continues to have a significant impact on engineering research and practice. Optimization algorithms form the core tools for (1) experimental design, parameter estimation, model development, and statistical analysis; (2) process synthesis analysis, design, and retrofit; (3) model predictive control and real-time optimization; and (4) planning, scheduling, and the integration of process operations into the supply chain.

As shown in Fig. 3-53, optimization problems that arise in chemical engineering can be classified in terms of continuous and discrete variables. For the former, nonlinear programming (NLP) problems form the most general case, and widely applied specializations include linear programming (LP) and quadratic programming (QP). An important distinction for NLP is whether the optimization problem is convex or nonconvex. The latter NLP problem may have multiple local optima, and an important question is whether a global solution is required for the NLP. Another important distinction is whether the problem is assumed to be differentiable or not.

Mixed integer problems also include discrete variables. These can be written as mixed integer nonlinear programs (MINLP), or as mixed integer linear programs (MILP), if all variables appear linearly in the constraint and objective functions. For the latter an important case occurs when all the variables are integer; this gives rise to an integer programming (IP) problem. IP problems can be further classified into many special problems (e.g., assignment, traveling salesperson, etc.), which are not shown in Fig. 3-53. Similarly, the MINLP problem also gives rise to special problem classes, although here the main distinction is whether its relaxation is convex or nonconvex.

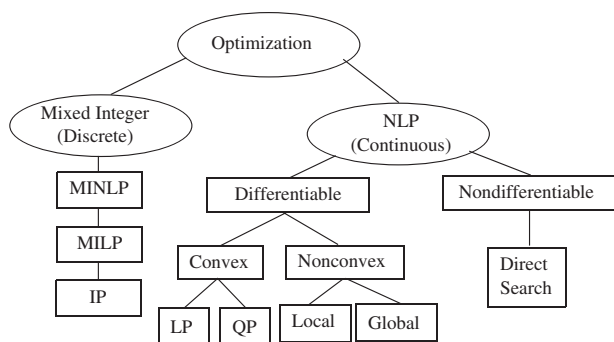


FIG. 3-53 Classes of optimization problems and algorithms.

The ingredients of formulating optimization problems include a mathematical model of the system, an objective function that quantifies a criterion to be extremized, variables that can serve as decisions, and, optionally, inequality constraints on the system. When represented in algebraic form, the general formulation of discrete/continuous optimization problems can be written as the following mixed integer optimization problem:

$$\begin{aligned} \text{Min } & f(x, y) \\ \text{subject to } & h(x, y) = 0 \\ & g(x, y) \leq 0 \\ & x \in \mathfrak{R}_n, y \in \{0, 1\} \end{aligned} \quad (3-84)$$

where $f(x, y)$ is the objective function (e.g., cost, energy consumption, etc.), $h(x, y) = 0$ are the equations that describe the performance of the system (e.g., material balances, production rates), and the inequality constraints $g(x, y) \leq 0$ can define process specifications or constraints for feasible plans and schedules. Note that the operator $\max f(x)$ is equivalent to $\min [-f(x)]$. We define the real n vector x to represent the continuous variables while the t vector y represents the discrete variables, which, without loss of generality, are often restricted to take values of 0 or 1 to define logical or discrete decisions, such as assignment of equipment and sequencing of tasks. (These variables can also be formulated to take on other integer values as well.) Problem (3-84) corresponds to a mixed integer nonlinear program when any of the functions involved are nonlinear. If all functions are linear, it corresponds to a mixed integer linear program. If there are no 0–1 variables, then problem (3-84) reduces to a nonlinear program (3-85) or linear program (3-97) depending on whether the functions are linear.

We start with continuous variable optimization and consider in the next section the solution of NLP problems with differentiable objective and constraint functions. If only local solutions are required for the NLP problem, then very efficient large-scale methods can be considered. This is followed by methods that are not based on local optimality criteria; we consider direct search optimization methods that do not require derivatives as well as deterministic global optimization methods. Following this, we consider the solution of mixed integer problems and outline the main characteristics of algorithms for their solution. Finally, we conclude with a discussion of optimization modeling software and its implementation on engineering models.

GRADIENT-BASED NONLINEAR PROGRAMMING

For continuous variable optimization we consider (3-84) without discrete variable y . The general NLP problem (3-85) is presented here:

$$\begin{aligned} \text{Min } & f(x) \\ \text{subject to } & h(x) = 0 \\ & g(x) \leq 0 \end{aligned} \quad (3-85)$$

and we assume that the functions $f(x)$, $h(x)$, and $g(x)$ have continuous first and second derivatives. A key characteristic of (3-85) is whether the problem is convex or not, i.e., whether it has a convex objective function and a convex feasible region. A function $\phi(x)$ of x in some domain X is convex if and only if for all points $x_1, x_2 \in X$

$$\phi[\alpha x_1 + (1 - \alpha)x_2] \leq \alpha\phi[x_1] + (1 - \alpha)\phi(x_2) \quad (3-86)$$

holds for all $\alpha \in (0, 1)$. [Strict convexity requires that the inequality (3-86) be strict.] Convex feasible regions require $g(x)$ to be a convex function and $h(x)$ to be linear. If (3-85) is a convex problem, then any local solution is guaranteed to be a global solution to (3-85). Moreover, if the objective function is strictly convex, then this solution x^* is unique. On the other hand, nonconvex problems may have multiple local solutions, i.e., feasible solutions that minimize the objective function within some neighborhood about the solution.

We consider first methods that find only local solutions to nonconvex problems, as more difficult (and expensive) search procedures are required to find a global solution. Local methods are currently very

efficient and have been developed to deal with very large NLP problems. Moreover, by considering the structure of convex NLP problems (including LP and QP problems), even more powerful methods can be applied. To study these methods, we first consider conditions for local optimality.

Local Optimality Conditions: A Kinematic Interpretation Instead of a formal development of conditions that define a local optimum, we present a more intuitive kinematic illustration. Consider the contour plot of the objective function $f(x)$, given in Fig. 3-54, as a smooth valley in space of the variables x_1 and x_2 . For the contour plot of this unconstrained problem $\text{Min } f(x)$, consider a ball rolling in this valley to the lowest point of $f(x)$, denoted by x^* . This point is at least a local minimum and is defined by a point with a zero gradient and at least nonnegative curvature in all (nonzero) directions p . We use the first-derivative (*gradient*) vector $\nabla f(x)$ and second-derivative (*Hessian*) matrix $\nabla_{xx} f(x)$ to state the necessary first- and second-order conditions for unconstrained optimality:

$$\nabla_x f(x^*) = 0 \quad p^T \nabla_{xx} f(x^*) p \geq 0 \quad \text{for all } p \neq 0 \quad (3-87)$$

These necessary conditions for local optimality can be strengthened to sufficient conditions by making the inequality in (3-87) strict (i.e., positive curvature in all directions). Equivalently, the sufficient (necessary) curvature conditions can be stated as follows: $\nabla_{xx} f(x^*)$ has all positive (nonnegative) eigenvalues and is therefore defined as a positive (semidefinite) definite matrix.

Now consider the imposition of inequality $[g(x) \leq 0]$ and equality constraints $[h(x) = 0]$ in Fig. 3-55. Continuing the kinematic interpretation, the inequality constraints $g(x) \leq 0$ act as “fences” in the valley, and equality constraints $h(x) = 0$ act as “rails.” Consider now a ball, constrained on a rail and within fences, to roll to its lowest point. This stationary point occurs when the normal forces exerted by the fences $[-\nabla g(x^*)]$ and rails $[-\nabla h(x^*)]$ on the ball are balanced by the force of gravity $[-\nabla f(x^*)]$. This condition can be stated by the following *Karush-Kuhn-Tucker (KKT) necessary conditions* for constrained optimality:

Balance of Forces It is convenient to define the L function $L(x, \lambda, v) = f(x) + g(x)^T \lambda + h(x)^T v$, along with “weights” or multipliers λ and v for the constraints. The stationarity condition (balance of forces acting on the ball) is then given by

$$\nabla L(x, \lambda, v) = \nabla f(x) + \nabla h(x) \lambda + \nabla g(x) v = 0 \quad (3-88)$$

Feasibility Both inequality and equality constraints must be satisfied (ball must lie on the rail and within the fences):

$$h(x) = 0, \quad g(x) \leq 0 \quad (3-89)$$

Complementarity Inequality constraints are either strictly satisfied (active) or inactive, in which case they are irrelevant to the solu-

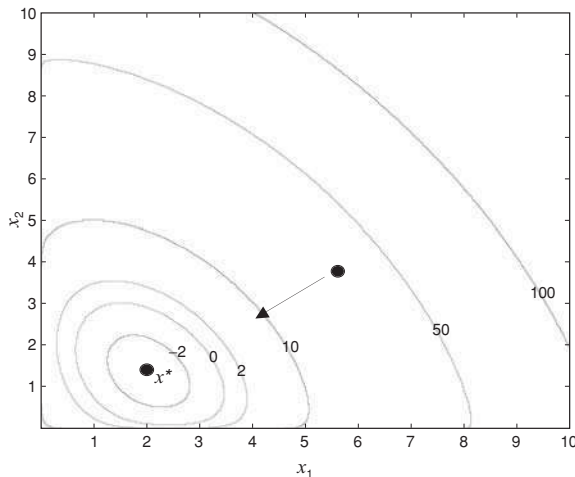


FIG. 3-54 Unconstrained minimum.

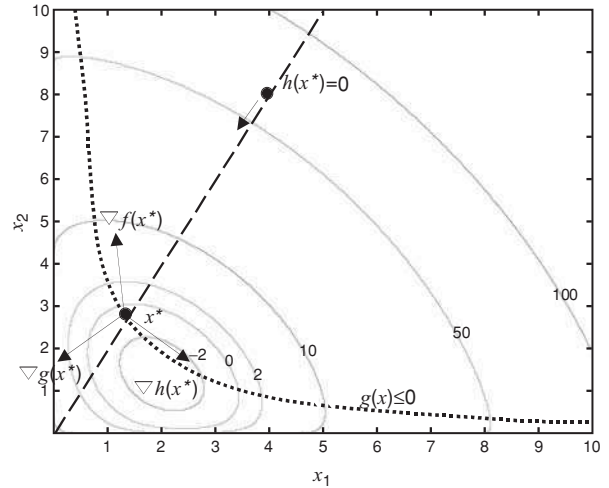


FIG. 3-55 Constrained minimum.

tion. In the latter case the corresponding KKT multiplier must be zero. This is written as

$$v^T g(x) = 0, \quad v \geq 0 \quad (3-90)$$

Constraint Qualification For a local optimum to satisfy the KKT conditions, an additional regularity condition is required on the constraints. This can be defined in several ways. A typical condition is that the active constraints at x^* be linearly independent; i.e., the matrix: $[\nabla h(x^*) | \nabla g_A(x^*)]$ is full column rank, where g_A is the vector of inequality constraints with elements that satisfy $g_{A,i}(x^*) = 0$. With this constraint qualification, the KKT multipliers (λ, v) are guaranteed to be unique at the optimal solution.

Second-Order Conditions As with unconstrained optimization, nonnegative (positive) curvature is necessary (sufficient) in all the allowable (i.e., constrained) nonzero directions p . The necessary second-order conditions can be stated as

$$p^T \nabla_{xx} L(x^*) p \geq 0 \quad \text{for all } p \neq 0, \nabla h(x^*)^T p = 0, \nabla g(x^*)^T p \geq 0, \nabla g_A(x^*)^T p = 0 \quad (3-91)$$

and the corresponding sufficient conditions require first the inequality in (3-91) to be strict. Note that for the example in Fig. 3-54, the allowable directions p span the entire space for x while in Fig. 3-55 there are no allowable directions p .

Example To illustrate the KKT conditions, consider the following unconstrained NLP problem:

$$\text{Min } (x_1)^2 - 4x_1 + 3/2 (x_2)^2 - 7x_2 + x_1 x_2 + 9 - \ln x_1 - \ln x_2 \quad (3-92)$$

corresponding to the contour plot in Fig. 3-54. The optimal solution can be found by solving for the first-order conditions (3-87):

$$\nabla f(x) = \begin{bmatrix} 2x_1 - 4 + x_2 - 1/x_1 \\ 3x_2 - 7 + x_1 - 1/x_2 \end{bmatrix} = 0 \Rightarrow x^* = \begin{bmatrix} 1.3475 \\ 2.0470 \end{bmatrix} \quad (3-93)$$

and $f(x^*) = -2.8742$. Checking the second-order conditions leads to

$$\begin{aligned} \nabla_{xx} f(x^*) &= \begin{bmatrix} 2 + 1/(x_1^*)^2 & 1 \\ 1 & 3 + 1/(x_2^*)^2 \end{bmatrix} \\ \Rightarrow \nabla_{xx} f(x^*) &= \begin{bmatrix} 2.5507 & 1 \\ 1 & 3.2387 \end{bmatrix} \quad (\text{positive definite}) \end{aligned} \quad (3-94)$$

Now consider the constrained NLP problems

$$\begin{aligned} \text{Min } (x_1^*)^2 - 4x_1 + 3/2 (x_2^*)^2 - 7x_2 + x_1 x_2 + 9 - \ln x_1 - \ln x_2 \\ \text{subject to } 4 - x_1 x_2 \leq 0 \\ 2x_1 - x_2 = 0 \end{aligned} \quad (3-95)$$

that correspond to the plot in Fig. 3-54. The optimal solution can be found by

applying the first-order KKT conditions (3-88) to (3-90):

$$\begin{aligned} \nabla L(x, \lambda, v) &= \nabla f(x) + \nabla h(x)\lambda + \nabla g(x)v = \begin{bmatrix} 2x_1 - 4 + x_2 - 1/x_1 \\ 3x_2 - 7 + x_1 - 1/x_2 \end{bmatrix} \\ &+ \begin{bmatrix} 2 \\ -1 \end{bmatrix} \lambda + \begin{bmatrix} -x_2 \\ -x_1 \end{bmatrix} v = 0 \\ g(x) &= 4 - x_1 x_2 \leq 0, \quad h(x) = 2x_1 - x_2 = 0 \\ g(x)v &= (4 - x_1 x_2)v, \quad v \geq 0 \end{aligned} \quad (3-96)$$

$$\Downarrow$$

$$x^* = \begin{bmatrix} 1.4142 \\ 2.8284 \end{bmatrix}, \quad \lambda^* = 1.036, \quad v^* = 1.068$$

and $f(x^*) = -1.8421$. Checking the second-order conditions (3-91) leads to

$$\begin{aligned} \nabla_{xx} L(x^*, \lambda^*, v^*) &= \nabla_{xx} [f(x^*) + h(x^*)\lambda^* + g(x^*)v^*] = \begin{bmatrix} 2 + 1/(x_1^2) & 1 - v \\ 1 - v & 3 + 1/(x_2^2) \end{bmatrix} \\ &= \begin{bmatrix} 2.5 & 0.068 \\ 0.068 & 3.125 \end{bmatrix} \\ [\nabla h(x^*) \mid \nabla g_A(x^*)]^T p &= \begin{bmatrix} 2 & -2.8284 \\ -1 & -1.4142 \end{bmatrix}^T p = 0, \quad p \neq 0 \end{aligned}$$

However, note that because $[\nabla h(x^*) \mid \nabla g_A(x^*)]$ is nonsingular, there are no nonzero vectors p that satisfy the allowable directions. Hence, the sufficient second-order conditions $[p^T \nabla_{xx} L(x^*, \lambda^*, v^*) p > 0, \text{ for all allowable } p]$ are *vacuously satisfied* for this problem.

Convex Cases of NLP Problems Linear programs and quadratic programs are special cases of (3-85) that allow for more efficient solution, based on application of KKT conditions (3-88) through (3-91). Because these are convex problems, any locally optimal solution is a global solution. In particular, if the objective and constraint functions in (3-85) are linear, then the following linear program (LP)

$$\begin{aligned} \text{Min } c^T x \\ \text{subject to } Ax = b \\ Cx \leq d \end{aligned} \quad (3-97)$$

can be solved in a finite number of steps, and the optimal solution lies at a vertex of the polyhedron described by the linear constraints. This is shown in Fig. 3-56, and in so-called primal degenerate cases, multiple vertices can be alternate optimal solutions, with the same values of the objective function. The standard method to solve (3-97) is the simplex method, developed in the late 1940s (see Dantzig, 1963) although, starting from Karmarkar's discovery in 1984, interior point methods have become quite advanced and competitive for highly constrained problems [Wright, S. J., *Primal-Dual Interior Point Methods*,

SIAM, Philadelphia (1996)]. The simplex method proceeds by moving successively from vertex to vertex with improved objective function values. Methods to solve (3-97) are well implemented and widely used, especially in planning and logistical applications. They also form the basis for MILP methods discussed later. Currently, state-of-the-art LP solvers can handle millions of variables and constraints, and the application of further decomposition methods leads to the solution of problems that are two or three orders of magnitude larger than this. See the general references of Hillier and Lieberman (1974) and Edgar et al. (2002) for more details. Also, the interior point method is described below from the perspective of more general NLP problems.

Quadratic programs (QPs) represent a slight modification of (3-97) and can be stated as

$$\begin{aligned} \text{Min } c^T x + 1/2 x^T Q x \\ \text{subject to } Ax = b \\ Cx \leq d \end{aligned} \quad (3-98)$$

If the matrix Q is positive semidefinite (positive definite) when projected into the null space of the active constraints, then (3-98) is (strictly) convex and the QP is a global (and unique) minimum. Otherwise, local solutions exist for (3-98), and more extensive global optimization methods are needed to obtain the global solution. Like LPs, convex QPs can be solved in a finite number of steps. However, as seen in Fig. 3-57, these optimal solutions can lie on a vertex, on a constraint boundary, or in the interior. A number of active set strategies have been created that solve the KKT conditions of the QP and incorporate efficient updates of active constraints. Popular methods include null space algorithms, range space methods, and Schur complement methods. As with LPs, QP problems can also be solved with interior point methods [see Wright (1996)].

Solving the General NLP Problem Solution techniques for (3-85) deal with satisfaction of the KKT conditions (3-88) through (3-91). Many NLP solvers are based on successive quadratic programming (SQP) as it allows the construction of a number of NLP algorithms based on the Newton-Raphson method for equation solving (see "Numerical Analysis" section). SQP solvers have been shown to require the fewest function evaluations to solve NLP problems [Schittkowski, K., *Lecture Notes in Economics and Mathematical Systems*, no. 282, Springer-Verlag, Berlin (1987)], and they can be tailored to a broad range of process engineering problems with different structure.

The SQP strategy applies the equivalent of a Newton step to the KKT conditions of the nonlinear programming problem, and this leads to a fast rate of convergence. By adding slack variables s , the first-order KKT conditions can be rewritten as

$$\nabla f(x) + \nabla h(x)\lambda + \nabla g(x)v = 0 \quad (3-99a)$$

$$h(x) = 0 \quad (3-99b)$$

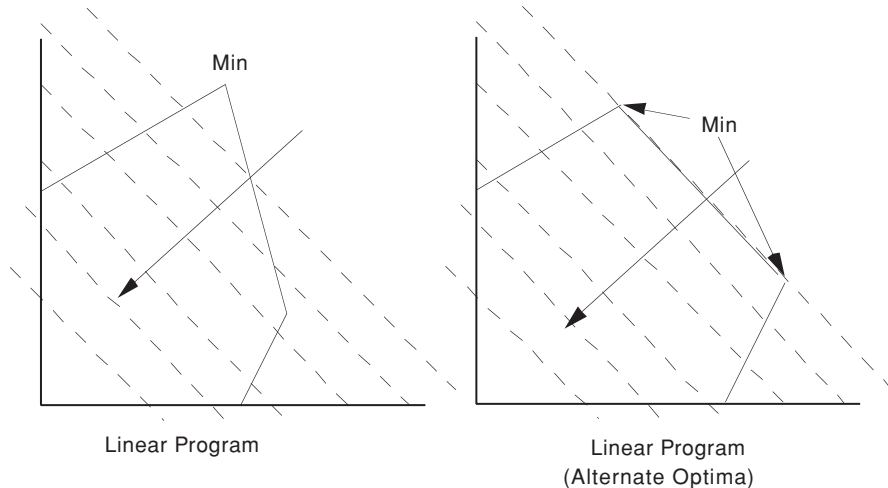


FIG. 3-56 Contour plots of linear programs.

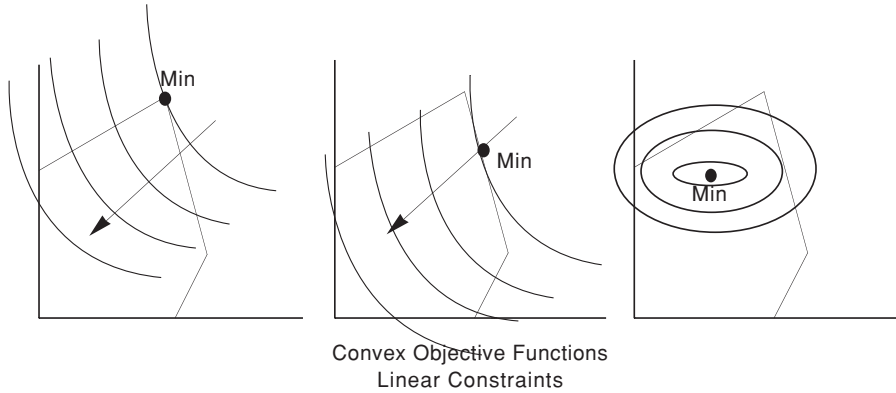


FIG. 3-57 Contour plots of convex quadratic programs.

$$g(x) + s = 0 \quad (3-99c)$$

$$SVe = 0 \quad (3-99d)$$

$$(s, v) \geq 0 \quad (3-99e)$$

where $e = [1, 1, \dots, 1]^T$, $S = \text{diag}\{s\}$, and $V = \text{diag}\{v\}$. SQP methods find solutions that satisfy (3-99) by generating Newton-like search directions at iteration k . However, Eqs. (3-99d) and active bounds (3-99e) are dependent at the solution and serve to make the KKT system ill-conditioned near the solution. SQP algorithms treat these conditions in two ways. In the *active set strategy*, discrete decisions are made regarding the active constraint set $i \in I = \{i | g_i(x^*) = 0\}$ and (3-99d) is replaced by $s_i = 0$, $i \in I$, and $v_i = 0$, $i \notin I$. Determining the active set is a combinatorial problem, and a straightforward way to determine an estimate of the active set [and to satisfy (3-99e)] is to formulate and solve, at a point x^k , the following QP at iteration k :

$$\begin{aligned} &\text{Min } \nabla f(x^k)^T p + 1/2 p^T \nabla_{xx} L(x^k, \lambda^k, v^k) p \\ &\text{subject to } h(x^k) + \nabla h(x^k)^T p = 0 \\ &\quad g(x^k) + \nabla g(x^k)^T p + s = 0, \quad s \geq 0 \end{aligned} \quad (3-100)$$

The KKT conditions of (3-100) are given by

$$\nabla f(x^k) + \nabla^2 L(x^k, \lambda^k, v^k) p + \nabla h(x^k) \lambda + \nabla g(x^k) v = 0 \quad (3-101a)$$

$$h(x^k) + \nabla h(x^k)^T p = 0 \quad (3-101b)$$

$$g(x^k) + \nabla g(x^k)^T p + s = 0 \quad (3-101c)$$

$$SVe = 0 \quad (3-101d)$$

$$(s, v) \geq 0 \quad (3-101e)$$

where the Hessian of the Lagrange function $\nabla_{xx} L(x, \lambda, v) = \nabla_{xx} [f(x) + h(x)^T \lambda + g(x)^T v]$ is calculated directly or through a quasi-Newton approximation (created by differences of gradient vectors). It is easy to show that (3-101a) through (3-101c) correspond to a Newton-Raphson step for (3-99a) through (3-99c) applied at iteration k . Also, selection of the active set is now handled at the QP level by satisfying the conditions (3-101d) and (3-101e). To evaluate and change candidate active sets, QP algorithms apply inexpensive matrix updating strategies to the KKT matrix associated with (3-100). Details of this approach can be found in Nocedal and Wright (1999) and Fletcher (1987).

As alternatives that avoid the combinatorial problem of selecting the active set, interior point (or barrier) methods modify the NLP problem (3-85) to form

$$\begin{aligned} &\text{Min } f(x^k) - \mu \sum_i \ln s_i \\ &\text{subject to } h(x^k) = 0 \\ &\quad g(x^k) + s = 0 \end{aligned} \quad (3-102)$$

where the solution to (3-103) has $s > 0$ for the penalty parameter $\mu > 0$, and decreasing μ to 0 leads to solution of problem (3-85). The KKT conditions for this problem can be written as

$$\nabla f(x^*) + \nabla h(x^*) \lambda + \nabla g(x^*) v = 0$$

$$\begin{aligned} &h(x^*) = 0 \\ &g(x^*) + s = 0 \\ &SVe = \mu e \end{aligned} \quad (3-103)$$

and for $\mu > 0$, $s > 0$, and $v > 0$, Newton steps generated to solve (3-103) are well behaved and analogous to (3-101), with a modification on the right-hand side of (3-101d). A detailed description of this algorithm, called IPOPT, can be found in Wächter and Biegler [*Math. Prog.* **106**(1), 25–57 (2006)].

Both active set and interior point methods possess clear tradeoffs. Interior point methods may require more iterations to solve (3-102) for various values of μ , while active set methods require the solution of the more expensive QP subproblem (3-100). Thus, if there are few inequality constraints or an active set is known (say from a good starting guess, or a known QP solution from a previous iteration), then solving (3-100) is not expensive and the active set method is favored. On the other hand, for problems with many inequality constraints, interior point methods are often faster, as they avoid the combinatorial problem of selecting the active set. This is especially true for large-scale problems and when a large number of bounds are active. Examples that demonstrate the performance of these approaches include the solution of model predictive control (MPC) problems [Rao et al., *J. Optim. Theory Appl.* **99**:723 (1998); Albuquerque et al., *Comp. Chem. Eng.* **23**:283 (1997)] and the solution of large optimal control problems using barrier NLP solvers. For instance, IPOPT allows the solution of problems with more than 1,000,000 variables and up to 50,000 degrees of freedom [see Biegler et al., *Chem. Eng. Sci.* **57**(4):575–593 (2002); Laird et al., *ASCE J. Water Resource Management and Planning* **131**(2):125 (2005)].

Other Gradient-Based NLP Solvers In addition to SQP methods, a number of NLP solvers have been developed and adapted for large-scale problems. Generally these methods require more function evaluations than of SQP methods, but they perform very well when interfaced to optimization modeling platforms, where function evaluations are cheap. All these can be derived from the perspective of applying Newton steps to portions of the KKT conditions.

LANCELOT (Conn et al., 2000) is based on the solution of bound-constrained subproblems. Here an augmented Lagrangian is formed from (3-85) and the following subproblem is solved:

$$\begin{aligned} &\text{Min } f(x) + \lambda^T h(x) + v^T [g(x) + s] + 1/2 \rho \|h(x), g(x) + s\|^2 \\ &\text{subject to } s \geq 0 \end{aligned} \quad (3-104)$$

The above subproblem can be solved very efficiently for fixed values of the multipliers λ and v and penalty parameter ρ . Here a gradient projection trust region method is applied. Once subproblem (3-104) is solved, the multipliers and penalty parameter are updated in an outer loop and the cycle repeats until the KKT conditions for (3-85) are satisfied. LANCELOT works best when exact second derivatives are available. This promotes a fast convergence rate in solving each

subproblem and allows a bound-constrained trust region method to exploit directions of negative curvature in the Hessian matrix.

Reduced gradient methods are active set strategies that rely on partitioning the variables and solving (3-99) in a nested manner. Without loss of generality, problem (3-85) can be rewritten as $\text{Min } f(z)$ subject to $c(z) = 0$, $a \leq z \leq b$. Variables are partitioned as nonbasic variables (those fixed to their bounds), basic variables (those that can be solved from the equality constraints), and superbasic variables (those remaining variables between bounds that serve to drive the optimization); this leads to $z^T = [z_N^T, z_B^T, z_S^T]$. This partition is derived from local information and may change over the course of the optimization iterations. The corresponding KKT conditions can be written as

$$\nabla_N f(z) + \nabla_N c(z)\gamma = \beta_a - \beta_b \quad (3-105a)$$

$$\nabla_B f(z) + \nabla_B c(z)\gamma = 0 \quad (3-105b)$$

$$\nabla_S f(z) + \nabla_S c(z)\gamma = 0 \quad (3-105c)$$

$$c(z) = 0 \quad (3-105d)$$

$$z_{N_j} = a_j \text{ or } b_j, \quad \beta_{a_j} \geq 0, \quad \beta_{b_j} = 0 \text{ or } \beta_{b_j} \geq 0, \quad \beta_{a_j} = 0 \quad (3-105e)$$

where γ and β are the KKT multipliers for the equality and bound constraints, respectively, and (3-105e) replaces the complementarity conditions (3-90). Reduced gradient methods work by nesting equations (3-105b,d) within (3-105a,c). At iteration k , for fixed values of z_N^k and z_S^k , we can solve for z_B by using (3-105d) and for γ by using (3-105b). Moreover, linearization of these equations leads to sensitivity information (i.e., constrained derivatives or reduced gradients) that indicates how z_B changes with respect to z_S and z_N . The algorithm then proceeds by updating z_S by using reduced gradients in a Newton-type iteration to solve Eq. (3-105c). Following this, bound multipliers β are calculated from (3-105a). Over the course of the iterations, if the variables z_B or z_S exceed their bounds or if some bound multipliers β become negative, then the variable partition needs to be changed and Eqs. (3-105) are reconstructed. These reduced gradient methods are embodied in the popular GRG2, CONOPT, and SOLVER codes (Edgar et al., 2002). The SOLVER code has been incorporated into Microsoft Excel. CONOPT [Drud, A., *ORSA J. Computing* 6: 207–216 (1994)] is an efficient and widely used code in several optimization modelling environments.

MINOS (Murtagh and Saunders, Technical Report SOL 83-20R, Stanford University, 1987) is a well-implemented package that offers a variation on reduced gradient strategies. At iteration k , Eq. (3-105d) is replaced by its linearization

$$c(z_N^k, z_B^k, z_S^k) + \nabla_B c(z^k)^T (z_B - z_B^k) + \nabla_S c(z^k)^T (z_S - z_S^k) = 0 \quad (3-106)$$

and Eqs. (3-105a–c, e) are solved with (3-106) as a subproblem by using concepts from the reduced gradient method. At the solution of this subproblem, constraints (3-105d) are relinearized and the cycle repeats until the KKT conditions of (3-105) are satisfied. The augmented lagrangian function from (3-104) is used to penalize movement away from the feasible region. For problems with few degrees of freedom, the resulting approach leads to an extremely efficient method even for very large problems. MINOS has been interfaced to a number of modelling systems and enjoys widespread use. It performs especially well on large problems with few nonlinear constraints. However, on highly nonlinear problems it is usually less reliable than other reduced gradient methods.

Algorithmic Details for NLP Methods All the above NLP methods incorporate concepts from the Newton-Raphson method for equation solving. Essential features of these methods are that they provide (1) accurate derivative information to solve for the KKT conditions, (2) stabilization strategies to promote convergence of the Newton-like method from poor starting points, and (3) regularization of the Jacobian matrix in Newton's method (the so-called KKT matrix) if it becomes singular or ill-conditioned.

1. *NLP methods provide first and second derivatives.* The KKT conditions require first derivatives to define stationary points, so accurate first derivatives are essential to determine locally optimal solutions for differentiable NLPs. Moreover, Newton-Raphson methods that are applied to the KKT conditions, as well as the task of checking second-order KKT conditions, necessarily require second-

derivative information. (Note that second-order conditions are not checked by methods that do not use second derivatives.) With the recent development of automatic differentiation tools, many modeling and simulation platforms can provide exact first and second derivatives for optimization. When second derivatives are available for the objective or constraint functions, they can be used directly in LANCELOT as well as SQP and reduced gradient methods. Otherwise, on problems with few superbasic variables, both reduced gradient methods and SQP methods [with reduced gradient methods applied to the QP subproblem (3-100)] can benefit from positive definite quasi-Newton approximations [Nocedal and Wright (1999)] applied to reduced second-derivative quantities (the so-called reduced Hessian). Finally, for problems with least squares functions (see “Statistics” subsection), as in data reconciliation, parameter estimation, and model predictive control, one can often assume that the values of the objective function and its gradient at the solution are vanishingly small. Under these conditions, one can show that the multipliers (λ, v) also vanish and $\nabla_{xx} L(x, \lambda, v)$ can be substituted by $\nabla_{xx} f(x^*)$. This *Gauss-Newton approximation* has been shown to be very efficient for the solution of least squares problems [see Nocedal and Wright (1999)].

2. *Line search and trust region methods promote convergence from poor starting points.* These are commonly used with the search directions calculated from NLP subproblems such as (3-100). In a *trust region approach*, the constraint $\|p\| \leq \Delta$ is added and the iteration step is taken if there is sufficient reduction of some merit function (e.g., the objective function weighted with some measure of the constraint violations). The size of the trust region Δ is adjusted based on the agreement of the reduction of the actual merit function compared to its predicted reduction from the subproblem (see Conn et al., 2000, for details). Such methods have strong global convergence properties and are especially appropriate for ill-conditioned NLPs. This approach has been applied in the KNITRO code [Byrd, Hribar, and Nocedal, *SIAM J. Optimization* 9(4):877 (1999)]. *Line search methods* can be more efficient on problems with reasonably good starting points and well-conditioned subproblems, as in real-time optimization. Typically, once a search direction is calculated from (3-100), or other related subproblem, a step size $\alpha \in (0, 1)$ is chosen so that $x^k + \alpha p$ leads to a sufficient decrease of a merit function. As a recent alternative, a novel *filter stabilization strategy* (for both line search and trust region approaches) has been developed based on a bicriterion minimization, with the objective function and constraint infeasibility as competing objectives [Fletcher et al., *SIAM J. Optimization* 13(3):635 (2002)]. This method often leads to better performance than that based on merit functions.

3. *Regularization of the KKT matrix for the NLP subproblem is essential for good performance of general-purpose algorithms.* For instance, to obtain a unique solution to (3-100), active constraint gradients must be full rank and the Hessian matrix, when projected into the null space of the active constraint gradients, must be positive definite. These properties may not hold far from the solution, and corrections to the Hessian in SQP may be necessary (see Fletcher, 1987). Regularization methods ensure that subproblems such as (3-100) remain well-conditioned; they include addition of positive constants to the diagonal of the Hessian matrix to ensure its positive definiteness, judicious selection of active constraint gradients to ensure that they are linearly independent, and scaling the subproblem to reduce the propagation of numerical errors. Often these strategies are heuristics built into particular NLP codes. While quite effective, most of these heuristics do not provide convergence guarantees for general NLPs.

From the conceptual descriptions as well as algorithmic details given above, it is clear that NLP solvers are complex algorithms that have required considerable research and development to turn them into reliable and efficient software tools. Practitioners who are confronted with engineering optimization problems should therefore leverage these efforts, rather than write their own codes. Table 3-4 presents a sampling of available NLP codes that represent the above classifications. Much more information on these and other codes can be found on the NEOS server (www-neos.mcs.anl.gov) and the NEOS Software Guide: <http://www-fp.mcs.anl.gov/otc/Guide/SoftwareGuide>.

TABLE 3-4 Representative NLP Solvers

Method	Algorithm type	Stabilization	Second-order information
CONOPT (Drud, 1994)	Reduced gradient	Line search	Exact and quasi-Newton
GRC2 (Edgar et al., 2002)	Reduced gradient	Line search	Quasi-Newton
IPOPT	SQP, barrier	Line search	Exact
KNITRO (Byrd et al., 1997)	SQP, barrier	Trust region	Exact and quasi-Newton
LANCELOT	Augmented lagrangian, bound constrained	Trust region	Exact and quasi-Newton
LOQO	SQP, barrier	Line search	Exact
MINOS	Reduced gradient, augmented lagrangian	Line search	Quasi-Newton
NPSOL	SQP, active set	Line search	Quasi-Newton
SNOPT	Reduced space SQP, active set	Line search	Quasi-Newton
SOCS	SQP, active set	Line search	Exact
SOLVER	Reduced gradient	Line search	Quasi-Newton
SRQP	Reduced space SQP, active set	Line search	Quasi-Newton

OPTIMIZATION METHODS WITHOUT DERIVATIVES

A broad class of optimization strategies does not require derivative information. These methods have the advantage of easy implementation and little prior knowledge of the optimization problem. In particular, such methods are well suited for "quick and dirty" optimization studies that explore the scope of optimization for new problems, prior to investing effort for more sophisticated modeling and solution strategies. Most of these methods are derived from heuristics that naturally spawn numerous variations. As a result, a very broad literature describes these methods. Here we discuss only a few important trends in this area.

Classical Direct Search Methods Developed in the 1960s and 1970s, these methods include one-at-a-time search and methods based on experimental designs (EVOP). At that time, these direct search methods were the most popular optimization methods in chemical engineering. Methods that fall into this class include the pattern search of Hooke and Jeeves [*J. ACM* **8**:212 (1961)], the conjugate direction method of Powell (1964), simplex and complex searches, in particular Nelder-Mead [*Comput. J.* **7**: 308 (1965)], and the adaptive random search methods of Luus-Jaakola [*AIChE J.* **19**: 760 (1973)], Goulcher and Cesares Long [*Comp. Chem. Engr.* **2**: 23 (1978)] and

Banga et al. [in *State of the Art in Global Optimization*, C. Floudas and P. Pardalos (eds.), Kluwer, Dordrecht, p. 563 (1996)]. All these methods require only objective function values for unconstrained minimization. Associated with these methods are numerous studies on a wide range of process problems. Moreover, many of these methods include heuristics that prevent premature termination (e.g., directional flexibility in the complex search as well as random restarts and direction generation). To illustrate these methods, Fig. 3-58 illustrates the performance of a pattern search method as well as a random search method on an unconstrained problem.

Simulated Annealing This strategy is related to random search methods and derives from a class of heuristics with analogies to the motion of molecules in the cooling and solidification of metals [Laarhoven and Aarts, *Simulated Annealing: Theory and Applications*, Reidel Publishing, Dordrecht (1987)]. Here a temperature parameter θ can be raised or lowered to influence the probability of accepting points that do not improve the objective function. The method starts with a base point x and objective value $f(x)$. The next point x' is chosen at random from a distribution. If $f(x') < f(x)$, the move is accepted with x' as the new point. Otherwise, x' is accepted with probability $p(\theta, x', x) = \exp\{-[f(x') - f(x)]/\theta\}$ and the Glauber distribution, $p(\theta, x, x') = \exp\{-[f(x') - f(x)]/\theta\} / (1 + \exp\{-[f(x') - f(x)]/\theta\})$. The θ parameter is then reduced, and the method continues until no further progress is made.

Genetic Algorithms This approach, first proposed in Holland, J. H., *Adaptations in Natural and Artificial Systems* [University of Michigan Press, Ann Arbor (1975)], is based on the analogy of improving a population of solutions through modifying their gene pool. It also has similar performance characteristics as random search methods and simulated annealing. Two forms of genetic modification, crossover or mutation, are used and the elements of the optimization vector x are represented as binary strings. Crossover deals with random swapping of vector elements (among parents with highest objective function values or other rankings of population) or any linear combinations of two parents. Mutation deals with the addition of a random variable to elements of the vector. Genetic algorithms (GAs) have seen widespread use in process engineering, and a number of codes are available. Edgar et al. (2002) describe a related GA that is available in MS Excel.

Derivative-free Optimization (DFO) In the past decade, the availability of parallel computers and faster computing hardware and the need to incorporate complex simulation models within optimization studies have led a number of optimization researchers to reconsider classical direct search approaches. In particular, Dennis and Torczon [*SIAM J. Optim.* **1**: 448 (1991)] developed a multidimensional search algorithm that extends the simplex approach of Nelder

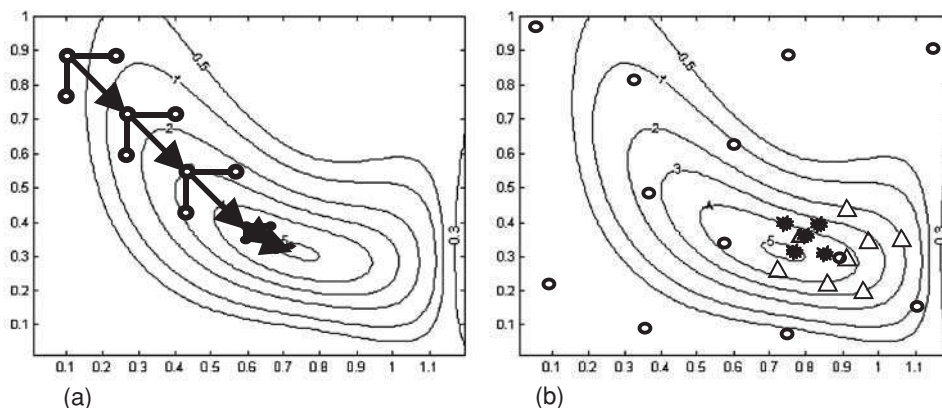


FIG. 3-58 Examples of optimization methods without derivatives. (a) Pattern search method. (b) Random search method. ○, first phase; △, second phase; ◻, third phase.

and Mead (1965). They note that the Nelder-Mead algorithm fails as the number of variables increases, even for very simple problems. To overcome this, their multidimensional pattern search approach combines reflection, expansion, and contraction steps that act as line search algorithms for a number of linear independent search directions. This approach is easily adapted to parallel computation, and the method can be tailored to the number of processors available. Moreover, this approach converges to locally optimal solutions for unconstrained problems and observes an unexpected performance synergy when multiple processors are used. The work of Dennis and Torczon (1991) has spawned considerable research on the analysis and code development for DFO methods. For instance, Conn et al. [*Math. Programming*, Series B, **79**(3): 397 (1997)] constructed a multivariable DFO algorithm that uses a surrogate model for the objective function within a trust region method. Here points are sampled to obtain a well-defined quadratic interpolation model, and descent conditions from trust region methods enforce convergence properties. A number of trust region methods that rely on this approach are reviewed in Conn et al. (1997). Moreover, a number of DFO codes have been developed that lead to black box optimization implementations for large, complex simulation models [see Audet and Dennis, *SIAM J. Optim.* **13**: 889 (2003); Kolda et al., *SIAM Rev.* **45**(3): 385 (2003)]. These include the DAKOTA package at Sandia National Lab (Eldred, 2002; <http://endo.sandia.gov/DAKOTA/software.html>) and FOCUS developed at Boeing Corporation (Booker et al., CRPC Technical Report 98739, Rice University, February 1998).

Direct search methods are easy to apply to a wide variety of problem types and optimization models. Moreover, because their termination criteria are not based on gradient information and stationary points, they are more likely to favor the search for globally optimal rather than locally optimal solutions. These methods can also be adapted easily to include integer variables. However, no rigorous convergence properties to globally optimal solutions have yet been discovered. Also, these methods are best suited for unconstrained problems or for problems with simple bounds. Otherwise, they may have difficulties with constraints, as the only options open for handling constraints are equality constraint elimination and addition of penalty functions for inequality constraints. Both approaches can be unreliable and may lead to failure of the optimization algorithm. Finally, the performance of direct search methods scales poorly (and often exponentially) with the number of decision variables. While performance can be improved with the use of parallel computing, these methods are rarely applied to problems with more than a few dozen decision variables.

GLOBAL OPTIMIZATION

Deterministic optimization methods are available for nonconvex nonlinear programming problems of the form (3-85) that guarantee convergence to the global optimum. More specifically, one can show under mild conditions that they converge to an ϵ distance to the global optimum in a finite number of steps. These methods are generally more expensive than local NLP methods, and they require the exploitation of the structure of the nonlinear program.

Because there are no optimality conditions like the KKT conditions for global optimization, these methods work by first partitioning the problem domain (i.e., containing the feasible region) into subregions. Upper bounds on the objective function are computed over all subregions of the problem. In addition, lower bounds can be derived from convex relaxations of the objective function and constraints for each subregion. The algorithm then proceeds to eliminate all subregions that have infeasible constraint relaxations or lower bounds that are greater than the least upper bound. After this, the remaining regions are further partitioned to create new subregions, and the cycle continues until the upper and lower bounds converge.

This basic concept leads to a wide variety of global algorithms, with the following features that can exploit different problem classes. *Bounding* strategies relate to the calculation of upper and lower bounds. For the former, any feasible point or, preferably, a locally optimal point in the subregion can be used. For the lower bound, convex relaxations of the objective and constraint functions are derived.

The *refining* step deals with the construction of partitions in the domain and further partitioning them during the search process. Finally, the *selection* step decides on the order of exploring the open subregions.

For simplicity, consider the problem $\text{Min } f(x)$ subject to $g(x) \leq 0$ where each function can be defined by additive terms. Convex relaxations for $f(x)$ and $g(x)$ can be derived in the following ways:

- Convex additive terms remain unmodified in these functions.
- Concave additive unary terms are replaced by a linear underestimating function that matches the term at the bounds of the subregion.
- Nonconvex polynomial terms can be replaced by a set of binary terms, with new variables introduced to define the higher-order polynomials.
- Binary terms can be relaxed by using the McCormick underestimator; e.g., the binary term xz is replaced by a new variable w and linear inequality constraints

$$\begin{aligned} w &\geq x_l \bar{z} + z_l \bar{x} - x_l \bar{z}_l \\ w &\geq x_u \bar{z} + z_u \bar{x} - x_u \bar{z}_u \\ w &\leq x_l \bar{z} + z_l \bar{x} - x_l \bar{z}_l \\ w &\leq x_u \bar{z} + z_u \bar{x} - x_u \bar{z}_u \end{aligned} \quad (3-107)$$

where the subregions are defined by $x_l \leq x \leq x_u$ and $z_l \leq z \leq z_u$. Thus the feasible region and the objective function are replaced by convex envelopes to form relaxed problems.

Solving these convex relaxed problems leads to global solutions that are lower bounds to the NLP in the particular subregion. Finally, we see that gradient-based NLP solvers play an important role in global optimization algorithms, as they often yield the lower and upper bounds for the subregions. The *spatial branch and bound* global optimization algorithm can therefore be given by the following steps:

0. Initialize algorithm: calculate upper and lower bounds over the entire (relaxed) feasible region.

For iteration k with a set of partitions M_{kj} and bounds in each subregion f_{lj} and f_{uj} :

1. *Bound*. Define best upper bound: $f_u = \min_j f_{uj}$ and delete (fathom) all subregions j with lower bounds $f_{lj} \geq f_u$. If the remaining subregions satisfy $f_{lj} \geq f_u - \epsilon$, stop.

2. *Refine*. Divide the remaining active subregions into partitions M_{k+1} and M_{k+2} . (Many branching rules are available for this step.)

3. *Select*. Solve the convex relaxed NLP in the new partitions to obtain f_{lj+1} and f_{uj+2} . Delete the partition if there is no feasible solution.

4. *Update*. Obtain upper bounds f_{uj+1} and f_{uj+2} to new partitions, if present. Set $k = k+1$, update partition sets, and go to step 1.

Example To illustrate the spatial branch and bound algorithm, consider the global solution of

$$\text{Min } f(x) = \frac{5}{2}x^4 - 20x^3 + 55x^2 - 57x, \quad \text{subject to } 0.5 \leq x \leq 2.5 \quad (3-108)$$

As seen in Fig. 3-59, this problem has local solutions at $x^* = 2.5$ and at $x^* = 0.8749$. The latter is also the global solution with $f(x^*) = -19.7$. To find the global solution, we note that all but the $-20x^3$ term in (3-108) are convex, so we replace this term by a new variable and a linear underestimator within a particular subregion, i.e.,

$$\begin{aligned} \text{Min } f_L(x) &= \frac{5}{2}x^4 - 20w + 55x^2 - 57x \\ \text{subject to } x_l &\leq x \leq x_u \\ w &= x_l^3 \frac{x_u - x}{x_u - x_l} + x_u^3 \frac{x - x_l}{x_u - x_l} \end{aligned} \quad (3-109)$$

In Fig. 3-59 we also propose subregions that are created by simple bisection partitioning rules, and we use a “loose” bounding tolerance of $\epsilon = 0.2$. In each partition the lower bound f_L is determined by (3-109) and the upper bound f_U is determined by the local solution of the original problem in the subregion. Figure 3-60 shows the progress of the spatial branch and bound algorithm as the partitions are refined and the bounds are updated. In Fig. 3-60, note the definitions of the partitions for the nodes, and the sequence numbers in each node that show the order in which the partitions are

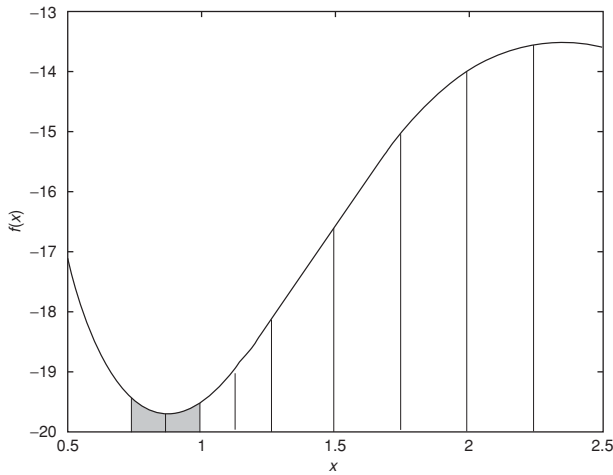


FIG. 3-59 Global optimization example with partitions.

processed. The grayed partitions correspond to the deleted subregions, and at termination of the algorithm we see that $f_{Lj} \geq f_U - \epsilon$ (that is, $-19.85 \geq -19.7 - 0.2$), with the gray subregions in Fig. 3-59 still active. Further partitioning in these subregions will allow the lower and upper bounds to converge to a tighter tolerance.

Note that a number of improvements can be made to the bounding, refinement, and selection strategies in the algorithm that accelerate the convergence of this method. A comprehensive discussion of all these options can be found in Floudas (2000), Tawarlamani and Sahinidis (2002), and Horst and Tuy (1993). Also, a number of efficient global optimization codes have recently been developed, including α BB, BARON, LGO, and OQNLP. An interesting numerical comparison of these and other codes can be found in Neumaier et al., *Math. Prog. B* 103(2): 335 (2005).

MIXED INTEGER PROGRAMMING

Mixed integer programming deals with both discrete and continuous decision variables. For the purpose of illustration we consider discrete decisions as binary variables, that is, $y_i = 0$ or 1 , and we consider the mixed integer problem (3-84). Unlike in local optimization methods, there are no optimality conditions, such as the KKT conditions, that can be applied directly. Instead, as in the global optimization methods, a systematic search of the solution space, coupled with upper and lower bounding information, is applied. As with global optimization problems, large mixed integer programs can be expensive to solve, and some care is needed in the problem formulation.

Mixed Integer Linear Programming If the objective and constraint functions are all linear, then (3-84) becomes a mixed integer linear programming problem given by

$$\begin{aligned} \text{Min } Z &= a^T x + c^T y \\ \text{subject to } Ax + By &\leq b \\ x &\geq 0, y \in \{0, 1\}^l \end{aligned} \quad (3-110)$$

Note that if we relax the t binary variables by the inequalities $0 \leq y_i \leq 1$, then (3-110) becomes a linear program with a (global) solution that is a lower bound to the MILP (3-110). There are specific MILP classes where the LP relaxation of (3-110) has the same solution as the MILP. Among these problems is the well-known *assignment problem*. Other MILPs that can be solved with efficient special-purpose methods are the *knapsack problem*, the *set covering* and *set partitioning* problems, and the *traveling salesperson problem*. See Nemhauser and Wolsey (1988) for a detailed treatment of these problems.

More generally, MILPs are solved with branch and bound algorithms, similar to the spatial branch and bound method of the previous section, that explore the search space. As seen in Fig. 3-61, binary variables are used to define the search tree, and a number of bounding properties can be noted from the structure of (3-110).

Upper bounds on the objective function can be found from any feasible solution to (3-110), with y set to integer values. These can be found at the bottom or “leaf” nodes of a branch and bound tree (and sometimes at intermediate nodes as well). The top, or root, node in

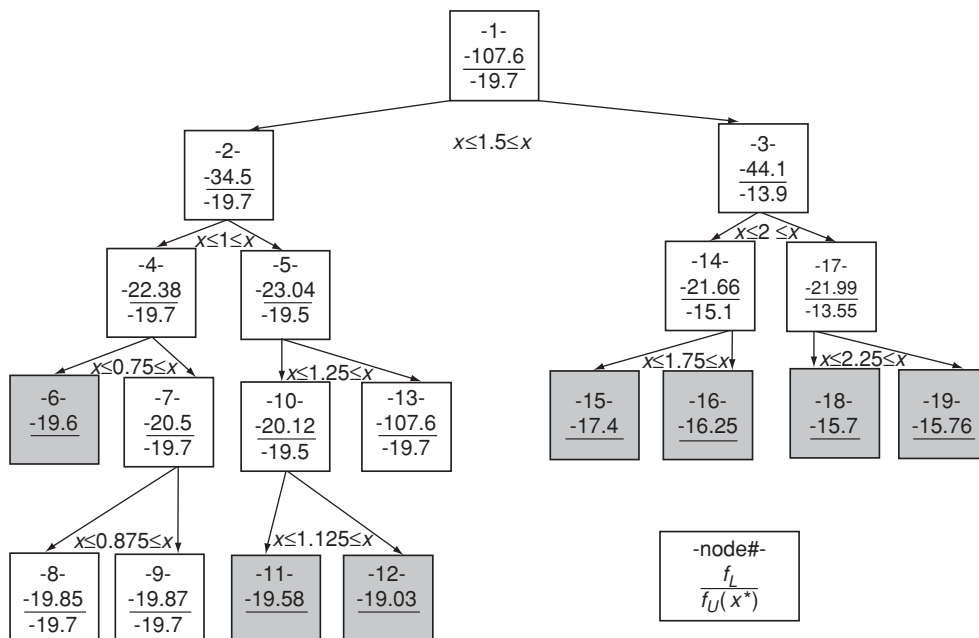


FIG. 3-60 Spatial branch and bound sequence for global optimization example.

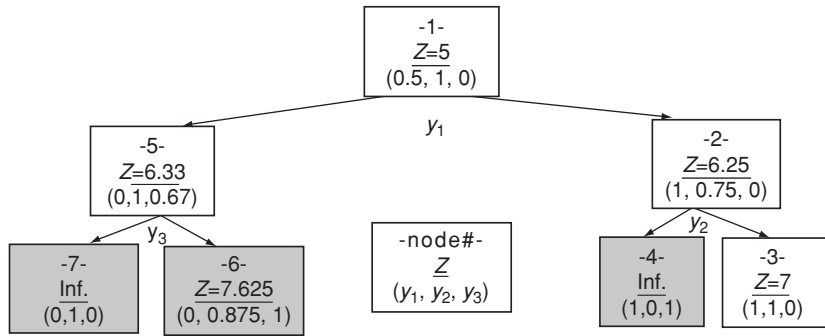


FIG. 3-61 Branch and bound sequence for MILP example.

the tree is the solution to the linear programming relaxation of (3-110); this is a lower bound to (3-110). On the other hand, as one proceeds down the tree with a partial assignment of the binary variables, a lower bound for any leaf node in that branch can be found from solution of the linear program at this intermediate node with the remaining binary variables relaxed. This leads to the following properties:

- Any intermediate node with an infeasible LP relaxation has infeasible leaf nodes and can be fathomed (i.e., all remaining children of this node can be eliminated).
- If the LP solution at an intermediate node is not less than an existing integer solution, then the node can be fathomed.

These properties lead to pruning of the search tree. Branching then continues in the tree until the upper and lower bounds converge.

This basic concept leads to a wide variety of MILP algorithms with the following features. *LP solutions* at intermediate nodes are relatively easy to calculate with the simplex method. If the solution of the parent node is known, multiplier information from this solution can be used to calculate (via a very efficient pivoting operation) the LP solution at the child node. *Branching strategies* to navigate the tree take a number of forms. More common *depth-first strategies* expand the most recent node to a leaf node or infeasible node and then backtrack to other branches in the tree. These strategies are simple to program and require little storage of past nodes. On the other hand, *breadth-first strategies* expand all the nodes at each level of the tree, select the node with the lowest objective function, and then proceed until the leaf nodes are reached. Here, more storage is required but generally fewer nodes are evaluated than in depth-first search. In addition, *selection of binary variable for branching* is based on a number of criteria, including choosing the variable with the relaxed value closest to 0 or 1, or the one leading to the largest change in the objective. Additional description of these strategies can be found in Biegler et al. (1997) and Nemhauser and Wolsey (1988).

Example To illustrate the branch and bound approach, we consider the MILP:

$$\begin{aligned}
 \text{Min } Z &= x + y_1 + 2y_2 + 3y_3 \\
 \text{subject to } &-x + 3y_1 + y_2 + 2y_3 \leq 0 \\
 &-4y_1 - 8y_2 - 3y_3 \leq -10 \\
 &x \geq 0, y_1, y_2, y_3 = \{0, 1\}
 \end{aligned} \quad (3-111)$$

The solution to (3-111) is given by $x = 4, y_1 = 1, y_2 = 1, y_3 = 0$, and $Z = 7$. Here we use a depth-first strategy and branch on the variables closest to 0 or 1. Fig. 3-61 shows the progress of the branch and bound algorithm as the binary variables are selected and the bounds are updated. The sequence numbers for each node in Fig. 3-61 show the order in which they are processed. The grayed partitions correspond to the deleted nodes, and at termination of the algorithm we see that $Z = 7$ and an integer solution is obtained at an intermediate node where coincidentally $y_3 = 0$.

A number of improvements that can be made to the branching rules will accelerate the convergence of this method. A comprehensive discussion of all these options can be found in Nemhauser and Wolsey (1988). Also, a number of efficient MILP codes have recently been developed, including CPLEX, OSL, XPRESS, and ZOOM. All these serve as excellent large-scale optimization codes as well. A detailed description and availability of these and other MILP solvers

can be found in the NEOS Software Guide: <http://www-fp.mcs.anl.gov/otc/Guide/SoftwareGuide>.

Mixed Integer Nonlinear Programming Without loss of generality, we can rewrite the MINLP in (3-84) as:

$$\begin{aligned}
 \text{Min } Z &= f(x) + c^T y \\
 \text{subject to } &g(x) + B y \leq b \\
 &x \geq 0, y \in \{0, 1\}^l
 \end{aligned} \quad (3-112)$$

where the binary variables are kept as separate linear terms. We develop several MINLP solution strategies by drawing from the material in the preceding sections. MINLP strategies can be classified into two types. The first deals with nonlinear extensions of the branch and bound method discussed above for MILPs. The second deals with outer approximation decomposition strategies that provide lower and upper bounding information for convergence.

Nonlinear Branch and Bound The MINLP (3-112) can be solved in a similar manner to (3-110). If the functions $f(x)$ and $g(x)$ in (3-112) are convex, then direct extensions to the branch and bound method can be made. A relaxed NLP can be solved at the root node, upper bounds to the solution of (3-112) can be found at the leaf nodes, and the bounding properties due to NLP solutions at intermediate nodes still hold. However, this approach is more expensive than the corresponding MILP method. First, NLPs are more expensive than LPs to solve. Second, unlike with relaxed LP solutions, NLP solutions at child nodes cannot be updated directly from solutions at parent nodes. Instead, the NLP needs to be solved again (but one hopes with a better starting guess). The NLP branch and bound method is used in the SBB code interfaced to GAMS. In addition, Leyffer, [Comput. Optim. Appl. 18: 295 (2001)] proposed a hybrid MINLP strategy nested within an SQP algorithm. At each iteration, a mixed integer quadratic program is formed, and a branch and bound algorithm is executed to solve it.

If $f(x)$ and $g(x)$ are nonconvex, additional difficulties can occur. In this case, nonunique, local solutions can be obtained at intermediate nodes, and consequently lower bounding properties would be lost. In addition, the nonconvexity in $g(x)$ can lead to locally infeasible problems at intermediate nodes, even if feasible solutions can be found in the corresponding leaf node. To overcome problems with nonconvexities, global solutions to relaxed NLPs can be solved at the intermediate nodes. This preserves the lower bounding information and allows nonlinear branch and bound to inherit the convergence properties from the linear case. However, as noted above, this leads to much more expensive solution strategies.

Outer Approximation Decomposition Methods Again, we consider the MINLP (3-112) with convex $f(x)$ and $g(x)$. Note that the NLP with binary variables fixed at $\bar{y}$

$$\begin{aligned}
 \text{Min } Z &= f(x) + c^T \bar{y} \\
 \text{subject to } &g(x) + B \bar{y} \leq b \\
 &x \geq 0
 \end{aligned} \quad (3-113)$$

if feasible, leads to a solution that is an upper bound on the MINLP

solution. In addition, linearizations of convex functions lead to underestimation of the function itself:

$$\phi(x) \geq \phi(x^k) + \nabla\phi(x^k)^T(x - x^k) \quad (3-114)$$

Consequently, linearization of (3-112) at a point x^k , to form the problem

$$\begin{aligned} \text{Min } Z &= f(x^k) + \nabla f(x^k)^T(x - x^k) + c^T y \\ \text{subject to } &g(x^k) + \nabla g(x^k)^T(x - x^k) + B y \leq b \\ &x \geq 0, y \in \{0, 1\}^l \end{aligned} \quad (3-115)$$

leads to overapproximation of the feasible region and underapproximation of the objective function in (3-112). Consequently, solution of (3-115) is a lower bound to the solution of (3-112). Adding more linearizations from other points does not change the bounding property, so for a set of points $x^l, l = 1, \dots, k$, the problem

$$\begin{aligned} \text{Min } Z &= \alpha \\ \text{subject to } &\alpha \geq f(x^l) + \nabla f(x^l)^T(x - x^l) + c^T y \\ &g(x^l) + \nabla g(x^l)^T(x - x^l) + B y \leq b \\ &x \geq 0, y \in \{0, 1\}^l \end{aligned} \quad l = 1, \dots, k \quad (3-116)$$

where α is a scalar variable, still has a solution that is a lower bound to (3-112). The outer approximation strategy is depicted in Fig. 3-62.

The outer approximation algorithm [Duran, M., and I. E. Grossmann, *Math. Programming* **36**: 307 (1986)] begins by initializing the problem, either with a predetermined starting guess or by solving a relaxed NLP based on (3-112). An upper bound to the solution is then generated by fixing the binary variables to their current values y^k and solving the NLP (3-113). This solution determines the continuous variable values x^k for the MILP (3-116). [If (3-113) is an infeasible problem, any point may be chosen for x^k , or the linearizations could be omitted.] Note that this MILP also contains linearizations from previous solutions of (3-113). Finally, the integer cut $\sum |y_l - y^k| \geq 1$ is added to (3-116) to avoid revisiting previously encountered values of binary variables. (In convex problems the integer cut is not needed, but it helps to accelerate solution of the MILP.) Solution of (3-116) yields new values of y and (without the integer cut) must lead to a lower bound to the solution of (3-112). Consequently, if the objective function of the lower bounding MILP is greater than the least upper bound determined in solutions of (3-113), then the algorithm terminates. Otherwise, the new values of y are used to solve the next NLP (3-113).

Compared to nonlinear branch and bound, the outer approximation algorithm usually requires very few solutions of the MILP and NLP subproblems. This is especially advantageous on problems where the NLPs are large and expensive to solve. Moreover, there are two variations of outer approximation that may be suitable for particular problem types:

In *Generalized Benders Decomposition* (GBD) [see Sahinidis and Grossmann, *Computers and Chem. Eng.* **15**: 481 (1991)], the lower

bounding problem in Fig. 3-62 is replaced by the following MILP:

$$\begin{aligned} \text{Min } \alpha \\ \text{subject to } &\alpha \geq f(x^l) + c^T y + v^l [g(x^l) + B y] \\ &\sum |y_l - y^l| \geq 1 \end{aligned} \quad l = 1, \dots, k \quad (3-117)$$

where v^l is the vector of KKT multipliers from the solution of (3-113) at iteration l . This MILP can be derived through a reformulation of the MILP used in Fig. 3-62 with the *inactive constraints* from (3-113) dropped. Solution of (3-117) leads to a weaker lower bound than (3-116), and consequently, more solutions of the NLP and MILP subproblems are needed to converge to the solution. However, (3-117) contains only a single continuous variable and far fewer inequality constraints and is much less expensive to solve than (3-116). Thus, GBD is favored over outer approximation if (3-113) is relatively inexpensive to solve or solution of (3-116) is too expensive.

The *extended cutting plane* (ECP) algorithm [Westerlund and Pettersson, *Computers and Chem. Engng.* **19**: S131 (1995)] is complementary to GBD. While the lower bounding problem in Fig. 3-62 remains essentially the same, the continuous variables x^k are chosen from the MILP solution and the NLP (3-113) is replaced by a simple evaluation of the objective and constraint functions. As a result, only MILP problems [(3-116) plus integer cuts] need be solved. Consequently, the ECP approach has weaker upper bounds than outer approximation and requires more MILP solutions. It has advantages over outer approximation when the NLP (3-113) is expensive to solve.

Additional difficulties arise for the outer approximation algorithm and its GBD and ECP extensions when neither $f(x)$ nor $g(x)$ is convex. Under these circumstances, the lower bounding properties resulting from the linearization and formulation of the MILP subproblem are lost, and the MILP solution may actually exclude the solution of (3-112). Hence, these algorithms need to be applied with care to nonconvex problems. To deal with nonconvexities, one can relax the linearizations in (3-116) through the introduction of additional *deviation variables* that can be penalized in the objective function. Alternately, the linearizations in (3-116) can be replaced by valid underestimating functions, such as those derived for global optimization [e.g., (3-107)]. However, this requires knowledge of structural information for (3-112) and may lead to weak lower bounds in the resulting MILP.

Finally, the performance of both MILP and MINLP algorithms is strongly dependent on the problem formulations (3-110) and (3-112). In particular, the efficiency of the approach is impacted by the lower bounds produced by the relaxation of the binary variables and subsequent solution of the linear program in the branch and bound tree. A number of approaches have been proposed to improve the quality of the lower bounds, including these:

- Logic-based methods such as *generalized disjunctive programming* (GDP) can be used to formulate MINLPs with fewer discrete variables that have tighter relaxations. Moreover, the imposition of logic-based constraints prevents the generation of unsuitable alternatives, leading to a less expensive search. In addition, constrained logic programming (CLP) methods offer an efficient search alternative to MILP solvers for highly combinatorial problems. See Raman and Grossmann, *Computers and Chem. Engng.* **18**(7): 563 (1994) for more details.
- Convex hull formulations of MILPs and MINLPs lead to relaxed problems that have much tighter lower bounds. This leads to the examination of far fewer nodes in the branch and bound tree. See Grossmann and Lee, *Comput. Optim. Applic.* **26**: 83 (2003) for more details.
- *Reformulation and preprocessing* strategies including bound tightening of the variables, coefficient reduction, lifting facets, and special ordered set constraints frequently lead to improved lower bounds and significant performance improvements in mixed integer programming algorithms.

A number of efficient codes are available for the solution of MINLPs. These include the AlphaECP, BARON, DICOPT, MINLP, and SBB solvers; descriptions of these can be found on the <http://www.gamsworld.org/minlp/solvers.htm> web site.

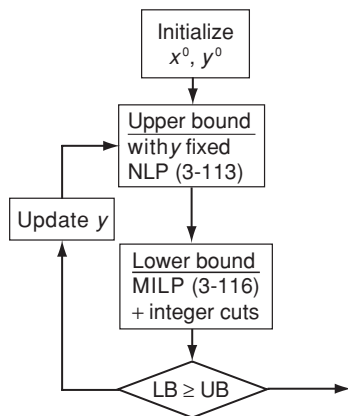


FIG. 3-62 Outer approximation MINLP algorithm.

DEVELOPMENT OF OPTIMIZATION MODELS

The most important aspect to a successful optimization study is the formulation of the optimization model. These models must reflect the real-world problem so that meaningful optimization results are obtained; they also must satisfy the properties of the problem class. For instance, NLPs addressed by gradient-based methods need to have functions that are defined in the variable domain and have bounded and continuous first and second derivatives. In mixed integer problems, proper formulations are also needed to yield good lower bounds for efficient search. With increased understanding of optimization methods and the development of efficient and reliable optimization codes, optimization practitioners now focus on the *formulation* of optimization models that are realistic, well posed, and inexpensive to solve. Finally, convergence properties of NLP, MILP, and MINLP solvers require accurate first (and often second) derivatives from the optimization model. If these contain numerical errors (say, through finite difference approximations), then the performance of these solvers can deteriorate considerably. As a result of these characteristics, modeling platforms are essential for the formulation task. These are classified into two broad areas: optimization modeling platforms and simulation platforms with optimization.

Optimization modeling platforms provide general-purpose interfaces for optimization algorithms and remove the need for the user to interface to the solver directly. These platforms allow the general formulation for all problem classes discussed above with direct interfaces to state-of-the-art optimization codes. Three representative platforms are the GAMS (General Algebraic Modeling Systems), AMPL (A Mathematical Programming Language), and AIMMS (Advanced Integrated Multidimensional Modeling Software). All three require problem model input via a declarative modeling language and provide exact gradient and Hessian information through automatic differentiation strategies. Although it is possible, these platforms were not designed to handle externally added procedural models. As a result, these platforms are best applied on optimization models that can be developed entirely within their modeling framework. Nevertheless, these platforms are widely used for large-scale research and industrial applications. In addition, the MATLAB platform allows the flexible formulation of optimization models as well, although it currently has only limited capabilities for automatic differentiation and limited optimization solvers. More information on these and other modeling platforms can be found at <http://www-fp.mcs.anl.gov/otc/Guide/SoftwareGuide>.

Simulation platforms with optimization are often dedicated, application-specific modeling tools to which optimization solvers have been interfaced. These lead to very useful optimization studies, but because they were not originally designed for optimization models, they need to be used with some caution. In particular, most of these platforms do not provide exact derivatives to the optimization solver; often they are approximated through finite differences. In addition, the models themselves are constructed and calculated through numerical procedures, instead of through an open declarative language. Examples of these include widely used process simulators such as Aspen/Plus, PRO/II, and Hysys. Also note that more recent platforms such as Aspen Custom Modeler and gPROMS include declarative models and exact derivatives.

For optimization tools linked to procedural models, note that reliable and efficient automatic differentiation tools are available that link to models written, say, in FORTRAN and C, and calculate exact first (and often second) derivatives. Examples of these include ADIFOR, ADOL-C, GRESS, Odyssey, and PADRE. When used with care, these can be applied to existing procedural models and, when linked to modern NLP and MINLP algorithms, can lead to powerful optimization capabilities. More information on these and other automatic differentiation tools can be found at http://www-unix.mcs.anl.gov/autodiff/AD_Tools/.

For optimization problems that are derived from (ordinary or partial) differential equation models, a number of advanced optimization strategies can be applied. Most of these problems are posed as NLPs, although recent work has also extended these models to MINLPs and global optimization formulations. For the optimization of profiles in time and space, *indirect methods* can be applied based on the optimality conditions of the infinite-dimensional problem using, for instance, the calculus of variations. However, these methods become difficult to apply if inequality constraints and discrete decisions are part of the optimization problem. Instead, current methods are based on NLP and MINLP formulations and can be divided into two classes:

- *Simultaneous formulations* are based on converting the differential equation model to algebraic constraints via discretization (say, with a Runge-Kutta method) and solving a large-scale NLP or MINLP. This approach requires efficient large-scale optimization solvers but also leads to very flexible and efficient problem formulations, particularly for optimal control and state-constrained problems. It also allows the treatment of unstable dynamics. A review of these strategies can be found in Biegler et al. [*Chem. Engng. Sci.* **57**(4): 575 (2002)] and Betts [*Practical Methods for Optimal Control Using Nonlinear Programming*, SIAM, Philadelphia (2001)].
- *Sequential formulations* require the linkage of NLP or MINLP solvers to embedded ODE or PDE solvers. This leads to smaller optimization problems only in the decision (or control) variables. However, an important aspect to this formulation is the need to recover accurate gradient information from the differential equation solver. Here, methods based on the solution of *direct* and *adjoint sensitivity* equations have been derived and automated [see Cao et al., *SIAM J. Sci. Comp.* **24**(3): 1076 (2003) for a description of these sensitivity methods]. For the former, commercial tools such as Aspen Custom Modeler and gPROMS have been developed.

Finally, the availability of automatic differentiation and related sensitivity tools for differential equation models allows for considerable flexibility in the formulation of optimization models. In van Bloemen Waanders et al. (Sandia Technical Report SAND2002-3198, October 2002), a seven-level modeling hierarchy is proposed that matches optimization algorithms with models that range from completely open (fully declarative model) to fully closed (entirely procedural without sensitivities). At the lowest, fully procedural level, only derivative-free optimization methods are applied, while the highest, declarative level allows the application of an efficient large-scale solver that uses first and second derivatives. This report notes that, based on the modeling level, optimization solver performance can vary by several orders of magnitude.

STATISTICS

REFERENCES: Baird, D. C., *Experimentation: An Introduction to Measurement Theory and Experiment Design*, 3d ed., Prentice-Hall, Englewood Cliffs, N.J. (1995); Box, G. P., J. S. Hunter, and W. G. Hunter, *Statistics for Experimenters: Design, Innovation, and Discovery*, 2d ed., Wiley, New York (2005); Cropley, J. B., "Heuristic Approach to Complex Kinetics," pp. 292–302 in *Chemical Reaction Engineering—Houston*, ACS Symposium Series 65, American Chemical Society, Washington (1978); Lipschutz, S., and J. J. Schiller, Jr., *Schaum's Outline of Theory and Problems of Introduction to Probability and Statistics*, McGraw-Hill, New York (1988); Moore, D. S., and G. P. McCabe, *Introduction to the Practice of Statistics*, 4th ed., Freeman, San Francisco (2003); Montgomery, D. C., and G. C. Runger, *Applied Statistics and Probability for Engineers*, 3d ed., Wiley, New York

(2002); and Montgomery, D. C., G. C. Runger, and N. F. Hubele, *Engineering Statistics*, 3d ed., Wiley, New York (2004).

INTRODUCTION

Statistics represents a body of knowledge which enables one to deal with quantitative data reflecting any degree of uncertainty. There are six basic aspects of applied statistics. These are:

1. Type of data
2. Random variables

3. Models
4. Parameters
5. Sample statistics
6. Characterization of chance occurrences

From these can be developed strategies and procedures for dealing with (1) estimation and (2) inferential statistics. The following has been directed more toward inferential statistics because of its broader utility.

Detailed illustrations and examples are used throughout to develop basic statistical methodology for dealing with a broad area of applications. However, in addition to this material, there are many specialized topics as well as some very subtle areas which have not been discussed. The references should be used for more detailed information. Section 8 discusses the use of statistics in statistical process control (SPC).

Type of Data In general, statistics deals with two types of data: counts and measurements. Counts represent the number of discrete outcomes, such as the number of defective parts in a shipment, the number of lost-time accidents, and so forth. Measurement data are treated as a continuum. For example, the tensile strength of a synthetic yarn theoretically could be measured to any degree of precision. A subtle aspect associated with count and measurement data is that some types of count data can be dealt with through the application of techniques which have been developed for measurement data alone. This ability is due to the fact that some simplified measurement statistics serve as an excellent approximation for the more tedious count statistics.

Random Variables Applied statistics deals with quantitative data. In tossing a fair coin the successive outcomes would tend to be different, with heads and tails occurring randomly over a period of time. Given a long strand of synthetic fiber, the tensile strength of successive samples would tend to vary significantly from sample to sample.

Counts and measurements are characterized as random variables, that is, observations which are susceptible to chance. Virtually all quantitative data are susceptible to chance in one way or another.

Models Part of the foundation of statistics consists of the mathematical models which characterize an experiment. The models themselves are mathematical ways of describing the probability, or relative likelihood, of observing specified values of random variables. For example, in tossing a coin once, a random variable x could be defined by assigning to x the value 1 for a head and 0 for a tail. Given a fair coin, the probability of observing a head on a toss would be a .5, and similarly for a tail. Therefore, the mathematical model governing this experiment can be written as

x	$P(x)$
0	.5
1	.5

where $P(x)$ stands for what is called a probability function. This term is reserved for count data, in that probabilities can be defined for particular outcomes.

The probability function that has been displayed is a very special case of the more general case, which is called the binomial probability distribution.

For measurement data which are considered continuous, the term *probability density* is used. For example, consider a spinner wheel which conceptually can be thought of as being marked off on the circumference infinitely precisely from 0 up to, but not including, 1. In spinning the wheel, the probability of the wheel's stopping at a specified marking point at any particular x value, where $0 \leq x < 1$, is zero, for example, stopping at the value $x = \sqrt{.5}$. For the spinning wheel, the probability density function would be defined by $f(x) = 1$ for $0 \leq x < 1$. Graphically, this is shown in Fig. 3-63. The relative-probability concept refers to the fact that density reflects the relative likelihood of occurrence; in this case, each number between 0 and 1 is equally likely. For measurement data, probability is defined by the area under the curve between specified limits. A density function always must have a total area of 1.

Example For the density of Fig. 3-63 the

$$P[0 \leq x \leq .4] = .4$$

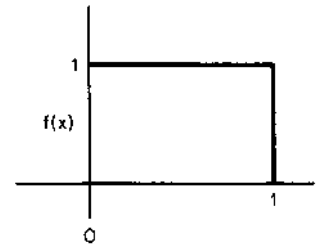


FIG. 3-63 Density function.

$$P[.2 \leq x \leq .9] = .7$$

$$P[.6 \leq x < 1] = .4$$

and so forth. Since the probability associated with any particular point value is zero, it makes no difference whether the limit point is defined by a closed interval ($\leq$ or $\geq$) or an open interval ($<$ or $>$).

Many different types of models are used as the foundation for statistical analysis. These models are also referred to as **populations**.

Parameters As a way of characterizing probability functions and densities, certain types of quantities called parameters can be defined. For example, the center of gravity of the distribution is defined to be the population mean, which is designated as μ . For the coin toss $\mu = .5$, which corresponds to the average value of x ; i.e., for half of the time x will take on a value 0 and for the other half a value 1. The average would be .5. For the spinning wheel, the average value would also be .5.

Another parameter is called the **standard deviation**, which is designated as σ . The square of the standard deviation is used frequently and is called the popular **variance**, σ^2 . Basically, the standard deviation is a quantity which measures the spread or dispersion of the distribution from its mean μ . If the spread is broad, then the standard deviation will be larger than if it were more constrained.

For specified probability and density functions, the respective means and variances are defined by the following:

Probability functions	Probability density functions
$E(x) = \mu = \sum_x x P(x)$	$E(x) = \mu = \int_x x f(x) dx$
$\text{Var}(x) = \sigma^2 = \sum_x (x - \mu)^2 P(x)$	$\text{Var}(x) = \sigma^2 = \int_x (x - \mu)^2 f(x) dx$

where $E(x)$ is defined to be the expected or average value of x .

Sample Statistics Many types of sample statistics will be defined. Two very special types are the **sample mean**, designated as $\bar{x}$, and the sample standard deviation, designated as s . These are, by definition, random variables. Parameters like μ and σ are not random variables; they are fixed constants.

Example In an experiment, six random numbers (rounded to four decimal places) were observed from the uniform distribution $f(x) = 1$ for $0 \leq x < 1$:

.1009
.3754
.0842
.9901
.1280
.6606

The sample mean corresponds to the arithmetic average of the observations, which will be designated as x_1 through x_6 , where

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \text{ with } n = 6, \quad \bar{x} = 0.3899$$

The sample standard deviation s is defined by the computation

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

$$= \sqrt{\frac{n \sum x_i^2 - (\sum x_i)^2}{n(n-1)}} \quad (3-118)$$

In effect, this represents the root of a statistical average of the squares. The divisor quantity $(n-1)$ will be referred to as the degrees of freedom. The sample value of the standard deviation for the data given is .3686.

The value of $n-1$ is used in the denominator because the deviations from the sample average must total zero, or

$$\sum (x_i - \bar{x}) = 0$$

Thus knowing $n-1$ values of $x_i - \bar{x}$ permits calculation of the n th value of $x_i - \bar{x}$.

The sample mean and sample standard deviation are obtained by using Microsoft Excel with the commands AVERAGE(B2:B7) and STDEV(B2:B7) when the observations are in cells B2 to B7.

In effect, the standard deviation quantifies the relative magnitude of the deviation numbers, i.e., a special type of "average" of the distance of points from their center. In statistical theory, it turns out that the corresponding variance quantities s^2 have remarkable properties which make possible broad generalizations for sample statistics and therefore also their counterparts, the standard deviations.

For the corresponding population, the parameter values are $\mu = .50$ and $\sigma = .2887$, which are obtained by calculating the integrals defined above with $f(x) = 1$ and $0 \leq x \leq 1$. If, instead of using individual observations only, averages of six were reported, then the corresponding population parameter values would be $\mu = .50$ and $\sigma_{\bar{x}} = \sigma/\sqrt{6} = .1179$. The corresponding variance for an average will be written occasionally as $\text{Var}(\bar{x}) = \text{var}(x)/n$. In effect, the variance of an average is inversely proportional to the sample size n , which reflects the fact that sample averages will tend to cluster about μ much more closely than individual observations. This is illustrated in greater detail under "Measurement Data and Sampling Densities."

Characterization of Chance Occurrences To deal with a broad area of statistical applications, it is necessary to characterize the way in which random variables will vary by chance alone. The basic foundation for this characteristic is laid through a density called the gaussian, or normal, distribution.

Determining the area under the normal curve is a very tedious procedure. However, by standardizing a random variable that is normally distributed, it is possible to relate all normally distributed random variables to one table. The standardization is defined by the identity $z = (x - \mu)/\sigma$, where z is called the unit normal. Further, it is possible to standardize the sampling distribution of averages $\bar{x}$ by the identity $z = (\bar{x} - \mu)/(\sigma/\sqrt{n})$.

A remarkable property of the normal distribution is that, almost regardless of the distribution of x , sample averages $\bar{x}$ will approach the gaussian distribution as n gets large. Even for relatively small values of n , of about 10, the approximation in most cases is quite close. For example, sample averages of size 10 from the uniform distribution will have essentially a gaussian distribution.

Also, in many applications involving count data, the normal distribution can be used as a close approximation. In particular, the approximation is quite close for the binomial distribution within certain guidelines.

The normal probability distribution function can be obtained in Microsoft Excel by using the NORMDIST function and supplying the desired mean and standard deviation. The cumulative value can also be determined. In MATLAB, the corresponding command is `randn`.

ENUMERATION DATA AND PROBABILITY DISTRIBUTIONS

Introduction Many types of statistical applications are characterized by enumeration data in the form of counts. Examples are the number of lost-time accidents in a plant, the number of defective items in a sample, and the number of items in a sample that fall within several specified categories.

The sampling distribution of count data can be characterized through probability distributions. In many cases, count data are appropriately

interpreted through their corresponding distributions. However, in other situations analysis is greatly facilitated through distributions which have been developed for measurement data. Examples of each will be illustrated in the following subsections.

Binomial Probability Distribution

Nature Consider an experiment in which each outcome is classified into one of two categories, one of which will be defined as a success and the other as a failure. Given that the probability of success p is constant from trial to trial, then the probability of observing a specified number of successes x in n trials is defined by the binomial distribution. The sequence of outcomes is called a **Bernoulli process**,

Nomenclature

n = total number of trials

x = number of successes in n trials

p = probability of observing a success on any one trial

$\hat{p} = x/n$, the proportion of successes in n trials

Probability Law

$$P(x) = P\left(\frac{x}{n}\right) = \binom{n}{x} p^x (1-p)^{n-x} \quad x = 0, 1, 2, \dots, n$$

$$\text{where } \binom{n}{x} = \frac{n!}{x!(n-x)!}$$

$$\begin{array}{ll} \text{Properties} & E(x) = np \quad \text{Var}(x) = np(1-p) \\ & E(\hat{p}) = p \quad \text{Var}(\hat{p}) = p(1-p)/n \end{array}$$

Geometric Probability Distribution

Nature Consider an experiment in which each outcome is classified into one of two categories, one of which will be defined as a success and the other as a failure. Given that the probability of success p is constant from trial to trial, then the probability of observing the first success on the x th trial is defined by the geometric distribution.

Nomenclature

p = probability of observing a success on any one trial

x = the number of trials to obtain the first success

Probability Law

$$P(x) = p(1-p)^{x-1} \quad x = 1, 2, 3, \dots$$

$$\begin{array}{ll} \text{Properties} & E(x) = 1/p \quad \text{Var}(x) = (1-p)/p^2 \end{array}$$

Poisson Probability Distribution

Nature In monitoring a moving threadline, one criterion of quality would be the frequency of broken filaments. These can be identified as they occur through the threadline by a broken-filament detector mounted adjacent to the threadline. In this context, the random occurrences of broken filaments can be modeled by the Poisson distribution. This is called a Poisson process and corresponds to a probabilistic description of the frequency of defects or, in general, what are called arrivals at points on a continuous line or in time. Other examples include:

1. The number of cars (arrivals) that pass a point on a high-speed highway between 10:00 and 11:00 A.M. on Wednesdays
2. The number of customers arriving at a bank between 10:00 and 10:10 A.M.
3. The number of telephone calls received through a switchboard between 9:00 and 10:00 A.M.
4. The number of insurance claims that are filed each week
5. The number of spinning machines that break down during 1 day at a large plant.

Nomenclature

x = total number of arrivals in a total length L or total period T

a = average rate of arrivals for a unit length or unit time

$\lambda = aL$ = expected or average number of arrivals for the total length L

$\lambda = aT$ = expected or average number of arrivals for the total time T

Probability Law Given that a is constant for the total length L or period T , the probability of observing x arrivals in some period L or T

is given by

$$P(x) = \frac{\lambda^x}{x!} e^{-\lambda} \quad x = 0, 1, 2, \dots$$

Properties $E(x) = \lambda$ $\text{Var}(x) = \lambda$

Example The number of broken filaments in a threadline has been averaging .015 per yard. What is the probability of observing exactly two broken filaments in the next 100 yd? In this example, $a = .015/\text{yd}$ and $L = 100$ yd; therefore $\lambda = (.015)(100) = 1.5$:

$$P(x = 2) = \frac{(1.5)^2}{2!} e^{-1.5} = .2510$$

Example A commercial item is sold in a retail outlet as a unit product. In the past, sales have averaged 10 units per month with no seasonal variation. The retail outlet must order replacement items 2 months in advance. If the outlet starts the next 2-month period with 25 items on hand, what is the probability that it will stock out before the end of the second month?

Given $a = 10/\text{month}$, then $\lambda = 10 \times 2 = 20$ for the total period of 2 months:

$$\begin{aligned} P(x \geq 26) &= \sum_{x=26}^{\infty} P(x) = 1 - \sum_{x=0}^{25} P(x) \\ &= \sum_{x=0}^{25} \frac{20^x}{x!} e^{-20} = e^{-20} \left[1 + \frac{20}{1} + \frac{20^2}{2!} + \dots + \frac{20^{25}}{25!} \right] \\ &= .887815 \end{aligned}$$

Therefore $P(x \geq 26) = .112185$ or roughly an 11 percent chance of a stockout.

Hypergeometric Probability Distribution

Nature In an experiment in which one samples from a relatively small group of items, each of which is classified in one of two categories, *A* or *B*, the hypergeometric distribution can be defined. One example is the probability of drawing two red and two black cards from a deck of cards. The hypergeometric distribution is the analog of the binomial distribution when successive trials are not independent, i.e., when the total group of items is not infinite. This happens when the drawn items are not replaced.

Nomenclature

N = total group size

n = sample group size

X = number of items in the total group with a specified attribute *A*

$N - X$ = number of items in the total group with the other attribute *B*

x = number of items in the sample with a specified attribute *A*

$n - x$ = number of items in the sample with the other attribute *B*

	Population	Sample
Category <i>A</i>	X	x
Category <i>B</i>	$N - X$	$n - x$
Total	N	n

Probability Law

$$P(x) = \frac{\binom{N-X}{n-x} \binom{X}{x}}{\binom{N}{n}}$$

$$E(x) = \frac{nX}{N}$$

$$\text{var}(x) = nP(1-P) \frac{N-n}{N-1}$$

Example What is the probability that an appointed special committee of 4 has no female members when the members are randomly selected from a candidate group of 10 males and 7 females?

$$P(x = 0) = \frac{\binom{10}{4} \binom{7}{0}}{\binom{17}{4}} = .0882$$

Example A bin contains 300 items, of which 240 are good and 60 are defective. In a sample of 6 what is the probability of selecting 4 good and 2 defective items by chance?

$$P(x) = \frac{\binom{240}{4} \binom{60}{2}}{\binom{300}{6}} = .2478$$

Multinomial Distribution

Nature For an experiment in which successive outcomes can be classified into two or more categories and the probabilities associated with the respective outcomes remain constant, then the experiment can be characterized through the multinomial distribution.

Nomenclature

n = total number of trials

k = total number of distinct categories

p_j = probability of observing category *j* on any one trial, $j = 1, 2, \dots, k$

x_j = total number of occurrences in category *j* in n trials

Probability Law

$$P(x_1, x_2, \dots, x_k) = \frac{n!}{x_1! x_2! \dots x_k!} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k}$$

Example In tossing a die 12 times, what is the probability that each face value will occur exactly twice?

$$p(2, 2, 2, 2, 2, 2) = \frac{12!}{2!2!2!2!2!2!} \left(\frac{1}{6}\right)^2 \left(\frac{1}{6}\right)^2 \left(\frac{1}{6}\right)^2 \left(\frac{1}{6}\right)^2 \left(\frac{1}{6}\right)^2 \left(\frac{1}{6}\right)^2 = .003438$$

To compute these probabilities in Microsoft Excel, put the value of x in cell B2, say, and use the functions

Binomial distribution: = BINOMDIST(B2, n , p , 0)

Poisson distribution: = POISSON(B2, λ , 0)

Hypergeometric distribution: = HYPGEOMDIST(B2, n , X , N)

To generate a table of values with these probability distributions, in Microsoft Excel use the following functions:

Bernoulli random values: = CRITBINOM(1, p , RAND())

Binomial random values: = CRITBINOM(n , p , RAND())

The factorial function is FACT(n) in Microsoft Excel and factorial(n) in MATLAB. Be sure that x is an integer.

MEASUREMENT DATA AND SAMPLING DENSITIES

Introduction The following example data are used throughout this subsection to illustrate concepts. Consider, for the purpose of illustration, that five synthetic-yarn samples have been selected randomly from a production line and tested for tensile strength on each of 20 production days. For this, assume that each group of five corresponds to a day, Monday through Friday, for a period of 4 weeks:

Monday 1	Tuesday 2	Wednesday 3	Thursday 4	Friday 5	Groups of 25 pooled
36.48	38.06	35.28	36.34	36.73	
35.33	31.86	36.58	36.25	37.17	
35.92	33.81	38.81	30.46	33.07	
32.28	30.30	33.31	37.37	34.27	
31.61	35.27	33.88	37.52	36.94	
$\bar{x} = 34.32$	33.86	35.57	35.59	35.64	35.00
$s = 2.22$	3.01	2.22	2.92	1.85	2.40
6	7	8	9	10	
38.67	36.62	35.03	35.80	36.82	
32.08	33.05	36.22	33.16	36.49	
33.79	35.43	32.71	35.19	32.83	
32.85	36.63	32.52	32.91	32.43	
35.22	31.46	27.23	35.44	34.16	
$\bar{x} = 34.52$	34.64	32.74	34.50	34.54	34.19
$s = 2.60$	2.30	3.46	1.36	2.03	2.35

11	12	13	14	15	
39.63	34.52	36.05	36.64	31.57	
34.38	37.39	35.36	31.18	36.21	
36.51	34.16	35.00	36.13	33.84	
30.00	35.76	33.61	37.51	35.01	
39.64	37.63	36.98	39.05	34.95	
$\bar{x} = 36.03$	35.89	35.40	36.10	34.32	35.55
$s = 4.04$	1.59	1.25	2.96	1.75	2.42
Monday 16	Tuesday 17	Wednesday 18	Thursday 19	Friday 20	Groups of 25 pooled
37.68	35.97	33.71	35.61	36.65	
36.38	35.92	32.34	37.13	37.91	
38.43	36.51	33.29	31.37	42.18	
39.07	33.89	32.81	35.89	39.25	
33.06	36.01	37.13	36.33	33.32	
$\bar{x} = 36.92$	35.66	33.86	35.27	37.86	35.91
$s = 2.38$	1.02	1.90	2.25	3.27	2.52

Pooled sample of 100: $\bar{x} = 35.16$ $s = 2.47$

Even if the process were at steady state, tensile strength, a key property would still reflect some variation. Steady state, or stable operation of any process, has associated with it a characteristic variation. Superimposed on this is the testing method, which is itself a process with its own characteristic variation. The observed variation is a composite of these two variations.

Assume that the table represents "typical" production-line performance. The numbers themselves have been generated on a computer and represent random observations from a population with $\mu = 35$ and a population standard deviation $\sigma = 2.45$. The sample values reflect the way in which tensile strength can vary by chance alone. In practice, a production supervisor unschooled in statistics but interested in high tensile performance would be despondent on the eighth day and exuberant on the twentieth day. If the supervisor were more concerned with uniformity, the lowest and highest points would have been on the eleventh and seventeenth days.

An objective of statistical analysis is to serve as a guide in decision making in the context of normal variation. In the case of the production supervisor, it is to make a decision, with a high probability of being correct, that something has in fact changed the operation.

Suppose that an engineering change has been made in the process and five new tensile samples have been tested with the results:

36.81	
38.34	$\bar{x} = 37.14$
34.87	$s = 1.85$
39.58	
36.12	

In this situation, management would inquire whether the product has been improved by increased tensile strength. To answer this question, in addition to a variety of analogous questions, it is necessary to have some type of scientific basis upon which to draw a conclusion.

A scientific basis for the evaluation and interpretation of data is contained in the accompanying table descriptions. These tables characterize the way in which sample values will vary by chance alone in the context of individual observations, averages, and variances.

Table number	Designated symbol	Variable	Sampling distribution of
3-5	z	$\frac{x - \mu}{\sigma}$	Observations*
3-5	z	$\frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$	Averages
3-6	t	$\frac{\bar{x} - \mu}{s/\sqrt{n}}$	Averages when σ is unknown*
3-7	χ^2	$(s^2/\sigma^2)(df)$	Variances*
3-8	F	s_1^2/s_2^2	Ratio of two independent sample variances*

*When sampling from a gaussian distribution.

Normal Distribution of Observations Many types of data follow what is called the gaussian, or bell-shaped, curve; this is especially true of averages. Basically, the gaussian curve is a purely mathematical function which has very special properties. However, owing to some mathematically intractable aspects primary use of the function is restricted to tabulated values.

Basically, the tabled values represent area (proportions or probability) associated with a scaling variable designated by Z in Fig. 3-64. The normal curve is centered at 0, and for particular values of Z , designated as z , the tabulated numbers represent the corresponding area under the curve between 0 and z . For example, between 0 and 1 the area is .3413. (Get this number from Table 3-5. The value of A includes the area on both sides of zero. Thus we want $A/2$. For $z = 1$, $A = 0.6827$, $A/2 = 0.3413$. For $z = 2$, $A/2 = 0.4772$.) The area between 0 and 2 is .4772; therefore, the area between 1 and 2 is .4772 - .3413 = .1359.

Also, since the normal curve is symmetric, areas to the left can be determined in exactly the same way. For example, the area between -2 and +1 would include the area between -2 and 0, .4772 (the same as 0 to 2), plus the area between 0 and 1, .3413, or a total area of .8185.

Any types of observation which are applicable to the normal curve can be transformed to Z values by the relationship $z = (x - \mu)/\sigma$ and, conversely, Z values to x values by $x = \mu + \sigma z$, as shown in Fig. 3-64. For example, for tensile strength, with $\mu = 35$ and $\sigma = 2.45$, this would dictate $z = (x - 35)/2.45$ and $x = 35 + 2.45z$.

Example What proportion of tensile values will fall between 34 and 36?

$$z_1 = (34 - 35)/2.45 = -.41 \quad z_2 = (36 - 35)/2.45 = .41$$

$$P[-.41 \leq z \leq .41] = .3182, \text{ or roughly } 32 \text{ percent}$$

The value 0.3182 is interpolated from Table 3-4 using $z = \pm 0.40$, $A = 0.3108$, and $z = \pm 0.45$, $A = 0.3473$.

Example What midrange of tensile values will include 95 percent of the sample values? Since $P[-1.96 \leq z \leq 1.96] = .95$, the corresponding values of x are

$$x_1 = 35 - 1.96(2.45) = 30.2$$

$$x_2 = 35 + 1.96(2.45) = 39.8$$

or

$$P[30.2 \leq x \leq 39.8] = .95$$

Normal Distribution of Averages An examination of the tensile-strength data previously tabulated would show that the range (largest minus the smallest) of tensile strength within days averages 5.12. The average range in $\bar{x}$ values within each week is 2.37, while the range in the four weekly averages is 1.72. This reflects the fact that averages tend to be less variable in a predictable way. Given that the variance of x is $\text{var}(x) = \sigma^2$, then the variance of $\bar{x}$ based on n observations is $\text{var}(\bar{x}) = \sigma^2/n$.

For averages of n observations, the corresponding relationship for the Z -scale relationship is

$$z = (\bar{x} - \mu) / \sigma / \sqrt{n} \quad \text{or} \quad \bar{x} = \mu + \frac{\sigma}{\sqrt{n}} z$$

The Microsoft Excel function NORMDIST($X, \mu, \sigma, 1$) gives the probability that $x \leq \mu$.

The command CONFIDENCE(α, σ, n) gives the confidence interval about the mean for a sample size n . To obtain 95 percent confidence limits, use $\alpha = .025$; $2\alpha = 1 - A$.

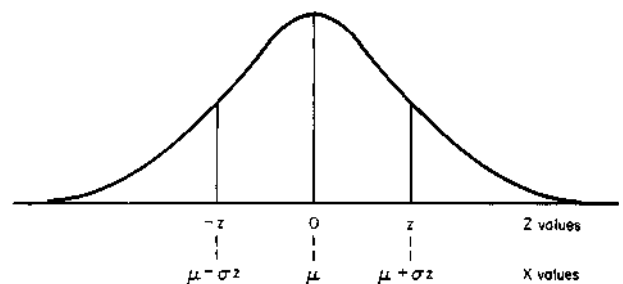


FIG. 3-64 Transformation of z values.

TABLE 3-5 Ordinates and Areas between Abscissa Values $-z$ and $+z$ of the Normal Distribution CurveA = area between $\mu - \sigma_z$ and $\mu + \sigma_z$

z	X	Y	A	$1 - A$	z	X	Y	A	$1 - A$
0	μ	0.399	0.0000	1.0000	± 1.50	$\mu \pm 1.50\sigma$	0.1295	0.8664	0.1336
$\pm .10$	$\mu \pm .10\sigma$	.397	.0797	.9203	± 1.60	$\mu \pm 1.60\sigma$	.1109	.8904	.1096
$\pm .20$	$\mu \pm .20\sigma$	.391	.1585	.8415	± 1.70	$\mu \pm 1.70\sigma$	.0940	.9109	.0891
$\pm .30$	$\mu \pm .30\sigma$	.381	.2358	.7642	± 1.80	$\mu \pm 1.80\sigma$	.0790	.9281	.0719
$\pm .40$	$\mu \pm .40\sigma$	.368	.3108	.6892	± 1.90	$\mu \pm 1.90\sigma$	.0656	.9446	.0574
$\pm .50$	$\mu \pm .50\sigma$	.352	.3829	.6171	± 2.00	$\mu \pm 2.00\sigma$	.0540	.9545	.0455
$\pm .60$	$\mu \pm .60\sigma$	.333	.4515	.5485	± 2.10	$\mu \pm 2.10\sigma$	.0440	.9643	.0357
$\pm .70$	$\mu \pm .70\sigma$	.312	.5161	.4839	± 2.20	$\mu \pm 2.20\sigma$	.0335	.9722	.0278
$\pm .80$	$\mu \pm .80\sigma$	.290	.5763	.4237	± 2.30	$\mu \pm 2.30\sigma$	.0283	.9786	.0214
$\pm .90$	$\mu \pm .90\sigma$	.266	.6319	.3681	± 2.40	$\mu \pm 2.40\sigma$	.0224	.9836	.0164
± 1.00	$\mu \pm 1.00\sigma$	.242	.6827	.3173	± 2.50	$\mu \pm 2.50\sigma$	.0175	.9876	.0124
± 1.10	$\mu \pm 1.10\sigma$	.218	.7287	.2713	± 2.60	$\mu \pm 2.60\sigma$	.0136	.9907	.0093
± 1.20	$\mu \pm 1.20\sigma$	.194	.7699	.2301	± 2.70	$\mu \pm 2.70\sigma$	.0104	.9931	.0069
± 1.30	$\mu \pm 1.30\sigma$	.171	.8064	.1936	± 2.80	$\mu \pm 2.80\sigma$	.0079	.9949	.0051
± 1.40	$\mu \pm 1.40\sigma$	.150	.8385	.1615	± 2.90	$\mu \pm 2.90\sigma$	.0060	.9963	.0037
± 1.50	$\mu \pm 1.50\sigma$	.130	.8664	.1336	± 3.00	$\mu \pm 3.00\sigma$	.0044	.9973	.0027
					± 4.00	$\mu \pm 4.00\sigma$	.0001	.99994	.00006
					± 5.00	$\mu \pm 5.00\sigma$	.000001	.9999994	.0000006
± 0.000	μ	0.3989	.0000	1.0000	± 1.036	$\mu \pm 1.036\sigma$	0.2331	0.7000	0.3000
$\pm .126$	$\mu \pm 0.126\sigma$	.3958	.1000	0.9000	± 1.282	$\mu \pm 1.282\sigma$	.1755	.8000	.2000
$\pm .253$	$\mu \pm .253\sigma$	.3863	.2000	.8000	± 1.645	$\mu \pm 1.645\sigma$	.1031	.9000	.1000
$\pm .385$	$\mu \pm .385\sigma$	.3704	.3000	.7000	± 1.960	$\mu \pm 1.960\sigma$	.0584	.9500	.0500
$\pm .524$	$\mu \pm .524\sigma$	.3477	.4000	.6000	± 2.576	$\mu \pm 2.576\sigma$	.0145	.9900	.0100
$\pm .674$	$\mu \pm .674\sigma$	.3178	.5000	.5000	± 3.291	$\mu \pm 3.291\sigma$	.0018	.9990	.0010
$\pm .842$	$\mu \pm .842\sigma$	.2800	.6000	.4000	± 3.891	$\mu \pm 3.891\sigma$	.0002	.9999	.0001

This table is obtained in Microsoft Excel with the function $Y = \text{NORMDIST}(X, \mu, \sigma, 0)$.
 If $X > \mu$, $A = 2 \text{ NORMDIST}(X, \mu, \sigma, 1) - 1$. If $X < \mu$, $A = 1 - 2 \text{ NORMDIST}(X, \mu, \sigma, 1)$

Example What proportion of daily tensile averages will fall between 34 and 36? Using Table 3-5

$$z_1 = (34 - 35)/(2.45/\sqrt{5}) = -.91 \quad z_2 = (36 - 35)/(2.45/\sqrt{5}) = .91$$

$$P[-.91 \leq z \leq .91] = .637, \text{ or roughly } 64 \text{ percent}$$

Using Microsoft Excel, the more precise answer is

$$P[34 \leq \bar{x} \leq 36] = 2 \text{ NORMDIST}(36, 35, 2.45/\sqrt{5}, 1) - 1 \\ = 1 - 2 \text{ NORMDIST}(34, 35, 2.45/\sqrt{5}, 1) = .6386$$

Example What midrange of daily tensile averages will include 95 percent of the sample values? Using Table 3-5

$$x_1 = 35 - 1.96(2.45/\sqrt{5}) = 32.85$$

$$x_2 = 35 + 1.96(2.45/\sqrt{5}) = 37.15$$

$$P[32.85 \leq \bar{x} \leq 37.15] = .95$$

or using Microsoft Excel,

$$35 \pm \text{CONFIDENCE}(.05, 2.45, 5) = 35 \pm 2.15$$

t Distribution of Averages The normal curve relies on a knowledge of σ , or in special cases, when it is unknown, s can be used with the normal curve as an approximation when $n > 30$. For example, with $n > 30$ the intervals $\bar{x} \pm s$ and $\bar{x} \pm 2s$ will include roughly 68 and 95 percent of the sample values respectively when the distribution is normal.

In applications sample sizes are usually small and σ unknown. In these cases, the t distribution can be used where

$$t = (\bar{x} - \mu)/(s/\sqrt{n}) \quad \text{or} \quad \bar{x} = \mu + ts/\sqrt{n}$$

The t distribution is also symmetric and centered at zero. It is said to be robust in the sense that even when the individual observations x are not normally distributed, sample averages of x have distributions which tend toward normality as n gets large. Even for small n of 5

through 10, the approximation is usually relatively accurate. It is sometimes called the Student's t distribution.

In reference to the tensile-strength table, consider the summary statistics $\bar{x}$ and s by days. For each day, the t statistic could be computed. If this were repeated over an extensive simulation and the resultant t quantities plotted in a frequency distribution, they would match the corresponding distribution of t values summarized in Table 3-6.

Since the t distribution relies on the sample standard deviation s , the resultant distribution will differ according to the sample size n . To designate this difference, the respective distributions are classified according to what are called the degrees of freedom and abbreviated as df. In simple problems, the df are just the sample size minus 1. In more complicated applications the df can be different. In general, degrees of freedom are the number of quantities minus the number of constraints. For example, four numbers in a square which must have row and column sums equal to zero have only one df, i.e., four numbers minus three constraints (the fourth constraint is redundant). The Microsoft Excel function $\text{TDIST}(X, df, 1)$ gives the right-tail probability, and $\text{TDIST}(X, df, 2)$ gives twice that. The probability that $t \leq X$ is $1 - \text{TDIST}(X, df, 1)$ when $X \geq 0$ and $\text{TDIST}(\text{abs}(X), df, 1)$ when $X < 0$. The probability that $-X \leq t \leq X$ is $1 - 2\text{TDIST}(X, df, 2)$.

Example For a sample size $n = 5$, what values of t define a midarea of 90 percent? For 4 df the tabled value of t corresponding to a midarea of 90 percent is 2.132; i.e., $P[-2.132 \leq t \leq 2.132] = .90$.

Using Microsoft Excel, $\text{TINV}(.1, 4) = 2.132$.

Example For a sample size $n = 25$, what values of t define a midarea of 95 percent? For 24 df the tabled value of t corresponding to a midarea of 95 percent is 2.064; i.e., $P[-2.064 \leq t \leq 2.064] = .95$.

Example Also, $1 - \text{TDIST}(2.064, 24, 2) = .9500$.

Using Microsoft Excel, $\text{TINV}(.05, 24) = 2.064$.

TABLE 3-6 Values of t

df	$t_{.40}$	$t_{.30}$	$t_{.20}$	$t_{.10}$	$t_{.05}$	$t_{.025}$	$t_{.01}$	$t_{.005}$
1	0.325	0.727	1.376	3.078	6.314	12.706	31.821	63.657
2	.289	.617	1.061	1.886	2.920	4.303	6.965	9.925
3	.277	.584	.978	1.638	2.353	3.182	4.541	5.841
4	.271	.569	.941	1.533	2.132	2.776	3.747	4.604
5	.267	.559	.920	1.476	2.015	2.571	3.365	4.032
6	.265	.553	.906	1.440	1.943	2.447	3.143	3.707
7	.263	.549	.896	1.415	1.895	2.365	2.998	3.499
8	.262	.546	.889	1.397	1.860	2.306	2.896	3.355
9	.261	.543	.883	1.383	1.833	2.262	2.821	3.250
10	.260	.542	.879	1.372	1.812	2.228	2.764	3.169
11	.260	.540	.876	1.363	1.796	2.201	2.718	3.106
12	.259	.539	.873	1.356	1.782	2.179	2.681	3.055
13	.259	.538	.870	1.350	1.771	2.160	2.650	3.012
14	.258	.537	.868	1.345	1.761	2.145	2.624	2.977
15	.258	.536	.866	1.341	1.753	2.131	2.602	2.947
16	.258	.535	.865	1.337	1.746	2.120	2.583	2.921
17	.257	.534	.863	1.333	1.740	2.110	2.567	2.898
18	.257	.534	.862	1.330	1.734	2.101	2.552	2.878
19	.257	.533	.861	1.328	1.729	2.093	2.539	2.861
20	.257	.533	.860	1.325	1.725	2.086	2.528	2.845
21	.257	.532	.859	1.323	1.721	2.080	2.518	2.831
22	.256	.532	.858	1.321	1.717	2.074	2.508	2.819
23	.256	.532	.858	1.319	1.714	2.069	2.500	2.807
24	.256	.531	.857	1.318	1.711	2.064	2.492	2.797
25	.256	.531	.856	1.316	1.708	2.060	2.485	2.787
26	.256	.531	.856	1.315	1.706	2.056	2.479	2.779
27	.256	.531	.855	1.314	1.703	2.052	2.473	2.771
28	.256	.530	.855	1.313	1.701	2.048	2.467	2.763
29	.256	.530	.854	1.311	1.699	2.045	2.462	2.756
30	.256	.530	.854	1.310	1.697	2.042	2.457	2.750
40	.255	.529	.851	1.303	1.684	2.021	2.423	2.704
60	.254	.527	.848	1.296	1.671	2.000	2.390	2.660
120	.254	.526	.845	1.289	1.658	1.980	2.358	2.617
∞	.253	.524	.842	1.282	1.645	1.960	2.326	2.576

Above values refer to a single tail outside the indicated limit of t . For example, for 95 percent of the area to be between $-t$ and $+t$ in a two-tailed t distribution, use the values for $t_{0.025}$ or 2.5 percent for each tail. This table is obtained in Microsoft Excel using the function $TINV(\alpha, df)$, where α is .05.

Example What is the sample value of t for the first day of tensile data?

$$\text{Sample } t = (34.32 - 35)/(2.22/\sqrt{5}) = -6.68$$

Note that on the average 90 percent of all such sample values would be expected to fall within the interval ± 2.132 .

t Distribution for the Difference in Two Sample Means with Equal Variances The t distribution can be readily extended to the difference in two sample means when the respective populations have the same variance σ :

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s_p \sqrt{1/n_1 + 1/n_2}} \quad (3-119)$$

where s_p^2 is a pooled variance defined by

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{(n_1 - 1) + (n_2 - 1)} \quad (3-120)$$

In this application, the t distribution has $(n_1 + n_2 - 2)$ df.

t Distribution for the Difference in Two Sample Means with Unequal Variances When population variances are unequal, an approximate t quantity can be used:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{a + b}}$$

with

$$a = s_1^2/n_1 \quad b = s_2^2/n_2$$

and

$$df = \frac{(a + b)^2}{a^2/(n_1 - 1) + b^2/(n_2 - 1)}$$

Chi-Square Distribution For some industrial applications, product uniformity is of primary importance. The sample standard deviation s is most often used to characterize uniformity. In dealing with this problem, the chi-square distribution can be used where $\chi^2 = (s^2/\sigma^2)(df)$. The chi-square distribution is a family of distributions which are defined by the degrees of freedom associated with the sample variance. For most applications, df is equal to the sample size minus 1.

The probability distribution function is

$$p(y) = y_0 y^{df-2} \exp \left[\frac{-(df)^2}{2} \right]$$

where y_0 is chosen such that the integral of $p(y)$ over all y is one.

In terms of the tensile-strength table previously given, the respective chi-square sample values for the daily, weekly, and monthly figures could be computed. The corresponding df would be 4, 24, and 99 respectively. These numbers would represent sample values from the respective distributions which are summarized in Table 3-7.

In a manner similar to the use of the t distribution, chi square can be interpreted in a direct probabilistic sense corresponding to a midarea of $(1 - \alpha)$:

$$P[\chi_1^2 \leq (s^2/\sigma^2)(df) \leq \chi_2^2] = 1 - \alpha$$

where χ_1^2 corresponds to a lower-tail area of $\alpha/2$ and χ_2^2 an upper-tail area of $\alpha/2$.

The basic underlying assumption for the mathematical derivation of chi square is that a random sample was selected from a normal distribution with variance σ^2 . When the population is not normal but skewed, chi-square probabilities could be substantially in error.

Example On the basis of a sample size $n = 5$, what midrange of values will include the sample ratio s/σ with a probability of 95 percent?

Use Table 3-7 for 4 df and read $\chi_1^2 = 0.484$ for a lower tail area of 0.05/2, 2.5 percent, and read $\chi_2^2 = 11.1$ for an upper tail area of 97.5 percent.

$$P[.484 \leq (s^2/\sigma^2)(4) \leq 11.1] = .95$$

or

$$P[.35 \leq s/\sigma \leq 1.66] = .95$$

The Microsoft Excel functions $CHIINV(.025, 4)$ and $CHIINV(.975, 4)$ give the same values.

Example On the basis of a sample size $n = 25$, what midrange of values will include the sample ratio s/σ with a probability of 95 percent?

$$P[12.4 \leq (s^2/\sigma^2)(24) \leq 39.4] = .95$$

or

$$P[.72 \leq s/\sigma \leq 1.28] = .95$$

This states that the sample standard deviation will be at least 72 percent and not more than 128 percent of the population variance 95 percent of the time. Conversely, 10 percent of the time the standard deviation will underestimate or overestimate the population standard deviation by the corresponding amount. Even for samples as large as 25, the relative reliability of a sample standard deviation is poor.

The chi-square distribution can be applied to other types of application which are of an entirely different nature. These include applications which are discussed under "Goodness-of-Fit Test" and "Two-Way Test for Independence of Count Data." In these applications, the mathematical formulation and context are entirely different, but they do result in the same table of values.

F Distribution In reference to the tensile-strength table, the successive pairs of daily standard deviations could be ratioed and squared. These ratios of variance would represent a sample from a distribution called the F distribution or F ratio. In general, the F ratio is defined by the identity

$$F(\gamma_1, \gamma_2) = s_1^2/s_2^2$$

where γ_1 and γ_2 correspond to the respective df's for the sample variances. In statistical applications, it turns out that the primary area of interest is found when the ratios are greater than 1. For this reason, most tabled values are defined for an upper-tail area. However,

TABLE 3-7 Percentiles of the χ^2 Distribution

df	Percent									
	0.5	1	2.5	5	10	90	95	97.5	99	99.5
1	0.000039	0.00016	0.00098	0.0039	0.0158	2.71	3.84	5.02	6.63	7.88
2	.0100	.0201	.0506	.1026	.2107	4.61	5.99	7.38	9.21	10.60
3	.0717	.115	.216	.352	.584	6.25	7.81	9.35	11.34	12.84
4	.207	.297	.484	.711	1.064	7.78	9.49	11.14	13.28	14.86
5	.412	.554	.831	1.15	1.61	9.24	11.07	12.83	15.09	16.75
6	.676	.872	1.24	1.64	2.20	10.64	12.59	14.45	16.81	18.55
7	.989	1.24	1.69	2.17	2.83	12.02	14.07	16.01	18.48	20.28
8	1.34	1.65	2.18	2.73	3.49	13.36	15.51	17.53	20.09	21.96
9	1.73	2.09	2.70	3.33	4.17	14.68	16.92	19.02	21.67	23.59
10	2.16	2.56	3.25	3.94	4.87	15.99	18.31	20.48	23.21	25.19
11	2.60	3.05	3.82	4.57	5.58	17.28	19.68	21.92	24.73	26.76
12	3.07	3.57	4.40	5.23	6.30	18.55	21.03	23.34	26.22	28.30
13	3.57	4.11	5.01	5.89	7.04	19.81	22.36	24.74	27.69	29.82
14	4.07	4.66	5.63	6.57	7.79	21.06	23.68	26.12	29.14	31.32
15	4.60	5.23	6.26	7.26	8.55	22.31	25.00	27.49	30.58	32.80
16	5.14	5.81	6.91	7.96	9.31	23.54	26.30	28.85	32.00	34.27
18	6.26	7.01	8.23	9.39	10.86	25.99	28.87	31.53	34.81	37.16
20	7.43	8.26	9.59	10.85	12.44	28.41	31.41	34.17	37.57	40.00
24	9.89	10.86	12.40	13.85	15.66	33.20	36.42	39.36	42.98	45.56
30	13.79	14.95	16.79	18.49	20.60	40.26	43.77	46.98	50.89	53.67
40	20.71	22.16	24.43	26.51	29.05	51.81	55.76	59.34	63.69	66.77
60	35.53	37.48	40.48	43.19	46.46	74.40	79.08	83.30	88.38	91.95
120	83.85	86.92	91.58	95.70	100.62	140.23	146.57	152.21	158.95	163.64

For large values of degrees of freedom the approximate formula

$$\chi_a^2 = n \left(1 - \frac{2}{9n} + z_a \sqrt{\frac{2}{9n}} \right)^3$$

where z_a is the normal deviate and n is the number of degrees of freedom, may be used. For example, $\chi_{.99}^2 = 60[1 - 0.00370 + 2.326(0.06086)]^3 = 60(1.1379)^3 = 88.4$ for the 99th percentile for 60 degrees of freedom. The Microsoft Excel function CHIDIST(X , df), where X is the table value, gives $1 - \text{Percent}$. The function CHIINV($1 - \text{Percent}$, df) gives the table value.

defining F_2 to be that value corresponding to an upper-tail area of $\alpha/2$, then F_1 for a lower-tail area of $\alpha/2$ can be determined through the identity

$$F_1(\gamma_1, \gamma_2) = 1/F_2(\gamma_2, \gamma_1)$$

The F distribution, similar to the chi square, is sensitive to the basic assumption that sample values were selected randomly from a normal distribution. The Microsoft Excel function FDIST(X , df_1 , df_2) gives the upper percent points of Table 3-8, where X is the tabular value. The function FINV(Percent, df_1 , df_2) gives the table value.

Example For two sample variances with 4 df each, what limits will bracket their ratio with a midarea probability of 90 percent?

Use Table 3-8 with 4 df in the numerator and denominator and upper 5 percent points (to get both sides totaling 10 percent). The entry is 6.39. Thus:

$$P[1/6.39 \leq s_1^2/s_2^2 \leq 6.39] = .90$$

$$\text{or} \quad P[.40 \leq s_1/s_2 \leq 2.53] = .90$$

The Microsoft Excel function FINV(.05, 4, 4) gives 6.39, too.

Confidence Interval for a Mean For the daily sample tensile-strength data with 4 df it is known that $P[-2.132 \leq t \leq 2.132] = .90$. This states that 90 percent of all samples will have sample t values which fall within the specified limits. In fact, for the 20 daily samples exactly 16 do fall within the specified limits (note that the binomial with $n = 20$ and $p = .90$ would describe the likelihood of exactly none through 20 falling within the prescribed limits—the sample of 20 is only a sample).

Consider the new daily sample (with $n = 5$, $\bar{x} = 37.14$, and $s = 1.85$) which was observed after a process change. In this case, the same probability holds. However, in this instance the sample value of t cannot be computed, since the new μ , under the process change, is not known. Therefore $P[-2.132 \leq (37.14 - \mu)/(1.85/\sqrt{5}) \leq 2.132] = .90$. In effect, this identity limits the magnitude of possible values for

μ . The magnitude of μ can be only large enough to retain the t quantity above -2.132 and small enough to retain the t quantity below $+2.132$. This can be found by rearranging the quantities within the bracket; i.e., $P[35.38 \leq \mu \leq 38.90] = .90$. This states that we are 90 percent sure that the interval from 35.38 to 38.90 includes the unknown parameter μ .

In general,

$$P\left[\bar{x} - t \frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + t \frac{s}{\sqrt{n}}\right] = 1 - \alpha$$

where t is defined for an upper-tail area of $\alpha/2$ with $(n - 1)$ df. In this application, the interval limits $(\bar{x} \pm t s/\sqrt{n})$ are random variables which will cover the unknown parameter μ with probability $(1 - \alpha)$. The converse, that we are $100(1 - \alpha)$ percent sure that the parameter value is within the interval, is not correct. This statement defines a probability for the parameter rather than the probability for the interval.

Example What values of t define the midarea of 95 percent for weekly samples of size 25, and what is the sample value of t for the second week?

$$P[-2.064 \leq t \leq 2.064] = .95$$

and

$$(34.19 - 35)/(2.35/\sqrt{25}) = 1.72.$$

Example For the composite sample of 100 tensile strengths, what is the 90 percent confidence interval for μ ?

Use Table 3-6 for $t_{.05}$ with $df = \infty$.

$$P\left[35.16 - 1.645 \frac{2.47}{\sqrt{100}} < \mu < 35.16 + 1.645 \frac{2.47}{\sqrt{100}}\right] = .90$$

or

$$P[34.75 \leq \mu \leq 35.57] = .90$$

Confidence Interval for the Difference in Two Population Means The confidence interval for a mean can be extended to

TABLE 3-8 F Distribution

		Upper 5% Points ($F_{.95}$)																		
		Degrees of freedom for numerator																		
		1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
Degrees of freedom for denominator	1	161	200	216	225	230	234	237	239	241	242	244	246	248	249	250	251	252	253	254
	2	18.5	19.0	19.2	19.2	19.3	19.3	19.4	19.4	19.4	19.4	19.4	19.4	19.4	19.5	19.5	19.5	19.5	19.5	19.5
	3	10.1	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
	4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
	5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.37
	6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
	7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
	8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
	9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
	10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
	11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
	12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
	13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
	14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
	15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
	16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
	17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
	18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
	19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
	20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
	21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
	22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
	23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
	24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
	25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
	30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
	40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
	60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
	120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.50	1.43	1.35	1.25
	∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

Interpolation should be performed using reciprocals of the degrees of freedom.

include the difference between two population means. This interval is based on the assumption that the respective populations have the same variance σ^2 :

$$(\bar{x}_1 - \bar{x}_2) - t_{s_p} \sqrt{1/n_1 + 1/n_2} \leq \mu_1 - \mu_2 \leq (\bar{x}_1 - \bar{x}_2) + t_{s_p} \sqrt{1/n_1 + 1/n_2}$$

Example Compute the 95 percent confidence interval based on the original 100-point sample and the subsequent 5-point sample:

$$s_p = \frac{99(2.47)^2 + 4(1.85)^2}{103} = 5.997$$

or $s_p = 2.45$

With 103 df and $\alpha = .05$, $t = \pm 1.96$ using $t_{.025}$ in Table 3-6. Therefore

$$(35.16 - 37.14) \pm 1.96(2.45) \sqrt{1/100 + 1/5} = -1.98 \pm 2.20$$

or $-4.18 \leq (\mu_1 - \mu_2) \leq .22$

Note that if the respective samples had been based on 52 observations each rather than 100 and 5, the uncertainty factor would have been $\pm .94$ rather than the observed ± 2.20 . The interval width tends to be minimum when $n_1 = n_2$.

Confidence Interval for a Variance The chi-square distribution can be used to derive a confidence interval for a population variance σ^2 when the parent population is normally distributed. For a $100(1 - \alpha)$ percent confidence interval

$$\frac{(df)s^2}{\chi^2_2} \leq \sigma^2 \leq \frac{(df)s^2}{\chi^2_1}$$

where χ^2_1 corresponds to a lower-tail area of $\alpha/2$ and χ^2_2 to an upper-tail area of $\alpha/2$.

Example For the first week of tensile-strength samples compute the 90 percent confidence interval for σ^2 ($df = 24$, corresponding to $n = 25$, using 5 percent and 95 percent in Table 3-7):

$$\frac{24(2.40)^2}{36.4} \leq \sigma^2 \leq \frac{24(2.40)^2}{13.8}$$

$$3.80 \leq \sigma^2 \leq 10.02$$

or

$$1.95 \leq \sigma \leq 3.17$$

TESTS OF HYPOTHESIS

General Nature of Tests The general nature of tests can be illustrated with a simple example. In a court of law, when a defendant is charged with a crime, the judge instructs the jury initially to presume that the defendant is innocent of the crime. The jurors are then presented with evidence and counterargument as to the defendant's guilt or innocence. If the evidence suggests beyond a reasonable doubt that the defendant did, in fact, commit the crime, they have been instructed to find the defendant guilty; otherwise, not guilty. The burden of proof is on the prosecution.

Jury trials represent a form of decision making. In statistics, an analogous procedure for making decisions falls into an area of statistical inference called **hypothesis testing**.

Suppose that a company has been using a certain supplier of raw materials in one of its chemical processes. A new supplier approaches the company and states that its material, at the same cost, will increase the process yield. If the new supplier has a good reputation, the company might be willing to run a limited test. On the basis of the test results it would then make a decision to change suppliers or not. Good management would dictate that an improvement must be demonstrated

(beyond a reasonable doubt) for the new material. That is, the burden of proof is tied to the new material. In setting up a test of hypothesis for this application, the initial assumption would be defined as a null hypothesis and symbolized as H_0 . The null hypothesis would state that yield for the new material is no greater than for the conventional material. The symbol μ_0 would be used to designate the known current level of yield for the standard material and μ for the unknown population yield for the new material. Thus, the null hypothesis can be symbolized as $H_0: \mu \leq \mu_0$.

The alternative to H_0 is called the alternative hypothesis and is symbolized as $H_1: \mu > \mu_0$.

Given a series of tests with the new material, the average yield $\bar{x}$ would be compared with μ_0 . If $\bar{x} < \mu_0$, the new supplier would be dismissed. If $\bar{x} > \mu_0$, the question would be: Is it sufficiently greater in the light of its corresponding reliability, i.e., beyond a reasonable doubt? If the confidence interval for μ included μ_0 , the answer would be no, but if it did not include μ_0 , the answer would be yes. In this simple application, the formal test of hypothesis would result in the same conclusion as that derived from the confidence interval. However, the utility of tests of hypothesis lies in their generality, whereas confidence intervals are restricted to a few special cases.

Test of Hypothesis for a Mean Procedure

Nomenclature

- μ = mean of the population from which the sample has been drawn
- σ = standard deviation of the population from which the sample has been drawn
- μ_0 = base or reference level
- H_0 = null hypothesis
- H_1 = alternative hypothesis
- α = significance level, usually set at .10, .05, or .01
- t = tabulated t value corresponding to the significance level α . For a two-tailed test, each corresponding tail would have an area of $\alpha/2$, and for a one-tailed test, one tail area would be equal to α . If σ^2 is known, then z would be used rather than the t .
- $t = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$ = sample value of the test statistic.

Assumptions

1. The n observations $x_1, x_2, \dots, x_n$ have been selected randomly.
2. The population from which the observations were obtained is normally distributed with an unknown mean μ and standard deviation σ . In actual practice, this is a robust test, in the sense that in most types of problems it is not sensitive to the normality assumption when the sample size is 10 or greater.

Test of Hypothesis

1. Under the null hypothesis, it is assumed that the sample came from a population whose mean μ is equivalent to some base or reference designated by μ_0 . This can take one of three forms:

Form 1	Form 2	Form 3
$H_0: \mu = \mu_0$	$H_0: \mu \leq \mu_0$	$H_0: \mu \geq \mu_0$
$H_1: \mu \neq \mu_0$	$H_1: \mu > \mu_0$	$H_1: \mu < \mu_0$
Two-tailed test	Upper-tailed test	Lower-tailed test

2. If the null hypothesis is assumed to be true, say, in the case of a two-sided test, form 1, then the distribution of the test statistic t is known. Given a random sample, one can predict how far its sample value of t might be expected to deviate from zero (the midvalue of t) by chance alone. If the sample value of t does, in fact, deviate too far from zero, then this is defined to be sufficient evidence to refute the assumption of the null hypothesis. It is consequently rejected, and the converse or alternative hypothesis is accepted.

3. The rule for accepting H_0 is specified by selection of the α level as indicated in Fig. 3-65. For forms 2 and 3 the α area is defined to be in the upper or the lower tail respectively. The parameter α is the probability of rejecting the null hypothesis when it is actually true.

4. The decision rules for each of the three forms are defined as follows: If the sample t falls within the acceptance region, accept H_0 for lack of contrary evidence. If the sample t falls in the critical region, reject H_0 at a significance level of 100α percent.

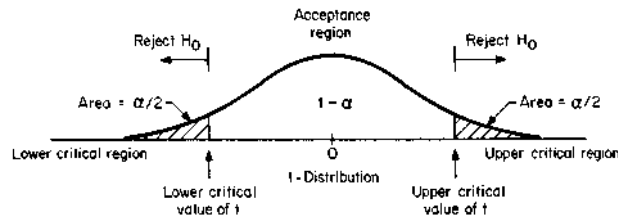


FIG. 3-65 Acceptance region for two-tailed test. For a one-tailed test, area = α on one side only.

Example

Application. In the past, the yield for a chemical process has been established at 89.6 percent with a standard deviation of 3.4 percent. A new supplier of raw materials will be used and tested for 7 days.

Procedure

1. The standard of reference is $\mu_0 = 89.6$ with a known $\sigma = 3.4$.
2. It is of interest to demonstrate whether an increase in yield is achieved with the new material; H_0 says it has not; therefore,

$$H_0: \mu \leq 89.6 \quad H_1: \mu > 89.6$$

3. Select $\alpha = .05$, and since σ is known (the new material would not affect the day-to-day variability in yield), the test statistic would be z with a corresponding critical value $cv(z) = 1.645$ (Table 3-6, $df = \infty$).

4. The decision rule:

Accept H_0 if sample $z < 1.645$

Reject H_0 if sample $z > 1.645$

5. A 7-day test was carried out, and daily yields averaged 91.6 percent with a sample standard deviation $s = 3.6$ (this is not needed for the test of hypothesis).

6. For the data sample $z = (91.6 - 89.6)/(3.4/\sqrt{7}) = 1.56$.

7. Since the sample $z < cv(z)$, accept the null hypothesis for lack of contrary evidence; i.e., an improvement has not been demonstrated beyond a reasonable doubt.

Example

Application. In the past, the break strength of a synthetic yarn has averaged 34.6 lb. The first-stage draw ratio of the spinning machines has been increased. Production management wants to determine whether the break strength has changed under the new condition.

Procedure

1. The standard of reference is $\mu_0 = 34.6$.
2. It is of interest to demonstrate whether a change has occurred; therefore,

$$H_0: \mu = 34.6 \quad H_1: \mu \neq 34.6$$

3. Select $\alpha = .05$, and since with the change in draw ratio the uniformity might change, the sample standard deviation would be used, and therefore t would be the appropriate test statistic.

4. A sample of 21 ends was selected randomly and tested on an Instron with the results $\bar{x} = 35.55$ and $s = 2.041$.

5. For 20 df and a two-tailed α level of 5 percent, the critical values of $t_{0.025}$ (two tailed) are given by ± 2.086 with a decision rule (Table 3-6, $t_{0.025}$, $df = 20$):

Accept H_0 if $-2.086 < \text{sample } t < 2.086$

Reject H_0 if sample $t < -2.086$ or > 2.086

6. For the data sample $t = (35.55 - 34.6)/(2.041/\sqrt{21}) = 2.133$.

7. Since $2.133 > 2.086$, reject H_0 and accept H_1 . It has been demonstrated that an improvement in break strength has been achieved.

Two-Population Test of Hypothesis for Means

Nature Two samples were selected from different locations in a plastic-film sheet and measured for thickness. The thickness of the respective samples was measured at 10 close but equally spaced points in each of the samples. It was of interest to compare the average thickness of the respective samples to detect whether they were significantly different. That is, was there a significant variation in thickness between locations?

From a modeling standpoint statisticians would define this problem as a two-population test of hypothesis. They would define the respective sample sheets as two populations from which 10 sample thickness determinations were measured for each.

In order to compare populations based on their respective samples, it is necessary to have some basis of comparison. This basis is predicated on the distribution of the t statistic. In effect, the t statistic characterizes

the way in which two sample means from two separate populations will tend to vary by chance alone when the population means and variances are equal. Consider the following:

Population 1		Population 2	
Normal	Sample 1	Normal	Sample 2
μ_1	n_1	μ_2	n_2
	$\bar{x}_1$		$\bar{x}_2$
σ_1^2	s_1^2	σ_2^2	s_2^2

Consider the hypothesis $\mu_1 = \mu_2$. If, in fact, the hypothesis is correct, i.e., $\mu_1 = \mu_2$ (under the condition $\sigma_1^2 = \sigma_2^2$), then the sampling distribution of $(\bar{x}_1 - \bar{x}_2)$ is predictable through the t distribution. The observed sample values then can be compared with the corresponding t distribution. If the sample values are reasonably close (as reflected through the α level), that is, $\bar{x}_1$ and $\bar{x}_2$ are not "too different" from each other on the basis of the t distribution, the null hypothesis would be accepted. Conversely, if they deviate from each other "too much" and the deviation is therefore not ascribable to chance, the conjecture would be questioned and the null hypothesis rejected.

Example

Application. Two samples were selected from different locations in a plastic-film sheet. The thickness of the respective samples was measured at 10 close but equally spaced points.

Procedure

1. Demonstrate whether the thicknesses of the respective sample locations are significantly different from each other; therefore,

$$H_0: \mu_1 = \mu_2 \quad H_1: \mu_1 \neq \mu_2$$

2. Select $\alpha = .05$.

3. Summarize the statistics for the respective samples:

Sample 1		Sample 2	
1.473	1.367	1.474	1.417
1.484	1.276	1.501	1.448
1.484	1.485	1.485	1.469
1.425	1.462	1.435	1.474
1.448	1.439	1.348	1.452
$\bar{x}_1 = 1.434$	$s_1 = .0664$	$\bar{x}_2 = 1.450$	$s_2 = .0435$

4. As a first step, the assumption for the standard t test, that $\sigma_1^2 = \sigma_2^2$, can be tested through the F distribution. For this hypothesis, $H_0: \sigma_1^2 = \sigma_2^2$ would be tested against $H_1: \sigma_1^2 \neq \sigma_2^2$. Since this is a two-tailed test and conventionally only the upper tail for F is published, the procedure is to use the largest ratio and the corresponding ordered degrees of freedom. This achieves the same end result through one table. However, since the largest ratio is arbitrary, it is necessary to define the true α level as twice the value of the tabled value. Therefore, by using Table 3-8 with $\alpha = .05$ the corresponding critical value for $F(9,9) = 3.18$ would be for a true $\alpha = .10$. For the sample,

$$\text{Sample } F = (.0664/.0435)^2 = 2.33$$

Therefore, the ratio of sample variances is no larger than one might expect to observe when in fact $\sigma_1^2 = \sigma_2^2$. There is not sufficient evidence to reject the null hypothesis that $\sigma_1^2 = \sigma_2^2$.

5. For 18 df and a two-tailed α level of 5 percent the critical values of t are given by ± 2.101 (Table 3-6, $t_{0.025}$, $df = 18$).

6. The decision rule:

Accept H_0 if $-2.101 \leq \text{sample } t \leq 2.101$

Reject H_0 otherwise

7. For the sample the pooled variance estimate is given by Eq. (3-120).

$$s_p^2 = \frac{9(.0664)^2 + 9(.0435)^2}{9 + 9} = \frac{(.0664)^2 + (.0435)^2}{2} = .00315$$

or

$$s_p = .056$$

8. The sample statistic value of t is given by Eq. (3-119).

$$\text{Sample } t = \frac{1.434 - 1.450}{.056\sqrt{1/10 + 1/10}} = -.64$$

9. Since the sample value of t falls within the acceptance region, accept H_0 for lack of contrary evidence; i.e., there is insufficient evidence to demonstrate that thickness differs between the two selected locations.

Test of Hypothesis for Paired Observations

Nature In some types of applications, associated pairs of observations are defined. For example, (1) pairs of samples from two populations are treated in the same way, or (2) two types of measurements are made on the same unit. For applications of this type, it is not only more effective but necessary to define the random variable as the difference between the pairs of observations. The difference numbers can then be tested by the standard t distribution.

Examples of the two types of applications are as follows:

1. *Sample treatment*

a. Two types of metal specimens buried in the ground together in a variety of soil types to determine corrosion resistance

b. Wear-rate test with two different types of tractor tires mounted in pairs on n tractors for a defined period of time

2. *Same unit*

a. Blood-pressure measurements made on the same individual before and after the administration of a stimulus

b. Smoothness determinations on the same film samples at two different testing laboratories

Test of Hypothesis for Matched Pairs: Procedure

Nomenclature

d_i = sample difference between the i th pair of observations

s = sample standard deviation of differences

μ = population mean of differences

σ = population standard deviation of differences

μ_0 = base or reference level of comparison

H_0 = null hypothesis

H_1 = alternative hypothesis

α = significance level

t = tabled value with $(n - 1)$ df

$t = (\bar{d} - \mu_0)/(s/\sqrt{n})$, the sample value of t

Assumptions

1. The n pairs of samples have been selected and assigned for testing in a random way.

2. The population of differences is normally distributed with a mean μ and variance σ^2 . As in the previous application of the t distribution, this is a robust procedure, i.e., not sensitive to the normality assumption if the sample size is 10 or greater in most situations.

Test of Hypothesis

1. Under the null hypothesis, it is assumed that the sample came from a population whose mean μ is equivalent to some base or reference level designated by μ_0 . For most applications of this type, the value of μ_0 is defined to be zero; that is, it is of interest generally to demonstrate a difference not equal to zero. The hypothesis can take one of three forms:

Form 1	Form 2	Form 3
$H_0: \mu = \mu_0$	$H_0: \mu \leq \mu_0$	$H_0: \mu \geq \mu_0$
$H_1: \mu \neq \mu_0$	$H_1: \mu > \mu_0$	$H_1: \mu < \mu_0$
Two-tailed test	Upper-tailed test	Lower-tailed test

2. If the null hypothesis is assumed to be true, say, in the case of a lower-tailed test, form 3, then the distribution of the test statistic t is known under the null hypothesis that limits $\mu = \mu_0$. Given a random sample, one can predict how far its sample value of t might be expected to deviate from zero by chance alone when $\mu = \mu_0$. If the sample value of t is too small, as in the case of a negative value, then this would be defined as sufficient evidence to reject the null hypothesis.

3. Select α .

4. The critical values or value of t would be defined by the tabled value of t with $(n - 1)$ df corresponding to a tail area of α . For a two-tailed test, each tail area would be $\alpha/2$, and for a one-tailed test there would be an upper-tail or a lower-tail area of α corresponding to forms 2 and 3 respectively.

5. The decision rule for each of the three forms would be to reject the null hypothesis if the sample value of t fell in that area of the t distribution defined by α , which is called the critical region.

Otherwise, the alternative hypothesis would be accepted for lack of contrary evidence.

Example

Application. Pairs of pipes have been buried in 11 different locations to determine corrosion on nonbituminous pipe coatings for underground use. One type includes a lead-coated steel pipe and the other a bare steel pipe.

Procedure

1. The standard of reference is taken as $\mu_0 = 0$, corresponding to no difference in the two types.
2. It is of interest to demonstrate whether either type of pipe has a greater corrosion resistance than the other. Therefore,

$$H_0: \mu = 0 \quad H_1: \mu \neq 0$$

3. Select $\alpha = .05$. Therefore, with $n = 11$ the critical values of t with 10 df are defined by $t = \pm 2.228$ (Table 3.5, $t_{.025}$).

4. The decision rule:

Accept H_0 if $-2.228 \leq \text{sample } t \leq 2.228$

Reject H_0 otherwise

5. The sample of 11 pairs of corrosion determinations and their differences are as follows:

Soil type	Lead-coated steel pipe	Bare steel pipe	$d = \text{difference}$
A	27.3	41.4	-14.1
B	18.4	18.9	-0.5
C	11.9	21.7	-9.8
D	11.3	16.8	-5.5
E	14.8	9.0	5.8
F	20.8	19.3	1.5
G	17.9	32.1	-14.2
H	7.8	7.4	0.4
I	14.7	20.7	-6.0
J	19.0	34.4	-15.4
K	65.3	76.2	-10.9

6. The sample statistics, Eq. (3-118)

$$\bar{d} = -6.245 \quad s^2 = \frac{11 \sum d^2 - (\sum d)^2}{11 \times 10} = 52.59$$

or

$$s = 7.25$$

$$\text{Sample } t = (-6.245 - 0) / (7.25 / \sqrt{11}) = -2.86$$

7. Since the sample t of $-2.86 < \text{tabled } t$ of -2.228 , reject H_0 and accept H_1 ; that is, it has been demonstrated that, on the basis of the evidence, lead-coated steel pipe has a greater corrosion resistance than bare steel pipe.

Example

Application. A stimulus was tested for its effect on blood pressure. Ten men were selected randomly, and their blood pressure was measured before and after the stimulus was administered. It was of interest to determine whether the stimulus had caused a significant increase in the blood pressure.

Procedure

1. The standard of reference was taken as $\mu_0 \leq 0$, corresponding to no increase.
2. It was of interest to demonstrate an increase in blood pressure if in fact an increase did occur. Therefore,

$$H_0: \mu_0 \leq 0 \quad H_1: \mu_0 > 0$$

3. Select $\alpha = .05$. Therefore, with $n = 10$ the critical value of t with 9 df is defined by $t = 1.833$ (Table 3-6, $t_{.05}$, one-sided).

4. The decision rule:

Accept H_0 if sample $t < 1.833$

Reject H_0 if sample $t > 1.833$

5. The sample of 10 pairs of blood pressure and their differences were as follows:

Individual	Before	After	$d = \text{difference}$
1	138	146	8
2	116	118	2
3	124	120	-4
4	128	136	8

5	155	174	19
6	129	133	4
7	130	129	-1
8	148	155	7
9	143	148	5
10	159	155	-4

6. The sample statistics:

$$\bar{d} = 4.4 \quad s = 6.85$$

$$\text{Sample } t = (4.4 - 0) / (6.85 / \sqrt{10}) = 2.03$$

7. Since the sample $t = 2.03 > \text{critical } t = 1.833$, reject the null hypothesis. It has been demonstrated that the population of men from which the sample was drawn tend, as a whole, to have an increase in blood pressure after the stimulus has been given. The distribution of differences d seems to indicate that the degree of response varies by individuals.

Test of Hypothesis for a Proportion

Nature Some types of statistical applications deal with counts and proportions rather than measurements. Examples are (1) the proportion of workers in a plant who are out sick, (2) lost-time worker accidents per month, (3) defective items in a shipment lot, and (4) preference in consumer surveys.

The procedure for testing the significance of a sample proportion follows that for a sample mean. In this case, however, owing to the nature of the problem the appropriate test statistic is Z . This follows from the fact that the null hypothesis requires the specification of the goal or reference quantity p_0 , and since the distribution is a binomial proportion, the associated variance is $[p_0(1 - p_0)]n$ under the null hypothesis. The primary requirement is that the sample size n satisfy normal approximation criteria for a binomial proportion, roughly $np > 5$ and $n(1 - p) > 5$.

Test of Hypothesis for a Proportion: Procedure

Nomenclature

p = mean proportion of the population from which the sample has been drawn

p_0 = base or reference proportion

$[p_0(1 - p_0)]/n$ = base or reference variance

$\hat{p} = x/n$ = sample proportion, where x refers to the number of observations out of n which have the specified attribute

H_0 = assumption or null hypothesis regarding the population proportion

H_1 = alternative hypothesis

α = significance level, usually set at .10, .05, or .01

z = tabled Z value corresponding to the significance level α . The sample sizes required for the z approximation according to the magnitude of p_0 are given in Table 3-6.

$z = (\hat{p} - p_0) / \sqrt{p_0(1 - p_0)/n}$, the sample value of the test statistic

Assumptions

1. The n observations have been selected randomly.
2. The sample size n is sufficiently large to meet the requirement for the Z approximation.

Test of Hypothesis

1. Under the null hypothesis, it is assumed that the sample came from a population with a proportion p_0 of items having the specified attribute. For example, in tossing a coin the population could be thought of as having an unbounded number of potential tosses. If it is assumed that the coin is fair, this would dictate $p_0 = 1/2$ for the proportion number of heads in the population. The null hypothesis can take one of three forms:

Form 1	Form 2	Form 3
$H_0: p = p_0$	$H_0: p \leq p_0$	$H_0: p \geq p_0$
$H_1: p \neq p_0$	$H_1: p > p_0$	$H_1: p < p_0$
Two-tailed test	Upper-tailed test	Lower-tailed test

2. If the null hypothesis is assumed to be true, then the sampling distribution of the test statistic Z is known. Given a random sample, it is possible to predict how far the sample proportion x/n might deviate from its assumed population proportion p_0 through the Z distribution. When the sample proportion deviates too far, as defined by the significance level α , this serves as the justification for rejecting the assumption, that is, rejecting the null hypothesis.

3. The decision rule is given by

Form 1: Accept H_0 if lower critical $z < \text{sample } z < \text{upper critical } z$

Reject H_0 otherwise

Form 2: Accept H_0 if sample $z < \text{upper critical } z$

Reject H_0 otherwise

Form 3: Accept H_0 if lower critical $z < \text{sample } z$

Reject H_0 otherwise

Example

Application. A company has received a very large shipment of rivets. One product specification required that no more than 2 percent of the rivets have diameters greater than 14.28 mm. Any rivet with a diameter greater than this would be classified as defective. A random sample of 600 was selected and tested with a go-no-go gauge. Of these, 16 rivets were found to be defective. Is this sufficient evidence to conclude that the shipment contains more than 2 percent defective rivets?

Procedure

1. The quality goal is $p \leq .02$. It would be assumed initially that the shipment meets this standard; i.e., $H_0: p \leq .02$.

2. The assumption in step 1 would first be tested by obtaining a random sample. Under the assumption that $p \leq .02$, the distribution for a sample proportion would be defined by the z distribution. This distribution would define an upper bound corresponding to the upper critical value for the sample proportion. It would be unlikely that the sample proportion would rise above that value if, in fact, $p \leq .02$. If the observed sample proportion exceeds that limit, corresponding to what would be a very unlikely chance outcome, this would lead one to question the assumption that $p \leq .02$. That is, one would conclude that the null hypothesis is false. To test, set

$$H_0: p \leq .02 \quad H_1: p > .02$$

3. Select $\alpha = .05$.

4. With $\alpha = .05$, the upper critical value of $Z = 1.645$ (Table 3-6, $t_{.05}$, $df = \infty$, one-sided).

5. The decision rule:

Accept H_0 if sample $z < 1.645$

Reject H_0 if sample $z > 1.645$

6. The sample z is given by

$$\begin{aligned} \text{Sample } z &= \frac{(16/600) - .02}{\sqrt{(.02)(.98)/600}} \\ &= 1.17 \end{aligned}$$

7. Since the sample $z < 1.645$, accept H_0 for lack of contrary evidence; there is not sufficient evidence to demonstrate that the defect proportion in the shipment is greater than 2 percent.

Test of Hypothesis for Two Proportions

Nature In some types of engineering and management-science problems, we may be concerned with a random variable which represents a proportion, for example, the proportional number of defective items per day. The method described previously relates to a single proportion. In this subsection two proportions will be considered.

A certain change in a manufacturing procedure for producing component parts is being considered. Samples are taken by using both the existing and the new procedures in order to determine whether the new procedure results in an improvement. In this application, it is of interest to demonstrate statistically whether the population proportion p_2 for the new procedure is less than the population proportion p_1 for the old procedure on the basis of a sample of data.

Test of Hypothesis for Two Proportions: Procedure

Nomenclature

p_1 = population 1 proportion

p_2 = population 2 proportion

n_1 = sample size from population 1

n_2 = sample size from population 2

x_1 = number of observations out of n_1 that have the designated attribute

x_2 = number of observations out of n_2 that have the designated attribute

$\hat{p}_1 = x_1/n_1$, the sample proportion from population 1

$\hat{p}_2 = x_2/n_2$, the sample proportion from population 2

α = significance level

H_0 = null hypothesis

H_1 = alternative hypothesis

z = tabled Z value corresponding to the stated significance level α

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_1(1 - \hat{p}_1)/n_1 + \hat{p}_2(1 - \hat{p}_2)/n_2}}, \text{ the sample value of } Z$$

Assumptions

1. The respective two samples of n_1 and n_2 observations have been selected randomly.

2. The sample sizes n_1 and n_2 are sufficiently large to meet the requirement for the Z approximation; i.e., $x_1 > 5$, $x_2 > 5$.

Test of Hypothesis

1. Under the null hypothesis, it is assumed that the respective two samples have come from populations with equal proportions $p_1 = p_2$. Under this hypothesis, the sampling distribution of the corresponding Z statistic is known. On the basis of the observed data, if the resultant sample value of Z represents an unusual outcome, that is, if it falls within the critical region, this would cast doubt on the assumption of equal proportions. Therefore, it will have been demonstrated statistically that the population proportions are in fact not equal. The various hypotheses can be stated:

Form 1	Form 2	Form 3
$H_0: p_1 = p_2$	$H_0: p_1 \leq p_2$	$H_0: p_1 \geq p_2$
$H_1: p_1 \neq p_2$	$H_1: p_1 > p_2$	$H_1: p_1 < p_2$
Two-tailed test	Upper-tailed test	Lower-tailed test

2. The decision rule for form 1 is given by

Accept H_0 if lower critical $z < \text{sample } z < \text{upper critical } z$

Reject H_0 otherwise

Example

Application. A change was made in a manufacturing procedure for component parts. Samples were taken during the last week of operations with the old procedure and during the first week of operations with the new procedure. Determine whether the proportional numbers of defects for the respective populations differ on the basis of the sample information.

Procedure

1. The hypotheses are

$$H_0: p_1 = p_2 \quad H_1: p_1 \neq p_2$$

2. Select $\alpha = .05$. Therefore, the critical values of z are ± 1.96 (Table 3-5, $A = 0.9500$).

3. For the samples, 75 out of 1720 parts from the previous procedure and 80 out of 2780 parts under the new procedure were found to be defective; therefore,

$$\hat{p}_1 = 75/1720 = .0436 \quad \hat{p}_2 = 80/2780 = .0288$$

4. The decision rule:

Accept H_0 if $-1.96 \leq \text{sample } z \leq 1.96$

Reject H_0 otherwise

5. The sample statistic:

$$\begin{aligned} \text{Sample } z &= \frac{.0436 - .0288}{\sqrt{(.0436)(.9564)/1720 + (.0288)(.9712)/2780}} \\ &= 2.53 \end{aligned}$$

6. Since the sample z of 2.53 > tabled z of 1.96, reject H_0 and conclude that the new procedure has resulted in a reduced defect rate.

Goodness-of-Fit Test

Nature A standard die has six sides numbered from 1 to 6. If one were really interested in determining whether a particular die was well balanced, one would have to carry out an experiment. To do this, it might be decided to count the frequencies of outcomes, 1 through 6, in tossing the die N times. On the assumption that the die is perfectly balanced,

one would expect to observe $N/6$ occurrences each for 1, 2, 3, 4, 5, and 6. However, chance dictates that exactly $N/6$ occurrences each will not be observed. For example, given a perfectly balanced die, the probability is only 1 chance in 65 that one will observe 1 outcome each, for 1 through 6, in tossing the die 6 times. Therefore, an outcome different from 1 occurrence each can be expected. Conversely, an outcome of six 3s would seem to be too unusual to have occurred by chance alone.

Some industrial applications involve the concept outlined here. The basic idea is to test whether or not a group of observations follows a pre-conceived distribution. In the case cited, the distribution is uniform; i.e., each face value should *tend* to occur with the same frequency.

Goodness-of-Fit Test: Procedure

Nomenclature Each experimental observation can be classified into one of r possible categories or cells.

r = total number of cells

O_j = number of observations occurring in cell j

E_j = expected number of observations for cell j based on the pre-conceived distribution

N = total number of observations

f = degrees of freedom for the test. In general, this will be equal to $(r - 1)$ minus the number of statistical quantities on which the E_j s are based (see the examples which follow for details).

Assumptions

1. The observations represent a sample selected randomly from a population which has been specified.

2. The number of expectation counts E_j within each category should be roughly 5 or more. If an E_j count is significantly less than 5, that cell should be pooled with an adjacent cell.

Computation for E_j On the basis of the specified population, the probability of observing a count in cell j is defined by p_j . For a sample of size N , corresponding to N total counts, the expected frequency is given by $E_j = Np_j$.

Test Statistics: Chi Square

$$\chi^2 = \sum_{j=1}^r \frac{(O_j - E_j)^2}{E_j} \quad \text{with } f \text{ df}$$

Test of Hypothesis

1. H_0 : The sample came from the specified theoretical distribution

H_1 : The sample did not come from the specified theoretical distribution

2. For a stated level of α ,

Reject H_0 if sample $\chi^2 > \text{tabled } \chi^2$
Accept H_0 if sample $\chi^2 < \text{tabled } \chi^2$

Example

Application A production-line product is rejected if one of its characteristics does not fall within specified limits. The standard goal is that no more than 2 percent of the production should be rejected.

Computation

1. Of 950 units produced during the day, 28 units were rejected.

2. The hypotheses:

H_0 : the process is in control

H_1 : the process is not in control

3. Assume that $\alpha = .05$; therefore, the critical value of $\chi^2(1) = 3.84$ (Table 3-7, 95 percent, $df = 1$). One degree of freedom is defined since $(r - 1) = 1$, and no statistical quantities have been computed for the data.

4. The decision rule:

Reject H_0 if sample $\chi^2 > 3.84$
Accept H_0 otherwise

5. Since it is assumed that $p = .02$, this would dictate that in a sample of 950 there would be on the average $(.02)(950) = 19$ defective items and 931 acceptable items:

Category	Observed O_j	Expectation $E_j = 950p_j$
Acceptable	922	931
Not acceptable	28	19
Total	950	950

$$\begin{aligned} \text{Sample } \chi^2 &= \frac{(922 - 931)^2}{931} + \frac{(28 - 19)^2}{19} \\ &= 4.35 \text{ with critical } \chi^2 = 3.84 \end{aligned}$$

6. Conclusion. Since the sample value exceeds the critical value, it would be concluded that the process is not in control.

Example

Application A frequency count of workers was tabulated according to the number of defective items that they produced. An unresolved question is whether the observed distribution is a Poisson distribution. That is, do observed and expected frequencies agree within chance variation?

Computation

1. The hypotheses:

H_0 : there are no significant differences, in number of defective units, between workers

H_1 : there are significant differences

2. Assume that $\alpha = .05$.

3. Test statistic:

No. of defective units	O_j	E_j
0	3	2.06
1	7	6.64
2	9	10.73
3	12	11.55
4	9	9.33
5	6	6.03
6	3	3.24
7	2	1.50
8	0	.60
9	1	.22
≥ 10	0	.10
Sum	52	52

The expectation numbers E_j were computed as follows: For the Poisson distribution, $\lambda = E(x)$; therefore, an estimate of λ is the average number of defective units per worker, i.e., $\lambda = (1/52)(0 \times 3 + 1 \times 7 + \dots + 9 \times 1) = 3.23$. Given this approximation, the probability of no defective units for a worker would be $(3.23)^0/0! = e^{-3.23} = .0396$. For the 52 workers, the number of workers producing no defective units would have an expectation $E = 52(0.0396) = 2.06$, and so forth.

The sample chi-square value is computed from

$$\begin{aligned} \chi^2 &= \frac{(0 - 2.06)^2}{2.06} + \frac{(7 - 6.64)^2}{6.64} + \dots + \frac{(6 - 3.24)^2}{3.24} \\ &= .522 \end{aligned}$$

4. The critical value of χ^2 would be based on four degrees of freedom. This corresponds to $(r - 1) - 1$, since one statistical quantity λ was computed from the sample and used to derive the expectation numbers.

5. The critical value of $\chi^2(4) = 9.49$ (Table 3-7) with $\alpha = .05$; therefore, accept H_0 .

Two-Way Test for Independence for Count Data

Nature When individuals or items are observed and classified according to two different criteria, the resultant counts can be statistically analyzed. For example, a market survey may examine whether a new product is preferred and if it is preferred due to a particular characteristic.

Count data, based on a random selection of individuals or items which are classified according to two different criteria, can be statistically analyzed through the χ^2 distribution. The purpose of this analysis is to determine whether the respective criteria are dependent. That is, is the product preferred because of a particular characteristic?

Two-Way Test for Independence for Count Data: Procedure

Nomenclature

1. Each observation is classified according to two categories:

a. The first one into 2, 3, . . . , or r categories

b. The second one into 2, 3, . . . , or c categories

2. O_{ij} = number of observations (observed counts) in cell (i, j) with

$$i = 1, 2, \dots, r$$

$$j = 1, 2, \dots, c$$

3. N = total number of observations
4. E_{ij} = computed number for cell (ij) which is an expectation based on the assumption that the two characteristics are independent
5. R_i = subtotal of counts in row i
6. C_j = subtotal of counts in column j
7. α = significance level
8. H_0 = null hypothesis
9. H_1 = alternative hypothesis
10. χ^2 = critical value of χ^2 corresponding to the significance level α and $(r-1)(c-1)$ df
11. Sample $\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$

Assumptions

1. The observations represent a sample selected randomly from a large total population.
2. The number of expectation counts E_{ij} within each cell should be approximately 2 or more for arrays 3×3 or larger. If any cell contains a number smaller than 2, appropriate rows or columns should be combined to increase the magnitude of the expectation count. For arrays 2×2 , approximately 4 or more are required. If the number is less than 4, the exact Fisher test should be used.

Test of Hypothesis Under the null hypothesis, the classification criteria are assumed to be independent, i.e.,

H_0 : the criteria are independent

H_1 : the criteria are not independent

For the stated level of α ,

Reject H_0 if sample $\chi^2 >$ tabled χ^2

Accept H_0 otherwise

Computation for E_{ij} Compute E_{ij} across rows or down columns by using either of the following identities:

$$E_{ij} = C_j \left(\frac{R_i}{N} \right) \text{ across rows}$$

$$E_{ij} = R_i \left(\frac{C_j}{N} \right) \text{ down columns}$$

Sample χ^2 Value

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

In the special case of $r = 2$ and $c = 2$, a more accurate and simplified formula which does not require the direct computation of E_{ij} can be used:

$$\chi^2 = \frac{[O_{11}O_{22} - O_{12}O_{21}]^2}{N} \cdot \frac{1}{R_1 R_2 C_1 C_2}$$

Example

Application A market research study was carried out to relate the subjective "feel" of a consumer product to consumer preference. In other words, is the consumer's preference for the product associated with the feel of the product, or is the preference independent of the product feel?

Procedure

1. It was of interest to demonstrate whether an association exists between feel and preference; therefore, assume

H_0 : feel and preference are independent

H_1 : they are not independent

2. A sample of 200 people was asked to classify the product according to two criteria:

- a. Liking for this product
- b. Liking for the feel of the product

		Like feel		
		Yes	No	R_i
Like product	Yes	114	13	= 127
	No	55	18	= 73
		169	31	200
		C_j		

3. Select $\alpha = .05$; therefore, with $(r-1)(c-1) = 1$ df, the critical value of χ^2 is 3.84 (Table 3-7, 95 percent).

4. The decision rule:

Accept H_0 if sample $\chi^2 < 3.84$

Reject H_0 otherwise

5. The sample value of χ^2 by using the special formula is

$$\text{Sample } \chi^2 = \frac{[114 \times 18 - 13 \times 55]^2}{(169)(31)(127)(73)} = 6.30$$

6. Since the sample χ^2 of 6.30 > tabled χ^2 of 3.84, reject H_0 and accept H_1 . The relative proportionality of $E_{11} = 169(127/200) = 107.3$ to the observed 114 compared with $E_{22} = 31(73/200) = 11.3$ to the observed 18 suggests that when the consumer likes the feel, the consumer tends to like the product, and conversely for not liking the feel. The proportions $169/200 = 84.5$ percent and $127/200 = 63.5$ percent suggest further that there are other attributes of the product which tend to nullify the beneficial feel of the product.

LEAST SQUARES

When experimental data is to be fit with a mathematical model, it is necessary to allow for the fact that the data has errors. The engineer is interested in finding the parameters in the model as well as the uncertainty in their determination. In the simplest case, the model is a linear equation with only two parameters, and they are found by a least-squares minimization of the errors in fitting the data. Multiple regression is just linear least squares applied with more terms. Non-linear regression allows the parameters of the model to enter in a non-linear fashion. See Press et al. (1986); for a description of maximum likelihood as it applies to both linear and nonlinear least squares.

In a least squares parameter estimation, it is desired to find parameters that minimize the sum of squares of the deviation between the experimental data and the theoretical equation.

$$\chi^2 = \sum_{i=1}^N [y_i - y(x_i; a_1, a_2, \dots, a_M)]^2$$

where y_i is the i th experimental data point for the value x_i , $y(x_i; a_1, a_2, \dots, a_M)$ is the theoretical equation at x_i , and the parameters $\{a_1, a_2, \dots, a_M\}$ are to be determined to minimize χ^2 . This will also minimize the variance of the curve fit

$$\sigma^2 = \sum_{i=1}^N \frac{[y_i - y(x_i; a_1, a_2, \dots, a_M)]^2}{N}$$

Linear Least Squares When the model is a straight line, one is minimizing

$$\chi^2 = \sum_{i=1}^N (y_i - a - bx_i)^2$$

The linear correlation coefficient r is defined by

$$r = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^N (y_i - \bar{y})^2}}$$

and

$$\chi^2 = (1 - r^2) \sum_{i=1}^N (y_i - \bar{y})^2$$

where $\bar{y}$ is the average of the y_i values. Values of r near 1 indicate a positive correlation; r near -1 means a negative correlation, and r near 0 means no correlation. These parameters are easily found by using standard programs, such as Microsoft Excel.

Example

Application. Brenner (*Magnetic Method for Measuring the Thickness of Non-magnetic Coatings on Iron and Steel*, National Bureau of Standards, RP1081, March 1938) suggests an alternative way of measuring the thickness of nonmagnetic coatings of galvanized zinc on iron and steel. This procedure is based on a nondestructive magnetic method as a substitute for the standard

destructive stripping method. A random sample of 11 pieces was selected and measured by both methods.

Nomenclature. The calibration between the magnetic and the stripping methods can be determined through the model

$$y = a + bx + \epsilon$$

where x = strip-method determination

y = magnetic-method determination

Sample data

Thickness, 10^{-5} in

Stripping method, x	Magnetic method, y
104	85
114	115
116	105
129	127
132	120
139	121
174	155
312	250
338	310
465	443
720	630

This example is solved by using Microsoft Excel. Put the data into columns A and B as shown, using rows 1 through 11. Then the commands

SLOPE(B1:B11, A1:A11), INTERCEPT(B1:B11, A1:A11),
RSQ(B1:B11, A1:A11)

give the slope b , the intercept a , and the value of r^2 . Here they are 3.20, 0.884, and 0.9928, respectively, so that $r = 0.9964$. By choosing Insert/Chart and Scatter Plot, the data are plotted. Once that is done, place the cursor on a data point and right-click; choose Format trendline with options selected to display the equation and r^2 , and you get Fig. 3-66. On the Macintosh, use CTRL-click.

Polynomial Regression In polynomial regression, one expands the function in a polynomial in x .

$$y(x) = \sum_{j=1}^M a_j x^{j-1}$$

Example

Application. Merriman ("The Method of Least Squares Applied to a Hydraulic Problem," *J. Franklin Inst.*, 233-241, October 1877) reported on a

study of stream velocity as a function of relative depth of the stream.

Sample data

Depth*	Velocity, y , ft/s
0	3.1950
.1	3.2299
.2	3.2532
.3	3.2611
.4	3.2516
.5	3.2282
.6	3.1807
.7	3.1266
.8	3.0594
.9	2.9759

*As a fraction of total depth.

The model is taken as a quadratic function of position:

$$\text{Velocity} = a + bx + cx^2$$

The parameters are easily determined by using computer software. In Microsoft Excel, the data are put into columns A and B and the graph is created as for a linear curve fit. This time, though, when adding the trendline, choose the polynomial icon and use 2 (which gives powers up to and including x^2). The result is

$$\text{Velocity} = 3.195 + 0.4425x - 0.7653x^2$$

The value of r^2 is .9993.

Multiple Regression In multiple regression, any set of functions can be used, not just polynomials, such as

$$y(x) = \sum_{j=1}^M a_j f_j(x)$$

where the set of functions $\{f_j(x)\}$ is known and specified. Note that the unknown parameters $\{a_j\}$ enter the equation linearly. In this case, the spreadsheet can be expanded to have a column for x and then successive columns for $f_j(x)$. In Microsoft Excel, you choose Regression under Tools/Data Analysis, and complete the form. In addition to the actual correlation, you get the expected variance of the unknowns, which allows you to assess how accurately they were determined. In the example above, by creating a column for x and x^2 , one obtains an intercept of 3.195 with a standard error of .0039, $b = .4416$ with a standard error of .018, and $c = -.7645$ with a standard error of .018.

Nonlinear Least Squares There are no analytic methods for determining the most appropriate model for a particular set of data.

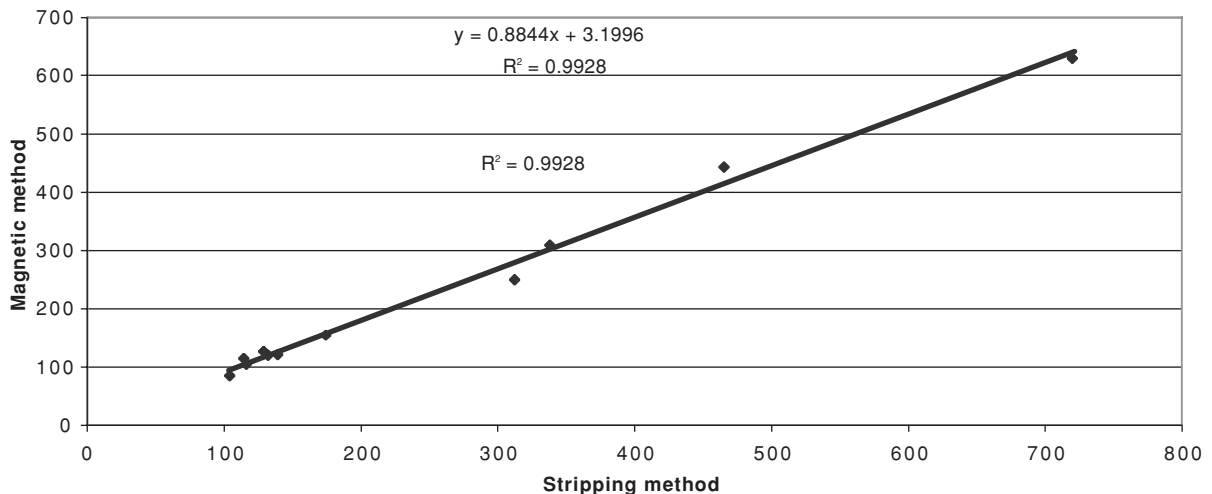


FIG. 3-66 Plot of data and correlating line.

In many cases, however, the engineer has some basis for a model. If the parameters occur in a nonlinear fashion, then the analysis becomes more difficult. For example, in relating the temperature to the elapsed time of a fluid cooling in the atmosphere, a model that has an asymptotic property would be the appropriate model ($\text{temp} = a + b \exp(-c \text{ time})$), where a represents the asymptotic temperature corresponding to $t \rightarrow \infty$. In this case, the parameter c appears nonlinearly. The usual practice is to concentrate on model development and computation rather than on statistical aspects. In general, nonlinear regression should be applied only to problems in which there is a well-defined, clear association between the two variables; therefore, a test of hypothesis on the significance of the fit would be somewhat ludicrous. In addition, the generalization of the theory for the associate confidence intervals for nonlinear coefficients is not well developed.

Example

Application. Data were collected on the cooling of water in the atmosphere as a function of time.

Sample data

Time x	Temperature y
0	92.0
1	85.5
2	79.5
3	74.5
5	67.0
7	60.5
10	53.5
15	45.0
20	39.5

Model. MATLAB can be used to find the best fit of the data to the formula $y = a + be^{cx}$: $a = 33.54$, $b = 57.89$, $c = 0.11$. The value of χ^2 is 1.83. Using an alternative form, $y = a + b/(c + x)$, gives $a = 9.872$, $b = 925.7$, $c = 11.27$, and $\chi = 0.19$. Since this model had a smaller value of χ^2 it might be the chosen one, but it is only a fit of the specified data and may not be generalized beyond that. Both forms give equivalent plots.

ERROR ANALYSIS OF EXPERIMENTS

Consider the problem of assessing the accuracy of a series of measurements. If measurements are for independent, identically distributed observations, then the errors are independent and uncorrelated. Then $\bar{y}$, the experimentally determined mean, varies about $E(y)$, the true mean, with variance σ^2/n , where n is the number of observations in $\bar{y}$. Thus, if one measures something several times today, and each day, and the measurements have the same distribution, then the variance of the means decreases with the number of samples in each day's measurement, n . Of course, other factors (weather, weekends) may make the observations on different days *not* distributed identically.

Consider next the problem of estimating the error in a variable that cannot be measured directly but must be calculated based on results of other measurements. Suppose the computed value Y is a linear combination of the measured variables $\{y_i\}$, $Y = \alpha_1 y_1 + \alpha_2 y_2 + \dots$. Let the random variables $y_1, y_2, \dots$ have means $E(y_1), E(y_2), \dots$ and variances $\sigma^2(y_1), \sigma^2(y_2), \dots$. The variable Y has mean

$$E(Y) = \alpha_1 E(y_1) + \alpha_2 E(y_2) + \dots$$

and variance (Cropley, 1978)

$$\sigma^2(Y) = \sum_{i=1}^n \alpha_i^2 \sigma^2(y_i) + 2 \sum_{i=1}^n \sum_{j=i+1}^n \alpha_i \alpha_j \text{Cov}(y_i, y_j)$$

If the variables are uncorrelated and have the same variance, then

$$\sigma^2(Y) = \left(\sum_{i=1}^n \alpha_i \right)^2 \sigma^2$$

Next suppose the model relating Y to $\{y_i\}$ is nonlinear, but the errors are small and independent of one another. Then a change in Y is related to changes in y_i by

$$dY = \frac{\partial Y}{\partial y_1} dy_1 + \frac{\partial Y}{\partial y_2} dy_2 + \dots$$

If the changes are indeed small, then the partial derivatives are constant among all the samples. Then the expected value of the change, $E(dY)$, is zero. The variances are given by the following equation (Baird, 1995; Box et al., 2005):

$$\sigma^2(dY) = \sum_{i=1}^N \left(\frac{\partial Y}{\partial y_i} \right)^2 \sigma_i^2$$

Thus, the variance of the desired quantity Y can be found. This gives an independent estimate of the errors in measuring the quantity Y from the errors in measuring each variable it depends upon.

Example Suppose one wants to measure the thermal conductivity of a solid (k). To do this, one needs to measure the heat flux (q), the thickness of the sample (d), and the temperature difference across the sample (ΔT). Each measurement has some error. The heat flux (q) may be the rate of electrical heat input ($\dot{Q}$) divided by the area (A), and both quantities are measured to some tolerance. The thickness of the sample is measured with some accuracy, and the temperatures are probably measured with a thermocouple to some accuracy. These measurements are combined, however, to obtain the thermal conductivity, and it is desired to know the error in the thermal conductivity. The formula is

$$k = -\frac{d}{A \Delta T} \dot{Q}$$

The variance in the thermal conductivity is then

$$\sigma_k^2 = \left(\frac{k}{d} \right)^2 \sigma_d^2 + \left(\frac{k}{\dot{Q}} \right)^2 \sigma_{\dot{Q}}^2 + \left(\frac{k}{A} \right)^2 \sigma_A^2 + \left(\frac{k}{\Delta T} \right)^2 \sigma_{\Delta T}^2$$

FACTORIAL DESIGN OF EXPERIMENTS AND ANALYSIS OF VARIANCE

Statistically designed experiments consider, of course, the effect of primary variables, but they also consider the effect of extraneous variables and the interactions between variables, and they include a measure of the random error. Primary variables are those whose effect you wish to determine. These variables can be quantitative or qualitative. The quantitative variables are ones you may fit to a model in order to determine the model parameters (see the section "Least Squares"). Qualitative variables are ones you wish to know the effect of, but you do not try to quantify that effect other than to assign possible errors or magnitudes. Qualitative variables can be further subdivided into Type I variables, whose effect you wish to determine directly, and Type II variables, which contribute to the performance variability and whose effect you wish to average out. For example, if you are studying the effect of several catalysts on yield in a chemical reactor, each different type of catalyst would be a Type I variable because you would like to know the effect of each. However, each time the catalyst is prepared, the results are slightly different due to random variations; thus, you may have several batches of what purports to be the same catalyst. The variability between batches is a Type II variable. Since the ultimate use will require using different batches, you would like to know the overall effect including that variation, since knowing precisely the results from one batch of one catalyst might not be representative of the results obtained from all batches of the same catalyst. A randomized block design, incomplete block design, or Latin square design (Box et al., *ibid.*), for example, all keep the effect of experimental error in the blocked variables from influencing the effect of the primary variables. Other uncontrolled variables are accounted for by introducing randomization in parts of the experimental design. To study all variables and their interaction requires a factorial design, involving all possible combinations of each variable, or a fractional factorial design, involving only a selected set. Statistical techniques are then used to determine which are the important variables, what are the important interactions, and what the error is in estimating these effects. The discussion here is only a brief overview of the excellent book by Box et al. (2005).

Suppose we have two methods of preparing some product and we wish to see which treatment is best. When there are only two treatments, then the sampling analysis discussed in the section "Two-Population Test of Hypothesis for Means" can be used to deduce if the means of the two treatments differ significantly. When there are more treatments, the analysis is more detailed. Suppose the experimental results are arranged

as shown in the table: several measurements for each treatment. The goal is to see if the treatments differ significantly from each other; that is, whether their means are different when the samples have the same variance. The hypothesis is that the treatments are all the same, and the null hypothesis is that they are different. The statistical validity of the hypothesis is determined by an analysis of variance.

Estimating the Effect of Four Treatments

	Treatment			
	1	2	3	4
	—	—	—	—
	—	—	—	—
	—	—	—	—
	—	—	—	—
Treatment average	—	—	—	—
Grand average	—	—	—	—

The data for $k = 4$ treatments is arranged in the table. For each treatment, there are n_t experiments and the outcome of the i th experiment with treatment t is called y_{ti} . Compute the treatment average

$$\bar{y}_t = \frac{\sum_{i=1}^{n_t} y_{ti}}{n_t}$$

Also compute the grand average

$$\bar{y} = \frac{\sum_{t=1}^k n_t \bar{y}_t}{N}, \quad N = \sum_{t=1}^k n_t$$

Next compute the sum of squares of deviations from the average within the t th treatment

$$S_t = \sum_{i=1}^{n_t} (y_{ti} - \bar{y}_t)^2$$

Since each treatment has n_t experiments, the number of degrees of freedom is $n_t - 1$. Then the sample variances are

$$s_t^2 = \frac{S_t}{n_t - 1}$$

The within-treatment sum of squares is

$$S_R = \sum_{t=1}^k S_t$$

and the within-treatment sample variance is

$$s_R^2 = \frac{S_R}{N - k}$$

Now, if there is no difference between treatments, a second estimate of σ^2 could be obtained by calculating the variation of the treatment averages about the grand average. Thus compute the between-treatment mean square

$$s_T^2 = \frac{S_T}{k - 1}, \quad S_T = \sum_{t=1}^k n_t (\bar{y}_t - \bar{y})^2$$

Basically the test for whether the hypothesis is true or not hinges on a comparison of the within-treatment estimate s_R^2 (with $v_R = N - k$ degrees of freedom) with the between-treatment estimate s_T^2 (with $v_T = k - 1$ degrees of freedom). The test is made based on the F distribution for v_R and v_T degrees of freedom (Table 3-8).

Next consider the case that uses randomized blocking to eliminate the effect of some variable whose effect is of no interest, such as the batch-to-batch variation of the catalysts in the chemical reactor example. Suppose there are k treatments and n experiments in each treatment. The results from nk experiments can be arranged as shown in the block design table; within each block, the various treatments are applied in a random order. Compute the block average, the treatment average, as well as the grand average as before.

Block Design with Four Treatments and Five Blocks

Treatment	1	2	3	4	Block average
Block 1	—	—	—	—	—
Block 2	—	—	—	—	—
Block 3	—	—	—	—	—
Block 4	—	—	—	—	—
Block 5	—	—	—	—	—

The following quantities are needed for the analysis of variance table.

Name	Formula	dof
average	$S_A = nk\bar{y}^2$	1
blocks	$S_B = k \sum_{i=1}^n (\bar{y}_i - \bar{y})^2$	$n - 1$
treatments	$S_T = n \sum_{t=1}^k (\bar{y}_t - \bar{y})^2$	$k - 1$
residuals	$S_R = \sum_{t=1}^k \sum_{i=1}^n (y_{ti} - \bar{y}_t - \bar{y}_i + \bar{y})^2$	$(n - 1)(k - 1)$
total	$S = \sum_{t=1}^k \sum_{i=1}^n y_{ti}^2$	$N = nk$

The key test is again a statistical one, based on the value of

$$\frac{s_T^2}{s_R^2}, \quad s_T^2 = \frac{S_T}{k - 1}, \quad s_R^2 = \frac{S_R}{(n - 1)(k - 1)}$$

and the F distribution for v_R and v_T degrees of freedom (Table 3-8). The assumption behind the analysis is that the variations are linear. See Box et al. (2005). There are ways to test this assumption as well as transformations to make if it is not true. Box et al. also give an excellent example of how the observations are broken down into a grand average, a block deviation, a treatment deviation, and a residual. For two-way factorial design in which the second variable is a real one rather than one you would like to block out, see Box et al.

To measure the effects of variables on a single outcome a factorial design is appropriate. In a two-level factorial design, each variable is considered at two levels only, a high and low value, often designated as a + and -. The two-level factorial design is useful for indicating trends, showing interactions, and it is also the basis for a fractional factorial design. As an example, consider a 2^3 factorial design with 3 variables and 2 levels for each. The experiments are indicated in the factorial design table.

Two-Level Factorial Design with Three Variables

Run	Variable		
	1	2	3
1	—	—	—
2	+	—	—
3	—	+	—
4	+	+	—
5	—	—	+
6	+	—	+
7	—	+	+
8	+	+	+

The main effects are calculated by calculating the difference between results from all high values of a variable and all low values of a variable; the result is divided by the number of experiments at each level. For example, for the first variable:

$$\text{Effect of variable 1} = \frac{[(y_2 + y_4 + y_6 + y_8) - (y_1 + y_3 + y_5 + y_7)]}{4}$$

Note that all observations are being used to supply information on each of the main effects and each effect is determined with the precision of a fourfold replicated difference. The advantage of a one-at-a-time experiment is the gain in precision if the variables are additive and the measure of nonadditivity if it occurs (Box et al., 2005).

Interaction effects between variables 1 and 2 are obtained by calculating the difference between the results obtained with the high and low value of 1 at the low value of 2 compared with the results obtained

with the high and low value 1 at the high value of 2. The 12-interaction is

$$12\text{-interaction} = \frac{[(y_4 - y_3 + y_8 - y_7) - (y_2 - y_1 + y_6 - y_5)]}{2}$$

The key step is to determine the errors associated with the effect of each variable and each interaction so that the significance can be determined. Thus, standard errors need to be assigned. This can be done by repeating the experiments, but it can also be done by using higher-order interactions (such as 123 interactions in a 2^4 factorial design). These are assumed negligible in their effect on the mean but can be used to estimate the standard error. Then, calculated

effects that are large compared with the standard error are considered important, while those that are small compared with the standard error are considered to be due to random variations and are unimportant.

In a fractional factorial design one does only part of the possible experiments. When there are k variables, a factorial design requires 2^k experiments. When k is large, the number of experiments can be large; for $k = 5$, $2^5 = 32$. For a k this large, Box et al. (2005) do a fractional factorial design. In the fractional factorial design with $k = 5$, only 16 experiments are done. Cropley (1978) gives an example of how to combine heuristics and statistical arguments in application to kinetics mechanisms in chemical engineering.

DIMENSIONAL ANALYSIS

Dimensional analysis allows the engineer to reduce the number of variables that must be considered to model experiments or correlate data. Consider a simple example in which two variables F_1 and F_2 have the units of force and two additional variables L_1 and L_2 have the units of length. Rather than having to deduce the relation of one variable on the other three, $F_1 = \text{fn}(F_2, L_1, L_2)$, dimensional analysis can be used to show that the relation must be of the form $F_1/F_2 = \text{fn}(L_1/L_2)$. Thus considerable experimentation is saved. Historically, dimensional analysis can be done using the Rayleigh method or the Buckingham pi method. This brief discussion is equivalent to the Buckingham pi method but uses concepts from linear algebra; see Amundson, N. R., *Mathematical Methods in Chemical Engineering*, Prentice-Hall, Englewood Cliffs, N.J. (1966), p. 54, for further information.

The general problem is posed as finding the minimum number of variables necessary to define the relationship between n variables. Let $\{Q_i\}$ represent a set of fundamental units, like length, time, force, and so on. Let $[P_i]$ represent the dimensions of a physical quantity P_i ; there are n physical quantities. Then form the matrix α_{ij}

	$[P_1]$	$[P_2]$	...	$[P_n]$
Q_1	α_{11}	α_{12}	...	α_{1n}
Q_2	α_{21}	α_{22}	...	α_{2n}
...				
Q_m	α_{m1}	α_{m2}	...	α_{mn}

in which the entries are the number of times each fundamental unit appears in the dimensions $[P_i]$. The dimensions can then be expressed as follows.

$$[P_i] = Q_1^{\alpha_{i1}} Q_2^{\alpha_{i2}} \cdots Q_m^{\alpha_{im}}$$

Let m be the rank of the α matrix. Then $p = n - m$ is the number of dimensionless groups that can be formed. One can choose m variables $\{P_i\}$ to be the basis and express the other p variables in terms of them, giving p dimensionless quantities.

Example: Buckingham Pi Method—Heat-Transfer Film Coefficient It is desired to determine a complete set of dimensionless groups with which to correlate experimental data on the film coefficient of heat transfer between the walls of a straight conduit with circular cross section and a fluid flowing in that conduit. The variables and the dimensional constant believed to be involved and their dimensions in the engineering system are given below:

- Film coefficient $= h = (F/L\theta T)$
- Conduit internal diameter $= D = (L)$
- Fluid linear velocity $= V = (L/\theta)$
- Fluid density $= \rho = (M/L^3)$
- Fluid absolute viscosity $= \mu = (M/L\theta)$
- Fluid thermal conductivity $= k = (F/\theta T)$
- Fluid specific heat $= c_p = (FL/MT)$
- Dimensional constant $= g_c = (ML/F\theta^2)$

The matrix α in this case is as follows.

	$[P_i]$							
	h	D	V	ρ	μ	k	C_p	g_c
F	1	0	0	0	0	1	1	-1
M	0	0	0	1	1	0	-1	1
L	-1	1	1	-3	-1	0	1	1
θ	-1	0	-1	0	-1	-1	0	-2
T	-1	0	0	0	0	-1	-1	0

Here $m \leq 5$, $n = 8$, $p \geq 3$. Choose D , V , μ , k , and g_c as the primary variables. By examining the 5×5 matrix associated with those variables, we can see that its determinant is not zero, so the rank of the matrix is $m = 5$; thus, $p = 3$. These variables are thus a possible basis set. The dimensions of the other three variables h , ρ , and C_p must be defined in terms of the primary variables. This can be done by inspection, although linear algebra can be used, too.

$$[h] = D^{-1}k^{-1}; \text{ thus } \frac{h}{D^{-1}k} = \frac{hD}{k} \text{ is a dimensionless group}$$

$$[\rho] = \mu^{-1}V^{-1}D^{-1}; \text{ thus } \frac{\rho}{\mu^{-1}V^{-1}D^{-1}} = \frac{\rho VD}{\mu} \text{ is a dimensionless group}$$

$$[C_p] = k^{-1}\mu^{-1}; \text{ thus } \frac{C_p}{k^{-1}\mu^{-1}} = \frac{C_p \mu}{k} \text{ is a dimensionless group}$$

Thus, the dimensionless groups are

$$\frac{[P_i]}{Q_1^{\alpha_{i1}} Q_2^{\alpha_{i2}} \cdots Q_m^{\alpha_{im}}}; \frac{hD}{k}, \frac{\rho VD}{\mu}, \frac{C_p \mu}{k}$$

The dimensionless group hD/k is called the Nusselt number, N_{Nu} , and the group $C_p \mu/k$ is the Prandtl number, N_{Pr} . The group $DV\rho/\mu$ is the familiar Reynolds number, N_{Re} , encountered in fluid-friction problems. These three dimensionless groups are frequently used in heat-transfer-film-coefficient correlations. Functionally, their relation may be expressed as

$$\Phi(N_{Nu}, N_{Pr}, N_{Re}) = 0 \quad (3-121)$$

or as

$$N_{Nu} = \Phi_1(N_{Pr}, N_{Re})$$

TABLE 3-9 Dimensionless Groups in the Engineering System of Dimensions

Biot number	N_{Bi}	hL/k
Condensation number	N_{Co}	$(h/k)(\mu^2/\rho^2 g)^{1/3}$
Number used in condensation of vapors	N_{Cv}	$L^3 \rho^2 g / \mu k \Delta t$
Euler number	N_{Eu}	$g_c(-dp)/\rho V^2$
Fourier number	N_{Fo}	$k\theta/\rho c L^2$
Froude number	N_{Fr}	V^2/Lg
Graetz number	N_{Gz}	wc/kL
Grashof number	N_{Gr}	$L^3 \rho^2 \beta g \Delta t / \mu^2$
Mach number	N_{Ma}	V/V_a
Nusselt number	N_{Nu}	hD/k
Peclet number	N_{Pe}	$DV\rho c/k$
Prandtl number	N_{Pr}	$c\mu/k$
Reynolds number	N_{Re}	$DV\rho/\mu$
Schmidt number	N_{Sc}	$\mu/\rho D_v$
Stanton number	N_{St}	$h/cV\rho$
Weber number	N_{We}	$LV^2\rho/\sigma g_c$

It has been found that these dimensionless groups may be correlated well by an equation of the type

$$hD/k = K(c_p \mu/k)^a (DV\rho/\mu)^b$$

in which K , a , and b are experimentally determined dimensionless constants. However, any other type of algebraic expression or perhaps simply a graphical relation among these three groups that accurately fits the experimental data would be an equally valid manner of expressing Eq. (3-121).

Naturally, other dimensionless groups might have been obtained in the example by employing a different set of five repeating quantities

PROCESS SIMULATION

REFERENCES: Dimian, A., *Chem. Eng. Prog.* **90**: 58-66 (Sept. 1994); Kister, H. Z., "Can We Believe the Simulation Results?" *Chem. Eng. Prog.*, pp. 52-58 (Oct. 2002); Krieger, J. H., *Chem. Eng. News* **73**: 50-61 (Mar. 27, 1995); Mah, R. S. H., *Chemical Process Structure and Information Flows*, Butterworths (1990); Seader, J. D., *Computer Modeling of Chemical Processes*, AIChE Monograph Series no. 15 (1985); Seider, W. D., J. D. Seader, and D. R. Lewin, *Product and Process Design Principles: Synthesis, Analysis, and Evaluation*, 2d ed., Wiley, New York (2004).

CLASSIFICATION

Process simulation refers to the activity in which mathematical systems of chemical processes and refineries are modeled with equations, usually on the computer. The usual distinction must be made between steady-state models and transient models, following the ideas presented in the introduction to this section. In a chemical process, of course, the process is nearly always in a transient mode, at some level of precision, but when the time-dependent fluctuations are below some value, a steady-state model can be formulated. This subsection presents briefly the ideas behind steady-state process simulation (also called flowsheeting), which are embodied in commercial codes. The transient simulations are important for designing the start-up of plants and are especially useful for the operation of chemical plants.

THERMODYNAMICS

The most important aspect of the simulation is that the thermodynamic data of the chemicals be modeled correctly. It is necessary to decide what equation of state to use for the vapor phase (ideal gas, Redlich-Kwong-Soave, Peng-Robinson, etc.) and what model to use for liquid activity coefficients [ideal solutions, solubility parameters, Wilson equation, nonrandom two liquid (NRTL), UNIFAC, etc.]. See Sec. 4, "Thermodynamics." It is necessary to consider mixtures of chemicals, and the interaction parameters must be predictable. The best case is to determine them from data, and the next-best case is to use correlations based on the molecular weight, structure, and normal boiling point. To validate the model, the computer results of vapor-liquid equilibria could be checked against experimental data to ensure their validity before the data are used in more complicated computer calculations.

PROCESS MODULES OR BLOCKS

At the first level of detail, it is not necessary to know the internal parameters for all the units, since what is desired is just the overall performance. For example, in a heat exchanger design, it suffices to know the heat duty, the total area, and the temperatures of the output streams; the details such as the percentage baffle cut, tube layout, or baffle spacing can be specified later when the details of the proposed plant are better defined. It is important to realize the level of detail modeled by a commercial computer program. For example, a chemical reactor could be modeled as an equilibrium reactor, in which the input stream is brought to a new temperature and pressure and the

that would not form a dimensionless group among themselves. Some of these groups may be found among those presented in Table 3-9. Such a complete set of three dimensionless groups might consist of Stanton, Reynolds, and Prandtl numbers or of Stanton, Peclet, and Prandtl numbers. Also, such a complete set different from that obtained in the preceding example will result from a multiplication of appropriate powers of the Nusselt, Prandtl, and Reynolds numbers. For such a set to be complete, however, it must satisfy the condition that each of the three dimensionless groups be independent of the other two.

output stream is in chemical equilibrium at those new conditions. Or, it may suffice to simply specify the conversion, and the computer program will calculate the outlet compositions. In these cases, the model equations are algebraic ones, and you do not learn the volume of the reactor. A more complicated reactor might be a stirred tank reactor, and then you would have to specify kinetic information so that the simulation can be made, and one output would be either the volume of the reactor or the conversion possible in a volume you specify. Such models are also composed of sets of algebraic equations. A plug flow reactor is modeled as a set of ordinary differential equations as initial-value problems, and the computer program must use numerical methods to integrate them. See "Numerical Solution of Ordinary Differential Equations as Initial Value Problems." Kinetic information must be specified, and one learns the conversion possible in a given reactor volume, or, in some cases, the volume reactor that will achieve a given conversion. The simulation engineer determines what a reactor of a given volume will do for the specified kinetics and reactor volume. The design engineer, though, wants to achieve a certain result and wants to know the volume necessary. Simulation packages are best suited for the simulation engineer, and the design engineer must vary specifications to achieve the desired output.

Distillation simulations can be based on shortcut methods, using correlations based on experience, but more rigorous methods involve solving for the vapor-liquid equilibrium on each tray. The shortcut method uses relatively simple equations, and the rigorous method requires solution of huge sets of nonlinear equations. The computation time of the latter is significant, but the rigorous method may be necessary when the chemicals you wish to distill are not well represented in the correlations. Then the designer must specify the number of trays and determine the separation that is possible. This, of course, is not what he or she wants: the number of trays needed to achieve a specified objective. Thus, again, some adjustment of parameters is necessary in a design situation.

Absorption columns can be modeled in a plate-to-plate fashion (even if it is a packed bed) or as a packed bed. The former model is a set of nonlinear algebraic equations, and the latter model is an ordinary differential equation. Since streams enter at both ends, the differential equation is a two-point boundary value problem, and numerical methods are used (see "Numerical Solution of Ordinary Differential Equations as Initial-Value Problems").

If one wants to model a process unit that has significant flow variation, and possibly some concentration distributions as well, one can consider using computational fluid dynamics (CFD) to do so. These calculations are very time-consuming, though, so that they are often left until the mechanical design of the unit. The exception would occur when the flow variation and concentration distribution had a significant effect on the output of the unit so that mass and energy balances couldn't be made without it.

The process units are described in greater detail in other sections of the Handbook. In each case, parameters of the unit are specified (size, temperature, pressure, area, and so forth). In addition, in a computer simulation, the computer program must be able to take any input to the unit and calculate the output for those parameters. Since the entire calculation is done iteratively, there is no assurance that the

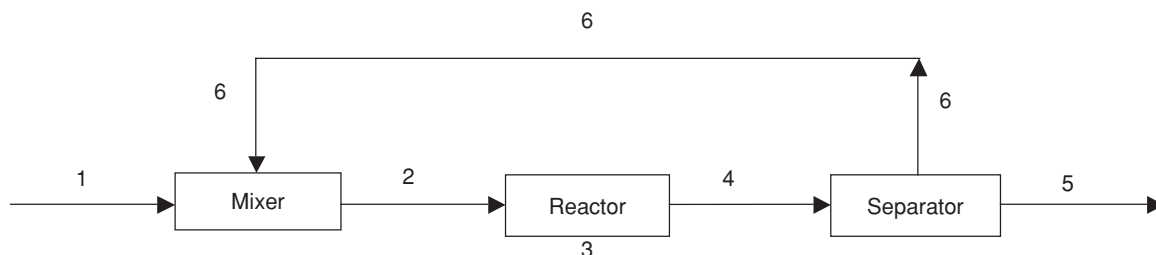


FIG. 3-67 Prototype flowsheet.

input stream is a “reasonable” one, so that the computer codes must be written to give some sort of output even when the input stream is unreasonable. This difficulty makes the iterative process even more complicated.

PROCESS TOPOLOGY

A chemical process usually consists of a series of units, such as distillation towers, reactors, and so forth (see Fig. 3-67). If the feed to the process is known and the operating parameters of the units are specified by the user, then one can begin with the first unit, take the process input, calculate the unit output, carry that output to the input of the next unit, and continue the process. However, if the process involves a recycle stream, as nearly all chemical processes do, then when the calculation is begun, it is discovered that the recycle stream is unknown. This situation leads to an iterative process: the flow rates, temperature, and pressure of the unknown recycle stream are guessed, and the calculations proceed as before. When one reaches the end of the process, where the recycle stream is formed to return to the first unit, it is necessary to check to see if the recycle stream is the same as assumed. If not, an iterative procedure must be used to cause convergence. Possible techniques are described in “Numerical Solutions of Nonlinear Equations in One Variable” and “Numerical Solution of Simultaneous Equations.” The direct method (or successive substitution method) just involves calculating around the process over and over. The Wegstein method accelerates convergence for a single variable, and Broyden’s method does the same for multiple variables. The Newton method can be used provided there is some way to calculate the derivatives (possibly by using a numerical derivative). Optimization methods can also be used (see “Optimization” in this section). In the description given here, the recycle stream is called the tear stream: this is the stream that must be guessed to begin the calculation. When there are multiple recycle streams, convergence is even more difficult, since more guesses are necessary, and what happens in one recycle stream may cause difficulties for the guesses in other recycle streams. See Seader (1985) and Mah (1990).

It is sometimes desired to control some stream by varying an operating parameter. For example, in a reaction/separation system, if there is an impurity that must be purged, a common objective is to set the purge fraction so that the impurity concentration into the reactor is kept at some moderate value. Commercial packages contain procedures for

doing this using what are often called control blocks. However, this can also make the solution more difficult to find.

An alternative method of solving the equations is to solve them as simultaneous equations. In that case, one can specify the design variables and the desired specifications and let the computer figure out the process parameters that will achieve those objectives. It is possible to overspecify the system or to give impossible conditions. However, the biggest drawback to this method of simulation is that large sets (tens of thousands) of nonlinear algebraic equations must be solved simultaneously. As computers become faster, this is less of an impediment, provided efficient software is available.

Dynamic simulations are also possible, and these require solving differential equations, sometimes with algebraic constraints. If some parts of the process change extremely quickly when there is a disturbance, that part of the process may be modeled in the steady state for the disturbance at any instant. Such situations are called stiff, and the methods for them are discussed in “Numerical Solution of Ordinary Differential Equations as Initial-Value Problems.” It must be realized, though, that a dynamic calculation can also be time-consuming, and sometimes the allowable units are lumped-parameter models that are simplifications of the equations used for the steady-state analysis. Thus, as always, the assumptions need to be examined critically before accepting the computer results.

COMMERCIAL PACKAGES

Computer programs are provided by many companies, and the models range from empirical models to deterministic models. For example, if one wanted to know the pressure drop in a piping network, one would normally use a correlation for friction factor as a function of Reynolds number to calculate the pressure drop in each segment. A sophisticated turbulence model of fluid flow is not needed in that case. As computers become faster, however, more and more models are deterministic. Since the commercial codes have been used by many customers, the data in them have been verified, but possibly not for the case you want to solve. Thus, you must test the thermodynamics correlations carefully. In 2005, there were a number of computer codes, but the company names change constantly. Here are a few of them for process simulation: Aspen Tech (Aspen Plus), Chemstations (CHEMCAD), Honeywell (UniSim Design), ProSim (ProSimPlus), and SimSci-Esseor (Pro II). The CAPE-OPEN project is working to make details as transferable as possible.

SECTION 4

Thermodynamics

PERRY'S CHEMICAL ENGINEERS' HANDBOOK

8TH EDITION



HENDRICK C. VAN NESS
MICHAEL M. ABBOTT

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Thermodynamics

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*Dr. Abbott died on May 31, 2006. This, his final contribution to the literature of chemical engineering, is deeply appreciated, as are his earlier contributions to the handbook.

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Nomenclature and Units

Correlation- and application-specific symbols are not shown.

Symbol	Definition	SI units	U.S. Customary System units	Symbol	Definition	SI units	U.S. Customary System units
A	Molar (or unit-mass) Helmholtz energy	J/mol [J/kg]	Btu/lb mol [Btu/lbm]	P_i^{sat}	Saturation or vapor pressure of species i	kPa	psi
A	Cross-sectional area in flow	m ²	ft ²	Q	Heat	J	Btu
$\hat{a}_i$	Activity of species i in solution	Dimensionless	Dimensionless	q	Volumetric flow rate	m ³ /s	ft ³ /s
$\bar{a}_i$	Partial parameter, cubic equation of state			$\dot{Q}$	Rate of heat transfer	J/s	Btu/s
B	2d virial coefficient, density expansion	cm ³ /mol	cm ³ /mol	R	Universal gas constant	J/(mol·K)	Btu/(lb-mol·R)
$\bar{B}_i$	Partial molar second virial coefficient	cm ³ /mol	cm ³ /mol	S	Molar (or unit-mass) entropy	J/(mol·K)	Btu/(lb-mol·R)
$\hat{B}$	Reduced second virial coefficient					[J/(kg·K)]	[Btu/(lbm·R)]
C	3d virial coefficient, density expansion	cm ⁶ /mol ²	cm ⁶ /mol ²	$\dot{S}_G$	Rate of entropy generation, Eq. (4-151)	J/(K·s)	Btu/(R·s)
$\hat{C}$	Reduced third virial coefficient			T	Absolute temperature	K	R
D	4th virial coefficient, density expansion	cm ⁹ /mol ³	cm ⁹ /mol ³	T_c	Critical temperature	K	R
B'	2d virial coefficient, pressure expansion	kPa ⁻¹	kPa ⁻¹	U	Molar (or unit-mass) internal energy	J/mol [J/kg]	Btu/(lb-mol) [Btu/lbm]
C'	3d virial coefficient, pressure expansion	kPa ⁻²	kPa ⁻²	u	Fluid velocity	m/s	ft/s
D'	4th virial coefficient, pressure expansion	kPa ⁻³	kPa ⁻³	V	Molar (or unit-mass) volume	m ³ /mol [m ³ /kg]	ft ³ /(lb-mol) [ft ³ /lbm]
B_{ij}	Interaction 2d virial coefficient	cm ³ /mol	cm ³ /mol	W	Work	J	Btu
C_{ijk}	Interaction 3d virial coefficient	cm ⁶ /mol ²	cm ⁶ /mol ²	W_s	Shaft work for flow process	J	Btu
C_p	Heat capacity at constant pressure	J/(mol·K)	Btu/(lb-mol·R)	$\dot{W}_s$	Shaft power for flow process	J/s	Btu/s
C_v	Heat capacity at constant volume	J/(mol·K)	Btu/(lb-mol·R)	x_i	Mole fraction in general		
f_i	Fugacity of pure species i	kPa	psi	x_i	Mole fraction of species i in liquid phase		
f_i	Fugacity of species i in solution	kPa	psi	y_i	Mole fraction of species i in vapor phase		
$\bar{G}$	Molar (or unit-mass) Gibbs energy	J/mol [J/kg]	Btu/(lb-mol) [Btu/lbm]	Z	Compressibility factor	Dimensionless	Dimensionless
g	Acceleration of gravity	m/s ²	ft/s ²	z	Elevation above a datum level	m	ft
$\frac{g}{RT}$	$\equiv C_p^0/RT$	Dimensionless	Dimensionless	Superscripts			
H	Molar (or unit-mass) enthalpy	J/mol [J/kg]	Btu/(lb-mol) [Btu/lbm]	E	Denotes excess thermodynamic property		
K_i	Equilibrium K value, y_i/x_i	Dimensionless	Dimensionless	id	Denotes value for an ideal solution		
K_j	Equilibrium constant for chemical reaction j	Dimensionless	Dimensionless	ig	Denotes value for an ideal gas		
k_1	Henry's constant for solute species 1	kPa	psi	l	Denotes liquid phase		
M	Molar or unit-mass solution property (A, G, H, S, U, V)			lv	Denotes phase transition, liquid to vapor		
$\mathbf{M}$	Mach number	Dimensionless	Dimensionless	R	Denotes residual thermodynamic property		
M_i	Molar or unit-mass pure-species property ($A_i, G_i, H_i, S_i, U_i, V_i$)			t	Denotes total value of property		
$\bar{M}_i$	Partial property of species i in solution ($A_i, \bar{G}_i, \bar{H}_i, \bar{S}_i, \bar{U}_i, \bar{V}_i$)			v	Denotes vapor phase		
M^R	Residual thermodynamic property ($A^R, G^R, H^R, S^R, \bar{U}^R, V^R$)			∞	Denotes value at infinite dilution		
M^E	Excess thermodynamic property ($A^E, G^E, H^E, S^E, U^E, V^E$)			Subscripts			
$\bar{M}_i^E$	Partial molar excess thermodynamic property			c	Denotes value for the critical state		
ΔM	Property change of mixing ($\Delta A, \Delta G, \Delta H, \Delta S, \Delta U, \Delta V$)			cv	Denotes the control volume		
ΔM_j^0	Standard property change of reaction j ($\Delta G_j^0, \Delta H_j^0, \Delta C_p^0$)			fs	Denotes flowing streams		
m	Mass	kg	lbm	n	Denotes the normal boiling point		
$\dot{m}$	Mass flow rate	kg/s	lbm/s	r	Denotes a reduced value		
n	Number of moles			rev	Denotes a reversible process		
$\dot{n}$	Molar flow rate			Greek Letters			
n_i	Number of moles of species i			α, β	As superscripts, identify phases		
P	Absolute pressure	kPa	psi	β	Volume expansivity	K ⁻¹	°R ⁻¹
				ϵ_j	Reaction coordinate for reaction j	mol	lb-mol
				$\Gamma_i(T)$	Defined by Eq. (4-196)	J/mol	Btu/(lb-mol)
				γ	Heat capacity ratio C_p/C_v	Dimensionless	Dimensionless
				γ_i	Activity coefficient of species i in solution	Dimensionless	Dimensionless
				κ	Isothermal compressibility	kPa ⁻¹	psi ⁻¹
				μ_i	Chemical potential of species i	J/mol	Btu/(lb-mol)
				ν_{sj}	Stoichiometric number of species i in reaction j	Dimensionless	Dimensionless
				ρ	Molar density	mol/m ³	lb-mol/ft ³
				σ	As subscript, denotes a heat reservoir		
				Φ_i	Defined by Eq. (4-304)	Dimensionless	Dimensionless
				ϕ_i	Fugacity coefficient of pure species i	Dimensionless	Dimensionless
				$\hat{\phi}_i$	Fugacity coefficient of species i in solution	Dimensionless	Dimensionless
				ω	Acentric factor	Dimensionless	Dimensionless

GENERAL REFERENCES: Abbott, M. M., and H. C. Van Ness, *Schaum's Outline of Theory and Problems of Thermodynamics*, 2d ed., McGraw-Hill, New York, 1989. Poling, B. E., J. M. Prausnitz, and J. P. O'Connell, *The Properties of Gases and Liquids*, 5th ed., McGraw-Hill, New York, 2001. Prausnitz, J. M., R. N. Lichtenthaler, and E. G. de Azevedo, *Molecular Thermodynamics of Fluid-Phase Equilibria*, 3d ed., Prentice-Hall PTR, Upper Saddle River, N.J., 1999. Sandler, S. I., *Chemical and Engineering Thermodynamics*, 3d ed.,

Wiley, New York, 1999. Smith, J. M., H. C. Van Ness, and M. M. Abbott, *Introduction to Chemical Engineering Thermodynamics*, 7th ed., McGraw-Hill, New York, 2005. Tester, J. W., and M. Modell, *Thermodynamics and Its Applications*, 3d ed., Prentice-Hall PTR, Upper Saddle River, N.J., 1997. Van Ness, H. C., and M. M. Abbott, *Classical Thermodynamics of Nonelectrolyte Solutions: With Applications to Phase Equilibria*, McGraw-Hill, New York, 1982.

INTRODUCTION

Thermodynamics is the branch of science that lends substance to the principles of energy transformation in macroscopic systems. The general restrictions shown by experience to apply to all such transformations are known as the laws of thermodynamics. These laws are primitive; they cannot be derived from anything more basic.

The first law of thermodynamics states that energy is conserved, that although it can be altered in form and transferred from one place to another, the total quantity remains constant. Thus the first law of thermodynamics depends on the concept of energy, but conversely energy is an *essential* thermodynamic function because it allows the first law to be formulated. This coupling is characteristic of the primitive concepts of thermodynamics.

The words *system* and *surroundings* are similarly coupled. A system can be an object, a quantity of matter, or a region of space, selected for study and set apart (mentally) from everything else, which is called the surroundings. An envelope, imagined to enclose the system and to separate it from its surroundings, is called the *boundary* of the system.

Attributed to this boundary are special properties which may serve either to *isolate* the system from its surroundings or to provide for *interaction* in specific ways between the system and surroundings. An isolated system exchanges neither matter nor energy with its surroundings. If a system is not isolated, its boundaries may permit exchange of matter or energy or both with its surroundings. If the exchange of matter is allowed, the system is said to be *open*; if only energy and not matter may be exchanged, the system is *closed* (but not isolated), and its mass is constant.

When a system is isolated, it cannot be affected by its surroundings. Nevertheless, changes may occur within the system that are detectable with measuring instruments such as thermometers and pressure gauges. However, such changes cannot continue indefinitely, and the system must eventually reach a final static condition of *internal equilibrium*.

For a closed system which interacts with its surroundings, a final static condition may likewise be reached such that the system is not only internally at equilibrium but also in *external equilibrium* with its surroundings.

The concept of equilibrium is central in thermodynamics, for associated with the condition of internal equilibrium is the concept of *state*. A system has an identifiable, reproducible state when all its *properties*, such as temperature T , pressure P , and molar volume V , are fixed. The concepts of *state* and *property* are again coupled. One can equally well say that the properties of a system are fixed by its state. Although the properties T , P , and V may be detected with measuring instruments, the existence of the *primitive* thermodynamic properties (see postulates 1 and 3 following) is recognized much more indirectly. The number of properties for which values must be specified in order to fix the state of a system depends on the nature of the system, and is ultimately determined from experience.

When a system is displaced from an equilibrium state, it undergoes a *process*, a change of state, which continues until its properties attain new equilibrium values. During such a process, the system may be caused to interact with its surroundings so as to interchange energy in the forms of heat and work and so to produce in the system changes considered desirable for one reason or another. A process that proceeds so that the system is never displaced more than differentially from an equilibrium state is said to be *reversible*, because such a process can be reversed at any point by an infinitesimal change in external conditions, causing it to retrace the initial path in the opposite direction.

Thermodynamics finds its origin in experience and experiment, from which are formulated a few postulates that form the foundation of the subject. The first two deal with energy.

POSTULATE 1

*There exists a form of energy, known as **internal energy**, which for systems at internal equilibrium is an intrinsic property of the system, functionally related to the measurable coordinates that characterize the system.*

POSTULATE 2 (FIRST LAW OF THERMODYNAMICS)

*The **total energy** of any system and its surroundings is conserved.*

Internal energy is quite distinct from such external forms as the kinetic and potential energies of macroscopic bodies. Although it is a macroscopic property, characterized by the macroscopic coordinates T and P , internal energy finds its origin in the kinetic and potential energies of molecules and submolecular particles. In applications of the first law of thermodynamics, all forms of energy must be considered, including the internal energy. It is therefore clear that postulate 2 depends on postulate 1. For an isolated system the first law requires that its energy be constant. For a closed (but not isolated) system, the first law requires that energy changes of the system be exactly compensated by energy changes in the surroundings. For such systems energy is exchanged between a system and its surroundings in two forms: heat and work.

Heat is energy crossing the system boundary under the influence of a temperature difference or gradient. A quantity of heat Q represents an amount of energy in transit between a system and its surroundings, and is not a property of the system. The convention with respect to sign makes numerical values of Q positive when heat is added to the system and negative when heat leaves the system.

Work is again energy in transit between a system and its surroundings, but resulting from the displacement of an external force acting on the system. Like heat, a quantity of work W represents an amount of energy, and is not a property of the system. The sign convention, analogous to that for heat, makes numerical values of W positive when work is done on the system by the surroundings and negative when work is done on the surroundings by the system.

When applied to closed (constant-mass) systems in which only internal-energy changes occur, the first law of thermodynamics is expressed mathematically as

$$dU^t = dQ + dW \quad (4-1)$$

where U^t is the total internal energy of the system. Note that dQ and dW , differential *quantities* representing energy exchanges between the system and its surroundings, serve to account for the energy change of the surroundings. On the other hand, dU^t is directly the differential *change* in internal energy of the system. Integration of Eq. (4-1) gives for a finite process

$$\Delta U^t = Q + W \quad (4-2)$$

where ΔU^t is the finite change given by the difference between the final and initial values of U^t . The heat Q and work W are finite quantities of heat and work; they are not properties of the system or functions of the thermodynamic coordinates that characterize the system.

POSTULATE 3

There exists a property called **entropy**, which for systems at internal equilibrium is an intrinsic property of the system, functionally related to the measurable coordinates that characterize the system. For **reversible** processes, changes in this property may be calculated by the equation

$$dS' = \frac{dQ_{\text{rev}}}{T} \quad (4-3)$$

where S' is the total entropy of the system and T is the absolute temperature of the system.

POSTULATE 4 (SECOND LAW OF THERMODYNAMICS)

The entropy change of any system and its surroundings, **considered together**, resulting from any real process is positive, approaching zero when the process approaches reversibility.

In the same way that the first law of thermodynamics cannot be formulated without the prior recognition of internal energy as a property, so also the second law can have no complete and quantitative expression without a prior assertion of the existence of entropy as a property.

The second law requires that the entropy of an isolated system either increase or, in the limit where the system has reached an equilibrium state, remain constant. For a closed (but not isolated) system it requires that any entropy decrease in either the system or its surroundings be more than compensated by an entropy increase in the other part, or that in the limit where the process is reversible, the total entropy of the system plus its surroundings be constant.

The fundamental thermodynamic properties that arise in connection with the first and second laws of thermodynamics are internal energy and entropy. These properties together with the two laws for which they are essential apply to all types of systems. However, different types of systems are characterized by different sets of measurable coordinates or variables. The type of system most commonly encountered in chemical technology is one for which the primary characteristic variables are temperature T , pressure P , molar volume V , and composition, not all of which are necessarily independent. Such systems are usually made up of fluids (liquid or gas) and are called **PVT** systems.

For closed systems of this kind the work of a **reversible** process may always be calculated from

$$dW_{\text{rev}} = -PdV^t \quad (4-4)$$

where P is the absolute pressure and V^t is the total volume of the system. This equation follows directly from the definition of mechanical work.

POSTULATE 5

The macroscopic properties of homogeneous PVT systems at internal equilibrium can be expressed as functions of temperature, pressure, and composition only.

This postulate imposes an idealization, and is the basis for all subsequent property relations for PVT systems. The PVT system serves as a satisfactory model in an enormous number of practical applications. In accepting this model one assumes that the effects of fields (e.g., electric, magnetic, or gravitational) are negligible and that surface and viscous shear effects are unimportant.

Temperature, pressure, and composition are thermodynamic coordinates representing conditions imposed upon or exhibited by the system, and the functional dependence of the thermodynamic properties on these conditions is determined by experiment. This is quite direct for molar or specific volume V , which can be measured, and leads immediately to the conclusion that there exists an *equation of state* relating molar volume to temperature, pressure, and composition for any particular homogeneous PVT system. The equation of state is a primary tool in applications of thermodynamics.

Postulate 5 affirms that the other molar or specific thermodynamic properties of PVT systems, such as internal energy U and entropy S , are also functions of temperature, pressure, and composition. These molar or unit-mass properties, represented by the plain symbols V , U , and S , are independent of system size and are called *intensive*. Temperature, pressure, and the composition variables, such as mole fraction, are also intensive. Total-system properties (V^t , U^t , S^t) do depend on system size and are *extensive*. For a system containing n mol of fluid, $M^t = nM$, where M is a molar property.

Applications of the thermodynamic postulates necessarily involve the abstract quantities of internal energy and entropy. The solution of any problem in applied thermodynamics is therefore found through these quantities.

VARIABLES, DEFINITIONS, AND RELATIONSHIPS

Consider a single-phase closed system in which there are no chemical reactions. Under these restrictions the composition is fixed. If such a system undergoes a differential, reversible process, then by Eq. (4-1)

$$dU^t = dQ_{\text{rev}} + dW_{\text{rev}}$$

Substitution for dQ_{rev} and dW_{rev} by Eqs. (4-3) and (4-4) gives

$$dU^t = T dS' - P dV^t$$

Although derived for a **reversible** process, this equation relates properties only and is valid for *any* change between equilibrium states in a closed system. It is equally well written as

$$d(nU) = T d(nS) - P d(nV) \quad (4-5)$$

where n is the number of moles of fluid in the system and is constant for the special case of a closed, nonreacting system. Note that

$$n \equiv n_1 + n_2 + n_3 + \cdots = \sum_i n_i$$

where i is an index identifying the chemical species present. When U , S , and V represent *specific* (unit-mass) properties, n is replaced by m .

Equation (4-5) shows that for a single-phase, nonreacting, closed system, $nU = u(nS, nV)$.

Then
$$d(nU) = \left[\frac{\partial(nU)}{\partial(nS)} \right]_{nV,n} d(nS) + \left[\frac{\partial(nU)}{\partial(nV)} \right]_{nS,n} d(nV)$$

where subscript n indicates that all mole numbers n_i (and hence n) are held constant. Comparison with Eq. (4-5) shows that

$$\left[\frac{\partial(nU)}{\partial(nS)} \right]_{nV,n} = T \quad \text{and} \quad \left[\frac{\partial(nU)}{\partial(nV)} \right]_{nS,n} = -P$$

For an **open** single-phase system, we assume that $nU = \mathcal{U}(nS, nV, n_1, n_2, n_3, \dots)$. In consequence,

$$d(nU) = \left[\frac{\partial(nU)}{\partial(nS)} \right]_{nV,n} d(nS) + \left[\frac{\partial(nU)}{\partial(nV)} \right]_{nS,n} d(nV) + \sum_i \left[\frac{\partial(nU)}{\partial n_i} \right]_{nS,nV,n_j} dn_i$$

where the summation is over all species present in the system and subscript n_j indicates that all mole numbers are held constant except the i th. Define

$$\mu_i \equiv \left[\frac{\partial(nU)}{\partial n_i} \right]_{nS,nV,n_j}$$

The expressions for T and $-P$ of the preceding paragraph and the definition of μ_i allow replacement of the partial differential coefficients in the preceding equation by T , $-P$, and μ_i . The result is Eq. (4-6) of Table 4-1, where important equations of this section are collected. Equation (4-6) is the **fundamental property relation** for single-phase PVT systems, from which all other equations connecting properties of

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TABLE 4-1 Mathematical Structure of Thermodynamic Property Relations

Primary thermodynamic functions	Fundamental property relations	For homogeneous systems of constant composition	Maxwell equations
$U = TS - PV + \sum_i x_i \mu_i$ (4-7)	$d(nU) = T d(nS) - P d(nV) + \sum_i \mu_i dn_i$ (4-6)	$dU = T dS - P dV$ (4-14)	$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V$ (4-18)
$H \equiv U + PV$ (4-8)	$d(nH) = T d(nS) + nV dP + \sum_i \mu_i dn_i$ (4-11)	$dH = T dS + V dP$ (4-15)	$\left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P$ (4-19)
$A \equiv U - TS$ (4-9)	$d(nA) = -nS dT - P d(nV) + \sum_i \mu_i dn_i$ (4-12)	$dA = -S dT - P dV$ (4-16)	$\left(\frac{\partial P}{\partial T}\right)_V = \left(\frac{\partial S}{\partial V}\right)_T$ (4-20)
$G \equiv H - TS$ (4-10)	$d(nG) = -nS dT + nV dP + \sum_i \mu_i dn_i$ (4-13)	$dG = -S dT + V dP$ (4-17)	$\left(\frac{\partial V}{\partial T}\right)_P = -\left(\frac{\partial S}{\partial P}\right)_T$ (4-21)
$U, H, \text{ and } S \text{ as functions of } T \text{ and } P \text{ or } T \text{ and } V$		Partial derivatives	Total derivatives
$dH = \left(\frac{\partial H}{\partial T}\right)_P dT + \left(\frac{\partial H}{\partial P}\right)_T dP$ (4-22)		$\left(\frac{\partial H}{\partial T}\right)_P = T \left(\frac{\partial S}{\partial T}\right)_P = C_P$ (4-28)	$dH = C_P dT + \left[V - T \left(\frac{\partial V}{\partial T}\right)_P\right] dP$ (4-32)
$dS = \left(\frac{\partial S}{\partial T}\right)_P dT + \left(\frac{\partial S}{\partial P}\right)_T dP$ (4-23)		$\left(\frac{\partial H}{\partial P}\right)_T = T \left(\frac{\partial S}{\partial P}\right)_T + V = V - T \left(\frac{\partial V}{\partial T}\right)_P$ (4-29)	$dS = \frac{C_P}{T} dT - \left(\frac{\partial V}{\partial T}\right)_P dP$ (4-33)
$dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV$ (4-24)		$\left(\frac{\partial U}{\partial T}\right)_V = T \left(\frac{\partial S}{\partial T}\right)_V = C_V$ (4-30)	$dU = C_V dT + \left[T \left(\frac{\partial P}{\partial T}\right)_V - P\right] dV$ (4-34)
$dS = \left(\frac{\partial S}{\partial T}\right)_V dT + \left(\frac{\partial S}{\partial V}\right)_T dV$ (4-25)		$\left(\frac{\partial U}{\partial V}\right)_T = T \left(\frac{\partial S}{\partial V}\right)_T - P = T \left(\frac{\partial P}{\partial T}\right)_V - P$ (4-31)	$dS = \frac{C_V}{T} dT + \left(\frac{\partial P}{\partial T}\right)_V dV$ (4-35)

U = Internal energy; H = enthalpy; A = Helmholtz energy; G = Gibbs energy.

such systems are derived. The quantity μ_i is called the *chemical potential* of species i , and it plays a vital role in the thermodynamics of phase and chemical equilibria.

Additional property relations follow directly from Eq. (4-6). Because $n_i = x_i n$, where x_i is the mole fraction of species i , this equation may be rewritten as

$$d(nU) - T d(nS) + P d(nV) - \sum_i \mu_i d(x_i n) = 0$$

Expansion of the differentials and collection of like terms yield

$$\left[dU - T dS + P dV - \sum_i \mu_i dx_i \right] n + \left[U - TS + PV - \sum_i x_i \mu_i \right] dn = 0$$

Because n and dn are independent and arbitrary, the terms in brackets must separately be zero. This provides two useful equations:

$$dU = T dS - P dV + \sum_i \mu_i dx_i \quad U = TS - PV + \sum_i x_i \mu_i$$

The first is similar to Eq. (4-6). However, Eq. (4-6) applies to a system of n mol where n may vary. Here, however, n is unity and invariant. It is therefore subject to the constraints $\sum_i x_i = 1$ and $\sum_i dx_i = 0$. Mole fractions are not independent of one another, whereas the mole numbers in Eq. (4-6) are.

The second of the preceding equations dictates the possible combinations of terms that may be defined as additional primary functions. Those in common use are shown in Table 4-1 as Eqs. (4-7) through (4-10). Additional thermodynamic properties are related to these and arise by arbitrary definition.

Multiplication of Eq. (4-8) of Table 4-1 by n and differentiation yield the general expression

$$d(nH) = d(nU) + P d(nV) + nV dP$$

Substitution for $d(nU)$ by Eq. (4-6) reduces this result to Eq. (4-11). The total differentials of nA and nG are obtained similarly and are expressed by Eqs. (4-12) and (4-13). These equations and Eq. (4-6) are equivalent forms of the fundamental property relation, and appear under that heading in Table 4-1. Each expresses a total property— nU , nH , nA , and nG —as a function of a particular set of independent

variables, called the *canonical variables* for the property. The choice of which equation to use in a particular application is dictated by convenience. However, the Gibbs energy G is special, because of its relation to the canonical variables T , P , and $\{n_i\}$, the variables of primary interest in chemical processing. Another set of equations results from the substitutions $n = 1$ and $n_i = x_i$. The resulting equations are of course less general than their parents. Moreover, because the mole fractions are not independent, mathematical operations requiring their independence are invalid.

CONSTANT-COMPOSITION SYSTEMS

For 1 mol of a homogeneous fluid of constant composition, Eqs. (4-6) and (4-11) through (4-13) simplify to Eqs. (4-14) through (4-17) of Table 4-1. Because these equations are *exact differential expressions*, application of the reciprocity relation for such expressions produces the common Maxwell relations as described in the subsection “Multi-variable Calculus Applied to Thermodynamics” in Sec. 3. These are Eqs. (4-18) through (4-21) of Table 4-1, in which the partial derivatives are taken with composition held constant.

$U, H, \text{ and } S$ as Functions of T and P or T and V At constant composition, molar thermodynamic properties can be considered functions of T and P (postulate 5). Alternatively, because V is related to T and P through an equation of state, V can serve rather than P as the second independent variable. The useful equations for the total differentials of U , H , and S that result are given in Table 4-1 by Eqs. (4-22) through (4-25). The obvious next step is substitution for the partial differential coefficients in favor of measurable quantities. This purpose is served by definition of two *heat capacities*, one at constant pressure and the other at constant volume:

$$C_P \equiv \left(\frac{\partial H}{\partial T}\right)_P \quad (4-26)$$

$$C_V \equiv \left(\frac{\partial U}{\partial T}\right)_V \quad (4-27)$$

Both are properties of the material and functions of temperature, pressure, and composition.

Equation (4-15) of Table 4-1 may be divided by dT and restricted to constant P , yielding $(\partial H/\partial T)_P$ as given by the first equality of Eq. (4-28). Division of Eq. (4-15) by dP and restriction to constant T yield $(\partial H/\partial P)_T$ as given by the first equality of Eq. (4-29). Equation (4-28) is completed by Eq. (4-26), and Eq. (4-29) is completed by Eq. (4-21). Similarly, equations for $(\partial U/\partial T)_V$ and $(\partial U/\partial V)_T$ derive from Eq. (4-14), and these with Eqs. (4-27) and (4-20) yield Eqs. (4-30) and (4-31) of Table 4-1.

Equations (4-22), (4-26), and (4-29) combine to yield Eq. (4-32); Eqs. (4-23), (4-28), and (4-21) to yield Eq. (4-33); Eqs. (4-24), (4-27), and (4-31) to yield Eq. (4-34); and Eqs. (4-25), (4-30), and (4-20) to yield Eq. (4-35).

Equations (4-32) and (4-33) are general expressions for the enthalpy and entropy of homogeneous fluids *at constant composition* as functions of T and P . Equations (4-34) and (4-35) are general expressions for the internal energy and entropy of homogeneous fluids *at constant composition* as functions of temperature and molar volume. The coefficients of dT , dP , and dV are all composed of measurable quantities.

The Ideal Gas Model An ideal gas is a model gas comprising imaginary molecules of zero volume that do not interact. Its PVT behavior is represented by the simplest of equations of state $PV^{ig} = RT$, where R is a universal constant, values of which are given in Table 1-9. The following partial derivatives, all taken at constant composition, are obtained from this equation:

$$\left(\frac{\partial P}{\partial T}\right)_V = \frac{R}{V^{ig}} = \frac{P}{T} \quad \left(\frac{\partial V^{ig}}{\partial T}\right)_P = \frac{R}{P} = \frac{V^{ig}}{T} \quad \left(\frac{\partial P}{\partial V}\right)_T = -\frac{P}{V^{ig}}$$

The first two of these relations when substituted appropriately into Eqs. (4-29) and (4-31) of Table 4-1 lead to very simple expressions for ideal gases:

$$\left(\frac{\partial U^{ig}}{\partial V}\right)_T = \left(\frac{\partial H^{ig}}{\partial P}\right)_T = 0 \quad \left(\frac{\partial S^{ig}}{\partial P}\right)_T = -\frac{R}{P} \quad \left(\frac{\partial S^{ig}}{\partial V}\right)_T = \frac{R}{V^{ig}}$$

Moreover, Eqs. (4-32) through (4-35) become

$$dH^{ig} = C_P^{ig} dT \quad dS^{ig} = \frac{C_P^{ig}}{T} dT - \frac{R}{P} dP$$

$$dU^{ig} = C_V^{ig} dT \quad dS^{ig} = \frac{C_V^{ig}}{T} dT + \frac{R}{V^{ig}} dV$$

In these equations V^{ig} , U^{ig} , C_V^{ig} , H^{ig} , C_P^{ig} , and S^{ig} are ideal gas state values—the values that a PVT system would have were the ideal gas equation the true equation of state. They apply equally to pure species and to constant-composition mixtures, and they show that U^{ig} , C_V^{ig} , H^{ig} , and C_P^{ig} are functions of temperature only, independent of P and V . The entropy, however, is a function of both T and P or of both T and V . Regardless of composition, the ideal gas volume is given by $V^{ig} = RT/P$, and it provides the basis for comparison with true molar volumes through the *compressibility factor* Z . By definition,

$$Z \equiv \frac{V}{V^{ig}} = \frac{V}{RT/P} = \frac{PV}{RT} \quad (4-36)$$

The ideal gas state properties of mixtures are directly related to the ideal gas state properties of the constituent pure species. For those properties that are independent of P — U^{ig} , H^{ig} , C_V^{ig} , and C_P^{ig} —the mixture property is the sum of the properties of the pure constituent species, each weighted by its mole fraction:

$$M^{ig} = \sum_i y_i M_i^{ig} \quad (4-37)$$

where M^{ig} can represent any of the properties listed. For the entropy, which is a function of both T and P , an additional term is required to account for the difference in partial pressure of a species between its pure state and its state in a mixture:

$$S^{ig} = \sum_i y_i S_i^{ig} - R \sum_i y_i \ln y_i \quad (4-38)$$

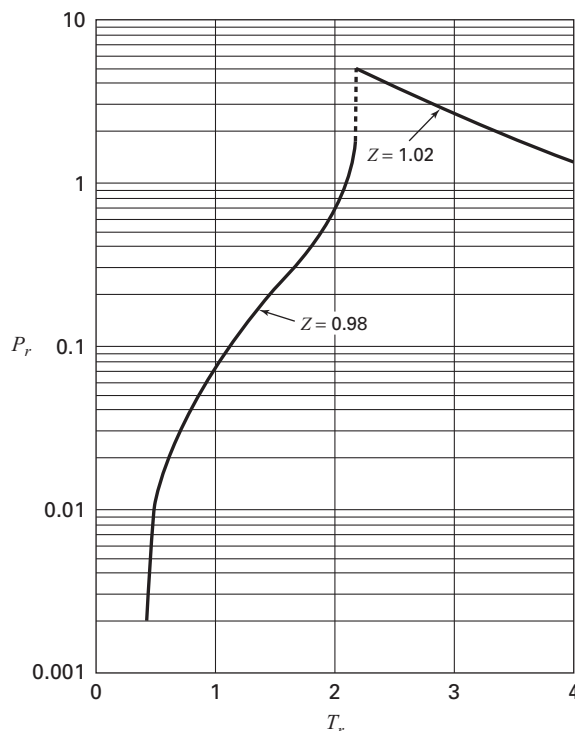


FIG. 4-1 Region where Z lies between 0.98 and 1.02, and the ideal-gas equation is a reasonable approximation. [Smith, Van Ness, and Abbott, *Introduction to Chemical Engineering Thermodynamics*, 7th ed., p. 104, McGraw-Hill, New York (2005).]

For the Gibbs energy, $G^{ig} = H^{ig} - TS^{ig}$; whence by Eqs. (4-37) and (4-38):

$$G^{ig} = \sum_i y_i G_i^{ig} + RT \sum_i y_i \ln y_i \quad (4-39)$$

The ideal gas model may serve as a reasonable approximation to reality under conditions indicated by Fig. 4-1.

Residual Properties The differences between true and ideal gas state properties are defined as residual properties M^R :

$$M^R \equiv M - M^{ig} \quad (4-40)$$

where M is the molar value of an extensive thermodynamic property of a fluid in its actual state and M^{ig} is its corresponding ideal gas state value at the same T , P , and composition. Residual properties depend on interactions *between* molecules and not on characteristics of individual molecules. Because the ideal gas state presumes the absence of molecular interactions, residual properties reflect deviations from ideality. The most commonly used residual properties are as follows:

$$\text{Residual volume } V^R \equiv V - V^{ig} \quad \text{Residual enthalpy } H^R \equiv H - H^{ig}$$

$$\text{Residual entropy } S^R \equiv S - S^{ig} \quad \text{Residual Gibbs energy } G^R \equiv G - G^{ig}$$

Useful relations connecting these residual properties derive from Eq. (4-17), an alternative form of which follows from the mathematical identity:

$$d\left(\frac{G}{RT}\right) \equiv \frac{1}{RT} dG - \frac{G}{RT^2} dT$$

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Substitution for dG by Eq. (4-17) and for G by Eq. (4-10) gives, after algebraic reduction,

$$d\left(\frac{G}{RT}\right) = \frac{V}{RT} dP - \frac{H}{RT^2} dT \quad (4-41)$$

This equation may be written for the special case of an ideal gas and subtracted from Eq. (4-41) itself, yielding

$$d\left(\frac{G^R}{RT}\right) = \frac{V^R}{RT} dP - \frac{H^R}{RT^2} dT \quad (4-42)$$

As a consequence,
$$\frac{V^R}{RT} = \left[\frac{\partial(G^R/RT)}{\partial P} \right]_T \quad (4-43)$$

and
$$\frac{H^R}{RT} = -T \left[\frac{\partial(G^R/RT)}{\partial T} \right]_P \quad (4-44)$$

Equation (4-43) provides a direct link to PVT correlations through the compressibility factor Z as given by Eq. (4-36). Thus, with $V = ZRT/P$,

$$V^R \equiv V - V^{ig} = \frac{ZRT}{P} - \frac{RT}{P} = \frac{RT}{P} (Z - 1)$$

This equation in combination with a rearrangement of Eq. (4-43) yields

$$d\left(\frac{G^R}{RT}\right) = \frac{V^R}{RT} dP = (Z - 1) \frac{dP}{P} \quad (\text{constant } T)$$

Integration from $P = 0$ to arbitrary pressure P gives

$$\frac{G^R}{RT} = \int_0^P (Z - 1) \frac{dP}{P} \quad (\text{constant } T) \quad (4-45)$$

Smith, Van Ness, and Abbott [*Introduction to Chemical Engineering Thermodynamics*, 7th ed., pp. 210–211, McGraw-Hill, New York (2005)] show that it is permissible here to set the lower limit of integration $(G^R/RT)_{P=0}$ equal to zero. Note also that the integrand $(Z - 1)/P$ remains finite as $P \rightarrow 0$. Differentiation of Eq. (4-45) with respect to T in accord with Eq. (4-44) gives

$$\frac{H^R}{RT} = -T \int_0^P \left(\frac{\partial Z}{\partial T} \right)_P \frac{dP}{P} \quad (\text{constant } T) \quad (4-46)$$

Because $G = H - TS$ and $G^{ig} = H^{ig} - TS^{ig}$, then by difference, $G^R = H^R - TS^R$, and

$$\frac{S^R}{R} = \frac{H^R}{RT} - \frac{G^R}{RT} \quad (4-47)$$

Equations (4-45) through (4-47) provide the basis for calculation of residual properties from PVT correlations. They may be put into *generalized* form by substitution of the relationships

$$\begin{aligned} P &= P_c P_r & T &= T_c T_r \\ dP &= P_c dP_r & dT &= T_c dT_r \end{aligned}$$

The resulting equations are

$$\frac{G^R}{RT} = \int_0^{P_r} (Z - 1) \frac{dP_r}{P_r} \quad (4-48)$$

$$\frac{H^R}{RT_c} = -T_r^2 \int_0^{P_r} \left(\frac{\partial Z}{\partial T_r} \right)_{P_r} \frac{dP_r}{P_r} \quad (4-49)$$

The terms on the right sides of these equations depend only on the upper limit P_r of the integrals and on the reduced temperature at which they are evaluated. Thus, values of G^R/RT and H^R/RT_c may be determined once and for all at any reduced temperature and pressure from generalized compressibility factor data.

PROPERTY CALCULATIONS FOR GASES AND VAPORS

The most satisfactory calculation procedure for the thermodynamic properties of gases and vapors is based on ideal gas state heat capacities and residual properties. Of primary interest are the enthalpy and entropy; these are given by rearrangement of the residual property definitions:

$$H = H^{ig} + H^R \quad \text{and} \quad S = S^{ig} + S^R$$

These are simple sums of the ideal gas and residual properties, evaluated separately.

EVALUATION OF ENTHALPY AND ENTROPY IN THE IDEAL GAS STATE

For the ideal gas state at constant composition:

$$dH^{ig} = C_P^{ig} dT \quad \text{and} \quad dS^{ig} = C_P^{ig} \frac{dT}{T} - R \frac{dP}{P}$$

Integration from an initial ideal gas *reference state* at conditions T_0 and P_0 to the ideal gas state at T and P gives

$$\begin{aligned} H^{ig} &= H_0^{ig} + \int_{T_0}^T C_P^{ig} dT \\ S^{ig} &= S_0^{ig} + \int_{T_0}^T C_P^{ig} \frac{dT}{T} - R \ln \frac{P}{P_0} \end{aligned}$$

Substitution into the equations for H and S yields

$$H = H_0^{ig} + \int_{T_0}^T C_P^{ig} dT + H^R \quad (4-50)$$

$$S = S_0^{ig} + \int_{T_0}^T C_P^{ig} \frac{dT}{T} - R \ln \frac{P}{P_0} + S^R \quad (4-51)$$

The reference state at T_0 and P_0 is arbitrarily selected, and the values assigned to H_0^{ig} and S_0^{ig} are also arbitrary. In practice, only *changes* in H and S are of interest, and fixed reference state values ultimately cancel in their calculation.

The ideal gas state heat capacity C_P^{ig} is a function of T but not of P . For a mixture the heat capacity is simply the molar average $\sum y_i C_{P,i}^{ig}$. Empirical equations relating C_P^{ig} to T are available for many pure gases; a common form is

$$\frac{C_P^{ig}}{R} = A + BT + CT^2 + DT^{-2} \quad (4-52)$$

where A , B , C , and D are constants characteristic of the particular gas, and either C or D is zero. The ratio C_P^{ig}/R is dimensionless; thus the units of C_P^{ig} are those of R . Data for ideal gas state heat capacities are given for many substances in Table 2-155.

Evaluation of the integrals $\int C_P^{ig} dT$ and $\int (C_P^{ig}/T) dT$ is accomplished by substitution for C_P^{ig} , followed by integration. For temperature limits of T_0 and T and with $\tau \equiv T/T_0$, the equations that follow from Eq. (4-52) are

$$\int_{T_0}^T \frac{C_P^{ig}}{R} dT = AT_0(\tau - 1) + \frac{B}{2} T_0^2(\tau^2 - 1) + \frac{C}{3} T_0^3(\tau^3 - 1) + \frac{D}{T_0} \left(\frac{\tau - 1}{\tau} \right) \quad (4-53)$$

$$\int_{T_0}^T \frac{C_P^{ig}}{RT} dT = A \ln \tau + \left[BT_0 + \left(CT_0^2 + \frac{D}{\tau^2 T_0^2} \right) \left(\frac{\tau + 1}{2} \right) \right] (\tau - 1) \quad (4-54)$$

Equations (4-50) and (4-51) may sometimes be advantageously expressed in alternative form through use of mean heat capacities:

$$H = H_0^{ig} + \langle C_P^{ig} \rangle_H (T - T_0) + H^R \quad (4-55)$$

$$S = S_0^{ig} + \langle C_P^{ig} \rangle_S \ln \frac{T}{T_0} - R \ln \frac{P}{P_0} + S^R \quad (4-56)$$

where $\langle C_P^{ig} \rangle_H$ and $\langle C_P^{ig} \rangle_S$ are mean heat capacities specific, respectively, for enthalpy and entropy calculations. They are given by the following equations:

$$\frac{\langle C_P^{ig} \rangle_H}{R} = A + \frac{B}{2} T_0 (\tau + 1) + \frac{C}{3} T_0^2 (\tau^2 + \tau + 1) + \frac{D}{\tau T_0^2} \quad (4-57)$$

$$\frac{\langle C_P^{ig} \rangle_S}{R} = A + \left[B T_0 + \left(C T_0^2 + \frac{D}{\tau^2 T_0^2} \right) \left(\frac{\tau + 1}{2} \right) \right] \frac{\tau - 1}{\ln \tau} \quad (4-58)$$

RESIDUAL ENTHALPY AND ENTROPY FROM PVT CORRELATIONS

The residual properties of gases and vapors depend on their PVT behavior. This is often expressed through correlations for the compressibility factor Z , defined by Eq. (4-36). Analytical expressions for Z as functions of T and P or T and V are known as *equations of state*. They may also be reformulated to give P as a function of T and V or V as a function of T and P .

Virial Equations of State The virial equation in *density* is an infinite series expansion of the compressibility factor Z in powers of molar density ρ (or reciprocal molar volume V^{-1}) about the real gas state at zero density (zero pressure):

$$Z = 1 + B\rho + C\rho^2 + D\rho^3 + \dots \quad (4-59)$$

The density series virial coefficients B , C , D , $\dots$ depend on temperature and composition only. In practice, truncation is to two or three terms. The composition dependencies of B and C are given by the exact *mixing rules*

$$B = \sum_i \sum_j y_i y_j B_{ij} \quad (4-60)$$

$$C = \sum_i \sum_j \sum_k y_i y_j y_k C_{ijk} \quad (4-61)$$

where y_i , y_j , and y_k are mole fractions for a gas mixture and i , j , and k identify species.

The coefficient B_{ij} characterizes a bimolecular interaction between molecules i and j , and therefore $B_{ij} = B_{ji}$. Two kinds of second virial coefficient arise: B_{ii} and B_{jj} , wherein the subscripts are the same ($i=j$), and B_{ij} , wherein they are different ($i \neq j$). The first is a virial coefficient for a pure species; the second is a mixture property, called a *cross coefficient*. Similarly for the third virial coefficients: C_{iii} , C_{jjj} , and C_{kkk} are for the pure species, and $C_{ijj} = C_{jji} = C_{jij} \dots$ are cross coefficients.

Although the virial equation itself is easily rationalized on empirical grounds, the mixing rules of Eqs. (4-60) and (4-61) follow rigorously from the methods of statistical mechanics. The temperature derivatives of B and C are given exactly by

$$\frac{dB}{dT} = \sum_i \sum_j y_i y_j \frac{dB_{ij}}{dT} \quad (4-62)$$

$$\frac{dC}{dT} = \sum_i \sum_j \sum_k y_i y_j y_k \frac{dC_{ijk}}{dT} \quad (4-63)$$

An alternative form of the virial equation expresses Z as an expansion in powers of pressure about the real gas state at zero pressure (zero density):

$$Z = 1 + B'P + C'P^2 + D'P^3 + \dots \quad (4-64)$$

Equation (4-64) is the virial equation in *pressure*, and B' , C' , D' , $\dots$ are the pressure series virial coefficients. Again, truncation is to two

or three terms, with B' and C' depending on temperature and composition only. Moreover, the two sets of coefficients are related:

$$B' = B/RT \quad (4-65)$$

$$C' = (C - B^2)/(RT)^2 \quad (4-66)$$

Values can often be found for B , but not so often for C . Generalized correlations for both B and C are given by Meng, Duan, and Li [*Fluid Phase Equilibria* **226**: 109–120 (2004)].

For pressures up to several bars, the two-term expansion in pressure, with B' given by Eq. (4-65), is usually preferred:

$$Z = 1 + B'P = 1 + BP/RT \quad (4-67)$$

For supercritical temperatures, it is satisfactory to ever higher pressures as the temperature increases. For pressures above the range where Eq. (4-67) is useful, but below the critical pressure, the virial expansion in density truncated to three terms is usually suitable:

$$Z = 1 + B\rho + C\rho^2 \quad (4-68)$$

Equations for residual enthalpy and entropy may be developed from each of these expressions. Consider first Eq. (4-67), which is explicit in volume. Equations (4-45) and (4-46) are therefore applicable. Direct substitution for Z in Eq. (4-45) gives

$$\frac{G^R}{RT} = \frac{BP}{RT} \quad (4-69)$$

Differentiation of Eq. (4-67) yields

$$\left(\frac{\partial Z}{\partial T} \right)_P = \left(\frac{dB}{dT} - \frac{B}{T} \right) \frac{P}{RT}$$

By Eq. (4-46),

$$\frac{H^R}{RT} = \frac{P}{R} \left(\frac{B}{T} - \frac{dB}{dT} \right) \quad (4-70)$$

and by Eq. (4-47),

$$\frac{S^R}{R} = - \frac{P}{R} \frac{dB}{dT} \quad (4-71)$$

An extensive set of three-parameter corresponding-states correlations has been developed by Pitzer and coworkers [Pitzer, *Thermodynamics*, 3d ed., App. 3, McGraw-Hill, New York (1995)]. Particularly useful is the one for the second virial coefficient. The basic equation is

$$\frac{BP_c}{RT_c} = B^0 + \omega B^1 \quad (4-72)$$

with the *acentric factor* defined by Eq. (2-17). For pure chemical species B^0 and B^1 are functions of reduced temperature only. Substitution for B in Eq. (4-67) by this expression gives

$$Z = 1 + (B^0 + \omega B^1) \frac{P_r}{T_r} \quad (4-73)$$

By differentiation,

$$\left(\frac{\partial Z}{\partial T_r} \right)_{P_r} = P_r \left(\frac{dB^0/dT_r}{T_r} - \frac{B^0}{T_r^2} \right) + \omega P_r \left(\frac{dB^1/dT_r}{T_r} - \frac{B^1}{T_r^2} \right)$$

Upon substitution of these equations into Eqs. (4-48) and (4-49), integration yields

$$\frac{G^R}{RT} = (B^0 + \omega B^1) \frac{P_r}{T_r} \quad (4-74)$$

$$\frac{H^R}{RT_c} = P_r \left[B^0 - T_r \frac{dB^0}{dT_r} + \omega \left(B^1 - T_r \frac{dB^1}{dT_r} \right) \right] \quad (4-75)$$

The residual entropy follows from Eq. (4-47):

$$\frac{S^R}{R} = - P_r \left(\frac{dB^0}{dT_r} + \omega \frac{dB^1}{dT_r} \right) \quad (4-76)$$

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In these equations, B^0 and B^1 and their derivatives are well represented by Abbott's correlations [Smith and Van Ness, *Introduction to Chemical Engineering Thermodynamics*, 3d ed., p. 87, McGraw-Hill, New York (1975)]:

$$B^0 = 0.083 - \frac{0.422}{T_r^{1.6}} \quad (4-77)$$

$$B^1 = 0.139 - \frac{0.172}{T_r^{4.2}} \quad (4-78)$$

$$\frac{dB^0}{dT_r} = \frac{0.675}{T_r^{2.6}} \quad (4-79)$$

$$\frac{dB^1}{dT_r} = \frac{0.722}{T_r^{5.2}} \quad (4-80)$$

Although limited to pressures where the two-term virial equation in pressure has approximate validity, these correlations are applicable for most chemical processing conditions. As with all generalized correlations, they are least accurate for polar and associating molecules.

Although developed for pure materials, these correlations can be extended to gas or vapor mixtures. Basic to this extension are the mixing rules for the second virial coefficient and its temperature derivative as given by Eqs. (4-60) and (4-62). Values for the cross coefficients B_{ij} , with $i \neq j$, and their derivatives are provided by Eq. (4-72) written in extended form:

$$B_{ij} = \frac{RT_{cij}}{P_{cij}} (B^0 + \omega_{ij} B^1) \quad (4-81)$$

where B^0 , B^1 , dB^0/dT_r , and dB^1/dT_r are the same functions of T_r as given by Eqs. (4-77) through (4-80). Differentiation produces

$$\begin{aligned} \frac{dB_{ij}}{dT} &= \frac{RT_{cij}}{P_{cij}} \left(\frac{dB^0}{dT} + \omega_{ij} \frac{dB^1}{dT} \right) \\ \frac{dB_{ij}}{dT} &= \frac{R}{P_{cij}} \left(\frac{dB^0}{dT_{rij}} + \omega_{ij} \frac{dB^1}{dT_{rij}} \right) \end{aligned} \quad (4-82)$$

where $T_{rij} = T/T_{cij}$. The following combining rules for ω_{ij} , T_{cij} , and P_{cij} are given by Prausnitz, Lichtenthaler, and de Azevedo [*Molecular Thermodynamics of Fluid-Phase Equilibria*, 2d ed., pp. 132 and 162, Prentice-Hall, Englewood Cliffs, N.J. (1986)]:

$$\omega_{ij} = \frac{\omega_i + \omega_j}{2} \quad (4-83)$$

$$T_{cij} = (T_{ci} T_{cj})^{1/2} (1 - k_{ij}) \quad (4-84)$$

$$P_{cij} = \frac{Z_{cij} RT_{cij}}{V_{cij}} \quad (4-85)$$

$$\text{with} \quad Z_{cij} = \frac{Z_{ci} + Z_{cj}}{2} \quad (4-86)$$

$$\text{and} \quad V_{cij} = \left(\frac{V_{ci}^{1/3} + V_{cj}^{1/3}}{2} \right)^3 \quad (4-87)$$

In Eq. (4-84), k_{ij} is an empirical interaction parameter specific to an $i-j$ molecular pair. When $i=j$ and for chemically similar species, $k_{ij}=0$. Otherwise, it is a small (usually) positive number evaluated from minimal PVT data or, absence data, set equal to zero.

When $i=j$, all equations reduce to the appropriate values for a pure species. When $i \neq j$, these equations define a set of interaction parameters without physical significance. For a mixture, values of B_{ij} and dB_{ij}/dT from Eqs. (4-81) and (4-82) are substituted into Eqs. (4-60) and (4-62) to provide values of the mixture second virial coefficient

B and its temperature derivative. Values of H^R and S^R are then given by Eqs. (4-70) and (4-71).

A primary virtue of Abbott's correlations for second virial coefficients is simplicity. More complex correlations of somewhat wider applicability include those by Tsonopoulos [*AIChE J.* **20**: 263–272 (1974); *ibid.*, **21**: 827–829 (1975); *ibid.*, **24**: 1112–1115 (1978); *Adv. in Chemistry Series* 182, pp. 143–162 (1979)] and Hayden and O'Connell [*Ind. Eng. Chem. Proc. Des. Dev.* **14**: 209–216 (1975)]. For aqueous systems see Bishop and O'Connell [*Ind. Eng. Chem. Res.*, **44**: 630–633 (2005)].

Because Eq. (4-68) is explicit in P , it is incompatible with Eqs. (4-45) and (4-46), and they must be transformed to make V (or molar density ρ) the variable of integration. The resulting equations are given by Smith, Van Ness, and Abbott [*Introduction to Chemical Engineering Thermodynamics*, 7th ed., pp. 216–217, McGraw-Hill, New York (2005)]:

$$\frac{G^R}{RT} = Z - 1 - \ln Z + \int_0^P (Z - 1) \frac{d\rho}{\rho} \quad (4-88)$$

$$\frac{H^R}{RT} = Z - 1 - T \int_0^P \left(\frac{\partial Z}{\partial T} \right)_\rho \frac{d\rho}{\rho} \quad (4-89)$$

By differentiation of Eq. (4-68),

$$\left(\frac{\partial Z}{\partial T} \right)_\rho = \frac{dB}{dT} \rho + \frac{dC}{dT} \rho^2$$

Substituting in Eqs. (4-88) and (4-89) for Z by Eq. (4-68) and in Eq. (4-89) for the derivative yields, upon integration and reduction,

$$\frac{G^R}{RT} = 2B\rho + \frac{3}{2}C\rho^2 - \ln Z \quad (4-90)$$

$$\frac{H^R}{RT} = \left(B - T \frac{dB}{dT} \right) \rho + \left(C - \frac{T}{2} \frac{dC}{dT} \right) \rho^2 \quad (4-91)$$

The residual entropy is given by Eq. (4-47).

In a process calculation, T and P , rather than T and ρ (or T and V), are usually the favored independent variables. Applications of Eqs. (4-90) and (4-91) therefore require prior solution of Eq. (4-68) for Z or ρ . With $Z = P/\rho RT$, Eq. (4-68) may be written in two equivalent forms:

$$Z^3 - Z^2 - \left(\frac{BP}{RT} \right) Z - \frac{CP^2}{(RT)^2} = 0 \quad (4-92)$$

$$\rho^3 + \left(\frac{B}{C} \right) \rho^2 + \left(\frac{1}{C} \right) \rho - \frac{P}{CRT} = 0 \quad (4-93)$$

In the event that three real roots obtain for these equations, only the largest Z (smallest ρ), appropriate for the vapor phase, has physical significance, because the virial equations are suitable only for vapors and gases.

Data for third virial coefficients are often lacking, but generalized correlations are available. Equation (4-68) may be rewritten in reduced form as

$$Z = 1 + \hat{B} \frac{P_r}{T_r Z} + \hat{C} \left(\frac{P_r}{T_r Z} \right)^2 \quad (4-94)$$

where $\hat{B}$ is the reduced second virial coefficient given by Eq. (4-72). Thus by definition,

$$\hat{B} \equiv \frac{BP_c}{RT_c} = B^0 + \omega B^1 \quad (4-95)$$

The reduced third virial coefficient $\hat{C}$ is defined as

$$\hat{C} \equiv \frac{CP_c^2}{R^2 T_c^2} \quad (4-96)$$

A Pitzer-type correlation for $\hat{C}$ is then written as

$$\hat{C} = C^0 + \omega C^1 \quad (4-97)$$

Correlations for C^0 and C^1 with reduced temperature are

$$C^0 = 0.01407 + \frac{0.02432}{T_r} - \frac{0.00313}{T_r^{10.5}} \quad (4-98)$$

$$C^1 = -0.02676 + \frac{0.05539}{T_r^{2.7}} - \frac{0.00242}{T_r^{10.5}} \quad (4-99)$$

The first is given by, and the second is inspired by, Orbey and Vera [AIChE J. **29**: 107–113 (1983)].

Equation (4-94) is cubic in Z ; with T_r and P_r specified, solution for Z is by iteration. An initial guess of $Z = 1$ on the right side usually leads to rapid convergence.

Another class of equations, known as *extended virial equations*, was introduced by Benedict, Webb, and Rubin [J. Chem. Phys. **8**: 334–345 (1940); **10**: 747–758 (1942)]. This equation contains eight parameters, all functions of composition. It and its modifications, despite their complexity, find application in the petroleum and natural gas industries for light hydrocarbons and a few other commonly encountered gases [see Lee and Kesler, AIChE J., **21**: 510–527 (1975)].

Cubic Equations of State The modern development of cubic equations of state started in 1949 with publication of the Redlich-Kwong (RK) equation [Chem. Rev., **44**: 233–244 (1949)], and many others have since been proposed. An extensive review is given by Valderrama [Ind. Eng. Chem. Res. **42**: 1603–1618 (2003)]. Of the equations published more recently, the two most popular are the Soave-Redlich-Kwong (SRK) equation, a modification of the RK equation [Chem. Eng. Sci. **27**: 1197–1203 (1972)] and the Peng-Robinson (PR) equation [Ind. Eng. Chem. Fundam. **15**: 59–64 (1976)]. All are encompassed by a *generic* cubic equation of state, written as

$$P = \frac{RT}{V - b} - \frac{a(T)}{(V + \epsilon b)(V + \sigma b)} \quad (4-100)$$

For a specific form of this equation, ϵ and σ are pure numbers, the same for all substances, whereas parameters $a(T)$ and b are substance-dependent. Suitable estimates of the parameters in cubic equations of state are usually found from values for the critical constants T_c and P_c . The procedure is discussed by Smith, Van Ness, and Abbott [Introduction to Chemical Engineering Thermodynamics, 7th ed., pp. 93–94, McGraw-Hill, New York (2005)], and for Eq. (4-100) the appropriate equations are given as

$$a(T) = \Psi \frac{\alpha(T_r) R^2 T_c^2}{P_c} \quad (4-101)$$

$$b = \Omega \frac{RT_c}{P_c} \quad (4-102)$$

Function $\alpha(T_r)$ is an empirical expression, specific to a particular form of the equation of state. In these equations Ψ and Ω are pure numbers, independent of substance and determined for a particular equation of state from the values assigned to ϵ and σ .

As an equation cubic in V , Eq. (4-100) has three volume roots, of which two may be complex. Physically meaningful values of V are always real, positive, and greater than parameter b . When $T > T_c$, solution for V at any positive value of P yields only one real positive root. When $T = T_c$, this is also true, except at the critical pressure, where three roots exist, all equal to V_c . For $T < T_c$, only one real positive (liquidlike) root exists at high pressures, but for a range of lower pressures there are three. Here, the middle root is of no significance; the smallest root is a liquid or liquidlike volume, and the largest root is a vapor or vaporlike volume.

Equation (4-100) may be rearranged to facilitate its solution either for a vapor or vaporlike volume or for a liquid or liquidlike volume.

$$\text{Vapor:} \quad V = \frac{RT}{P} + b - \frac{a(T)}{P} \frac{V - b}{(V + \epsilon b)(V + \sigma b)} \quad (4-103a)$$

$$\text{Liquid:} \quad V = b + (V + \epsilon b)(V + \sigma b) \left[\frac{RT - bP - VP}{a(T)} \right] \quad (4-103b)$$

Solution for V is most convenient with the solve routine of a software package. An initial estimate for V in Eq. (4-103a) is the ideal gas value RT/P ; for Eq. (4-103b) it is $V = b$. In either case, iteration is initiated by substituting the estimate on the right side. The resulting value of V on the left is returned to the right side, and the process continues until the change in V is suitably small.

Equations for Z equivalent to Eqs. (4-103) are obtained by substituting $V = ZRT/P$.

$$\text{Vapor:} \quad Z = 1 + \beta - q\beta \frac{Z - \beta}{(Z + \epsilon\beta)(Z + \sigma\beta)} \quad (4-104a)$$

$$\text{Liquid:} \quad Z = \beta + (Z + \epsilon b)(Z + \sigma b) \left(\frac{1 + \beta - Z}{q\beta} \right) \quad (4-104b)$$

$$\text{where by definition} \quad \beta = \frac{bP}{RT} \quad (4-105)$$

$$\text{and} \quad q = \frac{a(T)}{bRT} \quad (4-106)$$

These dimensionless quantities provide simplification, and when combined with Eqs. (4-101) and (4-102), they yield

$$\beta = \Omega \frac{P_r}{T_r} \quad (4-107)$$

$$q = \frac{\Psi \alpha(T_r)}{\Omega T_r} \quad (4-108)$$

In Eq. (4-104a) the initial estimate is $Z = 1$; in Eq. (4-104b) it is $Z = \beta$. Iteration follows the same pattern as for Eqs. (4-103). The final value of Z yields the volume root through $V = ZRT/P$.

Equations of state, such as the Redlich-Kwong (RK) equation, which expresses Z as a function of T_r and P_r only, yield two-parameter corresponding-states correlations. The SRK equation and the PR equation, in which the acentric factor ω enters through function $\alpha(T_r; \omega)$ as an additional parameter, yield three-parameter corresponding-states correlations. The numerical assignments for parameters ϵ , σ , Ω , and Ψ are given in Table 4-2. Expressions are also given for $\alpha(T_r; \omega)$ for the SRK and PR equations.

As shown by Smith, Van Ness, and Abbott [Introduction to Chemical Engineering Thermodynamics, 7th ed., pp. 218–219, McGraw-Hill, New York (2005)], Eqs. (4-104) in conjunction with Eqs. (4-88), (4-89), and (4-47) lead to

$$\frac{G^R}{RT} = Z - 1 - \ln(Z - \beta) - qI \quad (4-109)$$

$$\frac{H^R}{RT} = Z - 1 + \left[\frac{d \ln \alpha(T_r)}{d \ln T_r} - 1 \right] qI \quad (4-110)$$

TABLE 4-2 Parameter Assignments for Cubic Equations of State*

For use with Eqs. (4-104) through (4-106)

Eq. of state	$\alpha(T_r)$	σ	ϵ	Ω	Ψ
RK (1949)	$T_r^{-1/2}$	1	0	0.08664	0.42748
SRK (1972)	$\alpha_{\text{SRK}}(T_r; \omega)^{\dagger}$	1	0	0.08664	0.42748
PR (1976)	$\alpha_{\text{PR}}(T_r; \omega)^{\ddagger}$	$1 + \sqrt{2}$	$1 - \sqrt{2}$	0.07780	0.45724

*Smith, Van Ness, and Abbott, *Introduction to Chemical Engineering Thermodynamics*, 7th ed., p. 98, McGraw-Hill, New York (2005).

$^{\dagger}\alpha_{\text{SRK}}(T_r; \omega) = [1 + (0.480 + 1.574\omega - 0.176\omega^2)(1 - T_r^{1/2})]^2$

$^{\ddagger}\alpha_{\text{PR}}(T_r; \omega) = [1 + (0.37464 + 1.54226\omega - 0.26992\omega^2)(1 - T_r^{1/2})]^2$

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$$\frac{S^R}{R} = \ln(Z - \beta) + \frac{d \ln \alpha(T_r)}{d \ln T_r} qI \quad (4-111)$$

$$\text{where} \quad I = \frac{1}{\sigma - \epsilon} \ln \left(\frac{Z + \sigma\beta}{Z + \epsilon\beta} \right) \quad (4-112)$$

Preliminary to application of these equations Z is found by solution of either Eq. (4-104a) or (4-104b).

Cubic equations of state may be applied to mixtures through expressions that give the parameters as functions of composition. No established theory prescribes the form of this dependence, and empirical *mixing rules* are often used to relate mixture parameters to pure-species parameters. The simplest realistic expressions are a linear mixing rule for parameter b and a quadratic mixing rule for parameter a

$$b = \sum_i x_i b_i \quad (4-113)$$

$$a = \sum_i \sum_j x_i x_j a_{ij} \quad (4-114)$$

with $a_{ij} = a_{ji}$. The a_{ij} are of two types: pure-species parameters (like subscripts) and interaction parameters (unlike subscripts). Parameter b_i is for pure species i . The interaction parameter a_{ij} is often evaluated from pure-species parameters by a geometric mean *combining rule*

$$a_{ij} = (a_i a_j)^{1/2} \quad (4-115)$$

These traditional equations yield mixture parameters solely from parameters for the pure constituent species. They are most likely to be satisfactory for mixtures comprised of simple and chemically similar molecules.

Pitzer's Generalized Correlations In addition to the corresponding-states correlation for the second virial coefficient, Pitzer and coworkers [*Thermodynamics*, 3d ed., App. 3, McGraw-Hill, New York (1995)] developed a full set of generalized correlations. They have as their basis an equation for the compressibility factor, as given by Eq. (2-63):

$$Z = Z^0 + \omega Z^1 \quad (2-63)$$

where Z^0 and Z^1 are each functions of reduced temperature T_r and reduced pressure P_r . Acentric factor ω is defined by Eq. (2-17). Correlations for Z appear in Sec. 2.

Generalized correlations are developed here for the residual enthalpy and residual entropy from Eqs. (4-48) and (4-49). Substitution for Z by Eq. (2-63) puts Eq. (4-48) into generalized form:

$$\frac{G^R}{RT} = \int_0^{P_r} (Z^0 - 1) \frac{dP_r}{P_r} + \omega \int_0^{P_r} Z^1 \frac{dP_r}{P_r} \quad (4-116)$$

Differentiation of Eq. (2-63) yields

$$\left(\frac{\partial Z}{\partial T_r} \right)_{P_r} = \left(\frac{\partial Z^0}{\partial T_r} \right)_{P_r} + \omega \left(\frac{\partial Z^1}{\partial T_r} \right)_{P_r}$$

Substitution for $(\partial Z / \partial T_r)_{P_r}$ in Eq. (4-49) gives

$$\frac{H^R}{RT_c} = -T_r^2 \int_0^{P_r} \left(\frac{\partial Z^0}{\partial T_r} \right)_{P_r} \frac{dP_r}{P_r} - \omega T_r^2 \int_0^{P_r} \left(\frac{\partial Z^1}{\partial T_r} \right)_{P_r} \frac{dP_r}{P_r} \quad (4-117)$$

By Eq. (4-47),

$$\frac{S^R}{R} = \frac{1}{T_r} \left(\frac{H^R}{RT_c} \right) - \frac{G^R}{RT}$$

Combination of Eqs. (4-116) and (4-117) leads to

$$\frac{S^R}{R} = - \int_0^{P_r} \left[T_r \left(\frac{\partial Z^0}{\partial T_r} \right)_{P_r} + Z^0 - 1 \right] \frac{dP_r}{P_r} - \omega \int_0^{P_r} \left[T_r \left(\frac{\partial Z^1}{\partial T_r} \right)_{P_r} + Z^1 \right] \frac{dP_r}{P_r}$$

If the first terms on the right sides of Eq. (4-117) and of this equation (including the minus signs) are represented by $(H^R)^0/RT_c$ and $(S^R)^0/R$ and if the second terms, excluding ω but including the minus signs, are represented by $(H^R)^1/RT_c$ and $(S^R)^1/R$, then

$$\frac{H^R}{RT_c} = \frac{(H^R)^0}{RT_c} + \omega \frac{(H^R)^1}{RT_c} \quad (4-118)$$

$$\frac{S^R}{R} = \frac{(S^R)^0}{R} + \omega \frac{(S^R)^1}{R} \quad (4-119)$$

Pitzer's original correlations for Z and the derived quantities were determined graphically and presented in tabular form. Since then, analytical refinements to the tables have been developed, with extended range and accuracy. The most popular Pitzer-type correlation is that of Lee and Kesler [*AIChE J.* **21**: 510–527 (1975); see also Smith, Van Ness, and Abbott, *Introduction to Chemical Engineering Thermodynamics*, 5th, 6th, and 7th eds., App. E, McGraw-Hill, New York (1996, 2001, 2005)]. These tables cover both the liquid and gas phases and span the ranges $0.3 \leq T_r \leq 4.0$ and $0.01 \leq P_r \leq 10.0$. They list values of Z^0 , Z^1 , $(H^R)^0/RT_c$, $(H^R)^1/RT_c$, $(S^R)^0/R$, and $(S^R)^1/R$.

Lee and Kesler also included a Pitzer-type correlation for vapor pressures:

$$\ln P_r^{\text{sat}}(T_r) = \ln P_r^0(T_r) + \omega \ln P_r^1(T_r) \quad (4-120)$$

$$\text{where} \quad \ln P_r^0(T_r) = 5.92714 - \frac{6.09648}{T_r} - 1.28862 \ln T_r + 0.169347 T_r^6 \quad (4-121)$$

$$\text{and} \quad \ln P_r^1(T_r) = 15.2518 - \frac{15.6875}{T_r} - 13.4721 \ln T_r + 0.43577 T_r^6 \quad (4-122)$$

The value of ω to be used with Eq. (4-120) is found from the correlation by requiring that it reproduce the normal boiling point; that is, ω for a particular substance is determined from

$$\omega = \frac{\ln P_{r_s}^{\text{sat}} - \ln P_r^0(T_{r_s})}{\ln P_r^1(T_{r_s})} \quad (4-123)$$

where T_{r_s} is the reduced normal boiling point and $P_{r_s}^{\text{sat}}$ is the reduced vapor pressure corresponding to 1 standard atmosphere (1.01325 bar).

Although the tables representing the Pitzer correlations are based on data for pure materials, they may also be used for the calculation of mixture properties. A set of recipes is required relating the parameters T_c , P_c , and ω for a mixture to the pure-species values and to composition. One such set is given by Eqs. (2-80) through (2-82) in the Seventh Edition of *Perry's Chemical Engineers' Handbook* (1997). These equations define *pseudoparameters*, so called because the defined values of T_{pc} , P_{pc} , and ω have no physical significance for the mixture.

The Lee-Kesler correlations provide reliable data for nonpolar and slightly polar gases; errors of less than 3 percent are likely. Larger errors can be expected in applications to highly polar and associating gases.

The quantum gases (e.g., hydrogen, helium, and neon) do not conform to the same corresponding-states behavior as do normal fluids. Prausnitz, Lichtenthaler, and de Azevedo [*Molecular Thermodynamics of Fluid-Phase Equilibria*, 3d ed., pp. 172–173, Prentice-Hall PTR, Upper Saddle River, N.J. (1999)] propose the use of temperature-dependent *effective* critical parameters. For hydrogen, the quantum gas most commonly found in chemical processing, the recommended equations are

$$\frac{T_c}{\text{K}} = \frac{43.6}{1 + 21.8/2.016T} \quad (\text{for H}_2) \quad (4-124)$$

$$\frac{P_c}{\text{bar}} = \frac{20.5}{1 + 44.2/2.016T} \quad (\text{for H}_2) \quad (4-125)$$

$$\frac{V_c}{\text{cm}^3 \text{mol}^{-1}} = \frac{51.5}{1 - 9.91/2.016T} \quad (\text{for H}_2) \quad (4-126)$$

where T is absolute temperature in kelvins. Use of these *effective* critical parameters for hydrogen requires the further specification that $\omega = 0$.

OTHER PROPERTY FORMULATIONS

LIQUID PHASE

Although residual properties have formal reality for liquids as well as for gases, their advantageous use as *small* corrections to ideal gas state properties is lost. Calculation of property changes for the liquid state are usually based on alternative forms of Eqs. (4-32) through (4-35), shown in Table 4-1. Useful here are the definitions of two liquid-phase properties—the volume expansivity β and the isothermal compressibility κ :

$$\beta \equiv \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P \quad (4-127)$$

$$\kappa \equiv -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_T \quad (4-128)$$

$$\text{For } V = f(T, P), \quad dV = \left(\frac{\partial V}{\partial T} \right)_P dT + \left(\frac{\partial V}{\partial P} \right)_T dP$$

This equation in combination with Eqs. (4-127) and (4-128) becomes

$$\frac{dV}{V} = \beta dT - \kappa dP \quad (4-129)$$

$$\text{If } V \text{ is constant,} \quad \left(\frac{\partial P}{\partial T} \right)_V = \frac{\beta}{\kappa} \quad (4-130)$$

Because liquid-phase isotherms of P versus V are very steep and closely spaced, both β and κ are small. Moreover (outside the critical region), they are weak functions of T and P and are often assumed constant at average values. Integration of Eq. (4-129) then gives

$$\ln \frac{V_2}{V_1} = \beta(T_2 - T_1) - \kappa(P_2 - P_1) \quad (4-131)$$

Substitution for the partial derivatives in Eqs. (4-32) through (4-35) by Eqs. (4-127) and (4-130) yields

$$dH = C_P dT + (1 - \beta T) V dP \quad (4-132)$$

$$dS = C_P \frac{dT}{T} - \beta V dP \quad (4-133)$$

$$dU = C_V dT + \left(\frac{\beta}{\kappa} T - P \right) dV \quad (4-134)$$

$$dS = \frac{C_V}{T} dT + \frac{\beta}{\kappa} dV \quad (4-135)$$

Integration of these equations is most common from the saturated-liquid state to the state of compressed liquid at constant T . For example, Eqs. (4-132) and (4-133) in integral form become

$$H = H^{\text{sat}} + \int_{P^{\text{sat}}}^P (1 - \beta T) V dP \quad (4-136)$$

$$S = S^{\text{sat}} - \int_{P^{\text{sat}}}^P \beta V dP \quad (4-137)$$

Again, β and V are weak functions of pressure for liquids, and are often assumed constant at the values for the saturated liquid at temperature T . An alternative treatment of V comes from Eq. (4-131), which for this application can be written

$$V = V^{\text{sat}} \exp[-\kappa(P - P^{\text{sat}})]$$

LIQUID/VAPOR PHASE TRANSITION

The isothermal vaporization of a pure liquid results in a phase change from saturated liquid to saturated vapor at vapor pressure P^{sat} . The

treatment of this transition is facilitated by definition of property changes of vaporization ΔM^{lv} :

$$\Delta M^{lv} \equiv M^v - M^l \quad (4-138)$$

where M^l and M^v are molar properties for states of saturated liquid and saturated vapor. Some experimental values of the enthalpy change of vaporization ΔH^{lv} , usually called the latent heat of vaporization, are listed in Table 2-150.

The enthalpy change and entropy change of vaporization are directly related:

$$\Delta H^{lv} = T \Delta S^{lv} \quad (4-139)$$

This equation follows from Eq. (4-15), because vaporization at the vapor pressure P^{sat} occurs at constant T .

As shown by Smith, Van Ness, and Abbott [*Introduction to Chemical Engineering Thermodynamics*, 7th ed., p. 221, McGraw-Hill, New York (2005)] the heat of vaporization is directly related to the slope of the vapor-pressure curve.

$$\Delta H^{lv} = T \Delta V^{lv} \frac{dP^{\text{sat}}}{dT} \quad (4-140)$$

Known as the Clapeyron equation, this exact thermodynamic relation provides the connection between the properties of the liquid and vapor phases.

In application an empirical vapor pressure versus temperature relation is required. The simplest such equation is

$$\ln P^{\text{sat}} = A - \frac{B}{T} \quad (4-141)$$

where A and B are constants for a given chemical species. This equation approximates P^{sat} over its entire temperature range from triple point to critical point. It is also a sound basis for interpolation between reasonably spaced values of T . More satisfactory for general use is the Antoine equation

$$\ln P^{\text{sat}} = A - \frac{B}{T + C} \quad (4-142)$$

The Wagner equation is useful for accurate representation of vapor pressure data over a wide temperature range. It expresses the reduced vapor pressure as a function of reduced temperature

$$\ln P_r^{\text{sat}} = \frac{A\tau + B\tau^{1.5} + C\tau^3 + D\tau^6}{1 - \tau} \quad (4-143)$$

where

$$\tau \equiv 1 - T_r$$

and A , B , C , and D are constants. Values of the constants for either the Wagner equation or the Antoine equation are given for many species by Poling, Prausnitz, and O'Connell [*The Properties of Gases and Liquids*, 5th ed., App. A, McGraw-Hill, New York (2001)].

Latent heats of vaporization are functions of temperature, and experimental values at a particular temperature are often not available. Recourse is then made to approximate methods. *Trouton's rule* of 1884 provides a simple check on whether values calculated by other methods are reasonable:

$$\frac{\Delta H_n^{lv}}{RT_n} \sim 10$$

Here, T_n is the absolute temperature of the normal boiling point, and ΔH_n^{lv} is the latent heat at this temperature. The units of ΔH_n^{lv} , R , and T_n must be chosen so that $\Delta H_n^{lv}/RT_n$ is dimensionless.

A much more accurate equation is that of Riedel [*Chem. Ing. Tech.* **26**: 679-683 (1954)]:

$$\frac{\Delta H_n^{lv}}{RT_n} = \frac{1.092(\ln P_c - 1.013)}{0.930 - T_{r_c}} \quad (4-144)$$

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where P_c is the critical pressure in bars and T_r is the reduced temperature at T_r . This equation provides reasonable approximations; errors rarely exceed 5 percent.

Estimates of the latent heat of vaporization of a pure liquid at any temperature from the known value at a single temperature may be based on an experimental value or on a value estimated by Eq. (4-144).

THERMODYNAMICS OF FLOW PROCESSES

The thermodynamics of flow encompasses mass, energy, and entropy balances for open systems, i.e., for systems whose boundaries allow the inflow and outflow of fluids. The common measures of flow are as follows:

Mass flow rate $\dot{m}$ molar flow rate $\dot{n}$ volumetric flow rate q velocity u
Also $\dot{m} = M\dot{n}$ and $q = uA$

where M is molar mass. Mass flow rate is related to velocity by

$$\dot{m} = \rho u A \quad (4-146)$$

where A is the cross-sectional area of a conduit and ρ is mass density. If ρ is molar density, then this equation yields molar flow rate. Flow rates $\dot{m}$, $\dot{n}$, and q measure quantity per unit of time. Although velocity u does not represent quantity of flow, it is an important design parameter.

MASS, ENERGY, AND ENTROPY BALANCES FOR OPEN SYSTEMS

Mass and energy balances for an open system are written with respect to a region of space known as a *control volume*, bounded by an imaginary *control surface* that separates it from the surroundings. This surface may follow fixed walls or be arbitrarily placed; it may be rigid or flexible.

Mass Balance for Open Systems Because mass is conserved, the time rate of change of mass within the control volume equals the net rate of flow of mass into the control volume. The flow is positive when directed into the control volume and negative when directed out. The mass balance is expressed mathematically by

$$\frac{dm_{cv}}{dt} + \Delta(\dot{m})_{fs} = 0 \quad (4-147)$$

The operator Δ signifies the difference between exit and entrance flows, and the subscript fs indicates that the term encompasses all flowing streams. When the mass flow rate $\dot{m}$ is given by Eq. (4-146),

$$\frac{dm_{cv}}{dt} + \Delta(\rho u A)_{fs} = 0 \quad (4-148)$$

This form of the mass balance equation is often called the *continuity equation*. For the special case of *steady-state* flow, the control volume contains a constant mass of fluid, and the first term of Eq. (4-148) is zero.

General Energy Balance Because energy, like mass, is conserved, the time rate of change of energy within the control volume equals the net rate of energy transfer into the control volume. Streams flowing into and out of the control volume have associated with them energy in its internal, potential, and kinetic forms, and all contribute to the energy change of the system. Energy may also flow across the control surface as heat and work. Smith, Van Ness, and Abbott [*Introduction to Chemical Engineering Thermodynamics*, 7th ed., pp. 47–48, McGraw-Hill, New York (2005)] show that the general energy balance for flow processes is

$$\frac{d(mU)_{cv}}{dt} + \Delta \left[\left(H + \frac{1}{2}u^2 + zg \right) \dot{m} \right]_{fs} = \dot{Q} + \dot{W} \quad (4-149)$$

The work rate $\dot{W}$ may be of several forms. Most commonly there is shaft work $\dot{W}_s$. Work may be associated with expansion or contraction of the control volume, and there may be stirring work. The velocity u in the kinetic energy term is the bulk mean velocity as defined by the

Watson's equation [*Ind. Eng. Chem.* **35**: 398–406 (1943)] has found wide acceptance:

$$\frac{\Delta H_{12}^{lg}}{\Delta H_1^{lg}} = \left(\frac{1 - T_{r2}}{1 - T_{r1}} \right)^{0.38} \quad (4-145)$$

This equation is simple and fairly accurate.

equation $u = \dot{m}/\rho A$; z is elevation above a datum level, and g is the local acceleration of gravity.

Energy Balances for Steady-State Flow Processes Flow processes for which the first term of Eq. (4-149) is zero are said to occur at *steady state*. As discussed with respect to the mass balance, this means that the mass of the system within the control volume is constant; it also means that no changes occur with time in the properties of the fluid within the control volume or at its entrances and exits. No expansion of the control volume is possible under these circumstances. The only work of the process is shaft work, and the general energy balance, Eq. (4-149), becomes

$$\Delta \left[\left(H + \frac{1}{2}u^2 + zg \right) \dot{m} \right]_{fs} = \dot{Q} + \dot{W}_s \quad (4-150)$$

Entropy Balance for Open Systems An entropy balance differs from an energy balance in a very important way—*entropy is not conserved*. According to the second law, the entropy changes in the system and surroundings as the result of any process must be positive, with a limiting value of zero for a reversible process. Thus, the entropy changes resulting from the process sum not to zero, but to a positive quantity called the *entropy generation* term. The statement of balance, expressed as rates, is therefore

$$\left\{ \begin{array}{l} \text{Net rate of} \\ \text{change in} \\ \text{entropy of} \\ \text{flowing streams} \end{array} \right\} + \left\{ \begin{array}{l} \text{Time rate of} \\ \text{change of} \\ \text{entropy} \\ \text{in control} \\ \text{volume} \end{array} \right\} + \left\{ \begin{array}{l} \text{Time rate of} \\ \text{change of} \\ \text{entropy in} \\ \text{surroundings} \end{array} \right\} = \left\{ \begin{array}{l} \text{Total rate} \\ \text{of entropy} \\ \text{generation} \end{array} \right\}$$

The equivalent *equation of entropy balance* is

$$\Delta(S\dot{m})_{fs} + \frac{d(mS)_{cv}}{dt} + \frac{dS_{surr}}{dt} = \dot{S}_G \geq 0 \quad (4-151)$$

where $\dot{S}_G$ is the entropy generation term. In accord with the second law, it must be positive, with zero as a limiting value. This equation is the general *rate* form of the entropy balance, applicable at any instant. The three terms on the left are the net rate of gain in entropy of the flowing streams, the time rate of change of the entropy of the fluid contained within the control volume, and the time rate of change of the entropy of the surroundings.

The entropy change of the surroundings results from heat transfer between system and surroundings. Let $\dot{Q}_j$ represent the heat-transfer rate at a particular location on the control surface associated with a surroundings temperature $T_{\sigma,j}$. In accord with Eq. (4-3), the rate of entropy change in the surroundings as a result of this transfer is $-\dot{Q}_j/T_{\sigma,j}$. The minus sign converts $\dot{Q}_j$, defined with respect to the system, to a heat rate with respect to the surroundings. The third term in Eq. (4-151) is therefore the sum of all such quantities, and Eq. (4-151) can be written

$$\Delta(S\dot{m})_{fs} + \frac{d(mS)_{cv}}{dt} - \sum_j \frac{\dot{Q}_j}{T_{\sigma,j}} = \dot{S}_G \geq 0 \quad (4-152)$$

For any process, the two kinds of irreversibility are (1) those *internal* to the control volume and (2) those resulting from heat transfer across finite temperature differences that may exist between the

TABLE 4-3 Equations of Balance

General equations of balance	Balance equations for steady-flow processes	Balance equations for single-stream steady-flow processes
$\frac{dm_{cv}}{dt} + \Delta(\dot{m})_{fs} = 0$ (4-147)	$\Delta(\dot{m})_{fs} = 0$ (4-153)	$\dot{m}_1 = \dot{m}_2 = \dot{m}$ (4-154)
$\frac{d(mU)_{cv}}{dt} + \Delta\left[\left(H + \frac{1}{2}u^2 + zg\right)\dot{m}\right]_{fs} = \dot{Q} + \dot{W}$ (4-149)	$\Delta\left[\left(H + \frac{1}{2}u^2 + zg\right)\dot{m}\right]_{fs} = \dot{Q} + \dot{W}_s$ (4-150)	$\Delta H + \frac{\Delta u^2}{2} + g\Delta z = Q + W_s$ (4-155)
$\frac{d(mS)_{cv}}{dt} + \Delta(S\dot{m})_{fs} - \sum_j \frac{\dot{Q}_j}{T_{\sigma,j}} = \dot{S}_G \geq 0$ (4-152)	$\Delta(S\dot{m})_{fs} - \sum_j \frac{\dot{Q}_j}{T_{\sigma,j}} = \dot{S}_G \geq 0$ (4-156)	$\Delta S - \sum_j \frac{Q_j}{T_{\sigma,j}} = S_G \geq 0$ (4-157)

system and surroundings. In the limiting case where $\dot{S}_G = 0$, the process is *completely reversible*, implying that

- The process is internally reversible within the control volume.
- Heat transfer between the control volume and its surroundings is reversible.

Summary of Equations of Balance for Open Systems Only the most general equations of mass, energy, and entropy balance appear in the preceding sections. In each case important applications require less general versions. The most common restricted case is for steady flow processes, wherein the mass and thermodynamic properties of the fluid within the control volume are not time-dependent. A further simplification results when there is but one entrance and one exit to the control volume. In this event, $\dot{m}$ is the same for both streams, and the equations may be divided through by this rate to put them on the basis of a unit amount of fluid flowing through the control volume. Summarized in Table 4-3 are the basic equations of balance and their important restricted forms.

APPLICATIONS TO FLOW PROCESSES

Duct Flow of Compressible Fluids Thermodynamics provides equations interrelating pressure changes, velocity, duct cross-sectional area, enthalpy, entropy, and specific volume within a flowing stream. Considered here is the adiabatic, steady-state, one-dimensional flow of a compressible fluid in the absence of shaft work and changes in potential energy. The appropriate energy balance is Eq. (4-155). With Q , W_s , and Δz all set equal to zero,

$$\Delta H + \frac{\Delta u^2}{2} = 0$$

$$\text{In differential form,} \quad dH = -u du \quad (4-158)$$

The continuity equation given by Eq. (4-148) here becomes $d(\rho u A) = d(uA/V) = 0$, whence

$$\frac{dV}{V} - \frac{du}{u} - \frac{dA}{A} = 0 \quad (4-159)$$

Smith, Abbott, and Van Ness [*Introduction to Chemical Engineering Thermodynamics*, 7th ed., pp. 255–258, McGraw-Hill, New York (2005)] show that these basic equations in combination with Eq. (4-15) and other property relations yield two very general equations

$$V(1 - \mathbf{M}^2) \frac{dP}{dx} + T \left(1 + \frac{\beta u^2}{C_p} \right) \frac{dS}{dx} - \frac{u^2}{A} \frac{dA}{dx} = 0 \quad (4-160)$$

$$u \frac{du}{dx} - T \left(\frac{\beta u^2 / C_p + \mathbf{M}^2}{1 - \mathbf{M}^2} \right) \frac{dS}{dx} + \left(\frac{1}{1 - \mathbf{M}^2} \right) \frac{u^2}{A} \frac{dA}{dx} = 0 \quad (4-161)$$

Mach number $\mathbf{M}$ is the ratio of the speed of fluid in the duct to the speed of sound in the fluid. The derivatives in these equations are rates of change with length as the fluid passes through a duct. Equation (4-160) relates the pressure derivative, and Eq. (4-161), the velocity derivative, to the entropy and area derivatives. According to

the second law, the irreversibilities of fluid friction in adiabatic flow cause an entropy increase in the fluid in the direction of flow. In the limit as the flow approaches reversibility, this increase approaches zero. In general, then, $dS/dx \geq 0$.

Pipe Flow For a pipe of constant cross-sectional area, $dA/dx = 0$, and Eqs. (4-160) and (4-161) reduce to

$$\frac{dp}{dx} = -\frac{T}{V} \left(\frac{1 + \beta u^2 / C_p}{1 - \mathbf{M}^2} \right) \frac{ds}{dx} \quad u \frac{du}{dx} = T \left(\frac{\beta u^2 / C_p + \mathbf{M}^2}{1 - \mathbf{M}^2} \right) \frac{ds}{dx}$$

When flow is subsonic, $\mathbf{M}^2 < 1$; all terms on the right in these equations are then positive, and $dP/dx < 0$ and $du/dx > 0$. Pressure therefore decreases and velocity increases in the direction of flow. The velocity increase is, however, limited, because these inequalities would reverse if the velocity were to become supersonic. This is not possible in a pipe of constant cross-sectional area, and the maximum fluid velocity obtainable is the speed of sound, reached at the *exit* of the pipe. Here, dS/dx reaches its limiting value of zero. For a discharge pressure low enough, the flow becomes sonic and lengthening the pipe does not alter this result; the mass rate of flow decreases so that the sonic velocity is still obtained at the outlet of the lengthened pipe.

According to the equations for supersonic pipe flow, pressure increases and velocity decreases in the direction of flow. However, this flow regime is unstable, and a supersonic stream entering a pipe of constant cross section undergoes a compression shock, the result of which is an abrupt and finite increase in pressure and decrease in velocity to a subsonic value.

Nozzles Nozzle flow is quite different from pipe flow. In a properly designed nozzle, its cross-sectional area changes with length in such a way as to make the flow nearly frictionless. The limit is reversible flow, for which the rate of entropy increase is zero. In this event $dS/dx = 0$, and Eqs. (4-160) and (4-161) become

$$\frac{dP}{dx} = \frac{u^2}{VA} \left(\frac{1}{1 - \mathbf{M}^2} \right) \frac{dA}{dx} \quad \frac{du}{dx} = -\frac{u}{A} \left(\frac{1}{1 - \mathbf{M}^2} \right) \frac{dA}{dx}$$

The characteristics of flow depend on whether the flow is subsonic ($\mathbf{M} < 1$) or supersonic ($\mathbf{M} > 1$). The possibilities are summarized in Table 4-4. Thus, for subsonic flow in a converging nozzle, the velocity increases and the pressure decreases as the cross-sectional area

TABLE 4-4 Nozzle Characteristics

	Subsonic: $\mathbf{M} < 1$		Supersonic: $\mathbf{M} > 1$	
	Converging	Diverging	Converging	Diverging
$\frac{dA}{dx}$	–	+	–	+
$\frac{dP}{dx}$	–	+	+	–
$\frac{du}{dx}$	+	–	–	+

diminishes. The maximum possible fluid velocity is the speed of sound, reached at the exit. A converging subsonic nozzle can therefore deliver a constant flow rate into a region of variable pressure.

Supersonic velocities characterize the diverging section of a properly designed converging/diverging nozzle. Sonic velocity is reached at the throat, where $dA/dx = 0$, and a further increase in velocity and decrease in pressure require a diverging cross-sectional area to accommodate the increasing volume of flow. The pressure at the throat must be low enough for the velocity to become sonic. If this is not the case, the diverging section acts as a diffuser—the pressure rises and the velocity decreases in the conventional behavior of subsonic flow in a diverging section.

An analytical expression relating velocity to pressure in an isentropic nozzle is readily derived for an ideal gas with constant heat capacities. Combination of Eqs. (4-15) and (4-159) for isentropic flow gives

$$u du = -V dP$$

Integration, with nozzle entrance and exit conditions denoted by 1 and 2, yields

$$u_2^2 - u_1^2 = -2 \int_{P_1}^{P_2} V dP = \frac{2\gamma P_1 V_1}{\gamma - 1} \left[1 - \left(\frac{P_2}{P_1} \right)^{(\gamma-1)/\gamma} \right] \quad (4-162)$$

where the final term is obtained upon elimination of V by $PV^\gamma = \text{const}$, an equation valid for ideal gases with constant heat capacities. Here, $\gamma \equiv C_P/C_V$.

Throttling Process Fluid flowing through a restriction, such as an orifice, without appreciable change in kinetic or potential energy undergoes a finite pressure drop. This *throttling process* produces no shaft work, and in the absence of heat transfer, Eq. (4-155) reduces to $\Delta H = 0$ or $H_2 = H_1$. The process therefore occurs at constant enthalpy.

The temperature of an ideal gas is not changed by a throttling process, because its enthalpy depends on temperature only. For most real gases at moderate conditions of T and P , a reduction in pressure at constant enthalpy results in a decrease in temperature, although the effect is usually small. Throttling of a wet vapor to a sufficiently low pressure causes the liquid to evaporate and the vapor to become superheated. This results in a considerable temperature drop because of the evaporation of liquid.

If a saturated liquid is throttled to a lower pressure, some of the liquid vaporizes or *flashes*, producing a mixture of saturated liquid and saturated vapor at the lower pressure. Again, the large temperature drop results from evaporation of liquid.

Turbines (Expanders) High-velocity streams from nozzles impinging on blades attached to a rotating shaft form a turbine (or expander) through which vapor or gas flows in a steady-state expansion process which converts internal energy of a high-pressure stream into shaft work. The motive force may be provided by steam (turbine) or by a high-pressure gas (expander).

In any properly designed turbine, heat transfer and changes in potential and kinetic energy are negligible. Equation (4-155) therefore reduces to

$$W_s = \Delta H = H_2 - H_1 \quad (4-163)$$

The rate form of this equation is

$$\dot{W}_s = \dot{m} \Delta H = \dot{m} (H_2 - H_1) \quad (4-164)$$

When inlet conditions T_1 and P_1 and discharge pressure P_2 are known, the value of H_1 is fixed. In Eq. (4-163) both H_2 and W_s are unknown, and the energy balance alone does not allow their calculation. However, if the fluid expands *reversibly and adiabatically*, i.e., isentropically, in the turbine, then $S_2 = S_1$. This second equation establishes the final state of the fluid and allows calculation of H_2 . Equation (4-164) then gives the isentropic work:

$$W_s(\text{isentropic}) = (\Delta H)_s \quad (4-165)$$

The absolute value $|W_s|(\text{isentropic})$ is the *maximum* work that can be produced by an adiabatic turbine with given inlet conditions and given

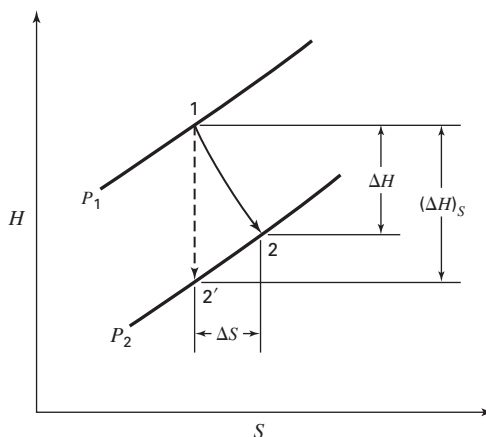


FIG. 4-2 Adiabatic expansion process in a turbine or expander. [Smith, Van Ness, and Abbott, *Introduction to Chemical Engineering Thermodynamics*, 7th ed., p. 269, McGraw-Hill, New York (2005).]

discharge pressure. Because the actual expansion process is irreversible, *turbine efficiency* is defined as

$$\eta \equiv \frac{W_s}{W_s(\text{isentropic})}$$

where W_s is the actual shaft work. By Eqs. (4-163) and (4-165),

$$\eta = \frac{\Delta H}{(\Delta H)_s} \quad (4-166)$$

Values of η usually range from 0.7 to 0.8.

The HS diagram of Fig. 4-2 compares the path of an actual expansion in a turbine with that of an isentropic expansion for the same intake conditions and the same discharge pressure. The isentropic path is the dashed vertical line from point 1 at intake pressure P_1 to point 2' at P_2 . The irreversible path (solid line) starts at point 1 and terminates at point 2 on the isobar for P_2 . The process is adiabatic, and irreversibilities cause the path to be directed toward increasing entropy. The greater the irreversibility, the farther point 2 lies to the right on the P_2 isobar, and the lower the value of η .

Compression Processes Compressors, pumps, fans, blowers, and vacuum pumps are all devices designed to bring about pressure increases. Their energy requirements for steady-state operation are of interest here. Compression of gases may be accomplished in rotating equipment (high-volume flow) or for high pressures in cylinders with reciprocating pistons. The energy equations are the same; indeed, based on the same assumptions, they are the same as for turbines or expanders. Thus, Eqs. (4-159) through (4-161) apply to adiabatic compression.

The isentropic work of compression, as given by Eq. (4-165), is the *minimum* shaft work required for compression of a gas from a given initial state to a given discharge pressure. A compressor efficiency is defined as

$$\eta \equiv \frac{W_s(\text{isentropic})}{W_s}$$

In view of Eqs. (4-163) and (4-165), this becomes

$$\eta \equiv \frac{(\Delta H)_s}{\Delta H} \quad (4-167)$$

Compressor efficiencies are usually in the range of 0.7 to 0.8.

The compression process is shown on an HS diagram in Fig. 4-3. The vertical dashed line rising from point 1 to point 2' represents the reversible adiabatic (isentropic) compression process from P_1 to P_2 .

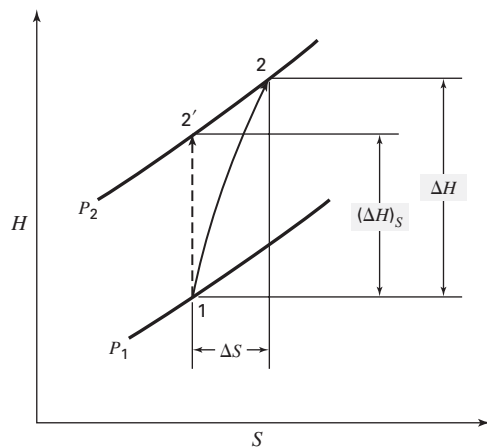


FIG. 4-3 Adiabatic compression process. [Smith, Van Ness, and Abbott, Introduction to Chemical Engineering Thermodynamics, 7th ed., p. 274, McGraw-Hill, New York (2005).]

The actual irreversible compression process follows the solid line from point 1 upward and to the right in the direction of increasing entropy, terminating at point 2. The more irreversible the process, the farther this point lies to the right on the P_2 isobar, and the lower the efficiency η of the process.

Liquids are moved by pumps, usually by rotating equipment. The same equations apply to adiabatic pumps as to adiabatic compressors. Thus, Eqs. (4-163) through (4-165) and Eq. (4-167) are valid. However, application of Eq. (4-163) requires values of the enthalpy of compressed (subcooled) liquids, and these are seldom available. The fundamental property relation, Eq. (4-15), provides an alternative. For an isentropic process,

$$dH = V dP \quad (\text{constant } S)$$

Combining this with Eq. (4-165) yields

$$W_s(\text{isentropic}) = (\Delta H)_S = \int_{P_1}^{P_2} V dP$$

The usual assumption for liquids (at conditions well removed from the critical point) is that V is independent of P . Integration then gives

$$W_s(\text{isentropic}) = (\Delta H)_S = V(P_2 - P_1) \quad (4-168)$$

Also useful are Eqs. (4-132) and (4-133). Because temperature changes in the pumped fluid are very small and because the properties of liquids are insensitive to pressure (again at conditions not close to the critical point), these equations are usually integrated on the assumption that C_p , V , and β are constant, usually at initial values. Thus, to a good approximation

$$\Delta H = C_p \Delta T + V(1 - \beta T) \Delta P \quad (4-169)$$

$$\Delta S = C_p \ln \frac{T_2}{T_1} - \beta V \Delta P \quad (4-170)$$

Example 1: LNG Vaporization and Compression A port facility for unloading liquefied natural gas (LNG) is under consideration. The LNG arrives by ship, stored as saturated liquid at 115 K and 1.325 bar, and is unloaded at the rate of 450 kg s⁻¹. It is proposed to vaporize the LNG with heat discarded from a heat engine operating between 300 K, the temperature of atmospheric air, and 115 K, the temperature of the vaporizing LNG. The saturated-vapor LNG so produced is compressed adiabatically to 20 bar, using the work produced by the heat engine to supply part of the compression work. Estimate the work to be supplied from an external source.

For estimation purposes we need not be concerned with the design of the heat engine, but assume that a suitable engine can be built to deliver 30 percent of the work of a Carnot engine operating between the temperatures of 300 and 115 K. The equations that apply to Carnot engines can be found in any thermodynamics text.

By the first law: $|W| = |Q_H| - |Q_C|$

By the second law: $\frac{|Q_H|}{|Q_C|} = \frac{T_H}{T_C}$

In combination: $|W| = |Q_C| \left(\frac{T_H}{T_C} - 1 \right)$

Here, $|W|$ is the work produced by the Carnot engine; $|Q_C|$ is the heat transferred at the cold temperature, i.e., to vaporize the LNG; T_H and T_C are the hot and cold temperatures of the heat reservoirs between which the heat engine operates, or 300 and 115 K, respectively. LNG is essentially pure methane, and enthalpy values from Table 2-281 of the Seventh Edition of *Perry's Chemical Engineers' Handbook* provide its heat of vaporization:

$$\Delta H^b = H^v - H^l = 802.5 - 297.7 = 504.8 \text{ kJ kg}^{-1}$$

For a flow rate of 450 kg s⁻¹,

$$|Q_C| = (450)(504.8) = 227,160 \text{ kJ s}^{-1}$$

The equation for work gives

$$|W| = (227,160) \left(\frac{300}{115} - 1 \right) = 3.654 \times 10^5 \text{ kJ s}^{-1} = 3.654 \times 10^5 \text{ kW}$$

This is the reversible work of a Carnot engine. The assumption is that the actual power produced is 30 percent of this, or $1.096 \times 10^5 \text{ kW}$.

The enthalpy and entropy of saturated vapor at 115 K and 1.325 bar are given in Table 2-281 of the Seventh Edition of *Perry's* as

$$H^v = 802.5 \text{ kJ kg}^{-1} \quad \text{and} \quad S^v = 9.436 \text{ kJ kg}^{-1} \text{ K}^{-1}$$

Isentropic compression of saturated vapor at 1.325 to 20 bar produces superheated vapor with an entropy of 9.436 kJ kg⁻¹ K⁻¹. Interpolation in Table 2-282 at 20 bar yields an enthalpy of $H = 1026.2 \text{ kJ kg}^{-1}$ at 234.65 K. The enthalpy change of isentropic compression is then

$$\Delta H_S = 1026.2 - 802.5 = 223.7 \text{ kJ kg}^{-1}$$

For a compressor efficiency of 75 percent, the actual enthalpy change of compression is

$$\Delta H = \frac{\Delta H_S}{\eta} = \frac{223.7}{0.75} = 298.3 \text{ kJ kg}^{-1}$$

The actual enthalpy of superheated LNG at 20 bar is then

$$H = 802.5 + 298.3 = 1100.8 \text{ kJ kg}^{-1}$$

Interpolation in Table 2-282 of the Seventh Edition of *Perry's* indicates an actual temperature of 265.9 K for the compressed LNG, which is quite suitable for its entry into the distribution system.

The work of compression is

$$|W| = m \Delta H = (450 \text{ kg s}^{-1})(298.3 \text{ kJ kg}^{-1}) = 1.342 \times 10^5 \text{ kJ s}^{-1} = 1.342 \times 10^5 \text{ kW}$$

The estimated power which must be supplied from an external source is

$$\dot{W} = 1.342 \times 10^5 - 1.096 \times 10^5 = 24,600 \text{ kW}$$

SYSTEMS OF VARIABLE COMPOSITION

The composition of a system may vary because the system is open or because of chemical reactions even in a closed system. The equations developed here apply regardless of the cause of composition changes.

PARTIAL MOLAR PROPERTIES

For a homogeneous PVT system comprised of any number of chemical species, let symbol M represent the molar (or unit-mass) value of an

extensive thermodynamic property, say, U , H , S , A , or G . A total-system property is then nM , where $n = \sum_i n_i$ and i is the index identifying chemical species. One might expect the solution property M to be related solely to the properties M_i of the pure chemical species which comprise the solution. However, no such generally valid relation is known, and the connection must be established experimentally for every specific system.

Although the chemical species which make up a solution do not have their own individual properties, a solution property may be arbitrarily

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apportioned among the individual species. Once an apportioning recipe is adopted, the assigned property values are quite logically treated as though they were indeed properties of the individual species in solution, and reasoning on this basis leads to valid conclusions.

For a homogeneous PVT system, postulate 5 requires that

$$nM = \mathcal{M}(T, P, n_1, n_2, n_3, \dots)$$

The total differential of nM is therefore

$$d(nM) = \left[\frac{\partial(nM)}{\partial T} \right]_{P,n} dT + \left[\frac{\partial(nM)}{\partial P} \right]_{T,n} dP + \sum_i \left[\frac{\partial(nM)}{\partial n_i} \right]_{T,P,n_j} dn_i$$

where subscript n indicates that all mole numbers n_i are held constant, and subscript n_j signifies that all mole numbers are held constant except the i th. This equation may also be written

$$d(nM) = n \left(\frac{\partial M}{\partial T} \right)_{P,x} dT + n \left(\frac{\partial M}{\partial P} \right)_{T,x} dP + \sum_i \left[\frac{\partial(nM)}{\partial n_i} \right]_{T,P,n_j} dn_i$$

where subscript x indicates that all mole fractions are held constant. The derivatives in the summation are called *partial molar properties*. They are given the generic symbol $\bar{M}_i$ and are defined by

$$\bar{M}_i \equiv \left[\frac{\partial(nM)}{\partial n_i} \right]_{T,P,n_j} \quad (4-171)$$

The basis for calculation of partial properties from solution properties is provided by this equation. Moreover,

$$d(nM) = n \left(\frac{\partial M}{\partial T} \right)_{P,x} dT + n \left(\frac{\partial M}{\partial P} \right)_{T,x} dP + \sum_i \bar{M}_i dn_i \quad (4-172)$$

This equation, valid for any equilibrium phase, either closed or open, attributes changes in total property nM to changes in T and P and to mole-number changes resulting from mass transfer or chemical reaction.

The following are mathematical identities:

$$d(nM) = n dM + M dn \quad dn_i = d(x_i n) = x_i dn + n dx_i$$

Combining these expressions with Eq. (4-172) and collecting like terms give

$$\left[dM - \left(\frac{\partial M}{\partial T} \right)_{P,x} dT - \left(\frac{\partial M}{\partial P} \right)_{T,x} dP - \sum_i \bar{M}_i dx_i \right] n + \left[M - \sum_i \bar{M}_i x_i \right] dn = 0$$

Because n and dn are independent and arbitrary, the terms in brackets must separately be zero, whence

$$dM = \left(\frac{\partial M}{\partial T} \right)_{P,x} dT + \left(\frac{\partial M}{\partial P} \right)_{T,x} dP + \sum_i \bar{M}_i dx_i \quad (4-173)$$

and

$$M = \sum_i x_i \bar{M}_i \quad (4-174)$$

The first of these equations is merely a special case of Eq. (4-172); however, Eq. (4-174) is a vital new relation. Known as the *summability equation*, it provides for the calculation of solution properties from partial properties, a purpose opposite to that of Eq. (4-171). Thus a solution property apportioned according to the recipe of Eq. (4-171) may be recovered simply by adding the properties attributed to the individual species, each weighted by its mole fraction in solution. The equations for partial molar properties are valid also for partial specific properties, in which case m replaces n and $\{x_i\}$ are mass fractions. Equation (4-171) applied to the definitions of Eqs. (4-8) through (4-10) yields the partial-property relations

$$\bar{H}_i = \bar{U}_i + P\bar{V}_i \quad \bar{A}_i = \bar{U}_i - T\bar{S}_i \quad \bar{G}_i = \bar{H}_i - T\bar{S}_i$$

These equations illustrate the parallelism that exists between the equations for a constant-composition solution and those for the corresponding partial properties. This parallelism exists whenever the

solution properties in the parent equation are related linearly (in the algebraic sense).

Gibbs-Duhem Equation Differentiation of Eq. (4-174) yields

$$dM = \sum_i x_i d\bar{M}_i + \sum_i \bar{M}_i dx_i$$

Because this equation and Eq. (4-173) are both valid in general, their right sides can be equated, yielding

$$\left(\frac{\partial M}{\partial T} \right)_{P,x} dT + \left(\frac{\partial M}{\partial P} \right)_{T,x} dP - \sum_i x_i d\bar{M}_i = 0 \quad (4-175)$$

This general result, the Gibbs-Duhem equation, imposes a constraint on how the partial properties of any phase may vary with temperature, pressure, and composition. For the special case where T and P are constant,

$$\sum_i x_i d\bar{M}_i = 0 \quad (\text{constant } T, P) \quad (4-176)$$

Symbol M may represent the molar value of any extensive thermodynamic property, say, V , U , H , S , or G . When $M \equiv H$, the derivatives $(\partial H/\partial H)_P$ and $(\partial H/\partial P)_T$ are given by Eqs. (4-28) and (4-29), and Eqs. (4-173), (4-174), and (4-175) specialize to

$$dH = C_P dT + \left[V - T \left(\frac{\partial V}{\partial T} \right)_{P,x} \right] dP + \sum_i \bar{H}_i dx_i \quad (4-177)$$

$$H = \sum_i x_i \bar{H}_i \quad (4-178)$$

$$C_P dT + \left[V - T \left(\frac{\partial V}{\partial T} \right)_{P,x} \right] dP - \sum_i x_i d\bar{H}_i = 0 \quad (4-179)$$

Similar equations are readily derived when M takes on other identities.

Equation (4-171), which defines a partial molar property, provides a general means by which partial-property values may be determined. However, for a *binary* solution an alternative method is useful. Equation (4-174) for a binary solution is

$$M = x_1 \bar{M}_1 + x_2 \bar{M}_2$$

Moreover, the Gibbs-Duhem equation for a solution at given T and P , Eq. (4-176), becomes

$$x_1 d\bar{M}_1 + x_2 d\bar{M}_2 = 0$$

These two equations combine to yield

$$\bar{M}_1 = M + x_2 \frac{dM}{dx_1} \quad (4-180a)$$

$$\bar{M}_2 = M - x_1 \frac{dM}{dx_1} \quad (4-180b)$$

Thus for a binary solution, the partial properties are given directly as functions of composition for given T and P . For multicomponent solutions such calculations are complex, and direct use of Eq. (4-171) is appropriate.

Partial Molar Equation-of-State Parameters The parameters in equations of state as applied to mixtures are related to composition by mixing rules. For the second virial coefficient

$$B = \sum_i \sum_j y_i y_j B_{ij} \quad (4-60)$$

The partial molar second virial coefficient is by definition

$$\bar{B}_i \equiv \left[\frac{\partial(nB)}{\partial n_i} \right]_{T,n_j} \quad (4-181)$$

Because B is independent of P , this is in accord with Eq. (4-171). These two equations lead through derivation to useful expressions for $\bar{B}_i$, as shown in detail by Van Ness and Abbott [*Classical Thermodynamics of Nonelectrolyte Solutions: With Applications to Phase*

Equilibria, pp. 137–140, McGraw-Hill, New York (1982)]. The simplest result is

$$\bar{B}_i = 2 \sum_k y_k B_{ki} - B \quad (4-182)$$

An analogous expression follows from Eq. (4-114) for parameter a in the generic cubic equation of state given by Eqs. (4-100), (4-103), and (4-104):

$$\bar{a}_i = 2 \sum_k y_k a_{ki} - a \quad (4-183)$$

This expression is independent of the combining rule [e.g., Eq. (4-114)] used for a_{ki} . For the *linear* mixing rule of Eq. (4-113) for b , the result of derivation is simply $b_i = b_i$.

Partial Molar Gibbs Energy Implicit in Eq. (4-13) is the relation

$$\mu_i = \left[\frac{\partial(nG)}{\partial n_i} \right]_{T,P,n_j}$$

Comparison with Eq. (4-171) indicates the following identity:

$$\mu_i = \bar{G}_i \quad (4-184)$$

The reciprocity relation for an exact differential applied to Eq. (4-13) produces not only the Maxwell relation, Eq. (4-21), but also two other useful equations:

$$\left(\frac{\partial \mu_i}{\partial P} \right)_{T,n} = \left[\frac{\partial(nV)}{\partial n_i} \right]_{T,P,n_j} = \bar{V}_i \quad (4-185)$$

$$\left(\frac{\partial \mu_i}{\partial T} \right)_{P,n} = - \left[\frac{\partial(nS)}{\partial n_i} \right]_{T,P,n_j} = -\bar{S}_i \quad (4-186)$$

Because $\mu = f(T, P)$,

$$d\mu_i = d\bar{G}_i = \left(\frac{\partial \mu_i}{\partial T} \right)_{P,n} dT + \left(\frac{\partial \mu_i}{\partial P} \right)_{T,n} dP$$

$$\text{and} \quad d\bar{G}_i = -\bar{S}_i dT + \bar{V}_i dP \quad (4-187)$$

Similarly, in view of Eqs. (4-14), (4-15), and (4-16),

$$d\bar{U}_i = T d\bar{S}_i - P d\bar{V}_i \quad (4-188)$$

$$d\bar{H}_i = T d\bar{S}_i + \bar{V}_i dP \quad (4-189)$$

$$d\bar{A}_i = -\bar{S}_i dT - P d\bar{V}_i \quad (4-190)$$

These equations again illustrate the fact that for every equation providing a *linear* relation among the thermodynamic properties of a *constant-composition* solution there exists a parallel relationship for the partial properties of the species in solution.

The following equation is a mathematical identity:

$$d\left(\frac{nG}{RT}\right) \equiv \frac{1}{RT} d(nG) - \frac{nG}{RT^2} dT$$

Substitution for $d(nG)$ by Eq. (4-13), with $\mu_i = \bar{G}_i$, and for G by Eq. (4-10) gives, after algebraic reduction,

$$d\left(\frac{nG}{RT}\right) = \frac{nV}{RT} dP - \frac{nH}{RT^2} dT + \sum_i \frac{\bar{G}_i}{RT} dn_i \quad (4-191)$$

This result is a useful alternative to the fundamental property relation given by Eq. (4-13). All terms in this equation have units of moles; moreover, the enthalpy rather than the entropy appears on the right side.

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Ideal Gas Mixture Model The ideal gas mixture model is useful because it is molecularly based, is analytically simple, is realistic in the

limit of zero pressure, and provides a conceptual basis upon which to build the structure of solution thermodynamics. Smith, Van Ness, and Abbott [*Introduction to Chemical Engineering Thermodynamics*, 7th ed., pp. 391–394, McGraw-Hill, New York (2005)] develop the following property relations for the ideal gas model.

$$\bar{V}_i^{ig} = V_i^{ig} = \frac{RT}{P} \quad (4-192)$$

Because the enthalpy is independent of pressure,

$$\bar{H}_i^{ig} = H_i^{ig} \quad (4-193)$$

where S_i^{ig} is evaluated at the *mixture* T and P . The entropy of an ideal gas *does* depend on pressure, and here

$$\bar{S}_i^{ig} = S_i^{ig} - R \ln y_i \quad (4-194)$$

where S_i^{ig} is evaluated at the mixture T and P .

From the definition of the Gibbs energy, $\bar{G}_i^{ig} = \bar{H}_i^{ig} - T\bar{S}_i^{ig}$. In combination with Eqs. (4-193) and (4-194), this becomes

$$\bar{G}_i^{ig} = H_i^{ig} - TS_i^{ig} + RT \ln y_i$$

$$\text{or} \quad \mu_i^{ig} \equiv \bar{G}_i^{ig} = G_i^{ig} + RT \ln y_i \quad (4-195)$$

Elimination of G_i^{ig} from this equation is accomplished through Eq. (4-17), written for pure species i as an ideal gas:

$$dG_i^{ig} = V_i^{ig} dP = \frac{RT}{P} dP = RT d \ln P \quad (\text{constant } T)$$

$$\text{Integration gives} \quad G_i^{ig} = \Gamma_i(T) + RT \ln P \quad (4-196)$$

where integration constant $\Gamma_i(T)$ is a function of temperature only. Equation (4-195) now becomes

$$\mu_i^{ig} = \bar{G}_i^{ig} = \Gamma_i(T) + RT \ln (y_i P) \quad (4-197)$$

$$\text{By Eq. (4-172)} \quad G^{id} = \sum_i y_i \Gamma_i(T) + RT \sum_i \ln (y_i P) \quad (4-198)$$

A dimensional ambiguity is apparent with Eqs. (4-196) through (4-198) in that P has units, whereas $\ln P$ must be dimensionless. In practice this is of no consequence, because only *differences* in Gibbs energy appear, along with *ratios* of the quantities with units of pressure in the arguments of the logarithm. Consistency in the units of pressure is, of course, required.

Fugacity and Fugacity Coefficient The chemical potential μ_i plays a vital role in both phase and chemical reaction equilibria. However, the chemical potential exhibits certain unfortunate characteristics that discourage its use in the solution of practical problems. The Gibbs energy, and hence μ_i , is defined in relation to the internal energy and entropy, both primitive quantities for which absolute values are unknown. Moreover, μ_i approaches negative infinity when either P or y_i approaches zero. While these characteristics do not preclude the use of chemical potentials, the application of equilibrium criteria is facilitated by introduction of the *fugacity*, a quantity that takes the place of μ_i but that does not exhibit its less desirable characteristics.

The origin of the fugacity concept resides in Eq. (4-196), an equation valid only for pure species i in the ideal gas state. For a *real* fluid, an analogous equation is written as

$$G_i \equiv \Gamma_i(T) + RT \ln f_i \quad (4-199)$$

in which a new property f_i replaces the pressure P . This equation serves as a partial definition of the *fugacity* f_i .

Subtraction of Eq. (4-196) from Eq. (4-199), both written for the same temperature and pressure, gives

$$G_i - G_i^{ig} = RT \ln \frac{f_i}{P}$$

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According to the definition of Eq. (4-40), $G_i - G_i^{ig}$ is the residual Gibbs energy G_i^R . The dimensionless ratio f_i/P is another new property called the *fugacity coefficient* ϕ_i . Thus,

$$G_i^R = RT \ln \phi_i \quad (4-200)$$

where

$$\phi_i \equiv \frac{f_i}{P} \quad (4-201)$$

The definition of fugacity is completed by setting the ideal gas state fugacity of pure species i equal to its pressure, $f_i^{ig} = P$. Thus for the special case of an ideal gas, $G_i^R = 0$, $\phi_i = 1$, and Eq. (4-196) is recovered from Eq. (4-199).

The definition of the fugacity of a species in solution is parallel to the definition of the pure-species fugacity. An equation analogous to the ideal gas expression, Eq. (4-197), is written for species i in a fluid mixture

$$\mu_i \equiv \Gamma_i(T) + RT \ln \hat{f}_i \quad (4-202)$$

where the partial pressure $y_i P$ is replaced by $\hat{f}_i$, the fugacity of species i in solution. Because it is not a partial property, it is identified by a circumflex rather than an overbar.

Subtracting Eq. (4-197) from Eq. (4-202), both written for the same temperature, pressure, and composition, yields

$$\mu_i - \mu_i^{ig} = RT \ln \frac{\hat{f}_i}{y_i P}$$

The residual Gibbs energy of a mixture is defined by $G^R \equiv G - G^{ig}$, and the analogous definition of a partial molar residual Gibbs energy is $G_i^R \equiv G_i - G_i^{ig} = \mu_i - \mu_i^{ig}$. Therefore

$$\bar{G}_i^R = RT \ln \hat{\phi}_i \quad (4-203)$$

where by definition

$$\hat{\phi}_i \equiv \frac{\hat{f}_i}{y_i P} \quad (4-204)$$

The dimensionless ratio $\hat{\phi}_i$ is called the *fugacity coefficient of species i in solution*.

Equation (4-203) is the analog of Eq. (4-200), which relates ϕ_i to G_i^R . For an ideal gas, G_i^R is necessarily zero; therefore $\hat{\phi}_i^{ig} = 1$ and $\hat{f}_i^{ig} = y_i P$. Thus the fugacity of species i in an ideal gas mixture is equal to its partial pressure.

Evaluation of Fugacity Coefficients Combining Eq. (4-200) with Eq. (4-45) gives

$$\ln \phi = \frac{G^R}{RT} = \int_0^P (Z - 1) \frac{dP}{P} \quad (4-205)$$

Subscript i is omitted, with the understanding that ϕ here is for a pure species. Clearly, all correlations for G^R/RT are also correlations for $\ln \phi$.

Equation (4-200) with Eqs. (4-48) and (4-73) yields

$$\ln \phi = \int_0^{P_r} (Z - 1) \frac{dP_r}{P_r} = (B^0 + \omega B^1) \frac{P_r}{T_r} \quad (4-206)$$

This equation, used in conjunction with Eqs. (4-77) and (4-78), provides a useful generalized correlation for the fugacity coefficients of pure species.

A more comprehensive generalized correlation results from Eqs. (4-200) and (4-116):

$$\ln \phi = \int_0^{P_r} (Z^0 - 1) \frac{dP_r}{P_r} + \omega \int_0^{P_r} Z^1 \frac{dP_r}{P_r}$$

An alternative form is $\ln \phi = \ln \phi^0 + \omega \ln \phi^1$ (4-207)

where $\ln \phi^0 \equiv \int_0^{P_r} (Z^0 - 1) \frac{dP_r}{P_r}$ and $\ln \phi^1 \equiv \int_0^{P_r} Z^1 \frac{dP_r}{P_r}$

By Eq. (4-207), $\phi = (\phi^0)(\phi^1)^\omega$ (4-208)

Correlations may therefore be presented for ϕ^0 and ϕ^1 , as was done by Lee and Kesler [AIChE J. **21**: 510-527 (1975)].

Ideal Solution Model The ideal gas model is useful as a standard of comparison for real gas behavior. This is formalized through residual properties. The *ideal solution* is similarly useful as a standard to which real solution behavior may be compared.

The partial molar Gibbs energy or chemical potential of species i in an ideal gas mixture is given by Eq. (4-195), written as

$$\mu_i^{ig} = \bar{G}_i^{ig} = G_i^{ig}(T, P) + RT \ln y_i$$

This equation takes on new meaning when $G_i^{ig}(T, P)$ is replaced by $G_i(T, P)$, the Gibbs energy of pure species i in its *real physical state* of gas, liquid, or solid at the mixture T and P . The ideal solution is therefore *defined* as one for which

$$\mu_i^{id} = \bar{G}_i^{id} \equiv G_i(T, P) + RT \ln x_i \quad (4-209)$$

where superscript *id* denotes an ideal solution property and x_i represents the mole fraction because application is usually to liquids.

This equation is the basis for development of expressions for all other thermodynamic properties of an ideal solution. Equations (4-185) and (4-186), applied to an ideal solution with μ_i replaced by $\bar{G}_i$, are written as

$$\bar{V}_i^{id} = \left(\frac{\partial \bar{G}_i^{id}}{\partial P} \right)_{T,x} \quad \text{and} \quad \bar{S}_i^{id} = - \left(\frac{\partial \bar{G}_i^{id}}{\partial T} \right)_{P,x}$$

Differentiation of Eq. (4-209) yields

$$\left(\frac{\partial \bar{G}_i^{id}}{\partial P} \right)_{T,x} = \left(\frac{\partial G_i}{\partial P} \right)_T \quad \text{and} \quad \left(\frac{\partial \bar{G}_i^{id}}{\partial T} \right)_{P,x} = \left(\frac{\partial G_i}{\partial T} \right)_P + R \ln x_i$$

Equation (4-17) implies that

$$\left(\frac{\partial G_i}{\partial P} \right)_T = V_i \quad \text{and} \quad \left(\frac{\partial G_i}{\partial T} \right)_P = -S_i$$

In combination these sets of equations provide

$$\bar{V}_i^{id} = V_i \quad (4-210)$$

and

$$\bar{S}_i^{id} = S_i - R \ln x_i \quad (4-211)$$

Because $\bar{H}_i^{id} = \bar{G}_i^{id} + T\bar{S}_i^{id}$, substitutions by Eqs. (4-209) and (4-211) yield

$$\bar{H}_i^{id} = H_i \quad (4-212)$$

The summability relation, Eq. (4-174), written for the special case of an ideal solution, may be applied to Eqs. (4-209) through (4-212):

$$G^{id} = \sum_i x_i G_i + RT \sum_i x_i \ln x_i \quad (4-213)$$

$$V^{id} = \sum_i x_i V_i \quad (4-214)$$

$$S^{id} = \sum_i x_i S_i - R \sum_i x_i \ln x_i \quad (4-215)$$

$$H^{id} = \sum_i x_i H_i \quad (4-216)$$

A simple equation for the fugacity of a species in an ideal solution follows from Eq. (4-209). For the special case of species i in an ideal solution, Eq. (4-202) becomes

$$\mu_i^{id} = \bar{G}_i^{id} = \Gamma_i(T) + RT \ln \hat{f}_i^{id}$$

When this equation and Eq. (4-199) are combined with Eq. (4-209), $\Gamma_i(T)$ is eliminated, and the resulting expression reduces to

$$\hat{f}_i^{id} = x_i f_i \quad (4-217)$$

This equation, known as the *Lewis-Randall rule*, shows that the fugacity of each species in an ideal solution is proportional to its mole fraction; the proportionality constant is the fugacity of *pure* species i in the same physical state as the solution and at the same T and P . Division of both sides of Eq. (4-217) by $x_i P$ and substitution of $\hat{\phi}_i^{id}$ for $\hat{f}_i^{id}/x_i P$ [Eq. (4-204)] and of ϕ_i for f_i/P [Eq. (4-201)] give the alternative form

$$\hat{\phi}_i^{id} = \phi_i \quad (4-218)$$

Thus the fugacity coefficient of species i in an ideal solution equals the fugacity coefficient of *pure* species i in the same physical state as the solution and at the same T and P .

Ideal solution behavior is often approximated by solutions comprised of molecules not too different in size and of the same chemical nature. Thus, a mixture of isomers conforms very closely to ideal solution behavior. So do mixtures of adjacent members of a homologous series.

Excess Properties An excess property M^E is defined as the difference between the actual property value of a solution and the value it would have as an ideal solution at the same T , P , and composition. Thus,

$$M^E \equiv M - M^{id} \quad (4-219)$$

where M represents the molar (or unit-mass) value of any extensive thermodynamic property (say, V , U , H , S , G). This definition is analogous to the definition of a residual property as given by Eq. (4-40). However, excess properties have no meaning for pure species, whereas residual properties exist for pure species as well as for mixtures. Partial molar excess properties $\bar{M}_i^E$ are defined analogously:

$$\bar{M}_i^E \equiv \bar{M}_i - \bar{M}_i^{id} \quad (4-220)$$

Of particular interest is the partial molar excess Gibbs energy. Equation (4-202) may be written as

$$\bar{G}_i = \Gamma_i(T) + RT \ln \hat{f}_i$$

In accord with Eq. (4-217) for an ideal solution, this becomes

$$\bar{G}_i^{id} = \Gamma_i(T) + RT \ln x_i \hat{f}_i$$

By difference

$$\bar{G}_i - \bar{G}_i^{id} = RT \ln \frac{\hat{f}_i}{x_i \hat{f}_i}$$

The left side is the partial excess Gibbs energy $\bar{G}_i^E$; the dimensionless ratio $\hat{f}_i/x_i \hat{f}_i$ on the right is the *activity coefficient of species i in solution*, given the symbol γ_i , and *by definition*,

$$\gamma_i \equiv \frac{\hat{f}_i}{x_i \hat{f}_i} \quad (4-221)$$

Thus,

$$\bar{G}_i^E = RT \ln \gamma_i \quad (4-222)$$

Comparison with Eq. (4-203) shows that Eq. (4-222) relates γ_i to $\bar{G}_i^E$ exactly as Eq. (4-203) relates $\hat{\phi}_i$ to $\bar{G}_i^R$. For an ideal solution, $\bar{G}_i^E = 0$, and therefore $\gamma_i = 1$.

Property Changes of Mixing A property change of mixing is defined by

$$\Delta M \equiv M - \sum_i x_i M_i \quad (4-223)$$

where M represents a molar thermodynamic property of a homogeneous solution and M_i is the molar property of pure species i at the T and P of the solution and in the same physical state. Applications are usually to liquids.

Each of Eqs. (4-213) through (4-216) is an expression for an ideal solution property, and each may be combined with the defining equation for an excess property [Eq. (4-219)], yielding the equations of the first column of Table 4-5. In view of Eq. (4-223) these may be written as shown in the second column of Table 4-5, where ΔG , ΔV ,

ΔS , and ΔH are the Gibbs energy change of mixing, the volume change of mixing, the entropy change of mixing, and the enthalpy change of mixing. For an ideal solution, each excess property is zero, and for this special case the equations reduce to those shown in the third column of Table 4-5.

Property changes of mixing and excess properties are easily calculated one from the other. The most common property changes of mixing are the volume change of mixing ΔV and the enthalpy change of mixing ΔH , commonly called the *heat of mixing*. These properties are identical to the corresponding excess properties. Moreover, they are directly measurable, providing an experimental entry into the network of equations of solution thermodynamics.

FUNDAMENTAL PROPERTY RELATIONS BASED ON THE GIBBS ENERGY

Of the four fundamental property relations shown in the second column of Table 4-1, only Eq. (4-13) has as its special or canonical variables T , P , and $\{n_i\}$. It is therefore the basis for extension to several useful supplementary thermodynamic properties. Indeed an alternative form has been developed as Eq. (4-191). These equations are the first two entries in the upper left quadrant of Table 4-6, which is now to be filled out with important derived relationships.

Fundamental Residual-Property Relation Equation (4-191) is general and may be written for the special case of an ideal gas

$$d\left(\frac{nG^{ig}}{RT}\right) = \frac{nV^{ig}}{RT} dP - \frac{nH^{ig}}{RT^2} dT + \sum_i \frac{\bar{G}_i^{ig}}{RT} dn_i$$

Subtraction of this equation from Eq. (4-191) gives

$$d\left(\frac{nG^R}{RT}\right) = \frac{nV^R}{RT} dP - \frac{nH^R}{RT^2} dT + \sum_i \frac{\bar{G}_i^R}{RT} dn_i \quad (4-236)$$

where the definitions $G^R \equiv G - G^{ig}$ and $\bar{G}_i^R \equiv \bar{G}_i - \bar{G}_i^{ig}$ have been imposed. Equation (4-236) is the *fundamental residual-property relation*. An alternative form follows by introduction of the fugacity coefficient given by Eq. (4-203). The result is listed as Eq. (4-237) in the upper left quadrant of Table 4-6.

Limited forms of this equation are particularly useful. Division by dP and restriction to constant T and composition lead to

$$\frac{V^R}{RT} = \left[\frac{\partial(G^R/RT)}{\partial P} \right]_{T,x} \quad (4-238)$$

Similarly, the result of division by dT and restriction to constant P and composition is

$$\frac{H^R}{RT} = -T \left[\frac{\partial(G^R/RT)}{\partial T} \right]_{P,x} \quad (4-239)$$

Also implicit in Eq. (4-237) is the relation

$$\ln \hat{\phi}_i = \left[\frac{\partial(nG^R/RT)}{\partial n_i} \right]_{T,P,n_j} \quad (4-240)$$

This equation demonstrates that $\ln \hat{\phi}_i$ is a partial property with respect to G^R/RT . The summability relation therefore applies, and

$$\frac{G^R}{RT} = \sum_i x_i \ln \hat{\phi}_i \quad (4-241)$$

TABLE 4-5 Relations Connecting Property Changes of Mixing and Excess Properties

M^E in relation to M	M^E in relation to ΔM	Expressions for ΔM^{id}
$G^E = G - \sum_i x_i G_i - RT \sum_i x_i \ln x_i$ (4-224)	$G^E = \Delta G - RT \sum_i x_i \ln x_i$ (4-228)	$\Delta G^{id} = RT \sum_i x_i \ln x_i$ (4-232)
$V^E = V - \sum_i x_i V_i$ (4-225)	$V^E = \Delta V$ (4-229)	$\Delta V^{id} = 0$ (4-233)
$S^E = S - \sum_i x_i S_i + R \sum_i x_i \ln x_i$ (4-226)	$S^E = \Delta S + R \sum_i x_i \ln x_i$ (4-230)	$\Delta S^{id} = -R \sum_i x_i \ln x_i$ (4-234)
$H^E = H - \sum_i x_i H_i$ (4-227)	$H^E = \Delta H$ (4-231)	$\Delta H^{id} = 0$ (4-235)

TABLE 4-6 Fundamental Property Relations for the Gibbs Energy and Related Properties

General equations for an open system	Equations for 1 mol (constant composition)
$d(nG) = nV dP - nS dT + \sum_i \mu_i dn_i$ (4-13)	$dG = V dP - S dT$ (4-17)
$d\left(\frac{nG}{RT}\right) = \frac{nV}{RT} dP - \frac{nH}{RT^2} dT + \sum_i \frac{\bar{G}_i}{RT} dn_i$ (4-191)	$d\left(\frac{G}{RT}\right) = \frac{V}{RT} dP - \frac{H}{RT^2} dT$ (4-253)
$d\left(\frac{nG^R}{RT}\right) = \frac{nV^R}{RT} dP - \frac{nH^R}{RT^2} dT + \sum_i \ln \hat{\phi}_i dn_i$ (4-237)	$d\left(\frac{G^R}{RT}\right) = \frac{V^R}{RT} dP - \frac{H^R}{RT^2} dT$ (4-254)
$d\left(\frac{nG^E}{RT}\right) = \frac{nV^E}{RT} dP - \frac{nH^E}{RT^2} dT + \sum_i \ln \gamma_i dn_i$ (4-248)	$d\left(\frac{G^E}{RT}\right) = \frac{V^E}{RT} dP - \frac{H^E}{RT^2} dT$ (4-255)
Equations for partial molar properties (constant composition)	Gibbs-Duhem equations
$d\bar{G}_i = d\mu_i = \bar{V}_i dP - \bar{S}_i dT$ (4-256)	$V dP - S dT = \sum_i x_i d\mu_i$ (4-260)
$d\left(\frac{\bar{G}_i}{RT}\right) = d\left(\frac{\mu_i}{RT}\right) = \frac{\bar{V}_i}{RT} dP - \frac{\bar{H}_i}{RT^2} dT$ (4-257)	$\frac{V}{RT} dP - \frac{H}{RT^2} dT = \sum_i x_i d\left(\frac{\bar{G}_i}{RT}\right)$ (4-261)
$d\left(\frac{\bar{G}_i^R}{RT}\right) = d \ln \hat{\phi}_i = \frac{\bar{V}_i^R}{RT} dP - \frac{\bar{H}_i^R}{RT^2} dT$ (4-258)	$\frac{V^R}{RT} dP - \frac{H^R}{RT^2} dT = \sum_i x_i d \ln \hat{\phi}_i$ (4-262)
$d\left(\frac{\bar{G}_i^E}{RT}\right) = d \ln \gamma_i = \frac{\bar{V}_i^E}{RT} dP - \frac{\bar{H}_i^E}{RT^2} dT$ (4-259)	$\frac{V^E}{RT} dP - \frac{H^E}{RT^2} dT = \sum_i x_i d \ln \gamma_i$ (4-263)

Application of Eq. (4-240) to an expression giving G^R as a function of composition yields an equation for $\ln \hat{\phi}_i$. In the simplest case of a gas mixture for which the virial equation [Eq. (4-67)] is appropriate, Eq. (4-69) provides the relation

$$\frac{nG^R}{RT} = \frac{P}{RT}(nB)$$

Differentiation in accord with Eqs. (4-240) and (4-181) yields

$$\ln \hat{\phi}_i = \frac{P}{RT} \bar{B}_i \quad (4-242)$$

where $\bar{B}_i$ is given by Eq. (4-182). For a binary system these equations reduce to

$$\ln \hat{\phi}_1 = \frac{P}{RT} (B_{11} + y_2^2 \delta_{12}) \quad (4-243a)$$

$$\ln \hat{\phi}_2 = \frac{P}{RT} (B_{22} + y_1^2 \delta_{12}) \quad (4-243b)$$

$$\text{where } \delta_{12} \equiv 2B_{12} - B_{11} - B_{22}$$

For the special case of pure species i , these equations reduce to

$$\ln \phi_i = \frac{P}{RT} B_{ii} \quad (4-244)$$

For the generic cubic equation of state [Eqs. (4-104)], G^R/RT is given by Eq. (4-109), which in view of Eq. (4-200) for a pure species becomes

$$\ln \phi_i = Z_i - 1 - \ln(Z_i - \beta_i) - q_i I_i \quad (4-245)$$

For species i in solution Smith, Van Ness, and Abbott [*Introduction to Chemical Engineering Thermodynamics*, 7th ed., pp. 562–563, McGraw-Hill, New York (2005)] show that

$$\ln \hat{\phi}_i = \frac{\bar{b}_i}{b} (Z - 1) - \ln(Z - \beta) - \bar{q}_i I \quad (4-246)$$

Symbols without subscripts represent mixture properties, and I is given by Eq. (4-112).

Fundamental Excess-Property Relation Equations for excess properties are developed in much the same way as those for residual properties. For the special case of an ideal solution, Eq. (4-191) becomes

$$d\left(\frac{nG^{id}}{RT}\right) = \frac{nV^{id}}{RT} dP - \frac{nH^{id}}{RT^2} dT + \sum_i \frac{\bar{G}_i^{id}}{RT} dn_i$$

Subtraction of this equation from Eq. (4-191) yields

$$d\left(\frac{nG^E}{RT}\right) = \frac{nV^E}{RT} dP - \frac{nH^E}{RT^2} dT + \sum_i \frac{\bar{G}_i^E}{RT} dn_i \quad (4-247)$$

where the definitions $G^E \equiv G - G^{id}$ and $\bar{G}_i^E \equiv \bar{G}_i - \bar{G}_i^{id}$ have been imposed. Equation (4-247) is the *fundamental excess-property relation*. An alternative form follows by introduction of the activity coefficient as given by Eq. (4-222). This result is listed as Eq. (4-248) in the upper left quadrant of Table 4-6.

The following equations are in complete analogy to those for residual properties.

$$\frac{V^E}{RT} = \left[\frac{\partial(G^E/RT)}{\partial P} \right]_{T,x} \quad (4-249)$$

$$\frac{H^E}{RT} = -T \left[\frac{\partial(G^E/RT)}{\partial T} \right]_{P,x} \quad (4-250)$$

$$\ln \gamma_i = \left[\frac{\partial(nG^E/RT)}{\partial n_i} \right]_{T,P,n_j} \quad (4-251)$$

This last equation demonstrates that $\ln \gamma_i$ is a partial property with respect to G^E/RT , implying also the validity of the summability relation

$$\frac{G^E}{RT} = \sum_i x_i \ln \gamma_i \quad (4-252)$$

The equations of the upper left quadrant of Table 4-6 reduce to those of the upper right quadrant for $n = 1$ and $dn_i = 0$. Each equation in the upper left quadrant has a partial-property analog, as shown in the lower left quadrant. Each equation of the upper left quadrant is a special case of Eq. (4-172) and therefore has associated with it a Gibbs-Duhem equation of the form of Eq. (4-173). These are shown in the lower right quadrant. The equations of Table 4-6 store an enormous amount of information, but they are so general that their direct

application is seldom appropriate. However, by inspection one can write a vast array of relations valid for particular applications. For example, one sees immediately from Eqs. (4-258) and (4-259) that

$$\left(\frac{\partial \ln \hat{\phi}_i}{\partial P}\right)_{T,x} = \frac{\bar{V}_i^R}{RT} \quad (4-264)$$

$$\left(\frac{\partial \ln \hat{\phi}_i}{\partial T}\right)_{P,x} = -\frac{\bar{H}_i^R}{RT^2} \quad (4-265)$$

$$\left(\frac{\partial \ln \gamma_i}{\partial P}\right)_{T,x} = \frac{\bar{V}_i^E}{RT} \quad (4-266)$$

$$\left(\frac{\partial \ln \gamma_i}{\partial T}\right)_{P,x} = -\frac{\bar{H}_i^E}{RT^2} \quad (4-267)$$

MODELS FOR THE EXCESS GIBBS ENERGY

Excess properties find application in the treatment of liquid solutions. Of primary importance for engineering calculations is the excess Gibbs energy G^E , because its canonical variables are T , P , and composition, the variables usually specified or sought in design calculations. Knowing G^E as a function of T , P , and composition, one can in principle compute from it all other excess properties.

The excess volume for liquid mixtures is usually small, and in accord with Eq. (4-249) the pressure dependence of G^E is usually ignored. Thus, engineering efforts to model G^E center on representing its composition and temperature dependence. For binary systems at constant T , G^E becomes a function of just x_1 , and the quantity most conveniently represented by an equation is G^E/x_1x_2RT . The simplest procedure is to express this quantity as a power series in x_1 :

$$\frac{G^E}{x_1x_2RT} = a + bx_1 + cx_1^2 + \dots \quad (\text{constant } T)$$

An equivalent power series with certain advantages is the *Redlich-Kister expansion* [Redlich, Kister, and Turnquist, *Chem. Eng. Progr. Symp. Ser. No. 2*, **48**: 49–61 (1952)]:

$$\frac{G^E}{x_1x_2RT} = A + B(x_1 - x_2) + C(x_1 - x_2)^2 + \dots$$

In application, different truncations of this expansion are appropriate, and for each truncation specific expressions for $\ln \gamma_1$ and $\ln \gamma_2$ result from application of Eq. (4-251). When all parameters are zero, $G^E/RT = 0$, and the solution is ideal. If $B = C = \dots = 0$, then

$$\frac{G^E}{x_1x_2RT} = A$$

where A is a constant for a given temperature. The corresponding equations for $\ln \gamma_1$ and $\ln \gamma_2$ are

$$\ln \gamma_1 = Ax_2^2 \quad (4-268a)$$

$$\ln \gamma_2 = Ax_1^2 \quad (4-268b)$$

The symmetric nature of these relations is evident. The infinite dilution values of the activity coefficients are $\ln \gamma_1^\infty = \ln \gamma_2^\infty = A$.

If $C = \dots = 0$, then

$$\frac{G^E}{x_1x_2RT} = A + B(x_1 - x_2) = A + B(2x_1 - 1)$$

and in this case G^E/x_1x_2RT is linear in x_1 . The substitutions $A + B = A_{21}$ and $A - B = A_{12}$ transform this expression to the *Margules equation*:

$$\frac{G^E}{x_1x_2RT} = A_{21}x_1 + A_{12}x_2 \quad (4-269)$$

Application of Eq. (4-251) yields

$$\ln \gamma_1 = x_2^2 [A_{12} + 2(A_{21} - A_{12})x_1] \quad (4-270a)$$

$$\ln \gamma_2 = x_1^2 [A_{21} + 2(A_{12} - A_{21})x_2] \quad (4-270b)$$

When $x_1 = 0$, $\ln \gamma_1^\infty = A_{12}$; when $x_2 = 0$, $\ln \gamma_2^\infty = A_{21}$.

An alternative equation is obtained when the reciprocal quantity x_1x_2RT/G^E is expressed as a linear function of x_1 :

$$\frac{x_1x_2}{G^E/RT} = A' + B'(x_1 - x_2) = A' + B'(2x_1 - 1)$$

This may also be written as

$$\frac{x_1x_2}{G^E/RT} = A'(x_1 + x_2) + B'(x_1 - x_2) = (A' + B')x_1 + (A' - B')x_2$$

The substitutions $A' + B' = 1/A'_{21}$ and $A' - B' = 1/A'_{12}$ ultimately produce

$$\frac{G^E}{x_1x_2RT} = \frac{A'_{12}A'_{21}}{A'_{12}x_1 + A'_{21}x_2} \quad (4-271)$$

The activity coefficients implied by this equation are given by

$$\ln \gamma_1 = A'_{12} \left(1 + \frac{A'_{12}x_1}{A'_{21}x_2} \right)^{-2} \quad (4-272a)$$

$$\ln \gamma_2 = A'_{21} \left(1 + \frac{A'_{21}x_2}{A'_{12}x_1} \right)^{-2} \quad (4-272b)$$

These are the *van Laar equations*. When $x_1 = 0$, $\ln \gamma_1^\infty = A'_{12}$; when $x_2 = 0$, $\ln \gamma_2^\infty = A'_{21}$.

The Redlich-Kister expansion, the Margules equations, and the van Laar equations are all special cases of a very general treatment based on rational functions, i.e., on equations for G^E given by ratios of polynomials [Van Ness and Abbott, *Classical Thermodynamics of Nonelectrolyte Solutions: With Applications to Phase Equilibria*, Sec. 5-7, McGraw-Hill, New York (1982)]. Although providing great flexibility in the fitting of VLE data for binary systems, they are without theoretical foundation, with no basis in theory for their extension to multi-component systems. Nor do they incorporate an explicit temperature dependence for the parameters.

Theoretical developments in the molecular thermodynamics of liquid solution behavior are often based on the concept of *local composition*, presumed to account for the short-range order and nonrandom molecular orientations resulting from differences in molecular size and intermolecular forces. Introduced by G. M. Wilson [*J. Am. Chem. Soc.* **86**: 127–130 (1964)] with the publication of a model for G^E , this concept prompted the development of alternative local composition models, most notably the NRTL (Non-Random Two-Liquid) equation of Renon and Prausnitz [*AIChE J.* **14**: 135–144 (1968)] and the UNIQUAC (UNIversal QUasi-Chemical) equation of Abrams and Prausnitz [*AIChE J.* **21**: 116–128 (1975)].

The Wilson equation, like the Margules and van Laar equations, contains just two parameters for a binary system (Λ_{12} and Λ_{21}) and is written as

$$\frac{G^E}{RT} = -x_1 \ln(x_1 + x_2\Lambda_{12}) - x_2 \ln(x_2 + x_1\Lambda_{21}) \quad (4-273)$$

$$\ln \gamma_1 = -\ln(x_1 + x_2\Lambda_{12}) + x_2 \left(\frac{\Lambda_{12}}{x_1 + x_2\Lambda_{12}} - \frac{\Lambda_{21}}{x_2 + x_1\Lambda_{21}} \right) \quad (4-274a)$$

$$\ln \gamma_2 = -\ln(x_2 + x_1\Lambda_{21}) - x_1 \left(\frac{\Lambda_{12}}{x_1 + x_2\Lambda_{12}} - \frac{\Lambda_{21}}{x_2 + x_1\Lambda_{21}} \right) \quad (4-274b)$$

At infinite dilution,

$$\ln \gamma_1^\infty = -\ln \Lambda_{12} + 1 - \Lambda_{21} \quad \ln \gamma_2^\infty = -\ln \Lambda_{21} + 1 - \Lambda_{12}$$

Both Λ_{12} and Λ_{21} must be positive numbers.

4-24 THERMODYNAMICS

The NRTL equation contains three parameters for a binary system and is written as

$$\frac{G^E}{x_1 x_2 RT} = \frac{G_{21} \tau_{21}}{x_1 + x_2 G_{21}} + \frac{G_{12} \tau_{12}}{x_2 + x_1 G_{12}} \quad (4-275)$$

$$\ln \gamma_1 = x_2^2 \left[\tau_{21} \left(\frac{G_{21}}{x_1 + x_2 G_{21}} \right)^2 + \frac{G_{12} \tau_{12}}{(x_2 + x_1 G_{12})^2} \right] \quad (4-276a)$$

$$\ln \gamma_2 = x_1^2 \left[\tau_{12} \left(\frac{G_{12}}{x_2 + x_1 G_{12}} \right)^2 + \frac{G_{21} \tau_{21}}{(x_1 + x_2 G_{21})^2} \right] \quad (4-276b)$$

Here $G_{12} = \exp(-\alpha \tau_{12})$ $G_{21} = \exp(-\alpha \tau_{21})$

and $\tau_{12} = \frac{b_{12}}{RT}$ $\tau_{21} = \frac{b_{21}}{RT}$

where α , b_{12} , and b_{21} , parameters specific to a particular pair of species, are independent of composition and temperature. The infinite dilution values of the activity coefficients are

$$\ln \gamma_1^\infty = \tau_{21} + \tau_{12} \exp(-\alpha \tau_{12}) \quad \ln \gamma_2^\infty = \tau_{12} + \tau_{21} \exp(-\alpha \tau_{21})$$

The local composition models have limited flexibility in the fitting of data, but they are adequate for most engineering purposes. Moreover, they are implicitly generalizable to multicomponent systems without the introduction of any parameters beyond those required to describe the constituent binary systems. For example, the Wilson equation for multicomponent systems is written as

$$\frac{G^E}{RT} = - \sum_i x_i \ln \left(\sum_j x_j \Lambda_{ij} \right) \quad (4-277)$$

and

$$\ln \gamma_i = 1 - \ln \left(\sum_j x_j \Lambda_{ij} \right) - \sum_k \frac{x_k \Lambda_{ki}}{\sum_j x_j \Lambda_{kj}} \quad (4-278)$$

where $\Lambda_{ij} = 1$ for $i = j$, etc. All indices in these equations refer to the same species, and all summations are over *all* species. For each ij pair there are two parameters, because $\Lambda_{ij} \neq \Lambda_{ji}$. For example, in a ternary system the three possible ij pairs are associated with the parameters Λ_{12} , Λ_{21} ; Λ_{13} , Λ_{31} ; and Λ_{23} , Λ_{32} .

The temperature dependence of the parameters is given by

$$\Lambda_{ij} = \frac{V_j}{V_i} \exp \frac{-a_{ij}}{RT} \quad i \neq j \quad (4-279)$$

where V_j and V_i are the molar volumes of pure liquids j and i and a_{ij} is a constant independent of composition and temperature. Molar volumes V_j and V_i , themselves weak functions of temperature, form ratios that in practice may be taken as independent of T , and are usually evaluated at or near 25°C.

The Wilson parameters Λ_{ij} and NRTL parameters G_{ij} inherit a Boltzmann-type T dependence from the origins of the expressions for G^E , but it is only approximate. Computations of properties sensitive to this dependence (e.g., heats of mixing and liquid/liquid solubility) are in general only qualitatively correct. However, all parameters are found from data for binary (in contrast to multicomponent) systems, and this makes parameter determination for the local composition models a task of manageable proportions.

The UNIQUAC equation treats $g \equiv G^E/RT$ as made up of two additive parts, a *combinatorial* term g^C , accounting for molecular size and shape differences, and a *residual* term g^R (not a residual property), accounting for molecular interactions:

$$g = g^C + g^R \quad (4-280)$$

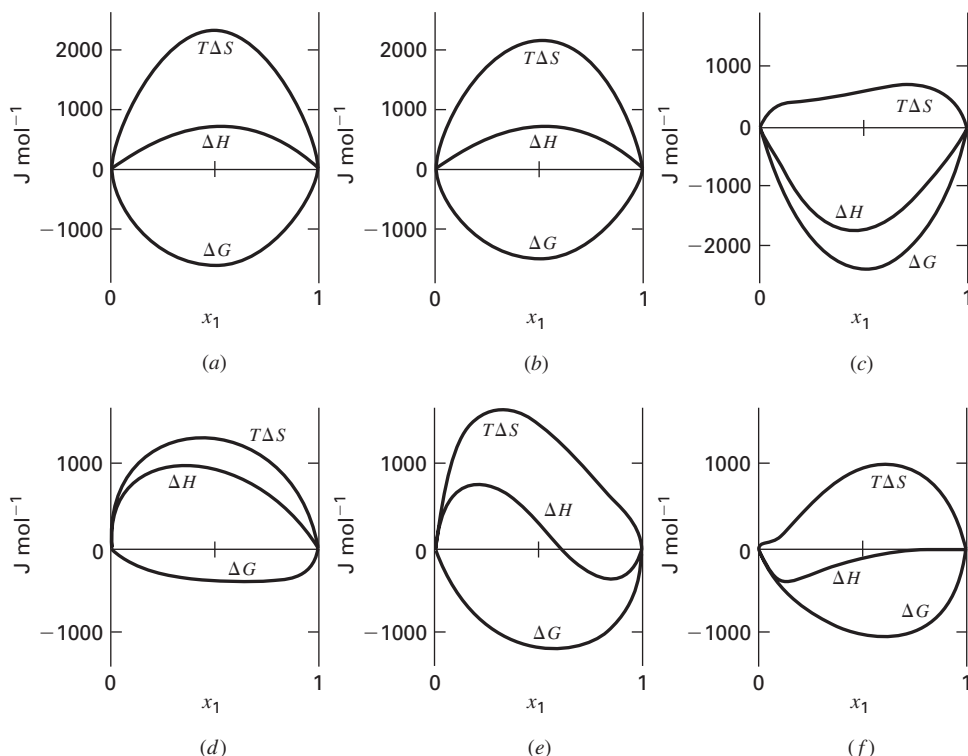


FIG. 4-4 Property changes of mixing at 50°C for six binary liquid systems: (a) chloroform(1)/n-heptane(2); (b) acetone(1)/methanol(2); (c) acetone(1)/chloroform(2); (d) ethanol(1)/n-heptane(2); (e) ethanol(1)/chloroform(2); (f) ethanol(1)/water(2). [Smith, Van Ness, and Abbott, *Introduction to Chemical Engineering Thermodynamics*, 7th ed., p. 455, McGraw-Hill, New York (2005).]

Function g^C contains pure-species parameters only, whereas function g^R incorporates two *binary* parameters for each pair of molecules. For a multicomponent system,

$$g^C = \sum_i x_i \ln \frac{\Phi_i}{x_i} + 5 \sum_i q_i x_i \ln \frac{\theta_i}{\Phi_i} \quad (4-281)$$

$$g^R = - \sum_i q_i x_i \ln \left(\sum_j \theta_j \tau_{ji} \right) \quad (4-282)$$

where

$$\Phi_i \equiv \frac{x_i r_i}{\sum_j x_j r_j} \quad (4-283)$$

and

$$\theta_i \equiv \frac{x_i q_i}{\sum_j x_j q_j} \quad (4-284)$$

Subscript i identifies species, and j is a dummy index; all summations are over all species. Note that $\tau_{ii} \neq \tau_{ij}$; however, when $i = j$, then $\tau_{ii} = \tau_{jj} = 1$. In these equations r_i (a relative molecular volume) and q_i (a relative molecular surface area) are pure-species parameters. The influence of temperature on g enters through the interaction parameters τ_{ji} of Eq. (4-282), which are temperature-dependent:

$$\tau_{ji} = \exp \frac{-(u_{ji} - u_{ii})}{RT} \quad (4-285)$$

Parameters for the UNIQUAC equation are therefore values of $u_{ji} - u_{ii}$.

An expression for $\ln \gamma_i$ is found by application of Eq. (4-251) to the UNIQUAC equation for g [Eqs. (4-280) through (4-282)]. The result is given by the following equations:

$$\ln \gamma_i = \ln \gamma_i^C + \ln \gamma_i^R \quad (4-286)$$

$$\ln \gamma_i^C = 1 - J_i + \ln J_i - 5q_i \left(1 - \frac{J_i}{L_i} + \ln \frac{J_i}{L_i} \right) \quad (4-287)$$

$$\ln \gamma_i^R = q_i \left(1 - \ln s_i - \sum_j \theta_j \frac{\tau_{ij}}{s_j} \right) \quad (4-288)$$

where in addition to Eqs. (4-284) and (4-285),

$$J_i = \frac{r_i}{\sum_j r_j x_j} \quad (4-289)$$

$$L_i = \frac{q_i}{\sum_j q_j x_j} \quad (4-290)$$

$$s_i = \sum_l \theta_l \tau_{li} \quad (4-291)$$

Again subscript i identifies species, and j and l are dummy indices.

Values for the parameters of the commonly used models for the excess Gibbs energy are given by Gmehling, Onken, and Arlt [*Vapor-Liquid Equilibrium Data Collection*, Chemistry Data Series, vol. 1, parts 1–8, DECHEMA, Frankfurt/Main (1974–1990)].

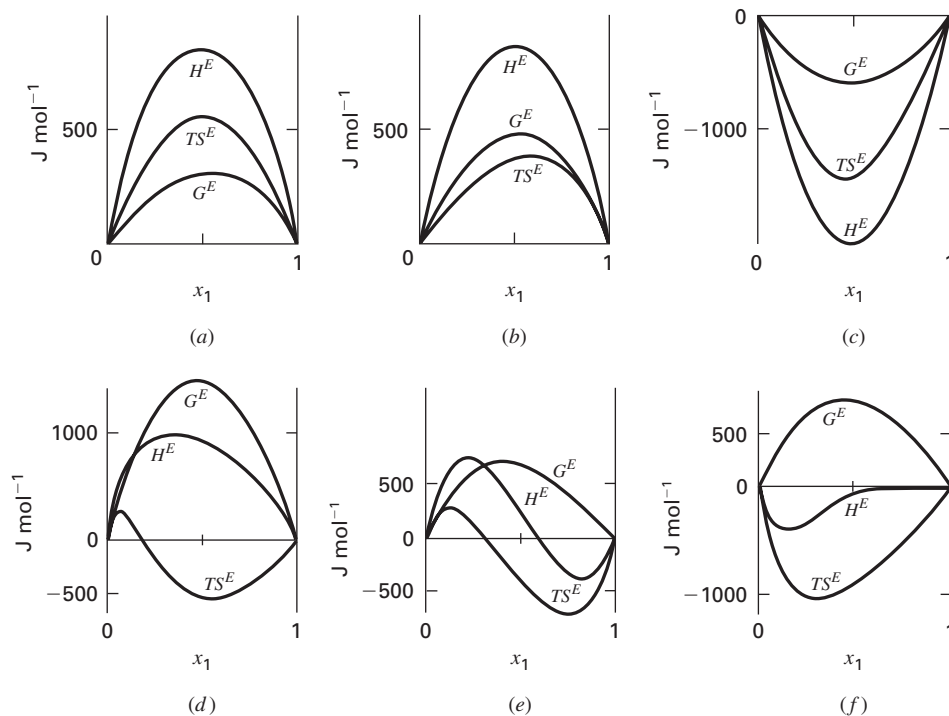


FIG. 4-5 Excess properties at 50°C for six binary liquid systems: (a) chloroform(1)/n-heptane(2); (b) acetone(1)/methanol(2); (c) acetone(1)/chloroform(2); (d) ethanol(1)/n-heptane(2); (e) ethanol(1)/chloroform(2); (f) ethanol(1)/water(2). [Smith, Van Ness, and Abbott, *Introduction to Chemical Engineering Thermodynamics*, 7th ed., p. 420, McGraw-Hill, New York (2005).]

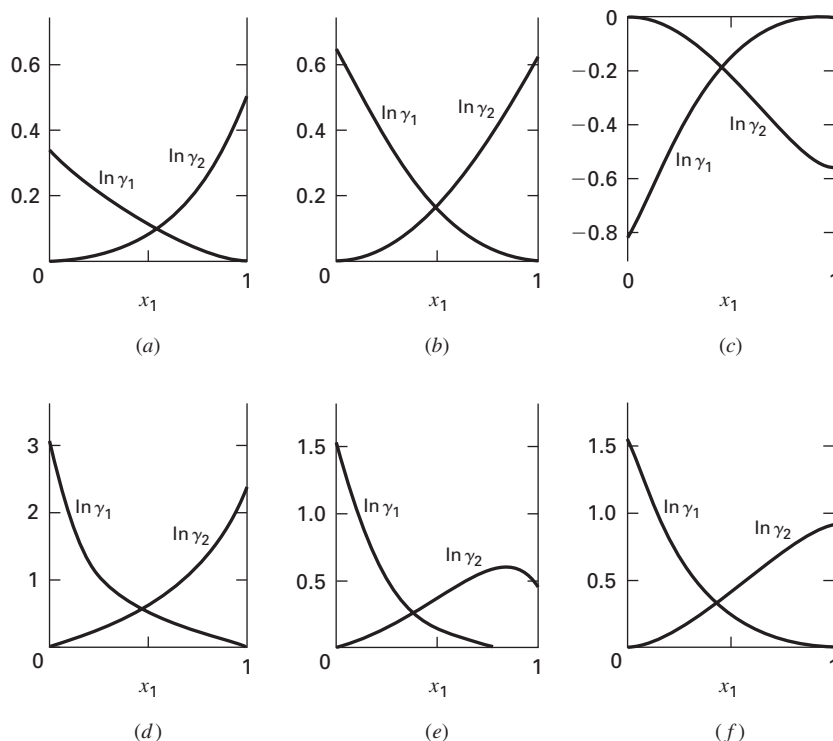


FIG. 4-6 Activity coefficients at 50°C for six binary liquid systems: (a) chloroform(1)/n-heptane(2); (b) acetone(1)/methanol(2); (c) acetone(1)/chloroform(2); (d) ethanol(1)/n-heptane(2); (e) ethanol(1)/chloroform(2); (f) ethanol(1)/water(2). [Smith, Van Ness, and Abbott, *Introduction to Chemical Engineering Thermodynamics*, 7th ed., p. 445, McGraw-Hill, New York (2005).]

Behavior of Binary Liquid Solutions Property changes of mixing and excess properties find greatest application in the description of liquid mixtures at low reduced temperatures, i.e., at temperatures well below the critical temperature of each constituent species. The properties of interest to the chemical engineer are V^E ($\equiv \Delta V$), H^E ($\equiv \Delta H$), S^E , ΔS , G^E , and ΔG . The activity coefficient is also of special importance because of its application in phase equilibrium calculations.

The volume change of mixing ($V^E = \Delta V$), the heat of mixing ($H^E = \Delta H$), and the excess Gibbs energy G^E are experimentally accessible, ΔV and ΔH by direct measurement and G^E (or $\ln \gamma_i$) indirectly by reduction of vapor/liquid equilibrium data. Knowledge of H^E and

G^E allows calculation of S^E by Eq. (4-10), written for excess properties as

$$S^E = \frac{H^E - G^E}{T} \quad (4-292)$$

with ΔS then given by Eq. (4-230).

Figure 4-4 displays plots of ΔH , ΔS , and ΔG as functions of composition for six binary solutions at 50°C. The corresponding excess properties are shown in Fig. 4-5; the activity coefficients, derived from Eq. (4-251), appear in Fig. 4-6. The properties shown here are insensitive to pressure and for practical purposes represent solution properties at 50°C and low pressure ($P \approx 1$ bar).

EQUILIBRIUM

CRITERIA

The equations developed in preceding sections are for *PVT* systems in states of internal equilibrium. The criteria for internal thermal and mechanical equilibrium simply require uniformity of temperature and pressure throughout the system. The criteria for phase and chemical reaction equilibria are less obvious.

If a closed *PVT* system of uniform T and P , either homogeneous or heterogeneous, is in thermal and mechanical equilibrium with its surroundings, but is not at internal equilibrium with respect to mass transfer or chemical reaction, then changes in the system are irreversible and necessarily bring the system closer to an equilibrium state. The first and second laws written for the entire system are

$$dU^t = dQ + dW \quad dS^t \geq \frac{dQ}{T}$$

Combination gives

$$dU^t - dW - TdS^t \leq 0$$

Because mechanical equilibrium is assumed, $dW = -PdV^t$, whence

$$dU^t + PdV^t - TdS^t \leq 0$$

The inequality applies to all incremental changes toward the equilibrium state, whereas the equality holds at the equilibrium state where change is reversible.

Constraints put on this expression produce alternative criteria for the directions of irreversible processes and for the condition of equilibrium. For example, $dU_{S,V}^t \leq 0$. Particularly important is fixing T and P ; this produces

$$d(U^t + PV^t - TS^t)_{T,P} \leq 0 \quad \text{or} \quad dG_{T,P}^t \leq 0$$

This expression shows that all irreversible processes occurring at constant $\bar{T}$ and P proceed in a direction such that the total Gibbs energy of the system decreases. Thus the equilibrium state of a closed system is the state with the minimum total Gibbs energy attainable at the given T and P . At the equilibrium state, differential variations may occur in the system at constant T and P without producing a change in G^t . This is the meaning of the equilibrium criterion

$$dG_{T,P}^t = 0 \quad (4-293)$$

This equation may be applied to a closed, nonreactive, two-phase system. Each phase taken separately is an *open* system, capable of exchanging mass with the other; Eq. (4-13) is written for each phase:

$$\begin{aligned} d(nG)' &= -(nS)' dT + (nV)' dP + \sum_i \mu_i' dn_i' \\ d(nG)'' &= -(nS)'' dT + (nV)'' dP + \sum_i \mu_i'' dn_i'' \end{aligned}$$

where the primes and double primes denote the two phases; the presumption is that T and P are uniform throughout the two phases. The change in the Gibbs energy of the two-phase system is the sum of these equations. When each total-system property is expressed by an equation of the form $nM = (nM)' + (nM)''$, this sum is given by

$$d(nG) = (nV)dP - (nS)dT + \sum_i \mu_i' dn_i' + \sum_i \mu_i'' dn_i''$$

If the two-phase system is at equilibrium, then application of Eq. (4-293) yields

$$dG_{T,P}^t \equiv d(nG)_{T,P} = \sum_i \mu_i' dn_i' + \sum_i \mu_i'' dn_i'' = 0$$

The system is closed and without chemical reaction; material balances therefore require that $dn_i'' = -dn_i'$, reducing the preceding equation to

$$\sum_i (\mu_i' - \mu_i'') dn_i' = 0$$

Because the dn_i' are independent and arbitrary, it follows that $\mu_i' = \mu_i''$. This is the criterion of two-phase equilibrium. It is readily generalized to multiple phases by successive application to pairs of phases. The general result is

$$\mu_i' = \mu_i'' = \mu_i''' = \dots \quad (4-294)$$

Substitution for each μ_i by Eq. (4-202) produces the equivalent result:

$$\hat{f}_i' = \hat{f}_i'' = \hat{f}_i''' = \dots \quad (4-295)$$

These are the criteria of phase equilibrium applied in the solution of practical problems.

For the case of equilibrium with respect to chemical reaction within a single-phase closed system, combination of Eqs. (4-13) and (4-293) leads immediately to

$$\sum_i \mu_i dn_i = 0 \quad (4-296)$$

For a system in which both phase and chemical reaction equilibrium prevail, the criteria of Eqs. (4-295) and (4-296) are superimposed.

PHASE RULE

The *intensive* state of a PVT system is established when its temperature and pressure and the compositions of all phases are fixed. However, for equilibrium states not all these variables are independent, and fixing a limited number of them automatically establishes the others. This number of independent variables is given by the phase rule, and it is called the *number of degrees of freedom* of the system. It is the number of variables that may be arbitrarily specified and that must be so specified in order to fix the *intensive* state of a system at equilibrium. This number is the difference between the number of variables needed to characterize the system and the number of equations that may be written connecting these variables.

For a system containing N chemical species distributed at equilibrium among π phases, the phase rule variables are T and P , presumed uniform throughout the system, and $N - 1$ mole fractions in each

phase. The number of these variables is $2 + (N - 1)\pi$. The masses of the phases are not phase rule variables, because they have nothing to do with the intensive state of the system.

The equilibrium equations that may be written express chemical potentials or fugacities as functions of T , P , and the phase compositions, the phase rule variables:

1. Equation (4-295) for each species, giving $(\pi - 1)N$ phase equilibrium equations

2. Equation (4-296) for each independent chemical reaction, giving r equations

The total number of independent equations is therefore $(\pi - 1)N + r$. Because the degrees of freedom of the system F is the difference between the number of variables and the number of equations,

$$F = 2 + (N - 1)\pi - (\pi - 1)N - r$$

or

$$F = 2 - \pi + N - r \quad (4-297)$$

The number of independent chemical reactions r can be determined as follows:

1. Write *formation* reactions from the elements for each chemical compound present in the system.

2. Combine these reaction equations so as to eliminate from the set all elements not present as elements in the system. A systematic procedure is to select one equation and combine it with each of the other equations of the set so as to eliminate a particular element. This usually reduces the set by one equation for each element eliminated, although two or more elements may be simultaneously eliminated.

The resulting set of r equations is a complete set of independent reactions. More than one such set is often possible, but all sets number r and are equivalent.

Example 2: Application of the Phase Rule

a. For a system of two miscible nonreacting species in vapor/liquid equilibrium,

$$F = 2 - \pi + N - r = 2 - 2 + 2 - 0 = 2$$

The 2 degrees of freedom for this system may be satisfied by setting T and P , or T and y_1 , or P and x_1 , or x_1 and y_1 , etc., at fixed values. Thus for equilibrium at a particular T and P , this state (if possible at all) exists only at one liquid and one vapor composition. Once the 2 degrees of freedom are used up, no further specification is possible that would restrict the phase rule variables. For example, one cannot *in addition* require that the system form an azeotrope (assuming this is possible), for this requires $x_1 = y_1$, an equation not taken into account in the derivation of the phase rule. Thus the requirement that the system form an azeotrope imposes a special constraint, making $F = 1$.

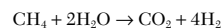
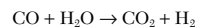
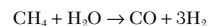
b. For a gaseous system consisting of CO, CO₂, H₂, H₂O, and CH₄ in chemical reaction equilibrium,

$$F = 2 - \pi + N - r = 2 - 1 + 5 - 2 = 4$$

The value of $r = 2$ is found from the formation reactions:



Systematic elimination of C and O₂ from this set of chemical equations reduces the set to two. Three possible pairs of equations may result, depending on how the combination of equations is effected. Any *pair* of the following three equations represents a complete set of independent reactions, and all pairs are equivalent.



The result, $F = 4$, means that one is free to specify, for example, T , P , and two mole fractions in an equilibrium mixture of these five chemical species, provided nothing else is arbitrarily set. Thus it cannot simultaneously be required that the system be prepared from specified amounts of particular constituent species.

Duhem's Theorem Because the phase rule treats only the intensive state of a system, it applies to both closed and open systems. Duhem's theorem, on the other hand, is a rule relating to closed systems only: *For any closed system formed initially from given masses of prescribed chemical species, the equilibrium state is completely*

determined by any two properties of the system, provided only that the two properties are independently variable at the equilibrium state. The meaning of completely determined is that both the intensive and extensive states of the system are fixed; not only are T , P , and the phase compositions established, but so also are the masses of the phases.

VAPOR/LIQUID EQUILIBRIUM

Vapor/liquid equilibrium (VLE) relationships (as well as other inter-phase equilibrium relationships) are needed in the solution of many engineering problems. The required data can be found by experiment, but measurements are seldom easy, even for binary systems, and they become ever more difficult as the number of species increases. This is the incentive for application of thermodynamics to the calculation of phase equilibrium relationships.

The general VLE problem treats a multicomponent system of N constituent species for which the independent variables are T , P , $N - 1$ liquid-phase mole fractions, and $N - 1$ vapor-phase mole fractions. (Note that $\sum x_i = 1$ and $\sum y_i = 1$, where x_i and y_i represent liquid and vapor mole fractions, respectively.) Thus there are $2N$ independent variables, and application of the phase rule shows that exactly N of these variables must be fixed to establish the intensive state of the system. This means that once N variables have been specified, the remaining N variables can be determined by simultaneous solution of the N equilibrium relations

$$\hat{f}_i^l = \hat{f}_i^v \quad i = 1, 2, \dots, N \quad (4-298)$$

where superscripts l and v denote the liquid and vapor phases, respectively.

In practice, either T or P and either the liquid-phase or vapor-phase composition are specified, thus fixing $1 + (N - 1) = N$ independent variables. The remaining N variables are then subject to calculation, provided that sufficient information is available to allow determination of all necessary thermodynamic properties.

Gamma/Phi Approach For many VLE systems of interest, the pressure is low enough that a relatively simple equation of state, such as the two-term virial equation, is satisfactory for the vapor phase. Liquid-phase behavior, on the other hand, is described by an equation for the excess Gibbs energy, from which activity coefficients are derived. The fugacity of species i in the liquid phase is given by Eq. (4-221), and the vapor-phase fugacity is given by Eq. (4-204). These are here written as

$$\hat{f}_i^l = \gamma_i x_i \hat{f}_i^v \quad \text{and} \quad \hat{f}_i^v = \hat{\phi}_i^v y_i P$$

$$\text{By Eq. (4-298),} \quad \gamma_i x_i \hat{f}_i^v = \hat{\phi}_i^v y_i P \quad i = 1, 2, \dots, N \quad (4-299)$$

Identifying superscripts l and v are omitted here with the understanding that γ_i and $\hat{f}_i$ are liquid-phase properties, whereas $\hat{\phi}_i$ is a vapor-phase property. Applications of Eq. (4-299) represent what is known as the *gamma/phi* approach to VLE calculations.

Evaluation of $\hat{\phi}_i$ is usually by Eq. (4-243), based on the two-term virial equation of state. The activity coefficient γ_i is ultimately based on Eq. (4-251) applied to an expression for G^E/RT , as described in the section "Models for the Excess Gibbs Energy."

The fugacity $\hat{f}_i$ of pure compressed liquid i must be evaluated at the T and P of the equilibrium mixture. This is done in two steps. First, one calculates the fugacity coefficient of saturated vapor $\hat{\phi}_i^* = \hat{\phi}_i^{\text{sat}}$ by an integrated form of Eq. (4-205), most commonly by Eq. (4-242) evaluated for pure species i at temperature T and the corresponding vapor pressure $P = P_i^{\text{sat}}$. Equation (4-298) written for pure species i becomes

$$\hat{f}_i^v = \hat{f}_i^l = \hat{f}_i^{\text{sat}} \quad (4-300)$$

where $\hat{f}_i^{\text{sat}}$ indicates the value both for saturated liquid and for saturated vapor. Division by P_i^{sat} yields corresponding fugacity coefficients:

$$\frac{\hat{f}_i^{\text{sat}}}{P_i^{\text{sat}}} = \frac{\hat{f}_i^v}{P_i^{\text{sat}}} = \frac{\hat{f}_i^l}{P_i^{\text{sat}}}$$

$$\text{or} \quad \hat{\phi}_i^v = \hat{\phi}_i^l = \hat{\phi}_i^{\text{sat}} \quad (4-301)$$

The second step is the evaluation of the change in fugacity of the liquid with a change in pressure to a value above or below P_i^{sat} . For this isothermal change of state from saturated liquid at P_i^{sat} to liquid at pressure P , Eq. (4-17) is integrated to give

$$G_i - G_i^{\text{sat}} = \int_{P_i^{\text{sat}}}^P V_i dP$$

Equation (4-199) is then written twice: for G_i and for G_i^{sat} . Subtraction provides another expression for $G_i - G_i^{\text{sat}}$:

$$G_i - G_i^{\text{sat}} = RT \ln \frac{\hat{f}_i}{\hat{f}_i^{\text{sat}}}$$

Equating the two expressions for $G_i - G_i^{\text{sat}}$ yields

$$\ln \frac{\hat{f}_i}{\hat{f}_i^{\text{sat}}} = \frac{1}{RT} \int_{P_i^{\text{sat}}}^P V_i dP$$

Because V_i , the liquid-phase molar volume, is a very weak function of P at temperatures well below T_c , an excellent approximation is usually obtained when evaluation of the integral is based on the assumption that V_i is constant at the value for saturated liquid V_i^l :

$$\ln \frac{\hat{f}_i}{\hat{f}_i^{\text{sat}}} = \frac{V_i^l (P - P_i^{\text{sat}})}{RT}$$

Substituting $\hat{f}_i^{\text{sat}} = \hat{\phi}_i^{\text{sat}} P_i^{\text{sat}}$ and solving for $\hat{f}_i$ give

$$\hat{f}_i = \hat{\phi}_i^{\text{sat}} P_i^{\text{sat}} \exp \frac{V_i^l (P - P_i^{\text{sat}})}{RT} \quad (4-302)$$

The exponential is known as the Poynting factor.

Equation (4-299) may now be written as

$$y_i P \hat{\Phi}_i = x_i \gamma_i P_i^{\text{sat}} \quad i = 1, 2, \dots, N \quad (4-303)$$

where

$$\hat{\Phi}_i = \frac{\hat{\phi}_i}{\hat{\phi}_i^{\text{sat}}} \exp \frac{-V_i^l (P - P_i^{\text{sat}})}{RT} \quad (4-304)$$

If evaluation of $\hat{\phi}_i^{\text{sat}}$ and $\hat{\phi}_i$ is by Eqs. (4-244) and (4-243), this reduces to

$$\hat{\Phi}_i = \exp \frac{\overline{B}_i - P_i^{\text{sat}} \overline{B}_i - V_i^l (P - P_i^{\text{sat}})}{RT} \quad (4-305)$$

where $\overline{B}_i$ is given by Eq. (4-182).

The N equations represented by Eq. (4-303) in conjunction with Eq. (4-305) may be solved for N unknown phase equilibrium variables. For a multicomponent system the calculation is formidable, but well suited to computer solution.

When Eq. (4-303) is applied to VLE for which the vapor phase is an ideal gas and the liquid phase is an ideal solution, it reduces to a very simple expression. For ideal gases, fugacity coefficients $\hat{\phi}_i$ and $\hat{\phi}_i^{\text{sat}}$ are unity, and the right side of Eq. (4-304) reduces to the Poynting factor. For the systems of interest here, this factor is always very close to unity, and for practical purposes $\hat{\Phi}_i = 1$. For ideal solutions, the activity coefficients γ_i are also unity, and Eq. (4-303) reduces to

$$y_i P = x_i P_i^{\text{sat}} \quad i = 1, 2, \dots, N \quad (4-306)$$

an equation which expresses *Raoult's law*. It is the simplest possible equation for VLE and as such fails to provide a realistic representation of real behavior for most systems. Nevertheless, it is useful as a standard of comparison.

Modified Raoult's Law Of the qualifications that lead to Raoult's law, the one least often reasonable is the supposition of solution ideality for the liquid phase. Real solution behavior is reflected by values of activity coefficients that differ from unity. When γ_i of Eq. (4-303) is retained in the equilibrium equation, the result is the *modified Raoult's law*:

$$y_i P = x_i \gamma_i P_i^{\text{sat}} \quad i = 1, 2, \dots, N \quad (4-307)$$

This equation is often adequate when applied to systems at low to moderate pressures and is therefore widely used. Bubble point and dew point calculations are only a bit more complex than the same calculations with Raoult's law.

Activity coefficients are functions of temperature and liquid-phase composition and are correlated through equations for the excess Gibbs energy. When an appropriate correlating equation for C^E is not available, suitable estimates of activity coefficients may often be obtained from a group contribution correlation. This is the "solution of groups" approach, wherein activity coefficients are found as sums of contributions from the structural groups that make up the molecules of a solution. The most widely applied such correlations are based on the UNIQUAC equation, and they have their origin in the UNIFAC method (UNIQUAC Functional-group Activity Coefficients), proposed by Fredenslund, Jones, and Prausnitz [*AIChE J.* **21**: 1086–1099 (1975)], and given detailed treatment by Fredenslund, Gmehling, and Rasmussen [*Vapor-Liquid Equilibrium Using UNIFAC*, Elsevier, Amsterdam (1977)].

Subsequent development has led to a variety of applications, including liquid/liquid equilibria [Magnussen, Rasmussen, and Fredenslund, *Ind. Eng. Chem. Process Des. Dev.* **20**: 331–339 (1981)], solid/liquid equilibria [Anderson and Prausnitz, *Ind. Eng. Chem. Fundam.* **17**: 269–273 (1978)], solvent activities in polymer solutions [Oishi and Prausnitz, *Ind. Eng. Chem. Process Des. Dev.* **17**: 333–339 (1978)], vapor pressures of pure species [Jensen, Fredenslund, and Rasmussen, *Ind. Eng. Chem. Fundam.* **20**: 239–246 (1981)], gas solubilities [Sander, Skjold-Jørgensen, and Rasmussen, *Fluid Phase Equilib.* **11**: 105–126 (1983)], and excess enthalpies [Dang and Tassios, *Ind. Eng. Chem. Process Des. Dev.* **25**: 22–31 (1986)].

The range of applicability of the original UNIFAC model has been greatly extended and its reliability enhanced. Its most recent revision and extension is treated by Wittig, Lohmann, and Gmehling [*Ind. Eng. Chem. Res.* **42**: 183–188 (2003)], wherein are cited earlier pertinent papers. Because it is based on temperature-independent parameters, its application is largely restricted to 0 to 150°C.

Two modified versions of the UNIFAC model, based on temperature-dependent parameters, have come into use. Not only do they provide a wide temperature range of applicability, but also they allow correlation of various kinds of property data, including phase equilibria, infinite dilution activity coefficients, and excess properties. The most recent revision and extension of the modified UNIFAC (Dortmund) model is provided by Gmehling et al. [*Ind. Eng. Chem. Res.* **41**: 1678–1688 (2002)]. An extended UNIFAC model called KT-UNIFAC is described in detail by Kang et al. [*Ind. Eng. Chem. Res.* **41**: 3260–3273 (2003)]. Both papers contain extensive literature citations.

The UNIFAC model has also been combined with the predictive Soave-Redlich-Kwong (PSRK) equation of state. The procedure is most completely described (with background literature citations) by Horstmann et al. [*Fluid Phase Equilibria* **227**: 157–164 (2005)].

Because $\sum y_i = 1$, Eq. (4-307) may be summed over all species to yield

$$P = \sum_i x_i \gamma_i P_i^{\text{sat}} \quad (4-308)$$

Alternatively, Eq. (4-307) may be solved for x_i , in which case summing over all species yields

$$P = \frac{1}{\sum_i y_i / \gamma_i P_i^{\text{sat}}} \quad (4-309)$$

Example 3: Dew and Bubble Point Calculations As indicated by Example 2a, a binary system in vapor/liquid equilibrium has 2 degrees of freedom. Thus of the four phase rule variables T , P , x_1 , and y_1 , two must be fixed to allow calculation of the other two, regardless of the formulation of the equilibrium equations. Modified Raoult's law [Eq. (4-307)] may therefore be applied to the calculation of any pair of phase rule variables, given the other two.

The necessary vapor pressures and activity coefficients are supplied by data correlations. For the system acetone(1)/*n*-hexane(2), vapor pressures are given by Eq. (4-142), the Antoine equation:

$$\ln P_i^{\text{sat}}/\text{kPa} = A_i - \frac{B_i}{T/\text{K} + C_i} \quad i = 1, 2 \quad (A)$$

with parameters

i	A_i	B_i	C_i
1	14.3145	2756.22	-45.090
2	13.8193	2696.04	-48.833

Activity coefficients are given by Eq. (4-274), the Wilson equation:

$$\ln \gamma_1 = -\ln(x_1 + x_2 \Lambda_{12}) + x_2 \lambda \quad (B)$$

$$\ln \gamma_2 = -\ln(x_2 + x_1 \Lambda_{21}) - x_1 \lambda \quad (C)$$

where

$$\lambda = \frac{\Lambda_{12}}{x_1 + x_2 \Lambda_{12}} - \frac{\Lambda_{21}}{x_2 + x_1 \Lambda_{21}}$$

By Eq. (4-279)

$$\Lambda_{ij} = \frac{V_j}{V_i} \exp \frac{-a_{ij}}{RT} \quad i \neq j$$

with parameters [Gmehling et al., *Vapor-Liquid Data Collection*, Chemistry Data Series, vol. 1, part 3, DECHEMA, Frankfurt/Main (1983)]

$\frac{a_{12}}{\text{cal mol}^{-1}}$	$\frac{a_{21}}{\text{cal mol}^{-1}}$	$\frac{V_1}{\text{cm}^3 \text{mol}^{-1}}$	$\frac{V_2}{\text{cm}^3 \text{mol}^{-1}}$
985.05	453.57	74.05	131.61

When T and x_1 are given, the calculation is direct, with final values for vapor pressures and activity coefficients given immediately by Eqs. (A), (B), and (C). In all other cases either T or x_1 or both are initially unknown, and calculations require trial or iteration.

a. BUBL P calculation: Find y_1 and P , given x_1 and T . Calculation here is direct. For $x_1 = 0.40$ and $T = 325.15 \text{ K}$ (52°C), Eqs. (A), (B), and (C) yield the values listed in the table on the following page. Equations (4-308) and (4-307) then become

$$P = x_1 \gamma_1 P_1^{\text{sat}} + x_2 \gamma_2 P_2^{\text{sat}} = (0.40)(1.8053)(87.616) + (0.60)(1.2869)(58.105) = 108.134 \text{ kPa}$$

$$y_1 = \frac{x_1 \gamma_1 P_1^{\text{sat}}}{P} = \frac{(0.40)(1.8053)(87.616)}{108.134} = 0.5851$$

b. DEW P calculation: Find x_1 and P , given y_1 and T . With x_1 an unknown, the activity coefficients cannot be immediately calculated. However, an iteration scheme based on Eqs. (4-309) and (4-307) is readily devised, and is part of any solve routine of a software package. Starting values result from setting each $\gamma_i = 1$. For $y_1 = 0.4$ and $T = 325.15 \text{ K}$ (52°C), results are listed in the accompanying table.

c. BUBL T calculation: Find y_1 and T , given x_1 and P . With T unknown, neither the vapor pressures nor the activity coefficients can be initially calculated. An iteration scheme or a solve routine with starting values for the unknowns is required. Results for $x_1 = 0.32$ and $P = 80 \text{ kPa}$ are listed in the accompanying table.

d. DEW T calculation: Find x_1 and T , given y_1 and P . Again, an iteration scheme or a solve routine with starting values for the unknowns is required. For $y_1 = 0.60$ and $P = 101.33 \text{ kPa}$, results are listed in the accompanying table.

e. Azeotrope calculations: As noted in Example 1a, only a single degree of freedom exists for this special case. The most sensitive quantity for identifying the azeotropic state is the *relative volatility*, defined as

$$\alpha_{12} = \frac{y_1/x_1}{y_2/x_2}$$

Because $y_i = x_i$ for the azeotropic state, $\alpha_{12} = 1$. Substitution for the two ratios by Eq. (4-307) provides an equation for calculation of α_{12} from the thermodynamic functions:

$$\alpha_{12} = \frac{\gamma_1 P_1^{\text{sat}}}{\gamma_2 P_2^{\text{sat}}}$$

Because α_{12} is a monotonic function of x_1 , the test of whether an azeotrope exists at a given T or P is provided by values of α_{12} in the limits of $x_1 = 0$ and $x_1 = 1$. If *both* values are either > 1 or < 1 , no azeotrope exists. But if one value is < 1 and the other > 1 , an azeotrope necessarily exists at the given T or P . Given T , the azeotropic composition and pressure is found by seeking the value of P that makes $x_1 = y_1$ or that makes $\alpha_{12} = 1$. Similarly, given P , one finds the azeotropic composition and temperature. Shown in the accompanying table are calculated azeotropic states for a temperature of 46°C and for a pressure of 101.33 kPa. At 46°C, the limiting values of α_{12} are 8.289 at $x_1 = 0$ and 0.223 at $x_1 = 1$.

4-30 THERMODYNAMICS

	T/K	$P_1^{\text{sat}}/\text{kPa}$	$P_2^{\text{sat}}/\text{kPa}$	γ_1	γ_2	x_1	y_1	P/kPa
a.	325.15	87.616	58.105	1.8053	1.2869	0.4000	0.5851	108.134
b.	325.15	87.616	58.105	3.5535	1.0237	0.1130	0.4000	87.939
c.	317.24	65.830	43.591	2.1286	1.1861	0.3200	0.5605	80.000
d.	322.98	81.125	53.779	1.6473	1.3828	0.4550	0.6000	101.330
e.	319.15	70.634	46.790	1.2700	1.9172	0.6445 = 0.6445	0.6445	89.707
f.	322.58	79.986	53.021	1.2669	1.9111	0.6454 = 0.6454	0.6454	101.330

Given values are italic; calculated results are boldface.

Data Reduction Correlations for G^E and the activity coefficients are based on VLE data taken at low to moderate pressures. Group-contribution methods, such as UNIFAC, depend for validity on parameters evaluated from a large base of such data. The process of finding a suitable analytic relation for g ($\equiv G^E/RT$) as a function of its independent variables T and x_1 , thus producing a correlation of VLE data, is known as data reduction. Although g is in principle also a function of P , the dependence is so weak as to be universally and properly neglected. Given here is a brief description of the treatment of data taken for binary systems under *isothermal* conditions. A more comprehensive development is given by Van Ness [J. Chem. Thermodyn. **27**: 113–134 (1995); Pure & Appl. Chem. **67**: 859–872 (1995)].

Presumed in all that follows is the existence of an equation inherently capable of correlating values of G^E for the liquid phase as a function of x_1 :

$$g \equiv G^E/RT = \mathcal{G}(x_1; \alpha, \beta, \dots) \quad (4-310)$$

where α, β , etc., represent adjustable parameters.

The measured variables of binary VLE are x_1, y_1, T , and P . Experimental values of the activity coefficient of species i in the liquid are related to these variables by Eq. (4-303), written as

$$\gamma_i^* = \frac{y_i^* P^*}{x_i P^{\text{sat}}} \Phi_i \quad i = 1, 2 \quad (4-311)$$

where Φ_i is given by Eq. (4-305) and the asterisks denote experimental values. A simple summability relation analogous to Eq. (4-252) defines an experimental value of g^* :

$$g^* \equiv x_1 \ln \gamma_1^* + x_2 \ln \gamma_2^* \quad (4-312)$$

Moreover, Eq. (4-263), the Gibbs-Duhem equation, may be written for experimental values in a binary system at constant T and P as

$$x_1 \frac{d \ln \gamma_1^*}{dx_1} + x_2 \frac{d \ln \gamma_2^*}{dx_1} = 0 \quad (4-313)$$

Because experimental measurements are subject to systematic error, sets of values of $\ln \gamma_1^*$ and $\ln \gamma_2^*$ may not satisfy, i.e., may not be *consistent* with, the Gibbs-Duhem equation. Thus Eq. (4-313) applied to sets of experimental values becomes a test of the thermodynamic consistency of the data, rather than a valid general relationship.

Values of g provided by the equation used to correlate the data, as represented by Eq. (4-310), are called *derived* values, and produce derived values of the activity coefficients by Eqs. (4-180) with $M \equiv g$:

$$\ln \gamma_1 = g + x_2 \frac{dg}{dx_1} \quad (4-314a)$$

$$\ln \gamma_2 = g - x_1 \frac{dg}{dx_1} \quad (4-314b)$$

These two equations combine to yield

$$\frac{dg}{dx_1} = \ln \frac{\gamma_1}{\gamma_2} \quad (4-315)$$

This equation is valid for *derived* property values. The corresponding *experimental* values are given by differentiation of Eq. (4-312):

$$\frac{dg^*}{dx_1} = x_1 \frac{d \ln \gamma_1^*}{dx_1} + \ln \gamma_1^* + x_2 \frac{d \ln \gamma_2^*}{dx_1} - \ln \gamma_2^*$$

$$\text{or} \quad \frac{dg^*}{dx_1} = \ln \frac{\gamma_1^*}{\gamma_2^*} + x_1 \frac{d \ln \gamma_1^*}{dx_1} + x_2 \frac{d \ln \gamma_2^*}{dx_1} \quad (4-316)$$

Subtraction of Eq. (4-316) from Eq. (4-315) gives

$$\frac{dg}{dx_1} - \frac{dg^*}{dx_1} = \ln \frac{\gamma_1}{\gamma_2} - \ln \frac{\gamma_1^*}{\gamma_2^*} - \left(x_1 \frac{d \ln \gamma_1^*}{dx_1} + x_2 \frac{d \ln \gamma_2^*}{dx_1} \right)$$

The differences between like terms represent *residuals* between derived and experimental values. Defining these residuals as

$$\delta g \equiv g - g^* \quad \text{and} \quad \delta \ln \frac{\gamma_i}{\gamma_j} \equiv \ln \frac{\gamma_i}{\gamma_j} - \ln \frac{\gamma_i^*}{\gamma_j^*}$$

puts this equation into the form

$$\frac{d\delta g}{dx_1} = \delta \ln \frac{\gamma_1}{\gamma_2} - \left(x_1 \frac{d \ln \gamma_1^*}{dx_1} + x_2 \frac{d \ln \gamma_2^*}{dx_1} \right)$$

If a data set is reduced so as to yield parameters— α, β , etc.—that make the δg residuals scatter about zero, then the derivative on the left is effectively zero, and the preceding equation becomes

$$\delta \ln \frac{\gamma_1}{\gamma_2} = x_1 \frac{d \ln \gamma_1^*}{dx_1} + x_2 \frac{d \ln \gamma_2^*}{dx_1} \quad (4-317)$$

The right side of this equation is the quantity required by Eq. (4-313), the Gibbs-Duhem equation, to be zero for consistent data. The residual on the left is therefore a direct measure of deviations from the Gibbs-Duhem equation. The extent to which values of this residual fail to scatter about zero measures the departure of the data from consistency with respect to this equation.

The data reduction procedure just described provides parameters in the correlating equation for g that make the δg residuals scatter about zero. This is usually accomplished by finding the parameters that minimize the sum of squares of the residuals. Once these parameters are found, they can be used for the calculation of derived values of both the pressure P and the vapor composition y_1 . Equation (4-303) is solved for $y_1 P$ and written for species 1 and for species 2. Adding the two equations gives

$$P = \frac{x_1 \gamma_1 P_1^{\text{sat}}}{\Phi_1} + \frac{x_2 \gamma_2 P_2^{\text{sat}}}{\Phi_2} \quad (4-318)$$

$$\text{whence by Eq. (4-303),} \quad y_1 = \frac{x_1 \gamma_1 P_1^{\text{sat}}}{\Phi_1 P} \quad (4-319)$$

These equations allow calculation of the *primary* residuals:

$$\delta P \equiv P - P^* \quad \text{and} \quad \delta y_1 \equiv y_1 - y_1^*$$

If the experimental values P^* and y_1^* are closely reproduced by the correlating equation for g , then these residuals, evaluated at the experimental values of x_1 , scatter about zero. This is the result obtained when the data approach thermodynamic consistency. When they do not, these residuals fail to scatter about zero and the correlation for g does not properly reproduce the experimental values P^* and y_1^* .

Such a correlation is unnecessarily divergent. An alternative is to base data reduction on just the P - x_1 data subset; this is possible because the full P - x_1 - y_1 data set includes redundant information. Assuming that the correlating equation is appropriate to the data, one merely searches for values of the parameters α, β , etc., that yield pressures by Eq. (4-318) that are as close as possible to the measured values. The usual procedure is to minimize the sum of squares of the residuals δP . Known as *Barker's method* [Austral. J. Chem. **6**: 207–210 (1953)], it provides the best possible fit of the experimental pressures. When experimental y_1^* values are not consistent with the P - x_1 data, Barker's method cannot lead to calculated y_1 values that closely match the experimental y_1^* values. With experimental error usually concentrated in the y_1^* values, the calculated y_1 values are likely to be more nearly correct. Because Barker's method requires only the P - x_1 data subset, the measurement of y_1^* values is not usually worth the extra effort, and the correlating parameters α, β , etc., are usually best determined without them. Hence, many P - x_1 data subsets appear in the literature; they are of course not subject to a test for consistency by the Gibbs-Duhem equation.

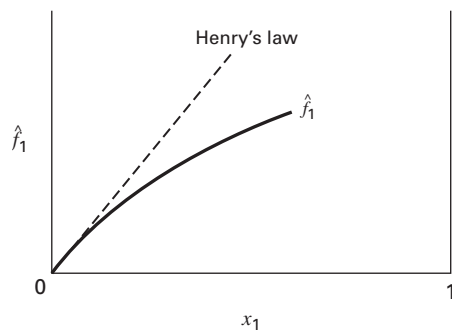


FIG. 4-7 Plot of solute fugacity $\hat{f}_1$ versus solute mole fraction. [Smith, Van Ness, and Abbott, *Introduction to Chemical Engineering Thermodynamics*, 7th ed., p. 555, McGraw-Hill, New York (2005).]

The world's store of VLE data has been compiled by Gmehling et al. [*Vapor-Liquid Equilibrium Data Collection*, Chemistry Data Series, vol. 1, parts 1–8, DECHEMA, Frankfurt am Main (1979–1990)].

Solute/Solvent Systems The gamma/phi approach to VLE calculations presumes knowledge of the vapor pressure of each species at the temperature of interest. For certain binary systems species 1, designated the solute, is either unstable at the system temperature or is *supercritical* ($T > T_c$). Its vapor pressure cannot be measured, and its fugacity as a pure liquid at the system temperature f_1 cannot be calculated by Eq. (4-302).

Equations (4-303) and (4-304) are applicable to species 2, designated the solvent, but not to the solute, for which an alternative approach is required. Figure 4-7 shows a typical plot of the liquid-phase fugacity of the solute $\hat{f}_1$ versus its mole fraction x_1 at constant temperature. Since the curve representing $\hat{f}_1$ does not extend all the way to $x_1 = 1$, the location of f_1 , the liquid-phase fugacity of pure species 1, is not established. The tangent line at the origin, representing *Henry's law*, provides alternative information. The slope of the tangent line is *Henry's constant*, defined as

$$k_1 \equiv \lim_{x_1 \rightarrow 0} \frac{\hat{f}_1}{x_1} \quad (4-320)$$

This is the definition of k_1 for temperature T and for a pressure equal to the vapor pressure of the pure solvent P_2^{sat} .

The activity coefficient of the solute at infinite dilution is

$$\lim_{x_1 \rightarrow 0} \gamma_1 = \lim_{x_1 \rightarrow 0} \frac{\hat{f}_1}{x_1 f_1} = \frac{1}{f_1} \lim_{x_1 \rightarrow 0} \frac{\hat{f}_1}{x_1}$$

In view of Eq. (4-320), this becomes $\gamma_1^\infty = k_1/f_1$, or

$$f_1 = \frac{k_1}{\gamma_1^\infty} \quad (4-321)$$

where γ_1^∞ represents the infinite dilution value of the activity coefficient of the solute. Because both k_1 and γ_1^∞ are evaluated at P_2^{sat} , this pressure also applies to f_1 . However, the effect of P on a liquid-phase fugacity, given by a Poynting factor, is very small and for practical purposes may usually be neglected. The activity coefficient of the solute then becomes

$$\gamma_1 \equiv \frac{\hat{f}_1}{x_1 f_1} = \frac{y_1 P \hat{\phi}_1}{x_1 f_1} = \frac{y_1 P \hat{\phi}_1 \gamma_1^\infty}{x_1 k_1}$$

For the solute, this equation takes the place of Eqs. (4-303) and (4-304). Solution for y_1 gives

$$y_1 = \frac{x_1 (\gamma_1 / \gamma_1^\infty) k_1}{\hat{\phi}_1 P} \quad (4-322)$$

For the solvent, species 2, the analog of Eq. (4-319) is

$$y_2 = \frac{x_2 \gamma_2 P_2^{\text{sat}}}{\hat{\Phi}_2 P} \quad (4-323)$$

$$\text{Because } y_1 + y_2 = 1, \quad P = \frac{x_1 (\gamma_1 / \gamma_1^\infty) k_1}{\hat{\phi}_1} + \frac{x_2 \gamma_2 P_2^{\text{sat}}}{\hat{\Phi}_2} \quad (4-324)$$

The same correlation that provides for the evaluation of γ_1 also allows evaluation of γ_1^∞ .

There remains the problem of finding Henry's constant from the available VLE data.

$$\text{For equilibrium} \quad \hat{f}_1 = \hat{f}_1^v = \hat{f}_1^g = y_1 P \hat{\phi}_1$$

$$\text{Division by } x_1 \text{ gives} \quad \frac{\hat{f}_1}{x_1} = P \hat{\phi}_1 \frac{y_1}{x_1}$$

Henry's constant is defined as the limit as $x_1 \rightarrow 0$ of the ratio on the left; therefore

$$k_1 = P_2^{\text{sat}} \hat{\phi}_1^\infty \lim_{x_1 \rightarrow 0} \frac{y_1}{x_1}$$

The limiting value of y_1/x_1 can be found by plotting y_1/x_1 versus x_1 and extrapolating to zero.

K Values, VLE, and Flash Calculations A measure of the distribution of a chemical species between liquid and vapor phases is the K value, defined as the equilibrium ratio:

$$K_i \equiv \frac{y_i}{x_i} \quad (4-325)$$

It has no thermodynamic content, but may make for computational convenience through elimination of one set of mole fractions in favor of the other. It does characterize "lightness" of a constituent species. A "light" species, with $K > 1$, tends to concentrate in the vapor phase whereas a "heavy" species, with $K < 1$, tends to concentrate in the liquid phase.

The rigorous evaluation of a K value follows from Eq. (4-299):

$$K_i \equiv \frac{y_i}{x_i} = \frac{\gamma_i \hat{f}_i}{\hat{\phi}_i P} \quad (4-326)$$

When Raoult's law applies, Eq. (4-326) reduces to $K_i = P_i^{\text{sat}}/P$. For modified Raoult's law, $K_i = \gamma_i P_i^{\text{sat}}/P$. With $K_i = y_i/x_i$, these are alternative expressions of Raoult's law and modified Raoult's law. Were Raoult's valid, K values could be correlated as functions of just T and P . However, Eq. (4-326) shows that they are in general functions of T , P , $\{x_i\}$, and $\{y_i\}$, making convenient and accurate correlation impossible. Those correlations that do exist are approximate and severely limited in application. The nomographs for K values of light hydrocarbons as functions of T and P , prepared by DePriester [*Chem. Eng. Progr. Symp. Ser. No. 7*, **49**: 1–43 (1953)], do allow for an *average* effect of composition, but their essential basis is Raoult's law.

The defining equation for K can be rearranged as $y_i = K_i x_i$. The sum $\sum y_i = 1$ then yields

$$\sum_i K_i x_i = 1 \quad (4-327)$$

With the alternative rearrangement $x_i = y_i/K_i$, the sum $\sum x_i = 1$ yields

$$\sum_i \frac{y_i}{K_i} = 1 \quad (4-328)$$

Thus for bubble point calculations, where the x_i are known, the problem is to find the set of K values that satisfies Eq. (4-327), whereas for dew point calculations, where the y_i are known, the problem is to find the set of K values that satisfies Eq. (4-328).

The *flash calculation* is a very common application of VLE. Considered here is the P , T flash, in which are calculated the quantities and compositions of the vapor and liquid phases in equilibrium at known T , P , and *overall* composition. This problem is determinate on the basis of Duhem's theorem: *For any closed system formed initially from*

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given masses of prescribed chemical species, the equilibrium state is completely determined when any two independent variables are fixed. The independent variables are here T and P , and systems are formed from given masses of nonreacting chemical species.

For 1 mol of a system with overall composition represented by the set of mole fractions $\{z_i\}$, let $\mathcal{L}$ represent the molar fraction of the system that is liquid (mole fractions $\{x_i\}$) and let $\mathcal{V}$ represent the molar fraction that is vapor (mole fractions $\{y_i\}$). The material balance equations are

$$\mathcal{L} + \mathcal{V} = 1 \quad \text{and} \quad z_i = x_i \mathcal{L} + y_i \mathcal{V} \quad i = 1, 2, \dots, N$$

Combining these equations to eliminate $\mathcal{L}$ gives

$$z_i = x_i(1 - \mathcal{V}) + y_i \mathcal{V} \quad i = 1, 2, \dots, N \quad (4-329)$$

Substitute $x_i = y_i/K_i$ and solve for y_i :

$$y_i = \frac{z_i K_i}{1 + \mathcal{V}(K_i - 1)} \quad i = 1, 2, \dots, N$$

Because $\sum y_i = 1$, this equation, summed over all species, yields

$$\sum_i \frac{z_i K_i}{1 + \mathcal{V}(K_i - 1)} = 1 \quad (4-330)$$

The initial step in solving a P, T flash problem is to find the value of $\mathcal{V}$ which satisfies this equation. Note that $\mathcal{V} = 1$ is always a *trivial* solution.

Example 4: Flash Calculation The system of Example 3 has the overall composition $z_1 = 0.4000$ at $T = 325.15$ K and $P = 101.33$ kPa. Determine $\mathcal{V}$, x_1 , and y_1 .

The *BUBL P* and *DEW P* calculations at $T = 325.15$ K of Example 3a and 3b show that for $x_1 = z_1$, $P_{\text{bubl}} = 108.134$ kPa, and for $y_1 = z_1$, $P_{\text{dew}} = 87.939$ kPa. Because P here lies between these values, the system is in two-phase equilibrium, and a flash calculation is appropriate.

The modified Raoult's law K values are given by

$$K_1 = \frac{(\gamma_1)(P_1^{\text{sat}})}{P} \quad \text{and} \quad K_2 = \frac{(\gamma_2)(P_2^{\text{sat}})}{P}$$

Equation (4-329) may be solved for $\mathcal{V}$: $\mathcal{V} = \frac{z_1 - x_1}{y_1 - x_1}$

Equation (4-330) here becomes

$$\frac{(z_1)(K_1)}{1 + \mathcal{V}(K_1 - 1)} + \frac{(z_2)(K_2)}{1 + \mathcal{V}(K_2 - 1)} = 1$$

A trial calculation illustrates the nature of the solution. Vapor pressures are taken from Example 3a or 3b; a trial value of x_1 then allows calculation of γ_1 and γ_2 by Eqs. (B) and (C) of Example 3. The values of K_1 , K_2 , and $\mathcal{V}$ that result are substituted into the summation equation. In the unlikely event that the sum is indeed unity, the chosen value of x_1 is correct. If not, then successive trials easily lead to this value. Note that the trivial solution giving $\mathcal{V} = 1$ must be avoided. More elegant solution procedures can of course be employed. The answers are

$$x_1 = 0.2373 \quad y_1 = 0.5190 \quad \mathcal{V} = 0.5775$$

with $\gamma_1 = 2.5297 \quad \gamma_2 = 1.0997 \quad K_1 = 2.1873 \quad K_2 = 0.6306$

Equation-of-State Approach Although the gamma/phi approach to VLE is in principle generally applicable to systems comprised of subcritical species, in practice it has found use primarily where pressures are no more than a few bars. Moreover, it is most satisfactory for correlation of constant-temperature data. A temperature dependence for the parameters in expressions for G^E is included only for the local composition equations, and it is at best approximate.

A generally applicable alternative to the gamma/phi approach results when both the liquid and vapor phases are described by the same equation of state. The defining equation for the fugacity coefficient, Eq. (4-204), may be applied to each phase:

$$\text{Liquid: } \hat{f}_i^L = \hat{\phi}_i^L x_i P \quad \text{Vapor: } \hat{f}_i^V = \hat{\phi}_i^V y_i P$$

By Eq. (4-298), $x_i \hat{\phi}_i^L = y_i \hat{\phi}_i^V \quad i = 1, 2, \dots, N \quad (4-331)$

This introduces compositions x_i and y_i into the equilibrium equations, but neither is explicit, because the $\hat{\phi}_i$ are functions, not only of T and

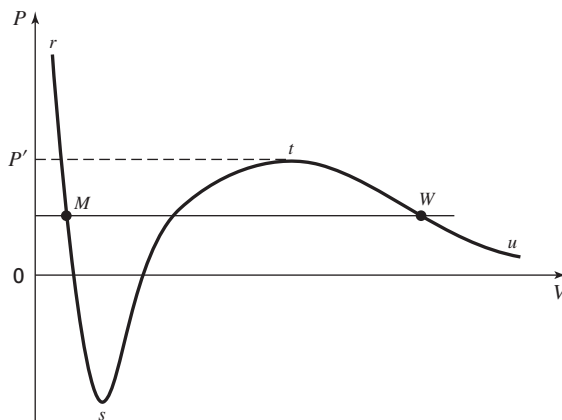


FIG. 4-8 A subcritical isotherm on a PV diagram for a pure fluid. [Smith, Van Ness, and Abbott, *Introduction to Chemical Engineering Thermodynamics*, 7th ed., p. 557, McGraw-Hill, New York (2005).]

P , but of composition. Thus, Eq. (4-331) represents N complex relationships connecting T , P , $\{x_i\}$, and $\{y_i\}$.

Two widely used cubic equations of state appropriate for VLE calculations, both special cases of Eq. (4-100) [with Eqs. (4-101) and (4-102)], are the Soave-Redlich-Kwong (SRK) equation and the Peng-Robinson (PR) equation. The present treatment is applicable to both. The pure numbers ϵ , σ , Ψ , and Ω and expressions for $\alpha(T_i)$ specific to these equations are listed in Table 4-2. The associated expression for ϕ_i is given by Eq. (4-246).

The simplest application of equations of state in vapor/liquid equilibrium is to the calculation of vapor pressures P_i^{sat} of pure liquids. Vapor pressures can of course be measured, but values are also implicit in cubic equations of state.

A subcritical PV isotherm, generated by a cubic equation of state, is shown in Fig. 4-8. Three segments are evident. The very steep one on the left (rs) is characteristic of liquids. Note that as $P \rightarrow \infty, V \rightarrow b$, where b is a constant in the cubic equation. The gently sloping segment on the right (tu) is characteristic of vapors; here $P \rightarrow 0$ as $V \rightarrow \infty$. The middle segment (st), with both a minimum (note $P < 0$) and a maximum, provides a transition from liquid to vapor, but has no physical meaning. The actual transition occurs along a horizontal line, such as connects points M and W .

For pure species i , Eq. (4-331) reduces to $\phi_i^L = \phi_i^V$, which may be written as

$$\ln \phi_i^L = \ln \phi_i^V \quad (4-332)$$

For given T , line MW lies at the vapor pressure P_i^{sat} if and only if the fugacity coefficients for points M and W satisfy Eq. (4-332). These points then represent *saturated* liquid and vapor phases in equilibrium at temperature T .

The fugacity coefficients in Eq. (4-332) are given by Eq. (4-245):

$$\ln \phi_i^p = Z_i^p - 1 - \ln(Z_i^p - \beta_i) - q_i I_i^p \quad p = l, v \quad (4-333)$$

Expressions for Z_i^v and Z_i^l come from Eqs. (4-104):

$$Z_i^v = 1 + \beta_i - q_i \beta_i \frac{Z_i^v - \beta_i}{(Z_i^v + \epsilon \beta_i)(Z_i^v + \sigma \beta_i)} \quad (4-334)$$

$$Z_i^l = \beta_i + (Z_i^l + \epsilon \beta_i)(Z_i^l + \sigma \beta_i) \left(\frac{1 + \beta_i - Z_i^l}{q_i \beta_i} \right) \quad (4-335)$$

and I_i^p comes from Eq. (4-112):

$$I_i^p = \frac{1}{\sigma - \epsilon} \ln \frac{Z_i^p + \sigma \beta_i}{Z_i^p + \epsilon \beta_i} \quad p = l, v \quad (4-336)$$

The equation-of-state parameters are independent of phase. As defined by Eq. (4-105), β_i is a function of P and here becomes

$$\beta_i \equiv \frac{b_i P_i^{\text{sat}}}{RT} \quad (4-337)$$

The remaining equation-of-state parameters, given by Eqs. (4-101), (4-102), and (4-106), are functions of T only and are written here as

$$a_i(T) = \psi \frac{\alpha(T_{c_i}) R^2 T_{c_i}^2}{P_{c_i}} \quad (4-338)$$

$$b_i = \Omega \frac{RT_{c_i}}{P_{c_i}} \quad (4-339)$$

$$q_i \equiv \frac{a_i(T)}{b_i RT} \quad (4-340)$$

The eight equations (4-332) through (4-337) may be solved for the eight unknowns P_i^{sat} , β_i , Z_i^l , Z_i^v , I_i^l , I_i^v , $\ln \phi_i^l$, and $\ln \phi_i^v$.

Perhaps more useful is the reverse calculation whereby an equation-of-state parameter is evaluated from a known vapor pressure. Thus, Eqs. (4-332) and (4-333) may be combined and solved for q_i , yielding

$$q_i = \frac{Z_i^v - Z_i^l + \ln [(Z_i^l - \beta_i)/(Z_i^v - \beta_i)]}{I_i^v - I_i^l} \quad (4-341)$$

Expressions for Z_i^l , Z_i^v , I_i^l , I_i^v , and β_i are given by Eqs. (4-334) through (4-337). Because Z_i^l and Z_i^v depend on q_i , an iterative procedure is indicated, with a starting value for q_i from a generalized correlation as given by Eqs. (4-338), (4-339), and (4-340).

For mixtures the presumption is that the equation of state has exactly the same form as when written for pure species. Equations (4-104) are therefore applicable, with parameters β and q given by Eqs. (4-105) and (4-106). Here, these parameters, and therefore b and $a(T)$, are functions of composition. Liquid and vapor mixtures in equilibrium in general have different compositions. The PV isotherms generated by an equation of state for these different compositions are represented in Fig. 4-9 by two similar lines: the solid line for the liquid-phase composition and the dashed line for the vapor-phase composition. They are displaced from each other because the equation-of-state parameters are different for the two compositions.

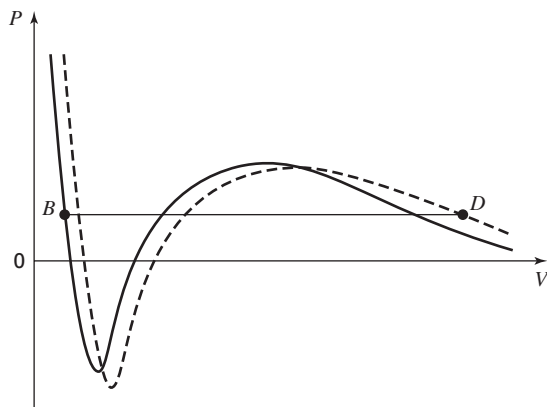


FIG. 4-9 Two PV isotherms at the same T for mixtures. The solid line is for a liquid-phase composition; the dashed line is for a vapor-phase composition. Point B represents a bubble point with the liquid-phase composition; point D represents a dew point with the vapor-phase composition. When these points lie at the same P (as shown), they represent phases in equilibrium. [Smith, Van Ness, and Abbott, *Introduction to Chemical Engineering Thermodynamics*, 7th ed., p. 560, McGraw-Hill, New York (2005).]

Each line includes three segments as described for the isotherm of Fig. 4-8: the leftmost segment representing a liquid phase and the rightmost segment, a vapor phase, both with the same composition. Each left segment contains a bubble point (saturated liquid), and each right segment contains a dew point (saturated vapor). Because these points for a given line are for the same composition, they do not represent phases in equilibrium and do not lie at the same pressure. Shown in Fig. 4-9 is a bubble point B on the solid line and a dew point D on the dashed line. Because they lie at the same P , they represent phases in equilibrium, and the lines are characterized by the liquid and vapor compositions.

For a $BUBL P$ calculation, the temperature and the liquid composition are known, and this fixes the location of the PV isotherm for the composition of the liquid phase (solid line). The problem then is to locate a second (dashed) line for a vapor composition such that the line contains a dew point D on its vapor segment that lies at the pressure of the bubble point B on the liquid segment of the solid line. This pressure is the phase equilibrium pressure, and the composition for the dashed line is that of the equilibrium vapor. This equilibrium condition is shown by Fig. 4-9.

In the absence of a theory to prescribe the composition dependence of parameters for cubic equations of state, empirical mixing rules are used to relate mixture parameters to pure-species parameters. The simplest realistic expressions are a linear mixing rule for parameter b and a quadratic mixing rule for parameter a , as shown by Eqs. (4-113) and (4-114). A common combining rule is given by Eq. (4-115). The general mole fraction variable x_i is used here because application is to both liquid and vapor mixtures. These equations, known as van der Waals prescriptions, provide for the evaluation of mixture parameters solely from parameters for the pure constituent species. They find application primarily for mixtures comprised of simple and chemically similar molecules.

Useful in the application of cubic equations of state to mixtures are partial equation-of-state parameters. For the parameters of the generic cubic, represented by Eqs. (4-104), (4-105), and (4-106), the definitions are

$$\bar{a}_i \equiv \left[\frac{\partial(na)}{\partial n_i} \right]_{T, n_j} \quad (4-342)$$

$$\bar{b}_i \equiv \left[\frac{\partial(nb)}{\partial n_i} \right]_{T, n_j} \quad (4-343)$$

$$\bar{q}_i \equiv \left[\frac{\partial(nq)}{\partial n_i} \right]_{T, n_j} \quad (4-344)$$

These are general equations, valid regardless of the particular mixing or combining rules adopted for the composition dependence of mixture parameters.

Parameter q is defined in relation to parameters a and b by Eq. (4-106). Thus,

$$nq = \frac{n(na)}{RT(nb)}$$

$$\text{whence} \quad \bar{q}_i \equiv \left[\frac{\partial(nq)}{\partial n_i} \right]_{T, n_j} = q \left(1 + \frac{\bar{a}_i}{a} - \frac{\bar{b}_i}{b} \right) \quad (4-345)$$

Any two of the three partial parameters form an independent pair, and any one of them can be found from the other two. Because q , a , and b are not linearly related, $\bar{q}_i \neq \bar{a}_i/\bar{b}_i RT$.

Values of ϕ_i^l and ϕ_i^v as given by Eq. (4-246) are implicit in an equation of state and with Eq. (4-331) allow calculation of mixture VLE. Although more complex, the same basic principle applies as for pure-species VLE. With ϕ_i^l a function of T , P , and $\{x_i\}$, and ϕ_i^v a function of T , P , and $\{y_i\}$, Eq. (4-331) represents N relations among the $2N$ variables: T , P , $(N-1) x_i$ s, and $(N-1) y_i$ s. Thus, specification of N of these variables, usually either T or P and either the liquid- or vapor-phase composition, allows solution for the remaining N variables by $BUBL P$, $DEW P$, $BUBL T$, and $DEW T$ calculations.

Because of limitations inherent in empirical mixing and combining rules, such as those given by Eqs. (4-113) through (4-115), the equation-of-state approach has found primary application to systems exhibiting modest deviations from ideal solution behavior in the liquid phase, e.g., to systems containing hydrocarbons and cryogenic fluids. However, since 1990, extensive research has been devoted to developing mixing rules that incorporate the excess Gibbs energy or activity coefficient data available for many systems. The extensive literature on this subject is reviewed by Valderrama [*Ind. Eng. Chem. Res.* **42**: 1603–1618 (2003)] and by Twu, Sim, and Tassone [*Chem. Eng. Progress* **98**(11): 58–65 (Nov. 2002)].

The idea here is to exploit the connection between fugacity coefficients and activity coefficients provided by their definitions:

$$\gamma_i \equiv \frac{\hat{f}_i}{x_i f_i} = \frac{\hat{f}_i/x_i P}{f_i/P} = \frac{\hat{\phi}_i}{\phi_i}$$

Therefore, $\ln \gamma_i = \ln \hat{\phi}_i - \ln \phi_i$ (4-346)

Because γ_i is a liquid-phase property, this equation is written for the liquid phase. Substituting for $\ln \hat{\phi}_i$ and $\ln \phi_i$ by Eqs. (4-245) and Eq. (4-246) gives

$$\ln \gamma_i = \frac{b_i}{b} (Z - 1) - Z_i + 1 - \ln \frac{Z - \beta}{Z_i - \beta_i} - \bar{q}_i I + q_i I_i$$

Symbols without subscripts are mixture properties. Solution for $\bar{q}_i$ yields

$$\bar{q}_i = \frac{1}{I} \left[1 - Z_i + \frac{b_i}{b} (Z - 1) - \ln \frac{Z - \beta}{Z_i - \beta_i} + q_i I_i - \ln \gamma_i \right] \quad (4-347)$$

Because $\bar{q}_i$ is a partial property, the summability equation provides an exact mixing rule:

$$q = \sum_i x_i \bar{q}_i \quad (4-348)$$

Application of this equation in the solution of VLE problems is illustrated by Smith, Van Ness, and Abbott [*Introduction to Chemical Engineering Thermodynamics*, 7th ed., pp. 569–572, McGraw-Hill, New York (2005)].

Extrapolation of Data with Temperature Liquid-phase excess-property data for binary systems at near-ambient temperatures appear in the literature. They provide for the extrapolation of G^E correlations with temperature. The key relations are Eq. (4-250), written as

$$d\left(\frac{G^E}{RT}\right) = -\frac{H^E}{RT^2} dT \quad \text{constant } P, x$$

and the excess-property analog of Eq. (4-26):

$$dH^E = C_P^E dT \quad \text{constant } P, x$$

Integration of the first equation from T_0 to T gives

$$\frac{G^E}{RT} = \left(\frac{G^E}{RT}\right)_{T_0} - \int_{T_0}^T \frac{H^E}{RT^2} dT \quad (4-349)$$

Integration of the second equation from T_1 to T yields

$$H^E = H_1^E + \int_{T_1}^T C_P^E dT \quad (4-350)$$

In addition,

$$dC_P^E = \left(\frac{\partial C_P^E}{\partial T}\right)_{P,x} dT$$

Integrate from T_2 to T :

$$C_P^E = C_{P_2}^E + \int_{T_2}^T \left(\frac{\partial C_P^E}{\partial T}\right)_{P,x} dT$$

Combining this equation with Eqs. (4-349) and (4-350) leads to

$$\begin{aligned} \frac{G^E}{RT} = & \left(\frac{G^E}{RT}\right)_{T_0} - \left(\frac{H^E}{RT}\right)_{T_1} \left(\frac{T}{T_0} - 1\right) \frac{T_1}{T} \\ & - \frac{C_{P_2}^E}{R} \left[\ln \frac{T}{T_0} - \left(\frac{T}{T_0} - 1\right) \frac{T_1}{T} \right] - I \end{aligned} \quad (4-351)$$

where

$$I \equiv \int_{T_0}^T \frac{1}{RT^2} \int_{T_1}^T \left(\frac{\partial C_P^E}{\partial T}\right)_{P,x} dT dT$$

This general equation employs excess Gibbs energy data at temperature T_0 , excess enthalpy (heat-of-mixing) data at T_1 , and excess heat capacity data at T_2 . Integral I depends on the temperature dependence of C_P^E . Excess heat capacity data are uncommon, and the T dependence is rarely known. Assuming C_P^E independent of T makes the integral zero, and the closer T_0 and T_1 are to T , the less the influence of this assumption. When no information is available for C_P^E and excess enthalpy data are available at only a single temperature, C_P^E must be assumed zero. In this case only the first two terms on the right side of Eq. (4-351) are retained, and it more rapidly becomes imprecise as T increases.

For application of Eq. (4-351) to binary systems at *infinite dilution* of one of the constituent species, it is divided by the product $x_1 x_2$.

$$\begin{aligned} \frac{G^E}{x_1 x_2 RT} = & \left(\frac{G^E}{x_1 x_2 RT}\right)_{T_0} - \left(\frac{H^E}{x_1 x_2 RT}\right)_{T_1} \left(\frac{T}{T_0} - 1\right) \frac{T_1}{T} \\ & - \frac{C_{P_2}^E}{x_1 x_2 R} \left[\ln \frac{T}{T_0} - \left(\frac{T}{T_0} - 1\right) \frac{T_1}{T} \right] \end{aligned}$$

The assumption here is that C_P^E is independent of T , making $I = 0$. As shown by Smith, Van Ness and Abbott [*Introduction to Chemical Engineering Thermodynamics*, 7th ed., p. 437, McGraw-Hill, New York (2005)],

$$\left(\frac{G^E}{x_1 x_2 RT}\right)_{x_i=0} \equiv \ln \gamma_i^\infty$$

The preceding equation may therefore be written as

$$\begin{aligned} \ln \gamma_i^\infty = & (\ln \gamma_i^\infty)_{T_0} - \left(\frac{H^E}{x_1 x_2 RT}\right)_{T_1, x_i=0} \left(\frac{T}{T_0} - 1\right) \frac{T_1}{T} \\ & - \left(\frac{C_{P_2}^E}{x_1 x_2 R}\right)_{x_i=0} \left[\ln \frac{T}{T_0} - \left(\frac{T}{T_0} - 1\right) \frac{T_1}{T} \right] \end{aligned} \quad (4-352)$$

Example 5: VLE at Several Temperatures For the methanol(1)/acetone(2) system at a base temperature of $T_0 = 323.15$ K (50°C), both VLE data [Van Ness and Abbott, *Int. DATA Ser., Ser. A, Sel. Data Mixtures*, **1978**: 67 (1978)] and excess enthalpy data [Morris et al., *J. Chem. Eng. Data* **20**: 403–405 (1975)] are available. The VLE data are well correlated by the Margules equations. As noted in connection with Eq. (4-270), parameters A_{12} and A_{21} relate directly to infinite dilution values of the activity coefficients. Thus, we have from the VLE data at 323.15 K:

$$A_{12} = \ln \gamma_1^\infty = 0.6281 \quad \text{and} \quad A_{21} = \ln \gamma_2^\infty = 0.6557$$

These values allow calculation of equilibrium pressures through Eqs. (4-270) and (4-308) for comparison with the measured pressures of the data set. Values of P_i^{sat} required in Eq. (4-308) are the measured values reported with the data set. The root-mean-square (rms) value of the pressure differences is given in Table 4-7 as 0.08 kPa, thus confirming the suitability of the Margules equation for this system. Vapor-phase mole fractions were not reported; hence no value can be given for rms δy_i .

Experimental VLE data at 372.8 and 397.7 K are given by Wilsak et al. [*Fluid Phase Equilib.* **28**: 13–37 (1986)]. Values of $\ln \gamma_i^\infty$ and hence of the Margules parameters for these higher temperatures are found from Eq. (4-352) with $C_P^E = 0$. The required excess enthalpy values at T_0 are

$$\left(\frac{H^E}{x_1 x_2 RT}\right)_{T_0, x_i=0} = 1.3636 \quad \text{and} \quad \left(\frac{H^E}{x_1 x_2 RT}\right)_{T_0, x_2=0} = 1.0362$$

TABLE 4-7 VLE Results for Methanol(1)/Acetone(2)

T, K	$A_{12} = \ln \gamma_1^\infty$	$A_{21} = \ln \gamma_2^\infty$	RMS $\delta P/\text{kPa}$	RMS % δP	RMS δy_1
323.15	0.6281	0.6557	0.08	0.12	
372.8	0.4465 (0.4607)	0.5177 (0.5271)	0.85 (0.83)	0.22	0.004 (0.006)
397.7	0.3725 (0.3764)	0.4615 (0.4640)	2.46 (1.39)	0.32	0.014 (0.013)

Results of calculations with the Margules equations are displayed as the primary entries at each temperature in Table 4-7. The values in parentheses are from the gamma/phi approach as reported in the papers cited.

Results for the higher temperatures indicate the quality of predictions based only on vapor-pressure data for the pure species and on mixture data at 323.15 K. Extrapolations based on the same data to still higher temperatures can be expected to become progressively less accurate.

Only the Wilson, NRTL, and UNIQUAC equations are suited to the treatment of multicomponent systems. For such systems, the parameters are determined for pairs of species exactly as for a binary system.

LIQUID/LIQUID AND VAPOR/LIQUID/LIQUID EQUILIBRIA

Equation (4-295) is the basis for both liquid/liquid equilibria (LLE) and vapor/liquid/liquid equilibria (VLLE). Thus for LLE with superscripts α and β denoting the two phases, Eq. (4-295) is written as

$$\hat{f}_i^\alpha = \hat{f}_i^\beta \quad i = 1, 2, \dots, N \quad (4-353)$$

Eliminating fugacities in favor of activity coefficients gives

$$x_i^\alpha \gamma_i^\alpha = x_i^\beta \gamma_i^\beta \quad i = 1, 2, \dots, N \quad (4-354)$$

For most LLE applications, the effect of pressure on the γ_i can be ignored, and Eq. (4-354) then constitutes a set of N equations relating equilibrium compositions to one another and to temperature. For a given temperature, solution of these equations requires a single expression for the composition dependence of G^E suitable for both liquid phases. Not all expressions for G^E suffice, even in principle, because some cannot represent liquid/liquid phase splitting. The UNIQUAC equation is suitable, and therefore prediction is possible by UNIFAC models. A special table of parameters for LLE calculations is given by Magnussen et al. [*Ind. Eng. Chem. Process Des. Dev.* **20**: 331–339 (1981)].

A comprehensive treatment of LLE is given by Sorensen et al. [*Fluid Phase Equilib.* **2**: 297–309 (1979); **3**: 47–82 (1979); **4**: 151–163 (1980)]. Data for LLE are collected in a three-part set compiled by Sorensen and Arlt [*Liquid-Liquid Equilibrium Data Collection*, Chemistry Data Series, vol. 5, parts 1–3, DECHEMA, Frankfurt am Main (1979–1980)].

For vapor/liquid/liquid equilibria, Eq. (4-295) becomes

$$\hat{f}_i^\alpha = \hat{f}_i^\beta = \hat{f}_i^v \quad i = 1, 2, \dots, N \quad (4-355)$$

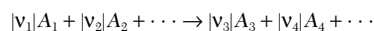
where α and β designate the two liquid phases. With activity coefficients applied to the liquid phases and fugacity coefficients to the vapor phase, the $2N$ equilibrium equations for subcritical VLLE are

$$\left. \begin{aligned} x_i^\alpha \gamma_i^\alpha \hat{f}_i^\alpha &= y_i \hat{\phi}_i P \\ x_i^\beta \gamma_i^\beta \hat{f}_i^\beta &= y_i \hat{\phi}_i P \end{aligned} \right\} \quad \text{all } i \quad (4-356)$$

As for LLE, an expression for G^E capable of representing liquid/liquid phase splitting is required; as for VLE, a vapor-phase equation of state for computing the $\hat{\phi}_i$ is also needed.

CHEMICAL REACTION STOICHIOMETRY

For a phase in which a chemical reaction occurs according to the equation



the $|v_i|$ are stoichiometric coefficients and the A_i stand for chemical formulas. The v_i themselves are called stoichiometric numbers, and associated with them is a sign convention such that the value is positive for a product and negative for a reactant. More generally, for a system containing N chemical species, any or all of which can participate in r chemical reactions, the reactions are represented by the equations

$$0 = \sum_i v_{ij} A_i \quad j = \text{I, II, } \dots, r \quad (4-357)$$

where $\text{sign}(v_{ij}) = \begin{cases} - & \text{for a reactant species} \\ + & \text{for a product species} \end{cases}$

If species i does not participate in reaction j , then $v_{ij} = 0$.

The stoichiometric numbers provide relations among the changes in mole numbers of chemical species which occur as the result of chemical reaction. Thus, for reaction j

$$\frac{\Delta n_{1j}}{v_{1j}} = \frac{\Delta n_{2j}}{v_{2j}} = \dots = \frac{\Delta n_{Nj}}{v_{Nj}} \quad (4-358)$$

All these terms are equal, and they can be equated to the change in a single quantity ϵ_j , called the *reaction coordinate* for reaction j , thereby giving

$$\Delta n_{ij} = v_{ij} \Delta \epsilon_j \quad \begin{aligned} i &= 1, 2, \dots, N \\ j &= \text{I, II, } \dots, r \end{aligned} \quad (4-359)$$

Because the total change in mole number Δn_i is just the sum of the changes Δn_{ij} resulting from the various reactions,

$$\Delta n_i = \sum_j \Delta n_{ij} = \sum_j v_{ij} \Delta \epsilon_j \quad i = 1, 2, \dots, N \quad (4-360)$$

If the initial number of moles of species i is n_{i0} and if the convention is adopted that $\epsilon_j = 0$ for each reaction in this initial state, then

$$n_i = n_{i0} + \sum_j v_{ij} \epsilon_j \quad i = 1, 2, \dots, N \quad (4-361)$$

Equation (4-361) is the basic expression of material balance for a closed system in which r chemical reactions occur. It shows for a reacting system that at most r mole-number-related quantities ϵ_j are capable of independent variation. It is not an equilibrium relation, but merely an accounting scheme, valid for tracking the progress of the reactions to arbitrary levels of conversion. The reaction coordinate has units of moles. A change in ϵ_j of 1 mol signifies a *mole of reaction*, meaning that reaction j has proceeded to such an extent that the change in mole number of each reactant and product is equal to its stoichiometric number.

CHEMICAL REACTION EQUILIBRIA

The general criterion of chemical reaction equilibria is given by Eq. (4-296). For a system in which just a single reaction occurs, Eq. (4-361) becomes

$$n_i = n_{i0} + v_i \epsilon \quad \text{whence} \quad dn_i = v_i d\epsilon$$

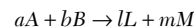
Substitution for dn_i in Eq. (4-296) leads to

$$\sum_i v_i \mu_i = 0 \quad (4-362)$$

Generalization of this result to multiple reactions produces

$$\sum_i v_{ij} \mu_i = 0 \quad j = \text{I, II, } \dots, r \quad (4-363)$$

Standard Property Changes of Reaction For the reaction



a *standard* property change is defined as the property change resulting when a mol of A and b mol of B in their *standard states at temperature*

T react to form l mol of L and m mol of M in their *standard states also at temperature T* . A *standard state* of species i is its real or hypothetical state as a pure species *at temperature T* and at a standard state pressure P° . The standard property change of reaction j is given the symbol ΔM_j° and its general mathematical definition is

$$\Delta M_j^\circ = \sum_i \nu_{i,j} M_i^\circ \quad (4-364)$$

For species present as gases in the actual reactive system, the standard state is the pure *ideal gas* at pressure P° . For liquids and solids, it is usually the state of pure real liquid or solid at P° . The standard state pressure P° is fixed at 100 kPa. Note that the standard states may represent different physical states for different species; any or all the species may be gases, liquids, or solids.

The most commonly used standard property changes of reaction are

$$\Delta G_j^\circ = \sum_i \nu_{i,j} G_i^\circ = \sum_i \nu_{i,j} \mu_i^\circ \quad (4-365)$$

$$\Delta H_j^\circ = \sum_i \nu_{i,j} H_i^\circ \quad (4-366)$$

$$\Delta C_{p,j}^\circ = \sum_i \nu_{i,j} C_{p,i}^\circ \quad (4-367)$$

The standard Gibbs energy change of reaction ΔG_j° is used in the calculation of equilibrium compositions. The standard heat of reaction ΔH_j° is used in the calculation of the heat effects of chemical reaction, and the standard heat capacity change of reaction is used for extrapolating ΔH_j° and ΔG_j° with T . Numerical values for ΔH_j° and ΔG_j° are computed from tabulated formation data, and $\Delta C_{p,j}^\circ$ is determined from empirical expressions for the T dependence of the $C_{p,i}^\circ$ [see, e.g., Eq. (4-52)].

Equilibrium Constants For practical application, Eq. (4-363) must be reformulated. The initial step is elimination of the μ_i in favor of fugacities. Equation (4-199) for species i in its standard state is subtracted from Eq. (4-202) for species i in the equilibrium mixture, giving

$$\mu_i = G_i^\circ + RT \ln \hat{a}_i \quad (4-368)$$

where by definition $\hat{a}_i \equiv \hat{f}_i/f_i^\circ$ and is called an *activity*. Substitution of this equation into Eq. (4-364) yields, upon rearrangement,

$$\sum_i [\nu_{i,j} (G_i^\circ + RT \ln \hat{a}_i)] = 0$$

$$\text{or} \quad \sum_i \nu_{i,j} G_i^\circ + RT \sum_i \ln \hat{a}_i^{\nu_{i,j}} = 0$$

$$\text{or} \quad \ln \prod_i \hat{a}_i^{\nu_{i,j}} = \frac{-\sum_i \nu_{i,j} G_i^\circ}{RT}$$

The right side of this equation is a function of temperature only for given reactions and given standard states. Convenience suggests setting it equal to $\ln K_j$, whence

$$\prod_i \hat{a}_i^{\nu_{i,j}} = K_j \quad \text{all } j \quad (4-369)$$

$$\text{where, by definition,} \quad K_j = \exp\left(\frac{-\Delta G_j^\circ}{RT}\right) \quad (4-370)$$

Quantity K_j is the chemical reaction equilibrium constant for reaction j , and ΔG_j° is the corresponding standard Gibbs energy change of reaction [see Eq. (4-365)]. Although called a “constant,” K_j is a function of T , but only of T .

The activities in Eq. (4-369) provide the connection between the *equilibrium* states of interest and the *standard* states of the constituent species, for which data are presumed available. The standard states are always at the equilibrium temperature. Although the standard state need not be the same for all species, for a *particular* species it must be the state represented by both G_i° and the f_i° upon which activity $\hat{a}_i$ is based.

The application of Eq. (4-369) requires explicit introduction of composition variables. For gas-phase reactions this is accomplished through the fugacity coefficient

$$\hat{a}_i \equiv \frac{\hat{f}_i}{f_i^\circ} = \frac{y_i \hat{\phi}_i P}{f_i^\circ}$$

However, the standard state for gases is the ideal gas state at the standard state pressure, for which $f_i^\circ = P^\circ$. Therefore,

$$\hat{a}_i = \frac{y_i \hat{\phi}_i P}{P^\circ}$$

and Eq. (4-369) becomes

$$\prod_i (y_i \hat{\phi}_i)^{\nu_{i,j}} \left(\frac{P}{P^\circ}\right)^{\sum_i \nu_{i,j}} = K_j \quad \text{all } j \quad (4-371)$$

where $\nu_j \equiv \sum_i \nu_{i,j}$ and P° is the standard state pressure of 100 kPa, expressed in the same units used for P . The y_i 's may be eliminated in favor of equilibrium values of the reaction coordinates ϵ_j (see Example 6). Then, for fixed temperature Eqs. (4-371) relate the ϵ_j to P . In principle, specification of the pressure allows solution for the ϵ_j . However, the problem may be complicated by the dependence of the $\hat{\phi}_i$ on composition, i.e., on the ϵ_j . If the equilibrium mixture is assumed an ideal solution, then [Eq. (4-218)] each $\hat{\phi}_i$ becomes ϕ_i , the fugacity coefficient of pure species i at the mixture T and P . This quantity does not depend on composition and may be determined from experimental data, from a generalized correlation, or from an equation of state.

An important special case of Eq. (4-371) results for gas-phase reactions when the phase is assumed an ideal gas. In this event $\hat{\phi}_i = 1$, and

$$\prod_i (y_i)^{\nu_{i,j}} \left(\frac{P}{P^\circ}\right)^{\nu_j} = K_j \quad \text{all } j \quad (4-372)$$

In the general case the evaluation of the $\hat{\phi}_i$ requires an iterative process. An initial step is to set each $\hat{\phi}_i$ equal to unity and to solve the problem by Eq. (4-372). This provides a set of y_i values, allowing evaluation of the $\hat{\phi}_i$ by, for example, Eq. (4-243) or (4-246). Equation (4-371) can then be solved for a new set of y_i values, with the process continued to convergence.

For liquid-phase reactions, Eq. (4-369) is modified by introduction of the activity coefficient $\gamma_i \equiv \hat{f}_i/x_i f_i^\circ$, where x_i is the liquid-phase mole fraction. The activity is then

$$\hat{a}_i \equiv \frac{\hat{f}_i}{f_i^\circ} = \gamma_i x_i \left(\frac{f_i}{f_i^\circ}\right)$$

Both f_i and f_i° represent fugacity of pure liquid i at temperature T , but at pressures P and P° , respectively. Except in the critical region, pressure has little effect on the properties of liquids, and the ratio f_i/f_i° is often taken as unity. When this is not acceptable, this ratio is evaluated by the equation

$$\ln \frac{f_i}{f_i^\circ} = \frac{1}{RT} \int_{P^\circ}^P V_i dP \approx \frac{V_i(P - P^\circ)}{RT}$$

When the ratio f_i/f_i° is taken as unity, $\hat{a}_i = \gamma_i x_i$, and Eq. (4-369) becomes

$$\prod_i (\gamma_i x_i)^{\nu_{i,j}} = K_j \quad \text{all } j \quad (4-373)$$

Here the difficulty is to determine the γ_i 's, which depend on the x_i 's. This problem has not been solved for the general case. Two courses are open: the first is experiment; the second, assumption of solution ideality. In the latter case, $\gamma_i = 1$, and Eq. (4-373) reduces to

$$\prod_i (x_i)^{\nu_{i,j}} = K_j \quad \text{all } j \quad (4-374)$$

the “law of mass action.” The significant feature of Eqs. (4-372) and (4-374), the simplest expressions for gas- and liquid-phase reaction

equilibrium, is that the temperature-, pressure-, and composition-dependent terms are distinct and separate.

The effect of temperature on the equilibrium constant follows from Eq. (4-41) written for pure species j in its standard state (wherein the pressure P° is fixed):

$$\frac{d(G_j^\circ/RT)}{dT} = \frac{-H_j^\circ}{RT^2}$$

With Eqs. (4-365) and (4-366) this equation easily extends to relate standard property changes of reaction:

$$\frac{d(\Delta G_j^\circ/RT)}{dT} = \frac{-\Delta H_j^\circ}{RT^2} \quad (4-375)$$

In view of Eq. (4-370) this may also be written as

$$\frac{d \ln K_j}{dT} = \frac{\Delta H_j^\circ}{RT^2} \quad (4-376)$$

For an endothermic reaction, ΔH_j° is positive and K_j increases with increasing T ; for an exothermic reaction, it is negative and K_j decreases with increasing T .

Because the standard state pressure is constant, Eq. (4-28) may be extended to relate standard properties of reaction, yielding

$$d\Delta H_j^\circ = \Delta C_{P,j}^\circ dT \quad \text{and} \quad d\Delta S_j^\circ = \Delta C_{P,j}^\circ \frac{dT}{T}$$

Integration of these equations from reference temperature T_0 (usually 298.15 K) to temperature T gives

$$\Delta H^\circ = \Delta H_0^\circ + R \int_{T_0}^T \frac{\Delta C_P^\circ}{R} dT \quad (4-377)$$

$$\Delta S^\circ = \Delta S_0^\circ + R \int_{T_0}^T \frac{\Delta C_P^\circ}{R} \frac{dT}{T} \quad (4-378)$$

where for simplicity subscript j has been suppressed. The definition of G leads directly to $\Delta G^\circ = \Delta H^\circ - T \Delta S^\circ$. Combining this equation with Eqs. (4-370), (4-377), and (4-378) yields

$$\ln K = \frac{-\Delta G^\circ}{RT} = \frac{-\Delta H_0^\circ}{RT} - \frac{1}{T} \int_{T_0}^T \frac{\Delta C_P^\circ}{R} dT + \frac{\Delta S_0^\circ}{R} + \int_{T_0}^T \frac{\Delta C_P^\circ}{R} \frac{dT}{T}$$

Substituting $\Delta S_0^\circ = (\Delta G_0^\circ - \Delta H_0^\circ)/T_0$, rearranging, and defining $\tau \equiv T/T_0$ give finally

$$\ln K = \frac{-\Delta G_0^\circ}{RT_0} + \frac{\Delta H_0^\circ}{RT_0} \left(\frac{\tau - 1}{\tau} \right) - \frac{1}{T} \int_{T_0}^T \frac{\Delta C_P^\circ}{R} dT + \int_{T_0}^T \frac{\Delta C_P^\circ}{R} \frac{dT}{T} \quad (4-379)$$

When heat capacity equations have the form of Eq. (4-52), the integrals are evaluated by equations of exactly the form of Eqs. (4-53) and (4-54), but with parameters A , B , C , and D replaced by ΔA , ΔB , ΔC , and ΔD , in accord with Eq. (4-364). Thus for the ideal gas standard state

$$\begin{aligned} \int_{T_0}^T \frac{\Delta C_P^\circ}{R} dT &= \Delta A T_0 (\tau - 1) + \frac{\Delta B}{2} T_0^2 (\tau^2 - 1) + \frac{\Delta C}{3} T_0^3 (\tau^3 - 1) \\ &\quad + \frac{\Delta D}{T_0} \left(\frac{\tau - 1}{\tau} \right) \end{aligned} \quad (4-380)$$

$$\int_{T_0}^T \frac{\Delta C_P^\circ}{R} \frac{dT}{T} = \Delta A \ln \tau + \left[\Delta B T_0 + \left(\Delta C T_0^2 + \frac{\Delta D}{\tau^2 T_0^2} \right) \left(\frac{\tau + 1}{2} \right) \right] (\tau - 1) \quad (4-381)$$

Equations (4-379) through (4-381) together allow an equation to be written for $\ln K$ as a function of T for any reaction for which appropriate data are available.

In the more extensive compilations of data, values of ΔG° and ΔH° for formation reactions are given for a wide range of temperatures, rather than just at the reference temperature $T_0 = 298.15$ K. [See in particular *TRC Thermodynamic Tables—Hydrocarbons* and *TRC Thermodynamic Tables—Non-hydrocarbons*, serial publications of the Thermodynamics Research Center, Texas A & M Univ. System, College Station, Tex.; "The NBS Tables of Chemical Thermodynamic Properties," *J. Phys. Chem. Ref. Data* **11**, supp. 2 (1982).] Where data are lacking, methods of estimation are available; these are reviewed by Poling, Prausnitz, and O'Connell, *The Properties of Gases and Liquids*, 5th ed., chap. 6, McGraw-Hill, New York, 2000. For an estimation procedure based on molecular structure, see Constantinou and Gani, *Fluid Phase Equilib.* **103**: 11–22 (1995). See also Sec. 2.

Example 6: Single-Reaction Equilibrium The hydrogenation of benzene to produce cyclohexane by the reaction



is carried out over a catalyst formulated to repress side reactions. Operating conditions cover a pressure range from 10 to 35 bar and a temperature range from 450 to 670 K. Reaction rate increases with increasing T , but because the reaction is exothermic the equilibrium conversion decreases with increasing T . A comprehensive study of the effect of operating variables on the chemical equilibrium of this reaction has been published by J. Carrero-Mantilla and M. Llano-Restrepo, *Fluid Phase Equilib.* **219**: 181–193 (2004). Presented here are calculations for a single set of operating conditions, namely, $T = 600$ K, $P = 15$ bar, and a molar feed ratio $\text{H}_2/\text{C}_6\text{H}_6 = 3$, the stoichiometric value. For these conditions we determine the fractional conversion of benzene to cyclohexane. Carrero-Mantilla and Llano-Restrepo express $\ln K$ as a function of T by an equation which for 600 K yields the value $K = 0.02874$.

A feed stream containing 3 mol H_2 for each 1 mol C_6H_6 is the basis of calculation, and for this single reaction, Eq. (4-361) becomes $n_i = n_{i0} + v_{i0}$, yielding

$$\begin{aligned} n_B &= 1 - \epsilon && \text{benzene} \\ n_H &= 3 - 3\epsilon && \text{hydrogen} \\ n_C &= \epsilon && \text{cyclohexane} \\ \sum_i n_i &= 4 - 3\epsilon \end{aligned}$$

Each mole fraction is therefore given by $y_i = n_i/(4 - 3\epsilon)$.

Assume first that the equilibrium mixture is an ideal gas, and apply Eq. (4-372), written for a single reaction, with subscript j omitted and $v = -3$:

$$\prod_i y_i^{v_i} \left(\frac{P}{P^\circ} \right)^v = \frac{\epsilon}{4 - 3\epsilon} \left(\frac{15}{1} \right)^{-3} = K = 0.02874$$

$$\text{whence} \quad \frac{\epsilon}{1 - \epsilon} \left(\frac{4 - 3\epsilon}{3 - 3\epsilon} \right)^3 (15)^{-3} = 0.02874 \quad \text{and} \quad \epsilon = 0.815$$

Thus, the assumption of ideal gases leads to a calculated conversion of 81.5 percent.

An alternative assumption is that the equilibrium mixture is an ideal solution. This requires application of Eq. (4-371). However, in the case of an ideal solution Eq. (4-218) indicates that $\phi_i^* = \phi_i$, in which case Eq. (4-371) for a single reaction becomes

$$\prod_i (y_i \phi_i)^{v_i} \left(\frac{P}{P^\circ} \right)^v = K$$

For purposes of illustration we evaluate the pure-species fugacity coefficients by Eq. (4-206), written here as

$$\phi_i = \exp(B_i^0 + \omega B_i^1) \frac{P_i}{T_i}$$

The following table shows values for the various quantities in this equation. Note that T_i and P_i for hydrogen are effective values as calculated by Eqs. (4-124) and (4-125) and used with $\omega = 0$.

	T_c	T_r	P_c	P_r	ω	B^0	B^1	ϕ
C_6H_6	562.2	1.067	48.98	0.306	0.21	-0.2972	0.008	0.919
H_2	42.8	14.009	19.78	0.758	0.00	0.0768	0.139	1.004
C_6H_{12}	553.6	1.084	40.73	0.368	0.21	-0.2880	0.016	0.908

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The equilibrium equation now becomes:

$$\prod_i (y_i \phi_i)^{v_i} \left(\frac{P}{P^\circ} \right)^v = \frac{\frac{\epsilon}{4-3\epsilon} \cdot 0.919}{\left(\frac{1-\epsilon}{4-3\epsilon} \cdot 0.908 \right) \left(\frac{3-3\epsilon}{4-3\epsilon} \cdot 1.004 \right)^3} \left(\frac{15}{1} \right)^{-3} = K = 0.02874$$

Solution yields

$$\epsilon = 0.816$$

This result is hardly different from that based on the ideal gas assumption. The fugacity coefficients in the equilibrium equation clearly cancel one another. This is not uncommon in reaction equilibrium calculations, as there are always products and reactions, making the ideal gas assumption far more useful than might be expected.

Carrero-Mantilla and Llano-Restrep present results for a wide range of conditions, both for the ideal gas assumption and for calculations wherein ϕ_i values are determined from the Soave-Redlich-Kwong equation of state. In no case are these calculated conversions significantly divergent.

Complex Chemical Reaction Equilibria When the composition of an equilibrium mixture is determined by a number of simultaneous reactions, calculations based on equilibrium constants become complex and tedious. A more direct procedure (and one suitable for general computer solution) is based on minimization of the total Gibbs energy G^t in accord with Eq. (4-293). The treatment here is limited to gas-phase reactions for which the problem is to find the equilibrium composition for given T and P and for a given initial feed.

1. Formulate the constraining material-balance equations, based on conservation of the total number of atoms of each *element* in a system comprised of w elements. Let subscript k identify a particular atom, and define A_k as the total number of atomic masses of the k th element in the feed. Further, let a_{ik} be the number of atoms of the k th element present in each molecule of chemical species i . The material balance for element k is then

$$\sum_i n_i a_{ik} = A_k \quad k = 1, 2, \dots, w \quad (4-382)$$

$$\text{or} \quad \sum_i n_i a_{ik} - A_k = 0 \quad k = 1, 2, \dots, w$$

2. Multiply each element balance by λ_k , a Lagrange multiplier:

$$\lambda_k \left(\sum_i n_i a_{ik} - A_k \right) = 0 \quad k = 1, 2, \dots, w$$

Summed over k , these equations give

$$\sum_k \lambda_k \left(\sum_i n_i a_{ik} - A_k \right) = 0$$

3. Form a function F by addition of this sum to G^t :

$$F = G^t + \sum_k \lambda_k \left(\sum_i n_i a_{ik} - A_k \right)$$

Function F is identical with G^t , because the summation term is zero. However, the partial derivatives of F and G^t with respect to n_i are different, because function F incorporates the constraints of the material balances.

4. The minimum value of both F and G^t is found when the partial derivatives of F with respect to n_i are set equal to zero:

$$\left(\frac{\partial F}{\partial n_i} \right)_{T,P,n_j} = \left(\frac{\partial G^t}{\partial n_i} \right)_{T,P,n_j} + \sum_k \lambda_k a_{ik} = 0$$

The first term on the right is the definition of the chemical potential; therefore,

$$\mu_i + \sum_k \lambda_k a_{ik} = 0 \quad i = 1, 2, \dots, N \quad (4-383)$$

However, the chemical potential is given by Eq. (4-368); for gas-phase reactions and standard states as the pure ideal gases at P° , this equation becomes

$$\mu_i = G_i^\circ + RT \ln \frac{\hat{f}_i}{P^\circ}$$

If G_i° is arbitrarily set equal to zero for all *elements* in their standard states, then for compounds $G_i^\circ = \Delta G_{f,i}^\circ$, the standard Gibbs energy change of formation of species i . In addition, the fugacity is eliminated—in favor of the fugacity coefficient by Eq. (4-204), $\hat{f}_i = y_i \phi_i P$. With these substitutions, the equation for μ_i becomes

$$\mu_i = \Delta G_{f,i}^\circ + RT \ln \frac{y_i \phi_i P}{P^\circ}$$

Combination with Eq. (4-383) gives

$$\Delta G_{f,i}^\circ + RT \ln \frac{y_i \phi_i P}{P^\circ} + \sum_k \lambda_k a_{ik} = 0 \quad i = 1, 2, \dots, N \quad (4-384)$$

If species i is an element, $\Delta G_{f,i}^\circ$ is zero. There are N equilibrium equations [Eqs. (4-384)], one for each chemical species, and there are w material balance equations [Eqs. (4-382)], one for each element—a total of $N + w$ equations. The unknowns in these equations are the n_i 's (note that $y_i = n_i / \sum n_i$), of which there are N , and the λ_k 's, of which there are w —a total of $N + w$ unknowns. Thus the number of equations is sufficient for the determination of all unknowns.

Equation (4-384) is derived on the presumption that the set $\{\phi_i\}$ is known. If the phase is an ideal gas, then each ϕ_i is unity. If the phase is an ideal solution, each ϕ_i becomes ϕ_i and can at least be estimated. For real gases, each ϕ_i is a function of the set $\{y_i\}$, the quantities being calculated. Thus an iterative procedure is indicated, initiated with each ϕ_i set equal to unity. Solution of the equations then provides a preliminary set $\{y_i\}$. For low pressures or high temperatures this result is usually adequate. Where it is not satisfactory, an equation of state with the preliminary set $\{y_i\}$ gives a new and more nearly correct set $\{\phi_i\}$ for use in Eq. (4-384). Then a new set $\{y_i\}$ is determined. The process is repeated to convergence. All calculations are well suited to computer solution.

In this procedure, the question of what chemical reactions are involved never enters directly into any of the equations. However, the choice of a set of species is entirely equivalent to the choice of a set of independent reactions among the species. In any event, a set of species or an equivalent set of independent reactions must always be assumed, and different assumptions produce different results.

A detailed example of a complex gas-phase equilibrium calculation is given by Smith, Van Ness, and Abbott [*Introduction to Chemical Engineering Thermodynamics*, 5th ed., Example 15.13, pp. 602–604; 6th ed., Example 13.14, pp. 511–513; 7th ed., Example 13.14, pp. 527–528, McGraw-Hill, New York (1996, 2001, 2005)]. General application of the method to multicomponent, multiphase systems is treated by Iglesias-Silva et al. [*Fluid Phase Equilib.* **210**: 229–245 (2003)] and by Sotyan, Ghajar, and Gasem [*Ind. Eng. Chem. Res.* **42**: 3786–3801 (2003)].

THERMODYNAMIC ANALYSIS OF PROCESSES

Real irreversible processes can be subjected to thermodynamic analysis. The goal is to calculate the efficiency of energy use or production and to show how wasted energy is apportioned among the steps of a process. The treatment here is limited to steady-state steady-flow processes, because of their predominance in chemical technology.

CALCULATION OF IDEAL WORK

In any steady-state steady-flow process *requiring* work, a minimum amount must be expended to bring about a specific change of state in the flowing fluid. In a process *producing* work, a maximum amount is

attainable for a specific change of state in the flowing fluid. In either case, the limiting value obtains when the specific change of state is accomplished *completely reversibly*. The implications of this requirement are that

1. The process is internally reversible within the control volume.
2. Heat transfer external to the control volume is reversible.

The second item means that heat exchange between system and surroundings must occur at the temperature of the surroundings, presumed to constitute a heat reservoir at a constant and uniform temperature T_σ . This may require Carnot engines or heat pumps internal to the system that provide for the reversible transfer of heat from the temperatures of the flowing fluid to that of the surroundings. Because Carnot engines and heat pumps are cyclic, they undergo no net change of state.

These conditions are implicit in the entropy balance of Eq. (4-156) when $\dot{S}_G = 0$. If in addition there is but a single surroundings temperature T_σ , this equation becomes

$$\Delta(\dot{S}\dot{m})_{fs} - \frac{\dot{Q}}{T_\sigma} = 0 \quad (4-385)$$

The energy balance for a steady-state steady-flow process as given by Eq. (4-150) is

$$\Delta\left[\left(H + \frac{1}{2}u^2 + zg\right)\dot{m}\right]_{fs} = \dot{Q} + \dot{W}_s \quad (4-150)$$

Combining this equation with Eq. (4-385) to eliminate $\dot{Q}$ yields

$$\Delta\left[\left(H + \frac{1}{2}u^2 + zg\right)\dot{m}\right]_{fs} = T_\sigma \Delta(\dot{S}\dot{m})_{fs} + \dot{W}_s(\text{rev})$$

where $\dot{W}_s(\text{rev})$ indicates that the shaft work is for a completely reversible process. This work is called the *ideal work* $\dot{W}_{\text{ideal}}$. Thus

$$\dot{W}_{\text{ideal}} = \Delta\left[\left(H + \frac{1}{2}u^2 + zg\right)\dot{m}\right]_{fs} - T_\sigma \Delta(\dot{S}\dot{m})_{fs} \quad (4-386)$$

In most applications to chemical processes, the kinetic and potential energy terms are negligible compared with the others; in this event Eq. (4-386) is written as

$$\dot{W}_{\text{ideal}} = \Delta(H\dot{m})_{fs} - T_\sigma \Delta(\dot{S}\dot{m})_{fs} \quad (4-387)$$

For the special case of a single stream flowing through the system, Eq. (4-387) becomes

$$\dot{W}_{\text{ideal}} = \dot{m}(\Delta H - T_\sigma \Delta S) \quad (4-388)$$

Division by $\dot{m}$ puts this equation on a unit-mass basis:

$$w_{\text{ideal}} = \Delta H - T_\sigma \Delta S \quad (4-389)$$

A completely reversible process is hypothetical, devised solely to find the ideal work associated with a given change of state. Its only connection with an actual process is that it brings about the same change of state as the actual process, allowing comparison of the actual work of a process with the work of the hypothetical reversible process.

Equations (4-386) through (4-389) give the work of a completely reversible process associated with given property changes in the flowing streams. When the same property changes occur in an actual process, the actual work $\dot{W}_s$ (or W_s) is given by an energy balance, and comparison can be made of the actual work with the ideal work. When $\dot{W}_{\text{ideal}}$ (or W_{ideal}) is positive, it is the *minimum work required* to bring about a given change in the properties of the flowing streams, and it is smaller than $\dot{W}_s$. In this case a thermodynamic efficiency η_t is defined as the ratio of the ideal work to the actual work:

$$\eta_t(\text{work required}) = \frac{\dot{W}_{\text{ideal}}}{\dot{W}_s} \quad (4-390)$$

When $\dot{W}_{\text{ideal}}$ (or W_{ideal}) is negative, $|\dot{W}_{\text{ideal}}|$ is the *maximum work obtainable* from a given change in the properties of the flowing streams, and it is larger than $|\dot{W}_s|$. In this case, the thermodynamic efficiency is defined as the ratio of the actual work to the ideal work:

$$\eta_t(\text{work produced}) = \frac{\dot{W}_s}{\dot{W}_{\text{ideal}}} \quad (4-391)$$

LOST WORK

Work that is wasted as the result of irreversibilities in a process is called *lost work* $\dot{W}_{\text{lost}}$, and it is defined as the difference between the actual work of a process and the ideal work for the process. Thus by definition,

$$W_{\text{lost}} \equiv W_s - W_{\text{ideal}} \quad (4-392)$$

$$\text{The rate form is} \quad \dot{W}_{\text{lost}} \equiv \dot{W}_s - \dot{W}_{\text{ideal}} \quad (4-393)$$

The actual work rate comes from Eq. (4-150):

$$\dot{W}_s = \Delta\left[\left(H + \frac{1}{2}u^2 + zg\right)\dot{m}\right]_{fs} - \dot{Q}$$

Subtracting the ideal work rate as given by Eq. (4-386) yields

$$\dot{W}_{\text{lost}} = T_\sigma \Delta(\dot{S}\dot{m})_{fs} - \dot{Q} \quad (4-394)$$

For the special case of a single stream flowing through the control volume,

$$\dot{W}_{\text{lost}} = \dot{m}T_\sigma \Delta S - \dot{Q} \quad (4-395)$$

Division of this equation by $\dot{m}$ gives

$$w_{\text{lost}} = T_\sigma \Delta S - Q \quad (4-396)$$

where the basis is now a unit amount of fluid flowing through the control volume.

The total rate of entropy generation (in both system and surroundings) as a result of a process is

$$\dot{S}_G = \Delta(\dot{S}\dot{m})_{fs} - \frac{\dot{Q}}{T_\sigma} \quad (4-397)$$

Division by $\dot{m}$ provides an equation based on a unit amount of fluid flowing through the control volume:

$$S_G = \Delta S - \frac{Q}{T_\sigma} \quad (4-398)$$

Equations (4-397) and (4-398) are special cases of Eqs. (4-156) and (4-157).

Multiplication of Eq. (4-397) by T_σ gives

$$T_\sigma \dot{S}_G = T_\sigma \Delta(\dot{S}\dot{m})_{fs} - \dot{Q}$$

Because the right sides of this equation and of Eq. (4-394) are identical, it follows that

$$\dot{W}_{\text{lost}} = T_\sigma \dot{S}_G \quad (4-399)$$

For flow on the basis of a unit amount of fluid, this becomes

$$w_{\text{lost}} = T_\sigma S_G \quad (4-400)$$

Because the second law of thermodynamics requires

$$\dot{S}_G \geq 0 \quad \text{and} \quad S_G \geq 0$$

therefore $\dot{W}_{\text{lost}} \geq 0$ and $w_{\text{lost}} \geq 0$

When a process is completely reversible, the equality holds and the lost work is zero. For irreversible processes the inequality holds, and the lost work, i.e., the energy that becomes unavailable for work, is positive. The engineering significance of this result is clear: The greater the irreversibility of a process, the greater the rate of entropy generation and the greater the amount of energy that becomes unavailable for work. Thus every irreversibility carries with it a price.

ANALYSIS OF STEADY-STATE STEADY-FLOW PROCESSES

Many processes consist of a number of steps, and lost-work calculations are then made for each step separately. Writing Eq. (4-399) for each step of the process and summing give

$$\sum \dot{W}_{\text{lost}} = T_\sigma \sum \dot{S}_G$$

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Dividing Eq. (4-399) by this result yields

$$\frac{\dot{W}_{\text{lost}}}{\sum \dot{W}_{\text{lost}}} = \frac{\dot{S}_G}{\sum \dot{S}_G}$$

Thus an analysis of the lost work, made by calculation of the fraction that each individual lost-work term represents of the total lost work, is the same as an analysis of the rate of entropy generation, made by expressing each individual entropy generation term as a fraction of the sum of all entropy generation terms.

An alternative to the lost-work or entropy generation analysis is a work analysis. This is based on Eq. (4-393), written as

$$\sum \dot{W}_{\text{lost}} = \dot{W}_s - \dot{W}_{\text{ideal}} \quad (4-401)$$

For a work-requiring process, all these work quantities are positive and $\dot{W}_s > \dot{W}_{\text{ideal}}$. The preceding equation is then expressed as

$$\dot{W}_s = \dot{W}_{\text{ideal}} + \sum \dot{W}_{\text{lost}} \quad (4-402)$$

A work analysis then gives each of the individual work terms in the summation on the right as a fraction of $\dot{W}_s$.

For a work-producing process, $\dot{W}_s$ and $\dot{W}_{\text{ideal}}$ are negative, and $|\dot{W}_{\text{ideal}}| > |\dot{W}_s|$. Equation (4-401) in this case is best written as

$$|\dot{W}_{\text{ideal}}| = |\dot{W}_s| + \sum \dot{W}_{\text{lost}} \quad (4-403)$$

A work analysis here expresses each of the individual work terms on the right as a fraction of $|\dot{W}_{\text{ideal}}|$. A work analysis cannot be carried out in the case where a process is so inefficient that $\dot{W}_{\text{ideal}}$ is negative, indicating that the process should produce work; but $\dot{W}_s$ is positive, indicating that the process in fact requires work. A lost-work or entropy generation analysis is always possible.

Example 7: Lost-Work Analysis A work analysis follows for a simple Linde system for the separation of air into gaseous oxygen and nitrogen, as depicted in Fig. 4-10. Table 4-8 lists a set of operating conditions for the numbered points of the diagram. Heat leaks into the column of 147 J/mol of entering air and into the exchanger of 70 J/mol of entering air have been assumed. Take $T_a = 300$ K.

The basis for analysis is 1 mol of entering air, assumed to contain 79 mol % N_2 and 21 mol % O_2 . By a material balance on the nitrogen, $0.79 = 0.9148x$, whence

$$\begin{aligned} x &= 0.8636 \text{ mol of nitrogen product} \\ 1 - x &= 0.1364 \text{ mol of oxygen product} \end{aligned}$$

Calculation of ideal work: If changes in kinetic and potential energies are neglected, Eq. (4-387) is applicable. From the tabulated data,

$$\begin{aligned} \Delta(H\dot{m})_s &= (13,460)(0.1364) + (12,074)(0.8636) - (12,407)(1) = -144 \text{ J} \\ \Delta(S\dot{m})_s &= (118.48)(0.1364) + (114.34)(0.8636) - (117.35)(1) = -2.4453 \text{ J/K} \end{aligned}$$

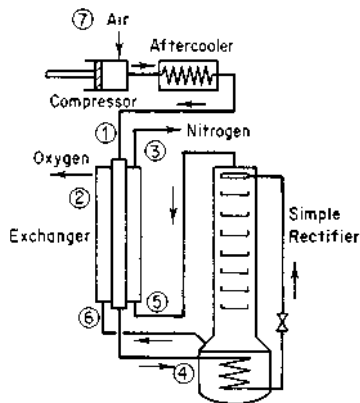


FIG. 4-10 Diagram of simple Linde system for air separation.

TABLE 4-8 States and Values of Properties for the Process of Fig. 4-10*

Point	P , bar	T , K	Composition	State	H , J/mol	S , J/(mol·K)
1	55.22	300	Air	Superheated	12,046	82.98
2	1.01	295	Pure O_2	Superheated	13,460	118.48
3	1.01	295	91.48% N_2	Superheated	12,074	114.34
4	55.22	147.2	Air	Superheated	5,850	52.08
5	1.01	79.4	91.48% N_2	Saturated vapor	5,773	75.82
6	1.01	90	Pure O_2	Saturated vapor	7,485	83.69
7	1.01	300	Air	Superheated	12,407	117.35

*Properties on the basis of Miller and Sullivan, U.S. Bur. Mines Tech. Pap. 424 (1928).

Thus, by Eq. (4-387),

$$\dot{W}_{\text{ideal}} = -144 - (300)(-2.4453) = 589.6 \text{ J}$$

Calculation of actual work of compression: For simplicity, the work of compression is calculated by the equation for an ideal gas in a three-stage reciprocating machine with complete intercooling and with isentropic compression in each stage. The work so calculated is assumed to represent 80 percent of the actual work. The following equation may be found in any number of textbooks on thermodynamics:

$$\dot{W}_s = \frac{n\gamma RT_1}{0.8(\gamma - 1)} \left[\left(\frac{P_2}{P_1} \right)^{(\gamma - 1)/n\gamma} - 1 \right]$$

where n = number of stages, here taken as 3

γ = ratio of heat capacities, here taken as 1.4

T_1 = initial absolute temperature, equal to 300 K

P_2/P_1 = overall pressure ratio, equal to 54.5

R = universal gas constant, equal to 8.314 J/(mol·K)

The efficiency factor of 0.8 is already included in the equation. Substitution of the remaining values gives

$$\dot{W}_s = \frac{(3)(1.4)(8.314)(300)}{(0.8)(0.4)} \left[(54.5)^{0.4/(3)(1.4)} - 1 \right] = 15,171 \text{ J}$$

The heat transferred to the surroundings during compression as a result of intercooling and aftercooling to 300 K is found from the first law:

$$\dot{Q} = \dot{m}(\Delta H) - \dot{W}_s = (12,046 - 12,407) - 15,171 = -15,532 \text{ J}$$

Calculation of lost work: Equation (4-394) may be applied to each of the major units of the process. For the compressor/cooler,

$$\begin{aligned} \dot{W}_{\text{lost}} &= (300)[(82.98)(1) - (117.35)(1)] - (-15,532) \\ &= 5221.0 \text{ J} \end{aligned}$$

For the exchanger,

$$\begin{aligned} \dot{W}_{\text{lost}} &= (300)[(118.48)(0.1364) + (114.34)(0.8636) + (52.08)(1) \\ &\quad - (75.82)(0.8636) - (83.69)(0.1364) - (82.98)(1)] - 70 \\ &= 2063.4 \text{ J} \end{aligned}$$

Finally, for the rectifier,

$$\dot{W}_{\text{lost}} = (300)[(75.82)(0.8636) + (83.69)(0.1364) - (52.08)(1)] - 147 = 7297.0 \text{ J}$$

Work analysis: Because the process requires work, Eq. (4-402) is appropriate for a work analysis. The various terms of this equation appear as entries in the following table and are on the basis of 1 mol of entering air.

			% of $\dot{W}_s$
$\dot{W}_{\text{ideal}}$		589.6 J	3.9
$\dot{W}_{\text{lost}}$	Compressor/cooler	5,221.0 J	34.4
$\dot{W}_{\text{lost}}$	Exchanger	2,063.4 J	13.6
$\dot{W}_{\text{lost}}$	Rectifier	7,297.0 J	48.1
$\dot{W}_s$		15,171.0 J	100.0

The thermodynamic efficiency of this process as given by Eq. (4-390) is only 3.9 percent. Significant inefficiencies reside with each of the primary units of the process.

SECTION 5

Heat and Mass
Transfer

PERRY'S CHEMICAL ENGINEERS' HANDBOOK

8TH EDITION



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Heat and Mass Transfer*

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*The contribution of James G. Knudsen, Ph.D., coeditor of this section in the seventh edition, is acknowledged.

[†]Professor H. C. Hottel was the principal author of the radiation section in this *Handbook*, from the first edition in 1934 through the seventh edition in 1997. His classic zone method remains the basis for the current revision.

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HEAT TRANSFER

GENERAL REFERENCES: Arpaci, *Conduction Heat Transfer*, Addison-Wesley, 1966. Arpaci, *Convection Heat Transfer*, Prentice-Hall, 1984. Arpaci, *Introduction to Heat Transfer*, Prentice-Hall, 1999. Baehr and Stephan, *Heat and Mass Transfer*, Springer, Berlin, 1998. Bejan, *Convection Heat Transfer*, Wiley, 1995. Carslaw and Jaeger, *Conduction of Heat in Solids*, Oxford University Press, 1959. Edwards, *Radiation Heat Transfer Notes*, Hemisphere Publishing, 1981. Hottel and Sarofim, *Radiative Transfer*, McGraw-Hill, 1967. Incropera and DeWitt, *Fundamentals of Heat and Mass Transfer*, 5th ed., Wiley, 2002. Kays and Crawford, *Convective Heat and Mass Transfer*, 3d ed., McGraw-Hill, 1993. Mills, *Heat Transfer*, 2d ed., Prentice-Hall, 1999. Modest, *Radiative Heat Transfer*, McGraw-Hill, 1993. Patankar, *Numerical Heat Transfer and Fluid Flow*, Taylor and Francis, London, 1980. Pletcher, Anderson, and Tannehill, *Computational Fluid Mechanics and Heat Transfer*, 2d ed., Taylor and Francis, London, 1997. Rohsenow, Hartnett, and Cho, *Handbook of Heat Transfer*, 3d ed., McGraw-Hill, 1998. Siegel and Howell, *Thermal Radiation Heat Transfer*, 4th ed., Taylor and Francis, London, 2001.

MODES OF HEAT TRANSFER

Heat is energy transferred due to a difference in temperature. There are three modes of heat transfer: conduction, convection, and radiation. All three may act at the same time. Conduction is the transfer of energy between adjacent particles of matter. It is a local phenomenon and can only occur through matter. Radiation is the transfer of energy from a point of higher temperature to a point of lower energy by electromagnetic radiation. Radiation can act at a distance through transparent media and vacuum. Convection is the transfer of energy by conduction and radiation in moving, fluid media. The motion of the fluid is an essential part of convective heat transfer.

HEAT TRANSFER BY CONDUCTION

FOURIER'S LAW

The heat flux due to conduction in the x direction is given by Fourier's law

$$\dot{Q} = -kA \frac{dT}{dx} \quad (5-1)$$

where $\dot{Q}$ is the rate of heat transfer (W), k is the thermal conductivity [W/(m·K)], A is the area perpendicular to the x direction, and T is temperature (K). For the homogeneous, one-dimensional plane shown in Fig. 5-1a, with constant k , the integrated form of (5-1) is

$$\dot{Q} = kA \frac{T_1 - T_2}{\Delta x} \quad (5-2)$$

where Δx is the thickness of the plane. Using the thermal circuit shown in Fig. 5-1b, Eq. (5-2) can be written in the form

$$\dot{Q} = \frac{T_1 - T_2}{\Delta x / kA} = \frac{T_1 - T_2}{R} \quad (5-3)$$

where R is the thermal resistance (K/W).

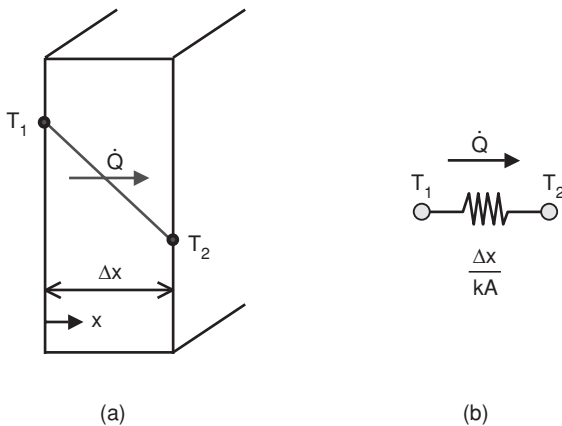


FIG. 5-1 Steady, one-dimensional conduction in a homogeneous planar wall with constant k . The thermal circuit is shown in (b) with thermal resistance $\Delta x/(kA)$.

THERMAL CONDUCTIVITY

The thermal conductivity k is a transport property whose value for a variety of gases, liquids, and solids is tabulated in Sec. 2. Section 2 also provides methods for predicting and correlating vapor and liquid thermal conductivities. The thermal conductivity is a function of temperature, but the use of constant or averaged values is frequently sufficient. Room temperature values for air, water, concrete, and copper are 0.026, 0.61, 1.4, and 400 W/(m·K). Methods for estimating contact resistances and the thermal conductivities of composites and insulation are summarized by Gebhart, *Heat Conduction and Mass Diffusion*, McGraw-Hill, 1993, p. 399.

STEADY-STATE CONDUCTION

One-Dimensional Conduction In the absence of energy source terms, $\dot{Q}$ is constant with distance, as shown in Fig. 5-1a. For steady conduction, the integrated form of (5-1) for a planar system with constant k and A is Eq. (5-2) or (5-3). For the general case of variables k (k is a function of temperature) and A (cylindrical and spherical systems with radial coordinate r ; as sketched in Fig. 5-2), the average heat-transfer area and thermal conductivity are defined such that

$$\dot{Q} = \bar{k}A \frac{T_1 - T_2}{\Delta x} = \frac{T_1 - T_2}{R} \quad (5-4)$$

For a thermal conductivity that depends linearly on T ,

$$k = k_0(1 + \gamma T) \quad (5-5)$$

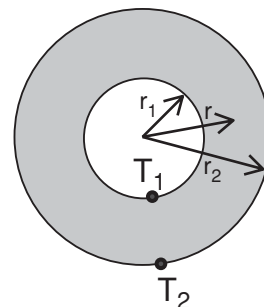


FIG. 5-2 The hollow sphere or cylinder.

5-4 HEAT AND MASS TRANSFER

Nomenclature and Units—Heat Transfer by Conduction, by Convection, and with Phase Change

Symbol	Definition	SI units	Symbol	Definition	SI units
A	Area for heat transfer	m^2	Ra_x	Rayleigh number, $\beta \Delta T g x^3 / \nu \alpha$	
A_c	Cross-sectional area	m^2	Re_D	Reynolds number, GD/μ	
A_f	Area for heat transfer for finned portion of tube	m^2	S	Volumetric source term	W/m^3
A_i	Inside area of tube		S	Cross-sectional area	m^2
A_o	External area of bare, unfinned tube	m^2	S_1	Fourier spatial function	
A_{of}	External area of tube before tubes are attached = A_o	m^2	t	Time	s
A_T	Total external area of finned tube	m^2	t_{sv}	Saturated-vapor temperature	K
A_{uf}	Area for heat transfer for unfinned portion of finned tube	m^2	T_s	Surface temperature	K
A_1	First Fourier coefficient		T	Temperature	K or °C
a_c	Cross-sectional area of fin	m^2	T_b	Bulk or mean temperature at a given cross section	K
b	Geometry; $b = 1$, plane; $b = 2$, cylinder; $b = 3$, sphere		$\bar{T}_b$	Bulk mean temperature, $(T_{b,in} + T_{b,out})/2$	K
b_f	Height of fin	m	T_C	Temperature of cold surface in enclosure	K
B_1	First Fourier coefficient		T_f	Film temperature, $(T_s + T_\infty)/2$	K
Bi	Biot number, hR/k		T_H	Temperature of hot surface in enclosure	K
c	Specific heat	$\text{J}/(\text{kg} \cdot \text{K})$	T_i	Initial temperature	K
c_p	Specific heat, constant pressure	$\text{J}/(\text{kg} \cdot \text{K})$	T_e	Temperature of free stream	K
D	Diameter	m	T_s	Temperature of surface	K
D_i	Inner diameter	m	T_∞	Temperature of fluid in contact with a solid surface	K
D_o	Outer diameter	m	U	Overall heat-transfer coefficient	$\text{W}/(\text{m}^2 \cdot \text{K})$
f	Fanning friction factor		V	Volume	m^3
Fo	Dimensionless time or Fourier number, $\alpha t/R^2$		V_F	Velocity of fluid approaching a bank of finned tubes	m/s
g_c	Conversion factor	$1.0 \text{ kg} \cdot \text{m}/(\text{N} \cdot \text{s}^2)$	V_∞	Velocity upstream of tube bank	m/s
g	Acceleration of gravity, 9.81 m/s^2	m/s^2	W_F	Total rate of vapor condensation on one tube	kg/s
$\bar{G}$	Mass velocity, $\dot{m}/A_c$; G_e for vapor mass velocity	$\text{kg}/(\text{m}^2 \cdot \text{s})$	x	Cartesian coordinate direction, characteristic dimension of a surface, or distance from entrance	m
G_{max}	Mass velocity through minimum free area between rows of tubes normal to the fluid stream	$\text{kg}/(\text{m}^2 \cdot \text{s})$	x	Vapor quality, x_i for inlet and x_o for outlet	
Gz	Graetz number = $Re \text{ Pr}$		$\tilde{z}_p$	Distance (perimeter) traveled by fluid across fin	m
$\bar{h}$	Heat-transfer coefficient	$\text{W}/(\text{m}^2 \cdot \text{K})$	Greek Symbols		
h	Average heat-transfer coefficient	$\text{W}/(\text{m}^2 \cdot \text{K})$	α	Thermal diffusivity, $k/(\rho c)$	m^2/s
h_f	Heat-transfer coefficient for finned-tube exchangers based on total external surface	$\text{W}/(\text{m}^2 \cdot \text{K})$	β	Volumetric coefficient of expansion	K^{-1}
h_f	Outside heat-transfer coefficient calculated for a bare tube for use with Eq. (5-73)	$\text{W}/(\text{m}^2 \cdot \text{K})$	β'	Contact angle between a bubble and a surface	°
h_{fi}	Effective outside heat-transfer coefficient based on inside area of a finned tube	$\text{W}/(\text{m}^2 \cdot \text{K})$	Γ	Mass flow rate per unit length perpendicular to flow	$\text{kg}/(\text{m} \cdot \text{s})$
h_i	Heat-transfer coefficient at inside tube surface	$\text{W}/(\text{m}^2 \cdot \text{K})$	ΔP	Pressure drop	Pa
h_o	Heat-transfer coefficient at outside tube surface	$\text{W}/(\text{m}^2 \cdot \text{K})$	Δt	Temperature difference	K
h_{am}	Heat-transfer coefficient for use with ΔT_{am} ; see Eq. (5-33)	$\text{W}/(\text{m}^2 \cdot \text{K})$	ΔT	Temperature difference	K
h_{im}	Heat-transfer coefficient for use with ΔT_{im} ; see Eq. (5-32)	$\text{W}/(\text{m}^2 \cdot \text{K})$	ΔT_{am}	Arithmetic mean temperature difference, see Eq. (5-32)	K
$\bar{k}$	Thermal conductivity	$\text{W}/(\text{m} \cdot \text{K})$	ΔT_{lm}	Logarithmic mean temperature difference, see Eq. (5-33)	K
k	Average thermal conductivity	$\text{W}/(\text{m} \cdot \text{K})$	Δx	Thickness of plane wall for conduction	m
L	Length of cylinder or length of flat plate in direction of flow or downstream distance. Length of heat-transfer surface	m	δ_1	First dimensionless eigenvalue	
m	Fin parameter defined by Eq. (5-75).		$\delta_{1,0}$	First dimensionless eigenvalue as Bi approaches 0	
$\dot{m}$	Mass flow rate	kg/s	$\delta_{1,\infty}$	First dimensionless eigenvalue as Bi approaches ∞	
Nu_D	Nusselt number based on diameter D , hD/k		δ_s	Correction factor, ratio of nonnewtonian to newtonian shear rates	
$\bar{Nu}_D$	Average Nusselt number based on diameter D , $\bar{h}D/k$		ε	Emissivity of a surface	
Nu_{im}	Nusselt number based on h_{im}		ζ	Dimensionless distance, r/R	
n'	Flow behavior index for nonnewtonian fluids		θ/θ_i	Dimensionless temperature, $(T - T_\infty)/(T_i - T_\infty)$	
p	Perimeter	m	λ	Latent heat (enthalpy) of vaporization (condensation)	J/kg
p_f	Fin perimeter	m	μ	Viscosity; μ_L , viscosity of liquid; μ_G , μ_g , μ_v , viscosity of gas or vapor	$\text{kg}/(\text{m} \cdot \text{s})$
p'	Center-to-center spacing of tubes in a bundle	m	ν	Kinematic viscosity, μ/ρ	m^2/s
P	Absolute pressure; P_c for critical pressure	kPa	ρ	Density; ρ_L , ρ_l for density of liquid; ρ_G , ρ_g for density of vapor	kg/m^3
Pr	Prandtl number, ν/α		σ	Stefan-Boltzmann constant, 5.67×10^{-8}	$\text{W}/(\text{m}^2 \cdot \text{K}^4)$
q	Rate of heat transfer	W	σ	Surface tension between and liquid and its vapor	N/m
Q	Amount of heat transfer	J	τ	Time constant, time scale	s
$\dot{Q}$	Rate of heat transfer	W	Ω	Efficiency of fin	
Q/Q_i	Heat loss fraction, $Q/[\rho c V (T_i - T_\infty)]$				
r	Distance from center in plate, cylinder, or sphere	m			
R	Thermal resistance or radius	K/W or m			

and the average heat thermal conductivity is

$$\bar{k} = k_0(1 + \gamma \bar{T}) \quad (5-6)$$

where $\bar{T} = 0.5(T_1 + T_2)$.

For cylinders and spheres, A is a function of radial position (see Fig. 5-2): $2\pi rL$ and $4\pi r^2$, where L is the length of the cylinder. For constant k , Eq. (5-4) becomes

$$\dot{Q} = \frac{T_1 - T_2}{[\ln(r_2/r_1)]/(2\pi kL)} \quad \text{cylinder} \quad (5-7)$$

and

$$\dot{Q} = \frac{T_1 - T_2}{(r_2 - r_1)/(4\pi k r_1 r_2)} \quad \text{sphere} \quad (5-8)$$

Conduction with Resistances in Series A steady-state temperature profile in a planar composite wall, with three constant thermal conductivities and no source terms, is shown in Fig. 5-3a. The corresponding thermal circuit is given in Fig. 5-3b. The rate of heat transfer through each of the layers is the same. The total resistance is the sum of the individual resistances shown in Fig. 5-3b:

$$Q = \frac{T_1 - T_2}{\frac{\Delta X_A}{k_A A} + \frac{\Delta X_B}{k_B A} + \frac{\Delta X_C}{k_C A}} = \frac{T_1 - T_2}{R_A + R_B + R_C} \quad (5-9)$$

Additional resistances in the series may occur at the surfaces of the solid if they are in contact with a fluid. The rate of convective heat transfer, between a surface of area A and a fluid, is represented by Newton's law of cooling as

$$\dot{Q} = hA(T_{\text{surface}} - T_{\text{fluid}}) = \frac{T_{\text{surface}} - T_{\text{fluid}}}{1/(hA)} \quad (5-10)$$

where $1/(hA)$ is the resistance due to convection (K/W) and the heat-transfer coefficient is $h[\text{W}/(\text{m}^2\cdot\text{K})]$. For the cylindrical geometry shown in Fig. 5-2, with convection to inner and outer fluids at temperatures T_i and T_o , with heat-transfer coefficients h_i and h_o , the steady-state rate of heat transfer is

$$\dot{Q} = \frac{T_i - T_o}{\frac{1}{2\pi r_1 L h_i} + \frac{\ln(r_2/r_1)}{2\pi k L} + \frac{1}{2\pi r_2 L h_o}} = \frac{T_i - T_o}{R_i + R_1 + R_o} \quad (5-11)$$

where resistances R_i and R_o are the convective resistances at the inner and outer surfaces. The total resistance is again the sum of the resistances in series.

Example 1: Conduction with Resistances in Series and Parallel Figure 5-4 shows the thermal circuit for a furnace wall. The outside surface has a known temperature $T_2 = 625$ K. The temperature of the surroundings

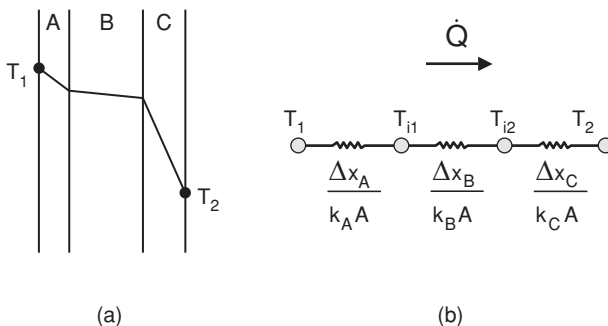


FIG. 5-3 Steady-state temperature profile in a composite wall with constant thermal conductivities k_A , k_B , and k_C and no energy sources in the wall. The thermal circuit is shown in (b). The total resistance is the sum of the three resistances shown.

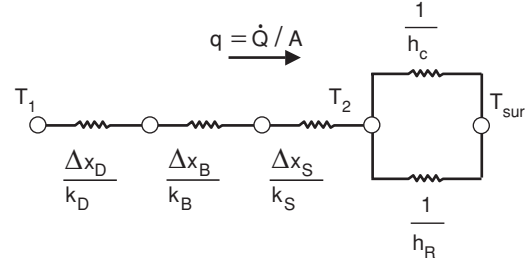


FIG. 5-4 Thermal circuit for Example 1. Steady-state conduction in a furnace wall with heat losses from the outside surface by convection (h_c) and radiation (h_r) to the surroundings at temperature T_{sur} . The thermal conductivities k_D , k_B , and k_S are constant, and there are no sources in the wall. The heat flux q has units of W/m^2 .

T_{sur} is 290 K. We want to estimate the temperature of the inside wall T_1 . The wall consists of three layers: deposit [$k_D = 1.6$ W/(m·K), $\Delta x_D = 0.080$ m], brick [$k_B = 1.7$ W/(m·K), $\Delta x_B = 0.15$ m], and steel [$k_S = 45$ W/(m·K), $\Delta x_S = 0.00254$ m]. The outside surface loses heat by two parallel mechanisms—convection and radiation. The convective heat-transfer coefficient $h_c = 5.0$ W/(m²·K). The radiative heat-transfer coefficient $h_r = 16.3$ W/(m²·K). The latter is calculated from

$$h_r = \epsilon_2 \sigma (T_2^2 + T_{\text{sur}}^2)(T_2 + T_{\text{sur}}) \quad (5-12)$$

where the emissivity of surface 2 is $\epsilon_2 = 0.76$ and the Stefan-Boltzmann constant $\sigma = 5.67 \times 10^{-8}$ W/(m²·K⁴).

Referring to Fig. 5-4, the steady-state heat flux q (W/m²) through the wall is

$$q = \frac{\dot{Q}}{A} = \frac{T_1 - T_2}{\frac{\Delta x_D}{k_D} + \frac{\Delta x_B}{k_B} + \frac{\Delta x_S}{k_S}} = (h_c + h_r)(T_2 - T_{\text{sur}})$$

Solving for T_1 gives

$$T_1 = T_2 + \left(\frac{\Delta x_D}{k_D} + \frac{\Delta x_B}{k_B} + \frac{\Delta x_S}{k_S} \right) (h_c + h_r)(T_2 - T_{\text{sur}})$$

and

$$T_1 = 625 + \left(\frac{0.080}{1.6} + \frac{0.15}{1.7} + \frac{0.00254}{45} \right) (5.0 + 16.3)(625 - 290) = 1610 \text{ K}$$

Conduction with Heat Source Application of the law of conservation of energy to a one-dimensional solid, with the heat flux given by (5-1) and volumetric source term S (W/m³), results in the following equations for steady-state conduction in a flat plate of thickness $2R$ ($b = 1$), a cylinder of diameter $2R$ ($b = 2$), and a sphere of diameter $2R$ ($b = 3$). The parameter b is a measure of the curvature. The thermal conductivity is constant, and there is convection at the surface, with heat-transfer coefficient h and fluid temperature T_{∞} .

$$\begin{aligned} \frac{d}{dr} \left(r^{b-1} \frac{dT}{dr} \right) + \frac{S}{k} r^{b-1} &= 0 \\ \frac{dT(0)}{dr} &= 0 \quad (\text{symmetry condition}) \\ -k \frac{dT}{dr} &= h[T(R) - T_{\infty}] \end{aligned} \quad (5-13)$$

The solutions to (5-13), for uniform S , are

$$\frac{T(r) - T_{\infty}}{SR^2/k} = \frac{1}{2b} \left[1 - \left(\frac{r}{R} \right)^2 \right] + \frac{1}{b\text{Bi}} \begin{cases} b = 1, \text{ plate, thickness } 2R \\ b = 2, \text{ cylinder, diameter } 2R \\ b = 3, \text{ sphere, diameter } 2R \end{cases} \quad (5-14)$$

where $\text{Bi} = hR/k$ is the Biot number. For $\text{Bi} \ll 1$, the temperature in the solid is uniform. For $\text{Bi} \gg 1$, the surface temperature $T(R) = T_{\infty}$.

Two- and Three-Dimensional Conduction Application of the law of conservation of energy to a three-dimensional solid, with the

5-6 HEAT AND MASS TRANSFER

heat flux given by (5-1) and volumetric source term S (W/m^3), results in the following equation for steady-state conduction in rectangular coordinates.

$$\frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + S = 0 \quad (5-15)$$

Similar equations apply to cylindrical and spherical coordinate systems. Finite difference, finite volume, or finite element methods are generally necessary to solve (5-15). Useful introductions to these numerical techniques are given in the General References and Sec. 3. Simple forms of (5-15) (constant k , uniform S) can be solved analytically. See Arpaci, *Conduction Heat Transfer*, Addison-Wesley, 1966, p. 180, and Carslaw and Jaeger, *Conduction of Heat in Solids*, Oxford University Press, 1959. For problems involving heat flow between two surfaces, each isothermal, with all other surfaces being adiabatic, the shape factor approach is useful (Mills, *Heat Transfer*, 2d ed., Prentice-Hall, 1999, p. 164).

UNSTEADY-STATE CONDUCTION

Application of the law of conservation of energy to a three-dimensional solid, with the heat flux given by (5-1) and volumetric source term S (W/m^3), results in the following equation for unsteady-state conduction in rectangular coordinates.

$$\rho c \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + S \quad (5-16)$$

The energy storage term is on the left-hand side, and ρ and c are the density (kg/m^3) and specific heat [$\text{J}/(\text{kg} \cdot \text{K})$]. Solutions to (5-16) are generally obtained numerically (see General References and Sec. 3). The one-dimensional form of (5-16), with constant k and no source term, is

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} \quad (5-17)$$

where $\alpha = k/(\rho c)$ is the thermal diffusivity (m^2/s).

One-Dimensional Conduction: Lumped and Distributed Analysis The one-dimensional transient conduction equations in rectangular ($b = 1$), cylindrical ($b = 2$), and spherical ($b = 3$) coordinates, with constant k , initial uniform temperature T_i , $S = 0$, and convection at the surface with heat-transfer coefficient h and fluid temperature T_∞ , are

$$\frac{\partial T}{\partial t} = \frac{\alpha}{r^{b-1}} \frac{\partial}{\partial r} \left(r^{b-1} \frac{\partial T}{\partial r} \right) \quad \begin{cases} b = 1, \text{ plate, thickness } 2R \\ b = 2, \text{ cylinder, diameter } 2R \\ b = 3, \text{ sphere, diameter } 2R \end{cases}$$

$$\text{for } t < 0, \quad T = T_i \quad (\text{initial temperature}) \quad (5-18)$$

$$\text{at } r = 0, \quad \frac{\partial T}{\partial r} = 0 \quad (\text{symmetry condition})$$

$$\text{at } r = R, \quad -k \frac{\partial T}{\partial r} = h(T - T_\infty)$$

The solutions to (5-18) can be compactly expressed by using dimensionless variables: (1) temperature $\theta/\theta_i = [T(r,t) - T_\infty]/(T_i - T_\infty)$; (2) heat loss fraction $Q/Q_i = Q/[\rho c V (T_i - T_\infty)]$, where V is volume; (3) distance from center $\zeta = r/R$; (4) time $\text{Fo} = \alpha t/R^2$; and (5) Biot number $\text{Bi} = hR/k$. The temperature and heat loss are functions of ζ , Fo , and Bi .

When the Biot number is small, $\text{Bi} < 0.2$, the temperature of the solid is nearly uniform and a lumped analysis is acceptable. The solution to the lumped analysis of (5-18) is

$$\frac{\theta}{\theta_i} = \exp \left(-\frac{hA}{\rho c V} t \right) \quad \text{and} \quad \frac{Q}{Q_i} = 1 - \exp \left(-\frac{hA}{\rho c V} t \right) \quad (5-19)$$

where A is the active surface area and V is the volume. The time scale for the lumped problem is

$$\tau = \frac{\rho c V}{hA} \quad (5-20)$$

TABLE 5-1 Fourier Coefficients and Spatial Functions for Use in Eqs. (5-21)

Geometry	A_1	B_1	S_1
Plate	$\frac{2 \sin \delta_1}{\delta_1 + \sin \delta_1 \cos \delta_1}$	$\frac{2 \text{Bi}^2}{\delta_1^2 (\text{Bi}^2 + \text{Bi} + \delta_1^2)}$	$\cos(\delta_1 \zeta)$
Cylinder	$\frac{2 J_1(\delta_1)}{\delta_1 [J_0'(\delta_1) + J_1'(\delta_1)]}$	$\frac{4 \text{Bi}^2}{\delta_1^2 (\delta_1^2 + \text{Bi}^2)}$	$J_0(\delta_1 \zeta)$
Sphere	$\frac{2 \text{Bi} [\delta_1^2 + (\text{Bi} - 1)^2]^{1/2}}{\delta_1^2 + \text{Bi}^2 - \text{Bi}}$	$\frac{6 \text{Bi}^2}{\delta_1^2 (\delta_1^2 + \text{Bi}^2 - \text{Bi})}$	$\frac{\sin \delta_1 \zeta}{\delta_1 \zeta}$

The time scale is the time required for most of the change in θ/θ_i or Q/Q_i to occur. When $t = \tau$, $\theta/\theta_i = \exp(-1) = 0.368$ and roughly two-thirds of the possible change has occurred.

When a lumped analysis is not valid ($\text{Bi} > 0.2$), the single-term solutions to (5-18) are convenient:

$$\frac{\theta}{\theta_i} = A_1 \exp(-\delta_1^2 \text{Fo}) S_1(\delta_1 \zeta) \quad \text{and} \quad \frac{Q}{Q_i} = 1 - B_1 \exp(-\delta_1^2 \text{Fo}) \quad (5-21)$$

where the first Fourier coefficients A_1 and B_1 and the spatial functions S_1 are given in Table 5-1. The first eigenvalue δ_1 is given by (5-22) in conjunction with Table 5-2. The one-term solutions are accurate to within 2 percent when $\text{Fo} > \text{Fo}_c$. The values of the critical Fourier number Fo_c are given in Table 5-2.

The first eigenvalue is accurately correlated by (Yovanovich, Chap. 3 of Rohsenow, Hartnett, and Cho, *Handbook of Heat Transfer*, 3d ed., McGraw-Hill, 1998, p. 3.25)

$$\delta_1 = \frac{\delta_{1,\infty}}{[1 + (\delta_{1,\infty} \delta_{1,0})^n]^{1/n}} \quad (5-22)$$

Equation (5-22) gives values of δ_1 that differ from the exact values by less than 0.4 percent, and it is valid for all values of Bi . The values of $\delta_{1,\infty}$, $\delta_{1,0}$, n , and Fo_c are given in Table 5-2.

Example 2: Correlation of First Eigenvalues by Eq. (5-22) As an example of the use of Eq. (5-22), suppose that we want δ_1 for the flat plate with $\text{Bi} = 5$. From Table 5-2, $\delta_{1,\infty} = \pi/2$, $\delta_{1,0} = \sqrt{\text{Bi}} = \sqrt{5}$, and $n = 2.139$. Equation (5-22) gives

$$\delta_1 = \frac{\pi/2}{[1 + (\pi/2\sqrt{5})^{2.139}]^{1/2.139}} = 1.312$$

The tabulated value is 1.3138.

Example 3: One-Dimensional, Unsteady Conduction Calculation As an example of the use of Eq. (5-21), Table 5-1, and Table 5-2, consider the cooking time required to raise the center of a spherical, 8-cm-diameter dumpling from 20 to 80°C. The initial temperature is uniform. The dumpling is heated with saturated steam at 95°C. The heat capacity, density, and thermal conductivity are estimated to be $c = 3500 \text{ J}/(\text{kg} \cdot \text{K})$, $\rho = 1000 \text{ kg}/\text{m}^3$, and $k = 0.5 \text{ W}/(\text{m} \cdot \text{K})$, respectively.

Because the heat-transfer coefficient for condensing steam is of order 10^4 , the $\text{Bi} \rightarrow \infty$ limit in Table 5-2 is a good choice and $\delta_1 = \pi$. Because we know the desired temperature at the center, we can calculate θ/θ_i and then solve (5-21) for the time.

$$\frac{\theta}{\theta_i} = \frac{T(0,t) - T_\infty}{T_i - T_\infty} = \frac{80 - 95}{20 - 95} = 0.200$$

For $\text{Bi} \rightarrow \infty$, A_1 in Table 5-1 is 2 and for $\zeta = 0$, S_1 in Table 5-1 is 1. Equation (5-21) becomes

$$\frac{\theta}{\theta_i} = 2 \exp(-\pi^2 \text{Fo}) = 2 \exp \left(-\frac{\pi^2 \alpha t}{R^2} \right)$$

TABLE 5-2 First Eigenvalues for $\text{Bi} \rightarrow 0$ and $\text{Bi} \rightarrow \infty$ and Correlation Parameter n

The single-term approximations apply only if $\text{Fo} \geq \text{Fo}_c$.

Geometry	$\text{Bi} \rightarrow 0$	$\text{Bi} \rightarrow \infty$	n	Fo_c
Plate	$\delta_1 \rightarrow \sqrt{\text{Bi}}$	$\delta_1 \rightarrow \pi/2$	2.139	0.24
Cylinder	$\delta_1 \rightarrow \sqrt{2 \text{Bi}}$	$\delta_1 \rightarrow 2.4048255$	2.238	0.21
Sphere	$\delta_1 \rightarrow \sqrt{3 \text{Bi}}$	$\delta_1 \rightarrow \pi$	2.314	0.18

Solving for t gives the desired cooking time.

$$t = -\frac{R^2}{\alpha\pi^2} \ln \frac{\theta}{2\theta_i} = -\frac{(0.04 \text{ m})^2}{1.43 \times 10^{-7} (\text{m}^2/\text{s})\pi^2} \ln \frac{0.2}{2} = 43.5 \text{ min}$$

Example 4: Rule of Thumb for Time Required to Diffuse a Distance R A general rule of thumb for estimating the time required to diffuse a distance R is obtained from the one-term approximations. Consider the equation for the temperature of a flat plate of thickness $2R$ in the limit as $\text{Bi} \rightarrow \infty$. From Table 5-2, the first eigenvalue is $\delta_1 = \pi/2$, and from Table 5-1,

$$\frac{\theta}{\theta_i} = A_1 \exp\left[-\left(\frac{\pi}{2}\right)^2 \frac{\alpha t}{R^2}\right] \cos \delta_1 \zeta$$

When $t = R^2/\alpha$, the temperature ratio at the center of the plate ($\zeta = 0$) has decayed to $\exp(-\pi^2/4)$, or 8 percent of its initial value. We conclude that *diffusion through a distance R takes roughly R^2/α units of time*, or alternatively, *the distance diffused in time t is about $(\alpha t)^{1/2}$* .

One-Dimensional Conduction: Semi-infinite Plate Consider a semi-infinite plate with an initial uniform temperature T_i . Suppose that the temperature of the surface is suddenly raised to T_∞ ; that is, the heat-transfer coefficient is infinite. The unsteady temperature of the plate is

$$\frac{T(x,t) - T_\infty}{T_i - T_\infty} = \text{erf}\left(\frac{x}{2\sqrt{\alpha t}}\right) \quad (5-23)$$

where $\text{erf}(z)$ is the error function. The depth to which the heat penetrates in time t is approximately $(12\alpha t)^{1/2}$.

If the heat-transfer coefficient is finite,

$$\frac{T(x,t) - T_\infty}{T_i - T_\infty} = \text{erfc}\left(\frac{x}{2\sqrt{\alpha t}}\right) - \exp\left(\frac{hx}{k} + \frac{h^2\alpha t}{k^2}\right) \text{erfc}\left(\frac{x}{2\sqrt{\alpha t}} + \frac{h\sqrt{\alpha t}}{k}\right) \quad (5-24)$$

where $\text{erfc}(z)$ is the complementary error function. Equations (5-23) and (5-24) are both applicable to finite plates provided that their half-thickness is greater than $(12\alpha t)^{1/2}$.

Two- and Three-Dimensional Conduction The one-dimensional solutions discussed above can be used to construct solutions to multidimensional problems. The unsteady temperature of a rectangular, solid box of height, length, and width $2H$, $2L$, and $2W$, respectively, with governing equations in each direction as in (5-18), is

$$\left(\frac{\theta}{\theta_i}\right)_{2H \times 2L \times 2W} = \left(\frac{\theta}{\theta_i}\right)_{2H} \left(\frac{\theta}{\theta_i}\right)_{2L} \left(\frac{\theta}{\theta_i}\right)_{2W} \quad (5-25)$$

Similar products apply for solids with other geometries, e.g., semi-infinite, cylindrical rods.

HEAT TRANSFER BY CONVECTION

CONVECTIVE HEAT-TRANSFER COEFFICIENT

Convection is the transfer of energy by conduction and radiation in moving, fluid media. The motion of the fluid is an essential part of convective heat transfer. A key step in calculating the rate of heat transfer by convection is the calculation of the heat-transfer coefficient. This section focuses on the estimation of heat-transfer coefficients for natural and forced convection. The conservation equations for mass, momentum, and energy, as presented in Sec. 6, can be used to calculate the rate of convective heat transfer. Our approach in this section is to rely on correlations.

In many cases of industrial importance, heat is transferred from one fluid, through a solid wall, to another fluid. The transfer occurs in a heat exchanger. Section 11 introduces several types of heat exchangers, design procedures, overall heat-transfer coefficients, and mean temperature differences. Section 3 introduces dimensional analysis and the dimensionless groups associated with the heat-transfer coefficient.

Individual Heat-Transfer Coefficient The local rate of convective heat transfer between a surface and a fluid is given by Newton's law of cooling

$$q = h(T_{\text{surface}} - T_{\text{fluid}}) \quad (5-26)$$

where h [$\text{W}/(\text{m}^2 \cdot \text{K})$] is the local heat-transfer coefficient and q is the energy flux (W/m^2). The definition of h is arbitrary, depending on whether the bulk fluid, centerline, free stream, or some other temperature is used for T_{fluid} . The heat-transfer coefficient may be defined on an average basis as noted below.

Consider a fluid with bulk temperature T_i flowing in a cylindrical tube of diameter D , with constant wall temperature T_s . An energy balance on a short section of the tube yields

$$c_p \dot{m} \frac{dT}{dx} = \pi D h (T_s - T) \quad (5-27)$$

where c_p is the specific heat at constant pressure [$\text{J}/(\text{kg} \cdot \text{K})$], $\dot{m}$ is the mass flow rate (kg/s), and x is the distance from the inlet. If the temperature of the fluid at the inlet is T_{in} , the temperature of the fluid at a downstream distance L is

$$\frac{T(L) - T_s}{T_{\text{in}} - T_s} = \exp\left(-\frac{\bar{h}\pi DL}{\dot{m}c_p}\right) \quad (5-28)$$

The average heat-transfer coefficient $\bar{h}$ is defined by

$$\bar{h} = \frac{1}{L} \int_0^L h \, dx \quad (5-29)$$

Overall Heat-Transfer Coefficient and Heat Exchangers A local, overall heat-transfer coefficient U for the cylindrical geometry shown in Fig. 5-2 is defined by using Eq. (5-11) as

$$\frac{\dot{Q}}{\Delta x} = \frac{T_i - T_o}{\frac{1}{2\pi r_1 h_i} + \frac{\ln(r_2/r_1)}{2\pi k} + \frac{1}{2\pi r_2 h_o}} = 2\pi r_1 U (T_i - T_o) \quad (5-30)$$

where Δx is a short length of tube in the axial direction. Equation (5-30) defines U by using the inside perimeter $2\pi r_1$. The outer perimeter can also be used. Equation (5-30) applies to clean tubes. Additional resistances are present in the denominator for dirty tubes (see Sec. 11).

For counterflow and parallel flow heat exchanges, with high- and low-temperature fluids (T_H and T_C) and flow directions as defined in Fig. 5-5, the total heat transfer for the exchanger is given by

$$\dot{Q} = UA \Delta T_{\text{lm}} \quad (5-31)$$

where A is the area for heat exchange and the log mean temperature difference ΔT_{lm} is defined as

$$\Delta T_{\text{lm}} = \frac{(T_H - T_C)_L - (T_H - T_C)_0}{\ln[(T_H - T_C)_L / (T_H - T_C)_0]} \quad (5-32)$$

Equation (5-32) applies to both counterflow and parallel flow exchangers with the nomenclature defined in Fig. 5-5. Correction factors to ΔT_{lm} for various heat exchanger configurations are given in Sec. 11.

In certain applications, the log mean temperature difference is replaced with an arithmetic mean difference:

$$\Delta T_{\text{am}} = \frac{(T_H - T_C)_L + (T_H - T_C)_0}{2} \quad (5-33)$$

Average heat-transfer coefficients are occasionally reported based on Eqs. (5-32) and (5-33) and are written as h_{lm} and h_{am} .

Representation of Heat-Transfer Coefficients Heat-transfer coefficients are usually expressed in two ways: (1) dimensionless relations and (2) dimensional equations. Both approaches are used below. The dimensionless form of the heat-transfer coefficient is the Nusselt

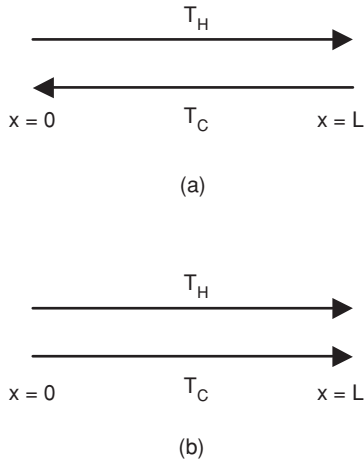


FIG. 5-5 Nomenclature for (a) counterflow and (b) parallel flow heat exchangers for use with Eq. (5-32).

number. For example, with a cylinder of diameter D in cross flow, the local Nusselt number is defined as $Nu_D = hD/k$, where k is the thermal conductivity of the fluid. The subscript D is important because different characteristic lengths can be used to define Nu . The average Nusselt number is written $\overline{Nu}_D = \overline{h}D/k$.

NATURAL CONVECTION

Natural convection occurs when a fluid is in contact with a solid surface of different temperature. Temperature differences create the density gradients that drive natural or free convection. In addition to the Nusselt number mentioned above, the key dimensionless parameters for natural convection include the Rayleigh number $Ra_L = \beta \Delta T g L^3 / \nu \alpha$ and the Prandtl number $Pr = \nu / \alpha$. The properties appearing in Ra and Pr include the volumetric coefficient of expansion β (K^{-1}); the difference ΔT between the surface (T_s) and free stream (T_∞) temperatures (K or $^\circ C$); the acceleration of gravity g (m/s^2); a characteristic dimension L of the surface (m); the kinematic viscosity ν (m^2/s); and the thermal diffusivity α (m^2/s). The volumetric coefficient of expansion for an ideal gas is $\beta = 1/T$, where T is absolute temperature. For a given geometry,

$$\overline{Nu}_L = f(Ra_L, Pr) \quad (5-34)$$

External Natural Flow for Various Geometries For vertical walls, Churchill and Chu [Int. J. Heat Mass Transfer, **18**, 1323 (1975)] recommend, for laminar and turbulent flow on isothermal, vertical walls with height L ,

$$\overline{Nu}_L = \left\{ 0.825 + \frac{0.387 Ra_L^{1/6}}{[1 + (0.492/Pr)^{9/16}]^{8/27}} \right\}^2 \quad (5-35)$$

where the fluid properties for Eq. (5-35) and $\overline{Nu}_L = \overline{h}L/k$ are evaluated at the film temperature $T_f = (T_s + T_\infty)/2$. This correlation is valid for all Pr and Ra_L . For vertical cylinders with boundary layer thickness much less than their diameter, Eq. (5-35) is applicable. An expression for uniform heating is available from the same reference.

For laminar and turbulent flow on isothermal, horizontal cylinders of diameter D , Churchill and Chu [Int. J. Heat Mass Transfer, **18**, 1049 (1975)] recommend

$$\overline{Nu}_L = \left\{ 0.60 + \frac{0.387 Ra_D^{1/6}}{[1 + (0.559/Pr)^{9/16}]^{8/27}} \right\}^2 \quad (5-36)$$

Fluid properties for (5-36) should be evaluated at the film temperature $T_f = (T_s + T_\infty)/2$. This correlation is valid for all Pr and Ra_D .

For horizontal flat surfaces, the characteristic dimension for the correlations is [Goldstein, Sparrow, and Jones, Int. J. Heat Mass Transfer, **16**, 1025–1035 (1973)]

$$L = \frac{A}{p} \quad (5-37)$$

where A is the area of the surface and p is the perimeter. With hot surfaces facing upward, or cold surfaces facing downward [Lloyd and Moran, ASME Paper 74-WA/HT-66 (1974)],

$$\overline{Nu}_L = \begin{cases} 0.54 Ra_L^{1/4} & 10^4 < Ra_L < 10^7 \\ 0.15 Ra_L^{1/3} & 10^7 < Ra_L < 10^{10} \end{cases} \quad (5-38)$$

and for hot surfaces facing downward, or cold surfaces facing upward,

$$\overline{Nu}_L = 0.27 Ra_L^{1/4} \quad 10^5 < Ra_L < 10^{10} \quad (5-40)$$

Fluid properties for Eqs. (5-38) to (5-40) should be evaluated at the film temperature $T_f = (T_s + T_\infty)/2$.

Simultaneous Heat Transfer by Radiation and Convection Simultaneous heat transfer by radiation and convection is treated per the procedure outlined in Examples 1 and 5. A radiative heat-transfer coefficient h_R is defined by (5-12).

Mixed Forced and Natural Convection Natural convection is commonly assisted or opposed by forced flow. These situations are discussed, e.g., by Mills (Heat Transfer, 2d ed., Prentice-Hall, 1999, p. 340) and Raithby and Hollands (Chap. 4 of Rohsenow, Hartnett, and Cho, Handbook of Heat Transfer, 3d ed., McGraw-Hill, 1998, p. 4.73).

Enclosed Spaces The rate of heat transfer across an enclosed space is described in terms of a heat-transfer coefficient based on the temperature difference between two surfaces:

$$\overline{h} = \frac{\dot{Q}/A}{T_H - T_C} \quad (5-41)$$

For rectangular cavities, the plate spacing between the two surfaces L is the characteristic dimension that defines the Nusselt and Rayleigh numbers. The temperature difference in the Rayleigh number, $Ra_L = \beta \Delta T g L^3 / \nu \alpha$ is $\Delta T = T_H - T_C$.

For a horizontal rectangular cavity heated from below, the onset of advection requires $Ra_L > 1708$. Globe and Dropkin [J. Heat Transfer, **81**, 24–28 (1959)] propose the correlation

$$\overline{Nu}_L = 0.069 Ra_L^{1/3} Pr^{0.074} \quad 3 \times 10^5 < Ra_L < 7 \times 10^9 \quad (5-42)$$

All properties in (5-42) are calculated at the average temperature $(T_H + T_C)/2$.

For vertical rectangular cavities of height H and spacing L , with $Pr \approx 0.7$ (gases) and $40 < H/L < 110$, the equation of Shewen et al. [J. Heat Transfer, **118**, 993–995 (1996)] is recommended:

$$\overline{Nu}_L = \left\{ 1 + \left[\frac{0.0665 Ra_L^{1/3}}{1 + (9000/Ra_L)^{1.4}} \right]^2 \right\}^{1/2} \quad Ra_L < 10^6 \quad (5-43)$$

All properties in (5-43) are calculated at the average temperature $(T_H + T_C)/2$.

Example 5: Comparison of the Relative Importance of Natural Convection and Radiation at Room Temperature

Estimate the heat losses by natural convection and radiation for an undraped person standing in still air. The temperatures of the air, surrounding surfaces, and skin are 19, 15, and 35 $^\circ C$, respectively. The height and surface area of the person are 1.8 m and 1.8 m 2 . The emissivity of the skin is 0.95.

We can estimate the Nusselt number by using (5-35) for a vertical, flat plate of height $L = 1.8$ m. The film temperature is $(19 + 35)/2 = 27^\circ C$. The Rayleigh number, evaluated at the film temperature, is

$$Ra_L = \frac{\beta \Delta T g L^3}{\nu \alpha} = \frac{(1/300)(35 - 19)9.81(1.8)^3}{1.589 \times 10^{-5}(2.25 \times 10^{-5})} = 8.53 \times 10^9$$

From (5-35) with $Pr = 0.707$, the Nusselt number is 240 and the average heat-transfer coefficient due to natural convection is

$$\overline{h} = \frac{k}{L} \overline{Nu}_L = \frac{0.0263}{1.8} (240) = 3.50 \frac{W}{m^2 \cdot K}$$

The radiative heat-transfer coefficient is given by (5-12):

$$h_R = \epsilon_{\text{skin}} \sigma (T_{\text{skin}}^2 + T_{\text{sur}}^2)(T_{\text{skin}} + T_{\text{sur}}) \\ = 0.95(5.67 \times 10^{-8})(308^2 + 288^2)(308 + 288) = 5.71 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

The total rate of heat loss is

$$\dot{Q} = \bar{h}A(T_{\text{skin}} - T_{\text{air}}) + \bar{h}_R A(T_{\text{skin}} - T_{\text{sur}}) \\ = 3.50(1.8)(35 - 19) + 5.71(1.8)(35 - 15) = 306 \text{ W}$$

At these conditions, radiation is nearly twice as important as natural convection.

FORCED CONVECTION

Forced convection heat transfer is probably the most common mode in the process industries. Forced flows may be internal or external. This subsection briefly introduces correlations for estimating heat-transfer coefficients for flows in tubes and ducts; flows across plates, cylinders, and spheres; flows through tube banks and packed beds; heat transfer to nonevaporating falling films; and rotating surfaces. Section 11 introduces several types of heat exchangers, design procedures, overall heat-transfer coefficients, and mean temperature differences.

Flow in Round Tubes In addition to the Nusselt ($\text{Nu}_D = hD/k$) and Prandtl ($\text{Pr} = \nu/\alpha$) numbers introduced above, the key dimensionless parameter for forced convection in round tubes of diameter D is the Reynolds number $\text{Re} = GD/\mu$, where G is the mass velocity $G = \dot{m}/A_c$ and A_c is the cross-sectional area $A_c = \pi D^2/4$. For internal flow in a tube or duct, the heat-transfer coefficient is defined as

$$q = h(T_b - T_s) \quad (5-44)$$

where T_b is the bulk or mean temperature at a given cross section and T_s is the corresponding surface temperature.

For laminar flow ($\text{Re}_D < 2100$) that is fully developed, both hydrodynamically and thermally, the Nusselt number has a constant value. For a uniform wall temperature, $\text{Nu}_D = 3.66$. For a uniform heat flux through the tube wall, $\text{Nu}_D = 4.36$. In both cases, the thermal conductivity of the fluid in Nu_D is evaluated at T_b . The distance x required for a fully developed laminar velocity profile is given by $[(x/D)/\text{Re}_D] \approx 0.05$. The distance x required for fully developed velocity and thermal profiles is obtained from $[(x/D)/(\text{Re}_D \text{Pr})] \approx 0.05$.

For a constant wall temperature, a fully developed laminar velocity profile, and a developing thermal profile, the average Nusselt number is estimated by [Hausen, *Allg. Waermetech.*, **9**, 75 (1959)]

$$\overline{\text{Nu}}_D = 3.66 + \frac{0.0668(D/L) \text{Re}_D \text{Pr}}{1 + 0.04[(D/L) \text{Re}_D \text{Pr}]^{1/3}} \quad (5-45)$$

For large values of L , Eq. (5-45) approaches $\text{Nu}_D = 3.66$. Equation (5-45) also applies to developing velocity and thermal profiles conditions if $\text{Pr} \gg 1$. The properties in (5-45) are evaluated at the bulk mean temperature

$$\overline{T}_b = (T_{b,\text{in}} + T_{b,\text{out}})/2 \quad (5-46)$$

For a constant wall temperature with developing laminar velocity and thermal profiles, the average Nusselt number is approximated by [Sieder and Tate, *Ind. Eng. Chem.*, **28**, 1429 (1936)]

$$\overline{\text{Nu}}_D = 1.86 \left(\frac{D}{L} \text{Re}_D \text{Pr} \right)^{1/3} \left(\frac{\mu_b}{\mu_s} \right)^{0.14} \quad (5-47)$$

The properties, except for μ_s , are evaluated at the bulk mean temperature per (5-46) and $0.48 < \text{Pr} < 16,700$ and $0.0044 < \mu_b/\mu_s < 9.75$.

For fully developed flow in the transition region between laminar and turbulent flow, and for fully developed turbulent flow, Gnielinski's [Int. Chem. Eng., **16**, 359 (1976)] equation is recommended:

$$\text{Nu}_D = \frac{(f/2)(\text{Re}_D - 1000)(\text{Pr})}{1 + 12.7(f/2)^{1/2}(\text{Pr}^{2/3} - 1)} K \quad (5-48)$$

where $0.5 < \text{Pr} < 10^5$, $2300 < \text{Re}_D < 10^6$, $K = (\text{Pr}_b/\text{Pr}_s)^{0.11}$ for liquids ($0.05 < \text{Pr}_b/\text{Pr}_s < 20$), and $K = (T_b/T_s)^{0.45}$ for gases ($0.5 < T_b/T_s < 1.5$). The factor K corrects for variable property effects. For smooth tubes, the Fanning friction factor f is given by

$$f = 0.25(0.790 \ln \text{Re}_D - 1.64)^{-2} \quad 2300 < \text{Re}_D < 10^6 \quad (5-49)$$

TABLE 5-3 Effect of Entrance Configuration on Values of C and n in Eq. (5-53) for $\text{Pr} \approx 1$ (Gases and Other Fluids with Pr about 1)

Entrance configuration	C	n
Long calming section	0.9756	0.760
Open end, 90° edge	2.4254	0.676
180° return bend	0.9759	0.700
90° round bend	1.0517	0.629
90° elbow	2.0152	0.614

For rough pipes, approximate values of Nu_D are obtained if f is estimated by the Moody diagram of Sec. 6. Equation (5-48) is corrected for entrance effects per (5-53) and Table 5-3. Sieder and Tate [Ind. Eng. Chem., **28**, 1429 (1936)] recommend a simpler but less accurate equation for fully developed turbulent flow

$$\text{Nu}_D = 0.027 \text{Re}_D^{4/5} \text{Pr}^{1/3} \left(\frac{\mu_b}{\mu_s} \right)^{0.14} \quad (5-50)$$

where $0.7 < \text{Pr} < 16,700$, $\text{Re}_D < 10,000$, and $L/D > 10$. Equations (5-48) and (5-50) apply to both constant temperature and uniform heat flux along the tube. The properties are evaluated at the bulk temperature T_b , except for μ_s , which is at the temperature of the tube. For L/D greater than about 10, Eqs. (5-48) and (5-50) provide an estimate of Nu_D . In this case, the properties are evaluated at the bulk mean temperature per (5-46). More complicated and comprehensive predictions of fully developed turbulent convection are available in Churchill and Žajic [AIChE J., **48**, 927 (2002)] and Yu, Ozoe, and Churchill [Chem. Eng. Science, **56**, 1781 (2001)].

For fully developed turbulent flow of liquid metals, the Nusselt number depends on the wall boundary condition. For a constant wall temperature [Notter and Sleicher, Chem. Eng. Science, **27**, 2073 (1972)],

$$\text{Nu}_D = 4.8 + 0.0156 \text{Re}_D^{0.85} \text{Pr}^{0.93} \quad (5-51)$$

while for a uniform wall heat flux,

$$\text{Nu}_D = 6.3 + 0.0167 \text{Re}_D^{0.85} \text{Pr}^{0.93} \quad (5-52)$$

In both cases the properties are evaluated at T_b and $0.004 < \text{Pr} < 0.01$ and $10^4 < \text{Re}_D < 10^6$.

Entrance effects for turbulent flow with simultaneously developing velocity and thermal profiles can be significant when $L/D < 10$. Shah and Bhatti correlated entrance effects for gases ($\text{Pr} \approx 1$) to give an equation for the average Nusselt number in the entrance region (in Kaka, Shah, and Aung, eds., *Handbook of Single-Phase Convective Heat Transfer*, Chap. 3, Wiley-Interscience, 1987).

$$\frac{\overline{\text{Nu}}_D}{\text{Nu}_D} = 1 + \frac{C}{(x/D)^n} \quad (5-53)$$

where Nu_D is the fully developed Nusselt number and the constants C and n are given in Table 5-3 (Ebadian and Dong, Chap. 5 of Rohsenow, Hartnett, and Cho, *Handbook of Heat Transfer*, 3d ed., McGraw-Hill, 1998, p. 5.31). The tube entrance configuration determines the values of C and n as shown in Table 5-3.

Flow in Noncircular Ducts The length scale in the Nusselt and Reynolds numbers for noncircular ducts is the hydraulic diameter, $D_h = 4A_c/p$, where A_c is the cross-sectional area for flow and p is the wetted perimeter. Nusselt numbers for fully developed laminar flow in a variety of noncircular ducts are given by Mills (*Heat Transfer*, 2d ed., Prentice-Hall, 1999, p. 307). For turbulent flows, correlations for round tubes can be used with D replaced by D_h .

For annular ducts, the accuracy of the Nusselt number given by (5-48) is improved by the following multiplicative factors [Petukhov and Roizen, *High Temp.*, **2**, 65 (1964)].

$$\text{Inner tube heated} \quad 0.86 \left(\frac{D_i}{D_o} \right)^{-0.16}$$

$$\text{Outer tube heated} \quad 1 - 0.14 \left(\frac{D_i}{D_o} \right)^{0.6}$$

where D_i and D_o are the inner and outer diameters, respectively.

Example 6: Turbulent Internal Flow Air at 300 K, 1 bar, and 0.05 kg/s enters a channel of a plate-type heat exchanger (Mills, *Heat Transfer*, 2d ed., Prentice-Hall, 1999) that measures 1 cm wide, 0.5 m high, and 0.8 m long. The walls are at 600 K, and the mass flow rate is 0.05 kg/s. The entrance has a 90° edge. We want to estimate the exit temperature of the air.

Our approach will use (5-48) to estimate the average heat-transfer coefficient, followed by application of (5-28) to calculate the exit temperature. We assume ideal gas behavior and an exit temperature of 500 K. The estimated bulk mean temperature of the air is, by (5-46), 400 K. At this temperature, the properties of the air are $Pr = 0.690$, $\mu = 2.301 \times 10^{-5}$ kg/(m·s), $k = 0.0338$ W/(m·K), and $c_p = 1014$ J/(kg·K).

We start by calculating the hydraulic diameter $D_h = 4A_c/p$. The cross-sectional area for flow A_c is 0.005 m², and the wetted perimeter p is 1.02 m. The hydraulic diameter $D_h = 0.01961$ m. The Reynolds number is

$$Re_D = \frac{\dot{m} D_h}{A_c \mu} = \frac{0.05(0.01961)}{0.005(2.301 \times 10^{-5})} = 8521$$

The flow is in the transition region, and Eqs. (5-49) and (5-48) apply:

$$f = 0.25(0.790 \ln Re_D - 1.64)^{-2} = 0.25(0.790 \ln 8521 - 1.64)^{-2} = 0.008235$$

$$Nu_D = \frac{(f/2)(Re_D - 1000)(Pr)}{1 + 12.7(f/2)^{1/2}(Pr^{2/3} - 1)} K$$

$$= \frac{(0.008235/2)(8521 - 1000)(0.690)}{1 + 12.7(0.008235/2)^{1/2}(0.690^{2/3} - 1)} \left(\frac{400}{600} \right)^{0.45} = 21.68$$

Entrance effects are included by using (5-53) for an open end, 90° edge:

$$\overline{Nu}_D = \left[1 + \frac{C}{(x/D)^n} \right] Nu_D = \left[1 + \frac{2.4254}{(0.8/0.01961)^{0.676}} \right] (21.68) = 25.96$$

The average heat-transfer coefficient becomes

$$\bar{h} = \frac{k}{D_h} \overline{Nu}_D = \frac{0.0338}{0.01961} (25.96) = 44.75 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

The exit temperature is calculated from (5-28):

$$T(L) = T_s - (T_s - T_m) \exp \left(- \frac{\bar{h} p L}{\dot{m} c_p} \right)$$

$$= 600 - (600 - 300) \exp \left[- \frac{44.75(1.02)(0.8)}{0.05(1014)} \right] = 450 \text{ K}$$

We conclude that our estimated exit temperature of 500 K is too high. We could repeat the calculations, using fluid properties evaluated at a revised bulk mean temperature of 375 K.

Coiled Tubes For turbulent flow inside helical coils, with tube inside radius a and coil radius R , the Nusselt number for a straight tube Nu_s is related to that for a coiled tube Nu_c by (Rohsenow, Hartnett, and Cho, *Handbook of Heat Transfer*, 3d ed., McGraw-Hill, 1998, p. 5.90)

$$\frac{Nu_c}{Nu_s} = 1.0 + 3.6 \left(1 - \frac{a}{R} \right) \left(\frac{a}{R} \right)^{0.8} \quad (5-54)$$

where $2 \times 10^4 < Re_D < 1.5 \times 10^5$ and $5 < R/a < 84$. For lower Reynolds numbers ($1.5 \times 10^3 < Re_D < 2 \times 10^4$), the same source recommends

$$\frac{Nu_c}{Nu_s} = 1.0 + 3.4 \frac{a}{R} \quad (5-55)$$

External Flows For a single cylinder in cross flow, Churchill and Bernstein recommend [J. *Heat Transfer*, **99**, 300 (1977)]

$$\overline{Nu}_D = 0.3 + \frac{0.62 Re_D^{1/2} Pr^{1/3}}{[1 + (0.4/Pr)^{1/4}]^{1/4}} \left[1 + \left(\frac{Re_D}{282,000} \right)^{5/8} \right]^{4/5} \quad (5-56)$$

where $\overline{Nu}_D = \bar{h} D/k$. Equation (5-56) is for all values of Re_D and Pr , provided that $Re_D Pr > 0.4$. The fluid properties are evaluated at the film temperature $(T_e + T_s)/2$, where T_e is the free-stream temperature and T_s is the surface temperature. Equation (5-56) also applies to the uniform heat flux boundary condition provided $\bar{h}$ is based on the perimeter-averaged temperature difference between T_s and T_e .

For an isothermal spherical surface, Whitaker recommends [AIChE, **18**, 361 (1972)]

$$\overline{Nu}_D = 2 + (0.4 Re_D^{1/2} + 0.06 Re_D^{2/3}) Pr^{0.4} \left(\frac{\mu_e}{\mu_s} \right)^{1/4} \quad (5-57)$$

This equation is based on data for $0.7 < Pr < 380$, $3.5 < Re_D < 8 \times 10^4$, and $1 < (\mu_e/\mu_s) < 3.2$. The properties are evaluated at the free-stream temperature T_e , with the exception of μ_s , which is evaluated at the surface temperature T_s .

The average Nusselt number for laminar flow over an isothermal flat plate of length x is estimated from [Churchill and Ozoe, J. *Heat Transfer*, **95**, 416 (1973)]

$$\overline{Nu}_x = \frac{1.128 Pr^{1/2} Re_x^{1/2}}{[1 + (0.0468/Pr)^{2/3}]^{1/4}} \quad (5-58)$$

This equation is valid for all values of Pr as long as $Re_x > 100$ and $Re_x < 5 \times 10^5$. The fluid properties are evaluated at the film temperature $(T_e + T_s)/2$, where T_e is the free-stream temperature and T_s is the surface temperature. For a uniformly heated flat plate, the local Nusselt number is given by [Churchill and Ozoe, J. *Heat Transfer*, **95**, 78 (1973)]

$$Nu_x = \frac{0.886 Pr^{1/2} Re_x^{1/2}}{[1 + (0.0207/Pr)^{2/3}]^{1/4}} \quad (5-59)$$

where again the properties are evaluated at the film temperature.

The average Nusselt number for turbulent flow over a smooth, isothermal flat plate of length x is given by (Mills, *Heat Transfer*, 2d ed., Prentice-Hall, 1999, p. 315)

$$\overline{Nu}_x = 0.664 Re_{cr}^{1/2} Pr^{1/3} + 0.036 Re_x^{0.8} Pr^{0.43} \left[1 - \left(\frac{Re_{cr}}{Re_x} \right)^{0.5} \right] \quad (5-60)$$

The critical Reynolds number Re_{cr} is typically taken as 5×10^5 , $Re_{cr} < Re_x < 3 \times 10^7$, and $0.7 < Pr < 400$. The fluid properties are evaluated at the film temperature $(T_e + T_s)/2$, where T_e is the free-stream temperature and T_s is the surface temperature. Equation (5-60) also applies to the uniform heat flux boundary condition provided $\bar{h}$ is based on the average temperature difference between T_s and T_e .

Flow-through Tube Banks Aligned and staggered tube banks are sketched in Fig. 5-6. The tube diameter is D , and the transverse and longitudinal pitches are S_T and S_L , respectively. The fluid velocity upstream

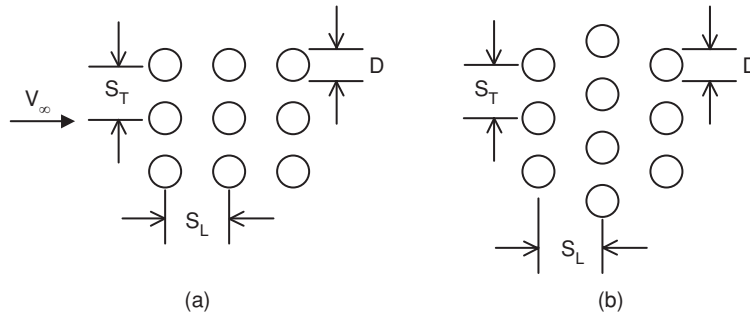


FIG. 5-6 (a) Aligned and (b) staggered tube bank configurations. The fluid velocity upstream of the tubes is V_∞ .

of the tubes is V_∞ . To estimate the overall heat-transfer coefficient for the tube bank, Mills proceeds as follows (*Heat Transfer*, 2d ed., Prentice-Hall, 1999, p. 348). The Reynolds number for use in (5-56) is recalculated with an effective average velocity in the space between adjacent tubes:

$$\frac{\bar{V}}{V_\infty} = \frac{S_T}{S_T - (\pi/4)D} \quad (5-61)$$

The heat-transfer coefficient increases from row 1 to about row 5 of the tube bank. The average Nusselt number for a tube bank with 10 or more rows is

$$\overline{Nu}_D^{10+} = \Phi \overline{Nu}_D \quad (5-62)$$

where Φ is an arrangement factor and $\overline{Nu}_D$ is the Nusselt number for the first row, calculated by using the velocity in (5-61). The arrangement factor is calculated as follows. Define dimensionless pitches as $P_T = S_T/D$ and P_L/D and calculate a factor ψ as follows.

$$\psi = \begin{cases} 1 - \frac{\pi}{4P_T} & \text{if } P_L \geq 1 \\ 1 - \frac{\pi}{4P_T P_L} & \text{if } P_L < 1 \end{cases} \quad (5-63)$$

The arrangement factors are

$$\Phi_{\text{aligned}} = 1 + \frac{0.7}{\psi^{1.5}} \frac{S_L/S_T - 0.3}{(S_L/S_T + 0.7)^2} \quad (5-64)$$

$$\Phi_{\text{staggered}} = 1 + \frac{2}{3P_L} \quad (5-65)$$

If there are fewer than 10 rows,

$$\overline{Nu}_D = \frac{1 + (N-1)\Phi}{N} \overline{Nu}_D \quad (5-66)$$

where N is the number of rows.

The fluid properties for gases are evaluated at the average mean film temperature $[(T_{\text{in}} + T_{\text{out}})/2 + T_s]/2$. For liquids, properties are evaluated at the bulk mean temperature $(T_{\text{in}} + T_{\text{out}})/2$, with a Prandtl number correction $(Pr_b/Pr_s)^{0.11}$ for cooling and $(Pr_b/Pr_s)^{0.25}$ for heating.

Falling Films When a liquid is distributed uniformly around the periphery at the top of a vertical tube (either inside or outside) and allowed to fall down the tube wall by the influence of gravity, the fluid does not fill the tube but rather flows as a thin layer. Similarly, when a liquid is applied uniformly to the outside and top of a horizontal tube, it flows in layer form around the periphery and falls off the bottom. In both these cases the mechanism is called gravity flow of liquid layers or falling films.

For the turbulent flow of **water** in layer form down the walls of **vertical tubes** the dimensional equation of McAdams, Drew, and Bays [*Trans. Am. Soc. Mech. Eng.*, **62**, 627 (1940)] is recommended:

$$h_{\text{lm}} = b\Gamma^{1/3} \quad (5-67)$$

where $b = 9150$ (SI) or 120 (U.S. Customary) and is based on values of $\Gamma = W_F = \dot{M}/\pi D$ ranging from 0.25 to 6.2 kg/(m·s) [600 to 15,000 lb/(h·ft)] of wetted perimeter. This type of water flow is used in vertical vapor-in-shell ammonia condensers, acid coolers, cycle water coolers, and other process-fluid coolers.

The following dimensional equations may be used for **any liquid** flowing in layer form down **vertical surfaces**:

$$\text{For } \frac{4\Gamma}{\mu} > 2100 \quad h_{\text{lm}} = 0.01 \left(\frac{k^3 \rho^2 g}{\mu^2} \right)^{1/3} \left(\frac{c\mu}{k} \right)^{1/3} \left(\frac{4\Gamma}{\mu} \right)^{1/3} \quad (5-68a)$$

$$\text{For } \frac{4\Gamma}{\mu} < 2100 \quad h_{\text{am}} = 0.50 \left(\frac{k^3 \rho^2 c g^{2/3}}{L\mu^{1/3}} \right)^{1/3} \left(\frac{\mu}{\mu_w} \right)^{1/4} \left(\frac{4\Gamma}{\mu} \right)^{1/9} \quad (5-68b)$$

Equation (5-68b) is based on the work of Bays and McAdams [*Ind. Eng. Chem.*, **29**, 1240 (1937)]. The significance of the term L is not clear. When $L = 0$, the coefficient is definitely not infinite. When L is large and the fluid temperature has not yet closely approached the wall temperature, it does not appear that the coefficient should

necessarily decrease. Within the finite limits of 0.12 to 1.8 m (0.4 to 6 ft), this equation should give results of the proper order of magnitude.

For falling films applied to the **outside of horizontal tubes**, the Reynolds number rarely exceeds 2100. Equations may be used for falling films on the outside of the tubes by substituting $\pi D/2$ for L .

For **water** flowing over a **horizontal tube**, data for several sizes of pipe are roughly correlated by the dimensional equation of McAdams, Drew, and Bays [*Trans. Am. Soc. Mech. Eng.*, **62**, 627 (1940)].

$$h_{\text{am}} = b(\Gamma/D_0)^{1/3} \quad (5-69)$$

where $b = 3360$ (SI) or 65.6 (U.S. Customary) and Γ ranges from 0.94 to 4 kg/(m·s) [100 to 1000 lb/(h·ft)].

Falling films are also used for evaporation in which the film is both entirely or partially evaporated (juice concentration). This principle is also used in crystallization (freezing).

The advantage of high coefficient in falling-film exchangers is partially offset by the difficulties involved in distribution of the film, maintaining complete wettability of the tube, and pumping costs required to lift the liquid to the top of the exchanger.

Finned Tubes (Extended Surface) When the heat-transfer coefficient on the outside of a metal tube is much lower than that on the inside, as when steam condensing in a pipe is being used to heat air, externally finned (or extended) heating surfaces are of value in increasing substantially the rate of heat transfer per unit length of tube. The data on extended heating surfaces, for the case of air flowing outside and at right angles to the axes of a bank of finned pipes, can be represented approximately by the dimensional equation derived from

$$h_f = b \frac{V_F^{0.6}}{D_0^{0.4}} \left(\frac{p'}{p' - D_0} \right)^{0.6} \quad (5-70)$$

where $b = 5.29$ (SI) or $(5.39)(10^{-3})$ (U.S. Customary); h_f is the coefficient of heat transfer on the air side; V_F is the face velocity of the air; p' is the center-to-center spacing, m, of the tubes in a row; and D_0 is the outside diameter, m, of the bare tube (diameter at the root of the fins).

In atmospheric air-cooled finned tube exchangers, the air-film coefficient from Eq. (5-70) is sometimes converted to a value based on outside bare surface as follows:

$$h_{fo} = h_f \frac{A_f + A_{uf}}{A_{of}} = h_f \frac{A_T}{A_o} \quad (5-71)$$

in which h_{fo} is the air-film coefficient based on external bare surface; h_f is the air-film coefficient based on total external surface; A_T is total external surface, and A_o is external bare surface of the unfinned tube; A_f is the area of the fins; A_{uf} is the external area of the unfinned portion of the tube; and A_{of} is area of tube before fins are attached.

Fin efficiency is defined as the ratio of the mean temperature difference from surface to fluid divided by the temperature difference from fin to fluid at the base or root of the fin. Graphs of fin efficiency for extended surfaces of various types are given by Gardner [*Trans. Am. Soc. Mech. Eng.*, **67**, 621 (1945)].

Heat-transfer coefficients for finned tubes of various types are given in a series of papers [*Trans. Am. Soc. Mech. Eng.*, **67**, 601 (1945)].

For flow of air normal to fins in the form of **short strips or pins**, Norris and Spofford [*Trans. Am. Soc. Mech. Eng.*, **64**, 489 (1942)] correlate their results for air by the dimensionless equation of Pohlhausen:

$$\frac{h_m}{c_p G_{\text{max}}} \left(\frac{c_p \mu}{k} \right)^{2/3} = 1.0 \left(\frac{z_p G_{\text{max}}}{\mu} \right)^{-0.5} \quad (5-72)$$

for values of $z_p G_{\text{max}}/\mu$ ranging from 2700 to 10,000.

For the general case, the treatment suggested by Kern (*Process Heat Transfer*, McGraw-Hill, New York, 1950, p. 512) is recommended. Because of the wide variations in fin-tube construction, it is convenient to convert all coefficients to values based on the inside bare surface of the tube. Thus to convert the coefficient based on outside area (finned side) to a value based on inside area Kern gives the following relationship:

$$h_{fi} = (\Omega A_f + A_o)(h_f/A_i) \quad (5-73)$$

5-12 HEAT AND MASS TRANSFER

in which h_f is the effective outside coefficient based on the inside area, h_f is the outside coefficient calculated from the applicable equation for bare tubes, A_f is the surface area of the fins, A_o is the surface area on the outside of the tube which is not finned, A_i is the inside area of the tube, and Ω is the fin efficiency defined as

$$\Omega = (\tanh mb_f)/mb_f \quad (5-74)$$

in which

$$m = (h_f p_f / ka_f)^{1/2} \quad \text{m}^{-1} \text{ (ft}^{-1}\text{)} \quad (5-75)$$

and b_f = height of fin. The other symbols are defined as follows: p_f is the perimeter of the fin, a_f is the cross-sectional area of the fin, and k is the thermal conductivity of the material from which the fin is made.

Fin efficiencies and fin dimensions are available from manufacturers. Ratios of finned to inside surface are usually available so that the terms A_f , A_o , and A_i may be obtained from these ratios rather than from the total surface areas of the heat exchangers.

JACKETS AND COILS OF AGITATED VESSELS

See Secs. 11 and 18.

NONNEWTONIAN FLUIDS

A wide variety of nonnewtonian fluids are encountered industrially. They may exhibit Bingham-plastic, pseudoplastic, or dilatant behavior

and may or may not be thixotropic. For design of equipment to handle or process nonnewtonian fluids, the properties must usually be measured experimentally, since no generalized relationships exist to predict the properties or behavior of the fluids. Details of handling nonnewtonian fluids are described completely by Skelland (*Non-Newtonian Flow and Heat Transfer*, Wiley, New York, 1967). The generalized shear-stress rate-of-strain relationship for nonnewtonian fluids is given as

$$n' = \frac{d \ln (D \Delta P / 4L)}{d \ln (8V/D)} \quad (5-76)$$

as determined from a plot of shear stress versus velocity gradient.

For **circular tubes**, $Gz > 100$, $n' > 0.1$, and laminar flow

$$\text{Nu}_m = 1.75 \delta_s^{1/3} Gz^{1/3} \quad (5-77)$$

where $\delta_s = (3n' + 1)/4n'$. When natural convection effects are considered, Metzner and Gluck [*Chem. Eng. Sci.*, **12**, 185 (1960)] obtained the following for **horizontal tubes**:

$$\text{Nu}_m = 1.75 \delta_s^{1/3} \left[Gz + 12.6 \left(\frac{\text{PrGrD}}{L} \right)^{1/3} \right]^{1/3} \left(\frac{\gamma_b}{\gamma_w} \right)^{0.14} \quad (5-78)$$

where properties are evaluated at the wall temperature, i.e., $\gamma = g_r K' 8n'^{-1}$ and $\tau_w = K' (8V/D)^{n'}$.

Metzner and Friend [*Ind. Eng. Chem.*, **51**, 879 (1959)] present relationships for turbulent heat transfer with nonnewtonian fluids. Relationships for heat transfer by natural convection and through laminar boundary layers are available in Skelland's book (op. cit.).

HEAT TRANSFER WITH CHANGE OF PHASE

In any operation in which a material undergoes a change of phase, provision must be made for the addition or removal of heat to provide for the latent heat of the change of phase plus any other sensible heating or cooling that occurs in the process. Heat may be transferred by any one or a combination of the three modes—conduction, convection, and radiation. The process involving change of phase involves mass transfer simultaneous with heat transfer.

CONDENSATION

Condensation Mechanisms Condensation occurs when a saturated vapor comes in contact with a surface whose temperature is below the saturation temperature. Normally a film of condensate is formed on the surface, and the thickness of this film, per unit of breadth, increases with increase in extent of the surface. This is called **film-type condensation**.

Another type of condensation, called **dropwise**, occurs when the wall is not uniformly wetted by the condensate, with the result that the condensate appears in many small droplets at various points on the surface. There is a growth of individual droplets, a coalescence of adjacent droplets, and finally a formation of a rivulet. Adhesive force is overcome by gravitational force, and the rivulet flows quickly to the bottom of the surface, capturing and absorbing all droplets in its path and leaving dry surface in its wake.

Film-type condensation is more common and more dependable. Dropwise condensation normally needs to be promoted by introducing an impurity into the vapor stream. Substantially higher (6 to 18 times) coefficients are obtained for dropwise condensation of steam, but design methods are not available. Therefore, the development of equations for condensation will be for the film type only.

The physical properties of the liquid, rather than those of the vapor, are used for determining the coefficient for condensation. Nusselt [*Z. Ver. Dtsch. Ing.*, **60**, 541, 569 (1916)] derived theoretical relationships for predicting the coefficient of heat transfer for condensation of a pure saturated vapor. A number of simplifying assumptions were used in the derivation.

The **Reynolds number** of the condensate film (falling film) is $4\Gamma/\mu$, where Γ is the weight rate of flow (loading rate) of condensate per unit perimeter $\text{kg}/(\text{s}\cdot\text{m})$ [$\text{lb}/(\text{h}\cdot\text{ft})$]. The thickness of the condensate film for Reynolds number less than 2100 is $(3\mu\Gamma/\rho^2g)^{1/3}$.

Condensation Coefficients

Vertical Tubes For the following cases Reynolds number < 2100 and is calculated by using $\Gamma = W_F/\pi D$. The **Nusselt equation** for the heat-transfer coefficient for condensate films may be written in the following ways (using liquid physical properties and where L is the cooled length and Δt is $t_{sv} - t_s$):

Nusselt type:

$$\frac{hL}{k} = 0.943 \left(\frac{L^3 \rho^2 g \lambda}{k \mu \Delta t} \right)^{1/4} = 0.925 \left(\frac{L^3 \rho^2 g}{\mu \Gamma} \right)^{1/3} \quad (5-79)^{\circ}$$

Dimensional:

$$h = b(k^3 \rho^2 D / \mu_b W_F)^{1/3} \quad (5-80)^{\circ}$$

where $b = 127$ (SI) or 756 (U.S. Customary). For steam at atmospheric pressure, $k = 0.682 \text{ J}/(\text{m}\cdot\text{s}\cdot\text{K})$ [$0.394 \text{ Btu}/(\text{h}\cdot\text{ft}\cdot^{\circ}\text{F})$], $\rho = 960 \text{ kg}/\text{m}^3$ ($60 \text{ lb}/\text{ft}^3$), $\mu_b = (0.28)(10^{-3}) \text{ Pa}\cdot\text{s}$ (0.28 cP),

$$h = b(D/W_F)^{1/3} \quad (5-81)$$

where $b = 2954$ (SI) or 6978 (U.S. Customary). For organic vapors at normal boiling point, $k = 0.138 \text{ J}/(\text{m}\cdot\text{s}\cdot\text{K})$ [$0.08 \text{ Btu}/(\text{h}\cdot\text{ft}\cdot^{\circ}\text{F})$], $\rho = 720 \text{ kg}/\text{m}^3$ ($45 \text{ lb}/\text{ft}^3$), $\mu_b = (0.35)(10^{-3}) \text{ Pa}\cdot\text{s}$ (0.35 cP),

$$h = b(D/W_F)^{1/3} \quad (5-82)$$

where $b = 457$ (SI) or 1080 (U.S. Customary).

Horizontal Tubes For the following cases Reynolds number < 2100 and is calculated by using $\Gamma = W_F/2L$.

[°] If the vapor density is significant, replace ρ^2 with $\rho(\rho_l - \rho_v)$.

No.	Substance
10	Acetic Acid
6	Acetone
1	Ammonia
5	Aniline
12	Benzene
8	Carbon Disulfide
14	Carbon Tetrachloride
9	Ethyl Acetate
4	Ethyl Alcohol
13	Ethyl Ether
3	Methyl Alcohol
11	Nitrobenzene
7	n-Propyl Alcohol
2	Water

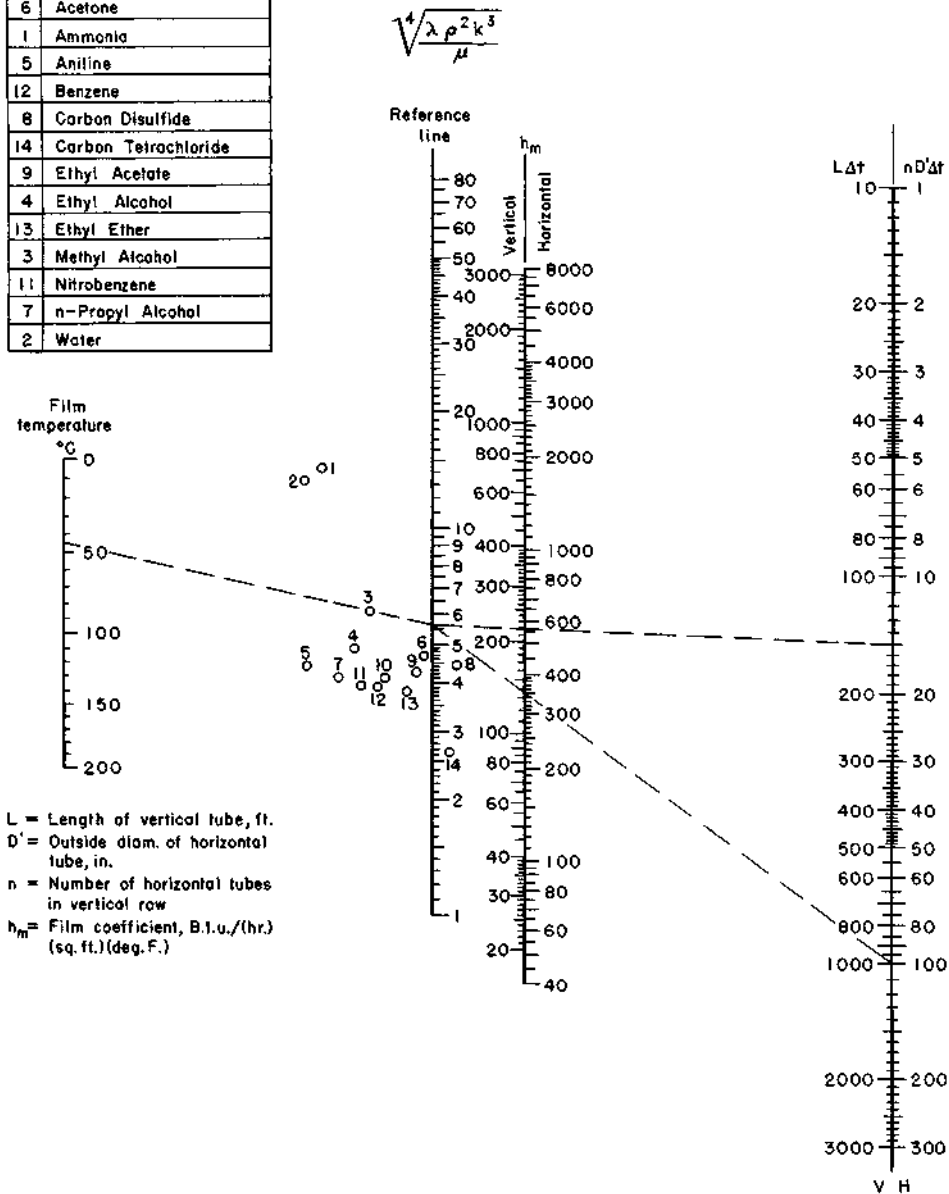


FIG. 5-7 Chart for determining heat-transfer coefficient h_m for film-type condensation of pure vapor, based on Eqs. (5-79) and (5-83). For vertical tubes multiply h_m by 1.2. If $4\Gamma/\mu$ exceeds 2100, use Fig. 5-8. $\sqrt{\lambda \rho^2 k^3}/\mu$ is in U.S. Customary units; to convert feet to meters, multiply by 0.3048; to convert inches to centimeters, multiply by 2.54; and to convert British thermal units per hour-square foot-degrees Fahrenheit to watts per square meter-kelvins, multiply by 5.6780.

Nusselt type:

$$\frac{hD}{k} = 0.73 \left(\frac{D^3 \rho^2 g \lambda}{k \mu \Delta t} \right)^{1/4} = 0.76 \left(\frac{D^3 \rho^2 g}{\mu \Gamma} \right)^{1/3} \quad (5-83)^{\circ}$$

Dimensional:

$$h = b(k^3 \rho^2 L / \mu_b W_F)^{1/3} \quad (5-84)^{\circ}$$

where $b = 205.4$ (SI) or 534 (U.S. Customary). For steam at atmospheric pressure

^o If the vapor density is significant, replace ρ^2 with $\rho(\rho_l - \rho_v)$.

$$h = b(L/W_F)^{1/3} \quad (5-85)$$

where $b = 2080$ (SI) or 4920 (U.S. Customary). For organic vapors at normal boiling point

$$h = b(L/W_F)^{1/3} \quad (5-86)$$

where $b = 324$ (SI) or 766 (U.S. Customary).

Figure 5-7 is a nomograph for determining coefficients of heat transfer for condensation of pure vapors.

Banks of Horizontal Tubes ($Re < 2100$) In the idealized case of N tubes in a vertical row where the total condensate flows smoothly from one tube to the one beneath it, without splashing, and still in laminar flow on the tube, the mean condensing coefficient h_N for the entire row of N tubes is related to the condensing coefficient for the top tube h_1 by

$$h_N = h_1 N^{-1/4} \quad (5-87)$$

Dukler Theory The preceding expressions for condensation are based on the classical Nusselt theory. It is generally known and conceded that the film coefficients for steam and organic vapors calculated by the Nusselt theory are conservatively low. Dukler [*Chem. Eng. Prog.*, **55**, 62 (1959)] developed equations for velocity and temperature distribution in thin films on vertical walls based on expressions of Deissler (NACA Tech. Notes 2129, 1950; 2138, 1952; 3145, 1959) for the eddy viscosity and thermal conductivity near the solid boundary. According to the Dukler theory, three fixed factors must be known to establish the value of the average film coefficient: the terminal Reynolds number, the Prandtl number of the condensed phase, and a dimensionless group N_d defined as follows:

$$N_d = (0.250 \mu_L^{1.173} \mu_G^{0.16}) / (g^{2/3} D^{2/3} \rho_L^{0.553} \rho_G^{0.78}) \quad (5-88)$$

Graphical relationships of these variables are available in Document 6058, ADI Auxiliary Publications Project, Library of Congress, Washington. If rigorous values for condensing-film coefficients are desired, especially if the value of N_d in Eq. (5-88) exceeds $(1)(10^{-5})$, it is suggested that these graphs be used. For the case in which interfacial shear is zero, Fig. 5-8 may be used. It is interesting to note that, according to the Dukler development, there is no definite transition Reynolds number; deviation from Nusselt theory is less at low Reynolds numbers; and when the Prandtl number of a fluid is less than 0.4 (at Reynolds number above 1000), the predicted values for film coefficient are lower than those predicted by the Nusselt theory.

The Dukler theory is applicable for condensate films on horizontal tubes and also for falling films, in general, i.e., those not associated with condensation or vaporization processes.

Vapor Shear Controlling For vertical in-tube condensation with vapor and liquid flowing concurrently downward, if gravity controls, Figs. 5-7 and 5-8 may be used. If vapor shear controls, the Carpenter-Colburn correlation [*General Discussion on Heat Transfer*, London, 1951, ASME, New York, p. 20] is applicable:

$$h \mu_L / k_f \rho^{1/2} = 0.065 (Pr)^{1/2} F_{ce}^{1/2} \quad (5-89a)$$

$$\text{where } F_{ce} = f G_{cm}^2 / 2 \rho_v \quad (5-89b)$$

$$G_{cm} = \left(\frac{G_{vi}^2 + G_{vi} G_{co} + G_{co}^2}{3} \right)^{1/2} \quad (5-89c)$$

and f is the Fanning friction factor evaluated at

$$(Re)_{cm} = D_i G_{cm} / \mu_v \quad (5-89d)$$

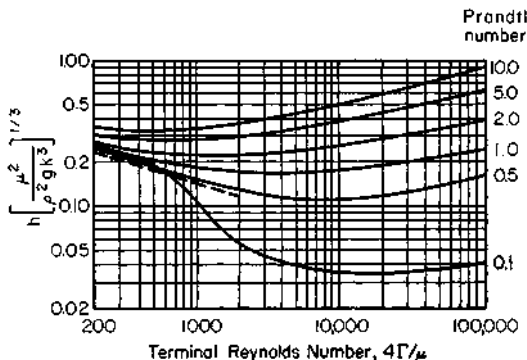


FIG. 5-8 Dukler plot showing average condensing-film coefficient as a function of physical properties of the condensate film and the terminal Reynolds number. (Dotted line indicates Nusselt theory for Reynolds number < 2100 .) [Reproduced by permission from *Chem. Eng. Prog.*, **55**, 64 (1959).]

and the subscripts vi and vo refer to the vapor inlet and outlet, respectively. An alternative formulation, directly in terms of the friction factor, is

$$h = 0.065 (c_p k_f / 2 \mu_v)^{1/2} G_{cm} \quad (5-89e)$$

expressed in consistent units.

Another correlation for vapor-shear-controlled condensation is the Boyko-Kruzhilin correlation [*Int. J. Heat Mass Transfer*, **10**, 361 (1967)], which gives the mean condensing coefficient for a stream between inlet quality x_i and outlet quality x_o :

$$\frac{h D_i}{k_l} = 0.024 \left(\frac{D_i G_T}{\mu_l} \right)^{0.8} (Pr)_l^{0.43} \frac{\sqrt{(\rho/\rho_m)_i} + \sqrt{(\rho/\rho_m)_o}}{2} \quad (5-90a)$$

where G_T = total mass velocity in consistent units

$$\left(\frac{\rho}{\rho_m} \right)_i = 1 + \frac{\rho_l - \rho_v}{\rho_v} x_i \quad (5-90b)$$

$$\text{and } \left(\frac{\rho}{\rho_m} \right)_o = 1 + \frac{\rho_l - \rho_v}{\rho_v} x_o \quad (5-90c)$$

For horizontal in-tube condensation at low flow rates Kern's modification (*Process Heat Transfer*, McGraw-Hill, New York, 1950) of the Nusselt equation is valid:

$$h_m = 0.761 \left[\frac{L k_l^3 \rho_l (\rho_l - \rho_v) g}{W_F \mu_l} \right]^{1/3} = 0.815 \left[\frac{k_l^3 \rho_l (\rho_l - \rho_v) g \lambda}{\pi \mu_l D_i \Delta t} \right]^{1/4} \quad (5-91)$$

where W_F is the total vapor condensed in one tube and Δt is $t_{sc} - t_s$. A more rigorous correlation has been proposed by Chaddock [*Refriger. Eng.*, **65**(4), 36 (1957)]. Use consistent units.

At high condensing loads, with vapor shear dominating, tube orientation has no effect, and Eq. (5-90a) may also be used for horizontal tubes.

Condensation of pure vapors under laminar conditions in the presence of noncondensable gases, interfacial resistance, superheating, variable properties, and diffusion has been analyzed by Minkowycz and Sparrow [*Int. J. Heat Mass Transfer*, **9**, 1125 (1966)].

BOILING (VAPORIZATION) OF LIQUIDS

Boiling Mechanisms Vaporization of liquids may result from various mechanisms of heat transfer, singly or combinations thereof. For example, vaporization may occur as a result of heat absorbed, by radiation and convection, at the surface of a pool of liquid; or as a result of heat absorbed by natural convection from a hot wall beneath the disengaging surface, in which case the vaporization takes place when the superheated liquid reaches the pool surface. Vaporization also occurs from falling films (the reverse of condensation) or from the flashing of liquids superheated by forced convection under pressure.

Pool boiling refers to the type of boiling experienced when the heating surface is surrounded by a relatively large body of fluid which is not flowing at any appreciable velocity and is agitated only by the motion of the bubbles and by natural-convection currents. Two types of pool boiling are possible: subcooled pool boiling, in which the bulk fluid temperature is below the saturation temperature, resulting in collapse of the bubbles before they reach the surface, and saturated pool boiling, with bulk temperature equal to saturation temperature, resulting in net vapor generation.

The general shape of the curve relating the heat-transfer coefficient to Δt_b , the temperature driving force (difference between the wall temperature and the bulk fluid temperature) is one of the few parametric relations that are reasonably well understood. The familiar boiling curve was originally demonstrated experimentally by Nukiyama [*J. Soc. Mech. Eng. (Japan)*, **37**, 367 (1934)]. This curve points out one of the great dilemmas for boiling-equipment designers. They are faced with at least six heat-transfer regimes in pool boiling: natural convection (+), incipient nucleate boiling (+), nucleate boiling (+), transition to film boiling (-), stable film boiling (+), and film boiling with increasing radiation (+). The signs indicate the sign of the derivative $d(q/A)/d \Delta t_b$. In the transition to film boiling, heat-transfer rate decreases with driving force. The regimes of greatest commercial interest are the nucleate-boiling and stable-film-boiling regimes.

Heat transfer by **nucleate boiling** is an important mechanism in the vaporization of liquids. It occurs in the vaporization of liquids in

kettle-type and natural-circulation reboilers commonly used in the process industries. High rates of heat transfer per unit of area (heat flux) are obtained as a result of bubble formation at the liquid-solid interface rather than from mechanical devices external to the heat exchanger. There are available several expressions from which reasonable values of the film coefficients may be obtained.

The boiling curve, particularly in the nucleate-boiling region, is significantly affected by the temperature driving force, the total system pressure, the nature of the boiling surface, the geometry of the system, and the properties of the boiling material. In the nucleate-boiling regime, heat flux is approximately proportional to the cube of the temperature driving force. Designers in addition must know the minimum Δt (the point at which nucleate boiling begins), the critical Δt (the Δt above which transition boiling begins), and the maximum heat flux (the heat flux corresponding to the critical Δt). For designers who do not have experimental data available, the following equations may be used.

Boiling Coefficients For the nucleate-boiling coefficient the Mostinski equation [Teplenergetika, **4**, 66 (1963)] may be used:

$$h = b P_c^{0.69} \left(\frac{q}{A} \right)^{0.7} \left[1.8 \left(\frac{P}{P_c} \right)^{0.17} + 4 \left(\frac{P}{P_c} \right)^{1.2} + 10 \left(\frac{P}{P_c} \right)^{10} \right] \quad (5-92)$$

where $b = (3.75)(10^{-5})$ (SI) or $(2.13)(10^{-4})$ (U.S. Customary), P_c is the critical pressure and P the system pressure, q/A is the heat flux, and h is the nucleate-boiling coefficient. The McNelly equation [J. Imp. Coll. Chem. Eng. Soc., 7(18), (1953)] may also be used:

$$h = 0.225 \left(\frac{q c_l}{A \lambda} \right)^{0.69} \left(\frac{P k_l}{\sigma} \right)^{0.31} \left(\frac{P_l}{P_c} - 1 \right)^{0.33} \quad (5-93)$$

where c_l is the liquid heat capacity, λ is the latent heat, P is the system pressure, k_l is the thermal conductivity of the liquid, and σ is the surface tension.

An equation of the Nusselt type has been suggested by Rohsenow [Trans. Am. Soc. Mech. Eng., **74**, 969 (1952)].

$$h D / k = C_r (D G / \mu)^{2/3} (c \mu / k)^{-0.7} \quad (5-94a)$$

in which the variables assume the following form:

$$\frac{h \beta'}{k} \left[\frac{g_c \sigma}{g(\rho_l - \rho_v)} \right]^{1/2} = C_r \left[\frac{\beta'}{\mu} \left(\frac{g_c \sigma}{g(\rho_l - \rho_v)} \right)^{1/2} \frac{W}{A} \right]^{2/3} \left(\frac{c \mu}{k} \right)^{-0.7} \quad (5-94b)$$

The coefficient C_r is not truly constant but varies from 0.006 to 0.015.* It is possible that the nature of the surface is partly responsible for the variation in the constant. The only factor in Eq. (5-94b) not readily available is the value of the contact angle β' .

Another Nusselt-type equation has been proposed by Forster and Zuber:†

$$Nu = 0.0015 Re^{0.62} Pr^{1/3} \quad (5-95)$$

which takes the following form:

$$\frac{c \rho_L \sqrt{\pi \alpha}}{k \rho_v} \frac{W}{A} \left(\frac{2 \sigma}{\Delta p} \right)^{1/2} \left(\frac{\rho_L}{\Delta p g_c} \right)^{1/4} = 0.0015 \left[\frac{\rho_L}{\mu} \left(\frac{c \rho_L \Delta T \sqrt{\pi \alpha}}{\lambda \rho_v} \right)^2 \right]^{0.62} \left(\frac{c \mu}{k} \right)^{1/2} \quad (5-96)$$

where $\alpha = k/\rho c$ (all liquid properties)

Δp = pressure of the vapor in a bubble minus saturation pressure of a flat liquid surface

Equations (5-94b) and (5-96) have been arranged in dimensional form by Westwater.

The numerical constant may be adjusted to suit any particular set of data if one desires to use a certain criterion. However, surface conditions vary so greatly that deviations may be as large as ± 25 percent from results obtained.

The **maximum heat flux** may be predicted by the Kutateladse-Zuber [Trans. Am. Soc. Mech. Eng., **80**, 711 (1958)] relationship, using consistent units:

$$\left(\frac{q}{A} \right)_{\max} = 0.18 g_c^{1/4} \rho_v \lambda \left[\frac{(\rho_l - \rho_v) \sigma g}{\rho_v^2} \right]^{1/4} \quad (5-97)$$

Alternatively, Mostinski presented an equation which approximately represents the Cichelli-Bonilla [Trans. Am. Inst. Chem. Eng., **41**, 755 (1945)] correlation:

$$\frac{(q/A)_{\max}}{P_c} = b \left(\frac{P}{P_c} \right)^{0.35} \left(1 - \frac{P}{P_c} \right)^{0.9} \quad (5-98)$$

where $b = 0.368$ (SI) or 5.58 (U.S. Customary); P_c is the critical pressure, Pa absolute; P is the system pressure; and $(q/A)_{\max}$ is the maximum heat flux.

The lower limit of applicability of the nucleate-boiling equations is from 0.1 to 0.2 of the maximum limit and depends upon the magnitude of natural-convection heat transfer for the liquid. The best method of determining the lower limit is to plot two curves: one of h versus Δt for natural convection, the other of h versus Δt for nucleate boiling. The intersection of these two curves may be considered the lower limit of applicability of the equations.

These equations apply to single tubes or to flat surfaces in a large pool. In tube bundles the equations are only approximate, and designers must rely upon experiment. Palen and Small [Hydrocarbon Process., **43**(11), 199 (1964)] have shown the effect of tube-bundle size on maximum heat flux.

$$\left(\frac{q}{A} \right)_{\max} = b \frac{P}{D_o \sqrt{N_T}} \rho_v \lambda \left[\frac{g \sigma (\rho_l - \rho_v)}{\rho_v^2} \right]^{1/4} \quad (5-99)$$

where $b = 0.43$ (SC) or 61.6 (U.S. Customary), p is the tube pitch, D_o is the tube outside diameter, and N_T is the number of tubes (twice the number of complete tubes for U-tube bundles).

For **film boiling**, Bromley's [Chem. Eng. Prog., **46**, 221 (1950)] correlation may be used:

$$h = b \left[\frac{k_v^3 (\rho_l - \rho_v) \rho_v g}{\mu_v D_o \Delta t_b} \right]^{1/4} \quad (5-100)$$

where $b = 4.306$ (SI) or 0.620 (U.S. Customary). Katz, Myers, and Balekjian [Pet. Refiner, **34**(2), 113 (1955)] report boiling heat-transfer coefficients on finned tubes.

HEAT TRANSFER BY RADIATION

GENERAL REFERENCES: Baukal, C. E., ed., *The John Zink Combustion Handbook*, CRC Press, Boca Raton, Fla., 2001. Blokh, A. G., *Heat Transfer in Steam Boiler Furnaces*, 3d ed., Taylor & Francis, New York, 1987. Brewster, M. Quinn, *Thermal Radiation Heat Transfer and Properties*, Wiley, New York, 1992. Goody, R. M., and Y. L. Yung, *Atmospheric Radiation—Theoretical Basis*, 2d ed., Oxford University Press, 1995. Hottel, H. C., and A. F. Sarofim, *Radiative Transfer*, McGraw-Hill, New York, 1967. Modest, Michael F., *Radiative Heat Transfer*, 2d ed., Academic Press, New York, 2003. Noble, James J., "The Zone

Method: Explicit Matrix Relations for Total Exchange Areas," *Int. J. Heat Mass Transfer*, **18**, 261–269 (1975). Rhine, J. M., and R. J. Tucker, *Modeling of Gas-Fired Furnaces and Boilers*, British Gas Association with McGraw-Hill, 1991. Siegel, Robert, and John R. Howell, *Thermal Radiative Heat Transfer*, 4th ed., Taylor & Francis, New York, 2001. Sparrow, E. M., and R. D. Cess, *Radiation Heat Transfer*, 3d ed., Taylor & Francis, New York, 1988. Stultz, S. C., and J. B. Kitto, *Steam: Its Generation and Use*, 40th ed., Babcock and Wilcox, Barksdon, Ohio, 1992.

* Reported by Westwater in Drew and Hoopes, *Advances in Chemical Engineering*, vol. I, Academic, New York, 1956, p. 15.

† Forster, J. *Appl. Phys.*, **25**, 1067 (1954); Forster and Zuber, J. *Appl. Phys.*, **25**, 474 (1954); Forster and Zuber, Conference on Nuclear Engineering, University of California, Los Angeles, 1955; excellent treatise on boiling of liquids by Westwater in Drew and Hoopes, *Advances in Chemical Engineering*, vol. I, Academic, New York, 1956.

INTRODUCTION

Heat transfer by thermal radiation involves the transport of electromagnetic (EM) energy from a source to a sink. In contrast to other modes of heat transfer, radiation does not require the presence of an intervening medium, e.g., as in the irradiation of the earth by the sun. Most industrially important applications of radiative heat transfer occur in the *near infrared* portion of the EM spectrum (0.7 through 25 μm) and may extend into the *far infrared* region (25 to 1000 μm). For very high temperature sources, such as solar radiation, relevant wavelengths encompass the entire *visible region* (0.4 to 0.7 μm) and may extend down to 0.2 μm in the *ultraviolet* (0.01- to 0.4- μm) portion of the EM spectrum. Radiative transfer can also exhibit unique *action-at-a-distance* phenomena which do not occur in other modes of heat transfer. Radiation differs from conduction and convection not only with regard to mathematical characterization but also with regard to its fourth power dependence on temperature. Thus it is usually dominant in high-temperature combustion applications. The temperature at which radiative transfer accounts for roughly one-half of the total heat loss from a surface in air depends on such factors as surface **emissivity** and the convection coefficient. For pipes in free convection, radiation is important at ambient temperatures. For fine wires of low emissivity it becomes important at temperatures associated with bright red heat (1300 K). Combustion gases at furnace temperatures typically lose more than 90 percent of their energy by radiative emission from constituent carbon dioxide, water vapor, and particulate matter. Radiative transfer methodologies are important in myriad engineering applications. These include semiconductor processing, illumination theory, and gas turbines and rocket nozzles, as well as furnace design.

THERMAL RADIATION FUNDAMENTALS

In a vacuum, the wavelength λ , frequency, ν and wavenumber η for electromagnetic radiation are interrelated by $\lambda = c/\nu = 1/\eta$, where c is the speed of light. Frequency is independent of the index of refraction of a medium n , but both the speed of light and the wavelength in the medium vary according to $c_m = c/n$ and $\lambda_m = \lambda/n$. When a radiation beam passes into a medium of different refractive index, not only does its wavelength change but so does its direction (Snell's law) as well as the magnitude of its intensity. In most engineering heat-transfer calculations, wavelength is usually employed to characterize radiation while wave number is often used in gas spectroscopy. For a vacuum, air at ambient conditions, and most gases, $n \approx 1.0$. For this reason this presentation sometimes does not distinguish between λ and λ_m . *Dielectric materials* exhibit $1.4 < n < 4$, and the speed of light decreases considerably in such media.

In radiation heat transfer, the **monochromatic intensity** $I_\lambda \equiv I_\lambda(\vec{r}, \vec{\Omega}, \lambda)$, is a *fundamental* (scalar) *field variable* which characterizes EM energy transport. Intensity defines the radiant energy flux passing through an infinitesimal area dA , oriented *normal* to a radiation beam of arbitrary direction $\vec{\Omega}$. At steady state, the monochromatic intensity is a function of position $\vec{r}$, direction $\vec{\Omega}$, and wavelength and has units of $\text{W}/(\text{m}^2 \cdot \text{sr} \cdot \mu\text{m})$. In the general case of an *absorbing-emitting* and *scattering* medium, characterized by some absorption coefficient $K(\text{m}^{-1})$, intensity in the direction $\vec{\Omega}$ will be modified by attenuation and by *scattering* of radiation into and out of the beam. For the special case of a nonabsorbing (transparent), nonscattering, medium of constant refractive index, the radiation intensity is *constant and independent of position in a given direction* $\vec{\Omega}$. This circumstance arises in *illumination theory* where the light intensity in a room is constant in a *given direction* but may vary with respect to *all other directions*. The basic conservation law for radiation intensity is termed the *equation of transfer* or *radiative transfer equation*. The equation of transfer is a *directional energy balance* and mathematically is an *integrodifferential equation*. The relevance of the transport equation to radiation heat transfer is discussed in many sources; see, e.g., Modest, M. F., *Radiative Heat Transfer*, 2d ed., Academic Press, 2003, or Siegel, R., and J. R. Howell, *Thermal Radiative Heat Transfer*, 4th ed., Taylor & Francis, New York, 2001.

Introduction to Radiation Geometry Consider a homogeneous medium of constant refractive index n . A pencil of radiation

originates at differential area element dA_i and is incident on differential area element dA_j . Designate $\vec{n}_i$ and $\vec{n}_j$ as the unit vectors normal to dA_i and dA_j , and let r_j with unit direction vector $\vec{\Omega}$, define the distance of separation between the area elements. Moreover, ϕ_i and ϕ_j denote the *confined angles* between $\vec{\Omega}$ and $\vec{n}_i$ and $\vec{n}_j$, respectively [i.e., $\cos\phi_i \equiv \cos(\vec{\Omega}, \vec{r}_i)$ and $\cos\phi_j \equiv \cos(\vec{\Omega}, \vec{r}_j)$]. As the beam travels toward dA_j , it will *diverge* and subtend a solid angle

$$d\Omega_j = \frac{\cos\phi_j}{r_j^2} dA_j \text{ sr}$$

at dA_j . Moreover, the projected area of dA_i in the direction of $\vec{\Omega}$ is given by $\cos(\vec{\Omega}, \vec{r}_i) dA_i = \cos\phi_i dA_i$. Multiplication of the intensity $I_\lambda \equiv I_\lambda(\vec{r}, \vec{\Omega}, \lambda)$ by $d\Omega_j$ and the apparent area of dA_i then yields an expression for the (differential) net *monochromatic radiant energy flux* dQ_{ij} originating at dA_i and intercepted by dA_j .

$$dQ_{ij} = I_\lambda(\vec{r}, \vec{\Omega}, \lambda) \cos\phi_i \cos\phi_j dA_i dA_j / r_j^2 \quad (5-101)$$

The **hemispherical emissive power**^{*} E is defined as the radiant flux density (W/m^2) associated with emission from an element of surface area dA into a surrounding unit hemisphere whose base is coplanar with dA . If the monochromatic intensity $I_\lambda(\vec{\Omega}, \lambda)$ of emission from the surface is *isotropic* (independent of the angle of emission, $\vec{\Omega}$), Eq. (5-101) may be integrated over the 2π sr of the surrounding unit hemisphere to yield the simple relation $E_\lambda = \pi I_\lambda$, where $E_\lambda \equiv E_\lambda(\lambda)$ is defined as the monochromatic or *spectral* hemispherical emissive power.

Blackbody Radiation Engineering calculations involving thermal radiation normally employ the **hemispherical blackbody emissive power** as the thermal driving force analogous to temperature in the cases of conduction and convection. A **blackbody** is a theoretical idealization for a *perfect theoretical radiator*; i.e., it absorbs all incident radiation without reflection and emits isotropically. In practice, soot-covered surfaces sometimes approximate blackbody behavior. Let $E_{b,\lambda} = E_{b,\lambda}(T, \lambda)$ denote the monochromatic blackbody hemispherical emissive power *frequency function* defined such that $E_{b,\lambda}(T, \lambda) d\lambda$ represents the fraction of blackbody energy lying in the wavelength region from λ to $\lambda + d\lambda$. The function $E_{b,\lambda} = E_{b,\lambda}(T, \lambda)$ is given by **Planck's law**

$$\frac{E_{b,\lambda}(T, \lambda)}{n^2 T^5} = \frac{c_1 (\lambda T)^{-5}}{e^{c_2/\lambda T} - 1} \quad (5-102)$$

where $c_1 = 2\pi^5 h c^2 / 15$ and $c_2 = hc/k$ are defined as Planck's first and second constants, respectively.

Integration of Eq. (5-102) over all wavelengths yields the **Stefan-Boltzman law** for the hemispherical blackbody emissive power

$$E_b(T) = \int_{\lambda=0}^{\infty} E_{b,\lambda}(T, \lambda) d\lambda = n^2 \sigma T^4 \quad (5-103)$$

where $\sigma = c_1(\pi/c_2)^4/15$ is the Stephan-Boltzman constant. Since a blackbody is an *isotropic emitter*, it follows that the intensity of blackbody emission is given by the simple formula $I_b = E_b/\pi = n^2 \sigma T^4/\pi$. The intensity of radiation emitted over all wavelengths by a blackbody is thus uniquely determined by its temperature. In this presentation, all references to hemispherical emissive power shall be to the *blackbody* emissive power, and the subscript b may be suppressed for expediency.

For short wavelengths $\lambda T \rightarrow 0$, the asymptotic form of Eq. (5-102) is known as the **Wien equation**

$$\frac{E_{b,\lambda}(T, \lambda)}{n^2 T^5} \equiv c_1 (\lambda T)^{-5} e^{-c_2/\lambda T} \quad (5-104)$$

The error introduced by use of the Wien equation is less than 1 percent when $\lambda T < 3000 \mu\text{m} \cdot \text{K}$. The Wien equation has significant practical value in optical pyrometry for $T < 4600 \text{ K}$ when a red filter ($\lambda = 0.65 \mu\text{m}$) is employed. The long-wavelength asymptotic approximation for Eq. (5-102) is known as the **Rayleigh-Jeans formula**, which is accurate to within 1 percent for $\lambda T > 778,000 \mu\text{m} \cdot \text{K}$. The Rayleigh-Jeans formula is of limited engineering utility since a blackbody emits over 99.9 percent of its total energy below the value of $\lambda T = 53,000 \mu\text{m} \cdot \text{K}$.

^{*}In the literature the emissive power is variously called the emittance, total hemispherical intensity, or radiant flux density.

Nomenclature and Units—Radiative Transfer

$a, a_g, a_{g,1}$	WSGG spectral model clear plus gray weighting constants
$\overline{C}_p, \overline{C}_{p,prod}$	Heat capacity per unit mass, J·kg ⁻¹ ·K ⁻¹
$ij = s_i s_j$	Shorthand notation for direct exchange area
A, A_i	Area of enclosure or zone i , m ²
c	Speed of light in vacuum, m/s
c_1, c_2	Planck's first and second constants, W·m ² and m·K
d_p, r_p	Particle diameter and radius, μ m
$E_{b,\lambda} = E_{b,\lambda}(T, \lambda)$	Monochromatic, blackbody emissive power, W/(m ² · μ m)
$E_n(x)$	Exponential integral of order n , where $n = 1, 2, 3, \dots$
E	Hemispherical emissive power, W/m ²
$E_b = n^2 \sigma T^4$	Hemispherical blackbody emissive power, W/m ²
f_c	Volumetric fraction of soot
$F_b(\lambda T)$	Blackbody fractional energy distribution
F_{ij}	Direct view factor from surface zone i to surface zone j
$\overline{F}_{ij}$	Refractory augmented black view factor; $\overline{F}$ -bar
$\mathcal{F}_{ij}$	Total view factor from surface zone i to surface zone j
h	Planck's constant, J·s
h_i	Heat-transfer coefficient, W/(m ² ·K)
H_i	Incident flux density for surface zone i , W/m ²
H	Enthalpy rate, W
$\dot{H}_F$	Enthalpy feed rate, W
$\dot{H}_k = I_k(\vec{r}, \vec{\Omega}, \lambda)$	Monochromatic radiation intensity, W/(m ² · μ m·sr)
k	Boltzmann's constant, J/K
$k_{\lambda,p}$	Monochromatic line absorption coefficient, (atm·m) ⁻¹
K	Gas absorption coefficient, m ⁻¹
L_M, L_{M0}	Average and optically thin mean beam lengths, m
$\dot{m}$	Mass flow rate, kg/h ⁻¹
n	Index of refraction
M, N	Number of surface and volume zones in enclosure
p_k	Partial pressure of species k , atm
P	Number of WSGG gray gas spectral windows
Q_i	Total radiative flux originating at surface zone i , W
Q_{ij}	Net radiative flux between zone i and zone j , W
T	Temperature, K
U	Overall heat-transfer coefficient in WSCC model
V	Enclosure volume, m ³
W	Leaving flux density (radiosity), W/m ²
Greek Characters	
$\alpha, \alpha_{1,2}$	Surface absorptivity or absorptance; subscript 1 refers to the surface temperature while subscript 2 refers to the radiation source
$\alpha_{g,1}, \epsilon_g, \tau_{g,1}$	Gas absorptivity, emissivity, and transmissivity
β	Dimensionless constant in mean beam length equation, $L_M = \beta \cdot L_{M0}$
$\Delta T_{gr} = T_g - T_e$	Adjustable temperature fitting parameter for WSCC model, K
ϵ	Gray diffuse surface emissivity
$\epsilon_g(T, r)$	Gas emissivity with path length r
$\epsilon_{\lambda}(T, \Omega, \lambda)$	Monochromatic, unidirectional, surface emissivity
$\eta = 1/\lambda$	Wave number in vacuum, cm ⁻¹
$\lambda = c/\nu$	Wavelength in vacuum, μ m
ν	Frequency, Hz
$\rho = 1 - \epsilon$	Diffuse reflectivity
σ	Stefan-Boltzmann constant, W/(m ² ·K ⁴)
Σ	Number of unique direct surface-to-surface direct exchange areas
$\tau_g = 1 - \epsilon_g$	Gas transmissivity
Ω	Solid angle, sr (steradians)
Φ	Equivalence ratio of fuel and oxidant
$\Psi^{(3)}(x)$	Pentagramma function of x
ω	Albedo for single scatter
Dimensionless Quantities	
$D_{\text{eff}} = \frac{N_{FD}}{(S_1 G_R / A_1) + N_{CR}}$	Effective firing density
$N_{CR} = \frac{h}{4\sigma T_{g,1}^3}$	Convection-radiation number
$N_{FD} = \dot{H}_F / \sigma T_{\text{Ref}}^4 \cdot A_1$	Dimensionless firing density
η_g	Gas-side furnace efficiency
$\eta_{gr} = \eta_g(1 - \Theta_0)$	Reduced furnace efficiency
$\Theta_i = T_i / T_{\text{Ref}}$	Dimensionless temperature
Vector Notation	
$\vec{n}_i$ and $\vec{n}_j$	Unit vectors normal to differential area elements dA_i and dA_j
$\vec{r}$	Position vector
$\vec{\Omega}$	Arbitrary unit direction vector

Matrix Notation	
$\mathbf{I}_M$	Column vector; all of whose elements are unity, $[M \times 1]$
$\mathbf{I} = [\delta_{ij}]$	Identity matrix, where δ_{ij} is the Kronecker delta; i.e., $\delta_{ij} = 1$ for $i = j$ and $\delta_{ij} = 0$ for $i \neq j$.
$\mathbf{aI}$	Diagonal matrix of WSGG gray gas surface zone a -weighting factors $[M \times M]$
$\mathbf{a}_g \mathbf{I}$	Diagonal matrix of gray gas WSGG volume zone a -weighting factors $[N \times N]$
$\mathbf{A} = [A_{ij}]$	Arbitrary nonsingular square matrix
$\mathbf{A}^T = [A_{ji}]$	Transpose of $\mathbf{A}$
$\mathbf{A}^{-1} = [A_{ij}]^{-1}$	Inverse of $\mathbf{A}$
$\mathbf{DI} = [D_i \cdot \delta_{ij}]$	Arbitrary diagonal matrix
$\mathbf{DI}^{-1} = [\delta_{ij} / D_i]$	Inverse of diagonal matrix
$\mathbf{CDI}$	$\mathbf{CI} \cdot \mathbf{DI} = [C_i \cdot D_i \cdot \delta_{ij}]$, product of two diagonal matrices
$\mathbf{AI} = [A_i \cdot \delta_{ij}]$	Diagonal matrix of surface zone areas, m ² $[M \times M]$
$\epsilon \mathbf{I} = [\epsilon_i \cdot \delta_{ij}]$	Diagonal matrix of diffuse zone emissivities $[M \times M]$
$\rho \mathbf{I} = [\rho_i \cdot \delta_{ij}]$	Diagonal matrix of diffuse zone reflectivities $[M \times M]$
$\mathbf{E} = [E_i] = [\sigma T_i^4]$	Column vector of surface blackbody hemispherical emissive powers, W/m ² $[M \times 1]$
$\mathbf{EI} = [E_i \cdot \delta_{ij}] = [\sigma T_i^4 \cdot \delta_{ij}]$	Diagonal matrix of surface blackbody emissive powers, W/m ² $[M \times M]$
$\mathbf{E}_g = [E_{g,i}] = [\sigma T_{g,i}^4]$	Column vector of gas blackbody hemispherical emissive powers, W/m ² $[N \times 1]$
$\mathbf{E}_g \mathbf{I} = [E_i \cdot \delta_{ij}] = [\sigma T_i^4 \cdot \delta_{ij}]$	Diagonal matrix of gas blackbody emissive powers, W/m ² $[N \times N]$
$\mathbf{H} = [H_i]$	Column vector of surface zone incident flux densities, W/m ² $[M \times 1]$
$\mathbf{W} = [W_i]$	Column vector of surface zone leaving flux densities, W/m ² $[M \times 1]$
$\mathbf{Q} = [Q_i]$	Column vector of surface zone fluxes, W $[M \times 1]$
$\mathbf{R} = [\mathbf{AI} - \mathbf{ss} \cdot \rho \mathbf{I}]^{-1}$	Inverse multiple-reflection matrix, m ⁻² $[M \times M]$
$\mathbf{KI}_p = [\delta_{ij} / K_{p,i}]$	Diagonal matrix of WSGG $K_{p,i}$ values for the i th zone and p th gray gas component, m ⁻¹ $[N \times N]$
$\overline{\mathbf{KI}}$	Diagonal matrix of WSGG-weighted gray gas absorption coefficients, m ⁻¹ $[N \times N]$
$\mathbf{S}'$	Column vector for net volume absorption, W $[N \times 1]$
$\overline{\mathbf{ss}} = [s_i s_j]$	Array of direct surface-to-surface exchange areas, m ² $[M \times M]$
$\overline{\mathbf{sg}} = [s_i g_j] = \overline{\mathbf{gs}}^T$	Array of direct gas-to-surface exchange areas, m ² $[M \times N]$
$\overline{\mathbf{gg}} = [g_i g_j]$	Array of direct gas-to-gas exchange areas, m ² $[N \times N]$
$\overline{\mathbf{SS}} = [\overline{S_i S_j}]$	Array of total surface-to-surface exchange areas, m ² $[M \times M]$
$\overline{\mathbf{SG}} = [\overline{S_i G_j}]$	Array of total gas-to-surface exchange areas, m ² $[M \times N]$
$\overline{\mathbf{GS}} = \overline{\mathbf{GS}}^T$	Array of total surface-to-gas exchange areas, m ² $[N \times M]$
$\overline{\mathbf{GG}} = [\overline{G_i G_j}]$	Array of total gas-to-gas exchange areas, m ² $[N \times N]$
$\overline{\mathbf{SS}} = [\overline{S_i S_i}]$	Array of directed surface-to-surface exchange areas, m ² $[M \times M]$
$\overline{\mathbf{SG}} = [\overline{S_i G_i}]$	Array of directed gas-to-surface exchange areas, m ² $[M \times N]$
$\overline{\mathbf{GS}} \neq \overline{\mathbf{SG}}^T$	Array of directed surface-to-gas exchange areas, m ² $[N \times M]$
$\overline{\mathbf{GG}} = [\overline{G_i G_i}]$	Array of directed gas-to-gas exchange areas, m ² $[N \times N]$
$\mathbf{VI} = [V_i \cdot \delta_{ij}]$	Diagonal matrix of zone volumes, m ³ $[N \times N]$
Subscripts	
b	Blackbody or denotes a black surface zone
f	Denotes flux surface zone
h	Denotes hemispherical surface emissivity
i, j	Zone number indices
n	Denotes normal component of surface emissivity
p	Index for p th gray gas window
r	Denotes refractory surface zone
s	Denotes source-sink surface zone
λ	Denotes monochromatic variable
Ref	Denotes reference quantity
Abbreviations	
CFD	Computational fluid dynamics
DO, FV	Discrete ordinate and finite volume methods
EM	Electromagnetic
RTE	Radiative transfer equation; equation of transfer
LPFF	Long plug flow furnace model
SSR	Source-sink refractory model
WSCC	Well-stirred combustion chamber model
WSGG	Weighted sum of gray gases spectral model

The blackbody fractional energy distribution function is defined by

$$F_b(\lambda T) = \frac{\int_0^\lambda E_{b,\lambda}(T, \lambda) d\lambda}{\int_0^\infty E_{b,\lambda}(T, \lambda) d\lambda} \quad (5-105)$$

The function $F_b(\lambda T)$ defines the fraction of total energy in the blackbody spectrum which lies below λT and is a unique function of λT . For purposes of digital computation, the following series expansion for $F_b(\lambda T)$ proves especially useful.

$$F_b(\lambda T) = \frac{15}{\pi^4} \sum_{k=1}^{\infty} \frac{e^{-k\xi}}{k} \left(\xi^3 + \frac{3\xi^2}{k} + \frac{6\xi}{k^2} + \frac{6}{k^3} \right) \text{ where } \xi = \frac{c_2}{\lambda T} \quad (5-106)$$

Equation (5-106) converges rapidly and is due to Lowan [1941] as referenced in Chang and Rhee [Int. Comm. Heat Mass Transfer, 11, 451-455 (1984)].

Numerically, in the preceding, $h = 6.6260693 \times 10^{-34}$ J-s is the Planck constant; $c = 2.99792458 \times 10^8$ m/s is the velocity of light in vacuum; and $k = 1.3806505 \times 10^{-23}$ J/K is the Boltzmann constant. These data lead to the following values of Planck's first and second constants: $c_1 = 3.741771 \times 10^{-16}$ W-m² and $c_2 = 1.438775 \times 10^{-2}$ m-K, respectively. Numerical values of the Stephan-Boltzmann constant σ in several systems of units are as follows: 5.67040×10^{-8} W/(m²-K⁴); 1.3544×10^{-12} cal/(cm²-s-K⁴); 4.8757×10^{-8} kcal/(m²-h-K⁴); 9.9862×10^{-9} CHU/(ft²-h-K⁴); and 0.17123×10^{-5} Btu/(ft²-h-°R⁴) (CHU = centigrade heat unit; 1.0 CHU = 1.8 Btu.)

Blackbody Displacement Laws The blackbody energy spectrum

$$\text{is plotted logarithmically in Fig. 5-9 as } \frac{E_{b,\lambda}(\lambda T)}{n^2 T^5} \times 10^{13} \frac{\text{W}}{\text{m}^2 \cdot \mu\text{m} \cdot \text{K}^5}$$

versus λT μm-K. For comparison a companion inset is provided in Cartesian coordinates. The upper abscissa of Fig. 5-9 also shows the blackbody energy distribution function $F_b(\lambda T)$. Figure 5-9 indicates that the wavelength-temperature product for which the maximum intensity occurs is $\lambda_{\max} T = 2898$ μm-K. This relationship is known as **Wien's displacement law**, which indicates that the wavelength for maximum intensity is *inversely* proportional to the absolute temperature. Blackbody displacement laws are useful in engineering practice to estimate wavelength intervals appropriate to relevant system temperatures. The Wien displacement law can be misleading, however, because the wavelength for maximum intensity depends on whether the intensity is defined in terms of frequency or wavelength interval. Two additional useful displacement laws are defined in terms of either the value of λT corresponding to the maximum energy per unit *fractional change* in wavelength or frequency, that is, $\lambda T = 3670$ μm-K, or to the value of λT corresponding to one-half the blackbody energy, that is, $\lambda T = 4107$ μm-K. Approximately one-half of the blackbody energy lies within the twofold λT range *geometrically centered* on $\lambda T = 3670$ μm-K, that is, $3670/\sqrt{2} < \lambda T < 3670\sqrt{2}$ μm-K. Some 95 percent of the blackbody energy lies in the interval $1662.6 < \lambda T < 16,295$ μm-K. It thus follows that for the temperature range between ambient (300 K) and flame temperatures (2000 K or

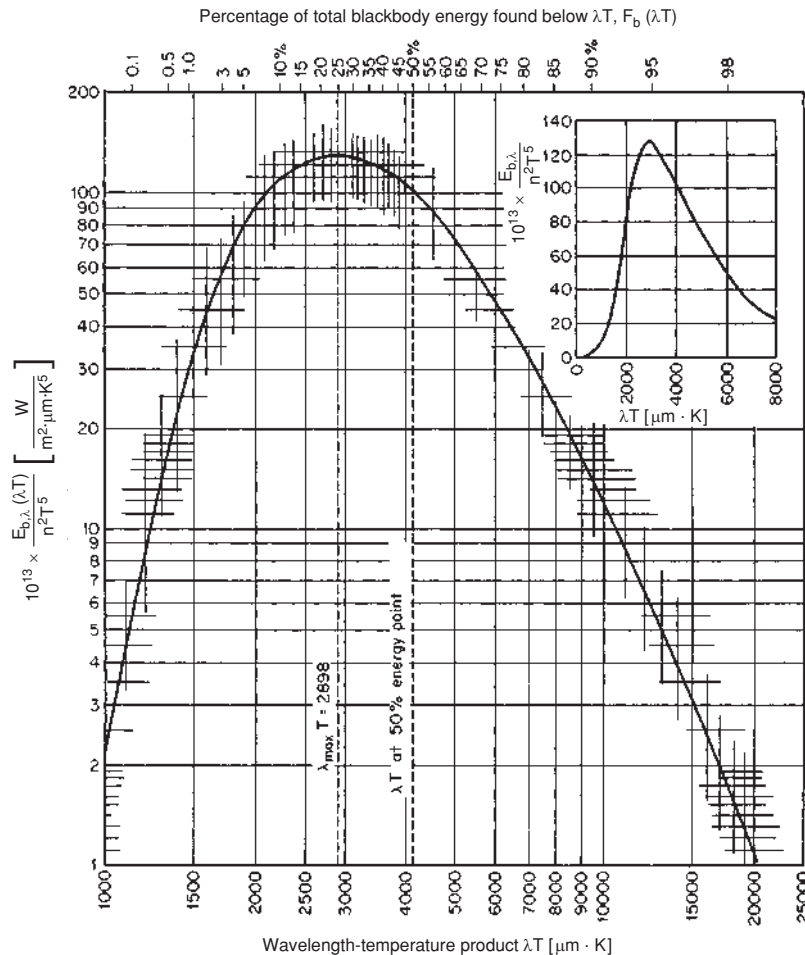


FIG. 5-9 Spectral dependence of monochromatic blackbody hemispherical emissive power.

3140°F), wavelengths of engineering heat-transfer importance are bounded between 0.83 and 54.3 μm .

RADIATIVE PROPERTIES OF OPAQUE SURFACES

Emittance and Absorptance The ratio of the total radiating power of any surface to that of a black surface at the same temperature is called the **emittance** or **emissivity**, ϵ of the surface.* In general, the monochromatic emissivity is a function of temperature, direction, and wavelength, that is, $\epsilon_\lambda = \epsilon_\lambda(T, \Omega, \lambda)$. The subscripts n and h are sometimes used to denote the normal and hemispherical values, respectively, of the emittance or emissivity. If radiation is incident on a surface, the fraction absorbed is called the **absorptance** (**absorptivity**). Two subscripts are usually appended to the absorptance $\alpha_{1,2}$ to distinguish between the temperature of the absorbing surface T_1 and the spectral energy distribution of the emitting surface T_2 . According to **Kirchhoff's law**, the emissivity and absorptivity of a surface exposed to surroundings at its own temperature are the same for both monochromatic and total radiation. When the temperatures of the surface and its surroundings differ, the total emissivity and absorptivity of the surface are often found to be unequal; but because the absorptivity is substantially independent of irradiation density, the monochromatic emissivity and absorptivity of surfaces are equal for all practical purposes. The difference between *total* emissivity and absorptivity depends on the variation of ϵ_λ with wavelength and on the difference between the temperature of the surface and the effective temperature of the surroundings.

Consider radiative exchange between a real surface of area A_1 at temperature T_1 with black surroundings at temperature T_2 . The *net* radiant interchange is given by

$$Q_{1,2} = A_1 \int_{\lambda=0}^{\infty} [\epsilon_\lambda(T_1, \lambda) \cdot E_{b,\lambda}(T_1, \lambda) - \alpha_\lambda(T_1, \lambda) \cdot E_{b,\lambda}(T_2, \lambda)] d\lambda \quad (5-107a)$$

$$\text{or} \quad Q_{1,2} = A_1 (\epsilon_1 \sigma T_1^4 - \alpha_{1,2} \sigma T_2^4) \quad (5-107b)$$

$$\text{where} \quad \epsilon_1(T_1) = \int_{\lambda=0}^{\infty} \epsilon_\lambda(T_1, \lambda) \cdot \frac{E_{b,\lambda}(T_1, \lambda)}{E_b(T_1)} d\lambda \quad (5-108)$$

$$\text{and since} \quad \alpha_\lambda(T, \lambda) = \epsilon_\lambda(T, \lambda),$$

$$\alpha_{1,2}(T_1, T_2) = \int_{\lambda=0}^{\infty} \epsilon_\lambda(T_1, \lambda) \cdot \frac{E_{b,\lambda}(T_2, \lambda)}{E_b(T_2)} d\lambda \quad (5-109)$$

For a *gray surface* $\epsilon_1 = \alpha_{1,2} = \epsilon_\lambda$. A *selective surface* is one for which $\epsilon_\lambda(T, \lambda)$ exhibits a strong dependence on wavelength. If the wavelength dependence is *monotonic*, it follows from Eqs. (5-107) to (5-109) that ϵ_1 and $\alpha_{1,2}$ can differ markedly when T_1 and T_2 are widely separated. For example, in solar energy applications, the nominal temperature of the earth is $T_1 = 294$ K, and the sun may be represented as a blackbody with radiation temperature $T_2 = 5800$ K. For these temperature conditions, a *white paint* can exhibit $\epsilon_1 = 0.9$ and $\alpha_{1,2} = 0.1$ to 0.2. In contrast, a *thin layer of copper oxide on bright aluminum* can exhibit ϵ_1 as low as 0.12 and $\alpha_{1,2}$ greater than 0.9.

The effect of radiation source temperature on low-temperature absorptivity for a number of representative materials is shown in Fig. 5-10. Polished aluminum (curve 15) and anodized (surface-oxidized) aluminum (curve 13) are representative of metals and nonmetals, respectively. Figure 5-10 thus demonstrates the generalization that metals and nonmetals respond in *opposite directions* with regard to changes in the radiation source temperature. Since the effective solar temperature is 5800 K (10,440°F), the extreme right-hand side of Fig. 5-10 provides surface absorptivity data relevant to solar energy applications. The dependence of emittance and absorptance on the real and imaginary components of the refractive index and on the geometric

structure of the surface layer is quite complex. However, a number of generalizations concerning the radiative properties of opaque surfaces are possible. These are summarized in the following discussion.

Polished Metals

1. In the infrared region, the magnitude of the monochromatic emissivity ϵ_λ is small and is dependent on free-electron contributions. Emissivity is also a function of the ratio of resistivity to wavelength r/λ , as depicted in Fig. 5-11. At shorter wavelengths, bound-electron contributions become significant, ϵ_λ is larger in magnitude, and it sometimes exhibits a maximum value. In the visible spectrum, common values for ϵ_λ are 0.4 to 0.8 and ϵ_λ decreases slightly as temperature increases. For $0.7 < \lambda < 1.5$ μm , ϵ_λ is approximately independent of temperature. For $\lambda > 8$ μm , ϵ_λ is approximately proportional to the square root of temperature since $\epsilon_\lambda \propto \sqrt{r}$ and $r \propto T$. Here the Drude or Hagen-Rubens relation applies, that is, $\epsilon_{\lambda,n} \approx 0.0365 \sqrt{r/\lambda}$, where r has units of ohm-meters and λ is measured in micrometers.

2. Total emittance is substantially proportional to absolute temperature, and at moderate temperatures $\epsilon_n = 0.058T \sqrt{rT}$, where T is measured in kelvins.

3. The total absorptance of a metal at temperature T_1 with respect to radiation from a black or gray source at temperature T_2 is equal to the emissivity evaluated at the geometric mean of T_1 and T_2 . Figure 5-11 gives values of ϵ_λ and $\epsilon_{\lambda,n}$, and their ratio, as a function of the product rT (solid lines). Although Fig. 5-11 is based on free-electron

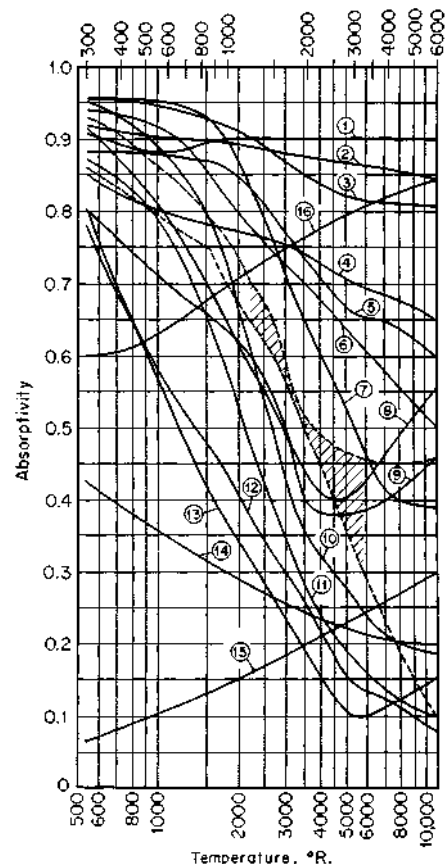


FIG. 5-10 Variation of absorptivity with temperature of radiation source. (1) Slate composition roofing. (2) Linoleum, red brown. (3) Asbestos slate. (4) Soft rubber, gray. (5) Concrete. (6) Porcelain. (7) Vitreous enamel, white. (8) Red brick. (9) Cork. (10) White dutch tile. (11) White chamotte. (12) MgO, evaporated. (13) Anodized aluminum. (14) Aluminum paint. (15) Polished aluminum. (16) Graphite. The two dashed lines bound the limits of data on gray paving brick, asbestos paper, wood, various cloths, plaster of paris, lithopone, and paper. To convert degrees Rankine to kelvins, multiply by $(5.556)/(10^{-3})$.

*In the literature, *emittance* and *emissivity* are often used interchangeably. NIST (the National Institute of Standards and Technology) recommends use of the suffix *-ivity* for pure materials with optically smooth surfaces, and *-ance* for rough and contaminated surfaces. Most real engineering materials fall into the latter category.

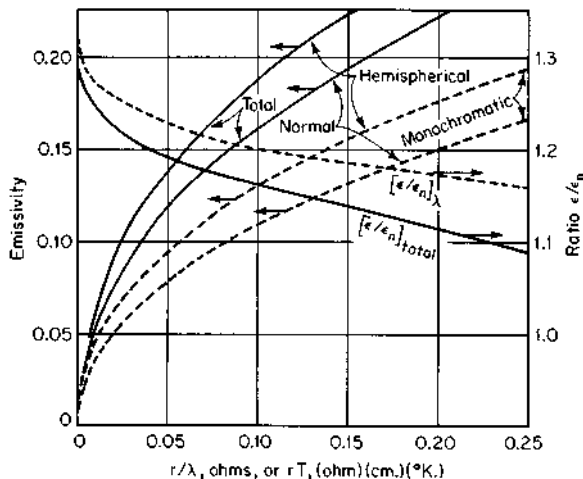


FIG. 5-11 Hemispherical and normal emissivities of metals and their ratio. Dashed lines: monochromatic (spectral) values versus r/λ . Solid lines: total values versus r/T . To convert ohm-centimeter-kelvins to ohm-meter-kelvins, multiply by 10^{-2} .

contributions to emissivity in the far infrared, the relations for total emissivity are remarkably good even at high temperatures. Unless extraordinary efforts are taken to prevent oxidation, a metallic surface may exhibit an emissance or absorptance which may be several times that of a polished specimen. For example, the emissance of iron and steel depends strongly on the degree of oxidation and roughness. Clean iron and steel surfaces have an emissance from 0.05 to 0.45 at ambient temperatures and 0.4 to 0.7 at high temperatures. Oxidized and/or roughened iron and steel surfaces have values of emissance ranging from 0.6 to 0.95 at low temperatures to 0.9 to 0.95 at high temperatures.

Refractory Materials For refractory materials, the dependence of emissance and absorptance on grain size and impurity concentrations is quite important.

1. Most refractory materials are characterized by $0.8 < \epsilon_\lambda < 1.0$ for the wavelength region $2 < \lambda < 4 \mu\text{m}$. The monochromatic emissivity ϵ_λ decreases rapidly toward shorter wavelengths for materials that are white in the visible range but demonstrates high values for black materials such as FeO and Cr_2O_3 . Small concentrations of FeO and Cr_2O_3 or other colored oxides, can cause marked increases in the emissance of materials that are normally white. The sensitivity of the emissance of refractory oxides to small additions of absorbing materials is demonstrated by the results of calculations presented in Fig. 5-12. Figure 5-12 shows the emissance of a semi-infinite absorbing-scattering medium as a function of its *albedo* $\omega \equiv K_s/(K_a + K_s)$, where K_a and K_s are the scatter and absorption coefficients, respectively. These results are relevant to the radiative properties of fibrous materials, paints, oxide coatings, refractory materials, and other *particulate* media. They demonstrate that over the relatively small range $1 - \omega = 0.005$ to 0.1 , the hemispherical emissance ϵ_h increases from approximately 0.15 to 1.0. For refractory materials, ϵ_λ varies little with temperature, with the exception of some white oxides which at high temperatures become good emitters in the visible spectrum as a consequence of the induced electronic transitions.

2. For refractory materials at ambient temperatures, the total emissance ϵ is generally high (0.7 to 1.0). Total refractory emissance decreases with increasing temperature, such that a temperature increase from 1000 to 1570°C may result in a 20 to 30 percent reduction in ϵ .

3. Emissance and absorptance increase with increase in grain size over a grain size range of 1 to 200 μm .

4. The ratio ϵ_h/ϵ_n of hemispherical to normal emissivity of polished surfaces varies with refractive index n ; e.g., the ratio decreases from a value of 1.0 when $n = 1.0$ to a value of 0.93 when $n = 1.5$ (common glass) and increases back to 0.96 at $n = 3.0$.

5. As shown in Fig. 5-12, for a surface composed of particulate

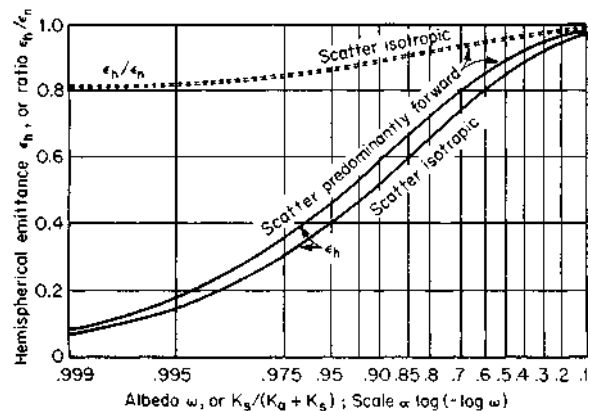


FIG. 5-12 Hemispherical emissance ϵ_h and the ratio of hemispherical to normal emissance ϵ_h/ϵ_n for a semi-infinite absorbing-scattering medium.

matter which scatters isotropically, the ratio ϵ_h/ϵ_n varies from 1.0 when $\omega < 0.1$ to about 0.8 when $\omega = 0.999$.

6. The total absorptance exhibits a decrease with an increase in temperature of the radiation source similar to the decrease in emissance with an increase in the emitter temperature.

Figure 5-10 shows a regular variation of $\alpha_{1,2}$ with T_2 . When T_2 is not very different from T_1 , $\alpha_{1,2} = \epsilon_1(T_2/T_1)^m$. It may be shown that Eq. (5-107b) is then approximated by

$$Q_{1,2} = (1 + m/4)\epsilon_{av} A_1 \sigma (T_1^4 - T_2^4) \quad (5-110)$$

where ϵ_{av} is evaluated at the arithmetic mean of T_1 and T_2 . For metals $m \approx 0.5$ while for nonmetals m is small and negative.

Table 5-4 illustrates values of emissance for materials encountered in engineering practice. It is based on a critical evaluation of early emissivity data. Table 5-4 demonstrates the wide variation possible in the emissivity of a particular material due to variations in surface roughness and thermal pretreatment. With few exceptions the data in Table 5-4 refer to emissances ϵ_n normal to the surface. The hemispherical emissance ϵ_h is usually slightly smaller, as demonstrated by the ratio ϵ_h/ϵ_n depicted in Fig. 5-12. More recent data support the range of emissance values given in Table 5-4 and their dependence on surface conditions. An extensive compilation is provided by Goldsmith, Waterman, and Hirschorn (*Thermophysical Properties of Matter*, Purdue University, Touloukian, ed., Plenum, 1970-1979).

For opaque materials the reflectance ρ is the complement of the absorptance. The directional distribution of the reflected radiation depends on the material, its degree of roughness or grain size, and, if a metal, its state of oxidation. Polished surfaces of homogeneous materials are specular reflectors. In contrast, the intensity of the radiation reflected from a perfectly diffuse or *Lambert surface* is independent of direction. The directional distribution of reflectance of many oxidized metals, refractory materials, and natural products approximates that of a perfectly diffuse reflector. A better model, adequate for many calculation purposes, is achieved by assuming that the total reflectance is the sum of diffuse and specular components ρ_D and ρ_S , as discussed in a subsequent section.

VIEW FACTORS AND DIRECT EXCHANGE AREAS

Consider radiative interchange between two *finite* black surface area elements A_1 and A_2 separated by a transparent medium. Since they are black, the surfaces emit isotropically and totally absorb all incident radiant energy. It is desired to compute the fraction of radiant energy, per unit emissive power E_1 , leaving A_1 in all directions which is intercepted and absorbed by A_2 . The required quantity is defined as the **direct view factor** and is assigned the notation $F_{1,2}$. Since the *net radiant energy interchange* $Q_{1,2} \equiv A_1 F_{1,2} E_1 - A_2 F_{2,1} E_2$ between surfaces A_1 and A_2 must be zero when their temperatures are equal, it follows

TABLE 5-4 Normal Total Emissivity of Various Surfaces

A. Metals and Their Oxides					
Surface	$t, ^\circ\text{F}^\circ$	Emissivity*	Surface	$t, ^\circ\text{F}^\circ$	Emissivity*
Aluminum			Sheet steel, strong rough oxide layer	75	0.80
Highly polished plate, 98.3% pure	440–1070	0.039–0.057	Dense shiny oxide layer	75	0.82
Polished plate	73	0.040	Cast plate:		
Rough plate	78	0.055	Smooth	73	0.80
Oxidized at 1110°F	390–1110	0.11–0.19	Rough	73	0.82
Aluminum-surfaced roofing	100	0.216	Cast iron, rough, strongly oxidized	100–480	0.95
Calorized surfaces, heated at 1110°F.			Wrought iron, dull oxidized	70–680	0.94
Copper	390–1110	0.18–0.19	Steel plate, rough	100–700	0.94–0.97
Steel	390–1110	0.52–0.57	High temperature alloy steels (see Nickel Alloys).		
Brass			Molten metal		
Highly polished:			Cast iron	2370–2550	0.29
73.2% Cu, 26.7% Zn	476–674	0.028–0.031	Mild steel	2910–3270	0.28
62.4% Cu, 36.8% Zn, 0.4% Pb, 0.3% Al	494–710	0.033–0.037	Lead		
82.9% Cu, 17.0% Zn	530	0.030	Pure (99.96%), unoxidized	260–440	0.057–0.075
Hard rolled, polished:			Gray oxidized	75	0.281
But direction of polishing visible	70	0.038	Oxidized at 390°F.	390	0.63
But somewhat attacked	73	0.043	Mercury	32–212	0.09–0.12
But traces of stearin from polish left on	75	0.053	Molybdenum filament	1340–4700	0.096–0.292
Polished	100–600	0.096	Monel metal, oxidized at 1110°F	390–1110	0.41–0.46
Rolled plate, natural surface	72	0.06	Nickel		
Rubbed with coarse emery	72	0.20	Electroplated on polished iron, then polished	74	0.045
Dull plate	120–660	0.22	Technically pure (98.9% Ni, + Mn), polished	440–710	0.07–0.087
Oxidized by heating at 1110°F	390–1110	0.61–0.59	Electroplated on pickled iron, not polished	68	0.11
Chromium; see Nickel Alloys for Ni-Cr steels	100–1000	0.08–0.26	Wire	368–1844	0.096–0.186
Copper			Plate, oxidized by heating at 1110°F	390–1110	0.37–0.48
Carefully polished electrolytic copper	176	0.018	Nickel oxide	1200–2290	0.59–0.86
Commercial, emiered, polished, but pits remaining	66	0.030	Nickel alloys		
Commercial, scraped shiny but not mirror-like	72	0.072	Chromnickel	125–1894	0.64–0.76
Polished	242	0.023	Nickelin (18–32 Ni; 55–68 Cu; 20 Zn), gray oxidized	70	0.262
Plate, heated long time, covered with thick oxide layer	77	0.78	KA-2S alloy steel (8% Ni; 18% Cr), light silvery, rough, brown, after heating After 42 hr. heating at 980°F.	420–914	0.44–0.36
Plate heated at 1110°F	390–1110	0.57	NCT-3 alloy (20% Ni; 25% Cr.), brown, splotted, oxidized from service	420–980	0.62–0.73
Cuprous oxide	1470–2010	0.66–0.54	NCT-6 alloy (60% Ni; 12% Cr), smooth, black, firm adhesive oxide coat from service	420–980	0.90–0.97
Molten copper	1970–2330	0.16–0.13	Platinum		
Gold			Pure, polished plate	440–1160	0.054–0.104
Pure, highly polished	440–1160	0.018–0.035	Strip	1700–2960	0.12–0.17
Iron and steel			Filament	80–2240	0.036–0.192
Metallic surfaces (or very thin oxide layer):			Wire	440–2510	0.073–0.182
Electrolytic iron, highly polished	350–440	0.052–0.064	Silver		
Polished iron	800–1880	0.144–0.377	Polished, pure	440–1160	0.0198–0.0324
Iron freshly emiered	68	0.242	Polished	100–700	0.0221–0.0312
Cast iron, polished	392	0.21	Steel, see Iron.		
Wrought iron, highly polished	100–480	0.28	Tantalum filament	2420–5430	0.194–0.31
Cast iron, newly turned	72	0.435	Tin—bright tinned iron sheet	76	0.043 and 0.064
Polished steel casting	1420–1900	0.52–0.56	Tungsten		
Ground sheet steel	1720–2010	0.55–0.61	Filament, aged	80–6000	0.032–0.35
Smooth sheet iron	1650–1900	0.55–0.60	Filament	6000	0.39
Cast iron, turned on lathe	1620–1810	0.60–0.70	Zinc		
Oxidized surfaces:			Commercial, 99.1% pure, polished	440–620	0.045–0.053
Iron plate, pickled, then rusted red	68	0.612	Oxidized by heating at 750°F.	750	0.11
Completely rusted	67	0.685	Galvanized sheet iron, fairly bright	82	0.228
Rolled sheet steel	70	0.657	Galvanized sheet iron, gray oxidized	75	0.276
Oxidized iron	212	0.736			
Cast iron, oxidized at 1100°F	390–1110	0.64–0.78			
Steel, oxidized at 1100°F	390–1110	0.79			
Smooth oxidized electrolytic iron	260–980	0.78–0.82			
Iron oxide	930–2190	0.85–0.89			
Rough ingot iron	1700–2040	0.87–0.95			
B. Refractories, Building Materials, Paints, and Miscellaneous					
Asbestos			Carbon		
Board	74	0.96	T-carbon (Gebr. Siemens) 0.9% ash (this started with emissivity at 260°F. of 0.72, but on heating changed to values given)	260–1160	0.81–0.79
Paper	100–700	0.93–0.945	Carbon filament	1900–2560	0.526
Brick			Candle soot	206–520	0.952
Red, rough, but no gross irregularities	70	0.93	Lampblack-waterglass coating	209–362	0.959–0.947
Silica, unglazed, rough	1832	0.80			
Silica, glazed, rough	2012	0.85			
Grog brick, glazed	2012	0.75			
See Refractory Materials below.					

TABLE 5-4 Normal Total Emissivity of Various Surfaces (Concluded)

B. Refractories, Building Materials, Paints, and Miscellaneous					
Surface	$t, ^\circ\text{F}^*$	Emissivity*	Surface	$t, ^\circ\text{F}^*$	Emissivity*
Same	260–440	0.957–0.952	Oil paints, sixteen different, all colors	212	0.92–0.96
Thin layer on iron plate	69	0.927	Aluminum paints and lacquers		
Thick coat	68	0.967	10% Al, 22% lacquer body, on rough or smooth surface	212	0.52
Lampblack, 0.003 in. or thicker	100–700	0.945	26% Al, 27% lacquer body, on rough or smooth surface	212	0.3
Enamel, white fused, on iron	66	0.897	Other Al paints, varying age and Al content	212	0.27–0.67
Glass, smooth	72	0.937	Al lacquer, varnish binder, on rough plate	70	0.39
Gypsum, 0.02 in. thick on smooth or blackened plate	70	0.903	Al paint, after heating to 620°F.	300–600	0.35
Marble, light gray, polished	72	0.931	Paper, thin	66	0.924
Oak, planed	70	0.895	Pasted on tinned iron plate	66	0.929
Oil layers on polished nickel (lube oil)	68		On rough iron plate	66	0.944
Polished surface, alone		0.045	On black lacquered plate	66	0.91
+0.001-in. oil		0.27	Plaster, rough lime	50–190	0.924
+0.002-in. oil		0.46	Porcelain, glazed	72	0.932
+0.005-in. oil		0.72	Quartz, rough, fused	70	
Infinitely thick oil layer		0.82	Refractory materials, 40 different poor radiators	1110–1830	$\left[\begin{array}{l} 0.65 \\ 0.70 \\ 0.80 \\ 0.85 \end{array} \right] - \left[\begin{array}{l} 0.75 \\ 0.85 \\ 0.90 \end{array} \right]$
Oil layers on aluminum foil (linseed oil)			good radiators		
Al foil	212	0.087†	Roofing paper	69	0.91
+1 coat oil	212	0.561	Rubber		
+2 coats oil	212	0.574	Hard, glossy plate	74	0.945
Paints, lacquers, varnishes			Soft, gray, rough (reclaimed)	76	0.859
Snowwhite enamel varnish or rough iron plate	73	0.906	Serpentine, polished	74	0.900
Black shiny lacquer, sprayed on iron	76	0.875	Water	32–212	0.95–0.963
Black shiny shellac on tinned iron sheet	70	0.821			
Black matte shellac	170–295	0.91			
Black lacquer	100–200	0.80–0.95			
Flat black lacquer	100–200	0.96–0.98			
White lacquer	100–200	0.80–0.95			

*When two temperatures and two emissivities are given, they correspond, first to first and second to second, and linear interpolation is permissible. $^\circ\text{C} = (^\circ\text{F} - 32)/1.8$.

†Although this value is probably high, it is given for comparison with the data by the same investigator to show the effect of oil layers. See Aluminum, Part A of this table.

thermodynamically that $A_1 F_{1,2} = A_2 F_{2,1}$. The product of area and view factor $\overline{s_i s_j} \equiv A_1 F_{1,2}$, which has the dimensions of area, is termed the **direct surface-to-surface exchange area** for finite black surfaces. Clearly, direct exchange areas are symmetric with respect to their subscripts, that is, $\overline{s_i s_j} = \overline{s_j s_i}$, but view factors are not symmetric unless the associated surface areas are equal. This property is referred to as the **symmetry or reciprocity relation** for direct exchange areas. The shorthand notation $\overline{s_i s_j} \equiv 12 = 21$ for direct exchange areas is often found useful in mathematical developments.

Equation (5-101) may also be restated as

$$\frac{\partial^2 \overline{s_i s_j}}{\partial A_i \partial A_j} = \frac{\cos \phi_i \cos \phi_j}{\pi r^2} \quad (5-111)$$

which leads directly to the required definition of the direct exchange area as a double surface integral

$$\overline{s_i s_j} = \iint_{A_i} \iint_{A_j} \frac{\cos \phi_i \cos \phi_j}{\pi r^2} dA_j dA_i \quad (5-112)$$

All terms in Eq. (5-112) have been previously defined.

Suppose now that Eq. (5-112) is integrated over the entire confining surface of an **enclosure** which has been subdivided into M finite area elements. Each of the M surface **zones** must then satisfy certain **conservation relations** involving *all* the direct exchange areas in the enclosure

$$\sum_{j=1}^M \overline{s_i s_j} = A_i \quad \text{for } 1 \leq i \leq M \quad (5-113a)$$

or in terms of view factors

$$\sum_{j=1}^M F_{ij} = 1 \quad \text{for } 1 \leq i \leq M \quad (5-113b)$$

Contour integration is commonly used to simplify the evaluation of Eq. (5-112) for specific geometries; see Modest (op. cit., Chap. 4)

or Siegel and Howell (op. cit., Chap. 5). The formulas for two particularly useful view factors involving perpendicular rectangles of area xz and yz with common edge z and equal parallel rectangles of area xy and distance of separation z are given for **perpendicular rectangles** with common dimension z

$$\begin{aligned} (\pi \cdot X) \cdot F_{XY} = & X \tan^{-1} \frac{1}{X} + Y \tan^{-1} \frac{1}{Y} - \sqrt{X^2 + Y^2} \tan^{-1} \sqrt{\frac{1}{X^2 + Y^2}} \\ & + \frac{1}{4} \ln \left\{ \frac{(1 + X^2)(1 + Y^2)}{1 + X^2 + Y^2} \left[\frac{X^2(1 + X^2 + Y^2)}{(1 + X^2)(X^2 + Y^2)} \right]^{X^2} \left[\frac{Y^2(1 + X^2 + Y^2)}{(1 + Y^2)(X^2 + Y^2)} \right]^{Y^2} \right\} \end{aligned} \quad (5-114a)$$

and for **parallel rectangles**, separated by distance z ,

$$\begin{aligned} \left(\frac{\pi \cdot X \cdot Y}{2} \right) \cdot F_{XY} = & \ln \left[\frac{(1 + X^2)(1 + Y^2)}{1 + X^2 + Y^2} \right]^{1/2} + X \sqrt{1 + Y^2} \tan^{-1} \frac{X}{\sqrt{1 + Y^2}} \\ & + Y \sqrt{1 + X^2} \tan^{-1} \frac{Y}{\sqrt{1 + X^2}} - X \tan^{-1} X - Y \tan^{-1} Y \end{aligned} \quad (5-114b)$$

In Eqs. (5-114) X and Y are normalized whereby $X = x/z$ and $Y = y/z$ and the corresponding **dimensional** direct surface areas are given by $\overline{s_x s_y} = xz F_{XY}$ and $\overline{s_y s_x} = yz F_{XY}$, respectively.

The exchange area between any two area elements of a sphere is independent of their relative shape and position and is simply the product of the areas, divided by the area of the entire sphere; i.e., any spot on a sphere has equal views of all other spots.

Figure 5-13, curves 1 through 4, shows view factors for selected parallel opposed disks, squares, and 2:1 rectangles and parallel rectangles with one infinite dimension as a function of the ratio of the

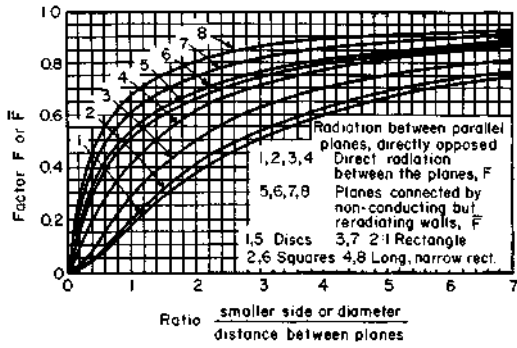


FIG. 5-13 Radiation between parallel planes, directly opposed.

smaller diameter or side to the distance of separation. Curves 2 through 4 of Fig. 5-13, for opposed rectangles, can be computed with Eq. (5-114b). The view factors for two finite coaxial coextensive cylinders of radii $r \leq R$ and height L are shown in Fig. 5-14. The direct view factors for an infinite plane parallel to a system of rows of parallel tubes (see Fig. 5-16) are given as curves 1 and 3 of Fig. 5-15. The view factors for this two-dimensional geometry can be readily calculated by using the **crossed-strings method**.

The crossed-strings method, due to Hottel (*Radiative Transfer*, McGraw-Hill, New York, 1967), is stated as follows: "The exchange area for two-dimensional surfaces, A_1 and A_2 , per unit length (in the infinite dimension) is given by the sum of the lengths of crossed strings from the ends of A_1 to the ends of A_2 less the sum of the uncrossed strings from and to the same points all divided by 2." The strings must be drawn so that all the flux from one surface to the other must cross each of a pair of crossed strings and neither of the pair of uncrossed strings. If one surface can see the other around both sides of an obstruction, two more pairs of strings are involved. The calculation procedure is demonstrated by evaluation of the tube-to-tube view factor for one row of a tube bank, as illustrated in Example 7.

Example 7: The Crossed-Strings Method Figure 5-16 depicts the transverse cross section of two infinitely long, parallel circular tubes of diameter D and center-to-center distance of separation C . Use the crossed-strings method to formulate the tube-to-tube direct exchange area and view factor $\bar{s}_1\bar{s}_2$ and F_{12} , respectively.

Solution: The circumferential area of each tube is $A_i = \pi D$ per unit length in the infinite dimension for this two-dimensional geometry. Application of the crossed-strings procedure then yields simply

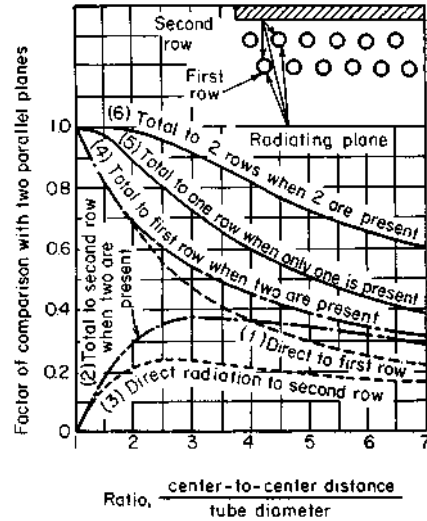


FIG. 5-15 Distribution of radiation to rows of tubes irradiated from one side. Dashed lines: direct view factor F from plane to tubes. Solid lines: total view factor F for black tubes backed by a refractory surface.

$$\bar{s}_1\bar{s}_2 = \frac{2(EFGH - HJ)}{2} = D[\sin^{-1}(1/R) + \sqrt{R^2 - 1} - R]$$

and

$$F_{12} = \bar{s}_1\bar{s}_2/A_1 = [\sin^{-1}(1/R) + \sqrt{R^2 - 1} - R]/\pi$$

where $EFGH$ and $HJ = C$ are the indicated line segments and $R \equiv C/D \geq 1$. Curve 1 of Fig. 5-15, denoted by $F_{p,1}$, is a function of F_{12} , that is, $F_{p,1} = (\pi/R)(\frac{1}{2} - F_{12})$.

The **Yamauti principle** [Yamauti, *Res. Electrotech. Lab. (Tokyo)*, 148 (1924); 194 (1927); 250 (1929)] is stated as follows; *The exchange areas between two pairs of surfaces are equal when there is a one-to-one correspondence for all sets of symmetrically positioned pairs of differential elements in the two surface combinations.* Figure 5-17 illustrates the Yamauti principle applied to surfaces in perpendicular planes having a common edge. With reference to Fig. 5-17, the Yamauti principle states that the diagonally opposed exchange areas are equal, that is, $(1)(4) = (2)(3)$. Figure 5-17 also shows a more complex geometric construction for displaced cylinders for which the Yamauti principle also applies. Collectively the three terms *reciprocity* or *symmetry principle*, *conservation*

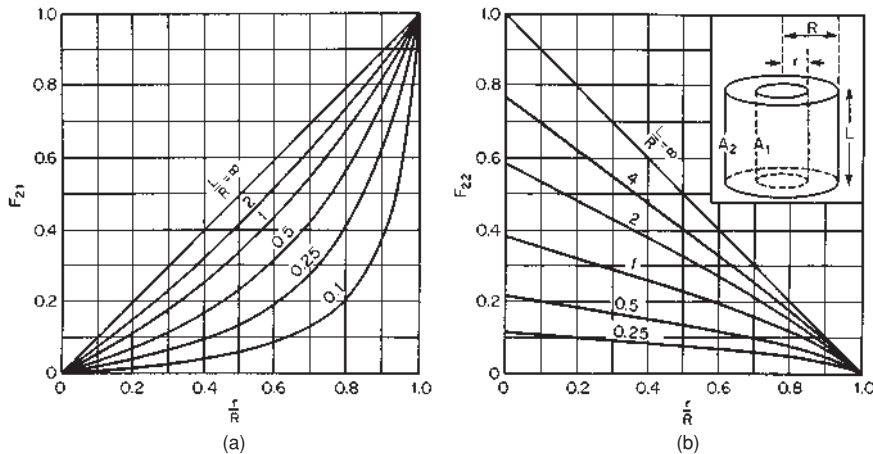


FIG. 5-14 View factors for a system of two concentric coaxial cylinders of equal length. (a) Inner surface of outer cylinder to inner cylinder. (b) Inner surface of outer cylinder to itself.

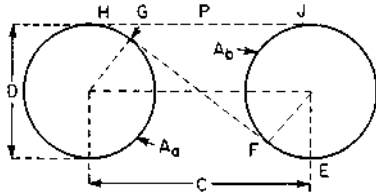


FIG. 5-16 Direct exchange between parallel circular tubes.

principle, and Yamauti principle are referred to as **view factor or exchange area algebra**.

Example 8: Illustration of Exchange Area Algebra Figure 5-17 shows a graphical construction depicting four perpendicular opposed rectangles with a common edge. Numerically evaluate the direct exchange areas and view factors for the diagonally opposed (shaded) rectangles A_1 and A_4 , that is, $(1)(4)$, as well as $(1)(3+4)$. The dimensions of the rectangular construction are shown in Fig. 5-17 as $x=3$, $y=2$, and $z=1$.

Solution: Using shorthand notation for direct exchange areas, the conservation principle yields

$$(1+2)(3+4) = (1+2)(3) + (1+2)(4) = (1)(3) + (2)(3) + (1)(4) + (2)(4)$$

Now by the Yamauti principle we have $(1)(4) = (2)(3)$. Combination of these two relations yields the first result $(1)(4) = [(1+2)(3+4) - (1)(3) - (2)(4)]/2$. For $(1)(3+4)$, again conservation yields $(1)(3+4) = (1)(3) + (1)(4)$, and substitution of the expression for $(1)(4)$ just obtained yields the second result, that is, $(1)(3+4) = [(1+2)(3+4) + (1)(3) - (2)(4)]/2.0$. All three required direct exchange areas in these two relations are readily evaluated from Eq. (5-114a). Moreover, these equations apply to opposed *parallel* rectangles as well as rectangles with a common edge oriented at *any angle*. Numerically it follows from Eq. (5-114a) that for $X = \frac{1}{3}$, $Y = \frac{2}{3}$, and $z = 3$ that $(1+2)(3+4) = 0.95990$; for $X = 1$, $Y = 2$, and $z = 1$ that $(1)(3) = 0.23285$; and for $X = \frac{1}{2}$, $Y = 1$, and $z = 2$ that $(2)(4) = 0.585747$. Since $A_1 = 1.0$, this leads to $s_1s_4 = F_{1,4} = (0.95990 - 0.23285 - 0.584747)/2.0 = 0.07115$ and $s_1s_{3+4} = F_{1,3+4} = (0.95990 + 0.23285 - 0.584747)/2.0 = 0.30400$.

Many literature sources document closed-form algebraic expressions for view factors. Particularly comprehensive references include the compendia by Modest (op. cit., App. D) and Siegel and Howell (op. cit., App. C). The appendices for both of these textbooks also provide a wealth of resource information for radiative transfer. Appendix F of Modest, e.g., references an extensive listing of Fortran computer codes for a variety of radiation calculations which include view factors. These codes are archived in the dedicated Internet web site maintained by the publisher. The textbook by Siegel and Howell also includes an extensive database of view factors archived on a CD-ROM and includes a reference to an author-maintained Internet web site. Other historical sources for view factors include Hottel and Sarofim (op. cit., Chap. 2) and Hamilton and Morgan (NACA-TN 2836, December 1952).

RADIATIVE EXCHANGE IN ENCLOSURES—THE ZONE METHOD

Total Exchange Areas When an enclosure contains *reflective* surface zones, allowance must be made for not only the radiant energy transferred *directly* between any two zones but also the additional transfer attendant to *however many multiple reflections* which occur among the intervening reflective surfaces. Under such circumstances,

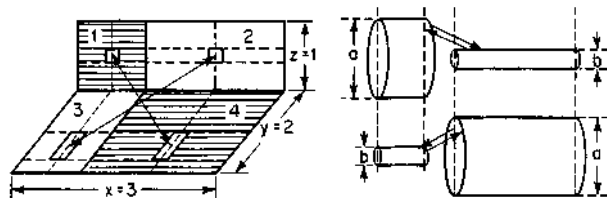


FIG. 5-17 Illustration of the Yamauti principle.

it can be shown that the *net* radiative flux Q_{ij} between all such surface zone pairs A_i and A_j , making full allowance for all multiple reflections, may be computed from

$$Q_{ij} = \sigma(A_i F_{ij} T_i^4 - A_j F_{ji} T_j^4) \quad (5-115)$$

Here, F_{ij} is defined as the **total surface-to-surface view factor** from A_i to A_j , and the quantity $S_i S_j \equiv A_i F_{ij}$ is defined as the corresponding **total surface-to-surface exchange area**. In analogy with the direct exchange areas, the total surface-to-surface exchange areas are also symmetric and thus obey reciprocity, that is, $A_i F_{ij} = A_j F_{ji}$ or $S_i S_j = S_j S_i$. When applied to an enclosure, total exchange areas and view factors also must satisfy appropriate conservation relations. Total exchange areas are functions of the *geometry and radiative properties of the entire enclosure*. They are also independent of temperature if all surfaces and any radiatively participating media are gray. The following subsection presents a general **matrix method** for the *explicit* evaluation of total exchange areas from direct exchange areas and other enclosure parameters.

In what follows, conventional matrix notation is strictly employed as in $\mathbf{A} = [a_{ij}]$ wherein the scalar subscripts always denote the row and column indices, respectively, and all **matrix entities** defined here are denoted by **boldface** notation. Section 3 of this handbook, "Mathematics," provides an especially convenient reference for introductory matrix algebra and matrix computations.

General Matrix Formulation The **zone method** is perhaps the simplest numerical quadrature of the governing *integral equations* for radiative transfer. It may be derived from first principles by starting with the *equation of transfer* for radiation intensity. The zone method *always conserves radiant energy* since the spatial discretization utilizes macroscopic energy balances involving *spatially averaged* radiative flux quantities. Because large sets of linear algebraic equations can arise in this process, *matrix algebra* provides the most compact notation and the most expeditious methods of solution. The mathematical approach presented here is a matrix generalization of the original (*scalar*) development of the zone method due to Hottel and Sarofim (op. cit.). The present matrix development is abstracted from that introduced by Noble [Noble, J. J., *Int. J. Heat Mass Transfer*, **18**, 261–269 (1975)].

Consider an arbitrary three-dimensional enclosure of total volume V and surface area A which confines an absorbing-emitting medium (gas). Let the enclosure be subdivided (zoned) into M finite surface area and N finite volume elements, each small enough that all such *zones* are substantially isothermal. The mathematical development in this section is restricted by the following conditions and/or assumptions:

1. The gas temperatures are given a priori.
2. Allowance is made for gas-to-surface radiative transfer.
3. Radiative transfer with respect to the confined gas is either monochromatic or gray. The gray gas absorption coefficient is denoted here by $K(\text{m}^{-1})$. In subsequent sections the monochromatic absorption coefficient is denoted by $K_\lambda(\lambda)$.
4. All surface emissivities are assumed to be gray and thus independent of temperature.
5. Surface emission and reflection are isotropic or diffuse.
6. The gas does not scatter.

Noble (op. cit.) has extended the present matrix methodology to the case where the gaseous absorbing-emitting medium also *scatters isotropically*.

In matrix notation the blackbody emissive powers for all surface and volume zones comprising the zoned enclosure are designated as $\mathbf{E} = [E_i]$, an $M \times 1$ vector, and $\mathbf{E}_g = [E_{g,i}] = [\sigma T_{g,i}^4]$, an $N \times 1$ vector, respectively. Moreover, all surface zones are characterized by three $M \times M$ diagonal matrices for zone area $\mathbf{A} \mathbf{I} = [A_i \delta_{ij}]$, diffuse emissivity $\mathbf{e} \mathbf{I} = [e_i \delta_{ij}]$, and diffuse reflectivity, $\rho \mathbf{I} = [(1 - e_i) \delta_{ij}]$, respectively. Here δ_{ij} is the **Kronecker delta** (that is, $\delta_{ij} = 1$ for $i = j$ and $\delta_{ij} = 0$ for $i \neq j$).

Two arrays of direct exchange areas are now defined; i.e., the matrix $\mathbf{ss} = [s_i s_j]$ is the $M \times M$ array of **direct surface-to-surface exchange areas**, and the matrix $\mathbf{sg} = [s_i g_j]$ is the $M \times N$ array of **direct gas-to-surface exchange areas**. Here the scalar elements of $\mathbf{ss}$ and $\mathbf{sg}$ are computed from the integrals

$$s_i s_j = \iint_{A_i} \iint_{A_j} \frac{e^{-Kr}}{\pi r^2} \cos \phi_i \cos \phi_j dA_i dA_j \quad (5-116a)$$

$$\overline{s_i g_j} = \iint_{A_i} \iint_{A_j} K \frac{e^{-K r}}{\pi r^2} \cos \phi_i dV_j dA_i \quad (5-116b)$$

Equation (5-116a) is a generalization of Eq. (5-112) for the case $K \neq 0$ while $\overline{s_i g_j}$ is a new quantity, which arises *only* for the case $K \neq 0$.

Matrix characterization of the radiative energy balance at each surface zone is facilitated via definition of three $M \times 1$ vectors; the radiative surface fluxes $\mathbf{Q} = [Q_i]$, with units of watts; and the vectors $\mathbf{H} = [H_i]$ and $\mathbf{W} = [W_i]$ both having units of W/m^2 . The arrays $\mathbf{H}$ and $\mathbf{W}$ define the *incident* and *leaving* flux densities, respectively, at each surface zone. The variable $\mathbf{W}$ is also referred to in the literature as the **radiosity** or **exitance**. Since $\mathbf{W} \equiv \epsilon \mathbf{I} \cdot \mathbf{E} + \rho \mathbf{I} \cdot \mathbf{H}$, the radiative flux at each surface zone is also defined in terms of $\mathbf{E}$, $\mathbf{H}$, and $\mathbf{W}$ by three equivalent matrix relations, namely,

$$\mathbf{Q} = \mathbf{AI} \cdot [\mathbf{W} - \mathbf{H}] = \epsilon \mathbf{AI} \cdot [\mathbf{E} - \mathbf{H}] = \rho \mathbf{I}^{-1} \cdot \epsilon \mathbf{AI} \cdot [\mathbf{E} - \mathbf{W}] \quad (5-117)$$

where the third form is valid *if and only if* the matrix inverse $\rho \mathbf{I}^{-1}$ exists. Two other ancillary matrix expressions are

$$\epsilon \mathbf{AI} \cdot \mathbf{E} = \rho \mathbf{I} \cdot \mathbf{Q} + \epsilon \mathbf{AI} \cdot \mathbf{W} \quad \text{and} \quad \mathbf{AI} \cdot \mathbf{H} = \overline{\mathbf{SS}} \cdot \mathbf{W} + \overline{\mathbf{SG}} \cdot \mathbf{E}_g \quad (5-117a,b)$$

which lead to

$$\epsilon \mathbf{I} \cdot \mathbf{E} = [\mathbf{I} - \rho \mathbf{I} \cdot \mathbf{AI}^{-1} \cdot \overline{\mathbf{SS}}] \cdot \mathbf{W} - \rho \mathbf{I} \cdot \mathbf{AI}^{-1} \cdot \overline{\mathbf{SG}} \cdot \mathbf{E}_g \quad (5-117c)$$

The latter relation is especially useful in radiation pyrometry where true wall temperatures must be computed from wall radiosities.

Explicit Matrix Solution for Total Exchange Areas For gray or monochromatic transfer, the **primary working relation** for zoning calculations via the matrix method is

$$\mathbf{Q} = \epsilon \mathbf{I} \cdot \mathbf{AI} \cdot \mathbf{E} - \overline{\mathbf{SS}} \cdot \mathbf{E} - \overline{\mathbf{SG}} \cdot \mathbf{E}_g \quad [M \times 1] \quad (5-118)$$

Equation (5-118) makes full allowance for multiple reflections in an enclosure of any degree of complexity. To apply Eq. (5-118) for design or simulation purposes, the gas temperatures must be known and surface boundary conditions must be specified for each and every surface zone in the form of *either* E_i or Q_i . In application of Eq. (5-118), *physically impossible values* of E_i may well result if *physically unrealistic* values of Q_i are specified.

In Eq. (5-118), $\overline{\mathbf{SS}}$ and $\overline{\mathbf{SG}}$ are defined as the *required* arrays of **total surface-to-surface exchange areas** and **total gas-to-surface exchange areas**, respectively. The matrices for total exchange areas are calculated *explicitly* from the corresponding arrays of direct exchange areas and the other enclosure parameters by the following matrix formulas:

$$\text{Surface-to-surface exchange} \quad \overline{\mathbf{SS}} = \epsilon \mathbf{I} \cdot \mathbf{AI} \cdot \mathbf{R} \cdot \overline{\mathbf{ss}} \cdot \epsilon \mathbf{I} \quad [M \times M] \quad (5-118a)$$

$$\text{Gas-to-surface exchange} \quad \overline{\mathbf{SG}} = \epsilon \mathbf{I} \cdot \mathbf{AI} \cdot \mathbf{R} \cdot \overline{\mathbf{sg}} \quad [M \times N] \quad (5-118b)$$

where in Eqs. (5-118), $\mathbf{R}$ is the explicit **inverse reflectivity matrix**, defined as

$$\mathbf{R} = [\mathbf{AI} - \overline{\mathbf{ss}} \cdot \rho \mathbf{I}]^{-1} \quad [M \times M] \quad (5-118c)$$

While the $\mathbf{R}$ matrix is generally not symmetric, the matrix product $\rho \mathbf{I} \cdot \mathbf{R}$ is *always* symmetric. This fact proves useful for error checking.

The most computationally significant aspect of the matrix method is that the inverse reflectivity matrix $\mathbf{R}$ *always exists* for any physically meaningful enclosure problem. More precisely, $\mathbf{R}$ always exists provided that $K \neq 0$. For a transparent medium, $\mathbf{R}$ exists provided that there *formally* exists at least one surface zone A_i such that $\epsilon_i \neq 0$. An important computational corollary of this statement for *transparent media* is that the matrix $[\mathbf{AI} - \overline{\mathbf{ss}}]$ is *always singular* and demonstrates *matrix rank* $M - 1$ (Noble, op. cit.).

Finally, the four matrix arrays $\overline{\mathbf{ss}}$, $\overline{\mathbf{gs}}$, $\overline{\mathbf{SS}}$, and $\overline{\mathbf{SG}}$ of direct and total exchange areas must satisfy **matrix conservation relations**, i.e.,

$$\text{Direct exchange areas} \quad \mathbf{AI} \cdot \mathbf{I}_M = \overline{\mathbf{ss}} \cdot \mathbf{I}_M + \overline{\mathbf{sg}} \cdot \mathbf{I}_N \quad (5-119a)$$

$$\text{Total exchange areas} \quad \epsilon \mathbf{I} \cdot \mathbf{AI} \cdot \mathbf{I}_M = \overline{\mathbf{SS}} \cdot \mathbf{I}_M + \overline{\mathbf{SG}} \cdot \mathbf{I}_N \quad (5-119b)$$

Here $\mathbf{I}_M$ is an $M \times 1$ column vector all of whose elements are unity. If $\epsilon \mathbf{I} = \mathbf{I}$ or equivalently, $\rho \mathbf{I} = \mathbf{0}$, Eq. (5-118c) reduces to $\mathbf{R} = \mathbf{AI}^{-1}$ with

the result that Eqs. (5-118a) and (5-118b) degenerate to simply $\overline{\mathbf{SS}} = \overline{\mathbf{ss}}$ and $\overline{\mathbf{SG}} = \overline{\mathbf{sg}}$. Further, while the array $\overline{\mathbf{SS}}$ is always symmetric, the array $\overline{\mathbf{SG}}$ is generally not square.

For purposes of digital computation, it is good practice to enter all data for direct exchange surface-to-surface areas $\overline{\mathbf{ss}}$ with a precision of *at least* five significant figures. This need arises because all the scalar elements of $\overline{\mathbf{sg}}$ can be calculated *arithmetically* from appropriate direct surface-to-surface exchange areas by using view factor algebra rather than via the definition of the defining integral, Eq. (5-116b). This process often involves small arithmetic differences between two numbers of nearly equal magnitude, and numerical significance is easily lost.

Computer implementation of the matrix method proves straightforward, given the availability of modern software applications. In particular, several especially user-friendly GUI mathematical utilities are available that perform matrix computations using *essentially algebraic notation*. Many simple zoning problems may be solved with spreadsheets. For large M and N , the matrix method can involve management of a large amount of data. Error checks based on symmetry and conservation by calculation of the row sums of the four arrays of direct and total exchange areas then prove indispensable.

Zone Methodology and Conventions For a transparent medium, no more than $\Sigma = M(M - 1)/2$ of the M^2 elements of the $\overline{\mathbf{ss}}$ array are *unique*. Further, surface zones are characterized into two generic types. **Source-sink zones** are defined as those for which temperature is specified and whose radiative flux Q_i is to be determined. For **flux zones**, conversely, these conditions are reversed. When both types of zones are present in an enclosure, Eq. (5-118) may be *partitioned* to produce a more efficient computational algorithm. Let $M = M_s + M_f$ represent the total number of surface zones where M_s is the number of source-sink zones and M_f is the number of flux zones. The flux zones are the *last to be numbered*. Equation (5-118) is then partitioned as follows:

$$\begin{bmatrix} \mathbf{Q}_1 \\ \mathbf{Q}_2 \end{bmatrix} = \begin{bmatrix} \epsilon \mathbf{AI}_{1,1} & \mathbf{0} \\ \mathbf{0} & \epsilon \mathbf{AI}_{2,2} \end{bmatrix} \cdot \begin{bmatrix} \mathbf{E}_1 \\ \mathbf{E}_2 \end{bmatrix} - \begin{bmatrix} \overline{\mathbf{SS}}_{1,1} & \overline{\mathbf{SS}}_{1,2} \\ \overline{\mathbf{SS}}_{2,1} & \overline{\mathbf{SS}}_{2,2} \end{bmatrix} \cdot \begin{bmatrix} \mathbf{E}_1 \\ \mathbf{E}_2 \end{bmatrix} - \begin{bmatrix} \overline{\mathbf{SG}}_1 \\ \overline{\mathbf{SG}}_2 \end{bmatrix} \cdot \mathbf{E}_g \quad (5-120)$$

Here the dimensions of the submatrices $\epsilon \mathbf{AI}_{1,1}$ and $\overline{\mathbf{SS}}_{1,1}$ are both $M_s \times M_s$ and $\overline{\mathbf{SG}}_1$ has dimensions $M_s \times N$. Partition algebra then yields the following two matrix equations for $\mathbf{Q}_1$, the $M_s \times 1$ vector of unknown source-sink fluxes and $\mathbf{E}_2$, the $M_f \times 1$ vector of unknown emissive powers for the flux zones, i.e.,

$$\mathbf{E}_2 = [\epsilon \mathbf{AI}_{2,2} - \overline{\mathbf{SS}}_{2,2}]^{-1} \cdot [\mathbf{Q}_2 + \overline{\mathbf{SS}}_{2,1} \cdot \mathbf{E}_1 + \overline{\mathbf{SG}}_2 \cdot \mathbf{E}_g] \quad (5-120a)$$

$$\mathbf{Q}_1 = \epsilon \mathbf{AI}_{1,1} \cdot \mathbf{E}_1 - \overline{\mathbf{SS}}_{1,1} \cdot \mathbf{E}_1 - \overline{\mathbf{SS}}_{1,2} \cdot \mathbf{E}_2 - \overline{\mathbf{SG}}_1 \cdot \mathbf{E}_g \quad (5-120b)$$

The inverse matrix in Eq. (5-120a) formally does not exist if there is at least one flux zone such that $\epsilon_i = 0$. However, well-behaved results are usually obtained with Eq. (5-120a) by utilizing a *notional zero*, say, $\epsilon_i \approx 10^{-5}$, to simulate $\epsilon_i = 0$. Computationally, $\mathbf{E}_2$ is first obtained from Eq. (5-120a) and then substituted into either Eq. (5-120b) or Eq. (5-118).

Surface zones need not be *contiguous*. For example, in a symmetric enclosure, zones on opposite sides of the plane of symmetry may be "lumped" into a single zone for computational purposes. Lumping nonsymmetrical zones is also possible as long as the zone temperatures and emissivities are equal.

An **adiabatic refractory surface** of area A_r and emissivity ϵ_r , for which $Q_r = 0$, proves quite important in practice. A nearly radiatively adiabatic refractory surface occurs when differences between internal conduction and convection and external heat losses through the refractory wall are small compared with the *magnitude* of the *incident* and *leaving* radiation fluxes. For *any* surface zone, the radiant flux is given by $Q = A(W - H) = \epsilon A(E - H)$ and $Q = \epsilon A/p(E - W)$ (if $\rho \neq 0$). These equations then lead to the result that if $Q_r = 0$, $E_r = H_r = W_r$ for all $0 \leq \epsilon_r \leq 1$. *Sufficient conditions* for modeling an adiabatic refractory zone are thus *either* to put $\epsilon_r = 0$ or to specify *directly* that $Q_r = 0$ with $\epsilon_r \neq 0$. If $\epsilon_r = 0$, $S_r S_j = 0$ for all $1 \leq j \leq M$ which leads *directly by definition* to $Q_r = 0$. For $\epsilon_r = 0$, the refractory emissive power E_r *never enters the zoning calculations*. For the special case of $K = 0$ and $M_r = 1$, a single (lumped) refractory, with $Q_r = 0$ and $\epsilon_r \neq 0$, $S_r S_j \neq 0$ and the refractory emissive power may be calculated from Eq. (5-120a) as a weighted sum of all *other known blackbody emissive powers* which

characterize the enclosure, i.e.,

$$E_r = \frac{\sum_{j=1}^{M_s} \overline{S_r S_j} E_j}{\sum_{j=1}^{M_s} \overline{S_r S_j}} \quad \text{with } j \neq r \quad (5-121)$$

Equation (5-121) specifically *includes* those zones which may not have a *direct* view of the refractory. When $Q_r = 0$, the refractory surface is said to be in **radiative equilibrium** with the entire enclosure. Equation (5-121) is indeterminate if $\epsilon_r = 0$. If $\epsilon_r = 0$, E_r does indeed exist and may be evaluated with use of the statement $E_r = H_r = W_r$. It transpires, however, that E_r is *independent of ϵ_r for all $0 \leq \epsilon_r \leq 1$* . Moreover, since $W_r = H_r$ when $Q_r = 0$, for all $0 \leq \epsilon_r \leq 1$, the value specified for ϵ_r is irrelevant to radiative transfer in the *entire enclosure*. In particular it follows that if $Q_r = 0$, then the *vectors $\mathbf{W}$, $\mathbf{H}$, and $\mathbf{Q}$ for the entire enclosure* are also independent of all $0 \leq \epsilon_r \leq 1.0$. A surface zone for which $\epsilon_i = 0$ is termed a **perfect diffuse mirror**. A perfect diffuse mirror is thus also an adiabatic surface zone. The matrix method automatically deals with *all* options for flux and adiabatic refractory surfaces.

The Limiting Case of a Transparent Medium For the special case of a transparent medium, $K = 0$, many practical engineering applications can be modeled with the zone method. These include combustion-fired muffle furnaces and electrical resistance furnaces. When $K \rightarrow 0$, $\overline{\mathbf{sg}} \rightarrow \mathbf{0}$ and $\overline{\mathbf{SG}} \rightarrow \mathbf{0}$. Equations (5-118) through (5-119) then reduce to three simple matrix relations

$$\mathbf{Q} = \epsilon \mathbf{I} \cdot \mathbf{A} \cdot \mathbf{E} - \overline{\mathbf{SS}} \cdot \mathbf{E} \quad (5-122a)$$

$$\overline{\mathbf{SS}} = \epsilon \mathbf{I} \cdot \mathbf{A} \cdot \mathbf{R} \cdot \overline{\mathbf{SS}} \cdot \epsilon \mathbf{I} \quad (5-122b)$$

with again

$$\mathbf{R} \equiv [\mathbf{A} \mathbf{I} - \overline{\mathbf{SS}} \cdot \rho \mathbf{I}]^{-1} \quad (5-122c)$$

The radiant surface flux vector $\mathbf{Q}$, as computed from Eq. (5-122a), *always* satisfies the (scalar) conservation condition $\mathbf{1}_M^T \cdot \mathbf{Q} = \mathbf{0}$ or $\sum_{i=1}^M Q_i = 0$, which is a statement of the overall *radiant* energy balance.

The matrix conservation relations also simplify to

$$\mathbf{A} \mathbf{I} \cdot \mathbf{I}_M = \overline{\mathbf{SS}} \cdot \mathbf{I}_M \quad (5-123a)$$

$$\epsilon \mathbf{I} \cdot \mathbf{A} \cdot \mathbf{I}_M = \overline{\mathbf{SS}} \cdot \mathbf{I}_M \quad (5-123b)$$

And the $M \times M$ arrays for *all* the direct and total view factors can be readily computed from

$$\mathbf{F} = \mathbf{A} \mathbf{I}^{-1} \cdot \overline{\mathbf{SS}} \quad (5-124a)$$

and

$$\mathcal{F} = \mathbf{A} \mathbf{I}^{-1} \cdot \overline{\mathbf{SS}} \quad (5-124b)$$

where the following matrix conservation relations must also be satisfied,

$$\mathbf{F} \cdot \mathbf{I}_M = \mathbf{I}_M \quad (5-125a)$$

and

$$\mathcal{F} \cdot \mathbf{I}_M = \epsilon \mathbf{I} \cdot \mathbf{I}_M \quad (5-125b)$$

The Two-Zone Enclosure Figure 5-18 depicts four simple enclosure geometries which are particularly useful for engineering calculations characterized by only *two* surface zones. For $M = 2$, the reflectivity matrix $\mathbf{R}$ is readily evaluated in *closed form* since an explicit algebraic inversion formula is available for a 2×2 matrix. In this case knowledge of only $\Sigma = 1$ direct exchange area is required. Direct evaluation of Eqs. (5-122) then leads to

$$\overline{\mathbf{SS}} = \begin{bmatrix} \epsilon_1 A_1 - \overline{S_1 S_2} & \overline{S_1 S_2} \\ \overline{S_1 S_2} & \epsilon_2 A_2 - \overline{S_1 S_2} \end{bmatrix} \quad (5-126)$$

where

$$\overline{S_1 S_2} = \frac{1}{\left(\frac{1}{\overline{s_1 s_2}} + \frac{\rho_1}{\epsilon_1 A_1} + \frac{\rho_2}{\epsilon_2 A_2} \right)} \quad (5-127)$$

Equation (5-127) is of general utility for any two-zone system for which $\epsilon_i \neq 0$.

The total exchange areas for the four geometries shown in Fig. 5-18 follow directly from Eqs. (5-126) and (5-127).

1. A planar surface A_1 completely surrounded by a second surface $A_2 > A_1$. Here $F_{1,1} = 0$, $F_{1,2} = 1$, and $\overline{s_1 s_2} = A_1$, resulting in

$$\overline{\mathbf{SS}} = \begin{bmatrix} \epsilon_1 \rho_2 A_1^2 + \epsilon_2 \rho_1 A_1 A_2 & \epsilon_1 \epsilon_2 A_1 A_2 \\ \epsilon_1 \epsilon_2 A_1 A_2 & \epsilon_2 A_2^2 + \epsilon_1 (\rho_2 - \epsilon_2) A_1 A_2 \end{bmatrix} / [\epsilon_1 \rho_2 A_1 + \epsilon_2 A_2]$$

and in particular $\overline{S_1 S_2} = \frac{A_1}{1/\epsilon_1 + (A_1/A_2)(\rho_2/\epsilon_2)}$ (5-127a)

In the limiting case, where A_1 has no negative curvature and is completely surrounded by a very much larger surface A_2 such that $A_1 \ll A_2$, Eq. (5-127a) leads to the even simpler result that $\overline{S_1 S_2} = \epsilon_1 \cdot A_1$.

2. Two parallel plates of equal area which are large compared to their distance of separation (infinite parallel plates). Case 2 is a limiting form of case 1 with $A_1 = A_2$. Algebraic manipulation then results in

$$\overline{\mathbf{SS}} = \begin{bmatrix} (\epsilon_1 + \epsilon_2 - 2\epsilon_1 \epsilon_2) \cdot A_1 & \epsilon_1 \epsilon_2 A_1 \\ \epsilon_1 \epsilon_2 A_1 & (\epsilon_1 + \epsilon_2 - 2\epsilon_1 \epsilon_2) A_1 \end{bmatrix} / [\epsilon_1 + \epsilon_2 - \epsilon_1 \epsilon_2]$$

and in particular

$$\overline{S_1 S_2} = \frac{A_1}{1/\epsilon_1 + 1/\epsilon_2 - 1} \quad (5-127b)$$

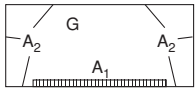
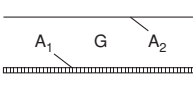
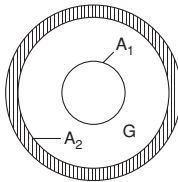
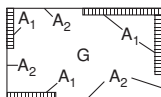
Case 1	Case 2	Case 3	Case 4
			
A planar surface A_1 completely surrounded by a second surface $A_2 > A_1$.	Two infinite parallel plates where $A_1 = A_2$.	Concentric spheres or infinite cylinders where $A_1 < A_2$. Identical to Case 1.	A speckled enclosure with two surface zones.
$\mathbf{F} = \begin{bmatrix} 0 & 1 \\ A_1/A_2 & 1 - A_1/A_2 \end{bmatrix}$	$\mathbf{F} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$\mathbf{F} = \begin{bmatrix} 0 & 1 \\ A_1/A_2 & 1 - A_1/A_2 \end{bmatrix}$	$\mathbf{F} = \frac{1}{(A_1 + A_2)} \begin{bmatrix} A_1 & A_2 \\ A_1 & A_2 \end{bmatrix}$

FIG. 5-18 Four enclosure geometries characterized by two surface zones and one volume zone. (Marks' Standard Handbook for Mechanical Engineers, McGraw-Hill, New York, 1999, p. 4-73, Table 4.3.5.)

3. Concentric spheres or cylinders where $A_2 > A_1$. Case 3 is mathematically identical to case 1.

4. A **speckled enclosure** with two surface zones. Here $\mathbf{F} = \frac{1}{A_1 + A_2} \begin{bmatrix} A_1 & A_2 \\ A_1 & A_2 \end{bmatrix}$ such that $\overline{\mathbf{SS}} = \frac{1}{A_1 + A_2} \begin{bmatrix} A_1^2 & A_1 A_2 \\ A_1 A_2 & A_2^2 \end{bmatrix}$ and Eqs.

(5-126) and (5-127) then produce

$$\overline{\mathbf{SS}} = \begin{bmatrix} \epsilon_1^2 A_1^2 & \epsilon_1 \epsilon_2 A_1 A_2 \\ \epsilon_1 \epsilon_2 A_1 A_2 & \epsilon_2^2 A_2^2 \end{bmatrix} / (\epsilon_1 A_1 + \epsilon_2 A_2)$$

with the particular result

$$\overline{S_1 S_2} = \frac{1}{1/(\epsilon_1 A_1) + 1/(\epsilon_2 A_2)} \quad (5-127c)$$

Physically, a two-zone speckled enclosure is characterized by the fact that the view factor from any point on the enclosure surface to the sink zone is identical to that from any other point on the bounding surface. This is only possible when the two zones are "intimately mixed." The seemingly simplistic concept of a speckled enclosure provides a surprisingly useful *default option* in engineering calculations when the *actual* enclosure geometries are quite complex.

Multizone Enclosures [$M \geq 3$] Again assume $K = 0$. The major numerical effort involved in implementation of the zone method is the evaluation of the inverse reflection matrix $\mathbf{R}$. For $M = 3$, explicit closed-form algebraic formulas *do indeed exist* for the nine scalar elements of the inverse of any arbitrary nonsingular matrix. These formulas are so algebraically complex, however, that it proves impractical to present universal closed-form expressions for the total exchange areas, as has been done for the case $M = 2$. Fortunately, many practical furnace configurations can be idealized with zoning such that only relatively simple *hand calculation procedures are required*. Here the enclosure is modeled with only $M = 3$ surface zones, e.g., a single source, a single sink, and a lumped adiabatic refractory zone. This approach is sometimes termed the **SSR model**. The SSR model assumes that all adiabatic refractory surfaces are *perfect diffuse mirrors*. To implement the SSR procedure, it is necessary to develop *specialized algebraic formulas* and to define a *third black view factor* $\overline{F_{ij}}$ with an overbar as follows.

Refractory Augmented Black View Factors $\overline{F_{ij}}$ Let $M = M_r + M_b$, where M_b is the number of black surface zones and M_r is the number of adiabatic refractory zones. Assume $\epsilon_r = 0$ or $\rho_r = 1$ or, equivalently, that all adiabatic refractory surfaces are *perfect diffuse mirrors*. The view factor $\overline{F_{ij}}$ is then defined as the **refractory augmented black view factor**, i.e., the direct view factor between any two black source-sink zones, A_i and A_j , with full allowance for reflections from all intervening refractory surfaces. The quantity $\overline{F_{ij}}$ shall be referred to as **$\mathbf{F}$ -bar**, for expediency.

Consider the special situation where $M_b = 2$, with any number of refractory zones $M_r \geq 1$. By use of appropriate row and column reduction of the reflectivity matrix $\mathbf{R}$, an especially useful relation can be derived that allows computation of the conventional total exchange area $\overline{S_i S_j}$ from the corresponding refractory augmented black view factor $\overline{F_{ij}}$

$$\overline{S_1 S_2} = \frac{1}{\left(\frac{\rho_1}{\epsilon_1 A_1} + \frac{\rho_2}{\epsilon_2 A_2} + \frac{1}{A_1 \overline{F_{1,2}}} \right)} \quad (5-128a)$$

or

$$\mathcal{F}_{1,2} = \frac{1}{\left(\frac{\rho_1}{\epsilon_1} + \left(\frac{A_1}{A_2} \right) \cdot \frac{\rho_2}{\epsilon_2} + \frac{1}{\overline{F_{1,2}}} \right)} \quad (5-128b)$$

where $\epsilon_i \neq 0$. Notice that Eq. (5-128a) appears *deceptively similar* to Eq. (5-127). Collectively, Eqs. (5-128) along with various formulas to compute $\overline{F_{ij}}$ ($\mathbf{F}$ -bar) are sometimes called the three-zone source/sink/refractory **SSR model**.

The following formulas permit the calculation of $\overline{F_{ij}}$ from requisite direct exchange areas. For the special case where the enclosure is divided into any number of black source-sink zones, $M_b \geq 2$, and the remainder of the enclosure is lumped together into a single refractory

zone ($M_r = 1$) with total area A_r and uniform average temperature T_r , then the direct refractory augmented exchange area for the black zone pairs is given by

$$A_i \overline{F_{ij}} = A_j \overline{F_{ji}} = \overline{s_i s_j} + \frac{\overline{s_i s_r} \cdot \overline{s_r s_j}}{A_r - \overline{s_r s_r}} \quad \text{for } 1 \leq i, j \leq M_b \quad (5-129)$$

For the special case $M_b = 2$ and $M_r = 1$, Eq. (5-129) then simplifies to

$$A_1 \overline{F_{1,2}} = A_2 \overline{F_{2,1}} = \overline{s_1 s_2} + \frac{1}{1/\overline{s_1 s_r} + 1/\overline{s_2 s_r}} \quad (5-130)$$

and if $\overline{s_1 s_r} = \overline{s_2 s_r} = 0$, Eq. (5-130) further reduces to

$$A_1 \overline{F_{1,2}} = \overline{s_1 s_2} + \frac{1}{1/(A_1 - \overline{s_1 s_2}) + 1/(A_2 - \overline{s_2 s_1})} = \frac{A_1 A_2 - (\overline{s_1 s_2})^2}{A_1 + A_2 - 2\overline{s_1 s_2}} \quad (5-131)$$

which necessitates the evaluation of only one direct exchange area. Let the M_r refractory zones be numbered *last*. Then the $M_b \times M_b$ array of refractory augmented direct exchange areas $[A_i \overline{F_{ij}}]$ is symmetric and satisfies the conservation relation

$$[A_i \overline{F_{ij}}] \cdot \mathbf{1}_{M_b} = \mathbf{A} \mathbf{I} \cdot \mathbf{1}_{M_b} \quad (5-132a)$$

with

$$\overline{\mathbf{F}} \cdot \mathbf{1}_{M_b} = \mathbf{1}_{M_b} \quad (5-132b)$$

Temporarily denote $\overline{S_1 S_2}_R$ as the value of $\overline{S_1 S_2}$ computed from Eq. (5-128a) which assumes $\epsilon_r = 0$. It remains to demonstrate the relationship between $\overline{S_1 S_2}_R$ and the total exchange area $\overline{S_1 S_2}$ computed from the matrix method for $M = 3$ when zone 3 is an adiabatic refractory for which $Q_3 = 0$ and $\epsilon_3 \neq 0$. Let $\Theta_i = (E_i - E_2)/(E_1 - E_2)$ denote the dimensionless emissive power where $E_1 > E_2$ such that $\Theta_1 = 1$ and $\Theta_2 = 0$. The dimensionless refractory emissive power may then be calculated from Eq. (5-121) as $\Theta_3 = S_3 S_1 / [S_3 S_1 + S_3 S_2]$, which when substituted into Eq. (5-122a) leads to $\overline{S_1 S_2}_R = \overline{S_1 S_2} + \overline{S_2 S_3} \Theta_3 = \overline{S_1 S_2} + \overline{S_1 S_3} \cdot \overline{S_3 S_2} / [\overline{S_3 S_1} + \overline{S_3 S_2}]$. Thus $\overline{S_1 S_2}_R$ is clearly the *refractory-aided* total exchange area between zone 1 and zone 2 and *not* $\overline{S_1 S_2}$ as calculated by the matrix method in general. That is, $\overline{S_1 S_2}_R$ includes not only the radiant energy originating at zone 1 and arriving at zone 2 directly and by reflection from zones 2 and 3, but also radiation originating at zone 1 that is *absorbed* by zone 3 and then wholly reemitted to zone 2; that is, $H_3 = W_3 = E_3$.

Evaluation of any total view factor $\mathcal{F}_{ij}$ using the requisite refractory augmented black view factor $\overline{F_{ij}}$ obviously requires that the latter be readily available and/or capable of calculation. The refractory augmented view factor $\overline{F_{ij}}$ is documented for a few geometrically simple cases and can be calculated or approximated for others. If A_1 and A_2 are equal parallel disks, squares, or rectangles, connected by nonconducting but reradiating refractory surfaces, then $\overline{F_{ij}}$ is given by Fig. 5-13 in curves 5 to 8. Let A_1 represent an infinite plane and A_2 represent one or two rows of infinite parallel tubes. If the only other surface is an adiabatic refractory surface located behind the tubes, $\overline{F_{2,1}}$ is then given by curve 5 or 6 of Fig. 5-15.

Experience has shown that the simple SSR model can yield quite useful results for a host of practical engineering applications without resorting to digital computation. The error due to representation of the source and sink by single zones is often small, even if the views of the enclosure from different parts of the same zone are dissimilar, provided the surface emissivities are near unity. The error is also small if the temperature variation of the refractory is small. Any degree of accuracy can, of course, be obtained via the matrix method for arbitrarily large M and N by using a digital computer. From a computational viewpoint, when $M \geq 4$, the matrix method *must be used*. The matrix method must also be used for finer-scale calculations such as more detailed wall temperature and flux density profiles.

The Electrical Network Analog At each surface zone the total radiant flux is proportional to the difference between E_i and W_i as indicated by the equation $Q_i = (\epsilon_i A_i)(E_i - W_i)$. The net flux between zones i and j is also given by $Q_{ij} = s_i s_j (W_i - W_j)$, where $Q_i = \sum_{j=1}^M Q_{ij}$, for all $1 \leq i \leq M$, is the total heat flux leaving each zone. These relations

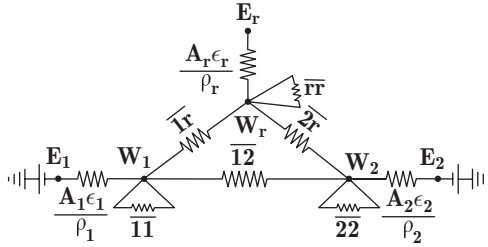


FIG. 5-19 Generalized electrical network analog for a three-zone enclosure. Here A_1 and A_2 are gray surfaces and A_r is a radiatively adiabatic surface. (Hottel, H. C., and A. F. Sarofim, Radiative Transfer, McGraw-Hill, New York, 1967, p. 91.)

suggest a *visual* electrical analog in which E_i and W_i are analogous to voltage potentials. The quantities $\epsilon_i A_i / \rho_i$ and $s_i s_j$ are analogous to conductances (reciprocal impedances), and Q_i or Q_{ij} is analogous to electric currents. Such an electrical analog has been developed by Oppenheim [Oppenheim, A. K., *Trans. ASME*, **78**, 725–735 (1956)].

Figure 5-19 illustrates a generalized electrical network analog for a three-zone enclosure consisting of one refractory zone and two gray zones A_1 and A_2 . The potential points E_i and W_i are separated by conductances $\epsilon_i A_i / \rho_i$. The emissive powers E_1 , E_2 represent potential sources or sinks, while W_1 , W_2 , and W_r are internal node points. In this construction the nodal point representing each surface is connected to that of every other surface it can see directly. Figure 5-19 can be used to formulate the total exchange area $S_1 S_2$ for the SSR model virtually by inspection. The refractory zone is first characterized by a floating potential such that $E_r = W_r$. Next, the resistance for the parallel “current paths” between the internal nodes W_1 and W_2 is defined

by $\frac{1}{A_1 F_{12}} \equiv \frac{1}{s_1 s_2 + 1/(1/s_1 s_r + 1/s_2 s_r)}$ which is identical to Eq. (5-130).

Finally, the overall impedance between the source E_1 and the sink E_2 is represented simply by three resistors in series and is thus given by

$$\frac{1}{S_1 S_2} = \frac{\rho_1}{\epsilon_1 A_1} + \frac{1}{A_1 F_{12}} + \frac{\rho_2}{\epsilon_2 A_2}$$

$$\text{or} \quad \bar{S}_1 \bar{S}_2 = \frac{1}{\frac{\rho_1}{\epsilon_1 A_1} + \frac{\rho_2}{\epsilon_2 A_2} + \frac{1}{A_1 F_{12}}} \quad (5-133)$$

This result is identically that for the SSR model as obtained previously in Eq. (5-128a). This equation is also valid for $M_r \geq 1$ as long as $M_b = 2$. The electrical network analog methodology can be generalized for enclosures having $M > 3$.

Some Examples from Furnace Design The theory of the past several subsections is best understood in the context of two engineering examples involving furnace modeling. The engineering idealization of the **equivalent gray plane concept** is introduced first. Figure 5-20 depicts a common furnace configuration in which the

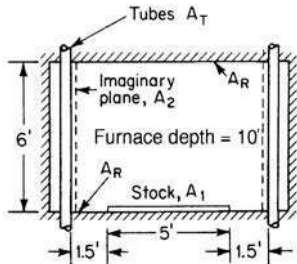


FIG. 5-20 Furnace chamber cross section. To convert feet to meters, multiply by 0.3048.

heating source is two refractory-backed, internally fired tube banks. Clearly the overall geometry for even this common furnace configuration is too complex to be modeled in an expeditious manner by anything other than a *simple engineering idealization*. Thus the furnace shown in Fig. 5-20 is modeled in Example 10, by *partitioning* the entire enclosure into two subordinate furnace compartments. The approach first defines an imaginary gray plane A_2 , located on the inward-facing side of the tube assemblies. Second, the total exchange area between the tubes to this *equivalent* gray plane is calculated, making full allowance for the reflection from the refractory tube backing. The plane-to-tube view factor is then *defined* to be the emissivity of the required equivalent gray plane whose temperature is further assumed to be that of the tubes. This procedure guarantees continuity of the radiant flux *into* the interior radiant portion of the furnace arising from a moderately complicated *external* source.

Example 9 demonstrates classical zoning calculations for radiation pyrometry in furnace applications. Example 10 is a classical furnace design calculation via zoning an enclosure with a *diathermanous* atmosphere and $M = 4$. The latter calculation can only be addressed with the matrix method. The results of Example 10 demonstrate the relative insensitivity of zoning to $M > 3$ and the engineering utility of the SSR model.

Example 9: Radiation Pyrometry A long tunnel furnace is heated by electrical resistance coils embedded in the ceiling. The stock travels on a floor-mounted conveyor belt and has an estimated emissivity of 0.7. The sidewalls are unheated refractories with emissivity 0.55, and the ceiling emissivity is 0.8. The furnace cross section is rectangular with height 1 m and width 2 m. A total radiation pyrometer is sighted on the walls and indicates the following apparent temperatures: ceiling, 1340°C; sidewall readings average about 1145°C; and the load indicates about 900°C. (a) What are the true temperatures of the furnace walls and stock? (b) What is the net heat flux at each surface? (c) How do the matrix method and SSR models compare?

Three-zone model, $M = 3$:

Zone 1: Source (top)

Zone 2: Sink (bottom)

Zone 3: Refractory (lumped sides)

Physical constants:

$$T_0 \equiv 273.15 \text{ K} \quad \sigma \equiv 5.670400 \times 10^{-8} \frac{\text{W}}{\text{m}^2 \cdot \text{K}^4}$$

Enclosure input parameters:

$$H_e := 1 \text{ m} \quad W_e := 2 \text{ m} \quad L_e := 1 \text{ m} \quad A_1 := W_e \cdot L_e \quad A_2 := A_1 \quad A_3 := 2H_e \cdot L_e$$

$$\epsilon_1 := .8 \quad \epsilon_2 := .7 \quad \epsilon_3 := .55 \quad \rho_1 := 1 - \epsilon_1 \quad \rho_2 := 1 - \epsilon_2 \quad \rho_3 := 1 - \epsilon_3$$

$$\mathbf{AI} := \begin{bmatrix} A_1 & 0 & 0 \\ 0 & A_2 & 0 \\ 0 & 0 & A_3 \end{bmatrix} \quad \mathbf{eI} := \begin{bmatrix} \epsilon_1 & 0 & 0 \\ 0 & \epsilon_2 & 0 \\ 0 & 0 & \epsilon_3 \end{bmatrix} \quad \mathbf{\rho I} := \begin{bmatrix} \rho_1 & 0 & 0 \\ 0 & \rho_2 & 0 \\ 0 & 0 & \rho_3 \end{bmatrix}$$

$$\mathbf{AI} := \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \text{ m}^2 \quad \mathbf{eI} := \begin{bmatrix} 0.8 & 0 & 0 \\ 0 & 0.7 & 0 \\ 0 & 0 & 0.55 \end{bmatrix} \quad \mathbf{\rho I} := \begin{bmatrix} 0.2 & 0 & 0 \\ 0 & 0.3 & 0 \\ 0 & 0 & 0.45 \end{bmatrix}$$

Compute direct exchange areas by using crossed strings ($\Sigma = 3$):

$$ss_{11} := 0 \quad ss_{22} := 0 \quad ss_{33} := 2(\sqrt{H_e^2 + W_e^2} - W_e) L_e \quad ss_{33} := 0.4721 \text{ m}^2$$

From symmetry and conservation, there are three linear simultaneous results for the off-diagonal elements of $\mathbf{ss}$:

$$\begin{bmatrix} 12 \\ 13 \\ 23 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} A_1 - \frac{11}{2} \\ A_2 - \frac{22}{2} \\ A_3 - \frac{33}{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

$$\times \begin{bmatrix} A_1 - \frac{11}{2} \\ A_2 - \frac{22}{2} \\ A_3 - \frac{33}{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} +A_1 + A_2 - A_3 - \frac{11}{2} - \frac{22}{2} + \frac{33}{2} \\ +A_1 - A_2 + A_3 - \frac{11}{2} + \frac{22}{2} - \frac{33}{2} \\ -A_1 + A_2 + A_3 + \frac{11}{2} - \frac{22}{2} - \frac{33}{2} \end{bmatrix}$$

Thus

$$ss_{12} := 0.5 (A_1 + A_2 - A_3 + ss_{33}) \quad ss_{13} := 0.5 (A_1 - A_2 + A_3 - ss_{33}) \quad ss_{23} := 0.5 (-A_1 + A_2 + A_3 - ss_{33})$$

$$\mathbf{ss} := \begin{bmatrix} ss_{11} & ss_{12} & ss_{13} \\ ss_{12} & ss_{22} & ss_{23} \\ ss_{13} & ss_{23} & ss_{33} \end{bmatrix} \quad \mathbf{ss} := \begin{bmatrix} 0 & 1.2361 & 0.7639 \\ 1.2361 & 0 & 0.7639 \\ 0.7639 & 0.7639 & 0.4721 \end{bmatrix} \text{ m}^2 \quad (\mathbf{AI} - \mathbf{ss}) \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ m}^2$$

Compute radiosities W from pyrometer temperature readings:

$$\mathbf{T}_{wc} := \begin{bmatrix} 1340.0 \\ 900.0 \\ 1145.0 \end{bmatrix} \text{ C} \quad \mathbf{W} := \sigma \left(\mathbf{T}_{wc} \frac{\text{K}}{\text{C}} + T_0 \right)^4 \quad \mathbf{W} := \begin{bmatrix} 384.0 \\ 107.4 \\ 229.4 \end{bmatrix} \frac{\text{kW}}{\text{m}^2}$$

Matrix wall flux density relations and heat flux calculations based on $\mathbf{W}$:

$$\mathbf{H} := \mathbf{AI}^{-1} \mathbf{ss} \cdot \mathbf{W} \quad \mathbf{Q} := \mathbf{AI}(\mathbf{W} - \mathbf{H}) \quad \mathbf{E} := (\mathbf{eI} - \mathbf{AI})^{-1} \cdot \mathbf{Q} + \mathbf{H}$$

$$\mathbf{H} := \begin{bmatrix} 154.0 \\ 324.9 \\ 241.8 \end{bmatrix} \frac{\text{kW}}{\text{m}^2} \quad \mathbf{Q} := \begin{bmatrix} 460.0 \\ -435.0 \\ -25.0 \end{bmatrix} \text{ kW} \quad \mathbf{E} := \begin{bmatrix} 441.5 \\ 14.2 \\ 219.1 \end{bmatrix} \frac{\text{kW}}{\text{m}^2}$$

The sidewalls act as near-adiabatic surfaces since the heat loss through each sidewall is only about 2.7 percent of the total heat flux originating at the source.

Actual temperatures versus pyrometer readings:

$$\mathbf{T} := \left[\left(\frac{\mathbf{E}}{\sigma} \right)^{0.25} - T_0 \right] \frac{\text{C}}{\text{K}} \quad \mathbf{T} = \begin{bmatrix} 1397.3 \\ 434.1 \\ 1128.9 \end{bmatrix} \text{ C} \quad \text{versus} \quad \mathbf{T}_{wc} = \begin{bmatrix} 1340.0 \\ 900.0 \\ 1145.0 \end{bmatrix} \text{ C}$$

Compare SSR model versus matrix method [use Eqs. (5-128a) and (5-130)]: From Eq. (5-130)

$$\text{ssbar}_{12} := \text{ss}_{1,2} + \frac{1}{\frac{1}{\text{ss}_{1,3}} + \frac{1}{\text{ss}_{2,3}}} \quad \text{ssbar}_{12} = 1.6180 \text{ m}^2$$

And from Eq. (5-128a)

$$\text{SSR}_{12} := \frac{1}{\frac{\rho_1}{\epsilon_1 A_1} + \frac{\rho_2}{\epsilon_2 A_2} + \frac{1}{\text{ssbar}_{12}}} \quad \text{SSR}_{12} = 1.0446 \text{ m}^2$$

With the numerical result $Q_{12} := \text{SSR}_{12}(E_1 - E_2) \quad Q_{12} = 446.3 \text{ kW}$

Thus the SSR model produces $Q_{12} = 446.3 \text{ kW}$ versus the measured value $Q_1 = 460.0 \text{ kW}$ or a discrepancy of about 3.0 percent. Mathematically the SSR model assumes a value of $\epsilon_3 = 0.0$, which precludes the sidewall heat loss of $Q_3 = -25.0 \text{ kW}$. This assumption accounts for all of the difference between the two values. It remains to compare SSR_{12} and $\text{SS}_{1,2}$ computed by the matrix method.

Compute total exchange areas ($\epsilon_3 = 0.55$):

$$\mathbf{R} := (\mathbf{AI} - \mathbf{ss} \cdot \mathbf{pI})^{-1}$$

$$\mathbf{ss} := \mathbf{eI} - \mathbf{AI} - \mathbf{R} \cdot \mathbf{ss} \cdot \mathbf{eI} \quad \mathbf{ss} = \begin{bmatrix} 0.2948 & 0.8284 & 0.4769 \\ 0.8284 & 0.1761 & 0.3955 \\ 0.4769 & 0.3955 & 0.2277 \end{bmatrix} \text{ m}^2 \quad (\mathbf{eI} - \mathbf{AI} - \mathbf{ss}) \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ m}^2$$

Clearly SSR_{21} and $\text{SS}_{1,2}$ are unequal. But if

$$\Theta_3 := \frac{\text{SS}_{3,1}}{\text{SS}_{3,1} + \text{SS}_{3,2}}$$

define

$$\text{SSA}_{12} := \text{SS}_{1,2} + \text{SS}_{2,3} \cdot \Theta_3$$

and

$$\Theta_3 := 0.5466 \quad \text{SSA}_{12} = 1.0446 \text{ m}^2 \quad E_r := E_2 + (E_1 - E_2) \cdot \Theta_3 \quad E_r = 247.8 \frac{\text{kW}}{\text{m}^2}$$

Numerically the matrix method predicts $\text{SSA}_{12} = 1.0446 \text{ m}^2$ for $Q_3 = 0$ and $\epsilon_3 = 0.55$, which is identical to $\text{SSR}_{1,2}$ for the SSR model. Thus $\text{SSR}_{1,2} = \text{SSA}_{12}$ is the refractory-aided total exchange area between zone 1 and zone 2. The SSR model also predicts $E_r = 247.8 \text{ kW/m}^2$ versus the experimental value $E_3 = 219.1 \text{ kW/m}^2$ (1172.6C vs. 1128.9C), which is also a consequence of the actual 25.0-kW refractory heat loss.

(This example was developed as a MATHCAD 14® worksheet. Mathcad is a registered trademark of Parametric Technology Corporation.)

Example 10: Furnace Simulation via Zoning The furnace chamber depicted in Fig. 5-20 is heated by combustion gases passing through 20 vertical radiant tubes which are backed by refractory sidewalls. The tubes have an outside diameter of $D = 5 \text{ in}$ (12.7 cm) mounted on $C = 12 \text{ in}$ (4.72 cm) centers and a gray body emissivity of 0.8. The interior (radiant) portion of the furnace is a $6 \times 8 \times 10 \text{ ft}$ rectangular parallelepiped with a total surface area of 376 ft^2 (34.932 m^2). A 50-ft² (4.645- m^2) sink is positioned centrally on the floor of the furnace. The tube and sink temperatures are measured with embedded thermocouples as 1500 and 1200°F, respectively. The gray refractory emissivity may be taken as 0.5. While all other refractories are assumed to be radiatively adia-

batic, the roof of the furnace is estimated to lose heat to the surroundings with a flux density (W/m^2) equal to 5 percent of the source and sink emissive power difference. An estimate of the radiant flux arriving at the sink is required, as well as estimates for the roof and average refractory temperatures in consideration of refractory service life.

Part (a): Equivalent Gray Plane Emissivity Algebraically compute the equivalent gray plane emissivity for the refractory-backed tube bank idealized by the imaginary plane A_2 , depicted in Fig. 5-15.

Solution: Let zone 1 represent one tube and zone 2 represent the effective plane 2, that is, the unit cell for the tube bank. Thus $A_1 = \pi D$ and $A_2 = C$ are the corresponding zone areas, respectively (per unit vertical dimension). This notation is consistent with Example 3. Also put $\epsilon_1 = 0.8$ with $\epsilon_2 = 1.0$ and define $R = C/D = 12/5 = 2.4$. The gray plane effective emissivity is then calculated as the total view factor for the effective plane to tubes, that is, $\mathcal{F}_{2,1} \equiv \epsilon_2$. For $R = 2.4$, Fig. 5-15, curve 5, yields the refractory augmented view factor $\bar{\mathcal{F}}_{2,1} \approx 0.81$. Then $\mathcal{F}_{2,1}$ is calculated from Eq. (5-128b) as $\mathcal{F}_{2,1} = \frac{1}{0/1 + (2.4/\pi) \cdot 0.2/0.8 + 1/0.81} \approx 0.702$.

A more accurate value is obtained via the matrix method as $\mathcal{F}_{2,1} = 0.70295$.

Part (b): Radiant Furnace Chamber with Heat Loss

Four-zone model, $M = 4$: Use matrix method.

Zone 1: Sink (floor)

Zone 2: Source (lumped sides)

Zone 3: Refractory (roof)

Zone 4: Refractory (ends and floor strips)

Physical constants:

$$T_0 \equiv 273.15 \text{ K} \quad \sigma \equiv 5.6704 \times 10^{-8} \frac{\text{W}}{\text{m}^2 \cdot \text{K}^4}$$

Enclosure input parameters:

$A_1 := 50 \text{ ft}^2 \quad A_2 := 120 \text{ ft}^2 \quad A_3 := 80 \text{ ft}^2 \quad A_4 := 126 \text{ ft}^2 \quad D := 5 \text{ in} \quad H := 6 \text{ ft}$

$\epsilon_1 := .9 \quad \epsilon_2 := .70295 \quad \epsilon_3 := .5 \quad \epsilon_4 := .5 \quad \mathbf{I}_4 := \text{identity}(4)$

$$\mathbf{eI} = \begin{bmatrix} 0.9 & 0 & 0 & 0 \\ 0 & 0.7029 & 0 & 0 \\ 0 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & 0.5 \end{bmatrix} \quad \mathbf{pI} := \mathbf{I}_4 - \mathbf{eI} \quad \mathbf{pI} = \begin{bmatrix} 0.1 & 0 & 0 & 0 \\ 0 & 0.2971 & 0 & 0 \\ 0 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & 0.5 \end{bmatrix}$$

Compute direct exchange areas: There are $\Sigma = 6$ unique direct exchange areas. These are obtained from Eqs. (5-114) and view factor algebra. The final array of direct exchange areas is:

$$\mathbf{ss} = \begin{bmatrix} 0 & 1.559 & 1.575 & 1.511 \\ 1.559 & 2.078 & 2.839 & 4.673 \\ 1.575 & 2.839 & 0 & 3.018 \\ 1.511 & 4.673 & 3.018 & 2.503 \end{bmatrix} \text{ m}^2 \quad \mathbf{AI} = \begin{bmatrix} 4.645 & 0 & 0 & 0 \\ 0 & 11.148 & 0 & 0 \\ 0 & 0 & 7.432 & 0 \\ 0 & 0 & 0 & 11.706 \end{bmatrix} \text{ m}^2$$

Compute total exchange areas:

$$\mathbf{R} := (\mathbf{AI} - \mathbf{ss} \cdot \mathbf{pI})^{-1}$$

$$\mathbf{ss} := \mathbf{eI} - \mathbf{AI} - \mathbf{R} \cdot \mathbf{ss} \cdot \mathbf{eI} \quad \mathbf{ss} = \begin{bmatrix} 0.405 & 1.669 & 0.955 & 1.152 \\ 1.669 & 2.272 & 1.465 & 2.431 \\ 0.955 & 1.465 & 0.234 & 1.063 \\ 1.152 & 2.431 & 1.063 & 1.207 \end{bmatrix} \text{ m}^2$$

Check matrix conservation via row-sums:

$$(\mathbf{AI} - \mathbf{ss}) \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \text{ m}^2 \quad (\mathbf{eI} - \mathbf{AI} - \mathbf{ss}) \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \text{ m}^2$$

Emissive power and wall flux input data:

$$\mathbf{T}_F := \begin{bmatrix} 1200 \\ 1500 \\ 32 \\ 32 \end{bmatrix} \text{ F} \quad \mathbf{T} := \frac{5}{9} (\mathbf{T}_F - 32 \text{ F}) \frac{\text{C}}{\text{F}} \quad \mathbf{E} := \sigma \left(\mathbf{T} \frac{\text{K}}{\text{C}} + T_0 \right)^4$$

$$\mathbf{T} = \begin{bmatrix} 648.9 \\ 815.6 \\ 0.0 \\ 0.0 \end{bmatrix} \text{ C} \quad \mathbf{E} = \begin{bmatrix} 40.98 \\ 79.66 \\ 0.32 \\ 0.32 \end{bmatrix} \frac{\text{kW}}{\text{m}^2}$$

$$Q_3 := -0.05 \cdot A_3 (E_2 - E_1) \quad Q_3 = -14.37 \text{ kW} \quad Q_4 := 0$$

5-30 HEAT AND MASS TRANSFER

Compute refractory emissive powers from known flux inputs Q_3 and Q_4 using partitioned matrix equations [Eq. (5-120b)]:

$$\mathbf{E}_R := \begin{bmatrix} \epsilon_3 A_3 - SS_{3,3} & -SS_{3,4} \\ -SS_{4,3} & \epsilon_4 A_4 - SS_{4,4} \end{bmatrix}^{-1} \cdot \begin{bmatrix} Q_3 + SS_{3,1} E_1 + SS_{3,2} E_2 \\ Q_4 + SS_{4,1} E_1 + SS_{4,2} E_2 \end{bmatrix}$$

$$\mathbf{E}_R = \begin{bmatrix} 60.68 \\ 65.73 \end{bmatrix} \frac{\text{kW}}{\text{m}^2} \quad \mathbf{E} := \begin{bmatrix} E_1 \\ E_2 \\ E_{R_1} \\ E_{R_2} \end{bmatrix} \quad \mathbf{E} = \begin{bmatrix} 40.98 \\ 79.66 \\ 60.68 \\ 65.73 \end{bmatrix} \frac{\text{kW}}{\text{m}^2}$$

Compute flux values and final zone temperatures:

$$\mathbf{Q} := \epsilon \mathbf{I} \mathbf{A} \mathbf{E} - \overline{SS} \mathbf{E} \quad \mathbf{T} := \left[\left(\frac{\mathbf{E}}{\sigma} \right)^{0.25} - T_0 \right] \frac{\text{C}}{\text{K}}$$

$$\mathbf{Q} = \begin{bmatrix} -111.87 \\ 126.24 \\ -14.37 \\ 0.00 \end{bmatrix} \text{kW} \quad \mathbf{T} = \begin{bmatrix} 648.9 \\ 815.6 \\ 743.9 \\ 764.5 \end{bmatrix} \text{C}$$

Auxiliary calculations for tube area and effective tube emissivity:

$$A_{\text{Tubes}} := 20\pi \cdot D \cdot H \quad \epsilon_{\text{Tubes}} := \frac{Q_2}{A_{\text{Tubes}}(E_2 - E_1)} \quad A_{\text{Tubes}} = 14.59 \text{ m}^2 \quad \epsilon_{\text{Tubes}} = 0.2237$$

Notes: (1) Results for Q and T here are independent of ϵ_3 and ϵ_4 with the exception of T_3 , which is indeed a function of ϵ_3 . (2) The total surface area of the tubes is $A_{\text{Tubes}} = 14.59 \text{ m}^2$. Suppose the tubes were totally surrounded by a black enclosure at the temperature of the sink. The hypothetical emissivity of the tubes would then be $\epsilon_{\text{Tubes}} = 0.224$. (3) A 5 percent roof heat loss is consistent with practical measurement errors. A sensitivity test was performed with $M = 3, 4$, and 5 with and without roof heat loss. The SSR model corresponds to $M = 3$ with zero heat loss. For $M = 5$, zone 4 corresponded to the furnace ends and zone 5 corresponded to the floor strips. The results are summarized in the following table. With the exception of the temperature of the floor strips, the computed results are seen to be remarkably insensitive to M .

Effect of Zone Number M on Computed Results

	Zero roof heat loss			5 percent roof heat loss		
	$M = 3$	$M = 4$	$M = 5$	$M = 3$	$M = 4$	$M = 5$
Temperature, °C						
T_3	765.8	762.0	762.4	756.2	743.9	744.3
T_4	NA	768.4	765.0	NA	764.5	761.1
T_5	NA	NA	780.9	NA	NA	776.7
Heat flux, kW						
Q_1	-117.657	-117.275	-116.251	-112.601	-111.870	-110.877
Q_2	117.657	117.275	116.251	126.975	126.244	125.251
Q_3	0.000	0.000	0.000	-14.374	-14.374	-14.374
Q_4	NA	0.000	0.000	NA	0.00	0.00
Q_5	NA	NA	0.000	NA	NA	0.00

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Allowance for Specular Reflection If the assumption that all surface zones are diffuse emitters and reflectors is relaxed, the zoning equations become much more complex. Here, all surface parameters become functions of the *angles of incidence and reflection* of the radiation beams at each surface. In practice, such details of reflectance and emission are seldom known. When they are, the Monte Carlo method of tracing a large number of beams emitted from random positions and in random initial directions is probably the best method of obtaining a solution. Siegel and Howell (op. cit., Chap. 10) and Modest (op. cit., Chap. 20) review the utilization of the Monte Carlo approach to a variety of radiant transfer applications. Among these is the Monte Carlo calculation of direct exchange areas for very complex geometries. Monte Carlo techniques are generally *not used* in practice for simpler engineering applications.

A simple *engineering* approach to specular reflection is the so-called **diffuse plus specular reflection model**. Here the total reflectivity $\rho_i = 1 - \epsilon_i = \rho_{si} + \rho_{di}$ is represented as the sum of a *diffuse component*

ρ_{si} and a *specular component* ρ_{di} . The method yields analytical results for a number of two surface zone geometries. In particular, the following equation is obtained for exchange between concentric spheres or infinitely long coaxial cylinders for which $A_1 < A_2$:

$$\overline{S_1 S_2} = \frac{A_1}{\left(\frac{1}{\epsilon_1} + \frac{\rho_2}{\epsilon_2} \frac{A_1}{A_2} + \frac{\rho_{s2}}{(1 - \rho_{s2})} [1 - A_1/A_2] \right)} \quad (5-134)$$

For $\rho_{D1} = \rho_{D2} = 0$ (or equivalently $\rho_1 = \rho_{s1}$ with $\rho_2 = \rho_{s2}$), Eq. (5-134) yields the limiting case for wholly specular reflection, i.e.

$$\text{Specular limit} \quad \overline{S_1 S_2} = \frac{A_1}{\left(\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1 \right)} \quad (5-134a)$$

which is independent of the area ratio, A_1/A_2 . It is important to notice that Eq. (5-124a) is *similar* to Eq. (5-127b) but the emissivities here are defined as $\epsilon_1 \equiv 1 - \rho_{s1}$ and $\epsilon_2 \equiv 1 - \rho_{s2}$. When surface reflection is wholly diffuse [$\rho_{s1} = \rho_{s2} = 0$ or $\rho_1 = \rho_{D1}$ with $\rho_2 = \rho_{D2}$], Eq. (5-134) results in a formula *identical* to Eq. (5-127a), viz.

$$\text{Diffuse limit} \quad \overline{S_1 S_2} = \frac{A_1}{\left[\frac{1}{\epsilon_1} + \left(\frac{A_1}{A_2} \right) \frac{\rho_2}{\epsilon_2} \right]} \quad (5-134b, 5-127a)$$

For the case of (infinite) parallel flat plates where $A_1 = A_2$, Eq. (5-134) leads to a general formula *similar* to Eq. (5-134a) but with the stipulation here that $\epsilon_1 \equiv 1 - \rho_{D1} - \rho_{s1}$ and $\epsilon_2 \equiv 1 - \rho_{D2} - \rho_{s2}$.

Another particularly interesting limit of Eq. (5-134) occurs when $A_2 \gg A_1$, which might represent a small sphere irradiated by an infinite surroundings which can reflect radiation originating at A_1 back to A_1 . That is to say, even though $A_2 \rightarrow \infty$, the “self” total exchange area does not necessarily vanish, to wit

$$\overline{S_1 S_1} = \frac{\epsilon_1^2 \rho_{s2} \cdot A_1}{[1 - \rho_1 \cdot \rho_{s2}]} \quad \text{and} \quad \overline{S_1 S_2} = \frac{\epsilon_1 (1 - \rho_{s2}) \cdot A_1}{[1 - \rho_1 \cdot \rho_{s2}]} \quad (5-134c,d)$$

which again exhibit diffuse and specular limits. The diffuse plus specular reflection model becomes significantly more complex for geometries with $M \geq 3$ where digital computation is usually required.

An Exact Solution to the Integral Equations—The Hohlraum Exact solutions of the fundamental *integral equations* for radiative transfer are available for only a few simple cases. One of these is the evaluation of the emittance from a small aperture, of area A_1 , in the surface of an isothermal spherical cavity of radius R . In German, this geometry is termed a *hohlraum* or hollow space. For this special case the radiosity W is constant over the inner surface of the cavity. It then follows that the ratio W/E is given by

$$W/E = \frac{\epsilon}{1 - \rho[1 - A_1/(4\pi R^2)]} \quad (5-135)$$

where ϵ and $\rho = 1 - \epsilon$ are the diffuse emissivity and reflectivity of the interior cavity surface, respectively. The ratio W/E is the effective emittance of the aperture as sensed by an external narrow-angle receiver (radiometer) viewing the cavity interior. Assume that the cavity is constructed of a rough material whose (diffuse) emissivity is $\epsilon = 0.5$. As a point of reference, if the cavity is to simulate a blackbody emitter to better than 98 percent of an ideal theoretical blackbody, Eq. (5-135) then predicts that the ratio of the aperture to sphere areas $A_1/(4\pi R^2)$ must be less than 2 percent. Equation (5-135) has practical utility in the experimental design of calibration standards for laboratory radiometers.

RADIATION FROM GASES AND SUSPENDED PARTICULATE MATTER

Introduction Flame radiation originates as a result of emission from water vapor and carbon dioxide in the hot gaseous combustion

products and from the presence of particulate matter. The latter includes emission from burning of microscopic and submicroscopic soot particles, and from large suspended particles of coal, coke, or ash. Thermal radiation owing to the presence of water vapor and carbon dioxide is not visible. The characteristic blue color of clean natural gas flames is due to chemiluminescence of the excited intermediates in the flame which contribute negligibly to the radiation from combustion products.

Gas Emissivities Radiant transfer in a gaseous medium is characterized by three quantities; the gas emissivity, gas absorptivity, and gas transmissivity. Gas emissivity refers to radiation originating within a gas volume which is incident on some reference surface. Gas absorptivity and transmissivity, however, refer to the absorption and transmission of radiation from some *external* surface radiation source characterized by some radiation temperature T_1 . The sum of the gas absorptivity and transmissivity must, by definition, be unity. Gas absorptivity may be calculated from an appropriate gas emissivity. The gas emissivity is a function only of the gas temperature T_g while the absorptivity and transmissivity are functions of both T_g and T_1 .

The **standard hemispherical monochromatic gas emissivity** is defined as the direct volume-to-surface exchange area for a hemispherical gas volume to an infinitesimal area element located at the center of the planar base. Consider monochromatic transfer in a *black* hemispherical enclosure of radius R that confines an isothermal volume of gas at temperature T_g . The temperature of the bounding surfaces is T_1 . Let A_2 denote the area of the finite hemispherical surface and dA_1 denote an infinitesimal element of area located at the center of the planar base. The (dimensionless) monochromatic direct exchange area for exchange between the finite hemispherical surface A_2 and dA_1 then follows from direct integration of Eq. (5-116a) as

$$\frac{\partial(\overline{s_1 s_2})_\lambda}{\partial A_1} = \int_{\phi=0}^{\pi/2} \frac{e^{-K_g R}}{\pi R^2} \cos \phi_1 \, 2\pi R^2 \sin \phi_1 \, d\phi_1 = e^{-K_g R} \quad (5-136a)$$

and from conservation there results

$$\frac{\partial(\overline{s_1 \overline{g}})_\lambda}{\partial A_1} = 1 - e^{-K_g R} \quad (5-136b)$$

Note that Eq. (5-136b) is identical to the expression for the gas emissivity for a column of path length R . In Eqs. (5-136) the gas absorption coefficient is a function of gas temperature, composition, and wavelength, that is, $K_\lambda = K_{g,\lambda}(T, \lambda)$. The net monochromatic radiant flux density at dA_1 due to irradiation from the gas volume is then given by

$$q_{1,g,\lambda} = \frac{\partial(\overline{s_1 \overline{g}})_\lambda}{\partial A_1} (E_{1,\lambda} - E_{g,\lambda}) \equiv \alpha_{g,1,\lambda} E_{1,\lambda} - \epsilon_{g,\lambda} E_{g,\lambda} \quad (5-137)$$

In Eq. (5-137), $\epsilon_{g,\lambda}(T, \lambda) = 1 - \exp(-K_g R)$ is defined as the monochromatic or *spectral gas emissivity* and $\alpha_{g,\lambda}(T, \lambda) = \epsilon_{g,\lambda}(T, \lambda)$.

If Eq. (5-137) is integrated with respect to wavelength over the entire EM spectrum, an expression for the *total* flux density is obtained

$$q_{1,g} = \alpha_{g,1} E_1 - \epsilon_g E_g \quad (5-138)$$

where
$$\epsilon_g(T_g) = \int_{\lambda=0}^{\infty} \epsilon_{g,\lambda}(T_g, \lambda) \cdot \frac{E_{b,\lambda}(T_g, \lambda)}{E_b(T_g)} d\lambda \quad (5-138a)$$

and
$$\alpha_{g,1}(T_1, T_g) = \int_{\lambda=0}^{\infty} \alpha_{g,\lambda}(T_g, \lambda) \cdot \frac{E_{b,\lambda}(T_1, \lambda)}{E_b(T_1)} d\lambda \quad (5-138b)$$

define the *total* gas emissivity and absorptivity, respectively. The notation used here is analogous to that used for surface emissivity and absorptivity as previously defined. For a *real* gas $\epsilon_g = \alpha_{g,1}$ only if $T_1 = T_g$, while for a *gray* gas mass of arbitrarily shaped volume $\epsilon_g = \alpha_{g,1} = \partial(\overline{s_1 \overline{g}})/\partial A_1$ is independent of temperature. Because $K_\lambda(T, \lambda)$ is also a function of the composition of the radiating species, it is necessary in what follows to define a second absorption coefficient $k_{p,\lambda}$, where $K_\lambda = k_{p,\lambda} p$. Here p is the partial pressure of the radiating species, and $k_{p,\lambda}$ with units of $(\text{atm-m})^{-1}$, is referred to as the *monochromatic line absorption coefficient*.

Mean Beam Lengths It is always possible to represent the emissivity of an arbitrarily shaped volume of gray gas (and thus the corre-

sponding direct gas-to-surface exchange area) with an equivalent sphere of radius $R = L_M$. In this context the hemispherical radius $R = L_M$ is referred to as the **mean beam length** of the arbitrary gas volume. Consider, e.g., an isothermal gas layer at temperature T_g confined by two infinite parallel plates separated by distance L . Direct integration of Eq. (5-116a) and use of conservation yield a closed-form expression for the requisite surface-gas direct exchange area

$$\frac{\partial(\overline{s_1 \overline{g}})}{\partial A_1} = [1 - 2E_3(KL)] \quad (5-139a)$$

where $E_n(z) = \int_{z=1}^{\infty} \frac{e^{-zt}}{t^n} dt$ is defined as the ***n*th-order exponential**

integral which is readily available. Employing the definition of gas emissivity, the **mean beam length** between the plates L_M is then defined by the expression

$$\epsilon_g = [1 - 2E_3(KL)] \equiv 1 - e^{-KL_M} \quad (5-139b)$$

Solution of Eq. (5-139b) yields $KL_M = -\ln[2E_3(KL)]$, and it is apparent that KL_M is a function of KL . Since $E_n(0) = 1/(n-1)$ for $n > 1$, the mean beam length approximation also correctly predicts the gas emissivity as zero when $K = 0$ and $K \rightarrow \infty$.

In the limit $K \rightarrow 0$, power series expansion of both sides of the Eq. (5-139b) leads to $KL_M \rightarrow 2KL \equiv KL_{M0}$, where $L_M \equiv L_{M0} = 2L$. Here L_{M0} is defined as the **optically thin mean beam length** for radiant transfer from the entire *infinite* planar gas layer to a differential element of surface area on one of the plates. The optically thin mean beam length for two infinite parallel plates is thus simply *twice* the plate spacing L . In a similar manner it may be shown that for a sphere of diameter D , $L_{M0} = \frac{3}{2}D$, and for an infinitely long cylinder $L_{M0} = D$. A useful default formula for an arbitrary enclosure of volume V and area A is given by $L_{M0} = 4V/A$. This expression predicts $L_{M0} = \frac{3}{2}R$ for the standard hemisphere of radius R because the optically thin mean beam length is averaged over the *entire* hemispherical enclosure.

Use of the optically thin value of the mean beam length yields values of gas emissivities or exchange areas that are too high. It is thus necessary to introduce a dimensionless constant $\beta \leq 1$ and define some new **average mean beam length** such that $KL_M \equiv \beta KL_{M0}$. For the case of parallel plates, we now require that the mean beam length exactly predict the gas emissivity for a *third* value of KL . In this example we find $\beta = -\ln[2E_3(KL)]/2KL$ and for $KL = 0.193095$ there results $\beta = 0.880$. The value $\beta = 0.880$ is not wholly arbitrary. It also happens to minimize the error defined by the so-called shape correction factor $\phi = [\partial(\overline{s_1 \overline{g}})/\partial A_1]/(1 - e^{-KL_M})$ for all $KL > 0$. The required *average* mean beam length for all $KL > 0$ is then taken simply as $L_M = 0.88L_{M0} = 1.76L$. The error in this approximation is less than 5 percent.

For an arbitrary geometry, the average mean beam length is defined as the radius of a hemisphere of gas which predicts values of the direct exchange area $\overline{s_1 \overline{g}}/A_1 = [1 - \exp(-KL_M)]$, subject to the optimization condition indicated above. It has been found that the error introduced by using average mean beam lengths to approximate direct exchange areas is sufficiently small to be appropriate for many engineering calculations. When $\beta = L_M/L_{M0}$ is evaluated for a large number of geometries, it is found that $0.8 < \beta < 0.95$. It is recommended here that $\beta = 0.88$ be employed in lieu of any further geometric information. For a single-gas zone, all the requisite direct exchange areas can be approximated for engineering purposes in terms of a *single* appropriately defined average mean beam length.

Emissivities of Combustion Products Absorption or emission of radiation by the constituents of gaseous combustion products is determined primarily by vibrational and rotational transitions between the energy levels of the gaseous molecules. Changes in both vibrational and rotational energy states gives rise to *discrete* spectral lines. Rotational lines accompanying vibrational transitions usually overlap, forming a so-called vibration-rotation band. These bands are thus associated with the major vibrational frequencies of the molecules.

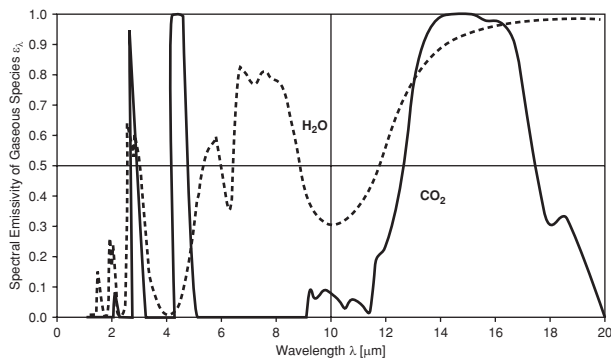


FIG. 5-21 Spectral emissivities for carbon dioxide and water vapor after RADCAL. $p_c L = p_w L = 0.36 \text{ atm}\cdot\text{m}$, $T_g = 1500 \text{ K}$.

Each spectral line is characterized by an absorption coefficient $k_{p,\lambda}$ which exhibits a maximum at some central characteristic wavelength or wave number $\eta_0 = 1/\lambda_0$ and is described by a Lorentz² probability distribution. Since the widths of spectral lines are dependent on collisions with other molecules, the absorption coefficient will also depend upon the composition of the combustion gases and the total system pressure. This brief discussion of gas spectroscopy is intended as an introduction to the factors controlling absorption coefficients and thus the factors which govern the empirical correlations to be presented for gas emissivities and absorptivities.

Figure 5-21 shows computed values of the spectral emissivity $\epsilon_{g,\lambda} \equiv \epsilon_{g,\lambda}(T, pL, \lambda)$ as a function of wavelength for an equimolar mixture of carbon dioxide and water vapor for a gas temperature of 1500 K, partial pressure of 0.18 atm, and a path length $L = 2 \text{ m}$. Three principal absorption-emission bands for CO_2 are seen to be centered on 2.7, 4.3, and 15 μm . Two weaker bands at 2 and 9.7 μm are also evident. Three principal absorption-emission bands for water vapor are also identified near 2.7, 6.6, and 20 μm with lesser bands at 1.17, 1.36, and 1.87 μm . The total emissivity ϵ_g and absorptivity $\alpha_{g,1}$ are calculated by integration with respect to wavelength of the spectral emissivities, using Eqs. (5-138) in a manner similar to the development of total surface properties.

Spectral Emissivities Highly resolved spectral emissivities can be generated at ambient temperatures from the HITRAN database (high-resolution transmission molecular absorption) that has been developed for atmospheric models [Rothman, L. S., Chance, K., and Goldman, A., eds., *J. Quant. Spectroscopy & Radiative Trans.*, **82** (1–4), 2003]. This database includes the chemical species: H_2O , CO_2 , O_3 , N_2O , CO , CH_4 , O_2 , NO , SO_2 , NO_2 , NH_3 , HNO_3 , OH , HF , HCl , HBr , ClO , OCS , H_2CO , HOCl , N_2 , HCN , CH_3C , HCl , H_2O_2 , C_2H_2 , C_2H_6 , PH_3 , COF_2 , SF_6 , H_2S , and HCO_2H . These data have been extended to high temperature for CO_2 and H_2O , allowing for the changes in the population of different energy levels and in the line half width [Denison, M. K., and Webb, B. W., *Heat Transfer*, **2**, 19–24 (1994)]. The resolution in the single-line models of emissivities is far greater than that needed in engineering calculations. A number of models are available that average the emissivities over narrow-wavelength regimes or over the entire band. An extensive set of measurements of narrowband parameters performed at NASA (Ludwig, C., et al., *Handbook of Infrared Radiation from Combustion Gases*, NASA SP-3080, 1973) has been used to develop the RADCAL computer code to obtain spectral emissivities for CO_2 , H_2O , CH_4 , CO , and soot (Grosshandler,

W. L., “RADCAL,” NIST Technical Note 1402, 1993). The exponential wideband model is available for emissions averaged over a band for H_2O , CO_2 , CO , CH_4 , NO , SO_2 , N_2O , NH_3 , and C_2H_2 [Edwards, D. K., and Menard, W. A., *Appl. Optics*, **3**, 621–625 (1964)]. The line and band models have the advantages of being able to account for complexities in determining emissivities of line broadening due to changes in composition and pressure, exchange with spectrally selective walls, and greater accuracy in formulating fluxes in gases with temperature gradients. These models can be used to generate the total emissivities and absorptivities that will be used in this chapter. RADCAL is a command-line FORTRAN code which is available in the public domain on the Internet.

Total Emissivities and Absorptivities Total emissivities and absorptivities for water vapor and carbon dioxide at present are still based on data embodied in the classical **Hottel emissivity charts**. These data have been adjusted with the more recent measurements in RADCAL and used to develop the correlations of emissivities given in Table 5-5. Two empirical correlations which permit hand calculation of emissivities for water vapor, carbon dioxide, and four mixtures of the two gases are presented in Table 5-5. The first section of Table 5-5 provides data for the two constants b and n in the empirical relation

$$\overline{\epsilon_g T_g} = b[pL - 0.015]^n \quad (5-140a)$$

while the second section of Table 5-5 utilizes the four constants in the empirical correlation

$$\log(\overline{\epsilon_g T_g}) = a_0 + a_1 \log(pL) + a_2 \log^2(pL) + a_3 \log^3(pL) \quad (5-140b)$$

In both cases the empirical constants are given for the three temperatures of 1000, 1500, and 2000 K. Table 5-5 also includes some six values for the partial pressure ratios p_w/p_c of water vapor to carbon dioxide, namely, 0, 0.5, 1.0, 2.0, 3.0, and ∞ . These ratios correspond to composition values of $p_c/(p_c + p_w) = 1/(1 + p_w/p_c)$ of 0, 1/3, 1/2, 2/3, 3/4, and unity. For emissivity calculations at other temperatures and mixture compositions, linear interpolation of the constants is recommended.

The absorptivity can be obtained from the emissivity with aid of Table 5-5 by using the following functional equivalence.

$$\overline{\alpha_{g,1} T_1} = [\overline{\epsilon_g T_1}(pL, T_1/T_g)] \left(\frac{T_g}{T_1} \right)^{0.5} \quad (5-141)$$

Verbally, the absorptivity computed from Eq. (5-141) by using the correlations in Table 5-5 is based on a value for gas emissivity ϵ_g calculated at a temperature T_1 and at a partial-pressure path-length product of $(p_c + p_w)LT_1/T_g$. The absorptivity is then equal to this value of gas emissivity multiplied by $(T_g/T_1)^{0.5}$. It is recommended that spectrally based models such as RADCAL (loc. cit.) be used particularly when extrapolating beyond the temperature, pressure, or partial-pressure-length product ranges presented in Table 5-5.

A comparison of the results of the predictions of Table 5-5 with values obtained via the integration of the spectral results calculated from the narrowband model in RADCAL is provided in Fig. 5-22. Here calculations are shown for $p_c L = p_w L = 0.12 \text{ atm}\cdot\text{m}$ and a gas temperature of 1500 K. The RADCAL predictions are 20 percent higher than the measurements at low values of pL and are 5 percent higher at the large values of pL . An extensive comparison of different sources of emissivity data shows that disparities up to 20 percent are to be expected at the current time [Lallemant, N., Sayre, A., and Weber, R., *Prog. Energy Combust. Sci.*, **22**, 543–574, (1996)]. However, smaller errors result for the range of the total emissivity measurements presented in the Hottel emissivity tables. This is demonstrated in Example 11.

Example 11: Calculations of Gas Emissivity and Absorptivity Consider a slab of gas confined between two infinite parallel plates with a distance of separation of $L = 1 \text{ m}$. The gas pressure is 101.325 kPa (1 atm), and the gas temperature is 1500 K (2240°F). The gas is an equimolar mixture of CO_2 and H_2O , each with a partial pressure of 12 kPa ($p_c = p_w = 0.12 \text{ atm}$). The radiative flux to one of its bounding surfaces has been calculated by using RADCAL for two cases. For case (a) the flux to the bounding surface is 68.3 kW/m^2 when the emitting gas is backed by a black surface at an ambient temperature of 300 K (80°F). This (cold) back surface contributes less than 1 percent to the flux. In case (b), the flux is calculated as 106.2 kW/m^2 when the gas is backed by a black surface at a temperature of 1000 K (1340°F). In this example, gas emissivity and

*Spectral lines are conventionally described in terms of wave number $\eta = 1/\lambda$, with each line having a peak absorption at wave number η_0 . The Lorentz distribution is defined as $k_\eta/S = \frac{b_c}{\pi[b_c^2 + (\eta - \eta_0)^2]}$ where S is the integral of k_η over all

wave numbers. The parameter S is known as the integrated line intensity, and b_c is defined as the collision line half-width, i.e., the half-width of the line is one-half of its peak centerline value. The units of k_η are $\text{m}^{-1} \text{ atm}^{-1}$.

TABLE 5-5 Emissivity-Temperature Product for CO₂-H₂O Mixtures, $\epsilon_g T_g$

Limited range for furnaces, valid over 25-fold range of $p_{w+c}L$, 0.046–1.15 m-atm (0.15–3.75 ft-atm)												
p_w/p_c	0	$\frac{1}{2}$	1		2		3		∞			
$\frac{p_w}{p_w + p_c}$	0	$\frac{1}{2}(0.3\text{--}0.42)$	$\frac{1}{2}(0.42\text{--}0.5)$		$\frac{2}{3}(0.6\text{--}0.7)$		$\frac{3}{4}(0.7\text{--}0.8)$		1			
	CO ₂ only	Corresponding to (CH ₂) ₂ , covering coal, heavy oils, pitch	Corresponding to (CH ₂) ₂ , covering distillate oils, paraffins, olefines		Corresponding to CH ₄ , covering natural gas and refinery gas		Corresponding to (CH ₄) ₂ , covering future high H ₂ fuels		H ₂ O only			
Section 1												

NOTE: $p_w/(p_w + p_c)$ of $\frac{1}{2}$, $\frac{1}{2}$, $\frac{2}{3}$, and $\frac{3}{4}$ may be used to cover the ranges 0.2–0.4, 0.4–0.6, 0.6–0.7, and 0.7–0.8, respectively, with a maximum error in ϵ_g of 5 percent at $pL = 6.5$ m-atm, less at lower pL 's. Linear interpolation reduces the error generally to less than 1 percent. Linear interpolation or extrapolation on T introduces an error generally below 2 percent, less than the accuracy of the original data.

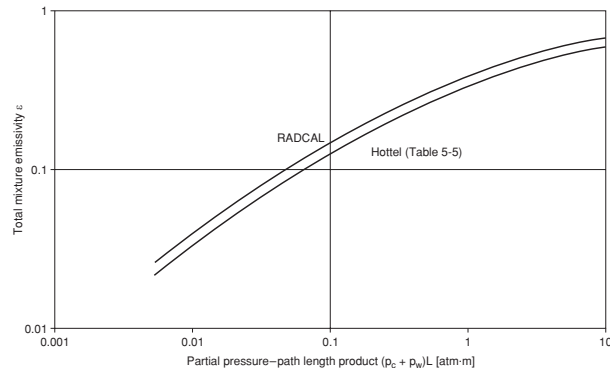


FIG. 5-22 Comparison of Hottel and RADCAL total gas emissivities. Equimolar gas mixture of CO₂ and H₂O with $p_c = p_w = 0.12$ atm and $T_g = 1500$ K.

absorptivity are to be computed from these flux values and compared with values obtained by using Table 5-5.

Case (a): The flux incident on the surface is equal to $\epsilon_g \sigma T_g^4 = 68.3 \text{ kW/m}^2$; therefore, $\epsilon_g = 68,300 / (5.6704 \times 10^{-8} \cdot 1500^4) = 0.238$. To utilize Table 5-5, the mean beam length for the gas is calculated from the relation $L_M = 0.88 L_{M0} = 0.88 \cdot 2L = 1.76$ m. For $T_g = 1500$ K and $(p_c + p_w)L_M = 0.24(1.76) = 0.422$ atm-m, the two-constant correlation in Table 5-5 yields $\epsilon_g = 0.230$ and the four-constant correlation yields $\epsilon_g = 0.234$. These results are clearly in excellent agreement with the predicted value of $\epsilon_g = 0.238$ obtained from RADCAL.

Case (b): The flux incident on the surface (106.2 kW/m^2) is the sum of that contributed by (1) gas emission $\epsilon_g \sigma T_g^4 = 68.3 \text{ kW/m}^2$ and (2) emission from the opposing surface corrected for absorption by the intervening gas using the gas transmissivity, that is, $\tau_{g,1} \sigma T_1^4$ where $\tau_{g,1} = 1 - \alpha_{g,1}$. Therefore $\alpha_{g,1} = [1 - (106,200 - 68,300) / (5.6704 \times 10^{-8} \cdot 1000^4)] = 0.332$. Using Table 5-5, the two-constant and four-constant gas emissivities evaluated at $T_1 = 1000$ K and $pL = 0.4224 \cdot (1000/1500) = 0.282$ atm-m are $\epsilon_g = 0.2654$ and $\epsilon_g = 0.2707$, respectively. Multiplication by the factor $(T_g/T_1)^{0.5} = (1500/1000)^{0.5} = 1.225$ produces the final values of the two corresponding gas absorptivities $\alpha_{g,1} = 0.325$ and $\alpha_{g,1} = 0.332$, respectively. Again the agreement with RADCAL is excellent.

Other Gases The most extensive available data for gas emissivity are those for carbon dioxide and water vapor because of their importance in the radiation from the products of fossil fuel combustion. Selected data for other species present in combustion gases are provided in Table 5-6.

TABLE 5-6 Total Emissivities of Some Gases

Temperature $P_z L$, atm-ft	1000°R			1600°R			2200°R			2800°R		
	0.01	0.1	1.0	0.01	0.1	1.0	0.01	0.1	1.0	0.01	0.1	1.0
NH ₃ ^a	0.047	0.20	0.61	0.020	0.120	0.44	0.0057	0.051	0.25	(0.001)	(0.015)	(0.14)
SO ₂ ^b	0.020	0.13	0.28	0.013	0.090	0.32	0.0085	0.051	0.27	0.0058	0.043	0.20
CH ₄ ^c	0.0116	0.0518	0.1296	0.0111	0.0615	0.1880	0.0087	0.0608	0.2004	0.00622	0.04702	0.1525
CO ^d	0.0052	0.0167	0.0403	0.0055	0.0196	0.0517	0.0036	0.0145	0.0418	0.00224	0.00986	0.02855
NO ^d	0.0046	0.018	0.060	0.0046	0.021	0.070	0.0019	0.010	0.040	0.0078	0.004	0.025
HCl ^e	0.00022	0.00079	0.0020	0.00036	0.0013	0.0033	0.00037	0.0014	0.0036	0.00029	0.0010	0.0027

NOTE: Figures in this table are taken from plots in Hottel and Sarofim, *Radiative Transfer*, McGraw-Hill, New York, 1967, chap. 6. Values in parentheses are extrapolated. To convert degrees Rankine to kelvins, multiply by $(5.556)(10^{-1})$. To convert atmosphere-feet to kilopascal-meters, multiply by 30.89.

^aTotal-radiation measurements of Port (Sc.D. thesis in chemical engineering, MIT, 1940) at 1-atm total pressure, $L = 1.68$ ft, T to 2000°R.

^bCalculations of Guerrieri (S.M. thesis in chemical engineering, MIT, 1932) from room-temperature absorption measurements of Coblentz (*Investigations of Infrared Spectra*, Carnegie Institution, Washington, 1905) with poor allowance for temperature.

^cEstimated using Grosshandler, W.L., "RADCAL: A Narrow-Band Model for Radial Calculations in a Combustion Environment," NIST Technical Note 1402, 1993.

^dCalculations of Malkmus and Thompson [*J. Quant. Spectros. Radiat. Transfer*, **2**, 16 (1962)], to $T = 5400^\circ\text{R}$ and $PL = 30$ atm-ft.

^eCalculations of Malkmus and Thompson [*J. Quant. Spectros. Radiat. Transfer*, **2**, 16 (1962)], to $T = 5400^\circ\text{R}$ and $PL = 300$ atm-ft.

Flames and Particle Clouds

Luminous Flames Luminosity conventionally refers to soot radiation. At atmospheric pressure, soot is formed in locally fuel-rich portions of flames in amounts that usually correspond to less than 1 percent of the carbon in the fuel. Because soot particles are small relative to the wavelength of the radiation of interest in flames (primary particle diameters of soot are of the order of 20 nm compared to wavelengths of interest of 500 to 8000 nm), the incident radiation permeates the particles, and the absorption is proportional to the volume of the particles. In the limit of $r_p/\lambda \ll 1$, the Rayleigh limit, the monochromatic emissivity ϵ_λ is given by

$$\epsilon_\lambda = 1 - \exp(-K \cdot f_v \cdot L/\lambda) \quad (5-142)$$

where f_v is the volumetric soot concentration, L is the path length in the same units as the wavelength λ , and K is dimensionless. The value K will vary with fuel type, experimental conditions, and the temperature history of the soot. The values of K for a wide range of systems are within a factor of about 2 of one another. The single most important variable governing the value of K is the hydrogen/carbon ratio of the soot, and the value of K increases as the H/C ratio decreases. A value of $K = 9.9$ is recommended on the basis of seven studies involving 29 fuels [Mulholland, G. W., and Croarkin, C., *Fire and Materials*, **24**, 227–230 (2000)].

The total emissivity of soot ϵ_s can be obtained by substituting ϵ_λ from Eq. (5-142) for ϵ_λ in Eq. (5-138a) to yield

$$\epsilon_s = \int_{\lambda=0}^{\infty} \epsilon_\lambda \frac{E_{b,\lambda}(T_g, \lambda)}{E_b(T_g)} d\lambda = 1 - \frac{15}{4} [\Psi^{(3)}(1 + K \cdot f_v \cdot L \cdot T/c_2)] \quad (5-143)$$

$$\equiv (1 + K \cdot f_v \cdot L \cdot T/c_2)^{-4}$$

Here $\Psi^{(3)}(x)$ is defined as the pentagamma function of x and c_2 (m·K) is again Planck's second constant. The approximate relation in Eq. (5-143) is accurate to better than 1 percent for arguments yielding values of $\epsilon_s < 0.7$. At present, the largest uncertainty in estimating total soot emissivities is in the estimation of the soot volume fraction f_v . Soot forms in the fuel-rich zones of flames. Soot formation rates are a function of fuel type, mixing rate, local equivalence ratio Φ , temperature, and pressure. The equivalence ratio is defined as the quotient of the actual to stoichiometric fuel-to-oxidant ratio $\Phi = [F/O]_{\text{act}}/[F/O]_{\text{stoich}}$. Soot formation increases with the aromaticity or C/H ratio of fuels with benzene, α -methyl naphthalene, and acetylene having a high propensity to form soot and methane having a low soot formation propensity. Oxygenated fuels, such as alcohols, emit little soot. In practical turbulent diffusion flames, soot forms on the fuel side of the flame front. In premixed flames, at a given temperature, the rate of soot formation increases rapidly for $\Phi > 2$. For temperatures above

1500 K, soot burns out rapidly (in less than 0.1s) under fuel-lean conditions, $\Phi < 1$. Because of this rapid soot burnout, soot is usually localized in a relatively small fraction of a furnace or combustor volume. Long, poorly mixed diffusion flames promote soot formation while highly backmixed combustors can burn soot-free. In a typical flame at atmospheric pressure, maximum volumetric soot concentrations are found to be in the range $10^{-7} < f_v < 10^{-6}$. This corresponds to a soot formation of about 1.5 to 15 percent of the carbon in the fuel. When f_v is to be calculated at high pressures, allowance must be made for the significant increase in soot formation with pressure and for the inverse proportionality of f_v with respect to pressure. Great progress is being made in the ability to calculate soot in premixed flames. For example, predicted and measured soot concentration have been compared in a well-stirred reactor operated over a wide range of temperatures and equivalence ratios [Brown, N.J., Revzan, K. L., Frenklach, M., *Twenty-seventh Symposium (International) on Combustion*, pp. 1573–1580, 1998]. Moreover, CFD (computational fluid dynamics) and population dynamics modeling have been used to simulate soot formation in a turbulent non-premixed ethylene-air flame [Zucca, A., Marchisio, D. L., Barresi, A. A., Fox, R. O., *Chem. Eng. Sci.*, 2005]. The importance of soot radiation varies widely between combustors. In large boilers the soot is confined to small volumes and is of only local importance. In gas turbines, cooling the combustor liner is of primary importance so that only small incremental soot radiation is of concern. In high-temperature glass tanks, the presence of soot adds 0.1 to 0.2 to emissivities of oil-fired flames. In natural gas-fired flames, efforts to augment flame emissivities with soot generation have generally been unsuccessful. The contributions of soot to the radiation from pool fires often dominates, and thus the presence of soot in such flames directly impacts the safe separation distances from dikes around oil tanks and the location of flares with respect to oil rigs.

Clouds of Large Black Particles The emissivity ϵ_M of a cloud of black particles with a large perimeter-to-wavelength ratio is

$$\epsilon_M = 1 - \exp[-(a/v)L] \quad (5-144)$$

where a/v is the projected area of the particles per unit volume of space. If the particles have no negative curvature (the particle does not "see" any of itself) and are randomly oriented, $a = A'/4$, where A' is the actual surface area. If the particles are uniform, $a/v = cA = cA'/4$, where A and A' are the projected and total areas of each particle and c is the number concentration of particles. For spherical particles this leads to

$$\epsilon_M = 1 - \exp[-(\pi/4)cd_p^2 L] = 1 - \exp(-1.5f_v L/d_p) \quad (5-145)$$

As an example, consider a heavy fuel oil (CH_{1.5}, specific gravity, 0.95) atomized to a mean surface particle diameter of d_p burned with

20 percent excess air to produce coke-residue particles having the original drop diameter and suspended in combustion products at 1204°C (2200°F). The flame emissivity due to the particles along a path of L m, with d_p measured in micrometers, is

$$\epsilon_M = 1 - \exp(-24.3L/d_p) \quad (5-146)$$

For 200- μ m particles and $L = 3.05$ m, the particle contribution to emissivity is calculated as 0.31.

Clouds of Nonblack Particles For nonblack particles, emissivity calculations are complicated by multiple scatter of the radiation reflected by each particle. The emissivity ϵ_M of a cloud of gray particles of individual emissivity ϵ_1 can be estimated by the use of a simple modification Eq. (5-144), i.e.,

$$\epsilon_M = 1 - \exp[-\epsilon_1(a/v)L] \quad (5-147)$$

Equation (5-147) predicts that $\epsilon_M \rightarrow 1$ as $L \rightarrow \infty$. This is *impossible* in a scattering system, and use of Eq. (5-147) is restricted to values of the optical thickness $(a/v)L < 2$. Instead, the asymptotic value of ϵ_M is obtained from Fig. 5-12 as $\epsilon_M = \epsilon_b$ ($\lim L \rightarrow \infty$), where the albedo ω is replaced by the particle-surface reflectance $\omega = 1 - \epsilon_1$. Particles with perimeter-to-wavelength ratios of 0.5 to 5.0 can be analyzed, with significant mathematical complexity, by use of the *Mie equations* (Bohren, C. F., and Huffman, D. R., *Absorption and Scattering of Light by Small Particles*, Wiley, 1998).

Combined Gas, Soot, and Particulate Emission In a mixture of emitting species, the emission of each constituent is attenuated on its way to the system boundary by absorption by all other constituents. The transmissivity of a mixture is the product of the transmissivities of its component parts. This statement is a corollary of Beer's law. For present purposes, the transmissivity of "species k " is defined as $\tau_k = 1 - \epsilon_k$. For a mixture of combustion products consisting of carbon dioxide, water vapor, soot, and oil coke or char particles, the total emissivity ϵ_T at any wavelength can therefore be obtained from

$$(1 - \epsilon_T)_\lambda = (1 - \epsilon_C)_\lambda(1 - \epsilon_W)_\lambda(1 - \epsilon_S)_\lambda(1 - \epsilon_M)_\lambda \quad (5-148)$$

where the subscripts denote the four flame species. The total emissivity is then obtained by integrating Eq. (5-148) over the entire EM energy spectrum, taking into account the variability of ϵ_C , ϵ_W , and ϵ_S with respect to wavelength. In Eq. (5-148), ϵ_M is independent of wavelength because absorbing char or coke particles are effectively black-body absorbers. Computer programs for spectral emissivity, such as RADCAL (loc. cit.), perform the integration with respect to wavelength for obtaining total emissivity. Corrections for the overlap of vibration-rotation bands of CO_2 and H_2O are automatically included in the correlations for ϵ_g for mixtures of these gases. The monochromatic soot emissivity is higher at shorter wavelengths, resulting in higher attenuations of the bands at 2.7 μ m for CO_2 and H_2O than at longer wavelengths. The following equation is recommended for calculating the emissivity ϵ_{g+s} of a mixture of CO_2 , H_2O , and soot

$$\epsilon_{g+s} = \epsilon_g + \epsilon_s - M \cdot \epsilon_g \epsilon_s \quad (5-149)$$

where M can be represented with acceptable error by the dimensionless function

$$M = 1.12 - 0.27 \cdot (T/1000) + 2.7 \times 10^{-5} f_c \cdot L \quad (5-150)$$

In Eq. (5-150), T has units of kelvins and L is measured in meters. Since coke or char emissivities are gray, their addition to those of the CO_2 , H_2O , and soot follows simply from Eq. (5-148) as

$$\epsilon_T = \epsilon_{g+s} + \epsilon_M - \epsilon_{g+s} \epsilon_M \quad (5-151)$$

with the definition $1 - \epsilon_{g+s} \equiv (1 - \epsilon_C)(1 - \epsilon_W)(1 - \epsilon_S)$.

RADIATIVE EXCHANGE WITH PARTICIPATING MEDIA

Energy Balances for Volume Zones—The Radiation Source Term Reconsider a generalized enclosure with N volume zones confining a gray gas. When the N gas temperatures are *unknown*, an *additional* set of N equations is required in the form of radiant energy

balances for each volume zone. These N equations are given by the *definition* of the N -vector for the **net radiant volume absorption** $\mathbf{S}' = [\mathbf{S}'_j]$ for each volume zone

$$\mathbf{S}' = \overline{\mathbf{GS}} \cdot \mathbf{E} + \overline{\mathbf{GG}} \cdot \mathbf{E}_g - 4K\mathbf{VI} \cdot \mathbf{E}_g \quad [N \times 1] \quad (5-152)$$

The radiative source term is a discretized formulation of the *net radiant absorption* for each volume zone which may be incorporated as a *source term* into numerical approximations for the generalized *energy equation*. As such, it permits formulation of energy balances on each zone that may include conductive and convective heat transfer. For $K \rightarrow 0$, $\overline{\mathbf{GS}} \rightarrow 0$, and $\overline{\mathbf{GG}} \rightarrow 0$ leading to $\mathbf{S}' \rightarrow \mathbf{0}_N$. When $K \neq 0$ and $\mathbf{S}' = \mathbf{0}_N$, the gas is said to be in a state of *radiative equilibrium*. In the notation usually associated with the discrete ordinate (DO) and finite volume (FV) methods, see Modest (op. cit., Chap. 16), one would write $S'_j/V_j = K[G - 4E_g] = -\nabla \cdot \vec{q}_r$. Here $H_g = G/4$ is the average flux density incident on a given volume zone from all other surface and volume zones. The DO and FV methods are currently available options as "RTE-solvers" in complex simulations of combustion systems using *computational fluid dynamics* (CFD).*

Implementation of Eq. (5-152) necessitates the definition of two additional *symmetric* $N \times N$ arrays of exchange areas, namely, $\overline{\mathbf{gg}} = [\overline{g_i g_j}]$ and $\overline{\mathbf{GG}} = [\overline{G_i G_j}]$. In Eq. (5-152) $\mathbf{VI} = [V_j \delta_{ij}]$ is an $N \times N$ diagonal matrix of zone volumes. The total exchange areas in Eq. (5-151) are *explicit* functions of the direct exchange areas as follows:

$$\begin{aligned} \text{Surface-to-gas exchange} \\ \overline{\mathbf{GS}} = \mathbf{SG}^T \quad [N \times M] \end{aligned} \quad (5-153a)$$

$$\begin{aligned} \text{Gas-to-gas exchange} \\ \overline{\mathbf{GG}} = \overline{\mathbf{gg}} + \mathbf{sg} \cdot \rho \mathbf{I} \cdot \mathbf{R} \cdot \overline{\mathbf{sg}} \quad [N \times M] \end{aligned} \quad (5-153b)$$

The matrices $\overline{\mathbf{gg}} = [\overline{g_i g_j}]$ and $\overline{\mathbf{GG}} = [\overline{G_i G_j}]$ must also satisfy the following **matrix conservation relations**:

$$\text{Direct exchange areas: } 4K\mathbf{VI} \cdot \mathbf{1}_N = \overline{\mathbf{gs}} \cdot \mathbf{1}_M + \overline{\mathbf{gg}} \cdot \mathbf{1}_N \quad (5-154a)$$

$$\text{Total exchange areas: } 4K\mathbf{VI} \cdot \mathbf{1}_N = \overline{\mathbf{GS}} \cdot \mathbf{1}_M + \overline{\mathbf{GG}} \cdot \mathbf{1}_N \quad (5-154b)$$

The formal integral definition of the direct gas-gas exchange area is

$$\overline{g_i g_j} = \iiint_{V_i} \iiint_{V_j} K^2 \frac{e^{-K_r}}{\pi^2} dV_j dV_i \quad (5-155)$$

Clearly, when $K = 0$, the two direct exchange areas involving a gas zone $\overline{\mathbf{gs}}$ and $\overline{\mathbf{gg}}$ vanish. Computationally it is *never* necessary to make resort to Eq. (5-155) for calculation of $\overline{g_i g_j}$. This is so because $\overline{g_i g_j}$, $\overline{g_i g_s}$, and $\overline{g_s g_j}$ may *all* be calculated *arithmetically* from appropriate values of $\overline{s_i s_j}$ by using associated conservation relations and view factor algebra.

Weighted Sum of Gray Gas (WSGG) Spectral Model Even in simple engineering calculations, the assumption of a gray gas is almost never a good one. The zone method is now further generalized to make allowance for nongray radiative transfer via incorporation of the **weighted sum of gray gas** (WSGG) spectral model. Hottel has shown that the emissivity $\epsilon_{g,T,L}$ of an absorbing-emitting gas mixture containing CO_2 and H_2O of known composition can be approximated by a *weighted sum of P gray gases*

$$\epsilon_g(T, L) \approx \sum_{p=1}^P a_p(T)(1 - e^{-K_p L}) \quad (5-156a)$$

where

$$\sum_{p=1}^P a_p(T) = 1.0 \quad (5-156b)$$

In Eqs. (5-156), K_p is some gray gas absorption coefficient and L is some appropriate path length. In practice, Eqs. (5-156) usually yield acceptable accuracy for $P \leq 3$. For $P = 1$, Eqs. (5-156) degenerate to the case of a single gray gas.

*To further clarify the mathematical differences between zoning and the DO and FV methods recognize that (neglecting scatter) the matrix expressions $\mathbf{H} = \mathbf{AI}^{-1} \cdot \mathbf{ss} \cdot \mathbf{W} + \mathbf{AI}^{-1} \cdot \mathbf{sg} \cdot \mathbf{E}_g$ and $4K\mathbf{VI} \cdot \mathbf{E}_g = \mathbf{VI}^{-1} \cdot \mathbf{gs} \cdot \mathbf{W} + \mathbf{VI}^{-1} \cdot \mathbf{gg} \cdot \mathbf{E}_g$ represent spatial discretizations of the integral form(s) of the RTE applied at any point (zone) on the boundary or interior of an enclosure, respectively, for a gray gas.

The Clear plus Gray Gas WSGG Spectral Model In principle, the emissivity of all gases approaches unity for infinite path length L . In practice, however, the gas emissivity may fall considerably short of unity for representative values of pL . This behavior results because of the *band* nature of real gas spectral absorption and emission whereby there is usually no significant overlap between dominant absorption bands. Mathematically, this physical phenomenon is modeled by defining one of the gray gas components in the WSGG spectral model to be *transparent*.

For $P = 2$ and path length L_M , Eqs. (5-156) yield the following expression for the gas emissivity

$$\epsilon_g = a_1(1 - e^{-K_1 L_M}) + a_2(1 - e^{-K_2 L_M}) \quad (5-157)$$

In Eq. (5-157) if $K_1 = 0$ and $a_2 \neq 0$, the limiting value of gas emissivity is $\epsilon_g(T, \infty) \rightarrow a_2$. Put $K_1 = 0$ in Eq. (5-157), $a_g = a_2$, and define $\tau_g = e^{-K_2 L_M}$ as the gray gas transmissivity. Equation (5-157) then simplifies to

$$\epsilon_g = a_g(1 - \tau_g) \quad (5-158)$$

It is important to note in Eq. (5-158) that $0 \leq a_g, \tau_g \leq 1.0$ while $0 \leq \epsilon_g \leq a_g$.

Equation (5-158) constitutes a two-parameter model which may be fitted with only *two* empirical emissivity data points. To obtain the constants a_g and τ_g in Eq. (5-158) at fixed composition and temperature, denote the two emissivity data points as $\epsilon_{g,2} = \epsilon_g(2pL) > \epsilon_{g,1} = \epsilon_g(pL)$ and recognize that $\epsilon_{g,1} = a_g(1 - \tau_g)$ and $\epsilon_{g,2} = a_g(1 - \tau_g^2) = a_g(1 - \tau_g)(1 + \tau_g) = \epsilon_{g,1}(1 + \tau_g)$. These relations lead directly to the final emissivity fitting equations

$$\tau_g = \frac{\epsilon_{g,2}}{\epsilon_{g,1}} - 1 \quad (5-159a)$$

and

$$a_g = \frac{\epsilon_{g,1}}{2 - \epsilon_{g,2}/\epsilon_{g,1}} \quad (5-159b)$$

The clear plus gray WSGG spectral model also readily leads to values for gas *absorptivity* and *transmissivity*, with respect to some appropriate surface radiation source at temperature T_1 , for example,

$$\alpha_{g,1} = a_{g,1}(1 - \tau_g) \quad (5-160a)$$

and

$$\tau_{g,1} = a_{g,1} \cdot \tau_g \quad (5-160b)$$

In Eqs. (5-160) the *gray* gas transmissivity τ_g is taken to be identical to that obtained for the gas emissivity ϵ_g . The constant $a_{g,1}$ in Eq. (5-160a) is then obtained with knowledge of *one additional empirical value* for $\alpha_{g,1}$ which may also be obtained from the correlations in Table 5-5. Notice further in the definitions of the three parameters ϵ_g , $\alpha_{g,1}$, and $\tau_{g,1}$ that *all* the temperature dependence is forced into the *two* WSGG constants a_g and $a_{g,1}$.

The *three* clear plus gray WSGG constants a_g , $a_{g,1}$, and τ_g are functions of total pressure, temperature, and mixture composition. It is not necessary to ascribe any particular physical significance to them. Rather, they may simply be visualized as three constants that happen to fit the gas emissivity data. It is noteworthy that *three* constants are far fewer than the number required to calculate gas emissivity data from fundamental spectroscopic data. The two constants a_g and $a_{g,1}$ defined in Eqs. (5-158) and (5-160) can, however, be interpreted *physically* in a particularly simple manner. Suppose the gas absorption spectrum is idealized by *many* absorption bands (boxes), *all* of which are characterized by the *identical* absorption coefficient K . The a 's might then be calculated from the total blackbody energy fraction $F_b(\lambda T)$ defined in Eqs. (5-105) and (5-106). That is, a_g simply represents the total *energy fraction* of the blackbody energy distribution in which the gas *absorbs*. This concept may be further generalized to real gas absorption spectra via the *wideband stepwise gray spectral box model* (Modest, op. cit., Chap. 14).

When $P \geq 3$, exponential curve-fitting procedures for the WSGG spectral model become significantly more difficult for hand computation but are quite routine with the aid of a variety of readily available

mathematical software utilities. The clear plus gray WSGG fitting procedure is demonstrated in Example 8.

The Zone Method and Directed Exchange Areas Spectral dependence of real gas spectral properties is now introduced into the zone method via the WSGG spectral model. It is still assumed, however, that all surface zones are gray isotropic emitters and absorbers.

General Matrix Representation We first define a new set of four **directed exchange areas** $\overleftarrow{\mathbf{SS}}$, $\overleftarrow{\mathbf{SG}}$, $\overleftarrow{\mathbf{GS}}$, and $\overleftarrow{\mathbf{GG}}$ which are denoted by an overarrow. The directed exchange areas are obtained from the total exchange areas for gray gases by simple matrix multiplication using weighting factors derived from the WSGG spectral model. The directed exchange areas are denoted by an overarrow to indicate the "sending" and "receiving" zone. The a -weighting factors for transfer originating at a gas zone $a_{g,i}$ are derived from WSGG gas *emissivity* calculations, while those for transfers originating at a surface zone, a_i are derived from appropriate WSGG gas *absorptivity* calculations. Let $\mathbf{a}\mathbf{g}\mathbf{I}_p = [a_{p,g,i}\delta_{ij}]$ and $\mathbf{a}\mathbf{I}_p = [a_{p,i}\delta_{ij}]$ represent the P [$M \times M$] and [$N \times N$] **diagonal** matrices comprised of the appropriate WSGG a constants. The *directed* exchange areas are then computed from the associated *total* gray gas exchange areas via simple **diagonal** matrix multiplication.

$$\overleftarrow{\mathbf{SS}} = \sum_{p=1}^P \overleftarrow{\mathbf{SS}}_p \cdot \mathbf{a}\mathbf{I}_p \quad [M \times M] \quad (5-161a)$$

$$\overleftarrow{\mathbf{SG}} = \sum_{p=1}^P \overleftarrow{\mathbf{SG}}_p \cdot \mathbf{a}\mathbf{g}\mathbf{I}_p \quad [M \times N] \quad (5-161b)$$

$$\overleftarrow{\mathbf{GS}} = \sum_{p=1}^P \overleftarrow{\mathbf{GS}}_p \cdot \mathbf{a}\mathbf{I}_p \quad [M \times N] \quad (5-161c)$$

$$\overleftarrow{\mathbf{GG}} = \sum_{p=1}^P \overleftarrow{\mathbf{GG}}_p \cdot \mathbf{a}\mathbf{g}\mathbf{I}_p \quad [N \times N] \quad (5-161d)$$

$$\text{with} \quad \overleftarrow{\mathbf{KI}} = \sum_{p=1}^P \mathbf{K}\mathbf{I}_p \cdot \mathbf{a}\mathbf{g}\mathbf{I}_p \quad [N \times N] \quad (5-161e)$$

In contrast to the *total* exchange areas which are always independent of temperature, the four *directed* arrays $\overleftarrow{\mathbf{SS}}$, $\overleftarrow{\mathbf{SG}}$, $\overleftarrow{\mathbf{GS}}$, and $\overleftarrow{\mathbf{GG}}$ are dependent on the temperatures of each and every zone, i.e., as in $a_{p,i} = a_p(T_i)$. Moreover, in contrast to *total* exchange areas, the *directed* arrays $\overleftarrow{\mathbf{SS}}$ and $\overleftarrow{\mathbf{GG}}$ are generally not symmetric and $\overleftarrow{\mathbf{GS}} \neq \overleftarrow{\mathbf{SG}}^T$. Finally, since the directed exchange areas are temperature-dependent, iteration may be required to update the $\mathbf{a}\mathbf{I}_p$ and $\mathbf{a}\mathbf{g}\mathbf{I}_p$ arrays during the course of a calculation. There is a great deal of latitude with regard to fitting the WSGG a constants in these matrix equations, especially if $N > 1$ and composition variations are to be allowed for in the gas. An extensive discussion of a fitting for $N > 1$ is beyond the scope of this presentation. Details of the fitting procedure, however, are presented in Example 12 in the context of a single-gas zone.

Having formulated the directed exchange areas, the governing matrix equations for the radiative flux equations at each surface zone and the radiant source term are then given as follows:

$$\mathbf{Q} = \epsilon \mathbf{A}\mathbf{I} \cdot \mathbf{E} - \overleftarrow{\mathbf{SS}} \cdot \mathbf{E} - \overleftarrow{\mathbf{SG}} \cdot \mathbf{E}_g \quad (5-162a)$$

$$\mathbf{S}' = \overleftarrow{\mathbf{GG}} \cdot \mathbf{E}_g + \overleftarrow{\mathbf{GS}} \cdot \mathbf{E} - 4\overleftarrow{\mathbf{KI}} \cdot \mathbf{V}\mathbf{I} \cdot \mathbf{E}_g \quad (5-162b)$$

or the alternative forms

$$\mathbf{Q} = [\mathbf{E}\mathbf{I} \cdot \overleftarrow{\mathbf{SS}} - \overleftarrow{\mathbf{SS}} \cdot \mathbf{E}\mathbf{I}] \cdot \mathbf{I}_M + [\mathbf{E}\mathbf{I} \cdot \overleftarrow{\mathbf{SG}} - \overleftarrow{\mathbf{SG}} \cdot \mathbf{E}_g \mathbf{I}] \cdot \mathbf{I}_N \quad (5-163a)$$

$$\mathbf{S}' = -[\mathbf{E}_g \mathbf{I} \cdot \overleftarrow{\mathbf{GS}} - \overleftarrow{\mathbf{GS}} \cdot \mathbf{E}\mathbf{I}] \cdot \mathbf{I}_M - [\mathbf{E}_g \mathbf{I} \cdot \overleftarrow{\mathbf{GG}} - \overleftarrow{\mathbf{GG}} \cdot \mathbf{E}_g \mathbf{I}] \cdot \mathbf{I}_N \quad (5-163b)$$

It may be *proved* that the $\mathbf{Q}$ and $\mathbf{S}'$ vectors computed from Eqs. (5-162) and (5-163) always *exactly* satisfy the overall (scalar) *radiant* energy balance $\mathbf{I}_M^T \mathbf{Q} = \mathbf{I}_N^T \mathbf{S}'$. In words, the total radiant gas emission for all gas zones in the enclosure must always exactly equal the total radiant energy received at all surface zones which comprise the enclosure. In Eqs. (5-162) and (5-163), the following definitions are employed for the *four forward-directed* exchange areas

$$\overrightarrow{\mathbf{SS}} = \overleftarrow{\mathbf{SS}}^T \quad \overrightarrow{\mathbf{SG}} = \overleftarrow{\mathbf{SG}}^T \quad \overrightarrow{\mathbf{GS}} = \overleftarrow{\mathbf{GS}}^T \quad \overrightarrow{\mathbf{GG}} = \overleftarrow{\mathbf{GG}}^T \quad (5-64a,b,c,d)$$

such that formally there are some *eight* matrices of directed exchange areas. The four *backward-directed* arrays of directed exchange areas must satisfy the following conservation relations

$$\overleftarrow{\mathbf{SS}} \cdot \mathbf{I}_M + \overleftarrow{\mathbf{SG}} \cdot \mathbf{I}_N = \epsilon \mathbf{I} \cdot \mathbf{AI} \cdot \mathbf{I}_M \quad (5-165a)$$

$$4\overleftarrow{\mathbf{KI}} \cdot \mathbf{VI} \cdot \mathbf{I}_N = \overleftarrow{\mathbf{GS}} \cdot \mathbf{I}_M + \overleftarrow{\mathbf{GG}} \cdot \mathbf{I}_N \quad (5-165b)$$

Subject to the restrictions of no scatter and diffuse surface emission and reflection, the above equations are the most general matrix statement possible for the zone method. When $P = 1$, the *directed* exchange areas all reduce to the *total* exchange areas for a single gray gas. If, in addition, $K = 0$, the much simpler case of radiative transfer in a transparent medium results. If, in addition, all surface zones are *black*, the direct, total, and directed exchange areas are all identical.

Allowance for Flux Zones As in the case of a transparent medium, we now distinguish between source and flux surface zones. Let $M = M_s + M_f$ represent the total number of surface zones where M_s is the number of source-sink zones and M_f is the number of flux zones. The flux zones are the *last to be numbered*. To accomplish this, partition the

surface emissive power and flux vectors as $\mathbf{E} = \begin{bmatrix} \mathbf{E}_1 \\ \mathbf{E}_2 \end{bmatrix}$ and $\mathbf{Q} = \begin{bmatrix} \mathbf{Q}_1 \\ \mathbf{Q}_2 \end{bmatrix}$,

where the subscript 1 denotes surface source/sink zones whose emissive power $\mathbf{E}_1$ is specified a priori, and subscript 2 denotes surface flux zones of unknown emissive power vector $\mathbf{E}_2$ and known radiative flux vector $\mathbf{Q}_2$. Suppose the radiative source vector $\mathbf{S}'$ is *known*. Appropriate partitioning of Eqs. (5-162) then produces

$$\begin{bmatrix} \mathbf{Q}_1 \\ \mathbf{Q}_2 \end{bmatrix} = \begin{bmatrix} \epsilon \mathbf{AI}_{1,1} & \mathbf{0} \\ \mathbf{0} & \epsilon \mathbf{AI}_{2,2} \end{bmatrix} \cdot \begin{bmatrix} \mathbf{E}_1 \\ \mathbf{E}_2 \end{bmatrix} + \begin{bmatrix} \overleftarrow{\mathbf{SS}}_{1,2} & \overleftarrow{\mathbf{SS}}_{1,2} \\ \overleftarrow{\mathbf{SS}}_{2,1} & \overleftarrow{\mathbf{SS}}_{2,2} \end{bmatrix} \cdot \begin{bmatrix} \mathbf{E}_1 \\ \mathbf{E}_2 \end{bmatrix} - \begin{bmatrix} \overleftarrow{\mathbf{SG}} \\ \overleftarrow{\mathbf{SG}} \end{bmatrix} \cdot \mathbf{E}_g \quad (5-166a)$$

and

$$\mathbf{S}' = \overleftarrow{\mathbf{GG}} \cdot \mathbf{E}_g + [\overleftarrow{\mathbf{GS}}_1 \quad \overleftarrow{\mathbf{GS}}_2] \begin{bmatrix} \mathbf{E}_1 \\ \mathbf{E}_2 \end{bmatrix} - 4\overleftarrow{\mathbf{KI}} \cdot \mathbf{VI} \cdot \mathbf{E}_g \quad (5-166b)$$

where the definitions of the matrix partitions follow the conventions with respect to Eq. (5-120). Simultaneous solution of the *two* unknown vectors in Eqs. (5-166) then yields

$$\mathbf{E}_2 = \mathbf{RP} \cdot [\overleftarrow{\mathbf{SS}}_{2,1} + \overleftarrow{\mathbf{SG}}_2 \cdot \mathbf{PP} \cdot \overleftarrow{\mathbf{GS}}_1] \cdot \mathbf{E}_1 + \mathbf{RP} \cdot [\mathbf{Q}_2 - \overleftarrow{\mathbf{SG}}_2 \cdot \mathbf{PP} \cdot \mathbf{S}'] \quad (5-167a)$$

and

$$\mathbf{E}_g = \mathbf{PP} \cdot [\overleftarrow{\mathbf{GS}}_1 \quad \overleftarrow{\mathbf{GS}}_2] \begin{bmatrix} \mathbf{E}_1 \\ \mathbf{E}_2 \end{bmatrix} - \mathbf{PP} \cdot \mathbf{S}' \quad (5-167b)$$

where two auxiliary inverse matrices $\mathbf{RP}$ and $\mathbf{PP}$ are defined as

$$\mathbf{PP} = [4\overleftarrow{\mathbf{KI}} \cdot \mathbf{VI} - \overleftarrow{\mathbf{GG}}]^{-1} \quad (5-168a)$$

$$\mathbf{RP} = [\epsilon \mathbf{AI}_{2,2} - \overleftarrow{\mathbf{SS}}_{2,2} - \overleftarrow{\mathbf{SG}}_2 \cdot \mathbf{PP} \cdot \overleftarrow{\mathbf{GS}}_2]^{-1} \quad (5-168b)$$

The emissive power vectors $\mathbf{E}$ and $\mathbf{E}_g$ are then both *known* quantities for purposes of subsequent calculation.

Algebraic Formulas for a Single Gas Zone As shown in Fig. 5-10, the three-zone system with $M = 2$ and $N = 1$ can be employed to simulate a surprisingly large number of useful engineering geometries. These include two infinite parallel plates confining an absorbing-emitting medium; *any* two-surface zone system where a nonconvex surface zone is completely surrounded by a second zone (this includes concentric spheres and cylinders), and the speckled two-surface enclosure. As in the case of a transparent medium, the inverse reflectivity matrix $\mathbf{R}$ is capable of explicit matrix inversion for $M = 2$. This allows derivation of *explicit* algebraic equations for all the required *directed* exchange areas for the clear plus gray WSGG spectral model with $M = 1$ and $N = 1$.

The Limiting Case $M = 1$ and $N = 1$ The directed exchange areas for this special case correspond to a single well-mixed gas zone completely surrounded by a single surface zone A_1 . Here the reflectivity matrix is a 1×1 scalar quantity which follows directly from the

general matrix equations as $\mathbf{R} = [1/(A_1 - \overline{s_1 s_1} \cdot \rho_1)]$. There are *two* WSGG clear plus gray constants a_1 and a_g , and only *one* unique direct exchange area which satisfies the conservation relation $\overline{s_1 s_1} + \overline{s_1 g} = A_1$. The only two physically meaningful directed exchange areas are those between the surface zone A_1 and the gas zone

$$\overleftarrow{S_1 G} = \frac{a_g \cdot \epsilon_1 A_1 \cdot \overline{s_1 g}}{\epsilon_1 \cdot A_1 + \rho_1 \cdot \overline{s_1 g}} \quad (5-169a)$$

$$\overleftarrow{GS_1} = \frac{a_1 \cdot \epsilon_1 A_1 \cdot \overline{s_1 g}}{\epsilon_1 \cdot A_1 + \rho_1 \cdot \overline{s_1 g}} \quad (5-169b)$$

The total radiative flux Q_1 at surface A_1 and the radiative source term $Q_1 = S$ are given by

$$Q_1 = \overleftarrow{GS_1} \cdot E_1 - \overleftarrow{S_1 G} \cdot E_g \quad (5-169)$$

Directed Exchange Areas for $M = 2$ and $N = 1$ For this case there are *four* WSGG constants, i.e., a_1 , a_2 , a_g , and τ_g . There is one required value of K that is readily obtained from the equation $K = -\ln(\tau_g)/L_{ab}$, where $\tau_g = \exp(-KL_{ab})$. For an enclosure with $M = 2$, $N = 1$, and $K \neq 0$, only *three unique* directed exchange areas are required because conservation stipulates $A_1 = \overline{s_1 s_2} + \overline{s_1 s_2} + \overline{s_1 g}$ and $A_2 = \overline{s_1 s_2} + \overline{s_2 s_2} + \overline{s_2 g}$. For $M = 2$ and $N = 1$, the matrix Eqs. (5-118) readily lead to the *general* gray gas matrix solution for $\overleftarrow{\mathbf{SS}}$ and $\overleftarrow{\mathbf{SG}}$ with $K \neq 0$ as

$$\overleftarrow{\mathbf{SS}} = \begin{bmatrix} \epsilon_1 A_1 - \overline{S_1 S_2} - \overline{S_1 G} & \overline{S_1 S_2} \\ \overline{S_1 S_2} & \epsilon_2 A_2 - \overline{S_1 S_2} - \overline{S_2 G} \end{bmatrix} \quad (5-170a)$$

where

$$\overline{S_1 S_2} = \epsilon_1 \epsilon_2 A_1 A_2 \overline{s_1 s_2} / \det \mathbf{R}^{-1} \quad (5-170b)$$

$$\text{and } \overleftarrow{\mathbf{SG}} = \begin{bmatrix} \epsilon_1 A_1 [(A_2 - \rho_2 \cdot \overline{s_2 s_2}) \cdot \overline{s_1 g} + \rho_2 \cdot \overline{s_1 s_2} \cdot \overline{s_2 g}] \\ \epsilon_2 A_2 [(A_1 - \rho_1 \cdot \overline{s_1 s_1}) \cdot \overline{s_2 g} + \rho_1 \cdot \overline{s_1 s_2} \cdot \overline{s_1 g}] \end{bmatrix} / \det \mathbf{R}^{-1} \quad (5-170c)$$

with $\overleftarrow{\mathbf{GS}} = \overleftarrow{\mathbf{SG}}^T$ and the indicated determinate of $\mathbf{R}^{-1}$ is evaluated algebraically as

$$\det \mathbf{R}^{-1} = (A_1 - \overline{s_1 s_1} \cdot \rho_1) \cdot (A_2 - \overline{s_2 s_2} \cdot \rho_2) - \rho_1 \cdot \rho_2 \cdot \overline{s_1 s_2}^2 \quad (5-170d)$$

For the WSGG clear gas components we denote $\overleftarrow{\mathbf{SS}}|_{K=0} \equiv \overleftarrow{\mathbf{SS}}_0$ and $\overleftarrow{\mathbf{SG}}|_{K=0} \equiv \overleftarrow{\mathbf{SG}}_0 = \mathbf{0}$. Finally the WSGG arrays of directed exchange areas are computed simply from *a*-weighted sums of the gray gas total exchange areas as

$$\overleftarrow{\mathbf{SS}} = \overleftarrow{\mathbf{SS}}_0 \cdot \begin{bmatrix} 1-a_1 & 0 \\ 0 & 1-a_2 \end{bmatrix} + \overleftarrow{\mathbf{SS}} \cdot \begin{bmatrix} a_1 & 0 \\ 0 & a_2 \end{bmatrix}$$

$$\overleftarrow{\mathbf{SG}} = \overleftarrow{\mathbf{SG}} \cdot a_g \quad (5-171a,b,c)$$

$$\overleftarrow{\mathbf{GS}} = \overleftarrow{\mathbf{GS}} \cdot \begin{bmatrix} a_1 & 0 \\ 0 & a_2 \end{bmatrix} \neq \overleftarrow{\mathbf{SG}}^T$$

and finally

$$\overleftarrow{\mathbf{GG}} = a_g \cdot 4KV - \overleftarrow{\mathbf{GS}} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad (5-171d)$$

The results of this development may be further expanded into algebraic form with the aid of Eq. (5-127) to yield the following

$$\overleftarrow{S_2 S_1} = \frac{\epsilon_1 \epsilon_2 A_1 A_2 \cdot \overline{s_2 s_1}|_0 (1 - a_1)}{\epsilon_1 \epsilon_1 A_1 A_2 + (\epsilon_1 A_1 \rho_2 + \epsilon_2 A_2 \rho_1) \cdot \overline{s_2 s_1}|_0} + \frac{\epsilon_1 \epsilon_2 A_1 A_2 \cdot \overline{s_2 s_1} \cdot a_1}{\det \mathbf{R}^{-1}} \quad (5-171e)$$

$$\overleftarrow{\mathbf{SG}} = \begin{bmatrix} \epsilon_1 A_1 [(A_2 - \rho_2 \cdot \overline{s_2 s_2}) \cdot \overline{s_1 g} + \rho_2 \cdot \overline{s_1 s_2} \cdot \overline{s_2 g}] a_g \\ \epsilon_2 A_2 [(A_1 - \rho_1 \cdot \overline{s_1 s_1}) \cdot \overline{s_2 g} + \rho_1 \cdot \overline{s_1 s_2} \cdot \overline{s_1 g}] a_g \end{bmatrix} / \det \mathbf{R}^{-1} \quad (5-171f)$$

and

$$\overleftarrow{\mathbf{GS}} = [\overleftarrow{GS_1} \quad \overleftarrow{GS_2}] \quad (5-171g)$$

whose matrix elements are given by $\overleftarrow{GS}_1 \equiv \epsilon_1 A_1 [(A_2 - \rho_2 \overleftarrow{s_1 s_1}) \cdot \overleftarrow{s_1 g} + \rho_2 \overleftarrow{s_1 s_2} \cdot \overleftarrow{s_2 g}] / \det \mathbf{R}^{-1}$ and $\overleftarrow{GS}_2 \equiv \epsilon_2 A_2 [(A_1 - \rho_1 \overleftarrow{s_1 s_1}) \cdot \overleftarrow{s_2 g} + \rho_1 \overleftarrow{s_1 s_2} \cdot \overleftarrow{s_1 g}] / \det \mathbf{R}^{-1}$. Derivation of the scalar (algebraic) forms for the directed exchange areas here is done primarily for pedagogical purposes. Computationally, the only advantage is to obviate the need for a digital computer to evaluate a $[2 \times 2]$ matrix inverse.

Allowance for an Adiabatic Refractory with $N = 1$ and $M = 2$
Put $N = 1$ and $M = 2$, and let zone 2 represent the refractory surface. Let $Q_2 = 0$ and $\epsilon_2 \neq 0$, and it then follows that we may define a *refractory-aided* directed exchange area $\overleftarrow{S_1 G_R}$ by

$$\overleftarrow{S_1 G_R} = \overleftarrow{S_1 G} + \frac{\overleftarrow{S_1 S_2} \cdot \overleftarrow{S_2 G}}{\overleftarrow{S_1 S_2} + \overleftarrow{S_2 G}} \quad (5-172a)$$

Assuming radiative equilibrium, the emissive power of the refractory may also be calculated from the companion equation

$$E_2 = \frac{\overleftarrow{S_2 S_1} \cdot E_1 + \overleftarrow{S_2 G} \cdot E_g}{\overleftarrow{S_2 S_1} + \overleftarrow{S_2 G}} \quad (5-172b)$$

In this circumstance, *all* the radiant energy originating in the gas volume is transferred to the *sole* sink zone A_1 . Equation (5-172a) is thus tantamount to the statement that $Q_1 = S'$ or that the net emission from the source ultimately must arrive at the sink. Notice that if $\epsilon_1 = 0$, Eq. (5-172a) leads to a physically incongruous statement since *all* the directed exchange areas would vanish and *no sink* would exist. Even for the simple case of $M = 2$, $N = 1$, the algebraic complexity of Eqs. (5-171) suggests that *numerical matrix manipulation* of directed exchange areas is to be preferred rather than calculations using algebraic formulas.

Engineering Approximations for Directed Exchange Areas
Use of the preceding equations for directed exchange areas with $M = 2$, $N = 1$ and the WSGG clear plus gray gas spectral approximation requires knowledge of three independent direct exchange areas. It also formally requires evaluation of three WSGG weighting constants a_1 , a_2 , and a_g with respect to the three temperatures T_1 , T_2 , and T_g . Further simplifications may be made by assuming that radiant transfer for the *entire* enclosure is characterized by the *single* mean beam length $L_M = 0.88 \cdot 4 \cdot V/A$. The requisite direct exchange areas are then approximated by

$$\overleftarrow{ss} = \tau_g \begin{bmatrix} A_1 \cdot F_{1,1} & A_1 \cdot F_{1,2} \\ A_2 \cdot F_{2,1} & A_2 \cdot F_{2,2} \end{bmatrix} \quad (5-173a)$$

$$\text{with} \quad \overleftarrow{sg} = (1 - \tau_g) \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} \quad (5-173b)$$

and for the particular case of a speckled enclosure

$$\overleftarrow{ss} = \frac{\tau_g}{A_1 + A_2} \begin{bmatrix} A_1^2 & A_1 \cdot A_2 \\ A_1 \cdot A_2 & A_2^2 \end{bmatrix} \quad (5-174a)$$

$$\text{also with} \quad \overleftarrow{sg} = (1 - \tau_g) \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} \quad (5-174b)$$

where again τ_g is obtained from the WSGG fit of gas emissivity. These approximate formulas clearly obviate the need for *exact* values of the direct exchange areas and may be used in conjunction with Eqs. (5-171).

For engineering calculations, an additional simplification is sometimes warranted. Again characterize the system by a single mean beam length $L_M = 0.88 \cdot 4 \cdot V/A$ and employ the *identical* value of $\tau_g = KL_M$ for all surface-gas transfers. The three a constants *might* then be obtained by a WSGG data-fitting procedure for gas emissivity and gas absorptivity which utilizes the three different temperatures T_g , T_1 , and T_2 . For engineering purposes we choose a simpler method, however. First calculate values of ϵ_g and $\alpha_{g,1}$ for gas temperature T_g with respect to the *dominant* (sink) temperature T_1 . The *net* radiative flux between an isothermal gas mass at temperature T_g and a black isothermal bounding surface A_1 at temperature T_1 (the sink) is given by Eq. (5-138) as

$$Q_{1,g} = A_1 \sigma (\alpha_{g,1} T_1^4 - \epsilon_g T_g^4) \quad (5-175)$$

It is clear that transfer from the gas to the surface and transfer from the surface into the gas are characterized by two different constants of proportionality, ϵ_g and $\alpha_{g,1}$. To allow for the difference between gas emissivity and absorptivity, it proves convenient to introduce a single *mean gas emissivity* defined by

$$\sigma [\epsilon_g T_g^4 - \alpha_{g,1} T_1^4] = \epsilon_m \sigma (T_g^4 - T_1^4) \quad (5-176a)$$

$$\text{or} \quad \epsilon_m \equiv \frac{\epsilon_g - \alpha_{g,1} (T_1/T_g)^4}{1 - (T_1/T_g)^4} \quad (5-176b)$$

The calculation then proceeds by computing *two* values of ϵ_m at the given T_g and T_1 temperature pair and the two values of pL_M and $2pL_M$. We thereby obtain the expression $\epsilon_m = a_m (1 - \tau_m)$. It is then assumed that $a_1 = a_2 = a_g = a_m$ for use in Eqs. (5-171). This simplification may be used for $M > 2$ as long as $N = 1$. This simplification is illustrated in Example 12.

Example 12: WSGG Clear plus Gray Gas Emissivity Calculations
Methane is burned to completion with 20 percent excess air (50 percent relative humidity at 298 K or 0.0088 mol water/mol dry air) in a furnace chamber of floor dimensions 3×10 m and height 5 m. The entire surface area of the enclosure is a gray sink with emissivity of 0.8 at temperature 1000 K. The confined gas is well stirred at a temperature of 1500 K. Evaluate the clear plus gray WSGG constants and the mean effective gas emissivity, and calculate the average radiative flux density to the enclosure surface.

Two-zone model, $M = 1$, $N = 1$: A single volume zone completely surrounded by a single sink surface zone.

Function definitions:

$$\text{Gas emissivity:} \quad \epsilon_g F(T_g, pL, b, n) := b \cdot (pL - 0.015)^n + T_g \quad \text{Eq. (5-140a)}$$

$$\text{Gas absorptivity:} \quad \alpha_{g1} F(T_g, T_1, pL, b, n) \quad \text{Eq. (5-141)}$$

$$:= \frac{\epsilon_g F(T_1, pL \cdot T_1 + T_g, b, n) \cdot T_1 \cdot (T_g + T_1)^{0.5}}{T_1}$$

$$\text{Mean effective gas emissivity:} \quad \epsilon_{gm}(\epsilon_g, \alpha_g, T_g, T_1) := \frac{\epsilon_g - \alpha_g (T_1 + T_g)^4}{1 - (T_1 + T_g)^4} \quad \text{Eq. (5-176a)}$$

$$\text{Physical constants:} \quad \sigma \equiv 5.670400 \times 10^{-8} \frac{\text{W}}{\text{m}^2 \cdot \text{K}^4}$$

Enclosure input parameters:

$$T_g := 1500 \text{ K} \quad T_1 := 1000 \text{ K} \quad A_1 := 190 \text{ m}^2 \quad V := 150 \text{ m}^3$$

$$\epsilon_1 := 0.8 \quad \rho_1 := 1 - \epsilon_1 \quad \rho_1 = 0.2$$

$$E_1 := \sigma T_1^4 \quad E_g := \sigma T_g^4 \quad E_1 = 56.70 \frac{\text{kW}}{\text{m}^2} \quad E_g = 287.06 \frac{\text{kW}}{\text{m}^2}$$

Stoichiometry yields the following mole table:

Mole Table: Basis 1.0 mol Methane

Species	MW	Moles in	Mass in	Moles out	Y out
CH ₄	16.04	1.00000	16.04	0.00000	0.00000
O ₂	32.00	2.40000	76.80	0.40000	0.03193
N ₂	28.01	9.02857	252.93	9.02857	0.72061
CO ₂	44.01	0.00000	0.00	1.00000	0.07981
H ₂ O	18.02	0.10057	1.81	2.10057	0.16765
Totals	27.742	12.52914	347.58	1.52914	1.00000

$$p_w := .16765 \text{ atm} \quad p_c := 0.07981 \text{ atm} \quad p := p_w + p_c \quad p = 0.2475 \text{ atm}$$

$$p_w + p_c = 2.101$$

The mean beam length is approximated by

$$L_M := 0.88 \cdot 4 \cdot V + A_1 \quad L_M = 2.7789 \text{ m}$$

$$\text{and} \quad pL_M := p \cdot L_M \quad pL_M = 0.6877 \text{ atm} \cdot \text{m} \quad pL_M = 0.6877$$

The gas emissivities and absorptivities are then calculated from the two constant correlation in Table 5-5 (column 5 with $p_w/p_c = 2.0$) as follows:

$$\epsilon_{g1} := \epsilon_g F(1500, pL_M, 540, .42) \quad \epsilon_{g1} = 0.3048$$

$$\begin{aligned}\epsilon_{g2} &:= \epsilon_g F(1500, 2pL_M, 540, .42) & \epsilon_{g2} &= 0.4097 \\ \alpha_{g11} &:= \alpha_g 1F(1500, 1000, pL_M, 444, .34) & \alpha_{g11} &= 0.4124 \\ \alpha_{g12} &:= \alpha_g 1F(1500, 1000, 2pL_M, 444, .34) & \alpha_{g12} &= 0.5250\end{aligned}$$

Case (a): Compute Flux Density Using Exact Values of the WSGG Constants

$$\begin{aligned}\tau_g &:= \frac{\epsilon_{g2}}{\epsilon_{g1}} - 1 & a_g &:= \frac{\epsilon_{g1}}{1 - \tau_g} & \epsilon_g &:= a_g \cdot (1 - \tau_g) & a_{g1} &:= \frac{\alpha_{g11}}{1 - \tau_g} \\ \tau_g &= 0.3442 & a_g &= 0.4647 & \epsilon_g &= 0.3048 & a_{g1} &= 0.6289\end{aligned}$$

and the WSGG gas absorption coefficient (which is necessary for calculation of direct exchange areas) is calculated as $K_1 := \frac{-(\ln \tau_g)}{L_M}$ or $K_1 = 0.3838 \frac{1}{m}$

Compute directed exchange areas:
Eqs. (5-169)

$$\begin{aligned}s1g &:= (1 - \tau_g) \cdot A_1 & s1g &= 124.61 \text{ m}^2 \\ DS1G &:= \frac{a_g \cdot \epsilon_1 \cdot A_1 \cdot s1g}{\epsilon_1 \cdot A_1 + \rho_1 \cdot s1g} & DGS1 &:= \frac{a_{g1} \cdot \epsilon_1 \cdot A_1 \cdot s1g}{\epsilon_1 \cdot A_1 + \rho_1 \cdot s1g} \\ DS1G &= 49.75 \text{ m}^2 & DGS1 &= 67.32 \text{ m}^2\end{aligned}$$

And finally the gas to sink flux density is computed as

$$Q_1 := DGS1 \cdot E_1 - DS1G \cdot E_g \quad Q_1 = -10464.0 \text{ kW} \quad \frac{Q_1}{A_1} = -55.07 \frac{\text{kW}}{\text{m}^2}$$

Case (b): Compute the Flux Density Using Mean Effective Gas Emissivity Approximation

$$\begin{aligned}\epsilon_{gm1} &:= \epsilon_{gm}(\epsilon_{g1}, \alpha_{g11}, T_g, T_1) & \epsilon_{gm1} &= 0.2783 \\ \epsilon_{gm2} &:= \epsilon_{gm}(\epsilon_{g2}, \alpha_{g12}, T_g, T_1) & \epsilon_{gm2} &= 0.3813 \\ \tau_m &:= \frac{\epsilon_{gm2}}{\epsilon_{gm1}} - 1 & a_m &:= \frac{\epsilon_{gm1}}{1 - \tau_m} & \epsilon_{gm} &:= a_m \cdot (1 - \tau_m) \\ \tau_m &= 0.3701 & a_m &= 0.4418 & \epsilon_{gm} &= 0.2783 & s1g_m &:= (1 - \tau_m) \cdot A_1 \\ S1G_m &:= \frac{\epsilon_1 \cdot a_m \cdot s1g_m \cdot A_1}{\epsilon_1 \cdot A_1 + \rho_1 \cdot s1g_m} & S1G_m &= 45.68 \text{ m}^2 & s1g_m &= 119.67 \text{ m}^2 \\ q_{1m} &:= \frac{S1G_m \cdot (E_1 - E_g)}{A_1} & q_{1m} &= -55.38 \frac{\text{kW}}{\text{m}^2} \\ \text{compared with } \frac{Q_1}{A_1} &= -55.07 \frac{\text{kW}}{\text{m}^2}\end{aligned}$$

The computed flux densities are nearly equal because there is a *single* sink zone A_1 .
(This example was developed as a MATHCAD 14® worksheet. Mathcad is a registered trademark of Parametric Technology Corporation.)

ENGINEERING MODELS FOR FUEL-FIRED FURNACES

Modern digital computation has evolved methodologies for the design and simulation of fuel-fired combustion chambers and furnaces which incorporate virtually *all* the transport phenomena, chemical kinetics, and thermodynamics studied by chemical engineers. Nonetheless, there still exist many furnace design circumstances where such computational sophistication is not always appropriate. Indeed, a practical need still exists for simple engineering models for purposes of conceptual process design, cost estimation, and the correlation of test performance data. In this section, the zone method is used to develop perhaps the simplest **computational template** available to address some of these practical engineering needs.

Input/Output Performance Parameters for Furnace Operation The term **firing density** is typically used to define the basic *operational input parameter* for fuel-fired furnaces. In practice, firing density is often defined as the input fuel feed rate per unit area (or volume) of furnace heat-transfer surface. Thus defined, the firing density is a *dimensional quantity*. Since the feed enthalpy rate H_f is

proportional to the feed rate, we employ the sink area A_1 to define a **dimensionless firing density** as $\bar{N}_{FD} = H_f / \sigma T_{\text{Ref}}^4 A_1$ where T_{Ref} is some characteristic reference temperature. In practice, gross furnace *output performance* is often described by using one of several *furnace efficiencies*. The most common is the **gas or gas-side furnace efficiency** η_g , defined as the total enthalpy transferred to furnace internals divided by the total available feed enthalpy. Here the total available feed enthalpy is defined to include the lower heating value (LHV) of the fuel plus any air preheat above an arbitrary ambient datum temperature. Under certain conditions the definition of furnace efficiency reduces to some variant of the simple equation $\eta_g = (T_{\text{Ref}} - T_{\text{out}}) / (T_{\text{Ref}} - T_0)$ where again T_{Ref} is some reference temperature appropriate to the system in question.

The Long Plug Flow Furnace (LPFF) Model If a combustion chamber of cross-sectional area A_{Duct} and perimeter P_{Duct} is sufficiently long in the direction of flow, compared to its mean hydraulic radius, $L \gg R_h = A_{\text{Duct}} / P_{\text{Duct}}$, the radiative flux from the gas to the bounding surfaces can sometimes be adequately characterized by the *local* gas temperature. The physical rationale for this is that the *magnitudes* of the opposed upstream and downstream radiative fluxes through a cross section transverse to the direction of flow are sufficiently large as to substantially balance each other. Such a situation is not unusual in engineering practice and is referred to as the *long furnace approximation*. As a result, the radiative flux from the gas to the bounding surface may then be approximated using *two-dimensional* directed exchange

areas, $\overleftarrow{S_1 G} / A_1 \equiv \frac{\partial(S_1 G)}{\partial A_1}$, calculated using methods as described previously.

Consider a duct of length L and perimeter P , and assume plug flow in the direction of flow z . Further assume high-intensity mixing at the entrance end of the chamber such that combustion is *complete* as the combustion products enter the duct. The duct then acts as a long heat exchanger in which heat is transferred to the walls at constant temperature T_1 by the combined effects of radiation and convection. Subject to the long furnace approximation, a differential energy balance on the duct then yields

$$\dot{m} \bar{C}_p \frac{dT_g}{dz} = P \left[\frac{\overleftarrow{S_1 G}}{A_1} \sigma (T_g^4 - T_1^4) + h(T_g - T_1) \right] \quad (5-177)$$

where $\dot{m}$ is the mass flow rate and $\bar{C}_p$ is the heat capacity per unit mass. Equation (5-177) is nonlinear with respect to temperature. To solve Eq. (5-177), first linearize the convective heat-transfer term in the right-hand side with the approximation $\Delta T = T_2 - T_1 \approx (T_2^3 - T_1^3) / 4\bar{T}_{1,2}^2$ where $\bar{T}_{1,2} = (T_1 + T_2) / 2$. This linearization underestimates ΔT by no more than 5 percent when $T_2 / T_1 < 1.59$. Integration of Eq. (5-177) then leads to the solution

$$\ln \left[\frac{(T_{g,\text{out}} - T_1)(T_{g,\text{in}} + T_1)}{(T_{g,\text{out}} + T_1)(T_{g,\text{in}} - T_1)} \right] + 2.0 \tan^{-1} \left[\frac{(T_{g,\text{in}} - T_{g,\text{out}}) \cdot T_1}{(T_1^2 + T_{g,\text{in}} \cdot T_{g,\text{out}})} \right] = - \frac{4}{D_{\text{eff}}} \quad (5-178)$$

The LPFF model is described by only two dimensionless parameters, namely an *effective* firing density and a dimensionless sink temperature, viz.,

$$D_{\text{eff}} = \frac{N_{\text{FD}}}{\left(\frac{\overleftarrow{S_1 G}}{A_1} \right) + N_{\text{CR}}} \quad \text{and} \quad \Theta_1 = T_1 / T_{g,\text{in}} \quad (5-178a,b)$$

Here the dimensionless firing density, N_{FD} , and a dimensionless convection-radiation number N_{CR} are defined as

$$N_{\text{FD}} = \frac{\dot{m} \bar{C}_p}{\sigma T_1^3 A_1} \quad \text{and} \quad N_{\text{CR}} = \frac{h}{4\sigma \bar{T}_{g,1}^3} \quad (5-178c,d)$$

where $A_1 = PL$ is the duct surface area (the sink area), and $\bar{T}_{g,1} = (T_g + T_1) / 2$ is treated as a constant. This definition of the effective dimensionless firing density, D_{eff} , clearly delineates the relative

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roles of radiation and convective heat transfer since radiation and convection are identified as *parallel (electrical) conductances*.

In analogy with a conventional heat exchanger, Eq. (5-178) displays two asymptotic limits. First define

$$\eta_f = \frac{T_{g,in} - T_{g,out}}{T_{g,in} - T_1} = 1 - \frac{T_{g,out} - T_1}{T_{g,in} - T_1} \quad (5-179)$$

as the efficiency of the long furnace. The two asymptotic limits with respect to firing density are then given by

$$D_{\text{eff}} \ll 1, \quad T_{g,out} \rightarrow T_1 \quad \eta_f \rightarrow 1 \quad (5-179a)$$

and

$$D_{\text{eff}} \gg 1, \quad T_{g,out} \rightarrow T_{g,in}$$

$$\eta_f \rightarrow \frac{4}{D_{\text{eff}} \left[1 - \frac{R-1}{R+1} - 2 \frac{R-1}{R^2+1} \right]} \quad (5-179b)$$

where $R \equiv T_{g,in}/T_1 = 1/\Theta_1$.

For low firing rates, the exit temperature of the furnace gases approaches that of the sink; i.e., sufficient residence time is provided for nearly complete heat removal from the gases. When the combustion chamber is overfired, only a small fraction of the available feed enthalpy heat is removed within the furnace. The exit gas temperature then remains essentially that of the inlet temperature, and the furnace efficiency tends asymptotically to zero.

It is important to recognize that the two-dimensional exchange area $\frac{\bar{S}_1 \bar{G}}{A_1} \equiv \frac{\partial(\bar{S}_1 \bar{G})}{\partial A_1}$ in the definition of D_{eff} can represent a *lumped* two-

dimensional exchange area of somewhat arbitrary complexity. This quantity also contains *all* the information concerning furnace geometry and gas and surface emissivities. To compare the relative importance of radiation with respect to convection, suppose $h = 10 \text{ Btu}/(\text{hr} \cdot \text{ft}^2 \cdot ^\circ\text{R}) = 0.057 \text{ kW}/(\text{K} \cdot \text{m}^2)$ and $T_{g,1} = 1250 \text{ K}$, which leads to the numerical value $N_{\text{CR}} = 0.128$; or, in general, N_{CR} is of order 0.1 or less. The importance of the radiation contribution is estimated by bounding the magnitude of the dimensionless directed exchange area. For the case of a single gas zone completely surrounded by a black enclosure, Eq. (5-169) reduces to simply $\bar{S}_1 \bar{G}/A_1 = \epsilon_g \leq 1.0$, and it is evident that the magnitude of the radiation contribution never exceeds unity. At high temperatures, radiative effects can easily dominate other modes of heat transfer by an order of magnitude or more. When mean beam length calculations are employed, use $L_{\text{MD}}/D = 0.94$ for a cylindrical cross section of diameter D , and

$$L_{\text{M0}} = \frac{2H \cdot W}{H + W}$$

for a rectangular duct of height H and width W .

The Well-Stirred Combustion Chamber (WSCC) Model
Many combustion chambers utilize high-momentum feed conditions with associated high-intensity mixing. The **well-stirred combustion chamber (WSCC) model** assumes a single gas zone and high-intensity mixing. Moreover, combustion and heat transfer are visualized to occur *simultaneously* within the combustion chamber. The WSCC model is characterized by some *six* temperatures which are listed in rank order as T_0 , T_{air} , T_1 , T_e , T_g , and T_f . Even though the combustion chamber is well mixed, it is *arbitrarily* assumed that the gas temperature within the enclosure T_g is *not necessarily equal* to the gas exit temperature T_e . Rather the two temperatures are related by the simple relation $\Delta T_{ge} \equiv T_g - T_e$, where $\Delta T_{ge} \approx 170 \text{ K}$ (as a representative value) is introduced as an *adjustable parameter* for purposes of data fitting and to make allowance for nonideal mixing. In addition, T_0 is the ambient temperature, T_{air} is the air preheat temperature, and T_f is a *pseudoadiabatic flame temperature*, as shall be explained in the following development. The condition $\Delta T_{ge} \equiv 0$ is intended to simulate a perfect continuous well-stirred reactor (CSTR).

Dimensional WSCC Approach A macroscopic enthalpy balance on the well-stirred combustion chamber is written as

$$-\Delta H = H_{\text{in}} - H_{\text{out}} = Q_{\text{Rad}} + Q_{\text{Con}} + Q_{\text{Ref}} \quad (5-180)$$

Here $Q_{\text{Rad}} = \bar{S}_1 \bar{G}_R \sigma (T_g^4 - T_1^4)$ represents radiative heat transfer to the sink (with due allowance for the presence of any refractory surfaces). And the two terms $Q_{\text{Con}} = h_1 A_1 (T_g - T_1)$ and $Q_{\text{Ref}} = U A_R (T_g - T_0)$ formulate the convective heat transfer to the sink and *through* the refractory, respectively.

Formulation of the left-hand side of Eq. (5-180) requires representative thermodynamic data and information on the combustion stoichiometry. In particular, the former includes the lower heating value of the fuel, the temperature-dependent molar heat capacity of the inlet and outlet streams, and the air preheat temperature T_{air} . It proves especially convenient now to introduce the definition of a pseudoadiabatic flame temperature T_f , which is *not* the true adiabatic flame temperature, but rather is an adiabatic flame temperature based on the average heat capacity of the combustion products over the temperature interval $T_0 < T < T_e$. The calculation of T_f does not allow for dissociation of chemical species and is a surrogate for the total enthalpy content of the input fuel-air mixture. It also proves to be an especially convenient system *reference temperature*. Details for the calculation of T_f are illustrated in Example 13.

In terms of this particular definition of the pseudoadiabatic flame temperature T_f , the total enthalpy change and gas efficiency are given simply as

$$\Delta H = \dot{H}_f - \dot{m} \cdot \bar{C}_{P,\text{Prod}} (T_e - T_0) = \dot{m} \bar{C}_{P,\text{Prod}} (T_f - T_e) \quad (5-181a,b)$$

where $\dot{H}_f \equiv \dot{m} \cdot \bar{C}_{P,\text{Prod}} (T_f - T_0)$ and $T_e = T_g - \Delta T_{ge}$. This particular definition of T_f leads to an especially convenient formulation of furnace efficiency

$$\eta_g = Q/\dot{H}_f = \frac{\dot{m} \bar{C}_{P,\text{Prod}} (T_f - T_e)}{\dot{m} \bar{C}_{P,\text{Prod}} (T_f - T_0)} = \frac{T_f - T_e}{T_f - T_0} \quad (5-182)$$

In Eq. (5-182), $\dot{m}$ is the total mass flow rate and $\bar{C}_{P,\text{Prod}}$ [J/(kg·K)] is defined as the average heat capacity of the product stream over the temperature interval $T_0 < T < T_e$.

The final overall enthalpy balance is then written as

$$\dot{m} \bar{C}_{P,\text{Prod}} (T_f - T_e) = \bar{S}_1 \bar{G}_R \sigma (T_g^4 - T_1^4) + h_1 A_1 (T_g - T_1) + U A_R (T_g - T_0) \quad (5-183)$$

with $T_e = T_g - \Delta T_{ge}$.

Equation (5-183) is a nonlinear algebraic equation which may be solved by a variety of iterative methods. The *sole* unknown quantity, however, in Eq. (5-183) is the gas temperature T_g . It should be recognized, in particular, that T_f , T_e , $\bar{C}_{P,\text{Prod}}$, and the directed exchange area are all explicit functions of T_g . The method of solution of Eq. (5-183) is demonstrated in some detail in Example 13.

Dimensionless WSCC Approach In Eq. (5-183), assume the convective heat loss through the refractory is negligible, and linearize the convective heat transfer to the sink. These approximations lead to the result

$$\dot{m} \bar{C}_{P,\text{Prod}} (T_f - T_g + \Delta T_{ge}) = \bar{S}_1 \bar{G}_R \sigma (T_g^4 - T_1^4) + h_1 A_1 (T_g - T_1) / 4 \bar{T}_{g,1}^3 \quad (5-184)$$

where $\bar{T}_{g,1} = (T_g + T_1)/2$ is some characteristic average temperature which is taken as constant. Now normalize all temperatures based on the pseudoadiabatic temperature as in $\Theta_i = T_i/T_f$. Equation (5-184) then leads to the dimensionless equation

$$D_{\text{eff}} (1 - \Theta_g + \Delta^\circ) = (\Theta_g^4 - \Theta_1^4) \quad (5-185)$$

where again $D_{\text{eff}} = N_{\text{FD}} / (\bar{S}_1 \bar{G}_R / A_1 + N_{\text{CR}})$ is defined *exactly* as in the case of the LPFF model, with the proviso that the WSCC dimensionless firing density is defined here as $N_{\text{FD}} = \dot{m} \bar{C}_{P,\text{Prod}} / (\sigma T_f^3 A_1)$. The dimensionless furnace efficiency follows directly from Eq. (5-182) as

$$\eta_g = \frac{1 - \Theta_e}{1 - \Theta_0} = \frac{1 - \Theta_g + \Delta^\circ}{1 - \Theta_0} \quad (5-186a)$$

We also define a **reduced furnace efficiency** η'_g

$$\eta'_g \equiv (1 - \Theta_0)\eta_g = 1 - \Theta_g + \Delta^\circ \quad (5-186b)$$

Since Eq. (5-186b) may be rewritten as $\Theta_g = (1 + \Delta^\circ - \eta'_g)$, combination of Eqs. (5-185) and (5-186b) then yields the final result

$$D_{\text{eff}}\eta'_g = (1 + \Delta^\circ - \eta'_g)^4 - \Theta_1^4 \quad (5-187)$$

Equation (5-187) provides an *explicit* relation between the modified furnace efficiency and the effective firing density directly in which the gas temperature is eliminated.

Equation (5-187) has two asymptotic limits

$$\begin{aligned} D_{\text{eff}} &\ll 1 & \Theta_g &\rightarrow \Theta_1 \\ \eta'_g &\rightarrow 1 - \Theta_1 + \Delta^\circ \end{aligned} \quad (5-188a)$$

and

$$\begin{aligned} D_{\text{eff}} &\gg 1 & \Theta_g &\rightarrow 1 + \Delta^\circ & \text{and} & \Theta_e &\rightarrow 1 \\ \eta'_g &\rightarrow \frac{(1 + \Delta^\circ)^4 - \Theta_1^4}{D_{\text{eff}} + 4(1 + \Delta^\circ)^3} \end{aligned} \quad (5-188b)$$

Figure 5-23 is a plot of η'_g versus D_{eff} computed from Eq. (5-187) for the case $\Delta^\circ = 0$.

The asymptotic behavior of Eq. (5-189) mirrors that of the LPFF model. Here, however, for low firing densities, the exit temperature of the furnace exit gases approaches $\Theta_e = \Theta_1 - \Delta^\circ$ rather than the sink temperature. Moreover, for $D_{\text{eff}} \ll 1$ the reduced furnace efficiency adopts the constant value $\eta'_g = 1 - \Theta_e = 1 + \Delta^\circ - \Theta_1$. Again at very high firing rates, only a very small fraction of the available feed enthalpy heat is recovered within the furnace. Thus the exit gas temperature remains nearly unchanged from the pseudoadiabatic flame temperature $[T_e \approx T_f]$ and the gas-side efficiency necessarily approaches zero.

Example 13: WSCC Furnace Model Calculations Consider the furnace geometry and combustion stoichiometry described in Example 12. The end-fired furnace is 3 m wide, 5 m tall, and 10 m long. Methane at a firing rate of 2500 kg/h is burned to completion with 20 percent excess air which is preheated to 600°C. The speckled furnace model is to be used. The sink (zone 1)

occupies 60 percent of the total interior furnace area and is covered with two rows of 5-in (0.127-m) tubes mounted on equilateral centers with a center-to-center distance of twice the tube diameter. The sink temperature is 1000 K, and the tube emissivity is 0.7. Combustion products discharge from a 10-m² duct in the roof which is also tube-screen covered and is to be considered part of the sink. The refractory (zone 2) with emissivity 0.6 is radiatively adiabatic but demonstrates a small convective heat loss to be calculated with an overall heat transfer coefficient U . Compute all unknown furnace temperatures, the gas-side furnace efficiency, and the mean heat flux density through the tube surface. Use the *dimensional* solution approach for the well-stirred combustor model and compare computed results with the *dimensionless* WSCC and LPFF models. Computed values for mean equivalent gas emissivity obtained from Eq. (5-174b) and Table 5-5 for $T_g = 2000$ K for $L_M = 2.7789$ m and $T_1 = 1000$ K are found to be

$$\begin{aligned} T_g = 1500 \text{ K} & & a_m = 0.44181 & & \tau_m = 0.37014 & & \epsilon_m = 0.27828 \\ T_g = 2000 \text{ K} & & a_m = 0.38435 & & \tau_m = 0.41844 & & \epsilon_m = 0.22352 \end{aligned}$$

Over this temperature range the gas emissivity may be calculated by linear interpolation. Additional heat-transfer and thermodynamic data are supplied in context.

Three-zone speckled furnace model, $M = 2, N = 1$:

Zone 1: Sink (60 percent of total furnace area)

Zone 2: Refractory surface (40 percent of total furnace area)

Physical constants:

$$\sigma \equiv 5.670400 \times 10^{-8} \frac{\text{W}}{\text{m}^2 \cdot \text{K}^4}$$

Linear interpolation function for mean effective gas emissivity constants:

$$\text{LINT}(T, Y2, Y1) := \frac{Y1 \cdot (2000 \text{ K} - T) + Y2 \cdot (T - 1500 \text{ K})}{500 \text{ K}} \quad (1500 \text{ K} < T < 2000 \text{ K})$$

Enclosure input parameters:

$$V_{\text{tot}} := 150 \text{ m}^3 \quad A_{\text{tot}} := 190 \text{ m}^2 \quad C_1 := 0.6 \quad C_2 := 1 - C_1$$

$$A_1 := C_1 A_{\text{tot}} \quad A_2 := C_2 A_{\text{tot}} \quad D_{\text{tube}} := 0.127 \text{ m}$$

Direct exchange areas for WSGG clear gas component (temperature independent):

$$F := \begin{pmatrix} C_1 & C_2 \\ C_1 & C_2 \end{pmatrix} \quad AI := \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix} \quad ss1 := AI \cdot F \quad ss1 = \begin{pmatrix} 68.40 & 45.60 \\ 45.60 & 30.40 \end{pmatrix} \text{ m}^2$$

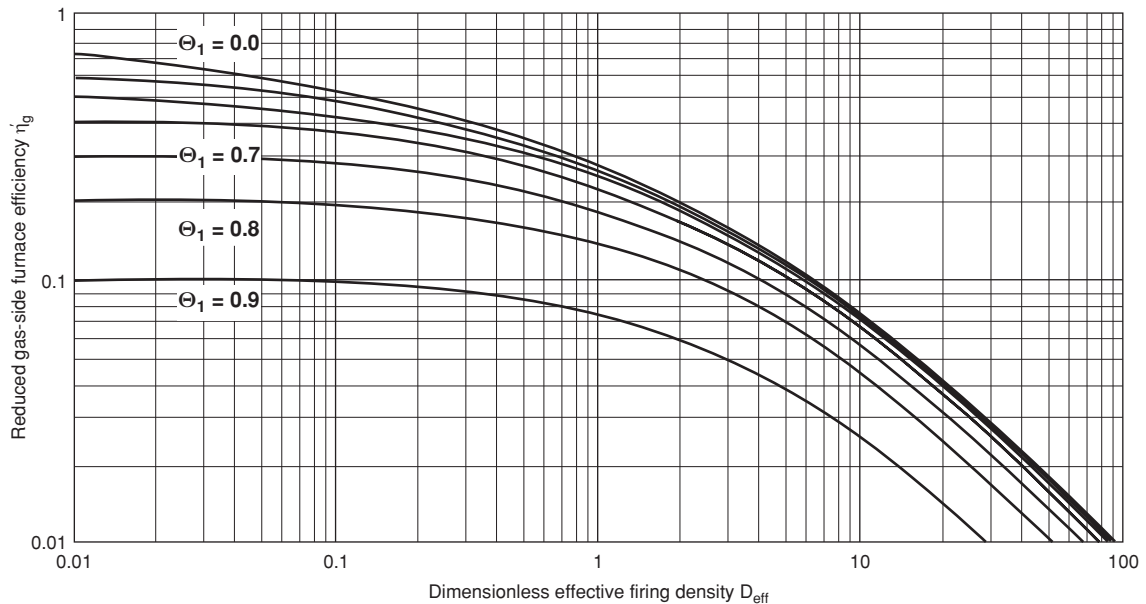


FIG. 5-23 Reduced gas-side furnace efficiency versus effective firing density for well-stirred combustion chamber model. $\Delta^\circ = 0$, $\Theta_1 = 0.0, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9$.

5-42 HEAT AND MASS TRANSFER

Equivalent gray plane emissivity calculations for sink:

$$\epsilon_{\text{tube}} := 0.7 \quad F_{\text{bar}} := 0.987 \quad (\text{from Fig. 5-13, curve 6 with ratio} = 2.0)$$

$$\epsilon_{1\text{eq}} := \frac{1}{\left(\frac{1}{\epsilon_1} - 1\right) + 2 \cdot \frac{D_{\text{tube}}}{2 \cdot \pi \cdot D_{\text{tube}}} \cdot \left(\frac{1}{\epsilon_{\text{tube}}} - 1\right) + \frac{1}{F_{\text{bar}}}} \quad \epsilon_{1\text{eq}} = 0.86988$$

$$\epsilon_1 := \epsilon_{1\text{eq}} \quad \epsilon_2 := 0.6 \quad \epsilon I := \begin{pmatrix} \epsilon_1 & 0 \\ 0 & \epsilon_2 \end{pmatrix} \quad \rho I := \text{identity}(2) - \epsilon I$$

Total exchange areas for WSGG clear gas component:

$$R1 := (AI - ss1 \cdot \rho I)^{-1} \quad SS1 := \epsilon I \cdot AI \cdot R1 \cdot ss1 \cdot \epsilon I \quad SS1 = \begin{pmatrix} 67.93 & 31.24 \\ 31.24 & 14.36 \end{pmatrix} \text{ m}^2$$

Temperature and emissive power input data:

$$T_1 := 1000.0 \text{ K} \quad T_{\text{air}} := 873.15 \text{ K}$$

$$T_0 := 298.15 \text{ K} \quad \Delta T_{\text{gr}} := 170 \text{ K} \quad E_1 := \sigma \cdot T_1^4 \quad E_1 = 56.704 \frac{\text{kJ}}{\text{m}^2}$$

Mean beam length calculations:

$$L_{M0} := 4 \cdot \frac{V_{\text{tot}}}{A_{\text{tot}}} \quad L_{M0} = 3.1579 \text{ m} \quad L_M := 0.88 L_{M0} \quad L_M = 2.7789 \text{ m}$$

Stoichiometric and thermodynamic input data:

$$\text{MCH4}_{\text{dot}} := 2500 \frac{\text{kg}}{\text{h}} \quad \text{MW}_{\text{CH4}} := 16.04 \frac{\text{g}}{\text{mol}} \quad \text{MW}_{\text{in}} := 27.742 \frac{\text{g}}{\text{mol}}$$

$$\Sigma \text{Mols} := 12.52914 \quad \text{LHV} := 191760 \frac{\text{cal}}{\text{mol}} \quad \text{MCP}_{\text{air}} := 7.31 \frac{\text{cal}}{\text{mol} \cdot \text{K}}$$

$$\text{MCP}(T) := \left[7.01 + 0.875 \left(\frac{T}{1000 \text{ K}} \right) \right] \frac{\text{cal}}{\text{mol} \cdot \text{K}} \quad (800 \text{ K} < T < 1200 \text{ K})$$

M_{dot} is the total mass flowrate and NCH4_{dot} is the molal flowrate of CH_4 .

$$M_{\text{dot}} := \frac{\text{MCH4}_{\text{dot}} \cdot \text{MW}_{\text{in}} \cdot \Sigma \text{Mols}}{\text{MW}_{\text{CH4}}} \quad \text{NCH4}_{\text{dot}} := \frac{\text{MCH4}_{\text{dot}}}{\text{MW}_{\text{CH4}}}$$

$$M_{\text{dot}} = 54174.5 \frac{\text{kg}}{\text{h}} \quad \text{NCH4}_{\text{dot}} = 155860.3 \frac{\text{mol}}{\text{h}}$$

Overall refractory heat-transfer coefficient:

$$D_r := 0.343 \text{ m} \quad k_r := 0.00050 \frac{\text{kJ}}{\text{m} \cdot \text{K}}$$

$$h_0 := 0.0114 \cdot \frac{\text{kJ}}{\text{m}^2 \cdot \text{K}} \quad h_0 = 2.0077 \frac{\text{Btu}}{\text{h} \cdot \text{ft}^2 \cdot \text{R}} \quad h_1 := 0.0170 \frac{\text{kJ}}{\text{m}^2 \cdot \text{K}}$$

$$h_1 = 2.9939 \frac{\text{Btu}}{\text{h} \cdot \text{ft}^2 \cdot \text{R}} \quad U := \frac{1}{\frac{1}{h_0} + \frac{1}{h_1} + \frac{D_r}{k_r}}$$

$$U = 0.001201 \frac{\text{kJ}}{\text{m}^2 \cdot \text{K}} \quad U = 0.2115 \frac{\text{Btu}}{\text{h} \cdot \text{ft}^2 \cdot \text{R}}$$

START OF ITERATION LOOP: Successive Substitution with T_g as the Trial Variable

$$\text{Assume } T_g := 1759.163322447 \text{ K} \quad T_r := T_g - \Delta T_{\text{gr}} \quad T_e = 1589.2 \text{ K}$$

$$E_g := \sigma \cdot T_g^4 \quad E_g = 543.05 \frac{\text{kJ}}{\text{m}^2}$$

Compute temperature-dependent mean effective gas emissivity via linear interpolation:

$$\tau_m := \text{LINTF}(T_g, 0.37014, 0.418442) \quad a_m := \text{LINTF}(T_g, 0.44181, 0.38435)$$

$$\tau_m = 0.3934 \quad a_m = 0.4141 \quad \epsilon_m := a_m \cdot (1 - \tau_m) \quad \epsilon_m = 0.2512$$

Compare interpolated value:

$$\epsilon_{\text{com}} := \text{LINTF}(T_g, 0.27828, 0.22352) \quad \epsilon_{\text{com}} = 0.2519$$

Direct and total exchange areas for WSGG gray gas component:

$$ss2 := \tau_m \cdot ss1 \quad sg2 := (AI - ss2) \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad R2 := (AI - ss2 \cdot \rho I)^{-1}$$

$$ss2 = \begin{pmatrix} 26.91 & 17.94 \\ 17.94 & 11.96 \end{pmatrix} \text{ m}^2 \quad sg2 = \begin{pmatrix} 69.15 \\ 46.10 \end{pmatrix} \text{ m}^2 \quad SS2 := \epsilon I \cdot AI \cdot R2 \cdot ss2 \cdot \epsilon I$$

Compute directed exchange areas:

$$DSS := (1 - a_m) \cdot SS1 + a_m \cdot SS2 \quad DSG := a_m \cdot \epsilon I \cdot AI \cdot R2 \cdot sg2$$

Refractory augmented directed gas-sink exchange area:

$$DS1GR := DSG_1 + \frac{DSS_{1,2} \cdot DSG_2}{DSS_{1,2} + DSG_2} \quad DS1GR = 35.59 \text{ m}^2 \quad \text{Eq. (5-172a)}$$

Compute refractory temperature (T_2): assume radiative equilibrium:

$$\text{Equation (5-172b): } E_2 := \frac{DSS_{2,1} \cdot E_1 + DSG_2 \cdot E_g}{DSS_{2,1} + DSG_2}$$

$$E_2 = 231.24 \frac{\text{kJ}}{\text{m}^2}$$

$$T_2 := \left(\frac{E_2}{\sigma} \right)^{0.25} \quad T_2 = 1421.1 \text{ K}$$

Enthalpy balance: Basis: 1 mole CH_4 :

$$h_{\text{in}} := \text{LHV} + (\Sigma \text{Mols} - 1) \cdot \text{MCP}_{\text{air}} \cdot (T_{\text{air}} - T_0) \quad h_{\text{in}} = 240219.9 \frac{\text{cal}}{\text{mol}}$$

Compute pseudoadiabatic flame temperature T_f :

$$T_f := T_0 + \frac{h_{\text{in}}}{\Sigma \text{Mols} \cdot \text{MCP}(T_e)} \quad h_{\text{out}} := \Sigma \text{Mols} \cdot \text{MCP}(T_e) \cdot (T_e - T_0)$$

$$T_f = 2580.5 \text{ K} \quad h_{\text{out}} = 135880.8 \frac{\text{cal}}{\text{mol}}$$

$$H_{\text{in}} := \text{NCH4}_{\text{dot}} \cdot h_{\text{in}} \quad H_{\text{out}} := \text{NCH4}_{\text{dot}} \cdot h_{\text{out}} \quad \Delta H := H_{\text{out}} - H_{\text{in}}$$

$$H_{\text{in}} = 43543.59 \text{ kW} \quad H_{\text{out}} = 24630.51 \text{ kW} \quad \Delta H = -18913.08 \text{ kW}$$

Overall enthalpy balance:

$$Q1_g := DS1GR \cdot (E_g - E_1) \quad Q1_g = 17308.45 \text{ kW}$$

$$Q1_{\text{com}} := h_1 \cdot A_1 \cdot (T_g - T_1) \quad Q1_{\text{com}} = 1471.26 \text{ kW}$$

$$Q2_{\text{com}} := U \cdot A_2 \cdot (T_g - T_0) \quad Q2_{\text{com}} = 133.37 \text{ kW}$$

$$\text{ERROR} := Q1_g + Q1_{\text{com}} + Q2_{\text{com}} + \Delta H \quad \% \text{ERROR1} := 100 \frac{\text{ERROR}}{\Delta H}$$

$$\text{ERROR} = -0.0000 \text{ kW} \quad \% \text{ERROR1} = 0.00000$$

$$T_{\text{gcalc}} := \left[\frac{DS1GR \cdot E_1 - (\Delta H + Q1_{\text{com}} + Q2_{\text{com}})}{DS1GR \cdot \sigma} \right]^{0.25} \quad T_{\text{gcalc}} = 1759.16332 \text{ K}$$

Average assumed and calculated temperatures for next iteration

$$T_{\text{gnew}} := \frac{T_g + T_{\text{gcalc}}}{2} \quad T_{\text{gnew}} = 1759.1633222935 \text{ K} \quad \text{Go to Start}$$

END OF ITERATION LOOP: Final Gas Temperature $T_g = 1759.16 \text{ K}$

$$\eta_g := \frac{H_{\text{in}} - H_{\text{out}}}{H_{\text{in}}} \quad \text{or} \quad \eta1_g := \frac{T_f - T_e}{T_f - T_0} \quad \eta_g = 0.43435 \quad \eta1_g = 0.43435$$

Heat flux density calculations:

$$q_1 := \frac{Q1_g + Q1_{\text{com}}}{A_1} \quad q_{\text{tube}} := q_1 \cdot \left(2 \cdot \frac{D_{\text{tube}}}{2 \cdot \pi \cdot D_{\text{tube}}} \right)$$

$$q_1 = 164.7 \frac{\text{kJ}}{\text{m}^2} \quad q_{\text{tube}} = 52.44 \frac{\text{kJ}}{\text{m}^2}$$

Note: This example was also solved with $\Delta T_{\text{gr}} = 0$. The results were as follows: $T_f = 2552.8 \text{ K}$, $T_e = T_r = 1707.1 \text{ K}$, $T_2 = 1381.1 \text{ K}$, $\eta_g = 37.51$ percent, $D_{\text{eff}} = 0.53371$, and $\Delta H = 16332.7 \text{ kW}$. The WSCC model with $\Delta T_{\text{gr}} = 0$ predicts a lower performance bound.

Compare dimensionless WSCC model:

$$\Theta_0 := \frac{T_0}{T_f} \quad \Theta_1 := \frac{T_1}{T_f} \quad \Delta T_{\text{star}} := \frac{\Delta T_{\text{gr}}}{T_f}$$

$$T_{g1} := \frac{T_1 + T_g}{2}$$

$$\Theta_0 = 0.1155 \quad \Theta_1 = 0.3875 \quad \Delta T_{\text{star}} = 0.06588 \quad T_{g1} = 1379.58 \text{ K}$$

$$C_{p\text{Prod}} := \frac{MCp(T_c)}{MW_{\text{in}}} \quad C_{p\text{Prod}} = 0.000352 \text{ kW} \cdot \frac{\text{h}}{\text{kg} \cdot \text{K}}$$

$$N_{\text{FD}} := M_{\text{dot}} \frac{C_{p\text{Prod}}}{\sigma T_{g1}^3 A_1} \quad N_{\text{CR}} := \frac{h_1}{4 \cdot \sigma \cdot T_{g1}^3} \quad D_{\text{eff}} := \frac{N_{\text{FD}}}{\frac{DS_{\text{IGR}}}{A_1} + N_{\text{CR}}}$$

$$N_{\text{FD}} = 0.17176 \quad N_{\text{CR}} = 0.02855 \quad D_{\text{eff}} = 0.50409 \quad \frac{DS_{\text{IGR}}}{A_1} = 0.31218$$

$$\eta_{\text{prime}} := \eta_g \cdot (1 - \Theta_0) \quad D1_{\text{eff}} := \frac{(1 + \Delta T_{\text{star}} - \eta_{\text{prime}})^4 - \Theta_1^4}{\eta_{\text{prime}}}$$

$$D1_{\text{eff}} = 0.50350 \quad \text{versus} \quad D_{\text{eff}} = 0.50409$$

This small discrepancy is due to linearization and neglect of convective refractory heat losses in the dimensionless WSCC model.

Compare dimensionless LPFF model:

$$R_{\text{in}} := \frac{1}{\Theta_1} \quad R_{\text{in}} = 2.58050$$

Trial and error calculation to match effective firing densities:

Assume:

$$T_{\text{out}} := 1000.13763 \text{ K} \quad R_{\text{out}} := \frac{T_{\text{out}}}{T_1} \quad R_{\text{out}} = 1.00014 \quad \frac{4}{D_{\text{eff}}} = 7.93513$$

$$C_{\text{Long}} := -\ln \left[\frac{(R_{\text{out}} - 1) \cdot (R_{\text{in}} + 1)}{(R_{\text{in}} - 1) \cdot (R_{\text{out}} + 1)} \right] - 2 \cdot \text{atan} \left(\frac{R_{\text{in}} - R_{\text{out}}}{1 + R_{\text{in}} \cdot R_{\text{out}}} \right) \quad C_{\text{Long}} = 7.93514$$

$$\eta_{L_f} := \frac{R_{\text{in}} - R_{\text{out}}}{R_{\text{in}} - 1} \quad \eta_{L_f} = 0.99991292 \quad \text{versus} \quad \eta_g = 0.43435$$

Note: The long plug flow furnace Model is so efficient that it would be grossly underfired using the computed WSCC effective firing density. Of the two models, the LPFF model always predicts an upper theoretical performance limit.

(This example was developed as a MATHCAD 14® worksheet. Mathcad is registered trademark of Parametric Technology Corporation.)

WSCC Model Utility and More Complex Zoning Models

Despite its simplicity, the WSCC construct has a wide variety of practical uses and is of significant pedagogical value. Here an engineering situation of inordinate complexity is described by the definition of only eight dimensionless quantities D_{eff} , N_{FD} , $\hat{S}_1 G_R / A_1$, N_{CR} , η_g , Δ^* , Θ_0 , and Θ_1 . The first three are related by the simple algebraic definition $D_{\text{eff}} = N_{\text{FD}} / (\hat{S}_1 G_R / A_1 + N_{\text{CR}})$. These dimensionless quantities con-

tain all the physical input information for the model, namely, furnace parameters and geometry, radiative properties of the combustion products, and the stoichiometry and thermodynamics of the combustion process. The WSCC model leads to a dimensionless two-dimensional plot of *reduced* effective furnace efficiency versus dimensionless effective firing density (Fig. 5-23), which is characterized by only two additional parameters, namely, Δ^* and Θ_1 .

Of the models presented here, the WSCC model with $\Delta T_{\text{gr}} = 0$ produces the lowest furnace efficiencies. The long furnace model usually produces the highest furnace efficiency. This is really not a fair statement because two distinctly different pieces of process equipment are compared. In this regard, a more appropriate definition of the dimensionless firing density for the LPFF model might be $N'_{\text{FD}} = \dot{m} \bar{C}_p / (\sigma \cdot T_{g,\text{in}}^3 \cdot A_1)$. It may be counterintuitive, but the WSCC and LPFF models *generally do not* characterize the extreme conditions for the performance of combustors as in the case of chemical reactors.

Figure 5-23 has been used to correlate furnace performance data for a multitude of industrial furnaces and combustors. Typical operational domains for a variety of fuel-fired industrial furnaces are summarized in Table 5-7. The WSCC approach (or "speckled" furnace model) is a classic contribution to furnace design methodology which was first due to Hottel [op. cit.]. The WSCC model provides a simple *furnace design template* which leads to a host of more complex furnace models. These models include an obvious extension to a tanks-in-series model as well as multizone models utilizing empirical cold-flow velocity patterns. For more information on practical furnace design models, reference is made to Hottel and Sarofim (op. cit., Chap. 14). Qualitative aspects of process equipment have been treated in some detail elsewhere (Baukal, C. E., ed., *The John Zink Combustion Handbook*, CRC Press, Boca Raton, Fla., 2001).

TABLE 5-7 Operational Domains for Representative Process Furnaces and Combustors

Domain	Furnace or combustor type	Dimensionless sink temperature	Dimensionless firing density
A	Oil processing furnaces; radiant section of oil tube stills and cracking coils	$\Theta_1 \approx 0.4$	$0.1 < D_{\text{eff}} < 1.0$
B	Domestic boiler combustion chambers	$\Theta_1 \approx 0.2$	$0.5 < D_{\text{eff}} < 1.1$
C	Glass furnaces	$0.7 < \Theta_1 < 0.8$	$0.035 < D_{\text{eff}} < 0.8$
D	Soaking pits	$\Theta_1 \approx 0.6$	$0.7 < D_{\text{eff}} < 1.1$
E	Gas-turbine combustors	$0.4 < \Theta_1 < 0.7$	$4.0 < D_{\text{eff}} < 25.0$

MASS TRANSFER

GENERAL REFERENCES: Bird, Stewart, and Lightfoot, *Transport Phenomena*, 2d ed., Wiley, New York, 2002. Cussler, *Diffusion: Mass Transfer in Fluid Systems*, 2d ed., Cambridge University Press, Cambridge, 1997. Danner and Daubert, *Manual for Predicting Chemical Process Design Data*, AIChE, New York, 1983. Daubert and Danner, *Physical and Thermodynamic Properties of Pure Chemicals*, Taylor and Francis, Bristol, Pa., 1989–1995. Gammon, Marsh, and Dewan, *Transport Properties and Related Thermodynamic Data of Binary Mixtures*, AIChE, New York. Part 1, 1993; Part 2, 1994. Geankoplis, *Transport Processes and Separation Process Principles*, 5th ed., Prentice-Hall PTR, Upper Saddle River, N.J., 2003. Kirwan, "Mass Transfer Principles," Chap. 2 in Rousseau, R. W. (ed.), *Handbook of Separation Process Technology*, Wiley, New York, 1987. McCabe, Smith, and Harriott, *Unit Operations of Chemical Engineering*, 7th ed., McGraw-Hill, New York, 2006. Middleman, *An Introduction to Mass and Heat Transfer*, Wiley, New York, 1997. Poling, Prausnitz, and O'Connell, *The Properties of Gases and Liquids*, 5th ed., McGraw-Hill, New York, 2001. Schwartzberg and Chao, *Food Technol.*, **36**(2), 73 (1982). Sherwood, Pigford, and Wilke, *Mass Transfer*, McGraw-Hill, New York, 1975. Skelland, *Diffusional Mass Transfer*, Wiley, New York, 1974. Taylor and Krishna, *Multicomponent Mass Transfer*, Wiley, New York, 1993. Thompson, *Introduction to Transport Phenomena*, Prentice-Hall PTR, Upper Saddle River, N.J., 1999. Treybal, *Mass-Transfer Operations*, 3d ed., McGraw-Hill, New York, 1980.

REFERENCES FOR DIFFUSIVITIES, TABLES 5-10, 5-13 TO 5-15

- Asfour and Dullien, *Chem. Eng. Sci.*, **41**, 1891 (1986).
- Blanc, *J. Phys.*, **7**, 825 (1908).
- Bosse and Bart, *Ind. Eng. Chem. Res.*, **45**, 1822 (2006).
- Brokaw, *Ind. Eng. Chem. Process Des. and Dev.*, **8**, 2, 240 (1969).
- Caldwell and Babb, *J. Phys. Chem.*, **60**, 51 (1956).
- Catchpole and King, *Ind. Eng. Chem. Res.*, **33**, 1828 (1994).
- Chen and Chen, *Chem. Eng. Sci.*, **40**, 1735 (1985).
- Cullinan, *AIChE J.*, **31**, 1740–1741 (1985).
- Cussler, *AIChE J.*, **26**, 1 (1980).
- Fuller, Schettler, and Giddings, *Ind. Eng. Chem.*, **58**, 18 (1966).
- Hayduk and Laudie, *AIChE J.*, **20**, 3, 611 (1974).
- Hayduk and Minhas, *Can. J. Chem. Eng.*, **60**, 195 (1982).
- He and Yu, *Ind. Eng. Chem. Res.*, **36**, 4430 (1997).
- Lee and Thodos, *Ind. Eng. Chem. Fundam.*, **22**, 17–26 (1983).
- Lee and Thodos, *Ind. Eng. Chem. Res.*, **27**, 992–997 (1988).
- Leffler and Cullinan, *Ind. Eng. Chem. Fundam.*, **9**, 84, 88 (1970).
- Liu and Ruckenstein, *Ind. Eng. Chem. Res.*, **36**, 888 (1997) and corrections **37**, 3524 (1998).
- Mathur and Thodos, *AIChE J.*, **11**, 613 (1965).
- Matthews and Akgerman, *AIChE J.*, **33**, 881 (1987).

20. Rathbun and Babb, *Ind. Eng. Chem. Proc. Des. Dev.*, **5**, 273 (1966).
 21. Riazi and Whitson, *Ind. Eng. Chem. Res.*, **32**, 3081 (1993).
 22. Siddiqi and Lucas, *Can. J. Chem. Eng.*, **64**, 839 (1986).
 23. Smith and Taylor, *Ind. Eng. Chem. Fundam.*, **22**, 97 (1983).
 24. Sridhar and Potter, *AIChE J.*, **23**, 4, 590 (1977).
 25. Sun and Chen, *Ind. Eng. Chem. Res.*, **26**, 815 (1987).
 26. Tyn and Calus, *J. Chem. Eng. Data*, **20**, 310 (1975).
 27. Umesi and Danner, *Ind. Eng. Chem. Process Des. Dev.*, **20**, 662 (1981).
 28. Vignes, *Ind. Eng. Chem. Fundam.*, **5**, 184 (1966).
 29. Wilke, *Chem. Eng. Prog.*, **46**, 2, 95 (1950).
 30. Wilke and Chang, *AIChE J.*, **1**, 164 (1955).
 31. Wilke and Lee, *Ind. Eng. Chem.*, **47**, 1253 (1955).
- REFERENCES FOR DIFFUSIVITIES IN POROUS SOLIDS, TABLE 5-16**
32. Ruthven, *Principles of Adsorption and Adsorption Processes*, Wiley, 1984.
 33. Satterfield, *Mass Transfer in Heterogeneous Catalysis*, MIT Press, 1970.
 34. Suzuki, *Adsorption Engineering*, Kodansha—Elsevier, 1990.
 35. Yang, *Gas Separation by Adsorption Processes*, Butterworths, 1987.
- REFERENCES FOR TABLES 5-17 TO 5-24**
36. Bahmanyar, et al., *Chem. Eng. Res. Des.*, **68**, 74 (1990).
 37. Baier et al., *Chem. Eng. Sci.*, **54**, 343 (1999).
 38. Beenackers and van Swaaij, *Chem. Eng. Sci.*, **48**, 3109 (1993).
 39. Bird, Stewart, and Lightfoot, *Transport Phenomena*, Wiley, 1960.
 40. Blatt et al. (eds.), (ed.), *Membrane Science and Technology*, **47**, Plenum, 1970.
 41. Bocquet et al., *AIChE J.*, **51**, 1067 (2005).
 42. Bolles and Fair, *Institution Chem. Eng. Symp. Ser.*, **56**, 3/35 (1979).
 43. Bolles and Fair, *Chem. Eng.*, **89**(14), 109 (July 12, 1982).
 44. Bravo and Fair, *Ind. Eng. Chem. Process Des. Dev.*, **21**, 162 (1982).
 45. Bravo, Rocha and Fair, *Hydrocarbon Processing*, 91 (Jan. 1985).
 46. Brian and Hales, *AIChE J.*, **15**, 419 (1969).
 47. Calderbank and Moo-Young, *Chem. Eng. Sci.*, **16**, 39 (1961).
 48. Cavatorta, Bohm, and Chiappori de del Giorgio, *AIChE J.*, **45**, 938 (1999).
 49. Chaumat et al., *Chem. Eng. Sci.*, **60**, 5930 (2005).
 50. Chen, Lin, and Liu, *Ind. Eng. Chem. Res.*, **44**, 7868 (2005).
 51. Chilton and Colburn, *Ind. Eng. Chem.*, **26**, 1183 (1934).
 52. Chowdhia, Foutch, and Lee, *Ind. Eng. Chem. Res.*, **42**, 1485 (2003).
 53. Colburn, *Trans. AIChE*, **29**, 174 (1933).
 54. Cornell, Knapp, and Fair, *Chem. Eng. Prog.*, **56**(7), 68 (1960).
 55. Cornet and Kaloo, *Proc. 3rd Int'l. Congr. Metallic Corrosion—Moscow*, **3**, 83 (1966).
 56. Crause and Nieuwoudt, *Ind. Eng. Chem. Res.*, **38**, 4928 (1999).
 57. Deckwer et al., *Biotech. Bioeng.* (1981).
 58. Dudukovic, Milosevic, and Pjanovic, *AIChE J.*, **42**, 269 (1996).
 59. Dwivedi and Upadhyay, *Ind. Eng. Chem. Process Des. Develop.*, **16**, 1657 (1977).
 60. Eisenberg, Tobias, and Wilke, *Chem. Eng. Prog. Symp. Sec.*, **51**(16), 1 (1955).
 61. El-Shazly et al., *Ind. Eng. Chem. Res.*, **41**, 5516 (2002).
 62. Elzinga and Banchero, *Chem. Eng. Progr. Symp. Ser.*, **55**(29), 149 (1959).
 63. Fair, "Distillation" in Rousseau (ed.), *Handbook of Separation Process Technology*, Wiley, 1987.
 64. Fan, Yang, and Wen, *AIChE J.*, **6**, 482 (1960).
 65. Garner and Suckling, *AIChE J.*, **4**, 114 (1958).
 66. Geankoplis, *Transport Processes and Unit Operations*, 4th ed., Prentice Hall, 2003.
 67. Gibilaro et al., *Chem. Eng. Sci.*, **40**, 1811 (1985).
 68. Gilliland and Sherwood, *Ind. Eng. Chem.*, **26**, 516 (1934).
 69. Costick et al., *Ind. Eng. Chem. Res.*, **42**, 3626 (2003).
 70. Griffith, *Chem. Eng. Sci.*, **12**, 198 (1960).
 71. Guedes de Carvalho and Alves, *AIChE J.*, **45**, 2495 (1999).
 72. Gupta and Thodos, *AIChE J.*, **9**, 751 (1963).
 73. Gupta and Thodos, *Ind. Eng. Chem. Fundam.*, **3**, 218 (1964).
 74. Harriott, *AIChE J.*, **8**, 93 (1962).
 75. Hassan et al., *Ind. Eng. Chem. Res.*, **44**, 5761 (2005).
 76. Heertjes, Holve, and Talsma, *Chem. Eng. Sci.*, **3**, 122 (1954).
 77. Hines and Maddox, *Mass Transfer: Fundamentals and Applications*, Prentice-Hall, 1985.
 78. Hoffer et al., *Chem. Eng. Sci.*, **59**, 259 (2004).
 79. Houwing, Billiet, and van der Wielen, *AIChE J.*, **49**, 1158 (2003).
 80. Hsiung and Thodos, *Int. J. Heat Mass Transfer*, **20**, 331 (1977).
 81. Hsu, Sato, and Sage, *Ind. Eng. Chem.*, **46**, 870 (1954).
 82. Hughmark, *Ind. Eng. Chem. Fundam.*, **6**, 408 (1967).
 83. Johnson, Besic, and Hamielec, *Can. J. Chem. Eng.*, **47**, 559 (1969).
 84. Johnstone and Pigford, *Trans. AIChE*, **38**, 25 (1942).
 85. Kafesjian, Plank, and Gerhard, *AIChE J.*, **7**, 463 (1961).
 86. Kelly and Swenson, *Chem. Eng. Prog.*, **52**, 263 (1956).
 87. King, *Separation Processes*, 2d ed., McGraw-Hill (1980).
 88. Kirwan, "Mass Transfer Principles" in Rousseau, *Handbook of Separation Process Technology*, Wiley, 1987.
 89. Klein, Ward, and Lacey, "Membrane Processes—Dialysis and Electro-Dialysis" in Rousseau, *Handbook of Separation Process Technology*, Wiley, 1987.
 90. Kohl, "Absorption and Stripping" in Rousseau, *Handbook of Separation Process Technology*, Wiley, 1987.
 91. Kojima et al., *J. Chem. Engng. Japan*, **20**, 104 (1987).
 92. Koloini, Sopcić, and Zumer, *Chem. Eng. Sci.*, **32**, 637 (1977).
 93. Kreutzer et al., *Ind. Eng. Chem. Res.*, **44**, 9646 (2005).
 94. Lancia, Musmarra, and Pepe, *Ind. Eng. Chem. Res.*, **36**, 3859 (1997).
 95. Larachi et al., *Ind. Eng. Chem. Res.*, **42**, 222 (2003).
 96. Lau et al., *Ind. Eng. Chem. Res.*, **43**, 1302 (2004).
 97. Lee, *Biochemical Engineering*, Prentice-Hall, 1992.
 98. Lee and Foster, *Appl. Catal.*, **63**, 1 (1990).
 99. Lee and Holder, *Ind. Eng. Chem. Res.*, **34**, 906 (1995).
 100. Lee and Lueptow, *Separ. Sci. Technol.*, **39**, 539 (2004).
 101. Levich, *Physicochemical Hydrodynamics*, Prentice-Hall, 1962.
 102. Levins and Gastonbury, *Trans. Inst. Chem. Eng.*, **50**, 32, 132 (1972).
 103. Linton and Sherwood, *Chem. Eng. Prog.*, **46**, 258 (1950).
 104. Ludwig, *Applied Process Design for Chemical and Petrochemical Plants*, 2d ed., vol. 2, Gulf Pub. Co., 1977.
 105. McCabe, Smith, and Harriott, *Unit Operations of Chemical Engineering*, 7th ed., McGraw-Hill, 2005.
 106. Nelson and Galloway, *Chem. Eng. Sci.*, **30**, 7 (1975).
 107. Notter and Schleicher, *Chem. Eng. Sci.*, **26**, 161 (1971).
 108. Ohashi et al., *J. Chem. Eng. Japan*, **14**, 433 (1981).
 109. Onda, Takeuchi, and Okumoto, *J. Chem. Eng. Japan*, **1**, 56 (1968).
 110. Pangarkar et al., *Ind. Eng. Chem. Res.*, **41**, 4141 (2002).
 111. Pasternak and Gauvin, *AIChE J.*, **7**, 254 (1961).
 112. Pasternak and Gauvin, *Can. J. Chem. Eng.*, **38**, 35 (April 1960).
 113. Patil, Deshmukh, and Joshi, *Ind. Eng. Chem. Res.*, **43**, 2765 (2004).
 114. Pekdemir, Davies, and Sara, *Ind. Eng. Chem. Res.*, **37**, 1560 (1998).
 115. Perez and Sandall, *AIChE J.*, **20**, 770 (1974).
 116. Petrovic and Thodos, *Ind. Eng. Chem. Fundam.*, **7**, 274 (1968).
 117. Pinczewski and Sideman, *Chem. Eng. Sci.*, **29**, 1969 (1974).
 118. Prasad and Sirkar, *AIChE J.*, **34**, 177 (1988).
 119. Rahman and Streat, *Chem. Eng. Sci.*, **36**, 293 (1981).
 120. Ramirez and Davis, *AIChE J.*, **45**, 1355 (1999).
 121. Ranz and Marshall, *Chem. Eng. Prog.*, **48**, 141, 173 (1952).
 122. Reiss, *Ind. Eng. Chem. Process Des. Develop.*, **6**, 486 (1967).
 123. Riet, *Ind. Eng. Chem. Process Des. Dev.*, **18**, 357 (1979).
 124. Rocha, Bravo, and Fair, *Ind. Eng. Chem. Res.*, **35**, 1660 (1996).
 125. Rowe, *Chem. Eng. Sci.*, **30**, 7 (1975).
 126. Rowe, Claxton, and Lewis, *Trans. Inst. Chem. Eng. London*, **43**, 14 (1965).
 127. Ruckenstein and Rajagopalan, *Chem. Eng. Commun.*, **4**, 15 (1980).
 128. Ruthven, *Principles of Adsorption and Adsorption Processes*, Wiley, 1984.
 129. Sanger and Deckwer, *Chem. Eng. J.*, **22**, 179 (1981).
 130. Satterfield, *AIChE J.*, **21**, 209 (1975).
 131. Schluter and Deckwer, *Chem. Eng. Sci.*, **47**, 2357 (1992).
 132. Schmitz, Steiff, and Weinspach, *Chem. Engng. Technol.*, **10**, 204 (1987).
 133. Schugert et al., *Adv. Biochem. Eng.*, **8**, 63 (1978).
 134. Scott and Lobato, *Ind. Eng. Chem. Res.*, **42**, 5697 (2003).
 135. Sedahmed, Zatout, and Zewail, *Ind. Eng. Chem. Res.*, **37**, 3481 (1998).
 136. Shah et al., *AIChE J.*, **28**, 353 (1982).
 137. Sherwood et al., *Ind. Eng. Chem. Fundam.*, **4**, 113 (1965).
 138. Sherwood, Pigford, and Wilke, *Mass Transfer*, McGraw-Hill, 1975.
 139. Siegel, Sparrow, and Hallman, *Appl. Sci. Res. Sec. A*, **7**, 386 (1958).
 140. Sissom and Pitts, *Elements of Transport Phenomena*, McGraw-Hill, 1972.
 141. Skelland, *Diffusional Mass Transfer*, Wiley (1974).
 142. Skelland and Cornish, *AIChE J.*, **9**, 73 (1963).
 143. Skelland and Moeti, *Ind. Eng. Chem. Res.*, **29**, 2258 (1990).
 144. Skelland and Tedder, "Extraction—Organic Chemicals Processing" in Rousseau, *Handbook of Separation Process Technology*, Wiley, 1987, pp. 405–466.
 145. Skelland and Wellek, *AIChE J.*, **10**, 491, 789 (1964).
 146. Slater, "Rate Coefficients in Liquid-Liquid Extraction Systems" in Godfrey and Slater, *Liquid-Liquid Extraction Equipment*, Wiley, 1994, pp. 45–94.
 147. Steinberger and Treybal, *AIChE J.*, **6**, 227 (1960).
 148. Taniguchi and Kimura, *AIChE J.*, **47**, 1967 (2000).
 149. Taylor and Krishna, *Multicomponent Mass Transfer*, Wiley, 1993.
 150. Tokarz, Millies, and Mewes, *AIChE J.*, **47**, 799 (2001).
 151. Tournie, Laguerie, and Couderc, *Chem. Eng. Sci.*, **34**, 1247 (1979).
 152. Treybal, *Mass Transfer Operations*, 3d ed., McGraw-Hill, 1980.

153. Vandu, Liu, and Krishna, *Chem. Eng. Sci.*, **60**, 6430 (2005).
154. Von Karman, *Trans. ASME*, **61**, 705 (1939).
155. Wakao and Funazkri, *Chem. Eng. Sci.*, **33**, 1375 (1978).
156. Wang, Yuan, and Yu, *Ind. Eng. Chem. Res.*, **44**, 8715 (2005).
157. Wankat, *Separation Process Engineering*, 2d ed., Prentice-Hall, 2007.
158. Wilson and Geankoplis, *Ind. Eng. Chem. Fundam.*, **5**, 9 (1966).
159. Wright and Glasser, *AIChE J.*, **47**, 474 (2001).
160. Yagi and Yoshida, *Ind. Eng. Chem. Process Des. Dev.*, **14**, 488 (1975).
161. Yapici and Ozbahar, *Ind. Eng. Chem. Res.*, **37**, 643 (1998).
162. Zaki, Nirdosh, and Sedahmed, *Ind. Eng. Chem. Res.*, **41**, 3307 (2002).

INTRODUCTION

This part of Sec. 5 provides a concise guide to solving problems in situations commonly encountered by chemical engineers. It deals with diffusivity and mass-transfer coefficient estimation and common flux equations, although material balances are also presented in typical coordinate systems to permit a wide range of problems to be formulated and solved.

Mass-transfer calculations involve transport properties, such as diffusivities, and other empirical factors that have been found to relate mass-transfer rates to measured "driving forces" in myriad geometries and conditions. The context of the problem dictates whether the fundamental or more applied coefficient should be used. One key distinction is that, whenever there is flow parallel to an interface through which mass transfer occurs, the relevant coefficient is an empirical combination of properties and conditions. Conversely, when diffusion occurs in stagnant media or in creeping flow without transverse velocity gradients, ordinary diffusivities may be suitable for solving the problem. In either case, it is strongly suggested to employ data, whenever available, instead of relying on correlations.

Units employed in diffusivity correlations commonly followed the cgs system. Similarly, correlations for mass transfer correlations used the cgs or English system. In both cases, only the most recent correlations employ SI units. Since most correlations involve other properties and physical parameters, often with mixed units, they are repeated here as originally stated. Common conversion factors are listed in Table 1-4.

Fick's First Law This law relates flux of a component to its composition gradient, employing a constant of proportionality called a diffusivity. It can be written in several forms, depending on the units and frame of reference. Three that are related but not identical are

$$vJ_A = -D_{AB} \frac{dc_A}{dz} \approx mJ_A = -cD_{AB} \frac{dx_A}{dz} \approx mJ_A = -\rho D_{AB} \frac{dw_A}{dz} \quad (5-189)$$

The first equality (on the left-hand side) corresponds to the molar flux with respect to the volume average velocity, while the equality in the center represents the molar flux with respect to the molar average velocity and the one on the right is the mass flux with respect to the mass average velocity. These must be used with consistent flux expressions for fixed coordinates and for N_C components, such as:

$$N_A = vJ_A + c_A \sum_{i=1}^{N_C} N_i \bar{V}_i = mJ_A + x_A \sum_{i=1}^{N_C} N_i = \frac{mJ_A + w_A \sum_{i=1}^{N_C} n_i}{M_A} \quad (5-190)$$

In each case, the term containing the summation accounts for *conveyance*, which is the amount of component A carried by the net flow in the direction of diffusion. Its impact on the total flux can be as much as 10 percent. In most cases it is much less, and it is frequently ignored. Some people refer to this as the "convective" term, but that conflicts with the other sense of convection which is promoted by flow perpendicular to the direction of flux.

Mutual Diffusivity, Mass Diffusivity, Interdiffusion Coefficient Diffusivity is denoted by D_{AB} and is defined by Fick's first law as the ratio of the flux to the concentration gradient, as in Eq. (5-189). It is analogous to the thermal diffusivity in Fourier's law and to the kinematic viscosity in Newton's law. These analogies are flawed because both heat and momentum are conveniently defined with respect to fixed coordinates, irrespective of the direction of transfer or its magnitude, while mass diffusivity most commonly requires infor-

mation about bulk motion of the medium in which diffusion occurs. For liquids, it is common to refer to the limit of infinite dilution of A in B using the symbol, D_{AB}° .

When the flux expressions are consistent, as in Eq. (5-190), the diffusivities in Eq. (5-189) are identical. As a result, experimental diffusivities are often measured under constant volume conditions but may be used for applications involving open systems. It turns out that the two versions are very nearly equivalent for gas-phase systems because there is negligible volume change on mixing. That is not usually true for liquids, however.

Self-Diffusivity Self-diffusivity is denoted by $D_{A'A}$ and is the measure of mobility of a species in itself; for instance, using a small concentration of molecules tagged with a radioactive isotope so they can be detected. Tagged and untagged molecules presumably do not have significantly different properties. Hence, the solution is ideal, and there are practically no gradients to "force" or "drive" diffusion. This kind of diffusion is presumed to be purely statistical in nature.

In the special case that A and B are similar in molecular weight, polarity, and so on, the self-diffusion coefficients of pure A and B will be approximately equal to the mutual diffusivity, D_{AB} . Second, when A and B are the less mobile and more mobile components, respectively, their self-diffusion coefficients can be used as rough lower and upper bounds of the mutual diffusion coefficient. That is, $D_{A'A} \leq D_{AB} \leq D_{B'B}$. Third, it is a common means for evaluating diffusion for gases at high pressure. Self-diffusion in liquids has been studied by many [Easteal, *AIChE J.* **30**, 641 (1984), Ertl and Dullien, *AIChE J.* **19**, 1215 (1973), and Vadovic and Colver, *AIChE J.* **18**, 1264 (1972)].

Tracer Diffusivity Tracer diffusivity, denoted by $D_{A'B}$ is related to both mutual and self-diffusivity. It is evaluated in the presence of a second component B, again using a tagged isotope of the first component. In the dilute range, tagging A merely provides a convenient method for indirect composition analysis. As concentration varies, tracer diffusivities approach mutual diffusivities at the dilute limit, and they approach self-diffusivities at the pure component limit. That is, at the limit of dilute A in B, $D_{A'B} \rightarrow D_{AB}^\circ$ and $D_{B'A} \rightarrow D_{B'B}^\circ$; likewise at the limit of dilute B in A, $D_{B'A} \rightarrow D_{BA}^\circ$ and $D_{A'B} \rightarrow D_{A'A}^\circ$.

Neither the tracer diffusivity nor the self-diffusivity has much practical value except as a means to understand ordinary diffusion and as order-of-magnitude estimates of mutual diffusivities. Darken's equation [Eq. (5-230)] was derived for tracer diffusivities but is often used to relate mutual diffusivities at moderate concentrations as opposed to infinite dilution.

Mass-Transfer Coefficient Denoted by k_c , k_x , K_c , and so on, the mass-transfer coefficient is the ratio of the flux to a concentration (or composition) difference. These coefficients generally represent rates of transfer that are much greater than those that occur by diffusion alone, as a result of convection or turbulence at the interface where mass transfer occurs. There exist several principles that relate that coefficient to the diffusivity and other fluid properties and to the intensity of motion and geometry. Examples that are outlined later are the film theory, the surface renewal theory, and the penetration theory, all of which pertain to idealized cases. For many situations of practical interest like investigating the flow inside tubes and over flat surfaces as well as measuring external flow through banks of tubes, in fixed beds of particles, and the like, correlations have been developed that follow the same forms as the above theories. Examples of these are provided in the subsequent section on mass-transfer coefficient correlations.

Problem Solving Methods Most, if not all, problems or applications that involve mass transfer can be approached by a systematic course of action. In the simplest cases, the unknown quantities are obvious. In more complex (e.g., multicomponent, multiphase, multi-dimensional, nonisothermal, and/or transient) systems, it is more subtle to resolve the known and unknown quantities. For example, in multicomponent systems, one must know the fluxes of the components before predicting their effective diffusivities and vice versa. More will be said about that dilemma later. Once the known and unknown quantities are resolved, however, a combination of conservation equations, definitions, empirical relations, and properties are

Nomenclature and Units—Mass Transfer

Symbols	Definition	SI units	U.S. Customary units
a	Effective interfacial mass transfer area per unit volume	m^2/m^3	ft^2/ft^3
A_{cs}	Cross-sectional area of vessel	m^2 or cm^2	ft^2
A'	Constant (see Table 5-24-K)		
a_p	See a		
c	Concentration = P/RT for an ideal gas	mol/m^3 or mol/l or gequiv/l	lbmol/ft^3
c_i	Concentration of component $i = x_i c$ at gas-liquid interface	mol/m^3 or mol/l or gequiv/l	lbmol/ft^3
c_p	Specific heat	$\text{kJ}/(\text{kg}\cdot\text{K})$	$\text{Btu}/(\text{lb}\cdot^\circ\text{F})$
d	Characteristic length	m or cm	ft
d_b	Bubble diameter	m	ft
d_c	Column diameter	m or cm	ft
d_{drop}	Sauter mean diameter	m	ft
d_{imp}	Impeller diameter	m	ft
d_{pore}	Pore diameter	m or cm	ft
$D_{A,A}$	Self-diffusivity ($= D_A$ at $x_A = 1$)	m^2/s or cm^2/s	ft^2/h
D_{AB}	Mutual diffusivity	m^2/s or cm^2/s	ft^2/h
D_{AB}°	Mutual diffusivity at infinite dilution of A in B	m^2/s or cm^2/s	ft^2/h
D_{eff}	Effective diffusivity within a porous solid $= \epsilon_p D/\tau$	m^2/s	ft^2/h
D_K	Knudson diffusivity for gases in small pores	m^2/s or cm^2/s	ft^2/h
D_L	Liquid phase diffusion coefficient	m^2/s	ft^2/h
D_s	Surface diffusivity	m^2/s or cm^2/s	ft^2/h
E	Energy dissipation rate/mass		
E_s	Activation energy for surface diffusion	J/mol or cal/mol	
f	Friction factor for fluid flow	Dimensionless	Dimensionless
F	Faraday's constant	$96,487 \text{ Coulomb}/\text{gequiv}$	
g	Acceleration due to gravity	m/s^2	ft/h^2
g_c	Conversion factor	1.0	$4.17 \times 10^8 \text{ lb ft}/(\text{lb}_f \cdot \text{h}^2)$
G	Gas-phase mass flux	$\text{kg}/(\text{s}\cdot\text{m}^2)$	$\text{lb}/(\text{h}\cdot\text{ft}^2)$
G_a	Dry air flux	$\text{kg}/(\text{s}\cdot\text{m}^2)$	$\text{lb}/(\text{h}\cdot\text{ft}^2)$
G_M	Molar gas-phase mass flux	$\text{kmol}/(\text{s}\cdot\text{m}^2)$	$(\text{lbmol})/(\text{h}\cdot\text{ft}^2)$
h'	Heat transfer coefficient	$\text{W}/(\text{m}^2\cdot\text{K}) = \text{J}/(\text{s}\cdot\text{m}^2\cdot\text{K})$	$\text{Btu}/(\text{h}\cdot\text{ft}^2\cdot^\circ\text{F})$
h_T	Total height of tower packing	m	ft
H	Compartment height	m	ft
H	Henry's law constant	$\text{kPa}/(\text{mole-fraction solute in liquid phase})$	$(\text{lb}_f/\text{in}^2)/(\text{mole-fraction solute in liquid phase})$
H'	Henry's law constant	$\text{kPa}/[\text{kmol}/(\text{m}^3 \text{ solute in liquid phase})]$	$(\text{lb}_f/\text{in}^2)/[(\text{lbmol})/(\text{ft}^3 \text{ solute in liquid phase})]$ or $\text{atm}/[(\text{lbmole})/(\text{ft}^3 \text{ solute in liquid phase})]$
H_G	Height of one transfer unit based on gas-phase resistance	m	ft
H_{OG}	Height of one overall gas-phase mass-transfer unit	m	ft
H_L	Height of one transfer unit based on liquid-phase resistance	m	ft
H_{OL}	Height of one overall liquid-phase mass-transfer unit	m	ft
HTU	Height of one transfer unit (general)	m	ft
j_D	Chilton-Colburn factor for mass transfer, Eq. (5-289)	Dimensionless	Dimensionless
j_H	Chilton-Colburn factor for heat transfer	Dimensionless	Dimensionless
j_M	See j_D		
mJ_A	Mass flux of A by diffusion with respect to the mean mass velocity	$\text{kmol}/(\text{m}^2\cdot\text{s})$ or $\text{mol}/(\text{cm}^2\cdot\text{s})$	$\text{lbmol}/(\text{ft}^2\cdot\text{h})$
MJ_A	Molar flux of A by diffusion with respect to mean molar velocity	$\text{kmol}/(\text{m}^2\cdot\text{s})$ or $\text{mol}/(\text{cm}^2\cdot\text{s})$	$\text{lbmol}/(\text{ft}^2\cdot\text{h})$
v_{JA}	Molar flux of A with respect to mean volume velocity	$\text{kmol}/(\text{m}^2\cdot\text{s})$	$\text{lbmol}/(\text{ft}^2\cdot\text{h})$
j_{si}	Molar flux by surface diffusion	$\text{kmol}/(\text{m}^2\cdot\text{s})$ or $\text{gmol}/(\text{cm}^2\cdot\text{s})$	$\text{lbmol}/(\text{ft}^2\cdot\text{h})$
k	Boltzmann's constant	$8.9308 \times 10^{-10} \text{ gequiv ohm/s}$	
k	Film mass transfer coefficient	m/s or cm/s	ft/hr
k	Thermal conductivity	$(\text{J}\cdot\text{m})/(\text{s}\cdot\text{m}^2\cdot\text{K})$	$\text{Btu}/(\text{h}\cdot\text{ft}\cdot^\circ\text{F})$
k'	Mass-transfer coefficient for dilute systems	$\text{kmol}/[(\text{s}\cdot\text{m}^3)(\text{kmol}/\text{m}^3)]$ or m/s	$\text{lbmol}/[(\text{h}\cdot\text{ft}^3)(\text{lbmol}/\text{ft}^3)]$ or ft/hr
k_G	Gas-phase mass-transfer coefficient for dilute systems	$\text{kmol}/[(\text{s}\cdot\text{m}^3)(\text{kPa solute partial pressure})]$	$\text{lbmol}/[(\text{h}\cdot\text{ft}^3)\text{lb}_f/\text{in}^2 \text{ solute partial pressure})]$
k'_G	Gas-phase mass-transfer coefficient for dilute systems	$\text{kmol}/[(\text{s}\cdot\text{m}^3)(\text{mole fraction in gas})]$	$\text{lbmol}/[(\text{h}\cdot\text{ft}^3)(\text{mole fraction in gas})]$
k_{Ga}	Volumetric gas-phase mass-transfer coefficient for concentrated systems	$\text{kmol}/[(\text{s}\cdot\text{m}^3)(\text{mole fraction})]$	$(\text{lbmol})/[(\text{h}\cdot\text{ft}^3)(\text{mole fraction})]$
k_{Ca}	Overall volumetric gas-phase mass-transfer coefficient for concentrated systems	$\text{kmol}/(\text{s}\cdot\text{m}^3)$	$\text{lbmol}/(\text{h}\cdot\text{ft}^3)$
k_L°	Liquid phase mass transfer coefficient for pure absorption (no reaction)	$\text{kmol}/(\text{s}\cdot\text{m}^2)$	$\text{lbmol}/(\text{h}\cdot\text{ft}^2)$
k_L	Liquid-phase mass-transfer coefficient for dilute systems	$\text{kmol}/[(\text{s}\cdot\text{m}^3)(\text{mole-fraction solution in liquid})]$	$(\text{lbmol})/[(\text{h}\cdot\text{ft}^3)(\text{mole-fraction solute in liquid})]$
k'_L	Liquid-phase mass-transfer coefficient for dilute systems	$\text{kmol}/[(\text{s}\cdot\text{m}^3)(\text{kmol}/\text{m}^3)]$ or m/s	$(\text{lbmol})/[(\text{h}\cdot\text{ft}^3)(\text{lbmol}/\text{ft}^3)]$ or ft/h
k_L	Liquid-phase mass-transfer coefficient for concentrated systems	$\text{kmol}/(\text{s}\cdot\text{m}^2)$	$\text{lbmol}/(\text{h}\cdot\text{ft}^2)$
k_{La}	Volumetric liquid-phase mass-transfer coefficient for dilute systems	$\text{kmol}/[(\text{s}\cdot\text{m}^3)(\text{mole fraction})]$	$(\text{lbmol})/[(\text{h}\cdot\text{ft}^3)(\text{mole fraction})]$
K	Overall mass transfer coefficient	m/s or cm/s	ft/h
K	α/R = specific conductance	ohm/cm	
K_G	Overall gas-phase mass-transfer coefficient for dilute systems	$\text{kmol}/[(\text{s}\cdot\text{m}^3)(\text{mole fraction})]$	$(\text{lbmol})/[(\text{h}\cdot\text{ft}^3)(\text{mole fraction})]$

Nomenclature and Units—Mass Transfer (Continued)

Symbols	Definition	SI units	U.S. Customary units
$\hat{K}_G$	Overall gas-phase mass-transfer coefficient for concentrated systems	kmol/(s·m ²)	lbmol/(h·ft ²)
K_{Ga}	Overall volumetric gas-phase mass-transfer dilute systems	kmol/[(s·m ³)(mole-fraction solute in gas)]	(lbmol)/[(h·ft ³)(mole-fraction solute in gas)]
K'_{Ga}	Overall volumetric gas-phase mass-transfer dilute systems	kmol/[(s·m ³)(kPa solute partial pressure)]	(lbmol)/[(h·ft ³)(lb f/in ² solute partial pressure)]
$(Ka)_H$	Overall enthalpy mass-transfer coefficient	kmol/[(s·m ²)(mole fraction)]	lb/[(h·ft ²)(lb water/lb dry air)]
K_L	Overall liquid-phase mass-transfer coefficient	kmol/[(s·m ²)(mole fraction)]	(lbmol)/[(h·ft ²)(mole fraction)]
$\hat{K}_L$	Liquid-phase mass-transfer coefficient for concentrated systems	kmol/(s·m ²)	(lbmol)/(h·ft ²)
K_{La}	Overall volumetric liquid-phase mass-transfer coefficient for dilute systems	kmol/[(s·m ³)(mole-fraction solute in liquid)]	(lbmol)/[(h·ft ³)(mole-fraction solute in liquid)]
$\hat{K}_{La}$	Overall volumetric liquid-phase mass-transfer coefficient for concentrated systems	kmol/(s·m ³)	(lbmol)/(h·ft ³)
L	Liquid-phase mass flux	kg/(s·m ²)	lb/(h·ft ²)
L_M	Molar liquid-phase mass flux	kmol/(s·m ²)	(lbmol)/(h·ft ²)
m	Slope of equilibrium curve = dy/dx (mole-fraction solute in gas)/(mole-fraction solute in liquid)	Dimensionless	Dimensionless
m	Molality of solute	mol/1000 g solvent	
M_i	Molecular weight of species i	kg/kmol or g/mol	lb/lbmol
M	Mass in a control volume V	kg or g	lb
$ n_+ , n_- $	Valences of cationic and anionic species	Dimensionless	Dimensionless
n'	See Table 5-24-K	Dimensionless	Dimensionless
n_A	Mass flux of A with respect to fixed coordinates	kg/(s·m ²)	lb/(h·ft ²)
N	Impeller speed	Revolution/s	Revolution/min
N'	Number deck levels	Dimensionless	Dimensionless
N_A	Interphase mass-transfer rate of solute A per interfacial area with respect to fixed coordinates	kmol/(s·m ²)	(lbmol)/(h·ft ²)
N_c	Number of components	Dimensionless	Dimensionless
N_{Fr}	Froude Number ($d_{imp} N^2/g$)	Dimensionless	Dimensionless
N_{Gr}	Grashof number $\left(\frac{g \beta \Delta T d^3}{\nu^2} \left(\frac{\rho_s}{\rho_f} - 1 \right) \right)$	Dimensionless	Dimensionless
N_{OG}	Number of overall gas-phase mass-transfer units	Dimensionless	Dimensionless
N_{OL}	Number of overall liquid-phase mass-transfer units	Dimensionless	Dimensionless
NTU	Number of transfer units (general)	Dimensionless	Dimensionless
N_{Kn}	Knudson number = l/d_{pore}	Dimensionless	Dimensionless
N_{Pr}	Prandtl number ($c_p \mu / k$)	Dimensionless	Dimensionless
N_{Re}	Reynolds number (Gd/μ_G)	Dimensionless	Dimensionless
N_{Sc}	Schmidt number ($\mu_G/\rho_G D_{AB}$) or ($\mu_L/\rho_L D_L$)	Dimensionless	Dimensionless
N_{Sh}	Sherwood number ($k_G RT d/D_{AB} p_T$), see also Tables 5-17 to 5-24	Dimensionless	Dimensionless
N_{St}	Stanton number (k_G/G_M) or (k_L/L_M)	Dimensionless	Dimensionless
N_{We}	Weber number ($\rho N^2 d_{imp}^3/\sigma$)	Dimensionless	Dimensionless
p	Solute partial pressure in bulk gas	kPa	lb f/in ²
$p_{B,M}$	Log mean partial pressure difference of stagnant gas B	Dimensionless	Dimensionless
p_i	Solute partial pressure at gas-liquid interface	kPa	lb f/in ²
p_T	Total system pressure	kPa	lb f/in ²
P	Pressure	Pa	lb f/in ² or atm
$\dot{P}$	Power	Watts	
P_c	Critical pressure	Pa	lb f/in ² or atm
Per	Perimeter/area	m ⁻¹	ft ⁻¹
Q	Volumetric flow rate	m ³ /s	ft ³ /h
r_A	Radius of dilute spherical solute	Å	
R	Gas constant	8.314 J/mol K = 8.314 Pa m ³ /(mol K) = 82.057 atm cm ³ /mol K	10.73 ft ³ psia/lbmol·h
R	Solution electrical resistance	ohm	
R_i	Radius of gyration of the component i molecule	Å	
s	Fractional surface-renewal rate	s ⁻¹	h ⁻¹
S	Tower cross-sectional area = $\pi d^2/4$	m ²	ft ²
t	Contact time	s	h
t_f	Formation time of drop	s	h
T	Temperature	K	°R
T_b	Normal boiling point	K	°R
T_c	Critical temperature	K	°R
T_r	Reduced temperature = T/T_c	Dimensionless	Dimensionless
u, v	Fluid velocity	m/s or cm/s	ft/h
u_o	Blowing or suction velocity	m/s	ft/h
u_∞	Velocity away from object	m/s	ft/h
u_L	Superficial liquid velocity in vertical direction	m/s	ft/h
v_s	Slip velocity	m/s	ft/h
v_T	Terminal velocity	m/s	ft/h
v_{TS}	Stokes law terminal velocity	m/s	ft/h
V	Packed volume in tower	m ³	ft ³
V	Control volume	m ³ or cm ³	ft ³
V_b	Volume at normal boiling point	m ³ /kmol or cm ³ /mol	ft ³ /lbmol
V_i	Molar volume of i at its normal boiling point	m ³ /kmol or cm ³ /mol	ft ³ /lbmol
$\bar{v}_i$	Partial molar volume of i	m ³ /kmol or cm ³ /mol	ft ³ /lbmol

Nomenclature and Units—Mass Transfer (Concluded)

Symbols	Definition	SI units	U.S. Customary units
V_{mli}	Molar volume of the liquid-phase component i at the melting point	m^3/kmol or cm^3/mol	ft^3/lbmol
V_{tower}	Tower volume per area	m^3/m^2	ft^3/ft^2
w	Width of film	m	ft
x	Length along plate	m	ft
x	Mole-fraction solute in bulk-liquid phase	$(\text{kmol solute})/(\text{kmol liquid})$	$(\text{lbmol solute})/(\text{lb mol liquid})$
x_A	Mole fraction of component A	$\text{kmole } A/\text{kmole fluid}$	$\text{lbmol } A/\text{lb mol fluid}$
x^o	Mole-fraction solute in bulk liquid in equilibrium with bulk-gas solute concentration y	$(\text{kmol solute})/(\text{kmol liquid})$	$(\text{lbmol solute})/(\text{lbmol liquid})$
x_{BM}	Logarithmic-mean solvent concentration between bulk liquid and interface values	$(\text{kmol solvent})/(\text{kmol liquid})$	$(\text{lbmol solvent})/(\text{lbmol liquid})$
x_{BM}^o	Logarithmic-mean inert-solvent concentration between bulk-liquid value and value in equilibrium with bulk gas	$(\text{kmol solvent})/(\text{kmol liquid})$	$(\text{lbmol solvent})/(\text{lbmol liquid})$
x_i	Mole-fraction solute in liquid at gas-liquid interface	$(\text{kmol solute})/(\text{kmol liquid})$	$(\text{lbmol solute})/(\text{lbmol liquid})$
y	Mole-fraction solute in bulk-gas phase	$(\text{kmol solute})/(\text{kmol gas})$	$(\text{lbmol solute})/(\text{lbmol gas})$
y_{BM}	Logarithmic-mean inert-gas concentration [Eq. (5-275)]	$(\text{kmol inert gas})/(\text{kmol gas})$	$(\text{lbmol inert gas})/(\text{lbmol gas})$
y_{BM}^o	Logarithmic-mean inert-gas concentration	$(\text{kmol inert gas})/(\text{kmol gas})$	$(\text{lbmol inert gas})/(\text{lbmol gas})$
y_i	Mole fraction solute in gas at interface	$(\text{kmol solute})/(\text{kmol gas})$	$(\text{lbmol solute})/(\text{lbmol gas})$
y_i^o	Mole-fraction solute in gas at interface in equilibrium with the liquid-phase interfacial solute concentration x_i	$(\text{kmol solute})/(\text{kmol gas})$	$(\text{lbmol solute})/(\text{lbmol gas})$
z	Direction of unidimensional diffusion	m	ft
Greek Symbols			
α	$1 + N_B/N_A$	Dimensionless	Dimensionless
α	Conductance cell constant (measured)	cm^{-1}	
β	$M_A^{1/2} p_c^{1/3} / T_c^{5/6}$	Dimensionless	Dimensionless
δ	Effective thickness of stagnant-film layer	m	ft
ϵ	Fraction of discontinuous phase in continuous phase for two-phase flow	Dimensionless	Dimensionless
ϵ	Void fraction available for gas flow or fractional gas holdup	m^3/m^3	ft^3/ft^3
ϵ_A	Characteristic Lennard-Jones energy	Dimensionless	Dimensionless
ϵ_{AB}	$(\epsilon_A \epsilon_B)^{1/2}$	Dimensionless	Dimensionless
γ_i	Activity coefficient of solute i	Dimensionless	Dimensionless
$\gamma_{\pm}$	Mean ionic activity coefficient of solute	Dimensionless	Dimensionless
λ_+ , λ_-	Infinite dilution conductance of cation and anion	$\text{cm}^2/(\text{equiv-ohm})$	
Λ	$1000 \text{ K/C} = \lambda_+ + \lambda_- = \Lambda_o + f(C)$	$\text{cm}^2/\text{ohm equiv}$	
Λ_o	Infinite dilution conductance	$\text{cm}^2/\text{equiv ohm}$	
μ_i	Dipole moment of i	Debyes	
μ_i	Viscosity of pure i	cP or Pa s	$\text{lb}/(\text{h}\cdot\text{ft})$
μ_G	Gas-phase viscosity	$\text{kg}/(\text{s}\cdot\text{m})$	$\text{lb}/(\text{h}\cdot\text{ft})$
μ_L	Liquid-phase viscosity	$\text{kg}/(\text{s}\cdot\text{m})$	$\text{lb}/(\text{h}\cdot\text{ft})$
ν	Kinematic viscosity $= \rho/\mu$	m^2/s	ft^2/h
ρ	Density of A	kg/m^3 or g/cm^3	lb/ft^3
ρ_c	Critical density of A	kg/m^3 or g/cm^3	lb/ft^3
ρ_c	Density continuous phase	kg/m^3	lb/ft^3
ρ_G	Gas-phase density	kg/m^3	lb/ft^3
ρ_L	Average molar density of liquid phase	kmol/m^3	$(\text{lbmol})/\text{ft}^3$
ρ_p	Particle density	kg/m^3 or g/cm^3	lb/ft^3
ρ_r	Reduced density $= \rho/\rho_c$	Dimensionless	Dimensionless
Ψ_i	Parachor of component $i = V_i \sigma^{1/4}$		
Ψ	Parameter, Table 5-24-G	Dimensionless	Dimensionless
σ	Interfacial tension	dyn/cm	lb/ft
σ_i	Characteristic length	$\text{\AA}$	
σ_i	Surface tension of component i	dyn/cm	
σ_{AB}	Binary pair characteristic length $= (\sigma_A + \sigma_B)/2$	$\text{\AA}$	
τ	Intraparticle tortuosity	Dimensionless	Dimensionless
ω	Pitzer's acentric factor $= -[1.0 + \log_{10}(P^o/P_c)]$	Dimensionless	
ω	Rotational velocity	Radians/s	
Ω	Diffusion collision integral $= f(kT/\epsilon_{AB})$	Dimensionless	Dimensionless
Subscripts			
A	Solute component in liquid or gas phase		
B	Inert-gas or inert-solvent component		
G	Gas phase		
m	Mean value		
L	Liquid phase		
super	Superficial velocity		
Superscript			
o	At equilibrium		

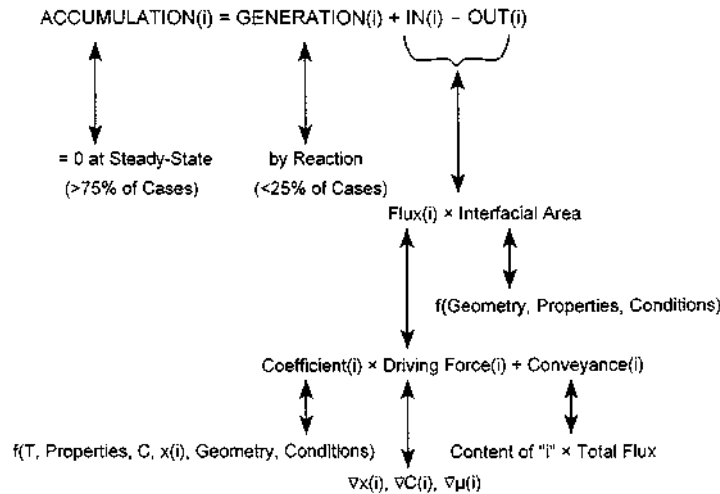


FIG. 5-24 Flowchart illustrating problem solving approach using mass-transfer rate expressions in the context of mass conservation.

applied to arrive at an answer. Figure 5-24 is a flowchart that illustrates the primary types of information and their relationships, and it applies to many mass-transfer problems.

CONTINUITY AND FLUX EXPRESSIONS

Material Balances Whenever mass-transfer applications involve equipment of specific dimensions, flux equations alone are inadequate to assess results. A material balance or continuity equation must also be used. When the geometry is simple, macroscopic balances suffice. The following equation is an overall mass balance for such a unit having N_m bulk-flow ports and N_n ports or interfaces through which diffusive flux can occur:

$$\frac{dM}{dt} = \sum_{i=1}^{N_m} m_i + \sum_{i=1}^{N_n} n_i A_{cs_i} \quad (5-191)$$

where M represents the mass in the unit volume V at any time t ; m_i is the mass flow rate through the i th port; and n_i is the mass flux through the i th port, which has a cross-sectional area of A_{cs_i} . The corresponding balance equation for individual components includes a reaction term:

$$\frac{dM_j}{dt} = \sum_{i=1}^{N_m} m_{ij} + \sum_{i=1}^{N_n} n_{ij} A_{cs_i} + r_j V \quad (5-192)$$

For the j th component, $m_{ij} = m_i w_{ij}$ is the component mass flow rate in stream i ; w_{ij} is the mass fraction of component j in stream i ; and r_j is the net reaction rate (mass generation minus consumption) per unit volume V that contains mass M . If it is inconvenient to measure mass flow rates, the product of density and volumetric flow rate is used instead.

In addition, most situations that involve mass transfer require material balances, but the pertinent area is ambiguous. Examples are packed columns for absorption, distillation, or extraction. In such

cases, flow rates through the discrete ports (nozzles) must be related to the mass-transfer rate in the packing. As a result, the mass-transfer rate is determined via flux equations, and the overall material balance incorporates the stream flow rates m_i and integrated fluxes. In such instances, it is common to begin with the most general, differential material balance equations. Then, by eliminating terms that are negligible, the simplest applicable set of equations remains to be solved. Table 5-8 provides material balances for Cartesian, cylindrical, and spherical coordinates. The generic form applies over a unit cross-sectional area and constant volume:

$$\frac{\partial \rho_j}{\partial t} = -\nabla \cdot n_j + r_j \quad (5-193a)$$

where $n_j = \rho v_j$. Applying Fick's law and expressing composition as concentration gives

$$\frac{\partial c_j}{\partial t} = -v \cdot \nabla c_j + D_j \nabla^2 c_j + r_j \quad (5-193b)$$

Flux Expressions: Simple Integrated Forms of Fick's First Law Simplified flux equations that arise from Eqs. (5-189) and (5-190) can be used for unidimensional, steady-state problems with binary mixtures. The boundary conditions represent the compositions x_{A_L} and x_{A_R} at the left-hand and right-hand sides of a hypothetical layer having thickness Δz . The principal restriction of the following equations is that the concentration and diffusivity are assumed to be constant. As written, the flux is positive from left to right, as depicted in Fig. 5-25.

1. Equimolar counterdiffusion ($N_A = -N_B$)

$$N_A = M J_A = -D_{AB} c \frac{dx_A}{dz} = \frac{D_{AB}}{\Delta z} c (x_{A_L} - x_{A_R}) \quad (5-197)$$

TABLE 5-8 Continuity Equation in Various Coordinate Systems

Coordinate System	Equation
Cartesian	$\frac{\partial \rho_j}{\partial t} = -\left(\frac{\partial n_{xj}}{\partial x} + \frac{\partial n_{yj}}{\partial y} + \frac{\partial n_{zj}}{\partial z} \right) + r_j \quad (5-194)$
Cylindrical	$\frac{\partial \rho_j}{\partial t} = -\left(\frac{1}{r} \frac{\partial m_{rj}}{\partial r} + \frac{1}{r} \frac{\partial n_{\theta j}}{\partial \theta} + \frac{\partial n_{zj}}{\partial z} \right) + r_j \quad (5-195)$
Spherical	$\frac{\partial \rho_j}{\partial t} = -\left(\frac{1}{r^2} \frac{\partial r^2 n_{rj}}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial n_{\theta j} \sin \theta}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial n_{\phi j}}{\partial \phi} \right) + r_j \quad (5-196)$

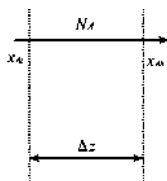


FIG. 5-25 Hypothetical film and boundary conditions.

2. Unimolar diffusion (
- $N_A \neq 0, N_B = 0$
-)

$$N_A = M J_A + x_A N_A = \frac{D_{AB}}{\Delta z} c \ln \frac{1 - x_{A_R}}{1 - x_{A_L}} \quad (5-198)$$

3. Steady state diffusion (
- $N_A \neq -N_B \neq 0$
-)

$$N_A = M J_A + x_A (N_A + N_B) = \frac{N_A}{N_A + N_B} \frac{D_{AB}}{\Delta z} c \ln \frac{\frac{N_A}{N_A + N_B} - x_{A_R}}{\frac{N_A}{N_A + N_B} - x_{A_L}} \quad (5-199)$$

The unfortunate aspect of the last relationship is that one must know a priori the ratio of the fluxes to determine the magnitudes. It is not possible to solve simultaneously the pair of equations that apply for components *A* and *B* because the equations are not independent.

Stefan-Maxwell Equations Following Eq. (5-190), a simple and intuitively appealing flux equation for applications involving N_c components is

$$N_i = -c D_{im} \nabla x_i + x_i \sum_{j=1}^{N_c} N_j \quad (5-200)$$

In the late 1800s, the development of the kinetic theory of gases led to a method for calculating multicomponent gas diffusion (e.g., the flux of each species in a mixture). The methods were developed simultaneously by Stefan and Maxwell. The problem is to determine the diffusion coefficient D_{im} . The Stefan-Maxwell equations are simpler in principle since they employ binary diffusivities:

$$\nabla x_i = \sum_{j=1}^{N_c} \frac{1}{c D_{ij}} (x_i N_j - x_j N_i) \quad (5-201)$$

If Eqs. (5-200) and (5-201) are combined, the multicomponent diffusion coefficient may be assessed in terms of binary diffusion coefficients [see Eq. (5-214)]. For gases, the values D_{ij} of this equation are approximately equal to the binary diffusivities for the *ij* pairs. The Stefan-Maxwell diffusion coefficients may be negative, and the method may be applied to liquids, even for electrolyte diffusion [Kraaijeveld, Wesselingh, and Kuiken, *Ind. Eng. Chem. Res.*, **33**, 750 (1994)]. Approximate solutions have been developed by linearization [Toor, H.L., *AIChE J.*, **10**, 448 and 460 (1964); Stewart and Prober, *Ind. Eng. Chem. Fundam.*, **3**, 224 (1964)]. Those differ in details but yield about the same accuracy. More recently, efficient algorithms for solving the equations exactly have been developed (see Taylor and Krishna, Krishnamurthy and Taylor [*Chem. Eng. J.*, **25**, 47 (1982)], and Taylor and Webb [*Comput. Chem. Eng.*, **5**, 61 (1981)]).

Useful studies of multicomponent diffusion were presented by Vrentas and Vrentas [*Ind. Eng. Chem. Res.*, **44**, 1112 (2005)], Curtis

and Bird [*Ind. Eng. Chem. Res.*, **38**, 2515 (1999)], and Amundson, Pan, and Paulson [*AIChE J.*, **48**, 813 (2003)]. Vrentas and Vrentas treated only ternary mixtures, such as restrictions due to the entropy inequality, application of the Onsager reciprocal relations, and stability. Curtis and Bird reconciled the multicomponent Fick's law approach with the more elegant Stefan-Maxwell theory. They also provided interrelationships of multicomponent diffusivities devised for various situations, i.e., binary, ternary, and quaternary mixtures. Amundson et al. presented numerical methods for coping with mixtures having four or more components, which are nearly intractable via the analytical S-M method, due to the difficult inversion. Related studies were performed by Ghorayeb and Firoozabadi [*AIChE J.*, **46**, 883 (2000)] and Firoozabadi, Ghorayeb, and Saukka [*AIChE J.*, **46**, 892 (2000)]. The former covered ordinary molecular diffusion as well as pressure and thermal diffusion for multicomponent mixtures. The latter covered thermal diffusion in multicomponent mixtures.

DIFFUSIVITY ESTIMATION—GASES

Whenever measured values of diffusivities are available, they should be used. Typically, measurement errors are less than those associated with predictions by empirical or even semitheoretical equations. A few general sources of data are Sec. 2 of this handbook; e.g., experimental values for gas mixtures are listed in Table 2-371. Estimation methods for some gaseous applications appear in Eqs. (2-150) through (2-154). Other pertinent references are Schwartzberg and Chao; Poling et al.; Gammon et al.; and Daubert and Danner. Many other more restricted sources are listed under specific topics later in this subsection.

Before using diffusivities from either data or correlations, it is a good idea to check their reasonableness with respect to values that have been commonly observed in similar situations. Table 5-9 is a compilation of several rules of thumb. These values are not authoritative; they simply represent guidelines based on experience.

Diffusivity correlations for gases are outlined in Table 5-10. Specific parameters for individual equations are defined in the specific text regarding each equation. References are given at the beginning of the "Mass Transfer" subsection. The errors reported for Eqs. (5-202) through (5-205) were compiled by Poling et al., who compared the predictions with 68 experimental values of D_{AB} . Errors cited for Eqs. (5-206) to (5-212) were reported by the authors.

Binary Mixtures—Low Pressure—Nonpolar Components

Many evaluations of correlations are available [Elliott and Watts, *Can. J. Chem.*, **50**, 31 (1972); Lugg, *Anal. Chem.*, **40**, 1072 (1968); Marrero and Mason, *AIChE J.*, **19**, 498 (1973)]. The differences in accuracy of the correlations are minor, and thus the major concern is ease of calculation. The Fuller-Schettler-Giddings equation is usually the simplest correlation to use and is recommended by Poling et al.

Chapman-Enskog (Bird et al.) and Wilke and Lee [31] The inherent assumptions of these equations are quite restrictive (i.e., low density, spherical atoms), and the intrinsic potential function is empirical. Despite that, they provide good estimates of D_{AB} for many polyatomic gases and gas mixtures, up to about 1000 K and a maximum of 70 atm. The latter constraint is because observations for many gases indicate that $D_{AB}P$ is constant up to 70 atm.

The characteristic length is $\sigma_{AB} = (\sigma_A + \sigma_B)/2$ in Å. In order to estimate Ω_D for Eq. (5-202) or (5-203), two empirical equations are

TABLE 5-9 Rules of Thumb for Diffusivities (See Cussler, Poling et al., Schwartzberg and Chao)

Continuous phase	D_i magnitude		D_i range		Comments
	m ² /s	cm ² /s	m ² /s	cm ² /s	
Gas at atmospheric pressure	10 ⁻⁵	0.1	10 ⁻⁴ –10 ⁻⁶	1–10 ⁻²	Accurate theories exist, generally within ±10%; $D_i P \approx \text{constant}$; $D_i \propto T^{1.66 \text{ to } 2.0}$
Liquid	10 ⁻⁹	10 ⁻⁵	10 ⁻⁸ –10 ⁻¹⁰	10 ⁻⁴ –10 ⁻⁶	Approximate correlations exist, generally within ±25%
Liquid occluded in solid matrix	10 ⁻¹⁰	10 ⁻⁶	10 ⁻⁸ –10 ⁻¹²	10 ⁻⁴ –10 ⁻⁸	Hard cell walls: $D_{\text{eff}}/D_i = 0.1$ to 0.2. Soft cell walls: $D_{\text{eff}}/D_i = 0.3$ to 0.9
Polymers and glasses	10 ⁻¹²	10 ⁻⁸	10 ⁻¹⁰ –10 ⁻¹⁴	10 ⁻⁶ –10 ⁻¹⁰	Approximate theories exist for dilute and concentrated limits; strong composition dependence
Solid	10 ⁻¹⁴	10 ⁻¹⁰	10 ⁻¹⁰ –10 ⁻³⁴	10 ⁻⁶ –10 ⁻³⁰	Approximate theories exist; strong temperature dependence

TABLE 5-10 Correlations of Diffusivities for Gases

Authors*	Equation	Error, %
1. Binary Mixtures—Low Pressure—Nonpolar		
Chapman-Enskog	$D_{AB} = \frac{0.001858 T^{3/2} M_{AB}^{1/2}}{P \sigma_{AB}^2 \Omega_D} \quad (5-202)$	7.3
Wilke-Lee [31]	$D_{AB} = \frac{(0.00217 - 0.0005 M_{AB}^{1/2}) T^{3/2} M_{AB}^{1/2}}{P \sigma_{AB}^2 \Omega_D} \quad (5-203)$	7.0
Fuller-Schettler-Giddings [10]	$D_{AB} = \frac{0.001 T^{1.75} M_{AB}^{1/2}}{P [(\sum v)_A^{1/3} + (\sum v)_B^{1/3}]^2} \quad (5-204)$	5.4
2. Binary Mixtures—Low Pressure—Polar		
Brokaw [4]	$D_{AB} = \frac{0.001858 T^{3/2} M_{AB}^{1/2}}{P \sigma_{AB}^2 \Omega_D} \quad (5-205)$	9.0
3. Self-Diffusivity		
Mathur-Thodos [18]	$D_{AA} = \frac{10.7 \times 10^{-5} T_r}{\beta \rho_r} \quad \{\rho_r \leq 1.5\} \quad (5-206)$	5
Lee-Thodos [14]	$D_{AA} = \frac{0.77 \times 10^{-5} T_r}{\rho \delta} \quad \{\rho_r \leq 1\} \quad (5-207)$	0.5
Lee-Thodos [15]	$D_{AA} = \frac{(0.007094 G + 0.001916)^{2.5} T_r}{\delta}, \quad [\rho_r > 1, G < 1] \quad (5-208)$	17
4. Supercritical Mixtures		
Sun and Chen [25]	$D_{AB} = \frac{1.23 \times 10^{-10} T}{\mu^{0.799} V_{CA}^{0.49}} \quad (5-209)$	5
Catchpole and King [6]	$D_{AB} = 5.152 D_c T_r \frac{(\rho^{-0.667} - 0.4510) (1 + M_A/M_B) R}{(1 + (V_{cB}/V_{cA})^{0.333})^2} \quad (5-210)$	10
Liu and Ruckenstein [17]	$D_{AB} = \frac{kT}{f\pi\mu_A\sigma_{AB}} \left[\frac{1}{1 + \frac{2}{3} \left(1 - \frac{1}{2} \theta_{AB}^{\infty}\right)^{1/2}} + \frac{\sigma_{AB}}{3\sigma_A} \right] \quad (5-211)$	5.7
	$D_{AB} = \alpha (V_A^k - \beta) \sqrt{\frac{T}{M_B}}, \quad \alpha = 10^{-5} \left[0.56392 + 2.1417 \exp\left(\frac{-0.95088 \sqrt{M_A V_{CA}}}{P_{CA}}\right) \right] \quad (5-212)$	6.9
	$\beta = 8.9061 + 0.93858 \frac{\sqrt{M_A V_{CA}}}{P_{CA}}$	
	$k = \frac{2}{3} [1 - 0.28 \exp(-0.3 \sqrt{M_A} \rho_{A^*})]$	

*References are listed at the beginning of the "Mass Transfer" subsection.

available. The first is:

$$\Omega_D = (44.54 T^{a-4.909} + 1.911 T^{a-1.575})^{0.10} \quad (5-213a)$$

where $T^* = kT/\epsilon_{AB}$ and $\epsilon_{AB} = (\epsilon_A \epsilon_B)^{1/2}$. Estimates for σ_i and ϵ_i are given in Table 5-11. This expression shows that Ω_D is proportional to temperature roughly to the -0.49 power at low temperatures and to the

-0.16 power at high temperature. Thus, gas diffusivities are proportional to temperatures to the 2.0 power and 1.66 power, respectively, at low and high temperatures. The second is:

$$\Omega_D = \frac{A}{T^{aB}} + \frac{C}{\exp(DT^*)} + \frac{E}{\exp(FT^*)} + \frac{G}{\exp(HT^*)} \quad (5-213b)$$

TABLE 5-11 Estimates for ϵ_i and σ_i (K, Å, atm, cm³, mol)

Critical point	$\epsilon/k = 0.75 T_c$	$\sigma = 0.841 V_c^{1/3}$ or $2.44 (T_c/P_c)^{1/3}$
Critical point	$\epsilon/k = 65.3 T_c \sigma_c^{-3.6}$	$\sigma = \frac{1.866 V_c^{1/3}}{\sigma_c^{1.2}}$
Normal boiling point	$\epsilon/k = 1.15 T_b$	$\sigma = 1.18 V_b^{1/3}$
Melting point	$\epsilon/k = 1.92 T_m$	$\sigma = 1.222 V_m^{1/3}$
Acentric factor	$\epsilon/k = (0.7915 + 0.1693 \omega) T_c$	$\sigma = (2.3551 - 0.087 \omega) \left(\frac{T_c}{P_c}\right)^{1/3}$

NOTE: These values may not agree closely, so usage of a consistent basis is suggested (e.g., data at the normal boiling point).

TABLE 5-12 Atomic Diffusion Volumes for Use in Estimating D_{AB} by the Method of Fuller, Schettler, and Giddings [10]

Atomic and Structural Diffusion-Volume Increments, v_i (cm ³ /mol)			
C	16.5	(Cl)	19.5
H	1.98	(S)	17.0
O	5.48	Aromatic ring	-20.2
(N)	5.69	Heterocyclic ring	-20.2
Diffusion Volumes for Simple Molecules, Σv_i (cm ³ /mol)			
H ₂	7.07	CO	18.9
D ₂	6.70	CO ₂	26.9
He	2.88	N ₂ O	35.9
N ₂	17.9	NH ₃	14.9
O ₂	16.6	H ₂ O	12.7
Air	20.1	(CCl ₂ F ₂)	114.8
Ar	16.1	(SF ₆)	69.7
Kr	22.8	(Cl ₂)	37.7
(Xe)	37.9	(Br ₂)	67.2
Ne	5.59	(SO ₂)	41.1

Parentheses indicate that the value listed is based on only a few data points.

where $A = 1.06036$, $B = 0.15610$, $C = 0.1930$, $D = 0.47635$, $E = 1.03587$, $F = 1.52996$, $G = 1.76474$, and $H = 3.89411$.

Fuller, Schettler, and Giddings [10] The parameters and constants for this correlation were determined by regression analysis of 340 experimental diffusion coefficient values of 153 binary systems. Values of Σv_i used in this equation are in Table 5-12.

Binary Mixtures—Low Pressure—Polar Components The Brokaw [4] correlation was based on the Chapman-Enskog equation, but σ_{AB}^* and Ω_D^* were evaluated with a modified Stockmayer potential for polar molecules. Hence, slightly different symbols are used. That potential model reduces to the Lennard-Jones 6-12 potential for interactions between nonpolar molecules. As a result, the method should yield accurate predictions for polar as well as nonpolar gas mixtures. Brokaw presented data for 9 relatively polar pairs along with the prediction. The agreement was good: an average absolute error of 6.4 percent, considering the complexity of some of the gas pairs [e.g., (CH₃)₂O and CH₃Cl]. Despite that, Poling, (op. cit.) found the average error was 9.0 percent for combinations of mixtures (including several polar-nonpolar gas pairs), temperatures and pressures. In this equation, Ω_D is calculated as described previously, and other terms are:

$$\begin{aligned}\Omega_{D^*} &= \Omega_D + 0.19 \delta_{AB}^2/T^* & T^* &= kT/\epsilon_{AB} \\ \sigma_{AB}^* &= (\sigma_A^* \sigma_B^*)^{1/2} & \sigma_i^* &= [1.585 V_{bi}/(1 + 1.3 \delta_i^2)]^{1/3} \\ \delta_{AB} &= (\delta_A \delta_B)^{1/2} & \delta_i &= 1.94 \times 10^3 \mu_i^2/V_{bi} T_{bi} \\ \epsilon_{AB}^* &= (\epsilon_A^* \epsilon_B^*)^{1/2} & \epsilon_i/k &= 1.18 (1 + 1.3 \delta_i^2) T_{bi}\end{aligned}$$

Binary Mixtures—High Pressure Of the various categories of gas-phase diffusion, this is the least studied. This is so because of the effects of diffusion being easily distorted by even a slight pressure gradient, which is difficult to avoid at high pressure. Harstad and Bellan [Ind. Eng. Chem. Res. **43**, 645 (2004)] developed a corresponding-states expression that extends the Chapman-Enskog method, covered earlier. They express the diffusivity at high pressure by accounting for the reduced temperature, and they suggest employing an equation of state and shifting from $D_{AB}' = f(T, P)$ to $D_{AB} = g(T, V)$.

Self-Diffusivity Self-diffusivity is a property that has little intrinsic value, e.g., for solving separation problems. Despite that, it reveals quite a lot about the inherent nature of molecular transport, because the effects of discrepancies of other physical properties are eliminated, except for those that constitute isotopic differences, which are necessary to ascertain composition differences. Self-diffusivity has been studied extensively under high pressures, e.g., greater than 70 atm. There are few accurate estimation methods for mutual diffusivities at such high pressures, because composition measurements are difficult.

The general observation for gas-phase diffusion $D_{AB} P = \text{constant}$, which holds at low pressure, is not valid at high pressure. Rather, $D_{AB} P$ decreases as pressure increases. In addition, composition effects, which frequently are negligible at low pressure, are very significant at high pressure.

Liu and Ruckenstein [Ind. Eng. Chem. Res. **36**, 3937 (1997)] studied self-diffusion for both liquids and gases. They proposed a semiempirical equation, based on hard-sphere theory, to estimate self-diffusivities. They extended it to Lennard-Jones fluids. The necessary energy parameter is estimated from viscosity data, but the molecular collision diameter is estimated from diffusion data. They compared their estimates to 26 pairs, with a total of 1822 data points, and achieved a relative deviation of 7.3 percent.

Zielinski and Hanley [AIChE J. **45**, 1 (1999)] developed a model to predict multicomponent diffusivities from self-diffusion coefficients and thermodynamic information. Their model was tested by estimated experimental diffusivity values for ternary systems, predicting drying behavior of ternary systems, and reconciling ternary self-diffusion data measured by pulsed-field gradient NMR.

Mathur and Thodos [18] showed that for reduced densities less than unity, the product $D_{AB} \rho$ is approximately constant at a given temperature. Thus, by knowing the value of the product at low pressure, it is possible to estimate its value at a higher pressure. They found at higher pressures the density increases, but the product $D_{AB} \rho$ decreases rapidly. In their correlation, $\beta = M_A^{1/2} P_C^{1/3} / T_C^{5/6}$.

Lee and Thodos [14] presented a generalized treatment of self-diffusivity for gases (and liquids). These correlations have been tested for more than 500 data points each. The average deviation of the first is 0.51 percent, and that of the second is 17.2 percent. $\delta = M_A^{1/2} / P_C^{1/2} V_C^{5/6}$, s/cm², and where $G = (X^* - X)/(X^* - 1)$, $X = \rho_r / T_r^{0.1}$, and $X^* = \rho_r / T_r^{0.1}$ evaluated at the solid melting point.

Lee and Thodos [15] expanded their earlier treatment of self-diffusivity to cover 58 substances and 975 data points, with an average absolute deviation of 5.26 percent. Their correlation is too involved to repeat here, but those interested should refer to the original paper.

Liu, Silva, and Macedo [Chem. Eng. Sci. **53**, 2403 (1998)] present a theoretical approach incorporating hard-sphere, square-well, and Lennard-Jones models. They compared their resulting estimates to estimates generated via the Lee-Thodos equation. For 2047 data points with nonpolar species, the Lee-Thodos equation was slightly superior to the Lennard-Jones fluid-based model, that is, 5.2 percent average deviation versus 5.5 percent, and much better than the square-well fluid-based model (10.6 percent deviation). For over 467 data points with polar species, the Lee-Thodos equation yielded 36 percent average deviation, compared with 25 percent for the Lennard-Jones fluid-based model, and 19 percent for the square-well fluid-based model.

Silva, Liu, and Macedo [Chem. Eng. Sci. **53**, 2423 (1998)] present an improved theoretical approach incorporating slightly different Lennard-Jones models. For 2047 data points with nonpolar species, their best model yielded 4.5 percent average deviation, while the Lee-Thodos equation yielded 5.2 percent, and the prior Lennard-Jones fluid-based model produced 5.5 percent. The new model was much better than all the other models for over 424 data points with polar species, yielding 4.3 percent deviation, while the Lee-Thodos equation yielded 34 percent and the Lennard-Jones fluid-based model yielded 23 percent.

Supercritical Mixtures Debenedetti and Reid [AIChE J., **32**, 2034 (1986) and **33**, 496 (1987)] showed that conventional correlations based on the Stokes-Einstein relation (for liquid phase) tend to overpredict diffusivities in the supercritical state. Nevertheless, they observed that the Stokes-Einstein group $D_{AB} \mu / T$ was constant. Thus, although no general correlation applies, only one data point is necessary to examine variations of fluid viscosity and/or temperature effects. They explored certain combinations of aromatic solids in SF₆ and CO₂.

Sun and Chen [25] examined tracer diffusion data of aromatic solutes in alcohols up to the supercritical range and found their data correlated with average deviations of 5 percent and a maximum deviation of 17 percent for their rather limited set of data.

Catchpole and King [6] examined binary diffusion data of near-critical fluids in the reduced density range of 1 to 2.5 and found that their data correlated with average deviations of 10 percent and a maximum deviation of 60 percent. They observed two classes of behavior. For the first, no correction factor was required ($R = 1$). That class was comprised of alcohols as solvents with aromatic or aliphatic solutes, or

carbon dioxide as a solvent with aliphatics except ketones as solutes, or ethylene as a solvent with aliphatics except ketones and naphthalene as solutes. For the second class, the correction factor was $R = X^{0.17}$. The class was comprised of carbon dioxide with aromatics; ketones and carbon tetrachloride as solutes; and aliphatics (propane, hexane, dimethyl butane), sulfur hexafluoride, and chlorotrifluoromethane as solvents with aromatics as solutes. In addition, sulfur hexafluoride combined with carbon tetrachloride, and chlorotrifluoromethane combined with 2-propanone were included in that class. In all cases, $X = (1 + (V_{CS}/V_{CA})^{1/3})^2 / (1 + M_A/M_B)$ was in the range of 1 to 10.

Liu and Ruckenstein [17] presented a semiempirical equation to estimate diffusivities under supercritical conditions that is based on the Stokes-Einstein relation and the long-range correlation, respectively. The parameter $2\theta_{AB}^*$ was estimated from the Peng-Robinson equation of state. In addition, $f = 2.72 - 0.3445 T_{CB}/T_{CA}$ for most solutes, but for C_5 through C_{14} linear alkanes, $f = 3.046 - 0.786 T_{CB}/T_{CA}$. In both cases T_{ci} is the species critical temperature. They compared their estimates to 33 pairs, with a total of 598 data points, and achieved lower deviations (5.7 percent) than the Sun-Chen correlation (13.3 percent) and the Catchpole-King equation (11.0 percent).

He and Yu [13] presented a semiempirical equation to estimate diffusivities under supercritical conditions that is based on hard-sphere theory. It is limited to $\rho_r \geq 0.21$, where the reduced density is $\rho_r = \rho_A(T, P)/\rho_{CA}$. They compared their estimates to 107 pairs, with a total of 1167 data points, and achieved lower deviations (7.8 percent) than the Catchpole-King equation (9.7 percent), which was restricted to $\rho_r \geq 1$.

Silva and Macedo [Ind. Eng. Chem. Res. **37**, 1490 (1998)] measured diffusivities of ethers in CO_2 under supercritical conditions and compared them to the Wilke-Chang [Eq. (5-218)], Tyn-Calus [Eq. (5-219)], Catchpole-King [Eq. (5-210)], and their own equations. They found that the Wilke-Chang equation provided the best fit.

Gonzalez, Bueno, and Medina [Ind. Eng. Chem. Res. **40**, 3711 (2001)] measured diffusivities of aromatic compounds in CO_2 under supercritical conditions and compared them to the Wilke-Chang [Eq. (5-218)], Hayduk-Minhas [Eq. (5-226)], and other equations. They recommended the Wilke-Chang equation (which yielded a relative error of 10.1 percent) but noted that the He-Yu equation provided the best fit (5.5 percent).

Low-Pressure/Multicomponent Mixtures These methods are outlined in Table 5-13. Stefan-Maxwell equations were discussed earlier. Smith and Taylor [23] compared various methods for predicting multicomponent diffusion rates and found that Eq. (5-214) was superior among the effective diffusivity approaches, though none is very good. They also found that linearized and exact solutions are roughly equivalent and accurate.

Blanc [3] provided a simple limiting case for dilute component i diffusing in a stagnant medium (i.e., $N \approx 0$), and the result, Eq. (5-215), is known as Blanc's law. The restriction basically means that the compositions of all the components, besides component i , are relatively large and uniform.

Wilke [29] obtained solutions to the Stefan-Maxwell equations. The first, Eq. (5-216), is simple and reliable under the same conditions as Blanc's law. This equation applies when component i diffuses through

a stagnant mixture. It has been tested and verified for diffusion of toluene in hydrogen + air + argon mixtures and for diffusion of ethyl propionate in hydrogen + air mixtures [Fairbanks and Wilke *Ind. Eng. Chem.*, **42**, 471 (1950)]. When the compositions vary from one boundary to the other, Wilke recommends that the arithmetic average mole fractions be used. Wilke also suggested using the Stefan-Maxwell equation, which applies when the fluxes of two or more components are significant. In this situation, the mole fractions are arithmetic averages of the boundary conditions, and the solution requires iteration because the ratio of fluxes is not known a priori.

DIFFUSIVITY ESTIMATION—LIQUIDS

Many more correlations are available for diffusion coefficients in the liquid phase than for the gas phase. Most, however, are restricted to binary diffusion at infinite dilution D_{AB}^∞ or to self-diffusivity D_{AA} . This reflects the much greater complexity of liquids on a molecular level. For example, gas-phase diffusion exhibits negligible composition effects and deviations from thermodynamic ideality. Conversely, liquid-phase diffusion almost always involves volumetric and thermodynamic effects due to composition variations. For concentrations greater than a few mole percent of A and B , corrections are needed to obtain the true diffusivity. Furthermore, there are many conditions that do not fit any of the correlations presented here. Thus, careful consideration is needed to produce a reasonable estimate. Again, if diffusivity data are available at the conditions of interest, then they are strongly preferred over the predictions of any correlations. Experimental values for liquid mixtures are listed in Table 2-325.

Stokes-Einstein and Free-Volume Theories The starting point for many correlations is the Stokes-Einstein equation. This equation is derived from continuum fluid mechanics and classical thermodynamics for the motion of large spherical particles in a liquid. For this case, the need for a molecular theory is cleverly avoided. The Stokes-Einstein equation is (Bird et al.)

$$D_{AB} = \frac{kT}{6\pi r_A \mu_B} \quad (5-217)$$

where A refers to the solute and B refers to the solvent. This equation is applicable to very large unhydrated molecules ($M > 1000$) in low-molecular-weight solvents or where the molar volume of the solute is greater than $500 \text{ cm}^3/\text{mol}$ (Reddy and Doraiswamy, *Ind. Eng. Chem. Fundam.*, **6**, 77 (1967); Wilke and Chang [30]). Despite its intellectual appeal, this equation is seldom used "as is." Rather, the following principles have been identified: (1) The diffusion coefficient is inversely proportional to the size $r_A \approx V_A^{1/3}$ of the solute molecules. Experimental observations, however, generally indicate that the exponent of the solute molar volume is larger than one-third. (2) The term $D_{AB}\mu_B/T$ is approximately constant only over a 10-to-15 K interval. Thus, the dependence of liquid diffusivity on properties and conditions does not generally obey the interactions implied by that grouping. For example, Robinson, Edmister, and Dullien [*Ind. Eng. Chem. Fundam.*, **5**, 75 (1966)] found that $\ln D_{AB} \propto -1/T$. (3) Finally, pressure does not affect liquid-phase diffusivity much, since μ_B and V_A are only weakly pressure-dependent. Pressure does have an impact at very high levels.

TABLE 5-13 Relationships for Diffusivities of Multicomponent Gas Mixtures at Low Pressure

Authors*	Equation
Stefan-Maxwell, Smith and Taylor [23]	$D_{im} = \left[1 - x_i \left(\sum_{j=1}^{N_C} N_j \right) / N_i \right] \left[\sum_{j=1}^{N_C} \left[\left(x_j - \frac{x_i N_i}{N_i} \right) / D_{ij} \right] \right]^{-1} \quad (5-214)$
Blanc [2]	$D_{im} = \left(\sum_{j=1}^{N_C} \frac{x_j}{D_{ij}} \right)^{-1} \quad (5-215)$
Wilke [29]	$D_{im} = \left(\sum_{j=1, j \neq i}^{N_C} \frac{x_j}{D_{ij}} \right)^{-1} \quad (5-216)$

*References are listed at the beginning of the "Mass Transfer" subsection.

Another advance in the concepts of liquid-phase diffusion was provided by Hildebrand [*Science*, **174**, 490 (1971)] who adapted a theory of viscosity to self-diffusivity. He postulated that $D_{A'A} = B(V - V_{ms})/V_{ms}$, where $D_{A'A}$ is the self-diffusion coefficient, V is the molar volume, and V_{ms} is the molar volume at which fluidity is zero (i.e., the molar volume of the solid phase at the melting temperature). The difference $(V - V_{ms})$ can be thought of as the free volume, which increases with temperature; and B is a proportionality constant.

Ertl and Dullien (ibid.) found that Hildebrand's equation could not fit their data with B as a constant. They modified it by applying an empirical exponent n (a constant greater than unity) to the volumetric ratio. The new equation is not generally useful, however, since there is no means for predicting n . The theory does identify the free volume as an important physical variable, since $n > 1$ for most liquids implies that diffusion is more strongly dependent on free volume than is viscosity.

Dilute Binary Nonelectrolytes: General Mixtures These correlations are outlined in Table 5-14.

Wilke-Chang [30] This correlation for D_{AB}° is one of the most widely used, and it is an empirical modification of the Stokes-Einstein equation. It is not very accurate, however, for water as the solute. Otherwise, it applies to diffusion of very dilute A in B . The average absolute error for 251 different systems is about 10 percent. ϕ_B is an association factor of solvent B that accounts for hydrogen bonding.

Component B	ϕ_B
Water	2.26
Methanol	1.9
Ethanol	1.5
Propanol	1.2
Others	1.0

The value of ϕ_B for water was originally stated as 2.6, although when the original data were reanalyzed, the empirical best fit was 2.26. Random comparisons of predictions with 2.26 versus 2.6 show no consistent advantage for either value, however. Kooijman [*Ind. Eng. Chem. Res.* **41**, 3326 (2002)] suggests replacing V_A with $\theta_A V_A$, in which $\theta_A = 1$ except when $A = \text{water}$, $\theta_A = 4.5$. This modification leads to an overall error of 8.7 percent for 41 cases he compared. He suggests retaining $\Phi_B = 2.6$ when $B = \text{water}$. It has been suggested to replace the exponent of 0.6 with 0.7 and to use an association factor of 0.7 for systems containing aromatic hydrocarbons. These modifications, however, are not recommended by Umes and Danner [27]. Lees and Sarram [*J. Chem. Eng. Data*, **16**, 41 (1971)] present a comparison of the association parameters. The average absolute error for 87 different solutes in water is 5.9 percent.

Tyn-Calus [26] This correlation requires data in the form of molar volumes and parachors $\psi_i = V_i \sigma^{1/4}$ (a property which, over moderate temperature ranges, is nearly constant), measured at the same temperature (not necessarily the temperature of interest). The parachors for the components may also be evaluated at different temperatures from each other. Quale [*Chem. Rev.* **53**, 439 (1953)] has compiled values of ψ_i for many chemicals. Group contribution methods are available for estimation purposes (Poling et al.). The following suggestions were made by Poling et al.: The correlation is constrained to cases in which $\mu_B < 30$ cP. If the solute is water or if the solute is an organic acid and the solvent is not water or a short-chain alcohol, dimerization of the solute A should be assumed for purposes of estimating its volume and parachor. For example, the appropriate values for water as solute at 25°C are $V_W = 37.4$ cm³/mol and $\psi_W = 105.2$ cm³/s^{1/4}/mol. Finally, if the solute is nonpolar, the solvent volume and parachor should be multiplied by $\delta\mu_B$. According to Kooijman (ibid.), if the Brock-Bird method (described in Poling et al.) is used to

TABLE 5-14 Correlations for Diffusivities of Dilute, Binary Mixtures of Nonelectrolytes in Liquids

Authors*	Equation	Error
1. General Mixtures		
Wilke-Chang [30]	$D_{AB}^\circ = \frac{7.4 \times 10^{-8} (\phi_B M_B)^{1/2} T}{\mu_B V_A^{0.6}} \quad (5-218)$	20%
Tyn-Calus [26]	$D_{AB}^\circ = \frac{8.93 \times 10^{-8} (V_A/V_B^2)^{1/6} (\psi_B/\psi_A)^{0.6} T}{\mu_B} \quad (5-219)$	10%
Umes-Danner [27]	$D_{AB}^\circ = \frac{2.75 \times 10^{-8} (R_B/R_A^{2/3}) T}{\mu_B} \quad (5-220)$	16%
Siddiqi-Lucas [22]	$D_{AB}^\circ = \frac{9.89 \times 10^{-8} V_B^{0.265} T}{V_A^{0.45} \mu_B^{0.907}} \quad (5-221)$	13%
2. Gases in Low Viscosity Liquids		
Sridhar-Potter [24]	$D_{AB}^\circ = D_{BB} \left(\frac{V_{cB}}{V_{cA}} \right)^{2/3} \left(\frac{V_B}{V_{mB}} \right) \quad (5-222)$	18%
Chen-Chen [7]	$D_{AB}^\circ = 2.018 \times 10^{-9} \frac{(\beta V_{cB})^{2/3} (RT_{cB})^{1/2}}{M_A^{1/6} (M_B V_{cA})^{1/3}} (V_r - 1) \left(\frac{T}{T_{cB}} \right)^{1/2} \quad (5-223)$	6%
3. Aqueous Solutions		
Hayduk-Laudie [11]	$D_{AW}^\circ = \frac{13.16 \times 10^{-5}}{\mu_w^{1.14} V_A^{0.589}} \quad (5-224)$	18%
Siddiqi-Lucas [22]	$D_{AW}^\circ = 2.98 \times 10^{-7} V_A^{-0.5473} \mu_w^{-1.026} T \quad (5-225)$	13%
4. Hydrocarbon Mixtures		
Hayduk-Minhas [12]	$D_{AB}^\circ = 13.3 \times 10^{-8} T^{1.47} \mu_B^{(10.2/V_A - 0.791)} V_A^{-0.71} \quad (5-226)$	5%
Matthews-Akgerman [19]	$D_{AB}^\circ = 32.88 M_A^{-0.61} V_D^{-1.04} T^{0.5} (V_B - V_D) \quad (5-227)$	5%
Riazi-Whitson [21]	$D_{AB} = 1.07 \frac{(\rho D_{AB})^\circ}{\rho} \left(\frac{\mu}{\mu^\circ} \right)^{-0.27 - 0.38 \omega + (-0.05 + 0.1 \omega) P_r} \quad (5-228)$	15%

*References are listed at the beginning of the "Mass Transfer" subsection.

estimate the surface tension, the error is only increased by about 2 percent, relative to employing experimentally measured values.

Umesi-Danner [27] They developed an equation for nonaqueous solvents with nonpolar and polar solutes. In all, 258 points were involved in the regression. R_i is the radius of gyration in Å of the component molecule, which has been tabulated by Passut and Danner [Chem. Eng. Progress Symp. Ser., **140**, 30 (1974)] for 250 compounds. The average absolute deviation was 16 percent, compared with 26 percent for the Wilke-Chang equation.

Siddiqi-Lucas [22] In an impressive empirical study, these authors examined 1275 organic liquid mixtures. Their equation yielded an average absolute deviation of 13.1 percent, which was less than that for the Wilke-Chang equation (17.8 percent). Note that this correlation does not encompass aqueous solutions; those were examined and a separate correlation was proposed, which is discussed later.

Binary Mixtures of Gases in Low-Viscosity, Nonelectrolyte Liquids Sridhar and Potter [24] derived an equation for predicting gas diffusion through liquid by combining existing correlations. Hildebrand had postulated the following dependence of the diffusivity for a gas in a liquid: $D_{AB}^0 = D_{BB}^0 (V_{cB}/V_{cA})^{2/3}$, where D_{BB}^0 is the solvent self-diffusion coefficient and V_{cA} is the critical volume of component i , respectively. To correct for minor changes in volumetric expansion, Sridhar and Potter multiplied the resulting equation by V_B/V_{mlB} , where V_{mlB} is the molar volume of the liquid B at its melting point and D_{BB}^0 can be estimated by the equation of Ertl and Dullien (see p. 5-54). Sridhar and Potter compared experimentally measured diffusion coefficients for twenty-seven data points of eleven binary mixtures. Their average absolute error was 13.5 percent, but Chen and Chen [7] analyzed about 50 combinations of conditions and 3 to 4 replicates each and found an average error of 18 percent. This correlation does not apply to hydrogen and helium as solutes. However, it demonstrates the usefulness of self-diffusion as a means to assess mutual diffusivities and the value of observable physical property changes, such as molar expansion, to account for changes in conditions.

Chen-Chen [7] Their correlation was based on diffusion measurements of 50 combinations of conditions with 3 to 4 replicates each and exhibited an average error of 6 percent. In this correlation, $V_r = V_B/[0.9724 (V_{mlB} + 0.04765)]$ and V_{mlB} is the liquid molar volume at the melting point, as discussed previously. Their association parameter β [which is different from the definition of that symbol in Eq. (5-229)] accounts for hydrogen bonding of the solvent. Values for acetonitrile and methanol are: $\beta = 1.58$ and 2.31 , respectively.

Dilute Binary Mixtures of a Nonelectrolyte in Water The correlations that were suggested previously for general mixtures, unless specified otherwise, may also be applied to diffusion of miscellaneous solutes in water. The following correlations are restricted to the present case, however.

Hayduk and Laudie [11] They presented a simple correlation for the infinite dilution diffusion coefficients of nonelectrolytes in water. It has about the same accuracy as the Wilke-Chang equation (about 5.9 percent). There is no explicit temperature dependence, but the 1.14 exponent on μ_w compensates for the absence of T in the numerator. That exponent was misprinted (as 1.4) in the original article and has been reproduced elsewhere erroneously.

Siddiqi and Lucas [227] These authors examined 658 aqueous liquid mixtures in an empirical study. They found an average absolute deviation of 19.7 percent. In contrast, the Wilke-Chang equation gave 35.0 percent and the Hayduk-Laudie correlation gave 30.4 percent.

Dilute Binary Hydrocarbon Mixtures Hayduk and Minhas [12] presented an accurate correlation for normal paraffin mixtures that was developed from 58 data points consisting of solutes from C_5 to C_{32} and solvents from C_5 to C_{16} . The average error was 3.4 percent for the 58 mixtures.

Matthews and Akgerman [19] The free-volume approach of Hildebrand was shown to be valid for binary, dilute liquid paraffin mixtures (as well as self-diffusion), consisting of solutes from C_8 to C_{16} and solvents of C_6 and C_{12} . The term they referred to as the "diffusion volume" was simply correlated with the critical volume, as $V_D = 0.308 V_c$. We can infer from Table 5-11 that this is approximately related to the volume at the melting point as $V_D = 0.945 V_m$. Their correlation was valid for diffusion of linear alkanes at temperatures

up to 300°C and pressures up to 3.45 MPa. Matthews, Rodden, and Akgerman [J. Chem. Eng. Data, **32**, 317 (1987)] and Erkey and Akgerman [AIChE J., **35**, 443 (1989)] completed similar studies of diffusion of alkanes, restricted to n -hexadecane and n -octane, respectively, as the solvents.

Riazi and Whitson [21] They presented a generalized correlation in terms of viscosity and molar density that was applicable to both gases and liquids. The average absolute deviation for gases was only about 8 percent, while for liquids it was 15 percent. Their expression relies on the Chapman-Enskog correlation [Eq. (5-202)] for the low-pressure diffusivity and the Stiel-Thodos [AIChE J., **7**, 234 (1961)] correlation for low-pressure viscosity:

$$\mu^0 = \frac{x_A \mu_A^0 M_A^{1/2} + x_B \mu_B^0 M_B^{1/2}}{x_A M_A^{1/2} + x_B M_B^{1/2}}$$

where $\mu_i^0 \xi_i = 3.4 \times 10^{-4} T_{r,i}^{0.94}$ for $T_{r,i} < 1.5$ or $\mu_i^0 \xi_i = 1.778 \times 10^{-4} (4.58 T_{r,i} - 1.67)^{0.58}$ for $T_{r,i} > 1.5$. In these equations, $\xi_i = T_{c,i}^{1/6} / P_{c,i}^{2/3} M_i^{1/2}$, and units are in cP, atm, K, and mol. For dense gases or liquids, the Chung et al. [Ind. Eng. Chem. Res., **27**, 671 (1988)] or Joshi-Stiel-Thodos [AIChE J., **8**, 59 (1962)] correlation may be used to estimate viscosity. The latter is:

$$(\mu - \mu^0) \xi + 10^{-4} = (0.1023 + 0.023364 p_r + 0.058533 p_r^2 - 0.040758 p_r^3 + 0.093324 p_r^4)$$

$$\text{where } \xi = \frac{(x_A T_{cA} + x_B T_{cB})^{1/6}}{(x_A M_A + x_B M_B)^{1/2} (x_A P_{cA} + x_B P_{cB})}$$

$$\text{and } p_r = (x_A V_{cA} + x_B V_{cB}) p_r$$

Dilute Binary Mixtures of Nonelectrolytes with Water as the Solute Olander [AIChE J., **7**, 175 (1961)] modified the Wilke-Chang equation to adapt it to the infinite dilution diffusivity of water as the solute. The modification he recommended is simply the division of the right-hand side of the Wilke-Chang equation by 2.3. Unfortunately, neither the Wilke-Chang equation nor that equation divided by 2.3 fit the data very well. A reasonably valid generalization is that the Wilke-Chang equation is accurate if water is very insoluble in the solvent, such as pure hydrocarbons, halogenated hydrocarbons, and nitro-hydrocarbons. On the other hand, the Wilke-Chang equation divided by 2.3 is accurate for solvents in which water is very soluble, as well as those that have low viscosities. Such solvents include alcohols, ketones, carboxylic acids, and aldehydes. Neither equation is accurate for higher-viscosity liquids, especially diols.

Dilute Dispersions of Macromolecules in Nonelectrolytes The Stokes-Einstein equation has already been presented. It was noted that its validity was restricted to large solutes, such as spherical macromolecules and particles in a continuum solvent. The equation has also been found to predict accurately the diffusion coefficient of spherical latex particles and globular proteins. Corrections to Stokes-Einstein for molecules approximating spheroids is given by Tanford *Physical Chemistry of Macromolecules*, Wiley, New York, (1961). Since solute-solute interactions are ignored in this theory, it applies in the dilute range only.

Hiss and Cussler [AIChE J., **19, 698 (1973)]** Their basis is the diffusion of a small solute in a fairly viscous solvent of relatively large molecules, which is the opposite of the Stokes-Einstein assumptions. The large solvent molecules investigated were not polymers or gels but were of moderate molecular weight so that the macroscopic and microscopic viscosities were the same. The major conclusion is that $D_{AB}^0 \mu^{2/3} = \text{constant}$ at a given temperature and for a solvent viscosity from 5×10^{-3} to 5 Pa·s or greater (5 to 5×10^3 cP). This observation is useful if D_{AB}^0 is known in a given high-viscosity liquid (oils, tars, etc.). Use of the usual relation of $D_{AB}^0 \propto 1/\mu$ for such an estimate could lead to large errors.

Concentrated, Binary Mixtures of Nonelectrolytes Several correlations that predict the composition dependence of D_{AB} are summarized in Table 5-15. Most are based on known values of D_{AB}^0 and D_{BA}^0 . In fact, a rule of thumb states that, for many binary systems, D_{AB}^0 and D_{BA}^0 bound the D_{AB} vs. x_A curve. Cullinan's [8] equation predicts diffusivities even in lieu of values at infinite dilution, but requires accurate density, viscosity, and activity coefficient data.

TABLE 5-15 Correlations of Diffusivities for Concentrated, Binary Mixtures of Nonelectrolyte Liquids

Authors*	Equation
Caldwell-Babb [5]	$D_{AB} = (x_A D_{BA}^\circ + x_B D_{AB}^\circ) \beta_A$ (5-231)
Rathbun-Babb [20]	$D_{AB} = (x_A D_{BA}^\circ + x_B D_{AB}^\circ) \beta_A^n$ (5-232)
Vignes [28]	$D_{AB} = D_{AB}^{\circ \text{ss}} D_{BA}^{\circ \text{ss}} \beta_A$ (5-233)
Leffler-Cullinan [16]	$D_{AB} \mu_{\text{mix}} = (D_{AB}^\circ \mu_B)^{x_B} (D_{BA}^\circ \mu_A)^{x_A} \beta_A$ (5-234)
Cussler [9]	$D_{AB} = D_0 \left[1 + \frac{K}{x_A x_B} \left(\frac{\partial \ln x_A}{\partial \ln a_A} - 1 \right) \right]^{-1/2}$ (5-235)
Cullinan [8]	$D_{AB} = \frac{kT}{2\pi \mu_{\text{mix}} (V/A)^{1/3}} \left[\frac{2\pi x_A x_B \beta_A}{1 + \beta_A (2\pi x_A x_B - 1)} \right]^{1/2}$ (5-236)
Asfour-Dullien [1]	$D_{AB} = \left(\frac{D_{AB}^\circ}{\mu_B} \right)^{x_B} \left(\frac{D_{BA}^\circ}{\mu_A} \right)^{x_A} \zeta \mu \beta_A$ (5-237)
Siddiqi-Lucas [22]	$D_{AB} = (C_B \bar{V}_B D_{AB}^\circ + C_A \bar{V}_A D_{BA}^\circ) \beta_A$ (5-238)
Bosse and Bart no. 1 [3]	$D_{AB} = (D_{AB}^\circ)^{x_B} (D_{BA}^\circ)^{x_A} \exp \left(-\frac{g^E}{RT} \right)$ (5-239)
Bosse and Bart no. 2 [3]	$\mu D_{AB} = (\mu_B D_{AB}^\circ)^{x_B} (\mu_A D_{BA}^\circ)^{x_A} \exp \left(-\frac{g^E}{RT} \right)$ (5-240)

Relative errors for the correlations in this table are very dependent on the components of interest and are cited in the text.

*See the beginning of the "Mass Transfer" subsection for references.

Since the infinite dilution values D_{AB}° and D_{BA}° are generally unequal, even a thermodynamically ideal solution like $\gamma_A = \gamma_B = 1$ will exhibit concentration dependence of the diffusivity. In addition, non-ideal solutions require a thermodynamic correction factor to retain the true "driving force" for molecular diffusion, or the gradient of the chemical potential rather than the composition gradient. That correction factor is:

$$\beta_A = 1 + \frac{\partial \ln \gamma_A}{\partial \ln x_A} \quad (5-229)$$

Caldwell and Babb [5] Darken [*Trans. Am. Inst. Mining Met. Eng.*, **175**, 184 (1948)] observed that solid-state diffusion in metallurgical applications followed a simple relation. His equation related the tracer diffusivities and mole fractions to the mutual diffusivity:

$$D_{AB} = (x_A D_B + x_B D_A) \beta_A \quad (5-230)$$

Caldwell and Babb used virtually the same equation to evaluate the mutual diffusivity for concentrated mixtures of common liquids.

Van Geet and Adamson [*J. Phys. Chem.*, **68**, 238 (1964)] tested that equation for the *n*-dodecane (*A*) and *n*-octane (*B*) system and found the average deviation of D_{AB} from experimental values to be -0.68 percent. In addition, that equation was tested for benzene + bromobenzene, *n*-hexane + *n*-dodecane, benzene + CCl_4 , octane + decane, heptane + cetane, benzene + diphenyl, and benzene + nitromethane with success. For systems that depart significantly from thermodynamic ideality, it breaks down, sometimes by a factor of eight. For example, in the binary systems acetone + CCl_4 , acetone + chloroform, and ethanol + CCl_4 , it is not accurate. Thus, it can be expected to be fairly accurate for nonpolar hydrocarbons of similar molecular weight but not for polar-polar mixtures. Siddiqi, Krahn, and Lucas [*J. Chem. Eng. Data*, **32**, 48 (1987)] found that this relation was superior to those of Vignes and Leffler and Cullinan for a variety of mixtures. Umesi and Danner [27] found an average absolute deviation of 13.9 percent for 198 data points.

Rathbun and Babb [20] suggested that Darken's equation could be improved by raising the thermodynamic correction factor β_A to a power, *n*, less than unity. They looked at systems exhibiting negative deviations from Raoult's law and found $n = 0.3$. Furthermore, for polar-nonpolar mixtures, they found $n = 0.6$. In a separate study, Siddiqi and Lucas [22] followed those suggestions and found an average absolute error of 3.3 percent for nonpolar-nonpolar mixtures, 11.0 percent for polar-nonpolar mixtures, and 14.6 percent for polar-polar mixtures. Siddiqi, Krahn, and Lucas (*ibid.*) examined a few other mixtures and

found that $n = 1$ was probably best. Thus, this approach is, at best, highly dependent on the type of components being considered.

Vignes [28] empirically correlated mixture diffusivity data for 12 binary mixtures. Later Ertl, Ghai, and Dollon [*AIChE J.*, **20**, 1 (1974)] evaluated 122 binary systems, which showed an average absolute deviation of only 7 percent. None of the latter systems, however, was very nonideal.

Leffler and Cullinan [16] modified Vignes' equation using some theoretical arguments to arrive at Eq. (5-234), which the authors compared to Eq. (5-233) for the 12 systems mentioned above. The average absolute maximum deviation was only 6 percent. Umesi and Danner [27], however, found an average absolute deviation of 11.4 percent for 198 data points. For normal paraffins, it is not very accurate. In general, the accuracies of Eqs. (5-233) and (5-234) are not much different, and since Vignes' is simpler to use, it is suggested. The application of either should be limited to nonassociating systems that do not deviate much from ideality ($0.95 < \beta_A < 1.05$).

Cussler [9] studied diffusion in concentrated associating systems and has shown that, in associating systems, it is the size of diffusing clusters rather than diffusing solutes that controls diffusion. D_0 is a reference diffusion coefficient discussed hereafter; a_A is the activity of component A; and K is a constant. By assuming that D_0 could be predicted by Eq. (5-233) with $\beta = 1$, K was found to be equal to 0.5 based on five binary systems and validated with a sixth binary mixture. The limitations of Eq. (5-235) using D_0 and K defined previously have not been explored, so caution is warranted. Gurkan [*AIChE J.*, **33**, 175 (1987)] showed that K should actually be closer to 0.3 (rather than 0.5) and discussed the overall results.

Cullinan [8] presented an extension of Cussler's cluster diffusion theory. His method accurately accounts for composition and temperature dependence of diffusivity. It is novel in that it contains no adjustable constants, and it relates transport properties and solution thermodynamics. This equation has been tested for six very different mixtures by Rollins and Knaebel [*AIChE J.*, **37**, 470 (1991)], and it was found to agree remarkably well with data for most conditions, considering the absence of adjustable parameters. In the dilute region (of either A or B), there are systematic errors probably caused by the breakdown of certain implicit assumptions (that nevertheless appear to be generally valid at higher concentrations).

Asfour and Dullien [1] developed a relation for predicting alkane diffusivities at moderate concentrations that employs:

$$\zeta = \left(\frac{V_m}{V_{fA} V_{fB}} \right)^{2/3} \frac{M_{xA} M_{xB}}{M_m} \quad (5-241)$$

where $V_{f_i} = V_{\beta_i}$; the fluid free volume is $V_{f_i} = V_i - V_{m,i}$ for $i = A, B$, and m , in which $V_{m,i}$ is the molar volume of the liquid at the melting point and

$$V_{m,m} = \left(\frac{x_A^2}{V_{m,A}} + \frac{2x_A x_B}{V_{m,AB}} + \frac{x_B^2}{V_{m,B}} \right)^{-1}$$

and

$$V_{m,AB} = \left[\frac{V_{m,A}^{1/3} + V_{m,B}^{1/3}}{2} \right]^3$$

and μ is the mixture viscosity; M_m is the mixture mean molecular weight; and β_A is defined by Eq. (5-229). The average absolute error of this equation is 1.4 percent, while the Vignes equation and the Lefler-Cullinan equation give 3.3 percent and 6.2 percent, respectively.

Siddiqi and Lucas [22] suggested that component volume fractions might be used to correlate the effects of concentration dependence. They found an average absolute deviation of 4.5 percent for nonpolar-nonpolar mixtures, 16.5 percent for polar-nonpolar mixtures, and 10.8 percent for polar-polar mixtures.

Bosse and Bart added a term to account for excess Gibbs free energy, involved in the activation energy for diffusion, which was previously omitted. Doing so yielded minor modifications of the Vignes and Lefler-Cullinan equations [Eqs. (5-233) and (5-234), respectively]. The UNIFAC method was used to assess the excess Gibbs free energy. Comparing predictions of the new equations with data for 36 pairs and 326 data points yielded relative deviations of 7.8 percent and 8.9 percent, respectively, but which were better than the closely related Vignes (12.8 percent) and Lefler-Cullinan (10.4 percent) equations.

Binary Electrolyte Mixtures When electrolytes are added to a solvent, they dissociate to a certain degree. It would appear that the solution contains at least three components: solvent, anions, and cations. If the solution is to remain neutral in charge at each point (assuming the absence of any applied electric potential field), the anions and cations diffuse effectively as a single component, as for molecular diffusion. The diffusion of the anionic and cationic species in the solvent can thus be treated as a binary mixture.

Nernst-Haskell The theory of dilute diffusion of salts is well developed and has been experimentally verified. For dilute solutions of a single salt, the well-known Nernst-Haskell equation (Poling et al.) is applicable:

$$D_{AB}^{\circ} = \frac{RT}{F^2} \frac{\left| \frac{1}{n_+} + \frac{1}{n_-} \right|}{\frac{1}{\lambda_+^{\circ}} + \frac{1}{\lambda_-^{\circ}}} = 8.9304 \times 10^{-10} T \frac{\left| \frac{1}{n_+} + \frac{1}{n_-} \right|}{\frac{1}{\lambda_+^{\circ}} + \frac{1}{\lambda_-^{\circ}}} \quad (5-242)$$

where D_{AB}° = diffusivity based on molarity rather than normality of dilute salt A in solvent B, cm^2/s .

The previous definitions can be interpreted in terms of ionic-species diffusivities and conductivities. The latter are easily measured and depend on temperature and composition. For example, the equivalent conductance Λ is commonly tabulated in chemistry handbooks as the limiting (infinite dilution) conductance Λ_0 and at standard concentrations, typically at 25°C. $\Lambda = 1000K/C = \lambda_+ + \lambda_- = \Lambda_0 + f(C)$, ($\text{cm}^2/\text{ohm gequiv}$); $K = \alpha/R$ = specific conductance, (ohm cm^{-1}); C = solution concentration, (gequiv/ℓ); α = conductance cell constant (measured), (cm^{-1}); R = solution electrical resistance, which is measured (ohm); and $f(C)$ = a complicated function of concentration. The resulting equation of the electrolyte diffusivity is

$$D_{AB} = \frac{|\lambda_+| + |\lambda_-|}{(|\lambda_-|/D_+) + (|\lambda_+|/D_-)} \quad (5-243)$$

where $|\lambda_{\pm}|$ represents the magnitude of the ionic charge and where the cationic or anionic diffusivities are $D_{\pm} = 8.9304 \times 10^{-10} T \lambda_{\pm} / |\lambda_{\pm}| \text{ cm}^2/\text{s}$. The coefficient is $kN_0/F^2 = R/F^2$. In practice, the equivalent conductance of the ion pair of interest would be obtained and supplemented with conductances of permutations of those ions and one independent cation and anion. This would allow determination of all the ionic conductances and hence the diffusivity of the electrolyte solution.

Gordon [J. Phys. Chem. **5**, 522 (1937)] Typically, as the concentration of a salt increases from infinite dilution, the diffusion coefficient

decreases rapidly from D_{AB}° . As concentration is increased further, however, D_{AB} rises steadily, often becoming greater than D_{AB}° . Gordon proposed the following empirical equation, which is applicable up to concentrations of 2N:

$$D_{AB} = D_{AB}^{\circ} \frac{1}{C_B \bar{V}_B} \frac{\mu_B}{\mu} \left(1 + \frac{\ln \gamma_{\pm}}{\ln m} \right) \quad (5-244)$$

where D_{AB}° is given by the Nernst-Haskell equation. References that tabulate $\gamma_{\pm}$ as a function of m , as well as other equations for D_{AB} , are given by Poling et al.

Morgan, Ferguson, and Scovazzo [Ind. Eng. Chem. Res. **44**, 4815 (2005)] They studied diffusion of gases in ionic liquids having moderate to high viscosity (up to about 1000 cP) at 30°C. Their range was limited, and the empirical equation they found was

$$D_{AB} = 3.7 \times 10^{-3} \left(\frac{1}{\mu_B^{0.59} V_{AP,B}^2} \right) \quad (5-245)$$

which yielded a correlation coefficient of 0.975. Of the estimated diffusivities 90 percent were within ± 20 percent of the experimental values. The exponent for viscosity approximately confirmed the observation of Hiss and Cussler (ibid).

Multicomponent Mixtures No simple, practical estimation methods have been developed for predicting multicomponent liquid-diffusion coefficients. Several theories have been developed, but the necessity for extensive activity data, pure component and mixture volumes, mixture viscosity data, and tracer and binary diffusion coefficients have significantly limited the utility of the theories (see Poling et al.).

The generalized Stefan-Maxwell equations using binary diffusion coefficients are not easily applicable to liquids since the coefficients are so dependent on conditions. That is, in liquids, each D_{ij} can be strongly composition dependent in binary mixtures and, moreover, the binary D_{ij} is strongly affected in a multicomponent mixture. Thus, the convenience of writing multicomponent flux equations in terms of binary coefficients is lost. Conversely, they apply to gas mixtures because each D_{ij} is practically independent of composition by itself and in a multicomponent mixture (see Taylor and Krishna for details).

One particular case of multicomponent diffusion that has been examined is the dilute diffusion of a solute in a homogeneous mixture (e.g., of A in B + C). Umesi and Danner [27] compared the three equations given below for 49 ternary systems. All three equations were equivalent, giving average absolute deviations of 25 percent.

Perkins and Geankoplis [Chem. Eng. Sci., **24**, 1035 (1969)]

$$D_{Am} \mu_m^{0.8} = \sum_{j=1}^n \sum_{j \neq A} x_j D_{Aj}^{\circ} \mu_j^{0.8} \quad (5-246)$$

Cullinan [Can. J. Chem. Eng. **45**, 377 (1967)] This is an extension of Vignes' equation to multicomponent systems:

$$D_{Am} = \prod_{j=1}^n (D_{Aj}^{\circ})^{x_j} \quad (5-247)$$

Lefler and Cullinan [16] They extended their binary relation to an arbitrary multicomponent mixture, as follows:

$$D_{Am} \mu_m = \prod_{j=1}^n (D_{Aj}^{\circ} \mu_j)^{x_j} \quad (5-248)$$

where D_{Aj} is the dilute binary diffusion coefficient of A in j; D_{Am} is the dilute diffusion of A through m; x_j is the mole fraction; μ_j is the viscosity of component j; and μ_m is the mixture viscosity.

Akita [Ind. Eng. Chem. Fundam., **10**, 89 (1981)] Another case of multicomponent dilute diffusion of significant practical interest is that of gases in aqueous electrolyte solutions. Many gas-absorption processes use electrolyte solutions. Akita presents experimentally tested equations for this case.

Graham and Dranoff [Ind. Eng. Chem. Fundam., **21**, 360 and 365 (1982)] They studied multicomponent diffusion of electrolytes in ion exchangers. They found that the Stefan-Maxwell interaction coefficients reduce to limiting ion tracer diffusivities of each ion.

Pinto and Graham [AIChE J. **32**, 291 (1986) and **33**, 436 (1987)] They studied multicomponent diffusion in electrolyte solutions. They focused on the Stefan-Maxwell equations and corrected for solvation effects. They achieved excellent results for 1-1 electrolytes in water at 25°C up to concentrations of 4M.

Anderko and Lencka [Ind. Eng. Chem. Res. **37**, 2878 (1998)] These authors present an analysis of self-diffusion in multicomponent aqueous electrolyte systems. Their model includes contributions of long-range (Coulombic) and short-range (hard-sphere) interactions. Their mixing rule was based on equations of nonequilibrium thermodynamics. The model accurately predicts self-diffusivities of ions and gases in aqueous solutions from dilute to about 30 mol/kg water. It makes it possible to take single-solute data and extend them to multicomponent mixtures.

DIFFUSION OF FLUIDS IN POROUS SOLIDS

Diffusion in porous solids is usually the most important factor controlling mass transfer in adsorption, ion exchange, drying, heterogeneous catalysis, leaching, and many other applications. Some of the applications of interest are outlined in Table 5-16. Applications of these equations are found in Secs. 16, 22, and 23.

Diffusion within the largest cavities of a porous medium is assumed to be similar to ordinary or bulk diffusion except that it is hindered by the pore walls (see Eq. 5-249). The tortuosity τ that expresses this hindrance has been estimated from geometric arguments. Unfortunately,

measured values are often an order of magnitude greater than those estimates. Thus, the effective diffusivity D_{eff} (and hence τ) is normally determined by comparing a diffusion model to experimental measurements. The normal range of tortuosities for silica gel, alumina, and other porous solids is $2 \leq \tau \leq 6$, but for activated carbon, $5 \leq \tau \leq 65$.

In small pores and at low pressures, the mean free path ℓ of the gas molecule (or atom) is significantly greater than the pore diameter d_{pore} . Its magnitude may be estimated from

$$\ell = \frac{3.2 \mu}{P} \left(\frac{RT}{2\pi M} \right)^{1/2} \text{ m}$$

As a result, collisions with the wall occur more frequently than with other molecules. This is referred to as the Knudsen mode of diffusion and is contrasted with ordinary or bulk diffusion, which occurs by intermolecular collisions. At intermediate pressures, both ordinary diffusion and Knudsen diffusion may be important [see Eqs. (5-252) and (5-253)].

For gases and vapors that adsorb on the porous solid, surface diffusion may be important, particularly at high surface coverage [see Eqs. (5-254) and (5-257)]. The mechanism of surface diffusion may be viewed as molecules hopping from one surface site to another. Thus, if adsorption is too strong, surface diffusion is impeded, while if adsorption is too weak, surface diffusion contributes insignificantly to the overall rate. Surface diffusion and bulk diffusion usually occur in parallel [see Eqs. (5-258) and (5-259)]. Although D_s is expected to be less than D_{eff} , the solute flux due to surface diffusion may be larger

TABLE 5-16 Relations for Diffusion in Porous Solids

Mechanism	Equation	Applies to	References*
Bulk diffusion in pores	$D_{\text{eff}} = \frac{\epsilon_p D}{\tau}$ (5-249)	Gases or liquids in large pores. $N_{Kn} = \ell/d_{\text{pore}} < 0.01$	[33]
Knudsen diffusion	$D_K = 48.5 d_{\text{pore}} \left(\frac{T}{M} \right)^{1/2}$ in m^2/s (5-250) $D_{K\text{eff}} = \frac{\epsilon_p D_K}{\tau}$	Dilute (low pressure) gases in small pores. $N_{Kn} = \ell/d_{\text{pore}} > 10$	Geankoplis, [34, 35]
Combined bulk and Knudsen diffusion	$N_i = -D_K \frac{dC_i}{dz}$ (5-251)	" " " "	Geankoplis, [32, 35]
	$D_{\text{eff}} = \left(\frac{1 - \alpha x_A}{D_{\text{eff}}} + \frac{1}{D_{K\text{eff}}} \right)^{-1}$ (5-252)	" " " "	
	$\alpha = 1 + \frac{N_B}{N_A}$	$N_A \neq N_B$	
	$D_{\text{eff}} = \left(\frac{1}{D_{\text{eff}}} + \frac{1}{D_{K\text{eff}}} \right)^{-1}$ (5-253)	$N_A = N_B$	
Surface diffusion	$J_{Si} = -D_{\text{Seff}} \rho_p \left(\frac{dq_i}{dz} \right)$ (5-254)	Adsorbed gases or vapors	[32, 34, 35]
	$D_{\text{Seff}} = \frac{\epsilon_p D_s}{\tau}$ (5-255)	" " " "	
	$D_{s0} = \frac{D_{s0=0}}{(1 - \theta)}$ (5-256)	θ = fractional surface coverage ≤ 0.6	
	$D_s = D'_s(q) \exp \left(\frac{-E_s}{RT} \right)$ (5-257)	" " " "	
Parallel bulk and surface diffusion	$J = - \left[D_{\text{eff}} \left(\frac{dp_i}{dz} \right) + D_{\text{Seff}} \rho_p \left(\frac{dq_i}{dz} \right) \right]$ (5-258)	" " " "	[34]
	$J = -D_{\text{app}} \left(\frac{dp_i}{dz} \right)$ (5-259)	" " " "	
	$D_{\text{app}} = D_{\text{eff}} + D_{\text{Seff}} \rho_p \left(\frac{dq_i}{dp_i} \right)$ (5-260)	" " " "	

*See the beginning of the "Mass Transfer" subsection for references.

than that due to bulk diffusion if $\partial q_i/\partial z \gg \partial C_i/\partial z$. This can occur when a component is strongly adsorbed and the surface coverage is high. For all that, surface diffusion is not well understood. The references in Table 5-16 should be consulted for further details.

INTERPHASE MASS TRANSFER

Transfer of material between phases is important in most separation processes in which two phases are involved. When one phase is pure, mass transfer in the pure phase is not involved. For example, when a pure liquid is being evaporated into a gas, only the gas-phase mass transfer need be calculated. Occasionally, mass transfer in one of the two phases may be neglected even though pure components are not involved. This will be the case when the resistance to mass transfer is much larger in one phase than in the other. Understanding the nature and magnitudes of these resistances is one of the keys to performing reliable mass transfer. In this section, mass transfer between gas and liquid phases will be discussed. The principles are easily applied to the other phases.

Mass-Transfer Principles: Dilute Systems When material is transferred from one phase to another across an interface that separates the two, the resistance to mass transfer in each phase causes a concentration gradient in each, as shown in Fig. 5-26 for a gas-liquid interface. The concentrations of the diffusing material in the two phases immediately adjacent to the interface generally are unequal, even if expressed in the same units, but usually are assumed to be related to each other by the laws of thermodynamic equilibrium. Thus, it is assumed that the thermodynamic equilibrium is reached at the gas-liquid interface almost immediately when a gas and a liquid are brought into contact.

For systems in which the solute concentrations in the gas and liquid phases are dilute, the rate of transfer may be expressed by equations which predict that the rate of mass transfer is proportional to the difference between the bulk concentration and the concentration at the gas-liquid interface. Thus

$$N_A = k'_G(p - p_i) = k'_L(c_i - c) \quad (5-261)$$

where N_A = mass-transfer rate, k'_G = gas-phase mass-transfer coefficient, k'_L = liquid-phase mass-transfer coefficient, p = solute partial pressure in bulk gas, p_i = solute partial pressure at interface, c = solute concentration in bulk liquid, and c_i = solute concentration in liquid at interface.

The mass-transfer coefficients k'_G and k'_L by definition are equal to the ratios of the molar mass flux N_A to the concentration driving forces $(p - p_i)$ and $(c_i - c)$ respectively. An alternative expression for the rate of transfer in dilute systems is given by

$$N_A = k_G(y - y_i) = k_L(x_i - x) \quad (5-262)$$

where N_A = mass-transfer rate, k_G = gas-phase mass-transfer coefficient, k_L = liquid-phase mass-transfer coefficient, y = mole-fraction

solute in bulk-gas phase, y_i = mole-fraction solute in gas at interface, x = mole-fraction solute in bulk-liquid phase, and x_i = mole-fraction solute in liquid at interface.

The mass-transfer coefficients defined by Eqs. (5-261) and (5-262) are related to each other as follows:

$$k_G = k'_G p_T \quad (5-263)$$

$$k_L = k'_L \bar{p}_L \quad (5-264)$$

where p_T = total system pressure employed during the experimental determinations of k'_G values and $\bar{p}_L$ = average molar density of the liquid phase. The coefficient k_G is relatively independent of the total system pressure and therefore is more convenient to use than k'_G , which is inversely proportional to the total system pressure.

The above equations may be used for finding the interfacial concentrations corresponding to any set of values of x and y provided the ratio of the individual coefficients is known. Thus

$$(y - y_i)/(x_i - x) = k_L/k_G = k'_L \bar{p}_L / k'_G p_T = L_M H_G / G_M H_L \quad (5-265)$$

where L_M = molar liquid mass velocity, G_M = molar gas mass velocity, H_L = height of one transfer unit based on liquid-phase resistance, and H_G = height of one transfer unit based on gas-phase resistance. The last term in Eq. (5-265) is derived from Eqs. (5-284) and (5-286).

Equation (5-265) may be solved graphically if a plot is made of the equilibrium vapor and liquid compositions and a point representing the bulk concentrations x and y is located on this diagram. A construction of this type is shown in Fig. 5-27, which represents a gas-absorption situation.

The interfacial mole fractions y_i and x_i can be determined by solving Eq. (5-265) simultaneously with the equilibrium relation $y_i^* = F(x_i)$ to obtain y_i and x_i . The rate of transfer may then be calculated from Eq. (5-262).

If the equilibrium relation $y_i^* = F(x_i)$ is sufficiently simple, e.g., if a plot of y_i^* versus x_i is a straight line, not necessarily through the origin, the rate of transfer is proportional to the difference between the bulk concentration in one phase and the concentration (in that same phase) which would be in equilibrium with the bulk concentration in the second phase. One such difference is $y - y^*$, and another is $x^* - x$. In this case, there is no need to solve for the interfacial compositions, as may be seen from the following derivation.

The rate of mass transfer may be defined by the equation

$$N_A = K_G(y - y^*) = k_G(y - y_i) = k_L(x_i - x) = K_L(x^* - x) \quad (5-266)$$

where K_G = overall gas-phase mass-transfer coefficient, K_L = overall liquid-phase mass-transfer coefficient, y^* = vapor composition in equilibrium with x , and x^* = liquid composition in equilibrium with vapor of composition y . This equation can be rearranged to the formula

$$\frac{1}{K_G} = \frac{1}{k_G} \left(\frac{y - y^*}{y - y_i} \right) = \frac{1}{k_G} + \frac{1}{k_G} \left(\frac{y_i - y^*}{y - y_i} \right) = \frac{1}{k_G} + \frac{1}{k_L} \left(\frac{y_i - y^*}{x_i - x} \right) \quad (5-267)$$

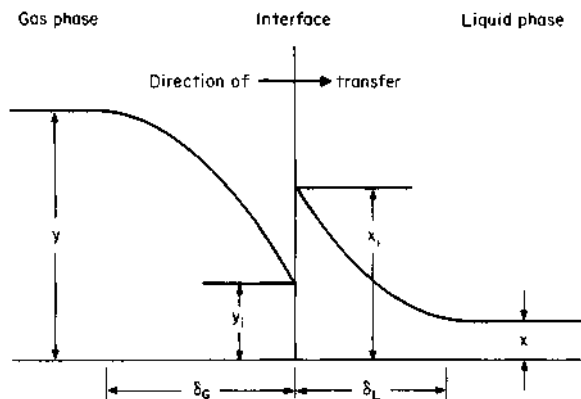


FIG. 5-26 Concentration gradients near a gas-liquid interface.

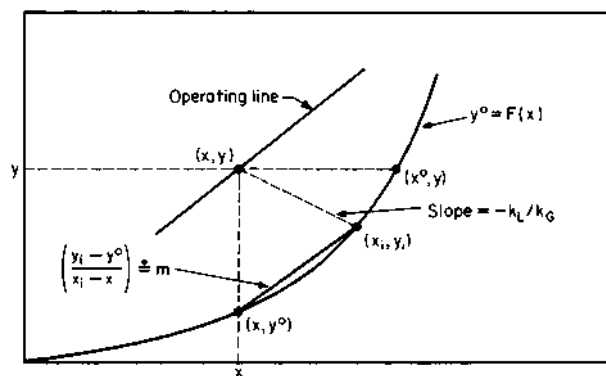


FIG. 5-27 Identification of concentrations at a point in a countercurrent absorption tower.

in view of Eq. (5-265). Comparison of the last term in parentheses with the diagram of Fig. 5-27 shows that it is equal to the slope of the chord connecting the points (x_i, y_i) and (x_i, y_i) . If the equilibrium curve is a straight line, then this term is the slope m . Thus

$$1/K_G = (1/k_G + m/k_L) \quad (5-268)$$

When Henry's law is valid ($p_A = Hx_A$ or $p_A = H'C_A$), the slope m can be computed according to the relationship

$$m = H/p_T = H'\bar{p}_L/p_T \quad (5-269)$$

where m is defined in terms of mole-fraction driving forces compatible with Eqs. (5-262) through (5-268), i.e., with the definitions of k_L , k_G , and K_G .

If it is desired to calculate the rate of transfer from the overall concentration difference based on bulk-liquid compositions ($x^\circ - x$), the appropriate overall coefficient K_L is related to the individual coefficients by the equation

$$1/K_L = [1/k_L + 1/(mk_G)] \quad (5-270)$$

Conversion of these equations to a k'_G, k'_L basis can be accomplished readily by direct substitution of Eqs. (5-263) and (5-264).

Occasionally one will find k'_L or K'_L values reported in units (SI) of meters per second. The correct units for these values are $\text{kmol}/[(\text{s}\cdot\text{m}^2)(\text{kmol}/\text{m}^3)]$, and Eq. (5-264) is the correct equation for converting them to a mole-fraction basis.

When k'_G and K'_G values are reported in units (SI) of $\text{kmol}/[(\text{s}\cdot\text{m}^2)(\text{kPa})]$, one must be careful in converting them to a mole-fraction basis to multiply by the total pressure actually employed in the original experiments and *not* by the total pressure of the system to be designed. This conversion is valid for systems in which Dalton's law of partial pressures ($p = y p_T$) is valid.

Comparison of Eqs. (5-268) and (5-270) shows that for systems in which the equilibrium line is straight, the overall mass transfer coefficients are related to each other by the equation

$$K_L = mK_G \quad (5-271)$$

When the equilibrium curve is not straight, there is no strictly logical basis for the use of an overall transfer coefficient, since the value of m will be a function of position in the apparatus, as can be seen from Fig. 5-27. In such cases the rate of transfer must be calculated by solving for the interfacial compositions as described above.

Experimentally observed rates of mass transfer often are expressed in terms of overall transfer coefficients even when the equilibrium lines are curved. This procedure is empirical, since the theory indicates that in such cases the rates of transfer may not vary in direct proportion to the overall bulk concentration differences $(y - y^\circ)$ and $(x^\circ - x)$ at all concentration levels even though the rates may be proportional to the concentration difference in each phase taken separately, i.e., $(x_i - x)$ and $(y - y_i)$.

In most types of separation equipment such as packed or spray towers, the interfacial area that is effective for mass transfer cannot be accurately determined. For this reason it is customary to report experimentally observed rates of transfer in terms of transfer coefficients based on a unit volume of the apparatus rather than on a unit of interfacial area. Such volumetric coefficients are designated as K_{Ga} , k_{La} , etc., where a represents the interfacial area per unit volume of the apparatus. Experimentally observed variations in the values of these volumetric coefficients with variations in flow rates, type of packing, etc., may be due as much to changes in the effective value of a as to changes in k . Calculation of the overall coefficients from the individual volumetric coefficients is made by means of the equations

$$1/K_{Ga} = (1/k_{Ga} + m/k_{La}) \quad (5-272)$$

$$1/K_{La} = (1/k_{La} + 1/mk_{Ga}) \quad (5-273)$$

Because of the wide variation in equilibrium, the variation in the values of m from one system to another can have an important effect on the overall coefficient and on the selection of the type of equipment to use. For example, if m is large, the liquid-phase part of the overall resistance might be extremely large where k_L might be relatively small. This kind of reasoning must be applied with caution, however, since species with different equilibrium characteristics are separated under different

operating conditions. Thus, the effect of changes in m on the overall resistance to mass transfer may partly be counterbalanced by changes in the individual specific resistances as the flow rates are changed.

Mass-Transfer Principles: Concentrated Systems When solute concentrations in the gas and/or liquid phases are large, the equations derived above for dilute systems no longer are applicable. The correct equations to use for concentrated systems are as follows:

$$N_A = \hat{k}_G(y - y_i)/y_{BM} = \hat{k}_L(x_i - x)/x_{BM} \\ = \hat{K}_G(y - y^\circ)/y_{BM} = \hat{K}_L(x^\circ - x)/x_{BM} \quad (5-274)$$

where ($N_B = 0$)

$$y_{BM} = \frac{(1 - y) - (1 - y_i)}{\ln [(1 - y)/(1 - y_i)]} \quad (5-275)$$

$$y_{BM}^\circ = \frac{(1 - y) - (1 - y^\circ)}{\ln [(1 - y)/(1 - y^\circ)]} \quad (5-276)$$

$$x_{BM} = \frac{(1 - x) - (1 - x_i)}{\ln [(1 - x)/(1 - x_i)]} \quad (5-277)$$

$$x_{BM}^\circ = \frac{(1 - x) - (1 - x^\circ)}{\ln [(1 - x)/(1 - x^\circ)]} \quad (5-278)$$

and where $\hat{k}_G$ and $\hat{k}_L$ are the gas-phase and liquid-phase mass-transfer coefficients for concentrated systems and $\hat{K}_G$ and $\hat{K}_L$ are the overall gas-phase and liquid-phase mass-transfer coefficients for concentrated systems. These coefficients are defined later in Eqs. (5-281) to (5-283).

The factors y_{BM} and x_{BM} arise from the fact that, in the diffusion of a solute through a second stationary layer of insoluble fluid, the resistance to diffusion varies in proportion to the concentration of the insoluble stationary fluid, approaching zero as the concentration of the insoluble fluid approaches zero. See Eq. (5-198).

The factors y_{BM}° and x_{BM}° cannot be justified on the basis of mass-transfer theory since they are based on overall resistances. These factors therefore are included in the equations by analogy with the corresponding film equations.

In dilute systems the logarithmic-mean insoluble-gas and nonvolatile-liquid concentrations approach unity, and Eq. (5-274) reduces to the dilute-system formula. For equimolar counter diffusion (e.g., binary distillation), these log-mean factors should be omitted. See Eq. (5-197).

Substitution of Eqs. (5-275) through (5-278) into Eq. (5-274) results in the following simplified formula:

$$N_A = \hat{k}_G \ln [(1 - y_i)/(1 - y)] \\ = \hat{K}_G \ln [(1 - y^\circ)/(1 - y)] \\ = \hat{k}_L \ln [(1 - x)/(1 - x_i)] \\ = \hat{K}_L \ln [(1 - x)/(1 - x^\circ)] \quad (5-279)$$

Note that the units of $\hat{k}_G$, $\hat{K}_G$, $\hat{k}_L$, and $\hat{K}_L$ are all identical to each other, i.e., $\text{kmol}/(\text{s}\cdot\text{m}^2)$ in SI units.

The equation for computing the interfacial gas and liquid compositions in concentrated systems is

$$(y - y_i)/(x_i - x) = \hat{k}_L y_{BM} / \hat{k}_G x_{BM} \\ = L_M H_G y_{BM} / G_M H_L x_{BM} = k_L / k_G \quad (5-280)$$

This equation is identical to the one for dilute systems since $\hat{k}_G = k_G y_{BM}$ and $\hat{k}_L = k_L x_{BM}$. Note, however, that when $\hat{k}_G$ and $\hat{k}_L$ are given, the equation must be solved by trial and error, since x_{BM} contains x_i and y_{BM} contains y_i .

The overall gas-phase and liquid-phase mass-transfer coefficients for concentrated systems are computed according to the following equations:

$$\frac{1}{\hat{K}_G} = \frac{y_{BM}}{y_{BM}^\circ} \frac{1}{\hat{k}_G} + \frac{x_{BM}}{y_{BM}^\circ} \frac{1}{\hat{k}_L} \left(\frac{y_i - y^\circ}{x_i - x} \right) \quad (5-281)$$

$$\frac{1}{\hat{K}_L} = \frac{x_{BM}}{x_{BM}^\circ} \frac{1}{\hat{k}_L} + \frac{y_{BM}}{x_{BM}^\circ} \frac{1}{\hat{k}_G} \left(\frac{x^\circ - x_i}{y - y_i} \right) \quad (5-282)$$

When the equilibrium curve is a straight line, the terms in parentheses can be replaced by the slope m or $1/m$ as before. In this case the

overall mass-transfer coefficients for concentrated systems are related to each other by the equation

$$\hat{K}_L = m \hat{K}_G (x_{BM}^{\circ}/y_{BM}^{\circ}) \quad (5-283)$$

All these equations reduce to their dilute-system equivalents as the inert concentrations approach unity in terms of mole fractions of inert concentrations in the fluids.

HTU (Height Equivalent to One Transfer Unit) Frequently the values of the individual coefficients of mass transfer are so strongly dependent on flow rates that the quantity obtained by dividing each coefficient by the flow rate of the phase to which it applies is more nearly constant than the coefficient itself. The quantity obtained by this procedure is called the height equivalent to one transfer unit, since it expresses in terms of a single length dimension the height of apparatus required to accomplish a separation of standard difficulty.

The following relations between the transfer coefficients and the values of HTU apply:

$$H_G = G_M/k_G a y_{BM} = G_M/\hat{K}_G a \quad (5-284)$$

$$H_{OG} = G_M/K_G a y_{BM}^{\circ} = G_M/\hat{K}_G a \quad (5-285)$$

$$H_L = L_M/k_L a x_{BM} = L_M/\hat{K}_L a \quad (5-286)$$

$$H_{OL} = L_M/K_L a x_{BM}^{\circ} = L_M/\hat{K}_L a \quad (5-287)$$

The equations that express the addition of individual resistances in terms of HTUs, applicable to either dilute or concentrated systems, are

$$H_{OG} = \frac{y_{BM}}{y_{BM}^{\circ}} H_G + \frac{m G_M}{L_M} \frac{x_{BM}}{y_{BM}^{\circ}} H_L \quad (5-288)$$

$$H_{OL} = \frac{x_{BM}}{x_{BM}^{\circ}} H_L + \frac{L_M}{m G_M} \frac{y_{BM}}{x_{BM}^{\circ}} H_G \quad (5-289)$$

These equations are strictly valid only when m , the slope of the equilibrium curve, is constant, as noted previously.

NTU (Number of Transfer Units) The NTU required for a given separation is closely related to the number of theoretical stages or plates required to carry out the same separation in a stagewise or plate-type apparatus. For equimolar counterdiffusion, such as in a binary distillation, the number of overall gas-phase transfer units N_{OG} required for changing the composition of the vapor stream from y_1 to y_2 is

$$N_{OG} = \int_{y_2}^{y_1} \frac{dy}{y - y^{\circ}} \quad (5-290)$$

When diffusion is in one direction only, as in the absorption of a soluble component from an insoluble gas,

$$N_{OG} = \int_{y_2}^{y_1} \frac{y_{BM}^{\circ} dy}{(1 - y)(y - y^{\circ})} \quad (5-291)$$

The total height of packing required is then

$$h_T = H_{OG} N_{OG} \quad (5-292)$$

When it is known that H_{OG} varies appreciably within the tower, this term must be placed inside the integral in Eqs. (5-290) and (5-291) for accurate calculations of h_T . For example, the packed-tower design equation in terms of the overall gas-phase mass-transfer coefficient for absorption would be expressed as follows:

$$h_T = \int_{y_2}^{y_1} \left[\frac{G_M}{K_G a y_{BM}^{\circ}} \right] \frac{y_{BM}^{\circ} dy}{(1 - y)(y - y^{\circ})} \quad (5-293)$$

where the first term under the integral can be recognized as the HTU term. Convenient solutions of these equations for special cases are discussed later.

Definitions of Mass-Transfer Coefficients $\hat{K}_G$ and $\hat{K}_L$ The mass-transfer coefficient is defined as the ratio of the molar mass flux N_A to the concentration driving force. This leads to many different ways of defining these coefficients. For example, gas-phase mass-transfer rates may be defined as

$$N_A = k_G(y - y_i) = k'_G(p - p_i) = \hat{K}_G(y - y_i)/y_{BM} \quad (5-294)$$

where the units (SI) of k_G are $\text{kmol}/[(\text{s}\cdot\text{m}^2)(\text{mole fraction})]$, the units of

k'_G are $\text{kmol}/[(\text{s}\cdot\text{m}^2)(\text{kPa})]$, and the units of $\hat{K}_G$ are $\text{kmol}/(\text{s}\cdot\text{m}^2)$. These coefficients are related to each other as follows:

$$k_G = k_C y_{BM} = k'_G p_T y_{BM} \quad (5-295)$$

where p_T is the total system pressure (it is assumed here that Dalton's law of partial pressures is valid).

In a similar way, liquid-phase mass-transfer rates may be defined by the relations

$$N_A = k_L(x_i - x) = k'_L(c_i - c) = \hat{K}_L(x_i - x)/x_{BM} \quad (5-296)$$

where the units (SI) of k_L are $\text{kmol}/[(\text{s}\cdot\text{m}^2)(\text{mole fraction})]$, the units of k'_L are $\text{kmol}/[(\text{s}\cdot\text{m}^2)(\text{kmol}/\text{m}^3)]$ or meters per second, and the units of $\hat{K}_L$ are $\text{kmol}/(\text{s}\cdot\text{m}^2)$. These coefficients are related as follows:

$$\hat{K}_L = k_L x_{BM} = k'_L \bar{\rho}_L x_{BM} \quad (5-297)$$

where $\bar{\rho}_L$ is the molar density of the liquid phase in units (SI) of kilomoles per cubic meter. Note that, for dilute solutions where $x_{BM} \approx 1$, k_L and $\hat{K}_L$ will have identical numerical values. Similarly, for dilute gases $k_G \approx k_C$.

Simplified Mass-Transfer Theories In certain simple situations, the mass-transfer coefficients can be calculated from first principles. The film, penetration, and surface-renewal theories are attempts to extend these theoretical calculations to more complex situations. Although these theories are often not accurate, they are useful to provide a physical picture for variations in the mass-transfer coefficient.

For the special case of steady-state unidirectional diffusion of a component through an inert-gas film in an ideal-gas system, the rate of mass transfer is derived as

$$N_A = \frac{D_{AB} p_T}{RT \delta_G} \frac{(y - y_i)}{y_{BM}} = \frac{D_{AB} p_T}{RT \delta_G} \ln \frac{1 - y_i}{1 - y} \quad (5-298)$$

where D_{AB} is the diffusion coefficient or "diffusivity," δ_G is the "effective" thickness of a stagnant-gas layer which would offer a resistance to molecular diffusion equal to the experimentally observed resistance, and R is the gas constant. [Nernst, *Z. Phys. Chem.*, **47**, 52 (1904); Whitman, *Chem. Mat. Eng.*, **29**, 149 (1923), and Lewis and Whitman, *Ind. Eng. Chem.*, **16**, 1215 (1924)].

The film thickness δ_G depends primarily on the hydrodynamics of the system and hence on the Reynolds number and the Schmidt number. Thus, various correlations have been developed for different geometries in terms of the following dimensionless variables:

$$N_{Sh} = \hat{K}_G R T d / D_{AB} p_T = f(N_{Re}, N_{Sc}) \quad (5-299)$$

where N_{Sh} is the Sherwood number, $N_{Re} (= Gd/\mu_G)$ is the Reynolds number based on the characteristic length d appropriate to the geometry of the particular system; and $N_{Sc} (= \mu_G/\rho_G D_{AB})$ is the Schmidt number.

According to this analysis one can see that for gas-absorption problems, which often exhibit unidirectional diffusion, the most appropriate driving-force expression is of the form $(y - y_i)/y_{BM}$, and the most appropriate mass-transfer coefficient is therefore $\hat{K}_G$. This concept is to be found in all the key equations for the design of mass-transfer equipment.

The Sherwood-number relation for gas-phase mass-transfer coefficients as represented by the film diffusion model in Eq. (5-299) can be rearranged as follows:

$$N_{Sh} = (\hat{K}_G/G_M) N_{Re} N_{Sc} = N_{St} N_{Re} N_{Sc} = f(N_{Re}, N_{Sc}) \quad (5-300)$$

where $N_{St} = \hat{K}_G/G_M = k'_G p_{BM}/G_M$ is known as the Stanton number. This equation can now be stated in the alternative functional forms

$$N_{St} = \hat{K}_G/G_M = g(N_{Re}, N_{Sc}) \quad (5-301)$$

$$j_D = N_{St} \cdot N_{Sc}^{2/3} \quad (5-302)$$

where j is the Chilton-Colburn "j factor" for mass transfer (discussed later).

The important point to note here is that the gas-phase mass-transfer coefficient $\hat{K}_G$ depends principally upon the transport properties of the fluid (N_{Sc}) and the hydrodynamics of the particular system involved (N_{Re}). It also is important to recognize that specific mass-transfer correlations can be derived only in conjunction with the

investigator's particular assumptions concerning the numerical values of the effective interfacial area a of the packing.

The stagnant-film model discussed previously assumes a steady state in which the local flux across each element of area is constant; i.e., there is no accumulation of the diffusing species within the film. Higbie [*Trans. Am. Inst. Chem. Eng.*, **31**, 365 (1935)] pointed out that industrial contactors often operate with repeated brief contacts between phases in which the contact times are too short for the steady state to be achieved. For example, Higbie advanced the theory that in a packed tower the liquid flows across each packing piece in laminar flow and is remixed at the points of discontinuity between the packing elements. Thus, a fresh liquid surface is formed at the top of each piece, and as it moves downward, it absorbs gas at a decreasing rate until it is mixed at the next discontinuity. This is the basis of penetration theory.

If the velocity of the flowing stream is uniform over a very deep region of liquid (total thickness, $\delta_T \gg \sqrt{Dt}$), the time-averaged mass-transfer coefficient according to penetration theory is given by

$$k'_L = 2\sqrt{D_L/\pi t} \quad (5-303)$$

where k'_L = liquid-phase mass-transfer coefficient, D_L = liquid-phase diffusion coefficient, and t = contact time.

In practice, the contact time t is not known except in special cases in which the hydrodynamics are clearly defined. This is somewhat similar to the case of the stagnant-film theory in which the unknown quantity is the thickness of the stagnant layer δ (in film theory, the liquid-phase mass-transfer coefficient is given by $k'_L = D_L/\delta$).

The penetration theory predicts that k'_L should vary by the square root of the molecular diffusivity, as compared with film theory, which predicts a first-power dependency on D . Various investigators have reported experimental powers of D ranging from 0.5 to 0.75, and the Chilton-Colburn analogy suggests a $2/3$ power.

Penetration theory often is used in analyzing absorption with chemical reaction because it makes no assumption about the depths of penetration of the various reacting species, and it gives a more accurate result when the diffusion coefficients of the reacting species are not equal. When the reaction process is very complex, however, penetration theory is more difficult to use than film theory, and the latter method normally is preferred.

Danckwerts [*Ind. Eng. Chem.*, **42**, 1460 (1951)] proposed an extension of the penetration theory, called the surface renewal theory, which allows for the eddy motion in the liquid to bring masses of fresh liquid continually from the interior to the surface, where they are exposed to the gas for finite lengths of time before being replaced. In his development, Danckwerts assumed that every element of fluid has an equal chance of being replaced regardless of its age. The Danckwerts model gives

$$k'_L = \sqrt{Ds} \quad (5-304)$$

where s = fractional rate of surface renewal.

Note that both the penetration and the surface-renewal theories predict a square-root dependency on D . Also, it should be recognized that values of the surface-renewal rate s generally are not available, which presents the same problems as do δ and t in the film and penetration models.

The predictions of correlations based on the film model often are nearly identical to predictions based on the penetration and surface-renewal models. Thus, in view of its relative simplicity, the film model normally is preferred for purposes of discussion or calculation. It should be noted that none of these theoretical models has proved adequate for making a priori predictions of mass-transfer rates in packed towers, and therefore empirical correlations such as those outlined later in Table 5-24 must be employed.

Mass-Transfer Correlations Because of the tremendous importance of mass transfer in chemical engineering, a very large number of studies have determined mass-transfer coefficients both empirically and theoretically. Some of these studies are summarized in Tables 5-17 to 5-24. Each table is for a specific geometry or type of contactor, starting with flat plates, which have the simplest geometry (Table 5-17); then wetted wall columns (Table 5-18); flow in pipes and ducts (Table 5-19); submerged objects (Table 5-20); drops and

bubbles (Table 5-21); agitated systems (Table 5-22); packed beds of particles for adsorption, ion exchange, and chemical reaction (Table 5-23); and finishing with packed bed two-phase contactors for distillation, absorption and other unit operations (Table 5-24). Although extensive, these tables are not meant to be encyclopedic, but a variety of different configurations are shown to provide a flavor of the range of correlations available. These correlations include transfer to or from one fluid and either a second fluid or a solid. Many of the correlations are for k_L and k_G values obtained from dilute systems where $x_{BM} \approx 1.0$ and $y_{BM} \approx 1.0$. The most extensive source for older mass-transfer correlations in a variety of geometries is Skelland (*Diffusional Mass Transfer*, 1974). The extensive review of bubble column systems (see Table 5-21) by Shah et al. [*AIChE J.*, **28**, 353 (1982)] includes estimation of bubble size, gas holdup, interfacial area $k_i a$, and liquid dispersion coefficient. For correlations for particle-liquid mass transfer in stirred tanks (part of Table 5-22) see the review by Pangarkar et al. [*Ind. Eng. Chem. Res.*, **41**, 4141 (2002)]. For mass transfer in distillation, absorption, and extraction in packed beds (Table 5-24), see also the appropriate sections in this handbook and the review by Wang, Yuan, and Yu [*Ind. Eng. Chem. Res.*, **44**, 8715 (2005)]. For simple geometries, one may be able to determine a theoretical (T) form of the mass-transfer correlation. For very complex geometries, only an empirical (E) form can be found. In systems of intermediate complexity, semiempirical (S) correlations where the form is determined from theory and the coefficients from experiment are often useful. Although the major limitations and constraints in use are usually included in the tables, obviously many details cannot be included in this summary form. Readers are strongly encouraged to check the references before using the correlations in important situations. Note that even authoritative sources occasionally have typographical errors in the fairly complex correlation equations. Thus, it is a good idea to check several sources, including the original paper. The references will often include figures comparing the correlations with data. These figures are very useful since they provide a visual picture of the scatter in the data.

Since there are often several correlations that are applicable, how does one choose the correlation to use? First, the engineer must determine which correlations are closest to the current situation. This involves recognizing the similarity of geometries, which is often challenging, and checking that the range of parameters in the correlation is appropriate. For example, the Bravo, Rocha, and Fair correlation for distillation with structured packings with triangular cross-sectional channels (Table 5-24-H) uses the Johnstone and Pigford correlation for rectification in vertical wetted wall columns (Table 5-18-F). Recognizing that this latter correlation pertains to a rather different application and geometry was a nontrivial step in the process of developing a correlation. If several correlations appear to be applicable, check to see if the correlations have been compared to each other and to the data. When a detailed comparison of correlations is not available, the following heuristics may be useful:

1. Mass-transfer coefficients are derived from models. They must be employed in a similar model. For example, if an arithmetic concentration difference was used to determine k , that k should only be used in a mass-transfer expression with an arithmetic concentration difference.
2. Semiempirical correlations are often preferred to purely empirical or purely theoretical correlations. Purely empirical correlations are dangerous to use for extrapolation. Purely theoretical correlations may predict trends accurately, but they can be several orders of magnitude off in the value of k .
3. Correlations with broader data bases are often preferred.
4. The analogy between heat and mass transfer holds over wider ranges than the analogy between mass and momentum transfer. Good heat transfer data (without radiation) can often be used to predict mass-transfer coefficients.
5. More recent data is often preferred to older data, since end effects are better understood, the new correlation often builds on earlier data and analysis, and better measurement techniques are often available.
6. With complicated geometries, the product of the interfacial area per volume and the mass-transfer coefficient is required. Correlations of ka , or of HTU are more accurate than individual correlations of k and

TABLE 5-17 Mass-Transfer Correlations for a Single Flat Plate or Disk—Transfer to or from Plate to Fluid

Situation	Correlation	Comments E = Empirical, S = Semiempirical, T = Theoretical	References ^a
A. Laminar, local, flat plate, forced flow	$N_{Sh,x} = \frac{k'x}{D} = 0.323(N_{Re,x})^{1/2}(N_{Sc})^{1/3}$ Coefficient 0.323 is a better fit.	[T] Low M.T. rates. Low mass-flux, constant property systems. $N_{Sh,x}$ is local k . Use with arithmetic difference in concentration. Coefficient 0.323 is Blasius' approximate solution. $N_{Re,x} = \frac{xu_{\infty}\rho}{\mu}$, x = length along plate	[77] p. 183 [87] p. 526 [138] p. 79 [140] p. 518 [141] p. 110
Laminar, average, flat plate, forced flow	$N_{Sh,avg} = \frac{k'_m L}{D} = 0.646(N_{Re,L})^{1/2}(N_{Sc})^{1/3}$ Coefficient 0.664 is a better fit. k'_m is mean mass-transfer coefficient for dilute systems.	$N_{Re,L} = \frac{Lu_{\infty}\rho}{\mu}$, 0.664 (Polhausen) is a better fit for $N_{Sc} > 0.6$, $N_{Re,x} < 3 \times 10^5$.	[91] p. 480
j -factors	$j_D = j_H = \frac{f}{2} = 0.664(N_{Re,L})^{-1/2}$	[S] Analogy. $N_{Sc} = 1.0$, f = drag coefficient. j_D is defined in terms of k'_m .	[141] p. 271
B. Laminar, local, flat plate, blowing or suction and forced flow	$N_{Sh,x} = \frac{k'x}{D} = (\text{Slope})_{y=0} (N_{Re,x})^{1/2} (N_{Sc})^{1/3}$	[T] Blowing is positive. Other conditions as above. $\frac{u_o}{u_{\infty}} \sqrt{N_{Re,x}}$ $\frac{u_{\infty}}{(\text{Slope})_{y=0}}$ 0.6 0.5 0.25 0.0 -2.5 0.01 0.06 0.17 0.332 1.64	[77] p. 185 [140] p. 271
C. Laminar, local, flat plate, natural convection vertical plate	$N_{Sh,x} = \frac{k'x}{D} = 0.508 N_{Sc}^{1/2} (0.952 + N_{Sc})^{-1/4} N_{Gr}^{1/4}$ $N_{Gr} = \frac{gx^3}{(\mu/\rho)^2} \left(\frac{\rho_{\infty}}{\rho_0} - 1 \right)$	[T] Low MT rates. Dilute systems, $\Delta\rho/\rho \ll 1$. $N_{Gr}, N_{Sc} < 10^8$. Use with arithmetic concentration difference. x = length from plate bottom.	[141] p. 120
D. Laminar, stationary disk	$N_{Sh} = \frac{k'd_{\text{disk}}}{D} = \frac{8}{\pi}$	[T] Stagnant fluid. Use arithmetic concentration difference.	[138] p. 240
Laminar, spinning disk	$N_{Sh} = \frac{k'd_{\text{disk}}}{D} = 0.879 N_{Re}^{1/2} N_{Sc}^{1/3}$ $N_{Re} < \sim 10^4$	[T] Asymptotic solution for large N_{Sc} . $u = \omega d_{\text{disk}}/2$, ω = rotational speed, rad/s. Rotating disks are often used in electrochemical research.	[101] p. 60 [138] p. 240
E. Laminar, inclined, plate	$N_{Sh,avg} = 0.783 N_{Re, \text{film}}^{1/3} N_{Sc}^{1/3} \left(\frac{x^3 \rho^2 g \sin \alpha}{\mu^2} \right)^{2/9}$ $N_{Re, \text{film}} = \frac{4Q\rho}{\mu^2} < 2000$ $N_{Sh,avg} = \frac{k'_m x}{D}$ $\delta_{\text{film}} = \left(\frac{3\mu Q}{w\rho g \sin \alpha} \right)^{1/3}$ = film thickness	[T] Constant-property liquid film with low mass-transfer rates. Use arithmetic concentration difference. Newtonian fluid. Solute does not penetrate past region of linear velocity profile. Differences between theory and experiment. w = width of plate, δ_f = film thickness, α = angle of inclination, x = distance from start soluble surface.	[141] p. 130 [138] p. 209
F. Turbulent, local flat plate, forced flow	$N_{Sh,x} = \frac{k'x}{D} = 0.0292 N_{Re,x}^{0.8}$	[S] Low mass-flux with constant property system. Use with arithmetic concentration difference. $N_{Sc} = 1.0$, $N_{Re,x} > 10^5$	[77] p. 191 [138] p. 201 [141] p. 221
Turbulent, average, flat plate, forced flow	$N_{Sh,avg} = \frac{k'_m L}{D} = 0.0365 N_{Re,L}^{0.8}$	Based on Prandtl's 1/7-power velocity law, $\frac{u}{u_{\infty}} = \left(\frac{y}{\delta} \right)^{1/7}$	
G. Laminar and turbulent, flat plate, forced flow	$j_D = j_H = \frac{f}{2} = 0.037 N_{Re,L}^{0.2}$ $j_D = (k_G/G_M) N_{Sc}^{2/3}$ $j_H = (h'/C_p G) N_{Pr}^{2/3}$	[E] Chilton-Colburn analogies, $N_{Sc} = 1.0$, (gases), f = drag coefficient. Corresponds to item 5-17-F and refers to same conditions. $8000 < N_{Re} < 300,000$. Can apply analogy, $j_D = f/2$, to entire plate (including laminar portion) if average values are used.	[77] p. 193 [88] p. 112 [138] p. 201 [141] p. 271 [80] [53]

TABLE 5-17 Mass-Transfer Correlations for a Single Flat Plate or Disk—Transfer to or from Plate to Fluid (Concluded)

Situation	Correlation	Comments E = Empirical, S = Semiempirical, T = Theoretical	References ^o
H. Laminar and turbulent, flat plate, forced flow	$N_{Sh,avg} = 0.037 N_{Re,L}^{1/3} (N_{Re,L}^{0.8} - 15,500)$ to $N_{Re,L} = 320,000$ $N_{Sh,avg} = 0.037 N_{Sc}^{1/3} \times \left(N_{Re,L}^{0.8} - N_{Re,Cr}^{0.8} + \frac{0.664}{0.037} N_{Re,Cr}^{1/2} \right)$ in range 3×10^5 to 3×10^6.	[E] Use arithmetic concentration difference. $N_{Sh,avg} = \frac{k_m L}{D}, N_{Sc} > 0.5$ Entrance effects are ignored. $N_{Re,Cr}$ is transition laminar to turbulent.	[88] p. 112 [138] p. 201
I. Turbulent, local flat plate, natural convection, vertical plate Turbulent, average, flat plate, natural convection, vertical plate	$N_{Sh,x} = \frac{k'x}{D} = 0.0299 N_{Gr}^{2/5} N_{Sc}^{7/15} \times (1 + 0.494 N_{Sc}^{2/3})^{-2/5}$ $N_{Sh,avg} = 0.0249 N_{Gr}^{2/5} N_{Sc}^{7/15} \times (1 + 0.494 N_{Sc}^{2/3})^{-2/5}$ $N_{Gr} = \frac{g x^3}{(\mu/\rho)^2} \left(\frac{\rho_\infty}{\rho_0} - 1 \right), \quad N_{Sh,avg} = \frac{k_m L}{D}$	[S] Low solute concentration and low transfer rates. Use arithmetic concentration difference. $N_{Gr} > 10^{10}$ Assumes laminar boundary layer is small fraction of total.	[141] p. 225
J. Perforated flat disk Perforated vertical plate. Natural convection.	$N_{Sh} = 0.059 N_{Sc}^{0.35} N_{Gr}^{0.75} \left(\frac{d_h}{d} \right)^{0.04}$ Characteristic length = disk diameter d $N_{Sh} = 0.1 N_{Sc}^{1/3} N_{Gr}^{1/3}$ Characteristic length = L, electrode height	[E] $6 \times 10^9 < N_{Sc} N_{Gr} < 10^{12}$ and $1943 < N_{Sc} < 2168$ d_h = hole diameter [E] $1 \times 10^{10} < N_{Sc} N_{Gr} < 5 \times 10^{13}$ and $1939 < N_{Sc} < 2186$ Average deviation $\pm 10\%$	[162]
K. Turbulent, vertical plate	$N_{Sh,avg} = \frac{k'_m x}{D} = 0.327 N_{Re,filin}^{2/9} N_{Sc}^{1/3} \left(\frac{x^3 \rho^2 g}{\mu^2} \right)^{2/9}$ $\delta_{filin} = 0.172 \left(\frac{Q^2}{w^2 g} \right)^{1/3}$	[E] See 5-17-E for terms. $N_{Re,filin} = \frac{4Q\rho}{w\mu^2} > 2360$ Solute remains in laminar sublayer.	[141] p. 229
L. Cross-corrugated plate (turbulence promoter for membrane systems)	$N_{Sh} = c N_{Re}^a N_{Sc}^{1/3}$	[E] Entrance turbulent channel For parallel flow and corrugations: $N_{Sc} = 1483, a = 0.56, c = 0.268$ $N_{Sc} = 4997, a = 0.50, c = 0.395$ Corrugations perpendicular to flow: $N_{Sc} = 1483, a = 0.57, c = 0.368$ $N_{Sc} = 4997, a = 0.52, c = 0.487$	[134]
M. Turbulent, spinning disk	$N_{Sh} = \frac{k'd_{disk}}{D} = 5.6 N_{Re}^{1.1} N_{Sc}^{1/3}$ $6 \times 10^5 < N_{Re} < 2 \times 10^6$ $120 < N_{Sc} < 1200$	[E] Use arithmetic concentration difference. $u = \omega d_{disk}/2$ where ω = rotational speed, radians/s. $N_{Re} = \rho \omega d^2/2\mu$.	[55] [138] p. 241
N. Mass transfer to a flat plate membrane in a stirred vessel	$N_{Sh} = \frac{k'd_{tank}}{D} = a N_{Re}^b N_{Sc}^c$ a depends on system. $a = 0.0443$ [40]; b is often 0.65–0.70 [89]. If $N_{Re} = \frac{\omega d_{tank}^2 \rho}{\mu}$	[E] Use arithmetic concentration difference. ω = stirrer speed, radians/s. Useful for laboratory dialysis, R.O., U.F., and microfiltration systems. $b = 0.785$ [40]. c is often 0.33 but other values have been reported [89].	[40] [89] p. 965
O. Spiral type RO (seawater desalination)	$N_{Sh} = 0.210 N_{Re}^{2/3} N_{Sc}^{1/4}$ Or with slightly larger error, $N_{Sh} = 0.080 N_{Re}^{0.875} N_{Sc}^{1/4}$	[E] Polyamide membrane. $p = 6.5$ MPa and TDS rejection = 99.8%. Recovery ratio 40%.	[148]

^oSee the beginning of the “Mass Transfer” subsection for references.

TABLE 5-18 Mass-Transfer Correlations for Falling Films with a Free Surface in Wetted Wall Columns—Transfer between Gas and Liquid

Situation	Correlation	Comments E = Empirical, S = Semiempirical, T = Theoretical	References*
A. Laminar, vertical wetted wall column	$N_{Sh,avg} = \frac{K'_m x}{D} \approx 3.41 \frac{x}{\delta_{film}}$ (first term of infinite series) $\delta_{film} = \left(\frac{3\mu Q}{w\rho g} \right)^{1/3} = \text{film thickness}$ $N_{Re,film} = \frac{4Q\rho}{w\mu} < 20$	[T] Low rates M.T. Use with log mean concentration difference. Parabolic velocity distribution in films. w = film width (circumference in column) Derived for flat plates, used for tubes if $r_{tube} \left(\frac{\rho g}{2\sigma} \right)^{1/2} > 3.0.$ σ = surface tension If $N_{Re,film} > 20$, surface waves and rates increase. An approximate solution $D_{apparent}$ can be used. Ripples are suppressed with a wetting agent good to $N_{Re} = 1200$.	[138] p. 78 [141] p. 137 [152] p. 50
B. Turbulent, vertical wetted wall column	$N_{Sh,avg} = \frac{K'_m d_t}{D} = 0.023 N_{Re}^{0.83} N_{Sc}^{0.44}$ A coefficient 0.0163 has also been reported using $N_{Re'}$, where $v = v$ of gas relative to liquid film.	[E] Use with log mean concentration difference for correlations in B and D. N_{Re} is for gas. N_{Sc} for vapor in gas. $2000 < N_{Re} \leq 35,000$, $0.6 \leq N_{Sc} \leq 2.5$. Use for gases, d_t = tube diameter.	[68] [77] p. 181 [138] p. 211 [141] p. 265 [149] p. 212 [152] p. 71 [58]
Better fit	$N_{Sh,avg} = 0.0318 N_{Re}^{0.790} N_{Sc}^{0.5}$	[S] Reevaluated data	
C. Turbulent, very short column	$N_{Sh} = 0.00283 N_{Re,g}^{0.5} N_{Sc,g}^{0.06} N_{Re,liq}^{0.06}$ $N_{Sh} = K_g(d_{tube} - 2\delta)/D$ $N_{Re,g} = \rho_g u_g (d_{tube} - 2\delta)/\mu_g$ $N_{Re,liq} = \rho_{liq} Q_{liq}/[\pi \mu (d_{tube} - 2\delta)]$	[E] Evaporation data $N_{Sh,g} = 11$ to 65, $N_{Re,g} = 2400$ to 9100 $N_{Re,liq} = 110$ to 480, $N_{Sc,g} = 0.62$ to 1.93 δ = film thickness	[56]
D. Turbulent, vertical wetted wall column with ripples	$N_{Sh,avg} = \frac{K'_m d_t}{D} = 0.00814 N_{Re}^{0.83} N_{Sc}^{0.44} \left(\frac{4Q\rho}{w\mu} \right)^{0.15}$ $30 \leq \left(\frac{4Q\rho}{w\mu} \right) < 1200$ $N_{Sh,avg} = \frac{K'_m d_t}{D} = 0.023 N_{Re}^{0.8} N_{Sc}^{1/3}$	[E] For gas systems with rippling. Fits 5-18-B for $\left(\frac{4Q\rho}{w\mu} \right) = 1000$ [E] "Rounded" approximation to include ripples. Includes solid-liquid mass-transfer data to find $1/3$ coefficient on N_{Sc}. May use $N_{Re}^{0.83}$. Use for liquids. See also Table 5-19.	[85] [138] p. 213
E. Turbulent, with ripples	$N_{Sh} = \left(\frac{2}{\pi} \right)^{0.5} N_{Re,\dot{\epsilon}}^{0.5} N_{Sc}^{0.5}$ $N_{Re,\dot{\epsilon}} = \dot{\epsilon} L^2/\nu$	[E] $\dot{\epsilon}$ = dilation rate of surface = $\frac{\partial v_x^s}{\partial x} + \frac{\partial v_y^s}{\partial y}$	[150]
F. Rectification in vertical wetted wall column with turbulent vapor flow, Johnstone and Pigford correlation	$N_{Sh,avg} = \frac{K'_G d_{col} p_{BM}}{D_e p} = 0.0328 (N'_{Re})^{0.77} N_{Sc}^{0.33}$ $3000 < N'_{Re} < 40,000$, $0.5 < N_{Sc} < 3$ $N'_{Re} = \frac{d_{col} v_{rel} \rho_v}{\mu_v}, v_{rel} = \text{gas velocity relative to liquid film} = \frac{3}{2} u_{avg} \text{ in film}$	[E] Use logarithmic mean driving force at two ends of column. Based on four systems with gas-side resistance only. p_{BM} = logarithmic mean partial pressure of nondiffusing species B in binary mixture. p = total pressure Modified form is used for structured packings (See Table 5-24-H).	[84] [138] p. 214 [156]

*See the beginning of the "Mass Transfer" subsection for references.

a_p since the measurements are simpler to determine the product ka_p or HTU.

7. Finally, if a mass-transfer coefficient looks too good to be true, it probably is incorrect.

To determine the mass-transfer rate, one needs the interfacial area in addition to the mass-transfer coefficient. For the simpler geometries, determining the interfacial area is straightforward. For packed beds of particles a , the interfacial area per volume can be estimated as shown in Table 5-23-A. For packed beds in distillation, absorption, and so on in Table 5-24, the interfacial area per volume is included with the mass-transfer coefficient in the correlations for HTU. For agitated liquid-liquid systems, the interfacial area can be estimated

from the dispersed phase holdup and mean drop size correlations. Godfrey, Obi, and Reeve [*Chem. Engr. Prog.* **85**, 61 (Dec. 1989)] summarize these correlations. For many systems, $\bar{d}_{drop}/d_{imp} = (\text{const})N_{We}^{-0.6}$ where $N_{We} = \rho_c N^2 d_{imp}^3 / \sigma$. Piché, Grandjean, and Larachi [*Ind. Eng. Chem. Res.* **41**, 4911 (2002)] developed two correlations for reconciling the gas-liquid mass-transfer coefficient and interfacial area in randomly packed towers. The correlation for the interfacial area was a function of five dimensionless groups, and yielded a relative error of 22.5 percent for 325 data points. That equation, when combined with a correlation for N_{Sh} as a function of four dimensionless groups, achieved a relative error of 24.4 percent, for 3455 data points for the product $k'_c a$.

TABLE 5-19 Mass-Transfer Correlations for Flow in Pipes and Ducts—Transfer Is from Wall to Fluid

Situation	Correlation	Comments E = Empirical, S = Semiempirical, T = Theoretical	References*																		
A. Tubes, laminar, fully developed parabolic velocity profile, developing concentration profile, constant wall concentration	$N_{Sh} = \frac{k'd_t}{D} = 3.66 + \frac{0.0668(d_t/x)N_{Re}N_{Sc}}{1 + 0.04[(d_t/x)N_{Re}N_{Sc}]^{2/3}}$	[T] Use log mean concentration difference. For $\frac{x/d_t}{N_{Re}N_{Sc}} < 0.10$, $N_{Re} < 2100$. x = distance from tube entrance. Good agreement with experiment at values $10^4 > \frac{\pi}{4} \frac{d_t}{x} N_{Re}N_{Sc} > 10$	[77] p. 176 [87] p. 525 [141] p. 159																		
Fully developed concentration profile	$N_{Sh} = \frac{k'd_t}{D} = 3.66$	[T] $\frac{x/d_t}{N_{Re}N_{Sc}} > 0.1$	[141] p. 165																		
B. Tubes, approximate solution	$N_{Sh,x} = \frac{k'd_t}{D} = 1.077 \left(\frac{d_t}{x} \right)^{1/3} (N_{Re}N_{Sc})^{1/3}$ $N_{Sh,avg} = \frac{k'd_t}{D} = 1.615 \left(\frac{d_t}{L} \right)^{1/3} (N_{Re}N_{Sc})^{1/3}$	[T] For arithmetic concentration difference. $\frac{W}{\rho D x} > 400$ Leveque's approximation: Concentration BL is thin. Assume velocity profile is linear. High mass velocity. Fits liquid data well.	[141] p. 166																		
C. Tubes, laminar, uniform plug velocity, developing concentration profile, constant wall concentration	$N_{Sh,avg} = \frac{1}{2} \frac{d_t}{L} N_{Re}N_{Sc} \left[\frac{1 - 4 \sum_{j=1}^{\infty} a_j^{-2} \exp \left(\frac{-2a_j^2(x/r_t)}{N_{Re}N_{Sc}} \right)}{1 + 4 \sum_{j=1}^{\infty} a_j^{-2} \exp \left(\frac{-2a_j^2(x/r_t)}{N_{Re}N_{Sc}} \right)} \right]$ Graetz solution for heat transfer written for M.T.	[T] Use arithmetic concentration difference. Fits gas data well, for $\frac{W}{D p x} < 50$ (fit is fortuitous). $N_{Sh,avg} = (k'_m d_t)/D$. $a_1 = 2.405$, $a_2 = 5.520$, $a_3 = 8.654$, $a_4 = 11.792$, $a_5 = 14.931$. Graphical solutions are in references.	[103] [141] p. 150																		
D. Laminar, fully developed parabolic velocity profile, constant mass flux at wall	$N_{Sh,x} = \left[\frac{11}{48} - \frac{1}{2} \sum_{j=1}^{\infty} \frac{\exp [-\lambda_j^2(x/r_t)/(N_{Re}N_{Sc})]}{C_j \lambda_j^4} \right]^{-1}$ <table><thead><tr><th>j</th><th>λ_j^2</th><th>C_j</th></tr></thead><tbody><tr><td>1</td><td>25.68</td><td>7.630×10^{-3}</td></tr><tr><td>2</td><td>83.86</td><td>2.058×10^{-3}</td></tr><tr><td>3</td><td>174.2</td><td>0.901×10^{-3}</td></tr><tr><td>4</td><td>296.5</td><td>0.487×10^{-3}</td></tr><tr><td>5</td><td>450.9</td><td>0.297×10^{-3}</td></tr></tbody></table>	j	λ_j^2	C_j	1	25.68	7.630×10^{-3}	2	83.86	2.058×10^{-3}	3	174.2	0.901×10^{-3}	4	296.5	0.487×10^{-3}	5	450.9	0.297×10^{-3}	[T] Use log mean concentration difference. $N_{Re} < 2100$ $N_{Sh,x} = \frac{k'd_t}{D}$ $N_{Re} = \frac{v d_t \rho}{\mu}$	[139] [141] p. 167
j	λ_j^2	C_j																			
1	25.68	7.630×10^{-3}																			
2	83.86	2.058×10^{-3}																			
3	174.2	0.901×10^{-3}																			
4	296.5	0.487×10^{-3}																			
5	450.9	0.297×10^{-3}																			
E. Laminar, alternate	$N_{Sh} = 4.36 + \frac{0.023(d_t/L)N_{Re}N_{Sc}}{1 + 0.0012(d_t/L)N_{Re}N_{Sc}}$	[T] $N_{sh} = \frac{k'd_t}{D}$, Use log mean concentration difference. $N_{Re} < 2100$	[77] p. 176																		
F. Laminar, fully developed concentration and velocity profile	$N_{Sh} = \frac{k'd_t}{D} = \frac{48}{11} = 4.3636$	[T] Use log mean concentration difference. $N_{Re} < 2100$	[141] p. 167																		
G. Vertical tubes, laminar flow, forced and natural convection	$N_{Sh,avg} = 1.62 N_{Gz}^{1/3} \left[1 \pm 0.0742 \frac{(N_{Gr}N_{Sc}d/L)^{3/4}}{N_{Gz}} \right]^{1/3}$	[T] Approximate solution. Use minus sign if forced and natural convection oppose each other. Good agreement with experiment. $N_{Gz} = \frac{N_{Re}N_{Sc}d}{L}$, $N_{Gr} = \frac{g \Delta \rho d^3}{\rho \nu^2}$	[127]																		
H. Hollow-fiber extraction inside fibers	$N_{Sh} = 0.5 N_{Gz}$, $N_{Gz} < 6$ $N_{Sh} = 1.62 N_{Gz}^{0.5}$, $N_{Gz} \geq 6$	[E] Use arithmetic concentration difference.	[41]																		
I. Tubes, laminar, RO systems	$N_{Sh,avg} = \frac{k'_m d_t}{D} = 1.632 \left(\frac{u d_t^2}{DL} \right)^{1/3}$	Use arithmetic concentration difference. Thin concentration polarization layer, not fully developed. $N_{Re} < 2000$, L = length tube.	[40]																		
J. Tubes and parallel plates, laminar RO	Graphical solutions for concentration polarization. Uniform velocity through walls.	[T]	[137]																		
K. Rotating annulus for reverse osmosis	For nonvortical flow: $N_{Sh} = 2.15 \left[N_{Ti} \left(\frac{d}{r_i} \right)^{0.5} \right]^{0.18} N_{Sc}^{1/3}$ For vortical flow: $N_{Sh} = 1.05 \left[N_{Ti} \left(\frac{d}{r_i} \right)^{0.5} \right] N_{Sc}^{1/3}$	[E,S] N_{Ti} = Taylor number = $r_i \omega d / \nu$ r_i = inner cylinder radius ω = rotational speed, rad/s d = gap width between cylinders	[100]																		

TABLE 5-19 Mass-Transfer Correlations for Flow in Pipes and Ducts—Transfer Is from Wall to Fluid (Continued)

Situation	Correlation	Comments E = Empirical, S = Semiempirical, T = Theoretical	References°
L. Parallel plates, laminar, parabolic velocity, developing concentration profile, constant wall concentration	Graphical solution	[T] Low transfer rates.	[141] p. 176
L'. 5-19-L, fully developed	$N_{Sh} = \frac{k'(2h)}{D} = 7.6$	[T] h = distance between plates. Use log mean concentration difference. $\frac{N_{Re}N_{Sc}}{x/(2h)} < 20$	[141] p. 177
M. Parallel plates, laminar, parabolic velocity, developing concentration profile, constant mass flux at wall	Graphical solution	[T] Low transfer rates.	[141] p. 176
N. 5-19-M, fully developed	$N_{Sh} = \frac{k'(2h)}{D} = 8.23$	[T] Use log mean concentration difference. $\frac{N_{Re}N_{Sc}}{x/(2h)} < 20$	[141] p. 177
O. Laminar flow, vertical parallel plates, forced and natural convection	$N_{Sh,avg} = 1.47N_{Gz}^{1/3} \left[1 \pm 0.0989 \frac{(N_{Gr}N_{Sc}h/L)^{3/4}}{N_{Gz}} \right]^{1/3}$	[T] Approximate solution. Use minus sign if forced and natural convection oppose each other. Good agreement with experiment. $N_{Gz} = \frac{N_{Re}N_{Sc}h}{L}$, $N_{Gr} = \frac{g\Delta\rho h^3}{\rho\nu^2}$	[127]
P. Parallel plates, laminar, RO systems	$N_{Sh,avg} = \frac{k'(2H_p)}{D} = 2.354 \left(\frac{uH_p^2}{DL} \right)^{1/3}$	Thin concentration polarization layer. Short tubes, concentration profile not fully developed. Use arithmetic concentration difference.	[40]
Q. Tubes, turbulent	$N_{Sh,avg} = \frac{k'_m d_t}{D} = 0.023N_{Re}^{0.83}N_{Sc}^{1/3}$ $2100 < N_{Re} < 35,000$ $0.6 < N_{Sc} < 3000$ $N_{Sh,avg} = \frac{k'_m d_t}{D} = 0.023N_{Re}^{0.83}N_{Sc}^{0.44}$ $2000 < N_{Re} < 35,000$ $0.6 < N_{Sc} < 2.5$	[E] Use with log mean concentration difference at two ends of tube. Good fit for liquids. From wetted wall column and dissolution data—see Table 5-18-B. [E] Evaporation of liquids. Use with log mean concentration difference. Better fit for gases.	[77] p. 181 [103] [152] p. 72 [68][77] p. 181 [88] p. 112 [138] p. 211
R. Tubes, turbulent	$N_{Sh} = \frac{k'd_t}{D} = 0.0096N_{Re}^{0.913}N_{Sc}^{0.346}$	[E] $430 < N_{Sc} < 100,000$. Dissolution data. Use for high N_{Sc} .	[105] p. 668
S. Tubes, turbulent, smooth tubes, Reynolds analogy	$N_{Sh} = \frac{k'd_t}{D} = \left(\frac{f}{2} \right) N_{Re}N_{Sc}$ f = Fanning friction factor	[T] Use arithmetic concentration difference. N_{Sc} near 1.0 Turbulent core extends to wall. Of limited utility.	[66] p. 474 [77] p. 171 [141] p. 239 [149] p. 250
T. Tubes, turbulent, smooth tubes, Chilton-Colburn analogy	$j_D = j_H \leq \frac{f}{2}$ If $\frac{f}{2} = 0.023N_{Re}^{-0.2}$, $j_D = \frac{N_{Sh}}{N_{Re}N_{Sc}^{1/3}} = 0.023N_{Re}^{-0.2}$ $N_{Sh} = \frac{k'd_t}{D}$, Sec. 5-17-G $j_D = j_H = f(N_{Re}, \text{geometry and B.C.})$	[E] Use log-mean concentration difference. Relating j_D to $f/2$ approximate. N_{Pr} and N_{Sc} near 1.0. Low concentration. Results about 20% lower than experiment. $3 \times 10^4 < N_{Re} < 10^6$ [E] Good over wide ranges.	[39] pp. 400, 647 [51][53] [141] p. 264 [149] p. 251 [66] p. 475 [39] p. 647 [51]
U. Tubes, turbulent, smooth tubes, constant surface concentration, Prandtl analogy	$N_{Sh} = \frac{k'd_t}{D} = \frac{(f/2)N_{Re}N_{Sc}}{1 + 5\sqrt{f/2}(N_{Sc} - 1)}$ $\frac{f}{2} = 0.04N_{Re}^{-0.25}$	[T] Use arithmetic concentration difference. Improvement over Reynolds analogy. Best for N_{Sc} near 1.0.	[77] p. 173 [141] p. 241

TABLE 5-19 Mass-Transfer Correlations for Flow in Pipes and Ducts—Transfer Is from Wall to Fluid (Concluded)

Situation	Correlation	Comments E = Empirical, S = Semiempirical, T = Theoretical	References°
V. Tubes, turbulent, smooth tubes, Constant surface concentration, Von Karman analogy	$N_{Sh} = \frac{(f/2)N_{Re}N_{Sc}}{1 + 5\sqrt{f/2} \left\{ (N_{Sc} - 1) + \ln \left[1 + \frac{5}{6}(N_{Sc} - 1) \right] \right\}}$ $\frac{f}{2} = 0.04N_{Re}^{-0.25}$	[T] Use arithmetic concentration difference. $N_{Sh} = k'd_t/D$. Improvement over Prandtl, $N_{Sc} < 25$.	[77] p. 173 [141] p. 243 [149] p. 250 [154]
W. Tubes, turbulent, smooth tubes, constant surface concentration	For $0.5 < N_{Sc} < 10$: $N_{Sh,avg} = 0.0097N_{Re}^{9/10}N_{Sc}^{1/2} \times (1.10 + 0.44N_{Sc}^{-1/3} - 0.70N_{Sc}^{-1/6})$ For $10 < N_{Sc} < 1000$: $N_{Sh,avg}$ $= \frac{0.0097N_{Re}^{9/10}N_{Sc}^{1/2}(1.10 + 0.44N_{Sc}^{-1/3} - 0.70N_{Sc}^{-1/6})}{1 + 0.064N_{Sc}^{1/2}(1.10 + 0.44N_{Sc}^{-1/3} - 0.70N_{Sc}^{-1/6})}$ For $N_{Sc} > 1000$: $N_{Sh,avg} = 0.0102N_{Re}^{9/10}N_{Sc}^{1/3}$	[S] Use arithmetic concentration difference. Based on partial fluid renewal and an infrequently replenished thin fluid layer for high N_{Sc} . Good fit to available data. $N_{Re} = \frac{u_{bulk} d_t}{\nu}$ $N_{Sh,avg} = \frac{k'_{avg} d_t}{D}$	[77] p. 179 [117]
X. Turbulent flow, tubes	$N_{St} = \frac{N_{Sh}}{N_{Pe}} = \frac{N_{Sh}}{N_{Re}N_{Sc}} = 0.0149N_{Re}^{-0.12}N_{Sc}^{-2/3}$	[E] Smooth pipe data. Data fits within 4% except at $N_{Sc} > 20,000$, where experimental data is underpredicted. $N_{Sc} > 100$, $10^5 > N_{Re} > 2100$	[107]
Y. Turbulent flow, noncircular ducts	Use correlations with $d_{eq} = \frac{4 \text{ cross-sectional area}}{\text{wetted perimeter}}$	Can be suspect for systems with sharp corners. Parallel plates: $d_{eq} = 4 \frac{2hw}{2w + 2h}$	[141] p. 289
Z. Decaying swirling flow in pipe	$N_{Sh,avg} = 0.3508N_{Re}^{1/2}N_{Sc}^{0.759}(x/d)^{-0.400} \times (1 + \tan\theta)^{0.271}$ $N_{Re} = 1730 \text{ to } 8650, N_{Sc} = 1692$	[E,S] x = axial distance, d = diameter, θ = vane angle (15° to 60°) Regression coefficient = 0.9793. Swirling increases mass transfer.	[161]

°See the beginning of the "Mass Transfer" subsection for references.

Effects of Total Pressure on $\hat{k}_C$ and $\hat{k}_L$ The influence of total system pressure on the rate of mass transfer from a gas to a liquid or to a solid has been shown to be the same as would be predicted from stagnant-film theory as defined in Eq. (5-298), where

$$\hat{k}_C = D_{AB}p_T/RT \delta_C \quad (5-305)$$

Since the quantity $D_{AB}p_T$ is known to be relatively independent of the pressure, it follows that the rate coefficients $\hat{k}_C$, $k_C y_{BM}$, and $k'_C p_T y_{BM}$ ($= k'_C p_{BM}$) do not depend on the total pressure of the system, subject to the limitations discussed later.

Investigators of tower packings normally report $k'_C a$ values measured at very low inlet-gas concentrations, so that $y_{BM} = 1$, and at total pressures close to 100 kPa (1 atm). Thus, the correct rate coefficient for use in packed-tower designs involving the use of the driving force $(y - y_i)/y_{BM}$ is obtained by multiplying the reported $k'_C a$ values by the value of p_T employed in the actual test unit (e.g., 100 kPa) and *not* the total pressure of the system to be designed.

From another point of view one can correct the reported values of $k'_C a$ in $\text{kmol}/[(s \cdot \text{m}^3)(\text{kPa})]$, valid for a pressure of 101.3 kPa (1 atm), to some other pressure by dividing the quoted values of $k'_C a$ by the design pressure and multiplying by 101.3 kPa, i.e., $(k'_C a \text{ at design pressure } p_T) = (k'_C a \text{ at } 1 \text{ atm}) \times 101.3/p_T$.

One way to avoid a lot of confusion on this point is to convert the experimentally measured $k'_C a$ values to values of $\hat{k}_C a$ straightaway, before beginning the design calculations. A design based on the rate coefficient $\hat{k}_C a$ and the driving force $(y - y_i)/y_{BM}$ will be independent of the total system pressure with the following limitations: caution should be employed in assuming that $\hat{k}_C a$ is independent of total pressure for systems having significant vapor-phase nonidealities, for systems that operate in the vicinity of the critical

point, or for total pressures higher than about 3040 to 4050 kPa (30 to 40 atm).

Experimental confirmations of the relative independence of $\hat{k}_C$ with respect to total pressure have been widely reported. Deviations do occur at extreme conditions. For example, Bretsznajder (*Prediction of Transport and Other Physical Properties of Fluids*, Pergamon Press, Oxford, 1971, p. 343) discusses the effects of pressure on the $D_{AB}p_T$ product and presents experimental data on the self-diffusion of CO_2 which show that the D - p product begins to decrease at a pressure of approximately 8100 kPa (80 atm). For reduced temperatures higher than about 1.5, the deviations are relatively modest for pressures up to the critical pressure. However, deviations are large near the critical point (see also p. 5-52). The effect of pressure on the gas-phase viscosity also is negligible for pressures below about 5060 kPa (50 atm).

For the liquid-phase mass-transfer coefficient $\hat{k}_L$, the effects of total system pressure can be ignored for all practical purposes. Thus, when using $\hat{k}_C$ and $\hat{k}_L$ for the design of gas absorbers or strippers, the primary pressure effects to consider will be those which affect the equilibrium curves and the values of m . If the pressure changes affect the hydrodynamics, then $\hat{k}_C$, $\hat{k}_L$, and a can all change significantly.

Effects of Temperature on $\hat{k}_C$ and $\hat{k}_L$ The Stanton-number relationship for gas-phase mass transfer in packed beds, Eq. (5-301), indicates that for a given system geometry the rate coefficient $\hat{k}_C$ depends only on the Reynolds number and the Schmidt number. Since the Schmidt number for a gas is approximately independent of temperature, the principal effect of temperature upon $\hat{k}_C$ arises from changes in the gas viscosity with changes in temperature. For normally encountered temperature ranges, these effects will be small owing to the fractional powers involved in Reynolds-number terms (see Tables 5-17 to 5-24). It thus can be concluded that for all

TABLE 5-20 Mass-Transfer Correlations for Flow Past Submerged Objects

Situation	Correlation	Comments E = Empirical, S = Semiempirical, T = Theoretical	References ^a
A. Single sphere	$N_{Sh} = \frac{k'_G p_{BLM} R T d_s}{P D} = \frac{2r}{r - r_s}$ $\frac{r/r_s}{N_{Sh}} \left \begin{array}{c c c c} 2 & 5 & 10 & 50 \\ \hline 4.0 & 2.5 & 2.22 & 2.04 \end{array} \right \infty \text{ (asymptotic limit)} \\ 2.0$	[T] Use with log mean concentration difference. r = distance from sphere, r_s , d_s = radius and diameter of sphere. No convection.	[141] p. 18
B. Single sphere, creeping flow with forced convection	$N_{Sh} = \frac{k'd}{D} = [4.0 + 1.21(N_{Re} N_{Sc})^{2/3}]^{1/2}$ $N_{Sh} = \frac{k'd}{D} = a(N_{Re} N_{Sc})^{1/3}$ $a = 1.00 \pm 0.01$	[T] Use with log mean concentration difference. Average over sphere. Numerical calculations. ($N_{Re} N_{Sc}$) < 10,000 N_{Re} < 1.0. Constant sphere diameter. Low mass-transfer rates. [T] Fit to above ignoring molecular diffusion. 1000 < ($N_{Re} N_{Sc}$) < 10,000.	[46][88] p. 114 [105] [138] p. 214 [101] p. 80 [138] p. 215
C. Single spheres, molecular diffusion, and forced convection, low flow rates	$N_{Sh} = 2.0 + A N_{Re}^{1/2} N_{Sc}^{1/3}$ $A = 0.5 \text{ to } 0.62$ $A = 0.60.$ $A = 0.95.$ $A = 0.95.$ $A = 0.544.$	[E] Use with log mean concentration difference. Average over sphere. Frössling Eq. ($A = 0.552$), $2 \leq N_{Re} \leq 800$, $0.6 \leq N_{Sc} \leq 2.7$. N_{Sh} lower than experimental at high N_{Re}. [E] Ranz and Marshall $2 \leq N_{Re} \leq 200$, $0.6 \leq N_{Sc} \leq 2.5$. Modifications recommended [110]. See also Table 5-23-O. [E] Liquids $2 \leq N_{Re} \leq 2,000$. Graph in Ref. 138, p. 217–218. [E] $100 \leq N_{Re} \leq 700$; $1,200 \leq N_{Sc} \leq 1525$. [E] Use with arithmetic concentration difference. $N_{Sc} = 1$; $50 \leq N_{Re} \leq 350$.	[39] [77], p. 194 [88] p. 114 [141] p. 276 [39] p. 409, 647 [121] [110] [138] p. 217 [141] p. 276 [65][66] p. 482 [138] p. 217 [126][141] p. 276 [81][141] p. 276
D. Same as 5-20-C	$N_{Sh} = \frac{k'd_s}{D} = 2.0 + 0.575 N_{Re}^{1/2} N_{Sc}^{0.35}$	[E] Use with log mean concentration difference. $N_{Sc} \leq 1$, $N_{Re} < 1$.	[70][141] p. 276
E. Same as 5-20-C	$N_{Sh} = \frac{k'd_s}{D} = 2.0 + 0.552 N_{Re}^{0.53} N_{Sc}^{1/3}$	[E] Use with log mean concentration difference. $1.0 < N_{Re} \leq 48,000$ Gases: $0.6 \leq N_{Sc} \leq 2.7$.	[66] p. 482
F. Single spheres, forced concentration, any flow rate	$N_{Sh} = \frac{k'_L d_s}{D} = 2.0 + 0.59 \left[\frac{E^{1/3} d_p^{4/3} \rho}{\mu} \right]^{0.57} N_{Sc}^{1/3}$ Energy dissipation rate per unit mass of fluid (ranges $570 < N_{Sc} < 1420$): $E = \left(\frac{C_{Dr}}{2} \right) \left(\frac{v_r^3}{d_p} \right) \frac{\text{m}^2}{\text{s}^3}$	[S] Correlates large amount of data and compares to published data. v_r = relative velocity between fluid and sphere, m/s. C_{Dr} = drag coefficient for single particle fixed in fluid at velocity v_r. See 5-23-F for calculation details and applications. $2 < \left(\frac{E^{1/3} d_p^{4/3} \rho}{\mu} \right) < 63,000$	[108]
G. Single spheres, forced convection, high flow rates, ignoring molecular diffusion	$N_{Sh} = \frac{k'd_s}{D} = 0.347 N_{Re}^{0.62} N_{Sc}^{1/3}$ $N_{Sh} = \frac{k'd_s}{D} = 0.33 N_{Re}^{0.6} N_{Sc}^{1/3}$ $N_{Sh} = \frac{k'd_s}{D} = 0.43 N_{Re}^{0.56} N_{Sc}^{1/3}$ $N_{Sh} = \frac{k'd_s}{D} = 0.692 N_{Re}^{0.514} N_{Sc}^{1/3}$	[E] Use with arithmetic concentration difference. Liquids, $2000 < N_{Re} < 17,000$. High N_{Sc}, graph in Ref. 138, p. 217–218. [E] $1500 \leq N_{Re} \leq 12,000$. [E] $200 \leq N_{Re} \leq 4 \times 10^4$, “air” $\leq N_{Sc} \leq$ “water.” [E] $500 \leq N_{Re} \leq 5000$.	[66] p. 482 [147] [138] p. 217 [141] p. 276 [141] p. 276 [112] [141] p. 276
H. Single sphere immersed in bed of smaller particles. For gases.	$N_{Sh,avg} = \frac{k d_1}{D'} = \epsilon \left[4 + \frac{4}{5} N_{Pe'}^{2/3} + \frac{4}{\pi} N_{Pe'} \right] \left(1 + \frac{1}{9} N_{Pe'} \right)^{1/2}$ Limit $N_{Pe'} \rightarrow 0$, $N_{Sh,avg} = 2\epsilon$	[T] Compared to experiment. $N_{Pe'} = \frac{u_o d_1}{D'}$, $D' = D/\tau$, D = molecular diffusivity, d_1 = diameter large particle, τ = tortuosity. Arithmetic conc. difference fluid flow in inert bed follows Darcy's law.	[71]
I. Single cylinders, perpendicular flow	$N_{Sh} = \frac{k'd_s}{D} = A N_{Re}^{1/2} N_{Sc}^{1/3}, A = 0.82$ $A = 0.74$ $A = 0.582$ $j_D = 0.600 (N_{Re})^{-0.487}$ $N_{Sh} = \frac{k'd_{cyl}}{D}$	[E] $100 < N_{Re} \leq 3500$, $N_{Sc} = 1560$. [E] $120 \leq N_{Re} \leq 6000$, $N_{Sc} = 2.44$. [E] $300 \leq N_{Re} \leq 7600$, $N_{Sc} = 1200$. [E] Use with arithmetic concentration difference. $50 \leq N_{Re} \leq 50,000$; gases, $0.6 \leq N_{Sc} \leq 2.6$; liquids; $1000 \leq N_{Sc} \leq 3000$. Data scatter $\pm 30\%$.	[141] p. 276 [141] p. 276 [142] [141] p. 276 [66] p. 486

5-70 HEAT AND MASS TRANSFER

TABLE 5-20 Mass-Transfer Correlations for Flow Past Submerged Objects (Concluded)

Situation	Correlation	Comments E = Empirical, S = Semiempirical, T = Theoretical	References*
J. Rotating cylinder in an infinite liquid, no forced flow	$j_D = \frac{k'}{v} N_{Sc}^{0.644} = 0.0791 N_{Re}^{-0.30}$ Results presented graphically to $N_{Re} = 241,000$. $N_{Re} = \frac{v d_{cyl} \rho}{\mu} \text{ where } v = \frac{\omega d_{cyl}}{2} = \text{peripheral velocity}$	[E] Used with arithmetic concentration difference. Useful geometry in electrochemical studies. $112 < N_{Re} \leq 100,000$, $835 < N_{Sc} < 11490$ k' = mass-transfer coefficient, cm/s; ω = rotational speed, radian/s.	[60] [138] p. 238
K. Stationary or rotating cylinder for air	Stationary: $N_{Sh,avg} = A N_{Re}^c S_c^{1/3}$ $2.0 \times 10^4 \leq N_{Re} \leq 2.5 \times 10^5$; $d/H = 0.3$, $Tu = 0.6\%$ $A = 0.0539$, $c = 0.771$ [114] A and c depend on geometry [37] Rotating in still air: $N_{Sh,avg} = 0.169 N_{Re, \omega}^{2/3}$ $1.0E4 \leq N_{Re, \omega} \leq 1.0E5$; $N_{Sc} \approx 2.0$; $N_{Gr} \approx 2.0 \times 10^6$	[E] Reasonable agreement with data of other investigators. d = diameter of cylinder, H = height of wind tunnel, Tu of = turbulence level, $N_{Re, \omega}$ = rotational Reynold's number = $u_\omega d \rho / \mu$, u_ω = cylinder surface velocity. Also correlations for two-dimensional slot jet flow [114]. For references to other correlations see [37].	[37] [114]
L. Oblate spheroid, forced convection	$j_D = \frac{N_{Sh}}{N_{Re} N_{Sc}^{1/3}} = 0.74 N_{Re}^{-0.5}$ $N_{Re} = \frac{d_{ch} v \rho}{\mu}, d_{ch} = \frac{\text{total surface area}}{\text{perimeter normal to flow}}$ e.g., for cube with side length a, $d_{ch} = 1.27a$. $N_{Sh} = \frac{k' d_{ch}}{D}$	[E] Used with arithmetic concentration difference. $120 \leq N_{Re} \leq 6000$; standard deviation 2.1%. Eccentricities between 1:1 (spheres) and 3:1. Oblate spheroid is often approximated by drops.	[141] p. 284 [142]
M. Other objects, including prisms, cubes, hemispheres, spheres, and cylinders; forced convection	$j_D = 0.692 N_{Re, p}^{-0.496}, N_{Re, p} = \frac{v d_{ch} \rho}{\mu}$ Terms same as in 5-20-J.	[E] Used with arithmetic concentration difference. Agrees with cylinder and oblate spheroid results, $\pm 15\%$. Assumes molecular diffusion and natural convection are negligible. $500 \leq N_{Re, p} \leq 5000$. Turbulent.	[88] p. 115 [141] p. 285 [111] [112]
N. Other objects, molecular diffusion limits	$N_{Sh} = \frac{k' d_{ch}}{D} = A$	[T] Use with arithmetic concentration difference. Hard to reach limits in experiments. Spheres and cubes $A = 2$, tetrahedrons $A = 2\sqrt{6}$ octahedrons $2\sqrt{2}$.	[88] p. 114
O. Shell side of microporous hollow fiber module for solvent extraction	$N_{Sh} = \beta [d_h (1 - \phi)/L] N_{Re}^{0.6} N_{Sc}^{0.33}$ $N_{Sh} = \frac{\bar{K} d_h}{D}$ $N_{Re} = \frac{d_h v \rho}{\mu}, \bar{K} = \text{overall mass-transfer coefficient}$ $\beta = 5.8$ for hydrophobic membrane. $\beta = 6.1$ for hydrophilic membrane.	[E] Use with logarithmic mean concentration difference. d_h = hydraulic diameter $= \frac{4 \times \text{cross-sectional area of flow}}{\text{wetted perimeter}}$ ϕ = packing fraction of shell side. L = module length. Based on area of contact according to inside or outside diameter of tubes depending on location of interface between aqueous and organic phases. Can also be applied to gas-liquid systems with liquid on shell side.	[118]

See Table 5-23 for flow in packed beds.

*See the beginning of the "Mass Transfer" subsection for references.

TABLE 5-21 Mass-Transfer Correlations for Drops, Bubbles, and Bubble Columns

Conditions	Correlations	Comments E = Empirical, S = Semiempirical, T = Theoretical	References*																																																	
A. Single liquid drop in immiscible liquid, drop formation, discontinuous (drop) phase coefficient	$\hat{k}_{df} = A \left(\frac{\rho_d}{M_d} \right)_{av} \left(\frac{D_d}{\pi t_f} \right)^{1/2}$ $A = \frac{24}{7}$ (penetration theory) $A = 1.31$ (semiempirical value) $A = \left[\frac{24}{7} (0.8624) \right]$ (extension by fresh surface elements)	[T,S] Use arithmetic mole fraction difference. Fits some, but not all, data. Low mass transfer rate. M_d = mean molecular weight of dispersed phase; t_f = formation time of drop. $k_{L,d}$ = mean dispersed liquid phase M.T. coefficient kmole/[s·m ² (mole fraction)].	[141] p. 399																																																	
B. Same as 5-21-A	$\hat{k}_{df} = 0.0432$ $\times \frac{d_p}{t_f} \left(\frac{\rho_d}{M_d} \right)_{av} \left(\frac{u_o}{d_p g} \right)^{0.089} \left(\frac{d_p^2}{t_f D_d} \right)^{-0.334} \left(\frac{\mu_d}{\sqrt{\rho_d d_p} \sigma_{gc}} \right)^{-0.601}$	[E] Use arithmetic mole fraction difference. Based on 23 data points for 3 systems. Average absolute deviation 26%. Use with surface area of drop after detachment occurs. u_o = velocity through nozzle; σ = interfacial tension.	[141] p. 401 [144] p. 434																																																	
C. Single liquid drop in immiscible liquid, drop formation, continuous phase coefficient	$\hat{k}_{cf} = 4.6 \left(\frac{\rho_c}{M_c} \right)_{av} \sqrt{\frac{D_c}{\pi t_f}}$	[T] Use arithmetic mole fraction difference. Based on rate of bubble growth away from fixed orifice. Approximately three times too high compared to experiments.	[141] p. 402																																																	
D. Same as 5-21-C	$k_{L,c} = 0.386$ $\times \left(\frac{\rho_c}{M_c} \right)_{av} \left(\frac{D_c}{t_f} \right)^{0.5} \left(\frac{\rho_c \sigma_{gc}}{\Delta \rho g t_f \mu_c} \right)^{0.407} \left(\frac{g t_f^2}{d_p} \right)^{0.148}$	[E] Average absolute deviation 11% for 20 data points for 3 systems.	[141] p. 402 [144] p. 434																																																	
E. Single liquid drop in immiscible liquid, free rise or fall, discontinuous phase coefficient, stagnant drops	$k_{L,d,m} = \frac{-d_p}{6t} \left(\frac{\rho_d}{M_d} \right)_{av} \ln \left\{ \frac{6}{\pi^2} \sum_{j=1}^{\infty} \frac{1}{j^2} \exp \left[\left(\frac{-D_d j^2 \pi^2 t}{(d_p/2)^2} \right) \right] \right\}$	[T] Use with log mean mole fraction differences based on ends of column. t = rise time. No continuous phase resistance. Stagnant drops are likely if drop is very viscous, quite small, or is coated with surface active agent. $k_{L,d,m}$ = mean dispersed liquid M.T. coefficient.	[141] p. 404 [144] p. 435																																																	
F. Same as 5-21-E	$\hat{k}_{L,d,m} = \frac{-d_p}{6t} \left(\frac{\rho_d}{M_d} \right)_{av} \ln \left[1 - \frac{\pi D_d^{1/2} t^{1/2}}{d_p/2} \right]$	[S] See 5-21-E. Approximation for fractional extractions less than 50%.	[141] p. 404 [144] p. 435																																																	
G. Same as 5-21-E, continuous phase coefficient, stagnant drops, spherical	$N_{Sh} = \frac{k_{L,c,m} d_c}{D_c} = 0.74 \left(\frac{\rho_c}{M_c} \right)_{av} N_{Re}^{1/2} (N_{Sc,c})^{1/3}$	[E] $N_{Re} = \frac{v_s d_p \rho_c}{\mu_c}$, v_s = slip velocity between drop and continuous phase.	[141] p. 407 [142][144] p. 436																																																	
H. Single bubble or drop with surfactant. Stokes flow.	$N_{Sh} = 2.0 + \alpha N_{Pe,s}$, $N_{Sh} = 2rk/D$ $\alpha = \frac{5.49}{A + 6.10} + \frac{A}{A + 28.64}$ $\beta = \frac{0.35A + 17.21}{A + 34.14}$ $2r = 2$ to $50 \mu\text{m}$, $A = 2.8\text{E}4$ to $7.0\text{E}5$ $0.0026 < N_{Pe,s} < 340$, $2.1 < N_{Ma} < 1.3\text{E}6$ $N_{Pe} = 1.0$ to 2.5×10^4 , $N_{Re} = 2.2 \times 10^{-6}$ to 0.034	[T] A = surface retardation parameter $A = B \Gamma_o r / \mu D_s = N_{Ma} N_{Pe,s}$ $N_{Ma} = B \Gamma_o / \mu u$ = Marangoni no. Γ = surfactant surface conc. $N_{Pe,s}$ = surface Peclet number = ur/D_s D_s = surface diffusivity N_{Pe} = bulk Peclet number For $A \gg 1$ acts like rigid sphere: $\beta \rightarrow 0.35$, $\alpha \rightarrow 1/2864 = 0.035$	[120]																																																	
I. 5-21-E, oblate spheroid	$N_{Sh} = \frac{k_{L,c,m} d_3}{D_c} = 0.74 \left(\frac{\rho_c}{M_c} \right)_{av} (N_{Re,3})^{1/2} (N_{Sc,c})^{1/3}$ $N_{Re,3} = \frac{v_s d_3 \rho_c}{\mu_c}$	[E] Used with log mean mole fraction. Differences based on ends of extraction column; 100 measured values $\pm 2\%$ deviation. Based on area oblate spheroid. v_s = slip velocity, $d_3 = \frac{\text{total drop surface area}}{\text{perimeter normal to flow}}$	[141] p. 285, 406, 407																																																	
J. Single liquid drop in immiscible liquid, Free rise or fall, discontinuous phase coefficient, circulating drops	$k_{d,circ} = -\frac{d_p}{6\theta} \ln \left[\frac{3}{8} \sum_{j=1}^{\infty} B_j^2 \exp \left(-\frac{\lambda_j 64 D_d \theta}{d_p^2} \right) \right]$ Eigenvalues for Circulating Drop <table><thead><tr><th>$k_d d_p / D_d$</th><th>λ_1</th><th>λ_2</th><th>λ_3</th><th>B_1</th><th>B_2</th><th>B_3</th></tr></thead><tbody><tr><td>3.20</td><td>0.262</td><td>0.424</td><td></td><td>1.49</td><td>0.107</td><td></td></tr><tr><td>10.7</td><td>0.680</td><td>4.92</td><td></td><td>1.49</td><td>0.300</td><td></td></tr><tr><td>26.7</td><td>1.082</td><td>5.90</td><td>15.7</td><td>1.49</td><td>0.495</td><td>0.205</td></tr><tr><td>107</td><td>1.484</td><td>7.88</td><td>19.5</td><td>1.39</td><td>0.603</td><td>0.384</td></tr><tr><td>320</td><td>1.60</td><td>8.62</td><td>21.3</td><td>1.31</td><td>0.583</td><td>0.391</td></tr><tr><td>∞</td><td>1.656</td><td>9.08</td><td>22.2</td><td>1.29</td><td>0.596</td><td>0.386</td></tr></tbody></table>	$k_d d_p / D_d$	λ_1	λ_2	λ_3	B_1	B_2	B_3	3.20	0.262	0.424		1.49	0.107		10.7	0.680	4.92		1.49	0.300		26.7	1.082	5.90	15.7	1.49	0.495	0.205	107	1.484	7.88	19.5	1.39	0.603	0.384	320	1.60	8.62	21.3	1.31	0.583	0.391	∞	1.656	9.08	22.2	1.29	0.596	0.386	[T] Use with arithmetic concentration difference. θ = drop residence time. A more complete listing of eigenvalues is given by Refs. 62 and 76. $k'_{L,d,circ}$ is m/s.	[62][76][141] p. 405 [152] p. 523
$k_d d_p / D_d$	λ_1	λ_2	λ_3	B_1	B_2	B_3																																														
3.20	0.262	0.424		1.49	0.107																																															
10.7	0.680	4.92		1.49	0.300																																															
26.7	1.082	5.90	15.7	1.49	0.495	0.205																																														
107	1.484	7.88	19.5	1.39	0.603	0.384																																														
320	1.60	8.62	21.3	1.31	0.583	0.391																																														
∞	1.656	9.08	22.2	1.29	0.596	0.386																																														

TABLE 5-21 Mass-Transfer Correlations for Drops, Bubbles, and Bubble Columns (Continued)

Conditions	Correlations	Comments E = Empirical, S = Semiempirical, T = Theoretical	References*
K. Same as 5-21-J	$\hat{k}_{L,d,circ} = -\frac{d_p}{6\theta} \left(\frac{\rho_d}{M_d} \right)_{av} \ln \left[1 - \frac{R^{1/2} \pi D_d^{1/2} \theta^{1/2}}{d_p/2} \right]$	[E] Used with mole fractions for extraction less than 50%, $R \approx 2.25$.	[141] p. 405
L. Same as 5-21-J	$N_{Sh} = \frac{\hat{k}_{L,d,circ} d_p}{D_d}$ $= 31.4 \left(\frac{\rho_d}{M_d} \right)_{av} \left(\frac{4D_d t}{d_p^2} \right)^{-0.34} N_{Sc,d}^{-0.125} \left(\frac{d_p v_s^2 \rho_c}{\sigma g_c} \right)^{-0.37}$	[E] Used with log mean mole fraction difference. d_p = diameter of sphere with same volume as drop. $856 \leq N_{Sc} \leq 79,800$, $2.34 \leq \sigma \leq 4.8$ dynes/cm.	[144] p. 435 [145]
M. Liquid drop in immiscible liquid, free rise or fall, continuous phase coefficient, circulating single drops	$N_{Sh,c} = \frac{k'_{L,c} d_p}{D_d}$ $= \left[2 + 0.463 N_{Re,drop}^{0.484} N_{Sc,c}^{0.339} \left(\frac{d_p g_c^{1/3}}{D_c^{2/3}} \right)^{0.072} \right] F$ $F = 0.281 + 1.615K + 3.73K^2 - 1.874K$ $K = N_{Re,drop}^{1/8} \left(\frac{\mu_c}{\mu_d} \right)^{1/4} \left(\frac{\mu_c v_s}{\sigma g_c} \right)^{1/6}$	[E] Used as an arithmetic concentration difference. $N_{Re,drop} = \frac{d_p v_s \rho_c}{\mu_c}$ Solid sphere form with correction factor F .	[82]
N. Same as 5-21-M, circulating, single drop	$N_{Sh} = \frac{k_{L,c} d_p}{D_c} = 0.6 \left(\frac{\rho_c}{M_c} \right)_{av} N_{Re,drop}^{1/2} N_{Sc,c}^{1/2}$	[E] Used as an arithmetic concentration difference. Low σ .	[141] p. 407
O. Same as 5-21-M, circulating swarm of drops	$k_{L,c} = 0.725 \left(\frac{\rho_c}{M_c} \right)_{av} N_{Re,drop}^{-0.43} N_{Sc,c}^{-0.58} v_s (1 - \phi_d)$	[E] Used as an arithmetic concentration difference. Low σ , disperse-phase holdup of drop swarm. ϕ_d = volume fraction dispersed phase.	[141] p. 407 [144] p. 436
P. Liquid drops in immiscible liquid, free rise or fall, discontinuous phase coefficient, oscillating drops	$N_{Sh} = \frac{k_{L,d,osc} d_p}{D_d}$ $= 0.32 \left(\frac{\rho_d}{M_d} \right)_{av} \left(\frac{4D_d t}{d_p^2} \right)^{-0.14} N_{Re,drop}^{0.68} \left(\frac{\sigma^3 g_c^2 \rho_c^2}{g \mu_c^4 \Delta \rho} \right)^{0.10}$	[E] Used with a log mean mole fraction difference. Based on ends of extraction column. $N_{Re,drop} = \frac{d_p v_s \rho_c}{\mu_c}$, $411 \leq N_{Re} \leq 3114$ d_p = diameter of sphere with volume of drop. Average absolute deviation from data, 10.5%. Low interfacial tension (3.5–5.8 dyn), $\mu_c < 1.35$ centipoise.	[141] p. 406 [144] p. 435 [145]
Q. Same as 5-21-P	$k_{L,d,osc} = \frac{0.00375 v_s}{1 + \mu_d / \mu_c}$	[T] Use with log mean concentration difference. Based on end of extraction column. No continuous phase resistance. $k_{L,d,osc}$ in cm/s, v_s = drop velocity relative to continuous phase.	[138] p. 228 [141] p. 405
R. Single liquid drop in immiscible liquid, range rigid to fully circulating	$N_{Sh,c,rigid} = \frac{k_c d_p}{D_c} = 2.43 + 0.774 N_{Re}^{0.5} N_{Sc}^{0.33}$ $+ 0.0103 N_{Re} N_{Sc}^{0.33}$ $N_{Sh,c,fully\ circular} = \left[\frac{2}{\pi^{0.5}} \right] N_{Pe,c}^{0.5}$ Drops in intermediate range: $\frac{N_{Sh,c} - N_{Sh,c,rigid}}{N_{Sh,c,fully\ circular} - N_{Sh,c,rigid}} = 1 - \exp [-(4.18 \times 10^{-3}) N_{Pe,c}^{0.42}]$	[E] Allows for slight effect of wake. Rigid drops: $10^4 < N_{Pe,c} < 10^6$ Circulating drops: $10 < N_{Re} < 1200$, $190 < N_{Sc} < 241,000$, $10^3 < N_{Pe,c} < 10^6$	[146] p. 58
S. Coalescing drops in immiscible liquid, discontinuous phase coefficient	$\hat{k}_{d,coal} = 0.173 \frac{d_p}{t_f} \left(\frac{\rho_d}{M_d} \right)_{av} \left(\frac{\mu_d}{\rho_d D_d} \right)^{-1.115}$ $\times \left(\frac{\Delta \rho g d_p^2}{\sigma g_c} \right)^{1.302} \left(\frac{v_s^2 t_f}{D_d} \right)^{0.146}$	[E] Used with log mean mole fraction difference. 23 data points. Average absolute deviation 25%. t_f = formation time.	[141] p. 408
T. Same as 5-21-S, continuous phase coefficient	$\hat{k}_{c,coal} = 5.959 \times 10^{-4} \left(\frac{\rho}{M} \right)_{av}$ $\times \left(\frac{D_c}{t_f} \right)^{0.5} \left(\frac{\rho_d u_s^3}{g \mu_c} \right)^{0.332} \left(\frac{d_p^2 \rho_c \rho_d v_s^3}{\mu_d \sigma g_c} \right)^{0.525}$	[E] Used with log mean mole fraction difference. 20 data points. Average absolute deviation 22%.	[141] p. 409

TABLE 5-21 Mass-Transfer Correlations for Drops, Bubbles, and Bubble Columns (Continued)

Conditions	Correlations	Comments E = Empirical, S = Semiempirical, T = Theoretical	References*
U. Single liquid drops in gas, gas side coefficient	$\frac{\hat{k}_g M_g d_p P}{D_{\text{gas}} \rho_g} = 2 + A N_{Re,g}^{1/2} N_{Sc,g}^{1/3}$ $A = 0.552 \text{ or } 0.60$ $N_{Re,g} = \frac{d_p \rho_g v_s}{\mu_g}$	[E] Used for spray drying (arithmetic partial pressure difference). v_s = slip velocity between drop and gas stream. Sometimes written with $M_g P / \rho_g = RT$.	[90] p. 388 [121]
V. Single water drop in air, liquid side coefficient	$k_L = 2 \left(\frac{D_L}{\pi t} \right)^{1/2}$, short contact times $k_L = 10 \frac{D_L}{d_p}$, long contact times	[T] Use arithmetic concentration difference. Penetration theory. t = contact time of drop. Gives plot for $k_G a$ also. Air-water system.	[90] p. 389
W. Single bubbles of gas in liquid, continuous phase coefficient, very small bubbles	$N_{Sh} = \frac{k'_c d_b}{D_c} = 1.0 (N_{Re} N_{Sc})^{1/3}$	[T] Solid-sphere Eq. (see Table 5-20-B). $d_b < 0.1 \text{ cm}$, k'_c is average over entire surface of bubble.	[105] [138] p. 214
X. Same as 5-21-W, medium to large bubbles	$N_{Sh} = \frac{k'_c d_b}{D_c} = 1.13 (N_{Re} N_{Sc})^{1/2}$	[T] Use arithmetic concentration difference. Droplet equation: $d_b > 0.5 \text{ cm}$.	[138] p. 231
Y. Same as 5-21-X	$N_{Sh} = \frac{k'_c d_b}{D_c} = 1.13 (N_{Re} N_{Sc})^{1/2} \left[\frac{d_b}{0.45 + 0.2 d_b} \right]$ $500 \leq N_{Re} \leq 8000$	[S] Use arithmetic concentration difference. Modification of above (X), $d_b > 0.5 \text{ cm}$. No effect SAA for $d_p > 0.6 \text{ cm}$.	[83][138] p. 231
Z. Taylor bubbles in single capillaries (square or circular)	$k_L a = 4.5 \left(\frac{D u_G}{L_{uc}} \right)^{1/2} \frac{1}{d_c}$ Applicable $\left(\frac{u_G + u_L}{L_{slug}} \right)^{0.5} > 3s^{-0.5}$	[E] Air-water L_{uc} = unit cell length, L_{slug} = slug length, d_c = capillary i.d. For most data $k_L a \pm 20\%$.	[153]
AA. Gas-liquid mass transfer in monoliths	$k_L a \approx 0.1 \left(\frac{P}{V} \right)^{1/4}$ P/V = power/volume (kW/m^3), range = 100 to 10,000	[E] Each channel in monolith is a capillary. Results are in expected order of magnitude for capillaries based on 5-21-Z. k_L is larger than in stirred tanks.	[93]
AB. Rising small bubbles of gas in liquid, continuous phase. Calderbank and Moo-Young correlation	$N_{Sh} = \frac{k'_c d_b}{D_c} = 2 + 0.31 (N_{Gr})^{1/3} N_{Sc}^{1/3}$, $d_b < 0.25 \text{ cm}$ $N_{Re} = \frac{d_b^3 \rho_G - \rho_L g}{\mu_L D_L} = \text{Raleigh number}$	[E] Use with arithmetic concentration difference. Valid for single bubbles or swarms. Independent of agitation as long as bubble size is constant. Recommended by [136]. Note that $N_{Re} = N_{Gr} N_{Sc}$.	[47][66] p. 451 [88] p. 119 [152] p. 156 [136]
AC. Same as 5-21-AB, large bubbles	$N_{Sh} = \frac{k'_c d_b}{D_c} = 0.42 (N_{Gr})^{1/3} N_{Sc}^{1/2}$, $d_b > 0.25 \text{ cm}$ $\frac{\text{Interfacial area}}{\text{volume}} = a = \frac{6 H_g}{d_b}$	[E] Use with arithmetic concentration difference. For large bubbles, k'_c is independent of bubble size and independent of agitation or liquid velocity. Resistance is entirely in liquid phase for most gas-liquid mass transfer. H_g = fractional gas holdup, volume gas/total volume.	[47][66] p. 452 [88] p. 119 [97] p. 249 [136]
AD. Bubbles in bubble columns. Hughmark correlation	$N_{Sh} = \frac{k_L d}{D} = 2 + b N_{Sc}^{0.546} N_{Re}^{0.779} \left(\frac{d_g^{1/3}}{D^{2/3}} \right)^{0.116}$ $b = 0.061$ single gas bubbles; $b = 0.0187$ swarms of bubbles, $V_s = \frac{V_g}{\Phi_G} - \frac{V_L}{1 - \Phi_G}$	[E] d = bubble diameter Air-liquid. Recommended by [136, 152]. For swarms, calculate N_{Re} with slip velocity V_s . Φ_G = gas holdup V_G = superficial gas velocity Col. diameter = 0.025 to 1.1 m $\rho'_L = 776 \text{ to } 1696 \text{ kg/m}^3$ $\mu_L = 0.0009 \text{ to } 0.152 \text{ Pa}\cdot\text{s}$	[55] [82] [152] p. 144
AE. Bubbles in bubble column	$k_L a = 0.00315 u_G^{0.59} \mu_{\text{eff}}^{-0.84}$	[E] Recommended by [136].	[57]

TABLE 5-21 Mass-Transfer Correlations for Drops, Bubbles, and Bubble Columns (Concluded)

Conditions	Correlations	Comments E = Empirical, S = Semiempirical, T = Theoretical	References*
AF. Bubbles in bubble column	$k_L = \frac{0.15D}{d_{Vs}} \left(\frac{v}{D} \right)^{1/2} N_{Re}^{3/4}$	[E] d_{Vs} = Sauter mean bubble diameter, $N_{Re} = d_{Vs} u_G \rho_L / \mu_L$. Recommended by [49] based on experiments in industrial system.	[49] [133]
AG. High-pressure bubble column	$k_L a = 1.77 \sigma^{-0.22} \exp(1.65 u_L - 65.3 \mu_L) \epsilon_g^{1.2}$ $790 < \rho_L < 1580 \text{ kg/m}^3$ $0.00036 < \mu_L < 0.0383 \text{ Pa}\cdot\text{s}$ $0.0232 < \sigma < 0.0726 \text{ N/m}$ $0.028 < u_g < 0.678 \text{ m/s}$ $0 < u_L < 0.00089 \text{ m/s}$	[E] Pressure up to 4.24 MPa. T up to 92°C. ϵ_g = gas holdup. Correlation to estimate ϵ_g is given. $0.045 < d_{col} < 0.45 \text{ m}$, $d_{col}/H_{col} > 5$ $0.97 < \rho_g < 33.4 \text{ kg/m}^3$	[96]
AH. Three phase (gas-liquid-solid) bubble column to solid spheres	$N_{sh} = \frac{k_L d_p}{D} = 2.0 + 0.545 N_{Sc}^{1/3} \left(\frac{e d_p^4}{\nu^3} \right)^{0.264}$ $N_{Sc} = 137 \text{ to } 50,000$ (very wide range) d_p = particle diameter (solids)	[E] e = local energy dissipation rate/unit mass, $e = u_g g$ $N_{Sc} = \mu_L / (\rho_L D)$ Recommended by [136].	[129] [136]

See Table 5-22 for agitated systems.

*See the beginning of the "Mass Transfer" subsection for references.

practical purposes $\hat{k}_C$ is independent of temperature and pressure in the normal ranges of these variables.

For modest changes in temperature the influence of temperature upon the interfacial area a may be neglected. For example, in experiments on the absorption of SO_2 in water, Whitney and Vivian [*Chem. Eng. Prog.*, **45**, 323 (1949)] found no appreciable effect of temperature upon $k'_C a$ over the range from 10 to 50°C.

With regard to the liquid-phase mass-transfer coefficient, Whitney and Vivian found that the effect of temperature upon $k_L a$ could be explained entirely by variations in the liquid-phase viscosity and diffusion coefficient with temperature. Similarly, the oxygen-desorption data of Sherwood and Holloway [*Trans. Am. Inst. Chem. Eng.*, **36**, 39 (1940)] show that the influence of temperature upon H_L can be explained by the effects of temperature upon the liquid-phase viscosity and diffusion coefficients (see Table 5-24-A).

It is important to recognize that the effects of temperature on the liquid-phase diffusion coefficients and viscosities can be very large and therefore must be carefully accounted for when using $\hat{k}_L$ or $\hat{H}_L$ data. For liquids the mass-transfer coefficient $\hat{k}_L$ is correlated as either the Sherwood number or the Stanton number as a function of the Reynolds and Schmidt numbers (see Table 5-24). Typically, the general form of the correlation for H_L is (Table 5-24)

$$H_L = b N_{Re}^a N_{Sc}^{1/2} \quad (5-306)$$

where b is a proportionality constant and the exponent a may range from about 0.2 to 0.5 for different packings and systems. The liquid-phase diffusion coefficients may be corrected from a base temperature T_1 to another temperature T_2 by using the Einstein relation as recommended by Wilke [*Chem. Eng. Prog.*, **45**, 218 (1949)]:

$$D_2 = D_1 (T_2/T_1) (\mu_1/\mu_2) \quad (5-307)$$

The Einstein relation can be rearranged to the following equation for relating Schmidt numbers at two temperatures:

$$N_{Sc2} = N_{Sc1} (T_1/T_2) (\rho_1/\rho_2) (\mu_2/\mu_1)^2 \quad (5-308)$$

Substitution of this relation into Eq. (5-306) shows that for a given geometry the effect of temperature on H_L can be estimated as

$$H_{L2} = H_{L1} (T_1/T_2)^{1/2} (\rho_1/\rho_2)^{1/2} (\mu_2/\mu_1)^{1-a} \quad (5-309)$$

In using these relations it should be noted that for equal liquid flow rates

$$H_{L2}/H_{L1} = (\hat{k}_{L1} a_1)/(\hat{k}_{L2} a_2) \quad (5-310)$$

Effects of System Physical Properties on $\hat{k}_C$ and $\hat{k}_L$ When designing packed towers for nonreacting gas-absorption systems for which no experimental data are available, it is necessary to make corrections for differences in composition between the existing test data and the system in question. The ammonia-water test data (see Table 5-24-B) can be used to estimate H_C , and the oxygen desorption data (see Table 5-24-A) can be used to estimate H_L . The method for doing this is illustrated in Table 5-24-E. There is some conflict on whether the value of the exponent for the Schmidt number is 0.5 or 2/3 [Yadav and Sharma, *Chem. Eng. Sci.*, **34**, 1423 (1979)]. Despite this disagreement, this method is extremely useful, especially for absorption and stripping systems.

It should be noted that the influence of substituting solvents of widely differing viscosities upon the interfacial area a can be very large. One therefore should be cautious about extrapolating $k_L a$ data to account for viscosity effects between different solvent systems.

Effects of High Solute Concentrations on $\hat{k}_C$ and $\hat{k}_L$ As discussed previously, the stagnant-film model indicates that $\hat{k}_C$ should be independent of y_{BM} and $\hat{k}_C$ should be inversely proportional to y_{BM} . The data of Vivian and Behrman [*Am. Inst. Chem. Eng. J.*, **11**, 656 (1965)] for the absorption of ammonia from an inert gas strongly suggest that the film model's predicted trend is correct. This is another indication that the most appropriate rate coefficient to use in concentrated systems is $\hat{k}_C$ and the proper driving-force term is of the form $(y - y_i)/y_{BM}$.

The use of the rate coefficient $\hat{k}_L$ and the driving force $(x_i - x)/x_{BM}$ is believed to be appropriate. For many practical situations the liquid-phase solute concentrations are low, thus making this assumption unimportant.

Influence of Chemical Reactions on $\hat{k}_C$ and $\hat{k}_L$ When a chemical reaction occurs, the transfer rate may be influenced by the chemical reaction as well as by the purely physical processes of diffusion and convection within the two phases. Since this situation is common in gas absorption, gas absorption will be the focus of this discussion. One must consider the impacts of chemical equilibrium and reaction kinetics on the absorption rate in addition to accounting for the effects of gas solubility, diffusivity, and system hydrodynamics.

There is no sharp dividing line between pure physical absorption and absorption controlled by the rate of a chemical reaction. Most cases fall in an intermediate range in which the rate of absorption is limited both by the resistance to diffusion and by the finite velocity of the reaction. Even in these intermediate cases the equilibria between the various diffusing species involved in the reaction may affect the rate of absorption.

TABLE 5-22 Mass-Transfer Correlations for Particles, Drops, and Bubbles in Agitated Systems

Situation	Correlation	Comments E = Empirical, S = Semiempirical, T = Theoretical	References*														
A. Solid particles suspended in agitated vessel containing vertical baffles, continuous phase coefficient	$\frac{k'_{LT} d_p}{D} = 2 + 0.6 N_{Re,T}^{1/2} N_{Sc}^{1/3}$ Replace v_{slip} with v_T = terminal velocity. Calculate Stokes' law terminal velocity $v_T = \frac{d_p^2 \rho_p - \rho_c g}{18 \mu_c}$ and correct: <table><tr><td>$\frac{N_{Re,T}}{v_T/v_T}$</td><td>1</td><td>10</td><td>100</td><td>1,000</td><td>10,000</td><td>100,000</td></tr><tr><td></td><td>0.9</td><td>0.65</td><td>0.37</td><td>0.17</td><td>0.07</td><td>0.023</td></tr></table> Approximate: $k'_L = 2k'_{LT}$	$\frac{N_{Re,T}}{v_T/v_T}$	1	10	100	1,000	10,000	100,000		0.9	0.65	0.37	0.17	0.07	0.023	[S] Use log mean concentration difference. Modified Frossling equation: $N_{Re,T} = \frac{v_T d_p \rho_c}{\mu_c}$ (Reynolds number based on Stokes' law.) $N_{Re,T} = \frac{v_T d_p \rho_c}{\mu_c}$ (terminal velocity Reynolds number.) k'_L almost independent of d_p . Harriott suggests different correction procedures. Range k'_L/k'_{LT} is 1.5 to 8.0.	[74][138] p. 220–222 [110] <
$\frac{N_{Re,T}}{v_T/v_T}$	1	10	100	1,000	10,000	100,000											
	0.9	0.65	0.37	0.17	0.07	0.023											

TABLE 5-22 Mass-Transfer Correlations for Particles, Drops, and Bubbles in Agitated Systems (Concluded)

Situation	Correlation	Comments E = Empirical, S = Semiempirical, T = Theoretical	References*
I. Gas bubble swarms in sparged tank reactors	$k_L' a \left(\frac{v}{g^2} \right)^{1/3} = C \left[\frac{P/V_L}{\rho(vg^4)^{1/3}} \right]^a \left[\frac{q_G}{V_L} \left(\frac{v}{g^2} \right) \right]^b$ Rushton turbines: $C = 7.94 \times 10^{-4}$, $a = 0.62$, $b = 0.23$. Intermig impellers: $C = 5.89 \times 10^{-4}$, $a = 0.62$, $b = 0.19$.	[E] Use arithmetic concentration difference. Done for biological system, O ₂ transfer. $h_{\text{tank}}/D_{\text{tank}} = 2.1$; P = power, kW. V_L = liquid volume, m ³ . q_G = gasing rate, m ³ /s. $k_L' a = s^{-1}$. Since $a = m^2/m^3$, v = kinematic viscosity, m ² /s. Low viscosity system. Better fit claimed with q_G/V_L than with u_G (see 5-22-J to N).	[131]
J. Same as 5-22-I	$k_L' a = 2.6 \times 10^{-2} \left(\frac{P}{V_L} \right)^{0.4} u_G^{0.5}$	[E] Use arithmetic concentration difference. Ion free water $V_L < 2.6$, u_G = superficial gas velocity in m/s. $500 < P/V_L < 10,000$. P/V_L = watts/m ³ , V_L = liquid volume, m ³ .	[98] [123]
K. Same as 5-22-J	$k_L' a = 2.0 \times 10^{-3} \left(\frac{P}{V_L} \right)^{0.7} u_G^{0.2}$	[E] Use arithmetic concentration difference. Water with ions. $0.002 < V_L < 4.4$, $500 < P/V_L < 10,000$. Same definitions as 5-22-I.	[98] [101]
L. Same as 5-22-I, baffled tank with standard blade Rushton impeller	$k_L' a = 93.37 \left(\frac{P}{V_L} \right)^{0.76} u_G^{0.45}$	[E] Air-water. Same definitions as 5-22-I. $0.005 < u_G < 0.025$, $3.83 < N < 8.33$, $400 < P/V_L < 7000$. $h = D_{\text{tank}} = 0.305$ or 0.610 m. V_G = gas volume, m ³ , N = stirrer speed, rpm. Method assumes perfect liquid mixing.	[67] [98]
M. Same as 5-22-L	$k_L' a \frac{d_{\text{imp}}^2}{D} = 7.57 \left[\frac{\mu_{\text{eff}}}{\rho D} \right]^{0.57} \left[\frac{\mu_G}{\mu_{\text{eff}}} \right]^{0.694} \times \left[\frac{d_{\text{imp}}^2 N \rho_L}{\mu_{\text{eff}}} \right]^{1.11} \left(\frac{u_G d}{\sigma} \right)^{0.447}$ d_{imp} = impeller diameter, m; D = diffusivity, m²/s	[E] Use arithmetic concentration difference. CO ₂ into aqueous carboxyl polymethylene. Same definitions as 5-22-L. μ_{eff} = effective viscosity from power law model, Pa·s. σ = surface tension liquid, N/m.	[98] [115]
N. Same as 5-22-L, bubbles	$\frac{k_L' a d_{\text{imp}}^2}{D} = 0.060 \left(\frac{d_{\text{imp}}^2 N \rho}{\mu_{\text{eff}}} \right) \left(\frac{d_{\text{imp}}^2 N^2}{g} \right)^{0.19} \left(\frac{\mu_{\text{eff}} u_G}{\sigma} \right)^{0.6}$	[E] Use arithmetic concentration difference. O ₂ into aqueous glycerol solutions. O ₂ into aqueous millet jelly solutions. Same definitions as 5-22-L.	[98] [160]
O. Gas bubble swarm in sparged stirred tank reactor with solids present	$\frac{k_L' a}{(k_L' a)_0} = 1 - 3.54(\epsilon_s - 0.03)$ $300 \leq P/V_{rx} < 10,000$ W/m³, $0.03 \leq \epsilon_s \leq 0.12$ $0.34 \leq u_G \leq 4.2$ cm/s, $5 < \mu_L < 75$ Pa·s	[E] Use arithmetic concentration difference. Solids are glass beads, $d_p = 320$ μm. ϵ_s = solids holdup m ³ /m ³ liquid. $(k_L' a)_0$ = mass transfer in absence of solids. Ionic salt solution—noncoalescing.	[38] [132]
P. Surface aerators for air-water contact	$\frac{k_L a}{N} = b N_p^{0.71} N_{Fr}^{0.48} N_{Re}^{0.82} \left(\frac{H}{d} \right)^{-0.54} \left(\frac{V}{d^3} \right)^{-1.08}$ $b = 7 \times 10^{-6}$, $N_p = P/(\rho N^3 d^5)$ $N_{Re} = N d^2 \rho_{liq}/\mu_{liq}$ $N_{Fr} = N^2 d/g$, $P/V = 90$ to 400 W/m³	[E] Three impellers: Pitched blade downflow turbine, pitched blade upflow turbine, standard disk turbine. Baffled cylindrical tanks 1.0- and 1.5-m ID and 8.2 × 8.2-m square tank. Submergence optimized all cases. Good agreement with data. N = impeller speed, s ⁻¹ ; d = impeller diameter, m; H = liquid height, m; V = liquid volume, m ³ ; $k_L a = s^{-1}$, g = acceleration gravity = 9.81 m/s ²	[113]
Q. Gas-inducing impeller for air-water contact	$k_L a V (v/g^2)^{1/3} d^3 = A N_{Fr}^B \left(\frac{V_A}{V} \right)^C$ Single impeller: $A = 0.00497$, $B = 0.56$, $C = 0.32$ Multiple impeller: $A = 0.00746$, $B = 0.54$, $C = 0.38$	[E] Same tanks and same definitions as in 5-22-P. V_A = active volume = $p/(\pi \rho g N d)$.	[113]
R. Gas-inducing impeller with dense solids	$Sh_{GL} = \frac{k_L a d_{st}^2}{D} = (1.26 \times 10^{-5}) N_{Re}^{1.8} N_{Sc}^{0.9} N_{We}^{-0.1}$ $N_{Re} = \rho N d_{st}^2 / \mu$, $N_{Sc} = \mu/(\rho D)$, $N_{We} = \rho N^2 d_{st}^3 / \sigma$	[E] Hydrogenation with Raney-type nickel catalyst in stirred autoclave. Used varying T , p , solvents. d_{st} = stirrer diameter.	[78]

See also Table 5-21.

*See the beginning of the "Mass Transfer" subsection for references.

TABLE 5-23 Mass-Transfer Correlations for Fixed and Fluidized Beds

Transfer is to or from particles

Situation	Correlation	Comments E = Empirical, S = Semiempirical, T = Theoretical	References°												
A. For gases, fixed and fluidized beds, Gupta and Thodos correlation	$j_H = j_D = \frac{2.06}{\epsilon N_{Re}^{0.575}}, 90 \leq N_{Re} \leq A$ Equivalent: $N_{Sh} = \frac{2.06}{\epsilon} N_{Re}^{0.425} N_{Sc}^{1/3}$ For other shapes: $\frac{\epsilon j_D}{(\epsilon j_D)_{\text{sphere}}} = 0.79 \text{ (cylinder) or } 0.71 \text{ (cube)}$	[E] For spheres. $N_{Re} = \frac{v_{\text{super}} d_p \rho}{\mu}$ $A = 2453$ [Ref. 141], $A = 4000$ [Ref. 77]. For $N_{Re} > 1900$, $j_H = 1.05 j_D$. Heat transfer result is in absence of radiation. $N_{Sh} = \frac{k' d_s}{D}$ Graphical results are available for N_{Re} from 1900 to 10,300. $a = \frac{\text{surface area}}{\text{volume}} = 6(1 - \epsilon)/d_p$ For spheres, d_p = diameter. For nonspherical: $d_p = 0.567 \sqrt{\text{Part. Surf. Area}}$	[72] [73] [77] p. 195 [141]												
B. For gases, for fixed beds, Petrovic and Thodos correlation	$N_{Sh} = \frac{0.357}{\epsilon} N_{Re}^{0.641} N_{Sc}^{1/3}$ $3 < N_{Re} < 900$ can be extrapolated to $N_{Re} < 2000$.	[E] Packed spheres, deep beds. Corrected for axial dispersion with axial Peclet number = 2.0. Prediction is low at low N_{Re} . N_{Re} defined as in 5-23-A.	[116][128] p. 214 [155]												
C. For gases and liquids, fixed and fluidized beds	$j_D = \frac{0.4548}{\epsilon N_{Re}^{0.4069}}, 10 \leq N_{Re} \leq 2000$ $j_D = \frac{N_{Sh}}{N_{Re} N_{Sc}^{1/3}}, N_{Sh} = \frac{k' d_s}{D}$	[E] Packed spheres, deep bed. Average deviation $\pm 20\%$. $N_{Re} = d_p v_{\text{super}} \rho / \mu$. Can use for fluidized beds. $10 \leq N_{Re} \leq 4000$.	[60][66] p. 484												
D. For gases, fixed beds	$j_D = \frac{0.499}{\epsilon N_{Re}^{0.382}}$	[E] Data on sublimation of naphthalene spheres dispersed in inert beds. $0.1 < N_{Re} < 100$, $N_{Sc} = 2.57$. Correlation coefficient = 0.978.	[80]												
E. For liquids, fixed bed, Wilson and Geankoplis correlation	$j_D = \frac{1.09}{\epsilon N_{Re}^{2/3}}, 0.0016 < N_{Re} < 55$ $165 \leq N_{Sc} \leq 70,600, 0.35 < \epsilon < 0.75$ Equivalent: $N_{Sh} = \frac{1.09}{\epsilon} N_{Re}^{1/3} N_{Sc}^{1/3}$ $j_D = \frac{0.25}{\epsilon N_{Re}^{0.31}}, 55 < N_{Re} < 1500, 165 \leq N_{Sc} \leq 10,690$ Equivalent: $N_{Sh} = \frac{0.25}{\epsilon} N_{Re}^{0.69} N_{Sc}^{1/3}$	[E] Beds of spheres, $N_{Re} = \frac{d_p V_{\text{super}} \rho}{\mu}$ Deep beds. $N_{Sh} = \frac{k' d_s}{D}$	[66] p. 484 [77] p. 195 [141] p. 287 [158]												
F. For liquids, fixed beds, Ohashi et al. correlation	$N_{Sh} = \frac{k' d_s}{D} = 2 + 0.51 \left(\frac{E^{1/3} d_p^{4/3} \rho}{\mu} \right)^{0.60} N_{Sc}^{1/3}$ E = Energy dissipation rate per unit mass of fluid $= 50(1 - \epsilon) \epsilon^2 C_{D0} \left(\frac{v_r^3}{d_p^3} \right), \text{ m}^3/\text{s}^3$ $= \left[\frac{50(1 - \epsilon) C_D}{\epsilon} \right] \left(\frac{v_{\text{super}}^3}{d_p^3} \right)$ General form: $N_{Sh} = 2 + K \left(\frac{E^{1/3} D_p^{4/3} \rho}{\mu} \right)^\alpha N_{Sc}^\beta$ applies to single particles, packed beds, two-phase tube flow, suspended bubble columns, and stirred tanks with different definitions of E .	[S] Correlates large amount of published data. Compares number of correlations, v_r = relative velocity, m/s. In packed bed, $v_r = v_{\text{super}}/\epsilon$. C_{D0} = single particle drag coefficient at v_{super} calculated from $C_{D0} = AN_{Re}^{-m}$. <table><tr><td>N_{Re}</td><td>A</td><td>m</td></tr><tr><td>0 to 5.8</td><td>24</td><td>1.0</td></tr><tr><td>5.8 to 500</td><td>10</td><td>0.5</td></tr><tr><td>>500</td><td>0.44</td><td>0</td></tr></table> Ranges for packed bed: $0.001 < N_{Re} < 1000, 505 < N_{Sc} < 70,600$, $0.2 < \frac{E^{1/3} d_p^{4/3} \rho}{\mu} < 4600$ Compares different situations versus general correlation. See also 5-20-F.	N_{Re}	A	m	0 to 5.8	24	1.0	5.8 to 500	10	0.5	>500	0.44	0	[108]
N_{Re}	A	m													
0 to 5.8	24	1.0													
5.8 to 500	10	0.5													
>500	0.44	0													

TABLE 5-23 Mass Transfer Correlations for Fixed and Fluidized Beds (Continued)

Situation	Correlation	Comments E = Empirical, S = Semiempirical, T = Theoretical	References*
G. Electrolytic system. Pall rings. Transfer from fluid to rings.	Full liquid upflow: $N_{sh} = k_L d_c / D = 4.1 N_{Re}^{0.39} N_{Sc}^{1/3}$ $N_{Re} d_c u / \nu = 80$ to 550 Irrigated liquid downflow (no gas flow): $N_{Sh} = 5.1 N_{Re}^{0.44} N_{Sc}^{1/3}$	[E] d_c = diameter of sphere with same surface area as Pall ring. Full liquid upflow agreed with literature values. Schmidt number dependence was assumed from literature values. In downflow, N_{Re} used superficial fluid velocity.	[69]
H. For liquids, fixed and fluidized beds	$\epsilon j_D = \frac{1.1068}{N_{Re}^{0.72}}, 1.0 < N_{Re} \leq 10$ $\epsilon j_D = \frac{N_{Sh}}{N_{Re} N_{Sc}^{1/3}}, N_{Sh} = \frac{k' d_s}{D}$	[E] Spheres: $N_{Re} = \frac{d_p v_{super} \rho}{\mu}$	[59][66] p. 484
I. For gases and liquids, fixed and fluidized beds, Dwivedi and Upadhyay correlation	$\epsilon j_D = \frac{0.765}{N_{Re}^{0.82}} + \frac{0.365}{N_{Re}^{0.386}}$ Gases: $10 \leq N_{Re} \leq 15,000$. Liquids: $0.01 \leq N_{Re} \leq 15,000$. $N_{Re} = \frac{d_p v_{super} \rho}{\mu}, N_{Sh} = \frac{k' d_s}{D}$	[E] Deep beds of spheres, $j_D = \frac{N_{Sh}}{N_{Re} N_{Sc}^{1/3}}$ Best fit correlation at low conc. [52] Based on 20 gas studies and 17 liquid studies. Recommended instead of 5-23-C or E.	[59] [77] p. 196 [52]
J. For gases and liquids, fixed bed	$j_D = 1.17 N_{Re}^{-0.415}, 10 \leq N_{Re} \leq 2500$ $j_D = \frac{k'}{v_{av}} \frac{p_{BM}}{P} N_{Sc}^{2/3}$	[E] Spheres: Variation in packing that changes ϵ not allowed for. Extensive data referenced. $0.5 < N_{Sc} < 15,000$. Comparison with other results are shown. $N_{Re} = \frac{d_p v_{super} \rho}{\mu}$	[138] p. 241
K. For liquids, fixed and fluidized beds, Rahman and Streat correlation	$N_{Sh} = \frac{0.86}{\epsilon} N_{Re}^{1/3} N_{Sc}^{1/3}, 2 \leq N_{Re} \leq 25$	[E] Can be extrapolated to $N_{Re} = 2000$. $N_{Re} = d_p v_{super} \rho / \mu$. Done for neutralization of ion exchange resin.	[119]
L. Size exclusion chromatography of proteins	$N_{Sh} = \frac{k_L d}{D} = \frac{1.903}{\epsilon} N_{Re}^{1/3} N_{Sc}^{1/3}$	[E] Slow mass transfer with large molecules. Aqueous solutions. Modest increase in N_{Sh} with increasing velocity.	[79]
M. Liquid-free convection with fixed bed Raschig rings. Electrochemical.	$N_{Sh} = kd/D = 0.15 (N_{Sc} N_{Gr})^{0.32}$ $N_{Gr} = \text{Grashof no.} = g d^3 \Delta \rho / (\nu^2 \rho)$ If forced convection superimposed, $N_{Sh, \text{overall}} = (N_{Sh, \text{forced}}^3 + N_{Sh, \text{free}}^3)^{1/3}$	[E] d = Raschig ring diameter, h = bed height $1810 < N_{Sc} < 2532, 0.17 < d/h < 1.0$ $10.6 \times 10^6 < N_{Sc} N_{Gr} < 21 \times 10^7$	[135]
N. Oscillating bed packed with Raschig rings. Dissolution of copper rings.	Batch (no net solution flow): $N_{Sh} = 0.76 N_{Sc}^{0.33} N_{Re, v}^{0.7} (d_c/h)^{0.35}$ $503 < N_{Re, v} < 2892$ $960 < N_{Sc} < 1364, 2.3 < d_c/h < 7.6$	[E] $N_{Sh} = kd/D, N_{Re, v}$ = vibrational $Re = \rho v_c d_c / \mu$ v_c = vibrational velocity (intensity) d_c = col. diameter, h = column height Average deviation is $\pm 12\%$.	[61]
O. For liquids and gases, Ranz and Marshall correlation	$N_{Sh} = \frac{k' d}{D} = 2.0 + 0.6 N_{Sc}^{1/3} N_{Re}^{1/2}$ $N_{Re} = \frac{d_p v_{super} \rho}{\mu}$	[E] Based on freely falling, evaporating spheres (see 5-20-C). Has been applied to packed beds, prediction is low compared to experimental data. Limit of 2.0 at low N_{Re} is too high. Not corrected for axial dispersion.	[121][128] p. 214 [155] [110]
P. For liquids and gases, Wakao and Funazkri correlation	$N_{Sh} = 2.0 + 1.1 N_{Sc}^{1/3} N_{Re}^{0.6}, 3 < N_{Re} < 10,000$ $N_{Sh} = \frac{k'_{lim} d_p}{D}, N_{Re} = \frac{\rho v_{super} \rho}{\mu}$ $\frac{\epsilon D_{axial}}{D} = 10 + 0.5 N_{Sc} N_{Re}$	[E] Correlate 20 gas studies and 16 liquid studies. Corrected for axial dispersion with: Graphical comparison with data shown [128], p. 215, and [155]. D_{axial} is axial dispersion coefficient.	[128] p. 214 [155]
Q. Acid dissolution of limestone in fixed bed	$N_{Sh} = 1.77 N_{Re}^{0.56} N_{Sc}^{1/3} (1 - \epsilon)^{0.44}$ $20 < N_{Re} < 6000$	[E] Best fit was to correlation of Chu et al., <i>Chem. Eng. Prog.</i> , 49 (3), 141(1953), even though no reaction in original.	[94]
R. Semifluidized or expanded bed. Liquid-solid transfer.	$N_{Sh} = \frac{k'_{lim} d_p}{D} = 2 + 1.5 (1 - \epsilon_L) N_{Re}^{1/3} N_{Sc}^{1/3}$ $N_{Re} = \rho_p d_p u / \mu \epsilon_L, N_{Sc} = \mu / \rho D$	[E] ϵ_L = liquid-phase void fraction, ρ_p = particle density, ρ = fluid density, d_p = particle diameter. Fits expanded bed chromatography in viscous liquids.	[64] [159]

TABLE 5-23 Mass Transfer Correlations for Fixed and Fluidized Beds (Concluded)

Situation	Correlation	Comments E = Empirical, S = Semiempirical, T = Theoretical	References ^a
S. Mass-transfer structured packing and static mixers. Liquid with or without fluidized particles. Electrochemical	Fixed bed: $j' = 0.927 N_{Re}^{0.572}$, $N_{Re} < 219$ $j' = 0.443 N_{Re}^{0.435}$, $219 < N_{Re} < 1360$ Fluidized bed with particles: $j = 6.02 N_{Re}^{0.885}$, or $j' = 16.40 N_{Re}^{0.950}$ Natural convection: $N_{Sh} = 0.252 (N_{Sc} N_{Gr})^{0.299}$ Bubble columns: Structured packing: $N_{St} = 0.105 (N_{Re} N_{Fr} N_{Sc}^2)^{-0.268}$ Static mixer: $N_{St} = 0.157 (N_{Re} N_{Fr} N_{Sc}^2)^{-0.298}$	[E] Sulzer packings, $j' = \frac{k \cos \beta}{v} N_{Sc}^{2/3}$, β = corrugation incline angle. $N_{Re} = v' d_p \rho / \mu$, $v' = v_{super} / (\epsilon \cos \beta)$, d_h = channel side width. Particles enhance mass transfer in laminar flow for natural convection. Good fit with correlation of Ray et al., <i>Intl. J. Heat Mass Transfer</i> , 41 , 1693 (1998). $N_{Gr} = g \Delta \rho Z' \rho / \mu^2$, Z = corrugated plate length. Bubble column results fit correlation of Neme et al., <i>Chem. Eng. Technol.</i> , 20 , 297 (1997) for structured packing. N_{St} = Stanton number = kZ/D N_{Fr} = Froude number = v_{super}^2 / gZ	[48]
T. Liquid fluidized beds	$N_{Sh} = \frac{2\xi/\epsilon^m + \left[\frac{(2\xi/\epsilon^m)(1-\epsilon)^{1/2}}{1 - (1-\epsilon)^{1/3}} - 2 \right] \tan h(\xi/\epsilon^m)}{\frac{\xi/\epsilon^m}{1 - (1-\epsilon)^{1/2}} - \tan h(\xi/\epsilon^m)}$ where $\xi = \left[\frac{1}{(1-\epsilon)^{1/3}} - 1 \right] \frac{\alpha}{2} N_{Sc}^{1/3} N_{Re}^{1/2}$ This simplifies to: $N_{Sh} = \frac{\epsilon^{1-2m}}{(1-\epsilon)^{1/3}} \left[\frac{1}{(1-\epsilon)^{1/3}} - 1 \right] \frac{\alpha^2}{2} N_{Re} N_{Sc}^{2/3} \quad (N_{Re} < 0.1)$	[S] Modification of theory to fit experimental data. For spheres, $m = 1$, $N_{Re} > 2$. $N_{Sh} = \frac{k_L' d_p}{D}, \quad N_{Re} = \frac{V_{super} d_p \xi}{\mu}$ $m = 1$ for $N_{Re} > 2$; $m = 0.5$ for $N_{Re} < 1.0$; ϵ = voidage; α = const. Best fit data is $\alpha = 0.7$. Comparison of theory and experimental ion exchange results in Ref. 92.	[92] [106] [125]
U. Liquid fluidized beds	$N_{Sh} = 0.250 N_{Re}^{0.023} N_{Ga}^{0.306} \left(\frac{\rho_s - \rho}{\rho} \right)^{0.282} N_{Sc}^{0.410} \quad (\epsilon < 0.85)$ $N_{Sh} = 0.304 N_{Re}^{-0.057} N_{Ga}^{0.332} \left(\frac{\rho_s - \rho}{\rho} \right)^{0.297} N_{Sc}^{0.404} \quad (\epsilon > 0.85)$ This can be simplified (with slight loss in accuracy at high ϵ) to $N_{Sh} = 0.245 N_{Ga}^{0.323} \left(\frac{\rho_s - \rho}{\rho} \right)^{0.300} N_{Sc}^{0.400}$	[E] Correlate amount of data from literature. Predicts very little dependence of N_{Sh} on velocity. Compare large number of published correlations. $N_{Sh} = \frac{k_L' d_p}{D}, \quad N_{Re} = \frac{d_p v_{super}}{\mu}, \quad N_{Ga} = \frac{d_p^3 \rho^2 g}{\mu^2}$ $N_{Sc} = \frac{\mu}{\rho D}$ $1.6 < N_{Re} < 1320, 2470 < N_{Ga} < 4.42 \times 10^6$ $0.27 < \frac{\rho_s - \rho}{\rho} < 1.114, 305 < N_{Sc} < 1595$	[151]
V. Liquid film flowing over solid particles with air present, trickle bed reactors, fixed bed	$N_{Sh} = \frac{k_L}{aD} = 1.8 N_{Re}^{1/2} N_{Sc}^{1/3}, 0.013 < N_{Re} < 12.6$ two-phases, liquid trickle, no forced flow of gas. $N_{Sh} = 0.8 N_{Re}^{1/2} N_{Sc}^{1/3}$, one-phase, liquid only.	[E] $N_{Re} = \frac{L}{a\mu}$, irregular granules of benzoic acid, $0.29 \leq d_p \leq 1.45$ cm. L = superficial liquid flow rate, kg/m ² s. a = surface area/col. volume, m ² /m ³ .	[130]
W. Supercritical fluids in packed bed	$\frac{N_{Sh}}{(N_{Sc} N_{Gr})^{1/4}} = 0.5265 \left(\frac{(N_{Re}^{1/2} N_{Sc}^{1/3})}{(N_{Sc} N_{Gr})^{1/4}} \right)^{1.6808}$ $+ 2.48 \left \left(\frac{N_{Re}^2 N_{Sc}^{1/3}}{N_{Gr}} \right)^{0.6439} - 0.8768 \right ^{1.553}$	[E] Natural and forced convection. $0.3 < N_{Re} < 135$.	[99]
X. Cocurrent gas-liquid flow in fixed beds	Downflow in trickle bed and upflow in bubble columns.	Literature review and meta-analysis. Analyzed both downflow and upflow. Recommendations for best mass- and heat-transfer correlations (see reference).	[95]
Y. Liquid-solid transfer. Electrochemical reaction. Lessing rings. Transfer from liquid to solid	Liquid only: $N_{Sh} = kd/D = 1.57 N_{Sc}^{1/3} N_{Re}^{0.46}$ $1390 < N_{Sc} < 4760, 166 < N_{Re} < 722$ Cocurrent two-phase (liquid and gas) in packed bubble column: $N_{Sh} = 1.93 N_{Sc}^{1/3} N_{Re}^{0.34} N_{Re, gas}^{0.11}$ $60 < N_{Re, gas} < 818, 144 < N_{Re} < 748$	[E] Electrochemical reactors only. d = Lessing ring diameter, $1 < d < 1.4$ cm, $N_{Re} = \rho v_{super} d / \mu$, Deviation $\pm 7\%$ for both cases. $N_{Re, gas} = \rho_{gas} V_{super, gas} d / \mu_{gas}$ Presence of gas enhances mass transfer.	[75]

NOTE: For $N_{Re} < 3$ convective contributions which are not included may become important. Use with logarithmic concentration difference (integrated form) or with arithmetic concentration difference (differential form).

^aSee the beginning of the "Mass Transfer" subsection for references.

TABLE 5-24 Mass-Transfer Correlations for Packed Two-Phase Contactors—Absorption, Distillation, Cooling Towers, and Extractors (Packing Is Inert)

Situation	Correlations	Comments E = Empirical, S = Semiempirical, T = Theoretical	References*																																																																																																
A. Absorption, counter-current, liquid-phase coefficient H_L , Sherwood and Holloway correlation for random packings	$H_L = a_L \left(\frac{L}{\mu_L} \right)^n N_{Sc,L}^{0.5}, L = \text{lb/hr ft}^2$ <table><thead><tr><th colspan="8">Ranges for 5-24-B (G and L)</th></tr><tr><th>Packing</th><th>a_G</th><th>b</th><th>c</th><th>G</th><th>L</th><th>a_L</th><th>n</th></tr></thead><tbody><tr><td colspan="8">Raschig rings</td></tr><tr><td>3/8 inch</td><td>2.32</td><td>0.45</td><td>0.47</td><td>200–500</td><td>500–1500</td><td>0.00182</td><td>0.46</td></tr><tr><td>1</td><td>7.00</td><td>0.39</td><td>0.58</td><td>200–800</td><td>400–500</td><td>0.010</td><td>0.22</td></tr><tr><td>1</td><td>6.41</td><td>0.32</td><td>0.51</td><td>200–600</td><td>500–4500</td><td>—</td><td>—</td></tr><tr><td>2</td><td>3.82</td><td>0.41</td><td>0.45</td><td>200–800</td><td>500–4500</td><td>0.0125</td><td>0.22</td></tr><tr><td colspan="8">Berl saddles</td></tr><tr><td>1/2 inch</td><td>32.4</td><td>0.30</td><td>0.74</td><td>200–700</td><td>500–1500</td><td>0.0067</td><td>0.28</td></tr><tr><td>1/2</td><td>0.811</td><td>0.30</td><td>0.24</td><td>200–800</td><td>400–4500</td><td>—</td><td>—</td></tr><tr><td>1</td><td>1.97</td><td>0.36</td><td>0.40</td><td>200–800</td><td>400–4500</td><td>0.0059</td><td>0.28</td></tr><tr><td>1.5</td><td>5.05</td><td>0.32</td><td>0.45</td><td>200–1000</td><td>400–4500</td><td>0.0062</td><td>0.28</td></tr></tbody></table>	Ranges for 5-24-B (G and L)								Packing	a_G	b	c	G	L	a_L	n	Raschig rings								3/8 inch	2.32	0.45	0.47	200–500	500–1500	0.00182	0.46	1	7.00	0.39	0.58	200–800	400–500	0.010	0.22	1	6.41	0.32	0.51	200–600	500–4500	—	—	2	3.82	0.41	0.45	200–800	500–4500	0.0125	0.22	Berl saddles								1/2 inch	32.4	0.30	0.74	200–700	500–1500	0.0067	0.28	1/2	0.811	0.30	0.24	200–800	400–4500	—	—	1	1.97	0.36	0.40	200–800	400–4500	0.0059	0.28	1.5	5.05	0.32	0.45	200–1000	400–4500	0.0062	0.28	[E] From experiments on desorption of sparingly soluble gases from water. Graphs [Ref. 138], p. 606. Equation is dimensional. A typical value of n is 0.3 [Ref. 66] has constants in kg, m, and s units for use in 5-24-A and B with $\hat{k}_G$ in kgmole/s m² and $\hat{k}_L$ in kgmole/s m² (kgmol/m³). Constants for other packings are given by Refs. 104, p. 187 and 152, p. 239. $H_L = \frac{L_M}{\hat{k}_{La}}$ $L_M = \text{lbmol/hr ft}^2, \hat{k}_L = \text{lbmol/hr ft}^2, a = \text{ft}^2/\text{ft}^3, \mu_L$ in lb/(hr ft). Range for 5-24-A is $400 < L < 15,000 \text{ lb/hr ft}^2$	[104] p. 187 [105] [138] p. 606 [157] [156]
Ranges for 5-24-B (G and L)																																																																																																			
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1.5	5.05	0.32	0.45	200–1000	400–4500	0.0062	0.28																																																																																												
B. Absorption counter-current, gas-phase coefficient H_G , for random packing	$H_G = \frac{G_M}{\hat{k}_G a} = \frac{a_G(G)^b N_{Sc,G}^{0.5}}{(L)^c}$	[E] Based on ammonia-water-air data in Fellingner's 1941 MIT thesis. Curves: Refs. 104, p. 186 and 138, p. 607. Constants given in 5-24-A. The equation is dimensional. $G = \text{lb/hr ft}^2, G_M = \text{lbmol/hr ft}^2, \hat{k}_G = \text{lbmol/hr ft}^2$.	[104] p. 189 [138] p. 607 [157]																																																																																																
C. Absorption and distillation, counter-current, gas and liquid individual coefficients and wetted surface area, Onda et al. correlation for random packings	$\frac{k'_G RT}{a_p D_G} = A \left(\frac{G}{a_p \mu_G} \right)^{0.7} N_{Sc,G}^{1/3} (a_p d_p')^{-2.0}$ $k'_L \left(\frac{\rho_L}{\mu_L g} \right)^{1/3} = 0.0051 \left(\frac{L}{a_w \mu_L} \right)^{2/3} N_{Sc,L}^{-1/2} (a_p d_p')^{0.4}$ $k'_L = \text{lbmol/hr ft}^2 \text{ (lbmol/ft}^3 \text{) [kgmol/s m}^2 \text{ (kgmol/m}^3 \text{)]}$ $\frac{a_w}{a_p} = 1 - \exp \left\{ \begin{aligned} & -1.45 \left(\frac{\sigma_c}{\sigma} \right)^{0.75} \left(\frac{L}{a_p \mu_L} \right)^{0.1} \\ & \times \left(\frac{L^2 a_p}{\rho_L^2 g} \right)^{-0.05} \left(\frac{L}{\rho_L \sigma a_p} \right)^{0.2} \end{aligned} \right\}$	[E] Gas absorption and desorption from water and organics plus vaporization of pure liquids for Raschig rings, saddles, spheres, and rods. d_p' = nominal packing size, a_p = dry packing surface area/volume, a_w = wetted packing surface area/volume. Equations are dimensionally consistent, so any set of consistent units can be used. σ = surface tension, dynes/cm. $A = 5.23$ for packing $\geq 1/2$ inch (0.012 m) $A = 2.0$ for packing $< 1/2$ inch (0.012 m) $\hat{k}'_G = \text{lbmol/hr ft}^2 \text{ atm [kg mol/s m}^2 \text{ (N/m}^2 \text{)]}$ Critical surface tensions, $\sigma_c = 61$ (ceramic), 75 (steel), 33 (polyethylene), 40 (PVC), 56 (carbon) dynes/cm. $4 < \frac{L}{a_w \mu_L} < 400$ $5 < \frac{G}{a_p \mu_G} < 1000$ Most data $\pm 20\%$ of correlation, some $\pm 50\%$. Graphical comparison with data in Ref. 109.	[44] [90] p. 380 [109][149] p. 355 [156]																																																																																																
D. Distillation and absorption, counter-current, random packings, modification of Onda correlation, Bravo and Fair correlation to determine interfacial area	Use Onda's correlations (5-24-C) for k'_G and k'_L. Calculate: $H_G = \frac{G}{k'_G a_e P M_G}, H_L = \frac{L}{k'_L a_e \rho_L}, H_{OG} = H_G + \lambda H_L$ $\lambda = \frac{m}{L_M/G_M}$ $a_e = 0.498 a_p \left(\frac{\sigma^{0.5}}{Z^{0.4}} \right) (N_{Ca,L} N_{Re,G})^{0.392}$ $N_{Re,G} = \frac{6G}{a_p \mu_G}, N_{Ca,L} = \frac{L \mu_L}{\rho_L \sigma g_c} \text{ (dimensionless)}$	[E] Use Bolles & Fair (Ref. 43) database to determine new effective area a_e to use with Onda et al. (Ref. 109) correlation. Same definitions as 5-24-C. P = total pressure, atm; M_G = gas, molecular weight; m = local slope of equilibrium curve; L_M/G_M = slope operating line; Z = height of packing in feet. Equation for a_e is dimensional. Fit to data for effective area quite good for distillation. Good for absorption at low values of $(N_{Ca,L} \times N_{Re,G})$, but correlation is too high at higher values of $(N_{Ca,L} \times N_{Re,G})$.	[44]																																																																																																

TABLE 5-24 Mass-Transfer Correlations for Packed Two-Phase Contactors—Absorption, Distillation, Cooling Towers, and Extractors (Packing Is Inert) (Continued)

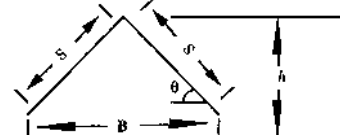
Situation	Correlations	Comments E = Empirical, S = Semiempirical, T = Theoretical	References*																														
E. Absorption and distillation, countercurrent gas-liquid flow, random and structured packing. Determine H_L and H_G	$H_G = \left(\frac{0.226}{f_p} \right) \left(\frac{N_{Sc}}{0.660} \right)^b \left(\frac{G_x}{6.782} \right)^{-0.5} \left(\frac{G_y}{0.678} \right)^{0.35}$ $H_L = \left(\frac{0.357}{f_p} \right) \left(\frac{N_{Sc}}{372} \right)^{0.5} \left(\frac{G_y/\mu}{6.782/0.0008937} \right)^{0.3}$ Relative transfer coefficients [91], f_p values are in table: <table><thead><tr><th>Size, in.</th><th>Ceramic Raschig rings</th><th>Ceramic Berl saddles</th><th>Metal Pall rings</th><th>Metal Intalox</th><th>Metal Hypac</th></tr></thead><tbody><tr><td>0.5</td><td>1.52</td><td>1.58</td><td>—</td><td>—</td><td>—</td></tr><tr><td>1.0</td><td>1.20</td><td>1.36</td><td>1.61</td><td>1.78</td><td>1.51</td></tr><tr><td>1.5</td><td>1.00</td><td>—</td><td>1.34</td><td>—</td><td>—</td></tr><tr><td>2.0</td><td>0.85</td><td>—</td><td>1.14</td><td>1.27</td><td>1.07</td></tr></tbody></table> Norton Intalox structured: 2T, $f_p = 1.98$; 3T, $f_p = 1.94$.	Size, in.	Ceramic Raschig rings	Ceramic Berl saddles	Metal Pall rings	Metal Intalox	Metal Hypac	0.5	1.52	1.58	—	—	—	1.0	1.20	1.36	1.61	1.78	1.51	1.5	1.00	—	1.34	—	—	2.0	0.85	—	1.14	1.27	1.07	[S] H_G based on NH_3 absorption data (5-28B) for which $H_{G,\text{base}} = 0.226$ m with $N_{Sc,\text{base}} = 0.660$ at $G_{x,\text{base}} = 6.782$ kg/(sm ²) and $G_{y,\text{base}} = 0.678$ kg/(sm ²) with 1½ in. ceramic Raschig rings. The exponent b on N_{Sc} is reported as either 0.5 or as ¾. $f_p = \frac{H_G \text{ for } \text{NH}_3 \text{ with } 1\frac{1}{2} \text{ Raschig rings}}{H_G \text{ for } \text{NH}_3 \text{ with desired packing}}$ H_L based on O_2 desorption data (5-24-A). Base viscosity, $\mu_{\text{base}} = 0.0008937$ kg/(ms). H_L in m. $G_y < 0.949$ kg/(sm ²), $0.678 < G_x < 6.782$ kg/(sm ²). Best use is for absorption and stripping. Limited use for organic distillation [156].	[66] p. 686, 659 [138] [156]
Size, in.	Ceramic Raschig rings	Ceramic Berl saddles	Metal Pall rings	Metal Intalox	Metal Hypac																												
0.5	1.52	1.58	—	—	—																												
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1.5	1.00	—	1.34	—	—																												
2.0	0.85	—	1.14	1.27	1.07																												
F. Absorption, cocurrent downward flow, random packings, Reiss correlation	Air-oxygen-water results correlated by $k'_L a = 0.12 E_L^{0.5}$. Extended to other systems. $k'_L a = 0.12 E_L^{0.5} \left(\frac{D_L}{2.4 \times 10^5} \right)^{0.5}$ $E_L = \left(\frac{\Delta p}{\Delta L} \right)_{2\text{-phase}} v_L$ $\frac{\Delta p}{\Delta L}$ = pressure loss in two-phase flow = lbf/ft ² ft $k'_G a = 2.0 + 0.91 E_G^{2/3}$ for NH_3 $E_G = \left(\frac{\Delta p}{\Delta L} \right)_{2\text{-phase}} v_g$ v_g = superficial gas velocity, ft/s	[E] Based on oxygen transfer from water to air 77°F. Liquid film resistance controls. ($D_{\text{water}} @ 77^\circ\text{F} = 2.4 \times 10^{-5}$). Equation is dimensional. Data was for thin-walled polyethylene Raschig rings. Correlation also fit data for spheres. Fit $\pm 25\%$. See [122] for graph. $k'_L a = \text{s}^{-1}$ $D_L = \text{cm/s}$ $E_L = \text{ft, lbf/s ft}^3$ $v_L = \text{superficial liquid velocity, ft/s}$ [E] Ammonia absorption into water from air at 70°F. Gas-film resistance controls. Thin-walled polyethylene Raschig rings and 1-inch Intalox saddles. Fit $\pm 25\%$. See [122] for fit. Terms defined as above.	[122] [130] p. 217																														
G. Absorption, stripping, distillation, counter-current, H_L , and H_G , random packings, Bolles and Fair correlation	For Raschig rings, Berl saddles, and spiral tile: $H_L = \frac{\phi C_{\text{flood}}}{3.28} N_{Sc,L}^{0.5} \left(\frac{Z}{3.05} \right)^{0.15}$ $C_{\text{flood}} = 1.0$ if below 40% flood—otherwise, use figure in [54] and [157]. $H_G = \frac{A \psi (d'_{\text{col}})^{m Z^{0.33} N_{Sc,G}^{0.5}}}{\left[L \left(\frac{\mu_L}{\mu_{\text{water}}} \right)^{0.16} \left(\frac{\rho_{\text{water}}}{\rho_L} \right)^{1.25} \left(\frac{\sigma_{\text{water}}}{\sigma_L} \right)^{0.8} \right]^n}$ Figures for ϕ and ψ in [42 and 43] Ranges: $0.02 < \phi < 0.300$; $25 < \psi < 190$ m.	[E] Z = packed height, m of each section with its own liquid distribution. The original work is reported in English units. Cornell et al. (Ref. 54) review early literature. Improved fit of Cornell's ϕ values given by Bolles and Fair (Refs. [42], [43]) and [157]. $A = 0.017$ (rings) or 0.029 (saddles) d'_{col} = column diameter in m (if diameter > 0.6 m, use $d'_{\text{col}} = 0.6$) $m = 1.24$ (rings) or 1.11 (saddles) $n = 0.6$ (rings) or 0.5 (saddles) L = liquid rate, kg/(sm ²), $\mu_{\text{water}} = 1.0$ Pa·s, $\rho_{\text{water}} = 1000$ kg/m ³ , $\sigma_{\text{water}} = 72.8$ mN/m (72.8 dyn/cm). H_G and H_L will vary from location to location. Design each section of packing separately.	[42, 43, 54] [77] p. 428 [90] p. 381 [141] p. 353 [157] [156]																														
H. Distillation and absorption. Counter-current flow. Structured packings. Gauze-type with triangular flow channels, Bravo, Rocha, and Fair correlation	Equivalent channel:  Use modified correlation for wetted wall column (See 5-18-F) $N_{Sh,e} = \frac{k'_L d_{eq}}{D_e} = 0.0338 \lambda_{Re,e}^{0.8} N_{Sc,e}^{0.333}$ $N_{Re,e} = \frac{d_{eq} \rho_e (U_{e,\text{eff}} + U_{L,\text{eff}})}{\mu_e}$ Calculate k'_L from penetration model (use time for liquid to flow distance s). $k'_L = 2(D_L U_{L,\text{eff}}/\pi s)^{1/2}$	[T] Check of 132 data points showed average deviation 14.6% from theory. Johnstone and Pigford [Ref. 84] correlation (5-18-F) has exponent on N_{Re} rounded to 0.8. Assume gauze packing is completely wet. Thus, $a_{\text{eff}} = a_p$ to calculate H_G and H_L . Same approach may be used generally applicable to sheet-metal packings, but they will not be completely wet and need to estimate transfer area. L = liquid flux, kg/s m ² , G = vapor flux, kg/s m ² . Fit to data shown in Ref. [45]. $H_G = \frac{G}{k'_L a_p \rho_e}, H_L = \frac{L}{k'_L a_p \rho_L}$ effective velocities $U_{e,\text{eff}} = \frac{U_{e,\text{super}}}{\epsilon \sin \theta}, U_{L,\text{eff}} = \frac{3\Gamma}{2\rho_L} \left(\frac{\rho_L^2 g}{3\mu_L \Gamma} \right)^{0.333}, \Gamma = \frac{L}{\text{Per}}$ $\text{Per} = \frac{\text{Perimeter}}{\text{Area}} = \frac{4S + 2B}{Bh}$	[45] [63] p. 310, 326 [149] p. 356, 362 [156]																														

TABLE 5-24 Mass-Transfer Correlations for Packed Two-Phase Contactors—Absorption, Distillation, Cooling Towers, and Extractors (Packing Is Inert) (Concluded)

Situation	Correlations	Comments E = Empirical, S = Semiempirical, T = Theoretical	References*																									
I. Distillation and absorption, counter-current flow. Structured packing with corrugations. Rocha, Bravo, and Fair correlation.	$N_{Sh,G} = \frac{k_g S}{D_g} = 0.054 N_{Re}^{0.8} N_{Sc}^{0.33}$ $u_{v,\text{eff}} = \frac{u_{g,\text{super}}}{\varepsilon (1 - h_L) \sin \theta}, \quad u_{L,\text{eff}} = \frac{u_{liL,\text{super}}}{\varepsilon h_L \sin \theta},$ $k_L = 2 \left(\frac{D_L C_E u_{L,\text{eff}}}{\pi S} \right)$ $H_{OG} = H_G + \lambda H_L = \frac{u_{g,\text{super}}}{k_g a_e} + \frac{\lambda u_{L,\text{super}}}{k_L a_e}$ Interfacial area: $\frac{a_e}{a_p} = F_{SE} \frac{29.12 (N_{We} N_{Fr})^{0.15} S^{0.359}}{N_{Re,L}^{0.2} \varepsilon^{0.6} (1 - 0.93 \cos \gamma) (\sin \theta)^{0.3}}$ Packing factors: <table><thead><tr><th></th><th>a_p</th><th>ε</th><th>F_{SE}</th><th>θ</th></tr></thead><tbody><tr><td>Flexi-pac 2</td><td>233</td><td>0.95</td><td>0.350</td><td>45°</td></tr><tr><td>Gempak 2A</td><td>233</td><td>0.95</td><td>0.344</td><td>45°</td></tr><tr><td>Intalox 2T</td><td>213</td><td>0.95</td><td>0.415</td><td>45°</td></tr><tr><td>Mellapak 350Y</td><td>350</td><td>0.93</td><td>0.350</td><td>45°</td></tr></tbody></table>		a_p	ε	F_{SE}	θ	Flexi-pac 2	233	0.95	0.350	45°	Gempak 2A	233	0.95	0.344	45°	Intalox 2T	213	0.95	0.415	45°	Mellapak 350Y	350	0.93	0.350	45°	[E, T] Modification of Bravo, Rocha, and Fair (5-24-H). Same definitions as in (5-24-H) unless defined differently here. Recommended [156]. h_L = fractional hold-up of liquid C_E = factor for slow surface renewal $C_E \sim 0.9$ a_e = effective area/volume (1/m) a_p = packing surface area/volume (1/m) F_{SE} = surface enhancement factor γ = contact angle; for sheet metal, $\cos \gamma = 0.9$ for $\sigma < 0.055$ N/m $\cos \gamma = 5.211 \times 10^{-16.8356}, \sigma > 0.055$ N/m $\lambda = \frac{m}{L/V}, m = \frac{dy}{dx}$ from equilibrium	[124], [156]
	a_p	ε	F_{SE}	θ																								
Flexi-pac 2	233	0.95	0.350	45°																								
Gempak 2A	233	0.95	0.344	45°																								
Intalox 2T	213	0.95	0.415	45°																								
Mellapak 350Y	350	0.93	0.350	45°																								
J. Rotating packed bed (Higee)	$\frac{k_L a d_p}{Da_p} \left(1 - 0.93 \frac{V_o}{V_i} - 1.13 \frac{V_i}{V_i} \right) = 0.65 N_{Sc}^{0.5}$ $\times \left(\frac{L}{a_p \mu} \right)^{0.17} \left(\frac{d_p^3 \rho^2 a_c}{\mu^2} \right)^{0.3} \left(\frac{L^2}{\rho a_p \sigma} \right)^{0.3}$ $500 \leq N_{Sc} \leq 1.2 \text{ E}5$; $0.0023 \leq L/(a_p \mu) \leq 8.7$ $120 \leq (d_p^3 \rho^2 a_c)/\mu^2 \leq 7.0 \text{ E}7$; $3.7 \text{ E} - 6 \leq L^2/(\rho a_p \sigma) \leq 9.4 \text{ E} - 4$ $9.12 \leq \frac{k_L a d_p}{Da_p} \leq 2540$	[E] Studied oxygen desorption from water into N₂. Packing 0.22-mm-diameter stainless-steel mesh. $\varepsilon = 0.954, a_p = 829$ (1/m), $h_{bed} = 2$ cm. a = gas-liquid area/vol (1/m) L = liquid mass flux, kg/(m²S) a_c = centrifugal accel, m²/S V_o, V_i, V_t = volumes inside inner radius, between outer radius and housing, and total, respectively, m³. Coefficient (0.3) on centrifugal acceleration agrees with literature values (0.3–0.38).	[50]																									
K. High-voidage packings, cooling towers, splash-grid packings	$\frac{(Ka)_H V_{tower}}{L} = 0.07 + A' N'^n \left(\frac{L}{G_a} \right)^{-n'}$ A' and n' depend on deck type (Ref. 86), $0.060 \leq A' \leq 0.135, 0.46 \leq n' \leq 0.62$. General form fits the graphical comparisons (Ref. 138).	[E] General form. $G_a = \text{lb dry air/hr ft}^2$. $L = \text{lb/h ft}^2, N' = \text{number of deck levels}$. $(Ka)_H$ = overall enthalpy transfer coefficient = $\text{lb/(h)(ft}^3) \left(\frac{\text{lb water}}{\text{lb dry air}} \right)$ V_{tower} = tower volume, ft³/ft². If normal packings are used, use absorption mass-transfer correlations.	[86][104] p. 220 [138] p. 286																									
L. Liquid-liquid extraction, packed towers	Use k values for drops (Table 5-21). Enhancement due to packing is at most 20%.	[E] Packing decreases drop size and increases interfacial area.	[146] p. 79																									
M. Liquid-liquid extraction in rotating-disc contactor (RDC)	$\frac{k_{c-RDC}}{k_c} = 1.0 + 2.44 \left(\frac{N}{N_{Cr}} \right)^{2.5}$ $N_{Cr} = 7.6 \times 10^{-4} \left(\frac{\sigma}{d_{drop} \mu_c} \right) \left(\frac{H}{D_{tank}} \right)$ $\frac{k_{d-RDC}}{k_d} = 1.0 + 1.825 \left(\frac{N}{N_{Cr}} \right) \frac{H}{D_{tank}}$	k_c, k_d are for drops (Table 5-21) Breakage occurs when $N > N_{Cr}$. Maximum enhancement before breakage was factor of 2.0. N = impeller speed H = compartment height, D_{tank} = tank diameter, σ = interfacial tension, N/m. Done in 0.152 and 0.600 m RDC.	[36][146] p. 79																									
N. Liquid-liquid extraction, stirred tanks	See Table 5-22-E, F, G, and H.	[E]																										

See also Sec. 14.

*See the beginning of the "Mass Transfer" subsection for references.

The gas-phase rate coefficient $\hat{k}_G$ is not affected by the fact that a chemical reaction is taking place in the liquid phase. If the liquid-phase chemical reaction is extremely fast and irreversible, the rate of absorption may be governed completely by the resistance to diffusion in the gas phase. In this case the absorption rate may be estimated by

knowing only the gas-phase rate coefficient $\hat{k}_G$ or else the height of one gas-phase transfer unit $H_G = G_M/(\hat{k}_G a)$.

It should be noted that the highest possible absorption rates will occur under conditions in which the liquid-phase resistance is negligible and the equilibrium back pressure of the gas over the solvent is zero.

Such situations would exist, for instance, for NH_3 absorption into an acid solution, for SO_2 absorption into an alkali solution, for vaporization of water into air, and for H_2S absorption from a dilute-gas stream into a strong alkali solution, provided there is a large excess of reagent in solution to consume all the dissolved gas. This is known as the gas-phase mass-transfer limited condition, when both the liquid-phase resistance and the back pressure of the gas equal zero. Even when the reaction is sufficiently reversible to allow a small back pressure, the absorption may be gas-phase-controlled, and the values of $\hat{k}_G$ and H_G that would apply to a physical-absorption process will govern the rate.

The liquid-phase rate coefficient k_L is strongly affected by fast chemical reactions and generally increases with increasing reaction rate. Indeed, the condition for zero liquid-phase resistance ($m/\hat{k}_L$) implies that either the equilibrium back pressure is negligible, or that $\hat{k}_L$ is very large, or both. Frequently, even though reaction consumes the solute as it is dissolving, thereby enhancing both the mass-transfer coefficient and the driving force for absorption, the reaction rate is slow enough that the liquid-phase resistance must be taken into account. This may be due either to an insufficient supply of a second reagent or to an inherently slow chemical reaction.

In any event the value of k_L in the presence of a chemical reaction normally is larger than the value found when only physical absorption occurs, $\hat{k}_L^0$. This has led to the presentation of data on the effects of chemical reaction in terms of the "reaction factor" or "enhancement factor" defined as

$$\phi = \hat{k}_L / \hat{k}_L^0 \geq 1 \quad (5-311)$$

where $\hat{k}_L$ = mass-transfer coefficient with reaction and $\hat{k}_L^0$ = mass-transfer coefficient for pure physical absorption.

It is important to understand that when chemical reactions are involved, this definition of $\hat{k}_L$ is based on the driving force defined as the difference between the concentration of *unreacted* solute gas at the interface and in the bulk of the liquid. A coefficient *smaller* than the total of both unreacted and reacted gas could have values *smaller* than the physical-absorption mass-transfer coefficient $\hat{k}_L^0$.

When liquid-phase resistance is important, particular care should be taken in employing any given set of experimental data to ensure that the equilibrium data used conform with those employed by the original author in calculating values of $\hat{k}_L$ or H_L . Extrapolation to widely different concentration ranges or operating conditions should be made with caution, since the mass-transfer coefficient $\hat{k}_L$ may vary in an unexpected fashion, owing to changes in the apparent chemical-reaction mechanism.

Generalized prediction methods for $\hat{k}_L$ and H_L do not apply when chemical reaction occurs in the liquid phase, and therefore one must use actual operating data for the particular system in question. A discussion of the various factors to consider in designing gas absorbers and strippers when chemical reactions are involved is presented by Astarita, Savage, and Bisio, *Gas Treating with Chemical Solvents*, Wiley (1983) and by Kohl and Nielsen, *Gas Purification*, 5th ed., Gulf (1997).

Effective Interfacial Mass-Transfer Area a In a packed tower of constant cross-sectional area S the differential change in solute flow per unit time is given by

$$-d(G_M S y) = N_A a dV = N_A a S dh \quad (5-312)$$

where a = interfacial area effective for mass transfer per unit of packed volume and V = packed volume. Owing to incomplete wetting of the packing surfaces and to the formation of areas of stagnation in the liquid film, the effective area normally is significantly less than the total external area of the packing pieces.

The effective interfacial area depends on a number of factors, as discussed in a review by Charpentier [*Chem. Eng. J.*, **11**, 161 (1976)]. Among these factors are (1) the shape and size of packing, (2) the packing material (for example, plastic generally gives smaller interfacial areas than either metal or ceramic), (3) the liquid mass velocity, and (4), for small-diameter towers, the column diameter.

Whereas the interfacial area generally increases with increasing liquid rate, it apparently is relatively independent of the superficial gas mass velocity below the flooding point. According to Charpentier's review, it appears valid to assume that the interfacial area is independent of the column height when specified in terms of unit packed volume (i.e., as a). Also, the existing data for chemically reacting gas-liquid systems (mostly aqueous electrolyte solutions) indicate that

the interfacial area is independent of the chemical system. However, this situation may not hold true for systems involving large heats of reaction.

Rizzuti et al. [*Chem. Eng. Sci.*, **36**, 973 (1981)] examined the influence of solvent viscosity upon the effective interfacial area in packed columns and concluded that for the systems studied the effective interfacial area a was proportional to the kinematic viscosity raised to the 0.7 power. Thus, the hydrodynamic behavior of a packed absorber is strongly affected by viscosity effects. Surface-tension effects also are important, as expressed in the work of Onda et al. (see Table 5-24-C).

In developing correlations for the mass-transfer coefficients $\hat{k}_G$ and $\hat{k}_L$, the various authors have assumed different but internally compatible correlations for the effective interfacial area a . It therefore would be inappropriate to mix the correlations of different authors unless it has been demonstrated that there is a valid area of overlap between them.

Volumetric Mass-Transfer Coefficients $\hat{K}_G a$ and $\hat{K}_L a$ Experimental determinations of the individual mass-transfer coefficients $\hat{k}_G$ and $\hat{k}_L$ and of the effective interfacial area a involve the use of extremely difficult techniques, and therefore such data are not plentiful. More often, column experimental data are reported in terms of overall volumetric coefficients, which normally are defined as follows:

$$K'_G a = n_A / (h_T S p_T \Delta y_{im}^o) \quad (5-313)$$

and

$$K_L a = n_A / (h_T S \Delta x_{im}^o) \quad (5-314)$$

where $K'_G a$ = overall volumetric gas-phase mass-transfer coefficient, $K_L a$ = overall volumetric liquid-phase mass-transfer coefficient, n_A = overall rate of transfer of solute A , h_T = total packed depth in tower, S = tower cross-sectional area, p_T = total system pressure employed during the experiment, and Δy_{im}^o and Δx_{im}^o are defined as

$$\Delta y_{im}^o = \frac{(y - y^o)_1 - (y - y^o)_2}{\ln [(y - y^o)_1 / (y - y^o)_2]} \quad (5-315)$$

and

$$\Delta x_{im}^o = \frac{(x^o - x)_2 - (x^o - x)_1}{\ln [(x^o - x)_2 / (x^o - x)_1]} \quad (5-316)$$

where subscripts 1 and 2 refer to the bottom and top of the tower respectively.

Experimental $K'_G a$ and $K_L a$ data are available for most absorption and stripping operations of commercial interest (see Sec. 14). The solute concentrations employed in these experiments normally are very low, so that $K_L a \doteq \hat{K}_L a$ and $K'_G a p_T \doteq \hat{K}_G a$, where p_T is the total pressure employed in the actual experimental-test system. Unlike the individual gas-film coefficient $\hat{k}_G a$, the overall coefficient $\hat{K}_G a$ will vary with the total system pressure except when the liquid-phase resistance is negligible (i.e., when either $m = 0$, or $\hat{k}_L a$ is very large, or both).

Extrapolation of $K'_G a$ data for absorption and stripping to conditions other than those for which the original measurements were made can be extremely risky, especially in systems involving chemical reactions in the liquid phase. One therefore would be wise to restrict the use of overall volumetric mass-transfer-coefficient data to conditions not too far removed from those employed in the actual tests. The most reliable data for this purpose would be those obtained from an operating commercial unit of similar design.

Experimental values of H_{OG} and H_{OL} for a number of distillation systems of commercial interest are also readily available. Extrapolation of the data or the correlations to conditions that differ significantly from those used for the original experiments is risky. For example, pressure has a major effect on vapor density and thus can affect the hydrodynamics significantly. Changes in flow patterns affect both mass-transfer coefficients and interfacial area.

Chilton-Colburn Analogy On occasion one will find that heat-transfer-rate data are available for a system in which mass-transfer-rate data are not readily available. The Chilton-Colburn analogy [90, 53] (see Tables 5-17-G and 5-19-T) provides a procedure for developing estimates of the mass-transfer rates based on heat-transfer data. Extrapolation of experimental j_M or j_H data obtained with gases to predict liquid systems (and vice versa) should be approached with caution, however. When pressure-drop or friction-factor data are available, one may be able to place an upper bound on the rates of heat and mass transfer of $f/2$. The Chilton-Colburn analogy can be used for simultaneous heat and mass transfer as long as the concentration and temperature fields are independent [Venkatesan and Fogler, *AIChE J.*, **50**, 1623 (2004)].

SECTION 6

Fluid and
Particle
Dynamics

PERRY'S CHEMICAL ENGINEERS' HANDBOOK

8TH EDITION



JAMES N. TILTON

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Fluid and Particle Dynamics

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FLUID DYNAMICS

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Nomenclature and Units*

In this listing, symbols used in this section are defined in a general way and appropriate SI units are given. Specific definitions, as denoted by subscripts, are stated at the place of application in the section. Some specialized symbols used in the section are defined only at the place of application. Some symbols have more than one definition; the appropriate one is identified at the place of application.

Symbol	Definition	SI units	U.S. customary units	Symbol	Definition	SI units	U.S. customary units
a	Pressure wave velocity	m/s	ft/s	s	Entropy per unit mass	J/(kg · K)	Btu/(lbm · R)
A	Area	m ²	ft ²	S	Slope	Dimensionless	Dimensionless
b	Wall thickness	m	in	S	Pumping speed	m ³ /s	ft ³ /s
b	Channel width	m	ft	S	Surface area per unit volume	l/m	l/ft
c	Acoustic velocity	m/s	ft/s	St	Strouhal number	Dimensionless	Dimensionless
c_f	Friction coefficient	Dimensionless	Dimensionless	t	Time	s	s
C	Conductance	m ³ /s	ft ³ /s	t	Force per unit area	Pa	lbf/in ²
Ca	Capillary number	Dimensionless	Dimensionless	T	Absolute temperature	K	R
C_0	Discharge coefficient	Dimensionless	Dimensionless	u	Internal energy per unit mass	J/kg	Btu/lbm
C_D	Drag coefficient	Dimensionless	Dimensionless	u	Velocity	m/s	ft/s
d	Diameter	m	ft	U	Velocity	m/s	ft/s
D	Diameter	m	ft	v	Velocity	m/s	ft/s
De	Dean number	Dimensionless	Dimensionless	V	Velocity	m/s	ft/s
D_{ij}	Deformation rate tensor components	1/s	1/s	V	Volume	m ³	ft ³
E	Elastic modulus	Pa	lbf/in ²	We	Weber number	Dimensionless	Dimensionless
$\dot{E}_v$	Energy dissipation rate	J/s	ft · lbf/s	$\dot{W}_s$	Rate of shaft work	J/s	Btu/s
Eo	Eotvos number	Dimensionless	Dimensionless	δW_s	Shaft work per unit mass	J/kg	Btu/lbm
f	Fanning friction factor	Dimensionless	Dimensionless	x	Cartesian coordinate	m	ft
f	Vortex shedding frequency	1/s	1/s	y	Cartesian coordinate	m	ft
F	Force	N	lbf	z	Cartesian coordinate	m	ft
F	Cumulative residence time distribution	Dimensionless	Dimensionless	z	Elevation	m	ft
Fr	Froude number	Dimensionless	Dimensionless	Greek Symbols			
g	Acceleration of gravity	m/s ²	ft/s ²	α	Velocity profile factor	Dimensionless	Dimensionless
G	Mass flux	kg/(m ² · s)	lbm/(ft ² · s)	α	Included angle	Radians	Radians
h	Enthalpy per unit mass	J/kg	Btu/lbm	β	Velocity profile factor	Dimensionless	Dimensionless
h	Liquid depth	m	ft	β	Bulk modulus of elasticity	Pa	lbf/in ²
k	Ratio of specific heats	Dimensionless	Dimensionless	$\dot{\gamma}$	Shear rate	1/s	1/s
k	Kinetic energy of turbulence	J/kg	ft · lbf/lbm	Γ	Mass flow rate per unit width	kg/(m · s)	lbm/(ft · s)
K	Power law coefficient	kg/(m · s ²⁻ⁿ)	lbm/(ft · s ²⁻ⁿ)	δ	Boundary layer or film thickness	m	ft
l_c	Viscous losses per unit mass	J/kg	ft · lbf/lbm	δ_{ij}	Kronecker delta	Dimensionless	Dimensionless
L	Length	m	ft	ϵ	Pipe roughness	m	ft
$\dot{m}$	Mass flow rate	kg/s	lbm/s	ϵ	Void fraction	Dimensionless	Dimensionless
M	Mass	kg	lbm	ϵ	Turbulent dissipation rate	J/(kg · s)	ft · lbf/(lbm · s)
M	Mach number	Dimensionless	Dimensionless	θ	Residence time	s	s
M	Morton number	Dimensionless	Dimensionless	θ	Angle	Radians	Radians
M_w	Molecular weight	kg/kgmole	lbm/lbmole	λ	Mean free path	m	ft
n	Power law exponent	Dimensionless	Dimensionless	μ	Viscosity	Pa · s	lbm/(ft · s)
N_b	Blend time number	Dimensionless	Dimensionless	ν	Kinematic viscosity	m ² /s	ft ² /s
N_D	Best number	Dimensionless	Dimensionless	ρ	Density	kg/m ³	lbm/ft ³
N_P	Power number	Dimensionless	Dimensionless	σ	Surface tension	N/m	lbf/ft
N_Q	Pumping number	Dimensionless	Dimensionless	σ	Cavitation number	Dimensionless	Dimensionless
p	Pressure	Pa	lbf/in ²	σ_{ij}	Components of total stress tensor	Pa	lbf/in ²
q	Entrained flow rate	m ³ /s	ft ³ /s	τ	Shear stress	Pa	lbf/in ²
Q	Volumetric flow rate	m ³ /s	ft ³ /s	τ	Time period	s	s
Q	Throughput (vacuum flow)	Pa · m ³ /s	lbf · ft ³ /s	τ_{ij}	Components of deviatoric stress tensor	Pa	lbf/in ²
δQ	Heat input per unit mass	J/kg	Btu/lbm	Φ	Energy dissipation rate per unit volume	J/(m ³ · s)	ft · lbf/(ft ³ · s)
r	Radial coordinate	m	ft	ϕ	Angle of inclination	Radians	Radians
R	Radius	m	ft	ω	Vorticity	1/s	1/s
R	Ideal gas universal constant	J/(kgmole · K)	Btu/(lbmole · R)				
R_i	Volume fraction of phase i	Dimensionless	Dimensionless				
Re	Reynolds number	Dimensionless	Dimensionless				
s	Density ratio	Dimensionless	Dimensionless				

* Note that with U.S. Customary units, the conversion factor g_c may be required to make equations in this section dimensionally consistent; $g_c = 32.17 \text{ (lbm} \cdot \text{ft)/(lbf} \cdot \text{s}^2\text{)}$.

FLUID DYNAMICS

GENERAL REFERENCES: Batchelor, *An Introduction to Fluid Dynamics*, Cambridge University, Cambridge, 1967; Bird, Stewart, and Lightfoot, *Transport Phenomena*, 2d ed., Wiley, New York, 2002; Brodkey, *The Phenomena of Fluid Motions*, Addison-Wesley, Reading, Mass., 1967; Denn, *Process Fluid Mechanics*, Prentice-Hall, Englewood Cliffs, N.J., 1980; Landau and Lifshitz, *Fluid Mechanics*, 2d ed., Pergamon, 1987; Govier and Aziz, *The Flow of Complex Mixtures in Pipes*, Van Nostrand Reinhold, New York, 1972; Krieger, Huntington, N.Y., 1977; Panton, *Incompressible Flow*, Wiley, New York, 1984; Schlichting, *Boundary Layer Theory*, 8th ed., McGraw-Hill, New York, 1987; Shames, *Mechanics of Fluids*, 3d ed., McGraw-Hill, New York, 1992; Streeter, *Handbook of Fluid Dynamics*, McGraw-Hill, New York, 1971; Streeter and Wylie, *Fluid Mechanics*, 8th ed., McGraw-Hill, New York, 1985; Vennard and Street, *Elementary Fluid Mechanics*, 5th ed., Wiley, New York, 1975; Whitaker, *Introduction to Fluid Mechanics*, Prentice-Hall, Englewood Cliffs, N.J., 1968; Krieger, Huntington, N.Y., 1981.

NATURE OF FLUIDS

Deformation and Stress A fluid is a substance which undergoes continuous deformation when subjected to a shear stress. Figure 6-1 illustrates this concept. A fluid is bounded by two large parallel plates, of area A , separated by a small distance H . The bottom plate is held fixed. Application of a force F to the upper plate causes it to move at a velocity U . The fluid continues to deform as long as the force is applied, unlike a solid, which would undergo only a finite deformation.

The force is directly proportional to the area of the plate; the shear stress is $\tau = F/A$. Within the fluid, a linear velocity profile $u = Uy/H$ is established; due to the **no-slip condition**, the fluid bounding the lower plate has zero velocity and the fluid bounding the upper plate moves at the plate velocity U . The velocity gradient $\dot{\gamma} = du/dy$ is called the **shear rate** for this flow. Shear rates are usually reported in units of reciprocal seconds. The flow in Fig. 6-1 is a **simple shear flow**.

Viscosity The ratio of shear stress to shear rate is the viscosity, μ .

$$\mu = \frac{\tau}{\dot{\gamma}} \quad (6-1)$$

The SI units of viscosity are $\text{kg}/(\text{m} \cdot \text{s})$ or $\text{Pa} \cdot \text{s}$ (pascal second). The cgs unit for viscosity is the poise; $1 \text{ Pa} \cdot \text{s}$ equals 10 poise or 1000 centipoise (cP) or $0.672 \text{ lbm}/(\text{ft} \cdot \text{s})$. The terms *absolute viscosity* and *shear viscosity* are synonymous with the viscosity as used in Eq. (6-1).

Kinematic viscosity $\nu \equiv \mu/\rho$ is the ratio of viscosity to density. The SI units of kinematic viscosity are m^2/s . The cgs stoke is $1 \text{ cm}^2/\text{s}$.

Rheology In general, fluid flow patterns are more complex than the one shown in Fig. 6-1, as is the relationship between fluid deformation and stress. Rheology is the discipline of fluid mechanics which studies this relationship. One goal of rheology is to obtain **constitutive equations** by which stresses may be computed from deformation rates. For simplicity, fluids may be classified into rheological types in reference to the simple shear flow of Fig. 6-1. Complete definitions require extension to multidimensional flow. For more information, several good references are available, including Bird, Armstrong, and Hassager (*Dynamics of Polymeric Liquids*, vol. 1: *Fluid Mechanics*, Wiley, New York, 1977); Metzner ("Flow of Non-Newtonian Fluids" in Streeter, *Handbook of Fluid Dynamics*, McGraw-Hill, New York, 1971); and Skelland (*Non-Newtonian Flow and Heat Transfer*, Wiley, New York, 1967).

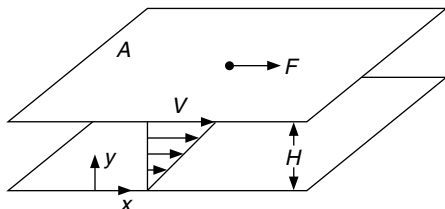


FIG. 6-1 Deformation of a fluid subjected to a shear stress.

Fluids without any solidlike elastic behavior do not undergo any reverse deformation when shear stress is removed, and are called **purely viscous** fluids. The shear stress depends only on the rate of deformation, and not on the extent of deformation (strain). Those which exhibit both viscous and elastic properties are called **viscoelastic** fluids.

Purely viscous fluids are further classified into time-independent and time-dependent fluids. For time-independent fluids, the shear stress depends only on the instantaneous shear rate. The shear stress for time-dependent fluids depends on the past history of the rate of deformation, as a result of structure or orientation buildup or breakdown during deformation.

A **rheogram** is a plot of shear stress versus shear rate for a fluid in simple shear flow, such as that in Fig. 6-1. Rheograms for several types of time-independent fluids are shown in Fig. 6-2. The **Newtonian** fluid rheogram is a straight line passing through the origin. The slope of the line is the viscosity. For a Newtonian fluid, the viscosity is independent of shear rate, and may depend only on temperature and perhaps pressure. By far, the Newtonian fluid is the largest class of fluid of engineering importance. Gases and low molecular weight liquids are generally Newtonian. Newton's law of viscosity is a rearrangement of Eq. (6-1) in which the viscosity is a constant:

$$\tau = \mu \dot{\gamma} = \mu \frac{du}{dy} \quad (6-2)$$

All fluids for which the viscosity varies with shear rate are **non-Newtonian fluids**. For non-Newtonian fluids the viscosity, defined as the ratio of shear stress to shear rate, is often called the **apparent viscosity** to emphasize the distinction from Newtonian behavior. Purely viscous, time-independent fluids, for which the apparent viscosity may be expressed as a function of shear rate, are called **generalized Newtonian fluids**.

Non-Newtonian fluids include those for which a finite stress τ_y is required before continuous deformation occurs; these are called **yield-stress materials**. The **Bingham plastic** fluid is the simplest yield-stress material; its rheogram has a constant slope μ_{∞} , called the *infinite shear viscosity*.

$$\tau = \tau_y + \mu_{\infty} \dot{\gamma} \quad (6-3)$$

Highly concentrated suspensions of fine solid particles frequently exhibit Bingham plastic behavior.

Shear-thinning fluids are those for which the slope of the rheogram decreases with increasing shear rate. These fluids have also been called *pseudoplastic*, but this terminology is outdated and discouraged. Many polymer melts and solutions, as well as some solids suspensions, are shear-thinning. Shear-thinning fluids without yield stresses typically obey a power law model over a range of shear rates.

$$\tau = K \dot{\gamma}^n \quad (6-4)$$

The apparent viscosity is

$$\mu = K \dot{\gamma}^{n-1} \quad (6-5)$$

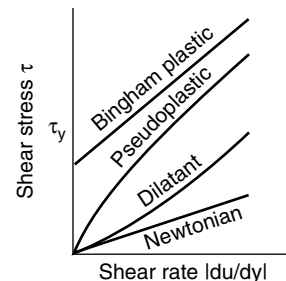


FIG. 6-2 Shear diagrams.

The factor K is the consistency index or power law coefficient, and n is the power law exponent. The exponent n is dimensionless, while K is in units of $\text{kg}/(\text{m} \cdot \text{s}^{-2-n})$. For shear-thinning fluids, $n < 1$. The power law model typically provides a good fit to data over a range of one to two orders of magnitude in shear rate; behavior at very low and very high shear rates is often Newtonian. Shear-thinning power law fluids with yield stresses are sometimes called *Herschel-Bulkley fluids*. Numerous other rheological model equations for shear-thinning fluids are in common use.

Dilatant, or shear-thickening, fluids show increasing viscosity with increasing shear rate. Over a limited range of shear rate, they may be described by the power law model with $n > 1$. Dilatancy is rare, observed only in certain concentration ranges in some particle suspensions (Govier and Aziz, pp. 33–34). Extensive discussions of dilatant suspensions, together with a listing of dilatant systems, are given by Green and Grisley (*Trans. Soc. Rheol.*, **12**[1], 13–25 [1968]); Grisley and Green (*AIChE J.*, **17**, 725–728 [1971]); and Bauer and Collins ("Thixotropy and Dilatancy," in Eirich, *Rheology*, vol. 4, Academic, New York, 1967).

Time-dependent fluids are those for which structural rearrangements occur during deformation at a rate too slow to maintain equilibrium configurations. As a result, shear stress changes with duration of shear. **Thixotropic** fluids, such as mayonnaise, clay suspensions used as drilling muds, and some paints and inks, show decreasing shear stress with time at constant shear rate. A detailed description of thixotropic behavior and a list of thixotropic systems is found in Bauer and Collins (*ibid.*).

Rheoplectic behavior is the opposite of thixotropy. Shear stress increases with time at constant shear rate. Rheoplectic behavior has been observed in bentonite sols, vanadium pentoxide sols, and gypsum suspensions in water (Bauer and Collins, *ibid.*) as well as in some polyester solutions (Steg and Katz, *J. Appl. Polym. Sci.*, **9**, 3, 177 [1965]).

Viscoelastic fluids exhibit elastic recovery from deformation when stress is removed. Polymeric liquids comprise the largest group of fluids in this class. A property of viscoelastic fluids is the *relaxation time*, which is a measure of the time required for elastic effects to decay. Viscoelastic effects may be important with sudden changes in rates of deformation, as in flow startup and stop, rapidly oscillating flows, or as a fluid passes through sudden expansions or contractions where accelerations occur. In many fully developed flows where such effects are absent, viscoelastic fluids behave as if they were purely viscous. In viscoelastic flows, normal stresses perpendicular to the direction of shear are different from those in the parallel direction. These give rise to such behaviors as the *Weissenberg effect*, in which fluid climbs up a shaft rotating in the fluid, and *die swell*, where a stream of fluid issuing from a tube may expand to two or more times the tube diameter.

A parameter indicating whether viscoelastic effects are important is the **Deborah number**, which is the ratio of the characteristic relaxation time of the fluid to the characteristic time scale of the flow. For small Deborah numbers, the relaxation is fast compared to the characteristic time of the flow, and the fluid behavior is purely viscous. For very large Deborah numbers, the behavior closely resembles that of an elastic solid.

Analysis of viscoelastic flows is very difficult. Simple constitutive equations are unable to describe all the material behavior exhibited by viscoelastic fluids even in geometrically simple flows. More complex constitutive equations may be more accurate, but become exceedingly difficult to apply, especially for complex geometries, even with advanced numerical methods. For good discussions of viscoelastic fluid behavior, including various types of constitutive equations, see Bird, Armstrong, and Hassager (*Dynamics of Polymeric Liquids*, vol. 1: *Fluid Mechanics*, vol. 2: *Kinetic Theory*, Wiley, New York, 1977); Middleman (*The Flow of High Polymers*, Interscience (Wiley) New York, 1968); or Astarita and Marrucci (*Principles of Non-Newtonian Fluid Mechanics*, McGraw-Hill, New York, 1974).

Polymer processing is the field which depends most on the flow of non-Newtonian fluids. Several excellent texts are available, including Middleman (*Fundamentals of Polymer Processing*, McGraw-Hill, New York, 1977) and Tadmor and Gogos (*Principles of Polymer Processing*, Wiley, New York, 1979).

There is a wide variety of instruments for measurement of Newtonian viscosity, as well as rheological properties of non-Newtonian fluids. They are described in Van Wazer, Lyons, Kim, and Colwell (*Viscosity and Flow Measurement*, Interscience, New York, 1963); Coleman, Markowitz, and Noll (*Viscometric Flows of Non-Newtonian Fluids*, Springer-Verlag, Berlin, 1966); Dealy and Wissbrun (*Melt Rheology and Its Role in Plastics Processing*, Van Nostrand Reinhold, 1990). Measurement of rheological behavior requires well-characterized flows. Such *rheometric* flows are thoroughly discussed by Astarita and Marrucci (*Principles of Non-Newtonian Fluid Mechanics*, McGraw-Hill, New York, 1974).

KINEMATICS OF FLUID FLOW

Velocity The term *kinematics* refers to the quantitative description of fluid motion or deformation. The rate of deformation depends on the distribution of velocity within the fluid. Fluid velocity $\mathbf{v}$ is a vector quantity, with three cartesian components v_x , v_y , and v_z . The velocity vector is a function of spatial position and time. A **steady** flow is one in which the velocity is independent of time, while in **unsteady** flow $\mathbf{v}$ varies with time.

Compressible and Incompressible Flow An incompressible flow is one in which the density of the fluid is constant or nearly constant. Liquid flows are normally treated as incompressible, except in the context of hydraulic transients (see following). Compressible fluids, such as gases, may undergo incompressible flow if pressure and/or temperature changes are small enough to render density changes insignificant. Frequently, compressible flows are regarded as flows in which the density varies by more than 5 to 10 percent.

Streamlines, Pathlines, and Streaklines These are curves in a flow field which provide insight into the flow pattern. *Streamlines* are tangent at every point to the local instantaneous velocity vector. A *pathline* is the path followed by a material element of fluid; it coincides with a streamline if the flow is steady. In unsteady flow the pathlines generally do not coincide with streamlines. *Streaklines* are curves on which are found all the material particles which passed through a particular point in space at some earlier time. For example, a streakline is revealed by releasing smoke or dye at a point in a flow field. For steady flows, streamlines, pathlines, and streaklines are indistinguishable. In two-dimensional incompressible flows, streamlines are contours of the **stream function**.

One-dimensional Flow Many flows of great practical importance, such as those in pipes and channels, are treated as one-dimensional flows. There is a single direction called the *flow direction*; velocity components perpendicular to this direction are either zero or considered unimportant. Variations of quantities such as velocity, pressure, density, and temperature are considered only in the flow direction. The fundamental conservation equations of fluid mechanics are greatly simplified for one-dimensional flows. A broader category of one-dimensional flow is one where there is only one nonzero velocity component, which depends on only one coordinate direction, and this coordinate direction may or may not be the same as the flow direction.

Rate of Deformation Tensor For general three-dimensional flows, where all three velocity components may be important and may vary in all three coordinate directions, the concept of deformation previously introduced must be generalized. The rate of deformation tensor D_{ij} has nine components. In Cartesian coordinates,

$$D_{ij} = \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right) \quad (6-6)$$

where the subscripts i and j refer to the three coordinate directions. Some authors define the deformation rate tensor as one-half of that given by Eq. (6-6).

Vorticity The relative motion between two points in a fluid can be decomposed into three components: rotation, dilatation, and deformation. The rate of deformation tensor has been defined. Dilatation refers to the volumetric expansion or compression of the fluid, and vanishes for incompressible flow. Rotation is described by a tensor $\omega_{ij} = \partial v_i / \partial x_j - \partial v_j / \partial x_i$. The vector of vorticity given by one-half the

6-6 FLUID AND PARTICLE DYNAMICS

curl of the velocity vector is another measure of rotation. In two-dimensional flow in the x - y plane, the vorticity ω is given by

$$\omega = \frac{1}{2} \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \quad (6-7)$$

Here ω is the magnitude of the vorticity vector, which is directed along the z axis. An **irrotational** flow is one with zero vorticity. Irrotational flows have been widely studied because of their useful mathematical properties and applicability to flow regions where viscous effects may be neglected. Such flows without viscous effects are called **inviscid** flows.

Laminar and Turbulent Flow, Reynolds Number These terms refer to two distinct types of flow. In *laminar flow*, there are smooth streamlines and the fluid velocity components vary smoothly with position, and with time if the flow is unsteady. The flow described in reference to Fig. 6-1 is laminar. In *turbulent flow*, there are no smooth streamlines, and the velocity shows chaotic fluctuations in time and space. Velocities in turbulent flow may be reported as the sum of a time-averaged velocity and a velocity fluctuation from the average. For any given flow geometry, a dimensionless **Reynolds number** may be defined for a Newtonian fluid as $Re = LU\rho/\mu$ where L is a characteristic length. Below a critical value of Re the flow is laminar, while above the critical value a transition to turbulent flow occurs. The geometry-dependent critical Reynolds number is determined experimentally.

CONSERVATION EQUATIONS

Macroscopic and Microscopic Balances Three postulates, regarded as laws of physics, are fundamental in fluid mechanics. These are *conservation of mass*, *conservation of momentum*, and *conservation of energy*. In addition, two other postulates, *conservation of moment of momentum* (angular momentum) and the *entropy inequality* (second law of thermodynamics) have occasional use. The conservation principles may be applied either to material systems or to control volumes in space. Most often, control volumes are used. The control volumes may be either of finite or differential size, resulting in either **algebraic** or **differential** conservation equations, respectively. These are often called **macroscopic** and **microscopic** balance equations.

Macroscopic Equations An arbitrary control volume of finite size V_a is bounded by a surface of area A_a with an outwardly directed unit normal vector $\mathbf{n}$. The control volume is not necessarily fixed in space. Its boundary moves with velocity $\mathbf{w}$. The fluid velocity is $\mathbf{v}$. Figure 6-3 shows the arbitrary control volume.

Mass Balance Applied to the control volume, the principle of conservation of mass may be written as (Whitaker, *Introduction to Fluid Mechanics*, Prentice-Hall, Englewood Cliffs, N.J., 1968, Krieger, Huntington, N.Y., 1981)

$$\frac{d}{dt} \int_{V_a} \rho dV + \int_{A_a} \rho(\mathbf{v} - \mathbf{w}) \cdot \mathbf{n} dA = 0 \quad (6-8)$$

This equation is also known as the **continuity** equation.

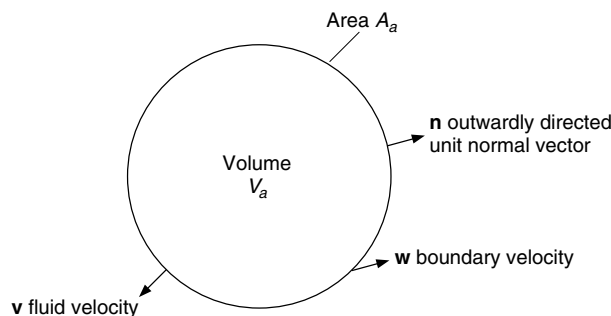


FIG. 6-3 Arbitrary control volume for application of conservation equations.

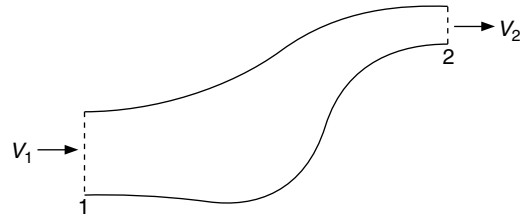


FIG. 6-4 Fixed control volume with one inlet and one outlet.

Simplified forms of Eq. (6-8) apply to special cases frequently found in practice. For a control volume fixed in space with one inlet of area A_1 through which an incompressible fluid enters the control volume at an average velocity V_1 , and one outlet of area A_2 through which fluid leaves at an average velocity V_2 , as shown in Fig. 6-4, the continuity equation becomes

$$V_1 A_1 = V_2 A_2 \quad (6-9)$$

The average velocity across a surface is given by

$$V = (1/A) \int_A v dA$$

where v is the local velocity component perpendicular to the inlet surface. The volumetric flow rate Q is the product of average velocity and the cross-sectional area, $Q = VA$. The average **mass velocity** is $G = \rho V$. For steady flows through fixed control volumes with multiple inlets and/or outlets, conservation of mass requires that the sum of inlet mass flow rates equals the sum of outlet mass flow rates. For incompressible flows through fixed control volumes, the sum of inlet flow rates (mass or volumetric) equals the sum of exit flow rates, whether the flow is steady or unsteady.

Momentum Balance Since momentum is a vector quantity, the momentum balance is a vector equation. Where gravity is the only body force acting on the fluid, the linear momentum principle, applied to the arbitrary control volume of Fig. 6-3, results in the following expression (Whitaker, *ibid.*).

$$\frac{d}{dt} \int_{V_a} \rho \mathbf{v} dV + \int_{A_a} \rho \mathbf{v}(\mathbf{v} - \mathbf{w}) \cdot \mathbf{n} dA = \int_{V_a} \rho \mathbf{g} dV + \int_{A_a} \mathbf{t}_n dA \quad (6-10)$$

Here $\mathbf{g}$ is the gravity vector and $\mathbf{t}_n$ is the force per unit area exerted by the surroundings on the fluid in the control volume. The integrand of the area integral on the left-hand side of Eq. (6-10) is nonzero only on the entrance and exit portions of the control volume boundary. For the special case of steady flow at a mass flow rate $\dot{m}$ through a control volume fixed in space with one inlet and one outlet (Fig. 6-4), with the inlet and outlet velocity vectors perpendicular to planar inlet and outlet surfaces, giving average velocity vectors $\mathbf{V}_1$ and $\mathbf{V}_2$, the momentum equation becomes

$$\dot{m}(\beta_2 \mathbf{V}_2 - \beta_1 \mathbf{V}_1) = -p_1 \mathbf{A}_1 - p_2 \mathbf{A}_2 + \mathbf{F} + M\mathbf{g} \quad (6-11)$$

where M is the total mass of fluid in the control volume. The factor β arises from the averaging of the velocity across the area of the inlet or outlet surface. It is the ratio of the area average of the square of velocity magnitude to the square of the area average velocity magnitude. For a uniform velocity, $\beta = 1$. For turbulent flow, β is nearly unity, while for laminar pipe flow with a parabolic velocity profile, $\beta = 4/3$. The vectors $\mathbf{A}_1$ and $\mathbf{A}_2$ have magnitude equal to the areas of the inlet and outlet surfaces, respectively, and are outwardly directed normal to the surfaces. The vector $\mathbf{F}$ is the force exerted on the fluid by the non-flow boundaries of the control volume. It is also assumed that the stress vector $\mathbf{t}_n$ is normal to the inlet and outlet surfaces, and that its magnitude may be approximated by the pressure p . Equation (6-11) may be generalized to multiple inlets and/or outlets. In such cases, the mass flow rates for all the inlets and outlets are not equal. A distinct flow rate $\dot{m}_i$ applies to each inlet or outlet i . To generalize the equation, $-p\mathbf{A}$ terms for each inlet and outlet, $-\dot{m}\beta\mathbf{V}$ terms for each inlet, and $\dot{m}\beta\mathbf{V}$ terms for each outlet are included.

Balance equations for angular momentum, or moment of momentum, may also be written. They are used less frequently than the linear momentum equations. See Whitaker (*Introduction to Fluid Mechanics*, Prentice-Hall, Englewood Cliffs, N.J., 1968, Krieger, Huntington, N.Y., 1981) or Shames (*Mechanics of Fluids*, 3d ed., McGraw-Hill, New York, 1992).

Total Energy Balance The total energy balance derives from the first law of thermodynamics. Applied to the arbitrary control volume of Fig. 6-3, it leads to an equation for the rate of change of the sum of internal, kinetic, and gravitational potential energy. In this equation, u is the internal energy per unit mass, v is the magnitude of the velocity vector $\mathbf{v}$, z is elevation, g is the gravitational acceleration, and $\mathbf{q}$ is the heat flux vector:

$$\frac{d}{dt} \int_{V_a} \rho \left(u + \frac{v^2}{2} + gz \right) dV + \int_{A_a} \rho \left(u + \frac{v^2}{2} + gz \right) (\mathbf{v} - \mathbf{w}) \cdot \mathbf{n} dA = \int_{A_a} (\mathbf{v} \cdot \mathbf{t}_n) dA - \int_{A_a} (\mathbf{q} \cdot \mathbf{n}) dA \quad (6-12)$$

The first integral on the right-hand side is the rate of work done on the fluid in the control volume by forces at the boundary. It includes both work done by moving solid boundaries and work done at flow entrances and exits. The work done by moving solid boundaries also includes that by such surfaces as pump impellers; this work is called **shaft work**; its rate is $\dot{W}_s$.

A useful simplification of the total energy equation applies to a particular set of assumptions. These are a control volume with fixed solid boundaries, except for those producing shaft work, steady state conditions, and mass flow at a rate $\dot{m}$ through a single planar entrance and a single planar exit (Fig. 6-4), to which the velocity vectors are perpendicular. As with Eq. (6-11), it is assumed that the stress vector $\mathbf{t}_n$ is normal to the entrance and exit surfaces and may be approximated by the pressure p . The **equivalent pressure**, $p + \rho gz$, is assumed to be uniform across the entrance and exit. The average velocity at the entrance and exit surfaces is denoted by V . Subscripts 1 and 2 denote the entrance and exit, respectively.

$$h_1 + \alpha_1 \frac{V_1^2}{2} + gz_1 = h_2 + \alpha_2 \frac{V_2^2}{2} + gz_2 - \delta Q - \delta W_s \quad (6-13)$$

Here, h is the enthalpy per unit mass, $h = u + p/\rho$. The shaft work per unit of mass flowing through the control volume is $\delta W_s = \dot{W}_s/\dot{m}$. Similarly, δQ is the heat input per unit of mass. The factor α is the ratio of the cross-sectional area average of the cube of the velocity to the cube of the average velocity. For a uniform velocity profile, $\alpha = 1$. In turbulent flow, α is usually assumed to equal unity; in turbulent pipe flow, it is typically about 1.07. For laminar flow in a circular pipe with a parabolic velocity profile, $\alpha = 2$.

Mechanical Energy Balance, Bernoulli Equation A balance equation for the sum of kinetic and potential energy may be obtained from the momentum balance by forming the scalar product with the velocity vector. The resulting equation, called the *mechanical energy balance*, contains a term accounting for the dissipation of mechanical energy into thermal energy by viscous forces. The mechanical energy equation is also derivable from the total energy equation in a way that reveals the relationship between the dissipation and entropy generation. The macroscopic mechanical energy balance for the arbitrary control volume of Fig. 6-3 may be written, with p = thermodynamic pressure, as

$$\frac{d}{dt} \int_{V_a} \rho \left(\frac{v^2}{2} + gz \right) dV + \int_{A_a} \rho \left(\frac{v^2}{2} + gz \right) (\mathbf{v} - \mathbf{w}) \cdot \mathbf{n} dA = \int_{V_a} p \nabla \cdot \mathbf{v} dV + \int_{A_a} (\mathbf{v} \cdot \mathbf{t}_n) dA - \int_{V_a} \Phi dV \quad (6-14)$$

The last term is the rate of viscous energy dissipation to internal energy, $\dot{E}_v = \int_{V_a} \Phi dV$, also called the rate of viscous losses. These losses are the origin of frictional pressure drop in fluid flow. Whitaker and Bird, Stewart, and Lightfoot provide expressions for the dissipation function Φ for Newtonian fluids in terms of the local velocity gradients. However, when using macroscopic balance equations the local velocity field within the control volume is usually unknown. For such

cases additional information, which may come from empirical correlations, is needed.

For the same special conditions as for Eq. (6-13), the mechanical energy equation is reduced to

$$\alpha_1 \frac{V_1^2}{2} + gz_1 + \delta W_s = \alpha_2 \frac{V_2^2}{2} + gz_2 + \int_{p_1}^{p_2} \frac{dp}{\rho} + l_v \quad (6-15)$$

Here $l_v = \dot{E}_v/\dot{m}$ is the energy dissipation per unit mass. This equation has been called the **engineering Bernoulli equation**. For an incompressible flow, Eq. (6-15) becomes

$$\frac{p_1}{\rho} + \alpha_1 \frac{V_1^2}{2} + gz_1 + \delta W_s = \frac{p_2}{\rho} + \alpha_2 \frac{V_2^2}{2} + gz_2 + l_v \quad (6-16)$$

The Bernoulli equation can be written for incompressible, inviscid flow along a streamline, where no shaft work is done.

$$\frac{p_1}{\rho} + \frac{V_1^2}{2} + gz_1 = \frac{p_2}{\rho} + \frac{V_2^2}{2} + gz_2 \quad (6-17)$$

Unlike the momentum equation (Eq. [6-11]), the Bernoulli equation is not easily generalized to multiple inlets or outlets.

Microscopic Balance Equations Partial differential balance equations express the conservation principles at a point in space. Equations for mass, momentum, total energy, and mechanical energy may be found in Whitaker (ibid.), Bird, Stewart, and Lightfoot (*Transport Phenomena*, Wiley, New York, 1960), and Slattery (*Momentum, Heat and Mass Transfer in Continua*, 2d ed., Krieger, Huntington, N.Y., 1981), for example. These references also present the equations in other useful coordinate systems besides the cartesian system. The coordinate systems are fixed in inertial reference frames. The two most used equations, for mass and momentum, are presented here.

Mass Balance, Continuity Equation The continuity equation, expressing conservation of mass, is written in cartesian coordinates as

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho v_x}{\partial x} + \frac{\partial \rho v_y}{\partial y} + \frac{\partial \rho v_z}{\partial z} = 0 \quad (6-18)$$

In terms of the **substantial derivative**, D/Dt ,

$$\frac{D\rho}{Dt} \equiv \frac{\partial \rho}{\partial t} + v_x \frac{\partial \rho}{\partial x} + v_y \frac{\partial \rho}{\partial y} + v_z \frac{\partial \rho}{\partial z} = -\rho \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right) \quad (6-19)$$

The substantial derivative, also called the **material derivative**, is the rate of change in a Lagrangian reference frame, that is, following a material particle. In vector notation the continuity equation may be expressed as

$$\frac{D\rho}{Dt} = -\rho \nabla \cdot \mathbf{v} \quad (6-20)$$

For incompressible flow,

$$\nabla \cdot \mathbf{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0 \quad (6-21)$$

Stress Tensor The stress tensor is needed to completely describe the stress state for microscopic momentum balances in multidimensional flows. The components of the stress tensor σ_{ij} give the force in the j direction on a plane perpendicular to the i direction, using a sign convention defining a positive stress as one where the fluid with the greater i coordinate value exerts a force in the positive i direction on the fluid with the lesser i coordinate. Several references in fluid mechanics and continuum mechanics provide discussions, to various levels of detail, of stress in a fluid (Denn; Bird, Stewart, and Lightfoot; Schlichting; Fung [*A First Course in Continuum Mechanics*, 2d ed., Prentice-Hall, Englewood Cliffs, N.J., 1977]; Truesdell and Toupin [in Flügge, *Handbuch der Physik*, vol. 3/1, Springer-Verlag, Berlin, 1960]; Slattery [*Momentum, Energy and Mass Transfer in Continua*, 2d ed., Krieger, Huntington, N.Y., 1981]).

The stress has an isotropic contribution due to fluid pressure and dilatation, and a **deviatoric** contribution due to viscous deformation effects. The deviatoric contribution for a Newtonian fluid is the three-dimensional generalization of Eq. (6-2):

$$\tau_{ij} = \mu D_{ij} \quad (6-22)$$

6-8 FLUID AND PARTICLE DYNAMICS

The total stress is

$$\sigma_{ij} = (-p + \lambda \nabla \cdot \mathbf{v}) \delta_{ij} + \tau_{ij} \quad (6-23)$$

The identity tensor δ_{ij} is zero for $i \neq j$ and unity for $i = j$. The coefficient λ is a material property related to the **bulk viscosity**, $\kappa = \lambda + 2\mu/3$. There is considerable uncertainty about the value of κ . Traditionally, Stokes' hypothesis, $\kappa = 0$, has been invoked, but the validity of this hypothesis is doubtful (Slattery, *ibid.*). For incompressible flow, the value of bulk viscosity is immaterial as Eq. (6-23) reduces to

$$\sigma_{ij} = -p \delta_{ij} + \tau_{ij} \quad (6-24)$$

Similar generalizations to multidimensional flow are necessary for non-Newtonian constitutive equations.

Cauchy Momentum and Navier-Stokes Equations The differential equations for conservation of momentum are called the **Cauchy momentum equations**. These may be found in general form in most fluid mechanics texts (e.g., Slattery [*ibid.*]; Denn; Whitaker; and Schlichting). For the important special case of an incompressible Newtonian fluid with constant viscosity, substitution of Eqs. (6-22) and (6-24) leads to the **Navier-Stokes equations**, whose three Cartesian components are

$$\begin{aligned} \rho \left(\frac{\partial v_x}{\partial t} + v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} + v_z \frac{\partial v_x}{\partial z} \right) \\ = -\frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2} \right) + \rho g_x \end{aligned} \quad (6-25)$$

$$\begin{aligned} \rho \left(\frac{\partial v_y}{\partial t} + v_x \frac{\partial v_y}{\partial x} + v_y \frac{\partial v_y}{\partial y} + v_z \frac{\partial v_y}{\partial z} \right) \\ = -\frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v_y}{\partial x^2} + \frac{\partial^2 v_y}{\partial y^2} + \frac{\partial^2 v_y}{\partial z^2} \right) + \rho g_y \end{aligned} \quad (6-26)$$

$$\begin{aligned} \rho \left(\frac{\partial v_z}{\partial t} + v_x \frac{\partial v_z}{\partial x} + v_y \frac{\partial v_z}{\partial y} + v_z \frac{\partial v_z}{\partial z} \right) \\ = -\frac{\partial p}{\partial z} + \mu \left(\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2} \right) + \rho g_z \end{aligned} \quad (6-27)$$

In vector notation,

$$\rho \frac{D\mathbf{v}}{Dt} = \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\nabla p + \mu \nabla^2 \mathbf{v} + \rho \mathbf{g} \quad (6-28)$$

The pressure and gravity terms may be combined by replacing the pressure p by the equivalent pressure $P = p + \rho g z$. The left-hand side terms of the Navier-Stokes equations are the **inertial terms**, while the terms including viscosity μ are the **viscous terms**. Limiting cases under which the Navier-Stokes equations may be simplified include **creeping flows** in which the inertial terms are neglected, **potential flows** (inviscid or irrotational flows) in which the viscous terms are neglected, and **boundary layer** and **lubrication** flows in which certain terms are neglected based on scaling arguments. Creeping flows are described by Happel and Brenner (*Low Reynolds Number Hydrodynamics*, Prentice-Hall, Englewood Cliffs, N.J., 1965); potential flows by Lamb (*Hydrodynamics*, 6th ed., Dover, New York, 1945) and Milne-Thompson (*Theoretical Hydrodynamics*, 5th ed., Macmillan, New York, 1968); boundary layer theory by Schlichting (*Boundary Layer Theory*, 8th ed., McGraw-Hill, New York, 1987); and lubrication theory by Batchelor (*An Introduction to Fluid Dynamics*, Cambridge University, Cambridge, 1967) and Denn (*Process Fluid Mechanics*, Prentice-Hall, Englewood Cliffs, N.J., 1980).

Because the Navier-Stokes equations are first-order in pressure and second-order in velocity, their solution requires one pressure boundary condition and two velocity boundary conditions (for each velocity component) to completely specify the solution. The **no slip** condition, which requires that the fluid velocity equal the velocity of any bounding solid surface, occurs in most problems. Specification of velocity is a type of boundary condition sometimes called a *Dirichlet condition*. Often boundary conditions involve stresses, and thus velocity gradients, rather

than the velocities themselves. Specification of velocity derivatives is a *Neumann boundary condition*. For example, at the boundary between a viscous liquid and a gas, it is often assumed that the liquid shear stresses are zero. In numerical solution of the Navier-Stokes equations, Dirichlet and Neumann, or **essential** and **natural**, boundary conditions may be satisfied by different means.

Fluid statics, discussed in Sec. 10 of the *Handbook* in reference to pressure measurement, is the branch of fluid mechanics in which the fluid velocity is either zero or is uniform and constant relative to an inertial reference frame. With velocity gradients equal to zero, the momentum equation reduces to a simple expression for the pressure field, $\nabla p = \rho \mathbf{g}$. Letting z be directed vertically upward, so that $g_z = -g$ where g is the gravitational acceleration (9.806 m/s²), the pressure field is given by

$$dp/dz = -\rho g \quad (6-29)$$

This equation applies to any incompressible or compressible static fluid. For an incompressible liquid, pressure varies linearly with depth. For compressible gases, p is obtained by integration accounting for the variation of ρ with z .

The **force exerted on a submerged planar surface** of area A is given by $F = p_c A$ where p_c is the pressure at the geometrical **centroid** of the surface. The **center of pressure**, the point of application of the net force, is always lower than the centroid. For details see, for example, Shames, where may also be found discussion of forces on **curved surfaces**, **buoyancy**, and **stability of floating bodies**.

Examples Four examples follow, illustrating the application of the conservation equations to obtain useful information about fluid flows.

Example 1: Force Exerted on a Reducing Bend An incompressible fluid flows through a reducing elbow (Fig. 6-5) situated in a horizontal plane. The inlet velocity V_1 is given and the pressures p_1 and p_2 are measured. Selecting the inlet and outlet surfaces 1 and 2 as shown, the continuity equation Eq. (6-9) can be used to find the exit velocity $V_2 = V_1 A_1 / A_2$. The mass flow rate is obtained by $\dot{m} = \rho V_1 A_1$.

Assume that the velocity profile is nearly uniform so that β is approximately unity. The force exerted on the fluid by the bend has x and y components; these can be found from Eq. (6-11). The x component gives

$$F_x = \dot{m}(V_{2x} - V_{1x}) + p_1 A_{1x} + p_2 A_{2x}$$

while the y component gives

$$F_y = \dot{m}(V_{2y} - V_{1y}) + p_1 A_{1y} + p_2 A_{2y}$$

The velocity components are $V_{1x} = V_1$, $V_{1y} = 0$, $V_{2x} = V_2 \cos \theta$, and $V_{2y} = V_2 \sin \theta$. The area vector components are $A_{1x} = -A_1$, $A_{1y} = 0$, $A_{2x} = A_2 \cos \theta$, and $A_{2y} = A_2 \sin \theta$. Therefore, the force components may be calculated from

$$F_x = \dot{m}(V_2 \cos \theta - V_1) - p_1 A_1 + p_2 A_2 \cos \theta$$

$$F_y = \dot{m} V_2 \sin \theta + p_2 A_2 \sin \theta$$

The force acting on the fluid is $\mathbf{F}$; the equal and opposite force exerted by the fluid on the bend is $-\mathbf{F}$.

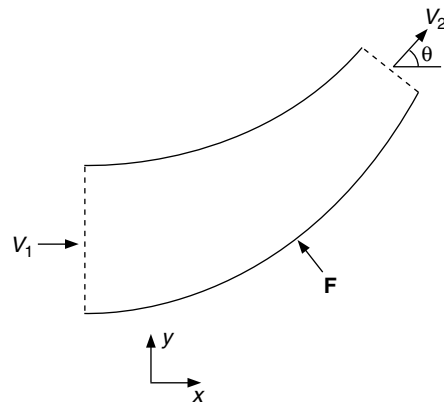


FIG. 6-5 Force at a reducing bend. $\mathbf{F}$ is the force exerted by the bend on the fluid. The force exerted by the fluid on the bend is $-\mathbf{F}$.

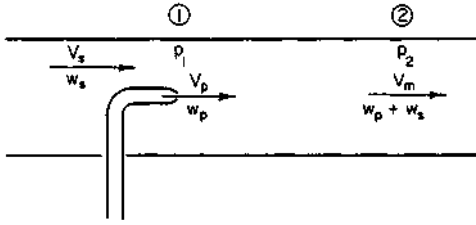


FIG. 6-6 Draft-tube ejector.

Example 2: Simplified Ejector Figure 6-6 shows a very simplified sketch of an ejector, a device that uses a high velocity primary fluid to pump another (secondary) fluid. The continuity and momentum equations may be applied on the control volume with inlet and outlet surfaces 1 and 2 as indicated in the figure. The cross-sectional area is uniform, $A_1 = A_2 = A$. Let the mass flow rates and velocities of the primary and secondary fluids be $\dot{m}_p$, $\dot{m}_s$, V_p and V_s . Assume for simplicity that the density is uniform. Conservation of mass gives $\dot{m}_2 = \dot{m}_p + \dot{m}_s$. The exit velocity is $V_2 = \dot{m}_2/(\rho A)$. The principle momentum exchange in the ejector occurs between the two fluids. Relative to this exchange, the force exerted by the walls of the device are found to be small. Therefore, the force term F is neglected from the momentum equation. Written in the flow direction, assuming uniform velocity profiles, and using the extension of Eq. (6-11) for multiple inlets, it gives the pressure rise developed by the device:

$$(p_2 - p_1)A = (\dot{m}_p + \dot{m}_s)V_2 - \dot{m}_p V_p - \dot{m}_s V_s$$

Application of the momentum equation to ejectors of other types is discussed in Lapple (*Fluid and Particle Dynamics*, University of Delaware, Newark, 1951) and in Sec. 10 of the *Handbook*.

Example 3: Venturi Flowmeter An incompressible fluid flows through the venturi flowmeter in Fig. 6-7. An equation is needed to relate the flow rate Q to the pressure drop measured by the manometer. This problem can be solved using the mechanical energy balance. In a well-made venturi, viscous losses are negligible, the pressure drop is entirely the result of acceleration into the throat, and the flow rate predicted neglecting losses is quite accurate. The inlet area is A and the throat area is a .

With control surfaces at 1 and 2 as shown in the figure, Eq. (6-17) in the absence of losses and shaft work gives

$$\frac{p_1}{\rho} + \frac{V_1^2}{2} = \frac{p_2}{\rho} + \frac{V_2^2}{2}$$

The continuity equation gives $V_2 = V_1 A/a$, and $V_1 = Q/A$. The pressure drop measured by the manometer is $p_1 - p_2 = (\rho_m - \rho)g\Delta z$. Substituting these relations into the energy balance and rearranging, the desired expression for the flow rate is found.

$$Q = \frac{1}{A} \sqrt{\frac{2(\rho_m - \rho)g\Delta z}{\rho[(A/a)^2 - 1]}}$$

Example 4: Plane Poiseuille Flow An incompressible Newtonian fluid flows at a steady rate in the x direction between two very large flat plates, as shown in Fig. 6-8. The flow is laminar. The velocity profile is to be found. This example is found in most fluid mechanics textbooks; the solution presented here closely follows Denn.

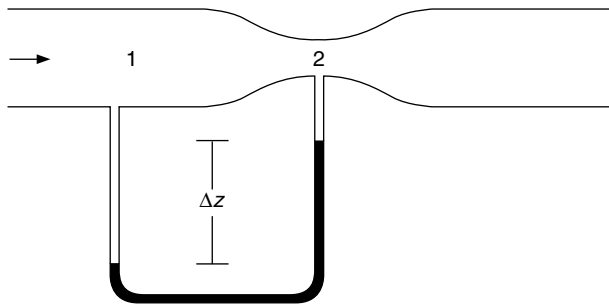


FIG. 6-7 Venturi flowmeter.

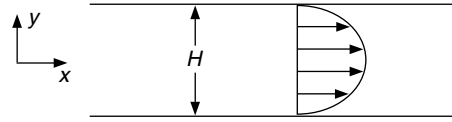


FIG. 6-8 Plane Poiseuille flow.

This problem requires use of the microscopic balance equations because the velocity is to be determined as a function of position. The boundary conditions for this flow result from the no-slip condition. All three velocity components must be zero at the plate surfaces, $y = H/2$ and $y = -H/2$.

Assume that the flow is **fully developed**, that is, all velocity derivatives vanish in the x direction. Since the flow field is infinite in the z direction, all velocity derivatives should be zero in the z direction. Therefore, velocity components are a function of y alone. It is also assumed that there is no flow in the z direction, so $v_z = 0$. The continuity equation Eq. (6-21), with $v_z = 0$ and $\partial v_x/\partial x = 0$, reduces to

$$\frac{dv_y}{dy} = 0$$

Since $v_y = 0$ at $y = \pm H/2$, the continuity equation integrates to $v_y = 0$. This is a direct result of the assumption of fully developed flow.

The Navier-Stokes equations are greatly simplified when it is noted that $v_y = v_z = 0$ and $\partial v_x/\partial x = \partial v_x/\partial z = \partial v_x/\partial t = 0$. The three components are written in terms of the equivalent pressure P :

$$0 = -\frac{\partial P}{\partial x} + \mu \frac{\partial^2 v_x}{\partial y^2}$$

$$0 = -\frac{\partial P}{\partial y}$$

$$0 = -\frac{\partial P}{\partial z}$$

The latter two equations require that P is a function only of x , and therefore $\partial P/\partial x = dP/dx$. Inspection of the first equation shows one term which is a function only of x and one which is only a function of y . This requires that both terms are constant. The pressure gradient $-dP/dx$ is constant. The x -component equation becomes

$$\frac{d^2 v_x}{dy^2} = \frac{1}{\mu} \frac{dP}{dx}$$

Two integrations of the x -component equation give

$$v_x = \frac{1}{2\mu} \frac{dP}{dx} y^2 + C_1 y + C_2$$

where the constants of integration C_1 and C_2 are evaluated from the boundary conditions $v_x = 0$ at $y = \pm H/2$. The result is

$$v_x = \frac{H^2}{8\mu} \left(-\frac{dP}{dx} \right) \left[1 - \left(\frac{2y}{H} \right)^2 \right]$$

This is a **parabolic** velocity distribution. The average velocity $V = (1/H) \int_{-H/2}^{H/2} v_x dy$ is

$$V = \frac{H^2}{12\mu} \left(-\frac{dP}{dx} \right)$$

This flow is one-dimensional, as there is only one nonzero velocity component, v_x , which, along with the pressure, varies in only one coordinate direction.

INCOMPRESSIBLE FLOW IN PIPES AND CHANNELS

Mechanical Energy Balance The mechanical energy balance, Eq. (6-16), for **fully developed** incompressible flow in a straight circular pipe of constant diameter D reduces to

$$\frac{p_1}{\rho} + gz_1 = \frac{p_2}{\rho} + gz_2 + l_v \quad (6-30)$$

In terms of the equivalent pressure, $P \equiv p + \rho gz$,

$$P_1 - P_2 = \rho l_v \quad (6-31)$$

The pressure drop due to frictional losses l_v is proportional to pipe length L for fully developed flow and may be denoted as the (positive) quantity $\Delta P \equiv P_1 - P_2$.

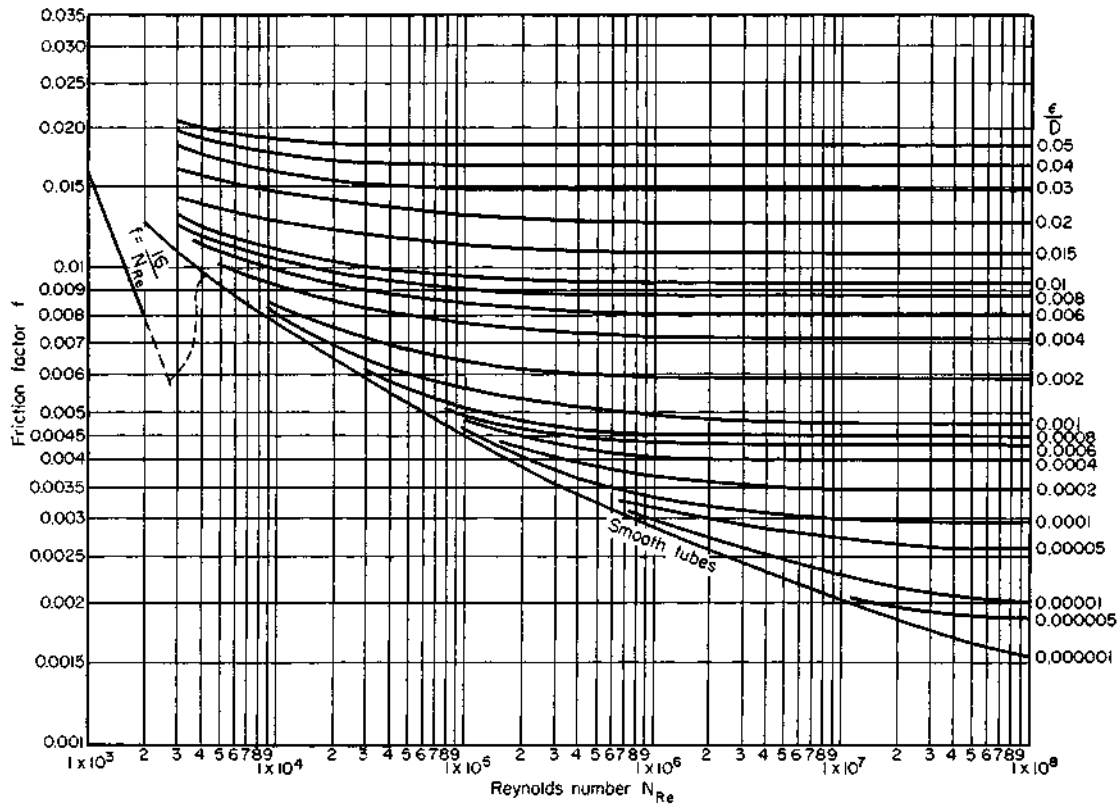


FIG. 6-9 Fanning Friction Factors. Reynolds number $Re = DV\rho/\mu$, where D = pipe diameter, V = velocity, ρ = fluid density, and μ = fluid viscosity. (Based on Moody, *Trans. ASME*, **66**, 671 [1944].)

Friction Factor and Reynolds Number For a Newtonian fluid in a smooth pipe, dimensional analysis relates the frictional pressure drop per unit length $\Delta P/L$ to the pipe diameter D , density ρ , viscosity μ , and average velocity V through two dimensionless groups, the **Fanning friction factor** f and the **Reynolds number** Re .

$$f = \frac{D\Delta P}{2\rho V^2 L} \quad (6-32)$$

$$Re = \frac{DV\rho}{\mu} \quad (6-33)$$

For smooth pipe, the friction factor is a function only of the Reynolds number. In rough pipe, the relative roughness ϵ/D also affects the friction factor. Figure 6-9 plots f as a function of Re and ϵ/D . Values of ϵ for various materials are given in Table 6-1. The Fanning friction factor should not be confused with the Darcy friction factor used by Moody (*Trans. ASME*, **66**, 671 [1944]), which is four times greater. Using the momentum equation, the stress at the wall of the pipe may be expressed in terms of the friction factor:

$$\tau_w = f \frac{\rho V^2}{2} \quad (6-34)$$

Laminar and Turbulent Flow Below a **critical Reynolds number** of about 2,100, the flow is laminar; over the range $2,100 < Re < 5,000$ there is a transition to turbulent flow. Reliable correlations for the friction factor in transitional flow are not available. For laminar flow, the Hagen-Poiseuille equation

$$f = \frac{16}{Re} \quad Re \leq 2,100 \quad (6-35)$$

TABLE 6-1 Values of Surface Roughness for Various Materials*

Material	Surface roughness ϵ , mm
Drawn tubing (brass, lead, glass, and the like)	0.00152
Commercial steel or wrought iron	0.0457
Asphalted cast iron	0.122
Galvanized iron	0.152
Cast iron	0.259
Wood stove	0.183–0.914
Concrete	0.305–3.05
Riveted steel	0.914–9.14

* From Moody, *Trans. Am. Soc. Mech. Eng.*, **66**, 671–684 (1944); *Mech. Eng.*, **69**, 1005–1006 (1947). Additional values of ϵ for various types or conditions of concrete wrought-iron, welded steel, riveted steel, and corrugated-metal pipes are given in Brater and King, *Handbook of Hydraulics*, 6th ed., McGraw-Hill, New York, 1976, pp. 6-12–6-13. To convert millimeters to feet, multiply by 3.281×10^{-3} .

may be derived from the Navier-Stokes equation and is in excellent agreement with experimental data. It may be rewritten in terms of volumetric flow rate, $Q = V\pi D^2/4$, as

$$Q = \frac{\pi \Delta P D^4}{128 \mu L} \quad Re \leq 2,100 \quad (6-36)$$

For turbulent flow in smooth tubes, the Blasius equation gives the friction factor accurately for a wide range of Reynolds numbers.

$$f = \frac{0.079}{Re^{0.25}} \quad 4,000 < Re < 10^5 \quad (6-37)$$

The Colebrook formula (Colebrook, *J. Inst. Civ. Eng. [London]*, **11**, 133–156 [1938–39]) gives a good approximation for the f - Re - (ϵ/D) data for rough pipes over the entire turbulent flow range:

$$\frac{1}{\sqrt{f}} = -4 \log \left[\frac{\epsilon}{3.7D} + \frac{1.256}{Re\sqrt{f}} \right] \quad Re > 4,000 \quad (6-38)$$

Equation (6-38) was used to construct the curves in the turbulent flow regime in Fig. 6-9.

An equation by Churchill (*Chem. Eng.*, **84**[24], 91–92 [Nov. 7, 1977]) approximating the Colebrook formula offers the advantage of being explicit in f :

$$\frac{1}{\sqrt{f}} = -4 \log \left[\frac{0.27\epsilon}{D} + \left(\frac{7}{Re} \right)^{0.9} \right] \quad Re > 4,000 \quad (6-39)$$

Churchill also provided a single equation that may be used for Reynolds numbers in laminar, transitional, and turbulent flow, closely fitting $f = 16/Re$ in the laminar regime, and the Colebrook formula, Eq. (6-38), in the turbulent regime. It also gives unique, reasonable values in the transition regime, where the friction factor is uncertain.

$$f = 2 \left[\left(\frac{8}{Re} \right)^{12} + \frac{1}{(A+B)^{3/2}} \right]^{1/12} \quad (6-40)$$

where

$$A = \left[2.457 \ln \frac{1}{(7/Re)^{0.9} + 0.27\epsilon/D} \right]^{16}$$

and

$$B = \left(\frac{37,530}{Re} \right)^{16}$$

In laminar flow, f is independent of ϵ/D . In turbulent flow, the friction factor for rough pipe follows the smooth tube curve for a range of Reynolds numbers (hydraulically smooth flow). For greater Reynolds numbers, f deviates from the smooth pipe curve, eventually becoming independent of Re . This region, often called *complete turbulence*, is frequently encountered in commercial pipe flows.

Two common pipe flow problems are calculation of pressure drop given the flow rate (or velocity) and calculation of flow rate (or velocity) given the pressure drop. When flow rate is given, the Reynolds number may be calculated directly to determine the flow regime, so that the appropriate relations between f and Re (or pressure drop and flow rate or velocity) can be selected. When flow rate is specified and the flow is turbulent, Eq. (6-39) or (6-40), being explicit in f , may be preferable to Eq. (6-38), which is implicit in f and pressure drop.

When the pressure drop is given and the velocity and flow rate are to be determined, the Reynolds number cannot be computed directly, since the velocity is unknown. Instead of guessing and checking the flow regime, it may be useful to observe that the quantity $Re\sqrt{f} = (D^{3/2}/\mu) \sqrt{\rho \Delta P/(2L)}$, appearing in the Colebrook equation (6-38), does not include velocity and so can be computed directly. The upper limit $Re = 2,100$ for laminar flow and use of Eq. (6-35) corresponds to $Re\sqrt{f} = 183$. For smooth pipe, the lower limit $Re = 4,000$ for the Colebrook equation corresponds to $Re\sqrt{f} = 400$. Thus, at least for smooth pipes, the flow regime can be determined without trial and error from $\Delta P/L$, μ , ρ , and D . When pressure drop is given, Eq. (6-38), being explicit in velocity, is preferable to Eqs. (6-39) and (6-40), which are implicit in velocity.

As Fig. 6-9 suggests, the friction factor is uncertain in the transition range, and a conservative choice should be made for design purposes.

Velocity Profiles In laminar flow, the solution of the Navier-Stokes equation, corresponding to the Hagen-Poiseuille equation, gives the velocity v as a function of radial position r in a circular pipe of radius R in terms of the average velocity $V = Q/A$. The **parabolic** profile, with centerline velocity twice the average velocity, is shown in Fig. 6-10.

$$v = 2V \left(1 - \frac{r^2}{R^2} \right) \quad (6-41)$$

In turbulent flow, the velocity profile is much more blunt, with most of the velocity gradient being in a region near the wall, described by a **universal** velocity profile. It is characterized by a **viscous sublayer**, a **turbulent core**, and a **buffer zone** in between.

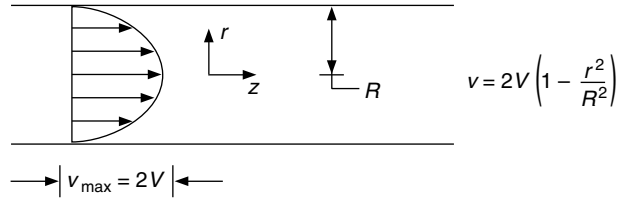


FIG. 6-10 Parabolic velocity profile for laminar flow in a pipe, with average velocity V .

Viscous sublayer

$$u_+ = y_+ \quad \text{for} \quad y_+ < 5 \quad (6-42)$$

Buffer zone

$$u_+ = 5.00 \ln y_+ - 3.05 \quad \text{for} \quad 5 < y_+ < 30 \quad (6-43)$$

Turbulent core

$$u_+ = 2.5 \ln y_+ + 5.5 \quad \text{for} \quad y_+ > 30 \quad (6-44)$$

Here, $u_+ = v/u_\tau$ is the dimensionless, time-averaged axial velocity, $u_\tau = \sqrt{\tau_w/\rho}$ is the **friction velocity** and $\tau_w = f\rho V^2/2$ is the wall stress. The friction velocity is of the order of the root mean square velocity fluctuation perpendicular to the wall in the turbulent core. The dimensionless distance from the wall is $y_+ = y u_\tau \rho / \mu$. The universal velocity profile is valid in the wall region for any cross-sectional channel shape. For incompressible flow in constant diameter circular pipes, $\tau_w = D\Delta P/4L$ where ΔP is the pressure drop in length L . In circular pipes, Eq. (6-44) gives a surprisingly good fit to experimental results over the entire cross section of the pipe, even though it is based on assumptions which are valid only near the pipe wall.

For rough pipes, the velocity profile in the turbulent core is given by

$$u_+ = 2.5 \ln y/\epsilon + 8.5 \quad \text{for} \quad y_+ > 30 \quad (6-45)$$

when the dimensionless roughness $\epsilon_+ = \epsilon u_\tau \rho / \mu$ is greater than 5 to 10; for smaller ϵ_+ , the velocity profile in the turbulent core is unaffected by roughness.

For velocity profiles in the transition region, see Patel and Head (*J. Fluid Mech.*, **38**, part 1, 181–201 [1969]) where profiles over the range $1,500 < Re < 10,000$ are reported.

Entrance and Exit Effects In the entrance region of a pipe, some distance is required for the flow to adjust from upstream conditions to the fully developed flow pattern. This distance depends on the Reynolds number and on the flow conditions upstream. For a uniform velocity profile at the pipe entrance, the computed length in laminar flow required for the centerline velocity to reach 99 percent of its fully developed value is (Dombrowski, Founeney, Ookawara, and Riza, *Can. J. Chem. Engr.*, **71**, 472–476 [1993])

$$L_{\text{ent}}/D = 0.370 \exp(-0.148Re) + 0.0550Re + 0.260 \quad (6-46)$$

In turbulent flow, the entrance length is about

$$L_{\text{ent}}/D = 40 \quad (6-47)$$

The frictional losses in the entrance region are larger than those for the same length of fully developed flow. (See the subsection, “Frictional Losses in Pipeline Elements,” following.) At the pipe exit, the velocity profile also undergoes rearrangement, but the exit length is much shorter than the entrance length. At low Re , it is about one pipe radius. At $Re > 100$, the exit length is essentially 0.

Residence Time Distribution For laminar Newtonian pipe flow, the cumulative residence time distribution $F(\theta)$ is given by

$$F(\theta) = 0 \quad \text{for} \quad \theta < \frac{\theta_{\text{avg}}}{2}$$

$$F(\theta) = 1 - \frac{1}{4} \left(\frac{\theta_{\text{avg}}}{\theta} \right)^2 \quad \text{for} \quad \theta \geq \frac{\theta_{\text{avg}}}{2} \quad (6-48)$$

where $F(\theta)$ is the fraction of material which resides in the pipe for less than time θ and θ_{avg} is the average residence time, $\theta = V/L$.

The residence time distribution in long transfer lines may be made narrower (more uniform) with the use of **flow inverters** or **static mixing elements**. These devices exchange fluid between the wall and central regions. Variations on the concept may be used to provide effective mixing of the fluid. See Godfrey ("Static Mixers," in Harnby, Edwards, and Nienow, *Mixing in the Process Industries*, 2d ed., Butterworth Heinemann, Oxford, 1992); Etchells and Meyer ("Mixing in Pipelines," in Paul, Atiemo-Obeng, and Kresta, *Handbook of Industrial Mixing*, Wiley Interscience, Hoboken, N.J., 2004).

A theoretically derived equation for laminar flow in helical pipe coils by Ruthven (*Chem. Eng. Sci.*, **26**, 1113–1121 [1971]; **33**, 628–629 [1978]) is given by

$$F(\theta) = 1 - \left(\frac{1}{4} \right) \left[\frac{\theta_{\text{avg}}}{\theta} \right]^{2.51} \quad \text{for} \quad 0.5 < \frac{\theta_{\text{avg}}}{\theta} < 1.63 \quad (6-49)$$

and was substantially confirmed by Trivedi and Vasudeva (*Chem. Eng. Sci.*, **29**, 2291–2295 [1974]) for $0.6 < \text{De} < 6$ and $0.0036 < D/D_c < 0.097$ where $\text{De} = \text{Re}\sqrt{D/D_c}$ is the Dean number and D_c is the diameter of curvature of the coil. Measurements by Saxena and Nigam (*Chem. Eng. Sci.*, **34**, 425–426 [1979]) indicate that such a distribution will hold for $\text{De} > 1$. The residence time distribution for helical coils is narrower than for straight circular pipes, due to the secondary flow which exchanges fluid between the wall and center regions.

In turbulent flow, axial mixing is usually described in terms of turbulent diffusion or dispersion coefficients, from which cumulative residence time distribution functions can be computed. Davies (*Turbulence Phenomena*, Academic, New York, 1972, p. 93) gives $D_L = 1.01\text{vRe}^{0.575}$ for the longitudinal dispersion coefficient. Levenspiel (*Chemical Reaction Engineering*, 2d ed., Wiley, New York, 1972, pp. 253–278) discusses the relations among various residence time distribution functions, and the relation between dispersion coefficient and residence time distribution.

Noncircular Channels Calculation of frictional pressure drop in noncircular channels depends on whether the flow is laminar or turbulent, and on whether the channel is full or open. For **turbulent flow in ducts running full**, the **hydraulic diameter** D_H should be substituted for D in the friction factor and Reynolds number definitions, Eqs. (6-32) and (6-33). The hydraulic diameter is defined as **four times the channel cross-sectional area divided by the wetted perimeter**. For example, the hydraulic diameter for a circular pipe is $D_H = D$, for an annulus of inner diameter d and outer diameter D , $D_H = D - d$, for a rectangular duct of sides a, b , $D_H = ab/[2(a + b)]$. The **hydraulic radius** R_H is defined as **one-fourth** of the hydraulic diameter.

With the hydraulic diameter substituted for D in f and Re , Eqs. (6-37) through (6-40) are good approximations. Note that V appearing in f and Re is the actual average velocity $V = Q/A$; for noncircular pipes; it is **not** $Q/(\pi D_H^2/4)$. The pressure drop should be calculated from the friction factor for noncircular pipes. Equations relating Q to ΔP and D for circular pipes **may not be used** for noncircular pipes with D replaced by D_H because $V \neq Q/(\pi D_H^2/4)$.

Turbulent flow in noncircular channels is generally accompanied by secondary flows perpendicular to the axial flow direction (Schlichting). These flows may cause the pressure drop to be slightly greater than that computed using the hydraulic diameter method. For data on pressure drop in annuli, see Brighton and Jones (*J. Basic Eng.*, **86**, 835–842 [1964]); Okiishi and Serovy (*J. Basic Eng.*, **89**, 823–836 [1967]); and Lawn and Elliot (*J. Mech. Eng. Sci.*, **14**, 195–204 [1972]). For rectangular ducts of large aspect ratio, Dean (*J. Fluids Mech.*, **100**, 215–233 [1978]) found that the numerator of the exponent in the Blasius equation (6-37) should be increased to 0.0868. Jones (*J. Fluids Mech.*, **98**, 173–181 [1976]) presents a method to improve the estimation of friction factors for rectangular ducts using a modification of the hydraulic diameter-based Reynolds number.

The hydraulic diameter method does not work well for **laminar flow** because the shape affects the flow resistance in a way that cannot be expressed as a function only of the ratio of cross-sectional area to wetted perimeter. For some shapes, the Navier-Stokes equations have been integrated to yield relations between flow rate and pressure drop. These relations may be expressed in terms of **equivalent diameters** D_E defined to make the relations reduce to the second form of the Hagen-Poiseuille equation, Eq. (6-36); that is, $D_E \equiv$

$(128Q\mu L/\pi\Delta P)^{1/4}$. **Equivalent diameters are not the same as hydraulic diameters**. Equivalent diameters yield the correct relation between flow rate and pressure drop when substituted into Eq. (6-36), but not Eq. (6-35) because $V \neq Q/(\pi D_E^2/4)$. Equivalent diameter D_E is not to be used in the friction factor and Reynolds number; $f \neq 16/\text{Re}$ using the equivalent diameters defined in the following. This situation is, by arbitrary definition, opposite to that for the hydraulic diameter D_H used for turbulent flow.

Ellipse, semiaxes a and b (Lamb, *Hydrodynamics*, 6th ed., Dover, New York, 1945, p. 587):

$$D_E = \left(\frac{32a^3b^3}{a^2 + b^2} \right)^{1/4} \quad (6-50)$$

Rectangle, width a , height b (Owen, *Trans. Am. Soc. Civ. Eng.*, **119**, 1157–1175 [1954]):

$$D_E = \left(\frac{128ab^3}{\pi K} \right)^{1/4} \quad (6-51)$$

$a/b =$	1	1.5	2	3	4	5	10	∞
$K =$	28.45	20.43	17.49	15.19	14.24	13.73	12.81	12

Annulus, inner diameter D_1 , outer diameter D_2 (Lamb, op. cit., p. 587):

$$D_E = \left\{ (D_2^2 - D_1^2) \left[D_2^2 + D_1^2 - \frac{D_2^2 - D_1^2}{\ln(D_2/D_1)} \right] \right\}^{1/4} \quad (6-52)$$

For isosceles triangles and regular polygons, see Sparrow (*AIChE J.*, **8**, 599–605 [1962]), Carlson and Irvine (*J. Heat Transfer*, **83**, 441–444 [1961]), Cheng (*Proc. Third Int. Heat Transfer Conf.*, New York, **1**, 64–76 [1966]), and Shih (*Can. J. Chem. Eng.*, **45**, 285–294 [1967]).

The critical Reynolds number for **transition from laminar to turbulent flow** in noncircular channels varies with channel shape. In rectangular ducts, $1,900 < \text{Re}_c < 2,800$ (Hanks and Ruo, *Ind. Eng. Chem. Fundam.*, **5**, 558–561 [1966]). In triangular ducts, $1,600 < \text{Re}_c < 1,800$ (Cope and Hanks, *Ind. Eng. Chem. Fundam.*, **11**, 106–117 [1972]; Bandopadhyay and Hinwood, *J. Fluid Mech.*, **59**, 775–783 [1973]).

Nonisothermal Flow For nonisothermal flow of **liquids**, the friction factor may be increased if the liquid is being cooled or decreased if the liquid is being heated, because of the effect of temperature on viscosity near the wall. In shell and tube heat-exchanger design, the recommended practice is to first estimate f using the bulk mean liquid temperature over the tube length. Then, in laminar flow, the result is divided by $(\mu_w/\mu_e)^{0.23}$ in the case of cooling or $(\mu_w/\mu_e)^{0.38}$ in the case of heating. For turbulent flow, f is divided by $(\mu_w/\mu_e)^{0.11}$ in the case of cooling or $(\mu_w/\mu_e)^{0.17}$ in case of heating. Here, μ_w is the viscosity at the average bulk temperature and μ_e is the viscosity at the average wall temperature (Seider and Tate, *Ind. Eng. Chem.*, **28**, 1429–1435 [1936]). In the case of rough commercial pipes, rather than heat-exchanger tubing, it is common for flow to be in the "complete" turbulence regime where f is independent of Re . In such cases, the friction factor should not be corrected for wall temperature. If the liquid density varies with temperature, the average bulk density should be used to calculate the pressure drop from the friction factor. In addition, a (usually small) correction may be applied for acceleration effects by adding the term $G^2[(1/\rho_2) - (1/\rho_1)]$ from the mechanical energy balance to the pressure drop $\Delta P = P_1 - P_2$, where G is the mass velocity. This acceleration results from small compressibility effects associated with temperature-dependent density. Christiansen and Gordon (*AIChE J.*, **15**, 504–507 [1969]) present equations and charts for frictional loss in laminar nonisothermal flow of Newtonian and non-Newtonian liquids heated or cooled with constant wall temperature.

Frictional dissipation of mechanical energy can result in significant heating of fluids, particularly for very viscous liquids in small channels. Under adiabatic conditions, the bulk liquid temperature rise is given by $\Delta T = \Delta P/C_p \rho$ for incompressible flow through a channel of constant cross-sectional area. For flow of polymers, this amounts to about 4°C per 10 MPa pressure drop, while for hydrocarbon liquids it is about

6°C per 10 MPa. The temperature rise in laminar flow is highly nonuniform, being concentrated near the pipe wall where most of the dissipation occurs. This may result in significant viscosity reduction near the wall, and greatly increased flow or reduced pressure drop, and a flattened velocity profile. Compensation should generally be made for the heat effect when ΔP exceeds 1.4 MPa (203 psi) for adiabatic walls or 3.5 MPa (508 psi) for isothermal walls (Gerard, Steidler, and Appeldoorn, *Ind. Eng. Chem. Fundam.*, **4**, 332–339 [1969]).

Open Channel Flow For flow in **open channels**, the data are largely based on experiments with water in turbulent flow, in channels of sufficient roughness that there is no Reynolds number effect. The hydraulic radius approach may be used to estimate a friction factor with which to compute friction losses. Under conditions of **uniform flow** where liquid depth and cross-sectional area do not vary significantly with position in the flow direction, there is a balance between gravitational forces and wall stress, or equivalently between frictional losses and potential energy change. The mechanical energy balance reduces to $l_v = g(z_1 - z_2)$. In terms of the friction factor and hydraulic diameter or hydraulic radius,

$$l_v = \frac{2fV^2L}{D_H} = \frac{fV^2L}{2R_H} = g(z_1 - z_2) \quad (6-53)$$

The hydraulic radius is the cross-sectional area divided by the wetted perimeter, where the wetted perimeter *does not include the free surface*. Letting $S = \sin \theta$ = channel slope (elevation loss per unit length of channel, θ = angle between channel and horizontal), Eq. (6-53) reduces to

$$V = \sqrt{\frac{2gSR_H}{f}} \quad (6-54)$$

The most often used friction correlation for open channel flows is due to Manning (*Trans. Inst. Civ. Engrs. Ireland*, **20**, 161 [1891]) and is equivalent to

$$f = \frac{29n^2}{R_H^{1/3}} \quad (6-55)$$

where n is the channel roughness, with dimensions of (length)^{1/6}. Table 6-2 gives roughness values for several channel types.

For gradual changes in channel cross section and liquid depth, and for slopes less than 10°, the momentum equation for a rectangular channel of width b and liquid depth h may be written as a differential equation in the flow direction x .

$$\frac{dh}{dx} (1 - Fr) - Fr \left(\frac{h}{b} \right) \frac{db}{dx} = S - \frac{fV^2(b + 2h)}{2gbh} \quad (6-56)$$

For a given fixed flow rate $Q = Vbh$, and channel width profile $b(x)$, Eq. (6-56) may be integrated to determine the liquid depth profile

TABLE 6-2 Average Values of n for Manning Formula, Eq. (6-55)

Surface	n , m ^{1/6}	n , ft ^{1/6}
Cast-iron pipe, fair condition	0.014	0.011
Riveted steel pipe	0.017	0.014
Vitrified sewer pipe	0.013	0.011
Concrete pipe	0.015	0.012
Wood-stave pipe	0.012	0.010
Planed-plank flume	0.012	0.010
Semicircular metal flumes, smooth	0.013	0.011
Semicircular metal flumes, corrugated	0.028	0.023
Canals and ditches		
Earth, straight and uniform	0.023	0.019
Winding sluggish canals	0.025	0.021
Dredged earth channels	0.028	0.023
Natural-stream channels		
Clean, straight bank, full stage	0.030	0.025
Winding, some pools and shoals	0.040	0.033
Same, but with stony sections	0.055	0.045
Sluggish reaches, very deep pools, rather weedy	0.070	0.057

SOURCE: Brater and King, *Handbook of Hydraulics*, 6th ed., McGraw-Hill, New York, 1976, p. 7-22. For detailed information, see Chow, *Open-Channel Hydraulics*, McGraw-Hill, New York, 1959, pp. 110–123.

$h(x)$. The dimensionless Froude number is $Fr = V^2/gh$. When $Fr = 1$, the flow is **critical**, when $Fr < 1$, the flow is **subcritical**, and when $Fr > 1$, the flow is **supercritical**. Surface disturbances move at a wave velocity $c = \sqrt{gh}$; they cannot propagate upstream in supercritical flows. The **specific energy** E_{sp} is nearly constant.

$$E_{sp} = h + \frac{V^2}{2g} \quad (6-57)$$

This equation is cubic in liquid depth. Below a minimum value of E_{sp} there are no real positive roots; above the minimum value there are two positive real roots. At this minimum value of E_{sp} the flow is critical; that is, $Fr = 1$, $V = \sqrt{gh}$, and $E_{sp} = (3/2)h$. Near critical flow conditions, wave motion and sudden depth changes called **hydraulic jumps** are likely. Chow (*Open Channel Hydraulics*, McGraw-Hill, New York, 1959) discusses the numerous surface profile shapes which may exist in nonuniform open channel flows.

For flow over a sharp-crested weir of width b and height L , from a liquid depth H , the flow rate is given approximately by

$$Q = \frac{2}{3} C_d b \sqrt{2g} (H - L)^{3/2} \quad (6-58)$$

where $C_d \approx 0.6$ is a discharge coefficient. Flow through notched weirs is described under flow meters in Sec. 10 of the *Handbook*.

Non-Newtonian Flow For **isothermal laminar flow** of time-independent non-Newtonian liquids, integration of the Cauchy momentum equations yields the fully developed velocity profile and flow rate–pressure drop relations. For the **Bingham plastic** fluid described by Eq. (6-3), in a pipe of diameter D and a pressure drop per unit length $\Delta P/L$, the flow rate is given by

$$Q = \frac{\pi D^3 \tau_w}{32\mu_w} \left[1 - \frac{4\tau_y}{3\tau_w} + \frac{\tau_y^4}{3\tau_w^4} \right] \quad (6-59)$$

where the wall stress is $\tau_w = D\Delta P/(4L)$. The velocity profile consists of a central nondeforming plug of radius $r_p = 2\tau_y/(\Delta P/L)$ and an annular deforming region. The velocity profile in the annular region is given by

$$v_z = \frac{1}{\mu_w} \left[\frac{\Delta P}{4L} (R^2 - r^2) - \tau_y (R - r) \right] \quad r_p \leq r \leq R \quad (6-60)$$

where r is the radial coordinate and R is the pipe radius. The velocity of the central, nondeforming plug is obtained by setting $r = r_p$ in Eq. (6-60). When Q is given and Eq. (6-59) is to be solved for τ_w and the pressure drop, multiple positive roots for the pressure drop may be found. The root corresponding to $\tau_w < \tau_y$ is physically unrealizable, as it corresponds to $r_p > R$ and the pressure drop is insufficient to overcome the yield stress.

For a **power law fluid**, Eq. (6-4), with constant properties K and n , the flow rate is given by

$$Q = \pi \left(\frac{\Delta P}{2KL} \right)^{1/n} \left(\frac{n}{1 + 3n} \right) R^{(1 + 3n)/n} \quad (6-61)$$

and the velocity profile by

$$v_z = \left(\frac{\Delta P}{2KL} \right)^{1/n} \left(\frac{n}{1 + n} \right) [R^{(1 + n)/n} - r^{(1 + n)/n}] \quad (6-62)$$

Similar relations for other non-Newtonian fluids may be found in Govier and Aziz and in Bird, Armstrong, and Hassager (*Dynamics of Polymeric Liquids*, vol. 1: *Fluid Mechanics*, Wiley, New York, 1977).

For steady-state laminar flow of any time-independent viscous fluid, at average velocity V in a pipe of diameter D , the Rabinowitsch-Mooney relations give a general relationship for the shear rate at the pipe wall.

$$\dot{\gamma}_w = \frac{8V}{D} \left(\frac{1 + 3n'}{4n'} \right) \quad (6-63)$$

where n' is the slope of a plot of $D\Delta P/(4L)$ versus $8V/D$ on logarithmic coordinates,

$$n' = \frac{d \ln [D\Delta P/(4L)]}{d \ln (8V/D)} \quad (6-64)$$

By plotting capillary viscometry data this way, they can be used directly for pressure drop design calculations, or to construct the rheogram for the fluid. For pressure drop calculation, the flow rate and diameter determine the velocity, from which $8V/D$ is calculated and $D\Delta P/(4L)$ read from the plot. For a Newtonian fluid, $n' = 1$ and the shear rate at the wall is $\dot{\gamma}_w = 8V/D$. For a power law fluid, $n' = n$. To construct a rheogram, n' is obtained from the slope of the experimental plot at a given value of $8V/D$. The shear rate at the wall is given by Eq. (6-63) and the corresponding shear stress at the wall is $\tau_w = D\Delta P/(4L)$ read from the plot. By varying the value of $8V/D$, the shear rate versus shear stress plot can be constructed.

The generalized approach of Metzner and Reed (*AIChE J.*, **1**, 434 [1955]) for time-independent non-Newtonian fluids defines a modified Reynolds number as

$$Re_{MR} \equiv \frac{D^n V^{2-n'} \rho}{K' 8^{n'-1}} \quad (6-65)$$

where K' satisfies

$$\frac{D\Delta P}{4L} = K' \left(\frac{8V}{D} \right)^{n'} \quad (6-66)$$

With this definition, $f = 16/Re_{MR}$ is automatically satisfied at the value of $8V/D$ where K' and n' are evaluated. Equation (6-66) may be obtained by integration of Eq. (6-64) only when n' is a constant, as, for example, the cases of Newtonian and power law fluids. For Newtonian fluids, $K' = \mu$ and $n' = 1$; for power law fluids, $K' = K[(1 + 3n)/(4n)]^n$ and $n' = n$. For Bingham plastics, K' and n' are variable, given as a function of τ_w (Metzner, *Ind. Eng. Chem.*, **49**, 1429–1432 [1957]).

$$K = \tau_w^{1-n'} \left[\frac{\mu_\infty}{1 - 4\tau_y/3\tau_w + (\tau_y/\tau_w)^{4/3}} \right]^{n'} \quad (6-67)$$

$$n' = \frac{1 - 4\tau_y/(3\tau_w) + (\tau_y/\tau_w)^{4/3}}{1 - (\tau_y/\tau_w)^{4/3}} \quad (6-68)$$

For laminar flow of power law fluids in channels of noncircular cross section, see Schecter (*AIChE J.*, **7**, 445–448 [1961]), Wheeler and Wissler (*AIChE J.*, **11**, 207–212 [1965]), Bird, Armstrong, and Hassager (*Dynamics of Polymeric Liquids*, vol. 1: *Fluid Mechanics*, Wiley, New York, 1977), and Skelland (*Non-Newtonian Flow and Heat Transfer*, Wiley, New York, 1967).

Steady-state, fully developed laminar flows of viscoelastic fluids in straight, constant-diameter pipes show no effects of viscoelasticity. The viscous component of the constitutive equation may be used to develop the flow rate–pressure drop relations, which apply downstream of the entrance region after viscoelastic effects have disappeared. A similar situation exists for time-dependent fluids.

The **transition to turbulent flow** begins at Re_{MR} in the range of 2,000 to 2,500 (Metzner and Reed, *AIChE J.*, **1**, 434 [1955]). For Bingham plastic materials, K' and n' must be evaluated for the τ_w condition in question in order to determine Re_{MR} and establish whether the flow is laminar. An alternative method for Bingham plastics is by Hanks (Hanks, *AIChE J.*, **9**, 306 [1963]; **14**, 691 [1968]; Hanks and Pratt, *Soc. Petrol. Engrs. J.*, **7**, 342 [1967]; and Govier and Aziz, pp. 213–215). The transition from laminar to turbulent flow is influenced by **viscoelastic** properties (Metzner and Park, *J. Fluid Mech.*, **20**, 291 [1964]) with the critical value of Re_{MR} increased to beyond 10,000 for some materials.

For **turbulent flow of non-Newtonian fluids**, the design chart of Dodge and Metzner (*AIChE J.*, **5**, 189 [1959]), Fig. 6-11, is most widely used. For Bingham plastic materials in turbulent flow, it is generally assumed that stresses greatly exceed the yield stress, so that the friction factor–Reynolds number relationship for Newtonian fluids applies, with μ_∞ substituted for μ . This is equivalent to setting $n' = 1$ and $\tau_y/\tau_w = 0$ in the Dodge-Metzner method, so that $Re_{MR} = DV\rho/\mu_\infty$. Wilson and Thomas (*Can. J. Chem. Eng.*, **63**, 539–546 [1985]) give friction factor equations for turbulent flow of power law fluids and Bingham plastic fluids.

Power law fluids:

$$\frac{1}{\sqrt{f}} = \frac{1}{\sqrt{f_N}} + 8.2 \frac{1-n}{1+n} + 1.77 \ln \left(\frac{1+n}{2} \right) \quad (6-69)$$

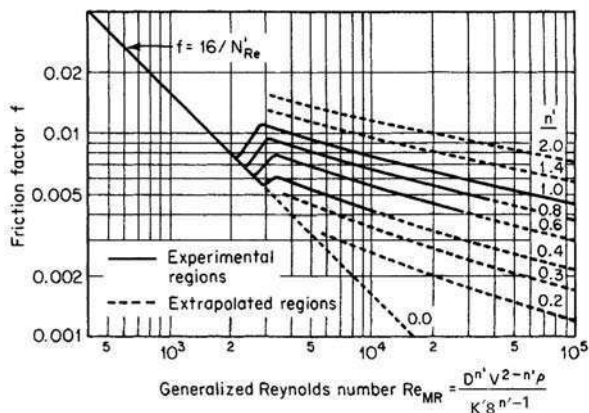


FIG. 6-11 Fanning friction factor for non-Newtonian flow. The abscissa is defined in Eq. (6-65). (From Dodge and Metzner, *Am. Inst. Chem. Eng. J.*, **5**, 189 [1959].)

where f_N is the friction factor for Newtonian fluid evaluated at $Re = DV\rho/\mu_{\text{eff}}$ where the effective viscosity is

$$\mu_{\text{eff}} = K \left(\frac{3n+1}{4n} \right)^{n-1} \left(\frac{8V}{D} \right)^{n-1} \quad (6-70)$$

Bingham fluids:

$$\frac{1}{\sqrt{f}} = \frac{1}{\sqrt{f_N}} + 1.77 \ln \left(\frac{(1-\xi)^2}{1+\xi} \right) + \xi(10 + 0.884\xi) \quad (6-71)$$

where f_N is evaluated at $Re = DV\rho/\mu_\infty$ and $\xi = \tau_y/\tau_w$. Iteration is required to use this equation since $\tau_w = f\rho V^2/2$.

Drag reduction in turbulent flow can be achieved by adding soluble high molecular weight polymers in extremely low concentration to Newtonian liquids. The reduction in friction is generally believed to be associated with the viscoelastic nature of the solutions effective in the wall region. For a given polymer, there is a minimum molecular weight necessary to initiate drag reduction at a given flow rate, and a critical concentration above which drag reduction will not occur (Kim, Little, and Ting, *J. Colloid Interface Sci.*, **47**, 530–535 [1974]). Drag reduction is reviewed by Hoyt (*J. Basic Eng.*, **94**, 258–285 [1972]); Little, et al. (*Ind. Eng. Chem. Fundam.*, **14**, 283–296 [1975]) and Virk (*AIChE J.*, **21**, 625–656 [1975]). At maximum possible drag reduction in smooth pipes,

$$\frac{1}{\sqrt{f}} = -19 \log \left(\frac{50.73}{Re\sqrt{f}} \right) \quad (6-72)$$

or, approximately,
$$f = \frac{0.58}{Re^{0.58}} \quad (6-73)$$

for 4,000 < Re < 40,000. The actual drag reduction depends on the polymer system. For further details, see Virk (*ibid.*).

Economic Pipe Diameter, Turbulent Flow The economic optimum pipe diameter may be computed so that the last increment of investment reduces the operating cost enough to produce the required minimum return on investment. For long cross-country pipelines, alloy pipes of appreciable length and complexity, or pipelines with control valves, detailed analyses of investment and operating costs should be made. Peters and Timmerhaus (*Plant Design and Economics for Chemical Engineers*, 4th ed., McGraw-Hill, New York, 1991) provide a detailed method for determining the economic optimum size. For pipelines of the lengths usually encountered in chemical plants and petroleum refineries, simplified selection charts are often adequate. In many cases there is an economic optimum velocity that is nearly independent of diameter, which may be used to estimate the economic diameter from the flow rate. For low-viscosity liquids in schedule 40 steel pipe, economic optimum velocity is typically in the range of 1.8 to 2.4 m/s (5.9 to 7.9 ft/s). For gases with density ranging

from 0.2 to 20 kg/m³ (0.013 to 1.25 lbm/ft³), the economic optimum velocity is about 40 m/s to 9 m/s (131 to 30 ft/s). Charts and rough guidelines for economic optimum size do not apply to multiphase flows.

Economic Pipe Diameter, Laminar Flow Pipelines for the transport of high-viscosity liquids are seldom designed purely on the basis of economics. More often, the size is dictated by operability considerations such as available pressure drop, shear rate, or residence time distribution. Peters and Timmerhaus (ibid., Chap. 10) provide an economic pipe diameter chart for laminar flow. For non-Newtonian fluids, see Skelland (*Non-Newtonian Flow and Heat Transfer*, Chap. 7, Wiley, New York, 1967).

Vacuum Flow When gas flows under high vacuum conditions or through very small openings, the continuum hypothesis is no longer appropriate if the channel dimension is not very large compared to the mean free path of the gas. When the mean free path is comparable to the channel dimension, flow is dominated by collisions of molecules with the wall, rather than by collisions between molecules. An approximate expression based on Brown, et al. (*J. Appl. Phys.*, **17**, 802–813 [1946]) for the mean free path is

$$\lambda = \left(\frac{2\mu}{p} \right) \sqrt{\frac{8RT}{\pi M_w}} \quad (6-74)$$

The Knudsen number Kn is the ratio of the mean free path to the channel dimension. For pipe flow, $Kn = \lambda/D$. **Molecular flow** is characterized by $Kn > 1.0$; continuum viscous (laminar or turbulent) flow is characterized by $Kn < 0.01$. **Transition or slip** flow applies over the range $0.01 < Kn < 1.0$.

Vacuum flow is usually described with flow variables different from those used for normal pressures, which often leads to confusion. **Pumping speed** S is the actual volumetric flow rate of gas through a flow cross section. **Throughput** Q is the product of pumping speed and absolute pressure. In the SI system, Q has units of Pa·m³/s.

$$Q = Sp \quad (6-75)$$

The mass flow rate w is related to the throughput using the ideal gas law.

$$w = \frac{M_w}{RT} Q \quad (6-76)$$

Throughput is therefore proportional to mass flow rate. For a given mass flow rate, throughput is independent of pressure. The relation between throughput and pressure drop $\Delta p = p_1 - p_2$ across a flow element is written in terms of the **conductance** C . **Resistance** is the reciprocal of conductance. Conductance has dimensions of volume per time.

$$Q = C\Delta p \quad (6-77)$$

The conductance of a series of flow elements is given by

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots \quad (6-78)$$

while for elements in parallel,

$$C = C_1 + C_2 + C_3 + \cdots \quad (6-79)$$

For a vacuum pump of speed S_p withdrawing from a vacuum vessel through a connecting line of conductance C , the pumping speed at the vessel is

$$S = \frac{S_p C}{S_p + C} \quad (6-80)$$

Molecular Flow Under molecular flow conditions, conductance is independent of pressure. It is proportional to $\sqrt{T/M_w}$, with the proportionality constant a function of geometry. For fully developed pipe flow,

$$C = \frac{\pi D^3}{8L} \sqrt{\frac{RT}{M_w}} \quad (6-81)$$

For an orifice of area A ,

$$C = 0.40A \sqrt{\frac{RT}{M_w}} \quad (6-82)$$

TABLE 6-3 Constants for Circular Annuli

D_2/D_1	K	D_2/D_1	K
0	1.00	0.707	1.254
0.259	1.072	0.866	1.430
0.500	1.154	0.966	1.675

Conductance equations for several other geometries are given by Ryans and Roper (*Process Vacuum System Design and Operation*, Chap. 2, McGraw-Hill, New York, 1986). For a circular annulus of outer and inner diameters D_1 and D_2 and length L , the method of Guthrie and Wakerling (*Vacuum Equipment and Techniques*, McGraw-Hill, New York, 1949) may be written

$$C = 0.42K \frac{(D_1 - D_2)^2(D_1 + D_2)}{L} \sqrt{\frac{RT}{M_w}} \quad (6-83)$$

where K is a dimensionless constant with values given in Table 6-3.

For a short pipe of circular cross section, the conductance as calculated for an orifice from Eq. (6-82) is multiplied by a correction factor K which may be approximated as (Kennard, *Kinetic Theory of Gases*, McGraw-Hill, New York, 1938, pp. 306–308)

$$K = \frac{1}{1 + (L/D)} \quad \text{for} \quad 0 \leq L/D \leq 0.75 \quad (6-84)$$

$$K = \frac{1 + 0.8(L/D)}{1 + 1.90(L/D) + 0.6(L/D)^2} \quad \text{for} \quad L/D > 0.75 \quad (6-85)$$

For $L/D > 100$, the error in neglecting the end correction by using the fully developed pipe flow equation (6-81) is less than 2 percent. For rectangular channels, see Normand (*Ind. Eng. Chem.*, **40**, 783–787 [1948]).

Yu and Sparrow (*J. Basic Eng.*, **70**, 405–410 [1970]) give a theoretically derived chart for slot seals with or without a sheet located in or passing through the seal, giving mass flow rate as a function of the ratio of seal plate thickness to gap opening.

Slip Flow In the transition region between molecular flow and continuum viscous flow, the conductance for fully developed pipe flow is most easily obtained by the method of Brown, et al. (*J. Appl. Phys.*, **17**, 802–813 [1946]), which uses the parameter

$$X = \sqrt{\frac{8}{\pi}} \left(\frac{\lambda}{D} \right) = \left(\frac{2\mu}{p_m D} \right) \sqrt{\frac{RT}{M}} \quad (6-86)$$

where p_m is the arithmetic mean absolute pressure. A correction factor F , read from Fig. 6-12 as a function of X , is applied to the conductance

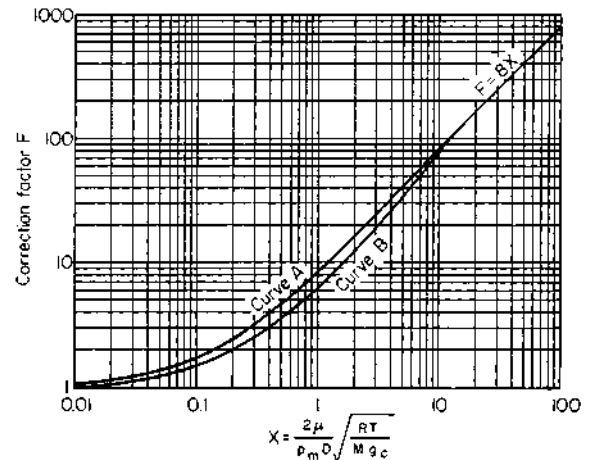


FIG. 6-12 Correction factor for Poiseuille's equation at low pressures. Curve A: experimental curve for glass capillaries and smooth metal tubes. (From Brown, et al., *J. Appl. Phys.*, **17**, 802 [1946].) Curve B: experimental curve for iron pipe (From Riggle, courtesy of E. I. du Pont de Nemours & Co.)

for viscous flow.

$$C = F \frac{\pi D^4 p_m}{128 \mu L} \quad (6-87)$$

For slip flow through **square channels**, see Milligan and Wilkerson (*J. Eng. Ind.*, **95**, 370–372 [1973]). For slip flow through **annuli**, see Maegley and Berman (*Phys. Fluids*, **15**, 780–785 [1972]).

The **pump-down time** θ for evacuating a vessel in the absence of air in-leakage is given approximately by

$$\theta = \left(\frac{V_t}{S_0} \right) \ln \left(\frac{p_1 - p_0}{p_2 - p_0} \right) \quad (6-88)$$

where V_t = volume of vessel plus volume of piping between vessel and pump; S_0 = system speed as given by Eq. (6-80), assumed independent of pressure; p_1 = initial vessel pressure; p_2 = final vessel pressure; and p_0 = lowest pump intake pressure attainable with the pump in question. See Dushman and Lafferty (*Scientific Foundations of Vacuum Technique*, 2d ed., Wiley, New York, 1962).

The amount of inerts which has to be removed by a pumping system after the pump-down stage depends on the in-leakage of air at the various fittings, connections, and so on. Air leakage is often correlated with system volume and pressure, but this approach introduces uncertainty because the number and size of leaks does not necessarily correlate with system volume, and leakage is sensitive to maintenance quality. Ryans and Roper (*Process Vacuum System Design and Operation*, McGraw-Hill, New York, 1986) present a thorough discussion of air leakage.

FRICIONAL LOSSES IN PIPELINE ELEMENTS

The viscous or frictional loss term in the mechanical energy balance for most cases is obtained experimentally. For many common fittings found in piping systems, such as expansions, contractions, elbows, and valves, data are available to estimate the losses. Substitution into the energy balance then allows calculation of pressure drop. A common error is to assume that pressure drop and frictional losses are equivalent. Equation (6-16) shows that in addition to frictional losses, other factors such as shaft work and velocity or elevation change influence pressure drop.

Losses l_v for incompressible flow in sections of straight pipe of constant diameter may be calculated as previously described using the Fanning friction factor:

$$l_v = \frac{\Delta P}{\rho} = \frac{2fV^2L}{D} \quad (6-89)$$

where ΔP = drop in equivalent pressure, $P = p + \rho gz$, with p = pressure, ρ = fluid density, g = acceleration of gravity, and z = elevation. Losses in the fittings of a piping network are frequently termed *minor losses* or *miscellaneous losses*. These descriptions are misleading because in process piping fitting losses are often much greater than the losses in straight piping sections.

Equivalent Length and Velocity Head Methods Two methods are in common use for estimating fitting loss. One, the **equivalent length** method, reports the losses in a piping element as the length of straight pipe which would have the same loss. For turbulent flows, the equivalent length is usually reported as a number of diameters of pipe of the same size as the fitting connection; L_e/D is given as

a fixed quantity, independent of D . This approach tends to be most accurate for a single fitting size and loses accuracy with deviation from this size. For laminar flows, L_e/D correlations normally have a size dependence through a Reynolds number term.

The other method is the **velocity head** method. The term $V^2/2g$ has dimensions of length and is commonly called a *velocity head*. Application of the Bernoulli equation to the problem of frictionless discharge at velocity V through a nozzle at the bottom of a column of liquid of height H shows that $H = V^2/2g$. Thus H is the liquid head corresponding to the velocity V . Use of the velocity head to scale pressure drops has wide application in fluid mechanics. Examination of the Navier-Stokes equations suggests that when the inertial terms dominate the viscous terms, pressure gradients are expected to be proportional to ρV^2 where V is a characteristic velocity of the flow.

In the velocity head method, the losses are reported as a number of velocity heads K . Then, the engineering Bernoulli equation for an incompressible fluid can be written

$$p_1 - p_2 = \alpha_2 \frac{\rho V_2^2}{2} - \alpha_1 \frac{\rho V_1^2}{2} + \rho g(z_2 - z_1) + K \frac{\rho V^2}{2} \quad (6-90)$$

where V is the reference velocity upon which the velocity head loss coefficient K is based. For a section of straight pipe, $K = 4fL/D$.

Contraction and Entrance Losses For a **sudden contraction** at a sharp-edged entrance to a pipe or sudden reduction in cross-sectional area of a channel, as shown in Fig. 6-13a, the loss coefficient based on the downstream velocity V_2 is given for **turbulent flow** in Crane Co. Tech Paper 410 (1980) approximately by

$$K = 0.5 \left(1 - \frac{A_2}{A_1} \right) \quad (6-91)$$

Example 5: Entrance Loss Water, $\rho = 1,000 \text{ kg/m}^3$, flows from a large vessel through a sharp-edged entrance into a pipe at a velocity in the pipe of 2 m/s. The flow is turbulent. Estimate the pressure drop from the vessel into the pipe.

With $A_2/A_1 \rightarrow 0$, the viscous loss coefficient is $K = 0.5$ from Eq. (6-91). The mechanical energy balance, Eq. (6-16) with $V_1 = 0$ and $z_2 - z_1 = 0$ and assuming uniform flow ($\alpha_2 = 1$) becomes

$$p_1 - p_2 = \frac{\rho V_2^2}{2} + 0.5 \frac{\rho V_2^2}{2} = 4,000 + 2,000 = 6,000 \text{ Pa}$$

Note that the total pressure drop consists of 0.5 velocity heads of frictional loss contribution, and 1 velocity head of velocity change contribution. The frictional contribution is a permanent loss of mechanical energy by viscous dissipation. The acceleration contribution is reversible; if the fluid were subsequently decelerated in a frictionless diffuser, a 4,000 Pa pressure rise would occur.

For a **trumpet-shaped** rounded entrance, with a radius of rounding greater than about 15 percent of the pipe diameter (Fig. 6-13b), the turbulent flow loss coefficient K is only about 0.1 (Vennard and Street, *Elementary Fluid Mechanics*, 5th ed., Wiley, New York, 1975, pp. 420–421). Rounding of the inlet prevents formation of the **vena contracta**, thereby reducing the resistance to flow.

For **laminar flow** the losses in sudden contraction may be estimated for area ratios $A_2/A_1 < 0.2$ by an equivalent additional pipe length L_e given by

$$L_e/D = 0.3 + 0.04\text{Re} \quad (6-92)$$

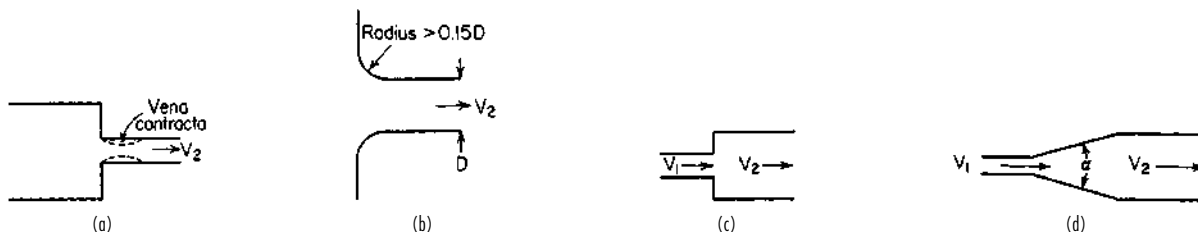


FIG. 6-13 Contractions and enlargements: (a) sudden contraction, (b) rounded contraction, (c) sudden enlargement, and (d) uniformly diverging duct.

where D is the diameter of the smaller pipe and Re is the Reynolds number in the smaller pipe. For laminar flow in the entrance to rectangular ducts, see Shah (*J. Fluids Eng.*, **100**, 177–179 [1978]) and Roscoe (*Philos. Mag.*, **40**, 338–351 [1949]). For creeping flow, $Re < 1$, of power law fluids, the entrance loss is approximately $L_e/D = 0.3/n$ (Boger, Gupta, and Tanner, *J. Non-Newtonian Fluid Mech.*, **4**, 239–248 [1978]). For viscoelastic fluid flow in circular channels with sudden contraction, a toroidal vortex forms upstream of the contraction plane. Such flows are reviewed by Boger (*Ann. Review Fluid Mech.*, **19**, 157–182 [1987]).

For creeping flow through **conical converging channels**, inertial acceleration terms are negligible and the viscous pressure drop $\Delta p = p_l$ may be computed by integration of the differential form of the Hagen-Poiseuille equation Eq. (6-36), provided the angle of convergence is small. The result for a power law fluid is

$$\Delta p = 4K \left(\frac{3n+1}{4n} \right)^n \left(\frac{8V_2}{D_2} \right)^n \left\{ \frac{1}{6n \tan(\alpha/2)} \left[1 - \left(\frac{D_2}{D_1} \right)^{3n} \right] \right\} \quad (6-93)$$

where D_1 = inlet diameter
 D_2 = exit diameter
 V_2 = velocity at the exit
 α = total included angle

Equation (6-93) agrees with experimental data (Kemblowski and Kiljanski, *Chem. Eng. J. (Lausanne)*, **9**, 141–151 [1975]) for $\alpha < 11^\circ$. For Newtonian liquids, Eq. (6-93) simplifies to

$$\Delta p = \mu \left(\frac{32V_2}{D_2} \right) \left\{ \frac{1}{6 \tan(\alpha/2)} \left[1 - \left(\frac{D_2}{D_1} \right)^3 \right] \right\} \quad (6-94)$$

For creeping flow through noncircular converging channels, the differential form of the Hagen-Poiseuille equation with equivalent diameter given by Eqs. (6-50) to (6-52) may be used, provided the convergence is gradual.

Expansion and Exit Losses For ducts of any cross section, the frictional loss for a **sudden enlargement** (Fig. 6-13c) with turbulent flow is given by the Borda-Carnot equation:

$$l_v = \frac{V_1^2 - V_2^2}{2} = \frac{V_1^2}{2} \left(1 - \frac{A_1}{A_2} \right)^2 \quad (6-95)$$

where V_1 = velocity in the smaller duct
 V_2 = velocity in the larger duct
 A_1 = cross-sectional area of the smaller duct
 A_2 = cross-sectional area of the larger duct

Equation (6-95) is valid for incompressible flow. For compressible flows, see Benedict, Wyler, Dudek, and Glead (*J. Eng. Power*, **98**, 327–334 [1976]). For an infinite expansion, $A_1/A_2 = 0$, Eq. (6-95) shows that the **exit loss** from a pipe is 1 velocity head. This result is easily deduced from the mechanical energy balance Eq. (6-90), noting that $p_1 = p_2$. This exit loss is due to the dissipation of the discharged jet; there is no pressure drop at the exit.

For creeping Newtonian flow ($Re < 1$), the frictional loss due to a sudden enlargement should be obtained from the same equation for a sudden contraction (Eq. [6-92]). Note, however, that Boger, Gupta, and Tanner (ibid.) give an exit friction equivalent length of 0.12 diameter, increasing for power law fluids as the exponent decreases. For laminar flows at higher Reynolds numbers, the pressure drop is twice that given by Eq. (6-95). This results from the velocity profile factor α in the mechanical energy balance being 2.0 for the parabolic laminar velocity profile.

If the transition from a small to a large duct of any cross-sectional shape is accomplished by a **uniformly diverging duct** (see Fig. 6-13d) with a straight axis, the total frictional pressure drop can be computed by integrating the differential form of Eq. (6-89), $dl_v/dx = 2fV^2/D$ over the length of the expansion, provided the total angle α between the diverging walls is less than 7° . For angles between 7° and 45° , the loss coefficient may be estimated as 2.6 $\sin(\alpha/2)$ times the loss coefficient for a sudden expansion; see Hooper (*Chem. Eng.*, Nov. 7, 1988). Gibson (*Hydraulics and Its Applications*, 5th ed., Constable, London 1952, p. 93) recommends multiplying the sudden enlargement loss by 0.13 for $5^\circ < \alpha < 7.5^\circ$ and by $0.0110\alpha^{1.22}$ for $7.5^\circ < \alpha <$

35° . For angles greater than 35° to 45° , the losses are normally considered equal to those for a sudden expansion, although in some cases the losses may be greater. Expanding flow through standard pipe reducers should be treated as sudden expansions.

Trumpet-shaped enlargements for turbulent flow designed for constant decrease in velocity head per unit length were found by Gibson (ibid., p. 95) to give 20 to 60 percent less frictional loss than straight taper pipes of the same length.

A special feature of expansion flows occurs when **viscoelastic** liquids are extruded through a die at a low Reynolds number. The extrudate may expand to a diameter several times greater than the die diameter, whereas for a Newtonian fluid the diameter expands only 10 percent. This phenomenon, called **die swell**, is most pronounced with short dies (Graessley, Glasscock, and Crawley, *Trans. Soc. Rheol.*, **14**, 519–544 [1970]). For velocity distribution measurements near the die exit, see Goulden and MacSporran (*J. Non-Newtonian Fluid Mech.*, **1**, 183–198 [1976]) and Whipple and Hill (*AIChE J.*, **24**, 664–671 [1978]). At high flow rates, the extrudate becomes distorted, suffering **melt fracture** at wall shear stresses greater than 10^5 N/m². This phenomenon is reviewed by Denn (*Ann. Review Fluid Mech.*, **22**, 13–34 [1990]). Ramamurthy (*J. Rheol.*, **30**, 337–357 [1986]) has found a dependence of apparent stick-slip behavior in melt fracture to be dependent on the material of construction of the die.

Fittings and Valves For **turbulent flow**, the frictional loss for fittings and valves can be expressed by the equivalent length or velocity head methods. As fitting size is varied, K values are relatively more constant than L_e/D values, but since fittings generally do not achieve geometric similarity between sizes, K values tend to decrease with increasing fitting size. Table 6-4 gives K values for many types of fittings and valves.

Manufacturers of valves, especially control valves, express valve capacity in terms of a flow coefficient C_v , which gives the flow rate through the valve in gal/min of water at 60°F under a pressure drop of 1 lbf/in². It is related to K by

$$C_v = \frac{C_1 d^2}{\sqrt{K}} \quad (6-96)$$

where C_1 is a dimensional constant equal to 29.9 and d is the diameter of the valve connections in inches.

For **laminar flow**, data for the frictional loss of valves and fittings are meager (Beck and Miller, *J. Am. Soc. Nav. Eng.*, **56**, 62–83 [1944]; Beck, ibid., **56**, 235–271, 366–388, 389–395 [1944]; De Craene, *Heat. Piping Air Cond.*, **27**[10], 90–95 [1955]; Karr and Schutz, *J. Am. Soc. Nav. Eng.*, **52**, 239–256 [1940]; and Kittredge and Rowley, *Trans. ASME*, **79**, 1759–1766 [1957]). The data of Kittredge and Rowley indicate that K is constant for Reynolds numbers above 500 to 2,000, but increases rapidly as Re decreases below 500. Typical values for K for laminar flow Reynolds numbers are shown in Table 6-5.

Methods to calculate losses for **tee and wye junctions** for dividing and combining flow are given by Miller (*Internal Flow Systems*, 2d ed., Chap. 13, BHRA, Cranfield, 1990), including effects of Reynolds number, angle between legs, area ratio, and radius. Junctions with more than three legs are also discussed. The sources of data for the loss coefficient charts are Blaisdell and Manson (*U.S. Dept. Agric. Res. Serv. Tech. Bull.* 1283 [August 1963]) for combining flow and Gardel (*Bull. Tech. Suisses Romande*, **85**[9], 123–130 [1957]; **85**[10], 143–148 [1957]) together with additional unpublished data for dividing flow.

Miller (*Internal Flow Systems*, 2d ed., Chap. 13, BHRA, Cranfield, 1990) gives the most complete information on losses in **bends and curved pipes**. For turbulent flow in circular cross-section bends of constant area, as shown in Fig. 6-14a, a more accurate estimate of the loss coefficient K than that given in Table 6-4 is

$$K = K^* C_{Re} C_{of} \quad (6-97)$$

where K^* , given in Fig. 6-14b, is the loss coefficient for a smooth-walled bend at a Reynolds number of 10^6 . The Reynolds number correction factor C_{Re} is given in Fig. 6-14c. For $0.7 < r/D < 1$ or for $K^* < 0.4$, use the C_{Re} value for $r/D = 1$. Otherwise, if $r/D < 1$, obtain C_{Re} from

$$C_{Re} = \frac{K^*}{K^* + 0.2(1 - C_{Re, r/D=1})} \quad (6-98)$$

TABLE 6-4 Additional Frictional Loss for Turbulent Flow through Fittings and Valves^a

Type of fitting or valve	Additional friction loss, equivalent no. of velocity heads, K
45° ell, standard ^{b,c,d,e,f}	0.35
45° ell, long radius ^c	0.2
90° ell, standard ^{b,c,e,f,g,h}	0.75
Long radius ^{b,c,d,e}	0.45
Square or miter ^h	1.3
180° bend, close return ^{b,c,e}	1.5
Tee, standard, along run, branch blanked off ⁱ	0.4
Used as ell, entering run ^{d,i}	1.0
Used as ell, entering branch ^{c,g,i}	1.0
Branching flow ^{j,k}	1 ^l
Coupling ^{c,e}	0.04
Union ^c	0.04
Gate valve, ^{b,c,m} open	0.17
3/4 open	0.9
1/2 open	4.5
1/4 open	24.0
Diaphragm valve, open	2.3
3/4 open	2.6
1/2 open	4.3
1/4 open	21.0
Globe valve, ^{c,m}	
Bevel seat, open	6.0
1/2 open	9.5
Composition seat, open	6.0
1/2 open	8.5
Plug disk, open	9.0
3/4 open	13.0
1/2 open	36.0
1/4 open	112.0
Angle valve, ^{b,e} open	2.0
Y or blowoff valve, ^{b,m} open	3.0
Plug cock	
θ = 5°	0.05
θ = 10°	0.29
θ = 20°	1.56
θ = 40°	17.3
θ = 60°	206.0
Butterfly valve	
θ = 5°	0.24
θ = 10°	0.52
θ = 20°	1.54
θ = 40°	10.8
θ = 60°	118.0
Check valve, ^{b,e,m} swing	2.0
Disk	10.0
Ball	70.0
Foot valve ^c	15.0
Water meter, ^h disk	7.0
Piston	15.0
Rotary (star-shaped disk)	10.0
Turbine-wheel	6.0

^aLapple, *Chem. Eng.*, **56**(5), 96–104 (1949), general survey reference.^b"Flow of Fluids through Valves, Fittings, and Pipe," Tech. Pap. 410, Crane Co., 1969.^cFreeman, *Experiments upon the Flow of Water in Pipes and Pipe Fittings*, American Society of Mechanical Engineers, New York, 1941.^dGiesecke, *J. Am. Soc. Heat. Vent. Eng.*, **32**, 461 (1926).^e*Pipe Friction Manual*, 3d ed., Hydraulic Institute, New York, 1961.^fIto, *J. Basic Eng.*, **82**, 131–143 (1960).^gGiesecke and Badgett, *Heat. Piping Air Cond.*, **4**(6), 443–447 (1932).^hSchoder and Dawson, *Hydraulics*, 2d ed., McGraw-Hill, New York, 1934, p. 213.ⁱHoopes, Isakoff, Clarke, and Drew, *Chem. Eng. Prog.*, **44**, 691–696 (1948).^jGilman, *Heat. Piping Air Cond.*, **27**(4), 141–147 (1955).^kMcNown, *Proc. Am. Soc. Civ. Eng.*, **79**, Separate 258, 1–22 (1953); discussion, *ibid.*, **80**, Separate 396, 19–45 (1954). For the effect of branch spacing on junction losses in dividing flow, see Hecker, Nystrom, and Qureshi, *Proc. Am. Soc. Civ. Eng., J. Hydraul. Div.*, **103**(HY3), 265–279 (1977).^lThis is pressure drop (including friction loss) between run and branch, based on velocity in the mainstream before branching. Actual value depends on the flow split, ranging from 0.5 to 1.3 if mainstream enters run and from 0.7 to 1.5 if mainstream enters branch.^mLansford, *Loss of Head in Flow of Fluids through Various Types of 1½-in. Valves*, Univ. Eng. Exp. Sta. Bull. Ser. 340, 1943.**TABLE 6-5 Additional Frictional Loss for Laminar Flow through Fittings and Valves**

Type of fitting or valve	Additional frictional loss expressed as K			
	Re = 1,000	500	100	50
90° ell, short radius	0.9	1.0	7.5	16
Gate valve	1.2	1.7	9.9	24
Globe valve, composition disk	11	12	20	30
Plug	12	14	19	27
Angle valve	8	8.5	11	19
Check valve, swing	4	4.5	17	55

SOURCE: From curves by Kittredge and Rowley, *Trans. Am. Soc. Mech. Eng.*, **79**, 1759–1766 (1957).

The correction C_o (Fig. 6-14d) accounts for the extra losses due to developing flow in the outlet tangent of the pipe, of length L_o . The total loss for the bend plus outlet pipe includes the bend loss K plus the straight pipe frictional loss in the outlet pipe $4fL_o/D$. Note that $C_o = 1$ for L_o/D greater than the termination of the curves on Fig. 6-14d, which indicate the distance at which fully developed flow in the outlet pipe is reached. Finally, the roughness correction is

$$C_f = \frac{f_{\text{rough}}}{f_{\text{smooth}}} \quad (6-99)$$

where f_{rough} is the friction factor for a pipe of diameter D with the roughness of the bend, at the bend inlet Reynolds number. Similarly, f_{smooth} is the friction factor for smooth pipe. For $\text{Re} > 10^6$ and $r/D \geq 1$, use the value of C_f for $\text{Re} = 10^6$.

Example 6: Losses with Fittings and Valves It is desired to calculate the liquid level in the vessel shown in Fig. 6-15 required to produce a discharge velocity of 2 m/s. The fluid is water at 20°C with $\rho = 1,000 \text{ kg/m}^3$ and $\mu = 0.001 \text{ Pa} \cdot \text{s}$, and the butterfly valve is at $\theta = 10^\circ$. The pipe is 2-in Schedule 40, with an inner diameter of 0.0525 m. The pipe roughness is 0.046 mm. Assuming the flow is turbulent and taking the velocity profile factor $\alpha = 1$, the engineering Bernoulli equation Eq. (6-16), written between surfaces 1 and 2, where the pressures are both atmospheric and the fluid velocities are 0 and $V = 2 \text{ m/s}$, respectively, and there is no shaft work, simplifies to

$$gZ = \frac{V^2}{2} + l_e$$

Contributing to l_e are losses for the entrance to the pipe, the three sections of straight pipe, the butterfly valve, and the 90° bend. Note that no exit loss is used because the discharged jet is outside the control volume. Instead, the $V^2/2$ term accounts for the kinetic energy of the discharging stream. The Reynolds number in the pipe is

$$\text{Re} = \frac{DV\rho}{\mu} = \frac{0.0525 \times 2 \times 1000}{0.001} = 1.05 \times 10^5$$

From Fig. 6-9 or Eq. (6-38), at $\epsilon/D = 0.046 \times 10^{-3}/0.0525 = 0.00088$, the friction factor is about 0.0054. The straight pipe losses are then

$$\begin{aligned} l_{e(\text{sp})} &= \left(\frac{4fL}{D} \right) \frac{V^2}{2} \\ &= \left(\frac{4 \times 0.0054 \times (1 + 1 + 1)}{0.0525} \right) \frac{V^2}{2} \\ &= 1.23 \frac{V^2}{2} \end{aligned}$$

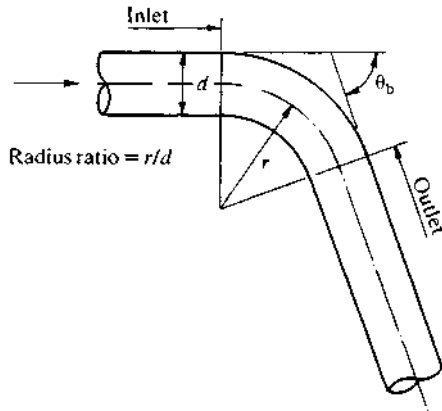
The losses from Table 6-4 in terms of velocity heads K are $K = 0.5$ for the sudden contraction and $K = 0.52$ for the butterfly valve. For the 90° standard radius ($r/D = 1$), the table gives $K = 0.75$. The method of Eq. (6-94), using Fig. 6-14, gives

$$\begin{aligned} K &= K^* C_{\text{re}} C_o C_f \\ &= 0.24 \times 1.24 \times 1.0 \times \left(\frac{0.0054}{0.0044} \right) \\ &= 0.37 \end{aligned}$$

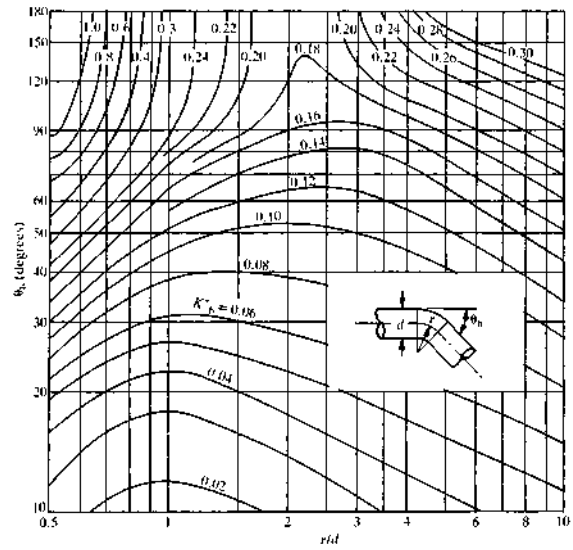
This value is more accurate than the value in Table 6-4. The value $f_{\text{smooth}} = 0.0044$ is obtainable either from Eq. (6-37) or Fig. 6-9.

The total losses are then

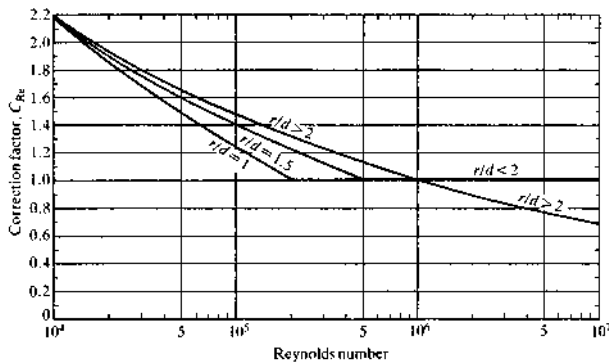
$$l_e = (1.23 + 0.5 + 0.52 + 0.37) \frac{V^2}{2} = 2.62 \frac{V^2}{2}$$



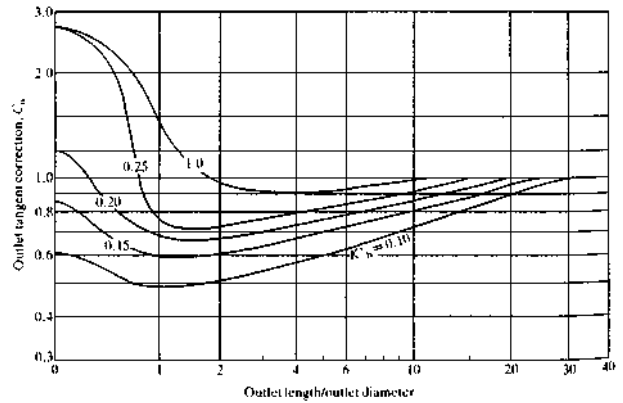
(a)



(b)



(c)



(d)

FIG. 6-14 Loss coefficients for flow in bends and curved pipes: (a) flow geometry, (b) loss coefficient for a smooth-walled bend at $Re = 10^6$, (c) Re correction factor, (d) outlet pipe correction factor. (From D. S. Miller; Internal Flow Systems, 2d ed., BHRA, Cranfield, U.K., 1990.)

and the liquid level Z is

$$Z = \frac{1}{g} \left(\frac{V^2}{2} + 2.62 \frac{V^2}{2} \right) = 3.62 \frac{V^2}{2g}$$

$$= \frac{3.62 \times 2^2}{2 \times 9.81} = 0.73 \text{ m}$$

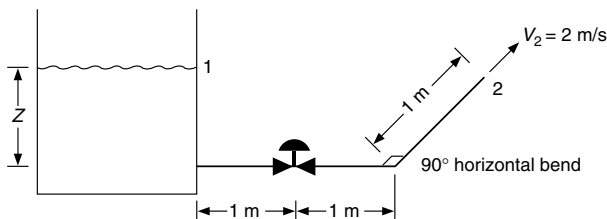


FIG. 6-15 Tank discharge example.

Curved Pipes and Coils For flow through curved pipe or coil, a secondary circulation perpendicular to the main flow called the **Dean effect** occurs. This circulation increases the friction relative to straight pipe flow and stabilizes laminar flow, delaying the transition Reynolds number to about

$$Re_{crit} = 2,100 \left(1 + 12 \sqrt{\frac{D}{D_c}} \right) \quad (6-100)$$

where D_c is the coil diameter. Equation (6-100) is valid for $10 < D_c/D < 250$. The **Dean number** is defined as

$$De = \frac{Re}{(D_c/D)^{1/2}} \quad (6-101)$$

In laminar flow, the friction factor for curved pipe f_c may be expressed in terms of the straight pipe friction factor $f = 16/Re$ as (Hart, *Chem. Eng. Sci.*, **43**, 775–783 [1988])

$$f_c/f = 1 + 0.090 \left(\frac{De^{1.5}}{70 + De} \right) \quad (6-102)$$

For turbulent flow, equations by Ito (*J. Basic Eng.*, **81**, 123 [1959]) and Srinivasan, Nandapurkar, and Holland (*Chem. Eng. [London]* no. 218, CE113-CE119 [May 1968]) may be used, with probable accuracy of ± 15 percent. Their equations are similar to

$$f_c = \frac{0.079}{\text{Re}^{0.25}} + \frac{0.0073}{\sqrt{(D_c/D)}} \quad (6-103)$$

The pressure drop for flow in **spirals** is discussed by Srinivasan, et al. (loc. cit.) and Ali and Seshadri (*Ind. Eng. Chem. Process Des. Dev.*, **10**, 328-332 [1971]). For friction loss in laminar flow through **semi-circular ducts**, see Masliyah and Nandakumar (*AIChE J.*, **25**, 478-487 [1979]); for curved channels of **square cross section**, see Cheng, Lin, and Ou (*J. Fluids Eng.*, **98**, 41-48 [1976]).

For **non-Newtonian (power law) fluids** in coiled tubes, Mashelkar and Devarajan (*Trans. Inst. Chem. Eng. (London)*, **54**, 108-114 [1976]) propose the correlation

$$f_c = (9.07 - 9.44n + 4.37n^2)(D/D_c)^{0.5}(\text{De}')^{-0.768 + 0.122n} \quad (6-104)$$

where De' is a modified Dean number given by

$$\text{De}' = \frac{1}{8} \left(\frac{6n+2}{n} \right)^n \text{Re}_{\text{MR}} \sqrt{\frac{D}{D_c}} \quad (6-105)$$

where Re_{MR} is the Metzner-Reed Reynolds number, Eq. (6-65). This correlation was tested for the range $\text{De}' = 70$ to 400, $D/D_c = 0.01$ to 0.135, and $n = 0.35$ to 1. See also Oliver and Asghar (*Trans. Inst. Chem. Eng. [London]*, **53**, 181-186 [1975]).

Screens The pressure drop for incompressible flow across a screen of fractional free area α may be computed from

$$\Delta p = K \frac{\rho V^2}{2} \quad (6-106)$$

where ρ = fluid density

V = superficial velocity based upon the gross area of the screen

K = velocity head loss

$$K = \left(\frac{1}{C^2} \right) \left(\frac{1 - \alpha^2}{\alpha^2} \right) \quad (6-107)$$

The discharge coefficient for the screen C with aperture D_s is given as a function of screen Reynolds number $\text{Re} = D_s(V/\alpha)\rho/\mu$ in Fig. 6-16 for **plain square-mesh screens**, $\alpha = 0.14$ to 0.79. This curve fits most of the data within ± 20 percent. In the laminar flow region, $\text{Re} < 20$, the discharge coefficient can be computed from

$$C = 0.1\sqrt{\text{Re}} \quad (6-108)$$

Coefficients greater than 1.0 in Fig. 6-16 probably indicate partial pressure recovery downstream of the minimum aperture, due to rounding of the wires.

Grootenhuis (*Proc. Inst. Mech. Eng. [London]*, **A168**, 837-846 [1954]) presents data which indicate that for a series of screens, the total pressure drop equals the number of screens times the pressure drop for one screen, and is not affected by the spacing between screens or their orientation with respect to one another, and presents a correlation for frictional losses across plain square-mesh screens and sintered gauzes. Armour and Cannon (*AIChE J.*, **14**, 415-420 [1968]) give a correlation based on a packed bed model for plain, twill, and "dutch" weaves. For losses through monofilament fabrics see Pederesen (*Filtr. Sep.*, **11**, 586-589 [1975]). For screens **inclined at an angle θ** , use the normal velocity component V'

$$V' = V \cos \theta \quad (6-109)$$

(Carothers and Baines, *J. Fluids Eng.*, **97**, 116-117 [1975]) in place of V in Eq. (6-106). This applies for $\text{Re} > 500$, $C = 1.26$, $\alpha \leq 0.97$, and $0 < \theta < 45^\circ$, for square-mesh screens and diamond-mesh netting. Screens inclined at an angle to the flow direction also experience a tangential stress.

For **non-Newtonian** fluids in slow flow, friction loss across a square-woven or full-twill-woven screen can be estimated by considering the screen as a set of parallel tubes, each of diameter equal to the average minimal opening between adjacent wires, and length twice the diameter, without entrance effects (Carley and Smith, *Polym. Eng. Sci.*, **18**, 408-415 [1978]). For screen stacks, the losses of individual screens should be summed.

JET BEHAVIOR

A **free jet**, upon leaving an outlet, will entrain the surrounding fluid, expand, and decelerate. To a first approximation, total momentum is conserved as jet momentum is transferred to the entrained fluid. For practical purposes, a jet is considered free when its cross-sectional area is less than one-fifth of the total cross-sectional flow area of the region through which the jet is flowing (Elrod, *Heat. Piping Air Cond.*, **26**[3], 149-155 [1954]), and the surrounding fluid is the same as the jet fluid. A **turbulent jet** in this discussion is considered to be a free jet with Reynolds number greater than 2,000. Additional discussion on the relation between Reynolds number and turbulence in jets is given by Elrod (ibid.). Abramowicz (*The Theory of Turbulent Jets*, MIT Press, Cambridge, 1963) and Rajaratnam (*Turbulent Jets*, Elsevier, Amsterdam, 1976) provide thorough discourses on turbulent jets. Hussein, et al. (*J. Fluid Mech.*, **258**, 31-75 [1994]) give extensive

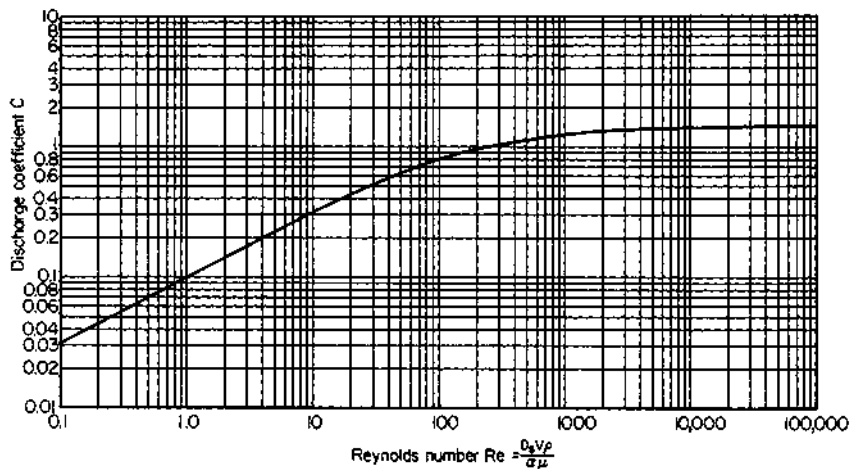


FIG. 6-16 Screen discharge coefficients, plain square-mesh screens. (Courtesy of E. I. du Pont de Nemours & Co.)

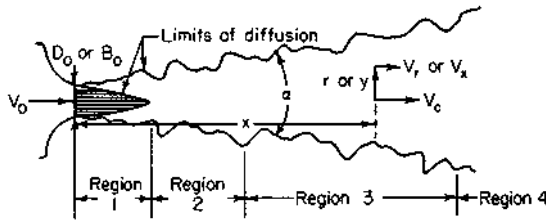


FIG. 6-17 Configuration of a turbulent free jet.

velocity data for a free jet, as well as an extensive discussion of free jet experimentation and comparison of data with momentum conservation equations.

A turbulent free jet is normally considered to consist of four flow regions (Tuve, *Heat. Piping Air Cond.*, **25**[1], 181–191 [1953]; Davies, *Turbulence Phenomena*, Academic, New York, 1972) as shown in Fig. 6-17:

1. Region of flow establishment—a short region whose length is about 6.4 nozzle diameters. The fluid in the conical core of the same length has a velocity about the same as the initial discharge velocity. The termination of this *potential core* occurs when the growing mixing or boundary layer between the jet and the surroundings reaches the centerline of the jet.
2. A transition region that extends to about 8 nozzle diameters.
3. Region of established flow—the principal region of the jet. In this region, the velocity profile transverse to the jet is self-preserving when normalized by the centerline velocity.
4. A terminal region where the residual centerline velocity reduces rapidly within a short distance. For air jets, the residual velocity will reduce to less than 0.3 m/s, (1.0 ft/s) usually considered still air.

Several references quote a length of 100 nozzle diameters for the length of the established flow region. However, this length is dependent on initial velocity and Reynolds number.

Table 6-6 gives characteristics of **rounded-inlet circular jets** and **rounded-inlet infinitely wide slot jets** (aspect ratio > 15). The information in the table is for a homogeneous, incompressible air system under isothermal conditions. The table uses the following nomenclature:

- B_0 = slot height
 D_0 = circular nozzle opening
 q = total jet flow at distance x
 q_0 = initial jet flow rate
 r = radius from circular jet centerline
 y = transverse distance from slot jet centerline
 V_c = centerline velocity
 V_r = circular jet velocity at r
 V_y = velocity at y

Witze (*Am. Inst. Aeronaut. Astronaut. J.*, **12**, 417–418 [1974]) gives equations for the centerline velocity decay of different types of subsonic and supersonic circular free jets. Entrainment of surrounding fluid in the region of flow establishment is lower than in the region of established flow (see Hill, *J. Fluid Mech.*, **51**, 773–779 [1972]). Data of Donald and Singer (*Trans. Inst. Chem. Eng. [London]*, **37**, 255–267 [1959]) indicate that jet angle and the coefficients given in Table 6-6 depend upon the fluids; for a water system, the jet angle for a circular jet is 14° and the entrainment ratio is about 70 percent of that for an air system. Most likely these variations are due to Reynolds number effects which are not taken into account in Table 6-6. Rushton (*AIChE J.*, **26**, 1038–1041 [1980]) examined available published results for circular jets and found that the centerline velocity decay is given by

$$\frac{V_c}{V_0} = 1.41 \text{Re}^{0.135} \left(\frac{D_0}{x} \right) \quad (6-110)$$

where $\text{Re} = D_0 V_0 \rho / \mu$ is the initial jet Reynolds number. This result corresponds to a jet angle $\tan \alpha/2$ proportional to $\text{Re}^{-0.135}$.

TABLE 6-6 Turbulent Free-Jet Characteristics

Where both jet fluid and entrained fluid are air

Rounded-inlet circular jet	
Longitudinal distribution of velocity along jet center line*†	
$\frac{V_c}{V_0} = K \frac{D_0}{x}$	for $7 < \frac{x}{D_0} < 100$
$K = 5$	for $V_0 = 2.5$ to 5.0 m/s
$K = 6.2$	for $V_0 = 10$ to 50 m/s
Radial distribution of longitudinal velocity†	
$\log \left(\frac{V_c}{V_r} \right) = 40 \left(\frac{r}{x} \right)^2$	for $7 < \frac{x}{D_0} < 100$
Jet angle°†	
$\alpha \approx 20^\circ$	for $\frac{x}{D_0} < 100$
Entrainment of surrounding fluid‡	
$\frac{q}{q_0} = 0.32 \frac{x}{D_0}$	for $7 < \frac{x}{D_0} < 100$
Rounded-inlet, infinitely wide slot jet	
Longitudinal distribution of velocity along jet centerline‡	
$\frac{V_c}{V_0} = 2.28 \left(\frac{B_0}{x} \right)^{0.5}$	for $5 < \frac{x}{B_0} < 2,000$ and $V_0 = 12$ to 55 m/s
Transverse distribution of longitudinal velocity‡	
$\log \left(\frac{V_c}{V_x} \right) = 18.4 \left(\frac{y}{x} \right)^2$	for $5 < \frac{x}{B_0} < 2,000$
Jet angle‡	
α is slightly larger than that for a circular jet	
Entrainment of surrounding fluid‡	
$\frac{q}{q_0} = 0.62 \left(\frac{x}{B_0} \right)^{0.5}$	for $5 < \frac{x}{B_0} < 2,000$

*Nottage, Slaby, and Gojsza, *Heat. Piping Air Cond.*, **24**(1), 165–176 (1952).

†Tuve, *Heat. Piping Air Cond.*, **25**(1), 181–191 (1953).

‡Albertson, Dai, Jensen, and Rouse, *Trans. Am. Soc. Civ. Eng.*, **115**, 639–664 (1950), and Discussion, *ibid.*, **115**, 665–697 (1950).

Characteristics of **rectangular jets** of various aspect ratios are given by Elrod (*Heat. Piping Air Cond.*, **26**[3], 149–155 [1954]). For **slot jets discharging into a moving fluid**, see Weinstein, Osterle, and Forstall (*J. Appl. Mech.*, **23**, 437–443 [1967]). **Coaxial jets** are discussed by Forstall and Shapiro (*J. Appl. Mech.*, **17**, 399–408 [1950]), and **double concentric jets** by Chigier and Beer (*J. Basic Eng.*, **86**, 797–804 [1964]). **Axisymmetric confined jets** are described by Barchilon and Curtet (*J. Basic Eng.*, **777–787** [1964]). **Restrained** turbulent jets of liquid discharging into air are described by Davies (*Turbulence Phenomena*, Academic, New York, 1972). These jets are inherently unstable and break up into drops after some distance. Lienhard and Day (*J. Basic Eng. Trans. ASME*, p. 515 [September 1970]) discuss the breakup of superheated liquid jets which flash upon discharge.

Density gradients affect the spread of a single-phase jet. A jet of lower density than the surroundings spreads more rapidly than a jet of the same density as the surroundings, and, conversely, a denser jet spreads less rapidly. Additional details are given by Keagy and Weller (*Proc. Heat Transfer Fluid Mech. Inst.*, ASME, pp. 89–98, June 22–24 [1949]) and Cleaves and Boelter (*Chem. Eng. Prog.*, **43**, 123–134 [1947]).

Few experimental data exist on **laminar jets** (see Gutfinger and Shinnar, *AIChE J.*, **10**, 631–639 [1964]). Theoretical analysis for velocity distributions and entrainment ratios are available in Schlichting and in Morton (*Phys. Fluids*, **10**, 2120–2127 [1967]).

Theoretical analyses of jet flows for power law **non-Newtonian fluids** are given by Vlachopoulos and Stournaras (*AIChE J.*, **21**, 385–388 [1975]), Mitwally (*J. Fluids Eng.*, **100**, 363 [1978]), and Sridhar and Rankin (*J. Fluids Eng.*, **100**, 500 [1978]).

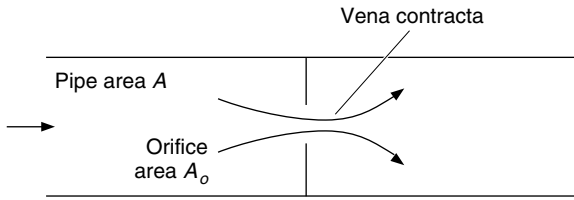


FIG. 6-18 Flow through an orifice.

FLOW THROUGH ORIFICES

Section 10 of this *Handbook* describes the use of orifice meters for flow measurement. In addition, **orifices** are commonly found within pipelines as flow-restricting devices, in perforated pipe distributing and return manifolds, and in perforated plates. Incompressible flow through an orifice in a pipeline, as shown in Fig. 6-18, is commonly described by the following equation for flow rate Q in terms of the pressures P_1 , P_2 , and P_3 ; the orifice area A_o ; the pipe cross-sectional area A ; and the density ρ .

$$Q = v_o A_o = C_o A_o \sqrt{\frac{2(P_1 - P_2)}{\rho[1 - (A_o/A)^2]}}$$

$$= C_o A_o \sqrt{\frac{2(P_1 - P_3)}{\rho(1 - A_o/A)[1 - (A_o/A)^2]}} \quad (6-111)$$

The velocity based on the hole area is v_o . The pressure P_1 is the pressure upstream of the orifice, typically about 1 pipe diameter upstream, the pressure P_2 is the pressure at the **vena contracta**, where the flow passes through a minimum area which is less than the orifice area, and the pressure P_3 is the pressure downstream of the vena contracta after pressure recovery associated with deceleration of the fluid. The velocity of approach factor $1 - (A_o/A)^2$ accounts for the kinetic energy approaching the orifice, and the **orifice coefficient** or **discharge coefficient** C_o accounts for the vena contracta. The location of the vena contracta varies with A_o/A , but is about 0.7 pipe diameter for $A_o/A < 0.25$. The factor $1 - A_o/A$ accounts for pressure recovery. Pressure recovery is complete by about 4 to 8 pipe diameters downstream of the orifice. The permanent pressure drop is $P_1 - P_3$. When the orifice is at the end of pipe, discharging directly into a large chamber, there is negligible pressure recovery, the permanent pressure drop is $P_1 - P_2$, and the last equality in Eq. (6-111) does not apply. Instead, $P_2 = P_3$. Equation (6-111) may also be used for flow across a perforated plate with open area A_o and total area A . The location of the vena contracta and complete recovery would scale not with the vessel or pipe diameter in which the plate is installed, but with the hole diameter and pitch between holes.

The orifice coefficient has a value of about 0.62 at large Reynolds numbers ($Re = D_o V_o \rho / \mu > 20,000$), although values ranging from 0.60 to 0.70 are frequently used. At lower Reynolds numbers, the orifice coefficient varies with both Re and with the area or diameter ratio. See Sec. 10 for more details.

When liquids discharge vertically downward from a pipe of diameter D_p through orifices into gas, gravity increases the discharge coefficient. Figure 6-19 shows this effect, giving the discharge coefficient in terms of a modified Froude number, $Fr = \Delta p / (\rho g D_p)$.

The orifice coefficient deviates from its value for sharp-edged orifices when the orifice wall thickness exceeds about 75 percent of the orifice diameter. Some pressure recovery occurs within the orifice and the orifice coefficient increases. Pressure drop across **segmental orifices** is roughly 10 percent greater than that for concentric circular orifices of the same open area.

COMPRESSIBLE FLOW

Flows are typically considered **compressible** when the density varies by more than 5 to 10 percent. In practice compressible flows are normally limited to gases, supercritical fluids, and multiphase flows

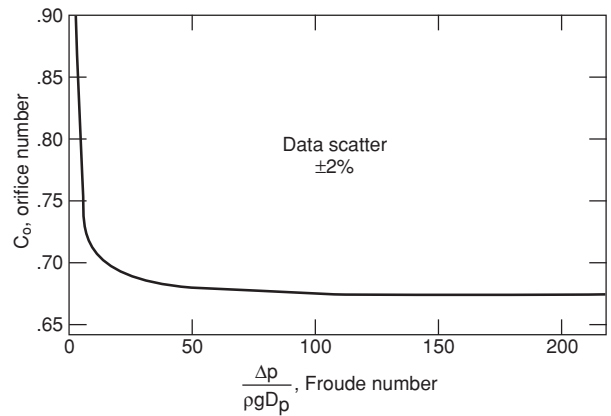


FIG. 6-19 Orifice coefficient vs. Froude number. (Courtesy E. I. duPont de Nemours & Co.)

containing gases. Liquid flows are normally considered incompressible, except for certain calculations involved in **hydraulic transient** analysis (see following) where compressibility effects are important even for nearly incompressible liquids with extremely small density variations. Textbooks on compressible gas flow include Shapiro (*Dynamics and Thermodynamics of Compressible Fluid Flow*, vols. I and II, Ronald Press, New York [1953]) and Zucrow and Hoffman (*Gas Dynamics*, vols. I and II, Wiley, New York [1976]).

In chemical process applications, one-dimensional gas flows through nozzles or orifices and in pipelines are the most important applications of compressible flow. Multidimensional external flows are of interest mainly in aerodynamic applications.

Mach Number and Speed of Sound The **Mach number** $M = V/c$ is the ratio of fluid velocity, V , to the **speed of sound** or **acoustic velocity**, c . The speed of sound is the propagation velocity of infinitesimal pressure disturbances and is derived from a momentum balance. The compression caused by the pressure wave is adiabatic and frictionless, and therefore isentropic.

$$c = \sqrt{\left(\frac{\partial p}{\partial \rho}\right)_s} \quad (6-112)$$

The derivative of pressure p with respect to density ρ is taken at constant entropy s . For an ideal gas,

$$\left(\frac{\partial p}{\partial \rho}\right)_s = \frac{kRT}{M_w}$$

where k = ratio of specific heats, C_p/C_v
 R = universal gas constant (8,314 J/kgmol K)
 T = absolute temperature
 M_w = molecular weight

Hence for an ideal gas,

$$c = \sqrt{\frac{kRT}{M_w}} \quad (6-113)$$

Most often, the Mach number is calculated using the speed of sound evaluated at the local pressure and temperature. When $M = 1$, the flow is **critical** or **sonic** and the velocity equals the local speed of sound. For **subsonic** flow $M < 1$ while **supersonic** flows have $M > 1$. Compressibility effects are important when the Mach number exceeds 0.1 to 0.2. A common error is to assume that compressibility effects are always negligible when the Mach number is small. The proper assessment of whether compressibility is important should be based on relative density changes, not on Mach number.

Isothermal Gas Flow in Pipes and Channels Isothermal compressible flow is often encountered in long transport lines, where there is sufficient heat transfer to maintain constant temperature. Velocities and Mach numbers are usually small, yet compressibility

effects are important when the total pressure drop is a large fraction of the absolute pressure. For an ideal gas with $\rho = pM_w/RT$, integration of the differential form of the momentum or mechanical energy balance equations, assuming a constant friction factor f over a length L of a channel of constant cross section and hydraulic diameter D_H , yields,

$$p_1^2 - p_2^2 = G^2 \frac{RT}{M_w} \left[\frac{4fL}{D_H} + 2 \ln \left(\frac{p_1}{p_2} \right) \right] \quad (6-114)$$

where the mass velocity $G = w/A = \rho V$ is the mass flow rate per unit cross-sectional area of the channel. The logarithmic term on the right-hand side accounts for the pressure change caused by acceleration of gas as its density decreases, while the first term is equivalent to the calculation of frictional losses using the density evaluated at the average pressure $(p_1 + p_2)/2$.

Solution of Eq. (6-114) for G and differentiation with respect to p_2 reveals a maximum mass flux $G_{\max} = p_2 \sqrt{M_w/(RT)}$ and a corresponding exit velocity $V_{2,\max} = \sqrt{RT/M_w}$ and exit Mach number $M_2 = 1/\sqrt{k}$. This apparent **choking** condition, though often cited, is not physically meaningful for isothermal flow because at such high velocities, and high rates of expansion, isothermal conditions are not maintained.

Adiabatic Frictionless Nozzle Flow In process plant pipelines, compressible flows are usually more nearly adiabatic than isothermal. Solutions for adiabatic flows through frictionless nozzles and in channels with constant cross section and constant friction factor are readily available.

Figure 6-20 illustrates adiabatic discharge of a **perfect** gas through a frictionless nozzle from a large chamber where velocity is effectively zero. A perfect gas obeys the ideal gas law $\rho = pM_w/RT$ and also has constant specific heat. The subscript 0 refers to the **stagnation** conditions in the chamber. More generally, stagnation conditions refer to the conditions which would be obtained by isentropically decelerating a gas flow to zero velocity. The minimum area section, or **throat**, of the nozzle is at the nozzle exit. The flow through the nozzle is isentropic because it is frictionless (reversible) and adiabatic. In terms of the exit Mach number M_1 and the upstream stagnation conditions, the flow conditions at the nozzle exit are given by

$$\frac{p_0}{p_1} = \left(1 + \frac{k-1}{2} M_1^2 \right)^{k/(k-1)} \quad (6-115)$$

$$\frac{T_0}{T_1} = 1 + \frac{k-1}{2} M_1^2 \quad (6-116)$$

$$\frac{\rho_0}{\rho_1} = \left(1 + \frac{k-1}{2} M_1^2 \right)^{1/(k-1)} \quad (6-117)$$

The mass velocity $G = w/A$, where w is the mass flow rate and A is the nozzle exit area, at the nozzle exit is given by

$$G = p_0 \sqrt{\frac{kM_w}{RT_0}} \frac{M_1}{\left(1 + \frac{k-1}{2} M_1^2 \right)^{(k+1)/2(k-1)}} \quad (6-118)$$

These equations are consistent with the isentropic relations for a per-

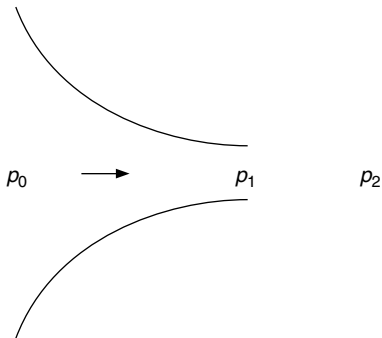


FIG. 6-20 Isentropic flow through a nozzle.

fect gas $p/p_0 = (\rho/\rho_0)^k$, $T/T_0 = (p/p_0)^{(k-1)/k}$. Equation (6-116) is valid for adiabatic flows with or without friction; it does not require isentropic flow. However, Eqs. (6-115) and (6-117) do require isentropic flow.

The exit Mach number M_1 may not exceed unity. At $M_1 = 1$, the flow is said to be **choked**, **sonic**, or **critical**. When the flow is choked, the pressure at the exit is greater than the pressure of the surroundings into which the gas flow discharges. The pressure drops from the exit pressure to the pressure of the surroundings in a series of shocks which are highly nonisentropic. Sonic flow conditions are denoted by * ; sonic exit conditions are found by substituting $M_1 = M_1^* = 1$ into Eqs. (6-115) to (6-118).

$$\frac{p^*}{p_0} = \left(\frac{2}{k+1} \right)^{k/(k-1)} \quad (6-119)$$

$$\frac{T^*}{T_0} = \frac{2}{k+1} \quad (6-120)$$

$$\frac{\rho^*}{\rho_0} = \left(\frac{2}{k+1} \right)^{1/(k-1)} \quad (6-121)$$

$$G^* = p_0 \sqrt{\left(\frac{2}{k+1} \right)^{(k+1)/(k-1)} \left(\frac{kM_w}{RT_0} \right)} \quad (6-122)$$

Note that under choked conditions, the exit velocity is $V = V^* = c^* = \sqrt{kRT^*/M_w}$, not $\sqrt{kRT_0/M_w}$. Sonic velocity must be evaluated at the exit temperature. For air, with $k = 1.4$, the critical pressure ratio p^*/p_0 is 0.5285 and the critical temperature ratio $T^*/T_0 = 0.8333$. Thus, for air discharging from 300 K, the temperature drops by 50 K (90 R). This large temperature decrease results from the conversion of internal energy into kinetic energy and is reversible. As the discharged jet decelerates in the external stagnant gas, it recovers its initial enthalpy.

When it is desired to determine the discharge rate through a nozzle from upstream pressure p_0 to external pressure p_2 , Equations (6-115) through (6-122) are best used as follows. The critical pressure is first determined from Eq. (6-119). If $p_2 > p^*$, then the flow is subsonic (subcritical, unchoked). Then $p_1 = p_2$ and M_1 may be obtained from Eq. (6-115). Substitution of M_1 into Eq. (6-118) then gives the desired mass velocity G . Equations (6-116) and (6-117) may be used to find the exit temperature and density. On the other hand, if $p_2 \leq p^*$, then the flow is choked and $M_1 = 1$. Then $p_1 = p^*$, and the mass velocity is G^* obtained from Eq. (6-122). The exit temperature and density may be obtained from Eqs. (6-120) and (6-121).

When the flow is choked, $G = G^*$ is independent of external downstream pressure. Reducing the downstream pressure will not increase the flow. The mass flow rate under choking conditions is directly proportional to the upstream pressure.

Example 7: Flow through Frictionless Nozzle Air at p_0 and temperature $T_0 = 293$ K discharges through a frictionless nozzle to atmospheric pressure. Compute the discharge mass flux G , the pressure, temperature, Mach number, and velocity at the exit. Consider two cases: (1) $p_0 = 7 \times 10^5$ Pa absolute, and (2) $p_0 = 1.5 \times 10^5$ Pa absolute.

1. $p_0 = 7.0 \times 10^5$ Pa. For air with $k = 1.4$, the critical pressure ratio from Eq. (6-119) is $p^*/p_0 = 0.5285$ and $p^* = 0.5285 \times 7.0 \times 10^5 = 3.70 \times 10^5$ Pa. Since this is greater than the external atmospheric pressure $p_2 = 1.01 \times 10^5$ Pa, the flow is choked and the exit pressure is $p_1 = 3.70 \times 10^5$ Pa. The exit Mach number is 1.0, and the mass flux is equal to G^* given by Eq. (6-118).

$$G^* = 7.0 \times 10^5 \times \sqrt{\left(\frac{2}{1.4+1} \right)^{(1.4+1)/(1.4-1)} \left(\frac{1.4 \times 29}{8314 \times 293} \right)} = 1,650 \text{ kg/m}^2 \cdot \text{s}$$

The exit temperature, since the flow is choked, is

$$T^* = \left(\frac{T^*}{T_0} \right) T_0 = \left(\frac{2}{1.4+1} \right) \times 293 = 244 \text{ K}$$

The exit velocity is $V = Mc = c^* = \sqrt{kRT^*/M_w} = 313$ m/s.

2. $p_0 = 1.5 \times 10^5$ Pa. In this case $p^* = 0.79 \times 10^5$ Pa, which is less than p_2 . Hence, $p_1 = p_2 = 1.01 \times 10^5$ Pa. The flow is unchoked (subsonic). Equation (6-115) is solved for the Mach number.

$$\frac{1.5 \times 10^5}{1.01 \times 10^5} = \left(1 + \frac{1.4-1}{2} M_1^2 \right)^{1.4/(1.4-1)}$$

$$M_1 = 0.773$$

Substitution into Eq. (6-118) gives G .

$$G = 1.5 \times 10^5 \times \sqrt{\frac{1.4 \times 29}{8,314 \times 293}} \times \frac{0.773}{\left(1 + \frac{(1.4-1)}{2} \times 0.773^2\right)^{(1.4+1)/2(1.4-1)}} = 337 \text{ kg/m}^2 \cdot \text{s}$$

The exit temperature is found from Eq. (6-116) to be 261.6 K or -11.5°C . The exit velocity is

$$V = Mc = 0.773 \times \sqrt{\frac{1.4 \times 8314 \times 261.6}{29}} = 250 \text{ m/s}$$

Adiabatic Flow with Friction in a Duct of Constant Cross Section Integration of the differential forms of the continuity, momentum, and total energy equations for a perfect gas, assuming a constant friction factor, leads to a tedious set of simultaneous algebraic equations. These may be found in Shapiro (*Dynamics and Thermodynamics of Compressible Fluid Flow*, vol. I, Ronald Press, New York, 1953) or Zucrow and Hoffman (*Gas Dynamics*, vol. I, Wiley, New York, 1976). Lapple's (*Trans. AICHE*, **39**, 395–432 [1943]) widely cited graphical presentation of the solution of these equations contained a subtle error, which was corrected by Levenspiel (*AICHE J.*, **23**, 402–403 [1977]). Levenspiel's graphical solutions are presented in Fig. 6-21. These charts refer to the physical situation illustrated in Fig. 6-22, where a perfect gas discharges from stagnation conditions in a large chamber through an isentropic nozzle followed by a duct of length L . The resistance parameter is $N = 4fL/D_H$, where f = Fanning friction factor and D_H = hydraulic diameter.

The exit Mach number M_2 may not exceed unity. $M_2 = 1$ corresponds to choked flow; sonic conditions may exist only at the pipe exit. The mass velocity G^* in the charts is the choked **mass flux for an isentropic nozzle** given by Eq. (6-118). For a pipe of finite length, the mass flux is less than G^* under choking conditions. The curves in Fig. 6-21 become vertical at the choking point, where flow becomes independent of downstream pressure.

The equations for nozzle flow, Eqs. (6-114) through (6-118), remain valid for the nozzle section even in the presence of the discharge pipe. Equations (6-116) and (6-120), for the temperature variation, may also be used for the pipe, with M_2 , p_2 replacing M_1 , p_1 since they are valid for adiabatic flow, with or without friction.

The graphs in Fig. 6-21 are based on accurate calculations, but are difficult to interpolate precisely. While they are quite useful for rough estimates, precise calculations are best done using the equations for one-dimensional adiabatic flow with friction, which are suitable for computer programming. Let subscripts 1 and 2 denote two points along a pipe of diameter D , point 2 being downstream of point 1. From a given point in the pipe, where the Mach number is M , the additional length of pipe required to accelerate the flow to sonic velocity ($M = 1$) is denoted $L_{\max}$ and may be computed from

$$\frac{4fL_{\max}}{D} = \frac{1-M^2}{kM^2} + \frac{k+1}{2k} \ln \left(\frac{\frac{k+1}{2} M^2}{1 + \frac{k-1}{2} M^2} \right) \quad (6-123)$$

With L = length of pipe between points 1 and 2, the change in Mach number may be computed from

$$\frac{4fL}{D} = \left(\frac{4fL_{\max}}{D} \right)_1 - \left(\frac{4fL_{\max}}{D} \right)_2 \quad (6-124)$$

Equations (6-116) and (6-113), which are valid for adiabatic flow with friction, may be used to determine the temperature and speed of sound at points 1 and 2. Since the mass flux $G = \rho v = \rho c M$ is constant, and $\rho = PM_w/RT$, the pressure at point 2 (or 1) can be found from G and the pressure at point 1 (or 2).

The additional frictional losses due to pipeline fittings such as elbows may be added to the velocity head loss $N = 4fL/D_H$ using the same velocity head loss values as for incompressible flow. This works well for fittings which do not significantly reduce the channel cross-sectional area, but may cause large errors when the flow area is greatly

reduced, as, for example, by restricting orifices. Compressible flow across restricting orifices is discussed in Sec. 10 of this *Handbook*. Similarly, elbows near the exit of a pipeline may choke the flow even though the Mach number is less than unity due to the nonuniform velocity profile in the elbow. For an abrupt contraction rather than rounded nozzle inlet, an additional 0.5 velocity head should be added to N . This is a reasonable approximation for G , but note that it allocates the additional losses to the pipeline, even though they are actually incurred in the entrance. It is an error to include one velocity head exit loss in N . The kinetic energy at the exit is already accounted for in the integration of the balance equations.

Example 8: Compressible Flow with Friction Losses Calculate the discharge rate of air to the atmosphere from a reservoir at 10^6 Pa gauge and 20°C through 10 m of straight 2-in Schedule 40 steel pipe (inside diameter = 0.0525 m), and 3 standard radius, flanged 90° elbows. Assume 0.5 velocity heads lost for the elbows.

For commercial steel pipe, with a roughness of 0.046 mm, the friction factor for fully rough flow is about 0.0047, from Eq. (6-38) or Fig. 6-9. It remains to be verified that the Reynolds number is sufficiently large to assume fully rough flow. Assuming an abrupt entrance with 0.5 velocity heads lost,

$$N = 4 \times 0.0047 \times \frac{10}{0.0525} + 0.5 + 3 \times 0.5 = 5.6$$

The pressure ratio p_3/p_0 is

$$\frac{1.01 \times 10^5}{(1 \times 10^6 + 1.01 \times 10^5)} = 0.092$$

From Fig. 6-21b at $N = 5.6$, $p_3/p_0 = 0.092$ and $k = 1.4$ for air, the flow is seen to be choked. At the choke point with $N = 5.6$ the critical pressure ratio p_2/p_0 is about 0.25 and G/G^* is about 0.48. Equation (6-122) gives

$$G^* = 1.101 \times 10^6 \times \sqrt{\left(\frac{2}{1.4+1}\right)^{(1.4+1)(1.4-1)} \left(\frac{1.4 \times 29}{8,314 \times 293.15}\right)} = 2,600 \text{ kg/m}^2 \cdot \text{s}$$

Multiplying by $G/G^* = 0.48$ yields $G = 1,250 \text{ kg/m}^2 \cdot \text{s}$. The discharge rate is $w = GA = 1,250 \times \pi \times 0.0525^2/4 = 2.7 \text{ kg/s}$.

Before accepting this solution, the Reynolds number should be checked. At the pipe exit, the temperature is given by Eq. (6-120) since the flow is choked. Thus, $T_2 = T^* = 244.6 \text{ K}$. The viscosity of air at this temperature is about $1.6 \times 10^{-5} \text{ Pa} \cdot \text{s}$. Then

$$\text{Re} = \frac{DV\rho}{\mu} = \frac{DG}{\mu} = \frac{0.0525 \times 1,250}{1.6 \times 10^{-5}} = 4.1 \times 10^6$$

At the beginning of the pipe, the temperature is greater, giving greater viscosity and a Reynolds number of 3.6×10^6 . Over the entire pipe length the Reynolds number is very large and the fully rough flow friction factor choice was indeed valid.

Once the mass flux G has been determined, Fig. 6-21a or 6-21b can be used to determine the pressure at any point along the pipe, simply by reducing $4fL/D_H$ and computing p_2 from the figures, given G , instead of the reverse. Charts for calculation between two points in a pipe with known flow and known pressure at either upstream or downstream locations have been presented by Loeb (*Chem. Eng.*, **76**[5], 179–184 [1969]) and for known downstream conditions by Powley (*Can. J. Chem. Eng.*, **36**, 241–245 [1958]).

Convergent/Divergent Nozzles (De Laval Nozzles) During frictionless adiabatic one-dimensional flow with changing cross-sectional area A the following relations are obeyed:

$$\frac{dA}{A} = \frac{dp}{\rho V^2} (1 - M^2) = \frac{1 - M^2}{M^2} \frac{dp}{p} = -(1 - M^2) \frac{dV}{V} \quad (6-125)$$

Equation (6-125) implies that in converging channels, subsonic flows are accelerated and the pressure and density decrease. In diverging channels, subsonic flows are decelerated as the pressure and density increase. In subsonic flow, the converging channels act as nozzles and diverging channels as diffusers. In supersonic flows, the opposite is true. Diverging channels act as nozzles accelerating the flow, while converging channels act as diffusers decelerating the flow.

Figure 6-23 shows a converging/diverging nozzle. When p_2/p_0 is less than the critical pressure ratio (p^*/p_0), the flow will be subsonic in the converging portion of the nozzle, sonic at the throat, and supersonic in the diverging portion. At the throat, where the flow is critical and the velocity is sonic, the area is denoted A^* . The cross-sectional

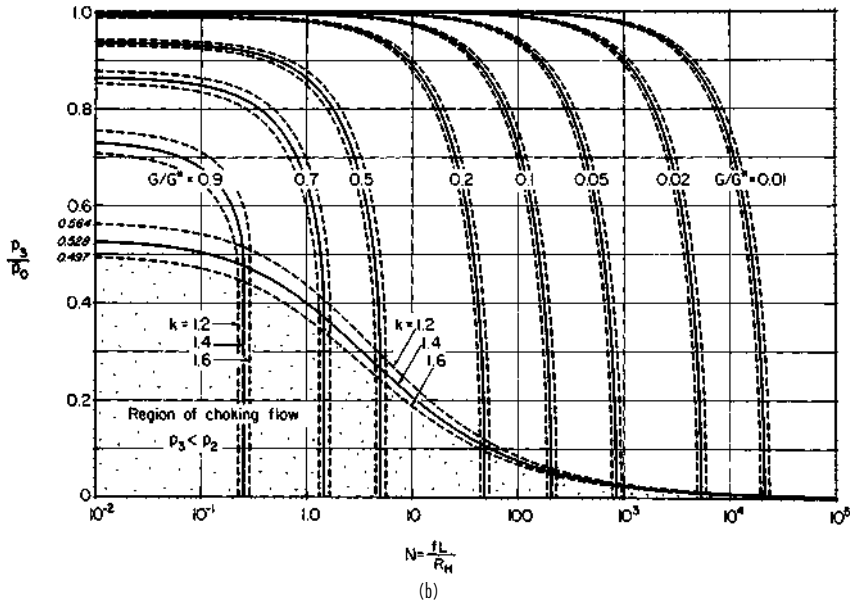
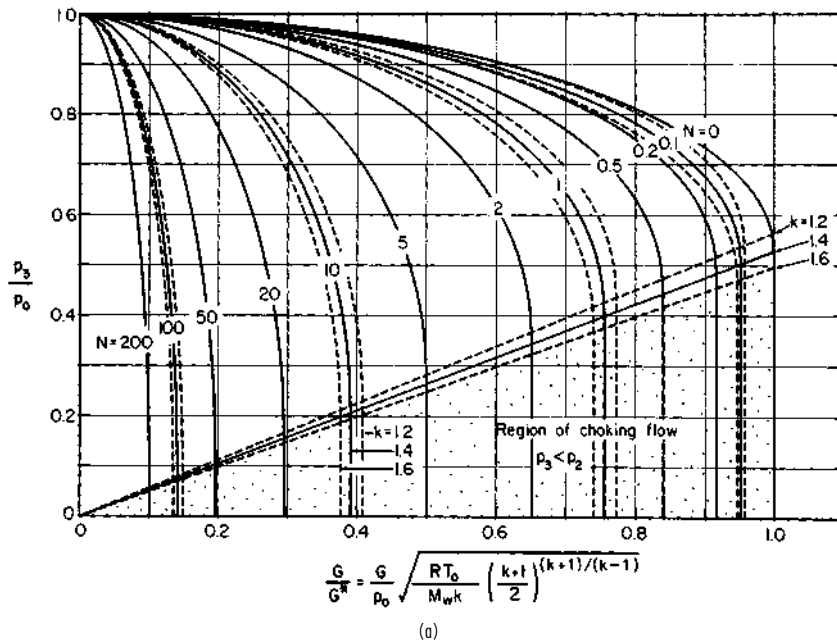


FIG. 6-21 Design charts for adiabatic flow of gases; (a) useful for finding the allowable pipe length for given flow rate; (b) useful for finding the discharge rate in a given piping system. (From Levenspiel, *Am. Inst. Chem. Eng. J.*, 23, 402 [1977].)

area and pressure vary with Mach number along the converging/diverging flow path according to the following equations for isentropic flow of a perfect gas:

$$\frac{A}{A^*} = \frac{1}{M} \left[\frac{2}{k+1} \left(1 + \frac{k-1}{2} M^2 \right) \right]^{\frac{k+1}{2(k-1)}} \quad (6-126)$$

$$\frac{p_0}{p} = \left(1 + \frac{k-1}{2} M^2 \right)^{\frac{k}{k-1}} \quad (6-127)$$

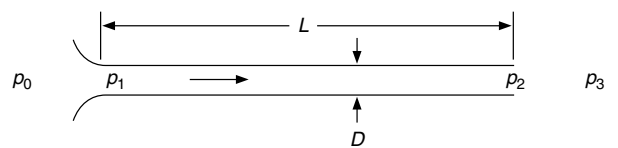


FIG. 6-22 Adiabatic compressible flow in a pipe with a well-rounded entrance.

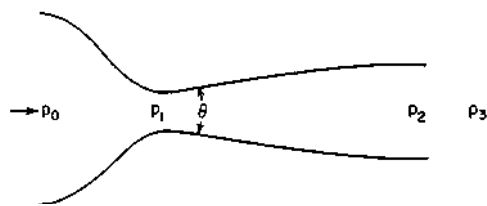


FIG. 6-23 Converging/diverging nozzle.

The temperature obeys the adiabatic flow equation for a perfect gas,

$$\frac{T_0}{T} = 1 + \frac{k-1}{2} M^2 \quad (6-128)$$

Equation (6-128) does not require frictionless (isentropic) flow. The sonic mass flux through the throat is given by Eq. (6-122). With A set equal to the nozzle exit area, the exit Mach number, pressure, and temperature may be calculated. Only if the exit pressure equals the ambient discharge pressure is the ultimate expansion velocity reached in the nozzle. Expansion will be incomplete if the exit pressure exceeds the ambient discharge pressure; shocks will occur outside the nozzle. If the calculated exit pressure is less than the ambient discharge pressure, the nozzle is overexpanded and compression shocks within the expanding portion will result.

The shape of the converging section is a smooth trumpet shape similar to the simple converging nozzle. However, special shapes of the diverging section are required to produce the maximum supersonic exit velocity. Shocks result if the divergence is too rapid and excessive boundary layer friction occurs if the divergence is too shallow. See Liepmann and Roshko (*Elements of Gas Dynamics*, Wiley, New York, 1957, p. 284). If the nozzle is to be used as a thrust device, the diverging section can be conical with a total included angle of 30° (Sutton, *Rocket Propulsion Elements*, 2d ed., Wiley, New York, 1956). To obtain large exit Mach numbers, slot-shaped rather than axisymmetric nozzles are used.

MULTIPHASE FLOW

Multiphase flows, even when restricted to simple pipeline geometry, are in general quite complex, and several features may be identified which make them more complicated than single-phase flow. Flow pattern description is not merely an identification of laminar or turbulent flow. The relative quantities of the phases and the topology of the interfaces must be described. Because of phase density differences, vertical flow patterns are different from horizontal flow patterns, and horizontal flows are not generally axisymmetric. Even when phase equilibrium is achieved by good mixing in two-phase flow, the changing equilibrium state as pressure drops with distance, or as heat is added or lost, may require that interphase mass transfer, and changes in the relative amounts of the phases, be considered.

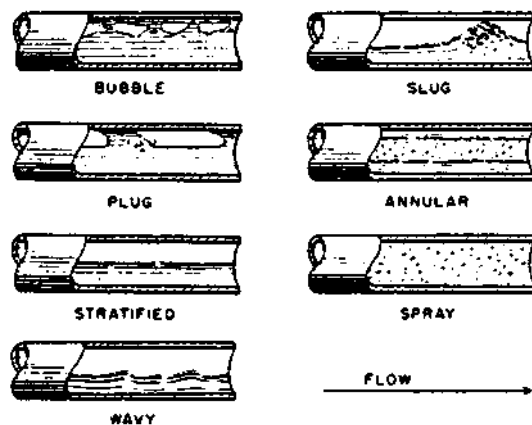
Wallis (*One-dimensional Two-phase Flow*, McGraw-Hill, New York, 1969) and Govier and Aziz present mass, momentum, mechanical energy, and total energy balance equations for two-phase flows. These equations are based on one-dimensional behavior for each phase. Such equations, for the most part, are used as a framework in which to interpret experimental data. Reliable prediction of multiphase flow behavior generally requires use of data or correlations. **Two-fluid modeling**, in which the full three-dimensional microscopic (partial differential) equations of motion are written for each phase, treating each as a continuum, occupying a volume fraction which is a continuous function of position, is a rapidly developing technique made possible by improved computational methods. For some relatively simple examples not requiring numerical computation, see Pearson (*Chem. Engr. Sci.*, **49**, 727–732 [1994]). Constitutive equations for two-fluid models are not yet sufficiently robust for accurate general-purpose two-phase flow computation, but may be quite good for particular classes of flows.

Liquids and Gases For cocurrent flow of liquids and gases in vertical (upflow), horizontal, and inclined pipes, a very large literature of experimental and theoretical work has been published, with less work on countercurrent and cocurrent vertical downflow. Much of the effort has been devoted to predicting flow patterns, pressure drop, and volume fractions of the phases, with emphasis on fully developed flow. In practice, many two-phase flows in process plants are not fully developed.

The most reliable methods for fully developed gas/liquid flows use **mechanistic models** to predict flow pattern, and use different pressure drop and void fraction estimation procedures for each flow pattern. Such methods are too lengthy to include here, and are well suited to incorporation into computer programs; commercial codes for gas/liquid pipeline flows are available. Some key references for mechanistic methods for flow pattern transitions and flow regime-specific pressure drop and void fraction methods include Taitel and Dukler (*AIChE J.*, **22**, 47–55 [1976]), Barnea, et al. (*Int. J. Multiphase Flow*, **6**, 217–225 [1980]), Barnea (*Int. J. Multiphase Flow*, **12**, 733–744 [1986]), Taitel, Barnea, and Dukler (*AIChE J.*, **26**, 345–354 [1980]), Wallis (*One-dimensional Two-phase Flow*, McGraw-Hill, New York, 1969), and Dukler and Hubbard (*Ind. Eng. Chem. Fundam.*, **14**, 337–347 [1975]). For preliminary or approximate calculations, **flow pattern maps** and flow regime-independent empirical correlations, are simpler and faster to use. Such methods for horizontal and vertical flows are provided in the following.

In **horizontal pipe**, flow patterns for fully developed flow have been reported in numerous studies. Transitions between flow patterns are gradual, and subjective owing to the visual interpretation of individual investigators. In some cases, statistical analysis of pressure fluctuations has been used to distinguish flow patterns. Figure 6-24 (Alves, *Chem. Eng. Progr.*, **50**, 449–456 [1954]) shows seven flow patterns for horizontal gas/liquid flow. **Bubble flow** is prevalent at high ratios of liquid to gas flow rates. The gas is dispersed as bubbles which move at velocity similar to the liquid and tend to concentrate near the top of the pipe at lower liquid velocities. **Plug flow** describes a pattern in which alternate plugs of gas and liquid move along the upper part of the pipe. In **stratified flow**, the liquid flows along the bottom of the pipe and the gas flows over a smooth liquid/gas interface. Similar to stratified flow, **wavy flow** occurs at greater gas velocities and has waves moving in the flow direction. When wave crests are sufficiently high to bridge the pipe, they form frothy slugs which move at much greater than the average liquid velocity. **Slug flow** can cause severe and sometimes dangerous vibrations in equipment because of impact of the high-velocity slugs against bends or other fittings. Slugs may also flood gas/liquid separation equipment.

In **annular flow**, liquid flows as a thin film along the pipe wall and gas flows in the core. Some liquid is entrained as droplets in the gas

FIG. 6-24 Gas/liquid flow patterns in horizontal pipes. (From Alves, *Chem. Eng. Progr.*, **50**, 449–456 [1954].)

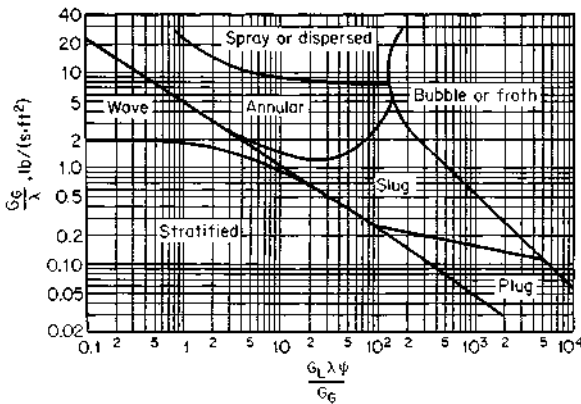


FIG. 6-25 Flow-pattern regions in cocurrent liquid/gas flow through horizontal pipes. To convert $\text{lbm}/(\text{ft}^2 \cdot \text{s})$ to $\text{kg}/(\text{m}^2 \cdot \text{s})$, multiply by 4.8824. (From Baker, *Oil Gas J.*, 53[12], 185–190, 192, 195 [1954].)

core. At very high gas velocities, nearly all the liquid is entrained as small droplets. This pattern is called **spray, dispersed, or mist flow**.

Approximate prediction of flow pattern may be quickly done using **flow pattern maps**, an example of which is shown in Fig. 6-25 (Baker, *Oil Gas J.*, 53[12], 185–190, 192–195 [1954]). The Baker chart remains widely used; however, for critical calculations the mechanistic model methods referenced previously are generally preferred for their greater accuracy, especially for large pipe diameters and fluids with physical properties different from air/water at atmospheric pressure. In the chart,

$$\lambda = (\rho'_G \rho'_L)^{1/2} \quad (6-129)$$

$$\psi = \frac{1}{\sigma'} \left[\frac{\mu'_L}{(\rho'_L)^2} \right]^{1/3} \quad (6-130)$$

G_L and G_G are the liquid and gas mass velocities, μ'_L is the ratio of liquid viscosity to water viscosity, ρ'_G is the ratio of gas density to air density, ρ'_L is the ratio of liquid density to water density, and σ' is the ratio of liquid surface tension to water surface tension. The reference properties are at 20°C (68°F) and atmospheric pressure, water density 1,000 kg/m^3 (62.4 lbm/ft^3), air density 1.20 kg/m^3 (0.075 lbm/ft^3), water viscosity 0.001 $\text{Pa} \cdot \text{s}$ (1.0 cP), and surface tension 0.073 N/m (0.0050 lb/ft). The empirical parameters λ and ψ provide a crude accounting for physical properties. The Baker chart is dimensionally inconsistent since the dimensional quantity G_G/λ is plotted against a dimensionless one, $G_L \lambda \psi / G_G$, and so must be used with G_G in $\text{lbm}/(\text{ft}^2 \cdot \text{s})$ units on the ordinate. To convert to $\text{kg}/(\text{m}^2 \cdot \text{s})$, multiply by 4.8824.

Rapid approximate predictions of **pressure drop** for fully developed, incompressible horizontal gas/liquid flow may be made using the method of Lockhart and Martinelli (*Chem. Eng. Prog.*, 45, 39–48 [1949]). First, the pressure drops that would be expected for each of the two phases as if flowing alone in single-phase flow are calculated. The Lockhart-Martinelli parameter X is defined in terms of the ratio of these pressure drops:

$$X = \left[\frac{(\Delta p/L)_L}{(\Delta p/L)_G} \right]^{1/2} \quad (6-131)$$

The two-phase pressure drop may then be estimated from either of the single-phase pressure drops, using

$$\left(\frac{\Delta p}{L} \right)_{TP} = Y_L \left(\frac{\Delta p}{L} \right)_L \quad (6-132)$$

$$\text{or} \quad \left(\frac{\Delta p}{L} \right)_{TP} = Y_G \left(\frac{\Delta p}{L} \right)_G \quad (6-133)$$

where Y_L and Y_G are read from Fig. 6-26 as functions of X . The curve labels refer to the flow regime (laminar or turbulent) found for each of

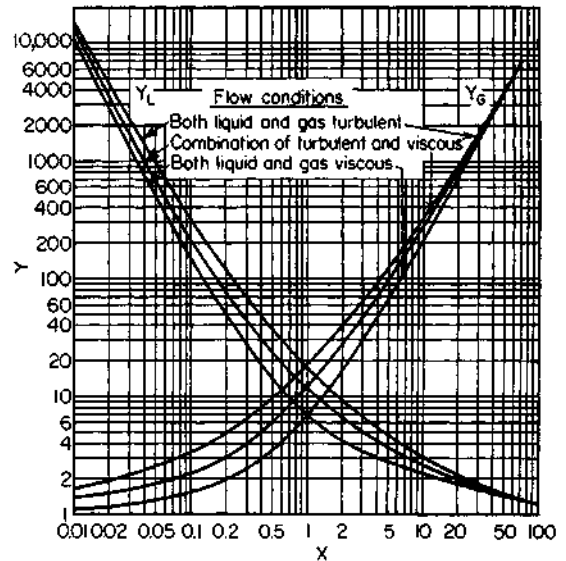


FIG. 6-26 Parameters for pressure drop in liquid/gas flow through horizontal pipes. (Based on Lockhart and Martinelli, *Chem. Eng. Prog.*, 45, 39 [1949].)

the phases flowing alone. The common turbulent-turbulent case is approximated well by

$$Y_L = 1 + \frac{20}{X} + \frac{1}{X^2} \quad (6-134)$$

Lockhart and Martinelli (ibid.) correlated pressure drop data from pipes 25 mm (1 in) in diameter or less within about ± 50 percent. In general, the predictions are high for stratified, wavy, and slug flows and low for annular flow. The correlation can be applied to pipe diameters up to about 0.1 m (4 in) with about the same accuracy.

The **volume fraction**, sometimes called **holdup**, of each phase in two-phase flow is generally not equal to its volumetric flow rate fraction, because of velocity differences, or **slip**, between the phases. For each phase, denoted by subscript i , the relations among superficial velocity V_i , in situ velocity v_i , volume fraction R_i , total volumetric flow rate Q_p , and pipe area A are

$$Q_i = V_i A = v_i R_i A \quad (6-135)$$

$$v_i = \frac{V_i}{R_i} \quad (6-136)$$

The **slip velocity** between gas and liquid is $v_s = v_G - v_L$. For two-phase gas/liquid flow, $R_L + R_G = 1$. A very common mistake in practice is to assume that in situ phase volume fractions are equal to input volume fractions.

For fully developed incompressible horizontal gas/liquid flow, a quick estimate for R_L may be obtained from Fig. 6-27, as a function of the Lockhart-Martinelli parameter X defined by Eq. (6-131). Indications are that liquid volume fractions may be overpredicted for liquids more viscous than water (Alves, *Chem. Eng. Prog.*, 50, 449–456 [1954]), and underpredicted for pipes larger than 25 mm diameter (Baker, *Oil Gas J.*, 53[12], 185–190, 192–195 [1954]).

A method for predicting pressure drop and volume fraction for **non-Newtonian fluids** in annular flow has been proposed by Eisenberg and Weinberger (*AIChE J.*, 25, 240–245 [1979]). Das, Biswas, and Matra (*Can. J. Chem. Eng.*, 70, 431–437 [1993]) studied holdup in both horizontal and vertical gas/liquid flow with non-Newtonian liquids. Farooqi and Richardson (*Trans. Inst. Chem. Engrs.*, 60, 292–305, 323–333 [1982]) developed correlations for holdup and pressure drop for gas/non-Newtonian liquid horizontal flow. They used a modified Lockhart-Martinelli parameter for non-Newtonian

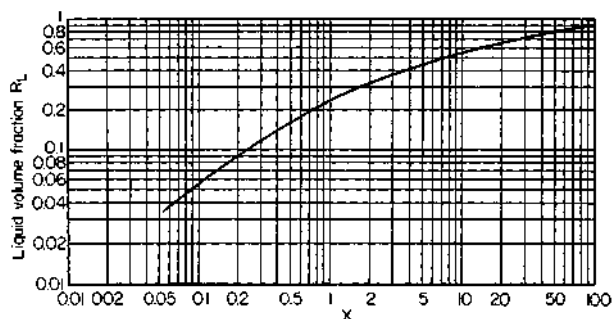


FIG. 6-27 Liquid volume fraction in liquid/gas flow through horizontal pipes. (From Lockhart and Martinelli, *Eng. Prog.*, **45**, 39 [1949].)

liquid holdup. They found that two-phase pressure drop may actually be less than the single-phase liquid pressure drop with shear thinning liquids in laminar flow.

Pressure drop data for a 1-in feed tee with the liquid entering the run and gas entering the branch are given by Alves (*Chem. Eng. Progr.*, **50**, 449–456 [1954]). Pressure drop and division of two-phase **annular flow in a tee** are discussed by Fouda and Rhodes (*Trans. Inst. Chem. Eng. [London]*, **52**, 354–360 [1974]). Flow through tees can result in unexpected flow splitting. Further reading on gas/liquid flow through tees may be found in Mudde, Groen, and van den Akker (*Int. J. Multiphase Flow*, **19**, 563–573 [1993]); Issa and Oliveira (*Computers and Fluids*, **23**, 347–372 [1993]) and Azzopardi and Smith (*Int. J. Multiphase Flow*, **18**, 861–875 [1992]).

Results by Chenoweth and Martin (*Pet. Refiner*, **34**[10], 151–155 [1955]) indicate that single-phase data for **fittings and valves** can be used in their correlation for two-phase pressure drop. Smith, Murdock, and Applebaum (*J. Eng. Power*, **99**, 343–347 [1977]) evaluated existing correlations for two-phase flow of steam/water and other gas/liquid mixtures through sharp-edged **orifices** meeting ASTM standards for flow measurement. The correlation of Murdock (*J. Basic Eng.*, **84**, 419–433 [1962]) may be used for these orifices. See also Collins and Gacesa (*J. Basic Eng.*, **93**, 11–21 [1971]), for measurements with steam and water beyond the limits of this correlation.

For pressure drop and holdup in **inclined pipe** with upward or downward flow, see Beggs and Brill (*J. Pet. Technol.*, **25**, 607–617 [1973]); the mechanistic model methods referenced above may also be applied to inclined pipes. Up to 10° from horizontal, upward pipe inclination has little effect on holdup (Gregory, *Can. J. Chem. Eng.*, **53**, 384–388 [1975]).

For fully developed incompressible **cocurrent upflow** of gases and liquids in **vertical pipes**, a variety of flow pattern terminologies and descriptions have appeared in the literature; some of these have been summarized and compared by Govier, Radford, and Dunn (*Can. J. Chem. Eng.*, **35**, 58–70 [1957]). One reasonable classification of patterns is illustrated in Fig. 6-28.

In **bubble flow**, gas is dispersed as bubbles throughout the liquid, but with some tendency to concentrate toward the center of the pipe. In **slug flow**, the gas forms large **Taylor bubbles** of diameter nearly equal to the pipe diameter. A thin film of liquid surrounds the Taylor bubble. Between the Taylor bubbles are liquid slugs containing some bubbles. **Froth** or **churn flow** is characterized by strong intermittency and intense mixing, with neither phase easily described as continuous or dispersed. There remains disagreement in the literature as to whether churn flow is a real fully developed flow pattern or is an indication of large entry length for developing slug flow (Zao and Dukler, *Int. J. Multiphase Flow*, **19**, 377–383 [1993]; Hewitt and Jayanti, *Int. J. Multiphase Flow*, **19**, 527–529 [1993]).

Ripple flow has an upward-moving wavy layer of liquid on the pipe wall; it may be thought of as a transition region to **annular**, **annular mist**, or **film flow**, in which gas flows in the core of the pipe while an annulus of liquid flows up the pipe wall. Some of the liquid is

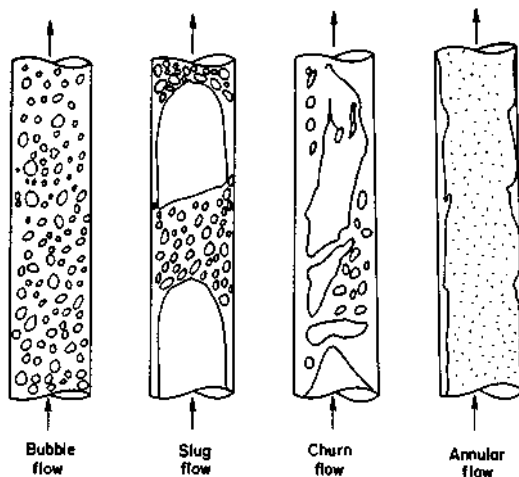


FIG. 6-28 Flow patterns in cocurrent upward vertical gas/liquid flow. (From Taitel, Barnea, and Dukler, *AIChE J.*, **26**, 345–354 [1980]. Reproduced by permission of the American Institute of Chemical Engineers © 1980 AIChE. All rights reserved.)

entrained as droplets in the gas core. **Mist flow** occurs when all the liquid is carried as fine drops in the gas phase; this pattern occurs at high gas velocities, typically 20 to 30 m/s (66 to 98 ft/s).

The correlation by Govier, et al. (*Can. J. Chem. Eng.*, **35**, 58–70 [1957]), Fig. 6-29, may be used for quick estimate of flow pattern.

Slip, or relative velocity between phases, occurs for vertical flow as well as for horizontal. No completely satisfactory, flow regime-independent correlation for volume fraction or holdup exists for vertical flow. Two frequently used flow regime-independent methods are those by Hughmark and Pressburg (*AIChE J.*, **7**, 677 [1961]) and Hughmark (*Chem. Eng. Prog.*, **58**[4], 62 [April 1962]). **Pressure drop in upflow** may be calculated by the procedure described in Hughmark (*Ind. Eng. Chem. Fundam.*, **2**, 315–321 [1963]). The mechanistic, flow regime-based methods are advisable for critical applications.

For **upflow in helically coiled tubes**, the flow pattern, pressure drop, and holdup can be predicted by the correlations of Banerjee, Rhodes, and Scott (*Can. J. Chem. Eng.*, **47**, 445–453 [1969]) and

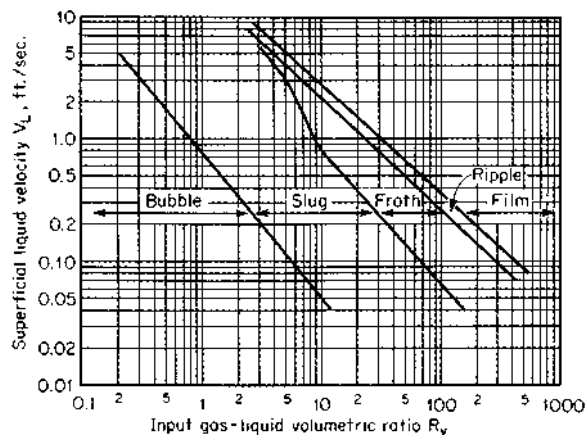


FIG. 6-29 Flow-pattern regions in cocurrent liquid/gas flow in upflow through vertical pipes. To convert ft/s to m/s, multiply by 0.3048. (From Govier, Radford, and Dunn, *Can. J. Chem. Eng.*, **35**, 58–70 [1957].)

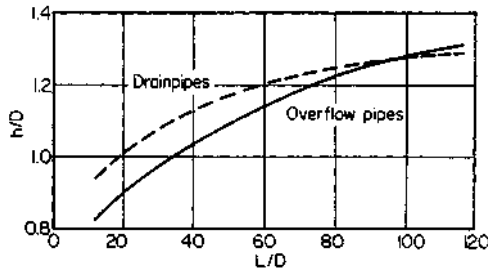


FIG. 6-30 Critical head for drain and overflow pipes. (From Kalinske, Univ. Iowa Stud. Eng., Bull. 26 [1939–1940].)

Akagawa, Sakaguchi, and Ueda (*Bull JSME*, **14**, 564–571 [1971]). Correlations for flow patterns in **downflow** in vertical pipe are given by Oshinowo and Charles (*Can. J. Chem. Eng.*, **52**, 25–35 [1974]) and Barnea, Shoham, and Taitel (*Chem. Eng. Sci.*, **37**, 741–744 [1982]). Use of **drift flux theory** for void fraction modeling in downflow is presented by Clark and Flemmer (*Chem. Eng. Sci.*, **39**, 170–173 [1984]). **Downward inclined** two-phase flow data and modeling are given by Barnea, Shoham, and Taitel (*Chem. Eng. Sci.*, **37**, 735–740 [1982]). Data for **downflow in helically coiled tubes** are presented by Casper (*Chem. Ing. Tech.*, **42**, 349–354 [1970]).

The entrance to a **drain** is flush with a horizontal surface, while the entrance to an **overflow** pipe is above the horizontal surface. When such pipes do not run full, considerable amounts of gas can be drawn down by the liquid. The amount of gas entrained is a function of pipe diameter, pipe length, and liquid flow rate, as well as the drainpipe outlet boundary condition. Extensive data on air entrainment and liquid head above the entrance as a function of water flow rate for pipe diameters from 43.9 to 148.3 mm (1.7 to 5.8 in) and lengths from about 1.22 to 5.18 m (4.0 to 17.0 ft) are reported by Kalinske (*Univ. Iowa Stud. Eng., Bull.* 26, pp. 26–40 [1939–1940]). For heads greater than the critical, the pipes will run full with no entrainment. The critical head h for flow of water in drains and overflow pipes is given in Fig. 6-30. Kalinske's results show little effect of the height of protrusion of overflow pipes when the protrusion height is greater than about one pipe diameter. For conservative design, McDuffie (*AIChE J.*, **23**, 37–40 [1977]) recommends the following relation for minimum liquid height to prevent entrainment.

$$Fr \leq 1.6 \left(\frac{h}{D} \right)^2 \quad (6-137)$$

where the Froude number is defined by

$$Fr \equiv \frac{V_L}{\sqrt{g(\rho_L - \rho_G)D/\rho_L}} \quad (6-138)$$

where g = acceleration due to gravity

V_L = liquid velocity in the drain pipe

ρ_L = liquid density

ρ_G = gas density

D = pipe inside diameter

h = liquid height

For additional information, see Simpson (*Chem. Eng.*, **75**[6], 192–214 [1968]). A critical Froude number of 0.31 to ensure vented flow is widely cited. Recent results (Thorpe, *3d Int. Conf. Multi-phase Flow*, The Hague, Netherlands, 18–20 May 1987, paper K2, and *4th Int. Conf. Multi-phase Flow*, Nice, France, 19–21 June 1989, paper K4) show hysteresis, with different critical Froude numbers for flooding and unflowing of drain pipes, and the influence of end effects. Wallis, Crowley, and Hagi (*Trans. ASME J. Fluids Eng.*, 405–413 [June 1977]) examine the conditions for horizontal discharge pipes to run full.

Flashing flow and **condensing flow** are two examples of multi-phase flow with **phase change**. Flashing flow occurs when pressure drops below the bubble point pressure of a flowing liquid. A frequently

used one-dimensional model for flashing flow through nozzles and pipes is the **homogeneous equilibrium model** which assumes that both phases move at the same in situ velocity, and maintain vapor/liquid equilibrium. It may be shown that a **critical flow** condition, analogous to sonic or critical flow during compressible gas flow, is given by the following expression for the mass flux G in terms of the derivative of pressure p with respect to mixture density ρ_m at constant entropy:

$$G_{crit} = \rho_m \sqrt{\left(\frac{\partial p}{\partial \rho_m} \right)_s} \quad (6-139)$$

The corresponding acoustic velocity $\sqrt{(\partial p / \partial \rho_m)_s}$ is normally much less than the acoustic velocity for gas flow. The mixture density is given in terms of the individual phase densities and the **quality** (mass flow fraction vapor) x by

$$\frac{1}{\rho_m} = \frac{x}{\rho_G} + \frac{1-x}{\rho_L} \quad (6-140)$$

Choked and unchoked flow situations arise in pipes and nozzles in the same fashion for homogeneous equilibrium flashing flow as for gas flow. For nozzle flow from stagnation pressure p_0 to exit pressure p_1 , the mass flux is given by

$$G^2 = -2\rho_{m1}^2 \int_{p_0}^{p_1} \frac{dp}{\rho_m} \quad (6-141)$$

The integration is carried out over an isentropic flash path: flashes at constant entropy must be carried out to evaluate ρ_m as a function of p . Experience shows that isenthalpic flashes provide good approximations unless the liquid mass fraction is very small. Choking occurs when G obtained by Eq. (6-141) goes through a maximum at a value of p_1 greater than the external discharge pressure. Equation (6-139) will also be satisfied at that point. In such a case the pressure at the nozzle exit equals the choking pressure and flashing shocks occur outside the nozzle exit.

For homogeneous flow in a pipe of diameter D , the differential form of the Bernoulli equation (6-15) rearranges to

$$\frac{dp}{\rho_m} + g dz + \frac{G^2}{\rho_m} d \frac{1}{\rho_m} + 2f \frac{dx'}{D} \frac{G^2}{\rho_m^2} = 0 \quad (6-142)$$

where x' is distance along the pipe. Integration over a length L of pipe assuming constant friction factor f yields

$$G^2 = \frac{-\int_{p_1}^{p_2} \rho_m dp - g \int_{z_1}^{z_2} \rho_m^2 dz}{\ln(\rho_{m1}/\rho_{m2}) + 2fL/D} \quad (6-143)$$

Frictional pipe flow is not isentropic. Strictly speaking, the flashes must be carried out at constant $h + V^2/2 + gz$, where h is the enthalpy per unit mass of the two-phase flashing mixture. The flash calculations are fully coupled with the integration of the Bernoulli equation; the velocity V must be known at every pressure p to evaluate ρ_m . Computational routines, employing the thermodynamic and material balance features of flowsheet simulators, are the most practical way to carry out such flashing flow calculations, particularly when multicomponent systems are involved. Significant simplification arises when the mass fraction liquid is large, for then the effect of the $V^2/2$ term on the flash splits may be neglected. If elevation effects are also negligible, the flash computations are decoupled from the Bernoulli equation integration. For many horizontal flashing flow calculations, this is satisfactory and the flash computations may be carried out first, to find ρ_m as a function of p from p_1 to p_2 , which may then be substituted into Eq. (6-143).

With flashes carried out along the appropriate thermodynamic paths, the formalism of Eqs. (6-139) through (6-143) applies to all homogeneous equilibrium compressible flows, including, for example, flashing flow, ideal gas flow, and nonideal gas flow. Equation (6-118), for example, is a special case of Eq. (6-141) where the quality $x = 1$ and the vapor phase is a perfect gas.

Various **nonequilibrium** and **slip flow** models have been proposed as improvements on the homogeneous equilibrium flow model. See, for example, Henry and Fauske (*Trans. ASME J. Heat Transfer*, 179–187 [May 1971]). Nonequilibrium and slip effects both increase

computed mass flux for fixed pressure drop, compared to homogeneous equilibrium flow. For flow paths greater than about 100 mm, homogeneous equilibrium behavior appears to be the best assumption (Fischer, et al., *Emergency Relief System Design Using DIERS Technology*, AIChE, New York [1992]). For shorter flow paths, the best estimate may sometimes be given by linearly interpolating (as a function of length) between **frozen flow** (constant quality, no flashing) at 0 length and equilibrium flow at 100 mm.

In a series of papers by Leung and coworkers (*AIChE J.*, **32**, 1743–1746 [1986]; **33**, 524–527 [1987]; **34**, 688–691 [1988]; *J. Loss Prevention Proc. Ind.*, **2**[2], 78–86 [April 1989]; **3**[1], 27–32 [January 1990]; *Trans. ASME J. Heat Transfer*, **112**, 524–528, 528–530 [1990]; **113**, 269–272 [1991]) approximate techniques have been developed for homogeneous equilibrium calculations based on pseudo-equation of state methods for flashing mixtures.

Relatively less work has been done on **condensing flows**. Slip effects are more important for condensing than for flashing flows. Soliman, Schuster, and Berenson (*J. Heat Transfer*, **90**, 267–276 [1968]) give a model for condensing vapor in **horizontal pipe**. They assume the condensate flows as an annular ring. The Lockhart-Martinelli correlation is used for the frictional pressure drop. To this pressure drop is added an acceleration term based on homogeneous flow, equivalent to the $G^2 d(1/\rho_m)$ term in Eq. (6-142). Pressure drop is computed by integration of the incremental pressure changes along the length of pipe.

For **condensing vapor in vertical downflow**, in which the liquid flows as a thin annular film, the frictional contribution to the pressure drop may be estimated based on the gas flow alone, using the friction factor plotted in Fig. 6-31, where Re_c is the Reynolds number for the gas flowing alone (Bergelin et al., *Proc. Heat Transfer Fluid Mech. Inst.*, ASME, June 22–24, 1949, pp. 19–28).

$$-\frac{dp}{dz} = \frac{2f_c \rho_c V_c^2}{D} \quad (6-144)$$

To this should be added the $G_c^2 d(1/\rho_c)/dx$ term to account for velocity change effects.

Gases and Solids The flow of gases and solids in **horizontal pipe** is usually classified as either **dilute phase** or **dense phase** flow. Unfortunately, there is no clear delineation between the two types of flow, and the **dense phase** description may take on more than one meaning, creating some confusion (Knowlton et al., *Chem. Eng. Progr.*, **90**[4], 44–54 [April 1994]). For dilute phase flow, achieved at low solids-to-gas weight ratios (loadings), and high gas velocities, the solids may be fully suspended and fairly uniformly dispersed over the pipe cross section (homogeneous flow), particularly for low-density or small particle size solids. At lower gas velocities, the solids may

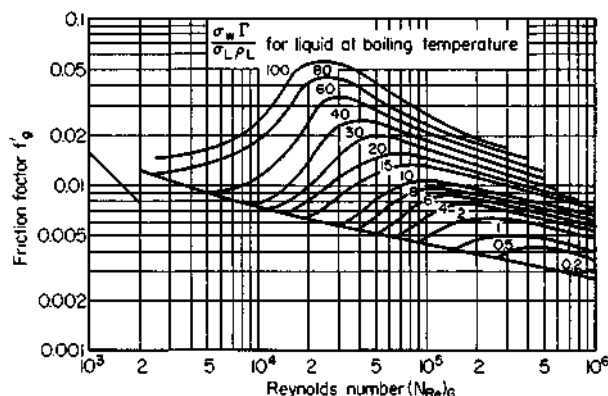


FIG. 6-31 Friction factors for condensing liquid/gas flow downward in vertical pipe. In this correlation $\Gamma/\rho L$ is in ft^3/h . To convert ft^3/h to m^3/s , multiply by 0.00155. (From Bergelin et al., *Proc. Heat Transfer Fluid Mech. Inst.*, ASME, 1949, p. 19.)

bounce along the bottom of the pipe. With higher loadings and lower gas velocities, the particles may settle to the bottom of the pipe, forming dunes, with the particles moving from dune to dune. In dense phase conveying, solids tend to concentrate in the lower portion of the pipe at high gas velocity. As gas velocity decreases, the solids may first form dense moving strands, followed by slugs. Discrete plugs of solids may be created intentionally by timed injection of solids, or the plugs may form spontaneously. Eventually the pipe may become blocked. For more information on flow patterns, see Coulson and Richardson (*Chemical Engineering*, vol. 2, 2d ed., Pergamon, New York, 1968, p. 583); Korn (*Chem. Eng.*, **57**[3], 108–111 [1950]); Patterson (*J. Eng. Power*, **81**, 43–54 [1959]); Wen and Simons (*AIChE J.*, **5**, 263–267 [1959]); and Knowlton et al. (*Chem. Eng. Progr.*, **90**[4], 44–54 [April 1994]).

For the **minimum velocity** required to prevent formation of dunes or settled beds in **horizontal flow**, some data are given by Zenz (*Ind. Eng. Chem. Fundam.*, **3**, 65–75 [1964]), who presented a correlation for the minimum velocity required to keep particles from depositing on the bottom of the pipe. This rather tedious estimation procedure may also be found in Govier and Aziz, who provide additional references and discussion on transition velocities. In practice, the actual conveying velocities used in systems with loadings less than 10 are generally over 15 m/s (49 ft/s) while for high loadings (>20) they are generally less than 7.5 m/s (24.6 ft/s) and are roughly twice the actual solids velocity (Wen and Simons, *AIChE J.*, **5**, 263–267 [1959]).

Total pressure drop for horizontal gas/solid flow includes acceleration effects at the entrance to the pipe and frictional effects beyond the entrance region. A great number of correlations for pressure gradient are available, none of which is applicable to all flow regimes. Govier and Aziz review many of these and provide recommendations on when to use them.

For **upflow** of gases and solids in **vertical pipes**, the **minimum conveying velocity** for low loadings may be estimated as twice the terminal settling velocity of the largest particles. Equations for terminal settling velocity are found in the “Particle Dynamics” subsection, following. **Choking** occurs as the velocity is dropped below the minimum conveying velocity and the solids are no longer transported, collapsing into solid plugs (Knowlton, et al., *Chem. Eng. Progr.*, **90**[4], 44–54 [April 1994]). See Smith (*Chem. Eng. Sci.*, **33**, 745–749 [1978]) for an equation to predict the onset of choking.

Total pressure drop for vertical upflow of gases and solids includes acceleration and frictional effects also found in horizontal flow, plus potential energy or hydrostatic effects. Govier and Aziz review many of the pressure drop calculation methods and provide recommendations for their use. See also Yang (*AIChE J.*, **24**, 548–552 [1978]).

Drag reduction has been reported for low loadings of small diameter particles (<60 μm diameter), ascribed to damping of turbulence near the wall (Rossettia and Pfeffer, *AIChE J.*, **18**, 31–39 [1972]).

For **dense phase transport in vertical pipes** of small diameter, see Sandy, Daubert, and Jones (*Chem. Eng. Progr.*, **66**, *Symp. Ser.*, 105, 133–142 [1970]).

The **flow of bulk solids through restrictions and bins** is discussed in symposium articles (*J. Eng. Ind.*, **91**[2] [1969]) and by Stepanoff (*Gravity Flow of Bulk Solids and Transportation of Solids in Suspension*, Wiley, New York, 1969). Some problems encountered in discharge from bins include (Knowlton et al., *Chem. Eng. Progr.*, **90**[4], 44–54 [April 1994]) flow stoppage due to **ratholing** or **arching**, **segregation** of fine and coarse particles, **flooding** upon collapse of ratholes, and poor **residence time distribution** when **funnel flow** occurs.

Solid and Liquids Slurry flow may be divided roughly into two categories based on settling behavior (see Etchells in Shamlou, *Processing of Solid-Liquid Suspensions*, Chap. 12, Butterworth-Heinemann, Oxford, 1993). **Nonsettling** slurries are made up of very fine, highly concentrated, or neutrally buoyant particles. These slurries are normally treated as pseudohomogeneous fluids. They may be quite viscous and are frequently non-Newtonian. Slurries of particles that tend to settle out rapidly are called **settling slurries** or **fast-settling slurries**. While in some cases positively buoyant solids are encountered, the present discussion will focus on solids which are more dense than the liquid.

For **horizontal flow of fast-settling slurries**, the following rough description may be made (Govier and Aziz). Ultrafine particles, 10 μm

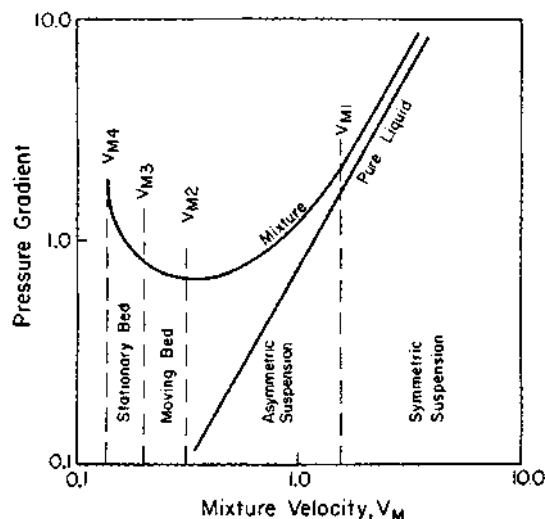


FIG. 6-32 Flow pattern regimes and pressure gradients in horizontal slurry flow. (From Govier and Aziz, *The Flow of Complex Mixtures in Pipes*, Van Nostrand Reinhold, New York, 1972.)

or smaller, are generally fully suspended and the particle distributions are not influenced by gravity. Fine particles 10 to 100 μm (3.3×10^{-5} to 33×10^{-5} ft) are usually fully suspended, but gravity causes concentration gradients. Medium-size particles, 100 to 1000 μm , may be fully suspended at high velocity, but often form a moving deposit on the bottom of the pipe. Coarse particles, 1,000 to 10,000 μm (0.0033 to 0.033 ft), are seldom fully suspended and are usually conveyed as a moving deposit. Ultracoarse particles larger than 10,000 μm (0.033 ft) are not suspended at normal velocities unless they are unusually light.

Figure 6-32, taken from Govier and Aziz, schematically indicates four flow pattern regions superimposed on a plot of pressure gradient vs. mixture velocity $V_M = V_L + V_S = (Q_L + Q_S)/A$ where V_L and V_S are the superficial liquid and solid velocities, Q_L and Q_S are liquid and solid volumetric flow rates, and A is the pipe cross-sectional area. V_{M4} is the transition velocity above which a bed exists in the bottom of the pipe, part of which is stationary and part of which moves by saltation, with the upper particles tumbling and bouncing over one another, often with formation of dunes. With a broad particle-size distribution, the finer particles may be fully suspended. Near V_{M4} , the pressure gradient rapidly increases as V_M decreases. Above V_{M3} , the entire bed moves. Above V_{M2} , the solids are fully suspended; that is, there is no deposit, moving or stationary, on the bottom of the pipe. However, the concentration distribution of solids is asymmetric. This flow pattern is the most frequently used for fast-settling slurry transport. Typical mixture velocities are in the range of 1 to 3 m/s (3.3 to 9.8 ft/s). The minimum in the pressure gradient is found to be near V_{M2} . Above V_{M1} , the particles are symmetrically distributed, and the pressure gradient curve is nearly parallel to that for the liquid by itself.

The most important transition velocity, often regarded as the minimum transport or conveying velocity for settling slurries, is V_{M2} . The Durand equation (Durand, Minnesota Int. Hydraulics Conf., *Proc.*, 89, Int. Assoc. for Hydraulic Research [1953]; Durand and Condolios, *Proc. Colloq. On the Hyd. Transport of Solids in Pipes*, Nat. Coal Board [UK], Paper IV, 39-35 [1952]) gives the minimum transport velocity as

$$V_{M2} = F_L [2gD(s-1)]^{0.5} \quad (6-145)$$

where g = acceleration of gravity
 D = pipe diameter
 $s = \rho_s/\rho_L$ = ratio of solid to liquid density
 F_L = a factor influenced by particle size and concentration

Probably F_L is a function of particle Reynolds number and concentration, but Fig. 6-33 gives Durand's empirical correlation for F_L as a function of particle diameter and the input, feed volume fraction

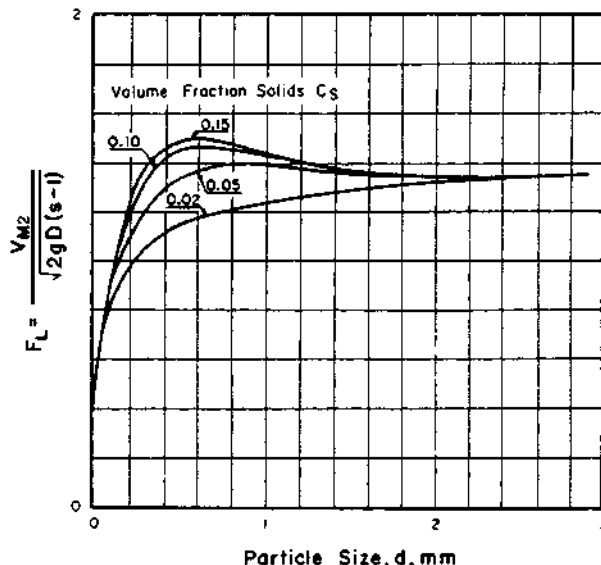


FIG. 6-33 Durand factor for minimum suspension velocity. (From Govier and Aziz, *The Flow of Complex Mixtures in Pipes*, Van Nostrand Reinhold, New York, 1972.)

solids, $C_s = Q_s/(Q_s + Q_L)$. The form of Eq. (6-145) may be derived from turbulence theory, as shown by Davies (*Chem. Eng. Sci.*, **42**, 1667-1670 [1987]).

No single correlation for **pressure drop** in horizontal solid/liquid flow has been found satisfactory for all particle sizes, densities, concentrations, and pipe sizes. However, with reference to Fig. 6-32, the following simplifications may be considered. The minimum pressure gradient occurs near V_{M2} and for conservative purposes it is generally desirable to exceed V_{M2} . When V_{M2} is exceeded, a rough guide for pressure drop is 25 percent greater than that calculated assuming that the slurry behaves as a pseudohomogeneous fluid with the density of the mixture and the viscosity of the liquid. Above the transition velocity to symmetric suspension, V_{M1} , the pressure drop closely approaches the pseudohomogeneous pressure drop. The following correlation by Spels (*Trans. Inst. Chem. Eng. [London]*, **33**, 79-84 [1955]) may be used for V_{M1} .

$$V_{M1}^2 = 0.075 \left(\frac{DV_{M1}\rho_M}{\mu} \right)^{0.775} gD_s(s-1) \quad (6-146)$$

where D = pipe diameter
 D_s = particle diameter (such that 85 percent by weight of particles are smaller than D_s)
 ρ_M = the slurry mixture density
 μ = liquid viscosity
 $s = \rho_s/\rho_L$ = ratio of solid to liquid density

Between V_{M2} and V_{M1} the concentration of solids gradually becomes more uniform in the vertical direction. This transition has been modeled by several authors as a concentration gradient where turbulent diffusion balances gravitational settling. See, for example, Karabelas (*AIChE J.*, **23**, 426-434 [1977]).

Published correlations for pressure drop are frequently very complicated and tedious to use, may not offer significant accuracy advantages over the simple guide given here, and many of them are applicable only for velocities above V_{M2} . One which does include the effect of sliding beds is due to Gaessler (Doctoral Dissertation, Technische Hochschule, Karlsruhe, Germany [1967]; reproduced by Govier and Aziz, pp. 668-669). Turian and Yuan (*AIChE J.*, **23**, 232-243 [1977]; see also Turian and Oroskar, *AIChE J.*, **24**, 1144 [1978]) segregated a large body of data into four flow regime groups

and developed empirical correlations for predicting pressure drop in each flow regime.

Pressure drop data for the flow of **paper stock** in pipes are given in the data section of *Standards of the Hydraulic Institute* (Hydraulic Institute, 1965). The flow behavior of fiber suspensions is discussed by Bobkovic and Gauvin (*Chem. Eng. Sci.*, **22**, 229–241 [1967]), Bugliarello and Daily (*TAPPI*, **44**, 881–893 [1961]), and Daily and Bugliarello (*TAPPI*, **44**, 497–512 [1961]).

In **vertical flow** of fast-settling slurries, the in situ concentration of solids with density greater than the liquid will exceed the feed concentration $C = Q_s/(Q_s + Q_L)$ for upflow and will be smaller than C for downflow. This results from slip between the phases. The **slip velocity**, the difference between the in situ average velocities of the two phases, is roughly equal to the terminal settling velocity of the solids in the liquid. Specification of the slip velocity for a pipe of a given diameter, along with the phase flow rates, allows calculation of in situ volume fractions, average velocities, and holdup ratios by simple material balances. Slip velocity may be affected by particle concentration and by turbulence conditions in the liquid. **Drift-flux theory**, a framework incorporating certain functional forms for empirical expressions for slip velocity, is described by Wallis (*One-Dimensional Two-Phase Flow*, McGraw-Hill, New York, 1969). **Minimum transport velocity** for upflow for design purposes is usually taken as twice the particle settling velocity. **Pressure drop** in vertical pipe flow includes the effects of kinetic and potential energy (elevation) changes and friction. Rose and Duckworth (*The Engineer*, **227**[5,903], 392 [1969]; **227**[5,904], 430 [1969]; **227**[5,905], 478 [1969]; see also Govier and Aziz, pp. 487–493) have developed a calculation procedure including all these effects, which may be applied not only to vertical solid/liquid flow, but also to gas/solid flow and to horizontal flow.

For fast-settling slurries, ensuring conveyance is usually the key design issue while pressure drop is somewhat less important. For **nonsettling slurries** conveyance is not an issue, because the particles do not separate from the liquid. Here, viscous and rheological behavior, which control pressure drop, take on critical importance.

Fine particles, often at high concentration, form nonsettling slurries for which useful design equations can be developed by treating them as homogeneous fluids. These fluids are usually very viscous and often non-Newtonian. Shear-thinning and Bingham plastic behavior are common; dilatancy is sometimes observed. Rheology of such fluids must in general be empirically determined, although theoretical results are available for some very limited circumstances. Further discussion of both fast-settling and nonsettling slurries may be found in Shook (in Shamlou, *Processing of Solid-Liquid Suspensions*, Chap. 11, Butterworth-Heinemann, Oxford, 1993).

FLUID DISTRIBUTION

Uniform fluid distribution is essential for efficient operation of chemical-processing equipment such as contactors, reactors, mixers, burners, heat exchangers, extrusion dies, and textile-spinning chimneys. To obtain optimum distribution, proper consideration must be given to flow behavior in the distributor, flow conditions upstream and downstream of the distributor, and the distribution requirements of the equipment. Even though the principles of fluid distribution have been well developed for more than three decades, they are frequently overlooked by equipment designers, and a significant fraction of process equipment needlessly suffers from maldistribution. In this subsection, guides for the design of various types of fluid distributors, taking into account only the flow behavior within the distributor, are given.

Perforated-Pipe Distributors The simple perforated pipe or sparger (Fig. 6-34) is a common type of distributor. As shown, the flow

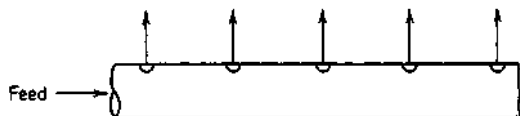


FIG. 6-34 Perforated-pipe distributor.

distribution is uniform; this is the case in which pressure recovery due to kinetic energy or momentum changes, frictional pressure drop along the length of the pipe, and pressure drop across the outlet holes have been properly considered. In typical turbulent flow applications, inertial effects associated with velocity changes may dominate frictional losses in determining the pressure distribution along the pipe, unless the length between orifices is large. Application of the momentum or mechanical energy equations in such a case shows that the pressure inside the pipe increases with distance from the entrance of the pipe. If the outlet holes are uniform in size and spacing, the discharge flow will be biased toward the closed end. Disturbances upstream of the distributor, such as pipe bends, may increase or decrease the flow to the holes at the beginning of the distributor. When frictional pressure drop dominates the inertial pressure recovery, the distribution is biased toward the feed end of the distributor.

For turbulent flow, with roughly uniform distribution, assuming a constant friction factor, the combined effect of friction and inertial (momentum) pressure recovery is given by

$$\Delta p = \left(\frac{4fL}{3D} - 2K \right) \frac{\rho V_i^2}{2} \quad (\text{discharge manifolds}) \quad (6-147)$$

where Δp = net pressure drop over the length of the distributor

L = pipe length

D = pipe diameter

f = Fanning friction factor

V_i = distributor inlet velocity

The factor K would be 1 in the case of full momentum recovery, or 0.5 in the case of negligible viscous losses in the portion of flow which remains in the pipe after the flow divides at a takeoff point (Denn, pp. 126–127). Experimental data (Van der Hegge Zijnen, *Appl. Sci. Res.*, **A3**, 144–162 [1951–1953]; and Bailey, *J. Mech. Eng. Sci.*, **17**, 338–347 [1975]), while scattered, show that K is probably close to 0.5 for discharge manifolds. For inertially dominated flows, Δp will be negative. For **return manifolds** the recovery factor K is close to 1.0, and the pressure drop between the first hole and the exit is given by

$$\Delta p = \left(\frac{4fL}{3D} + 2K \right) \frac{\rho V_e^2}{2} \quad (\text{return manifolds}) \quad (6-148)$$

where V_e is the pipe exit velocity.

One means to obtain a desired uniform distribution is to make the average pressure drop across the holes Δp_o large compared to the pressure variation over the length of pipe Δp . Then, the relative variation in pressure drop across the various holes will be small, and so will be the variation in flow. When the area of an individual hole is small compared to the cross-sectional area of the pipe, hole pressure drop may be expressed in terms of the discharge coefficient C_o and the velocity across the hole V_o as

$$\Delta p_o = \frac{1}{C_o^2} \frac{\rho V_o^2}{2} \quad (6-149)$$

Provided C_o is the same for all the holes, the **percent maldistribution**, defined as the percentage variation in flow between the first and last holes, may be estimated reasonably well for small maldistribution by (Senecal, *Ind. Eng. Chem.*, **49**, 993–997 [1957])

$$\text{Percent maldistribution} = 100 \left(1 - \sqrt{\frac{\Delta p_o - |\Delta p|}{\Delta p_o}} \right) \quad (6-150)$$

This equation shows that for 5 percent maldistribution, the pressure drop across the holes should be about 10 times the pressure drop over the length of the pipe. For discharge manifolds with $K = 0.5$ in Eq. (6-147), and with $4fL/3D \ll 1$, the pressure drop across the holes should be 10 times the inlet velocity head, $\rho V_i^2/2$ for 5 percent maldistribution. This leads to a simple design equation.

Discharge manifolds, $4fL/3D \ll 1$, 5% maldistribution:

$$\frac{V_o}{V_i} = \frac{A_p}{A_o} = \sqrt{10C_o} \quad (6-151)$$

Here A_p = pipe cross-sectional area and A_o is the *total* hole area of the distributor. Use of large hole velocity to pipe velocity ratios promotes perpendicular discharge streams. In practice, there are many cases where the $4fL/3D$ term will be less than unity but not close to zero. In such cases, Eq. (6-151) will be conservative, while Eqs. (6-147), (6-149), and (6-150) will give more accurate design calculations. In cases where $4fL/3D > 2$, friction effects are large enough to render Eq. (6-151) nonconservative. When significant variations in f along the length of the distributor occur, calculations should be made by dividing the distributor into small enough sections that constant f may be assumed over each section.

For return manifolds with $K = 1.0$ and $4fL/3D \ll 1$, 5 percent maldistribution is achieved when hole pressure drop is 20 times the pipe exit velocity head.

Return manifolds, $4fL/3D \ll 1$, 5% maldistribution:

$$\frac{V_o}{V_e} = \frac{A_p}{A_o} = \sqrt{20C_o} \quad (6-152)$$

When $4fL/3D$ is not negligible, Eq. (6-152) is not conservative and Eqs. (6-148), (6-149), and (6-150) should be used.

One common misconception is that good distribution is always provided by high pressure drop, so that increasing flow rate improves distribution by increasing pressure drop. Conversely, it is mistakenly believed that turndown of flow through a perforated pipe designed using Eqs. (6-151) and (6-152) will cause maldistribution. However, when the distribution is nearly uniform, decreasing the flow rate decreases Δp and Δp_o in the same proportion, and Eqs. (6-151) and (6-152) are still satisfied, preserving good distribution independent of flow rate, as long as friction losses remain small compared to inertial (velocity head change) effects. Conversely, increasing the flow rate through a distributor with severe maldistribution will not generally produce good distribution.

Often, the pressure drop required for design flow rate is unacceptably large for a distributor pipe designed for uniform velocity through uniformly sized and spaced orifices. Several measures may be taken in such situations. These include the following:

1. Taper the diameter of the distributor pipe so that the pipe velocity and velocity head remain constant along the pipe, thus substantially reducing pressure variation in the pipe.
2. Vary the hole size and/or the spacing between holes to compensate for the pressure variation along the pipe. This method may be sensitive to flow rate and a distributor optimized for one flow rate may suffer increased maldistribution as flow rate deviates from design rate.
3. Feed or withdraw from both ends, reducing the pipe flow velocity head and required hole pressure drop by a factor of 4.

The orifice discharge coefficient C_o is usually taken to be about 0.62. However, C_o is dependent on the ratio of hole diameter to pipe diameter, pipe wall thickness to hole diameter ratio, and pipe velocity to hole velocity ratio. As long as all these are small, the coefficient 0.62 is generally adequate.

Example 9: Pipe Distributor A 3-in schedule 40 (inside diameter 7.793 cm) pipe is to be used as a distributor for a flow of 0.010 m³/s of water ($\rho = 1,000 \text{ kg/m}^3$, $\mu = 0.001 \text{ Pa} \cdot \text{s}$). The pipe is 0.7 m long and is to have 10 holes of uniform diameter and spacing along the length of the pipe. The distributor pipe is submerged. Calculate the required hole size to limit maldistribution to 5 percent, and estimate the pressure drop across the distributor.

The inlet velocity computed from $V_i = Q/A = 4Q/(\pi D^2)$ is 2.10 m/s, and the inlet Reynolds number is

$$\text{Re} = \frac{DV_i\rho}{\mu} = \frac{0.07793 \times 2.10 \times 1,000}{0.001} = 1.64 \times 10^5$$

For commercial pipe with roughness $\epsilon = 0.046 \text{ mm}$, the friction factor is about 0.0043. Approaching the last hole, the flow rate, velocity, and Reynolds number are about one-tenth their inlet values. At $\text{Re} = 16,400$ the friction factor f is about 0.0070. Using an average value of $f = 0.0057$ over the length of the pipe, $4fL/3D$ is 0.068 and may reasonably be neglected so that Eq. (6-151) may be used. With $C_o = 0.62$,

$$\frac{V_o}{V_i} = \frac{A_p}{A_o} = \sqrt{10C_o} = \sqrt{10} \times 0.62 = 1.96$$

With pipe cross-sectional area $A_p = 0.00477 \text{ m}^2$, the total hole area is $0.00477/1.96 = 0.00243 \text{ m}^2$. The area and diameter of each hole are then

$0.00243/10 = 0.000243 \text{ m}^2$ and 1.76 cm. With $V_o/V_i = 1.96$, the hole velocity is $1.96 \times 2.10 = 4.12 \text{ m/s}$ and the pressure drop across the holes is obtained from Eq. (6-149).

$$\Delta p_o = \frac{1}{C_o^2} \frac{\rho V_o^2}{2} = \frac{1}{0.62^2} \times \frac{1,000(4.12)^2}{2} = 22,100 \text{ Pa}$$

Since the hole pressure drop is 10 times the pressure variation in the pipe, the total pressure drop from the inlet of the distributor may be taken as approximately 22,100 Pa.

Further detailed information on pipe distributors may be found in Senecal (*Ind. Eng. Chem.*, **49**, 993–997 [1957]). Much of the information on tapered manifold design has appeared in the pulp and paper literature (Spengos and Kaiser, *TAPPI*, **46**[3], 195–200 [1963]; Madeley, *Paper Technology*, **9**[1], 35–39 [1968]; Mardon, et al., *TAPPI*, **46**[3], 172–187 [1963]; Mardon, et al., *Pulp and Paper Magazine of Canada*, **72**[11], 76–81 [November 1971]; Trufitt, *TAPPI*, **58**[11], 144–145 [1975]).

Slot Distributors These are generally used in sheeting dies for extrusion of films and coatings and in air knives for control of thickness of a material applied to a moving sheet. A simple slotted pipe for turbulent flow conditions may give severe maldistribution because of nonuniform discharge velocity, but also because this type of design does not readily give perpendicular discharge (Koestel and Tuve, *Heat. Piping Air Cond.*, **20**[1], 153–157 [1948]; Senecal, *Ind. Eng. Chem.*, **49**, 993–997 [1957]; Koestel and Young, *Heat. Piping Air Cond.*, **23**[7], 111–115 [1951]). For slots in tapered ducts where the duct cross-sectional area decreases linearly to zero at the far end, the discharge angle will be constant along the length of the duct (Koestel and Young, *ibid.*). One way to ensure an almost perpendicular discharge is to have the ratio of the area of the slot to the cross-sectional area of the pipe equal to or less than 0.1. As in the case of perforated-pipe distributors, pressure variation within the slot manifold and pressure drop across the slot must be carefully considered.

In practice, the following methods may be used to keep the diameter of the pipe to a minimum consistent with good performance (Senecal, *Ind. Eng. Chem.*, **49**, 993–997 [1957]):

1. Feed from both ends.
2. Modify the cross-sectional design (Fig. 6-35); the slot is thus farther away from the influence of feed-stream velocity.
3. Increase pressure drop across the slot; this can be accomplished by lengthening the lips (Fig. 6-35).
4. Use screens (Fig. 6-35) to increase overall pressure drop across the slot.

Design considerations for air knives are discussed by Senecal (*ibid.*). Design procedures for extrusion dies when the flow is laminar, as with highly viscous fluids, are presented by Bernhardt (*Processing of Thermoplastic Materials*, Reinhold, New York, 1959, pp. 248–281).

Turning Vanes In applications such as ventilation, the discharge profile from slots can be improved by turning vanes. The tapered duct is the most amenable for turning vanes because the discharge angle remains constant. One way of installing the vanes is shown in Fig. 6-36. The vanes should have a depth twice the spacing (*Heating, Ventilating, Air Conditioning Guide*, vol. 38, American Society of Heating, Refrigerating and Air-Conditioning Engineers, 1960, pp. 282–283) and a curvature at the upstream end of the vanes of a circular arc which is tangent to the discharge angle θ of a slot without vanes and perpendicular at the downstream or discharge end of the vanes (Koestel and Young, *Heat. Piping Air Cond.*, **23**[7], 111–115 [1951]). Angle θ can be estimated from

$$\cot \theta = \frac{C_d A_s}{A_d} \quad (6-153)$$

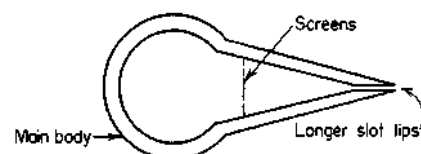


FIG. 6-35 Modified slot distributor.

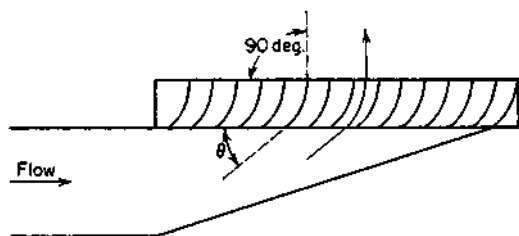


FIG. 6-36 Turning vanes in a slot distributor.

where A_s = slot area
 A_d = duct cross-sectional area at upstream end
 C_d = discharge coefficient of slot

Vanes may be used to improve velocity distribution and reduce frictional loss in bends, when the ratio of bend turning radius to pipe diameter is less than 1.0. For a miter bend with low-velocity flows, simple circular arcs (Fig. 6-37) can be used, and with high-velocity flows, vanes of special airfoil shapes are required. For additional details and references, see Ower and Pankhurst (*The Measurement of Air Flow*, Pergamon, New York, 1977, p. 102); Pankhurst and Holder (*Wind-Tunnel Technique*, Pitman, London, 1952, pp. 92–93); Rouse (*Engineering Hydraulics*, Wiley, New York, 1950, pp. 399–401); and Jorgensen (*Fan Engineering*, 7th ed., Buffalo Forge Co., Buffalo, 1970, pp. 111, 117, 118).

Perforated Plates and Screens A nonuniform velocity profile in turbulent flow through channels or process equipment can be smoothed out to any desired degree by adding sufficient uniform resistance, such as perforated plates or screens across the flow channel, as shown in Fig. 6-38. Stoker (*Ind. Eng. Chem.*, **38**, 622–624 [1946]) provides the following equation for the effect of a uniform resistance on velocity profile:

$$\frac{V_{2,\max}}{V} = \sqrt{\frac{(V_{1,\max}/V)^2 + \alpha_2 - \alpha_1 + \alpha_2 K}{1 + K}} \quad (6-154)$$

Here, V is the area average velocity, K is the number of velocity heads of pressure drop provided by the uniform resistance, $\Delta p = K\rho V^2/2$, and α is the velocity profile factor used in the mechanical energy balance, Eq. (6-13). It is the ratio of the area average of the cube of the velocity, to the cube of the area average velocity V . The shape of the exit velocity profile appears twice in Eq. (6-154), in $V_{2,\max}/V$ and α_2 . Typically, K is on the order of 10, and the desired exit velocity profile

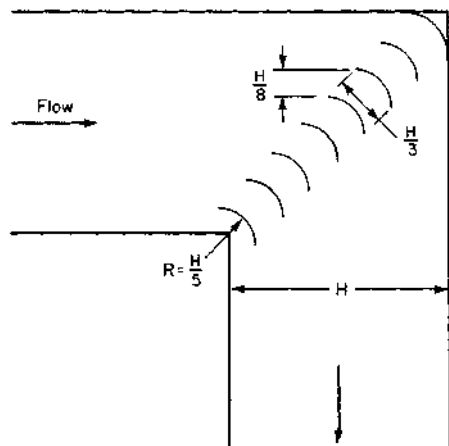


FIG. 6-37 Miter bend with vanes.

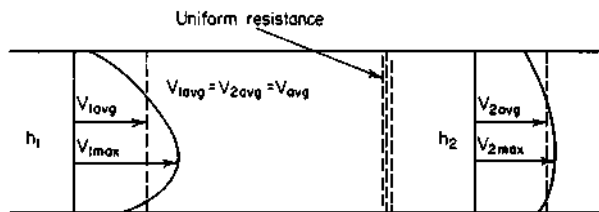


FIG. 6-38 Smoothing out a nonuniform profile in a channel.

is fairly uniform so that $\alpha_2 \sim 1.0$ may be appropriate. Downstream of the resistance, the velocity profile will gradually reestablish the fully developed profile characteristic of the Reynolds number and channel shape. The screen or perforated plate open area required to produce the resistance K may be computed from Eqs. (6-107) or (6-111).

Screens and other flow restrictions may also be used to suppress stream swirl and turbulence (Loehrke and Nagib, *J. Fluids Eng.*, **98**, 342–353 [1976]). Contraction of the channel, as in a venturi, provides further reduction in turbulence level and flow nonuniformity.

Beds of Solids A suitable depth of solids can be used as a fluid distributor. As for other types of distribution devices, a pressure drop of 10 velocity heads is typically used, here based on the superficial velocity through the bed. There are several substantial disadvantages to use of particle beds for flow distribution. Heterogeneity of the bed may actually worsen rather than improve distribution. In general, uniform flow may be found only downstream of the point in the bed where sufficient pressure drop has occurred to produce uniform flow. Therefore, inefficiency results when the bed also serves reaction or mass transfer functions, as in catalysts, adsorbents, or tower packings for gas/liquid contacting, since portions of the bed are bypassed. In the case of trickle flow of liquid downward through column packings, inlet distribution is critical since the bed itself is relatively ineffective in distributing the liquid. Maldistribution of flow through packed beds also arises when the ratio of bed diameter to particle size is less than 10 to 30.

Other Flow Straightening Devices Other devices designed to produce uniform velocity or reduce swirl, sometimes with reduced pressure drop, are available. These include both commercial devices of proprietary design and devices discussed in the literature. For pipeline flows, see the references under flow inverters and static mixing elements previously discussed in the “Incompressible Flow in Pipes and Channels” subsection. For large area changes, as at the entrance to a vessel, it is sometimes necessary to diffuse the momentum of the inlet jet discharging from the feed pipe in order to produce a more uniform velocity profile within the vessel. Methods for this application exist, but remain largely in the domain of proprietary, commercial design.

FLUID MIXING

Mixing of fluids is a discipline of fluid mechanics. Fluid motion is used to accelerate the otherwise slow processes of diffusion and conduction to bring about uniformity of concentration and temperature, blend materials, facilitate chemical reactions, bring about intimate contact of multiple phases, and so on. As the subject is too broad to cover fully, only a brief introduction and some references for further information are given here.

Several texts are available. These include Paul, Atiemo-Obeng, and Kresta (*Handbook of Industrial Mixing*, Wiley-Interscience, Hoboken N.J., 2004); Harnby, Edwards, and Nienow (*Mixing in the Process Industries*, 2d ed., Butterworths, London, 1992); Oldshue (*Fluid Mixing Technology*, McGraw-Hill, New York, 1983); Tatterson (*Fluid Mixing and Gas Dispersion in Agitated Tanks*, McGraw-Hill, New York, 1991); Uhl and Gray (*Mixing*, vols. I–III, Academic, New York, 1966, 1967, 1986); and Nagata (*Mixing: Principles and Applications*, Wiley, New York, 1975). A good overview of stirred tank agitation is given in the series of articles from *Chemical Engineering* (110–114, Dec. 8, 1975; 139–145, Jan. 5, 1976; 93–100, Feb. 2, 1976; 102–110,

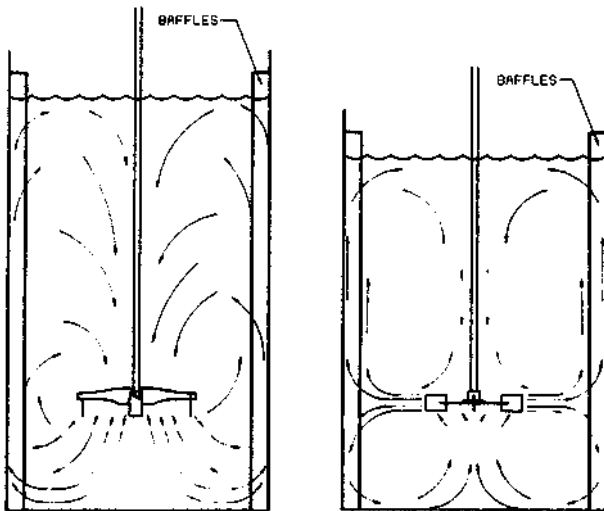


FIG. 6-39 Typical stirred tank configurations, showing time-averaged flow patterns for axial flow and radial flow impellers. (From Oldshue, *Fluid Mixing Technology*, McGraw-Hill, New York, 1983.)

Apr. 26, 1976; 144–150, May 24, 1976; 141–148, July 19, 1976; 89–94, Aug. 2, 1976; 101–108, Aug. 30, 1976; 109–112, Sept. 27, 1976; 119–126, Oct. 25, 1976; 127–133, Nov. 8, 1976).

Process mixing is commonly carried out in pipeline and vessel geometries. The terms **radial mixing** and **axial mixing** are commonly used. Axial mixing refers to mixing of materials which pass a given point at different times, and thus leads to **backmixing**. For example, backmixing or axial mixing occurs in stirred tanks where fluid elements entering the tank at different times are intermingled.

Mixing of elements initially at different axial positions in a pipeline is axial mixing. Radial mixing occurs between fluid elements passing a given point at the same time, as, for example, between fluids mixing in a pipeline tee.

Turbulent flow, by means of the chaotic eddy motion associated with velocity fluctuation, is conducive to rapid mixing and, therefore, is the preferred flow regime for mixing. **Laminar mixing** is carried out when high viscosity makes turbulent flow impractical.

Stirred Tank Agitation Turbine impeller agitators, of a variety of shapes, are used for stirred tanks, predominantly in turbulent flow. Figure 6-39 shows typical stirred tank configurations and time-averaged flow patterns for axial flow and radial flow impellers. In order to prevent formation of a **vortex**, four vertical baffles are normally installed. These cause top-to-bottom mixing and prevent mixing-ineffective swirling motion.

For a given impeller and tank geometry, the impeller Reynolds number determines the flow pattern in the tank:

$$Re_I = \frac{D^2 N \rho}{\mu} \quad (6-155)$$

where D = impeller diameter, N = rotational speed, and ρ and μ are the liquid density and viscosity. Rotational speed N is typically reported in revolutions per minute, or revolutions per second in SI units. Radians per second are almost never used. Typically, $Re_I > 10^4$ is required for fully turbulent conditions throughout the tank. A wide transition region between laminar and turbulent flow occurs over the range $10 < Re_I < 10^4$.

The power P drawn by the impeller is made dimensionless in a group called the power number:

$$N_p = \frac{P}{\rho N^3 D^5} \quad (6-156)$$

Figure 6-40 shows power number vs. impeller Reynolds number for a typical configuration. The similarity to the friction factor vs. Reynolds number behavior for pipe flow is significant. In laminar flow, the power number is inversely proportional to Reynolds number, reflecting the dominance of viscous forces over inertial forces. In

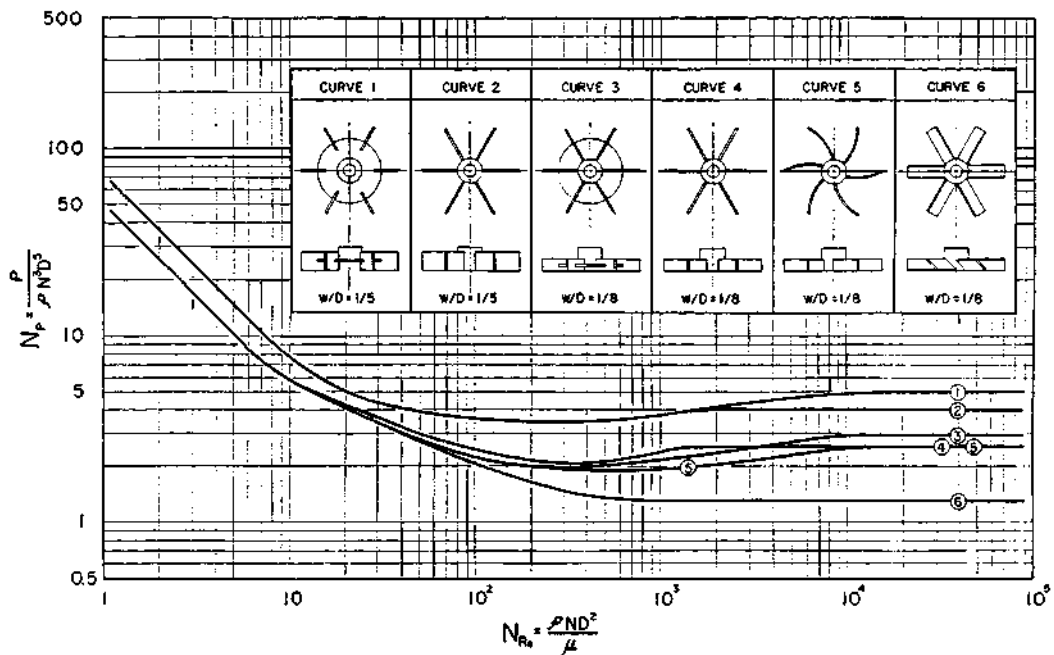


FIG. 6-40 Dimensionless power number in stirred tanks. (Reprinted with permission from Bates, Fondy, and Corpstein, *Ind. Eng. Chem. Process Design Develop.*, 2, 310 [1963].)

turbulent flow, where inertial forces dominate, the power number is nearly constant.

Impellers are sometimes viewed as pumping devices; the total volumetric flow rate Q discharged by an impeller is made dimensionless in a pumping number:

$$N_Q = \frac{Q}{ND^3} \quad (6-157)$$

Blend time t_b , the time required to achieve a specified maximum standard deviation of concentration after injection of a tracer into a stirred tank, is made dimensionless by multiplying by the impeller rotational speed:

$$N_b = t_b N \quad (6-158)$$

Dimensionless pumping number and blend time are independent of Reynolds number under fully turbulent conditions. The magnitude of concentration fluctuations from the final well-mixed value in batch mixing decays exponentially with time.

The design of mixing equipment depends on the desired process result. There is often a tradeoff between operating cost, which depends mainly on power, and capital cost, which depends on agitator size and torque. For some applications bulk flow throughout the vessel is desired, while for others high local turbulence intensity is required. Multiphase systems introduce such design criteria as solids suspension and gas dispersion. In very viscous systems, helical ribbons, extruders, and other specialized equipment types are favored over turbine agitators.

Pipeline Mixing Mixing may be carried out with **mixing tees**, **inline** or **motionless mixing elements**, or in empty pipe. In the latter case, large pipe lengths may be required to obtain adequate mixing. Coaxially injected streams require lengths on the order of 100 pipe diameters. Coaxial mixing in turbulent single-phase flow is characterized by the turbulent diffusivity (eddy diffusivity) D_E which determines the rate of radial mixing. Davies (*Turbulence Phenomena*, Academic, New York, 1972) provides an equation for D_E which may be rewritten as

$$D_E \sim 0.015 DV \text{Re}^{-0.125} \quad (6-159)$$

where D = pipe diameter
 V = average velocity
 Re = pipe Reynolds number, $DV\rho/\mu$
 ρ = density
 μ = viscosity

Properly designed tee mixers, with due consideration given to main stream and injected stream momentum, are capable of producing high degrees of uniformity in just a few diameters. Forney (*Jet Injection for Optimum Pipeline Mixing*, in "Encyclopedia of Fluid Mechanics," vol. 2., Chap. 25, Gulf Publishing, 1986) provides a thorough discussion of tee mixing. Inline or motionless mixers are generally of proprietary commercial design, and may be selected for viscous or turbulent, single or multiphase mixing applications. They substantially reduce required pipe length for mixing.

TUBE BANKS

Pressure drop across tube banks may not be correlated by means of a single, simple friction factor—Reynolds number curve, owing to the variety of tube configurations and spacings encountered, two of which are shown in Fig. 6-41. Several investigators have allowed for configuration and spacing by incorporating spacing factors in their friction factor expressions or by using multiple friction factor plots. Commercial computer codes for heat-exchanger design are available which include features for estimating pressure drop across tube banks.

Turbulent Flow The correlation by Grimison (*Trans. ASME*, **59**, 583–594 [1937]) is recommended for predicting pressure drop for turbulent flow ($\text{Re} \geq 2,000$) across staggered or in-line tube banks for tube spacings $[(a/D_t), (b/D_t)]$ ranging from 1.25 to 3.0. The pressure drop is given by

$$\Delta p = \frac{4fN_r \rho V_{\max}^2}{2} \quad (6-160)$$

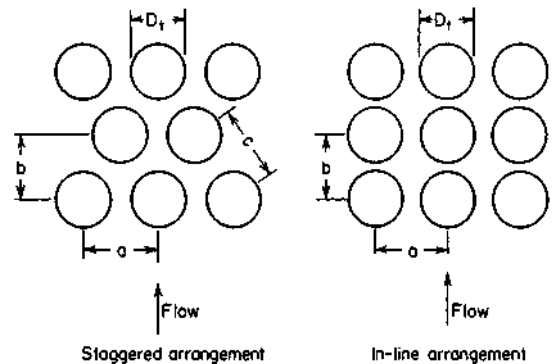


FIG. 6-41 Tube-bank configurations.

where f = friction factor

N_r = number of rows of tubes in the direction of flow

ρ = fluid density

$V_{\max}$ = fluid velocity through the minimum area available for flow

For banks of **staggered tubes**, the friction factor for isothermal flow is obtained from Fig. (6-42). Each "fence" (group of parametric curves) represents a particular Reynolds number defined as

$$\text{Re} = \frac{D_t V_{\max} \rho}{\mu} \quad (6-161)$$

where D_t = tube outside diameter and μ = fluid viscosity. The numbers along each fence represent the transverse and inflow-direction spacings. The upper chart is for the case in which the minimum area for flow is in the transverse openings, while the lower chart is for the case in which the minimum area is in the diagonal openings. In the latter case, $V_{\max}$ is based on the area of the diagonal openings and N_r is the number of rows in the direction of flow minus 1. A critical comparison of this method with all the data available at the time showed an average deviation of the order of ± 15 percent (Boucher and Lapple, *Chem. Eng. Prog.*, **44**, 117–134 [1948]). For tube spacings greater than 3 tube diameters, the correlation by Gunter and Shaw (*Trans. ASME*, **67**, 643–660 [1945]) can be used as an approximation. As an **approximation**, the pressure drop can be taken as 0.72 velocity head (based on $V_{\max}$ per row of tubes for tube spacings commonly encountered in practice (Lapple, et al., *Fluid and Particle Mechanics*, University of Delaware, Newark, 1954).

For banks of **in-line tubes**, f for isothermal flow is obtained from Fig. 6-43. Average deviation from available data is on the order of ± 15 percent. For tube spacings greater than $3D_t$, the charts of Gram, Mackey, and Monroe (*Trans. ASME*, **80**, 25–35 [1958]) can be used. As an **approximation**, the pressure drop can be taken as 0.32 velocity head (based on $V_{\max}$) per row of tubes (Lapple, et al., *Fluid and Particle Mechanics*, University of Delaware, Newark, 1954).

For turbulent flow through **shallow** tube banks, the average friction factor per row will be somewhat greater than indicated by Figs. 6-42 and 6-43, which are based on 10 or more rows depth. A 30 percent increase per row for 2 rows, 15 percent per row for 3 rows, and 7 percent per row for 4 rows can be taken as the maximum likely to be encountered (Boucher and Lapple, *Chem. Eng. Prog.*, **44**, 117–134 [1948]).

For a **single row of tubes**, the friction factor is given by Curve B in Fig. 6-44 as a function of tube spacing. This curve is based on the data of several experimenters, all adjusted to a Reynolds number of 10,000. The values should be substantially independent of Re for $1,000 < \text{Re} < 100,000$.

For **extended surfaces**, which include fins mounted perpendicularly to the tubes or spiral-wound fins, pin fins, plate fins, and so on, friction data for the specific surface involved should be used. For

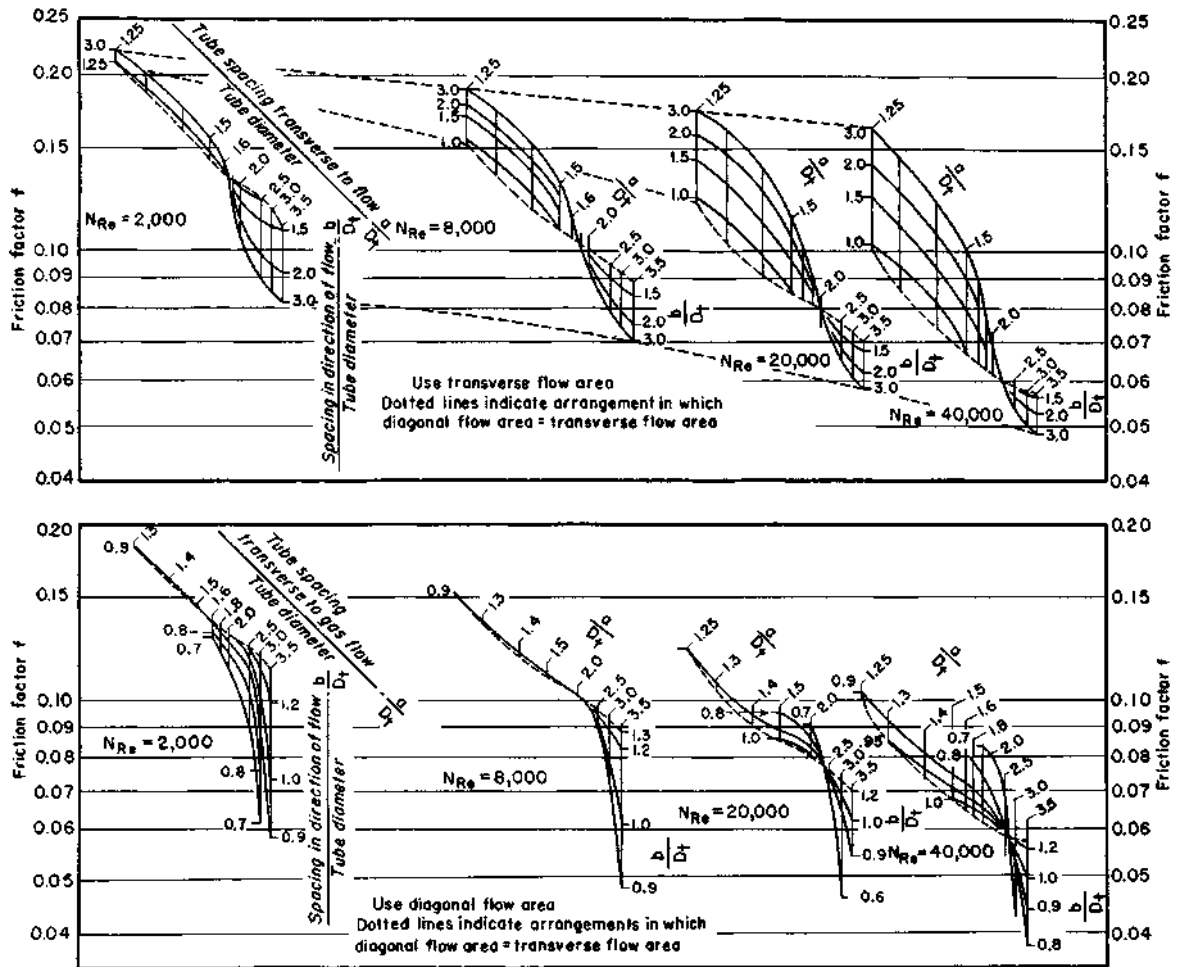


FIG. 6-42 Upper chart: Friction factors for staggered tube banks with minimum fluid flow area in transverse openings. Lower chart: Friction factors for staggered tube banks with minimum fluid flow area in diagonal openings. (From Grimson, *Trans. ASME*, 59, 583 [1937].)

details, see Kays and London (*Compact Heat Exchangers*, 2d ed., McGraw-Hill, New York, 1964). If specific data are unavailable, the correlation by Gunter and Shaw (*Trans. ASME*, 67, 643-660 [1945]) may be used as an approximation.

When a large temperature change occurs in a gas flowing across a tube bundle, gas properties should be evaluated at the mean temperature

$$T_m = T_i + K \Delta T_{lm} \quad (6-162)$$

where T_i = average tube-wall temperature

K = constant

ΔT_{lm} = log-mean temperature difference between the gas and the tubes

Values of K averaged from the recommendations of Chilton and Genereaux (*Trans. AIChE*, 29, 151-173 [1933]) and Grimson (*Trans. ASME*, 59, 583-594 [1937]) are as follows: for in-line tubes, 0.9 for cooling and -0.9 for heating; for staggered tubes, 0.75 for cooling and -0.8 for heating.

For nonisothermal flow of **liquids** across tube bundles, the friction factor is increased if the liquid is being cooled and decreased if the liquid is being heated. The factors previously given for nonisothermal

flow of liquids in pipes ("Incompressible Flow in Pipes and Channels") should be used.

For two-phase gas/liquid horizontal cross flow through tube banks, the method of Diehl and Unruh (*Pet. Refiner*, 37[10], 124-128 [1958]) is available.

Transition Region This region extends roughly over the range $200 < Re < 2,000$. Figure 6-45 taken from Bergelin, Brown, and Doberstein (*Trans. ASME*, 74, 953-960 [1952]) gives curves for friction factor f_T for five different configurations. Pressure drop for liquid flow is given by

$$\Delta p = \frac{4f_T N_r \rho V_{\max}^2}{2} \left(\frac{\mu_s}{\mu_b} \right)^{0.14} \quad (6-163)$$

where N_r = number of major restrictions encountered in flow through the bank (equal to number of rows when minimum flow area occurs in transverse openings, and to number of rows minus 1 when it occurs in the diagonal openings); ρ = fluid density; $V_{\max}$ = velocity through minimum flow area; μ_s = fluid viscosity at tube-surface temperature and μ_b = fluid viscosity at average bulk temperature. This method gives the friction factor within about ± 25 percent.

Laminar Region Bergelin, Colburn, and Hull (*Univ. Delaware Eng. Exp. Sta. Bull.*, 2 [1950]) recommend the following equations for

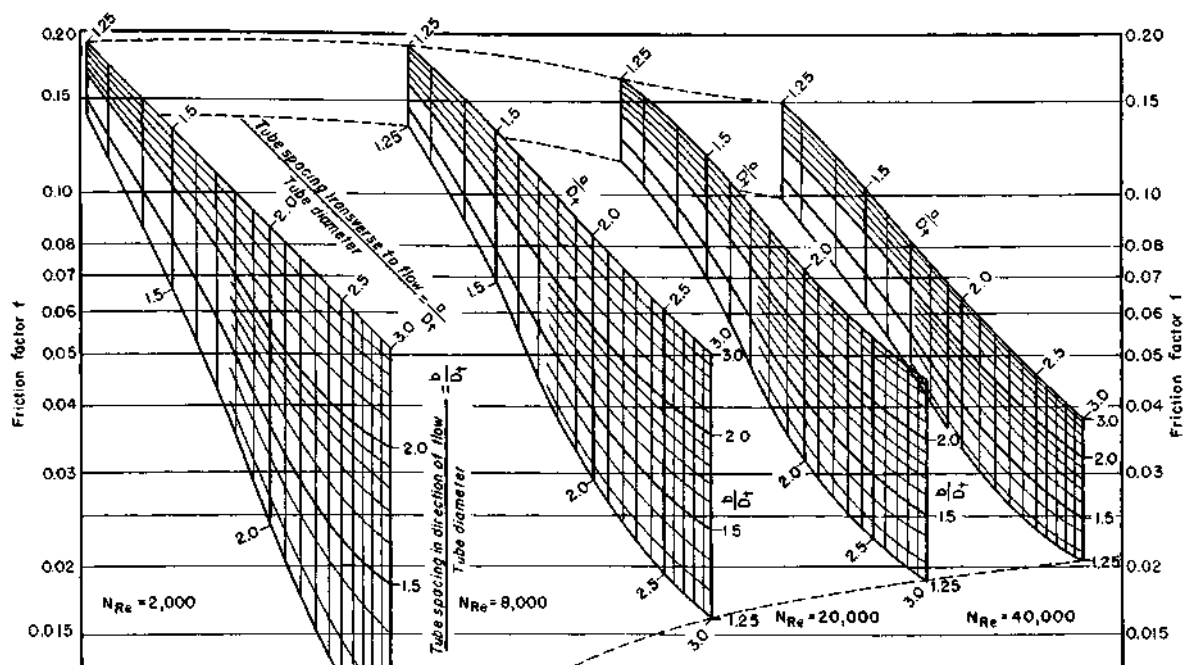


FIG. 6-43 Friction factors for in-line tube banks. (From Grimson, Trans. ASME, 59, 583 [1937].)

pressure drop with laminar flow ($Re_v < 100$) of liquids across banks of plain tubes with pitch ratios P/D_t of 1.25 and 1.50:

$$\Delta p = \frac{280N_r}{Re_v} \left(\frac{D_t}{P} \right)^{1.6} \left(\frac{\mu_s}{\mu_b} \right)^m \left(\frac{\rho V_{\max}^2}{2} \right) \quad (6-164)$$

$$m = \frac{0.57}{(Re_v)^{0.25}} \quad (6-165)$$

where $Re_v = D_v V_{\max} \rho / \mu_b$; D_v = volumetric hydraulic diameter $[(4 \times \text{free-bundle volume}) / (\text{exposed surface area of tubes})]$; P = pitch ($= a$ for in-line arrangements, $= a$ or c [whichever is smaller] for staggered

arrangements), and other quantities are as defined following Eq. (6-163). Bergelin, et al. (ibid.) show that pressure drop per row is independent of the number of rows in the bank with laminar flow. The pressure drop is predicted within about ± 25 percent.

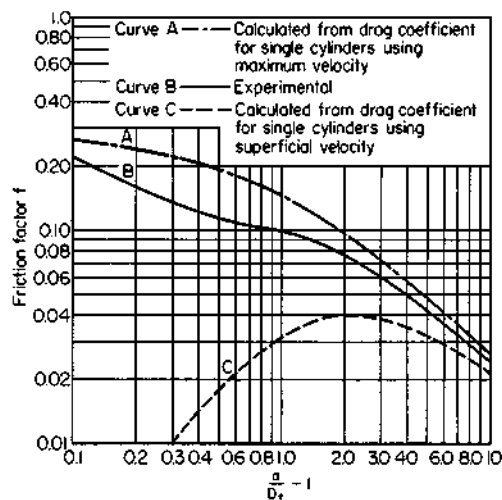
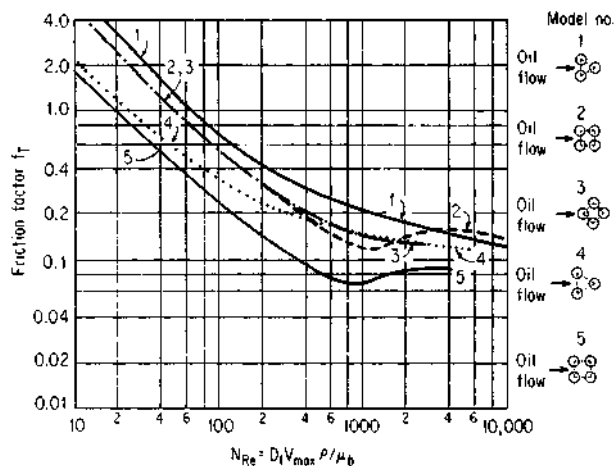


FIG. 6-44 Friction factors vs. transverse spacing for single row of tubes. (From Boucher and Lapple, Chem. Eng. Prog., 44, 117 [1948].)



Model	Rows	D_t , in	Pitch/ D_t
1	10	%	1.25
2	10	%	1.25
3	14	%	1.25
4	10	%	1.50
5	10	%	1.50

FIG. 6-45 Friction factors for transition region flow across tube banks. (Pitch is the minimum center-to-center tube spacing.) (From Bergelin, Brown, and Doberstein, Trans. ASME, 74, 953 [1952].)

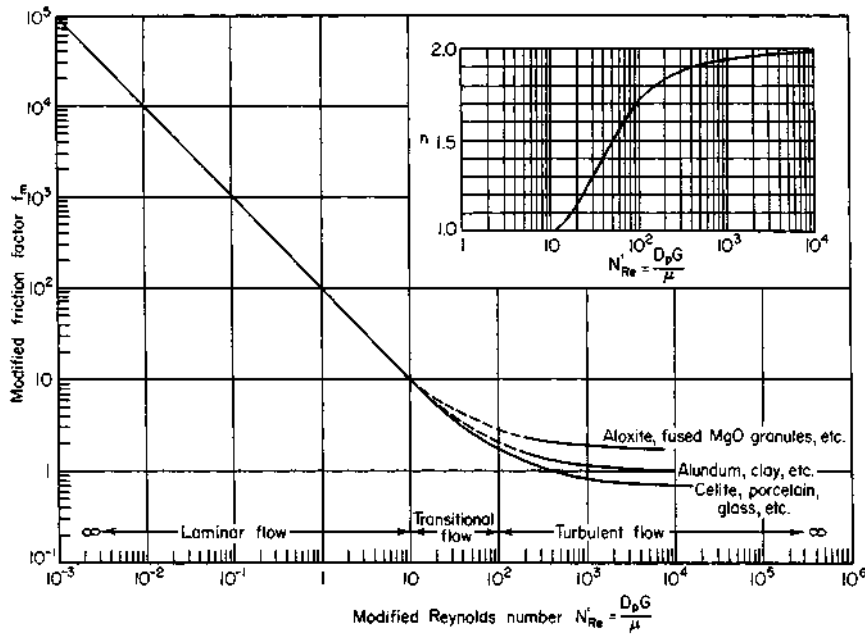


FIG. 6-46 Friction factor for beds of solids. (From Leva, *Fluidization*, McGraw-Hill, New York, 1959, p. 49.)

The validity of extrapolating Eq. (6-164) to pitch ratios larger than 1.50 is unknown. The correlation of Gunter and Shaw (*Trans. ASME*, **67**, 643–660 [1945]) may be used as an approximation in such cases.

For laminar flow of non-Newtonian fluids across tube banks, see Adams and Bell (*Chem. Eng. Prog.*, **64**, *Symp. Ser.*, **82**, 133–145 [1968]).

Flow-induced **tube vibration** occurs at critical fluid velocities through tube banks, and is to be avoided because of the severe damage that can result. Methods to predict and correct vibration problems may be found in Eisinger (*Trans. ASME J. Pressure Vessel Tech.*, **102**, 138–145 [May 1980]) and Chen (*J. Sound Vibration*, **93**, 439–455 [1984]).

BEDS OF SOLIDS

Fixed Beds of Granular Solids Pressure-drop prediction is complicated by the variety of granular materials and of their packing arrangement. For flow of a **single incompressible fluid** through an incompressible bed of granular solids, the pressure drop may be estimated by the correlation given in Fig. 6-46 (Leva, *Chem. Eng.*, **56**[5], 115–117 [1949]), or *Fluidization*, McGraw-Hill, New York, 1959). The modified friction factor and Reynolds number are defined by

$$f_m = \frac{D_p \rho \phi_s^{3-n} \epsilon^3 |\Delta p|}{2G^2 L (1 - \epsilon)^{3-n}} \quad (6-166)$$

$$Re' = \frac{D_p G}{\mu} \quad (6-167)$$

where $-\Delta p$ = pressure drop

L = depth of bed

D_p = average particle diameter, defined as the diameter of a sphere of the same volume as the particle

ϵ = void fraction

n = exponent given in Fig. 6-46 as a function of Re'

ϕ_s = shape factor defined as the area of sphere of diameter

D_p divided by the actual surface area of the particle

G = fluid superficial mass velocity based on the empty chamber cross section

ρ = fluid density

μ = fluid viscosity

As for any incompressible single-phase flow, the equivalent pressure $P = p + \rho g z$ where g = acceleration of gravity z = elevation, may be used in place of p to account for gravitational effects in flows with vertical components.

In creeping flow ($Re' < 10$),

$$f_m = \frac{100}{Re'} \quad (6-168)$$

At high Reynolds numbers the friction factor becomes nearly constant, approaching a value of the order of unity for most packed beds.

In terms of S , particle surface area per unit volume of bed,

$$D_p = \frac{6(1 - \epsilon)}{\phi_s S} \quad (6-169)$$

Porous Media Packed beds of granular solids are one type of the general class referred to as **porous media**, which include geological formations such as petroleum reservoirs and aquifers, manufactured materials such as sintered metals and porous catalysts, burning coal or char particles, and textile fabrics, to name a few. Pressure drop for incompressible flow across a porous medium has the same qualitative behavior as that given by Leva's correlation in the preceding. At low Reynolds numbers, viscous forces dominate and pressure drop is proportional to fluid viscosity and superficial velocity, and at high Reynolds numbers, pressure drop is proportional to fluid density and to the square of superficial velocity.

Creeping flow ($Re' < 1$) through porous media is often described in terms of the **permeability** k and Darcy's law:

$$\frac{-\Delta P}{L} = \frac{\mu}{k} V \quad (6-170)$$

where V = superficial velocity. The SI units for permeability are m^2 . Creeping flow conditions generally prevail in geological porous media. For **multidimensional flows** through **isotropic** porous media, the superficial velocity $\mathbf{V}$ and pressure gradient ∇P vectors replace the corresponding one-dimensional variables in Eq. (6-170).

$$\nabla P = - \frac{\mu}{k} \mathbf{V} \quad (6-171)$$

For isotropic homogeneous porous media (uniform permeability and porosity), the pressure for creeping incompressible single phase-flow may be shown to satisfy the Laplace equation:

$$\nabla^2 P = 0 \quad (6-172)$$

For **anisotropic** or **oriented** porous media, as are frequently found in geological media, permeability varies with direction and a **permeability tensor** $\mathbf{K}$, with nine components K_{ij} giving the velocity component in the i direction due to a pressure gradient in the j direction, may be introduced. For further information, see Slattery (*Momentum, Energy and Mass Transfer in Continua*, Krieger, Huntington, New York, 1981, pp. 193–218). See also Dullien (*Chem. Eng. J. [Lausanne]*, **10**, 1,034 [1975]) for a review of pressure-drop methods in single-phase flow. Solutions for Darcy's law for several geometries of interest in petroleum reservoirs and aquifers, for both incompressible and compressible flows, are given in Craft and Hawkins (*Applied Petroleum Reservoir Engineering*, Prentice-Hall, Englewood Cliffs, N.J., 1959). See also Todd (*Groundwater Hydrology*, 2nd ed., Wiley, New York, 1980).

For granular solids of **mixed sizes** the average particle diameter may be calculated as

$$\frac{1}{D_p} = \sum_i \frac{x_i}{D_{p,i}} \quad (6-173)$$

where x_i = weight fraction of particles of size $D_{p,i}$.

For **isothermal compressible flow** of a gas with constant compressibility factor Z through a packed bed of granular solids, an equation similar to Eq. (6-114) for pipe flow may be derived:

$$p_1^2 - p_2^2 = \frac{2ZRG^2T}{M_w} \left[\ln \frac{v_2}{v_1} + \frac{2f_m L(1-\epsilon)^{3-n}}{\phi_s^{3-n} \epsilon^3 D_p} \right] \quad (6-174)$$

where p_1 = upstream absolute pressure
 p_2 = downstream absolute pressure
 R = gas constant
 T = absolute temperature
 M_w = molecular weight
 v_1 = upstream specific volume of gas
 v_2 = downstream specific volume of gas

For creeping flow of **power law** non-Newtonian fluids, the method of Christopher and Middleton (*Ind. Eng. Chem. Fundam.*, **4**, 422–426 [1965]) may be used:

$$-\Delta p = \frac{150HLV^n(1-\epsilon)^2}{D_p^2 \phi_s^2 \epsilon^3} \quad (6-175)$$

$$H = \frac{K}{12} \left(9 + \frac{3}{n} \right)^n \left[\frac{D_p^2 \phi_s^2 \epsilon^4}{(1-\epsilon)^2} \right]^{(1-n)/2} \quad (6-176)$$

where $V = G/\rho$ = superficial velocity, K , n = power law material constants, and all other variables are as defined in Eq. (6-166). This correlation is supported by data from Christopher and Middleton (*ibid.*), Gregory and Griskey (*AIChE J.*, **13**, 122–125 [1967]), Yu, Wen, and Bailie (*Can. J. Chem. Eng.*, **46**, 149–154 [1968]), Siskovic, Gregory, and Griskey (*AIChE J.*, **17**, 176–187 [1978]), Kemblowski and Meril (*Chem. Eng. Sci.*, **29**, 213–223 [1974]), and Kemblowski and Dziubinski (*Rheol. Acta*, **17**, 176–187 [1978]). The measurements cover the range $n = 0.50$ to 1.60 , and modified Reynolds number $Re' = 10^{-8}$ to 10 , where

$$Re' = \frac{D_p V^{2-n} \rho}{H} \quad (6-177)$$

For the case $n = 1$ (Newtonian fluid), Eqs. (6-175) and (6-176) give a pressure drop 25 percent less than that given by Eqs. (6-166) through (6-168).

For **viscoelastic fluids** see Marshall and Metzner (*Ind. Eng. Chem. Fundam.*, **6**, 393–400 [1967]), Siskovic, Gregory, and Griskey (*AIChE J.*, **13**, 122–125 [1967]) and Kemblowski and Dziubinski (*Rheol. Acta*, **17**, 176–187 [1978]).

For gas flow through porous media with small pore diameters, the slip flow and molecular flow equations previously given (see the "Vacuum Flow" subsection) may be applied when the pore is of the same or

smaller order as the mean free path, as described by Monet and Vermeulen (*Chem. Eng. Prog.*, **55**, *Symp. Ser.*, **25** [1959]).

Tower Packings For the flow of a **single fluid** through a bed of tower packing, pressure drop may be estimated using the preceding methods. See also Sec. 14 of this *Handbook*. For **countercurrent gas/liquid flow** in commercial tower packings, both structured and unstructured, several sources of data and correlations for pressure drop and flooding are available. See, for example, Strigle (*Random Packings and Packed Towers, Design and Applications*, Gulf Publishing, Houston, 1989; *Chem. Eng. Prog.*, **89**[8], 79–83 [August 1993]), Hughmark (*Ind. Eng. Chem. Fundam.*, **25**, 405–409 [1986]), Chen (*Chem. Eng. Sci.*, **40**, 2139–2140 [1985]), Billet and Mackowiak (*Chem. Eng. Technol.*, **11**, 213–217 [1988]), Krehenwinkel and Knapp (*Chem. Eng. Technol.*, **10**, 231–242 [1987]), Mersmann and Deixler (*Ger. Chem. Eng.*, **9**, 265–276 [1986]), and Robbins (*Chem. Eng. Progr.*, **87**[5], 87–91 [May 1991]). Data and correlations for flooding and pressure drop for structured packings are given by Fair and Bravo (*Chem. Eng. Progr.*, **86**[1], 19–29 [January 1990]).

Fluidized Beds When gas or liquid flows upward through a vertically unconstrained bed of particles, there is a minimum fluid velocity at which the particles will begin to move. Above this minimum velocity, the bed is said to be **fluidized**. Fluidized beds are widely used, in part because of their excellent mixing and heat and mass transfer characteristics. See Sec. 17 of this *Handbook* for detailed information.

BOUNDARY LAYER FLOWS

Boundary layer flows are a special class of flows in which the flow far from the surface of an object is inviscid, and the effects of viscosity are manifest only in a thin region near the surface where steep velocity gradients occur to satisfy the no-slip condition at the solid surface. The thin layer where the velocity decreases from the inviscid, potential flow velocity to zero (relative velocity) at the solid surface is called the **boundary layer**. The thickness of the boundary layer is indefinite because the velocity asymptotically approaches the free-stream velocity at the outer edge. The boundary layer thickness is conventionally taken to be the distance for which the velocity equals 0.99 times the free-stream velocity. The boundary layer may be either laminar or turbulent. Particularly in the former case, the equations of motion may be simplified by scaling arguments. Schlichting (*Boundary Layer Theory*, 8th ed., McGraw-Hill, New York, 1987) is the most comprehensive source for information on boundary layer flows.

Flat Plate, Zero Angle of Incidence For flow over a wide, thin flat plate at zero angle of incidence with a uniform free-stream velocity, as shown in Fig. 6-47, the **critical Reynolds number** at which the boundary layer becomes turbulent is normally taken to be

$$Re_x = \frac{xV\rho}{\mu} = 500,000 \quad (6-178)$$

where V = free-stream velocity
 ρ = fluid density
 μ = fluid viscosity
 x = distance from leading edge of the plate

Uniform free-stream velocity

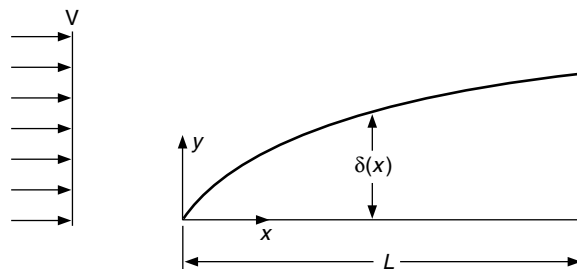


FIG. 6-47 Boundary layer on a flat plate at zero angle of incidence.

However, the transition Reynolds number depends on free-stream turbulence and may range from 3×10^5 to 3×10^6 . The **laminar boundary layer** thickness δ is a function of distance from the leading edge:

$$\delta \approx 5.0x\text{Re}_x^{-0.5} \quad (6-179)$$

The total drag on the plate of length L and width b for a laminar boundary layer, including the drag on both surfaces, is:

$$F_D = 1.328bL\rho V^2\text{Re}_L^{-0.5} \quad (6-180)$$

For **non-Newtonian power law fluids** (Acrivos, Shah, and Peterson, *AIChE J.*, **6**, 312–317 [1960]; Hsu, *AIChE J.*, **15**, 367–370 [1969]),

$$F_D = CbL\rho V^2\text{Re}_L^{-1/(1+n)} \quad (6-181)$$

$n =$	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
$C =$	2.075	1.958	1.838	1.727	1.627	1.538	1.460	1.390	1.328

where $\text{Re}_L = \rho V^{2-n}L^n/K$ and K and n are the power law material constants (see Eq. [6-4]).

For a **turbulent boundary layer**, the thickness may be estimated as

$$\delta \approx 0.37x\text{Re}_x^{-0.2} \quad (6-182)$$

and the total drag force on both sides of the plate of length L is

$$F_D = \left[\frac{0.455}{(\log \text{Re}_L)^{2.58}} - \frac{1,700}{\text{Re}_L} \right] \rho b L V^2 \quad 5 \times 10^5 < \text{Re}_L < 10^9 \quad (6-183)$$

Here the second term accounts for the laminar leading edge of the boundary layer and assumes that the critical Reynolds number is 500,000.

Cylindrical Boundary Layer Laminar boundary layers on cylindrical surfaces, with flow parallel to the cylinder axis, are described by Glauert and Lighthill (*Proc. R. Soc. [London]*, **230A**, 188–203 [1955]), Jaffe and Okamura (*Z. Angew. Math. Phys.*, **19**, 564–574 [1968]), and Stewartson (*Q. Appl. Math.*, **13**, 113–122 [1955]). For a turbulent boundary layer, the total drag may be estimated as

$$F_D = \bar{c}_f \pi r L \rho V^2 \quad (6-184)$$

where r = cylinder radius, L = cylinder length, and the average friction coefficient is given by (White, *J. Basic Eng.*, **94**, 200–206 [1972])

$$\bar{c}_f = 0.0015 + \left[0.30 + 0.015 \left(\frac{L}{r} \right)^{0.4} \right] \text{Re}_L^{-1/3} \quad (6-185)$$

for $\text{Re}_L = 10^6$ to 10^9 and $L/r < 10^6$.

Continuous Flat Surface Boundary layers on continuous surfaces drawn through a stagnant fluid are shown in Fig. 6-48. Figure 6-48a shows the continuous flat surface (Sakiadis, *AIChE J.*, **7**, 26–28,

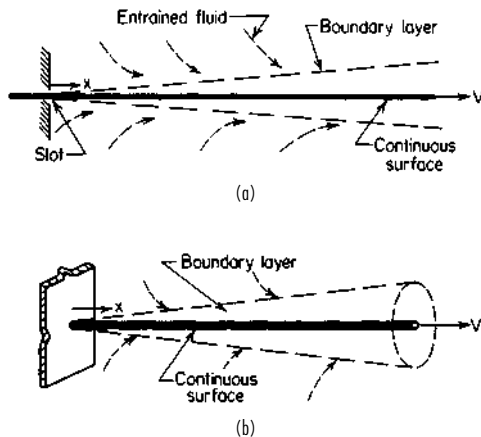


FIG. 6-48 Continuous surface: (a) continuous flat surface, (b) continuous cylindrical surface. (From Sakiadis, *Am. Inst. Chem. Eng. J.*, **7**, 221, 467 [1961].)

221–225, 467–472 [1961]). The critical Reynolds number for transition to turbulent flow may be greater than the 500,000 value for the finite flat-plate case discussed previously (Tsou, Sparrow, and Kurtz, *J. Fluid Mech.*, **26**, 145–161 [1966]). For a laminar boundary layer, the thickness is given by

$$\delta = 6.37x\text{Re}_x^{-0.5} \quad (6-186)$$

and the total drag exerted on the two surfaces is

$$F_D = 1.776bL\rho V^2\text{Re}_L^{-0.5} \quad (6-187)$$

The total flow rate of fluid entrained by the surface is

$$q = 3.232bL\rho V\text{Re}_L^{-0.5} \quad (6-188)$$

The theoretical velocity field was experimentally verified by Tsou, Sparrow, and Goldstein (*Int. J. Heat Mass Transfer*, **10**, 219–235 [1967]) and Szeri, Yates, and Hai (*J. Lubr. Technol.*, **98**, 145–156 [1976]). For **non-Newtonian power law fluids** see Fox, Erickson, and Fan (*AIChE J.*, **15**, 327–333 [1969]).

For a turbulent boundary layer, the thickness is given by

$$\delta = 1.01x\text{Re}_x^{-0.2} \quad (6-189)$$

and the total drag on both sides by

$$F_D = 0.056bL\rho V^2\text{Re}_L^{-0.2} \quad (6-190)$$

and the total entrainment by

$$q = 0.252bL\rho V\text{Re}_L^{-0.2} \quad (6-191)$$

When the laminar boundary layer is a significant part of the total length of the object, the total drag should be corrected by subtracting a calculated turbulent drag for the length of the laminar section and then adding the laminar drag for the laminar section. Tsou, Sparrow, and Goldstein (*Int. J. Heat Mass Transfer*, **10**, 219–235 [1967]) give an improved analysis of the turbulent boundary layer; their data indicate that Eq. (6-190) underestimates the drag by about 15 percent.

Continuous Cylindrical Surface The continuous surface shown in Fig. 6-48b is applicable, for example, for a wire drawn through a stagnant fluid (Sakiadis, *AIChE J.*, **7**, 26–28, 221–225, 467–472 [1961]; Vasudevan and Middleman, *AIChE J.*, **16**, 614 [1970]). The critical-length Reynolds number for transition is $\text{Re}_x = 200,000$. The laminar boundary layer thickness, total drag, and entrainment flow rate may be obtained from Fig. 6-49. The normalized boundary layer thickness and integral friction coefficient are from Vasudevan and Middleman, who used a similarity solution of the boundary layer equations. The drag force over a length x is given by

$$F_D = \bar{c}_f x \frac{\rho V^2}{2} 2\pi r_o \quad (6-192)$$

The entrainment flow rate is from Sakiadis, who used an integral momentum approximation rather than the exact similarity solution.

$$q = V\Delta \quad (6-193)$$

Further laminar boundary layer analysis is given by Crane (*Z. Angew. Math. Phys.*, **23**, 201–212 [1972]).

For a turbulent boundary layer, the total drag may be roughly estimated using Eqs. (6-184) and (6-185) for finite cylinders. Measured forces by Kwon and Prevorsek (*J. Eng. Ind.*, **101**, 73–79 [1979]) are greater than predicted this way.

The laminar boundary layer on **deforming continuous surfaces** with velocity varying with axial position is discussed by Vleggaar (*Chem. Eng. Sci.*, **32**, 1517–1525 [1977]) and Crane (*Z. Angew. Math. Phys.*, **26**, 619–622 [1975]).

VORTEX SHEDDING

When fluid flows past objects or through orifices or similar restrictions, vortices may periodically be shed downstream. Objects such as smokestacks, chemical-processing columns, suspended pipelines, and electrical transmission lines can be subjected to damaging vibrations and forces due to the vortices, especially if the shedding frequency is close to a natural vibration frequency of the object. The shedding can

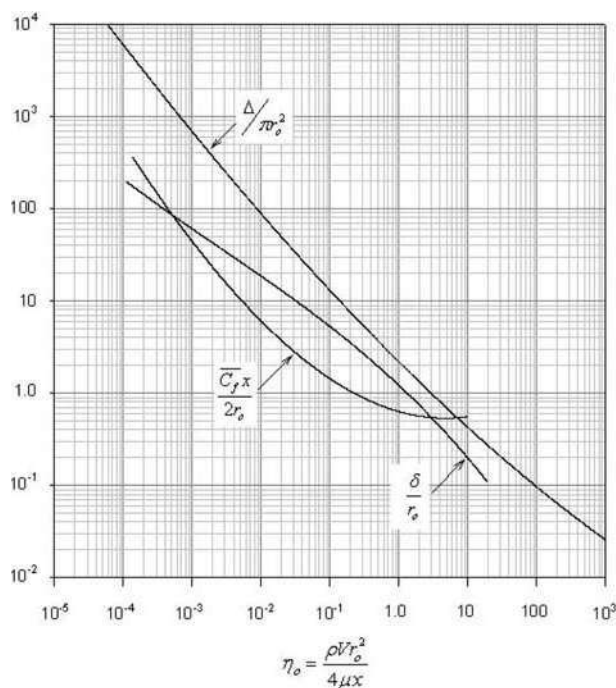


FIG 6-49 Boundary layer parameters for continuous cylindrical surfaces. ($\Delta/\pi r_o^2$ is from Sakiadis, *Am. Inst. Chem. Engr. J.*, **7**, 467 [1961]; $C_f x/2r_o$ and δ/r_o are from Vasudevan and Middleman, *Am. Inst. Chem. Eng. J.*, **16**, 614 [1970].)

also produce sound. See Krzywoblocki (*Appl. Mech. Rev.*, **6**, 393–397 [1953]) and Marris (*J. Basic Eng.*, **86**, 185–196 [1964]).

Development of a vortex street, or *von Kármán vortex street* is shown in Fig. 6-50. Discussions of the vortex street may be found in Pantón (pp. 387–393). The Reynolds number is

$$Re = \frac{DV\rho}{\mu} \quad (6-194)$$

where D = diameter of cylinder or effective width of object
 V = free-stream velocity
 ρ = fluid density
 μ = fluid viscosity

For flow past a cylinder, the vortex street forms at Reynolds numbers above about 40. The vortices initially form in the wake, the point of formation moving closer to the cylinder as Re is increased. At a Reynolds number of 60 to 100, the vortices are formed from eddies attached to the cylinder surface. The vortices move at a velocity slightly less than $\bar{V}$. The frequency of vortex shedding f is given in terms of the Strouhal number, which is approximately constant over a wide range of Reynolds numbers.

$$St \equiv \frac{fD}{V} \quad (6-195)$$

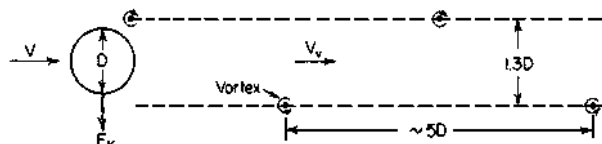


FIG. 6-50 Vortex street behind a cylinder.

For $40 < Re < 200$ the vortices are laminar and the Strouhal number has a nearly constant value of 0.2 for flow past a cylinder. Between $Re = 200$ and 400 the Strouhal number is no longer constant and the wake becomes irregular. Above about $Re = 400$ the vortices become turbulent, the wake is once again stable, and the Strouhal number remains constant at about 0.2 up to a Reynolds number of about 10^5 . Above $Re = 10^5$ the vortex shedding is difficult to see in flow visualization experiments, but velocity measurements still show a strong spectral component at $St = 0.2$ (Pantón, p. 392). Experimental data suggest that the vortex street disappears over the range $5 \times 10^5 < Re < 3.5 \times 10^6$, but is reestablished at above 3.5×10^6 (Schlichting).

Vortex shedding exerts alternating lateral forces on a cylinder, perpendicular to the flow direction. Such forces may lead to severe vibration or mechanical failure of cylindrical elements such as heat-exchanger tubes, transmission lines, stacks, and columns when the vortex shedding frequency is close to resonant bending frequency. According to Den Hartog (*Proc. Nat. Acad. Sci.*, **40**, 155–157 [1954]), the vortex shedding and cylinder vibration frequency will shift to the resonant frequency when the calculated shedding frequency is within 20 percent of the resonant frequency. The well-known Tacoma Narrows bridge collapse resulted from resonance between a torsional oscillation and vortex shedding (Pantón, p. 392). Spiral strakes are sometimes installed on tall stacks so that vortices at different axial positions are not shed simultaneously. The alternating lateral force F_K , sometimes called the *von Kármán force*, is given by (Den Hartog, *Mechanical Vibrations*, 4th ed., McGraw-Hill, New York, 1956, pp. 305–309):

$$F_K = C_K A \frac{\rho V^2}{2} \quad (6-196)$$

where C_K = von Kármán coefficient
 A = projected area perpendicular to the flow
 ρ = fluid density
 V = free-stream fluid velocity

For a cylinder, $C_K = 1.7$. For a vibrating cylinder, the effective projected area exceeds, but is always less than twice, the actual cylinder projected area (Rouse, *Engineering Hydraulics*, Wiley, New York, 1950).

The following references pertain to discussions of vortex shedding in specific structures: steel stacks (Osler and Smith, *Trans. ASME*, **78**, 1381–1391 [1956]; Smith and McCarthy, *Mech. Eng.*, **87**, 38–41 [1965]); chemical-processing columns (Freese, *J. Eng. Ind.*, **81**, 77–91 [1959]); heat exchangers (Eisinger, *Trans. ASME J. Pressure Vessel Tech.*, **102**, 138–145 [May 1980]; Chen, *J. Sound Vibration*, **93**, 439–455 [1984]; Gainsboro, *Chem. Eng. Prog.*, **64**[3], 85–88 [1968]; “Flow-Induced Vibration in Heat Exchangers,” *Symp. Proc.*, ASME, New York, 1970); suspended pipe lines (Baird, *Trans. ASME*, **77**, 797–804 [1955]); and suspended cable (Steidel, *J. Appl. Mech.*, **23**, 649–650 [1956]).

COATING FLOWS

In coating flows, liquid films are entrained on moving solid surfaces. For general discussions, see Ruschak (*Ann. Rev. Fluid Mech.*, **17**, 65–89 [1985]), Cohen and Gutoff (*Modern Coating and Drying Technology*, VCH Publishers, New York, 1992), and Middleman (*Fundamentals of Polymer Processing*, McGraw-Hill, New York, 1977). It is generally important to control the thickness and uniformity of the coatings.

In **dip coating**, or free withdrawal coating, a solid surface is withdrawn from a liquid pool, as shown in Fig. 6-51. It illustrates many of the features found in other coating flows, as well. Tallmadge and Gutfinger (*Ind. Eng. Chem.*, **59**[11], 19–34 [1967]) provide an early review of the theory of dip coating. The coating flow rate and film thickness are controlled by the withdrawal rate and the flow behavior in the meniscus region. For a withdrawal velocity V and an angle of inclination from the horizontal ϕ , the film thickness h may be

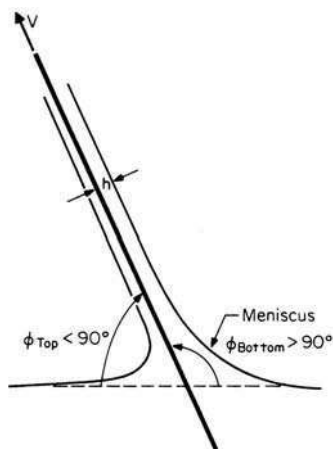


FIG. 6-51 Dip coating.

estimated for low withdrawal velocities by

$$h \left(\frac{\rho g}{\sigma} \right)^{1/2} = \frac{0.944}{(1 - \cos \phi)^{1/2}} Ca^{2/3} \quad (6-197)$$

where g = acceleration of gravity
 $Ca = \mu V / \sigma$ = capillary number
 μ = viscosity
 σ = surface tension

Equation (6-197) is asymptotically valid as $Ca \rightarrow 0$ and agrees with experimental data up to capillary numbers in the range of 0.01 to 0.03. In practice, where high production rates require high withdrawal speeds, capillary numbers are usually too large for Eq. (6-197) to apply. Approximate analytical methods for larger capillary numbers have been obtained by numerous investigators, but none appears wholly satisfactory, and some are based on questionable assumptions (Ruschak, *Ann. Rev. Fluid Mech.*, **17**, 65–89 [1985]). With the availability of high-speed computers and the development of the field of computational fluid dynamics, numerical solutions accounting for two-dimensional flow aspects, along with gravitational, viscous, inertial, and surface tension forces are now the most effective means to analyze coating flow problems.

Other common coating flows include premetered flows, such as **slide** and **curtain coating**, where the film thickness is an independent parameter that may be controlled within limits, and the curvature of the meniscus adjusts accordingly; the closely related **blade coating**; and **roll coating** and **extrusion coating**. See Ruschak (ibid.), Cohen and Gutoff (*Modern Coating and Drying Technology*, VCH Publishers, New York, 1992), and Middleman (*Fundamentals of Polymer Processing*, McGraw-Hill, New York, 1977). For dip coating of wires, see Taughy (*Int. J. Numerical Meth. Fluids*, **4**, 441–475 [1984]).

Many coating flows are subject to instabilities that lead to unacceptable coating defects. Three-dimensional flow instabilities lead to such problems as **ribbing**. Air entrainment is another common defect.

FALLING FILMS

Minimum Wetting Rate The minimum liquid rate required for complete wetting of a vertical surface is about 0.03 to 0.3 kg/m²·s for water at room temperature. The minimum rate depends on the geometry and nature of the vertical surface, liquid surface tension, and mass transfer between surrounding gas and the liquid. See Ponter, et al. (*Int. J. Heat Mass Transfer*, **10**, 349–359 [1967]; *Trans. Inst. Chem.*

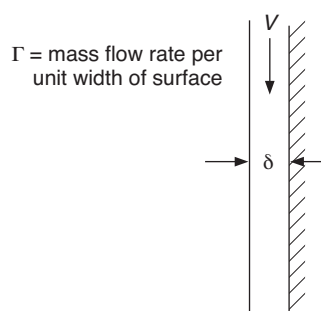


FIG. 6-52 Falling film.

Eng. [London], **45**, 345–352 [1967]), Stainthorp and Allen (*Trans. Inst. Chem. Eng. [London]*, **43**, 85–91 [1967]) and Watanabe, et al. (*J. Chem. Eng. [Japan]*, **8**[1], 75 [1975]).

Laminar Flow For films falling down **vertical flat surfaces**, as shown in Fig. 6-52, or vertical tubes with small film thickness compared to tube radius, laminar flow conditions prevail for Reynolds numbers less than about 2,000, where the Reynolds number is given by

$$Re = \frac{4\Gamma}{\mu} \quad (6-198)$$

where Γ = liquid mass flow rate per unit width of surface and μ = liquid viscosity. For a flat film surface, the following equations may be derived. The film thickness δ is

$$\delta = \left(\frac{3\Gamma\mu}{\rho^2 g} \right)^{1/3} \quad (6-199)$$

The average film velocity is

$$V = \frac{\Gamma}{\rho\delta} = \frac{g\rho\delta^2}{3\mu} \quad (6-200)$$

The downward velocity profile $u(x)$ where $x = 0$ at the solid surface and $x = \delta$ at the liquid/gas interface is given by

$$u = 1.5V \left[\frac{2x}{\delta} - \left(\frac{x}{\delta} \right)^2 \right] \quad (6-201)$$

These equations assume that there is no drag force at the gas/liquid interface, such as would be produced by gas flow. For a flat surface **inclined** at an angle θ with the horizontal, the preceding equations may be modified by replacing g by $g \sin \theta$. For films falling inside vertical tubes with film thickness up to and including the full pipe radius, see Jackson (*AIChE J.*, **1**, 231–240 [1955]).

These equations have generally given good agreement with experimental results for low-viscosity liquids (<0.005 Pa·s) (<5 cP) whereas Jackson (ibid.) found film thicknesses for higher-viscosity liquids (0.01 to 0.02 Pa·s (10 to 20 cP)) were significantly less than predicted by Eq. (6-197). At Reynolds numbers of 25 or greater, **surface waves** will be present on the liquid film. West and Cole (*Chem. Eng. Sci.*, **22**, 1388–1389 [1967]) found that the surface velocity $u(x = \delta)$ is still within ± 7 percent of that given by Eq. (6-201) even in wavy flow.

For laminar non-Newtonian film flow, see Bird, Armstrong, and Hassager (*Dynamics of Polymeric Liquids*, vol. 1: *Fluid Mechanics*, Wiley, New York, 1977, p. 215, 217), Astarita, Marrucci, and Palumbo (*Ind. Eng. Chem. Fundam.*, **3**, 333–339 [1964]) and Cheng (*Ind. Eng. Chem. Fundam.*, **13**, 394–395 [1974]).

Turbulent Flow In turbulent flow, $Re > 2,000$, for vertical surfaces, the film thickness may be estimated to within ± 25 percent using

$$\delta = 0.304 \left(\frac{\Gamma^{1.75} \mu^{0.25}}{\rho^2 g} \right)^{1/3} \quad (6-202)$$

Replace g by $g \sin \theta$ for a surface inclined at angle θ to the horizontal. The average film velocity is $V = \Gamma/\rho\delta$.

Tallmadge and Gutfinger (*Ind. Eng. Chem.*, **59**[11], 19–34 [1967]) discuss prediction of drainage rates from liquid films on flat and cylindrical surfaces.

Effect of Surface Traction If a drag is exerted on the surface of the film because of motion of the surrounding fluid, the film thickness will be reduced or increased, depending upon whether the drag acts with or against gravity. Thomas and Portalski (*Ind. Eng. Chem.*, **50**, 1081–1088 [1958]), Dukler (*Chem. Eng. Prog.*, **55**[10], 62–67 [1959]), and Kosky (*Int. J. Heat Mass Transfer*, **14**, 1220–1224 [1971]) have presented calculations of film thickness and film velocity. Film thickness data for falling water films with cocurrent and countercurrent air flow in pipes are given by Zhivaikin (*Int. Chem. Eng.*, **2**, 337–341 [1962]), Zabaras, Dukler, and Moalem-Maron (*AIChE J.*, **32**, 829–843 [1986]) and Zabaras and Dukler (*AIChE J.*, **34**, 389–396 [1988]) present studies of film flow in vertical tubes with both cocurrent and countercurrent gas flow, including measurements of film thickness, wall shear stress, wave velocity, wave amplitude, pressure drop, and flooding point for countercurrent flow.

Flooding With countercurrent gas flow, a condition is reached with increasing gas rate for which flow reversal occurs and liquid is carried upward. The mechanism for this flooding condition has been most often attributed to waves either bridging the pipe or reversing direction to flow upward at flooding. However, the results of Zabaras and Dukler (*ibid.*) suggest that flooding may be controlled by flow conditions at the liquid inlet and that wave bridging or upward wave motion does not occur, at least for the 50.8-mm diameter pipe used for their study. Flooding mechanisms are still incompletely understood. Under some circumstances, as when the gas is allowed to develop its normal velocity profile in a “calming length” of pipe beneath the liquid draw-off, the gas superficial velocity at flooding will be increased, and increases with decreasing length of wetted pipe (Hewitt, Lacy, and Nicholls, *Proc. Two-Phase Flow Symp.*, University of Exeter, paper 4H, AERE-4 4614 [1965]). A bevel cut at the bottom of the pipe with an angle 30° from the vertical will increase the flooding velocity in small-diameter tubes at moderate liquid flow rates. If the gas approaches the tube from the side, the taper should be oriented with the point facing the gas entrance. Figures 6-53 and 6-54 give correlations for flooding in tubes with square and slant bottoms (courtesy Holmes, DuPont Co.) The superficial mass velocities of gas and liquid G_G and G_L , and the physical property parameters λ and ψ are the same as those defined for the Baker chart (“Multiphase Flow” subsection, Fig. 6-25). For tubes larger than 50 mm (2 in), flooding velocity appears to be relatively insensitive to diameter and the flooding curves for 1.98-in diameter may be used.

HYDRAULIC TRANSIENTS

Many transient flows of liquids may be analyzed by using the full time-dependent equations of motion for incompressible flow. However,

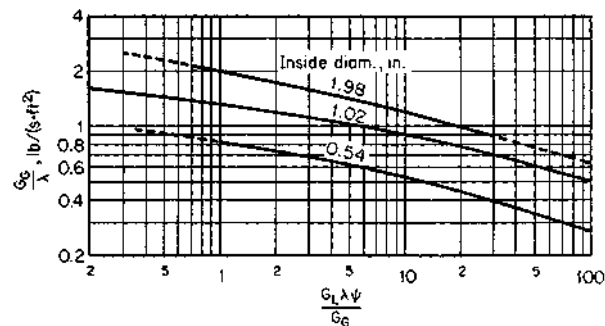


FIG. 6-53 Flooding in vertical tubes with square top and square bottom. To convert $\text{lbm}/(\text{ft}^2 \cdot \text{s})$ to $\text{kg}/(\text{m}^2 \cdot \text{s})$, multiply by 4.8824; to convert in to mm, multiply by 25.4. (Courtesy of E. I. du Pont de Nemours & Co.)

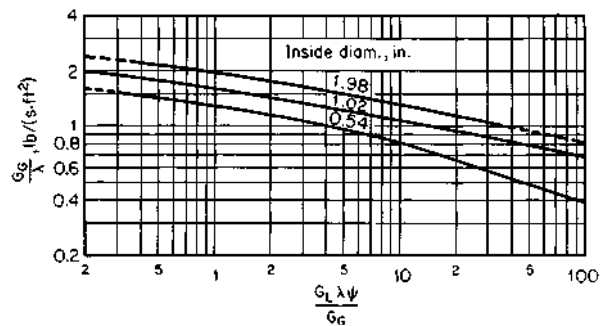


FIG. 6-54 Flooding in vertical tubes with square top and slant bottom. To convert $\text{lbm}/(\text{ft}^2 \cdot \text{s})$ to $\text{kg}/(\text{m}^2 \cdot \text{s})$, multiply by 4.8824; to convert in to mm, multiply by 25.4. (Courtesy of E. I. du Pont de Nemours & Co.)

there are some phenomena that are controlled by the small compressibility of liquids. These phenomena are generally called **hydraulic transients**.

Water Hammer When liquid flowing in a pipe is suddenly decelerated to zero velocity by a fast-closing valve, a pressure wave propagates upstream to the pipe inlet, where it is reflected; a pounding of the line commonly known as **water hammer** is often produced. For an instantaneous flow stoppage of a truly incompressible fluid in an inelastic pipe, the pressure rise would be infinite. Finite compressibility of the fluid and elasticity of the pipe limit the pressure rise to a finite value. The Joukowski formula gives the maximum pressure rise as

$$\Delta p = \rho a \Delta V \quad (6-203)$$

where ρ = liquid density
 ΔV = change in liquid velocity
 a = pressure wave velocity

The wave velocity is given by

$$a = \frac{\sqrt{\beta/\rho}}{\sqrt{1 + (\beta/E)(D/b)}} \quad (6-204)$$

where β = liquid bulk modulus of elasticity
 E = elastic modulus of pipe wall
 D = pipe inside diameter
 b = pipe wall thickness

The numerator gives the wave velocity for perfectly rigid pipe, and the denominator corrects for wall elasticity. This formula is for thin-walled pipes; for thick-walled pipes, the factor D/b is replaced by

$$2 \frac{D_o^3 + D_i^3}{D_o^3 - D_i^3}$$

where D_o = pipe outside diameter
 D_i = pipe inside diameter

Example 10: Response to Instantaneous Valve Closing Compute the wave speed and maximum pressure rise for instantaneous valve closing, with an initial velocity of 2.0 m/s, in a 4-in Schedule 40 steel pipe with elastic modulus 207×10^9 Pa. Repeat for a plastic pipe of the same dimensions, with $E = 1.4 \times 10^9$ Pa. The liquid is water with $\beta = 2.2 \times 10^9$ Pa and $\rho = 1,000$ kg/m^3 .

For the steel pipe, $D = 102.3$ mm, $b = 6.02$ mm, and the wave speed is

$$\begin{aligned} a &= \frac{\sqrt{\beta/\rho}}{\sqrt{1 + (\beta/E)(D/b)}} \\ &= \frac{\sqrt{2.2 \times 10^9/1000}}{\sqrt{1 + (2.2 \times 10^9/207 \times 10^9)(102.3/6.02)}} \\ &= 1365 \text{ m/s} \end{aligned}$$

The maximum pressure rise is

$$\begin{aligned}\Delta p &= \rho a \Delta V \\ &= 1,000 \times 1,365 \times 2.0 = 2.73 \times 10^6 \text{ Pa}\end{aligned}$$

For the plastic pipe,

$$\begin{aligned}a &= \frac{\sqrt{2.2 \times 10^9 / 1000}}{\sqrt{1 + (2.2 \times 10^9 / 1.4 \times 10^9)(102.3/6.02)}} \\ &= 282 \text{ m/s} \\ \Delta p &= \rho a \Delta V = 1,000 \times 282 \times 2.0 = 5.64 \times 10^5 \text{ Pa}\end{aligned}$$

The maximum pressure surge is obtained when the valve closes in less time than the period τ required for the pressure wave to travel from the valve to the pipe inlet and back, a total distance of $2L$.

$$\tau = \frac{2L}{a} \quad (6-205)$$

The pressure surge will be reduced when the time of flow stoppage exceeds the pipe period τ , due to cancellation between direct and reflected waves. Wood and Jones (*Proc. Am. Soc. Civ. Eng., J. Hydraul. Div.*, **99**, (HY1), 167–178 [1973]) present charts for reliable estimates of water-hammer pressure for different valve closure modes. Wylie and Streeter (*Hydraulic Transients*, McGraw-Hill, New York, 1978) describe several solution methods for hydraulic transients, including the method of characteristics, which is well suited to computer methods for accurate solutions. A rough approximation for the peak pressure for cases where the valve closure time t_c exceeds the pipe period τ is (Daugherty and Franzini, *Fluid Mechanics with Engineering Applications*, McGraw-Hill, New York, 1985):

$$\Delta p \approx \left(\frac{\tau}{t_c}\right) \rho a \Delta V \quad (6-206)$$

Successive reflections of the pressure wave between the pipe inlet and the closed valve result in alternating pressure increases and decreases, which are gradually attenuated by fluid friction and imperfect elasticity of the pipe. Periods of reduced pressure occur while the reflected pressure wave is traveling from inlet to valve. Degassing of the liquid may occur, as may vaporization if the pressure drops below the vapor pressure of the liquid. Gas and vapor bubbles decrease the wave velocity. Vaporization may lead to what is often called *liquid column separation*; subsequent collapse of the vapor pocket can result in pipe rupture.

In addition to water hammer induced by changes in valve setting, including closure, numerous other hydraulic transient flows are of interest, as, for example (Wylie and Streeter, *Hydraulic Transients*, McGraw-Hill, New York, 1978), those arising from starting or stopping of pumps; changes in power demand from turbines; reciprocating pumps; changing elevation of a reservoir; waves on a reservoir; turbine governor hunting; vibration of impellers or guide vanes in pumps, fans, or turbines; vibration of deformable parts such as valves; draft-tube instabilities due to vortexing; and unstable pump or fan characteristics. Tube failure in heat exchangers may be added to this list.

Pulsating Flow Reciprocating machinery (pumps and compressors) produces flow pulsations, which adversely affect flow meters and process control elements and can cause vibration and equipment failure, in addition to undesirable process results. Vibration and damage can result not only from the fundamental frequency of the pulse producer but also from higher harmonics. Multipiston double-acting units reduce vibrations. Pulsation dampeners are often added. Damping methods are described by M. W. Kellogg Co. (*Design of Piping Systems*, rev. 2d ed., Wiley, New York, 1965). For liquid phase pulsation damping, gas-filled surge chambers, also known as accumulators, are commonly used; see Wylie and Streeter (*Hydraulic Transients*, McGraw-Hill, New York, 1978).

Software packages are commercially available for simulation of hydraulic transients. These may be used to analyze piping systems to reveal unsatisfactory behavior, and they allow the assessment of design changes such as increases in pipe-wall thickness, changes in valve actuation, and addition of check valves, surge tanks, and pulsation dampeners.

Cavitation Loosely regarded as related to water hammer and hydraulic transients because it may cause similar vibration and equipment damage, **cavitation** is the phenomenon of collapse of vapor bubbles in flowing liquid. These bubbles may be formed anywhere the local liquid pressure drops below the vapor pressure, or they may be injected into the liquid, as when steam is sparged into water. Local low-pressure zones may be produced by local velocity increases (in accordance with the Bernoulli equation; see the preceding “Conservation Equations” subsection) as in eddies or vortices, or near boundary contours; by rapid vibration of a boundary; by separation of liquid during water hammer; or by an overall reduction in static pressure, as due to pressure drop in the suction line of a pump.

Collapse of vapor bubbles once they reach zones where the pressure exceeds the vapor pressure can cause objectionable noise and vibration and extensive erosion or pitting of the boundary materials. The critical cavitation number at inception of cavitation, denoted σ_c , is useful in correlating equipment performance data:

$$\sigma_c = \frac{p - p_v}{\rho V^2 / 2} \quad (6-207)$$

where p = static pressure in undisturbed flow
 p_v = vapor pressure
 ρ = liquid density
 V = free-stream velocity of the liquid

The value of the cavitation number for incipient cavitation for a specific piece of equipment is a characteristic of that equipment. Cavitation numbers for various head forms of cylinders, for disks, and for various hydrofoils are given by Holl and Wislicenus (*J. Basic Eng.*, **83**, 385–398 [1961]) and for various surface irregularities by Arndt and Ippen (*J. Basic Eng.*, **90**, 249–261 [1968]), Ball (*Proc. ASCE J. Constr. Div.*, **89**(C02), 91–110 [1963]), and Holl (*J. Basic Eng.*, **82**, 169–183 [1960]). As a guide only, for blunt forms the cavitation number is generally in the range of 1 to 2.5, and for somewhat streamlined forms the cavitation number is in the range of 0.2 to 0.5. Critical cavitation numbers generally depend on a characteristic length dimension of the equipment in a way that has not been explained. This renders scale-up of cavitation data questionable.

For cavitation in flow through orifices, Fig. 6-55 (Thorpe, *Int. J. Multiphase Flow*, **16**, 1023–1045 [1990]) gives the critical cavitation number for inception of cavitation. To use this cavitation number in Eq. (6-207), the pressure p is the orifice backpressure downstream of the vena contracta after full pressure recovery, and V is the average velocity through the orifice. Figure 6-55 includes data from Tullis and Govindarajan (*ASCE J. Hydraul. Div.*, **HY13**, 417–430 [1973]) modified to use the same cavitation number definition; their data also include critical cavitation numbers for 30.50- and 59.70-cm pipes

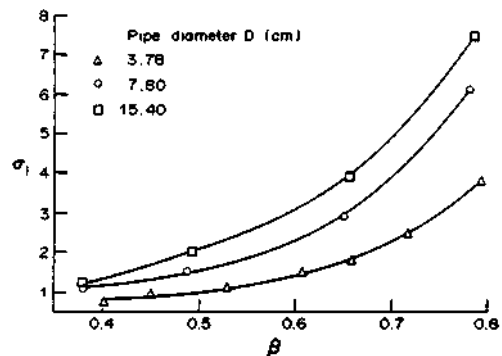


FIG. 6-55 Critical cavitation number vs. diameter ratio β . (Reprinted from Thorpe, “Flow regime transitions due to cavitation in the flow through an orifice,” *Int. J. Multiphase Flow*, **16**, 1023–1045. Copyright © 1990, with kind permission from Elsevier Science, Ltd., The Boulevard, Langford Lane, Kidlington OX5 1GB, United Kingdom.)

(12.00- to 23.50-in). Very roughly, compared with the 15.40-cm pipe, the cavitation number is about 20 percent greater for the 30.50-cm (12.01-in) pipe and about 40 percent greater for the 59.70-cm (23.50-in) diameter pipe. Inception of cavitation appears to be related to release of dissolved gas and not merely vaporization of the liquid. For further discussion of cavitation, see Eisenberg and Tulin (Streeter, *Handbook of Fluid Dynamics*, Sec. 12, McGraw-Hill, New York, 1961).

TURBULENCE

Turbulent flow occurs when the Reynolds number exceeds a critical value above which laminar flow is unstable; the critical Reynolds number depends on the flow geometry. There is generally a transition regime between the critical Reynolds number and the Reynolds number at which the flow may be considered fully turbulent. The transition regime is very wide for some geometries. In turbulent flow, variables such as velocity and pressure fluctuate chaotically; statistical methods are used to quantify turbulence.

Time Averaging In turbulent flows it is useful to define time-averaged and fluctuation values of flow variables such as velocity components. For example, the x -component velocity fluctuation v'_x is the difference between the actual instantaneous velocity v_x and the time-averaged velocity $\bar{v}_x$:

$$v'_x(x, y, z, t) = v_x(x, y, z, t) - \bar{v}_x(x, y, z) \quad (6-208)$$

The actual and fluctuating velocity components are, in general, functions of the three spatial coordinates $x, y,$ and z and of time t . The time-averaged velocity $\bar{v}_x$ is independent of time for a **stationary** flow. Nonstationary processes may be considered where averages are defined over time scales long compared to the time scale of the turbulent fluctuations, but short compared to longer time scales over which the time-averaged flow variables change due, for example, to time-varying boundary conditions. The time average over a time interval $2T$ centered at time t of a turbulently fluctuating variable $\zeta(t)$ is defined as

$$\bar{\zeta}(t) = \frac{1}{2T} \int_{t-T}^{t+T} \zeta(\tau) d\tau \quad (6-209)$$

where τ = dummy integration variable. For stationary turbulence, $\bar{\zeta}$ does not vary with time.

$$\bar{\zeta} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{t-T}^{t+T} \zeta(\tau) d\tau \quad (6-210)$$

The time average of a fluctuation $\bar{\zeta}' = \bar{\zeta} - \bar{\zeta} = 0$. Fluctuation magnitudes are quantified by root mean squares.

$$\bar{v}'_x = \sqrt{(\bar{v}'_x)^2} \quad (6-211)$$

In **isotropic** turbulence, statistical measures of fluctuations are equal in all directions.

$$\bar{v}'_x = \bar{v}'_y = \bar{v}'_z \quad (6-212)$$

In **homogeneous** turbulence, turbulence properties are independent of spatial position. The **kinetic energy of turbulence** k is given by

$$k = \frac{1}{2} (\bar{v}'_x^2 + \bar{v}'_y^2 + \bar{v}'_z^2) \quad (6-213)$$

Turbulent velocity fluctuations ultimately dissipate their kinetic energy through viscous effects. Macroscopically, this energy dissipation requires pressure drop, or velocity decrease. The **energy dissipation rate** per unit mass is usually denoted ϵ . For steady flow in a pipe, the average energy dissipation rate per unit mass is given by

$$\epsilon = \frac{2fV^3}{D} \quad (6-214)$$

where ρ = fluid density
 f = Fanning friction factor
 D = pipe inside diameter

When the continuity equation and the Navier-Stokes equations for incompressible flow are time averaged, equations for the time-averaged

velocities and pressures are obtained which appear identical to the original equations (6-18 through 6-28), except for the appearance of additional terms in the Navier-Stokes equations. Called **Reynolds stress** terms, they result from the nonlinear effects of momentum transport by the velocity fluctuations. In each i -component ($i = x, y, z$) Navier-Stokes equation, the following additional terms appear on the right-hand side:

$$\sum_{j=1}^3 \frac{\partial \tau_{ji}^{(t)}}{\partial x_j}$$

with j components also being x, y, z . The Reynolds stresses are given by

$$\tau_{ji}^{(t)} = -\rho \bar{v}'_i v'_j \quad (6-215)$$

The Reynolds stresses are nonzero because the velocity fluctuations in different coordinate directions are correlated so that $\bar{v}'_i v'_j$ in general is nonzero.

Although direct numerical simulations under limited circumstances have been carried out to determine (unaveraged) fluctuating velocity fields, in general the solution of the equations of motion for turbulent flow is based on the time-averaged equations. This requires semi-empirical models to express the Reynolds stresses in terms of time-averaged velocities. This is the **closure** problem of turbulence. In all but the simplest geometries, numerical methods are required.

Closure Models Many closure models have been proposed. A few of the more important ones are introduced here. Many employ the Boussinesq approximation, simplified here for incompressible flow, which treats the Reynolds stresses as analogous to viscous stresses, introducing a scalar quantity called the turbulent or eddy viscosity μ_t .

$$-\rho \bar{v}'_i v'_j = \mu_t \left(\frac{\partial \bar{v}_i}{\partial x_j} + \frac{\partial \bar{v}_j}{\partial x_i} \right) \quad (6-216)$$

An additional **turbulence pressure** term equal to $-2/3 k \delta_{ij}$, where k = turbulent kinetic energy and $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$, is sometimes included in the right-hand side. To solve the equations of motion using the Boussinesq approximation, it is necessary to provide equations for the single scalar unknown μ_t (and k , if used) rather than the nine unknown tensor components $\tau_{ji}^{(t)}$. With this approximation, and using the effective viscosity $\mu_{\text{eff}} = \mu + \mu_t$, the time-averaged momentum equation is similar to the original Navier-Stokes equation, with time-averaged variables and μ_{eff} replacing the instantaneous variables and molecular viscosity. However, solutions to the time-averaged equations for turbulent flow are not identical to those for laminar flow because μ_{eff} is not a constant.

The universal turbulent velocity profile near the pipe wall presented in the preceding subsection "Incompressible Flow in Pipes and Channels" may be developed using the Prandtl mixing length approximation for the eddy viscosity,

$$\mu_t = \rho l_p^2 \left| \frac{d\bar{v}_x}{dy} \right| \quad (6-217)$$

where l_p is the Prandtl mixing length. The turbulent core of the universal velocity profile is obtained by assuming that the mixing length is proportional to the distance from the wall. The proportionality constant is one of two constants adjusted to fit experimental data.

The Prandtl mixing length concept is useful for shear flows parallel to walls, but is inadequate for more general three-dimensional flows. A more complicated semiempirical model commonly used in numerical computations, and found in most commercial software for computational fluid dynamics (CFD; see the following subsection), is the k - ϵ model described by Launder and Spalding (*Lectures in Mathematical Models of Turbulence*, Academic, London, 1972). In this model the eddy viscosity is assumed proportional to the ratio k^2/ϵ .

$$\mu_t = \rho C_\mu \frac{k^2}{\epsilon} \quad (6-218)$$

where the value $C_\mu = 0.09$ is normally used. Semiempirical partial differential conservation equations for k and ϵ derived from the

Navier-Stokes equations with simplifying closure assumptions are coupled with the equations of continuity and momentum:

$$\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_i}(\rho \bar{v}_i k) = \frac{\partial}{\partial x_i} \left(\frac{\mu_t}{\sigma_k} \frac{\partial k}{\partial x_i} \right) + \mu_t \left(\frac{\partial \bar{v}_i}{\partial x_j} + \frac{\partial \bar{v}_j}{\partial x_i} \right) \frac{\partial \bar{v}_i}{\partial x_j} - \rho \epsilon \quad (6-219)$$

$$\frac{\partial}{\partial t}(\rho \epsilon) + \frac{\partial}{\partial x_i}(\rho \bar{v}_i \epsilon) = \frac{\partial}{\partial x_i} \left(\frac{\mu_t}{\sigma_\epsilon} \frac{\partial \epsilon}{\partial x_i} \right) + C_{1\epsilon} \frac{\epsilon \mu_t}{k} \left(\frac{\partial \bar{v}_i}{\partial x_j} + \frac{\partial \bar{v}_j}{\partial x_i} \right) \frac{\partial \bar{v}_i}{\partial x_j} - C_{2\epsilon} \frac{\rho \epsilon^2}{k} \quad (6-220)$$

In these equations summations over repeated indices are implied. The values for the empirical constants $C_{1\epsilon} = 1.44$, $C_{2\epsilon} = 1.92$, $\sigma_k = 1.0$, and $\sigma_\epsilon = 1.3$ are widely accepted (Lauder and Spaulding, *The Numerical Computation of Turbulent Flows*, Imperial Coll. Sci. Tech. London, NTIS N74-12066 [1973]). The k - ϵ model has proved reasonably accurate for many flows without highly curved streamlines or significant swirl. It usually underestimates flow separation and overestimates turbulence production by normal straining. The k - ϵ model is suitable for high Reynolds number flows. See Virendra, Patel, Rodi, and Scheuerer (AIAA J., **23**, 1308–1319 [1984]) for a review of low Reynolds number k - ϵ models.

More advanced models, more complex and computationally intensive, are being developed. For example, the renormalization group theory (Yakhot and Orszag, *J. Scientific Computing*, **1**, 1–51 [1986]; Yakhot, Orszag, Thangam, Gatski, and Speziale, *Phys. Fluids A*, **4**, 1510–1520 [1992]) modification of the k - ϵ model provides theoretical values of the model constants and provides substantial improvement in predictions of flows with stagnation, separation, normal straining, transient behavior such as vortex shedding, and relaminarization. Stress transport models provide equations for all nine Reynolds stress components, rather than introducing eddy viscosity. Algebraic closure equations for the Reynolds stresses are available, but are no longer in common use. Differential Reynolds stress models (e.g., Launder, Reece, and Rodi, *J. Fluid Mech.*, **68**, 537–566 [1975]) use differential conservation equations for all nine Reynolds stress components.

In **direct numerical simulation** of turbulent flows, the solution of the unaveraged equations of motion is sought. Due to the extreme computational intensity, solutions to date have been limited to relatively low Reynolds numbers in simple geometries. Since computational grids must be sufficiently fine to resolve even the smallest eddies, the computational difficulty rapidly becomes prohibitive as Reynolds number increases. **Large eddy simulations** use models for subgrid turbulence while solving for larger-scale fluctuations.

Eddy Spectrum The energy that produces and sustains turbulence is extracted from velocity gradients in the mean flow, principally through vortex stretching. At Reynolds numbers well above the critical value there is a wide spectrum of eddy sizes, often described as a cascade of energy from the largest down to the smallest eddies. The largest eddies are of the order of the equipment size. The smallest are those for which viscous forces associated with the eddy velocity fluctuations are of the same order as inertial forces, so that turbulent fluctuations are rapidly damped out by viscous effects at smaller length scales. Most of the turbulent kinetic energy is contained in the larger eddies, while most of the dissipation occurs in the smaller eddies. Large eddies, which extract energy from the mean flow velocity gradients, are generally anisotropic. At smaller length scales, the directionality of the mean flow exerts less influence, and **local isotropy** is approached. The range of eddy scales for which local isotropy holds is called the **equilibrium range**.

Davies (*Turbulence Phenomena*, Academic, New York, 1972) presents a good discussion of the spectrum of eddy lengths for well-developed isotropic turbulence. The smallest eddies, usually called **Kolmogorov eddies** (Kolmogorov, *Compt. Rend. Acad. Sci. URSS*, **30**, 301; **32**, 16 [1941]), have a characteristic velocity fluctuation $\bar{v}_k$ given by

$$\bar{v}_k = (\nu \epsilon)^{1/4} \quad (6-221)$$

where ν = kinematic viscosity and ϵ = energy dissipation per unit mass. The size of the Kolmogorov eddy scale is

$$l_k = (\nu^3/\epsilon)^{1/4} \quad (6-222)$$

The Reynolds number for the Kolmogorov eddy, $Re_k = l_k \bar{v}_k/\nu$, is equal to unity by definition. In the equilibrium range, which exists for well-developed turbulence and extends from the medium eddy sizes down to the smallest, the energy dissipation at the smaller length scales is supplied by turbulent energy drawn from the bulk flow and passed down the spectrum of eddy lengths according to the scaling rule

$$\epsilon = \frac{(\bar{v}')^3}{l} \quad (6-223)$$

which is consistent with Eqs. (6-221) and (6-222). For the medium, or energy-containing, eddy size,

$$\epsilon = \frac{(\bar{v}_e')^3}{l_e} \quad (6-224)$$

For turbulent pipe flow, the friction velocity $u_* = \sqrt{\tau_w/\rho}$ used earlier in describing the universal turbulent velocity profile may be used as an estimate for $\bar{v}_e'$. Together with the Blasius equation for the friction factor from which ϵ may be obtained (Eq. 6-214), this provides an estimate for the energy-containing eddy size in turbulent pipe flow:

$$l_e = 0.05 D Re^{-1/8} \quad (6-225)$$

where D = pipe diameter and Re = pipe Reynolds number. Similarly, the Kolmogorov eddy size is

$$l_k = 4 D Re^{-0.78} \quad (6-226)$$

Most of the energy dissipation occurs on a length scale about 5 times the Kolmogorov eddy size. The characteristic fluctuating velocity for these energy-dissipating eddies is about 1.7 times the Kolmogorov velocity.

The eddy spectrum is normally described using Fourier transform methods; see, for example, Hinze (*Turbulence*, McGraw-Hill, New York, 1975), and Tennekes and Lumley (*A First Course in Turbulence*, MIT Press, Cambridge, 1972). The spectrum $E(\kappa)$ gives the fraction of turbulent kinetic energy contained in eddies of wavenumber between κ and $\kappa + d\kappa$, so that $k = \int_0^\infty E(\kappa) d\kappa$. The portion of the equilibrium range excluding the smallest eddies, those which are affected by dissipation, is the **inertial subrange**. The Kolmogorov law gives $E(\kappa) \propto \kappa^{-5/3}$ in the inertial subrange.

Several texts are available for further reading on turbulent flow, including Pope (*Turbulent Flows*, Cambridge University Press, Cambridge, U.K., 2000), Tennekes and Lumley (ibid.), Hinze (*Turbulence*, McGraw-Hill, New York, 1975), Landau and Lifshitz (*Fluid Mechanics*, 2d ed., Chap. 3, Pergamon, Oxford, 1987) and Pantano (*Incompressible Flow*, Wiley, New York, 1984).

COMPUTATIONAL FLUID DYNAMICS

Computational fluid dynamics (CFD) emerged in the 1980s as a significant tool for fluid dynamics both in research and in practice, enabled by rapid development in computer hardware and software. Commercial CFD software is widely available. Computational fluid dynamics is the numerical solution of the equations of continuity and momentum (Navier-Stokes equations for incompressible Newtonian fluids) along with additional conservation equations for energy and material species in order to solve problems of nonisothermal flow, mixing, and chemical reaction.

Textbooks include Fletcher (*Computational Techniques for Fluid Dynamics*, vol. 1: *Fundamental and General Techniques*, and vol. 2: *Specific Techniques for Different Flow Categories*, Springer-Verlag, Berlin, 1988), Hirsch (*Numerical Computation of Internal and External Flows*, vol. 1: *Fundamentals of Numerical Discretization*, and vol. 2: *Computational Methods for Inviscid and Viscous Flows*, Wiley, New York, 1988), Peyret and Taylor (*Computational Methods for Fluid*

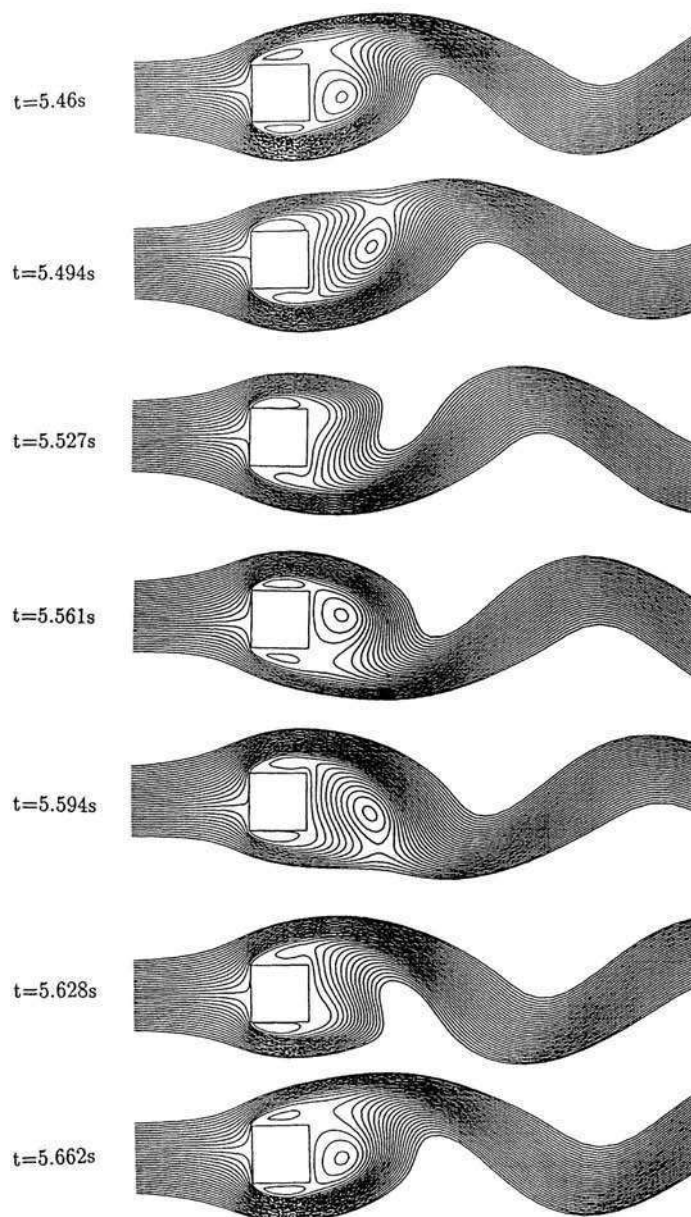


FIG. 6-56 Computational fluid dynamic simulation of flow over a square cylinder, showing one vortex shedding period. (From Choudhury *et al.*, Trans. ASME Fluids Div., TN-076 [1994].)

Flow, Springer-Verlag, Berlin, 1990), Canuto, Hussaini, Quarteroni, and Zang (*Spectral Methods in Fluid Dynamics*, Springer-Verlag, Berlin, 1988), Anderson, Tannehill, and Pletcher (*Computational Fluid Mechanics and Heat Transfer*, Hemisphere, New York, 1984), and Patankar (*Numerical Heat Transfer and Fluid Flow*, Hemisphere, Washington, D.C., 1980).

A wide variety of numerical methods has been employed, but three basic steps are common.

1. **Subdivision or discretization of the flow domain into cells or elements.** There are methods, called *boundary element methods*, in which the surface of the flow domain, rather than the volume, is discretized, but the vast majority of CFD work uses volume discretization. Discretization produces a set of **grid** lines or curves which define a **mesh** and a set of **nodes** at which the flow variables are to be calculated. The equations of motion are solved approximately on a domain defined by the grid. Curvilinear or **body-fitted** coordinate system grids may be used to ensure that the discretized domain accurately represents the true problem domain.

2. **Discretization of the governing equations.** In this step, the exact partial differential equations to be solved are replaced by approximate algebraic equations written in terms of the nodal values of the dependent variables. Among the numerous discretization methods, **finite difference**, **finite volume**, and **finite element** methods are the most common. The *finite difference* method estimates spatial derivatives in terms of the nodal values and spacing between nodes. The governing equations are then written in terms of the nodal unknowns at each interior node. *Finite volume* methods, related to finite difference methods, may be derived by a volume integration of the equations of motion, with application of the divergence theorem, reducing by one the order of the differential equations. Equivalently, macroscopic balance equations are written on each cell. *Finite element* methods are weighted residual techniques in which the unknown dependent variables are expressed in terms of **basis functions** interpolating among the nodal values. The basis functions are substituted into the equations of motion, resulting in error residuals which are multiplied by the weighting functions, integrated over the control volume, and set to zero to produce algebraic equations in terms of the nodal unknowns. Selection of the weighting functions defines the various finite element methods. For example, Galerkin's method uses the nodal interpolation basis functions as weighting functions. Each method also has its own method for implementing **boundary conditions**. The end result after discretization of the equations and application of the boundary conditions is a set of algebraic equations for the nodal unknown variables. Discretization in time is also required for the $\partial/\partial t$ time derivative terms in unsteady flow; finite differencing in time is often used. The discretized equations represent an approximation of the exact equations, and their solution gives an approximation for the flow variables. The accuracy of the solution improves as the grid is **refined**; that is, as the number of nodal points is increased.

3. **Solution of the algebraic equations.** For creeping flows with constant viscosity, the algebraic equations are linear and a linear matrix equation is to be solved. Both direct and iterative solvers have been used. For most flows, the nonlinear inertial terms in the momentum equation are important and the algebraic discretized equations are therefore nonlinear. Solution yields the nodal values of the unknowns.

A CFD method called the lattice Boltzmann method is based on modeling the fluid as a set of particles moving with discrete velocities on a discrete grid or lattice, rather than on discretization of the governing continuum partial differential equations. Lattice Boltzmann approximations can be constructed that give the same macroscopic behavior as the Navier-Stokes equations. The method is currently used mainly in academic and research codes, rather than in general-purpose commercial CFD codes. There appear to be significant computational advantages to the lattice Boltzmann method. Lattice Boltzmann simulations incorporating turbulence models, and of multiphase flows and flows with heat transfer, species diffusion, and reaction, have been carried out. For a review of the method, see Chen and Doolen [*Ann. Rev. Fluid Mech.*, **30**, 329 (1998)].

CFD solutions, especially for complex three-dimensional flows, generate very large quantities of solution data. Computer graphics have greatly improved the ability to examine CFD solutions and visualize flow.

CFD methods are used for incompressible and compressible, creeping, laminar and turbulent, Newtonian and non-Newtonian, and isothermal and nonisothermal flows. Chemically reacting flows, particularly in the field of combustion, have been simulated. Solution accuracy must be considered from several perspectives. These include convergence of the algorithms for solving the nonlinear discretized equations and convergence with respect to refinement of the mesh so that the discretized equations better approximate the exact equations and, in some cases, so that the mesh more accurately fits the true geometry. The possibility that steady-state solutions are unstable must always be considered. In addition to numerical sources of error, modeling errors are introduced in turbulent flow, where semiempirical closure models are used to solve time-averaged equations of motion, as discussed previously. Most commercial CFD codes include the $k-\epsilon$ turbulence model, which has been by far the most widely used. More accurate models, such as differential Reynolds stress and renormalization group theory models, are also becoming available. Significant solution error is known to result in some problems from inadequacy of the turbulence model. Closure models for nonlinear chemical reaction source terms may also contribute to inaccuracy.

Large eddy simulation (LES) methods for turbulent flow are available in some commercial CFD codes. LES methods are based on filtering fluctuating variables, so that lower-frequency eddies, with scales larger than the grid spacing, are resolved, while higher-frequency eddies, the subgrid fluctuations, are filtered out. The subgrid-scale Reynolds stress is estimated by a turbulence model. The Smagorinsky model, a one-equation mixing length model, is used in most commercial codes that offer LES options and is also used in many academic and research CFD codes. See Wilcox (*Turbulence Modeling for CFD*, 2d ed., DCW Industries, La Cañada, Calif., 1998).

In its general sense, multiphase flow is not currently solvable by computational fluid dynamics. However, in certain cases reasonable solutions are possible. These include well-separated flows where the phases are confined to relatively well-defined regions separated by one or a few interfaces and flows in which a second phase appears as discrete particles of known size and shape whose motion may be approximately computed with drag coefficient formulations, or rigorously computed with refined meshes applying boundary conditions at the particle surface. **Two-fluid modeling**, in which the phases are treated as overlapping continua, with each phase occupying a volume fraction that is a continuous function of position (and time) is a useful approximation which is becoming available in commercial software. See Elghobashi and Abou-Arab (*J. Physics Fluids*, **26**, 931-938 [1983]) for a $k-\epsilon$ model for two-fluid systems.

Figure 6-56 gives an example CFD calculation for time-dependent flow past a square cylinder at a Reynolds number of 22,000 (Choudhury, et al., *Trans. ASME Fluids Div.*, Lake Tahoe, Nev. [1994]). The computation was done with an implementation of the renormalization group theory $k-\epsilon$ model. The series of contour plots of stream function shows a sequence in time over about 1 vortex-shedding period. The calculated Strouhal number (Eq. [6-195]) is 0.146, in excellent agreement with experiment, as is the time-averaged drag coefficient, $C_D = 2.24$. Similar computations for a circular cylinder at $Re = 14,500$ have given excellent agreement with experimental measurements for St and C_D (*Introduction to the Renormalization Group Method and Turbulence Modeling*, Fluent, Inc., 1993).

DIMENSIONLESS GROUPS

For purposes of data correlation, model studies, and scale-up, it is useful to arrange variables into dimensionless groups. Table 6-7 lists many of the dimensionless groups commonly found in fluid mechanics problems, along with their physical interpretations and areas of application. More extensive tabulations may be found in Catchpole and Fulford (*Ind. Eng. Chem.*, **58**[3], 46-60 [1966]) and Fulford and Catchpole (*Ind. Eng. Chem.*, **60**[3], 71-78 [1968]).

TABLE 6-7 Dimensionless Groups and Their Significance

Name	Symbol	Formula	Physical interpretation	Comments
Archimedes number	Ar	$\frac{gL^3(\rho_p - \rho)\rho}{\mu^3}$	$\frac{\text{inertial forces} \times \text{buoyancy forces}}{(\text{viscous forces})^2}$	Particle settling
Bingham number	Bm	$\frac{\tau_y L}{\mu_\infty V}$	$\frac{\text{yield stress}}{\text{viscous stress}}$	Flow of Bingham plastics = yield number, Y
Bingham Reynolds number	Re _B	$\frac{LV\rho}{\mu_\infty}$	$\frac{\text{inertial force}}{\text{viscous force}}$	Flow of Bingham plastics
Blake number	B	$\frac{V\rho}{\mu(1 - \epsilon)s}$	$\frac{\text{inertial force}}{\text{viscous force}}$	Beds of solids
Bond number	Bo	$\frac{(\rho_L - \rho_G)L^2g}{\sigma}$	$\frac{\text{gravitational force}}{\text{surface-tension force}}$	Atomization = Eotvos number, Eo
Capillary number	Ca	$\frac{\mu V}{\sigma}$	$\frac{\text{viscous force}}{\text{surface-tension force}}$	Two-phase flows, free surface flows
Cauchy number	C	$\frac{\rho V^2}{\beta}$	$\frac{\text{inertial force}}{\text{compressibility force}}$	Compressible flow, hydraulic transients
Cavitation number	σ	$\frac{p - p_v}{\rho V^2/2}$	$\frac{\text{excess pressure above vapor pressure}}{\text{velocity head}}$	Cavitation
Dean number	De	$\frac{\text{Re}}{(Dc/D)^{1/2}}$	Reynolds number $\times \frac{\text{inertial force}}{\text{centrifugal force}}$	Flow in curved channels
Deborah number	De	$\lambda\omega$	$\frac{\text{fluid relaxation time}}{\text{flow characteristic time}}$	Viscoelastic flow
Drag coefficient	C_D	$\frac{F_D}{A\rho V^2/2}$	$\frac{\text{drag force}}{\text{projected area} \times \text{velocity head}}$	Flow around objects, particle settling
Elasticity number	El	$\frac{\lambda\mu}{\rho L^2}$	$\frac{\text{elastic force}}{\text{inertial force}}$	Viscoelastic flow
Euler number	Eu	$\frac{\Delta p}{\rho V^2}$	$\frac{\text{frictional pressure loss}}{2 \times \text{velocity head}}$	Fluid friction in conduits
Fanning friction factor	f	$\frac{D\Delta p}{2\rho V^2 L} = \frac{2\tau_w}{\rho V^2}$	$\frac{\text{wall shear stress}}{\text{velocity head}}$	Fluid friction in conduits Darcy friction factor = $4f$
Froude number	Fr	$\frac{V^2}{gL}$	$\frac{\text{inertial force}}{\text{gravity force}}$	Often defined as $\text{Fr} = V/\sqrt{gL}$
Densometric Froude number	Fr'	$\frac{\rho V^2}{(\rho_d - \rho)gL}$	$\frac{\text{inertial force}}{\text{gravity force}}$	or $\text{Fr}' = \frac{V}{\sqrt{(\rho_d - \rho)gL/\rho}}$
Hedstrom number	He	$\frac{L^2\tau_y\rho}{\mu_\infty^2}$	Bingham Reynolds number $\times$ Bingham number	Flow of Bingham plastics
Hodgson number	H	$\frac{V'\omega\Delta p}{\bar{q}\bar{p}}$	$\frac{\text{time constant of system}}{\text{period of pulsation}}$	Pulsating gas flow
Mach number	M	$\frac{V}{c}$	$\frac{\text{fluid velocity}}{\text{sonic velocity}}$	Compressible flow
Newton number	Ne	$\frac{\Delta PD}{\rho V^2 L}$	$2 \times$ Fanning friction factor	
Ohnesorge number	Z	$\frac{\mu}{(\rho L\sigma)^{1/2}}$	$\frac{\text{viscous force}}{(\text{inertial force} \times \text{surface tension force})^{1/2}}$	Atomization = $\frac{\text{Weber number}}{\text{Reynolds number}}$
Peclet number	Pe	$\frac{LV}{D}$	$\frac{\text{convective transport}}{\text{diffusive transport}}$	Heat, mass transfer, mixing
Pipeline parameter	Pn	$\frac{dV_o}{2gH}$	$\frac{\text{maximum water-hammer pressure rise}}{2 \times \text{static pressure}}$	Water hammer

TABLE 6-7 Dimensionless Groups and Their Significance (Concluded)

Name	Symbol	Formula	Physical interpretation		Comments	
Power number	Po	$\frac{P}{\rho N^3 L^5}$	$\frac{\text{impeller drag force}}{\text{inertial force}}$		Agitation	
Prandtl velocity ratio	v^+	$\frac{v}{(\tau_w/\rho)^{1/2}}$	velocity normalized by friction velocity		Turbulent flow near a wall, friction velocity = $\sqrt{\tau_w/\rho}$	
Reynolds number	Re	$\frac{LV\rho}{\mu}$	$\frac{\text{inertial force}}{\text{viscous force}}$			
Strouhal number	St	$\frac{f'L}{V}$	vortex shedding frequency $\times$ characteristic flow time scale		Vortex shedding, von Karman vortex streets	
Weber number	We	$\frac{\rho V^2 L}{\sigma}$	$\frac{\text{inertial force}}{\text{surface tension force}}$		Bubble, drop formation	
Nomenclature			SI Units	Nomenclature	SI Units	
a	Wave speed		m/s	$\bar{P}$	Power	Watts
A	Projected area		m	$\bar{q}$	Average volumetric flow rate	m ³ /s
c	Sonic velocity		m/s	s	Particle area/particle volume	1/m
D	Diameter of pipe		m	v	Local fluid velocity	m/s
D_c	Diameter of curvature		m	V	Characteristic or average fluid velocity	m/s
D'	Diffusivity		m ² /s	V'	System volume	m ³
f'	Vortex shedding frequency		1/s	β	Bulk modulus	Pa
F_D	Drag force		N	ϵ	Void fraction	m ³
g	Acceleration of gravity		m/s	λ	Fluid relaxation time	s
H	Static head		m	μ	Fluid viscosity	Pa · s
L	Characteristic length		m	μ_∞	Infinite shear viscosity (Bingham plastics)	Pa · s
N	Rotational speed		1/s	ρ	Fluid density	kg/m ³
p	Pressure		Pa	ρ_G, ρ_L	Gas, liquid densities	kg/m ³
p_v	Vapor pressure		Pa	ρ_d	Dispersed phase density	kg/m ³
$\bar{p}$	Average static pressure		Pa	σ	Surface tension	N/m
Δp	Frictional pressure drop		Pa	ω	Characteristic frequency or reciprocal time scale of flow	1/s

PARTICLE DYNAMICS

GENERAL REFERENCES: Brodkey, *The Phenomena of Fluid Motions*, Addison-Wesley, Reading, Mass., 1967; Clift, Grace, and Weber, *Bubbles, Drops and Particles*, Academic, New York, 1978; Govier and Aziz, *The Flow of Complex Mixtures in Pipes*, Van Nostrand Reinhold, New York, 1972; Krieger, Huntington, N.Y., 1977; Lapple, et al., *Fluid and Particle Mechanics*, University of Delaware, Newark, 1951; Levich, *Physicochemical Hydrodynamics*, Prentice-Hall, Englewood Cliffs, N.J., 1962; Orr, *Particulate Technology*, Macmillan, New York, 1966; Shook and Roco, *Slurry Flow*, Butterworth-Heinemann, Boston, 1991; Wallis, *One-dimensional Two-phase Flow*, McGraw-Hill, New York, 1969.

DRAG COEFFICIENT

Whenever relative motion exists between a particle and a surrounding fluid, the fluid will exert a drag upon the particle. In steady flow, the drag force on the particle is

$$F_D = \frac{C_D A_F \rho u^2}{2} \quad (6-227)$$

where F_D = drag force
 C_D = drag coefficient
 A_F = projected particle area in direction of motion
 ρ = density of surrounding fluid
 u = relative velocity between particle and fluid

The drag force is exerted in a direction parallel to the fluid velocity. Equation (6-227) defines the **drag coefficient**. For some solid bodies, such as aerofoils, a lift force component perpendicular to the liquid velocity is also exerted. For free-falling particles, lift

forces are generally not important. However, even spherical particles experience lift forces in shear flows near solid surfaces.

TERMINAL SETTLING VELOCITY

A particle falling under the action of gravity will accelerate until the drag force balances gravitational force, after which it falls at a constant **terminal or free-settling velocity** u_t , given by

$$u_t = \sqrt{\frac{2gm_p(\rho_p - \rho)}{\rho\rho_p A_F C_D}} \quad (6-228)$$

where g = acceleration of gravity
 m_p = particle mass
 ρ_p = particle density

and the remaining symbols are as previously defined.

Settling particles may undergo fluctuating motions owing to vortex shedding, among other factors. Oscillation is enhanced with increasing separation between the mass and geometric centers of the particle. Variations in mean velocity are usually less than 10 percent. The drag force on a particle fixed in space with fluid moving is somewhat lower than the drag force on a particle freely settling in a stationary fluid at the same relative velocity.

Spherical Particles For spherical particles of diameter d_p , Eq. (6-228) becomes

$$u_t = \sqrt{\frac{4gd_p(\rho_p - \rho)}{3\rho C_D}} \quad (6-229)$$

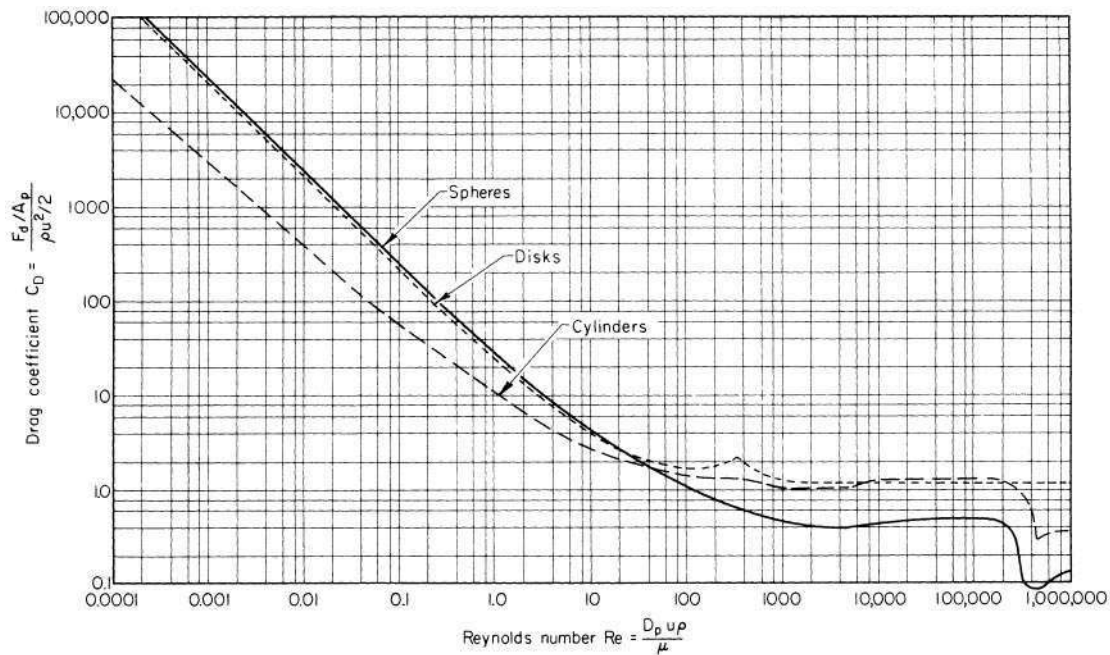


FIG. 6-57 Drag coefficients for spheres, disks, and cylinders: A_p = area of particle projected on a plane normal to direction of motion; C = overall drag coefficient, dimensionless; D_p = diameter of particle; F_d = drag or resistance to motion of body in fluid; Re = Reynolds number, dimensionless; u = relative velocity between particle and main body of fluid; μ = fluid viscosity; and ρ = fluid density. (From Lapple and Shepherd, Ind. Eng. Chem., **32**, 605 [1940].)

The drag coefficient for rigid spherical particles is a function of particle Reynolds number, $Re_p = d_p \rho u / \mu$ where μ = fluid viscosity, as shown in Fig. 6-57. At low Reynolds number, **Stokes' law** gives

$$C_D = \frac{24}{Re_p} \quad Re_p < 0.1 \quad (6-230)$$

which may also be written

$$F_D = 3\pi\mu u d_p \quad Re_p < 0.1 \quad (6-231)$$

and gives for the terminal settling velocity

$$u_t = \frac{g d_p^2 (\rho_p - \rho)}{18\mu} \quad Re_p < 0.1 \quad (6-232)$$

In the **intermediate regime** ($0.1 < Re_p < 1,000$), the drag coefficient may be estimated within 6 percent by

$$C_D = \left(\frac{24}{Re_p} \right) \left(1 + 0.14 Re_p^{0.70} \right) \quad 0.1 < Re_p < 1,000 \quad (6-233)$$

In the **Newton's law** regime, which covers the range $1,000 < Re_p < 350,000$, $C_D = 0.445$, within 13 percent. In this region, Eq. (6-227) becomes

$$u_t = 1.73 \sqrt{\frac{g d_p (\rho_p - \rho)}{\rho}} \quad 1,000 < Re_p < 350,000 \quad (6-234)$$

Between about $Re_p = 350,000$ and 1×10^6 , the drag coefficient drops dramatically in a **drag crisis** owing to the transition to turbulent flow in the boundary layer around the particle, which delays aft separation, resulting in a smaller wake and less drag. Beyond $Re = 1 \times 10^6$, the drag coefficient may be estimated from (Clift, Grace, and Weber):

$$C_D = 0.19 - \frac{8 \times 10^4}{Re_p} \quad Re_p > 1 \times 10^6 \quad (6-235)$$

Drag coefficients may be affected by turbulence in the free-stream flow; the drag crisis occurs at lower Reynolds numbers when the free

stream is turbulent. Torobin and Guvin (*AIChE J.*, **7**, 615–619 [1961]) found that the drag crisis Reynolds number decreases with increasing free-stream turbulence, reaching a value of 400 when the relative turbulence intensity, defined as $\sqrt{u'}/U_R$ is 0.4. Here $\sqrt{u'}$ is the rms fluctuating velocity and U_R is the relative velocity between the particle and the fluid.

For computing the terminal settling velocity, correlations for drag coefficient as a function of Archimedes number

$$Ar = \frac{g d_p^3 (\rho_p - \rho) \rho}{\mu^2} \quad (6-236)$$

may be more convenient than C_D - Re correlations, because the latter are implicit in terminal velocity, and the settling regime is unknown. Karamanev [*Chem. Eng. Comm.*, **147**, 75 (1996)] provided a correlation for drag coefficient for settling solid spheres in terms of Ar .

$$C_D = \frac{432}{Ar} (1 + 0.0470 Ar^{2/3}) + \frac{0.517}{1 + 154 Ar^{-1/3}} \quad (6-237)$$

This equation reduces to Stokes' law $C_D = 24/Re$ in the limit $Ar \rightarrow 0$ and is a fit to data up to about $Ar = 2 \times 10^{10}$, where it gives $C_D = 0.50$, slightly greater than the Newton's law value above. For rising light spheres, which exhibit more energy dissipating lateral motion than do falling dense spheres, Karamanev found that Eq. (6-237) is followed up to $Ar = 13,000$ and that for $Ar > 13,000$, the drag coefficient is $C_D = 0.95$.

For particles settling in **non-Newtonian** fluids, correlations are given by Dallon and Christiansen (Preprint 24C, *Symposium on Selected Papers*, part III, 61st Ann. Mtg. AIChE, Los Angeles, Dec. 1–5, 1968) for spheres settling in shear-thinning liquids, and by Ito and Kajuchi (*J. Chem. Eng. Japan*, **2**[1], 19–24 [1969]) and Pazwash and Robertson (*J. Hydraul. Res.*, **13**, 35–55 [1975]) for spheres settling in Bingham plastics. Beris, Tsamopoulos, Armstrong, and Brown (*J. Fluid Mech.*, **158** [1985]) present a finite element calculation for creeping motion of a sphere through a Bingham plastic.

Nonspherical Rigid Particles The drag on a nonspherical particle depends upon its shape and orientation with respect to the

TABLE 6-8 Free-Fall Orientation of Particles

Reynolds number*	Orientation
0.1–5.5	All orientations are stable when there are three or more perpendicular axes of symmetry.
5.5–200	Stable in position of maximum drag.
200–500	Unpredictable. Disks and plates tend to wobble, while fuller bluff bodies tend to rotate.
500–200,000	Rotation about axis of least inertia, frequently coupled with spiral translation.

SOURCE: From Becker, *Can. J. Chem. Eng.*, **37**, 85–91 (1959).

*Based on diameter of a sphere having the same surface area as the particle.

direction of motion. The orientation in free fall as a function of Reynolds number is given in Table 6-8.

The drag coefficients for **disks** (flat side perpendicular to the direction of motion) and for **cylinders** (infinite length with axis perpendicular to the direction of motion) are given in Fig. 6-57 as a function of Reynolds number. The effect of length-to-diameter ratio for cylinders in the Newton's law region is reported by Knudsen and Katz (*Fluid Mechanics and Heat Transfer*, McGraw-Hill, New York, 1958).

Pettyjohn and Christiansen (*Chem. Eng. Prog.*, **44**, 157–172 [1948]) present correlations for the effect of particle shape on free-settling velocities of **isometric particles**. For $Re < 0.05$, the terminal or free-settling velocity is given by

$$u_t = K_1 \frac{g d_s^2 (\rho_p - \rho)}{18\mu} \quad (6-238)$$

$$K_1 = 0.843 \log \left(\frac{\psi}{0.065} \right) \quad (6-239)$$

where ψ = sphericity, the surface area of a sphere having the same volume as the particle, divided by the actual surface area of the particle; d_s = equivalent diameter, equal to the diameter of the equivalent sphere having the same volume as the particle; and other variables are as previously defined.

In the **Newton's law region**, the terminal velocity is given by

$$u_t = \sqrt{\frac{4d_s(\rho_p - \rho)g}{3K_2\rho}} \quad (6-240)$$

$$K_2 = 5.31 - 4.88\psi \quad (6-241)$$

Equations (6-238) to (6-241) are based on experiments on cube-octahedrons, octahedrons, cubes, and tetrahedrons for which the sphericity ψ ranges from 0.906 to 0.670, respectively. See also Clift, Grace, and Weber. A graph of drag coefficient vs. Reynolds number with ψ as a parameter may be found in Brown, et al. (*Unit Operations*, Wiley, New York, 1950) and in Govier and Aziz.

For particles with $\psi < 0.67$, the correlations of Becker (*Can. J. Chem. Eng.*, **37**, 85–91 [1959]) should be used. Reference to this paper is also recommended for **intermediate region** flow. Settling characteristics of nonspherical particles are discussed by Clift, Grace, and Weber, Chaps. 4 and 6.

The terminal velocity of **axisymmetric particles in axial motion** can be computed from Bowen and Masliyah (*Can. J. Chem. Eng.*, **51**, 8–15 [1973]) for low-Reynolds number motion:

$$u_t = \frac{V'}{K_2} \frac{g D_s^2 (\rho_p - \rho)}{18\mu} \quad (6-242)$$

$$K_2 = 0.244 + 1.035\Sigma - 0.712\Sigma^2 + 0.441\Sigma^3 \quad (6-243)$$

where D_s = diameter of sphere with perimeter equal to maximum particle projected perimeter

V' = ratio of particle volume to volume of sphere with diameter D_s

Σ = ratio of surface area of particle to surface area of a sphere with diameter D_s

and other variables are as defined previously.

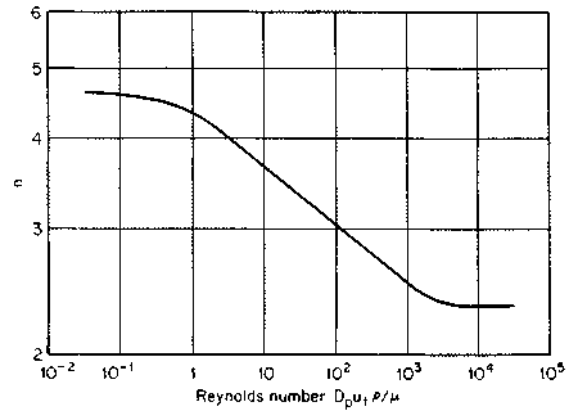


FIG. 6-58 Values of exponent n for use in Eq. (6-242). (From Maude and Whitmore, *Br. J. Appl. Phys.*, **9**, 481 [1958]. Courtesy of the Institute of Physics and the Physical Society.)

Hindered Settling When particle concentration increases, particle settling velocities decrease because of hydrodynamic interaction between particles and the upward motion of displaced liquid. The suspension viscosity increases. Hindered settling is normally encountered in sedimentation and transport of concentrated slurries. Below 0.1 percent volumetric particle concentration, there is less than a 1 percent reduction in settling velocity. Several expressions have been given to estimate the effect of particle volume fraction on settling velocity. Maude and Whitmore (*Br. J. Appl. Phys.*, **9**, 477–482 [1958]) give, for uniformly sized spheres,

$$u_t = u_{t0}(1 - c)^n \quad (6-244)$$

where u_t = terminal settling velocity

u_{t0} = terminal velocity of a single sphere (infinite dilution)

c = volume fraction solid in the suspension

n = function of Reynolds number $Re_p = d_p u_{t0} \rho / \mu$ as given

Fig. 6-58

In the Stokes' law region ($Re_p < 0.3$), $n = 4.65$ and in the Newton's law region ($Re_p > 1,000$), $n = 2.33$. Equation (6-244) may be applied to particles of any size in a polydisperse system, provided the volume fraction corresponding to all the particles is used in computing terminal velocity (Richardson and Shabi, *Trans. Inst. Chem. Eng. [London]*, **38**, 33–42 [1960]). The concentration effect is greater for nonspherical and angular particles than for spherical particles (Steinour, *Ind. Eng. Chem.*, **36**, 840–847 [1944]). Theoretical developments for low-Reynolds number flow assemblages of spheres are given by Happel and Brenner (*Low Reynolds Number Hydrodynamics*, Prentice-Hall, Englewood Cliffs, N.J., 1965) and Famularo and Happel (*AIChE J.*, **11**, 981 [1965]) leading to an equation of the form

$$u_t = \frac{u_{t0}}{1 + \gamma c^{1/3}} \quad (6-245)$$

where γ is about 1.3. As particle concentration increases, resulting in interparticle contact, hindered settling velocities are difficult to predict. Thomas (*AIChE J.*, **9**, 310 [1963]) provides an empirical expression reported to be valid over the range $0.08 < u_t/u_{t0} < 1$:

$$\ln \left(\frac{u_t}{u_{t0}} \right) = -5.9c \quad (6-246)$$

Time-dependent Motion The time-dependent motion of particles is computed by application of Newton's second law, equating the rate of change of particle motion to the net force acting on the particle. Rotation of particles may also be computed from the net torque. For large particles moving through low-density gases, it is usually sufficient to compute the force due to fluid drag from the

relative velocity and the drag coefficient computed for steady flow conditions. For two- and three-dimensional problems, the velocity appearing in the particle Reynolds number and the drag coefficient is the amplitude of the relative velocity. The drag force, not the relative velocity, is to be resolved into vector components to compute the particle acceleration components. Clift, Grace, and Weber (*Bubbles, Drops and Particles*, Academic, London, 1978) discuss the complexities that arise in the computation of transient drag forces on particles when the transient nature of the flow is important. Analytical solutions for the case of a single particle in creeping flow ($Re_p = 0$) are available. For example, the creeping motion of a spherical particle released from rest in a stagnant fluid is described by

$$\rho_p V \frac{dU}{dt} = g(\rho_p - \rho)V - 3\pi\mu d_p U - \frac{\rho}{2} V \frac{dU}{dt} - \left(\frac{3}{2}\right) d_p^2 \sqrt{\pi\mu} \int_0^t \frac{(dU/dt)_{t=s} ds}{\sqrt{t-s}} \quad (6-247)$$

Here, U = particle velocity, positive in the direction of gravity, and V = particle volume. The first term on the right-hand side is the net gravitational force on the particle, accounting for buoyancy. The second is the steady-state Stokes drag (Eq. 6-231). The third is the **added mass** or **virtual mass** term, which may be interpreted as the inertial effect of the fluid which is accelerated along with the particle. The volume of the added mass of fluid is half the particle volume. The last term, the **Basset force**, depends on the entire history of the transient motion, with past motions weighted inversely with the square root of elapsed time. Clift, et al. provide integrated solutions. In **turbulent flows**, particle velocity will closely follow fluid eddy velocities when (Clift et al.)

$$\tau_0 \gg \frac{d_p^2[(2\rho_p/\rho) + 1]}{36\nu} \quad (6-248)$$

where τ_0 = oscillation period or eddy time scale, the right-hand side expression is the **particle relaxation time**, and ν = kinematic viscosity.

Gas Bubbles Fluid particles, unlike rigid solid particles, may undergo deformation and internal circulation. Figure 6-59 shows rise velocity data for air bubbles in stagnant water. In the figure, Eo = Eotvos number, $g(\rho_L - \rho_G)d_e/\sigma$, where ρ_L = liquid density, ρ_G = gas density, d_e = bubble diameter, σ = surface tension, and the equivalent diameter d_e is the diameter of a sphere with volume equal to that of

the bubble. Small bubbles (<1-mm [0.04-in] diameter) remain spherical and rise in straight lines. The presence of surface active materials generally renders small bubbles rigid, and they rise roughly according to the drag coefficient and terminal velocity equations for spherical solid particles. Bubbles roughly in the range 2- to 8-mm (0.079- to 0.32-in) diameter assume flattened, ellipsoidal shape, and rise in a zig-zag or spiral pattern. This motion increases dissipation and drag, and the rise velocity may actually decrease with increasing bubble diameter in this region, characterized by rise velocities in the range of 20 to 30 cm/s (0.7 to 1.0 ft/s). Large bubbles, >8-mm (0.32-in) diameter, are greatly deformed, assuming a mushroomlike, spherical cap shape. These bubbles are unstable and may break into smaller bubbles. Carefully purified water, free of surface active materials, allows bubbles to freely circulate even when they are quite small. Under creeping flow conditions $Re_b = d_b \mu_r \rho_L / \mu_L < 1$, where u_r = bubble rise velocity and μ_L = liquid viscosity, the bubble rise velocity may be computed analytically from the Hadamard-Rybczynski formula (Levich, *Physicochemical Hydrodynamics*, Prentice-Hall, Englewood Cliffs, N.J., 1962, p. 402). When $\mu_G/\mu_L \ll 1$, which is normally the case, the rise velocity is 1.5 times the rigid sphere law velocity. However, in practice, most liquids, including ordinary distilled water, contain sufficient surface active materials to render small bubbles rigid. Larger bubbles undergo deformation in both purified and ordinary liquids; however, the variation in rise velocity for large bubbles with degree of purity is quite evident in Fig. 6-59. For additional discussion, see Clift, et al., Chap. 7.

Karamanev [op. cit.] provided equations for bubble rise velocity based on the Archimedes number and on use of the bubble projected diameter d_b in the drag coefficient and the bubble equivalent diameter in Ar . The Archimedes number is as defined in Eq. (6-236) except that the density difference is liquid density minus gas density, and d_p is replaced by d_e .

$$u_r = 40.3 \frac{d_e}{d_h} \sqrt{\frac{V^{1/3}}{C_D}} = 40.3 \frac{d_e}{d_h} \sqrt{\frac{(\pi d_e^3/6)^{1/3}}{C_D}} \quad (6-249)$$

$$C_D = \frac{432}{Ar} (1 + 0.0470 Ar^{2/3}) + \frac{0.517}{1 + 154 Ar^{-1/3}} \quad Ar < 13,000 \quad (6-250)$$

$$C_D = 0.95 \quad Ar > 13,000 \quad (6-251)$$

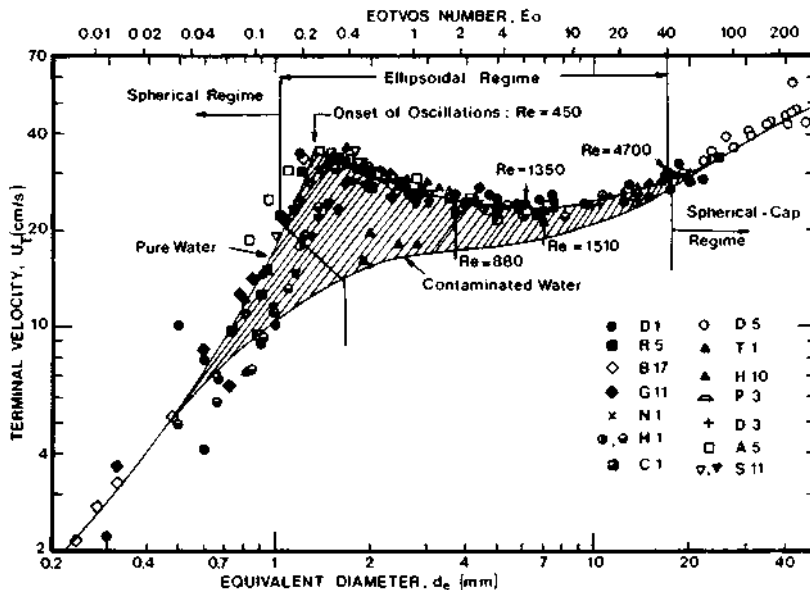


FIG. 6-59 Terminal velocity of air bubbles in water at 20°C. (From Clift, Grace, and Weber, *Bubbles, Drops and Particles*, Academic, New York, 1978).

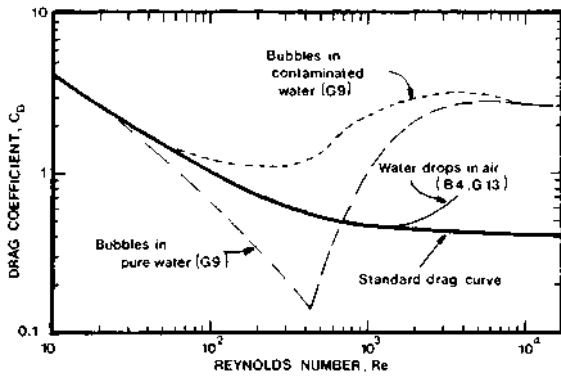


FIG. 6-60 Drag coefficient for water drops in air and air bubbles in water. Standard drag curve is for rigid spheres. (From Clift, Grace, and Weber, *Bubbles, Drops and Particles*, Academic, New York, 1978.)

$$\frac{d_e}{d_h} = (1 + 0.163Eo^{0.757})^{-1/3} \quad Eo < 40 \quad (6-252)$$

$$\frac{d_e}{d_h} = 0.62 \quad Eo > 40 \quad (6-253)$$

Applied to air bubbles in water, these expressions give reasonable agreement with the contaminated water curve in Fig. 6-59.

Figure 6-60 gives the drag coefficient as a function of bubble or drop Reynolds number for air bubbles in water and water drops in air, compared with the standard drag curve for rigid spheres. Information on bubble motion in **non-Newtonian** liquids may be found in Astarita and Apuzzo (*AIChE J.*, **11**, 815-820 [1965]); Calderbank, Johnson, and Loudon (*Chem. Eng. Sci.*, **25**, 235-256 [1970]); and Acharya, Mashelkar, and Ulbrecht (*Chem. Eng. Sci.*, **32**, 863-872 [1977]).

Liquid Drops in Liquids Very small liquid drops in immiscible liquids behave like rigid spheres, and the terminal velocity can be approximated by use of the drag coefficient for solid spheres up to a Reynolds number of about 10 (Warshaw, Bogusz, Johnson, and Kintner, *Can. J. Chem. Eng.*, **37**, 29-36 [1959]). Between Reynolds numbers of 10 and 500, the terminal velocity exceeds that for rigid spheres owing to internal circulation. In normal practice, the effect of drop phase viscosity is neglected. Grace, Waigeli, and Nguyen (*Trans. Inst. Chem. Eng.*, **54**, 167-173 [1976]; Clift, et al., op. cit., pp. 175-177) present a correlation for terminal velocity valid in the range

$$M < 10^{-3} \quad Eo < 40 \quad Re > 0.1 \quad (6-254)$$

where M = Morton number = $g\mu^4\Delta\rho/\rho^2\sigma^3$
 Eo = Eotvos number = $g\Delta\rho d^2/\sigma$
 Re = Reynolds number = $d\rho u/\mu$
 $\Delta\rho$ = density difference between the phases
 ρ = density of continuous liquid phase
 d = drop diameter
 μ = continuous liquid viscosity
 σ = surface tension
 u = relative velocity

The correlation is represented by

$$J = 0.94H^{0.757} \quad (2 < H \leq 59.3) \quad (6-255)$$

$$J = 3.42H^{0.441} \quad (H > 59.3) \quad (6-256)$$

where
$$H = \frac{4}{3} EoM^{-0.149} \left(\frac{\mu}{\mu_w} \right)^{-0.14} \quad (6-257)$$

$$J = ReM^{0.149} + 0.857 \quad (6-258)$$

Note that the terminal velocity may be evaluated explicitly from

$$u = \frac{\mu}{\rho d} M^{-0.149} (J + 0.857) \quad (6-259)$$

In Eq. (6-257), μ = viscosity of continuous liquid and μ_w = viscosity of water, taken as 0.9 cP (0.0009 Pa · s).

For drop velocities in non-Newtonian liquids, see Mhatre and Kintner (*Ind. Eng. Chem.*, **51**, 865-867 [1959]); Marrucci, Apuzzo, and Astarita (*AIChE J.*, **16**, 538-541 [1970]); and Mohan, et al. (*Can. J. Chem. Eng.*, **50**, 37-40 [1972]).

Liquid Drops in Gases Liquid drops falling in stagnant gases appear to remain spherical and follow the rigid sphere drag relationships up to a Reynolds number of about 100. Large drops will deform,

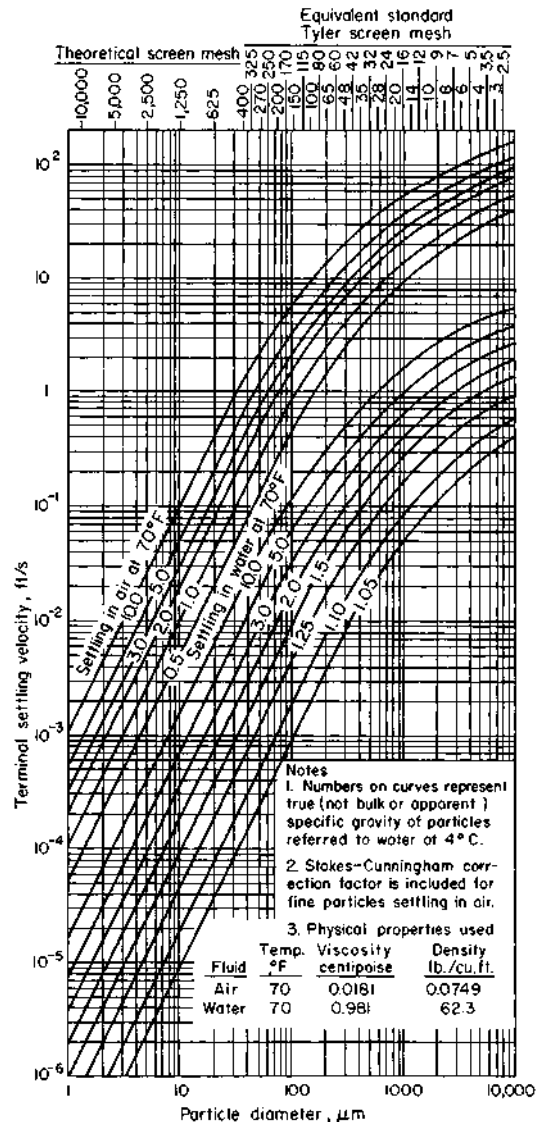


FIG. 6-61 Terminal velocities of spherical particles of different densities settling in air and water at 70°F under the action of gravity. To convert ft/s to m/s, multiply by 0.3048. (From Lapple, et al., *Fluid and Particle Mechanics*, University of Delaware, Newark, 1951, p. 292.)

with a resulting increase in drag, and in some cases will shatter. The largest water drop which will fall in air at its terminal velocity is about 8 mm (0.32 in) in diameter, with a corresponding velocity of about 9 m/s (30 ft/s). Drops shatter when the Weber number defined as

$$We = \frac{\rho_C u^2 d}{\sigma} \quad (6-260)$$

exceeds a critical value. Here, ρ_C = gas density, u = drop velocity, d = drop diameter, and σ = surface tension. A value of $We_c = 13$ is often cited for the critical Weber number.

Terminal velocities for water drops in air have been correlated by Berry and Prnager (*J. Appl. Meteorol.*, **13**, 108–113 [1974]) as

$$Re = \exp [-3.126 + 1.013 \ln N_D - 0.01912(\ln N_D)^2] \quad (6-261)$$

for $2.4 < N_D < 10^7$ and $0.1 < Re < 3,550$. The dimensionless group N_D (often called the *Best* number [Clift et al.]) is given by

$$N_D = \frac{4\rho\Delta\rho g d^3}{3\mu^2} \quad (6-262)$$

and is proportional to the similar Archimedes and Galileo numbers.

Figure 6-61 gives calculated settling velocities for solid spherical particles settling in air or water using the standard drag coefficient curve for spherical particles. For fine particles settling in air, the **Stokes-Cunningham correction** has been applied to account for particle size comparable to the mean free path of the gas. The correction is less than 1 percent for particles larger than 16 μm settling in air. Smaller particles are also subject to **Brownian motion**. Motion of particles smaller than 0.1 μm is dominated by Brownian forces and gravitational effects are small.

Wall Effects When the diameter of a settling particle is significant compared to the diameter of the container, the settling velocity is

TABLE 6-9 Wall Correction Factor for Rigid Spheres in Stokes' Law Region

β^*	k_w	β	k_w
0.0	1.000	0.4	0.279
0.05	0.885	0.5	0.170
0.1	0.792	0.6	0.0945
0.2	0.596	0.7	0.0468
0.3	0.422	0.8	0.0205

SOURCE: From Haberman and Sayre, *David W. Taylor Model Basin Report* 1143, 1958.

* β = particle diameter divided by vessel diameter.

reduced. For rigid spherical particles settling with $Re < 1$, the correction given in Table 6-9 may be used. The factor k_w is multiplied by the settling velocity obtained from Stokes' law to obtain the corrected settling rate. For values of diameter ratio β = particle diameter/vessel diameter less than 0.05, $k_w = 1/(1 + 2.1\beta)$ (Zenz and Othmer, *Fluidization and Fluid-Particle Systems*, Reinhold, New York, 1960, pp. 208–209). In the range $100 < Re < 10,000$, the computed terminal velocity for rigid spheres may be multiplied by k'_w to account for wall effects, where k'_w is given by (Harmathy, *AIChE J.*, **6**, 281 [1960])

$$k'_w = \frac{1 - \beta^2}{\sqrt{1 + \beta^4}} \quad (6-263)$$

For gas bubbles in liquids, there is little wall effect for $\beta < 0.1$. For $\beta > 0.1$, see Uto and Kintner (*AIChE J.*, **2**, 420–424 [1956]), Maneri and Mendelson (*Chem. Eng. Prog.*, **64**, *Symp. Ser.*, **82**, 72–80 [1968]), and Collins (*J. Fluid Mech.*, **28**, part 1, 97–112 [1967]).

SECTION 7

Reaction Kinetics

PERRY'S CHEMICAL ENGINEERS' HANDBOOK

8TH EDITION



TIBERIU M. LEIB
CARMO J. PEREIRA

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Reaction Kinetics*

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Nomenclature and Units

The component A is identified by the subscript a . Thus, the number of moles is n_a ; the fractional conversion is X_a ; the extent of reaction is ξ ; the partial pressure is p ; the rate of consumption is r_a ; the molar flow rate is N_a ; the volumetric flow rate is q ; reactor volume is V , or simply V for batch reactors; the volumetric concentration is $C_a = n_a/V$ or $C_a = N_a/q$; the total pressure is P ; and the temperature is T . Throughout this section, equations are presented without specification of units. Use of any consistent unit set is appropriate.

Following is a listing of typical nomenclature expressed in SI and U.S. Customary System units.

Symbol	Definition	SI units	U.S. Customary System units	Symbol	Definition	SI units	U.S. Customary System units
A, B, C, . . .	Names of substances			p_a	Partial pressure of substance A	Pa	psi
A	Free radical, as CH ₃			q	Volumetric flow rate	m ³ /s	ft ³ /s
a	Activity			Q	Electric charge	Coulomb	
BR	Batch reactor			$\bar{R}$	Radial position, radius, universal gas constant		
b	Estimate of kinetic parameters, vector			Re	Reynolds number		
C_a	Concentration of substance A	kg·mol/m ³	lb·mol/ft ³	RgSS	Regression sum of squares		
CSTR	Continuous stirred tank reactor			RSS	Residual sum of squares		
C_0	Initial concentration	kg·mol/m ³	lb·mol/ft ³	r_a	Rate of reaction of A per unit volume		
c_p	Heat capacity at constant pressure	kJ/(kg·K)	Btu/(lbm·°F)	S	Selectivity, stoichiometric matrix, objective function for parameter estimation		
Δc_p	Heat capacity change in a reaction	kJ/(kg·K)	Btu/(lbm·°F)	SBR	Semibatch reactor		
D	Diffusivity, dispersion coefficient	m ² /s	ft ² /s	Sc	Schmidt number		
D_e	Effective diffusivity	m ² /s	ft ² /s	Sh	Sherwood number		
D_K	Knudsen diffusivity	m ² /s	ft ² /s	ΔS	Entropy change	kJ/(kg·mol·K)	Btu/(lb·mol·R _e)
DP	Degree of polymerization			s	Estimate of variance		
E	Activation energy, enhancement factor for gas-liquid mass transfer with reaction, electrochemical cell potential			t	Time, t statistic		
F	Faraday constant, F statistic			u	Linear velocity	m/s	ft/s
f	Efficiency of initiation in polymerization			V	Volume of reactor, variance-covariance matrix		
f_a	C_a/C_{a0} or n_a/n_{a0} , fraction of A remaining unconverted			v	Molar volume	m ³ /kg·mol	ft ³ /lb·mol
Ha	Hatta number			WCLD	Weight chain length distribution		
He	Henry constant for absorption of gas in liquid			WMMD	Weight molecular weight distribution		
ΔG	Free energy change	kJ/kg·mol	Btu/lb·mol	X	Linear model matrix for parameter estimation, fractional conversion		
ΔH_r	Heat of reaction	kJ/kg·mol	Btu/lb·mol	X_a	$1 - f_a = 1 - C_a/C_{a0}$ or $1 - n_a/n_{a0}$, fraction of A converted		
I	Initiator for polymerization, modified Bessel functions, electric current			x	Axial position in a reactor, mole fraction in liquid	Variable	
j	Electric current density	A/m ²		Y	Yield; yield coefficient for biochemical reactions		
K_a	Adsorption constant			y	Mole fraction in gas, predicted dependent variable		
K_e	Chemical equilibrium constant			z	x/L , normalized axial position		
k	Specific rate constant of reaction, mass-transfer coefficient			Greek letters			
L	Length of path in reactor	m	ft	α	Fraction of initial catalyst activity, probability of propagation for chain polymerization, confidence level		
LFSS	Lack of fit sum of squares			β	r/R , normalized radial position, fraction of poisoned catalyst, kinetic parameter vector		
M	Average molecular weight in polymers, dead polymer species, monomer			δ	Film thickness or boundary layer thickness, relative change in number of moles by reaction		
m	Number of moles in electrochemical reaction	kg·mol	lb·mol	$\delta(t)$	Unit impulse input, Dirac function		
N	Molar flow rate, molar flux			ε	Fraction void space in a packed bed, relative change in number of moles by reaction, residual error, porosity, current efficiency		
NCLD	Number chain length distribution			Φ	Weisz Prater parameter		
NMWD	Number molecular weight distribution			ϕ	Thiele modulus		
n	Number of stages in a CSTR battery, reaction order, number of electrons in electrochemical reaction, number of experiments						
n_a	Number of moles of A present	kg·mol	lb·mol				
n_t	Total number of moles	kg·mol	lb·mol				
P	Total pressure, live polymer species						
PESS	Pure error sum of squares						
PFR	Plug flow reactor						
p	Number of kinetic parameters						

7-4 REACTION KINETICS

Nomenclature and Units (Concluded)

Symbol	Definition	SI units	U.S. Customary System units	Subscripts	
				Greek letters	
η	Effectiveness factor of porous catalyst, overpotential in electrochemical reactions			d	Deactivation
λ	Parameter for instantaneous gas-liquid reaction, moments in polymer chain length			e	Equilibrium
μ	Viscosity, biomass growth rate, average chain length in polymers			f	Forward reaction, final, formation
ν	μ/ρ , kinematic viscosity, stoichiometric coefficient, fraction of surface covered by adsorbed species			G	Gas
θ	Dimensionless time			i	Component i
ρ	Density	kg/m ³	lbm/ft ³	j	Reaction j
σ	Variance			L	Liquid
τ	Residence time, tortuosity factor			m	Based on mass, mass transfer
ζ	Extent of reaction			max	Maximum biomass growth, maximum extent of reaction
				n	Chain length in polymers
				o	Oxidized
act	Activation			obs	observed
anode	At anode			p	Particle
B	Bed			projected	Electrode projected area
cathode	At cathode			r	Reverse reaction, reduced
cell	Electrochemical cell			S	Substrate
current, j	Current, species j			s	Solid or catalyst, saturation, surface
D	Diffusion, dispersion			surf	Surface
				v	Based on volume
				x	Biomass
				0	At initial or inlet conditions, as in C_{a0} , n_{a0} , V_0 , at reference temperature
				1/2	Half-life
				Superscripts	
				eq	Equilibrium
				o	At reference temperature
				T	Transposed matrix

REFERENCES

GENERAL REFERENCES: Amundson, *Mathematical Methods in Chemical Engineering—Matrices and Their Application*, Prentice-Hall International, New York, 1966; Aris, *Elementary Chemical Reactor Analysis*, Prentice-Hall, 1969; Astarita, *Mass Transfer with Chemical Reaction*, Elsevier, New York, 1967; Bamford and Tipper (eds.), *Comprehensive Chemical Kinetics*, Elsevier, 1969; Bird, Stewart, and Lightfoot, *Transport Phenomena*, 2d ed., Wiley, New York, 2002; Boudart, *Kinetics of Chemical Processes*, Prentice-Hall, 1968; Boudart and Djega-Mariadassou, *Kinetics of Heterogeneous Catalytic Reactions*, Princeton University Press, Princeton, N.J., 1984; Brotz, *Fundamentals of Chemical Reaction Engineering*, Addison-Wesley, 1965; Butt, *Reaction Kinetics and Reactor Design*, Prentice-Hall, 1980; Butt and Petersen, *Activation, Deactivation and Poisoning of Catalysts*, Academic Press, 1988; Capello and Bielski, *Kinetic Systems: Mathematical Description of Kinetics in Solution*, Wiley, 1972; Carberry, *Chemical and Catalytic Reaction Engineering*, McGraw-Hill, 1976; Carberry and Varma (eds.), *Chemical Reaction and Reactor Engineering*, Dekker, 1987; Chen, *Process Reactor Design*, Allyn & Bacon, 1983; Churchill, *The Interpretation and Use of Rate Data: The Rate Concept*, McGraw-Hill, New York, 1974; Cooper and Jeffreys, *Chemical Kinetics and Reactor Design*, Prentice-Hall, 1971; Cremer and Watkins (eds.), *Chemical Engineering Practice*, vol. 8: *Chemical Kinetics*, Butterworths, 1965; Davis and Davis, *Fundamentals of Chemical Reaction Engineering*, McGraw-Hill, 2003; Delmon and Froment, *Catalyst Deactivation*, Elsevier, 1980; Denbigh and Turner, *Chemical Reactor Theory*, Cambridge, 1971; Denn, *Process Modeling*, Langman, New York, 1986; Fogler, *Elements of Chemical Reaction Engineering*, 4th ed., Prentice-Hall, 2006; Froment and Bischoff, *Chemical Reactor Analysis and Design*, Wiley, 1990; Froment and Hosten, "Catalytic Kinetics—Modeling," in *Catalysis—Science and Technology*, Springer Verlag, New York, 1981; Harriott, *Chemical Reactor Design*, Dekker, 2003; Hill, *An Introduction to Chemical Engineering Kinetics and Reactor Design*, 2d ed., Wiley, 1990; Holland and Anthony, *Fundamentals of Chemical Reaction Engineering*, Prentice-Hall, 1989; Kafarov, *Cybernetic Methods in Chemistry and Chemical Engineering*, Mir Publishers, 1976; Laidler, *Chemical Kinetics*, Harper & Row, 1987; Lapidus and Amundson (eds.), *Chemical Reactor Theory—A Review*, Prentice-Hall, 1977; Levenspiel, *Chemical Reaction Engineering*, 3d ed., Wiley, 1999; Lewis (ed.), *Techniques of Chemistry*, vol. 4: *Investigation of Rates and Mechanisms of Reactions*, Wiley, 1974; Masel, *Chemical Kinetics and*

Catalysis, Wiley, 2001; Naumann, *Chemical Reactor Design*, Wiley, 1987; Panchenkov and Lebedev, *Chemical Kinetics and Catalysis*, Mir Publishers, 1976; Petersen, *Chemical Reaction Analysis*, Prentice-Hall, 1965; Rase, *Chemical Reactor Design for Process Plants: Principles and Case Studies*, Wiley, 1977; Rose, *Chemical Reactor Design in Practice*, Elsevier, 1981; Satterfield, *Heterogeneous Catalysis in Practice*, McGraw-Hill, 1991; Schmidt, *The Engineering of Chemical Reactions*, Oxford University Press, 1998; Smith, *Chemical Engineering Kinetics*, McGraw-Hill, 1981; Steinfeld, Francisco, and Hasse, *Chemical Kinetics and Dynamics*, Prentice-Hall, 1989; Ulrich, *Guide to Chemical Engineering Reactor Design and Kinetics*, Ulrich, 1993; Van Santen and Neurock, *Molecular Heterogeneous Catalysis: A Conceptual and Computational Approach*, Wiley, 2006; Van Santen and Niemantsverdriet, *Chemical Kinetics and Catalysis, Fundamental and Applied Catalysis*, Plenum Press, New York, 1995; van't Riet and Tramper, *Basic Bioreactor Design*, Dekker, 1991; Walas, *Reaction Kinetics for Chemical Engineers*, McGraw-Hill, 1959; reprint, Butterworths, 1989; Walas, *Chemical Reaction Engineering Handbook of Solved Problems*, Gordon & Breach Publishers, 1995; Westerterp, van Swaaij, and Beenackers, *Chemical Reactor Design and Operation*, Wiley, 1984.

REFERENCES FOR LABORATORY REACTORS: Berty, Laboratory reactors for catalytic studies, in Leach, ed., *Applied Industrial Catalysis*, vol. 1, Academic, 1983, pp. 41–57; Berty, *Experiments in Catalytic Reaction Engineering*, Elsevier, 1999; Danckwerts, *Gas-Liquid Reactions*, McGraw-Hill, 1970; Hoffmann, Industrial Process Kinetics and parameter estimation, in *ACS Advances in Chemistry* 109:519–534 (1972); Hoffman, Kinetic data analysis and parameter estimation, in de Lasa (ed.), *Chemical Reactor Design and Technology*, Martinus Nijhoff, 1986, pp. 69–105; Horak and Pasek, *Design of Industrial Chemical Reactors from Laboratory Data*, Heiden, Philadelphia, 1978; Rase, *Chemical Reactor Design for Process Plants*, Wiley, 1977, pp. 195–259; Shah, *Gas-Liquid-Solid Reactor Design*, McGraw-Hill, 1979, pp. 149–179; Charpentier, *Mass Transfer Rates in Gas-Liquid Absorbers and Reactors*, in Drew et al., eds., *Advances in Chemical Engineering*, vol. 11, Academic Press, 1981.

The mechanism and corresponding kinetics provide the rate at which the chemical or biochemical species in the reactor system react at the prevailing conditions of temperature, pressure, composition, mixing, flow, heat, and mass transfer. Observable kinetics represent the true intrinsic chemical kinetics only when competing phenomena such as transport of mass and heat are not limiting the rates. The intrinsic chemical mechanism and kinetics are unique to the reaction system. Knowledge of the intrinsic kinetics therefore facilitates reactor selection, choice of optimal operating conditions, and reactor scale-up and design, when combined with understanding of the associated physical and transport phenomena for different reactor scales and types.

This section covers the following key aspects of reaction kinetics:

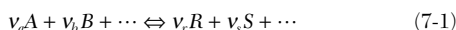
- Chemical mechanism of a reaction system and its relation to kinetics
- Intrinsic rate data using equations that can be correlative, lumped, or based on detailed elementary kinetics
- Catalytic kinetics
- Effect of mass transfer on kinetics in heterogeneous systems
- Intrinsic kinetic rates from experimental data and/or from theoretical calculations
- Kinetic parameter estimation

The use of reaction kinetics for analyzing and designing suitable reactors is discussed in Sec. 19.

BASIC CONCEPTS

MECHANISM

The mechanism describes the reaction steps and the relationship between the reaction rates of the chemical components. A single chemical reaction includes reactants $A, B, \dots$ and products $R, S, \dots$



where ν_i are the *stoichiometric coefficients* of components $A, B, \dots$, i.e., the relative number of molecules of $A, B, \dots$ that participate in the reaction. For instance, the HBr synthesis has the global stoichiometry $\text{H}_2 + \text{Br}_2 \rightleftharpoons 2\text{HBr}$.

The *stoichiometry* of the reaction defines the reaction elemental balance (atoms of H and Br, for instance) and therefore relates the number of molecules of reactants and products participating in the reaction. The stoichiometric coefficients are not unique for a given reaction, but their ratios are unique. For instance, for the HBr synthesis above we could have written the stoichiometric equation $\frac{1}{2}\text{H}_2 + \frac{1}{2}\text{Br}_2 \rightleftharpoons \text{HBr}$ as well.

Often several reactions occur simultaneously, resulting in a *network of reactions*. When the network is broken down into *elementary* or *single-event* steps (such as a single electron transfer), the network represents the *true mechanism* of the chemical transformations leading from initial reactants to final products through intermediates. The intermediates can be molecules, ions, free radicals, transition state complexes, and other moieties. A network of *global reactions*, with each reaction representing the combination of a number of elementary steps, does not represent the true mechanism of the chemical transformation but is still useful for global reaction rate calculations, albeit empirically. The stoichiometry can only be written in a unique manner for elementary reactions, since as shown later, the reaction rate for elementary reactions is determined directly by the stoichiometry through the concept of the law of mass action.

REACTION RATE

The *specific rate* of consumption or production of any reaction species i , r_i , is the rate of change of the number of molecules of species i with time per unit volume of reaction medium:

$$r_i = \frac{1}{V} \frac{dn_i}{dt} \quad (7-2)$$

The rate is negative when i represents a *reactant* (dn_i/dt is negative since n_i is decreasing with time) and positive when i represents a *product*

(dn_i/dt positive since n_i is increasing with time). The specific rate of a reaction, e.g., that in Eq. (7-1) is defined as

$$\begin{aligned} r &= -r_i/\nu_i && \text{for reactants} \\ r &= r_i/\nu_i && \text{for products} \end{aligned} \quad (7-3)$$

By this definition, the specific rate of reaction is uniquely defined, and its sign is always positive. Inversely, the rate of reaction of each component or species participating in the reaction is the specific reaction rate multiplied by the species' stoichiometric coefficient with the corrected sign (negative for reactants, positive for products).

CLASSIFICATION OF REACTIONS

Reactions can be classified in several ways. On the basis of *mechanism* they may be

1. *Irreversible*, i.e., the reverse reaction rate is negligible: $A + B \Rightarrow C + D$, e.g., CO oxidation $\text{CO} + \frac{1}{2}\text{O}_2 \Rightarrow \text{CO}_2$
2. *Reversible*: $A + B \rightleftharpoons C + D$, e.g., the water-gas shift $\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$
3. *Equilibrium*, a special case with zero net rate, i.e., with the forward and reverse reaction rates of a reversible reaction being equal. All reversible reactions, if left to go to completion, end in equilibrium.
4. *Networks of simultaneous reactions*, i.e., consecutive, parallel, complex (combination of consecutive and parallel reactions):



e.g., two-step hydrogenation of acetylene to ethane



A further classification is from the point of view of the number of reactant molecules participating in the reaction, or the *molecularity*:

1. *Unimolecular*: $A \Rightarrow B$, e.g., isomerization of ortho-xylene to para-xylene, O -xylene $\Rightarrow P$ -xylene, or $A \Rightarrow B + C$, e.g., decomposition $\text{CaCO}_3 \Rightarrow \text{CaO} + \text{CO}_2$
2. *Bimolecular*: $A + B \Rightarrow C$ or $2A \Rightarrow B$ or $A + B \Rightarrow C + D$, e.g., $\text{C}_2\text{H}_4 + \text{H}_2 \Rightarrow \text{C}_2\text{H}_6$
3. *Trimolecular*: $A + B + C \Rightarrow D$ or $3A \Rightarrow B$

This last classification has fundamental meaning only when considering elementary reactions, i.e., reactions that constitute a single chemical transformation or a single event, such as a single electron transfer. For elementary reactions, molecularity is rarely higher than 2. Often elementary reactions are not truly unimolecular, since in order for the reaction to occur, energy is required and it is obtained through collision with other molecules such as an inert solvent or gas.

7-6 REACTION KINETICS

Thus the unimolecular reaction $A \Rightarrow B$ could in reality be represented as a bimolecular reaction $A + X \Rightarrow B + X$, i.e., A collides with X to produce B and X , and thus no net consumption of X occurs.

Reactions can be further classified according to the *phases* present. Examples for the more common cases are

1. *Homogeneous gas*, e.g., methane combustion
2. *Homogeneous liquid*, e.g., acid/base reactions to produce soluble salts
3. *Heterogeneous gas-solid*, e.g., HCN synthesis from NH_3 , CH_4 , and air on a solid catalyst
4. *Heterogeneous gas-liquid*, e.g., absorption of CO_2 in amine solutions
5. *Heterogeneous liquid-liquid*, e.g., reaction in immiscible organic and aqueous phases such as synthesis of adipic acid from cyclohexanone and nitric acid
6. *Heterogeneous liquid-solid*, e.g., reaction of limestone with sulfuric acid to make gypsum
7. *Heterogeneous solid-solid*, e.g., self-propagating, high-temperature synthesis of inorganic pure oxides (SHS)
8. *Heterogeneous gas-liquid-solid*, e.g., catalytic Fischer-Tropsch synthesis of hydrocarbons from CO and H_2
9. *Heterogeneous gas-liquid-liquid*, e.g., oxidations or hydrogenations with phase transfer catalysts

Reactions can also be classified with respect to the *mode of operation* in the reaction system as

1. *Isothermal constant volume* (batch)
2. *Isothermal constant pressure* (continuous)
3. *Adiabatic*
4. *Nonisothermal* temperature-controlled (by cooling or heating), batch or continuous

EFFECT OF CONCENTRATION ON RATE

The *concentration* of the reaction components determines the rate of reaction. For instance, for the irreversible reaction



the rate can be represented *empirically* as a power law function of the reactant concentrations such as

$$r = kC_a^a C_b^b \quad C_i = \frac{n_i}{V} \quad (7-5)$$

The exponents a and b represent the *order of the reaction* with respect to components A and B , and the *sum* $a + b$ represents the *overall order of the reaction*. The order can be a positive, zero, or negative number indicating that the rate increases, is independent of, or decreases with an increase in a species concentration, respectively. The exponents can be whole (integral order) or fraction (fractional order). In Eq. (7-5) k is the *specific rate constant* of the reaction, and it is independent of concentrations for elementary reactions only. For global reactions consisting of several elementary steps, k may still be constant over a narrow range of compositions and operating conditions and therefore can be considered constant for limited practical purposes. A further complexity arises for nonideal chemical solutions where activities have to be used instead of concentrations. In this case the rate constant can be a function of composition even for elementary steps (see, for instance, Froment and Bischoff, *Chemical Reactor Analysis and Design*, Wiley, 1990).

When Eq. (7-4) represents a global reaction combining a number of elementary steps, then rate equation (7-5) represents an empirical correlation of the global or overall reaction rate. In this case exponents a and b have no clear physical meaning other than indicating the overall effect of the various concentrations on rate, and they do not have any obvious relationship to the stoichiometric coefficients p and q . This is not so for elementary reactions, as shown in the next subsection. Also, as shown later, power law and rate expressions other than power law (e.g., hyperbolic) can be developed for specific reactions by starting with the mechanism of the elementary steps and making simplifying assumptions that are valid under certain conditions.

LAW OF MASS ACTION

As indicated above, the dependence of rate on concentration can be shown to be of the general form

$$r = kf(C_a, C_b, \dots) \quad (7-6)$$

For elementary reactions, the *law of mass action* states that the rate is proportional to the concentrations of the reactants raised to the power of their respective molecularity. Thus for an elementary irreversible reaction such as (7-4) the rate equation is

$$r = kC_a^p C_b^q \quad (7-7)$$

Hence, the exponents p and q of Eq. (7-7) are the stoichiometric coefficients when the stoichiometric equation truly represents the mechanism of reaction, i.e., when the reactions are elementary. As discussed above, the exponents a and b in Eq. (7-5) identify the order of the reaction, while the stoichiometric coefficients p and q in Eq. (7-7) also identify the molecularity—for elementary reactions these are the same.

EFFECT OF TEMPERATURE

The *Arrhenius equation* relates the specific rate constant to the absolute temperature

$$k = k_0 \exp\left(-\frac{E}{RT}\right) \quad (7-8)$$

where E is called the *activation energy* and k_0 is the *preexponential factor*. As seen from Eq. (7-8), the rate can be a very strongly increasing (exponential) function of temperature, depending on the magnitude of the activation energy E . This equation works well for elementary reactions, and it also works reasonably well for global reactions over a relatively narrow range of temperatures in the absence of mass-transfer limitations. The Arrhenius form represents an energy barrier on the reaction pathway between reactants and products that has to be overcome by the reactant molecules.

The Arrhenius equation can be derived from theoretical considerations using either of two competing theories, the *collision theory* and the *transition state theory*. A more accurate form of Eq. (7-8) includes an additional temperature factor

$$k = k_0 T^m \exp\left(-\frac{E}{RT}\right) \quad 0 < m < 1 \quad (7-9)$$

but the T^m factor is often neglected because of the usually much stronger dependence on temperature of the exponential factor in Eq. (7-9), as m is usually small. When m is larger, as it can be for complex molecules, then the T^m term has to be taken into consideration. For more details, see Masel, *Chemical Kinetics and Catalysis*, Wiley, 2001; Levenspiel, *Chemical Reaction Engineering*, 3d ed., Wiley, 1999).

HEAT OF REACTION

Chemical reactions are accompanied by evolution or absorption of energy. The enthalpy change (difference between the total enthalpy of formation of the products and that of the reactants) is called the *heat of reaction* ΔH_r :

$$\Delta H_r = (v_r H_{fr} + v_s H_{fs} + \dots) - (v_a H_{fa} + v_b H_{fb} + \dots) \quad (7-10)$$

where H_{fi} are the *enthalpies of formation* of components i . The reaction is *exothermic* if heat is produced by the reaction (negative heat of reaction) and *endothermic* if heat is consumed (positive heat of reaction). The magnitude of the effective heat of reaction depends upon temperature and the phases of the reactants and product. To estimate the dependence of the heat of reaction on temperature relative to a

reference temperature T_0 , the following expression can be used, provided there is no phase change:

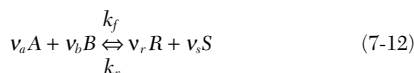
$$\Delta H_r(T) = \Delta H_r(T_0) + \int_{T_0}^T \Delta c_p dT \quad (7-11)$$

$$\Delta c_p = (v_r c_{pr} + v_s c_{ps} + \cdots) - (v_a c_{pa} + v_b c_{pb} + \cdots)$$

where c_{pi} are the *heat capacities* of component i . The heat of reaction can be measured by using calorimetry, or it can be calculated by using a variety of thermodynamic methods out of the scope of this chapter (see relevant sections of this handbook, thermodynamic texts, and Bird, Stewart, and Lightfoot, *Transport Phenomena*, 2d ed., John Wiley & Sons, New York, 2002). It is important to accurately capture the energy balance and its relation to the heat of reaction and heat capacities (see also Denn, *Process Modeling*, Langman, New York, 1986, for correct formulations). The coupling of the heat of reaction with the reaction rate through the Arrhenius equation often has a dominating effect on reactor selection and control, and on the laboratory reactor setup required to obtain accurate intrinsic kinetics and mechanism. More on this can be found in Sec. 19.

CHEMICAL EQUILIBRIUM

Often reactions or reaction steps in a network of reactions are at chemical equilibrium; i.e., the rate of the forward reaction equals the rate of the reverse reaction. For instance, for the reversible reaction



with mass action kinetics, the rate may be written as

$$r = r_f - r_r = k_f C_a^{v_a} C_b^{v_b} - k_r C_r^{v_r} C_s^{v_s} \quad (7-13)$$

At chemical equilibrium the forward and reverse reaction rates are equal according to the principle of *microscopic reversibility*:

$$r = r_f - r_r = 0 \quad (7-14)$$

The equilibrium constant K_e (based on volumetric concentrations) is defined as the ratio of the forward and reverse rate constants and is related to the composition at equilibrium as follows:

$$K_e = \frac{k_f}{k_r} = \frac{C_r^{v_r} C_s^{v_s}}{C_a^{v_a} C_b^{v_b}} \quad (7-15)$$

K_e can be calculated from the free energy change of the reaction. Using the van't Hoff relation, we obtain the dependence of K_e on temperature:

$$\frac{d(\ln K_e)}{dT} = \frac{\Delta H_r}{RT^2} \quad (7-16)$$

Integrating with respect to temperature, we obtain a form similar to the Arrhenius expression of the rate constant for a narrow range of temperature:

$$K_e = K_{e0} \exp\left(\frac{\Delta H_r}{RT}\right) \quad (7-17)$$

A more general integral form of Eq. (7-16) is

$$\ln K_e(T) = \ln K_e(T_0) + \frac{1}{R} T \int_{T_0}^T \frac{\Delta H_r(T_0) + \int_{T_0}^T \Delta c_p dT}{T^2} dt \quad (7-18)$$

When a reversible reaction is not at equilibrium, knowledge of K_e can be used to eliminate the rate constant of the reverse reaction by using Eq. (7-15) as follows:

$$r = k_f (C_a^{v_a} C_b^{v_b} - C_r^{v_r} C_s^{v_s} / K_e) \quad (7-19)$$

When several reversible reactions occur simultaneously, each reaction r_j is characterized by its equilibrium constant $K_{e,j}$. When the $K_{e,j}$ are known, the composition at equilibrium can be calculated from a set of equations such as Eq. (7-15) for each reaction.

CONVERSION, EXTENT OF REACTION, SELECTIVITY, AND YIELD

Conversion of a reactant is the number of moles converted per initial or feed moles of a reactant. Thus for component A

$$X_a = 1 - \frac{n_a}{n_{a0}} \quad (7-20)$$

A *limiting reactant* is a reactant whose concentration at the start of the reaction is the least of all reactants relative to the required stoichiometric amount needed for complete conversion. For instance, for the single reaction (7-12), A is the limiting reactant if the initial molar ratio of concentrations of A and B is less than the ratio of their stoichiometric coefficients:

$$\frac{n_{a0}}{n_{b0}} < \frac{v_a}{v_b} \quad (7-21)$$

Once the limiting reactant is depleted, the respective reaction stops even though other (nonlimiting) reactants may still be abundant.

For each reaction or each step in a network of reactions, a unique *extent of reaction* ξ that relates the composition of components that participate in the reaction to each other can be defined. For instance, for the single reaction (7-1):

$$\xi = \frac{n_{a0} - n_a}{v_a} = \frac{n_{b0} - n_b}{v_b} = \cdots = -\frac{n_{r0} - n_r}{v_r} = -\frac{n_{s0} - n_s}{v_s} = \cdots \quad (7-22)$$

The extent of reaction is related to conversion as follows:

$$\xi = \frac{X_a n_{a0}}{v_a} = \frac{X_b n_{b0}}{v_b} = \cdots \quad (7-23)$$

When A is the limiting reactant as in Eq. (7-21), the maximum extent of reaction (with A fully converted) is

$$\xi_{\max} = \frac{n_{a0}}{v_a} \quad (7-24)$$

For multiple reactions with reactants participating in more than one reaction, it is more difficult to determine the limiting reactant and often it is necessary to calculate the concentration as the reactions proceed to determine which reactant is consumed first. When the limiting reactant is depleted, all reactions that use this component as reactant stop, and the corresponding rates become zero.

Selectivity S of a product is the ratio of the rate of production of that product to the rate of production of all products combined. For a single reaction selectivity is trivial—if more than one product occurs, then the selectivity of each product is the ratio of the stoichiometric coefficient of that product to the sum of stoichiometric coefficients of all the products. Thus for reaction (7-1)

$$S_r = \frac{v_r}{v_r + v_s + \cdots} \quad (7-25)$$

7-8 REACTION KINETICS

The selectivity of product R for a network of reactions, with all the reactions making the various products included, is

$$S_r = \frac{r_r}{\sum_{\text{all products } i} r_i} \quad (7-26)$$

For instance, for the network of reactions $A + B \xrightarrow{1} C + D$
 $C + E \xrightarrow{2} F + G$, the selectivity to product C is

$$S_c = \frac{r_c}{r_c + r_d + r_f + r_g} = \frac{r_1 - r_2}{(r_1 - r_2) + r_1 + r_2 + r_2} = \frac{r_1 - r_2}{2r_1 + r_2}$$

The *yield* Y of a product R with respect to a reactant A is the ratio of the rate of production of R to that of consumption of A :

$$Y_r = \frac{r_r}{-r_a} \quad (7-27)$$

For a single reaction the yield is trivial and Eq. (7-27) simplifies to the ratio of the respective stoichiometric coefficients:

$$Y_r = \frac{v_r}{v_a} \quad (7-28)$$

The yield quantifies the efficiency of the respective reactant utilization to make the desired products.

CONCENTRATION TYPES

Different concentration types are used for different reaction systems. For gas-phase reactions, *volumetric concentration* or *partial pressures* are equally useful and can be related by the thermodynamic equation of state. For instance, for ideal gases (approximation valid for gases at very low pressure)

$$PV = nRT \quad (7-29)$$

When applied to individual components in a constant-volume system,

$$p_i V = n_i RT \quad (7-30)$$

Using Eq. (7-5), we obtain the relationship between the volumetric concentrations and partial pressures:

$$C_i = \frac{p_i}{RT} \quad (7-31)$$

For an ideal gas the total concentration is

$$C = \frac{P}{RT} \quad (7-32)$$

For higher pressure and nonideal gases, a *compressibility factor* z_i can be used:

$$p_i V = z_i n_i RT \quad \text{and} \quad C_i = \frac{p_i}{z_i RT} \quad (7-33)$$

Other relevant equations of state can also be used for both gases and liquids. This aspect is not in the scope of this section, and the reader is referred to the relevant section of this handbook.

Other concentration units include *mole fractions* for liquid x_i :

$$x_i = \frac{n_i}{\sum n_i} = \frac{C_i}{\sum C_i} = \frac{C_i}{C} \quad (7-34)$$

and for gas y_i

$$y_i = \frac{n_i}{\sum n_i} = \frac{C_i}{\sum C_i} = \frac{C_i}{C} = \frac{p_i}{\sum p_i} = \frac{p_i}{P} \quad (7-35)$$

The last two terms are only valid for an ideal gas.

STOICHIOMETRIC BALANCES

Single Reactions Equation (7-22) shows that for a single reaction, the number of moles and concentration of all other components can be calculated from the extent of reaction ξ or the conversion based on the limiting reactant, say A , X_a . In terms of number of moles n_i ,

$$\begin{aligned} n_a &= n_{a0} - v_a \xi = n_{a0}(1 - X_a) \\ n_b &= n_{b0} - v_b \xi = n_{b0}(1 - X_b) = n_{b0} - \frac{v_b}{v_a} n_{a0} X_a \\ &\dots \\ n_r &= n_{r0} + v_r \xi = n_{r0} - \frac{v_r}{v_a} n_{a0} X_a \\ n_s &= n_{s0} + v_s \xi = n_{s0} - \frac{v_s}{v_a} n_{a0} X_a \\ &\dots \end{aligned} \quad (7-36)$$

Similarly the number of moles of each component in terms of moles of A , n_a , is

$$\begin{aligned} n_b &= n_{b0} - \frac{v_b}{v_a} (n_{a0} - n_a) \\ &\dots \\ n_r &= n_{r0} + \frac{v_r}{v_a} (n_{a0} - n_a) \\ n_s &= n_{s0} + \frac{v_s}{v_a} (n_{a0} - n_a) \\ &\dots \end{aligned} \quad (7-37)$$

Change in number of moles by the reaction and change in temperature, pressure, and density affect the translation of stoichiometric balances from number of moles to volumetric concentrations. These relationships are different for gases and liquids. For instance, for *constant density* systems (such as many liquid-phase isothermal reactions) or for *constant temperature, constant pressure* gas reaction with no change in number of moles, Eqs. (7-36) and (7-37) can be changed to volumetric concentration C_i by dividing each equation by the constant reaction volume V (e.g., in a batch reactor) and using Eq. (7-5). For example, for the single reaction (7-4) with rate (7-5):

$$\begin{aligned} C_b &= C_{b0} - \frac{q}{p} (C_{a0} - C_a) \\ C_r &= C_{r0} + \frac{r}{p} (C_{a0} - C_a) \\ C_s &= C_{s0} + \frac{s}{p} (C_{a0} - C_a) \end{aligned} \quad (7-38)$$

$$r = k C_a^a \left[C_{b0} - \frac{q}{p} (C_{a0} - C_a) \right]^b = k [C_{a0}(1 - X_a)]^a \left(C_{b0} - \frac{q}{p} C_{a0} X_a \right)^b \quad (7-39)$$

It is best to represent all concentrations in terms of that of the limiting reactant.

Often there is a change in total number of moles due to reaction. Taking the general reaction (7-1), in the gas phase the change in

number of moles relative to moles of component A converted, δ_a , and the total number of moles can be calculated as follows:

$$\begin{aligned}\delta_a &= \frac{v_q + v_s + \dots - v_a - v_b - \dots}{v_a} \\ n_0 &= \sum n_{i0} \quad n = \sum n_i \\ \frac{n}{n_0} &= 1 + y_{a0}\delta_a X_a = 1 + \varepsilon_a X_a \\ \varepsilon_a &= y_{a0}\delta_a\end{aligned}\quad (7-40)$$

Using the ideal gas law, Eq. (7-29), the volume change depends on conversion as follows:

$$\frac{V}{V_0} = \frac{T}{T_0} \frac{P_0}{P} \frac{n}{n_0} = \frac{T}{T_0} \frac{P_0}{P} (1 + \varepsilon_a X_a) \quad (7-41)$$

Hence, for an isothermal constant-pressure ideal gas reaction system,

$$\begin{aligned}C_a &= \frac{n_a}{V} = \frac{C_{a0}(1 - X_a)}{1 + \varepsilon_a X_a} \\ C_b &= \frac{n_b}{V} = \frac{C_{b0} - \frac{v_b}{v_a} C_{a0} X_a}{1 + \varepsilon_a X_a} \\ \dots \\ C_r &= \frac{n_r}{V} = \frac{C_{r0} - \frac{v_r}{v_a} C_{a0} X_a}{1 + \varepsilon_a X_a} \\ C_s &= \frac{n_s}{V} = \frac{C_{s0} - \frac{v_s}{v_a} C_{a0} X_a}{1 + \varepsilon_a X_a} \\ \dots\end{aligned}\quad (7-42)$$

Applying this to reaction (7-4) and rate (7-5) gives

$$r = k \left[\frac{C_{a0}(1 - X_a)}{1 + \varepsilon_a X_a} \right]^a \left[\frac{C_{b0} - (q/p)C_{a0}X_a}{1 + \varepsilon_a X_a} \right]^b \quad (7-43)$$

Compare this to Eq. (7-39) where there is no change in number of moles.

Reaction Networks The analysis for single reactions can be extended to a network of reactions by defining an extent of reaction for each reaction, or by choosing a representative reactant concentration for each reaction step. For a complex network, the number of independent extents of reaction required to calculate the concentration of all components is equal to the number of *independent reactions*, which is less than or equal to the total number of reactions in the network. To calculate the number of independent reactions, and to form a set of independent reactions and corresponding independent set of concentrations or extents of reaction, we need to construct the *stoichiometric matrix* and determine its *rank*. The stoichiometric matrix is used to derive a relationship between the concentrations and the independent extents of reaction similar to that of a single reaction.

The stoichiometric matrix is the matrix of the stoichiometric coefficients of the reaction network with negative signs for reactants and positive signs for products. For instance, the hydrodechlorination of Freon 12 (CF_2Cl_2) can proceed with the following consecutive mechanism [Bonarowska et al., "Hydrodechlorination of CCl_2F_2 (CFC-12) over Silica-Supported Palladium-Gold Catalysts," *Appl. Catal. B: Environmental*, **30**:187–193, 2001]:



The stoichiometric matrix S for this network is

$$S = \begin{pmatrix} -1 & 1 & 0 & -1 & 1 \\ 0 & -1 & 1 & -1 & 1 \end{pmatrix}$$

The first row refers to the first reaction and the second row to the second reaction. The columns (species) are in the following order: 1- CF_2Cl_2 , 2- CF_2ClH , 3- CF_2H_2 , 4- H_2 , and 5-HCl. The rank of a matrix is the largest square submatrix obtained by deleting rows and columns, whose determinant is not zero. The rank equals the number of independent reactions. This is also equivalent to stating that there are reactions in the network that are linear combinations of the independent reactions. The rank of S above is 2, since the *determinant* of the first 2×2 submatrix is not zero (there are other 2×2 submatrices that are not zero as well but it is sufficient to have at least one that is not zero):

$$S_1 = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix} \quad \det(S_1) = 1 \neq 0$$

Hence the two reactions are independent. Now if we add another step, which converts Freon 12 directly into the final hydrofluorocarbon CF_2H_2 : $\text{CF}_2\text{Cl}_2 + 2\text{H}_2 \Rightarrow \text{CF}_2\text{H}_2 + 2\text{HCl}$, then the stoichiometric matrix becomes

$$S = \begin{pmatrix} -1 & 1 & 0 & -1 & 1 \\ 0 & -1 & 1 & -1 & 1 \\ -1 & 0 & 1 & -2 & 2 \end{pmatrix}$$

Since the last reaction is a linear combination of the first two (sum), it can be easily proved that the rank remains unchanged at 2. So to conclude, the concentrations of all components in this network can be expressed in terms of two, say H_2 and Freon 12, and the first two reactions form an independent reaction set. In case of more complicated networks it may be difficult to determine the independent reactions by observation alone. In this case the *Gauss-Jordan decomposition* leads to a set of independent reactions (see, e.g., Amundson, *Mathematical Methods in Chemical Engineering—Matrices and Their Application*, Prentice-Hall International, New York, 1966).

For a network of reactions the general procedure is as follows:

1. Generate the reaction network by including all known reaction steps.
2. Generate the corresponding stoichiometric matrix.
3. Calculate the rank of the stoichiometric matrix which equals the number of independent reactions and independent component concentrations required to calculate all the remaining component concentrations.
4. For relatively simple networks, observation allows selection of reactions that are independent—for more complex systems use the Gauss-Jordan elimination to reduce the network to a set of independent (nonzero rows) reactions.
5. Select the independent concentration variables and independent reactions, and use these to calculate all other concentrations and reaction rates.

CATALYSIS

A catalyst is a material that increases the rate of both the forward and reverse reactions of a reaction step, with no net consumption or generation of catalyst by the reaction. A catalyst does not affect the reaction thermodynamics, i.e., the equilibrium composition or the heat of reaction. It does, however, affect the temperature sensitivity of the reaction rate by lowering the activation energy or the energy barrier on the reaction pathway from reactants to products. This allows the reaction to occur faster than the corresponding uncatalyzed reaction at a given temperature. Alternatively, catalytic reactions can proceed at lower temperatures than the corresponding noncatalytic reactions. For a network of reactions, the catalyst is often used to speed up desired reactions and/or to slow down undesired reactions for improved selectivity. On the basis of catalysis, reactions can be further classified into

1. Noncatalytic reactions, e.g., free-radical gas-phase reactions such as combustion of hydrocarbons.

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2. Homogeneous catalytic reactions with the catalyst being dissolved in the same phase as the reactants and products in a homogeneous reaction medium. Here the catalyst is uniformly distributed throughout the system, e.g., the hydroformylation of olefins in the presence of dissolved Co or Rh carbonyls.

3. Heterogeneous catalytic reactions, with the catalyst, for instance, being a solid in contact with reactants and products in a gas-solid, gas-liquid-solid, or a liquid-solid reaction system. Here the catalyst is not uniformly distributed, and the reaction occurring on the catalyst surface requires, for instance, adsorption of reactants and desorption of products from the solid surface, e.g., the catalytic cracking of gasoil to gasoline and lighter hydrocarbons.

Table 7-1 illustrates the enhancement of the reaction rates by the catalyst—this enhancement can be of many orders of magnitude.

TABLE 7-1 The Rate of Enhancement of Some Reactions in the Presence of a Catalyst

Reaction	Catalyst	Rate enhancement	Temperature, K
Ortho $\text{H}_2 \Rightarrow$ para H_2	Pt (solid)	10^{40}	300
$2\text{NH}_3 \Rightarrow \text{N}_2 + 3\text{H}_2$	Mo (solid)	10^{20}	600
$\text{C}_2\text{H}_4 + \text{H}_2 \Rightarrow \text{C}_2\text{H}_6$	Pt (solid)	10^{42}	300
$\text{H}_2 + \text{Br}_2 \Rightarrow 2\text{HBr}$	Pt (solid)	1×10^8	300
$2\text{NO} + 2\text{H}_2 \Rightarrow \text{N}_2 + 2\text{H}_2\text{O}$	Ru (solid)	3×10^{16}	500
$\text{CH}_3\text{COH} \Rightarrow \text{CH}_4 + \text{CO}$	I_2 (gas)	4×10^6	500
$\text{CH}_3\text{CH}_3 \Rightarrow \text{C}_2\text{H}_4 + \text{H}_2$	NO_2 (gas)	1×10^9	750
$(\text{CH}_3)_3\text{COH} \Rightarrow (\text{CH}_3)_2\text{CH}_2\text{CH}_2 + \text{H}_2\text{O}$	HBr (gas)	3×10^8	750

SOURCE: Masel, *Chemical Kinetics and Catalysis*, Wiley, 2001, Table 12.1.

IDEAL REACTORS

Reactions occur in reactors, and in addition to the intrinsic kinetics, observed reaction rates depend on the reactor type, scale, geometry, mode of operation, and operating conditions. Similarly, understanding of the reactor system used in the kinetic experiments is required to determine the reaction mechanism and intrinsic kinetics. In this section we address the effect of reactor type on observed rates. In Sec. 19 the effect of reactor type on performance (rates, selectivity, yield) is discussed in greater detail.

Material, energy, and momentum balances are essential to fully describe the performance of reactors, and often simplifying assumptions and phenomenological assumptions are needed especially for energy and momentum terms, as indicated in greater detail in Sec. 19

(see also Bird, Stewart, and Lightfoot, *Transport Phenomena*, 2d ed., John Wiley & Sons, New York, 2002). Ideal reactors allow us to simplify the energy, momentum, and material balances, thus focusing the analysis on intrinsic kinetics. A useful classification of ideal reactor types is in terms of their concentration distributions versus reaction time and space. Three types of ideal reactors are considered in this section:

1. Ideal *batch reactors* (BRs) including *semibatch reactors* (SBRs)
2. Ideal *continuously stirred tank reactor* (CSTR), including single and multiple stages
3. *Plug flow reactor* (PFR) with and without *recycle*

Figure 7-1 shows these types of ideal reactors. Other types of ideal and nonideal reactors are treated in detail in Sec. 19.

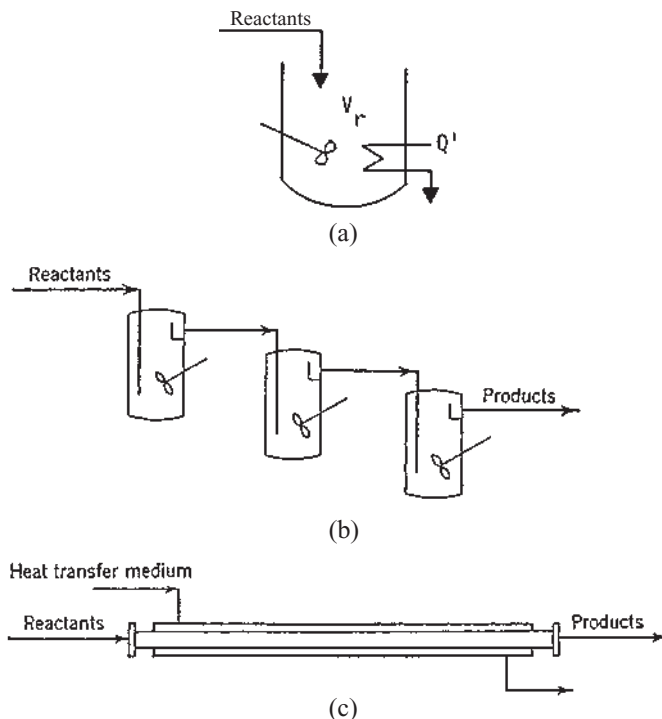


FIG. 7-1 Types of ideal reactors: (a) Batch or semibatch. (b) CSTR or series of CSTRs. (c) Plug flow.

The general form of a balance equation is

$$\text{Input} + \text{sources} - \text{outputs} = \text{accumulation} \quad (7-44)$$

IDEAL BATCH REACTOR

Batch Reactor (BR) Ideal batch reactors (Fig. 7-1a) are tanks provided with agitation for uniform composition and temperature at all times. An ideal batch reactor can be operated under isothermal conditions (constant temperature), temperature-programmed mode (by controlling cooling rate according to a protocol), or adiabatic mode with no heat crossing the reactor boundaries). In adiabatic mode the temperature is increasing, decreasing, or constant as the reaction proceeds for exothermic, endothermic, and thermally neutral reactions, respectively. In the ideal batch reactor, all the reactants are loaded into the reactor and well mixed by agitation before the conditions for reaction initiation (temperature and pressure) are reached; as the reaction proceeds, the concentration varies with time, but at any one time it is uniform throughout due to agitation.

Laboratory batch reactors can be single-phase (e.g., gas or liquid), multiphase (e.g., gas-liquid or gas-liquid-solid), and catalytic or non-catalytic. In this section we limit the discussion to operation at isothermal conditions. This eliminates the need to consider energy, and due to the uniform composition the component material balances are simple ordinary differential equations with time as the independent variable.

An ideal isothermal single-phase batch reactor in which a general reaction network takes place has the following general material balance equation:

$$\frac{dn_i}{dt} = V \sum_j \nu_{ij} r_j \quad n_i = n_{i0} \quad \text{at } t = 0 \quad (7-45)$$

The left-hand side is the accumulation term in moles per second of component i , and the right-hand side is the source term due to chemical reaction also in moles per second, which includes all reactions j that consume or produce component i , and the corresponding stoichiometric coefficients are represented in matrix form as ν_{ij} with a positive sign for products and a negative sign for reactants. This molar

balance is valid for each component since we can multiply each side of the equation by the component molecular weight to obtain the true mass balance equation. In terms of conversion, Eq. (7-45) can be rewritten as

$$\frac{n_{i0}}{dt} \frac{dX_i}{dt} = -V \sum_j \nu_{ij} r_j \quad n_i = n_{i0} \quad \text{at } t = 0 \quad (7-46)$$

and we can integrate this equation to get the *batch reaction time* or *batch residence time* τ_{BR} required to obtain a conversion X_i , starting with initial conversion X_{i0} and ending with final conversion X_{if} :

$$\tau_{BR} = -n_{i0} \int_{X_{i0}}^{X_{if}} \frac{dX_i}{V \sum_j \nu_{ij} r_j} \quad (7-47)$$

To integrate we need to represent all reaction rates r_j in terms of the conversion X_i . For a single reaction this is straightforward [see, e.g., Eq. (7-43)]. However, for a network of reactions, integration of a system of often nonlinear differential equations is required using implicit or semi-implicit integration. For references please see the relevant section of this handbook or any textbook on ordinary differential equations.

A special case of batch reactors is constant-volume or constant-density operation typical of liquid-phase reactions, with volume invariant with time:

$$\frac{dC_i}{dt} = \sum_j \nu_{ij} r_j \quad C_i = C_{i0} \quad \text{at } t = 0 \quad (7-48)$$

A typical concentration profile versus time for a reactant is shown in Fig. 7-2a. Integration of Eq. (7-48) gives the batch residence time

$$\tau_{BR} = \int_{C_{i0}}^{C_{if}} \frac{dC_i}{\sum_j \nu_{ij} r_j} \quad (7-49)$$

For instance, for a single reaction, Eq. (7-43) can be used to describe the reaction rate r_i in terms of one reactant concentration. For reaction networks integration of a system of ordinary differential equations is required.

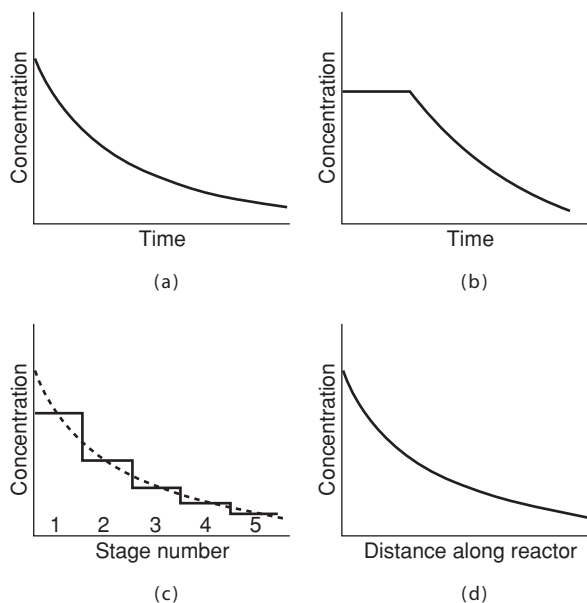


FIG. 7-2 Concentration profiles in batch and continuous flow: (a) Batch time profile. (b) Semibatch time profile. (c) Five-stage CSTRs distance profile. (d) Tubular flow distance profile.

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Semibatch Reactor (SBR) In semibatch operation, a gas of limited solubility or a liquid reactant may be fed in gradually as it is used up. An ideal isothermal single-phase semibatch reactor in which a general reaction network takes place has the following general material balance equation:

$$\frac{dn_i}{dt} = N_{i0} + V \sum v_{ij} r_j \quad n_i = n_{i0} \quad \text{at } t = 0$$

$$N_{i0} = N_{i0}(t) \quad \text{for } t_{s0i} \leq t \leq t_{s1i} \quad (7-50)$$

The first term on the right-hand side of Eq. (7-50) is the molar feed rate of the components, which can be different for each component, hence the subscript i , and can vary with time. A typical concentration profile versus time for a reactant whose concentration is kept constant initially by controlling the feed rate is shown in Fig. 7-2b. Knowledge of the reaction kinetics allows these ordinary differential equations to be integrated to obtain the reactor composition versus time.

IDEAL CONTINUOUS STIRRED TANK REACTOR (CSTR)

In an ideal continuous stirred tank reactor, composition and temperature are uniform throughout just as in the ideal batch reactor. But this reactor also has a continuous feed of reactants and a continuous withdrawal of products and unconverted reactants, and the effluent composition and temperature are the same as those in the tank (Fig. 7-1b). A CSTR can be operated under transient conditions (due to variation in feed composition, temperature, cooling rate, etc., with time), or it can be operated under steady-state conditions. In this section we limit the discussion to isothermal conditions. This eliminates the need to consider energy balance equations, and due to the uniform composition the component material balances are simple ordinary differential equations with time as the independent variable:

$$\frac{dn_i}{dt} = N_{i0} - N_i + V \sum v_{ij} r_j \quad n_i = n_{i0} \quad \text{at } t = 0 \quad (7-51)$$

At steady state the differential equations simplify to algebraic equations as indicated below:

$$V = - \frac{N_{i0} - N_i}{\sum v_{ij} r_j} \quad (7-52)$$

Equation (7-52) can be expressed in terms of volumetric concentration or in terms of conversions just as we did with the batch reactor. An *apparent residence time* based on feed conditions can be defined for a single-phase CSTR as follows:

$$\tau_{\text{CSTR}} = \frac{V}{q_0} \quad N_{i0} = q_0 C_{i0} \quad N_i = q C_i \quad (7-53)$$

In Eq. (7-53) the feed and effluent molar rates N_{i0} and N_i are expressed in terms of volumetric flow rates q_0 and q (inlet and outlet, respectively) and concentrations. Thus Eq. (7-52) can be rewritten as

$$\tau_{\text{CSTR}} = \frac{C_{i0} - (q/q_0)C_i}{\sum v_{ij} r_j} \quad (7-54)$$

Equation (7-54) allows calculation of the residence time required to achieve a given conversion or effluent composition. In the case of a network of reactions, knowing the reaction rates as a function of volumetric concentrations allows solution of the set of often nonlinear algebraic material balance equations using an implicit solver such as the multi variable Newton-Raphson method to determine the CSTR effluent concentration as a function of the residence time. As for batch reactors, for a single reaction all compositions can be expressed in terms of a component conversion or volumetric concentration, and Eq. (7-54) then becomes a single nonlinear algebraic equation solved by the Newton-Raphson method (for more details on this method see the relevant section this handbook).

A special case of Eq. (7-54) is a constant-density system (e.g., a liquid-phase reaction), with the true average residence time τ_{CSTR}

$$\tau_{\text{CSTR}} = \frac{C_{i0} - C_i}{\sum v_{ij} r_j} \quad q = q_0 \quad (7-55)$$

When a number of such CSTRs are employed in series, the concentration profile is step-shaped if the abscissa is the total residence time or the stage number as indicated by a typical reactant concentration profile in Fig. 7-2c.

PLUG FLOW REACTOR (PFR)

In a *plug flow reactor* all portions of the feed stream move with the same radially uniform velocity along parallel streamlines and therefore have the same residence time; that is, there is no mixing in the axial direction but complete mixing radially (Fig. 7-1c). As the reaction proceeds, the concentration falls off with distance. A PFR can be operated under either transient conditions or steady-state conditions. In this section we limit the discussion to steady-state conditions. This eliminates the need to consider partial differential equations in time and space. We further limit the discussion to isothermal operation, which together with the defined plug flow also eliminates the need for energy and momentum balance equations. Due to the radially uniform composition, the component material balances are simple ordinary differential equations with axial distance from inlet as the independent variable. An isothermal single-phase steady-state PFR in which a general reaction network takes place has the following general material balance equation:

$$\frac{dN_i}{dV} = \sum v_{ij} r_j \quad N_i = N_{i0} \quad \text{at } V = 0 \quad (7-56)$$

Note the similarity between the ideal batch and the plug flow reactors, Eqs. (7-45) and (7-56), respectively. In terms of conversion, Eq. (7-56) can be written as

$$N_{i0} \frac{dX_i}{dV} = - \sum v_{ij} r_j \quad (7-57)$$

Equation (7-57) can be integrated to calculate the reactor volume required to achieve a given conversion X_i :

$$V = -N_{i0} \int_{X_{i0}}^{X_{if}} \frac{dX_i}{\sum v_{ij} r_j} \quad (7-58)$$

An *apparent residence time* based on feed conditions can be defined for a single-phase PFR as follows:

$$\tau_{\text{PFR}} = \frac{V}{q_0} \quad (7-59)$$

Equation (7-58) becomes

$$\tau_{\text{PFR}} = -C_{i0} \int_{X_{i0}}^{X_{if}} \frac{dX_i}{\sum v_{ij} r_j} \quad (7-60)$$

Equation (7-60) is identical to that of the ideal batch reactor, Eq. (7-47), and the two reactor systems can be modeled in identical fashion.

For a constant-density system with no change in number of moles, with the true residence time τ_{PFR} :

$$\tau_{\text{PFR}} = \int_{C_{i0}}^{C_{if}} \frac{dC_i}{\sum v_{ij} r_j} \quad (7-61)$$

This is identical to the corresponding ideal batch reactor, Eq. (7-49).

Ideal Recycle Reactor All reactor modes can sometimes be advantageously operated with recycling part of the product or intermediate streams. Heated or cooled recycle streams serve to moderate undesirable temperature gradients, and they can be processed for changes in composition such as separating products to remove equilibrium limitations, before being returned. Say the recycle flow rate in

TABLE 7-2 Consecutive and Parallel First-Order Reactions in an Isothermal Constant-Volume Ideal Batch or Plug Flow Reactor.

The independent variable t is either the batch time or the plug flow residence time.

Reaction network	Material balances	Concentration profiles
$A \xrightarrow{1} B \xrightarrow{2} C$	$\frac{dC_a}{dt} = -k_1 C_a$	$C_a = C_{a0} e^{-k_1 t}$
	$\frac{dC_b}{dt} = k_1 C_a - k_2 C_b$	$C_b = C_{b0} e^{-k_2 t} + \frac{k_1 C_{a0}}{k_2 - k_1} (e^{-k_1 t} - e^{-k_2 t})$
	$\frac{dC_c}{dt} = k_2 C_b$	$C_c = C_{a0} + C_{b0} + C_{c0} - C_a - C_b$
$A \xrightarrow{1} B$ $A \xrightarrow{2} C$	$\frac{dC_a}{dt} = -(k_1 + k_2) C_a$	$C_a = C_{a0} e^{-(k_1 + k_2)t}$
	$\frac{dC_b}{dt} = k_1 C_a$	$C_b = C_{b0} + \frac{k_1 C_{a0}}{k_2 + k_1} (1 - e^{-(k_1 + k_2)t})$
	$\frac{dC_c}{dt} = k_2 C_a$	$C_c = C_{a0} + C_{b0} + C_{c0} - C_a - C_b$

a PFR is q_R and the fresh feed rate is q_0 . With a fresh feed concentration of C_0 and a product concentration of C_2 , the composite reactor feed concentration C_1 and the recycle ratio R are

$$C_1 = \frac{C_0 + RC_2}{1 + R} \quad R = \frac{q_R}{q_0} \quad (7-62)$$

The change in concentration across the reactor becomes

$$\Delta C = C_1 - C_2 = \frac{C_2 - C_0}{1 + R} \quad (7-63)$$

Accordingly, the change in concentration (or in temperature) across the reactor can be made as small as desired by increasing the recycle ratio. Eventually, the reactor can become a well-mixed unit with essentially constant concentration and temperature, while substantial differences in composition will concurrently arise between the fresh feed inlet and the product withdrawal outlet, similar to a CSTR. Such an operation is useful for obtaining experimental data for analysis of rate

equations. In the simplest case, where the product is recycled without change in composition, the flow reactor equation at constant density is

$$\tau_{\text{PFR}} = (1 + R) \int_{C_{i0}}^{C_{if}} \frac{dC_i}{\sum v_j r_j} \quad (7-64)$$

Hence, recycling increases the residence time or reactor size required to achieve a given conversion, since $1 + R > 1$.

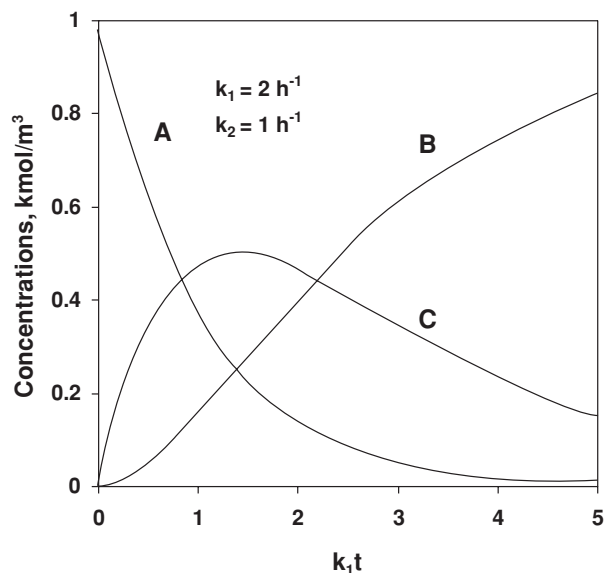
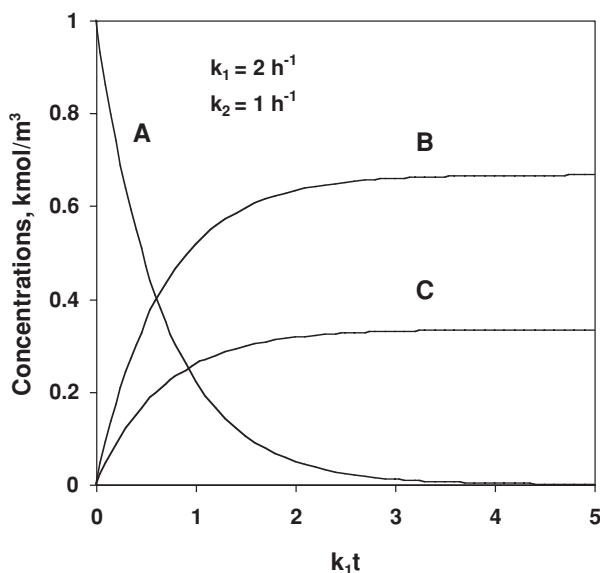
EXAMPLES FOR SOME SIMPLE REACTIONS

Table 7-2 and Figs. 7-3 and 7-4 show the analytical solution of the integrals for two simple first-order reaction systems in an isothermal constant-volume batch reactor or plug flow reactor. Table 7-3 shows the analytical solution for the same reaction systems in an isothermal constant-density CSTR.

Section 19 provides discussion about advantages and disadvantages of CSTRs versus PFR and BR for various reaction systems.

TABLE 7-3 Consecutive and Parallel First-Order Reactions in an Isothermal Constant-Volume Ideal CSTR

Reaction network	Material balances	Concentration profiles
$A \xrightarrow{1} B \xrightarrow{2} C$	$C_{a0} - C_a - \tau k_1 C_a = 0$	$C_a = \frac{C_{a0}}{1 + \tau k_1}$
	$C_{b0} - C_b + \tau(k_1 C_a - k_2 C_b) = 0$	$C_b = \frac{C_{b0}}{1 + \tau k_2} + \frac{\tau k_1 C_{a0}}{(1 + \tau k_1)(1 + \tau k_2)}$
	$C_{c0} - C_c + \tau k_2 C_b = 0$	$C_c = C_{a0} + C_{b0} + C_{c0} - C_a - C_b$
$A \xrightarrow{1} B$ $A \xrightarrow{2} C$	$C_{a0} - C_a - \tau(k_1 + k_2) C_a = 0$	$C_a = \frac{C_{a0}}{1 + \tau(k_1 + k_2)}$
	$C_{b0} - C_b + \tau k_1 C_a = 0$	$C_b = C_{b0} \frac{\tau k_1 C_{a0}}{1 + \tau(k_1 + k_2)}$
	$C_{c0} - C_c + \tau k_2 C_a = 0$	$C_c = C_{a0} + C_{b0} + C_{c0} - C_a - C_b$

FIG. 7-3 Concentration profiles for the reaction $A \rightarrow B \rightarrow C$.FIG. 7-4 Concentration profiles for the reaction $A \rightarrow B$ and $A \rightarrow C$.

KINETICS OF COMPLEX HOMOGENEOUS REACTIONS

Global or complex reactions are not usually well represented by mass action kinetics because the rate results from the combined effect of several simultaneous elementary reactions (each subject to mass action kinetics) that underlie the global reaction. The elementary steps include short-lived and unstable intermediate components such as free radicals, ions, molecules, transition complexes, etc.

The reason many global reactions between stable reactants and products have complex mechanisms is that these unstable intermediates have to be produced in order for the reaction to proceed at reasonable rates. Often simplifying assumptions lead to closed-form kinetic rate expressions even for very complex global reactions, but care must be taken when using these since the simplifying assumptions are valid over limited ranges of compositions, temperature, and pressure. These assumptions can fail completely—in that case the full elementary reaction network has to be considered, and no closed-form kinetics can be derived to represent the complex system as a global reaction.

Typical simplifying assumptions include these:

- *Pseudo-steady-state* approximation for the unstable intermediate; i.e., the concentration of these does not change during reaction
- Equilibrium for certain fast reversible reactions and completion of very fast irreversible steps
- *Rate-determining step(s)*; i.e., the global reaction rate is determined by the rate(s) of the slowest step(s) in the reaction network composing the overall or global reaction

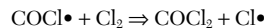
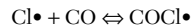
These simplifying assumptions allow elimination of some reaction steps, and representation of free radical and short-lived intermediates concentrations in terms of the concentration of the stable measurable components, resulting in complex non-mass action rate expressions.

Complex reactions can proceed through *chain* or *nonchain* mechanisms. In a *chain reaction*, the active unstable components are produced in an *initiation* step and are repeatedly regenerated through *propagation* steps, and only a small fraction of these are converted to stable components through a *termination* step. Free radicals are examples of such unstable components frequently encountered in chain reactions: free radicals are molecular fragments having one or more unpaired electrons, are usually short-lived (milliseconds), and are highly reactive. They are detectable spectroscopically, and some have been isolated. They occur as initiators and intermediates in such

basic phenomena as oxidation, combustion, photolysis, and polymerization. Several examples of free radical mechanisms possessing non-integral power law or hyperbolic rate equations are cited below. In a *nonchain reaction*, the unstable intermediate, such as an *activated complex* or *transition state complex*, reacts further to produce the products, and it is not regenerated through propagation but is continually made from reactants in stoichiometric quantities.

CHAIN REACTIONS

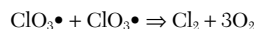
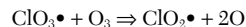
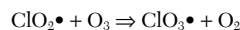
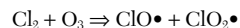
Phosgene Synthesis The global reaction $\text{CO} + \text{Cl}_2 \Rightarrow \text{COCl}_2$ proceeds through the following free radical mechanism:



Assuming the first two reactions are in equilibrium, expressions are found for the concentrations of the free radicals $\text{Cl}\bullet$ and $\text{COCl}\bullet$ in terms of the species CO , Cl_2 , and COCl_2 , and when these are substituted into the mass action rate expression of the third reaction, the rate becomes

$$r_{\text{COCl}_2} = k(\text{CO})(\text{Cl}_2)^{3/2} \quad (7-65)$$

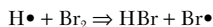
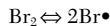
Ozone Conversion to Oxygen in Presence of Chlorine The global reaction $2\text{O}_3 \xrightarrow{\text{Cl}_2} 3\text{O}_2$ in the presence of Cl_2 proceeds through the following sequence:



The chain carriers $\text{ClO}\bullet$, $\text{ClO}_2\bullet$, and $\text{ClO}_3\bullet$ are assumed to attain pseudo-steady state. Then,

$$r_{\text{O}_3} = k(\text{Cl}_2)^{1/2}(\text{O}_3)^{3/2} \quad (7-66)$$

Hydrogen Bromide Synthesis The global reaction $\text{H}_2 + \text{Br}_2 \Rightarrow 2\text{HBr}$ proceeds through the following chain of reactions:

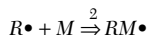


Assuming pseudo-steady state for the concentrations of the free radicals $\text{H}\bullet$ and $\text{Br}\bullet$, the global rate equation becomes

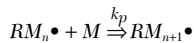
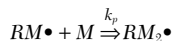
$$r_{\text{HBr}} = \frac{k_1(\text{H}_2)(\text{Br}_2)^{1/2}}{k_2 + \text{HBr}/\text{Br}_2} \quad (7-67)$$

Chain Polymerization For free radical polymerization, the following generic mechanism can be postulated:

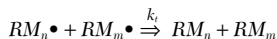
Initiation:



Propagation:



Termination:



The rates of formation of the free radicals $R\bullet$ and $M\bullet$ reach pseudo-steady states, resulting in the following polymerization rate:

$$r_p = k_p(M)(M^*) = k_p\left(\frac{k_i}{k_t}\right)^{1/2}(M)(I)^{1/2} \quad (7-68)$$

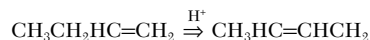
NONCHAIN REACTIONS

Nonchain reactions proceed through an active intermediate to the products. Many homogeneous nonchain reactions are also homogeneously catalyzed reactions, discussed below.

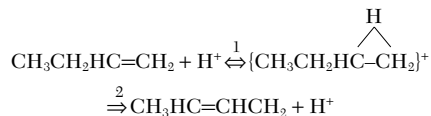
HOMOGENEOUS CATALYSIS

Homogeneous catalysts proceed through an activated or transition state complex between reactant(s) and catalysts, which decomposes into products. Homogeneous catalysts are dissolved in the homogeneous reaction mixture and include among others acids/bases, metal salts, radical initiators, solvents, and enzymes.

Acid-Catalyzed Isomerization of Butene-1 Butene-1 isomerizes to butene-2 in the presence of an acid according to the global reaction



Even though this appears to be a monomolecular reaction, it is not, as it proceeds through the following mechanism:



Assuming reaction 1 is in equilibrium, the reaction rate is

$$r = k_2K_1[\text{H}^+][\text{CH}_3\text{CH}_2\text{HC}=\text{CH}_2] \quad (7-69)$$

Enzyme Kinetics Enzymes are homogeneous catalysts for cellular and enzymatic reactions. The enzyme E and the reactant S are

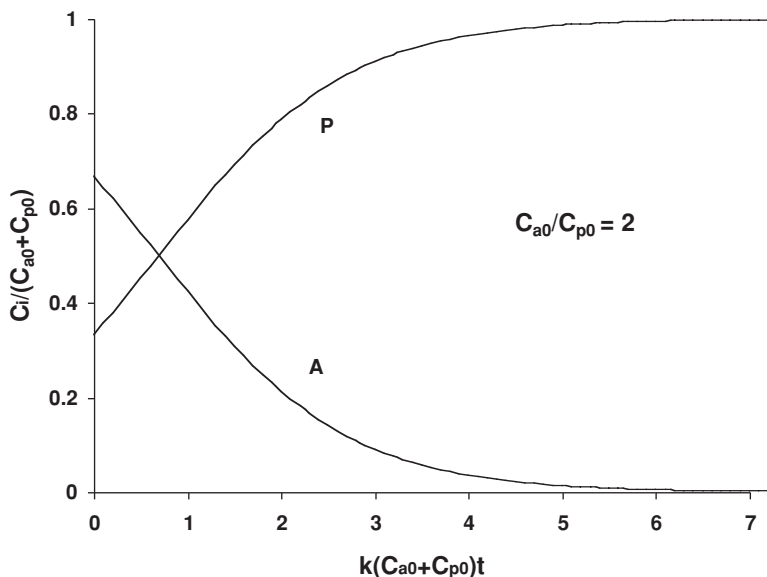
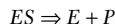
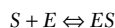


FIG. 7-5 Product concentration profile for the autocatalytic reaction $A + P \Rightarrow 2P$ with rate $r = kC_aC_p$.

7-16 REACTION KINETICS

assumed to form a complex ES that then dissociates into product P and releases the enzyme:



Assuming equilibrium for the first step results in the following rate, developed by Michaelis and Menten [*Biochem. Zeit.*, **49**:333 (1913)] and named Michaelis-Menten kinetics,

$$r_p = -r_s = \frac{k(S)}{K_m + (S)} \quad (7-70)$$

Here K_m is the inverse of the equilibrium constant for the first reaction.

AUTOCATALYSIS

In an autocatalytic reaction, a reactant reacts with a product to make more product. For the reaction to proceed, therefore, product must be present

initially in a batch or in the feed of a continuous reactor. Examples are cell growth in fermentation and combustion of fuels. For instance, the irreversible elementary reaction $A + P \Rightarrow 2P$ has the mass action kinetics

$$r = kC_a C_p \quad (7-71)$$

For an ideal batch reactor (see, e.g., Steinfeld, Francisco, and Hase, *Chemical Kinetics and Dynamics*, Prentice-Hall, 1989):

$$\frac{dC_a}{dt} = -kC_a C_p \quad \frac{dC_p}{dt} = kC_a C_p \quad C_a + C_p = C_{a0} + C_{p0} \quad (7-72)$$

Integration results in the following concentration profile:

$$C_p = \frac{C_{a0} + C_{p0}}{1 + (C_{a0}/C_{p0})e^{-k(C_{a0} + C_{p0})t}} \quad C_a = C_{a0} + C_{p0} - C_p \quad (7-73)$$

Figure 7-5 illustrates the dimensionless concentration profile for the reactant A and product P , $C_i/(C_{a0} + C_{p0})$, for $C_{a0}/C_{p0} = 2$, indicative of a maximum rate at the inflexion point (maximum in slope of the concentration-time curve), typical to autocatalytic reactions.

INTRINSIC KINETICS FOR FLUID-SOLID CATALYTIC REACTIONS

There are a large number of fluid-solid catalytic reactions, mostly gas-solid, including catalytic cracking, oxidation of polluting gases in automotive and power generation catalytic converters, partial oxidation synthesis reactions such as HCN synthesis, chemical vapor deposition, etc. (see, e.g., Sec. 19 for more examples). Examples of solid catalysts include, among others, supported metals, transition metal oxides and sulfides, solid acids and bases, and immobilized homogeneous catalysts and enzymes. Solid catalysts can be a fine powder (suspended in a liquid or fluidized by a flowing gas), cylindrical, spherical, and more-complex-shaped particles (in a packed bed), a thin layer of active components (on the walls of a monolith or a foam) and gauzes. The solid catalyst can be porous with active component distributed throughout the particle volume, or nonporous with active component present on the exposed catalyst external surface alone.

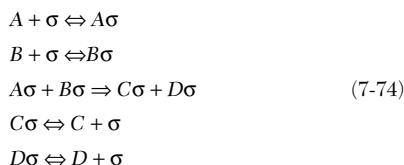
The analysis of Langmuir [*J. Am. Chem. Soc.* **40**:1361 (1918)] and Hinshelwood (*Kinetics of Chemical Change*, Oxford, 1940) form the basis for the simplified treatment of kinetics on heterogeneous catalysts. For a solid catalyzed reaction between gas phase reactants A and B , the postulated mechanism consists of the following steps in series:

1. The reactants from the gas adsorb to bond to active sites on the catalyst surface as molecules or dissociated atoms. The rate of adsorption is proportional to the partial pressure of reactants and to the fraction of uncovered surface sites ϑ_v . More than one type of active site can be present. The adsorption isotherms such as the Langmuir isotherm relate the partial pressure of an adsorbed species to its surface coverage, and the form of this relationship is indicative of the type of adsorption process taking place (see, for more details, Masel, *Chemical Kinetics and Catalysis*, Wiley, 2001).

2. The adsorbed species react on the surface to form adsorbed products. The rate of reaction between adsorbed species is proportional to their adsorbed concentrations on the surface.

3. The adsorbed products desorb into the gas. The rate of desorption of species A is proportional to the fraction of the surface covered by A , ϑ_a .

For instance, for the simple irreversible reaction $A + B \Rightarrow C + D$, the postulated mechanism is



$A\sigma$, $B\sigma$, $C\sigma$, and $D\sigma$ above are adsorbed species on the catalyst surface, and σ is an available active site. We will consider a variety of possible scenarios for this simple solid-catalyzed system. Note that the intrinsic specific reaction rate for such systems is often expressed on a unit mass catalyst basis (r_m) instead of unit reaction volume basis (r_v), and the latter is related to the former through the catalyst loading (mass catalyst/reaction volume) or bed density:

$$r_v = \frac{r_m M_{\text{cat}}}{V} = r_m \rho_B \quad (7-75)$$

ADSORPTION EQUILIBRIUM

Assuming equilibrium for all adsorption steps (e.g., the surface reaction is rate-limiting), the net rates of adsorption of reactants and product are all zero.

$$r_i = k_i p_i \vartheta_v - k_{-i} \vartheta_i \Rightarrow 0 \quad i = a, b, c, d \quad (7-76)$$

A material balance on all sites yields

$$\vartheta_v = 1 - \vartheta_a - \vartheta_b - \vartheta_c - \vartheta_d \quad (7-77)$$

and solving for the surface coverages gives

$$\begin{aligned} \vartheta_a &= \left(\frac{k_a}{k_{-a}} \right) p_a \vartheta_v = K_a p_a \vartheta_v & \vartheta_b &= \left(\frac{k_b}{k_{-b}} \right) p_b \vartheta_v = K_b p_b \vartheta_v \\ \vartheta_c &= \left(\frac{k_c}{k_{-c}} \right) p_c \vartheta_v = K_c p_c \vartheta_v & \vartheta_d &= \left(\frac{k_d}{k_{-d}} \right) p_d \vartheta_v = K_d p_d \vartheta_v \end{aligned} \quad (7-78)$$

The fraction of surface not covered is

$$\vartheta_v = \frac{1}{1 + K_a p_a + K_b p_b + K_c p_c + K_d p_d} \quad (7-79)$$

In the denominator, terms may be added for adsorbed inerts (e.g., $K_i p_i$) that may be present, and analogous expressions for the other participants. The rate of reaction or the rate-determining step is that between adsorbed reactant species:

$$r = k p_a p_b \vartheta_v^2 = \frac{k p_a p_b}{(1 + K_a p_a + K_b p_b + K_c p_c + K_d p_d)^2} \quad (7-80)$$

DISSOCIATION

A diatomic molecule A_2 may adsorb dissociatively as atoms



with the result

$$\theta_a = \frac{\sqrt{K_a p_a}}{1 + \sqrt{K_a p_a} + K_b p_b + \dots} = \sqrt{K_a p_a} \theta_v \quad (7-82)$$

and the rate-determining step and its rate are

**DIFFERENT SITES**

When A and B adsorb on chemically different sites σ_1 and σ_2 , the rate of the reaction, with surface reaction controlling,



is

$$r = \frac{k p_a p_b}{(1 + K_a p_a)(1 + K_b p_b)} \quad (7-85)$$

CHANGE IN NUMBER OF MOLES

When the numbers of moles of product is larger than that of the reactants, extra sites are required:

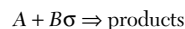


and the rate is

$$r = k \left(\theta_a \theta_v - \frac{\theta_m \theta_n}{K} \right) = \left(p_a - \frac{p_m p_n}{K} \right) \theta_v^2 = \frac{k(p_a - p_m p_n / K)}{(1 + K_a p_a + K_m p_m + K_n p_n)^2} \quad (7-88)$$

REACTANT IN THE GAS PHASE

When A in the gas phase reacts directly with adsorbed B :

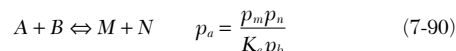


$$r = k p_a \theta_b = k p_a p_b \theta_v = \frac{k p_a p_b}{1 + \sum K_i p_i} \quad (7-89)$$

This mechanism is called the *Ely-Rideal kinetics*.

CHEMICAL EQUILIBRIUM IN GAS PHASE

When A is not in adsorptive equilibrium but it is in chemical equilibrium in the gas phase according to



this expression is substituted for p_a wherever it appears in the rate equation. If the rate-determining step is the surface reaction between adsorbed species, then

$$r = k p_a p_b \theta_v^2 = \frac{k p_m p_n / K_e}{1 + K_a (p_m p_n / K_e p_b) + K_b p_b + K_m p_m + K_n p_n} \quad (7-91)$$

Table 7-4 summarizes some examples of reactions where all substances are in adsorptive equilibrium and the surface reaction controls the rate. In Table 7-5, substance A is not in adsorptive equilibrium, and its adsorption rate is controlling. Details of the derivations of these and some other equations are presented by Yang and Hougen [*Chem. Eng.*

TABLE 7-4 Surface-Reaction Controlling.

Adsorptive equilibrium maintained for all participants.

Reaction	Special condition	Basic rate equation	Driving force	Adsorption term
1. $A \rightarrow M + N$ $A \rightarrow M + N$ $A \rightarrow M + N$	General case Sparsely covered surface Fully covered surface	$r = k \theta_a$ $r = k \theta_a$ $r = k \theta_a$	p_a p_a 1	$1 + K_a p_a + K_m p_m + K_n p_n$ 1 1
2. $A \rightleftharpoons M$		$r = k_1 \theta_a - k_{-1} \theta_m$	$p_a - \frac{p_m}{K}$	$1 + K_a p_a + K_m p_m$
3. $A \rightleftharpoons M + N$	Adsorbed A reacts with vacant site	$r = k_1 \theta_a \theta_v - k_{-1} \theta_m \theta_n$	$p_a - \frac{p_m p_n}{K}$	$(1 + K_a p_a + K_m p_m + K_n p_n)^2$
4. $A_2 \rightleftharpoons M$	Dissociation of A_2 upon adsorption	$r = k_1 \theta_a^2 - k_{-1} \theta_m \theta_v$	$p_a - \frac{p_m}{K}$	$(1 + \sqrt{K_a p_a} + K_m p_m)^2$
5. $A + B \rightarrow M + N$ $A + B \rightarrow M + N$	Adsorbed B reacts with A in gas but not with adsorbed A	$r = k \theta_a \theta_b$ $r = k p_a \theta_b$	$p_a p_b$ $p_a p_b$	$(1 + K_a p_a + K_b p_b + K_m p_m + K_n p_n)^2$ $1 + K_a p_a + K_b p_b + K_m p_m + K_n p_n$
6. $A + B \rightleftharpoons M$		$r = k_1 \theta_a \theta_b - k_{-1} \theta_m \theta_v$	$p_a p_b - \frac{p_m}{K}$	$(1 + K_a p_a + K_b p_b + K_m p_m)^2$
7. $A + B \rightleftharpoons M + N$		$r = k_1 \theta_a \theta_b - k_{-1} \theta_m \theta_n$	$p_a p_b - \frac{p_m p_n}{K}$	$(1 + K_a p_a + K_b p_b + K_m p_m + K_n p_n)^2$
8. $A_2 + B \rightleftharpoons M + N$	Dissociation of A_2 upon adsorption	$r = k_1 \theta_a^2 \theta_b - k_{-1} \theta_m \theta_v \theta_n$	$p_a p_b - \frac{p_m p_n}{K}$	$(1 + \sqrt{K_a p_a} + K_b p_b + K_m p_m + K_n p_n)^3$

NOTE: The rate equation is:

$$r = \frac{k (\text{driving force})}{\text{adsorption term}}$$

When an inert substance I is adsorbed, the term $K_I p_I$ is to be added to the adsorption term.

SOURCE: From Walas, *Reaction Kinetics for Chemical Engineers*, McGraw-Hill, 1959; Butterworths, 1989.

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TABLE 7-5 Adsorption-Rate Controlling (Rapid Surface Reaction)

Reaction	Special condition	Basic rate equation	Driving force	Adsorption term
1. $A \rightarrow M + N$		$r = kp_a\theta_c$	p_a	$1 + \frac{K_d p_m p_n}{K} + K_m p_m + K_n p_n$
2. $A \rightleftharpoons M$		$r = k \left(p_a \theta_c - \frac{\theta_a}{K_a} \right)$	$p_a - \frac{p_m}{K}$	$1 + \frac{K_d p_m}{K} + K_m p_m$
3. $A \rightleftharpoons M + N$		$r = k \left(p_a \theta_c - \frac{\theta_a}{K_a} \right)$	$p_a - \frac{p_m p_n}{K}$	$1 + \frac{K_d p_m p_n}{K} + K_m p_m + K_n p_n$
4. $A_2 \rightleftharpoons M$	Dissociation of A_2 upon adsorption	$r = k \left(p_a \theta_c^2 - \frac{\theta_a^2}{K_a} \right)$	$p_a - \frac{p_m}{K}$	$\left(1 + \sqrt{\frac{K_d p_m}{K} + K_m p_m} \right)^2$
5. $A + B \rightarrow M + N$	Unadsorbed A reacts with adsorbed B	$r = kp_a\theta_c$	p_a	$1 + \frac{K_d p_m p_n}{K p_b} + K_b p_b + K_m p_m + K_n p_n$
6. $A + B \rightleftharpoons M$		$r = k \left(p_a \theta_c - \frac{\theta_a}{K_a} \right)$	$p_a - \frac{p_m}{K p_b}$	$1 + \frac{K_d p_m}{K p_b} + K_b p_b + K_m p_m$
7. $A + B \rightleftharpoons M + N$		$r = k \left(p_a \theta_c - \frac{\theta_a}{K_a} \right)$	$p_a - \frac{p_m p_n}{K p_b}$	$1 + \frac{K_d p_m p_n}{K p_b} + K_b p_b + K_m p_m + K_n p_n$
8. $A_2 + B \rightleftharpoons M + N$	Dissociation of A_2 upon adsorption	$r = k \left(p_a \theta_c^2 - \frac{\theta_a^2}{K_a} \right)$	$p_a - \frac{p_m p_n}{K p_b}$	$\left(1 + \sqrt{\frac{K_d p_m p_n}{K p_b} + K_b p_b + K_m p_m + K_n p_n} \right)^2$

NOTES: The rate equation is:

$$r = \frac{k \text{ (driving force)}}{\text{adsorption term}}$$

Adsorption rate of substance A is controlling in each case. When an inert substance I is adsorbed, the term $K_I p_I$ is to be added to the adsorption term.

SOURCE: From Walas, *Reaction Kinetics for Chemical Engineers*, McGraw-Hill, 1959; Butterworths, 1989.

Prog. **46**:146 (1950)], Walas (*Reaction Kinetics for Chemical Engineers*, McGraw-Hill, 1959; Butterworths, 1989, pp. 153–164), and Rase (*Chemical Reactor Design for Process Plants*, vol. 1, Wiley, 1977, pp. 178–191).

NO RATE-CONTROLLING STEP

All the relations developed above assume that only one step is controlling. In a reaction system, changing the operating conditions may shift the control from one step to another. It is therefore also obvious that at certain conditions there is no single step controlling. In that case all the reactions and their respective rates have to be considered, and the adsorbed species cannot be eliminated from the rate expressions to obtain a single closed-form kinetic rate.

LIQUID-SOLID CATALYTIC REACTIONS

An analogous treatment for liquid-solid catalysis can be derived, with partial pressures replaced by liquid volumetric concentrations. Otherwise all the analyses hold.

BIOCATALYSIS

Biochemical reactions such as *aerobic* and *anaerobic fermentations* occur in the presence of living organisms or *cells*, such as bacteria, algae, and yeast. These reactions can be considered as *biocatalyzed* by the organism. Thus in a typical bioreactor a *substrate* (such as glucose) is fed into the *fermenter* or *bioreactor* in the presence of an initial amount of cells. The desired product can be the cells themselves or a secreted chemical called a *metabolite*. In either case the cells multiply in the presence of the substrate, and the rate of production of cells is proportional to the concentration of the cells—hence this process is autocatalytic. In a batch reactor with ample

supply of substrate, this results in *exponential growth* of the culture. A typical cell or biomass growth rate function, called the *Monod kinetics*, is identical in form to the Michaelis-Menten enzyme kinetics in Eq. (7-70):

$$\mu = \frac{\mu_{\max} C_s}{K_s + C_s} \quad (7-92)$$

In Eq. (7-92) μ is the *specific growth rate* of the culture. It is measured in units of reciprocal time (h^{-1}). Growth rate μ is related to the *volumetric growth rate* r_x of the culture: $r_x = C_x \mu$. This means that the true unit of μ is, e.g., (g biomass formed/h)/(g biomass present), where *g biomass* is the *dry-weight* (DW) of the biomass, obtained after evaporation of the water content of the cell (which constitutes about 80 percent of the *wet biomass weight*). Similarly C_s has the unit, e.g., (g DW)/(L medium volume). The variable C_s in Eq. (7-92) is the concentration of the *limiting substrate* in the medium (g/L). There are many substrates (including micronutrients) in the medium, but there is usually just one that determines the specific growth rate. This substrate is often a sugar (most likely glucose) but it could also be a metal ion (Mg^{2+} etc.), or PO_4^{3-} , NH_4^+ , . . . , or perhaps a hormone. The limiting substrate may easily change during a fermentation, and then the rate expression will change.

The two parameters in Eq. (7-92) are the *maximum specific growth rate* $\mu_{\max}$ (h^{-1}) and the *saturation constant* K_s (g substrate/L). The value of K_s is obtained as the substrate concentration at which $\mu = 1/2 \mu_{\max}$ (see Fig. 7-6). The form of Eq. (7-92) is entirely empirical, but it incorporates two important features: (1) At high substrate concentration the whole cell machinery is involved in cell synthesis, and the specific growth rate reaches a maximum $\mu_{\max}$; (2) at low substrate concentration formation of biomass is a first-order rate process (as in any other chemical reaction) and $\mu \rightarrow (\mu_{\max}/K_s)C_s$. Note that for many commonly used microorganisms K_s is much

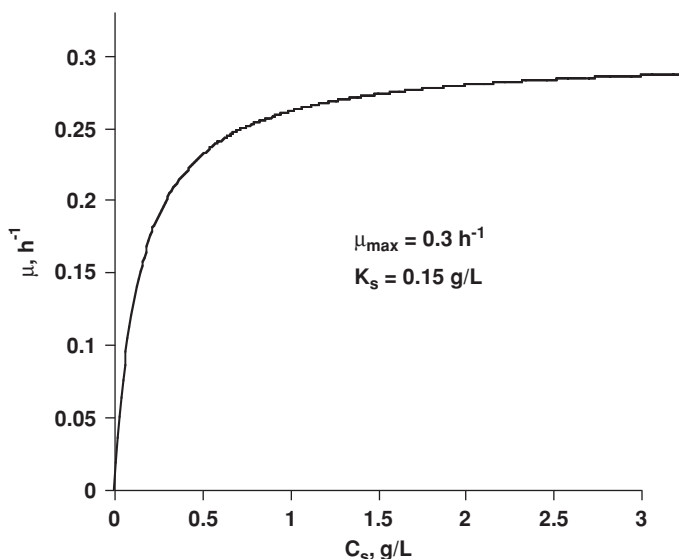


FIG. 7-6 The effect of substrate concentration on specific growth rate.

smaller than the substrate concentration C_s . Thus in batch cultivations K_s is several orders of magnitude smaller than C_s until the very end of the batch, and this is what gives the well-known *exponential growth* [see Eq. (7-93)]. Equation (7-93) applies for batch cultures after an initial *lag phase* when cell machinery is synthesized. Typical values for K_s are 150 mg/L (*Saccharomyces cerevisiae*), 5 to 10 mg/L (lactic bacteria and *E. coli*), and less than 1 mg/L (filamentous fungi).

$$\frac{dC_x}{dt} = \mu C_x \quad C_x = C_{x0} e^{\mu t} \quad (7-93)$$

Equation (7-93) may have to be modified by subtraction of a *death-rate* term $\mu_d C_x$. μ_d may well increase during the batch fermentation in which case the net growth rate of (viable) cells eventually becomes negative, and the concentration of (viable) cells will start to decrease.

FLUID-SOLID REACTIONS WITH MASS AND HEAT TRANSFER

GAS-SOLID CATALYTIC REACTIONS

The Langmuir-Hinshelwood mechanism of adsorption/reaction described above allowed us to relate the gas concentrations and partial pressures in the vicinity of the catalyst surface to the adsorbed species concentration at the active sites, which in turn determined the surface reaction rates. In practice, two additional mass-transfer processes may need to be considered:

1. Diffusion to and from the bulk gas to the external catalyst surface, represented as an external mass-transfer process across a film or boundary layer concentration gradient. For nonporous catalyst this is the only mass-transfer step.
2. Diffusion to and from the catalyst external surface through pores in a porous catalyst particle to active sites inside the catalyst particle where the adsorption and reaction occur, represented as intraparticle diffusion and modeled as a diffusion-reaction process.

External Mass Transfer In a reactor, the solid catalyst is deposited on the surface of narrow tubes (such as monolith or foams), is packed as particles in a tube, or is suspended in slurry or in a fluidized bed as fine particles. For these systems, the bulk concentration of the gas phase approaches that on the catalyst surface if the mass-transfer rate from bulk to surface is substantially larger than the reaction rates on the surface. This, however, is often not the case. The mechanism of mass transfer and reaction on the external catalyst surface includes the following consecutive steps:

1. Mass transfer of gas reactants from the bulk gas to the solid catalyst surface, also called external mass transfer
2. Adsorption, reaction on the surface, and desorption of products, e.g., Langmuir-Hinshelwood kinetics
3. Mass transfer of products from the catalyst surface to the bulk gas

At steady state all these rates are equal.

For example, for a first-order irreversible reaction $A \Rightarrow B$, the rate of mass transfer equals the rate of intrinsic reaction:

$$k_s a_s (C_a - C_{as}) = k C_{as} \quad (7-94)$$

Here a_s is, for instance, the external particle surface area/volume of reactor. Eliminating the surface concentration C_{as} in terms of the observable bulk gas concentration C_a yields the overall specific rate of consumption of A:

$$r_{\text{obs}} = k C_a = \frac{C_a}{1/(k_s a_s) + 1/k} \quad (7-95)$$

$$k_{\text{obs}} = \frac{1}{1/(k_s a_s) + 1/k}$$

Hence the observable overall rate constant k_{obs} is actually a combination of the mass-transfer coefficient and the intrinsic rate

7-20 REACTION KINETICS

coefficient; or in terms of resistances (in units of time) the overall resistance is the sum of the mass transfer and intrinsic kinetic resistance. For this first-order rate case, the overall behavior remains first order in bulk gas concentration. The two limiting cases are mass transfer and reaction control, respectively:

$$\begin{aligned} k_{\text{obs}} &= k_s a_s & \text{and } r_{\text{obs}} &= k_s a_s C_a & \text{for } k_s a_s \ll k & \text{Mass transfer control} \\ k_{\text{obs}} &= k & \text{and } r_{\text{obs}} &= k C_a & \text{for } k_s a_s \gg k & \text{Kinetic control} \end{aligned} \quad (7-96)$$

The mass-transfer coefficient depends on the geometry of the solid surface, on the hydrodynamic conditions in the vicinity of the catalyst (which are a function, e.g., of the reactor type, geometry, operating conditions, flow regime), and it also depends on the diffusivity of the gas species. Correlations for the mass-transfer coefficient are a large topic and outside the scope of this section. For more details see Bird, Stewart, and Lightfoot, *Transport Phenomena*, 2d ed., John Wiley & Sons, New York, 2002, and relevant sections in this handbook. For non-first-order kinetics a closed-form relationship such as the series of resistances cannot always be derived, but the steady-state assumption of the consecutive mass and reaction steps still applies.

Intraparticle Diffusion As indicated above, the larger the catalyst surface area per unit reaction volume a_s , the larger the overall reaction rate. For a fixed mass of catalyst, decreasing the particle size increases the total external surface area available for the reaction. Another way to increase the surface area is by providing a porous catalyst with lots of internal surface area. The internal structure of the catalyst determines how accessible these internal sites are to the gas-phase reactant and how easily can the products escape back to the gas. The analysis is based on the pseudo-homogeneous reaction diffusion equation, with the gas reactant diffusing through the pores and reacting at active sites inside the catalyst particle. For a first-order irreversible reaction of species A in an infinite slab geometry, the diffusion-reaction equations describe the decreasing reactant concentration from the external surface to the center of the slab:

$$\begin{aligned} D_{ea} \frac{d^2 C_{ay}}{dy^2} - k C_{ay} &= 0 & C_{ay}(L) &= C_{as} \\ \frac{dC_{ay}}{dy} (0) &= 0 \end{aligned} \quad (7-97)$$

The concept of effectiveness factor has been developed to calculate the overall reaction rate in terms of the concentration at the external surface C_{as} :

$$r = \eta k C_{as} \quad (7-98)$$

Hence the effectiveness factor is the ratio of the actual rate to that if the reactions were to occur at the external surface concentration, i.e., in absence of intraparticle diffusion resistance:

$$\eta = \frac{(1/L) \int_0^L r(C_{ay}) dy}{r(C_{as})} = \frac{\text{rate with pore diffusion resistance}}{\text{rate at external surface conditions}} \quad (7-99)$$

The effectiveness factor can be written as a function of a dimensionless independent variable called the Thiele modulus, which for a first-order reaction is defined below together with the corresponding effectiveness factor derived by integration of the corresponding diffusion-reaction equation (7-97):

$$\phi_{\text{slab}} = L \sqrt{\frac{k}{D_{ea}}} \quad \eta = \frac{\tanh \phi_{\text{slab}}}{\phi_{\text{slab}}} \quad (7-100)$$

Since the model is pseudo-homogeneous, there is no distinction

between the gas-filled pores and the solid parts of the particle. For most catalysts (except for straight channel monoliths), the diffusion path is not straight and has varying cross section. Hence, the effective diffusivity of A is defined based on the catalyst internal structure and the gas diffusivity of A in the gas mixture as follows:

$$D_{ea} = \frac{\epsilon_s}{\tau} D_a \quad (7-101)$$

The parameters that describe the pore structure are the porosity ϵ_s , accounting for the fact that diffusion only occurs through the gas-filled part of the particle, and the tortuosity τ accounting for the effect of diffusion path length and contraction/expansion of pores along the diffusion path. The diffusion regime depends on the diffusing molecule, pore size, and operating conditions (concentration, temperature, pressure), and this can be visualized in Fig. 7-7. As indicated, the effective diffusion coefficient ranges over many orders of magnitude from very low values in the configurational regime (e.g., in zeolites) to high values in the regular regime.

There is a large body of literature that deals with the proper definition of the diffusivity used in the intraparticle diffusion-reaction model, especially in multicomponent mixtures found in many practical reaction systems. The reader should consult references, e.g., Bird, Stewart, and Lightfoot, *Transport Phenomena*, 2d ed., John Wiley & Sons, New York, 2002; Taylor and Krishna, *Multicomponent Mass Transfer*, Wiley, 1993; and Cussler, *Diffusion Mass Transfer in Fluid Systems*, Cambridge University Press, 1997.

The larger the characteristic length L , the larger the Thiele modulus, the smaller the effectiveness factor, and the steeper the reactant concentration profile in the catalyst particle. A generalized characteristic length definition V_p/S_{px} (particle volume/external particle surface area) brings together the η - ϕ curves for a variety of particle shapes, as illustrated in Table 7-6 and Fig. 7-8 for for slabs, cylinders, and spheres. Here I_0 and I_1 are the corresponding modified Bessel functions of the first kind.

Further generalization of the Thiele modulus and effectiveness factor for a general global reaction and various shapes is

$$\phi = \frac{(V_p/S_{px}) r_a(C_{as})}{\sqrt{\frac{1}{2} \int_{C_{as}}^{C_{as}} D_{ea}(C_{ay}) r_a(C_{ay}) dC_{ay}}} \quad (7-102)$$

In Eq. (7-102) component A is the limiting reactant. For example, for an n th-order irreversible reaction

$$\phi = \frac{V_p}{S_{px}} \sqrt{\frac{(n+1)}{2} \frac{k C_{as}^{n-1}}{D_{ea}}} \quad (7-103)$$

This generalized Thiele modulus works well with the effectiveness factors for low and high values of the Thiele modulus, but it is not as accurate for intermediate values. However, these differences are not significant, given the uncertainties associated with measuring some of the other key parameters that go into the calculation of the Thiele modulus, e.g., the effective diffusivity and the intrinsic rate constant.

Effect of Intraparticle Diffusion on Observed Order and Activation Energy Taking the n th-order reaction case in the limit of intraparticle diffusion control, i.e., large Thiele modulus, the effectiveness factor is

$$\eta = \frac{1}{\phi} \quad (7-104)$$

the observed rate is

$$r_{\text{obs}} = \eta r = \frac{S_{px}}{V_p} \sqrt{\frac{2 D_{ea} k}{(n+1)}} C_{as}^{(n+1)/2} \quad (7-105)$$

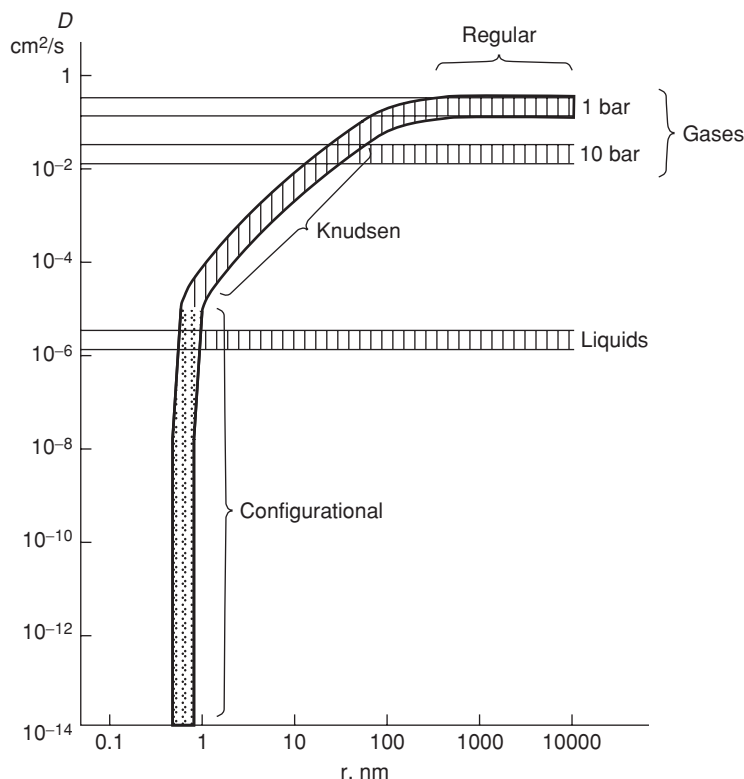


FIG. 7-7 Diffusion regimes in heterogeneous catalysts. [From Weisz, *Trans. Fara. Soc.* **69**: 1696–1705 (1973); Froment and Bischoff, *Chemical Reactor Analysis and Design*, Wiley, 1990, Figure 3.5.1-1.]

and the observed rate constant is

$$k_{\text{obs}} = \eta k = \frac{S_{px}}{V_p} \sqrt{\frac{2}{n+1}} D_{\text{eff}} e^{(E_D/RT)} k_0 e^{(E/RT)} \quad (7-106)$$

Hence, the observed order and activation energy differ from those of the intrinsic n th-order kinetics:

$$n_{\text{obs}} = \frac{n+1}{2} \quad E_{\text{obs}} = \frac{E + E_D}{2} \approx \frac{E}{2} \quad (7-107)$$

Here E_D is the activation energy for diffusion.

TABLE 7-6 Effectiveness Factors for Different Shapes for a First-Order Reaction

Shape	V_p/S_{px}	Effectiveness factor η
Infinite slab	R	$\frac{\tanh \phi}{\phi}$
Infinite cylinder	$R/2$	$\frac{I_1(2\phi)}{\phi I_0(2\phi)}$
Sphere	$R/3$	$\frac{1}{\phi} \left(\frac{3}{\tanh 3\phi} - \frac{1}{\phi} \right)$

The observed and intrinsic reaction order is the same under intraparticle diffusion control only for a first-order reaction. Weisz and Prater [“Interpretation of Measurements in Experimental Catalysis,” *Adv. Catal.* **6**: 144 (1954)] developed general estimates for the observed order and activation energy over the entire range of ϕ :

$$n_{\text{obs}} = n + \frac{n-1}{2} \frac{d \ln \eta}{d \ln \phi} \quad E_{\text{obs}} = E + \frac{E - E_D}{2} \frac{d \ln \eta}{d \ln \phi} \quad (7-108)$$

Weisz and Prater [“Interpretation of Measurements in Experimental Catalysis,” *Adv. Catal.* **6**: 144 (1954)] also developed a general criterion for diffusion limitations, which can guide the lab analysis of rate data:

$$\text{If } \Phi = \left(\frac{3V_p}{S_{px}} \right)^2 \frac{r_{\text{obs}}}{D_{\text{eff}} C_{\text{as}}} \gg 1 \quad \text{then diffusion-limited} \quad (7-109)$$

$$\ll 1 \quad \text{then no diffusional resistance}$$

Effect of Intraparticle Diffusion for Reaction Networks

For multiple reactions, intraparticle diffusion resistance can also affect the observed selectivity and yield. For example, for consecutive reactions intraparticle diffusion resistance reduces the yield of the intermediate (often desired) product if both reactions have the same order. For parallel reactions diffusion resistance reduces the selectivity to the higher-order reaction. For more details see, e.g., Carberry, *Chemical and Catalytic Reaction Engineering*, McGraw-Hill, 1976; and Levenspiel, *Chemical Reaction Engineering*, 3d ed., Wiley, 1999.

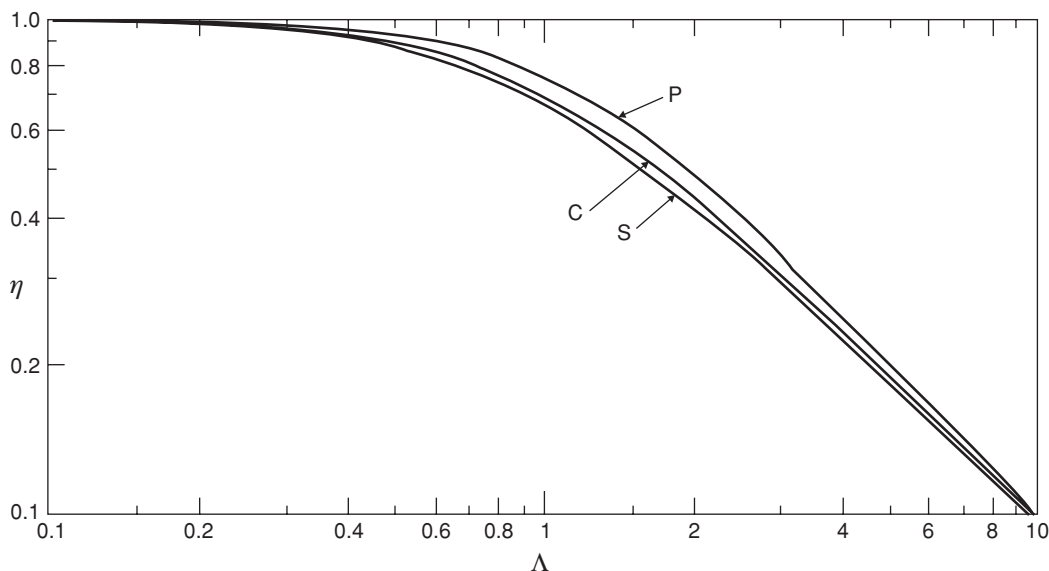


FIG. 7-8 Effectiveness factors for a slab (P), a cylinder (C), and a sphere (S). [Adapted from Fig. 1 in Aris and Rester, "The Effect of Shape on the Effectiveness Factor," Chem. Eng. Sci. **24**: 793 (1969).]

For more complex reactions, the effect of intraparticle diffusion resistance on rate, selectivity, and yield depends on the particulars of the network. Also, the use of the Thiele modulus–effectiveness factor relationships is not as easily applicable, and numerical solution of the diffusion-reaction equations may be required.

Intraparticle Diffusion and External Mass-Transfer Resistance For typical industrial conditions, external mass transfer is important only if there is substantial intraparticle diffusion resistance. This subject has been discussed by Luss, "Diffusion-Reaction Interactions in Catalyst Pellets," in Carberry and Varma (eds.), *Chemical Reaction and Reactor Engineering*, Dekker, 1987. This, however, may not be the case for laboratory conditions, and care must be exerted in including the proper data interpretation. For instance, for a spherical particle with both external and internal mass-transfer limitations and first-order reaction, an overall effectiveness factor η_t can be derived, indicating the series-of-resistances nature of external mass transfer followed by intraparticle diffusion-reaction:

$$\frac{1}{\eta_t} = \frac{1}{\eta} + \frac{\phi^2}{3Sh'} \quad (7-110)$$

$$Sh' = \frac{\epsilon k_p R}{D_{ea}} \quad (7-111)$$

As indicated above, intraparticle diffusion lowers the apparent activation energy. The apparent activation energy is even further lowered under external mass-transfer control. Figure 7-9 illustrates how the rate-controlling step changes with temperature, and as a result the dependence of the apparent first-order rate constant on temperature also changes, from a very strong dependence under kinetic control to virtual independence under external mass-transfer control.

Note that in the limit of external diffusion control, the activation energy $E_{obs} \rightarrow 0$, as can be shown when substituting Eq. (7-110) in Eq. (7-108). For more details on how to represent the combined effect of external and intraparticle diffusion on effectiveness factor for more complex systems, see Luss, "Diffusion-Reaction Interactions in Catalyst Pellets."

Heat-Transfer Resistances A similar analysis regarding external and intraparticle heat-transfer limitations leads to temperature

gradients which add further complexity to the behavior of heterogeneous catalytic systems, including steady-state multiplicity. More details are given in Sec. 19.

Catalyst Deactivation The catalyst life ranges from seconds to minutes to a few days to several years, as the active surface of a catalyst is degraded by chemical, thermal, or mechanical factors. *Chemical deactivation* occurs due to feed or product poisoning or masking. Poisoning may be due to compounds such as P, S, As, Na, and Bi that have free electron pairs and is generally considered irreversible. In some cases a reduced life is simply accepted, as in the case of slow accumulation of trace metals from feed to catalytic cracking; but in other cases the deactivation is too rapid. Sulfur and water are removed from feed to ammonia synthesis, sulfur from feed to platinum reforming, and arsenic from feed to SO_2 oxidation with platinum. Masking may be due to covering of the active sites by contaminants in either the feed or products. Examples of feed masking agents can include Si (from organic silicones) and rust. An example of product masking is coking. Reactivation sometimes is done in place;

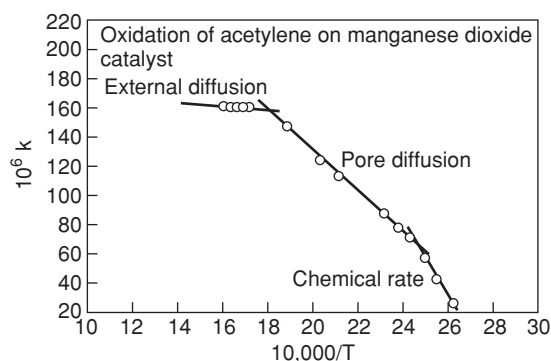


FIG. 7-9 Dependence of the rate-controlling step on temperature.

for instance, coke is burned off cracking catalyst or off nickel and nickel-molybdenum catalysts in a fluidized reactor/regenerator system. *Thermal deactivation* is primarily due to rearrangement of the active sites at high temperature due to *sintering*. Sintering results in agglomeration of active ingredients (lower dispersion). In most cases sintering is irreversible; however Pt/Al₂O₃ catalysts have been regenerated in place by Cl₂ treatment. The catalyst also can be modified by additives, for instance, chromia to nickel to prevent sintering, rhenium to platinum to reduce coking, and so on. *Mechanical deactivation* may be caused by attrition or erosion and subsequent loss of catalyst as fines. The attrition resistance of catalysts is related to the nature of the support and its porosity.

For additional references, see, e.g., Thomas, *Catalytic Processes and Proven Catalysts*, Academic Press, 1970; Butt and Petersen, *Activation, Deactivation and Poisoning of Catalysts*, Academic Press, 1988; and Delmon and Froment, *Catalyst Deactivation*, Elsevier, 1980.

The *activity* α at any time on stream may be simply defined as the ratio of the rate at time t to the rate with fresh catalyst

$$\alpha = \frac{r(t)}{r(t=0)} \quad (7-112)$$

The rate of destruction of active sites and pore structure can be expressed as a kinetic relation that can be generally first- or second-order. For instance, for a second-order dependence,

$$\frac{d\alpha}{dt} = -k_d \alpha^2 \quad (7-113)$$

the corresponding integral is

$$\alpha = \frac{1}{1 + k_d t} \quad (7-114)$$

This type of deactivation mechanism often applies catalyst sintering and coke deactivation. The deactivation rate constant is expected to have an Arrhenius dependence on temperature.

When the feedstock contains constant proportions of reactive impurities, the rate of decline may also depend on the concentration of the main reactant, e.g., for a power law rate

$$\frac{d\alpha}{dt} = -k_d \alpha^n C^q \quad (7-115)$$

Such a differential equation must be solved simultaneously with a rate equation for the main reactant.

The deactivation rate constants are estimated by methods like those for finding constants of any rate equation, given suitable (α , t) data. There are different chemical deactivation mechanisms—two of the most common are described below. For more details see Butt and Petersen, *Activation, Deactivation and Poisoning of Catalysts*, Academic Press, 1988; and Froment and Bischoff, *Chemical Reactor Analysis and Design*, Wiley, 1990. In *uniform deactivation*, the poison is distributed uniformly throughout the pellet and degrades it gradually. In *pore mouth (shell progressive) poisoning*, the poison is so effective that it kills the active site as it enters the pore; hence complete deactivation begins at the mouth and moves gradually inward.

Uniform Deactivation When uniform deactivation occurs, the specific rate declines by a factor $1 - \beta$, where β is the fractional poisoning. β is calculated from the poisoning rate, and it is often assumed to be proportional to the concentration of the poison in the bulk fluid. Then a power law rate equation becomes

$$r = k(1 - \beta)\eta C_s^n \quad (7-116)$$

The effectiveness depends on β through the Thiele modulus

$$\phi = L \sqrt{\frac{k(1 - \beta)C_s^{n-1}}{D_e}} \quad (7-117)$$

To find the effectiveness under poisoned conditions, this form of the Thiele modulus is substituted into the appropriate relation for effectiveness. For example, for a first-order reaction in slab geometry, the effectiveness factor is

$$\eta = \frac{\tanh \phi}{\phi} = \frac{\tanh[L\sqrt{k(1 - \beta)/D_e}]}{L\sqrt{k(1 - \beta)/D_e}} \quad (7-118)$$

Figure 7-10a shows the ratio of the effectiveness factor with uniform poisoning to that without poisoning versus the fraction poisoned for the above case of first-order reaction in a slab.

Pore Mouth (or Shell Progressive) Poisoning This mechanism occurs when the poisoning of a pore surface begins at the mouth of the pore and moves gradually inward. This is a moving boundary problem, and the pseudo-steady-state assumption is made that the boundary moves slowly compared with diffusion of poison and reactants and reaction on the active surface. β is the fraction of the pore that is deactivated. The poison diffuses through the dead zone and deposits at the interface between the dead and active zones. The reactants diffuse across the dead zone without reaction, followed by diffusion-reaction in the active zone.

Figure 7-10b shows simulation results for the ratio of the effectiveness factor with pore mouth poisoning to that without poisoning for a first-order reaction in a slab.

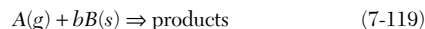
GAS-SOLID NONCATALYTIC REACTIONS

Examples of gas-solid noncatalytic reactions include production of iron from iron ores, roasting of sulfide oxides, combustion of solid fuels, chlorination of Ti ores to make TiCl₄ in the production of TiO₂ pigments, incineration of waste, decomposition of solids to produce gases, e.g., solid propellants and explosives. The kinetic treatment of these reactions has to take into consideration external mass transfer and intraparticle diffusion just as in the case of gas-solid catalytic reactions. However there are major differences, the primary one being consumption of the solid reactant, making the conditions inside the solid particle transient in nature, including change in unreacted particle size, particle density, porosity, etc. For more details see, e.g., Wen ["Noncatalytic Heterogeneous Solid-Fluid Reaction Models," *Ind. Eng. Chem.* **60**(9):34–54 (1968)], Szekey (in Lapidus and Amundson (eds.), *Chemical Reactor Theory—A Review*, Prentice-Hall, 1977), Doraiswamy and Kulkarni [in Carberry and Varma (eds.), *Chemical Reaction and Reactor Engineering*, Dekker, 1987], and Levenspiel (*Chemical Reaction Engineering*, 3d ed., Wiley, 1999).

The basic steps are identical to those of catalytic gas-solid reactions. However, as indicated above, the process is transient (non-steady-state) due to change in particle size and properties as the reaction progresses.

Several models that describe gas-solid noncatalytic reactions are summarized in Table 7-7. The first two, the sharp interface and volume reaction models, are pseudo-homogeneous, form part of the class of shrinking core models, and can be treated by using the Thiele modulus and effectiveness factor concept. The last three are heterogeneous models.

Sharp Interface Model For a first-order reaction in gas reactant,



a rate of conversion of the solid B per unit particle volume of

$$r_b = k C_{as} C_s \quad (7-120)$$

and assuming pseudo-steady state for the gas-phase component, it can be shown that

$$r_{\text{obs}} = \frac{C_{a0}}{\frac{1}{k_p a_p} + \frac{R_p^2}{D_{ea}} \left(\frac{R_p}{R_s} - 1 \right) + \frac{b}{k C_{s0}}} \quad (7-121)$$

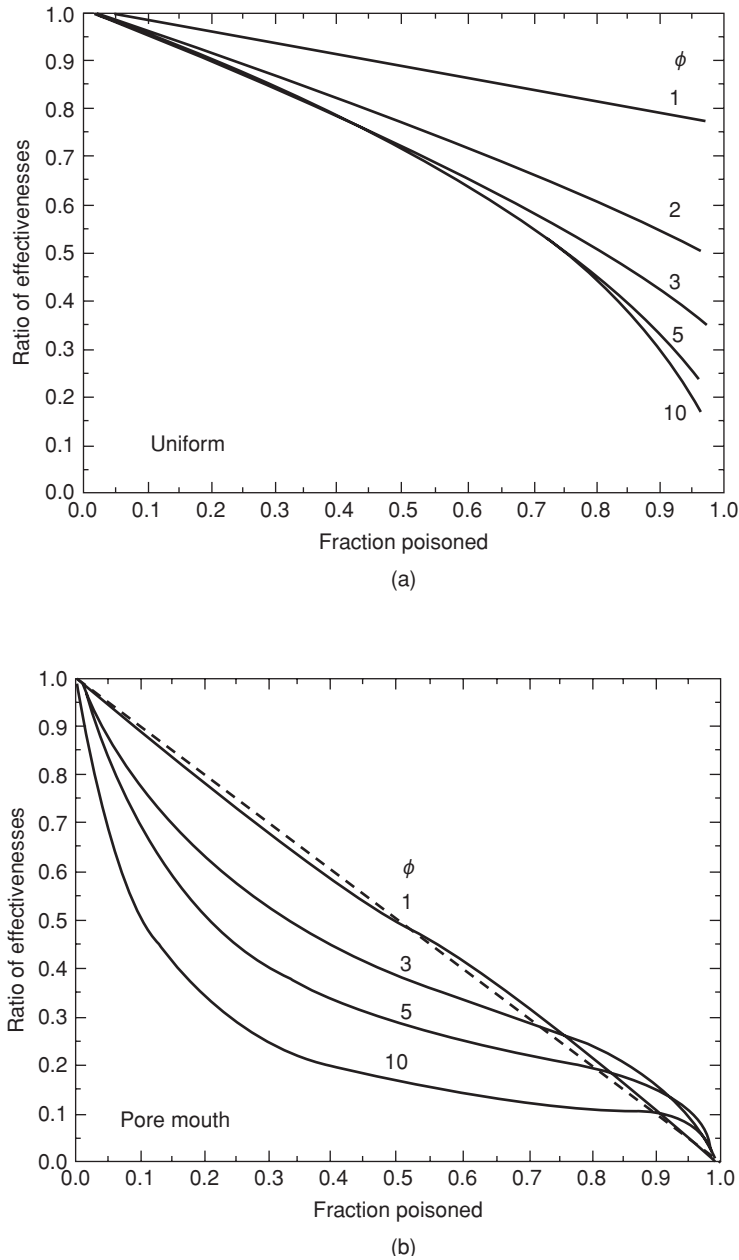


FIG. 7-10 Poisoning of a first-order reaction. (a) Uniform poisoning. (b) Pore mouth poisoning.

where R_p and R_s are the particle and solid core radii, a_p is the external particle diameter-based interfacial area per particle volume, and k is based on particle volume. Equation (7-121) represents three resistances in series—external mass transfer, diffusion in the reacted (ash) zone, and reaction at the unreacted solid-ash interface.

The conversion of the solid reactant B is obtained from integration of the pseudo-steady-state diffusion model with reaction at the boundary:

$$\begin{aligned} \frac{1}{r^2} \frac{\partial}{\partial r} \left(D_{ea} r^2 \frac{\partial C_{as}}{\partial r} \right) &= 0 \\ D_{ea} \frac{\partial C_{as}}{\partial r} &= \frac{1}{b} k C_{as} C_s \quad \text{at } r = R_s \\ D_{ea} \frac{\partial C_{as}}{\partial r} &= k_s (C_{a0} - C_{as}) \quad \text{at } r = R_p \end{aligned} \quad (7-122)$$

TABLE 7-7 Noncatalytic Gas-Solid Reaction-Diffusion Models

Model	Main features
Sharp interface model (SIM)	Reacting solid is nonporous. Reacted solid ash is porous. Reaction occurs at the ash-unreacted solid interface.
Volume reaction model	Reacting solid is also porous. Reaction occurs everywhere in the particle.
Grain model	Particle is divided into identical solid spherical grains. Each grain reacts according to the sharp interface model.
Crackling core model	Combination of SIM and grain model.
Nucleation model	Nucleation of metals in metal reduction reactions.

The moving boundary radius R_c is determined from a material balance that relates the unreacted solid volume to the reaction rate. Integration gives the time τ required to achieve a given conversion of the solid B , X_b :

$$\tau = \frac{R_p C_{s0}}{b C_{a0}} \left\{ \frac{1}{3} \left(\frac{1}{k_s} - \frac{R_p}{D_{ea}} \right) X_b + \frac{R_p}{2 D_{ea}} \left[1 - (1 - X_b)^{2/3} \right] + \frac{3b}{C_{s0} R_p k} \left[1 - (1 - X_b)^{1/3} \right] \right\} \quad X_b = 1 - \left(\frac{R_c}{R_p} \right)^3 \quad (7-123)$$

Similar solutions can be obtained for other shapes (Doraiswamy and Kulkarni, in Carberry and Varma (eds.), *Chemical Reaction and Reactor Engineering*, Dekker, 1987). Figure 7-11 shows typical concentration profiles for this case.

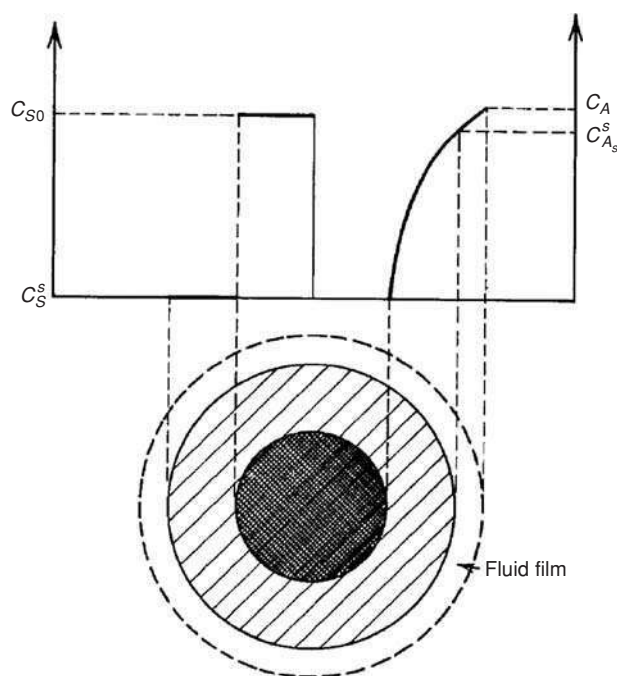


FIG. 7-11 Sharp interface model—concentration profiles. [From Wen, "Noncatalytic Heterogeneous Solid-Fluid Reaction Models," Ind. Eng. Chem. **60**(9): 34–54 (1968), Fig. 1.]

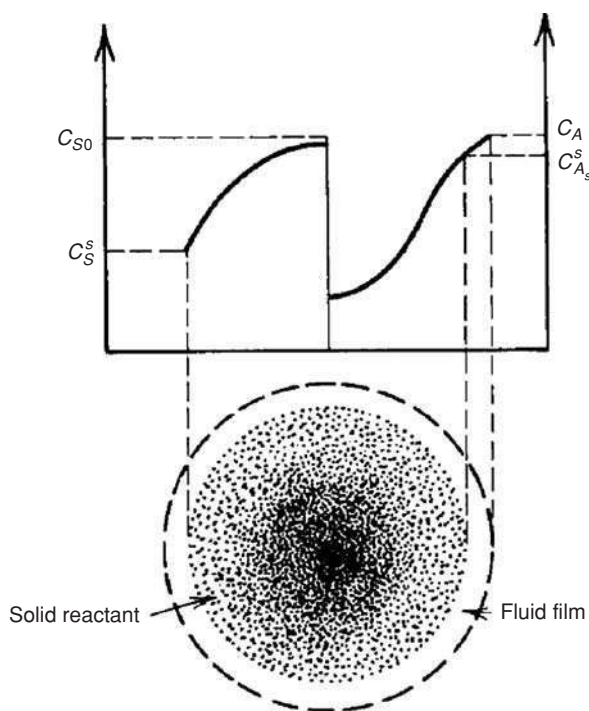


FIG. 7-12 Typical concentration profiles for the volume reaction model. [From Wen, "Noncatalytic Heterogeneous Solid-Fluid Reaction Models," Ind. Eng. Chem. **60**(9): 34–54 (1968), Fig. 3.]

Volume Reaction Model A typical concentration profile for the volume reaction model is shown in Fig. 7-12.

A general transient model of diffusion-reaction that uses the effective diffusivity concept described for gas-solid catalytic reactions can be derived here as well, e.g., for a spherical particle:

$$\frac{\partial(\epsilon_s C_{as})}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(D_{ea} r^2 \frac{\partial C_{as}}{\partial r} \right) - r_a \quad \frac{\partial C_s}{\partial t} = -r_s$$

$$C_{as} = C_{as0} \quad C_s = C_{s0} \quad \text{at } t = 0 \quad \frac{\partial C_{as}}{\partial r} = 0 \quad \text{at } r = 0$$

$$D_{ea} \frac{\partial C_{as}}{\partial r} \Big|_{r=R} = k_s (C_a - C_{as}|_{r=R}) \quad (7-124)$$

$$\epsilon_s = \epsilon_{s0} + C_{s0}(v_s - v_p) \left(1 - \frac{C_s}{C_{s0}} \right) \quad \frac{D_{ea}}{D_{ea0}} = \left(\frac{\epsilon_s}{\epsilon_{s0}} \right)^\beta, \quad \beta = 2 - 3 \quad (7-125)$$

Here the porosity and the diffusivity vary with conversion of solid; v_s and v_p are the reactant and product molar volumes. A Thiele modulus ϕ and dimensionless time θ can be defined, e.g., for a rate second-order in A and first-order in S:

$$r = k C_a^2 C_s \quad (7-126)$$

$$\phi = R \sqrt{\frac{k C_{as}(R_s) C_{s0}}{D_{ea0}}} \quad \theta = b k C_{as0}^2 t \quad (7-127)$$

For the given rate expression, equations (7-124) to (7-127) can be numerically integrated, e.g., in Fig. 7-13 for reaction control and Fig. 7-14 for intraparticle diffusion control, both with negligible external mass-transfer resistance; x is the fractional conversion.

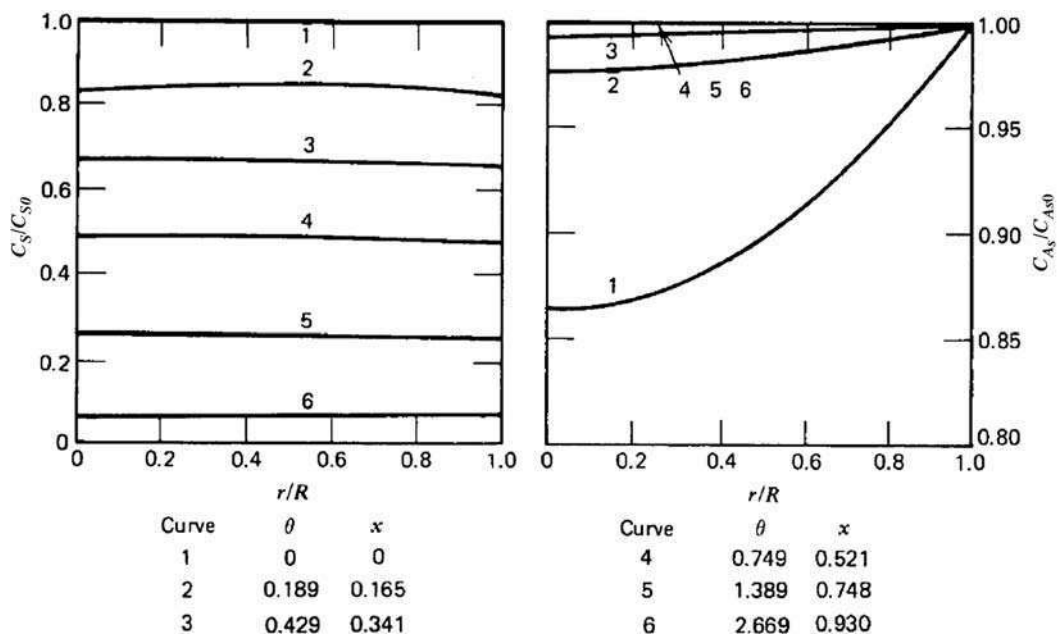


FIG. 7-13 Concentration profiles with reaction control $\phi = 1$, in absence of gas particle mass-transfer resistance. [From Wen, "Noncatalytic Heterogeneous Solid-Fluid Reaction Models," Ind. Eng. Chem. **60**(9): 34-54 (1968), Fig. 11.]

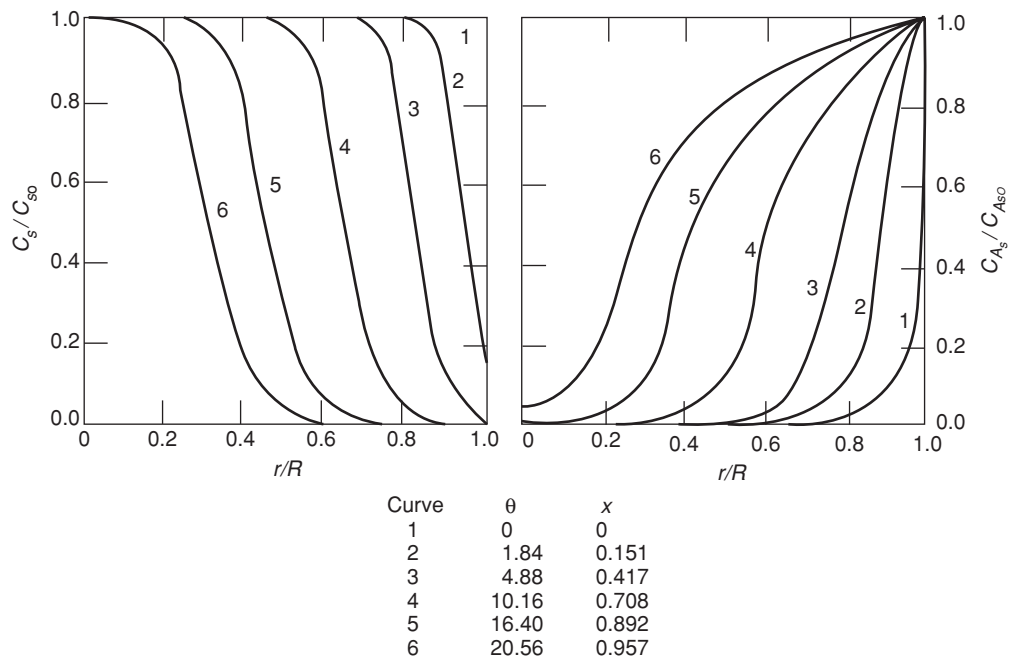


FIG. 7-14 Concentration profiles with intraparticle diffusion control, $\phi = 70$, in absence of gas particle mass-transfer resistance. [From Wen, "Noncatalytic Heterogeneous Solid-Fluid Reaction Models," Ind. Eng. Chem. **60**(9): 34-54 (1968), Fig. 12.]

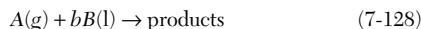
GAS-LIQUID REACTIONS

Many industrial processes employ gas-liquid reactions that can be either noncatalytic or homogeneously catalyzed. These include, for instance, absorption of acid gases (SO_3 , NO_2 , CO_2), chlorinations (benzene, dodecane, toluene), oxidations (P-Xylene to terephthalic acid, cyclohexane to cyclohexanone and cyclohexanol, acetaldehyde to acetic acid), hydrogenations (olefins, esters to fatty acids), and hydroformylation of olefins to alcohols, to name a few. See also Sec. 19 and Shah (*Gas-Liquid-Solid Reactor Design*, McGraw-Hill, 1979). These reactions include gas reactants dissolving in a liquid and reacting there with a liquid reactant. When determining the kinetics of such reactions from lab data, one needs to understand the mechanism and the controlling steps, just as was the case for heterogeneous gas-solid reactions. The simplest model is the two-film model, and it involves the following consecutive steps for the gaseous reactant:

1. Mass transfer of gas reactant from bulk gas to the gas-liquid interface across the gas film.
2. At the gas-liquid interface, the liquid and gas concentrations of the gaseous reactant are assumed to be at thermodynamic equilibrium.
3. Mass transfer of the dissolved gas reactant to the bulk liquid across the liquid film—if the reaction is fast, the reaction will occur both in the liquid film (in parallel with diffusion) and in the bulk liquid.

For a volatile liquid reactant or a volatile product, these steps are essentially reversed. For a nonvolatile liquid reactant or product, only the reaction and diffusion in the liquid take place. Figure 7-15 describes the absorbing gas concentration profiles in a gas-liquid system.

For a general gas-liquid reaction:



the two-film pseudo-steady-state model is described by the following fluxes across the interface for the gaseous reactant A:

$$\begin{aligned} N_{aG} &= \frac{D_{aG}}{\delta_G} (P_a - P_{ai}) = k_G (P_a - P_{ai}) = N_{aL} = \frac{D_{aL}}{\delta_L} (C_{Lai} - C_{La}) \\ &= k_L (C_{Lai} - C_{La}) \end{aligned} \quad (7-129)$$

Here the subscript L denotes liquid, G denotes gas, i denotes the gas-liquid interface (where the gas and liquid concentrations are in equilibrium). The thickness of the liquid and gas films is not a directly measurable quantity, and instead mass-transfer coefficients are

defined as indicated above. These depend on the diffusivity of the molecule, geometry, flow, and operating conditions; and typical values can be viewed in Sec. 19. In addition to the two-film steady-state model, other more accurate, non-steady-state models have also been developed such as the surface renewal and penetration models (see, e.g., Astarita, *Mass Transfer with Chemical Reaction*, Elsevier, 1967). In many industrial cases of interest, mass-transfer resistance in the gas-film is negligible, especially considering that gas-phase diffusivities are 2 to 3 orders of magnitude larger for the same species than those in the liquid. Hence we drop the subscripts L and G from the concentrations since the concentrations considered are in the liquid phase only.

REACTION-DIFFUSION REGIMES

Depending on the relative rates of diffusion and reaction, the following diffusion-reaction regimes occur:

$$\begin{aligned} t_D \ll t_r & \quad \text{slow reaction regime with reaction control} \\ t_D \gg t_r & \quad \text{fast reaction regime with diffusion control} \\ t_D \approx t_r & \quad \text{both reaction and diffusion are important} \end{aligned} \quad (7-130)$$

$$t_D = \frac{D_a}{k_L^2} \quad t_r = \frac{C_{ai} - C_{ae}}{r(C_{ai})}$$

Here t_D and t_r are the diffusion and reaction times, respectively, and k_L is the mass-transfer coefficient in the absence of reaction. For the *fast reaction regime*, diffusion and reaction occur in parallel in the liquid film, while for the *slow reaction regime*, there is no reaction in the liquid film and the mass transfer can be considered to occur independently of reaction in a consecutive manner. For the slow reaction regime, the following subregimes can be defined:

$$\begin{aligned} t_m \ll t_r & \quad \text{slow reaction kinetic control} \\ t_m \gg t_r & \quad \text{slow reaction mass-transfer control} \\ t_m \approx t_r & \quad \text{both reaction and mass transfer are important} \end{aligned} \quad (7-131)$$

$$t_m = \frac{1}{k_L a}$$

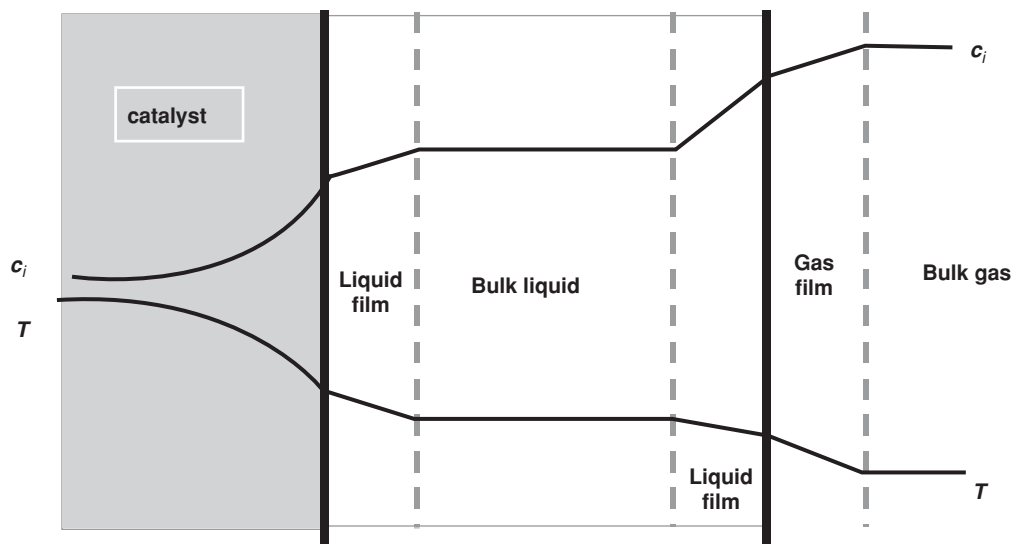


FIG. 7-15 Absorbing gas concentration and temperature profiles (exothermic reaction) in gas-liquid and gas-liquid-solid reactions.

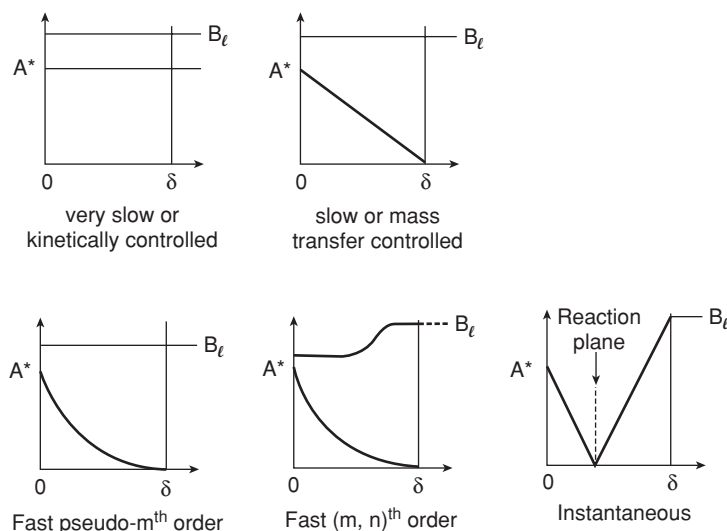


FIG. 7-16 Concentration profiles for the general reaction $A(g) + bB(l) \rightarrow \text{products}$ with the rate $r = kC_a^m C_b^n$. [Adapted from Mills, Ramachandran, and Chaudhari, "Multiphase Reaction Engineering for Fine Chemicals and Pharmaceuticals," Rev. Chem. Eng. 8(1-2):1 (1992), Figs. 19 and 20.]

Here t_m is the mass-transfer time. Only under *slow reaction kinetic control regime* can intrinsic kinetics be derived directly from lab data. Otherwise the intrinsic kinetics have to be extracted from the observed rate by using the mass-transfer and diffusion-reaction equations, in a manner similar to those defined for catalytic gas-solid reactions. For instance, in the *slow reaction regime*,

$$r_{a,obs} = \frac{C_{ai}}{He_a/k_G a + 1/k_L a + 1/k} \quad (7-132)$$

$$k_{obs} = \frac{1}{He_a/k_G a + 1/k_L a + 1/k} \quad (7-133)$$

Here He_a is the Henry constant for the solute a . For the fast reaction regime, instead of the effectiveness factor adjustment for the intrinsic reaction rate, it is customary to define an *enhancement factor* for mass-transfer enhancement by the reaction, defined as the ratio of mass transfer in presence of reaction in the liquid, to mass transfer in absence of reaction:

$$E = k_L/k_{L0} \quad (7-134)$$

Solving the diffusion-reaction equation in the liquid, the enhancement factor can be related to the *Hatta number* Ha , which is similar to the Thiele modulus defined for heterogeneous gas-solid catalysts. Thus, the Hatta number and its relation to the controlling regime are

$$Ha = \frac{t_D}{t_R} = \sqrt{\frac{\text{maximum reaction rate in the film}}{\text{maximum mass transfer rate through film}}}$$

$$Ha \ll 1 \quad \text{slow reaction regime} \quad (7-135)$$

$$Ha \gg 1 \quad \text{fast reaction regime}$$

For instance, for a first-order reaction in the gaseous reactant A (e.g., with large excess of liquid reactant B), the following relates the enhancement factor to the Hatta number:

$$Ha = \delta_L \sqrt{\frac{k}{D_a}} = \frac{\sqrt{k D_a}}{k_{L0}} \quad \text{for } C_b \gg C_{ai} \quad (7-136)$$

$$E = \frac{Ha}{\tanh Ha} \left(1 - \frac{C_a}{C_{ai}} \frac{1}{\cosh Ha} \right) > 1 \quad (7-137)$$

When both A and B have comparable concentrations, then the enhancement factor is an increasing function of an additional parameter:

$$\lambda = \frac{D_b C_b}{b D_a C_{ai}} \quad (7-138)$$

In the limit of an instantaneous reaction, the reaction occurs at a plane where the concentration of both reactants A and B is zero and the flux of A equals the flux of B . The criterion for an instantaneous reaction is

$$Ha^{1/2} \gg \frac{C_b}{b C_{ai}} \quad E_{\infty} = 1 + \lambda \gg 1 \quad (7-139)$$

Figure 7-16 illustrates typical concentration profiles of A and B for the various diffusion-reaction regimes.

GAS-LIQUID-SOLID REACTIONS

GAS-LIQUID-SOLID CATALYTIC REACTIONS

Many solid catalyzed reactions take place with one of the reactants absorbing from the gas phase into the liquid and reacting with a liquid reactant on the surface or inside the pores of a solid catalyst (see Fig. 7-15). Examples include the Fischer-Tropsch synthesis of hydrocarbons from synthesis gas (CO and H_2) in the presence of Fe or Co -

based heterogeneous catalysts, methanol synthesis from synthesis gas ($\text{H}_2 + \text{CO}$) in the presence of heterogeneous CuO/ZnO catalyst, and a large number of noble metal catalyzed hydrogenations among others. For a slow first-order reaction of a gaseous reactant, the concept of resistances in series can be expanded as follows, e.g., for a slurry reactor with fine catalyst powder:

$$r_{a,obs} = \frac{C_{ai}}{\frac{H_e}{k_{ca}} + \frac{1}{k_{la}} + \frac{1}{k_{as}} + \frac{1}{k}} \quad k_{obs} = \frac{1}{\frac{H_e}{k_{ca}} + \frac{1}{k_{la}} + \frac{1}{k_{as}} + \frac{1}{k}} \quad (7-140)$$

Intraparticle diffusion resistance may become important when the particles are larger than the powders used in slurry reactors, such as for catalytic packed beds operating in trickle flow mode (gas and liquid downflow), in upflow gas-liquid mode, or countercurrent gas-liquid mode. For these the effectiveness factor concept for intraparticle diffusion resistance has to be considered in addition to the other resistances present. See more details in Sec. 19.

POLYMERIZATION REACTIONS

Polymers are high-molecular-weight compounds assembled by the linking of small molecules called monomers. Most polymerization reactions involve two or three phases, as indicated below. There are several excellent references dealing with *polymerization kinetics* and reactors, including Ray in Lapidus and Amundson, (eds.), *Chemical Reactor Theory—A Review*, Prentice-Hall, 1977; Tirrel et al. in Carberry and Varma (eds.), *Chemical Reaction and Reactor Engineering*, Dekker, 1987; and Meyer and Keurentjes (eds.), *Handbook of Polymer Reaction Engineering*, Wiley, 2005.

Polymerization can be classified according to the main phase in which the reaction occurs as liquid (most polymerizations), vapor (e.g., Ziegler Natta polymerization of olefins), and solid phase (e.g., finishing of melt step polymerization). Polymerization reactions occur in liquid phase and can be further subclassified into

1. Bulk mass polymerization:
 - a. Polymer soluble in monomer
 - b. Polymer insoluble in monomer
 - c. Polymer swollen by monomer
2. Solution polymerization
 - a. Polymer soluble in solvent
 - b. Polymer insoluble in solvent
3. Suspension polymerization with initiator dissolved in monomer
4. Emulsion polymerization with initiator dissolved in dispersing medium

Polymerization can be catalytic or noncatalytic, and can be homogeneously or heterogeneously catalyzed. Polymers that form from the liquid phase may remain dissolved in the remaining monomer or solvent, or they may precipitate. Sometimes beads are formed and remain in suspension; sometimes emulsions form. In some processes solid polymers precipitate from a fluidized gas phase. Polymerization processes are also characterized by extremes in temperature, viscosity, and reaction times. For instance, many industrial polymers are mixtures with molecular weights of 10^4 to 10^7 . In polymerization of styrene the viscosity increased by a factor of 10^6 as conversion increased from 0 to 60 percent. The adiabatic reaction temperature for complete polymerization of ethylene is 1800 K (3240°C). Initiators of the chain reactions have concentration as low as 10^{-8} g·mol/L, so they are highly sensitive to small concentrations of poisons and impurities.

Polymerization mechanism and kinetics require special treatment and special mathematical tools due to the very large number of similar reaction steps. Some polymerization types are briefly described next.

Bulk Polymerization The monomer and initiators are reacted with or without mixing, e.g., without mixing to make useful shapes directly. Because of viscosity limitations, stirred bulk polymerization is not carried to completion. For instance, for addition polymerization conversions as low as 30 to 60 percent are achieved, with the remaining monomer stripped out and recycled (e.g., in the case of polystyrene).

Bead Polymerization Bulk reaction proceeds in droplets of 10- to 1000- μ m diameter suspended in water or other medium and insulated from each other by some colloid. A typical suspending agent is polyvinyl alcohol dissolved in water. The polymerization can be done to high conversion. Temperature control is easy because of the moderating thermal effect of the water and its low viscosity. The suspensions sometimes are unstable and agitation may be critical. Examples

are polyvinyl acetate in methanol, copolymers of acrylates and methacrylates, and polyacrylonitrile in aqueous $ZnCl_2$ solution.

Emulsion Polymerization Emulsions have particles of 0.05- to 5.0- μ m diameter. The product is a stable latex, rather than a filterable suspension. Some latexes are usable directly, as in paints, or they may be coagulated by various means to produce very high-molecular-weight polymers. Examples are polyvinyl chloride and butadiene-styrene rubber.

Solution Polymerization These processes may retain the polymer in solution or precipitate it. Examples include polyethylene, the copolymerization of styrene and acrylonitrile in methanol, the aqueous solution of acrylonitrile to precipitate polyacrylonitrile.

Polymer Characterization The physical properties of polymers depend largely on the *molecular weight distribution (MWD)*, which can cover a wide range. Since it is impractical to fractionate the products and reformulate them into desirable ranges of molecular weights, immediate attainment of desired properties must be achieved through the correct choice of reactor type and operating conditions, notably of distributions of residence time and temperature. High viscosities influence those factors. For instance, high viscosities prevalent in bulk and melt polymerizations can be avoided with solution, bead, or emulsion operations. The interaction between the flow pattern in the reactor and the type of reaction affects the MWD. If the period during which the molecule is growing is short compared with the residence time in the reactor, the MWD in a batch reactor is broader than in a CSTR. This situation holds for many free radical and ionic polymerization processes where the reaction intermediates are very short lived. In cases where the growth period is the same as the residence time in the reactor, the MWD is narrower in batch than in CSTR. Polymerizations that have no termination step—for instance, polycondensations—are of this type. This topic is treated by Denbigh [*J. Applied Chem.*, 1:227(1951)].

Four types of MWD can be defined: (1) The *number chain length distribution (NCLD)*, relating the chain length distribution to the number of molecules per unit volume; (2) the *weight chain length distribution (WCLD)* relating the chain length distribution to the weight of molecules per unit volume; (3) the *number molecular weight distribution (NMWD)* relating the chain length distribution to molecular weight; and (4) the *weight molecular weight distribution (WMWD)* relating the weight distribution to molecular weight. Two average molecular weights and corresponding average chain lengths are typically defined: the *number average molecular weight* M_n and the corresponding *number average chain length* μ_n ; and the *weight average molecular weight* M_w and the corresponding *weight average chain length* μ_w . Their ratio is called *polydispersity* and describes the width of the molecular weight distribution.

$$M_n = \frac{w \sum_{j=1}^{\infty} j P_j}{\sum_{j=1}^{\infty} P_j} \quad \mu_n = \frac{\sum_{j=1}^{\infty} j P_j}{\sum_{j=1}^{\infty} P_j} \quad M_w = \frac{w \sum_{j=1}^{\infty} j^2 P_j}{\sum_{j=1}^{\infty} j P_j} \quad \mu_w = \frac{\sum_{j=1}^{\infty} j^2 P_j}{\sum_{j=1}^{\infty} j P_j}$$

$$\text{polydispersity} = \frac{M_w}{M_n} = \frac{\mu_w}{\mu_n} \quad (7-141)$$

The average chain lengths can be related to the *moments* λ_k of the distribution as follows:

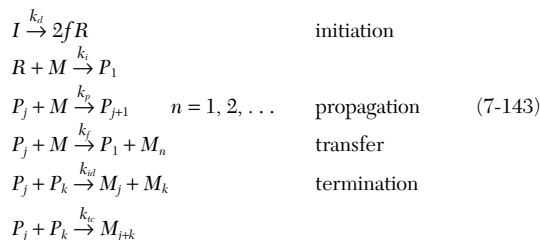
$$\mu_n = \frac{\lambda_1}{\lambda_0} \quad \mu_w = \frac{\lambda_2}{\lambda_1} \quad \text{polydispersity} = \frac{\lambda_0 \lambda_2}{\lambda_1^2} \quad \lambda_k = \sum_{j=1}^{\infty} j^k P_j \quad (7-142)$$

Here P_j is the concentration of the polymer with chain length j —the same symbol is also used for representing the polymer species P_j ; w is the molecular weight of the repeating unit in the chain.

A factor in addition to the residence time distribution and temperature distribution that affects the molecular weight distribution is the type of the chemical reaction (e.g., step or addition polymerization).

Two major polymerization mechanisms are considered: *chain growth* and *step growth*. In addition, polymerization can be *homopolymerization*—a single monomer is used—and *copolymerization* usually with two monomers with complementary functional groups.

Chain Homopolymerization Mechanism and Kinetics Free radical and ionic polymerizations proceed through this type of mechanism, such as styrene polymerization. Here one monomer molecule is added to the chain in each step. The general reaction steps and corresponding rates can be written as follows:



Here P_j is the *growing* or *live polymer*, and M_j is the *dead* or *product polymer*. Assuming reaction steps independent of chain length and assuming pseudo-steady-state approximation for the radicals lead to the following rates for monomer and initiator conversion and live polymer distribution. The growing chains distribution is the most probable distribution [see, e.g., Ray in Lapidus and Amundson (eds.), *Chemical Reactor Theory—A Review*, Prentice-Hall, 1977; Tirrel et al. in Carberry and Varma (eds.), *Chemical Reaction and Reactor Engineering*, Dekker, 1987]:

$$\begin{aligned}
 P_n &= (1 - \alpha)P\alpha^{n-1} && \alpha = \frac{k_p M}{(k_p + k_t)M + (k_{tc} + k_{td})P} \\
 P &= \left(\frac{2fk_d I}{k_{tc} + k_{td}} \right)^{1/2} && \frac{dI}{dt} = -k_d I \\
 r = \frac{dM}{dt} &= -k_p \left(\frac{2fk_d I}{k_{tc} + k_{td}} \right)^{1/2} I^{1/2} M && DP_n^{\text{inst}} = \frac{k_p M}{(0.5k_{tc} + k_{td})} P
 \end{aligned} \quad (7-144)$$

Here r is the rate of polymerization, α is the *probability of propagation*, DP_n^{inst} is the *instantaneous degree of polymerization*, i.e., the number of monomer units on the dead polymer, and f is the *initiation efficiency*. Compare r in Eq. (7-144) with the simpler Eq. (7-68). When chain transfer is the primary termination mechanism, such as in anionic polymerization, then the polydispersity is 2.

Mathematically, the infinite set of equations describing the rate of each chain length can be solved by using the z transform method (a discrete method), *continuous variable approximation method*, or the *method of moments* [see, e.g., Ray in Lapidus and Amundson (eds.), *Chemical Reactor Theory—A Review*, Prentice-Hall, 1977].

Typical ranges of the kinetic parameters for low conversion homopolymerization are given in Table 7-8. For more details see Hutchenson in Meyer and Keurentjes (eds.), *Handbook of Polymer Reaction Engineering*, Wiley, 2005.

Step Growth Homopolymerization Mechanism and Kinetics

Here any two growing chains can react with each other. The propagation mechanism is an infinite set of reactions:



TABLE 7-8 Typical Ranges of Kinetic Parameters

Coefficient/concentration	Typical range
k_d , 1/s	10^{-6} – 10^{-4}
f	0.4–0.9
k_p , L/(mol/s)	10^2 – 10^4
k_t , L/(mol/s)	10^6 – 10^8
k_{tr}/k_p	10^{-6} – 10^{-4}
I , mol/L	10^{-4} – 10^{-2}
M , mol/L	1–10

SOURCE: Hutchenson, "Typical Ranges of Kinetic Parameters," in *Handbook of Reaction Engineering*, Wiley, 2005, Table 4.1.

For instance, some nylons are produced through this mechanism. This is usually modeled under the simplifying assumption that the rate constants are independent of chain length. This assumption was proved pretty accurate, and by using the z transform it results in the *Flory distribution*:

$$\begin{aligned}
 P_n &= \frac{P_{10}(P_{10}\tau)^{n-1}}{(P_{10}\tau + 1)^{n+1}} \quad n \geq 1 \quad k_{pnm} = k_p \quad \text{for all } n, m \quad \tau = \int_0^t k_p M dt \\
 \mu_n &= \frac{2 - \alpha}{1 - \alpha} \quad \mu_w = \frac{2}{(1 - \alpha)(2 - \alpha)} \quad \text{polydispersity} = \frac{\mu_w}{\mu_n} = \frac{2}{(2 - \alpha)^2} \\
 \alpha &= \frac{P_{10}\tau}{P_{10}\tau + 1} \quad (7-146)
 \end{aligned}$$

Copolymerization Copolymerization involves more than one monomer, usually two comonomers, as opposed to the single monomer involved in the chain growth and step homopolymerization schemes above. Examples are some nylons, polyesters, and aramids. Here as well there are step growth and chain growth mechanisms, and these are much more complex [see, e.g., Ray in Lapidus and Amundson (eds.), *Chemical Reactor Theory—A Review*, Prentice-Hall, 1977].

BIOCHEMICAL REACTIONS

Mechanism and kinetics in biochemical systems describe the cellular reactions that occur in living cells. Biochemical reactions involve two or three phases. For example, aerobic fermentation involves gas (air), liquid (water and dissolved nutrients), and solid (cells), as described in the "Biocatalysis" subsection above. Bioreactions convert feeds called *substrates* into more *cells* or *biomass* (cell growth), proteins, and *metabolic* products. Any of these can be the desired product in a commercial *fermentation*. For instance, methane is converted to biomass in a commercial process to supply fish meal to the fish farming industry. Ethanol, a metabolic product used in transportation fuels, is obtained by fermentation of corn-based or sugar-cane-based sugars. There is a substantial effort to develop genetically modified biocatalysts that produce a desired metabolite at high yield.

Bioreactions follow the same general laws that govern conventional chemical reactions, but the complexity of the mechanism is higher due to the close coupling of bioreactions and *enzymes* that are turned on (*expressed*) or off (*repressed*) by the cell depending on the conditions in the fermenter and in the cell. Thus the rate expression (7-92) can mainly be used to design bioreaction processes when the culture is in *balanced growth*, i.e., for steady-state cultivations or batch growth for as long as the substrate concentration is much higher than C_s . After a sudden process upset (e.g., a sudden change in substrate concentration or pH), the *control network* of the cell that lies under the *mass flow network* is activated, and dramatic changes in the kinetics of product formation can occur. Table 7-9 summarizes key differences between biochemical and conventional chemical systems [see, e.g., Leib, Pereira, and Villadsen, "Bioreactors, A Chemical Engineering Perspective," *Chem. Eng. Sci.* **56**: 5485–5497 (2001)].

TABLE 7-9 Biological versus Chemical Systems

- There is tighter control on conditions (e.g., pH, temperature, substrate and product concentrations, dissolved O_2 concentration, avoidance of contamination by foreign organisms).
- Pathways can be turned on/off by the microorganism through expression of certain enzymes depending on the substrate type and concentration and operating conditions, leading to a richness of behavior unparalleled in chemical systems.
- The global stoichiometry changes with operating conditions and feed composition; kinetics and stoichiometry obtained from steady-state (chemostat) data cannot be used reliably over a wide range of conditions, unless fundamental models are employed.
- Long-term adaptations (mutations) may occur in response to environment changes that can alter completely the product distribution.
- Only the substrates that maximize biomass growth are utilized even in the presence of multiple substrates.
- Cell energy balance requirements pose additional constraints on the stoichiometry that can make it very difficult to predict flux limitations.

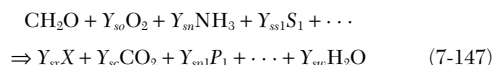
TABLE 7-10 Hierarchy of Kinetic Models in Biological Systems

- Stoichiometric black box models (similar to a single global chemical reaction) represent the biochemistry by a single global reaction with fixed stoichiometric or *yield coefficients* (limited to a narrow range of conditions). Black box models can be used over a wider range of conditions by establishing different sets of yield coefficient for different conditions. These are also needed to establish the quantitative amounts of various nutrients needed for the completion of the bioreaction.
- Unstructured models view the cell as a single component interacting with the fermentation medium, and each bioreaction is considered to be a global reaction, with a corresponding empirical rate expression.
- Structured models include information on individual reactions or groups of reactions occurring in the cell, and cell components such as DNA, RNA, and proteins are included in addition to the primary metabolites and substrates (see, e.g., the active cell model of Nielsen and Villadsen, *Bioreaction Engineering Principles*, 2d ed., Kluwer Academic/Plenum Press, 2003).
- Fundamental models include cell dimensions, transport of substrates and metabolites across the cell membrane, and the elementary cell bioreaction steps and their corresponding enzyme induction mechanism. In recent years further kinetic steps have been added to the above models which are based on the conversion of substrates to metabolites. Thus the kinetics of protein synthesis by *transcription* and *translation* from the *genome* add much further complexity to cell kinetics.

The network of bioreactions is called the *metabolic network*, the series of consecutive steps between key intermediates in the network are called *metabolic pathways*, and the determination of the mechanism and kinetics is called *metabolic flux analysis*. As for chemical systems, there are several levels of mechanistic and kinetic representation and analysis, listed in order of increasing complexity in Table 7-10.

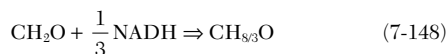
Additional complexity can be included through cell population balances that account for the distribution of cell generation present in the fermenter through use of stochastic models. In this section we limit the discussion to simple black box and unstructured models. For more details on bioreaction systems, see, e.g., Nielsen, Villadsen, and Liden, *Bioreaction Engineering Principles*, 2d ed., Kluwer, Academic/Plenum Press, 2003; Bailey and Ollis, *Biochemical Engineering Fundamentals*, 2d ed., McGraw-Hill, 1986; Blanch and Clark, *Biochemical Engineering*, Marcel Dekker, 1997; and Sec. 19.

Mechanism Stoichiometric balances are done on a C atom basis called *C-moles*, e.g., relative to the substrate (denoted by subscript *s*), and the corresponding stoichiometric coefficients Y_{si} (based on C-mole of the primary substrate) are called *yield coefficients*. For instance,



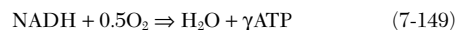
Here the reactants (substrates) are glucose (CH_2O), O_2 , NH_3 , and a sulfur-providing nutrient S_1 , and the products are biomass X, CO_2 , metabolic product P_1 , and H_2O .

The products of bioreactions can be reduced or oxidized, and all feasible pathways have to be *redox neutral*. There are several cofactors that transfer redox power in a pathway or between pathways, each equivalent to the reducing power of a molecule of H_2 , e.g., nicotinamide adenine dinucleotide (NADH), and these have to be included in the stoichiometric balances as H equivalents through redox balancing. For instance, for the reaction of glucose to glycerol ($\text{CH}_{3/3}\text{O}$), $\frac{1}{3}$ NADH equivalent is consumed:



The stoichiometry in the biochemical literature often does not show H_2O produced by the reaction; however, for complete elemental balance, water has to be included, and this is easily done once an O_2 requirement has been determined based on a redox balance. Likewise for simplicity, the other form of the cofactor [e.g., the oxidized form of the cofactor NADH in Eq. (7-148)] is usually left out. In

addition to C balances, for aerobic systems cell respiration has to be accounted for as well through a stoichiometric equation:



The associated free energy produced or consumed in each reaction is captured in units of adenosine triphosphate (ATP). The ATP stoichiometry is usually obtained from biochemical tables since the energy has to be also balanced for the cell. Thus for Eq. (7-148) the stoichiometric ATP requirement to convert one C-mole of glucose to one C-mole of glycerol is $\frac{1}{3}$. In calculations of the carbon flux distribution in different pathways this ATP requirement has to be added on the left-hand side of the equation. Again the other form of the cofactor ATP is usually left out to simplify the reaction equation.

There are several metabolic pathways that are repeated for many living cells, and these are split into two: *catabolic* or energy-producing and *anabolic* or energy-consuming, the later producing building blocks such as amino acids and the resulting macromolecules such as proteins. Of course the energy produced in catabolic steps has to be balanced by the energy consumed in anabolic steps. Catabolic pathways include the well-studied *glycolysis*, *TCA cycle*, *oxidative phosphorylation*, and *fermentative pathways*. For more details see Stephanopoulos, Aristidou, and Nielsen, *Metabolic Engineering: Principles and Methodologies*, Academic Press, 1998; and Nielsen, Villadsen, and Liden, *Bioreaction Engineering Principles*, 2d ed., Kluwer, Academic/Plenum Press, 2003; Bailey and Ollis, *Biochemical Engineering Fundamentals*, 2d ed., McGraw-Hill, 1986.

Monod-Type Empirical Kinetics Many bioreactions show increased biomass growth rate with increasing substrate concentration at low substrate concentration for the limiting substrate, but no effect of substrate concentration at high concentrations. This behavior can be represented by the Monod equation (7-92). Additional variations on the Monod equation are briefly illustrated below. For two essential substrates the Monod equation can be modified as

$$\mu = \frac{\mu_{\max} C_{s1} C_{s2}}{(K_{s1} + C_{s1})(K_{s2} + C_{s2})} \quad (7-150)$$

This type of rate expression is often used in models for water treatment, and many environmental factors can be included (the effect of, e.g., phosphate, ammonia, volatile fatty acids, etc.). The correlation between parameters in such complicated models is, however, severe, and very often a simple Monod model (7-92) with only one limiting substrate is sufficient.

When *substrate inhibition* occurs,

$$\mu = \frac{\mu_{\max} C_s}{K_s + C_s + K_i/C_s^2} \quad (7-151)$$

O_2 is typically a substrate that in high concentrations leads to substrate inhibition, but a high concentration of the carbon source can also be inhibiting (e.g., in bioremediation of toxic waste a high concentration of the organic substrate can well lead to severe inhibition or death of the microorganism).

When *product inhibition* is present,

$$\mu = \frac{\mu_{\max} C_s}{K_s + C_s} \left(1 - \frac{C_p}{C_{p\max}} \right) \quad (7-152)$$

Here the typical example is the inhibitor effect of ethanol on yeast growth. Considerable efforts are made by the biocompanies to develop yeast strains that are tolerant to high ethanol concentrations since this will give considerable savings in, e.g., production of biofuel by fermentation.

The various component reaction rates for a single reaction can be related to the growth rate by using the stoichiometric (yield) coefficients, e.g., from Eq. (7-147):

$$r_i = Y_{si} \mu C_x = \frac{Y_{si}}{Y_{sx}} \mu C_x \quad (7-153)$$

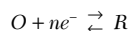
Chemostat with Empirical Kinetics Using the CSTR equation (7-54) for a constant-volume single reaction [Eq. (7-147)], the substrate, biomass, and product material balances are

$$\begin{aligned}\frac{1}{Y_{sx}} \mu C_x + D(C_{s0} - C_s) &= 0 \\ \mu C_x - DC_x &= 0 \rightarrow D = \mu \\ \frac{Y_{sp}}{Y_{sx}} \mu C_x - DC_p &= 0\end{aligned}\quad (7-154)$$

Here C_{s0} is the feed substrate concentration, and D is the *dilution rate*, which at steady-state constant volume is equal to both the feed and effluent volumetric flow rates and to the specific growth rate. The effluent concentrations of substrate, biomass, and products can be calculated by using a suitable expression for the specific growth rate μ such as one of the relevant variants of the Monod kinetics described above.

ELECTROCHEMICAL REACTIONS

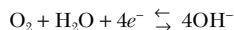
Electrochemical reactions involve coupling between chemical reactions and electric charge transfer and may have two or three phases, for instance, a gas (e.g., H_2 or O_2 evolved at the electrodes or fed as reactants), a liquid (the electrolyte solution), and solids (electrodes). Electrocatalysts may be employed to enhance the reaction for a particular desired product. Hence, electrochemical reactions are heterogeneous reactions that occur at the surface of electrodes and involve the transfer of charge in the form of electrons as part of a chemical reaction. The electrochemical reaction can produce a chemical change by passing an electric current through the system (e.g., electrolysis), or reversely a chemical change can produce electric energy (e.g., using a battery or fuel cell to power an appliance). There are a variety of practical electrochemical reactions, some occurring naturally, such as corrosion, and others used in production of chemicals (e.g., the decomposition of HCl to produce Cl_2 and H_2 , the production of caustic soda and chlorine, the smelting of aluminum), electroplating, and energy generation (e.g., fuel cells and photovoltaics). Electrochemical reactions are reversible and can be generally written as a *reduction-oxidation (redox)* couple:



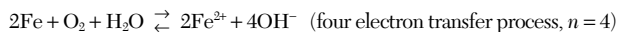
where O is an oxidized and R is a reduced species. For instance, the corrosion process includes oxidation at the *anode*:



and reduction at the *cathode*:



The overall electrochemical reaction is the stoichiometric sum of the anode and cathode reactions:



The anode and cathode reactions are close coupled in that the electric charge is conserved; therefore, the overall production rate is a direct function of the electric charge passed per unit time, the electric current I .

For references on electrochemical reaction kinetics and mechanism, see, e.g., Newman and Thomas-Alvea, *Electrochemical Systems*, 3d ed., Wiley Interscience, 2004; Bard and Faulkner, *Electrochemical Methods: Fundamentals and Applications*, 2d ed., Wiley, 2001; Bethune and Swendeman, "Table of Electrode Potentials and Temperature Coefficients," *Encyclopedia of Electrochemistry*, Van Nostrand Reinhold, New York 1964, pp. 414–424; and Bethune and Swendeman, *Standard Aqueous Electrode Potentials and Temperature Coefficients*, C. A. Hampel Publisher, 1964.

Faraday's law relates the charge transferred by ions in the electrolyte and electrons in the external circuit, to the moles of chemical species reacted (Newman and Thomas-Alvea, *Electrochemical Systems*, 3d ed., Wiley Interscience, 2004):

$$\begin{aligned}Q &= nmF & F &= 96,485 \text{ C/equiv} \\ I &= \frac{Q}{t} = \frac{\text{charge}}{\text{time}} & A\end{aligned}\quad (7-155)$$

where n is the number of equivalents per mole, m is the number of moles, F is the Faraday constant, Q is the charge, and t is time. The total current passed may represent several parallel electrochemical reactions; therefore, we designate a current efficiency for each chemical species. The chemical species production rate (mass/time) is related to the total current passed I , the species *current efficiency* $\epsilon_{\text{current},i}$, and the molecular weight of the chemical species MW_i :

$$\begin{aligned}\dot{m} &= \frac{I \times \epsilon_{\text{current},i} \times MW_i}{nF} = \frac{\text{mass}}{\text{time}} \\ j &= \frac{\text{current}}{\text{area}} = \frac{I}{A_{\text{projected}}}\end{aligned}\quad (7-156)$$

Since electrochemical reactions are heterogeneous at electrode surfaces, the current I is generally normalized by dividing it by the geometric or projected area of the electrode, resulting in the quantity known as the *current density* j , in units of kA/m^2 .

The overall electrochemical cell *equilibrium potential* E°_{cell} , as measured between the cathode and the anode, is related to the Gibbs free energy change for the overall electrochemical reaction:

$$\begin{aligned}\Delta G^\circ &= \Delta H^\circ - T \Delta S^\circ = -nFE^\circ_{\text{cell}} \\ E^\circ_{\text{cell}} &= -\frac{\Delta G^\circ}{nF} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}\end{aligned}\quad (7-157)$$

Each electrode reaction, anode and cathode, or half-cell reaction has an associated energy level or electrical potential (volts) associated with it. Values of the standard equilibrium electrode reduction potentials E° at unit activity and 25°C may be obtained from the literature (de Bethune and Swendeman Loud, *Encyclopedia of Electrochemistry*, Van Nostrand Reinhold, 1964). The overall electrochemical cell *equilibrium potential* either can be obtained from ΔG values or is equal to the cathode half-cell potential minus the anode half-cell potential, as shown above.

The *Nernst equation* allows one to calculate the *equilibrium potential* E^{eq} when the activity of the reactants or products is not at unity:

$$\begin{aligned}\sum_i \nu_i M_i^n &\rightarrow ne^- \\ E^{\text{eq}} &= E^\circ - \frac{RT}{nF} \ln(\Pi a_i^{\nu_i}) \\ \left(\frac{\partial E}{\partial T} \right)_P &= \frac{\Delta S}{nF}\end{aligned}\quad (7-158)$$

where ν_i is the stoichiometric coefficient of chemical species i (positive for products; negative for reactants), M_i is the symbol for species i , n_i is the charge number of the species, a_i is the activity of the chemical species, E is the *formal potential*, and Π represents the product of all respective activities raised to their stoichiometric powers as required by the reaction. Please note that if the value of the equilibrium potential is desired at another temperature, E° must also be evaluated at the new temperature as indicated.

Kinetic Control In 1905, Julius Tafel experimentally observed that when mass transport was not limiting, the *current density* j of electrochemical reactions exhibited the following behavior:

$$j = a'e^{\eta_{\text{act}}/b'} \quad \text{or} \quad \eta_{\text{act}} = a + b \log j$$

where the quantity η_{act} is known as the *activation overpotential* $E - E^{\text{eq}}$, and is the difference between the actual electrode potential

E and the reversible equilibrium potential of the electrochemical reaction E^{eq} . Thus the driving force for the electrochemical reaction is not the absolute potential; it is the activation overpotential η_{act} .

This relationship between the current density and activation overpotential has been further developed and resulted in the Butler-Volmer equation:

$$r = \frac{j}{nF} = k_f C_o - k_r C_r \quad j = j_0 (e^{-(\alpha_a F/RT)\eta_{\text{act}}} - e^{[(1-\alpha_c)F/RT]\eta_{\text{act}}})$$

$$\eta_{\text{act}} = E - E^{\text{eq}} \quad (7-159)$$

Here the reaction rate r is defined per unit electrode area, moles per area per time, j_0 is the *equilibrium exchange current* when $E = E^{\text{eq}}$, η_{act} is the *activation overpotential*, and α is the *transfer coefficient*. For large activation overpotentials, the *Tafel empirical equation* applies:

$$\eta_{\text{act}} = a + b \log j \quad \text{for } \eta_{\text{act}} > 100 \text{ mV, } b = \text{Tafel slope} \quad (7-160)$$

For small activation overpotentials, linearization gives

$$j = j_0 \frac{nF}{RT} \eta_{\text{act}} \quad (7-161)$$

Mass-Transfer Control The surface concentration at the electrodes differs significantly from the bulk electrolyte concentration. The Nernst equation applies to the surface concentrations (or activities in case of nonideal solutions):

$$E^{\text{eq}} = E^o - \frac{RT}{nF} \ln(\Pi a_{i,\text{surf}}^{\nu_i}) \quad (7-162)$$

If mass transfer is limiting, then a *limiting current* is obtained for each chemical species i :

$$j_{i,\text{lim}} = \frac{nFD_i C_i}{\delta} = nFk_{L,i} C_i \quad (7-163)$$

where D_i is the diffusion coefficient, δ is the boundary layer thickness, and $k_{L,i}$ is the mass-transfer coefficient of species i . The effect of mass transfer is included as follows:

$$j = j_0 \left[\left(1 - \frac{j}{j_{a,\text{lim}}} \right) e^{-(\alpha_a F/RT)\eta_{\text{act}}} - \left(1 - \frac{j}{j_{c,\text{lim}}} \right) e^{[(1-\alpha_c)F/RT]\eta_{\text{act}}} \right]$$

$$\frac{C_{i,\text{surf}}}{C_i} = \left(1 - \frac{j}{j_{i,\text{lim}}} \right) \quad i = o, r \quad (7-164)$$

Ohmic Control The overall electrochemical reactor cell voltage may be dependent on the kinetic and mass-transfer aspects of the electrochemical reactions; however, a third factor is the potential lost within the electrolyte as current is passing through this phase. The potential drops may become dominant and limit the electrochemical reactions requiring an external potential to be applied to drive the reactions or significantly lower the delivered electrical potential in power generation applications such as batteries and fuel cells.

Multiple Reactions With multiple reactions, the total current is the sum of the currents from the individual reactions with anodic currents positive and cathodic currents negative. This is called the *mixed potential principle*. For more details see Bard and Faulkner, *Electrochemical Methods: Fundamentals and Applications*, 2d ed., Wiley, 2001.

DETERMINATION OF MECHANISM AND KINETICS

Laboratory data are the predominant source for reaction mechanism and kinetics in industrial practice. However, often laboratory data intended for scoping and demonstration studies rather than for kinetic evaluation have to be used, thus reducing the effectiveness and accuracy of the resulting kinetic model. The following are the steps required to obtain kinetics from laboratory data:

1. Develop initial guesses on mechanism, reaction time scale, and potential kinetic models from the literature, scoping experiments, similar chemistries, and computational chemistry calculations, when possible.
2. Select a suitable laboratory reactor type and scale, and analytical tools for kinetic measurements.
3. Develop a priori factorial experimental design or sequential experimental design.
4. When possible, provide ideal reactor conditions, e.g., good mechanical agitation in batch and CSTR, high velocity flow in PFR.
5. Estimate the limiting diffusion-reaction regimes under the prevailing lab reactor conditions for heterogeneous reactions, and use the appropriate lab reactor model. When possible, operate the reactor under kinetic control.
6. Discriminate between competing mechanisms and kinetic rates by forcing maximum differentiation between competing hypotheses through the experimental design, and by obtaining the best fit of the kinetic data to the proposed kinetic forms.

LABORATORY REACTORS

Selection of the laboratory reactor type and size, and associated feed and product handling, control, and analytical schemes depends on the type of reaction, reaction time scales, and type of analytical methods required. The criteria for selection include equipment cost, ease of operation, ease of data analysis, accuracy, versatility, temperature uniformity, and controllability, suitability for mixed phases, and scale-up

feasibility. Many configurations of laboratory reactors have been employed. Rase (*Chemical Reactor Design for Process Plants*, Wiley, 1977) and Shah (*Gas-Liquid-Solid Reactor Design*, McGraw-Hill, 1979) each have about 25 sketches, and Shah's bibliography has 145 items classified into 22 categories of reactor types. Jankowski et al. [*Chemische Technik* 30: 441–446 (1978)] illustrate 25 different kinds of gradientless laboratory reactors for use with solid catalysts.

Laboratory reactors are of two main types:

1. Reactors used to obtain fundamental data on intrinsic chemical rates free of mass-transfer resistances or other complications. Some of the gas-liquid lab reactors, for instance, employ known interfacial areas, thus avoiding the uncertainty regarding the area for gas to liquid mass transfer. When ideal behavior cannot be achieved, intrinsic kinetic estimates need to account for mass- and heat-transfer effects.
2. Reactors used to obtain scale-up data due to their similarity to the reactor intended for the pilot or commercial plant scale. How to scale down from the conceptual commercial or pilot scale to lab scale is a difficult problem in itself, and it is not possible to maintain all key features while scaling down.

The first type is often the preferred one—once the intrinsic kinetics are obtained at “ideal” lab conditions, scale-up is done by using models or correlations that describe large-scale reactor hydrodynamics coupled with the intrinsic kinetics. However, in some cases ideal conditions cannot be achieved, and the laboratory reactor has to be adequately modeled to account for mass and heat transfer and non-ideal mixing effects to enable extraction of intrinsic kinetics. In addition, with homogeneous reactions, attention must be given to prevent wall-catalyzed reactions, which can result in observed kinetics that are fundamentally different from intrinsic homogeneous kinetics. This is a problem for scale-up, due to the high surface/volume ratio in small reactors versus the low surface/volume ratio in large-scale systems, resulting in widely different contributions of wall effects at different scales. Similar issues arise in bioreactors with the potential of

undesirable wall growth of the biocatalyst cells masking the homogeneous growth kinetics. In catalytic reactions certain reactor configurations may enhance undesirable homogeneous reactions, and the importance of these reactions may be different at larger scale, causing potential scale-up pitfalls.

The reaction rate is expressed in terms of chemical compositions of the reacting species, so ultimately the variation of composition with time or space must be found. The composition is determined in terms of a property that is measured by some instrument and calibrated. Among the measures that have been used are titration, pressure, refractive index, density, chromatography, spectrometry, polarimetry, conductimetry, absorbance, and magnetic resonance. Therefore, batch or semibatch data are converted to composition as a function of time (C, t), or to composition and temperature as functions of time (C, T, t), to prepare for kinetic analysis. In a steady CSTR and PFR, the rate and compositions in the effluent are observed as a function of residence time.

When a reaction has many reactive species (which may be the case even for apparently simple processes such as pyrolysis of ethane or synthesis of methanol), a factorial or sequential experimental design should be developed and the data can be subjected to a *response surface analysis* (Box, Hunter, and Hunter, *Statistics for Experimenters*, 2d ed., Wiley Interscience, 2005; Davies, *Design and Analysis of Industrial Experiments*, Oliver & Boyd, 1954). This can result in a black box correlation or statistical model, such as a quadratic (limited to first- and second-order effects) for the variables x_1, x_2 , and x_3 :

$$r = k_1x_1 + k_2x_2 + k_3x_3 + k_{12}x_1x_2 + k_{13}x_1x_3 + k_{23}x_2x_3$$

Analysis of such statistical correlations may reveal the significant variables and interactions and may suggest potential mechanisms and kinetic models, say, of the Langmuir-Hinshelwood type, that could be analyzed in greater detail by a regression process. The variables x_i could be various parameters of heterogeneous processes as well as concentrations. An application of this method to isomerization of *n*-pentane is given by Kittrel and Erjavec [*Ind. Eng. Chem. Proc. Des. Dev.* 7: 321 (1968)].

Table 7-11 summarizes laboratory reactor types that approach the three ideal concepts BR, CSTR and PFR, classified according to reaction types.

TABLE 7-11 Laboratory Reactors

Reaction	Reactor
Homogeneous gas	Isothermal U-tube in temperature-controlled batch
Homogeneous liquid	Mechanically agitated batch or CSTR with jacketed cooling/heating
Catalytic gas-solid	Packed tube in furnace Isothermal U-tube in temperature-controlled bath Rotating basket with jacketed cooling/heating Internal recirculation (Berty) reactor with jacketed cooling/heating
Noncatalytic gas-solid	Packed tube in furnace
Liquid-solid	Packed tube in furnace
Gas-liquid	CSTR with jacketed cooling/heating Fixed interface CSTR Wetted wall Laminar jet
Gas-liquid-solid	Slurry CSTR with jacketed cooling/heating Packed bed with downflow, upflow, or countercurrent
Solid-solid	Packed tube in furnace

For instance, Fig. 7-17 summarizes laboratory reactor types and hydrodynamics for gas-liquid reactions.

Batch Reactors In the simplest kind of investigation, reactants can be loaded into a number of sealed tubes, kept in a thermostatic bath for various periods, shaken mechanically to maintain uniform composition, and analyzed. In terms of cost and versatility, the stirred batch reactor is the unit of choice for homogeneous or heterogeneous slurry reactions including gas-liquid and gas-liquid-solid systems. For multiphase systems the reactants can be semibatch or continuous. The BR is especially suited to reactions with half-lives in excess of 10 min. Samples are taken at time intervals, and the reaction is stopped by cooling, by dilution, or by destroying a residual reactant such as an acid or base; analysis can then be made at a later time. Analytic methods that do not necessitate termination of reaction include nonintrusive measurements of (1) the amount of gas produced, (2) the gas pressure in a constant-volume vessel, (3) absorption of light, (4) electrical or thermal conductivity, (5) polarography, (6) viscosity of polymerization, (7) pH and DO probes, and so on. Operation may be isothermal, with the important effect of temperature determined from several isothermal runs, or the composition and temperature may be recorded simultaneously and the

TYPE	LAMINAR JET	CYLINDRICAL WETTED WALL	CONIC WETTED WALL	SPHERICAL WETTED WALL	STRING OF DISKS	ROTATING DRUM	STIRRED VESSEL	STIRRED VESSEL
SCHEME								
k_L cm/sec	0.016 -0.16	$3.6 \cdot 10^{-3}$ -0.016	$5 \cdot 10^{-3}$ -0.011	$5 \cdot 10^{-3}$ -0.016	$3.6 \cdot 10^{-3}$ -0.016	0.016 -0.36	$1.6 \cdot 10^{-3}$ -0.02	$2 \cdot 10^{-3}$ -0.02
CONTACT TIMES	10^{-3} - 10^{-1} sec	10^{-1} -2 sec	0.2-1 sec	0.1-1 sec	10^{-1} -2 sec	$2 \cdot 10^{-4}$ -10 sec	0.06-10 sec	0.08-10 sec
INTERFACIAL AREA	0.3-10 cm ² high precision	10-100 cm ² high precision	80 cm ² high precision	10-40 cm ² high precision	30-360 cm ² moderate precision	diameter 10 cm length 12 cm high precision	80 cm ² good precision	diameter 10 cm length 15 cm 2-30% open

FIG. 7-17 Principal types of laboratory reactors for gas-liquid reactions. [From Fig. 8 in J. C. Charpentier, "Mass Transfer Rates in Gas-Liquid Absorbers and Reactors," in Drew et al. (eds.), *Advances in Chemical Engineering*, vol. 11, Academic Press, 1981.]

data regressed. On the laboratory scale, it is essential to ensure that a BR is stirred to uniform composition, and for critical cases such as high viscosities this should be checked with tracer tests.

Flow Reactors CSTRs and other devices that require flow control are more expensive and difficult to operate. However, CSTRs and PFRs are the preferred laboratory reactors for steady operation. One of the benefits of CSTRs is their isothermicity and the fact that their mathematical representation is algebraic, involving no differential equations, thus making data analysis simpler. For laboratory research purposes, CSTRs are considered feasible for holding times of 1 to 4000 s, reactor volumes of 2 to 1000 cm³ (0.122 to 61 in³), and flow rates of 0.1 to 2.0 cm³/s. Fast reactions and those in the gas phase are generally done in tubular flow reactors, just as they are often done on the commercial scale. Usually it is not possible to measure compositions along a PFR, although temperatures can be measured using a thermowell with fixed or mobile thermocouple bundle. PFRs can be kept at nearly constant temperatures; small-diameter tubes immersed in a fluidized sand bed or molten salt can hold quite constant temperatures of a few hundred degrees. Other PFRs are operated at near adiabatic conditions by providing dual radial temperature control to minimize the radial heat flux, with multiple axial zones. A recycle unit can be operated as a differential reactor with arbitrarily small conversion and temperature change. Test work in a tubular flow unit may be desirable if the intended commercial unit is of that type.

Multiphase Reactors Reactions between gas-liquid, liquid-liquid, and gas-liquid-solid phases are often tested in CSTRs. Other laboratory types are suggested by the commercial units depicted in appropriate sketches in Sec. 19 and in Fig. 7-17 [Charpentier, *Mass Transfer Rates in Gas-Liquid Absorbers and Reactors*, in Drew et al. (eds.), *Advances in Chemical Engineering*, vol. 11, Academic Press, 1981]. Liquids can be reacted with gases of low solubilities in stirred vessels, with the liquid charged first and the gas fed continuously at the rate of reaction or dissolution. Some of these reactors are designed to have known interfacial areas. Most equipment for gas absorption without reaction is adaptable to absorption with reaction. The many types of equipment for liquid-liquid extraction also are adaptable to reactions of immiscible liquid phases.

Solid Catalysts Processes with solid catalysts are affected by diffusion of heat and mass (1) within the pores of the pellet, (2) between the fluid and the particle, and (3) axially and radially within the packed bed. Criteria in terms of various dimensionless groups have been developed to tell when these effects are appreciable, and some of these were discussed above. For more details see Mears [*Ind. Eng. Chem. Proc. Des. Devel.* **10**: 541–547 (1971); *Ind. Eng. Chem. Fund.* **15**: 20–23 (1976)] and Satterfield (*Heterogeneous Catalysis in Practice*, McGraw-Hill, 1991, p. 491). For catalytic investigations, the rotating basket or fixed basket with internal recirculation is the standard device, usually more convenient and less expensive than equipment with external recirculation. In the fixed-basket type, an internal recirculation rate of 10 to 15 or so times the feed rate effectively eliminates external diffusional resistance, and temperature gradients (see, e.g., Berty, *Experiments in Catalytic Reaction Engineering*, Elsevier, 1999). A unit holding 50 cm³ (3.05 in³) of catalyst can operate up to 800 K (1440°R) and 50 bar (725 psi). When deactivation occurs rapidly (in a few seconds during catalytic cracking, for instance), the fresh activity can be maintained with a transport reactor through which both reactants and fresh catalyst flow without slip and with short contact time. Since catalysts often are sensitive to traces of impurities, the time deactivation of the catalyst usually can be evaluated only with commercial feedstock. Physical properties of catalysts also may need to be checked periodically, including pellet size, specific surface, porosity, pore size and size distribution, effective diffusivity, and active metals content and dispersion. The effectiveness of a porous catalyst is found by measuring conversions with successively smaller pellets until no further change occurs. These topics are touched on by Satterfield (*Heterogeneous Catalysis in Industrial Practice*, McGraw-Hill, 1991).

To determine the deactivation kinetics, long-term deactivation studies at constant conditions and at different temperatures are required. In some cases, accelerated aging can be induced to reduce the time required for the experimental work, by either increasing the feed flow

rate (if the deactivation is a result of feed or product poisoning) or increasing the temperature above the standard reaction temperature. These require a good understanding of how the higher-temperature or rate-accelerated deactivation correlates with deactivation at the operating reaction temperature and rate.

Bioreactors There are several types of laboratory bioreactors used with live organisms as biocatalysts:

1. Mechanically agitated batch/semibatch with pH control and nutrients or other species either fed at the start or added continuously based on a recipe or protocol.

2. CSTR to maintain a constant *dilution rate* (the feed rate). These require some means to separate the biocatalyst from the product and recycle to the reactor, such as centrifuge or microfiltration:

- a. *Chemostat* controls the flow to maintain a constant fermentation volume.

- b. *Turbidostat* controls the biomass or cells concentration.

- c. *pH-auxostat* controls pH in the effluent (same as pH in reactor).

- d. *Productostat* controls the effluent concentration of one of the metabolic products.

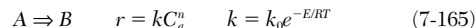
The preferred reactor for kinetics is the chemostat, but semibatch reactors are more often used owing to their simpler operation.

Calorimetry Another category of laboratory systems that can be used for kinetics includes *calorimeters*. These are primarily used to establish temperature effects and thermal runaway conditions, but can also be employed to determine reaction kinetics. Types of calorimeters are summarized in Table 7-12; for more details see Reid, "Differential Microcalorimeters," *J. Physics E: Scientific Instruments*, **9** (1976).

Additional methods of laboratory data acquisition are described in Masel, *Chemical Kinetics and Catalysis*, Wiley, 2001.

KINETIC PARAMETERS

The kinetic parameters are constants that appear in the intrinsic kinetic rate expressions and are required to describe the rate of a reaction or reaction network. For instance, for the simple global *n*th-order reaction with Arrhenius temperature dependence:



The kinetic parameters are k_0 , E , and n , and knowledge of these parameters and the prevailing concentration and temperature fully determines the reaction rate.

For a more complex expression such as the Langmuir-Hinshelwood rate for gas reaction on heterogeneous catalyst surface with equilibrium adsorption of reactants *A* and *B* on two different sites and non-adsorbing products, Eq. (7-85) can be rewritten as

$$r = \frac{k_0 e^{-E/RT} P_a P_b}{(1 + K_{a0} e^{-E_{a0}/RT})(1 + K_{b0} e^{-E_{b0}/RT})} \quad (7-166)$$

and the kinetic parameters are k_0 , E , K_{a0} , E_{a0} , K_{b0} , and E_{b0} .

A number of factors limit the accuracy with which parameters needed for the design of commercial equipment can be determined. The kinetic parameters may be affected by inaccurate accounting for laboratory reactor heat and mass transport, and hydrodynamics; correlations for these are typically determined under nonreacting conditions at ambient temperature and pressure and with nonreactive model fluids and may not be applicable or accurate at reaction conditions. Experimental uncertainty including errors in analysis, measurement,

TABLE 7-12 Calorimetric Methods

Adiabatic	Nonadiabatic
Accelerating rate calorimeter (ARC)	Reaction calorimeter (RC1) + IR
Vent sizing package (VSP)	Differential scanning calorimeter (DSC)
calorimeter	Thermal gravimetry (TG)
PHI-TEC	Isothermal calorimetry
Dewar	Differential thermal analysis (DTA)
Automatic pressure	Differential microcalorimeters
tracking adiabatic	Advanced reaction system
calorimeter (APTAC)	screening tool (ARSTST)

and control is also a contributing factor (see, e.g., Hoffman, "Kinetic Data Analysis and Parameter Estimation," in de Lasa (ed.), *Chemical Reactor Design and Technology*, Martinus Nijhoff, 1986).

DATA ANALYSIS METHODS

In this section we focus on the three main types of ideal reactors: BR, CSTR, and PFR. Laboratory data are usually in the form of concentrations or partial pressures versus batch time (batch reactors), concentrations or partial pressures versus distance from reactor inlet or residence time (PFR), or rates versus residence time (CSTR). Rates can also be calculated from batch and PFR data by differentiating the concentration versus time or distance data, usually by numerical curve fitting first. It follows that a general classification of experimental methods is based on whether the data measure rates directly (differential or direct method) or indirectly (integral of indirect method). Table 7-13 shows the pros and cons of these methods.

Some simple reaction kinetics are amenable to analytical solutions and graphical linearized analysis to calculate the kinetic parameters from rate data. More complex systems require numerical solution of nonlinear systems of differential and algebraic equations coupled with nonlinear parameter estimation or regression methods.

Differential Data Analysis As indicated above, the rates can be obtained either directly from differential CSTR data or by differentiation of integral data. A common way of evaluating the kinetic parameters is by rearrangement of the rate equation, to make it linear in parameters (or some transformation of parameters) where possible. For instance, using the simple n th-order reaction in Eq. (7-165) as an example, taking the natural logarithm of both sides of the equation results in a linear relationship between the variables $\ln r$, $1/T$, and $\ln C_a$:

$$\ln r = \ln k_0 - \frac{E}{RT} + n \ln C_a \quad (7-167)$$

Multilinear regression can be used to find the constants k_0 , E , and n . For constant-temperature (isothermal) data, Eq. (7-167) can be simplified by using the Arrhenius form as

$$\ln r = \ln k + n \ln C_a \quad (7-168)$$

and the kinetic parameters n and k can be determined as the intercept and slope of the best straight-line fit to the data, respectively, as shown in Fig. 7-18.

The preexponential k_0 and activation energy E can be obtained from multiple isothermal data sets at different temperatures by using the linearized form of the Arrhenius equation

$$\ln k = \ln k_0 - \frac{E}{RT} \quad (7-169)$$

as shown in Fig. 7-19.

Integral Data Analysis Integral data such as from batch and PFR relate concentration to time or distance. Integration of the BR equation for an n th-order homogeneous constant-volume reaction yields

TABLE 7-13 Comparison of Direct and Indirect Methods

Direct method	Indirect method
Advantages	Disadvantages
Get rate equation directly	Must infer rate equation
Easy to fit data to a rate law	Hard to analyze rate data
High confidence on final rate equation	Low confidence on final rate equation
Disadvantages	Advantages
Difficult experiment	Easier experiment
Need many runs	Can do a few runs and get important information
Not suitable for very fast or very slow reactions	Suitable for all reactions including very fast or very slow ones

SOURCE: Masel, *Chemical Kinetics and Catalysis*, Wiley, 2001, Table 3.2.

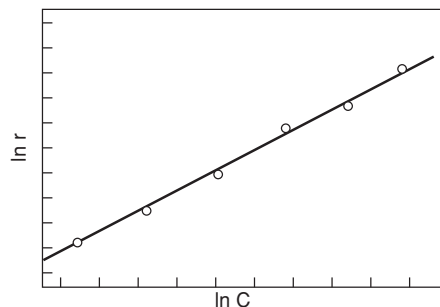


FIG. 7-18 Determination of the rate constant and reaction order.

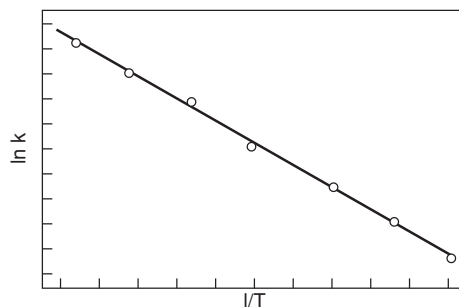


FIG. 7-19 Determination of the activation energy.

$$\ln \frac{C_{a0}}{C_a} = k\tau \quad \text{for } n = 1$$

$$\left(\frac{C_{a0}}{C_a} \right)^{n-1} = 1 + k\tau(n-1)C_{a0}^{n-1} \quad \text{for } n \neq 1 \quad (7-170)$$

For the first-order case, the rate constant k can be obtained directly from the slope of the graph of the left-hand side of Eq. (7-170) versus batch time, as shown in Fig. 7-20.

For orders other than first, plotting the natural log of Eq. (7-170) can at least indicate if the order is larger or smaller than 1, as shown in Fig. 7-21.

The Half-Life Method The half-life is the batch time required to get 50 percent conversion. For an n th-order reaction,

$$\tau_{1/2} = \frac{\ln 2}{k} \quad \text{for } n = 1$$

$$\tau_{1/2} = \frac{2^{n-1} - 1}{(n-1)kC_{a0}^{n-1}} \quad \text{for } n \neq 1 \quad (7-171)$$

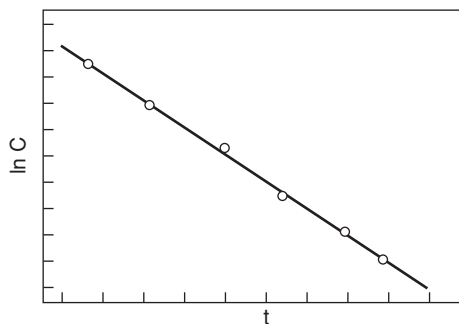


FIG. 7-20 Determination of first-order rate constant from integral data.

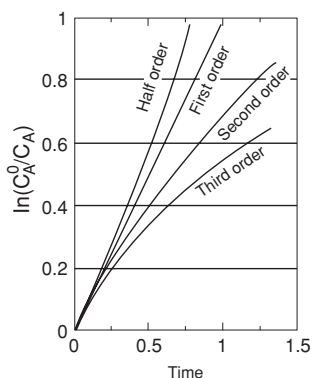


FIG. 7-21 Reaction behavior for n th-order reaction. (Masel, *Chemical Kinetics and Catalysis*, Wiley, 2001, Fig. 3.15.)

Thus for first-order reactions, the *half-life* is constant and independent of the initial reactant concentration and can be used directly to calculate the rate constant k . For non-first-order reactions, Eq. (7-171) can be linearized as follows:

$$\ln \tau_{1/2} = \ln \frac{2^{n-1} - 1}{(n-1)k} - (n-1) \ln C_{A0} \quad \text{for } n \neq 1 \quad (7-172)$$

The reaction order n can be obtained from the slope and the rate constant k from the intercept of the plot of Eq. (7-172), shown in Fig. 7-22.

Complex Rate Equations The examples above are for special cases amenable to simple treatment. Complex rate equations and reaction networks with complex kinetics require individual treatment, which often includes both numerical solvers for the differential and algebraic equations describing the laboratory reactor used to obtain the data and linear or nonlinear parameter estimation.

PARAMETER ESTIMATION

The straightforward method to obtain kinetic parameters from data is the numerical fitting of the concentration data (e.g., from BR or PFR) to integral equations, or the rate data (e.g., from a CSTR or from differentiation of BR or PFR) to rate equations. This is done by parameter estimation methods described here. An excellent reference for experimental design and parameter estimation (illustrated for heterogeneous gas-solid reactions) is the review paper of Froment and Hosten, "Catalytic Kinetics—Modeling," in *Catalysis—Science and Technology*, Springer-Verlag, New York, 1981. Two previous papers devoted to this topic by Hofmann [in *Chemical Reaction Engineering*, ACS Advances in Chemistry, **109**: 519–534 (1972); in de Lasa (ed.),

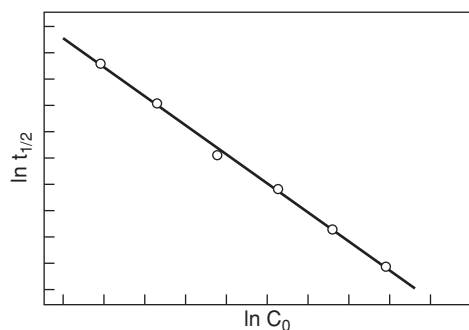


FIG. 7-22 Determination of reaction order and rate constant from half-life data.

Chemical Reactor Design and Technology, Martinus Nijhoff, 1985, pp. 69–105] are also very useful. As indicated above, the acquisition of kinetic data and parameter estimation can be a complex endeavor. It includes statistical design of experiments, laboratory equipment, computer-based data acquisition, complex analytical methods, and statistical evaluation of the data.

Regression is the procedure used to estimate the kinetic parameters by fitting kinetic model predictions to experimental data. When the parameters can be made to appear linear in the kinetic model (through transformations, grouping of parameters, and rearrangement), the regression is linear, and an accurate fit to data can be obtained, provided the form of the kinetic model represents well the reaction kinetics and the data provide enough width in temperature, pressure, and composition for statistically significant estimates. Often such linearization is not possible.

Linear Models in Parameters, Single Reaction We adopt the terminology from Froment and Hosten, "Catalytic Kinetics—Modeling," in *Catalysis—Science and Technology*, Springer-Verlag, New York, 1981. For n observations (experiments) of the concentration vector y for a model linear in the parameter vector β of length $p < n$, the *residual error* ϵ is the difference between the kinetic model-predicted values and the measured data values:

$$\epsilon = y - X\beta = y - \hat{y} \quad (7-173)$$

The linear model is represented as a linear transformation of the parameter vector β through the model matrix X . Estimates b of the true parameters β are obtained by minimizing the *objective function* $S(\beta)$, the *sum of squares of the residual errors*, while varying the values of the parameters:

$$S(\beta) = \epsilon^T \epsilon = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \rightarrow \text{Min} \quad (7-174)$$

This linear optimization problem, subject to constraints on the possible values of the parameters (e.g., requiring positive preexponentials, activation energies, etc.) can be solved to give the estimated parameters:

$$b = (X^T X)^{-1} X^T y \quad (7-175)$$

When the error is *normally distributed* and has zero mean and variance σ^2 , then the *variance-covariance matrix* $V(b)$ is defined as

$$V(b) = (X^T X)^{-1} \sigma^2 \quad (7-176)$$

An estimate for σ^2 , denoted s^2 , is

$$s^2 = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n - p} \quad (7-177)$$

When $V(b)$ is known from experimental observations, a *weighted objective function* should be used for optimization of the objective function:

$$S(\beta) = \epsilon^T V^{-1} \epsilon \rightarrow \text{Min} \quad (7-178)$$

and the estimates b are obtained as

$$b = (X^T V^{-1} X)^{-1} X^T V^{-1} y \quad (7-179)$$

The parameter fit is adequate if the *F test* is satisfied, that is, F_c , the calculated F , is larger than the tabulated statistical one at the *confidence level* of $1 - \alpha$:

$$F_c = \frac{\frac{\text{LFSS}}{n - p - n_e + 1}}{\frac{\text{PESS}}{n_e - 1}} \geq F(n - p - n_e + 1, n_e - 1; 1 - \alpha) \quad (7-180)$$

$$\text{LFSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 - \sum_{i=1}^{ne} (y_i - \bar{y}_i)^2 \quad \text{PESS} = \sum_{i=1}^{ne} (y_i - \bar{y}_i)^2$$

Here $\bar{y}_i$ are the averaged values of the data for replicates. Equation (7-180) is valid if there are n replicate experiments and the pure error sum of squares (PESS) is known. Without replicates,

$$F_c = \frac{\frac{\text{RgSS}}{p}}{\frac{\text{RSS}}{n-p}} \geq F(p, n-p; 1-\alpha) \quad \text{RgSS} = \sum_{i=1}^n \hat{y}_i^2$$

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (7-181)$$

The *bounds* on the parameter estimates are given by the *t* statistics:

$$b_i - t\left(n-p; 1 - \frac{\alpha}{2}\right) \leq \beta_i \leq b_i + t\left(n-p; 1 - \frac{\alpha}{2}\right) \quad (7-182)$$

An example of a linear model in parameters is Eq. (7-167), where the parameters are $\ln k_0$, E , and n , and the linear regression can be used directly to estimate these.

Nonlinear Models in Parameters, Single Reaction In practice, the parameters appear often in nonlinear form in the rate expressions, requiring nonlinear regression. Nonlinear regression does not guarantee optimal parameter estimates even if the kinetic model adequately represents the true kinetics and the data width is adequate. Further, the statistical tests of model adequacy apply rigorously only to models linear in parameters, and can only be considered approximate for nonlinear models.

For a general nonlinear model $f(x, \beta)$, where x is the vector of the independent model variables and β is the vector of parameters,

$$\varepsilon = y - f(x, \beta) \quad (7-183)$$

An example of a model nonlinear in parameters is Eq. (7-166). Here it is not possible through any number of transformations to obtain a linear form in all the parameters k_0 , E , K_{ab} , E_{aa} , K_{bb} , E_{ab} . Note that for some Langmuir-Hinshelwood rate expressions it is possible to linearize the model in parameters at isothermal conditions and obtain the kinetic constants for each temperature, followed by Arrhenius-type plots to obtain activation energies (see, e.g., Churchill, *The Interpretation and Use of Rate Data: The Rate Concept*, McGraw-Hill, 1974).

Minimization of the sum of squares of residuals does not result in a closed form for nonlinear parameter estimates as for the linear case; rather it requires an iterative numerical solution, and having a reasonable initial estimate for the parameter values and their feasible ranges is critical for success. Also, the minima in the residual sum of squares are local and not global. To obtain global minima that better represent the kinetics over a wide range of conditions, parameter estimation has to be repeated with a wide range of initial parameter guesses to increase the chance of reaching the global minimum. The nonlinear regression procedure typically involves a *steepest descent* optimization search combined with *Newton's linearization* method when a minimum is approached, enhancing the convergence speed [e.g., the *Marquardt-Levenberg* or *Newton-Gauss* method; Marquardt, *J. Soc. Ind. Appl. Math.* **2**: 431 (1963)].

An integral part of the parameter estimation methodology is *mechanism discrimination*, i.e., selection of the best mechanism that would result in the best kinetic model. Nonlinear parameter estimation is an extensive topic and will not be further discussed here. For more details see Froment and Hostén, "Catalytic Kinetics—Modeling," in *Catalysis—Science and Technology*, Springer-Verlag, New York, 1981.

Network of Reactions The statistical parameter estimation for multiple reactions is more complex than for a single reaction. As indicated before, a single reaction can be represented by a single con-

centration [e.g., Eq. 7-39]. With a network of reactions, there are a number of dependent variables equal to the number of stoichiometrically independent reactions, also called *responses*. In this case the objective function has to be modified. For details see Froment and Hostén, "Catalytic Kinetics—Modeling," in *Catalysis—Science and Technology*, Springer-Verlag, New York, 1981.

THEORETICAL METHODS

Prediction of Mechanism and Kinetics Reaction mechanisms for a variety of reaction systems can be predicted to some extent by following a set of heuristic rules derived from experience with a wide range of chemistries. For instance, Masel, *Chemical Kinetics and Catalysis*, Wiley, 2001, chapter 5, enumerates the rules for gas-phase chain and nonchain reactions including limits on activation energies for various elementary steps. Other reaction systems such as ionic reactions, and reactions on metal and acid surfaces, are also discussed by Masel, although these mechanisms are not as well understood. Nevertheless, the rules can lead to computer-generated mechanisms for complex systems such as homogeneous gas-phase combustion and partial oxidation of methane and higher hydrocarbons. Developments in computational chemistry methods allow, in addition to the derivation of most probable elementary mechanisms, prediction of thermodynamic and kinetic reaction parameters for relatively small molecules in homogeneous gas-phase and liquid-phase reactions, and even for some heterogeneous catalytic systems. This is especially useful for complex kinetics where there is no easily discernible rate-determining step, and therefore no simple closed-form global reaction rate can be determined. In particular, estimating a large number of kinetic parameters from laboratory data requires a large number of experiments and use of intermediate reaction components that are not stable or not readily available. The nonlinear parameter estimation with many parameters is difficult, with no assurance that global minima are actually obtained. For such complex systems, computational chemistry estimates are an attractive starting point, requiring experimental validation.

Computational chemistry includes a wide range of methods of varying accuracy and complexity, summarized in Table 7-14. Many of these methods have been implemented as software packages that require high-speed supercomputers or parallel computers to solve realistic reactions. For more details on computational chemistry, see, e.g., Cramer, *Essentials of Computational Chemistry: Theories and Models*, 2d ed., Wiley, 2004.

Lumping and Mechanism Reduction It is often useful to reduce complex reaction networks to a smaller reaction set which still maintains the key features of the detailed reaction network but with a much smaller number of representative species, reactions, and kinetic parameters. Simple examples were already given above for reducing simple networks into global reactions through assumptions such as pseudo-steady state, rate-limiting step, and equilibrium reactions.

In general, *mechanism reduction* can only be used over a limited range of conditions for which the simplified system simulates the original complete reaction network. This reduces the number of kinetic parameters that have to be either estimated from data or calculated by

TABLE 7-14 Computational Chemistry Methods

<i>Abinitio</i> methods (no empirical parameters)	
Electronic structure determination (time-independent Schrodinger equation)	
Hartree-Fock (HF) with corrections	
Quantum Monte Carlo (QMT)	
Density functional theory (DFT)	
Chemical dynamics determination (time-dependent Schrodinger equation)	
Split operator technique	
Multiconfigurational time-dependent Hartree-Fock method	
Semiclassical method	
Semiempirical methods (approximate parts of HF calculations such as two-electron integrals)	
Huckel	
Extended Huckel	
Molecular mechanics (avoids quantum mechanical calculations)	
Empirical methods (group contributions)	
Polanyi linear approximation of activation energy	

using computational chemistry. The simplified system also reduces the computation load for reactor scale-up, design, and optimization.

A type of mechanism reduction called *lumping* is typically performed on a reaction network that consists of a large number of similar reactions occurring between similar species, such as homologous series or molecules having similar functional groups. Such situations occur, for instance, in the oil refining industry, examples including catalytic reforming, catalytic cracking, hydrocracking, and hydrotreating. Lumping is done by grouping similar species, or molecules with similar functional groups, into pseudo components called lumped species. The behavior of the lumped system depends on the initial composition, the distribution of the rate constants in the detailed system, and the form of the rate equation. The two main issues in lumping are

1. Determination of the lump structure that simulates the detailed system over the required range of conditions
2. Determination of the kinetics of the lumped system from general knowledge about the type of kinetics and the overall range of parameters of the detailed system

Lumping has been applied extensively to first-order reaction networks [e.g., Wei and Kuo, "A Lumping Analysis in Monomolecular Reaction Systems," *I&EC Fundamentals* **8**(1): 114–123 (1969); Golikeri and Luss, "Aggregation of Many Coupled Consecutive First Order Reactions," *Chem. Eng. Sci.* **29**: 845–855 (1974)]. For instance, it has been shown that a lumped reaction network of first-order reactions can behave under certain conditions as a global second-order reaction. Where analytical solutions were not available, others, such as Golikeri and Luss, "Aggregation of Many Coupled Consecutive First Order Reactions," *Chem. Eng. Sci.* **29**: 845–855 (1974), developed bounds that bracketed the behavior of the lump for first-order reactions as a function of the initial composition and the rate constant distribution. Lumping has not been applied as successfully to nonlinear or higher-order kinetics. More recent applications of lumping were published, including structure-oriented lumping that lumps similar structural groups, by Quann and Jaffe, "Building Useful Models of Complex Reaction Systems in Petroleum Refining," *Chem. Eng. Sci.* **51**(10): 1615–1635 (1996).

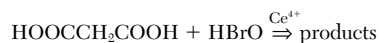
For other types of systems such as highly branched reaction networks for homogeneous gas-phase combustion and combined homogeneous and catalytic partial oxidation, mechanism reduction involves pruning branches and pathways of the reaction network that do not contribute significantly to the overall reaction. This pruning is done by using *sensitivity analysis*. See, e.g., Bui et al., "Hierarchical Reduced Models for Catalytic Combustion: H₂/Air Mixtures near Platinum Surfaces," *Combustion Sci. Technol.* **129**(1–6):243–275 (1997).

Multiple Steady States, Oscillations, and Chaotic Behavior
There are reaction systems whose steady-state behavior depends on

the initial or starting conditions; i.e., for different starting conditions, different steady states can be reached at the same final operating conditions. This behavior is called steady-state multiplicity and is often the result of the interaction of kinetic and transport phenomena having distinct time scales. For some cases, the cause of the multiplicity is entirely reaction-related, as shown below. Associated with steady-state multiplicity is hysteresis, and higher-order instabilities such as self-sustained oscillations and chaotic behavior. The existence of multiple steady states may be relevant to analysis of laboratory data, since faster or slower rates may be observed at the same conditions depending on how the lab reactor is started up.

For example, CO oxidation on heterogeneous Rh catalyst exhibits hysteresis and multiple steady states, and one of the explained causes is the existence of two crystal structures for Rh, each with a different reactivity (Masel, *Chemical Kinetics and Catalysis*, Wiley, 2001, p. 38).

Another well-known example of chemistry-related instability includes the oscillatory behavior of the Bhlousov-Zhabotinsky reaction of malonic acid and bromate in the presence of homogeneous Ce catalyst having the overall reaction



Ce can be in two oxidation states, Ce³⁺ and Ce⁴⁺, and there are competing reaction pathways. Complex kinetic models are required to predict the oscillatory behavior, the most well known being that of Noyes [e.g., Showalter, Noyes, and Bar-Eli, *J. Chem. Phys.* **69**(6): 2514–2524 (1978)].

A large body of work has been done to develop criteria that determine the onset of chemistry and transport chemistry-based instabilities. More details and transport-reaction coupling-related examples are discussed in Sec. 19.

SOFTWARE TOOLS

There are a number of useful software packages that enable efficient analysis of laboratory data for developing the mechanism and kinetics of reactions and for testing the kinetics by using simple reactor models. The reader is referred to search the Internet as some of these software packages change ownership or name. Worth mentioning are the Aspen Engineering Suite (Aspen), the Thermal Safety Software suite (Cheminform St. Petersburg), the Matlab suite (Mathworks), the Chemkin software suite (Reaction Design), the NIST Chemical Kinetics database (NIST), and Gepasi for biochemical kinetics (freeware). The user is advised to experiment and validate any software package with known data and kinetics to ensure robustness and reliability.

SECTION **8**

Process Control

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8TH EDITION



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Process Control

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8-4 PROCESS CONTROL

Nomenclature

Symbol	Definition	Symbol	Definition
A	Area	s	Laplace transform variable
A_a	Actuator area	s	Search direction
A_c	Output amplitude limits	S_i	Step response coefficient
A_v	Amplitude of controlled variable	t	Time
A_1	Cross-sectional area of tank	T	Temperature, target
b	Controller output bias	$T(s)$	Decoupler transfer function
B	Bottoms flow rate	T_b	Base temperature
B_i°	Limit on control	T_f	Exhaust temperature
c_A	Concentration of A	T_R	Reset time
C	Cumulative sum	U	Heat-transfer coefficient
C_d	Discharge coefficient	u, U	Manipulated variable, controller output
C_i	Inlet concentration	V	Volume
C_i°	Limit on control move	V_p	Product value
C_L	Specific heat of liquid	w	Mass flow rate
C_0	Integration constant	w_i	Weighting factor
C_p	Process capability	W	Steam flow rate
C_r	Heat capacity of reactants	x	Mass fraction
C_v	Valve flow coefficient	$\bar{x}$	Sample mean
D	Distillate flow rate, disturbance	x_i	Optimization variable
D_i°	Limit on output	x_T	Pressure drop ratio factor
e	Error	X	Transform of deviation variable
E	Economy of evaporator	y, Y	Process output, controlled variable, valve travel
f	Function of time	Y_{sp}	Set point
F, f	Feed flow rate	z	Controller tuning law, expansion factor
F_L	Pressure recovery factor	z_i	Feed mole fraction (distillation)
g_c	Unit conversion constant	Z	Compressibility factor
g_d	Algebraic inequality constraint	Greek Symbols	
G	Transfer function	α	Digital filter coefficient
G_c	Controller transfer function	a_T	Temperature coefficient of resistance
G_d	Disturbance transfer function	β	Resistance thermometer parameter
G_f	Feedforward controller transfer function	γ	Ratio of specific heats
G_m	Sensor transfer function	δ	Move suppression factor, shift in target value
G_p	Process transfer function	Δq	Load step change
G_t	Transmitter transfer function	Δt	Time step
G_v	Valve transfer function	ΔT	Temperature change
h_i	Algebraic equality constraints	Δu	Control move
h_1	Liquid head in tank	ε	Spectral emissivity, step size
H	Latent heat of vaporization, control limit or threshold	ζ	Damping factor (second-order system)
i	Summation index	θ	Time delay
I_i	Impulse response coefficient	λ	Relative gain array parameter, wavelength
j	Time index	Λ	Relative gain array
J	Objective function or performance index	ξ	Deviation variable
k	Time index	ρ	Density
k_f	Flow coefficient	σ	Stefan-Boltzmann constant, standard deviation
k_r	Kinetic rate constant	Σ_t	Total response time
K	Gain, slack parameter	τ	Time constant
K_c	Controller gain	τ_d	Natural period of closed loop, disturbance time constant
K_d	Disturbance transfer function gain	τ_D	Derivative time (PID controller)
K_m	Measurement gain	τ_F	Filter time constant
K_p	Process gain	τ_I	Integral time (PID controller)
K_u	Ultimate controller gain (stability)	τ_P	Process time constant
L	Load variable	τ_o	Period of oscillation
L_p	Sound pressure level	ϕ_{PI}	Phase lag
M	Manipulated variable	Subscripts	
m_c	Number of constraints	A	Species A
M_e	Mass flow	b	Best
M_r	Mass of reactants	c	Controller
M_w	Molecular weight	d	Disturbance
n	Number of data points, number of stages or effects	eff	Effective
N	Number of inputs/outputs, model horizon	f	Feedforward
p	Proportional band (%)	i	Initial, inlet
p_c	Vapor pressure	L	Load, disturbance
p_d	Actuator pressure	m	Measurement or sensor
p_i	Pressure	p	Process
p_u	Proportional band (ultimate)	s	Steady state
q	Radiated energy flux	sp	Set-point value
q_b	Energy flux to a black body	t	Transmitter
Q	Flow rate	u	Ultimate
r_c	Number of constraints	v	Valve
R	Equal-percentage valve characteristic		
R_T	Resistance in temperature sensor		
R_1	Valve resistance		

FUNDAMENTALS OF PROCESS DYNAMICS AND CONTROL

THE GENERAL CONTROL SYSTEM

A process is shown in Fig. 8-1 with a manipulated input U , a load input D , and a controlled output Y , which could be flow, pressure, liquid level, temperature, composition, or any other inventory, environmental, or quality variable that is to be held at a desired value identified as the set point Y_{sp} . The load may be a single variable or an aggregate of variables either acting independently or manipulated for other purposes, affecting the controlled variable much as the manipulated variable does. Changes in load may occur randomly as caused by changes in weather, diurnally with ambient temperature, manually when operators change production rate, stepwise when equipment is switched into or out of service, or cyclically as the result of oscillations in other control loops. Variations in load will drive the controlled variable away from the set point, requiring a corresponding change in the manipulated variable to bring it back. The manipulated variable must also change to move the controlled variable from one set point to another.

An open-loop system positions the manipulated variable either manually or on a programmed basis, without using any process measurements. This operation is acceptable for well-defined processes without disturbances. An automated transfer switch is provided to allow manual adjustment of the manipulated variable in case the process or the control system is not performing satisfactorily.

A closed-loop system uses the measurement of one or more process variables to move the manipulated variable to achieve control. Closed-loop systems may include feedforward, feedback, or both.

Feedback Control In a feedback control loop, the controlled variable is compared to the set point Y_{sp} , with the error E acted upon by the controller to move U in such a way as to minimize the error. This action is specifically negative feedback, in that an increase in error moves U so as to decrease the error. (Positive feedback would cause the error to expand rather than diminish and therefore does not regulate.) The action of the controller is selectable to allow use on process gains of both signs.

The controller has tuning parameters related to proportional, integral, derivative, lag, dead time, and sampling functions. A negative feedback loop will oscillate if the controller gain is too high; but if it is too low, control will be ineffective. The controller parameters must be properly related to the process parameters to ensure closed-loop stability while still providing effective control. This relationship is accomplished, first,

by the proper selection of control modes to satisfy the requirements of the process and, second, by the appropriate tuning of those modes.

Feedforward Control A feedforward system uses measurements of disturbance variables to position the manipulated variable in such a way as to minimize any resulting deviation. The disturbance variables could be either measured loads or the set point, the former being more common. The feedforward gain must be set precisely to reduce the deviation of the controlled variable from the set point.

Feedforward control is usually combined with feedback control to eliminate any offset resulting from inaccurate measurements and calculations and unmeasured load components. The feedback controller can be used as a bias on the feedforward controller or in a multiplicative form.

Computer Control Computers have been used to replace analog PID controllers, either by setting set points of lower-level controllers in supervisory control or by driving valves directly in direct digital control. Single-station digital controllers perform PID control in one or two loops, including computing functions such as mathematical operations, characterization, lags, and dead time, with digital logic and alarms. Distributed control systems provide all these functions, with the digital processor shared among many control loops; separate processors may be used for displays, communications, file servers, and the like. A host computer may be added to perform high-level operations such as scheduling, optimization, and multivariable control. More details on computer control are provided later in this section.

PROCESS DYNAMICS AND MATHEMATICAL MODELS

GENERAL REFERENCES: Seborg, Edgar, and Mellichamp, *Process Dynamics and Control*, Wiley, New York, 2004; Marlin, *Process Control*, McGraw-Hill, New York, 2000; Ogunnaike and Ray, *Process Dynamics Modeling and Control*, Oxford University Press, New York, 1994; Smith and Corripio, *Principles and Practices of Automatic Process Control*, Wiley, New York, 1997.

Open-Loop versus Closed-Loop Dynamics It is common in industry to manipulate coolant in a jacketed reactor in order to control conditions in the reactor itself. A simplified schematic diagram of such a reactor control system is shown in Fig. 8-2. Assume that the reactor temperature is adjusted by a controller that increases the coolant flow in proportion to the difference between the desired reactor temperature and the temperature that is measured. The proportionality constant is K_c . If a small change in the temperature of the inlet stream occurs, then

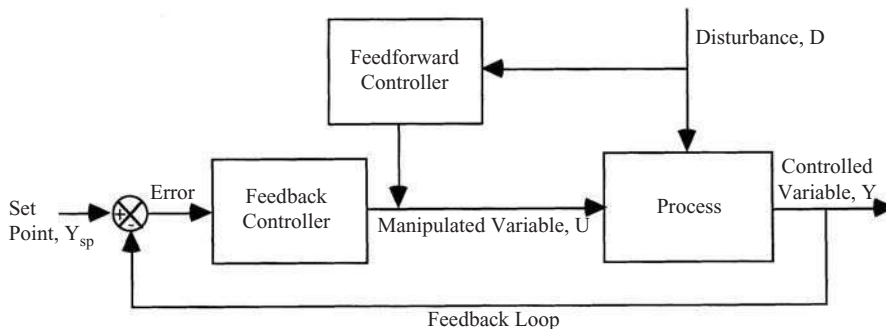


FIG. 8-1 Block diagram for feedforward and feedback control.

8-6 PROCESS CONTROL

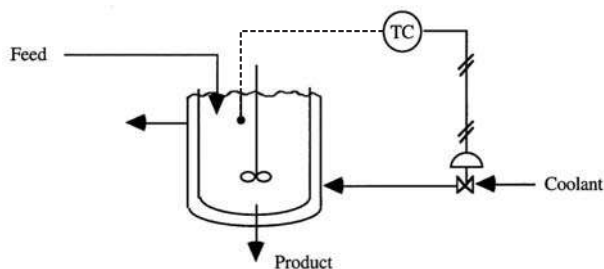


FIG. 8-2 Reactor control system.

depending on the value of K_c , one might observe the reactor temperature responses shown in Fig. 8-3. The top plot shows the case for no control ($K_c = 0$), which is called the open loop, or the normal dynamic response of the process by itself. As K_c increases, several effects can be noted. First, the reactor temperature responds faster and faster. Second, for the initial increases in K , the maximum deviation in the reactor temperature becomes smaller. Both of these effects are desirable so that disturbances from normal operation have as small an effect as possible on the process under study. As the gain is increased further, eventually a point is reached where the reactor temperature oscillates indefinitely, which is undesirable. This point is called the stability limit, where $K_c = K_u$, the ultimate controller gain. Increasing K_c further causes the magnitude of the oscillations to increase, with the result that the control valve will cycle between full open and closed.

The responses shown in Fig. 8-3 are typical of the vast majority of regulatory loops encountered in the process industries. Figure 8-3

shows that there is an optimal choice for K_c , somewhere between 0 (no control) and K_u (stability limit). If one has a dynamic model of a process, then this model can be used to calculate controller settings. In Fig. 8-3, no time scale is given, but rather the figure shows relative responses. A well-designed controller might be able to speed up the response of a process by a factor of roughly 2 to 4. Exactly how fast the control system responds is determined by the dynamics of the process itself.

Physical Models versus Empirical Models In developing a dynamic process model, there are two distinct approaches that can be taken. The first involves models based on first principles, called physical or first principles models, and the second involves empirical models. The conservation laws of mass, energy, and momentum form the basis for developing physical models. The resulting models typically involve sets of differential and algebraic equations that must be solved simultaneously. Empirical models, by contrast, involve postulating the form of a dynamic model, usually as a transfer function, which is discussed below. This transfer function contains a number of parameters that need to be estimated from data. For the development of both physical and empirical models, the most expensive step normally involves verification of their accuracy in predicting plant behavior.

To illustrate the development of a physical model, a simplified treatment of the reactor, shown in Fig. 8-2, is used. It is assumed that the reactor is operating isothermally and that the inlet and exit volumetric flows and densities are the same. There are two components, A and B, in the reactor, and a single first-order reaction of $A \rightarrow B$ takes place. The inlet concentration of A, which we call c_{Ai} , varies with time. A dynamic mass balance for the concentration of A, denoted c_A , can be written as follows:

$$V \frac{dc_A}{dt} = Fc_i - Fc_A - k_r V c_A \quad (8-1)$$

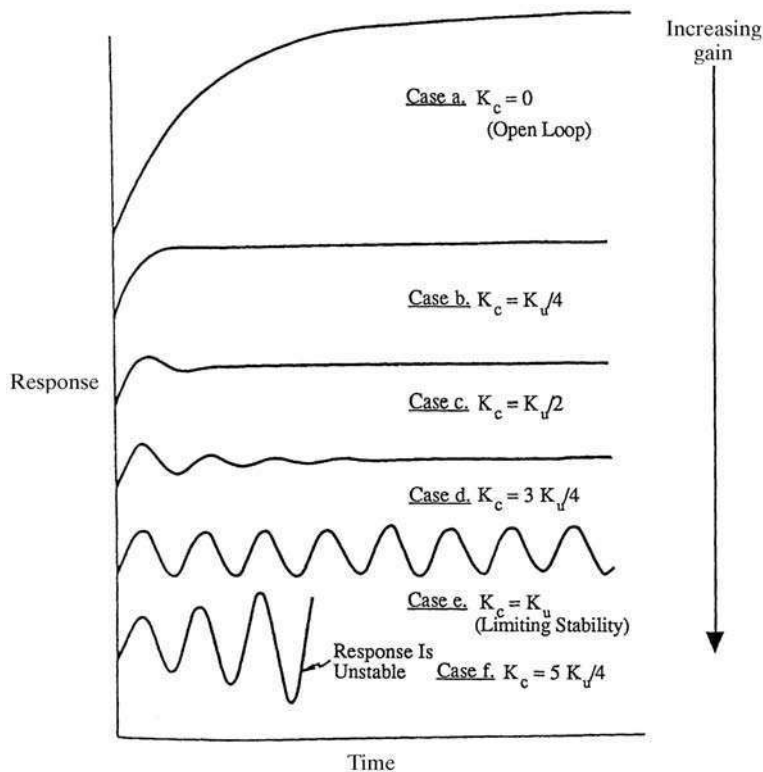


FIG. 8-3 Typical control system responses.

In Eq. (8-1), the flow in of component A is Fc_i , the flow out is Fc_A , and the loss via reaction is $k_r V c_A$, where V = reactor volume and k_r = kinetic rate constant. In this example, c_i is the input, or forcing, variable and c_A is the output variable. If V , F , and k_r are constant, Eq. (8-1) can be rearranged by dividing by $F + k_r V$ so that it contains only two groups of parameters. The result is

$$\tau \frac{dc_A}{dt} = Kc_i - c_A \quad (8-2)$$

where $\tau = V/(F + k_r V)$ and $K = F/(F + k_r V)$. For this example, the resulting model is a first-order differential equation in which τ is called the time constant and K the process gain.

As an alternative to deriving Eq. (8-2) from a dynamic mass balance, one could simply postulate a first-order differential equation to be valid (empirical modeling). Then it would be necessary to estimate values for τ and K so that the postulated model described the reactor's dynamic response. The advantage of the physical model over the empirical model is that the physical model gives insight into how reactor parameters affect the values of τ and K , which in turn affects the dynamic response of the reactor.

Nonlinear versus Linear Models If V , F , and k are constant, then Eq. (8-1) is an example of a linear differential equation model. In a linear equation, the output and input variables and their derivatives appear to only the first power. If the rate of reaction were second-order, then the resulting dynamic mass balance would be

$$V \frac{dc_A}{dt} = Fc_i - Fc_A - k_r V c_A^2 \quad (8-3)$$

Since c_A appears in this equation to the second power, the equation is nonlinear.

The difference between linear systems and nonlinear systems can be seen by considering the steady-state behavior of Eq. (8-1) compared to Eq. (8-3) (the left-hand side is zero; that is, $dc_A/dt = 0$). For a given change in c_i , Δc_i , the change in c_A calculated from Eq. (8-1), Δc_A , is always proportional to Δc_i , and the proportionality constant is K [see Eq. (8-2)]. The change in the output of a system divided by a change in the input to the system is called the *process gain*. Linear systems have constant process gains for all changes in the input. By contrast, Eq. (8-3) gives a Δc_A that varies with Δc_i , which is a function of the concentration levels in the reactor. Thus, depending on the reactor operating conditions, a change in c_i produces different changes in c_A . In this case, the process has a nonlinear gain. Systems with nonlinear gains are more difficult to control than linear systems that have constant gains.

Simulation of Dynamic Models Linear dynamic models are particularly useful for analyzing control system behavior. The insight gained through linear analysis is invaluable. However, accurate dynamic process models can involve large sets of nonlinear equations. Analytical solution of these models is not possible. Thus, in these cases, one must turn to simulation approaches to study process dynamics and the effect of process control. Equation (8-3) will be used to illustrate the simulation of nonlinear processes. If dc_A/dt on the left-hand side of Eq. (8-3) is replaced with its finite difference approximation, one gets

$$c_A(t + \Delta t) = \frac{c_A(t) + \Delta t \cdot [Fc_i(t) - Fc_A(t) - k_r V c_A^2(t)]}{V} \quad (8-4)$$

Starting with an initial value of c_A and given $c_i(t)$, Eq. (8-4) can be solved for $c_A(t + \Delta t)$. Once $c_A(t + \Delta t)$ is known, the solution process can be repeated to calculate $c_A(t + 2\Delta t)$, and so on. This approach is called the Euler integration method; while it is simple, it is not necessarily the best approach to numerically integrating nonlinear differential equations. As discussed in Sec. 3, more sophisticated approaches are available that allow much larger step sizes to be taken but require additional calculations. One widely used approach is the fourth-order Runge Kutta method, which involves the following calculations:

Define

$$f(c_A, t) = \frac{Fc_i(t) - Fc_A - k_r V c_A^2}{V} \quad (8-5)$$

then

$$c_A(t + \Delta t) = c_A(t) + \Delta t(m_1 + 2m_2 + 2m_3 + m_4) \quad (8-6)$$

with

$$m_1 = f[c_A(t), t] \quad (8-7)$$

$$m_2 = f\left[c_A(t) + 2\frac{m_1 \Delta t}{2}, t + \frac{\Delta t}{2}\right] \quad (8-8)$$

$$m_3 = f\left[c_A(t) + 2\frac{m_2 \Delta t}{2}, t + \frac{\Delta t}{2}\right] \quad (8-9)$$

$$m_4 = f[c_A(t) + m_3 \Delta t, t + \Delta t] \quad (8-10)$$

In this method, the m_i 's are calculated sequentially in order to take a step in time. Even though this method requires calculation of the four additional m_i values, for equivalent accuracy the fourth-order Runge Kutta method can result in a faster numerical solution, because it permits a larger step Δt to be taken. Increasingly sophisticated simulation packages are being used to calculate the dynamic behavior of processes and to test control system behavior. These packages have good user interfaces, and they can handle stiff systems where some variables respond on a time scale that is much much faster or slower than that of other variables. A simple Euler approach cannot effectively handle stiff systems, which frequently occur in chemical process models. See Sec. 3 of this handbook for more details.

Laplace Transforms When mathematical models are used to describe process dynamics in conjunction with control system analysis, the models generally involve linear differential equations. Laplace transforms are very effective for solving linear differential equations. The key advantage of using Laplace transforms is that they convert differential equations to algebraic equations. The resulting algebraic equations are easier to solve than the original differential equations. When the Laplace transform is applied to a linear differential equation in time, the result is an algebraic equation in a new variable s , called the Laplace variable. To get the solution to the original differential equation, one needs to invert the Laplace transform. Table 8-1 gives a number of useful Laplace transform pairs, and more extensive tables are available (Seborg, Edgar, and Mellichamp, *Process Dynamics and Control*, Wiley, New York, 2004).

To illustrate how Laplace transforms work, consider the problem of solving Eq. (8-2), subject to the initial condition that $c_A = c_i = 0$ at $t = 0$. If c_A were not initially zero, one would define a deviation variable between c_A and its initial value c_{A0} . Then the transfer function would be developed by using this deviation variable. If c_i changes from zero to $\bar{c}_i$, taking the Laplace transform of both sides of Eq. (8-2) gives

$$\mathcal{L}\left(\tau \frac{dc_A}{dt}\right) = \mathcal{L}(K\bar{c}_i) - \mathcal{L}(c_A) \quad (8-11)$$

TABLE 8-1 Frequently Used Laplace Transforms

Time function $f(t)$	Transform $F(s)$
A	A/s
At	A/s^2
Ae^{-at}	$A/(s + a)$
$A(1 - e^{-at})$	$A/[s(s + a)]$
$A \sin \omega t$	$A\omega/(s^2 + \omega^2)$
$f(t - \theta)$	$e^{-s\theta}F(s)$
df/dt	$sF(s) - f(0)$
$\int f(t)dt$	$F(s)/s$

8-8 PROCESS CONTROL

Denoting $\mathcal{L}(C_A)$ as $C_A(s)$ and using the relationships in Table 8-1 give

$$\tau s C_A(s) = \frac{K \bar{C}_i}{s} - C_A(s) \quad (8-12)$$

Equation (8-12) can be solved for C_A to give

$$C_A(s) = \frac{K \bar{C}_i / s}{\tau s + 1} \quad (8-13)$$

By using the entries in Table 8-1, Eq. (8-13) can be inverted to give the transient response of c_A as

$$c_A(t) = (K \bar{C}_i)(1 - e^{-t/\tau}) \quad (8-14)$$

Equation (8-14) shows that c_A starts from 0 and builds up exponentially to a final concentration of $K \bar{C}_i$. Note that to get Eq. (8-14), it was only necessary to solve the algebraic Eq. (8-12) and then find the inverse of $C_A(s)$ in Table 8-1. The original differential equation was not solved directly. In general, techniques such as partial fraction expansion must be used to solve higher-order differential equations with Laplace transforms.

Transfer Functions and Block Diagrams A very convenient and compact method of representing the process dynamics of linear systems involves the use of transfer functions and block diagrams. A transfer function can be obtained by starting with a physical model, as discussed previously. If the physical model is nonlinear, first it needs to be linearized around an operating point. The resulting linearized model is then approximately valid in a region around this operating point. To illustrate how transfer functions are developed, Eq. (8-2) will again be used. First, one defines deviation variables, which are the process variables minus their steady-state values at the operating point. For Eq. (8-2), there would be deviation variables for both c_A and c_b , and these are defined as

$$\xi = c_A - \bar{c}_A \quad (8-15)$$

$$\xi_i = c_i - \bar{c}_i \quad (8-16)$$

where the overbar stands for steady state. Substitution of Eqs. (8-15) and (8-16) into Eq. (8-2) gives

$$\tau \frac{d\xi}{dt} = K \xi_i - \xi + (K \bar{c}_i - \bar{c}_A) \quad (8-17)$$

The term in parentheses in Eq. (8-17) is zero at steady state, and thus it can be dropped. Next the Laplace transform is taken, and the resulting algebraic equation is solved.

By denoting $X(s)$ as the Laplace transform of ξ and $X_i(s)$ as the transform of ξ_i , the final transfer function can be written as

$$\frac{X}{X_i} = \frac{K}{\tau s + 1} \quad (8-18)$$

Equation (8-18) is an example of a first-order transfer function. As mentioned above, an alternative to formally deriving Eq. (8-18) involves simply postulating its form and then identifying its two parameters, the process gain K and time constant τ , to fit the process under study. In fitting the parameters, data can be generated by forcing the process. If step forcing is used, then the resulting response is called the process reaction curve. Often transfer functions are placed in block diagrams, as shown in Fig. 8-4. Block diagrams show how changes in an input variable affect an output variable. Block diagrams are a means of concisely representing the dynamics of a process under study. Since linearity is assumed in developing a block diagram, if more than one variable affects an output, the contributions from each can be added.

Continuous versus Discrete Models The preceding discussion has focused on systems where variables change continuously with

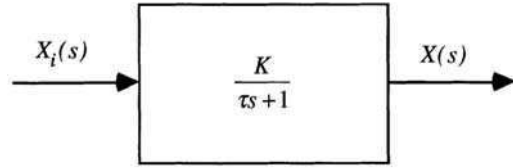


FIG. 8-4 First-order transfer function.

time. Most real processes have variables that are continuous, such as temperature, pressure, and flow. However, some processes involve discrete events, such as the starting or stopping of a pump. In addition, modern plants are controlled by digital computers, which are discrete. In controlling a process, a digital system samples variables at a fixed rate, and the resulting system is a sampled data system. From one sampling instant until the next, variables are assumed to remain fixed at their sampled values. Similarly, in controlling a process, a digital computer sends out signals to control elements, usually valves, at discrete instants of time. These signals remain fixed until the next sampling instant.

Figure 8-5 illustrates the concept of sampling a continuous function. At integer values of the sampling rate Δt , the value of the variable to be sampled is measured and held until the next sampling instant. To deal with sampled data systems, the z transform has been developed. The z transform of the function given in Fig. 8-5 is defined as

$$Z(f) = \sum_{n=0}^{\infty} f(n \Delta t) z^{-n} \quad (8-19)$$

In an analogous manner to Laplace transforms, one can develop transfer functions in the z domain as well as block diagrams. Tables of z transform pairs have been published (Seborg, Edgar, and Mellichamp, *Process Dynamics and Control*, Wiley, New York, 2004) so that the discrete transfer functions can be inverted back to the time domain. The inverse gives the value of the function at the discrete sampling instants. Sampling a continuous variable results in a loss of information. However, in practical applications, sampling is fast enough that the loss is typically insignificant and the difference between continuous and discrete modeling is small in terms of its effect on control. Increasingly, model predictive controllers that make use of discrete dynamic models are being used in the process industries. The purpose of these controllers is to guide a process to optimum operating points. These model predictive control algorithms are typically run at much slower sampling rates than are used for basic control loops such as flow control or pressure control. The discrete dynamic models used are normally developed from data generated

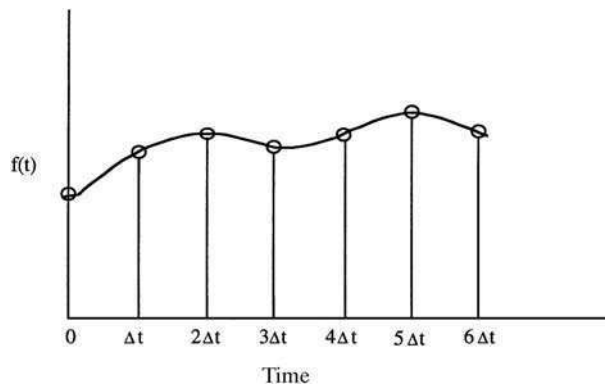


FIG. 8-5 Sampled data example.

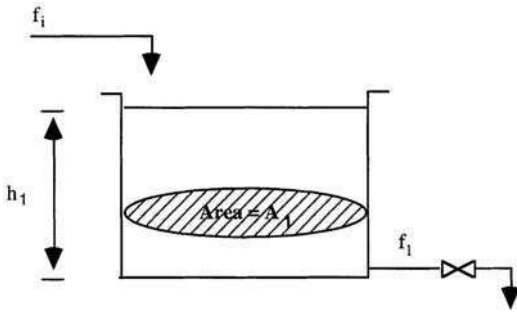


FIG. 8-6 Single tank with exit valve.

from plant testing, as discussed hereafter. For a detailed discussion of modeling sampled data systems, the interested reader is referred to textbooks on digital control (Astrom and Wittenmark, *Computer Controlled Systems*, Prentice-Hall, Englewood Cliffs, N.J., 1997).

Process Characteristics in Transfer Functions In many cases, process characteristics are expressed in the form of transfer functions. In the previous discussion, a reactor example was used to illustrate how a transfer function could be derived. Here, another system involving flow out of a tank, shown in Fig. 8-6, is considered.

Proportional Element First, consider the outflow through the exit valve on the tank. If the flow through the line is turbulent, then Bernoulli's equation can be used to relate the flow rate through the valve to the pressure drop across the valve as

$$f_1 = k_f A_v \sqrt{2g_c(h_1 - h_0)} \quad (8-20)$$

where f_1 = flow rate, k_f = flow coefficient, A_v = cross-sectional area of the restriction, g_c = constant, h_1 = liquid head in tank (pressure at the base of the tank), and h_0 = atmospheric pressure. This relationship between flow and pressure drop across the valve is nonlinear, and it can be linearized around a particular operating point to give

$$f_1 - \bar{f}_1 = \left(\frac{1}{R_1} \right) (h_1 - \bar{h}_1) \quad (8-21)$$

where $R_1 = \bar{f}_1 / (g_c k_f^2 A_v^2)$ is called the resistance of the valve in analogy with an electrical resistance. The transfer function relating changes in flow to changes in head is shown in Fig. 8-7, and it is an example of a pure gain system with no dynamics. In this case, the process gain is $K = 1/R_1$. Such a system has an instantaneous dynamic response, and for a step change in head, there is an immediate step change in flow, as shown in Fig. 8-8. The exact magnitude of the step in flow depends on the operating flow $\bar{f}_1$ as the definition of R_1 shows.

First-Order Lag (Time Constant Element) Next consider the system to be the tank itself. A dynamic mass balance on the tank gives

$$A_1 \frac{dh_1}{dt} = f_i - f_1 \quad (8-22)$$

where A_1 is the cross-sectional area of the tank and f_i is the inlet flow. By substituting Eq. (8-21) into Eq. (8-22) and following the approach

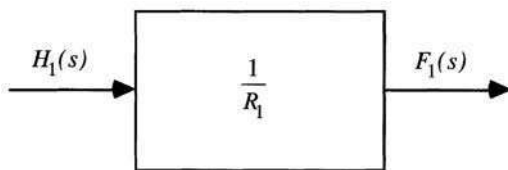


FIG. 8-7 Proportional element transfer function.

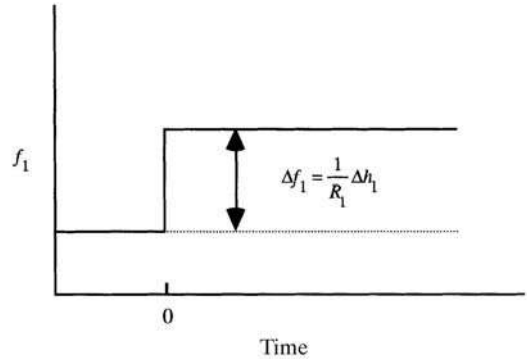


FIG. 8-8 Response of proportional element.

discussed above for deriving transfer functions, one can develop the transfer function relating changes in h_1 to changes in f_1 . The resulting transfer function is another example of a first-order system, shown in Fig. 8-4, and it has a gain $K = R_1$ and a time constant $\tau_1 = R_1 A_1$. For a step change in f_i , h_1 follows a decaying exponential response from its initial value $\bar{h}_1$ to a final value of $\bar{h}_1 + R_1 \Delta f_i$ (Fig. 8-9). At a time equal to τ_1 , the transient in h_1 is 63 percent finished; and at $3\tau_1$, the response is 95 percent finished. These percentages are the same for all first-order processes. Thus, knowledge of the time constant of a first-order process gives insight into how fast the process responds to sudden input changes.

Capacity Element Now consider the case where the valve in Fig. 8-7 is replaced with a pump. In this case, it is reasonable to assume that the exit flow from the tank is independent of the level in the tank. For such a case, Eq. (8-22) still holds, except that f_1 no longer depends on h_1 . For changes in f_i , the transfer function relating changes in h_1 to changes in f_i is shown in Fig. 8-10. This is an example of a pure capacity process, also called an integrating system. The cross-sectional area of the tank is the chemical process equivalent of an electrical capacitor. If the inlet flow is step forced while the outlet is held constant, then the level builds up linearly, as shown in Fig. 8-11. Eventually the liquid will overflow the tank.

Second-Order Element Because of their linear nature, transfer functions can be combined in a straightforward manner. Consider the two-tank system shown in Fig. 8-12. For tank 1, the transfer function relating changes in f_1 to changes in f_i is

$$\frac{F_1(s)}{F_i(s)} = \frac{1}{A_1 R_1 + 1} \quad (8-23)$$

Since f_1 is the inlet flow to tank 2, the transfer function relating changes in h_2 to changes in f_1 has the same form as that given in Fig. 8-4:

$$\frac{H_2(s)}{F_1(s)} = \frac{R_2}{A_2 R_2 s + 1} \quad (8-24)$$

Equations (8-23) and (8-24) can be multiplied to give the final transfer function relating changes in h_2 to changes in f_i , as shown in Fig. 8-13. This is an example of a second-order transfer function. This transfer function has a gain R_2 and two time constants $A_1 R_1$ and $A_2 R_2$. For two tanks with equal areas, a step change in f_i produces the S-shaped response in level in the second tank shown in Fig. 8-14.

General Second-Order Element Figure 8-3 illustrates the fact that closed-loop systems can exhibit oscillatory behavior. A general second-order transfer function that can exhibit oscillatory behavior is important for the study of automatic control systems. Such a transfer function is given in Fig. 8-15. For a unit step input, the transient responses shown in Fig. 8-16 result. As can be seen, when $\zeta < 1$, the response oscillates; and when $\zeta < 1$, the response is S-shaped. Few open-loop chemical processes exhibit an oscillating response; most exhibit an S-shaped step response.

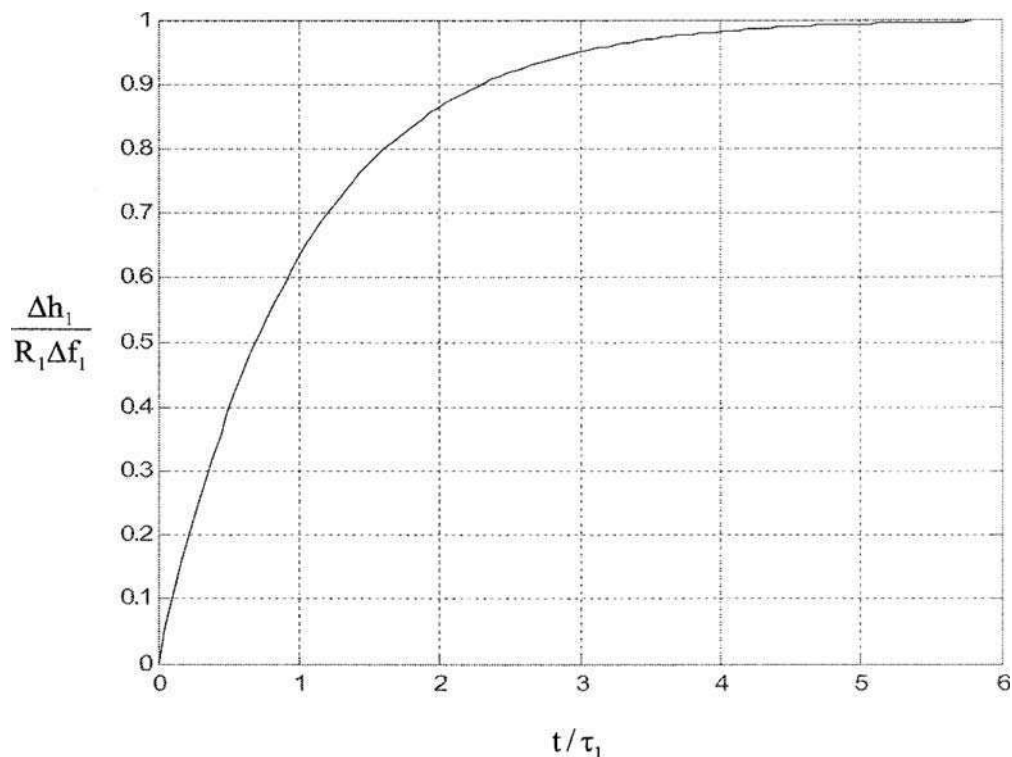


FIG. 8-9 Response of first-order system.

Distance-Velocity Lag (Dead-Time Element) The dead-time or time-delay element, commonly called a distance-velocity lag, is often encountered in process systems. For example, if a temperature-measuring element is located downstream from a heat exchanger, a time delay occurs before the heated fluid leaving the exchanger

arrives at the temperature measurement point. If some element of a system produces a dead time of θ time units, then an input to that unit $f(t)$ will be reproduced at the output as $f(t - \theta)$. The transfer function for a pure dead-time element is shown in Fig. 8-17, and the transient response of the element is shown in Fig. 8-18.

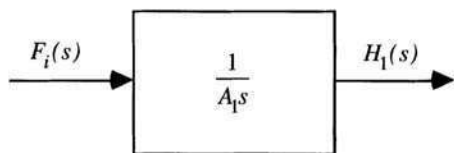


FIG. 8-10 Pure capacity or integrating transfer function.

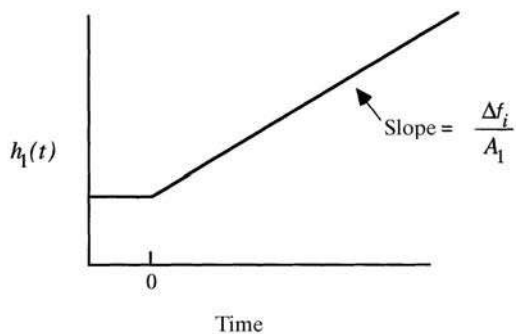


FIG. 8-11 Response of pure capacity system.

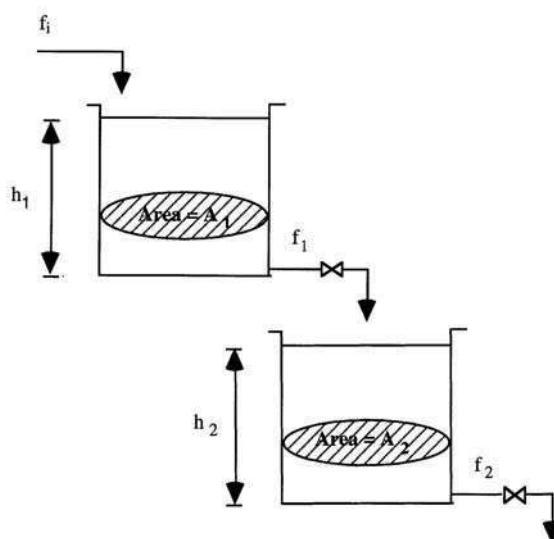


FIG. 8-12 Two tanks in series.

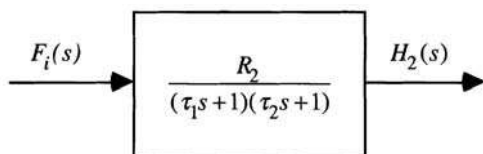


FIG. 8-13 Second-order transfer function.

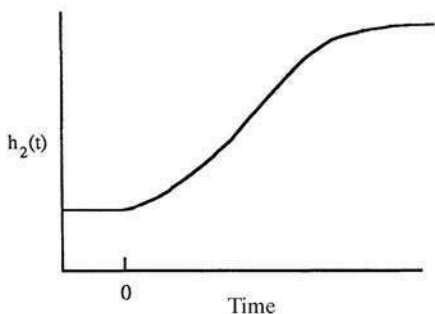


FIG. 8-14 Response of second-order system.

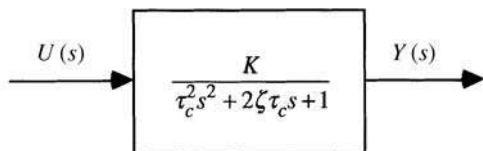


FIG. 8-15 General second-order transfer function.

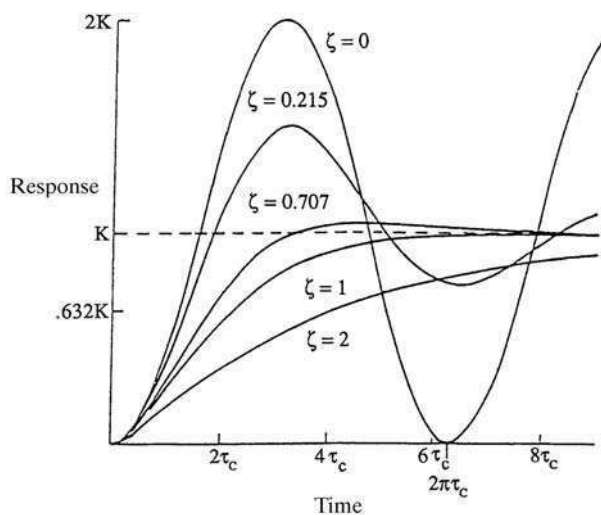


FIG. 8-16 Response of general second-order system.

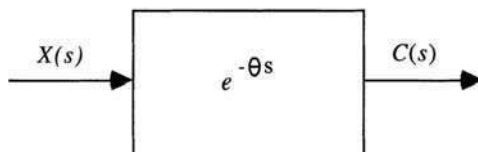


FIG. 8-17 Dead-time transfer function.

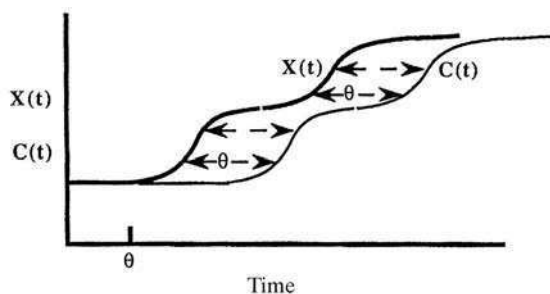
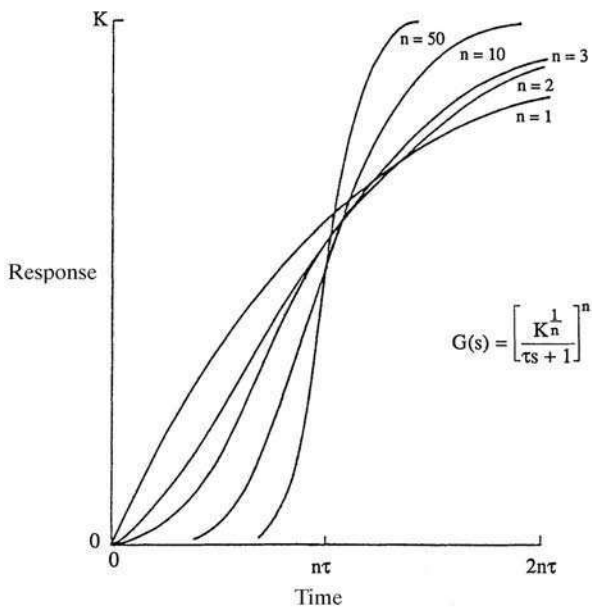


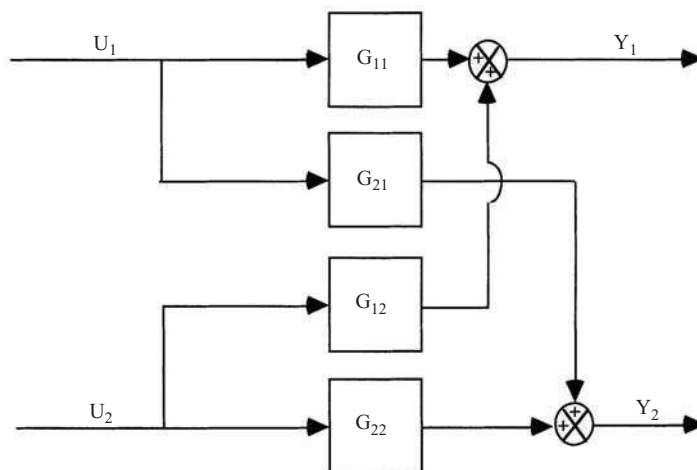
FIG. 8-18 Response of dead-time system.

Higher-Order Lags If a process is described by a series of n first-order lags, the overall system response becomes proportionally slower with each lag added. The special case of a series of n first-order lags with equal time constants has a transfer function given by

$$G(s) = \frac{K}{(\tau s + 1)^n} \quad (8-25)$$

The step response of this transfer function is shown in Fig. 8-19. Note that all curves reach about 60 percent of their final value at $t = n\tau$.


 FIG. 8-19 Response of n th-order lags.

FIG. 8-20 Example of 2×2 transfer function.

Higher-order systems can be approximated by a first- or second-order plus dead-time system for control system design.

Multi-input, Multioutput Systems The dynamic systems considered up to this point have been examples of single-input, single-output (SISO) systems. In chemical processes, one often encounters systems where one input can affect more than one output. For example, assume that one is studying a distillation tower in which both reflux and boil-up are manipulated for control purposes. If the output variables are the top and bottom product compositions, then each input affects both outputs. For this distillation example, the process is referred to as a 2×2 system to indicate the number of inputs and outputs. In general, multi-input, multioutput (MIMO) systems can have n inputs and m outputs with $n \neq m$, and they can be nonlinear. Such a system would be called an $n \times m$ system. An example of a transfer function for a 2×2 linear system is given in Fig. 8-20. Note that since linear systems are involved, the effects of the two inputs on each output are additive. In many process control systems, one input is selected to control one output in a MIMO system. For m outputs there would be m such selections. For this type of control strategy, one needs to consider which inputs and outputs to couple, and this problem is referred to as loop pairing. Another important issue that arises involves interaction between control loops. When one loop makes a change in its manipulated variable, the change affects the other loops in the system. These changes are the direct result of the multivariable nature of the process. In some cases, the interaction can be so severe that overall control system performance is drastically reduced. Finally, some of the modern approaches to process control tackle the MIMO problem directly, and they simultaneously use all manipulated variables to control all output variables rather than pair one input to one output (see later section on multivariable control).

Fitting Dynamic Models to Experimental Data In developing empirical transfer functions, it is necessary to identify model parameters from experimental data. There are a number of approaches to process identification that have been published. The simplest approach involves introducing a step test into the process and recording the response of the process, as illustrated in Fig. 8-21. The x 's in the figure represent the recorded data. For purposes of illustration, the process under study will be assumed to be first-order with dead time and have the transfer function

$$G(s) = \frac{Y(s)}{U(s)} = K \exp(-\theta s) / \tau s + 1 \quad (8-26)$$

The response $y(t)$, produced by Eq. (8-26) can be found by inverting the transfer function, and it is also shown in Fig. 8-21 for a set of

model parameters K , τ , and θ fitted to the data. These parameters are calculated by using optimization to minimize the squared difference between the model predictions and the data, i.e., a least squares approach. Let each measured data point be represented by y_j (measured response), t_j (time of measured response), $j = 1$ to n . Then the least squares problem can be formulated as

$$\min_{\tau, \theta, K} \sum_{j=1}^n [y_j - \hat{y}(t_j)]^2 \quad (8-27)$$

where $\hat{y}(t_j)$ is the predicted value of y at time t_j and n is the number of data points. This optimization problem can be solved to calculate the optimal values of K , τ , and θ . A number of software packages such as Excel Solver are available for minimizing Eq. (8-27).

One operational problem caused by step forcing is the fact that the process under study is moved away from its steady-state operating point. Plant managers may be reluctant to allow large steady-state changes, since normal production will be disturbed by the changes. As a result, alternative methods of forcing actual processes have been developed, and these included pulse testing and pseudo-random binary signal (PRBS) forcing, both of which are illustrated in Fig. 8-22. With pulse forcing, one introduces a step, and then after a period of time the input is returned to its original value. The result is that the process dynamics are excited, but after the forcing the process returns to its original steady state. PRBS forcing involves a series of pulses of fixed height and random duration, as shown in Fig. 8-22. The advantage of PRBS is that forcing can be concentrated on particular frequency ranges that are important for control system design.

Transfer function models are linear, but chemical processes are known to exhibit nonlinear behavior. One could use the same type of optimization objective as given in Eq. (8-27) to determine parameters in nonlinear first-principles models, such as Eq. (8-3) presented earlier. Also, nonlinear empirical models, such as neural network models, have recently been proposed for process applications. The key to the use of these nonlinear empirical models is to have high-quality process data, which allows the important nonlinearities to be identified.

FEEDBACK CONTROL SYSTEM CHARACTERISTICS

GENERAL REFERENCES: Shinskey, *Process Control Systems*, 4th ed., McGraw-Hill, New York, 1996; Seborg, Edgar, and Mellichamp, *Process Dynamics and Control*, Wiley, New York, 1989.

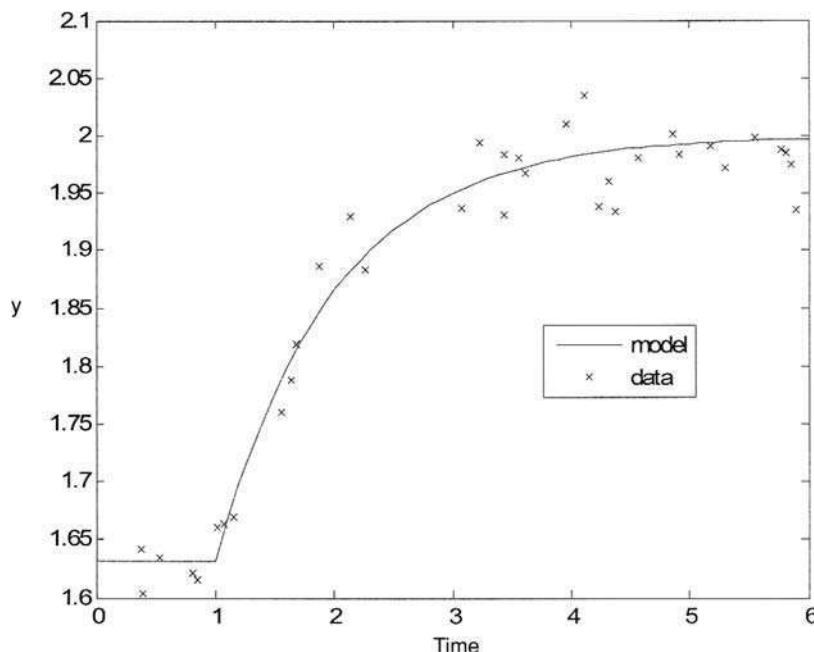


FIG. 8-21 Plot of experimental data and first-order model fit.

There are two objectives in applying feedback control: (1) regulate the controlled variable at the set point following changes in load and (2) respond to set-point changes, the latter called servo operation. In fluid processes, almost all control loops must contend with variations in load, and therefore regulation is of primary importance. While most loops will operate continuously at fixed set points, frequent changes in set points can occur in flow loops and in batch production. The most common mechanism for achieving both objectives is feedback control, because it is the simplest and most universally applicable approach to the problem.

Closing the Loop The simplest representation of the closed feedback loop is shown in Fig. 8-23. The load is shown entering the process at the same point as the manipulated variable because that is the most common point of entry, and because, lacking better information, the elements in the path of the manipulated variable are the best estimates of those in the load path. The load rarely impacts directly on the controlled variable without passing through the dominant lag in the process. Where the load is unmeasured, its current value can be observed as the controller output required to keep the controlled variable Y at set point Y_{sp} .

If the loop is opened—either by placing the controller in manual operation or by setting its gains to zero—the load will have complete influence over the controlled variable, and the set point will have none. Only by closing the loop with controller gain as high as possible will the influence of the load be minimized and that of the set point be maximized. There is a practical limit to the controller gain, however, at the point where the controlled variable develops a uniform oscillation. This is defined as the limit of stability, and it is reached when the product of gains in the loop $\Pi G = G_c G_p G_v$ is equal to 1.0 at the period of the oscillation. If the gain of any element in the loop increases from this condition, oscillations will expand, creating a dangerous situation where safe limits of operation could be exceeded in a few cycles. Consequently, control loops should be left in a condition where the loop gain is less than 1.0 by a safe margin that allows for possible variations in process parameters. Figure 8-24 describes a load response under PID (proportional-integral-derivative) control where the loop is well damped at a loop gain of 0.56; loop gain is then increased to 0.93 and to 1.05, creating a lightly damped and then an expanding cycle, respectively.

In controller tuning, a choice must be made between performance and robustness. Performance is a measure of how well a given con-

troller with certain parameter settings regulates a variable, relative to the best response that can be achieved for that particular process. Robustness is a measure of how small a change in a process parameter is required to bring the loop from its current state to the limit of stability ($\Pi G = 1.0$). The well-damped loop in Fig. 8-24 has a robustness of 79 percent, in that increasing the gain of any element in the loop by a factor of 1/0.56, or 1.79, would bring the loop to the limit of stability. Increasing controller performance by raising its gain can therefore be expected to decrease robustness. Both performance and robustness are functions of the dynamics of the process being controlled, the selection of the controller, and the tuning of the controller parameters.

On/Off Control An on/off controller is used for manipulated variables having only two states. They commonly control temperatures in homes, electric water heaters and refrigerators, and pressure and liquid level in pumped storage systems. On/off control is satisfactory where slow cycling is acceptable, because it always leads to cycling when the load lies between the two states of the manipulated variable. The cycle will be positioned symmetrically about the set point only if the load happens to be equidistant between the two states of the manipulated variable. The period of the symmetric cycle will be approximately 4θ , where θ is the dead time in the loop. If the load is not centered between the states of the manipulated variable, the period will tend to increase and the cycle will follow a sawtooth pattern.

Every on/off controller has some degree of dead band, also known as lockup, or differential gap. Its function is to prevent erratic switching between states, thereby extending the life of contacts and motors. Instead of changing states precisely when the controlled variable crosses the set point, the controller will change states at two different points for increasing and decreasing signals. The difference between these two switching points is the dead band (see Fig. 8-25); it increases the amplitude and period of the cycle, similar to the effects of dead time.

A three-state controller is used to drive either a pair of independent two-state actuators, such as heating and cooling valves, or a bidirectional motorized actuator. The controller is comprised of two on/off controllers, each with dead band, separated by a dead zone. While the controlled variable lies within the dead zone, neither output is energized. This controller can drive a motorized valve to the point where the manipulated variable matches the load, thereby avoiding cycling.

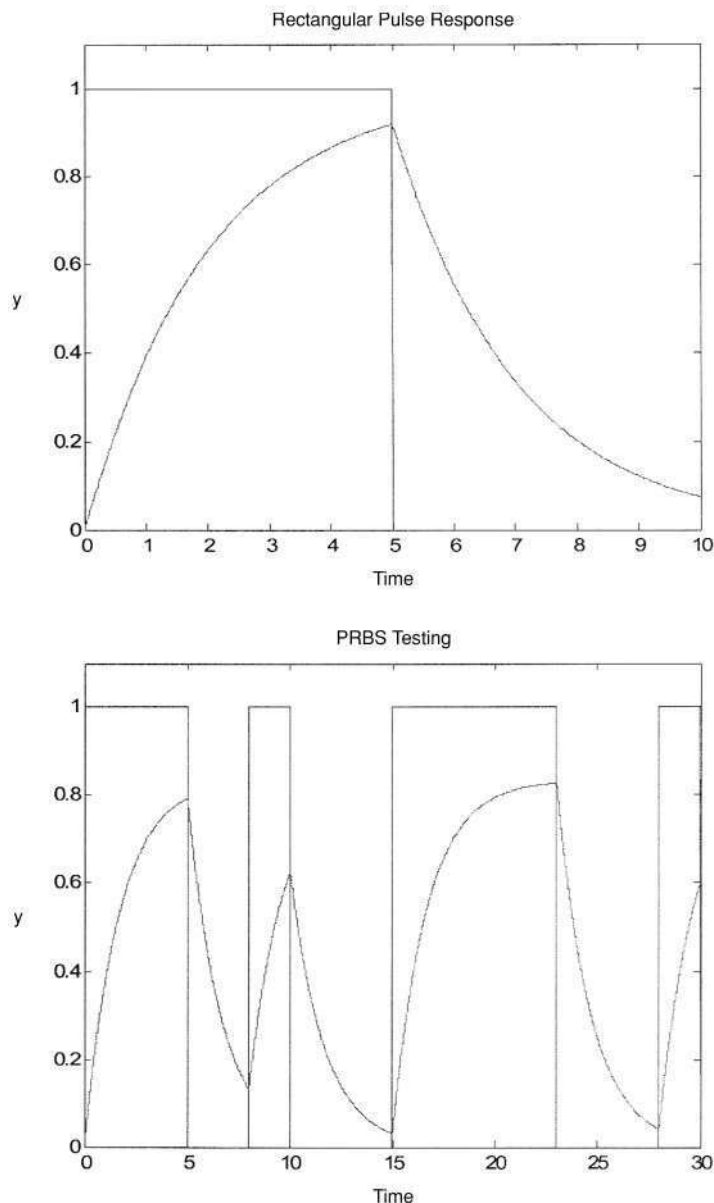


FIG. 8-22 Rectangular pulse response and PRBS testing.

Proportional Control A proportional controller moves its output proportional to the deviation e between the controlled variable y and its set point y_{sp} :

$$u = K_c e + b = \frac{100}{P} e + b \quad (8-28)$$

where $e = \pm(y - y_{sp})$, the sign selected to produce negative feedback. In some controllers, proportional gain K_c is introduced as a pure number; in others, it is set as $100/P$, where P is the proportional band in percent. The output bias b of the controller is also known as manual reset. The proportional controller is not a good regulator, because any change in output required to respond to a change in load results in a corresponding change in the controlled variable. To minimize the resulting offset, the bias

should be set at the best estimate of the load, and the proportional band set as low as possible. Processes requiring a proportional band of more than a few percent may control with unacceptably large values of offset.

Proportional control is most often used to regulate liquid level, where variations in the controlled variable carry no economic penalty and where other control modes can easily destabilize the loop. It is actually recommended for controlling the level in a surge tank when manipulating the flow of feed to a critical downstream process. By setting the proportional band just under 100 percent, the level is allowed to vary over the full range of the tank capacity as inflow fluctuates, thereby minimizing the resulting rate of change of manipulated outflow. This technique is called averaging level control.

Proportional-plus-Integral (PI) Control Integral action eliminates the offset described above by moving the controller output at a

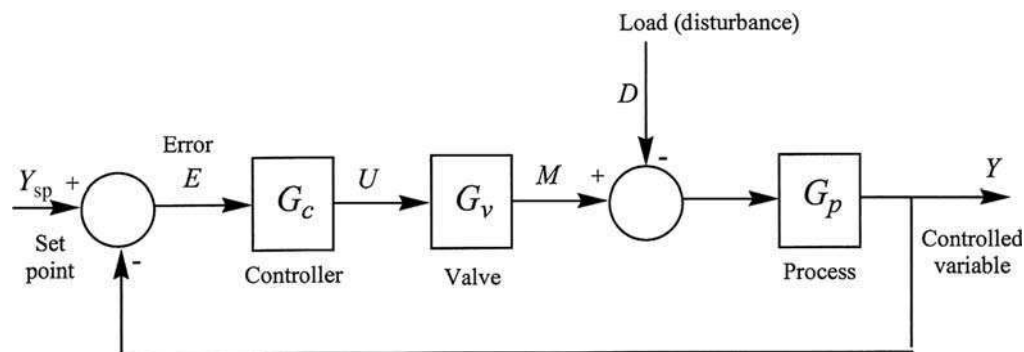


FIG. 8-23 Both load regulation and set-point response require high gains for the feedback controller.

rate proportional to the deviation from set point—the output will then not stop moving until the deviation is zero. Although available alone in an integral controller, it is most often combined with proportional action in a PI controller:

$$u = \frac{100}{P} \left(e + \frac{1}{\tau_i} \int e dt \right) + C_0 \quad (8-29)$$

where τ_i is the integral time constant in minutes; in some controllers it is introduced as integral gain or reset rate $1/\tau_i$ in repeats per minute. The last term in the equation is the constant of integration, the value of the controller output when integration begins. The PI controller is by far the most commonly used controller in the process industries.

Because the integral term lags the proportional term by 90° in phase, the PI controller then always produces a phase lag between 0° and 90° :

$$\phi_{PI} = -\tan^{-1} \frac{\tau_o}{2\pi\tau_i} \quad (8-30)$$

where τ_o is the period of oscillation of the loop. The phase angle should be kept between 15° for lag-dominant processes and 45° for dead-time-dominant processes for optimum results.

Proportional-plus-Integral-plus-Derivative (PID) Control The derivative mode moves the controller output as a function of the rate of change of the controlled variable, which adds phase lead to the controller, increasing its speed of response. It is normally combined with proportional and integral modes. The noninteracting or ideal form of the PID controller appears functionally as

$$u = \frac{100}{P} \left(e + \frac{1}{\tau_i} \int e dt \pm \tau_D \frac{dy}{dt} \right) + C_0 \quad (8-31)$$

where τ_D is the derivative time constant. Note that derivative action is applied to the controlled variable rather than to the deviation, as it should not be applied to the set point; the selection of the sign for the derivative term must be consistent with the action of the controller.

In some PID controllers, the integral and derivative terms are combined serially rather than in parallel, as done in the last equation. This results in interaction between these modes, such that the effective values of the controller parameters differ from their set values as follows:

$$\begin{aligned} \tau_{i,\text{eff}} &= \tau_i + \tau_D \\ \tau_{D,\text{eff}} &= \frac{1}{1/\tau_D + 1/\tau_i} \\ K_c &= \frac{100}{P} \left(1 + \frac{\tau_D}{\tau_i} \right) \end{aligned} \quad (8-32)$$

The performance of the interacting controller is almost as high as that of the noninteracting controller on most processes, but the tuning rules differ because of the above relationships. Both controllers are in common use in digital systems.

There is always a gain limit placed upon the derivative vector—a value of 10 is typical. However, interaction decreases the derivative

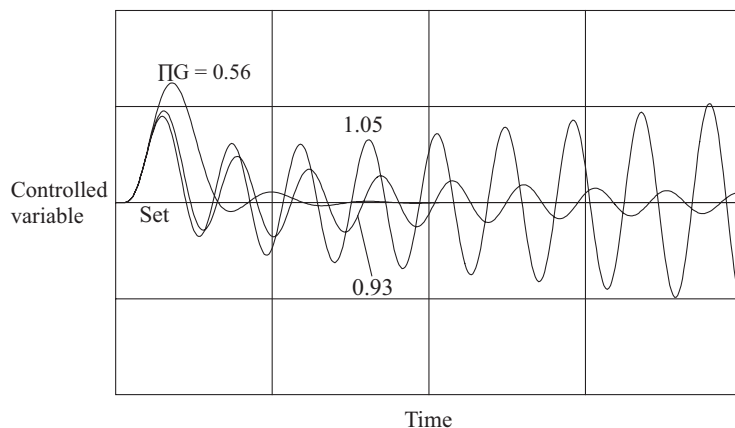


FIG. 8-24 Transition from well-damped load response to instability develops as loop gain increases.

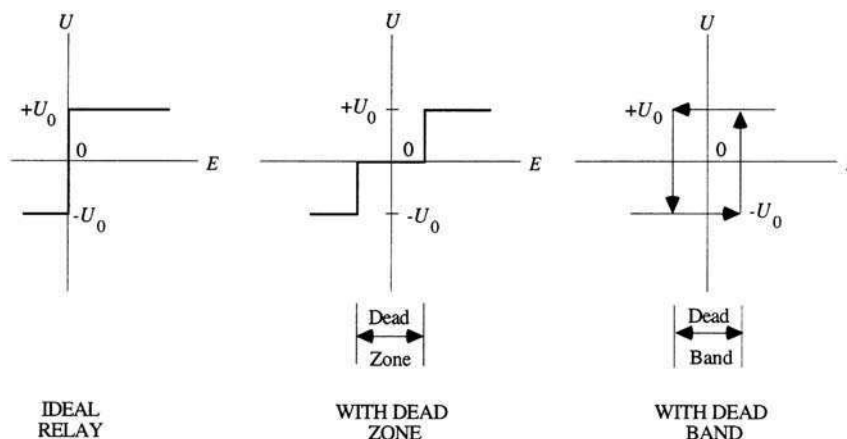


FIG. 8-25 On/off controller characteristics.

gain below this value by the factor $1 + \tau_D/\tau_I$, which is the reason for the decreased performance of the interacting PID controller. Sampling in a digital controller has a similar effect, limiting derivative gain to the ratio of derivative time to the sample interval of the controller. Noise on the controlled variable is amplified by the derivative gain, preventing its use in controlling flow and liquid level. Derivative action is recommended for control of temperature and composition in multiple-capacity processes with little measurement noise.

Controller Comparison Figure 8-26 compares the step load response of a distributed lag without control, and with P, PI, and interacting PID control. A distributed lag is a process whose resistance and capacity are distributed throughout its length—a heat exchanger is characteristic of this class, its heat-transfer surface and heat capacity being uniformly distributed. Other examples include imperfectly stirred tanks and distillation columns—both trayed and packed. The signature of a distributed lag is its open-loop (uncontrolled) step response, featuring a relatively short dead time followed by a dominant lag called $\Sigma\tau$, which is the time required to reach 63.2 percent complete response.

The proportional controller is unable to return the controlled variable to the set point following the step load change, as a deviation is required to sustain its output at a value different from its fixed bias b . The amount of proportional offset produced as a fraction of the uncontrolled offset is $1/(1 + KK_c)$, where K is the steady-state process

gain—in Fig. 8-26, that fraction is 0.13. Increasing K_c can reduce the offset, but with an accompanying loss in damping.

The PI and PID controller were tuned to produce a minimum *integrated absolute error* (IAE). Their response curves are similar in appearance to a gaussian distribution curve, but with a damped cycle in the trailing edge. The peak deviation of the PID response curve is only 0.12 times the uncontrolled offset, occurring at $0.36\Sigma\tau$; the peak deviation of the PI response curve is 0.21 times the uncontrolled offset, occurring at $0.48\Sigma\tau$. These values can be used to predict the load response of any distributed lag whose parameters K and $\Sigma\tau$ are known or can be estimated as described below.

CONTROLLER TUNING

The performance of a controller depends as much on its tuning as on its design. Tuning must be applied by the end user to fit the controller to the controlled process. There are many different approaches to controller tuning, based on the particular performance criteria selected, whether load or set-point changes are more important, whether the process is lag- or dead-time-dominant, and the availability of information about the process dynamics. The earliest definitive work in this field was done at the Taylor Instrument Company by Ziegler and Nichols (*Trans. ASME*, p. 759, 1942), tuning PI and interacting PID

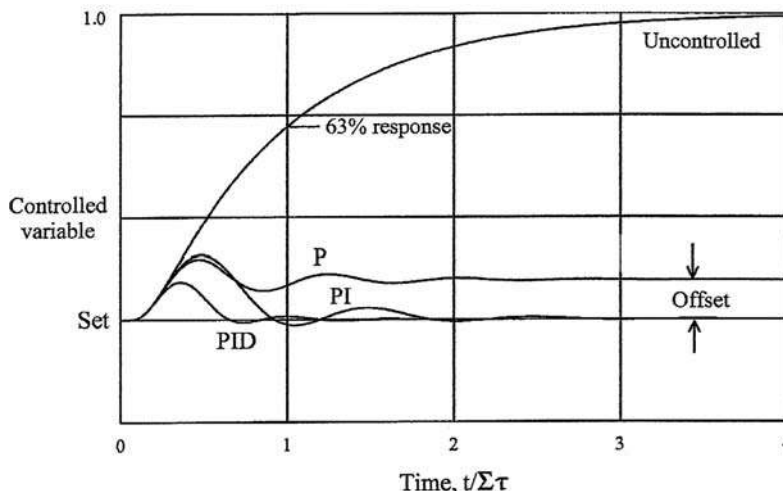


FIG. 8-26 Minimum-IAE tuning gives very satisfactory load response for a distributed lag.

controllers for optimum response to step load changes applied to lag-dominant processes. While these tuning rules are still in use, they do not apply to set-point changes, dead-time-dominant processes, or noninteracting PID controllers (Seborg, Edgar, and Mellichamp, *Process Dynamics and Control*, Wiley, New York, 2004).

Controller Performance Criteria The most useful measures of controller performance in an industrial setting are the maximum deviation in the controlled variable resulting from a disturbance, and its integral. The disturbance could be to the set point or to the load, depending on the variable being controlled and its context in the process. The size of the deviation and its integral are proportional to the size of the disturbance (if the loop is linear at the operating point). While actual disturbances arising in a plant may appear to be random, the controller needs a reproducible test to determine how well it is tuned. The disturbance of choice for test purposes is the step, because it can be applied manually, and by containing all frequencies including zero it exercises all modes of the controller. (The step actually has the same frequency distribution as integrated white noise, a “random walk.”) When tuned optimally for step disturbances, the controller will be optimally tuned for most other disturbances as well.

A step change in set point, however, may be a poor indicator of a loop’s load response. For example, a liquid-level controller does not have to integrate to follow a set-point change, as its steady-state output is independent of the set point. Stepping a flow controller’s set point is

an effective test of its tuning, however, as its steady-state output is proportional to its set point. Other loops should be load-tested: simulate a load change from a steady state at zero deviation by transferring the controller to manual and stepping its output, and then immediately transferring back to automatic before a deviation develops.

Figure 8-27*a* and *b* shows variations in the response of a distributed lag to a step change in load for different combinations of proportional and integral settings of a PI controller. The maximum deviation is the most important criterion for variables that could exceed safe operating levels, such as steam pressure, drum level, and steam temperature in a boiler. The same rule can apply to product quality if violating specifications causes it to be rejected. However, if the product can be accumulated in a downstream storage tank, its average quality is more important, and this is a function of the deviation integrated over the residence time of the tank. Deviation in the other direction, where the product is better than specification, is safe but increases production costs in proportion to the integrated deviation because quality is given away.

For a PI or PID controller, the integrated deviation—better known as integrated error IE—is related to the controller settings

$$IE = \Delta u \frac{P\tau_i}{100} \quad (8-33)$$

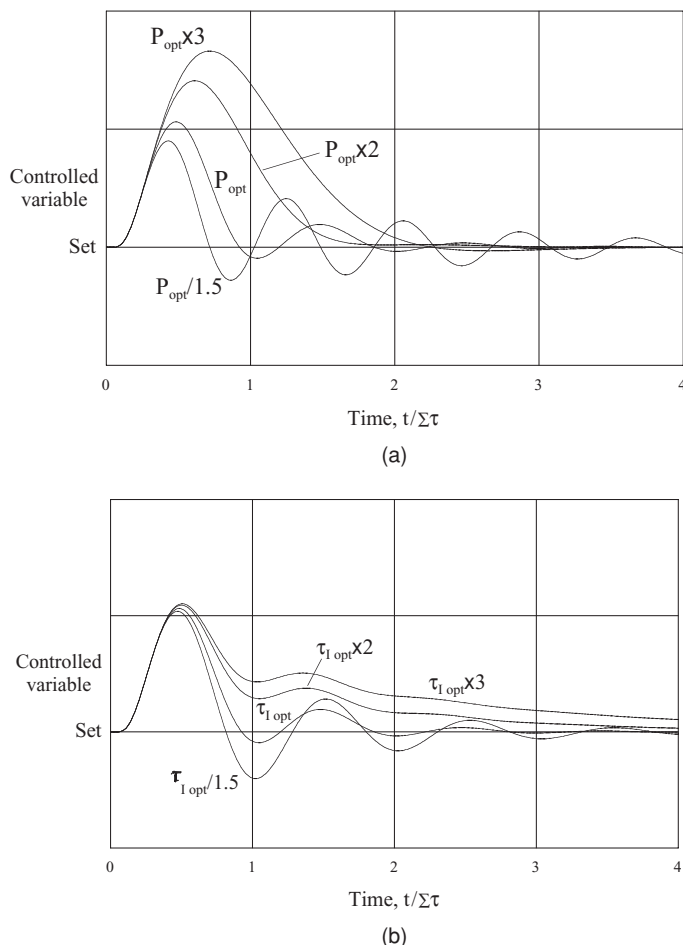


FIG. 8-27 The optimum settings produce minimum-IAE load response. (a) The proportional band primarily affects damping and peak deviation. (b) Integral time determines overshoot.

where Δu is the difference in controller outputs between two steady states, as required by a change in load or set point. The proportional band P and integral time τ_i are the indicated settings of the controller for PI and both interacting and noninteracting PID controllers. Although the derivative term does not appear in the relationship, its use typically allows a 50 percent reduction in integral time and therefore in IE. The integral time in the IE expression should be augmented by the sample interval if the controller is digital, the time constant of any filter used, and the value of any dead-time compensator.

It would appear, from the above, that minimizing IE is simply a matter of minimizing the P and τ_i settings of the controller. However, settings will be reached that produce excessive oscillations, such as shown in the lowest two response curves in Fig. 8-27a and b. It is preferable instead to find a combination of controller settings that minimizes integrated absolute error IAE, which for both load and set-point changes is a well-damped response with minimal overshoot. The curves designated P_{opt} and $\tau_{i,opt}$ in Fig. 8-27 are the same minimum-IAE response to a step change in load for a distributed-lag process under PI control. Because of the very small overshoot, the IAE will be only slightly larger than the IE. Loops that are tuned to minimize IAE tend to give responses that are close to minimum IE and with minimum peak deviation. The other curves in Fig. 8-27a and b describe the effects of individual adjustments to P and τ_i , respectively, around those optimum values and can serve as a guide to fine-tuning a PI controller.

The performance of a controller (and its tuning) must be based on what is achievable for a given process. The concept of best practical IE (IE_b) for a step change in load ΔL to a process consisting of dead time and one or two lags can be estimated (Shinskey, *Process Control Systems*, 4th ed., McGraw-Hill, New York, 1996)

$$IE_b = \Delta L K_L \tau_L (1 - e^{-\theta/\tau_L}) \quad (8-34)$$

where K_L is the gain and τ_L the primary time constant in the load path, and θ the dead time in the manipulated path to the controlled variable. If the load or its gain is unknown, Δu and $\bar{K}$ ($= K_L K_p$) may be substituted. If the process is non-self-regulating (i.e., an integrator), the relationship is

$$IE_b = \frac{\Delta L \theta^2}{\tau_i} \quad (8-35)$$

where τ_i is the time constant of the process integrator. The peak deviation of the best practical response curve is

$$e_b = \frac{IE_b}{\theta + \tau_2} \quad (8-36)$$

where τ_2 is the time constant of a common secondary lag (e.g., in the measuring device).

TABLE 8-2 Tuning Rules Using Known Process Parameters

Process	Controller	P	τ_i	τ_D
Dead-time-dominant	PI	250K	0.5 θ	
Lag-dominant	PI	106K θ/τ_m	4.0 θ	
	PID _n	77K θ/τ_m	1.8 θ	0.45 θ
	PID _i	106K θ/τ_m	1.5 θ	0.55 θ
Non-self-regulating	PI	106 θ/τ_i	4.0 θ	
	PID _n	78 θ/τ_i	1.9 θ	0.48 θ
	PID _i	108 θ/τ_i	1.6 θ	0.58 θ
Distributed lags	PI	20K	0.50 $\Sigma\tau$	
	PID _n	10K	0.30 $\Sigma\tau$	0.09 $\Sigma\tau$
	PID _i	15K	0.25 $\Sigma\tau$	0.10 $\Sigma\tau$

NOTE: n = noninteracting, i = interacting controller modes.

The performance of any controller can be measured against this standard by comparing the IE it achieves in responding to a load change with the best practical IE. Maximum performance levels for PI controllers on lag-dominant processes lie in the 20 to 30 percent range, while for PID controllers they fall between 40 and 60 percent, varying with secondary lags.

Tuning Methods Based on Known Process Models The most accurate tuning rules for controllers have been based on simulation, where the process parameters can be specified and IAE and IE can be integrated during the simulation as an indication of performance. Controller settings are then iterated until a minimum IAE is reached for a given disturbance. Next these optimum settings are related to the parameters of the simulated process in tables, graphs, or equations, as a guide to tuning controllers for processes whose parameters are known (Seborg, Edgar, and Mellichamp, *Process Dynamics and Control*, Wiley, New York, 2004). This is a multidimensional problem, however, in that the relationships change as a function of process type, controller type, and source of disturbance.

Table 8-2 summarizes these rules for minimum-IAE load response for the most common controllers. The process gain and time constant τ_m are obtained from the product of G_c and G_p in Fig. 8-23. Derivative action is not effective for dead-time-dominant processes. Any secondary lag, sampling interval, or filter time constant should be added to dead time θ .

The principal limitation to using these rules is that the true process parameters are often unknown. Steady-state gain K can be calculated from a process model, or determined from the steady-state results of a step test as $\Delta c/\Delta u$, as shown in Fig. 8-28. The test will not be viable, however if the time constant of the process τ_m is longer than a few minutes, since five time constants must elapse to approach a steady state within 1 percent, and unexpected disturbances may intervene. Estimated dead time θ is the time from the step to the intercept of a

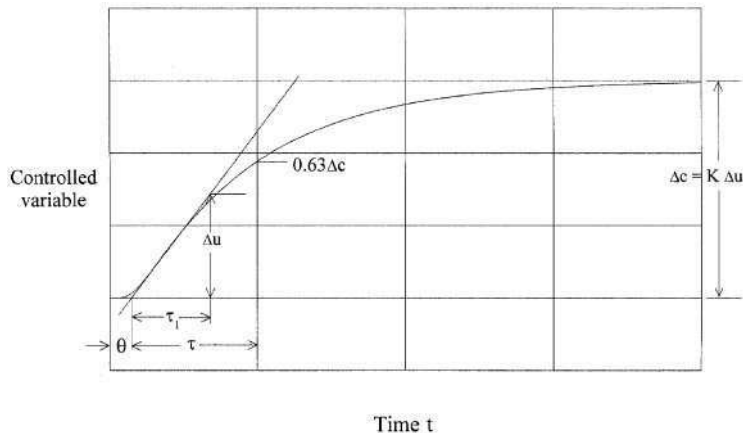


FIG. 8-28 If a steady state can be reached, gain K and time constant τ can be estimated from a step response; if not, use τ_i instead.

TABLE 8-3 Tuning Rules Using Slope and Intercept

Controller	P	τ_i	τ_D
PI	$150\theta/\tau_1$	3.5 θ	
PID _n	$75\theta/\tau_1$	2.1 θ	0.63 θ
PID _i	$113\theta/\tau_1$	1.8 θ	0.7 θ

NOTE: n = noninteracting, i = interacting controller modes.

straight line tangent to the steepest part of the response curve. The estimated time constant τ is the time from that point to 63 percent of the complete response. In the presence of a significant secondary lag, these results will not be completely accurate, however. The time for 63 percent response may be more accurately calculated as the residence time of the process: its volume divided by current volumetric flow rate.

Tuning Methods When Process Model Is Unknown Ziegler and Nichols developed two tuning methods for processes with unknown parameters. The open-loop method uses a step test without waiting for a steady state to be reached and is therefore applicable to very slow processes. Dead time is estimated from the intercept of the steepest tangent to the response curve in Fig. 8-28, whose slope is also used. If the process is non-self-regulating, the controlled variable will continue to follow this slope, changing by an amount equal to Δu in a time equal to its time constant τ_1 . This time estimate τ_1 is used along with θ to tune controllers according to Table 8-3, applicable to lag-dominant processes. If the process is known to be a distributed lag, such as a heat exchanger, distillation column, or stirred tank, then better results will be obtained by converting the estimated values of θ and τ_1 to K and $\Sigma\tau$ and using Table 8-2. The conversion factors are $K = 7.5\theta/\tau_1$ and $\Sigma\tau = 7.0\theta$.

The Ziegler and Nichols closed-loop method requires forcing the loop to cycle uniformly under proportional control, by setting the integral time to maximum and derivative time to zero and reducing the proportional band until a constant-amplitude cycle results. The natural period τ_n of the cycle (the proportional controller contributes no phase shift to alter it) is used to set the optimum integral and derivative time constants. The optimum proportional band is set relative to the undamped proportional band P_u which was found to produce the uniform oscillation. Table 8-4 lists the tuning rules for a lag-dominant process.

A uniform cycle can also be forced by using on-off control to cycle the manipulated variable between two limits. The period of the cycle will be close to τ_n if the cycle is symmetric; the peak-to-peak ampli-

TABLE 8-4 Tuning Rules Using Proportional Cycle

Controller	P	τ_i	τ_D
PI	$1.70P_u$	$0.81\tau_n$	
PID _n	$1.30P_u$	$0.48\tau_n$	$0.11\tau_n$
PID _i	$1.80P_u$	$0.38\tau_n$	$0.14\tau_n$

NOTE: n = noninteracting, i = interacting controller modes.

tude A_c of the controlled variable divided by the difference between the output limits A_u is a measure of process gain at that period and is therefore related to P_u for the proportional cycle:

$$P_u = 100 \frac{\pi}{4} \frac{A_c}{A_u} \quad (8-37)$$

The factor $\pi/4$ compensates for the square wave in the output. Tuning rules are given in Table 8-4.

Set-Point Response All the above tuning methods are intended to minimize IAE for step load changes. When applied to lag-dominant processes, the resulting controller settings produce excessive overshoot of set-point changes. This behavior has led to the practice of tuning to optimize set-point response, which unfortunately degrades the load response of lag-dominant loops. An option has been available with some controllers to remove proportional action from set-point changes, which eliminates set-point overshoot but lengthens settling time. A preferred solution to this dilemma is available in many modern controllers which feature an independent gain adjustment for the set point, through which set-point response can be optimized after the controller has been tuned to optimize load response.

Figure 8-29 shows set-point and load responses of a distributed lag for both set-point and load tuning, including the effects of fractional set-point gain K_r . The set point was stepped at time zero, and the load stepped at time 2.4. With full set-point gain, the PI controller was tuned for minimum-IAE set-point response with $P = 29K$ and $\tau_i = \Sigma\tau$, compared to $P = 20K$ and $\tau_i = 0.50\Sigma\tau$ for minimum-IAE load response. These settings increase its IE for load response by a factor of 2.9, and its peak deviation by 20 percent, over optimum load tuning. However, with optimum load tuning, that same set-point overshoot can be obtained with set-point gain $K_r = 0.54$. The effects of full set-point gain (1.0) and no set-point gain (0) are shown for comparison.

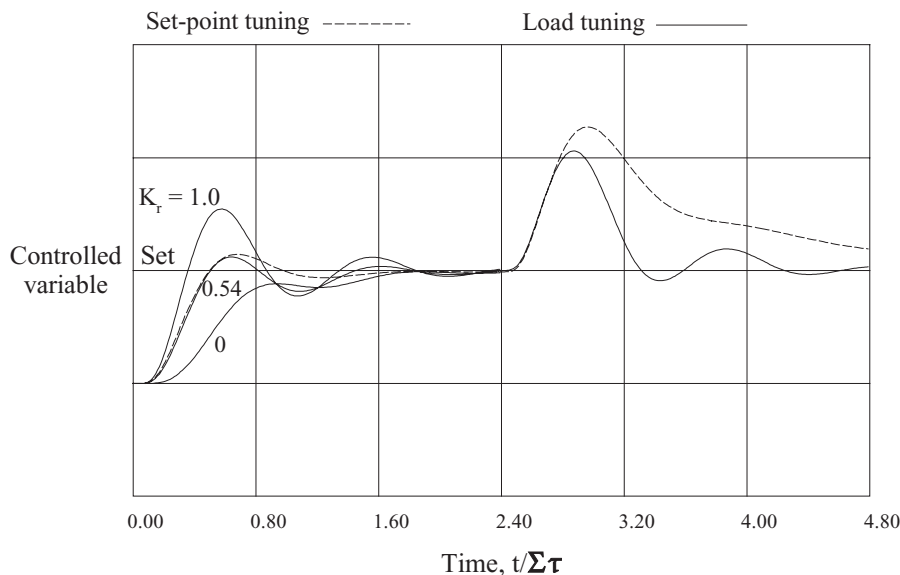


FIG. 8-29 Tuning proportional and integral settings to optimize set-point response degrades load response; using a separate set-point gain adjustment allows both responses to be optimized.

ADVANCED CONTROL SYSTEMS

BENEFITS OF ADVANCED CONTROL

The economics of most processes are determined by the steady-state operating conditions. Excursions from these steady-state conditions usually have a less important effect on the economics of the process, except when the excursions lead to off-specification products. To enhance the economic performance of a process, the steady-state operating conditions must be altered in a manner that leads to more efficient process operation.

The hierarchy shown in Fig. 8-30 indicates that process control activities consist of the following five levels:

- Level 1: Measurement devices and actuators
- Level 2: Safety, environmental/equipment protection
- Level 3: Regulatory control
- Level 4: Real-time optimization
- Level 5: Planning and scheduling

Levels 4 and 5 clearly affect the process economics, as both levels are directed to optimizing the process in some manner. In contrast, levels 1, 2, and 3 would appear to have no effect on process economics. Their direct effect is indeed minimal, although indirectly they can have a major effect. Basically, these levels provide the foundation for all higher levels. A process cannot be optimized until it can be operated consistently at the prescribed targets. Thus, satisfactory regulatory control must be the first goal of any automation effort. In turn, the measurements and actuators provide the process interface for regulatory control.

For most processes, the optimum operating point is determined by a constraint. The constraint might be a product specification (a product stream can contain no more than 2 percent ethane); violation of this constraint causes off-specification product. The constraint might be an equipment limit (vessel pressure rating is 300 psig); violation of this constraint causes the equipment protection mechanism (pressure

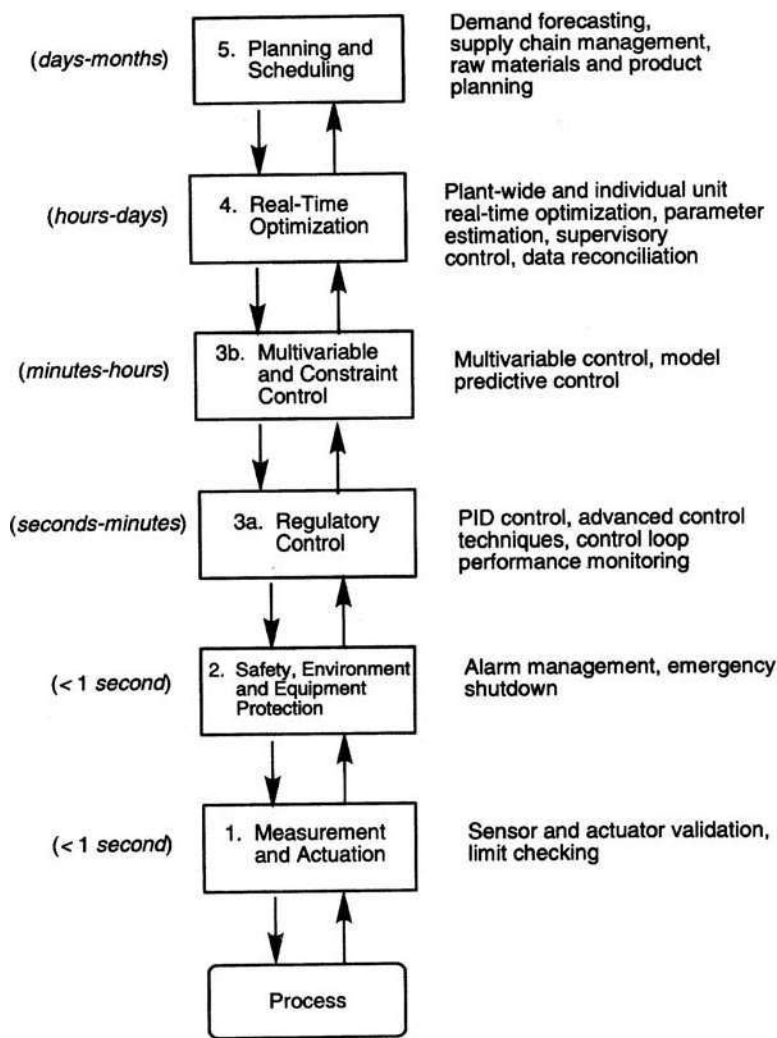


FIG. 8-30 The five levels of process control and optimization in manufacturing. Time scales are shown for each level. (Source: Seborg et al., *Process Dynamics and Control*, 2d ed., Wiley, New York, 2004.)

relief device) to activate. As the penalties are serious, violation of such constraints must be very infrequent.

If the regulatory control system were perfect, the target could be set exactly equal to the constraint (i.e., the target for the pressure controller could be set at the vessel relief pressure). However, no regulatory control system is perfect. Therefore, the value specified for the target must be on the safe side of the constraint, thus allowing the control system some operating margin. How much depends on the following:

1. The performance of the control system (i.e., how effectively it responds to disturbances). The faster the control system reacts to a disturbance, the closer the process can be operated to the constraint.

2. The magnitude of the disturbances to which the control system must respond. If the magnitude of the major disturbances can be reduced, the process can be operated closer to the constraint.

One measure of the performance of a control system is the variance of the controlled variable from the target. Both improving the control system and reducing the disturbances will lead to a lower variance in the controlled variable.

In a few applications, improving the control system leads to a reduction in off-specification product and thus improved process economics. However, in most situations, the process is operated sufficiently far from the constraint that very little, if any, off-specification product results from control system deficiencies. Management often places considerable emphasis on avoiding off-specification production, so consequently the target is actually set far more conservatively than it should be.

In most applications, simply improving the control system does not directly lead to improved process economics. Instead, the control system improvement must be accompanied by shifting the target closer to the constraint. There is always a cost of operating a process in a conservative manner. The cost may be a lower production rate, a lower process efficiency, a product giveaway, or other. When management places undue emphasis on avoiding off-specification production, the natural reaction is to operate very conservatively, thus incurring other costs.

The immediate objective of an advanced control effort is to reduce the variance in an important controlled variable. However, this effort must be coupled with a commitment to adjust the target for this controlled variable so that the process is operated closer to the constraint. In large-throughput (commodity) processes, very small shifts in operating targets can lead to large economic returns.

ADVANCED CONTROL TECHNIQUES

GENERAL REFERENCES: Seborg, Edgar, and Mellichamp, *Process Dynamics and Control*, Wiley, New York, 2004. Stephanopoulos, *Chemical Process Control: An Introduction to Theory and Practice*, Prentice-Hall, Englewood Cliffs, N.J., 1984. Shinskey, *Process Control Systems*, 4th ed., McGraw-Hill, New York, 1996. Ogunnaike and Ray, *Process Dynamics, Modeling, and Control*, Oxford University Press, New York, 1994.

While the single-loop PID controller is satisfactory in many process applications, it does not perform well for processes with slow dynamics, time delays, frequent disturbances, or multivariable interactions. We discuss several advanced control methods below that can be implemented via computer control, namely, feedforward control, cascade control, time-delay compensation, selective and override control, adaptive control, fuzzy logic control, and statistical process control.

Feedforward Control If the process exhibits slow dynamic response and disturbances are frequent, then the application of feedforward control may be advantageous. Feedforward (FF) control differs from feedback (FB) control in that the primary disturbance or load (D) is measured via a sensor and the manipulated variable (U) is adjusted so that deviations in the controlled variable from the set point are minimized or eliminated (see Fig. 8-31). By taking control action based on measured disturbances rather than controlled variable error, the controller can reject disturbances before they affect the controlled variable Y . To determine the appropriate settings for the manipulated variable, one must develop mathematical models that relate

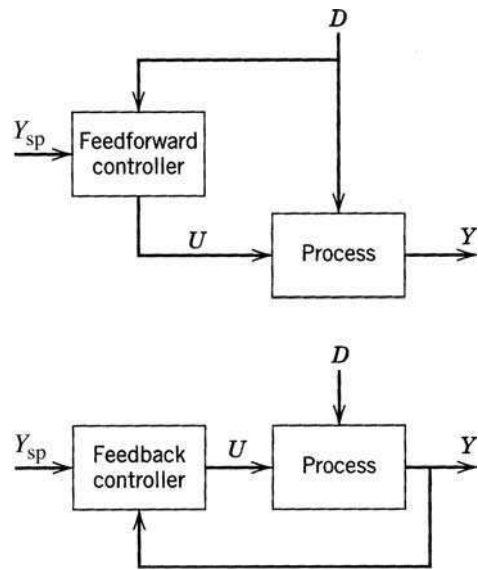


FIG. 8-31 Simplified block diagrams for feedforward and feedback control.

1. The effect of the manipulated variable U on the controlled variable Y
2. The effect of the disturbance D on the controlled variable Y

These models can be based on steady-state or dynamic analysis. The performance of the feedforward controller depends on the accuracy of both models. If the models are exact, then feedforward control offers the potential of perfect control (i.e., holding the controlled variable precisely at the set point at all times because of the ability to predict the appropriate control action). However, since most mathematical models are only approximate and since not all disturbances are measurable, it is standard practice to utilize feedforward control in conjunction with feedback control. Table 8-5 lists the relative advantages and disadvantages of feedforward and feedback control. By combining the two control methods, the strengths of both schemes can be utilized.

FF control therefore attempts to eliminate the effects of measurable disturbances, while FB control would correct for unmeasurable disturbances and modeling errors. This latter case is often referred to as feedback trim. These controllers have become widely accepted in the chemical process industries since the 1960s.

Design Based on Material and Energy Balances Consider a heat exchanger example (see Fig. 8-32) to illustrate the use of FF and FB control. The control objective is to maintain T_2 , the exit liquid temperature, at the desired value (or set point) T_{2sp} despite variations in the inlet liquid flow rate F and inlet liquid temperature T_1 . This is done by manipulating W , the steam flow rate. A feedback control scheme would entail measuring T_2 , comparing T_2 to T_{2sp} , and then adjusting W . A feedforward control scheme requires measuring F and T_1 , and adjusting W (knowing T_{2sp}), in order to control exit temperature T_2 .

Figure 8-33a and b shows the control system diagrams for FB and FF control. A feedforward control algorithm can be designed for the heat exchanger in the following manner. Using a steady-state energy balance and assuming no heat loss from the heat exchanger,

$$WH = FC(T_2 - T_1) \quad (8-38)$$

where H = latent heat of vaporization and C = specific heat of liquid

$$W = \frac{C}{H} F(T_2 - T_1) \quad (8-39)$$

or

$$W = K_1 F(T_2 - T_1) \quad (8-40)$$

TABLE 8-5 Relative Advantages and Disadvantages of Feedforward and Feedback

Advantages	Disadvantages
Feedforward	
• Acts before the effect of a disturbance has been felt by the system • Is good for systems with large time constant or dead time • Does not introduce instability in the closed-loop response	• Requires direct measurement of all possible disturbances • Cannot cope with unmeasured disturbances • Is sensitive to process/model error
Feedback	
• Does not require identification and measurement of any disturbance for corrective action • Does not require an explicit process model • Is possible to design controller to be robust to process/model errors	• Control action not taken until the effect of the disturbance has been felt by the system • Is unsatisfactory for processes with large time constants and frequent disturbances • May cause instability in the closed-loop response

with

$$K_1 = \frac{C_L}{H}$$

(8-41)

Replace T_2 by T_{2sp}

$$W = K_1 F (T_{set} - T_1)$$

(8-42)

Equation (8-42) can be used in the FF calculation, assuming one knows the physical properties C and H . Of course, it is probable that the model will contain errors (e.g., unmeasured heat losses, incorrect C or H). Therefore, K_1 can be designated as an adjustable parameter that can be tuned. The use of a physical model for FF control is desirable because it provides a physical basis for the control law and gives an a priori estimate of what the tuning parameters should be. Note that such a model could be nonlinear [e.g., in Eq. (8-42), F and T_{2sp} are multiplied].

Block Diagram Analysis One shortcoming of this feedforward design procedure is that it is based on the steady-state characteristics of the process and, as such, neglects process dynamics (i.e., how fast the controlled variable responds to changes in the load and manipulated variables). Thus, it is often necessary to include “dynamic compensation” in the feedforward controller. The most direct method of designing the FF dynamic compensator is to use a block diagram of a general process, as shown in Fig. 8-34, where G_d represents the disturbance transmitter, G_f is the feedforward controller, G_v relates the disturbance to the controlled variable, G_c is the valve, G_p is the process, G_m is the output transmitter, and G_e is the feedback controller. All blocks correspond to transfer functions (via Laplace transforms).

Using block diagram algebra and Laplace transform variables, the controlled variable $Y(s)$ is given by

$$Y(s) = \frac{G_f G_d L(s) + G_d D(s)}{1 + G_m G_c G_v G_p}$$

(8-43)

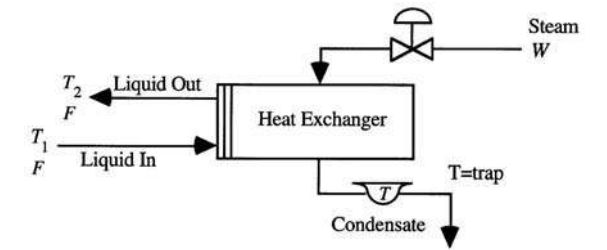
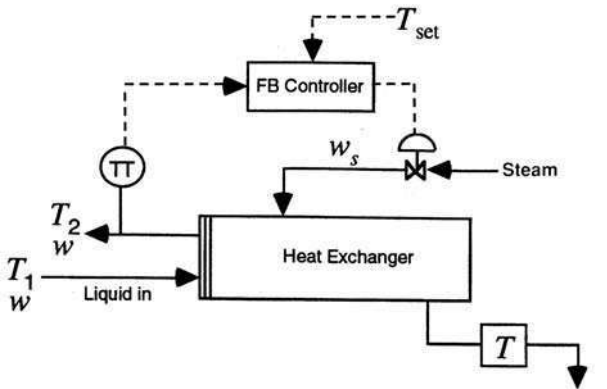


FIG. 8-32 A heat exchanger diagram.

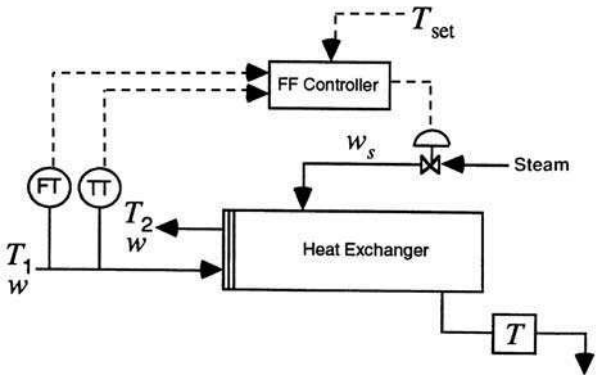
For disturbance rejection [$D(s) \neq 0$] we require that $Y(s) = 0$, or zero error. Solving Eq. (8-43) for G_f gives

$$G_f = \frac{-G_d}{G_c G_v G_p}$$

(8-44)



(a)



(b)

FIG. 8-33 (a) Feedback control of a heat exchanger. (b) Feedforward control of a heat exchanger.

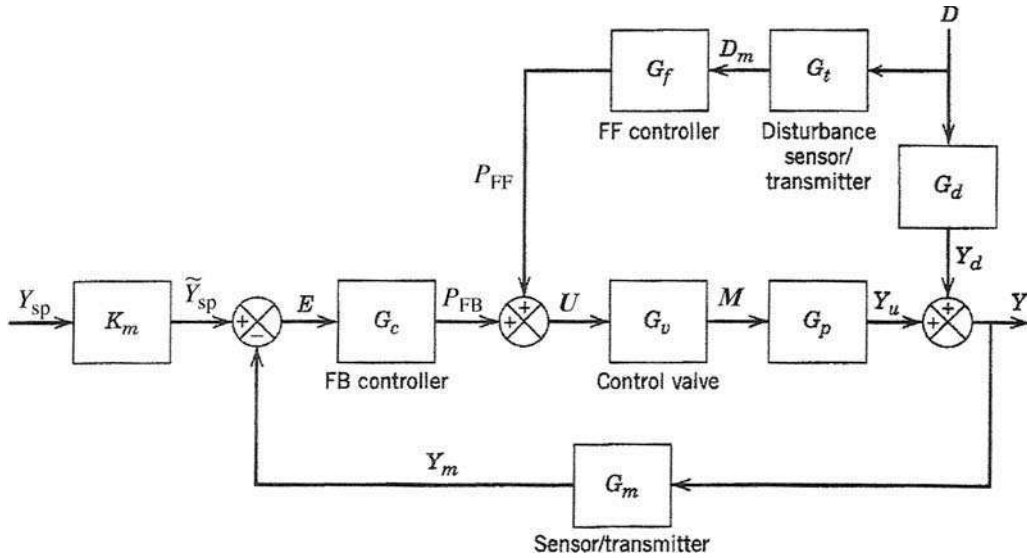


FIG. 8-34 A block diagram of a feedforward-feedback control system. (Source: Seborg *et al.*, *Process Dynamics and Control*, 2d ed., Wiley, New York, 2004.)

Suppose the dynamics of G_d and G_p are first-order; in addition, assume that $G_c = K_c$ and $G_t = K_t$ (constant gains for simplicity).

$$G_d(s) = \frac{K_d}{\tau_d s + 1} = \frac{Y(s)}{D(s)} \quad (8-45)$$

$$G_p(s) = \frac{K_p}{\tau_p s + 1} = \frac{Y(s)}{U(s)} \quad (8-46)$$

Using Eq. (8-44),

$$G_f(s) = \frac{\tau_p s + 1}{\tau_d s + 1} \cdot \frac{-K_d}{K_p K_c K_t} = \frac{-K(\tau_p s + 1)}{\tau_d s + 1} \quad (8-47)$$

where K is the overall ratio of the gains in Eq. (8-47).

The above FF controller can be implemented by using a digital computer. Figure 8-35*a* and *b* compares typical responses for PID FB control, steady-state FF control ($s = 0$), dynamic FF control, and combined FF/FB control. In practice, the engineer can tune K , τ_p , and τ_d in the field to improve the performance of the FF controller. The feedforward controller can also be simplified to provide steady-state feedforward control. This is done by setting $s = 0$ in $G_f(s)$. This might be appropriate if there is uncertainty in the dynamic models for G_d and G_p .

Other Considerations in Feedforward Control The tuning of feedforward and feedback control systems can be performed independently. In analyzing the block diagram in Fig. 8-34, note that G_f is chosen to cancel out the effects of the disturbance $D(s)$, as long as there are no model errors. For the feedback loop, therefore, the effects of $D(s)$ can also be ignored, which for the servo case is

$$\frac{Y(s)}{Y_{sp}(s)} = \frac{G_c G_p K_m}{1 + G_c G_c G_p G_m} \quad (8-48)$$

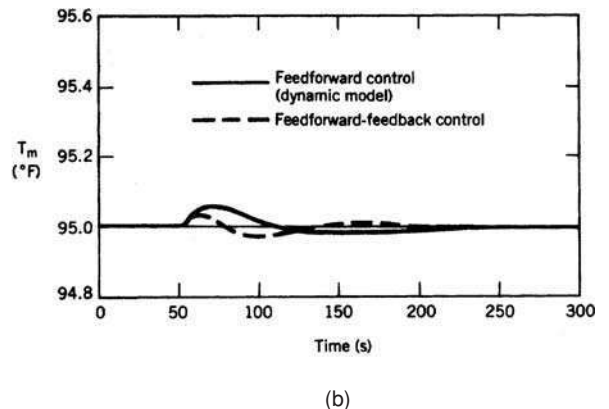
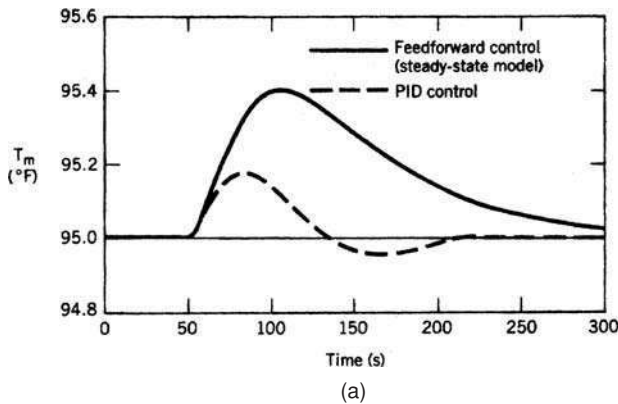


FIG. 8-35 (a) Comparison of FF (steady-state model) and PID FB control for disturbance change. (b) Comparison of FF (dynamic model) and combined FF/FB control.

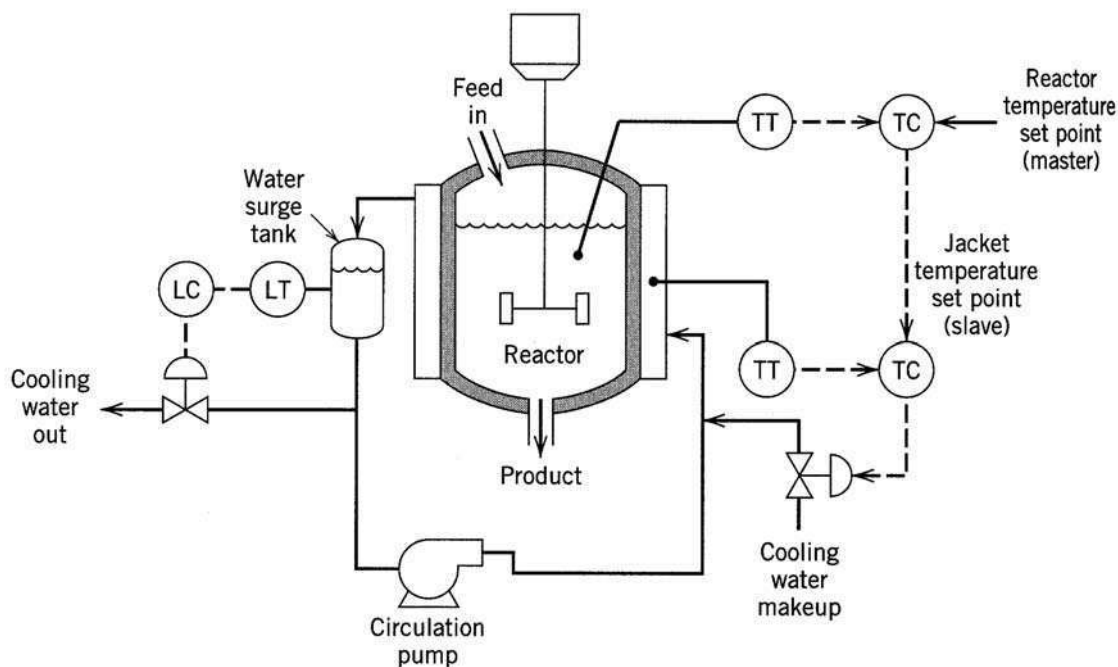


FIG. 8-36 Cascade control of an exothermic chemical reactor. (Source: Seborg et al., *Process Dynamics and Control*, 2d ed., Wiley, New York, 2004.)

Note that the characteristic equation will be unchanged for the FF + FB system; hence system stability will be unaffected by the presence of the FF controller. In general, the tuning of the FF controller can be less conservative than for the case of FB alone, because smaller excursions from the set point will result. This in turn would make the dynamic model $Y(s)$ more accurate.

The tuning of the controller in the feedback loop can be theoretically performed independently of the feedforward loop (i.e., the feedforward loop does not introduce instability in the closed-loop response). For more information on feedforward/feedback control applications and design of such controllers, refer to the general references.

Cascade Control One of the disadvantages of using conventional feedback control for processes with large time lags or delays is that disturbances are not recognized until after the controlled variable deviates from its set point. In these processes, correction by feedback control is generally slow and results in long-term deviation from the set point. One way to improve the dynamic response to load changes is by using a secondary measurement point and a secondary controller; the secondary measurement point is located so that it recognizes the upset condition before the primary controlled variable is affected.

One such approach is called cascade control, which is routinely used in most modern computer control systems. Consider a chemical reactor, where reactor temperature is to be controlled by coolant flow to the jacket of the reactor. The reactor temperature can be influenced by changes in disturbance variables such as feed rate or feed temperature; a feedback controller could be employed to compensate for such disturbances by adjusting a valve on the coolant flow to the reactor jacket. However, suppose an increase occurs in the coolant temperature as a result of changes in the plant coolant system. This will cause a change in the reactor temperature measurement, although such a change will not occur quickly, and the corrective action taken by the controller will be delayed.

Cascade control is one solution to this problem (see Fig. 8-36). Here the jacket temperature is measured, and an error signal is sent from this point to the coolant control valve; this reduces coolant flow,

maintaining the heat-transfer rate to the reactor at a constant level and rejecting the disturbance. The cascade control configuration will also adjust the setting of the coolant control valve when an error occurs in the reactor temperature. The cascade control scheme shown in Fig. 8-36 contains two controllers. The primary controller is the reactor temperature controller. It measures the reactor temperature, compares it to the set point, and computes an output, which is the set point for the coolant flow rate controller. The secondary controller compares this set point to the coolant temperature measurement and adjusts the valve. The principal advantage of cascade control is that the secondary measurement (jacket temperature) is located closer to a potential disturbance in order to improve the closed-loop response.

Figure 8-37 shows the block diagram for a general cascade control system. In tuning of a cascade control system, the secondary controller (in the inner loop) is tuned first with the primary controller in manual. Often only a proportional controller is needed for the secondary loop, because offset in the secondary loop can be treated by using proportional-plus-integral action in the primary loop. When the primary controller is transferred to automatic, it can be tuned by using the techniques described earlier in this section. For more information on theoretical analysis of cascade control systems, see the general references for a discussion of applications of cascade control.

Time-Delay Compensation Time delays are a common occurrence in the process industries because of the presence of recycle loops, fluid-flow distance lags, and dead time in composition measurements resulting from use of chromatographic analysis. The presence of a time delay in a process severely limits the performance of a conventional PID control system, reducing the stability margin of the closed-loop control system. Consequently, the controller gain must be reduced below that which could be used for a process without delay. Thus, the response of the closed-loop system will be sluggish compared to that of the system with no time delay.

To improve the performance of time-delay systems, special control algorithms have been developed to provide time-delay compensation. The Smith predictor technique is the best-known algorithm; a related method is called the analytical predictor. Various investigators have

> denotes a high selector and < a low selector. For example, a high selector can be used to determine the hot-spot temperature in a fixed-bed chemical reactor. In this case, the output from the high selector is the input to the temperature controller. In an exothermic catalytic reaction, the process may run away due to disturbances or changes in the reactor. Immediate action should be taken to prevent a dangerous rise in temperature. Because a hot spot may potentially develop at one of several possible locations in the reactor, multiple (redundant) measurement points should be employed. This approach minimizes the time required to identify when a temperature has risen too high at some point in the bed.

The use of high or low limits for process variables is another type of selective control, called an override. The feature of antireset windup in feedback controllers is a type of override. Another example is a distillation column with lower and upper limits on the heat input to the column reboiler. The minimum level ensures that liquid will remain on the trays, while the upper limit is determined by the onset of flooding. Overrides are also used in forced-draft combustion control systems to prevent an imbalance between airflow and fuel flow, which could result in unsafe operating conditions.

Other types of selective systems employ multiple final control elements or multiple controllers. In some applications, several manipulated variables are used to control a single process variable (also called split-range control). Typical examples include the adjustment of both inflow and outflow from a chemical reactor to control reactor pressure or the use of both acid and base to control pH in wastewater treatment. In this approach, the selector chooses among several controller outputs which final control element should be adjusted.

Adaptive Control Process control problems inevitably require online tuning of the controller constants to achieve a satisfactory degree of control. If the process operating conditions or the environment changes significantly, the controller may have to be retuned. If these changes occur quite frequently, then adaptive control techniques should be considered. An adaptive control system is one in which the controller parameters are adjusted automatically to compensate for changing process conditions.

The subject of adaptive control is one of current interest. New algorithms continue to be developed, but these need to be field-tested before industrial acceptance can be expected. An adaptive controller is inherently nonlinear and therefore more complicated than the conventional PID controller.

Fuzzy Logic Control The application of fuzzy logic to process control requires the concepts of fuzzy rules and fuzzy inference. A fuzzy rule, also known as a fuzzy IF-THEN statement, has the form

If x then y
 if input1 = high
 and input2 = low
 then output = medium

Three functions are required to perform logical inferencing with fuzzy rules. The fuzzy AND is the product of a rule's input membership values, generating a weight for the rule's output. The fuzzy OR is a normalized sum of the weights assigned to each rule that contributes to a particular decision. The third function used is defuzzification, which generates a crisp final output. In one approach, the crisp output is the weighted average of the peak element values.

With a single feedback control architecture, information that is readily available to the algorithm includes the error signal, difference between the process variable and the set-point variable, change in error from previous cycles to the current cycle, changes to the set-point variable, change of the manipulated variable from cycle to cycle, and change in the process variable from past to present. In addition, multiple combinations of the system response data are available. As long as the irregularity lies in that dimension wherein fuzzy decisions are being based or associated, the result should be enhanced performance. This enhanced performance should be demonstrated in both the transient and steady-state response. If the system tends to have

changing dynamic characteristics or exhibits nonlinearities, fuzzy logic control should offer a better alternative to using constant PID settings. Most fuzzy logic software begins building its information base during the autotune function. In fact, the majority of the information used in the early stages of system start-up comes from the autotune solutions.

In addition to single-loop process controllers, products that have benefited from the implementation of fuzzy logic are camcorders, elevators, antilock braking systems, and televisions with automatic color, brightness, and sound control. Sometimes fuzzy logic controllers are combined with pattern recognition software such as artificial neural networks (Passino and Yurkovich, *Fuzzy Control*, Addison-Wesley, Reading, Mass., 1998; Kosko, *Neural Networks and Fuzzy Systems*, Prentice-Hall, Englewood Cliffs, N.J., 1992).

EXPERT SYSTEMS

An expert system is a computer program that uses an expert's knowledge in a particular domain to solve a narrowly focused, complex problem. An offline system uses information entered manually and produces results in visual form to guide the user in solving the problem at hand. An online system uses information taken directly from process measurements to perform tasks automatically or instruct or alert operating personnel to the status of the plant.

Each expert system has a rule base created by the expert to respond as the expert would to sets of input information. Expert systems used for plant diagnostics and management usually have an open rule base, which can be changed and augmented as more experience accumulates and more tasks are automated. The system begins as an empty shell with an assortment of functions such as equation solving, logic, and simulation, as well as input and display tools to allow an expert to construct a proprietary rule base. The "expert" in this case would be the person or persons having the deepest knowledge about the process, its problems, its symptoms, and remedies. Converting these inputs to meaningful outputs is the principal task in constructing a rule base. First-principles models (deep knowledge) produce the most accurate results, although heuristics are always required to establish limits. Often modeling tools such as artificial neural nets are used to develop relationships among the process variables.

A number of process control vendors offer comprehensive, object-oriented software environments for building and deploying expert systems. Advantages of such software include transforming complex real-time data to useful information through knowledge-based reasoning and analysis, monitoring for potential problems before they adversely impact operations, diagnosing root causes of time-critical problems to speed up resolution, and recommending or taking corrective actions to help ensure successful recovery.

MULTIVARIABLE CONTROL

GENERAL REFERENCES: Shinskey, *Process Control Systems*, 4th ed., McGraw-Hill, New York, 1996. Seborg, Edgar, and Mellichamp, *Process Dynamics and Control*, 2d ed., Wiley, New York, 2004. McAvoy, *Interaction Analysis*, ISA, Research Triangle Park, N.C., 1983.

Process control books and journal articles tend to emphasize problems with a single controlled variable. In contrast, many processes require multivariable control with many process variables to be controlled. In fact, for virtually any important industrial process, at least two variables must be controlled: product quality and throughput. In this section, strategies for multivariable control are considered.

Three examples of simple multivariable control systems are shown in Fig. 8-39. The in-line blending system blends pure components A and B to produce a product stream with flow rate w and mass fraction of A, x . Adjusting either inlet flow rate w_A or w_B affects both of the controlled variables w and x . For the pH neutralization process in Fig. 8-39b, liquid level h and exit stream pH are to be controlled by adjusting the acid and base flow rates w_A and w_B . Each of the manipulated variables affects both of the controlled variables. Thus, both the blending system and the pH neutralization process are said to exhibit

strong process interactions. In contrast, the process interactions for the gas-liquid separator in Fig. 8-39c are not as strong because one manipulated variable, liquid flow rate L , has only a small and indirect effect on one of the controlled variables, pressure P .

Strong process interactions can cause serious problems if a conventional multiloop feedback control scheme (e.g., either PI or PID controllers) is employed. The process interactions can produce undesirable control loop interactions where the controllers fight each other. Also, it may be difficult to determine the best pairing of controlled and manipulated variables. For example, in the in-line blending process in Fig. 8-39a, should w be controlled with w_A and x with w_B , or vice versa?

Control Strategies for Multivariable Control If a conventional multiloop control strategy performs poorly due to control loop interactions, a number of solutions are available:

1. Detune one or more of the control loops.
2. Choose different controlled or manipulated variables (or their pairings).

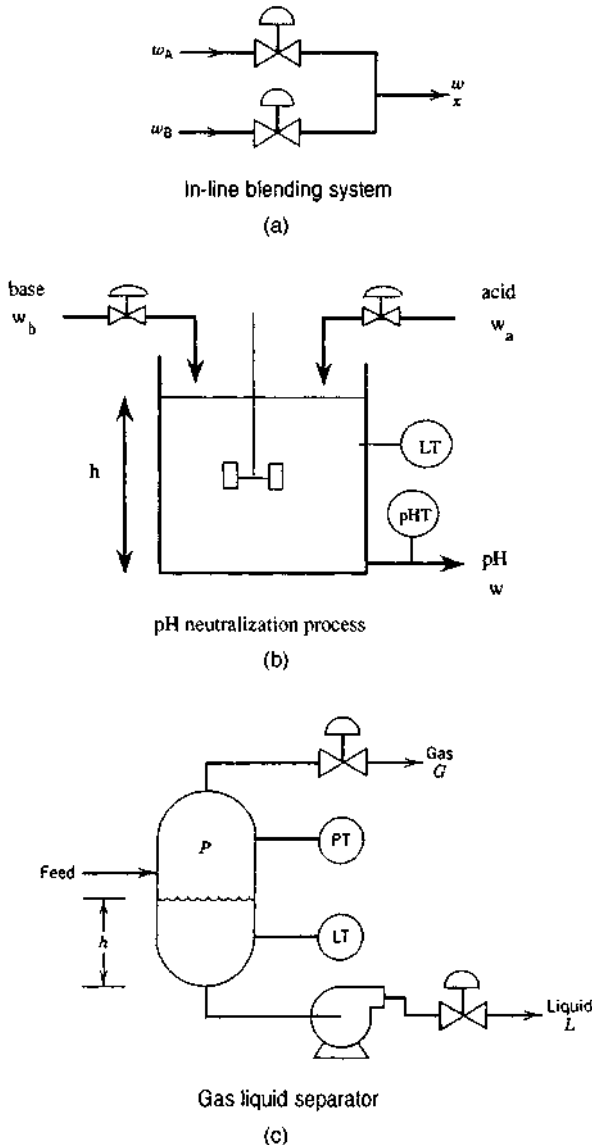


FIG. 8-39 Physical examples of multivariable control problems.

3. Use a decoupling control system.

4. Use a multivariable control scheme (e.g., model predictive control).

Detuning a controller (e.g., using a smaller controller gain or a larger reset time) tends to reduce control loop interactions by sacrificing the performance for the detuned loops. This approach may be acceptable if some of the controlled variables are faster or less important than others.

The selection of controlled and manipulated variables is of crucial importance in designing a control system. In particular, a judicious choice may significantly reduce control loop interactions. For the blending process in Fig. 8-39a, a straightforward control strategy would be to control x by adjusting w_A , and w by adjusting w_B . But physical intuition suggests that it would be better to control x by adjusting the ratio $w_A/(w_A + w_B)$ and to control product flow rate w by the sum $w_A + w_B$. Thus, the new manipulated variables would be $U_1 = w_A/(w_A + w_B)$ and $U_2 = w_A + w_B$. In this control scheme, U_1 affects only x , and U_2 affects only w . Thus, the control loop interactions have been eliminated. Similarly, for the pH neutralization process in Fig. 8-39b, the control loop interactions would be greatly reduced if pH were controlled by $U_1 = w_A/(w_A + w_B)$ and liquid level h were controlled by $U_2 = w_A + w_B$.

Decoupling Control Systems Decoupling control systems provide an alternative approach for reducing control loop interactions. The basic idea is to use additional controllers called decouplers to compensate for undesirable process interactions.

As an illustrative example, consider the simplified block diagram for a representative decoupling control system shown in Fig. 8-40. The two controlled variables Y_1 and Y_2 and two manipulated variables U_1 and U_2 are related by the four process transfer functions G_{p11} , G_{p12} , and so on. For example, G_{p11} denotes the transfer function between U_1 and Y_1 :

$$\frac{Y_1(s)}{U_1(s)} = G_{p11}(s) \quad (8-49)$$

Figure 8-40 includes two conventional feedback controllers: G_{c1} controls Y_1 by manipulating U_1 , and G_{c2} controls Y_2 by manipulating U_2 . The output signals from the feedback controllers serve as input signals to the two decouplers T_{12} and T_{21} . The block diagram is in a simplified form because the disturbance variables and transfer functions for the final control elements and sensors have been omitted.

The function of the decouplers is to compensate for the undesirable process interactions represented by G_{p12} and G_{p21} . Suppose that the process transfer functions are all known. Then the ideal design equations are

$$T_{12}(s) = -\frac{G_{p12}(s)}{G_{p11}(s)} \quad (8-50)$$

$$T_{21}(s) = -\frac{G_{p21}(s)}{G_{p22}(s)} \quad (8-51)$$

These decoupler design equations are very similar to the ones for feedforward control in an earlier section. In fact, decoupling can be interpreted as a type of feedforward control where the input signal is the output of a feedback controller rather than a measured disturbance variable.

In principle, ideal decoupling eliminates control loop interactions and allows the closed-loop system to behave as a set of independent control loops. But in practice, this ideal behavior is not attained for a variety of reasons, including imperfect process models and the presence of saturation constraints on controller outputs and manipulated variables. Furthermore, the ideal decoupler design equations in (8-50) and (8-51) may not be physically realizable and thus would have to be approximated.

In practice, other types of decouplers and decoupling control configurations have been employed. For example, in partial decoupling, only a single decoupler is employed (i.e., either T_{12} or T_{21} in Fig. 8-40

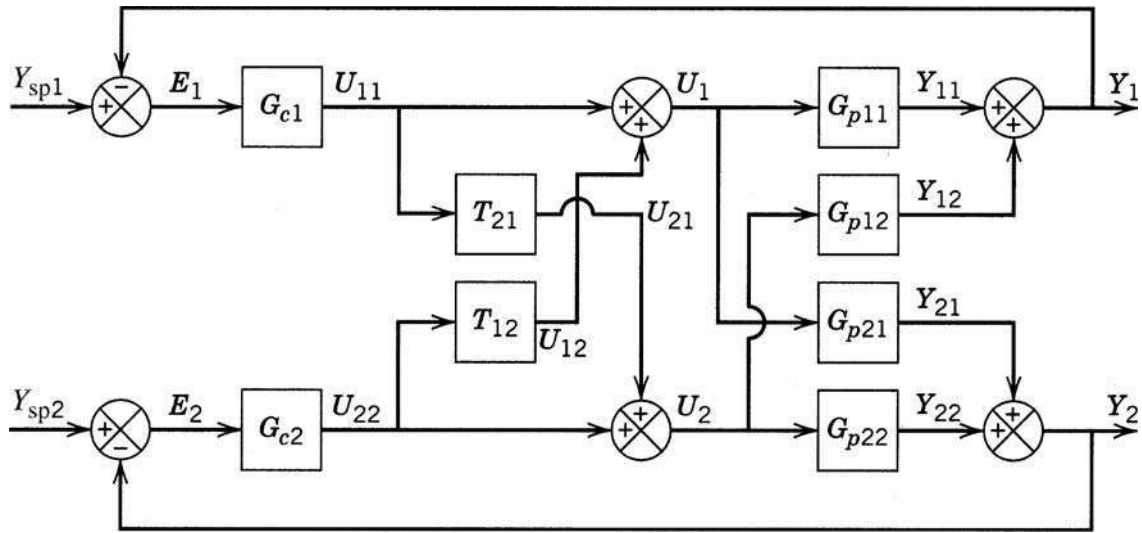


FIG. 8-40 A decoupling control system.

is set equal to zero). This approach tends to be more robust than complete decoupling and is preferred when one of the controlled variables is more important than the other. Static decouplers can be used to reduce the steady-state interactions between control loops. They can be designed by replacing the transfer functions in Eqs. (8-50) and (8-51) with the corresponding steady-state gains

$$T_{12}(s) = -\frac{K_{p12}}{K_{p11}} \quad (8-52)$$

$$T_{21}(s) = -\frac{K_{p21}}{K_{p22}} \quad (8-53)$$

The advantage of static decoupling is that less process information is required, namely, only steady-state gains. Nonlinear decouplers can be used when the process behavior is nonlinear.

Pairing of Controlled and Manipulated Variables A key decision in multiloop control system design is the pairing of manipulated and controlled variables. This is referred to as the controller pairing problem. Suppose there are N controlled variables and N manipulated variables. Then $N!$ distinct control configurations exist. For example, if $N = 5$, then there are 120 different multiloop control schemes. In practice, many would be rejected based on physical insight or previous experience. But a smaller number (say, 5 to 15) may appear to be feasible, and further analysis would be warranted. Thus, it is very useful to have a simple method for choosing the most promising control configuration.

The most popular and widely used technique for determining the best controller pairing is the relative gain array (RGA) method (Bristol, "On a New Measure of Process Interaction," *IEEE Trans. Auto. Control*, **AC-11**: 133, 1966). The RGA method provides two important items of information:

1. A measure of the degree of process interactions between the manipulated and controlled variables
2. A recommended controller pairing

An important advantage of the RGA method is that it requires minimal process information, namely, steady-state gains. Another advantage is that the results are independent of both the physical units used and the scaling of the process variables. The chief disadvantage of the RGA method is that it neglects process dynamics, which can be an important factor in the pairing decision. Thus, the RGA analysis

should be supplemented with an evaluation of process dynamics. Although extensions of the RGA method that incorporate process dynamics have been reported, these extensions have not been widely applied.

RGA Method for 2×2 Control Problems To illustrate the use of the RGA method, consider a control problem with two inputs and two outputs. The more general case of $N \times N$ control problems is considered elsewhere (McAvoy, *Interaction Analysis*, ISA, Research Triangle Park, N.C., 1983). As a starting point, it is assumed that a linear, steady-state process model in Eqs. (8-54) and (8-55) is available, where U_1 and U_2 are steady-state values of the manipulated inputs; Y_1 and Y_2 are steady-state values of the controlled outputs; and the K values are steady-state gains. The Y and U variables are deviation variables from nominal steady-state values. This process model could be obtained in a variety of ways, such as by linearizing a theoretical model or by calculating steady-state gains from experimental data or a steady-state simulation.

$$Y_1 = K_{11}U_1 + K_{12}U_2 \quad (8-54)$$

$$Y_2 = K_{21}U_1 + K_{22}U_2 \quad (8-55)$$

By definition, the relative gain λ_{ij} between the i th manipulated variable and the j th controlled variable is defined as

$$\lambda_{ij} = \frac{\text{open-loop gain between } Y_i \text{ and } U_j}{\text{closed-loop gain between } Y_i \text{ and } U_j} \quad (8-56)$$

where the open-loop gain is simply K_{ij} from Eqs. (8-54) and (8-55). The closed-loop gain is defined to be the steady-state gain between U_j and Y_i when the other control loop is closed and no offset occurs in the other controlled variable due to the presence of integral control action. The RGA for the 2×2 process is denoted by

$$\Lambda = \begin{bmatrix} \lambda_{11} & \lambda_{12} \\ \lambda_{21} & \lambda_{22} \end{bmatrix} \quad (8-57)$$

The RGA has the important normalization property that the sum of the elements in each row and each column is exactly 1. Consequently, the RGA in Eq. (8-57) can be written as

$$\Lambda = \begin{bmatrix} \lambda & 1 - \lambda \\ 1 - \lambda & \lambda \end{bmatrix} \quad (8-58)$$

where λ can be calculated from the following formula:

$$\lambda = \frac{1}{1 - K_{12}K_{21}/(K_{11}K_{22})} \quad (8-59)$$

Ideally, the relative gains that correspond to the proposed controller pairing should have a value of 1 because Eq. (8-56) implies that the open- and closed-loop gains are then identical. If a relative gain equals 1, the steady-state operation of this loop will not be affected when the other control loop is changed from manual to automatic, or vice versa. Consequently, the recommendation for the best controller pairing is to pair the controlled and manipulated variables so that the corresponding relative gains are positive and close to 1.

RGA Example To illustrate use of the RGA method, consider the following steady-state version of a transfer function model for a pilot-scale, methanol-water distillation column (Wood and Berry, "Terminal Composition Control of a Binary Distillation Column," *Chem. Eng. Sci.*, **28**: 1707, 1973): $K_{11} = 12.8$, $K_{12} = -18.9$, $K_{21} = 6.6$, and $K_{22} = -19.4$. It follows that $\lambda = 2$ and

$$\Lambda = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \quad (8-60)$$

Thus it is concluded that the column is fairly interacting and the recommended controller pairing is to pair Y_1 with U_1 and Y_2 with U_2 .

MODEL PREDICTIVE CONTROL

GENERAL REFERENCES: Qin and Badgwell, *Control Eng. Practice*, **11**: 773, 2003. Rawlings, *IEEE Control Systems Magazine*, **20**(3): 38, 2000). Camacho and Bordons, *Model Predictive Control*, 2d ed., Springer-Verlag, New York, 2004. Maciejowski, *Predictive Control with Constraints*, Prentice-Hall, Upper Saddle River, N.J., 2002. Seborg, Edgar, and Mellichamp, *Process Dynamics and Control*, 2d ed., Wiley, New York, 2004, Chap. 20.

The model-based control strategy that has been most widely applied in the process industries is *model predictive control (MPC)*. It is a general method that is especially well suited for difficult multi-input, multioutput (MIMO) control problems where there are significant interactions between the manipulated inputs and the controlled outputs. Unlike other model-based control strategies, MPC can easily accommodate inequality constraints on input and output variables such as upper and lower limits and rate-of-change limits.

A key feature of MPC is that future process behavior is predicted by using a dynamic model and available measurements. The controller outputs are calculated so as to minimize the difference between the predicted process response and the desired response. At each sampling instant, the control calculations are repeated and the predictions updated based on current measurements. In typical industrial applications, the set point and target values for the MPC calculations are updated by using online optimization based on a steady-state model of the process.

The current widespread interest in MPC techniques was initiated by pioneering research performed by two industrial groups in the 1970s. Shell Oil (Houston, Tex.) reported its Dynamic Matrix Control (DMC) approach in 1979, while a similar technique, marketed as IDCOM, was published by a small French company ADERSA in 1978. Since then, there have been thousands of applications of these and related MPC techniques in oil refineries and petrochemical plants around the world. Thus, MPC has had a substantial impact and is currently the method of choice for difficult multivariable control problems in these industries. However, relatively few applications have been reported in other process industries, even though MPC is a very general approach that is not limited to a particular industry.

Advantages and Disadvantages of MPC Model predictive control offers a number of important advantages in comparison with conventional multiloop PID control:

1. It is a general control strategy for MIMO processes with inequality constraints on input and output variables.

2. It can easily accommodate difficult or unusual dynamic behavior such as large time delays and inverse responses.

3. Because the control calculations are based on optimizing control system performance, MPC can be readily integrated with online optimization strategies to optimize plant performance.

4. The control strategy can be easily updated online to compensate for changes in process conditions, constraints, or performance criteria.

But current versions of MPC have significant disadvantages in comparison with conventional multiloop control:

1. The MPC strategy is very different from conventional multiloop control strategies and thus initially unfamiliar to plant personnel.

2. The MPC calculations can be relatively complicated [e.g., solving a linear programming (LP) or quadratic programming (QP) problem at each sampling instant] and thus require a significant amount of computer resources and effort. These optimization strategies are described in the next section.

3. The development of a dynamic model from plant data is time-consuming, typically requiring days, or even weeks, of around-the-clock plant tests.

4. Because empirical models are generally used, they are valid only over the range of conditions considered during the plant tests.

5. Theoretical studies have demonstrated that MPC can perform poorly for some types of process disturbances, especially when output constraints are employed.

Because MPC has been widely used and has had considerable impact, there is a broad consensus that its advantages far outweigh its disadvantages.

Economic Incentives for Automation Projects Industrial applications of advanced process control strategies such as MPC are motivated by the need for improvements regarding safety, product quality, environmental standards, and economic operation of the process. One view of the economics incentives for advanced automation techniques is illustrated in Fig. 8-41. Distributed control systems (DCS) are widely used for data acquisition and conventional single-loop (PID) control. The addition of advanced regulatory control systems such as selective controls, gain scheduling, and time-delay compensation can provide benefits for a modest incremental cost. But

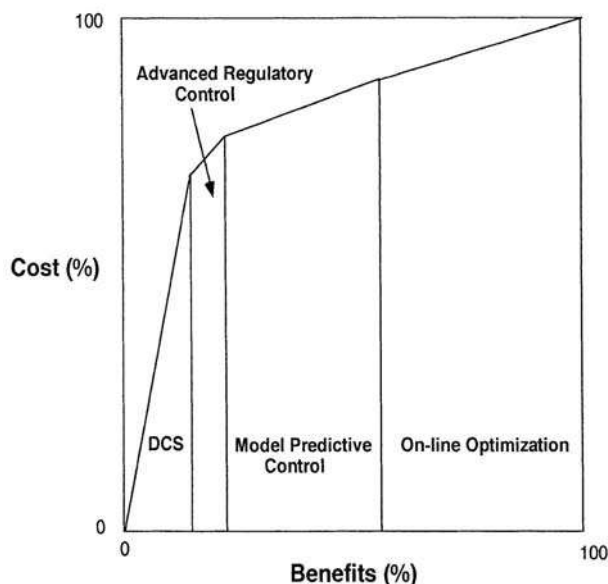


FIG. 8-41 Economic incentives for automation projects in the process industries.

experience has indicated that the major benefits can be obtained for relatively small incremental costs through a combination of MPC and online optimization. The results in Fig. 8-41 are shown qualitatively, rather than quantitatively, because the actual costs and benefits are application-dependent.

A key reason why MPC has become a major commercial and technical success is that there are numerous vendors who are licensed to market MPC products and install them on a turnkey basis. Consequently, even medium-sized companies are able to take advantage of this new technology. Payout times of 3 to 12 months have been widely reported.

Basic Features of MPC Model predictive control strategies have a number of distinguishing features:

1. A dynamic model of the process is used to predict the future outputs over a prediction horizon consisting of the next P sampling periods.
2. A reference trajectory is used to represent the desired output response over the prediction horizon.
3. Inequality constraints on the input and output variables can be included as an option.
4. At each sampling instant, a control policy consisting of the next M control moves is calculated. The control calculations are based on minimizing a quadratic or linear performance index over the prediction horizon while satisfying the constraints.
5. The performance index is expressed in terms of future control moves and the predicted deviations from the reference trajectory.
6. A *receding horizon approach* is employed. At each sampling instant, only the first control move (of the M moves that were calculated) is actually implemented.
7. Then the predictions and control calculations are repeated at the next sampling instant.

These distinguishing features of MPC will now be described in greater detail.

Dynamic Model A key feature of MPC is that a dynamic model of the process is used to predict future values of the controlled outputs. There is considerable flexibility concerning the choice of the dynamic model. For example, a physical model based on first principles (e.g., mass and energy balances) or an empirical model developed from data could be employed. Also, the empirical model could be a linear model (e.g., transfer function, step response model, or state space model) or a nonlinear model (e.g., neural net model). However, most industrial applications of MPC have relied on linear empirical models, which may include simple nonlinear transformations of process variables.

The original formulations of MPC (i.e., DMC and IDCOM) were based on empirical linear models expressed in either step response or impulse response form. For simplicity, we consider only a single-input, single-output (SISO) model. However, the SISO model can be easily generalized to the MIMO models that are used in industrial applications. The step response model relating a single controlled variable y and a single manipulated variable u can be expressed as

$$\hat{y}(k+1) = y_0 + \sum_{i=1}^{N-1} S_i \Delta u(k-i+1) + S_N u(k-N+1) \quad (8-61)$$

where $\hat{y}(k+1)$ is the predicted value of y at the $k+1$ sampling instant, $u(k)$ is the value of the manipulated input at time k , and the model parameters S_i are referred to as the *step response coefficients*. The initial value y_0 is assumed to be known. The change in the manipulated input from one sampling instant to the next is denoted by

$$\Delta u(k) = u(k) - u(k-1) \quad (8-62)$$

The step response model is also referred to as a discrete convolution model.

In principle, the step response coefficients can be determined from the output response to a step change in the input. A typical response to a unit step change in input u is shown in Fig. 8-42. The step response coefficients S_i are simply the values of the output variable at the sampling instants, after the initial value y_0 has been subtracted. Theoretically, they can be determined from a single step response, but, in practice, a num-

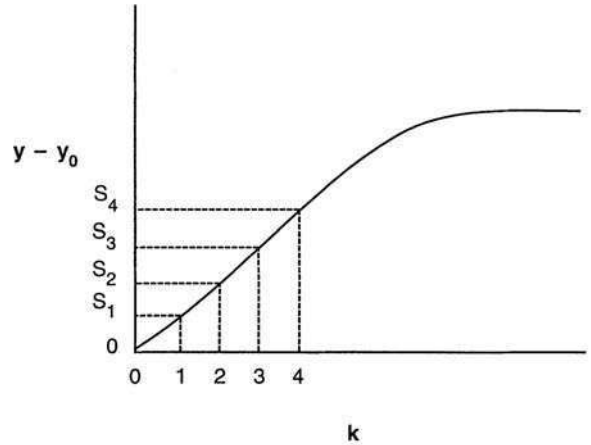


FIG. 8-42 Step response for a unit step change in the input.

ber of “bump tests” are required to compensate for unanticipated disturbances, process nonlinearities, and noisy measurements.

Horizons The step response model in Eq. (8-61) is equivalent to the following impulse response model:

$$\hat{y}(k) = \sum_{i=1}^N h_i u(k-i) + y_0 \quad (8-63)$$

where the impulse response coefficients h_i are related to the step response coefficients by $h_i = S_i - S_{i-1}$. Step and impulse response models typically contain a large number of parameters because the model horizon N is usually quite large ($30 < N < 120$). In fact, these models are often referred to as nonparametric models.

The receding horizon feature of MPC is shown in Fig. 8-43 with the current sampling instant denoted by k . Past and present input signals $[u(i) \text{ for } i \leq k]$ are used to predict the output at the next P sampling instants $[y(k+i) \text{ for } i = 1, 2, \dots, P]$. The control calculations are performed to generate an M -step control policy $[u(k), u(k+1), \dots, u(k+M-1)]$, which optimizes a performance index. The first control action $u(k)$ is implemented. Then at the next sampling instant $k+1$, the prediction and control calculations are repeated in order to determine $u(k+1)$. In Fig. 8-43 the reference trajectory (or target) is considered to be constant. Other possibilities include a gradual or step set-point change that can be generated by online optimization.

Performance Index The performance index for MPC applications is usually a linear or quadratic function of the predicted errors and calculated future control moves. For example, the following quadratic performance index has been widely used:

$$\min_{\Delta u(k)} J = \sum_{i=1}^P Q e^2(k+i) + \sum_{i=1}^M R_i \Delta u^2(k+i-1) \quad (8-64)$$

The value $e(k+i)$ denotes the predicted error at time $k+i$,

$$e(k+i) = y_{sp}(k+i) - \hat{y}(k+i) \quad (8-65)$$

where $y_{sp}(k+i)$ is the set point at time $k+i$ and $\Delta u(k)$ denotes the column vector of current and future control moves over the next M sampling instants:

$$\Delta u(k) = [\Delta u(k), \Delta u(k+1), \dots, \Delta u(k+M-1)]^T \quad (8-66)$$

Equation (8-64) contains two types of design parameters that can also be used for tuning purposes. Weighting factor R_i penalizes large control moves, while weighting factor Q_i allows the predicted errors to be weighed differently at each time step, if desired.

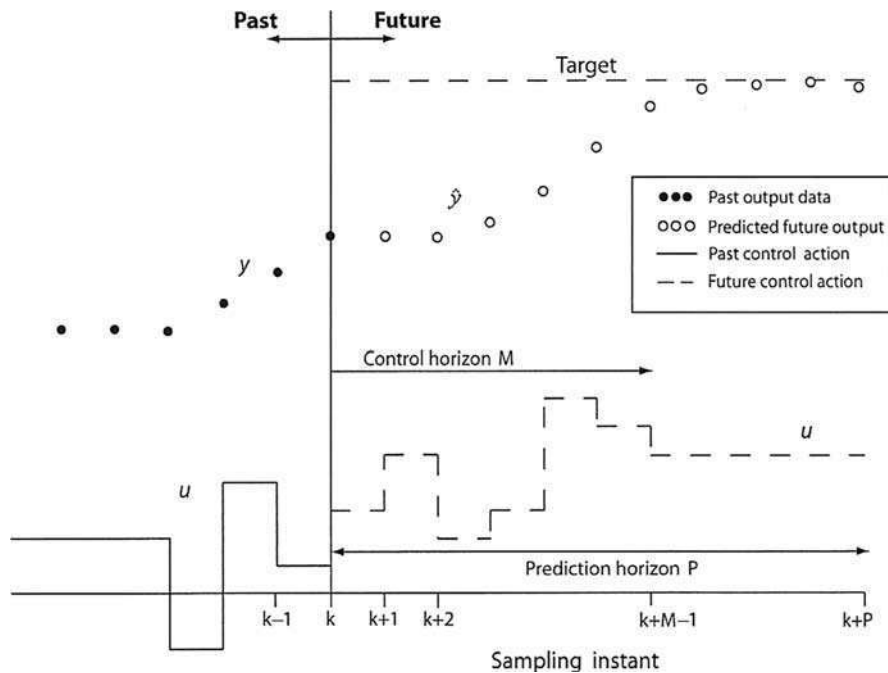


FIG. 8-43 The “moving horizon” approach of model predictive control. (Seborg, Edgar, and Mellichamp, *Process Dynamics and Control*, 2d ed., Wiley, New York, 2004.)

Inequality Constraints Inequality constraints on the future inputs or their rates of change are widely used in the MPC calculations. For example, if both upper and lower limits are required, the constraints could be expressed as

$$u^-(k) \leq u(k+j) \leq u^+(k) \quad \text{for } j = 0, 1, \dots, M-1 \quad (8-67)$$

$$\Delta u^-(k) \leq \Delta u(k+j) \leq \Delta u^+(k) \quad \text{for } j = 0, 1, \dots, M-1 \quad (8-68)$$

where B_j and C_j are constants. Constraints on the predicted outputs are sometimes included as well:

$$y^-(k) \leq \hat{y}(k+j) \leq y^+(k) \quad \text{for } j = 0, 1, \dots, P \quad (8-69)$$

The minimization of the quadratic performance index in Eq. (8-64), subject to the constraints in Eqs. (8-67) to (8-69) and the step response model in Eq. (8-61), can be formulated as a standard QP (quadratic programming) problem. Consequently, efficient QP solution techniques can be employed. When the inequality constraints in Eqs. (8-67) to (8-69) are omitted, the optimization problem has an analytical solution (Camacho and Bordons, *Model Predictive Control*, 2d ed., Springer-Verlag, New York, 2004; Maciejowski, *Predictive Control with Constraints*, Prentice-Hall, Upper Saddle River, N.J., 2002). If the quadratic terms in Eq. (8-64) are replaced by linear terms, an LP (linear programming) problem results that can also be solved by using standard methods. This MPC formulation for SISO control problems can easily be extended to MIMO problems.

Implementation of MPC For a new MPC application, a cost/benefit analysis is usually performed prior to project approval. Then the steps involved in the implementation of MPC can be summarized as follows (Hokanson and Gerstle, *Dynamic Matrix Control Multivariable Controllers*, in *Practical Distillation Control*, Luyben (ed.), Van Nostrand Reinhold, New York, 1992, p. 248; Qin and Badgwell, *Control Eng. Practice*, 11: 773, 2003).

Step 1: Initial Controller Design The first step in MPC design is to select the controlled, manipulated, and measured disturbance variables. These choices determine the structure of the MPC system and should be based on process knowledge and control objectives. In typical

applications the number of controlled variables ranges from 5 to 40, and the number of manipulated variables is typically between 5 and 20.

Step 2: Pretest Activity During the pretest activity, the plant instrumentation is checked to ensure that it is working properly and to decide whether additional sensors should be installed. The pretest data can be used to estimate the steady-state gain and approximate settling times for each input/output pair. This information is used to plan the full plant tests of step 3.

As part of the pretest, it is desirable to benchmark the performance of the existing control system for later comparison with MPC performance (step 8).

Step 3: Plant Tests The dynamic model for the MPC calculations is developed from data collected during special plant tests. The excitation for the plant tests usually consists of changing an input variable or a disturbance variable (if possible) from one value to another, using either a series of step changes with different durations or a pseudorandom binary sequence (PRBS). To ensure that sufficient data are obtained for model identification, each input variable is typically moved to a new value, 8 to 15 times during a plant test (Qin and Badgwell, *Control Eng. Practice*, 11: 773, 2003). Identification during closed-loop operation is becoming more common.

Step 4: Model Development The dynamic model is developed from the plant test data by selecting a model form (e.g., a step response model) and then estimating the model parameters. However, first it is important to eliminate periods of test data where plant upsets or other abnormal situations have occurred. Decisions to omit portions of the test data are based on visual inspection of the data, knowledge of the process, and experience. Parameter estimation is usually based on least squares estimation.

Step 5: Control System Design and Simulation The preliminary control system design from step 1 is critically evaluated and modified, if necessary. Then the MPC design parameters are selected including the sampling periods, weighting factors, and control and prediction horizons. Next, the closed-loop system is simulated, and the MPC design parameters are adjusted, if necessary, to obtain satisfactory control system performance and robustness over the specified range of operating conditions.

Step 6: Operator Interface Design and Operator Training Operator training is important because MPC concepts such as predictive control, multivariable interactions, and constraint handling are very different from conventional regulatory control concepts. Thus, understanding why the MPC system responds the way that it does, especially for unusual operating conditions, can be a challenge for both operators and engineers.

Step 7: Installation and Commissioning After an MPC control system is installed, it is first evaluated in a "prediction mode." Model predictions are compared with measurements, but the process continues to be controlled by the existing control system. After the output predictions are judged to be satisfactory, the calculated MPC control moves are evaluated to determine if they are reasonable. Finally, the MPC software is evaluated during closed-loop operation with the calculated control moves implemented as set points to the DCS control loops. The MPC design parameters are tuned, if necessary. The commissioning period typically requires some troubleshooting and can take as long as, or even longer than, the plant tests of step 3.

Step 8: Measuring Results and Monitoring Performance The evaluation of MPC system performance is not easy, and widely accepted metrics and monitoring strategies are not available. However, useful diagnostic information is provided by basic statistics such as the means and standard deviations for both measured variables and calculated quantities, such as control errors and model residuals. Another useful statistic is the relative amount of time that an input is saturated or a constraint is violated, expressed as a percentage of the total time the MPC system is in service.

Integration of MPC and Online Optimization As indicated in Fig. 8-41, significant potential benefits can be realized by using a combination of MPC and online optimization. At present, most commercial MPC packages integrate the two methodologies in a hierarchical configuration such as the one shown in Fig. 8-44. The MPC calculations

are performed quite often (e.g., every 1 to 10 min) and implemented as set points for PID control loops at the DCS level. The targets and constraints for the MPC calculations are generated by solving a steady-state optimization problem (LP or QP) based on a linear process model. These calculations may be performed as often as the MPC calculations. As an option, the targets and constraints for the LP or QP optimization can be generated from a nonlinear process model using a nonlinear optimization technique. These calculations tend to be performed less frequently (e.g., every 1 to 24 h) due to the complexity of the calculations and the process models.

The combination of MPC and frequent online optimization has been successfully applied in oil refineries and petrochemical plants around the world.

REAL-TIME PROCESS OPTIMIZATION

GENERAL REFERENCES: Biegler, Grossmann, and Westerberg, *Systematic Methods of Chemical Process Design*, Prentice-Hall, Upper Saddle River, N.J., 1997. Darby and White, *On-line Optimization of Complex Process Units*, *Chem. Engr. Prog.*, **84**(10): 51, 1998. Edgar, Himmelblau, and Lasdon, *Optimization of Chemical Processes*, 2d ed., McGraw-Hill, New York, 2001. Forbes, Marlin, and MacGregor, *Model Selection Criteria for Economics-Based Optimizing Control*, *Comp. Chem. Engng.*, **18**: 497, 1994; Marlin and Hrymak, *Real-Time Optimization of Continuous Processes*, *Chem. Proc. Conf. V, AIChE Symp. Ser.*, **93**(316): 156, 1997. Narashimhan and Jordache, *Data Reconciliation and Gross Error Detection*, Gulf Publishing, Houston, Tex., 2000. Nash and Sofer, *Linear and Nonlinear Programming*, McGraw-Hill, New York, 1996. Shobrys and White, *Planning, Scheduling, and Control Systems: Why They Cannot Work Together*, *Comp. Chem. Engng.*, **26**: 149, 2002. Timmons, Jackson, and White, *Distinguishing On-line Optimization Benefits from Those of Advanced Controls*, *Hydrocarb. Proc.*, **79**(6): 69, 2000.

The chemical industry has undergone significant changes during the past 20 years due to the increased cost of energy and raw materials, more stringent environmental regulations, and intense worldwide competition. Modifications of both plant design procedures and plant operating conditions have been implemented to reduce costs and meet constraints. One of the most important engineering tools that can be employed in such activities is optimization. As plant computers have become more powerful, the size and complexity of problems that can be solved by optimization techniques have correspondingly expanded. A wide variety of problems in the operation and analysis of chemical plants (as well as many other industrial processes) can be solved by optimization. Real-time optimization means that the process operating conditions (set points) are evaluated on a regular basis and optimized, as shown earlier in level 4 in Fig. 8-28. Sometimes this is called steady-state optimization or supervisory control. This section examines the basic characteristics of optimization problems and their solution techniques and describes some representative benefits and applications in the chemical and petroleum industries.

Typical problems in chemical engineering process design or plant operation have many possible solutions. Optimization is concerned with selecting the best among the entire set of solutions by efficient quantitative methods. Computers and associated software make the computations involved in the selection manageable and cost-effective. Engineers work to improve the initial design of equipment and strive for enhancements in the operation of the equipment once it is installed in order to realize the greatest production, the greatest profit, the maximum cost, the least energy usage, and so on. In plant operations, benefits arise from improved plant performance, such as improved yields of valuable products (or reduced yields of contaminants), reduced energy consumption, higher processing rates, and longer times between shutdowns. Optimization can also lead to reduced maintenance costs, less equipment wear, and better staff utilization. It is helpful to systematically identify the objective, constraints, and degrees of freedom in a process or a plant if such benefits as improved quality of designs, faster and more reliable troubleshooting, and faster decision making are to be achieved.

Optimization can take place at many levels in a company, ranging from a complex combination of plants and distribution facilities down through individual plants, combinations of units, individual pieces of equipment, subsystems in a piece of equipment, or even smaller entities.

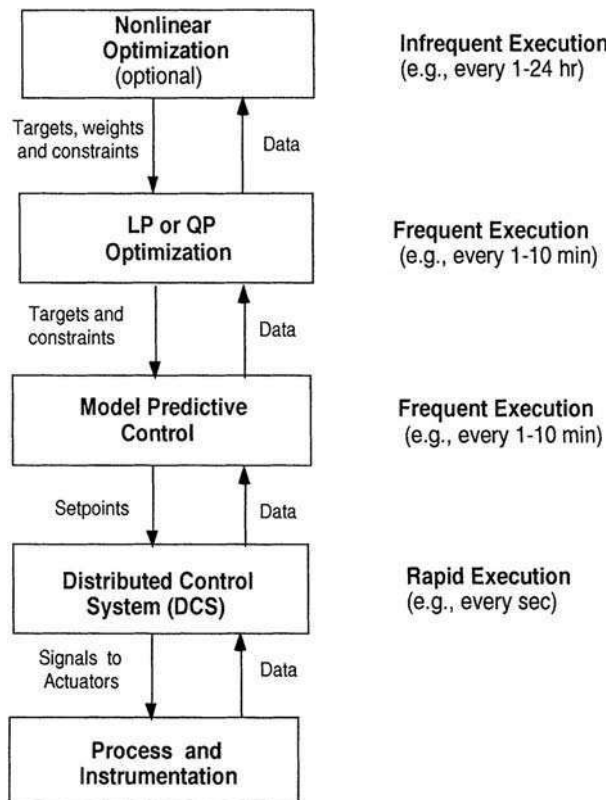


FIG. 8-44 Hierarchical control configuration for MPC and online optimization.

Problems that can be solved by optimization can be found at all these levels.

While process design and equipment specification are usually performed prior to the implementation of the process, optimization of operating conditions is carried out monthly, weekly, daily, hourly, or even every minute. Optimization of plant operations determines the set points for each unit at the temperatures, pressures, and flow rates that are the best in some sense. For example, the selection of the percentage of excess air in a process heater is quite critical and involves a balance on the fuel/air ratio to ensure complete combustion and at the same time make the maximum use of the heating potential of the fuel. Typical day-to-day optimization in a plant minimizes steam consumption or cooling water consumption, optimizes the reflux ratio in a distillation column, or allocates raw materials on an economic basis [Latour, *Hydro. Proc.*, **58**(6): 73, 1979, and **58**(7): 219, 1979].

A real-time optimization (RTO) system determines set-point changes and implements them via the computer control system without intervention from unit operators. The RTO system completes all data transfer, optimization calculations, and set-point implementation before unit conditions change that may invalidate the computed optimum. In addition, the RTO system should perform all tasks without upsetting plant operations. Several steps are necessary for implementation of RTO, including determination of the plant steady state, data gathering and validation, updating of model parameters (if necessary) to match current operations, calculation of the new (optimized) set points, and the implementation of these set points.

To determine if a process unit is at steady state, a program monitors key plant measurements (e.g., compositions, product rates, feed rates, and so on) and determines if the plant is close enough to steady state to start the sequence. Only when all the key measurements are within the allowable tolerances is the plant considered steady and the optimization sequence started. Tolerances for each measurement can be tuned separately. Measured data are then collected by the optimization computer. The optimization system runs a program to screen the measurements for unreasonable data (gross error detection). This validity checking automatically modifies the model updating calculation to reflect any bad data or when equipment is taken out of service. Data validation and reconciliation (online or offline) is an extremely critical part of any optimization system.

The optimization system then may run a parameter-fitting case that updates model parameters to match current plant operation. The integrated process model calculates such items as exchanger heat-transfer coefficients, reactor performance parameters, furnace efficiencies, and heat and material balances for the entire plant. Parameter fitting allows for continual updating of the model to account for plant deviations and degradation of process equipment. After completion of the parameter fitting, the information regarding the current plant constraints, control status data, and economic values for feed products, utilities, and other operating costs is collected. The economic values are updated by the planning and scheduling department on a regular basis. The optimization system then calculates the optimized set points. The steady-state condition of the plant is rechecked after the optimization case is successfully completed. If the plant is still steady, then the values of the optimization targets are transferred to the process control system for implementation. After a line-out period, the process control computer resumes the steady-state detection calculations, restarting the cycle.

Essential Features of Optimization Problems The solution of optimization problems involves the use of various tools of mathematics, which is discussed in detail in Sec. 3. The formulation of an optimization problem requires the use of mathematical expressions. From a practical viewpoint, it is important to mesh properly the problem statement with the anticipated solution technique. Every optimization problem contains three essential categories:

1. An objective function to be optimized (revenue function, cost function, etc.)
 2. Equality constraints (equations)
 3. Inequality constraints (inequalities)
- Categories 2 and 3 comprise the model of the process or equipment; category 1 is sometimes called the economic model.

TABLE 8-6 Six Steps Used to Solve Optimization Problems

- 1 Analyze the process itself so that the process variables and specific characteristics of interest are defined (i.e., make a list of all the variables).
- 2 Determine the criterion for optimization, and specify the objective function in terms of the above variables together with coefficients. This step provides the performance model (sometimes called the economic model, when appropriate).
- 3 Develop via mathematical expressions a valid process or equipment model that relates the input/output variables of the process and associated coefficients. Include both equality and inequality constraints. Use well-known physical principles (mass balances, energy balances), empirical relations, implicit concepts, and external restrictions. Identify the independent and dependent variables (number of degrees of freedom).
- 4 If the problem formulation is too large in scope, (a) break it up into manageable parts and/or (b) simplify the objective function and model.
- 5 Apply a suitable optimization technique to the mathematical statement of the problem.
- 6 Check the answers and examine the sensitivity of the result to changes in the coefficients in the problem and the assumptions.

No single method or algorithm of optimization exists that can be applied efficiently to all problems. The method chosen for any particular case will depend primarily on (1) the character of the objective function, (2) the nature of the constraints, and (3) the number of independent and dependent variables. Table 8-6 summarizes the six general steps for the analysis and solution of optimization problems (Edgar, Himmelblau, and Lasdon, *Optimization of Chemical Processes*, 2d ed., McGraw-Hill, New York, 2001). You do not have to follow the cited order exactly, but you should cover all the steps at some level of detail. Shortcuts in the procedure are allowable, and the easy steps can be performed first. Steps 1, 2, and 3 deal with the mathematical definition of the problem: identification of variables, specification of the objective function, and statement of the constraints. If the process to be optimized is very complex, it may be necessary to reformulate the problem so that it can be solved with reasonable effort. Later in this section, we discuss the development of mathematical models for the process and the objective function (the economic model) in typical RTO applications.

Step 5 in Table 8-6 involves the computation of the optimum point. Quite a few techniques exist to obtain the optimal solution for a problem. We describe several classes of methods below. In general, the solution of most optimization problems involves the use of a digital computer to obtain numerical answers. Over the past 15 years, substantial progress has been made in developing efficient and robust computational methods for optimization. Much is known about which methods are most successful. Virtually all numerical optimization methods involve iteration, and the effectiveness of a given technique can depend on a good first guess for the values of the variables at the optimal solution. After the optimum is computed, a sensitivity analysis for the objective function value should be performed to determine the effects of errors or uncertainty in the objective function, mathematical model, or other constraints.

Development of Process (Mathematical) Models Constraints in optimization problems arise from physical bounds on the variables, empirical relations, physical laws, and so on. The mathematical relations describing the process also comprise constraints. Two general categories of models exist:

1. Those based on physical theory
 2. Those based on strictly empirical descriptions
- Mathematical models based on physical and chemical laws (e.g., mass and energy balances, thermodynamics, chemical reaction kinetics) are frequently employed in optimization applications. These models are conceptually attractive because a general model for any system size can be developed before the system is constructed. On the other hand, an empirical model can be devised that simply correlates input/output data without any physiochemical analysis of the process. For these models, optimization is often used to fit a model to process data, using a procedure called parameter estimation. The well-known least squares curve-fitting procedure is based on optimization theory, assuming that the model parameters are contained linearly in the model. One example is the yield matrix, where the percentage yield of each product in a unit operation is estimated for each feed component

by using process data rather than employing a mechanistic set of chemical reactions.

Formulation of the Objective Function The formulation of objective functions is one of the crucial steps in the application of optimization to a practical problem. You must be able to translate the desired objective to mathematical terms. In the chemical process industries, the objective function often is expressed in units of currency per unit time (e.g., U.S. dollars per week, month, or year) because the normal industrial goal is to minimize costs or maximize profits subject to a variety of constraints.

A typical economic model involves the costs of raw materials, values of products, and costs of production as functions of operating conditions, projected sales figures, and the like. An objective function can be expressed in terms of these quantities; e.g., annual operating profit (\$/yr) might be expressed as

$$J = \sum_N F_s V_s - \sum_r F_r C_r - \text{OC} \quad (8-70)$$

where J = profit/time

$\sum_s F_s V_s$ = sum of product flow rates times respective product values (income)

$\sum_r F_r C_r$ = sum of feed flows times respective unit costs

OC = operating costs/time

Unconstrained Optimization *Unconstrained optimization* refers to the case where no inequality constraints are present and all equality constraints can be eliminated by solving for selected dependent variables followed by substitution for them in the objective function. Very few realistic problems in process optimization are unconstrained. However, the availability of efficient unconstrained optimization techniques is important because these techniques must be applied in real time, and iterative calculations may require excessive computer time. Two classes of unconstrained techniques are single-variable optimization and multivariable optimization.

Single-Variable Optimization Many real-time optimization problems can be reduced to the variation of a single-variable so as to maximize profit or some other overall process objective function. Some examples of single-variable optimization include optimizing the reflux ratio in a distillation column or the air/fuel ratio in a furnace. While most processes actually are multivariable processes with several operating degrees of freedom, often we choose to optimize only the most important variable in order to keep the strategy uncomplicated. One characteristic implicitly required in a single-variable optimization problem is that the objective function J be unimodal in the variable x .

The selection of a method for one-dimensional search is based on the tradeoff between the number of function evaluations and computer time. We can find the optimum by evaluating the objective function at many values of x , using a small grid spacing (Δx) over the allowable range of x values, but this method is generally inefficient. There are three classes of techniques that can be used efficiently for one-dimensional search: indirect, region elimination, and interpolation.

Indirect methods seek to solve the necessary condition $dJ/dx = 0$ by iteration, but these methods are not as popular as the second two classes. Region elimination methods include equal interval search, dichotomous search (or bisection), Fibonacci search, and golden section. These methods do not use information on the shape of the function (other than its being unimodal) and thus tend to be rather conservative. The third class of techniques uses repeated polynomial fitting to predict the optimum. These interpolation methods tend to converge rapidly to the optimum without being very complicated. Two interpolation methods, quadratic and cubic interpolation, are used in many optimization packages.

Multivariable Optimization The numerical optimization of general nonlinear multivariable objective functions requires that efficient and robust techniques be employed. Efficiency is important since iteration is employed. For example, in multivariable "grid" search for a

problem with four independent variables, an equally spaced grid for each variable is prescribed. For 10 values of each of the four variables, 10^4 total function evaluations would be required to find the best answer for the grid intersections, but this result may not be close enough to the true optimum and would require further search. A larger number of variables (say, 20) would require exponentially more computation, so grid search is a very inefficient method for most problems.

In multivariable optimization, the difficulty of dealing with multivariable functions is usually resolved by treating the problem as a series of one-dimensional searches. For a given starting point, a search direction s is specified, and the optimum is found by searching along that direction. The step size ϵ is the distance moved along s . Then a new search direction is determined, followed by another one-dimensional search. The algorithm used to specify the search direction depends on the optimization method selected.

There are two basic types of unconstrained optimization algorithms: (1) those requiring function derivatives and (2) those that do not. Here we give only an overview and refer the reader to Sec. 3 or the references for more details. The nonderivative methods are of interest in optimization applications because these methods can be readily adapted to the case in which experiments are carried out directly on the process. In such cases, an actual process measurement (such as yield) can be the objective function, and no mathematical model for the process is required. Methods that do not require derivatives are called direct methods and include sequential simplex (Nelder-Mead) and Powell's method. The sequential simplex method is quite satisfactory for optimization with two or three independent variables, is simple to understand, and is fairly easy to execute. Powell's method is more efficient than the simplex method and is based on the concept of conjugate search directions. This class of methods can be used in special cases but is not recommended for optimization involving more than 6 to 10 variables.

The second class of multivariable optimization techniques in principle requires the use of partial derivatives of the objective function, although finite difference formulas can be substituted for derivatives. Such techniques are called indirect methods and include the following classes:

1. Steepest descent (gradient) method
2. Conjugate gradient (Fletcher-Reeves) method
3. Newton's method
4. Quasi-newton methods

The steepest descent method is quite old and utilizes the intuitive concept of moving in the direction in which the objective function changes the most. However, it is clearly not as efficient as the other three. Conjugate gradient utilizes only first-derivative information, as does steepest descent, but generates improved search directions. Newton's method requires second-derivative information but is very efficient, while quasi-newton retains most of the benefits of Newton's method but utilizes only first-derivative information. All these techniques are also used with constrained optimization.

Constrained Optimization When constraints exist and cannot be eliminated in an optimization problem, more general methods must be employed than those described above, because the unconstrained optimum may correspond to unrealistic values of the operating variables. The general form of a nonlinear programming problem allows for a nonlinear objective function and nonlinear constraints, or

$$\begin{aligned} \text{Minimize} \quad & J(x_1, x_2, \dots, x_n) \\ \text{Subject to} \quad & h_i(x_1, x_2, \dots, x_n) = 0 \quad i = 1, r_c \\ & g_i(x_1, x_2, \dots, x_n) > 0 \quad i = 1, m_c \end{aligned} \quad (8-71)$$

In this case, there are n process variables with r_c equality constraints and m_c inequality constraints. Such problems pose a serious challenge to performing optimization calculations in a reasonable amount of time. Typical constraints in chemical process optimization include operating conditions (temperatures, pressures, and flows have limits), storage capacities, and product purity specifications.

An important class of constrained optimization problems is one in which both the objective function and the constraints are linear. The solution of these problems is highly structured and can be obtained

rapidly. The accepted procedure, linear programming (LP), has become quite popular in the past 20 years, solving a wide range of industrial problems. It is increasingly being used for online optimization. For processing plants, there are several different kinds of linear constraints that may arise, making the LP method of great utility.

1. Production limitation due to equipment throughput restrictions, storage limits, or market constraints
2. Raw material (feedstock) limitation
3. Safety restrictions on allowable operating temperatures and pressures
4. Physical property specifications placed on the composition of the final product. For blends of various products, we usually assume that a composite property can be calculated through the mass-averaging of pure-component physical properties
5. Material and energy balances of the steady-state model

The optimum in linear programming lies at the constraint intersections, which was generalized to any number of variables and constraints by George Dantzig. The simplex algorithm is a matrix-based numerical procedure for which many digital computer codes exist, for both mainframe and microcomputers (Edgar, Himmelblau, and Lasdon, *Optimization of Chemical Processes*, 2d ed., McGraw-Hill, New York, 2001; Nash and Sofer, *Linear and Nonlinear Programming*, McGraw-Hill, New York, 1996). The algorithm can handle virtually any number of inequality constraints and any number of variables in the objective function, and utilizes the observation that only the constraint boundaries need to be examined to find the optimum. In some instances, nonlinear optimization problems even with nonlinear constraints can be linearized so that the LP algorithm can be employed to solve them (called successive linear programming, or SLP). In the process industries, LP and SLP have been applied to a wide range of RTO problems, including refinery scheduling, olefins production, the optimal allocation of boiler fuel, and the optimization of a total plant.

Nonlinear Programming The most general case for optimization occurs when both the objective function and the constraints are nonlinear, a case referred to as nonlinear programming. While the ideas behind the search methods used for unconstrained multivariable problems are applicable, the presence of constraints complicates the solution procedure. All the methods discussed below have been utilized to solve nonlinear programming problems in the field of chemical engineering design and operations. Nonlinear programming is now used extensively in the area of real-time optimization.

One of the older and most accessible NLP algorithms uses iterative linearization and is called the *generalized reduced gradient (GRG) algorithm*. The GRG algorithm employs linear or linearized constraints and uses slack variables to convert all constraints to equality constraints. The GRG algorithm is used in the Excel Solver. CONOPT is a reduced gradient algorithm that works well for large-scale problems and nonlinear constraints. CONOPT and GRG work best for problems in which the number of degrees of freedom is small (the number of constraints is nearly equal to the number of variables).

Successive quadratic programming (SQP) solves a sequence of quadratic programs that approach the solution of the original NLP by linearizing the constraints and using a quadratic approximation to the objective function. Lagrange multipliers are introduced to handle constraints, and the search procedure generally employs some variation of Newton's method, a second-order method that approximates the hessian matrix using first derivatives (Biegler et al., 1997; Edgar et al., 2001). MINOS and NPSOL are software packages that are suitable for problems with large numbers of variables (more variables than equations) and constraints that are linear or nearly linear. *Successive linear programming (SLP)* is used less often for solving RTO problems. It requires linear approximations of both the objective function and the constraints but sometimes exhibits poor convergence to optima that are not located at constraint intersections.

One important class of nonlinear programming techniques is called *quadratic programming (QP)*, where the objective function is quadratic and the constraints are linear. While the solution is iterative, it can be obtained quickly as in linear programming. This is the basis for the newest type of constrained multivariable control algorithms called model predictive control, which is heavily used in the refining industry. See the earlier subsection on model predictive control for more details.

Figure 8-45 gives an overview of which optimization algorithms are appropriate for certain types of RTO problems. Software libraries such as GAMS (General Algebraic Modeling System) or NAG (Numerical Algorithms Group) offer one or more NLP algorithms, but rarely are all algorithms available from a single source. Also there are quite a few good optimization software programs that are free and can be found by a web search. No single NLP algorithm is best for every problem, so several solvers should be tested on a given application. See Edgar, Himmelblau, and Lasdon, *Optimization of Chemical Processes*, McGraw-Hill, New York, 2001.

Linear and nonlinear programming solvers have been interfaced to spreadsheet software for desktop computers. The spreadsheet has become a popular user interface for entering and manipulating numeric data. Spreadsheet software increasingly incorporates analytic tools that are accessible from the spreadsheet interface and permit access to external databases. For example, Microsoft Excel incorporates an optimization-based routine called Solver that operates on the values and formulas of a spreadsheet model. Current versions (4.0 and later) include LP and NLP solvers and mixed integer programming (MIP) capability for both linear and nonlinear problems. The user specifies a set of cell addresses to be independently adjusted (the decision variables), a set of formula cells whose values are to be constrained (the constraints), and a formula cell designated as the optimization objective.

Referring to Fig. 8-30, the highest level of process control, planning and scheduling, also employs optimization extensively, often with variables that are integer. Level 5 sets production goals to meet supply and logistics constraints and addresses time-varying capacity and workforce utilization decisions. Enterprise resource planning (ERP) and supply chain management (SCM) in level 5 refer to the links in a web of relationships involving retailing (sales), distribution, transportation, and manufacturing. Planning and scheduling usually operate over relatively long time scales and tend to be decoupled from the rest of the activities in lower levels. For example, all of the refineries owned by an oil company are usually included in a comprehensive planning and scheduling model. This model can be optimized to obtain target levels and prices for interrefinery transfers, crude oil and product allocations to each refinery, production targets, inventory targets, optimal operating conditions, stream allocations, and blends for each refinery.

Some planning and scheduling problems are mixed-integer optimization problems that involve both continuous and integer problems; whether or not to operate or use a piece of equipment is a binary (on/off) decision that arises in batch processing. Solution techniques for this type of problem include branch and bound methods and global search. This latter approach handles very complex problems with multiple optima by using algorithms such as tabu search, scatter search, simulated annealing, and genetic evolutionary algorithms (see Edgar, Himmelblau, and Lasdon).

STATISTICAL PROCESS CONTROL

In industrial plants, large numbers of process variables must be maintained within specified limits in order for the plant to operate properly. Excursions of key variables beyond these limits can have significant consequences for plant safety, the environment, product quality, and plant profitability. *Statistical process control (SPC)*, also called *statistical quality control (SQC)*, involves the application of statistical techniques to determine whether a process is operating normally or abnormally. Thus, SPC is a process monitoring technique that relies on *quality control charts* to monitor measured variables, especially product quality.

The basic SPC concepts and control chart methodology were introduced by Walter Shewhart in the 1930s. The current widespread interest in SPC techniques began in the 1950s when they were successfully applied first in Japan and then elsewhere. Control chart methodologies are now widely used to monitor product quality and other variables that are measured infrequently or irregularly. The basic SPC methodology is described in introductory statistics textbooks (e.g., Montgomery and Runger, *Applied Statistics and Probability for Engineers*, 3d ed., Wiley, New York, 2002) and some process control textbooks (e.g., Seborg, Edgar, and Mellichamp, *Process Dynamics and Control*, 2d ed., Wiley, New York, 2004).

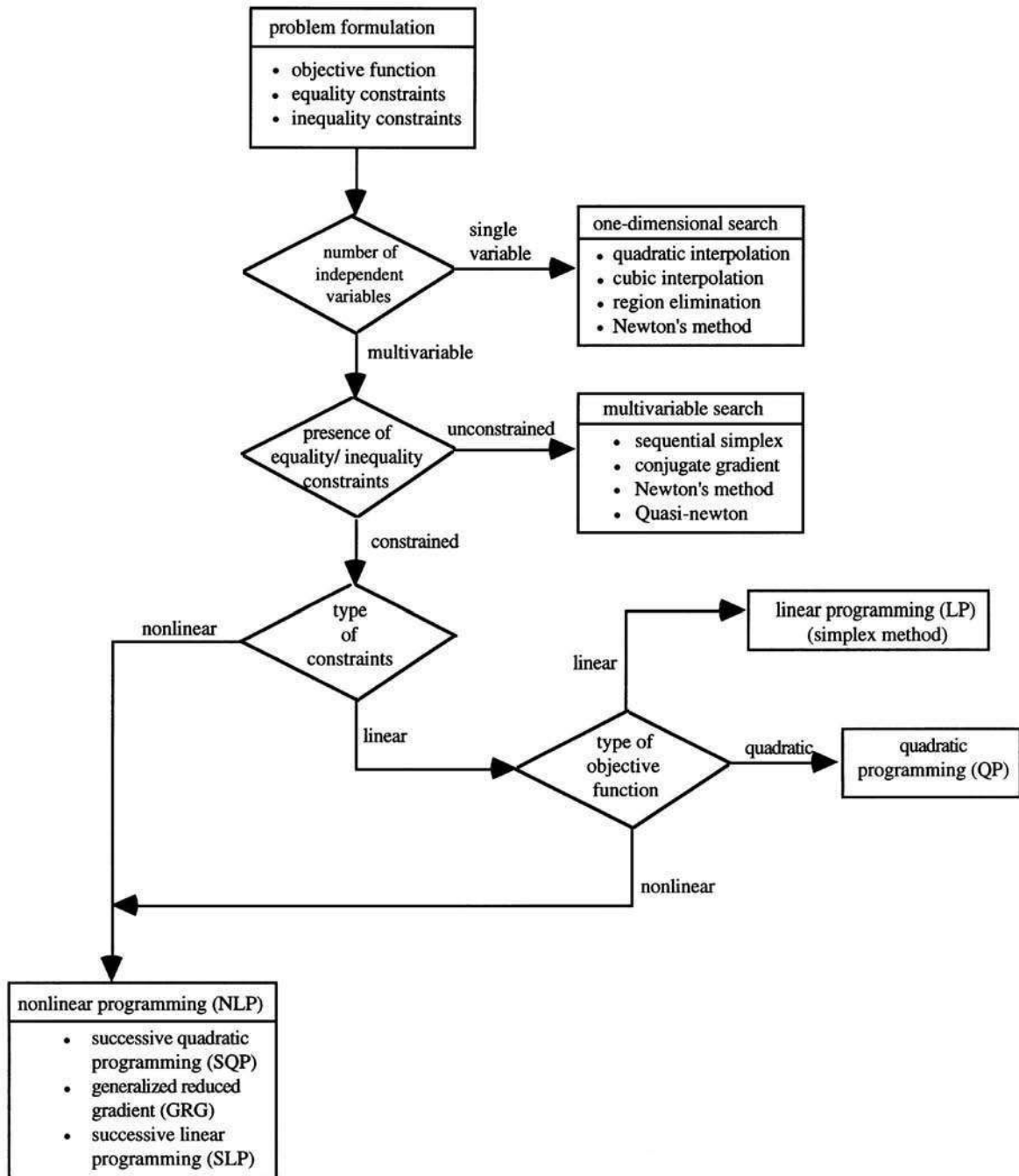


FIG. 8-45 Diagram for selection of optimization techniques with algebraic constraints and objective function.

An example of the most common control chart, the Shewhart chart, is shown in Fig. 8-46. It merely consists of measurements plotted versus sample number with control limits that indicate the range for normal process operation. The plotted data are either an individual measurement x or the sample mean $\bar{x}$ if more than one sample is measured at each sampling instant. The sample mean for k samples is calculated as,

culated as,

$$\bar{x} = \frac{1}{n} \sum_{j=1}^n x_j \quad (8-72)$$

The Shewhart chart in Fig. 8-46 has a *target* (T), an *upper control limit* (UCL), and a *lower control limit* (LCL). The target (or centerline) is the

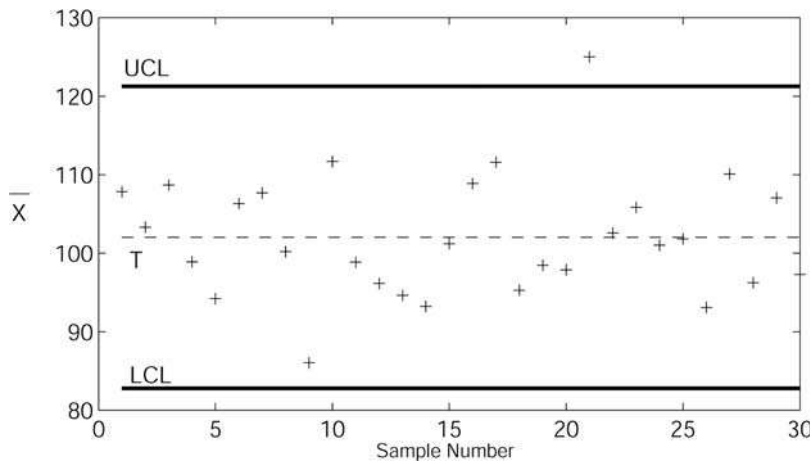


FIG. 8-46 Shewhart chart for sample mean $\bar{x}$. (Source: Seborg et al., Process Dynamics and Control, 2d ed., Wiley, New York, 2004.)

desired (or expected) value for $\bar{x}$ while the region between UCL and LCL defines the range of normal variability. If all the $\bar{x}$ data are within the control limits, the process operation is considered to be normal or “in a state of control.” Data points outside the control limits are considered to be abnormal, indicating that the process operation is out of control. This situation occurs for the 21st sample. A single measurement slightly beyond a control limit is not necessarily a cause for concern. But frequent or large chart violations should be investigated to determine the root cause.

The major objective in SPC is to use process data and statistical techniques to determine whether the process operation is normal or abnormal. The SPC methodology is based on the fundamental assumption that normal process operation can be characterized by random variations around a mean value. The random variability is caused by the cumulative effects of a number of largely unavoidable phenomena such as electrical measurement noise, turbulence, and random fluctuations in feedstock or catalyst preparation. If this situation exists, the process is said to be in a *state of statistical control* (or *in control*), and the control chart measurements tend to be normally distributed about the mean value. By contrast, frequent control chart violations would indicate abnormal process behavior or an *out-of-control* situation. Then a search would be initiated to attempt to identify the *assignable cause* or the *special cause* of the abnormal behavior.

The control limits in Fig. 8-46 (UCL and LCL) are based on the assumption that the measurements follow a normal distribution. Figure 8-47 shows the probability distribution for a normally distributed random variable x with mean μ and standard deviation σ . There is a very high probability (99.7 percent) that any measurement is within 3 standard deviations of the mean. Consequently, the control limits for x are typically chosen to be $T \pm 3\hat{\sigma}$, where $\hat{\sigma}$ is an estimate of σ . This estimate is usually determined from a set of representative data for a period of time when the process operation is believed to be typical. For the common situation in which the plotted variable is the sample mean, its standard deviation is estimated.

Shewhart control charts enable average process performance to be monitored, as reflected by the sample mean. It is also advantageous to monitor process variability. Process variability within a sample of k measurements can be characterized by its range, standard deviation, or sample variance. Consequently, control charts are often used for one of these three statistics.

Western Electric Rules Shewhart control charts can detect abnormal process behavior by comparing individual measurements with control chart limits. But the pattern of measurements can also provide useful information. For example, if 10 consecutive measurements are all increasing, then it is very unlikely that the process is in a

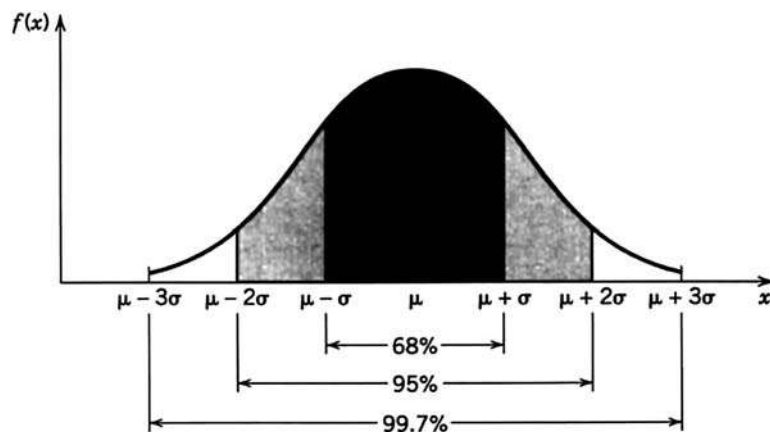


FIG. 8-47 Probabilities associated with the normal distribution. (Source: Montgomery and Runger, Applied Statistics and Probability for Engineers, 3d ed., Wiley, New York, 2002.)

state of control. A wide variety of *pattern tests* (also called *zone rules*) can be developed based on the properties of the normal distribution. For example, the following excerpts from the *Western Electric Rules* (Western Electric Company, *Statistical Quality Control Handbook*, Delmar Printing Company, Charlotte, N.C., 1956; Montgomery and Runger, *Applied Statistics and Probability for Engineers*, 3d ed., Wiley, New York, 2002) indicate that the process is out of control if one or more of the following conditions occur:

1. One data point outside the 3σ control limits
2. Two out of three consecutive data points beyond a 2σ limit
3. Four out of five consecutive data points beyond a 1σ limit and on one side of the centerline
4. Eight consecutive points on one side of the centerline

Note that the first condition is the familiar Shewhart chart limits. Pattern tests can be used to augment Shewhart charts. This combination enables out-of-control behavior to be detected earlier, but the false-alarm rate is higher than that for Shewhart charts alone.

CUSUM Control Charts Although Shewhart charts with 3σ limits can quickly detect large process changes, they are ineffective for small, sustained process changes (e.g., changes smaller than 1.5σ). Alternative control charts have been developed to detect small changes such as the CUSUM control chart. They are often used in conjunction with Shewhart charts. The *cumulative sum* (CUSUM) is defined to be a running summation of the deviations of the plotted variable from its target. If the sample mean is plotted, the cumulative sum at sampling instant k , $C(k)$, is

$$C(k) = \sum_{j=1}^k [\bar{x}(j) - T] \quad (8-73)$$

where T is the target for $\bar{x}$. During normal process operation, $C(k)$ fluctuates about zero. But if a process change causes a small shift in $\bar{x}$, $C(k)$ will drift either upward or downward.

The CUSUM control chart was originally developed using a graphical approach based on *V masks*. However, for computer calculations, it is more convenient to use an equivalent algebraic version that consists of two recursive equations

$$C^+(k) = \max[0, \bar{x}(k) - (T + K) + C^+(k-1)] \quad (8-74)$$

$$C^-(k) = \max[0, (T - K) - \bar{x}(k) + C^-(k-1)] \quad (8-75)$$

where C^+ and C^- denote the sums for the high and low directions, and K is a constant, the *slack parameter*. The CUSUM calculations are initialized by setting $C^+(0) = C^-(0) = 0$. A deviation from the target that is larger than K increases either C^+ or C^- . A control limit violation occurs when either C^+ or C^- exceeds a specified control limit (or *threshold*) H . After a limit violation occurs, that sum is reset to zero or to a specified value.

The selection of the threshold H can be based on considerations of *average run length* (ARL), the average number of samples required to detect a disturbance of specified magnitude. For example, suppose that the objective is to be able to detect if the sample mean $\bar{x}$ has shifted from the target by a small amount δ . The slack parameter K is usually specified as $K = 0.5\delta$. For the ideal situation (e.g., normally distributed, uncorrelated disturbances), ARL values have been tabulated for different values of δ , K , and H . Table 8-7 summarizes ARL

values for two values of H and different values of δ . (The values of δ are usually expressed as multiples of the standard deviation σ .) The ARL values indicate the average number of samples before a change of δ is detected. Thus the ARL values for $\delta = 0$ indicate the average time between false alarms, i.e., the average time between successive CUSUM alarms when no shift in $\bar{x}$ has occurred. Ideally, we would like the ARL value to be very large for $\delta = 0$ and small for $\delta \neq 0$. Table 8-7 shows that as the magnitude of the shift δ increases, ARL decreases and thus the CUSUM control chart detects the change faster. Increasing the value of H from 4σ to 5σ increases all the ARL values and thus provides a more conservative approach.

The relative performance of the Shewhart and CUSUM control charts is compared in Fig. 8-48 for a set of simulated data for the tensile strength of a resin. It is assumed that the tensile strength x is normally distributed with a mean of $\mu = 70$ MPa and a standard deviation of $\sigma = 3$ MPa. A single measurement is available at each sampling instant. A constant ($\sigma = 0.5\sigma = 1.5$) was added to $x(k)$ for $k \geq 10$ in order to evaluate each chart's ability to detect a small process shift. The CUSUM chart was designed using $K = 0.5\sigma$ and $H = 5\sigma$.

The Shewhart chart fails to detect the 0.5σ shift in x at $k = 10$. But the CUSUM chart quickly detects this change because a limit violation occurs at $k = 20$. The mean shift can also be detected by applying the Western Electric Rules in the previous section.

Process Capability Indices Also known as *process capability ratios*, these provide a measure of whether an in-control process is meeting its product specifications. Suppose that a quality variable x must have a volume between an *upper specification limit* (USL) and a *lower specification limit* (LSL) in order for product to satisfy customer requirements. The *capability index* C_p is defined as

$$C_p = \frac{USL - LSL}{6\sigma} \quad (8-76)$$

where σ is the standard deviation of x . Suppose that $C_p = 1$ and x is normally distributed. Based on the normal distribution, we would expect that 99.7 percent of the measurements satisfy the specification limits, or equivalently, we would expect that only 2700 out of 1 million measurements would lie outside the specification limits. If $C_p < 1$, the product specifications are satisfied; for $C_p > 1$, they are not. However, capability indices are applicable even when the data are not normally distributed.

A second capability index C_{pk} is based on average process performance $\bar{x}$ as well as process variability σ . It is defined as

$$C_{pk} = \frac{\min[\bar{x} - LSL, USL - \bar{x}]}{3\sigma} \quad (8-77)$$

Although both C_p and C_{pk} are used, we consider C_{pk} to be superior to C_p for the following reason. If $\bar{x} = T$, the process is said to be "centered" and $C_{pk} = C_p$. But for $\bar{x} \neq T$, C_p does not change, even though the process performance is worse, while C_{pk} does decrease. For this reason, C_{pk} is preferred.

If the standard deviation σ is not known, it is replaced by an estimate $\hat{\sigma}$ in Eqs. (8-76) and (8-77). For situations where there is only a single specification limit, either USL or LSL, the definitions of C_p and C_{pk} can be modified accordingly.

In practical applications, a common objective is to have a capability index of 2.0 while a value greater than 1.5 is considered to be acceptable. If the C_{pk} value is too low, it can be improved by making a change that either reduces process variability or causes $\bar{x}$ to move closer to the target. These improvements can be achieved in a number of ways that include better process control, better process maintenance, reduced variability in raw materials, improved operator training, and process changes.

Six-Sigma Approach Product quality specifications continue to become more stringent as a result of market demands and intense worldwide competition. Meeting quality requirements is especially difficult for products that consist of a very large number of components and for manufacturing processes that consist of hundreds of individual steps. For example, the production of a microelectronic device typically requires 100 to 300 batch processing steps. Suppose

TABLE 8-7 Average Run Lengths for CUSUM Control Charts

Shift from target (in multiples of σ)	ARL for $H = 4\sigma$	ARL for $H = 5\sigma$
0	168.	465.
0.25	74.2	139.
0.50	26.6	38.0
0.75	13.3	17.0
1.00	8.38	10.4
2.00	3.34	4.01
3.00	2.19	2.57

Adapted from Ryan, *Statistical Methods for Quality Improvement*, 2d ed., Wiley, New York, 2000.

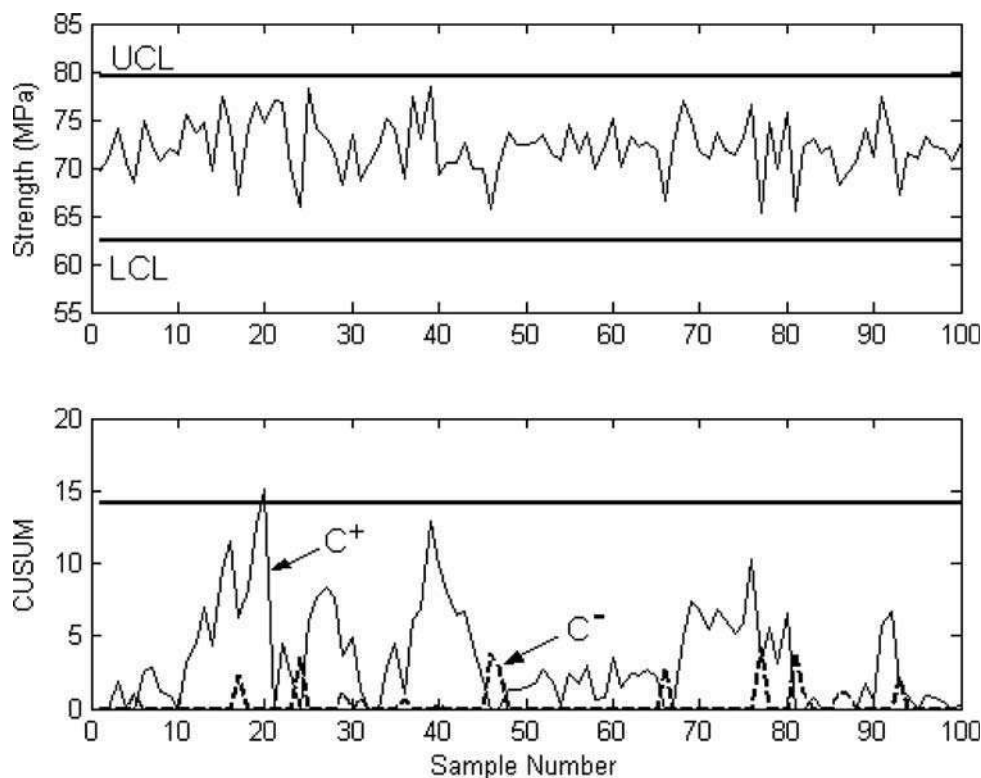


FIG. 8-48 Comparison of Shewhart and CUSUM control charts for the resin example. (Source: Seborg *et al.*, *Process Dynamics and Control*, 2d ed., Wiley, New York, 2004.)

that there are 200 steps and that each one must meet a quality specification for the final product to function properly. If each step is independent of the others and has a 99 percent success rate, the overall yield of satisfactory product is $(0.99)^{200} = 0.134$, or only 13.4 percent. This low yield is clearly unsatisfactory. Similarly, even when a processing step meets 3σ specifications (99.73 percent success rate), it will still result in an average of 2700 “defects” for every 1 million produced. Furthermore, the overall yield for this 200-step process is still only 58.2 percent.

The six-sigma approach was pioneered by the Motorola Corporation in the early 1980s as a strategy for achieving both six-sigma quality and continuous improvement. Since then, other large corporations have adopted companywide programs that apply the six-sigma approach to all their business operations, both manufacturing and nonmanufacturing. Thus, although the six-sigma approach is “data-driven” and based on statistical techniques, it has evolved into a broader management philosophy that has been implemented successfully by many large corpora-

tions. The six-sigma programs have also had a significant financial impact.

Multivariate Statistical Techniques For common SPC monitoring problems, two or more quality variables are important, and they can be highly correlated. For these situations, multivariable (or *multivariate*) SPC techniques can offer significant advantages over the single-variable methods discussed earlier. Multivariate monitoring based on the classical Hotelling’s T^2 statistic (Montgomery, *Introduction to Statistical Quality Control*, 4th ed., Wiley, New York, 2001) can be effective if the data are not highly correlated and the number of variables p is not large (for example, $p < 10$). Fortunately, alternative multivariate monitoring techniques such as *principal-component analysis* (PCA) and *partial least squares* (PLS) methods have been developed that are very effective for monitoring problems with large numbers of variables and highly correlated data [Piovoso and Hoo (eds.), Special Issue of *IEEE Control Systems Magazine*, 22(5), 2002].

UNIT OPERATIONS CONTROL

PIPING AND INSTRUMENTATION DIAGRAMS

GENERAL REFERENCES: Shinsky, *Process Control Systems*, 4th ed., McGraw-Hill, New York, 1996. Luyben, *Practical Distillation Control*, Van Nostrand Reinhold, New York, 1992. Luyben, Tyreus, and Luyben, *Plantwide Process Control*, McGraw-Hill, New York, 1998.

The piping and instrumentation (P&I) diagram provides a graphical representation of the control configuration of the process. P&I diagrams illustrate the measuring devices that provide inputs to the con-

trol strategy, the actuators that will implement the results of the control calculations, and the function blocks that provide the control logic. They may also include piping details such as line sizes and the location of hand valves and condensate traps.

The symbology for drawing P&I diagrams generally follows standards developed by the Instrumentation, Systems, and Automation Society (ISA). The chemicals, refining, and food industries generally follow this standard. The standards are updated from time to time, primarily because the continuing evolution in control system hardware and software provides additional capabilities for implementing

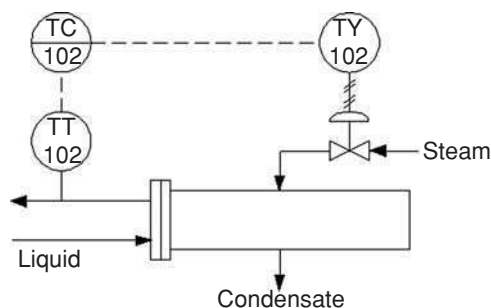


FIG. 8-49 Example of a simplified piping and instrumentation diagram.

control schemes. The ISA symbols are simple and represent a device or function as a circle containing its tag number and identifying the type of variable being controlled, e.g., pressure, and the function performed, e.g., control: PC-105. Examples of extensions of the ISA standard appear on the pages following.

Figure 8-49 presents a simplified P&I diagram for a temperature control loop that applies the ISA symbology. The measurement devices and most elements of the control logic are shown as circles:

1. TT102 is the temperature transmitter.
2. TC102 is the temperature controller.
3. TY102 is the current-to-pneumatic (I/P) transducer.

The symbol for the control valve in Fig. 8-49 is for a pneumatic modulating valve without a valve positioner.

Electronic (4- to 20-mA) signals are represented by dashed lines. In Fig. 8-49, these include the signal from the transmitter to the controller and the signal from the controller to the I/P transducer. Pneumatic signals are represented by solid lines with double crosshatching at regular intervals. The signal from the I/P transducer to the valve actuator is pneumatic.

The ISA symbology provides different symbols for different types of actuators. Furthermore, variations for the controller symbol distinguish control algorithms implemented in distributed control systems from those in panel-mounted single-loop controllers.

CONTROL OF HEAT EXCHANGERS

Steam-Heated Exchangers Steam, the most common heating medium, transfers its latent heat in condensing, causing heat flow to be proportional to steam flow. Thus a measurement of steam flow is essentially a measure of heat transfer. Consider raising a liquid from temperature T_1 to T_2 by condensing steam:

$$Q = WH = MC(T_2 - T_1) \quad (8-78)$$

where W and H are the mass flow of steam and its latent heat, M and C are the mass flow and specific heat of the liquid, and Q is the rate of heat transfer. The response of controlled temperature to steam flow is linear:

$$\frac{dT_2}{dW} = \frac{H}{MC} \quad (8-79)$$

However, the steady-state process gain described by this derivative varies inversely with liquid flow: adding a given increment of heat flow to a smaller flow of liquid produces a greater temperature rise.

Dynamically, the response of liquid temperature to a step in steam flow is that of a distributed lag, shown in Fig. 8-26 (uncontrolled). The time required to reach 63 percent complete response, $\Sigma\tau$, is essentially the residence time of the fluid in the exchanger, which is its volume divided by its flow. The residence time then varies inversely with flow. Table 8-2 gives optimum settings for PI and PID controllers for distributed lags, the proportional band varying directly with steady-state gain, and integral and derivative settings directly with $\Sigma\tau$. Since both these parameters vary inversely with liquid flow, fixed settings for the temperature controller are optimal at only one flow rate.

The variability of the process parameters with flow causes variability in load response, as shown in Fig. 8-50. The PID controller was tuned for optimum (minimum-IAE) load response at 50 percent flow. Each curve represents the response of exit temperature to a 10 percent step in liquid flow, culminating at the stated flow. The 60 percent curve is overdamped and the 40 percent curve is underdamped. The differences in gain are reflected in the amplitude of the deviation, and the differences in dynamics are reflected in the period of oscillation.

If steam flow is linear with controller output, as it is in Fig. 8-50, undamped oscillations will be produced when the flow decreases by

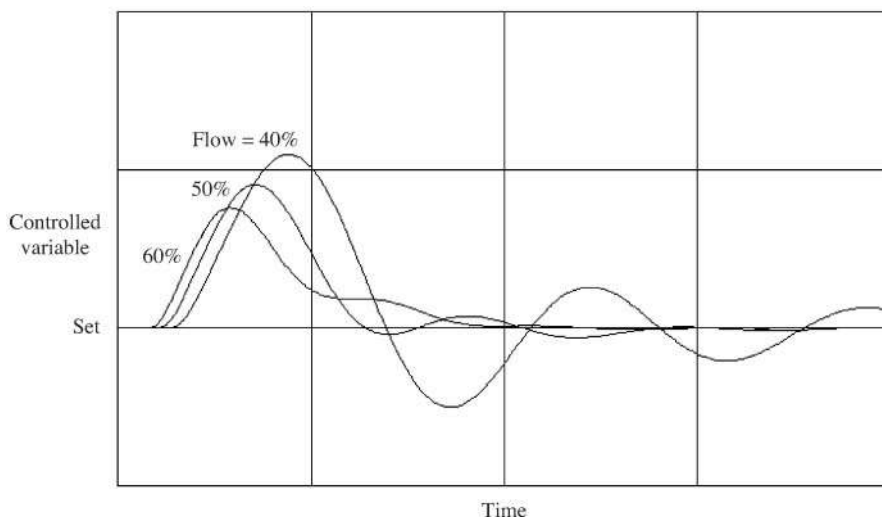


FIG. 8-50 The response of a heat exchanger varies with flow in both gain and dynamics; here the PID temperature controller was tuned for optimum response at 50 percent flow.

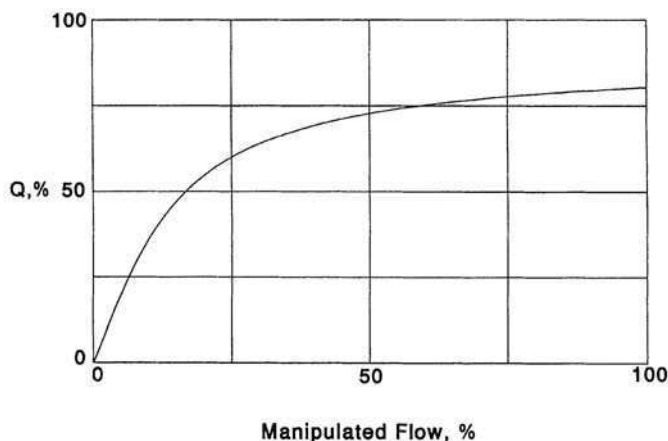


FIG. 8-51 Heat-transfer rate in sensible-heat exchange varies nonlinearly with flow of the manipulated fluid, requiring equal-percentage valve characterization.

one-third from the value at which the controller was optimally tuned—in this example at 33 percent flow. The stable operating range can be extended to one-half the original flow by using an equal-percentage (logarithmic) steam valve, whose gain varies directly with steam flow, thereby compensating for the variable process gain. Further extension requires increasing the integral setting and reducing the derivative setting from their optimum values. The best solution is to adapt all three PID settings to change inversely with measured flow, thereby keeping the controller optimally tuned for all flow rates.

Feedforward control can also be applied, as described previously under “Advanced Control Techniques.” The feedforward system solves Eq. (8-78) for the manipulated set point to the steam flow controller, first by subtracting inlet temperature T_1 from the output of the outlet temperature controller (in place of T_2), and then by multiplying the result by the dynamically compensated liquid flow measurement. If the inlet temperature is not subject to rapid or wide variation, it can be left out of the calculation. Feedforward is capable of a reduction in integrated error as much as a 100-fold, but requires the use of a steam flow loop and lead-lag compensator to approach this effectiveness. Multiplication of the controller output by liquid flow varies the feedback loop gain directly proportional to flow, extending the stable operating range of the feedback loop much as the equal-percentage steam valve did without feedforward. (This system is eminently applicable to control of fired heaters in oil refineries, which commonly provide a circulating flow of hot oil to several distillation columns and are therefore subject to frequent disturbances).

Steam flow is sometimes controlled by manipulating a valve in the condensate line rather than the steam line, because it is smaller and hence less costly. Heat transfer then is changed by raising or lowering the level of condensate flooding the heat-transfer surface, an operation that is slower than manipulating a steam valve. Protection also needs to be provided against an open condensate valve blowing steam into the condensate system.

Exchange of Sensible Heat When there is no change in phase, the rate of heat transfer is no longer linear with the flow of the manipulated stream, but is a function of the mean temperature difference ΔT_m :

$$Q = UA \Delta T_m = M_H C_H (T_{H1} - T_{H2}) = M_C C_C (T_{C2} - T_{C1}) \quad (8-79a)$$

where U and A are the overall heat-transfer coefficient and area and subscripts H and C refer to the hot and cold fluids, respectively. An example would be a countercurrent cooler, where the hot-stream outlet temperature is controlled. Using the logarithmic mean tempera-

ture difference and solving for T_{H2} give

$$T_{H2} = T_{C1} + (T_{H1} - T_{C1}) \frac{1 - M_H C_H / M_C C_C}{\epsilon - M_H C_H / M_C C_C} \quad (8-79b)$$

where

$$\epsilon = \exp \left[-UA \left(\frac{1}{M_C C_C} - \frac{1}{M_H C_H} \right) \right] \quad (8-79c)$$

At a given flow of hot fluid, the heat-transfer rate is plotted as a function of coolant flow in Fig. 8-51, as a percentage of its maximum value (corresponding to $T_{C2} = T_{C1}$). The extreme nonlinearity of this relationship requires the use of an equal-percentage coolant valve for gain compensation. The variable dynamics of the distributed lag also apply, limiting the stable operating range in the same way as for the steam-heated exchanger.

Sensible-heat exchangers are also subject to variations in the temperature of the manipulated stream, an increasingly common problem where heat is being recovered at variable temperatures for reuse. Figure 8-52 shows a temperature controller (TC) setting a heat flow controller (QC) in cascade. A measurement of the manipulated flow is multiplied by its temperature difference across the heat exchanger to calculate the current heat-transfer rate, by using the right side of Eq. (8-78). Variations in supply temperature then appear as variations in calculated heat-transfer rate, which the QC can quickly correct by adjusting the manipulated flow. An equal-percentage valve is still required to linearize the secondary loop, but the primary loop of temperature-setting heat flow is linear. Feedforward can be added by multiplying the dynamically compensated flow measurement of the other fluid by the output of the temperature controller.

When a stream is manipulated whose flow is independently determined, such as the flow of a product or of a heat-transfer fluid from a fired heater, a three-way valve is used to divert the required flow to the heat exchanger. This does not alter the linearity of the process or its sensitivity to supply variations, and it even adds the possibility of independent flow variations. The three-way valve should have equal-percentage characteristics, and heat flow control may be even more beneficial.

DISTILLATION COLUMN CONTROL

Distillation columns have four or more closed loops—increasing with the number of product streams and their specifications—all of which interact with one another to some extent. Because of this interaction, there are many possible ways to pair manipulated and controlled variables through controllers and other mathematical functions, with widely differing

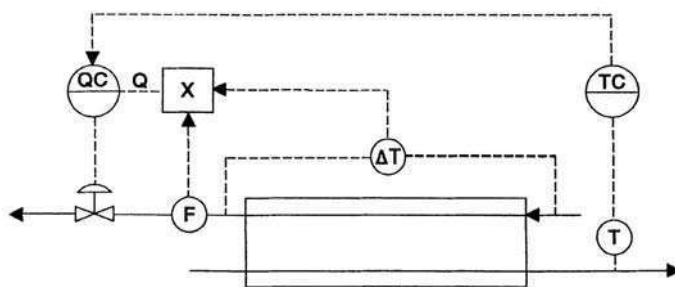


FIG. 8-52 Manipulating heat flow linearizes the loop and protects against variations in supply temperature.

degrees of effectiveness. Columns also differ from one another, so that no single rule of configuring control loops can be applied successfully to all. The following rules apply to the most common separations.

Controlling Quality of a Single Product If one of the products of a column is far more valuable than the other(s), its quality should be controlled to satisfy given specifications, and its recovery should be maximized by minimizing losses of its principal component in other streams. This is achieved by maximizing the reflux ratio consistent with flooding limits on trays, which means maximizing the flow of internal reflux or vapor, whichever is limiting. The same rule should be followed when heating and cooling have little value. A typical example is the separation of high-purity propylene from much lower-valued propane, usually achieved with the waste heat of quench water from the cracking reactors.

The most important factor affecting product quality is the material balance. In separating a feed stream F into distillate D and bottom B products, an overall mole flow balance must be maintained

$$F = D + B \quad (8-80)$$

as well as a balance on each component

$$Fz_i = Dy_i + Bx_i \quad (8-81)$$

where z , y , and x are mole fractions of component i in the respective streams. Combining these equations gives a relationship between the composition of the products and their relative portion of the feed:

$$\frac{D}{F} = 1 - \frac{B}{F} = \frac{z_i - x_i}{y_i - x_i} \quad (8-82)$$

From the above, it can be seen that control of either x_i or y_i requires both product flow rates to change with feed rate and feed composition.

Figure 8-53 shows a propylene-propane fractionator controlled at maximum boil-up by the differential pressure controller (DPC) across the trays. This loop is fast enough to reject upsets in the temperature

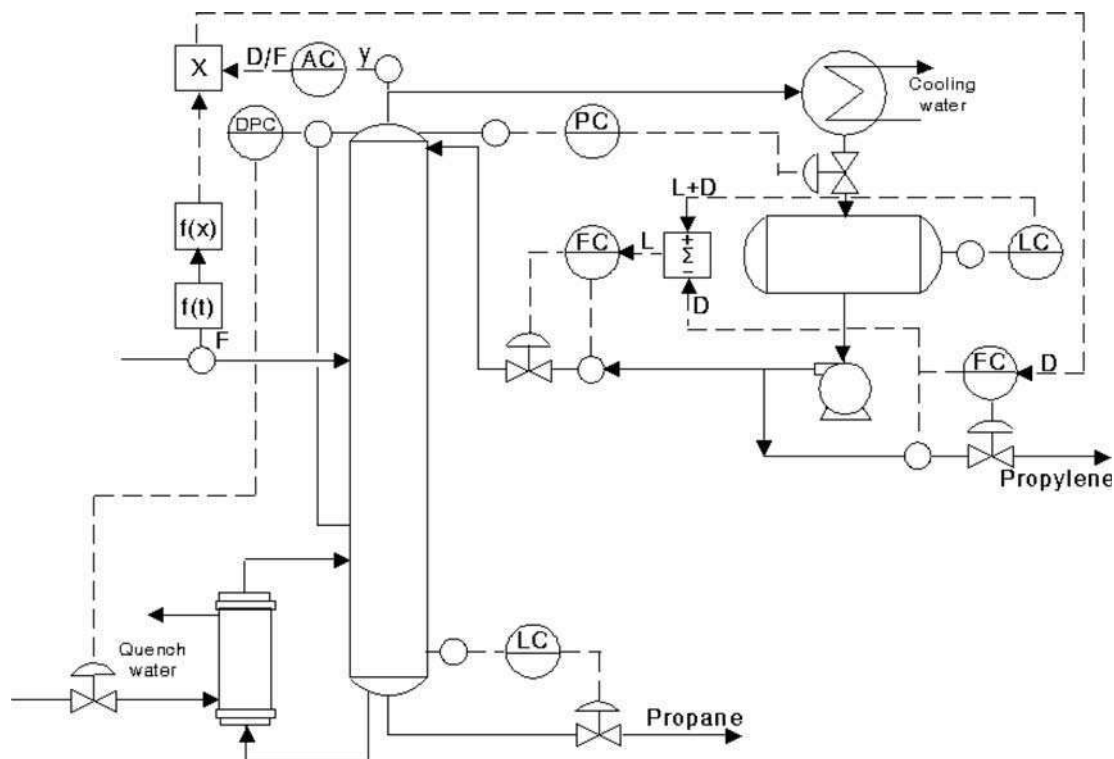


FIG. 8-53 The quality of high-purity propylene should be controlled by manipulating the material balance.

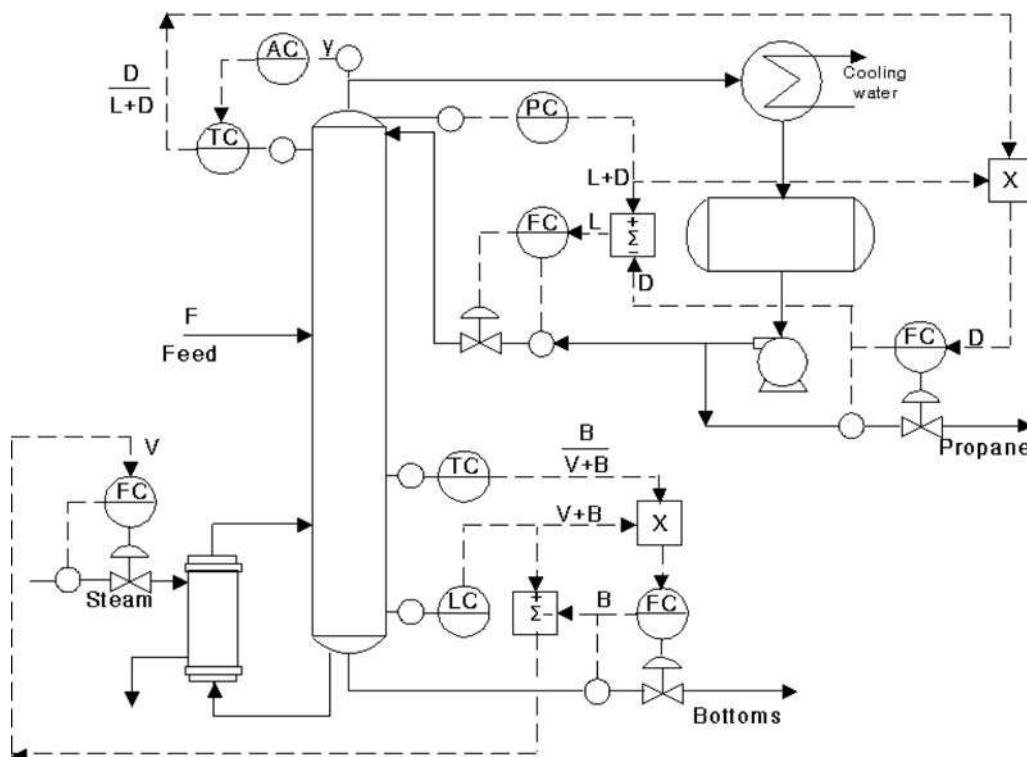


FIG. 8-54 Depropanizers require control of the quality of both products, here using reflux ratio and boil-up ratio manipulation.

of the quench water quite easily. Pressure is controlled by manipulating the heat-transfer surface in the condenser through flooding. If the condenser should become overloaded, pressure will rise above the set point, but this has no significant effect on the other control loops. Temperature measurements on this column are not helpful, as the difference between the component boiling points is too small. Propane content in the propylene distillate is measured by a chromatographic analyzer sampling the overhead vapor for fast response, and it is controlled by the analyzer controller (AC) manipulating the ratio of distillate to feed rates. The feedforward signal from feed rate is dynamically compensated by $f(t)$ and nonlinearly characterized by $f(x)$ to account for variations in propylene recovery as the feed rate changes. Distillate flow can be measured and controlled more accurately than reflux flow by a factor equal to the reflux ratio, which in this column is typically between 10 and 20. Therefore reflux flow is placed under accumulator level control (LC). Yet composition responds to the difference between internal vapor and reflux flow rates. To eliminate the lag inherent in the response of the accumulator level controller, reflux flow is driven by the subtractor in the direction opposite to distillate flow—this is essential to fast response of the composition loop. The gain of converting distillate flow changes to reflux flow changes can even be increased beyond -1 , thereby changing the accumulator level loop from a lag into a dominant lead.

Controlling Quality of Two Products Where the two products have similar values, or where heating and cooling costs are comparable to product losses, the compositions of both products should be controlled. This introduces the possibility of strong interaction between the two composition loops, as they tend to have similar speeds of response. Interaction in most columns can be minimized by controlling distillate composition with reflux ratio and bottom composition with boil-up, or preferably boil-up/bottom flow ratio. These loops are insensitive to variations in feed rate, eliminating the need for feedforward control, and they also reject heat balance upsets quite effectively.

Figure 8-54 shows a depropanizer controlled by reflux and boil-up ratios. The actual mechanism through which these ratios are manipulated is as $D/(L + D)$ and $B/(V + B)$, where L is reflux flow and V is vapor boil-up, which decouples the temperature loops from the liquid-level loops. Column pressure here is controlled by flooding both the condenser and accumulator; however, there is no level controller on the accumulator, so this arrangement will not function with an overloaded condenser. Temperatures are used as indications of composition in this column because of the substantial difference in boiling points between propane and butanes. However, off-key components such as ethane do affect the accuracy of the relationship, so that an analyzer controller is used to set the top temperature controller (TC) in cascade.

If the products from a column are especially pure, even this configuration may produce excessive interaction between the composition loops. Then the composition of the less pure product should be controlled by manipulating its own flow; the composition of the remaining product should be controlled by manipulating the reflux ratio if it is the distillate, or the boil-up ratio if it is the bottom product.

Most sidestream columns have a small flow dedicated to removing an off-key impurity entering the feed, and that stream must be manipulated to control its content in the major product. For example, an ethylene fractionator separates its feed into a high-purity ethylene sidestream, an ethane-rich bottom product, and a small flow of methane overhead. This small flow must be withdrawn to control the methane content in the ethylene product. The key impurities may then be controlled in the same way as in a two-product column.

Most volatile mixtures have a relative volatility that varies inversely with column pressure. Therefore, their separation requires less energy at lower pressure, and savings in the range of 20 to 40 percent have been achieved. Column pressure can be minimized by floating on the condenser, i.e., by operating the condenser with minimal or no restrictions. In some columns, such as the propylene-propane splitter, pressure can be left uncontrolled. Where it cannot, the set point of the

pressure controller can be indexed by an integral-only controller acting to slowly drive the pressure control valve toward a position just short of maximum cooling. In the case of a flooded condenser, the degree of reflux subcooling can be controlled in place of condenser valve position. Where column temperatures are used to indicate product composition, their measurements must be pressure-compensated.

CHEMICAL REACTORS

Composition Control The first requirement for successful control of a chemical reactor is to establish the proper stoichiometry, i.e., to control the flow rates of the reactants in the proportions needed to satisfy the reaction chemistry. In a continuous reactor, this begins by setting ingredient flow rates in ratio to one another. However, because of variations in the purity of the feed streams and inaccuracy in flow metering, some indication of excess reactant such as pH or a composition measurement should be used to trim the ratios. Many reactions are incomplete, leaving one or more reactants unconverted. They are separated from the products of the reaction and recycled to the reactor, usually contaminated with inert components. While reactants can be recycled to complete conversion (extinction), inerts can accumulate to the point of impeding the reaction and must be purged from the system. Inerts include noncondensable gases that must be vented and nonvolatiles from which volatile products must be stripped.

If one of the reactants differs in phase from the others and the product(s), it may be manipulated to close the material balance on that phase. For example, a gas reacting with liquids to produce a liquid product may be added as it is consumed to control reactor pressure; a gaseous purge would be necessary. Similarly, a liquid reacting with a gas to produce a gaseous product could be added as it is consumed to control the liquid level in the reactor; a liquid purge would be required. Where a large excess of one reactant *A* is used to minimize side reactions, the unreacted excess is sent to a storage tank for recycling. Its flow from the recycle storage tank is set in the desired ratio to the flow of reactant *B*, with the flow of fresh *A* manipulated to control the recycle tank level if the feed is a liquid, or tank pressure if it is a gas. Some catalysts travel with the reactants and must be recycled in the same way.

With batch reactors, it may be possible to add all reactants in their proper quantities initially, if the reaction rate can be controlled by injection of initiator or adjustment of temperature. In semibatch operation, one key ingredient is flow-controlled into the batch at a rate that sets the production. This ingredient should not be manipulated for temperature control of an exothermic reactor, as the loop includes two dominant lags—concentration of the reactant and heat capacity of the reaction mass—and can easily go unstable. It also presents the unfavorable dynamic of inverse response—increasing feed rate may lower temperature by its sensible heat before the increased reaction rate raises temperature.

Temperature Control Reactor temperature should always be controlled by heat transfer. Endothermic reactions require heat and therefore are eminently self-regulating. Exothermic reactions produce heat, which tends to raise reaction temperature, thereby increasing the reaction rate and producing more heat. This positive feedback is countered by negative feedback in the cooling system, which removes more heat as the reactor temperature rises. Most continuous reactors have enough heat-transfer surface relative to reaction mass that negative feedback dominates and they are self-regulating. But most batch reactors do not, and they are therefore steady-state unstable. Unstable reactors can be controlled if their temperature controller gain can be set high enough, and if their cooling system has enough margin to accommodate the largest expected disturbance in heat load. Stirred-tank reactors are lag-dominant, and their dynamics allow a high controller gain, but plug flow reactors are dead-time-dominant, preventing their temperature controller from providing enough gain to overcome steady-state instability. Therefore unstable plug flow reactors are also uncontrollable, their temperature tending to limit-cycle in a sawtooth wave. A stable reactor can become unstable as its heat-transfer surface fouls, or as the production rate is increased beyond a critical point (Shinsky, "Exothermic Reactors: The Stable, the Unstable, and the Uncontrollable," *Chem. Eng.*, pp. 54–59, March 2002).

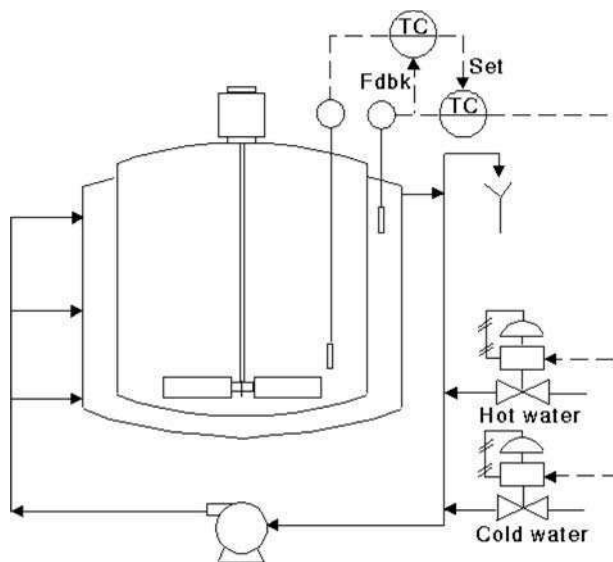


FIG. 8-55 The stirred-tank reactor temperature controller sets the coolant outlet temperature in cascade, with primary integral feedback taken from the secondary temperature measurement.

Figure 8-55 shows the recommended system for controlling the temperature of an exothermic stirred-tank reactor, either continuous or batch. The circulating pump on the coolant loop is absolutely essential to effective temperature control in keeping dead time minimum and constant—without it, dead time varies inversely with cooling load, causing limit-cycling at low loads. Heating is usually required to raise the temperature to reaction conditions, although it is often locked out of a batch reactor once the initiator is introduced. The valves are operated in split range, the heating valve opening from 50 to 100 percent of controller output and the cooling valve opening from 50 to 0 percent. The cascade system linearizes the reactor temperature response, speeds its response, and protects it from disturbances in the cooling system. The flow of heat removed per unit of coolant flow is directly proportional to the temperature rise of the coolant, which varies with both the temperature of the reactor and the rate of heat transfer from it. Using an equal-percentage cooling valve helps compensate for this nonlinearity, but incompletely.

The flow of heat across the heat-transfer surface is linear with both temperatures, leaving the primary loop with a constant gain. Using the coolant exit temperature as the secondary controlled variable as shown in Fig. 8-55 places the jacket dynamics in the secondary loop, thereby reducing the period of the primary loop. This is dynamically advantageous for a stirred-tank reactor because of the slow response of its large heat capacity. However, a plug flow reactor cooled by an external heat exchanger lacks this heat capacity, and requires the faster response of the coolant inlet temperature loop.

Performance and robustness are both improved by using the secondary temperature measurement as the feedback signal to the integral mode of the primary controller. (This feature may be available only with controllers that integrate by positive feedback.) This places the entire secondary loop in the integral path of the primary controller, effectively pacing its integral time to the rate at which the secondary temperature is able to respond. It also permits the primary controller to be left in the automatic mode at all times without integral windup.

The primary time constant of the reactor is

$$\tau_1 = \frac{M_r C_r}{UA} \quad (8-83)$$

where M_r and C_r are the mass and heat capacity of the reactants and U and A are the overall heat-transfer coefficient and area, respectively.

The control system of Fig. 8-55 was tested on a pilot reactor where the heat-transfer area and mass could both be changed by a factor of 2, changing τ_1 by a factor of 4 as confirmed by observations of the rates of temperature rise. Yet neither controller required retuning as τ_1 varied. The primary controller should be PID and the secondary controller at least PI in this system (if the secondary controller has no integral mode, the primary will control with offset). Set-point overshoot in batch reactor control can be avoided by setting the derivative time of the primary controller higher than its integral time, but this is effective only with interacting PID controllers.

CONTROLLING EVAPORATORS

The most important consideration in controlling the quality of concentrate from an evaporator is the forcing of the vapor withdrawal rate to match the flow of excess solvent entering the feed. The mass flow rates of solid material entering and leaving are equal in the steady state

$$M_0 x_0 = M_n x_n \quad (8-84)$$

where M_0 and x_0 are the mass flow and solid fraction of the feed and M_n and x_n are their values in the product after n effects of evaporation. The total solvent evaporated from all the effects must then be

$$\sum W = M_0 - M_n = M_0 \left(1 - \frac{x_0}{x_n} \right) \quad (8-85)$$

For a steam-heated evaporator, each unit of steam W_0 applied produces a known amount of evaporation, based on the number of effects and their fractional economy E :

$$\sum W = nEW_0 \quad (8-86)$$

(A comparable statement can be made with regard to the power applied to a mechanical recompression evaporator.) In summary, the steam flow required to increase the solid content of the feed from x_0

to x_n is

$$W_0 = \frac{M_0(1 - x_0/x_n)}{nE} \quad (8-87)$$

The usual measuring device for feed flow is a magnetic flowmeter, which is a volumetric device whose output F must be multiplied by density ρ to produce mass flow M_0 . For most aqueous solutions fed to evaporators, the product of density and the function of solid content appearing above is linear with density:

$$F\rho \left(1 - \frac{x_0}{x_n} \right) \approx F[1 - m(\rho - 1)] \quad (8-88)$$

where slope m is determined by the desired product concentration and density is in grams per milliliter. The required steam flow in pounds per hour for feed measured in gallons per minute is then

$$W_0 = \frac{500F[1 - m(\rho - 1)]}{nE} \quad (8-89)$$

where the factor of 500 converts gallons per minute of water to pounds per hour. The factor nE is about 1.74 for a double-effect evaporator and 2.74 for a triple-effect. Using a thermocompressor (ejector) driven with 150 lb/in² steam on a single-effect evaporator gives an nE of 2.05; it essentially adds the equivalent of one effect to the evaporator train.

A cocurrent evaporator train with its controls is illustrated in Fig. 8-56. The control system applies equally well to countercurrent or mixed-feed evaporators, the principal difference being the tuning of the dynamic compensator $f(t)$, which must be done in the field to minimize the short-term effects of changes in feed flow on product quality. Solid concentration in the product is usually measured as density; feedback trim is applied by the analyzer controller AC adjusting the slope m of the density function, which is the only term related to x_n . This recalibrates the system whenever x_n must move to a new set point.

The accuracy of the system depends on controlling heat flow; therefore if steam pressure varies, compensation must be applied to correct

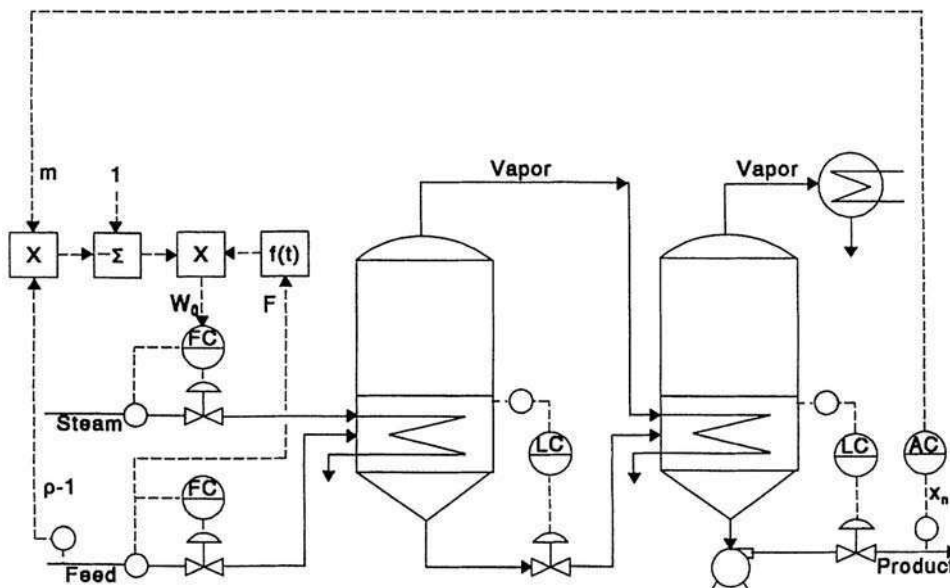


FIG. 8-56 Controlling the evaporators requires matching steam flow and evaporative load, here using feedforward control.

for both steam density and enthalpy as a function of pressure. Some evaporators must use unreliable sources of low-pressure steam. In this case, the measurement of pressure-compensated steam flow can be used to set feed flow by solving the last equation for F , using W_0 as a variable. The steam flow controller would be set for a given production rate, but the dynamically compensated steam flow measurement would be the input signal to calculate the feed flow set point. Both of these configurations are widely used in controlling corn syrup concentrators.

DRYING OPERATIONS

Controlling dryers is much different from controlling evaporators, because online measurements of feed rate and composition and product composition are rarely available. Most dryers transfer moisture from wet feed into hot dry air in a single pass. The process is generally very self-regulating, in that moisture becomes progressively harder to remove from the product as it dries: this is known as falling-rate drying. Controlling the temperature of the air leaving a cocurrent dryer tends to regulate the moisture in the product, as long as the rate and moisture content of the feed and air are reasonably constant. However, at constant outlet air temperature, product moisture tends to rise with all three of these disturbance variables.

In the absence of moisture analyzers, regulation of product quality can be improved by raising the temperature of the exhaust air in proportion to the evaporative load. The evaporative load can be estimated by the loss in temperature of the air passing through the dryer in the steady state. Changes in load are first observed in upsets in exhaust temperature at a given inlet temperature; the controller then responds by returning the exhaust air to its original temperature by changing that of the inlet air.

Figure 8-57 illustrates the simplest application of this principle as the linear relationship

$$T_0 = T_b + K \Delta T \quad (8-90)$$

where T_0 is the set point for exhaust temperature elevated above a base temperature T_b corresponding to zero-load operation and ΔT is the drop in air temperature from inlet to outlet. Coefficient K must be set to regulate product moisture over the expected range of evaporative load. If K is set too low, product moisture will increase with increasing load; if K is set too high, product moisture will decrease with increasing load. While K can be estimated from the model of a dryer, it does depend on the rate-of-drying curve for the product, its mean particle size, and whether the load variations are due primarily to changes in feed rate or feed moisture.

It is important to have the most accurate measurement of exhaust temperature attainable. Note that Fig. 8-57 shows the sensor inserted into the dryer upstream of the rotating seal, because air infiltration there could cause the temperature in the exhaust duct to read low—even lower than the wet-bulb temperature, an impossibility without either substantial heat loss or outside-air infiltration.

The calculation of the exhaust temperature set point forms a positive feedback loop capable of destabilizing the dryer. For example, an increase in evaporative load causes the controller to raise the inlet temperature, which will in turn raise the calculated set point, calling for a further increase in inlet temperature. The gain in the set-point loop K is typically well below the gain of the exhaust temperature measurement responding to the same change in inlet temperature. Negative feedback then dominates in the steady state, but the response of the exhaust temperature measurement is delayed by the dryer. A compensating lag $f(t)$ is shown inserted in the set-point loop to prevent positive feedback from dominating in the short term, which could cause cycling. Lag time can be safely set equal to the integral time of the outlet-air temperature controller.

If product moisture is measured offline, analytical results can be used to adjust K and T_b manually. If an online analyzer is used, the analyzer controller would be most effective in adjusting the bias T_b , as is done in the figure.

While a rotary dryer is shown, commonly used for grains and minerals, this control system has been successfully applied to fluid-bed drying of plastic pellets, air-lift drying of wood fibers, and spray drying of milk solids. The air may be steam-heated as shown or heated by direct combustion of fuel, provided that a representative measurement of inlet air temperature can be made. If it cannot, then evaporative load can be inferred from a measurement of fuel flow, which then would replace ΔT in the set-point calculation.

If the feed flows countercurrent to the air, as is the case when drying granulated sugar, the exhaust temperature does not respond to variations in product moisture. For these dryers, the moisture in the product can better be regulated by controlling its temperature at the point of discharge. Conveyor-type dryers are usually divided into a number of zones, each separately heated with recirculation of air, which raises its wet-bulb temperature. Only the last two zones may require indexing of exhaust air temperature as a function of ΔT .

Batch drying, used on small lots such as pharmaceuticals, begins operation by blowing air at constant inlet temperature through saturated product in constant-rate drying, where ΔT is constant at its maximum value ΔT_c . When product moisture reaches the point where falling-rate drying begins, the exhaust temperature begins to rise. The desired product moisture will be reached at a corresponding exhaust

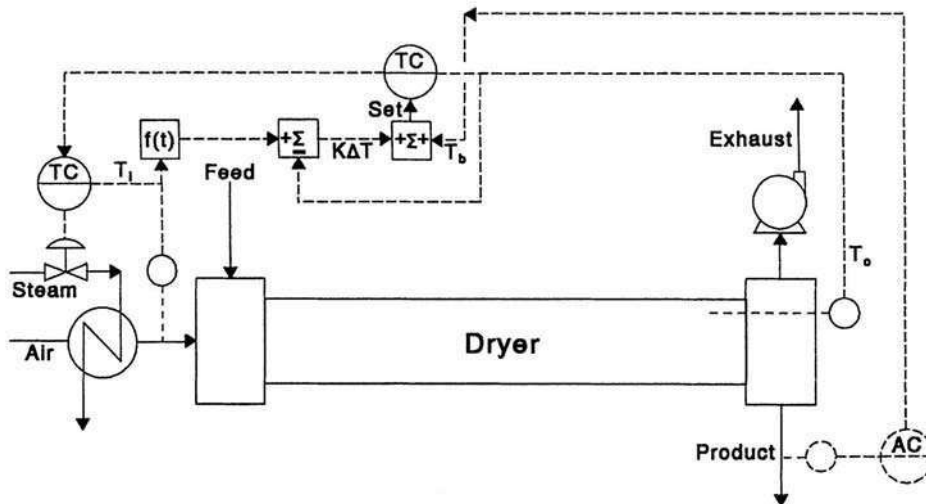


FIG. 8-57 Product moisture from a cocurrent dryer can be regulated through temperature control indexed to heat load.

temperature T_f , which is related to the temperature T_c observed during constant-rate drying, as well as to ΔT_c :

$$T_f = T_c + K \Delta T_c \quad (8-91)$$

The control system requires that the values of T_c and ΔT_c observed during the first minutes of operation be stored as the basis for the above calculation of endpoint. When the exhaust temperature then reaches

the calculated value of T_f , drying is terminated. Coefficient K can be estimated from models, but requires adjustment online to reach product specifications repeatedly. Products having different moisture specifications or particle size will require different settings of K , but the system does compensate for variations in feed moisture, batch size, air moisture, and inlet temperature. Some exhaust air may be recirculated to control the dew point of the inlet air, thereby conserving energy toward the end of the batch and when the ambient air is especially dry.

BATCH PROCESS CONTROL

GENERAL REFERENCES: Fisher, *Batch Control Systems: Design, Application, and Implementation*, ISA, Research Triangle Park, N.C., 1990. Rosenof and Ghosh, *Batch Process Automation*, Van Nostrand Reinhold, New York, 1987.

BATCH VERSUS CONTINUOUS PROCESSES

When one is categorizing process plants, the following two extremes can be identified:

1. *Commodity plants.* These plants are custom-designed to produce large amounts of a single product (or a primary product plus one or more secondary products). An example is a chlorine plant, where the primary product is chlorine and the secondary products are hydrogen and sodium hydroxide. Usually the margins (product value less manufacturing costs) for the products from commodity plants are small, so the plants must be designed and operated for best possible efficiencies. Although a few are batch, most commodity plants are continuous. Factors such as energy costs are life-and-death issues for such plants.

2. *Specialty plants.* These plants are capable of producing small amounts of a variety of products. Such plants are common in fine chemicals, pharmaceuticals, foods, and so on. In specialty plants, the margins are usually high, so factors such as energy costs are important but not life-and-death issues. As the production amounts are relatively small, it is not economically feasible to dedicate processing equipment to the manufacture of only one product. Instead, batch processing is utilized so that several products (perhaps hundreds) can be manufactured with the same process equipment. The key issue in such plants is to manufacture consistently each product in accordance with its specifications.

The above two categories represent the extremes in process configurations. The term *semibatch* designates plants in which some processing is continuous but other processing is batch. Even processes that are considered to be continuous can have a modest amount of batch processing. For example, the reformer unit within a refinery is thought of as a continuous process, but the catalyst regeneration is normally a batch process.

In a continuous process, the conditions within the process are largely the same from one day to the next. Variations in feed composition, plant utilities (e.g., cooling water temperature), catalyst activities, and other variables occur, but normally these changes either are about an average (e.g., feed compositions) or exhibit a gradual change over an extended period (e.g., catalyst activities). Summary data such as hourly averages, daily averages, and the like are meaningful in a continuous process.

In a batch process, the conditions within the process are continually changing. The technology for making a given product is contained in the product recipe that is specific to that product. Such recipes normally state the following:

1. *Raw material amounts.* This is the stuff needed to make the product.

2. *Processing instructions.* This is what must be done with the stuff to make the desired product.

This concept of a recipe is quite consistent with the recipes found in cookbooks. Sometimes the term *recipe* is used to designate only the raw material amounts and other parameters to be used in manufacturing a batch. Although appropriate for some batch processes, this concept is far too restrictive for others. For some products, the differ-

ences from one product to the next are largely physical as opposed to chemical. For such products, the processing instructions are especially important. The term *formula* is more appropriate for the raw material amounts and other parameters, with *recipe* designating the formula and the processing instructions. The above concept of a recipe permits the following three different categories of batch processes to be identified:

1. *Cyclical batch.* Both the formula and the processing instructions are the same from batch to batch. Batch operations within processes that are primarily continuous often fall into this category. The catalyst regenerator within a reformer unit is a cyclical batch process.

2. *Multigrade.* The processing instructions are the same from batch to batch, but the formula can be changed to produce modest variations in the product. In a batch PVC plant, the different grades of PVC are manufactured by changing the formula. In a batch pulp digester, the processing of each batch or cook is the same, but at the start of each cook, the process operator is permitted to change the formula values for chemical-to-wood ratios, cook time, cook temperature, and so on.

3. *Flexible batch.* Both the formula and the processing instructions can change from batch to batch. Emulsion polymerization reactors are a good example of a flexible batch facility. The recipe for each product must detail both the raw materials required and how conditions within the reactor must be sequenced to make the desired product.

Of these, the flexible batch is by far the most difficult to automate and requires a far more sophisticated control system than either the cyclical batch or the multigrade batch facility.

Batches and Recipes Each batch of product is manufactured in accordance with a product recipe, which contains all information (formula and processing instructions) required to make a batch of the product (see Fig. 8-58). For each batch of product, there will be one and only one product recipe. However, a given product recipe is normally used to make several batches of product. To uniquely identify a batch of product, each batch is assigned a unique identifier called the batch ID. Most companies adopt a convention for generating the batch ID, but this convention varies from one company to the next. In most batch facilities, more than one batch of product will be in some stage of production at any given time. The batches in progress may or may not be using the same recipe. The maximum number of batches that can be in progress at any given time is a function of the equipment configuration for the plant.

The existence of multiple batches in progress at a given time presents numerous opportunities for the process operator to make errors, such as charging a material to the wrong batch. Charging a material to the wrong batch is almost always detrimental to the batch to which the material is incorrectly charged. Unless this error is recognized quickly so that the proper charge can be made, the error is also detrimental to the batch to which the charge was supposed to have been made. Such errors usually lead to an off-specification batch, but the consequences could be more serious and could result in a hazardous condition.

Recipe management refers to the assumption of such duties by the control system. Each batch of product is tracked throughout its production, which may involve multiple processing operations on various pieces of processing equipment. Recipe management ensures that all actions specified in the product recipe are performed on each batch of product made in accordance with that recipe. As the batch proceeds from one piece of processing equipment to the next, recipe management

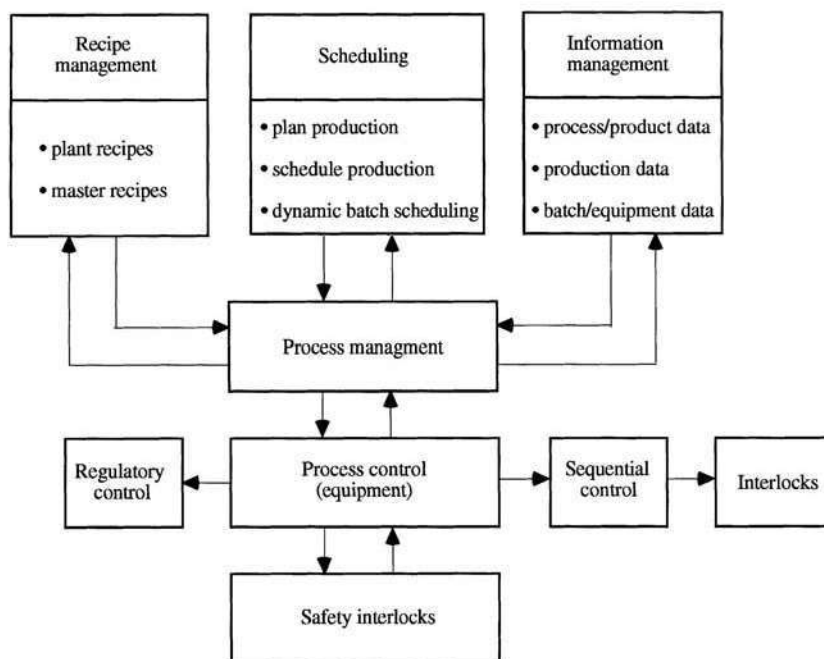


FIG. 8-58 Batch control overview.

is also responsible for ensuring that the proper type of process equipment is used and that this processing equipment is not currently in use by another batch.

By assuming such responsibilities, the control system greatly reduces the incidences where operator error results in off-specification batches. Such a reduction in error is essential to implement just-in-time production practices, where each batch of product is manufactured at the last possible moment. When a batch (or batches) is made today for shipment by overnight truck, there is insufficient time for producing another batch to make up for an off-specification batch.

Routing and Production Monitoring In some facilities, batches are individually scheduled. However, in most facilities, production is scheduled by product runs (also called process orders), where a run is the production of a stated quantity of a given product. From the stated quantity and the standard yield of each batch, the number of batches can be determined. As this is normally more than one batch of product, a production run is normally a sequence of some number of batches of the same product.

In executing a production run, the following issues must be addressed (see Fig. 8-58):

1. *Processing equipment must be dedicated to making the run.* More than one run is normally in progress at a given time. The maximum number of runs simultaneously in progress depends on the equipment configuration of the plant. Routing involves determining which processing equipment will be used for each production run.

2. *Raw material must be utilized.* When a production run is scheduled, the necessary raw materials must be allocated to the production run. As the individual batches proceed, the consumption of raw materials must be monitored for consistency with the allocation of raw materials to the production run.

3. *The production quantity for the run must be achieved by executing the appropriate number of batches.* The number of batches is determined from a standard yield for each batch. However, some batches may achieve yields higher than the standard yield, but other

batches may achieve yields lower than the standard yield. The actual yields from each batch must be monitored, and significant deviations from the expected yields must be communicated to those responsible for scheduling production.

The last two activities are key components of production monitoring, although production monitoring may also involve other activities such as tracking equipment utilization.

Production Scheduling In this regard, it is important to distinguish between scheduling runs (sometimes called long-term scheduling) and assigning equipment to runs (sometimes called routing or short-term scheduling). As used here production scheduling refers to scheduling runs and is usually a corporate-level as opposed to a plant-level function. Short-term scheduling or routing was previously discussed and is implemented at the plant level. The long-term scheduling is basically a material resources planning (MRP) activity involving the following:

1. *Forecasting.* Orders for long-delivery raw materials are issued at the corporate level based on the forecast for the demand for products. The current inventory of such raw materials is also maintained at the corporate level. This constitutes the resources from which products can be manufactured. Functions of this type are now incorporated into supply chain management.

2. *Orders for products.* Orders are normally received at the corporate level and then assigned to individual plants for production and shipment. Although the scheduling of some products is based on required product inventory levels, scheduling based on orders and shipping directly to the customer (usually referred to as just-in-time) avoids the costs associated with maintaining product inventories.

3. *Plant locations and capacities.* While producing a product at the nearest plant usually lowers transportation costs, plant capacity limitations sometimes dictate otherwise. Any company competing in the world economy needs the flexibility to accept orders on a worldwide basis and then assign them to individual plants to be filled. Such a function is logically implemented within the corporate-level information technology framework.

BATCH AUTOMATION FUNCTIONS

Automating a batch facility requires a spectrum of functions.

Interlocks Some of these are provided for safety and are properly called safety interlocks. However, others are provided to avoid mistakes in processing the batch. When safety is not involved, terms such as *permissives* and *process actions* are sometimes used in lieu of interlocks. Some understand the term *interlock* to have a connection to safety (interlock will be subsequently defined as a protective response initiated on the detection of a process hazard).

Discrete Device States Discrete devices such as two-position valves can be driven to either of two possible states. Such devices can be optionally outfitted with limit switches that indicate the state of the device. For two-position valves, the following combinations are possible:

1. No limit switches
2. One limit switch on the closed position
3. One limit switch on the open position
4. Two limit switches

In process control terminology, the discrete device driver is the software routine that generates the output to a discrete device such as a valve and also monitors the state feedback information to ascertain that the discrete device actually attains the desired state. Given the variety of discrete devices used in batch facilities, this logic must include a variety of capabilities. For example, valves do not instantly change states; instead each valve exhibits a travel time for the change from one state to another. To accommodate this characteristic of the field device, the processing logic within the discrete device driver must provide for a user-specified transition time for each field device. When equipped with limit switches, the potential states for a valve are as follows:

1. *Open*. The valve has been commanded to open, and the limit switch inputs are consistent with the open state.
2. *Closed*. The valve has been commanded to close, and the limit switch inputs are consistent with the closed state.
3. *Transition*. This is a temporary state that is only possible after the valve has been commanded to change state. The limit switch inputs are not consistent with the commanded state, but the transition time has not expired.

4. *Invalid*. The transition time has expired, and the limit switch inputs are not consistent with the commanded state for the valve.

The invalid state is an abnormal condition that is generally handled in a manner similar to process alarms. The transition state is not considered to be an abnormal state but may be implemented in either of the following ways:

1. *Drive and wait*. Further actions are delayed until the device attains its commanded state.
2. *Drive and proceed*. Further actions are initiated while the device is in the transition state.

The latter is generally necessary for devices with long travel times, such as flush-fitting reactor discharge valves that are motor-driven. Closing of such valves is normally done via drive and wait; however, drive and proceed is usually appropriate when opening the valve. Although two-state devices are most common, the need occasionally arises for devices with three or more states. For example, an agitator may be on high speed, on slow speed, or off.

Process States Batch processing usually involves imposing the proper sequence of states on the process. For example, a simple blending sequence might be as follows:

1. Transfer specified amount of material from tank A to tank R. The process state is "transfer from A."
2. Transfer specified amount of material from tank B to tank R. The process state is "transfer from B."
3. Agitate for specified time. The process state is "agitate without cooling."
4. Cool (with agitation) to specified target temperature. The process state is "agitate with cooling."

For each process state, the various discrete devices are expected to be in a specified device state. For process state "transfer from A," the device states might be as follows:

1. Tank A discharge valve: open
2. Tank R inlet valve: open

3. Tank A transfer pump: running
4. Tank R agitator: off
5. Tank R cooling valve: closed

For many batch processes, process state representations are a very convenient mechanism for representing the batch logic. A grid or table can be constructed, with the process states as rows and the discrete device states as columns (or vice versa). For each process state, the state of every discrete device is specified to be one of the following:

1. Device state 0, which may be valve closed, agitator off, and so on
2. Device state 1, which may be valve open, agitator on, and so on
3. No change or don't care

This representation is easily understandable by those knowledgeable about the process technology and is a convenient mechanism for conveying the process requirements to the control engineers responsible for implementing the batch logic.

Many batch software packages also recognize process states. A configuration tool is provided to define a process state. With such a mechanism, the batch logic does not need to drive individual devices but can simply command that the desired process state be achieved. The system software then drives the discrete devices to the device states required for the target process state. This normally includes the following:

1. Generating the necessary commands to drive each device to its proper state.
2. Monitoring the transition status of each device to determine when all devices have attained their proper states.

3. Continuing to monitor the state of each device to ensure that the devices remain in their proper states. Should any discrete device not remain in its target state, failure logic must be initiated.

Regulatory Control For most batch processes, the discrete logic requirements overshadow the continuous control requirements. For many batch processes, the continuous control can be provided by simple loops for flow, pressure, level, and temperature. However, very sophisticated advanced control techniques are occasionally applied. As temperature control is especially critical in reactors, the simple feedback approach is replaced by model-based strategies that rival, if not exceed, the sophistication of advanced control loops in continuous plants.

In some installations, alternative approaches for regulatory control may be required. Where a variety of products are manufactured, the reactor may be equipped with alternative heat removal capabilities, including the following:

1. Jacket filled with cooling water. Most such jackets are once-through, but some are recirculating.
2. Heat exchanger in a pump-around loop.
3. Reflux condenser.

The heat removal capability to be used usually depends on the product being manufactured. Therefore, regulatory loops must be configured for each possible option, and sometimes for certain combinations of the possible options. These loops are enabled and disabled depending on the product being manufactured.

The interface between continuous controls and sequence logic (discussed shortly) is also important. For example, a feed might be metered into a reactor at a variable rate, depending on another feed or possibly on reactor temperature. However, the product recipe calls for a specified quantity of this feed. The flow must be totalized (i.e., integrated), and when the flow total attains a specified value, the feed must be terminated. The sequence logic must have access to operational parameters such as controller modes. That is, the sequence logic must be able to switch a controller to manual, automatic, or cascade. Furthermore, the sequence logic must be able to force the controller output to a specified value.

Sequence Logic Sequence logic must not be confused with discrete logic. Discrete logic is especially suitable for interlocks or permissives; e.g., the reactor discharge valve must be closed for the feed valve to be opened. Sequence logic is used to force the process to attain the proper sequence of states. For example, a feed preparation might be to first charge A, then charge B, next mix, and finally cool. Although discrete logic can be used to implement sequence logic, other alternatives are often more attractive.

Sequence logic is often, but not necessarily, coupled with the concept of a process state. Basically, the sequence logic determines when the process should proceed from the current state to the next and sometimes what the next state should be.

Sequence logic must encompass both normal and abnormal process operations. Thus, sequence logic is often viewed as consisting of two distinct but related parts:

1. *Normal logic.* This sequence logic provides for the normal or expected progression from one process state to another.
2. *Failure logic.* This logic provides for responding to abnormal conditions, such as equipment failures.

Of these, the failure logic can easily be the most demanding. The simplest approach is to stop or hold on any abnormal condition and let the process operator sort things out. However, this is not always acceptable. Some failures lead to hazardous conditions that require immediate action; waiting for the operator to decide what to do is not acceptable. The appropriate response to such situations is best determined in conjunction with the process hazards analysis.

No single approach has evolved as the preferred way to implement sequence logic. The approaches utilized include the following:

1. *Discrete logic.* Although sequence logic is different from discrete logic, sequence logic can be implemented using discrete logic capabilities. Simple sequences are commonly implemented as ladder diagrams in programmable logic controllers (PLCs). Sequence logic can also be implemented using the boolean logic functions provided by a distributed control system (DCS), although this approach is now infrequently pursued.

2. *Programming languages.* Traditional procedural languages do not provide the necessary constructs for implementing sequence logic. This necessitates one of the following:

- a. *Special languages.* The necessary extensions for sequence logic are provided by extending the syntax of the programming language. This is the most common approach within distributed control systems. The early implementations used BASIC as the starting point for the extensions; the later implementations used C as the starting point. A major problem with this approach is portability, especially from one manufacturer to the next but sometimes from one product version to the next within the same manufacturer's product line.

- b. *Subroutine or function libraries.* The facilities for sequence logic are provided via subroutines or functions that can be referenced from programs written in FORTRAN or C. This requires a general-purpose program development environment and excellent facilities to trap the inevitable errors in such programs. Operating systems with such capabilities have long been available on the larger computers, but not for the microprocessors utilized within DCSs. However, such operating systems are becoming more common within DCSs.

3. *State machines.* This technology is commonly applied within the discrete manufacturing industries. However, its migration to process batch applications has been limited.

4. *Graphical implementations.* For sequence logic, the flowchart traditionally used to represent the logic of computer programs must be extended to provide parallel execution paths. Such extensions have been implemented in a graphical representation generally referred to as a sequential function chart, which is a derivative of an earlier technology known as Grafset. As process engineers have demonstrated a strong dislike for ladder logic, most PLC manufacturers now provide sequential function charts either in addition to or as an alternative to ladder logic. Many DCS manufacturers also provide sequential function charts either in addition to or as an alternative to special sequence languages.

INDUSTRIAL APPLICATIONS

An industrial example requiring simple sequence logic is the effluent tank with two sump pumps illustrated in Fig. 8-59. There are two sump pumps, A and B. The tank is equipped with three level switches, one for low level (LL), one for high level (LH), and one for high-high level (LHH). All level switches actuate on rising level. The logic is to be as follows:

1. When level switch LH actuates, start one sump pump. This must alternate between the sump pumps. If pump A is started on this occasion, then pump B must be started on the next occasion.

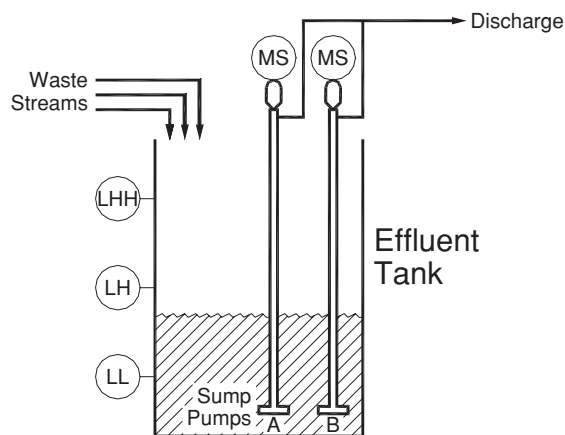


FIG. 8-59 Effluent tank process.

2. When level switch LHH actuates, start the other sump pump.
 3. When level switch LL deactuates, stop all sump pumps.
- Once a sump pump is started, it is not stopped until level switch LL deactuates. With this logic, one, both, or no sump pump may be running when the level is between LL and LH. Either one or both sump pumps may be running when the level is between LH and LHH.

Figure 8-60a presents the ladder logic implementation of the sequence logic. Ladder diagrams were originally developed for representing hardwired logic, but are now widely used in PLCs. The vertical bar on the left provides the source of power; the vertical bar on the right is ground. If a coil is connected between the power source and ground, the coil will be energized. If a circuit consisting of a set of contacts is inserted between the power source and the coil, the coil will be energized only if power can flow through the circuit. This will depend on the configuration of the circuit and the states of the contacts within the circuit. Ladder diagrams are constructed as rungs, with each rung consisting of a circuit of contacts and an output coil.

Contacts are represented as vertical bars. A vertical bar represents a normally open contact; power flows through this contact only if the device with which the contact is associated is actuated (energized). Vertical bars separated by a slash represent a normally closed contact; power flows through this contact only if the device with which the contact is associated is not actuated. The level switches actuate on rising level. If the vessel level is below the location of the switch, the normally open contact is open and the normally closed contact is closed. If the level is above the location of the switch, the normally closed contact is closed and the normally open contact is open.

The first rung in Fig. 8-60a is for pump A. It will run if one (or more) of the following conditions is true:

1. Level is above LH and pump A is the lead pump. A coil (designated as LeadIsB) will be subsequently provided to designate the pump to be started next (called the lead pump). If this coil is energized, pump B is the lead pump. Hence, pump A is to be started at LH if this coil is not energized, hence the use of the normally closed contact on coil LeadIsB in the rung of ladder logic for pump A.

2. Level is above LHH.

3. Pump A is running and the level is above LL.

The second rung is an almost identical circuit for pump B. The difference is the use of the normally open contact on the coil LeadIsB.

When implemented as hardwired logic, ladder diagrams are truly parallel logic; i.e., all circuits are active at all instants of time. But when ladder diagrams are implemented in PLCs, the behavior is slightly different. The ladder logic is scanned very rapidly (on the order of 100 times per second), which gives the appearance of parallel logic. But within a scan of ladder logic, the rungs are executed sequentially. This

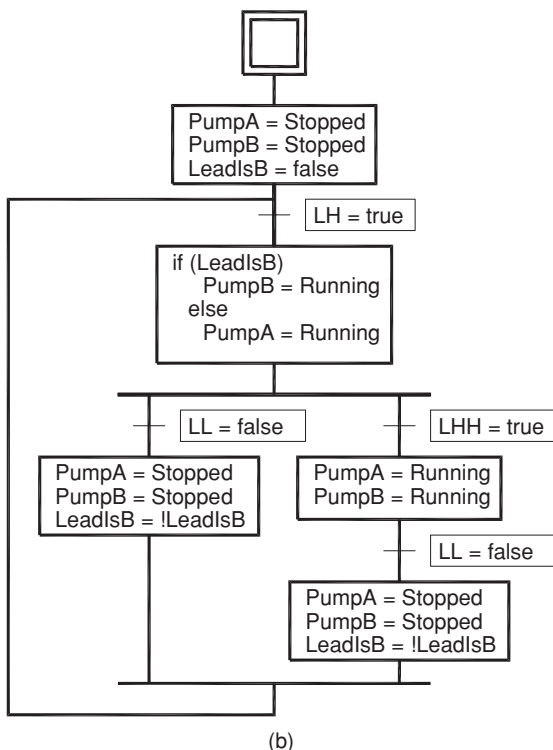
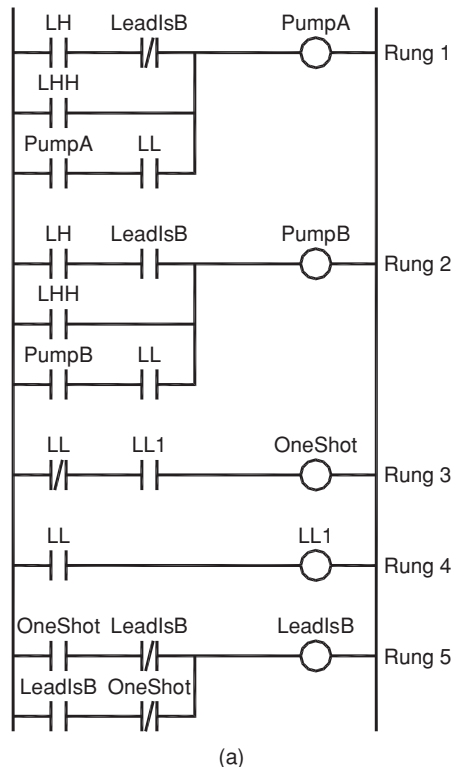


FIG. 8-60 (a) Ladder logic. (b) Sequence logic for effluent tank sump pumps.

permits constructs within ladder logic for PLCs that make no sense in hardwired circuits.

One such construct is for a “one-shot.” Some PLCs provide this as a built-in function, but here it will be presented in terms of separate components. The one-shot is generated by the third rung of ladder logic in Fig. 8-60a. But first examine the fourth rung. The input LL drives the output coil LL1. This coil provides the state of level switch LL on the previous scan of ladder logic. This is used in the third rung to produce the one-shot. Output coil OneShot is energized if

1. LL is not actuated on this scan of ladder logic (note the use of the normally closed contact for LL)
2. LL was actuated on the previous scan of ladder logic (note the use of the normally open contact for LL1)

When LL deactuates, coil OneShot is energized for one scan of ladder logic. OneShot does not energize when LL actuates (a slight modification of the circuit would give a one-shot when LL actuates).

The one-shot is used in the fifth rung of ladder logic to toggle the lead pump. The output coil LeadIsB is energized provided that

1. LeadIsB is energized and OneShot is not energized. Once LeadIsB is energized, it remains energized until the next “firing” of the one-shot.
2. LeadIsB is not energized and OneShot is energized. This causes coil LeadIsB to change states each time the one-shot fires.

Ladder diagrams are ideally suited for representing discrete logic, such as required for interlocks. Sequence logic can be implemented via ladder logic, but usually with some tricks or gimmicks (the one-shot in Fig. 8-60a is such a gimmick). These are well known to those “skilled in the art” of PLC programming. But to others, they can be quite confusing.

Figure 8-60b provides a sequential function chart for the pumps. Sequential function charts consist of steps and transitions. A step consists of actions to be performed, as represented by statements. A transition consists of a logical expression. As long as the logical expression is false, the sequence logic remains at the transition. When the logical expression is true, the sequence logic proceeds to the step following the transition.

The basic constructs of sequential function charts are presented in Fig. 8-61. The basic construct of a sequential function chart is the step-transition-step. But also note the constructs for OR and AND. At the divergent OR, the logic proceeds on only one of the possible paths, specifically, the one whose transition is the first to attain the true condition. At the divergent AND, the logic proceeds on all paths simultaneously, and all must complete to proceed beyond the convergent AND. This enables sequential function charts to provide parallel logic.

In the sequential function chart in Fig. 8-60b for the pumps, the logic is initiated with both pumps stopped and pump A as the lead pump. When LH actuates, the lead pump is started. A divergent OR is used to create two paths:

1. If LL deactuates, both pumps are stopped and the lead pump is swapped.
2. If LHH actuates, both pumps are started (one is already running). Both remain running until LL deactuates, at which time both are stopped. The logic then loops to the transition for LH actuating.

Although not illustrated here, programming languages (either custom sequence languages or traditional languages extended by libraries of real-time functions) are a viable alternative for implementing the logic for the pumps. Graphical constructs such as ladder logic and sequential function charts are appealing to those uncomfortable with traditional programming languages. But in reality, these are programming methodologies.

BATCH REACTOR CONTROL

The reactors in flexible batch chemical plants usually present challenges. Many reactors have multiple mechanisms for heating and/or cooling. The reactor in Fig. 8-62 has three mechanisms:

1. Heat with steam.
2. Cool with cooling tower water.
3. Cool with chilled water.

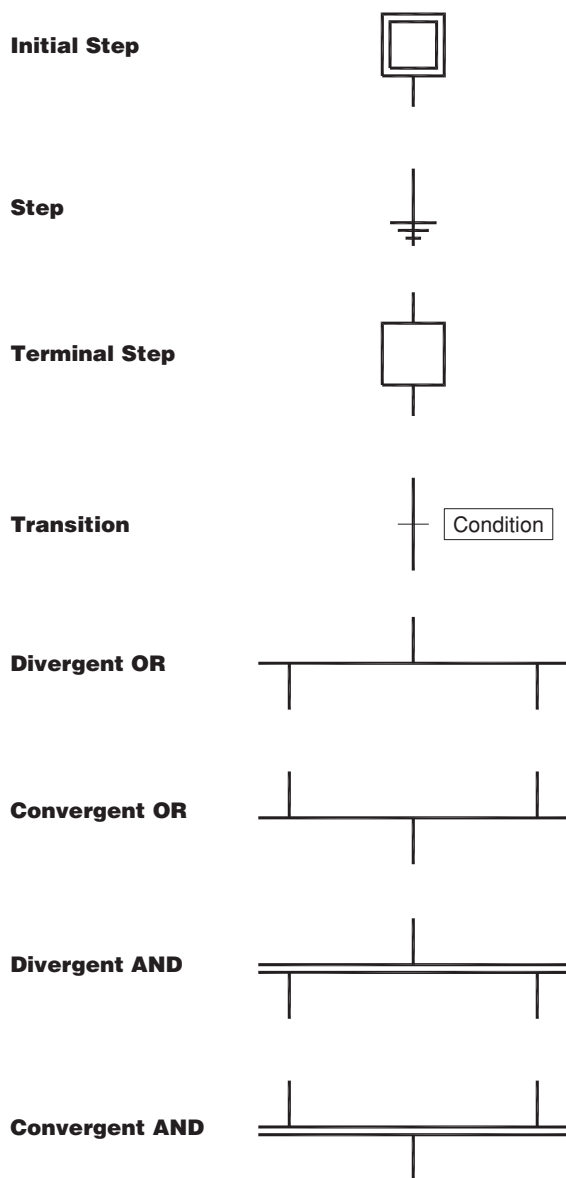


FIG. 8-61 Elements of sequential function charts.

Sometimes glycol is an option; occasionally liquid nitrogen is used to achieve even lower temperatures. Some jacket configurations require sequences for switching between the various modes for heating and cooling (the jacket has to be drained or depressurized before another medium can be admitted).

The reactor in Fig. 8-62 has three mechanisms for pressure control:

1. Vacuum
2. Atmospheric (using the vent and inert valves)
3. Pressure

Some reactors are equipped with multiple vacuum pumps, with different operating modes for moderate vacuum versus high vacuum. Sequence logic is usually required to start vacuum pumps and establish vacuum.

With three options for heating/cooling and three options for pressure, the reactor in Fig. 8-62 has nine combinations of operating modes.

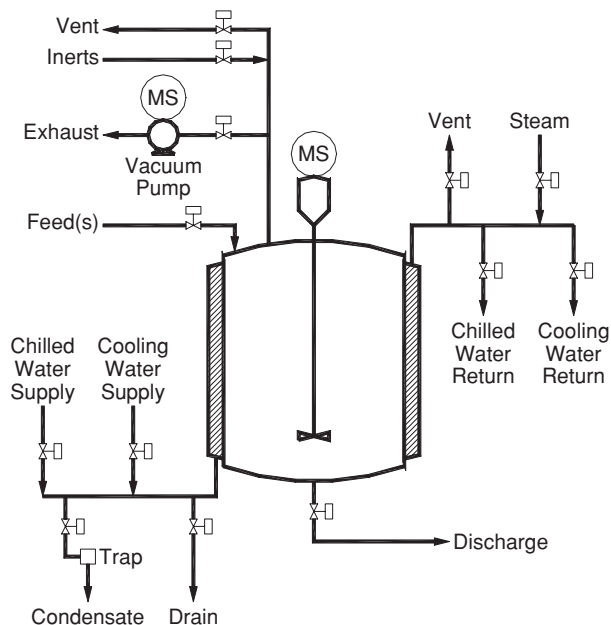


FIG. 8-62 Chemical reactor schematic.

In practice, this is actually a low number. This number increases with features such as

1. Recirculations or pump-arounds containing a heater and/or cooler
2. Reflux condensers that can be operated at total reflux (providing only cooling) or such that some component is removed from the reacting system

These further increase the number of possible combinations. Some combinations may not make sense, may not be used in current production operations, or otherwise can be eliminated. However, the net number of combinations that must be supported tends to be large.

The order in which the systems are activated usually depends on the product being manufactured. Sometimes heating/cooling and pressure control are established simultaneously; sometimes heating/cooling is established first and then pressure control, and sometimes pressure control is established first and then heating/cooling. One has to be very careful when imposing restrictions. Suppose no current products establish pressure control first and then establish heating/cooling. But what about the next product to be introduced? After all, this is a flexible batch facility.

Such challenging applications for recipe management and sequence logic require a detailed analysis of the production equipment and the operations conducted within that production equipment. While applications such as the sump pumps in the effluent tank can be pursued without it, a structured approach is essential in flexible batch facilities.

BATCH PRODUCTION FACILITIES

Especially for flexible batch applications, the batch logic must be properly structured in order to be implemented and maintained in a reasonable manner. An underlying requirement is that the batch process equipment be properly structured. The following structure is appropriate for most batch production facilities.

Plant A plant is the collection of production facilities at a geographic site. The production facilities at a site normally share warehousing, utilities, and the like.

Equipment Suite An equipment suite is the collection of equipment available for producing a group of products. Normally, this

group of products is similar in certain respects. For example, they might all be manufactured from the same major raw materials. Within the equipment suite, material transfer and metering capabilities are available for these raw materials. The equipment suite contains all the necessary types of processing equipment (reactors, separators, and so on) required to convert the raw materials to salable products. A plant may consist of only one suite of equipment, but large plants usually contain multiple equipment suites.

Process Unit or Batch Unit A process unit is a collection of processing equipment that can, at least at certain times, be operated in a manner completely independent of the remainder of the plant. A process unit normally provides a specific function in the production of a batch of product. For example, a process unit might be a reactor complete with all associated equipment (jacket, recirculation pump, reflux condenser, and so on). However, each feed preparation tank is usually a separate process unit. With this separation, preparation of the feed for the next batch can be started as soon as the feed tank is emptied for the current batch.

All but the very simplest equipment suites contain multiple process units. The minimum number of process units is one for each type of processing equipment required to make a batch of product. However, many equipment suites contain multiple process units of each type. In such equipment suites, multiple batches and multiple production runs can be in progress at a given time.

Item of Equipment An item of equipment is a hardware item that performs a specific purpose. Examples are pumps, heat exchangers, agitators, and the like. A process unit could consist of a single item of equipment, but most process units consist of several items of equipment that must be operated in harmony to achieve the function expected of the process unit.

Device A device is the smallest element of interest to batch logic. Examples of devices include measurement devices and actuators.

STRUCTURED BATCH LOGIC

Flexible batch applications must be pursued by using a structured approach to batch logic. In such applications, the same processing equipment is used to make a variety of products. In most facilities, little or no proprietary technology is associated with the equipment itself; the proprietary technology is how this equipment is used to produce each of the products.

The primary objective of the structured approach is to separate cleanly the following two aspects of the batch logic:

Product Technology Basically, this encompasses the product technology, such as how to mix certain molecules to make other molecules. This technology ultimately determines the chemical and physical properties of the final product. The product recipe is the principal source for the product technology.

Process Technology The process equipment permits certain processing operations (e.g., heat to a specified temperature) to be undertaken. Each processing operation will involve certain actions (e.g., opening appropriate valves).

The need to keep these two aspects separated is best illustrated by a situation where the same product is to be made at different plants. While it is possible that the processing equipment at the two plants is identical, this is rarely the case. Suppose one plant uses steam for heating its vessels, but the other uses a hot oil system as the source of heat. When a product recipe requires that material be heated to a specified temperature, each plant can accomplish this objective, but each will go about it in quite different ways. The ideal case for a product recipe is as follows:

1. It contains all the product technology required to make a product.
2. It contains no equipment-dependent information, i.e., no process technology.

In the previous example, such a recipe would simply state that the product must be heated to a specified temperature. Whether heating is undertaken with steam or hot oil is irrelevant to the product technology. By restricting the product recipe to a given product technology, the same product recipe can be used to make products at

different sites. At a given site, the specific approach to be used to heat a vessel is important. The traditional approach is for an engineer at each site to expand the product recipe into a document that explains in detail how the product is to be made at the specific site. This document goes by various names, although standard operating procedure or SOP is a common one. Depending on the level of detail to which it is written, the SOP could specify exactly which valves must be opened to heat the contents of a vessel. Thus, the SOP is site-dependent and contains both product technology and process technology.

In structuring the logic for a flexible batch application, the following organization permits product technology to be cleanly separated from process technology:

- A recipe consists of a formula and one or more processing operations. Ideally, only product technology is contained in a recipe.
- A processing operation consists of one or more phases. Ideally, only product technology is contained in a processing operation.
- A phase consists of one or more actions. Ideally, only process technology is contained in a phase.

In this structure, the recipe and processing operations would be the same at each site that manufactures the product. However, the logic that comprises each phase would be specific to a given site. In the heating example from above, each site would require a phase to heat the contents of the vessel. However, the logic within the phase at one site would accomplish the heating by opening the appropriate steam valves, while the logic at the other site would accomplish the heating by opening the appropriate hot oil valves.

Usually the critical part of structuring batch logic is the definition of the phases. There are two ways to approach this:

1. Examine the recipes for the current products for commonality, and structure the phases to reflect this commonality.

2. Examine the processing equipment to determine what processing capabilities are possible, and write phases to accomplish each possible processing capability.

There is the additional philosophical issue of whether to have a large number of simple phases with few options each, or a small number of complex phases with numerous options. The issues are analogous to structuring a complex computer program into subprograms. Each possible alternative has advantages and disadvantages.

As the phase contains no product technology, the implementation of a phase must be undertaken by those familiar with the process equipment. Furthermore, they should undertake this on the basis that the result will be used to make a variety of products, not just those that are initially contemplated. The development of the phase logic must also encompass all equipment-related safety issues. The phase should accomplish a clearly defined objective, so the implementers should be able to thoroughly consider all relevant issues in accomplishing this objective. The phase logic is defined in detail, implemented in the control system, and then thoroughly tested. Except when the processing equipment is modified, future modifications to the phase should be infrequent. The result should be a very dependable module that can serve as a building block for batch logic.

Even for flexible batch applications, a comprehensive menu of phases should permit most new products to be implemented by using currently existing phases. By reusing existing phases, numerous advantages accrue:

1. The engineering effort to introduce a new recipe at a site is reduced.
2. The product is more likely to be on-spec the first time, thus avoiding the need to dispose of off-spec product.
3. The new product can be supplied to customers sooner, hopefully before competitors can supply the product.

There is also a distinct advantage in maintenance. When a problem with a phase is discovered and the phase logic is corrected, the correction is effectively implemented in all recipes that use the phase. If a change is implemented in the processing equipment, the affected phases must be modified accordingly and then thoroughly tested. These modifications are also effectively implemented in all recipes that use these phases.

PROCESS MEASUREMENTS

GENERAL REFERENCES: Baker, *Flow Measurement Handbook*, Cambridge University Press, New York, 2000. Connell, *Process Instrumentation Applications Manual*, McGraw-Hill, New York, 1966. Dakin, and Culshaw (eds), *Optical Fiber Sensors: Applications, Analysis, and Future Trends*, vol. IV, Artech House, Norwood, Mass., 1997. Dolenc, "Choose the Right Flow Meter," *Chem. Engr. Prog.*, **92**(1): 22, 1996. Johnson, *Process Control Instrumentation Technology*, 6th ed., Prentice-Hall, Upper Saddle River, N.J., 2000. Liptak (ed.), *Instrument Engineers Handbook*, 3d ed., vol. 1: *Process Measurement*, Chilton Books, Philadelphia, 2000. Nichols, *On-Line Process Analyzers*, Wiley, New York, 1988. Seborg, Edgar, and Mellichamp, *Process Dynamics and Control*, Wiley, New York, 2004. Soloman, *Sensors Handbook*, McGraw-Hill, New York, 1999. Spitzer, *Flow Measurement*, 2d ed., ISA, Research Triangle Park, N.C., 2001.

GENERAL CONSIDERATIONS

Process measurements encompass the application of the principles of metrology to the process in question. The objective is to obtain values for the current conditions within the process and to make this information available in a form usable by the control system, process operators, or management information systems. The term *measured variable* or *process variable* designates the process condition that is being determined.

Process measurements fall into two categories:

1. *Continuous measurements.* An example of a continuous measurement is a level measurement device that determines the liquid level in a tank (e.g., in meters).
2. *Discrete measurements.* An example of a discrete measurement is a level switch that indicates the presence or absence of liquid at the location at which the level switch is installed.

In continuous processes, most process control applications rely on continuous measurements. In batch processes, many of the process control applications utilize discrete as well as continuous measurements. In both types of processes, the safety interlocks and process interlocks rely largely on discrete measurements.

Continuous Measurements In most applications, continuous measurements provide more information than discrete measurements. Basically, discrete measurements involve a yes/no decision, whereas continuous measurements may entail considerable signal processing.

The components of a typical continuous measurement device are as follows:

1. *Sensor.* This component produces a signal that is related in a known manner to the process variable of interest. The sensors in use today are primarily of the electrical analog variety, and the signal is in the form of a voltage, a resistance, a capacitance, or some other directly measurable electrical quantity. Prior to the mid-1970s, instruments tended to use sensors whose signal was mechanical and thus compatible with pneumatic technology. Since that time, the fraction of sensors that are digital has grown considerably, often eliminating the need for analog-to-digital conversion.
2. *Signal processing.* The signal from a sensor is usually related in a nonlinear fashion to the process variable of interest. For the output of the measurement device to be linear with respect to the process variable of interest, linearization is required. Furthermore, the signal from the sensor might be affected by variables other than the process variable. In this case, additional variables must be sensed, and the signal from the sensor compensated to account for the other variables. For example, reference junction compensation is required for thermocouples (except when used for differential temperature measurements).
3. *Transmitter.* The measurement device output must be a signal that can be transmitted over some distance. Where electronic analog transmission is used, the low range on the transmitter output is 4 mA, and the upper range is 20 mA. Microprocessor-based transmitters (often referred to as smart transmitters) are usually capable of transmitting the measured variable digitally in engineering units.

Accuracy and Repeatability Definitions of terminology pertaining to process measurements can be obtained from standards available from the Instrumentation, Systems, and Automation Society

(ISA) and from the Scientific Apparatus Makers Association [now Measurement, Control, and Automation Association (MCAA)], both of which are updated periodically. An appreciation of accuracy and repeatability is especially important. Some applications depend on the accuracy of the instrument, but other applications depend on repeatability. Excellent accuracy implies excellent repeatability; however, an instrument can have poor accuracy but excellent repeatability. In some applications, this is acceptable, as discussed below.

Range and Span A continuous measurement device is expected to provide credible values of the measured value between a lower range and an upper range. The difference between the upper range and the lower range is the span of the measurement device. The maximum value for the upper range and the minimum value for the lower range depend on the principles on which the measurement device is based and on the design chosen by the manufacturer of the measurement device. If the measured variable is greater than the upper range or less than the lower range, the measured variable is said to be out of range or the measurement device is said to be overranged.

Accuracy Accuracy refers to the difference between the measured value and the true value of the measured variable. Unfortunately, the true value is never known, so in practice accuracy refers to the difference between the measured value and an accepted standard value for the measured variable.

Accuracy can be expressed in four ways:

1. As an absolute difference in the units of the measured variable
2. As a percent of the current reading
3. As a percent of the span of the measured variable
4. As a percent of the upper range of the span

For process measurements, accuracy as a percent of span is the most common.

Manufacturers of measurement devices always state the accuracy of the instrument. However, these statements always specify specific or reference conditions at which the measurement device will perform with the stated accuracy, with temperature and pressure most often appearing in the reference conditions. When the measurement device is applied at other conditions, the accuracy is affected. Manufacturers usually also provide some statements on how accuracy is affected when the conditions of use deviate from the referenced conditions in the statement of accuracy. Although appropriate calibration procedures can minimize some of these effects, rarely can they be totally eliminated. It is easily possible for such effects to cause a measurement device with a stated accuracy of 0.25 percent of span at reference conditions to ultimately provide measured values with accuracies of 1 percent or less. Microprocessor-based measurement devices usually provide better accuracy than do the traditional electronic measurement devices.

In practice, most attention is given to accuracy when the measured variable is the basis for billing, such as in custody transfer applications. However, whenever a measurement device provides data to any type of optimization strategy, accuracy is very important.

Repeatability Repeatability refers to the difference between the measurements when the process conditions are the same. This can also be viewed from the opposite perspective. If the measured values are the same, repeatability refers to the difference between the process conditions.

For regulatory control, repeatability is of major interest. The basic objective of regulatory control is to maintain uniform process operation. Suppose that on two different occasions, it is desired that the temperature in a vessel be 800°C. The regulatory control system takes appropriate actions to bring the measured variable to 800°C. The difference between the process conditions at these two times is determined by the repeatability of the measurement device.

In the use of temperature measurement for control of the separation in a distillation column, repeatability is crucial but accuracy is not. Composition control for the overhead product would be based on a measurement of the temperature on one of the trays in the rectifying section. A target would be provided for this temperature. However, at

periodic intervals, a sample of the overhead product is analyzed in the laboratory and the information is provided to the process operator. Should this analysis be outside acceptable limits, the operator would adjust the set point for the temperature. This procedure effectively compensates for an inaccurate temperature measurement; however, the success of this approach requires good repeatability from the temperature measurement.

Dynamics of Process Measurements Especially where the measurement device is incorporated into a closed-loop control configuration, dynamics are important. The dynamic characteristics depend on the nature of the measurement device, and on the nature of components associated with the measurement device (e.g., thermowells and sample conditioning equipment). The term *measurement system* designates the measurement device and its associated components.

The following dynamics are commonly exhibited by measurement systems:

- **Time constants.** Where there is a capacity and a throughput, the measurement device will exhibit a time constant. For example, any temperature measurement device has a thermal capacity (mass times heat capacity) and a heat flow term (heat-transfer coefficient and area). Both the temperature measurement device and its associated thermowell will exhibit behavior typical of time constants.
- **Dead time.** Probably the best example of a measurement device that exhibits pure dead time (time delay) is the chromatograph, because the analysis is not available for some time after a sample is injected. Additional dead time results from the transportation lag within the sample system. Even continuous analyzer installations can exhibit dead time from the sample system.
- **Underdamped.** Measurement devices with mechanical components often have a natural harmonic and can exhibit underdamped behavior. The displacer type of level measurement device is capable of such behavior.

While the manufacturers of measurement devices can supply some information on the dynamic characteristics of their devices, interpretation is often difficult. Measurement device dynamics are quoted on varying bases, such as rise time, time to 63 percent response, settling time, etc. Even where the time to 63 percent response is quoted, it might not be safe to assume that the measurement device exhibits first-order behavior.

Where the manufacturer of the measurement device does not supply the associated equipment (thermowells, sample conditioning equipment, etc.), the user must incorporate the characteristics of these components to obtain the dynamics of the measurement system. An additional complication is that most dynamic data are stated for configurations involving reference materials such as water and air. The nature of the process material will affect the dynamic characteristics. For example, a thermowell will exhibit different characteristics when immersed in a viscous organic emulsion than when immersed in water. It is often difficult to extrapolate the available data to process conditions of interest.

Similarly, it is often impossible, or at least very difficult, to experimentally determine the characteristics of a measurement system under the conditions where it is used. It is certainly possible to fill an emulsion polymerization reactor with water and determine the dynamic characteristics of the temperature measurement system. However, it is not possible to determine these characteristics when the reactor is filled with the emulsion under polymerization conditions.

The primary impact of unfavorable measurement dynamics is on the performance of closed-loop control systems. This explains why most control engineers are very concerned with minimizing measurement dynamics, even though the factors considered in dynamics are often subjective.

Selection Criteria The selection of a measurement device entails a number of considerations given below, some of which are almost entirely subjective.

1. **Measurement span.** The measurement span required for the measured variable must lie entirely within the instrument's envelope of performance.

2. **Performance.** Depending on the application, accuracy, repeatability, or perhaps some other measure of performance is appropriate.

Where closed-loop control is contemplated, speed of response must be included.

3. **Reliability.** Data available from the manufacturers can be expressed in various ways and at various reference conditions. Often, previous experience with the measurement device within the purchaser's organization is weighted most heavily.

4. **Materials of construction.** The instrument must withstand the process conditions to which it is exposed. This encompasses considerations such as operating temperatures, operating pressures, corrosion, and abrasion. For some applications, seals or purges may be necessary.

5. **Prior use.** For the first installation of a specific measurement device at a site, training of maintenance personnel and purchases of spare parts might be necessary.

6. **Potential for releasing process materials to the environment.** Fugitive emissions are receiving ever-increasing attention. Exposure considerations, both immediate and long-term, for maintenance personnel are especially important when the process fluid is either corrosive or toxic.

7. **Electrical classification.** Article 500 of the National Electric Code provides for the classification of the hazardous nature of the process area in which the measurement device will be installed. If the measurement device is not inherently compatible with this classification, suitable enclosures must be purchased and included in the installation costs.

8. **Physical access.** Subsequent to installation, maintenance personnel must have physical access to the measurement device for maintenance and calibration. If additional structural facilities are required, they must be included in the installation costs.

9. **Invasive or noninvasive.** The insertion of a probe can result in fouling problems and a need for maintenance. Probe location must be selected carefully for good accuracy and minimal fouling.

10. **Cost.** There are two aspects of the cost:

a. Initial purchase and installation (capital cost).

b. Recurring costs (operational expense). This encompasses instrument maintenance, instrument calibration, consumables (e.g., titrating solutions must be purchased for automatic titrators), and any other costs entailed in keeping the measurement device in service.

Calibration Calibration entails the adjustment of a measurement device so that the value from the measurement device agrees with the value from a standard. The International Standards Organization (ISO) has developed a number of standards specifically directed to calibration of measurement devices. Furthermore, compliance with the ISO 9000 standards requires that the working standard used to calibrate a measurement device be traceable to an internationally recognized standard such as those maintained by the National Institute of Standards and Technology (NIST).

Within most companies, the responsibility for calibrating measurement devices is delegated to a specific department. Often, this department may also be responsible for maintaining the measurement device. The specific calibration procedures depend on the type of measurement device. The frequency of calibration is normally predetermined, but earlier action may be dictated if the values from the measurement device become suspect.

Calibration of some measurement devices involves comparing the measured value with the value from the working standard. Pressure and differential pressure transmitters are calibrated in this manner. Calibration of analyzers normally involves using the measurement device to analyze a specially prepared sample whose composition is known. These and similar approaches can be applied to most measurement devices.

Flow is an important measurement whose calibration presents some challenges. When a flow measurement device is used in applications such as custody transfer, provision is made to pass a known flow through the meter. However, such a provision is costly and is not available for most in-process flowmeters. Without such a provision, a true calibration of the flow element itself is not possible. For orifice meters, calibration of the flowmeter normally involves calibration of the differential pressure transmitter, and the orifice plate is usually only inspected for deformation, abrasion, etc. Similarly, calibration of a magnetic flowmeter normally involves calibration of the voltage

TABLE 8-8 Online Measurement Options for Process Control

Temperature	Flow	Pressure	Level	Composition
Thermocouple	Orifice	Liquid column	Float-activated	Gas-liquid chromatography (GLC)
Resistance temperature detector (RTD)	Venturi	Elastic element	Chain gauge	Mass spectrometry (MS)
Filled-system thermometer	Rotameter	Bourdon tube	Lever	Magnetic resonance analysis (MRA)
Bimetal thermometer	Turbine	Bellow	Magnetically coupled	Infrared (IR) spectroscopy
Pyrometer	Vortex-shedding	Diaphragm	Head devices	Raman spectroscopy
Total radiation	Ultrasonic	Strain gauges	Bubble tube	Ultraviolet (uv) spectroscopy
Photoelectric	Magnetic	Piezoresistive transducers	Electrical (conductivity)	Thermal conductivity
Ratio	Thermal mass	Piezoelectric transducers	Sonic	Refractive index (RI)
Laser	Coriolis	Optical fiber	Laser	Capacitance probe
Surface acoustic wave	Target		Radiation	Surface acoustic wave
Semiconductor			Radar	Electrophoresis
				Electrochemical
				Paramagnetic
				Chemi/bioluminescence
				Tunable diode laser absorption

measurement circuitry, which is analogous to calibration of the differential pressure transmitter for an orifice meter.

In the next section we cover the major types of measurement devices used in the process industries, principally the “big five” measurements: temperature, flow rate, pressure, level, and composition, along with online physical property measurement techniques. Table 8-8 summarizes the different options under each of the principal measurements.

TEMPERATURE MEASUREMENTS

Measurement of the hotness or coldness of a body or fluid is commonplace in the process industries. Temperature-measuring devices utilize systems with properties that vary with temperature in a simple, reproducible manner and thus can be calibrated against known references (sometimes called secondary thermometers). The three dominant measurement devices used in automatic control are thermocouples, resistance thermometers, and pyrometers, and they are applicable over different temperature regimes.

Thermocouples Temperature measurements using thermocouples are based on the discovery by Seebeck in 1821 that an electric current flows in a continuous circuit of two different metallic wires if the two junctions are at different temperatures. The thermocouple may be represented diagrammatically as shown in Fig. 8-63. There A and B are the two metals, and T_1 and T_2 are the temperatures of the junctions. Let T_1 and T_2 be the reference junction (cold junction) and the measuring junction, respectively. If the thermoelectric current i flows in the direction indicated in Fig. 8-63, metal A is customarily referred to as thermoelectrically positive to metal B. Metal pairs used for thermocouples include platinum-rhodium (the most popular and accurate), chromel-alumel, copper-constantan, and iron-constantan. The thermal emf is a measure of the difference in temperature between T_2 and T_1 . In control systems the reference junction is usually located at the emf-measuring device. The reference junction may be held at constant temperature such as in an ice bath or a thermostated oven, or it may be at ambient temperature but electrically compen-

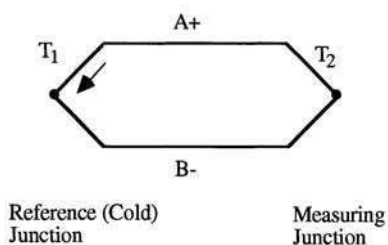


FIG. 8-63 Basic circuit of Seebeck effect.

sated (cold junction compensated circuit) so that it appears to be held at a constant temperature.

Resistance Thermometers The resistance thermometer depends upon the inherent characteristics of materials to change in electrical resistance when they undergo a change in temperature. Industrial resistance thermometers are usually constructed of platinum, copper, or nickel, and more recently semiconducting materials such as thermistors are being used. Basically, a resistance thermometer is an instrument for measuring electrical resistance that is calibrated in units of temperature instead of in units of resistance (typically ohms). Several common forms of bridge circuits are employed in industrial resistance thermometry, the most common being the Wheatstone bridge. A resistance thermometer detector (RTD) consists of a resistance conductor (metal) which generally shows an increase in resistance with temperature. The following equation represents the variation of resistance with temperature ($^{\circ}\text{C}$):

$$R_T = R_0(1 + a_1T + a_2T^2 + \dots + a_nT^n) \quad (8-92)$$

R_0 = resistance at 0°C

The temperature coefficient of resistance, α_T is expressed as

$$\alpha_T = \frac{1}{R_T} \frac{dR_T}{dT} \quad (8-93)$$

For most metals α_T is positive. For many pure metals, the coefficient is essentially constant and stable over large portions of their useful range. Typical resistance versus temperature curves for platinum, copper, and nickel are given in Fig. 8-64, with platinum usually the metal of choice. Platinum has a useful range of -200 to 800°C , while nickel (-80 to 320°C) and copper (-100 to 100°C) are more limited. Detailed resistance versus temperature tables are available from the National Institute of Standards and Technology (NIST) and suppliers of resistance thermometers. Table 8-9 gives recommended temperature measurement ranges for thermocouples and RTDs. Resistance thermometers are receiving increased usage because they are about 10 times more accurate than thermocouples.

Thermistors Thermistors are nonlinear temperature-dependent resistors, and normally only the materials with negative temperature coefficient of resistance (NTC type) are used. The resistance is related to temperature as

$$R_T = R_r \exp \left[\beta \left(\frac{1}{T} - \frac{1}{T_r} \right) \right] \quad (8-94)$$

where α_r is a reference temperature, which is generally 298 K. Thus

$$\alpha_r = \frac{1}{R_r} \frac{dR_r}{dT} \quad (8-95)$$

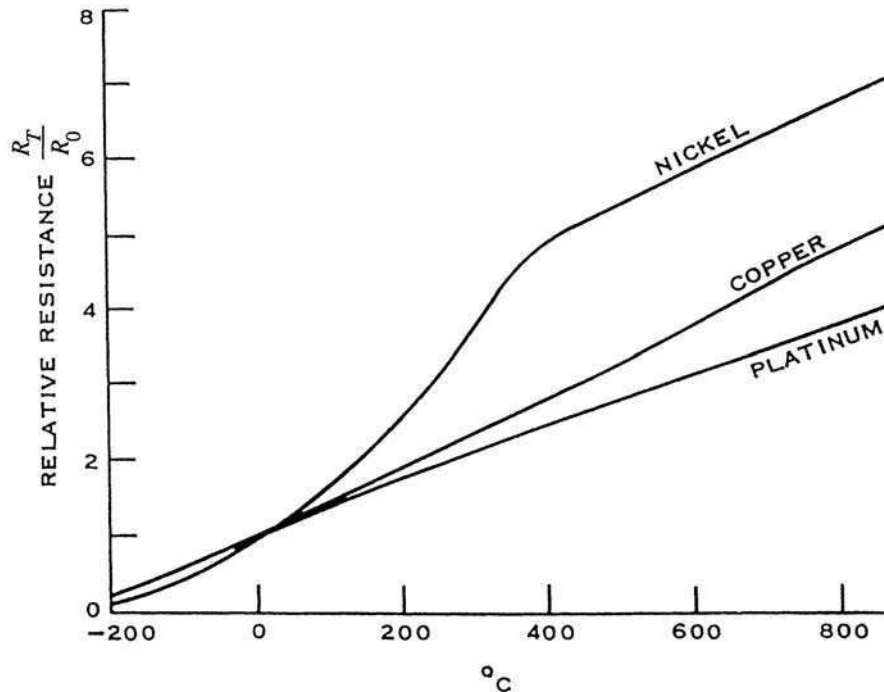


FIG. 8-64 Typical resistance thermometer curves for platinum, copper, and nickel wire, where R_T = resistance at temperature T and R_0 = resistance at 0°C.

The value of β is on the order of 4000, so at room temperature (298 K), $\alpha_T = -0.045$ for thermistor and 0.0035 for 100- Ω platinum RTD. Compared with RTDs, NTC-type thermistors are advantageous in that the detector dimension can be made small, resistance value is higher (less affected by the resistances of the connecting leads), and it has higher temperature sensitivity and low thermal inertia of the sensor. Disadvantages of thermistors to RTDs include nonlinear characteristics and low measuring temperature range.

Filled-System Thermometers The filled-system thermometer is designed to provide an indication of temperature some distance removed from the point of measurement. The measuring element (bulb) contains a gas or liquid that changes in volume, pressure, or vapor pressure with temperature. This change is communicated through a capillary tube to a Bourdon tube or other pressure- or volume-sensitive device. The Bourdon tube responds so as to provide a motion related to the bulb temperature. Those systems that respond to volume changes are completely filled with a liquid. Systems that respond to

pressure changes either are filled with a gas or are partially filled with a volatile liquid. Changes in gas or vapor pressure with changes in bulb temperatures are carried through the capillary to the Bourdon. The latter bulbs are sometimes constructed so that the capillary is filled with a nonvolatile liquid.

Fluid-filled bulbs deliver enough power to drive controller mechanisms and even directly actuate control valves. These devices are characterized by large thermal capacity, which sometimes leads to slow response, particularly when they are enclosed in a thermal well for process measurements. Filled-system thermometers are used extensively in industrial processes for a number of reasons. The simplicity of these devices allows rugged construction, minimizing the possibility of failure with a low level of maintenance, and inexpensive overall design of control equipment. In case of system failure, the entire unit must be replaced or repaired.

As they are normally used in the process industries, the sensitivity and percentage of span accuracy of these thermometers are generally the equal of those of other temperature-measuring instruments. Sensitivity and absolute accuracy are not the equal of those of short-span electrical instruments used in connection with resistance-thermometer bulbs. Also the maximum temperature is somewhat limited.

Bimetal Thermometers Thermostatic bimetal can be defined as a composite material made up of strips of two or more metals fastened together. This composite, because of the different expansion rates of its components, tends to change curvature when subjected to a change in temperature. With one end of a straight strip fixed, the other end deflects in proportion to the temperature change, the square of the length, and inversely as the thickness, throughout the linear portion of the deflection characteristic curve. If a bimetallic strip is wound into a helix or a spiral and one end is fixed, the other end will rotate when heat is applied. For a thermometer with uniform scale divisions, a bimetal must be designed to have linear deflection over the desired temperature range. Bimetal thermometers are used at temperatures ranging from 580 down to -180°C and lower. However, at the low temperatures

TABLE 8-9 Recommended Temperature Measurement Ranges for RTDs and Thermocouples

Resistance Thermometer Detectors (RTDs)	
100V Pt	-200 – $+850^\circ\text{C}$
120V Ni	-80 – $+320^\circ\text{C}$
Thermocouples	
Type B	700 – $+1820^\circ\text{C}$
Type E	-175 – $+1000^\circ\text{C}$
Type J	-185 – $+1200^\circ\text{C}$
Type K	-175 – $+1372^\circ\text{C}$
Type N	0 – $+1300^\circ\text{C}$
Type R	125 – $+1768^\circ\text{C}$
Type S	150 – $+1768^\circ\text{C}$
Type T	-170 – $+400^\circ\text{C}$

the rate of deflection drops off quite rapidly. Bimetal thermometers do not have long-time stability at temperatures above 430°C.

Pyrometers Planck's distribution law gives the radiated energy flux $q_b(\lambda, T)d\lambda$ in the wavelength range λ to $\lambda + d\lambda$ from a black surface:

$$q_b(\lambda, T) = \frac{C_1}{\lambda^5} \frac{1}{e^{C_2/\lambda T} - 1} \quad (8-96)$$

where $C_1 = 3.7418 \times 10^{10} \mu W \cdot \mu m^4 \cdot cm^{-2}$ and $C_2 = 14,388 \mu m \cdot K$.

If the target object is a black body and if the pyrometer has a detector which measures the specific wavelength signal from the object, the temperature of the object can be exactly estimated from Eq. (8-96). While it is possible to construct a physical body that closely approximates black body behavior, most real-world objects are not black bodies. The deviation from a black body can be described by the spectral emissivity

$$\epsilon_T = \frac{q(T)}{q_b(T)} \quad (8-97)$$

where $q(\lambda, T)$ is the radiated energy flux from a real body in the wavelength range λ to $\lambda + d\lambda$ and $0 < \epsilon_{\lambda, T} < 1$. Integrating Eq. (8-96) over all wavelengths gives the Stefan-Boltzmann equation

$$q_b(T) = \int_0^\infty q_b(\lambda, T) d\lambda = \sigma T^4 \quad (8-98)$$

where σ is the Stefan-Boltzmann constant. Similar to Eq. (8-93), the emissivity ϵ_T for the total radiation is

$$\epsilon_T = \frac{q(T)}{q_b(T)} \quad (8-99)$$

where $q(T)$ is the radiated energy flux from a real body with emissivity ϵ_T .

Total Radiation Pyrometers In total radiation pyrometers, the thermal radiation is detected over a large range of wavelengths from the object at high temperature. The detector is normally a thermopile, which is built by connecting several thermocouples in series to increase the temperature measurement range. The pyrometer is calibrated for black bodies, so the indicated temperature T_p should be converted for non-black body temperature.

Photoelectric Pyrometers Photoelectric pyrometers belong to the class of band radiation pyrometers. The thermal inertia of thermal radiation detectors does not permit the measurement of rapidly changing temperatures. For example, the smallest time constant of a thermal detector is about 1 ms while the smallest time constant of a photoelectric detector can be about 1 or 2 s. Photoelectric pyrometers may use photoconductors, photodiodes, photovoltaic cells, or vacuum photocells. Photoconductors are built from glass plates with thin film coatings of 1- μm thickness, using PbS, CdS, PbSe, or PbTe. When the incident radiation has the same wavelength as the materials are able to absorb, the captured incident photons free photoelectrons, which form an electric current. Photodiodes in germanium or silicon are operated with a reverse bias voltage applied. Under the influence of the incident radiation, their conductivity as well as their reverse saturation current is proportional to the intensity of the radiation within the spectral response band from 0.4 to 1.7 μm for Ge and from 0.6 to 1.1 μm for Si. Because of the above characteristics, the operating range of a photoelectric pyrometer can be either spectral or in specific band. Photoelectric pyrometers can be applied for a specific choice of the wavelength.

Disappearing Filament Pyrometers Disappearing filament pyrometers can be classified as spectral pyrometers. The brightness of a lamp filament is changed by adjusting the lamp current until the filament disappears against the background of the target, at which point the temperature is measured. Because the detector is the human eye, it is difficult to calibrate for online measurements.

Ratio Pyrometers The ratio pyrometer is also called the two-color pyrometer. Two different wavelengths are utilized for detecting

the radiated signal. If one uses Wien's law for small values of λT , the detected signals from spectral radiant energy flux emitted at wavelengths λ_1 and λ_2 with emissivities ϵ_{λ_1} and ϵ_{λ_2} are

$$S_{\lambda_1} = KC_1 \epsilon_{\lambda_1} \lambda_1^{-5} \exp\left(\frac{-C_2}{\lambda_1 T}\right) \quad (8-100)$$

$$S_{\lambda_2} = KC_1 \epsilon_{\lambda_2} \lambda_2^{-5} \exp\left(\frac{-C_2}{\lambda_2 T}\right) \quad (8-101)$$

The ratio of the signals S_{λ_1} and S_{λ_2} is

$$\frac{S_{\lambda_1}}{S_{\lambda_2}} = \frac{\epsilon_{\lambda_1}}{\epsilon_{\lambda_2}} \left(\frac{\lambda_2}{\lambda_1}\right)^5 \exp\left[\frac{C_2}{T} \left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1}\right)\right] \quad (8-102)$$

Nonblack or nongray bodies are characterized by the wavelength dependence of their spectral emissivity. Let T_c be defined as the temperature of the body corresponding to the temperature of a black body. If the ratio of its radiant intensities at wavelengths λ_1 and λ_2 equals the ratio of the radiant intensities of the non-black body, whose temperature is to be measured at the same wavelength, then Wien's law gives

$$\frac{\epsilon_{\lambda_1} \exp(-C_2/\lambda_1 T)}{\epsilon_{\lambda_2} \exp(-C_2/\lambda_2 T)} = \frac{\exp(-C_2/\lambda_1 T_c)}{\exp(-C_2/\lambda_2 T_c)} \quad (8-103)$$

where T is the true temperature of the body. Rearranging Eq. (8-103) gives

$$T = \left[\frac{\ln \epsilon_{\lambda_1}/\epsilon_{\lambda_2}}{C_2(1/\lambda_1 - 1/\lambda_2)} + \frac{1}{T_c} \right]^{-1} \quad (8-104)$$

For black or gray bodies, Eq. (8-104) reduces to

$$\frac{S_{\lambda_1}}{S_{\lambda_2}} = \left(\frac{\lambda_2}{\lambda_1}\right)^5 \exp\left[\frac{C_2}{T} \left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1}\right)\right] \quad (8-105)$$

Thus by measuring S_{λ_1} and S_{λ_2} , the temperature T can be estimated.

Accuracy of Pyrometers Most of the temperature estimation methods for pyrometers assume that the object is either a gray body or has known emissivity values. The emissivity of the non-black body depends on the internal state or the surface geometry of the objects. Also the medium through which the thermal radiation passes is not always transparent. These inherent uncertainties of the emissivity values make the accurate estimation of the temperature of the target objects difficult. Proper selection of the pyrometer and accurate emissivity values can provide a high level of accuracy.

PRESSURE MEASUREMENTS

Pressure, defined as force per unit area, is usually expressed in terms of familiar units of weight-force and area or the height of a column of liquid which produces a like pressure at its base. Process pressure-measuring devices may be divided into three groups: (1) those based on the measurement of the height of a liquid column, (2) those based on the measurement of the distortion of an elastic pressure chamber, and (3) electrical sensing devices.

Liquid-Column Methods Liquid-column pressure-measuring devices are those in which the pressure being measured is balanced against the pressure exerted by a column of liquid. If the density of the liquid is known, the height of the liquid column is a measure of the pressure. Most forms of liquid-column pressure-measuring devices are commonly called manometers. When the height of the liquid is

observed visually, the liquid columns are contained in glass or other transparent tubes. The height of the liquid column may be measured in length units or calibrated in pressure units. Depending on the pressure range, water and mercury are the liquids most frequently used. Because the density of the liquid used varies with temperature, the temperature must be taken into account for accurate pressure measurements.

Elastic Element Methods Elastic element pressure-measuring devices are those in which the measured pressure deforms some elastic material (usually metallic) within its elastic limit, the magnitude of the deformation being approximately proportional to the applied pressure. These devices may be loosely classified into three types: Bourdon tube, bellows, and diaphragm.

Bourdon Tube Elements Probably the most frequently used process pressure-indicating device is the C-spring Bourdon tube pressure gauge. Gauges of this general type are available in a wide variety of pressure ranges and materials of construction. Materials are selected on the basis of pressure range, resistance to corrosion by the process materials, and effect of temperature on calibration. Gauges calibrated with pressure, vacuum, compound (combination pressure and vacuum), and suppressed-zero ranges are available.

Bellows Element The bellows element is an axially elastic cylinder with deep folds or convolutions. The bellows may be used unopposed, or it may be restrained by an opposing spring. The pressure to be measured may be applied either to the inside or to the space outside the bellows, with the other side exposed to atmospheric pressure. For measurement of absolute pressure either the inside or the space outside of the bellows can be evacuated and sealed. Differential pressures may be measured by applying the pressures to opposite sides of a single bellows or to two opposing bellows.

Diaphragm Elements Diaphragm elements may be classified into two principal types: those that utilize the elastic characteristics of the diaphragm and those that are opposed by a spring or other separate elastic element. The first type usually consists of one or more capsules, each composed of two diaphragms bonded together by soldering, brazing, or welding. The diaphragms are flat or corrugated circular metallic disks. Metals commonly used in diaphragm elements include brass, phosphor bronze, beryllium copper, and stainless steel. Ranges are available from fractions of an inch of water to over 800 in (200 kPa) gauge. The second type of diaphragm is used for containing the pressure and exerting a force on the opposing elastic element. The diaphragm is a flexible or slack diaphragm of rubber, leather, impregnated fabric, or plastic. Movement of the diaphragm is opposed by a spring which determines the deflection for a given pressure. This type of diaphragm is used for the measurement of extremely low pressure, vacuum, or differential pressure.

Electrical Methods Electrical methods for pressure measurement include strain gauges, piezoresistive transducers, and piezoelectric transducers.

Strain Gauges When a wire or other electrical conductor is stretched elastically, its length is increased and its diameter is decreased. Both of these dimensional changes result in an increase in the electrical resistance of the conductor. Devices utilizing resistance-wire grids for measuring small distortions in elastically stressed materials are commonly called strain gauges. Pressure-measuring elements utilizing strain gauges are available in a wide variety of forms. They usually consist of one of the elastic elements described earlier to which one or more strain gauges have been attached to measure the deformation. There are two basic strain gauge forms: bonded and unbonded. Bonded strain gauges are bonded directly to the surface of the elastic element whose strain is to be measured. The unbonded strain gauge transducer consists of a fixed frame and an armature that moves with respect to the frame in response to the measured pressure. The strain gauge wire filaments are stretched between the armature and frame. The strain gauges are usually connected electrically in a Wheatstone bridge configuration.

Strain gauge pressure transducers are manufactured in many forms for measuring gauge, absolute, and differential pressures and vacuum. Full-scale ranges from 25.4 mm of water to 10,134 MPa are available. Strain gauges bonded directly to a diaphragm pressure-sensitive element usually have an extremely fast response time and are suitable for high-frequency dynamic pressure measurements.

Piezoresistive Transducers A variation of the conventional strain gauge pressure transducer uses bonded single-crystal semiconductor wafers, usually silicon, whose resistance varies with strain or distortion. Transducer construction and electrical configurations are similar to those using conventional strain gauges. A permanent magnetic field is applied perpendicular to the resonating sensor. An alternating current causes the resonator to vibrate, and the resonant frequency is a function of the pressure (tension) of the resonator. The principal advantages of piezoresistive transducers are a much higher bridge voltage output and smaller size. Full-scale output voltages of 50 to 100 mV/V of excitation are typical. Some newer devices provide digital rather than analog output.

Piezoelectric Transducers Certain crystals produce a potential difference between their surfaces when stressed in appropriate directions. Piezoelectric pressure transducers generate a potential difference proportional to a pressure-generated stress. Because of the extremely high electrical impedance of piezoelectric crystals at low frequency, these transducers are usually not suitable for measurement of static process pressures.

FLOW MEASUREMENTS

Flow, defined as volume per unit of time at specified temperature and pressure conditions, is generally measured by positive displacement or rate meters. The term *positive displacement meter* applies to a device in which the flow is divided into isolated measured volumes when the number of fillings of these volumes is counted in some manner. The term *rate meter* applies to all types of flowmeters through which the material passes without being divided into isolated quantities. Movement of the material is usually sensed by a primary measuring element that activates a secondary device. The flow rate is then inferred from the response of the secondary device by means of known physical laws or from empirical relationships.

The principal classes of flow-measuring instruments used in the process industries are variable-head, variable-area, positive-displacement, and turbine instruments; mass flowmeters; vortex-shedding and ultrasonic flowmeters; magnetic flowmeters; and more recently, Coriolis mass flowmeters. Head meters are covered in detail in Sec. 5.

Orifice Meter The most widely used flowmeter involves placing a fixed-area flow restriction (an orifice) in the pipe carrying the fluid. This flow restriction causes a pressure drop which can be related to flow rate. The sharp-edge orifice is popular because of its simplicity, low cost, and the large amount of research data on its behavior. For the orifice meter, the flow rate Q_a for a liquid is given by

$$Q_a = \frac{C_d A_2}{\sqrt{1 - (A_2/A_1)^2}} \cdot \sqrt{\frac{2(p_1 - p_2)}{\rho}} \quad (8-106)$$

where $p_1 - p_2$ is the pressure drop, ρ is the density, A_1 is the pipe cross-sectional area, A_2 is the orifice cross-sectional area, and C_d is the discharge coefficient. The discharge coefficient C_d varies with the Reynolds number at the orifice and can be calibrated with a single fluid, such as water (typically $C_d \approx 0.6$). If the orifice and pressure taps are constructed according to certain standard dimensions, quite accurate (about 0.4 to 0.8 percent error) values of C_d may be obtained. Also note that the standard calibration data assume no significant flow disturbances such as elbows and valves for a certain minimum distance upstream of the orifice. The presence of such disturbances close to the orifice can cause errors of as much as 15 percent. Accuracy in measurements limits the meter to a flow rate range of 3:1. The orifice has a relatively large permanent pressure loss that must be made up by the pumping machinery.

Venturi Meter The venturi tube operates on exactly the same principle as the orifice [see Eq. (8-102)]. Discharge coefficients of venturis are larger than those for orifices and vary from about 0.94 to 0.99. A venturi gives a definite improvement in power losses over an orifice and is often indicated for measuring very large flow rates, where power losses can become economically significant. The initial higher cost of a venturi over an orifice may thus be offset by reduced operating costs.

Rotameter A rotameter consists of a vertical tube with a tapered bore in which a float changes position with the flow rate through the tube. For a given flow rate, the float remains stationary because the vertical forces of differential pressure, gravity, viscosity, and buoyancy are balanced. The float position is the output of the meter and can be made essentially linear with flow rate by making the tube area vary linearly with the vertical distance.

Turbine Meter If a turbine wheel is placed in a pipe containing a flowing fluid, its rotary speed depends on the flow rate of the fluid. A turbine can be designed whose speed varies linearly with flow rate. The speed can be measured accurately by counting the rate at which turbine blades pass a given point, using magnetic pickup to produce voltage pulses. By feeding these pulses to an electronic pulse rate meter one can measure flow rate by summing the pulses during a timed interval. Turbine meters are available with full-scale flow rates ranging from about 0.1 to 30,000 gal/min for liquids and 0.1 to 15,000 ft³/min for air. Nonlinearity can be less than 0.05 percent in the larger sizes. Pressure drop across the meter varies with the square of flow rate and is about 3 to 10 psi at full flow. Turbine meters can follow flow transients quite accurately since their fluid/mechanical time constant is on the order of 2 to 10 ms.

Vortex-Shedding Flowmeters These flowmeters take advantage of vortex shedding, which occurs when a fluid flows past a non-streamlined object (a blunt body). The flow cannot follow the shape of the object and separates from it, forming turbulent vortices or eddies at the object's side surfaces. As the vortices move downstream, they grow in size and are eventually shed or detached from the object. Shedding takes place alternately at either side of the object, and the rate of vortex formation and shedding is directly proportional to the volumetric flow rate. The vortices are counted and used to develop a signal linearly proportional to the flow rate. The digital signals can easily be totaled over an interval of time to yield the flow rate. Accuracy can be maintained regardless of density, viscosity, temperature, or pressure when the Reynolds number is greater than 10,000. There is usually a low flow cutoff point below which the meter output is clamped at zero. This flowmeter is recommended for use with relatively clean, low-viscosity liquids, gases, and vapors, and rangeability of 10:1 to 20:1 is typical. A sufficient length of straight-run pipe is necessary to prevent distortion in the fluid velocity profile.

Ultrasonic Flowmeters An ultrasonic flowmeter is based upon the variable time delays of received sound waves which arise when a flowing liquid's rate of flow is varied. Two fundamental measurement techniques, depending upon liquid cleanliness, are generally used. In the first technique two opposing transducers are inserted in a pipe so that one transducer is downstream from the other. These transducers are then used to measure the difference between the velocity at which the sound travels with the direction of flow and the velocity at which it travels against the direction of flow. The differential velocity is measured either by (1) direct time delays using sound wave burst or (2) frequency shifts derived from beat-together, continuous signals. The frequency measurement technique is usually preferred because of its simplicity and independence of the liquid static velocity. A relatively clean liquid is required to preserve the uniqueness of the measurement path.

In the second technique, the flowing liquid must contain scatters in the form of particles or bubbles which will reflect the sound waves. These scatters should be traveling at the velocity of the liquid. A Doppler method is applied by transmitting sound waves along the flow path and measuring the frequency shift in the returned signal from the scatters in the process fluid. This frequency shift is proportional to liquid velocity.

Magnetic Flowmeters The principle behind these flowmeters is Faraday's law of electromagnetic inductance. The magnitude of the voltage induced in a conductive medium moving at right angles through a magnetic field is directly proportional to the product of the magnetic flux density, the velocity of the medium, and the path length between the probes. A minimum value of fluid conductivity is required to make this approach viable. The pressure of multiple phases or undissolved solids can affect the accuracy of the measurement if the velocities of the phases are different from that for straight-run pipe. Magmeters are very accurate over wide flow ranges and are especially

accurate at low flow rates. Typical applications include metering viscous fluids, slurries, or highly corrosive chemicals. Because magmeters should be filled with fluid, the preferred installation is in vertical lines with flow going upward. However, magmeters can be used in tight piping schemes where it is impractical to have long pipe runs, typically requiring lengths equivalent to five or more pipe diameters.

Coriolis Mass Flowmeters Coriolis mass flowmeters utilize a vibrating tube in which Coriolis acceleration of a fluid in a flow loop can be created and measured. They can be used with virtually any liquid and are extremely insensitive to operating conditions, with pressure ranges over 100:1. These meters are more expensive than volumetric meters and range in size from $\frac{1}{16}$ to 6 in. Due to the circuitous path of flow through the meter, Coriolis flowmeters exhibit higher than average pressure changes. The meter should be installed so that it will remain full of fluid, with the best installation in a vertical pipe with flow going upward. There is no Reynolds number limitation with this meter, and it is quite insensitive to velocity profile distortions and swirl, hence there is no requirement for straight piping upstream.

Thermal Mass Flowmeters The trend in the chemical process industries is toward increased usage of mass flowmeters that are independent of changes in pressure, temperature, viscosity, and density. Thermal mass meters are widely used in semiconductor manufacturing and in bioprocessing for control of low flow rates (called mass flow controllers, or MFCs). MFCs measure the heat loss from a heated element, which varies with flow rate, with an accuracy of ± 1 percent. Capacitance probes measure the dielectric constant of the fluid and are useful for flow measurements of slurries and other two-phase flows.

LEVEL MEASUREMENTS

The measurement of level can be defined as the determination of the location of the interface between two fluids, separable by gravity, with respect to a fixed reference plane. The most common level measurement is that of the interface between a liquid and a gas. Other level measurements frequently encountered are the interface between two liquids, between a granular or fluidized solid and a gas, and between a liquid and its vapor.

A commonly used basis for classification of level devices is as follows: float-actuated, displacer, and head devices, and a miscellaneous group which depends mainly on fluid characteristics.

Float-Actuated Devices Float-actuated devices are characterized by a buoyant member which floats at the interface between two fluids. Because a significant force is usually required to move the indicating mechanism, float-actuated devices are generally limited to liquid-gas interfaces. By properly weighting the float, they can be used to measure liquid-liquid interfaces. Float-actuated devices may be classified on the basis of the method used to couple the float motion to the indicating system, as discussed below.

Chain or Tape Float Gauge In these types of gauges, the float is connected to the indicating mechanism by means of a flexible chain or tape. These gauges are commonly used in large atmospheric storage tanks. The gauge-board type is provided with a counterweight to keep the tape or chain taut. The tape is stored in the gauge head on a spring-loaded reel. The float is usually a pancake-shaped hollow metal float, with guide wires from top to bottom of the tank to constrain it.

Lever and Shaft Mechanisms In pressurized vessels, float-actuated lever and shaft mechanisms are frequently used for level measurement. This type of mechanism consists of a hollow metal float and lever attached to a rotary shaft which transmits the float motion to the outside of the vessel through a rotary seal.

Magnetically Coupled Devices A variety of float-actuated level devices which transmit the float motion by means of magnetic coupling have been developed. Typical of this class of devices are magnetically operated level switches and magnetic-bond float gauges. A typical magnetic-bond float gauge consists of a hollow magnet-carrying float which rides along a vertical nonmagnetic guide tube. The follower magnet is connected and drives an indicating dial similar to that on a conventional tape float gauge. The float and guide tube are in contact with the measured fluid and come in a variety of materials for resistance to corrosion and to withstand high pressures or vacuum. Weighted floats for liquid-liquid interfaces are available.

Head Devices A variety of devices utilize hydrostatic head as a measure of level. As in the case of displacer devices, accurate level measurement by hydrostatic head requires an accurate knowledge of the densities of both heavier-phase and lighter-phase fluids. The majority of this class of systems utilize standard pressure and differential pressure measuring devices.

Bubble Tube Systems The commonly used bubble tube system sharply reduces restrictions on the location of the measuring element. To eliminate or reduce variations in pressure drop due to the gas flow rate, a constant differential regulator is commonly employed to maintain a constant gas flow rate. Because the flow of gas through the bubble tube prevents entry of the process liquid into the measuring system, this technique is particularly useful with corrosive or viscous liquids, liquids subject to freezing, and liquids containing entrained solids.

Electrical Methods Two electrical characteristics of fluids, conductivity and dielectric constant, are frequently used to distinguish between two phases for level measurement purposes. An application of electrical conductivity is the fixed-point level detection of a conductive liquid such as high and low water levels. A voltage is applied between two electrodes inserted into the vessel at different levels. When both electrodes are immersed in the liquid, a current flows. Capacitance-type level measurements are based on the fact that the electrical capacitance between two electrodes varies with the dielectric constant of the material between them. A typical continuous level measurement system consists of a rod electrode positioned vertically in a vessel, the other electrode usually being the metallic vessel wall. The electrical capacitance between the electrodes is a measure of the height of the interface along the rod electrode. The rod is usually conductively insulated from process fluids by a coating of plastic. The dielectric constants of most liquids and solids are markedly higher than those of gases and vapors (by a factor of 2 to 5). The dielectric constant of water and other polar liquids is 10 to 20 times that of hydrocarbons and other nonpolar liquids.

Thermal Methods Level-measuring systems may be based on the difference in thermal characteristics between the fluids, such as temperature or thermal conductivity. A fixed-point level sensor based on the difference in thermal conductivity between two fluids consists of an electrically heated thermistor inserted into the vessel. The temperature of the thermistor and consequently its electrical resistance increase as the thermal conductivity of the fluid in which it is immersed decreases. Because the thermal conductivity of liquids is markedly higher than that of vapors, such a device can be used as a point level detector for liquid-vapor interface.

Sonic Methods A fixed-point level detector based on sonic propagation characteristics is available for detection of a liquid-vapor interface. This device uses a piezoelectric transmitter and receiver, separated by a short gap. When the gap is filled with liquid, ultrasonic energy is transmitted across the gap, and the receiver actuates a relay. With a vapor filling the gap, the transmission of ultrasonic energy is insufficient to actuate the receiver.

Laser Level Transmitters These are designed for bulk solids, slurries, and opaque liquids. A laser near the vessel top fires a short pulse of light down to the surface of the process liquid, where it reflects back to a detector at the vessel top. A timing circuit measures the elapsed time and calculates the fluid depth. Lasers are attractive because lasers have no false echoes and can be directed through tight spaces.

Radar Level Transmitters Radar systems operate by beaming microwaves downward, from either a horn or parabolic dish located on top of the vessel. The signal reflects off the fluid surface back to the source after it detects a change in dielectric constant from the vapor to the fluid. The round-trip time is proportional to the distance to the fluid level. Guided-wave radar systems provide a rigid probe or flexible cable to guide the microwave down the height of the tank and back. Guided-wave radar is much more efficient than open-air radar because the guide provides a more focused energy path.

PHYSICAL PROPERTY MEASUREMENTS

Physical property measurements are sometimes equivalent to composition analyzers, because the composition can frequently be inferred from the measurement of a selected physical property.

Density and Specific Gravity For binary or pseudobinary mixtures of liquids or gases or a solution of a solid or gas in a solvent, the density is a function of the composition at a given temperature and pressure. Specific gravity is the ratio of the density of a noncompressible substance to the density of water at the same physical conditions. For nonideal solutions, empirical calibration will give the relationship between density and composition. Several types of measuring devices are described below.

Liquid Column Density may be determined by measuring the gauge pressure at the base of a fixed-height liquid column open to the atmosphere. If the process system is closed, then a differential pressure measurement is made between the bottom of the fixed-height liquid column and the vapor over the column. If vapor space is not always present, the differential pressure measurement is made between the bottom and top of a fixed-height column with the top measurement being made at a point below the liquid surface.

Displacement There are a variety of density measurement devices based on displacement techniques. A hydrometer is a constant-weight, variable-immersion device. The degree of immersion, when the weight of the hydrometer equals the weight of the displaced liquid, is a measure of the density. The hydrometer is adaptable to manual or automatic usage. Another modification includes a magnetic float suspended below a solenoid, the varying magnetic field maintaining the float at a constant distance from the solenoid. Change in position of the float, resulting from a density change, excites an electrical system which increases or decreases the current through the solenoid.

Direct Mass Measurement One type of densitometer measures the natural vibration frequency and relates the amplitude to changes in density. The density sensor is a V-shaped tube held stationary at its node points and allowed to vibrate at its natural frequency. At the curved end of the V is an electrochemical device that periodically strikes the tube. At the other end of the V the fluid is continuously passed through the tube. Between strikes, the tube vibrates at its natural frequency. The frequency changes directly in proportion to changes in density. A pickup device at the curved end of the V measures the frequency and electronically determines the fluid density. This technique is useful because it is not affected by the optical properties of the fluid. However, particulate matter in the process fluid can affect the accuracy.

Radiation-Density Gauges Gamma radiation may be used to measure the density of material inside a pipe or process vessel. The equipment is basically the same as for level measurement, except that here the pipe or vessel must be filled over the effective, irradiated sample volume. The source is mounted on one side of the pipe or vessel and the detector on the other side with appropriate safety radiation shielding surrounding the installation. Cesium 137 is used as the radiation source for path lengths under 610 mm (24 in) and cobalt 60 above 610 mm. The detector is usually an ionization gauge. The absorption of the gamma radiation is a function of density. Since the absorption path includes the pipe or vessel walls, an empirical calibration is used. Appropriate corrections must be made for the source intensity decay with time.

Viscosity Continuous viscometers generally measure either the resistance to flow or the drag or torque produced by movement of an element (moving surface) through the fluid. Each installation is normally applied over a narrow range of viscosities. Empirical calibration over this range allows use on both newtonian and nonnewtonian fluids. One such device uses a piston inside a cylinder. The hydrodynamic pressure of the process fluid raises the piston to a preset height. Then the inlet valve closes, the piston is allowed to free-fall, and the time of travel (typically a few seconds) is a measure of viscosity. Other geometries include the rotation of a spindle inside a sample chamber and a vibrating probe immersed in the fluid. Because viscosity depends on temperature, the viscosity measurement must be thermostated with a heater or cooler.

Refractive Index When light travels from one medium (e.g., air or glass) into another (e.g., a liquid), it undergoes a change of velocity and, if the angle of incidence is not 90°, a change of direction. For a given interface, angle, temperature, and wavelength of light, the amount of deviation or refraction will depend on the composition of

the liquid. If the sample is transparent, the normal method is to measure the refraction of light transmitted through the glass-sample interface. If the sample is opaque, the reflectance near the critical angle at a glass-sample interface is measured. In an online refractometer the process fluid is separated from the optics by a prism material. A beam of light is focused on a point in the fluid which creates a conic section of light at the prism, striking the fluid at different angles (greater than or less than the critical angle). The critical angle depends on the species concentrations; as the critical angle changes, the proportions of reflected and refracted light change. A photodetector produces a voltage signal proportional to the light refracted, when compared to a reference signal. Refractometers can be used with opaque fluids and in streams that contain particulates.

Dielectric Constant The dielectric constant of material represents its ability to reduce the electric force between two charges separated in space. This property is useful in process control for polymers, ceramic materials, and semiconductors. Dielectric constants are measured with respect to vacuum (1.0); typical values range from 2 (benzene) to 33 (methanol) to 80 (water). The value for water is higher than that for most plastics. A measuring cell is made of glass or some other insulating material and is usually doughnut-shaped, with the cylinders coated with metal, which constitute the plates of the capacitor.

Thermal Conductivity All gases and vapor have the ability to conduct heat from a heat source. At a given temperature and physical environment, radiation and convection heat losses will be stabilized, and the temperature of the heat source will mainly depend on the thermal conductivity and thus the composition of the surrounding gases. Thermal conductivity analyzers normally consist of a sample cell and a reference cell, each containing a combined heat source and detector. These cells are normally contained in a metal block with two small cavities in which the detectors are mounted. The sample flows through the sample cell cavity past the detector. The reference cell is an identical cavity with a detector through which a known gas flows. The combined heat source and detectors are normally either wire filaments or thermistors heated by a constant current. Because their resistance is a function of temperature, the sample detector resistance will vary with sample composition while the reference detector resistance will remain constant. The output from the detector bridge will be a function of sample composition.

CHEMICAL COMPOSITION ANALYZERS

Chemical composition is generally the most challenging online measurement. Before the era of online analyzers, messengers were required to deliver samples to the laboratory for analysis and to return the results to the control room. The long time delay involved prevented process adjustment from being made, affecting product quality. The development of online analyzers has automated this approach and reduced the analysis time. However, manual sampling is still frequently employed, especially in the specialty chemical industry where few instruments are commercially available. It is not unusual for a chemical composition analysis system to cost over \$100,000, so it is important to assess the payback of such an investment versus the cost of manual sampling. Potential quality improvements can be an important consideration.

A number of composition analyzers used for process monitoring and control require chemical conversion of one or more sample components preceding quantitative measurement. These reactions include formation of suspended solids for turbidimetric measurement, formation of colored materials for colorimetric detection, selective oxidation or reduction for electrochemical measurement, and formation of electrolytes for measurement by electrical conductance. Some nonvolatile materials may be separated and measured by gas chromatography after conversion to volatile derivatives.

Chromatographic Analyzers These analyzers are widely used for the separation and measurement of volatile compounds and of compounds that can be quantitatively converted to volatile derivatives. The compounds to be measured are separated by placing a portion of the sample in a chromatographic column and carrying the compounds through the column with a gas stream, called gas chroma-

tography, or GC. As a result of the different affinities of the sample components for the column packing, the compounds emerge successively as binary mixtures with the carrier gas. A detector at the column outlet measures a specific physical property that can be related to the concentrations of the compounds in the carrier gas. Both the concentration peak height and the peak height-time integral, i.e., peak area, can be related to the concentration of the compound in the original sample. The two detectors most commonly used for process chromatographs are the thermal conductivity detector and the hydrogen flame ionization detector. Thermal conductivity detectors, discussed earlier, require calibration for the thermal response of each compound. Hydrogen flame ionization detectors are more complicated than thermal conductivity detectors but are capable of 100 to 10,000 times greater sensitivity for hydrocarbons and organic compounds. For ultrasensitive detection of trace impurities, carrier gases must be specially purified.

Typically, all components can be analyzed in a 5- to 10-min time period (although miniaturized GCs are faster). High-performance liquid chromatography (HPLC) can be used to measure dissolved solute levels, including proteins.

Infrared Analyzers Many gaseous and liquid compounds absorb infrared radiation to some degree. The degree of absorption at specific wavelengths depends on molecular structure and concentration. There are two common detector types for nondispersive infrared analyzers. These analyzers normally have two beams of radiation, an analyzing and a reference beam. One type of detector consists of two gas-filled cells separated by a diaphragm. As the amount of infrared energy absorbed by the detector gas in one cell changes, the cell pressure changes. This causes movement in the diaphragm, which in turn causes a change in capacitance between the diaphragm and a reference electrode. This change in electrical capacitance is measured as the output. The second type of detector consists of two thermopiles or two bolometers, one in each of the two radiation beams. The infrared radiation absorbed by the detector is measured by a differential thermocouple output or a resistance thermometer (bolometer) bridge circuit.

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Ultraviolet and Visible-Radiation Analyzers Many gas and liquid compounds absorb radiation in the near-ultraviolet or visible region. For example, organic compounds containing aromatic and carbonyl structural groups are good absorbers in the ultraviolet region. Also many inorganic salts and gases absorb in the ultraviolet or visible region. In contrast, straight chain and saturated hydrocarbons, inert gases, air, and water vapor are essentially transparent. Process analyzers are designed to measure the absorbance in a particular wavelength band. The desired band is normally isolated by means of optical filters. When the absorbance is in the visible region, the term *colorimetry* is used. A phototube is the normal detector. Appropriate optical filters are used to limit the energy reaching the detector to the desired level and the desired wavelength region. Because absorption by the sample is logarithmic if a sufficiently narrow wavelength region is used, an exponential amplifier is sometimes used to compensate and produce a linear output.

Paramagnetism A few gases including O₂, NO, and NO₂ exhibit paramagnetic properties as a result of unpaired electrons. In a nonuniform magnetic field, paramagnetic gases, because of their magnetic susceptibility, tend to move toward the strongest part of the

field, thus displacing diamagnetic gases. Paramagnetic susceptibility of these gases decreases with temperature. These effects permit measurement of the concentration of the strongest paramagnetic gas, oxygen. An oxygen analyzer uses a dumbbell suspended in the magnetic field which is repelled or attracted toward the magnetic field depending on the magnetic susceptibility of the gas.

Other Analyzers Mass spectroscopy (MS) determines the partial pressures of gases in a mixture of directing ionized gases into a detector under a vacuum (10^{-6} torr), and the gas phase composition is then monitored more or less continuously based on the molecular weight of the species (Nichols, 1988). Sometimes GC is combined with MS to obtain a higher level of discrimination of the components present. Fiber-optic sensors are attractive options (although higher-cost) for acquiring measurements in harsh environments such as high temperature or pressure. The transducing technique used by these sensors is optical and does not involve electric signals, so they are immune to electromagnetic interference. Raman spectroscopy uses fiber optics and involves pulsed light scattering by molecules. It has a wide variety of applications in process control. Workman, Koch, and Veltkamp, *Anal. Chem.*, **75**: 2859, 2003.

Significant advances have occurred during the past decade to miniaturize the size of the measurement system in order to make online analysis economically feasible and to reduce the time delays that often are present in analyzers. Recently, chemical sensors have been placed on microchips, even those requiring multiple physical, chemical, and biochemical steps (such as electrophoresis) in the analysis. This device has been called *lab-on-a-chip*. The measurements of chemical composition can be direct or indirect, the latter case referring to applications where some property of the process stream is measured (such as refractive index) and then related to composition of a particular component.

ELECTROANALYTICAL INSTRUMENTS

Conductometric Analysis Solutions of electrolytes in ionizing solvents (e.g., water) conduct current when an electrical potential is applied across electrodes immersed in the solution. Conductance is a function of ion concentration, ionic charge, and ion mobility. Conductance measurements are ideally suited for measurement of the concentration of a single strong electrolyte in dilute solutions. At higher concentrations conductance becomes a complex, nonlinear function of concentration requiring suitable calibration for quantitative measurements.

Measurement of pH The primary detecting element in pH measurement is the glass electrode. A potential is developed at the pH-sensitive glass membrane as a result of differences in hydrogen ion activity in the sample and a standard solution contained within the electrode. This potential measured relative to the potential of the reference electrode gives a voltage which is expressed as pH. Instrumentation for pH measurement is among the most widely used process measurement devices. Rugged electrode systems and highly reliable electronic circuits have been developed for this use.

After installation, the majority of pH measurement problems are sensor-related, mostly on the reference side, including junction plugging, poisoning, and depletion of electrolyte. For the glass (measuring electrode), common difficulties are broken or cracked glass, coating, and etching or abrasion. Symptoms such as drift, sluggish response, unstable readings, and inability to calibrate are indications of measurement problems. Online diagnostics such as impedance measurements, wiring checks, and electrode temperature are now available in most instruments. Other characteristics that can be measured offline include efficiency or slope and asymmetry potential (offset), which indicate whether the unit should be cleaned or changed [Nichols, *Chem. Engr. Prog.*, **90**(12):64, 1994; McMillan, *Chem. Engr. Prog.*, **87**(12):30, 1991].

Specific-Ion Electrodes In addition to the pH glass electrode specific for hydrogen ions, a number of electrodes which are selective for the measurement of other ions have been developed. This selectivity is obtained through the composition of the electrode membrane (glass, polymer, or liquid-liquid) and the composition of the electrode. These electrodes are subject to interference from other ions, and the

response is a function of the total ionic strength of the solution. However, electrodes have been designed to be highly selective for specific ions, and when properly used, these provide valuable process measurements.

MOISTURE MEASUREMENT

Moisture measurements are important in the process industries because moisture can foul products, poison reactions, damage equipment, or cause explosions. Moisture measurements include both absolute moisture methods and relative-humidity methods. The absolute methods provide a primary output that can be directly calibrated in terms of dew point temperature, molar concentration, or weight concentration. Loss of weight on heating is the most familiar of these methods. The relative-humidity methods provide a primary output that can be more directly calibrated in terms of percentage of saturation of moisture.

Dew Point Method For many applications the dew point is the desired moisture measurement. When concentration is desired, the relation between water content and dew point is well known and available. The dew point method requires an inert surface whose temperature can be adjusted and measured, a sample gas stream flowing past the surface, a manipulated variable for adjusting the surface temperature to the dew point, and a means of detecting the onset of condensation.

Although the presence of condensate can be detected electrically, the original and most often used method is the optical detection of change in light reflection from an inert metallic-surface mirror. Some instruments measure the attenuation of reflected light at the onset of condensation. Others measure the increase of light dispersed and scattered by the condensate instead of, or in addition to, the reflected-light measurement. Surface cooling is obtained with an expendable refrigerant liquid, conventional mechanical refrigeration, or thermoelectric cooling. Surface-temperature measurement is usually made with a thermocouple or a thermistor.

Piezoelectric Method A piezoelectric crystal in a suitable oscillator circuit will oscillate at a frequency dependent on its mass. If the crystal has a stable hygroscopic film on its surface, the equivalent mass of the crystal varies with the mass of water sorbed in the film. Thus the frequency of oscillation depends on the water in the film. The analyzer contains two such crystals in matched oscillator circuits. Typically, valves alternately direct the sample to one crystal and a dry gas to the other on a 30-s cycle. The oscillator frequencies of the two circuits are compared electronically, and the output is the difference between the two frequencies. This output is then representative of the moisture content of the sample. The output frequency is usually converted to a variable dc voltage for meter readout and recording. Multiple ranges are provided for measurement from about 1 ppm to near saturation. The dry reference gas is preferably the same as the sample except for the moisture content of the sample. Other reference gases which are adsorbed in a manner similar to the dried sample gas may be used. The dry gas is usually supplied by an automatic dryer. The method requires a vapor sample to the detector. Mist striking the detector destroys the accuracy of measurement until it vaporizes or is washed off the crystals. Water droplets or mist may destroy the hygroscopic film, thus requiring crystal replacement. Vaporization or gas-liquid strippers may sometimes be used for the analysis of moisture in liquids.

Capacitance Method Several analyzers utilize the high dielectric constant of water for its detection in solutions. The alternating electric current through a capacitor containing all or part of the sample between the capacitor plates is measured. Selectivity and sensitivity are enhanced by increasing the concentration of moisture in the cell by filling the capacitor sample cell with a moisture-specific sorbent as part of the dielectric. This both increases the moisture content and reduces the amount of other interfering sample components. Granulated alumina is the most frequently used sorbent. These detectors may be cleaned and recharged easily and with satisfactory reproducibility if the sorbent itself is uniform.

Oxide Sensors Aluminum oxide can be used as a sensor for moisture analysis. A conductivity cell has one electrode node of aluminum,

which is anodized to form a thin film of aluminum oxide, followed by coating with a thin layer of gold (the opposite electrode). Moisture is selectively adsorbed through the gold layer and into the hygroscopic aluminum oxide layer, which in turn determines the electrical conductivity between gold and aluminum oxide. This value can be related to ppm water in the sample. This sensor can operate between near vacuum to several hundred atmospheres, and it is independent of flow rate (including static conditions). Temperature, however, must be carefully monitored. A similar device is based on phosphorous pentoxide. Moisture content influences the electric current between two inert metal electrodes, which are fabricated as a helix on the inner wall of a tubular nonconductive sample cell. For a constant dc voltage applied to the electrodes, a current flows that is proportional to moisture. The moisture is absorbed into the hygroscopic phosphorous pentoxide, where the current electrolyzes the water molecules into hydrogen and oxygen. This sensor will handle moisture up to 1000 ppm and 6-atm pressure. As with the aluminum oxide ion, temperature control is very important.

Photometric Moisture Analysis This analyzer requires a light source, a filter wheel rotated by a synchronous motor, a sample cell, a detector to measure the light transmitted, and associated electronics. Water has two absorption bands in the near-infrared region at 1400 and 1900 nm. This analyzer can measure moisture in liquid or gaseous samples at levels from 5 ppm up to 100 percent, depending on other chemical species in the sample. Response time is less than 1 s, and samples can be run up to 300°C and 400 psig.

OTHER TRANSDUCERS

Other types of transducers used in process measurements include mechanical drivers such as gear trains and electrical drivers such as a differential transformer or a Hall effect (semiconductor-based) sensor.

Gear Train Rotary motion and angular position are easily transmitted by various types of gear arrangements. A gear train in conjunction with a mechanical counter is a direct and effective way to obtain a digital readout of shaft rotations. The numbers on the counter can mean anything desired, depending on the gear ratio and the actuating device used to turn the shaft. A pointer attached to a gear train can be used to indicate a number of revolutions or a small fraction of a revolution for any specified pointer rotation.

Differential Transformer These devices produce an ac electrical output from linear movement of an armature. They are very versatile in that they can be designed for a full range of output with any range of armature travel up to several inches. The transformers have one or two primaries and two secondaries connected to oppose each other. With an ac voltage applied to the primary, the output voltage depends on the position of the armature and the coupling. Such devices produce accuracies of 0.5 to 1.0 percent of full scale and are used to transmit forces, pressures, differential pressures, or weights up to 1500 m. They can also be designed to transmit rotary motion.

Hall Effect Sensors Some semiconductor materials exhibit a phenomenon in the presence of a magnetic field which is adaptable to sensing devices. When a current is passed through one pair of wires attached to a semiconductor, such as germanium, another pair of wires properly attached and oriented with respect to the semiconductor will develop a voltage proportional to the magnetic field present and the current in the other pair of wires. Holding the exciting current constant and moving a permanent magnet near the semiconductor produce a voltage output proportional to the movement of the magnet. The magnet may be attached to a process variable measurement device which moves the magnet as the variable changes. Hall effect devices provide high speed of response, excellent temperature stability, and no physical contact.

SAMPLING SYSTEMS FOR PROCESS ANALYZERS

The sampling system consists of all the equipment required to present a process analyzer with a clean representative sample of a process stream and to dispose of that sample. When the analyzer is part of an automatic control loop, the reliability of the sampling system is as important as the reliability of the analyzer or the control equipment.

Sampling systems have several functions. The sample must be withdrawn from the process, transported, conditioned, introduced to the analyzer, and disposed. Probably the most common problem in sample system design is the lack of realistic information concerning the properties of the process material at the sampling point. Another common problem is the lack of information regarding the conditioning required so that the analyzer may utilize the sample without malfunction for long periods. Some samples require enough conditioning and treating that the sampling systems become equivalent to miniature online processing plants. These systems possess many of the same fabrication, reliability, and operating problems as small-scale pilot plants except that the sampling system must generally operate reliably for much longer periods of time.

Selecting the Sampling Point The selection of the sampling point is based primarily on supplying the analyzer with a sample whose composition or physical properties are pertinent to the control function to be performed. Other considerations include selecting locations that provide representative homogeneous samples with minimum transport delay, locations which collect a minimum of contaminating material, and locations which are accessible for test and maintenance procedures.

Sample Withdrawal from Process A number of considerations are involved in the design of sample withdrawal devices which will provide representative samples. For example, in a horizontal pipe carrying process fluid, a sample point on the bottom of the pipe will collect a maximum count of rust, scale, or other solid materials being carried along by the process fluid. In a gas stream, such a location will also collect a maximum amount of liquid contaminants. A sample point on the top side of a pipe will, for liquid streams, collect a maximum amount of vapor contaminants being carried along. Bends in the piping which produce swirls or cause centrifugal concentration of the denser phase may cause maximum contamination to be at unexpected locations. Two-phase process materials are difficult to sample for a total-composition representative sample.

A typical method for obtaining a sample of process fluid well away from vessel or pipe walls is an eduction tube inserted through a packing gland. This sampling method withdraws liquid sample and vaporizes it for transporting to the analyzer location. The transport lag time from the end of the probe to the vaporizer is minimized by using tubing having a small internal volume compared with pipe and valve volumes.

This sample probe may be removed for maintenance and reinstalled without shutting down the process. The eduction tube is made of material which will not corrode so that it will slide through the packing gland even after long periods of service. There may be a small amount of process fluid leakage until the tubing is withdrawn sufficiently to close the gate valve. A swaged ferrule on the end of the tube prevents accidental ejection of the eduction tube prior to removal of the packing gland. The section of pipe surrounding the eduction tube and extending into the process vessel provides mechanical protection for the eduction tube.

Sample Transport Transport time—the time elapsed between sample withdrawal from the process and its introduction into the analyzer—should be minimized, particularly if the analyzer is an automatic analyzer-controller. Any sample transport time in the analyzer-controller loop must be treated as equivalent to process dead time in determining conventional feedback controller settings or in evaluating controller performance. Reduction in transport time usually means transporting the sample in the vapor state.

Design considerations for sample lines are as follows:

1. The structural strength or protection must be compatible with the area through which the sample line runs.
2. Line size and length must be small enough to meet transport time requirements without excessive pressure drop or excessive bypass of sample at the analyzer input.
3. Line size and internal surface quality must be adequate to prevent clogging by the contaminants in the sample.
4. The prevention of a change of state of the sample may require installation, refrigeration, or heating of the sample line.
5. Sample line material must be such as to minimize corrosion due to sample or the environment.

Sample Conditioning Sample conditioning usually involves the removal of contaminants or some deleterious component from the sample mixture and/or the adjustment of temperature, pressure, and flow rate of the sample to values acceptable to the analyzer. Some of the more common contaminants which must be removed are rust, scale, corrosion products, deposits due to chemical reactions, and tar. In sampling some process streams, the material to be removed may include the primary process product such as polymer or the main constituent of the stream such as oil. In other

cases the material to be removed is present in trace quantities, e.g., water in an online chromatograph sample which can damage the chromatographic column packing. When contaminants or other materials which will hinder analysis represent a large percentage of the stream composition, their removal may significantly alter the integrity of the sample. In some cases, removal must be done as part of the analysis function so that removed material can be accounted for. In other cases, proper calibration of the analyzer output will suffice.

TELEMETERING AND TRANSMISSION

ANALOG SIGNAL TRANSMISSION

Modern control systems permit the measurement device, the control unit, and the final control element to be physically separated by several hundred meters, if necessary. This requires the transmission of the measured variable from the measurement device to the control unit, and the transmission of the controller output from the control unit to the final control element. In each case, transmission of a single value in only one direction is required. Such requirements can be met by analog signal transmission. A measurement range is defined for the value to be transmitted, and the value is basically transmitted as a percent of the span. For the measured variable, the logical span is the measurement span. For the controller output, the logical span is the range of the final control element (e.g., valve fully closed to valve fully open).

For pneumatic transmission systems, the signal range used for the transmission is 3 to 15 psig. In each pneumatic transmission system, there can be only one transmitter, but there can be any number of receivers. When most measurement devices were pneumatic, pneumatic transmission was the logical choice. However, with the displacement of pneumatic measurement devices by electronic devices, pneumatic transmission is now limited to applications where the unique characteristics of pneumatics make it the logical choice.

In order for electronic transmission systems to be less susceptible to interference from magnetic fields, current is used for the transmission signal instead of voltage. The signal range is 4 to 20 mA. In each circuit or "current loop," there can be only one transmitter. There can be more than one receiver, but not an unlimited number. For each receiver, a 250- Ω range resistor is inserted into the current loop, which provides a 1- to 5-V input to the receiving device. The number of receivers is limited by the power available from the transmitter.

Both pneumatic and electronic transmission use a *live zero*. This enables the receiver to distinguish a transmitted value of 0 percent of span from a transmitter or transmission system failure. Transmission of 0 percent of span provides a signal of 4 mA in electronic transmission. Should the transmitter or the transmission system fail (i.e., an open circuit in a current loop), the signal level would be 0 mA.

For most measurement variable transmissions, the lower range corresponds to 4 mA and the upper range corresponds to 20 mA. On an open circuit, the measured variable would fail to its lower range. In some applications, this is undesirable. For example, in a fired heater that is heating material to a target temperature, failure of the temperature measurement to its lower range value would drive the output of the combustion control logic to the maximum possible firing rate. In such applications, the analog transmission signal is normally inverted, with the upper range corresponding to 4 mA and the lower range corresponding to 20 mA. On an open circuit, the measured variable would fail to its upper range. For the fired heater, failure of the measured variable to its upper range would drive the output of the combustion control logic to the minimum firing rate.

DIGITAL SYSTEMS

With the advent of the microprocessor, digital technology began to be used for data collection, feedback control, and all other information processing requirements in production facilities. Such systems must

acquire data from a variety of measurement devices, and control systems must drive final control elements.

Analog Inputs and Outputs Analog inputs are generally divided into two categories:

1. *High level.* Where the source is a process transmitter, the range resistor in the current loop converts the 4- to 20-mA signal to a 1- to 5-V signal. The conversion equipment can be unipolar (i.e., capable of processing only positive voltages).

2. *Low level.* The most common low-level signals are inputs from thermocouples. These inputs rarely exceed 30 mV and could be zero or even negative. The conversion equipment must be bipolar (i.e., capable of processing positive and negative voltages).

Ultimately, such signals are converted to digital values via an analog-to-digital (A/D) converter. However, the A/D converter is normally preceded by two other components:

1. *Multiplexer.* This permits one A/D converter to service multiple analog inputs. The number of inputs to a multiplexer is usually between 8 and 256.

2. *Amplifier.* As A/D converters require high-level signals, a high-gain amplifier is required to convert low-level signals to high-level signals.

One of the important parameters for the A/D converter is its resolution. The resolution is stated in terms of the number of significant binary digits (bits) in the digital value. As the repeatability of most process transmitters is around 0.1 percent, the minimum acceptable resolution for a bipolar A/D converter is 12 bits, which translates to 11 data bits plus 1 bit for the sign. With this resolution, the analog input values can be represented to 1 part in 2^{11} , or 1 part in 2048. Normally, a 5-V input is converted to a digital value of 2000, which effectively gives a resolution of 1 part in 2000, or 0.05 percent. Very few process control systems utilize resolutions higher than 14 bits, which translates to a resolution of 1 part in 8000, or 0.0125 percent.

For 4- to 20-mA inputs, the resolution is not quite as good as stated above. For a 12-bit bipolar A/D converter, 1 V converts to a digital value of 400. Thus, the range for the digital value is 400 to 2000, making the effective input resolution 1 part in 1600, or 0.0625 percent. Sometimes this is expressed as a resolution of 7.4 bits, where $2^{7.4} = 1600$.

On the output side, dedicated digital-to-analog (D/A) converters are provided for each analog output. Outputs are normally unipolar and require a lower resolution than inputs. A 10-bit resolution is normally sufficient, giving a resolution of 1 part in 1000, or 0.1 percent.

Pulse Inputs Where the sensor within the measurement device is digital, analog-to-digital conversion can be avoided. For rotational devices, the rotational element can be outfitted with a shaft encoder that generates a known number of pulses per revolution. The digital system can process such inputs in any of the following ways:

1. Count the number of pulses over a fixed interval of time.
2. Determine the time for a specified number of pulses.
3. Determine the duration of time between the leading (or trailing) edges of successive pulses.

Of these, the first option is the most commonly used in process applications. Turbine flowmeters are probably the most common example where pulse inputs are used. Another example is a watt-hour meter. Basically any measurement device that involves a rotational element can be interfaced via pulses. Occasionally, a nonrotational measurement device can generate pulse outputs. One example is the

vortex-shedding meter, where a pulse can be generated when each vortex passes over the detector.

Serial Interfaces Some very important measurement devices cannot be reasonably interfaced via either analog or pulse inputs. Two examples are the following:

1. Chromatographs can perform a total composition analysis for a sample. It is possible but inconvenient to provide an analog input for each component. Furthermore, it is often desirable to capture other information, such as the time that the analysis was made (normally the time the sample was injected).

2. Load cells are capable of resolutions of 1 part in 100,000. A/D converters for analog inputs cannot even approach such resolutions.

One approach to interfacing with such devices involves serial interfaces. This has two aspects:

1. *Hardware interface.* The RS-232 interface standard is the basis for most serial interfaces.

2. *Protocol.* This is interpreting the sequence of characters transmitted by the measurement device. There are no standards for protocols, which means that custom software is required.

One advantage of serial interfaces is that two-way communication is possible. For example, a "tare" command can be issued to a load cell.

Microprocessor-Based Transmitters The cost of microprocessor technology has declined to the point where it is economically feasible to incorporate a microprocessor into each transmitter. Such microprocessor-based transmitters are often referred to as *smart* transmitters. As opposed to conventional or *dumb* transmitters, the smart transmitters offer the following capabilities:

1. They check on the internal electronics, such as verifying that the voltage levels of internal power supplies are within specifications.

2. They check on environmental conditions within the instruments, such as verifying that the case temperature is within specifications.

3. They perform compensation of the measured value for conditions within the instrument, such as compensating the output of a pressure transmitter for the temperature within the transmitter. Smart transmitters are much less affected by temperature and pressure variations than are conventional transmitters.

4. They perform compensation of the measured value for other process conditions, such as compensating the output of a capacitance level transmitter for variations in process temperature.

5. They linearize the output of the transmitter. Functions such as square root extraction of the differential pressure for a head-type flowmeter can be done within the instrument instead of within the control system.

6. They configure the transmitter from a remote location, such as changing the span of the transmitter output.

7. They do automatic recalibration of the transmitter. Although this is highly desired by users, the capabilities, if any, in this respect depend on the type of measurement.

Due to these capabilities, smart transmitters offer improved performance over conventional transmitters.

Transmitter/Actuator Networks With the advent of smart transmitters and smart actuators, the limitations of the 4- to 20-mA analog signal transmission retard the full utilization of the capabilities of smart devices. For smart transmitters, the following capabilities are required:

1. *Transmission of more than one value from a transmitter.* Information beyond the measured variable is available from the smart transmitter. For example, a smart pressure transmitter can also report the temperature within its housing. Knowing that this temperature is above normal values permits corrective action to be taken before the device fails. Such information is especially important during the initial commissioning of a plant.

2. *Bidirectional transmission.* Configuration parameters such as span, engineering units, and resolution must be communicated to the smart transmitter.

Similar capabilities are required for smart actuators.

Eventually this technology, generally referred to as field bus, will replace all other forms of transmission (analog, pulse, and serial). Its acceptance by the user community has been slow, mainly due to the absence of a standard. Smart transmitters and smart valves have gained widespread acceptance, but initially the signal transmission continued to be via current loops. During this interim, several manu-

facturers introduced products based on proprietary communications technologies. Most manufacturers released sufficient information for others to develop products using their communications technologies, but those considering doing so found the costs to be an impediment.

A standard is now available, specifically, IEC 61158, "Digital Data Communications for Measurement and Control." In most industries, a standard defines only one way of doing something, but not in the computer industry. IEC 61158 defines five different communications technologies. Several manufacturers had selected a direction, and standards are not immune to commercial considerations. The net result is that one person's "field bus" might not be the same as another's. Most plants select one technology for their field bus, although it is possible to have multiple field bus interfaces on a control system and not have all the same type.

Although there will be winners and losers, most companies find the appealing features of field bus technologies too great to continue with the older technologies. U.S. manufacturers have generally chosen a field bus technology known as Foundation Fieldbus. In Europe, a field bus technology known as Profibus has gained wide acceptance. Probably the major advantages of these technologies lie in installation and commissioning. There are fewer wires and connections in a field bus installation. In most plants, the cost of the wiring is proving to be a minor issue. With fewer wires and connections, errors in field bus installations are fewer and can be located more quickly. The major cost savings prove to be in configuring and commissioning the system. With time, field bus will largely replace analog, pulse, and serial signal transmissions within industrial facilities.

FILTERING AND SMOOTHING

A signal received from a process transmitter generally contains the following components distinguished by their frequency [frequencies are measured in hertz (Hz), with 60-cycle ac being a 60-Hz frequency]:

1. *Low-frequency process disturbances.* The control system is expected to react to these disturbances.

2. *High-frequency process disturbances.* The frequency of these disturbances is beyond the capability of the control system to effectively react.

3. *Measurement noise.*

4. *Stray electrical pickup, primarily 50- or 60-cycle ac.*

The objective of filtering and smoothing is to remove the last three components, leaving only the low-frequency process disturbances. Normally this has to be accomplished by using the proper combination of analog and digital filters. Sampling a continuous signal results in a phenomenon often referred to as aliasing or foldover. To represent a sinusoidal signal, a minimum of four samples are required during each cycle. Consequently, when a signal is sampled at a frequency ω_s , all frequencies higher than $(\pi/2)\omega_s$ cannot be represented at their original frequency. Instead, they are present in the sampled signal with their original amplitude but at a lower-frequency harmonic. Because of the aliasing or foldover issues, a combination of analog and digital filtering is usually required. The sampler (i.e., the A/D converter) must be preceded by an analog filter that rejects those high-frequency components such as stray electrical pickup that would result in foldover when sampled. In commercial products, analog filters are normally incorporated into the input processing hardware by the manufacturer. The software then permits the user to specify digital filtering to remove any undesirable low-frequency components.

On the analog side, the filter is often the conventional resistor-capacitor or RC filter. However, other possibilities exist. For example, one type of A/D converter is called an *integrating A/D* because the converter basically integrates the input signal over a fixed interval of time. By making the interval $\frac{1}{60}$ s, this approach provides excellent rejection of any 60-Hz electrical noise.

On the digital side, the input processing software generally provides for smoothing via the exponentially weighted moving average, which is the digital counterpart to the RC network analog filter. The smoothing equation is as follows:

$$y_i = \alpha x_i + (1 - \alpha)y_{i-1} \quad (8-107)$$

where x_i = current value of input

y_i = current output from filter

y_{i-1} = previous output from filter
 α = filter coefficient

The degree of smoothing is determined by the filter coefficient α , with $\alpha = 1$ being no smoothing and $\alpha = 0$ being infinite smoothing (no effect of new measurements). The filter coefficient α is related to the filter time constant τ_f and the sampling interval Δt by

$$\alpha = 1 - \exp\left(\frac{-\Delta t}{\tau_f}\right) \quad (8-108)$$

or by the approximation

$$\alpha = \frac{\Delta t}{\Delta t + \tau_f} \quad (8-109)$$

Another approach to smoothing is to use the arithmetic moving average, which is represented by

$$y_i = \frac{\sum_{j=1}^n x_{i+1-j}}{n} \quad (8-110)$$

The term *moving* is used because the filter software maintains a storage array with the previous n values of the input. When a new value is received, the oldest value in the storage array is replaced with the new value, and the arithmetic average is recomputed. This permits the filtered value to be updated each time a new input value is received.

In process applications, determining τ_f (or α) for the exponential filter and n for the moving average filter is often done merely by observing the behavior of the filtered value. If the filtered value is "bouncing," the degree of smoothing (that is, τ_f or n) is increased. This can easily lead to an excessive degree of filtering, which will limit the performance of any control system that uses the filtered value. The degree of filtering is best determined from the frequency spectrum of the measured input, but such information is rarely available for process measurements.

ALARMS

The purpose of an alarm is to alert the process operator to a process condition that requires immediate attention. An alarm is said to occur whenever the abnormal condition is detected and the alert is issued. An alarm is said to return to normal when the abnormal condition no longer exists. Analog alarms can be defined on measured variables, calculated variables, controller outputs, and the like. For analog alarms, the following possibilities exist:

1. *High/low alarms.* A high alarm is generated when the value is greater than or equal to the value specified for the high-alarm limit. A low alarm is generated when the value is less than or equal to the value specified for the low-alarm limit.

2. *Deviation alarms.* An alarm limit and a target are specified. A high-deviation alarm is generated when the value is greater than or equal to the target plus the deviation alarm limit. A low-deviation alarm is generated when the value is less than or equal to the target minus the deviation alarm limit.

3. *Trend or rate-of-change alarms.* A limit is specified for the maximum rate of change, usually specified as a change in the measured value per minute. A high-trend alarm is generated when the rate of change of the variable is greater than or equal to the value specified for the trend alarm limit. A low-trend alarm is generated when the rate of change of the variable is less than or equal to the negative of the value specified for the trend alarm limit.

Most systems permit multiple alarms of a given type to be configured for a given value. For example, configuring three high alarms provides a high alarm, a high-high alarm, and a high-high-high alarm.

One operational problem with analog alarms is that noise in the variable can cause multiple alarms whenever its value approaches a limit. This can be avoided by defining a dead band on the alarm. For example, a high alarm would be processed as follows:

1. *Occurrence.* The high alarm is generated when the value is greater than or equal to the value specified for the high-alarm limit.

2. *Return to normal.* The high-alarm return to normal is generated when the value is less than or equal to the high-alarm limit less the dead band.

As the degree of noise varies from one input to the next, the dead band must be individually configurable for each alarm.

Discrete alarms can be defined on discrete inputs, limit switch inputs from on/off actuators, and so on. For discrete alarms, the following possibilities exist:

1. *Status alarms.* An expected or normal state is specified for the discrete value. A status alarm is generated when the discrete value is other than its expected or normal state.

2. *Change-of-state alarm.* A change-of-state alarm is generated on any change of the discrete value.

The expected sequence of events on an alarm is basically as follows:

1. The alarm occurs. This usually activates an audible annunciator.

2. The alarm occurrence is acknowledged by the process operator.

When all alarms have been acknowledged, the audible annunciator is silenced.

3. Corrective action is initiated by the process operator.

4. The alarm condition returns to normal.

However, additional requirements are imposed at some plants. Sometimes the process operator must acknowledge the alarm's return to normal. Some plants require that the alarm occurrence be reissued if the alarm remains in the occurred state longer than a specified time. Consequently, some "personalization" of the alarming facilities is done.

When alarms were largely hardware-based (i.e., the panel alarm systems), the purchase and installation of the alarm hardware imposed a certain discipline on the configuration of alarms. With digital systems, the suppliers have made it extremely easy to configure alarms. In fact, it is sometimes easier to configure alarms on a measured value than not to configure the alarms. Furthermore, the engineer assigned the responsibility for defining alarms is often paranoid that an abnormal process condition will go undetected because an alarm has not been configured. When alarms are defined on every measured and calculated variable, the result is an excessive number of alarms, most of which are duplicative and unnecessary.

The accident at the Three Mile Island nuclear plant clearly demonstrated that an alarm system can be counterproductive. An excessive number of alarms can distract the operator's attention from the real problem that needs to be addressed. Alarms that merely tell the operator something that is already known do the same. In fact, a very good definition of a nuisance alarm is one that informs the operator of a situation of which the operator is already aware. The only problem with applying this definition lies determining what the operator already knows.

Unless some discipline is imposed, engineering personnel, especially where contractors are involved, will define far more alarms than plant operations require. This situation may be addressed by simply setting the alarm limits to values such that the alarms never occur. However, changes in alarms and alarm limits are changes from the perspective of the process safety management regulations. It is prudent to impose the necessary discipline to avoid an excessive number of alarms. Potential guidelines are as follows:

1. For each alarm, a specific action is expected from the process operator. Operator actions such as call maintenance are inappropriate with modern systems. If maintenance needs to know, modern systems can inform maintenance directly.

2. Alarms should be restricted to abnormal situations for which the process operator is responsible. A high alarm on the temperature in one of the control system cabinets should not be issued to the process operator. Correcting this situation is the responsibility of maintenance, not the process operator.

3. Process operators are expected to be exercising normal surveillance of the process. Therefore, alarms are not appropriate for situations known to the operator either through previous alarms or through normal process surveillance. The "sleeping operator" problem can be addressed by far more effective means than the alarm system.

4. When the process is operating normally, no alarms should be triggered. Within the electric utility industry, this design objective is

known as *darkboard*. Application of darkboard is especially important in batch plants, where much of the process equipment is operated intermittently.

Ultimately, guidelines such as those above will be taken seriously only if production management pays attention to the process alarms. The consequences of excessive and redundant alarms will be felt primarily by those responsible for production operations. Therefore, production management must make adequate resources available for reviewing and analyzing the proposed alarm configurations.

Another serious distraction to a process operator is the multiple-alarm event, where a single event within the process results in multiple alarms. When the operator must individually acknowledge each alarm, considerable time can be lost in silencing the obnoxious annunciator before the real problem is addressed. Air-handling systems are especially vulnerable to this, where any fluctuation in pressure (e.g., resulting from a blower trip) can cause a number of pressure alarms to occur.

Point alarms (high alarms, low alarms, status alarms, etc.) are especially vulnerable to the multiple-alarm event. This can be addressed in one of two ways:

1. *Ganging alarms*. Instead of individually issuing the point alarms, all alarms associated with a certain aspect of the process

are configured to give a single trouble alarm. The responsibility rests entirely with the operator to determine the nature of the problem.

2. *Intelligent alarms*. Logic is incorporated into the alarm system to determine the nature of the problem and then issue a single alarm to the process operator.

While the intelligent alarm approach is clearly preferable, substantial process analysis is required to support intelligent alarming. Meeting the following two objectives is quite challenging:

1. The alarm logic must consistently detect abnormal conditions within the process.
2. The alarm logic must not issue an alert to an abnormal condition when in fact none exists.

Often the latter case is more challenging than the former. Logically, the intelligent alarm effort must be linked to the process hazards analysis. Developing an effective intelligent alarming system requires substantial commitments of effort, involving process engineers, control systems engineers, and production personnel. Methodologies such as expert systems can facilitate the implementation of an intelligent alarming system, but they must still be based on a sound analysis of the potential process hazards.

DIGITAL TECHNOLOGY FOR PROCESS CONTROL

GENERAL REFERENCES: Auslander and Ridgely, *Design and Implementation of Real-Time Software for the Control of Mechanical Systems*, Prentice-Hall, Upper Saddle River, N.J., 2002. Herb, *Understanding Distributed Processor Systems for Control*, ISA, Research Triangle Park, N.C., 1999. Hughes, *Programmable Controllers*, ISA, Research Triangle Park, N.C., 1997. Johnson, *Process Control Instrumentation Technology*, 6th ed., Prentice-Hall, Upper Saddle River, N.J., 2000. Liptak, *Instrument Engineers Handbook*, Chilton Book Company, Philadelphia, 1995. Webb and Reis, *Programmable Logic Controllers*, 4th ed., Prentice-Hall, Upper Saddle River, N.J., 2002.

Since the 1970s, process controls have evolved from pneumatic analog technology to electronic analog technology to microprocessor-based controls. Electronic and pneumatic controllers have now virtually disappeared from process control systems, which are dominated by programmable electronic systems based on microprocessor technology.

HIERARCHY OF INFORMATION SYSTEMS

Coupling digital controls with networking technology permits information to be passed from level to level within a corporation at high rates of speed. This technology is capable of presenting the measured variable from a flow transmitter installed in a plant in a remote location anywhere in the world to the company headquarters in less than 1 s.

A hierarchical representation of the information flow within a company leads to a better understanding of how information is passed from one layer to the next. Such representations can be developed in varying degrees of detail, and most companies have developed one that describes their specific practices. The following hierarchy consists of five levels, as shown in Fig. 8-30.

Measurement Devices and Final Control Elements This lowest layer couples the control and information systems to the process. The measurement devices provide information on the current conditions within the process. The final control elements permit control decisions to be imposed on the process. Although traditionally analog, smart transmitters and smart valves based on microprocessor technology are now beginning to dominate this layer.

Safety and Environmental/Equipment Protection The level 2 functions play a critical role by ensuring that the process is operating safely and satisfies environmental regulations. Process safety relies on the principle of multiple protection layers that involve groupings of equipment and human actions. One layer includes process control functions, such as alarm management during abnormal situations, and safety instrumented systems for emergency shutdowns. The safety equipment (including sensors and block valves) operates independently of the regular instrumentation used for regulatory control in level 3. Sensor validation techniques can be employed to confirm that the sensors are functioning properly.

Regulatory Controls The objective of this layer is to operate the process at or near the targets supplied by a higher layer in the hierarchy. To achieve consistent process operations, a high degree of automatic control is required from the regulatory layer. The direct result is a reduction in variance in the key process variables. More uniform product quality is an obvious benefit. However, consistent process operation is a prerequisite for optimizing the process operations. To ensure success for the upper-level functions, the first objective of any automation effort must be to achieve a high degree of regulatory control.

Real-Time Optimization Determining the most appropriate targets for the regulatory layer is the responsibility of the RTO layer. Given the current production and quality targets for a unit, RTO determines how the process can be best operated to meet them. Usually this optimization has a limited scope, being confined to a single production unit or possibly even a single unit operation within a production unit. RTO translates changes in factors such as current process efficiencies, current energy costs, cooling medium temperatures, and so on to changes in process operating targets so as to optimize process operations.

Production Controls The nature of the production control logic differs greatly between continuous and batch plants. A good example of production control in a continuous process is refinery optimization. From the assay of the incoming crude oil, the values of the various possible refined products, the contractual commitments to deliver certain products, the performance measures of the various units within a refinery, and the like, it is possible to determine the mix of products that optimizes the economic return from processing this crude. The solution of this problem involves many relationships and constraints and is solved with techniques such as linear programming.

In a batch plant, production control often takes the form of routing or short-term scheduling. For a multiproduct batch plant, determining the long-term schedule is basically a manufacturing resource planning (MRP) problem, where the specific products to be manufactured and the amounts to be manufactured are determined from the outstanding orders, the raw materials available for production, the production capacities of the process equipment, and other factors. The goal of the MRP effort is the long-term schedule, which is a list of the products to be manufactured over a specified period of time (often one week). For each product on the list, a target amount is also specified. To manufacture this amount usually involves several batches. The term *production run* often refers to the sequence of batches required to make the target amount of product, so in effect the long-term schedule is a list of production runs.

Most multiproduct batch plants have more than one piece of equipment of each type. Routing refers to determining the specific pieces of equipment that will be used to manufacture each run on the long-term

production schedule. For example, the plant might have five reactors, eight neutralization tanks, three grinders, and four packing machines. For a given run, a rather large number of routes are possible. Furthermore, rarely is only one run in progress at a given time. The objective of routing is to determine the specific pieces of production equipment to be used for each run on the long-term production schedule. Given the dynamic nature of the production process (equipment failures, insertion/deletion of runs into the long-term schedule, etc.), the solution of the routing problem continues to be quite challenging.

Corporate Information Systems Terms such as *management information systems* (MIS), *enterprise resource planning* (ERP), *supply chain management* (SCM), and *information technology* (IT) are frequently used to designate the upper levels of computer systems within a corporation. From a control perspective, the functions performed at this level are normally long-term and/or strategic. For example, in a processing plant, long-term contracts are required with the providers of the feedstocks. A forecast must be developed for the demand for possible products from the plant. This demand must be translated to needed raw materials, and then contracts executed with the suppliers to deliver these materials on a relatively uniform schedule.

DIGITAL HARDWARE IN PROCESS CONTROL

Digital control technology was first applied to process control in 1959, using a single central computer (and analog backup for reliability). In the mid-1970s, a microcomputer-based process control architecture referred to as a *distributed control system* (DCS) was introduced and rapidly became a commercial success. A DCS consists of some number of microprocessor-based nodes that are interconnected by a digital communications network, often called a data highway. Today the DCS is still dominant, but there are other options for carrying out computer control, such as single-loop controllers, programmable logic controllers, and personal computer controllers. A brief review of each type of controller device is given below; see the following section "Controllers, Final Control Elements, and Regulators" for more details on controller hardware options.

Single-Loop Controllers The single-loop controller (SLC) is the digital equivalent of analog single-loop controllers. It is a self-contained microprocessor-based unit that can be rack-mounted. Many manufacturers produce single processor units that handle cascade control or multiple loops, typically 4, 8, or 16 loops per unit, and incorporate self-tuning or auto-tuning PID control algorithms.

Programmable Logic Controllers Programmable logic controllers (PLCs) are simple digital devices that are widely used to control sequential and batch processes, although they now have additional functions that implement PID control and other mathematical operations. PLCs can be utilized as stand-alone devices or in conjunction with digital computer control systems. Because the logical functions are stored in main memory, one measure of a PLC's capability is its memory scan rate. Most PLCs are equipped with an internal timing capability to delay an action by a prescribed amount of time, to execute an action at a prescribed time, and so on. Newer PLC models often are networked to serve as one component of a DCS, with operator I/O provided by a separate component in the network. A distinction is made between configurable and programmable PLCs. The term *configurable* implies that logical operations (performed on inputs to yield a desired output) are located in PLC memory, perhaps in the form of ladder diagrams by selecting from a PLC menu or by direct interrogation of the PLC. Most control engineers prefer the simplicity of configuring the PLC to the alternative of programming it. However, some batch applications, particularly those involving complex sequencing, are best handled by a programmable approach.

Personal Computer Controllers In comparison with PLCs, PCs have the advantages of lower purchase cost, graphics output, large memory, large selection of software products (including databases and development tools), more programming options (use of C or Java versus ladder logic), richer operating systems, and open networking. PLCs have the following advantages: lower maintenance cost, operating system and hardware optimized for control, fast boot times, ruggedness, low failure rate, longer support for product models, and self-contained units.

A number of vendors have introduced so-called scalable process control systems. Scalable means that the size of the control and instrumentation systems is easily expanded by simply adding more devices. This feature is possible because of the trend toward more openness (i.e., "plug and play" between devices), smaller size, lower cost, greater flexibility, and more off-the-shelf hardware and software in digital control systems. A typical system includes personal computers, an operating system, object-oriented database technology, modular field-mounted controllers, and plug-and-play integration of both system and intelligent field devices. New devices are automatically recognized and configured with the system. Advanced control algorithms can be executed at the PC level.

Distributed Control System Figure 8-65 depicts a representative distributed control system. The DCS consists of many commonly

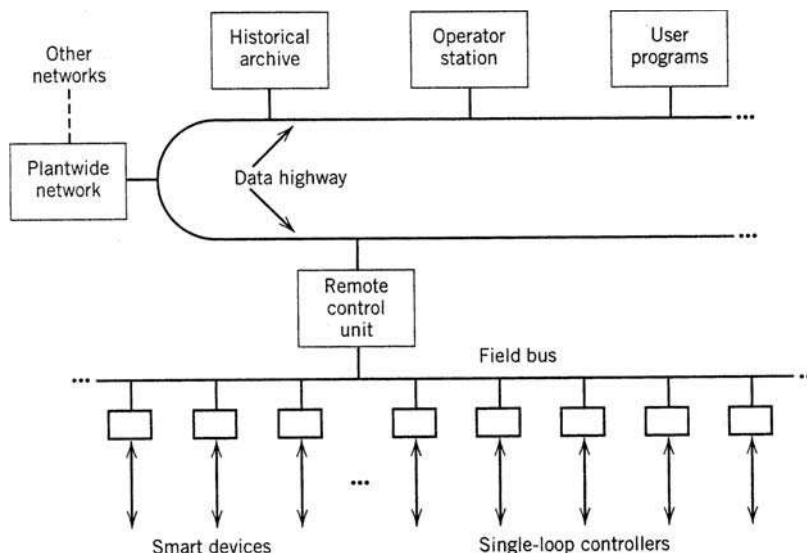


FIG. 8-65 A DCS using a broadband (high-bandwidth) data highway and field bus connected to a single remote control unit that operates smart devices and single-loop controllers.

used components, including multiplexers (MUXs), single-loop and multiple-loop controllers, PLCs, and smart devices. A system includes some of or all the following components:

1. *Control network.* The control network is the communication link between the individual components of a network. Coaxial cable and, more recently, fiber-optic cable have often been used, sometimes with Ethernet protocols. A redundant pair of cables (dual redundant highway) is normally supplied to reduce the possibility of link failure.

2. *Workstations.* Workstations are the most powerful computers in the system, capable of performing functions not normally available in other units. A workstation acts both as an arbitrator unit to route internodal communications and as the database server. An operator interface is supported, and various peripheral devices are coordinated through the workstations. Computationally intensive tasks, such as real-time optimization or model predictive control, are implemented in a workstation. Operators supervise and control processes from these workstations. Operator stations may be connected directly to printers for alarm logging, printing reports, or process graphics.

3. *Remote control units (RCUs).* These components are used to implement basic control functions such as PID control. Some RCUs may be configured to acquire or supply set points to single-loop controllers. Radio telemetry (wireless) may be installed to communicate with MUX units located at great distances.

4. *Application stations.* These separate computers run application software such as databases, spreadsheets, financial software, and simulation software via an OPC interface. OPC is an acronym for object linking and embedding for process control, a software architecture based on standard interfaces. These stations can be used for e-mail and as web servers, for remote diagnosis configuration, and even for operation of devices that have an IP (Internet Protocol) address. Applications stations can communicate with the main database contained in online mass storage systems. Typically hard disk drives are used to store active data, including online and historical databases and nonmemory resident programs. Memory resident programs are also stored to allow loading at system start-up.

5. *Field buses and smart devices.* An increasing number of field-mounted devices are available that support digital communication of the process I/O in addition to, or in place of, the traditional 4- to 20-mA current signal. These devices have greater functionality, resulting in reduced setup time, improved control, combined functionality of separate devices, and control valve diagnostic capabilities. Digital communication also allows the control system to become completely distributed where, e.g., a PID control algorithm could reside in a valve positioner or in a sensor/transmitter.

DISTRIBUTED DATABASE AND THE DATABASE MANAGER

A database is a centralized location for data storage. The use of databases enhances system performance by maintaining complex relations between data elements while reducing data redundancy. A database may be built based on the relational model, the entity relationship model, or some other model. The database manager is a system utility program or programs acting as the gatekeeper to the databases. All functions retrieving or modifying data must submit a request to the manager. Information required to access the database includes the tag name of the database entity, often referred to as a point, the attributes to be accessed, and the values, if modifying. The database manager maintains the integrity of the databases by executing a request only when not processing other conflicting requests. Although a number of functions may read the same data item at the same time, writing by a number of functions or simultaneous read and write of the same data item is not permitted.

To allow flexibility, the database manager must also perform point addition or deletion. However, the ability to create a point type or to add or delete attributes of a point type is not normally required because, unlike other data processing systems, a process control system normally involves a fixed number of point types and related attributes. For example, analog and binary input and output types are required for process I/O points. Related attributes for these point types include tag names, values, and hardware addresses. Different

system manufacturers may define different point types using different data structures. We will discuss other commonly used point types and attributes as they appear.

Data Historian A historical database is built similar to an online database. Unlike their online counterparts, the information stored in a historical database is not normally accessed directly by other subsystems for process control and monitoring. Periodic reports and long-term trends are generated based on the archived data. The reports are often used for planning and system performance evaluations such as statistical process (quality) control. The trends may be used to detect process drifts or to compare process variations at different times.

The historical data are sampled at user-specified intervals. A typical process plant contains a large number of data points, but it is not feasible to store data for all points at all times. The user determines if a data point should be included in the list of archive points. Most systems provide archive-point menu displays. The operators are able to add or delete data points to the archive point lists. The sampling periods are normally some multiples of their base scan frequencies. However, some systems allow historical data sampling of arbitrary intervals. This is necessary when intermediate virtual data points that do not have the scan frequency attribute are involved. The archive point lists are continuously scanned by the historical database software. Online databases are polled for data, and the times of data retrieval are recorded with the data obtained. To conserve storage space, different data compression techniques are employed by various manufacturers.

DIGITAL FIELD COMMUNICATIONS AND FIELD BUS

Microprocessor-based equipment, such as smart instruments and single-loop controllers with digital communications capability, are now used extensively in process plants. A field bus, which is a low-cost protocol, is necessary to perform efficient communication between the DCS and devices that may be obtained from different vendors. Figure 8-65 illustrates a LAN-based DCS with field buses and smart devices connected to a data highway.

Presently, there are several regional and industry-based field bus standards, including the French standard (FIP), the German standard (Profibus), and proprietary standards by DCS vendors, generally in the United States, led by the Fieldbus Foundation, a not-for-profit corporation. As of 2002, international standards organizations had adopted all these field bus standards rather than a single unifying standard. However, there will likely be further developments in field bus standards in the future. A benefit of standardizing the field bus is that it has encouraged third-party traditional equipment manufacturers to enter the smart equipment market, resulting in increased competition and improved equipment quality.

Several manufacturers have made available field bus controllers that reside in the final control element or measurement transmitter. A suitable communications modem is present in the device to interface with a proprietary PC-based, or hybrid analog/digital bus network. At present, field bus controllers are single-loop controllers containing 8- and 16-bit microprocessors that support the basic PID control algorithm as well as other functionalities. Case studies in implementing such digital systems have shown significant reductions in cost of installation (mostly cabling and connections) versus traditional analog field communication.

A general movement has also begun in the direction of using the high-speed Ethernet standard (100 Mbit/s or higher), allowing data transfer by TCP/IP that is used pervasively in computer networking. This would allow any smart device to communicate directly with others in the network or to be queried by the operator regarding its status and settings.

INTERNODAL COMMUNICATIONS

For a group of computers to become a network, intercomputer communication is required. Prior to the 1980s, each system vendor used a proprietary protocol to network its computers. Ad hoc approaches were sometimes used to connect third-party equipment but were not cost-effective with regard to system maintenance, upgrade, and expansion.

The recent introduction of standardized communication protocols has led to a decrease in capital cost. Most current DCS network protocol designs are based on the ISO-OSI seven-layer model with physical, data link, network, transport, session, presentation, and application layers.

An effort in standardizing communication protocols for plant automation was initiated by General Motors in the early 1980s. This work culminated in the Manufacturing Automation Protocol (MAP), which adopted the ISO-OSI standards as its basis. MAP specifies a broadband backbone local-area network (LAN) that incorporates a selection of existing standard protocols suitable for discrete component manufacturing. MAP was intended to address the integration of DCSs used in process control. Subsequently, TCP/IP (Transmission Control Protocol/Internet Protocol) was adopted for communication between nodes that have different operating systems.

Communication programs also act as links to the database manager. When data are requested from a remote node, the database manager transfers the request to the remote node database manager via the communication programs. The remote node communication programs then relay the request to the resident database manager and return the requested data. The remote database access and the existence of communications equipment and software are transparent to the user.

PROCESS CONTROL LANGUAGES

Originally, software for process control utilized high-level programming languages such as FORTRAN and BASIC. Some companies have incorporated libraries of software routines for these languages, but others have developed specialty languages characterized by natural language statements. The most widely adopted user-friendly approach is the fill-in-the-forms or table-driven process control languages (PCLs). Typical PCLs include function block diagrams, ladder logic, and programmable logic. The core of these languages is a num-

ber of basic function blocks or software modules, such as analog in, digital in, analog out, digital out, PID, summer, and splitter. Using a module is analogous to calling a subroutine in conventional FORTRAN or C programs.

In general, each module contains one or more inputs and an output. The programming involves connecting outputs of function blocks to inputs of other blocks via the graphical-user interface. Some modules may require additional parameters to direct module execution. Users are required to fill in templates to indicate the sources of input values, the destinations of output values, and the parameters for forms/tables prepared for the modules. The source and destination blanks may specify process I/O channels and tag names when appropriate. To connect modules, some systems require filling in the tag names of modules originating or receiving data.

Many DCSs allow users to write custom code (similar to BASIC) and attach it to data points, so that the code is executed each time the point is scanned. The use of custom code allows many tasks to be performed that cannot be carried out by standard blocks.

All process control languages contain PID control blocks of different forms. Other categories of function blocks include

1. *Logical operators.* AND, OR, and exclusive OR (XOR) functions.
2. *Calculations.* Algebraic operations such as addition, multiplication, square root extraction, or special function evaluation.
3. *Selectors.* Min and max functions, transferring data in a selected input to the output or the input to a selected output.
4. *Comparators.* Comparison of two analog values and transmission of a binary signal to indicate whether one analog value exceeds the other.
5. *Timers.* Delayed activation of the output for a programmed duration after activation by the input signal.
6. *Process dynamics.* Emulation of a first-order process lag (or lead) and time delay.

CONTROLLERS, FINAL CONTROL ELEMENTS, AND REGULATORS

GENERAL REFERENCES: Driskell, *Control-Valve Selection and Sizing*, ISA, 1983. McMillan, *Process/Industrial Instruments and Controls Handbook*, 5th ed., McGraw-Hill, 1999. Hammit, *Cavitation and Multiphase Flow Phenomena*, McGraw-Hill, 1980. ANSI/ISA-75.25.01, *Test Procedure for Control Valve Response Measurement from Step Inputs*, The Instrumentation Systems and Automation Society, 2000. Norton and Karczub, *Fundamentals of Noise and Vibration Analysis for Engineers*, 2d ed., Cambridge University Press, 2003. Ulanski, *Valve and Actuator Technology*, McGraw-Hill, 1991. Kinsler et al., *Fundamentals of Acoustics*, 4th ed., Wiley. *National Electrical Code Handbook*, 9th ed., National Fire Protection Association, Inc.

External control of the process is achieved by devices that are specially designed, selected, and configured for the intended process control application. The text that follows covers three very common function classifications of process control devices: controllers, final control elements, and regulators.

The process controller is the "master" of the process control system. It accepts a set point and other inputs and generates an output or outputs that it computes from a rule or set of rules that is part of its internal configuration. The controller output serves as an input to another controller or, more often, as an input to a final control element. The final control element typically is a device that affects the flow in the piping system of the process. The final control element serves as an interface between the process controller and the process. Control valves and adjustable speed pumps are the principal types discussed.

Regulators, though not controllers or final control elements, perform the combined function of these two devices (controller and final control element) along with the measurement function commonly associated with the process variable transmitter. The uniqueness, control performance, and widespread usage of the regulator make it deserving of a functional grouping of its own.

PNEUMATIC, ELECTRONIC, AND DIGITAL CONTROLLERS

Pneumatic Controllers The pneumatic controller is an automatic controller that uses variable air pressure as input and output signals. An air supply is also required to "power" the mechanical components of the controller and provide an air source for the controller output signal. Pneumatic controllers were first available in the early 1940s but are now rarely used for large-scale industrial control projects. Pneumatic controllers are still used where cost, ruggedness, or the installation requires an all-pneumatic solution.

Pneumatic process transmitters are used to produce a pressure signal that is proportional to the calibrated range of the measuring element. Of the transmitter range 0 to 100 percent is typically represented by a 0.2- to 1.0-bar (3- to 15-psig) pneumatic signal. This signal is sent through tubing to the pneumatic controller process variable feedback connection. The process variable feedback can also be sensed directly in cases where the sensing element has been incorporated into the controller design. Controllers with integral sensing elements are available that sense pressure, differential pressure, temperature, and level.

The pneumatic controller is designed so that 0 to 100 percent output is also represented by 0.2 to 1.0 bar (3 to 15 psig). The output signal is sent through tubing to the control valve or other final control element. Most pneumatic controllers provide a manual control mode where the output pressure is manually set by operating personnel. The controller design also provides a mechanism to adjust the set point.

Early controller designs required "balancing" of the controller output prior to switching to or from automatic and manual modes. This procedure minimized inadvertent disturbance to the process caused by potentially large differences between the automatic and manual output levels. Later designs featured "bumpless" or "procedureless" automatic-to-manual transfer.

Although the pneumatic controller is often used in single-loop control applications, cascade strategies can be implemented where the controller design supports input of external or remote set-point signals. A balancing procedure is typically required to align the remote set point with the local set point before the controller is switched into cascade mode.

Almost all pneumatic controllers include indicators for process variable, set point, and output. Many controller designs also feature integral chart recorders. There are versions of the pneumatic controller that support combinations of proportional, integral, and derivative actions.

The pneumatic controller can be installed into panel boards that are adjacent to the process being controlled or in a centrally located control room. Field-mountable controllers can be installed directly onto the control valve, a nearby pipestand, or wall in close proximity to the control valve and/or measurement transmitter.

If operated on clean, dry plant air, pneumatic controllers offer good performance and are extremely reliable. In many cases, however, plant air is neither clean nor dry. A poor-quality air supply will cause unreliable performance of pneumatic controllers, pneumatic field measurement devices, and final control elements. The main shortcoming of the pneumatic controller is its lack of flexibility when compared to modern electronic controller designs. Increased range of adjustability, choice of alternative control algorithms, the communication link to the control system, and other features and services provided by the electronic controller make it a superior choice in most of today's applications. Controller performance is also affected by the time delay induced by pneumatic tubing runs. For example, a 100-m run of 6.35-mm ($\frac{1}{4}$ -in) tubing will typically cause 5 s of apparent process dead time, which will limit the control performance of fast processes such as flows and pressures.

Pneumatic controllers continue to be used in areas where it would be hazardous to use electronic equipment, such as locations with flammable or explosive atmospheres or other locations where compressed air is available but where access to electrical services is limited or restricted.

Electronic (Digital) Controllers Almost all the electronic process controllers used today are microprocessor-based, digital devices. In the transition from pneumatic to electronic controllers, a number of analog controller designs were available. Due to the inflexible nature of the analog designs, these controllers have been almost completely replaced by digital designs. The microprocessor-based controllers contain, or have access to, input/output (I/O) interface electronics that allow various types of signals to enter and leave the controller's processor.

The resolution of the analog I/O channels of the controller varies by manufacturer and age of the design. The 12- to 14-bit conversion resolution of the analog input channels is quite common. Conversion resolution of the analog output channels is typically 10- to 12-bit. Newer designs support up to 16 bit input and output resolution. Although 10-bit output resolution had been considered satisfactory for many years, it has recently been identified as a limitation of control performance. This limitation has emerged as the performance of control valve actuators has improved and the use of other high-resolution field devices, such as variable-speed pump drives, has become more prevalent. These improvements have been driven by the need to deliver higher operating efficiencies and improved product specifications through enhanced process control performance.

Sample rates for the majority of digital controllers are adjustable and range from 1 sample every 5 s to 10 samples per second. Some controller designs have fixed sample rates that fall within the same range. Hardwired low-pass filters are usually installed on the analog inputs to the controller to help protect the sampler from aliasing errors.

The real advantage of digital controllers is the substantial flexibility offered by a number of different configuration schemes. The simplest form of configuration is seen in a controller design that features a number of user-selectable control strategies. These strategies are customized by setting "tunable" parameters within the strategy. Another common configuration scheme uses a library of function blocks that can be selected and combined to form the desired control strategy. Each function block has adjustable parameters. Additional configuration schemes include text-based scripting languages, higher-level languages such as Basic or C, and ladder logic.

Some digital controller designs allow the execution rates of control strategy elements to be set independently of each other and indepen-

dently of the I/O subsystem sample rate. Data passed from control element to subsystems that operate at slower sample or execution rates present additional opportunities for timing and aliasing errors.

Distributed Control Systems Some knowledge of the distributed control system (DCS) is useful in understanding electronic controllers. A DCS is a process control system with sufficient performance to support large-scale, real-time process applications. The DCS has (1) an operations workstation with input devices, such as a keyboard, mouse, track ball, or other similar device, and a display device, such as a CRT or LCD panel; (2) a controller subsystem that supports various types of controllers and controller functions; (3) an I/O subsystem for converting analog process input signals to digital data and digital data back to analog output signals; (4) a higher-level computing platform for performing process supervision, historical data trending and archiving functions, information processing, and analysis; and (5) communication networks to tie the DCS subsystems, plant areas, and other plant systems together. The component controllers used in the controller subsystem portion of the DCS can be of various types and include multiloop controllers, programmable logic controllers, personal computer controllers, single-loop controllers, and field bus controllers. The type of electronic controller utilized depends on the size and functional characteristic of the process application being controlled. Personal computers are increasingly being used as DCS operations workstations or interface stations in place of custom-built machines. This is due to the low cost and high performance of the PC. See the earlier section "Digital Technology for Process Control."

Multiloop Controllers The multiloop controller is a DCS network device that uses a single 32-bit microprocessor to provide control functions to many process loops. The controller operates independently of the other devices on the DCS network and can support from 20 to 500 loops. Data acquisition capability for 1000 analog and discrete I/O channels or more can also be provided by this controller. The I/O is typically processed through a subsystem that is connected to the controller through a dedicated bus or network interface. The multiloop controller contains a variety of function blocks (for example, PID, totalizer, lead/lag compensator, ratio control, alarm, sequencer, and boolean) that can be "soft-wired" together to form complex control strategies. The multiloop controller also supports additional configuration schemes including text-based scripting languages, higher-level languages such as Basic or C, and, to a limited extent, ladder logic. The multiloop controller, as part of a DCS, communicates with other controllers and human/machine interface (HMI) devices also on the DCS network.

Programmable Logic Controllers The programmable logic controller (PLC) originated as a solid-state, and far more flexible, replacement for the hardwired relay control panel and was first used in the automotive industry for discrete manufacturing control. Today, PLCs are primarily used to implement boolean logic functions, timers, and counters. Some PLCs offer a limited number of math functions and PID control. PLCs are often used with on/off input and output devices such as limit or proximity switches, solenoid-actuated process control valves, and motor switch gear. PLCs vary greatly in size with the smallest supporting less than 128 I/O channels and the largest supporting more than 1023 I/O channels. Very small PLCs combine processor, I/O, and communications functions into a single, self-contained unit. For larger PLC systems, hardware modules such as the power supply, processor module, I/O modules, communication module, and backplane are specified based on the application. These systems support multiple I/O backplanes that can be chained together to increase the I/O count available to the processor. Discrete I/O modules are available that support high-current motor loads and general-purpose voltage and current loads. Other modules support analog I/O and special-purpose I/O for servomotors, stepping motors, high-speed pulse counting, resolvers, decoders, displays, and keyboards. PLC I/O modules often come with indicators to determine the status of key I/O channels. When used as an alternative to a DCS, the PLC is programmed with a handheld or computer-based loader. The PLC is typically programmed with ladder logic or a high-level computer language such as BASIC, FORTRAN, or C. Programmable logic controllers use 16- or 32-bit microprocessors and offer some form of point-to-point serial communications such as RS-232C, RS-485, or

networked communication such as Ethernet with proprietary or open protocols. PLCs typically execute the boolean or ladder logic configuration at high rates; 10-ms execution intervals are common. This does not necessarily imply that the analog I/O or PID control functions are executed at the same rate. Many PLCs execute the analog program at a much slower rate. Manufacturers' specifications must be consulted.

Personal Computer Controller Because of its high performance at low cost and its unexcelled ease of use, application of the personal computer (PC) as a platform for process controllers is growing. When configured to perform scan, control, alarm, and data acquisition (SCADA) functions and combined with a spreadsheet or database management application, the PC controller can be a low-cost, basic alternative to the DCS or PLC. Using the PC for control requires installation of a board into the expansion slot in the computer, or the PC can be connected to an external I/O module by using a standard communication port on the PC. The communication is typically achieved through a serial interface (RS-232, RS-422, or IEEE-488), universal serial bus (USB), or Ethernet. The controller card/module supports 16- or 32-bit microprocessors. Standardization and high volume in the PC market have produced a large selection of hardware and software tools for PC controllers.

The PC can also be interfaced to a DCS to perform advanced control or optimization functions that are not available within the standard DCS function library.

Single-Loop Controller The single-loop controller (SLC) is a process controller that produces a single output. SLCs can be pneumatic, analog electronic, or microprocessor-based. Pneumatic SLCs are discussed in the pneumatic controller section, and analog electronic SLC is not discussed because it has been virtually replaced by the microprocessor-based design. The microprocessor-based SLC uses an 8- or 16-bit microprocessor with a small number of digital and analog process input channels with control logic for the I/O incorporated within the controller. Analog inputs and outputs are available in the standard ranges (1 to 5 V dc and 4 to 20 mA dc). Direct process inputs for temperature sensors (thermistor RTD and thermocouple types) are available. Binary outputs are also available. The face of the SLC has some form of visible display and pushbuttons that are used to view or adjust control values and configuration. SLCs are available for mounting in panel openings as small as 48×48 mm (1.9×1.9 in).

The processor-based SLC allows the user to select from a set of predefined control strategies or to create a custom control strategy by using a set of control function blocks. Control function blocks include PID, on/off, lead/lag, adder/subtractor, multiply/divide, filter functions, signal selector, peak detector, and analog track. SLCs feature auto/manual transfer switching, multi-set-point self-diagnostics, gain scheduling, and perhaps also time sequencing. Most processor-based SLCs have self-tuning or auto-tuning PID control algorithms. Sample times for the microprocessor-based SLCs vary from 0.1 to 0.5 s. Low-pass analog electronic filters are usually installed on the process inputs to minimize aliasing errors caused by high-frequency content in the process signal. Input filter time constants are typically in the range of 0.1 to 1 s. Microprocessor-based SLCs may be made part of a DCS by using the communication port (RS-485 is common) on the controller or may be operated in a stand-alone mode independently of the DCS.

Field Bus Controller Field bus technology is a network-based communications system that interconnects measurement and control equipment such as sensors, actuators, and controllers. Advanced field bus systems, intended for process control applications, such as Foundation Fieldbus, enable digital interoperability among these devices and have a built-in capability to distribute the control application across the network. Several manufacturers have made available Foundation Fieldbus devices that support process controller functionality. These controllers, known as field bus controllers, typically reside in the final control element or measurement transmitter, but can be designed into any field bus device. A suitable communications interface connects the Fieldbus segment to the distributed control system. When configuring the control strategy, all or part of the strategy may be loaded into the field bus devices. The remaining part of the control strategy would reside in the DCS itself. The distribution of the control function depends on the processing capacity of the field bus devices, the control strategy, and where it makes sense to perform these functions. Linearization of a control valve could be performed in the digital valve positioner (controller), for

example. Temperature and pressure compensation of a flow measurement could be performed in the flow transmitter processor. The capability of field bus devices varies greatly. Some devices will allow instances of control system function blocks to be loaded and executed, while other devices allow the use of only preconfigured function blocks. Field bus controllers are typically configured as single-loop PID controllers, but cascade or other complex control strategies can be configured depending on the capability of the field bus device. Field bus devices that have native support for process control functions do not necessarily implement the PID algorithm in the same way. It is important to understand these differences so that the controller tuning will deliver the desired closed-loop characteristics. The functionality of field bus devices is projected to increase as the controller market develops.

Controller Reliability and Application Trends Critical process control applications demand a high level of reliability from the electronic controller. Some methods that improve the reliability of electronic controllers include (1) focusing on robust circuit design using quality components; (2) using redundant circuits, modules, or subsystems where necessary; (3) using small backup systems when needed; (4) reducing repair time and using more powerful diagnostics; and (5) distributing functionality to more independent modules to limit the impact of a failed module. Currently, the trend in process control is away from centralized process control and toward an increased number of small distributed control or PLC systems. This trend will put emphasis on the evolution of the field bus controller and continued growth of the PC-based controller. Also, as hardware and software improve, the functionality of the controller will increase, and the supporting hardware will be physically smaller. Hence, the traditional lines between the DCS and the PLC will become less distinct as systems become capable of supporting either function set.

Controller Performance and Process Dynamics The design of a control loop must take the control objectives into account. What do you want this loop to do? And under what operating conditions? There may be control applications that require a single control objective and others that have several control objectives. Control objectives may include such requirements as minimum variance control at steady state, maximum speed of recovery from a major disturbance, maximum speed of set-point response where overshoot and ringing are acceptable, critically damped set-point response with no overshoot, robustness issues, and special start-up or shut-down requirements. The control objectives will define not only the tuning requirements of the controller, but also, to a large extent, the allowable dynamic parameters of the field instruments and process design. Process dynamics alone can prevent control objectives from being realized. Tuning of the controller cannot compensate for an incompatible process or unrealistic control objectives. For most controllers, the difference between the set-point and process feedback signal, the error, is the input to the PID algorithm. The calculated PID output is sent back to the final control element. Every component between the controller output and the process feedback is considered by the controller as the "process" and will directly affect the dynamics and ultimately the performance of the system. This includes not only the dynamics of the physical process, but also the dynamics of the field instruments, signal conditioning equipment, and controller signal processing elements such as filters, scaling, and linearization routines. The choice of final control element can significantly affect the dynamics of the system. If the process dynamics are relatively slow, with time constants of a few minutes or longer, most control valves are fast enough that their contribution to the overall process time response will be negligible. In cases where the process time constants are only a few seconds, the control valve dynamics may become the dominant lag in the overall response. Excessive filtering in the field-sensing devices may also mask the true process dynamics and potentially limit control performance. Often, the design of a control loop and the tuning of the controller are a compromise between a number of different control objectives. When a compromise is unacceptable, gain scheduling or other adaptive tuning routine may be necessary to match the controller response to the most appropriate control objective.

When one is tuning a controller, the form of the PID algorithm must be known. The three common forms of the PID algorithm are parallel

or noninteracting, classical or interacting, and the ISA Standard form. In most cases, a controller with any of these PID forms can be tuned to produce the desired closed-loop response. The actual tuning parameters will be different. The units of the tuning parameters also affect their value. The controller gain parameter is typically represented as a pure gain (K_c), acting on the error, or as proportional band (PB). In cases where the proportional band parameter is used, the equivalent controller gain is equal to 100 divided by the proportional band and represents the percent span that the error must traverse to produce a 100 percent change in the controller output. The proportional band is always applied to the controller error in terms of percent of span and percent of output. Controllers that use a gain tuning parameter commonly scale the error into percent span and use a percent output basis. In some controllers, the error is scaled by using a separate parameter into percent span prior to the PID algorithm. The gain parameter can also be applied to the error in engineering units. Even though most controller outputs are scaled as a percent, in cascade strategies the controller output may need to be scaled to the same span at the slave loop set point. In this case, the controller gain may in fact be required to calculate the controller output in terms of the slave loop engineering units.

The execution rate of a digital controller should be sufficiently fast, compared to the process dynamics, to produce a response that closely approximates that of an analog controller with the same tuning. A general rule of thumb is that the execution interval should be at least 3 times faster than the dominant lag of the process or about 10 times faster than the closed-loop time constant. The controller can be used when the sample rates are slower than this recommendation, but the controller output will consist of a series of coarse steps as compared to a smooth response. This may create stability problems. Some integral and derivative algorithms may not be accurate when the time-based tuning parameters approach the controller execution interval. The analog inputs of the controller are typically protected from aliasing errors through the use of one- or two-pole analog filters. Faster sample rates allow a smaller antialiasing filter and improved input response characteristics. Some controllers or I/O subsystems oversample the analog inputs with respect to the controller execution interval and then process the input data through a digital filter. This technique can produce a more representative measurement with less quantization noise.

Differences in the PID algorithm, controller parameters, units, and other fundamental control functions highlight the importance of understanding the structure of the controller and the requirement of sufficiently detailed documentation. This is especially important for the controller but is also important for the field instruments, final control elements, and device that have the potential to affect the signal characteristics.

CONTROL VALVES

A control valve consists of a valve, an actuator, and usually one or more valve control devices. The valves discussed in this section are applicable to throttling control (i.e., where flow through the valve is regulated to any desired amount between maximum and minimum limits). Other valves such as check, isolation, and relief valves are addressed in the next subsection. As defined, control valves are automatic control devices that modify the fluid flow rate as specified by the controller.

Valve Types Types of valves are categorized according to their design style. These styles can be grouped into type of stem motion—linear or rotary. The valve stem is the rod, shaft, or spindle that connects the actuator with the closure member (i.e., a movable part of the valve that is positioned in the flow path to modify the rate of flow). Movement of either type of stem is known as travel. The major categories are described briefly below.

Globe and Angle The most common linear stem-motion control valve is the globe valve. The name comes from the globular cavities around the port. In general, a port is any fluid passageway, but often the reference is to the passage that is blocked off by the closure member when the valve is closed. In globe valves, the closure member is called a plug. The plug in the valve shown in Fig. 8-66 is guided by a large-diameter port and moves within the port to provide the flow control orifice of the valve. A very popular alternate construction is a cage-guided plug, as illustrated in Fig. 8-67. In many such designs,

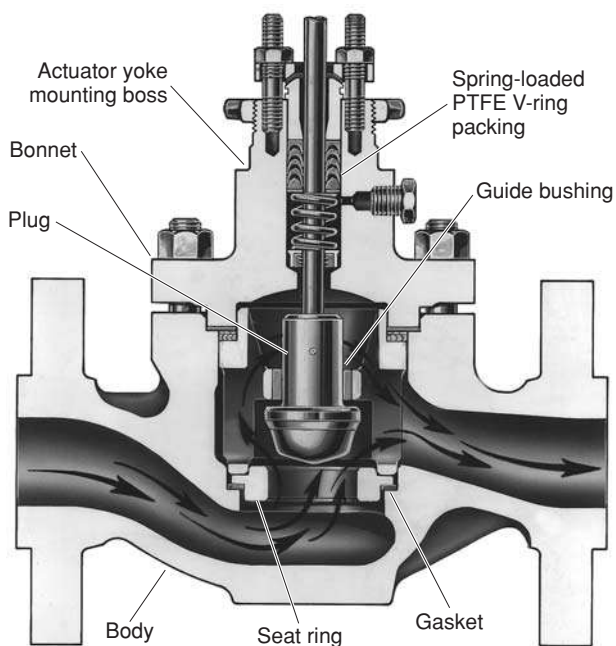


FIG. 8-66 Post-guided contour plug globe valve with metal seat and raised-face flange end connections. (Courtesy Fisher Controls International LLC.)

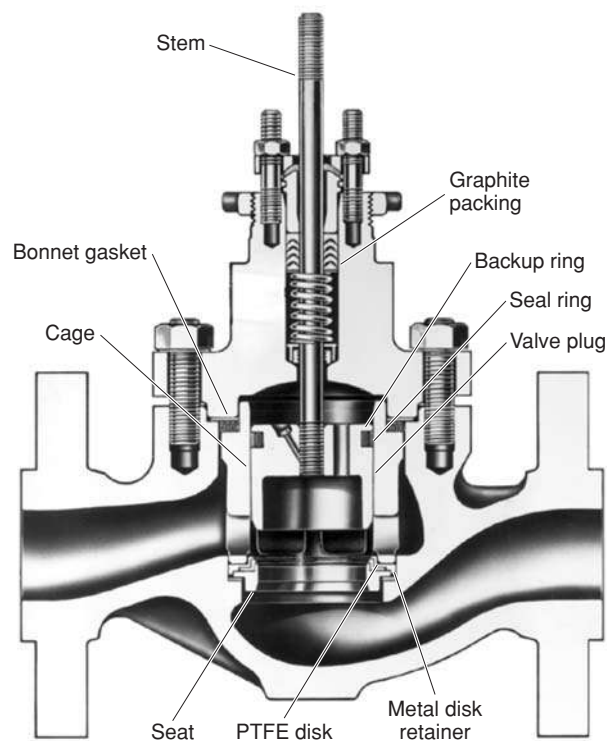


FIG. 8-67 Cage-guided balanced plug globe valve with polymer seat and plug seal. (Courtesy Fisher Controls International LLC.)

openings in the cage provide the flow control orifices. The valve seat is the zone of contact between the moving closure member and the stationary valve body, which shuts off the flow when the valve is closed. Often the seat in the body is on a replaceable part known as a seat ring. This stationary seat can also be designed as an integral part of the cage. Plugs may also be port-guided by wings or a skirt that fits snugly into the seat-ring bore.

One distinct advantage of cage guiding is the use of balanced plugs in single-port designs. The unbalanced plug depicted in Fig. 8-66 is subjected to a static pressure force equal to the port area times the valve pressure differential (plus the stem area times the downstream pressure) when the valve is closed. In the balanced design (Fig. 8-67), note that both the top and bottom of the plug are subjected to the same downstream pressure when the valve is closed. Leakage via the plug-to-cage clearance is prevented by a plug seal. Both plug types are subjected to hydrostatic force due to internal pressure acting on the stem area and to dynamic flow forces when the valve is flowing.

The plug, cage, seat ring, and associated seals are known as the trim. A key feature of globe valves is that they allow maintenance of the trim via a removable bonnet without removing the valve body from the line. Bonnets are typically bolted on but may be threaded in smaller sizes.

Angle valves are an alternate form of the globe valve. They often share the same trim options and have the top-entry bonnet style. Angle valves can eliminate the need for an elbow but are especially useful when direct impingement of the process fluid on the body wall is to be avoided. Sometimes it is not practical to package a long trim within a globe body, so an angle body is used. Some angle bodies are self-draining, which is an important feature for dangerous fluids.

Butterfly The classic design of butterfly valves is shown in Fig. 8-68. Its chief advantage is high capacity in a small package and a very low initial cost. Much of the size and cost advantage is due to the wafer body design, which is clamped between two pipeline flanges. In the simplest design, there is no seal as such, merely a small clearance gap between the disc OD and the body ID. Often a true seal is provided by a resilient material in the body that is engaged via an interference fit with the disc. In a lined butterfly valve, this material covers the entire body ID and extends around the body ends to eliminate the need for pipeline joint gaskets. In a fully lined valve, the disc is also coated to minimize corrosion or erosion.

A high-performance butterfly valve has a disc that is offset from the shaft centerline. This eccentricity causes the seating surface to move away from the seal once the disc is out of the closed position, reducing friction and seal wear. It is also known as an eccentric disc valve; the advantage of the butterfly valve is improved shutoff while maintaining

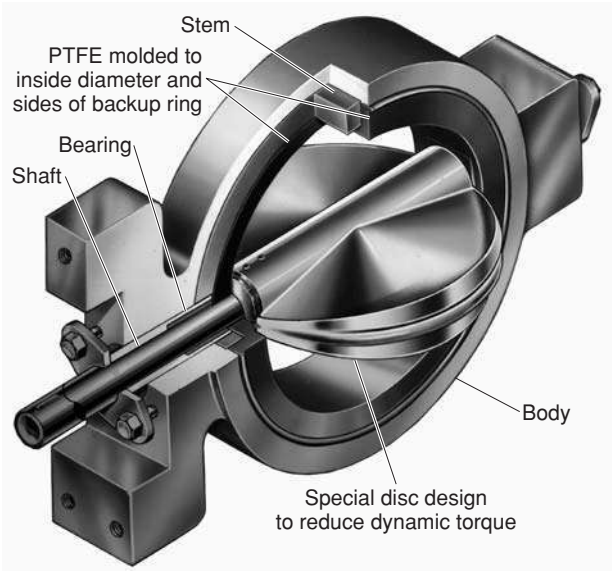


FIG. 8-68 Partial cutaway of wafer-style lined butterfly valve. (Courtesy Fisher Controls International LLC.)

high ultimate capacity at a reasonable cost. This cost advantage relative to other design styles is particularly true in sizes above 6-in nominal pipe size (NPS). Improved shutoff is due to advances in seal technologies, including polymer, flexing metal, combination metal with polymer inserts, and so on, many utilizing pressure assist.

Ball Ball valves get their name from the shape of the closure member. One version uses a full spherical member with a cylindrical bore through it. The ball is rotated one-quarter turn from the full-closed to the full-open position. If the bore is the same diameter as the mating-pipe fitting ID, the valve is referred to as full-bore. If the hole is undersized, the ball valve is considered to be a venturi style. A segmented ball is a portion of a hollow sphere that is large enough to block the port when closed. Segmented balls often have a V-shaped contour along one edge, which provides a desirable flow characteristic (see Fig. 8-69). Both full ball and segmented ball valves are known for

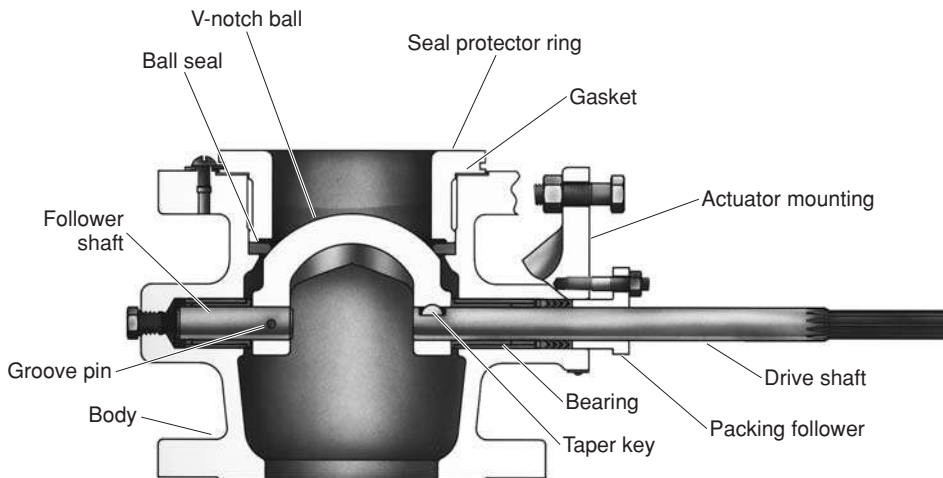


FIG. 8-69 Segmented ball valve. Partial view of actuator mounting shown 90° out of position. (Courtesy Fisher Controls International LLC.)

their low resistance to flow when full open. Shutoff leakage is minimized through the use of flexing or spring-loaded elastomeric or metal seals. Bodies are usually in two or three pieces or have a removable retainer to facilitate installing seals. End connections are usually flanged or threaded in small sizes, although segmented ball valves are offered in wafer style also.

Plug There are two substantially different rotary valve design categories referred to as plug valves. The first consists of a cylindrical or slightly conical plug with a port through it. The plug rotates to vary the flow much as a ball valve does. The body is top-entry but is geometrically simpler than a globe valve and thus can be lined with fluorocarbon polymer to protect against corrosion. These plug valves have excellent shutoff but are generally not for modulating service due to high friction. A variation of the basic design (similar to the eccentric butterfly disc) only makes sealing contact in the closed position and is used for control.

The other rotary plug design is portrayed in Fig. 8-70. The seating surface is substantially offset from the shaft, producing a ball-valve-like motion with the additional cam action of the plug into the seat when closing. In reverse flow, high-velocity fluid motion is directed inward, impinging on itself and only contacting the plug and seat ring.

Multiport This term refers to any valve or manifold of valves with more than one inlet or outlet. For throttling control, the three-way body is used for blending (two inlets, one outlet) or as a diverter (one inlet, two outlets). A three-way valve is most commonly a special globe-like body with special trim that allows flow both over and under the plug. Two rotary valves and a pipe tee can also be used. Special three-, four-, and five-way ball valve designs are used for switching applications.

Special Application Valves

Digital Valves True digital valves consist of discrete solenoid-operated flow ports that are sized according to binary weighing. The valve can be designed with sharp-edged orifices or with streamlined nozzles that can be used for flow metering. Precise control of the throttling control orifice is the strength of the digital valve. Digital valves are mechanically complicated and expensive, and they have considerably reduced maximum flow capacities compared to the globe and rotary valve styles.

Cryogenic Service Valves designed to minimize heat absorption for throttling liquids and gases below 80 K are called cryogenic service

valves. These valves are designed with small valve bodies to minimize heat absorption and long bonnets between the valve and actuator to allow for extra layers of insulation around the valve. For extreme cases, vacuum jacketing can be constructed around the entire valve to minimize heat influx.

High Pressure Valves used for pressures nominally above 760 bar (11,000 psi, pressures above ANSI Class 4500) are often custom-designed for specific applications. Normally, these valves are of the plug type and use specially hardened plug and seat assemblies. Internal surfaces are polished, and internal corners and intersecting bores are smoothed to reduce high localized stresses in the valve body. Steam loops in the valve body are available to raise the body temperature to increase ductility and impact strength of the body material.

High-Viscous Process Used most extensively by the polymer industry, the valve for high-viscous fluids is designed with smooth finished internal passages to prevent stagnation and polymer degradation. These valves are available with integral body passages through which a heat-transfer fluid is pumped to keep the valve and process fluid heated.

Pinch The industrial equivalent of controlling flow by pinching a soda straw is the pinch valve. Valves of this type use fabric-reinforced elastomer sleeves that completely isolate the process fluid from the metal parts in the valve. The valve is actuated by applying air pressure directly to the outside of the sleeve, causing it to contract or pinch. Another method is to pinch the sleeve with a linear actuator with a specially attached foot. Pinch valves are used extensively for corrosive material service and erosive slurry service. This type of valve is used in applications with pressure drops up to 10 bar (145 psi).

Fire-Rated Valves that handle flammable fluids may have additional safety-related requirements for minimal external leakage, minimal internal (downstream) leakage, and operability during and after a fire. Being fire-rated does not mean being totally impervious to fire, but a sample valve must meet particular specifications such as those of American Petroleum Institute (API) 607, Factory Mutual Research Corp. (FM) 7440, or the British Standard 5146 under a simulated fire test. Due to very high flame temperature, metal seating (either primary or as a backup to a burned-out elastomer) is mandatory.

Solids Metering The control valves described earlier are primarily used for the control of fluid (liquid or gas) flow. Sometimes these valves, particularly the ball, butterfly, or sliding gate valves, are used to throttle dry or slurry solids. More often, special throttling mechanisms such as venturi ejectors, conveyers, knife-type gate valves, or rotating vane valves are used. The particular solids-metering valve hardware depends on the volume, density, particle shape, and coarseness of the solids to be handled.

Actuators An actuator is a device that applies the force (torque) necessary to cause a valve's closure member to move. Actuators must overcome pressure and flow forces as well as friction from packing, bearings or guide surfaces, and seals; and must provide the seating force. In rotary valves, maximum friction occurs in the closed position, and the moment necessary to overcome it is referred to as breakout torque. The rotary valve shaft torque generated by steady-state flow and pressure forces is called dynamic torque. It may tend to open or close the valve depending on valve design and travel. Dynamic torque per unit pressure differential is largest in butterfly valves at roughly 70° open. In linear stem-motion valves, the flow forces should not exceed the available actuator force, but this is usually accounted for by default when the seating force is provided.

Actuators often provide a fail-safe function. In the event of an interruption in the power source, the actuator will place the valve in a predetermined safe position, usually either full-open or full-closed. Safety systems are often designed to trigger local fail-safe action at specific valves to cause a needed action to occur, which may not be a complete process or plant shutdown.

Actuators are classified according to their power source. The nature of these sources leads naturally to design features that make their performance characteristics distinct.

Pneumatic Despite the availability of more sophisticated alternatives, the pneumatically driven actuator is still by far the most popular type. Historically the most common has been the spring and diaphragm design (Fig. 8-71). The compressed air input signal fills a

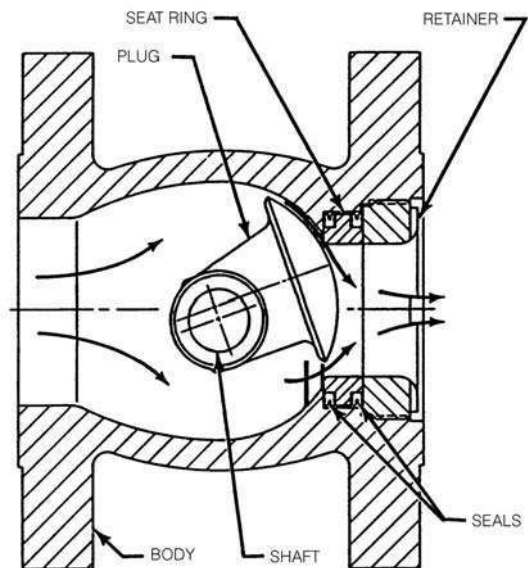


FIG. 8-70 Eccentric plug valve shown in erosion-resistant reverse flow direction. Shaded components can be made of hard metal or ceramic materials. (Courtesy Fisher Controls International LLC.)

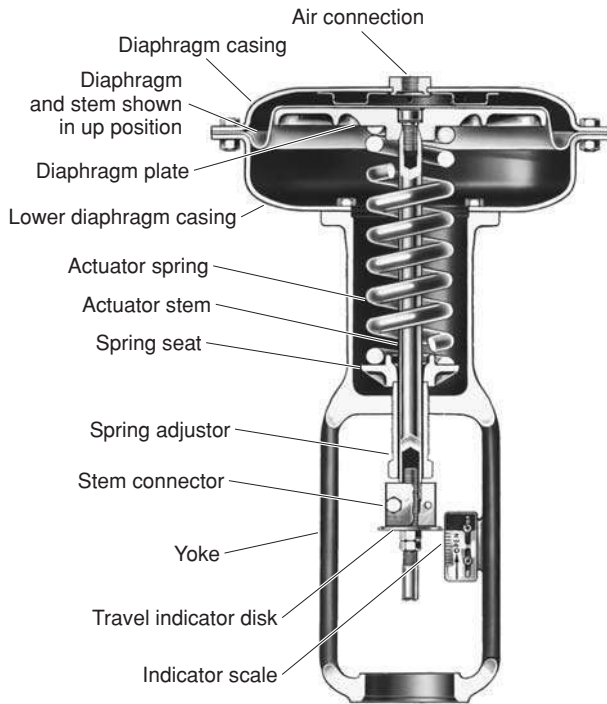


FIG. 8-71 Spring and diaphragm actuator with an “up” fail-safe mode. Spring adjuster allows slight alteration of bench set. (Courtesy Fisher Controls International LLC.)

chamber sealed by an elastomeric diaphragm. The pressure force on the diaphragm plate causes a spring to be compressed and the actuator stem to move. This spring provides the fail-safe function and contributes to the dynamic stiffness of the actuator. If the accompanying valve is “push down to close,” the actuator depicted in Fig. 8-71 will be described as “air to close” or synonymously as fail-open. A slightly different design yields “air to open” or fail-closed action. The spring is typically precompressed to provide a significant available force in the failed position (e.g., to provide seating load). The spring also provides a proportional relationship between the force generated by air pressure and stem position. The pressure range over which a spring and diaphragm actuator strokes in the absence of valve forces is known as the bench set. The chief advantages of spring and diaphragm actuators are their high reliability, low cost, adequate dynamic response, and fail-safe action—all of which are inherent in their simple design.

Alternately, the pressurized chamber can be formed by a circular piston with a seal on its outer edge sliding within a cylindrical bore. Higher operating pressure [6 bar (~ 90 psig) is typical] and longer strokes are possible. Piston actuators can be spring-opposed but many times are in a dual-acting configuration (i.e., compressed air is applied to both sides of the piston with the net force determined from the pressure difference—see Fig. 8-72). Dynamic stiffness is usually higher with piston designs than with spring and diaphragm actuators; see “Positioner/Actuator Stiffness.” Fail-safe action, if necessary, is achieved without a spring through the use of additional solenoid valves, trip valves, or relays. See “Valve Control Devices.”

Motion Conversion Actuator power units with translational output can be adapted to rotary valves that generally need 90° or less rotation. A lever is attached to the rotating shaft, and a link with pivoting means on the end connects to the linear output of the power unit, an arrangement similar to an internal combustion engine crankshaft, connecting rod, and piston. When the actuator piston, or more commonly the diaphragm plate, is designed to tilt, one pivot can be eliminated (see Fig. 8-72). Scotch yoke and rack-and-pinion arrangements are also commonly used, especially with piston power units.

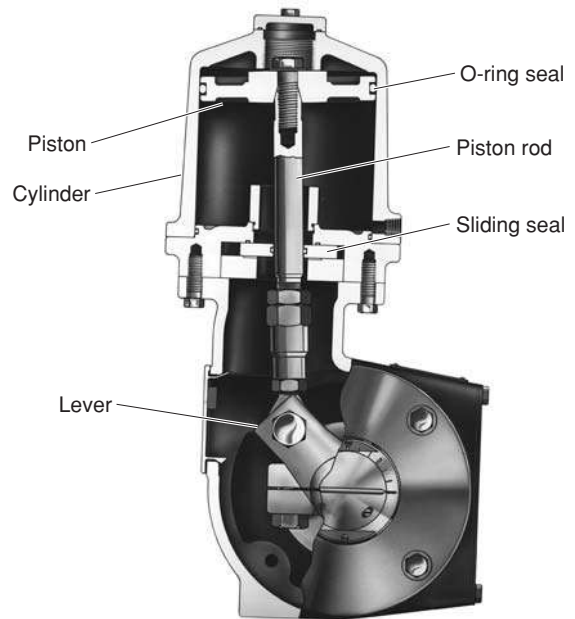


FIG. 8-72 Double-acting piston rotary actuator with lever and tilting piston for motion conversion. (Courtesy Fisher Controls International LLC.)

Friction and the changing mechanical advantage of these motion conversion mechanisms mean the available torque may vary greatly with travel. One notable exception is vane-style rotary actuators whose offset “piston” pivots, giving direct rotary output.

Hydraulic The design of typical hydraulic actuators is similar to that of double-acting piston pneumatic types. One key advantage is the high pressure [typically 35 to 70 bar (500 to 1000 psi)], which leads to high thrust in a smaller package. The incompressible nature of the hydraulic oil means these actuators have very high dynamic stiffness. The incompressibility and small chamber size connote fast stroking speed and good frequency response. The disadvantages include high initial cost, especially when considering the hydraulic supply. Maintenance is much more difficult than with pneumatics, especially on the hydraulic positioner.

Electrohydraulic actuators have similar performance characteristics and cost/maintenance ramifications. The main difference is that they contain their own electric-powered hydraulic pump. The pump may run continuously or be switched on when a change in position is required. Their main application is remote sites without an air supply when a fail-safe spring return is needed.

Electric The most common electric actuators use a typical motor—three-phase ac induction, capacitor-start split-phase induction, or dc. Normally the motor output passes through a large gear reduction and, if linear motion output is required, a ball screw or thread. These devices can provide large thrust, especially given their size. Lost motion in the gearing system does create backlash, but if not operating across a thrust reversal, this type of actuator has very high stiffness. Usually the gearing system is self-locking, which means that forces on the closure member cannot move it by spinning a nonenergized motor. This behavior is called a lock-in-last-position fail-safe mode. Some gear systems (e.g., low-reduction spur gears) can be backdriven. A solenoid-activated mechanical brake or locking current to motor field coils is added to provide lock-in-last-position fail-safe mode. A battery backup system for a dc motor can guard against power failures. Otherwise, an electric actuator is not acceptable if fail-open/closed action is mandatory. Using electric power requires environmental enclosures and explosion protection, especially in hydrocarbon processing facilities; see the full discussion in “Valve Control Devices.”

Unless sophisticated speed control power electronics is used, position modulation is achieved via bang-zero-bang control. Mechanical inertia causes overshoot, which is (1) minimized by braking and/or (2) hidden by adding dead band to the position control. Without these provisions, high starting currents would cause motors to overheat from constant "hunting" within the position loop. Travel is limited with power interruption switches or with force (torque) electro-mechanical cutouts when the closed position is against a mechanical stop (e.g., a globe valve). Electric actuators are often used for on/off service. Stepper motors can be used instead, and they, as their name implies, move in fixed incremental steps. Through gear reduction, the typical number of increments for 90° rotation ranges from 5000 to 10,000; hence positioning resolution at the actuator is excellent.

An electromagnetic solenoid can be used to directly actuate the plug on very small linear stem-motion valves. A solenoid is usually designed as a two-position device, so this valve control is on/off. Special solenoids with position feedback can provide proportional action for modulating control. Force requirements of medium-sized valves can be met with piloted plug designs, which use process pressure to assist the solenoid force. Piloted plugs are also used to minimize the size of common pneumatic actuators, especially when there is need for high seating load.

Manual A manually positioned valve is by definition not an automatic control valve, but it may be involved with process control. For rotary valves, the manual operator can be as simple as a lever, but a wheel driving a gear reduction is necessary in larger valves. Linear motion is normally created with a wheel turning a screw-type device. A manual override is usually available as an option for the powered actuators listed above. For spring-opposed designs, an adjustable travel stop will work as a one-way manual input. In more complex designs, the handwheel can provide loop control override via an engagement means. Some gear reduction systems of electric actuators allow the manual positioning to be independent of the automatic positioning without declutching.

OTHER PROCESS VALVES

In addition to the throttling control valve, other types of process valves can be used to manipulate a process.

Valves for On/Off Applications Valves are often required for service that is primarily nonthrottling. Valves in this category, depending on the service requirements, may be of the same design as the types used for throttling control or, as in the case of gate valves, different in design. Valves in this category usually have tight shutoff when they are closed and low pressure drops when they are wide open. The on/off valve can be operated manually, such as by hand-wheel or lever; or automatically, with pneumatic or electric actuators.

Batch Batch process operation is an application requiring on/off valve service. Here the valve is opened and closed to provide reactant, catalyst, or product to and from the batch reactor. Like the throttling control valve, the valve used in this service must be designed to open and close thousands of times. For this reason, valves used in this application are often the same valves used in continuous throttling applications. Ball valves are especially useful in batch operations. The ball valve has a straight-through flow passage that reduces pressure drop in the wide-open state and provides tight shutoff capability when closed. In addition, the segmented ball valve provides for shearing action between the ball and the ball seat that promotes closure in slurry service.

Isolation A means for pressure-isolating control valves, pumps, and other piping hardware for installation and maintenance is another common application for an on/off valve. In this application, the valve is required to have tight shutoff so that leakage is stopped when the piping system is under repair. As the need to cycle the valve in this application is far less than that of a throttling control valve, the wear characteristics of the valve are less important. Also, because many are required in a plant, the isolation valve needs to be reliable, simple in design, and simple in operation.

The gate valve, shown in Fig. 8-73, is the most widely used valve in this application. The gate valve is composed of a gatelike disc that moves perpendicular to the flow stream. The disc is moved up and

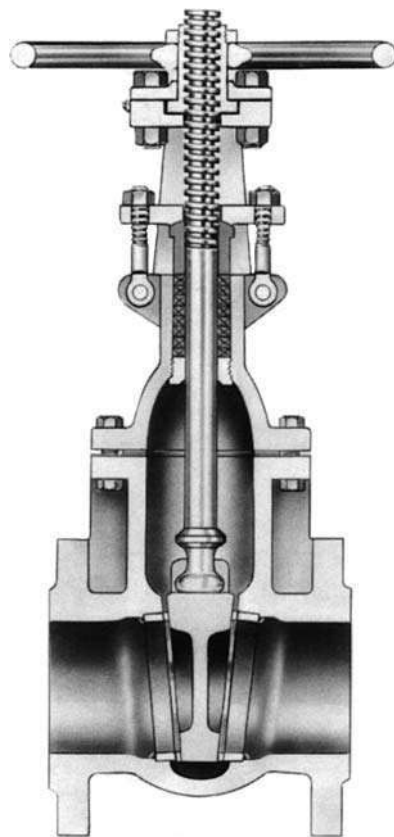


FIG. 8-73 Gate valve. (Courtesy Crane Valves.)

down by a threaded screw that is rotated to effect disc movement. Because the disc is large and at right angles to the process pressure, a large seat loading for tight shutoff is possible. Wear produced by high seat loading during the movement of the disc prohibits the use of the gate valve for throttling applications.

Pressure Relief Valves The pressure relief valve is an automatic pressure-relieving device designed to open when normal conditions are exceeded and to close again when normal conditions are restored. Within this class there are relief valves, pilot-operated pressure relief valves, and safety valves. Relief valves (see Fig. 8-74) have spring-loaded discs that close a main orifice against a pressure source. As pressure rises, the disc begins to rise off the orifice and a small amount of fluid passes through the valve. Continued rise in pressure above the opening pressure causes the disc to open the orifice in a proportional fashion. The main orifice reduces and closes when the pressure returns to the set pressure. Additional sensitivity to overpressure conditions can be improved by adding an auxiliary pressure relief valve (pilot) to the basic pressure relief valve. This combination is known as a pilot-operated pressure relief valve.

The safety valve is a pressure relief valve that is designed to open fully, or pop, with only a small amount of pressure over the rated limit. Where conventional safety valves are sensitive to downstream pressure and may have unsatisfactory operating characteristics in variable backpressure applications, pressure-balanced safety relief valve designs are available to minimize the effect of downstream pressure on performance.

Application and sizing of pressure relief valves, pilot-operated pressure relief valves, and safety valves for use on pressure vessels are found in the ASME Boiler and Pressure Vessel Code, Section VIII, Division 1, "Rules for Construction of Pressure Vessels," Paragraphs UG-125 through UG-137.

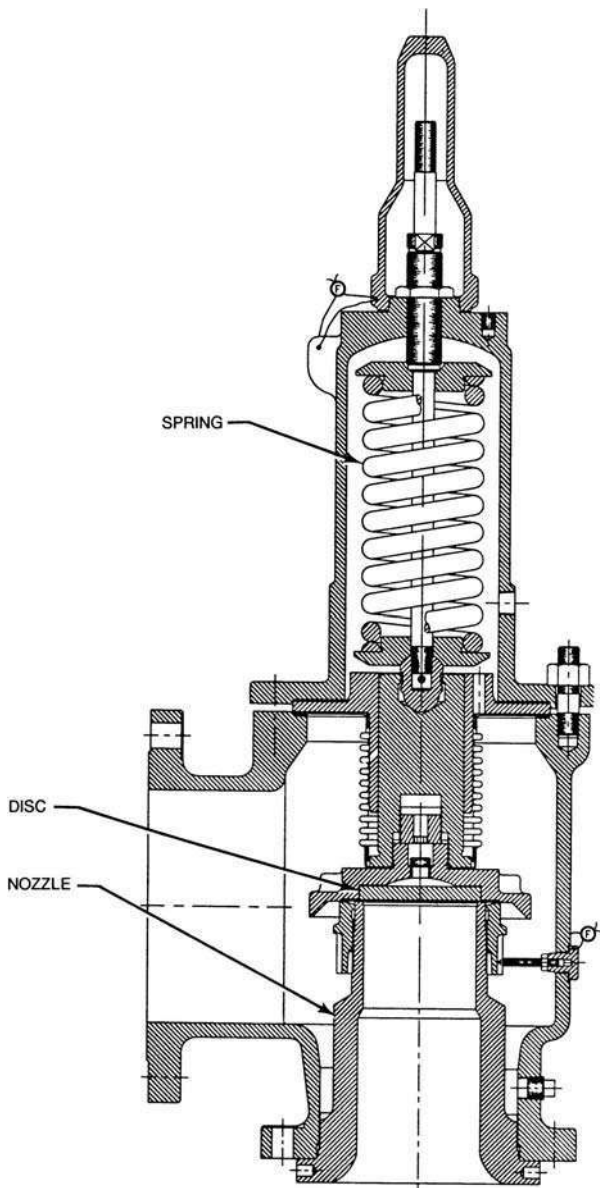


FIG. 8-74 Relief valve. (Courtesy of Teledyne Fluid Systems, Farris Engineering.)

Check Valves The purpose of a check valve is to allow relatively unimpeded flow in the desired direction but to prevent flow in the reverse direction. Two common designs are swing-type and lift-type check valves, the names of which denote the motion of the closure member. In the forward direction, flow forces overcome the weight of the member or a spring to open the flow passage. With reverse pressure conditions, flow forces drive the closure member into the valve seat, thus providing shutoff.

VALVE DESIGN CONSIDERATIONS

Functional requirements and the properties of the controlled fluid determine which valve and actuator types are best for a specific application. If demands are modest and no unique valve features are required, the valve design style selection may be determined solely by

cost. If so, general-purpose globe or angle valves provide exceptional value, especially in sizes less than 3-in NPS and hence are very popular. Beyond type selection, there are many other valve specifications that must be determined properly to ultimately yield improved process control.

Materials and Pressure Ratings Valves must be constructed from materials that are sufficiently immune to corrosive or erosive action by the process fluid. Common body materials are cast iron, steel, stainless steel, high-nickel alloys, and copper alloys such as bronze. Trim materials need better corrosion and erosion resistance due to the higher fluid velocity in the throttling region. High hardness is desirable in erosive and cavitating applications. Heat-treated and precipitation-hardened stainless steels are common. High hardness is also good for guiding, bearing, and seating surfaces; cobalt-chromium alloys are utilized in cast or wrought form and frequently as welded overlays called hard facing. In less stringent situations, chrome plating, heat-treated nickel coatings, and nitriding are used. Tungsten carbide and ceramic trim are warranted in extremely erosive services. See Sec. 25, "Materials of Construction," for specific material properties.

Since the valve body is a pressurized vessel, it is usually designed to comply with a standardized system of pressure ratings. Two common systems are described in the standards ASME B16.34 and EN 12516. Internal pressure limits under these standards are divided into broad classes, with specific limits being a function of material and temperature. Manufacturers also assign their own pressure ratings based on internal design rules. A common insignia is 250 WOG, which means a pressure rating of 250 psig (~17 bar) in water, oil, or gas at ambient temperature. "Storage and Process Vessels" in Sec. 10 provides introductory information on compliance of pressure vessel design to industry codes (e.g., ASME Boiler and Pressure Vessel Code, Section VIII; ASME B31.3 Chemical Plant and Petroleum Refinery Piping).

Valve bodies are also standardized to mate with common piping connections: flanged, butt-welded end, socket-welded end, and screwed end. Dimensional information for some of these joints and class pressure-temperature ratings are included in Sec. 10, "Process Plant Piping." Control valves have their own standardized face-to-face dimensions that are governed by ANSI/ISA Standards S75.08 and S75.22. Butterfly valves are also governed by API 609 and Manufacturers Standardization Society (MSS) SP-67 and SP-68.

Sizing Throttling control valves must be selected to pass the required flow rate, given expected pressure conditions. Sizing is not merely matching the end connection size with surrounding piping; it is a key step in ensuring that the process can be properly controlled. Sizing methods range from simple models based on elementary fluid mechanics to very complex models when unusual thermodynamics or nonideal behaviors occur. Basic sizing practices have been standardized (for example, ANSI-75.01.01) and are implemented as PC-based programs by manufacturers. The following is a discussion of very basic sizing equations and the associated physics.

Regardless of the particular process variable being controlled (e.g., temperature, level, pH), the output of a control valve is the flow rate. The throttling valve performs its function of manipulating the flow rate by virtue of being an adjustable resistance to flow. Flow rate and pressure conditions are normally known when a process is designed, and the valve resistance range must be matched accordingly. In the tradition of orifice and nozzle discharge coefficients, this resistance is embodied in the valve flow coefficient C_v . By applying the principles of conservation of mass and energy, the mass flow rate w kg/h is given for a liquid by

$$w = 27.3 C_v \sqrt{\rho(p_1 - p_2)} \quad (8-111)$$

where p_1 and p_2 are upstream and downstream static pressure, bar, respectively. The density of the fluid ρ is expressed in kilograms per cubic meter. This equation is valid for nonvaporizing, turbulent flow conditions for a valve with no attached fittings.

While the above equation gives the relationship between pressure and flow from a macroscopic point of view, it does not explain what is going on inside the valve. Valves create a resistance to flow by restricting the cross-sectional area of the flow passage and by forcing the fluid to change direction as it passes through the body and trim.

The conservation of mass principle dictates that, for steady flow, the product of density, average velocity, and cross-sectional area remain a constant. The average velocity of the fluid stream at the minimum restriction in the valve is therefore much higher than that at the inlet. Note that due to the abrupt nature of the flow contraction that forms the minimum passage, the main fluid stream may separate from the passage walls and form a jet that has an even smaller cross section, the so-called vena contracta. The ratio of minimum stream area to the corresponding passage area is called the contraction coefficient. As the fluid expands from the minimum cross-sectional area to the full passage area in the downstream piping, large amounts of turbulence are generated. Direction changes can also induce significant amounts of turbulence.

Figure 8-75 is an illustration of how the mean pressure changes as fluid moves through a valve. Some of the potential energy that was stored in the fluid by pressurizing it (e.g., the work done by a pump) is first converted to the kinetic energy of the fast-moving fluid at the vena contracta. Some of that kinetic energy turns into the kinetic energy of turbulence. As the turbulent eddies break down into smaller and smaller structures, viscous effects ultimately convert all the turbulent energy to heat. Therefore, a valve converts fluid energy from one form to another.

For many valve constructions, it is reasonable to approximate the fluid transition from the valve inlet to the minimum cross section of the flow stream as an isentropic or lossless process. Using this approximation, the minimum pressure p_{VC} can be estimated from the

Bernoulli relationship. See Sec. 6 ("Fluid and Particle Mechanics") for more background information. Downstream of the vena contracta, the flow is definitely not lossless due to all the turbulence that is generated. As the flow passage area increases and the fluid slows down, some of the kinetic energy of the fluid is converted back to pressure energy as pressure recovers. The energy that is permanently lost via turbulence accounts for the permanent pressure or head loss of the valve. The relative amount of pressure that is recouped determines whether the valve is considered to be high- or low-recovery. The flow passage geometry at and downstream of the vena contracta primarily determines the amount of recovery. The amount of recovery is quantified by the liquid pressure recovery factor F_L .

$$F_L = \sqrt{\frac{p_1 - p_2}{p_1 - p_{VC}}} \quad (8-112)$$

Under some operating conditions, sufficient pressure differential may exist across the valve to cause the vena contracta pressure to drop to the saturation pressure (also known as the *vapor pressure*) of the liquid. If this occurs, a portion of the liquid will vaporize, forming a two-phase, compressible mixture within the valve. If sufficient vapor forms, the flow may *choke*. When a flow is choked, any increase in pressure differential across the valve no longer produces an increase in flow through the valve.

The vena contracta condition at choked flow for pure liquids has been shown to be

$$p_{VC} = F_F p_v \quad (8-113)$$

where

$$F_F = 0.96 - 0.28 \sqrt{\frac{p_v}{p_c}} \quad (8-114)$$

and p_{VC} is the absolute vena contracta pressure under choked conditions, F_F is the liquid critical pressure ratio factor, p_v is the absolute vapor pressure of the liquid at inlet temperature, and p_c is the absolute thermodynamic critical pressure of the fluid.

Equations (8-112) and (8-113) can be used together to determine the pressure differential across the valve at the point where the flow chokes

$$\Delta p_{\text{choked}} = F_L^2 (p_1 - F_F p_v) \quad (8-115)$$

The pressure recovery factor is a constant for any given valve at a given opening. The value of this factor can be established by flow test and is published by the valve manufacturer. If the actual pressure differential across the valve is greater than the choked pressure differential of Eq. (8-115), then Δp_{choked} should be used in Eq. (8-111) to determine the correct valve size. A more complete presentation of sizing relationships is given in ANSI 75.01.01, including provisions for pipe reducers and Reynolds number effects.

Equations (8-111) to (8-115) are restricted to incompressible fluids. For gases and vapors, the fluid density is dependent on pressure. For convenience, compressible fluids are often assumed to follow the ideal gas law model. Deviations from ideal behavior are corrected for, to first order, with nonunity values of compressibility factor Z (see Sec. 2, "Physical and Chemical Data," for definitions and data for common fluids). For compressible fluids

$$w = 94.8 C_v p_1 Y \sqrt{\frac{x M_w}{T_1 Z}} \quad (8-116)$$

where p_1 is in bar absolute, T_1 is inlet temperature in kelvins, M_w is the molecular weight, and x is the dimensionless pressure drop ratio $(p_1 - p_2)/p_1$. The expansion factor Y accounts for changes in the fluid density as the fluid passes through the valve. It is dependent on pressure drop and valve geometry. Experimental data have shown that for

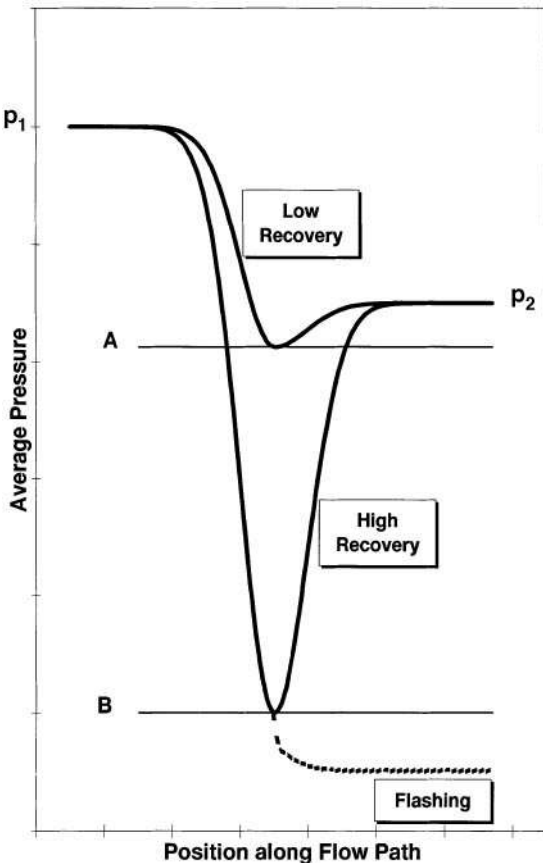


FIG. 8-75 Generic depictions of average pressure at subsequent cross sections throughout a control valve. The F_L values selected for illustration are 0.9 and 0.63 for low and high recovery, respectively. Internal pressure in the high-recovery valve is shown as a dashed line for flashing conditions ($p_2 < p_v$) with $p_v = B$.

small departures in the ratio of specific heat from that of air (1.4), a simple linear relationship can be used to represent the expansion factor:

$$Y = 1 - \frac{1.4x}{3x_T\gamma} \quad \text{for} \quad x \leq \frac{x_T\gamma}{1.4} \quad (8-117)$$

where γ is the ratio of specific heats and x_T is an experimentally determined factor for a specific valve and is the largest value of x that contributes to flow (i.e., values of x greater than x_T do not contribute to flow).

The terminal value of x , x_T , results from a phenomenon known as choking. Given a nozzle geometry with fixed inlet conditions, the mass flow rate will increase as p_2 is decreased up to a maximum amount at the critical pressure drop. The velocity at the vena contracta has reached sonic, and a standing shock has formed. This shock causes a step change in pressure as flow passes through it, and further reduction in p_2 does not increase mass flow. Thus x_T relates to the critical pressure drop ratio and also accounts for valve geometry effects. The value of x_T varies with flow path geometry and is supplied by the valve manufacturer. In the choked case,

$$x > \frac{x_T\gamma}{1.4} \quad \text{and} \quad Y = 0.67 \quad (8-118)$$

Noise Control Sound is a fluctuation of air pressure that can be detected by the human ear. Sound travels through any fluid (e.g., the air) as a compression/expansion wave. This wave travels radially outward in all directions from the sound source. The pressure wave induces an oscillating motion in the transmitting medium that is superimposed on any other net motion it may have. These waves are reflected, refracted, scattered, and absorbed as they encounter solid objects. Sound is transmitted through solids in a complex array of types of elastic waves. Sound is characterized by its amplitude, frequency, phase, and direction of propagation.

Sound strength is therefore location-dependent and is often quantified as a sound pressure level L_p in decibels based on the root mean square (rms) sound pressure value p_s , where

$$L_p = 10 \log \left(\frac{p_s}{p_{\text{reference}}} \right)^2 \quad (8-119)$$

For airborne sound, the reference pressure is 2×10^{-5} Pa (29×10^{-1} psi), which is nominally the human threshold of hearing at 1000 Hz. The corresponding sound pressure level is 0 dB. A voice in conversation is about 50 dB, and a jackhammer operator is subject to 100 dB. Extreme levels such as a jet engine at takeoff might produce 140 dB at a distance of 3 m, which is a pressure amplitude of 200 Pa (29×10^{-3} psi). These examples demonstrate both the sensitivity and the wide dynamic range of the human ear.

Traveling sound waves carry energy. Sound intensity I is a measure of the power passing through a unit area in a specified direction and is related to p_s . Measuring sound intensity in a process plant gives clues to the location of the source. As one moves away from the source, the fact that the energy is spread over a larger area requires that the sound pressure level decrease. For example, doubling one's distance from a point source reduces L_p by 6 dB. Viscous action from the induced fluid motion absorbs additional acoustic energy. However, in free air, this viscous damping is negligible over short distances (on the order of 1 m).

Noise is a group of sounds with many nonharmonic frequency components of varying amplitudes and random phase. The turbulence generated by a throttling valve creates noise. As a valve converts potential energy to heat, some of the energy becomes acoustic energy as an intermediate step. Valves handling large amounts of compressible fluid through a large pressure change create the most noise because more total power is being transformed. Liquid flows are noisy only under special circumstances, as will be seen in the next subsection. Due to the random nature of turbulence and the broad distribution of length and velocity scales of turbulent eddies, valve-generated sound is

usually random, broad-spectrum noise. The total sound pressure level from two such statistically uncorrelated sources is (in decibels)

$$L_p = 10 \log \frac{(p_{s1})^2 + (p_{s2})^2}{(p_{\text{reference}})^2} \quad (8-120)$$

For example, two sources of equal strength combine to create an L_p that is 3 dB higher.

While noise is annoying to listen to, the real reasons for being concerned about noise relate to its impact on people and equipment. Hearing loss can occur due to long-term exposure to moderately high, or even short exposure to very high, noise levels. The U.S. Occupational Safety and Health Act (OSHA) has specific guidelines for permissible levels and exposure times. The human ear has a frequency-dependent sensitivity to sound. When the effect on humans is the criterion, L_p measurements are weighted to account for the ear's response. This so-called A-weighted scale is defined in ANSI S1.4 and is commonly reported as L_{pA} . Figure 8-76 illustrates the difference between actual and perceived airborne sound pressure levels. At sufficiently high levels, noise and the associated vibration can damage equipment.

There are two approaches to fluid-generated noise control—source or path treatment. Path treatment means absorbing or blocking the transmission of noise after it has been created. The pipe itself is a barrier. The sound pressure level inside a standard schedule pipe is roughly 40 to 60 dB higher than on the outside. Thicker-walled pipe reduces levels somewhat more, and adding acoustical insulation on the outside of the pipe reduces ambient levels up to 10 dB per inch of thickness. Since noise propagates relatively unimpeded inside the

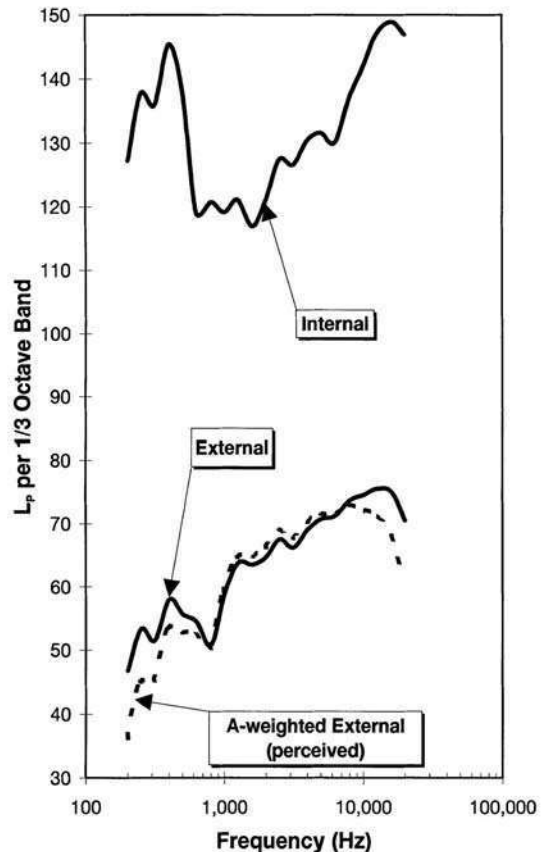


FIG. 8-76 Valve-generated sound pressure level spectra.

pipe, barrier approaches require the entire downstream piping system to be treated in order to be totally effective. In-line silencers place absorbent material inside the flow stream, thus reducing the level of the internally propagating noise. Noise reductions up to 25 dB can be achieved economically with silencers.

The other approach to valve noise problems is the use of quiet trim. Two basic strategies are used to reduce the initial production of noise—dividing the flow stream into multiple paths and using several flow resistances in series. Sound pressure level L_p is proportional to mass flow and is dependent on vena contracta velocity. If each path is an independent source, it is easy to show from Eq. (8-120) that p_v^2 is inversely proportional to the number of passages; additionally, smaller passage size shifts the predominate spectral content to higher frequencies, where structural resonance may be less of a problem. Series resistances or multiple stages can reduce maximum velocity and/or produce backpressure to keep jets issuing from multiple passages from acting independently. While some of the basic principles are understood, predicting noise for a particle flow passage requires some empirical data as a basis. Valve manufacturers have developed noise prediction methods for the valves they build. ANSI/ISA-75.17 is a public-domain methodology for standard (non-low-noise) valve types, although treatment of some multistage, multipath types is underway. Low-noise hardware consists of special cages in linear stem valves, perforated domes or plates and multichannel inserts in rotary valves, and separate devices that use multiple fixed restrictions.

Cavitation and Flashing From the discussion of pressure recovery it was seen that the pressure at the vena contracta can be much lower than the downstream pressure. If the pressure on a liquid falls below its vapor pressure p_v , the liquid will vaporize. Due to the effect of surface tension, this vapor phase will first appear as bubbles. These bubbles are carried downstream with the flow, where they collapse if the pressure recovers to a value above p_v . This pressure-driven process of vapor bubble formation and collapse is known as cavitation.

Cavitation has three negative side effects in valves—noise and vibration, material removal, and reduced flow. The bubble collapse process is a violent asymmetric implosion that forms a high-speed microjet and induces pressure waves in the fluid. This hydrodynamic noise and the mechanical vibration that it can produce are far stronger than other noise generation sources in liquid flows. If implosions occur adjacent to a solid component, minute pieces of material can be removed, which, over time, will leave a rough, cinderlike surface.

The presence of vapor in the vena contracta region puts an upper limit on the amount of liquid that will pass through a valve. A mixture of vapor and liquid has a lower density than that of the liquid alone. While Eq. (8-111) is not applicable to two-phase flows because pressure changes are redistributed due to varying density and the two phases do not necessarily have the same average velocity, it does suggest that lower density reduces the total mass flow rate. Figure 8-77 illustrates a typical flow rate/pressure drop relationship. As with compressible gas flow at a given p_1 , flow increases as p_2 is decreased until the flow chokes (i.e., no additional fluid will pass). The transition between incompressible and choked flow is gradual because, within the convoluted flow passages of valves, the pressure is actually an uneven distribution at each cross section and consequently vapor formation zones increase gradually. In fact, isolated zones of bubble formation or incipient cavitation often occur at pressure drops well below that at which a reduction in flow is noticeable. The similarity between liquid and gas choking is not serendipitous; it is surmised that the two-phase fluid is traveling at the mixture's sonic velocity in the throat when choked. Complex fluids with components having varying vapor pressures and/or entrained noncondensable gases (e.g., crude oil) will exhibit soft vaporization/implosion transitions.

There are several methods to reduce cavitation or at least its negative side effects. Material damage is slowed by using harder materials and by directing the cavitating stream away from passage walls (e.g., with an angle body flowing down). Sometimes the system can be designed to place the valve in a higher p_2 location or add downstream resistance, which creates backpressure. A low recovery valve has a higher minimum pressure for a given p_2 and so is a means to eliminate the cavitation itself, not just its side effects. In Fig. 8-75, if $p_v < B$, neither valve will cavitate substantially. For $p_v > B$ but $p_v < A$, the high

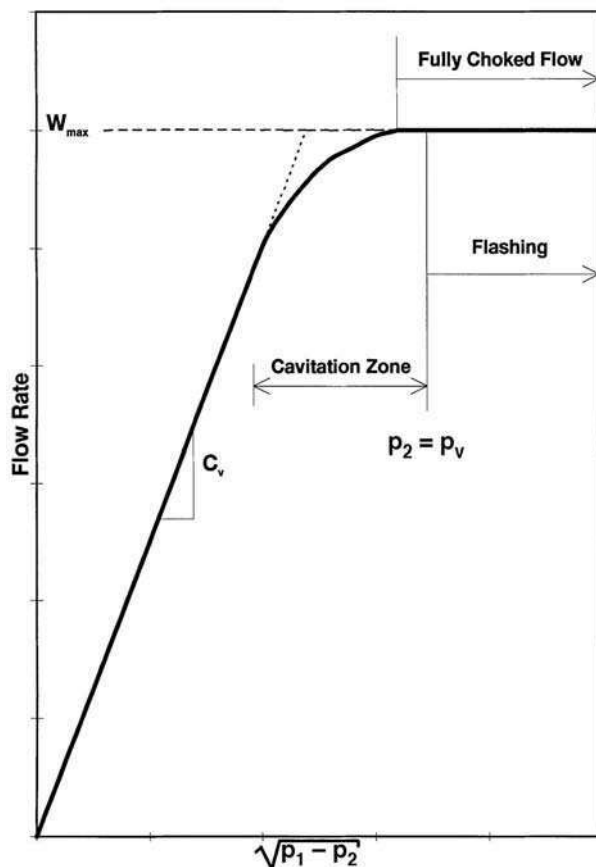


FIG. 8-77 Liquid flow rate versus pressure drop (assuming constant p_1 and p_v).

recovery valve will cavitate substantially, but the low recovery valve will not. Special anticavitation trims are available for globe and angle valves and more recently for some rotary valves. These trims use multiple contraction/expansion stages or other distributed resistances to boost F_L to values sometimes near unity.

If p_2 is below p_v , the two-phase mixture will continue to vaporize in the body outlet and/or downstream pipe until all liquid phase is gone, a condition known as flashing. The resulting huge increase in specific volume leads to high velocities, and any remaining liquid droplets acquire much of the higher vapor-phase velocity. Impingement of these droplets can produce material damage, but it differs from cavitation damage because it exhibits a smooth surface. Hard materials and directing the two-phase jets away from solid surfaces are means to avoid this damage.

Seals, Bearings, and Packing Systems In addition to their control function, valves often need to provide shutoff. FCI 70-2-1998 and IEC 60534-4 recognize six standard classifications and define their as-shipped qualification tests. Class I is an amount agreed to by user and supplier with no test needs. Classes II, III, and IV are based on an air test with maximum leakage of 0.5 percent, 0.1 percent, and 0.01 percent of rated capacity, respectively. Class V restricts leakage to 5×10^{-6} mL of water per second per millimeter of port diameter per bar differential. Class VI allows 0.15 to 6.75 mL/min of air to escape depending on port size; this class implies the need for interference-fit elastomeric seals. With the exception of class V, all classes are based on standardized pressure conditions that may not represent actual conditions. Therefore, it is difficult to estimate leakage in service. Leakage normally increases over time as seals and seating surfaces become nicked or worn. Leak passages across the seat-contact line, known as

wire drawing, may form and become worse over time—even in hard metal seats under sufficiently high-pressure differentials.

Polymers used for seat and plug seals and internal static seals include PTFE (polytetrafluoroethylene) and other fluorocarbons, polyethylene, nylon, polyether-ether-ketone, and acetal. Fluorocarbons are often carbon- or glass-filled to improve mechanical properties and heat resistance. Temperature and chemical compatibility with the process fluid are the key selection criteria. Polymer-lined bearings and guides are used to decrease friction, which lessens dead band and reduces actuator force requirements. See Sec. 25, “Materials of Construction,” for properties.

Packing forms the pressure-tight seal, where the stem protrudes through the pressure boundary. Packing is typically made from PTFE or, for high temperature, a bonded graphite. If the process fluid is toxic, more sophisticated systems such as dual packing, live-loaded, or a flexible metal bellows may be warranted. Packing friction can significantly degrade control performance. Pipe, bonnet, and internal-trim joint gaskets are typically a flat sheet composite. Gaskets intended to absorb dimensional mismatch are typically made from filled spiral-wound flat stainless-steel wire with PTFE or graphite filler. The use of asbestos in packing and gaskets has been largely eliminated.

Flow Characteristics The relationship between valve flow and valve travel is called the valve flow characteristic. The purpose of flow characterization is to make loop dynamics independent of load, so that a single controller tuning remains optimal for all loads. Valve gain is one factor affecting loop dynamics. In general, gain is the ratio of change in output to change in input. The input of a valve is travel y , and the output is flow w . Since pressure conditions at the valve can depend on flow (hence travel), valve gain is

$$\frac{dw}{dy} = \frac{\partial w}{\partial C_V} \frac{dC_V}{dy} + \frac{\partial w}{\partial p_1} \frac{dp_1}{dy} + \frac{\partial w}{\partial p_2} \frac{dp_2}{dy} \quad (8-121)$$

An inherent valve flow characteristic is defined as the relationship between flow rate and travel, under constant-pressure conditions. Since the rightmost two terms in Eq. (8-121) are zero in this case, the inherent characteristic is necessarily also the relationship between flow coefficient and travel.

Figure 8-78 shows three common inherent characteristics. A linear characteristic has a constant slope, meaning the inherent valve gain is a constant. The most popular characteristic is equal-percentage, which gets its name from the fact that equal changes in travel produce equal-percentage changes in the existing flow coefficient. In other words, the slope of the curve is proportional to C_V , or equivalently that inherent valve gain is proportional to flow. The equal-percentage characteristic can be expressed mathematically by

$$C_V(y) = (\text{rated } C_V) \exp \left[\left(\frac{y}{\text{rated } y} - 1 \right) \ln R \right] \quad (8-122)$$

This expression represents a set of curves parameterized by R . Note that $C_V(y = 0)$ equals $(\text{rated } C_V)/R$ rather than zero; real equal-percentage characteristics deviate from theory at some small travel to meet shutoff requirements. An equal-percentage characteristic provides perfect compensation for a process where the gain is inversely proportional to flow (e.g., liquid pressure). *Quick opening* does not have a standardized mathematical definition. Its shape arises naturally from high-capacity plug designs used in on/off service globe valves. Frequently, pressure conditions at the valve will change with flow rate. This so-called process influence [the rightmost two terms on the right-hand side of Eq. (8-121)] combines with inherent gain to express the installed valve gain. The flow versus travel relationship for a specific set of conditions is called the installed flow characteristic. Typically, valve Δp decreases with load, since pressure losses in the piping system increase with flow. Figure 8-79 illustrates how allocation of total system head to the valve influences the installed flow characteristics. For a linear or quick-opening characteristic, this transition toward a concave down shape would be more extreme. This effect of typical process pressure variation, which causes equal-percentage

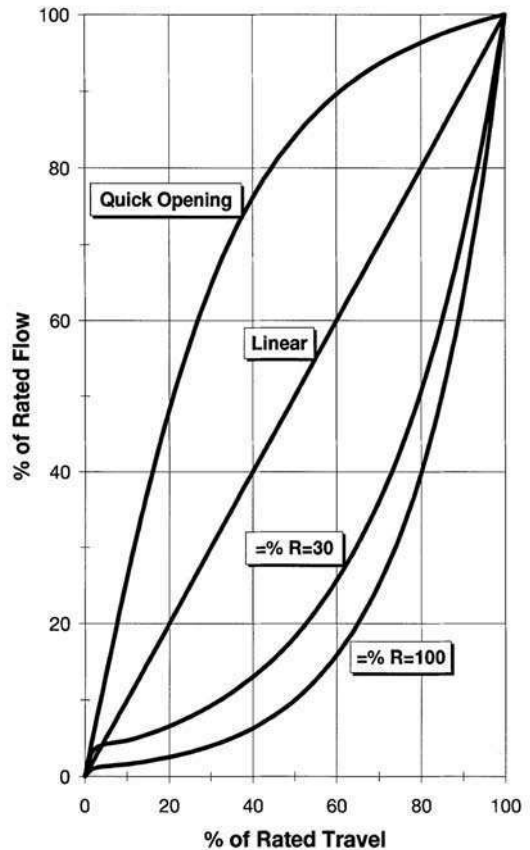


FIG. 8-78 Typical inherent flow characteristics.

characteristics to have fairly constant installed gain, is one reason the equal-percentage characteristic is the most popular.

Due to clearance flow, flow force gradients, seal friction, and the like, flow cannot be throttled to an arbitrarily small value. Installed rangeability is the ratio of maximum to minimum controllable flow. The actuator and positioner, as well as the valve, influence the installed rangeability. Inherent rangeability is defined as the ratio of the largest to the smallest C_V within which the characteristic meets specified criteria (see ISA 75.11). The R value in the equal-percentage definition is a theoretical rangeability only. While high installed rangeability is desirable, it is also important not to oversize a valve; otherwise, turndown (ratio of maximum normal to minimum controllable flow) will be limited.

Sliding stem valves are characterized by altering the contour of the plug when the port and plug determine the minimum (controlling) flow area. Passage area versus travel is also easily manipulated in characterized cage designs. Inherent rangeability varies widely, but typical values are 30 for contoured plugs and 20 to 50 for characterized cages. While these types of valves can be characterized, the degree to which manufacturers conform to the mathematical ideal is revealed by plotting measured C_V versus travel. Note that ideal equal-percentage will plot as a straight line on a semilog graph. Custom characteristics that compensate for a specific process are possible.

Rotary stem-valve designs are normally offered only in their naturally occurring characteristic, since it is difficult to appreciably alter this. If additional characterization is required, the positioner or controller may be characterized. However, these approaches are less direct, since it is possible for device nonlinearity and dynamics to distort the compensation.

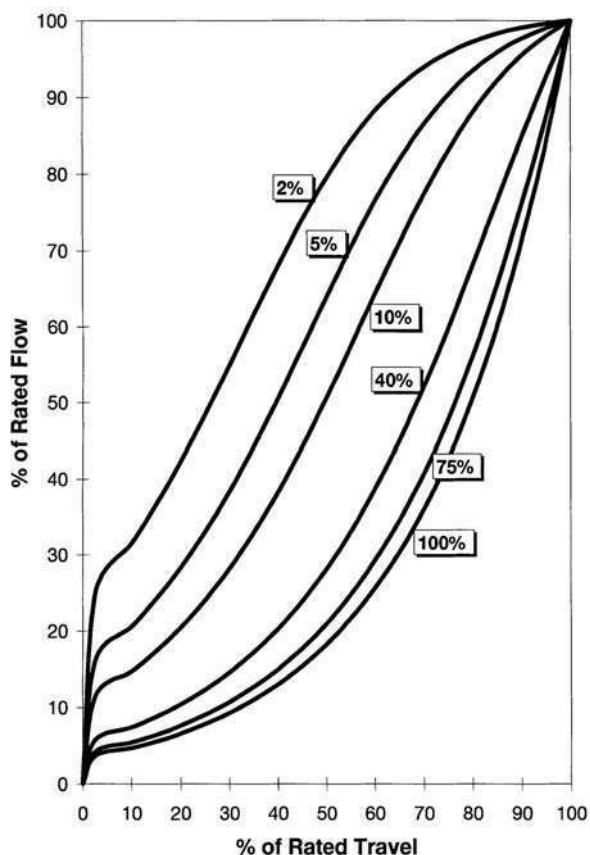


FIG. 8-79 Installed flow characteristic as a function of percent of total system head allocated to the control valve (assuming constant-head pump, no elevation head loss, and an R equal to 30 equal-percentage inherent characteristic).

VALVE CONTROL DEVICES

Devices mounted on the control valve that interface various forms of input signals, monitor and transmit valve position, or modify valve response are valve control devices. In some applications, several auxiliary devices are used together on the same control valve. For example, mounted on the control valve, one may find a current-to-pressure transducer, a valve positioner, a volume booster relay, a solenoid valve, a trip valve, a limit switch, a process controller, and/or a stem position transmitter. Figure 8-80 shows a valve positioner mounted on the yoke leg of a spring and diaphragm actuator.

As most throttling control valves are still operated by pneumatic actuators, the control valve device descriptions that follow relate primarily to devices that are used with pneumatic actuators. The functions of hydraulic and electrical counterparts are very similar. Specific details on a particular valve control device are available from the vendor of the device.

Valve Positioners The valve positioner, when combined with an appropriate actuator, forms a complete closed-loop valve position control system. This system makes the valve stem conform to the input signal coming from the process controller in spite of force loads that the actuator may encounter while moving the control valve. Usually, the valve positioner is contained in its own enclosure and is mounted on the control valve.

The key parts of the positioner/actuator system, shown in Fig. 8-81a, are (1) an input conversion network, (2) a stem position feedback network, (3) a summing junction, (4) an amplifier network, and (5) an actuator.



FIG. 8-80 Valve and actuator with valve positioner attached. (Courtesy Fisher Controls International LLC.)

The input conversion network shown is the interface between the input signal and the summer. This block converts the input current or pressure (from an I/P transducer or a pneumatic process controller) to a voltage, electric current, force, torque, displacement, or other particular variable that can be directly used by the summer. The input conversion usually contains a means to adjust the slope and offset of the block to provide for a means of spanning and zeroing the positioner during calibration. In addition, means for changing the sense (known as "action") of the input/output characteristic are often addressed in this block. Also exponential, logarithmic, or other predetermined characterization can be put in this block to provide a characteristic that is useful in offsetting or reinforcing a nonlinear valve or process characteristic.

The stem position feedback network converts stem travel to a useful form for the summer. This block includes the feedback linkage which varies with actuator type. Depending on positioner design, the stem position feedback network can provide span and zero and characterization functions similar to that described for the input conversion block.

The amplifier network provides signal conversion and suitable static and dynamic compensation for good positioner performance. Control from this block usually reduces to a form of proportional or proportional plus derivative control. The output from this block in the case of a pneumatic positioner is a single connection to the spring and diaphragm actuator or two connections for push/pull operation of a

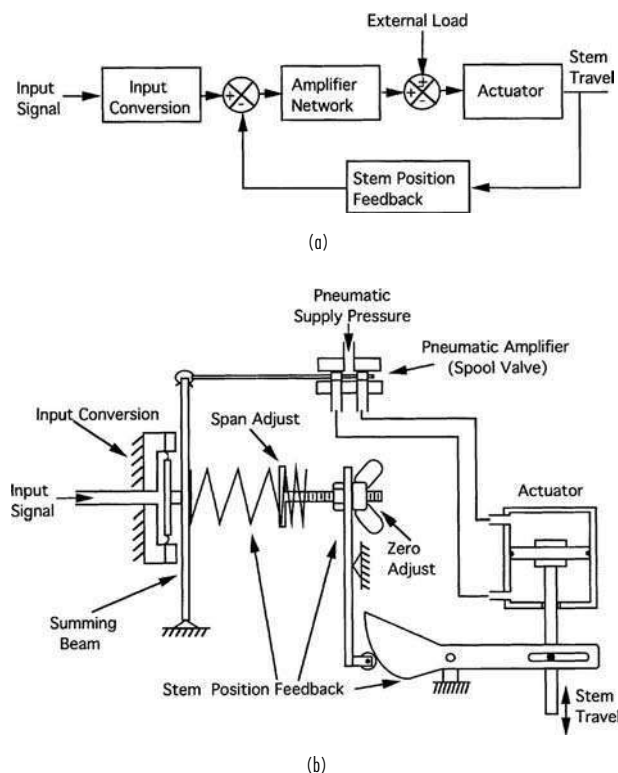


FIG. 8-81 Positioner/actuators. (a) Generic block diagram. (b) Example of a pneumatic positioner/actuator.

springless piston actuator. The action of the amplifier network and the action of the stem position feedback can be reversed together to provide for reversed positioner action.

By design, the gain of the amplifier network shown in Fig. 8-81a is made very large. Large gain in the amplifier network means that only a small proportional deviation will be required to position the actuator through its active range of travel. This means that the signals into the summer track very closely and that the gain of the input conversion block and the stem position feedback block determine the closed-loop relationship between the input signal and the stem travel.

Large amplifier gain also means that only a small amount of additional stem travel deviation will result when large external force loads are applied to the actuator stem. For example, if the positioner's amplifier network has a gain of 50 (and assuming that high packing box friction loads require 25 percent of the actuator's range of thrust to move the actuator), then only 25 percent/50 (or 0.5 percent deviation) between input signal and output travel will result due to valve friction.

Figure 8-81b is an example of a pneumatic positioner/actuator. The input signal is a pneumatic pressure that (1) moves the summing beam, which (2) operates the spool valve amplifier, which (3) provides flow to and from the piston actuator, which (4) causes the actuator to move and continue moving until (5) the feedback force returns the beam to its original position and stops valve travel at a new position. Typical positioner operation is thereby achieved.

Static performance measurements related to positioner/actuator operation include the conformity, measured accuracy, hysteresis, dead band, repeatability, and locked stem pressure gain. Definitions and standardized test procedures for determining these measurements can be found in ISA-S75.13, "Method of Evaluating the Performance of Positioners with Analog Input Signals and Pneumatic Output."

Dynamics of Positioner-Based Control Valve Assemblies

Control valve assemblies are complete, functional units that include the valve body, actuator, positioner, if so equipped, associated linkages, and any auxiliary equipment such as current to pneumatic signal transducers and air supply pressure regulators. Although performance information such as frequency response, sensitivity, and repeatability data may be available for a number of these components individually, it is the performance of the entire assembly that will ultimately determine how well the demand signal from the controller output is transferred through the control valve to the process. The valve body, actuator, and positioner combination is typically responsible for the majority of the control valve assembly's dynamic behavior. On larger actuators, the air supply pressure regulator capacity or other airflow restrictions may limit the control valve assembly's speed of response.

The control valve assembly response can usually be characterized quite well by using a first-order plus dead-time response model. The control valve assembly will also exhibit backlash, stiction, and other non-linear behavior. During normal operation of a control loop, the controller usually makes small output changes from one second to the next. Typically this change is less than 1 percent. With very small controller output changes, e.g., less than 0.1 percent, the control valve assembly may not move at all. As the magnitude of the controller output change increases, eventually the control valve will move. At the threshold of movement, the positional accuracy and repeatability of the control valve are usually quite poor. The speed of response may be quite slow and may occur after a number of seconds of dead time. This poor performance is due to the large backlash and stiction effects relative to the requested movement and the small output change of the positioner. With a further increase in the magnitude of the controller output steps, the behavior of the control valve typically becomes more repeatable and "linear." Dead time usually drops to only a fraction of a second, and the first-order time constant becomes faster. For much larger steps in the controller output, e.g., over 10 percent, the positioner and air supply equipment may be unable to deliver the necessary air volume to maintain the first-order response. In this case, the control valve will exhibit very little dead time, but will be rate-limited and will ramp toward the requested position. It is within the linear region of motion that the potential for the best control performance exists.

When one is specifying a control valve for process control applications, in addition to material, style, and size information, the dynamic response characteristics and maximum allowable dead band (sum of backlash, stiction, and hysteresis effects) *must* be stated. The requirement for the control valve assembly's speed of response is ultimately determined by the dynamic characteristics of the process and the control objectives. Typically, the equivalent first-order time constant specified for the control valve assembly should be at least 5 times faster than the desired controller closed-loop time constant. If this requirement is not met, the tuning of the control loop must be slowed down to accommodate the slow control valve response, otherwise, control robustness and stability may be compromised. The dead band of the control valve assembly is typically the determining factor for control resolution and frequently causes control instability in the form of a "limit" cycle. The controller output will typically oscillate across a range that is 1 to 2 times the magnitude of the control valve dead band. This is very dependent on the nature of the control valve nonlinearities, the process dynamics, and the controller tuning. The magnitude of the process limit cycle is determined by the size of the control valve dead band multiplied by the installed gain of the control valve. For this reason, a high-performance control valve assembly, e.g., with only 0.5 percent dead band, may cause an unacceptably large process limit cycle if the valve is oversized and has a high installed gain. For typical process control applications, the installed gain of the control valve should be in the range of 0.5 to 2 percent of the process variable span per percent of the controller output. The total dead band of the control valve assembly should be less than 1 percent. For applications that require more precise control, the dead band and possibly the installed gain of the control valve must be reduced. Specialized actuators are available that are accurate down to 0.1 percent or less. At this level of performance, however, the design of the valve body, bearings, linkages, and seals starts to become a significant source of dead band.

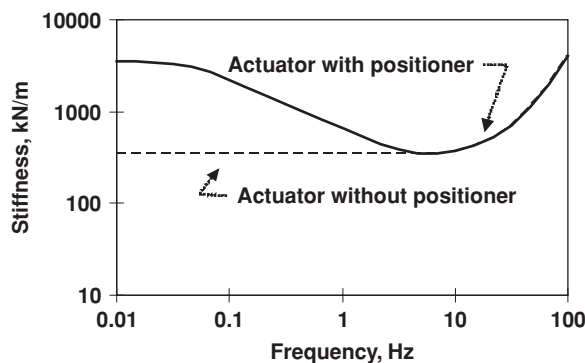


FIG. 8-82 Actuator stiffness as a function of frequency for a 69-in² spring and diaphragm pneumatic actuator. Actuator with positioner exhibits higher stiffness over the lower frequency range compared to that of the pneumatic actuator without a positioner.

Positioner/Actuator Stiffness Minimizing the effect of dynamic loads on valve stem travel is an important characteristic of the positioner/actuator. Stem position must be maintained in spite of changing reaction forces caused by valve throttling. These forces can be random (buffeting force) or can result from a negative-slope force/stem travel characteristic (negative gradient); either could result in valve stem instability and loss of control. To reduce and eliminate the effect of these forces, the effective stiffness of the positioner/actuator must be made sufficiently high to maintain stationary control of the valve stem.

The stiffness characteristic of the positioner/actuator varies with the forcing frequency. Figure 8-82 indicates the stiffness of the positioner/actuator is increased at low frequencies and is directly related to the locked-stem pressure gain provided by the positioner. As frequency increases, a dip in the stiffness curve results from dynamic gain attenuation in the pneumatic amplifiers in the positioner. The value at the bottom of the dip is the sum of the mechanical stiffness of the spring in the actuator and the air spring effect produced by air enclosed in the actuator casing. At yet higher frequencies, actuator inertia dominates and causes a corresponding rise in system stiffness.

The air spring effect results from adiabatic expansion and compression of air in the actuator casing. Numerically, the small perturbation value for air spring stiffness in newtons per meter is given by

$$\text{Air spring rate} = \frac{\gamma p_a A_a^2}{V} \quad (8-123)$$

where γ is the ratio of specific heats (1.4 for air), p_a is the actuator pressure in pascals absolute, A_a is the actuator pressure area in square meters, and V is the internal actuator volume in cubic meters.

Positioner Application Positioners are widely used on pneumatic valve actuators. Often they provide improved process loop control because they reduce valve-related nonlinearity. Dynamically, positioners maintain their ability to improve control valve performance for sinusoidal input frequencies up to about one-half of the positioner bandwidth. At input frequencies greater than this, the attenuation in the positioner amplifier network gets large, and valve nonlinearity begins to affect final control element performance more significantly. Because of this, the most successful use of the positioner occurs when the positioner response bandwidth is greater than twice that of the most dominant time lag in the process loop.

Some typical examples in which the dynamics of the positioner are sufficiently fast to improve process control are the following:

1. In a distributed control system (DCS) process loop with an electronic transmitter. The DCS controller and the electronic transmitter have time constants that are dominant over the positioner response. Positioner operation is therefore beneficial in reducing valve-related nonlinearity.

2. In a process loop with a pneumatic controller and a large process time constant. Here the process time constant is dominant, and the positioner will improve the linearity of the final control element. Some common processes with large time constants that benefit from positioner application are liquid level, temperature, large-volume gas pressure, and mixing.

3. Additional situations in which valve positioners are used:

- a. On springless actuators where the actuator is not usable for throttling control without position feedback.

- b. When split ranging is required to control two or more valves sequentially. In the case of two valves, the smaller control valve is calibrated to open in the lower half of the input signal range, and a larger valve is calibrated to open in the upper half of the input signal range. Calibrating the input command signal range in this way is known as split-range operation and increases the practical range of throttling process flows over that of a single valve.

- c. In open-loop control applications where best static accuracy is needed. On occasion, positioner use can degrade process control. Such is the case when the process controller, process, and process transmitter have time constants that are similar to or smaller than that of the positioner/actuator. This situation is characterized by low process controller proportional gain (gain < 0.5), and hunting or limit cycling of the process variable is observed.

Improvements here can be made by doing one of the following:

1. Install a dominant first-order, low-pass filter in the loop ahead of the positioner and retune the process loop. This should allow increased proportional gain in the process loop and reduce hunting. Possible means for adding the filter include adding it to the firmware of the DCS controller, by adding an external RC network on the output of the process controller or by enabling the filter function in the input of the positioner, if it is available. Also, some transducers, when connected directly to the actuator, form a dominant first-order lag that can be used to stabilize the process loop.

2. Select a positioner with a faster response characteristic.

Processor-Based Positioners When designed around an electronic microcontroller, the valve positioner [now commonly referred to as a *digital valve controller* (DVC)] takes on additional functionality that provides convenience and performance enhancements over the traditional design. The most common form of processor-based positioner, shown in Fig. 8-81, is a digitally communicating stem position controller that operates by using the fundamental blocks shown in Fig. 8-81a. A local display is part of the positioner and provides tag information, command input and travel, servo tuning parameters, and diagnostic information. Often auxiliary sensors are integrated into the device to provide increased levels of functionality and performance. Sensed variables can include actuator pressure, relay input pressure, relay valve position, board temperature, or a discrete input. A 4- to 20-mA valve travel readback circuit is also common. The travel sensor is based on a potentiometer or can be a noncontacting type such as a variable capacitance sensor, Hall effect sensor, or GMR device. Some positioners require a separate connection to an ac or dc supply voltage, but the majority of the designs are "loop-powered," which means that they receive power either through the current input (for positioners that require a 4- to 20-mA analog input signal) or through the digital communications link when the control signal is a digital signal.

Processor-based positioners support automatic travel calibration and automatic response tuning for quick commissioning of the final control element. Features of this type of valve positioner include compensators for improved static and dynamic travel response; diagnostics for evaluating positioner, actuator, and valve health; and the capability to be polled from remote locations through a PC-based application or through a handheld communicator attached to the field wiring. Capability to support custom firmware for special valve applications, such as emergency safety shutdown, is also a characteristic of the processor-based design.

Digital Field Communications To provide increased data transmission capability between valve-mounted devices and the host control system, manufacturers are providing digital network means in their devices. The field networks, commonly known as field buses, compete fiercely in the marketplace and have varying degrees of flexibility and specific application strengths. A prospective field bus

customer is advised to study the available bus technologies and to make a selection based on needs, and not be seduced by the technology itself.

Generally, a field bus protocol must be nonproprietary ("open") so that different vendors of valve devices can design their bus interface to operate properly on the selected field bus network. Users demand that the devices be "interoperable" so that the device will work with other devices on the same segment or can be substituted with a device from an alternate manufacturer. International standardization of some of the protocols is currently underway (for example, IEC 61158) whereas others are sponsored by user groups or foundations that provide democratic upgrades to the standard and provide network compliance testing.

The physical wiring typically used is the plant standard twisted-pair wiring for 4- to 20-mA instrumentation. Because of the networking capability of the bus, more than one device can be supported on a single pair of wires, and thus wiring requirements are reduced. Compared to a host level bus such as Ethernet, field buses exhibit slower communication rates, have longer transmission distance capability (1 to 2 km), use standard two-wire installation, are capable of multidrop busing, can support bus-powered devices, do not have redundant modes of bus operation, and are available for intrinsically safe installations. Devices on the field bus network may be either powered by the bus itself or powered separately.

The simplest digital networks available today support discrete sensors and on/off actuators, including limit switches and motor starters. Networks of this type have fast cycle times and are often used as an alternative to PLC discrete I/O. More sophisticated field networks are designed to support process automation, more complex process transmitters, and throttling valve actuators. These process-level networks are fundamentally continuous and analoglike in operation, and data computation is floating-point. They support communication of materials of construction, calibration and commissioning, device and loop level diagnostics (including information displays outlining corrective action), and unique manufacturer-specific functionality. Some process networks are able to automatically detect, identify, and assign an address to a new device added to the network, thus reducing labor, eliminating addressing errors, and indicating proper network function immediately after the connection is made. Final control elements operated by the process-level network include I/P transducers, motorized valves, digital valve controllers, and transmitters.

A particular field network protocol known as HART® (Highway Addressable Remote Transducer) is the most widely used field network protocol. It is estimated that as of 2004 there are more than 14 million HART-enabled devices installed globally and that 70 percent

of all processor-based process measurement and control instruments installed each year use HART communications. HART's popularity is based on its similarity to the traditional 4- to 20-mA field signaling and thus represents a safe, controlled transition to digital field communications without the risk often associated with an abrupt change to a totally digital field bus. With this protocol, the digital communications occur over the same two wires that provide the 4- to 20-mA process control signal without disrupting the process signal. The protocol uses the frequency-shift keying (FSK) technique (see Fig. 8-83) where two individual frequencies, one representing the mark and the other representing the space, are superimposed on the 4- to 20-mA current signal. As the average value of the signals used is zero, there is no dc offset value added to the 4- to 20-mA signal. The HART protocol is principally a master/slave protocol which means that a field device (slave) speaks only when requested by a master device. In this mode of operation, the slave can update the master at a rate of twice per second. An optional communication mode, *burst mode*, allows a HART slave device to continuously broadcast updates without stimulus requests from the master device. Update rates of 3 to 4 updates per second are typical in the burst mode of operation.

HART-enabled devices are provided by the valve device manufacturer at little or no additional cost. The HART network is compatible with existing 4- to 20-mA applications using current plant personnel and practices, provides for a gradual transition from analog to fully digital protocols, and is provided by the valve device manufacturer at little or no additional cost. Contact the HART Communication Foundation for additional information.

Wireless digital communication to and from the final control element is not yet commercially available but is presently being investigated by more than one device manufacturer. The positive attribute of a wireless field network is the reduced cost of a wireless installation compared to a wired installation. Hurdles for wireless transmissions include security from nonnetwork sources, transmission reliability in the plant environment, limited bus speed, and the conservative nature of the process industry relative to change. Initial installations of wireless networks will support secondary variables and diagnostics, then primary control of processes with large time constants, and finally general application to process control. Both point-to-point and mesh architectures are being evaluated for commercialization at the device level. Mesh architectures rely on the other transmitting devices in the area to receive and then pass on any data transmission, thus rerouting communications around sources of interference. Two unlicensed spread spectrum radio bands are the main focus for current wireless development: 900 MHz and 2.4 GHz. The 900-MHz band is unique to North America and has better propagation and penetrating properties than the 2.4-GHz band. The 2.4-GHz band is a worldwide band and has wider channels, allowing much higher data rates. The spread

*HART is a registered trademark of the HART Communication Foundation.

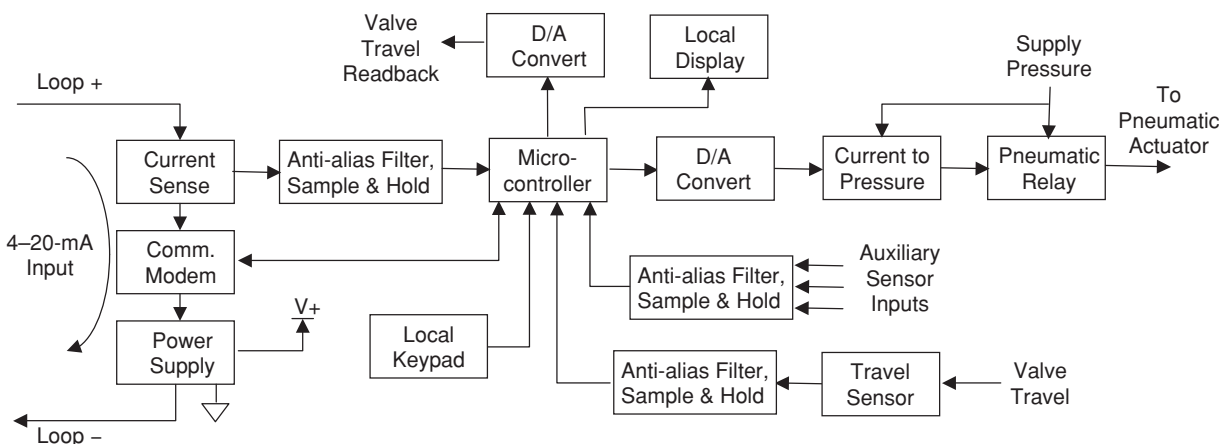


FIG. 8-83 Generic loop powered digital valve controller.

spectrum technique uses multiple frequencies within the radio band to transmit data. Spread spectrum is further divided into the direct sequence technique, where the device changes frequency many times per data bit, and the frequency-hopping technique, where the device transmits a data packet on one frequency and then changes to a different frequency. Because of the rapid growth expected in this decade, the prospective wireless customer is encouraged to review up-to-date literature to determine the state of field wireless commercialization as it applies to her or his specific application.

Diagnostic Capability The rapid proliferation of communicating, processor-based digital valve controllers over the last decade has led to a corresponding rise in diagnostic capability at the control valve. Diagnosing control valve health is critical to plant operation as maintenance costs can be reduced by identifying the valves that are candidates for repair. Less time is spent during plant shutdown repairing valves that do not need repair, which ultimately results in increased online operating time. Valve diagnostics can detect and flag a failed valve more quickly than by any other means, and can be configured to cause the valve to move to its fail-safe position on detection of specified fault conditions. The diagnostic-enabled positioner, when used with its host-based software application, can pinpoint exact components in a given final control element that have failed, and can recommend precise maintenance procedures to follow to remedy the fault condition.

The state variables that provide valve position control are used to diagnose the health of the final control element. In addition, some digital valve controller designs integrate additional sensors into their construction to provide increased diagnostic capability. For example, pressure sensors are provided to detect supply pressure, actuator pressure (upper and lower cylinder pressures in the case of a springless piston actuator), and internal pilot pressure. Also, the position of the pneumatic relay valve is available in some designs to provide quiescent flow data used for leak detection in the actuator.

Valve diagnostics are divided into two types: online and offline. Offline diagnostics are those diagnostics that occur when the control valve is bypassed or otherwise isolated from the process. The offline diagnostic routine manipulates the travel command to the valve and records the corresponding valve travel, actuator pressure, and servodrive value. These parameters are plotted in various combinations to provide hysteresis plus dead-band information, actuator operating pressure as a function of travel, valve friction, servodrive performance, valve seating load, positioner calibration endpoints, and dynamic response traces. Small- and large-amplitude step inputs as well as large slow ramps (exceeding 100 percent of the input range) are common offline test waveforms generated by the diagnostic as command inputs for offline diagnostic tests. Figure 8-84 is an example of one offline diagnostic test performed on a small globe valve actuated by a spring and diaphragm actuator. During this test the command input, travel, actuator pressure, and servodrive level are recorded and plotted as they result from a command input that is slowly ramped by the diagnostic routine (Fig. 8-85a). This diagnostic is extremely useful in detecting problems with the valve/actuator system and can flag potential problems with the final control element before catastrophic failure occurs. For example, Fig. 8-85b indicates the overall tracking capability of the control valve, and Fig. 8-85c indicates the pressure operating range of the actuator and the amount of frictional force resulting from the combined effects of valve packing and valve plug contact. Figure 8-85d displays the level of servodrive required to stroke the valve from one end of travel to the other. The composite operative health of the control valve is determined through comparison of the empirical levels presented in Fig. 8-85 with the manufacturers' recommendations. Recommended maintenance actions result from this comparison.

Online diagnostics are diagnostics that monitor and evaluate conditions at the control valve during normal throttling periods (i.e., during valve-in-service periods). Online diagnostics monitor mean levels and disturbances generated in the normal operation of the valve and typically do not force or generate disturbances on the valve's operation. For example, an online diagnostic can calculate travel deviation relative to the input command and flag a condition where the valve travel has deviated beyond a preset band. Such an event, if it exists for more than a short time, indicates that the valve has lost its ability to track the input

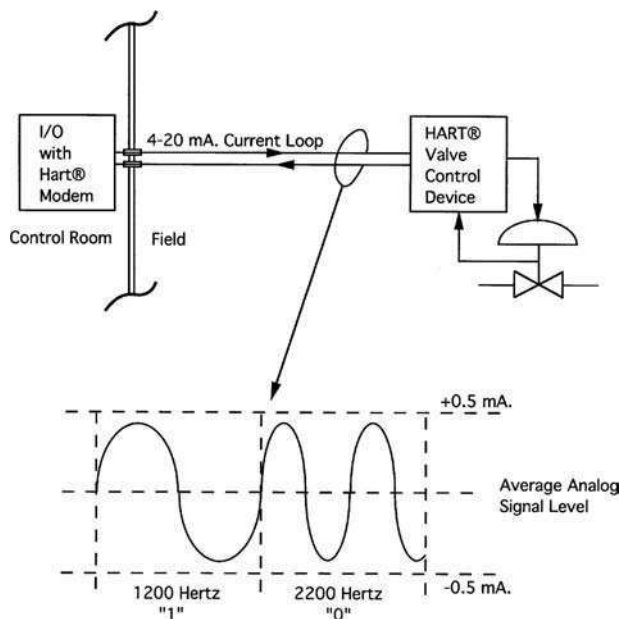


FIG. 8-84 Hybrid point-to-point communications between the control room and the control valve device.

command within specified limits. Additional diagnostics could suggest that the feedback linkage has ceased functioning, or that the valve has stuck, or that some other specific malfunction is the cause of excess travel deviation. The manufacturer of the positioner diagnostic incorporates default limits into the host software application that are used to determine the relative importance of a specific deviation. To quickly indicate the severity of a problem detected by a diagnostic routine, a red, yellow, or green, or "advise, maintenance now, or failed," indication is presented on the user-interface screen for the valve problem diagnosed. Help notes and recommended remedial action are available by pointing and clicking on the diagnostic icon presented on the user's display.

Event-triggered recording is an online diagnostic technique supported in digital valve controllers (DVCs). Functionally a triggering event, such as a valve coming off a travel stop or a travel deviation alert, starts a time-series recording of selected variables. A collection of variables such as the input command, stem travel, actuator pressure, and drive command are stored for several minutes before and after the triggered event. These variables are then plotted as time series for immediate inspection or are stored in memory for later review. Event-triggering diagnostics are particularly useful in diagnosing valves that are closed or full-open for extended periods. In this case the event-triggered diagnostic focuses on diagnostic rich data at the time the valve is actually in operation and minimizes the recording of flat-line data with little diagnostic content. Other online diagnostics detected by DVC manufacturers include excess valve friction, supply pressure failure, relay operation failure, broken actuator spring, current to pressure module failure, actuator diaphragm leaking, and shifted travel calibration.

Safety shutdown valves, which are normally wide open and operate infrequently, are expected to respond to a safety trip command reliably and without fault. To achieve the level of reliability required in this application, the safety valve must be periodically tested to ensure positive operation under safety trip conditions. To test the operation of the shutdown system without disturbing the process, the traditional method is to physically lock the valve stem in the wide-open position and then to electrically operate the pneumatic shutdown solenoid valve. Observing that the pneumatic solenoid valve has properly vented the actuator pressure to zero, the actuator is seen as capable of applying sufficient spring force to close the valve, and a positive safety valve test is indicated. The

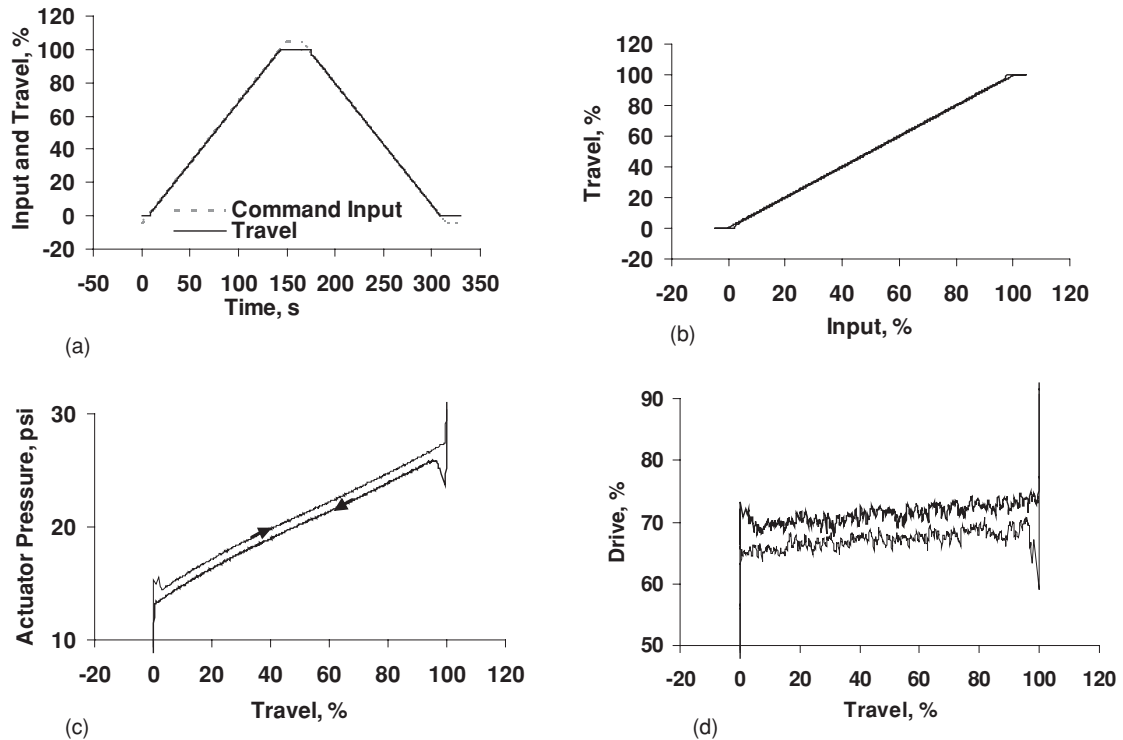


FIG. 8-85 Offline valve diagnostic scan showing results of a diagnostic ramp. (a) The command input and resulting travel. (b) The dynamic scan. (c) The valve signature. (d) The servodrive versus travel plot. The hysteresis shown in the valve signature results from sliding friction due to valve packing and valve plug contact.

pneumatic solenoid valve is then returned to its normal electrical state, the actuator pressure returns to full supply pressure, and the valve stem lock mechanism is removed. This procedure, though necessary to enhance process safety, is time-consuming and takes the valve out of service during the locked stem test. Digital valve controllers are able to validate the operation of a safety shutdown valve by using an online diagnostic referred to as a partial stroke test. The partial stroke test is substituted for the traditional test method described above and does not require the valve to be locked in the wide-open position to perform the test. In a fashion similar to that shown in Fig. 8-85a (the partial stroke diagnostic), the system physically ramps the command input to the positioner from the wide-open position to a new position, pauses at the new position for a few seconds, and then ramps the command input back to the wide-open position (see Fig. 8-86a). During this time, the valve travel measurement is monitored and compared to the input command. If the travel measurement deviates from the input by more than a fixed amount for the configured period of time, the valve is considered to have failed the test and a failed-test message is communicated to the host system. Also during this test, the actuator pressure required to move the valve is detected via a dedicated pressure sensor (see Fig. 8-86b). If the thrust (pressure) required to move the valve during the partial stroke test exceeds the predefined thrust limit for this test, the control valve is determined to have a serious sticking problem, the test is immediately aborted, and the valve is flagged as needing maintenance. The partial stroke test can be automated to perform on a periodic basis, for instance, once a week; or it can be initialized by operator request at any time. The amount of valve travel that occurs during the partial stroke test is typically limited to a minimum valve position of 70 percent open or greater. This limit is imposed to prevent the partial stroking of the safety valve from significantly affecting the process flow through the valve.

Comparison of partial stroke curves from past tests can indicate the gradual degradation of valve components. Use of “overlay” graphics, identification of unhealthy shifts in servodrive, increases in valve friction, and changes in dynamic response provide information leading to a diagnosis of needed maintenance.

In addition to device-level diagnostics, networked final control elements, process controllers, and transmitters can provide “loop” level diagnostics that can detect loops that are operating below expectations. Process variability, time in a limit (saturated) condition, and time in the wrong control mode are metrics used to detect problems in process loop operation.

Transducers The current-to-pressure transducer (I/P transducer) is a conversion interface that accepts a standard 4- to 20-mA input current from the process controller and converts it to a pneumatic output in a standard pneumatic pressure range [normally 0.2 to 1.0 bar (3 to 15 psig) or, less frequently, 0.4 to 2.0 bar (6 to 30 psig)]. The output pressure generated by the transducer is connected directly to the pressure connection on a spring-opposed diaphragm actuator or to the input of a pneumatic valve positioner.

Figure 8-87a is the schematic of a basic I/P transducer. The transducer shown is characterized by (1) an input conversion that generates an angular displacement of the beam proportional to the input current, (2) a pneumatic amplifier stage that converts the resulting angular displacement to pneumatic pressure, and (3) a pressure area that serves as a means to return the beam to very near its original position when the new output pressure is achieved. The result is a device that generates a pressure output that tracks the input current signal. The transducer shown in Fig. 8-88a is used to provide pressure to small load volumes (normally 4.0 in³ or less), such as a positioner or booster input. With only one stage of pneumatic amplification, the flow

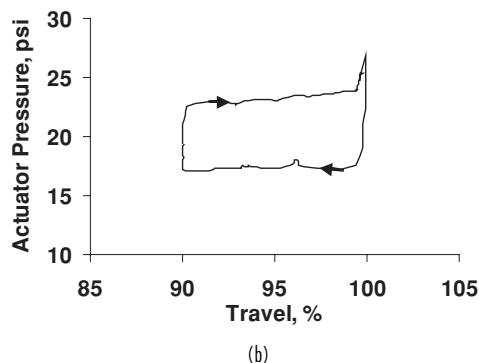
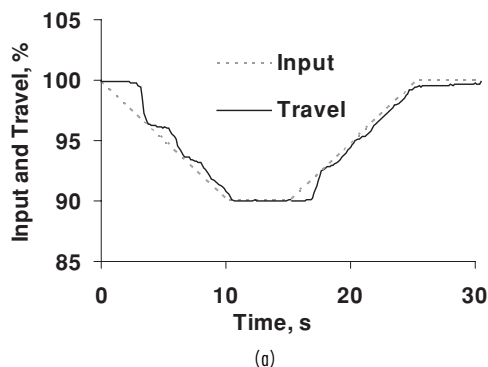


FIG. 8-86 Online partial stroke diagnostic used to validate the operability of a pneumatically operated safety shutdown valve. (a) Input command generated by the diagnostic and resulting travel. (b) Actuator pressure measured over the tested range of travel.

capacity of this transducer is limited and not sufficient to provide responsive load pressure directly to a pneumatic actuator.

The flow capacity of the transducer can be increased by adding a booster relay such as the one shown in Fig. 8-87b. The flow capacity of the booster relay is nominally 50 to 100 times that of the nozzle amplifier shown in Fig. 8-87a and makes the combined transducer/booster suitably responsive to operate pneumatic actuators. This type of transducer is stable for all sizes of load volume and produces measured accuracy (see ANSI/ISA-51.1, "Process Instrumentation Terminology," for the definition of measured accuracy) of 0.5 to 1.0 percent of span.

Better measured accuracy results from the transducer design shown in Fig. 8-87c. In this design, pressure feedback is taken at the output of the booster relay stage and fed back to the main summer. This allows the transducer to correct for errors generated in the pneumatic booster as well as errors in the I/P conversion stage. Also, particularly with the new analog electric and digital versions of this design, PID control is used in the transducer control network to give extremely good static accuracy, fast dynamic response, and reasonable stability into a wide range of load volumes (small instrument bellows to large actuators). Also environmental factors such as temperature change, vibration, and supply pressure fluctuation affect this type of transducer the least. Even a perfectly accurate I/P transducer cannot compensate for stem position errors generated by friction, backlash, and varying force loads coming from the actuator and valve. To do this compensation, a different control valve device—the valve positioner—is required.

Booster Relays The booster relay is a single-stage power amplifier having a fixed gain relationship between the input and output pressures. The device is packaged as a complete stand-alone unit with pipe thread connections for input, output, and supply pressure. The booster amplifier shown in Fig. 8-87b shows the basic construction of the booster relay. Enhanced versions are available that provide specific features such as (1) variable gain to split the output range of a pneumatic controller to operate more than one valve or to provide additional actuator force; (2) low hysteresis for relaying measurement and control signals; (3) high flow capacity for increased actuator stroking speed; and (4) arithmetic, logic, or other compensation functions for control system design.

A particular type of booster relay, called a dead-band booster, is shown in Fig. 8-88. This booster is designed to be used exclusively between the output of a valve positioner and the input to a pneumatic actuator. It is designed to provide extra flow capacity to stroke the actuator faster than with the positioner alone. The dead-band booster is designed intentionally with a large dead band (approximately 5 percent of the input span), elastomer seats for tight shutoff, and an adjustable bypass valve connected between the input and output of the booster. The bypass valve is tuned to provide the best compromise between increased actuator stroking speed and positioner/actuator stability.

With the exception of the dead-band booster, the application of booster relays has diminished somewhat by the increased use of current-to-pressure transducers, electropneumatic positioners, and electronic control systems. Transducers and valve positioners serve much the same functionality as the booster relay in addition to interfacing with the electronic process controller.

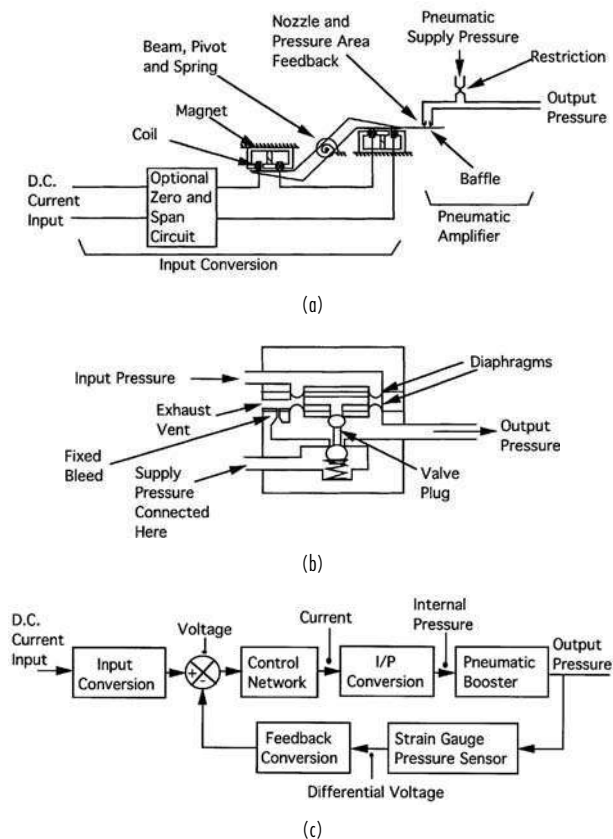


FIG. 8-87 Current-to-pressure transducer component parts. (a) Direct-current-pressure conversion. (b) Pneumatic booster amplifier (relay). (c) Block diagram of a modern I/P transducer.

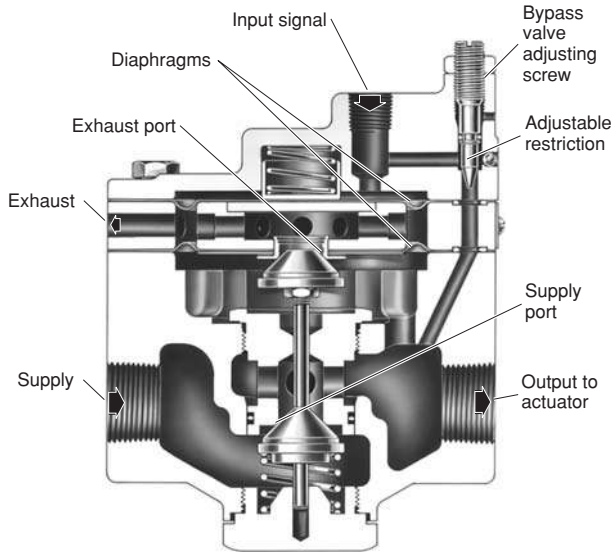


FIG. 8-88 Dead-band booster relay. (Courtesy Fisher Controls International LLC.)

Solenoid Valves The electric solenoid valve has two output states. When sufficient electric current is supplied to the coil, an internal armature moves against a spring to an extreme position. This motion causes an attached pneumatic or hydraulic valve to operate. When current is removed, the spring returns the armature and the attached solenoid valve to the deenergized position. An intermediate pilot stage is sometimes used when additional force is required to operate the main solenoid valve. Generally, solenoid valves are used to pressurize or vent the actuator casing for on/off control valve application and safety shutdown applications.

Trip Valves The trip valve is part of a system used where a specific valve action (i.e., fail up, fail down, or lock in last position) is required when pneumatic supply pressure to the control valve falls below a preset level. Trip systems are used primarily on springless piston actuators requiring fail-open or fail-closed action. An air storage or "volume" tank and a check valve are used with the trip valve to provide power to stroke the valve when supply pressure is lost. Trip valves are designed with hysteresis around the trip point to avoid instability when the trip pressure and the reset pressure settings are too close to the same value.

Limit Switches and Stem Position Transmitters Travel limit switches, position switches, and valve position transmitters are devices that detect the component's relative position, when mounted on the valve, actuator, damper, louver, or other throttling element. The switches are used to operate alarms, signal lights, relays, solenoid valves, or discrete inputs into the control system. The valve position transmitter generates a 4- to 20-mA output that is proportional to the position of the valve.

FIRE AND EXPLOSION PROTECTION

Electrical equipment and wiring methods can be sources of ignition in environments with combustible concentrations of gas, liquid, dust, fibers, or flyings. Most of the time it is possible to locate the electronic equipment away from these hazardous areas. However, where electric or electronic valve-mounted instruments must be used in areas where there is a hazard of fire or explosion, the equipment and installation must meet requirements for safety. Articles 500 through 504 of the National Electrical Code cover the definitions and requirements for electrical and electronic equipment used in the class I (flammable gases or vapors), divisions 1 and 2; class II (combustible dust), divisions 1 and 2; and class III (ignitable fibers or flyings), divisions 1 and 2. Division 1 locations are locations with hazardous concentrations of gases, vapors, or combustible dust under normal operating conditions;

hazardous concentration of gases, vapors, or combustible dust that occur frequently due to repair, maintenance, or leakage; or hazardous due to the presence of easily ignitable fibers or materials producing combustible flyings during handling, manufacturing, or use. Division 2 locations are locations that normally do not have ignitable concentrations of gases, vapors, or combustible dust. Division 2 locations might become hazardous through failure of ventilating equipment; adjacent proximity to a class I, division 1 location where ignitable concentrations of gases or vapors might occasionally exist; through dust accumulations on or in the vicinity of the electrical equipment sufficient to interfere with the safe dissipation of heat or by abnormal operation or failure of electrical equipment; or when easily ignitable fibers are stored or handled other than in the process of manufacture. An alternate method used for class I hazardous locations is the European "zone" method described in IEC 60079-10, "Electrical Apparatus for Explosive Gas Atmospheres." The zone designation for class I locations has been adapted by the NEC as an alternate method and is defined in Article 505 of the NEC.

Acceptable protection techniques for electrical and electronic valve accessories used in specific class and division locations include explosion-proof enclosures; intrinsically safe circuits; nonincendive circuits, equipment, and components; dust-ignition-proof enclosures; dusttight enclosures; purged and pressurized enclosures; oil immersion for current-interrupting contacts; and hermetically sealed equipment. Details of these techniques can be found in the *National Electrical Code Handbook*, available from the *National Fire Protection Association*.

Certified testing and approval for control valve devices used in hazardous locations is normally procured by the manufacturer of the device. The manufacturer typically goes to a third-party laboratory for testing and certification. Applicable approval standards are available from CSA, CENELEC, FM, SAA, and UL.

Environmental Enclosures Enclosures for valve accessories are sometimes required to provide protection from specific environmental conditions. The National Electrical Manufacturers Association (NEMA) provides descriptions and test methods for equipment used in specific environmental conditions in NEMA 250. IEC 60529, "Degrees of Protection Provided by Enclosures (IP Code)," describes the European system for classifying the degrees of protection provided by the enclosures of electrical equipment. Rain, windblown dust, hose-directed water, and external ice formation are examples of environmental conditions that are covered by these enclosure standards.

Of growing importance is the electronic control valve device's level of immunity to, and emission of, electromagnetic interference in the chemical valve environment. Electromagnetic compatibility (EMC) for control valve devices is presently mandatory in the European Community and is specified in International Electrotechnical Commission (IEC) 61326, "Electrical Equipment for Measurement Control and Laboratory Use—EMC Requirements." Test methods for EMC testing are found in the series IEC 61000-4, "EMC Compatibility (EMC), Testing and Measurement Techniques." Somewhat more stringent EMC guidelines are found in the German document NAMUR NE21, "Electromagnetic Compatibility of Industrial Process and Laboratory Control Equipment."

ADJUSTABLE-SPEED PUMPS

An alternative to throttling a process with a process control valve and a fixed-speed pump is by adjusting the speed of the process pump and not using a throttling control valve at all. Pump speed can be varied by using variable-speed prime movers such as turbines, motors with magnetic or hydraulic couplings, and electric motors. Each of these methods of modulating pump speed has its own strengths and weaknesses, but all offer energy savings and dynamic performance advantages over throttling with a control valve.

The centrifugal pump directly driven by a variable-speed electric motor is the most commonly used hardware combination for adjustable-speed pumping. The motor is operated by an electronic motor speed controller whose function is to generate the voltage or current waveform required by the motor to make the speed of the motor track the input command signal from the process controller.

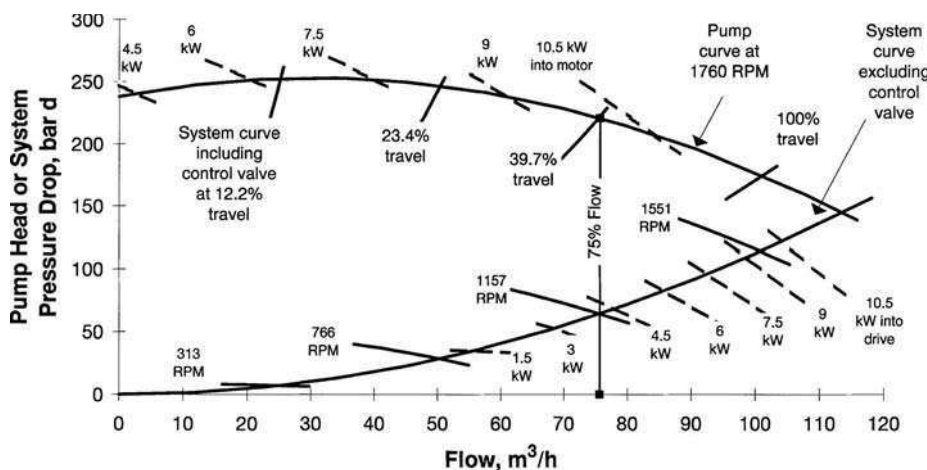


FIG. 8-89 Pressure, flow, and power for throttling a process using a control valve and a constant-speed pump compared to throttling with an adjustable-speed pump.

The most popular form of motor speed control for adjustable-speed pumping is the voltage-controlled pulse-width-modulated (PWM) frequency synthesizer and ac squirrel-cage induction motor combination. The flexibility of application of the PWM motor drive and its 90+ percent electrical efficiency along with the proven ruggedness of the traditional ac induction motor makes this combination popular.

From an energy consumption standpoint, the power required to maintain steady process flow with an adjustable-speed-pump system (three-phase PWM drive and a squirrel-cage induction motor driving a centrifugal pump on water) is less than that required with a conventional control valve and a fixed-speed pump. Figure 8-89 shows this to be the case for a system where 100 percent of the pressure loss is due to flow velocity losses. At 75 percent flow, the figure shows that using the constant-speed pump/control valve results in a 10.1-kW rate, while throttling with the adjustable-speed pump and not using a control valve results in a 4.1-kW rate. This trend of reduced energy consumption is true for the entire range of flows, although amounts vary.

From a dynamic response standpoint, the electronic adjustable-speed pump has a dynamic characteristic that is more suitable in process control applications than those characteristics of control valves. The small amplitude response of an adjustable-speed pump does not contain the dead band or the dead time commonly found in the small amplitude response of the control valve. Nonlinearities associated with friction in the valve and discontinuities in the pneumatic portion of the control valve instrumentation are not present with electronic variable-speed drive technology. As a result, process control with the adjustable-speed pump does not exhibit limit cycles, problems related to low controller gain, and generally degraded process loop performance caused by control valve nonlinearities.

Unlike the control valve, the centrifugal pump has poor or nonexistent shutoff capability. A flow check valve or an automated on/off valve may be required to achieve shutoff requirements. This requirement may be met by automating an existing isolation valve in retrofit applications.

REGULATORS

A regulator is a compact device that maintains the process variable at a specific value in spite of disturbances in load flow. It combines the functions of the measurement sensor, controller, and final control element into one self-contained device. Regulators are available to control pressure, differential pressure, temperature, flow, liquid level, and other basic process variables. They are used to control the differential across a filter press, heat exchanger, or orifice plate. Regulators are used for monitoring pressure variables for redundancy, flow check, and liquid surge relief.

Regulators may be used in gas blanketing systems to maintain a protective environment above any liquid stored in a tank or vessel as the liquid is pumped out. When the temperature of the vessel is suddenly cooled, the regulator maintains the tank pressure and protects the walls of the tank from possible collapse. Regulators are known for their fast dynamic response. The absence of time delay that often comes with more sophisticated control systems makes the regulator useful in applications requiring fast corrective action.

Regulators are designed to operate on the process pressures in the pipeline without any other sources of energy. Upstream and downstream pressures are used to supply and exhaust the regulator. Exhausting is connected back to the downstream piping so that no contamination or leakage to the external environment occurs. This makes regulators useful in remote locations where power is not available or where external venting is not allowed.

The regulator is limited to operating on processes with clean, non-slurry process fluids. The small orifice and valve assemblies contained in the regulator can plug and malfunction if the process fluid that operates the regulator is not sufficiently clean.

Regulators are normally not suited to systems that require constant set-point adjustment. Although regulators are available with capability to respond to remote set-point adjustment, this feature adds complexity to the regulator and may be better addressed by a control-valve-based system. In the simplest of regulators, tuning of the regulator for best control is accomplished by changing a spring, an orifice, or a nozzle.

Self-Operated Regulators Self-operated regulators are the simplest form of regulator. This regulator (see Fig. 8-90a) is composed of a main throttling valve, a diaphragm or piston to sense pressure, and a spring. The self-contained regulator is completely operated by the process fluid, and no outside control lines or pilot stage is used. In general, self-operated regulators are simple in construction, are easy to operate and maintain, and are usually stable devices. Except for some of the pitot-tube types, self-operated regulators have very good dynamic response characteristics. This is so because any change in the controlled variable registers directly and immediately upon the main diaphragm to produce a quick response to the disturbance.

The disadvantage of the self-operated regulator is that it is not generally capable of maintaining a set point as load flow is increased. Because of the proportional nature of the spring and diaphragm-throttling effect, offset from set point occurs in the controlled variable as flow increases. Figure 8-91 shows a typical regulation curve for the self-contained regulator.

Reduced set-point offset with increasing load flow can be achieved by adding a pitot tube to the self-operated regulator. The

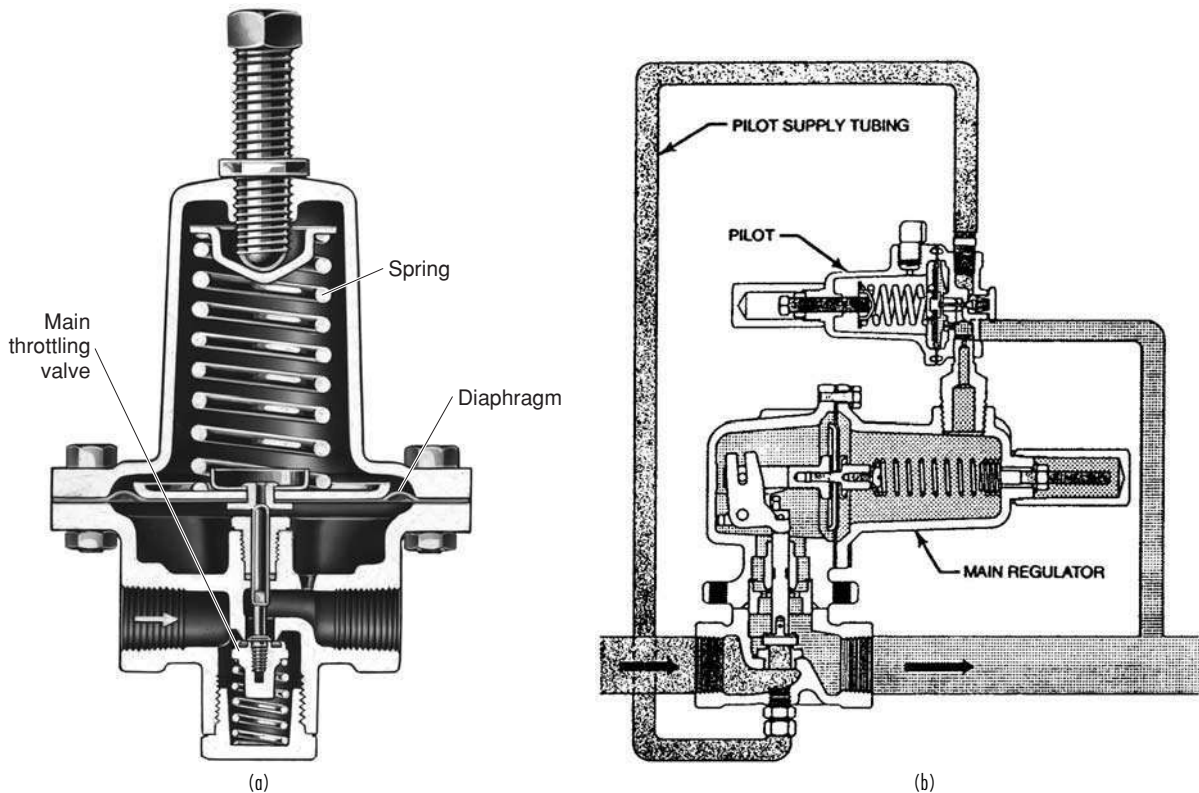


FIG. 8-90 Regulators. (a) Self-operated. (b) Pilot-operated. (Courtesy Fisher Controls International LLC.)

tube is positioned somewhere near the vena contracta of the main regulator valve. As flow through the valve increases, the measured feedback pressure from the pitot tube drops below the control pressure. This causes the main valve to open or boost more than it would if the static value of control pressure were acting on the diaphragm. The resultant effect keeps the control pressure closer to the set point and thus prevents a large drop in process pressure during high-load-flow conditions. Figure 8-91 shows the improvement that the pitot-

tube regulator provides over the regulator without the tube. A side effect of adding a pitot-tube method is that the response of the regulator can be slowed due to the restriction provided by the pitot tube.

Pilot-Operated Regulators Another category of regulators uses a pilot stage to provide the load pressure on the main diaphragm. This pilot is a regulator itself that has the ability to multiply a small change in downstream pressure into a large change in pressure applied to the regulator diaphragm. Due to this high-gain feature, pilot-operated

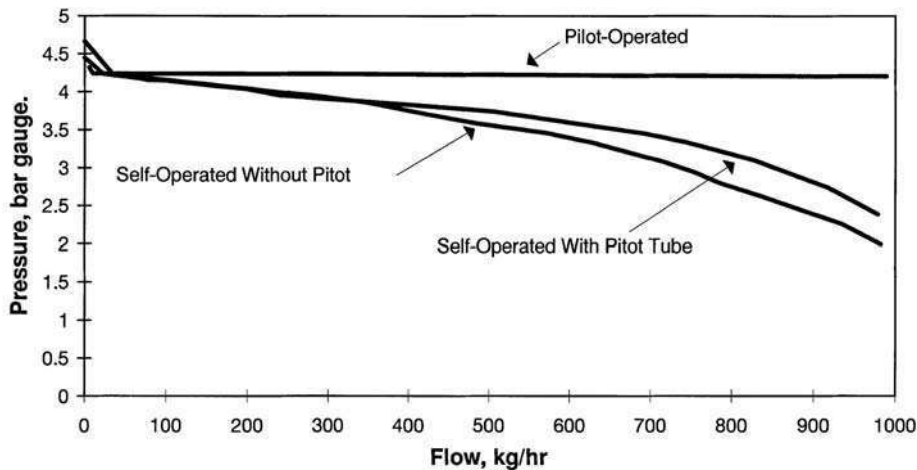


FIG. 8-91 Pressure regulation curves for three regulator types.

regulators can achieve a dramatic improvement in steady-state accuracy over that achieved with a self-operated regulator. Figure 8-91 shows for regulation at high flows the pilot-operated regulator is the best of the three regulators shown.

The main limitation of the pilot-operated regulator is stability. When the gain in the pilot amplifier is raised too much, the loop can become unstable and oscillate or hunt. The two-path pilot regulator (see Fig. 8-90b) is also available. This regulator combines the effects of self-operated and the pilot-operated styles and mathematically produces the equivalent of proportional plus reset control of the process pressure.

Overpressure Protection Figure 8-91 shows a characteristic rise in control pressure that occurs at low or zero flow. This lockup

tail is due to the effects of imperfect plug and seat alignment and the elastomeric effects of the main throttle valve. If, for some reason, the main throttle valve fails to completely shut off, or if the valve shuts off but the control pressure continues to rise for other reasons, the lockup tail could get very large, and the control pressure could rise to extremely high values. Damage to the regulator or the downstream pressure volume could occur.

To avoid this situation, some regulators are designed with a built-in overpressure relief mechanism. Overpressure relief circuits usually are composed of a spring-opposed diaphragm and valve assembly that vents the downstream piping when the control pressure rises above the set-point pressure.

PROCESS CONTROL AND PLANT SAFETY

GENERAL REFERENCE: *Guidelines for Safe Automation of Chemical Processes*, AIChE Center for Chemical Process Safety, New York, 1993.

Accidents in chemical plants make headline news, especially when there is loss of life or the general public is affected in even the slightest way. This increases the public's concern and may lead to government action. The terms *hazard* and *risk* are defined as follows:

- **Hazard.** A potential source of harm to people, property, or the environment.
- **Risk.** Possibility of injury, loss, or an environmental accident created by a hazard.

Safety is the freedom from hazards and thus the absence of any associated risks. Unfortunately, absolute safety cannot be realized.

The design and implementation of safety systems must be undertaken with a view to two issues:

- **Regulatory.** The safety system must be consistent with all applicable codes and standards as well as "generally accepted good engineering practices."
- **Technical.** Just meeting all applicable regulations and "following the crowd" do not relieve a company of its responsibilities. The safety system must work.

The regulatory environment will continue to change. As of this writing, the key regulatory instrument is OSHA 29 CFR 1910.119, "Process Safety Management of Highly Hazardous Chemicals," which pertains to process safety management within plants in which certain chemicals are present.

In addition to government regulation, industry groups and professional societies are producing documents ranging from standards to guidelines. Two applicable standards are IEC 61508, "Functional Safety of Electrical/Electronic/Programmable Electronic Safety-related Systems," and ANSI/ISA S84.01, "Application of Safety Instrumented Systems for the Process Industries." *Guidelines for Safe Automation of Chemical Processes* from the American Institute of Chemical Engineers' Center for Chemical Process Safety (1993) provides comprehensive coverage of the various aspects of safety; and although short on specifics, it is very useful to operating companies developing their own specific safety practices (i.e., it does not tell you what to do, but it helps you decide what is proper for your plant).

The ultimate responsibility for safety rests with the operating company; OSHA 1910.119 is clear on this. Each company is expected to develop (and enforce) its own practices in the design, installation, testing, and maintenance of safety systems. Fortunately, some companies make these documents public. Monsanto's *Safety System Design Practices* was published in its entirety in the proceedings of the International Symposium and Workshop on Safe Chemical Process Automation, Houston, Texas, September 27–29, 1994 (available from the American Institute of Chemical Engineers' Center for Chemical Process Safety).

ROLE OF AUTOMATION IN PLANT SAFETY

As microprocessor-based controls displaced hardwired electronic and pneumatic controls, the impact on plant safety has definitely been positive. When automated procedures replace manual procedures for routine operations, the probability of human errors leading to hazardous situations is lowered. The enhanced capability for presenting information to the process operators in a timely manner and in the most meaningful form increases the operator's awareness of current conditions in the process. Process operators are expected to exercise due diligence in the supervision of the process, and timely recognition of an abnormal situation reduces the likelihood that the situation will progress to the hazardous state. Figure 8-92 depicts the layers of safety protection in a typical chemical plant. Although microprocessor-based process controls enhance plant safety, their primary objective is efficient process operation. Manual operations are automated to reduce variability, to minimize the time required, to increase productivity, and so on. Remaining competitive in the world market demands that the plant be operated in the best manner possible, and microprocessor-based process controls provide numerous functions that make this possible. Safety is never compromised in the effort to increase competitiveness, but enhanced safety is a by-product of the process control function and is not a primary objective. By attempting to maintain process conditions at or near their design values, the process controls also attempt to prevent abnormal conditions from developing within the process.

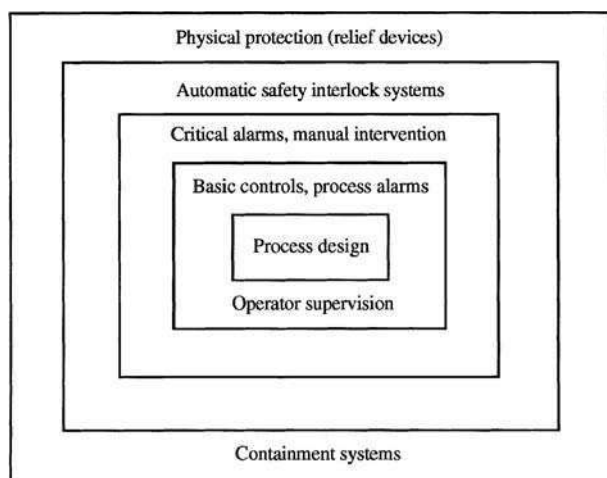


FIG. 8-92 Layers of safety protection in chemical plants.

Although process controls can be viewed as a protective layer, this is really a by-product and not the primary function. Where the objective of a function is specifically to reduce risk, the implementation is normally not within the process controls. Instead, the implementation is within a separate system specifically provided to reduce risk. This system is generally referred to as the safety interlock system.

As safety begins with the process design, an inherently safe process is the objective of modern plant designs. When this cannot be achieved, process hazards of varying severity will exist. Where these hazards put plant workers and/or the general public at risk, some form of protective system is required. Process safety management addresses the various issues, ranging from assessment of the process hazard to ensuring the integrity of the protective equipment installed to cope with the hazard. When the protective system is an automatic action, it is incorporated into the safety interlock system, not within the process controls.

INTEGRITY OF PROCESS CONTROL SYSTEMS

Ensuring the integrity of process controls involves hardware issues, software issues, and human issues. Of these, the hardware issues are usually the easiest to assess and the software issues the most difficult.

The hardware issues are addressed by providing various degrees of redundancy, by providing multiple sources of power and/or an uninterruptible power supply, and the like. The manufacturers of process controls provide a variety of configuration options. Where the process is inherently safe and infrequent shutdowns can be tolerated, nonredundant configurations are acceptable. For more-demanding situations, an appropriate requirement might be that no single component failure be able to render the process control system inoperable. For the very critical situations, triple-redundant controls with voting logic might be appropriate. The difficulty lies in assessing what is required for a given process.

Another difficulty lies in assessing the potential for human errors. If redundancy is accompanied with increased complexity, the resulting increased potential for human errors must be taken into consideration. Redundant systems require maintenance procedures that can correct problems in one part of the system while the remainder of the system is in full operation. When maintenance is conducted in such situations, the consequences of human errors can be rather unpleasant.

The use of programmable systems for process control presents some possibilities for failures that do not exist in hardwired electro-mechanical implementations. Probably of greatest concern are latent defects or "bugs" in the software, either the software provided by the supplier or the software developed by the user. The source of this problem is very simple. There is no methodology available that can be applied to obtain absolute assurance that a given set of software is completely free of defects. Increased confidence in a set of software is achieved via extensive testing, but no amount of testing results in absolute assurance that there are no defects. This is especially true of real-time systems, where the software can easily be exposed to a sequence of events that was not anticipated. Just because the software performs correctly for each event individually does not mean that it will perform correctly when two (or more) events occur at nearly the same time. This is further complicated by the fact that the defect may not be in the programming; it may be in how the software was designed to respond to the events.

The testing of any collection of software is made more difficult as the complexity of the software increases. Software for process control has become progressively complex, mainly because the requirements have become progressively demanding. To remain competitive in the world market, processes must be operated at higher production rates, within narrower operating ranges, closer to equipment limits, and so on. Demanding applications require sophisticated control strategies, which translate to more-complex software. Even with the best efforts of both supplier and user, complex software systems are unlikely to be completely free of defects.

CONSIDERATIONS IN IMPLEMENTATION OF SAFETY INTERLOCK SYSTEMS

Where hazardous conditions can develop within a process, a protective system of some type must be provided. Sometimes this is in the form of process hardware such as pressure relief devices. However, sometimes logic must be provided for the specific purpose of taking the process to a state where the hazardous condition cannot exist. The term *safety interlock system* is normally used to designate such logic.

The purpose of the logic within the safety interlock system is very different from that of the logic within the process controls. Fortunately, the logic within the safety interlock system is normally much simpler than the logic within the process controls. This simplicity means that a hardwired implementation of the safety interlock system is usually an option. Should a programmable implementation be chosen, this simplicity means that latent defects in the software are less likely to be present. Most safety systems only have to do simple things, but they must do them very, very well.

The difference in the nature of process controls and safety interlock systems leads to the conclusion that these two should be physically separated (see Fig. 8-92). That is, safety interlocks should not be piggybacked onto a process control system. Instead, the safety interlocks should be provided by equipment, either hardwired or programmable, that is dedicated to the safety functions. As the process controls become more complex, faults are more likely. Separation means that faults within the process controls have no consequences in the safety interlock system.

Modifications to the process controls are more frequent than modifications to the safety interlock system. Therefore, physically separating the safety interlock system from the process controls provides the following benefits:

1. The possibility of a change to the process controls leading to an unintentional change to the safety interlock system is eliminated.
2. The possibility of a human error in the maintenance of the process controls having consequences for the safety interlock system is eliminated.
3. Management of change is simplified.
4. Administrative procedures for software version control are more manageable.

Separation also applies to the measurement devices and actuators. Although the traditional point of reference for safety interlock systems is a hardwired implementation, a programmed implementation is an alternative. The potential for latent defects in software implementation is a definite concern. Another concern is that solid-state components are not guaranteed to fail to the safe state. The former is addressed by extensive testing; the latter is addressed by manufacturer-supplied and/or user-supplied diagnostics that are routinely executed by the processor within the safety interlock system. Although issues must be addressed in programmable implementations, the hardwired implementations are not perfect either.

Where a programmed implementation is deemed to be acceptable, the choice is usually a programmable logic controller that is dedicated to the safety function. PLCs are programmed with the traditional relay ladder diagrams used for hardwired implementations. The facilities for developing, testing, and troubleshooting PLCs are excellent. However, for PLCs used in safety interlock systems, administrative procedures must be developed and implemented to address the following issues:

1. Version controls for the PLC program must be implemented and rigidly enforced. Revisions to the program must be reviewed in detail and thoroughly tested before implementation in the PLC. The various versions must be clearly identified so that there can be no doubt as to what logic is provided by each version of the program.
2. The version of the program that is currently being executed by the PLC must be known with absolute certainty. It must be impossible for a revised version of the program undergoing testing to be downloaded to the PLC.

Constant vigilance is required to prevent lapses in such administrative procedures.

INTERLOCKS

An interlock is a protective response initiated on the detection of a process hazard. The interlock system consists of the measurement devices, logic solvers, and final control elements that recognize the hazard and initiate an appropriate response. Most interlocks consist of one or more logic conditions that detect out-of-limit process conditions and respond by driving the final control elements to the safe states. For example, one must specify that a valve fails open or fails closed.

The potential that the logic within the interlock could contain a defect or bug is a strong incentive to keep it simple. Within process plants, most interlocks are implemented with discrete logic, which means either hardwired electromechanical devices or programmable logic controllers.

The discrete logic within process plants can be broadly classified as follows:

1. *Safety interlocks.* These are designed to protect the public, the plant personnel, and possibly the plant equipment from process hazards. These are implemented within the safety interlock system.

2. *Process actions.* These are designed to prevent process conditions that would unduly stress equipment (perhaps leading to minor damage), lead to off-specification product, and so on. Basically, the process actions address hazards whose consequences essentially lead to a monetary loss, possibly even a short plant shutdown. Although sometimes referred to as interlocks, process actions address situations that are not deemed to be process hazards.

Implementation of process actions within process control systems is perfectly acceptable. Furthermore, it is also permissible (and probably advisable) for responsible operations personnel to be authorized to bypass or ignore a process action. Safety interlocks must be implemented within the separate safety interlock system. Bypassing or ignoring safety interlocks by operations personnel is simply not permitted. When this is necessary for actions such as verifying that the interlock continues to be functional, such situations must be infrequent and incorporated into the design of the interlock.

Safety interlocks are assigned to categories that reflect the severity of the consequences, should the interlock fail to perform as intended. The specific categories used within a company are completely at the discretion of the company. However, most companies use categories that distinguish among the following:

1. *Hazards that pose a risk to the public.* Complete redundancy is normally required.

2. *Hazards that could lead to injury of company personnel.* Partial redundancy is often required (e.g., redundant measurements but not redundant logic).

3. *Hazards that could result in major equipment damage and consequently lengthy plant downtime.* No redundancy is normally required for these, although redundancy is always an option.

Situations resulting in minor equipment damage that can be quickly repaired do not generally require a safety interlock; however, a process action might be appropriate.

A process hazards analysis is intended to identify the safety interlocks required for a process and to provide the following for each:

1. The hazard that is to be addressed by the safety interlock
2. The classification of the safety interlock
3. The logic for the safety interlock, including inputs from measurement devices and outputs to actuators

The process hazards analysis is conducted by an experienced, multi-disciplinary team that examines the process design, plant equipment, operating procedures, and so on, using techniques such as hazard and operability studies (HAZOP), failure mode and effect analysis (FMEA), and others. The process hazards analysis recommends appropriate measures to reduce the risk, including (but not limited to) the safety interlocks to be implemented in the safety interlock system.

Diversity is recognized as a useful approach to reduce the number of defects. The team that conducts the process hazards analysis does not implement the safety interlocks but provides the specifications for the safety interlocks to another organization for implementation. This organization reviews the specifications for each safety interlock, seeking clarifications as necessary from the process hazards analysis team and bringing any perceived deficiencies to the attention of the process hazards analysis team.

Diversity can be used to further advantage in redundant configurations. Where redundant measurement devices are required, different technology can be used for each. Where redundant logic is required, one can be programmed and one hardwired. Reliability of the interlock systems has two aspects:

1. It must react, should the hazard arise.
2. It must not react when there is no hazard.

Emergency shutdowns often pose risks in themselves, and therefore they should be undertaken only when truly appropriate. The need to avoid extraneous shutdowns is not motivated by a desire simply to avoid disruption in production operations.

Although safety interlocks can inappropriately initiate shutdowns, the process actions are usually the major source of problems. It is possible to configure so many process actions that it is not possible to operate the plant.

TESTING

As part of the detailed design of each safety interlock, written test procedures must be developed for the following purposes:

1. Ensure that the initial implementation complies with the requirements defined by the process hazards analysis team.

2. Ensure that the interlock (hardware, software, and I/O) continues to function as designed. The design must also determine the time interval over which this must be done. Often these tests must be done with the plant in full operation.

The former is the responsibility of the implementation team and is required for the initial implementation and following any modification to the interlock. The latter is the responsibility of plant maintenance, with plant management responsible for seeing that it is done at the specified interval of time.

Execution of each test must be documented, showing when it was done, by whom, and the results. Failures must be analyzed for possible changes in the design or implementation of the interlock. These tests must encompass the complete interlock system, from the measurement devices through the final control elements. Merely simulating inputs and checking the outputs is not sufficient. The tests must duplicate the process conditions and operating environments as closely as possible. The measurement devices and final control elements are exposed to process and ambient conditions and thus are usually the most likely to fail. Valves that remain in the same position for extended periods may stick in that position and not operate when needed. The easiest component to test is the logic; however, this is the least likely to fail.

SECTION 9

Process
Economics

PERRY'S CHEMICAL ENGINEERS' HANDBOOK

8TH EDITION



JAMES R. COUPER
DARRYL W. HERTZ
F. LEE SMITH

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Process Economics

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GLOSSARY

Nomenclature and Units

Symbol	Definition	Units	Symbol	Definition	Units
A_1	Annual conversion expense at production rate 1	\$	LIFO	Last in, last out (inventory)	lb
A_{TC}	Annual capital outlay	\$	M	Annual raw material expense	\$
B	Constant	Dimensionless	MACRS	Modified Accelerated Cost Recovery System	
C	Cost of equipment	\$	m	Number of interest periods per year	Varies
C_B	Base cost of carbon steel exchanger	\$	m, n, p, q	constants or exponent	Dimensionless
CE	Chemical Engineering cost index	Dimensionless	N	Annual labor requirements	Operators per shift per year
$(C_{FC})_{BL}$	Battery-limits fixed capital investment	\$	n	Number of years, depreciation	Years
$(C_{EQ})_{DEL}$	Delivered equipment cost	\$	P	Principal, present value, present worth	\$
C_{HE}	Purchased equipment cost, heat exchanger	\$	PC	Personal computer	
C_L	Cost of labor	\$	POP	Payout period (no interest)	Years
COE	Cash operating expenses	\$	POP + I	Payout period plus interest	Years
C_P	Equipment cost in base year	\$	Q	Energy transferred	Btu/h
cP	Viscosity	cP	$\dot{R}_1, \dot{R}_2$	Annual production rates	lb/yr
D	Depreciation	\$	S	Salvage value or equipment capacity	Various
DCFRROR	Discounted cash flow rate of return	%	SL	Straight-line depreciation	
EBIT	Earnings before interest and taxes	\$	Sp gr	Specific gravity	Dimensionless
e	Naperian logarithm base	2.718	TE	Total expenses	\$
F	Future value, future worth, future amount	\$	T_c	Combined incremental tax rate	%
F	Heat exchanger efficiency factor	Dimensionless	T_f	Incremental federal income tax rate	%
F_B	Heat exchanger design type	Dimensionless	T_s	Incremental state income tax rate	%
FCI	Fixed capital investment	\$	U	Annual utility expenses	\$
FE	Fixed expenses	\$	UAC	Uniform annual cost	\$
FEL	Front-end loading		U_D	Overall heat-transfer coefficient	Btu/(h·ft ² ·F)
FIFO	First in, first out (inventory)	lb	V_e	Asset value at end of year	\$
FOB	Free on board		V_i	Asset value at beginning of year	\$
F_M	Material of construction cost factor	Dimensionless	VE	Variable expenses	\$
F_P	Design pressure cost factor	Dimensionless	VIP	Value-improving practice	
f_1, f_2, f_3	Inflation factors for years 1, 2, and 3	Dimensionless	X	Plant capacity	tons/day
f	Declining-balance factor	Dimensionless	Y	Operating labor	operator-hour/ton per processing step
I	Investment	\$			
IRS	Internal Revenue Service				
i	Nominal interest	%			
i_{eff}	Effective interest	%			
K	Factor for cost index	Dimensionless			

GENERAL REFERENCES: Allen, D. H., *Economic Evaluation of Projects*, 3d ed., Institution of Chemical Engineers, Rugby, England, 1991. Baasel, W. D., *Chemical Engineering Plant Design*, 2d ed., Van Nostrand Reinhold, New York, 1989. Brown, T. R., *Hydrocarbon Processing*, October 2000, pp. 93–100. Canada, J. R., and J. A. White, *Capital Investment Decision: Analysis for Management and Engineering*, 2d ed., Prentice-Hall, Englewood Cliffs, N.J., 1980. Couper, J. R., and W. H. Rader, *Applied Finance and Economic Analysis for Scientists and Engineers*, Van Nostrand Reinhold, New York, 1986. Couper, J. R., and O. T. Beasley, *The Chemical Process Industries: Function and Economics*, Dekker, New York, 2001. Couper, J. R., *Process Engineering Economics*, Dekker, New York, 2003. Garrett, D. E., *Chemical Engineering Economics*, Van Nostrand Reinhold, New York, 1989. Grant, E. L., and W. G. Ireson, *Engineering Economy*, 2d ed., Wiley, New York, 1950. Grant, E. L., W. G. Ireson, and R. S. Leavenworth, *Engineering Economy*, 8th ed., Wiley, New York, 1990. Hackney, J. W., and K. K. Humphreys (eds.), *Control and Management of Capital Projects*, 2d ed., McGraw-Hill, New York, 1992. Hill, D. A., and L. E. Rockley, *Secrets of Successful Financial Management*, Heinemann, London, 1990. Holland, F. A., F. A. Watson, and J. E. Wilkerson, *Introduction to Process Economics*, 2d ed., Wiley, London, 1983. K. K. Humphreys, F. C. Jelen, and J. H. Black (eds.), *Cost and*

Optimization Engineering, 3d ed., McGraw-Hill, New York, 1991. A *Guide to Capital Cost Estimation*, 3d ed., Institution of Chemical Engineers, Rugby, England, 1990. Kharbanda, O. P., and E. A. Stallworthy, *Capital Cost Estimating in the Process Industries*, 2d ed., Butterworth-Heinemann, London, 1988. *How to Read an Annual Report*, Merrill Lynch, New York, 1997. Nickerson, C. B., *Accounting Handbook for Non Accountants*, 2d ed., CBI Publishing, Boston, 1979. Ostwald, P. F., *Engineering Cost Estimating*, 3d ed., Prentice-Hall, Englewood Cliffs, N.J., 1991. Park, W. R., and D. E. Jackson, *Cost Engineering Analysis*, 2d ed., Wiley, New York, 1984. Peters, M. S., and K. D. Timmerhaus, *Plant Design and Economics for Chemical Engineers*, 6th ed., McGraw-Hill, New York, 2003. Popper, H. (ed.), *Modern Cost Estimating Techniques*, McGraw-Hill, New York, 1970. Rose, L. M., *Engineering Investment Decisions: Planning under Uncertainty*, Elsevier, Amsterdam, 1976. Thorne, H. C., and J. B. Weaver (eds.), *Investment Appraisal for Chemical Engineers*, American Institute of Chemical Engineers, New York, 1991. Ulrich, G., and P. T. Vasudevan, *Chemical Engineering Process Design and Economics*, CRC Press, Boca Raton, Fla., 2004. Valle-Riestra, J. F., *Project Evaluation in the Chemical Process Industries*, McGraw-Hill, New York, 1983. Wells, G. L., *Process Engineering with Economic Objectives*, Wiley, New York, 1973. Woods, D. R., *Process Design and Engineering*, Prentice-Hall, Englewood Cliffs, N.J., 1993.

GENERAL COMMENTS

One of the most confusing aspects of process engineering economics is the nomenclature used by various authors and companies. In this part of Sec. 9, generic, descriptive terms have been used. Further, an attempt has been made to bring together most of the methods currently in use for project evaluation and to present them in such a way as to make them amenable to modern computational techniques. Most of the calculations can be performed on handheld calculators equipped with scientific function keys. For calculations requiring greater sophistication than that of handheld calculators, algorithms may be solved by using such programs as MATHCAD, TKSOLVER, etc. Spreadsheets are also used whenever the solution to a problem lends itself to this technique.

The nomenclature in process economics has been developed by accountants, engineers, and others such that there is no one correct set of nomenclature. Often it seems confusing, but one must question

what is meant by a certain term since companies have adopted their own language. A glossary of terms is included at the end of this section to assist the reader in understanding the nomenclature. Further, abbreviations of terms such as DCFRR (discounted cash flow rate of return) are used to reduce the wordiness. The number of letters and numbers used to define a variable has been limited to five. The parentheses are removed whenever the letter group is used to define a variable for a computer. Also, a general symbol is defined for a type variable and is modified by mnemonic subscript, e.g., an annual cash quantity, annual capital outlay A_{TC} , \$/year. Wherever a term like this is introduced, it is defined in the text.

It is impossible to allow for all possible variations of equation requirements, but it is hoped that the nomenclature presented will prove adequate for most purposes and will be capable of logical extension to other more specialized requirements.

ACCOUNTING AND FINANCIAL CONSIDERATIONS

PRINCIPLES OF ACCOUNTING

Accounting has been defined as the art of recording business transactions in a systematic manner. It is the language of business and is used to communicate financial information. Conventions that govern accounting are fairly simple, but their application is complex. In this section, the basic principles are illustrated by a simple example and applied to analyzing a company report. The fair allocation of costs requires considerable technical knowledge of operations, so a close liaison between process engineers and accountants in a company is desirable.

In simplest terms, assets that are the economic resources of a company are balanced against equities that are claims against the firm. In equation form,

$$\text{Assets} = \text{Equities}$$

or

$$\text{Assets} = \text{Liabilities} + \text{Owners' Equity}$$

This dual aspect has led to the *double-entry* bookkeeping system in use today. Any transaction that takes place causes changes in the accounting equation. An increase in assets must be accompanied by one of the following:

- An increase in liabilities
- An increase in stockholders' equity
- An increase in assets

A change in one part of the equation due to an economic transaction must be accompanied by an equal change in another place—therefore, the term *double-entry* bookkeeping. On a page of an account, the left-hand side is designated the *debit* side and the right-hand side is the *credit* side. This convention holds regardless of the type of account. Therefore, for every economic transaction, there is an entry on the debit side balanced by the same entry on the credit side.

All transactions in their original form (receipts and invoices) are recorded chronologically in a *journal*. The date of the transaction together with an account title and a brief description of the transaction is entered. Table 9-1 is an example of a typical journal page for a company. Journal entries are transferred to a *ledger* in a process called *posting*. Separate ledger accounts, such as a revenue account, expense account, liability account, or asset account, may be set up for each major transaction. Table 9-2 shows an example of a typical ledger page. The number of ledger accounts depends on the information that management needs to make decisions. Periodically, perhaps on a monthly basis but certainly on a yearly basis, the ledger sheets are closed and balanced. The ledger sheets are then intermediate documents between journal records and balance sheets, income statements, and retained earnings statements, and they provide information for management and various government reports. For example, a consolidated income statement can be prepared for the ledger, revenue, and expense accounts. In like manner, the asset and liability accounts provide information for balance sheets.

TABLE 9-1 Typical Journal Page

Date	Explanation	LP	Debit	Credit
200X				
Mar 1	Cash	1	\$95,000	
	J. Jones, Capital	2		\$95,000
Mar 4	Property	4	5,000	
	Cash	1		3,000
	Mortgage	3		2,000
Mar 11	Remodeling Bldg.	5	7,800	
	Cash	1		7,800
Mar 13	Equipment	6	62,300	
	Cash	1		10,000
	Note Payable	3		52,300
Apr 4	To J. Jones	2	2,500	
	Cash	1		2,500

SOURCE: J. R. Couper, *Process Engineering Economics*, Dekker, New York, 2003. By permission of Taylor & Francis Books, Inc., Boca Raton, Fla.

FINANCIAL STATEMENTS

A basic knowledge of accounting and financial statements is necessary for a chemical professional to be able to analyze a firm's operation and to communicate with accountants, financial personnel, and managers. Financial reports of a company are important sources of information used by management, owners, creditors, investment bankers, and financial analysts. All publicly held companies are required to submit annual reports to the Securities and Exchange Commission. As with any field a certain basic nomenclature is used to be able to understand the financial operation of a company. It should be emphasized that companies may also have their own internal nomenclature, but some terms are universally accepted. In this section, the common terminology is used.

A financial report contains two important documents—the *balance sheet* and the *income statement*. Two other documents that appear in the financial report are the *accumulated retained earnings* and the *changes in working capital*. All these documents are discussed in the following sections using a fictitious company.

Balance Sheet The balance sheet represents an accounting view of the financial status of a company on a particular date. Table 9-3 is an example of a balance sheet for a company. The date frequently used by corporations is December 31 of any given year, although some companies are now using June 30 or September 30 as the closing date. It is as if the company's operation were frozen in time on that date. The term *consolidated* means that all the balance sheet and income statement data include information from the parent as well as subsidiary operations. The balance sheet consists of two parts: *assets* are the items that the company owns, and *liabilities and stockholders' equity* are what the

company owes to creditors and stockholders. Although the balance sheet has two sides, it is not part of the double-entry accounting system. The balance sheet is not an account but a statement of claims against company assets on the date of the reporting period. The claims are the creditors and the stockholders. Therefore, the total assets must equal the total liabilities plus the stockholders' equity.

Assets are classified as current, fixed, or intangibles. *Current assets* include cash, cash equivalents, marketable securities, accounts receivable, inventories, and prepaid expenses. Cash and cash equivalents are those items that can be easily converted to cash. Marketable securities are securities that a company holds that also may be converted to cash. Accounts receivable are the amounts due a company from customers from material that has been delivered but has not been collected as yet. Customers are given 30, 60, or 90 days in which to pay; however, some customers fail to pay bills on time or may not be able to pay at all. An allowance is made for doubtful accounts. The amount is deducted from the accounts receivables. Inventories include the cost of raw materials, goods in process, and product on hand. Prepaid expenses include insurance premiums paid, charges for leased equipment, and charges for advertising that are paid prior to the receipt of the benefit from these items. The sum of all the above items is the *total current assets*. The term *current* refers to the fact that these assets are easily converted within a year, or more likely in a shorter time, say, 90 days.

Fixed assets are items that have a relatively long life such as land, buildings, and manufacturing equipment. The sum of these items is the *total property, plant, and equipment*. From this total, *accumulated depreciation* is subtracted and the result is *net property and equipment*. Last, an item referred to as *intangibles* includes a variety of items such as patents, licenses, intellectual capital, and goodwill. Intangibles are difficult to evaluate since they have no physical existence; e.g., goodwill is the value of the company's name and reputation. The sum of the *total current assets*, *net property*, and *intangibles* is the *total assets*.

Liabilities are the obligations that the company owes to creditors and stockholders. *Current liabilities* are obligations that come due within a year and include *accounts payable* (money owed to creditors for goods and services), *notes payable* (money owed to banks, corporations, or other lenders), *accrued expenses* (salaries and wages to employees, interest on borrowed funds, fees due to professionals, etc.), *income taxes payable*, *current part of long-term debt*, and *other current liabilities* due within the year.

Long-term liabilities are the amounts due after 1 year from date of the financial report. They include *deferred income taxes* that a company is permitted to postpone due to accelerated depreciation to encourage investment, (but they must be paid sometime in the future) and *bonds and notes* that do not have to be paid within the year but at some later date. The sum of the *current* and *long-term liabilities* is the *total liabilities*.

TABLE 9-2 Typical Ledger Page

Cash: Account 01						
200X						
Mar 1 Capital	J-1	\$95,000	Mar 1	Property	J-1	\$3,000
			Mar 11	Remodeling	J-1	7,800
			Mar 13	Equipment	J-1	10,000
			Apr 4	J. Jones	J-1	2,500
Capital: Account 02						
Apr 4 Cash to J. Jones	J-1	\$2,500	Mar 1	Capital	J-1	\$95,000
Accounts Payable: Account 03						
			Mar 4	Mortgage	J-1	\$2,000
			Mar 13	Note Payable	J-1	52,300
Property and Building: Account 04						
Mar 4	J-1	\$5,000				
Mar 11	J-1	7,800				
Equipment: Account 05						
Mar 13	J-1	\$62,300				

SOURCE: J. R. Couper, *Process Engineering Economics*, Dekker, New York, 2003. By permission of Taylor & Francis Books, Inc., Boca Raton, Fla.

9-6 PROCESS ECONOMICS

TABLE 9-3 Consolidated Balance Sheet^a (December 31)

Assets	2005	2004
Current assets		
Cash	\$63,000	\$51,000
Marketable securities	41,000	39,000
Accounts receivable ^b	135,000	126,000
Inventories	149,000	153,000
Prepaid expenses	3,200	2,500
Total current assets	\$391,200	\$371,500
Fixed assets		
Land	35,000	35,000
Buildings	101,000	97,500
Machinery	278,000	221,000
Office equipment	24,000	19,000
Total fixed assets	\$438,000	\$372,500
Less accumulated depreciation	128,000	102,000
Net fixed assets	\$310,000	\$270,500
Intangibles	4,500	4,500
Total assets	\$705,700	\$646,500
Liabilities	2005	2004
Current liabilities		
Accounts payable	\$92,300	\$81,300
Notes payable	67,500	59,500
Accrued expenses payable	23,200	26,300
Federal income taxes payable	18,500	17,500
Total current liabilities	\$201,500	\$184,600
Long-term liabilities		
Debenture bonds, 10.3% due in 2015	110,000	110,000
Debenture bonds, 11.5% due in 2007	125,000	125,000
Deferred income taxes	11,600	10,000
Total liabilities	\$448,100	\$429,600
Stockholder's equity		
Preferred stock, 5% cumulative \$5 par value—200,000 shares	\$10,000	\$10,000
Common stock, \$1 par value 2000 28,000,000 shares 2000X 32,000,000 shares	32,000	28,000
Capital surplus	8,000	6,000
Accumulated retained earnings	207,600	172,900
Total stockholder's equity	\$257,600	\$216,900
Total liabilities and stockholder's equity	\$705,700	\$646,500

^aAll amounts in thousands of dollars.

^bIncludes an allowance for doubtful accounts.

SOURCE: J. R. Couper, *Process Engineering Economics*, Dekker, New York, 2003. By permission of Taylor & Francis Books, Inc., Boca Raton, Fla.

Stockholders' equity is the interest that all stockholders have in a company and is a liability with respect to the company. This category includes *preferred* and *common stock* as well as additional paid-in capital (the amount that stockholders paid above the par value of the stock) and *retained earnings*. These are earnings from accumulated profit that a company earns and are used for reinvestment in the company. The sum of these items is the *stockholders' equity*.

On a balance sheet, the sum of the *total liabilities* and the *stockholders' equity* must equal the *total assets*, hence the term *balance sheet*. Comparing balance sheets for successive years, one can follow changes in various items that will indicate how well the company manages its assets and meets its obligations.

Income Statement An income statement shows the revenue and the corresponding expenses for the year and serves as a guide for how the company may do in the future. Often income statements may show how the company performed for the last two or three years. Table 9-4 is an example of a consolidated income statement.

Net sales are the primary source of revenue from goods and services. This figure includes the amount reported after returned goods, discounts, and allowances for price reductions are taken into account. *Cost of sales* represents all the expenses to convert raw materials to finished products. The major components of these expenses are direct material, direct labor, and overhead. If the cost of sales is subtracted from net sales, the result is the *gross margin*. One of the most important items on the income statement is *depreciation* and *amortization*. Depreciation is an allowance the federal government permits for the

TABLE 9-4 Consolidated Income Statement (December 31)

	2005	2004
Net sales (revenue)	\$932,000	\$850,000
Cost of sales and operating expenses		
Cost of goods sold	692,000	610,000
Depreciation and amortization	40,000	36,000
Sales, general, and administrative expenses	113,500	110,000
Operating profit	\$86,500	\$94,000
Other income (expenses)		
Dividends and interest income	10,000	7,000
Interest expense	(22,000)	(22,000)
Income before provision for income taxes	\$74,500	\$79,000
Provision for federal income taxes	24,500	26,000
Net profit for year	\$50,000	\$53,000

SOURCE: J. R. Couper, *Process Engineering Economics*, Dekker, New York, 2003. By permission of Taylor & Francis Books, Inc., Boca Raton, Fla.

wear and tear as well as the obsolescence of plant and equipment and is treated as an expense. Amortization is the decline in value of intangible assets such as patents, franchises, and goodwill. *Selling, general, and administrative expenses* include the marketing salaries, advertising expenses, travel, executive salaries, as well as office and payroll expenses. When depreciation, amortization, and the sales and administrative expenses are subtracted from the gross margin, the result is the *operating income*. *Dividends and interest income* received by the company are then added. Next *interest expense* earned by the stockholders and *income taxes* are subtracted, yielding the term *income before extraordinary loss*. It is the expenses a company may incur for unusual and infrequent occasions. When all the above items are added or subtracted from the operating income, *net income* (or *loss*) is obtained. This latter term is the "bottom line" often referred to in various reports.

Accumulated Retained Earnings This is an important part of the financial report because it shows how much money has been retained for growth and how much has been paid as dividends to stockholders. When the accumulated retained earnings increase, the company has greater value. The calculation of this value of the retained earnings begins with the previous year's balance. To that figure add the net profit after taxes for the year. Dividends paid to stockholders are then deducted, and the result is the accumulated retained earnings for the year. See Table 9-5.

Concluding Remarks One of the most important sections of an annual report is the "notes." These contain any liabilities that a company may have due to impending litigation that could result in charges or expenses not included in the annual report.

OTHER FINANCIAL TERMS

Profit margin is the ratio of net income to total sales, expressed as a percentage or sometimes quoted as the ratio of profit before interest and taxes to sales, expressed as a percentage. *Operating margin* is obtained by subtracting operating expenses from gross profit expressed as a percentage of sales. *Net worth* is the difference between total assets and total liabilities plus stockholders' equity. *Working capital* is the difference between total current assets and current liabilities.

TABLE 9-5 Accumulated Retained Earnings Statement^a (December 31)

	2005	2004
Balance as of January 1	\$172,900	\$141,850
Net profit for year	50,000	53,000
Total for year	\$222,900	\$194,850
Less dividends paid on:		
Preferred stock	700	700
Common stock	14,600	21,250
Balance December 31	\$207,600	\$172,900

^aAll amounts in thousands of dollars.

SOURCE: J. R. Couper, *Process Engineering Economics*, Dekker, New York, 2003. By permission of Taylor & Francis Books, Inc., Boca Raton, Fla.

FINANCIAL RATIOS

There are many financial ratios of interest to financial analysts. A brief discussion of some of these ratios follows; however, a more complete discussion may be found in Couper (2003).

Liquidity ratios are a measure of a company's ability to pay its short-term debts. *Current ratio* is obtained by dividing the current assets by the current liabilities. Depending on the economic climate, this ratio is 1.5 to 2.0 for the chemical process industries, but some companies operate closer to 1.0. The *quick ratio* is another measure of liquidity and is cash plus marketable securities divided by the current liabilities and is slightly greater than 1.0.

Leverage ratios are an indication of the company's overall debt burden. The *debt/total assets ratio* is determined by dividing the total debt by total assets expressed as a percentage. The industry average is 35 percent. *Debt/equity ratio* is another such ratio. The higher these ratios, the greater the financial risk since if an economic downturn did occur, it might be difficult for a company to meet the creditors' demands. The *times interest earned* is a measure of the extent to which profit could decline before a company is unable to pay interest charges. The ratio is calculated by dividing the *earnings before interest and taxes (EBIT)* by interest charges. The *fixed-charge coverage* is obtained by dividing the income available for meeting fixed charges by the fixed charges.

Activity ratios are a measure of how effectively a firm manages its assets. There are two inventory/turnover ratios in common use today. The *inventory/sales ratio* is found by dividing the inventory by the sales. Another method is to divide the cost of sales by inventory. The *average collection period* measures the number of days that customers' invoices remain unpaid. *Fixed assets* and *total assets turnover* indicate how well the fixed and total assets of the firm are being used.

Profitability ratios are used to determine how well income is being managed. The *gross profit margin* is found by dividing the gross profits by the net sales, expressed as a percentage. The *net operating margin* is equal to the earnings before interest and taxes divided by net sales. Another measure, the *profit margin on sales*, is calculated by dividing the net profit after taxes by net sales. The *return on total assets* ratio is the net profit after taxes divided by the total assets expressed as a percentage. The *return on equity* ratio is the net income after taxes and interest divided by stockholders' equity.

Table 9-6 shows the financial ratios for Tables 9-3 and 9-4. Table 9-7 is a summary of selected financial ratios and industry averages.

RELATIONSHIP BETWEEN BALANCE SHEETS AND INCOME STATEMENTS

There is a relationship between these two documents because information obtained from each is used to calculate the returns on assets and equity. Figure 9-1 is an operating profitability tree for a fictitious

TABLE 9-6 Financial Ratios for Tables 9-3 and 9-4

Liquidity	
Current ratio	= \$391,200/\$201,500 = 1.94
Cash ratio	= \$391,200 - 149,000/\$201,500 = 1.20
Leverage	
Debt/assets ratio	= [(\$448,100 - 201,500)/\$705,700] × 100 = 35%
Times interest earned	= \$74,500 - 22,000/\$22,000 = 4.39
Fixed-charge coverage	= \$86,500/\$22,000 = 3.93
Activity	
Inventory turnover	= \$932,000/\$149,000 = 6.25
Average collection period	= \$135,000/(\$932,000/365) = 52.8 days
Fixed-assets turnover	= \$932,000/\$438,000 = 2.13
Total-assets turnover	= \$932,000/\$705,700 = 1.32
Profitability	
Gross profit margin	= [(\$932,000 - 692,000)/\$932,000] × 100 = 25.8%
Net operating margin	= \$74,500/\$932,000 × 100 = 7.99%
Profit margin on sales	= \$50,000/\$932,000 × 100 = 5.36%
Return on net worth (return on equity)	= [\$50,000/(\$705,700 - 448,100)] × 100 = 19.4%
Return on total assets	= (\$50,000/\$705,700) × 100 = 7.09%

company and contains the fixed and variable expenses as reported on internal company reports, such as the manufacturing expense sheet. Figure 9-2 is a financial family tree for the same company depicting the relationship between values in the income statement and the balance sheet.

FINANCING ASSETS BY DEBT AND/OR EQUITY

The various options for obtaining funds to finance new projects are not a simple matter. Significant factors such as the state of the economy, inflation, a company's present indebtedness, and the cost of capital will affect the decision. Should a company incur more long-term debt, or should it seek new venture capital from equity sources? A simple yes or no answer will not suffice because the financial decision is complex. One consideration is the company's position with respect to leverage. If a company has a large proportion of its debt in bonds and preferred stock, the common stock is highly leveraged. Should the earnings decline, say, by 10 percent, the dividends available to common stockholders might be wiped out. The company also might not be able to cover the interest on its bonds without dipping into the accumulated earnings. A high debt/equity ratio illustrates the fundamental weakness of companies with a large amount of debt. When low-interest financing is available, such as for large government projects, the return-on-equity evaluations are used. Such leveraging is tantamount to transferring money from one pocket to another; or, to

TABLE 9-7 Selected Financial Ratios

Item	Equation for calculation	Industry average
Liquidity		
Current ratio	Current assets/current liabilities	1.5–2.0
Cash ratio	Current assets – inventory/current liabilities	1.0–1.5
Leverage		
Debt to total assets	Total debt/total assets	30–40%
Times interest earned	Profit before taxes plus interest charges/interest charges	7.0–8.0
Fixed-charge coverage	Income available for meeting fixed charges/fixed charges	6.0
Activity		
Inventory turnover	Sales or revenue/inventory	7.0
Average collection period	Receivables/sales per day	40–60 days
Fixed assets turnover	Sales/fixed assets	2–4
Total assets turnover	Sales/total assets	1–2
Profitability		
Gross profit margin	Net sales – cost of goods sold/sales	25–40%
Net operating margin	Net operating profit before taxes/sales	10–15%
Profit margin on sales	Net profit after taxes/sales	5–8%
Return on net worth (return on equity)	Net profit after taxes/net worth	15%
Return on total assets	Net profit after taxes/total assets	7–10%

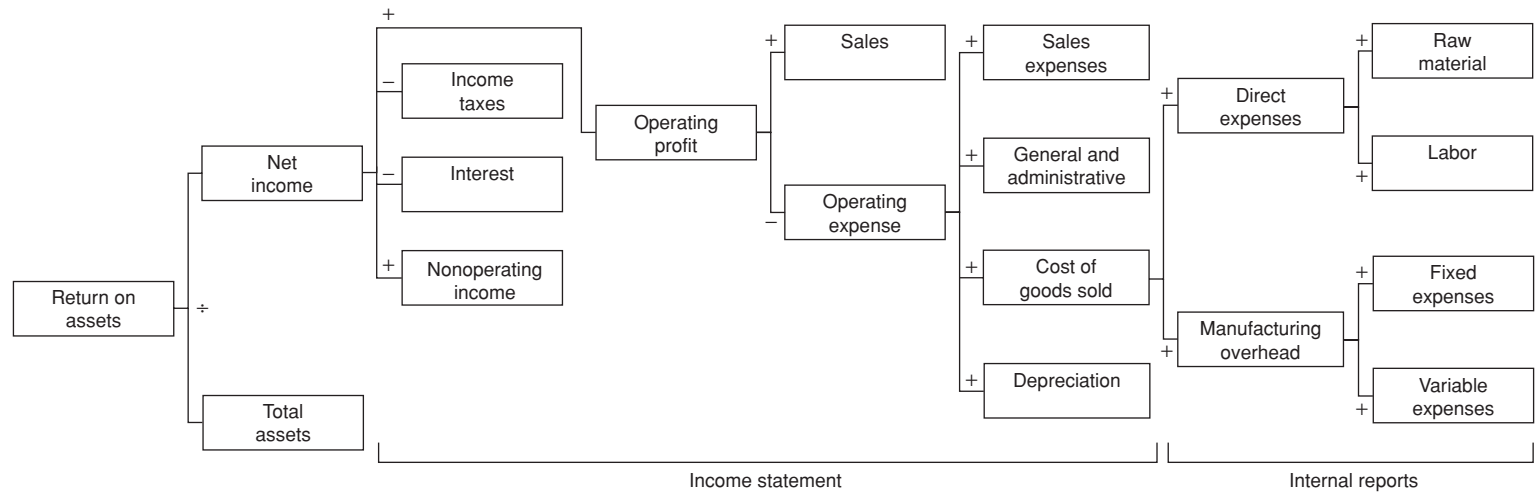


FIG. 9-1 Operating profitability tree. (Source: Adapted from Couper, 2003.)

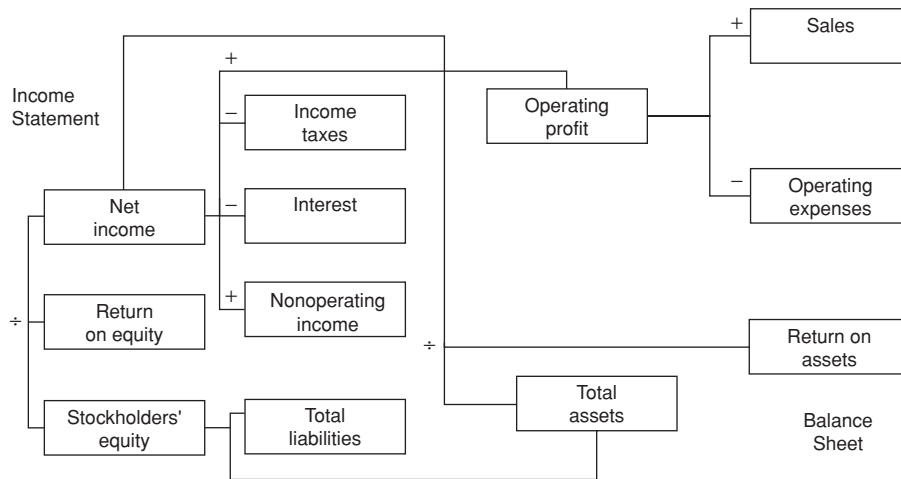


FIG. 9-2 Financial family tree. (Source: Adapted from Couper, 2003.)

put it another way, a company may find itself borrowing from itself. In the chemical process industries, debt/equity ratios of 0.3 to 0.5 are common for industries that are capital-intensive (Couper et al., 2001). Much has been written on the strategies of financing a corporate venture. The correct strategy has to be evaluated from the standpoint of what is best for the company. It must maintain a debt/equity ratio similar to those of successful companies in the same line of business.

COST OF CAPITAL

The cost of capital is what it costs a company to borrow money from all sources, such as loans, bonds, and preferred and common stock. It is an important consideration in determining a company's minimum acceptable rate of return on an investment. A company must make more than the cost of capital to pay its debts and make a profit. From profits, a company pays dividends to the stockholders. If a company ignores the cost of capital to increase dividends to the stockholders, then management is not meeting its obligations to pay off outstanding debts.

A sample calculation of the after-tax weighted cost of capital is found in Table 9-8. Each debt item is divided by the total debt, and

that result is multiplied by the after-tax yield to maturity that equals the after-tax weighted average cost of that debt item contributing to the cost of capital. The information to estimate the cost of capital may be obtained from the annual report, the 10K, or the 10Q reports.

WORKING CAPITAL

The accounting definition of *working capital* is total current assets minus total current liabilities. This information can be found from the balance sheet. Current assets consist chiefly of cash, marketable securities, accounts receivable, and inventories; current liabilities include accounts payable, short-term debts, and the part of the long-term debt currently due. The accounting definition is in terms of the entire company.

For economic evaluation purposes, another definition of working capital is used. It is the funds, in addition to the fixed capital, that a company must contribute to a project. It must be adequate to get the plant in operation and to meet subsequent obligations when they come due. Working capital is not a one-time investment that is known at the project inception, but varies with the sales level and other factors. The relationship of working capital to other project elements may be viewed in the cash flow model (see Fig. 9-9). Estimation of an adequate amount of working capital is found in the section "Capital Investment."

TABLE 9-8 Cost of Capital Illustration

Balance sheet 12/31/XX	Debt, \$M	After-tax yield to maturity, %	After-tax weighted average cost, %
Long-term debt			
Revolving account	5.0	4.5	0.02
4 $\frac{3}{8}$ % debentures	12.0	4.0	0.05
6 $\frac{1}{8}$ % debentures	3.4	4.7	0.02
6 $\frac{3}{4}$ % debentures	9.4	4.2	0.04
7 $\frac{1}{8}$ % debentures	74.5	4.2	0.30
9 $\frac{3}{8}$ % loan	125.0	4.4	0.53
Other	23.2	4.4	0.10
Total long-term debt	252.5		1.06
Deferred taxes	67.7	0.0	0
Reserves	16.1	0.0	0
Preferred stock	50.0	8.6	0.42
Shareholders' equity	653.9	15.6	9.80
Total debt	1,040.2		11.28

Each debt item in \$M divided by the total debt times the after-tax yield to maturity equals the after-tax weighted average cost contributing to the cost of capital.

SOURCE: Private communication.

INVENTORY EVALUATION AND COST CONTROL

Under ordinary circumstances, inventories are priced (valued) at some form of cost. The problem in valuating inventory lies in "determining what costs are to be identified with inventories in a given situation" (Nickerson, 1979).

Valuation of materials can be made by using the

- Cost of a specific lot
- Average cost
- Standard cost

Under "cost of a specific lot," those lots to be valued must be identified by referring to related invoices. Many companies use the average cost for valuating inventories. The average used should be weighted by the quantities purchased rather than by an average purchase price. Average cost method tends to spread the effects of short-run price changes and has a tendency to level out profits in those industries that use raw materials whose prices are volatile. For many manufacturing companies, inventory valuation is an important

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consideration varying in degree of importance. Inventories that are large are subject to significant fluctuations from time to time in size and mix and in prices, costs, and values.

Materials are valued in accordance with their acquisition. Some companies use the first-in, first-out (FIFO) basis. Materials are used in order of their acquisition to minimize losses from deterioration. Another method is last-in, first-out (LIFO) in which materials coming in are the first to leave storage for use. The method used depends on a number of factors. Accounting texts discuss the pros and cons of each method, often giving numerical examples. Some items to consider are income tax considerations and cash flow that motivate management to adopt conservative valuation policies. Tax savings may accrue using one method compared to the other, but they may not be permanent. Whatever method is selected, consistency is important so that comparability of reported figures may be maintained from one time period to another. It is management's responsibility to make the decision regarding the method used. In some countries, government regulations control the method to be used. There are several computer software programs that permit the user to organize, store, search, and manage inventory from a desktop computer.

BUDGETS AND COST CONTROL

A budget is an objective expressed in monetary terms for planning and controlling the resources of a company. Budgeted numbers are

objectives, not achievements. A comparison of actual expenses with budgeted (cost standards) figures is used for control at the company, plant, departmental, or project level. A continuing record of performance should be maintained to provide the data for preparing future budgets (Nickerson, 1979). Often when a company compares actual results with cost standards or budgeted figures, a need for improving operations will surface. For example, if repairs to equipment continuously exceed the budgeted amount, perhaps it is time to consider replacement of that older equipment with a newer, more efficient model. Budgets are usually developed for a 1-year period; however, budgets for various time frames are frequently prepared. For example, in planning future operations, an intermediate time period of, say, 5 years may be appropriate, or for long-range planning the time period selected may be 10 years.

A cost control system is used

- To provide early warning of uneconomical or excessive costs in operations
- To provide relevant feedback to the personnel responsible for developing budgets
- To develop cost standards
- To promote a sense of cost consciousness
- To summarize progress

Budgetary models based upon mathematical equations are available to determine the effect of changes in variables. There are numerous sources extant in the literature for these models.

CAPITAL COST ESTIMATION

TOTAL CAPITAL INVESTMENT

The *total capital investment* includes funds required to purchase land, design and purchase equipment, structures, and buildings as well as to bring the facility into operation (Couper, 2003). The following is a list of items constituting the total capital investment:

- Land
- Fixed capital investment
- Offsite capital
- Allocated capital
- Working capital
- Start-up expenses
- Other capital items (interest on borrowed funds prior to start-up; catalysts and chemicals; patents, licenses, and royalties; etc.)

Land Land is often acquired by a company some time prior to the building of a manufacturing facility. When a project is committed to be built on this land, the value of the land becomes part of that facility's capital investment.

Fixed Capital Investment When a firm considers the manufacture of a product, a capital cost estimate is prepared. These estimates are required for a variety of reasons such as feasibility studies, the selection of alternative processes or equipment, etc., to provide information for planning capital appropriations or to enable a contractor to bid on a project. Included in the fixed capital investment is the cost of purchasing, delivery, and installation of manufacturing equipment, piping, automatic controls, buildings, structures, insulation, painting, site preparation, environmental control equipment, and engineering and construction costs. The fixed capital investment is significant in developing the economics of a process since this figure is used in estimating operating expenses and calculating depreciation, cash flow, and project profitability. The estimating method used should be the best, most accurate means consistent with the time and money available to prepare the estimate.

Classification of Estimates There are two broad classes of estimates: grass roots and battery limits. Grass-roots estimates include the entire facility, starting with site preparation, buildings and structures, processing equipment, utilities, services, storage facilities, railroad yards, docks, and plant roads. A battery-limits estimate is one in which an imaginary boundary is drawn around the proposed facility to be estimated. It is assumed that all materials, utilities, and services are available in the quality and quantity required to manufacture a product. Only costs within the boundary are estimated.

Quality of Estimates Capital cost estimation is more art than science. An estimator must use considerable judgment in preparing the estimate, and as the estimator gains experience, the accuracy of the estimate improves. There are several types of fixed capital cost estimates:

- *Order-of-magnitude (ratio estimate)*. Rule-of-thumb methods based on cost data from similar-type plants are used. The probable accuracy is -30 percent to +50 percent.
- *Study estimate (factored estimate)*. This type requires knowledge of preliminary material and energy balances as well as major equipment items. It has a probable accuracy of -25 to +30 percent.
- *Preliminary estimate (budget authorization estimate)*. More details about the process and equipment, e.g., design of major plant items, are required. The accuracy is probably -20 to +25 percent.
- *Definitive estimate (project control estimate)*. The data needed for this type of estimate are more detailed than those for a preliminary estimate and include the preparation of specifications and drawings. The probable accuracy is -10 to +15 percent.
- *Detailed estimate (firm estimate)*. Complete specifications, drawings, and site surveys for the plant construction are required, and the estimate has an accuracy of -5 to +10 percent.

Detailed information requirements for each type of estimate may be found in Fig. 9-3.

In periods of high inflation, the results of various estimates and accuracy may overlap. At such times, four categories may be more suitable, namely, study, preliminary, definitive, and detailed categories. At present, some companies employing the *front-end loading (FEL)* process for project definition and execution use three categories:

Project stage	Accuracy
Conceptual	±40%
Feasibility	±25%
Definition	±10%

For more information on the FEL process, see "Capital Project Execution and Analysis" near the end of Sec 9.

Scope The scope is a document that defines a project. It contains words, drawings, and costs. A scope should answer the following questions clearly:

What product is being manufactured?

How much is being produced?

ESTIMATING INFORMATION GUIDE					
Information Either Required or Available					
Estimate types	Detailed (firm)				
	Definitive (project control)				
	Preliminary (budget authorization)				
	Study (factored)				
	Order of magnitude (ratio)				
Site	Location		•	•	•
	General description		•	•	•
Process flow	Site survey			•	•
	Geotechnical report			•	•
Equipment	Site plot plan and contours			•	•
	Well-developed site facilities				•
Buildings and structures	Rough sketches		•	•	•
	Preliminary			•	•
Utilities and services	Engineered				•
	Rough sizes and construction		•	•	•
Piping and insulation	Engineered specifications			•	•
	Vessel data sheets			•	•
Instrumentation	General arrangement			•	•
	Final arrangement				•
Electrical	Rough sizes and construction		•	•	•
	Foundation sketches			•	•
Work-hours	Architectural and construction			•	•
	Preliminary structural design			•	•
Project scope	General arrangements and elevations			•	•
	Detailed drawings				•
Project scope	Rough quantities		•	•	•
	Preliminary heat balance			•	•
Project scope	Preliminary flow sheets			•	•
	Engineered heat balance			•	•
Project scope	Engineered flow sheets			•	•
	Detailed drawings				•
Project scope	Preliminary flow sheets		•	•	•
	Engineered flow sheets			•	•
Project scope	Piping layouts and schedules			•	•
	Insulation rough specifications			•	•
Project scope	Insulation applications			•	•
	Insulation details				•
Project scope	Preliminary list			•	•
	Engineered list			•	•
Project scope	Detail drawings				•
	Rough motor list and sizes		•	•	•
Project scope	Engineered list and sizes			•	•
	Substation number and size			•	•
Project scope	Preliminary specifications			•	•
	Distribution specifications			•	•
Project scope	Preliminary interlocks and controls			•	•
	Engineered single-line diagrams			•	•
Project scope	Detailed drawings				•
	Engineering and drafting		•	•	•
Project scope	Construction supervision			•	•
	Craft labor			•	•
Project scope	Product, capacity, location, utilities, and services	•	•	•	•
	Building requirements, process, storage, and handling				

FIG. 9-3 Information guide for preparing estimates. (Source: Perry's Chemical Engineers' Handbook, 5th ed., McGraw-Hill, New York, 1973.)

What is the quality of the product?
 Where is the product to be produced?
 What is the quality of the estimate?
 What is the basis for the estimate?
 What are the knowns and unknowns with respect to the project?
 Before an estimate can be prepared, it is essential to prepare a scope. It may be as simple as a single page, such as for an order-of-

magnitude estimate, or several large manuals, for a detailed estimate. As the project moves forward from inception to a detailed estimate, the scope must be revised and updated to provide the latest information. Changes during the progress of a project are inevitable, but a well-defined scope prepared in advance can help minimize costly changes. If a scope is properly defined, the following results:

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An understanding between those who prepared the scope (engineering) and those who accept it (management)

A document that indicates clearly what is provided in terms of technology, quality, schedule, and cost

A basis in enough detail to be used in controlling the project and its costs to permit proper evaluation of any proposed changes

A device to permit subsequent evaluation of the performance compared to the intended performance

A document to control the detailed estimate for the final design, construction, and design

Equipment Cost Data The foundation of a fixed capital investment estimate is the equipment cost data. From this information, through the application of factors or percentages based upon the estimator's experience, the fixed capital investment is developed.

These cost data are reported as purchased, delivered, or installed cost. Purchased cost is the price of the equipment FOB at the manufacturer's plant, whereas delivered cost is the purchased price plus the delivery charge to the purchaser's plant FOB. Installed cost means the equipment has been purchased, delivered, uncrated, and placed on a foundation in the purchaser's operating department but does not include piping, electrical, instrumentation, insulation, etc., costs. Perhaps a better name might be *set-in-place* cost.

It is essential to have reliable cost data since the engineer producing the estimate starts with this information and develops the fixed capital cost estimate. The estimator must know the source of the data, the basis for the data, its date, potential errors, and the range over which the data apply. There are many sources of graphical equipment cost data in the literature, but some are old and the latest published data were in the early 1990s. There have been no significant cost data published recently. To obtain current cost data, one should solicit bids from vendors; however, it is essential to impress on the vendor that the information is to be used for preliminary estimates. A disadvantage of using vendor sources is that there is a chance of compromising proprietary information.

Cost-capacity plots of equipment indicate a straight-line relationship on a log-log plot. Figure 9-4 is an example of such a plot. A convenient method of presenting these data is in equation format:

$$C_2 = C_1 \left(\frac{S_2}{S_1} \right)^n \quad (9-1)$$

where C_1 = cost of equipment of capacity S_1
 C_2 = cost of equipment of capacity S_2

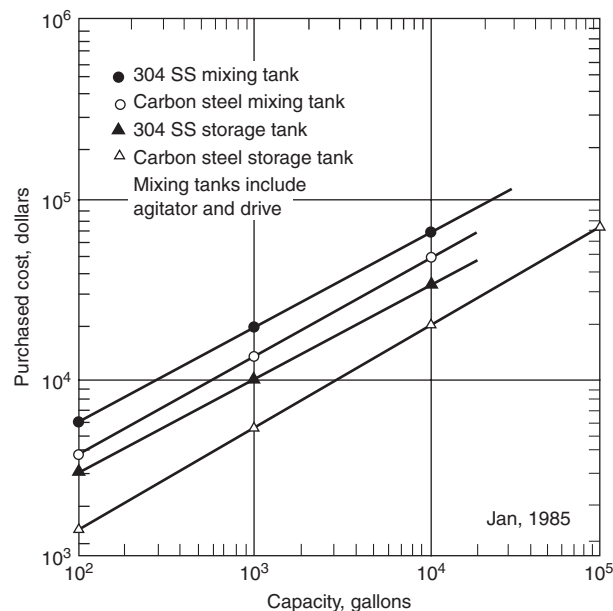


FIG. 9-4 Cost-capacity plot.

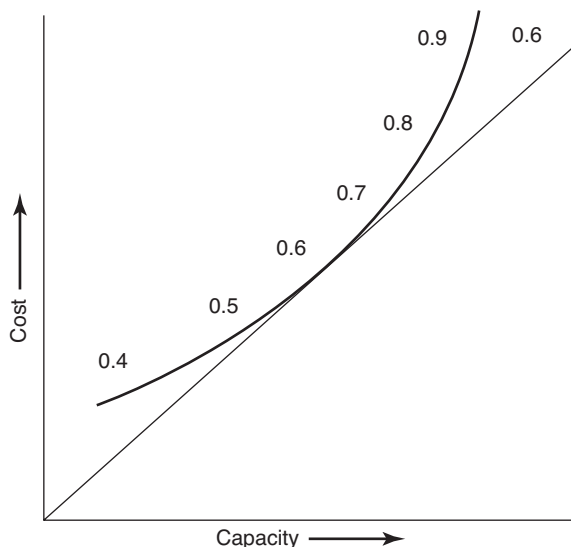


FIG. 9-5 Variation of n on cost-capacity plot.

n = exponent that may vary between 0.4 and 1.2 depending on type of equipment

Equation (9-1) is known as the *six-tenths rule* since the average value for all equipment is about 0.6. D. S. Remer and L. H. Chai (*Chemical Engineering Progress*, August 1990, pp. 77-82) published an extensive list of six-tenths data. Figure 9-5 shows how the exponent may vary from 0.4 to 0.9 for a given equipment item. Data accuracy is the highest in the narrow, middle-range of capacity, but at either end of the plot, the error is great. These errors occur when one correlates cost data with one independent variable when more than one variable is necessary to represent the data, or when pressure, temperature, materials of construction, or design features vary considerably.

A convenient way to display cost-capacity data is by algorithms. They are readily adaptable for computerized cost estimation programs. Algorithm modifiers in equation format may be used to account for temperature, pressure, material of construction, equipment type, etc. Equation (9-2) is an example of obtaining the cost of a shell-and-tube heat exchanger by using such modifiers.

$$C_{HE} = K C_B F_D F_M F_P \quad (9-2)$$

where C_{HE} = purchased equipment cost
 K = factor for cost index based upon a base year
 C_B = base cost of a carbon-steel floating-head exchanger, 150-psig design pressure
 F_D = design-type cost factor if different from that in C_B
 F_M = material-of-construction cost factor
 F_P = design pressure cost factor

Each cost factor is obtained from equations or tables from Couper, 2003, App. C, and have been updated to third-quarter 2002.

Cost Indices Cost data are given as of a specific date and can be converted to more recent costs through the use of cost indices. In general, the indices are based upon constant dollars in a base year and actual dollars in a specific year. In this way, with the proper application of the index, the effect of inflation (or deflation) and price increases by multiplying the historical cost by the ratio of the present cost index divided by the index applicable in the historical year. Labor, material, construction costs, energy prices, and product prices all change at different rates. Most cost indices represent national averages, and local averages may vary considerably. Table 9-9 is a list of selected values of three cost indices of significance in the chemical process industries.

TABLE 9-9 Selected Cost Indices

Year Base	M&S Index ^{(1)a} 1926 = 100	CE index ⁽¹⁾ 1957 – 1959 = 100	Nelson-Farrar index ⁽²⁾ 1946 = 100
1990	915.1	357.6	1225.7
1992	943.1	358.2	1277.3
1994	964.2	368.1	1349.7
1996	1039.1	381.7	1418.9
1998	1061.9	389.5	1477.6
2000	1089.0	394.1	1542.7
2001	1093.9	394.3	1579.7
2002	1104.2	395.6	1642.2
2003	1123.6	402.0	1710.4
2004 (3Q)	1124.7	457.4	1856.1

(1) From 1990 onward, the M&S and CE indices are from *Chemical Engineering* magazine.

(2) The Nelson-Farrar indices from 1990 onward are found in *Oil and Gas Journal*.

^aProcess industry average instead of all industry average.

The chemical engineering (CE) index and the Marshall and Swift index are found in each issue of the magazine *Chemical Engineering*. The *Oil and Gas Journal* reports the *Nelson-Farrar Refinery* indices in the first issue of each quarter. The base years selected for each index are generally periods of low inflation so that the index is stable. The derivation of base values is referred to in the respective publications.

A cost index is used to project a cost from a base year to another selected year. The following equation is used:

$$\text{Cost at } \Theta_2 = \text{cost at } \Theta_1 \left(\frac{\text{index at } \Theta_2}{\text{index at } \Theta_1} \right) \quad (9-3)$$

Example 1: Use of Cost Index A centrifuge cost \$95,000 in 1999. What is the cost of the same centrifuge in third quarter of 2004? Use the CE index.

Solution:

CE index in 1999 = 390.6

CE index in 3d quarter 2004 = 457.4

Cost in 2004 = cost in 1999 (CE index in 3d quarter 2004/ CE index in 1999)

$$= \$95,000 \left(\frac{457.4}{390.6} \right) = \$111,200$$

Inflation When costs are to be projected into the future due to inflation, it is a highly speculative exercise, but it is necessary for estimating investment costs, operating expenses, etc. Inflation is the increase in price of goods without a corresponding increase in productivity. A method for estimating an inflated cost is

$$C_i = (1 + f_1) (1 + f_2) (1 + f_3) C_P \quad (9-4)$$

where C_i = inflated cost

f_1 = inflation rate the first year

f_2 = inflation rate the second year

f_3 = inflation rate the third year

C_P = cost in a base year

The assumed inflation factors f are obtained from federal economic reports, financial sources such as banks and investment houses, and news media. These factors must be reviewed periodically to update estimates.

Example 2: Inflation A dryer today costs \$475,000. The projected inflation rates for the next 3 years are 3, 4.2, and 4.7 percent. Calculate the projected cost in 3 years.

Solution:

$$C_i = (1 + f_1) (1 + f_2) (1 + f_3) C_P$$

$$= (1.030) (1.042) (1.047) (\$475,000) = \$533,800$$

Equipment Sizing Before equipment costs can be obtained, it is necessary to determine equipment size from material and energy balances. For preliminary estimates, rules of thumb may be used; but for definitive and detailed estimates, detailed equipment calculations must be made.

Example 3: Equipment Sizing and Costing Oil at 490,000 lb/h is to be heated from 100 to 170°F with 145,000 lb/h of kerosene initially at 390°F from another section of a plant. The oil enters at 20 psig and the kerosene at 25 psig. The physical properties are

Oil—0.85 sp gr, 3.5 cP at 135°F, 0.49 sp ht

Kerosene—0.82 sp gr, 0.45 cP, 0.61 sp ht

Estimate the cost of an all-carbon-steel exchanger in late 2004. Assume a counterflow shell-and-tube exchanger.

Solution:

Energy required to heat oil stream $(490,000)(0.49)(170 - 100) = 16,807,000$ Btu/h

$$\text{Exit kerosene temperature } T = 390 - \left(\frac{490,000}{145,000} \right) \left(\frac{0.49}{0.61} \right) (170 - 100)$$

$$= 200^\circ \text{F}$$

$$\text{LMTD} = \frac{220 - 100}{\ln 2.2} = 152^\circ \text{F}$$

Calculate the exchanger efficiency factor, F .

$$P = \frac{170 - 100}{390 - 100} = 0.241$$

$$R = \frac{390 - 200}{170 - 100} = 2.71$$

From Perry $F = 0.88$. Since the factor must be greater than 0.75, the exchanger is satisfactory. Therefore, $\Delta T = (F)(\text{LMTD}) = (0.88)(152) = 134^\circ \text{F}$. Assume $U_D = 50$ Btu/(h·ft²·°F).

$$Q = U_D A \Delta T = 16,800,000 = (50)(A)(134)$$

$$A = 2510 \text{ ft}^2$$

Use the cost algorithm cited above.

$$C_B = \exp [8.821 - 0.30863 \ln A + 0.0681(\ln A)^2]$$

$$= \exp [8.821 - 0.30863(7.83) + 0.0681(61.3)] = \$39,300 \text{ base cost}$$

$$F_D = 1.0 \quad F_M = \text{for cs/cs material} = 1.0$$

$$F_P = 1.00 \quad \text{since this exchange is operating below 4 bar}$$

$$K = 1.218 \left(\frac{\text{CE index 4th qtr 2004/CE index 1st qtr 2003}}{463/406} \right) = 1.389$$

$$\text{Therefore, } C_{HE} = K C_B F_D F_M F_P = (1.389)(\$39,300)(1.0)(1.0)(1.0) = \$54,600.$$

Estimation of Fixed Capital Investment

Order-of-Magnitude Methods The **ratio method** will give the fixed capital investment per gross annual sales; however, most of these data are from the 1960s, and no recent data have been published. The ratio above is called the *capital ratio*, often used by financial analysts. The reciprocal of the capital ratio is the *turnover ratio* that for various businesses ranges from 4 to 0.3. The chemical industry has an average of about 0.4 to 0.5. The ratio method of obtaining fixed capital investment is rapid but suitable only for order-of-magnitude estimates.

The **exponential method** may be used to obtain a rapid capital cost for a plant based upon existing company data or from published sources such as those of D. S. Remer and L. H. Chai, *Chemical Engineering*,

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April 1990, pp. 138–175. In the method known as the *seven-tenths rule*, the cost-capacity data for process plants may be correlated by a logarithmic plot similar to the six-tenths plot for equipment. Remer and Chai compiled exponents for a variety of processes and found that the exponents ranged from 0.6 to 0.8. When the data are used to obtain a capital cost for a different-size plant, the estimated capital must be for the same process.

The equation is

$$\text{Cost of plant B} = \text{cost of plant A} \left(\frac{\text{capacity of plant B}}{\text{capacity of plant A}} \right)^{0.7} \quad (9-5)$$

Cost indices may be used to correct costs for time changes.

Example 4: Seven-Tenths Rule A company is considering the manufacture of 150,000 tons annually of ethylene oxide by the direct oxidation of ethylene. According to Remer and Chai (1990), the cost capacity exponent for such a plant is 0.67. A subsidiary of the company built a 100,000-ton annual capacity plant for \$70 million fixed capital investment in 1996. Using the seven-tenths rule, estimate the cost of the proposed new facility in the third quarter 2004.

Solution:

$$\begin{aligned} \text{Cost}_{150} &= \text{Cost}_{100} \left(\frac{\text{Cap}_{150}}{\text{Cap}_{100}} \right)^{0.67} \left(\frac{\text{CE index 3Q 2004}}{\text{CE index 1996}} \right) \\ &= (\$70,000,000) \left(\frac{150,000}{100,000} \right)^{0.67} \left(\frac{457.4}{381.7} \right) = \$110,000,000 \end{aligned}$$

Study Method The single-factor method begins with collecting the delivered cost of various items of equipment and applying one factor to obtain the battery-limits (BL) fixed capital (FC) investment or total capital investment as follows:

$$(C_{\text{FC}})_{\text{BL}} = f \sum (C_{\text{EQ}})_{\text{DEL}} \quad (9-6)$$

where $(C_{\text{FC}})_{\text{BL}}$ = battery-limits fixed capital investment or total capital investment
 $(C_{\text{EQ}})_{\text{DEL}}$ = delivered equipment costs

The single factors include piping, automatic controls, insulation, painting, electrical work, engineering costs, etc. (Couper, 2003). Table 9-10 shows the Lang factors for various types of processing plants. The boundaries between the classifications are not clear-cut, and considerable judgment is required in the selection of the appropriate factors.

Preliminary Estimate Methods A refinement of the Lang factor method is the Hand method. The Hand factors are found in Table 9-11. Equipment is grouped in categories, such as heat exchangers and pumps, and then a factor is applied to each group to obtain the installed cost; finally the groups are summed to give the battery-limits installed cost. Wroth compiled a more detailed list of installation factors; a selection of these can be found in Table 9-12. The Lang and Hand methods start with purchased equipment costs whereas the Wroth method begins with delivered equipment costs, so delivery charges must be included in the Lang and Hand methods. At best the Lang and Hand methods will yield study quality estimates, and the Wroth method might yield a preliminary quality estimate.

Example 5: Fixed Capital Investment Using the Lang, Hand, and Wroth Methods The following is a list of the purchased equipment costs for a proposed processing unit:

Heat exchangers	\$620,000
Distillation towers and internals	975,000
Receivers	320,000

TABLE 9-10 Lang Factors

Type of plant	Lang factors	
	Fixed capital investment	Total capital investment
Solid processing	4.0	4.7
Solid-fluid processing	4.3	5.0
Fluid processing	5.0	6.0

Adapted from M. S. Peters, K. D. Timmerhaus, and R. West, *Plant Design and Economics for Chemical Engineers*, 5th ed., McGraw-Hill, New York, 2004.

TABLE 9-11 Hand Factors

Equipment type	Factor
Fractionating columns	4.0
Pressure vessels	4.0
Heat exchangers	3.5
Fired heaters	2.0
Pumps	4.0
Compressors	2.5
Instruments	4.0
Miscellaneous equipment	2.5

Adapted from W. E. Hand, *Petroleum Refiner*, September 1958, pp. 331–334.

Accumulator drum	125,000
Pumps and motors	220,000
Automatic controls	275,000
Miscellaneous equipment	150,000

Assume delivery charges are 5 percent of the purchased price. Estimate the fixed capital investment 2 years into the future, using the Lang, Hand, and Wroth methods. The inflation rates are 3.5 percent for the first year and 4.0 percent for the second.

Solution:

Equipment	Purchased equipment cost	Delivered equipment cost
Heat exchangers	\$620,000	\$651,000
Distillation towers, internals	975,000	1,024,000
Receivers	320,000	336,000
Accumulator drum	125,000	131,000
Pumps and motors	220,000	231,000
Automatic controls	275,000	289,000
Miscellaneous equipment	150,000	158,000
Total	\$2,685,000	\$2,820,000

Lang method: The Lang factor for a fluid processing unit starting with purchased equipment costs is 5.0. Therefore, fixed capital investment is \$2,820,000 $\times$ 5.0 $\times$ 1.035 $\times$ 1.040 = \$15,177,000.

Hand method: The Hand method begins with purchased equipment costs, and factors are applied from Table 9-11.

TABLE 9-12 Selected Wroth Factors

Equipment	Factor
Blender	2.0
Blowers and fans	2.5
Centrifuge	2.0
Compressors	
Centrifugal (motor-driven)	2.0
Centrifugal (steam-driven, including turbine)	2.0
Reciprocating (steam and gas)	2.3
Reciprocating (motor-driven less motor)	2.3
Ejectors, vacuum	2.5
Furnaces (packaged units)	2.0
Heat exchangers	4.8
Instruments	4.1
Motors, electric	3.5
Pumps	
Centrifugal (motor-driven less motor)	7.0
Centrifugal (steam-driven including turbine)	6.5
Positive-displacement (less motor)	5.0
Reactors (factor as appropriate, equivalent-type equipment)	—
Refrigeration (packaged units)	2.5
Tanks	
Process	4.1
Storage	3.5
Fabricated and field-erected 50,000+ gal	2.0
Towers (columns)	4.0

Abstracted from W. F. Wroth, *Chemical Engineering*, October 17, 1960, p. 204.

Hand method:

Equipment	Purchased equipment cost	Hand factor	Purchased equipment installed cost
Heat exchangers	\$620,000	3.5	\$2,170,000
Distillation towers, internals	975,000	4.0	3,900,000
Receivers	320,000	2.5°	800,000
Accumulator drum	125,000	2.5°	313,000
Pumps and motors	220,000	4.0	880,000
Automatic controls	275,000	4.0	1,100,000
Miscellaneous	150,000	2.5	375,000
TOTAL	\$2,685,000		\$9,538,000

The asterisk on the receivers and accumulators indicates that if these vessels are pressure vessels, a factor of 4.0 should be used instead of 2.5. The total purchased equipment installed is \$9,538,000 for non-pressure vessels and the delivered cost is \$10,015,000. Therefore, the fixed capital investment installed would be $\$10,015,000 \times 1.035 \times 1.040 = \$10,780,000$. Using pressure vessels increases the total purchased equipment cost \$667,000; therefore, the fixed capital investment for this case including inflation would be $\$10,780,000 \times 1.05 \times 1.035 \times 1.04 = \$11,534,000$.

Wroth method:

Equipment	Delivered equipment cost	Wroth factor	Delivered equipment installed cost
Heat exchangers	\$651,000	4.8	\$3,125,000
Distillation towers, internals	1,024,000	4.0	4,096,000
Receivers	336,000	3.5	1,176,000
Accumulator drum	131,000	3.5	459,000
Pumps and motors	231,000	7.0	1,617,000
Automatic controls	289,000	4.1	1,185,000
Miscellaneous	158,000	4.0 (assumed)	632,000
TOTAL	\$2,820,000		\$12,290,000

The total delivered installed equipment cost is the fixed capital investment and, corrected for 2 years of inflation, will be $\$12,290,000 \times 1.035 \times 1.040 = \$13,229,000$.

Therefore, the summary of the fixed capital investment by the various methods is

Lang	\$15,177,000
Hand	11,534,000
Wroth	13,229,000

Experience has shown that the fixed capital investment by the Lang method is generally higher than that of the other methods. Whatever figure is reported to management, it is advisable to state the potential accuracy of these methods.

Brown developed guidelines for the preparation of order-of-magnitude and study capital cost estimates based upon the Lang and Hand methods. Brown modified Lang and Hand methods for materials of construction, instrumentation, and location factors. He found that the modified Hand and Garrett module factor methods gave results within 3.5 percent.

Other multiple-factor methods that have been published in the past are those by C. E. Chilton, *Cost Estimation in the Process Industries*, McGraw-Hill, New York, 1960; M. S. Peters, K. D. Timmerhaus, and R. E. West, *Plant Design and Economics for Chemical Engineers*, 5th ed., McGraw-Hill, New York, 2003; C. A. Miller, *Chemical Engineering*, Sept. 13, 1965, pp. 226–236; and F. A. Holland, F. A. Watson, and V. K. Wilkinson, *Chem. Eng.*, Apr. 1, 1974, pp. 71–76. These methods produced preliminary quality estimates. Most companies have developed their own in-house multiple-factor methods for preliminary cost estimation.

Step-counting methods are based upon a number of processing steps or “functional units.” The concept was first introduced by H. E. Wessel, *Chem. Eng.*, 1952, p. 209. Subsequently, R. D. Hill, *Petrol. Refin.*, **35**(8):106–110, August 1956; F. C. Zevnik and R. L. Buchanan, *Chem. Eng. Progress*, **59**(2):70–77, February 1963; and J. H. Taylor, *Eng. Process Econ.*, **2**:259–267, 1977, further developed the step-counting method.

A step or functional unit is a significant process step including all process equipment and ancillary equipment necessary for operating the unit. A functional unit may be a unit operation, unit process, or separation in which mass and energy are transferred. The sum of all functional units is the total fixed capital investment. Pumping and heat exchangers are considered as part of a functional unit. In-process storage is generally ignored except for raw materials, intermediates, or products. Difficulties are encountered in applying the method due to defining a step. This takes practice and experience. If equipment has been omitted from a step, the resulting estimate is seriously affected. These methods are reported to yield estimates of study quality or at best preliminary quality.

Definitive Estimate Methods Modular methods are an extension of the multiple-factor methods and have been proposed by several authors. One of the most comprehensive methods and one of the earliest was that of K. M. Guthrie, *Chem. Eng.*, **76**:114–142, Mar. 24, 1969. It began with equipment FOB equipment costs, and through the use of factors listed in Table 9-13, the module material cost was obtained. Labor for erection and setting equipment was added to the material cost as well as indirect costs for freight, insurance, engineering, and field expenses to give a total module cost. Such items as contingencies, contractors’ fees, auxiliaries, site development land, and industrial buildings were added if applicable. Since any plant consists of equipment modules, these are summed to give the total fixed capital investment. Unfortunately, the factors and data are old but the concept is useful. Garrett (1989) developed a similar method based upon a variety of equipment modules, starting with purchased equipment costs obtained from plots and applying factors for materials of construction, instrumentation, and plant location. The method provides for all supporting and connecting equipment to make the equipment installation operational. See Table 9-14. T. R. Brown, *Hydrocarbon Processing*, October 2000, pp. 93–100, made modifications to the Garrett method.

Another method, called the *discipline method*, mentioned by L. R. Dysert, *Cost Eng.* **45**(6), June 6, 2003, is similar to the models of

TABLE 9-13 Guthrie Method Factors*

Details	Exchangers			Vessels		Pump and driver	Compressor and driver	Tanks
	Furnaces	Shell and tube	Air-cooled	Vertical	Horizontal			
FOB equipment	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Piping	0.18	0.46	0.18	0.61	0.42	0.30	0.21	
Concrete	0.10	0.05	0.02	0.10	0.06	0.04	0.12	
Steel		0.03		0.08				
Instruments	0.04	0.10	0.05	0.12	0.06	0.03	0.08	
Electrical	0.02	0.02	0.12	0.05	0.05	0.31	0.16	
Insulation		0.05		0.08	0.05	0.03	0.03	
Paint			0.01	0.01	0.01	0.01	0.01	
Total materials = <i>M</i>	1.34	1.71	1.38	2.05	1.65	1.72	1.61	1.20
Erection and setting (<i>L</i>)	0.30	0.63	0.38	0.95	0.59	0.70	0.58	0.13
χ , excluding site preparation and auxiliaries (<i>M</i> + <i>L</i>)	1.64	2.34	1.76	3.00	2.24	2.42	2.19	1.33
Freight, insurance, taxes, engineering, home office, construction		0.08		0.08	0.08	0.08	0.08	0.08
Overhead or field expense	0.60	0.95	0.70	1.12	0.92	0.97	0.97	
Total module factor	2.24	3.37	2.46	4.20	3.24	3.47	3.24	1.41

*From K. M. Guthrie, *Chem. Eng.*, **76**, 114–142 (Mar. 24, 1969). Based on FOB equipment cost = 100 (carbon steel).

TABLE 9-14 Selected Garrett Module Factors

Equipment type (carbon steel unless otherwise noted)	Module factor
Agitators: dual-bladed turbines/single-blade propellers	2.0
Agitated tanks	2.5
Air conditioning	1.46
Blender, ribbon	2.0
Blowers, centrifugal	2.5
Centrifuges: solid-bowl, screen-bowl, pusher, stainless steel	2.0
Columns: distillation, absorption, etc.	
Horizontal	3.05
Vertical	4.16
Compressors: low-, medium-, high-pressure	2.6 avg.
Coolers, quenchers	2.7
Crystallizers	2.6 avg.
Drives/motors	
Electric, for fans, compressors, pumps	1.5
Electric for other units	2.0
Gasoline	2.0
Turbine: gas and steam	3.5
Dryers	
Fluid bed, spray	2.7
Rotary	2.3
Dust collectors	
Bag filters	2.2
Cyclones, multiclones	3.0
Evaporators, single-effect stainless steel	
Falling film	2.3
Forced circulation	2.9
Fans	2.2
Filters	
Belt, rotary drum and leaf, tilting pan	2.4
Others	2.8
Furnaces	2.1
Heat exchangers	
Air-cooled	2.2
Double-pipe	1.8
Shell-and-tube	3.2
Mills	
Hammer	2.8
Ball, rod	2.3 avg.
Pumps	
Centrifugal	5.0
Reciprocating	3.3
Turbine	1.8
Reactors, jacketed, no agitator	
304 SS	1.8
Glass-lined	2.1
Mild steel	2.3
Vacuum equipment	2.2

SOURCE: Adapted from Garrett (1989).

Guthrie and Garrett. It uses equipment factors to generate separate costs for each of the “disciplines” associated with the installation of equipment, such as installation labor, concrete, structural steel, and piping, to obtain direct field costs for each type of equipment, e.g., heat exchangers, towers, and reactors.

Modular methods, depending on the amount of detail provided, will yield preliminary quality estimates.

Detailed Estimate Method For estimates in the detailed category, a code of account needs to be used to prevent oversight of certain significant items in the capital cost. See Table 9-15. Each item in the code is estimated and provides the capital cost estimate; then this estimate serves for cost control during the construction phase of a project.

Comments on Significant Cost Items

Piping This cost includes the cost of the pipe, installation labor, valves, fittings, supports, and miscellaneous items necessary for complete installation of all pipes in the process. The accuracy of the estimates can be seriously in error by the improper application of estimating techniques to this component. Many pipe estimating methods are extant in the literature.

Two general methods have been used to estimate piping costs when detailed flow sheets are not available. One method is to use a percentage of the FOB equipment costs or a percentage of the fixed capital investment. Typical figures are 80 to 100 percent of the FOB equipment costs or 20 to 30 percent of the fixed capital investment. This method is used

TABLE 9-15 Code of Accounts

Category number	Direct capital cost account titles
010	Equipment items
020	Instrument items
030	Setting and testing equipment
040	Setting and testing instruments
050	Piling
060	Excavation
070	Foundations
080	Supports, platforms, and structures
090	Other building items
100	Fire protection and sprinklers
110	Piping
120	Ductwork
130	Electrical and wiring
140	Site preparation
150	Sewers, drains, and plumbing
160	Underground piping
170	Yards, roads, and fencing
180	Railroads
190	Insulation
200	Painting
210	Walls, masonry, roofs, and roofing
220	Spares
230	Lump-sum contracts
Distributives	
500	Site burden
510	Direct labor burden
530	Construction equipment, tools, and supplies
550	Rental and servicing construction equipment and tools
580	Premium wages and overtime—contractor
670	Temporary facilities
740	Cancellation charges
750	Abandoned design
760	Self-insured losses
790	Unvouchered liabilities
800	In-house engineering
810	Outside engineering
870	Undeveloped design allowances
880	Distributives transferred to expense
890	Contingencies—capital items
Expense	
900	Dismantling
910	Sales and use taxes
920	Repairs expense
930	Relocation and modification expense
940	Start-up relocation and modification expense
990	Contingencies

SOURCE: Private communication.

for preliminary estimates. Another group of methods such as the Dickson “N” method (R. A. Dickson, *Chem. Eng.*, **57**:123–135, Nov. 1947), estimating by weight, estimating by cost per joint, etc., requires a detailed piping takeoff from either PID or piping drawings with piping specifications, material costs, labor expenses, etc. These methods are used for definitive or detailed estimates where accuracy of 10 to 15 percent is required. The takeoff methods must be employed with great care and accuracy by an experienced engineer. A detailed breakdown by plant type for process piping costs is presented in Peters et al. (2003) and in *Perry's Chemical Engineers' Handbook*, 6th ed., 1984.

Electrical This item consists of transformers, wiring, switching gear, as well as instrumentation and control wiring. The installed costs of the electrical items may be estimated as 20 to 40 percent of the delivered equipment costs or 5 to 10 percent of the fixed capital investment for preliminary estimates. As with piping estimation, the process design must be well along toward completion before detailed electrical takeoffs can be made.

Buildings and Structures The cost of the erection of buildings and structures in a chemical process plant as well as the plumbing, heating and ventilation, and miscellaneous building service items may be estimated as 20 to 50 percent of delivered equipment costs or as 10 to 20 percent of the fixed capital investment for a preliminary estimate.

Yards, Railroad Sidings, Roads, etc. This investment includes roads, railroad spurs, docks, and fences. A reasonable figure for preliminary estimates is 15 to 20 percent of the FOB equipment cost or 3 to 7 percent of the fixed capital investment for a preliminary estimate.

Service Facilities For a process plant, utility services such as steam, water, electric power, fuel, compressed air, shop facilities, and a cafeteria require capital expenditures. The cost of these facilities lumped together may be 10 to 20 percent of the fixed capital investment for a preliminary estimate. (Note: Buildings, yards, and service facilities must be well defined to obtain a definitive or detailed estimate.)

Environmental Control and Waste Disposal These items are treated as a separate expenditure and are difficult to estimate due to the variety and complexity of the process requirements. Pollution control equipment is generally included as part of the process design. Couper (2003) and Peters et al. (2003) mention that at present there are no general guidelines for estimating these expenditures.

Computerized Cost Estimation With the advent of powerful personal computers (PCs) and software packages, capital cost estimates advanced from large mainframe computers to the PCs. The reasons for using computer cost estimation and economic evaluation packages are time saved on repetitive calculations and reduction in mathematical errors. Numerous computer simulation software packages have been developed over the past two decades. Examples of such software are those produced by ASPEN, ICARUS, CHEMCAD, SUPERPRO, PRO II, HYSYS, etc.; but most do not contain cost estimation software packages. ICARUS developed a PC cost estimation and economic evaluation package called Questimate. This system built a cost estimate from design and equipment cost modules, bulk items, site construction, piping and ductwork, buildings, electrical equipment, instruments, etc., developing worker-hours for engineering and fieldwork costs. This process is similar to quantity takeoff methods to which unit costs are applied. A code of accounts is also provided.

ASPEN acquired ICARUS in 2000 and developed Process Evaluator based on Questimate that is used for conceptual design, known as front-end loading (FEL). More information on FEL and value-improving process (VIP) is found later in Sec. 9. Basic and detailed estimates are coupled with a business decision framework in ASPEN-TECH ICARUS 2000.

EstPro is a process plant cost estimation package for conceptual cost estimation for conceptual design only. It may be obtained from Gulf Publishing, Houston, Tex.

Many companies have developed their own factored estimates using computer spreadsheets based upon their in-house experience and cost database information that they have built from company project history. For detailed estimates, the job is outsourced to design-construction companies that have the staff to perform those estimates.

Whatever package is used, it is recommended that computer-generated costs be spot-checked for reasonable results using a hand-held calculator since errors do occur. Some commercial software companies will develop cost estimation databases in cooperation with a company for site-specific costs.

Contingency This is a provision for unforeseen events that experience has demonstrated are likely to occur. Contingencies are of two types: process and project contingency. In the former, there are uncertainties in

- Equipment and performance
- Integration of old and new process steps
- Scaling up to a large-scale plant size
- Accurate definition of certain process parameters, such as severity of process conditions, number of recycles, process blocks and equipment, multiphase streams, and unusual separations
- No matter how much time and effort are spent preparing estimates, there is a chance of errors occurring due to
 - Engineering errors and omissions
 - Cost and labor rate changes
 - Construction problems
 - Estimating inaccuracies
 - Miscellaneous "unforeseens"
 - Weather-related problems
 - Strikes by fabricators, transportation, and construction personnel

For preliminary estimates, a 15 to 20 percent project contingency should be applied if the process information is firm. As the quality of the estimate moves to definitive and detailed, the contingency value may be lowered to 10 to 15 percent and 5 to 10 percent, respectively. Experience has shown that the smaller the dollar value of the project, the higher the contingency should be.

Offsite Capital These facilities include all structures, equipment, and services that do not enter into the manufacture of a product but are important to the functioning of the plant. Such capital items might be steam-generating and electrical-generating and distribution facilities, well-water cooling tower, and pumping stations for water distribution, etc. Service capital might be auxiliary buildings, such as warehouses, service roads, railroad spurs, material storage, fire protection equipment, and security systems. For estimating purposes, the following percentages of the fixed capital investment might be used:

- Small modification of offsites, 1 to 5 percent
- Restructuring of offsites, 5 to 15 percent
- Major expansion of offsites, 15 to 45 percent
- Grass-roots plants, 45 to 150 percent

Allocated Capital This is capital that is shared due to its proportionate share use in a new facility. Such items include intermediate chemicals, utilities, services and sales, administration, research, and engineering overhead.

Working Capital Working capital is the funds necessary to conduct day-to-day company business. These are funds required to purchase raw materials, supplies, etc. It is continuously liquidated and rejuvenated from the sale of products or services. If an adequate amount of working capital is available, management has the necessary flexibility to cover expenses in case of strikes, delays, fires, etc. Several methods are available for estimating an adequate amount of working capital. They may be broadly classified into percentage and inventory methods. The percentage methods are satisfactory for study and preliminary capital estimates. The percentage methods are of two types: percentage based on capital investment and percentage based upon sales. In the former method, 15 to 25 percent of the total capital investment may be sufficient for preliminary estimates. In the case of certain specialty chemicals where the raw materials are expensive, it is perhaps better to use the percentage of sales method. Such chemicals as flavors, fragrances, perfumes, etc., are in this category. Experience has shown that 15 to 45 percent of sales has been used with 30 to 35 percent being a reasonable average value.

Start-up Expenses Start-up expenses are defined as the total costs directly related to bringing a new manufacturing facility onstream. Start-up time is the time span between the end of construction and the beginning of normal operation. Normal operation is operation at a certain percentage of design capacity or a specified number of days of continuous operation or the ability to make product of a specified purity. Start-up costs are part of the total capital investment and include labor, materials, and overhead for design modifications or changes due to errors on the part of engineering, contractors, costs of tests, final alterations and adjustments. These items cannot be included as contingency because it is known that such work will be necessary before the project is completed. Experience has shown that start-up costs are a percentage of the battery-limits fixed capital investment of the order on average of 3 percent.

Depending on the tax laws in effect, not all start-up costs can be expensed and a portion must be capitalized. Start-up costs can reduce the after-tax earnings during the early years of a project because of a delay in the start-up of production causing a loss of earnings. Construction changes are items of capital cost, and production start-up costs are expensed as an operating expense.

Other Capital Items Paid-up royalties and licenses are considered part of the capital investment since these are replacements for capital to perform process research and development. The initial catalyst and chemical charge, especially for noble metal catalysts and/or in electrolytic processes, is a large amount. These materials are considered to have a life of 1 year. If funds must be borrowed for a new facility, then the interest on borrowed funds during the construction period is capitalized; otherwise, the interest is part of the operating expense.

MANUFACTURING-OPERATING EXPENSES

The estimation of manufacturing expenses has received less attention in the open literature than the estimation of capital requirements. Operating expenses are estimated from proprietary company files. In this section, methods for estimating the elements that constitute operating expenses are presented. Operating expenses consist of the expense of manufacturing a product, packaging and shipping, as well as general overhead expense. These are described later in this section. Figure 9-6 shows an example of a typical manufacturing expense sheet.

RAW MATERIAL EXPENSE

Estimates of the amount of raw material consumed can be obtained from the process material balance. Normally, the raw material expense is the largest expense item in the manufacture of a product. Since yields in a chemical reaction determine the quantity of raw materials consumed, assumed yields may be used to obtain approximate exploratory estimates if possible ranges are given. The prices of the raw materials are published in various trade journals that list material according to form, grade, method of delivery, unit of measure, and cost per unit. The *Chemical Marketing Reporter* is a typical source of these prices. The prices are generally higher than quotations from suppliers, and these latter should be used whenever possible. It may be possible for a company to negotiate the price of a raw material based upon large-quantity use on a long-term basis. With the amount of material used from the material balance and the price of the raw material, the following information can be obtained: annual material consumption, annual material expense, as well as the consumption and expense per unit of product.

Occasionally, by-products may be produced, and if there is a market for these materials, a credit can be given. By-products are treated in the same manner as raw materials and are entered into the manufacturing expense sheet as a credit. If by-products are intermediates for which no market exists, they may be credited to downstream or subsequent operations at a value equivalent to their value as a replacement, or no credit may be obtained.

DIRECT EXPENSES

These are the expenses that are directly associated with the manufacture of a product, e.g., utilities, labor, and maintenance.

Utilities The utility requirements are obtained from the material and energy balances. Utilities include steam, electricity, cooling water, fuel, compressed air, and refrigeration. The current utility prices can be obtained from company plant accounting or from the plant utility supervisor. This person might be able to provide information concerning rate prices for the near future. As requirements increase, the unit cost declines. If large incremental amounts are required, e.g., electricity, it may be necessary to tie the company's utility line to a local utility as a floating source.

With the current energy demands increasing, the unit costs of all utilities are increasing. Any prices quoted need to be reviewed periodically to determine their effect on plant operations. A company utility supervisor is a good source of future price trends. Unfortunately, there are no shortcuts for estimating and projecting utility prices. Utilities are the third largest expense item in the manufacture of a product, behind raw materials and labor.

Operating Labor The most reliable method for estimating labor requirements is to prepare a table of shift, weekend, and vacation coverage. For round-the-clock operation of a continuous process, one operator per shift requires 4.2 operators, if it is assumed that 21 shifts cover the operation and each operator works five, 8-h shifts per week. For batch or semicontinuous operation, it is advisable to prepare a labor table, listing the number of tasks and the number of operators required per task, paying particular attention to primary processing steps such as filtration and distillation that may have several items of equipment per step.

Labor rates may be obtained from the union contract or from a company labor relation supervisor. This person will know the current

labor rates and any potential labor rate increases in the near future. One should not forget shift differential and overtime charges. Once the number of operators per shift has been established, the annual labor expense and unit expense may be estimated. Wessel (*Chem. Eng.*, 59:209-210, July 1952) developed a method for estimating labor requirements for various types of chemical processes in the United States. The equation is applicable for a production rate of 2 to 2000 tons/day (2000 lb/ton).

$$\log Y = -0.783 \log X + 1.252 + B \quad (9-7)$$

where Y = operating labor, operator h/ton per processing step
 X = plant capacity, tons/day
 B = constant depending upon type of process
 + 0.132 (for batch operations that have minimum labor requirements)
 + 0 (for operations with average labor requirements)
 - 0.167 (for a well-instrumented continuous process)

A processing step is one in which a unit operation occurs; e.g., a filtration step might consist of a feed (precoat) tank, pump, filter, and receiver so a processing step may have several items of equipment. By using a flow sheet, the number of processing steps may be counted. The Wessel equation does not take into account changes in labor productivity, but this information can be obtained from each issue of *Chemical Engineering*. Labor productivity varies widely in various sections of this country but even more widely in foreign countries.

Ulrich (1984) developed a table for estimating labor requirements from flow sheets and drawings of the process. Consideration is given to the type and arrangement of equipment, multiplicity of units, and amount of process control equipment. This method is easier to use than the Wessel method and has been updated in a new edition of the original text.

Supervision The approximate expense for supervision of operations depends on process complexity, but 15 to 30 percent of the operating labor expense is reasonable.

Payroll Charges This item includes workers' compensation, social security premiums, unemployment taxes, paid vacations, holidays, and some part of health and dental insurance premiums. The figure has steadily declined from 1980 and now is 30 to 40 percent of operating labor plus supervision expenses.

Maintenance The maintenance expense consists of two components, namely, materials and labor, approximately 60 and 40 percent, respectively. Company records are the best information sources, however, a value of 6 to 10 percent of the fixed capital investment is a reasonable figure. Processes with a large amount of rotating equipment or that operate at extremes of temperature and/or pressure have higher maintenance requirements.

Miscellaneous Direct Expenses These items include operating supplies, clothing and laundry, laboratory expenses, royalties, environmental control expenses, etc.

Item	Basis	Percentage
Operating supplies	Operating labor	5-7
Clothing and laundry	Operating labor	10-15
Laboratory expenses	Operating labor	10-20
Royalties and patents	Sales	1-5

Environmental Control Expense Wastes from manufacturing operations must be disposed of in an environmentally acceptable manner. This direct expense is borne by each manufacturing department. Some companies have their own disposal facilities, or they may contract with a firm that handles the disposal operation. However the wastes are handled, there is an expense. Published data are found in the open literature, some of which have been published by Couper (2003).

TOTAL OPERATING EXPENSES				
=====				
PRODUCT:		PLASTICIZER X		
TOTAL SALES (\$/YR):		7200000		
RATED CAPACITY (MM LBS/YR):		12		
LOCATION:				
FIXED CAPITAL INVESTMENT:		800000		
LAND		25000		
WORKING CAPITAL		120000		
OPERATING HOURS (HRS/YR):				
DATE:				
BY:				

RAW MATERIALS:				
MATERIAL	UNIT	ANNUAL QUANTITY	\$/UNIT	\$/YEAR
A AND B	LB	12000000	.23	2760000
				0
				0
				0
GROSS MATERIAL EXPENSE		2760000		

BY-PRODUCTS:	UNIT	ANNUAL QUANTITY	\$/UNIT	\$/YEAR
				0
				0
BY-PRODUCT CREDIT		0		

NET MATERIAL EXPENSE		2760000		

DIRECT EXPENSES:				
	UNIT	ANNUAL QUANTITY	\$/UNIT	\$/YEAR
UTILITIES:				
steam, low pressure				0
steam, medium pressure				0
steam, high pressure	LB	60000000	.003	180000
GROSS STEAM EXPENSE				180000
STEAM CREDIT				0
NET STEAM EXPENSE		180000		
electricity	KWH	3000000	.035	105000
cooling water	GALLONS	72000000	.000045	3240
fuel gas				0
other:				0
city water	GALLONS	360000000	.0002	72000
TOTAL UTILITIES COST		540240		
LABOR:				
men per shift	4			
annual labor rate per shift	25000			25000
TOTAL LABOR COSTS		100000		
SUPERVISION:				
% total of labor expense				0
SUPERVISION EXPENSE=		18000		
PAYROLL CHARGES, FRINGE BENEFITS:				
% total of labor expense	40			
PAYROLL EXPENSE		47200		
MAINTENANCE				
% of fixed capital investment	8			
MAINTENANCE EXPENSE		64000		
SUPPLIES:				
% of operating labor				0
SUPPLIES EXPENSE		1800		
LABORATORY:				
laboratory hours per year	900			
cost per hour	30			
TOTAL LABORATORY EXPENSE		27000		

FIG. 9-6 Total operating expense sheet.

ROYALTIES		
WASTE DISPOSAL:		0
tons per year		
waste charge per ton		0
WASTE DISPOSAL EXPENSE		0
OTHER:		
laundry	6000	6000
TOTAL DIRECT EXPENSE		804240
TOTAL DIRECT + NET MATERIAL COSTS		3564240
INDIRECT EXPENSES:		
DEPRECIATION		
% of fixed capital investment	100	
life of project (yrs)	7	
DEPRECIATION		114000
PLANT INDIRECT EXPENSES		
% of fixed capital investment	5	
PLANT INDIRECT EXPENSES		40000
TOTAL INDIRECT EXPENSES		154000
TOTAL MANUFACTURING EXPENSE:		3718240
PACKAGING, SHIPPING EXPENSE		
rated capacity per	12000000	
dollars per unit	.005	
PACKAGING AND SHIPPING EXPENSE		60000
TOTAL PRODUCTION EXPENSE		3778240
GENERAL OVERHEAD EXPENSES		
percent of annual sales	5	
GENERAL OVERHEAD EXPENSES		360000
TOTAL OPERATING EXPENSE		4138240

FIG. 9-6 (Continued)

INDIRECT EXPENSES

These indirect expenses consist of two major items; depreciation and plant indirect expenses.

Depreciation The Internal Revenue Service allows a deduction for the "exhaustion, wear and tear and normal obsolescence of equipment used in the trade or business." (This topic is treated more fully later in this section.) Briefly, for manufacturing expense estimates, straight-line depreciation is used, and accelerated methods are employed for cash flow analysis and profitability calculations.

Plant Indirect Expenses These expenses cover a wide range of items such as property taxes, personal and property liability insurance premiums, fire protection, plant safety and security, maintenance of plant roads, yards and docks, plant personnel staff, and cafeteria expenses (if one is available). A quick estimate of these expenses based upon company records is on the order of 2 to 4 percent of the fixed capital investment. Hackney presented a method for estimating these expenses based upon a capital investment factor, and a labor factor, but the result is high.

TOTAL MANUFACTURING EXPENSE

The total manufacturing expense for a product is the sum of the raw materials and direct and indirect expenses.

PACKAGING AND SHIPPING EXPENSES

The packaging expense depends on how the product is sold. The package may vary from small containers to fiberpacks to leverpacks, or the product may be shipped via tank truck, tank car, or pipeline. Each product must be considered and the expense of the container included on a case-by-case basis. The shipping expense includes the in-plant movement to warehousing facilities. Product delivery expenses are difficult to estimate because products are shipped in

various amounts to numerous destinations. Often these expenses come under the heading of freight allowed in the sale of a product.

TOTAL PRODUCT EXPENSE

The sum of the total manufacturing expense and the packaging and in-plant shipping expense is the total product expense.

GENERAL OVERHEAD EXPENSE

This expense is often separated from the manufacturing expenses. It includes the expense of maintaining sales offices throughout the country, staff engineering departments, and research and development facilities and administrative offices. All manufacturing departments are expected to share in these expenses so an appropriate charge is made for each product varying between 6 and 15 percent of the product's annual revenue. The wide range in percentage will vary depending on the amount of customer service required due to the nature of the product.

TOTAL OPERATING EXPENSE

The sum of the total product expense and the general overhead expense is the total operating expense. This item ultimately becomes part of the operating expense on the income statement.

RAPID MANUFACTURING EXPENSE ESTIMATION

Holland et al. (1953) developed an expression for estimating annual manufacturing expenses for production rates other than the base case based upon fixed capital investment, labor requirements, and utility expense.

$$A_1 = mC_{fd} + nc_L N_1 + pU_1 \quad (9-8)$$

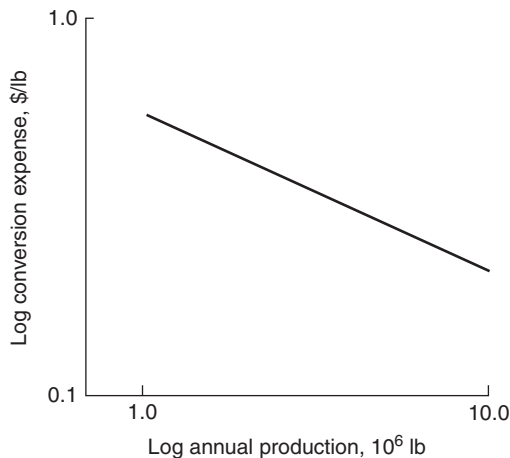


FIG. 9-7 Annual conversion expense as a function of production rate.

where $C_{\text{fc}} =$ fixed capital investment, \$
 $C_L =$ cost of labor, \$ per operator per shift
 $N_1 =$ annual labor requirements, operators/shift/year at rate 1
 $U_1 =$ annual utility expenses at production rate 1
 $A_1 =$ annual conversion expense at rate 1
 $m, n, p =$ constants obtained from company records in consistent units

Equation (9-8) can be modified to include raw materials by adding a term qM_1 , where $q =$ a constant and $M_1 =$ annual raw material expense at rate 1.

SCALE-UP OF MANUFACTURING EXPENSES

If it is desired to estimate the annual manufacturing expense at some rate other than a base case, the following modification may be made:

TABLE 9-16 Typical Labor Requirements for Various Equipment

Equipment	Laborers per unit per shift
Blowers and compressor	0.1–0.2
Centrifuge	0.25–0.50
Crystallizer, mechanical	0.16
Dryers	
Rotary	0.5
Spray	1.0
Tray	0.5
Evaporator	0.25
Filters	
Vacuum	0.125–0.25
Plate and frame	1.0
Rotary and belt	0.1
Heat exchangers	0.1
Process vessels, towers (including auxiliary pumps and exchangers)	0.2–0.5
Reactors	
Batch	1.0
Continuous	0.5

Adapted from G. D. Ulrich, *A Guide to Chemical Engineering Process Design and Economics*, Wiley, New York, 1984.

$$A_2 = mC_{\text{fc}} \left(\frac{R_2}{R_1} \right)^{0.7} + nC_L N_1 \left(\frac{R_2}{R_1} \right)^{0.25} + pU_1 \left(\frac{R_2}{R_1} \right) + qM_1 \left(\frac{R_2}{R_1} \right) \quad (9-9)$$

where $A_2 =$ annual manufacturing expense at production rate 2
 $R_1 =$ production rate 1
 $R_2 =$ production rate 2

Equation (9-9) may also be used to calculate data for a plot of manufacturing expense as a function of annual production rate, as shown in Fig. 9-7. Plots of these data show that the manufacturing expense per unit of production decreases with increasing plant size. The first term in Eq. (9-9) reflects the increase in the capital investment by using the 0.7 power for variations in production rates. Labor varies as the 0.25 power for continuous operations based upon experience. Utilities and raw materials are essentially in direct proportion to the amount of product manufactured, so the exponent of these terms is unity.

FACTORS THAT AFFECT PROFITABILITY

DEPRECIATION

According to the Internal Revenue Service (IRS), *depreciation* is defined as an allowance for the decrease in value of a property over a period of time due to wear and tear, deterioration, and normal obsolescence. The intent is to recover the cost of an asset over a period of time. It begins when a property is placed in a business or trade for the production of income and ends when the asset is retired from service or when the cost of the asset is fully recovered, whichever comes first. Depreciation and taxes are irrevocably tied together. It is essential to be aware of the latest tax law changes because the rules governing depreciation will probably change. Over the past 70 years, there have been many changes in the tax laws of which depreciation is a major component. Couper (2003) discussed the history and development of depreciation accounting. Accelerated depreciation was introduced in the early 1950s to stimulate investment and the economy. It allowed greater depreciation rates in the early years of a project when markets were not well established, manufacturing facilities were coming onstream, and expenses were high due to bringing the facility up to design capacity.

The current methods for determining annual depreciation charges are the *straight-line depreciation* and the *Modified Accelerated Cost Recovery System (MACRS)*. In the straight-line method, the cost of an asset is distributed over its expected useful life such that the annual charge is

$$D = \frac{I + S}{n} \quad \text{\$/year} \quad (9-10)$$

where $D =$ annual depreciation charge
 $I =$ investment
 $n =$ number of years
 $S =$ salvage value

The MACRS went into effect in January 1987 (Couper, 2003) with six asset recovery periods: 3, 5, 7, 10, 15, and 20 years. It is based upon the declining-balance method. The equation for the declining-balance method is

$$V_e = V_i (1 - f) \quad (9-11)$$

where $V_i =$ value of asset at beginning of year
 $V_e =$ value of asset at end of year
 $f =$ declining-balance factor

For 150 percent declining balance $f = 1.5$, and for 200 percent $f = 2.0$. These factors are applied to the previous year's remaining balance. It is evident that the declining-balance method will not recover the asset that the IRS permits. Therefore, a combination of the declining-balance and straight-line methods forms the basis for the MACRS.

Class lives for selected industries are found in Couper (2003), but most chemical processing equipment falls in the 5-year category and petroleum processing equipment in the 7-year category. For those assets with class lives less than 10 years, a 200 percent declining-balance

TABLE 9-17 Depreciation Class Lives and MACRS Recovery Periods

Asset class	Description of asset	Class life, yr	MACRS recovery period, yr
00.12	Information systems	6	5
00.4	Industrial steam and electric generation and/or distribution systems	22	15
13.3	Petroleum refining	16	10
20.3	Manufacture of vegetable oils and vegetable oil products	18	10
20.5	Manufacture of food and beverages	4	3
22.4	Manufacture of textile yarns	8	5
22.5	Manufacture of nonwoven fabrics	10	7
26.1	Manufacture of pulp and paper	13	7
28.0	Manufacture of chemicals and allied products	9.5	5
30.1	Manufacture of rubber products	14	7
30.2	Manufacture of finished plastic products	11	7
32.1	Manufacture of glass products	14	7
32.2	Manufacture of cement	20	15
32.3	Manufacture of other stone and clay products	15	7
33.2	Manufacture of primary nonferrous metals	14	7
32.4	Manufacture of primary steel mill products	15	7
49.223	Substitute natural gas-coal gasification	18	10
49.25	Liquefied natural gas plant	22	15

SOURCE: "How to Depreciate Property," Publication 946, Internal Revenue Service, U.S. Department of Treasury, Washington, 1999.

method with a switch to straight-line in the later years is used. The IRS adopted a half-year convention for both depreciation methods. Under this convention, a property placed in service is considered to be only one-half year irrespective of when during the year the property was placed in service. Table 9-17 is a listing of the class lives, and Table 9-18 contains factors with the half-year convention for both the MACRS and straight-line methods.

Depreciation is entered as an indirect expense on the manufacturing expense sheet based upon the straight-line method. However, when one is determining the after-tax cash flow, straight-line depreciation is removed from the manufacturing expense and the MACRS depreciation is entered. This is illustrated under the section on cash flow.

There are certain terms that apply to depreciation:

- *Depreciation reserve* is the accumulated depreciation at a specific time.
- *Book value* is the original investment minus the accumulated depreciation.
- *Service life* is the time period during which an asset is in service and is economically feasible.
- *Salvage value* is the net amount of money obtained from the sale of a used property over and above any charges involved in the removal and sale of the property.
- *Scrap value* implies that the asset has no further useful life and is sold for the amount of scrap material in it.
- *Economic life* is the most likely period of successful operation before a need arises for subsequent investment in additional equipment as the result of product or process obsolescence or equipment due to wear and tear.

DEPLETION

Depletion is concerned with the diminution of natural resources. Generally depletion does not enter into process economic studies. Rules for determining the amount of depletion are found in the IRS Publication 535.

AMORTIZATION

Amortization is the ratable deduction for the cost of an intangible property over its useful life, perhaps a 15-year life, via straight-line calculations. An example of an intangible property is a franchise, patent, trademark, etc. Two IRS publications, Form 4562 and Publication 535 (1999), established the regulations regarding amortization.

TAXES

Corporations pay an income tax based upon gross earnings, as shown in Table 9-19. Most major corporations pay the federal tax rate of 34 percent on their annual gross earnings. In addition, some states have a stepwise corporate income tax rate. State income tax is deductible as an expense item before the calculation of the federal tax. If T_s is the incremental tax rate and T_f is the incremental federal tax, both expressed as decimals, then the combined incremental rate T_c is

TABLE 9-18 Depreciation Rates for Straight-Line and MACRS Methods

Straight-line half-year convention							MACRS ^a half-year convention						
Year	3	5	7	10	15	20	Year	3	5	7	10	15	20
1	16.67%	10.00%	7.14%	5.0%	3.33%	2.5%	1	33.33%	20.00%	14.29%	10.00%	5.00%	3.750%
2	33.33	20.00	14.29	10.0	6.67	5.0	2	44.45	32.00	24.49	18.00	9.50	7.219
3	33.33	20.00	14.29	10.0	6.67	5.0	3	14.81	19.20	17.49	14.40	8.55	6.677
4	16.67	20.00	14.28	10.0	6.67	5.0	4	7.41	11.52	12.49	11.52	7.70	6.177
5		20.00	14.29	10.0	6.67	5.0	5		11.52	8.93	9.22	6.93	5.713
6		10.00	14.28	10.0	6.67	5.0	6		5.76	8.92	7.37	6.23	5.285
7			14.29	10.0	6.67	5.0	7			8.93	6.55	5.90	4.888
8			7.14	10.0	6.66	5.0	8			4.46	6.55	5.90	4.522
9				10.0	6.67	5.0	9				6.56	5.91	4.462
10				10.0	6.66	5.0	10				6.55	5.90	4.461
11				5.0	6.67	5.0	11				3.28	5.91	4.462
12					6.66	5.0	12					5.90	4.461
13					6.67	5.0	13					5.91	4.462
14					6.66	5.0	14					5.90	4.461
15					6.67	5.0	15					5.91	4.462
16					3.33	5.0	16					2.95	4.461
17						5.0	17						4.462
18						5.0	18						4.461
19						5.0	19						4.462
20						5.0	20						4.461
21						2.5	21						2.231

^aGeneral depreciation system. Declining-balance switching to straight-line. Recovery periods 3, 5, 7, 10, 15, and 20 years.

SOURCE: "How to Depreciate Property," Publication 946, Internal Revenue Service, U.S. Department of Treasury, Washington, 1999.